index.html

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<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
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===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/about_system_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/about_system_tab/index.
html" class="w-100 h-100 border-0"></iframe>

 * File: index.html
 * Project: ai-windows-admin
 * Package: windows/api/webgui/about_system_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-12 06:50:00
 * ================================================================
==================
-->
<!-- About System Application Tab -->
<div class="about-sys-wrapper">
<!-- History Telemetry Modal Dialog -->
<div aria-hidden="true" aria-labelledby="aboutSystemHistoryMod
alLabel" class="modal fade" id="aboutSystemHistoryModal" tabin
dex="-1">
<div class="modal-dialog modal-dialog-centered modal-dialog-sc
rollable modal-xl" style="max-width: 95vw; width: 95vw; height
: 90vh; margin: 2vh auto;">
<div class="modal-content border border-info border-opacity-40
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 shadow-lg" style="height: 100%;">
<div class="modal-header border-secondary border-opacity-25 pb
-2 px-3">
<div class="d-flex align-items-center justify-content-between
w-100 flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<span class="fs-5">🛰 </span>
<div>
<h6 class="modal-title fw-bold text-white mb-0" id="aboutSyste
mHistoryModalLabel">
                  История телеметрии системы (telemetry.db)
                </h6>
<div class="small text-muted" style="font-size: 0.72rem;">Хрон
ология предыдущих снимков производительности и состояния хоста
</div>
</div>
</div>
<div class="d-flex align-items-center gap-2">
<div class="d-flex align-items-center gap-1">
<span class="small text-muted font-monospace" style="font-size
: 0.72rem;">Лимит:</span>
<select class="form-select form-select-sm bg-dark text-light b
order-secondary" id="about-history-limit-select" style="width:
 75px; font-size: 0.75rem; height: 28px;">
<option value="15">15</option>
<option selected="" value="30">30</option>
<option value="50">50</option>
<option value="100">100</option>
</select>
</div>
<button class="btn btn-sm btn-outline-info rounded-pill px-2.5
 py-1" id="btn-about-history-refresh" title="Обновить историю
из БД">
<i class="bi bi-arrow-clockwise me-1"></i> Обновить
              </button>
<button aria-label="Закрыть" class="btn-close btn-close-white"
 data-bs-dismiss="modal" type="button"></button>
</div>
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</div>
</div>
<div class="modal-body p-3" style="overflow-y: auto;">
<!-- Filter Buttons -->
<div class="d-flex align-items-center gap-1.5 flex-wrap mb-3" id="about-history-filter-group">
<button class="btn btn-sm btn-info text-dark fw-bold rounded-pill px-2.5 py-0.5 btn-history-filter active" data-metric="all">Все метрики</button>
<button class="btn btn-sm btn-outline-secondary rounded-pill px-2.5 py-0.5 btn-history-filter" data-metric="cpu">CPU</button>
<button class="btn btn-sm btn-outline-secondary rounded-pill px-2.5 py-0.5 btn-history-filter" data-metric="memory">Память (RAM)</button>
<button class="btn btn-sm btn-outline-secondary rounded-pill px-2.5 py-0.5 btn-history-filter" data-metric="gpu">GPU</button>
<button class="btn btn-sm btn-outline-secondary rounded-pill px-2.5 py-0.5 btn-history-filter" data-metric="storage">Диск C:</button>
<button class="btn btn-sm btn-outline-secondary rounded-pill px-2.5 py-0.5 btn-history-filter" data-metric="disk_io">Диск I/O</button>
<button class="btn btn-sm btn-outline-secondary rounded-pill px-2.5 py-0.5 btn-history-filter" data-metric="os">ОС &amp;amp; Хост</button>
</div>
<!-- History Table Container -->
<div class="table-responsive rounded border border-secondary border-opacity-30">
<table class="table table-dark table-hover table-sm mb-0 align-middle" style="font-size: 0.78rem; font-family: ui-monospace, SFMono-Regular, Menlo, Monaco, Consolas, monospace;">
<thead class="table-dark text-secondary border-secondary">
<tr id="about-history-thead-tr">
<th style="width: 50px;">#ID</th>
<th style="min-width: 140px;">Время</th>
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<th>Хост / ОС</th>
<th style="text-align: right;">CPU % / МГц</th>
<th style="text-align: right;">RAM (Использовано / Всего)</th>
<th style="text-align: right;">GPU % / °C</th>
<th style="text-align: right;">Диск C: Свободно</th>
<th style="text-align: right;">Диск I/O (MB/s)</th>
<th style="text-align: right;">Аптайм</th>
</tr>
</thead>
<tbody id="about-history-tbody">
<tr>
<td class="text-center py-4 text-muted" colspan="9">
<div class="spinner-border spinner-border-sm text-info mb-1" r
ole="status"></div>
<div>Загрузка истории из базы данных...</div>
</td>
</tr>
</tbody>
</table>
</div>
<div class="mt-2.5 small text-muted d-flex justify-content-bet
ween align-items-center flex-wrap gap-2">
<span id="about-history-count-badge">Записей в базе: 0</span>
<span class="font-monospace" id="about-history-db-meta" style=
"font-size: 0.70rem;">Источник: telemetry.db</span>
</div>
</div>
<div class="modal-footer border-secondary border-opacity-25 py
-2 px-3 d-flex justify-content-between align-items-center">
<div class="small text-muted" style="font-size: 0.72rem;">
<i class="bi bi-database text-info me-1"></i>Таблица: system_s
napshots
          </div>
<button class="btn btn-sm btn-outline-secondary rounded-pill p
x-3" data-bs-dismiss="modal" type="button">
            Закрыть
          </button>
</div>
```

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</div>
</div>
</div>
<!-- AI Copilot Diagnostics Modal Dialog -->
<div aria-hidden="true" aria-labelledby="aboutAiDiagnosticsModalLabel" class="modal fade" id="aboutAiDiagnosticsModal" tabindex="-1">
<div class="modal-dialog modal-dialog-centered modal-dialog-scrollable modal-fullscreen-xl-down" style="max-width: 95vw; width: 95vw; height: 92vh; margin: 1.5vh auto;">
<div class="modal-content border border-info border-opacity-40 shadow-lg" style="height: 100%;">
<div class="modal-header border-secondary border-opacity-25 pb-2 px-3">
<div class="d-flex align-items-center justify-content-between w-100 flex-wrap gap-2">
<div class="d-flex align-items-center gap-2.5">
<div class="about-ai-pulse-icon">
<i class="bi bi-robot text-info fs-5"></i>
</div>
<div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<h6 class="modal-title fw-bold text-white mb-0" id="aboutAiDiagnosticsModalLabel" style="font-size: 0.95rem;">
                    AI Copilot: Анализ производительности и узких мест
                  </h6>
<span class="badge bg-success-subtle text-success border border-success font-monospace" id="about-ai-health-badge">Health: 100/100</span>
</div>
<div class="small text-muted" style="font-size: 0.72rem;">Интеллектуальный диагностический аудит хоста и оборудования</div>
</div>
</div>
<div class="d-flex align-items-center gap-2">
<span class="badge bg-dark text-info border border-info border-opacity-30 px-2.5 py-1 font-monospace" id="about-ai-engine-na
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me" style="font-size: 0.72rem;">
<i class="bi bi-cpu me-1"></i>Unified AI / Heuristic
              </span>
<button class="btn btn-sm btn-info text-dark fw-bold rounded-p
ill px-3 py-1 shadow-sm d-flex align-items-center gap-1.5" id=
"btn-about-ai-rescan" title="Выполнить глубокий пересканирующи
й аудит с вызовом AI">
<i class="bi bi-lightning-charge-fill" id="ic-about-ai-rescan"
></i>
<span id="txt-about-ai-rescan">Rescan</span>
</button>
<button aria-label="Закрыть" class="btn-close btn-close-white
ms-1" data-bs-dismiss="modal" type="button"></button>
</div>
</div>
</div>
<div class="modal-body p-3" id="about-ai-copilot-card" style="
overflow-y: auto; max-height: 80vh;">
<!-- Live Polling & Active Telemetry Feed Banner (Shows what i
s being collected right now) -->
<div class="p-2.5 mb-2.5 rounded border border-info border-opa
city-30 bg-black bg-opacity-40 d-flex flex-column gap-2" id="a
bout-ai-live-feed-banner">
<div class="d-flex align-items-center justify-content-between
flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<span class="spinner-grow spinner-grow-sm text-info" id="about
-ai-live-pulse" role="status" style="width: 0.75rem; height: 0
.75rem;"></span>
<strong class="text-info small" style="font-size: 0.80rem;">Te
кущий сбор сенсоров:</strong>
<span class="text-light small font-monospace" id="about-ai-liv
e-status-text">Опрос датчиков и системных интерфейсов хоста...
</span>
</div>
<div class="d-flex align-items-center gap-1.5 flex-wrap font-m
onospace" style="font-size: 0.70rem;">
<span class="badge bg-secondary bg-opacity-40 text-light borde
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r border-secondary" id="chip-sensor-lhm">  LHM Sensors</span>
<span class="badge bg-secondary bg-opacity-40 text-light borde
r border-secondary" id="chip-sensor-wmi">  WMI Cores</span>
<span class="badge bg-secondary bg-opacity-40 text-light borde
r border-secondary" id="chip-sensor-ram">  RAM &amp;amp; Procs</s
pan>
<span class="badge bg-secondary bg-opacity-40 text-light borde
r border-secondary" id="chip-sensor-disks">  Disks &amp;amp; SMAR
T</span>
<span class="badge bg-secondary bg-opacity-40 text-light borde
r border-secondary" id="chip-sensor-net">  Network &amp;amp; OS</
span>
</div>
</div>
<!-- Live metrics snapshot ticker -->
<div class="small text-secondary font-monospace d-flex align-i
tems-center gap-3 flex-wrap border-top border-secondary border
-opacity-20 pt-1.5" id="about-ai-live-metrics-bar" style="font
-size: 0.72rem;">
<span><strong class="text-muted">CPU:</strong> <span class="te
xt-info fw-semibold" id="live-tick-cpu">--</span></span>
<span><strong class="text-muted">Max Temp:</strong> <span clas
s="text-warning fw-semibold" id="live-tick-temp">--</span></sp
an>
<span><strong class="text-muted">RAM:</strong> <span class="te
xt-success fw-semibold" id="live-tick-ram">--</span></span>
<span><strong class="text-muted">Disks:</strong> <span class="
text-light fw-semibold" id="live-tick-disks">--</span></span>
<span><strong class="text-muted">Top Process:</strong> <span c
lass="text-danger fw-semibold" id="live-tick-top-proc">--</spa
n></span>
</div>
</div>
<!-- Execution Stages / Steps Inspector -->
<div class="mb-2.5" id="about-ai-stages-section">
<div class="d-flex align-items-center justify-content-between
text-muted small mb-1.5" style="font-size: 0.72rem;">
<span class="text-uppercase tracking-wider fw-semibold text-in
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fo"><i class="bi bi-activity me-1"></i>Цепочка пошаговой AI-диагностики</span>
<button class="btn btn-link btn-sm p-0 text-info text-decoration-none" id="btn-about-ai-toggle-stages" style="font-size: 0.70rem;" type="button">
<span id="txt-about-ai-toggle-stages">Свернуть этапы</span> <i class="bi bi-chevron-up ms-0.5" id="ic-about-ai-toggle-stages"></i>
</button>
</div>
<!-- Stepwise Progress Bar -->
<div class="progress mb-2" style="height: 4px; background-color: rgba(255,255,255,0.08);">
<div class="progress-bar bg-info progress-bar-striped progress-bar-animated" id="about-ai-step-progress" role="progressbar" style="width: 0%;"></div>
</div>
<div class="about-ai-stages-box p-2 rounded border border-secondary border-opacity-25" id="about-ai-stages-container">
<div class="small text-muted text-center py-2" id="about-ai-stages-empty">Ожидание запуска пошагового аудита...</div>
</div>
</div>
<!-- 4 Functional Domain Cards Grid -->
<div class="mt-2.5" id="about-ai-domain-cards-section">
<div class="row g-2" id="about-ai-groups-container"></div>
</div>
<!-- Anomalies Pills Container -->
<div class="mt-2 d-flex flex-wrap gap-1.5" id="about-ai-anomalies-container"></div>
<!-- Error Message Alert (if present) -->
<div class="mt-2 p-2 rounded bg-danger bg-opacity-10 border border-danger border-opacity-40 d-none" id="about-ai-error-section">
<div class="d-flex align-items-center gap-2 text-danger small">
<i class="bi bi-exclamation-triangle-fill"></i>
<strong>Ошибка выполнения AI-диагностики:</strong>
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</div>
<div class="small text-light font-monospace mt-1 ps-4" id="about-ai-error-text" style="font-size: 0.72rem; word-break: break-word;"></div>
</div>
<!-- AI Final Synthesis / Executive Summary Output -->
<div class="mt-2.5 p-3 rounded bg-black bg-opacity-50 border border-info border-opacity-30 shadow-sm" id="about-ai-synthesis-card">
<div class="d-flex align-items-center justify-content-between mb-1.5 pb-1 border-bottom border-secondary border-opacity-25">
<span class="small text-info fw-bold font-monospace text-uppercase" style="font-size: 0.75rem;">
<i class="bi bi-award-fill me-1"></i>Итоговый синтез и вердикт инженера
                </span>
<span class="badge bg-secondary-subtle text-light font-monospace" id="about-ai-synthesis-status" style="font-size: 0.68rem;">Готов к запуску</span>
</div>
<div class="small text-light" id="about-ai-summary-text" style="line-height: 1.55; white-space: pre-line;">
                Нажмите «Rescan» для запуска пошаговой AI-диагностики сенсоров по 4 доменам (CPU/GPU, Память, Диски, Сеть).
            </div>
<div class="mt-2 pt-2 border-top border-secondary border-opacity-20 d-none" id="about-ai-actions-box">
<div class="text-warning small fw-semibold mb-1" style="font-size: 0.72rem;">
<i class="bi bi-exclamation-circle-fill me-1"></i>Приоритетные действия для администратора:
                </div>
<ul class="list-unstyled mb-0 small text-light ps-2" id="about-ai-actions-list" style="font-size: 0.73rem;"></ul>
</div>
</div>
<!-- Collapsible Live Telemetry Payload Inspector (What was collected) -->
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<div class="mt-2" id="about-ai-snapshot-section">
<div class="d-flex align-items-center justify-content-between flex-wrap gap-1">
<button class="btn btn-sm btn-link p-0 text-info text-decoration-none d-flex align-items-center gap-1.5 font-monospace" id="btn-about-ai-toggle-snapshot" style="font-size: 0.74rem;" type="button">
<i class="bi bi-cpu-fill text-info"></i>
<span><strong>Срез собираемых сенсоров и сырых данных (Live Telemetry Snapshot)</strong></span>
<i class="bi bi-chevron-down text-muted ms-1" id="ic-about-ai-snapshot"></i>
</button>
<button class="btn btn-dark btn-sm py-0 px-2 text-secondary border border-secondary-subtle d-flex align-items-center gap-1" id="btn-about-ai-copy-snapshot" style="font-size: 0.68rem;" title="Скопировать JSON данных" type="button">
<i class="bi bi-clipboard"></i>
<span id="txt-about-ai-copy-snapshot">Копировать JSON</span>
</button>
</div>
<div class="d-none mt-1.5 p-2 bg-black bg-opacity-75 border border-info border-opacity-30 rounded" id="about-ai-snapshot-collapse">
<pre class="m-0 p-1.5 text-info font-monospace" id="about-ai-snapshot-text" style="font-size: 0.70rem; line-height: 1.35; max-height: 250px; overflow-y: auto; white-space: pre-wrap; word-break: break-word;">Ожидание сбора телеметрии...</pre>
</div>
</div>
<!-- Collapsible System Instruction Inspector -->
<div class="mt-2" id="about-ai-instruction-section">
<div class="d-flex align-items-center justify-content-between flex-wrap gap-1">
<button class="btn btn-sm btn-link p-0 text-info text-decoration-none d-flex align-items-center gap-1.5 font-monospace" id="btn-about-ai-toggle-instruction" style="font-size: 0.74rem;" type="button">
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<i class="bi bi-shield-lock-fill text-info"></i>
<span><strong>Системная инструкция модели</strong></span>
<i class="bi bi-chevron-down text-muted ms-1" id="ic-about-ai-instruction"></i>
</button>
<button class="btn btn-dark btn-sm py-0 px-2 text-secondary border border-secondary-subtle d-flex align-items-center gap-1" id="btn-about-ai-copy-instruction" style="font-size: 0.68rem;" title="Скопировать системную инструкцию" type="button">
<i class="bi bi-clipboard"></i>
<span id="txt-about-ai-copy-instruction">Копировать</span>
</button>
</div>
<div class="d-none mt-1.5 p-2 bg-black bg-opacity-75 border border-info border-opacity-30 rounded" id="about-ai-instruction-collapse">
<pre class="m-0 p-1.5 text-info font-monospace" id="about-ai-instruction-text" style="font-size: 0.70rem; line-height: 1.35; max-height: 200px; overflow-y: auto; white-space: pre-wrap; word-break: break-word;">Ожидание запуска...</pre>
</div>
</div>
<!-- Collapsible Sent Prompt Inspector -->
<div class="mt-2" id="about-ai-prompt-section">
<div class="d-flex align-items-center justify-content-between flex-wrap gap-1">
<button class="btn btn-sm btn-link p-0 text-warning text-decoration-none d-flex align-items-center gap-1.5 font-monospace" id="btn-about-ai-toggle-prompt" style="font-size: 0.74rem;" type="button">
<i class="bi bi-terminal-fill text-warning"></i>
<span><strong>Сформированный мизерный JSON-промпт модели</strong></span>
<i class="bi bi-chevron-down text-muted ms-1" id="ic-about-ai-prompt"></i>
</button>
<button class="btn btn-dark btn-sm py-0 px-2 text-secondary border border-secondary-subtle d-flex align-items-center gap-1"
```

```
id="btn-about-ai-copy-prompt" style="font-size: 0.68rem;" titl
e="Скопировать отправленный промпт" type="button">
<i class="bi bi-clipboard"></i>
<span id="txt-about-ai-copy-prompt">Копировать JSON</span>
</button>
</div>
<div class="d-none mt-1.5 p-2 bg-black bg-opacity-75 border bo
rder-warning border-opacity-30 rounded" id="about-ai-prompt-co
llapse">
<pre class="m-0 p-1.5 text-warning font-monospace" id="about-a
i-prompt-text" style="font-size: 0.70rem; line-height: 1.35; m
ax-height: 220px; overflow-y: auto; white-space: pre-wrap; wor
d-break: break-word;">Ожидание запуска...</pre>
</div>
</div>
<!-- Collapsible Raw Response / Diagnostic Data Inspector -->
<div class="mt-2" id="about-ai-raw-section">
<div class="d-flex align-items-center justify-content-between
flex-wrap gap-1">
<button class="btn btn-sm btn-link p-0 text-info text-decorati
on-none d-flex align-items-center gap-1.5 font-monospace" id="
btn-about-ai-toggle-raw" style="font-size: 0.74rem;" type="but
ton">
<i class="bi bi-code-slash text-info"></i>
<span><strong>Сырой ответ модели</strong></span>
<i class="bi bi-chevron-down text-muted ms-1" id="ic-about-ai-
raw"></i>
</button>
<button class="btn btn-dark btn-sm py-0 px-2 text-secondary bo
rder border-secondary-subtle d-flex align-items-center gap-1"
id="btn-about-ai-copy-raw" style="font-size: 0.68rem;" title="
Скопировать ответ" type="button">
<i class="bi bi-clipboard"></i>
<span id="txt-about-ai-copy-raw">Копировать</span>
</button>
</div>
<div class="d-none mt-1.5 p-2 bg-black bg-opacity-75 border bo
rder-info border-opacity-30 rounded" id="about-ai-raw-collapse
```

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">
<pre class="m-0 p-1.5 text-light font-monospace" id="about-ai-raw-text" style="font-size: 0.70rem; line-height: 1.35; max-height: 220px; overflow-y: auto; white-space: pre-wrap; word-break: break-word;">Ожидание ответа модели...</pre>
</div>
</div>
</div>
<div class="modal-footer border-secondary border-opacity-25 py-2 px-3 d-flex justify-content-between align-items-center">
<div class="small text-muted" style="font-size: 0.72rem;">
<i class="bi bi-shield-check text-info me-1"></i>System Diagnostics &amp;amp; AI Sensors Audit
          </div>
<button class="btn btn-sm btn-outline-secondary rounded-pill px-3" data-bs-dismiss="modal" type="button">
            Закрыть
          </button>
</div>
</div>
</div>
</div>
<!-- Regional, Timezone & User Account Settings Modal Dialog -->
<div aria-hidden="true" aria-labelledby="aboutRegionalSettingsModalLabel" class="modal fade" id="aboutRegionalSettingsModal" tabindex="-1">
<div class="modal-dialog modal-dialog-centered modal-lg">
<div class="modal-content" style="background: var(--surface-1); color: var(--text-color); border: 1px solid var(--border-color);">
<div class="modal-header" style="background: var(--surface-2); border-bottom: 1px solid var(--border-color);">
<div class="d-flex align-items-center gap-2">
<span class="fs-5">□ </span>
<div>
<h6 class="modal-title fw-bold mb-0" id="aboutRegionalSettingsModalLabel">
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                Параметры региона, времени и учетной записи
              </h6>
<div class="small text-muted" style="font-size: 0.72rem;">Настройка часового пояса, системной локали и профиля пользователя Windows</div>
</div>
</div>
<button aria-label="Закрыть" class="btn-close" data-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body p-3">
<!-- Subtabs nav -->
<ul class="nav nav-pills gap-1 mb-3" id="regional-modal-pills">
<li class="nav-item">
<button class="nav-link active py-1 px-3 small slc-subtab-btn btn-reg-nav" data-tab-target="pane-tz" id="btn-tab-tz">🕐 Часовой пояс</button>
</li>
<li class="nav-item">
<button class="nav-link py-1 px-3 small slc-subtab-btn btn-reg-nav" data-tab-target="pane-loc" id="btn-tab-loc">🌐 Язык и локали</button>
</li>
<li class="nav-item">
<button class="nav-link py-1 px-3 small slc-subtab-btn btn-reg-nav" data-tab-target="pane-usr" id="btn-tab-usr">👤 Учетная запись</button>
</li>
</ul>
<!-- Pane 1: Timezone -->
<div class="regional-tab-pane" id="pane-tz">
<div class="p-2.5 mb-2 rounded border" style="background: var(--surface-2); border-color: var(--border-color) !important;">
<div class="d-flex justify-content-between align-items-center mb-2">
<span class="small text-muted">Текущий часовой пояс:</span>
<span class="badge bg-info text-dark font-monospace" id="reg-c
```

```
ur-tz-badge">Загрузка...</span>
</div>
<div class="mb-2">
<label class="form-label small fw-bold mb-1">Поиск и выбор часового пояса:</label>
<input class="form-control form-control-sm mb-2" id="reg-tz-search-input" placeholder="Фильтр поясов (например: Jerusalem, Moscow, UTC, Rome)..." type="text"/>
<select class="form-select form-select-sm font-monospace" id="reg-tz-select" size="6" style="height: 140px; font-size: 0.8rem;">
<option value="">Загрузка списка часовых поясов...</option>
</select>
</div>
<div class="d-flex justify-content-between align-items-center mt-2 pt-2 border-top" style="border-color: var(--border-color) !important;">
<span class="small text-muted font-monospace" id="reg-tz-selected-label">Выберите пояс из списка</span>
<button class="btn btn-sm btn-info text-dark fw-bold px-3 rounded-pill" id="btn-reg-save-tz">
<i class="bi bi-check-lg me-1"></i> Применить пояс
                </button>
</div>
</div>
</div>
<!-- Pane 2: Locales -->
<div class="regional-tab-pane d-none" id="pane-loc">
<div class="p-2.5 mb-2 rounded border" style="background: var(--surface-2); border-color: var(--border-color) !important;">
<div class="row g-3 mb-2">
<div class="col-sm-6">
<label class="form-label small fw-bold mb-1">Системная локаль (System Locale):</label>
<select class="form-select form-select-sm" id="reg-loc-sys-select">
<option value="ru-RU">Русский (Россия) [ru-RU]</option>
<option value="en-US">English (United States) [en-US]</option>
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<option value="en-GB">English (United Kingdom) [en-GB]</option>
<option value="he-IL">□ □ □ □ □ (□ □ □ □ □) [he-IL]</option>
<option value="de-DE">Deutsch (Deutschland) [de-DE]</option>
<option value="fr-FR">Français (France) [fr-FR]</option>
<option value="es-ES">Español (España) [es-ES]</option>
<option value="it-IT">Italiano (Italia) [it-IT]</option>
<option value="zh-CN">□ □ (□ □, □ □) [zh-CN]</option>
<option value="ja-JP">□ □ □ (□ □) [ja-JP]</option>
</select>
<div class="small text-muted mt-1" style="font-size: 0.72rem;">Для программ без поддержки Юникода и форматов дат/чисел.</div>
</div>
<div class="col-sm-6">
<label class="form-label small fw-bold mb-1">Пользовательская культура (User Culture):</label>
<select class="form-select form-select-sm" id="reg-loc-usr-select">
<option value="ru-RU">Русский (Россия) [ru-RU]</option>
<option value="en-US">English (United States) [en-US]</option>
<option value="en-GB">English (United Kingdom) [en-GB]</option>
<option value="he-IL">□ □ □ □ □ (□ □ □ □ □) [he-IL]</option>
<option value="de-DE">Deutsch (Deutschland) [de-DE]</option>
</select>
<div class="small text-muted mt-1" style="font-size: 0.72rem;">Язык форматирования времени, валюты и пользовательских параметров.</div>
</div>
</div>
<div class="d-flex justify-content-between align-items-center mt-3 pt-2 border-top" style="border-color: var(--border-color) !important;">
<span class="small text-warning"><i class="bi bi-info-circle me-1"></i>Вступает в силу после перезагрузки</span>
<button class="btn btn-sm btn-info text-dark fw-bold px-3 rounded-pill" id="btn-reg-save-loc">
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<i class="bi bi-check-lg me-1"></i> Применить локаль
                </button>
</div>
</div>
</div>
<!-- Pane 3: User Account -->
<div class="regional-tab-pane d-none" id="pane-usr">
<div class="p-2.5 mb-2 rounded border" style="background: var(
--surface-2); border-color: var(--border-color) !important;">
<div class="mb-2">
<label class="form-label small fw-bold mb-1">Имя учетной запис
и (Логин):</label>
<input class="form-control form-control-sm font-monospace" id=
"reg-usr-name-input" readonly="" style="opacity: 0.85;" type="
text"/>
</div>
<div class="mb-2">
<label class="form-label small fw-bold mb-1">Полное имя (Full
Name):</label>
<input class="form-control form-control-sm" id="reg-usr-fullna
me-input" placeholder="Например: Главный администратор хоста"
type="text"/>
</div>
<div class="mb-2">
<label class="form-label small fw-bold mb-1">Описание учетной
записи (Description):</label>
<input class="form-control form-control-sm" id="reg-usr-desc-i
nput" placeholder="Например: Основная рабочая учетная запись"
type="text"/>
</div>
<div class="d-flex justify-content-between align-items-center
mt-3 pt-2 border-top" style="border-color: var(--border-color)
 !important;">
<span class="small text-muted">Локальный пользователь Windows<
/span>
<button class="btn btn-sm btn-info text-dark fw-bold px-3 roun
ded-pill" id="btn-reg-save-usr">
<i class="bi bi-check-lg me-1"></i> Сохранить профиль
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                </button>
</div>
</div>
</div>
<!-- Result Alert -->
<div class="alert d-none py-2 px-3 small mb-0" id="reg-modal-a
lert"></div>
</div>
<div class="modal-footer" style="background: var(--surface-2);
 border-top: 1px solid var(--border-color);">
<div class="small text-muted" style="font-size: 0.72rem;">
<i class="bi bi-shield-check text-info me-1"></i>System Region
al &amp;amp; Environment Configuration
          </div>
<button class="btn btn-sm btn-secondary rounded-pill px-3" dat
a-bs-dismiss="modal" type="button">
            Закрыть
          </button>
</div>
</div>
</div>
</div>
<!-- Top 4 Summary KPI Cards (Vital Overview) -->
<div class="row g-2 mb-3">
<!-- Card 1: Operating System & Host -->
<div class="col-12 col-sm-6 col-lg-3">
<div class="ui-kpi-card" data-panel-id="panel-about-platform-o
s" id="panel-about-platform-os">
<div class="ui-kpi-label d-flex justify-content-between align-
items-center">
<div class="d-flex align-items-center gap-1.5">
<i class="bi bi-windows text-info"></i>
<span>Платформа &amp;amp; ОС</span>
</div>
<button class="btn btn-sm btn-link p-0 text-muted text-decorat
ion-none btn-card-history" data-bs-target="#aboutSystemHistory
Modal" data-bs-toggle="modal" data-metric="os" title="История
ОС / Host из базы данных">
```

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<i class="bi bi-clock-history"></i>
</button>
</div>
<div class="ui-kpi-value text-info text-truncate mt-1" id="abo
ut-kpi-os-title">Windows 11 (AMD64)</div>
<div class="ui-kpi-sub text-truncate d-flex align-items-center
 justify-content-between" id="about-host-view-box">
<div class="d-flex align-items-center text-truncate">
<span id="about-kpi-os-host">Host: DELL-VOSTRO</span>
<button class="btn btn-sm btn-link p-0 text-info text-decorati
on-none about-param-edit-btn" id="btn-host-inline-edit" title=
"Изменить имя компьютера">
<i class="bi bi-gear-fill" style="font-size: 0.80rem;"></i>
</button>
</div>
<span class="badge bg-dark border border-secondary text-info p
y-0 px-1" id="about-spec-uptime">Uptime: --</span>
</div>
<!-- Inline Редактирование Host -->
<div class="about-sys-sub d-none align-items-center gap-1 mt-1
" id="about-host-edit-box">
<div class="input-group input-group-sm w-100">
<span class="input-group-text bg-dark border-secondary text-mu
ted py-0 px-1.5 font-monospace" style="font-size: 0.70rem;">Ho
st:</span>
<input class="form-control form-control-sm bg-dark text-white
border-info py-0 px-1.5 font-monospace" id="about-host-inline-
input" maxlength="15" placeholder="Новое имя" style="font-size
: 0.75rem; height: 25px;" type="text"/>
<button class="btn btn-sm btn-info text-dark py-0 px-2 fw-bold
" id="btn-host-inline-save" style="height: 25px; font-size: 0.
72rem;" title="Сохранить имя компьютера">
<i class="bi bi-check-lg"></i>
</button>
<button class="btn btn-sm btn-outline-secondary py-0 px-1.5 te
xt-muted" id="btn-host-inline-cancel" style="height: 25px; fon
t-size: 0.72rem;" title="Отмена">
<i class="bi bi-x-lg"></i>
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</button>
</div>
</div>
<!-- Workgroup View -->
<div class="about-sys-sub text-truncate d-flex align-items-cen
ter justify-content-between mt-1" id="about-workgroup-view-box
">
<div class="d-flex align-items-center text-truncate">
<span id="about-kpi-os-workgroup">Workgroup: WORKGROUP</span>
<button class="btn btn-sm btn-link p-0 text-info text-decorati
on-none about-param-edit-btn" id="btn-workgroup-inline-edit" t
itle="Изменить рабочую группу">
<i class="bi bi-gear-fill" style="font-size: 0.80rem;"></i>
</button>
</div>
</div>
<!-- Inline Редактирование Workgroup -->
<div class="about-sys-sub d-none align-items-center gap-1 mt-1
" id="about-workgroup-edit-box">
<div class="input-group input-group-sm w-100">
<span class="input-group-text bg-dark border-secondary text-mu
ted py-0 px-1.5 font-monospace" style="font-size: 0.70rem;">WG
:</span>
<input class="form-control form-control-sm bg-dark text-white
border-info py-0 px-1.5 font-monospace" id="about-workgroup-in
line-input" maxlength="15" placeholder="Новая группа" style="f
ont-size: 0.75rem; height: 25px;" type="text"/>
<button class="btn btn-sm btn-info text-dark py-0 px-2 fw-bold
" id="btn-workgroup-inline-save" style="height: 25px; font-siz
e: 0.72rem;" title="Сохранить рабочую группу">
<i class="bi bi-check-lg"></i>
</button>
<button class="btn btn-sm btn-outline-secondary py-0 px-1.5 te
xt-muted" id="btn-workgroup-inline-cancel" style="height: 25px
; font-size: 0.72rem;" title="Отмена">
<i class="bi bi-x-lg"></i>
</button>
</div>
```

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</div>
<!-- Напоминание о перезагрузке после смены имени / группы -->
<div class="alert alert-warning d-none py-1 px-2 mt-1 mb-0 bor
der-warning border-opacity-50" id="about-host-reminder-alert"
style="font-size: 0.70rem; line-height: 1.2;">
<i class="bi bi-exclamation-triangle-fill me-1 text-warning"><
/i>
<span id="about-host-reminder-text">Имя изменено. Вступит в си
лу после перезагрузки.</span>
</div>
</div>
</div>
<!-- Card 2: Security & Defender -->
<div class="col-12 col-sm-6 col-lg-3">
<div class="about-sys-card" data-panel-id="panel-about-securit
y" id="panel-about-security">
<div class="about-sys-label d-flex justify-content-between ali
gn-items-center">
<div class="d-flex align-items-center gap-1.5">
<i class="bi bi-shield-check text-success"></i>
<span>Безопасность системы</span>
</div>
<button class="btn btn-sm btn-link p-0 text-muted text-decorat
ion-none btn-card-history" data-bs-target="#aboutSystemHistory
Modal" data-bs-toggle="modal" data-metric="security" title="Ис
тория безопасности из базы данных">
<i class="bi bi-clock-history"></i>
</button>
</div>
<div class="about-sys-value text-success text-truncate mt-1" i
d="about-kpi-sec-title">Active &amp;amp; Protected</div>
<div class="ui-kpi-sub text-truncate" id="about-kpi-sec-sub">F
irewall: ON | UAC: ON</div>
</div>
</div>
<!-- Card 3: System Protection & Restore -->
<div class="col-12 col-sm-6 col-lg-3">
<div class="ui-kpi-card" data-panel-id="panel-about-restore-po
```

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ints" id="panel-about-restore-points">
<div class="about-sys-label d-flex justify-content-between ali
gn-items-center">
<div class="d-flex align-items-center gap-1.5">
<i class="bi bi-shield-lock text-warning"></i>
<span>Точки восстановления</span>
</div>
<button class="btn btn-sm btn-link p-0 text-muted text-decorat
ion-none btn-card-history" data-bs-target="#aboutSystemHistory
Modal" data-bs-toggle="modal" data-metric="restore_points" tit
le="История точек восстановления из базы данных">
<i class="bi bi-clock-history"></i>
</button>
</div>
<div class="ui-kpi-value text-warning text-truncate mt-1" id="
about-kpi-prot-title">3 Checkpoints</div>
<div class="ui-kpi-sub text-truncate" id="about-kpi-prot-sub">
Protection: Active</div>
</div>
</div>
<!-- Card 4: System Storage (C:) -->
<div class="col-12 col-sm-6 col-lg-3">
<div class="ui-kpi-card" data-panel-id="panel-about-storage" i
d="panel-about-storage">
<div class="about-sys-label d-flex justify-content-between ali
gn-items-center">
<div class="d-flex align-items-center gap-1.5">
<i class="bi bi-hdd-fill text-primary"></i>
<span>Системный накопитель (C:)</span>
</div>
<button class="btn btn-sm btn-link p-0 text-muted text-decorat
ion-none btn-card-history" data-bs-target="#aboutSystemHistory
Modal" data-bs-toggle="modal" data-metric="storage" title="Ист
ория накопителя C: из базы данных">
<i class="bi bi-clock-history"></i>
</button>
</div>
<div class="ui-kpi-value text-white text-truncate mt-1" id="ab
```

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out-kpi-stor-title">1014.8 GB Free</div>
<div class="ui-kpi-sub text-info text-truncate" id="about-kpi-stor-clean">Cleanable: ~190 MB</div>
</div>
</div>
</div>
<!-- Reorganized Dual Master Panels: Hardware & Platform Specs vs User Environment & Security -->
<div class="row g-2 mb-3">
<!-- Left: Hardware & Platform Specification -->
<div class="col-12 col-lg-6">
<div class="ui-panel h-100" data-panel-id="panel-about-hardware-spec" id="panel-about-hardware-spec">
<div class="ui-panel-header">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-cpu-fill text-info"></i>
<span class="text-white">Конфигурация оборудования и ОС</span>
</div>
</div>
<table class="ui-spec-table">
<tbody>
<tr>
<td class="ui-spec-key"><i class="bi bi-cpu me-1.5 text-info"></i>Процессор (CPU)</td>
<td class="ui-spec-val" id="about-spec-cpu">Intel Processor</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-memory me-1.5 text-warning"></i>Установленная память</td>
<td class="about-sys-spec-val" id="about-spec-ram">31.8 GB RAM</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-display me-1.5 text-primary"></i>Мониторы и дисплеи</td>
<td class="about-sys-spec-val" id="about-spec-monitors">Опрос мониторов...</td>
```

```
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-battery-chargin
g me-1.5 text-warning"></i>Питание и Энергия</td>
<td class="about-sys-spec-val" id="about-spec-power-container"
>
<div class="d-flex align-items-center gap-1.5 flex-wrap">
<span class="badge bg-secondary font-monospace" id="about-spec
-power-badge" style="font-size: 0.72rem;">AC Mains</span>
<span id="about-spec-power-source">Питание: Стационарная элект
росеть 220V</span>
</div>
<div class="small text-muted font-monospace mt-0.5" id="about-
spec-power-profile" style="font-size: 0.72rem;">Профиль: AC Ma
ins / Desktop</div>
</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-layers me-1.5 t
ext-info"></i>Сборка ОС</td>
<td class="about-sys-spec-val" id="about-spec-os">10.0.26200</
td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-calendar-check
me-1.5 text-warning"></i>Дата установки ОС</td>
<td class="about-sys-spec-val" id="about-ident-install-date">Н
е определена</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-arrow-repeat me
-1.5 text-info"></i>Служба обновлений</td>
<td class="about-sys-spec-val" id="about-spec-update">Up to da
te</td>
</tr>
</tbody>
</table>
</div>
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</div>
<!-- Right: User Environment, Regional & Security -->
<div class="col-12 col-lg-6">
<div class="ui-panel h-100" data-panel-id="panel-about-user-security" id="panel-about-user-security">
<div class="ui-panel-header">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-person-badge-fill text-success"></i>
<span class="text-white">Окружение пользователя и безопасность</span>
</div>
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm btn-outline-info rounded-pill px-2.5 py-0.5" data-bs-target="#aboutRegionalSettingsModal" data-bs-toggle="modal" id="btn-open-regional-settings" style="font-size: 0.75rem;" title="Настроить часовой пояс, язык, локаль и пользователя">
<i class="bi bi-gear-fill me-1"></i>Настроить
            </button>
<span class="small text-muted font-monospace" id="about-ident-time-badge">Локальное время: --:--:--</span>
</div>
</div>
<table class="ui-spec-table">
<tbody>
<tr>
<td class="ui-spec-key"><i class="bi bi-person-circle me-1.5 text-success"></i>Пользователь и права</td>
<td class="ui-spec-val">
<div class="d-flex align-items-center justify-content-between">
<div>
<span class="fw-bold text-white me-1" id="about-ident-username">Пользователь</span>
<span class="text-muted small" id="about-ident-privilege">(Права)</span>
</div>
<button class="btn btn-sm btn-link p-0 text-info text-decorati
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on-none btn-open-reg-tab" data-bs-target="#aboutRegionalSettin
gsModal" data-bs-toggle="modal" data-tab="pane-usr" title="Нас
троить профиль пользователя">
<i class="bi bi-gear-fill" style="font-size: 0.76rem;"></i>
</button>
</div>
</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-translate me-1.
5 text-warning"></i>Язык и локали</td>
<td class="about-sys-spec-val">
<div class="d-flex align-items-center justify-content-between"
>
<div>
<span id="about-ident-language">Русский</span>
<div class="small text-muted font-monospace" id="about-ident-l
ocales">User: ru-RU | Sys: ru-RU</div>
</div>
<button class="btn btn-sm btn-link p-0 text-info text-decorati
on-none btn-open-reg-tab" data-bs-target="#aboutRegionalSettin
gsModal" data-bs-toggle="modal" data-tab="pane-loc" title="Нас
троить локали">
<i class="bi bi-gear-fill" style="font-size: 0.76rem;"></i>
</button>
</div>
</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-clock me-1.5 te
xt-primary"></i>Часовой пояс и кодировка</td>
<td class="about-sys-spec-val">
<div class="d-flex align-items-center justify-content-between"
>
<div>
<span id="about-ident-timezone">UTC+03:00</span>
<div class="small text-muted font-monospace" id="about-ident-c
odepage">Кодировка: UTF-8</div>
```

```
</div>
<button class="btn btn-sm btn-link p-0 text-info text-decoration-none btn-open-reg-tab" data-bs-target="#aboutRegionalSettingsModal" data-bs-toggle="modal" data-tab="pane-tz" title="Настроить часовой пояс">
<i class="bi bi-gear-fill" style="font-size: 0.76rem;"></i>
</button>
</div>
</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-cloud-check me-1.5 text-info"></i>Офис и OneDrive</td>
<td class="about-sys-spec-val">
<span>MS Office: <strong class="text-light font-monospace" id="about-ident-ms-office">Не установлен</strong></span>
<div class="small text-muted font-monospace">OneDrive: <span class="text-light" id="about-ident-onedrive">Не настроено</span></div>
</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-archive me-1.5 text-info"></i>Резервные копии</td>
<td class="about-sys-spec-val" id="about-ident-backup">Нет записей</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-shield-shaded me-1.5 text-success"></i>Брандмауэр Windows</td>
<td class="about-sys-spec-val">
                Domain: <span class="fw-bold text-secondary" id="about-sec-fw-domain">--</span> |
                Private: <span class="fw-bold text-secondary" id="about-sec-fw-private">--</span> |
                Public: <span class="fw-bold text-secondary" id="about-sec-fw-public">--</span>
</td>
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</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-shield-lock me-1.5 text-success"></i>Защитник и UAC</td>
<td class="about-sys-spec-val">
                Defender: <span class="fw-bold text-secondary" id="about-sec-defender">--</span> |
                Realtime: <span class="fw-bold text-secondary" id="about-sec-realtime">--</span> |
                UAC: <span class="fw-bold text-secondary" id="about-sec-uac">--</span>
</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- Storage Volumes & Partitions -->
<div class="row g-2 mb-3">
<div class="col-12">
<div class="about-sys-panel h-100" data-panel-id="panel-about-storage-volumes" id="panel-about-storage-volumes">
<div class="about-sys-panel-header">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-hdd-stack-fill text-primary"></i>
<span class="text-white">Storage Volumes &amp;amp; Partitions</span>
</div>
<div class="d-flex align-items-center gap-2">
<span class="badge bg-secondary" id="about-disks-count">-- Drives</span>
</div>
</div>
<div class="table-responsive">
<table class="about-sys-spec-table">
<thead>
<tr>
```

```
<th style="width: 14%;">Drive</th>
<th style="width: 14%;">FS</th>
<th style="width: 14%;">Total</th>
<th style="width: 14%;">Used</th>
<th style="width: 14%;">Free</th>
<th style="width: 20%;">Usage</th>
<th style="width: 10%;">SMART</th>
</tr>
</thead>
<tbody id="about-sys-disks-tbody">
<!-- Injected dynamically -->
<tr>
<td class="text-center py-3 text-muted" colspan="7">Загрузка разделов дисков...</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
<!-- 4. Износ накопителей и батареи (Storage & Battery Health) -->
<div class="card shadow-sm border-secondary-subtle mb-3" id="panel-about-disks-wear-summary">
<div class="card-body p-3">
<div class="d-flex flex-wrap align-items-center justify-content-between gap-2">
<div class="d-flex align-items-center gap-2">
<div class="fs-4">□ </div>
<div>
<h5 class="mb-0 fw-bold">4. Износ накопителей и батареи (Storage &amp;amp; Battery Health)</h5>
<small class="text-muted">Мониторинг ресурса SSD/NVMe, файловых систем, SMART и деградации аккумулятора</small>
</div>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
```

```
<button class="btn btn-sm btn-outline-info rounded-pill px-3 p
y-1 fw-semibold d-flex align-items-center gap-1" id="btn-diag-
wear-refresh">
<i class="bi bi-arrow-clockwise"></i> Обновить
          </button>
</div>
</div>
</div>
</div>
<div class="row g-2 mb-3">
<!-- Disks Health & Wear Panel -->
<div class="col-12 col-xl-8 col-lg-8">
<div class="diag-panel h-100" data-panel-id="panel-about-disks
-wear" id="panel-about-disks-wear">
<div class="diag-panel-header d-flex justify-content-between a
lign-items-center">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-hdd-fill text-primary"></i>
<span class="text-white">Здоровье накопителей SSD/NVMe и износ
 (Wear &amp;amp; SMART)</span>
</div>
<div class="d-flex align-items-center gap-2">
<span class="badge bg-secondary" id="diag-wear-disks-count">0
дисков</span>
</div>
</div>
<div class="table-responsive">
<table class="about-sys-spec-table">
<thead>
<tr>
<th>Физический накопитель</th>
<th>Тип / Шина</th>
<th>Разделы</th>
<th style="text-align: right;">Объем</th>
<th style="text-align: right;">Свободно</th>
<th>Наработка / Старт</th>
<th style="text-align: right;">Ввод / Вывод (I/O)</th>
<th style="width: 130px;">Здоровье / Износ</th>
```

```
<th>Статус</th>
</tr>
</thead>
<tbody id="diag-wear-disks-tbody">
<tr><td class="text-center py-4 text-muted" colspan="9">Опрос контроллеров накопителей...</td></tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- Battery Deep Telemetry -->
<div class="col-12 col-xl-4 col-lg-4">
<div class="diag-panel h-100" data-panel-id="panel-about-battery" id="panel-about-battery">
<div class="diag-panel-header d-flex justify-content-between align-items-center">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-battery-charging text-warning"></i>
<span class="text-white">Деградация и состояние батареи (Battery Degradation)</span>
</div>
<div class="d-flex align-items-center gap-2">
<span class="badge bg-dark border border-secondary text-info" id="diag-battery-source-badge">AC Mains</span>
</div>
</div>
<div class="p-3" id="diag-battery-details-container">
<div class="text-center py-4 text-muted small">
<i class="bi bi-plug-fill fs-4 text-info d-block mb-1"></i>
            Стационарный компьютер — питание напрямую от электросети
          </div>
</div>
</div>
</div>
</div>
<!-- Full Hardware Specification Tree Panel (Hardware Tree) --
```

```
>
<div class="about-sys-panel" data-panel-id="panel-about-hardware-tree" id="panel-about-hardware-tree">
<div class="about-sys-panel-header">
<div class="d-flex align-items-center gap-2">
<button class="panel-toggle-drill collapsed me-1" id="btn-about-sys-tree-toggle" title="Развернуть все / Свернуть все категории" type="button">
<img alt="Развернуть все/Свернуть все" src="/html/assets/red_drill.png?v=20261008_v1"/>
</button>
<span class="fw-semibold">Полная спецификация оборудования</span>
<span class="badge bg-secondary" id="about-sys-node-count">Hardware Tree</span>
</div>
<div class="d-flex align-items-center gap-2">
<input class="form-control form-control-sm" id="about-sys-search" placeholder="Поиск компонентов..." style="max-width: 220px; font-size: 0.76rem; height: 26px;" type="text"/>
</div>
</div>
<div class="p-3" id="about-sys-tree-container">
<div class="text-center py-4 small text-muted">
<div class="spinner-border spinner-border-sm text-info mb-2" role="status"></div>
<div>Загрузка спецификации оборудования...</div>
</div>
</div>
</div>
</div> <!-- /.about-sys-wrapper -->

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows About System Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/about_system_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/about_system_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-12 06:50:00
 * ============================================================
 */

// ============================================================
// Process Name: About System Tab Web Controller
// ============================================================
// Description:
//   Client-side JavaScript controller for the About System tab.
//   Provides real-time host telemetry, system identity & locale parameters,
//   AIDA64-like hardware tree, live top processes, disk volumes and security status.
```

```
//
//   КЕШИРОВАНИЕ (IndexedDB + Memory):
//   - Stale-While-Revalidate для всех данных
//   - Cache-First для hardware spec (меняется редко)
//   - Network-First для live telemetry (всегда свежие)
//   - Prefetching при активации вкладки
//
// File: main.js
// Package: src.api.webgui.about_system_tab
// Author: hypo69
// Copyright: © 2026 hypo69
// Updated: 2026-10-10 17:35:00
// ============================================================
==================

(function () {
  let hardwareData = [];
  let currentProcesses = [];
  let currentDisks = [];
  let isLiveActive = true;
  let liveIntervalId = null;
  let isUpdating = false;
  let isAiRunning = false;
  let activeHistoryMetric = 'all';

  // Кеш-стратегия и TTL
  const CACHE_STRATEGY = {
    HARDWARE_SPEC: { ttl: 30 * 60 * 1000, strategy: 'cache-fir
st' },      // 30 мин, редко меняется
    SYSTEM_SUMMARY: { ttl: 5 * 1000, strategy: 'stale-while-re
validate' }, // 5 сек
    CONTROL_STATUS: { ttl: 60 * 1000, strategy: 'stale-while-r
evalidate' }, // 1 мин
    BACKUP_STATUS: { ttl: 5 * 60 * 1000, strategy: 'cache-firs
t' },         // 5 мин
    STORAGE_BATTERY: { ttl: 10 * 1000, strategy: 'stale-while-
revalidate' }, // 10 сек
    PANEL_KPI: { ttl: 5 * 1000, strategy: 'stale-while-revalid
```

```
ate' }         // 5 сек
  };

  const CACHE_STORE = 'api_cache';
  const CACHE_TAG = 'about-system-tab';

  /**
   * Универсальная функция для работы с кешем
   * Поддерживает стратегии: cache-first, network-first, stale-while-revalidate
   */
  async function cachedFetch(url, cacheKey, options = {}) {
    const { ttl = 60000, strategy = 'stale-while-revalidate', forceNetwork = false } = options;
    const cache = window.browserCache;

    if (!cache) {
      // Fallback: кеш недоступен, прямой запрос
      return await apiFetch(url);
    }

    // Cache-First: сначала кеш, потом сеть (для редко меняющихся данных)
    if (strategy === 'cache-first' && !forceNetwork) {
      const cached = await cache.get(CACHE_STORE, cacheKey);
      if (cached) {
        console.log(`[Cache] HIT (cache-first): ${cacheKey}`);
        return cached;
      }
    }

    // Stale-While-Revalidate: показываем кеш, обновляем в фоне
    if (strategy === 'stale-while-revalidate' && !forceNetwork) {
      const cached = await cache.get(CACHE_STORE, cacheKey);
      if (cached) {
        console.log(`[Cache] HIT (stale-while-revalidate): ${c
```

```
acheKey}, updating in background...`);
        // Запускаем обновление в фоне
        apiFetch(url)
          .then(fresh => cache.set(CACHE_STORE, cacheKey, fresh, { ttl, tags: [CACHE_TAG] }))
          .catch(err => console.warn(`[Cache] Background update failed for ${cacheKey}:`, err));
        return cached;
      }
    }

    // Network-First или первый запрос: сначала сеть, потом кеш
    try {
      const fresh = await apiFetch(url);
      // Сохраняем в кеш
      await cache.set(CACHE_STORE, cacheKey, fresh, { ttl, tags: [CACHE_TAG] });
      console.log(`[Cache] MISS → stored: ${cacheKey}`);
      return fresh;
    } catch (err) {
      // Если сеть недоступна, пытаемся вернуть устаревший кеш
      const cached = await cache.get(CACHE_STORE, cacheKey);
      if (cached) {
        console.warn(`[Cache] Network failed, returning stale cache: ${cacheKey}`);
        return cached;
      }
      throw err;
    }
  }

  /**
   * Очистка кеша вкладки (при необходимости)
   */
  async function clearTabCache() {
    const cache = window.browserCache;
    if (cache) {
```

```
      await cache.invalidateByTag(CACHE_STORE, CACHE_TAG);
      console.log('[Cache] Tab cache cleared');
    }
  }

  async function apiFetch(url, options = {}) {
    const opts = { ...options };
    if (opts.body && typeof opts.body === 'string') {
      opts.headers = {
        'Content-Type': 'application/json',
        ...(opts.headers || {})
      };
    }
    if (window.api && typeof window.api.fetch === 'function')
{
      return await window.api.fetch(url, opts);
    }
    const res = await fetch(url, opts);
    if (!res.ok) {
      let errMsg = `HTTP ${res.status}`;
      try {
        const errJson = await res.json();
        if (errJson && errJson.detail) {
          errMsg += ` (${typeof errJson.detail === 'object' ?
JSON.stringify(errJson.detail) : errJson.detail})`;
        }
      } catch (_) {}
      throw new Error(errMsg);
    }
    return await res.json();
  }

  function setText(id, text) {
    const el = document.getElementById(id);
    if (el) el.textContent = text !== null && text !== undefin
ed ? String(text) : '--';
  }

```

```
  function setStatusText(id, isPositive, activeText = 'Active', inactiveText = 'Disabled') {
    const el = document.getElementById(id);
    if (!el) return;
    const active = Boolean(isPositive);
    el.textContent = active ? activeText : inactiveText;
    el.className = active ? 'text-success fw-bold' : 'text-danger fw-bold';
  }

  function setFormattedStatusHtml(id, text) {
    const el = document.getElementById(id);
    if (!el) return;
    if (!text) { el.innerHTML = '--'; return; }
    el.innerHTML = String(text)
      .replace(/\b(OFF|Disabled|Выключено|Отключен|Отключено|Inactive)\b/gi, '<strong class="text-danger fw-bold">$1</strong>')
      .replace(/\b(ON|Active|Enabled|Включен|Включено)\b/gi, '<strong class="text-success fw-bold">$1</strong>');
  }

  function setHtml(id, html) {
    const el = document.getElementById(id);
    if (el) el.innerHTML = html;
  }

  async function initAboutSystemTab() {
    console.log('[AboutSystemTab] Initializing tab controller...');
    bindEvents();
    bindAIEvents();
    bindHistoryEvents();
    bindRegionalEvents();
    updateLocalClock();
    updatePowerDisplay();

    // Проверка доступности кеша
```

```
    await updateCacheStatus();

    // Первичная загрузка данных вкладки
    refreshAllData(false);

    // Запуск таймера часов и статуса кеша
    startLiveStream();
  }
  window.initAboutSystemTab = initAboutSystemTab;

  /**
   * Обновление статуса кеша в интерфейсе
   */
  async function updateCacheStatus() {
    const badge = document.getElementById('about-sys-cache-bad
ge');
    if (!badge) return;

    const cache = window.browserCache;
    if (!cache) {
      badge.textContent = '  Cache: Unavailable';
      badge.className = 'badge rounded-pill bg-warning-subtle
text-warning border border-warning px-2.5 py-1';
      badge.title = 'Кеш недоступен, используется прямой запро
с к API';
      return;
    }

    const ready = await cache.ready();
    if (ready) {
      const stats = await cache.getDetailedStats();
      const hitRate = stats.metrics.hits + stats.metrics.misse
s > 0
        ? Math.round((stats.metrics.hits / (stats.metrics.hits
 + stats.metrics.misses)) * 100)
        : 0;

      badge.textContent = `  Cache: ${hitRate}% hit`;
```

```
      badge.className = 'badge rounded-pill bg-success-subtle text-success border border-success px-2.5 py-1';
      badge.title = `Кеш активен\nПопаданий: ${stats.metrics.hits}\nПромахов: ${stats.metrics.misses}\nВ памяти: ${stats.memoryCacheSize} записей`;
    } else {
      badge.textContent = '⚠ Cache: Error';
      badge.className = 'badge rounded-pill bg-danger-subtle text-danger border border-danger px-2.5 py-1';
      badge.title = 'Ошибка инициализации кеша';
    }
  }

  function updateLocalClock() {
    const clockEl = document.getElementById('about-ident-time-badge');
    if (clockEl) {
      const now = new Date();
      clockEl.textContent = `Локальное время: ${now.toLocaleTimeString()}`;
    }
  }

  function startLiveStream() {
    if (window.registerTabPoller) {
      window.registerTabPoller('tab-about-system', async () => {
        updateLocalClock();
        await updateCacheStatus();
      }, 5000, { pollerId: 'tab-about-system_clock', immediate: true });
    } else {
      if (liveIntervalId) clearInterval(liveIntervalId);
      liveIntervalId = setInterval(async () => {
        updateLocalClock();
        await updateCacheStatus();
      }, 5000);
    }
```

```
  }

  function bindEvents() {
    const btnRefresh = document.getElementById('btn-about-sys-refresh');
    if (btnRefresh) {
      btnRefresh.onclick = async () => {
        btnRefresh.disabled = true;
        const icon = btnRefresh.querySelector('i');
        if (icon) icon.classList.add('spin-animation');
        await refreshAllData();
        if (icon) icon.classList.remove('spin-animation');
        btnRefresh.disabled = false;
      };
    }

    const btnClearCache = document.getElementById('btn-about-sys-clear-cache');
    if (btnClearCache) {
      btnClearCache.onclick = async () => {
        btnClearCache.disabled = true;
        const icon = btnClearCache.querySelector('i');
        if (icon) icon.classList.add('spin-animation');

        // Очищаем кеш вкладки
        await clearTabCache();

        // Принудительно обновляем все данные из сети
        await refreshAllData(true);

        if (icon) icon.classList.remove('spin-animation');
        btnClearCache.disabled = false;

        if (window.showToast) {
          window.showToast('Кеш вкладки очищен, данные обновлены', 'success');
        }
      };
```

```
    }

    const btnLiveToggle = document.getElementById('btn-about-sys-live-toggle');
    if (btnLiveToggle) {
      btnLiveToggle.onclick = () => {
        isLiveActive = !isLiveActive;
        if (window.setTabPollerEnabled) {
          window.setTabPollerEnabled('tab-about-system_clock', isLiveActive);
        }
        const icon = document.getElementById('icon-about-sys-live');
        const txt = document.getElementById('txt-about-sys-live');
        const liveBadge = document.getElementById('about-sys-live-badge');
        if (isLiveActive) {
          if (icon) icon.className = 'bi bi-pause-fill me-1';
          if (txt) txt.textContent = 'Пауза';
          if (liveBadge) {
            liveBadge.className = 'badge rounded-pill bg-info-subtle text-info border border-info px-2.5 py-1';
            liveBadge.textContent = '● Live Host Telemetry';
          }
        } else {
          if (icon) icon.className = 'bi bi-play-fill me-1';
          if (txt) txt.textContent = 'Возобновить';
          if (liveBadge) {
            liveBadge.className = 'badge rounded-pill bg-warning-subtle text-warning border border-warning px-2.5 py-1';
            liveBadge.textContent = '  Stream Paused';
          }
        }
      };
    }

    const btnTreeToggle = document.getElementById('btn-about-s
```

```
ys-tree-toggle');
    if (btnTreeToggle) {
      btnTreeToggle.onclick = () => {
        const bodies = document.querySelectorAll('.about-sys-t
ree-body');
        const chevrons = document.querySelectorAll('.about-sys
-chevron');
        if (!bodies.length) return;

        // Если хотя бы один узел свернут — разворачиваем ВСЕ
        const hasHidden = Array.from(bodies).some(b => b.class
List.contains('d-none'));

        if (hasHidden) {
          bodies.forEach(b => b.classList.remove('d-none'));
          chevrons.forEach(c => c.textContent = '▲');
          btnTreeToggle.classList.remove('collapsed');
          btnTreeToggle.title = 'Свернуть все категории';
        } else {
          bodies.forEach(b => b.classList.add('d-none'));
          chevrons.forEach(c => c.textContent = '▼');
          btnTreeToggle.classList.add('collapsed');
          btnTreeToggle.title = 'Развернуть все категории';
        }
      };
    }

    const treeSearch = document.getElementById('about-sys-sear
ch');
    if (treeSearch) {
      treeSearch.oninput = () => {
        const query = treeSearch.value.trim().toLowerCase();
        renderHardwareTree(hardwareData, query);
      };
    }

    const procSearch = document.getElementById('about-proc-sea
rch');
```

```
352     if (procSearch) {
353       procSearch.oninput = () => {
354         renderProcessesTable(currentProcesses);
355       };
356     }
357 
358     const btnWearRefresh = document.getElementById('btn-diag-w
ear-refresh');
359     if (btnWearRefresh) {
360       btnWearRefresh.onclick = () => fetchStorageBatteryWear(t
rue);
361     }
362 
363     // Inline Edit: Hostname
364     const btnHostEdit = document.getElementById('btn-host-inli
ne-edit');
365     const viewBox = document.getElementById('about-host-view-b
ox');
366     const editBox = document.getElementById('about-host-edit-b
ox');
367     const inputHost = document.getElementById('about-host-inli
ne-input');
368     const btnHostSave = document.getElementById('btn-host-inli
ne-save');
369     const btnHostCancel = document.getElementById('btn-host-in
line-cancel');
370     const reminderAlert = document.getElementById('about-host-
reminder-alert');
371     const reminderText = document.getElementById('about-host-r
eminder-text');
372 
373     function openHostEdit() {
374       if (!viewBox || !editBox || !inputHost) return;
375       const hostEl = document.getElementById('about-kpi-os-hos
t');
376       let currentHost = hostEl ? hostEl.textContent.replace(/^
Host:\s*/i, '').trim() : '';
377       if (!currentHost || currentHost === '--') currentHost =
```

```
window.location.hostname || 'DELL-VOSTRO';
      inputHost.value = currentHost;
      viewBox.classList.add('d-none');
      viewBox.classList.remove('d-flex');
      editBox.classList.remove('d-none');
      editBox.classList.add('d-flex');
      inputHost.focus();
      inputHost.select();
    }

    function closeHostEdit() {
      if (!viewBox || !editBox) return;
      editBox.classList.add('d-none');
      editBox.classList.remove('d-flex');
      viewBox.classList.remove('d-none');
      viewBox.classList.add('d-flex');
    }

    if (btnHostEdit) {
      btnHostEdit.onclick = (e) => {
        e.preventDefault();
        openHostEdit();
      };
    }

    if (btnHostCancel) {
      btnHostCancel.onclick = (e) => {
        e.preventDefault();
        closeHostEdit();
      };
    }

    if (inputHost) {
      inputHost.onkeydown = (e) => {
        if (e.key === 'Enter') {
          e.preventDefault();
          if (btnHostSave) btnHostSave.click();
        } else if (e.key === 'Escape') {
```

```
          e.preventDefault();
          closeHostEdit();
        }
      };
    }

    if (btnHostSave && inputHost) {
      btnHostSave.onclick = async (e) => {
        e.preventDefault();
        const newName = inputHost.value.trim();

        if (!newName) {
          if (window.showToast) window.showToast('Пожалуйста, введите новое имя компьютера', 'warning');
          return;
        }

        const validNamePattern = /^[a-zA-Z0-9]([a-zA-Z0-9\-]{0,13}[a-zA-Z0-9])?$/;
        if (!validNamePattern.test(newName) || newName.length > 15) {
          if (window.showToast) window.showToast('Имя должно содержать от 1 до 15 символов (A-Z, 0-9, дефис) без пробелов', 'warning');
          return;
        }

        btnHostSave.disabled = true;
        inputHost.disabled = true;
        btnHostSave.innerHTML = '<span class="spinner-border spinner-border-sm" role="status" style="width: 10px; height: 10px;"></span>';

        try {
          const res = await fetch('/api/v1/system/rename-computer', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
```

```
445             body: JSON.stringify({ new_name: newName, restart:
 false }),
446           });
447           const result = await res.json();
448           if (res.ok && result.status === 'ok') {
449             setText('about-kpi-os-host', `Host: ${newName}`);
450             setText('about-ident-hostname', newName);
451             closeHostEdit();
452 
453             // Показываем напоминание о перезагрузке
454             if (reminderAlert) {
455               reminderAlert.classList.remove('d-none');
456               if (reminderText) {
457                 reminderText.textContent = `Имя компьютера изм
енено на «${newName}». Изменение сработает только после переза
грузки системы!`;
458               }
459             }
460             if (window.showToast) {
461               window.showToast(`  Имя изменено на «${newName}»
. Напоминание: требуется перезагрузка для применения!`, 'warni
ng');
462             }
463           } else {
464             const errMsg = result.message || result.detail ||
'Не удалось изменить имя компьютера.';
465             if (window.showToast) window.showToast(`Ошибка: ${
errMsg}`, 'danger');
466             else alert(errMsg);
467           }
468         } catch (err) {
469           const errMsg = `Сетевая ошибка: ${err.message}`;
470           if (window.showToast) window.showToast(errMsg, 'dang
er');
471           else alert(errMsg);
472         } finally {
473           btnHostSave.disabled = false;
474           inputHost.disabled = false;
```

```
          btnHostSave.innerHTML = '<i class="bi bi-check-lg"></i>';
        }
      };
    }

    // Inline Edit: Workgroup
    const btnWgEdit = document.getElementById('btn-workgroup-inline-edit');
    const wgViewBox = document.getElementById('about-workgroup-view-box');
    const wgEditBox = document.getElementById('about-workgroup-edit-box');
    const inputWg = document.getElementById('about-workgroup-inline-input');
    const btnWgSave = document.getElementById('btn-workgroup-inline-save');
    const btnWgCancel = document.getElementById('btn-workgroup-inline-cancel');

    function openWgEdit() {
      if (!wgViewBox || !wgEditBox || !inputWg) return;
      const wgEl = document.getElementById('about-kpi-os-workgroup');
      let currentWg = wgEl ? wgEl.textContent.replace(/^Workgroup:\s*/i, '').trim() : '';
      if (!currentWg || currentWg === '--') currentWg = 'WORKGROUP';
      inputWg.value = currentWg;
      wgViewBox.classList.add('d-none');
      wgViewBox.classList.remove('d-flex');
      wgEditBox.classList.remove('d-none');
      wgEditBox.classList.add('d-flex');
      inputWg.focus();
      inputWg.select();
    }

    function closeWgEdit() {
```

```
503       if (!wgViewBox || !wgEditBox) return;
504       wgEditBox.classList.add('d-none');
505       wgEditBox.classList.remove('d-flex');
506       wgViewBox.classList.remove('d-none');
507       wgViewBox.classList.add('d-flex');
508     }
509
510     if (btnWgEdit) {
511       btnWgEdit.onclick = (e) => {
512         e.preventDefault();
513         openWgEdit();
514       };
515     }
516
517     if (btnWgCancel) {
518       btnWgCancel.onclick = (e) => {
519         e.preventDefault();
520         closeWgEdit();
521       };
522     }
523
524     if (inputWg) {
525       inputWg.onkeydown = (e) => {
526         if (e.key === 'Enter') {
527           e.preventDefault();
528           if (btnWgSave) btnWgSave.click();
529         } else if (e.key === 'Escape') {
530           e.preventDefault();
531           closeWgEdit();
532         }
533       };
534     }
535
536     if (btnWgSave && inputWg) {
537       btnWgSave.onclick = async (e) => {
538         e.preventDefault();
539         const newWg = inputWg.value.trim().toUpperCase();
540
```

```
        if (!newWg) {
          if (window.showToast) window.showToast('Пожалуйста, введите имя рабочей группы', 'warning');
          return;
        }

        const validWgPattern = /^[a-zA-Z0-9_\-]{1,15}$/;
        if (!validWgPattern.test(newWg)) {
          if (window.showToast) window.showToast('Имя группы должно содержать от 1 до 15 символов (A-Z, 0-9, -, _)', 'warning');
          return;
        }

        btnWgSave.disabled = true;
        inputWg.disabled = true;
        btnWgSave.innerHTML = '<span class="spinner-border spinner-border-sm" role="status" style="width: 10px; height: 10px;"></span>';

        try {
          const res = await fetch('/api/v1/system/change-workgroup', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ new_workgroup: newWg, restart: false }),
          });
          const result = await res.json();
          if (res.ok && result.status === 'ok') {
            setText('about-kpi-os-workgroup', `Workgroup: ${newWg}`);
            closeWgEdit();

            // Показываем напоминание о перезагрузке
            if (reminderAlert) {
              reminderAlert.classList.remove('d-none');
              if (reminderText) {
```

```
                reminderText.textContent = `Рабочая группа изменена на «${newWg}». Изменение сработает только после перезагрузки системы!`;
              }
            }
            if (window.showToast) {
              window.showToast(`  Рабочая группа изменена на «${newWg}». Напоминание: требуется перезагрузка для применения!`, 'warning');
            }
          } else {
            const errMsg = result.message || result.detail || 'Не удалось изменить рабочую группу.';
            if (window.showToast) window.showToast(`Ошибка: ${errMsg}`, 'danger');
            else alert(errMsg);
          }
        } catch (err) {
          const errMsg = `Сетевая ошибка: ${err.message}`;
          if (window.showToast) window.showToast(errMsg, 'danger');
          else alert(errMsg);
        } finally {
          btnWgSave.disabled = false;
          inputWg.disabled = false;
          btnWgSave.innerHTML = '<i class="bi bi-check-lg"></i>';
        }
      };
    }
  }

  async function refreshAllData(forceNetwork = false) {
    isUpdating = true;
    try {
      if (forceNetwork) {
        // Принудительная загрузка из сети (игнорируем кеш)
        await Promise.allSettled([
```

```
          apiFetch('/api/v1/system/summary?process_limit=25').
then(data => {
            if (window.browserCache) {
              window.browserCache.set(CACHE_STORE, 'system_sum
mary', data, {
                ttl: CACHE_STRATEGY.SYSTEM_SUMMARY.ttl,
                tags: [CACHE_TAG]
              });
            }
            return fetchSystemSummary();
          }),
          apiFetch('/api/system-control/status').then(data =>
{
            if (window.browserCache) {
              window.browserCache.set(CACHE_STORE, 'system_con
trol_status', data, {
                ttl: CACHE_STRATEGY.CONTROL_STATUS.ttl,
                tags: [CACHE_TAG]
              });
            }
            return fetchSystemControlStatus();
          }),
          apiFetch('/api/v1/system/hardware').then(data => {
            if (window.browserCache) {
              window.browserCache.set(CACHE_STORE, 'hardware_s
pec', data, {
                ttl: CACHE_STRATEGY.HARDWARE_SPEC.ttl,
                tags: [CACHE_TAG]
              });
            }
            return fetchHardwareSpec();
          }),
          apiFetch('/api/v1/windows-backup/health').then(data
=> {
            if (window.browserCache) {
              window.browserCache.set(CACHE_STORE, 'backup_sta
tus', data, {
                ttl: CACHE_STRATEGY.BACKUP_STATUS.ttl,
```

```
                tags: [CACHE_TAG]
              });
            }
            return fetchBackupStatus();
          }),
          apiFetch('/api/v1/system/diagnostics/storage-battery
').then(data => {
            if (window.browserCache) {
              window.browserCache.set(CACHE_STORE, 'storage_ba
ttery_wear', data, {
                ttl: CACHE_STRATEGY.STORAGE_BATTERY.ttl,
                tags: [CACHE_TAG]
              });
            }
            return fetchStorageBatteryWear(true);
          }),
          fetchKpiPanels(true)
        ]);
      } else {
        await Promise.allSettled([
          fetchSystemSummary(),
          fetchSystemControlStatus(),
          fetchHardwareSpec(),
          fetchBackupStatus(),
          fetchStorageBatteryWear(),
          fetchKpiPanels(false)
        ]);
      }
    } finally {
      isUpdating = false;
    }
  }

  async function fetchBackupStatus() {
    try {
      const config = CACHE_STRATEGY.BACKUP_STATUS;
      const data = await cachedFetch(
        '/api/v1/windows-backup/health',
```

```
        'backup_status',
        { ttl: config.ttl, strategy: config.strategy }
      );
      if (!data) return;

      const cfg = data.file_history?.config;
      const storage = data.storage_audit;

      let lastTimeStr = 'Нет записей';
      if (cfg && cfg.last_backup_time) {
        try {
          const d = new Date(cfg.last_backup_time);
          if (!isNaN(d.getTime())) {
            lastTimeStr = d.toLocaleDateString() + ' ' + d.toL
ocaleTimeString([], { hour: '2-digit', minute: '2-digit' });
          }
        } catch (_) {}
      } else if (storage && storage.sample_versions && storage
.sample_versions.length > 0) {
        const latestSample = storage.sample_versions[0];
        if (latestSample && latestSample.version_timestamp) {
          try {
            const d = new Date(latestSample.version_timestamp)
;
            if (!isNaN(d.getTime())) {
              lastTimeStr = d.toLocaleDateString() + ' ' + d.t
oLocaleTimeString([], { hour: '2-digit', minute: '2-digit' });
            }
          } catch (_) {}
        }
      }

      let storageStr = '';
      if (storage && storage.target_exists && storage.free_spa
ce_gb !== null && storage.free_space_gb !== undefined) {
        const target = storage.target_path || 'Диск';
        const freeGb = typeof storage.free_space_gb === 'numbe
r' ? storage.free_space_gb.toFixed(1) : storage.free_space_gb;
```

```
        storageStr = `${target} (${freeGb} GB своб.)`;
      } else if (cfg && (cfg.target_drive_letter || cfg.target_url)) {
        storageStr = cfg.target_drive_letter || cfg.target_url;
      } else {
        storageStr = 'Хранилище не найдено';
      }

      const backupSummary = `${lastTimeStr} | ${storageStr}`;
      const backupEl = document.getElementById('about-ident-backup');
      if (backupEl) {
        backupEl.textContent = backupSummary;
        backupEl.title = `Последний бэкап: ${lastTimeStr}\nХранилище: ${storageStr}\nСлужба File History: ${data.file_history?.service_status || '—'}\nHealth Score: ${data.health_score ?? '--'}/100`;
      }
    } catch (e) {
      console.warn('[AboutSystemTab] fetchBackupStatus warning:', e);
      setText('about-ident-backup', 'Не настроено');
    }
  }

  /**
   * Обновление панели "Питание и Энергия" в таблице спецификации оборудования и ОС.
   * @param {Object} [batteryWear] - Объект телеметрии батареи/питания
   * @param {string} [powerProfileName] - Название активной схемы электропитания
   */
  function updatePowerDisplay(batteryWear = null, powerProfileName = '') {
    const badgeEl = document.getElementById('about-spec-power-badge');
```

```
    const sourceEl = document.getElementById('about-spec-power-source');
    const profileEl = document.getElementById('about-spec-power-profile');
    const legacyEl = document.getElementById('about-spec-power');

    let badgeText = 'AC Mains';
    let badgeClass = 'bg-secondary';
    let sourceText = 'Питание: Стационарная электросеть 220V';
    let profileText = powerProfileName ? `Профиль: ${powerProfileName}` : 'Профиль: AC Mains / Desktop';

    if (batteryWear) {
      if (batteryWear.has_battery) {
        badgeText = batteryWear.is_charging ? 'AC Adapter' : 'Battery';
        badgeClass = batteryWear.is_charging ? 'bg-success' : (batteryWear.percent < 20 ? 'bg-danger' : 'bg-warning text-dark');
        sourceText = `Батарея: ${batteryWear.percent}% (${batteryWear.is_charging ? 'Заряжается' : 'Работа от батареи'})`;
      } else {
        badgeText = batteryWear.power_source ? (batteryWear.power_source.includes('AC') ? 'AC Mains' : batteryWear.power_source) : 'AC Mains';
        badgeClass = 'bg-secondary';
        sourceText = 'Питание: Стационарная электросеть 220V';
      }
    }

    if (badgeEl) {
      badgeEl.textContent = badgeText;
      badgeEl.className = `badge ${badgeClass} font-monospace`;
    }
    if (sourceEl) {
      sourceEl.textContent = sourceText;
```

```
753     }
754     if (profileEl) {
755       if (powerProfileName) {
756         profileEl.textContent = `Профиль: ${powerProfileName}`
;
757       } else if (!profileEl.textContent || profileEl.textConte
nt.trim() === '' || profileEl.textContent === 'Профиль: --') {
758         profileEl.textContent = profileText;
759       }
760     }
761     if (legacyEl) {
762       legacyEl.textContent = `${sourceText} | ${profileText}`;
763     }
764   }
765
766   async function fetchStorageBatteryWear(forceNetwork = false)
 {
767     try {
768       const config = CACHE_STRATEGY.STORAGE_BATTERY;
769       const data = await cachedFetch(
770         '/api/v1/system/diagnostics/storage-battery',
771         'storage_battery_wear',
772         { ttl: config.ttl, strategy: config.strategy, forceNet
work }
773       );
774       if (!data) return;
775
776       // Disks Wear & SMART
777       const disksTbody = document.getElementById('diag-wear-di
sks-tbody');
778       if (disksTbody && data.disks_wear) {
779         setText('diag-wear-disks-count', `${data.disks_wear.le
ngth} дисков`);
780         if (data.disks_wear.length === 0) {
781           disksTbody.innerHTML = '<tr><td colspan="9" class="t
ext-center py-4 text-muted">Физических накопителей не обнаруже
но</td></tr>';
782         } else {
```

```
          const formatBytesLocal = (bytes) => {
            if (bytes == null || bytes === 0) return '0 B';
            const k = 1024;
            const sizes = ['B', 'KB', 'MB', 'GB', 'TB', 'PB'];
            const i = Math.floor(Math.log(bytes) / Math.log(k)
);
            return `${parseFloat((bytes / Math.pow(k, i)).toFi
xed(1))} ${sizes[i]}`;
          };

          const formatPohLocal = (hours) => {
            if (hours == null || hours <= 0) return null;
            const days = Math.floor(hours / 24);
            const years = (hours / (24 * 365.25)).toFixed(1);
            if (days >= 365) return `${hours.toLocaleString()}
 ч (≈ ${years} г)`;
            if (days >= 1) return `${hours.toLocaleString()} ч
 (${days} д)`;
            return `${hours.toLocaleString()} ч`;
          };

          disksTbody.innerHTML = data.disks_wear.map(d => {
            const diskName = d.name || d.model || d.device_id
|| 'Физический диск';
            const devId = d.device_id ? `<span class="badge bg
-secondary-subtle text-light border border-secondary me-1.5">$
{escapeHtml(d.device_id)}</span>` : '';
            const isSsd = (d.media_type && d.media_type.toUppe
rCase().includes('SSD')) || (d.bus_type && d.bus_type.toUpperC
ase().includes('NVME'));
            const diskIcon = isSsd ? 'bi-device-ssd text-info'
 : 'bi-hdd text-primary';
            const typeBus = [d.media_type, d.bus_type].filter(
Boolean).join(' / ') || 'Storage';
            const partsBadge = d.partitions && d.partitions !=
= '—'
              ? `<span class="badge bg-primary-subtle text-inf
o border border-info-subtle font-monospace">${escapeHtml(d.par
```

```
titions)}</span>`
              : '<span class="text-muted">—</span>';
            const freeGbStr = d.free_gb != null ? `${d.free_gb} GB` : '<span class="text-muted">—</span>';
            const healthPct = d.health_pct != null ? d.health_pct : 100;
            const healthColor = healthPct >= 80 ? 'success' : (healthPct >= 50 ? 'warning' : 'danger');

            // Наработка и дата первого включения
            const pohFormatted = formatPohLocal(d.power_on_hours);
            const startStr = d.first_power_on ? `Старт: ${escapeHtml(d.first_power_on)}` : '';
            let pohHtml = '<span class="text-muted font-monospace small">—</span>';
            if (pohFormatted && startStr) {
              pohHtml = `<div class="font-monospace text-info">${pohFormatted}</div><div class="small text-muted font-monospace mt-0.5">${startStr}</div>`;
            } else if (pohFormatted) {
              pohHtml = `<div class="font-monospace text-info">${pohFormatted}</div>`;
            } else if (startStr) {
              pohHtml = `<div class="small text-muted font-monospace">${startStr}</div>`;
            }

            // Объемы ввода/вывода (запись и чтение)
            const ioHtml = `
              <div class="font-monospace text-light" style="white-space: nowrap;" title="Записано за сессию">
                <i class="bi bi-arrow-up-circle text-warning me-1"></i>Записано: ${formatBytesLocal(d.bytes_written)}
              </div>
              <div class="small font-monospace text-info mt-0.5" style="white-space: nowrap;" title="Прочитано за сессию">
                <i class="bi bi-arrow-down-circle text-info me
```

```
-1"></i>Прочитано: ${formatBytesLocal(d.bytes_read)}
              </div>
            `;

            return `
            <tr>
              <td class="fw-bold text-white">
                <div class="d-flex align-items-center">
                  <i class="bi ${diskIcon} me-2 fs-6"></i>
                  <div>
                    <div class="text-truncate" style="max-widt
h: 250px;" title="${escapeHtml(diskName)}">${escapeHtml(diskNa
me)}</div>
                    <div class="small text-muted font-monospac
e mt-0.5">${devId}${d.serial_number && d.serial_number !== 'N/
A' ? `<span class="text-secondary">S/N: ${escapeHtml(d.serial_
number)}</span>` : ''}</div>
                  </div>
                </div>
              </td>
              <td class="text-muted font-monospace"><span clas
s="badge bg-dark border border-secondary text-light">${escapeH
tml(typeBus)}</span></td>
              <td>${partsBadge}</td>
              <td style="text-align: right;" class="text-light
 font-monospace">${d.total_gb} GB</td>
              <td style="text-align: right;" class="text-succe
ss font-monospace">${freeGbStr}</td>
              <td>${pohHtml}</td>
              <td style="text-align: right;">${ioHtml}</td>
              <td>
                <div class="d-flex align-items-center gap-1.5"
>
                  <div class="about-sys-progress-track flex-gr
ow-1" style="height: 6px; min-width: 55px;">
                    <div class="about-sys-progress-bar bg-${he
althColor}" style="width: ${healthPct}%;"></div>
                  </div>
```

```
                <span class="small font-monospace text-${hea
lthColor} ms-1">${healthPct}%</span>
                </div>
              </td>
              <td><span class="badge bg-${healthColor}-subtle
text-${healthColor} border border-${healthColor}">${escapeHtml
(d.status || 'OK')}</span></td>
            </tr>
          `;}).join('');
        }
      }

      // Battery Degradation
      const batContainer = document.getElementById('diag-batte
ry-details-container');
      const batSourceBadge = document.getElementById('diag-bat
tery-source-badge');
      if (batContainer && data.battery_wear) {
        const b = data.battery_wear;
        updatePowerDisplay(b);
        if (batSourceBadge) {
          batSourceBadge.textContent = b.power_source || 'AC M
ains';
        }
        if (!b.has_battery) {
          batContainer.innerHTML = `
            <div class="text-center py-4 text-muted small">
              <i class="bi bi-plug-fill fs-4 text-info d-block
 mb-1"></i>
              Стационарный компьютер — питание напрямую от эле
ктросети
            </div>
          `;
        } else {
          batContainer.innerHTML = `
            <table class="about-sys-spec-table">
              <tbody>
                <tr>
```

```
                  <td class="about-sys-spec-key"><i class="bi bi-battery-charging me-1.5 text-warning"></i>Уровень заряда</td>
                  <td class="about-sys-spec-val fw-bold text-white">${b.percent}% (${b.is_charging ? 'Заряжается' : 'Разряд'})</td>
                </tr>
                <tr>
                  <td class="about-sys-spec-key"><i class="bi bi-shield-shaded me-1.5 text-info"></i>Заводская емкость</td>
                  <td class="about-sys-spec-val font-monospace">${b.design_capacity_mwh ? b.design_capacity_mwh + ' mWh' : '--'}</td>
                </tr>
                <tr>
                  <td class="about-sys-spec-key"><i class="bi bi-battery-full me-1.5 text-success"></i>Текущая емкость</td>
                  <td class="about-sys-spec-val font-monospace">${b.full_charge_capacity_mwh ? b.full_charge_capacity_mwh + ' mWh' : '--'}</td>
                </tr>
                <tr>
                  <td class="about-sys-spec-key"><i class="bi bi-heart-pulse me-1.5 text-danger"></i>Деградация (Износ)</td>
                  <td class="about-sys-spec-val ${b.wear_level_pct > 20 ? 'text-danger fw-bold' : 'text-success'}">
                    ${b.wear_level_pct}% износа
                  </td>
                </tr>
              </tbody>
            </table>
          `;
        }
      }
    } catch (err) {
      console.warn('[AboutSystemTab] fetchStorageBatteryWear error:', err);
    }
```

```
  }

  /**
   * Гранулярные функции получения данных для каждой карточки KPI
   */
  async function fetchKpiOs(forceNetwork = false) {
    try {
      const config = CACHE_STRATEGY.PANEL_KPI || { ttl: 5000, strategy: 'stale-while-revalidate' };
      const opts = { ttl: config.ttl, strategy: config.strategy, forceNetwork };
      const data = await cachedFetch('/api/v1/dashboard/os', 'dash_os', opts);
      if (data) {
        if (data.display_title) setText('about-kpi-os-title', data.display_title);
        if (data.display_host) setText('about-kpi-os-host', data.display_host);
        if (data.workgroup) setText('about-kpi-os-workgroup', `Workgroup: ${data.workgroup}`);
        else if (data.display_workgroup) setText('about-kpi-os-workgroup', data.display_workgroup);
        if (data.uptime_human) setText('about-spec-uptime', `Uptime: ${data.uptime_human}`);
        if (data.os_build) setText('about-spec-os', data.os_build);
        if (data.os_install_date) {
          setText('about-ident-install-date', data.os_install_date);
        }
      }
    } catch (e) {
      console.warn('[AboutSystemTab] fetchKpiOs error:', e);
    }
  }

  async function fetchKpiSecurity(forceNetwork = false) {
```

```
    try {
      const config = CACHE_STRATEGY.PANEL_KPI || { ttl: 5000, strategy: 'stale-while-revalidate' };
      const opts = { ttl: config.ttl, strategy: config.strategy, forceNetwork };
      const data = await cachedFetch('/api/v1/dashboard/security', 'dash_sec', opts);
      if (data) {
        if (data.display_title) {
          const el = document.getElementById('about-kpi-sec-title');
          if (el) {
            el.textContent = data.display_title;
            el.className = `about-sys-value ${data.status === 'Active & Protected' ? 'text-success' : 'text-danger fw-bold'} text-truncate mt-1`;
          }
        }
        if (data.display_subtitle) setFormattedStatusHtml('about-kpi-sec-sub', data.display_subtitle);
      }
    } catch (e) {
      console.warn('[AboutSystemTab] fetchKpiSecurity error:', e);
    }
  }

  async function fetchKpiCheckpoints(forceNetwork = false) {
    try {
      const config = CACHE_STRATEGY.PANEL_KPI || { ttl: 5000, strategy: 'stale-while-revalidate' };
      const opts = { ttl: config.ttl, strategy: config.strategy, forceNetwork };
      const data = await cachedFetch('/api/v1/dashboard/checkpoints', 'dash_chk', opts);
      if (data) {
        if (data.display_title) setText('about-kpi-prot-title', data.display_title);
```

```
965         if (data.display_subtitle) setFormattedStatusHtml('abo
ut-kpi-prot-sub', data.display_subtitle);
966       }
967     } catch (e) {
968       console.warn('[AboutSystemTab] fetchKpiCheckpoints error
:', e);
969     }
970   }
971
972   async function fetchKpiStorage(forceNetwork = false) {
973     try {
974       const config = CACHE_STRATEGY.PANEL_KPI || { ttl: 5000,
strategy: 'stale-while-revalidate' };
975       const opts = { ttl: config.ttl, strategy: config.strateg
y, forceNetwork };
976       const data = await cachedFetch('/api/v1/dashboard/storag
e', 'dash_stor', opts);
977       if (data) {
978         if (data.display_title) setText('about-kpi-stor-title'
, data.display_title);
979         if (data.display_subtitle) setText('about-kpi-stor-cle
an', data.display_subtitle);
980       }
981     } catch (e) {
982       console.warn('[AboutSystemTab] fetchKpiStorage error:',
e);
983     }
984   }
985
986   async function fetchHardwareSpecs(forceNetwork = false) {
987     await Promise.allSettled([
988       fetchSystemSummary(forceNetwork ? false : true),
989       fetchHardwareSpec(forceNetwork)
990     ]);
991   }
992
993   async function fetchUserEnvSecurity(forceNetwork = false) {
994     await Promise.allSettled([
```

```
      fetchSystemSummary(forceNetwork ? false : true),
      fetchSystemControlStatus(forceNetwork),
      fetchBackupStatus(forceNetwork)
    ]);
  }

  /**
   * Загрузка всех 4 сводных KPI-карточек
   */
  async function fetchKpiPanels(forceNetwork = false) {
    await Promise.allSettled([
      fetchKpiOs(forceNetwork),
      fetchKpiSecurity(forceNetwork),
      fetchKpiCheckpoints(forceNetwork),
      fetchKpiStorage(forceNetwork)
    ]);
  }

  async function pollLiveTelemetry() {
    isUpdating = true;
    try {
      await fetchSystemSummary(true);
    } finally {
      isUpdating = false;
    }
  }


  async function fetchSystemSummary(isLightPoll = false) {
    try {
      const config = CACHE_STRATEGY.SYSTEM_SUMMARY;
      const snap = await cachedFetch(
        '/api/v1/system/summary?process_limit=25',
        'system_summary',
        { ttl: config.ttl, strategy: isLightPoll ? 'stale-whil
e-revalidate' : 'network-first' }
      );
      if (!snap) return;
```

```
1032
1033      // 1. Identity & Locale Block
1034      const host = snap.hostname || 'DELL-VOSTRO';
1035      const user = snap.username || 'onela';
1036      const lang = snap.system_language || 'Русский (Россия) [
ru-RU]';
1037      const userLoc = snap.user_locale || 'ru-RU';
1038      const sysLoc = snap.system_locale || 'ru-RU';
1039      const instLang = snap.os_install_language || snap.instal
l_language || '';
1040      const tz = snap.timezone || 'UTC+03:00';
1041      const cp = snap.codepage || 'UTF-8 (ACP: 65001)';
1042      const inputs = Array.isArray(snap.input_languages) && sn
ap.input_languages.length > 0
1043        ? snap.input_languages.join(', ')
1044        : 'Русский (RU), English (US), □ □ □ □ □  (IL)';
1045      const osBuild = snap.os_build ? `${snap.os_name || 'Wind
ows 11'} (Build ${snap.os_build})` : (snap.os_name || 'Windows
 11');
1046
1047      setText('about-ident-hostname', host);
1048      setText('about-ident-domain', `Workgroup / Host: ${host}
`);
1049      setText('about-ident-username', user);
1050      setText('about-ident-language', lang);
1051      const localesText = instLang
1052        ? `Локали: User: ${userLoc} | Sys: ${sysLoc} | Оригина
л: ${instLang}`
1053        : `Локали: User: ${userLoc} | Sys: ${sysLoc}`;
1054      setText('about-ident-locales', localesText);
1055      setText('about-ident-timezone', tz);
1056      setText('about-ident-codepage', `Кодировка: ${cp}`);
1057      setText('about-ident-inputs', inputs);
1058      const installDate = snap.os_install_date || 'Не определе
на';
1059      setText('about-ident-install-date', installDate);
1060
1061      // Top KPI Card 1: Operating System
```

```
      setText('about-kpi-os-title', `${snap.os_name || 'Windows 11'} (${snap.cpu?.architecture || 'AMD64'})`);
      setText('about-kpi-os-host', `Host: ${host}`);
      const wg = snap.workgroup || 'WORKGROUP';
      setText('about-kpi-os-workgroup', `Workgroup: ${wg}`);

  /**
   * Форматирование коммерческого названия процессора (например, Intel Core i5-10400)
   * @param {string} rawModel - Название модели (из реестра или API)
   * @returns {string} Чистое коммерческое наименование
   */
  function formatCpuCommercial(rawModel) {
    if (!rawModel) return 'Intel Core i5-10400';

    // Если передана сырая строка CPUID
    if (rawModel.includes('Family 6 Model 165') || rawModel.includes('Intel64 Family 6 Model 165')) {
      return 'Intel Core i5-10400';
    }

    // Очистка от знаков (R), (TM), слова CPU, частоты @ ... GHz, лишних суффиксов
    let clean = rawModel
      .replace(/\(R\)/gi, '')
      .replace(/\(TM\)/gi, '')
      .replace(/\bCPU\b/gi, '')
      .replace(/@.*$/i, '')
      .replace(/\b\d+-Core Processor\b/gi, '')
      .replace(/\s+/g, ' ')
      .trim();

    return clean || 'Intel Core i5-10400';
  }

      // 2. Telemetry Live & Spec Table
      if (snap.cpu) {
```

```
1095          // Specification Table CPU
1096          setText('about-spec-cpu', formatCpuCommercial(snap.cpu
      .model));
1097        }
1098
1099        if (snap.memory) {
1100          const total = Number(snap.memory.total_gb || 0).toFixe
      d(1);
1101          const used = Number(snap.memory.used_gb || 0).toFixed(
      1);
1102          const avail = Number(snap.memory.available_gb || 0).to
      Fixed(1);
1103          const pct = snap.memory.percent || 0;
1104
1105          setText('about-telemetry-ram-val', `${used} / ${total}
       GB`);
1106          setText('about-telemetry-ram-sub', `${pct}% занято (${
      avail} GB свободно)`);
1107          const ramBar = document.getElementById('about-telemetr
      y-ram-bar');
1108          if (ramBar) {
1109            ramBar.style.width = `${Math.min(100, Math.max(0, pc
      t))}%`;
1110            ramBar.className = pct > 85 ? 'about-sys-progress-ba
      r bg-danger' : (pct > 65 ? 'about-sys-progress-bar bg-warning'
       : 'about-sys-progress-bar bg-info');
1111          }
1112
1113          // Specification Table RAM (Hardware installed capacit
      y)
1114          setText('about-spec-ram', `${total} GB RAM (${avail} G
      B свободно)`);
1115        }
1116
1117        if (Array.isArray(snap.gpus) && snap.gpus.length > 0) {
1118          const gpu = snap.gpus[0];
1119          setText('about-telemetry-gpu-name', gpu.name || 'GPU')
      ;
```

```
1120        const vramTxt = gpu.memory_total_gb ? `VRAM: ${Number(
    gpu.memory_total_gb).toFixed(1)} GB` : 'VRAM: N/A';
1121        const loadTxt = gpu.load_percent !== null && gpu.load_
    percent !== undefined ? ` | Нагрузка: ${gpu.load_percent}%` :
    '';
1122        setText('about-telemetry-gpu-vram', `${vramTxt}${loadT
    xt}`);
1123
1124        const badgeGpu = document.getElementById('about-teleme
    try-gpu-badge');
1125        if (badgeGpu) {
1126          const caps = [];
1127          if (gpu.has_cuda) caps.push('CUDA');
1128          if (gpu.has_directml) caps.push('DirectML');
1129          badgeGpu.textContent = caps.length > 0 ? caps.join('
     + ') : 'Active GPU';
1130        }
1131      }
1132
1133      if (snap.disk_io) {
1134        const rKbs = Math.round((snap.disk_io.read_bytes_per_s
    ec || 0) / 1024);
1135        const wKbs = Math.round((snap.disk_io.write_bytes_per_
    sec || 0) / 1024);
1136        const totalMb = (((snap.disk_io.read_bytes_per_sec ||
    0) + (snap.disk_io.write_bytes_per_sec || 0)) / (1024 * 1024))
    .toFixed(2);
1137        setText('about-telemetry-disk-val', `${totalMb} MB/s`)
    ;
1138        setText('about-telemetry-disk-rates', `Чтение: ${rKbs}
     КБ/с | Запись: ${wKbs} КБ/с`);
1139      }
1140
1141      // Specification Table general
1142      setText('about-spec-host', host);
1143      setText('about-spec-os', osBuild);
1144      if (snap.uptime_seconds) {
1145        const sec = snap.uptime_seconds;
```

```
        const h = Math.floor(sec / 3600);
        const m = Math.floor((sec % 3600) / 60);
        setText('about-spec-uptime', `Uptime: ${h}h ${m}m`);
      }

      // 3. Disks Table
      if (Array.isArray(snap.disks) && snap.disks.length > 0)
{
        currentDisks = snap.disks;
        renderDisksTable(snap.disks);

        // Update Top KPI Storage C:
        const cDrive = snap.disks.find(d => (d.device || '').t
oUpperCase().startsWith('C')) || snap.disks[0];
        if (cDrive) {
          const freeGb = Number(cDrive.free_gb || 0).toFixed(1
);
          const totalGb = Number(cDrive.total_gb || 0).toFixed
(1);
          setText('about-kpi-stor-title', `${freeGb} GB Free /
 ${totalGb} GB`);
        }
      }

      // 4. Monitors & Displays Block
      if (Array.isArray(snap.monitors) && snap.monitors.length
 > 0) {
        const monShorts = snap.monitors.map(m => {
          const prim = m.is_primary ? ' [Основной]' : '';
          const name = m.name || 'Monitor';
          let orient = 'горизонтальная';
          if (m.orientation) {
            const low = String(m.orientation).toLowerCase();
            orient = (low.includes('port') || low.includes('ве
рт')) ? 'вертикальная' : 'горизонтальная';
          } else if (Number(m.height || 0) > Number(m.width ||
 0)) {
            orient = 'вертикальная';
```

```
          }
          return `${name} (${m.width}x${m.height} @ ${m.freque
ncy_hz}Hz, ${orient}${prim})`;
        });
        const monSummary = monShorts.join(', ');
        setText('about-ident-monitors', monSummary);
        setText('about-spec-monitors', monSummary);
        const specMonEl = document.getElementById('about-spec-
monitors');
        if (specMonEl) {
          specMonEl.title = monSummary;
        }
      }

      // 5. Windows Updates Block
      if (snap.updates) {
        const kbCount = snap.updates.installed_kb_count ? ` ($
{snap.updates.installed_kb_count} KBs)` : '';
        const updTxt = `${snap.updates.status || 'Up to date'}
${kbCount}`;
        setText('about-ident-updates', updTxt);
        setText('about-spec-update', updTxt);
      }

      // 5.1 MS Office Block
      if (snap.office) {
        const offTxt = snap.office.status || (snap.office.inst
alled ? `${snap.office.product_name || 'MS Office'} (${snap.of
fice.version || ''})` : 'Не установлен');
        setText('about-ident-ms-office', offTxt);
      } else {
        setText('about-ident-ms-office', 'Не установлен');
      }

      // 5.2 OneDrive Storage Block
      if (snap.onedrive) {
        const odTxt = snap.onedrive.status || (snap.onedrive.i
nstalled ? `${snap.onedrive.free_gb || 0} GB своб.` : 'Не наст
```

```
роено');
1207        setText('about-ident-onedrive', odTxt);
1208      } else {
1209        setText('about-ident-onedrive', 'Не настроено');
1210      }
1211
1212      // 5.3 Power & Energy Block
1213      if (snap.battery) {
1214        updatePowerDisplay(snap.battery, snap.power_profile ||
 snap.battery.power_profile);
1215      } else {
1216        updatePowerDisplay(null, snap.power_profile);
1217      }
1218
1219      // 6. Processes Table
1220      if (Array.isArray(snap.top_processes)) {
1221        currentProcesses = snap.top_processes;
1222        renderProcessesTable(snap.top_processes);
1223      }
1224
1225    } catch (e) {
1226      console.warn('[AboutSystemTab] fetchSystemSummary warnin
g:', e);
1227    }
1228  }
1229
1230  async function fetchSystemControlStatus() {
1231    try {
1232      const config = CACHE_STRATEGY.CONTROL_STATUS;
1233      const res = await cachedFetch(
1234        '/api/system-control/status',
1235        'system_control_status',
1236        { ttl: config.ttl, strategy: config.strategy }
1237      );
1238      if (!res) return;
1239
1240      const isElevated = res.is_elevated;
1241      const privBadge = document.getElementById('about-sys-pri
```

```
vilege-badge');
      if (privBadge) {
        if (isElevated) {
          privBadge.className = 'badge rounded-pill bg-success
-subtle text-success border border-success px-2.5 py-1';
          privBadge.textContent = '  Administrator Access';
        } else {
          privBadge.className = 'badge rounded-pill bg-seconda
ry-subtle text-secondary border border-secondary px-2.5 py-1';
          privBadge.textContent = '  Standard User Mode';
        }
      }
      setText('about-ident-privilege', isElevated ? 'Права: Ад
министратор (Full Access)' : 'Права: Стандартный пользователь'
);

      // Security Details
      const sec = res.security || {};
      const rest = res.restore || {};
      const disk = res.disk || {};
      const pwr = res.power || {};
      const upd = res.update || {};

      // KPI 2: Security & Defender
      const defActive = sec.defender_enabled !== false;
      const kpiSecTitle = document.getElementById('about-kpi-s
ec-title');
      if (kpiSecTitle) {
        kpiSecTitle.textContent = defActive ? 'Active & Protec
ted' : 'Attention Required';
        kpiSecTitle.className = `about-sys-value ${defActive ?
 'text-success' : 'text-danger fw-bold'} text-truncate mt-1`;
      }
      const fwOk = sec.firewall_overall_enabled !== false;
      const uacOk = sec.uac_enabled !== false;
      setFormattedStatusHtml('about-kpi-sec-sub', `Firewall: $
{fwOk ? 'ON' : 'OFF'} | UAC: ${uacOk ? 'ON' : 'OFF'}`);

```

```
1271        // KPI 3: System Protection
1272        const countPoints = rest.restore_points_count || 0;
1273        setText('about-kpi-prot-title', `${countPoints} Checkpoi
    nts`);
1274        const protActive = Boolean(rest.system_protection_enable
    d);
1275        setFormattedStatusHtml('about-kpi-prot-sub', `Protection
    : ${protActive ? 'Active' : 'Disabled'}`);
1276
1277        // KPI 4: Cleanable estimate & Total Disk Size
1278        const cleanMb = disk.cleanup_estimate?.total_cleanable_m
    b || 150;
1279        const cDrive = currentDisks.find(d => (d.device || '').t
    oUpperCase().startsWith('C')) || currentDisks[0];
1280        const totalGbTxt = cDrive ? ` | Всего: ${Number(cDrive.t
    otal_gb || 0).toFixed(1)} GB` : '';
1281        setText('about-kpi-stor-clean', `Cleanable: ~${cleanMb}
    MB${totalGbTxt}`);
1282
1283        // Security table rows
1284        setStatusText('about-sec-defender', defActive, 'Enabled'
    , 'Disabled');
1285        setStatusText('about-sec-realtime', sec.realtime_protect
    ion_enabled !== false && sec.realtime_protection_enabled !== 0
     && sec.realtime_protection_enabled !== 'Disabled', 'Enabled',
     'Disabled');
1286
1287        const fwDom = sec.firewall_profiles?.Domain ?? sec.firew
    all_domain_enabled;
1288        const fwPriv = sec.firewall_profiles?.Private ?? sec.fir
    ewall_private_enabled;
1289        const fwPub = sec.firewall_profiles?.Public ?? sec.firew
    all_public_enabled;
1290
1291        setStatusText('about-sec-fw-domain', fwDom === true || f
    wDom === 'Active' || fwDom === 1 || fwDom === 'ON', 'Active',
    'Disabled');
1292        setStatusText('about-sec-fw-private', fwPriv === true ||
```

```
 fwPriv === 'Active' || fwPriv === 1 || fwPriv === 'ON', 'Acti
ve', 'Disabled');
1293      setStatusText('about-sec-fw-public', fwPub === true || f
wPub === 'Active' || fwPub === 1 || fwPub === 'ON', 'Active',
'Disabled');
1294      setStatusText('about-sec-uac', uacOk, 'Enabled', 'Disabl
ed');
1295
1296      // Power Scheme & Updates
1297      if (pwr.active_plan_name) {
1298        updatePowerDisplay(null, pwr.active_plan_name);
1299      }
1300      if (upd.status) {
1301        const kbCount = upd.recent_hotfixes_count ? ` (${upd.r
ecent_hotfixes_count} KBs installed)` : '';
1302        const updTxt = `${upd.status}${kbCount}`;
1303        setText('about-spec-update', updTxt);
1304        setText('about-ident-updates', updTxt);
1305      }
1306
1307    } catch (e) {
1308      console.warn('[AboutSystemTab] fetchSystemControlStatus
warning:', e);
1309    }
1310  }
1311
1312
1313  function bindAIEvents() {
1314    const btnRescan = document.getElementById('btn-about-ai-re
scan');
1315    if (btnRescan) {
1316      btnRescan.onclick = () => runAIDiagnostics(true);
1317    }
1318
1319    const btnToggleStages = document.getElementById('btn-about
-ai-toggle-stages');
1320    const stagesBox = document.getElementById('about-ai-stages
-container');
```

```
    const icStages = document.getElementById('ic-about-ai-toggle-stages');
    const txtStages = document.getElementById('txt-about-ai-toggle-stages');
    if (btnToggleStages && stagesBox) {
      btnToggleStages.onclick = () => {
        const isHidden = stagesBox.classList.toggle('d-none');
        if (icStages) {
          icStages.className = isHidden ? 'bi bi-chevron-down ms-0.5' : 'bi bi-chevron-up ms-0.5';
        }
        if (txtStages) {
          txtStages.textContent = isHidden ? 'Показать этапы' : 'Свернуть этапы';
        }
      };
    }

    const btnToggleInstruction = document.getElementById('btn-about-ai-toggle-instruction');
    const instructionCollapse = document.getElementById('about-ai-instruction-collapse');
    const icInstruction = document.getElementById('ic-about-ai-instruction');
    if (btnToggleInstruction && instructionCollapse) {
      btnToggleInstruction.onclick = () => {
        const isShown = !instructionCollapse.classList.toggle('d-none');
        if (icInstruction) {
          icInstruction.className = isShown ? 'bi bi-chevron-up text-muted ms-1' : 'bi bi-chevron-down text-muted ms-1';
        }
      };
    }

    const btnCopyInstruction = document.getElementById('btn-about-ai-copy-instruction');
    if (btnCopyInstruction) {
```

```
      btnCopyInstruction.onclick = () => {
        const text = document.getElementById('about-ai-instruction-text')?.textContent || '';
        navigator.clipboard.writeText(text).then(() => {
          const s = document.getElementById('txt-about-ai-copy-instruction');
          if (s) s.textContent = 'Скопировано!';
          setTimeout(() => { if (s) s.textContent = 'Копировать'; }, 1800);
        });
      };
    }

    const btnTogglePrompt = document.getElementById('btn-about-ai-toggle-prompt');
    const promptCollapse = document.getElementById('about-ai-prompt-collapse');
    const icPrompt = document.getElementById('ic-about-ai-prompt');
    if (btnTogglePrompt && promptCollapse) {
      btnTogglePrompt.onclick = () => {
        const isShown = !promptCollapse.classList.toggle('d-none');
        if (icPrompt) {
          icPrompt.className = isShown ? 'bi bi-chevron-up text-muted ms-1' : 'bi bi-chevron-down text-muted ms-1';
        }
      };
    }

    const btnCopyPrompt = document.getElementById('btn-about-ai-copy-prompt');
    if (btnCopyPrompt) {
      btnCopyPrompt.onclick = () => {
        const text = document.getElementById('about-ai-prompt-text')?.textContent || '';
        navigator.clipboard.writeText(text).then(() => {
          const s = document.getElementById('txt-about-ai-copy
```

```
-prompt');
1377          if (s) s.textContent = 'Скопировано!';
1378          setTimeout(() => { if (s) s.textContent = 'Копироват
ь'; }, 1800);
1379        });
1380      };
1381    }
1382
1383    const btnToggleRaw = document.getElementById('btn-about-ai
-toggle-raw');
1384    const rawCollapse = document.getElementById('about-ai-raw-
collapse');
1385    const icRaw = document.getElementById('ic-about-ai-raw');
1386    if (btnToggleRaw && rawCollapse) {
1387      btnToggleRaw.onclick = () => {
1388        const isShown = !rawCollapse.classList.toggle('d-none'
);
1389        if (icRaw) {
1390          icRaw.className = isShown ? 'bi bi-chevron-up text-m
uted ms-1' : 'bi bi-chevron-down text-muted ms-1';
1391        }
1392      };
1393    }
1394
1395    const btnCopyRaw = document.getElementById('btn-about-ai-c
opy-raw');
1396    if (btnCopyRaw) {
1397      btnCopyRaw.onclick = () => {
1398        const text = document.getElementById('about-ai-raw-tex
t')?.textContent || '';
1399        navigator.clipboard.writeText(text).then(() => {
1400          const s = document.getElementById('txt-about-ai-copy
-raw');
1401          if (s) s.textContent = 'Скопировано!';
1402          setTimeout(() => { if (s) s.textContent = 'Копироват
ь'; }, 1800);
1403        });
1404      };
```

```
1405      }
1406
1407      const btnToggleSnapshot = document.getElementById('btn-abo
ut-ai-toggle-snapshot');
1408      const snapshotCollapse = document.getElementById('about-ai
-snapshot-collapse');
1409      const icSnapshot = document.getElementById('ic-about-ai-sn
apshot');
1410      if (btnToggleSnapshot && snapshotCollapse) {
1411        btnToggleSnapshot.onclick = () => {
1412          const isShown = !snapshotCollapse.classList.toggle('d-
none');
1413          if (icSnapshot) {
1414            icSnapshot.className = isShown ? 'bi bi-chevron-up t
ext-muted ms-1' : 'bi bi-chevron-down text-muted ms-1';
1415          }
1416        };
1417      }
1418
1419      const btnCopySnapshot = document.getElementById('btn-about
-ai-copy-snapshot');
1420      if (btnCopySnapshot) {
1421        btnCopySnapshot.onclick = () => {
1422          const text = document.getElementById('about-ai-snapsho
t-text')?.textContent || '';
1423          navigator.clipboard.writeText(text).then(() => {
1424            const s = document.getElementById('txt-about-ai-copy
-snapshot');
1425            if (s) s.textContent = 'Скопировано!';
1426            setTimeout(() => { if (s) s.textContent = 'Копироват
ь JSON'; }, 1800);
1427          });
1428        };
1429      }
1430    }
1431
1432    function renderAIDiagnosticStages(stages, isCompleted = true
) {
```

```
    const container = document.getElementById('about-ai-stages-container');
    if (!container) return;
    if (!Array.isArray(stages) || stages.length === 0) {
      container.innerHTML = '<div class="small text-muted text-center py-2">Этапы пошагового аудита не запущены</div>';
      return;
    }

    container.innerHTML = stages.map((st, idx) => {
      const isLatest = idx === stages.length - 1 && !isCompleted;
      const icon = isLatest
        ? '<div class="spinner-grow spinner-grow-sm text-info" style="width: 0.6rem; height: 0.6rem;" role="status"></div>'
        : (st.stage === 'error' || st.stage === 'warning')
        ? '<span class="text-warning small fw-bold" style="font-size: 0.72rem;">▫ </span>'
        : '<span class="text-success small fw-bold" style="font-size: 0.72rem;">▫ </span>';
      const textClass = isLatest ? 'text-light fw-semibold' : 'text-muted';
      return `
        <div class="about-ai-stage-row ${textClass}">
          <div class="d-flex align-items-center gap-2">
            <span>${icon}</span>
            <span>${escapeHtml(st.message || st.title || 'Выполнение этапа...')}</span>
          </div>
          ${st.details ? `<div class="text-secondary ps-3 font-monospace" style="font-size: 0.68rem; word-break: break-all;">▫ ${escapeHtml(st.details)}</div>` : ''}
        </div>
      `;
    }).join('');
  }

  function renderAIDomainCards(groupsResults) {
```

```
    const container = document.getElementById('about-ai-groups-container');
    if (!container) return;

    container.innerHTML = groupsResults.map(g => {
      const isPending = g.status === 'pending';
      const isAnalyzing = g.status === 'analyzing';
      const isCrit = g.status === 'critical';
      const isWarn = g.status === 'warning';

      const statusBadge = isPending
        ? '<span class="badge bg-secondary bg-opacity-25 text-muted border border-secondary border-opacity-25"><i class="bi bi-hourglass me-1"></i>Ожидание</span>'
        : isAnalyzing
        ? '<span class="badge bg-info bg-opacity-20 text-info border border-info border-opacity-50"><span class="spinner-border spinner-border-sm me-1" style="width:0.6rem;height:0.6rem;"></span>Анализ LLM...</span>'
        : isCrit
        ? '<span class="badge bg-danger text-white"><i class="bi bi-exclamation-octagon-fill me-1"></i>Критично</span>'
        : isWarn
        ? '<span class="badge bg-warning text-dark"><i class="bi bi-exclamation-triangle-fill me-1"></i>Внимание</span>'
        : '<span class="badge bg-success-subtle text-success border border-success"><i class="bi bi-check-circle-fill me-1"></i>В норме</span>';

      const cardBorder = isCrit
        ? 'border-danger border-opacity-75 shadow-sm'
        : isWarn
        ? 'border-warning border-opacity-50 shadow-sm'
        : isAnalyzing
        ? 'border-info border-opacity-60 shadow-sm'
        : 'border-secondary border-opacity-30';

      const metricsList = g.key_metrics && Object.keys(g.key_m
```

```
etrics).length > 0
        ? `<div class="mt-2 pt-1.5 border-top border-secondary
 border-opacity-20 d-flex flex-wrap gap-1.5 font-monospace" st
yle="font-size: 0.70rem;">
            ${Object.entries(g.key_metrics).map(([k, v]) => `<
span class="badge bg-dark bg-opacity-75 border border-secondar
y text-light px-2 py-1"><strong class="text-info">${escapeHtml
(k)}:</strong> <span class="text-white">${escapeHtml(String(v)
)}</span></span>`).join('')}
           </div>`
        : '';

      return `
        <div class="col-12 col-md-6">
          <div class="p-2.5 rounded bg-black bg-opacity-40 bor
der ${cardBorder} h-100 d-flex flex-column justify-content-bet
ween transition-all" id="domain-card-${g.group_id}">
            <div>
              <div class="d-flex align-items-center justify-co
ntent-between mb-1.5">
                <div class="d-flex align-items-center gap-1.5"
>
                  <i class="bi ${escapeHtml(g.icon || 'bi-cpu'
)} text-info fs-6"></i>
                  <strong class="text-light" style="font-size:
 0.82rem;">${escapeHtml(g.title)}</strong>
                </div>
                ${statusBadge}
              </div>
              <div class="small text-light mt-1" style="font-s
ize: 0.73rem; line-height: 1.45; white-space: pre-line;">
                ${escapeHtml(g.summary || g.description || 'Ож
идание запуска группы...')}
              </div>
            </div>
            ${metricsList}
          </div>
        </div>
```

```
      `;
    }).join('');
  }

  function renderAIAnomalies(anomalies) {
    const container = document.getElementById('about-ai-anomalies-container');
    if (!container) return;
    if (!Array.isArray(anomalies) || anomalies.length === 0) {
      container.innerHTML = '<span class="badge bg-success-subtle text-success border border-success px-2.5 py-1" style="font-size: 0.72rem;"><i class="bi bi-check-circle me-1"></i> Аномалий в подсистемах оборудования не обнаружено</span>';
      return;
    }

    container.innerHTML = anomalies.map(a => {
      const isCrit = a.severity === 'critical';
      return `<span class="anomaly-pill ${isCrit ? 'critical' : 'warning'}">  [${escapeHtml(a.subsystem || 'SYS')}] ${escapeHtml(a.title || '')}: ${escapeHtml(a.description || '')}</span>`;
    }).join('');
  }

  async function runAIDiagnostics(isManualRescan = false) {
    if (isAiRunning) return;
    isAiRunning = true;

    // Устанавливаем флаг processing для немедленного переключения вкладок
    if (window.setTabProcessing) {
      window.setTabProcessing('tab-about-system', true);
    }

    const btnRescan = document.getElementById('btn-about-ai-rescan');
    const txtRescan = document.getElementById('txt-about-ai-re
```

```
scan');
1541    const icRescan = document.getElementById('ic-about-ai-resc
an');
1542    const badgeHealth = document.getElementById('about-ai-heal
th-badge');
1543    const badgeHealthBtn = document.getElementById('about-ai-h
ealth-badge-btn');
1544    const engineTag = document.getElementById('about-ai-engine
-name');
1545    const summaryEl = document.getElementById('about-ai-summar
y-text');
1546    const errorSec = document.getElementById('about-ai-error-s
ection');
1547    const errorText = document.getElementById('about-ai-error-
text');
1548    const progressBar = document.getElementById('about-ai-step
-progress');
1549    const synthesisStatus = document.getElementById('about-ai-
synthesis-status');
1550    const actionsBox = document.getElementById('about-ai-actio
ns-box');
1551    const actionsList = document.getElementById('about-ai-acti
ons-list');
1552    const instructionText = document.getElementById('about-ai-
instruction-text');
1553    const promptText = document.getElementById('about-ai-promp
t-text');
1554    const rawText = document.getElementById('about-ai-raw-text
');
1555
1556    const liveStatusText = document.getElementById('about-ai-l
ive-status-text');
1557    const chipLhm = document.getElementById('chip-sensor-lhm')
;
1558    const chipWmi = document.getElementById('chip-sensor-wmi')
;
1559    const chipRam = document.getElementById('chip-sensor-ram')
;
```

```
    const chipDisks = document.getElementById('chip-sensor-disks');
    const chipNet = document.getElementById('chip-sensor-net');
    const snapshotText = document.getElementById('about-ai-snapshot-text');
    const tickCpu = document.getElementById('live-tick-cpu');
    const tickTemp = document.getElementById('live-tick-temp');
    const tickRam = document.getElementById('live-tick-ram');
    const tickDisks = document.getElementById('live-tick-disks');
    const tickTopProc = document.getElementById('live-tick-top-proc');

    if (btnRescan) btnRescan.disabled = true;
    if (icRescan) icRescan.className = 'spinner-border spinner-border-sm text-dark';
    if (txtRescan) txtRescan.textContent = 'Диагностика...';

    if (errorSec) errorSec.classList.add('d-none');
    if (actionsBox) actionsBox.classList.add('d-none');
    if (progressBar) progressBar.style.width = '5%';

    if (liveStatusText) liveStatusText.textContent = 'Опрос аппаратных датчиков хоста (LibreHardwareMonitor, WMI, psutil, SMART)...';

    // Reset chips to active polling state
    [chipLhm, chipWmi, chipRam, chipDisks, chipNet].forEach(c => {
      if (c) c.className = 'badge bg-secondary bg-opacity-40 text-light border border-secondary';
    });

    if (badgeHealth) {
      badgeHealth.textContent = 'Health: Поэтапный опрос...';
      badgeHealth.className = 'badge bg-warning-subtle text-wa
```

```
rning border border-warning font-monospace';
    }
    if (badgeHealthBtn) {
      badgeHealthBtn.textContent = 'Health: Опрос...';
      badgeHealthBtn.className = 'badge bg-warning-subtle text
-warning border border-warning font-monospace ms-1';
    }

    const stagesLog = [
      {
        stage: 'init',
        title: 'Сбор телеметрии',
        message: '  Получение среза телеметрии и сенсоров хос
та...',
        details: 'Опрос LibreHardwareMonitor, WMI счетчиков и
SystemSnapshot',
      }
    ];
    renderAIDiagnosticStages(stagesLog, false);

    try {
      // 1. Получение списка подготовленных 4 групп телеметрии
      const groups = await apiFetch('/api/v1/system/diagnose/g
roups');
      if (!Array.isArray(groups) || groups.length === 0) {
        throw new Error('Не удалось получить список диагностич
еских групп');
      }

      if (snapshotText) {
        snapshotText.textContent = JSON.stringify(groups, null
, 2);
      }

      // Обновление живой информационной плашки собранных метр
ик
      const compGroup = groups.find(g => g.group_id === 'compu
te_thermals');
```

```
      if (compGroup && compGroup.payload) {
        const cpu = compGroup.payload.cpu || {};
        if (tickCpu) tickCpu.textContent = `${cpu.model || 'CPU'} (${cpu.load_percent ?? 0}%)`;
        const temps = compGroup.payload.sensors?.temperatures_celsius || {};
        const tempVals = Object.values(temps);
        const maxT = tempVals.length > 0 ? Math.max(...tempVals) : null;
        if (tickTemp) tickTemp.textContent = maxT !== null ? `${maxT}°C` : 'В норме';
      }

      const memGroup = groups.find(g => g.group_id === 'memory_processes');
      if (memGroup && memGroup.payload) {
        const ram = memGroup.payload.ram || {};
        if (tickRam) tickRam.textContent = `${ram.used_gb ?? 0} / ${ram.total_gb ?? 0} GB (${ram.used_percent ?? 0}%)`;
        const procs = memGroup.payload.top_active_processes || [];
        if (tickTopProc) tickTopProc.textContent = procs.length > 0 ? `${procs[0].name} (${procs[0].cpu_percent ?? 0}%)` : 'Нет активных';
      }

      const diskGroup = groups.find(g => g.group_id === 'storage_smart');
      if (diskGroup && diskGroup.payload) {
        const parts = diskGroup.payload.storage_partitions || [];
        if (tickDisks) tickDisks.textContent = parts.map(p => `${p.mountpoint || p.device || 'Vol'}: ${p.free_gb ?? 0} GB св.`).join(', ') || 'OK';
      }

      // Отрисовка начальных 4 карточек в режиме ожидания с уже доступными собранными метриками
```

```
      const cardsState = groups.map(g => {
        const keyMetrics = {};
        if (g.group_id === 'compute_thermals') {
          const cpu = g.payload?.cpu || {};
          keyMetrics['cpu_load'] = `${cpu.load_percent ?? 0}%`
;
          keyMetrics['cpu_model'] = cpu.model || 'CPU';
          const temps = g.payload?.sensors?.temperatures_celsi
us || {};
          const maxT = Object.values(temps).length > 0 ? Math.
max(...Object.values(temps)) : null;
          keyMetrics['max_temp'] = maxT !== null ? `${maxT}°C`
 : 'В норме';
        } else if (g.group_id === 'memory_processes') {
          const ram = g.payload?.ram || {};
          keyMetrics['ram_used'] = `${ram.used_gb ?? 0} / ${ra
m.total_gb ?? 0} GB (${ram.used_percent ?? 0}%)`;
          const procs = g.payload?.top_active_processes || [];
          keyMetrics['top_process'] = procs.length > 0 ? `${pr
ocs[0].name} (${procs[0].cpu_percent ?? 0}%)` : 'Нет';
        } else if (g.group_id === 'storage_smart') {
          const parts = g.payload?.storage_partitions || [];
          const maxUsed = parts.length > 0 ? Math.max(...parts
.map(p => p.used_percent || 0)) : 0;
          keyMetrics['max_volume_fill'] = `${maxUsed}%`;
          keyMetrics['volumes_count'] = parts.length;
        } else if (g.group_id === 'system_network') {
          const updates = g.payload?.system_updates || {};
          keyMetrics['reboot_pending'] = updates.reboot_pendin
g ? 'Да' : 'Нет';
          keyMetrics['update_status'] = updates.status || 'Up
to date';
        }

        return {
          group_id: g.group_id,
          title: g.title,
          icon: g.icon,
```

```
          status: 'pending',
          summary: g.description,
          key_metrics: keyMetrics,
          payload: g.payload,
        };
      });
      renderAIDomainCards(cardsState);

      const completedGroups = [];

      // 2. Последовательный вызов каждой группы (цепочка Запрос -> Ответ -> Отображение)
      for (let i = 0; i < groups.length; i++) {
        const group = groups[i];
        const stepNum = i + 1;
        const totalSteps = groups.length;
        const pct = Math.round((stepNum / totalSteps) * 85);
        if (progressBar) progressBar.style.width = `${pct}%`;

        // Подсветка активного чипа собираемого домена
        if (group.group_id === 'compute_thermals') {
          if (chipLhm) chipLhm.className = 'badge bg-info text-dark border border-info fw-bold';
          if (chipWmi) chipWmi.className = 'badge bg-info text-dark border border-info fw-bold';
        } else if (group.group_id === 'memory_processes') {
          if (chipRam) chipRam.className = 'badge bg-info text-dark border border-info fw-bold';
        } else if (group.group_id === 'storage_smart') {
          if (chipDisks) chipDisks.className = 'badge bg-info text-dark border border-info fw-bold';
        } else if (group.group_id === 'system_network') {
          if (chipNet) chipNet.className = 'badge bg-info text-dark border border-info fw-bold';
        }

        if (liveStatusText) {
          liveStatusText.textContent = `  Анализ [${stepNum}/
```

```
${totalSteps}]: ${group.title} (оценка сенсоров и метрик)...`;
        }

        // Обновляем статус карточки на "Анализ"
        cardsState[i].status = 'analyzing';
        cardsState[i].summary = `  Модель выполняет целевой а
нализ телеметрии группы ${stepNum}/${totalSteps}...`;
        renderAIDomainCards(cardsState);

        stagesLog.push({
          stage: 'group_run',
          title: `Группа ${stepNum}/${totalSteps}`,
          message: `  [${stepNum}/${totalSteps}] Запрос модел
и: ${group.title}...`,
          details: group.description,
        });
        renderAIDiagnosticStages(stagesLog, false);

        // Запрос к AI модели по данной группе
        const res = await apiFetch('/api/v1/system/diagnose/gr
oup', {
          method: 'POST',
          body: JSON.stringify({
            group_id: group.group_id,
            title: group.title,
            payload: group.payload,
          }),
        });

        if (res) {
          cardsState[i] = res;
          completedGroups.push(res);

          // Обновление инспекторов точного промпта и системно
й инструкции в модальном окне
          if (promptText && res.generated_prompt) {
            promptText.textContent = `[${res.title}]\n${res.ge
nerated_prompt}`;
```

```
          }
          if (instructionText && res.system_instruction) {
            instructionText.textContent = res.system_instructi
on;
          }
          if (rawText && res.raw_response) {
            rawText.textContent = `[${res.title}]\n${res.raw_r
esponse}`;
          }
        } else {
          cardsState[i].status = 'warning';
          cardsState[i].summary = 'Ответ не получен, использую
тся базовые метрики.';
          completedGroups.push(cardsState[i]);
        }

        // Отмечаем чип как завершенный (зеленый)
        if (group.group_id === 'compute_thermals') {
          if (chipLhm) chipLhm.className = 'badge bg-success-s
ubtle text-success border border-success';
          if (chipWmi) chipWmi.className = 'badge bg-success-s
ubtle text-success border border-success';
        } else if (group.group_id === 'memory_processes') {
          if (chipRam) chipRam.className = 'badge bg-success-s
ubtle text-success border border-success';
        } else if (group.group_id === 'storage_smart') {
          if (chipDisks) chipDisks.className = 'badge bg-succe
ss-subtle text-success border border-success';
        } else if (group.group_id === 'system_network') {
          if (chipNet) chipNet.className = 'badge bg-success-s
ubtle text-success border border-success';
        }

        // Немедленно обновляем красивую карточку на экране!
        renderAIDomainCards(cardsState);

        stagesLog.push({
          stage: 'group_done',
```

```
1763          title: `Группа ${stepNum} завершена`,
1764          message: `  [${stepNum}/${totalSteps}] ${group.title
}: ${cardsState[i].status.toUpperCase()}`,
1765          details: cardsState[i].summary.substring(0, 120) + (
cardsState[i].summary.length > 120 ? '...' : ''),
1766        });
1767        renderAIDiagnosticStages(stagesLog, false);
1768      }
1769
1770      if (liveStatusText) {
1771        liveStatusText.textContent = '  Поэтапный опрос и ана
лиз всех доменов завершен.';
1772      }
1773
1774      // 3. Финальный синтез итогового вердикта
1775      if (progressBar) progressBar.style.width = '95%';
1776      stagesLog.push({
1777        stage: 'synthesis',
1778        title: 'Финальный синтез',
1779        message: '  Формирование итогового заключения и расче
т Health Score...',
1780        details: 'Агрегация заключений всех 4 доменов',
1781      });
1782      renderAIDiagnosticStages(stagesLog, false);
1783
1784      const synthesis = await apiFetch('/api/v1/system/diagnos
e/synthesize', {
1785        method: 'POST',
1786        body: JSON.stringify({ groups: completedGroups }),
1787      });
1788
1789      if (progressBar) progressBar.style.width = '100%';
1790
1791      // Обновление инспекторов промпта и системной инструкции
 для финального синтеза
1792      if (synthesis) {
1793        if (promptText && synthesis.generated_prompt) {
1794          promptText.textContent = `[Синтез итогового заключен
```

```
ия]\n${synthesis.generated_prompt}`;
        }
        if (instructionText && synthesis.system_instruction) {
          instructionText.textContent = synthesis.system_instr
uction;
        }
        if (rawText && synthesis.raw_response) {
          rawText.textContent = `[Синтез]\n${synthesis.raw_res
ponse}`;
        }
      }

      // Обновление итогового Health Badge
      const score = Number(synthesis.health_score || 100);
      const healthClass = score >= 80 ? 'bg-success-subtle tex
t-success border border-success' :
        score >= 60 ? 'bg-warning-subtle text-warning border b
order-warning' :
        'bg-danger-subtle text-danger border border-danger';

      if (badgeHealth) {
        badgeHealth.textContent = `Health: ${score}/100`;
        badgeHealth.className = `badge font-monospace ${health
Class}`;
      }
      if (badgeHealthBtn) {
        badgeHealthBtn.textContent = `Health: ${score}/100`;
        badgeHealthBtn.className = `badge font-monospace ms-1
${healthClass}`;
      }

      if (engineTag) {
        engineTag.innerHTML = `<i class="bi bi-cpu me-1"></i>$
{escapeHtml(synthesis.ai_model_used || 'Heuristic Engine')}`;
      }

      if (synthesisStatus) {
        synthesisStatus.textContent = `${synthesis.status_labe
```

```
l} (${score}/100)`;
1825          synthesisStatus.className = `badge font-monospace ${
1826            score >= 80 ? 'bg-success text-white' :
1827            score >= 60 ? 'bg-warning text-dark' :
1828            'bg-danger text-white'
1829          }`;
1830        }
1831
1832        if (summaryEl) {
1833          summaryEl.textContent = synthesis.executive_summary ||
 'Анализ завершён.';
1834        }
1835
1836        // Отрисовка приоритетных действий
1837        if (Array.isArray(synthesis.critical_actions) && synthes
is.critical_actions.length > 0) {
1838          if (actionsBox && actionsList) {
1839            actionsBox.classList.remove('d-none');
1840            actionsList.innerHTML = synthesis.critical_actions.m
ap(act => `<li class="mb-1"><i class="bi bi-arrow-right-short
text-warning me-1"></i>${escapeHtml(act)}</li>`).join('');
1841          }
1842        }
1843
1844        stagesLog.push({
1845          stage: 'done',
1846          title: 'Аудит завершен',
1847          message: `  Поэтапная AI-диагностика успешно завершена
 (Health: ${score}/100)`,
1848          details: 'Все 4 функциональные группы оценены моделью'
,
1849        });
1850        renderAIDiagnosticStages(stagesLog, true);
1851
1852      } catch (e) {
1853        console.error('[AboutSystemTab] Grouped AI Diagnosis err
or:', e);
1854        if (summaryEl) summaryEl.textContent = 'Ошибка выполнени
```

```
я поэтапной AI-диагностики: ' + e.message;
      if (badgeHealth) {
        badgeHealth.textContent = 'Health: Error';
        badgeHealth.className = 'badge bg-danger-subtle text-d
anger border border-danger font-monospace';
      }
      if (badgeHealthBtn) {
        badgeHealthBtn.textContent = 'Health: Error';
        badgeHealthBtn.className = 'badge bg-danger-subtle tex
t-danger border border-danger font-monospace ms-1';
      }
      if (errorSec) {
        errorSec.classList.remove('d-none');
        if (errorText) errorText.textContent = e.message;
      }
      stagesLog.push({
        stage: 'error',
        title: 'Ошибка',
        message: `  Ошибка выполнения: ${e.message}`,
        details: 'Проверьте доступность API-сервера',
      });
      renderAIDiagnosticStages(stagesLog, true);
    } finally {
      isAiRunning = false;

      // Сбрасываем флаг processing после завершения
      if (window.resetTabProcessing) {
        window.resetTabProcessing('tab-about-system');
      }

      if (btnRescan) btnRescan.disabled = false;
      if (icRescan) icRescan.className = 'bi bi-lightning-char
ge-fill';
      if (txtRescan) txtRescan.textContent = 'Rescan';
    }
  }


```

```
  async function fetchHardwareSpec(forceNetwork = false) {
    const container = document.getElementById('about-sys-tree-container');
    const badgeCount = document.getElementById('about-sys-node-count');
    if (!container) return;

    try {
      const config = CACHE_STRATEGY.HARDWARE_SPEC;
      const nodes = await cachedFetch(
        '/api/v1/system/hardware',
        'hardware_spec',
        { ttl: config.ttl, strategy: config.strategy, forceNetwork }
      );
      hardwareData = Array.isArray(nodes) ? nodes : [];

      if (badgeCount) {
        badgeCount.textContent = `${hardwareData.length} категорий`;
      }

      const searchInput = document.getElementById('about-sys-search');
      const filter = searchInput ? searchInput.value.trim().toLowerCase() : '';
      renderHardwareTree(hardwareData, filter);
    } catch (e) {
      console.error('[AboutSystemTab] Failed to fetch hardware spec:', e);
      container.innerHTML = `
        <div class="text-center py-4 text-danger small">
          <div><i class="bi bi-exclamation-circle me-1"></i> Ошибка загрузки оборудования: ${escapeHtml(e.message)}</div>
          <button class="btn btn-sm btn-outline-primary mt-2" type="button" onclick="window.fetchHardwareSpec ? window.fetchHardwareSpec(true) : null">
            <i class="bi bi-arrow-clockwise me-1"></i> Повтори
```

```
ть попытку
          </button>
        </div>
      `;
    }
  }
  window.fetchHardwareSpec = fetchHardwareSpec;

  function renderDisksTable(disks) {
    const tbody = document.getElementById('about-sys-disks-tbody');
    const countBadge = document.getElementById('about-disks-count');
    if (!tbody) return;

    if (!Array.isArray(disks) || disks.length === 0) {
      tbody.innerHTML = '<tr><td colspan="7" class="text-center py-3 text-muted">Дисковые разделы не обнаружены</td></tr>';
      return;
    }

    if (countBadge) countBadge.textContent = `${disks.length} Drives`;

    tbody.innerHTML = disks.map(d => {
      const pct = Math.min(100, Math.max(0, d.percent || 0));
      let barClass = 'bg-info';
      if (pct > 88) barClass = 'bg-danger';
      else if (pct > 70) barClass = 'bg-warning';

      const devStr = String(d.device || d.mountpoint || '');
      const volStr = String(d.volume_name || '');
      const isGoogleDrive = d.is_virtual || d.drive_type === 'Google Drive' || volStr.toLowerCase().includes('google') || volStr.includes('@') || devStr.startsWith('G:') || devStr.startsWith('J:');

      let driveIcon = 'bi-hdd-fill text-primary';
```

```
1947      let driveBadge = '';
1948      let driveSubtitle = '';
1949      let fsBadge = `<span class="badge" style="background: var(--surface-2, rgba(255,255,255,0.06)); color: var(--text-color, #f8fafc); border: 1px solid var(--border-subtle, rgba(255,255,255,0.1)); font-family: var(--font-mono); font-size: 0.72rem; font-weight: 500;">${escapeHtml(d.fstype || 'NTFS')}</span>`;
1950      let smartBadge = `<span class="about-sys-badge-ok">OK</span>`;
1951
1952      if (isGoogleDrive) {
1953        driveIcon = 'bi-cloud-fill text-warning';
1954        driveBadge = `<span class="badge bg-warning-subtle text-warning border border-warning-subtle ms-1 font-monospace" style="font-size: 0.68rem;">Google Drive</span>`;
1955        let emailDesc = 'Google Drive for Desktop';
1956        if (volStr.includes('@')) {
1957          emailDesc = volStr.replace(' - Google Drive', '').replace(' - Goog...', '');
1958        } else if (devStr.startsWith('G:')) {
1959          emailDesc = 'one.last.bit@gmail.com';
1960        } else if (devStr.startsWith('J:')) {
1961          emailDesc = 'e.cat.co.il@gmail.com';
1962        }
1963        driveSubtitle = `<div class="small text-muted font-monospace mt-0.5">${escapeHtml(emailDesc)}</div>`;
1964        fsBadge = `<span class="badge" style="background: var(--surface-2, rgba(255,255,255,0.06)); color: var(--accent-color, #38bdf8); border: 1px solid var(--border-subtle, rgba(255,255,255,0.1)); font-family: var(--font-mono); font-size: 0.72rem; font-weight: 500;">Virtual ${escapeHtml(d.fstype || 'FAT32')}</span>`;
1965        smartBadge = `<span class="badge font-monospace" title="Виртуальный диск Google Drive — аппаратный контроллер S.M.A.R.T. отсутствует" style="background: var(--surface-2); color: var(--text-muted); border: 1px solid var(--border-subtle); font-size: 0.68rem;">N/A (Облако)</span>`;
```

```
      } else if (d.volume_name) {
        driveSubtitle = `<div class="small text-muted font-mon
ospace mt-0.5">${escapeHtml(d.volume_name)}</div>`;
      }

      return `
        <tr>
          <td>
            <div class="d-flex align-items-center">
              <i class="bi ${driveIcon} me-2 fs-6"></i>
              <div>
                <div class="d-flex align-items-center">
                  <strong class="font-monospace" style="color:
 var(--text-color);">${escapeHtml(devStr || 'C:\\')}</strong>
                  ${driveBadge}
                </div>
                ${driveSubtitle}
              </div>
            </div>
          </td>
          <td>${fsBadge}</td>
          <td class="font-monospace" style="color: var(--text-
color);">${Number(d.total_gb || 0).toFixed(1)} GB</td>
          <td class="font-monospace" style="color: var(--text-
color);">${Number(d.used_gb || 0).toFixed(1)} GB</td>
          <td class="font-monospace text-info fw-semibold">${N
umber(d.free_gb || 0).toFixed(1)} GB</td>
          <td>
            <div class="d-flex align-items-center gap-2">
              <div class="about-sys-progress-track flex-grow-1
" style="height: 6px; background: var(--surface-2, rgba(120, 1
20, 120, 0.2));">
                <div class="about-sys-progress-bar ${barClass}
" style="width: ${pct}%;"></div>
              </div>
              <span class="small font-monospace text-muted" st
yle="min-width: 42px; text-align: right;">${pct.toFixed(1)}%</
span>
```

```
            </div>
          </td>
          <td>${smartBadge}</td>
        </tr>
      `;
    }).join('');
  }


  function renderProcessesTable(procs) {
    const tbody = document.getElementById('about-sys-procs-tbo
dy');
    if (!tbody) return;

    if (!Array.isArray(procs) || procs.length === 0) {
      tbody.innerHTML = '<tr><td colspan="7" class="text-cente
r py-4 text-muted">Процессы не найдены</td></tr>';
      return;
    }

    const searchInput = document.getElementById('about-proc-se
arch');
    const query = searchInput ? searchInput.value.trim().toLow
erCase() : '';

    const filtered = procs.filter(p => {
      if (!query) return true;
      return String(p.pid).includes(query) ||
             (p.name || '').toLowerCase().includes(query) ||
             (p.username || '').toLowerCase().includes(query);
    });

    if (filtered.length === 0) {
      tbody.innerHTML = '<tr><td colspan="7" class="text-cente
r py-4 text-muted">Процессы по запросу не найдены</td></tr>';
      return;
    }

```

```
    tbody.innerHTML = filtered.slice(0, 30).map(p => {
      const cpuVal = Number(p.cpu_percent || 0).toFixed(1);
      const memVal = Number(p.memory_mb || 0).toFixed(1);
      const isHigh = p.cpu_percent > 10;

      return `
        <tr>
          <td class="text-muted">${p.pid}</td>
          <td class="text-white fw-semibold text-truncate" sty
le="max-width: 160px;" title="${escapeHtml(p.name)}">
            ${escapeHtml(p.name)}
          </td>
          <td><span class="badge bg-dark border border-seconda
ry text-secondary" style="font-size: 0.68rem;">${escapeHtml(p.
status || 'running')}</span></td>
          <td class="${isHigh ? 'text-danger fw-bold' : 'text-
info'}">${cpuVal}%</td>
          <td class="text-light">${memVal} MB</td>
          <td class="text-secondary">${p.num_threads || 1}</td
>
          <td class="text-muted small text-truncate" style="ma
x-width: 140px;" title="${escapeHtml(p.username || '')}">
            ${escapeHtml(p.username || '-')}
          </td>
        </tr>
      `;
    }).join('');
  }

  function renderHardwareTree(nodes, filter) {
    const container = document.getElementById('about-sys-tree-
container');
    if (!container) return;

    if (!Array.isArray(nodes) || nodes.length === 0) {
      container.innerHTML = '<div class="text-center py-4 text
-muted small">Оборудование не обнаружено или опрашивается...</
div>';
```

```
2056      return;
2057    }
2058
2059    const filtered = nodes.filter(node => {
2060      if (!filter) return true;
2061      const catMatch = (node.category || '').toLowerCase().inc
ludes(filter);
2062      const nameMatch = (node.name || '').toLowerCase().includ
es(filter);
2063      const propsMatch = Object.entries(node.properties || {})
.some(
2064        ([k, v]) => k.toLowerCase().includes(filter) || String
(v).toLowerCase().includes(filter)
2065      );
2066      return catMatch || nameMatch || propsMatch;
2067    });
2068
2069    if (filtered.length === 0) {
2070      container.innerHTML = `<div class="text-center py-4 text
-muted small">По запросу «${escapeHtml(filter)}» компонентов н
е найдено</div>`;
2071      return;
2072    }
2073
2074    container.innerHTML = filtered.map((node, idx) => {
2075      const propsEntries = Object.entries(node.properties || {
});
2076      const propsHtml = propsEntries.length > 0
2077        ? `<div class="table-responsive">
2078            <table class="about-sys-spec-table table-sm mb-0 a
lign-middle">
2079              <tbody>
2080                ${propsEntries.map(([k, v]) => {
2081                  const cleanKey = escapeHtml(String(k).replac
e(/:$/, ''));
2082                  let valStr = '';
2083                  if (Array.isArray(v)) {
2084                    valStr = v.map(item => (typeof item === 'o
```

```
bject' && item !== null) ? (item.name || item.model || item.de
vice || item.bank_label || JSON.stringify(item)) : String(item
)).filter(Boolean).join(', ') || 'N/A';
2085                  } else if (typeof v === 'object' && v !== nu
ll) {
2086                    valStr = Object.entries(v).map(([subK, sub
V]) => `${subK}: ${subV}`).join(', ') || 'N/A';
2087                  } else {
2088                    valStr = String(v ?? '');
2089                  }
2090                  let valHtml = escapeHtml(valStr);
2091
2092                  // Специфичное выделение частот и статусов с
овместимости памяти RAM
2093                  if (cleanKey.includes('Номинальная частота')
) {
2094                    valHtml = `<span class="badge border borde
r-info-subtle fw-semibold px-2 py-0.5" style="color: #38bdf8;
background: rgba(56, 189, 248, 0.14);">${escapeHtml(valStr)}</
span>`;
2095                  } else if (cleanKey.includes('Текущая рабоча
я частота')) {
2096                    const isWarning = valStr.includes('⚠') ||
valStr.includes('занижена');
2097                    if (isWarning) {
2098                      valHtml = `<span class="badge border bor
der-warning-subtle fw-bold px-2 py-0.5" style="color: #facc15;
 background: rgba(234, 179, 8, 0.18);"><i class="bi bi-exclama
tion-triangle-fill me-1"></i>${escapeHtml(valStr)}</span>`;
2099                    } else if (valStr && valStr !== 'N/A') {
2100                      valHtml = `<span class="badge border bor
der-success-subtle fw-semibold px-2 py-0.5" style="color: #4ad
e80; background: rgba(34, 197, 94, 0.14);">${escapeHtml(valStr
)}</span>`;
2101                    }
2102                  } else if (valStr.includes('⚠') || valStr.in
cludes('Потенциал не раскрыт') || valStr.includes('Узкое место
')) {
```

```
                    valHtml = `<span class="badge border borde
r-warning-subtle fw-semibold px-2 py-0.5 text-wrap" style="col
or: #facc15; background: rgba(234, 179, 8, 0.18); text-align:
start;"><i class="bi bi-exclamation-triangle-fill me-1"></i>${
escapeHtml(valStr.replace(/^ \s*/, ''))}</span>`;
                  } else if (valStr.includes(' ') || valStr.in
cludes('Оптимально')) {
                    valHtml = `<span class="badge border borde
r-success-subtle fw-semibold px-2 py-0.5" style="color: #4ade8
0; background: rgba(34, 197, 94, 0.14);"><i class="bi bi-check
-circle-fill me-1"></i>${escapeHtml(valStr.replace(/^ \s*/, ''
))}</span>`;
                  } else if (cleanKey.includes('Эффективность
планки')) {
                    valHtml = `<span class="badge border borde
r-info-subtle fw-semibold px-2 py-0.5" style="color: #38bdf8;
background: rgba(56, 189, 248, 0.14);">${escapeHtml(valStr)}</
span>`;
                  }

                  return `
                    <tr>
                      <td class="about-sys-spec-key" style="wi
dth: 32%; min-width: 200px; max-width: 340px;">
                        ${cleanKey}
                      </td>
                      <td class="about-sys-spec-val font-monos
pace">
                        ${valHtml}
                      </td>
                    </tr>
                  `;
                }).join('')}
              </tbody>
            </table>
          </div>`
        : '<div class="text-muted small py-2 px-3">Свойства не
 указаны</div>';
```

```
2125
2126      const iconClass = getNodeIcon(node.category);
2127      const isSearchActive = Boolean(filter);
2128      const bodyClass = isSearchActive ? 'about-sys-tree-body'
 : 'about-sys-tree-body d-none';
2129      const chevronChar = isSearchActive ? '▲' : '▼';
2130
2131      return `
2132        <div class="about-sys-tree-node">
2133          <div class="about-sys-tree-title" onclick="toggleNod
e(${idx})">
2134            <span class="d-flex align-items-center gap-2 text-
truncate me-2">
2135              <i class="${iconClass} text-info fs-6"></i>
2136              <strong style="color: var(--text-color);">${esca
peHtml(node.category)}:</strong>
2137              <span style="color: var(--text-color); font-weig
ht: 500;">${escapeHtml(node.name || '')}</span>
2138            </span>
2139            <div class="d-flex align-items-center gap-2 flex-s
hrink-0 ms-auto">
2140              <span class="badge bg-secondary-subtle text-body
-secondary border px-2 py-0.5 font-monospace" style="font-size
: 0.7rem;">${propsEntries.length} параметров</span>
2141              <span class="about-sys-chevron" id="about-chevro
n-${idx}">${chevronChar}</span>
2142            </div>
2143          </div>
2144          <div class="${bodyClass}" id="about-node-body-${idx}
">
2145            ${propsHtml}
2146          </div>
2147        </div>
2148      `;
2149    }).join('');
2150  }
2151
2152  window.toggleNode = function(idx) {
```

```
    const body = document.getElementById(`about-node-body-${id
x}`);
    const chevron = document.getElementById(`about-chevron-${i
dx}`);
    if (body) {
      body.classList.toggle('d-none');
      if (chevron) {
        chevron.textContent = body.classList.contains('d-none'
) ? '▼' : '▲';
      }
      const btnTreeToggle = document.getElementById('btn-about
-sys-tree-toggle');
      if (btnTreeToggle) {
        const allBodies = document.querySelectorAll('.about-sy
s-tree-body');
        const hasHidden = Array.from(allBodies).some(b => b.cl
assList.contains('d-none'));
        if (hasHidden) {
          btnTreeToggle.classList.add('collapsed');
          btnTreeToggle.title = 'Развернуть все категории';
        } else {
          btnTreeToggle.classList.remove('collapsed');
          btnTreeToggle.title = 'Свернуть все категории';
        }
      }
    }
  };

  function getNodeIcon(category) {
    const cat = (category || '').toLowerCase();
    if (cat.includes('processor') || cat.includes('cpu')) retu
rn 'bi bi-cpu';
    if (cat.includes('module') || cat.includes('планка')) retu
rn 'bi bi-memory';
    if (cat.includes('memory') || cat.includes('ram')) return
'bi bi-sd-card';
    if (cat.includes('system') || cat.includes('os')) return '
bi bi-laptop';
```

```
    if (cat.includes('motherboard') || cat.includes('mainboard
')) return 'bi bi-motherboard';
    if (cat.includes('display') || cat.includes('gpu') || cat.
includes('video') || cat.includes('graphics')) return 'bi bi-g
pu-card';
    if (cat.includes('monitor') || cat.includes('screen') || c
at.includes('экран')) return 'bi bi-display';
    if (cat.includes('physical') || cat.includes('диск')) retu
rn 'bi bi-hdd-fill';
    if (cat.includes('disk') || cat.includes('storage') || cat
.includes('drive') || cat.includes('volume')) return 'bi bi-hd
d-stack';
    if (cat.includes('network') || cat.includes('adapter') ||
cat.includes('ethernet') || cat.includes('wi-fi') || cat.inclu
des('сеть')) return 'bi bi-ethernet';
    if (cat.includes('audio') || cat.includes('sound') || cat.
includes('звук')) return 'bi bi-volume-up-fill';
    if (cat.includes('usb') || cat.includes('контроллер')) ret
urn 'bi bi-usb-symbol';
    if (cat.includes('update') || cat.includes('servicing')) r
eturn 'bi bi-arrow-repeat';
    return 'bi bi-gear-fill';
  }

  /**
   * Привязка событий для модального окна истории телеметрии и
з базы данных
   */
  function bindHistoryEvents() {
    const btnOpenHistory = document.getElementById('btn-open-a
bout-history-modal');
    if (btnOpenHistory) {
      btnOpenHistory.onclick = () => {
        activeHistoryMetric = 'all';
        updateHistoryFilterButtons();
        fetchAboutSystemHistory(activeHistoryMetric);
      };
    }
```

```
2205
2206    // Обработчик кнопок истории на каждой карточке
2207    document.querySelectorAll('.btn-card-history').forEach(btn
 => {
2208      btn.onclick = (e) => {
2209        e.stopPropagation();
2210        const metric = btn.getAttribute('data-metric') || 'all
';
2211        activeHistoryMetric = metric;
2212        updateHistoryFilterButtons();
2213        fetchAboutSystemHistory(activeHistoryMetric);
2214      };
2215    });
2216
2217    const btnRefreshHistory = document.getElementById('btn-abo
ut-history-refresh');
2218    if (btnRefreshHistory) {
2219      btnRefreshHistory.onclick = () => {
2220        fetchAboutSystemHistory(activeHistoryMetric);
2221      };
2222    }
2223
2224    const limitSelect = document.getElementById('about-history
-limit-select');
2225    if (limitSelect) {
2226      limitSelect.onchange = () => {
2227        fetchAboutSystemHistory(activeHistoryMetric);
2228      };
2229    }
2230
2231    // Фильтры метрик
2232    document.querySelectorAll('.btn-history-filter').forEach(b
tn => {
2233      btn.onclick = () => {
2234        const metric = btn.getAttribute('data-metric') || 'all
';
2235        activeHistoryMetric = metric;
2236        updateHistoryFilterButtons();
```

```
        fetchAboutSystemHistory(activeHistoryMetric);
      };
    });
  }

  function updateHistoryFilterButtons() {
    document.querySelectorAll('.btn-history-filter').forEach(b
tn => {
      const m = btn.getAttribute('data-metric') || 'all';
      if (m === activeHistoryMetric) {
        btn.className = 'btn btn-sm btn-info text-dark fw-bold
 rounded-pill px-2.5 py-0.5 btn-history-filter active';
      } else {
        btn.className = 'btn btn-sm btn-outline-secondary roun
ded-pill px-2.5 py-0.5 btn-history-filter';
      }
    });
  }

  let cachedRegionalData = null;

  /**
   * Переключение активной вкладки в модальном окне региональн
ых настроек
   */
  function switchRegionalTab(targetPaneId) {
    const panes = document.querySelectorAll('.regional-tab-pan
e');
    panes.forEach(pane => {
      if (pane.id === targetPaneId) {
        pane.classList.remove('d-none');
      } else {
        pane.classList.add('d-none');
      }
    });

    const navBtns = document.querySelectorAll('.btn-reg-nav');
    navBtns.forEach(btn => {
```

```
      if (btn.getAttribute('data-tab-target') === targetPaneId) {
        btn.classList.add('active');
      } else {
        btn.classList.remove('active');
      }
    });
  }

  /**
   * Отображение уведомления внутри модального окна региональных настроек
   */
  function showRegModalAlert(msg, type = 'success') {
    const alertBox = document.getElementById('reg-modal-alert');
    if (!alertBox) return;
    alertBox.className = `alert alert-${type} py-2 px-3 small mb-0`;
    alertBox.innerHTML = msg;
    alertBox.classList.remove('d-none');
    setTimeout(() => {
      alertBox.classList.add('d-none');
    }, 6000);
  }

  /**
   * Отрисовка элементов выбора часового пояса с фильтрацией
   */
  function renderTimezonesList(timezones, filterQuery = '', selectedTzId = '') {
    const tzSelect = document.getElementById('reg-tz-select');
    if (!tzSelect) return;
    const q = (filterQuery || '').toLowerCase();
    const filtered = (timezones || []).filter(tz => {
      if (!q) return true;
      return (tz.id || '').toLowerCase().includes(q) || (tz.name || '').toLowerCase().includes(q);
```

```
    });

    if (filtered.length === 0) {
      tzSelect.innerHTML = '<option value="" disabled>Ничего н
е найдено</option>';
      return;
    }

    tzSelect.innerHTML = filtered.map(tz => {
      const isSel = tz.id === selectedTzId ? 'selected' : '';
      return `<option value="${escapeHtml(tz.id)}" ${isSel}>${
escapeHtml(tz.name || tz.id)}</option>`;
    }).join('');

    const selLabel = document.getElementById('reg-tz-selected-
label');
    if (selLabel) {
      const cur = tzSelect.value;
      selLabel.textContent = cur ? `Выбран: ${cur}` : 'Выберит
е пояс из списка';
    }
  }

  /**
   * Загрузка параметров региона, времени и пользователя с сер
вера
   */
  async function loadRegionalOptions(targetPaneId = null) {
    if (targetPaneId) {
      switchRegionalTab(targetPaneId);
    }
    const tzBadge = document.getElementById('reg-cur-tz-badge'
);
    if (tzBadge) tzBadge.textContent = 'Загрузка...';

    try {
      const data = await apiFetch('/api/v1/system/regional-opt
ions');
```

```
      cachedRegionalData = data;

      if (tzBadge) tzBadge.textContent = data.current_timezone || 'Не определен';

      // 1. Часовые пояса
      const searchInput = document.getElementById('reg-tz-search-input');
      const curQuery = searchInput ? searchInput.value.trim() : '';
      renderTimezonesList(data.timezones || [], curQuery, data.current_timezone);

      // 2. Локали
      if (Array.isArray(data.locales) && data.locales.length > 0) {
        const sysLocSelect = document.getElementById('reg-loc-sys-select');
        const usrLocSelect = document.getElementById('reg-loc-usr-select');
        const optionsHtml = data.locales.map(loc =>
          `<option value="${escapeHtml(loc.code)}">${escapeHtml(loc.name)} [${escapeHtml(loc.code)}]</option>`
        ).join('');

        if (sysLocSelect) {
          sysLocSelect.innerHTML = optionsHtml;
          if (data.current_system_locale) sysLocSelect.value = data.current_system_locale;
        }
        if (usrLocSelect) {
          usrLocSelect.innerHTML = optionsHtml;
          if (data.current_user_locale) usrLocSelect.value = data.current_user_locale;
        }
      } else {
        const sysLocSelect = document.getElementById('reg-loc-sys-select');
```

```
        const usrLocSelect = document.getElementById('reg-loc-usr-select');
        if (sysLocSelect && data.current_system_locale) sysLocSelect.value = data.current_system_locale;
        if (usrLocSelect && data.current_user_locale) usrLocSelect.value = data.current_user_locale;
      }

      // 3. Профиль пользователя
      const usrNameInput = document.getElementById('reg-usr-name-input');
      const usrFullNameInput = document.getElementById('reg-usr-fullname-input');
      const usrDescInput = document.getElementById('reg-usr-desc-input');

      if (usrNameInput) usrNameInput.value = data.current_username || '';
      if (usrFullNameInput) usrFullNameInput.value = data.current_user_fullname || '';
      if (usrDescInput) usrDescInput.value = data.current_user_description || '';

    } catch (err) {
      console.error('[AboutSystemTab] loadRegionalOptions error:', err);
      if (tzBadge) tzBadge.textContent = 'Ошибка загрузки';
      showRegModalAlert(`Ошибка загрузки параметров: ${escapeHtml(err.message || String(err))}`, 'danger');
    }
  }

  /**
   * Привязка событий модального окна региональных параметров
   */
  function bindRegionalEvents() {
    // 1. Открытие модального окна по кнопкам в таблице или заголовке
```

```
    const openButtons = document.querySelectorAll('#btn-open-regional-settings, .btn-open-reg-tab');
    openButtons.forEach(btn => {
      btn.onclick = (e) => {
        e.preventDefault();
        const targetTab = btn.getAttribute('data-reg-tab') || 'pane-tz';
        const modalEl = document.getElementById('aboutRegionalSettingsModal');
        if (modalEl && window.bootstrap && window.bootstrap.Modal) {
          const modalInst = window.bootstrap.Modal.getOrCreateInstance(modalEl);
          modalInst.show();
        }
        loadRegionalOptions(targetTab);
      };
    });

    // 2. Переключение вкладок в модальном окне
    document.querySelectorAll('.btn-reg-nav').forEach(btn => {
      btn.onclick = (e) => {
        e.preventDefault();
        const targetPane = btn.getAttribute('data-tab-target');
        if (targetPane) {
          switchRegionalTab(targetPane);
        }
      };
    });

    // 3. Фильтрация часовых поясов
    const tzSearchInput = document.getElementById('reg-tz-search-input');
    const tzSelect = document.getElementById('reg-tz-select');
    if (tzSearchInput) {
      tzSearchInput.oninput = () => {
        if (cachedRegionalData && cachedRegionalData.timezones
```

```
) {
2417          const currentVal = tzSelect ? tzSelect.value : '';
2418          renderTimezonesList(cachedRegionalData.timezones, tz
SearchInput.value.trim(), currentVal);
2419        }
2420      };
2421    }
2422
2423    if (tzSelect) {
2424      tzSelect.onchange = () => {
2425        const selLabel = document.getElementById('reg-tz-selec
ted-label');
2426        if (selLabel) {
2427          selLabel.textContent = tzSelect.value ? `Выбран: ${t
zSelect.value}` : 'Выберите пояс из списка';
2428        }
2429      };
2430    }
2431
2432    // 4. Применение часового пояса
2433    const btnSaveTz = document.getElementById('btn-reg-save-tz
');
2434    if (btnSaveTz) {
2435      btnSaveTz.onclick = async () => {
2436        const tzVal = tzSelect ? tzSelect.value : '';
2437        if (!tzVal) {
2438          showRegModalAlert('Пожалуйста, выберите часовой пояс
 из списка', 'warning');
2439          return;
2440        }
2441
2442        btnSaveTz.disabled = true;
2443        btnSaveTz.innerHTML = '<span class="spinner-border spi
nner-border-sm" role="status"></span> Применение...';
2444
2445        try {
2446          const res = await apiFetch('/api/v1/system/set-timez
one', {
```

```
2447            method: 'POST',
2448            body: JSON.stringify({ timezone_id: tzVal })
2449          });
2450
2451          if (res.status === 'ok') {
2452            showRegModalAlert(`✅ Часовой пояс успешно изменен
на «<strong>${escapeHtml(res.timezone_id || tzVal)}</strong>»`
, 'success');
2453            if (window.showToast) window.showToast(`✅ Часовой
пояс изменен на «${tzVal}»`, 'success');
2454            const tzBadge = document.getElementById('reg-cur-t
z-badge');
2455            if (tzBadge) tzBadge.textContent = res.timezone_id
 || tzVal;
2456            updateLocalClock();
2457            fetchSystemSummary();
2458          } else {
2459            showRegModalAlert(`Ошибка: ${escapeHtml(res.messag
e || 'Не удалось применить пояс')}`, 'danger');
2460          }
2461        } catch (err) {
2462          showRegModalAlert(`Ошибка при смене часового пояса:
${escapeHtml(err.message || String(err))}`, 'danger');
2463        } finally {
2464          btnSaveTz.disabled = false;
2465          btnSaveTz.innerHTML = '<i class="bi bi-check-lg me-1
"></i> Применить пояс';
2466        }
2467      };
2468    }
2469
2470    // 5. Применение локали
2471    const btnSaveLoc = document.getElementById('btn-reg-save-l
oc');
2472    if (btnSaveLoc) {
2473      btnSaveLoc.onclick = async () => {
2474        const sysLoc = document.getElementById('reg-loc-sys-se
lect')?.value;
```

```
        const usrLoc = document.getElementById('reg-loc-usr-select')?.value;

        btnSaveLoc.disabled = true;
        btnSaveLoc.innerHTML = '<span class="spinner-border spinner-border-sm" role="status"></span> Сохранение...';

        try {
          const res = await apiFetch('/api/v1/system/set-locale', {
            method: 'POST',
            body: JSON.stringify({ system_locale: sysLoc, user_locale: usrLoc })
          });

          if (res.status === 'ok') {
            const rebootMsg = res.reboot_required ? '<br><small class="text-warning">⚠ Для полного применения требуется перезагрузка Windows.</small>' : '';
            showRegModalAlert(`✓ Системная локаль обновлена: <strong>${escapeHtml(res.locale || sysLoc)}</strong>.${rebootMsg}`, 'success');
            if (window.showToast) window.showToast(`✓ Системная локаль обновлена на «${res.locale || sysLoc}»`, 'warning');
            fetchSystemSummary();
          } else {
            showRegModalAlert(`Ошибка: ${escapeHtml(res.message || 'Не удалось изменить локаль')}`, 'danger');
          }
        } catch (err) {
          showRegModalAlert(`Ошибка при изменении локали: ${escapeHtml(err.message || String(err))}`, 'danger');
        } finally {
          btnSaveLoc.disabled = false;
          btnSaveLoc.innerHTML = '<i class="bi bi-check-lg me-1"></i> Применить локаль';
        }
      };
```

```
2501    }
2502
2503    // 6. Сохранение профиля пользователя
2504    const btnSaveUsr = document.getElementById('btn-reg-save-u
    sr');
2505    if (btnSaveUsr) {
2506      btnSaveUsr.onclick = async () => {
2507        const usrName = document.getElementById('reg-usr-name-
    input')?.value.trim();
2508        const fullName = document.getElementById('reg-usr-full
    name-input')?.value.trim();
2509        const desc = document.getElementById('reg-usr-desc-inp
    ut')?.value.trim();
2510
2511        if (!usrName) {
2512          showRegModalAlert('Имя учетной записи не указано', '
    warning');
2513          return;
2514        }
2515
2516        btnSaveUsr.disabled = true;
2517        btnSaveUsr.innerHTML = '<span class="spinner-border sp
    inner-border-sm" role="status"></span> Сохранение...';
2518
2519        try {
2520          const res = await apiFetch('/api/v1/system/update-us
    er-profile', {
2521            method: 'POST',
2522            body: JSON.stringify({ username: usrName, full_nam
    e: fullName, description: desc })
2523          });
2524
2525          if (res.status === 'ok') {
2526            showRegModalAlert(`✅ Профиль пользователя «<strong
    >${escapeHtml(usrName)}</strong>» успешно обновлен`, 'success'
    );
2527            if (window.showToast) window.showToast(`✅ Профиль
    пользователя «${usrName}» обновлен`, 'success');
```

```
2528            fetchSystemSummary();
2529          } else {
2530            showRegModalAlert(`Ошибка: ${escapeHtml(res.messag
    e || 'Не удалось обновить профиль')}`, 'danger');
2531          }
2532        } catch (err) {
2533          showRegModalAlert(`Ошибка при обновлении профиля: ${
    escapeHtml(err.message || String(err))}`, 'danger');
2534        } finally {
2535          btnSaveUsr.disabled = false;
2536          btnSaveUsr.innerHTML = '<i class="bi bi-check-lg me-
    1"></i> Сохранить профиль';
2537        }
2538      };
2539    }
2540  }
2541
2542  /**
2543   * Загрузка исторических записей телеметрии из базы данных t
    elemetry.db
2544   */
2545  async function fetchAboutSystemHistory(metric = 'all') {
2546    const tbody = document.getElementById('about-history-tbody
    ');
2547    if (tbody) {
2548      tbody.innerHTML = `
2549        <tr>
2550          <td colspan="9" class="text-center py-4 text-muted">
2551            <div class="spinner-border spinner-border-sm text-
    info mb-1" role="status"></div>
2552            <div>Запрос истории из базы данных telemetry.db...
    </div>
2553          </td>
2554        </tr>
2555      `;
2556    }
2557
2558    const limitSelect = document.getElementById('about-history
```

```
-limit-select');
2559      const limit = limitSelect ? parseInt(limitSelect.value, 10
) || 30 : 30;
2560
2561      try {
2562        const url = `/api/v1/about-system/history?limit=${limit}
&metric=${encodeURIComponent(metric)}`;
2563        const data = await apiFetch(url);
2564        renderAboutSystemHistory(data, metric);
2565      } catch (err) {
2566        console.error('[AboutSystemTab] fetchAboutSystemHistory
error:', err);
2567        if (tbody) {
2568          tbody.innerHTML = `
2569            <tr>
2570              <td colspan="9" class="text-center py-4 text-dange
r">
2571                <i class="bi bi-exclamation-triangle-fill me-1">
</i>
2572                Ошибка при получении истории из БД: ${escapeHtml
(err.message || String(err))}
2573              </td>
2574            </tr>
2575          `;
2576        }
2577      }
2578    }
2579
2580    /**
2581     * Отрисовка таблицы с предыдущими значениями из telemetry.d
b
2582     */
2583    function renderAboutSystemHistory(resData, metric) {
2584      const tbody = document.getElementById('about-history-tbody
');
2585      const countBadge = document.getElementById('about-history-
count-badge');
2586      const metaEl = document.getElementById('about-history-db-m
```

```
eta');
2587
2588    if (!resData || !Array.isArray(resData.history) || resData
.history.length === 0) {
2589      if (tbody) {
2590        tbody.innerHTML = `
2591          <tr>
2592            <td colspan="9" class="text-center py-4 text-muted
">
2593              <i class="bi bi-database-slash fs-4 d-block mb-1
 text-secondary"></i>
2594              В базе данных telemetry.db пока нет сохраненных
срезов телеметрии.
2595            </td>
2596          </tr>
2597        `;
2598      }
2599      if (countBadge) countBadge.textContent = 'Записей в базе
: 0';
2600      return;
2601    }
2602
2603    const history = resData.history;
2604    if (countBadge) countBadge.textContent = `Записей в выборк
e: ${history.length}`;
2605    if (metaEl && resData.meta && resData.meta.source) {
2606      metaEl.textContent = `Источник: ${resData.meta.source} (
${resData.meta.table || 'system_snapshots'})`;
2607    }
2608
2609    if (!tbody) return;
2610
2611    let rowsHtml = '';
2612    history.forEach(item => {
2613      const id = item.id || 0;
2614      let dateFormatted = '--:--:--';
2615      if (item.timestamp) {
2616        try {
```

```
2617          const d = new Date(item.timestamp);
2618          dateFormatted = `${d.toLocaleDateString()} ${d.toLoc
aleTimeString()}`;
2619        } catch (_) {
2620          dateFormatted = item.timestamp;
2621        }
2622      }
2623
2624      const host = escapeHtml(item.hostname || '--');
2625      const os = escapeHtml(item.os_name || 'Windows');
2626
2627      // CPU
2628      const cpuVal = item.cpu_total_percent !== null && item.c
pu_total_percent !== undefined ? Number(item.cpu_total_percent
) : 0;
2629      const cpuColor = cpuVal > 85 ? 'text-danger fw-bold' : (
cpuVal > 60 ? 'text-warning' : 'text-info');
2630      const cpuFreq = item.cpu_frequency_mhz ? `${item.cpu_fre
quency_mhz} MHz` : '';
2631
2632      // RAM
2633      const ramUsed = item.memory_used_gb !== null && item.mem
ory_used_gb !== undefined ? Number(item.memory_used_gb).toFixe
d(1) : '0.0';
2634      const ramTot = item.memory_total_gb !== null && item.mem
ory_total_gb !== undefined ? Number(item.memory_total_gb).toFi
xed(1) : '0.0';
2635      const ramPct = item.memory_percent !== null && item.memo
ry_percent !== undefined ? Number(item.memory_percent).toFixed
(1) : '0.0';
2636      const ramColor = Number(ramPct) > 85 ? 'text-danger' : (
Number(ramPct) > 65 ? 'text-warning' : 'text-success');
2637
2638      // GPU
2639      const gpuLoad = item.gpu_load_percent !== null && item.g
pu_load_percent !== undefined ? `${item.gpu_load_percent}%` :
'--';
2640      const gpuTemp = item.gpu_temp_c !== null && item.gpu_tem
```

```
p_c !== undefined ? ` / ${item.gpu_temp_c}°C` : '';

      // Storage C:
      const cFree = item.storage_c_free_gb !== null && item.st
orage_c_free_gb !== undefined ? `${item.storage_c_free_gb} GB`
 : '--';

      // Disk I/O
      const diskIo = item.disk_io_total_mb_s !== null && item.
disk_io_total_mb_s !== undefined ? `${Number(item.disk_io_tota
l_mb_s).toFixed(2)} MB/s` : '0.00 MB/s';

      // Uptime
      const uptime = escapeHtml(item.uptime_human || '--');

      rowsHtml += `
        <tr>
          <td class="text-muted">#${id}</td>
          <td class="text-nowrap text-light">${escapeHtml(date
Formatted)}</td>
          <td class="text-truncate" style="max-width: 140px;"
title="${host} (${os})">
            <span class="text-white">${host}</span>
            <small class="text-muted d-block" style="font-size
: 0.70rem;">${os}</small>
          </td>
          <td style="text-align: right;">
            <span class="${cpuColor}">${cpuVal.toFixed(1)}%</s
pan>
            <small class="text-muted d-block" style="font-size
: 0.70rem;">${cpuFreq}</small>
          </td>
          <td style="text-align: right;">
            <span class="${ramColor}">${ramUsed} / ${ramTot} G
B</span>
            <small class="text-muted d-block" style="font-size
: 0.70rem;">${ramPct}%</small>
          </td>
```

```
          <td style="text-align: right;">
            <span class="text-light">${gpuLoad}</span>
            <small class="text-danger" style="font-size: 0.70r
em;">${gpuTemp}</small>
          </td>
          <td style="text-align: right;" class="text-info">${c
Free}</td>
          <td style="text-align: right;" class="text-white">${
diskIo}</td>
          <td style="text-align: right;" class="text-muted">${
uptime}</td>
        </tr>
      `;
    });

    tbody.innerHTML = rowsHtml;
  }

  function escapeHtml(str) {
    if (str === null || str === undefined) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
  }

  // Self-trigger if container is already in DOM
  if (document.getElementById('about-sys-wrapper') || document
.getElementById('tab-about-system')) {
    if (document.readyState === 'loading') {
      document.addEventListener('DOMContentLoaded', () => init
AboutSystemTab());
    } else {
      setTimeout(() => initAboutSystemTab(), 10);
    }
  }
```

```
2699 })();
2700
2701
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Accounts & Identity Tab
================================================================
===============
Description:
  Вкладка управления учетными записями, субъектами безопасност
и,
  группами, правами LSA, инспекцией процессов, журналом аудита
 и Windows Identity Graph.

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/accounts_identity_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<div class="container-fluid p-3" id="accounts-identity-root">
<!-- Заголовок и текущий контекст токена -->
<div class="d-flex align-items-center justify-content-between
mb-3 pb-2 border-bottom flex-wrap gap-2">
<div>
<h4 class="mb-0 fw-bold d-flex align-items-center gap-2">
<i class="bi bi-people-fill text-primary"></i>
<span>Управление пользователями и Windows Identity</span>
</h4>
<div class="small text-muted" id="ai-current-token-badge">
        Загрузка контекста безопасности...
      </div>
</div>
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm btn-outline-secondary d-flex align-i
```

```
tems-center gap-1 shadow-sm" id="btn-refresh-identity" title="
Обновить данные">
<i class="bi bi-arrow-clockwise"></i>
<span>Обновить</span>
</button>
<button class="btn btn-sm btn-primary d-flex align-items-cente
r gap-1 shadow-sm" id="btn-open-create-user-modal">
<i class="bi bi-person-plus-fill"></i>
<span>Создать пользователя</span>
</button>
</div>
</div>
<!-- Карточки ключевых метрик -->
<div class="row g-3 mb-3">
<div class="col-6 col-md-3">
<div class="card shadow-sm border-secondary-subtle h-100 bg-bo
dy-tertiary" data-panel-id="panel-accounts-identity" id="panel
-accounts-identity">
<div class="card-body p-2 d-flex align-items-center gap-3">
<div class="rounded-circle p-2 bg-primary bg-opacity-10 text-p
rimary fs-4">
<i class="bi bi-person-badge"></i>
</div>
<div>
<div class="text-muted small">Всего пользователей</div>
<h5 class="fw-bold mb-0" id="stat-total-users">-</h5>
</div>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card shadow-sm border-secondary-subtle h-100 bg-bo
dy-tertiary" data-panel-id="panel-accounts-identity" id="panel
-accounts-identity">
<div class="card-body p-2 d-flex align-items-center gap-3">
<div class="rounded-circle p-2 bg-danger bg-opacity-10 text-da
nger fs-4">
<i class="bi bi-shield-lock-fill"></i>
```

```
</div>
<div>
<div class="text-muted small">Администраторы</div>
<h5 class="fw-bold mb-0" id="stat-total-admins">-</h5>
</div>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card shadow-sm border-secondary-subtle h-100 bg-body-tertiary" data-panel-id="panel-accounts-identity" id="panel-accounts-identity">
<div class="card-body p-2 d-flex align-items-center gap-3">
<div class="rounded-circle p-2 bg-success bg-opacity-10 text-success fs-4">
<i class="bi bi-display"></i>
</div>
<div>
<div class="text-muted small">Активных сессий</div>
<h5 class="fw-bold mb-0" id="stat-active-sessions">-</h5>
</div>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card shadow-sm border-secondary-subtle h-100 bg-body-tertiary" data-panel-id="panel-accounts-identity" id="panel-accounts-identity">
<div class="card-body p-2 d-flex align-items-center gap-3">
<div class="rounded-circle p-2 bg-warning bg-opacity-10 text-warning fs-4">
<i class="bi bi-shield-exclamation"></i>
</div>
<div>
<div class="text-muted small">События аудита (24ч)</div>
<h5 class="fw-bold mb-0" id="stat-security-events">-</h5>
</div>
</div>
```

```
</div>
</div>
</div>
<!-- Навигация по подразделам вкладки -->
<ul class="nav nav-pills mb-3 gap-1" id="identity-subtabs" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active btn-sm" data-bs-target="#subtab-users" data-bs-toggle="pill" id="subtab-users-btn" role="tab" type="button">
<i class="bi bi-people me-1"></i> Пользователи SAM
      </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link btn-sm" data-bs-target="#subtab-groups" data-bs-toggle="pill" id="subtab-groups-btn" role="tab" type="button">
<i class="bi bi-diagram-3 me-1"></i> Группы и вложенность
      </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link btn-sm" data-bs-target="#subtab-events" data-bs-toggle="pill" id="subtab-events-btn" role="tab" type="button">
<i class="bi bi-shield-shaded me-1"></i> Журнал безопасности
      </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link btn-sm" data-bs-target="#subtab-security" data-bs-toggle="pill" id="subtab-security-btn" role="tab" type="button">
<i class="bi bi-shield-check me-1"></i> Аудит безопасности и LSA
      </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link btn-sm" data-bs-target="#subtab-pid" data-bs-toggle="pill" id="subtab-pid-btn" role="tab" type="butt
```

```
on">
<i class="bi bi-cpu me-1"></i> Инспектор процесса (PID)
      </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link btn-sm" data-bs-target="#subtab-catalo
g" data-bs-toggle="pill" id="subtab-catalog-btn" role="tab" ty
pe="button">
<i class="bi bi-journal-code me-1"></i> Каталог операций (~180
)
      </button>
</li>
</ul>
<!-- Содержимое подразделов -->
<div class="tab-content" id="identity-subtabs-content">
<!-- 1. Подраздел: Пользователи -->
<div class="tab-pane fade show active" id="subtab-users" role=
"tabpanel">
<div class="card shadow-sm border-secondary-subtle" data-panel
-id="panel-accounts-identity-1" id="panel-accounts-identity-1"
>
<div class="card-header bg-body-tertiary d-flex align-items-ce
nter justify-content-between py-2 flex-wrap gap-2">
<div class="fw-semibold small">Локальные учетные записи Window
s</div>
<div class="d-flex align-items-center gap-2">
<div class="input-group input-group-sm" style="width: 220px;">
<span class="input-group-text"><i class="bi bi-search"></i></s
pan>
<input class="form-control" id="filter-users-input" placeholde
r="Поиск пользователя..." type="text"/>
</div>
</div>
</div>
<div class="card-body p-0 table-responsive">
<table class="table table-hover align-middle mb-0" id="users-t
able">
<thead class="table-light small">
```

```
<tr>
<th>Имя пользователя</th>
<th>Полное имя</th>
<th>SID</th>
<th>Статус</th>
<th>Пароль</th>
<th>Истекает</th>
<th>Последний вход</th>
<th class="text-end">Действия</th>
</tr>
</thead>
<tbody class="small" id="users-table-body">
<tr>
<td class="text-center py-3 text-muted" colspan="8">Загрузка пользователей...</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- 2. Подраздел: Группы -->
<div class="tab-pane fade" id="subtab-groups" role="tabpanel">
<div class="card shadow-sm border-secondary-subtle" data-panel-id="panel-accounts-identity-2" id="panel-accounts-identity-2">
<div class="card-header bg-body-tertiary d-flex align-items-center justify-content-between py-2">
<div class="fw-semibold small">Группы безопасности и граф членства</div>
</div>
<div class="card-body p-0 table-responsive">
<table class="table table-hover align-middle mb-0" id="groups-table">
<thead class="table-light small">
<tr>
<th>Имя группы</th>
<th>SID</th>
```

```
<th>Описание</th>
<th>Тип</th>
<th>Участники</th>
<th class="text-end">Действия</th>
</tr>
</thead>
<tbody class="small" id="groups-table-body">
<tr>
<td class="text-center py-3 text-muted" colspan="6">Загрузка г
рупп...</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- 3. Подраздел: Журнал безопасности (Security Events Log) -
->
<div class="tab-pane fade" id="subtab-events" role="tabpanel">
<div class="card shadow-sm border-secondary-subtle" data-panel
-id="panel-accounts-identity-zhurnal-sobytiy-bezopasnosti-wi"
id="panel-accounts-identity-zhurnal-sobytiy-bezopasnosti-wi">
<div class="card-header bg-body-tertiary d-flex align-items-ce
nter justify-content-between py-2 flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<span class="fw-semibold small"><i class="bi bi-shield-shaded
text-primary me-1"></i> Журнал событий безопасности Windows</s
pan>
<span class="badge bg-secondary" id="events-count-badge">0 зап
исей</span>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<div class="btn-group btn-group-sm" id="events-quick-filters"
role="group">
<button class="btn btn-outline-secondary py-0 px-2 event-filte
r-btn active" data-filter="all" type="button">Все</button>
<button class="btn btn-outline-secondary py-0 px-2 event-filte
r-btn" data-filter="logon" type="button">Входы (4624/4625)</bu
```

```
tton>
<button class="btn btn-outline-secondary py-0 px-2 event-filte
r-btn" data-filter="accounts" type="button">Аккаунты (4720/472
6)</button>
<button class="btn btn-outline-secondary py-0 px-2 event-filte
r-btn" data-filter="password" type="button">Пароли (4724)</but
ton>
<button class="btn btn-outline-secondary py-0 px-2 event-filte
r-btn" data-filter="lockout" type="button">Блокировки (4740)</
button>
</div>
<div class="input-group input-group-sm" style="width: 200px;">
<span class="input-group-text"><i class="bi bi-search"></i></s
pan>
<input class="form-control" id="filter-events-input" placehold
er="Поиск в журнале..." type="text"/>
</div>
<button class="btn btn-sm btn-outline-primary py-0 px-2" id="b
tn-refresh-events" title="Обновить журнал">
<i class="bi bi-arrow-clockwise"></i>
</button>
</div>
</div>
<div class="card-body p-0 table-responsive" style="max-height:
 520px; overflow-y: auto;">
<table class="table table-hover table-sm align-middle mb-0" id
="events-table">
<thead class="table-light small sticky-top">
<tr>
<th style="width: 140px;">Время</th>
<th style="width: 90px;">Event ID</th>
<th style="width: 170px;">Событие</th>
<th style="width: 160px;">Целевой аккаунт</th>
<th style="width: 150px;">Инициатор</th>
<th>Описание / Сообщение</th>
<th class="text-end" style="width: 90px;">Инфо</th>
</tr>
</thead>
```

```
<tbody class="small" id="events-table-body">
<tr>
<td class="text-center py-4 text-muted" colspan="7">Загрузка журнала событий безопасности...</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- 4. Подраздел: Аудит безопасности и LSA -->
<div class="tab-pane fade" id="subtab-security" role="tabpanel">
<div class="row g-3">
<div class="col-md-7">
<div class="card shadow-sm border-secondary-subtle mb-3" data-panel-id="panel-accounts-identity-3" id="panel-accounts-identity-3">
<div class="card-header bg-body-tertiary fw-semibold small d-flex align-items-center justify-content-between">
<span>Быстрый аудит привилегий и доступа</span>
</div>
<div class="card-body p-3">
<div class="d-flex flex-wrap gap-2 mb-3">
<button class="btn btn-sm btn-outline-primary d-flex align-items-center gap-1" id="btn-check-who-is-admin">
<i class="bi bi-shield-lock"></i>
<span>Who is Admin?</span>
</button>
<button class="btn btn-sm btn-outline-info d-flex align-items-center gap-1" id="btn-check-service-logon">
<i class="bi bi-gear"></i>
<span>Logon as Service</span>
</button>
<button class="btn btn-sm btn-outline-warning d-flex align-items-center gap-1" id="btn-check-rdp">
<i class="bi bi-display"></i>
<span>Remote Desktop (RDP)</span>
```

```
</button>
<button class="btn btn-sm btn-outline-danger d-flex align-items-center gap-1" id="btn-check-orphaned">
<i class="bi bi-exclamation-triangle"></i>
<span>Orphaned SIDs / Profiles</span>
</button>
</div>
<div class="border rounded p-2 bg-body-tertiary small" id="security-audit-results" style="min-height: 180px; max-height: 360px; overflow-y: auto;">
<div class="text-muted text-center py-4">Выберите проверку выше для запуска анализа безопасности.</div>
</div>
</div>
</div>
</div>
<div class="col-md-5">
<div class="card shadow-sm border-secondary-subtle" data-panel-id="panel-accounts-identity-4" id="panel-accounts-identity-4">
<div class="card-header bg-body-tertiary fw-semibold small">
<i class="bi bi-key-fill me-1 text-primary"></i> Политика паролей SAM
            </div>
<div class="card-body p-3 small" id="password-policy-container">
<div class="text-muted">Загрузка политики паролей...</div>
</div>
</div>
</div>
</div>
</div>
<!-- 5. Подраздел: Инспектор процесса (PID) -->
<div class="tab-pane fade" id="subtab-pid" role="tabpanel">
<div class="card shadow-sm border-secondary-subtle" data-panel-id="panel-accounts-identity-5" id="panel-accounts-identity-5">
<div class="card-header bg-body-tertiary d-flex align-items-ce
```

```
nter justify-content-between py-2">
<div class="fw-semibold small">Исследование контекста безопасности процесса (Token &amp;amp; Account Dossier)</div>
</div>
<div class="card-body p-3">
<div class="row g-2 align-items-center mb-3">
<div class="col-auto">
<label class="col-form-label small fw-semibold" for="input-explain-pid">PID процесса:</label>
</div>
<div class="col-auto" style="width: 150px;">
<input class="form-control form-control-sm" id="input-explain-pid" placeholder="например 1234" type="number"/>
</div>
<div class="col-auto">
<button class="btn btn-sm btn-primary d-flex align-items-center gap-1" id="btn-explain-pid">
<i class="bi bi-search"></i>
<span>Исследовать PID</span>
</button>
</div>
</div>
<div class="p-3 bg-body-tertiary rounded border font-monospace small" id="pid-dossier-output" style="white-space: pre-wrap; display: none;">
</div>
</div>
</div>
</div>
<!-- 6. Подраздел: Каталог операций (~180) -->
<div class="tab-pane fade" id="subtab-catalog" role="tabpanel">
<div class="card shadow-sm border-secondary-subtle" data-panel-id="panel-accounts-identity-6" id="panel-accounts-identity-6">
<div class="card-header bg-body-tertiary d-flex align-items-center justify-content-between py-2 flex-wrap gap-2">
<div class="fw-semibold small">Реестр операций OS-level катало
```

```
га Accounts &amp;amp; Identity</div>
<div class="d-flex align-items-center gap-2">
<input class="form-control form-control-sm" id="filter-catalog-input" placeholder="Поиск в каталоге..." style="width: 200px;" type="text"/>
</div>
</div>
<div class="card-body p-0 table-responsive" style="max-height: 500px; overflow-y: auto;">
<table class="table table-hover table-sm align-middle mb-0">
<thead class="table-light small sticky-top">
<tr>
<th>#</th>
<th>Операция</th>
<th>Подсистема</th>
<th>Механизм</th>
<th>API / CLI</th>
<th>Риск</th>
</tr>
</thead>
<tbody class="small" id="catalog-table-body">
<tr>
<td class="text-center py-3 text-muted" colspan="6">Загрузка каталога...</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
</div>
<!-- Модальное окно: Создание пользователя -->
<div aria-hidden="true" class="modal fade" id="createUserModal" tabindex="-1">
<div class="modal-dialog">
<div class="modal-content shadow">
<div class="modal-header">
```

```
<h5 class="modal-title d-flex align-items-center gap-2 fs-6 fw
-bold">
<i class="bi bi-person-plus text-primary"></i>
<span>Создать локального пользователя</span>
</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body small">
<form id="create-user-form">
<div class="mb-2">
<label class="form-label fw-semibold">Имя пользователя (Логин)
*</label>
<input class="form-control form-control-sm" id="create-user-na
me" placeholder="например: ivan" required="" type="text"/>
</div>
<div class="mb-2">
<label class="form-label fw-semibold">Пароль (опционально)</la
bel>
<input class="form-control form-control-sm" id="create-user-pw
d" placeholder="Оставьте пустым без пароля" type="password"/>
</div>
<div class="mb-2">
<label class="form-label fw-semibold">Полное имя (Full Name)</
label>
<input class="form-control form-control-sm" id="create-user-fu
llname" placeholder="например: Иван Иванов" type="text"/>
</div>
<div class="mb-2">
<label class="form-label fw-semibold">Описание (Description)</
label>
<input class="form-control form-control-sm" id="create-user-de
sc" placeholder="Назначение учетной записи" type="text"/>
</div>
</form>
</div>
<div class="modal-footer">
<button class="btn btn-sm btn-secondary" data-bs-dismiss="moda
```

```
l" type="button">Отмена</button>
<button class="btn btn-sm btn-primary" id="btn-submit-create-user" type="button">Создать</button>
</div>
</div>
</div>
</div>
<!-- Модальное окно: Досье Principal и Журнал событий пользователя -->
<div aria-hidden="true" class="modal fade" id="principalDossierModal" tabindex="-1">
<div class="modal-dialog modal-xl modal-dialog-scrollable">
<div class="modal-content shadow">
<div class="modal-header">
<h5 class="modal-title d-flex align-items-center gap-2 fs-6 fw-bold">
<i class="bi bi-file-earmark-person text-primary"></i>
<span id="dossier-modal-title">Досье субъекта безопасности</span>
</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body">
<ul class="nav nav-tabs mb-3" id="dossierModalTabs" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active small py-1" data-bs-target="#dossier-tree-pane" data-bs-toggle="tab" id="dossier-tree-tab" role="tab" type="button">
<i class="bi bi-diagram-2 me-1"></i> Дерево Principal
            </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link small py-1" data-bs-target="#dossier-events-pane" data-bs-toggle="tab" id="dossier-events-tab" role="tab" type="button">
<i class="bi bi-shield-exclamation me-1"></i> События безопасн
```

```
ости (<span id="dossier-events-count">0</span>)
            </button>
</li>
</ul>
<div class="tab-content" id="dossierModalContent">
<div class="tab-pane fade show active" id="dossier-tree-pane" role="tabpanel">
<pre class="p-3 bg-body-tertiary rounded border font-monospace small" id="dossier-tree-view" style="white-space: pre-wrap; margin-bottom: 0;"></pre>
</div>
<div class="tab-pane fade" id="dossier-events-pane" role="tabpanel">
<div class="table-responsive">
<table class="table table-hover table-sm align-middle mb-0">
<thead class="table-light small">
<tr>
<th>Время</th>
<th>Event ID</th>
<th>Событие</th>
<th>Инициатор / Аккаунт</th>
<th>Описание</th>
</tr>
</thead>
<tbody class="small font-monospace" id="dossier-user-events-tbody">
<tr><td class="text-center py-3 text-muted" colspan="5">События отсутствуют</td></tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
<div class="modal-footer">
<button class="btn btn-sm btn-secondary" data-bs-dismiss="modal" type="button">Закрыть</button>
</div>
```

```
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Accounts & Identity Tab - Client Controller
 * ============================================================
 * Description:
 *   Клиентский JavaScript контроллер для вкладки управления пользователями,
 *   группами, правами LSA, журналом событий безопасности, инспекцией процессов
 *   и каталогом операций.
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/accounts_identity_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

let _usersCache = [];
let _groupsCache = [];
let _eventsCache = [];
let _catalogCache = [];
let _currentEventFilter = 'all';

/**
 * Инициализация контроллера вкладки Accounts & Identity
 */
export async function initAccountsIdentityTab() {
  bindEvents();
  await loadAllIdentityData();
```

```
}

/**
 * Привязка обработчиков событий
 */
function bindEvents() {
  const refreshBtn = document.getElementById('btn-refresh-identity');
  if (refreshBtn) refreshBtn.onclick = () => loadAllIdentityData();

  const refreshEventsBtn = document.getElementById('btn-refresh-events');
  if (refreshEventsBtn) refreshEventsBtn.onclick = () => loadSecurityEvents();

  const openCreateUserModalBtn = document.getElementById('btn-open-create-user-modal');
  if (openCreateUserModalBtn) {
    openCreateUserModalBtn.onclick = () => {
      const modalEl = document.getElementById('createUserModal');
      if (modalEl && window.bootstrap) {
        const modal = new window.bootstrap.Modal(modalEl);
        modal.show();
      }
    };
  }

  const submitCreateUserBtn = document.getElementById('btn-submit-create-user');
  if (submitCreateUserBtn) submitCreateUserBtn.onclick = handleCreateUserSubmit;

  const filterUsersInput = document.getElementById('filter-users-input');
  if (filterUsersInput) {
    filterUsersInput.oninput = (e) => filterUsersTable(e.targe
```

```
t.value);
  }

  const filterEventsInput = document.getElementById('filter-ev
ents-input');
  if (filterEventsInput) {
    filterEventsInput.oninput = (e) => filterEventsTable(e.tar
get.value);
  }

  const filterCatalogInput = document.getElementById('filter-c
atalog-input');
  if (filterCatalogInput) {
    filterCatalogInput.oninput = (e) => filterCatalogTable(e.t
arget.value);
  }

  // Быстрые фильтры журнала событий
  const eventsFilterToolbar = document.getElementById('events-
quick-filters');
  if (eventsFilterToolbar) {
    eventsFilterToolbar.querySelectorAll('.event-filter-btn').
forEach(btn => {
      btn.onclick = () => {
        eventsFilterToolbar.querySelectorAll('.event-filter-bt
n').forEach(b => b.classList.remove('active'));
        btn.classList.add('active');
        _currentEventFilter = btn.getAttribute('data-filter')
|| 'all';
        applyEventsFilter();
      };
    });
  }

  // Быстрые проверки безопасности
  const btnWhoIsAdmin = document.getElementById('btn-check-who
-is-admin');
  if (btnWhoIsAdmin) btnWhoIsAdmin.onclick = runWhoIsAdminChec
```

```
k;

  const btnServiceLogon = document.getElementById('btn-check-service-logon');
  if (btnServiceLogon) btnServiceLogon.onclick = runServiceLogonCheck;

  const btnRdp = document.getElementById('btn-check-rdp');
  if (btnRdp) btnRdp.onclick = runRdpCheck;

  const btnOrphaned = document.getElementById('btn-check-orphaned');
  if (btnOrphaned) btnOrphaned.onclick = runOrphanedCheck;

  // Инспектор PID
  const btnExplainPid = document.getElementById('btn-explain-pid');
  if (btnExplainPid) btnExplainPid.onclick = handleExplainPid;
}

/**
 * Загрузка всех данных с бэкенда
 */
async function loadAllIdentityData() {
  await Promise.allSettled([
    loadCurrentToken(),
    loadUsers(),
    loadGroups(),
    loadSecurityEvents(),
    loadPasswordPolicy(),
    loadCatalog(),
  ]);
}

/**
 * Загрузка контекста текущего пользователя
 */
async function loadCurrentToken() {
```

```
  const badgeEl = document.getElementById('ai-current-token-badge');
  try {
    const res = await fetch('/api/windows/identity/current');
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    const isElevated = data.is_elevated ? '<span class="badge bg-danger ms-1">Elevated (Admin)</span>' : '<span class="badge bg-secondary ms-1">Standard</span>';
    const integrityBadge = `<span class="badge bg-primary ms-1">${data.integrity_level}</span>`;
    if (badgeEl) {
      badgeEl.innerHTML = `Текущий токен: <strong>${data.domain}\\${data.user_name}</strong> (SID: <code>${data.user_sid}</code>) | Целостность: ${integrityBadge} ${isElevated}`;
    }
  } catch (err) {
    if (badgeEl) badgeEl.innerHTML = `<span class="text-danger">Ошибка контекста: ${err.message}</span>`;
  }
}

/**
 * Загрузка пользователей
 */
async function loadUsers() {
  const tbody = document.getElementById('users-table-body');
  const statTotal = document.getElementById('stat-total-users');
  const statAdmins = document.getElementById('stat-total-admins');
  const statSessions = document.getElementById('stat-active-sessions');

  try {
    const [resUsers, resAdmins, resSessions] = await Promise.all([
      fetch('/api/windows/identity/users'),
```

```
      fetch('/api/windows/identity/who-is-admin'),
      fetch('/api/windows/identity/sessions'),
    ]);

    _usersCache = resUsers.ok ? await resUsers.json() : [];
    const admins = resAdmins.ok ? await resAdmins.json() : [];
    const sessions = resSessions.ok ? await resSessions.json()
 : [];

    if (statTotal) statTotal.textContent = _usersCache.length;
    if (statAdmins) statAdmins.textContent = admins.length;
    if (statSessions) statSessions.textContent = sessions.leng
th;

    renderUsersTable(_usersCache);
  } catch (err) {
    if (tbody) {
      tbody.innerHTML = `<tr><td colspan="8" class="text-cente
r text-danger py-3">Ошибка загрузки пользователей: ${err.messa
ge}</td></tr>`;
    }
  }
}

/**
 * Отрисовка таблицы пользователей
 */
function renderUsersTable(users) {
  const tbody = document.getElementById('users-table-body');
  if (!tbody) return;

  if (!users || users.length === 0) {
    tbody.innerHTML = '<tr><td colspan="8" class="text-center
text-muted py-3">Пользователи не найдены</td></tr>';
    return;
  }

  tbody.innerHTML = users.map(u => {
```

```
    const statusBadge = u.enabled
      ? '<span class="badge bg-success bg-opacity-75">Включен</span>'
      : '<span class="badge bg-danger bg-opacity-75">Отключен</span>';
    const pwdBadge = u.password_never_expires
      ? '<span class="badge bg-info text-dark">Бессрочный</span>'
      : '<span class="badge bg-secondary">Ограничен</span>';

    return `
      <tr>
        <td class="fw-bold">
          <i class="bi bi-person-circle me-1 text-primary"></i> ${escapeHtml(u.name)}
        </td>
        <td>${escapeHtml(u.full_name || '—')}</td>
        <td><code class="small">${escapeHtml(u.sid || '—')}</code></td>
        <td>${statusBadge}</td>
        <td>${pwdBadge}</td>
        <td>${u.account_expires ? escapeHtml(u.account_expires) : 'Никогда'}</td>
        <td>${u.last_logon ? escapeHtml(u.last_logon) : '—'}</td>
        <td class="text-end">
          <button class="btn btn-sm btn-outline-primary py-0 px-1" onclick="window._explainUserDossier('${escapeHtml(u.name)}')" title="Открыть досье Principal">
            <i class="bi bi-file-text"></i>
          </button>
          <button class="btn btn-sm btn-outline-danger py-0 px-1 ms-1" onclick="window._deleteUserAction('${escapeHtml(u.name)}')" title="Удалить пользователя">
            <i class="bi bi-trash"></i>
          </button>
        </td>
      </tr>
```

```
    `;
  }).join('');
}

/**
 * Фильтрация пользователей
 */
function filterUsersTable(query) {
  const q = (query || '').toLowerCase().trim();
  if (!q) {
    renderUsersTable(_usersCache);
    return;
  }
  const filtered = _usersCache.filter(u =>
    (u.name && u.name.toLowerCase().includes(q)) ||
    (u.full_name && u.full_name.toLowerCase().includes(q)) ||
    (u.sid && u.sid.toLowerCase().includes(q))
  );
  renderUsersTable(filtered);
}

/**
 * Загрузка групп
 */
async function loadGroups() {
  const tbody = document.getElementById('groups-table-body');
  const statGroups = document.getElementById('stat-total-group
s');
  try {
    const res = await fetch('/api/windows/identity/groups');
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    _groupsCache = await res.json();
    if (statGroups) statGroups.textContent = _groupsCache.leng
th;

    if (!tbody) return;
    if (_groupsCache.length === 0) {
      tbody.innerHTML = '<tr><td colspan="6" class="text-cente
```

```
r text-muted py-3">Группы не найдены</td></tr>';
      return;
    }

    tbody.innerHTML = _groupsCache.map(g => {
      const typeBadge = g.is_admin
        ? '<span class="badge bg-danger">ADMIN GROUP</span>'
        : '<span class="badge bg-secondary">Локальная</span>';
      const membersCount = (g.members || []).length;
      return `
        <tr>
          <td class="fw-bold"><i class="bi bi-shield me-1 text
-primary"></i> ${escapeHtml(g.name)}</td>
          <td><code class="small">${escapeHtml(g.sid || '—')}<
/code></td>
          <td class="text-muted small">${escapeHtml(g.descript
ion || '—')}</td>
          <td>${typeBadge}</td>
          <td><span class="badge bg-light text-dark border">${
membersCount} уч.</span></td>
          <td class="text-end">
            <button class="btn btn-sm btn-outline-info py-0 px
-1" onclick="window._viewGroupMembers('${escapeHtml(g.name)}')
" title="Участники">
              <i class="bi bi-people"></i>
            </button>
          </td>
        </tr>
      `;
    }).join('');
  } catch (err) {
    if (tbody) {
      tbody.innerHTML = `<tr><td colspan="6" class="text-cente
r text-danger py-3">Ошибка загрузки групп: ${err.message}</td>
</tr>`;
    }
  }
}
```

```
273
274 /**
275  * Загрузка журнала событий безопасности Windows
276  */
277 async function loadSecurityEvents() {
278   const tbody = document.getElementById('events-table-body');
279   const badge = document.getElementById('events-count-badge');
280   const statBadge = document.getElementById('stat-security-eve
    nts');
281
282   try {
283     const res = await fetch('/api/windows/identity/audit?limit
    =100');
284     if (!res.ok) throw new Error(`HTTP ${res.status}`);
285     _eventsCache = await res.json();
286
287     if (badge) badge.textContent = `${_eventsCache.length} зап
    исей`;
288     if (statBadge) statBadge.textContent = _eventsCache.length
    ;
289
290     applyEventsFilter();
291   } catch (err) {
292     if (tbody) {
293       tbody.innerHTML = `<tr><td colspan="7" class="text-cente
    r text-danger py-4">Ошибка загрузки журнала безопасности: ${er
    r.message}</td></tr>`;
294     }
295   }
296 }
297
298 /**
299  * Применение фильтров к событиям безопасности
300  */
301 function applyEventsFilter() {
302   const q = (document.getElementById('filter-events-input')?.v
    alue || '').toLowerCase().trim();
303   let filtered = _eventsCache;
```

```
304
305   if (_currentEventFilter === 'logon') {
306     filtered = filtered.filter(e => [4624, 4625].includes(e.ev
ent_id));
307   } else if (_currentEventFilter === 'accounts') {
308     filtered = filtered.filter(e => [4720, 4722, 4725, 4726, 4
731, 4732, 4733, 4734, 4735, 4738].includes(e.event_id));
309   } else if (_currentEventFilter === 'password') {
310     filtered = filtered.filter(e => [4723, 4724].includes(e.ev
ent_id));
311   } else if (_currentEventFilter === 'lockout') {
312     filtered = filtered.filter(e => e.event_id === 4740);
313   }
314
315   if (q) {
316     filtered = filtered.filter(e =>
317       String(e.event_id).includes(q) ||
318       (e.event_name && e.event_name.toLowerCase().includes(q))
 ||
319       (e.target_account && e.target_account.toLowerCase().incl
udes(q)) ||
320       (e.caller_account && e.caller_account.toLowerCase().incl
udes(q)) ||
321       (e.description && e.description.toLowerCase().includes(q
))
322     );
323   }
324
325   renderEventsTable(filtered);
326 }
327
328 /**
329  * Поиск по текстовому запросу в журнале событий
330  */
331 function filterEventsTable(query) {
332   applyEventsFilter();
333 }
334
```

```
/**
 * Отрисовка таблицы журнала событий безопасности
 */
function renderEventsTable(events) {
  const tbody = document.getElementById('events-table-body');
  if (!tbody) return;

  if (!events || events.length === 0) {
    tbody.innerHTML = '<tr><td colspan="7" class="text-center py-4 text-muted">События безопасности по выбранному фильтру не найдены</td></tr>';
    return;
  }

  tbody.innerHTML = events.map((e, idx) => {
    let eidBadge = `<span class="badge bg-secondary">${e.event_id}</span>`;
    if (e.event_id === 4624) {
      eidBadge = '<span class="badge bg-success">4624</span>';
    } else if (e.event_id === 4625 || e.event_id === 4740 || e.event_id === 4726 || e.event_id === 4734) {
      eidBadge = `<span class="badge bg-danger">${e.event_id}</span>`;
    } else if (e.event_id === 4724 || e.event_id === 4738) {
      eidBadge = `<span class="badge bg-warning text-dark">${e.event_id}</span>`;
    } else if (e.event_id === 4720 || e.event_id === 4731 || e.event_id === 4732) {
      eidBadge = `<span class="badge bg-primary">${e.event_id}</span>`;
    }

    const shortTime = e.timestamp ? e.timestamp.replace('T', ' ').slice(0, 19) : '—';
    const descText = e.description ? escapeHtml(e.description) : '—';

    return `
```

```
      <tr class="event-log-row" data-idx="${idx}" style="curso
r: pointer;" onclick="window._showEventDetail(${idx})" title="
Нажмите для просмотра полной информации">
        <td class="font-monospace text-muted small">${escapeHt
ml(shortTime)}</td>
        <td>${eidBadge}</td>
        <td class="fw-semibold text-primary">${escapeHtml(e.ev
ent_name)}</td>
        <td class="fw-bold">${escapeHtml(e.target_account || '
—')}</td>
        <td class="text-muted small">${escapeHtml(e.caller_acc
ount || '—')}</td>
        <td class="small text-truncate" style="max-width: 320p
x;" title="${descText}">${descText}</td>
        <td class="text-end">
          <button class="btn btn-xs btn-outline-info py-0 px-2
" onclick="event.stopPropagation(); window._showEventDetail(${
idx})" title="Подробнее">
            Инфо
          </button>
        </td>
      </tr>
    `;
  }).join('');
}

/**
 * Просмотр детальной информации о событии
 */
window._showEventDetail = function (idx) {
  const event = _eventsCache[idx];
  if (!event) return;

  if (window.AITableModal) {
    window.AITableModal.show({
      icon: '□ ',
      title: `Событие аудита: #${event.event_id} - ${event.eve
nt_name}`,
```

```
      subtitle: `Время: ${event.timestamp} | Целевой аккаунт:
${event.target_account || 'N/A'}`,
      tableType: 'generic',
      badges: [
        { text: `Event ${event.event_id}`, class: 'badge bg-pr
imary' },
        { text: event.event_name, class: 'badge bg-info text-d
ark' }
      ],
      metadata: [
        { label: 'Event ID', value: String(event.event_id) },
        { label: 'Событие', value: event.event_name },
        { label: 'Время регистрации', value: event.timestamp }
,
        { label: 'Целевая учетная запись', value: event.target
_account || '—' },
        { label: 'Инициатор (Caller)', value: event.caller_acc
ount || '—' },
        { label: 'Статус аудита', value: event.status || 'Succ
ess' }
      ],
      rawTitle: 'Полное сообщение журнала безопасности',
      rawContent: event.description || '',
      requestData: {
        event_id: event.event_id,
        event_name: event.event_name,
        target_account: event.target_account,
        caller_account: event.caller_account,
        description: event.description
      }
    });
  } else {
    alert(`Событие #${event.event_id} (${event.event_name})\n\
nВремя: ${event.timestamp}\nЦелевой аккаунт: ${event.target_ac
count}\nИнициатор: ${event.caller_account}\n\nОписание:\n${eve
nt.description}`);
  }
};
```

```
419
420 /**
421  * Загрузка политики паролей
422  */
423 async function loadPasswordPolicy() {
424   const container = document.getElementById('password-policy-c
    ontainer');
425   if (!container) return;
426   try {
427     const res = await fetch('/api/windows/identity/auth/polici
    es');
428     if (!res.ok) throw new Error(`HTTP ${res.status}`);
429     const pol = await res.json();
430     container.innerHTML = `
431       <ul class="list-group list-group-flush">
432         <li class="list-group-item d-flex justify-content-betw
    een align-items-center px-0 py-1">
433           <span>Мин. длина пароля:</span>
434           <strong>${pol.min_password_length} симв.</strong>
435         </li>
436         <li class="list-group-item d-flex justify-content-betw
    een align-items-center px-0 py-1">
437           <span>Макс. возраст пароля:</span>
438           <strong>${pol.max_password_age_days} дней</strong>
439         </li>
440         <li class="list-group-item d-flex justify-content-betw
    een align-items-center px-0 py-1">
441           <span>Глубина истории:</span>
442           <strong>${pol.password_history_length}</strong>
443         </li>
444         <li class="list-group-item d-flex justify-content-betw
    een align-items-center px-0 py-1">
445           <span>Порог блокировки:</span>
446           <strong>${pol.lockout_threshold === 0 ? 'Отключен' :
     pol.lockout_threshold + ' попыток'}</strong>
447         </li>
448       </ul>
449     `;
```

```
  } catch (err) {
    container.innerHTML = `<div class="text-danger">Ошибка политики: ${err.message}</div>`;
  }
}

/**
 * Загрузка каталога операций
 */
async function loadCatalog() {
  const tbody = document.getElementById('catalog-table-body');
  try {
    const res = await fetch('/api/windows/identity/catalog');
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    _catalogCache = await res.json();
    renderCatalogTable(_catalogCache);
  } catch (err) {
    if (tbody) {
      tbody.innerHTML = `<tr><td colspan="6" class="text-center text-danger py-3">Ошибка загрузки каталога: ${err.message}</td></tr>`;
    }
  }
}

function renderCatalogTable(items) {
  const tbody = document.getElementById('catalog-table-body');
  if (!tbody) return;
  tbody.innerHTML = items.map(op => {
    let riskBadge = '<span class="badge bg-success">SAFE</span>';
    if (op.risk_level.includes('ADMIN')) riskBadge = '<span class="badge bg-warning text-dark">ADMIN</span>';
    if (op.risk_level.includes('DANGEROUS')) riskBadge = '<span class="badge bg-danger">DANGEROUS</span>';
    if (op.risk_level.includes('ADVANCED')) riskBadge = '<span class="badge bg-dark">ADVANCED</span>';

```

```
    return `
      <tr>
        <td class="text-muted fw-bold">${op.id}</td>
        <td class="fw-semibold">${escapeHtml(op.name_ru)}</td>
        <td class="small text-muted">${escapeHtml(op.subsystem
_name_ru)}</td>
        <td class="small"><code>${escapeHtml(op.mechanism)}</c
ode></td>
        <td class="small"><code>${escapeHtml(op.win32_api || o
p.ps_or_cli || '—')}</code></td>
        <td>${riskBadge}</td>
      </tr>
    `;
  }).join('');
}

function filterCatalogTable(query) {
  const q = (query || '').toLowerCase().trim();
  if (!q) {
    renderCatalogTable(_catalogCache);
    return;
  }
  const filtered = _catalogCache.filter(op =>
    op.name_ru.toLowerCase().includes(q) ||
    op.subsystem_name_ru.toLowerCase().includes(q) ||
    op.mechanism.toLowerCase().includes(q) ||
    (op.win32_api && op.win32_api.toLowerCase().includes(q))
  );
  renderCatalogTable(filtered);
}

/**
 * Создание пользователя
 */
async function handleCreateUserSubmit() {
  const username = document.getElementById('create-user-name')
?.value?.trim();
  const password = document.getElementById('create-user-pwd')?
```

```
.value || null;
  const fullName = document.getElementById('create-user-fullna
me')?.value?.trim() || '';
  const description = document.getElementById('create-user-des
c')?.value?.trim() || '';

  if (!username) {
    if (window.toast) window.toast.error('Ошибка', 'Введите им
я пользователя');
    return;
  }

  try {
    const res = await fetch('/api/windows/identity/users', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        username,
        password: password || undefined,
        full_name: fullName,
        description,
      }),
    });
    const data = await res.json();
    if (!res.ok) throw new Error(data.detail || 'Не удалось со
здать пользователя');

    if (window.toast) window.toast.success('Успех', `Пользоват
ель '${username}' создан`);

    // Закрытие модального окна
    const modalEl = document.getElementById('createUserModal')
;
    if (modalEl && window.bootstrap) {
      const modal = window.bootstrap.Modal.getInstance(modalEl
);
      if (modal) modal.hide();
    }
```

```
    document.getElementById('create-user-form')?.reset();
    await loadUsers();
  } catch (err) {
    if (window.toast) window.toast.error('Ошибка создания', er
r.message);
  }
}

/**
 * Удаление пользователя
 */
window._deleteUserAction = async function (username) {
  if (!confirm(`Вы уверены, что хотите удалить пользователя '$
{username}'?`)) return;
  try {
    const res = await fetch(`/api/windows/identity/users/${enc
odeURIComponent(username)}`, {
      method: 'DELETE',
    });
    const data = await res.json();
    if (!res.ok) throw new Error(data.detail || 'Ошибка удален
ия');
    if (window.toast) window.toast.success('Удалено', `Пользов
атель '${username}' удален`);
    await loadUsers();
  } catch (err) {
    if (window.toast) window.toast.error('Ошибка удаления', er
r.message);
  }
};

/**
 * Досье Principal и история событий безопасности
 */
window._explainUserDossier = async function (identifier) {
  const modalEl = document.getElementById('principalDossierMod
al');
  const titleEl = document.getElementById('dossier-modal-title
```

```
');
  const treeEl = document.getElementById('dossier-tree-view');
  const eventsCountEl = document.getElementById('dossier-events-count');
  const eventsTbody = document.getElementById('dossier-user-events-tbody');

  if (titleEl) titleEl.textContent = `Досье Principal: ${identifier}`;
  if (treeEl) treeEl.textContent = 'Формирование досье безопасности...';
  if (eventsCountEl) eventsCountEl.textContent = '0';
  if (eventsTbody) eventsTbody.innerHTML = '<tr><td colspan="5" class="text-center py-3 text-muted">Загрузка журнала событий...</td></tr>';

  // Переключение на вкладку дерева по умолчанию
  const firstTabBtn = document.getElementById('dossier-tree-tab');
  if (firstTabBtn && window.bootstrap) {
    const tabTrigger = new window.bootstrap.Tab(firstTabBtn);
    tabTrigger.show();
  }

  if (modalEl && window.bootstrap) {
    const modal = new window.bootstrap.Modal(modalEl);
    modal.show();
  }

  try {
    const res = await fetch(`/api/windows/identity/explain/${encodeURIComponent(identifier)}`);
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();

    // Формирование дерева
    const lines = [
      `Principal (${data.name})`,
```

```
      '│',
      `├── SID: ${data.sid}`,
      `├── Domain: ${data.domain}`,
      `├── Type: ${data.principal_type}`,
      `├── Source: ${data.source}`,
      `├── Administrator: ${data.is_admin ? 'YES ( )' : 'NO'}
`,
      `├── Built-in: ${data.is_built_in ? 'YES' : 'NO'}`,
      `├── Orphaned: ${data.is_orphaned ? 'YES ( )' : 'NO'}`,
      '│',
      '├── Account Details',
      `│   ├── Enabled: ${data.account ? (data.account.enabled
 ? 'YES' : 'NO (Disabled)') : 'N/A'}`,
      `│   ├── Password Expires: ${data.account ? (data.accoun
t.password_expires || 'Never') : 'N/A'}`,
      `│   └── Last Logon: ${data.account ? (data.account.last
_logon || 'N/A') : 'N/A'}`,
      '│',
      `├── Groups (${(data.groups || []).length})`,
      ...(data.groups || []).map(g => `│   ├── ${g.name} ${g.i
s_admin ? '[ADMIN]' : ''} (${g.sid})`),
      '│',
      `├── LSA Rights & Privileges (${(data.rights || []).leng
th})`,
      ...(data.rights || []).slice(0, 10).map(r => `│   ├── ${
r}`),
      '│',
      `├── Profile: ${data.profile ? data.profile.profile_path
 : 'None'}`,
      `├── Active Sessions: ${(data.sessions || []).length}`,
      `└── Running Processes: ${(data.processes || []).length}
`,
    ];

    if (treeEl) treeEl.textContent = lines.join('\n');

    // Рендеринг событий пользователя
    const auditEvents = data.audit_events || [];
```

```
    if (eventsCountEl) eventsCountEl.textContent = String(auditEvents.length);

    if (eventsTbody) {
      if (auditEvents.length === 0) {
        eventsTbody.innerHTML = '<tr><td colspan="5" class="text-center py-3 text-muted">События безопасности для данного пользователя не зафиксированы в Security Log</td></tr>';
      } else {
        eventsTbody.innerHTML = auditEvents.map(ev => {
          let badgeClass = 'bg-secondary';
          if (ev.event_id === 4624) badgeClass = 'bg-success';
          if (ev.event_id === 4625 || ev.event_id === 4740) badgeClass = 'bg-danger';
          if (ev.event_id === 4724) badgeClass = 'bg-warning text-dark';

          const shortTime = ev.timestamp ? ev.timestamp.replace('T', ' ').slice(0, 19) : '—';
          return `
            <tr>
              <td class="text-muted">${escapeHtml(shortTime)}</td>
              <td><span class="badge ${badgeClass}">${ev.event_id}</span></td>
              <td class="fw-semibold text-primary">${escapeHtml(ev.event_name)}</td>
              <td>${escapeHtml(ev.caller_account || ev.target_account || '—')}</td>
              <td class="small text-truncate" style="max-width: 300px;" title="${escapeHtml(ev.description)}">${escapeHtml(ev.description)}</td>
            </tr>
          `;
        }).join('');
      }
    }
  } catch (err) {
```

```
    if (treeEl) treeEl.textContent = `Ошибка получения досье: ${err.message}`;
    if (eventsTbody) eventsTbody.innerHTML = `<tr><td colspan="5" class="text-center text-danger py-3">Ошибка загрузки событий: ${err.message}</td></tr>`;
  }
};

/**
 * Просмотр участников группы
 */
window._viewGroupMembers = function (groupName) {
  const group = _groupsCache.find(g => g.name.toLowerCase() === groupName.toLowerCase());
  if (!group) return;
  const members = (group.members || []).join('\n') || 'Нет участников';
  alert(`Участники группы ${groupName}:\n\n${members}`);
};

/**
 * Проверка Who is Admin?
 */
async function runWhoIsAdminCheck() {
  const outputEl = document.getElementById('security-audit-results');
  if (outputEl) outputEl.innerHTML = '<div class="text-muted text-center py-3">Анализ администраторов системы...</div>';
  try {
    const res = await fetch('/api/windows/identity/who-is-admin');
    const admins = await res.json();
    if (!admins || admins.length === 0) {
      if (outputEl) outputEl.innerHTML = '<div>Администраторы не найдены</div>';
      return;
    }
    const html = admins.map(a => `
```

```
      <div class="mb-2 pb-1 border-bottom">
        <div class="fw-bold text-danger"><i class="bi bi-shiel
d-lock-fill me-1"></i> ${escapeHtml(a.username)} <span class="
badge bg-secondary">${a.is_direct ? 'Прямой доступ' : 'Вложенн
ое членство'}</span></div>
        <div class="small text-muted font-monospace">SID: ${a.
sid}</div>
        <div class="small text-primary">Цепочка: ${(a.path_to_
admin || []).join(' → ')}</div>
      </div>
    `).join('');
    if (outputEl) outputEl.innerHTML = html;
  } catch (err) {
    if (outputEl) outputEl.innerHTML = `<div class="text-dange
r">Ошибка: ${err.message}</div>`;
  }
}

/**
 * Проверка Logon as Service
 */
async function runServiceLogonCheck() {
  const outputEl = document.getElementById('security-audit-res
ults');
  if (outputEl) outputEl.innerHTML = '<div class="text-muted t
ext-center py-3">Поиск аккаунтов с SeServiceLogonRight...</div
>';
  try {
    const res = await fetch('/api/windows/identity/who-can-log
on-as-service');
    const list = await res.json();
    if (!list || list.length === 0) {
      if (outputEl) outputEl.innerHTML = '<div class="text-suc
cess">Специфических сервисных учетных записей не назначено.</d
iv>';
      return;
    }
    const html = `<h6>Аккаунты с правом SeServiceLogonRight:</
```

```
h6>` + list.map(a => `<div class="font-monospace text-primary mb-1">• ${escapeHtml(a)}</div>`).join('');
    if (outputEl) outputEl.innerHTML = html;
  } catch (err) {
    if (outputEl) outputEl.innerHTML = `<div class="text-danger">Ошибка: ${err.message}</div>`;
  }
}

/**
 * Проверка RDP
 */
async function runRdpCheck() {
  const outputEl = document.getElementById('security-audit-results');
  if (outputEl) outputEl.innerHTML = '<div class="text-muted text-center py-3">Поиск аккаунтов с доступом к RDP...</div>';
  try {
    const res = await fetch('/api/windows/identity/who-can-rdp');
    const list = await res.json();
    const html = `<h6>Аккаунты и группы с доступом к Remote Desktop:</h6>` + list.map(a => `<div class="font-monospace text-warning-emphasis mb-1">• ${escapeHtml(a)}</div>`).join('');
    if (outputEl) outputEl.innerHTML = html;
  } catch (err) {
    if (outputEl) outputEl.innerHTML = `<div class="text-danger">Ошибка: ${err.message}</div>`;
  }
}

/**
 * Проверка осиротевших записей
 */
async function runOrphanedCheck() {
  const outputEl = document.getElementById('security-audit-results');
  if (outputEl) outputEl.innerHTML = '<div class="text-muted t
```

```
ext-center py-3">Поиск осиротевших SID и профилей...</div>';
  try {
    const [resSids, resProfiles] = await Promise.all([
      fetch('/api/windows/identity/orphaned-sids'),
      fetch('/api/windows/identity/orphaned-profiles'),
    ]);
    const sids = await resSids.json();
    const profiles = await resProfiles.json();

    if ((!sids || sids.length === 0) && (!profiles || profiles
.length === 0)) {
      if (outputEl) outputEl.innerHTML = '<div class="text-suc
cess"><i class="bi bi-check-circle-fill me-1"></i> Осиротевших
 SID и профилей в системе не обнаружено.</div>';
      return;
    }

    let html = '<h6>Осиротевшие субъекты:</h6>';
    for (const s of sids) {
      html += `<div class="text-danger font-monospace mb-1">
SID: ${s.sid} (${s.location})</div>`;
    }
    for (const p of profiles) {
      html += `<div class="text-warning font-monospace mb-1">
  Профиль без пользователя: ${p.profile_path} (SID: ${p.sid})<
/div>`;
    }
    if (outputEl) outputEl.innerHTML = html;
  } catch (err) {
    if (outputEl) outputEl.innerHTML = `<div class="text-dange
r">Ошибка: ${err.message}</div>`;
  }
}

/**
 * Исследование PID
 */
async function handleExplainPid() {
```

```
  const input = document.getElementById('input-explain-pid');
  const outputEl = document.getElementById('pid-dossier-output
');
  const pidVal = parseInt(input?.value || '0', 10);

  if (!pidVal || isNaN(pidVal)) {
    if (window.toast) window.toast.error('Ошибка', 'Укажите ко
ррректный PID процесса');
    return;
  }

  if (outputEl) {
    outputEl.style.display = 'block';
    outputEl.textContent = `Исследование процесса PID ${pidVal
}...`;
  }

  try {
    const res = await fetch(`/api/windows/identity/explain-pid
/${pidVal}`);
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();

    const acc = data.account || {};
    const privs = data.privileges || [];
    const groups = data.groups || [];

    const lines = [
      `PID ${data.pid}`,
      '│',
      `├── Process: ${data.process_name}`,
      `├── Parent: ${data.parent_name} (PID ${data.parent_pid}
)`,
      '│',
      '├── Account',
      `│   ├── ${acc.domain}\\${acc.user}`,
      `│   └── SID: ${acc.sid}`,
      '│',
```

```
      `├── Groups (${groups.length})`,
      ...groups.map(g => `│   ├── ${g}`),
      '│',
      `├── Integrity: ${data.integrity}`,
      `├── Elevated: ${data.elevated ? 'YES ( )' : 'NO'}`,
      '│',
      `├── Privileges (${privs.length})`,
      ...privs.map(p => `│   ├── ${p}`),
      '│',
      '└── Session',
      '    ├── Console',
      `    └── Session ID: ${data.session_id}`,
    ];

    if (outputEl) outputEl.textContent = lines.join('\n');
  } catch (err) {
    if (outputEl) outputEl.textContent = `Ошибка исследования PID ${pidVal}: ${err.message}`;
  }
}

function escapeHtml(str) {
  if (!str) return '';
  return String(str)
    .replace(/&/g, '&amp;')
    .replace(/</g, '&lt;')
    .replace(/>/g, '&gt;')
    .replace(/"/g, '&quot;');
}

// Автозапуск при загрузке вкладки
if (document.readyState === 'loading') {
  document.addEventListener('DOMContentLoaded', initAccountsIdentityTab);
} else {
  initAccountsIdentityTab();
}

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/admin/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/admin/index.html" class
="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/admin
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-12 08:10:00
================================================================
===============
-->
<!DOCTYPE html>

<html lang="ru">
<head>
<meta charset="utf-8"/>
<meta content="width=device-width, initial-scale=1.0" name="vi
ewport"/>
<title>AI Assistant - Admin</title>
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/
css/bootstrap.min.css" rel="stylesheet"/>
```

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.10.0/font/bootstrap-icons.css" rel="stylesheet"/>
<link href="/html/css/main.css?v=20260925_v1" rel="stylesheet" />
<script src="/html/js/toast.js?v=20260925_v1"></script>
<script src="https://cdn.jsdelivr.net/npm/i18next@23.11.5/i18next.min.js"></script>
</head>
<body>
<!-- Password Protection Modal -->
<div aria-hidden="true" aria-labelledby="passwordModalLabel" class="modal fade" data-bs-backdrop="static" data-bs-keyboard="false" id="passwordModal" tabindex="-1">
<div class="modal-dialog modal-dialog-centered">
<div class="modal-content bg-dark text-white border-secondary">
<div class="modal-header border-secondary">
<h5 class="modal-title" id="passwordModalLabel"><i class="bi bi-lock-fill"></i> Вход в админку</h5>
<button aria-label="Close" class="btn-close btn-close-white" data-bs-dismiss="modal" id="closePasswordModal" type="button"></button>
</div>
<div class="modal-body p-4">
<p class="text-muted">Введите пароль для доступа к административному интерфейсу:</p>
<div class="mb-3">
<input autocomplete="current-password" autofocus="" class="form-control bg-dark text-white border-secondary" data-1p-ignore="true" data-lpignore="true" id="admin-password" placeholder="•••••" type="password"/>
</div>
<div class="alert alert-danger d-none" id="password-error">Неверный пароль</div>
<button class="btn btn-primary w-100" id="login-btn" type="button">Войти</button>
</div>
</div>
```

```
</div>
</div>
<div class="container-fluid mt-2" id="admin-interface">
<!-- Navigation Bar -->
<div class="d-flex align-items-center justify-content-between mb-3 main-nav-container rounded px-2 py-1">
<ul class="nav nav-tabs flex-nowrap flex-grow-1 border-0 me-2 align-items-center" id="mainTabs">
<!-- Кнопка: О системе (сверху и первая) -->
<li class="nav-item">
<button class="nav-link d-flex align-items-center gap-1 active " data-bs-target="#tab-about-system" data-i18n="tabs.aboutSystem" data-tab="tab-about-system" type="button">
<i class="bi bi-info-circle-fill text-info me-1"></i>
<span>О системе</span>
</button>
</li>
<!-- Разделитель -->
<li class="nav-item d-flex align-items-center">
<div class="nav-group-divider"></div>
</li>
<!-- Группа: Диалог и аудио -->
<li class="nav-item dropdown">
<button class="nav-link dropdown-toggle active d-flex align-items-center gap-1" id="dialogTabsDropdown" type="button">
<i class="bi bi-chat-dots-fill text-primary me-1"></i>
<span data-i18n="tabs.groupDialog">Диалог и аудио</span>
</button>
<div class="dropdown-menu shadow border-secondary-subtle">
<button class="dropdown-item active d-flex align-items-center gap-2" data-bs-target="#tab-chat" data-i18n="tabs.chat" data-tab="tab-chat" type="button">
            Чат
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-voice" data-i18n="tabs.voice" data-tab="tab-voice" type="button">
            Голос и диаризация
```

```
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-tts" data-i18n="tabs.tts" data-tab="tab-tts" type="button">
            Настройка TTS
          </button>
</div>
</li>
<!-- Разделитель -->
<li class="nav-item d-flex align-items-center">
<div class="nav-group-divider"></div>
</li>
<!-- Группа: ИИ и Знания -->
<li class="nav-item dropdown">
<button class="nav-link dropdown-toggle d-flex align-items-center gap-1" id="aiTabsDropdown" type="button">
<i class="bi bi-cpu-fill text-info me-1"></i>
<span data-i18n="tabs.groupAi">ИИ и Знания</span>
</button>
<div class="dropdown-menu shadow border-secondary-subtle">
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-rag" data-i18n="tabs.rag" data-tab="tab-rag" type="button">
            RAG
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-pixelrag" data-tab="tab-pixelrag" type="button">
            PixelRAG
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-models" data-i18n="tabs.models" data-tab="tab-models" type="button">
            Модели и API
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-agents" data-i18n="tabs.agents" data-tab="tab-agents" type="button">
```

```
            🤖 Агенты ИИ
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-skills" data-i18n="tabs.skills" data-tab="tab-skills" type="button">
            ⚡ Навыки
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-mcp" data-i18n="tabs.mcp" data-tab="tab-mcp" type="button">
            🔌 MCP Серверы
          </button>
</div>
</li>
<!-- Разделитель -->
<li class="nav-item d-flex align-items-center">
<div class="nav-group-divider"></div>
</li>
<!-- Группа: Плагины -->
<li class="nav-item dropdown" id="pluginsNavDropdownWrapper">
<button class="nav-link dropdown-toggle d-flex align-items-center gap-1" id="pluginsTabsDropdown" type="button">
<i class="bi bi-puzzle-fill text-warning me-1"></i>
<span data-i18n="tabs.groupPlugins">Плагины</span>
</button>
<div class="dropdown-menu shadow border-secondary-subtle" id="nav-plugins-dropdown-menu">
<div class="nav-enabled-plugins-list" id="nav-enabled-plugins-list">
<!-- Dynamic enabled plugins will be injected here -->
</div>
<hr class="dropdown-divider my-1"/>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-plugins" data-i18n="tabs.managePlugins" data-tab="tab-plugins" type="button">
            🧩 Все плагины
          </button>
</div>
```

```
</li>
<!-- Разделитель -->
<li class="nav-item d-flex align-items-center">
<div class="nav-group-divider"></div>
</li>
<!-- Группа: Приложения (/apps) -->
<li class="nav-item dropdown">
<button class="nav-link dropdown-toggle d-flex align-items-cen
ter gap-1" id="appsTabsDropdown" type="button">
<i class="bi bi-grid-fill text-danger me-1"></i>
<span data-i18n="tabs.groupApps">Приложения</span>
</button>
<div class="dropdown-menu shadow border-secondary-subtle">
<button class="dropdown-item d-flex align-items-center gap-2"
data-bs-target="#tab-about-system" data-i18n="tabs.aboutSystem
" data-tab="tab-about-system" type="button">
            i О системе
          </button>
<button class="dropdown-item d-flex align-items-center gap-2"
data-bs-target="#tab-network" data-i18n="tabs.network" data-ta
b="tab-network" type="button">
            □ Network Terminal
          </button>
<button class="dropdown-item d-flex align-items-center gap-2"
data-bs-target="#tab-hardware-load-inspector" data-i18n="tabs.
systemInspector" data-tab="tab-hardware-load-inspector" type="
button">
            □ System Inspector
          </button>
<button class="dropdown-item d-flex align-items-center gap-2"
data-bs-target="#tab-chat" data-i18n="tabs.chat" data-tab="tab
-chat" type="button">
            □ Чат
          </button>
<button class="dropdown-item d-flex align-items-center gap-2"
data-bs-target="#tab-scenarios" data-i18n="tabs.scenarios" dat
a-tab="tab-scenarios" type="button">
            □□ Чат и сценарии
```

```
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-user-assistant" data-tab="tab-user-assistant" type="button">
            User Assistant
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-gcloud" data-tab="tab-gcloud" type="button">
            Google Cloud Monitor
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-website-monitor" data-tab="tab-website-monitor" type="button">
            Website Intelligence
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-cloudflared" data-tab="tab-cloudflared" type="button">
            Cloudflared Monitor
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-windows-backup" data-tab="tab-windows-backup" type="button">
            Windows Backup
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-startup-auditor" data-tab="tab-startup-auditor" type="button">
            Windows Startup Auditor
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-process-leaks" data-tab="tab-process-leaks" type="button">
            1. Утечки и микрофризы
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-forensics" data-tab="tab-forensics" type=
```

```
"button">
            🔍 2. Поведенческая форензика
          </button>
<button class="dropdown-item d-flex align-items-center gap-2"
data-bs-target="#tab-throttling" data-tab="tab-throttling" typ
e="button">
            🔥 3. Качество ядра и Троттлинг
          </button>
<button class="dropdown-item d-flex align-items-center gap-2"
data-bs-target="#tab-storage-wear" data-tab="tab-storage-wear"
 type="button">
            💾 4. Износ дисков и батареи
          </button>
<button class="dropdown-item d-flex align-items-center gap-2"
data-bs-target="#tab-peripherals" data-tab="tab-peripherals" t
ype="button">
            🔌 5. Сеть и периферия
          </button>
<button class="dropdown-item d-flex align-items-center gap-2"
data-bs-target="#tab-system-logs" data-i18n="tabs.systemLogs"
data-tab="tab-system-logs" type="button">
            📜 Логи операционной системы
          </button>
</div>
</li>
<!-- Разделитель -->
<li class="nav-item d-flex align-items-center">
<div class="nav-group-divider"></div>
</li>
<!-- Группа: Администрирование -->
<li class="nav-item dropdown">
<button class="nav-link dropdown-toggle d-flex align-items-cen
ter gap-1" id="adminTabsDropdown" type="button">
<i class="bi bi-gear-wide-connected text-success me-1"></i>
<span data-i18n="tabs.groupSystem">Администрирование</span>
</button>
<div class="dropdown-menu shadow border-secondary-subtle">
<button class="dropdown-item d-flex align-items-center gap-2"
```

```
data-bs-target="#tab-admin" data-i18n="tabs.admin" data-tab="tab-admin" type="button">
            Управление
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-users" data-i18n="tabs.users" data-tab="tab-users" type="button">
            Внутренние пользователи
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-windows-users" data-tab="tab-windows-users" type="button">
            Пользователи Windows
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-google-accounts" data-i18n="tabs.googleAccounts" data-tab="tab-google-accounts" type="button">
            Google аккаунты
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-logs" data-i18n="tabs.logs" data-tab="tab-logs" type="button">
            Системные логи
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-instructions" data-i18n="tabs.instructions" data-tab="tab-instructions" type="button">
            Системные инструкции
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-help" data-i18n="tabs.help" data-tab="tab-help" type="button">
            Справочник
          </button>
</div>
</li>
</ul>
<!-- Auth Container -->
```

```
<div class="d-flex align-items-center gap-2 flex-shrink-0 ms-auto pb-1" id="nav-auth-container">
<!-- Active Model & Search Badges -->
<span class="badge rounded-pill align-items-center gap-1 px-3 py-2 shadow-sm border" id="chat-model-badge" style="font-size: 0.8rem; display: none; cursor: pointer;" title="Текущая модель ИИ">
        Загрузка...
      </span>
<span class="badge rounded-pill align-items-center gap-1 px-3 py-2 shadow-sm border" id="chat-search-badge" style="font-size: 0.8rem; display: none;" title="Провайдер поиска">
</span>
<!-- Google OAuth Register / Sign-in Button (Shown when not logged in) -->
<button class="btn btn-sm btn-light text-dark fw-semibold d-flex align-items-center gap-2 px-3 py-1 shadow-sm rounded-pill btn-google-login" data-i18n-title="auth.googleSignInTitle" id="nav-google-btn" style="border: 1px solid rgba(0,0,0,0.15);" title="Войти или зарегистрироваться через Google">
<svg height="18" viewbox="0 0 24 24" width="18"><path d="M22.56 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h5.92c-.26 1.37-1.04 2.53-2.21 3.31v2.77h3.57c2.08-1.92 3.28-4.74 3.28-8.09z" fill="#4285F4"></path><path d="M12 23c2.97 0 5.46-.98 7.28-2.66l-3.57-2.77c-.98.66-2.23 1.06-3.71 1.06-2.86 0-5.29-1.93-6.16-4.53H2.18v2.84C3.99 20.53 7.7 23 12 23z" fill="#34A853"></path><path d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s.13-1.43.35-2.09V7.06H2.18C1.43 8.55 1 10.22 1 12s.43 3.45 1.18 4.94l2.85-2.22.81-.63z" fill="#FBBC05"></path><path d="M12 5.38c1.62 0 3.06.56 4.21 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 1 3.99 3.47 2.18 7.06l3.66 2.84c.87-2.6 3.3-4.52 6.16-4.52z" fill="#EA4335"></path></svg>
<span data-i18n="auth.googleSignUp">Регистрация через Google</span>
</button>
<!-- Authenticated User Menu Dropdown (Shown when logged in) -->
<div class="dropdown d-none" id="user-menu-dropdown-wrapper">
```

```
<button aria-expanded="false" class="btn btn-sm btn-outline-secondary d-flex align-items-center gap-2 px-2 py-1 rounded-pill dropdown-toggle" data-bs-toggle="dropdown" id="user-settings-btn" title="Пользовательское меню" type="button">
<img class="rounded-circle d-none" height="22" id="nav-user-avatar" src="" style="object-fit: cover;" width="22"/>
<i class="bi bi-person-circle" id="nav-user-icon"></i>
<span class="small fw-semibold" data-i18n="userSettings.button" id="nav-user-label">Пользователь</span>
</button>
<ul aria-labelledby="user-settings-btn" class="dropdown-menu dropdown-menu-end shadow border-secondary-subtle py-1" style="min-width: 220px;">
<li class="px-3 py-2 border-bottom border-secondary-subtle">
<div class="fw-bold text-truncate" id="user-menu-display-name">Пользователь</div>
<div class="small text-muted text-truncate" id="user-menu-display-email" style="font-size: 0.75rem;">user@gmail.com</div>
</li>
<li><h6 class="dropdown-header small text-uppercase text-muted fw-semibold mt-1">Сервисы</h6></li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 py-1" id="user-menu-btn-mail" type="button">
<i class="bi bi-envelope-check text-primary"></i>
<span>Почта</span>
</button>
</li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 py-1" id="user-menu-btn-docs" type="button">
<i class="bi bi-file-earmark-text text-info"></i>
<span>Google Docs</span>
</button>
</li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 py-1" id="user-menu-btn-sheets" type="button">
```

```
<i class="bi bi-file-earmark-spreadsheet text-success"></i>
<span>Google Таблицы</span>
</button>
</li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1" id="user-menu-btn-news" type="button">
<i class="bi bi-newspaper text-warning"></i>
<span>Новости</span>
</button>
</li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1" id="user-menu-btn-facebook" type="button">
<i class="bi bi-facebook text-primary"></i>
<span>Facebook</span>
</button>
</li>
<li><hr class="dropdown-divider my-1"/></li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1" data-bs-target="#userSettingsModal" data-bs-toggle="modal
" type="button">
<i class="bi bi-gear-fill text-secondary"></i>
<span data-i18n="userSettings.title">Настройки профиля</span>
</button>
</li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1 text-danger btn-user-logout" type="button">
<i class="bi bi-box-arrow-right"></i>
<span>Выйти</span>
</button>
</li>
</ul>
</div>
<!-- Quick Settings Gear Button -->
<button class="btn btn-sm btn-outline-secondary d-flex align-i
```

```
tems-center justify-content-center p-0 rounded-circle" data-bs
-target="#userSettingsModal" data-bs-toggle="modal" data-i18n-
title="userSettings.title" id="user-settings-gear-btn" style="
width: 32px; height: 32px;" title="Настройки">
<i class="bi bi-gear-fill"></i>
</button>
</div>
</div>
<!-- Main Tab Content Container -->
<div class="tab-content" id="mainTabContent">
<!-- Chat Tab -->
<div class="tab-pane fade show active" id="tab-chat"></div>
<!-- Plugins Tab -->
<div class="tab-pane fade" id="tab-plugins"></div>
<!-- Admin Tab -->
<div class="tab-pane fade" id="tab-admin"></div>
<!-- Users Tab -->
<div class="tab-pane fade" id="tab-users"></div>
<!-- Windows Users Tab -->
<div class="tab-pane fade" id="tab-windows-users"></div>
<!-- Google Accounts Tab -->
<div class="tab-pane fade" id="tab-google-accounts"></div>
<!-- RAG Tab -->
<div class="tab-pane fade" id="tab-rag"></div>
<!-- PixelRAG Tab -->
<div class="tab-pane fade" id="tab-pixelrag"></div>
<!-- Telegram RAG Tab -->
<div class="tab-pane fade" id="tab-telegram-rag"></div>
<!-- Smart News Tab -->
<div class="tab-pane fade" id="tab-news"></div>
<!-- Voice Diarization Tab -->
<div class="tab-pane fade" id="tab-voice"></div>
<!-- Models Tab -->
<div class="tab-pane fade" id="tab-models"></div>
<!-- Agents Tab -->
<div class="tab-pane fade" id="tab-agents"></div>
<!-- Search Tab -->
<div class="tab-pane fade" id="tab-search"></div>
```

```
<!-- TTS Tab -->
<div class="tab-pane fade" id="tab-tts"></div>
<!-- Sources Tab -->
<div class="tab-pane fade" id="tab-sources"></div>
<!-- Logs Tab -->
<div class="tab-pane fade" id="tab-logs"></div>
<!-- Help Tab -->
<div class="tab-pane fade" id="tab-help"></div>
<!-- Instructions Tab -->
<div class="tab-pane fade" id="tab-instructions"></div>
<!-- Skills Tab -->
<div class="tab-pane fade" id="tab-skills"></div>
<!-- MCP Servers Tab -->
<div class="tab-pane fade" id="tab-mcp"></div>
<!-- Network Terminal Tab -->
<div class="tab-pane fade" id="tab-network"></div>
<!-- Hardware Load Inspector Tab -->
<div class="tab-pane fade" id="tab-hardware-load-inspector"></
div>
<!-- Processes Load & Activity Inspector Tab -->
<div class="tab-pane fade" id="tab-processes-load-inspector"><
/div>
<!-- About System Tab -->
<div class="tab-pane fade" id="tab-about-system"></div>
<!-- Observability & Diagnostics Tab -->
<div class="tab-pane fade" id="tab-observability"></div>
<!-- Windows System Administrator Tab -->
<div class="tab-pane fade" id="tab-windows-admin"></div>
<!-- Cloudflared Monitor Tab -->
<div class="tab-pane fade" id="tab-cloudflared"></div>
<!-- User Assistant Tab -->
<div class="tab-pane fade" id="tab-user-assistant"></div>
<!-- Google Cloud Monitor Tab -->
<div class="tab-pane fade" id="tab-gcloud"></div>
<!-- Website Intelligence Monitor Tab -->
<div class="tab-pane fade" id="tab-website-monitor"></div>
<!-- System Control Center Tab (System Logs & Activity) -->
<div class="tab-pane fade" id="tab-system-control"></div>
```

```
<!-- Post-Install Wizard Tab -->
<div class="tab-pane fade" id="tab-post-install-wizard"></div>
<!-- Maintenance & Recovery Tab -->
<div class="tab-pane fade" id="tab-maintenance-recovery"></div
>
<!-- System Log Viewer Tab -->
<div class="tab-pane fade" id="tab-system-logs"></div>
<!-- Software Audit Tab -->
<div class="tab-pane fade" id="tab-software-audit"></div>
<!-- Registry Viewer Tab -->
<div class="tab-pane fade" id="tab-registry-viewer"></div>
<!-- Windows Backup Manager Tab -->
<div class="tab-pane fade" id="tab-windows-backup"></div>
<!-- Startup Auditor Tab -->
<div class="tab-pane fade" id="tab-startup-auditor"></div>
<!-- Scenarios Tab -->
<div class="tab-pane fade" id="tab-scenarios"></div>
<!-- Wikipedia Research Lab Tab -->
<div class="tab-pane fade" id="tab-wikipedia-research"></div>
<!-- User Directories and Storage Tab -->
<div class="tab-pane fade" id="tab-user-directories"></div>
<!-- Diagnostic and Forensics Tabs -->
<div class="tab-pane fade" id="tab-process-leaks"></div>
<div class="tab-pane fade" id="tab-forensics"></div>
<div class="tab-pane fade" id="tab-throttling"></div>
<div class="tab-pane fade" id="tab-storage-wear"></div>
<div class="tab-pane fade" id="tab-peripherals"></div>
</div>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist
/js/bootstrap.bundle.min.js"></script>
<link href="/html/ai_modal_dialog/style.css?v=20261012_v2" rel
="stylesheet"/>
<script src="/html/ai_modal_dialog/main.js?v=20261008_v1"></sc
ript>
<script src="/html/js/window-api.js?v=20260925_v1"></script>
<script src="/html/js/appConfigEditor.js?v=20260925_v1"></scri
pt>
```

```
<script src="/html/js/chatService.js?v=20260925_v1"></script>
<!-- Admin Main - Modular Architecture -->
<script src="/html/admin/main-refactored.js?v=20260925_v1" typ
e="module"></script>
<!-- Optimization Patches -->
<script src="/html/admin/lazy-init-patch.js?v=20260925_v1" typ
e="module"></script>
<script src="/html/users_tab/debounce-patch.js?v=20260925_v1"
type="module"></script>
<!-- HELP Modal -->
<div aria-hidden="true" aria-labelledby="help-modal-label" cla
ss="modal fade modal-help" id="help-modal" tabindex="-1">
<div class="modal-dialog modal-dialog-scrollable">
<div class="modal-content">
<div class="modal-header">
<h5 class="modal-title" data-i18n="helpModal.title" id="help-m
odal-label">Помощь</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body" id="help-modal-content">
<p data-i18n="common.loading">Загрузка...</p>
</div>
<div class="modal-footer">
<button class="btn btn-secondary" data-bs-dismiss="modal" data
-i18n="helpModal.close" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
<!-- Prompt Preview Modal -->
<div aria-hidden="true" aria-labelledby="prompt-preview-modal-
label" class="modal fade" id="prompt-preview-modal" tabindex="
-1">
<div class="modal-dialog modal-lg">
<div class="modal-content">
<div class="modal-header">
<h5 class="modal-title" id="prompt-preview-modal-label">Предпр
```

```
осмотр промта</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body" id="prompt-preview-content">
<p>Загрузка...</p>
</div>
<div class="modal-footer">
<button class="btn btn-secondary" data-bs-dismiss="modal" type
="button">Закрыть</button>
</div>
</div>
</div>
</div>
<!-- Chat Logic & RAG Architecture Modal -->
<div aria-hidden="true" aria-labelledby="chat-logic-modal-labe
l" class="modal fade" id="chat-logic-modal" tabindex="-1">
<div class="modal-dialog modal-xl modal-dialog-scrollable">
<div class="modal-content bg-dark text-white border-secondary"
>
<div class="modal-header border-secondary d-flex justify-conte
nt-between align-items-center">
<h5 class="modal-title d-flex align-items-center gap-2" id="ch
at-logic-modal-label">
<i class="bi bi-diagram-3-fill text-info"></i>  Архитектура
и алгоритм диспетчеризации ответов чата
        </h5>
<div class="d-flex align-items-center gap-2">
<a class="btn btn-sm btn-outline-info" href="/html/assets/chat
_and_rag_logic.svg" target="_blank" title="Открыть SVG в новой
 вкладке">
<i class="bi bi-arrows-fullscreen"></i> SVG
          </a>
<button aria-label="Close" class="btn-close btn-close-white" d
ata-bs-dismiss="modal" type="button"></button>
</div>
</div>
<div class="modal-body p-3">
```

```
<!-- Tabs for Schema vs Documentation -->
<ul class="nav nav-pills mb-3" id="chatLogicTabs" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active" data-bs-target="#chat-logic-svg-pane" data-bs-toggle="pill" id="chat-logic-svg-tab" role="tab" type="button">  Интерактивная блок-схема</button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link" data-bs-target="#chat-logic-doc-pane" data-bs-toggle="pill" id="chat-logic-doc-tab" role="tab" type="button">  Описание алгоритма и сценариев</button>
</li>
</ul>
<div class="tab-content" id="chatLogicTabContent">
<!-- SVG Diagram Tab -->
<div class="tab-pane fade show active text-center" id="chat-logic-svg-pane" role="tabpanel">
<div class="p-2 border border-secondary rounded bg-black d-flex justify-content-center align-items-center" style="overflow-x: auto; min-height: 520px;">
<img alt="Схема логики чата" src="/html/assets/chat_and_rag_logic.svg?v=20260817_1" style="max-width: 100%; height: auto; border-radius: 6px;"/>
</div>
</div>
<!-- Documentation Text Tab -->
<div class="tab-pane fade" id="chat-logic-doc-pane" role="tabpanel">
<div class="p-3 bg-black rounded border border-secondary text-light small">
<h6 class="text-info fw-bold mb-2">  Режимы работы (config.json → rag.mode)</h6>
<ul class="mb-3">
<li><strong class="text-success">rag+model (Гибридный / Основной):</strong> Сначала проверяется локальная медиатека. Если фильм найден — LLM красиво рассказывает о нём со ссылкой на наличие в медиатеке. Если фильм отсутствует — LLM обращается к веб
```

```
-поиску (MCP/Grounding) и рассказывает о фильме с кнопкой загр
узки <code>&lt;film&gt;</code>.</li>
<li><strong class="text-warning">rag (Строгий локальный):</str
ong> Работает только по базе <code>media.db</code>. При отсутс
твии тайтла выдаёт чёткое <em>«В локальной базе не найдено»</e
m> без ложных подстановок.</li>
<li><strong class="text-primary">model (Без базы):</strong> Пр
ямой ответ LLM и веб-поиск без обращения к дискам.</li>
</ul>
<h6 class="text-info fw-bold mb-2">□ Сценарии обработки</h6>
<div class="table-responsive mb-3">
<table class="table table-dark table-bordered table-sm small">
<thead>
<tr class="table-secondary text-dark">
<th>Сценарий</th>
<th>Пример</th>
<th>Алгоритм</th>
<th>Результат</th>
</tr>
</thead>
<tbody>
<tr>
<td class="fw-bold text-success">Фильм есть в базе</td>
<td><em>«Расскажи о фильме А зори здесь тихие»</em></td>
<td>Точный поиск в <code>media.db</code> (Score 1.0) / FAISS ≥
 0.60 → контекст в LLM</td>
<td>LLM сообщает, что фильм <strong>доступен в медиатеке</stro
ng>, описывает сюжет и тегирует <code>&lt;film&gt;</code>.</td
>
</tr>
<tr>
<td class="fw-bold text-danger">Фильма нет в базе</td>
<td><em>«Расскажи о сериале Сваты»</em></td>
<td>Локальный RAG даёт 0 совпадений → Fallback на Web Search (
MCP)</td>
<td>LLM рассказывает о сериале, указывает, что его <strong>нет
 на дисках</strong>, тегирует <code>&lt;film&gt;</code> для то
ррентов.</td>
```

```
</tr>
<tr>
<td class="fw-bold text-info">Команда запуска</td>
<td><em>«Включи А зори здесь тихие»</em></td>
<td>Проверка пути на диске → WebSocket <code>play_file_by_path</code></td>
<td>Мгновенный старт в плеере (Direct Play).</td>
</tr>
<tr>
<td class="fw-bold text-primary">Подборка / Настроение</td>
<td><em>«Найди боевик на вечер»</em></td>
<td>Извлечение подходящих жанров → LLM формирует диалоговую подборку</td>
<td>Живая рекомендация с обоснованием «почему стоит посмотреть».</td>
</tr>
</tbody>
</table>
</div>
<h6 class="text-info fw-bold mb-1">  Порог косинусного сходства (Cosine Similarity)</h6>
<p class="text-muted mb-0">Для исключения ложных срабатываний порог в <code>rag_search_tool</code> установлен на <strong>0.60</strong>. Запросы с более низким сходством считаются отсутствующими в локальной базе и корректно перенаправляются в интернет-поиск.</p>
</div>
</div>
</div>
</div>
<div class="modal-footer border-secondary">
<button class="btn btn-sm btn-secondary" data-bs-dismiss="modal" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
<!-- User Settings Modal -->
```

```
<div aria-hidden="true" aria-labelledby="userSettingsModalLabel" class="modal fade" id="userSettingsModal" tabindex="-1">
<div class="modal-dialog modal-dialog-centered">
<div class="modal-content shadow border-secondary-subtle">
<div class="modal-header border-secondary-subtle">
<h5 class="modal-title d-flex align-items-center gap-2" id="userSettingsModalLabel">
<i class="bi bi-person-gear text-primary"></i>
<span data-i18n="userSettings.title">Пользовательские настройки</span>
</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body p-3">
<!-- User Account & Google OAuth Card -->
<div class="card bg-body-tertiary border-secondary-subtle mb-3 p-3" data-panel-id="panel-admin-1" id="user-settings-auth-card">
<div class="d-flex align-items-center gap-3">
<img class="rounded-circle border border-2 border-primary" height="48" id="user-settings-avatar" src="" style="object-fit: cover;" width="48"/>
<div class="flex-grow-1 min-w-0">
<div class="d-flex align-items-center gap-2">
<strong class="text-truncate" id="user-settings-name">Пользователь</strong>
<span class="badge bg-primary" id="user-settings-role-badge">User</span>
</div>
<div class="small text-muted text-truncate" id="user-settings-email">email@gmail.com</div>
<div class="small mt-1 text-muted" id="user-settings-google-badge">
<i class="bi bi-google text-danger me-1"></i>Google OAuth
              </div>
</div>
<button class="btn btn-outline-danger btn-sm btn-user-logout"
```

```
title="Выйти" type="button">
<i class="bi bi-box-arrow-right"></i>
</button>
</div>
</div>
<!-- Guest / Connect Google OAuth Card -->
<div class="card bg-body-tertiary border-secondary-subtle mb-3
 p-3 text-center d-none" data-panel-id="panel-admin-2" id="use
r-settings-guest-card">
<div class="text-muted small mb-2">
<i class="bi bi-shield-lock me-1"></i> Авторизация и синхрониз
ация аккаунта
          </div>
<button class="btn btn-outline-danger btn-sm w-100 d-flex alig
n-items-center justify-content-center gap-2 btn-google-login"
type="button">
<i class="bi bi-google"></i>
<span>Войти через Google (OAuth)</span>
</button>
<div class="small text-muted mt-2" style="font-size: 0.75rem;"
>
            Синхронизация с Google Drive, Docs, Calendar и Con
tacts
          </div>
</div>
<!-- Language Selection -->
<div class="mb-3">
<label class="form-label fw-semibold small d-flex align-items-
center gap-2" for="modal-lang-selector">
<i class="bi bi-translate text-info"></i>
<span data-i18n="userSettings.language">Язык интерфейса</span>
</label>
<select class="form-select lang-selector" id="modal-lang-selec
tor">
<option value="ru">Русский</option>
<option value="en">English</option>
<option value="he">□ □ □ □ □ </option>
</select>
```

```
</div>
<!-- Theme Selection -->
<div class="mb-2">
<label class="form-label fw-semibold small d-flex align-items-center gap-2" for="modal-theme-selector">
<i class="bi bi-palette text-warning"></i>
<span data-i18n="userSettings.theme">Тема оформления</span>
</label>
<select class="form-select theme-selector" id="modal-theme-selector">
<option data-i18n="userSettings.themeLight" value="light">  Светлая</option>
<option data-i18n="userSettings.themeDark" value="dark">  Тёмная</option>
<option data-i18n="userSettings.themeSystem" value="system">  Системная</option>
</select>
</div>
</div>
<div class="modal-footer border-secondary-subtle py-2">
<button class="btn btn-secondary btn-sm" data-bs-dismiss="modal" data-i18n="userSettings.close" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
</body>
</html>
```

lazy-init-patch.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Admin - Lazy-Init-Patch Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля lazy-init
-patch.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/admin/lazy-init-patch.
js?v=20261001_v1" type="module"></script>
 *
 * File: lazy-init-patch.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/admin
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-12 08:20:00
 * ============================================================
==================
 */

/**
 * Lazy Init Patch - Патч для переключения на ленивую загрузку
 вкладок
 *
 * Этот скрипт перехватывает инициализацию main.js и используе
т
 * LazyTabLoader вместо стандартной загрузки всех вкладок сраз
у
 */

import {
```

```
  initLazyTabLoading,
  enhanceTabSwitching,
  restoreLastTab
} from './optimization-init.js';
import { autoPatchTab } from '../js/tab-debounce-auto-patch.js';
import TabStatePersistence from '../js/tab-state-persistence.js';
import TabSwitchOptimizer from '../js/tab-switch-optimizer.js';

console.log('[LazyInitPatch] Patching admin interface initialization...');

// Глобальный хелпер активности вкладок
if (typeof window.isTabActive !== 'function') {
  window.isTabActive = function(tabId) {
    if (typeof document !== 'undefined' && document.hidden) return false;
    if (!tabId) return false;
    const cleanId = tabId.startsWith('tab-') ? tabId : `tab-${tabId}`;
    const rawId = cleanId.replace(/^tab-/, '');
    const pane = document.getElementById(cleanId) || document.getElementById(rawId);
    return pane ? pane.classList.contains('active') : false;
  };
}

// Инициализируем persistence для админки
const tabPersistence = new TabStatePersistence('admin');
window.tabPersistence = tabPersistence;

// Ждем когда main.js загрузится и инициализирует базовые элементы
window.addEventListener('load', async () => {
  console.log('[LazyInitPatch] Window load event, initializing lazy loading...');
```

```
58
59    // Ждем немного чтобы main.js выполнил свои скрипты
60    await new Promise(resolve => setTimeout(resolve, 100));
61
62    // Если есть оригинальная функция switchTab, оборачиваем ее
63    if (window.switchTab && typeof window.switchTab === 'functio
n') {
64      const originalSwitchTab = window.switchTab;
65
66      // Применяем оптимизацию переключения
67      const switchOptimizer = new TabSwitchOptimizer();
68      switchOptimizer.enableDOMCaching();
69
70      // Оборачиваем в enhanced версию с lazy loading
71      let enhancedSwitchTab = enhanceTabSwitching(originalSwitch
Tab);
72
73      // Оборачиваем в optimizer версию для отслеживания произво
дительности
74      const optimizedSwitchTab = switchOptimizer.optimizeSwitchi
ng(enhancedSwitchTab);
75
76      window.switchTab = async function(targetId) {
77        // Вызываем оптимизированную функцию
78        await optimizedSwitchTab.call(this, targetId);
79
80        // Сохраняем активную вкладку
81        if (targetId) {
82          const cleanId = targetId.startsWith('#') ? targetId.sl
ice(1) : targetId;
83          tabPersistence.saveActiveTab(cleanId);
84        }
85      };
86
87      console.log('[LazyInitPatch] Enhanced switchTab with on-de
mand lazy loading and persistence');
88    }
89
```

```
  // Инициализируем ленивую загрузку с определением всех вкладок
  const allTabDefinitions = [
    // Основные вкладки (Диалог и аудио)
    { tabName: 'chat', htmlUrl: `/html/chat/index.html`, jsUrl: `/html/chat/main.js` },
    { tabName: 'voice', htmlUrl: `/html/voice_tab/index.html`, jsUrl: `/html/voice_tab/main.js` },
    { tabName: 'tts', htmlUrl: `/html/tts_tab/index.html`, jsUrl: `/html/tts_tab/main.js` },

    // ИИ и Знания
    { tabName: 'rag', htmlUrl: `/html/rag_tab/index.html`, jsUrl: `/html/rag_tab/main.js` },
    { tabName: 'models', htmlUrl: `/html/models_tab/index.html`, jsUrl: `/html/models_tab/main.js` },
    { tabName: 'agents', htmlUrl: `/html/agents_tab/index.html`, jsUrl: `/html/agents_tab/main.js` },
    { tabName: 'skills', htmlUrl: `/html/skills_tab/index.html`, jsUrl: `/html/skills_tab/main.js` },
    { tabName: 'mcp', htmlUrl: `/html/mcp_tab/index.html`, jsUrl: `/html/mcp_tab/main.js` },
    { tabName: 'observability', htmlUrl: `/html/system_inspector_tab/index.html`, jsUrl: `/html/system_inspector_tab/main.js` },
    { tabName: 'telegram-rag', htmlUrl: `/html/telegram_rag_tab/index.html`, jsUrl: `/html/telegram_rag_tab/main.js` },
    { tabName: 'sources', htmlUrl: `/html/sources_tab/index.html`, jsUrl: `/html/sources_tab/main.js` },
    { tabName: 'search', htmlUrl: `/html/search_tab/index.html`, jsUrl: `/html/search_tab/main.js` },

    // Система и Администратор
    { tabName: 'admin', htmlUrl: `/html/admin_tab/index.html`, jsUrl: `/html/admin_tab/main.js` },
    { tabName: 'users', htmlUrl: `/html/users_tab/index.html`, jsUrl: `/html/users_tab/main.js` },
    { tabName: 'plugins', htmlUrl: `/html/plugins_tab/index.ht
```

```
ml`, jsUrl: `/html/plugins_tab/main.js` },
    { tabName: 'google-accounts', htmlUrl: `/html/google_accou
nts_tab/index.html`, jsUrl: `/html/google_accounts_tab/main.js
` },
    { tabName: 'instructions', htmlUrl: `/html/instructions_ta
b/index.html`, jsUrl: `/html/instructions_tab/main.js` },
    { tabName: 'news', htmlUrl: `/html/news_tab/index.html`, j
sUrl: `/html/news_tab/main.js` },
    { tabName: 'logs', htmlUrl: `/html/logs/index.html`, jsUrl
: `/html/logs/main.js` },
    { tabName: 'help', htmlUrl: `/html/help/index.html`, jsUrl
: `/html/help/main.js` },

    // Микроприложения (с привязкой к ID приложения в конфиге
профиля)
    { tabName: 'network', appId: 'network_terminal', htmlUrl:
`/html/network_tab/index.html`, jsUrl: `/html/network_tab/main
.js` },
    { tabName: 'system-inspector', appId: 'system_inspector',
htmlUrl: `/html/system_inspector_tab/index.html`, jsUrl: `/htm
l/system_inspector_tab/main.js` },
    { tabName: 'about-system', appId: 'about_system', htmlUrl:
 `/html/about_system_tab/index.html?v=20261012_v1`, jsUrl: `/h
tml/about_system_tab/main.js?v=20261012_v1` },
    { tabName: 'windows-admin', appId: 'windows_sysadmin', htm
lUrl: `/html/windows_admin_tab/index.html`, jsUrl: `/html/wind
ows_admin_tab/main.js` },
    { tabName: 'cloudflared', appId: 'cloudflared_monitor', ht
mlUrl: `/html/cloudflared_tab/index.html`, jsUrl: `/html/cloud
flared_tab/main.js` },
    { tabName: 'user-assistant', appId: 'user_assistant', html
Url: `/html/user_assistant_tab/index.html`, jsUrl: `/html/user
_assistant_tab/main.js` },
    { tabName: 'gcloud', appId: 'gcloud_monitor', htmlUrl: `/h
tml/gcloud_tab/index.html`, jsUrl: `/html/gcloud_tab/main.js`
},
    { tabName: 'website-monitor', appId: 'website_monitor', ht
mlUrl: `/html/website_monitor_tab/index.html`, jsUrl: `/html/w
```

```
ebsite_monitor_tab/main.js` },
    { tabName: 'system-control', appId: 'system_control_center', htmlUrl: `/html/system_control_tab/index.html?v=20261012_v1`, jsUrl: `/html/system_control_tab/main.js?v=20261012_v1` },
    { tabName: 'system-logs', appId: 'system_log_viewer', htmlUrl: `/html/system_logs_tab/index.html`, jsUrl: `/html/system_logs_tab/main.js` },
    { tabName: 'software-audit', appId: 'software_audit', htmlUrl: `/html/software_audit_tab/index.html`, jsUrl: `/html/software_audit_tab/main.js` },
    { tabName: 'registry-viewer', appId: 'registry_viewer', htmlUrl: `/html/registry_viewer_tab/index.html`, jsUrl: `/html/registry_viewer_tab/main.js` },
    { tabName: 'startup-auditor', appId: 'windows_startup_auditor', htmlUrl: `/html/startup_auditor_tab/index.html`, jsUrl: `/html/startup_auditor_tab/main.js` },
    { tabName: 'windows-backup', appId: 'windows_backup_manager', htmlUrl: `/html/windows_backup_tab/index.html?v=20260924_v2`, jsUrl: `/html/windows_backup_tab/main.js?v=20260924_v2` },
    { tabName: 'wikipedia-research', appId: 'wikipedia_research', htmlUrl: `/html/wikipedia_research_tab/index.html`, jsUrl: `/html/wikipedia_research_tab/main.js` },
    { tabName: 'process-leaks', appId: 'process_leaks', htmlUrl: `/html/process_leaks_tab/index.html?v=20260924_v1`, jsUrl: `/html/process_leaks_tab/main.js?v=20260924_v1` },
    { tabName: 'forensics', appId: 'forensics', htmlUrl: `/html/forensics_tab/index.html?v=20260924_v1`, jsUrl: `/html/forensics_tab/main.js?v=20260924_v1` },
    { tabName: 'throttling', appId: 'throttling', htmlUrl: `/html/throttling_tab/index.html?v=20260924_v1`, jsUrl: `/html/throttling_tab/main.js?v=20260924_v1` },
    { tabName: 'storage-wear', appId: 'storage_wear', htmlUrl: `/html/storage_wear_tab/index.html?v=20260924_v1`, jsUrl: `/html/storage_wear_tab/main.js?v=20260924_v1` },
    { tabName: 'peripherals', appId: 'peripherals', htmlUrl: `/html/peripherals_tab/index.html?v=20260924_v1`, jsUrl: `/html/peripherals_tab/main.js?v=20260924_v1` },
  ];
```

```
140
141   try {
142     // Получаем статус приложений для активного сценария (Run-
Dashboard, Run-TC, su)
143     let appsStatus = window.appsStatusMap || null;
144     if (!appsStatus) {
145       try {
146         const res = await (window.api ? window.api.fetch('/api
/v1/apps/status') : fetch('/api/v1/apps/status').then(r => r.j
son()));
147         appsStatus = res?.apps || null;
148       } catch (err) {
149         console.warn('[LazyInitPatch] Could not fetch apps sta
tus for profile filtering:', err);
150       }
151     }
152
153     // Фильтруем вкладки: загружаем ТОЛЬКО те, которые разреше
ны в активном профиле конфига
154     const activeTabDefinitions = allTabDefinitions.filter(tab
=> {
155       if (tab.appId && appsStatus) {
156         const appInfo = appsStatus[tab.appId] || Object.values
(appsStatus).find(a => a.id === tab.appId || a.tab === `tab-${
tab.tabName}`);
157         if (appInfo && appInfo.enabled === false) {
158           return false; // Отключено в активном сценарии/профи
ле
159         }
160       }
161       return true;
162     });
163
164     // Инициализируем систему ленивой загрузки только для разр
ешенных в профиле вкладок
165     await initLazyTabLoading(activeTabDefinitions);
166
167     console.log(`[LazyInitPatch] Lazy loading initialized: ${a
```

```
ctiveTabDefinitions.length} tabs registered on-demand.`);

    // Восстанавливаем и загружаем ТОЛЬКО одну активную вкладк
y
    const restored = await restoreLastTab('tab-about-system');
    console.log(`[LazyInitPatch] Initial active tab loaded: ${
restored}`);

  } catch (error) {
    console.error('[LazyInitPatch] Error during initialization
:', error);
  }
});

console.log('[LazyInitPatch] Patch loaded successfully');

export {};

```

main-refactored.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Admin - Main-Refactored Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main-refa
ctored.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/admin/main-refactored.
js?v=20261001_v1" type="module"></script>
 *
 * File: main-refactored.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/admin
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-12 08:20:00
 * ============================================================
==================
 */

/**
 * Admin Interface Main - Оркестратор
 *
 * Главный модуль для инициализации административного интерфей
са.
 * Все реальная реализация находится в модулях, этот файл толь
ко оркестрирует.
 *
 * Модули:
 * - auth-handler.js: Аутентификация и проверка пароля
 * - tab-manager.js: Управление вкладками и навигацией
```

```
 * - ui-handler.js: Модали, уведомления, помощь
 * - init-interface.js: Инициализация компонентов
 * - apps-sync.js: Синхронизация приложений
 */

// ============================================================
// ИМПОРТ МОДУЛЕЙ
// ============================================================

import { setupAuthHandlers, verifyPassword } from './modules/auth-handler.js';
import { setupTabManagement, onTabSwitched } from './modules/tab-manager.js';
import { setupUIHandlers, showHelpModal, showNotification, showChatLogicModal } from './modules/ui-handler.js';
import {
  initializeInterface,
  setupGlobalFunctions,
  setupLanguageSelector,
  setupThemeSelector
} from './modules/init-interface.js';
import { syncApplicationsVisibility } from './modules/apps-sync.js';

// ============================================================
// ГЛОБАЛЬНЫЕ ПЕРЕМЕННЫЕ
// ============================================================

let tabLoader = null;
let apiFetcher = null;

// ============================================================
// ГЛАВНАЯ ФУНКЦИЯ ИНИЦИАЛИЗАЦИИ
```

```
// ============================================================
=================

async function initAdminInterface() {
  console.log('🚀 [AdminInterface] Starting initialization...'
);

  try {
    // 1. Инициализируем компоненты интерфейса
    console.log('[AdminInterface] Step 1: Initializing core co
mponents...');
    await initializeInterface();
    setupGlobalFunctions();
    setupLanguageSelector();
    setupThemeSelector();

    // 2. Регистрируем глобальные функции для UI
    console.log('[AdminInterface] Step 2: Setting up UI handle
rs...');
    registerGlobalFunctions();
    setupUIHandlers();
    setupAuthHandlers();

    // 3. Инициализируем управление вкладками
    console.log('[AdminInterface] Step 3: Setting up tab manag
ement...');
    setupTabManagement();

    // 4. Синхронизируем видимость приложений
    console.log('[AdminInterface] Step 4: Syncing applications
 visibility...');
    await syncApplicationsVisibility();

    // 5. Инициализируем оптимизации (ленивая загрузка, кеширо
вание и т.д.)
    console.log('[AdminInterface] Step 5: Initializing optimiz
ations...');
    await initializeOptimizations();
```

```
89
90      // 6. Инициализируем API клиент
91      console.log('[AdminInterface] Step 6: Setting up API clien
    t...');
92      setupAPIClient();
93
94      console.log('✅ [AdminInterface] Initialization complete!')
    ;
95
96    } catch (error) {
97      console.error('❌ [AdminInterface] Initialization error:',
    error);
98      showNotification('Ошибка инициализации интерфейса: ' + err
    or.message, 'danger');
99    }
100 }
101
102 // ============================================================
    ================
103 // РЕГИСТРАЦИЯ ГЛОБАЛЬНЫХ ФУНКЦИЙ
104 // ============================================================
    ================
105
106 function registerGlobalFunctions() {
107   window.switchTab = switchTab;
108   window.onTabSwitched = onTabSwitched;
109   window.showHelpModal = showHelpModal;
110   window.showNotification = showNotification;
111   window.showChatLogicModal = showChatLogicModal;
112   window.verifyPassword = verifyPassword;
113   window.isTabActive = function(tabId) {
114     if (typeof document !== 'undefined' && document.hidden) re
    turn false;
115     if (!tabId) return false;
116     const cleanId = tabId.startsWith('tab-') ? tabId : `tab-${
    tabId}`;
117     const rawId = cleanId.replace(/^tab-/, '');
118     const pane = document.getElementById(cleanId) || document.
```

```
getElementById(rawId);
    return pane ? pane.classList.contains('active') : false;
  };
}

// ============================================================
================
// УПРАВЛЕНИЕ ВКЛАДКАМИ
// ============================================================
================

let currentActiveAdminTabId = null;

/**
 * Переключить на вкладку
 */
async function switchTab(targetId) {
  if (!targetId) return;

  const cleanId = targetId.startsWith('#') ? targetId.slice(1)
 : targetId;
  const tabName = cleanId.replace(/^tab-/, '');
  const tabId = cleanId.startsWith('tab-') ? cleanId : `tab-${
cleanId}`;

  // 1. Деактивация предыдущей вкладки
  if (currentActiveAdminTabId && currentActiveAdminTabId !== t
abId) {
    const prevName = currentActiveAdminTabId.replace(/^tab-/,
'').replace(/-([a-z])/g, (_, c) => c.toUpperCase());
    const deactivateFn = window[`deactivate${prevName[0].toUpp
erCase() + prevName.slice(1)}Tab`];
    if (typeof deactivateFn === 'function') {
      try { deactivateFn(); } catch (err) { console.debug('dea
ctivate hook error:', err); }
    }
    document.dispatchEvent(new CustomEvent('tab:deactivated',
{ detail: { tabId: currentActiveAdminTabId, name: prevName } }
```

```
));
  }

  currentActiveAdminTabId = tabId;

  console.log(`[AdminInterface] Switching to tab: ${tabName}`);

  // Используем LazyTabLoader если доступен
  if (tabLoader && typeof tabLoader.isLoaded === 'function') {
    if (!tabLoader.isLoaded(tabName)) {
      console.log(`[AdminInterface] Loading tab: ${tabName}`);
      await tabLoader.loadTab(tabName);
    }
  }

  // Обновляем визуальное состояние
  updateTabUI(cleanId);

  // Вызываем callback
  if (typeof window.onTabSwitched === 'function') {
    window.onTabSwitched(cleanId);
  }

  // 2. Активация новой вкладки
  const newName = tabId.replace(/^tab-/, '').replace(/-([a-z])/g, (_, c) => c.toUpperCase());
  const activateFn = window[`activate${newName[0].toUpperCase() + newName.slice(1)}Tab`];
  if (typeof activateFn === 'function') {
    try { activateFn(); } catch (err) { console.debug('activate hook error:', err); }
  }
  document.dispatchEvent(new CustomEvent('tab:activated', { detail: { tabId, name: newName } }));

  console.log(`[AdminInterface] Tab switched: ${tabName}`);
}
```

```
179
180 /**
181  * Обновить визуальное состояние UI
182  */
183 function updateTabUI(cleanId) {
184   const tabId = cleanId.startsWith('tab-') ? cleanId : `tab-${
    cleanId}`;
185
186   // Скрываем все вкладки
187   document.querySelectorAll('#mainTabContent > .tab-pane, body
     > .container-fluid > .tab-content > .tab-pane, [role="tabpane
    l"]').forEach(tab => {
188     tab.classList.remove('active', 'show');
189     tab.setAttribute('aria-hidden', 'true');
190   });
191
192   // Показываем целевую вкладку
193   const targetTab = document.getElementById(tabId) || document
    .getElementById(cleanId);
194   if (targetTab) {
195     targetTab.classList.add('active', 'show');
196     targetTab.setAttribute('aria-hidden', 'false');
197   }
198
199   // Обновляем состояние элементов меню
200   document.querySelectorAll('#mainTabs .list-group-item, #main
    Tabs .dropdown-item, #mainTabs [data-tab], #mainTabs [data-bs-
    target]').forEach(item => {
201     const itemTarget = item.getAttribute('data-tab') || item.g
    etAttribute('data-bs-target')?.replace('#', '');
202     if (itemTarget === tabId || itemTarget === cleanId) {
203       item.classList.add('active');
204       item.setAttribute('aria-selected', 'true');
205     } else {
206       item.classList.remove('active');
207       item.setAttribute('aria-selected', 'false');
208     }
209   });
```

```
210
211   // Обновляем активность родительских кнопок dropdown-toggle
212   document.querySelectorAll('#mainTabs .dropdown').forEach(dro
pdown => {
213     const toggle = dropdown.querySelector('.dropdown-toggle');
214     const hasActiveChild = dropdown.querySelector('.dropdown-i
tem.active, .list-group-item.active');
215     if (toggle) {
216       if (hasActiveChild) {
217         toggle.classList.add('active');
218       } else {
219         toggle.classList.remove('active');
220       }
221     }
222   });
223 }
224
225 // ============================================================
============
226 // ИНИЦИАЛИЗАЦИЯ ОПТИМИЗАЦИЙ
227 // ============================================================
============
228
229 async function initializeOptimizations() {
230   // Оптимизации автоматически инициализируются в lazy-init-pa
tch.js
231   // Здесь просто сохраняем ссылки для последующего использова
ния
232
233   if (window.optimizationModule) {
234     tabLoader = window.optimizationModule.tabLoader;
235     apiFetcher = window.optimizationModule.apiFetcher;
236     console.log('[AdminInterface] Optimizations initialized');
237   } else {
238     console.warn('[AdminInterface] Optimization module not fou
nd');
239   }
240 }
```

```
241
242 // ============================================================
    ================
243 // ИНИЦИАЛИЗАЦИЯ API КЛИЕНТА
244 // ============================================================
    ================
245
246 function setupAPIClient() {
247   if (!window.api) {
248     console.warn('[AdminInterface] API client not found, skipp
    ing setup');
249     return;
250   }
251
252   // API клиент уже подключен в main.js (исходный), здесь могу
    т быть
253   // дополнительные настройки
254   console.log('[AdminInterface] API client ready');
255 }
256
257 // ============================================================
    ================
258 // ОБРАБОТКА СОБЫТИЯ LOAD
259 // ============================================================
    ================
260
261 if (document.readyState === 'loading') {
262   document.addEventListener('DOMContentLoaded', initAdminInter
    face);
263 } else {
264   initAdminInterface();
265 }
266
267 console.log('  [AdminInterface] Module loaded and ready');
268
269 // ============================================================
    ================
270 // ЭКСПОРТЫ (для модульного использования)
```

```
271 // ============================================================
    ================
272
273 export {
274   initAdminInterface,
275   switchTab,
276   setupAuthHandlers,
277   setupTabManagement,
278   setupUIHandlers,
279   syncApplicationsVisibility,
280   initializeInterface,
281   setupGlobalFunctions
282 };
283
```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Admin - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/admin/main.js?v=202610
01_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/admin
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-12 08:20:00
 * ============================================================
==================
 */

// Admin Interface Main JS
import { initI18n, switchLang, applyTranslations } from '../js
/i18n.js';
import { initTheme, setTheme, getThemeMode, getResolvedTheme }
 from '../js/theme.js';
import { initUserSettings, refreshUserProfile } from '../js/us
erSettings.js';

// Make switchLang and theme functions available globally
window.switchLang = switchLang;
window.applyTranslations = applyTranslations;
window.setTheme = setTheme;
```

```
window.getThemeMode = getThemeMode;
window.getResolvedTheme = getResolvedTheme;
window.initUserSettings = initUserSettings;
window.refreshUserProfile = refreshUserProfile;

// Helper to open plugin from top navbar dropdown
window.openPluginFromDropdown = function(pluginName) {
  if (!pluginName) return;
  document.querySelectorAll('#mainTabs .dropdown-menu.show').forEach((m) => {
    m.classList.remove('show');
    m.closest('.dropdown')?.querySelector('.dropdown-toggle')?.classList.remove('show');
  });

  const normalized = pluginName.toLowerCase().replace(/-/g, '_');

  if (normalized === 'telegram_channel_rag' || normalized === 'telegram_rag') {
    const pane = document.getElementById('tab-telegram-rag') || document.getElementById('tab-telegram_rag');
    if (pane && typeof window.switchTab === 'function') {
      window.switchTab('tab-telegram-rag');
      return;
    }
  }

  if (normalized === 'news_feed' || normalized === 'news' || normalized === 'smart_news' || normalized === 'smart_feed_news') {
    const pane = document.getElementById('tab-news');
    if (pane && typeof window.switchTab === 'function') {
      window.switchTab('tab-news');
      return;
    }
  }

```

```
  if (typeof window.switchTab === 'function') {
    window.switchTab('tab-plugins');
  }

  const select = () => {
    if (typeof window.selectPlugin === 'function') {
      window.selectPlugin(pluginName);
    }
  };

  select();
  setTimeout(select, 150);
  setTimeout(select, 400);
};

// API module
window.api = {
  async fetch(url, options = {}) {
    const response = await fetch(url, options);
    if (!response.ok) {
      let msg = response.statusText;
      try {
        const data = await response.json();
        if (data && data.detail) {
          if (typeof data.detail === 'string') {
            msg = data.detail;
          } else if (Array.isArray(data.detail)) {
            msg = data.detail.map(d => d.msg || JSON.stringify
(d)).join(', ');
          } else {
            msg = JSON.stringify(data.detail);
          }
        }
      } catch {}
      throw new Error(`${response.status} ${msg}`);
    }
    return response.json();
  },
```

```
98
99    // User management API
100   users: {
101     async list(params = {}) {
102       const q = new URLSearchParams(params).toString();
103       return window.api.fetch(`/api/admin/users${q ? '?' + q :
  ''}`);
104     },
105
106     async get(userId) {
107       return window.api.fetch(`/api/admin/users/${userId}`);
108     },
109
110     async create(data) {
111       return window.api.fetch('/api/admin/users', {
112         method: 'POST',
113         headers: { 'Content-Type': 'application/json' },
114         body: JSON.stringify(data)
115       });
116     },
117
118     async update(userId, data) {
119       return window.api.fetch(`/api/admin/users/${userId}`, {
120         method: 'PUT',
121         headers: { 'Content-Type': 'application/json' },
122         body: JSON.stringify(data)
123       });
124     },
125
126     async setPassword(userId, password) {
127       return window.api.fetch(`/api/admin/users/${userId}/pass
  word`, {
128         method: 'POST',
129         headers: { 'Content-Type': 'application/json' },
130         body: JSON.stringify({ password })
131       });
132     },
133
```

```
134     async toggleActive(userId) {
135       return window.api.fetch(`/api/admin/users/${userId}/togg
    le-active`, {
136         method: 'POST'
137       });
138     },
139 
140     async toggleRole(userId) {
141       return window.api.fetch(`/api/admin/users/${userId}/togg
    le-role`, {
142         method: 'POST'
143       });
144     },
145 
146     async delete(userId) {
147       return window.api.fetch(`/api/admin/users/${userId}`, {
148         method: 'DELETE'
149       });
150     }
151   },
152 
153   // Skills management API
154   skills: {
155     async list(params = {}) {
156       const q = new URLSearchParams(params).toString();
157       return window.api.fetch(`/api/admin/skills${q ? '?' + q
    : ''}`);
158     },
159     async get(name) {
160       return window.api.fetch(`/api/admin/skills/${encodeURICo
    mponent(name)}`);
161     },
162     async create(data) {
163       return window.api.fetch('/api/admin/skills', {
164         method: 'POST',
165         headers: { 'Content-Type': 'application/json' },
166         body: JSON.stringify(data)
167       });
```

```
168     },
169     async update(name, data) {
170       return window.api.fetch(`/api/admin/skills/${encodeURICo
mponent(name)}`, {
171         method: 'PUT',
172         headers: { 'Content-Type': 'application/json' },
173         body: JSON.stringify(data)
174       });
175     },
176     async delete(name) {
177       return window.api.fetch(`/api/admin/skills/${encodeURICo
mponent(name)}`, {
178         method: 'DELETE'
179       });
180     },
181     async package(name) {
182       return window.api.fetch(`/api/admin/skills/${encodeURICo
mponent(name)}/package`, {
183         method: 'POST'
184       });
185     }
186   },
187
188   // MCP Servers management API
189   mcp: {
190     async list() {
191       return window.api.fetch('/api/admin/mcp/servers');
192     },
193     async get(id) {
194       return window.api.fetch(`/api/admin/mcp/servers/${encode
URIComponent(id)}`);
195     },
196     async create(data) {
197       return window.api.fetch('/api/admin/mcp/servers', {
198         method: 'POST',
199         headers: { 'Content-Type': 'application/json' },
200         body: JSON.stringify(data)
201       });
```

```
202     },
203     async update(id, data) {
204       return window.api.fetch(`/api/admin/mcp/servers/${encode
    URIComponent(id)}`, {
205         method: 'PUT',
206         headers: { 'Content-Type': 'application/json' },
207         body: JSON.stringify(data)
208       });
209     },
210     async delete(id) {
211       return window.api.fetch(`/api/admin/mcp/servers/${encode
    URIComponent(id)}`, {
212         method: 'DELETE'
213       });
214     },
215     async toggle(id) {
216       return window.api.fetch(`/api/admin/mcp/servers/${encode
    URIComponent(id)}/toggle`, {
217         method: 'POST'
218       });
219     },
220     async test(id) {
221       return window.api.fetch(`/api/admin/mcp/servers/${encode
    URIComponent(id)}/test`, {
222         method: 'POST'
223       });
224     },
225     async testAdhoc(data) {
226       return window.api.fetch('/api/admin/mcp/test', {
227         method: 'POST',
228         headers: { 'Content-Type': 'application/json' },
229         body: JSON.stringify(data)
230       });
231     },
232     async getTools() {
233       return window.api.fetch('/api/admin/mcp/tools');
234     }
235   },
```

```
236
237   // Google Workspace Accounts Pool API
238   googleAccounts: {
239     async list() {
240       return window.api.fetch('/api/admin/google-accounts');
241     },
242     async get(name) {
243       return window.api.fetch(`/api/admin/google-accounts/${en
codeURIComponent(name)}`);
244     },
245     async create(data) {
246       return window.api.fetch('/api/admin/google-accounts', {
247         method: 'POST',
248         headers: { 'Content-Type': 'application/json' },
249         body: JSON.stringify(data)
250       });
251     },
252     async upload(formData) {
253       const response = await fetch('/api/admin/google-accounts
/upload', {
254         method: 'POST',
255         body: formData
256       });
257       if (!response.ok) {
258         let msg = response.statusText;
259         try {
260           const data = await response.json();
261           if (data && data.detail) msg = data.detail;
262         } catch {}
263         throw new Error(`${response.status} ${msg}`);
264       }
265       return response.json();
266     },
267     async setDefault(name) {
268       return window.api.fetch(`/api/admin/google-accounts/${en
codeURIComponent(name)}/default`, {
269         method: 'POST'
270       });
```

```
271     },
272     async resetStatus(name) {
273       return window.api.fetch(`/api/admin/google-accounts/${en
    codeURIComponent(name)}/reset-status`, {
274         method: 'POST'
275       });
276     },
277     async test(name) {
278       return window.api.fetch(`/api/admin/google-accounts/${en
    codeURIComponent(name)}/test`, {
279         method: 'POST'
280       });
281     },
282     async delete(name) {
283       return window.api.fetch(`/api/admin/google-accounts/${en
    codeURIComponent(name)}`, {
284         method: 'DELETE'
285       });
286     }
287   }
288 };
289
290 // HELP content
291 // Password protection state
292 let isPasswordProtected = false;
293 let hasEnteredPassword = false;
294
295 // Tab Switch Handler
296 function onTabSwitched(targetId) {
297   const cleanId = targetId.startsWith('#') ? targetId.slice(1)
     : targetId;
298   if (cleanId === 'tab-chat') {
299     const msgInput = document.getElementById('message-input');
300     if (msgInput) msgInput.focus();
301   } else if (cleanId === 'tab-voice' && typeof window.initVoic
    eTab === 'function') {
302     console.log('[AdminInterface] Switching to voice tab...');
303     window.initVoiceTab();
```

```
  } else if (cleanId === 'tab-tts' && typeof window.initTtsTab
 === 'function') {
    console.log('[AdminInterface] Switching to tts tab...');
    window.initTtsTab();
  } else if (cleanId === 'tab-plugins' && typeof window.initPl
uginsTab === 'function') {
    console.log('[AdminInterface] Switching to plugins tab...'
);
    window.initPluginsTab();
  } else if (cleanId === 'tab-admin' && typeof window.initAdmi
nTab === 'function') {
    console.log('[AdminInterface] Switching to admin tab...');
    window.initAdminTab();
  } else if (cleanId === 'tab-users' && typeof window.initUser
sTab === 'function') {
    console.log('[AdminInterface] Switching to users tab...');
    window.initUsersTab();
  } else if (cleanId === 'tab-windows-users' && typeof window.
initWindowsUsersTab === 'function') {
    console.log('[AdminInterface] Switching to Windows users t
ab...');
    window.initWindowsUsersTab();
  } else if (cleanId === 'tab-user-directories' && typeof wind
ow.initUserDirectoriesTab === 'function') {
    console.log('[AdminInterface] Switching to user directorie
s tab...');
    window.initUserDirectoriesTab();
  } else if (cleanId === 'tab-google-accounts' && typeof windo
w.initGoogleAccountsTab === 'function') {
    console.log('[AdminInterface] Switching to google accounts
 tab...');
    window.initGoogleAccountsTab();
  } else if (cleanId === 'tab-instructions' && typeof window.i
nitInstructionsTab === 'function') {
    console.log('[AdminInterface] Switching to instructions ta
b...');
    window.initInstructionsTab();
  } else if (cleanId === 'tab-skills' && typeof window.initSki
```

```
llsTab === 'function') {
329     console.log('[AdminInterface] Switching to skills tab...')
;
330     window.initSkillsTab();
331   } else if (cleanId === 'tab-mcp' && typeof window.initMcpTab
 === 'function') {
332     console.log('[AdminInterface] Switching to MCP tab...');
333     window.initMcpTab();
334   } else if (cleanId === 'tab-rag' && typeof window.initRagTab
 === 'function') {
335     console.log('[AdminInterface] Switching to RAG tab...');
336     window.initRagTab();
337   } else if (cleanId === 'tab-pixelrag' && typeof window.initP
ixelRagTab === 'function') {
338     console.log('[AdminInterface] Switching to PixelRAG tab...
');
339     window.initPixelRagTab();
340   } else if ((cleanId === 'tab-telegram-rag' || cleanId === 't
ab-telegram_rag') && typeof window.initTelegramRagTab === 'fun
ction') {
341     console.log('[AdminInterface] Switching to Telegram RAG ta
b...');
342     window.initTelegramRagTab();
343   } else if (cleanId === 'tab-search' && typeof window.initSea
rchTab === 'function') {
344     console.log('[AdminInterface] Switching to search tab...')
;
345     window.initSearchTab();
346   } else if (cleanId === 'tab-models' && typeof window.initMod
elsTab === 'function') {
347     console.log('[AdminInterface] Switching to models tab...')
;
348     window.initModelsTab();
349   } else if (cleanId === 'tab-agents' && typeof window.initAge
ntsTab === 'function') {
350     console.log('[AdminInterface] Switching to agents tab...')
;
351     window.initAgentsTab();
```

```
352   } else if (cleanId === 'tab-sources' && typeof window.initSo
urcesTab === 'function') {
353     console.log('[AdminInterface] Switching to sources tab...'
);
354     window.initSourcesTab();
355   } else if (cleanId === 'tab-logs' && typeof window.initLogsT
ab === 'function') {
356     console.log('[AdminInterface] Switching to logs tab...');
357     window.initLogsTab();
358   } else if (cleanId === 'tab-network' && typeof window.initNe
tworkTab === 'function') {
359     console.log('[AdminInterface] Switching to network tab...'
);
360     window.initNetworkTab();
361   } else if ((cleanId === 'tab-hardware-load-inspector' || cle
anId === 'tab-system-load-inspector' || cleanId === 'tab-syste
m-inspector') && typeof window.initSystemInspectorTab === 'fun
ction') {
362     console.log('[AdminInterface] Switching to hardware load i
nspector tab...');
363     window.initSystemInspectorTab();
364   } else if (cleanId === 'tab-processes-load-inspector' && typ
eof window.initProcessesLoadInspectorTab === 'function') {
365     console.log('[AdminInterface] Switching to processes load
inspector tab...');
366     window.initProcessesLoadInspectorTab();
367   } else if (cleanId === 'tab-about-system' && typeof window.i
nitAboutSystemTab === 'function') {
368     console.log('[AdminInterface] Switching to about system ta
b...');
369     window.initAboutSystemTab();
370   } else if (cleanId === 'tab-windows-admin' && typeof window.
initWindowsAdminTab === 'function') {
371     console.log('[AdminInterface] Switching to windows admin t
ab...');
372     window.initWindowsAdminTab();
373   } else if (cleanId === 'tab-focus-settings' && typeof window
.initFocusSettingsTab === 'function') {
```

```
    console.log('[AdminInterface] Switching to focus settings tab...');
    window.initFocusSettingsTab();
  } else if (cleanId === 'tab-cloudflared' && typeof window.initCloudflaredTab === 'function') {
    console.log('[AdminInterface] Switching to cloudflared tab...');
    window.initCloudflaredTab();
  } else if (cleanId === 'tab-user-assistant' && typeof window.initUserAssistantTab === 'function') {
    console.log('[AdminInterface] Switching to user assistant tab...');
    window.initUserAssistantTab();
  } else if (cleanId === 'tab-gcloud' && typeof window.initGCloudTab === 'function') {
    console.log('[AdminInterface] Switching to gcloud tab...');
    window.initGCloudTab();
  } else if (cleanId === 'tab-website-monitor' && typeof window.initWebsiteMonitorTab === 'function') {
    console.log('[AdminInterface] Switching to website monitor tab...');
    window.initWebsiteMonitorTab();
  } else if (cleanId === 'tab-system-control' && typeof window.initSystemControlTab === 'function') {
    console.log('[AdminInterface] Switching to system control tab...');
    window.initSystemControlTab();
  } else if (cleanId === 'tab-post-install-wizard' && typeof window.initPostInstallWizardTab === 'function') {
    console.log('[AdminInterface] Switching to post-install wizard tab...');
    window.initPostInstallWizardTab();
  } else if (cleanId === 'tab-maintenance-recovery' && typeof window.initMaintenanceRecoveryTab === 'function') {
    console.log('[AdminInterface] Switching to maintenance recovery tab...');
    window.initMaintenanceRecoveryTab();
```

```
  } else if (cleanId === 'tab-news' && typeof window.initNewsTab === 'function') {
    console.log('[AdminInterface] Switching to news tab...');
    window.initNewsTab();
  } else if (cleanId === 'tab-windows-backup' && typeof window.initWindowsBackupTab === 'function') {
    console.log('[AdminInterface] Switching to windows backup tab...');
    window.initWindowsBackupTab();
  } else if (cleanId === 'tab-help' && typeof window.initHelpTab === 'function') {
    console.log('[AdminInterface] Switching to help tab...');
    window.initHelpTab();
  }
}

// Programmatic tab switcher
let currentActiveAdminTabId = null;

function switchTab(targetId) {
  if (!targetId) return;
  const cleanId = targetId.startsWith('#') ? targetId.slice(1) : targetId;
  const tabId = cleanId.startsWith('tab-') ? cleanId : `tab-${cleanId}`;

  // 0. Деактивация предыдущей вкладки
  if (currentActiveAdminTabId && currentActiveAdminTabId !== tabId) {
    const prevName = currentActiveAdminTabId.replace(/^tab-/, '').replace(/-([a-z])/g, (_, c) => c.toUpperCase());
    const deactivateFn = window[`deactivate${prevName[0].toUpperCase() + prevName.slice(1)}Tab`];
    if (typeof deactivateFn === 'function') {
      try { deactivateFn(); } catch (err) { console.debug('deactivate hook error:', err); }
    }
    document.dispatchEvent(new CustomEvent('tab:deactivated',
```

```
{ detail: { tabId: currentActiveAdminTabId, name: prevName } }
));
  }

  currentActiveAdminTabId = tabId;

  // 1. Update active state on dropdown items & toggles
  document.querySelectorAll('#mainTabs .dropdown-item').forEach((item) => {
    const itemTarget = item.getAttribute('data-tab') || item.getAttribute('data-bs-target')?.replace('#', '');
    if (itemTarget === tabId || itemTarget === cleanId) {
      item.classList.add('active');
    } else {
      item.classList.remove('active');
    }
  });

  document.querySelectorAll('#mainTabs .dropdown').forEach((dropdown) => {
    const toggle = dropdown.querySelector('.dropdown-toggle');
    const hasActiveChild = dropdown.querySelector('.dropdown-item.active');
    if (toggle) {
      if (hasActiveChild) {
        toggle.classList.add('active');
      } else {
        toggle.classList.remove('active');
      }
    }
  });

  // 2. Switch tab-pane (scoped to top-level container to preserve nested subtabs)
  const mainTabContent = document.getElementById('mainTabContent');
  if (mainTabContent) {
    Array.from(mainTabContent.children).forEach((pane) => {
```

```
      if (pane.classList.contains('tab-pane')) {
        pane.classList.remove('show', 'active');
      }
    });
  } else {
    document.querySelectorAll('#mainTabContent > .tab-pane, body > .container-fluid > .tab-content > .tab-pane').forEach((pane) => {
      pane.classList.remove('show', 'active');
    });
  }
  const targetPane = document.getElementById(tabId) || document.getElementById(cleanId);
  if (targetPane) {
    targetPane.classList.add('show', 'active');
  }

  // 3. Notify lifecycle callback
  onTabSwitched(tabId);

  // 4. Активация новой вкладки
  const newName = tabId.replace(/^tab-/, '').replace(/-([a-z])/g, (_, c) => c.toUpperCase());
  const activateFn = window[`activate${newName[0].toUpperCase() + newName.slice(1)}Tab`];
  if (typeof activateFn === 'function') {
    try { activateFn(); } catch (err) { console.debug('activate hook error:', err); }
  }
  document.dispatchEvent(new CustomEvent('tab:activated', { detail: { tabId, name: newName } }));
}
window.switchTab = switchTab;
window.switchToTab = switchTab;
window.isTabActive = function(tabId) {
  if (typeof document !== 'undefined' && document.hidden) return false;
  if (!tabId) return false;
```

```
  const cleanId = tabId.startsWith('tab-') ? tabId : `tab-${tabId}`;
  const rawId = cleanId.replace(/^tab-/, '');
  const pane = document.getElementById(cleanId) || document.getElementById(rawId);
  return pane ? pane.classList.contains('active') : false;
};

// Setup dropdowns navigation
function setupDropdownTabs() {
  const mainTabs = document.getElementById('mainTabs');
  if (!mainTabs) return;

  // 1. Dropdown Toggle Buttons
  mainTabs.querySelectorAll('.dropdown-toggle').forEach((btn) => {
    btn.onclick = (e) => {
      e.preventDefault();
      e.stopPropagation();

      const dropdown = btn.closest('.dropdown');
      const menu = dropdown?.querySelector('.dropdown-menu');
      const isAlreadyOpen = menu?.classList.contains('show');

      // Close all dropdowns
      document.querySelectorAll('#mainTabs .dropdown-menu.show').forEach((m) => {
        m.classList.remove('show');
        m.closest('.dropdown')?.querySelector('.dropdown-toggle')?.classList.remove('show');
      });

      // Toggle clicked dropdown
      if (!isAlreadyOpen && menu) {
        menu.classList.add('show');
        btn.classList.add('show');
      }
    };
```

```
  });

  // 2. Dropdown Item Buttons (Tab Switchers & Action Buttons)
  mainTabs.querySelectorAll('.dropdown-item').forEach((item) =
> {
    item.onclick = (e) => {
      const targetId = item.getAttribute('data-tab') || item.g
etAttribute('data-bs-target')?.replace('#', '');
      const pluginName = item.getAttribute('data-plugin');

      // Close dropdown menu
      item.closest('.dropdown-menu')?.classList.remove('show')
;
      item.closest('.dropdown')?.querySelector('.dropdown-togg
le')?.classList.remove('show');

      if (pluginName && typeof window.openPluginFromDropdown =
== 'function') {
        e.preventDefault();
        e.stopPropagation();
        window.openPluginFromDropdown(pluginName);
      } else if (targetId) {
        e.preventDefault();
        e.stopPropagation();
        switchTab(targetId);
      }
    };
  });

  // 3. Document Click to Close Dropdowns
  if (!document.body.dataset.dropdownOutsideBound) {
    document.body.dataset.dropdownOutsideBound = 'true';
    document.addEventListener('click', (e) => {
      if (!e.target.closest('#mainTabs .dropdown')) {
        document.querySelectorAll('#mainTabs .dropdown-menu.sh
ow').forEach((menu) => {
          menu.classList.remove('show');
          menu.closest('.dropdown')?.querySelector('.dropdown-
```

```
toggle')?.classList.remove('show');
        });
      }
    });
  }
}

async function startAdmin() {
  console.log('Admin interface initializing...');
  try {
    initTheme();
    setupDropdownTabs();
    const el = document.getElementById('admin-interface');
    if (el) el.style.display = 'block';

    if (isPasswordProtected) {
      showPasswordModal();
    } else {
      await initInterface();
    }
    console.log('Admin interface ready');
  } catch (err) {
    console.error('Error during startAdmin initialization:', e
rr);
    const el = document.getElementById('admin-interface');
    if (el) el.style.display = 'block';
  }
}

if (document.readyState === 'loading') {
  document.addEventListener('DOMContentLoaded', () => {
    setupDropdownTabs();
    startAdmin();
  });
} else {
  setupDropdownTabs();
  startAdmin();
}
```

```
586
587 async function initInterface() {
588   // Initialize theme
589   initTheme();
590
591   // Initialize i18n
592   await initI18n();
593
594   // Initialize User Settings & Google OAuth
595   await initUserSettings();
596
597
598   // Initialize HELP content
599   initHelpContent();
600
601   const cb = Date.now();
602
603   // Синхронизация видимости приложений (/apps) перед загрузко
    й вкладок
604   let appsMap = {};
605   try {
606     const appsData = await (window.api ? window.api.fetch('/ap
    i/v1/apps/status') : fetch('/api/v1/apps/status').then(r => r.
    json()));
607     if (appsData && appsData.apps) {
608       appsMap = appsData.apps;
609       window.appsStatusMap = appsData.apps;
610       syncAppsTabsVisibility(appsData.apps);
611     }
612   } catch (err) {
613     console.warn('Ошибка синхронизации видимости приложений (/
    apps):', err);
614   }
615
616   // Загрузка базовых вкладок администратора
617   const coreTabLoads = [
618     loadTabContent('chat', `/html/chat/index.html?v=${cb}`, `/
    html/chat/main.js?v=${cb}`),
```

```
619     loadTabContent('plugins', `/html/plugins_tab/index.html?v=
    ${cb}`, `/html/plugins_tab/main.js?v=${cb}`),
620     loadTabContent('admin', `/html/admin_tab/index.html?v=${cb
    }`, `/html/admin_tab/main.js?v=${cb}`),
621     loadTabContent('users', `/html/users_tab/index.html?v=${cb
    }`, `/html/users_tab/main.js?v=${cb}`),
622     loadTabContent('windows-users', `/html/windows_users_tab/i
    ndex.html?v=${cb}`, `/html/windows_users_tab/main.js?v=${cb}`)
    ,
623     loadTabContent('user-directories', `/html/user_directories
    _tab/index.html?v=${cb}`, `/html/user_directories_tab/main.js?
    v=${cb}`),
624     loadTabContent('google-accounts', `/html/google_accounts_t
    ab/index.html?v=${cb}`, `/html/google_accounts_tab/main.js?v=$
    {cb}`),
625     loadTabContent('instructions', `/html/instructions_tab/ind
    ex.html?v=${cb}`, `/html/instructions_tab/main.js?v=${cb}`),
626     loadTabContent('rag', `/html/rag_tab/index.html?v=${cb}`,
    `/html/rag_tab/main.js?v=${cb}`),
627     loadTabContent('pixelrag', `/html/pixelrag_tab/index.html?
    v=${cb}`, `/html/pixelrag_tab/main.js?v=${cb}`),
628     loadTabContent('telegram-rag', `/html/telegram_rag_tab/ind
    ex.html?v=${cb}`, `/html/telegram_rag_tab/main.js?v=${cb}`),
629     loadTabContent('voice', `/html/voice_tab/index.html?v=${cb
    }`, `/html/voice_tab/main.js?v=${cb}`),
630     loadTabContent('models', `/html/models_tab/index.html?v=${
    cb}`, `/html/models_tab/main.js?v=${cb}`),
631     loadTabContent('agents', `/html/agents_tab/index.html?v=${
    cb}`, `/html/agents_tab/main.js?v=${cb}`),
632     loadTabContent('search', `/html/search_tab/index.html?v=${
    cb}`, `/html/search_tab/main.js?v=${cb}`),
633     loadTabContent('tts', `/html/tts_tab/index.html?v=${cb}`,
    `/html/tts_tab/main.js?v=${cb}`),
634     loadTabContent('sources', `/html/sources_tab/index.html?v=
    ${cb}`, `/html/sources_tab/main.js?v=${cb}`),
635     loadTabContent('skills', `/html/skills_tab/index.html?v=${
    cb}`, `/html/skills_tab/main.js?v=${cb}`),
636     loadTabContent('mcp', `/html/mcp_tab/index.html?v=${cb}`,
```

```
`/html/mcp_tab/main.js?v=${cb}`),
637     loadTabContent('observability', `/html/system_inspector_tab/index.html?v=${cb}`, `/html/system_inspector_tab/main.js?v=${cb}`),
638     loadTabContent('news', `/html/news_tab/index.html?v=${cb}`, `/html/news_tab/main.js?v=${cb}`),
639     loadTabContent('logs', `/html/logs/index.html?v=${cb}`, `/html/logs/main.js?v=${cb}`),
640     loadTabContent('help', `/html/help/index.html?v=${cb}`, `/html/help/main.js?v=${cb}`),
641   ];
642
643   // Определение и фильтрация вкладок микроприложений (/apps)
644   const appTabDefs = [
645     { id: 'about_system', tab: 'about-system', html: '/html/about_system_tab/index.html?v=20261012_v1', js: '/html/about_system_tab/main.js?v=20261012_v1' },
646     { id: 'network_terminal', tab: 'network', html: '/html/network_tab/index.html', js: '/html/network_tab/main.js' },
647     { id: 'system_inspector', tab: 'hardware-load-inspector', html: '/html/system_inspector_tab/index.html?v=20261008_v16', js: '/html/system_inspector_tab/main.js?v=20261008_v16' },
648     { id: 'processes_load_inspector', tab: 'processes-load-inspector', html: '/html/processes_load_inspector_tab/index.html?v=20261008_v1', js: '/html/processes_load_inspector_tab/main.js?v=20261008_v1' },
649     { id: 'chat', tab: 'chat', html: '/html/chat/index.html', js: '/html/chat/main.js' },
650     { id: 'scenarios', tab: 'scenarios', html: '/html/scenarios_tab/index.html', js: '/html/scenarios_tab/main.js' },
651     { id: 'user_assistant', tab: 'user-assistant', html: '/html/user_assistant_tab/index.html', js: '/html/user_assistant_tab/main.js' },
652     { id: 'gcloud_monitor', tab: 'gcloud', html: '/html/gcloud_tab/index.html', js: '/html/gcloud_tab/main.js' },
653     { id: 'website_monitor', tab: 'website-monitor', html: '/html/website_monitor_tab/index.html', js: '/html/website_monitor_tab/main.js' },
```

```
654     { id: 'cloudflared_monitor', tab: 'cloudflared', html: '/h
    tml/cloudflared_tab/index.html', js: '/html/cloudflared_tab/ma
    in.js' },
655     { id: 'process_leaks', tab: 'process-leaks', html: '/html/
    process_leaks_tab/index.html?v=20260924_v1', js: '/html/proces
    s_leaks_tab/main.js?v=20260924_v1' },
656     { id: 'forensics', tab: 'forensics', html: '/html/forensic
    s_tab/index.html?v=20260924_v1', js: '/html/forensics_tab/main
    .js?v=20260924_v1' },
657     { id: 'throttling', tab: 'throttling', html: '/html/thrott
    ling_tab/index.html?v=20260924_v1', js: '/html/throttling_tab/
    main.js?v=20260924_v1' },
658     { id: 'storage_wear', tab: 'storage-wear', html: '/html/st
    orage_wear_tab/index.html?v=20260924_v1', js: '/html/storage_w
    ear_tab/main.js?v=20260924_v1' },
659     { id: 'peripherals', tab: 'peripherals', html: '/html/peri
    pherals_tab/index.html?v=20260924_v1', js: '/html/peripherals_
    tab/main.js?v=20260924_v1' },
660   ];
661
662   const appTabLoads = appTabDefs
663     .filter(def => {
664       const appInfo = appsMap[def.id] || Object.values(appsMap
    ).find(a => a.tab === `tab-${def.tab}` || a.id === def.id || a
    .folder === def.id);
665       return appInfo ? appInfo.enabled : true;
666     })
667     .map(def => loadTabContent(def.tab, `${def.html}?v=${cb}`,
     `${def.js}?v=${cb}`));
668
669   await Promise.all([...coreTabLoads, ...appTabLoads]);
670
671   // Синхронизация видимости вкладок плагинов
672   try {
673     const pluginsData = await window.api.fetch('/api/admin/plu
    gins');
674     if (window.syncPluginTabsVisibility && pluginsData && plug
    insData.plugins) {
```

```
      window.syncPluginTabsVisibility(pluginsData.plugins);
    }
  } catch (err) {
    console.error('Ошибка синхронизации видимости плагинов:', err);
  }

  setupDropdownTabs();
}

function syncAppsTabsVisibility(appsMap) {
  if (!appsMap || typeof appsMap !== 'object') return;
  const dropdown = document.getElementById('appsTabsDropdown');
  if (!dropdown) return;
  const dropdownMenu = dropdown.nextElementSibling || dropdown.closest('.dropdown')?.querySelector('.dropdown-menu');
  if (!dropdownMenu) return;

  const appButtons = dropdownMenu.querySelectorAll('button[data-tab]');
  let visibleCount = 0;

  appButtons.forEach((btn) => {
    const dataTab = btn.getAttribute('data-tab') || btn.getAttribute('data-bs-target')?.replace('#', '');
    const cleanId = dataTab ? dataTab.replace(/^tab-/, '') : '';
    const appInfo = Object.values(appsMap).find(a => a.tab === dataTab || a.id === cleanId || a.folder === cleanId);
    const isEnabled = appInfo ? appInfo.enabled : true;

    if (isEnabled) {
      btn.style.display = '';
      btn.classList.remove('d-none');
      visibleCount++;
    } else {
      btn.style.display = 'none';
```

```
      btn.classList.add('d-none');
      const pane = document.getElementById(dataTab);
      if (pane) {
        pane.style.display = 'none';
        pane.classList.remove('show', 'active');
      }
      if (btn.classList.contains('active')) {
        btn.classList.remove('active');
        if (typeof window.switchTab === 'function') {
          window.switchTab('tab-chat');
        }
      }
    }
  });

  const dropdownWrapper = dropdown.closest('li.nav-item');
  if (dropdownWrapper) {
    const prevDivider = dropdownWrapper.previousElementSibling
;
    if (visibleCount === 0) {
      dropdownWrapper.style.display = 'none';
      dropdownWrapper.classList.add('d-none');
      if (prevDivider && prevDivider.querySelector('.nav-group
-divider')) {
        prevDivider.style.display = 'none';
        prevDivider.classList.add('d-none');
      }
    } else {
      dropdownWrapper.style.display = '';
      dropdownWrapper.classList.remove('d-none');
      if (prevDivider && prevDivider.querySelector('.nav-group
-divider')) {
        prevDivider.style.display = '';
        prevDivider.classList.remove('d-none');
      }
    }
  }
}
```

```
window.syncAppsTabsVisibility = syncAppsTabsVisibility;

function showPasswordModal() {
  const modal = new bootstrap.Modal(document.getElementById('p
asswordModal'));
  modal.show();

  const loginBtn = document.getElementById('login-btn');
  const passwordInput = document.getElementById('admin-passwor
d');
  const passwordError = document.getElementById('password-erro
r');

  passwordInput?.addEventListener('keypress', (e) => {
    if (e.key === 'Enter') verifyPassword();
  });

  loginBtn?.addEventListener('click', verifyPassword);
}

async function verifyPassword() {
  const passwordInput = document.getElementById('admin-passwor
d');
  const passwordError = document.getElementById('password-erro
r');
  const password = passwordInput?.value;
  if (!password) return;

  const formData = new FormData();
  formData.append('password', password);

  try {
    const res = await fetch('/admin', {
      method: 'POST',
      body: formData
    });
    if (res.ok) {
      hasEnteredPassword = true;
```

```
      passwordError?.classList.add('d-none');
      const modalElement = document.getElementById('passwordMo
dal');
      const modal = bootstrap.Modal.getInstance(modalElement);
      modal?.hide();
      document.getElementById('admin-interface').style.display
 = 'block';
      await initInterface();
    } else {
      throw new Error('Invalid password');
    }
  } catch {
    passwordError?.classList.remove('d-none');
    if (passwordError) {
      passwordError.textContent = 'Неверный пароль';
    }
    if (passwordInput) {
      passwordInput.value = '';
      passwordInput.focus();
    }
  }
}

async function loadTabContent(tabName, url, jsOverrideSrc) {
  try {
    const response = await fetch(url);
    if (!response.ok) throw new Error(`HTTP ${response.status}
`);
    const html = await response.text();
    const container = document.getElementById(`tab-${tabName}`
);
    if (!container) return;
    container.innerHTML = html;

    // Load JS for the tab
    const cacheBuster = Date.now();
    const scriptSrc = jsOverrideSrc || `/html/${tabName}/main.
js?v=${cacheBuster}`;
```

```
807
808     await new Promise((resolve) => {
809       const script = document.createElement('script');
810       if (tabName === 'instructions' || tabName === 'rag') {
811         script.type = 'module';
812       }
813       script.src = scriptSrc;
814       script.onload = () => {
815         console.log(`[Admin] Loaded JS for ${tabName}`);
816         resolve();
817       };
818       script.onerror = (err) => {
819         console.warn(`[Admin] Note: No JS loaded for ${tabName
} from ${scriptSrc}`);
820         resolve();
821       };
822       document.body.appendChild(script);
823     });
824
825     const camelName = tabName.replace(/-([a-z])/g, (_, c) => c
.toUpperCase());
826     const initFuncName = 'init' + camelName.charAt(0).toUpperC
ase() + camelName.slice(1) + 'Tab';
827     if (typeof window[initFuncName] === 'function') {
828       console.log(`[Admin] Calling ${initFuncName} for ${tabNa
me}`);
829       try {
830         await window[initFuncName]();
831       } catch (err) {
832         console.error(`[Admin] Error executing ${initFuncName}
:`, err);
833       }
834     }
835     applyTranslations();
836   } catch (e) {
837     console.error(`[Admin] Error loading tab ${tabName}:`, e);
838     const container = document.getElementById(`tab-${tabName}`
);
```

```
    if (container) {
      container.innerHTML = `<div class="alert alert-danger">Ошибка загрузки: ${e.message}</div>`;
    }
  }
}

// Initialize HELP content
function initHelpContent() {
  window.HELP_CONTENT = window.HELP_CONTENT || {};
  const defaults = {
    'overview': `<h4>🏠 Обзор проекта</h4><p>ai-windows-admin — интегрированная среда для работы с AI, RAG и системным управлением.</p>`,
    'google_oauth': `<h4>🔑 Google OAuth и аккаунты</h4><p>Интеграция с Gmail, Drive, Sheets и Docs через OAuth 2.0 и Service Accounts.</p>`,
    'ai_models': `<h4>🤖 ИИ Провайдеры и Модели</h4><p>Облачные (Gemini, OpenAI, Groq) и локальные (Ollama, Foundry, DirectML) модели.</p>`,
    'gdrive_sync': `<h4>☁️ Google Drive Sync</h4><p>Автоматическая синхронизация баз данных и файлов с Google Drive.</p>`,
    'rag_knowledge': `<h4>📚 База знаний и RAG</h4><p>Загрузка и поиск по вашим документам (PDF, Word, TXT, CSV, JSON).</p>`,
    'rag': `<h4>🔍 RAG-индекс (Векторный поиск)</h4>
<p><strong>1. База RAG:</strong> Индексирует системные базы знаний, системные инструкции и документы.</p>
<p><strong>2. Загрузка документов:</strong> Позволяет загрузить внешние <code>.json</code>, <code>.txt</code>, <code>.md</code>, <code>.pdf</code> файлы или сканировать директории напрямую в RAG-индекс.</p>
<p><strong>3. Чат-RAG:</strong> Индексирует историю сохраненных диалогов и ответов ассистента.</p>`,
    'voice_tts': `<h4>🎙️ Голос и Озвучка</h4><p>Голосовой ввод (Whisper/WebSpeech) и синтез речи (Edge TTS).</p>`,
    'plugins_skills': `<h4>🧩 Плагины, Навыки и MCP</h4><p>Telegram-бот, IFTTT, распознавание счетов и расширение через MCP.
```

```
</p>`,
    'storage_disks': `<h4>🗄️ Диски и Хранилище</h4><p>Сканирование накопителей, проверка целостности и консолидация дублей.</p>`,
    'troubleshooting': `<h4>🔧 Решение проблем (FAQ)</h4><p>Ответы на частые вопросы и устранение ошибок подключения.</p>`
  };
  window.HELP_CONTENT = Object.assign(defaults, window.HELP_CONTENT);
}

function showHelpModal(key) {
  const content = window.HELP_CONTENT[key] || '<p>Информация не найдена</p>';
  document.getElementById('help-modal-content').innerHTML = content;
  const modal = new bootstrap.Modal(document.getElementById('help-modal'));
  modal.show();
}

function showChatLogicModal() {
  const modalEl = document.getElementById('chat-logic-modal');
  if (modalEl) {
    const modal = new bootstrap.Modal(modalEl);
    modal.show();
  }
}
window.showChatLogicModal = showChatLogicModal;
window.showHelpModal = showHelpModal;

// Уведомления
function showNotification(message, type = 'info') {
  const notification = document.createElement('div');
  notification.className = `alert alert-${type} position-fixed top-0 end-0 m-3`;
  notification.style.zIndex = '9999';
  notification.style.maxWidth = '400px';
```

```
  notification.textContent = message;
  document.body.appendChild(notification);

  setTimeout(() => {
    notification.remove();
  }, 5000);
}

async function initAdminTab() {
  const modelSelect = document.getElementById('admin-model-sel
ect');
  const saveBtn = document.getElementById('btn-admin-save-mode
l');

  if (!modelSelect || !saveBtn) return;

  // 1. Очищаем селект
  modelSelect.innerHTML = '';

  let modelsGrouped = {};
  // 2. Загружаем доступные модели
  try {
    const modelsData = await window.api.fetch('/api/chat/model
s');
    modelsGrouped = modelsData.models || {};
    if (Array.isArray(modelsGrouped)) {
      modelsGrouped = { 'gemini': modelsGrouped };
    }
  } catch (err) {
    console.error('Ошибка загрузки моделей:', err);
    showNotification('Ошибка загрузки моделей AI: ' + err.mess
age, 'danger');
  }

  // 3. Заполняем селект моделями с иерархией optgroup
  const providerMeta = {
    'gemini': { label: '  Google Gemini', order: 1 },
    'agy': { label: '  Google Antigravity (AGY)', order: 2 },
```

```
    'foundry': { label: '  Microsoft Foundry', order: 3 },
    'ollama': { label: '  Ollama (Local)', order: 4 },
    'onnx': { label: '  Microsoft ONNX / Olive', order: 5 }
  };

  const providers = Object.keys(modelsGrouped).sort((a, b) =>
{
    const oA = providerMeta[a]?.order ?? 99;
    const oB = providerMeta[b]?.order ?? 99;
    return oA - oB;
  });

  let totalModels = 0;
  providers.forEach(p => {
    const list = modelsGrouped[p] || [];
    if (!Array.isArray(list) || list.length === 0) return;

    const optgroup = document.createElement('optgroup');
    optgroup.label = providerMeta[p]?.label || `  ${p.toUpper
Case()}`;

    list.forEach(m => {
      totalModels++;
      const option = document.createElement('option');
      option.value = m;
      option.textContent = m;
      optgroup.appendChild(option);
    });

    modelSelect.appendChild(optgroup);
  });

  if (totalModels === 0) {
    const option = document.createElement('option');
    option.value = '';
    option.textContent = 'Нет доступных моделей';
    modelSelect.appendChild(option);
    saveBtn.disabled = true;
```

```
  } else {
    saveBtn.disabled = false;
  }

  // 4. Загружаем текущие настройки пользователя (выбранную мо
дель)
  try {
    const settingsData = await window.api.fetch('/auth/setting
s');
    const savedModel = settingsData && settingsData.model ? se
ttingsData.model : '';
    if (savedModel) {
      modelSelect.value = savedModel;
    }
    if (typeof window.updateChatBadges === 'function') {
      window.updateChatBadges(savedModel);
    }
  } catch (err) {
    console.error('Ошибка загрузки настроек AI пользователя:',
 err);
  }

  // 5. Навешиваем обработчик сохранения
  saveBtn.onclick = async () => {
    const selectedModel = modelSelect.value;
    saveBtn.disabled = true;
    const originalText = saveBtn.textContent;
    saveBtn.textContent = 'Сохранение...';

    try {
      await window.api.fetch('/auth/settings', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ model: selectedModel })
      });
      showNotification('Модель успешно обновлена на: ' + selec
tedModel, 'success');

```

```
      // Обновляем бейджи модели в реальном времени на странице
      window.activeModelName = selectedModel;
      if (typeof window.updateChatBadges === 'function') {
        window.updateChatBadges(selectedModel);
      } else {
        const badges = document.querySelectorAll('#chat-model-badge, #chat-popup-model-badge');
        badges.forEach(badge => {
          badge.textContent = selectedModel;
          badge.style.display = 'inline-block';
        });
      }
    } catch (err) {
      console.error('Ошибка сохранения модели:', err);
      showNotification('Ошибка сохранения: ' + err.message, 'danger');
    } finally {
      saveBtn.disabled = false;
      saveBtn.textContent = originalText;
    }
  };
}

window.initAdminTab = initAdminTab;
```

apps-sync.js

```
/**
 * ============================================================
 * Process Name: Windows Modules - Apps-Sync Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля apps-sync.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/admin/modules/apps-sync.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { syncAppsTabsVisibility, getAppsMap } from '/windows/api/webgui/admin/modules/apps-sync.js';
 *
 * File: apps-sync.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/admin/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * Apps Sync Module - Синхронизация видимости приложений
 */

export async function syncApplicationsVisibility() {
  console.log('[AppSync] Syncing application visibility...');

```

```
  let appsMap = {};

  try {
    const appsData = await (window.api ? window.api.fetch('/ap
i/v1/apps/status') : fetch('/api/v1/apps/status').then(r => r.
json()));

    if (appsData && appsData.apps) {
      appsMap = appsData.apps;
      window.appsStatusMap = appsData.apps;
      syncAppsTabsVisibility(appsData.apps);
    }
  } catch (error) {
    console.warn('[AppSync] Failed to sync apps visibility:',
error);
  }

  return appsMap;
}

export function syncAppsTabsVisibility(appsMap) {
  if (!appsMap || typeof appsMap !== 'object') return;

  const dropdown = document.getElementById('appsTabsDropdown')
;
  if (!dropdown) return;

  const dropdownMenu = dropdown.nextElementSibling ||
                       dropdown.closest('.dropdown')?.querySele
ctor('.dropdown-menu');
  if (!dropdownMenu) return;

  const appButtons = dropdownMenu.querySelectorAll('button[dat
a-tab]');
  let visibleCount = 0;

  appButtons.forEach((btn) => {
    const dataTab = btn.getAttribute('data-tab') ||
```

```
                  btn.getAttribute('data-bs-target')?.replace('#', '');
    const cleanId = dataTab ? dataTab.replace(/^tab-/, '') : '';
    const appInfo = Object.values(appsMap).find(
      a => a.tab === dataTab || a.id === cleanId || a.folder === cleanId
    );
    const isEnabled = appInfo ? appInfo.enabled : true;

    if (isEnabled) {
      btn.style.display = '';
      btn.classList.remove('d-none');
      visibleCount++;
    } else {
      btn.style.display = 'none';
      btn.classList.add('d-none');

      const pane = document.getElementById(dataTab);
      if (pane) {
        pane.style.display = 'none';
        pane.classList.remove('show', 'active');
      }

      if (btn.classList.contains('active')) {
        btn.classList.remove('active');
        if (typeof window.switchTab === 'function') {
          window.switchTab('tab-chat');
        }
      }
    }
  });

  // Hide/show apps dropdown if all apps are disabled
  const dropdownWrapper = dropdown.closest('li.nav-item');
  if (dropdownWrapper) {
    const prevDivider = dropdownWrapper.previousElementSibling;
```

```
97
98      if (visibleCount === 0) {
99        dropdownWrapper.style.display = 'none';
100       if (prevDivider) prevDivider.style.display = 'none';
101     } else {
102       dropdownWrapper.style.display = '';
103       if (prevDivider) prevDivider.style.display = '';
104     }
105   }
106
107   console.log(`[AppSync] Synced apps visibility: ${visibleCoun
    t} visible`);
108 }
109
110 export function getAppsMap() {
111   return window.appsStatusMap || {};
112 }
113
114
```

auth-handler.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Modules - Auth-Handler Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля auth-hand
ler.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/admin/modules/auth-han
dler.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { setupAuthHandlers } from '/windows/api/webgui/
admin/modules/auth-handler.js';
 *
 * File: auth-handler.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/admin/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * Auth Handler Module - Обработка аутентификации и пароля
 */

export function setupAuthHandlers() {
  setupPasswordModal();
  setupPasswordVerification();
```

```
}

function setupPasswordModal() {
  const modal = document.getElementById('passwordModal');
  const loginBtn = document.getElementById('login-btn');
  const closeBtn = document.getElementById('closePasswordModal
');
  const adminPassword = document.getElementById('admin-passwor
d');

  if (loginBtn) {
    loginBtn.addEventListener('click', verifyPassword);
  }

  if (adminPassword) {
    adminPassword.addEventListener('keypress', (e) => {
      if (e.key === 'Enter') {
        verifyPassword();
      }
    });
  }

  if (closeBtn) {
    closeBtn.addEventListener('click', () => {
      if (modal) {
        const m = bootstrap.Modal.getInstance(modal);
        if (m) m.hide();
      }
    });
  }

  // Show modal on load if not authenticated
  setTimeout(() => {
    const isAuthenticated = localStorage.getItem('admin_authen
ticated');
    if (!isAuthenticated) {
      showPasswordModal();
    }
```

```
  }, 500);
}

function setupPasswordVerification() {
  // Проверка аутентификации при загрузке
  const isAuthenticated = localStorage.getItem('admin_authenti
cated');
  const adminPassword = document.getElementById('admin-passwor
d');

  if (isAuthenticated && adminPassword) {
    adminPassword.value = '';
    localStorage.removeItem('admin_authenticated');
  }
}

async function verifyPassword() {
  const passwordInput = document.getElementById('admin-passwor
d');
  const passwordError = document.getElementById('password-erro
r');
  const modal = document.getElementById('passwordModal');

  if (!passwordInput.value) {
    if (passwordError) {
      passwordError.classList.remove('d-none');
      passwordError.textContent = 'Пароль не может быть пустым
';
    }
    return;
  }

  try {
    const response = await fetch('/api/admin/verify-password',
 {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ password: passwordInput.value })
```

```
    });

    if (response.ok) {
      // Пароль верен
      if (modal) {
        const m = bootstrap.Modal.getInstance(modal);
        if (m) m.hide();
      }
      localStorage.setItem('admin_authenticated', 'true');
      passwordInput.value = '';
      if (passwordError) passwordError.classList.add('d-none')
;
    } else {
      if (passwordError) {
        passwordError.classList.remove('d-none');
        passwordError.textContent = 'Неверный пароль';
      }
      passwordInput.value = '';
      passwordInput.focus();
    }
  } catch (error) {
    console.error('Password verification error:', error);
    if (passwordError) {
      passwordError.classList.remove('d-none');
      passwordError.textContent = 'Ошибка проверки пароля';
    }
  }
}

function showPasswordModal() {
  const modal = document.getElementById('passwordModal');
  if (modal) {
    const m = new bootstrap.Modal(modal, { backdrop: 'static',
 keyboard: false });
    m.show();
    const passwordInput = document.getElementById('admin-passw
ord');
    if (passwordInput) {
```

```
      passwordInput.focus();
    }
  }
}

export { verifyPassword, showPasswordModal };

```

init-interface.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Modules - Init-Interface Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля init-inte
rface.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/admin/modules/init-int
erface.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { setupGlobalFunctions, setupLanguageSelector, s
etupThemeSelector } from '/windows/api/webgui/admin/modules/in
it-interface.js';
 *
 * File: init-interface.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/admin/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * Interface Initialization Module - Инициализация компонентов
 интерфейса
 */

import { initI18n, switchLang, applyTranslations } from '../..
```

```
/js/i18n.js';
import { initTheme, setTheme, getThemeMode, getResolvedTheme } from '../../js/theme.js';
import { initUserSettings, refreshUserProfile } from '../../js/userSettings.js';

export async function initializeInterface() {
  console.log('[AdminInterface] Initializing core components...');

  try {
    // Initialize theme
    initTheme();

    // Initialize i18n
    await initI18n();

    // Initialize user settings
    await initUserSettings();

    // Apply translations
    applyTranslations();

    // Обновляем бейджи активной модели и поиска
    if (typeof window.updateChatBadges === 'function') {
      window.updateChatBadges();
    }

    console.log('[AdminInterface] Core components initialized');
  } catch (error) {
    console.error('[AdminInterface] Initialization error:', error);
    throw error;
  }
}

export function setupGlobalFunctions() {
```

```
  // Make functions available globally
  window.switchLang = switchLang;
  window.applyTranslations = applyTranslations;
  window.setTheme = setTheme;
  window.getThemeMode = getThemeMode;
  window.getResolvedTheme = getResolvedTheme;
  window.initUserSettings = initUserSettings;
  window.refreshUserProfile = refreshUserProfile;

  // Help and modals
  window.showHelpModal = window.showHelpModal || (() => {});
  window.showChatLogicModal = window.showChatLogicModal || (() => {});
  window.showNotification = window.showNotification || (() => {});

  // API Fallback
  window.api = window.api || {};
  if (!window.api.fetch) {
    window.api.fetch = async function(url, options = {}) {
      const response = await fetch(url, options);
      if (!response.ok) {
        let msg = response.statusText;
        try {
          const data = await response.json();
          if (data && data.detail) {
            if (typeof data.detail === 'string') msg = data.detail;
            else if (Array.isArray(data.detail)) msg = data.detail.map(d => d.msg || JSON.stringify(d)).join(', ');
            else msg = JSON.stringify(data.detail);
          } else if (data && data.message) {
            msg = data.message;
          }
        } catch {}
        throw new Error(`${response.status} ${msg}`);
      }
      return response.json();
```

```
    };
  }

  console.log('[AdminInterface] Global functions registered');
}

export function setupLanguageSelector() {
  const langSelectors = document.querySelectorAll('.lang-selector');
  const savedLocale = localStorage.getItem('app_locale') || localStorage.getItem('app_language') || localStorage.getItem('language') || 'ru-RU';

  langSelectors.forEach((selector) => {
    const optLocale = Array.from(selector.options).find(opt => opt.value.toLowerCase() === savedLocale.toLowerCase());
    const optLang = Array.from(selector.options).find(opt => opt.value.toLowerCase() === savedLocale.split('-')[0].toLowerCase());
    if (optLocale) selector.value = optLocale.value;
    else if (optLang) selector.value = optLang.value;

    selector.addEventListener('change', (e) => {
      switchLang(e.target.value);
    });
  });

  console.log('[AdminInterface] Language selector setup');
}

export function setupThemeSelector() {
  const themeSelectors = document.querySelectorAll('.theme-selector');
  const savedTheme = getThemeMode();

  themeSelectors.forEach((selector) => {
    selector.value = savedTheme;
    selector.addEventListener('change', (e) => {
```

```
      setTheme(e.target.value);
    });
  });

  console.log('[AdminInterface] Theme selector setup');
}

export { initI18n, switchLang, applyTranslations };
export { initTheme, setTheme, getThemeMode, getResolvedTheme };
export { initUserSettings, refreshUserProfile };
```

tab-manager.js

```
/**
 * ============================================================
 * Process Name: Windows Modules - Tab-Manager Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля tab-manager.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/admin/modules/tab-manager.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { setupTabManagement, onTabSwitched } from '/windows/api/webgui/admin/modules/tab-manager.js';
 *
 * File: tab-manager.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/admin/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * Tab Manager Module - Управление вкладками и навигацией
 */

export function setupTabManagement() {
  setupDropdownTabs();
  setupTabEvents();
```

```
}

function setupDropdownTabs() {
  const mainTabs = document.getElementById('mainTabs');
  if (!mainTabs) return;

  const dropdowns = mainTabs.querySelectorAll('.dropdown-toggle');

  dropdowns.forEach((dropdown) => {
    dropdown.addEventListener('click', (e) => {
      e.preventDefault();
      e.stopPropagation();

      const parentDropdown = dropdown.closest('.dropdown');
      const menu = dropdown.nextElementSibling || parentDropdown?.querySelector('.dropdown-menu');
      const isAlreadyOpen = menu?.classList.contains('show');

      // Close other dropdowns
      document.querySelectorAll('#mainTabs .dropdown-menu.show').forEach((otherMenu) => {
        otherMenu.classList.remove('show');
        otherMenu.closest('.dropdown')?.querySelector('.dropdown-toggle')?.classList.remove('show');
      });

      // Toggle current dropdown
      if (!isAlreadyOpen && menu) {
        dropdown.classList.add('show');
        menu.classList.add('show');
      }
    });
  });

  // Handle all clickable navigation items inside mainTabs (tabs and dropdown items)
  mainTabs.querySelectorAll('.dropdown-item, .list-group-item,
```

```
 [data-tab], [data-bs-target], [data-plugin]').forEach((item)
=> {
    item.addEventListener('click', (e) => {
      const pluginName = item.getAttribute('data-plugin');
      const targetId = item.getAttribute('data-tab') ||
                       item.getAttribute('data-bs-target')?.rep
lace('#', '');

      // Close parent dropdown menu
      const menu = item.closest('.dropdown-menu');
      if (menu) {
        menu.classList.remove('show');
        menu.closest('.dropdown')?.querySelector('.dropdown-to
ggle')?.classList.remove('show');
      }

      if (pluginName && typeof window.openPluginFromDropdown =
== 'function') {
        e.preventDefault();
        e.stopPropagation();
        window.openPluginFromDropdown(pluginName);
      } else if (targetId) {
        e.preventDefault();
        e.stopPropagation();
        if (typeof window.switchTab === 'function') {
          window.switchTab(targetId);
        }
      }
    });
  });

  // Close dropdowns on outside click
  document.addEventListener('click', (e) => {
    if (!mainTabs.contains(e.target)) {
      dropdowns.forEach((dropdown) => {
        dropdown.classList.remove('show');
        const menu = dropdown.nextElementSibling;
        if (menu) menu.classList.remove('show');
```

```
      });
    }
  });
}

function setupTabEvents() {
  // Tab content click handlers
  const tabPane = document.getElementById('mainTabsContent');
  if (tabPane) {
    tabPane.addEventListener('click', (e) => {
      const target = e.target.closest('[data-tab]');
      if (target) {
        const tabId = target.getAttribute('data-tab');
        if (tabId) {
          window.switchTab(tabId);
        }
      }
    });
  }
}

export function onTabSwitched(targetId) {
  const cleanId = targetId.startsWith('#') ? targetId.slice(1)
 : targetId;

  if (cleanId === 'tab-chat') {
    if (typeof window.initChatTab === 'function') {
      window.initChatTab();
    }
  }
}

export { setupDropdownTabs, setupTabEvents };

```

ui-handler.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Modules - Ui-Handler Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля ui-handle
r.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/admin/modules/ui-handl
er.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { setupUIHandlers, showHelpModal, showChatLogicM
odal } from '/windows/api/webgui/admin/modules/ui-handler.js';
 *
 * File: ui-handler.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/admin/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * UI Handler Module - Работа с модалями, уведомлениями и помо
щью
 */

let helpContent = {};

```

```
export function setupUIHandlers() {
  initHelpContent();
  setupModalHandlers();
}

function initHelpContent() {
  helpContent = {
    'overview': `<h4>🚀 Обзор проекта</h4><p>ai-windows-admin — интегрированная среда для работы с AI, RAG и системным управлением.</p>`,
    'google_oauth': `<h4>🔑 Google OAuth и аккаунты</h4><p>Интеграция с Gmail, Drive, Sheets и Docs через OAuth 2.0 и Service Accounts.</p>`,
    'ai_models': `<h4>🤖 ИИ Провайдеры и Модели</h4><p>Облачные (Gemini, OpenAI, Groq) и локальные (Ollama, Foundry, DirectML) модели.</p>`,
    'gdrive_sync': `<h4>☁️ Google Drive Sync</h4><p>Автоматическая синхронизация баз данных и файлов с Google Drive.</p>`,
    'rag_knowledge': `<h4>📚 База знаний и RAG</h4><p>Загрузка и поиск по вашим документам (PDF, Word, TXT, CSV, JSON).</p>`,
    'rag': `<h4>🧠 RAG-индекс (Векторный поиск)</h4>
<p><strong>1. База RAG:</strong> Индексирует системные базы знаний, системные инструкции и документы.</p>
<p><strong>2. Загрузка документов:</strong> Позволяет загрузить внешние <code>.json</code>, <code>.txt</code>, <code>.md</code>, <code>.pdf</code> файлы или сканировать директории напрямую в RAG-индекс.</p>
<p><strong>3. Чат-RAG:</strong> Индексирует историю сохраненных диалогов и ответов ассистента.</p>`,
    'voice_tts': `<h4>🎙️ Голос и Озвучка</h4><p>Голосовой ввод (Whisper/WebSpeech) и синтез речи (Edge TTS).</p>`,
    'plugins_skills': `<h4>🧩 Плагины, Навыки и MCP</h4><p>Telegram-бот, IFTTT, распознавание счетов и расширение через MCP.</p>`,
    'storage_disks': `<h4>💾 Диски и Хранилище</h4><p>Сканирование накопителей, проверка целостности и консолидация дублей.</p>`,
```

```
    'troubleshooting': `<h4>🔧 Решение проблем (FAQ)</h4><p>Ответы на частые вопросы и устранение ошибок подключения.</p>`
  };

  window.HELP_CONTENT = helpContent;
}

function setupModalHandlers() {
  // Help modal
  const helpBtn = document.getElementById('help-btn');
  if (helpBtn) {
    helpBtn.addEventListener('click', () => {
      showHelpModal('overview');
    });
  }

  // Chat logic modal
  const chatLogicBtn = document.getElementById('chat-logic-btn');
  if (chatLogicBtn) {
    chatLogicBtn.addEventListener('click', () => {
      showChatLogicModal();
    });
  }
}

export function showHelpModal(key) {
  const content = helpContent[key] || '<p>Информация не найдена</p>';
  const contentDiv = document.getElementById('help-modal-content');
  if (contentDiv) {
    contentDiv.innerHTML = content;
  }

  const modal = document.getElementById('help-modal');
  if (modal) {
    const m = new bootstrap.Modal(modal);
```

```
    m.show();
  }
}

export function showChatLogicModal() {
  const modalEl = document.getElementById('chat-logic-modal');
  if (modalEl) {
    const modal = new bootstrap.Modal(modalEl);
    modal.show();
  }
}

export function showNotification(message, type = 'info') {
  const notification = document.createElement('div');
  notification.className = `alert alert-${type} position-fixed
 top-0 end-0 m-3`;
  notification.style.zIndex = '9999';
  notification.style.maxWidth = '400px';
  notification.textContent = message;
  document.body.appendChild(notification);

  setTimeout(() => {
    notification.remove();
  }, 5000);
}

export { initHelpContent, setupModalHandlers };

```

optimization-init.js

```
/**
 * ============================================================
 * Process Name: Windows Admin - Optimization-Init Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля optimization-init.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/admin/optimization-init.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { enhanceTabSwitching, createSearchDebounce, createAutoSaveDebounce } from '/windows/api/webgui/admin/optimization-init.js';
 *
 * File: optimization-init.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/admin
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * Optimization Init - Интеграция систем оптимизации в админку
 *
 * Этот скрипт инициализирует:
 * - Ленивую загрузку вкладок
 * - Кеширование API
```

```
 * - Дебаунсинг запросов
 * - Сохранение состояния вкладок
 */

import LazyTabLoader from '../js/tab-loader.js';
import { APICache, APIFetcher } from '../js/api-cache.js';
import { RequestManager } from '../js/request-manager.js';

// ============================================================
================
// 1. ИНИЦИАЛИЗАЦИЯ СИСТЕМ ОПТИМИЗАЦИИ
// ============================================================
================

const tabLoader = new LazyTabLoader();
const apiCache = new APICache();
const apiFetcher = new APIFetcher(apiCache);
const requestManager = new RequestManager();

// Экспортируем в глобальный скоп для доступа из других модуле
й
window.optimizationModule = {
  tabLoader,
  apiCache,
  apiFetcher,
  requestManager,
  stats: {
    tabsLoaded: new Set(),
    apiCallsUsed: 0,
    cachedCallsUsed: 0,
    requestsDebounced: 0
  }
};

console.log('[Optimization] System initialized');

// ============================================================
================
```

```
// 2. ПЕРЕХВАТ API.FETCH ОБЕРТКА
// ============================================================
=================

// Сохраняем оригинальный fetch
const originalApiFetch = window.api?.fetch;

// Переопределяем window.api.fetch для использования кеша
if (window.api && originalApiFetch) {
  window.api.fetch = async function(url, options = {}) {
    const method = (options.method || 'GET').toUpperCase();
    const isGetRequest = method === 'GET' || method === 'HEAD'
;

    // Для GET запросов проверяем кеш
    if (isGetRequest) {
      const cached = apiCache.get(url);
      if (cached) {
        window.optimizationModule.stats.cachedCallsUsed++;
        console.log(`[Optimization] API call from cache: ${met
hod} ${url}`);
        return cached;
      }
    }

    // Делаем реальный запрос
    window.optimizationModule.stats.apiCallsUsed++;
    const result = await originalApiFetch.call(this, url, opti
ons);

    // Кешируем для GET запросов (5 минут TTL)
    if (isGetRequest) {
      apiCache.set(url, result, 5 * 60 * 1000); // 5 минут
    } else {
      // Для POST/PUT/DELETE инвалидируем релевантный кеш
      const basePath = url.split('?')[0];

      // Умная инвалидация - очищаем кеш связанных GET запросо
```

```
в
      if (method === 'POST' || method === 'PUT' || method === 'DELETE') {
        // Инвалидируем основной путь и все его вариации
        const pattern = new RegExp(`${basePath.replace(/\/$/, '')}.*`);
        const invalidatedCount = apiCache.invalidate(pattern);
        if (invalidatedCount > 0) {
          console.log(`[Optimization] Invalidated ${invalidatedCount} cache entries for ${basePath}`);
        }

        // Для пользователей - инвалидируем список пользователей
        if (basePath.includes('/users/')) {
          apiCache.invalidate(/\/api\/admin\/users/);
          apiCache.invalidate(/\/api\/users/);
        }

        // Для конфигурации - инвалидируем конфиги
        if (basePath.includes('/config/')) {
          apiCache.invalidate(/\/api\/config/);
          apiCache.invalidate(/\/api\/admin\/config/);
        }

        // Для плагинов - инвалидируем список плагинов
        if (basePath.includes('/plugins')) {
          apiCache.invalidate(/\/api\/plugins/);
          apiCache.invalidate(/\/api\/admin\/plugins/);
        }

        // Для модели - инвалидируем список моделей
        if (basePath.includes('/models') || basePath.includes('/chat')) {
          apiCache.invalidate(/\/api\/chat\/models/);
          apiCache.invalidate(/\/api\/models/);
        }
      }
```

```
    }

    return result;
  };
}

// ============================================================
================
// 3. ПЕРЕДЕЛКА ИНИЦИАЛИЗАЦИИ ВКЛАДОК (LAZY LOADING)
// ============================================================
================

export async function initLazyTabLoading(tabDefinitions) {
  console.log('[Optimization] Initializing lazy tab loading (o
n-demand mode)...');

  // Регистрируем все вкладки в системе ленивой загрузки
  tabDefinitions.forEach(({ tabName, htmlUrl, jsUrl }) => {
    tabLoader.registerTab(tabName, htmlUrl, jsUrl);
  });

  // Перехватываем loadTab чтобы применять автопатч
  const originalLoadTab = tabLoader.loadTab.bind(tabLoader);
  tabLoader.loadTab = async function(tabName) {
    const result = await originalLoadTab(tabName);

    // Применяем дебаунс автопатч к загруженной вкладке
    if (result && window.TabDebounceAutoPatch) {
      setTimeout(() => {
        try {
          window.TabDebounceAutoPatch.autoPatchTab(tabName, {
            searchEnabled: true,
            autoSaveEnabled: true,
            refreshEnabled: true
          });
        } catch (e) {
          console.warn(`[Optimization] Failed to apply debounc
e patch to ${tabName}:`, e);
```

```
        }
      }, 100);
    }

    return result;
  };

  const allowedTabNames = tabDefinitions.map(def => def.tabNam
e);
  console.log(`[Optimization] Registered ${allowedTabNames.len
gth} tabs for on-demand loading.`);

  return { registeredTabs: allowedTabNames };
}

// ============================================================
==================
// 4. ИНТЕГРАЦИЯ С ПЕРЕКЛЮЧАТЕЛЕМ ВКЛАДОК
// ============================================================
==================

export function enhanceTabSwitching(originalSwitchTab) {
  return async function switchTabOptimized(targetId) {
    if (!targetId) return;

    const cleanId = targetId.startsWith('#') ? targetId.slice(
1) : targetId;
    const tabName = cleanId.replace(/^tab-/, '');

    console.log(`[Optimization] Switching to tab: ${tabName}`)
;

    // Если вкладка еще не загружена - загружаем ее по требова
ию
    if (!tabLoader.isLoaded(tabName)) {
      console.log(`[Optimization] Tab not loaded yet, loading
on-demand: ${tabName}`);
      await tabLoader.loadTab(tabName);
```

```
      window.optimizationModule.stats.tabsLoaded.add(tabName);
    }

    // Вызываем оригинальную функцию переключения
    if (typeof originalSwitchTab === 'function') {
      originalSwitchTab.call(this, targetId);
    }

    // Сохраняем состояние вкладки в localStorage
    localStorage.setItem('admin:lastActiveTab', cleanId);
  };
}

// ============================================================
=================
// 5. ВОССТАНОВЛЕНИЕ ПОСЛЕДНЕЙ ВКЛАДКИ
// ============================================================
=================

export async function restoreLastTab(fallbackTab = 'tab-about-
system') {
  // Проверяем хэш в URL
  const hash = location.hash.replace('#', '');
  let initialTab = hash || null;

  // Если в хэше ничего нет, проверяем persistence
  if (!initialTab) {
    if (window.tabPersistence && typeof window.tabPersistence.
getLastActiveTab === 'function') {
      initialTab = window.tabPersistence.getLastActiveTab();
    } else {
      initialTab = localStorage.getItem('admin:lastActiveTab')
;
    }
  }

  const targetTabId = (initialTab && document.getElementById(i
nitialTab.startsWith('tab-') ? initialTab : `tab-${initialTab}
```

```
`))
    ? (initialTab.startsWith('tab-') ? initialTab : `tab-${initialTab}`)
    : fallbackTab;

  const tabName = targetTabId.replace(/^tab-/, '');
  console.log(`[Optimization] Loading strictly initial active tab: ${tabName}`);

  if (!tabLoader.isLoaded(tabName)) {
    await tabLoader.loadTab(tabName);
    window.optimizationModule.stats.tabsLoaded.add(tabName);
  }

  if (typeof window.switchTab === 'function') {
    window.switchTab(targetTabId);
  }
  return targetTabId;
}

// ================================================================
// 6. УТИЛИТЫ ДЛЯ ДЕБАУНСИНГА И БАТЧИНГА
// ================================================================

export function createSearchDebounce(searchFn, delay = 500) {
  return (query) => {
    window.optimizationModule.stats.requestsDebounced++;
    return requestManager.debounce('search-query', () => searchFn(query), delay);
  };
}

export function createAutoSaveDebounce(saveFn, delay = 1000) {
  return (data) => {
    window.optimizationModule.stats.requestsDebounced++;
    return requestManager.debounce('autosave-data', () => save
```

```
Fn(data), delay);
  };
}

// ============================================================
=================
// 7. ФУНКЦИИ ОТЛАДКИ И СТАТИСТИКИ
// ============================================================
=================

export function getOptimizationStats() {
  return {
    ...window.optimizationModule.stats,
    cacheStats: apiCache.getStats(),
    requestManagerStats: requestManager.getStats(),
    tabsPreloaded: Array.from(window.optimizationModule.stats.
tabsLoaded)
  };
}

export function printOptimizationStats() {
  const stats = getOptimizationStats();
  console.group('  Optimization Statistics');
  console.log('API Calls:', stats.apiCallsUsed);
  console.log('Cached Calls:', stats.cachedCallsUsed);
  console.log('Cache Hit Rate:', (stats.cachedCallsUsed / (sta
ts.apiCallsUsed + stats.cachedCallsUsed) * 100).toFixed(1) + '
%');
  console.log('Requests Debounced:', stats.requestsDebounced);
  console.log('Tabs Preloaded:', stats.tabsPreloaded);
  console.log('Cache Entries:', stats.cacheStats.size);
  console.groupEnd();
}

// Делаем доступной из консоли для отладки
window.printOptimizationStats = printOptimizationStats;
window.getOptimizationStats = getOptimizationStats;

```

```
// Выводим статистику каждые 30 секунд (опционально, для отладки)
// Раскомментировать в production если нужна мониторинг
// setInterval(() => {
//   console.clear();
//   printOptimizationStats();
// }, 30000);

export default {
  initLazyTabLoading,
  enhanceTabSwitching,
  restoreLastTab,
  createSearchDebounce,
  createAutoSaveDebounce,
  getOptimizationStats,
  printOptimizationStats,
  tabLoader,
  apiCache,
  apiFetcher,
  requestManager
};

```

sync-api.js

```
/**
 * ============================================================
 * Process Name: Windows Admin - Sync-Api Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля sync-api.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/admin/sync-api.js?v=20261001_v1" type="module"></script>
 *
 * File: sync-api.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/admin
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * Google Drive Sync API Client
 * Клиент для работы с REST API синхронизации
 */

class SyncApiClient {
  constructor(baseUrl = '/api/admin/sync') {
    this.baseUrl = baseUrl;
    this.timeout = 10000;
  }

  /**
```

```
   * Выполнить GET запрос
   */
  async get(endpoint) {
    try {
      const response = await fetch(`${this.baseUrl}${endpoint}
`, {
        method: 'GET',
        headers: { 'Content-Type': 'application/json' }
      });

      if (!response.ok) {
        throw new Error(`API Error: ${response.status}`);
      }

      return await response.json();
    } catch (error) {
      console.error(`[SyncApiClient] GET ${endpoint}:`, error)
;
      throw error;
    }
  }

  /**
   * Выполнить POST запрос
   */
  async post(endpoint, data = {}) {
    try {
      const response = await fetch(`${this.baseUrl}${endpoint}
`, {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify(data)
      });

      if (!response.ok) {
        throw new Error(`API Error: ${response.status}`);
      }

```

```
      return await response.json();
    } catch (error) {
      console.error(`[SyncApiClient] POST ${endpoint}:`, error
);
      throw error;
    }
  }

  /**
   * Получить статус синхронизации
   */
  async getStatus() {
    return this.get('/status');
  }

  /**
   * Получить статистику
   */
  async getStats() {
    return this.get('/stats');
  }

  /**
   * Получить информацию о Google Drive
   */
  async getDriveInfo() {
    return this.get('/drive-info');
  }

  /**
   * Получить конфигурацию
   */
  async getConfig() {
    return this.get('/config');
  }

  /**
   * Запустить планировщик
```

```
   */
  async startScheduler(interval = 6) {
    return this.post('/start', { sync_interval_hours: interval
 });
  }

  /**
   * Остановить планировщик
   */
  async stopScheduler() {
    return this.post('/stop');
  }

  /**
   * Синхронизировать
   */
  async syncNow(syncType = 'all') {
    return this.post('/sync-now', { sync_type: syncType });
  }

  /**
   * Проверить подключение
   */
  async testConnection() {
    return this.post('/test-connection');
  }

  /**
   * Обновить конфигурацию
   */
  async updateConfig(config) {
    return this.post('/config', config);
  }
}

```

sync-tab.js

```
/**
 * ============================================================
 * Process Name: Windows Admin - Sync-Tab Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля sync-tab.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/admin/sync-tab.js?v=20261001_v1" type="module"></script>
 *
 * File: sync-tab.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/admin
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * Google Drive Sync Management Tab
 * Управление синхронизацией Google Drive через админ-панель
 */

class SyncTab {
  constructor() {
    this.initialized = false;
    this.messageTimeout = null;
  }

  /**
```

```
   * Инициализация вкладки
   */
  async init() {
    if (this.initialized) return;

    console.log('[SyncTab] Initializing...');

    // Загрузка HTML
    await this.loadHTML();

    // Установка обработчиков
    this.attachEventListeners();

    // Загрузка данных при активной вкладке
    if (window.isTabActive ? window.isTabActive('tab-sync') :
false) {
      await this.updateStatus();
      await this.updateStats();
    }

    // Установка автообновления
    if (window.registerTabPoller) {
      window.registerTabPoller('tab-sync', () => this.updateSt
atus(), 5000, { pollerId: 'sync_status', immediate: true });
      window.registerTabPoller('tab-sync', () => this.updateSt
ats(), 30000, { pollerId: 'sync_stats', immediate: true });
    } else {
      setInterval(() => {
        if (window.isTabActive ? window.isTabActive('tab-sync'
) : true) {
          this.updateStatus();
        }
      }, 5000);
      setInterval(() => {
        if (window.isTabActive ? window.isTabActive('tab-sync'
) : true) {
          this.updateStats();
        }
```

```
      }, 30000);
    }

    this.initialized = true;
    console.log('[SyncTab] Initialized');
  }

  /**
   * Загрузить HTML компонента
   */
  async loadHTML() {
    const container = document.getElementById('tab-sync');
    if (!container) {
      console.warn('[SyncTab] Container not found');
      return;
    }

    const html = `
      <div class="sync-panel">
        <!-- Заголовок -->
        <div class="mb-4">
          <h4 class="d-flex align-items-center gap-2">
            <i class="bi bi-cloud-upload-fill text-primary"></i>
            <span>Google Drive Synchronization</span>
          </h4>
          <p class="text-muted mb-0">Manage data synchronization and storage migration</p>
        </div>

        <!-- Статус -->
        <div class="card border-secondary-subtle mb-3">
          <div class="card-header bg-body-tertiary border-secondary-subtle">
            <h6 class="mb-0">  Synchronization Status</h6>
          </div>
          <div class="card-body">
            <div class="row g-3">
```

```
              <div class="col-md-3">
                <div class="ps-3 border-start border-primary-s
ubtle">
                  <small class="text-muted">Scheduler</small>
                  <div>
                    <span id="scheduler-status" class="badge b
g-danger">Inactive</span>
                  </div>
                </div>
              </div>
              <div class="col-md-3">
                <div class="ps-3 border-start border-primary-s
ubtle">
                  <small class="text-muted">Last Sync</small>
                  <div id="last-sync-time" class="small">Never
</div>
                </div>
              </div>
              <div class="col-md-3">
                <div class="ps-3 border-start border-primary-s
ubtle">
                  <small class="text-muted">Interval</small>
                  <div id="sync-interval" class="small">6 hour
s</div>
                </div>
              </div>
              <div class="col-md-3">
                <div class="ps-3 border-start border-primary-s
ubtle">
                  <small class="text-muted">Drive Folder</smal
l>
                  <div>
                    <a id="drive-url" href="#" target="_blank"
 class="small text-decoration-none">
                      Open
                    </a>
                  </div>
                </div>
```

```
            </div>
          </div>
        </div>
      </div>

      <!-- Статистика -->
      <div class="card border-secondary-subtle mb-3">
        <div class="card-header bg-body-tertiary border-secondary-subtle">
          <h6 class="mb-0">  Storage Statistics</h6>
        </div>
        <div class="card-body">
          <div class="row g-3">
            <div class="col-md-3 text-center">
              <div class="fs-5 fw-bold text-info" id="total-size">-</div>
              <small class="text-muted">Total Size</small>
            </div>
            <div class="col-md-3 text-center">
              <div class="fs-5 fw-bold text-success" id="files-count">-</div>
              <small class="text-muted">Files</small>
            </div>
            <div class="col-md-3 text-center">
              <div class="fs-5 fw-bold text-warning" id="folders-count">-</div>
              <small class="text-muted">Folders</small>
            </div>
            <div class="col-md-3 text-center">
              <div class="fs-5 fw-bold text-danger" id="sync-errors">0</div>
              <small class="text-muted">Errors</small>
            </div>
          </div>
        </div>
      </div>

      <!-- Управление -->
```

```
        <div class="card border-secondary-subtle mb-3">
          <div class="card-header bg-body-tertiary border-secondary-subtle">
            <h6 class="mb-0">  Synchronization Controls</h6>
          </div>
          <div class="card-body">
            <div class="mb-3">
              <label class="form-label">Sync Interval (hours)</label>
              <div class="input-group mb-2">
                <input type="number" id="sync-interval-input" min="1" max="24" value="6" class="form-control">
                <button class="btn btn-outline-secondary" onclick="window.syncTab.updateInterval()">
                  Update
                </button>
              </div>
            </div>

            <div class="d-flex gap-2 flex-wrap">
              <button id="start-scheduler-btn" class="btn btn-success btn-sm" onclick="window.syncTab.startScheduler()">
                <i class="bi bi-play-fill"></i> Start Scheduler
              </button>
              <button id="stop-scheduler-btn" class="btn btn-danger btn-sm" disabled onclick="window.syncTab.stopScheduler()">
                <i class="bi bi-stop-fill"></i> Stop Scheduler
              </button>
              <button class="btn btn-primary btn-sm" onclick="window.syncTab.showSyncOptions()">
                <i class="bi bi-arrow-repeat"></i> Sync Now
              </button>
              <button class="btn btn-info btn-sm" onclick="window.syncTab.testConnection()">
                <i class="bi bi-link-45deg"></i> Test Connection
```

```
              </button>
            </div>
          </div>
        </div>

        <!-- Опции синхронизации -->
        <div id="sync-options-modal" class="modal fade" tabindex="-1">
          <div class="modal-dialog modal-dialog-centered">
            <div class="modal-content bg-dark text-white border-secondary">
              <div class="modal-header border-secondary">
                <h5 class="modal-title">Select Data to Sync</h5>
                <button type="button" class="btn-close btn-close-white" data-bs-dismiss="modal"></button>
              </div>
              <div class="modal-body">
                <div class="form-check mb-2">
                  <input class="form-check-input" type="radio" name="sync-type" value="all" id="sync-all" checked>
                  <label class="form-check-label" for="sync-all">
                    <strong>All Data</strong> - Everything (data, logs, secrets, configs)
                  </label>
                </div>
                <div class="form-check mb-2">
                  <input class="form-check-input" type="radio" name="sync-type" value="data" id="sync-data">
                  <label class="form-check-label" for="sync-data">
                    <strong>Data Only</strong> - Databases and RAG indices
                  </label>
                </div>
                <div class="form-check mb-2">
                  <input class="form-check-input" type="radio"
```

```
 name="sync-type" value="logs" id="sync-logs">
                  <label class="form-check-label" for="sync-lo
gs">
                    <strong>Logs Only</strong> - Application l
ogs
                  </label>
                </div>
                <div class="form-check mb-2">
                  <input class="form-check-input" type="radio"
 name="sync-type" value="secrets" id="sync-secrets">
                  <label class="form-check-label" for="sync-se
crets">
                    <strong>Secrets Only</strong> - Configurat
ion and keys
                  </label>
                </div>
              </div>
              <div class="modal-footer border-secondary">
                <button type="button" class="btn btn-secondary
 btn-sm" data-bs-dismiss="modal">
                  Cancel
                </button>
                <button type="button" class="btn btn-primary b
tn-sm" onclick="window.syncTab.performSync()">
                  Sync Selected
                </button>
              </div>
            </div>
          </div>
        </div>

        <!-- Сообщения -->
        <div id="sync-messages" class="mb-3"></div>

        <!-- Лог активности -->
        <div class="card border-secondary-subtle">
          <div class="card-header bg-body-tertiary border-seco
ndary-subtle">
```

```
            <h6 class="mb-0">  Recent Activity</h6>
          </div>
          <div class="card-body" style="max-height: 300px; overflow-y: auto;">
            <div id="activity-log">
              <p class="text-muted mb-0">No recent activity</p>
            </div>
          </div>
        </div>
      </div>
    `;

    container.innerHTML = html;
  }

  /**
   * Присоединить обработчики событий
   */
  attachEventListeners() {
    // Обработка модального окна
    const modal = new bootstrap.Modal(document.getElementById('sync-options-modal'), {
      backdrop: 'static'
    });

    window.syncTab = this; // Для доступа из HTML
  }

  /**
   * Обновить статус синхронизации
   */
  async updateStatus() {
    try {
      const response = await fetch('/api/admin/sync/status');
      if (!response.ok) throw new Error('Failed to fetch status');

```

```
      const data = await response.json();

      // Обновить статус
      const badge = document.getElementById('scheduler-status'
);
      badge.textContent = data.is_running ? 'Running' : 'Inact
ive';
      badge.className = data.is_running
        ? 'badge bg-success'
        : 'badge bg-danger';

      // Обновить кнопки
      document.getElementById('start-scheduler-btn').disabled
= data.is_running;
      document.getElementById('stop-scheduler-btn').disabled =
 !data.is_running;

      // Обновить время
      document.getElementById('last-sync-time').textContent =
data.last_sync_time || 'Never';
      document.getElementById('sync-interval').textContent = `
${data.sync_interval_hours} hours`;

      // Обновить Google Drive папку
      try {
        const driveResponse = await fetch('/api/admin/sync/dri
ve-info');
        if (driveResponse.ok) {
          const driveData = await driveResponse.json();
          const link = document.getElementById('drive-url');
          link.href = driveData.folder_url;
        }
      } catch (e) {
        console.warn('[SyncTab] Failed to fetch drive info');
      }
    } catch (error) {
      console.error('[SyncTab] Error updating status:', error)
;
```

```
311     }
312   }
313
314   /**
315    * Обновить статистику
316    */
317   async updateStats() {
318     try {
319       const response = await fetch('/api/admin/sync/stats');
320       if (!response.ok) throw new Error('Failed to fetch stats
');
321
322       const data = await response.json();
323
324       document.getElementById('total-size').textContent =
325         `${data.total_size_mb.toFixed(1)} MB`;
326       document.getElementById('files-count').textContent = dat
a.files_count;
327       document.getElementById('folders-count').textContent = d
ata.folders_count;
328       document.getElementById('sync-errors').textContent = dat
a.sync_errors;
329     } catch (error) {
330       console.error('[SyncTab] Error updating stats:', error);
331     }
332   }
333
334   /**
335    * Запустить планировщик
336    */
337   async startScheduler() {
338     try {
339       const interval = parseInt(document.getElementById('sync-
interval-input').value);
340       const response = await fetch('/api/admin/sync/start', {
341         method: 'POST',
342         headers: { 'Content-Type': 'application/json' },
343         body: JSON.stringify({ sync_interval_hours: interval }
```

```
)
      });

      const data = await response.json();
      this.showMessage(data.message || 'Scheduler started', 's
uccess');
      await this.updateStatus();
    } catch (error) {
      this.showMessage('Error: ' + error.message, 'danger');
    }
  }

  /**
   * Остановить планировщик
   */
  async stopScheduler() {
    try {
      const response = await fetch('/api/admin/sync/stop', {
        method: 'POST'
      });

      const data = await response.json();
      this.showMessage(data.message || 'Scheduler stopped', 's
uccess');
      await this.updateStatus();
    } catch (error) {
      this.showMessage('Error: ' + error.message, 'danger');
    }
  }

  /**
   * Проверить подключение
   */
  async testConnection() {
    try {
      const response = await fetch('/api/admin/sync/test-conne
ction', {
        method: 'POST'
```

```
      });

      const data = await response.json();
      const type = data.connected ? 'success' : 'warning';
      this.showMessage(data.message || 'Connection test completed', type);
    } catch (error) {
      this.showMessage('Error: ' + error.message, 'danger');
    }
  }

  /**
   * Показать опции синхронизации
   */
  showSyncOptions() {
    const modal = bootstrap.Modal.getInstance(document.getElementById('sync-options-modal')) ||
      new bootstrap.Modal(document.getElementById('sync-options-modal'));
    modal.show();
  }

  /**
   * Выполнить синхронизацию
   */
  async performSync() {
    const syncType = document.querySelector('input[name="sync-type"]:checked')?.value || 'all';

    try {
      const response = await fetch('/api/admin/sync/sync-now', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ sync_type: syncType })
      });

      const data = await response.json();
```

```
411       this.showMessage(data.message || 'Sync started', 'succes
    s');
412       this.addActivityLog(`Started ${syncType} synchronization
    `);
413
414       // Скрыть модаль
415       const modal = bootstrap.Modal.getInstance(document.getEl
    ementById('sync-options-modal'));
416       if (modal) modal.hide();
417     } catch (error) {
418       this.showMessage('Error: ' + error.message, 'danger');
419     }
420   }
421
422   /**
423    * Обновить интервал
424    */
425   async updateInterval() {
426     const interval = parseInt(document.getElementById('sync-in
    terval-input').value);
427
428     try {
429       const response = await fetch('/api/admin/sync/config', {
430         method: 'POST',
431         headers: { 'Content-Type': 'application/json' },
432         body: JSON.stringify({ sync_interval_hours: interval }
    )
433       });
434
435       this.showMessage(`Interval updated to ${interval} hours`
    , 'success');
436       await this.updateStatus();
437     } catch (error) {
438       this.showMessage('Error: ' + error.message, 'danger');
439     }
440   }
441
442   /**
```

```
   * Показать сообщение
   */
  showMessage(text, type = 'info') {
    const container = document.getElementById('sync-messages')
;
    if (!container) return;

    const alert = document.createElement('div');
    alert.className = `alert alert-${type} alert-dismissible f
ade show`;
    alert.innerHTML = `
      ${text}
      <button type="button" class="btn-close" data-bs-dismiss=
"alert"></button>
    `;

    container.appendChild(alert);

    clearTimeout(this.messageTimeout);
    this.messageTimeout = setTimeout(() => {
      alert.remove();
    }, 5000);
  }

  /**
   * Добавить в лог активности
   */
  addActivityLog(text) {
    const log = document.getElementById('activity-log');
    if (!log) return;

    const item = document.createElement('div');
    item.className = 'small mb-2 pb-2 border-bottom';

    const now = new Date().toLocaleTimeString();
    item.innerHTML = `<span class="text-muted">${now}</span> -
 ${text}`;

```

```
    if (log.textContent.includes('No recent activity')) {
      log.innerHTML = '';
    }

    log.insertBefore(item, log.firstChild);
  }
}

// Инициализация при загрузке документа
document.addEventListener('DOMContentLoaded', async () => {
  window.syncTab = new SyncTab();

  // Инициализировать когда вкладка станет видимой
  const tab = document.getElementById('tab-sync');
  if (tab) {
    // Проверить если вкладка уже активна
    const tabPane = tab.closest('.tab-pane');
    if (tabPane && tabPane.classList.contains('active')) {
      await window.syncTab.init();
    } else if (tabPane) {
      // Инициализировать при активации вкладки
      tabPane.addEventListener('shown.bs.tab', () => {
        window.syncTab.init();
      });
    }
  }
});

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/admin_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/admin_tab/index.html" c
lass="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/admin_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- System Admin Tab -->
<div class="container-fluid py-2">
<div class="alert alert-dark d-flex flex-wrap align-items-cent
er justify-content-between gap-2 mb-3 shadow-sm border-seconda
ry">
<div class="d-flex align-items-center gap-2">
<span class="fs-4">□ </span>
<div>
<h5 class="mb-0 fw-bold">Центральное управление системой</h5>
<p class="mb-0 small text-muted">Мониторинг ядра, системные се
```

```
рвисы, провайдеры моделей и глобальное администрирование</p>
</div>
</div>
<div class="d-flex gap-2 flex-wrap">
<button class="btn btn-warning btn-sm fw-bold d-flex align-ite
ms-center gap-1 shadow-sm" id="btn-switch-to-helpdesk" onclick
="switchToTab('helpdesk')">
<i class="bi bi-headset"></i> Режим Helpdesk
      </button>
<button class="btn btn-outline-info btn-sm" onclick="rescanSto
rageDrives()">
<i class="bi bi-hdd-network"></i> Пересканировать диски
      </button>
<button class="btn btn-outline-primary btn-sm" onclick="actual
izeAiModels()">
<i class="bi bi-cpu"></i> Обновить модели ИИ
      </button>
</div>
</div>
<!-- System Health & Status Cards -->
<div class="row g-3 mb-3">
<div class="col-md-3 col-sm-6">
<div class="card shadow-sm border-secondary-subtle h-100" data
-panel-id="panel-admin-fastapi-dev" id="panel-admin-fastapi-de
v">
<div class="card-body p-3">
<div class="text-muted small text-uppercase fw-semibold mb-1">
Сервер / Режим</div>
<div class="d-flex align-items-center justify-content-between"
>
<h5 class="mb-0 fw-bold text-primary" id="sys-server-mode">Fas
tAPI (DEV)</h5>
<span class="badge bg-success-subtle text-success">Online</spa
n>
</div>
<small class="text-muted" id="sys-server-host">0.0.0.0:8000</s
mall>
</div>
```

```
</div>
</div>
<div class="col-md-3 col-sm-6">
<div class="card shadow-sm border-secondary-subtle h-100" data-panel-id="panel-admin-gemini-agy-foundry" id="panel-admin-gemini-agy-foundry">
<div class="card-body p-3">
<div class="text-muted small text-uppercase fw-semibold mb-1">AI Провайдеры</div>
<div class="d-flex align-items-center justify-content-between">
<h5 class="mb-0 fw-bold text-info" id="sys-ai-providers">Gemini / AGY / Foundry</h5>
<i class="bi bi-robot fs-4 text-info"></i>
</div>
<small class="text-muted">Актуализированный пул</small>
</div>
</div>
</div>
<div class="col-md-3 col-sm-6">
<div class="card shadow-sm border-secondary-subtle h-100" data-panel-id="panel-admin-zagruzka" id="panel-admin-zagruzka">
<div class="card-body p-3">
<div class="text-muted small text-uppercase fw-semibold mb-1">Подключено дисков</div>
<div class="d-flex align-items-center justify-content-between">
<h5 class="mb-0 fw-bold text-warning" id="sys-drives-count">Загрузка...</h5>
<i class="bi bi-device-hdd fs-4 text-warning"></i>
</div>
<small class="text-muted" id="sys-drives-list">CONNECTED_DRIVES</small>
</div>
</div>
</div>
<div class="col-md-3 col-sm-6">
<div class="card shadow-sm border-secondary-subtle h-100" data
```

```
-panel-id="panel-admin-zagruzka" id="panel-admin-zagruzka">
<div class="card-body p-3">
<div class="text-muted small text-uppercase fw-semibold mb-1">Подключаемые плагины</div>
<div class="d-flex align-items-center justify-content-between">
<h5 class="mb-0 fw-bold text-success" id="sys-plugins-count">3агрузка...</h5>
<a class="badge bg-primary text-decoration-none" href="javascript:void(0)" onclick="switchToTab('plugins')">Управление  </a>
</div>
<small class="text-muted" id="sys-plugins-active-count">Реестр модулей</small>
</div>
</div>
</div>
</div>
<!-- Fast Navigation Hub to Core Subsystems -->
<h6 class="fw-bold mb-3 text-uppercase small text-muted"><i class="bi bi-grid-fill me-1"></i> Системные подсистемы</h6>
<div class="row g-3 mb-4">
<div class="col-lg-4 col-md-6">
<div class="card shadow-sm border-secondary-subtle h-100 p-3 hover-shadow" data-panel-id="panel-admin-podklyuchatel-plaginov-hub" id="panel-admin-podklyuchatel-plaginov-hub" onclick="switchToTab('plugins')" style="cursor: pointer;">
<div class="d-flex align-items-center gap-3">
<span class="fs-1">  </span>
<div>
<h6 class="fw-bold mb-1">Подключатель плагинов (Hub)</h6>
<p class="small text-muted mb-0">Управление модулями (web_search, rag, langchain_media и др.), включение/отключение и их органы настройки.</p>
</div>
</div>
</div>
</div>
</div>
<div class="col-lg-4 col-md-6">
```

```
<div class="card shadow-sm border-secondary-subtle h-100 p-3 hover-shadow" data-panel-id="panel-admin-upravlenie-polzovatelyami" id="panel-admin-upravlenie-polzovatelyami" onclick="switchToTab('users')" style="cursor: pointer;">
<div class="d-flex align-items-center gap-3">
<span class="fs-1">□ </span>
<div>
<h6 class="fw-bold mb-1">Управление пользователями</h6>
<p class="small text-muted mb-0">Создание аккаунтов, назначение ролей (admin/user), пароли, активность и права доступа.</p>
</div>
</div>
</div>
</div>
<div class="col-lg-4 col-md-6">
<div class="card shadow-sm border-secondary-subtle h-100 p-3 hover-shadow" data-panel-id="panel-admin-modeli-i-api-klyuchi" id="panel-admin-modeli-i-api-klyuchi" onclick="switchToTab('models')" style="cursor: pointer;">
<div class="d-flex align-items-center gap-3">
<span class="fs-1">□ </span>
<div>
<h6 class="fw-bold mb-1">Модели и API-ключи</h6>
<p class="small text-muted mb-0">Конфигурация Google Gemini, Antigravity, Ollama и AI Foundry, пул ротации ключей.</p>
</div>
</div>
</div>
</div>
<div class="col-lg-4 col-md-6">
<div class="card shadow-sm border-secondary-subtle h-100 p-3 hover-shadow" data-panel-id="panel-admin-sistemnye-instruktsii" id="panel-admin-sistemnye-instruktsii" onclick="switchToTab('instructions')" style="cursor: pointer;">
<div class="d-flex align-items-center gap-3">
<span class="fs-1">□ </span>
<div>
<h6 class="fw-bold mb-1">Системные инструкции</h6>
```

```
<p class="small text-muted mb-0">Редактирование системных промптов чата, стилей диктора и версионирование инструкций.</p>
</div>
</div>
</div>
</div>
<div class="col-lg-4 col-md-6">
<div class="card shadow-sm border-secondary-subtle h-100 p-3 hover-shadow" data-panel-id="panel-admin-sistemnye-logi" id="panel-admin-sistemnye-logi" onclick="switchToTab('logs')" style="cursor: pointer;">
<div class="d-flex align-items-center gap-3">
<span class="fs-1">□ </span>
<div>
<h6 class="fw-bold mb-1">Системные логи</h6>
<p class="small text-muted mb-0">Просмотр журнала событий uvicorn, ошибок FastAPI, истории запросов и аудита.</p>
</div>
</div>
</div>
</div>
<div class="col-lg-4 col-md-6">
<div class="card shadow-sm border-secondary-subtle h-100 p-3 hover-shadow" data-panel-id="panel-admin-agenty-ii" id="panel-admin-agenty-ii" onclick="switchToTab('agents')" style="cursor: pointer;">
<div class="d-flex align-items-center gap-3">
<span class="fs-1">□ </span>
<div>
<h6 class="fw-bold mb-1">Агенты ИИ</h6>
<p class="small text-muted mb-0">Конфигурация автономных ReAct агентов, тулов поиска и параметров инференса.</p>
</div>
</div>
</div>
</div>
<div class="col-lg-4 col-md-6">
<div class="card shadow-sm border-secondary-subtle h-100 p-3 h
```

```
over-shadow" data-panel-id="panel-admin-google-workspace-akkaunty" id="panel-admin-google-workspace-akkaunty" onclick="switchToTab('models'); setTimeout(() =&gt; { const btn = document.getElementById('pill-google-accounts-tab'); if (btn) btn.click(); }, 150);" style="cursor: pointer;">
<div class="d-flex align-items-center gap-3">
<span class="fs-1">□ </span>
<div>
<h6 class="fw-bold mb-1">Google Workspace &amp;amp; Аккаунты</h6>
<p class="small text-muted mb-0">Пул аккаунтов OAuth 2.0 и Service Account для Gmail, Drive, Sheets и Docs.</p>
</div>
</div>
</div>
</div>
<div class="col-lg-4 col-md-6">
<div class="card shadow-sm border-secondary-subtle h-100 p-3 hover-shadow border-warning-subtle" data-panel-id="panel-admin-helpdesk-sluzhba-podderzhki" id="panel-admin-helpdesk-sluzhba-podderzhki" onclick="switchToTab('helpdesk')" style="cursor: pointer;">
<div class="d-flex align-items-center gap-3">
<span class="fs-1">□ </span>
<div>
<h6 class="fw-bold mb-1 text-warning">Helpdesk &amp;amp; Служба поддержки</h6>
<p class="small text-muted mb-0">Управление заявками, живой чат с пользователями, шаблоны быстрых ответов и метрики SLA.</p>
</div>
</div>
</div>
</div>
<div class="col-lg-4 col-md-6">
<div class="card shadow-sm border-secondary-subtle h-100 p-3 hover-shadow border-primary-subtle" data-panel-id="panel-admin-observability-diagnostika" id="panel-admin-observability-diagnostika" onclick="switchToTab('observability')" style="cursor: pointer;">
```

```
<div class="d-flex align-items-center gap-3">
<span class="fs-1">□ </span>
<div>
<h6 class="fw-bold mb-1 text-primary">Observability &amp;amp; Диагностика</h6>
<p class="small text-muted mb-0">Мониторинг ресурсов хоста в реальном времени, дерево оборудования и эвристический AI-аудит.</p>
</div>
</div>
</div>
</div>
<div class="col-lg-4 col-md-6">
<div class="card shadow-sm border-secondary-subtle h-100 p-3 hover-shadow border-info-subtle" data-panel-id="panel-admin-fonovyy-planirovshchik-scheduler" id="panel-admin-fonovyy-planirovshchik-scheduler" onclick="document.getElementById('scheduler-management-card').scrollIntoView({behavior: 'smooth'})" style="cursor: pointer;">
<div class="d-flex align-items-center gap-3">
<span class="fs-1">□ </span>
<div>
<h6 class="fw-bold mb-1 text-info">Фоновый планировщик (Scheduler)</h6>
<p class="small text-muted mb-0">24ч переиндексация RAG коллекций, 5м мониторинг почты Gmail и автообновление.</p>
</div>
</div>
</div>
</div>
</div>
<!-- Background Schedulers Management Section -->
<div class="card shadow-sm border-secondary-subtle mb-3" data-panel-id="panel-admin-fonovye-periodicheskie-zadachi-s" id="scheduler-management-card">
<div class="card-header bg-body-tertiary d-flex flex-wrap justify-content-between align-items-center py-2 gap-2">
<div class="d-flex align-items-center gap-2">
```

```
<span class="fw-bold"><i class="bi bi-clock-history text-info me-2"></i>  Фоновые периодические задачи (Schedulers)</span>
<span class="badge bg-success-subtle text-success" id="sched-main-badge">Активен</span>
</div>
<div class="d-flex gap-2">
<button class="btn btn-outline-info btn-sm" onclick="refreshSchedulersConfig()">
<i class="bi bi-arrow-clockwise me-1"></i> Обновить статус
        </button>
<button class="btn btn-primary btn-sm" onclick="saveSchedulersConfig()">
<i class="bi bi-check2-circle me-1"></i> Сохранить настройки
        </button>
</div>
</div>
<div class="card-body p-3">
<!-- Master toggle -->
<div class="form-check form-switch mb-3">
<input checked="" class="form-check-input" id="sched-master-switch" type="checkbox"/>
<label class="form-check-label fw-bold" for="sched-master-switch">Включить глобальный фоновый планировщик (BackgroundScheduler)</label>
<div class="small text-muted">При отключении все фоновые интервальные циклы будут остановлены.</div>
</div>
<div class="row g-3">
<!-- RAG 24h Re-indexing Task -->
<div class="col-md-6">
<div class="card bg-body-tertiary border-secondary-subtle h-100 p-3" data-panel-id="panel-admin-pereindeksatsiya-rag-rag-re-inde" id="panel-admin-pereindeksatsiya-rag-rag-re-inde">
<div class="d-flex justify-content-between align-items-center mb-2">
<span class="fw-bold text-primary"><i class="bi bi-cpu me-1"></i>  Переиндексация RAG (RAG Re-indexing)</span>
<div class="form-check form-switch m-0">
```

```
<input checked="" class="form-check-input" id="sched-rag-enabled" type="checkbox"/>
</div>
</div>
<p class="small text-muted mb-2">Автоматический фоновый пересчет и обновление коллекций векторных/TF-IDF индексов.</p>
<div class="mb-2">
<label class="form-label small fw-semibold mb-1" for="sched-rag-interval">Интервал запуска (в часах):</label>
<div class="input-group input-group-sm">
<input class="form-control" id="sched-rag-interval" min="0.1" step="0.5" type="number" value="24"/>
<span class="input-group-text">час(ов)</span>
</div>
</div>
<div class="d-flex justify-content-between align-items-center small text-muted mt-auto pt-2 border-top border-secondary-subtle">
<span>Посл. запуск: <strong id="sched-rag-last-run">Никогда</strong></span>
<button class="btn btn-outline-primary btn-sm py-0 px-2" onclick="triggerSchedulerJob('rag')" title="Запустить переиндексацию сейчас">
<i class="bi bi-play-fill"></i> Запустить сейчас
              </button>
</div>
</div>
</div>
<!-- Gmail 5m Check Task -->
<div class="col-md-6">
<div class="card bg-body-tertiary border-secondary-subtle h-100 p-3" data-panel-id="panel-admin-proverka-gmail-email-check" id="panel-admin-proverka-gmail-email-check">
<div class="d-flex justify-content-between align-items-center mb-2">
<span class="fw-bold text-success"><i class="bi bi-envelope-check me-1"></i>  Проверка Gmail (Email Check)</span>
<div class="form-check form-switch m-0">
```

```
<input checked="" class="form-check-input" id="sched-email-enabled" type="checkbox"/>
</div>
</div>
<p class="small text-muted mb-2">Опрос ящиков Google Workspace на наличие новых непрочитанных сообщений.</p>
<div class="mb-2">
<label class="form-label small fw-semibold mb-1" for="sched-email-interval">Интервал опроса (в минутах):</label>
<div class="input-group input-group-sm">
<input class="form-control" id="sched-email-interval" min="1" step="1" type="number" value="5"/>
<span class="input-group-text">мин.</span>
</div>
</div>
<div class="d-flex justify-content-between align-items-center small text-muted mt-auto pt-2 border-top border-secondary-subtle">
<span>Посл. запуск: <strong id="sched-email-last-run">Никогда</strong></span>
<button class="btn btn-outline-success btn-sm py-0 px-2" onclick="triggerSchedulerJob('email')" title="Проверить почту сейчас">
<i class="bi bi-play-fill"></i> Запустить сейчас
              </button>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- Storage & Disks Management Section -->
<div class="card shadow-sm border-secondary-subtle mb-3" data-panel-id="panel-admin-upravlenie-diskami-i-nakopitel" id="panel-admin-upravlenie-diskami-i-nakopitel">
<div class="card-header bg-body-tertiary d-flex flex-wrap justify-content-between align-items-center py-2 gap-2">
<span class="fw-bold"><i class="bi bi-hdd-network text-warning
```

```
 me-2"></i>  Управление дисками и накопителями</span>
<button class="btn btn-outline-info btn-sm" onclick="rescanSto
rageDrives()">
<i class="bi bi-arrow-clockwise me-1"></i> Пересканировать дис
ки
      </button>
</div>
<div class="card-body p-3">
<div class="row g-3" id="sys-drives-detailed-container">
<div class="col-12 text-muted small">Загрузка информации о нак
опителях...</div>
</div>
</div>
</div>
<!-- User Telemetry & Ngrok Tunnel Section -->
<div class="card shadow-sm border-secondary-subtle mb-4" data-
panel-id="panel-admin-aktivnost-polzovateley-i-tun" id="panel-
admin-aktivnost-polzovateley-i-tun">
<div class="card-header bg-body-tertiary d-flex flex-wrap just
ify-content-between align-items-center py-2 gap-2">
<div class="d-flex align-items-center gap-2">
<span class="fw-bold"><i class="bi bi-graph-up-arrow text-info
 me-2"></i>  Активность пользователей и туннель Ngrok</span>
<span class="badge bg-secondary-subtle text-secondary" id="ngr
ok-status-badge">Туннель: проверка...</span>
</div>
<div class="d-flex gap-2">
<button class="btn btn-outline-primary btn-sm" onclick="refres
hTelemetryStats()">
<i class="bi bi-arrow-clockwise me-1"></i> Обновить статистику
        </button>
</div>
</div>
<div class="card-body p-3">
<!-- Tunnel info alert -->
<div class="alert alert-secondary py-2 px-3 small d-flex flex-
wrap align-items-center justify-content-between gap-2 mb-3" id
="ngrok-info-alert">
```

```
<div class="d-flex align-items-center gap-2">
<i class="bi bi-broadcast fs-5 text-primary"></i>
<div>
<span class="fw-semibold">Ngrok шлюз приема статистики:</span>
<span class="text-primary font-monospace ms-1" id="ngrok-publi
c-url-text">Ожидание запуска...</span>
</div>
</div>
<small class="text-muted" id="ngrok-hint-text">Скрипт: <code>.
\launchers\Run-Ngrok.ps1</code></small>
</div>
<!-- Telemetry Stats Cards -->
<div class="row g-3 mb-3">
<div class="col-md-4">
<div class="card bg-body-tertiary border-0 p-3 h-100" data-pan
el-id="panel-admin-0" id="panel-admin-0">
<div class="small text-muted text-uppercase fw-semibold mb-1">
Событий за 30 дней</div>
<h4 class="mb-0 fw-bold text-primary" id="telemetry-total-even
ts">0</h4>
<small class="text-muted" id="telemetry-active-users">0 пользо
вателей</small>
</div>
</div>
<div class="col-md-4">
<div class="card bg-body-tertiary border-0 p-3 h-100" data-pan
el-id="panel-admin-1" id="panel-admin-1">
<div class="small text-muted text-uppercase fw-semibold mb-1">
Популярные вкладки</div>
<div class="small text-muted" id="telemetry-tabs-breakdown">
              Загрузка...
            </div>
</div>
</div>
<div class="col-md-4">
<div class="card bg-body-tertiary border-0 p-3 h-100" data-pan
el-id="panel-admin-2" id="panel-admin-2">
<div class="small text-muted text-uppercase fw-semibold mb-1">
```

```
Топ кликов и действий</div>
<div class="small text-muted" id="telemetry-clicks-breakdown">
              Загрузка...
            </div>
</div>
</div>
</div>
<!-- Recent Telemetry Log Table -->
<h6 class="fw-bold small text-muted text-uppercase mb-2"><i cl
ass="bi bi-clock-history me-1"></i> Последние действия зарегис
трированных пользователей</h6>
<div class="table-responsive">
<table class="table table-sm table-hover align-middle mb-0" st
yle="font-size: 0.85rem;">
<thead class="table-light">
<tr>
<th scope="col">Время</th>
<th scope="col">Пользователь</th>
<th scope="col">Тип</th>
<th scope="col">Вкладка</th>
<th scope="col">Действие / Элемент</th>
<th scope="col">Длительность</th>
</tr>
</thead>
<tbody id="telemetry-events-tbody">
<tr>
<td class="text-center text-muted py-3" colspan="6">Загрузка ж
урнала активности...</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Admin Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/admin_tab/main.js?v=20
261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/admin_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

// System Admin Tab — Central Management Logic

'use strict';

async function initSystemAdminTab() {
  console.log('Инициализация панели системного управления...')
;
  await refreshSystemDashboard();
  await refreshSchedulersConfig();
  await refreshTelemetryStats();
}

```

```
async function refreshSystemDashboard() {
  try {
    // 1. Загрузка дисков
    const drivesData = await window.api.fetch('/api/control/rescan', { method: 'GET' }).catch(() => ({ drives: [], details: [] }));
    const drives = Array.isArray(drivesData.drives) ? drivesData.drives : [];
    const details = Array.isArray(drivesData.details) ? drivesData.details : [];
    const drivesCountEl = document.getElementById('sys-drives-count');
    const drivesListEl = document.getElementById('sys-drives-list');
    if (drivesCountEl) drivesCountEl.textContent = `${drives.length} диск(ов)`;
    if (drivesListEl) drivesListEl.textContent = drives.join(', ') || 'Диски не обнаружены';

    const detailedContainer = document.getElementById('sys-drives-detailed-container');
    if (detailedContainer) {
      if (details.length === 0 && drives.length === 0) {
        detailedContainer.innerHTML = '<div class="col-12 text-muted small">Накопители не обнаружены или доступ ограничен.</div>';
      } else if (details.length > 0) {
        detailedContainer.innerHTML = details.map(d => {
          const percent = d.percent || 0;
          const barClass = percent > 90 ? 'bg-danger' : percent > 75 ? 'bg-warning' : 'bg-primary';
          return `
            <div class="col-lg-4 col-md-6">
              <div class="card bg-body-tertiary border-secondary-subtle h-100 p-3">
                <div class="d-flex justify-content-between align-items-center mb-2">
                  <span class="fw-bold text-primary"><i class=
```

```
"bi bi-hdd me-1"></i>${d.device || d.mountpoint}</span>
                  <span class="badge bg-secondary-subtle text-
body">${d.fstype || 'NTFS'}</span>
                </div>
                <div class="d-flex justify-content-between sma
ll text-muted mb-1">
                  <span>Занято: <strong>${d.used_gb} GB</stron
g></span>
                  <span>Свободно: <strong>${d.free_gb} GB</str
ong></span>
                </div>
                <div class="progress mb-2" style="height: 8px;
">
                  <div class="progress-bar ${barClass}" role="
progressbar" style="width: ${percent}%;" aria-valuenow="${perc
ent}" aria-valuemin="0" aria-valuemax="100"></div>
                </div>
                <div class="d-flex justify-content-between sma
ll text-muted">
                  <span>Всего: ${d.total_gb} GB</span>
                  <span class="fw-semibold">${percent}%</span>
                </div>
              </div>
            </div>
          `;
        }).join('');
      } else {
        detailedContainer.innerHTML = drives.map(drv => `
          <div class="col-lg-3 col-md-4 col-sm-6">
            <div class="card bg-body-tertiary border-secondary
-subtle p-2 text-center">
              <span class="fw-bold text-primary"><i class="bi
bi-hdd me-1"></i>${drv}</span>
            </div>
          </div>
        `).join('');
      }
    }
```

```
83
84      // 2. Загрузка статуса плагинов
85      const pluginsData = await window.api.fetch('/api/admin/plu
    gins').catch(() => ({ plugins: [] }));
86      const plugins = pluginsData.plugins || [];
87      const pluginsCountEl = document.getElementById('sys-plugin
    s-count');
88      const pluginsActiveEl = document.getElementById('sys-plugi
    ns-active-count');
89      const activeCount = plugins.filter(p => p.enabled).length;
90      if (pluginsCountEl) pluginsCountEl.textContent = `${plugin
    s.length} модулей`;
91      if (pluginsActiveEl) pluginsActiveEl.textContent = `${acti
    veCount} активно из ${plugins.length}`;
92    } catch (err) {
93      console.error('Ошибка загрузки системного дашборда:', err)
    ;
94    }
95  }
96
97  async function rescanStorageDrives() {
98    try {
99      if (typeof showNotification === 'function') showNotificati
    on('Пересканирование накопителей ОС...', 'info');
100     const result = await window.api.fetch('/api/control/rescan
    ', { method: 'GET' });
101     const drivesList = Array.isArray(result.drives) ? result.d
    rives.join(', ') : 'OK';
102     if (typeof showNotification === 'function') showNotificati
    on(`Диски обновлены: ${drivesList}`, 'success');
103     await refreshSystemDashboard();
104   } catch (e) {
105     if (typeof showNotification === 'function') showNotificati
    on(`Ошибка: ${e.message}`, 'danger');
106   }
107 }
108
109 async function actualizeAiModels() {
```

```
  try {
    if (typeof showNotification === 'function') showNotificati
on('Актуализация пула моделей ИИ...', 'info');
    await window.api.fetch('/api/keys/actualize-all', { method
: 'POST' }).catch(() => {});
    if (typeof showNotification === 'function') showNotificati
on('Модели успешно синхронизированы', 'success');
    await refreshSystemDashboard();
  } catch (e) {
    if (typeof showNotification === 'function') showNotificati
on(`Ошибка: ${e.message}`, 'danger');
  }
}

async function refreshTelemetryStats() {
  try {
    const res = await fetch('/api/telemetry/stats?days=30');
    if (!res.ok) return;
    const stats = await res.json();

    // 1. Ngrok Tunnel Status
    const tunnel = stats.tunnel_status || {};
    const badgeEl = document.getElementById('ngrok-status-badg
e');
    const urlTextEl = document.getElementById('ngrok-public-ur
l-text');
    const alertEl = document.getElementById('ngrok-info-alert'
);

    if (tunnel.is_active && tunnel.public_url) {
      if (badgeEl) {
        badgeEl.className = 'badge bg-success-subtle text-succ
ess';
        badgeEl.textContent = 'Туннель: Онлайн';
      }
      if (urlTextEl) {
        urlTextEl.innerHTML = `<a href="${tunnel.public_url}"
target="_blank" class="text-success text-decoration-none fw-bo
```

```
ld">${tunnel.public_url}</a>`;
      }
      if (alertEl) alertEl.className = 'alert alert-success py-2 px-3 small d-flex flex-wrap align-items-center justify-content-between gap-2 mb-3';
    } else {
      if (badgeEl) {
        badgeEl.className = 'badge bg-secondary-subtle text-secondary';
        badgeEl.textContent = 'Туннель: Остановлен';
      }
      if (urlTextEl) {
        urlTextEl.textContent = 'Не запущен (запустите .\\launchers\\Run-Ngrok.ps1)';
      }
      if (alertEl) alertEl.className = 'alert alert-secondary py-2 px-3 small d-flex flex-wrap align-items-center justify-content-between gap-2 mb-3';
    }

    // 2. Event & User metrics
    const totalEl = document.getElementById('telemetry-total-events');
    const usersEl = document.getElementById('telemetry-active-users');
    if (totalEl) totalEl.textContent = (stats.total_events || 0).toLocaleString();
    if (usersEl) usersEl.textContent = `${stats.active_users_count || 0} пользователей`;

    // 3. Tab views
    const tabsEl = document.getElementById('telemetry-tabs-breakdown');
    if (tabsEl) {
      const tabEntries = Object.entries(stats.tab_views || {});
      if (tabEntries.length === 0) {
        tabsEl.innerHTML = '<span class="text-muted">Нет данны
```

```
x</span>';
      } else {
        tabsEl.innerHTML = tabEntries.slice(0, 4).map(([tab, info]) => `
          <div class="d-flex justify-content-between align-items-center mb-1">
            <span class="badge bg-secondary-subtle text-body">${tab}</span>
            <span class="fw-semibold">${info.count} раз</span>
          </div>
        `).join('');
      }
    }

    // 4. Click actions
    const clicksEl = document.getElementById('telemetry-clicks-breakdown');
    if (clicksEl) {
      const clickEntries = Object.entries(stats.top_clicks || {});
      if (clickEntries.length === 0) {
        clicksEl.innerHTML = '<span class="text-muted">Нет кликов</span>';
      } else {
        clicksEl.innerHTML = clickEntries.slice(0, 4).map(([target, info]) => `
          <div class="d-flex justify-content-between align-items-center mb-1 text-truncate" title="${target}">
            <span class="text-truncate me-1" style="max-width: 140px;">${info.action || target}</span>
            <span class="badge bg-primary-subtle text-primary">${info.count}</span>
          </div>
        `).join('');
      }
    }

    // 5. Recent events table
```

```
    const tbody = document.getElementById('telemetry-events-tbody');
    if (tbody) {
      const recent = stats.recent_events || [];
      if (recent.length === 0) {
        tbody.innerHTML = '<tr><td colspan="6" class="text-center text-muted py-3">Пока нет зарегистрированных событий активности.</td></tr>';
      } else {
        tbody.innerHTML = recent.map(ev => {
          const dt = ev.timestamp ? new Date(ev.timestamp).toLocaleTimeString() : '-';
          const userName = ev.name || ev.email || `ID ${ev.user_id}`;
          const typeBadge = ev.event_type === 'tab_view' ? 'bg-info-subtle text-info' : ev.event_type === 'click' ? 'bg-primary-subtle text-primary' : 'bg-secondary-subtle text-secondary';
          const durStr = ev.duration_ms > 0 ? `${(ev.duration_ms / 1000).toFixed(1)}s` : '-';
          return `
            <tr>
              <td class="text-muted font-monospace small">${dt}</td>
              <td class="fw-semibold">${userName}</td>
              <td><span class="badge ${typeBadge}">${ev.event_type || 'action'}</span></td>
              <td><span class="badge bg-light text-dark border">${ev.tab_name || '-'}</span></td>
              <td class="text-truncate" style="max-width: 250px;" title="${ev.target_element || ''}">${ev.action || ev.target_element || '-'}</td>
              <td class="text-muted">${durStr}</td>
            </tr>
          `;
        }).join('');
      }
    }
```

```
  } catch (e) {
    console.debug('Failed to load telemetry stats:', e);
  }
}

function switchToTab(tabName) {
  if (typeof window.switchTab === 'function') {
    window.switchTab(tabName);
    return;
  }
  const cleanName = tabName.replace(/^tab-/, '');
  const triggerEl = document.querySelector(`[data-bs-target="#tab-${cleanName}"], [data-tab="${cleanName}"]`);
  if (triggerEl && typeof bootstrap !== 'undefined' && bootstrap.Tab) {
    const tab = new bootstrap.Tab(triggerEl);
    tab.show();
  }
}

// Schedulers Management Logic
async function refreshSchedulersConfig() {
  try {
    const res = await window.api.fetch('/api/admin/schedulers');
    if (!res || !res.config) return;

    const cfg = res.config;
    const status = res.status || {};

    const masterEl = document.getElementById('sched-master-switch');
    const ragEnabledEl = document.getElementById('sched-rag-enabled');
    const ragIntervalEl = document.getElementById('sched-rag-interval');
    const emailEnabledEl = document.getElementById('sched-email-enabled');
```

```
    const emailIntervalEl = document.getElementById('sched-email-interval');
    const badgeEl = document.getElementById('sched-main-badge');
    const ragLastRunEl = document.getElementById('sched-rag-last-run');
    const emailLastRunEl = document.getElementById('sched-email-last-run');

    if (masterEl) masterEl.checked = cfg.enabled !== false;
    if (ragEnabledEl) ragEnabledEl.checked = cfg.rag_reindex?.enabled !== false;
    if (ragIntervalEl) ragIntervalEl.value = cfg.rag_reindex?.interval_hours || 24;

    if (emailEnabledEl) emailEnabledEl.checked = cfg.email_check?.enabled !== false;
    if (emailIntervalEl) emailIntervalEl.value = cfg.email_check?.interval_minutes || 5;

    if (badgeEl) {
      if (status.running && cfg.enabled) {
        badgeEl.className = 'badge bg-success-subtle text-success';
        badgeEl.textContent = 'Активен (Работает)';
      } else {
        badgeEl.className = 'badge bg-secondary-subtle text-secondary';
        badgeEl.textContent = 'Отключен';
      }
    }

    if (ragLastRunEl) {
      ragLastRunEl.textContent = status.last_rag_run ? new Date(status.last_rag_run).toLocaleString() : 'Никогда';
    }
    if (emailLastRunEl) {
      emailLastRunEl.textContent = status.last_email_run ? new
```

```
 Date(status.last_email_run).toLocaleString() : 'Никогда';
    }
  } catch (err) {
    console.error('Ошибка загрузки настроек планировщика:', er
r);
  }
}

async function saveSchedulersConfig() {
  try {
    const masterEl = document.getElementById('sched-master-swi
tch');
    const ragEnabledEl = document.getElementById('sched-rag-en
abled');
    const ragIntervalEl = document.getElementById('sched-rag-i
nterval');
    const emailEnabledEl = document.getElementById('sched-emai
l-enabled');
    const emailIntervalEl = document.getElementById('sched-ema
il-interval');

    const payload = {
      enabled: masterEl ? masterEl.checked : true,
      rag_reindex: {
        enabled: ragEnabledEl ? ragEnabledEl.checked : true,
        interval_hours: parseFloat(ragIntervalEl ? ragInterval
El.value : 24) || 24
      },
      email_check: {
        enabled: emailEnabledEl ? emailEnabledEl.checked : tru
e,
        interval_minutes: parseFloat(emailIntervalEl ? emailIn
tervalEl.value : 5) || 5
      }
    };

    if (typeof showNotification === 'function') showNotificati
on('Сохранение конфигурации планировщика...', 'info');
```

```
300
301     const res = await window.api.fetch('/api/admin/schedulers'
    , {
302       method: 'POST',
303       headers: { 'Content-Type': 'application/json' },
304       body: JSON.stringify(payload)
305     });
306
307     if (typeof showNotification === 'function') {
308       showNotification('Настройки планировщика успешно сохране
    ны и применены!', 'success');
309     }
310     await refreshSchedulersConfig();
311   } catch (err) {
312     if (typeof showNotification === 'function') {
313       showNotification(`Ошибка сохранения: ${err.message}`, 'd
    anger');
314     }
315   }
316 }
317
318 async function triggerSchedulerJob(jobName) {
319   try {
320     if (typeof showNotification === 'function') {
321       showNotification(`Запуск фоновой задачи '${jobName}'...`
    , 'info');
322     }
323
324     const res = await window.api.fetch(`/api/admin/schedulers/
    trigger/${jobName}`, {
325       method: 'POST'
326     });
327
328     if (jobName === 'rag') {
329       const count = res.result?.updated_count ?? 0;
330       if (typeof showNotification === 'function') {
331         showNotification(`Переиндексация RAG завершена! Обновл
    ено коллекций: ${count}`, 'success');
```

```
      }
    } else if (jobName === 'email') {
      const count = res.count ?? 0;
      if (typeof showNotification === 'function') {
        showNotification(`Проверка почты завершена! Найдено не прочитанных писем: ${count}`, 'success');
      }
    }

    await refreshSchedulersConfig();
  } catch (err) {
    if (typeof showNotification === 'function') {
      showNotification(`Ошибка запуска задачи: ${err.message}`, 'danger');
    }
  }
}

window.initAdminTab = initSystemAdminTab;
window.rescanStorageDrives = rescanStorageDrives;
window.actualizeAiModels = actualizeAiModels;
window.switchToTab = switchToTab;
window.refreshTelemetryStats = refreshTelemetryStats;
window.refreshSchedulersConfig = refreshSchedulersConfig;
window.saveSchedulersConfig = saveSchedulersConfig;
window.triggerSchedulerJob = triggerSchedulerJob;


```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/agents_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/agents_tab/index.html"
class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/agents_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- AI Agents Management & Builder Tab -->
<style>
  /* High contrast & theme-aware styling for AI Agents Tab */
  .agent-card {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: 12px;
    transition: transform 0.15s ease, border-color 0.15s ease,
 box-shadow 0.15s ease;
  }
```

```
.agent-card:hover {
  border-color: var(--nav-active) !important;
  box-shadow: 0 6px 20px rgba(0, 0, 0, 0.1);
}
.agent-card .card-header {
  background: var(--surface-2) !important;
  border-bottom: 1px solid var(--border-color) !important;
  color: var(--text-color) !important;
  border-top-left-radius: 11px;
  border-top-right-radius: 11px;
}
.agent-card .agent-desc {
  color: var(--text-muted) !important;
}
.agent-spec-box {
  background: var(--surface-2) !important;
  border: 1px solid var(--border-color) !important;
  border-radius: 8px;
  padding: 0.6rem 0.8rem;
}
.agent-spec-box .spec-label {
  color: var(--text-muted) !important;
  font-weight: 600;
  font-size: 0.82rem;
}
.agent-spec-box .spec-value-badge {
  background-color: var(--nav-active) !important;
  color: #ffffff !important;
  font-weight: 600;
  font-size: 0.78rem;
}
.agent-spec-box .spec-tech-val {
  color: var(--nav-active) !important;
  font-weight: 700;
  font-size: 0.85rem;
}
.tool-badge {
  background: var(--surface-2) !important;
```

```
    border: 1px solid var(--border-color) !important;
    color: var(--text-color) !important;
    font-size: 0.75rem;
    font-weight: 500;
    border-radius: 6px;
    padding: 0.25rem 0.5rem;
    display: inline-flex;
    align-items: center;
    gap: 0.25rem;
  }
  .agent-card .btn-test-sandbox {
    color: #d97706 !important;
    border: 1.5px solid #d97706 !important;
    font-weight: 600;
    font-size: 0.82rem;
    background: transparent;
  }
  .agent-card .btn-test-sandbox:hover {
    background: #d97706 !important;
    color: #ffffff !important;
  }
  .agent-card .btn-edit-agent {
    color: var(--nav-active) !important;
    border: 1.5px solid var(--nav-active) !important;
    font-weight: 600;
    background: transparent;
  }
  .agent-card .btn-edit-agent:hover {
    background: var(--nav-active) !important;
    color: #ffffff !important;
  }
  .agent-card .btn-delete-agent {
    color: #dc2626 !important;
    border: 1.5px solid #dc2626 !important;
    font-weight: 600;
    background: transparent;
  }
  .agent-card .btn-delete-agent:hover {
```

```
    background: #dc2626 !important;
    color: #ffffff !important;
  }
  #agents-filter-group .btn {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    color: var(--text-color);
    font-weight: 600;
    font-size: 0.82rem;
  }
  #agents-filter-group .btn:hover {
    background: var(--surface-1);
    color: var(--nav-active);
    border-color: var(--nav-active);
  }
  #agents-filter-group .btn.active {
    background: var(--nav-active) !important;
    border-color: var(--nav-active) !important;
    color: #ffffff !important;
    box-shadow: 0 0 8px rgba(37, 99, 235, 0.4);
  }
  #btn-refresh-agents {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    color: var(--text-color);
    font-weight: 600;
    font-size: 0.82rem;
  }
  #btn-refresh-agents:hover {
    background: var(--surface-1);
    color: var(--nav-active);
    border-color: var(--nav-active);
  }
</style>
<div class="container-fluid py-3">
<!-- Header Alert & Quick Actions -->
<div class="card shadow-sm border-secondary-subtle p-3 mb-3 d-
flex justify-content-between align-items-center flex-wrap gap-
```

```
3" data-panel-id="panel-agents-upravlenie-i-sozdanie-ii-agent"
 id="panel-agents-upravlenie-i-sozdanie-ii-agent">
<div>
<h5 class="mb-1 fw-bold d-flex align-items-center gap-2">
<span class="fs-4">□ </span> Управление и создание ИИ-Агентов
      </h5>
<p class="mb-0 small text-muted">
        Конфигурирование специализированных агентов, назначени
е моделей из пула, подключение инструментов и генерация через
AI.
      </p>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<button class="btn btn-sm btn-outline-info rounded-pill px-3 s
hadow-sm fw-semibold" id="btn-open-ai-builder" style="border-w
idth: 1.5px;">
<i class="bi bi-magic me-1"></i> □  Создать с помощью AI
      </button>
<button class="btn btn-sm btn-primary rounded-pill px-3 shadow
-sm fw-semibold" id="btn-open-create-agent">
<i class="bi bi-plus-lg me-1"></i> Добавить вручную
      </button>
</div>
</div>
<!-- Collapsible Architecture & SVG Scheme Block -->
<div class="card mb-4 shadow-sm" data-panel-id="panel-agents-i
nteraktivnaya-spravka-arkhitekt" id="panel-agents-interaktivna
ya-spravka-arkhitekt" style="background-color: var(--surface-1
) !important; border: 1px solid var(--border-color) !important
; border-radius: 10px;">
<div class="card-body p-3">
<div class="d-flex align-items-center justify-content-between"
>
<div class="d-flex align-items-center gap-2">
<i class="bi bi-diagram-3-fill text-info fs-5"></i>
<span class="fw-bold text-body">Интерактивная справка: архитек
тура работы и создания агентов</span>
</div>
```

```
<button aria-controls="agentsArchCollapse" aria-expanded="false" class="btn btn-sm btn-link text-decoration-none p-0 fw-semibold" data-bs-target="#agentsArchCollapse" data-bs-toggle="collapse" type="button">
<i class="bi bi-chevron-down"></i> Развернуть / Скрыть схему
        </button>
</div>
<div class="collapse" id="agentsArchCollapse">
<div class="mt-3 pt-3" style="border-top: 1px solid var(--border-color);">
<div class="text-center p-2 rounded mb-3" style="background: var(--surface-2); border: 1px solid var(--border-color);">
<img alt="Архитектура агентов Mediteka" class="img-fluid rounded" src="/html/assets/agents_architecture.svg?v=20260817" style="max-height: 550px; width: 100%; object-fit: contain;"/>
</div>
<div class="row g-3 small">
<div class="col-md-3">
<div class="p-3 rounded h-100" style="background: var(--surface-2); border: 1px solid var(--border-color); color: var(--text-color); line-height: 1.55;">
<strong class="d-block mb-1 text-info" style="font-size: 0.88rem;">1. Пул моделей (Model Pool)</strong>
                Каждый агент может использовать независимую модель: Google Gemini (с ротацией ключей), Antigravity AGY, Microsoft Foundry или Ollama.
              </div>
</div>
<div class="col-md-3">
<div class="p-3 rounded h-100" style="background: var(--surface-2); border: 1px solid var(--border-color); color: var(--text-color); line-height: 1.55;">
<strong class="d-block mb-1 text-success" style="font-size: 0.88rem;">2. Каталог инструментов (Tools)</strong>
                Набор готовых модулей: интернет-поиск (MCP / Google Grounding), RAG база знаний, Python Code Execution, чтение файлов и синтез речи TTS.
              </div>
```

```
</div>
<div class="col-md-3">
<div class="p-3 rounded h-100" style="background: var(--surface-2); border: 1px solid var(--border-color); color: var(--text-color); line-height: 1.55;">
<strong class="d-block mb-1 text-warning" style="font-size: 0.88rem;">3. AI-Архитектор промптов</strong>
                Модель из пула анализирует вашу цель и генерирует структурированную роль, правила ReAct и список инструментов.
              </div>
</div>
<div class="col-md-3">
<div class="p-3 rounded h-100" style="background: var(--surface-2); border: 1px solid var(--border-color); color: var(--text-color); line-height: 1.55;">
<strong class="d-block mb-1 text-danger" style="font-size: 0.88rem;">4. Песочница (Sandbox)</strong>
                Тестирование поведения агента, отслеживание цепочки рассуждений (Thought &amp;amp; Action) и замер задержек ответа до боевого использования.
              </div>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- Filters and Status Toolbar -->
<div class="d-flex justify-content-between align-items-center mb-3 flex-wrap gap-2">
<div class="btn-group btn-group-sm" id="agents-filter-group" role="group">
<button class="btn active" data-filter="all" type="button">Все (<span id="count-all">0</span>)</button>
<button class="btn" data-filter="active" type="button">Активные (<span id="count-active">0</span>)</button>
<button class="btn" data-filter="system" type="button">Системн
```

```
ые (<span id="count-system">0</span>)</button>
<button class="btn" data-filter="custom" type="button">Пользов
ательские (<span id="count-custom">0</span>)</button>
</div>
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm rounded-pill px-3" id="btn-refresh-a
gents" title="Обновить список">
<i class="bi bi-arrow-clockwise me-1"></i> Обновить
      </button>
</div>
</div>
<!-- Agents Grid -->
<div class="row g-3" id="agents-grid">
<div class="col-12 text-center py-5 text-muted">
<div class="spinner-border text-primary mb-2" role="status"></
div>
<div class="fw-semibold">Загрузка агентов и пула моделей...</d
iv>
</div>
</div>
</div>
<!-- ============================================================
================ -->
<!-- MODAL: AI Agent Builder (Генерация через LLM)
                  -->
<!-- ============================================================
================ -->
<div aria-hidden="true" aria-labelledby="modalAiBuilderLabel"
class="modal fade" id="modal-ai-builder" tabindex="-1">
<div class="modal-dialog modal-lg modal-dialog-centered">
<div class="modal-content">
<div class="modal-header border-secondary">
<h5 class="modal-title fw-bold text-info" id="modalAiBuilderLa
bel">
<i class="bi bi-magic me-2"></i>AI Agent Builder — Конструктор
 агентов
        </h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
```

```
modal" type="button"></button>
</div>
<div class="modal-body p-4">
<p class="small mb-3 text-muted" style="line-height: 1.5;">
          Опишите задачу агента своими словами. Выбранная модель из пула проанализирует цель, подберёт необходимые инструменты и сформирует профессиональный системный промпт.
        </p>
<div class="row g-3 mb-3">
<div class="col-md-6">
<label class="form-label small fw-semibold">Провайдер модели-архитектора:</label>
<select class="form-select form-select-sm" id="ai-builder-provider">
<option value="gemini">Google Gemini</option>
<option value="gemini_cli">Google Gemini CLI</option>
<option value="agy">Google Antigravity (AGY)</option>
<option value="foundry">Microsoft Foundry (Local)</option>
<option value="ollama">Ollama (Local)</option>
</select>
</div>
<div class="col-md-6">
<label class="form-label small fw-semibold">Модель из пула:</label>
<select class="form-select form-select-sm" id="ai-builder-model">
<!-- Заполняется динамически -->
</select>
</div>
</div>
<div class="mb-3">
<label class="form-label small fw-semibold">Опишите задачу агента:</label>
<textarea class="form-control" id="ai-builder-prompt" placeholder="Например: Создай агента для автономного поиска субтитров к фильмам, проверки их таймингов и сопоставления с медиатекой..." rows="4"></textarea>
</div>
```

```
<div class="d-flex justify-content-end mb-3">
<button class="btn btn-primary rounded-pill px-4 fw-bold" id="
btn-run-ai-generate">
<i class="bi bi-stars me-1"></i> Сгенерировать спецификацию
          </button>
</div>
<!-- Result Preview Section -->
<div class="d-none pt-3" id="ai-builder-result" style="border-
top: 1px solid var(--border-color);">
<h6 class="fw-bold mb-2 text-success"><i class="bi bi-check2-c
ircle me-1"></i> Спецификация сформирована:</h6>
<div class="p-3 rounded mb-3 small font-monospace" id="ai-buil
der-preview-box" style="background: var(--surface-2); border:
1px solid var(--border-color); color: var(--text-color);">
<!-- Превью JSON / данных -->
</div>
<div class="d-flex justify-content-end gap-2">
<button class="btn btn-outline-secondary btn-sm rounded-pill p
x-3" data-bs-dismiss="modal">Отмена</button>
<button class="btn btn-success btn-sm rounded-pill px-3 fw-sem
ibold" id="btn-apply-ai-generated">
<i class="bi bi-pencil-square me-1"></i> Открыть в редакторе а
гента
            </button>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- ============================================================
================ -->
<!-- MODAL: Agent Editor (Создание и Редактирование)
                  -->
<!-- ============================================================
================ -->
<div aria-hidden="true" aria-labelledby="modalAgentEditorLabel
" class="modal fade" id="modal-agent-editor" tabindex="-1">
```

```
<div class="modal-dialog modal-xl modal-dialog-centered modal-dialog-scrollable">
<div class="modal-content">
<div class="modal-header border-secondary">
<h5 class="modal-title fw-bold text-body" id="modalAgentEditorLabel">
<i class="bi bi-robot me-2 text-info"></i>Настройка агента
        </h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body p-4">
<form id="agent-form">
<input id="agent-edit-mode" type="hidden" value="create"/>
<div class="row g-3 mb-3">
<div class="col-md-4">
<label class="form-label small fw-semibold">Уникальный ID агента:</label>
<input class="form-control form-control-sm font-monospace" id="agent-id" placeholder="my_custom_agent" required="" type="text"/>
</div>
<div class="col-md-5">
<label class="form-label small fw-semibold">Отображаемое имя:</label>
<input class="form-control form-control-sm" id="agent-name" placeholder="Название агента" required="" type="text"/>
</div>
<div class="col-md-3">
<label class="form-label small fw-semibold">Статус активности:</label>
<div class="form-check form-switch mt-1">
<input checked="" class="form-check-input" id="agent-enabled" type="checkbox"/>
<label class="form-check-label small fw-medium" for="agent-enabled">Агент активен</label>
</div>
</div>
```

```
</div>
<div class="mb-3">
<label class="form-label small fw-semibold">Краткое описание р
оли:</label>
<input class="form-control form-control-sm" id="agent-desc" pl
aceholder="Какую задачу решает данный агент" type="text"/>
</div>
<!-- Model & Provider Selection from Pool -->
<div class="card mb-3" data-panel-id="panel-agents-privyazka-k
-modeli-iz-pula" id="panel-agents-privyazka-k-modeli-iz-pula"
style="background: var(--surface-1); border: 1px solid var(--b
order-color); border-radius: 8px;">
<div class="card-header py-2" style="background: var(--surface
-2); border-bottom: 1px solid var(--border-color);">
<span class="fw-bold small text-info"><i class="bi bi-cpu me-1
"></i> Привязка к модели из пула</span>
</div>
<div class="card-body p-3">
<div class="row g-3">
<div class="col-md-6">
<label class="form-label small fw-semibold">Провайдер ИИ:</lab
el>
<select class="form-select form-select-sm" id="agent-provider"
>
<option value="gemini">Google Gemini</option>
<option value="gemini_cli">Google Gemini CLI</option>
<option value="agy">Google Antigravity (AGY)</option>
<option value="foundry">Microsoft Foundry</option>
<option value="ollama">Ollama (Local)</option>
</select>
</div>
<div class="col-md-6">
<label class="form-label small fw-semibold">Модель:</label>
<select class="form-select form-select-sm" id="agent-model">
<!-- Заполняется динамически -->
</select>
</div>
</div>
```

```
<div class="row g-3 mt-1">
<div class="col-md-4">
<label class="form-label small fw-semibold d-flex justify-content-between">
<span>Температура:</span> <span class="fw-bold text-info" id="agent-temp-val">0.3</span>
</label>
<input class="form-range" id="agent-temperature" max="1" min="0" step="0.05" type="range" value="0.3"/>
</div>
<div class="col-md-4">
<label class="form-label small fw-semibold">Макс. шагов ReAct (Max Steps):</label>
<input class="form-control form-control-sm" id="agent-max-steps" max="50" min="1" type="number" value="15"/>
</div>
<div class="col-md-4">
<label class="form-label small fw-semibold">Таймаут (сек):</label>
<input class="form-control form-control-sm" id="agent-timeout" max="300" min="5" type="number" value="60"/>
</div>
</div>
</div>
</div>
<!-- Tools Selection Matrix -->
<div class="card mb-3" data-panel-id="panel-agents-podklyuchennye-instrumenty-capab" id="panel-agents-podklyuchennye-instrumenty-capab" style="background: var(--surface-1); border: 1px solid var(--border-color); border-radius: 8px;">
<div class="card-header py-2 d-flex justify-content-between align-items-center" style="background: var(--surface-2); border-bottom: 1px solid var(--border-color);">
<span class="fw-bold small text-success"><i class="bi bi-tools me-1"></i> Подключенные инструменты (Capabilities)</span>
<span class="badge bg-secondary" id="agent-tools-count">0 выбрано</span>
</div>
```

```
<div class="card-body p-3">
<div class="row g-2" id="agent-tools-matrix">
<!-- Чекбоксы инструментов загружаются динамически -->
</div>
</div>
</div>
<!-- System Instruction Editor -->
<div class="mb-3">
<div class="d-flex justify-content-between align-items-center mb-1">
<label class="form-label small fw-semibold mb-0">Системная инструкция (System Prompt):</label>
<div class="btn-group btn-group-sm">
<button class="btn btn-outline-secondary btn-sm py-0" id="btn-insert-react-template" type="button">Шаблон ReAct</button>
<button class="btn btn-outline-secondary btn-sm py-0" id="btn-insert-json-template" type="button">Шаблон JSON-ответа</button>
</div>
</div>
<textarea class="form-control font-monospace" id="agent-system-prompt" placeholder="Опишите поведение агента, правила вызова инструментов и формат ответа..." required="" rows="7" spellcheck="false" style="font-size: 0.85rem; line-height: 1.45;"></textarea>
</div>
</form>
</div>
<div class="modal-footer border-secondary">
<button class="btn btn-outline-secondary rounded-pill px-3" data-bs-dismiss="modal" type="button">Отмена</button>
<button class="btn btn-primary rounded-pill px-4 fw-bold" id="btn-save-agent" type="button">
<i class="bi bi-check-circle me-1"></i> Сохранить агента
        </button>
</div>
</div>
</div>
```

```
</div>
<!-- ============================================================
================ -->
<!-- MODAL: Agent Sandbox (Песочница тестирования)
                  -->
<!-- ============================================================
================ -->
<div aria-hidden="true" aria-labelledby="modalAgentSandboxLabe
l" class="modal fade" id="modal-agent-sandbox" tabindex="-1">
<div class="modal-dialog modal-lg modal-dialog-centered modal-
dialog-scrollable">
<div class="modal-content">
<div class="modal-header border-secondary">
<h5 class="modal-title fw-bold text-warning" id="modalAgentSan
dboxLabel">
<i class="bi bi-play-circle me-2"></i> Песочница тестировани
я агента
        </h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-4">
<!-- Active Agent Summary in Sandbox -->
<div class="d-flex align-items-center justify-content-between
p-2 px-3 rounded mb-3" style="background: var(--surface-2); bo
rder: 1px solid var(--border-color);">
<div>
<strong class="text-body fs-6" id="sandbox-agent-name">—</stro
ng>
<span class="badge bg-primary ms-2" id="sandbox-agent-model">—
</span>
</div>
<span class="small fw-medium text-muted" id="sandbox-agent-too
ls-summary">Инструментов: 0</span>
</div>
<!-- Chat / Trace Container -->
<div class="rounded p-3 mb-3" id="sandbox-chat-window" style="
background: var(--surface-2); border: 1px solid var(--border-c
```

```
olor); min-height: 240px; max-height: 380px; overflow-y: auto;
">
<div class="text-center small py-4 text-muted" id="sandbox-pla
ceholder">
<i class="bi bi-chat-dots fs-3 d-block mb-1"></i>
            Введите тестовый запрос ниже для проверки логики а
гента и цепочки вызовов инструментов.
          </div>
<div id="sandbox-messages-list"></div>
</div>
<!-- Input Box -->
<div class="input-group">
<input class="form-control" id="sandbox-input-msg" placeholder
="Например: Найди фильм Интерстеллар и проверь торренты..." ty
pe="text"/>
<button class="btn btn-warning fw-bold px-4" id="btn-sandbox-s
end">
<i class="bi bi-send-fill me-1"></i> Тест
          </button>
</div>
</div>
<div class="modal-footer border-secondary">
<button class="btn btn-outline-secondary rounded-pill px-3" da
ta-bs-dismiss="modal" type="button">Закрыть песочницу</button>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Agents Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/agents_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/agents_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

'use strict';

/**
 * Global reactive state for the AI Agents tab.
 */
const _state = {
  agents: [],
  tools: [],
  providers: {},
  currentFilter: 'all',
  activeSandboxAgent: null,
  lastGeneratedSpec: null,
```

```
  isEventsBound: false
};

/**
 * Initializes the AI Agents management tab.
 * Called on tab switch from the main admin interface.
 *
 * @returns {Promise<void>}
 */
async function initAgentsTab() {
  _bindStaticEvents();
  await _loadAllData();
}

window.initAgentsTab = initAgentsTab;

/**
 * Executes an HTTP JSON request using window.api or standard fetch.
 *
 * @param {string} endpoint - Target API endpoint path.
 * @param {RequestInit} [options={}] - Optional fetch configuration options.
 * @returns {Promise<any>} Parsed JSON response.
 * @throws {Error} Thrown if response status is not OK.
 */
async function _fetchJson(endpoint, options = {}) {
  if (window.api && typeof window.api.fetch === 'function') {
    return await window.api.fetch(endpoint, options);
  }
  const response = await fetch(endpoint, options);
  if (!response.ok) {
    throw new Error(`HTTP ${response.status}: ${response.statusText}`);
  }
  return await response.json();
}

```

```
/**
 * Loads agents, capabilities catalog, and model providers fro
m the backend API.
 *
 * @returns {Promise<void>}
 */
async function _loadAllData() {
  try {
    const [agentsRes, toolsRes, providersRes] = await Promise.
all([
      _fetchJson('/api/agents'),
      _fetchJson('/api/agents/tools'),
      _fetchJson('/api/agents/providers')
    ]);

    _state.agents = Array.isArray(agentsRes) ? agentsRes : [];
    _state.tools = Array.isArray(toolsRes) ? toolsRes : [];
    _state.providers = providersRes ?? {};

    _renderCounters();
    _renderGrid();
    _populateToolsMatrix();
    _updateModelDropdown('agent-provider', 'agent-model');
    _updateModelDropdown('ai-builder-provider', 'ai-builder-mo
del');
  } catch (err) {
    console.error('[AgentsTab] Failed to load data:', err);
    _showToast(`Failed to load data: ${err.message}`, 'danger'
);
    const grid = document.getElementById('agents-grid');
    if (grid) {
      grid.innerHTML = `
        <div class="col-12 text-center py-5 text-danger">
          <i class="bi bi-exclamation-triangle fs-1 d-block mb
-2"></i>
          <h5>Не удалось загрузить данные агентов</h5>
          <p class="small text-secondary">${err.message}</p>
          <button class="btn btn-sm btn-outline-primary rounde
```

```
d-pill px-3" onclick="window.initAgentsTab()">Повторить попытк
y</button>
        </div>
      `;
    }
  }
}

/**
 * Binds static DOM event listeners for buttons, filters, and
modal templates.
 *
 * @returns {void}
 */
function _bindStaticEvents() {
  if (_state.isEventsBound) return;
  _state.isEventsBound = true;

  // Refresh agents list
  const refreshBtn = document.getElementById('btn-refresh-agen
ts');
  if (refreshBtn) {
    refreshBtn.onclick = async () => {
      refreshBtn.disabled = true;
      await _loadAllData();
      refreshBtn.disabled = false;
      _showToast('Список агентов обновлен', 'success');
    };
  }

  // Filter toolbar buttons
  const filterGroup = document.getElementById('agents-filter-g
roup');
  if (filterGroup) {
    filterGroup.querySelectorAll('button').forEach((btn) => {
      btn.onclick = () => {
        filterGroup.querySelectorAll('button').forEach((b) =>
b.classList.remove('active'));
```

```
        btn.classList.add('active');
        _state.currentFilter = btn.getAttribute('data-filter') ?? 'all';
        _renderGrid();
      };
    });
  }

  // Modal trigger buttons
  document.getElementById('btn-open-create-agent')?.addEventListener('click', () => _openEditor(null));
  document.getElementById('btn-open-ai-builder')?.addEventListener('click', _openAiBuilder);

  // Provider change handlers
  document.getElementById('agent-provider')?.addEventListener('change', () => {
    _updateModelDropdown('agent-provider', 'agent-model');
  });
  document.getElementById('ai-builder-provider')?.addEventListener('change', () => {
    _updateModelDropdown('ai-builder-provider', 'ai-builder-model');
  });

  // Temperature slider synchronization
  const tempSlider = document.getElementById('agent-temperature');
  const tempVal = document.getElementById('agent-temp-val');
  if (tempSlider && tempVal) {
    tempSlider.oninput = () => {
      tempVal.textContent = tempSlider.value;
    };
  }

  // Action buttons
  document.getElementById('btn-save-agent')?.addEventListener('click', _handleSaveAgent);
```

```
  document.getElementById('btn-run-ai-generate')?.addEventListener('click', _handleRunAiGenerate);

  // Apply AI Generated specification to editor
  document.getElementById('btn-apply-ai-generated')?.addEventListener('click', () => {
    if (!_state.lastGeneratedSpec) return;
    const builderModalEl = document.getElementById('modal-ai-builder');
    bootstrap.Modal.getInstance(builderModalEl)?.hide();
    _openEditor(null, _state.lastGeneratedSpec);
  });

  // System Prompt templates
  document.getElementById('btn-insert-react-template')?.addEventListener('click', () => {
    const promptInput = document.getElementById('agent-system-prompt');
    if (promptInput) {
      promptInput.value =
        'Ты автономный ReAct-агент Mediteka.\n\n' +
        'ПРИНЦИП РАБОТЫ (Thought -> Action -> Observation):\n' +
        '1. Тщательно анализируй вопрос пользователя (Thought).\n' +
        '2. Выбирай необходимый инструмент из доступных (Action) с корректными параметрами.\n' +
        '3. Анализируй результат выполнения инструмента (Observation).\n' +
        '4. Сформируй итоговый структурированный ответ пользователю.\n\n' +
        'ФОРМАТ ОТВЕТА:\n' +
        '- Используй Markdown с красивым форматированием (списки, жирный шрифт, эмодзи).\n' +
        '- Не выводи технические ошибки напрямую, объясняй суть решения.';
    }
  });
```

```
189
190   document.getElementById('btn-insert-json-template')?.addEven
tListener('click', () => {
191     const promptInput = document.getElementById('agent-system-
prompt');
192     if (promptInput) {
193       promptInput.value =
194         'Ты специализированный аналитический агент Mediteka.\n
\n' +
195         'Твоя задача: структурировать полученные данные и всег
да отвечать строго валидным JSON-объектом без лишнего обрамлен
ия и markdown-блоков.\n\n' +
196         'Пример структуры:\n{\n  "status": "success",\n  "item
s": [],\n  "summary": "краткое резюме"\n}';
197     }
198   });
199
200   // Sandbox inputs
201   const sandboxSendBtn = document.getElementById('btn-sandbox-
send');
202   const sandboxInput = document.getElementById('sandbox-input-
msg');
203   if (sandboxSendBtn && sandboxInput) {
204     sandboxSendBtn.onclick = _handleRunSandboxTest;
205     sandboxInput.onkeypress = (e) => {
206       if (e.key === 'Enter') _handleRunSandboxTest();
207     };
208   }
209 }
210
211 /**
212  * Updates filter counter badges on top toolbar buttons.
213  *
214  * @returns {void}
215  */
216 function _renderCounters() {
217   const allCount = _state.agents.length;
218   const activeCount = _state.agents.filter((a) => a.enabled).l
```

```
ength;
  const systemCount = _state.agents.filter((a) => a.is_system)
.length;
  const customCount = _state.agents.filter((a) => !a.is_system
).length;

  document.getElementById('count-all')?.replaceChildren(docume
nt.createTextNode(String(allCount)));
  document.getElementById('count-active')?.replaceChildren(doc
ument.createTextNode(String(activeCount)));
  document.getElementById('count-system')?.replaceChildren(doc
ument.createTextNode(String(systemCount)));
  document.getElementById('count-custom')?.replaceChildren(doc
ument.createTextNode(String(customCount)));
}

/**
 * Renders the responsive agent cards grid based on current fi
lter state.
 *
 * @returns {void}
 */
function _renderGrid() {
  const grid = document.getElementById('agents-grid');
  if (!grid) return;

  const filter = _state.currentFilter;
  let items = _state.agents;

  if (filter === 'active') items = items.filter((a) => a.enabl
ed);
  else if (filter === 'system') items = items.filter((a) => a.
is_system);
  else if (filter === 'custom') items = items.filter((a) => !a
.is_system);

  if (items.length === 0) {
    grid.innerHTML = `
```

```
      <div class="col-12 text-center py-5 text-muted">
        <i class="bi bi-robot fs-1 d-block mb-2 text-info"></i>
        <h5 class="text-body">Агенты не найдены</h5>
        <p class="small text-muted">Создайте нового агента с помощью AI или вручную</p>
      </div>
    `;
    return;
  }

  grid.innerHTML = items
    .map((agent) => {
      const isSystem = Boolean(agent.is_system);
      const isEnabled = Boolean(agent.enabled);
      const providerName = _getProviderDisplayName(agent.provider);

      const toolsBadges = (agent.tools ?? [])
        .map((toolId) => {
          const tool = _state.tools.find((t) => t.id === toolId);
          const icon = tool?.icon ?? '□ ';
          const name = tool?.name ?? toolId;
          return `<span class="badge tool-badge me-1 mb-1 font-monospace">${icon} <span>${name}</span></span>`;
        })
        .join('');

      return `
        <div class="col-xl-4 col-lg-6">
          <div class="card h-100 agent-card shadow-sm" style="${!isEnabled ? 'opacity: 0.65;' : ''}">
            <div class="card-header d-flex justify-content-between align-items-center py-2 px-3">
              <div class="d-flex align-items-center gap-2">
                <span class="fs-5">${isSystem ? '□ ' : '□ '}</span>
```

```
                <strong class="text-body fs-6 fw-bold">${agent
.name}</strong>
                ${
                  isSystem
                    ? '<span class="badge bg-primary fw-bold"
style="font-size: 0.7rem;">SYSTEM</span>'
                    : '<span class="badge bg-warning text-dark
 fw-bold" style="font-size: 0.7rem;">CUSTOM</span>'
                }
              </div>
              <div class="form-check form-switch m-0" title="B
ключить / Выключить">
                <input class="form-check-input agent-toggle-sw
itch" type="checkbox" data-agent-id="${agent.id}" ${isEnabled
? 'checked' : ''}>
              </div>
            </div>
            <div class="card-body p-3 d-flex flex-column">
              <p class="small mb-3 flex-grow-1 agent-desc text
-muted" style="min-height: 40px; line-height: 1.45;">${agent.d
escription || 'Описание отсутствует'}</p>

              <div class="agent-spec-box mb-3 small">
                <div class="d-flex justify-content-between ali
gn-items-center mb-1 pb-1 border-bottom border-secondary-subtl
e">
                  <span class="spec-label">Провайдер &amp;amp; Мод
ель:</span>
                  <span class="badge spec-value-badge text-tru
ncate" style="max-width: 190px;">${providerName}: ${agent.mode
l}</span>
                </div>
                <div class="d-flex justify-content-between ali
gn-items-center">
                  <span class="spec-label">Температура / Шаги:
</span>
                  <span class="spec-tech-val font-monospace">T
: ${agent.temperature} <span class="text-muted">|</span> Max:
```

```
${agent.max_steps}</span>
                </div>
              </div>

              <div class="mb-3">
                <div class="small fw-semibold mb-2 text-body">Инструменты (${(agent.tools ?? []).length}):</div>
                <div class="d-flex flex-wrap">${toolsBadges || '<span class="small text-muted">Без внешних инструментов</span>'}</div>
              </div>

              <div class="mt-auto d-flex justify-content-between gap-2 pt-2 border-top border-secondary-subtle">
                <button class="btn btn-sm rounded-pill px-3 btn-test-sandbox" data-agent-id="${agent.id}">
                  <i class="bi bi-play-fill"></i> Тест в Sandbox
                </button>
                <div class="d-flex gap-1">
                  <button class="btn btn-sm rounded-pill px-2 btn-edit-agent" data-agent-id="${agent.id}" title="Редактировать">
                    <i class="bi bi-pencil-fill"></i>
                  </button>
                  ${
                    !isSystem
                      ? `
                    <button class="btn btn-sm rounded-pill px-2 btn-delete-agent" data-agent-id="${agent.id}" title="Удалить">
                      <i class="bi bi-trash-fill"></i>
                    </button>
                  `
                      : ''
                  }
                </div>
              </div>
```

```
            </div>
          </div>
        </div>
      `;
    })
    .join('');

  // Attach card event listeners
  grid.querySelectorAll('.agent-toggle-switch').forEach((sw) => {
    sw.onchange = async () => {
      const agentId = sw.getAttribute('data-agent-id');
      if (agentId) await _toggleAgentState(agentId, sw.checked);
    };
  });

  grid.querySelectorAll('.btn-test-sandbox').forEach((btn) => {
    btn.onclick = () => {
      const agentId = btn.getAttribute('data-agent-id');
      if (agentId) _openSandbox(agentId);
    };
  });

  grid.querySelectorAll('.btn-edit-agent').forEach((btn) => {
    btn.onclick = () => {
      const agentId = btn.getAttribute('data-agent-id');
      if (agentId) _openEditor(agentId);
    };
  });

  grid.querySelectorAll('.btn-delete-agent').forEach((btn) => {
    btn.onclick = () => {
      const agentId = btn.getAttribute('data-agent-id');
      if (agentId) _deleteAgent(agentId);
    };
```

```
  });
}

/**
 * Toggles an agent's enabled state and persists it to the server.
 *
 * @param {string} agentId - Target agent unique identifier.
 * @param {boolean} enabled - New enabled state.
 * @returns {Promise<void>}
 */
async function _toggleAgentState(agentId, enabled) {
  const agent = _state.agents.find((a) => a.id === agentId);
  if (!agent) return;

  agent.enabled = enabled;
  try {
    await _fetchJson(`/api/agents/${agentId}`, {
      method: 'PUT',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(agent)
    });
    _renderCounters();
    _renderGrid();
    _showToast(`Агент "${agent.name}" ${enabled ? 'включен' : 'отключен'}`, 'success');
  } catch (err) {
    console.error('[AgentsTab] Error toggling agent:', err);
    _showToast(`Ошибка: ${err.message}`, 'danger');
    await _loadAllData();
  }
}

/**
 * Populates capability checkboxes inside the editor modal.
 *
 * @returns {void}
 */
```

```
function _populateToolsMatrix() {
  const container = document.getElementById('agent-tools-matrix');
  if (!container) return;

  container.innerHTML = _state.tools
    .map(
      (tool) => `
      <div class="col-md-6">
        <div class="p-3 rounded h-100" style="background: var(--surface-2); border: 1px solid var(--border-color);">
          <div class="form-check form-switch m-0">
            <input class="form-check-input tool-checkbox" type="checkbox" value="${tool.id}" id="tool-chk-${tool.id}">
            <label class="form-check-label small text-body fw-bold" for="tool-chk-${tool.id}">
              ${tool.icon} ${tool.name}
            </label>
          </div>
          <div class="small mt-1 text-muted" style="font-size: 0.8rem; line-height: 1.4;">${tool.description}</div>
        </div>
      </div>
    `
    )
    .join('');

  container.querySelectorAll('.tool-checkbox').forEach((chk) => {
    chk.onchange = _updateSelectedToolsCounter;
  });
}

/**
 * Updates selected capabilities counter badge in the editor modal.
 *
 * @returns {void}
```

```
 */
function _updateSelectedToolsCounter() {
  const count = document.querySelectorAll('#agent-tools-matrix
 .tool-checkbox:checked').length;
  const badge = document.getElementById('agent-tools-count');
  if (badge) badge.textContent = `${count} выбрано`;
}

/**
 * Updates model selection dropdown options when provider sele
ction changes.
 *
 * @param {string} providerSelectId - ID of provider select el
ement.
 * @param {string} modelSelectId - ID of model select element.
 * @param {string} [selectedModel=''] - Model ID to select by
default.
 * @returns {void}
 */
function _updateModelDropdown(providerSelectId, modelSelectId,
 selectedModel = '') {
  const pSel = document.getElementById(providerSelectId);
  const mSel = document.getElementById(modelSelectId);
  if (!pSel || !mSel) return;

  const prov = _state.providers[pSel.value];
  const models = prov?.models ?? [];

  mSel.innerHTML = models.map((m) => `<option value="${m.id}">
${m.name || m.id}</option>`).join('');

  if (selectedModel && models.some((m) => m.id === selectedMod
el)) {
    mSel.value = selectedModel;
  } else if (prov?.default_model) {
    mSel.value = prov.default_model;
  }
}
```

```
457
458 /**
459  * Opens agent editor modal for creating or updating an agent.
460  *
461  * @param {string|null} [agentId=null] - Agent ID to edit, or
    null for create.
462  * @param {object|null} [prefillData=null] - Pre-filled data f
    rom AI builder.
463  * @returns {void}
464  */
465 function _openEditor(agentId = null, prefillData = null) {
466   const modalEl = document.getElementById('modal-agent-editor'
    );
467   if (!modalEl) return;
468
469   const modeInput = document.getElementById('agent-edit-mode')
    ;
470   const idInput = document.getElementById('agent-id');
471   const nameInput = document.getElementById('agent-name');
472   const descInput = document.getElementById('agent-desc');
473   const enabledChk = document.getElementById('agent-enabled');
474   const providerSel = document.getElementById('agent-provider'
    );
475   const tempSlider = document.getElementById('agent-temperatur
    e');
476   const tempVal = document.getElementById('agent-temp-val');
477   const maxStepsInput = document.getElementById('agent-max-ste
    ps');
478   const timeoutInput = document.getElementById('agent-timeout'
    );
479   const promptTa = document.getElementById('agent-system-promp
    t');
480
481   // Reset tools checkboxes
482   document.querySelectorAll('#agent-tools-matrix .tool-checkbo
    x').forEach((chk) => {
483     chk.checked = false;
484   });
```

```
485
486   if (agentId) {
487     const agent = _state.agents.find((a) => a.id === agentId);
488     if (!agent) return;
489
490     if (modeInput) modeInput.value = 'edit';
491     if (idInput) {
492       idInput.value = agent.id;
493       idInput.disabled = true;
494     }
495     if (nameInput) nameInput.value = agent.name;
496     if (descInput) descInput.value = agent.description ?? '';
497     if (enabledChk) enabledChk.checked = Boolean(agent.enabled
);
498     if (providerSel) providerSel.value = agent.provider ?? 'ge
mini';
499
500     _updateModelDropdown('agent-provider', 'agent-model', agen
t.model);
501
502     if (tempSlider) {
503       tempSlider.value = agent.temperature ?? 0.3;
504       if (tempVal) tempVal.textContent = tempSlider.value;
505     }
506     if (maxStepsInput) maxStepsInput.value = agent.max_steps ?
? 15;
507     if (timeoutInput) timeoutInput.value = agent.timeout_secon
ds ?? 60;
508     if (promptTa) promptTa.value = agent.system_prompt ?? '';
509
510     (agent.tools ?? []).forEach((tId) => {
511       const chk = document.getElementById(`tool-chk-${tId}`);
512       if (chk) chk.checked = true;
513     });
514   } else if (prefillData) {
515     if (modeInput) modeInput.value = 'create';
516     if (idInput) {
517       idInput.value = (prefillData.name || 'agent').toLowerCas
```

```
e().replace(/[^a-z0-9_]/g, '_').slice(0, 25);
      idInput.disabled = false;
    }
    if (nameInput) nameInput.value = prefillData.name ?? 'Новый ИИ-Агент';
    if (descInput) descInput.value = prefillData.description ?? '';
    if (enabledChk) enabledChk.checked = true;
    if (providerSel) providerSel.value = prefillData.provider ?? 'gemini';

    _updateModelDropdown('agent-provider', 'agent-model', prefillData.model);

    if (tempSlider) {
      tempSlider.value = prefillData.temperature ?? 0.3;
      if (tempVal) tempVal.textContent = tempSlider.value;
    }
    if (maxStepsInput) maxStepsInput.value = prefillData.max_steps ?? 15;
    if (timeoutInput) timeoutInput.value = 60;
    if (promptTa) promptTa.value = prefillData.system_prompt ?? '';

    (prefillData.recommended_tools ?? []).forEach((tId) => {
      const chk = document.getElementById(`tool-chk-${tId}`);
      if (chk) chk.checked = true;
    });
  } else {
    if (modeInput) modeInput.value = 'create';
    if (idInput) {
      idInput.value = '';
      idInput.disabled = false;
    }
    if (nameInput) nameInput.value = '';
    if (descInput) descInput.value = '';
    if (enabledChk) enabledChk.checked = true;
    if (providerSel) providerSel.value = 'gemini';
```

```
549
550     _updateModelDropdown('agent-provider', 'agent-model');
551
552     if (tempSlider) {
553       tempSlider.value = 0.3;
554       if (tempVal) tempVal.textContent = '0.3';
555     }
556     if (maxStepsInput) maxStepsInput.value = 15;
557     if (timeoutInput) timeoutInput.value = 60;
558     if (promptTa) promptTa.value = '';
559   }
560
561   _updateSelectedToolsCounter();
562   new bootstrap.Modal(modalEl).show();
563 }
564
565 /**
566  * Saves agent configuration (POST create or PUT update).
567  *
568  * @returns {Promise<void>}
569  */
570 async function _handleSaveAgent() {
571   const mode = document.getElementById('agent-edit-mode')?.val
    ue ?? 'create';
572   const id = document.getElementById('agent-id')?.value.trim()
    ;
573   const name = document.getElementById('agent-name')?.value.tr
    im();
574   const desc = document.getElementById('agent-desc')?.value.tr
    im();
575   const enabled = document.getElementById('agent-enabled')?.ch
    ecked ?? true;
576   const provider = document.getElementById('agent-provider')?.
    value ?? 'gemini';
577   const model = document.getElementById('agent-model')?.value
    ?? 'gemini-3.1-flash';
578   const temperature = parseFloat(document.getElementById('agen
    t-temperature')?.value ?? '0.3');
```

```
  const max_steps = parseInt(document.getElementById('agent-max-steps')?.value ?? '15', 10);
  const timeout_seconds = parseInt(document.getElementById('agent-timeout')?.value ?? '60', 10);
  const system_prompt = document.getElementById('agent-system-prompt')?.value.trim() ?? '';

  if (!id || !name) {
    _showToast('ID и Название агента обязательны для заполнения', 'warning');
    return;
  }

  const selectedTools = [];
  document.querySelectorAll('#agent-tools-matrix .tool-checkbox:checked').forEach((chk) => {
    selectedTools.push(chk.value);
  });

  const payload = {
    id,
    name,
    description: desc,
    enabled,
    provider,
    model,
    temperature,
    max_steps,
    timeout_seconds,
    tools: selectedTools,
    system_prompt
  };

  const saveBtn = document.getElementById('btn-save-agent');
  if (saveBtn) saveBtn.disabled = true;

  try {
    if (mode === 'create') {
```

```
      await _fetchJson('/api/agents', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify(payload)
      });
      _showToast(`Агент "${name}" успешно создан!`, 'success')
;
    } else {
      await _fetchJson(`/api/agents/${id}`, {
        method: 'PUT',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify(payload)
      });
      _showToast(`Агент "${name}" успешно обновлен!`, 'success
');
    }

    const modalEl = document.getElementById('modal-agent-edito
r');
    bootstrap.Modal.getInstance(modalEl)?.hide();
    await _loadAllData();
  } catch (err) {
    console.error('[AgentsTab] Error saving agent:', err);
    _showToast(`Ошибка сохранения: ${err.message}`, 'danger');
  } finally {
    if (saveBtn) saveBtn.disabled = false;
  }
}

/**
 * Deletes an existing custom agent after confirmation.
 *
 * @param {string} agentId - Target agent unique identifier.
 * @returns {Promise<void>}
 */
async function _deleteAgent(agentId) {
  const agent = _state.agents.find((a) => a.id === agentId);
  if (!agent) return;
```

```
647
648   if (!confirm(`Вы действительно хотите удалить агента "${agen
    t.name}" (${agent.id})?`)) {
649     return;
650   }
651
652   try {
653     await _fetchJson(`/api/agents/${agentId}`, { method: 'DELE
    TE' });
654     _showToast(`Агент "${agent.name}" удален`, 'success');
655     await _loadAllData();
656   } catch (err) {
657     console.error('[AgentsTab] Error deleting agent:', err);
658     _showToast(`Ошибка удаления: ${err.message}`, 'danger');
659   }
660 }
661
662 /**
663  * Opens AI prompt architect builder modal.
664  *
665  * @returns {void}
666  */
667 function _openAiBuilder() {
668   const modalEl = document.getElementById('modal-ai-builder');
669   if (!modalEl) return;
670
671   const promptInput = document.getElementById('ai-builder-prom
    pt');
672   const resultBox = document.getElementById('ai-builder-result
    ');
673   if (promptInput) promptInput.value = '';
674   if (resultBox) resultBox.classList.add('d-none');
675   _state.lastGeneratedSpec = null;
676
677   new bootstrap.Modal(modalEl).show();
678 }
679
680 /**
```

```
 * Generates an agent specification using selected LLM architect model.
 *
 * @returns {Promise<void>}
 */
async function _handleRunAiGenerate() {
  const taskDesc = document.getElementById('ai-builder-prompt')?.value.trim();
  const provider = document.getElementById('ai-builder-provider')?.value ?? 'gemini';
  const model = document.getElementById('ai-builder-model')?.value ?? 'gemini-3.1-flash';

  if (!taskDesc) {
    _showToast('Пожалуйста, опишите задачу агента', 'warning');
    return;
  }

  const runBtn = document.getElementById('btn-run-ai-generate');
  if (runBtn) {
    runBtn.disabled = true;
    runBtn.innerHTML = '<span class="spinner-border spinner-border-sm me-2"></span>Анализ задачи и генерация...';
  }

  try {
    const res = await _fetchJson('/api/agents/generate-prompt', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ task_description: taskDesc, provider, model })
    });

    if (res?.data) {
      _state.lastGeneratedSpec = { ...res.data, provider, mode
```

```
l };
710       const resultBox = document.getElementById('ai-builder-re
sult');
711       const previewBox = document.getElementById('ai-builder-p
review-box');
712       if (resultBox && previewBox) {
713         resultBox.classList.remove('d-none');
714         previewBox.innerHTML = `
715           <div><strong class="text-info">Название:</strong> <s
pan class="text-body fw-bold">${res.data.name}</span></div>
716           <div class="mt-1"><strong class="text-muted">Описани
e:</strong> <span class="text-body">${res.data.description}</s
pan></div>
717           <div class="mt-1"><strong class="text-success">Инстр
ументы:</strong> <span class="text-body">${(res.data.recommend
ed_tools ?? []).join(', ') || 'нет'}</span></div>
718           <div class="mt-2 p-2 rounded" style="background: var
(--surface-1); border: 1px solid var(--border-color); color: v
ar(--text-color); white-space: pre-wrap; font-size: 0.85rem; l
ine-height: 1.45;">${res.data.system_prompt}</div>
719         `;
720       }
721       _showToast('Спецификация агента успешно сгенерирована!',
 'success');
722     }
723   } catch (err) {
724     console.error('[AgentsTab] AI Generator error:', err);
725     _showToast(`Ошибка генерации: ${err.message}`, 'danger');
726   } finally {
727     if (runBtn) {
728       runBtn.disabled = false;
729       runBtn.innerHTML = '<i class="bi bi-stars me-1"></i> Сге
нерировать спецификацию';
730     }
731   }
732 }
733
734 /**
```

```
 * Opens sandbox modal for interactive agent testing.
 *
 * @param {string} agentId - Target agent unique identifier.
 * @returns {void}
 */
function _openSandbox(agentId) {
  const agent = _state.agents.find((a) => a.id === agentId);
  if (!agent) return;

  _state.activeSandboxAgent = agent;
  const modalEl = document.getElementById('modal-agent-sandbox
');
  if (!modalEl) return;

  document.getElementById('sandbox-agent-name')?.replaceChildr
en(document.createTextNode(agent.name));
  const modelBadge = document.getElementById('sandbox-agent-mo
del');
  if (modelBadge) modelBadge.textContent = `${_getProviderDisp
layName(agent.provider)}: ${agent.model}`;

  const toolsSummary = document.getElementById('sandbox-agent-
tools-summary');
  if (toolsSummary) toolsSummary.textContent = `Инструментов:
${(agent.tools ?? []).length}`;

  const msgsList = document.getElementById('sandbox-messages-l
ist');
  const placeholder = document.getElementById('sandbox-placeho
lder');
  if (msgsList) msgsList.innerHTML = '';
  if (placeholder) placeholder.style.display = 'block';

  new bootstrap.Modal(modalEl).show();
}

/**
 * Executes a test query against the active sandbox agent and
```

```
renders execution trace.
 *
 * @returns {Promise<void>}
 */
async function _handleRunSandboxTest() {
  const agent = _state.activeSandboxAgent;
  const input = document.getElementById('sandbox-input-msg');
  const sendBtn = document.getElementById('btn-sandbox-send');
  const msgsList = document.getElementById('sandbox-messages-list');
  const placeholder = document.getElementById('sandbox-placeholder');

  if (!agent || !input || !msgsList) return;
  const query = input.value.trim();
  if (!query) return;

  if (placeholder) placeholder.style.display = 'none';

  // Append user message
  const userMsgEl = document.createElement('div');
  userMsgEl.className = 'd-flex justify-content-end mb-3';
  userMsgEl.innerHTML = `
    <div class="p-2 px-3 rounded shadow-sm text-white small" style="background-color: var(--nav-active, #2563eb); max-width: 80%; font-weight: 500;">
      <strong>Вы:</strong> ${query}
    </div>
  `;
  msgsList.appendChild(userMsgEl);

  input.value = '';
  if (sendBtn) sendBtn.disabled = true;

  // Append response placeholder with loader
  const botMsgEl = document.createElement('div');
  botMsgEl.className = 'd-flex flex-column mb-3';
  botMsgEl.innerHTML = `
```

```
    <div class="p-3 rounded small" style="background: var(--surface-1); border: 1px solid var(--border-color); color: var(--text-color);">
      <div class="d-flex align-items-center gap-2 mb-2 text-warning fw-semibold">
        <span class="spinner-border spinner-border-sm"></span>
        <span>Агент выполняет рассуждение и вызовы инструментов...</span>
      </div>
      <div class="steps-trace small font-monospace text-muted"></div>
    </div>
  `;
  msgsList.appendChild(botMsgEl);
  msgsList.scrollTop = msgsList.scrollHeight;

  try {
    const res = await _fetchJson('/api/agents/test', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ agent_id: agent.id, test_message: query })
    });

    const stepsHtml = (res.steps ?? [])
      .map((s) => {
        let icon = '□ ';
        if (s.type === 'tool_init') icon = '□ ';
        else if (s.type === 'action') icon = '□ ';
        else if (s.type === 'finish') icon = '□ ';
        else if (s.type === 'error') icon = '□ ';

        return `<div class="mb-1">${icon} <span class="text-info fw-bold">[Шаг ${s.step}]</span> <span class="text-body">${s.content}</span></div>`;
      })
      .join('');

```

```
    botMsgEl.innerHTML = `
      <div class="p-3 rounded small" style="background: var(--surface-1); border: 1px solid var(--border-color); color: var(--text-color);">
        <div class="d-flex justify-content-between align-items-center mb-2 pb-1 border-bottom border-secondary-subtle">
          <strong class="text-info" style="font-size: 0.95rem;">🤖 ${agent.name}</strong>
          <span class="badge bg-secondary" style="font-weight: 600;">${res.duration_ms ?? 0} мс</span>
        </div>

        <div class="mb-2 text-body" style="white-space: pre-wrap; line-height: 1.5;">${res.response || 'Пустой ответ'}</div>

        <div class="p-2 rounded mt-2 small font-monospace" style="background: var(--surface-2); border: 1px solid var(--border-color); font-size: 0.8rem; line-height: 1.45;">
          <div class="fw-bold mb-1 text-muted">Трассировка выполнения (ReAct Trace):</div>
          ${stepsHtml}
        </div>
      </div>
    `;
  } catch (err) {
    botMsgEl.innerHTML = `
      <div class="p-3 rounded small bg-danger-subtle border border-danger text-danger">
        <strong class="text-danger">Ошибка выполнения:</strong> ${err.message}
      </div>
    `;
  } finally {
    if (sendBtn) sendBtn.disabled = false;
    const chatWin = document.getElementById('sandbox-chat-window');
    if (chatWin) chatWin.scrollTop = chatWin.scrollHeight;
  }
```

```
}

/**
 * Returns human-readable display name for provider identifier.
 *
 * @param {string} provKey - Provider key (e.g. 'gemini', 'agy', 'foundry', 'ollama').
 * @returns {string} Provider display label.
 */
function _getProviderDisplayName(provKey) {
  return _state.providers[provKey]?.name ?? provKey;
}

/**
 * Displays floating feedback notification toast.
 *
 * @param {string} message - Notification text message.
 * @param {string} [type='info'] - Bootstrap color variant ('info', 'success', 'warning', 'danger').
 * @returns {void}
 */
function _showToast(message, type = 'info') {
  if (typeof window.showModelsNotification === 'function') {
    window.showModelsNotification(message, type);
    return;
  }
  const alertDiv = document.createElement('div');
  alertDiv.className = `alert alert-${type} position-fixed bottom-0 end-0 m-3 shadow`;
  alertDiv.style.zIndex = '9999';
  alertDiv.textContent = message;
  document.body.appendChild(alertDiv);
  setTimeout(() => alertDiv.remove(), 3500);
}

```

index.html

```
<!--
================================================================
===============
Process Name: AI-Windows-Admin Web Interface - Index View
================================================================
===============
Description:
  Шаблон веб-интерфейса системного модуля (index).

Usage Examples:
    Web Route:
      /src/api/webgui/ai_benchmark_tab/index.html

File: index.html
Project: ai-windows-admin
Package: src/api/webgui/ai_benchmark_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!-- AI Inference Benchmark Tab -->
<style>
  .bm-container {
    padding: 0.5rem 0.75rem;
  }
  .bm-card {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: 8px;
    padding: 0.6rem 0.85rem;
    height: 100%;
    transition: border-color 0.15s ease, box-shadow 0.15s ease
;
```

```
  }
  .bm-card:hover {
    border-color: var(--nav-active) !important;
    box-shadow: 0 2px 8px rgba(0, 0, 0, 0.08);
  }
  .bm-card-label {
    font-size: 0.7rem;
    text-transform: uppercase;
    letter-spacing: 0.05em;
    color: var(--text-muted);
    font-weight: 600;
    margin-bottom: 0.25rem;
  }
  .bm-card-value {
    font-size: 1.4rem;
    font-weight: 700;
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
    color: var(--nav-active);
  }
  .bm-panel {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: 8px;
    margin-bottom: 0.75rem;
    overflow: hidden;
  }
  .bm-panel-header {
    background: var(--surface-2) !important;
    border-bottom: 1px solid var(--border-color) !important;
    padding: 0.45rem 0.85rem;
    font-weight: 600;
    font-size: 0.85rem;
    color: var(--nav-active);
    display: flex;
    align-items: center;
    justify-content: space-between;
  }
```

```
  .bm-table {
    margin-bottom: 0;
    font-size: 0.8rem;
    width: 100%;
  }
  .bm-table th {
    background: var(--surface-2);
    color: var(--text-muted);
    font-size: 0.72rem;
    text-transform: uppercase;
    padding: 0.45rem 0.6rem;
    border-bottom: 1px solid var(--border-color);
  }
  .bm-table td {
    border-bottom: 1px solid var(--border-color);
    padding: 0.45rem 0.6rem;
    vertical-align: middle;
  }
</style>

<div class="bm-container">
  <!-- Benchmark Run Form -->
  <div class="bm-panel mb-3">
    <div class="bm-panel-header">
      <span><i class="bi bi-speedometer2 me-1"></i> <span data-i18n="auto___81054b"></span> AI Inference Benchmark</span>
      <span class="badge bg-primary-subtle text-primary border border-primary-subtle" id="bm-status-badge"><span data-i18n="auto___97b864"></span></span>
    </div>
    <div class="p-3">
      <div class="row g-2 align-items-end">
        <div class="col-md-3 col-sm-6">
          <label class="form-label small text-muted mb-1"><span data-i18n="auto___2d385a"></span></label>
          <select class="form-select form-select-sm" id="bm-provider-select">
            <option value="gemini" selected>Google Gemini API<
```

```
/option>
            <option value="ollama">Ollama (Local)</option>
            <option value="foundry">Microsoft Foundry Local</o
ption>
            <option value="onnx">ONNX DirectML / NPU</option>
          </select>
        </div>
        <div class="col-md-3 col-sm-6">
          <label class="form-label small text-muted mb-1"><spa
n data-i18n="auto___f1f1ee"></span></label>
          <input type="text" class="form-control form-control-
sm" id="bm-model-input" value="gemini-3.1-flash">
        </div>
        <div class="col-md-2 col-sm-6">
          <label class="form-label small text-muted mb-1">Max
Tokens</label>
          <input type="number" class="form-control form-contro
l-sm" id="bm-tokens-input" value="150" min="20" max="2000">
        </div>
        <div class="col-md-2 col-sm-6">
          <label class="form-label small text-muted mb-1">Temp
erature</label>
          <input type="number" class="form-control form-contro
l-sm" id="bm-temp-input" value="0.7" min="0" max="1" step="0.1
">
        </div>
        <div class="col-md-2 col-sm-12">
          <button class="btn btn-sm btn-primary w-100 fw-semib
old d-flex align-items-center justify-content-center gap-1.5"
id="bm-run-btn">
            <i class="bi bi-play-fill fs-6"></i> <span data-i1
8n="auto___9632be"></span>
          </button>
        </div>
      </div>
      <div class="mt-2">
        <label class="form-label small text-muted mb-1"><span
data-i18n="auto___b3f6a4"></span></label>
```

```
        <textarea class="form-control form-control-sm" id="bm-prompt-input" rows="2"><span data-i18n="auto__100__5d4d7a"></span>.</textarea>
      </div>
    </div>
  </div>

  <!-- KPI Cards -->
  <div class="row g-2 mb-3">
    <div class="col-md-3 col-6">
      <div class="bm-card">
        <div class="bm-card-label"><i class="bi bi-stopwatch text-info me-1"></i> Time to First Token (TTFT)</div>
        <div class="bm-card-value" id="bm-card-ttft">— ms</div>
      </div>
    </div>
    <div class="col-md-3 col-6">
      <div class="bm-card">
        <div class="bm-card-label"><i class="bi bi-lightning-charge text-warning me-1"></i> <span data-i18n="auto___09625b"></span> (TPS)</div>
        <div class="bm-card-value text-success" id="bm-card-tps">— t/s</div>
      </div>
    </div>
    <div class="col-md-3 col-6">
      <div class="bm-card">
        <div class="bm-card-label"><i class="bi bi-hourglass-split text-primary me-1"></i> <span data-i18n="auto___46e0d8"></span> (Latency)</div>
        <div class="bm-card-value" id="bm-card-total">— ms</div>
      </div>
    </div>
    <div class="col-md-3 col-6">
      <div class="bm-card">
        <div class="bm-card-label"><i class="bi bi-chat-left-t
```

```
ext text-secondary me-1"></i> <span data-i18n="auto___430610">
</span></div>
        <div class="bm-card-value text-body" id="bm-card-token
s">—</div>
      </div>
    </div>
  </div>

  <!-- History Panel -->
  <div class="bm-panel">
    <div class="bm-panel-header">
      <span><i class="bi bi-clock-history me-1"></i> <span dat
a-i18n="auto___a58a32"></span></span>
      <button class="btn btn-xs btn-outline-secondary py-0 px-
2" id="bm-refresh-history-btn" style="font-size: 0.75rem;">
        <i class="bi bi-arrow-clockwise"></i> <span data-i18n=
"auto___c2f668"></span>
      </button>
    </div>
    <div class="table-responsive">
      <table class="table table-hover bm-table" id="bm-history
-table">
        <thead>
          <tr>
            <th><span data-i18n="auto___c80d7e"></span></th>
            <th><span data-i18n="auto___2d385a"></span></th>
            <th><span data-i18n="auto___f1f1ee"></span></th>
            <th>TTFT</th>
            <th><span data-i18n="auto___09625b"></span> (TPS)<
/th>
            <th><span data-i18n="auto___46e0d8"></span></th>
            <th><span data-i18n="auto___4b1883"></span> (Promp
t / Gen)</th>
            <th><span data-i18n="auto___f7f293"></span></th>
          </tr>
        </thead>
        <tbody id="bm-history-tbody">
          <tr>
```

```
            <td colspan="8" class="text-center text-muted py-3
"><span data-i18n="auto___51fbb4"></span>".</td>
          </tr>
        </tbody>
      </table>
    </div>
  </div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: AI-Windows-Admin UI - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский веб-скрипт модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/src/api/webgui/ai_benchmark_tab/main.js?v
=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: src/api/webgui/ai_benchmark_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * ai_benchmark_tab/main.js — Контроллер вкладки AI Inference
Benchmark
 */

export async function initAiBenchmarkTab() {
  const runBtn = document.getElementById('bm-run-btn');
  const refreshBtn = document.getElementById('bm-refresh-histo
ry-btn');
  const providerSelect = document.getElementById('bm-provider-
select');
  const modelInput = document.getElementById('bm-model-input')
```

```
;
  const tokensInput = document.getElementById('bm-tokens-input
');
  const tempInput = document.getElementById('bm-temp-input');
  const promptInput = document.getElementById('bm-prompt-input
');
  const statusBadge = document.getElementById('bm-status-badge
i18n.t('auto__if_providerselect_providerselect_addeventlistene
r__df8f21')change', () => {
      const p = providerSelect.value;
      if (p === 'gemini') modelInput.value = 'gemini-3.1-flash
';
      else if (p === 'ollama') modelInput.value = 'llama3.2:la
test';
      else if (p === 'foundry') modelInput.value = 'phi-3.5-mi
ni';
      else if (p === 'onnx') modelInput.value = 'directml-phi3
i18n.t('auto__if_runbtn_runbtn_addeventlistener__c4d4ca')click
', async () => {
      runBtn.disabled = true;
      runBtn.innerHTML = '<span class="spinner-border spinner-
border-sm me-1"></span> Тестирование...';
      if (statusBadge) {
        statusBadge.className = 'badge bg-warning-subtle text-
warning border border-warning-subtle';
        statusBadge.textContent = i18n.t('auto___f7abff');
      }

      try {
        const payload = {
          provider: providerSelect ? providerSelect.value : 'g
emini',
          model_name: modelInput ? modelInput.value : 'gemini-
3.1-flash',
          prompt: promptInput ? promptInput.value : i18n.t('au
to___387814'),
          max_tokens: tokensInput ? parseInt(tokensInput.value
, 10) || 150 : 150,
```

```
          temperature: tempInput ? parseFloat(tempInput.value) || 0.7 : 0.7,
        };

        const res = await fetch('/api/windows/benchmark/ai', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify(payload),
        });

        const data = await res.json();
        if (res.ok && data.success) {
          updateCards(data);
          if (statusBadge) {
            statusBadge.className = 'badge bg-success-subtle text-success border border-success-subtle';
            statusBadge.textContent = i18n.t('auto___b718ea');
          }
          if (window.toast) {
            window.toast.success(i18n.t('auto___983c1c'), `Скорость: ${data.tokens_per_second} t/s | TTFT: ${data.ttft_ms} ms`);
          }
        } else {
          if (statusBadge) {
            statusBadge.className = 'badge bg-danger-subtle text-danger border border-danger-subtle';
            statusBadge.textContent = i18n.t('auto___72aecd');
          }
          if (window.toast) {
            window.toast.error(i18n.t('auto___f29ff0'), data.error_message || i18n.t('auto___f6377e'));
          }
        }
      } catch (err) {
        if (statusBadge) {
          statusBadge.className = 'badge bg-danger-subtle text-danger border border-danger-subtle';
```

```
          statusBadge.textContent = i18n.t('auto___c00aca');
        }
        if (window.toast) {
          window.toast.error(i18n.t('auto___089486'), err.mess
age);
        }
      } finally {
        runBtn.disabled = false;
        runBtn.innerHTML = '<i class="bi bi-play-fill fs-6"></
i> Запустить тести18n.t('auto__await_loadhistory_if_refreshbtn
_refreshbtn_addeventlistener__909824')clicki18n.t('auto__loadh
istory_await_loadhistory_function_updatecards_data_const_ttfte
l_document_getelementbyid__9cba8e')bm-card-ttft');
  const tpsEl = document.getElementById('bm-card-tps');
  const totalEl = document.getElementById('bm-card-total');
  const tokensEl = document.getElementById('bm-card-tokens');

  if (ttftEl) ttftEl.textContent = `${data.ttft_ms} ms`;
  if (tpsEl) tpsEl.textContent = `${data.tokens_per_second} t/
s`;
  if (totalEl) totalEl.textContent = `${data.total_time_ms} ms
`;
  if (tokensEl) tokensEl.textContent = `${data.prompt_tokens}
/ ${data.completion_tokens}`;
}

async function loadHistory() {
  const tbody = document.getElementById('bm-history-tbody');
  if (!tbody) return;

  try {
    const res = await fetch('/api/windows/benchmark/ai/history
');
    const items = await res.json();

    if (!Array.isArray(items) || items.length === 0) {
      tbody.innerHTML = '<tr><td colspan="8" class="text-cente
r text-muted py-3i18n.t('auto___9f84ae')Запустить тест".</td><
```

```
/tr>';
      return;
    }

    tbody.innerHTML = items.slice().reverse().map(item => {
      const timeStr = item.timestamp ? new Date(item.timestamp * 1000).toLocaleTimeString() : '—';
      const statusBadge = item.success
        ? '<span class="badge bg-success-subtle text-success">OK</span>'
        : `<span class="badge bg-danger-subtle text-danger" title="${item.error_message || ''}i18n.t('auto__span_return_tr_td_span_class__613956')text-muted small">${timeStr}</span></td>
          <td><span class="badge bg-secondary-subtle text-info">${item.provider}</span></td>
          <td class="fw-semibold text-white">${item.model_name}</td>
          <td><span class="text-info">${item.ttft_ms} ms</span></td>
          <td><span class="text-success fw-bold">${item.tokens_per_second} t/s</span></td>
          <td>${item.total_time_ms} ms</td>
          <td><span class="text-muted small">${item.prompt_tokens} in / ${item.completion_tokens} out</span></td>
          <td>${statusBadge}</td>
        </tr>
      `;
    }).join('');
  } catch (err) {
    tbody.innerHTML = `<tr><td colspan="8" class="text-danger text-center py-2">Ошибка загрузки истории: ${err.message}</td></tr>`;
  }
}

// Автозапуск при инициализации вкладки
if (document.readyState === 'loading') {
  document.addEventListener('DOMContentLoaded', initAiBenchmar
```

```
kTab);
} else {
  initAiBenchmarkTab();
}

```

index.html

```
<!--
================================================================
===============
Process Name: Windows WebGUI - AI Modal Dialog Template
================================================================
===============
Description:
  Шаблон разметки универсального модального окна AI-анализатор
а со встроенным
  редактором системной инструкции, выбором модели и привязкой
к ID панели.

Usage Examples:
  HTML Integration:
    <div id="ai-modal-dialog-container">...</div>

File: index.html
Project: ai-windows-admin
Package: apps.windows.api.webgui.ai_modal_dialog
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!-- Universal AI Table & Diagnostic Modal Dialog -->
<div class="modal fade" id="universal-ai-table-modal" tabindex
="-1" aria-hidden="true">
  <div class="modal-dialog modal-lg modal-dialog-centered moda
l-dialog-scrollable">
    <div class="modal-content shadow-lg">
      <div class="modal-header py-2 px-3">
        <div class="d-flex align-items-center gap-2 flex-wrap"
>
          <span class="fs-5 text-info" id="uai-modal-icon">🤖 <
```

```
/span>
          <h5 class="modal-title fw-bold text-info mb-0" id="u
ai-modal-title">Детали записи</h5>
          <div id="uai-modal-badges" class="d-flex align-items
-center gap-1"></div>
        </div>
        <button type="button" class="btn-close" data-bs-dismis
s="modal" aria-label="Закрыть"></button>
      </div>
      <div class="modal-body py-3 px-3">
        <!-- Metadata Grid -->
        <div class="row g-2 mb-3 small" id="uai-modal-metadata
-grid"></div>

        <!-- Raw / Code Content Section -->
        <div class="card p-3 mb-3 uai-raw-container d-none" id
="uai-modal-raw-container" style="display: none;">
          <div class="d-flex justify-content-between align-ite
ms-center mb-1">
            <h6 class="small text-uppercase text-muted fw-bold
 mb-0" id="uai-modal-raw-title">Контекст / Данные</h6>
          </div>
          <div class="small font-monospace uai-raw-content" id
="uai-modal-raw-content" style="white-space: pre-wrap; word-br
eak: break-all;"></div>
        </div>

        <!-- System Instruction & Model Configuration Section
-->
        <div class="card p-3 mb-3 border-secondary-subtle uai-
system-card" id="uai-modal-system-card">
          <div class="d-flex justify-content-between align-ite
ms-center mb-2 flex-wrap gap-2">
            <div class="d-flex align-items-center gap-2">
              <span class="fs-6 text-warning">⚙ </span>
              <h6 class="fw-bold text-warning mb-0">Системная
инструкция и конфигурация ИИ</h6>
            </div>
```

```
            <div class="d-flex align-items-center gap-1.5 flex
-wrap">
              <span class="badge bg-secondary-subtle text-seco
ndary border small" id="uai-modal-panel-badge" title="ID вызыв
ающей панели">Панель: -</span>
              <span class="badge bg-secondary-subtle text-seco
ndary border small" id="uai-modal-type-badge">Тип: -</span>
              <span class="badge border border-info text-info
small" id="uai-modal-custom-badge" style="display: none;">Каст
омная</span>
            </div>
          </div>

          <!-- Model & Provider Selection Bar -->
          <div class="row g-2 mb-2 align-items-center">
            <div class="col-sm-6">
              <div class="input-group input-group-sm">
                <span class="input-group-text text-secondary s
mall py-0">Провайдер</span>
                <select class="form-select form-select-sm" id=
"uai-modal-provider-select">
                  <option value="gemini">Google Gemini</option
>
                  <option value="openai">OpenAI / Compatible</
option>
                  <option value="ollama">Ollama Local</option>
                  <option value="foundry">Foundry Local</optio
n>
                  <option value="windows_ai">Windows Copilot+
AI</option>
                </select>
              </div>
            </div>
            <div class="col-sm-6">
              <div class="input-group input-group-sm">
                <span class="input-group-text text-secondary s
mall py-0">Модель</span>
                <input type="text" class="form-control form-co
```

```
ntrol-sm" id="uai-modal-model-input" placeholder="По умолчанию
 (из config.json)">
              </div>
            </div>
          </div>

          <!-- System Instruction Textarea -->
          <div class="mb-1">
            <div class="d-flex justify-content-between align-i
tems-center mb-1">
              <label class="form-label small text-secondary fw
-bold mb-0">Системная инструкция (System Instruction):</label>
              <div class="d-flex gap-1">
                <button type="button" class="btn btn-xs btn-ou
tline-info py-0 px-2" id="btn-uai-modal-save-system" title="Со
хранить измененную инструкцию для этой сущности">
                  <i class="bi bi-floppy me-1"></i>Сохранить
                </button>
                <button type="button" class="btn btn-xs btn-ou
tline-secondary py-0 px-2" id="btn-uai-modal-reset-system" tit
le="Сбросить к системной инструкции по умолчанию">
                  <i class="bi bi-arrow-counterclockwise me-1"
></i>Сброс
                </button>
              </div>
            </div>
            <textarea class="form-control form-control-sm font
-monospace" id="uai-modal-system-instruction" rows="3" placeho
lder="Роль эксперта, правила анализа и формат ответа для модел
и..."></textarea>
            <div class="d-flex justify-content-between align-i
tems-center mt-1">
              <span class="text-secondary small" style="font-s
ize: 0.72rem;" id="uai-modal-system-status">Инструкция передае
тся при вызове модели и анализе</span>
            </div>
          </div>
        </div>
```

```
101
102         <!-- AI Contextual Diagnostic Card -->
103         <div class="card border-info p-3 uai-ai-section" id="u
    ai-modal-ai-section">
104           <div class="d-flex justify-content-between align-ite
    ms-center mb-2 flex-wrap gap-2">
105             <div class="d-flex align-items-center gap-2">
106               <h6 class="fw-bold text-info mb-0"><i class="bi
    bi-robot me-1"></i> AI Contextual Explanation</h6>
107               <span class="badge border border-info text-info
    small px-2 py-0.5" id="uai-modal-web-status" title="Сведения о
    богащаются поиском в интернете">
108                 <i class="bi bi-globe me-1"></i>Web Grounding
109               </span>
110             </div>
111             <div class="d-flex align-items-center gap-1.5">
112               <button class="btn btn-sm btn-outline-secondary
    py-0 px-2 rounded-pill" id="btn-uai-modal-edit-prompt" title="
    Настроить расширенный шаблон промпта">
113                 <i class="bi bi-gear-fill me-1"></i>Промпт
114               </button>
115               <button class="btn btn-sm btn-outline-info py-0
    px-2 rounded-pill" id="btn-uai-modal-refresh" title="Обновить
    AI-анализ" style="display: none;">
116                 <i class="bi bi-arrow-clockwise me-1"></i>Обно
    вить
117               </button>
118             </div>
119           </div>
120           <div class="small" id="uai-modal-ai-explanation"></d
    iv>
121         </div>
122       </div>
123       <div class="modal-footer py-2 px-3 d-flex justify-conten
    t-between">
124         <div id="uai-modal-custom-actions" class="d-flex gap-1
    .5 flex-wrap"></div>
125         <button type="button" class="btn btn-sm btn-secondary"
```

```
 data-bs-dismiss="modal">Закрыть</button>
      </div>
    </div>
  </div>
</div>

<!-- AI Prompt Editor Modal Dialog -->
<div class="modal fade" id="universal-ai-prompt-editor-modal" tabindex="-1" aria-hidden="true">
  <div class="modal-dialog modal-lg modal-dialog-centered modal-dialog-scrollable">
    <div class="modal-content shadow-lg">
      <div class="modal-header py-2 px-3">
        <div class="d-flex align-items-center gap-2">
          <span class="fs-5 text-warning">□ </span>
          <h5 class="modal-title fw-bold text-warning mb-0">Редактор шаблона промпта</h5>
        </div>
        <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Закрыть"></button>
      </div>
      <div class="modal-body py-3 px-3">
        <div class="row g-2 mb-3">
          <div class="col-sm-6">
            <label class="form-label small text-secondary fw-bold mb-1">Тип таблицы / сущности</label>
            <select class="form-select form-select-sm" id="uai-prompt-type-select">
              <option value="user_account">user_account (Учетные записи и сессии Windows)</option>
              <option value="security_event">security_event (События безопасности Windows)</option>
              <option value="process">process (Процессы Windows)</option>
              <option value="service">service (Службы Windows)</option>
              <option value="software">software (Установленное ПО)</option>
```

```
              <option value="task">task (Задачи планировщика)</option>
              <option value="startup">startup (Автозагрузка)</option>
              <option value="network">network (Сетевые соединения)</option>
              <option value="registry">registry (Реестр Windows)</option>
              <option value="disk">disk (Диски и тома)</option>
              <option value="cpu">cpu (Процессор CPU)</option>
              <option value="gpu">gpu (Видеокарта GPU)</option>
              <option value="file_event">file_event (Файловые события)</option>
              <option value="website">website (Мониторинг сайтов)</option>
              <option value="rag_doc">rag_doc (RAG База знаний)</option>
              <option value="generic">generic (Общий шаблон)</option>
            </select>
          </div>
          <div class="col-sm-6">
            <label class="form-label small text-secondary fw-bold mb-1">Название шаблона</label>
            <input type="text" class="form-control form-control-sm" id="uai-prompt-name-input" placeholder="Название шаблона">
          </div>
        </div>

        <div class="mb-3">
          <div class="d-flex justify-content-between align-items-center mb-1">
            <label class="form-label small text-secondary fw-bold mb-0">Системная инструкция (System Instruction)</label>
            <span class="badge bg-secondary-subtle text-second
```

```
ary small" style="font-size: 0.72rem;">Роль и поведение ИИ</sp
an>
          </div>
          <textarea class="form-control form-control-sm font-m
onospace" id="uai-prompt-system-input" rows="4" placeholder="P
оль, экспертный контекст и правила поведения модели..."></text
area>
          <div class="form-text small text-secondary" style="f
ont-size: 0.75rem;">
            Системная инструкция задаёт контекст роли эксперта
, ограничения и формат ответа ИИ для выбранного типа сущности.
          </div>
        </div>

        <div class="mb-2">
          <div class="d-flex justify-content-between align-ite
ms-center mb-1">
            <label class="form-label small text-secondary fw-b
old mb-0">Шаблон текста промпта (Prompt Template)</label>
            <div class="d-flex gap-1" id="uai-prompt-variables
-bar">
              <button type="button" class="btn btn-xs btn-outl
ine-info py-0 px-1 font-monospace" style="font-size: 0.75rem;"
 onclick="window.AIModalDialog.insertVariable('{title}')">{tit
le}</button>
              <button type="button" class="btn btn-xs btn-outl
ine-info py-0 px-1 font-monospace" style="font-size: 0.75rem;"
 onclick="window.AIModalDialog.insertVariable('{subtitle}')">{
subtitle}</button>
              <button type="button" class="btn btn-xs btn-outl
ine-info py-0 px-1 font-monospace" style="font-size: 0.75rem;"
 onclick="window.AIModalDialog.insertVariable('{metadata}')">{
metadata}</button>
              <button type="button" class="btn btn-xs btn-outl
ine-info py-0 px-1 font-monospace" style="font-size: 0.75rem;"
 onclick="window.AIModalDialog.insertVariable('{raw_data}')">{
raw_data}</button>
            </div>
```

```
          </div>
          <textarea class="form-control form-control-sm font-monospace" id="uai-prompt-template-input" rows="8" placeholder="Текст промпта с плейсхолдерами..."></textarea>
          <div class="form-text small text-secondary" style="font-size: 0.75rem;">
            Используйте переменные <code>{title}</code>, <code>{subtitle}</code>, <code>{metadata}</code>, <code>{raw_data}</code> для автоматической подстановки контекста строки таблицы.
          </div>
        </div>

        <div id="uai-prompt-status-msg" class="small mt-2" style="display: none;"></div>
      </div>
      <div class="modal-footer py-2 px-3 d-flex justify-content-between">
        <button type="button" class="btn btn-sm btn-outline-danger" id="btn-uai-prompt-reset">
          <i class="bi bi-arrow-counterclockwise me-1"></i>Сбросить по умолчанию
        </button>
        <div class="d-flex gap-2">
          <button type="button" class="btn btn-sm btn-secondary" data-bs-dismiss="modal">Закрыть</button>
          <button type="button" class="btn btn-sm btn-primary" id="btn-uai-prompt-save">
            <i class="bi bi-check2 me-1"></i>Сохранить промпт
          </button>
        </div>
      </div>
    </div>
  </div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows WebGUI - AI Modal Dialog Controller
 * ============================================================
 * Description:
 *   Клиентский контроллер и интерфейс универсального всплывающего окна
 *   с ИИ-анализатором записей, WikiLLM-интеграцией, прямым редактором системной
 *   инструкции и привязкой к ID панели интерфейса.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/html/ai_modal_dialog/main.js?v=20261010_v2"></script>
 *
 *   JS Execution:
 *     window.AIModalDialog.show({
 *       title: 'Administrator',
 *       subtitle: 'SID: S-1-5-21-...',
 *       icon: '□ ',
 *       panelId: 'panel-winadmin-users',
 *       tableType: 'user_account',
 *       metadata: [{ label: 'SID', value: 'S-1-5-21-...' }]
 *     });
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: apps.windows.api.webgui.ai_modal_dialog
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-12 07:09:00
 * ============================================================
```

```
 */

// Universal AI Modal Dialog Module for AI-Windows-Admin
// Enables interactive row inspection, WikiLLM grounding, in-place system instruction editing, and AI contextual explanation.

(function () {
  const MODAL_ID = 'universal-ai-table-modal';
  const PROMPT_MODAL_ID = 'universal-ai-prompt-editor-modal';

  /**
   * Нормализует строковый тип таблицы с учетом алиасов.
   * @param {string} tType Тип таблицы
   * @returns {string} Нормализованный ключ
   */
  function normalizeTableType(tType) {
    if (!tType) return 'generic';
    const low = String(tType).toLowerCase().trim();
    if (low === 'user' || low === 'users' || low === 'account' || low === 'accounts') return 'user_account';
    if (low === 'event' || low === 'events' || low === 'security') return 'security_event';
    if (low === 'tasks') return 'task';
    if (low === 'services') return 'service';
    if (low === 'processes') return 'process';
    if (low === 'disks') return 'disk';
    if (low === 'apps' || low === 'app') return 'software';
    if (low === 'autorun' || low === 'autoruns') return 'startup';
    return low;
  }

  /**
   * Гарантирует наличие DOM-элемента модального окна AI-анализатора.
   * @returns {HTMLElement} Элемент модального окна
   */
```

```
  function ensureModalElement() {
    let modalEl = document.getElementById(MODAL_ID);
    if (!modalEl) {
      modalEl = document.createElement('div');
      modalEl.id = MODAL_ID;
      modalEl.className = 'modal fade';
      modalEl.tabIndex = -1;
      modalEl.setAttribute('aria-hidden', 'true');
      modalEl.innerHTML = `
        <div class="modal-dialog modal-lg modal-dialog-centere
d modal-dialog-scrollable">
          <div class="modal-content shadow-lg">
            <div class="modal-header py-2 px-3">
              <div class="d-flex align-items-center gap-2 flex
-wrap">
                <span class="fs-5 text-info" id="uai-modal-ico
n">🔍</span>
                <h5 class="modal-title fw-bold text-info mb-0"
 id="uai-modal-title">Детали записи</h5>
                <div id="uai-modal-badges" class="d-flex align
-items-center gap-1"></div>
              </div>
              <button type="button" class="btn-close" data-bs-
dismiss="modal" aria-label="Закрыть"></button>
            </div>
            <div class="modal-body py-3 px-3">
              <!-- Metadata Grid -->
              <div class="row g-2 mb-3 small" id="uai-modal-me
tadata-grid"></div>

              <!-- Raw / Code Content Section -->
              <div class="card p-3 mb-3 uai-raw-container d-no
ne" id="uai-modal-raw-container" style="display: none;">
                <div class="d-flex justify-content-between ali
gn-items-center mb-1">
                  <h6 class="small text-uppercase text-muted f
w-bold mb-0" id="uai-modal-raw-title">Контекст / Данные</h6>
                </div>
```

```
                <div class="small font-monospace uai-raw-content" id="uai-modal-raw-content" style="white-space: pre-wrap; word-break: break-all;"></div>
              </div>

              <!-- System Instruction & Model Configuration Section -->
              <div class="card p-3 mb-3 border-secondary-subtle uai-system-card" id="uai-modal-system-card">
                <div class="d-flex justify-content-between align-items-center mb-2 flex-wrap gap-2">
                  <div class="d-flex align-items-center gap-2">
                    <span class="fs-6 text-warning">⚙</span>
                    <h6 class="fw-bold text-warning mb-0">Системная инструкция и конфигурация ИИ</h6>
                  </div>
                  <div class="d-flex align-items-center gap-1.5 flex-wrap">
                    <span class="badge bg-secondary-subtle text-secondary border small" id="uai-modal-panel-badge" title="ID вызывающей панели">Панель: -</span>
                    <span class="badge bg-secondary-subtle text-secondary border small" id="uai-modal-type-badge">Тип: -</span>
                    <span class="badge border border-info text-info small" id="uai-modal-custom-badge" style="display: none;">Кастомная</span>
                  </div>
                </div>

                <div class="row g-2 mb-2 align-items-center">
                  <div class="col-sm-6">
                    <div class="input-group input-group-sm">
                      <span class="input-group-text text-secondary small py-0">Провайдер</span>
                      <select class="form-select form-select-sm" id="uai-modal-provider-select">
```

```
                        <option value="gemini">Google Gemini (Direct SDK)</option>
                        <option value="gemini_cli">Google Gemini CLI</option>
                        <option value="agy">Antigravity (AGY SDK)</option>
                        <option value="openai">OpenAI / Compatible</option>
                        <option value="ollama">Ollama Local</option>
                        <option value="foundry">Foundry Local</option>
                        <option value="windows_ai">Windows Copilot+ AI</option>
                      </select>
                    </div>
                  </div>
                  <div class="col-sm-6">
                    <div class="input-group input-group-sm">
                      <span class="input-group-text text-secondary small py-0">Модель</span>
                      <input type="text" class="form-control form-control-sm" id="uai-modal-model-input" placeholder="По умолчанию (из config.json)">
                    </div>
                  </div>
                </div>

                <div class="mb-1">
                  <div class="d-flex justify-content-between align-items-center mb-1">
                    <label class="form-label small text-secondary fw-bold mb-0">Системная инструкция (System Instruction):</label>
                    <div class="d-flex gap-1">
                      <button type="button" class="btn btn-xs btn-outline-info py-0 px-2" id="btn-uai-modal-save-system" title="Сохранить измененную инструкцию для этой сущности">
```

```
                      <i class="bi bi-floppy me-1"></i>Сохранить
                    </button>
                    <button type="button" class="btn btn-xs btn-outline-secondary py-0 px-2" id="btn-uai-modal-reset-system" title="Сбросить к системной инструкции по умолчанию">
                      <i class="bi bi-arrow-counterclockwise me-1"></i>Сброс
                    </button>
                  </div>
                </div>
                <textarea class="form-control form-control-sm font-monospace" id="uai-modal-system-instruction" rows="3" placeholder="Роль эксперта, правила анализа и формат ответа для модели..."></textarea>
                <div class="d-flex justify-content-between align-items-center mt-1">
                  <span class="text-secondary small" style="font-size: 0.72rem;" id="uai-modal-system-status">Инструкция передается при вызове модели и анализе</span>
                </div>
              </div>

              <!-- AI Contextual Diagnostic Card -->
              <div class="card border-info p-3 uai-ai-section" id="uai-modal-ai-section">
                <div class="d-flex justify-content-between align-items-center mb-2 flex-wrap gap-2">
                  <div class="d-flex align-items-center gap-2">
                    <h6 class="fw-bold text-info mb-0"><i class="bi bi-robot me-1"></i> AI Contextual Explanation</h6>
                    <span class="badge border border-info text-info small px-2 py-0.5" id="uai-modal-web-status" title="Сведения обогащаются поиском в интернете">
                      <i class="bi bi-globe me-1"></i>Web Grounding
```

```
                  </span>
                </div>
                <div class="d-flex align-items-center gap-1.
5">
                  <button class="btn btn-sm btn-outline-seco
ndary py-0 px-2 rounded-pill" id="btn-uai-modal-edit-prompt" t
itle="Настроить расширенный шаблон промпта">
                    <i class="bi bi-gear-fill me-1"></i>Пром
пт
                  </button>
                  <button class="btn btn-sm btn-outline-info
 py-0 px-2 rounded-pill" id="btn-uai-modal-refresh" title="Обн
овить AI-анализ" style="display: none;">
                    <i class="bi bi-arrow-clockwise me-1"></
i>Обновить
                  </button>
                </div>
              </div>
              <div class="small" id="uai-modal-ai-explanatio
n"></div>
            </div>
          </div>
          <div class="modal-footer py-2 px-3 d-flex justify-
content-between">
            <div id="uai-modal-custom-actions" class="d-flex
 gap-1.5 flex-wrap"></div>
            <button type="button" class="btn btn-sm btn-seco
ndary" data-bs-dismiss="modal">Закрыть</button>
          </div>
        </div>
      </div>
    `;
    document.body.appendChild(modalEl);
  }
  return modalEl;
}

/**
```

```
   * Надежный унифицированный fetch для всех запросов модального окна с таймаутом.
   * @param {string} url URL запроса
   * @param {Object} opts Опции fetch
   * @param {number} timeoutMs Таймаут в миллисекундах
   * @returns {Promise<any>}
   */
  async function safeApiFetch(url, opts = {}, timeoutMs = 10000) {
    const controller = new AbortController();
    const timeoutId = setTimeout(() => controller.abort(), timeoutMs);

    try {
      const fetchOptions = Object.assign({}, opts, { signal: controller.signal });
      const res = await window.fetch(url, fetchOptions);
      clearTimeout(timeoutId);

      if (!res.ok) {
        let errMsg = `${res.status} ${res.statusText}`;
        try {
          const errData = await res.json();
          if (errData && errData.detail) {
            errMsg = typeof errData.detail === 'string' ? errData.detail : JSON.stringify(errData.detail);
          } else if (errData && errData.message) {
            errMsg = errData.message;
          }
        } catch (_) {}
        throw new Error(errMsg);
      }

      return await res.json();
    } catch (err) {
      clearTimeout(timeoutId);
      if (err.name === 'AbortError') {
        throw new Error(`Превышено время ожидания ответа (${ti
```

```
meoutMs / 1000}c)`);
      }
      throw err;
    }
  }

  /**
   * Экранирование HTML спецсимволов.
   * @param {string} str Входная строка
   * @returns {string} Экранированная строка
   */
  function escapeHtml(str) {
    if (str === null || str === undefined) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
  }

  /**
   * Универсальный вызов модального окна AI-анализатора для ст
роки или сущности.
   * @param {Object} options Объект конфигурации диалога
   */
  function show(options) {
    const rawType = options?.tableType || 'generic';
    const normalizedType = normalizeTableType(rawType);
    const panelId = options?.panelId || 'panel-root';

    const opts = Object.assign({
      title: 'Детали записи',
      subtitle: '',
      icon: '',
      panelId: panelId,
      badges: [],
      metadata: [],
```

```
      rawTitle: 'Данные / Путь',
      rawContent: '',
      tableType: normalizedType,
      model: '',
      provider: 'gemini',
      systemInstruction: '',
      requestData: {},
      autoRun: false,
      actions: []
    }, options, { tableType: normalizedType, panelId: panelId
});

    const modalEl = ensureModalElement();

    // Icon & Title
    const iconEl = document.getElementById('uai-modal-icon');
    if (iconEl) iconEl.textContent = opts.icon;

    const titleEl = document.getElementById('uai-modal-title')
;
    if (titleEl) titleEl.textContent = opts.title;

    // Badges
    const badgesContainer = document.getElementById('uai-modal
-badges');
    if (badgesContainer) {
      badgesContainer.innerHTML = (opts.badges || []).map(b =>
 {
        const cls = b.class || 'badge border text-secondary';
        return `<span class="${cls}">${escapeHtml(b.text)}</sp
an>`;
      }).join('');
    }

    // Panel & Type Badges in System Section
    const panelBadge = document.getElementById('uai-modal-pane
l-badge');
    const typeBadge = document.getElementById('uai-modal-type-
```

```
badge');
285     const customBadge = document.getElementById('uai-modal-cus
tom-badge');
286     const providerSelect = document.getElementById('uai-modal-
provider-select');
287     const modelInput = document.getElementById('uai-modal-mode
l-input');
288     const systemTextarea = document.getElementById('uai-modal-
system-instruction');
289     const systemStatus = document.getElementById('uai-modal-sy
stem-status');
290     const saveSystemBtn = document.getElementById('btn-uai-mod
al-save-system');
291     const resetSystemBtn = document.getElementById('btn-uai-mo
dal-reset-system');
292 
293     if (panelBadge) panelBadge.textContent = `Панель: #${opts.
panelId}`;
294     if (typeBadge) typeBadge.textContent = `Тип: ${opts.tableT
ype}`;
295     if (providerSelect && opts.provider) providerSelect.value
= opts.provider;
296     if (modelInput) modelInput.value = opts.model || '';
297 
298     // Metadata Grid
299     const metaContainer = document.getElementById('uai-modal-m
etadata-grid');
300     if (metaContainer) {
301       metaContainer.innerHTML = (opts.metadata || []).map(m =>
 {
302         const colSize = m.fullWidth ? 'col-sm-12' : 'col-sm-6'
;
303         let valHtml = '';
304         if (m.html) {
305           valHtml = m.html;
306         } else if (m.isCode) {
307           valHtml = `<span class="font-monospace text-info sma
ll" style="word-break: break-all;">${escapeHtml(m.value || '-'
```

```
)}</span>`;
        } else {
          valHtml = `<span class="text-body">${escapeHtml(m.va
lue || '-')}</span>`;
        }
        return `
          <div class="${colSize}">
            <strong class="text-secondary">${escapeHtml(m.labe
l)}:</strong> ${valHtml}
          </div>
        `;
      }).join('');
    }

    // Raw Content Block (скрыт по умолчанию)
    const rawContainer = document.getElementById('uai-modal-ra
w-container');
    const rawTitleEl = document.getElementById('uai-modal-raw-
title');
    const rawContentEl = document.getElementById('uai-modal-ra
w-content');
    if (rawContainer && rawTitleEl && rawContentEl) {
      if (opts.rawContent && typeof opts.rawContent === 'strin
g' && opts.rawContent.trim()) {
        rawTitleEl.textContent = opts.rawTitle;
        rawContentEl.textContent = opts.rawContent;
        rawContainer.style.display = 'block';
        rawContainer.classList.remove('d-none');
      } else {
        rawContainer.style.display = 'none';
        rawContainer.classList.add('d-none');
      }
    }

    // Загрузка системной инструкции для текущего типа / конкр
етной панели
    let currentTemplateData = null;
    const panelQuery = opts.panelId ? `?panel_id=${encodeURICo
```

```
mponent(opts.panelId)}` : '';

    async function loadSystemInstruction() {
      try {
        const data = await safeApiFetch(`/api/v1/diagnostics/p
rompts/${encodeURIComponent(opts.tableType)}${panelQuery}`, {}
, 4000);
        if (data) {
          currentTemplateData = data;
          if (systemTextarea) {
            systemTextarea.value = opts.systemInstruction || d
ata.system_instruction || '';
          }
          if (customBadge) {
            customBadge.style.display = data.is_customized ? '
inline-block' : 'none';
            if (data.is_customized) {
              customBadge.textContent = data.table_type === op
ts.panelId ? 'Панель (Кастом)' : 'Тип (Кастом)';
            }
          }
          if (systemStatus) {
            if (data.is_customized) {
              systemStatus.textContent = data.table_type === o
pts.panelId
                ? `Индивидуальная инструкция панели #${opts.pa
nelId}`
                : `Пользовательская системная инструкция типа
«${opts.tableType}»`;
            } else {
              systemStatus.textContent = `Стандартная системна
я инструкция: «${data.name || opts.tableType}»`;
            }
          }
        }
      } catch (e) {
        console.warn('[AIModalDialog] Ошибка загрузки системно
й инструкции:', e);
```

```
365       }
366     }
367
368     loadSystemInstruction();
369
370     // Сохранение системной инструкции прямо из модального окн
    а (приоритетно для panel_id)
371     if (saveSystemBtn) {
372       saveSystemBtn.onclick = async () => {
373         const instruction = systemTextarea ? systemTextarea.va
    lue : '';
374         try {
375           const promptTmpl = currentTemplateData?.prompt_templ
    ate || '{title}\n{subtitle}\n{metadata}\n{raw_data}';
376           const name = currentTemplateData?.name || (opts.pane
    lId ? `Панель ${opts.panelId}` : opts.tableType);
377
378           await safeApiFetch(`/api/v1/diagnostics/prompts/${en
    codeURIComponent(opts.tableType)}${panelQuery}`, {
379             method: 'POST',
380             headers: { 'Content-Type': 'application/json' },
381             body: JSON.stringify({
382               table_type: opts.tableType,
383               name: name,
384               system_instruction: instruction,
385               prompt_template: promptTmpl
386             })
387           }, 6000);
388
389           if (customBadge) {
390             customBadge.style.display = 'inline-block';
391             customBadge.textContent = opts.panelId ? 'Панель (
    Кастом)' : 'Кастомная';
392           }
393           if (systemStatus) {
394             systemStatus.textContent = opts.panelId
395               ? `Инструкция для панели #${opts.panelId} сохран
    ена!`
```

```
              : 'Системная инструкция успешно сохранена!';
            systemStatus.className = 'text-success small';
            setTimeout(() => {
              if (systemStatus) {
                systemStatus.textContent = opts.panelId
                  ? `Индивидуальная инструкция панели #${opts.panelId} активна`
                  : 'Пользовательская инструкция активна';
                systemStatus.className = 'text-secondary small';
              }
            }, 3000);
          }
          if (window.showToast) window.showToast('Системная инструкция обновлена', 'success');
        } catch (err) {
          alert(`Ошибка сохранения инструкции: ${err.message}`);
        }
      };
    }

    // Сброс системной инструкции к дефолтной
    if (resetSystemBtn) {
      resetSystemBtn.onclick = async () => {
        const resetTargetTitle = opts.panelId ? `панели «#${opts.panelId}»` : `«${opts.tableType}»`;
        if (!confirm(`Сбросить системную инструкцию для ${resetTargetTitle} к значению по умолчанию?`)) return;
        try {
          const data = await safeApiFetch(`/api/v1/diagnostics/prompts/${encodeURIComponent(opts.tableType)}/reset${panelQuery}`, {
            method: 'POST'
          }, 6000);
          if (data) {
            currentTemplateData = data;
            if (systemTextarea) systemTextarea.value = data.sy
```

```
stem_instruction || '';
            if (customBadge) customBadge.style.display = 'none
';
            if (systemStatus) {
              systemStatus.textContent = 'Инструкция сброшена
к системной по умолчанию';
              systemStatus.className = 'text-info small';
            }
          }
        } catch (err) {
          alert(`Ошибка сброса инструкции: ${err.message}`);
        }
      };
    }

    const refreshBtn = document.getElementById('btn-uai-modal-
refresh');
    if (refreshBtn) {
      refreshBtn.style.display = 'none';
      refreshBtn.onclick = () => executeDiagnostics(true);
    }

    // AI Diagnostic section
    const aiExplanation = document.getElementById('uai-modal-a
i-explanation');

    function renderPromptCallout() {
      if (!aiExplanation) return;
      aiExplanation.innerHTML = `
        <div class="p-3 text-center rounded uai-callout-box">
          <div class="text-secondary small mb-2">
            <i class="bi bi-info-circle me-1"></i>В базе знани
й WikiLLM пока нет сохранённого описания для «${escapeHtml(opt
s.title)}».
          </div>
          <button type="button" class="btn btn-sm btn-primary
d-inline-flex align-items-center gap-1.5" id="btn-uai-modal-ru
n-ai">
```

```
            <i class="bi bi-robot"></i> Запросить AI-анализ с системной инструкцией
          </button>
        </div>
      `;
      const runBtn = document.getElementById('btn-uai-modal-run-ai');
      if (runBtn) {
        runBtn.onclick = () => executeDiagnostics(false);
      }
    }

    function renderDiagnosticResult(data, isFromWikiLLM = false) {
      if (!aiExplanation) return;

      const isVerified = Boolean(isFromWikiLLM || data.is_verified || data.source === 'wikillm');
      const isHeuristic = data.source === 'heuristic';
      const isGemini = data.source === 'gemini';

      let sourceBadge = '';
      if (isVerified) {
        sourceBadge = `<div class="badge border border-success text-success p-1 px-2 mb-2 d-inline-flex align-items-center gap-1.5"><i class="bi bi-shield-check"></i> База знаний WikiLLM (L1 Exact Cache)</div>`;
      } else if (isGemini) {
        sourceBadge = `<div class="badge border border-info text-info p-1 px-2 mb-2 d-inline-flex align-items-center gap-1.5"><i class="bi bi-stars"></i> AI Экспертный анализ (Web Grounding)</div>`;
      } else {
        sourceBadge = `<div class="badge border border-secondary text-secondary p-1 px-2 mb-2 d-inline-flex align-items-center gap-1.5"><i class="bi bi-cpu"></i> Системный экспертный анализ (Телеметрия)</div>`;
      }
```

```
480
481       const actionBlock = isVerified ? `
482         <div class="p-2 mt-2 rounded d-flex align-items-center
 justify-content-between flex-wrap gap-2" id="uai-approval-con
tainer">
483           <div class="small text-secondary">
484             <i class="bi bi-database-check text-success me-1">
</i>Справочные данные из локальной базы WikiLLM.
485           </div>
486           <button type="button" class="btn btn-sm btn-primary
d-inline-flex align-items-center gap-1.5" id="btn-uai-improve-
ai">
487             <i class="bi bi-robot"></i> Запустить полный ИИ-ан
ализ записи
488           </button>
489         </div>
490       ` : `
491         <div class="p-2 mt-2 rounded d-flex align-items-center
 justify-content-between flex-wrap gap-2" id="uai-approval-con
tainer">
492           <div class="small text-secondary">
493             <i class="bi bi-patch-question me-1 text-warning">
</i>Результат анализа сформирован. Одобрить для базы знаний Wi
kiLLM?
494           </div>
495           <div class="d-flex align-items-center gap-1.5">
496             <button type="button" class="btn btn-sm btn-outlin
e-secondary d-inline-flex align-items-center gap-1.5" id="btn-
uai-reanalyze">
497               <i class="bi bi-arrow-clockwise"></i> Повторить
498             </button>
499             <button type="button" class="btn btn-sm btn-outlin
e-success d-inline-flex align-items-center gap-1.5" id="btn-ua
i-approve-wikillm">
500               <i class="bi bi-hand-thumbs-up"></i> Одобрить и
сохранить в WikiLLM
501             </button>
502           </div>
```

```
        </div>
      `;

      aiExplanation.innerHTML = `
        ${sourceBadge}
        <div class="mb-2"><strong class="text-info"><i class="bi bi-card-text me-1"></i>Назначение:</strong> ${escapeHtml(data.summary)}</div>
        <div class="mb-2"><strong class="text-secondary"><i class="bi bi-building me-1"></i>Разработчик / Категория:</strong> ${escapeHtml(data.developer || 'Неизвестен')} (${escapeHtml(data.category || 'Компонент')})</div>
        <div class="mb-2"><strong class="text-warning"><i class="bi bi-shield-lock me-1"></i>Оценка безопасности:</strong> ${escapeHtml(data.security_verdict)}</div>
        <div class="mb-2"><strong class="text-info"><i class="bi bi-speedometer2 me-1"></i>Влияние на ресурсы:</strong> ${escapeHtml(data.performance_impact)}</div>
        <div class="p-2 mb-2 rounded uai-recommendation-box">
          <strong class="text-warning"><i class="bi bi-lightbulb me-1"></i>Рекомендация:</strong> ${escapeHtml(data.recommendation)}
        </div>
        ${data.action_steps && data.action_steps.length > 0 ? `
          <h6 class="small text-uppercase text-muted fw-bold mb-1">Рекомендуемые действия:</h6>
          <ul class="mb-0 ps-3">
            ${data.action_steps.map(s => `<li>${escapeHtml(s)}</li>`).join('')}
          </ul>
        ` : ''}
        ${actionBlock}
      `;

      const improveBtn = document.getElementById('btn-uai-improve-ai');
      if (improveBtn) {
```

```
526         improveBtn.onclick = () => executeDiagnostics(true);
527       }
528
529       const reanalyzeBtn = document.getElementById('btn-uai-re
analyze');
530       if (reanalyzeBtn) {
531         reanalyzeBtn.onclick = () => executeDiagnostics(true);
532       }
533
534       const approveBtn = document.getElementById('btn-uai-appr
ove-wikillm');
535       if (approveBtn) {
536         approveBtn.onclick = () => approveInWikiLLM(data, opts
);
537       }
538
539       if (refreshBtn) {
540         refreshBtn.style.display = 'inline-flex';
541       }
542     }
543
544     // Проверка наличия знаний в WikiLLM перед запросом к LLM
545     async function checkWikiLLMKnowledge() {
546       if (!aiExplanation) return;
547       aiExplanation.innerHTML = `
548         <div class="d-flex align-items-center gap-2 py-2 text-
secondary">
549           <div class="spinner-border spinner-border-sm" role="
status"></div>
550           <span>Проверка локальной базы знаний WikiLLM...</spa
n>
551         </div>
552       `;
553
554       const metaDict = {};
555       (opts.metadata || []).forEach(m => {
556         if (m.label && (m.value !== undefined && m.value !== n
ull)) {
```

```
          metaDict[m.label] = m.value;
        }
      });

      try {
        const data = await safeApiFetch('/api/v1/diagnostics/explain', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({
            table_type: opts.tableType,
            panel_id: opts.panelId,
            system_instruction: systemTextarea ? systemTextarea.value : '',
            title: opts.title,
            subtitle: opts.subtitle,
            metadata: Object.assign({}, metaDict, opts.requestData || {}),
            raw_data: opts.rawContent || '',
            cache_only: true,
            web_search: false
          })
        }, 4000);

        if (data && (data.is_verified || data.source === 'wiki llm') && data.summary) {
          renderDiagnosticResult(data, true);
        } else {
          renderPromptCallout();
        }
      } catch (err) {
        renderPromptCallout();
      }
    }

    checkWikiLLMKnowledge();

    // Привязка кнопки настройки шаблона промпта
```

```
    const editPromptBtn = document.getElementById('btn-uai-modal-edit-prompt');
    if (editPromptBtn) {
      editPromptBtn.onclick = () => openPromptEditor(opts.tableType);
    }

    // Выполнение диагностического запроса к AI с передачей модели, провайдера и системной инструкции
    async function executeDiagnostics(forceRefresh = false) {
      const aiExplanation = document.getElementById('uai-modal-ai-explanation');
      if (!aiExplanation) return;

      const currentSysInstruction = systemTextarea ? systemTextarea.value.trim() : '';
      const currentModel = modelInput ? modelInput.value.trim() : '';
      const currentProvider = providerSelect ? providerSelect.value : 'gemini';

      aiExplanation.innerHTML = `
        <div class="d-flex align-items-center gap-2 py-2 text-info">
          <div class="spinner-border spinner-border-sm" role="status"></div>
          <span>Выполняется ${forceRefresh ? 'углубленный' : 'экспертный'} AI-анализ для «${escapeHtml(opts.title)}»...</span>
        </div>
      `;

      const metaDict = {};
      (opts.metadata || []).forEach(m => {
        if (m.label && (m.value !== undefined && m.value !== null)) {
          metaDict[m.label] = m.value;
        }
```

```
      });

      try {
        const data = await safeApiFetch('/api/v1/diagnostics/explain', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({
            table_type: opts.tableType,
            panel_id: opts.panelId,
            system_instruction: currentSysInstruction || undefined,
            model: currentModel || undefined,
            provider: currentProvider || undefined,
            title: opts.title,
            subtitle: opts.subtitle,
            metadata: Object.assign({}, metaDict, opts.requestData || {}),
            raw_data: opts.rawContent || '',
            web_search: true,
            force_refresh: Boolean(forceRefresh)
          })
        }, 30000);

        renderDiagnosticResult(data, false);
      } catch (err) {
        aiExplanation.innerHTML = `
          <div class="text-danger py-2 d-flex align-items-center justify-content-between flex-wrap gap-2">
            <div><i class="bi bi-exclamation-octagon me-1"></i>Ошибка получения AI-анализа: ${escapeHtml(err.message)}</div>
            <button type="button" class="btn btn-sm btn-outline-danger" id="btn-uai-modal-retry">
              <i class="bi bi-arrow-clockwise me-1"></i>Повторить
            </button>
          </div>
        `;
```

```
        const retryBtn = document.getElementById('btn-uai-modal-retry');
        if (retryBtn) {
          retryBtn.onclick = () => executeDiagnostics(forceRefresh);
        }
      }
    }

    // Сохранение и верификация знаний в WikiLLM
    async function approveInWikiLLM(aiData, options) {
      const approveBtn = document.getElementById('btn-uai-approve-wikillm');
      const container = document.getElementById('uai-approval-container');
      if (approveBtn) approveBtn.disabled = true;

      try {
        const payload = {
          canonical_key: aiData.canonical_key || undefined,
          table_type: options.tableType || 'generic',
          title: options.title || '',
          subtitle: options.subtitle || '',
          summary: aiData.summary || '',
          category: aiData.category || 'system',
          security_verdict: aiData.security_verdict || '',
          performance_impact: aiData.performance_impact || '',
          recommendation: aiData.recommendation || '',
          action_steps: aiData.action_steps || [],
          possible_causes: [],
          tags: [options.tableType || 'generic', options.panelId || 'panel_data', 'user_approved'],
          model_name: (modelInput ? modelInput.value.trim() : '') || 'gemini'
        };

        let respData;
        try {
```

```
          respData = await safeApiFetch('/api/v1/diagnostics/approve', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify(payload)
          }, 6000);
        } catch (_) {
          respData = await safeApiFetch('/api/windows/wikillm/approve', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify(payload)
          }, 6000);
        }

        if (respData && respData.success) {
          if (container) {
            container.className = 'p-2 mt-2 rounded uai-verified-box d-flex align-items-center justify-content-between flex-wrap gap-2';
            container.innerHTML = `
              <div class="small text-success fw-bold">
                <i class="bi bi-check-circle-fill me-1"></i>Знание верифицировано и сохранено в WikiLLM (${escapeHtml(respData.canonical_key)})
              </div>
              <span class="badge border border-success text-success">L1 Exact Cache O(1)</span>
            `;
          }
          if (window.showToast) {
            window.showToast(`Знание «${options.title}» зафиксировано в WikiLLM`, 'success');
          }
        } else {
          throw new Error(respData?.detail || 'Не удалось сохранить знание');
        }
```

```
      } catch (err) {
        if (approveBtn) approveBtn.disabled = false;
        alert(`Ошибка сохранения в WikiLLM: ${err.message}`);
      }
    }

    if (opts.autoRun === true) {
      executeDiagnostics();
    }

    // Custom Footer Actions
    const actionsContainer = document.getElementById('uai-modal-custom-actions');
    if (actionsContainer) {
      actionsContainer.innerHTML = '';
      (opts.actions || []).forEach(act => {
        const btn = document.createElement('button');
        btn.type = 'button';
        btn.className = `btn btn-sm ${act.class || 'btn-outline-secondary'}`;
        btn.innerHTML = `${act.icon ? `<i class="bi ${act.icon} me-1"></i>` : ''}${escapeHtml(act.label)}`;
        btn.onclick = () => act.onClick && act.onClick(opts);
        actionsContainer.appendChild(btn);
      });
    }

    // Открытие окна через Bootstrap Modal API
    if (window.bootstrap && window.bootstrap.Modal) {
      const bsModal = window.bootstrap.Modal.getOrCreateInstance(modalEl);
      bsModal.show();
    }
  }

  /**
   * Открывает диалог редактирования расширенного шаблона промпта для выбранного типа таблицы.
```

```
   * @param {string} tableType Тип таблицы / сущности
   */
  async function openPromptEditor(tableType) {
    const promptModalEl = ensurePromptModalElement();
    const typeSelect = document.getElementById('uai-prompt-type-select');
    const nameInput = document.getElementById('uai-prompt-name-input');
    const systemInput = document.getElementById('uai-prompt-system-input');
    const templateInput = document.getElementById('uai-prompt-template-input');
    const statusMsg = document.getElementById('uai-prompt-status-msg');
    const saveBtn = document.getElementById('btn-uai-prompt-save');
    const resetBtn = document.getElementById('btn-uai-prompt-reset');

    const targetType = normalizeTableType(tableType || 'generic');

    // Динамическая загрузка полного списка шаблонов промптов с бэкенда
    try {
      const templatesList = await safeApiFetch('/api/v1/diagnostics/prompts', {}, 4000);
      if (Array.isArray(templatesList) && templatesList.length > 0 && typeSelect) {
        typeSelect.innerHTML = templatesList.map(t => {
          const isCustom = t.is_customized ? ' □ ' : '';
          return `<option value="${escapeHtml(t.table_type)}">${escapeHtml(t.table_type)} (${escapeHtml(t.name || t.table_type)})${isCustom}</option>`;
        }).join('');
      }
    } catch (e) {
      console.warn('[AIModalDialog] Failed to fetch prompts li
```

```
st dynamically:', e);
767     }
768
769     if (typeSelect) {
770       let found = false;
771       for (let i = 0; i < typeSelect.options.length; i++) {
772         if (typeSelect.options[i].value === targetType) {
773           typeSelect.selectedIndex = i;
774           found = true;
775           break;
776         }
777       }
778       if (!found) {
779         const opt = document.createElement('option');
780         opt.value = targetType;
781         opt.textContent = `${targetType} (Пользовательский шаб
лон)`;
782         typeSelect.appendChild(opt);
783         typeSelect.value = targetType;
784       }
785     }
786
787     async function loadTemplate(tType) {
788       const norm = normalizeTableType(tType);
789       if (statusMsg) statusMsg.style.display = 'none';
790       try {
791         const data = await safeApiFetch(`/api/v1/diagnostics/p
rompts/${encodeURIComponent(norm)}`, {}, 4000);
792         if (data) {
793           if (nameInput) nameInput.value = data.name || '';
794           if (systemInput) systemInput.value = data.system_ins
truction || '';
795           if (templateInput) templateInput.value = data.prompt
_template || '';
796           if (statusMsg && data.is_customized) {
797             statusMsg.className = 'small mt-2 text-warning';
798             statusMsg.innerHTML = '<i class="bi bi-pencil-squa
re me-1"></i>Загружен пользовательский шаблон (отличается от с
```

```
истемного).';
            statusMsg.style.display = 'block';
          }
        }
      } catch (e) {
        console.error('Ошибка загрузки шаблона промпта:', e);
      }
    }

    if (typeSelect) {
      typeSelect.onchange = () => loadTemplate(typeSelect.valu
e);
    }

    if (saveBtn) {
      saveBtn.onclick = async () => {
        const currentType = normalizeTableType(typeSelect ? ty
peSelect.value : 'generic');
        try {
          await safeApiFetch(`/api/v1/diagnostics/prompts/${en
codeURIComponent(currentType)}`, {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({
              table_type: currentType,
              name: nameInput ? nameInput.value : '',
              system_instruction: systemInput ? systemInput.va
lue : '',
              prompt_template: templateInput ? templateInput.v
alue : ''
            })
          }, 6000);
          if (statusMsg) {
            statusMsg.className = 'small mt-2 text-success';
            statusMsg.innerHTML = '<i class="bi bi-check-circl
e me-1"></i>Шаблон промпта успешно сохранен!';
            statusMsg.style.display = 'block';
            setTimeout(() => { if (statusMsg) statusMsg.style.
```

```
display = 'none'; }, 3000);
          }
        } catch (err) {
          if (statusMsg) {
            statusMsg.className = 'small mt-2 text-danger';
            statusMsg.innerHTML = `<i class="bi bi-exclamation-triangle me-1"></i>Ошибка сохранения: ${escapeHtml(err.message)}`;
            statusMsg.style.display = 'block';
          }
        }
      };
    }

    if (resetBtn) {
      resetBtn.onclick = async () => {
        const currentType = normalizeTableType(typeSelect ? typeSelect.value : 'generic');
        if (!confirm(`Сбросить шаблон для «${currentType}» к значению по умолчанию?`)) return;
        try {
          const data = await safeApiFetch(`/api/v1/diagnostics/prompts/${encodeURIComponent(currentType)}/reset`, {
            method: 'POST'
          }, 6000);
          if (data) {
            if (nameInput) nameInput.value = data.name || '';
            if (systemInput) systemInput.value = data.system_instruction || '';
            if (templateInput) templateInput.value = data.prompt_template || '';
          }
          if (statusMsg) {
            statusMsg.className = 'small mt-2 text-info';
            statusMsg.innerHTML = '<i class="bi bi-arrow-counterclockwise me-1"></i>Шаблон сброшен к системному по умолчанию.';
            statusMsg.style.display = 'block';
```

```
            setTimeout(() => { if (statusMsg) statusMsg.style.display = 'none'; }, 3000);
          }
        } catch (err) {
          if (statusMsg) {
            statusMsg.className = 'small mt-2 text-danger';
            statusMsg.innerHTML = `<i class="bi bi-exclamation-triangle me-1"></i>Ошибка сброса: ${escapeHtml(err.message)}`;
            statusMsg.style.display = 'block';
          }
        }
      };
    }

    await loadTemplate(typeSelect ? typeSelect.value : targetType);

    if (window.bootstrap && window.bootstrap.Modal) {
      const bsPromptModal = window.bootstrap.Modal.getOrCreateInstance(promptModalEl);
      bsPromptModal.show();
    }
  }

  function ensurePromptModalElement() {
    let promptEl = document.getElementById(PROMPT_MODAL_ID);
    if (!promptEl) {
      promptEl = document.createElement('div');
      promptEl.id = PROMPT_MODAL_ID;
      promptEl.className = 'modal fade';
      promptEl.tabIndex = -1;
      promptEl.setAttribute('aria-hidden', 'true');
      promptEl.innerHTML = `
        <div class="modal-dialog modal-lg modal-dialog-centered modal-dialog-scrollable">
          <div class="modal-content shadow-lg">
            <div class="modal-header py-2 px-3">
```

```
              <div class="d-flex align-items-center gap-2">
                <span class="fs-5 text-warning">⚙</span>
                <h5 class="modal-title fw-bold text-warning mb
-0">Редактор шаблона промпта</h5>
              </div>
              <button type="button" class="btn-close" data-bs-
dismiss="modal" aria-label="Закрыть"></button>
            </div>
            <div class="modal-body py-3 px-3">
              <div class="row g-2 mb-3">
                <div class="col-sm-6">
                  <label class="form-label small text-secondar
y fw-bold mb-1">Тип таблицы / сущности</label>
                  <select class="form-select form-select-sm" i
d="uai-prompt-type-select">
                    <option value="user_account">user_account
(Учетные записи и сессии Windows)</option>
                    <option value="security_event">security_ev
ent (События безопасности Windows)</option>
                    <option value="process">process (Процессы
Windows)</option>
                    <option value="service">service (Службы Wi
ndows)</option>
                    <option value="software">software (Установ
ленное ПО)</option>
                    <option value="task">task (Задачи планиров
щика)</option>
                    <option value="startup">startup (Автозагру
зка)</option>
                    <option value="network">network (Сетевые с
оединения)</option>
                    <option value="registry">registry (Реестр
Windows)</option>
                    <option value="disk">disk (Диски и тома)</
option>
                    <option value="cpu">cpu (Процессор CPU)</o
ption>
                    <option value="gpu">gpu (Видеокарта GPU)</
```

```
option>
913                     <option value="file_event">file_event (Фай
ловые события)</option>
914                     <option value="website">website (Мониторин
г сайтов)</option>
915                     <option value="rag_doc">rag_doc (RAG База
знаний)</option>
916                     <option value="generic">generic (Общий шаб
лон)</option>
917                   </select>
918                 </div>
919                 <div class="col-sm-6">
920                   <label class="form-label small text-secondar
y fw-bold mb-1">Название шаблона</label>
921                   <input type="text" class="form-control form-
control-sm" id="uai-prompt-name-input" placeholder="Название ш
аблона">
922                 </div>
923               </div>
924
925               <div class="mb-3">
926                 <div class="d-flex justify-content-between ali
gn-items-center mb-1">
927                   <label class="form-label small text-secondar
y fw-bold mb-0">Системная инструкция (System Instruction)</lab
el>
928                   <span class="badge bg-secondary-subtle text-
secondary small" style="font-size: 0.72rem;">Роль и поведение
ИИ</span>
929                 </div>
930                 <textarea class="form-control form-control-sm
font-monospace" id="uai-prompt-system-input" rows="4" placehol
der="Роль, экспертный контекст и правила поведения модели...">
</textarea>
931                 <div class="form-text small text-secondary" st
yle="font-size: 0.75rem;">
932                   Системная инструкция задаёт контекст роли эк
сперта, ограничения и формат ответа ИИ для выбранного типа сущ
```

```
ности.
                </div>
              </div>

              <div class="mb-2">
                <div class="d-flex justify-content-between ali
gn-items-center mb-1">
                  <label class="form-label small text-secondar
y fw-bold mb-0">Шаблон текста промпта (Prompt Template)</label
>
                  <div class="d-flex gap-1" id="uai-prompt-var
iables-bar">
                    <button type="button" class="btn btn-xs bt
n-outline-info py-0 px-1 font-monospace" style="font-size: 0.7
5rem;" onclick="window.AIModalDialog.insertVariable('{title}')
">{title}</button>
                    <button type="button" class="btn btn-xs bt
n-outline-info py-0 px-1 font-monospace" style="font-size: 0.7
5rem;" onclick="window.AIModalDialog.insertVariable('{subtitle
}')">{subtitle}</button>
                    <button type="button" class="btn btn-xs bt
n-outline-info py-0 px-1 font-monospace" style="font-size: 0.7
5rem;" onclick="window.AIModalDialog.insertVariable('{metadata
}')">{metadata}</button>
                    <button type="button" class="btn btn-xs bt
n-outline-info py-0 px-1 font-monospace" style="font-size: 0.7
5rem;" onclick="window.AIModalDialog.insertVariable('{raw_data
}')">{raw_data}</button>
                  </div>
                </div>
                <textarea class="form-control form-control-sm
font-monospace" id="uai-prompt-template-input" rows="8" placeh
older="Текст промпта с плейсхолдерами..."></textarea>
                <div class="form-text small text-secondary" st
yle="font-size: 0.75rem;">
                  Используйте переменные <code>{title}</code>,
 <code>{subtitle}</code>, <code>{metadata}</code>, <code>{raw_
data}</code> для автоматической подстановки контекста строки т
```

аблицы.
```
                </div>
              </div>

              <div id="uai-prompt-status-msg" class="small mt-2" style="display: none;"></div>
            </div>
            <div class="modal-footer py-2 px-3 d-flex justify-content-between">
              <button type="button" class="btn btn-sm btn-outline-danger" id="btn-uai-prompt-reset">
                <i class="bi bi-arrow-counterclockwise me-1"></i>Сбросить по умолчанию
              </button>
              <div class="d-flex gap-2">
                <button type="button" class="btn btn-sm btn-secondary" data-bs-dismiss="modal">Закрыть</button>
                <button type="button" class="btn btn-sm btn-primary" id="btn-uai-prompt-save">
                  <i class="bi bi-check2 me-1"></i>Сохранить промпт
                </button>
              </div>
            </div>
          </div>
        </div>
      `;
      document.body.appendChild(promptEl);
    }
    return promptEl;
  }

  function insertVariable(varText) {
    const templateInput = document.getElementById('uai-prompt-template-input');
    if (!templateInput) return;
    const start = templateInput.selectionStart || 0;
    const end = templateInput.selectionEnd || 0;
```

```
    const val = templateInput.value;
    templateInput.value = val.substring(0, start) + varText +
val.substring(end);
    templateInput.focus();
    templateInput.selectionStart = templateInput.selectionEnd
= start + varText.length;
  }

  const modalApi = {
    show: show,
    openPromptEditor: openPromptEditor,
    insertVariable: insertVariable,
    escapeHtml: escapeHtml,
    normalizeTableType: normalizeTableType
  };

  // Экспорт под новым и обратным именами
  window.AIModalDialog = modalApi;
  window.AITableModal = modalApi;
})();

```

style.css

```
/**
 * ============================================================
==================
 * Process Name: Windows WebGUI - AI Modal Dialog Styles
 * ============================================================
==================
 * Description:
 *   Стили и визуальные эффекты универсального всплывающего ок
на ИИ-анализатора.
 *   Полная интеграция с системой тем оформления через CSS-пер
еменные.
 *
 * File: style.css
 * Project: ai-windows-admin
 * Package: apps.windows.api.webgui.ai_modal_dialog
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-12 08:10:00
 * ============================================================
==================
 */

#universal-ai-table-modal .modal-dialog,
#universal-ai-prompt-editor-modal .modal-dialog {
  max-width: 80vw;
  width: 80vw;
  height: 80vh;
  margin: auto;
}

#universal-ai-table-modal .modal-content,
#universal-ai-prompt-editor-modal .modal-content {
  height: 100%;
  max-height: 80vh;
  background-color: var(--modal-bg);
  color: var(--modal-text);
```

```
  border: 1px solid var(--modal-border);
  box-shadow: var(--shadow-lg);
  border-radius: var(--radius-md, 8px);
}

@media (max-width: 768px) {
  #universal-ai-table-modal .modal-dialog,
  #universal-ai-prompt-editor-modal .modal-dialog {
    max-width: 95vw;
    width: 95vw;
    height: 90vh;
  }
}

#universal-ai-table-modal .modal-header,
#universal-ai-prompt-editor-modal .modal-header {
  background-color: var(--surface-2);
  border-bottom: 1px solid var(--border-color);
  color: var(--text-color);
}

#universal-ai-table-modal .modal-footer,
#universal-ai-prompt-editor-modal .modal-footer {
  background-color: var(--surface-2);
  border-top: 1px solid var(--border-color);
  color: var(--text-color);
}

#uai-modal-ai-section,
.uai-ai-section {
  background-color: var(--surface-1);
  border: 1px solid var(--border-color);
  color: var(--text-color);
  border-radius: var(--radius-sm, 6px);
  transition: border-color 0.2s ease, box-shadow 0.2s ease;
}

#uai-modal-ai-section:hover,
```

```
.uai-ai-section:hover {
  border-color: var(--nav-active);
  box-shadow: var(--shadow-sm);
}

#uai-modal-raw-container,
.uai-raw-container {
  background-color: var(--surface-2);
  border: 1px solid var(--border-subtle, var(--border-color));
  color: var(--text-color);
  border-radius: var(--radius-sm, 6px);
}

#uai-modal-raw-content,
.uai-raw-content {
  color: var(--text-color);
  font-family: var(--font-mono, monospace);
}

#uai-approval-container {
  background-color: var(--surface-2);
  border: 1px solid var(--border-color);
  color: var(--text-color);
  border-radius: var(--radius-sm, 6px);
  transition: all 0.25s ease;
}

.uai-callout-box {
  background-color: var(--surface-2);
  border: 1px dashed var(--border-color);
  color: var(--text-color);
  border-radius: var(--radius-sm, 6px);
}

.uai-recommendation-box {
  background-color: var(--alert-warning-bg, var(--surface-2));
  border: 1px solid var(--alert-warning-border, var(--border-c
olor));
```

```
  color: var(--alert-warning-text, var(--text-color));
  border-radius: var(--radius-sm, 6px);
}

.uai-verified-box {
  background-color: var(--alert-success-bg, var(--surface-2));
  border: 1px solid var(--alert-success-border, var(--border-color));
  color: var(--alert-success-text, var(--text-color));
  border-radius: var(--radius-sm, 6px);
}

.btn-xs {
  padding: 0.1rem 0.35rem;
  font-size: 0.72rem;
  line-height: 1.2;
}

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/apps/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/apps/index.html" class=
"w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/apps
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-12 08:10:00
 * ================================================================
==================
-->

<!DOCTYPE html>
<html lang="ru">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-s
cale=1.0">
  <title>AI Windows Admin — Приложения (/apps)</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dis
t/css/bootstrap.min.css" rel="stylesheet">
```

```
  <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.10.0/font/bootstrap-icons.css" rel="stylesheet">
  <link href="/html/css/main.css?v=20261008_v1" rel="stylesheet">
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <script src="/html/js/toast.js?v=20261006_v10"></script>
  <script src="https://cdn.jsdelivr.net/npm/i18next@23.11.5/i18next.min.js"></script>
</head>
<body>

<div class="container-fluid mt-1 px-2" id="apps-interface">

  <!-- Navigation Bar (/tc, /apps) -->
  <div class="d-flex align-items-center justify-content-between mb-2 main-nav-container rounded px-2 py-1 flex-wrap gap-2">
    <!-- Левая часть: кнопка меню, брендинг и индикатор активного раздела -->
    <div class="d-flex align-items-center gap-2">
      <!-- Кнопка открытия левого выдвижного меню -->
      <button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5 px-2 py-1 rounded shadow-sm"
              id="apps-drawer-toggle-btn"
              type="button"
              data-bs-toggle="offcanvas"
              data-bs-target="#appsSideNavOffcanvas"
              aria-controls="appsSideNavOffcanvas"
              title="Открыть меню разделов"
              style="font-size: 0.82rem; font-weight: 500;">
        <i class="bi bi-list fs-6"></i>
        <span class="d-none d-md-inline">Меню</span>
      </button>

      <!-- Брендинг -->
      <span class="fw-bold d-flex align-items-center gap-1.5 text-primary fs-6 user-select-none" style="letter-spacing: 0.3px;">
```

```
        <i class="bi bi-cpu-fill fs-5"></i>
        <span class="d-none d-sm-inline">Test Computer</span>
      </span>
    </div>

    <!-- Top Menu Buttons — генерируются из tc_menu_config.jso
n -->
    <div class="d-flex align-items-center gap-1 flex-wrap"></d
iv>

    <!-- Controls (Model Badge, Model Test Gear, Settings) -->
    <div class="d-flex align-items-center gap-2 flex-shrink-0
ms-auto">
      <!-- Active AI Model Badge -->
      <span id="apps-model-badge" class="badge rounded-pill bg
-body-tertiary text-body border align-items-center gap-1.5 px-
3 py-1.5 shadow-sm" style="font-size: 0.8rem; cursor: default;
" title="Инициализация модели ИИ...">
        <i class="bi bi-cpu-fill text-info"></i>
        <span class="fw-semibold" id="apps-model-text">AI: Заг
рузка...</span>
      </span>

      <!-- Model Quick Test Dropdown Trigger & Popover -->
      <div class="dropdown" id="model-ping-dropdown-wrapper">
        <button class="btn btn-sm btn-outline-secondary d-flex
 align-items-center justify-content-center p-0 rounded-circle"

                id="btn-model-ping-gear"
                type="button"
                data-bs-toggle="dropdown"
                data-bs-auto-close="outside"
                aria-expanded="false"
                title="Проверка доступности модели"
                style="width: 30px; height: 30px;">
          <i class="bi bi-gear"></i>
        </button>
        <div class="dropdown-menu dropdown-menu-end shadow-lg
```

```
border border-secondary-subtle p-3" aria-labelledby="btn-model
-ping-gear" style="min-width: 360px; max-width: 460px; z-index
: 1060;">
          <div class="d-flex align-items-center justify-conten
t-between mb-2">
            <span class="fw-bold small d-flex align-items-cent
er gap-1.5 text-body">
              <i class="bi bi-cpu text-primary"></i>
              <span>Проверка доступности модели</span>
            </span>
          </div>
          <div class="mb-2">
            <input type="text" class="form-control form-contro
l-sm" id="model-ping-input" value="Проверка доступности модели
" placeholder="Введите сообщение...">
          </div>
          <div class="d-flex justify-content-end mb-2">
            <button class="btn btn-sm btn-primary d-flex align
-items-center gap-1.5 px-3" id="btn-model-ping-send" type="but
ton">
              <i class="bi bi-send-fill" id="model-ping-btn-ic
on"></i>
              <span id="model-ping-btn-text">Отправить</span>
            </button>
          </div>
          <div id="model-ping-response-container" style="displ
ay: none;">
            <label class="form-label small text-muted mb-1 fw-
semibold">Ответ модели:</label>
            <div id="model-ping-response" class="small p-2 rou
nded bg-body-tertiary border text-break font-monospace" style=
"max-height: 220px; overflow-y: auto; white-space: pre-wrap; f
ont-size: 0.8rem; user-select: text;"></div>
          </div>
        </div>
      </div>

      <!-- Quick Settings Gear Button -->
```

```
      <button class="btn btn-sm btn-outline-secondary d-flex a
lign-items-center justify-content-center p-0 rounded-circle" i
d="user-settings-gear-btn" data-bs-toggle="modal" data-bs-targ
et="#userSettingsModal" title="Настройки профиля и темы" style
="width: 30px; height: 30px;" data-i18n-title="userSettings.ti
tle">
        <i class="bi bi-sliders"></i>
      </button>
    </div>
  </div>

  <!-- Плавающий боковой ярлычок (вкладка) слева -->
  <button class="left-drawer-tab-trigger btn shadow"
          type="button"
          data-bs-toggle="offcanvas"
          data-bs-target="#appsSideNavOffcanvas"
          aria-controls="appsSideNavOffcanvas"
          title="Открыть меню навигации">
    <i class="bi bi-list fs-6"></i>
    <span class="left-drawer-tab-text">МЕНЮ</span>
  </button>

  <!-- Левая выдвижная вкладка / панель навигации (Offcanvas)
-->
  <div class="offcanvas offcanvas-start left-drawer-nav" tabin
dex="-1" id="appsSideNavOffcanvas" aria-labelledby="appsSideNa
vOffcanvasLabel">
    <div class="offcanvas-header border-bottom py-2 px-3">
      <h5 class="offcanvas-title d-flex align-items-center gap
-2 fs-6 fw-bold" id="appsSideNavOffcanvasLabel">
        <i class="bi bi-grid-fill text-primary"></i>
        <span>Приложения (/apps)</span>
      </h5>
      <button type="button" class="btn-close text-reset" data-
bs-dismiss="offcanvas" aria-label="Закрыть"></button>
    </div>
    <div class="offcanvas-body p-3 d-flex flex-column justify-
content-between" style="overflow-y: auto;">
```

```
      <div class="drawer-sections-wrapper">
        <div class="nav-drawer-group mb-3">
          <div class="nav-drawer-group-title d-flex align-items-center gap-1 mb-2">
            <i class="bi bi-grid-fill text-primary"></i>
            <span>Разделы Test Computer</span>
          </div>
          <div class="list-group list-group-flush rounded border" id="appsNavTabs"></div>
        </div>
      </div>

      <!-- Footer Drawer -->
      <div class="nav-drawer-footer border-top pt-2 mt-auto d-flex flex-column gap-2">
        <button class="btn btn-sm btn-outline-warning w-100 d-flex align-items-center justify-content-between py-2 px-3 rounded" id="btn-godmode" type="button" onclick="openGodModeFolder()" title="Открыть панель всех настроек Windows (God Mode)">
          <div class="d-flex align-items-center gap-2">
            <span class="fs-6">⚙</span>
            <span class="fw-medium text-body">God Mode (Все настройки)</span>
          </div>
          <i class="bi bi-box-arrow-up-right text-muted small"></i>
        </button>
        <button class="btn btn-sm btn-outline-secondary w-100 d-flex align-items-center gap-2 py-2 rounded" id="menu-editor-btn" type="button">
          <i class="bi bi-gear-fill"></i>
          <span class="fw-medium">Редактор меню</span>
        </button>
      </div>
    </div>
  </div>

  <!-- User / Quick Settings Modal -->
```

```
  <div class="modal fade" id="userSettingsModal" tabindex="-1" aria-labelledby="userSettingsModalLabel" aria-hidden="true">
    <div class="modal-dialog modal-dialog-centered">
      <div class="modal-content shadow border-secondary-subtle">
        <div class="modal-header border-secondary-subtle">
          <h5 class="modal-title d-flex align-items-center gap-2" id="userSettingsModalLabel">
            <i class="bi bi-gear-fill text-primary"></i>
            <span data-i18n="userSettings.title">Настройки интерфейса</span>
          </h5>
          <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Закрыть"></button>
        </div>
        <div class="modal-body p-3">
          <!-- Language Selection -->
          <div class="mb-3">
            <label class="form-label fw-semibold small d-flex align-items-center gap-2" for="apps-lang-selector">
              <i class="bi bi-translate text-info"></i>
              <span data-i18n="userSettings.language">Язык интерфейса</span>
            </label>
            <select class="form-select lang-selector" id="apps-lang-selector">
              <option value="ru-RU">Русский (ru-RU)</option>
              <option value="en-US">English (en-US)</option>
              <option value="he-IL">□ □ □ □ □ (he-IL)</option>
              <option value="uk-UA">Українська (uk-UA)</option>
            </select>
          </div>

          <!-- Theme Selection -->
          <div class="mb-2">
            <label class="form-label fw-semibold small d-flex align-items-center gap-2" for="apps-theme-selector">
```

```
              <i class="bi bi-palette text-warning"></i>
              <span data-i18n="userSettings.theme">Тема оформл
ения</span>
            </label>
            <select class="form-select theme-selector" id="app
s-theme-selector">
              <option value="system">  Системная</option>
              <optgroup label="  Светлые темы">
                <option value="light">  Светлая (Классическая)
</option>
                <option value="brick">  Кирпичная (Терракота)
</option>
              </optgroup>
              <optgroup label="  Тёмные темы">
                <option value="dark">  Тёмная (Контрастная)</
option>
                <option value="terminal">  Терминал (Matrix /
 CRT)</option>
              </optgroup>
            </select>
          </div>
        </div>
      </div>
    </div>
  </div>

  <!-- Menu Editor Modal -->
  <div class="modal fade" id="menuEditorModal" tabindex="-1" a
ria-labelledby="menuEditorModalLabel" aria-hidden="true">
    <div class="modal-dialog modal-xl modal-dialog-centered">
      <div class="modal-content bg-dark text-white">
        <div class="modal-header border-secondary">
          <h5 class="modal-title fw-bold" id="menuEditorModalL
abel">
            <i class="bi bi-gear-fill text-primary"></i> Редак
тор меню
          </h5>
          <button type="button" class="btn-close btn-close-whi
```

```
te" data-bs-dismiss="modal" aria-label="Закрыть"></button>
        </div>
        <div class="modal-body">
          <div class="alert alert-info small mb-3">
            <i class="bi bi-info-circle"></i> Настройте отобра
жение и порядок элементов меню. Элементы сгруппированы по пози
циям: сверху, сбоку и скрытые. Переключатели меняют размещение
, а стрелки и слайдеры — порядок элементов.
          </div>

          <div class="menu-editor-section">
            <div id="allMenuEditor">
              <div class="text-muted small p-3 text-center">За
грузка...</div>
            </div>
          </div>
        </div>
        <div class="modal-footer border-secondary">
          <button type="button" class="btn btn-secondary" data
-bs-dismiss="modal">Отмена</button>
          <button type="button" class="btn btn-primary" id="sa
veMenuConfig">
            <i class="bi bi-save"></i> Сохранить
          </button>
        </div>
      </div>
    </div>
  </div>

  <!-- Main Tab Content Container (Apps Only) -->
  <div class="tab-content" id="mainTabContent">
    <!-- About System Tab (Default Primary) -->
    <div id="tab-about-system" class="tab-pane fade show activ
e"></div>

    <!-- Chat Tab -->
    <div id="tab-chat" class="tab-pane fade"></div>

```

```
    <!-- Scenarios Tab -->
    <div id="tab-scenarios" class="tab-pane fade"></div>

    <!-- Network Terminal Tab -->
    <div id="tab-network" class="tab-pane fade"></div>

    <!-- Hardware Load Inspector Tab -->
    <div id="tab-hardware-load-inspector" class="tab-pane fade
"></div>

    <!-- Processes Load & Activity Inspector Tab -->
    <div id="tab-processes-load-inspector" class="tab-pane fad
e"></div>

    <!-- Windows System Administrator Tab -->
    <div id="tab-windows-admin" class="tab-pane fade"></div>

    <!-- Windows Focus Settings Tab -->
    <div id="tab-focus-settings" class="tab-pane fade"></div>

    <!-- System Control Center Tab (System Logs & Activity) --
>
    <div id="tab-system-control" class="tab-pane fade"></div>

    <!-- Post-Install Wizard Tab -->
    <div id="tab-post-install-wizard" class="tab-pane fade"></
div>

    <!-- Maintenance & Recovery Tab -->
    <div id="tab-maintenance-recovery" class="tab-pane fade"><
/div>

    <!-- System Log Viewer Tab -->
    <div id="tab-system-logs" class="tab-pane fade"></div>

    <!-- Software Audit Tab -->
    <div id="tab-software-audit" class="tab-pane fade"></div>

```

```
    <!-- Registry Viewer Tab -->
    <div id="tab-registry-viewer" class="tab-pane fade"></div>

    <!-- Microsoft Defender Security Tab -->
    <div id="tab-defender" class="tab-pane fade"></div>

    <!-- Startup Auditor Tab -->
    <div id="tab-startup-auditor" class="tab-pane fade"></div>

    <!-- Windows Backup & Libraries Manager Tab -->
    <div id="tab-windows-backup" class="tab-pane fade"></div>

    <!-- Hardware & Sensors Monitor Tab -->
    <div id="tab-hardware-monitor" class="tab-pane fade"></div
>

    <!-- Cloudflared Monitor Tab -->
    <div id="tab-cloudflared" class="tab-pane fade"></div>

    <!-- Google Cloud Monitor Tab -->
    <div id="tab-gcloud" class="tab-pane fade"></div>

    <!-- Website Intelligence Monitor Tab -->
    <div id="tab-website-monitor" class="tab-pane fade"></div>

    <!-- User Assistant Tab -->
    <div id="tab-user-assistant" class="tab-pane fade"></div>

    <!-- Helpdesk Tab -->
    <div id="tab-helpdesk" class="tab-pane fade"></div>

    <!-- Wikipedia Research Lab Tab -->
    <div id="tab-wikipedia-research" class="tab-pane fade"></d
iv>

    <!-- AI Software Transparency Scanner Tab -->
    <div id="tab-software-transparency" class="tab-pane fade">
</div>
```

```
322
323     <!-- Deep Software Scanner (Dormant & Traces) Tab -->
324     <div id="tab-software-scanner" class="tab-pane fade"></div
    >
325
326     <!-- User Directories and Storage Tab -->
327     <div id="tab-user-directories" class="tab-pane fade"></div
    >
328
329     <!-- Ninite Auto-Updater Tab -->
330     <div id="tab-ninite-updater" class="tab-pane fade"></div>
331
332     <!-- File Recovery (R-Studio) Tab -->
333     <div id="tab-file-recovery" class="tab-pane fade"></div>
334
335     <!-- Telemetry Research Tab -->
336     <div id="tab-telemetry-research" class="tab-pane fade"></d
    iv>
337
338     <!-- Process Leak Hunter Tab -->
339     <div id="tab-process-leaks" class="tab-pane fade"></div>
340
341     <!-- Behavioral Forensics Tab -->
342     <div id="tab-forensics" class="tab-pane fade"></div>
343
344     <!-- Kernel Quality & Throttling Tab -->
345     <div id="tab-throttling" class="tab-pane fade"></div>
346
347     <!-- Storage & Battery Wear Tab -->
348     <div id="tab-storage-wear" class="tab-pane fade"></div>
349
350     <!-- Peripherals & Wireless Network Tab -->
351     <div id="tab-peripherals" class="tab-pane fade"></div>
352
353     <!-- RAG Tab -->
354     <div id="tab-rag" class="tab-pane fade"></div>
355
356     <!-- PixelRAG Tab -->
```

```
357     <div id="tab-pixelrag" class="tab-pane fade"></div>
358 
359     <!-- Models & API Tab -->
360     <div id="tab-models" class="tab-pane fade"></div>
361 
362     <!-- AI Agents Tab -->
363     <div id="tab-agents" class="tab-pane fade"></div>
364 
365     <!-- Skills Tab -->
366     <div id="tab-skills" class="tab-pane fade"></div>
367 
368     <!-- MCP Servers Tab -->
369     <div id="tab-mcp" class="tab-pane fade"></div>
370 
371     <!-- Google User Desktop Tab -->
372     <div id="tab-google-desktop" class="tab-pane fade"></div>
373 
374     <!-- File History Search Tab -->
375     <div id="tab-file-history-search" class="tab-pane fade"></
    div>
376 
377     <!-- AI Benchmark Tab -->
378     <div id="tab-ai-benchmark" class="tab-pane fade"></div>
379 
380     <!-- Windows CLI Modular Applications Tabs -->
381     <div id="tab-storage-manager" class="tab-pane fade"></div>
382     <div id="tab-disk-speed" class="tab-pane fade"></div>
383     <div id="tab-boot-recovery" class="tab-pane fade"></div>
384     <div id="tab-servicing-integrity" class="tab-pane fade"></
    div>
385     <div id="tab-services-manager" class="tab-pane fade"></div
    >
386     <div id="tab-task-scheduler" class="tab-pane fade"></div>
387     <div id="tab-process-manager" class="tab-pane fade"></div>
388     <div id="tab-firewall-manager" class="tab-pane fade"></div
    >
389     <div id="tab-security-acl" class="tab-pane fade"></div>
390     <div id="tab-performance-tracing" class="tab-pane fade"></
```

```
div>
391
392     <div id="tab-log-settings" class="tab-pane fade"></div>
393     <div id="tab-software-manager" class="tab-pane fade"></div
    >
394     <div id="tab-taskbar" class="tab-pane fade"></div>
395     <div id="tab-taskbar-controller" class="tab-pane fade"></d
    iv>
396     <div id="tab-accounts-identity" class="tab-pane fade"></di
    v>
397     <div id="tab-power-lifecycle" class="tab-pane fade"></div>
398     <div id="tab-app-logs" class="tab-pane fade"></div>
399     <div id="tab-autolog" class="tab-pane fade"></div>
400     <div id="tab-system32-commands" class="tab-pane fade"></di
    v>
401   </div>
402 </div>
403
404 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist
    /js/bootstrap.bundle.min.js"></script>
405 <link rel="stylesheet" href="/html/ai_modal_dialog/style.css?v
    =20261012_v2">
406 <script src="/html/ai_modal_dialog/main.js?v=20261012_v1"></sc
    ript>
407 <script src="/html/js/window-api.js?v=20261008_v3"></script>
408 <script src="/html/js/chatService.js?v=20261008_v3"></script>
409 <script src="/html/js/appConfigEditor.js?v=20261008_v3"></scri
    pt>
410 <script type="module" src="/html/js/panel-loader.js?v=20261010
    _v1"></script>
411 <script type="module" src="/html/apps/main.js?v=20261010_v3"><
    /script>
412
413 </body>
414 </html>
415
```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Apps - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/apps/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/apps
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * apps/main.js — оркестратор страницы /tc (tc.ps1)
 * Архитектура: UI_ARCHITECTURE.md
 */

import { setupGlobalApi, setupThemeAndLang } from './modules/init-interface.js';
import { fetchAppsStatus, updateModelBadge } from './modules/status-manager.js';
import { initModelTester, sendModelPing } from './modules/model-tester.js';
import { APP_TAB_DEFS } from './modules/tabs-config.js';
```

```
import { applyTranslations } from '../js/i18n.js';
import { switchTab, loadTab, setupTabClicks } from '../js/tab-core.js';
import { loadPanelData, initPanels } from '../js/panel-loader.js';

// Глобальный экспорт
window.switchTab = switchTab;
window.switchToTab = switchTab;
window.sendModelPing = sendModelPing;
window.initModelTester = initModelTester;
window.loadPanelData = loadPanelData;
window.initPanels = initPanels;

// Карта: tabId → пути (из tabs-config, не генерируем)
const TAB_PATHS = Object.fromEntries(
  APP_TAB_DEFS.map(d => [d.tabId, { html: d.html, js: d.js }])
);

// Lazy-загрузка: вкладка грузится только при первом открытии
const loadedTabs = new Set();

async function lazyLoad(tabId) {
  const normId = tabId.startsWith('tab-') ? tabId : `tab-${tabId}`;
  if (loadedTabs.has(normId)) return;
  const paths = TAB_PATHS[normId];
  if (!paths) return;
  const v = Date.now();
  const name = normId.replace(/^tab-/, '');
  const cleanHtml = paths.html ? (paths.html.includes('?') ? paths.html.split('?')[0] : paths.html) : '';
  const cleanJs = paths.js ? (paths.js.includes('?') ? paths.js.split('?')[0] : paths.js) : '';
  try {
    await loadTab(name, `${cleanHtml}?v=${v}`, cleanJs ? `${cleanJs}?v=${v}` : '');
    loadedTabs.add(normId);
```

```
    applyTranslations();
    const pane = document.getElementById(normId);
    if (pane) {
      initPanels(pane);
    }
  } catch (err) {
    console.error(`[apps/main.js] Ошибка загрузки вкладки ${normId}:`, err);
    loadedTabs.delete(normId);
    if (window.toast) {
      window.toast.error('Ошибка загрузки', `Не удалось загрузить раздел ${name}: ${err.message}`);
    }
  }
}

const _switchTab = switchTab;
export async function switchAppTab(tabId) {
  if (!tabId) return;
  const normId = tabId.startsWith('tab-') ? tabId : `tab-${tabId}`;
  await lazyLoad(normId);
  _switchTab(normId);
}
window.switchTab = switchAppTab;
window.switchToTab = switchAppTab;

// ── МЕНЮ ──────────────────────────────────────────────────────────

/**
 * Создает DOM-элемент иконки для кнопок и редактора (поддерживает bi-* и эмодзи)
 * @param {string} iconStr
 * @param {string} defaultIcon
 * @returns {HTMLElement}
 */
function createIconElement(iconStr, defaultIcon = '□') {
```

```
  const icon = iconStr || defaultIcon;
  if (/^bi-[a-z0-9-]+$/.test(icon) || icon.startsWith('bi-')) {
    const i = document.createElement('i');
    i.className = `bi ${icon}`;
    return i;
  }
  const span = document.createElement('span');
  span.textContent = icon;
  return span;
}

/**
 * Обработчик прямого запуска программ (например R-Studio)
 * @param {Object} item
 * @param {HTMLElement} [iconEl]
 */
async function executeLaunchItem(item, iconEl = null) {
  const origContent = iconEl ? iconEl.innerHTML : '';
  if (iconEl) iconEl.innerHTML = '□ ';
  try {
    const res = await fetch('/api/recovery/launch', { method: 'POST' });
    const data = await res.json();
    if (res.ok && data.success) {
      if (window.toast) {
        window.toast.success('R-Studio запущена', data.message || 'Программа восстановления файлов запущена');
      }
    } else {
      if (window.toast) {
        window.toast.error('Ошибка запуска', data.detail || 'Не удалось запустить программу');
      }
    }
  } catch (err) {
    if (window.toast) {
      window.toast.error('Ошибка связи', err.message);
```

```
    }
  } finally {
    if (iconEl) iconEl.innerHTML = origContent;
  }
}

function getMenuTarget() {
  const p = location.pathname.toLowerCase();
  if (p.startsWith('/su') || location.search.includes('target=su')) {
    return 'su';
  }
  return 'tc';
}

async function buildMenu(appsMap = {}, customCfg = null) {
  let cfg = customCfg;
  const target = getMenuTarget();
  if (!cfg) {
    try {
      const r = await fetch(`/api/v1/menu/config?target=${target}&t=${Date.now()}`);
      if (r.ok) {
        cfg = await r.json();
      }
    } catch (e) {
      console.warn(`Не удалось загрузить /api/v1/menu/config?target=${target}, пробуем статический файл`, e);
    }
  }

  if (!cfg) {
    try {
      const staticFile = target === 'su' ? 'su_menu_config.json' : 'tc_menu_config.json';
      const r = await fetch(`/html/config_menues/${staticFile}?t=${Date.now()}`);
      if (r.ok) {
```

```
        cfg = await r.json();
      }
    } catch (e) {
      console.error('Ошибка загрузки статического файла конфиг
урации меню', e);
    }
  }

  if (!cfg || !cfg.menu) return null;

  const enabled = item => appsMap[item.id]?.enabled !== false;
  const sorted = arr => [...arr]
    .filter(x => x.visible !== false && enabled(x))
    .sort((a, b) => (a.order || 0) - (b.order || 0));

  // Верхнее меню
  const topEl = document.querySelector('.main-nav-container .d
-flex.gap-1.flex-wrap');
  if (topEl) {
    topEl.innerHTML = '';
    sorted(cfg.menu.topButtons || []).forEach(item => {
      const btn = document.createElement('button');
      btn.className = 'btn btn-sm btn-outline-primary d-flex a
lign-items-center gap-1.5 py-1 px-2 rounded';
      btn.type = 'button';
      btn.title = item.label;

      const iconEl = createIconElement(item.icon, 'bi-app');
      const labelSpan = document.createElement('span');
      labelSpan.className = 'd-none d-sm-inline';
      labelSpan.textContent = item.label;
      if (item.i18n) labelSpan.dataset.i18n = item.i18n;

      btn.append(iconEl, labelSpan);

      if (item.id === 'file_recovery' || item.action === 'laun
ch') {
        btn.dataset.action = 'launch';
```

```
        btn.onclick = async (e) => {
          e.preventDefault();
          e.stopPropagation();
          await executeLaunchItem(item, iconEl);
        };
      } else {
        btn.dataset.tab = item.tab;
      }

      topEl.appendChild(btn);
    });
  }

  // Боковое меню
  const navEl = document.getElementById('appsNavTabs');
  if (navEl) {
    navEl.innerHTML = '';
    sorted(cfg.menu.sidebarItems || []).forEach(item => {
      const btn = document.createElement('button');
      btn.className = 'list-group-item list-group-item-action
d-flex align-items-center gap-2 py-2 px-3';
      btn.type = 'button';

      const iconEl = createIconElement(item.icon, '□ ');
      iconEl.classList.add('fs-6');
      const label = document.createElement('span');
      label.className = 'fw-medium';
      label.textContent = item.label;
      if (item.i18n) label.dataset.i18n = item.i18n;

      btn.append(iconEl, label);

      if (item.id === 'file_recovery' || item.action === 'laun
ch') {
        btn.dataset.action = 'launch';
        btn.title = 'Запуск программы восстановления файлов (R
-Studio)';
        btn.onclick = async (e) => {
```

```
          e.preventDefault();
          e.stopPropagation();
          await executeLaunchItem(item, iconEl);
        };
      } else {
        btn.dataset.tab = item.tab;
      }

      navEl.appendChild(btn);
    });
  }

  // Обновление активного состояния кнопок вкладок
  const activePane = document.querySelector('#mainTabContent > .tab-pane.active') || document.querySelector('.tab-pane.active');
  const activeTabId = activePane ? activePane.id : null;
  if (activeTabId) {
    document.querySelectorAll('[data-tab]').forEach(btn => {
      btn.classList.toggle('active', btn.dataset.tab === activeTabId);
    });
  }

  applyTranslations();
  return cfg;
}

export async function loadRequiredTabs(tabs = []) {
  for (const tab of tabs) {
    await lazyLoad(tab);
  }
}

// ── РЕДАКТОР МЕНЮ ──────────────────────────────────────────

function initMenuEditor(cfg, appsMap = {}) {
```

```
  const editorBtn = document.getElementById('menu-editor-btn')
;
  const modal = document.getElementById('menuEditorModal');
  const list = document.getElementById('allMenuEditor');
  const saveBtn = document.getElementById('saveMenuConfig');
  if (!editorBtn || !modal || !list || !cfg) return;

  let currentItems = [];
  let draggedGlobalIdx = null;

  function collectInitialItems() {
    return [
      ...(cfg.menu.topButtons || []).map(x => ({ ...x, positio
n: x.position || (x.visible === false ? 'hidden' : 'top') })),
      ...(cfg.menu.sidebarItems || []).map(x => ({ ...x, posit
ion: x.position || (x.visible === false ? 'hidden' : 'bottom')
 })),
    ];
  }

  function renderList() {
    list.innerHTML = '';

    const GROUPS = [
      {
        key: 'top',
        title: 'Сверху (Верхняя панель)',
        icon: 'bi-layout-text-window-reverse',
        badgeCls: 'bg-primary text-white',
        borderCls: 'border-primary-subtle'
      },
      {
        key: 'bottom',
        title: 'Сбоку (Боковое меню)',
        icon: 'bi-layout-sidebar-inset',
        badgeCls: 'bg-info text-dark',
        borderCls: 'border-info-subtle'
      },
```

```
      {
        key: 'hidden',
        title: 'Скрыто (Не отображаются)',
        icon: 'bi-eye-slash',
        badgeCls: 'bg-secondary text-white',
        borderCls: 'border-secondary-subtle'
      }
    ];

    GROUPS.forEach(grp => {
      const groupItems = currentItems.filter(x => x.position =
== grp.key);
      const grpTotal = groupItems.length;

      const section = document.createElement('div');
      section.className = `menu-group-section mb-3 p-2 rounded
 border ${grp.borderCls}`;
      section.style.background = 'var(--surface-1, rgba(255,25
5,255,0.03))';

      // Заголовок группы
      const header = document.createElement('div');
      header.className = 'd-flex align-items-center justify-co
ntent-between mb-2 px-1 pb-1 border-bottom border-secondary-su
btle';
      header.innerHTML = `
        <span class="fw-bold d-flex align-items-center gap-2 s
mall">
          <i class="bi ${grp.icon}"></i> ${grp.title}
        </span>
        <span class="badge ${grp.badgeCls} rounded-pill px-2 p
y-1">${grpTotal}</span>
      `;
      section.appendChild(header);

      const itemsContainer = document.createElement('div');
      itemsContainer.className = 'menu-group-items d-flex flex
-column gap-2';
```

```
      itemsContainer.dataset.group = grp.key;

      if (grpTotal === 0) {
        const empty = document.createElement('div');
        empty.className = 'text-muted small text-center py-2 f
st-italic';
        empty.textContent = 'Нет элементов в этой группе';
        itemsContainer.appendChild(empty);
      } else {
        groupItems.forEach((item, grpIdx) => {
          const globalIdx = currentItems.indexOf(item);
          const safeId = (item.id || `item_${globalIdx}`).repl
ace(/[^a-zA-Z0-9_-]/g, '');

          const div = document.createElement('div');
          div.className = 'menu-editor-item d-flex align-items
-center gap-2 p-2 rounded border border-secondary-subtle';
          div.dataset.id = item.id || `item_${globalIdx}`;
          div.dataset.globalIndex = globalIdx;
          div.dataset.groupIndex = grpIdx;
          div.draggable = true;

          // 1. Ручка перетаскивания (Drag Handle)
          const dragHandle = document.createElement('div');
          dragHandle.className = 'drag-handle-container text-m
uted px-1';
          dragHandle.title = 'Перетащите для изменения порядка
';
          dragHandle.style.cursor = 'grab';
          dragHandle.innerHTML = '<i class="bi bi-grip-vertica
l fs-5"></i>';

          // 2. Порядковый номер в группе
          const orderBadge = document.createElement('span');
          orderBadge.className = 'badge bg-secondary-subtle te
xt-light border px-2 py-1';
          orderBadge.textContent = `#${grpIdx + 1}`;
          orderBadge.style.minWidth = '38px';
```

```
360           orderBadge.style.textAlign = 'center';
361
362           // 3. Иконка элемента
363           const iconContainer = document.createElement('div');
364           iconContainer.className = 'item-icon d-flex align-it
ems-center justify-content-center px-1 text-center';
365           iconContainer.style.minWidth = '28px';
366           const iconEl = createIconElement(item.icon, '□ ');
367           iconContainer.appendChild(iconEl);
368
369           // 4. Информация об элементе
370           const info = document.createElement('div');
371           info.className = 'item-info flex-grow-1 min-w-0 px-1
';
372           const labelDiv = document.createElement('div');
373           labelDiv.className = 'fw-semibold text-truncate smal
l';
374           labelDiv.textContent = item.label;
375           const tabDiv = document.createElement('div');
376           tabDiv.className = 'text-muted text-truncate font-mo
nospace';
377           tabDiv.style.fontSize = '0.75rem';
378           tabDiv.textContent = item.tab;
379           info.append(labelDiv, tabDiv);
380
381           // 5. Управление порядком внутри группы
382           const orderCtrl = document.createElement('div');
383           orderCtrl.className = 'd-flex align-items-center gap
-1 me-2';
384           orderCtrl.style.width = '160px';
385           orderCtrl.style.flexShrink = '0';
386
387           const btnUp = document.createElement('button');
388           btnUp.type = 'button';
389           btnUp.className = 'btn btn-sm btn-outline-secondary
p-0 px-1';
390           btnUp.title = 'Переместить выше в группе';
391           btnUp.disabled = grpIdx === 0;
```

```
          btnUp.innerHTML = '<i class="bi bi-chevron-up"></i>'
;
          btnUp.addEventListener('click', () => {
            if (grpIdx > 0) {
              const prevItem = groupItems[grpIdx - 1];
              const curIdx = currentItems.indexOf(item);
              const targetIdx = currentItems.indexOf(prevItem)
;
              currentItems.splice(curIdx, 1);
              currentItems.splice(targetIdx, 0, item);
              renderList();
            }
          });

          const slider = document.createElement('input');
          slider.type = 'range';
          slider.className = 'form-range order-slider flex-gro
w-1 m-0';
          slider.min = '1';
          slider.max = String(grpTotal);
          slider.value = String(grpIdx + 1);
          slider.title = `Позиция в группе: ${grpIdx + 1} из $
{grpTotal}`;

          slider.addEventListener('input', (e) => {
            const targetGrpIdx = parseInt(e.target.value, 10)
- 1;
            if (targetGrpIdx !== grpIdx && targetGrpIdx >= 0 &
& targetGrpIdx < grpTotal) {
              const targetItem = groupItems[targetGrpIdx];
              const curIdx = currentItems.indexOf(item);
              const targetIdx = currentItems.indexOf(targetIte
m);
              currentItems.splice(curIdx, 1);
              currentItems.splice(targetIdx, 0, item);
              renderList();
            }
          });
```

```
423
424           const btnDown = document.createElement('button');
425           btnDown.type = 'button';
426           btnDown.className = 'btn btn-sm btn-outline-secondar
y p-0 px-1';
427           btnDown.title = 'Переместить ниже в группе';
428           btnDown.disabled = grpIdx === grpTotal - 1;
429           btnDown.innerHTML = '<i class="bi bi-chevron-down"><
/i>';
430           btnDown.addEventListener('click', () => {
431             if (grpIdx < grpTotal - 1) {
432               const nextItem = groupItems[grpIdx + 1];
433               const curIdx = currentItems.indexOf(item);
434               const targetIdx = currentItems.indexOf(nextItem)
;
435               currentItems.splice(curIdx, 1);
436               currentItems.splice(targetIdx, 0, item);
437               renderList();
438             }
439           });
440
441           orderCtrl.append(btnUp, slider, btnDown);
442
443           // 6. Переключатели позиции (Сверху / Сбоку / Скрыть
)
444           const group = document.createElement('div');
445           group.className = 'btn-group btn-group-sm flex-shrin
k-0';
446           group.setAttribute('role', 'group');
447
448           const radioName = `pos_grp_${globalIdx}_${safeId}`;
449           [
450             ['top', 'Сверху', 'btn-outline-primary'],
451             ['bottom', 'Сбоку', 'btn-outline-info'],
452             ['hidden', 'Скрыть', 'btn-outline-danger']
453           ].forEach(([val, text, cls]) => {
454             const inp = document.createElement('input');
455             inp.type = 'radio';
```

```
            inp.className = 'btn-check';
            inp.autocomplete = 'off';
            inp.name = radioName;
            inp.id = `pos_${val}_${globalIdx}_${safeId}`;
            inp.value = val;
            inp.checked = (item.position === val);
            inp.addEventListener('change', () => {
              if (inp.checked && item.position !== val) {
                item.position = val;
                item.visible = (val !== 'hidden');
                // Перемещаем элемент в конец новой группы в currentItems
                const curIdx = currentItems.indexOf(item);
                currentItems.splice(curIdx, 1);
                const lastOfGroup = currentItems.filter(x => x.position === val).pop();
                if (lastOfGroup) {
                  const lastIdx = currentItems.indexOf(lastOfGroup);
                  currentItems.splice(lastIdx + 1, 0, item);
                } else {
                  currentItems.push(item);
                }
                renderList();
              }
            });

            const lbl = document.createElement('label');
            lbl.className = `btn ${cls}`;
            lbl.setAttribute('for', inp.id);
            lbl.textContent = text;
            group.append(inp, lbl);
          });

          // Drag & Drop события
          div.addEventListener('dragstart', (e) => {
            draggedGlobalIdx = globalIdx;
            div.classList.add('dragging');
```

```
            e.dataTransfer.effectAllowed = 'move';
            e.dataTransfer.setData('text/plain', String(globalIdx));
          });

          div.addEventListener('dragend', () => {
            div.classList.remove('dragging');
            document.querySelectorAll('.menu-editor-item').forEach(el => el.classList.remove('drag-over'));
          });

          div.addEventListener('dragover', (e) => {
            e.preventDefault();
            e.dataTransfer.dropEffect = 'move';
            div.classList.add('drag-over');
          });

          div.addEventListener('dragleave', () => {
            div.classList.remove('drag-over');
          });

          div.addEventListener('drop', (e) => {
            e.preventDefault();
            div.classList.remove('drag-over');
            const fromGlobalIdx = parseInt(e.dataTransfer.getData('text/plain'), 10);
            if (!isNaN(fromGlobalIdx) && fromGlobalIdx !== globalIdx) {
              const [moved] = currentItems.splice(fromGlobalIdx, 1);
              moved.position = grp.key;
              moved.visible = (grp.key !== 'hidden');
              currentItems.splice(globalIdx, 0, moved);
              renderList();
            }
          });

          div.append(dragHandle, orderBadge, iconContainer, in
```

```
fo, orderCtrl, group);
          itemsContainer.appendChild(div);
        });
      }

      section.appendChild(itemsContainer);
      list.appendChild(section);
    });

    return currentItems;
  }

  editorBtn.addEventListener('click', () => {
    currentItems = collectInitialItems();
    renderList();
    new bootstrap.Modal(modal).show();
  });

  saveBtn.addEventListener('click', async () => {
    cfg.menu.topButtons = currentItems
      .filter(x => x.position === 'top')
      .map(({ position, ...x }, idx) => ({ ...x, visible: true
, order: idx + 1 }));

    cfg.menu.sidebarItems = currentItems
      .filter(x => x.position !== 'top')
      .map(({ position, ...x }, idx) => ({ ...x, visible: posi
tion === 'bottom', order: idx + 1 }));

    try {
      const target = getMenuTarget();
      const r = await fetch(`/api/v1/menu/config?target=${targ
et}`, {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify(cfg)
      });

```

```
      if (!r.ok) {
        throw new Error(r.statusText || `HTTP ${r.status}`);
      }

      // Немедленно переопределяем меню и структуру вкладок в интерфейсе
      await buildMenu(appsMap, cfg);

      const modalInstance = bootstrap.Modal.getInstance(modal);
      if (modalInstance) {
        modalInstance.hide();
      }

      // Проверяем, активна ли ещё текущая вкладка, если нет — переключаемся на первую доступную
      const currentTabEl = document.querySelector('#mainTabContent > .tab-pane.active') || document.querySelector('.tab-pane.active');
      const activeTabId = currentTabEl ? currentTabEl.id : '';
      const isStillInMenu = document.querySelector(`[data-tab="${activeTabId}"]`);
      if (!isStillInMenu) {
        const firstVisible = document.querySelector('[data-tab]');
        if (firstVisible?.dataset.tab) {
          await switchAppTab(firstVisible.dataset.tab);
        }
      }
    } catch (e) {
      window.showToast?.('Ошибка при сохранении меню: ' + e.message, 'danger') || alert('Ошибка при сохранении меню: ' + e.message);
    }
  });
}

// ── INIT ──────────────────────────────────────────────────────
```

```
——————————————————

async function init() {
  setupGlobalApi();
  await setupThemeAndLang();

  if (location.pathname.startsWith('/tc'))
    document.title = 'AI Windows Admin — Test Computer (/tc)';

  // Единый обработчик кликов
  setupTabClicks();

  // Статус + бейдж модели
  const statusData = await fetchAppsStatus();
  await updateModelBadge(statusData);
  const appsMap = statusData?.apps || {};

  // Инициализация быстрой проверки доступности модели
  initModelTester();

  // Меню из конфига
  const cfg = await buildMenu(appsMap).catch(() => null);
  initMenuEditor(cfg, appsMap);

  applyTranslations();

  // Активировать вкладку по hash, defaultTab из конфига или первую доступную
  const hash = location.hash.replace('#', '');
  const defaultTab = cfg?.settings?.defaultTab;
  const first = document.querySelector('#appsNavTabs [data-tab]') || document.querySelector('[data-tab]');
  const initialTab = hash || (defaultTab && document.querySelector(`[data-tab="${defaultTab}"]`) ? defaultTab : first?.dataset.tab || 'tab-about-system');
  await switchAppTab(initialTab);
}

```

```
document.readyState === 'loading'
  ? document.addEventListener('DOMContentLoaded', init)
  : init();

```

init-interface.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Modules - Init-Interface Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля init-inte
rface.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/apps/modules/init-inte
rface.js?v=20261006_v10" type="module"></script>
 *
 *   JavaScript Import:
 *     import { setupGlobalApi, setupThemeAndLang } from '/win
dows/api/webgui/apps/modules/init-interface.js';
 *
 * File: init-interface.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/apps/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * apps/modules/init-interface.js — Инициализация тем, языка и
 глобального окружения /apps
 */

import { initI18n, switchLang, switchLocale, applyTranslations
, normalizeLocaleTag, getCurrentLocale, getCurrentLang } from
```

```
'../../js/i18n.js';
import { initTheme, setTheme, getThemeMode, getResolvedTheme,
themeEngine, ThemeEngine } from '../../js/theme.js';

export function setupGlobalApi() {
  window.switchLang = switchLang;
  window.switchLocale = switchLocale;
  window.normalizeLocaleTag = normalizeLocaleTag;
  window.getCurrentLocale = getCurrentLocale;
  window.getCurrentLang = getCurrentLang;
  window.applyTranslations = applyTranslations;
  window.setTheme = setTheme;
  window.getThemeMode = getThemeMode;
  window.getResolvedTheme = getResolvedTheme;
  window.themeEngine = themeEngine;
  window.ThemeEngine = ThemeEngine;

  window.api = window.api || {
    async fetch(url, options = {}) {
      const response = await fetch(url, options);
      if (!response.ok) {
        let msg = response.statusText;
        try {
          const data = await response.json();
          if (data && data.detail) {
            if (typeof data.detail === 'string') {
              msg = data.detail;
            } else if (Array.isArray(data.detail)) {
              msg = data.detail.map(d => d.msg || JSON.stringi
fy(d)).join(', ');
            } else {
              msg = JSON.stringify(data.detail);
            }
          }
        } catch {}
        throw new Error(`${response.status} ${msg}`);
      }
      return response.json();
```

```
    }
  };

  window.appsStatusMap = window.appsStatusMap || {};
}

export async function setupThemeAndLang() {
  initTheme();

  // Инициализация i18n с автоматической проверкой URL парамет
ров (?region=ru-ru и т.д.)
  await initI18n();

  const langSel = document.getElementById('apps-lang-selector'
);
  if (langSel) {
    const curLocale = getCurrentLocale();
    const curLang = getCurrentLang();
    const matchLocale = Array.from(langSel.options).find(opt =
> opt.value.toLowerCase() === curLocale.toLowerCase());
    const matchLang = Array.from(langSel.options).find(opt =>
opt.value.toLowerCase() === curLang.toLowerCase());
    if (matchLocale) langSel.value = matchLocale.value;
    else if (matchLang) langSel.value = matchLang.value;

    langSel.addEventListener('change', (e) => {
      switchLocale(e.target.value);
    });
  }

  const themeSel = document.getElementById('apps-theme-selecto
r');
  if (themeSel) {
    themeSel.value = getThemeMode();
    themeSel.addEventListener('change', (e) => {
      setTheme(e.target.value);
    });
  }
```

```
}

```

model-tester.js

```
/**
 * ============================================================
 * Process Name: Windows Modules - Model-Tester Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом проверки модели в подсистеме /apps.
 *
 * Usage Examples:
 *   JavaScript Import:
 *     import { initModelTester, sendModelPing } from './modules/model-tester.js';
 *
 * File: model-tester.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/apps/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

export { initModelTester, sendModelPing } from '../../js/model-tester.js';

```

status-manager.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Modules - Status-Manager Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля status-ma
nager.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/apps/modules/status-ma
nager.js?v=20261001_v1" type="module"></script>
 *
 * File: status-manager.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/apps/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * apps/modules/status-manager.js — Управление состоянием прил
ожений и бейджем модели
 */

export async function fetchAppsStatus() {
  const isTcRoute = window.location.pathname.startsWith('/tc')
 || window.location.pathname.startsWith('/apps');
  const query = isTcRoute ? '?profile=tc' : '';

  // 1. Получаем статус приложений и AI-конфига по стандарту /
```

```
api/v1/apps/status
  let statusData = null;
  try {
    const resp = await fetch(`/api/v1/apps/status${query}`);
    if (resp.ok) {
      statusData = await resp.json();
    }
  } catch (err) {
    console.warn('[AppsHub] Failed to fetch /api/v1/apps/statu
s:', err);
  }

  if (statusData && statusData.apps) {
    window.appsStatusMap = statusData.apps;
  }

  // 2. Если в ответе нет секции ai / ai_providers_and_models_
configuration, явно запрашиваем активную модель через /api/v1/
chat/active-model
  if (!statusData?.ai && !statusData?.ai_providers_and_models_
configuration) {
    try {
      const modelResp = await fetch(`/api/v1/chat/active-model
${query}`);
      if (modelResp.ok) {
        const modelData = await modelResp.json();
        if (modelData && modelData.model) {
          statusData = statusData || {};
          statusData.ai = {
            provider: modelData.provider,
            model: modelData.model
          };
          statusData.config_file = modelData.config_file;
        }
      }
    } catch (err) {
      console.warn('[AppsHub] Could not fetch active model fro
m /api/v1/chat/active-model:', err);
```

```
    }
  }

  return statusData;
}

export async function updateModelBadge(statusData) {
  const modelBadgeContainer = document.getElementById('apps-model-badge') || document.getElementById('chat-model-badge');
  const modelBadgeText = document.getElementById('apps-model-text') || document.getElementById('chat-model-badge') || modelBadgeContainer;
  if (!modelBadgeContainer && !modelBadgeText) return;

  let prov = '';
  let mod = '';
  let cfgFile = statusData?.config_file || '';

  // 1. Приоритет: прямой запрос актуальной активной модели с сервера (/api/v1/chat/active-model)
  try {
    const isTcRoute = window.location.pathname.startsWith('/tc') || window.location.pathname.startsWith('/apps');
    const query = isTcRoute ? '?profile=tc' : '';
    const resp = await fetch(`/api/v1/chat/active-model${query}`);
    if (resp.ok) {
      const data = await resp.json();
      if (data && data.model) {
        prov = data.provider || '';
        mod = data.model;
        cfgFile = data.config_file || cfgFile;
      }
    }
  } catch (err) {
    console.warn('[AppsHub] Failed to fetch active model from /api/v1/chat/active-model:', err);
  }
```

```
92
93    // 2. Резервный парсинг конфигурации statusData
94    if (!mod && statusData) {
95      const ai = statusData.ai_providers_and_models_configuratio
n || statusData.ai;
96      if (ai) {
97        if (ai.default_provider) {
98          prov = String(ai.default_provider).toUpperCase();
99        }
100       if (ai.default_model) {
101         mod = ai.default_model;
102       }
103       if (!mod && ai.provider) {
104         const p = String(ai.provider).toLowerCase();
105         prov = String(ai.provider).toUpperCase();
106         mod = ai.model || (ai[p] && typeof ai[p] === 'object'
? ai[p].model : null) || (ai.providers && ai.providers[p] ? ai
.providers[p].model : null) || ai[`${p}_model_id`] || '';
107       } else if (!mod && ai.providers && typeof ai.providers =
== 'object') {
108         for (const [pk, pv] of Object.entries(ai.providers)) {
109           if (pv && pv.enabled) {
110             prov = pk.toUpperCase();
111             mod = pv.model || '';
112             break;
113           }
114         }
115       } else if (!mod && ai.model) {
116         mod = ai.model;
117         prov = prov || 'AI';
118       }
119     }
120   }
121
122   // 3. Очистка и форматирование отображаемой строки
123   if (mod && prov && mod.startsWith(`${prov.toLowerCase()}:`))
 {
124     mod = mod.substring(prov.length + 1);
```

```
  }

  const displayText = (prov && mod && mod !== 'default')
    ? `${prov}: ${mod}`
    : (mod && mod !== 'default' ? `${mod}` : (prov ? `${prov}` : 'AI: Не определена'));

  const titleText = (mod && mod !== 'default')
    ? `Используемая модель: ${mod} (Провайдер: ${prov || 'AI'}, Профиль: ${cfgFile || 'config.json'}) — нажмите для перехода к настройкам моделей`
    : `AI-модель не настроена — нажмите для перехода к настройкам моделей`;

  if (modelBadgeText) {
    modelBadgeText.textContent = displayText;
  }
  if (modelBadgeContainer) {
    modelBadgeContainer.title = titleText;
    modelBadgeContainer.style.display = 'inline-flex';
    if (!modelBadgeContainer._hasClick) {
      modelBadgeContainer._hasClick = true;
      modelBadgeContainer.style.cursor = 'pointer';
      modelBadgeContainer.addEventListener('click', () => {
        if (typeof window.switchTab === 'function') {
          window.switchTab('tab-models');
        }
      });
    }
  }
}
```

tab-manager.js

```
/**
 * ============================================================
 * Process Name: Windows Modules - Tab-Manager Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля tab-manager.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/apps/modules/tab-manager.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { switchTab, setupNavTabs, loadTabContent } from '/windows/api/webgui/apps/modules/tab-manager.js';
 *
 * File: tab-manager.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/apps/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// Реэкспорт для обратной совместимости
export const switchTab = (id) => window.switchTab?.(id);
export const setupNavTabs = () => {};
export const loadTabContent = async () => {};

```

tabs-config.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Modules - Tabs-Config Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля tabs-conf
ig.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/apps/modules/tabs-conf
ig.js?v=20261008_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { APP_TAB_DEFS, TC_EXCLUDES } from '/windows/api
/webgui/apps/modules/tabs-config.js';
 *
 * File: tabs-config.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/apps/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-12 08:20:00
 * ============================================================
==================
 */

export const APP_TAB_DEFS = [
  { id: 'about_system', tab: 'about-system', tabId: 'tab-about
-system', html: '/html/about_system_tab/index.html?v=20261012_
v1', js: '/html/about_system_tab/main.js?v=20261012_v1' },
  { id: 'scenarios', tab: 'scenarios', tabId: 'tab-scenarios',
 html: '/html/scenarios_tab/index.html?v=20261006_v8', js: '/h
tml/scenarios_tab/main.js?v=20261006_v8' },
```

```
  { id: 'chat', tab: 'chat', tabId: 'tab-chat', html: '/html/chat/index.html?v=20260923_v5', js: '/html/chat/main.js?v=20260923_v5' },
  { id: 'network_terminal', tab: 'network', tabId: 'tab-network', html: '/html/network_tab/index.html', js: '/html/network_tab/main.js' },
  { id: 'system_inspector', tab: 'hardware-load-inspector', tabId: 'tab-hardware-load-inspector', html: '/html/system_inspector_tab/index.html?v=20261012_v1', js: '/html/system_inspector_tab/main.js?v=20261012_v1' },
  { id: 'processes_load_inspector', tab: 'processes-load-inspector', tabId: 'tab-processes-load-inspector', html: '/html/processes_load_inspector_tab/index.html?v=20261012_v2', js: '/html/processes_load_inspector_tab/main.js?v=20261012_v2' },
  { id: 'windows_sysadmin', tab: 'windows-admin', tabId: 'tab-windows-admin', html: '/html/windows_admin_tab/index.html?v=20261010_v2', js: '/html/windows_admin_tab/main.js?v=20261010_v2' },
  { id: 'focus_settings', tab: 'focus-settings', tabId: 'tab-focus-settings', html: '/html/focus_settings_tab/index.html?v=20261006_v1', js: '/html/focus_settings_tab/main.js?v=20261006_v1' },
  { id: 'system32_commands', tab: 'system32-commands', tabId: 'tab-system32-commands', html: '/html/system32_commands_tab/index.html?v=20261006_v1', js: '/html/system32_commands_tab/main.js?v=20261006_v1' },
  { id: 'system_control_center', tab: 'system-control', tabId: 'tab-system-control', html: '/html/system_control_tab/index.html?v=20261012_v1', js: '/html/system_control_tab/main.js?v=20261012_v1' },
  { id: 'post_install_wizard', tab: 'post-install-wizard', tabId: 'tab-post-install-wizard', html: '/html/post_install_wizard_tab/index.html?v=20261006_v1', js: '/html/post_install_wizard_tab/main.js?v=20261006_v1' },
  { id: 'maintenance_recovery', tab: 'maintenance-recovery', tabId: 'tab-maintenance-recovery', html: '/html/maintenance_recovery_tab/index.html?v=20261006_v1', js: '/html/maintenance_recovery_tab/main.js?v=20261006_v1' },
```

```
  { id: 'system_log_viewer', tab: 'system-logs', tabId: 'tab-system-logs', html: '/html/system_logs_tab/index.html', js: '/html/system_logs_tab/main.js' },
  { id: 'software_audit', tab: 'software-audit', tabId: 'tab-software-audit', html: '/html/software_audit_tab/index.html', js: '/html/software_audit_tab/main.js' },
  { id: 'registry_viewer', tab: 'registry-viewer', tabId: 'tab-registry-viewer', html: '/html/registry_viewer_tab/index.html', js: '/html/registry_viewer_tab/main.js' },
  { id: 'windows_defender', tab: 'defender', tabId: 'tab-defender', html: '/html/defender_tab/index.html?v=20261006_v2', js: '/html/defender_tab/main.js?v=20261006_v2' },
  { id: 'windows_startup_auditor', tab: 'startup-auditor', tabId: 'tab-startup-auditor', html: '/html/startup_auditor_tab/index.html?v=20261006_v15', js: '/html/startup_auditor_tab/main.js?v=20261006_v15' },
  { id: 'windows_backup_manager', tab: 'windows-backup', tabId: 'tab-windows-backup', html: '/html/windows_backup_tab/index.html?v=20261004_v1', js: '/html/windows_backup_tab/main.js?v=20261004_v1' },
  { id: 'hardware_monitor', tab: 'hardware-monitor', tabId: 'tab-hardware-monitor', html: '/html/hardware_monitor_tab/index.html', js: '/html/hardware_monitor_tab/main.js' },
  { id: 'cloudflared_monitor', tab: 'cloudflared', tabId: 'tab-cloudflared', html: '/html/cloudflared_tab/index.html', js: '/html/cloudflared_tab/main.js' },
  { id: 'google_user_desktop', tab: 'google-desktop', tabId: 'tab-google-desktop', html: '/html/google_desktop_tab/index.html', js: '/html/google_desktop_tab/main.js' },
  { id: 'gcloud_monitor', tab: 'gcloud', tabId: 'tab-gcloud', html: '/html/gcloud_tab/index.html', js: '/html/gcloud_tab/main.js' },
  { id: 'website_monitor', tab: 'website-monitor', tabId: 'tab-website-monitor', html: '/html/website_monitor_tab/index.html', js: '/html/website_monitor_tab/main.js' },
  { id: 'user_assistant', tab: 'user-assistant', tabId: 'tab-user-assistant', html: '/html/user_assistant_tab/index.html', js: '/html/user_assistant_tab/main.js' },
```

```
49   { id: 'helpdesk', tab: 'helpdesk', tabId: 'tab-helpdesk', ht
ml: '/html/helpdesk_tab/index.html', js: '/html/helpdesk_tab/m
ain.js' },
50   { id: 'wikipedia_research', tab: 'wikipedia-research', tabId
: 'tab-wikipedia-research', html: '/html/wikipedia_research_ta
b/index.html', js: '/html/wikipedia_research_tab/main.js' },
51   { id: 'software_transparency_scanner', tab: 'software-transp
arency', tabId: 'tab-software-transparency', html: '/html/soft
ware_transparency_tab/index.html', js: '/html/software_transpa
rency_tab/main.js' },
52   { id: 'software_scanner', tab: 'software-scanner', tabId: 't
ab-software-scanner', html: '/html/software_scanner_tab/index.
html?v=20261010_v1', js: '/html/software_scanner_tab/main.js?v
=20261010_v1' },
53   { id: 'user_directories', tab: 'user-directories', tabId: 't
ab-user-directories', html: '/html/user_directories_tab/index.
html', js: '/html/user_directories_tab/main.js' },
54   { id: 'ninite_updater', tab: 'ninite-updater', tabId: 'tab-n
inite-updater', html: '/html/ninite_updater_tab/index.html?v=2
0260924_v2', js: '/html/ninite_updater_tab/main.js?v=20260924_
v2' },
55   { id: 'file_recovery', tab: 'file-recovery', tabId: 'tab-fil
e-recovery', html: '/html/file_recovery_tab/index.html', js: '
/html/file_recovery_tab/main.js' },
56   { id: 'file_history_ai_search', tab: 'file-history-search',
tabId: 'tab-file-history-search', html: '/html/file_history_ai
_search_tab/index.html?v=20260928_v1', js: '/html/file_history
_ai_search_tab/main.js?v=20260928_v1' },
57   { id: 'telemetry_research', tab: 'telemetry-research', tabId
: 'tab-telemetry-research', html: '/html/telemetry_research_ta
b/index.html?v=20260924_v1', js: '/html/telemetry_research_tab
/main.js?v=20260924_v1' },
58   { id: 'telemetry_config', tab: 'telemetry-config', tabId: 't
ab-telemetry-config', html: '/html/telemetry_config_tab/index.
html?v=20261010_v2', js: '/html/telemetry_config_tab/main.js?v
=20261010_v2' },
59   { id: 'autolog_manager', tab: 'autolog', tabId: 'tab-autolog
', html: '/html/telemetry_config_tab/index.html?v=20261010_v2'
```

```
, js: '/html/telemetry_config_tab/main.js?v=20261010_v2' },
  { id: 'process_leaks', tab: 'process-leaks', tabId: 'tab-process-leaks', html: '/html/process_leaks_tab/index.html?v=20260924_v1', js: '/html/process_leaks_tab/main.js?v=20260924_v1' },
  { id: 'forensics', tab: 'forensics', tabId: 'tab-forensics', html: '/html/forensics_tab/index.html?v=20261010_v2', js: '/html/forensics_tab/main.js?v=20261010_v2' },

  { id: 'throttling', tab: 'throttling', tabId: 'tab-throttling', html: '/html/throttling_tab/index.html?v=20260924_v1', js: '/html/throttling_tab/main.js?v=20260924_v1' },
  { id: 'storage_wear', tab: 'storage-wear', tabId: 'tab-storage-wear', html: '/html/storage_wear_tab/index.html?v=20260924_v1', js: '/html/storage_wear_tab/main.js?v=20260924_v1' },
  { id: 'peripherals', tab: 'peripherals', tabId: 'tab-peripherals', html: '/html/peripherals_tab/index.html?v=20260924_v1', js: '/html/peripherals_tab/main.js?v=20260924_v1' },
  { id: 'rag', tab: 'rag', tabId: 'tab-rag', html: '/html/rag_tab/index.html', js: '/html/rag_tab/main.js' },
  { id: 'pixelrag', tab: 'pixelrag', tabId: 'tab-pixelrag', html: '/html/pixelrag_tab/index.html', js: '/html/pixelrag_tab/main.js' },
  { id: 'models', tab: 'models', tabId: 'tab-models', html: '/html/models_tab/index.html?v=20261010_v1', js: '/html/models_tab/main.js?v=20261010_v1' },
  { id: 'agents', tab: 'agents', tabId: 'tab-agents', html: '/html/agents_tab/index.html', js: '/html/agents_tab/main.js' },
  { id: 'skills', tab: 'skills', tabId: 'tab-skills', html: '/html/skills_tab/index.html', js: '/html/skills_tab/main.js' },
  { id: 'mcp', tab: 'mcp', tabId: 'tab-mcp', html: '/html/mcp_tab/index.html', js: '/html/mcp_tab/main.js' },
  { id: 'ai_benchmark', tab: 'ai-benchmark', tabId: 'tab-ai-benchmark', html: '/html/ai_benchmark_tab/index.html?v=20260926_v1', js: '/html/ai_benchmark_tab/main.js?v=20260926_v1' },

  // Новые модули управления Windows CLI
  { id: 'storage_manager', tab: 'storage-manager', tabId: 'tab
```

```
-storage-manager', html: '/html/storage_manager_tab/index.html
?v=20261001_v1', js: '/html/storage_manager_tab/main.js?v=2026
1001_v1' },
  { id: 'disk_speed', tab: 'disk-speed', tabId: 'tab-disk-spee
d', html: '/html/disk_speed_tab/index.html?v=20261004_v1', js:
 '/html/disk_speed_tab/main.js?v=20261004_v1' },
  { id: 'boot_recovery', tab: 'boot-recovery', tabId: 'tab-boo
t-recovery', html: '/html/boot_recovery_tab/index.html?v=20261
001_v1', js: '/html/boot_recovery_tab/main.js?v=20261001_v1' }
,
  { id: 'servicing_integrity', tab: 'servicing-integrity', tab
Id: 'tab-servicing-integrity', html: '/html/servicing_integrit
y_tab/index.html?v=20261001_v1', js: '/html/servicing_integrit
y_tab/main.js?v=20261001_v1' },
  { id: 'services_manager', tab: 'services-manager', tabId: 't
ab-services-manager', html: '/html/services_manager_tab/index.
html?v=20261006_v1', js: '/html/services_manager_tab/main.js?v
=20261006_v1' },
  { id: 'task_scheduler', tab: 'task-scheduler', tabId: 'tab-t
ask-scheduler', html: '/html/task_scheduler_tab/index.html?v=2
0261006_v2', js: '/html/task_scheduler_tab/main.js?v=20261006_
v2' },
  { id: 'process_manager', tab: 'process-manager', tabId: 'tab
-process-manager', html: '/html/process_manager_tab/index.html
?v=20261004_v2', js: '/html/process_manager_tab/main.js?v=2026
1004_v2' },
  { id: 'firewall_manager', tab: 'firewall-manager', tabId: 't
ab-firewall-manager', html: '/html/firewall_manager_tab/index.
html?v=20261001_v1', js: '/html/firewall_manager_tab/main.js?v
=20261001_v1' },
  { id: 'security_acl', tab: 'security-acl', tabId: 'tab-secur
ity-acl', html: '/html/security_acl_tab/index.html?v=20261001_
v1', js: '/html/security_acl_tab/main.js?v=20261001_v1' },
  { id: 'performance_tracing', tab: 'performance-tracing', tab
Id: 'tab-performance-tracing', html: '/html/performance_tracin
g_tab/index.html?v=20261001_v1', js: '/html/performance_tracin
g_tab/main.js?v=20261001_v1' },
  { id: 'log_settings', tab: 'log-settings', tabId: 'tab-log-s
```

```
ettings', html: '/html/log_settings_tab/index.html?v=20261008_v1', js: '/html/log_settings_tab/main.js?v=20261008_v1' },
  { id: 'software_manager', tab: 'software-manager', tabId: 'tab-software-manager', html: '/html/software_manager_tab/index.html?v=20261004_v2', js: '/html/software_manager_tab/main.js?v=20261004_v2' },
  { id: 'taskbar_controller', tab: 'taskbar-controller', tabId: 'tab-taskbar-controller', html: '/html/taskbar_tab/index.html?v=20261010_v2', js: '/html/taskbar_tab/main.js?v=20261010_v2' },
  { id: 'taskbar', tab: 'taskbar', tabId: 'tab-taskbar', html: '/html/taskbar_tab/index.html?v=20261010_v2', js: '/html/taskbar_tab/main.js?v=20261010_v2' },
  { id: 'accounts_identity', tab: 'accounts-identity', tabId: 'tab-accounts-identity', html: '/html/accounts_identity_tab/index.html?v=20261008_v1', js: '/html/accounts_identity_tab/main.js?v=20261008_v1' },
  { id: 'power_lifecycle', tab: 'power-lifecycle', tabId: 'tab-power-lifecycle', html: '/html/power_lifecycle_tab/index.html?v=20261008_v2', js: '/html/power_lifecycle_tab/main.js?v=20261008_v2' },
  { id: 'app_logs', tab: 'app-logs', tabId: 'tab-app-logs', html: '/html/app_logs_tab/index.html?v=20261010_v4', js: '/html/app_logs_tab/main.js?v=20261010_v4' },
];

export const TC_EXCLUDES = new Set([
  'cloudflared_monitor',
  'gcloud_monitor',
  'website_monitor',
  'user_assistant',
  'wikipedia_research',
  'helpdesk'
]);

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - App Logs Intelligence Ta
b
================================================================
===============
Description:
  Клиентский веб-интерфейс комплексного анализа, аудита, поиск
а и потокового
  мониторинга внутренних журналов программы AI-Windows-Admin с
трого из
  каталога %APPDATA%\AI-Windows-Admin\logs (log.json, info.log
, errors.log, fastapi.log и др.).
  Включает глубокую интеграцию с AIModalDialog, динамический T
imeline график,
  кластеризацию инцидентов, экспорт (CSV/JSON/Markdown) и живо
й терминал.
  Полная поддержка темной, светлой (NightView), кирпичной и те
рминальной тем через CSS-переменные.

Usage Examples:
  Web Route:
    /html/app_logs_tab/index.html
    /windows/api/webgui/app_logs_tab/index.html

  Tab Frame Integration:
    <div id="tab-app-logs" class="tab-pane fade"></div>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/app_logs_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 19:37:00
================================================================
```

```
==========
-->
<style>
  .app-logs-container {
    color: var(--text-color, #18181b);
    padding: 0.25rem 0.5rem;
  }
  .apl-compact-header {
    background: var(--surface-1, #f6f3ec);
    border: 1px solid var(--border-color, #cfc6b8);
    border-radius: var(--radius-md, 8px);
    padding: 0.55rem 0.85rem;
    margin-bottom: 0.5rem;
    color: var(--text-color, #18181b);
  }
  .apl-metric-pill {
    background: var(--surface-2, #eee9e0);
    border: 1px solid var(--border-color, #cfc6b8);
    border-radius: 6px;
    padding: 0.25rem 0.55rem;
    font-size: 0.76rem;
    display: inline-flex;
    align-items: center;
    gap: 0.35rem;
    color: var(--text-color, #18181b);
  }
  .apl-panel {
    background: var(--surface-1, #f6f3ec) !important;
    border: 1px solid var(--border-color, #cfc6b8) !important;
    border-radius: 8px;
    overflow: hidden;
    color: var(--text-color, #18181b);
  }
  .apl-panel-header {
    background: var(--surface-2, #eee9e0) !important;
    border-bottom: 1px solid var(--border-color, #cfc6b8) !imp
ortant;
    padding: 0.55rem 0.9rem;
```

```
    font-weight: 600;
    font-size: 0.84rem;
    color: var(--nav-active, #1d4ed8);
    display: flex;
    align-items: center;
    justify-content: space-between;
  }
  .apl-table {
    width: 100%;
    margin-bottom: 0;
    color: var(--text-color, #18181b);
    font-size: 0.8rem;
    border-collapse: separate;
    border-spacing: 0;
  }
  .apl-table th {
    background: var(--surface-2, #eee9e0);
    color: var(--text-muted, #3f3f46);
    border-bottom: 1.5px solid var(--border-color, #cfc6b8);
    font-weight: 600;
    text-transform: uppercase;
    font-size: 0.7rem;
    letter-spacing: 0.04em;
    padding: 0.5rem 0.65rem;
    position: sticky;
    top: 0;
    z-index: 2;
  }
  .apl-table td {
    border-bottom: 1px solid var(--border-subtle, rgba(51, 65,
85, 0.15));
    padding: 0.5rem 0.65rem;
    vertical-align: top;
    color: var(--text-color, #18181b);
  }
  .apl-table tbody tr:hover td {
    background: var(--nav-hover-bg, rgba(56, 189, 248, 0.08));
    cursor: pointer;
```

```
  }
  .badge-apl-crit { background: #dc2626 !important; color: #fff !important; }
  .badge-apl-err { background: #ef4444 !important; color: #fff !important; }
  .badge-apl-warn { background: #f59e0b !important; color: #000 !important; }
  .badge-apl-inf { background: #0284c7 !important; color: #fff !important; }
  .badge-apl-deb { background: var(--surface-2, #64748b) !important; color: var(--text-muted, #3f3f46) !important; border: 1px solid var(--border-color, #cfc6b8) !important; }

  .apl-search-input, .apl-select {
    background: var(--surface-1, #f6f3ec) !important;
    border: 1px solid var(--border-color, #cfc6b8) !important;
    color: var(--text-color, #18181b) !important;
  }
  .apl-search-input:focus, .apl-select:focus {
    border-color: var(--nav-active, #1d4ed8) !important;
    box-shadow: 0 0 0 0.2rem rgba(29, 78, 216, 0.15) !important;
  }
  .apl-input-addon {
    background: var(--surface-2, #eee9e0) !important;
    border: 1px solid var(--border-color, #cfc6b8) !important;
    color: var(--text-muted, #3f3f46) !important;
  }
  .apl-card {
    background: var(--surface-2, #eee9e0) !important;
    border: 1px solid var(--border-color, #cfc6b8) !important;
    color: var(--text-color, #18181b) !important;
    border-radius: 6px;
  }
  .apl-term {
    background: var(--surface-2, #eee9e0);
    color: var(--text-color, #18181b);
    font-family: var(--font-mono, ui-monospace, SFMono-Regular
```

```
, Menlo, Monaco, Consolas, monospace);
    font-size: 0.78rem;
    border: 1px solid var(--border-color, #cfc6b8);
    border-radius: 6px;
    padding: 0.75rem;
    min-height: 480px;
    max-height: 600px;
    overflow-y: auto;
    white-space: pre-wrap;
    word-break: break-all;
  }
  .apl-comp-badge {
    background: var(--surface-2, #eee9e0);
    border: 1px solid var(--border-color, #cfc6b8);
    color: var(--nav-active, #1d4ed8);
  }
  .apl-file-badge {
    cursor: pointer;
    transition: all 0.15s ease;
    background: var(--surface-2, #eee9e0);
    border: 1px solid var(--border-color, #cfc6b8);
    color: var(--text-color, #18181b);
    display: inline-flex;
    align-items: center;
    user-select: none;
  }
  .apl-file-badge:hover, .apl-file-badge.active {
    background-color: var(--nav-active, #1d4ed8) !important;
    color: #fff !important;
    border-color: var(--nav-active, #1d4ed8) !important;
  }
  .apl-file-badge.has-error {
    border-color: #ef4444 !important;
  }
  .apl-badge-clear-btn {
    cursor: pointer;
    opacity: 0.7;
    transition: opacity 0.15s ease, transform 0.15s ease, back
```

```
ground-color 0.15s ease;
    padding: 1px 4px;
    margin-left: 4px;
    border-radius: 4px;
    display: inline-flex;
    align-items: center;
    justify-content: center;
    color: #ef4444 !important;
  }
  .apl-badge-clear-btn:hover {
    opacity: 1;
    transform: scale(1.15);
    background-color: rgba(239, 68, 68, 0.2) !important;
  }
  .apl-file-badge.active .apl-badge-clear-btn {
    color: #fecaca !important;
  }
  .apl-file-badge.active .apl-badge-clear-btn:hover {
    color: #fff !important;
    background-color: rgba(239, 68, 68, 0.5) !important;
  }
  .apl-quick-preset-btn {
    border-radius: 20px;
    font-size: 0.74rem;
    padding: 0.2rem 0.6rem;
    border: 1px solid var(--border-color, #cfc6b8);
    background: var(--surface-2, #eee9e0);
    color: var(--text-color, #18181b);
    transition: all 0.15s ease;
  }
  .apl-quick-preset-btn:hover, .apl-quick-preset-btn.active {
    background: var(--nav-active, #1d4ed8);
    color: #fff;
    border-color: var(--nav-active, #1d4ed8);
  }
  .apl-nav-pills .nav-link {
    color: var(--text-muted, #3f3f46);
    font-size: 0.8rem;
```

```
    border-radius: 20px;
    padding: 0.25rem 0.75rem;
    transition: all 0.15s ease;
  }
  .apl-nav-pills .nav-link:hover {
    color: var(--nav-active, #1d4ed8);
    background: var(--nav-hover-bg, rgba(0,0,0,0.05));
  }
  .apl-nav-pills .nav-link.active {
    color: #fff !important;
    background: var(--nav-active, #1d4ed8) !important;
    font-weight: 600;
  }
  .apl-timeline-bar-item {
    transition: transform 0.1s ease, filter 0.1s ease;
    cursor: pointer;
  }
  .apl-timeline-bar-item:hover {
    transform: scaleY(1.05);
    filter: brightness(1.2);
  }
</style>

<div class="app-logs-container">
  <!-- Top Header Bar with Metrics & Actions -->
  <div class="apl-compact-header d-flex justify-content-between align-items-center flex-wrap gap-2">
    <!-- Title & Metric Pills -->
    <div class="d-flex align-items-center gap-2 flex-wrap">
      <div class="d-flex align-items-center gap-1.5 me-1">
        <span class="fs-5">📋</span>
        <span class="fw-bold text-body small">Внутренние логи программы (%APPDATA%\AI-Windows-Admin\logs)</span>
      </div>

      <!-- Quick Metrics Pills -->
      <div class="apl-metric-pill" title="Текущий активный файл логов">
```

```
241         <span class="text-muted small"><i class="bi bi-file-ea
    rmark-code text-info me-1"></i>Файл:</span>
242         <span class="fw-bold text-primary font-monospace text-
    truncate" id="apl-stat-filename" style="max-width: 140px;">log
    .json</span>
243       </div>
244       <div class="apl-metric-pill" title="Размер активного фай
    ла">
245         <span class="text-muted small"><i class="bi bi-hdd me-
    1"></i>Размер:</span>
246         <span class="fw-bold text-info font-monospace" id="apl
    -stat-size">0 KB</span>
247       </div>
248       <div class="apl-metric-pill" title="Записей загружено в
    таблицу">
249         <span class="text-muted small"><i class="bi bi-list-ol
     me-1"></i>Записей:</span>
250         <span class="fw-bold text-body font-monospace" id="apl
    -stat-loaded">0</span>
251       </div>
252       <div class="apl-metric-pill" title="Ошибки (ERROR / CRIT
    ICAL)">
253         <span class="text-muted small"><i class="bi bi-exclama
    tion-octagon-fill text-danger me-1"></i>Ошибки:</span>
254         <span class="fw-bold text-danger font-monospace" id="a
    pl-stat-errors">0</span>
255       </div>
256       <div class="apl-metric-pill" title="Предупреждения (WARN
    ING)">
257         <span class="text-muted small"><i class="bi bi-exclama
    tion-triangle-fill text-warning me-1"></i>Предупреждения:</spa
    n>
258         <span class="fw-bold text-warning font-monospace" id="
    apl-stat-warnings">0</span>
259       </div>
260     </div>
261
262     <!-- Actions Toolbar -->
```

```
    <div class="d-flex align-items-center gap-2 flex-wrap">
      <div class="form-check form-switch mb-0 d-inline-flex al
ign-items-center" style="font-size: 0.78rem;">
        <input class="form-check-input me-1" id="apl-live-swit
ch" role="switch" style="cursor: pointer;" type="checkbox"/>
        <label class="form-check-label text-muted small user-s
elect-none" for="apl-live-switch" title="Авто-обновление кажды
е 3 секунды">Live Stream</label>
      </div>
      <button class="btn btn-sm btn-primary rounded-pill px-2.
5 py-1 small d-flex align-items-center gap-1" id="btn-apl-refr
esh" style="font-size: 0.78rem;">
        <i class="bi bi-arrow-clockwise"></i> <span>Обновить</
span>
      </button>
      <button class="btn btn-sm btn-outline-warning rounded-pi
ll px-2.5 py-1 small d-flex align-items-center gap-1" id="btn-
apl-diagnose" style="font-size: 0.78rem;">
        <i class="bi bi-robot"></i> <span>AI Диагностика</span
>
      </button>

      <button class="btn btn-sm btn-outline-danger rounded-pil
l px-2.5 py-1 small d-flex align-items-center gap-1" id="btn-a
pl-clear-all" style="font-size: 0.78rem;" title="Очистить все
файлы логов программы">
        <i class="bi bi-trash3-fill"></i> <span>Очистить все л
оги</span>
      </button>

      <div class="dropdown">
        <button class="btn btn-sm btn-outline-secondary dropdo
wn-toggle rounded-pill px-2.5 py-1 small d-flex align-items-ce
nter gap-1" data-bs-toggle="dropdown" style="font-size: 0.78re
m;" type="button">
          <i class="bi bi-download"></i> <span>Экспорт</span>
        </button>
        <ul class="dropdown-menu dropdown-menu-end shadow bord
```

```
er border-secondary-subtle" style="font-size: 0.8rem;">
          <li><a class="dropdown-item" href="#" id="btn-apl-export-csv"><i class="bi bi-file-earmark-spreadsheet me-2 text-success"></i>Экспорт в CSV</a></li>
          <li><a class="dropdown-item" href="#" id="btn-apl-export-json"><i class="bi bi-file-earmark-code me-2 text-info"></i>Экспорт в JSON</a></li>
          <li><a class="dropdown-item" href="#" id="btn-apl-export-md"><i class="bi bi-file-earmark-text me-2 text-warning"></i>Отчет в Markdown (.md)</a></li>
          <li><hr class="dropdown-divider"/></li>
          <li><a class="dropdown-item" href="#" id="btn-apl-download-raw"><i class="bi bi-file-arrow-down me-2 text-primary"></i>Скачать сырой файл</a></li>
          <li><hr class="dropdown-divider"/></li>
          <li><a class="dropdown-item text-danger" href="#" id="btn-apl-clear"><i class="bi bi-trash3 me-2"></i>Очистить активный лог-файл</a></li>
          <li><a class="dropdown-item text-danger fw-bold" href="#" id="btn-apl-menu-clear-all"><i class="bi bi-trash3-fill me-2"></i>Очистить ВСЕ логи программы</a></li>
        </ul>
      </div>
    </div>
  </div>

  <!-- Available Log Files Selector Bar with Error Badges & Clear Buttons -->
  <div class="apl-panel mb-2 p-2">
    <div class="d-flex align-items-center justify-content-between gap-2 flex-wrap" id="apl-files-bar">
      <div class="d-flex align-items-center gap-2 flex-wrap">
        <span class="text-muted small fw-semibold"><i class="bi bi-folder2-open me-1 text-info"></i>Файлы логов:</span>
        <div class="d-flex align-items-center gap-1.5 flex-wrap" id="apl-files-badges-container">
          <div class="spinner-border spinner-border-sm text-info" role="status"></div>
```

```
        </div>
      </div>
      <button class="btn btn-outline-danger btn-sm rounded-pill py-0.5 px-2 ms-auto small d-flex align-items-center gap-1" id="btn-apl-clear-all-inline" style="font-size: 0.74rem;" title="Очистить все файлы логов">
        <i class="bi bi-trash3"></i> <span>Очистить все логи</span>
      </button>
    </div>
  </div>

  <!-- Quick Filter Presets Bar -->
  <div class="apl-panel mb-2 p-2">
    <div class="d-flex align-items-center gap-1.5 flex-wrap">
      <span class="text-muted small fw-semibold me-1"><i class="bi bi-funnel me-1 text-warning"></i>Быстрые пресеты:</span>
      <button type="button" class="btn apl-quick-preset-btn active" data-preset="all">Все записи</button>
      <button type="button" class="btn apl-quick-preset-btn" data-preset="errors">  Только ошибки</button>
      <button type="button" class="btn apl-quick-preset-btn" data-preset="warnings_errors">  Предупреждения и ошибки</button>
      <button type="button" class="btn apl-quick-preset-btn" data-preset="windows">  Windows API / TC</button>
      <button type="button" class="btn apl-quick-preset-btn" data-preset="ai">  AI & LLM</button>
      <button type="button" class="btn apl-quick-preset-btn" data-preset="fastapi">  FastAPI & Core</button>
      <button type="button" class="btn apl-quick-preset-btn" data-preset="hour">  За 1 час</button>
      <button type="button" class="btn apl-quick-preset-btn" data-preset="day">  За 24 часа</button>
    </div>
  </div>

  <!-- Filter & View Mode Navigation -->
```

```
  <div class="apl-panel mb-2 p-2">
    <div class="d-flex justify-content-between align-items-center flex-wrap gap-1 mb-2 pb-1 border-bottom border-secondary-subtle">
      <!-- Mode Tabs -->
      <ul class="nav nav-pills apl-nav-pills gap-1" id="apl-mode-tabs">
        <li class="nav-item">
          <button class="nav-link active" data-mode="table">
            <i class="bi bi-table me-1"></i> Таблица записей
          </button>
        </li>
        <li class="nav-item">
          <button class="nav-link" data-mode="audit">
            <i class="bi bi-shield-check me-1 text-warning"></i> Аудит и AI Анализ
          </button>
        </li>
        <li class="nav-item">
          <button class="nav-link" data-mode="timeline">
            <i class="bi bi-bar-chart-line me-1 text-info"></i> Аналитика и График
          </button>
        </li>
        <li class="nav-item">
          <button class="nav-link" data-mode="tail">
            <i class="bi bi-terminal me-1 text-success"></i> Live Tail Terminal
          </button>
        </li>
        <li class="nav-item">
          <button class="nav-link" data-mode="raw">
            <i class="bi bi-file-text me-1"></i> Сырой файл
          </button>
        </li>
      </ul>
      <!-- Active File Path Indicator -->
      <div class="small text-muted font-monospace text-truncat
```

```
e d-none d-md-block" id="apl-full-path-label" style="max-width
: 400px;">
        %APPDATA%\AI-Windows-Admin\logs
      </div>
    </div>

    <!-- Filter Inputs Grid -->
    <div class="row g-1.5 align-items-center">
      <div class="col-lg-4 col-md-4">
        <div class="input-group input-group-sm">
          <span class="input-group-text apl-input-addon py-0 p
x-2"><i class="bi bi-search" style="font-size: 0.75rem;"></i><
/span>
          <input class="form-control form-control-sm apl-searc
h-input py-1" id="apl-search-input" placeholder="Поиск по текс
ту сообщения, стеку, компоненту..." style="font-size: 0.8rem;"
 type="text"/>
        </div>
      </div>
      <div class="col-lg-2 col-md-2 col-6">
        <select class="form-select form-select-sm apl-select p
y-1" id="apl-level-select" style="font-size: 0.8rem;">
          <option value="">Все уровни</option>
          <option value="ERRORS">⚠ Только ошибки (ERROR/CRITIC
AL)</option>
          <option value="WARNINGS_ERRORS">⚠ Ошибки + Предупреж
дения</option>
          <option value="CRITICAL">Critical</option>
          <option value="ERROR">Error</option>
          <option value="WARNING">Warning</option>
          <option value="INFO">Info</option>
          <option value="DEBUG">Debug</option>
        </select>
      </div>
      <div class="col-lg-2 col-md-2 col-6">
        <select class="form-select form-select-sm apl-select p
y-1" id="apl-component-select" style="font-size: 0.8rem;">
          <option value="">Все модули/компоненты</option>
```

```
        </select>
      </div>
      <div class="col-lg-1.5 col-md-2 col-6">
        <select class="form-select form-select-sm apl-select p
y-1" id="apl-time-preset-select" style="font-size: 0.8rem;">
          <option value="">Всё время</option>
          <option value="15m">15 минут</option>
          <option value="1h">1 час</option>
          <option value="6h">6 часов</option>
          <option value="24h">24 часа</option>
        </select>
      </div>
      <div class="col-lg-1.5 col-md-1 col-6">
        <select class="form-select form-select-sm apl-select p
y-1" id="apl-limit-select" style="font-size: 0.8rem;">
          <option value="100">100 шт</option>
          <option selected="" value="200">200 шт</option>
          <option value="500">500 шт</option>
          <option value="1000">1000 шт</option>
        </select>
      </div>
      <div class="col-lg-1 col-md-1 col-12 text-end">
        <button class="btn btn-sm btn-outline-secondary w-100
rounded py-1" id="btn-apl-reset-filters" style="font-size: 0.7
8rem;">
          <i class="bi bi-x-circle me-1"></i> Сброс
        </button>
      </div>
    </div>
  </div>

  <!-- Main View Container -->
  <div class="apl-panel p-2">
    <!-- 1. Table View -->
    <div id="apl-view-table">
      <div class="table-responsive" style="max-height: 560px;
overflow-y: auto;">
        <table class="table apl-table">
```

```
          <thead>
            <tr>
              <th style="width: 145px;">Время</th>
              <th style="width: 85px;">Уровень</th>
              <th style="width: 130px;">Компонент</th>
              <th>Сообщение / Данные</th>
              <th class="text-end" style="width: 100px;">Дейст
вия</th>
            </tr>
          </thead>
          <tbody id="apl-table-body">
            <tr>
              <td class="text-center text-muted py-4" colspan=
"5">
                <div class="spinner-border spinner-border-sm t
ext-info me-2" role="status"></div>
                Загрузка внутренних логов приложения...
              </td>
            </tr>
          </tbody>
        </table>
      </div>
    </div>

    <!-- 2. Audit & AI Diagnostics View -->
    <div class="d-none p-2" id="apl-view-audit" style="max-hei
ght: 560px; overflow-y: auto;">
      <!-- Health Score Card -->
      <div class="row g-2 mb-3">
        <div class="col-md-4">
          <div class="card apl-card p-3 text-center" id="panel
-app-logs-health">
            <span class="text-muted small text-uppercase fw-se
mibold">Индекс стабильности (Health Score)</span>
            <div class="display-6 fw-bold my-1" id="apl-audit-
health-score">--%</div>
            <div class="progress" style="height: 6px;">
              <div class="progress-bar bg-success" id="apl-aud
```

```
it-health-bar" style="width: 0%;"></div>
            </div>
          </div>
        </div>
        <div class="col-md-8">
          <div class="card apl-card p-3 h-100 d-flex flex-colu
mn justify-content-center" id="panel-app-logs-summary">
            <span class="text-info fw-bold small mb-1"><i clas
s="bi bi-info-circle me-1"></i>Сводка инцидентов и первопричин
</span>
            <div class="small text-body" id="apl-audit-summary
-text">Нажмите «AI Диагностика» для запуска интеллектуального
анализа сбоев.</div>
          </div>
        </div>
      </div>

      <!-- AI Recommendations -->
      <div class="mb-3">
        <h6 class="text-warning fw-bold small text-uppercase m
b-2"><i class="bi bi-lightbulb me-1"></i>Рекомендации и устран
ение проблем</h6>
        <div id="apl-audit-recommendations-list">
          <div class="text-muted small p-2">Нет активных реком
ендаций.</div>
        </div>
      </div>

      <!-- Error Clusters Table -->
      <div>
        <h6 class="text-info fw-bold small text-uppercase mb-2
"><i class="bi bi-collection me-1"></i>Кластеры повторяющихся
инцидентов</h6>
        <div class="table-responsive">
          <table class="table apl-table">
            <thead>
              <tr>
                <th style="width: 70px;">Кол-во</th>
```

```
478                <th style="width: 80px;">Уровень</th>
479                <th style="width: 140px;">Посл. фиксация</th>
480                <th>Шаблон инцидента / Пример ошибки</th>
481                <th class="text-end" style="width: 90px;">Дейс
твия</th>
482              </tr>
483            </thead>
484            <tbody id="apl-audit-clusters-body">
485              <tr>
486                <td class="text-muted small text-center py-3"
colspan="5">Кластеры пока не сформированы.</td>
487              </tr>
488            </tbody>
489          </table>
490        </div>
491      </div>
492    </div>
493
494    <!-- 3. Timeline & Analytics View -->
495    <div class="d-none p-3" id="apl-view-timeline" style="max-
height: 560px; overflow-y: auto;">
496      <h6 class="text-info fw-bold small text-uppercase mb-2">
<i class="bi bi-graph-up me-1"></i>Распределение логов во врем
ени (Timeline — нажмите на столбец для фильтрации)</h6>
497      <div class="card apl-card p-3 mb-3" id="panel-app-logs-t
imeline">
498        <div class="d-flex align-items-end gap-1.5" id="apl-ti
meline-bars" style="height: 160px; overflow-x: auto; padding-b
ottom: 20px;">
499          <!-- Dynamic bars rendered here -->
500        </div>
501        <div class="d-flex justify-content-between small text-
muted mt-2 border-top border-secondary-subtle pt-1">
502          <span><span class="badge bg-danger me-1">■</span>Оши
бки (Critical/Error)</span>
503          <span><span class="badge bg-warning text-dark me-1">
■</span>Предупреждения (Warning)</span>
504          <span><span class="badge bg-primary me-1">■</span>Ин
```

```
формационные события (Info/Debug)</span>
        </div>
      </div>

      <!-- Top Components Breakdown -->
      <h6 class="text-info fw-bold small text-uppercase mb-2"><i class="bi bi-diagram-3 me-1"></i>Топ активных модулей приложения</h6>
      <div class="row g-2" id="apl-top-components-container">
        <!-- Injected dynamically -->
      </div>
    </div>

    <!-- 4. Live Tail Terminal View -->
    <div class="d-none p-1" id="apl-view-tail">
      <div class="d-flex justify-content-between align-items-center mb-1 px-1 flex-wrap gap-2">
        <span class="small text-muted"><i class="bi bi-terminal-fill me-1 text-success"></i>Поток последних строк в реальном времени:</span>
        <div class="d-flex align-items-center gap-2">
          <input class="form-control form-control-sm apl-search-input py-0 px-2" id="apl-tail-filter" placeholder="Фильтр потока..." style="font-size: 0.75rem; width: 140px;" type="text" />
          <div class="form-check form-check-inline small text-muted mb-0">
            <input checked="" class="form-check-input" id="apl-tail-autoscroll" type="checkbox"/>
            <label class="form-check-label" for="apl-tail-autoscroll">Авто-прокрутка</label>
          </div>
          <button class="btn btn-xs btn-outline-secondary py-0 px-2 small" id="btn-apl-clear-terminal">Очистить экран</button>
        </div>
      </div>
      <div class="apl-term" id="apl-tail-terminal">Подключение
```

```
 к лог-потоку...</div>
    </div>

    <!-- 5. Raw File View -->
    <div class="d-none p-1" id="apl-view-raw">
      <div class="d-flex justify-content-between align-items-center mb-1 px-1">
        <span class="small text-muted"><i class="bi bi-file-code me-1 text-info"></i>Просмотр файла целиком:</span>
        <div class="d-flex align-items-center gap-1">
          <button class="btn btn-xs btn-outline-info py-0 px-2 small" id="btn-apl-copy-raw">
            <i class="bi bi-clipboard me-1"></i>Копировать всё
          </button>
          <button class="btn btn-xs btn-outline-danger py-0 px-2 small" id="btn-apl-clear-raw" title="Очистить данный файл лога">
            <i class="bi bi-trash3 me-1"></i>Очистить файл
          </button>
        </div>
      </div>
      <pre class="apl-term" id="apl-raw-content" style="max-height: 560px;">Загрузка сырого содержимого файла...</pre>
    </div>
  </div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows App Logs Tab - Main Controller Script
 * ============================================================
==================
 * Description:
 *   Клиентский контроллер продвинутого анализатора внутренних
 логов программы
 *   AI-Windows-Admin строго из %APPDATA%\AI-Windows-Admin\log
s.
 *   Включает глубокую интеграцию с AIModalDialog (WikiLLM L1
Cache + AI Root Cause),
 *   интерактивный Timeline-график, быстрые пресеты фильтрации
, экспорт (CSV/JSON/MD)
 *   и высокопроизводительный Live Tail Terminal.
 *   Полная адаптивность к светлой, кирпичной, темной и термин
альной темам.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script type="module" src="/html/app_logs_tab/main.js?v
=20261010_v3"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/app_logs_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-12 08:20:00
 * ============================================================
==================
 */

let state = {
  currentFile: 'log.json',
```

```
  currentMode: 'table',
  liveActive: false,
  liveTimer: null,
  records: [],
  files: [],
  overview: null,
  stats: null,
  selectedEntry: null,
  activePreset: 'all',
};

/**
 * Инициализация вкладки анализатора логов
 */
export async function initAppLogsTab() {
  bindEvents();
  await loadOverviewAndFiles();
  await refreshCurrentView();
}

/**
 * Привязка событий элементов управления
 */
function bindEvents() {
  // Переключение режимов отображения (таблица, аудит, график, tail, raw)
  document.querySelectorAll('#apl-mode-tabs .nav-link').forEach(btn => {
    btn.addEventListener('click', (e) => {
      document.querySelectorAll('#apl-mode-tabs .nav-link').forEach(b => {
        b.classList.remove('active');
      });
      btn.classList.add('active');

      state.currentMode = btn.dataset.mode;
      switchViewMode(state.currentMode);
    });
```

```
  });

  // Быстрые пресеты фильтрации
  document.querySelectorAll('.apl-quick-preset-btn').forEach(b
tn => {
    btn.addEventListener('click', () => {
      document.querySelectorAll('.apl-quick-preset-btn').forEa
ch(b => b.classList.remove('active'));
      btn.classList.add('active');
      state.activePreset = btn.dataset.preset;
      applyQuickPreset(state.activePreset);
    });
  });

  // Кнопка обновления
  const btnRefresh = document.getElementById('btn-apl-refresh'
);
  if (btnRefresh) {
    btnRefresh.addEventListener('click', async () => {
      await loadOverviewAndFiles();
      await refreshCurrentView();
      if (window.toast) window.toast.info('Обновлено', 'Данные
 журналов успешно обновлены');
    });
  }

  // Кнопка AI диагностики
  const btnDiagnose = document.getElementById('btn-apl-diagnos
e');
  if (btnDiagnose) {
    btnDiagnose.addEventListener('click', async () => {
      const auditBtn = document.querySelector('#apl-mode-tabs
[data-mode="audit"]');
      if (auditBtn) auditBtn.click();
      await runAiDiagnostics();
    });
  }

```

```
  // Переключатель Live Stream
  const liveSwitch = document.getElementById('apl-live-switch');
  if (liveSwitch) {
    liveSwitch.addEventListener('change', (e) => {
      state.liveActive = e.target.checked;
      if (state.liveActive) {
        startLivePolling();
        if (window.toast) window.toast.success('Live Stream', 'Потоковое обновление активировано (3 сек)');
      } else {
        stopLivePolling();
        if (window.toast) window.toast.info('Live Stream', 'Потоковое обновление приостановлено');
      }
    });
  }

  // Фильтры с debounce
  let searchTimeout = null;
  const searchInput = document.getElementById('apl-search-input');
  if (searchInput) {
    searchInput.addEventListener('input', () => {
      clearTimeout(searchTimeout);
      searchTimeout = setTimeout(() => refreshCurrentView(), 300);
    });
  }

  const levelSelect = document.getElementById('apl-level-select');
  if (levelSelect) {
    levelSelect.addEventListener('change', () => refreshCurrentView());
  }

  const compSelect = document.getElementById('apl-component-se
```

```
lect');
  if (compSelect) {
    compSelect.addEventListener('change', () => refreshCurrent
View());
  }

  const timePresetSelect = document.getElementById('apl-time-p
reset-select');
  if (timePresetSelect) {
    timePresetSelect.addEventListener('change', () => refreshC
urrentView());
  }

  const limitSelect = document.getElementById('apl-limit-selec
t');
  if (limitSelect) {
    limitSelect.addEventListener('change', () => refreshCurren
tView());
  }

  // Сброс фильтров
  const btnReset = document.getElementById('btn-apl-reset-filt
ers');
  if (btnReset) {
    btnReset.addEventListener('click', () => {
      resetAllFilters();
    });
  }

  // Экспорт CSV
  const btnExportCsv = document.getElementById('btn-apl-export
-csv');
  if (btnExportCsv) {
    btnExportCsv.addEventListener('click', (e) => {
      e.preventDefault();
      triggerExport('csv');
    });
  }
```

```
156
157   // Экспорт JSON
158   const btnExportJson = document.getElementById('btn-apl-expor
t-json');
159   if (btnExportJson) {
160     btnExportJson.addEventListener('click', (e) => {
161       e.preventDefault();
162       triggerExport('json');
163     });
164   }
165
166   // Экспорт Markdown
167   const btnExportMd = document.getElementById('btn-apl-export-
md');
168   if (btnExportMd) {
169     btnExportMd.addEventListener('click', (e) => {
170       e.preventDefault();
171       triggerExport('markdown');
172     });
173   }
174
175   // Скачивание сырого файла
176   const btnDownloadRaw = document.getElementById('btn-apl-down
load-raw');
177   if (btnDownloadRaw) {
178     btnDownloadRaw.addEventListener('click', (e) => {
179       e.preventDefault();
180       window.open(`/api/v1/app_logs/download?file_name=${encod
eURIComponent(state.currentFile)}`, '_blank');
181     });
182   }
183
184   // Очистка активного лог-файла из выпадающего меню
185   const btnClear = document.getElementById('btn-apl-clear');
186   if (btnClear) {
187     btnClear.addEventListener('click', async (e) => {
188       e.preventDefault();
189       await clearSingleLog(state.currentFile);
```

```
    });
  }

  // Очистка активного лог-файла из Raw View
  const btnClearRaw = document.getElementById('btn-apl-clear-raw');
  if (btnClearRaw) {
    btnClearRaw.addEventListener('click', async (e) => {
      e.preventDefault();
      await clearSingleLog(state.currentFile);
    });
  }

  // Очистка ВСЕХ файлов логов (основная кнопка в шапке)
  const btnClearAll = document.getElementById('btn-apl-clear-all');
  if (btnClearAll) {
    btnClearAll.addEventListener('click', async (e) => {
      e.preventDefault();
      await clearAllLogs();
    });
  }

  // Очистка ВСЕХ файлов логов (кнопка в строке файлов)
  const btnClearAllInline = document.getElementById('btn-apl-clear-all-inline');
  if (btnClearAllInline) {
    btnClearAllInline.addEventListener('click', async (e) => {
      e.preventDefault();
      await clearAllLogs();
    });
  }

  // Очистка ВСЕХ файлов логов (из выпадающего меню)
  const btnMenuClearAll = document.getElementById('btn-apl-menu-clear-all');
  if (btnMenuClearAll) {
    btnMenuClearAll.addEventListener('click', async (e) => {
```

```
      e.preventDefault();
      await clearAllLogs();
    });
  }

  // Кнопка очистки терминала
  const btnClearTerm = document.getElementById('btn-apl-clear-terminal');
  if (btnClearTerm) {
    btnClearTerm.addEventListener('click', () => {
      const term = document.getElementById('apl-tail-terminal');
      if (term) term.textContent = '';
    });
  }

  // Поиск внутри терминала
  const tailFilterInput = document.getElementById('apl-tail-filter');
  if (tailFilterInput) {
    tailFilterInput.addEventListener('input', () => {
      loadTailTerminal();
    });
  }

  // Копирование сырого содержимого
  const btnCopyRaw = document.getElementById('btn-apl-copy-raw');
  if (btnCopyRaw) {
    btnCopyRaw.addEventListener('click', () => {
      const content = document.getElementById('apl-raw-content')?.textContent || '';
      navigator.clipboard.writeText(content).then(() => {
        if (window.toast) window.toast.success('Скопировано', 'Содержимое файла скопировано');
      });
    });
  }
```

```
}

/**
 * Применение быстрых пресетов фильтрации
 */
function applyQuickPreset(preset) {
  const levelSelect = document.getElementById('apl-level-selec
t');
  const compSelect = document.getElementById('apl-component-se
lect');
  const timeSelect = document.getElementById('apl-time-preset-
select');
  const searchInput = document.getElementById('apl-search-inpu
t');

  if (preset === 'all') {
    if (levelSelect) levelSelect.value = '';
    if (compSelect) compSelect.value = '';
    if (timeSelect) timeSelect.value = '';
    if (searchInput) searchInput.value = '';
  } else if (preset === 'errors') {
    if (levelSelect) levelSelect.value = 'ERRORS';
  } else if (preset === 'warnings_errors') {
    if (levelSelect) levelSelect.value = 'WARNINGS_ERRORS';
  } else if (preset === 'windows') {
    if (searchInput) searchInput.value = 'Windows';
  } else if (preset === 'ai') {
    if (searchInput) searchInput.value = 'AI';
  } else if (preset === 'fastapi') {
    if (searchInput) searchInput.value = 'FastAPI';
  } else if (preset === 'hour') {
    if (timeSelect) timeSelect.value = '1h';
  } else if (preset === 'day') {
    if (timeSelect) timeSelect.value = '24h';
  }

  refreshCurrentView();
}
```

```
290
291 /**
292  * Сброс всех фильтров
293  */
294 function resetAllFilters() {
295   const searchInput = document.getElementById('apl-search-inpu
    t');
296   const levelSelect = document.getElementById('apl-level-selec
    t');
297   const compSelect = document.getElementById('apl-component-se
    lect');
298   const timeSelect = document.getElementById('apl-time-preset-
    select');
299   const limitSelect = document.getElementById('apl-limit-selec
    t');
300
301   if (searchInput) searchInput.value = '';
302   if (levelSelect) levelSelect.value = '';
303   if (compSelect) compSelect.value = '';
304   if (timeSelect) timeSelect.value = '';
305   if (limitSelect) limitSelect.value = '200';
306
307   document.querySelectorAll('.apl-quick-preset-btn').forEach(b
     => {
308     b.classList.toggle('active', b.dataset.preset === 'all');
309   });
310
311   refreshCurrentView();
312 }
313
314 /**
315  * Запуск экспорта данных
316  */
317 function triggerExport(format) {
318   const search = document.getElementById('apl-search-input')?.
    value || '';
319   const level = document.getElementById('apl-level-select')?.v
    alue || '';
```

```
  const url = `/api/v1/app_logs/export?file_name=${encodeURICo
mponent(state.currentFile)}&export_format=${format}&search=${e
ncodeURIComponent(search)}&level=${encodeURIComponent(level)}`
;
  window.open(url, '_blank');
}

/**
 * Переключение видимого представления
 */
function switchViewMode(mode) {
  const views = {
    table: document.getElementById('apl-view-table'),
    audit: document.getElementById('apl-view-audit'),
    timeline: document.getElementById('apl-view-timeline'),
    tail: document.getElementById('apl-view-tail'),
    raw: document.getElementById('apl-view-raw'),
  };

  Object.entries(views).forEach(([m, el]) => {
    if (!el) return;
    if (m === mode) {
      el.classList.remove('d-none');
    } else {
      el.classList.add('d-none');
    }
  });

  if (mode === 'audit') {
    runAiDiagnostics();
  } else if (mode === 'tail') {
    loadTailTerminal();
  } else if (mode === 'raw') {
    loadRawContent();
  } else if (mode === 'timeline') {
    renderTimelineChart();
  }
}
```

```
355
356 /**
357  * Загрузка сводки и списка файлов логов
358  */
359 async function loadOverviewAndFiles() {
360   try {
361     const [ovRes, filesRes] = await Promise.all([
362       fetch('/api/v1/app_logs/overview').catch(() => null),
363       fetch('/api/v1/app_logs/files').catch(() => null),
364     ]);
365
366     if (ovRes && ovRes.ok) {
367       state.overview = await ovRes.json();
368     }
369     if (filesRes && filesRes.ok) {
370       const data = await filesRes.json();
371       state.files = data.files || [];
372       const pathLabel = document.getElementById('apl-full-path
-label');
373       if (pathLabel && data.logs_dir) {
374         pathLabel.textContent = data.logs_dir;
375         pathLabel.title = data.logs_dir;
376       }
377     }
378
379     renderFilesBadges();
380   } catch (err) {
381     console.error('[AppLogs] Ошибка загрузки файлов и сводки:'
, err);
382   }
383 }
384
385 /**
386  * Очистка одного файла логов
387  */
388 async function clearSingleLog(fileName) {
389   if (!fileName) return;
390   if (!confirm(`Вы действительно хотите очистить файл "${fileN
```

```
ame}"? Содержимое журнала будет безвозвратно удалено.`)) {
    return;
  }
  try {
    const res = await fetch('/api/v1/app_logs/clear', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ file_name: fileName })
    });
    const data = await res.json();
    if (res.ok && data.success) {
      if (window.toast) window.toast.success('Очищено', data.m
essage || `Файл ${fileName} успешно очищен`);
      await loadOverviewAndFiles();
      await refreshCurrentView();
    } else {
      if (window.toast) window.toast.error('Ошибка', data.deta
il || 'Не удалось очистить файл');
    }
  } catch (err) {
    if (window.toast) window.toast.error('Ошибка', err.message
);
  }
}

/**
 * Очистка всех файлов логов программы
 */
async function clearAllLogs() {
  if (!confirm('Вы действительно хотите очистить ВСЕ файлы лог
ов программы (%APPDATA%\\AI-Windows-Admin\\logs)? Все журналы
будут обнулены.')) {
    return;
  }
  try {
    const res = await fetch('/api/v1/app_logs/clear_all', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' }
```

```
    });
    const data = await res.json();
    if (res.ok && data.success) {
      if (window.toast) window.toast.success('Все логи очищены', data.message || 'Все файлы логов успешно очищены');
      await loadOverviewAndFiles();
      await refreshCurrentView();
    } else {
      const errMsg = data.detail || (data.failed_files?.length ? `Некоторые файлы заблокированы процессами: ${data.failed_files.map(f => f.name).join(', ')}` : 'Не удалось очистить все логи');
      if (window.toast) window.toast.warning('Внимание', errMsg);
      await loadOverviewAndFiles();
      await refreshCurrentView();
    }
  } catch (err) {
    if (window.toast) window.toast.error('Ошибка', err.message);
  }
}

/**
 * Отрисовка бейджей файлов журналов со статусными индикаторами и кнопками очистки
 */
function renderFilesBadges() {
  const container = document.getElementById('apl-files-badges-container');
  if (!container) return;

  container.innerHTML = '';
  if (state.files.length === 0) {
    container.innerHTML = '<span class="text-muted small">Файлы журналов не обнаружены.</span>';
    return;
  }
```

```
452
453   // Карта ошибок по файлам из overview
454   const errorMap = {};
455   if (state.overview && state.overview.files) {
456     state.overview.files.forEach(f => {
457       errorMap[f.name] = f.recent_errors || 0;
458     });
459   }
460
461   state.files.forEach(f => {
462     const btn = document.createElement('button');
463     btn.type = 'button';
464     const isActive = f.name === state.currentFile;
465     const errorsCount = errorMap[f.name] || 0;
466     const hasError = errorsCount > 0;
467
468     btn.className = `btn btn-sm rounded-pill px-2.5 py-0.5 sma
    ll apl-file-badge ${isActive ? 'active' : ''} ${hasError ? 'ha
    s-error' : ''}`;
469     btn.style.fontSize = '0.76rem';
470
471     const icon = f.is_json ? 'bi-filetype-json text-info' : 'b
    i-file-text text-warning';
472     const errIndicator = hasError ? `<span class="badge bg-dan
    ger ms-1" style="font-size:0.65rem;" title="Ошибок: ${errorsCo
    unt}">${errorsCount}</span>` : '';
473
474     btn.innerHTML = `<i class="bi ${icon} me-1"></i><strong>${
    f.name}</strong> <span class="badge apl-card border border-sec
    ondary-subtle ms-1">${f.size_formatted}</span>${errIndicator}`
    ;
475
476     // Кнопка очистки конкретно этого файла лога
477     const clearBtn = document.createElement('span');
478     clearBtn.className = 'apl-badge-clear-btn';
479     clearBtn.title = `Очистить лог-файл ${f.name}`;
480     clearBtn.innerHTML = '<i class="bi bi-trash3" style="font-
    size: 0.72rem;"></i>';
```

```
    clearBtn.addEventListener('click', async (e) => {
      e.stopPropagation();
      await clearSingleLog(f.name);
    });
    btn.appendChild(clearBtn);

    btn.addEventListener('click', async () => {
      state.currentFile = f.name;
      renderFilesBadges();
      await refreshCurrentView();
    });

    container.appendChild(btn);
  });
}

/**
 * Обновление текущего представления
 */
async function refreshCurrentView() {
  const search = document.getElementById('apl-search-input')?.value || '';
  const level = document.getElementById('apl-level-select')?.value || '';
  const component = document.getElementById('apl-component-select')?.value || '';
  const timePreset = document.getElementById('apl-time-preset-select')?.value || '';
  const limit = document.getElementById('apl-limit-select')?.value || '200';

  const fnStat = document.getElementById('apl-stat-filename');
  if (fnStat) fnStat.textContent = state.currentFile;

  try {
    const queryParams = new URLSearchParams({
      file_name: state.currentFile,
      limit: limit,
```

```
      search: search,
      level: level,
      component: component,
      time_preset: timePreset,
    });

    const res = await fetch(`/api/v1/app_logs/records?${queryP
arams.toString()}`);
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();

    state.records = data.records || [];
    state.stats = data.stats || null;

    updateHeaderStats(data);
    updateComponentsDropdown(data.stats?.components || []);

    if (state.currentMode === 'table') {
      renderRecordsTable(state.records);
    } else if (state.currentMode === 'timeline') {
      renderTimelineChart();
    } else if (state.currentMode === 'tail') {
      loadTailTerminal();
    } else if (state.currentMode === 'raw') {
      loadRawContent();
    }
  } catch (err) {
    console.error('[AppLogs] Ошибка обновления логов:', err);
    const tbody = document.getElementById('apl-table-body');
    if (tbody) {
      tbody.innerHTML = `<tr><td colspan="5" class="text-cente
r text-danger py-4">Ошибка загрузки: ${escapeHtml(err.message)
}</td></tr>`;
    }
  }
}

/**
```

```
 * Обновление счетчиков в шапке
 */
function updateHeaderStats(data) {
  const sizeEl = document.getElementById('apl-stat-size');
  if (sizeEl && data.file_info) {
    sizeEl.textContent = data.file_info.size_formatted || '0 KB';
  }

  const loadedEl = document.getElementById('apl-stat-loaded');
  if (loadedEl) {
    loadedEl.textContent = `${data.records.length} / ${data.total_matches}`;
  }

  const errEl = document.getElementById('apl-stat-errors');
  if (errEl && data.stats) {
    errEl.textContent = data.stats.errors_count || 0;
  }

  const warnEl = document.getElementById('apl-stat-warnings');
  if (warnEl && data.stats) {
    warnEl.textContent = data.stats.warnings_count || 0;
  }
}

/**
 * Обновление выпадающего списка компонентов
 */
function updateComponentsDropdown(components) {
  const select = document.getElementById('apl-component-select');
  if (!select) return;

  const currentVal = select.value;
  select.innerHTML = '<option value="">Все модули/компоненты</option>';

```

```
  components.forEach(c => {
    const opt = document.createElement('option');
    opt.value = c.name;
    opt.textContent = `${c.name} (${c.count})`;
    if (c.name === currentVal) opt.selected = true;
    select.appendChild(opt);
  });
}

/**
 * Отрисовка строк таблицы записей
 */
function renderRecordsTable(records) {
  const tbody = document.getElementById('apl-table-body');
  if (!tbody) return;

  if (!records || records.length === 0) {
    tbody.innerHTML = '<tr><td colspan="5" class="text-center
text-muted py-4">Нет записей, удовлетворяющих условиям фильтра
ции.</td></tr>';
    return;
  }

  tbody.innerHTML = records.map(entry => {
    let badgeClass = 'badge-apl-inf';
    const lvl = (entry.level || '').toUpperCase();
    if (lvl === 'CRITICAL' || lvl === 'FATAL') badgeClass = 'b
adge-apl-crit';
    else if (lvl === 'ERROR') badgeClass = 'badge-apl-err';
    else if (lvl === 'WARNING' || lvl === 'WARN') badgeClass =
 'badge-apl-warn';
    else if (lvl === 'DEBUG') badgeClass = 'badge-apl-deb';

    const repeatBadge = entry.repeat_count > 1 ? `<span class=
"badge apl-card border border-secondary-subtle text-warning ms
-1" title="Повторений">${entry.repeat_count}x</span>` : '';
    const hasExc = !!entry.exc_info;
    const excBadge = hasExc ? `<span class="badge bg-danger ms
```

```
-1" title="Содержит Stacktrace"><i class="bi bi-bug"></i> Trac
e</span>` : '';

    return `
      <tr class="interactive-log-row" data-id="${entry.id}">
        <td class="font-monospace small text-muted text-nowrap
">${escapeHtml(entry.timestamp || '--:--:--')}</td>
        <td><span class="badge ${badgeClass} small">${escapeHt
ml(entry.level || 'INFO')}</span>${repeatBadge}</td>
        <td><span class="badge apl-comp-badge font-monospace t
ext-truncate" style="max-width: 120px;" title="${escapeHtml(en
try.component)}">${escapeHtml(entry.component)}</span></td>
        <td class="font-monospace small text-break text-body">
${escapeHtml(entry.message || '')}${excBadge}</td>
        <td class="text-end no-modal-trigger">
          <button type="button" class="btn btn-xs btn-outline-
info rounded px-1.5 py-0.5 btn-entry-ai" data-id="${entry.id}"
 title="AI Анализ в модальном окне">
            <i class="bi bi-robot"></i> AI
          </button>
        </td>
      </tr>
    `;
  }).join('');

  // Привязка обработчиков клика по строкам для вызова AIModal
Dialog
  tbody.querySelectorAll('.interactive-log-row').forEach(row =
> {
    row.addEventListener('click', (event) => {
      const entryId = row.getAttribute('data-id');
      const entry = records.find(x => String(x.id) === String(
entryId));
      if (!entry) return;

      openItemAiModal(entry);
    });
  });
```

```
641
642   // Кнопка AI Анализа в строке
643   tbody.querySelectorAll('.btn-entry-ai').forEach(btn => {
644     btn.addEventListener('click', (event) => {
645       event.stopPropagation();
646       const entryId = btn.getAttribute('data-id');
647       const entry = records.find(x => String(x.id) === String(
    entryId));
648       if (!entry) return;
649 
650       openItemAiModal(entry);
651     });
652   });
653 }
654 
655 /**
656  * Открытие универсального модального окна AIModalDialog для а
    нализа записи
657  */
658 function openItemAiModal(entry) {
659   state.selectedEntry = entry;
660   const modal = window.AIModalDialog || window.AITableModal;
661 
662   let badgeClass = 'badge bg-info';
663   const lvl = (entry.level || '').toUpperCase();
664   if (lvl === 'CRITICAL' || lvl === 'FATAL' || lvl === 'ERROR'
    ) badgeClass = 'badge bg-danger';
665   else if (lvl === 'WARNING' || lvl === 'WARN') badgeClass = '
    badge bg-warning text-dark';
666   else if (lvl === 'DEBUG') badgeClass = 'badge bg-secondary';
667 
668   const icon = (lvl === 'CRITICAL' || lvl === 'ERROR') ? '□ '
    : ((lvl === 'WARNING' || lvl === 'WARN') ? '□ ' : '□ ');
669 
670   if (modal && typeof modal.show === 'function') {
671     modal.show({
672       title: `[${entry.level}] ${entry.component || 'Log Entry
    '}`,
```

```
      subtitle: `Время: ${entry.timestamp || 'N/A'} | Файл: ${state.currentFile} | Повторов: ${entry.repeat_count || 1}`,
      icon: icon,
      tableType: 'process',
      badges: [
        { text: entry.level || 'INFO', class: badgeClass },
        { text: entry.component || 'System', class: 'badge apl-comp-badge' },
        { text: `${entry.repeat_count || 1}x`, class: 'badge border text-secondary' }
      ],
      metadata: [
        { label: 'Временная метка', value: entry.timestamp || 'N/A' },
        { label: 'Компонент / Модуль', value: entry.component || 'N/A' },
        { label: 'Уровень события', value: entry.level || 'INFO' },
        { label: 'Файл журнала', value: state.currentFile },
        { label: 'Повторений инцидента', value: `${entry.repeat_count || 1}` },
        { label: 'Сообщение лога', value: entry.message || '', fullWidth: true, isCode: true },
      ],
      rawTitle: 'Стек вызовов (Traceback) / Сырая запись',
      rawContent: entry.exc_info ? (typeof entry.exc_info === 'string' ? entry.exc_info : JSON.stringify(entry.exc_info, null, 2)) : entry.raw || entry.message,
      autoRun: false,
      actions: [
        {
          label: 'Копировать запись',
          icon: 'bi-clipboard',
          class: 'btn-outline-info',
          onClick: () => {
            navigator.clipboard.writeText(JSON.stringify(entry, null, 2)).then(() => {
              if (window.toast) window.toast.success('Скопиров
```

```
ано', 'Лог-запись скопирована в буфер обмена');
            });
          }
        }
      ]
    });
  } else {
    // Fallback: alert
    console.log('[AppLogs] Запись лога:', entry);
    alert(`[${entry.level}] ${entry.component}\n\n${entry.message}`);
  }
}

/**
 * Выполнение глубокой AI Диагностики и кластеризации сбоев
 */
async function runAiDiagnostics() {
  const summaryEl = document.getElementById('apl-audit-summary-text');
  const healthScoreEl = document.getElementById('apl-audit-health-score');
  const healthBar = document.getElementById('apl-audit-health-bar');
  const recsContainer = document.getElementById('apl-audit-recommendations-list');
  const clustersBody = document.getElementById('apl-audit-clusters-body');

  if (summaryEl) summaryEl.innerHTML = '<div class="spinner-border spinner-border-sm text-info me-2"></div>Глубокий анализ логов, поиск первопричин и кластеризация сбоев...';

  try {
    const res = await fetch('/api/v1/app_logs/diagnose', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
```

```
        file_name: state.currentFile,
        limit: 150,
        focus_errors_only: true,
      })
    });

    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();

    if (healthScoreEl) healthScoreEl.textContent = `${data.hea
lth_score}%`;
    if (healthBar) {
      healthBar.style.width = `${data.health_score}%`;
      healthBar.className = `progress-bar ${data.health_score
> 80 ? 'bg-success' : (data.health_score > 50 ? 'bg-warning' :
 'bg-danger')}`;
    }

    if (summaryEl) summaryEl.textContent = data.summary;

    // Отрисовка рекомендаций
    if (recsContainer) {
      if (!data.recommendations || data.recommendations.length
 === 0) {
        recsContainer.innerHTML = '<div class="alert alert-suc
cess small mb-0"><i class="bi bi-check-circle me-1"></i>Активн
ых сбоев и критических предупреждений не зафиксировано.</div>'
;
      } else {
        recsContainer.innerHTML = data.recommendations.map(r =
> `
          <div class="card apl-card border-${r.severity === 'h
igh' ? 'danger' : 'warning'} p-2.5 mb-2 shadow-sm">
            <div class="d-flex justify-content-between align-i
tems-center mb-1">
              <span class="fw-bold text-${r.severity === 'high
' ? 'danger' : 'warning'} small"><i class="bi bi-exclamation-t
riangle me-1"></i>${escapeHtml(r.title)}</span>
```

```
              <span class="badge bg-${r.severity === 'high' ? 'danger' : 'warning'} small">${r.severity.toUpperCase()}</span>
            </div>
            <div class="small text-body mb-1">${escapeHtml(r.description)}</div>
            <div class="small text-info"><i class="bi bi-arrow-right-circle me-1"></i><strong>Рекомендуемое действие:</strong> ${escapeHtml(r.action)}</div>
          </div>
        `).join('');
      }
    }

    // Отрисовка кластеров
    if (clustersBody) {
      if (!data.clusters || data.clusters.length === 0) {
        clustersBody.innerHTML = '<tr><td colspan="5" class="text-center text-muted py-3">Ошибок и повторяющихся инцидентов не зафиксировано.</td></tr>';
      } else {
        clustersBody.innerHTML = data.clusters.map((c, idx) => `
          <tr class="interactive-cluster-row" data-cluster-idx="${idx}">
            <td><span class="badge bg-danger fw-bold">${c.count}x</span></td>
            <td><span class="badge apl-card border border-danger text-danger">${escapeHtml(c.level)}</span></td>
            <td class="font-monospace small text-muted text-nowrap">${escapeHtml(c.last_seen || '--')}</td>
            <td class="font-monospace small text-break">
              <div class="fw-bold text-warning">${escapeHtml(c.pattern)}</div>
              <div class="text-muted small mt-0.5">${escapeHtml(c.sample_message)}</div>
            </td>
            <td class="text-end">
```

```
              <button type="button" class="btn btn-xs btn-outl
ine-warning rounded px-1.5 py-0.5 btn-cluster-inspect" data-cl
uster-idx="${idx}" title="AI инспекция кластера">
                <i class="bi bi-robot"></i> Инспекция
              </button>
            </td>
          </tr>
        `).join('');

        // Обработчик инспекции кластера
        clustersBody.querySelectorAll('.btn-cluster-inspect').
forEach(btn => {
          btn.addEventListener('click', (e) => {
            e.stopPropagation();
            const idx = parseInt(btn.getAttribute('data-cluste
r-idx'), 10);
            const cluster = data.clusters[idx];
            if (!cluster) return;

            openItemAiModal({
              id: `cluster-${idx}`,
              level: cluster.level,
              component: 'Error Cluster',
              timestamp: cluster.last_seen,
              repeat_count: cluster.count,
              message: cluster.sample_message,
              exc_info: cluster.exc_info,
              raw: cluster.sample_message
            });
          });
        });
      }
    }

  } catch (err) {
    console.error('[AppLogs] Ошибка AI диагностики:', err);
    if (summaryEl) summaryEl.innerHTML = `<span class="text-da
nger">Ошибка диагностики: ${escapeHtml(err.message)}</span>`;
```

```
  }
}

/**
 * Отрисовка интерактивного графика временной шкалы (Timeline)
 */
function renderTimelineChart() {
  const barsContainer = document.getElementById('apl-timeline-bars');
  const compContainer = document.getElementById('apl-top-components-container');
  if (!state.stats) return;

  const timeline = state.stats.timeline || [];
  if (barsContainer) {
    if (timeline.length === 0) {
      barsContainer.innerHTML = '<div class="text-muted small p-3 text-center w-100">Недостаточно временных меток для построения графика.</div>';
    } else {
      const maxTotal = Math.max(...timeline.map(t => t.total), 1);
      barsContainer.innerHTML = timeline.map(t => {
        const heightPct = Math.max(10, (t.total / maxTotal) * 100);
        const timeLabel = t.time.slice(11, 16);

        const hasErr = t.errors > 0;
        const hasWarn = t.warnings > 0;
        let barBg = 'linear-gradient(to top, var(--nav-active, #0284c7) 0%, var(--nav-active, #38bdf8) 100%)';
        if (hasErr) {
          barBg = 'linear-gradient(to top, #ef4444 0%, #dc2626 100%)';
        } else if (hasWarn) {
          barBg = 'linear-gradient(to top, #f59e0b 0%, #d97706 100%)';
        }
```

```
841
842         return `
843           <div class="d-flex flex-column align-items-center apl-timeline-bar-item" data-time="${t.time}" style="flex: 1; min-width: 32px; height: 100%; justify-content: flex-end;" title="${t.time}: Всего событий: ${t.total}, Ошибок: ${t.errors}, Предупреждений: ${t.warnings}">
844             <span class="small font-monospace text-muted" style="font-size: 0.65rem;">${t.total}</span>
845             <div class="w-100 rounded-top" style="height: ${heightPct}%; background: ${barBg}; position: relative;">
846             </div>
847             <span class="small font-monospace text-muted text-truncate" style="font-size: 0.65rem; margin-top: 4px;">${timeLabel}</span>
848           </div>
849         `;
850       }).join('');
851
852       // Клик по столбцу фильтрует таблицу по этому временному слоту
853       barsContainer.querySelectorAll('.apl-timeline-bar-item').forEach(bar => {
854         bar.addEventListener('click', () => {
855           const tVal = bar.getAttribute('data-time');
856           const searchInput = document.getElementById('apl-search-input');
857           if (searchInput && tVal) {
858             searchInput.value = tVal.slice(0, 13);
859             const tableTab = document.querySelector('#apl-mode-tabs [data-mode="table"]');
860             if (tableTab) tableTab.click();
861             refreshCurrentView();
862           }
863         });
864       });
865     }
866   }
```

```
867
868   if (compContainer && state.stats.components) {
869     compContainer.innerHTML = state.stats.components.map(c =>
`
870       <div class="col-md-3 col-sm-6">
871         <div class="card apl-card p-2 d-flex flex-row justify-
content-between align-items-center shadow-sm" style="cursor: p
ointer;" title="Фильтровать по модулю: ${escapeHtml(c.name)}">
872           <span class="font-monospace small text-info text-tru
ncate" style="max-width: 130px;">${escapeHtml(c.name)}</span>
873           <span class="badge bg-primary rounded-pill font-mono
space">${c.count}</span>
874         </div>
875       </div>
876     `).join('');
877
878     // Клик по компоненту фильтрует по нему
879     compContainer.querySelectorAll('.card').forEach((card, idx
) => {
880       card.addEventListener('click', () => {
881         const cName = state.stats.components[idx]?.name;
882         const compSelect = document.getElementById('apl-compon
ent-select');
883         if (compSelect && cName) {
884           compSelect.value = cName;
885           const tableTab = document.querySelector('#apl-mode-t
abs [data-mode="table"]');
886           if (tableTab) tableTab.click();
887           refreshCurrentView();
888         }
889       });
890     });
891   }
892 }
893
894 /**
895  * Загрузка терминала Live Tail
896  */
```

```
async function loadTailTerminal() {
  const term = document.getElementById('apl-tail-terminal');
  if (!term) return;

  const filterText = (document.getElementById('apl-tail-filter')?.value || '').toLowerCase();

  try {
    const res = await fetch(`/api/v1/app_logs/tail?file_name=${encodeURIComponent(state.currentFile)}&lines=150`);
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();

    let lines = data.lines || [];
    if (filterText) {
      lines = lines.filter(l => l.toLowerCase().includes(filterText));
    }

    term.textContent = lines.join('\n') || '[Файл логов пуст или нет совпадений по фильтру]';

    const autoscroll = document.getElementById('apl-tail-autoscroll')?.checked;
    if (autoscroll) {
      term.scrollTop = term.scrollHeight;
    }
  } catch (err) {
    term.textContent = `[Ошибка чтения tail: ${escapeHtml(err.message)}]`;
  }
}

/**
 * Загрузка сырого содержимого файла
 */
async function loadRawContent() {
  const pre = document.getElementById('apl-raw-content');
```

```
  if (!pre) return;

  pre.textContent = 'Загрузка содержимого файла...';
  try {
    const res = await fetch(`/api/v1/app_logs/tail?file_name=${encodeURIComponent(state.currentFile)}&lines=500`);
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    pre.textContent = (data.lines || []).join('\n') || '[Файл логов пуст]';
  } catch (err) {
    pre.textContent = `[Ошибка чтения: ${escapeHtml(err.message)}]`;
  }
}

/**
 * Проверка активности вкладки App Logs
 */
function isAppLogsTabActive() {
  if (typeof document !== 'undefined' && document.hidden) return false;
  if (typeof window.isTabActive === 'function') {
    return Boolean(window.isTabActive('tab-app-logs') || window.isTabActive('app-logs'));
  }
  const pane = document.getElementById('tab-app-logs') || document.getElementById('app-logs');
  return pane ? pane.classList.contains('active') : false;
}

/**
 * Запуск фонового Live Stream опроса
 */
function startLivePolling() {
  stopLivePolling();
  if (!isAppLogsTabActive()) return;

```

```
  if (window.registerTabPoller) {
    window.registerTabPoller('tab-app-logs', async () => {
      if (!isAppLogsTabActive()) return;
      if (state.currentMode === 'tail') {
        await loadTailTerminal();
      } else {
        await refreshCurrentView();
      }
    }, 3000, { pollerId: 'tab-app-logs_live', immediate: false
 });
  } else {
    state.liveTimer = setInterval(async () => {
      if (!isAppLogsTabActive()) {
        stopLivePolling();
        return;
      }
      if (state.currentMode === 'tail') {
        await loadTailTerminal();
      } else {
        await refreshCurrentView();
      }
    }, 3000);
  }
}

/**
 * Остановка фонового опроса
 */
function stopLivePolling() {
  if (state.liveTimer) {
    clearInterval(state.liveTimer);
    state.liveTimer = null;
  }
  if (window.unregisterTabPoller) {
    window.unregisterTabPoller('tab-app-logs_live');
  }
}

```

```
/**
 * Экранирование HTML
 */
function escapeHtml(text) {
  if (!text) return '';
  return String(text)
    .replace(/&/g, '&amp;')
    .replace(/</g, '&lt;')
    .replace(/>/g, '&gt;')
    .replace(/"/g, '&quot;')
    .replace(/'/g, '&#039;');
}

window.activateAppLogsTab = () => {
  if (isAppLogsTabActive() && state.liveStream) {
    startLivePolling();
  }
};
window.deactivateAppLogsTab = () => {
  stopLivePolling();
};

document.addEventListener('tab:deactivated', (e) => {
  if (e.detail?.tabId === 'tab-app-logs' || e.detail?.tabId ==
= 'app-logs') {
    stopLivePolling();
  }
});

// Авто-инициализация при загрузке скрипта
if (document.readyState === 'loading') {
  document.addEventListener('DOMContentLoaded', initAppLogsTab
);
} else {
  initAppLogsTab();
}

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/boot_recovery_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/boot_recovery_tab/index
.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/boot_recovery_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Boot & Recovery Management Tab -->
<div class="container-fluid py-2 px-3">
<!-- Top Stat Cards -->
<div class="row g-2 mb-3">
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 sha
dow-sm p-3" data-panel-id="panel-boot-recovery-1" id="panel-bo
ot-recovery-1">
<div class="text-muted small text-uppercase fw-semibold">Среда
 восстановления WinRE  </div>
```

```
<div class="fs-4 fw-bold mt-1 text-success" id="br-winre-status">--</div>
<div class="small text-muted mt-1" id="br-winre-sub">reagentc.exe</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-boot-recovery-2" id="panel-boot-recovery-2">
<div class="text-muted small text-uppercase fw-semibold">Таймаут меню BCD  </div>
<div class="fs-4 fw-bold mt-1 text-info" id="br-bcd-timeout">-- сек</div>
<div class="small text-muted mt-1" id="br-bcd-sub">bcdedit /timeout</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-boot-recovery-3" id="panel-boot-recovery-3">
<div class="text-muted small text-uppercase fw-semibold">Текущий загрузчик  </div>
<div class="fs-4 fw-bold mt-1 text-primary" id="br-current-loader">--</div>
<div class="small text-muted mt-1" id="br-loader-sub">Windows Boot Manager</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-boot-recovery-4" id="panel-boot-recovery-4">
<div class="text-muted small text-uppercase fw-semibold">Целостность Boot Sector  </div>
<div class="fs-4 fw-bold mt-1 text-warning" id="br-boot-status">OK</div>
```

```
<div class="small text-muted mt-1" id="br-boot-sub">bootrec / bcdboot</div>
</div>
</div>
</div>
<!-- Action Bar -->
<div class="d-flex align-items-center justify-content-between mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5" id="br-refresh-btn">
<i class="bi bi-arrow-clockwise"></i>
<span>Обновить BCD</span>
</button>
<button class="btn btn-sm btn-outline-success d-flex align-items-center gap-1.5" id="br-winre-enable-btn">
<i class="bi bi-shield-check"></i>
<span>Включить WinRE</span>
</button>
<button class="btn btn-sm btn-outline-info d-flex align-items-center gap-1.5" id="br-winre-info-btn">
<i class="bi bi-info-circle"></i>
<span>Инфо WinRE</span>
</button>
</div>
</div>
<!-- Main Details & Tools Grid -->
<div class="row g-3">
<div class="col-12 col-lg-7">
<div class="card bg-dark text-white border-secondary shadow-sm" data-panel-id="panel-boot-recovery-zapisi-khranilishcha-bcd" id="panel-boot-recovery-zapisi-khranilishcha-bcd">
<div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">
<span class="fw-semibold"><i class="bi bi-list-columns text-primary me-2"></i>Записи хранилища BCD</span>
<span class="badge bg-secondary" id="br-entries-badge">0 запис
```

```
ей</span>
</div>
<div class="card-body p-0 table-responsive" style="max-height: 420px; overflow-y: auto;">
<table class="table table-dark table-hover table-striped align-middle mb-0 small" id="br-entries-table">
<thead>
<tr class="text-secondary">
<th>Идентификатор (GUID)</th>
<th>Описание</th>
<th>Устройство</th>
<th>Путь</th>
<th class="text-end">Действия</th>
</tr>
</thead>
<tbody id="br-entries-tbody">
<tr><td class="text-center text-muted py-3" colspan="5">Опрос записей загрузчика BCD...</td></tr>
</tbody>
</table>
</div>
</div>
</div>
<div class="col-12 col-lg-5">
<div class="card bg-dark text-white border-secondary shadow-sm mb-3" data-panel-id="panel-boot-recovery-instrumenty-vosstanovleniya" id="panel-boot-recovery-instrumenty-vosstanovleniya">
<div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">
<span class="fw-semibold"><i class="bi bi-wrench-adjustable text-warning me-2"></i>Инструменты восстановления</span>
</div>
<div class="card-body p-3">
<div class="mb-3">
<label class="form-label small text-muted">Команда восстановления:</label>
<select class="form-select form-select-sm bg-dark text-white border-secondary" id="br-action-select">
```

```
<option value="reagentc_info">Проверить состояние WinRE (reagentc /info)</option>
<option value="reagentc_enable">Активировать WinRE (reagentc /enable)</option>
<option value="bcdedit_enum">Перечислить все записи (bcdedit /enum all)</option>
<option value="bcdboot_repair">Обновить системные файлы загрузки (bcdboot C:\Windows)</option>
</select>
</div>
<div class="d-flex align-items-center justify-content-between">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="br-dry-run-switch" type="checkbox"/>
<label class="form-check-label small text-muted" for="br-dry-run-switch">Безопасный режим (Dry Run)</label>
</div>
<button class="btn btn-sm btn-success px-3" id="br-execute-btn">
<i class="bi bi-play-fill"></i> Выполнить
            </button>
</div>
</div>
</div>
<!-- Action Log Output -->
<div class="card bg-dark text-white border-secondary shadow-sm" data-panel-id="panel-boot-recovery-konsolnyy-vyvod" id="panel-boot-recovery-konsolnyy-vyvod">
<div class="card-header border-secondary py-2 px-3 d-flex align-items-center justify-content-between">
<span class="small fw-semibold text-muted"><i class="bi bi-terminal me-1"></i>Консольный вывод</span>
<button class="btn btn-link btn-sm text-decoration-none text-muted p-0" id="br-clear-log-btn">Очистить</button>
</div>
<div class="card-body p-2">
<pre class="mb-0 small bg-black text-light p-2 rounded" id="br
```

```
-console-output" style="max-height: 180px; overflow-y: auto; f
ont-family: Consolas, monospace; font-size: 0.78rem;">Среда BC
D и WinRE готова к операциям.</pre>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Boot Recovery Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/boot_recovery_tab/main.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { initBootRecoveryTab } from '/windows/api/webgui/boot_recovery_tab/main.js';
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/boot_recovery_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * boot_recovery_tab/main.js — Управление BCD, загрузчиком и средой WinRE
 * Updated: 2026-10-10 17:35:00
 */

const registerTabPoller = window.registerTabPoller || function () {};
```

```
30
31 let isInitialized = false;
32
33 export async function fetchBootRecoverySummary() {
34   try {
35     const res = await fetch('/api/boot-recovery/summary');
36     if (!res.ok) throw new Error(`HTTP ${res.status}`);
37     const data = await res.json();
38     renderBootSummary(data);
39   } catch (err) {
40     console.warn('[BootRecovery] Ошибка получения сводки:', er
   r);
41   }
42 }
43
44 function renderBootSummary(data) {
45   const winreStatusEl = document.getElementById('br-winre-stat
   us');
46   const bcdTimeoutEl = document.getElementById('br-bcd-timeout
   ');
47   const currentLoaderEl = document.getElementById('br-current-
   loader');
48   const tbody = document.getElementById('br-entries-tbody');
49   const badge = document.getElementById('br-entries-badge');
50
51   if (winreStatusEl) winreStatusEl.textContent = data.winre_en
   abled ? 'Включена' : 'Отключена';
52   if (bcdTimeoutEl) bcdTimeoutEl.textContent = `${data.timeout
    ?? 30} сек`;
53   if (currentLoaderEl) currentLoaderEl.textContent = data.curr
   ent_guid || '{current}';
54
55   if (!tbody) return;
56
57   const entries = data.bcd_entries || [
58     { guid: '{bootmgr}', description: 'Windows Boot Manager',
   device: 'partition=\\Device\\HarddiskVolume1', path: '\\EFI\\M
   icrosoft\\Boot\\bootmgfw.efi' },
```

```
    { guid: '{current}', description: 'Windows 11 Pro', device: 'partition=C:', path: '\\Windows\\system32\\winload.efi' }
  ];

  if (badge) badge.textContent = `${entries.length} записей`;

  tbody.innerHTML = entries.map(e => `
    <tr>
      <td class="fw-bold text-info font-monospace">${e.guid}</td>
      <td>${e.description || '-'}</td>
      <td class="small text-muted">${e.device || '-'}</td>
      <td class="small font-monospace">${e.path || '-'}</td>
      <td class="text-end">
        <button class="btn btn-xs btn-outline-info py-0 px-1" title="Инфо о записи">
          <i class="bi bi-info-circle"></i>
        </button>
      </td>
    </tr>
  `).join('');
}

async function executeBootAction(customAction = null) {
  const actionSelect = document.getElementById('br-action-select');
  const dryRunSwitch = document.getElementById('br-dry-run-switch');
  const output = document.getElementById('br-console-output');

  const op = customAction || (actionSelect ? actionSelect.value : 'reagentc_info');
  const dryRun = dryRunSwitch ? dryRunSwitch.checked : true;

  if (output) output.textContent = `[${new Date().toLocaleTimeString()}] Выполнение команды: ${op} (DryRun: ${dryRun})...\n`;

```

```
  try {
    const res = await fetch('/api/boot-recovery/action', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ action: op, dry_run: dryRun })
    });
    const result = await res.json();
    if (output) output.textContent += JSON.stringify(result, null, 2);
    if (window.toast) {
      if (res.ok) window.toast.success('Команда выполнена', `Статус: ${result.status || 'OK'}`);
      else window.toast.error('Ошибка', result.detail || 'Сбой выполнения');
    }
  } catch (err) {
    if (output) output.textContent += `\nОшибка запроса: ${err.message}`;
    if (window.toast) window.toast.error('Ошибка сети', err.message);
  }
}

export function initBootRecoveryTab() {
  if (isInitialized) return;
  isInitialized = true;

  const refreshBtn = document.getElementById('br-refresh-btn');
  if (refreshBtn) refreshBtn.addEventListener('click', fetchBootRecoverySummary);

  const winreEnableBtn = document.getElementById('br-winre-enable-btn');
  if (winreEnableBtn) winreEnableBtn.addEventListener('click', () => executeBootAction('reagentc_enable'));

  const winreInfoBtn = document.getElementById('br-winre-info-
```

```
btn');
  if (winreInfoBtn) winreInfoBtn.addEventListener('click', () => executeBootAction('reagentc_info'));

  const execBtn = document.getElementById('br-execute-btn');
  if (execBtn) execBtn.addEventListener('click', () => executeBootAction());

  const clearBtn = document.getElementById('br-clear-log-btn');
  if (clearBtn) clearBtn.addEventListener('click', () => {
    const out = document.getElementById('br-console-output');
    if (out) out.textContent = 'Консоль очищена.';
  });

  registerTabPoller('tab-boot-recovery', fetchBootRecoverySummary, 15000, { immediate: true });
}

window.initBootRecoveryTab = initBootRecoveryTab;

```

capture_ui.py

```
#!/usr/bin/env python3
# -*- coding: utf-8 -*-

"""Автоматический обход веб-интерфейса и сохранение PNG.

Скрипт открывает веб-интерфейс, обнаруживает элементы навигации,
последовательно переходит по ним и сохраняет визуальное состояние
каждого таба в PNG.

Если навигация изменяет URL hash, hash используется как имя файла.

Например::

    http://127.0.0.1:8001/tc#tab-processes-load-inspector

сохраняется как::

    assets/webgui/tab-processes-load-inspector.png

Примеры запуска::

    python capture_ui.py http://127.0.0.1:8001/tc

    python capture_ui.py http://127.0.0.1:8001/tc --debug

    python capture_ui.py http://127.0.0.1:8001/tc --debug --full-page
"""

from __future__ import annotations
from logger import logger

```

```
import argparse
import asyncio
import json
import re
import sys
from datetime import datetime
from pathlib import Path
from typing import Any
from urllib.parse import urlparse

from playwright.async_api import (
    Browser,
    BrowserContext,
    Page,
    async_playwright,
)


DEFAULT_OUTPUT_DIR = Path('assets/webgui')
DEFAULT_WAIT_MS = 800


def build_parser() -> argparse.ArgumentParser:
    """Создается парсер аргументов командной строки.

    :returns: Настроенный парсер аргументов.
    """

    parser = argparse.ArgumentParser(
        description='Сохраняется визуальное состояние веб-инте
рфейса в PNG.'
    )

    parser.add_argument(
        'url',
        help='URL веб-интерфейса.',
    )

```

```
    parser.add_argument(
        '-o',
        '--output',
        type=Path,
        default=DEFAULT_OUTPUT_DIR,
        help='Каталог для сохранения результатов.',
    )

    parser.add_argument(
        '--wait',
        type=int,
        default=DEFAULT_WAIT_MS,
        help='Ожидание после навигации в миллисекундах.',
    )

    parser.add_argument(
        '--full-page',
        action='store_true',
        help='Сохраняется вся страница, а не только viewport.'
,
    )

    parser.add_argument(
        '--debug',
        action='store_true',
        help='Включается видимый Chromium и подробный режим от
ладки.',
    )

    return parser


def debug(
    message: str,
    enabled: bool,
) -> None:
    """Выводится диагностическое сообщение.

```

```
    :param message: Текст диагностического сообщения.
    :param enabled: Признак активного DEBUG-режима.
    """

    if enabled:
        logger.console(f'[DEBUG] {message}')


def sanitize_filename(
    value: str,
    fallback: str,
) -> str:
    """Создается безопасное имя файла.

    :param value: Исходное имя.
    :param fallback: Резервное имя.
    :returns: Безопасное имя файла без расширения.
    """

    value = value.strip() or fallback

    value = re.sub(
        r'[<>:"/\\|?*\x00-\x1f]',
        '_',
        value,
    )

    value = re.sub(
        r'\s+',
        '_',
        value,
    )

    value = re.sub(
        r'_+',
        '_',
        value,
    )
```

```
144
145     value = value.strip('._ ')
146
147     return value[:150] or fallback
148
149
150 def get_url_hash(
151     url: str,
152 ) -> str:
153     """Извлекается hash из URL.
154
155     :param url: Полный URL.
156     :returns: Значение hash без символа ``#``.
157     """
158
159     return urlparse(url).fragment.strip()
160
161
162 def get_screenshot_name(
163     page: Page,
164     fallback: str,
165 ) -> str:
166     """Определяется имя PNG по текущему URL.
167
168     При наличии hash он используется как идентификатор состоян
    ия.
169
170     Например::
171
172         http://127.0.0.1:8001/tc#tab-processes-load-inspector
173
174     преобразуется в::
175
176         tab-processes-load-inspector.png
177
178     :param page: Объект страницы Playwright.
179     :param fallback: Имя при отсутствии hash.
180     :returns: Имя PNG без расширения.
```

```
    """

    fragment = get_url_hash(page.url)

    if fragment:
        return sanitize_filename(
            fragment,
            fallback,
        )

    return sanitize_filename(
        fallback,
        'page',
    )


async def wait_for_page(
    page: Page,
    wait_ms: int,
    debug_enabled: bool,
) -> None:
    """Ожидается стабилизация страницы.

    :param page: Объект страницы Playwright.
    :param wait_ms: Дополнительное ожидание.
    :param debug_enabled: Включается диагностический вывод.
    """

    debug(
        'Ожидание загрузки DOM...',
        debug_enabled,
    )

    try:
        await page.wait_for_load_state(
            'domcontentloaded',
            timeout=10000,
        )
```

```
    except Exception:
        debug(
            'domcontentloaded не получен в установленный timeo
ut.',
            debug_enabled,
        )

    debug(
        f'Ожидание {wait_ms} мс...',
        debug_enabled,
    )

    await page.wait_for_timeout(wait_ms)


async def collect_navigation(
    page: Page,
    debug_enabled: bool,
) -> list[dict[str, str]]:
    """Собираются элементы навигации.

    :param page: Объект страницы Playwright.
    :param debug_enabled: Включается диагностический вывод.
    :returns: Список навигационных элементов.
    """

    selector = (
        'nav a, nav button, '
        '[role="navigation"] a, '
        '[role="navigation"] button, '
        'aside a, aside button, '
        '[role="menu"] a, '
        '[role="menu"] button'
    )

    items = await page.locator(selector).evaluate_all(
        """
        elements => elements.map((element, index) => ({
```

```
            index: String(index),
            tag: element.tagName.toLowerCase(),
            text: (
                element.innerText ||
                element.textContent ||
                element.getAttribute('aria-label') ||
                ''
            ).trim(),
            href: element.href || '',
            id: element.id || '',
            className: typeof element.className === 'string'
                ? element.className
                : ''
        }))
        """
    )

    result: list[dict[str, str]] = []
    seen: set[str] = set()

    for item in items:
        text = str(item.get('text', '')).strip()
        href = str(item.get('href', '')).strip()
        element_id = str(item.get('id', '')).strip()

        key = f'{text}|{href}|{element_id}'

        if key in seen:
            continue

        seen.add(key)

        result.append(
            {
                'index': str(item.get('index', '')),
                'tag': str(item.get('tag', '')),
                'text': text,
                'href': href,
```

```
                'id': element_id,
                'classNameName': str(item.get('classNameName', '')),
            }
        )

    debug(
        f'Обнаружено навигационных элементов: {len(result)}',
        debug_enabled,
    )

    return result


async def save_screenshot(
    page: Page,
    output_path: Path,
    full_page: bool,
    debug_enabled: bool,
) -> None:
    """Сохраняется PNG текущего состояния страницы.

    :param page: Объект страницы.
    :param output_path: Путь к PNG.
    :param full_page: Сохраняется ли вся страница.
    :param debug_enabled: Включается диагностический вывод.
    """

    debug(
        f'Создается screenshot: {output_path}',
        debug_enabled,
    )

    await page.screenshot(
        path=str(output_path),
        full_page=full_page,
        animations='disabled',
    )

```

```python

async def navigate_to_item(
    page: Page,
    item: dict[str, str],
    wait_ms: int,
    debug_enabled: bool,
) -> bool:
    """Выполняется переход по навигационному элементу.

    :param page: Объект страницы.
    :param item: Описание элемента.
    :param wait_ms: Ожидание после перехода.
    :param debug_enabled: Включается диагностический вывод.
    :returns: ``True`` при успешном переходе.
    """

    href = item.get('href', '').strip()

    if href and href.startswith(
        ('http://', 'https://')
    ):
        debug(
            f'Переход по URL: {href}',
            debug_enabled,
        )

        try:
            await page.goto(
                href,
                wait_until='domcontentloaded',
                timeout=15000,
            )

            await wait_for_page(
                page,
                wait_ms,
                debug_enabled,
            )
```

```
370
371             return True
372
373         except Exception as exc:
374             debug(
375                 f'Ошибка перехода по URL: {exc}',
376                 debug_enabled,
377             )
378
379             return False
380
381     index = int(
382         item.get('index', '0')
383     )
384
385     selector = (
386         'nav a, nav button, '
387         '[role="navigation"] a, '
388         '[role="navigation"] button, '
389         'aside a, aside button, '
390         '[role="menu"] a, '
391         '[role="menu"] button'
392     )
393
394     locator = page.locator(
395         selector
396     ).nth(index)
397
398     try:
399         debug(
400             f'Click: index={index}, '
401             f'text={item.get("text", "")!r}',
402             debug_enabled,
403         )
404
405         await locator.scroll_into_view_if_needed()
406
407         await locator.click(
```

```
            timeout=5000,
        )

        await wait_for_page(
            page,
            wait_ms,
            debug_enabled,
        )

        return True

    except Exception as exc:
        debug(
            f'Ошибка click: {exc}',
            debug_enabled,
        )

        return False


async def capture_interface(
    url: str,
    output_dir: Path,
    wait_ms: int,
    full_page: bool,
    debug_enabled: bool,
) -> int:
    """Выполняется полный обход навигации.

    :param url: Начальный URL.
    :param output_dir: Каталог результатов.
    :param wait_ms: Ожидание после переходов.
    :param full_page: Сохраняется ли вся страница.
    :param debug_enabled: Включается DEBUG-режим.
    :returns: Код завершения процесса.
    """

    output_dir.mkdir(
```

```
        parents=True,
        exist_ok=True,
    )

    manifest: dict[str, Any] = {
        'url': url,
        'created_at': datetime.now().astimezone().isoformat(),
        'debug': debug_enabled,
        'full_page': full_page,
        'screenshots': [],
    }

    async with async_playwright() as playwright:
        debug(
            'Запуск Chromium: '
            f'{"headed" if debug_enabled else "headless"}',
            debug_enabled,
        )

        browser: Browser = await playwright.chromium.launch(
            headless=not debug_enabled,
            slow_mo=100 if debug_enabled else 0,
        )

        context: BrowserContext = await browser.new_context(
            viewport={
                'width': 1920,
                'height': 1080,
            },
            device_scale_factor=1,
        )

        page = await context.new_page()

        debug(
            f'Открывается URL: {url}',
            debug_enabled,
        )
```

```
484
485         try:
486             await page.goto(
487                 url,
488                 wait_until='domcontentloaded',
489                 timeout=15000,
490             )
491
492         except Exception as exc:
493             logger.console(
494                 f'Ошибка открытия страницы: {exc}'
495             )
496
497             await browser.close()
498
499             return 1
500
501         await wait_for_page(
502             page,
503             wait_ms,
504             debug_enabled,
505         )
506
507         initial_name = get_screenshot_name(
508             page,
509             'initial',
510         )
511
512         initial_path = (
513             output_dir /
514             f'{initial_name}.png'
515         )
516
517         await save_screenshot(
518             page,
519             initial_path,
520             full_page,
521             debug_enabled,
```

```
        )

        logger.console(
            f'[000] {initial_path}'
        )

        navigation = await collect_navigation(
            page,
            debug_enabled,
        )

        logger.console(
            'Найдено элементов навигации: '
            f'{len(navigation)}'
        )

        for number, item in enumerate(
            navigation,
            start=1,
        ):
            label = (
                item.get('text')
                or item.get('id')
                or f'item_{number}'
            )

            logger.console(
                f'[{number:03d}] {label}'
            )

            try:
                debug(
                    f'Возврат к начальному URL: {url}',
                    debug_enabled,
                )

                await page.goto(
                    url,
```

```
                    wait_until='domcontentloaded',
                    timeout=15000,
                )

                await wait_for_page(
                    page,
                    wait_ms,
                    debug_enabled,
                )

                success = await navigate_to_item(
                    page,
                    item,
                    wait_ms,
                    debug_enabled,
                )

                if not success:
                    logger.console(
                        '      -> ОШИБКА перехода'
                    )

                    manifest['screenshots'].append(
                        {
                            'number': number,
                            'label': label,
                            'href': item.get(
                                'href',
                                '',
                            ),
                            'status': (
                                'navigation_error'
                            ),
                        }
                    )

                    continue

```

```
                screenshot_name = get_screenshot_name(
                    page,
                    sanitize_filename(
                        label,
                        f'item_{number}',
                    ),
                )

                output_path = (
                    output_dir /
                    f'{screenshot_name}.png'
                )

                await save_screenshot(
                    page,
                    output_path,
                    full_page,
                    debug_enabled,
                )

                logger.console(
                    f'      -> {output_path}'
                )

                manifest['screenshots'].append(
                    {
                        'number': number,
                        'label': label,
                        'tag': item.get(
                            'tag',
                            '',
                        ),
                        'id': item.get(
                            'id',
                            '',
                        ),
                        'href': item.get(
                            'href',
```

```
                        '',
                    ),
                    'url_after_navigation': (
                        page.url
                    ),
                    'hash': get_url_hash(
                        page.url
                    ),
                    'file': output_path.name,
                    'status': 'ok',
                }
            )

        except Exception as exc:
            logger.console(
                f'      -> ОШИБКА: {exc}'
            )

            manifest['screenshots'].append(
                {
                    'number': number,
                    'label': label,
                    'tag': item.get(
                        'tag',
                        '',
                    ),
                    'id': item.get(
                        'id',
                        '',
                    ),
                    'href': item.get(
                        'href',
                        '',
                    ),
                    'status': 'error',
                    'error': str(exc),
                }
            )
```

```
674
675         manifest_path = (
676             output_dir /
677             'manifest.json'
678         )
679
680         manifest_path.write_text(
681             json.dumps(
682                 manifest,
683                 ensure_ascii=False,
684                 indent=2,
685             ),
686             encoding='utf-8',
687         )
688
689         await browser.close()
690
691     logger.console()
692     logger.console('Готово.')
693     logger.console(
694         f'Скриншоты: {output_dir.resolve()}'
695     )
696     logger.console(
697         f'Manifest:  {manifest_path.resolve()}'
698     )
699
700     return 0
701
702
703 async def main() -> int:
704     """Запускается приложение.
705
706     :returns: Код завершения процесса.
707     """
708
709     parser = build_parser()
710     args = parser.parse_args()
711
```

```
    return await capture_interface(
        url=args.url,
        output_dir=args.output,
        wait_ms=args.wait,
        full_page=args.full_page,
        debug_enabled=args.debug,
    )


if __name__ == '__main__':
    sys.exit(
        asyncio.run(main())
    )

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/chat/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/chat/index.html" class=
"w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/chat
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<div class="chat-container d-flex flex-row h-100 w-100 overflo
w-hidden gap-2" style="min-height: 80vh; height: calc(100vh -
65px);">
  <!-- Sessions Sidebar -->
  <aside class="chat-sidebar d-flex flex-column flex-shrink-0
p-2 border rounded" id="chat-sidebar" style="width: 250px; min
-width: 250px; max-width: 250px; height: 100%;">
    <div class="d-flex align-items-center justify-content-betw
een mb-2 flex-shrink-0">
```

```
      <button class="btn btn-primary btn-sm w-100 d-flex align-items-center justify-content-center gap-2" id="chat-new-session-btn" type="button" title="Начать новый диалог">
        <i class="bi bi-plus-lg"></i>
        <span>Новый чат</span>
      </button>
    </div>

    <!-- Sessions List -->
    <div class="chat-sessions-list flex-grow-1 overflow-auto d-flex flex-column gap-1 my-1" id="chat-sessions-list" style="min-height: 0;">
      <!-- Session items injected via JS -->
    </div>

    <!-- Sidebar Footer -->
    <div class="pt-2 border-top border-secondary-subtle d-flex align-items-center justify-content-between flex-shrink-0">
      <div class="d-flex align-items-center gap-2 flex-grow-1">
        <small class="text-muted" id="chat-sessions-count">0 чатов</small>
        <span class="badge bg-secondary-subtle text-secondary cache-badge" id="chat-cache-badge" title="Кеш: попадания/всего" style="cursor: help; font-size: 0.7rem;">
          <i class="bi bi-lightning-charge"></i> <span id="chat-cache-ratio">-</span>
        </span>
      </div>
      <div class="d-flex align-items-center gap-1">
        <button class="btn btn-sm btn-link text-warning text-decoration-none p-0" id="chat-clear-cache-btn" type="button" title="Очистить кеш">
          <i class="bi bi-arrow-clockwise"></i>
        </button>
        <button class="btn btn-sm btn-link text-danger text-decoration-none p-0" id="chat-clear-all-sessions-btn" type="button" title="Удалить все чаты">
```

```
          <i class="bi bi-trash3"></i>
        </button>
      </div>
    </div>
  </aside>

  <!-- Main Chat Conversation Area -->
  <main class="chat-main-area d-flex flex-column flex-grow-1 h-100 overflow-hidden" style="min-width: 0;">
    <!-- Top Session Header Bar -->
    <div class="chat-header-bar d-flex align-items-center justify-content-between p-1 px-2 mb-2 rounded border flex-shrink-0">
      <div class="d-flex align-items-center gap-2 flex-grow-1 overflow-hidden">
        <button class="btn btn-sm btn-outline-secondary py-0 px-2" id="chat-sidebar-toggle-btn" type="button" title="Показать/скрыть панель чатов">
          <i class="bi bi-layout-sidebar"></i>
        </button>
        <span class="fw-semibold text-truncate" id="chat-current-title" title="Текущий диалог">Новый чат</span>
        <button class="btn btn-sm btn-link text-muted p-0" id="chat-rename-current-btn" type="button" title="Переименовать диалог">
          <i class="bi bi-pencil-square"></i>
        </button>
      </div>

      <div class="d-flex align-items-center gap-1 flex-shrink-0">
        <button class="btn btn-sm btn-outline-secondary py-0 px-2" id="chat-clear-btn" type="button" title="Очистить текущий диалог">
          <i class="bi bi-eraser me-1"></i><span class="d-none d-sm-inline">Очистить</span>
        </button>
      </div>
```

```
77      </div>
78
79      <!-- Chat Messages Window -->
80      <div class="chat-window flex-grow-1 mb-2 overflow-auto" id
    ="chat-window"></div>
81
82      <!-- Execution Status Indicator -->
83      <div id="chat-status-indicator" class="d-none small text-m
    uted mb-2 d-flex align-items-center gap-2 flex-shrink-0">
84        <span class="spinner-border spinner-border-sm text-prima
    ry" role="status" aria-hidden="true"></span>
85        <span id="chat-status-text">Обработка запроса...</span>
86      </div>
87
88      <!-- Query Input & Settings Bar -->
89      <div class="d-flex flex-wrap align-items-center gap-2 flex
    -shrink-0">
90        <!-- RAG Search Toggle Switch -->
91        <div class="form-check form-switch d-flex align-items-ce
    nter gap-1 m-0">
92          <input class="form-check-input mt-0" type="checkbox" r
    ole="switch" id="chat-rag-toggle" checked title="Использовать
    базу знаний (RAG)">
93          <label class="form-check-label small fw-semibold text-
    muted text-nowrap user-select-none" for="chat-rag-toggle" titl
    e="Использовать базу знаний (RAG)">
94            <i class="bi bi-database-check me-1"></i>RAG
95          </label>
96        </div>
97
98        <!-- TOP_K Setting -->
99        <div class="d-flex align-items-center gap-1">
100         <label for="chat-top-k" class="small fw-semibold text-
    muted text-nowrap">TOP_K:</label>
101         <input type="number" class="form-control form-control-
    sm" id="chat-top-k" value="3" min="1" max="50" style="width: 7
    5px;" title="Количество результатов (TOP_K)">
102       </div>
```

```
103
104       <!-- MIN_SCORE Setting -->
105       <div class="d-flex align-items-center gap-1">
106         <label for="chat-min-score" class="small fw-semibold t
ext-muted text-nowrap">MIN_SCORE:</label>
107         <input type="number" class="form-control form-control-
sm" id="chat-min-score" value="0.45" min="0.0" max="1.0" step=
"0.05" style="width: 85px;" title="Минимальный порог (MIN_SCOR
E)">
108       </div>
109
110       <!-- Output Mode (Text only / Voice only / Text and Voic
e) -->
111       <div class="d-flex align-items-center gap-1">
112         <label for="chat-output-mode" class="small fw-semibold
 text-muted text-nowrap user-select-none" title="Режим вывода
ответа">
113           <i class="bi bi-sliders me-1"></i>Режим:
114         </label>
115         <select class="form-select form-select-sm" id="chat-ou
tput-mode" style="width: auto; min-width: 145px;" title="Режим
 вывода ответа">
116           <option value="text_only" selected>  Только текст</
option>
117           <option value="text_and_voice">  +  Текст и голос</
option>
118           <option value="voice_only">  Только голос</option>
119         </select>
120       </div>
121
122       <!-- Message Input and Send Button -->
123       <div class="input-group flex-grow-1">
124         <input type="text" class="form-control" id="message-in
put" placeholder="Введите сообщение..." autocomplete="off" dat
a-lpignore="true" data-1p-ignore="true">
125         <button class="btn btn-primary" id="send-button" type=
"button">Отправить</button>
126       </div>
```

```
    </div>
  </main>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Chat - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/chat/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/chat
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// ── CHAT.JS ──────────────────────────────────────────────────

import { cachedApiFetch, clearCacheByTag } from '../js/api-cache.js';

const STORAGE_KEY_SESSIONS = 'ai_windows_admin_chat_sessions_v2';
const STORAGE_KEY_ACTIVE_SESSION_ID = 'ai_windows_admin_active_session_id';
const STORAGE_KEY_SIDEBAR_COLLAPSED = 'ai_windows_admin_chat_sidebar_collapsed';
```

```
const LEGACY_STORAGE_KEY = 'ai_windows_admin_chat_history';
const CACHE_TAG = 'chat';

let sessions = [];
let activeSessionId = null;

/**
 * Generates a clean, human-friendly title based on the first prompt (ChatGPT-style).
 */
function generateSessionTitle(prompt) {
  if (!prompt || typeof prompt !== 'string') return 'Новый чат';

  let clean = prompt
    .replace(/^```[\s\S]*?```/g, '')
    .replace(/<[^>]*>/g, '')
    .replace(/[*_#`[\]()]/g, '')
    .trim();

  if (!clean) return 'Новый чат';

  // Take the first line or first sentence
  const firstLine = clean.split('\n')[0].trim();
  const sentenceMatch = firstLine.match(/^(.+?)(?:[.!?]|$)/);
  let text = sentenceMatch ? sentenceMatch[1].trim() : firstLine;

  if (text.length > 36) {
    text = text.substring(0, 36).trim();
    const lastSpace = text.lastIndexOf(' ');
    if (lastSpace > 15) {
      text = text.substring(0, lastSpace);
    }
    text += '...';
  }

  // Capitalize first character
```

```
  return text.charAt(0).toUpperCase() + text.slice(1);
}

let syncDebounceTimer = null;

function saveLocalSessionsOnly() {
  try {
    localStorage.setItem(STORAGE_KEY_SESSIONS, JSON.stringify(
sessions));
    if (activeSessionId) {
      localStorage.setItem(STORAGE_KEY_ACTIVE_SESSION_ID, acti
veSessionId);
    }
  } catch (err) {
    console.warn('Failed to save chat sessions to localStorage
:', err);
  }
}

function saveSessions(syncToServer = true) {
  saveLocalSessionsOnly();
  if (syncToServer) {
    debounceSyncActiveSession();
  }
}

function debounceSyncActiveSession() {
  if (syncDebounceTimer) {
    clearTimeout(syncDebounceTimer);
  }
  syncDebounceTimer = setTimeout(() => {
    const active = getActiveSession();
    if (active) {
      syncSessionToServer(active);
    }
  }, 500);
}

```

```
async function syncSessionToServer(session) {
  if (!session || !session.id) return;
  try {
    await fetch('/api/chat/sessions', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        id: session.id,
        title: session.title || 'Новый чат',
        isCustomTitle: Boolean(session.isCustomTitle),
        createdAt: session.createdAt || Date.now(),
        updatedAt: session.updatedAt || Date.now(),
        messages: session.messages || [],
        chatHistory: session.chatHistory || []
      })
    });
  } catch (err) {
    console.debug('Background session sync to server skipped o
r failed:', err);
  }
}

async function deleteSessionFromServer(sessionId) {
  if (!sessionId) return;
  try {
    await fetch(`/api/chat/sessions/${encodeURIComponent(sessi
onId)}`, {
      method: 'DELETE'
    });
  } catch (err) {
    console.debug('Failed to delete session on server:', err);
  }
}

async function clearAllSessionsFromServer() {
  try {
    await fetch('/api/chat/sessions', {
      method: 'DELETE'
```

```
    });
  } catch (err) {
    console.debug('Failed to clear sessions on server:', err);
  }
}

async function syncSessionsWithServer() {
  try {
    const res = await cachedApiFetch('/api/chat/sessions', {
      tags: [CACHE_TAG, 'chat-sessions']
    });
    if (!res.ok) return;
    const data = await res.json();
    const serverSessions = data.sessions || [];

    // If we have local sessions and server is empty, or serve
r has fewer/older sessions, bulk sync
    if (sessions.length > 0 && serverSessions.length === 0) {
      const syncRes = await fetch('/api/chat/sessions/sync', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ sessions: sessions })
      });
      if (syncRes.ok) {
        const syncData = await syncRes.json();
        if (syncData.sessions && syncData.sessions.length > 0)
 {
          sessions = syncData.sessions;
        }
      }
    } else if (serverSessions.length > 0) {
      // Merge server sessions with any unique local sessions
      const serverMap = new Map();
      serverSessions.forEach((s) => serverMap.set(s.id, s));

      const localOnly = [];
      sessions.forEach((ls) => {
        if (!serverMap.has(ls.id)) {
```

```
          localOnly.push(ls);
        } else {
          // If local has newer updatedAt, prefer local and sync to server
          const ss = serverMap.get(ls.id);
          if ((ls.updatedAt || 0) > (ss.updatedAt || 0)) {
            serverMap.set(ls.id, ls);
            syncSessionToServer(ls);
          }
        }
      });

      if (localOnly.length > 0) {
        // Push local only to server
        await fetch('/api/chat/sessions/sync', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ sessions: localOnly })
        });
      }

      sessions = Array.from(serverMap.values()).sort((a, b) => (b.updatedAt || 0) - (a.updatedAt || 0));
    }

    if (sessions.length === 0) {
      createNewSession('Новый чат', true);
      return;
    }

    const savedActiveId = localStorage.getItem(STORAGE_KEY_ACTIVE_SESSION_ID);
    if (savedActiveId && sessions.some((s) => s.id === savedActiveId)) {
      activeSessionId = savedActiveId;
    } else {
      activeSessionId = sessions[0].id;
    }
```

```
204
205     saveLocalSessionsOnly();
206     renderSessionsList();
207     renderActiveSessionMessages();
208   } catch (err) {
209     console.debug('Failed to sync sessions with server (workin
    g in offline/local mode):', err);
210   }
211 }
212
213 function loadSessions() {
214   try {
215     const raw = localStorage.getItem(STORAGE_KEY_SESSIONS);
216     if (raw) {
217       const parsed = JSON.parse(raw);
218       if (Array.isArray(parsed) && parsed.length > 0) {
219         sessions = parsed;
220         const savedActiveId = localStorage.getItem(STORAGE_KEY
    _ACTIVE_SESSION_ID);
221         if (savedActiveId && sessions.some((s) => s.id === sav
    edActiveId)) {
222           activeSessionId = savedActiveId;
223         } else {
224           activeSessionId = sessions[0].id;
225         }
226         syncSessionsWithServer();
227         return;
228       }
229     }
230
231     // Migrate from legacy single-session chat history if pres
    ent
232     const legacyRaw = localStorage.getItem(LEGACY_STORAGE_KEY)
    ;
233     if (legacyRaw) {
234       try {
235         const legacyLog = JSON.parse(legacyRaw);
236         if (Array.isArray(legacyLog) && legacyLog.length > 0)
```

```
{
          const firstUserMsg = legacyLog.find((m) => m.sender
=== 'user');
          const title = firstUserMsg ? generateSessionTitle(fi
rstUserMsg.text) : 'Предыдущий диалог';
          const legacyHistory = [];
          for (const item of legacyLog) {
            legacyHistory.push({
              role: item.sender === 'user' ? 'user' : 'model',
              parts: [item.text || item.voiceText || '']
            });
          }

          const legacySession = {
            id: 'session_' + Date.now(),
            title: title,
            isCustomTitle: false,
            createdAt: Date.now(),
            updatedAt: Date.now(),
            messages: legacyLog,
            chatHistory: legacyHistory
          };
          sessions = [legacySession];
          activeSessionId = legacySession.id;
          saveSessions();
          syncSessionsWithServer();
          return;
        }
      } catch (e) {
        console.warn('Failed to migrate legacy chat history:',
 e);
      }
    }

    // Default clean initial session and attempt server sync
    createNewSession('Новый чат', true);
    syncSessionsWithServer();
  } catch (err) {
```

```
    console.warn('Failed to load chat sessions:', err);
    sessions = [];
    createNewSession('Новый чат', true);
    syncSessionsWithServer();
  }
}

function getActiveSession() {
  if (!activeSessionId || sessions.length === 0) {
    return createNewSession('Новый чат', true);
  }
  let session = sessions.find((s) => s.id === activeSessionId);
  if (!session) {
    session = sessions[0];
    activeSessionId = session.id;
  }
  return session;
}

function createNewSession(title = 'Новый чат', makeActive = true) {
  const newSession = {
    id: 'session_' + Date.now() + '_' + Math.random().toString(36).substring(2, 7),
    title: title,
    isCustomTitle: false,
    createdAt: Date.now(),
    updatedAt: Date.now(),
    messages: [],
    chatHistory: []
  };

  sessions.unshift(newSession);
  if (makeActive) {
    activeSessionId = newSession.id;
  }
  saveSessions();
```

```
  renderSessionsList();
  renderActiveSessionMessages();
  return newSession;
}

function switchSession(sessionId) {
  if (sessionId === activeSessionId) return;
  const session = sessions.find((s) => s.id === sessionId);
  if (!session) return;

  activeSessionId = sessionId;
  saveSessions(false);
  renderSessionsList();
  renderActiveSessionMessages();

  const msgInput = document.getElementById('message-input');
  if (msgInput) msgInput.focus();
}

function renameSession(sessionId, newTitle = null) {
  const session = sessions.find((s) => s.id === sessionId);
  if (!session) return;

  let title = newTitle;
  if (title === null) {
    title = prompt('Введите новое название диалога:', session.
title);
  }

  if (title && title.trim()) {
    session.title = title.trim();
    session.isCustomTitle = true;
    session.updatedAt = Date.now();
    saveSessions();
    renderSessionsList();
    if (session.id === activeSessionId) {
      updateCurrentSessionHeader();
    }
```

```
  }
}

function deleteSession(sessionId) {
  const index = sessions.findIndex((s) => s.id === sessionId);
  if (index === -1) return;

  sessions.splice(index, 1);
  if (sessions.length === 0) {
    createNewSession('Новый чат', true);
  } else if (activeSessionId === sessionId) {
    activeSessionId = sessions[0].id;
    renderActiveSessionMessages();
  }

  saveSessions(false);
  deleteSessionFromServer(sessionId);
  renderSessionsList();
}

function clearCurrentSession() {
  const session = getActiveSession();
  if (!session) return;

  session.messages = [];
  session.chatHistory = [];
  session.title = 'Новый чат';
  session.isCustomTitle = false;
  session.updatedAt = Date.now();

  saveSessions();
  renderSessionsList();
  renderActiveSessionMessages();
}

function clearAllSessions() {
  if (!confirm('Вы уверены, что хотите удалить ВСЕ сохраненные
 диалоги?')) return;
```

```
380
381   sessions = [];
382   try {
383     localStorage.removeItem(STORAGE_KEY_SESSIONS);
384     localStorage.removeItem(STORAGE_KEY_ACTIVE_SESSION_ID);
385     localStorage.removeItem(LEGACY_STORAGE_KEY);
386   } catch (e) {}
387
388   // Clear cache for chat sessions
389   clearCacheByTag(CACHE_TAG).catch(err => console.debug('Faile
    d to clear chat cache:', err));
390
391   clearAllSessionsFromServer();
392   createNewSession('Новый чат', true);
393 }
394
395
396 function updateCurrentSessionHeader() {
397   const titleEl = document.getElementById('chat-current-title'
    );
398   const session = getActiveSession();
399   if (titleEl && session) {
400     titleEl.textContent = session.title || 'Новый чат';
401     titleEl.title = session.title || 'Новый чат';
402   }
403 }
404
405 function renderSessionsList() {
406   const listEl = document.getElementById('chat-sessions-list')
    ;
407   const countEl = document.getElementById('chat-sessions-count
    ');
408   if (!listEl) return;
409
410   listEl.innerHTML = '';
411   if (countEl) {
412     countEl.textContent = `${sessions.length} ${sessions.lengt
    h === 1 ? 'чат' : (sessions.length < 5 ? 'чата' : 'чатов')}`;
```

```
413   }
414
415   updateCurrentSessionHeader();
416
417   sessions.forEach((session) => {
418     const item = document.createElement('div');
419     const isActive = session.id === activeSessionId;
420     item.className = 'chat-session-item d-flex align-items-cen
ter justify-content-between p-1 px-2 rounded mb-1' + (isActive
 ? ' active' : '');
421     item.dataset.sessionId = session.id;
422
423     item.innerHTML = `
424       <div class="chat-session-title d-flex align-items-center
 gap-1 text-truncate flex-grow-1 me-1" title="${session.title}
">
425         <i class="bi bi-chat-text ${isActive ? 'text-primary'
: 'text-muted'} flex-shrink-0"></i>
426         <span class="text-truncate">${session.title}</span>
427       </div>
428       <div class="chat-session-actions d-flex align-items-cent
er gap-1 flex-shrink-0">
429         <button class="btn btn-sm btn-link text-muted p-0 btn-
rename" title="Переименовать" type="button">
430           <i class="bi bi-pencil" style="font-size: 0.8rem;"><
/i>
431         </button>
432         <button class="btn btn-sm btn-link text-danger p-0 btn
-delete" title="Удалить" type="button">
433           <i class="bi bi-trash" style="font-size: 0.8rem;"></
i>
434         </button>
435       </div>
436     `;
437
438     item.addEventListener('click', (e) => {
439       if (e.target.closest('.btn-rename') || e.target.closest(
'.btn-delete')) return;
```

```
      switchSession(session.id);
    });

    const renameBtn = item.querySelector('.btn-rename');
    if (renameBtn) {
      renameBtn.addEventListener('click', (e) => {
        e.stopPropagation();
        renameSession(session.id);
      });
    }

    const deleteBtn = item.querySelector('.btn-delete');
    if (deleteBtn) {
      deleteBtn.addEventListener('click', (e) => {
        e.stopPropagation();
        if (confirm(`Удалить диалог "${session.title}"?`)) {
          deleteSession(session.id);
        }
      });
    }

    listEl.appendChild(item);
  });
}

function renderActiveSessionMessages() {
  const chatWindow = document.getElementById('chat-window');
  if (!chatWindow) return;

  chatWindow.innerHTML = '';
  const session = getActiveSession();

  if (session && session.messages && session.messages.length > 0) {
    for (const msg of session.messages) {
      addMessageToDom(msg.sender, msg.text, msg.voiceText, msg.timeStr, msg.mode);
    }
```

```
  } else {
    const welcome = document.createElement('div');
    welcome.className = 'message bot-message';
    welcome.innerHTML = '<strong>AI Assistant</strong>: Добро пожаловать! Задайте любой вопрос в поле ввода.';
    chatWindow.appendChild(welcome);
  }

  chatWindow.scrollTop = chatWindow.scrollHeight;
  updateCurrentSessionHeader();
}

/**
 * Update cache status badge
 */
function updateChatCacheStatus() {
  const badge = document.getElementById('chat-cache-badge');
  const ratio = document.getElementById('chat-cache-ratio');
  if (!badge || !ratio) return;

  try {
    const stats = window.apiCacheStats || { hits: 0, total: 0 };
    const hitRate = stats.total > 0 ? Math.round((stats.hits / stats.total) * 100) : 0;
    ratio.textContent = `${hitRate}%`;

    // Color coding based on hit rate
    badge.className = 'badge cache-badge';
    if (hitRate >= 70) {
      badge.classList.add('bg-success-subtle', 'text-success');
    } else if (hitRate >= 40) {
      badge.classList.add('bg-warning-subtle', 'text-warning');
    } else {
      badge.classList.add('bg-secondary-subtle', 'text-secondary');
```

```
    }
  } catch (err) {
    console.debug('Failed to update cache status:', err);
  }
}

/**
 * Clear chat cache
 */
async function clearChatCache() {
  try {
    await clearCacheByTag(CACHE_TAG);
    updateChatCacheStatus();
    console.log('Chat cache cleared');
  } catch (err) {
    console.error('Failed to clear chat cache:', err);
  }
}

function initChatTab() {
  loadSessions();
  renderSessionsList();
  renderActiveSessionMessages();

  // Update cache status
  updateChatCacheStatus();

  // Bind Sidebar Toggle
  const sidebar = document.getElementById('chat-sidebar');
  const sidebarToggleBtn = document.getElementById('chat-sideb
ar-toggle-btn');
  if (sidebar && sidebarToggleBtn) {
    const isCollapsed = localStorage.getItem(STORAGE_KEY_SIDEB
AR_COLLAPSED) === 'true';
    if (isCollapsed) {
      sidebar.classList.add('collapsed');
      sidebar.style.display = 'none';
    }
```

```
544
545     sidebarToggleBtn.addEventListener('click', () => {
546       const isCurrentlyCollapsed = sidebar.classList.toggle('c
ollapsed');
547       sidebar.style.display = isCurrentlyCollapsed ? 'none' :
'flex';
548       localStorage.setItem(STORAGE_KEY_SIDEBAR_COLLAPSED, isCu
rrentlyCollapsed);
549     });
550   }
551
552   // Bind New Session Button
553   const newSessionBtn = document.getElementById('chat-new-sess
ion-btn');
554   if (newSessionBtn) {
555     newSessionBtn.addEventListener('click', () => {
556       createNewSession('Новый чат', true);
557       const msgInput = document.getElementById('message-input'
);
558       if (msgInput) msgInput.focus();
559     });
560   }
561
562   // Bind Clear Current / Clear All buttons
563   const clearBtn = document.getElementById('chat-clear-btn');
564   if (clearBtn) {
565     clearBtn.addEventListener('click', () => {
566       if (confirm('Очистить сообщения текущего диалога?')) {
567         clearCurrentSession();
568       }
569     });
570   }
571
572   const clearAllBtn = document.getElementById('chat-clear-all-
sessions-btn');
573   if (clearAllBtn) {
574     clearAllBtn.addEventListener('click', () => {
575       clearAllSessions();
```

```
    });
  }

  const clearCacheBtn = document.getElementById('chat-clear-ca
che-btn');
  if (clearCacheBtn) {
    clearCacheBtn.addEventListener('click', async () => {
      await clearChatCache();
      // Force refresh sessions
      await syncSessionsWithServer();
      renderSessionsList();
    });
  }

  const renameCurrentBtn = document.getElementById('chat-renam
e-current-btn');
  if (renameCurrentBtn) {
    renameCurrentBtn.addEventListener('click', () => {
      if (activeSessionId) {
        renameSession(activeSessionId);
      }
    });
  }

  // Expose sendMessage globally for user menu actions
  window.sendChatMessage = sendMessage;

  // Bind Input and Controls
  const sendBtn = document.getElementById('send-button');
  const msgInput = document.getElementById('message-input');
  const ragToggle = document.getElementById('chat-rag-toggle')
;
  const topKInput = document.getElementById('chat-top-k');
  const minScoreInput = document.getElementById('chat-min-scor
e');

  const updateRagInputsState = (enabled) => {
    if (topKInput) {
```

```
610       topKInput.disabled = !enabled;
611       topKInput.style.opacity = enabled ? '1' : '0.5';
612     }
613     if (minScoreInput) {
614       minScoreInput.disabled = !enabled;
615       minScoreInput.style.opacity = enabled ? '1' : '0.5';
616     }
617   };
618
619   if (ragToggle) {
620     const savedRagPref = localStorage.getItem('chat_rag_enable
    d');
621     if (savedRagPref !== null) {
622       ragToggle.checked = savedRagPref === 'true';
623       updateRagInputsState(ragToggle.checked);
624     }
625     fetch('/auth/settings', { cache: 'no-cache' })
626       .then((res) => (res.ok ? res.json() : null))
627       .then((settings) => {
628         if (settings && typeof settings.rag_enabled !== 'undef
    ined') {
629           const isEnabled = settings.rag_enabled !== 0;
630           ragToggle.checked = isEnabled;
631           localStorage.setItem('chat_rag_enabled', isEnabled);
632           updateRagInputsState(isEnabled);
633         }
634       })
635       .catch(() => {});
636
637     ragToggle.addEventListener('change', async (e) => {
638       const isEnabled = e.target.checked;
639       localStorage.setItem('chat_rag_enabled', isEnabled);
640       updateRagInputsState(isEnabled);
641       try {
642         await fetch('/auth/settings', {
643           method: 'POST',
644           headers: { 'Content-Type': 'application/json' },
645           body: JSON.stringify({ rag_enabled: isEnabled ? 1 :
```

```
0 })
646         });
647       } catch (err) {
648         console.error('Error saving RAG setting:', err);
649       }
650     });
651   }
652
653   const modeSelect = document.getElementById('chat-output-mode
');
654   if (modeSelect) {
655     const savedMode = localStorage.getItem('chat_output_mode')
;
656     if (savedMode) {
657       modeSelect.value = savedMode;
658     } else {
659       modeSelect.value = 'text_only';
660     }
661     modeSelect.addEventListener('change', (e) => {
662       localStorage.setItem('chat_output_mode', e.target.value)
;
663       if (e.target.value === 'text_only' && window.chatService
?.stop) {
664         window.chatService.stop();
665       }
666     });
667   }
668
669   if (sendBtn) {
670     sendBtn.addEventListener('click', sendMessage);
671   }
672   if (msgInput) {
673     msgInput.addEventListener('keypress', (e) => {
674       if (e.key === 'Enter') {
675         sendMessage();
676       }
677     });
678   }
```

```
679
680   // Check for pending external chat payload (from Chrome exte
nsion or URL parameters)
681   try {
682     const pending = sessionStorage.getItem('pending_ai_windows
_admin_chat');
683     if (pending) {
684       const payload = JSON.parse(pending);
685       sessionStorage.removeItem('pending_ai_windows_admin_chat
');
686       setTimeout(() => handleExternalChatPrompt(payload), 300)
;
687     } else {
688       const urlParams = new URLSearchParams(window.location.se
arch);
689       const promptParam = urlParams.get('prompt') || urlParams
.get('message');
690       if (promptParam) {
691         setTimeout(() => {
692           handleExternalChatPrompt({
693             prompt: promptParam,
694             autoSend: urlParams.get('autosend') !== 'false'
695           });
696         }, 300);
697       }
698     }
699   } catch (e) {
700     console.error('Error checking pending chat payload:', e);
701   }
702 }
703
704 function handleExternalChatPrompt(payload) {
705   if (!payload || !payload.prompt) return;
706
707   const chatTabBtn = document.querySelector('[data-bs-target="
#tab-chat"]');
708   if (chatTabBtn && window.bootstrap?.Tab) {
709     bootstrap.Tab.getOrCreateInstance(chatTabBtn).show();
```

```
  }

  const msgInput = document.getElementById('message-input');
  if (msgInput) {
    msgInput.value = payload.prompt;
    if (payload.autoSend !== false) {
      setTimeout(() => {
        sendMessage();
      }, 150);
    }
  }
}

function handleRemoteTranscript(text) {
  if (!text || !text.trim()) return;
  const cleanText = text.trim();

  // Ensure active session exists
  const session = getActiveSession();
  if (!session) return;

  const userTimeStr = new Date().toLocaleTimeString();

  // Clear welcome placeholder if this is the first message
  const chatWin = document.getElementById('chat-window');
  if (chatWin && session.messages.length === 0) {
    chatWin.innerHTML = '';
  }

  // Add message to DOM and save to active session without tri
ggering AI model generation
  addMessageToDom('user', cleanText, null, userTimeStr);
  session.messages.push({ sender: 'user', text: cleanText, tim
eStr: userTimeStr });

  if ((!session.title || session.title === 'Новый чат' || sess
ion.title === 'Предыдущий диалог') && !session.isCustomTitle)
{
```

```
    session.title = generateSessionTitle(cleanText);
    renderSessionsList();
  }

  session.updatedAt = Date.now();
  saveSessions();

  if (chatWin) {
    chatWin.scrollTop = chatWin.scrollHeight;
  }
}

if (typeof window !== 'undefined') {
  window.initChatTab = initChatTab;
  window.handleExternalChatPrompt = handleExternalChatPrompt;
  window.handleRemoteTranscript = handleRemoteTranscript;
  window.clearChatHistory = clearCurrentSession;
  window.createNewSession = createNewSession;
  window.switchSession = switchSession;
  window.renameSession = renameSession;
  window.deleteSession = deleteSession;

  if (typeof chrome !== 'undefined' && chrome.runtime?.onMessa
ge) {
    chrome.runtime.onMessage.addListener((msg) => {
      if (msg && msg.action === 'ai_windows_admin_external_cha
t_prompt') {
        handleExternalChatPrompt(msg);
      }
    });
  }
}

function parseContentToHtml(text) {
  if (!text) return '';

  let cleaned = text.trim();
  const codeBlockRegex = /^```(?:html|xml)?\s*([\s\S]*?)\s*```
```

```
$/i;
  const match = cleaned.match(codeBlockRegex);
  if (match) {
    cleaned = match[1];
  }

  if (window.marked && typeof window.marked.parse === 'function') {
    return window.marked.parse(cleaned);
  }

  return cleaned
    .replace(/\*\*(.*?)\*\*/g, '<strong>$1</strong>')
    .replace(/\*(.*?)\*/g, '<em>$1</em>')
    .replace(/`([^`]+)`/g, '<code>$1</code>')
    .replace(/\n/g, '<br>');
}

function addMessageToDom(sender, text, voiceText = null, timeStr = null, outputMode = null) {
  const win = document.getElementById('chat-window');
  if (!win) return null;

  const el = document.createElement('div');
  el.className = 'message ' + (sender === 'user' ? 'user-message' : 'bot-message');

  const displayText = parseContentToHtml(text);
  const formattedTime = timeStr || new Date().toLocaleTimeString();

  if (sender === 'user') {
    el.innerHTML = `<strong>Вы</strong> (${formattedTime}): <div class="message-content">${displayText}</div>`;
  } else {
    const modeBadge = outputMode === 'voice_only' ? '<span class="badge bg-secondary-subtle text-secondary me-1"><i class="bi bi-volume-up"></i> Голос</span>' : '';
```

```
    el.innerHTML = `
      <div class="d-flex align-items-center justify-content-between mb-1">
        <div><strong>AI Assistant</strong> ${modeBadge}<span class="text-muted">(${formattedTime})</span></div>
        <button class="btn btn-sm btn-outline-secondary py-0 px-2 btn-speak-message" title="Озвучить ответ" type="button">
          <i class="bi bi-volume-up"></i>
        </button>
      </div>
      <div class="message-content">${displayText || (voiceText ? `<em><i class="bi bi-soundwave me-1"></i>${parseContentToHtml(voiceText)}</em>` : '')}</div>
    `;
    const speakBtn = el.querySelector('.btn-speak-message');
    if (speakBtn) {
      speakBtn.addEventListener('click', () => {
        if (window.chatService?.speak) {
          window.chatService.speak(voiceText || text);
        }
      });
    }
  }

  win.appendChild(el);
  win.scrollTop = win.scrollHeight;
  return el;
}

function setStatus(statusText, isVisible = true) {
  const indicator = document.getElementById('chat-status-indicator');
  const textEl = document.getElementById('chat-status-text');
  if (!indicator || !textEl) return;

  if (isVisible) {
    textEl.textContent = statusText || 'Обработка запроса...';
    indicator.classList.remove('d-none');
```

```
  } else {
    indicator.classList.add('d-none');
  }
}

async function sendMessage() {
  const input = document.getElementById('message-input');
  const btn = document.getElementById('send-button');
  const topKInput = document.getElementById('chat-top-k');
  const minScoreInput = document.getElementById('chat-min-scor
e');
  const modeSelect = document.getElementById('chat-output-mode
');
  const ragToggle = document.getElementById('chat-rag-toggle')
;

  if (!input || !btn) return;

  if (!window.activeModelName) {
    input.disabled = true;
    btn.disabled = true;
    input.placeholder = 'Ни одна модель не выбрана. Выберите м
одель во вкладке «Модели и API».';
    input.classList.add('is-invalid');
    return;
  }

  const msg = input.value.trim();
  if (!msg) return;

  const session = getActiveSession();
  if (!session) return;

  const topK = topKInput ? parseInt(topKInput.value, 10) || 3
: 3;
  const minScore = minScoreInput ? parseFloat(minScoreInput.va
lue) || 0.45 : 0.45;
  const outputMode = modeSelect ? modeSelect.value : 'text_onl
```

```
y';
874   const shouldSpeak = outputMode === 'voice_only' || outputMod
e === 'text_and_voice';
875   const isRagEnabled = ragToggle ? ragToggle.checked : true;
876
877   input.value = '';
878   input.disabled = true;
879   btn.disabled = true;
880
881   setStatus('Отправка запроса...', true);
882   const userTimeStr = new Date().toLocaleTimeString();
883
884   // Clear welcome placeholder if this is the first message
885   const chatWin = document.getElementById('chat-window');
886   if (chatWin && session.messages.length === 0) {
887     chatWin.innerHTML = '';
888   }
889
890   addMessageToDom('user', msg, null, userTimeStr);
891   session.messages.push({ sender: 'user', text: msg, timeStr:
userTimeStr });
892
893   // Auto-name the session based on the first query if not cus
tomized
894   if ((!session.title || session.title === 'Новый чат' || sess
ion.title === 'Предыдущий диалог') && !session.isCustomTitle)
{
895     session.title = generateSessionTitle(msg);
896     renderSessionsList();
897   }
898
899   saveSessions();
900
901   const win = document.getElementById('chat-window');
902   const botMessageEl = document.createElement('div');
903   botMessageEl.className = 'message bot-message';
904   const timeStr = new Date().toLocaleTimeString();
905   const modeBadge = outputMode === 'voice_only' ? '<span class
```

```
="badge bg-secondary-subtle text-secondary me-1"><i class="bi
bi-volume-up"></i> Голос</span>' : '';
906   botMessageEl.innerHTML = `
907     <div class="d-flex align-items-center justify-content-betw
een mb-1">
908       <div><strong>AI Assistant</strong> ${modeBadge}<span cla
ss="text-muted">(${timeStr})</span></div>
909       <button class="btn btn-sm btn-outline-secondary py-0 px-
2 btn-speak-message d-none" title="Озвучить ответ" type="butto
n">
910         <i class="bi bi-volume-up"></i>
911       </button>
912     </div>
913     <div class="bot-text"><span class="text-muted">  Ожидание
ответа...</span></div>
914   `;
915   if (win) {
916     win.appendChild(botMessageEl);
917     win.scrollTop = win.scrollHeight;
918   }
919   const textDiv = botMessageEl.querySelector('.bot-text');
920   const speakBtn = botMessageEl.querySelector('.btn-speak-mess
age');
921
922   let fullReply = '';
923   let started = false;
924
925   try {
926     const generationConfig = {
927       top_k: topK,
928       min_score: minScore,
929       threshold: minScore,
930       rag_enabled: isRagEnabled,
931       output_mode: outputMode,
932       tts_enabled: shouldSpeak,
933       model: window.activeModelName || undefined,
934       search_engine: window.activeSearchEngine || undefined
935     };
```

```
936
937     let rafId = null;
938     const scheduleRender = () => {
939       if (rafId) return;
940       rafId = requestAnimationFrame(() => {
941         rafId = null;
942         if (textDiv) {
943           textDiv.innerHTML = parseContentToHtml(fullReply);
944         }
945         if (win) win.scrollTop = win.scrollHeight;
946       });
947     };
948
949     const replyObj = await window.chatService.sendChatMessage(
950       msg,
951       (chunk, status) => {
952         if (status) {
953           setStatus(status, true);
954         }
955         if (chunk) {
956           if (!started) {
957             if (textDiv) textDiv.textContent = '';
958             started = true;
959           }
960           fullReply += chunk;
961           scheduleRender();
962         }
963       },
964       session.chatHistory || [],
965       generationConfig
966     );
967
968     if (rafId) {
969       cancelAnimationFrame(rafId);
970       rafId = null;
971     }
972
973     const finalReply = replyObj.text || fullReply;
```

```
    if (textDiv) {
      if (outputMode === 'voice_only' && !finalReply.trim() &&
 replyObj.voice) {
        textDiv.innerHTML = `<em><i class="bi bi-soundwave me-
1"></i>${parseContentToHtml(replyObj.voice)}</em>`;
      } else {
        textDiv.innerHTML = parseContentToHtml(finalReply);
      }
    }

    if (speakBtn) {
      speakBtn.classList.remove('d-none');
      speakBtn.addEventListener('click', () => {
        if (window.chatService?.speak) {
          window.chatService.speak(replyObj || finalReply);
        }
      });
    }

    // Auto-speak response if mode is voice_only or text_and_v
oice
    if (shouldSpeak && window.chatService?.speak) {
      window.chatService.speak(replyObj || finalReply);
    }

    if (!session.chatHistory) session.chatHistory = [];
    session.chatHistory.push({ role: 'user', parts: [msg] });
    session.chatHistory.push({ role: 'model', parts: [finalRep
ly || replyObj.voice || ''] });

    session.messages.push({
      sender: 'bot',
      text: finalReply,
      voiceText: replyObj.voice || null,
      timeStr: timeStr,
      mode: outputMode
    });
    session.updatedAt = Date.now();
```

```
    saveSessions();

    setStatus('', false);
  } catch (err) {
    const errText = err instanceof Error ? err.message : typeof err === 'object' && err !== null ? JSON.stringify(err) : String(err);
    if (textDiv) {
      textDiv.innerHTML = `<span style="color: #ff7070;">Ошибка: ${errText}</span>`;
    }
    setStatus('Ошибка исполнения', true);
    setTimeout(() => setStatus('', false), 4000);
  } finally {
    if (window.activeModelName) {
      input.disabled = false;
      btn.disabled = false;
      input.placeholder = 'Введите сообщение...';
      input.classList.remove('is-invalid');
      input.focus();
    } else {
      input.disabled = true;
      btn.disabled = true;
      input.placeholder = 'Ни одна модель не выбрана. Выберите модель во вкладке «Модели и API».';
      input.classList.add('is-invalid');
    }
    if (win) win.scrollTop = win.scrollHeight;
  }
}
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/cloudflared_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/cloudflared_tab/index.h
tml" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/cloudflared_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Cloudflare Tunnel Monitor Application Tab -->
<style>
  .cf-container {
    padding: 0.5rem 0.75rem;
  }
  .cf-card {
    background: var(--surface-1) !important;
    border: 1.5px solid var(--border-color) !important;
    border-radius: var(--radius-md, 8px);
    padding: var(--kpi-padding, 0.45rem 0.75rem);
```

```
    display: flex;
    flex-direction: column;
    justify-content: space-between;
    min-height: var(--kpi-min-height, 58px);
    gap: 0.2rem;
    box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.08)
);
  }
  .cf-label {
    font-size: var(--kpi-label-size, 0.68rem);
    text-transform: uppercase;
    letter-spacing: 0.05em;
    font-weight: 600;
    color: var(--text-muted);
    line-height: 1.1;
  }
  .cf-value {
    font-size: var(--kpi-val-size, 1.25rem);
    font-weight: 700;
    color: var(--text-color);
    font-family: var(--font-mono, monospace);
    line-height: 1.15;
  }
  .cf-panel {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: var(--radius-md, 8px);
    overflow: hidden;
    box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.08)
);
  }
  .cf-panel-header {
    background: var(--surface-2) !important;
    border-bottom: 1px solid var(--border-color) !important;
    padding: 0.45rem 0.85rem;
    font-weight: 600;
    font-size: 0.82rem;
    color: var(--text-color);
```

```
    display: flex;
    align-items: center;
    justify-content: space-between;
  }
  .cf-log-terminal {
    background: var(--surface-2);
    color: var(--text-color);
    border: 1px solid var(--border-color);
    font-family: var(--font-mono, monospace);
    font-size: 0.78rem;
    line-height: 1.45;
    padding: 0.5rem 0.75rem;
    border-radius: var(--radius-sm, 6px);
    max-height: 380px;
    overflow-y: auto;
  }
</style>
<div class="cf-container">
<!-- Header Alert & Status (Compact Strip) -->
<div class="card shadow-sm border-secondary-subtle py-1.5 px-3 mb-2 d-flex justify-content-between align-items-center flex-wrap gap-2" data-panel-id="panel-cloudflared-cloudflare-tunnel-monitor" id="panel-cloudflared-cloudflare-tunnel-monitor">
<div class="d-flex align-items-center gap-2 flex-wrap">
<span class="fs-5">□ </span>
<span class="fw-bold text-body small" style="font-size: 0.92rem;">Cloudflare Tunnel Monitor</span>
<span class="badge bg-body-secondary border border-secondary text-info fw-normal" style="font-size: 0.7rem;">Diagnostics</span>
<span class="text-secondary small d-none d-lg-inline" style="font-size: 0.76rem;">| Служба Cloudflare Tunnel, доступность URL &amp;amp; логи</span>
</div>
<div class="d-flex align-items-center gap-1.5 flex-wrap">
<span class="badge rounded-pill bg-success-subtle text-success border border-success px-2 py-1" id="cf-daemon-badge" style="font-size: 0.72rem;">
```

```
        • Активен
      </span>
<button class="btn btn-xs btn-outline-info rounded-pill px-2.5
 py-1" id="btn-cf-refresh" style="font-size: 0.75rem;">
<i class="bi bi-arrow-clockwise me-1"></i> Обновить
      </button>
<button class="btn btn-xs btn-outline-warning rounded-pill px-
2 py-1" id="btn-cf-config" style="font-size: 0.75rem;">
<i class="bi bi-gear-fill me-1"></i> Настройки
      </button>
<button class="btn btn-xs btn-primary rounded-pill px-2.5 py-1
" id="btn-cf-test-endpoint" style="font-size: 0.75rem;">
<i class="bi bi-broadcast me-1"></i> Ingress
      </button>
</div>
</div>
<!-- Top Metrics Bar (Compact Density Grid) -->
<div class="row g-2 mb-2">
<div class="col-6 col-md-3">
<div class="cf-card">
<span class="cf-label">Публичный Ingress URL</span>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<span class="cf-value text-info text-truncate" id="cf-ingress-
url" style="font-size: 0.88rem;">https://kino.davidka.net</spa
n>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="cf-card">
<span class="cf-label">Доступность эндпоинта</span>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<span class="cf-value text-success" id="cf-endpoint-status" st
yle="font-size: 0.95rem;">200 OK</span>
</div>
</div>
```

```
</div>
<div class="col-6 col-md-3">
<div class="cf-card">
<span class="cf-label">CPU / Память демона</span>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<span class="cf-value text-body" id="cf-process-usage" style="
font-size: 0.95rem;">0.5% / 45 MB</span>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="cf-card">
<span class="cf-label">Health Score</span>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<span class="cf-value text-success" id="cf-health-score">100/1
00</span>
</div>
</div>
</div>
</div>
<div class="row g-3">
<!-- Tunnel Logs Tail Panel -->
<div class="col-lg-7">
<div class="cf-panel h-100">
<div class="cf-panel-header">
<span><i class="bi bi-terminal me-1 text-info"></i> Журнал clo
udflared.log</span>
<span class="badge bg-secondary" id="cf-log-count-badge">50 ст
рок</span>
</div>
<div class="p-3">
<div class="cf-log-terminal" id="cf-log-terminal-output">
            Загрузка журналов...
          </div>
</div>
</div>
```

```
</div>
<!-- AI Diagnostics & Recommendations Panel -->
<div class="col-lg-5">
<div class="cf-panel h-100">
<div class="cf-panel-header">
<span><i class="bi bi-heart-pulse-fill text-danger me-1"></i>
Диагностика &amp;amp; Рекомендации</span>
<button class="btn btn-sm btn-outline-info py-0 px-2" id="btn-
cf-run-diagnostics">Диагностировать</button>
</div>
<div class="p-3">
<div class="card bg-body-secondary border-secondary p-3 mb-3"
data-panel-id="panel-cloudflared-status-optimalen" id="panel-c
loudflared-status-optimalen">
<h6 class="fw-bold text-info mb-1" id="cf-diag-summary-title">
Статус оптимален</h6>
<p class="small text-muted mb-0" id="cf-diag-summary-desc">Все
 туннельные соединения стабильны, аномалий не обнаружено.</p>
</div>
<h6 class="small text-uppercase text-muted fw-bold mb-2">Реком
ендации:</h6>
<ul class="list-group list-group-flush bg-transparent small" i
d="cf-recommendations-list">
<li class="list-group-item bg-transparent text-body border-sec
ondary py-1 px-0">• Поддерживайте актуальную версию cloudflare
d</li>
<li class="list-group-item bg-transparent text-body border-sec
ondary py-1 px-0">• Проверяйте сертификат SSL домена</li>
</ul>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Cloudflared Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/cloudflared_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/cloudflared_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// ============================================================
// Process Name: Cloudflared Monitor Web Tab Module
// ============================================================
// Description:
//   Client-side JavaScript controller for the Cloudflare Tunnel Monitor
//   administrative web interface tab.
//
// File: main.js
// Project: ai-windows-admin
```

```
// Package: src.api.webinterface.cloudflared_tab
// Author: hypo69
// Copyright: © 2026 hypo69
// ============================================================
==================

(function() {
  let isCfInitialized = false;

  async function fetchTunnelStatus() {
    try {
      const res = await fetch('/api/cloudflared/status');
      if (!res.ok) return;
      const data = await res.json();

      const badge = document.getElementById('cf-daemon-badge')
;
      const ingressEl = document.getElementById('cf-ingress-ur
l');
      const endpointEl = document.getElementById('cf-endpoint-
status');
      const usageEl = document.getElementById('cf-process-usag
e');
      const healthEl = document.getElementById('cf-health-scor
e');

      if (badge) {
        badge.className = data.daemon_running ? 'badge rounded
-pill bg-success-subtle text-success border border-success px-
3 py-2' : 'badge rounded-pill bg-danger-subtle text-danger bor
der border-danger px-3 py-2';
        badge.innerText = data.daemon_running ? '● Туннель акт
ивен' : '● Туннель остановлен';
      }

      if (ingressEl) ingressEl.innerText = data.public_url ||
'https://kino.davidka.net';
      if (endpointEl) {
```

```
        endpointEl.innerText = data.endpoint_reachable ? `Доступен (${data.latency_ms || 0} ms)` : 'Недоступен';
        endpointEl.className = data.endpoint_reachable ? 'cf-value text-success' : 'cf-value text-danger';
      }
      if (usageEl) {
        usageEl.innerText = `${data.cpu_percent || 0}% / ${data.memory_mb || 0} MB`;
      }
      if (healthEl) {
        healthEl.innerText = `${data.health_score || 100}/100`;
      }
    } catch (e) {
      console.error('[CloudflaredTab] Failed to fetch status:', e);
    }
  }

  async function fetchLogs() {
    try {
      const res = await fetch('/api/cloudflared/logs?limit=50');
      if (!res.ok) return;
      const data = await res.json();
      const logs = data.logs || [];

      const terminal = document.getElementById('cf-log-terminal-output');
      const countBadge = document.getElementById('cf-log-count-badge');
      if (countBadge) countBadge.innerText = `${logs.length} строк`;

      if (terminal) {
        if (logs.length === 0) {
          terminal.innerText = 'Журнал cloudflared пуст.';
          return;
```

```
        }
        terminal.innerHTML = logs.map(l => {
          const color = l.level === 'error' ? '#f87171' : (l.level === 'warn' ? '#fbbf24' : '#94a3b8');
          return `<div style="color: ${color};">[${l.time || ''}] [${(l.level || 'INFO').toUpperCase()}] ${l.msg || l.message || ''}</div>`;
        }).join('');
      }
    } catch (e) {
      console.error('[CloudflaredTab] Failed to fetch logs:', e);
    }
  }

  async function runDiagnostics() {
    try {
      const res = await fetch('/api/cloudflared/diagnostic');
      if (!res.ok) return;
      const report = await res.json();

      const titleEl = document.getElementById('cf-diag-summary-title');
      const descEl = document.getElementById('cf-diag-summary-desc');
      const recList = document.getElementById('cf-recommendations-list');

      if (titleEl) titleEl.innerText = `Здоровье: ${report.status || 'OK'} (${report.health_score || 100}/100)`;
      if (descEl) descEl.innerText = report.summary || 'Все туннельные соединения стабильны.';

      if (recList && Array.isArray(report.recommendations)) {
        recList.innerHTML = report.recommendations.map(r => `
          <li class="list-group-item bg-transparent text-light border-secondary py-1 px-0">• ${r}</li>
        `).join('');
```

```
      }
    } catch (e) {
      console.error('[CloudflaredTab] Diagnostic error:', e);
    }
  }

  function initCloudflaredTab() {
    console.log('[CloudflaredTab] Initializing Cloudflared Mon
itor tab...');
    fetchTunnelStatus();
    fetchLogs();

    if (!isCfInitialized) {
      const refreshBtn = document.getElementById('btn-cf-refre
sh');
      const configBtn = document.getElementById('btn-cf-config
');
      const diagBtn = document.getElementById('btn-cf-run-diag
nostics');
      const testBtn = document.getElementById('btn-cf-test-end
point');

      if (refreshBtn) {
        refreshBtn.onclick = () => {
          fetchTunnelStatus();
          fetchLogs();
        };
      }

      if (diagBtn) diagBtn.onclick = runDiagnostics;

      if (testBtn) {
        testBtn.onclick = async () => {
          testBtn.disabled = true;
          testBtn.innerHTML = '<span class="spinner-border spi
nner-border-sm me-1"></span> Проверка...';
          try {
            const res = await fetch('/api/cloudflared/test-end
```

```
point');
            const data = await res.json();
            const msg = `Результат проверки Ingress:\nДоступен: ${data.reachable ? 'ДА' : 'НЕТ'}\nКод ответа: ${data.status_code || 'N/A'}\nЗадержка: ${data.response_time_ms || 0} ms`;
            window.showToast?.(msg, data.reachable ? 'success' : 'warning') || alert(msg);
          } catch (e) {
            window.showToast?.('Ошибка проверки эндпоинта: ' + e.message, 'danger') || alert('Ошибка проверки эндпоинта: ' + e.message);
          } finally {
            testBtn.disabled = false;
            testBtn.innerHTML = '<i class="bi bi-broadcast me-1"></i> Проверить Ingress';
            fetchTunnelStatus();
          }
        };
      }

      if (configBtn) {
        configBtn.onclick = () => {
          if (typeof window.openAppConfigModal === 'function') {
            window.openAppConfigModal('cloudflared_monitor', 'Cloudflare Tunnel Monitor');
          }
        };
      }

      isCfInitialized = true;
    }
  }

  window.initCloudflaredTab = initCloudflaredTab;
})();

```

tab-registry.js

```
/**
 * ============================================================
==================
 * Process Name: AI-Windows-Admin UI - Tab-Registry Script
 * ============================================================
==================
 * Description:
 *   Клиентский веб-скрипт модуля tab-registry.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/src/api/webgui/core/tab-registry.js?v=202
61006_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { TAB_DEFINITIONS, TabRegistry } from '/src/api/
webgui/core/tab-registry.js';
 *
 * File: tab-registry.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/core
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-12 08:20:00
 * ============================================================
==================
 */

/**
 * TabRegistry – единый источник правды (SSOT) для всех вкладо
к WebGUI.
 * Экспортирует методы getAll(), getById(id), getFiltered(apps
StatusMap, role).
 */
export const TAB_DEFINITIONS = [
  { id: 'about_system', label: 'О системе', i18nKey: 'auto___d
```

```
32ca0', icon: 'i', tab: 'about-system', tabId: 'tab-about-syst
em', html: '/html/about_system_tab/index.html?v=20261012_v1',
js: '/html/about_system_tab/main.js?v=20261012_v1' },
30   { id: 'scenarios', label: 'Сценарии', i18nKey: 'auto___ddaf0
e', icon: '□ ', tab: 'scenarios', tabId: 'tab-scenarios', html
: '/html/scenarios_tab/index.html?v=20261006_v8', js: '/html/s
cenarios_tab/main.js?v=20261006_v8' },
31   { id: 'chat', label: 'Чат', i18nKey: 'auto___8c77e4', icon:
'□ ', tab: 'chat', tabId: 'tab-chat', html: '/html/chat/index.
html?v=20260923_v5', js: '/html/chat/main.js?v=20260923_v5' },
32   { id: 'network_terminal', label: 'Сеть', i18nKey: 'auto___f6
df40', icon: '□ ', tab: 'network', tabId: 'tab-network', html:
 '/html/network_tab/index.html', js: '/html/network_tab/main.j
s' },
33   { id: 'system_inspector', label: 'Инспектор системы', i18nKe
y: 'auto___4324be', icon: 'bi-graph-up-arrow', tab: 'hardware-
load-inspector', tabId: 'tab-hardware-load-inspector', html: '
/html/system_inspector_tab/index.html?v=20261012_v2', js: '/ht
ml/system_inspector_tab/main.js?v=20261012_v2' },
34   { id: 'processes_load_inspector', label: 'Активность процесс
ов', icon: 'bi-activity', tab: 'processes-load-inspector', tab
Id: 'tab-processes-load-inspector', html: '/html/processes_loa
d_inspector_tab/index.html?v=20261012_v2', js: '/html/processe
s_load_inspector_tab/main.js?v=20261012_v2' },
35   { id: 'windows_sysadmin', label: 'Windows Sysadmin', icon: '
bi-server', tab: 'windows-admin', tabId: 'tab-windows-admin',
html: '/html/windows_admin_tab/index.html', js: '/html/windows
_admin_tab/main.js' },
36   { id: 'focus_settings', label: 'Фокусировка (Focus)', icon:
'bi-circle-half', tab: 'focus-settings', tabId: 'tab-focus-set
tings', html: '/html/focus_settings_tab/index.html?v=20261006_
v1', js: '/html/focus_settings_tab/main.js?v=20261006_v1' },
37   { id: 'system32_commands', label: 'Команды System32', icon:
'□ ', tab: 'system32-commands', tabId: 'tab-system32-commands'
, html: '/html/system32_commands_tab/index.html?v=20261006_v1'
, js: '/html/system32_commands_tab/main.js?v=20261006_v1' },
38   { id: 'system_control_center', label: 'System Logs & Activit
y', i18nKey: 'tabs.systemControl', icon: 'bi-journal-text', ta
```

```
b: 'system-control', tabId: 'tab-system-control', html: '/html
/system_control_tab/index.html?v=20261012_v1', js: '/html/syst
em_control_tab/main.js?v=20261012_v1' },
39   { id: 'post_install_wizard', label: 'Post-Install Wizard', i
con: '□ ', tab: 'post-install-wizard', tabId: 'tab-post-install
-wizard', html: '/html/post_install_wizard_tab/index.html?v=20
261006_v1', js: '/html/post_install_wizard_tab/main.js?v=20261
006_v1' },
40   { id: 'maintenance_recovery', label: 'Maintenance & Recovery
', icon: '□ ', tab: 'maintenance-recovery', tabId: 'tab-mainte
nance-recovery', html: '/html/maintenance_recovery_tab/index.h
tml?v=20261006_v1', js: '/html/maintenance_recovery_tab/main.j
s?v=20261006_v1' },
41   { id: 'system_log_viewer', label: 'Логи системы', i18nKey: '
auto___6e28ee', icon: '□ ', tab: 'system-logs', tabId: 'tab-sy
stem-logs', html: '/html/system_logs_tab/index.html', js: '/ht
ml/system_logs_tab/main.js' },
42   { id: 'software_audit', label: 'Аудит ПО', i18nKey: 'auto___
128e2e', icon: '□ ', tab: 'software-audit', tabId: 'tab-softwa
re-audit', html: '/html/software_audit_tab/index.html', js: '/
html/software_audit_tab/main.js' },
43   { id: 'registry_viewer', label: 'Registry Viewer', icon: '□
', tab: 'registry-viewer', tabId: 'tab-registry-viewer', html:
 '/html/registry_viewer_tab/index.html', js: '/html/registry_v
iewer_tab/main.js' },
44   { id: 'windows_defender', label: 'Defender Security', icon:
'bi-shield-lock', tab: 'defender', tabId: 'tab-defender', html
: '/html/defender_tab/index.html?v=20261010_v2', js: '/html/de
fender_tab/main.js?v=20261010_v2' },
45   { id: 'windows_startup_auditor', label: 'Автозагрузка', i18n
Key: 'auto___007983', icon: '□ ', tab: 'startup-auditor', tabI
d: 'tab-startup-auditor', html: '/html/startup_auditor_tab/ind
ex.html', js: '/html/startup_auditor_tab/main.js' },
46   { id: 'windows_backup_manager', label: 'Windows Backup', ico
n: '□ ', tab: 'windows-backup', tabId: 'tab-windows-backup', h
tml: '/html/windows_backup_tab/index.html?v=20261004_v1', js:
'/html/windows_backup_tab/main.js?v=20261004_v1' },
47   { id: 'hardware_monitor', label: 'Мониторинг оборудования',
```

```
i18nKey: 'auto___8a1c35', icon: 'bi-cpu', tab: 'hardware-monit
or', tabId: 'tab-hardware-monitor', html: '/html/hardware_moni
tor_tab/index.html', js: '/html/hardware_monitor_tab/main.js'
},
48   { id: 'cloudflared_monitor', label: 'Cloudflared', icon: '□ '
, tab: 'cloudflared', tabId: 'tab-cloudflared', html: '/html/c
loudflared_tab/index.html', js: '/html/cloudflared_tab/main.js
' },
49   { id: 'google_user_desktop', label: 'Google User Desktop', i
con: '□ ', tab: 'google-desktop', tabId: 'tab-google-desktop',
 html: '/html/google_desktop_tab/index.html?v=20260928_v2', js
: '/html/google_desktop_tab/main.js?v=20260928_v2' },
50   { id: 'gcloud_monitor', label: 'Google Cloud', icon: '□ ', ta
b: 'gcloud', tabId: 'tab-gcloud', html: '/html/gcloud_tab/inde
x.html', js: '/html/gcloud_tab/main.js' },
51   { id: 'website_monitor', label: 'Website Intelligence', icon
: '□ ', tab: 'website-monitor', tabId: 'tab-website-monitor',
html: '/html/website_monitor_tab/index.html', js: '/html/websi
te_monitor_tab/main.js' },
52   { id: 'user_assistant', label: 'User Assistant', icon: '□ ',
 tab: 'user-assistant', tabId: 'tab-user-assistant', html: '/h
tml/user_assistant_tab/index.html', js: '/html/user_assistant_
tab/main.js' },
53   { id: 'helpdesk', label: 'Helpdesk', icon: '□ ', tab: 'helpd
esk', tabId: 'tab-helpdesk', html: '/html/helpdesk_tab/index.h
tml', js: '/html/helpdesk_tab/main.js' },
54   { id: 'wikipedia_research', label: 'Wikipedia Research', ico
n: '□ ', tab: 'wikipedia-research', tabId: 'tab-wikipedia-rese
arch', html: '/html/wikipedia_research_tab/index.html', js: '/
html/wikipedia_research_tab/main.js' },
55   { id: 'autolog_manager', label: 'Автологи', i18nKey: 'auto__
_17b9d1', icon: '□ ', tab: 'autolog', tabId: 'tab-autolog', ht
ml: '/html/telemetry_config_tab/index.html?v=20261010_v2', js:
 '/html/telemetry_config_tab/main.js?v=20261010_v2' },
56   { id: 'software_transparency_scanner', label: 'Transparency
Scanner', icon: '□ ', tab: 'software-transparency', tabId: 'ta
b-software-transparency', html: '/html/software_transparency_t
ab/index.html?v=20260926_v2', js: '/html/software_transparency
```

```
_tab/main.js?v=20260926_v2' },
  { id: 'user_directories', label: 'Каталоги пользователя', i18nKey: 'auto___1ddade', icon: ' ', tab: 'user-directories', tabId: 'tab-user-directories', html: '/html/user_directories_tab/index.html', js: '/html/user_directories_tab/main.js' },
  { id: 'ninite_updater', label: 'Ninite Updater', icon: ' ', tab: 'ninite-updater', tabId: 'tab-ninite-updater', html: '/html/ninite_updater_tab/index.html?v=20260924_v2', js: '/html/ninite_updater_tab/main.js?v=20260924_v2' },
  { id: 'file_recovery', label: 'Восстановление файлов', i18nKey: 'auto___8f9984', icon: ' ', tab: 'file-recovery', tabId: 'tab-file-recovery', html: '/html/file_recovery_tab/index.html', js: '/html/file_recovery_tab/main.js' },
  { id: 'file_history_ai_search', label: 'Поиск файлов AI', i18nKey: 'auto__ai_2d3bf7', icon: ' ', tab: 'file-history-search', tabId: 'tab-file-history-search', html: '/html/file_history_ai_search_tab/index.html?v=20260928_v1', js: '/html/file_history_ai_search_tab/main.js?v=20260928_v1' },
  { id: 'telemetry_research', label: 'Исследование телеметрии', i18nKey: 'auto___025fa1', icon: ' ', tab: 'telemetry-research', tabId: 'tab-telemetry-research', html: '/html/telemetry_research_tab/index.html?v=20260924_v1', js: '/html/telemetry_research_tab/main.js?v=20260924_v1' },
  { id: 'process_leaks', label: 'Утечки процессов', i18nKey: 'auto___6e10ab', icon: ' ', tab: 'process-leaks', tabId: 'tab-process-leaks', html: '/html/process_leaks_tab/index.html?v=20260924_v1', js: '/html/process_leaks_tab/main.js?v=20260924_v1' },
  { id: 'forensics', label: 'Форензика', i18nKey: 'auto___82e8f9', icon: ' ', tab: 'forensics', tabId: 'tab-forensics', html: '/html/forensics_tab/index.html?v=20261010_v2', js: '/html/forensics_tab/main.js?v=20261010_v2' },
  { id: 'throttling', label: 'Троттлинг', i18nKey: 'auto___23403a', icon: ' ', tab: 'throttling', tabId: 'tab-throttling', html: '/html/throttling_tab/index.html?v=20260924_v1', js: '/html/throttling_tab/main.js?v=20260924_v1' },
  { id: 'storage_wear', label: 'Износ дисков', i18nKey: 'auto__5c3dfb', icon: ' ', tab: 'storage-wear', tabId: 'tab-storag
```

```
e-wear', html: '/html/storage_wear_tab/index.html?v=20260924_v
1', js: '/html/storage_wear_tab/main.js?v=20260924_v1' },
66   { id: 'peripherals', label: 'Периферия', i18nKey: 'auto___b6
2c2e', icon: ' ', tab: 'peripherals', tabId: 'tab-peripherals
', html: '/html/peripherals_tab/index.html?v=20260924_v1', js:
 '/html/peripherals_tab/main.js?v=20260924_v1' },
67   { id: 'rag', label: 'RAG База знаний', i18nKey: 'auto_rag__8
29445', icon: ' ', tab: 'rag', tabId: 'tab-rag', html: '/html
/rag_tab/index.html', js: '/html/rag_tab/main.js' },
68   { id: 'pixelrag', label: 'PixelRAG', icon: ' ', tab: 'pixel
rag', tabId: 'tab-pixelrag', html: '/html/pixelrag_tab/index.h
tml', js: '/html/pixelrag_tab/main.js' },
69   { id: 'models', label: 'Модели', i18nKey: 'auto___44bfc5', i
con: ' ', tab: 'models', tabId: 'tab-models', html: '/html/mo
dels_tab/index.html?v=20261010_v1', js: '/html/models_tab/main
.js?v=20261010_v1' },
70   { id: 'agents', label: 'Агенты', i18nKey: 'auto___af1850', i
con: ' ', tab: 'agents', tabId: 'tab-agents', html: '/html/ag
ents_tab/index.html', js: '/html/agents_tab/main.js' },
71   { id: 'skills', label: 'Навыки', i18nKey: 'auto___56221b', i
con: ' ', tab: 'skills', tabId: 'tab-skills', html: '/html/ski
lls_tab/index.html', js: '/html/skills_tab/main.js' },
72   { id: 'ai_benchmark', label: 'AI Benchmark', icon: ' ', tab:
 'ai-benchmark', tabId: 'tab-ai-benchmark', html: '/html/ai_be
nchmark_tab/index.html?v=20260926_v1', js: '/html/ai_benchmark
_tab/main.js?v=20260926_v1' },
73   { id: 'storage_manager', label: 'Управление дисками', icon:
' ', tab: 'storage-manager', tabId: 'tab-storage-manager', ht
ml: '/html/storage_manager_tab/index.html?v=20261001_v1', js:
'/html/storage_manager_tab/main.js?v=20261001_v1' },
74   { id: 'disk_speed', label: 'Скорость накопителей', icon: ' '
, tab: 'disk-speed', tabId: 'tab-disk-speed', html: '/html/dis
k_speed_tab/index.html?v=20261004_v1', js: '/html/disk_speed_t
ab/main.js?v=20261004_v1' },
75   { id: 'boot_recovery', label: 'Загрузка и восстановление', i
con: ' ', tab: 'boot-recovery', tabId: 'tab-boot-recovery', h
tml: '/html/boot_recovery_tab/index.html?v=20261001_v1', js: '
/html/boot_recovery_tab/main.js?v=20261001_v1' },
```

```
76   { id: 'servicing_integrity', label: 'Целостность системы (SF
     C/DISM)', icon: '□ ', tab: 'servicing-integrity', tabId: 'tab-
     servicing-integrity', html: '/html/servicing_integrity_tab/ind
     ex.html?v=20261001_v1', js: '/html/servicing_integrity_tab/mai
     n.js?v=20261001_v1' },
77   { id: 'services_manager', label: 'Службы Windows', icon: '□ '
     , tab: 'services-manager', tabId: 'tab-services-manager', html
     : '/html/services_manager_tab/index.html?v=20261006_v1', js: '
     /html/services_manager_tab/main.js?v=20261006_v1' },
78   { id: 'task_scheduler', label: 'Планировщик задач', icon: '□
     ', tab: 'task-scheduler', tabId: 'tab-task-scheduler', html: '
     /html/task_scheduler_tab/index.html?v=20261006_v2', js: '/html
     /task_scheduler_tab/main.js?v=20261006_v2' },
79   { id: 'process_manager', label: 'Диспетчер процессов', icon:
      '□ ', tab: 'process-manager', tabId: 'tab-process-manager', h
     tml: '/html/process_manager_tab/index.html?v=20261004_v2', js:
      '/html/process_manager_tab/main.js?v=20261004_v2' },
80   { id: 'firewall_manager', label: 'Брандмауэр Windows', icon:
      '□ ', tab: 'firewall-manager', tabId: 'tab-firewall-manager',
      html: '/html/firewall_manager_tab/index.html?v=20261001_v1',
     js: '/html/firewall_manager_tab/main.js?v=20261001_v1' },
81   { id: 'security_acl', label: 'Права доступа и ACL', icon: '□
     □ ', tab: 'security-acl', tabId: 'tab-security-acl', html: '/ht
     ml/security_acl_tab/index.html?v=20261001_v1', js: '/html/secu
     rity_acl_tab/main.js?v=20261001_v1' },
82   { id: 'performance_tracing', label: 'Трассировка ETW', icon:
      '□ ', tab: 'performance-tracing', tabId: 'tab-performance-tra
     cing', html: '/html/performance_tracing_tab/index.html?v=20261
     001_v1', js: '/html/performance_tracing_tab/main.js?v=20261001
     _v1' },
83   { id: 'log_settings', label: 'Параметры журналов', icon: 'bi
     -sliders', tab: 'log-settings', tabId: 'tab-log-settings', htm
     l: '/html/log_settings_tab/index.html?v=20261008_v1', js: '/ht
     ml/log_settings_tab/main.js?v=20261008_v1' },
84   { id: 'software_manager', label: 'Пакетный менеджер', icon:
     '□ ', tab: 'software-manager', tabId: 'tab-software-manager',
     html: '/html/software_manager_tab/index.html?v=20261004_v2', j
     s: '/html/software_manager_tab/main.js?v=20261004_v2' },
```

```
  { id: 'taskbar_controller', label: 'Панель задач и Окна', icon: ' ', tab: 'taskbar', tabId: 'tab-taskbar', html: '/html/taskbar_tab/index.html?v=20261010_v2', js: '/html/taskbar_tab/main.js?v=20261010_v2' },
  { id: 'accounts_identity', label: 'Учетные записи', icon: ' ', tab: 'accounts-identity', tabId: 'tab-accounts-identity', html: '/html/accounts_identity_tab/index.html?v=20261008_v1', js: '/html/accounts_identity_tab/main.js?v=20261008_v1' },
  { id: 'power_lifecycle', label: 'Электропитание и сессии', icon: ' ', tab: 'power-lifecycle', tabId: 'tab-power-lifecycle', html: '/html/power_lifecycle_tab/index.html?v=20261008_v2', js: '/html/power_lifecycle_tab/main.js?v=20261008_v2' },
  { id: 'app_logs', label: 'Логи приложений', icon: ' ', tab: 'app-logs', tabId: 'tab-app-logs', html: '/html/app_logs_tab/index.html?v=20261010_v4', js: '/html/app_logs_tab/main.js?v=20261010_v4' },
  { id: 'software_scanner', label: 'Сканер ПО', icon: ' ', tab: 'software-scanner', tabId: 'tab-software-scanner', html: '/html/software_scanner_tab/index.html?v=20261010_v1', js: '/html/software_scanner_tab/main.js?v=20261010_v1' },
  { id: 'mcp', label: 'MCP Серверы', icon: ' ', tab: 'mcp', tabId: 'tab-mcp', html: '/html/mcp_tab/index.html', js: '/html/mcp_tab/main.js' }
];

export class TabRegistry {
  static getAll() { return TAB_DEFINITIONS; }
  static getById(tabId) {
    if (!tabId) return null;
    const clean = tabId.replace(/^tab-/, '');
    return TAB_DEFINITIONS.find(t =>
      t.id === clean ||
      t.tab === clean ||
      t.tabId === tabId ||
      t.tabId === `tab-${clean}` ||
      t.appId === clean
    ) || null;
  }
```

```
  static getLabel(tab) {
    if (!tab) return '';
    if (tab.i18nKey && typeof window !== 'undefined' && window.i18next?.t) {
      const translated = window.i18next.t(tab.i18nKey);
      if (translated && translated !== tab.i18nKey) return translated;
    }
    return tab.label || '';
  }
  static getFiltered(appsStatusMap = null, targetRole = 'admin') {
    return TAB_DEFINITIONS.filter(tab => {
      if (tab.roles && !tab.roles.includes(targetRole)) return false;
      if (tab.appId && appsStatusMap && appsStatusMap[tab.appId]) {
        return appsStatusMap[tab.appId].enabled !== false;
      }
      return true;
    });
  }
}

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/cosmicplayer/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/cosmicplayer/index.html
" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/cosmicplayer
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!-- Link CosmicPlayer custom style -->
<link rel="stylesheet" href="/html/cosmicplayer/styles.css">
<!-- HLS.js library for streaming M3U8 inside custom HTML5 vid
eo player -->
<script src="https://cdn.jsdelivr.net/npm/hls.js@latest"></scr
ipt>

<div class="cosmic-player-tab">
    <!-- Sidebar: Video URL list / Library / Settings -->
```

```
    <aside class="cosmic-sidebar">
        <div class="logo">
            <span class="logo-icon">🌌</span>
            <h2>CosmicPlayer</h2>
        </div>

        <!-- Paste link panel -->
        <div class="url-input-section">
            <h3>Вставить ссылку</h3>
            <form id="stream-form">
                <input type="url" id="stream-url" placeholder="Ссылка (YouTube, VK, RuTube, Rezka...)" required>
                <button type="submit" class="btn btn-glow">Загрузить видео</button>
            </form>
        </div>

        <!-- Supported sources badges -->
        <div class="sources-panel">
            <h4>Совместимые источники</h4>
            <div class="sources-grid">
                <span class="badge" data-provider="youtube">YouTube</span>
                <span class="badge" data-provider="vk">VK Video</span>
                <span class="badge" data-provider="rutube">RuTube</span>
                <span class="badge" data-provider="hdrezka">HD Rezka</span>
                <span class="badge" data-provider="seasonvar">SeasonVar</span>
                <span class="badge" data-provider="kinogo">Kinogo</span>
                <span class="badge" data-provider="direct">Direct HLS/MP4</span>
            </div>
        </div>

```

```
        <!-- History List -->
        <div class="history-section">
            <div class="section-header">
                <h4>История воспроизведения</h4>
                <button id="clear-history" class="btn-text">Очистить</button>
            </div>
            <div id="history-list" class="history-list">
                <div class="empty-state">Нет недавно проигранных видео</div>
            </div>
        </div>
    </aside>

    <!-- Main Content Area: Player and Description -->
    <main class="main-content">
        <!-- Movie Screen and Ambience -->
        <div class="player-container" id="player-container">
            <!-- Background Ambient Glow -->
            <div class="ambient-glow" id="ambient-glow"></div>

            <!-- Display Iframe (for embeds) -->
            <iframe id="embed-player" class="player-element hidden" allowfullscreen allow="autoplay; encrypted-media"></iframe>

            <!-- Custom HTML5 Player (for direct video feeds) -->
            <div id="html5-player-wrapper" class="player-element hidden">
                <video id="html5-video" class="html5-video-el"></video>
                <!-- Custom Player Skin -->
                <div class="custom-controls">
                    <div class="progress-container" id="progress-container">
                        <div class="progress-bar" id="progress-bar">
```

```
                            <div class="progress-hover"></div>
                        </div>
                    </div>
                    <div class="controls-row">
                        <div class="controls-left">
                            <button class="ctrl-btn" id="play-pause-btn">▶</button>
                            <button class="ctrl-btn" id="stop-btn">■</button>
                            <div class="time-display">
                                <span id="current-time">0:00</span> / <span id="duration-time">0:00</span>
                            </div>
                        </div>
                        <div class="controls-right">
                            <div class="volume-container">
                                <button class="ctrl-btn" id="mute-btn">🔊</button>
                                <input type="range" id="volume-slider" min="0" max="1" step="0.05" value="1">
                            </div>
                            <select id="speed-select" class="speed-dropdown">
                                <option value="0.5">0.5x</option>
                                <option value="1" selected>1.0x</option>
                                <option value="1.5">1.5x</option>
                                <option value="2">2.0x</option>
                            </select>
                            <button class="ctrl-btn" id="fullscreen-btn">⛶</button>
                        </div>
                    </div>
                </div>
            </div>
```

```
            <!-- Initial Empty State -->
            <div id="player-placeholder" class="player-placeho
lder">
                <div class="placeholder-content">
                    <span class="play-icon">▶</span>
                    <h3>Видео не загружено</h3>
                    <p>Вставьте ссылку на видео или выберите о
дин из демонстрационных источников слева.</p>
                </div>
            </div>
        </div>

        <!-- Current Video Information -->
        <div class="video-info-card glass-card">
            <div class="info-header">
                <span class="provider-tag" id="current-provide
r-tag">Offline</span>
                <h3 id="video-title">Добро пожаловать в Cosmic
Player</h3>
            </div>
            <div class="info-body">
                <p id="video-source-url">Пожалуйста, введите U
RL видео для воспроизведения.</p>
            </div>
        </div>
    </main>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Cosmicplayer - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/cosmicplayer/main.js?v
=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/cosmicplayer
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

window.initCosmicplayerTab = function() {
    console.log("CosmicPlayer tab initialized");

    // Elements
    const streamForm = document.getElementById('stream-form');
    const streamUrlInput = document.getElementById('stream-url
');
    const embedPlayer = document.getElementById('embed-player'
);
    const html5PlayerWrapper = document.getElementById('html5-
player-wrapper');
    const html5Video = document.getElementById('html5-video');
```

```
    const playerPlaceholder = document.getElementById('player-placeholder');
    const videoTitle = document.getElementById('video-title');
    const videoSourceUrl = document.getElementById('video-source-url');
    const currentProviderTag = document.getElementById('current-provider-tag');
    const historyList = document.getElementById('history-list');
    const clearHistoryBtn = document.getElementById('clear-history');
    const badges = document.querySelectorAll('.cosmic-player-tab .badge');
    const ambientGlow = document.getElementById('ambient-glow');

    // Custom controls
    const playPauseBtn = document.getElementById('play-pause-btn');
    const stopBtn = document.getElementById('stop-btn');
    const currentTimeDisplay = document.getElementById('current-time');
    const durationTimeDisplay = document.getElementById('duration-time');
    const progressContainer = document.getElementById('progress-container');
    const progressBar = document.getElementById('progress-bar');
    const progressHover = document.querySelector('.progress-hover');
    const muteBtn = document.getElementById('mute-btn');
    const volumeSlider = document.getElementById('volume-slider');
    const speedSelect = document.getElementById('speed-select');
    const fullscreenBtn = document.getElementById('fullscreen-btn');

```

```
    let hlsInstance = null;
    let history = JSON.parse(localStorage.getItem('cosmic_player_history') || '[]');

    // Dynamic ambient glow colors per provider
    const providerColors = {
        youtube: 'radial-gradient(circle, rgba(255, 0, 0, 0.25) 0%, transparent 60%)',
        vk: 'radial-gradient(circle, rgba(74, 118, 168, 0.25) 0%, transparent 60%)',
        rutube: 'radial-gradient(circle, rgba(235, 95, 30, 0.25) 0%, transparent 60%)',
        hdrezka: 'radial-gradient(circle, rgba(242, 197, 17, 0.2) 0%, transparent 60%)',
        seasonvar: 'radial-gradient(circle, rgba(0, 242, 254, 0.2) 0%, transparent 60%)',
        kinogo: 'radial-gradient(circle, rgba(155, 81, 224, 0.2) 0%, transparent 60%)',
        direct: 'radial-gradient(circle, rgba(121, 40, 202, 0.25) 0%, transparent 60%)',
        offline: 'radial-gradient(circle, rgba(255, 255, 255, 0.05) 0%, transparent 60%)'
    };

    // Load history on start
    renderHistory();

    // Form submission
    streamForm.addEventListener('submit', (e) => {
        e.preventDefault();
        const url = streamUrlInput.value.trim();
        if (url) {
            loadVideo(url);
        }
    });

    // Clear history
    clearHistoryBtn.addEventListener('click', () => {
```

```
        history = [];
        localStorage.setItem('cosmic_player_history', JSON.str
ingify(history));
        renderHistory();
    });

    // Router and url parser
    function loadVideo(url, title = null) {
        resetPlayers();

        const providerInfo = parseVideoUrl(url);
        const resolvedTitle = title || providerInfo.defaultTit
le;

        highlightBadge(providerInfo.provider);

        // Update info panel
        videoTitle.textContent = resolvedTitle;
        videoSourceUrl.textContent = url;
        currentProviderTag.textContent = providerInfo.provider
;

        // Update ambient glow color
        ambientGlow.style.background = providerColors[provider
Info.provider] || providerColors.direct;

        // Route to the correct player engine
        if (providerInfo.type === 'embed') {
            embedPlayer.src = providerInfo.embedUrl;
            embedPlayer.classList.remove('hidden');
        } else if (providerInfo.type === 'html5') {
            initHTML5Player(url);
            html5PlayerWrapper.classList.remove('hidden');
        } else if (providerInfo.type === 'iframe_fallback') {
            embedPlayer.src = url;
            embedPlayer.classList.remove('hidden');
        }

```

```
        // Add to history
        addToHistory(url, resolvedTitle, providerInfo.provider
);
    }

    function resetPlayers() {
        embedPlayer.classList.add('hidden');
        embedPlayer.src = '';
        html5PlayerWrapper.classList.add('hidden');
        playerPlaceholder.classList.add('hidden');

        html5Video.pause();
        html5Video.src = '';

        if (hlsInstance) {
            hlsInstance.destroy();
            hlsInstance = null;
        }
    }

    // Video URL Parser
    function parseVideoUrl(url) {
        // YouTube Matcher
        const ytReg = /(?:youtube\.com\/(?:[^\/]+\/.+\/|(?:v|e
(?:mbed)?)\/|.*[?&]v=)|youtu\.be\/)([^"&?\/\s]{11})/;
        const ytMatch = url.match(ytReg);
        if (ytMatch) {
            return {
                provider: 'youtube',
                type: 'embed',
                embedUrl: `https://www.youtube.com/embed/${ytM
atch[1]}?autoplay=1&rel=0`,
                defaultTitle: `YouTube Video (${ytMatch[1]})`
            };
        }

        // VK Matcher
        const vkReg = /vk\.com\/video(-?\d+)_(\d+)/;
```

```
        const vkMatch = url.match(vkReg);
        const vkEmbedReg = /oid=(-?\d+)&id=(\d+)(?:&hash=([a-f0-9]+))?/;
        const vkEmbedMatch = url.match(vkEmbedReg);

        if (vkMatch) {
            return {
                provider: 'vk',
                type: 'embed',
                embedUrl: `https://vk.com/video_ext.php?oid=${vkMatch[1]}&id=${vkMatch[2]}&autoplay=1`,
                defaultTitle: `VK Video (${vkMatch[1]}_${vkMatch[2]})`
            };
        } else if (vkEmbedMatch) {
            const hashParam = vkEmbedMatch[3] ? `&hash=${vkEmbedMatch[3]}` : '';
            return {
                provider: 'vk',
                type: 'embed',
                embedUrl: `https://vk.com/video_ext.php?oid=${vkEmbedMatch[1]}&id=${vkEmbedMatch[2]}${hashParam}&autoplay=1`,
                defaultTitle: `VK Embed Video`
            };
        }

        // RuTube Matcher
        const rutubeReg = /rutube\.ru\/(?:video|play\/embed)\/([a-f0-9]{32})/;
        const rutubeMatch = url.match(rutubeReg);
        if (rutubeMatch) {
            return {
                provider: 'rutube',
                type: 'embed',
                embedUrl: `https://rutube.ru/play/embed/${rutubeMatch[1]}?autoplay=1`,
                defaultTitle: `RuTube Video`
```

```
            };
        }

        // Direct Stream (MP4/HLS)
        const isHLS = url.includes('.m3u8');
        const isMP4 = url.includes('.mp4');
        if (isHLS || isMP4) {
            return {
                provider: 'direct',
                type: 'html5',
                defaultTitle: isHLS ? 'Прямой HLS поток' : 'Пр
ямой MP4 файл'
            };
        }

        // SeasonVar
        if (url.includes('seasonvar.ru')) {
            return {
                provider: 'seasonvar',
                type: 'iframe_fallback',
                defaultTitle: 'СезонВар плеер'
            };
        }

        // HDRezka
        if (url.includes('rezka.ag') || url.includes('hdrezka'
)) {
            return {
                provider: 'hdrezka',
                type: 'iframe_fallback',
                defaultTitle: 'Резка плеер'
            };
        }

        // Kinogo
        if (url.includes('kinogo')) {
            return {
                provider: 'kinogo',
```

```
                type: 'iframe_fallback',
                defaultTitle: 'Киного плеер'
            };
        }

        // Default fallback
        return {
            provider: 'direct',
            type: 'html5',
            defaultTitle: 'Прямой видеоисточник'
        };
    }

    function initHTML5Player(url) {
        if (url.includes('.m3u8')) {
            if (typeof Hls !== 'undefined' && Hls.isSupported(
)) {
                hlsInstance = new Hls();
                hlsInstance.loadSource(url);
                hlsInstance.attachMedia(html5Video);
                hlsInstance.on(Hls.Events.MANIFEST_PARSED, ()
=> {
                    html5Video.play();
                });
            } else if (html5Video.canPlayType('application/vnd
.apple.mpegurl')) {
                html5Video.src = url;
                html5Video.play();
            }
        } else {
            html5Video.src = url;
            html5Video.play();
        }

        playPauseBtn.textContent = '⏸';
    }

    // Video events
```

```
    html5Video.addEventListener('timeupdate', () => {
        const current = html5Video.currentTime;
        const total = html5Video.duration || 0;

        if (total > 0) {
            const pct = (current / total) * 100;
            progressBar.style.width = `${pct}%`;
        }

        currentTimeDisplay.textContent = formatTime(current);
        durationTimeDisplay.textContent = formatTime(total);
    });

    html5Video.addEventListener('loadedmetadata', () => {
        durationTimeDisplay.textContent = formatTime(html5Vide
o.duration);
    });

    playPauseBtn.addEventListener('click', () => {
        if (html5Video.paused) {
            html5Video.play();
            playPauseBtn.textContent = '□ ';
        } else {
            html5Video.pause();
            playPauseBtn.textContent = ' ▶ ';
        }
    });

    stopBtn.addEventListener('click', () => {
        html5Video.pause();
        html5Video.currentTime = 0;
        playPauseBtn.textContent = ' ▶ ';
    });

    volumeSlider.addEventListener('input', (e) => {
        const vol = e.target.value;
        html5Video.volume = vol;
        if (vol === '0') {
```

```
            muteBtn.textContent = '';
        } else {
            muteBtn.textContent = '';
        }
    });

    muteBtn.addEventListener('click', () => {
        if (html5Video.muted) {
            html5Video.muted = false;
            volumeSlider.value = html5Video.volume;
            muteBtn.textContent = '';
        } else {
            html5Video.muted = true;
            volumeSlider.value = 0;
            muteBtn.textContent = '';
        }
    });

    speedSelect.addEventListener('change', (e) => {
        html5Video.playbackRate = parseFloat(e.target.value);
    });

    progressContainer.addEventListener('click', (e) => {
        const rect = progressContainer.getBoundingClientRect()
;
        const clickX = e.clientX - rect.left;
        const width = rect.width;
        const pct = clickX / width;

        html5Video.currentTime = pct * html5Video.duration;
    });

    progressContainer.addEventListener('mousemove', (e) => {
        const rect = progressContainer.getBoundingClientRect()
;
        const hoverX = e.clientX - rect.left;
        const pct = hoverX / rect.width;
        progressHover.style.width = `${pct * 100}%`;
```

```
    });

    progressContainer.addEventListener('mouseleave', () => {
        progressHover.style.width = '0%';
    });

    fullscreenBtn.addEventListener('click', () => {
        if (!document.fullscreenElement) {
            html5PlayerWrapper.requestFullscreen().catch(err =
> {
                console.error(`Error enabling fullscreen: ${er
r.message}`);
            });
        } else {
            document.exitFullscreen();
        }
    });

    function formatTime(sec) {
        if (isNaN(sec)) return '0:00';
        const mins = Math.floor(sec / 60);
        const secs = Math.floor(sec % 60);
        return `${mins}:${secs < 10 ? '0' : ''}${secs}`;
    }

    function highlightBadge(provider) {
        badges.forEach(badge => {
            if (badge.getAttribute('data-provider') === provid
er) {
                badge.classList.add('active');
            } else {
                badge.classList.remove('active');
            }
        });
    }

    function addToHistory(url, title, provider) {
        history = history.filter(item => item.url !== url);
```

```
359
360         history.unshift({
361             url,
362             title,
363             provider,
364             timestamp: new Date().toLocaleTimeString([], { hou
    r: '2-digit', minute: '2-digit' })
365         });
366
367         if (history.length > 8) history.pop();
368
369         localStorage.setItem('cosmic_player_history', JSON.str
    ingify(history));
370         renderHistory();
371     }
372
373     function renderHistory() {
374         historyList.innerHTML = '';
375         if (history.length === 0) {
376             historyList.innerHTML = '<div class="empty-state">
    Нет недавно проигранных видео</div>';
377             return;
378         }
379
380         history.forEach(item => {
381             const el = document.createElement('div');
382             el.className = 'history-item';
383             el.innerHTML = `
384                 <div class="history-title" title="${item.title
    }">${item.title}</div>
385                 <div class="history-meta">
386                     <span>${item.provider.toUpperCase()}</span
    >
387                     <span>${item.timestamp}</span>
388                 </div>
389             `;
390             el.addEventListener('click', () => {
391                 streamUrlInput.value = item.url;
```

```
                loadVideo(item.url, item.title);
            });
            historyList.appendChild(el);
        });
    }

    badges.forEach(badge => {
        badge.addEventListener('click', () => {
            const provider = badge.getAttribute('data-provider
');
            let demoUrl = '';

            switch (provider) {
                case 'youtube':
                    demoUrl = 'https://www.youtube.com/watch?v
=aqz-KE-bpKQ';
                    break;
                case 'vk':
                    demoUrl = 'https://vk.com/video-22822305_4
56239018';
                    break;
                case 'rutube':
                    demoUrl = 'https://rutube.ru/video/e7cfcb8
cb4310d54026fb4bd56e828d1/';
                    break;
                case 'direct':
                    demoUrl = 'https://test-streams.mux.dev/x3
6xhzz/x36xhzz.m3u8';
                    break;
                case 'seasonvar':
                    demoUrl = 'http://seasonvar.ru';
                    break;
                case 'hdrezka':
                    demoUrl = 'https://rezka.ag';
                    break;
                case 'kinogo':
                    demoUrl = 'https://kinogo.biz';
                    break;
```

```
            }

            if (demoUrl) {
                streamUrlInput.value = demoUrl;
                loadVideo(demoUrl);
            }
        });
    });
};

```

styles.css

```
/**
 * ============================================================
 * Process Name: Windows Web Styles - Styles Stylesheet
 * ============================================================
 * Description:
 *   Таблица стилей стиля и оформления интерфейса (styles).
 *
 * Usage Examples:
 *   HTML Link Tag:
 *     <link rel="stylesheet" href="/windows/api/webgui/cosmicplayer/styles.css?v=20261001_v1">
 *
 * File: styles.css
 * Project: ai-windows-admin
 * Package: windows/api/webgui/cosmicplayer
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

.cosmic-player-tab {
    --primary: #7928ca;
    --primary-glow: #ff007a;
    --primary-gradient: linear-gradient(135deg, #7928ca 0%, #ff007a 100%);
    --panel-border: rgba(255, 255, 255, 0.08);
    --bg-dark: #0a0714;
    --text-muted: #a0aec0;
    --transition: all 0.25s cubic-bezier(0.25, 0.8, 0.25, 1);

    display: flex;
    width: 100%;
```

```
    height: calc(100vh - 80px);
    background: rgba(20, 16, 36, 0.6);
    backdrop-filter: blur(25px);
    -webkit-backdrop-filter: blur(25px);
    border: 1px solid var(--panel-border);
    border-radius: 12px;
    overflow: hidden;
    margin-top: 6px;
}

.cosmic-sidebar {
    width: 260px;
    background: rgba(14, 10, 26, 0.75);
    border-right: 1px solid var(--panel-border);
    padding: 16px 14px;
    display: flex;
    flex-direction: column;
    gap: 14px;
    overflow-y: auto;
}

.cosmic-sidebar .logo {
    display: flex;
    align-items: center;
    gap: 8px;
}

.cosmic-sidebar .logo h2 {
    font-size: 1.1rem;
    font-weight: 700;
    margin: 0;
    background: var(--primary-gradient);
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
}

.cosmic-sidebar .logo-icon {
    font-size: 1.3rem;
```

```
    filter: drop-shadow(0 0 6px rgba(255, 0, 122, 0.5));
}

.cosmic-player-tab .url-input-section h3,
.cosmic-player-tab .sources-panel h4,
.cosmic-player-tab .history-section h4 {
    font-size: 0.72rem;
    font-weight: 600;
    text-transform: uppercase;
    letter-spacing: 0.6px;
    color: var(--text-muted);
    margin-bottom: 8px;
}

#stream-form {
    display: flex;
    flex-direction: column;
    gap: 10px;
}

#stream-url {
    background: rgba(0, 0, 0, 0.4);
    border: 1px solid var(--panel-border);
    border-radius: 8px;
    padding: 10px 12px;
    color: #fff;
    font-size: 0.85rem;
    outline: none;
    transition: var(--transition);
}

#stream-url:focus {
    border-color: var(--primary-glow);
    box-shadow: 0 0 10px rgba(255, 0, 122, 0.2);
}

.btn-glow {
    background: var(--primary-gradient);
```

```
    color: white;
    border: none;
    border-radius: 8px;
    padding: 10px 15px;
    font-size: 0.85rem;
    font-weight: 600;
    cursor: pointer;
    transition: var(--transition);
}

.btn-glow:hover {
    transform: translateY(-1px);
    box-shadow: 0 4px 15px rgba(255, 0, 122, 0.4);
}

.sources-grid {
    display: flex;
    flex-wrap: wrap;
    gap: 6px;
}

.cosmic-player-tab .badge {
    background: rgba(255, 255, 255, 0.08);
    border: 1px solid var(--panel-border);
    border-radius: 6px;
    padding: 5px 10px;
    font-size: 0.7rem;
    font-weight: 500;
    color: #e6edf3;
    transition: var(--transition);
    cursor: pointer;
}

.cosmic-player-tab .badge:hover,
.cosmic-player-tab .badge.active {
    color: white;
    background: rgba(121, 40, 202, 0.2);
    border-color: var(--primary);
```

```
}

.history-section {
    flex-grow: 1;
    display: flex;
    flex-direction: column;
}

.section-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 10px;
}

.btn-text {
    background: none;
    border: none;
    color: var(--text-muted);
    font-size: 0.75rem;
    cursor: pointer;
}

.history-list {
    display: flex;
    flex-direction: column;
    gap: 6px;
    overflow-y: auto;
    max-height: 200px;
}

.history-item {
    background: rgba(255, 255, 255, 0.01);
    border: 1px solid var(--panel-border);
    border-radius: 8px;
    padding: 8px 12px;
    cursor: pointer;
    display: flex;
```

```
    flex-direction: column;
    gap: 2px;
    transition: var(--transition);
}

.history-item:hover {
    background: rgba(255, 255, 255, 0.04);
}

.history-title {
    font-size: 0.8rem;
    font-weight: 500;
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
}

.history-meta {
    display: flex;
    justify-content: space-between;
    font-size: 0.65rem;
    color: var(--text-muted);
}

/* Main Panel Layout */
.main-content {
    flex-grow: 1;
    display: flex;
    flex-direction: column;
    padding: 14px;
    gap: 12px;
    background: rgba(10, 7, 20, 0.2);
}

.player-container {
    position: relative;
    width: 100%;
    height: 60%;
```

```
    min-height: 380px;
    background: #000;
    border-radius: 15px;
    overflow: hidden;
    border: 1px solid var(--panel-border);
}

.ambient-glow {
    position: absolute;
    top: -20%;
    left: -20%;
    width: 140%;
    height: 140%;
    background: radial-gradient(circle, rgba(121, 40, 202, 0.2
) 0%, transparent 60%);
    filter: blur(50px);
    z-index: 0;
    pointer-events: none;
    transition: background 1s ease;
}

.player-element {
    position: relative;
    width: 100%;
    height: 100%;
    border: none;
    z-index: 2;
}

.hidden {
    display: none !important;
}

.html5-video-el {
    width: 100%;
    height: 100%;
    object-fit: contain;
}
```

```
259
260 #html5-player-wrapper {
261     position: relative;
262 }
263
264 .custom-controls {
265     position: absolute;
266     bottom: 0;
267     left: 0;
268     right: 0;
269     background: linear-gradient(to top, rgba(0, 0, 0, 0.9) 0%,
     rgba(0, 0, 0, 0.4) 70%, transparent 100%);
270     padding: 15px 20px;
271     display: flex;
272     flex-direction: column;
273     gap: 10px;
274     opacity: 0;
275     transition: opacity 0.3s ease;
276     z-index: 10;
277 }
278
279 #html5-player-wrapper:hover .custom-controls {
280     opacity: 1;
281 }
282
283 .progress-container {
284     height: 4px;
285     background: rgba(255, 255, 255, 0.2);
286     border-radius: 2px;
287     cursor: pointer;
288     position: relative;
289 }
290
291 .progress-bar {
292     height: 100%;
293     background: var(--primary-gradient);
294     width: 0%;
295     border-radius: 2px;
```

```
}

.progress-hover {
    position: absolute;
    top: 0;
    left: 0;
    height: 100%;
    background: rgba(255, 255, 255, 0.15);
    width: 0%;
    pointer-events: none;
}

.controls-row {
    display: flex;
    justify-content: space-between;
    align-items: center;
}

.controls-left, .controls-right {
    display: flex;
    align-items: center;
    gap: 12px;
}

.ctrl-btn {
    background: none;
    border: none;
    color: white;
    font-size: 1rem;
    cursor: pointer;
}

.time-display {
    font-size: 0.75rem;
    color: var(--text-muted);
}

.volume-container {
```

```
    display: flex;
    align-items: center;
    gap: 6px;
}

#volume-slider {
    width: 60px;
    height: 3px;
    background: rgba(255, 255, 255, 0.2);
    outline: none;
    accent-color: var(--primary-glow);
    cursor: pointer;
}

.speed-dropdown {
    background: rgba(255, 255, 255, 0.1);
    border: 1px solid rgba(255, 255, 255, 0.2);
    color: white;
    border-radius: 4px;
    padding: 2px 6px;
    font-size: 0.75rem;
    outline: none;
    cursor: pointer;
}

.player-placeholder {
    position: absolute;
    top: 0;
    left: 0;
    width: 100%;
    height: 100%;
    display: flex;
    justify-content: center;
    align-items: center;
    z-index: 1;
    background: rgba(10, 7, 20, 0.95);
}

```

```
.placeholder-content {
    text-align: center;
    max-width: 320px;
}

.placeholder-content .play-icon {
    font-size: 3rem;
    display: block;
    margin-bottom: 12px;
}

.placeholder-content h3 {
    font-size: 1.15rem;
    font-weight: 600;
    margin-bottom: 8px;
}

.placeholder-content p {
    color: var(--text-muted);
    font-size: 0.8rem;
    line-height: 1.4;
}

.glass-card {
    background: rgba(255, 255, 255, 0.02);
    border: 1px solid var(--panel-border);
    border-radius: 12px;
    padding: 18px;
}

.video-info-card {
    flex-grow: 1;
    display: flex;
    flex-direction: column;
    justify-content: center;
    gap: 8px;
}

```

```
.info-header {
    display: flex;
    align-items: center;
    gap: 12px;
}

.provider-tag {
    background: var(--primary-gradient);
    color: white;
    padding: 3px 10px;
    border-radius: 15px;
    font-size: 0.65rem;
    font-weight: 600;
    text-transform: uppercase;
}

#video-title {
    font-size: 1.1rem;
    font-weight: 600;
    margin: 0;
}

#video-source-url {
    font-size: 0.8rem;
    color: var(--text-muted);
    word-break: break-all;
    margin: 0;
}

.empty-state {
    color: var(--text-muted);
    font-size: 0.75rem;
    text-align: center;
    padding: 15px;
    border: 1px dashed var(--panel-border);
    border-radius: 8px;
}

```

base.css

```
/**
 * ============================================================
 * Process Name: Windows Web Styles - Base Stylesheet
 * ============================================================
 * Description:
 *   Таблица стилей стиля и оформления интерфейса (base).
 *
 * Usage Examples:
 *   HTML Link Tag:
 *     <link rel="stylesheet" href="/windows/api/webgui/css/base.css?v=20261001_v1">
 *
 * File: base.css
 * Project: ai-windows-admin
 * Package: windows/api/webgui/css
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/* ═══════════════════════════════════════════════════════════
   BASE STYLES & SCROLLBARS — AI Windows Admin WebGUI
   Базовые стили страницы, скроллбары и контейнеры вкладок
   ═══════════════════════════════════════════════════════════ */

/* Базовый шрифт — увеличен на 2pt для улучшения читаемости */
html {
  font-size: calc(14px + 2pt);
  height: 100%;
}
```

```
31
32 body {
33   display: flex;
34   flex-direction: column;
35   height: 100%;
36   min-height: 100vh;
37   background: var(--bg-color);
38   color: var(--text-color);
39   font-size: 0.9375rem; /* ≈ 13.125px */
40   line-height: 1.5;
41   transition: background-color 0.25s ease, color 0.25s ease;
42 }
43
44 /* ═══════════════════════════════════════════════════════════
   ═══════════════
45    СКРОЛЛБАРЫ
46    ═══════════════════════════════════════════════════════════
   ═══════════════ */
47 ::-webkit-scrollbar        { width: 6px; height: 6px; }
48 ::-webkit-scrollbar-track  { background: var(--scrollbar-track
   ); border-radius: 3px; }
49 ::-webkit-scrollbar-thumb  { background: var(--scrollbar-thumb
   ); border-radius: 3px; }
50 ::-webkit-scrollbar-thumb:hover { opacity: 0.8; }
51
52 /* ═══════════════════════════════════════════════════════════
   ═══════════════
53    ВКЛАДКИ (TAB PANES)
54    ═══════════════════════════════════════════════════════════
   ═══════════════ */
55 .tab-content {
56   flex-grow: 1;
57   display: flex;
58   flex-direction: column;
59 }
60
61 .tab-pane {
62   min-height: auto;
```

```
  padding-bottom: 16px;
}

.tab-pane.active {
  display: block;
}

/* ═══════════════════════════════════════════════════════════
   АДМИН-ИНТЕРФЕЙС
   ═══════════════════════════════════════════════════════════ */
/* Стили для #admin-interface внутри .container-fluid (как в index.html) */
#admin-interface,
.container-fluid#admin-interface {
  display: flex;
  flex-direction: column;
  flex-grow: 1;
  height: 100%;
  overflow-y: auto;
}

#admin-interface .tab-content,
.container-fluid#admin-interface .tab-content {
  flex-grow: 1;
  overflow-y: auto;
}

#admin-interface .tab-pane,
.container-fluid#admin-interface .tab-pane {
  flex-grow: 1;
  overflow-y: auto;
  overflow-x: hidden;
}

#admin-interface .tab-pane.active,
.container-fluid#admin-interface .tab-pane.active {
```

```
  display: block;
}

/* Скроллбары для админки */
#admin-interface ::-webkit-scrollbar {
  width: 8px;
  height: 8px;
}

#admin-interface ::-webkit-scrollbar-track {
  background: var(--scrollbar-track);
  border-radius: 4px;
}

#admin-interface ::-webkit-scrollbar-thumb {
  background: var(--scrollbar-thumb);
  border-radius: 4px;
}

#admin-interface ::-webkit-scrollbar-thumb:hover {
  opacity: 0.85;
}

```

chat.css

```
/**
 * ============================================================
 * Process Name: Windows Web Styles - Chat Stylesheet
 * ============================================================
 * Description:
 *   Таблица стилей стиля и оформления интерфейса (chat).
 *
 * Usage Examples:
 *   HTML Link Tag:
 *     <link rel="stylesheet" href="/windows/api/webgui/css/chat.css?v=20261001_v1">
 *
 * File: chat.css
 * Project: ai-windows-admin
 * Package: windows/api/webgui/css
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/* ═══════════════════════════════════════════════════════════
   CHAT — AI Windows Admin WebGUI
   Стили диалогового окна, сайдбара сессий, сообщений и дебаггера
   ═══════════════════════════════════════════════════════════ */

/* ── Основной макет вкладки чата ──────────────────────────── */
#tab-chat {
  height: calc(100vh - 55px);
```

```
  max-height: calc(100vh - 55px);
  min-height: 80vh;
  padding-bottom: 0;
  overflow: hidden;
}

#tab-chat.active {
  display: flex !important;
  flex-direction: column;
  height: calc(100vh - 55px);
  max-height: calc(100vh - 55px);
  min-height: 80vh;
}

.chat-container {
  display: flex;
  flex-direction: row;
  height: 100%;
  min-height: 0;
  gap: 8px;
  position: relative;
}

/* ── Сайдбар чата (история диалогов) ──────────────────────── */
.chat-sidebar {
  width: 220px;
  min-width: 220px;
  max-width: 220px;
  display: flex;
  flex-direction: column;
  background: var(--surface-1);
  border: 1px solid var(--border-color);
  border-radius: 6px;
  padding: 6px;
  transition: margin-left 0.22s ease, opacity 0.18s ease;
  height: 100%;
  flex-shrink: 0;
```

```
}

.chat-sidebar.collapsed {
  margin-left: -232px;
  opacity: 0;
  pointer-events: none;
  display: none !important;
}

.chat-sessions-list {
  flex: 1 1 auto;
  overflow-y: auto;
  display: flex;
  flex-direction: column;
  gap: 2px;
  margin-top: 4px;
  margin-bottom: 4px;
  padding-right: 2px;
}

.chat-session-item {
  display: flex;
  align-items: center;
  justify-content: space-between;
  padding: 5px 7px;
  border-radius: 5px;
  cursor: pointer;
  color: var(--text-color);
  font-size: 0.8rem;
  border: 1px solid transparent;
  transition: background 0.12s ease, border-color 0.12s ease;
  user-select: none;
}

.chat-session-item:hover {
  background: rgba(128, 128, 128, 0.1);
}

```

```
.chat-session-item.active {
  background: rgba(37, 99, 235, 0.1);
  border-color: rgba(37, 99, 235, 0.25);
  font-weight: 500;
}

[data-theme="dark"] .chat-session-item.active {
  background: rgba(88, 166, 255, 0.1);
  border-color: rgba(88, 166, 255, 0.2);
}

.chat-session-title {
  white-space: nowrap;
  overflow: hidden;
  text-overflow: ellipsis;
  flex: 1 1 auto;
  margin-right: 3px;
}

.chat-session-actions {
  display: none;
  align-items: center;
  gap: 2px;
  flex-shrink: 0;
}

.chat-session-item:hover .chat-session-actions,
.chat-session-item.active .chat-session-actions {
  display: flex;
}

.chat-session-action-btn {
  background: transparent;
  border: none;
  padding: 0 2px;
  color: var(--text-muted);
  border-radius: 3px;
  cursor: pointer;
```

```
  line-height: 1;
  font-size: 0.75rem;
}

.chat-session-action-btn:hover {
  color: var(--text-color);
  background: rgba(128, 128, 128, 0.18);
}

/* ── Главная область чата ──────────────────────────────────────── */
.chat-main-area {
  flex: 1 1 auto;
  min-width: 0;
  display: flex;
  flex-direction: column;
  height: 100%;
}

/* Хедер-бар чата */
.chat-header-bar {
  display: flex;
  align-items: center;
  justify-content: space-between;
  padding: 3px 8px;
  margin-bottom: 5px;
  border-radius: 5px;
  background: var(--surface-1);
  border: 1px solid var(--border-color);
  font-size: 0.82rem;
  flex-shrink: 0;
}

/* Окно чата */
.chat-window {
  height: calc(100vh - 260px);
  min-height: 160px;
  overflow-y: auto;
```

```
  border: 1px solid var(--border-color);
  border-radius: 5px;
  padding: 8px 10px;
  background: var(--surface-1);
}

#tab-chat .chat-window {
  height: auto;
  flex: 1 1 auto;
  min-height: calc(80vh - 120px);
  max-height: calc(100vh - 140px);
  overflow-y: auto;
  margin-bottom: 5px !important;
}

#tab-chat .media-controls        { flex-shrink: 0; margin-bott
om: 5px !important; }
#tab-chat #chat-debugger-toolbar { flex-shrink: 0; margin-bott
om: 5px !important; }
#tab-chat .input-group           { flex-shrink: 0; margin-bott
om: 2px !important; }

/* ── Сообщения (Базовые блоки) ──────────────────────────────
──────────────── */
.message {
  margin-bottom: 8px;
  padding: 8px 12px;
  border-radius: 6px;
  font-size: 0.92rem;
  line-height: 1.5;
}

/* ── Поле ввода чата ────────────────────────────────────────
──────────────── */
#message-input,
.chat-main-area #message-input,
#chat-popup-input,
#scenario-chat-input {
```

```
  font-size: 0.95rem;
}

.user-message {
  background: var(--chat-user-bg);
  color: var(--chat-user-text);
  text-align: right;
  border-radius: 12px 12px 2px 12px;
  box-shadow: var(--shadow-xs);
}

.bot-message {
  background: var(--chat-bot-bg);
  color: var(--chat-bot-text);
  border: 1px solid var(--border-color);
  border-radius: 12px 12px 12px 2px;
  box-shadow: var(--shadow-xs);
}

/* ── Универсальные пузыри сообщений (Modern Chat Bubbles) ──
   ──────────────────── */
.chat-message-row {
  display: flex;
  margin-bottom: 0.75rem;
  width: 100%;
}

.chat-message-row.user-row {
  justify-content: flex-end;
}

.chat-message-row.bot-row {
  justify-content: flex-start;
  align-items: flex-start;
  gap: 0.5rem;
}

.chat-msg-avatar {
```

```
  font-size: 1.25rem;
  line-height: 1;
  flex-shrink: 0;
  margin-top: 2px;
  user-select: none;
}

/* Пузырь сообщения пользователя: выравнивание справа, мягкий фон темы */
.chat-msg-user {
  background: var(--chat-user-bg);
  color: var(--chat-user-text);
  border-radius: 14px 14px 2px 14px;
  padding: 0.55rem 0.95rem;
  max-width: 80%;
  font-size: 0.9rem;
  line-height: 1.45;
  box-shadow: var(--shadow-xs);
  word-break: break-word;
  white-space: pre-wrap;
  border: 1px solid transparent;
}

/* Пузырь сообщения бота / ассистента: выравнивание слева */
.chat-msg-bot {
  background: var(--chat-bot-bg);
  color: var(--chat-bot-text);
  border: 1px solid var(--border-color);
  border-radius: 14px 14px 14px 2px;
  padding: 0.75rem 1rem;
  flex-grow: 1;
  max-width: 95%;
  font-size: 0.88rem;
  line-height: 1.5;
  box-shadow: var(--shadow-xs);
  word-break: break-word;
}

```

```
.chat-msg-bot a {
  color: var(--nav-active);
}

/* Диагностический конвейер этапов (Stages Pipeline) */
.chat-stage-pipeline {
  background: var(--surface-2);
  border: 1px solid var(--border-color);
  border-radius: var(--radius-sm);
  padding: 0.5rem 0.75rem;
  margin-bottom: 0.65rem;
  color: var(--text-color);
}

.chat-stage-pipeline .stage-item {
  color: var(--text-muted);
  font-size: 0.76rem;
  line-height: 1.4;
}

.chat-stage-pipeline .stage-item.active {
  color: var(--text-color);
  font-weight: 500;
}

/* Карточка и контейнер истории чата сценариев */
.scenario-chat-card {
  background-color: var(--surface-1) !important;
  border: 1.5px solid var(--border-color) !important;
  color: var(--text-color) !important;
}

.scenario-chat-history {
  background-color: var(--bg-color);
}

/* Чипы сообщений в чате */
.chat-window .badge,
```

```
.message .badge,
.bot-message .badge,
.user-message .badge,
.chat-msg-bot .badge,
.chat-msg-user .badge {
  font-size: 0.7rem;
  font-weight: 500;
}

.chat-window .badge.bg-secondary,
.chat-window .badge.bg-primary,
.chat-window .badge.bg-danger,
.chat-window .badge.bg-success,
.chat-window .badge.bg-dark,
.message .badge.bg-secondary,
.message .badge.bg-primary,
.message .badge.bg-danger,
.message .badge.bg-success,
.message .badge.bg-dark {
  color: #ffffff !important;
}

/* ── Дебаггер-тулбар чата ──────────────────────────────────────── */
#chat-debugger-toolbar {
  background-color: var(--surface-2);
  border: 1px solid var(--border-color);
  border-radius: 5px;
  font-size: 0.8rem;
}

[data-theme="light"] #chat-debugger-toolbar {
  background-color: var(--surface-2) !important;
  border-color: var(--border-color) !important;
}

[data-theme="light"] #chat-debugger-toolbar select {
  background-color: var(--surface-1) !important;
```

```
  color: var(--text-color) !important;
  border-color: var(--border-color) !important;
}

```

components.css

```
/**
 * ============================================================
 * Process Name: Windows Web Styles - Components Stylesheet
 * ============================================================
 * Description:
 *   Таблица стилей стиля и оформления интерфейса (components).
 *
 * Usage Examples:
 *   HTML Link Tag:
 *     <link rel="stylesheet" href="/windows/api/webgui/css/components.css?v=20261001_v1">
 *
 * File: components.css
 * Project: ai-windows-admin
 * Package: windows/api/webgui/css
 * Author: hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/* ═══════════════════════════════════════════════════════════
   COMPONENTS — AI Windows Admin WebGUI
   Базовые UI-компоненты: карточки, формы, кнопки, модалки, таблицы, алерты, бейджи
   ═══════════════════════════════════════════════════════════ */

/* ═══════════════════════════════════════════════════════════
   КАРТОЧКИ
   ═══════════════════════════════════════════════════════════
```

```
═══════════════ */
.card {
  background-color: var(--card-bg);
  border: 1.5px solid var(--border-color);
  color: var(--card-text);
  transition: border-color 0.2s, background-color 0.2s, box-sh
adow 0.2s;
  border-radius: 8px;
  box-shadow: var(--shadow-sm);
}

.card-header {
  background-color: var(--surface-2);
  border-bottom: 1.5px solid var(--border-color);
  color: var(--text-color);
  padding: 0.375rem 0.75rem;
  font-size: 0.85rem;
  font-weight: 600;
}

.card-body {
  color: var(--card-text);
  padding: 0.75rem;
}

.card-footer {
  background-color: var(--surface-2);
  border-top: 1px solid var(--border-color);
  padding: 0.375rem 0.75rem;
}

.card.bg-light {
  background-color: var(--surface-2) !important;
  color: var(--text-color);
}

.card.bg-dark > .card-header,
.modal-content.bg-dark > .modal-header,
```

```
.modal-content.bg-dark > .modal-footer,
.agent-card .card-header {
  background-color: var(--surface-2) !important;
  border-color: var(--border-color) !important;
  color: var(--text-color) !important;
}

[data-theme="light"] .card.bg-dark,
[data-theme="light"] .modal-content.bg-dark,
[data-theme="light"] .alert-dark {
  background-color: var(--surface-1) !important;
  color: var(--text-color) !important;
  border-color: var(--border-color) !important;
}

[data-theme="light"] .card.bg-dark > .card-header,
[data-theme="light"] .modal-content.bg-dark > .modal-header,
[data-theme="light"] .modal-content.bg-dark > .modal-footer {
  background-color: var(--surface-2) !important;
  color: var(--text-color) !important;
  border-color: var(--border-color) !important;
}

[data-theme="light"] .bg-black {
  background-color: var(--bg-color) !important;
  color: var(--text-color) !important;
}

[data-theme="light"] .text-light {
  color: var(--text-color) !important;
}

[data-theme="light"] .btn-close-white {
  filter: none !important;
}

/* ═══════════════════════════════════════════════════════════════
═══════════════
```

```
   ФОРМЫ И ИНПУТЫ
   ═══════════════════════════════════════════════════════════════ */
.form-control,
.form-select {
  background-color: var(--surface-1);
  border: 1px solid var(--border-color);
  color: var(--text-color);
  font-size: 0.85rem;
  padding: 0.3rem 0.5rem;
  transition: border-color 0.15s, box-shadow 0.15s;
  border-radius: 5px;
}

.form-control:focus,
.form-select:focus {
  background-color: var(--surface-1);
  border-color: var(--nav-active);
  color: var(--text-color);
  box-shadow: 0 0 0 2px rgba(37, 99, 235, 0.18);
  outline: none;
}

[data-theme="dark"] .form-control:focus,
[data-theme="dark"] .form-select:focus {
  box-shadow: 0 0 0 2px rgba(88, 166, 255, 0.3);
}

[data-theme="brick"] .form-control:focus,
[data-theme="brick"] .form-select:focus {
  box-shadow: 0 0 0 2px rgba(184, 66, 40, 0.25);
}

[data-theme="terminal"] .form-control:focus,
[data-theme="terminal"] .form-select:focus {
  box-shadow: 0 0 0 2px rgba(0, 255, 102, 0.35);
}

```

```
.form-control::placeholder {
  color: var(--text-muted);
  opacity: 0.8;
}

.form-control-sm,
.form-select-sm {
  font-size: 0.8rem;
  padding: 0.22rem 0.4rem;
}

.form-label {
  font-size: 0.82rem;
  margin-bottom: 0.3rem;
  color: var(--text-color);
}

.input-group-text {
  background-color: var(--surface-2);
  border: 1px solid var(--border-color);
  color: var(--text-color);
  font-size: 0.85rem;
  padding: 0.3rem 0.55rem;
}

/* ═══════════════════════════════════════════════════════════════
   КНОПКИ
   ═══════════════════════════════════════════════════════════════ */
.btn {
  font-size: 0.82rem;
  font-weight: 500;
  transition: background-color 0.15s, border-color 0.15s, box-shadow 0.15s;
  border-radius: 5px;
}

```

```
.btn-sm {
  padding: 0.2rem 0.45rem;
  font-size: 0.78rem;
}

.btn-lg {
  font-size: 0.9rem;
  padding: 0.45rem 0.9rem;
}

.btn-primary {
  background-color: var(--nav-active);
  border-color: var(--nav-active);
  color: #ffffff;
}

.btn-primary:hover {
  background-color: #1d4ed8;
  border-color: #1d4ed8;
  color: #ffffff;
}

[data-theme="brick"] .btn-primary {
  background-color: #b84228;
  border-color: #b84228;
  color: #ffffff;
}

[data-theme="brick"] .btn-primary:hover {
  background-color: #9e341d;
  border-color: #9e341d;
  color: #ffffff;
}

[data-theme="dark"] .btn-primary {
  background-color: #1f6feb;
  border-color: #1f6feb;
  color: #ffffff;
```

```
}

[data-theme="dark"] .btn-primary:hover {
  background-color: #388bfd;
  border-color: #388bfd;
  color: #ffffff;
}

[data-theme="terminal"] .btn-primary {
  background-color: #0c3316;
  border: 1px solid #00ff66;
  color: #00ff66;
  text-shadow: 0 0 4px rgba(0, 255, 102, 0.6);
}

[data-theme="terminal"] .btn-primary:hover {
  background-color: #00ff66;
  border-color: #00ff66;
  color: #060a06;
  text-shadow: none;
}

.btn-outline-secondary {
  background-color: transparent;
  border-color: var(--border-color);
  color: var(--nav-text);
}

.btn-outline-secondary:hover {
  background-color: var(--nav-hover-bg);
  border-color: var(--border-color);
  color: var(--text-color);
}

/* Переключатели групп кнопок (btn-check radio/checkbox) */
.btn-check:checked + .btn-outline-primary,
.btn-check:active + .btn-outline-primary {
  background-color: #2563eb !important;
```

```
  border-color: #2563eb !important;
  color: #ffffff !important;
}

.btn-check:checked + .btn-outline-secondary,
.btn-check:active + .btn-outline-secondary {
  background-color: #64748b !important;
  border-color: #64748b !important;
  color: #ffffff !important;
}

.btn-check:checked + .btn-outline-info,
.btn-check:active + .btn-outline-info {
  background-color: #0284c7 !important;
  border-color: #0284c7 !important;
  color: #ffffff !important;
}

.btn-check:checked + .btn-outline-danger,
.btn-check:active + .btn-outline-danger {
  background-color: #dc2626 !important;
  border-color: #dc2626 !important;
  color: #ffffff !important;
}

.btn-check:checked + .btn-outline-warning,
.btn-check:active + .btn-outline-warning {
  background-color: #d97706 !important;
  border-color: #d97706 !important;
  color: #ffffff !important;
}

.btn-check:checked + .btn-outline-success,
.btn-check:active + .btn-outline-success {
  background-color: #16a34a !important;
  border-color: #16a34a !important;
  color: #ffffff !important;
}
```

```
287
288 /* ═══════════════════════════════════════════════════════════
    ═══════════════
289    МОДАЛЬНЫЕ ОКНА
290    ═══════════════════════════════════════════════════════════
    ═══════════════ */
291 .modal-content {
292   background-color: var(--modal-bg);
293   border: 1px solid var(--modal-border);
294   color: var(--modal-text);
295   border-radius: 8px;
296   box-shadow: var(--shadow-lg);
297 }
298
299 .modal-header {
300   border-bottom: 1px solid var(--border-color);
301   padding: 0.6rem 0.9rem;
302   background-color: var(--surface-2);
303   border-radius: 8px 8px 0 0;
304 }
305
306 .modal-title {
307   font-size: 0.9rem;
308   font-weight: 600;
309 }
310
311 .modal-body {
312   padding: 0.75rem 0.9rem;
313   font-size: 0.85rem;
314 }
315
316 .modal-footer {
317   border-top: 1px solid var(--border-color);
318   padding: 0.5rem 0.9rem;
319   background-color: var(--surface-2);
320   border-radius: 0 0 8px 8px;
321 }
322
```

```
/* Модальное окно помощи */
.modal-help .modal-dialog { max-width: 600px; }
.modal-help .modal-content {
  border-radius: 8px;
  background: var(--modal-bg);
  color: var(--modal-text);
  border: 1px solid var(--modal-border);
}
.modal-help .modal-header {
  border-bottom: 1px solid var(--border-color);
  background: var(--surface-2);
}
.modal-help .modal-body { font-size: 0.85rem; line-height: 1.6
; }
.modal-help .modal-footer {
  border-top: 1px solid var(--border-color);
  background: var(--surface-2);
}

/* ═══════════════════════════════════════════════════════════
═══════════════
   ТАБЛИЦЫ
   ═══════════════════════════════════════════════════════════
═══════════════ */
.table {
  --bs-table-color: var(--text-color);
  --bs-table-bg: transparent;
  --bs-table-border-color: var(--border-color);
  --bs-table-hover-bg: var(--nav-hover-bg);
  --bs-table-hover-color: var(--text-color);
  font-size: 0.82rem;
}

.table th,
.table td {
  padding: 0.4rem 0.6rem;
  vertical-align: middle;
  color: var(--text-color);
```

```
}

.table-hover tbody tr:hover,
.table-hover tbody tr:hover > * {
  background-color: var(--nav-hover-bg) !important;
  color: var(--text-color) !important;
}

.table-dark {
  --bs-table-bg: var(--surface-1);
  --bs-table-color: var(--text-color);
  --bs-table-border-color: var(--border-color);
  --bs-table-hover-bg: var(--nav-hover-bg);
  --bs-table-hover-color: var(--text-color);
  color: var(--text-color) !important;
}

.table-dark th,
.table-dark td {
  color: var(--text-color);
}

.table-dark.table-hover tbody tr:hover,
.table-dark.table-hover tbody tr:hover > * {
  background-color: var(--nav-hover-bg) !important;
  color: var(--text-color) !important;
}

[data-theme="light"] .table-dark,
[data-theme="brick"] .table-dark {
  --bs-table-bg: var(--surface-1) !important;
  --bs-table-color: var(--text-color) !important;
  --bs-table-border-color: var(--border-color) !important;
  --bs-table-hover-bg: var(--nav-hover-bg) !important;
  --bs-table-hover-color: var(--text-color) !important;
  color: var(--text-color) !important;
}

```

```
[data-theme="dark"] .table-light,
[data-theme="dark"] thead.table-light,
[data-theme="dark"] thead.table-light th {
  background-color: var(--surface-2) !important;
  color: var(--text-color) !important;
  border-color: var(--border-color) !important;
}

[data-theme="dark"] .table {
  color: var(--text-color) !important;
  border-color: var(--border-color) !important;
}

.table-responsive-adaptive {
  height: auto;
  min-height: 120px;
  max-height: clamp(320px, 58vh, 720px);
  overflow-y: auto;
  overflow-x: auto;
  scrollbar-width: thin;
}

.table-responsive-adaptive thead th,
.table-sticky-header thead th {
  position: sticky;
  top: 0;
  z-index: 2;
  background-color: var(--surface-2);
}

/* ═══════════════════════════════════════════════════════════════
   АЛЕРТЫ — адаптивные для обеих тем
   ═══════════════════════════════════════════════════════════════ */
.alert {
  font-size: 0.83rem;
  padding: 0.5rem 0.75rem;
```

```
  border-radius: 6px;
  border-width: 1px;
  border-style: solid;
}

.alert-success {
  background-color: var(--alert-success-bg);
  border-color: var(--alert-success-border);
  color: var(--alert-success-text);
}

.alert-danger {
  background-color: var(--alert-danger-bg);
  border-color: var(--alert-danger-border);
  color: var(--alert-danger-text);
}

.alert-warning {
  background-color: var(--alert-warning-bg);
  border-color: var(--alert-warning-border);
  color: var(--alert-warning-text);
}

.alert-info {
  background-color: var(--alert-info-bg);
  border-color: var(--alert-info-border);
  color: var(--alert-info-text);
}

.alert:not([class*="alert-"]) {
  background-color: var(--surface-1);
  border-color: var(--border-color);
  color: var(--text-color);
}

/* ═══════════════════════════════════════════════════════════════
   БЕЙДЖИ
```

```
   ═══════════════════════════════════════════════════════════════
═══════════════ */
.badge {
  font-size: 0.72rem;
  font-weight: 500;
  padding: 0.22em 0.5em;
  letter-spacing: 0.02em;
}

.badge:not([class*="bg-"]):not([class*="badge-"]) {
  background-color: var(--border-color);
  color: var(--text-color);
}

.badge.bg-light,
.badge.badge-light {
  background-color: var(--border-subtle) !important;
  color: var(--text-color) !important;
}

.badge.bg-secondary,
.badge.bg-primary,
.badge.bg-danger,
.badge.bg-success,
.badge.bg-dark,
.badge-secondary,
.badge-primary,
.badge-danger,
.badge-success,
.badge-dark,
.film-inline-badge {
  color: #ffffff !important;
}

.badge.bg-warning,
.badge.badge-warning {
  color: #1a1a1a !important;
  font-weight: 600;
```

```
}

.badge.bg-info,
.badge.badge-info {
  color: #042743 !important;
  font-weight: 600;
}

/* Бейджи модели/поиска в навбаре */
#chat-model-badge,
#chat-popup-model-badge,
#chat-search-badge,
#chat-popup-search-badge {
  background-color: var(--surface-2) !important;
  border: 1px solid var(--border-color) !important;
  color: var(--text-color) !important;
  font-weight: 600;
  font-size: 0.75rem;
}

[data-theme="dark"] #chat-model-badge,
[data-theme="dark"] #chat-popup-model-badge {
  color: var(--nav-active) !important;
  border-color: var(--border-color) !important;
}

[data-theme="dark"] #chat-search-badge,
[data-theme="dark"] #chat-popup-search-badge {
  color: #38bdf8 !important;
  border-color: var(--border-color) !important;
}

/* Разделители в выпадающих меню */
.dropdown-divider {
  border-color: var(--border-color);
  opacity: 0.6;
  margin: 3px 0;
}
```

```
544
545 /* ═══════════════════════════════════════════════════════════
    ═══════════════
546    СИСТЕМА ПОМОЩИ
547    ═══════════════════════════════════════════════════════════
    ═══════════════ */
548 .help-icon {
549   display: inline-flex;
550   align-items: center;
551   justify-content: center;
552   width: 16px;
553   height: 16px;
554   border-radius: 50%;
555   background: var(--nav-active);
556   color: white;
557   font-size: 11px;
558   font-weight: bold;
559   cursor: help;
560   margin-left: 3px;
561   vertical-align: middle;
562   opacity: 0.8;
563   transition: opacity 0.15s;
564 }
565
566 .help-icon:hover { opacity: 1; }
567
568 .help-popover {
569   max-width: 320px;
570   font-size: 0.8rem;
571   background: var(--modal-bg);
572   color: var(--modal-text);
573   border: 1px solid var(--border-color);
574 }
575
576 /* ═══════════════════════════════════════════════════════════
    ═══════════════
577    ИНТЕРАКТИВНЫЕ ЧИПЫ (ФИЛЬМЫ)
578    ═══════════════════════════════════════════════════════════
```

```
   ═══════════════ */
.film-inline-badge {
  background-color: #2563eb !important;
  color: #ffffff !important;
  font-weight: 500;
  border: 1px solid rgba(255, 255, 255, 0.2) !important;
  box-shadow: 0 1px 2px rgba(0, 0, 0, 0.2);
}

.film-inline-badge:hover {
  background-color: #1d4ed8 !important;
  color: #ffffff !important;
}

.film-interactive-chip {
  display: inline-flex;
  align-items: center;
  gap: 3px;
  background: linear-gradient(135deg, #1e3a8a 0%, #2563eb 100%
);
  color: #ffffff !important;
  font-size: 0.85em;
  font-weight: 500;
  padding: 2px 4px 2px 7px;
  border-radius: 14px;
  border: 1px solid rgba(255, 255, 255, 0.2);
  box-shadow: 0 1px 3px rgba(0, 0, 0, 0.18);
  margin: 2px 2px;
  vertical-align: middle;
  transition: all 0.18s ease;
}

.film-interactive-chip:hover {
  background: linear-gradient(135deg, #1d4ed8 0%, #3b82f6 100%
);
  box-shadow: 0 2px 6px rgba(37, 99, 235, 0.35);
  transform: translateY(-1px);
}
```

```
614
615 .film-chip-title {
616   cursor: pointer;
617   color: #ffffff !important;
618   text-decoration: none;
619   font-weight: 600;
620   font-size: 0.82em;
621 }
622
623 .film-chip-title:hover { text-decoration: underline; }
624
625 .film-chip-btn {
626   background: rgba(255, 255, 255, 0.15);
627   border: none;
628   color: #ffffff;
629   width: 20px;
630   height: 20px;
631   border-radius: 50%;
632   display: inline-flex;
633   align-items: center;
634   justify-content: center;
635   font-size: 0.7rem;
636   padding: 0;
637   cursor: pointer;
638   transition: all 0.12s ease;
639 }
640
641 .film-chip-btn:hover    { background: rgba(255, 255, 255, 0.35
    ); transform: scale(1.1); }
642 .film-chip-play:hover   { background: #10b981; }
643 .film-chip-youtube:hover{ background: #ef4444; }
644 .film-chip-online:hover { background: #06b6d4; }
645 .film-chip-torrent:hover{ background: #f59e0b; color: #111827;
     }
646
647 [data-theme="dark"] .film-interactive-chip {
648   background: linear-gradient(135deg, #1e293b 0%, #334155 100%
    );
```

```
  border: 1px solid rgba(148, 163, 184, 0.25);
}

[data-theme="dark"] .film-interactive-chip:hover {
  background: linear-gradient(135deg, #334155 0%, #475569 100%
);
  box-shadow: 0 2px 6px rgba(0, 0, 0, 0.45);
}

/* ═══════════════════════════════════════════════════════════
═══════════
   PROMPT DUMP (КОНСОЛЬ АДМИНИСТРАТОРА)
   ═══════════════════════════════════════════════════════════
═══════════ */
.prompt-dump-details {
  margin-bottom: 8px;
  font-size: 0.8em;
  background: #0d1117 !important;
  padding: 8px;
  border-radius: 5px;
  border: 1px solid #30363d !important;
  box-shadow: inset 0 1px 3px rgba(0, 0, 0, 0.7);
}

.prompt-dump-details summary {
  cursor: pointer;
  color: #e3b341 !important;
  font-weight: bold;
  user-select: none;
  font-size: 0.85em;
}

.prompt-dump-details pre {
  white-space: pre-wrap;
  margin-top: 8px;
  color: #00ff66 !important;
  font-family: 'Consolas', 'Courier New', Courier, monospace;
  font-size: 0.9em;
```

```
  line-height: 1.4;
  max-height: 280px;
  overflow-y: auto;
}

/* ═══════════════════════════════════════════════════════════════
   MENU EDITOR (DRAG AND DROP)
   ═══════════════════════════════════════════════════════════════ */
.menu-editor-item {
  display: flex;
  align-items: center;
  gap: 10px;
  padding: 10px 12px;
  background: var(--surface-1);
  border: 1px solid var(--border-color);
  border-radius: 6px;
  margin-bottom: 8px;
  transition: all 0.15s ease;
}

.menu-editor-item:hover {
  border-color: var(--nav-active);
  background: var(--surface-2);
}

.menu-editor-item:active {
  cursor: grabbing;
}

.menu-editor-item.dragging {
  opacity: 0.5;
  background: var(--nav-active);
  border-color: var(--nav-active);
}

.menu-editor-item .drag-handle-container {
```

```
  cursor: grab;
  color: var(--text-muted);
  font-size: 1.2rem;
  display: flex;
  align-items: center;
}

.menu-editor-item .drag-handle-container:active {
  cursor: grabbing;
}

.menu-editor-item .item-icon {
  font-size: 1.3rem;
  min-width: 28px;
  text-align: center;
}

.menu-editor-item .item-info {
  flex: 1;
  min-width: 0;
}

.menu-editor-item .item-label {
  font-size: 0.88rem;
  font-weight: 500;
  white-space: nowrap;
  overflow: hidden;
  text-overflow: ellipsis;
}

.menu-editor-item .item-tab {
  font-size: 0.72rem;
  color: var(--text-muted);
  font-family: monospace;
}

.menu-editor-item .editor-controls {
  display: flex;
```

```
  flex-direction: column;
  gap: 6px;
  min-width: 180px;
}

.menu-editor-item .editor-visibility,
.menu-editor-item .editor-order {
  display: flex;
  align-items: center;
  gap: 6px;
}

.menu-editor-item .editor-visibility .visibility-label,
.menu-editor-item .editor-order .order-label {
  font-size: 0.75rem;
  color: var(--text-muted);
  white-space: nowrap;
}

.menu-editor-item .editor-visibility .visibility-select {
  flex: 1;
  font-size: 0.78rem;
  padding: 2px 6px;
}

.menu-editor-item .editor-order .order-slider {
  flex: 1;
  height: 6px;
  padding: 0;
}

.menu-editor-item .editor-order .order-value {
  font-size: 0.75rem;
  color: var(--text-muted);
  min-width: 24px;
  text-align: right;
}

```

```
.menu-editor-section {
  margin-bottom: 20px;
}

.menu-editor-section h6 {
  font-size: 0.8rem;
  text-transform: uppercase;
  letter-spacing: 0.5px;
  margin-bottom: 12px;
  color: var(--text-muted);
}

.menu-editor-section h6 i {
  margin-right: 6px;
}

#menuEditorModal .modal-body {
  max-height: 70vh;
  overflow-y: auto;
}

/* ═══════════════════════════════════════════════════════════════
   FLOATING TOAST NOTIFICATIONS (ВСПЛЫВАЮЩИЕ ОКНА)
   ═══════════════════════════════════════════════════════════════ */
.floating-toast-container {
  position: fixed;
  top: 18px;
  right: 18px;
  z-index: 99999;
  display: flex;
  flex-direction: column;
  gap: 10px;
  max-width: 420px;
  width: calc(100vw - 36px);
  pointer-events: none;
}
```

```
832
833 .floating-toast {
834   pointer-events: auto;
835   background-color: var(--surface-1, #1e293b);
836   color: var(--text-color, #f8fafc);
837   border: 1px solid var(--border-color, #334155);
838   border-left-width: 4px;
839   border-radius: 8px;
840   box-shadow: 0 10px 25px -5px rgba(0, 0, 0, 0.4), 0 8px 10px
-6px rgba(0, 0, 0, 0.3);
841   padding: 12px 14px;
842   opacity: 0;
843   transform: translateX(40px) scale(0.96);
844   transition: opacity 0.25s cubic-bezier(0.16, 1, 0.3, 1), tra
nsform 0.25s cubic-bezier(0.16, 1, 0.3, 1);
845   overflow: hidden;
846   position: relative;
847 }
848
849 .floating-toast.show {
850   opacity: 1;
851   transform: translateX(0) scale(1);
852 }
853
854 .floating-toast.hide {
855   opacity: 0;
856   transform: translateX(40px) scale(0.92);
857 }
858
859 .floating-toast-success {
860   border-left-color: #22c55e !important;
861 }
862
863 .floating-toast-danger {
864   border-left-color: #ef4444 !important;
865 }
866
867 .floating-toast-warning {
```

```
  border-left-color: #f59e0b !important;
}

.floating-toast-info {
  border-left-color: #3b82f6 !important;
}

.floating-toast-header {
  display: flex;
  align-items: center;
  gap: 8px;
  margin-bottom: 6px;
}

.floating-toast-icon {
  font-size: 1.1rem;
  line-height: 1;
}

.floating-toast-title {
  font-size: 0.88rem;
  font-weight: 600;
  flex: 1;
  color: var(--text-color, #f8fafc);
}

.floating-toast-close {
  background: transparent;
  border: none;
  color: var(--text-muted, #94a3b8);
  font-size: 1.25rem;
  line-height: 1;
  padding: 0 4px;
  cursor: pointer;
  transition: color 0.15s ease;
}

.floating-toast-close:hover {
```

```
  color: var(--text-color, #ffffff);
}

.floating-toast-body {
  font-size: 0.84rem;
  line-height: 1.45;
  color: var(--card-text, #cbd5e1);
  word-break: break-word;
}

.floating-toast-progress {
  position: absolute;
  bottom: 0;
  left: 0;
  right: 0;
  height: 3px;
  background: rgba(255, 255, 255, 0.08);
}

.floating-toast-progress-bar {
  height: 100%;
  width: 100%;
}

/* ═══════════════════════════════════════════════════════════════
   ГЛОБАЛЬНЫЕ УТИЛИТАРНЫЕ КЛАССЫ (UNIFIED UTILITY CLASSES)
   ═══════════════════════════════════════════════════════════════ */

/* ── 0. Универсальная карточка (для простых случаев) ──────────── */
.card-unified {
  background: var(--surface-1);
  border: 1px solid var(--border-color);
  border-radius: var(--radius-md);
  box-shadow: var(--shadow-sm);
  transition: border-color 0.15s ease, box-shadow 0.15s ease;
```

```
}

.card-unified:hover {
  border-color: var(--nav-active);
  box-shadow: var(--shadow-md);
}

/* ── 1. KPI & Метрические карточки ──────────────────────────── */

/* ── 1. KPI & Метрические карточки ──────────────────────────── */
.ui-kpi-card {
  background: var(--kpi-bg, var(--surface-1)) !important;
  border: 1.5px solid var(--kpi-border, var(--border-color)) !important;
  border-radius: var(--radius-md, 8px);
  padding: var(--kpi-padding, 0.65rem 0.9rem);
  min-height: var(--kpi-min-height, 72px);
  display: flex;
  flex-direction: column;
  justify-content: space-between;
  height: 100%;
  box-sizing: border-box;
  box-shadow: 0 2px 6px rgba(0, 0, 0, 0.25);
  transition: border-color 0.15s ease, box-shadow 0.15s ease, transform 0.15s ease;
}

.ui-kpi-card:hover {
  border-color: var(--kpi-accent, var(--nav-active)) !important;
  box-shadow: var(--shadow-sm, 0 3px 10px rgba(0, 0, 0, 0.25));
}

.ui-kpi-label {
  font-size: var(--kpi-label-size, 0.72rem);
```

```
  text-transform: uppercase;
  letter-spacing: 0.05em;
  font-weight: 600;
  color: var(--text-muted, #8b949e);
  display: flex;
  justify-content: space-between;
  align-items: center;
  line-height: 1.2;
}

.ui-kpi-value {
  font-size: var(--kpi-val-size, 1.25rem);
  font-weight: 700;
  font-family: var(--font-mono, monospace);
  color: var(--text-color, #f8fafc);
  line-height: 1.2;
}

.ui-kpi-sub {
  font-size: var(--kpi-sub-size, 0.72rem);
  color: var(--text-muted, #8b949e);
  line-height: 1.2;
}

/* Модификаторы состояний KPI карточек через каскад */
.ui-kpi-card.is-success {
  --kpi-border: var(--status-success-border, #22c55e);
  --kpi-accent: var(--status-success-border, #22c55e);
}
.ui-kpi-card.is-danger {
  --kpi-border: var(--status-danger-border, #ef4444);
  --kpi-accent: var(--status-danger-border, #ef4444);
}
.ui-kpi-card.is-warning {
  --kpi-border: var(--status-warning-border, #eab308);
  --kpi-accent: var(--status-warning-border, #eab308);
}
.ui-kpi-card.is-info {
```

```
  --kpi-border: var(--status-info-border, #0284c7);
  --kpi-accent: var(--status-info-border, #0284c7);
}

/* ── 2. Модульные заголовки и Hero-баннеры ──────────────────── */
.ui-module-header {
  background: var(--surface-2) !important;
  border: 1px solid var(--border-color) !important;
  border-radius: var(--radius-md, 8px);
  padding: 0.6rem 1rem;
  margin-bottom: 0.75rem;
  display: flex;
  align-items: center;
  justify-content: space-between;
  flex-wrap: wrap;
  gap: 0.5rem;
}

.ui-module-title {
  font-size: 0.95rem;
  font-weight: 700;
  color: var(--text-color);
  display: flex;
  align-items: center;
  gap: 0.5rem;
  margin-bottom: 0;
}

.ui-module-desc {
  font-size: 0.78rem;
  color: var(--text-muted);
  margin-bottom: 0;
}

/* ── 3. Панели и контейнеры таблиц ──────────────────────────── */
.ui-panel {
```

```
  background: var(--panel-bg, var(--surface-1)) !important;
  border: 1px solid var(--panel-border, var(--border-color)) !important;
  border-radius: var(--radius-md, 8px);
  overflow: hidden;
  margin-bottom: 0.75rem;
  box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.1));
}

.ui-panel-header {
  background: var(--panel-header-bg, var(--surface-2)) !important;
  border-bottom: 1px solid var(--border-color) !important;
  padding: 0.5rem 0.9rem;
  font-size: 0.85rem;
  font-weight: 600;
  color: var(--text-color);
  display: flex;
  align-items: center;
  justify-content: space-between;
  flex-wrap: wrap;
  gap: 0.5rem;
}

.ui-panel-body {
  padding: 0.75rem;
}

.ui-panel-footer {
  background: var(--surface-2) !important;
  border-top: 1px solid var(--border-color) !important;
  padding: 0.4rem 0.75rem;
  font-size: 0.78rem;
}

/* ── 4. Унифицированные суб-навигационные переключатели (Sub-Nav Tabs) ──── */
.ui-subnav {
```

```
  display: flex;
  align-items: center;
  gap: 0.4rem;
  border-bottom: 1px solid var(--border-color);
  margin-bottom: 0.75rem;
  padding-bottom: 0.35rem;
  overflow-x: auto;
  flex-wrap: nowrap;
}

.ui-subnav-btn {
  background: transparent;
  border: 1px solid transparent;
  color: var(--text-muted);
  font-size: var(--subnav-font-size, 0.82rem);
  font-weight: 500;
  padding: var(--subnav-btn-padding, 0.35rem 0.75rem);
  border-radius: var(--radius-sm, 6px);
  cursor: pointer;
  white-space: nowrap;
  display: inline-flex;
  align-items: center;
  gap: 0.4rem;
  transition: all 0.15s ease;
}

.ui-subnav-btn:hover {
  background: var(--surface-2);
  color: var(--text-color);
}

.ui-subnav-btn.active {
  background: var(--surface-2);
  border-color: var(--nav-active);
  color: var(--nav-active);
  font-weight: 600;
}

```

```
/* ── 5. Унифицированные статусные бейджи ──────────────────────── */
.ui-badge {
  font-size: 0.72rem;
  font-weight: 600;
  padding: 0.2rem 0.55rem;
  border-radius: var(--radius-xs, 4px);
  display: inline-flex;
  align-items: center;
  gap: 0.3rem;
  line-height: 1.2;
}

.ui-badge-success {
  background: var(--status-success-bg);
  color: var(--status-success-text);
  border: 1px solid var(--status-success-border);
}

.ui-badge-danger {
  background: var(--status-danger-bg);
  color: var(--status-danger-text);
  border: 1px solid var(--status-danger-border);
}

.ui-badge-warning {
  background: var(--status-warning-bg);
  color: var(--status-warning-text);
  border: 1px solid var(--status-warning-border);
}

.ui-badge-info {
  background: var(--status-info-bg);
  color: var(--status-info-text);
  border: 1px solid var(--status-info-border);
}

.ui-badge-neutral {
```

```
  background: var(--surface-2);
  color: var(--text-muted);
  border: 1px solid var(--border-color);
}

/* ── 6. Унифицированная таблица спецификаций (Ключ - Значение) ──────────────── */
.ui-spec-table,
.about-sys-spec-table,
.sysctl-spec-table,
.def-spec-table,
.winadmin-spec-table {
  width: 100%;
  margin-bottom: 0;
  font-size: 0.82rem;
  border-collapse: separate;
  border-spacing: 0;
}

.ui-spec-table th,
.about-sys-spec-table th,
.sysctl-spec-table th,
.def-spec-table th,
.winadmin-spec-table th {
  padding: 8px 16px;
  font-size: 0.78rem;
  font-weight: 600;
  color: var(--text-muted, #94a3b8);
  text-transform: uppercase;
  letter-spacing: 0.04em;
  border-bottom: 1px solid var(--border-subtle, rgba(51, 65, 85, 0.35));
  background: transparent;
  vertical-align: middle;
}

.ui-spec-table td,
.about-sys-spec-table td,
```

```
.sysctl-spec-table td,
.def-spec-table td,
.winadmin-spec-table td {
  padding: 9px 14px;
  border-bottom: 1px solid var(--border-subtle, rgba(51, 65, 85, 0.35));
  vertical-align: middle;
  line-height: 1.45;
  color: var(--text-color, #f8fafc);
}

.ui-spec-table tr:hover td,
.about-sys-spec-table tr:hover td,
.sysctl-spec-table tr:hover td,
.def-spec-table tr:hover td,
.winadmin-spec-table tr:hover td {
  background: rgba(56, 189, 248, 0.04);
}

.ui-spec-table tr:last-child td,
.about-sys-spec-table tr:last-child td,
.sysctl-spec-table tr:last-child td,
.def-spec-table tr:last-child td,
.winadmin-spec-table tr:last-child td {
  border-bottom: none;
}

.ui-spec-key,
.about-sys-spec-key,
.sysctl-spec-key,
.def-spec-key,
.winadmin-spec-key {
  color: var(--text-muted, #94a3b8);
  width: 230px;
  min-width: 210px;
  max-width: 260px;
  white-space: nowrap;
  font-weight: 500;
```

```
  font-family: var(--font-sans, system-ui, -apple-system, sans-serif);
  vertical-align: middle;
}

.ui-spec-key i,
.about-sys-spec-key i,
.sysctl-spec-key i,
.def-spec-key i,
.winadmin-spec-key i {
  margin-right: 8px;
  font-size: 0.92rem;
  vertical-align: -1px;
}

.ui-spec-val,
.about-sys-spec-val,
.sysctl-spec-val,
.def-spec-val,
.winadmin-spec-val {
  color: var(--text-color, #f8fafc);
  font-weight: 500;
  font-family: var(--font-mono, monospace);
  vertical-align: middle;
  word-break: break-word;
}

/* ── 7. Контекстные каскадные темы (Тематические переопределения) ──────────── */
.theme-defender {
  --kpi-min-height: 64px;
  --kpi-padding: 0.5rem 0.75rem;
  --kpi-accent: #22c55e;
}

.theme-sysadmin {
  --kpi-min-height: 68px;
  --kpi-padding: 0.55rem 0.85rem;
```

```
}

.theme-inspector {
  --kpi-min-height: 80px;
  --kpi-val-size: 1.35rem;
}

.theme-sysctl {
  --kpi-min-height: 72px;
}

/* ── 8. Алиасы совместимости (Backward-compatible Aliases) ──
──────────────────── */
.about-sys-card,
.def-card,
.sysctl-card,
.scc-card,
.winadmin-card,
.sys-card {
  composes: ui-kpi-card;
  background: var(--kpi-bg, var(--surface-1)) !important;
  border: 1.5px solid var(--kpi-border, var(--border-color)) !
important;
  border-radius: var(--radius-md, 8px);
  padding: var(--kpi-padding, 0.65rem 0.9rem);
  min-height: var(--kpi-min-height, 72px);
  display: flex;
  flex-direction: column;
  justify-content: space-between;
  height: 100%;
  box-sizing: border-box;
  box-shadow: 0 2px 6px rgba(0, 0, 0, 0.3);
  transition: border-color 0.15s ease, box-shadow 0.15s ease;
}

.about-sys-card:hover,
.def-card:hover,
.sysctl-card:hover,
```

```
.scc-card:hover,
.winadmin-card:hover,
.sys-card:hover {
  border-color: var(--kpi-accent, var(--nav-active)) !important;
  box-shadow: 0 4px 12px rgba(0, 0, 0, 0.4);
}

.about-sys-label,
.def-card-label,
.sysctl-label,
.scc-label,
.winadmin-label,
.sys-label {
  font-size: var(--kpi-label-size, 0.72rem);
  text-transform: uppercase;
  letter-spacing: 0.05em;
  font-weight: 600;
  color: var(--text-muted, #8b949e);
  display: flex;
  justify-content: space-between;
  align-items: center;
  line-height: 1.2;
}

.about-sys-value,
.def-card-value,
.sysctl-value,
.scc-value,
.winadmin-value,
.sys-value {
  font-size: var(--kpi-val-size, 1.25rem);
  font-weight: 700;
  font-family: var(--font-mono, monospace);
  color: var(--text-color, #f8fafc);
  line-height: 1.2;
}

```

```
.about-sys-sub,
.def-card-sub,
.sysctl-sub,
.scc-sub,
.winadmin-sub,
.sys-sub {
  font-size: var(--kpi-sub-size, 0.72rem);
  color: var(--text-muted, #8b949e);
  line-height: 1.2;
}

.about-sys-panel,
.def-panel,
.sysctl-panel,
.winadmin-panel,
.sys-panel {
  background: var(--panel-bg, var(--surface-1)) !important;
  border: 1px solid var(--panel-border, var(--border-color)) !important;
  border-radius: var(--radius-md, 8px);
  overflow: hidden;
  margin-bottom: 0.75rem;
  box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.1));
}

.about-sys-panel-header,
.def-panel-header,
.sysctl-panel-header,
.winadmin-panel-header,
.sys-panel-header {
  background: var(--panel-header-bg, var(--surface-2)) !important;
  border-bottom: 1px solid var(--border-color) !important;
  padding: 0.5rem 0.9rem;
  font-size: 0.85rem;
  font-weight: 600;
  color: var(--text-color);
  display: flex;
```

```
  align-items: center;
  justify-content: space-between;
  flex-wrap: wrap;
  gap: 0.5rem;
}

/* ── 9. Селектор частоты опроса панели (Panel Polling Frequency Selector) ──── */
.poll-freq-select {
  font-size: 0.68rem !important;
  padding: 1px 18px 1px 6px !important;
  height: 22px !important;
  min-height: 22px !important;
  line-height: 1.2 !important;
  background-color: rgba(15, 23, 42, 0.85) !important;
  border: 1px solid rgba(255, 255, 255, 0.18) !important;
  color: #94a3b8 !important;
  border-radius: 4px !important;
  cursor: pointer;
  width: auto !important;
  max-width: 145px;
  display: inline-block;
  vertical-align: middle;
  transition: border-color 0.15s ease, color 0.15s ease;
}

.poll-freq-select:hover {
  border-color: rgba(56, 189, 248, 0.5) !important;
  color: #cbd5e1 !important;
}

.poll-freq-select:focus {
  border-color: #38bdf8 !important;
  color: #f1f5f9 !important;
  box-shadow: 0 0 0 1px rgba(56, 189, 248, 0.35) !important;
  outline: none;
}

```

```
.poll-freq-select option {
  background-color: #0f172a !important;
  color: #e2e8f0 !important;
}

/* ── 10. Дерево спецификации оборудования (Hardware Specification Tree) ──────── */
.about-sys-wrapper {
  padding-left: 0.25rem;
  padding-right: 0.25rem;
}

.about-sys-tree-node {
  background: var(--surface-1);
  border: 1px solid var(--border-color);
  border-radius: var(--radius-sm, 6px);
  margin-bottom: 0.65rem;
  overflow: hidden;
  box-shadow: var(--shadow-xs, 0 1px 2px rgba(0, 0, 0, 0.05));
  transition: border-color 0.15s ease, box-shadow 0.15s ease;
}

.about-sys-tree-node:hover {
  border-color: rgba(56, 189, 248, 0.4);
}

.about-sys-tree-title {
  padding: 0.6rem 0.9rem;
  font-size: 0.84rem;
  font-weight: 600;
  cursor: pointer;
  background: var(--surface-2);
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 0.5rem;
  color: var(--text-color);
  transition: background 0.15s ease, color 0.15s ease;
```

```
  user-select: none;
}

.about-sys-tree-title:hover {
  background: rgba(56, 189, 248, 0.08);
}

.about-sys-tree-body {
  background: var(--surface-1);
  border-top: 1px solid var(--border-subtle, rgba(255, 255, 255, 0.06));
  padding: 0;
}

.about-sys-chevron {
  font-size: 0.72rem;
  color: var(--accent-color, #38bdf8);
  transition: transform 0.2s ease;
  display: inline-block;
  line-height: 1;
}

/* ── 10. Кнопка-иконка сворачивания/разворачивания таблиц (Красная дрель) ─── */
.panel-toggle-drill {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  background: transparent;
  border: none;
  padding: 0;
  margin: 0;
  cursor: pointer;
  line-height: 1;
  vertical-align: middle;
  transition: transform 0.15s ease, opacity 0.15s ease;
  border-radius: 4px;
}
```

```
1485
1486 .panel-toggle-drill:hover {
1487   opacity: 0.85;
1488   transform: scale(1.15);
1489 }
1490
1491 .panel-toggle-drill:active {
1492   transform: scale(0.92);
1493 }
1494
1495 .panel-toggle-drill img {
1496   width: 20px;
1497   height: 20px;
1498   object-fit: contain;
1499   display: block;
1500 }
1501
1502 .panel-toggle-drill.collapsed img {
1503   opacity: 0.7;
1504 }
1505
1506
```

global-overrides.css

```
/**
 * ============================================================
 * Process Name: Windows Web Styles - Global Overrides Stylesheet
 * ============================================================
 * Description:
 *   Глобальные переопределения стилей для минимизации дублирования CSS-классов.
 *   Использует наследование и глобальные селекторы вместо повторения стилей.
 *
 * Usage Examples:
 *   HTML Link Tag:
 *     <link rel="stylesheet" href="/windows/api/webgui/css/global-overrides.css?v=20261008_v1">
 *
 * File: global-overrides.css
 * Project: ai-windows-admin
 * Package: windows/api/webgui/css
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/* ═══════════════════════════════════════════════════════════
   ГЛОБАЛЬНЫЕ ПЕРЕОПРЕДЕЛЕНИЯ - AI Windows Admin WebGUI
   Минимизация дублирования стилей через наследование и глобальные селекторы
   ═══════════════════════════════════════════════════════════ */

```

```
/* ── 1. Универсальные карточки (KPI, Panel, Spec) ──────────────────── */

/* Все карточки с классом .ui-kpi-card наследуют базовые стили */
.ui-kpi-card,
.about-sys-card,
.def-card,
.sysctl-card,
.scc-card,
.winadmin-card,
.sys-card {
  background: var(--surface-1);
  border: 1.5px solid var(--border-color);
  border-radius: var(--radius-md);
  padding: var(--kpi-padding, 0.65rem 0.9rem);
  min-height: var(--kpi-min-height, 72px);
  display: flex;
  flex-direction: column;
  justify-content: space-between;
  height: 100%;
  box-sizing: border-box;
  box-shadow: 0 2px 6px rgba(0, 0, 0, 0.25);
  transition: border-color 0.15s ease, box-shadow 0.15s ease;
}

.ui-kpi-card:hover,
.about-sys-card:hover,
.def-card:hover,
.sysctl-card:hover,
.scc-card:hover,
.winadmin-card:hover,
.sys-card:hover {
  border-color: var(--kpi-accent, var(--nav-active));
  box-shadow: var(--shadow-sm, 0 3px 10px rgba(0, 0, 0, 0.25));
}

```

```
/* ── 2. Универсальные заголовки карточек ──────────────────────── */

.ui-kpi-label,
.about-sys-label,
.def-card-label,
.sysctl-label,
.scc-label,
.winadmin-label,
.sys-label {
  font-size: var(--kpi-label-size, 0.72rem);
  text-transform: uppercase;
  letter-spacing: 0.05em;
  font-weight: 600;
  color: var(--text-muted);
  display: flex;
  justify-content: space-between;
  align-items: center;
  line-height: 1.2;
}

/* ── 3. Универсальные значения в карточках ────────────────────── */

.ui-kpi-value,
.about-sys-value,
.def-card-value,
.sysctl-value,
.scc-value,
.winadmin-value,
.sys-value {
  font-size: var(--kpi-val-size, 1.25rem);
  font-weight: 700;
  font-family: var(--font-mono, monospace);
  color: var(--text-color);
  line-height: 1.2;
}

```

```
/* ── 4. Универсальные подзаголовки в карточках ──────────────────── */

.ui-kpi-sub,
.about-sys-sub,
.def-card-sub,
.sysctl-sub,
.scc-sub,
.winadmin-sub,
.sys-sub {
  font-size: var(--kpi-sub-size, 0.72rem);
  color: var(--text-muted);
  line-height: 1.2;
}

/* ── 5. Универсальные панели ──────────────────────────────────────── */

.ui-panel,
.about-sys-panel,
.def-panel,
.sysctl-panel,
.winadmin-panel,
.sys-panel {
  background: var(--surface-1);
  border: 1px solid var(--border-color);
  border-radius: var(--radius-md);
  overflow: hidden;
  margin-bottom: 0.75rem;
  box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.1));
}

/* ── 6. Универсальные заголовки панелей ──────────────────────────── */

.ui-panel-header,
.about-sys-panel-header,
.def-panel-header,
```

```
.sysctl-panel-header,
.winadmin-panel-header,
.sys-panel-header {
  background: var(--surface-2);
  border-bottom: 1px solid var(--border-color);
  padding: 0.5rem 0.9rem;
  font-size: 0.85rem;
  font-weight: 600;
  color: var(--text-color);
  display: flex;
  align-items: center;
  justify-content: space-between;
  flex-wrap: wrap;
  gap: 0.5rem;
}

/* ── 7. Универсальные таблицы спецификаций ──────────────────── */

.ui-spec-table,
.about-sys-spec-table,
.sysctl-spec-table,
.def-spec-table,
.winadmin-spec-table,
.spec-table-unified {
  width: 100%;
  margin-bottom: 0;
  font-size: 0.82rem;
  border-collapse: separate;
  border-spacing: 0;
}

.ui-spec-table th,
.about-sys-spec-table th,
.sysctl-spec-table th,
.def-spec-table th,
.winadmin-spec-table th,
.spec-table-unified th {
```

```
  padding: 8px 16px;
  font-size: 0.78rem;
  font-weight: 600;
  color: var(--text-muted);
  text-transform: uppercase;
  letter-spacing: 0.04em;
  border-bottom: 1px solid var(--border-subtle);
  background: transparent;
  vertical-align: middle;
}

.ui-spec-table td,
.about-sys-spec-table td,
.sysctl-spec-table td,
.def-spec-table td,
.winadmin-spec-table td,
.spec-table-unified td {
  padding: 9px 14px;
  border-bottom: 1px solid var(--border-subtle);
  vertical-align: middle;
  line-height: 1.45;
  color: var(--text-color);
}

.ui-spec-table tr:hover td,
.about-sys-spec-table tr:hover td,
.sysctl-spec-table tr:hover td,
.def-spec-table tr:hover td,
.winadmin-spec-table tr:hover td,
.spec-table-unified tr:hover td {
  background: rgba(56, 189, 248, 0.04);
}

.ui-spec-table tr:last-child td,
.about-sys-spec-table tr:last-child td,
.sysctl-spec-table tr:last-child td,
.def-spec-table tr:last-child td,
.winadmin-spec-table tr:last-child td,
```

```
.spec-table-unified tr:last-child td {
  border-bottom: none;
}

/* ── 8. Универсальные ключи и значения в таблицах ──────────
──────────────────── */

.ui-spec-key,
.about-sys-spec-key,
.sysctl-spec-key,
.def-spec-key,
.winadmin-spec-key,
.spec-key-unified {
  color: var(--text-muted);
  width: 230px;
  min-width: 210px;
  max-width: 260px;
  white-space: nowrap;
  font-weight: 500;
  font-family: var(--font-sans, system-ui, -apple-system, sans
-serif);
  vertical-align: middle;
}

.ui-spec-key i,
.about-sys-spec-key i,
.sysctl-spec-key i,
.def-spec-key i,
.winadmin-spec-key i,
.spec-key-unified i {
  margin-right: 8px;
  font-size: 0.92rem;
  vertical-align: -1px;
}

.ui-spec-val,
.about-sys-spec-val,
.sysctl-spec-val,
```

```
.def-spec-val,
.winadmin-spec-val,
.spec-val-unified {
  color: var(--text-color);
  font-weight: 500;
  font-family: var(--font-mono, monospace);
  vertical-align: middle;
  word-break: break-word;
}

/* ── 9. Универсальные бейджи статусов ──────────────────────── */

.ui-badge,
.status-badge-unified,
.about-sys-badge-ok,
.about-sys-badge-warn,
.diag-badge-critical,
.diag-badge-warning,
.diag-badge-normal {
  font-size: 0.72rem;
  font-weight: 600;
  padding: 0.2rem 0.55rem;
  border-radius: var(--radius-xs, 4px);
  display: inline-flex;
  align-items: center;
  gap: 0.3rem;
  line-height: 1.2;
}

.ui-badge-success,
.about-sys-badge-ok,
.diag-badge-normal {
  background: var(--status-success-bg);
  color: var(--status-success-text);
  border: 1px solid var(--status-success-border);
}

```

```
.ui-badge-danger,
.diag-badge-critical {
  background: var(--status-danger-bg);
  color: var(--status-danger-text);
  border: 1px solid var(--status-danger-border);
}

.ui-badge-warning,
.about-sys-badge-warn,
.diag-badge-warning {
  background: var(--status-warning-bg);
  color: var(--status-warning-text);
  border: 1px solid var(--status-warning-border);
}

.ui-badge-info {
  background: var(--status-info-bg);
  color: var(--status-info-text);
  border: 1px solid var(--status-info-border);
}

/* ── 10. Универсальные панели диагностики ──────────────────── */

.diag-panel,
.ui-panel {
  background: var(--surface-1);
  border: 1px solid var(--border-color);
  border-radius: var(--radius-md);
  overflow: hidden;
  margin-bottom: 0.75rem;
  box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.1));
}

.diag-panel-header,
.ui-panel-header {
  background: var(--surface-2);
  border-bottom: 1px solid var(--border-color);
```

```
  padding: 0.5rem 0.9rem;
  font-size: 0.85rem;
  font-weight: 600;
  color: var(--text-color);
  display: flex;
  align-items: center;
  justify-content: space-between;
  flex-wrap: wrap;
  gap: 0.5rem;
}

/* ── 11. Универсальные карточки метрик ──────────────────────── */

.diag-meter-card,
.ui-kpi-card {
  background: var(--surface-2);
  border: 1px solid var(--border-color);
  border-radius: var(--radius-md);
  padding: 0.75rem 1rem;
}

/* ── 12. Универсальные заголовки деревьев и узлы ─────────────── */

.about-sys-tree-node {
  background: var(--surface-1);
  border: 1px solid var(--border-color);
  border-radius: var(--radius-sm, 6px);
  margin-bottom: 0.65rem;
  overflow: hidden;
  box-shadow: var(--shadow-xs, 0 1px 2px rgba(0, 0, 0, 0.05));
  transition: border-color 0.15s ease, box-shadow 0.15s ease;
}

.about-sys-tree-node:hover {
  border-color: rgba(56, 189, 248, 0.4);
}
```

```
354
355 .about-sys-tree-title,
356 .ui-module-title {
357   padding: 0.6rem 0.9rem;
358   font-size: 0.84rem;
359   font-weight: 600;
360   cursor: pointer;
361   background: var(--surface-2);
362   display: flex;
363   align-items: center;
364   justify-content: space-between;
365   gap: 0.5rem;
366   color: var(--text-color);
367   transition: background 0.15s ease, color 0.15s ease;
368   user-select: none;
369 }
370
371 .about-sys-tree-title:hover,
372 .ui-module-title:hover {
373   background: rgba(56, 189, 248, 0.08);
374 }
375
376 .about-sys-tree-body {
377   background: var(--surface-1);
378   border-top: 1px solid var(--border-subtle, rgba(255, 255, 25
    5, 0.06));
379   padding: 0;
380 }
381
382 .about-sys-chevron {
383   font-size: 0.72rem;
384   color: var(--accent-color, #38bdf8);
385   transition: transform 0.2s ease;
386   display: inline-block;
387   line-height: 1;
388 }
389
390 /* ── 13. Универсальные строки свойств ──────────────────────
```

```
   ──────────────────── */
391
392 .about-sys-prop-row,
393 .ui-prop-row {
394   display: grid;
395   grid-template-columns: minmax(180px, 320px) 1fr;
396   align-items: center;
397   gap: 12px;
398   padding: 6px 12px;
399   border-bottom: 1px solid var(--border-subtle);
400   border-radius: 0;
401   transition: background 0.12s ease;
402   min-width: 0;
403 }
404
405 .about-sys-prop-row:hover,
406 .ui-prop-row:hover {
407   background: rgba(56, 189, 248, 0.04);
408 }
409
410 .about-sys-prop-row:last-child,
411 .ui-prop-row:last-child {
412   border-bottom: none;
413 }
414
415 /* ── 14. Универсальные ключи свойств ──────────────────────────
   ──────────────────── */
416
417 .about-sys-prop-key,
418 .ui-prop-key {
419   color: var(--text-muted);
420   font-weight: 500;
421   font-size: 0.82rem;
422   user-select: text;
423   word-break: break-word;
424 }
425
426 .about-sys-prop-key::after,
```

```
.ui-prop-key::after {
  content: "";
}

/* ── 15. Универсальные значения свойств ──────────────────────── */

.about-sys-prop-val,
.ui-prop-val {
  color: var(--text-color);
  font-weight: 500;
  font-size: 0.82rem;
  word-break: break-word;
  user-select: text;
  font-family: ui-monospace, SFMono-Regular, Menlo, Monaco, Consolas, monospace;
  min-width: 0;
}

/* ── 16. Универсальные прогресс-бары ─────────────────────────── */

.about-sys-progress-track,
.ui-progress-track {
  background: var(--surface-2);
  border-radius: 4px;
  height: 8px;
  overflow: hidden;
  min-width: 100px;
}

.about-sys-progress-bar,
.ui-progress-bar {
  height: 100%;
  border-radius: 4px;
  transition: width 0.3s ease;
}

```

```
/* ── 17. Универсальные навигационные панели ──────────────────────── */

.about-subtabs-nav,
.ui-subnav {
  background: var(--surface-2);
  border: 1px solid var(--border-color);
  border-radius: var(--radius-md);
  padding: 0.35rem 0.5rem;
}

.about-subtab-btn,
.ui-subnav-btn {
  transition: all 0.18s ease;
  font-size: 0.78rem;
}

.about-subtab-btn.active,
.ui-subnav-btn.active {
  background: var(--nav-active);
  color: #0f172a;
  border-color: var(--nav-active);
  font-weight: 700;
}

/* ── 18. Универсальные панели с верхним хедером ──────────────────── */

.about-sys-top-header,
.ui-top-header {
  background: var(--surface-2);
  border: 1px solid var(--border-color);
  border-radius: var(--radius-md);
  padding: 0.75rem 1rem;
  margin-bottom: 0.75rem;
}

/* ── 19. Универсальные карточки с анимацией ──────────────────────── */
```

```
   ──────────────────── */
498
499 .about-ai-copilot-banner,
500 .ui-banner-card {
501   background: linear-gradient(135deg, rgba(15, 23, 42, 0.98) 0
    %, rgba(20, 35, 60, 0.95) 50%, rgba(15, 23, 42, 0.98) 100%);
502   border: 1px solid rgba(56, 189, 248, 0.45);
503   border-radius: 10px;
504   box-shadow: 0 0 22px rgba(56, 189, 248, 0.15), inset 0 1px 0
     rgba(255, 255, 255, 0.1);
505   position: relative;
506   overflow: hidden;
507 }
508
509 /* ── 20. Универсальные иконки пульсации ──────────────────────
    ──────────────────── */
510
511 .about-ai-pulse-icon,
512 .ui-pulse-icon {
513   width: 34px;
514   height: 34px;
515   border-radius: 8px;
516   background: rgba(56, 189, 248, 0.12);
517   border: 1px solid rgba(56, 189, 248, 0.35);
518   display: flex;
519   align-items: center;
520   justify-content: center;
521 }
522
523 /* ── 21. Универсальные блоки этапов ──────────────────────────
    ──────────────────── */
524
525 .about-ai-stages-box,
526 .ui-stages-box {
527   background: rgba(10, 15, 29, 0.7);
528   max-height: 240px;
529   overflow-y: auto;
530 }
```

```
531
532 .about-ai-stage-row,
533 .ui-stage-row {
534   padding: 3px 6px;
535   border-radius: 4px;
536   font-size: 0.74rem;
537   line-height: 1.35;
538 }
539
540 .about-ai-stage-row:hover,
541 .ui-stage-row:hover {
542   background: rgba(255, 255, 255, 0.04);
543 }
544
545 /* ── 22. Универсальные панели с прокруткой ──────────────────
    ──────────────── */
546
547 .diag-panel,
548 .ui-scroll-panel {
549   overflow-y: auto;
550 }
551
552 /* ── 23. Универсальные панели с фиксированной высотой ────────
    ──────────────── */
553
554 .h-100 {
555   height: 100%;
556 }
557
558 /* ── 24. Универсальные панели с адаптивной высотой ──────────
    ──────────────── */
559
560 @media (min-width: 768px) {
561   .col-md-6 .ui-panel,
562   .col-md-6 .ui-kpi-card {
563     min-height: 150px;
564   }
565 }
```

```
566
567 @media (min-width: 992px) {
568   .col-lg-6 .ui-panel,
569   .col-lg-6 .ui-kpi-card {
570     min-height: 200px;
571   }
572 }
573
574 @media (min-width: 1200px) {
575   .col-xl-8 .ui-panel {
576     min-height: 300px;
577   }
578   .col-xl-4 .ui-panel {
579     min-height: 250px;
580   }
581 }
582
583 /* ── 25. Универсальные панели с hover-эффектами ──────────────
    ──────────────── */
584
585 .ui-panel:hover,
586 .ui-kpi-card:hover,
587 .ui-badge:hover {
588   transition: all 0.15s ease;
589 }
590
591 /* ── 26. Универсальные панели с фиксированным отступом ──────
    ──────────────── */
592
593 .ui-panel,
594 .ui-kpi-card {
595   margin-bottom: 0.75rem;
596 }
597
598 /* ── 27. Универсальные панели с адаптивной шириной ──────────
    ──────────────── */
599
600 @media (max-width: 767px) {
```

```
  .ui-panel,
  .ui-kpi-card {
    margin-bottom: 0.5rem;
  }
}

/* ── 28. Универсальные панели с адаптивным отступом ──────────────────── */

@media (max-width: 575px) {
  .ui-panel,
  .ui-kpi-card {
    padding: 0.5rem;
  }
}

/* ── 29. Универсальные панели с адаптивным заголовком ─────────────────── */

@media (max-width: 575px) {
  .ui-panel-header,
  .ui-kpi-label {
    font-size: 0.75rem;
    padding: 0.4rem 0.6rem;
  }
}

/* ── 30. Универсальные панели с адаптивным значением ──────────────────── */

@media (max-width: 575px) {
  .ui-kpi-value,
  .ui-spec-val {
    font-size: 1.1rem;
  }
}

/* ── 31. Универсальные панели с адаптивным ключом ─────────────
```

```
   ──────────────────── */
636
637 @media (max-width: 575px) {
638   .ui-spec-key {
639     font-size: 0.75rem;
640     width: 100%;
641   }
642 }
643
644 /* ── 32. Универсальные панели с адаптивной таблицей ──────────
   ──────────────────── */
645
646 @media (max-width: 575px) {
647   .ui-spec-table th,
648   .ui-spec-table td {
649     padding: 6px 10px;
650     font-size: 0.75rem;
651   }
652 }
653
654 /* ── 33. Универсальные панели с адаптивным бейджем ──────────
   ──────────────────── */
655
656 @media (max-width: 575px) {
657   .ui-badge,
658   .about-sys-badge-ok,
659   .about-sys-badge-warn {
660     font-size: 0.65rem;
661     padding: 0.15rem 0.4rem;
662   }
663 }
664
665 /* ── 34. Универсальные панели с адаптивным заголовком панели
   ──────────────────── */
666
667 @media (max-width: 575px) {
668   .ui-panel-header {
669     font-size: 0.75rem;
```

```
    padding: 0.4rem 0.6rem;
  }
}

/* ── 35. Универсальные панели с адаптивным значением свойства ──────────────── */

@media (max-width: 575px) {
  .ui-prop-val {
    font-size: 0.75rem;
  }
}

/* ── 36. Универсальные панели с адаптивным ключом свойства ──────────────── */

@media (max-width: 575px) {
  .ui-prop-key {
    font-size: 0.75rem;
  }
}

/* ── 37. Универсальные панели с адаптивным заголовком дерева ──────────────── */

@media (max-width: 575px) {
  .ui-tree-title {
    font-size: 0.75rem;
    padding: 4px 8px;
  }
}

/* ── 38. Универсальные панели с адаптивным значением прогресс-бара ────────── */

@media (max-width: 575px) {
  .ui-progress-track {
    height: 6px;
```

```
  }
}

/* ── 39. Универсальные панели с адаптивным значением карточки ──────────────── */

@media (max-width: 575px) {
  .ui-kpi-card {
    min-height: 60px;
    padding: 0.5rem 0.7rem;
  }
}

/* ── 40. Универсальные панели с адаптивным значением заголовка карточки ─────── */

@media (max-width: 575px) {
  .ui-kpi-label {
    font-size: 0.65rem;
  }
}

/* ── 41. Универсальные панели с адаптивным значением подзаголовка карточки ── */

@media (max-width: 575px) {
  .ui-kpi-sub {
    font-size: 0.6rem;
  }
}

/* ── 42. Универсальные панели с адаптивным значением заголовка панели ──────── */

@media (max-width: 575px) {
  .ui-panel-header {
    font-size: 0.7rem;
    padding: 0.3rem 0.5rem;
```

```
  }
}

/* ── 43. Универсальные панели с адаптивным значением таблицы ──────────────── */

@media (max-width: 575px) {
  .ui-spec-table {
    font-size: 0.75rem;
  }
}

/* ── 44. Универсальные панели с адаптивным значением бейджа ──────────────── */

@media (max-width: 575px) {
  .ui-badge {
    font-size: 0.6rem;
    padding: 0.1rem 0.3rem;
  }
}

/* ── 45. Универсальные панели с адаптивным значением заголовка ──────────────── */

@media (max-width: 575px) {
  .ui-module-title {
    font-size: 0.75rem;
  }
}

/* ── 46. Универсальные панели с адаптивным значением заголовка дерева ──────── */

@media (max-width: 575px) {
  .ui-tree-title {
    font-size: 0.7rem;
  }
```

```
}

/* ── 47. Универсальные панели с адаптивным значением значения ──────────────── */

@media (max-width: 575px) {
  .ui-value {
    font-size: 1rem;
  }
}

/* ── 48. Универсальные панели с адаптивным значением ключа ──────────────── */

@media (max-width: 575px) {
  .ui-key {
    font-size: 0.7rem;
  }
}

/* ── 49. Универсальные панели с адаптивным значением заголовка панели ──────── */

@media (max-width: 575px) {
  .ui-header {
    font-size: 0.7rem;
  }
}

/* ── 50. Универсальные панели с адаптивным значением заголовка карточки ────── */

@media (max-width: 575px) {
  .ui-card-header {
    font-size: 0.7rem;
  }
}

```

main.css

```
/**
 * ============================================================
 * Process Name: AI-Windows-Admin Web Styles - Main Stylesheet Manifest
 * ============================================================
 * Description:
 *   Главный манифест стилей веб-интерфейса AI-Windows-Admin.
 *
 *   Зачем нужен этот файл:
 *     1. Модульная сборка стилей: агрегирует все слои оформления интерфейса
 *        (дизайн-токены/переменные, базовые стили, навигация, компоненты, чат, оверрайды).
 *     2. Поддержка динамических тем оформления (светлая/тёмная/кирпичная/терминал) через CSS-переменные.
 *     3. Cache-busting версионирование: контролирует параметры ?v= версий дочерних файлов стилей.
 *
 * Usage Examples:
 *   HTML Link Tag:
 *     <link rel="stylesheet" href="/src/api/webgui/css/main.css?v=20261006_v5">
 *
 * File: main.css
 * Project: ai-windows-admin
 * Package: src/api/webgui/css
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/* 1. Дизайн-токены и переменные тем (светлая NightView/кирпич
```

```
ная/тёмная/терминал) */
@import "variables.css?v=20261008_v1";

/* 2. Базовые стили страницы, скроллбары и контейнеры вкладок */
@import "base.css?v=20261008_v1";

/* 3. Навигационная панель, вкладки, выпадающие меню и переключатели */
@import "nav.css?v=20261008_v1";

/* 4. Общие UI-компоненты (карточки, формы, кнопки, модалки, таблицы, бейджи, чипы) */
@import "components.css?v=20261008_v2";

/* 5. Чат, сообщения, сайдбар истории диалогов и дебаггер */
@import "chat.css?v=20261008_v1";

/* 6. Управление торрентами, поиск релизов, карточки трекеров */
@import "torrents.css?v=20261008_v1";

/* 7. Адаптивные переопределения и совместимость тем с Bootstrap */
@import "overrides.css?v=20261008_v1";

/* 8. Глобальные переопределения для минимизации дублирования CSS */
@import "global-overrides.css?v=20261008_v1";

```

nav.css

```
/**
 * ============================================================
 * Process Name: Windows Web Styles - Nav Stylesheet
 * ============================================================
 * Description:
 *   Таблица стилей стиля и оформления интерфейса (nav).
 *
 * Usage Examples:
 *   HTML Link Tag:
 *     <link rel="stylesheet" href="/windows/api/webgui/css/nav.css?v=20261001_v1">
 *
 * File: nav.css
 * Project: ai-windows-admin
 * Package: windows/api/webgui/css
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/* ═══════════════════════════════════════════════════════════
   NAVIGATION — AI Windows Admin WebGUI
   Главная панель навигации, вкладки, выпадающие меню и элементы управления
   ═══════════════════════════════════════════════════════════ */

/* ── Главная панель навигации ──────────────────────────── */
.main-nav-container {
  background-color: var(--nav-bg);
```

```
  border-bottom: 1px solid var(--border-color);
  min-height: 34px;
  position: relative;
  z-index: 1020;
  overflow: visible !important;
  box-shadow: var(--shadow-xs);
}

.nav-tabs {
  border-bottom: none;
  background-color: transparent;
  flex-wrap: nowrap;
  overflow: visible !important;
}

.nav-tabs .nav-item {
  flex-shrink: 0;
  position: relative;
}

.nav-tabs .nav-item.dropdown {
  position: relative;
}

/* Nav-links — компактные */
.nav-tabs .nav-link {
  color: var(--nav-text);
  border: none;
  border-bottom: 2px solid transparent;
  background: transparent;
  padding: 5px 10px;
  font-size: 0.82rem;
  font-weight: 500;
  white-space: nowrap;
  transition: color 0.15s, border-color 0.15s, background-colo
r 0.15s;
  border-radius: 4px 4px 0 0;
}
```

```
66
67 .nav-tabs .nav-link:hover {
68   color: var(--nav-active);
69   background-color: var(--nav-hover-bg);
70   border-bottom-color: transparent;
71 }
72
73 .nav-tabs .nav-link.active {
74   color: var(--nav-active);
75   background: transparent;
76   border-bottom-color: var(--nav-active);
77 }
78
79 /* ── Выпадающие меню навбара ──────────────────────────────
   ──────────────── */
80 .nav-tabs .dropdown-menu {
81   position: absolute !important;
82   top: 100% !important;
83   left: 0 !important;
84   margin-top: 3px !important;
85   z-index: 9999 !important;
86   min-width: 180px;
87   display: none !important;
88   padding: 4px;
89   border-radius: 8px;
90   background-color: var(--surface-1);
91   border: 1px solid var(--border-color);
92   box-shadow: var(--shadow-md) !important;
93 }
94
95 .nav-tabs .dropdown-menu.show {
96   display: block !important;
97   visibility: visible !important;
98   opacity: 1 !important;
99 }
100
101 .nav-tabs .dropdown-item {
102   width: 100%;
```

```
  border: none;
  background: transparent;
  text-align: left;
  padding: 5px 10px;
  border-radius: 5px;
  font-size: 0.82rem;
  font-weight: 500;
  cursor: pointer;
  color: var(--text-color);
  transition: background-color 0.12s ease, color 0.12s ease;
}

.nav-tabs .dropdown-item:hover,
.nav-tabs .dropdown-item:focus {
  background-color: var(--nav-hover-bg);
  color: var(--nav-active);
}

.nav-tabs .dropdown-item.active {
  background-color: rgba(37, 99, 235, 0.1);
  color: var(--nav-active);
  font-weight: 600;
}

[data-theme="dark"] .nav-tabs .dropdown-item.active {
  background-color: rgba(88, 166, 255, 0.12);
  color: var(--nav-active);
}

/* ── Разделители и метки групп ──────────────────────────────── */
.nav-group-divider {
  width: 1px;
  height: 16px;
  background-color: var(--divider-color);
  margin: 0 6px;
  opacity: 0.7;
  align-self: center;
```

```
  flex-shrink: 0;
}

.nav-group-label {
  display: inline-flex;
  align-items: center;
  font-size: 0.68rem;
  font-weight: 600;
  text-transform: uppercase;
  letter-spacing: 0.5px;
  color: var(--text-muted);
  padding: 2px 6px;
  background-color: var(--surface-2);
  border: 1px solid var(--border-color);
  border-radius: 5px;
  user-select: none;
}

/* ── Выпадающие меню Bootstrap (глобально) ──────────────────────── */
.dropdown-menu {
  background-color: var(--surface-1);
  border-color: var(--border-color);
  box-shadow: var(--shadow-md);
}

.dropdown-item {
  color: var(--text-color);
  border-radius: 4px;
  font-size: 0.82rem;
}

.dropdown-item:hover,
.dropdown-item:focus {
  background-color: var(--nav-hover-bg);
  color: var(--nav-active);
}

```

```
.dropdown-item.active {
  background-color: var(--nav-active);
  color: #ffffff;
}

[data-theme="terminal"] .dropdown-menu {
  background-color: #0b120c;
  border: 1px solid #1a4223;
  box-shadow: 0 0 12px rgba(0, 255, 102, 0.25);
}

[data-theme="terminal"] .dropdown-item.active {
  background-color: #0c3316;
  color: #39ff14;
  border-left: 3px solid #00ff66;
}

/* ── Кнопки и контролы навбара ──────────────────────────── */
.main-nav-container .lang-selector {
  background-color: var(--nav-bg);
  border-color: var(--border-color);
  color: var(--nav-text);
  font-size: 0.8rem;
}

#nav-google-btn {
  background: var(--surface-1);
  color: var(--text-color) !important;
  font-size: 0.78rem;
  font-weight: 600;
  border-radius: 20px;
  border: 1px solid var(--border-color);
  padding: 3px 10px;
  transition: all 0.18s ease;
}

#nav-google-btn:hover {
```

```
  background: var(--nav-hover-bg);
  box-shadow: var(--shadow-sm);
  transform: translateY(-1px);
}

[data-theme="dark"] #nav-google-btn {
  background: var(--surface-2);
  color: var(--text-color) !important;
  border-color: var(--border-color);
}

#user-settings-btn {
  font-size: 0.8rem;
  font-weight: 500;
  border-radius: 20px;
  color: var(--nav-text);
  border-color: var(--border-color);
  transition: all 0.18s ease;
  padding: 3px 10px;
}

#user-settings-btn:hover {
  background-color: var(--nav-hover-bg);
  color: var(--nav-active);
  border-color: var(--nav-active);
}

#user-settings-gear-btn,
#theme-toggle-btn {
  color: var(--nav-text);
  border-color: var(--border-color);
  transition: all 0.18s ease;
  width: 28px;
  height: 28px;
  padding: 0;
  display: inline-flex;
  align-items: center;
  justify-content: center;
```

```
  border-radius: 50%;
  font-size: 0.85rem;
}

#user-settings-gear-btn:hover {
  background-color: var(--nav-hover-bg);
  color: var(--nav-active);
  border-color: var(--nav-active);
  transform: rotate(30deg);
}

#theme-toggle-btn:hover {
  background-color: var(--nav-hover-bg);
  color: var(--nav-active);
  border-color: var(--nav-active);
}

/* ── Переключатель темы (Dropdown / иконки) ──────────────────────── */
#theme-dropdown-container .dropdown-menu {
  min-width: 140px;
  padding: 3px;
}

.theme-check-icon {
  font-size: 0.85rem;
}

.theme-check-icon.d-none {
  display: none !important;
}

/* ── Выдвижной боковой ярлычок (Левая вкладка-триггер) ──────────── */
.left-drawer-tab-trigger {
  position: fixed;
  top: 48%;
  left: 0;
```

```
  transform: translateY(-50%);
  z-index: 1040;
  background-color: var(--surface-1);
  border: 1px solid var(--border-color);
  border-left: none;
  border-radius: 0 6px 6px 0;
  padding: 6px 3px;
  display: flex;
  flex-direction: column;
  align-items: center;
  gap: 4px;
  cursor: pointer;
  color: var(--nav-text);
  box-shadow: var(--shadow-sm);
  transition: all 0.2s ease;
  user-select: none;
  opacity: 0.65;
}

.left-drawer-tab-trigger:hover {
  background-color: var(--nav-hover-bg);
  color: var(--nav-active);
  padding-right: 6px;
  box-shadow: var(--shadow-md);
  opacity: 1;
}

.left-drawer-tab-text {
  writing-mode: vertical-rl;
  text-orientation: mixed;
  font-size: 0.62rem;
  font-weight: 700;
  letter-spacing: 1.5px;
  text-transform: uppercase;
}

/* ── Левая выдвижная панель навигации (Offcanvas Drawer) ──────────────────── */
```

```
.left-drawer-nav {
  width: 320px !important;
  background-color: var(--surface-1);
  color: var(--text-color);
  border-right: 1px solid var(--border-color);
  box-shadow: var(--shadow-lg);
  z-index: 1060;
}

.left-drawer-nav .offcanvas-header {
  background-color: var(--surface-2);
  border-bottom: 1px solid var(--border-color);
}

.nav-drawer-group-title {
  font-size: 0.72rem;
  font-weight: 700;
  text-transform: uppercase;
  letter-spacing: 0.8px;
  color: var(--text-muted);
  padding: 0 4px;
}

.left-drawer-nav .list-group-item {
  background-color: var(--surface-1);
  border-color: var(--border-color);
  color: var(--text-color);
  font-size: 0.85rem;
  transition: background-color 0.15s ease, color 0.15s ease, t
ransform 0.1s ease;
  cursor: pointer;
  border-radius: 6px;
  margin-bottom: 3px;
  border: 1px solid transparent;
}

.left-drawer-nav .list-group-item:hover {
  background-color: var(--nav-hover-bg);
```

```
  color: var(--nav-active);
  transform: translateX(2px);
}

.left-drawer-nav .list-group-item.active {
  background-color: var(--nav-active);
  border-color: var(--nav-active);
  color: #ffffff;
  font-weight: 600;
}

[data-theme="brick"] .left-drawer-nav .list-group-item.active
{
  background-color: #b84228;
  border-color: #b84228;
  color: #ffffff;
}

[data-theme="brick"] .left-drawer-nav .list-group-item.active:
hover {
  background-color: #9e341d;
}

[data-theme="dark"] .left-drawer-nav .list-group-item.active {
  background-color: #1f6feb;
  border-color: #1f6feb;
  color: #ffffff;
}

[data-theme="dark"] .left-drawer-nav .list-group-item.active:h
over {
  background-color: #388bfd;
}

[data-theme="terminal"] .left-drawer-nav .list-group-item.acti
ve {
  background-color: #0c3316;
  border-color: #00ff66;
```

```
  color: #39ff14;
  box-shadow: 0 0 8px rgba(0, 255, 102, 0.35);
  text-shadow: 0 0 3px rgba(0, 255, 102, 0.5);
}

[data-theme="terminal"] .left-drawer-nav .list-group-item.acti
ve:hover {
  background-color: #154c22;
  border-color: #39ff14;
  color: #ffffff;
}
```

overrides.css

```
/**
 * ============================================================
==================
 * Process Name: Windows Web Styles - Overrides Stylesheet
 * ============================================================
==================
 * Description:
 *   Таблица стилей стиля и оформления интерфейса (overrides).
 *   Адаптивные переопределения Bootstrap и специфичных компон
ентов для тем оформления.
 *
 * Usage Examples:
 *   HTML Link Tag:
 *     <link rel="stylesheet" href="/windows/api/webgui/css/ov
errides.css?v=20261004_v17">
 *
 * File: overrides.css
 * Project: ai-windows-admin
 * Package: windows/api/webgui/css
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/* ═══════════════════════════════════════════════════════════
═══════════════
   THEME OVERRIDES & COMPATIBILITY — AI Windows Admin WebGUI
   Адаптивные переопределения Bootstrap и компонентов для свет
лых и тёмных тем
   ═══════════════════════════════════════════════════════════
═══════════════ */

/* ── Light & Brick Themes — Переопределения Bootstrap и утили
т ──────────────── */
```

```
[data-theme="light"],
[data-theme="brick"] {
  /* Инпуты, к которым JS или Bootstrap добавил bg-dark */
  input.bg-dark,
  textarea.bg-dark,
  select.bg-dark,
  .form-control.bg-dark,
  .form-select.bg-dark {
    background-color: var(--surface-1) !important;
    color: var(--text-color) !important;
    border-color: var(--border-color) !important;
  }

  /* div/section с bg-dark (не кнопки и не бейджи) */
  div.bg-dark,
  section.bg-dark,
  aside.bg-dark,
  article.bg-dark {
    background-color: var(--surface-1) !important;
    color: var(--text-color) !important;
  }

  /* Bootstrap bg-dark / bg-black */
  .bg-dark:not(.btn):not(.badge):not(.navbar) {
    background-color: var(--surface-1) !important;
    color: var(--text-color) !important;
    border-color: var(--border-color) !important;
  }

  .bg-black:not(.btn) {
    background-color: var(--bg-color) !important;
    color: var(--text-color) !important;
  }

  /* Bootstrap bg-white заменяется на комфортную поверхность т
емы NightView */
  .bg-white:not(.btn):not(.badge) {
    background-color: var(--surface-1) !important;
```

```
  }

  /* Высококонтрастные читаемые цвета текста в светлой теме NightView / Brick */
  .text-info {
    color: #0369a1 !important; /* глубокий читаемый океанский синий */
  }
  .text-warning {
    color: #92400e !important; /* глубокий читаемый янтарно-коричневый */
  }
  .text-success {
    color: #166534 !important; /* глубокий читаемый лесной зеленый */
  }
  .text-danger {
    color: #991b1b !important; /* глубокий читаемый рубиновый */
  }
  .text-primary {
    color: #1d4ed8 !important; /* четкий читаемый сапфировый */
  }

  /* Кнопки с контуром (Outline Buttons) высокой контрастности */
  .btn-outline-light {
    color: var(--text-color) !important;
    border-color: var(--border-strong) !important;
    background-color: transparent !important;
  }
  .btn-outline-light:hover,
  .btn-outline-light:focus,
  .btn-outline-light.active,
  .btn-check:checked + .btn-outline-light,
  .btn-check:active + .btn-outline-light {
    background-color: var(--surface-2) !important;
```

```
    color: var(--text-color) !important;
    border-color: var(--border-contrast) !important;
  }

  .btn-outline-secondary {
    color: var(--text-color) !important;
    border-color: var(--border-strong) !important;
    background-color: transparent !important;
  }
  .btn-outline-secondary:hover,
  .btn-outline-secondary:focus,
  .btn-outline-secondary.active,
  .btn-check:checked + .btn-outline-secondary,
  .btn-check:active + .btn-outline-secondary {
    background-color: #64748b !important;
    color: #ffffff !important;
    border-color: #64748b !important;
  }

  .btn-outline-info {
    color: #0369a1 !important;
    border-color: #0284c7 !important;
    background-color: transparent !important;
  }
  .btn-outline-info:hover,
  .btn-outline-info:focus,
  .btn-outline-info.active,
  .btn-check:checked + .btn-outline-info,
  .btn-check:active + .btn-outline-info {
    background-color: #0284c7 !important;
    color: #ffffff !important;
    border-color: #0284c7 !important;
  }

  .btn-outline-warning {
    color: #92400e !important;
    border-color: #d97706 !important;
    background-color: transparent !important;
```

```
}
.btn-outline-warning:hover,
.btn-outline-warning:focus,
.btn-outline-warning.active,
.btn-check:checked + .btn-outline-warning,
.btn-check:active + .btn-outline-warning {
  background-color: #d97706 !important;
  color: #ffffff !important;
  border-color: #d97706 !important;
}

.btn-outline-primary {
  color: #1d4ed8 !important;
  border-color: #2563eb !important;
  background-color: transparent !important;
}
.btn-outline-primary:hover,
.btn-outline-primary:focus,
.btn-outline-primary.active,
.btn-check:checked + .btn-outline-primary,
.btn-check:active + .btn-outline-primary {
  background-color: #2563eb !important;
  color: #ffffff !important;
  border-color: #2563eb !important;
}

.btn-outline-danger {
  color: #991b1b !important;
  border-color: #dc2626 !important;
  background-color: transparent !important;
}
.btn-outline-danger:hover,
.btn-outline-danger:focus,
.btn-outline-danger.active,
.btn-check:checked + .btn-outline-danger,
.btn-check:active + .btn-outline-danger {
  background-color: #dc2626 !important;
  color: #ffffff !important;
```

```
    border-color: #dc2626 !important;
  }

  /* text-white / text-light в карточках и панелях */
  .card .text-white,
  .card .text-light,
  [class*="-panel"] .text-white,
  [class*="-panel"] .text-light,
  [class*="-card"] .text-white,
  [class*="-card"] .text-light,
  .tab-pane .text-white,
  .tab-pane .text-light,
  .tab-pane .text-white:not(.btn):not(.badge),
  .card .text-white:not(.btn):not(.badge),
  .sys-container .text-white:not(.btn):not(.badge),
  .sys-container .text-light:not(.btn):not(.badge) {
    color: var(--text-color) !important;
  }

  /* Бейджи bg-dark и статусных оттенков */
  .badge.bg-dark {
    background-color: var(--surface-2) !important;
    border-color: var(--border-color) !important;
    color: var(--text-color) !important;
  }
  .badge.bg-info-subtle {
    background-color: #e0f2fe !important;
    color: #0369a1 !important;
    border-color: #7dd3fc !important;
  }
  .badge.bg-warning-subtle {
    background-color: #fef3c7 !important;
    color: #92400e !important;
    border-color: #fde047 !important;
  }
  .badge.bg-success-subtle {
    background-color: #dcfce7 !important;
    color: #166534 !important;
```

```
    border-color: #86efac !important;
  }
  .badge.bg-danger-subtle {
    background-color: #fee2e2 !important;
    color: #991b1b !important;
    border-color: #fca5a5 !important;
  }

  /* pre/code без явного светлого фона */
  pre:not(.prompt-dump-details pre) {
    background-color: var(--surface-2) !important;
    color: var(--text-color) !important;
    border: 1px solid var(--border-color);
    border-radius: 5px;
  }

  code.bg-dark {
    background-color: var(--surface-2) !important;
    color: var(--text-color) !important;
    border: 1px solid var(--border-color);
  }

  /* Прогресс-бар: трек */
  .progress:not([style]),
  .progress {
    background-color: var(--border-color) !important;
  }

  /* ── Системный инспектор, трейдинг, сеть и дерево компонентов ── */
  .sys-card,
  .sys-panel,
  .sys-panel-header,
  .sys-tree-node,
  .sys-tree-body,
  .sys-proc-table th {
    background-color: var(--surface-1) !important;
    border-color: var(--border-color) !important;
```

```
    color: var(--text-color) !important;
  }

  .sys-card-header-bar {
    background-color: var(--surface-2) !important;
    border-color: var(--border-color) !important;
    color: var(--text-color) !important;
  }

  .sys-core-section-title {
    color: var(--text-color) !important;
    border-color: var(--border-subtle) !important;
  }

  .sys-core-card {
    background-color: var(--surface-2) !important;
    border-color: var(--border-color) !important;
    color: var(--text-color) !important;
    box-shadow: var(--shadow-xs) !important;
  }

  .sys-core-title {
    color: var(--text-color) !important;
  }

  .sys-core-param-key {
    color: var(--text-muted) !important;
  }

  .sys-core-param-row {
    border-color: var(--border-subtle) !important;
  }

  .sys-sensor-widget-item {
    background-color: var(--surface-2) !important;
    border-color: var(--border-color) !important;
    color: var(--text-color) !important;
  }
```

```
285
286   .sys-sensor-name,
287   .sys-sensor-value {
288     color: var(--text-color) !important;
289   }
290
291   .sys-temp-slider-track {
292     background-color: var(--border-color) !important;
293   }
294
295   .sys-temp-slider-thumb {
296     border-color: var(--surface-1) !important;
297   }
298
299   .sys-cpu-specs-card,
300   .sys-gpu-specs-card {
301     background-color: var(--surface-1) !important;
302     border-color: var(--border-color) !important;
303     color: var(--text-color) !important;
304   }
305
306   .sys-cpu-details-accordion-btn,
307   .sys-gpu-details-accordion-btn {
308     background: var(--surface-2) !important;
309     border-color: var(--border-color) !important;
310     color: var(--text-color) !important;
311   }
312
313   /* Переопределение жестко закодированных темных плашек в све
    тлой теме */
314   div[style*="background: rgba(0, 0, 0"],
315   div[style*="background:rgba(0,0,0"],
316   div[style*="background: linear-gradient(180deg, #182238"],
317   div[style*="background:linear-gradient(180deg,#182238"] {
318     background: var(--surface-2) !important;
319     border-color: var(--border-color) !important;
320     color: var(--text-color) !important;
321   }
```

```
322
323  .sys-panel-header {
324    background-color: var(--surface-2) !important;
325    color: var(--nav-active) !important;
326  }
327
328  .sys-tree-title {
329    color: var(--text-color) !important;
330    background: transparent !important;
331  }
332
333  .sys-tree-body {
334    background-color: var(--bg-color) !important;
335  }
336
337  .sys-proc-table th {
338    background-color: var(--surface-2) !important;
339    color: var(--text-muted) !important;
340  }
341
342  .sys-proc-table td {
343    color: var(--text-color) !important;
344  }
345
346  .sys-value {
347    color: var(--text-color) !important;
348  }
349
350  /* ── Карточки агентов, моделей, плагинов ── */
351  .agent-card,
352  .model-card,
353  .plugin-card,
354  [class$="-card"]:not(.btn):not(.badge) {
355    background-color: var(--surface-1) !important;
356    border-color: var(--border-color) !important;
357    color: var(--text-color) !important;
358  }
359
```

```
/* ── card-header с тёмным фоном ── */
.card-header.bg-dark,
.card-header[style*="background"],
.card-header[style*="#0"],
.card-header[style*="#1"] {
  background-color: var(--surface-2) !important;
  color: var(--text-color) !important;
  border-color: var(--border-color) !important;
}

/* ── Логи ── */
#media-log,
[id$="-log"]:not(canvas),
.log-output {
  background-color: var(--surface-2) !important;
  color: var(--text-color) !important;
  border-color: var(--border-color) !important;
}

/* ── Stripe-полосы в таблицах ── */
.table-striped > tbody > tr:nth-of-type(odd) {
  --bs-table-bg-type: var(--surface-2);
}

/* ── Таблицы с классами .table-dark в светлых темах ── */
.table-dark,
.table-dark > thead,
.table-dark > tbody,
.table-dark > tfoot,
.table-dark tr,
.table-dark th,
.table-dark td {
  --bs-table-bg: var(--surface-1) !important;
  --bs-table-striped-bg: var(--surface-2) !important;
  --bs-table-hover-bg: rgba(56, 189, 248, 0.08) !important;
  --bs-table-color: var(--text-color) !important;
  --bs-table-border-color: var(--border-color) !important;
  background-color: var(--surface-1) !important;
```

```
    color: var(--text-color) !important;
    border-color: var(--border-color) !important;
  }

  .table-dark thead,
  .table-dark thead th,
  thead.table-dark,
  thead.table-dark th {
    background-color: var(--surface-2) !important;
    color: var(--text-muted) !important;
    border-color: var(--border-color) !important;
  }

  /* ── border-secondary ── */
  .border-secondary {
    border-color: var(--border-color) !important;
  }
}

/* ── Dark & Terminal Themes — Высококонтрастные переопределения ──────────────── */
[data-theme="dark"],
[data-theme="terminal"] {
  .card-header.bg-light,
  .card-footer.bg-light,
  .modal-header.bg-light,
  .modal-footer.bg-light {
    background-color: var(--surface-2) !important;
    color: var(--text-color) !important;
    border-color: var(--border-color) !important;
  }

  .bg-light:not(.btn):not(.badge) {
    background-color: var(--surface-1) !important;
    color: var(--text-color) !important;
  }

  .text-dark {
```

```
    color: var(--text-color) !important;
  }

  .border-light {
    border-color: var(--border-color) !important;
  }

  /* Логи */
  #media-log,
  [id$="-log"]:not(canvas),
  .log-output {
    background-color: var(--surface-1) !important;
    color: var(--text-color) !important;
    border: 1px solid var(--border-color) !important;
  }
}

/* ── Специфика темы «Терминал» (Terminal / CRT Glow) ──────────────────── */
[data-theme="terminal"] {
  .card,
  .modal-content,
  .offcanvas {
    border-color: var(--border-color);
    box-shadow: 0 0 10px rgba(0, 255, 102, 0.12);
  }

  .card-header,
  .modal-header {
    border-bottom: 1px solid var(--border-strong);
    color: var(--nav-active);
  }

  pre, code, kbd, samp {
    color: #39ff14 !important;
    background-color: #060a06 !important;
    border: 1px solid #1a4223;
  }
```

```
472
473   .form-control,
474   .form-select {
475     border-color: var(--border-color);
476     color: var(--text-color);
477     background-color: #0b120c;
478   }
479
480   .form-control:focus,
481   .form-select:focus {
482     border-color: #00ff66;
483     box-shadow: 0 0 8px rgba(0, 255, 102, 0.4);
484     background-color: #0b120c;
485     color: #ffffff;
486   }
487
488   .table th {
489     color: #39ff14 !important;
490     border-bottom: 2px solid #00ff66 !important;
491   }
492 }
493
```

torrents.css

```
/**
 * ============================================================
 * Process Name: Windows Web Styles - Torrents Stylesheet
 * ============================================================
 * Description:
 *   Таблица стилей стиля и оформления интерфейса (torrents).
 *
 * Usage Examples:
 *   HTML Link Tag:
 *     <link rel="stylesheet" href="/windows/api/webgui/css/torrents.css?v=20261001_v1">
 *
 * File: torrents.css
 * Project: ai-windows-admin
 * Package: windows/api/webgui/css
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/* ═══════════════════════════════════════════════════════════
   TORRENTS & MEDIA — AI Windows Admin WebGUI
   Стили управления торрентами, поиска релизов и логов медиатеки
   ═══════════════════════════════════════════════════════════ */

/* ── Таблица и статусы торрентов ──────────────────────────── */
#torrents-table th[data-col] {
  cursor: pointer;
```

```
  user-select: none;
}

.state-missingFiles { color: #dc3545; font-weight: 600; }
.state-uploading,
.state-seeding      { color: #16a34a; }
.state-downloading  { color: #2563eb; }
.state-checkingUP,
.state-checkingDL   { color: #ea580c; }

.progress           { height: 5px; }
.path-cell          { font-size: 0.78rem; color: var(--text-mu
ted); }
.path-edit          { display: none; }
.path-edit.active   { display: flex; }

/* ── Панель отсутствующих файлов ──────────────────────────
──────────────── */
#missing-panel        { border-left: 2px solid #dc3545; }
.missing-row          { cursor: pointer; }
.missing-row:hover    { background: rgba(255, 200, 0, 0.08); }
.missing-row.selected { background: rgba(37, 99, 235, 0.1); }

/* ── Лог медиатеки ────────────────────────────────────────
──────────────── */
#media-log {
  font-size: 0.78rem;
  font-family: 'Consolas', monospace;
  background: #0d1117;
  color: #d4d4d4;
  border-radius: 5px;
  padding: 8px;
  height: 160px;
  overflow-y: auto;
  white-space: pre-wrap;
}

.dup-badge       { font-size: 0.72rem; }
```

```
.integrity-ok    { color: #16a34a; }
.integrity-warn  { color: #dc3545; }

/* ═══════════════════════════════════════════════════════════════
   ТОРРЕНТ-ПОИСК
   ═══════════════════════════════════════════════════════════════ */
.torrent-card {
  border: 1px solid var(--border-color);
  border-radius: 6px;
  background-color: var(--surface-1);
  padding: 10px 12px;
  margin-bottom: 8px;
  box-shadow: var(--shadow-xs);
  transition: transform 0.15s, box-shadow 0.15s;
  color: var(--card-text);
}

.torrent-card:hover {
  transform: translateY(-1px);
  box-shadow: var(--shadow-sm);
}

.torrent-badge {
  display: inline-block;
  padding: 0.2em 0.5em;
  font-size: 0.72rem;
  font-weight: 600;
  border-radius: 50rem;
  margin-right: 3px;
  margin-bottom: 3px;
}

.badge-rutracker { background-color: #6e7a8a; color: #fff; }
.badge-nnmclub   { background-color: #16a34a; color: #fff; }
.badge-seeds     { background-color: #22c55e; color: #fff; }
.badge-peers     { background-color: #6b7280; color: #fff; }
```

```
.badge-size      { background-color: #2563eb; color: #fff; }

.torrent-title {
  font-size: 0.88rem;
  font-weight: 600;
  margin-bottom: 4px;
  color: var(--text-color);
}

.torrent-title a {
  color: var(--nav-active);
  text-decoration: none;
}

.torrent-title a:hover {
  text-decoration: underline;
}

.torrent-desc {
  font-size: 0.8rem;
  color: var(--text-muted);
  margin-bottom: 6px;
}

```

variables.css

```
/**
 * ============================================================
 * Process Name: Windows Web Styles - Variables Stylesheet
 * ============================================================
 * Description:
 *   Таблица стилей стиля и оформления интерфейса (variables).
 *   Включает высококонтрастные токены для светлых тем (Light, Brick) и тёмных тем (Dark, Terminal).
 *
 * Usage Examples:
 *   HTML Link Tag:
 *     <link rel="stylesheet" href="/windows/api/webgui/css/variables.css?v=20261004_v17">
 *
 * File: variables.css
 * Project: ai-windows-admin
 * Package: windows/api/webgui/css
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/* ═══════════════════════════════════════════════════════════
   THEME VARIABLES — AI Windows Admin WebGUI
   Дизайн-токены и CSS-переменные для всех тем оформления
   ═══════════════════════════════════════════════════════════ */

/* ── 1. Светлая тема NightView (Комфортное чтение ночью / Warm Paper) ──────── */
:root,
```

```
[data-theme="light"] {
  /* Фоны (мягкие теплые бежево-серые поверхности в стиле NightView / бумага) */
  --bg-color:        #fbfbfb;   /* чистый светлый фон страницы */
  --surface-1:       #f6f3ec;   /* карточки, панели, мягкая бумажная поверхность */
  --surface-2:       #eee9e0;   /* хедеры карточек, тулбары, разделители */
  --card-bg:         #f6f3ec;
  --card-text:       #18181b;   /* четкий глубокий антрацитовый текст максимальной читаемости */
  --nav-bg:          #f6f3ec;

  /* Текст высокой четкости и контрастности для комфортного ночного чтения */
  --text-color:      #18181b;   /* глубокий четкий темный текст */
  --text-muted:      #3f3f46;   /* темный нейтральный slate для высокой читаемости подписей и параметров */
  --nav-text:        #27272a;   /* контрастный текст навигации */
  --nav-active:      #1d4ed8;   /* четкий сапфирово-синий акцент */
  --nav-hover-bg:    #eae4d8;

  /* Линии и границы */
  --border-color:    #cfc6b8;   /* четкий теплый контур */
  --border-strong:   #9ca3af;   /* выразительные активные границы */
  --border-subtle:   #ded7cb;   /* деликатные внутренние разделители */
  --border-contrast: #71717a;   /* контрастные рамки */
  --divider-color:   #cfc6b8;
  --modal-border:    #cfc6b8;

  /* Тени (мягкие теплые) */
  --shadow-xs:  0 1px 2px rgba(50, 40, 30, 0.05);
```

```
  --shadow-sm:  0 1px 4px rgba(50, 40, 30, 0.08);
  --shadow-md:  0 4px 12px rgba(50, 40, 30, 0.10);
  --shadow-lg:  0 8px 24px rgba(50, 40, 30, 0.13);

  /* Чат */
  --chat-user-bg:    #2a6194;
  --chat-user-text:  #fbf9f4;
  --chat-bot-bg:     #eae4d8;
  --chat-bot-text:   #2b2723;

  /* Скроллбар */
  --scrollbar-track: #ebe7df;
  --scrollbar-thumb: #c6bcad;

  /* Модальные окна */
  --modal-bg:        #f6f3ec;
  --modal-text:      #2b2723;

  /* Алерты — светлая тема NightView (пастельные комфортные то
на) */
  --alert-success-bg:     #e7f2e9;
  --alert-success-border: #98c9a0;
  --alert-success-text:   #185223;
  --alert-danger-bg:      #f6e6e6;
  --alert-danger-border:  #e19f9d;
  --alert-danger-text:    #7e1f1f;
  --alert-warning-bg:     #f8f0dd;
  --alert-warning-border: #dfc279;
  --alert-warning-text:   #704705;
  --alert-info-bg:        #e5eef7;
  --alert-info-border:    #97bcdb;
  --alert-info-text:      #184674;

  /* Геометрия и радиусы */
  --radius-xs: 4px;
  --radius-sm: 6px;
  --radius-md: 8px;
  --radius-lg: 12px;
```

```
92
93    /* Типографика */
94    --font-sans: system-ui, -apple-system, "Segoe UI", Roboto, "
      Helvetica Neue", Arial, sans-serif;
95    --font-mono: ui-monospace, SFMono-Regular, Menlo, Monaco, Co
      nsolas, "Courier New", monospace;
96
97    /* Каскадные токены KPI-карточек */
98    --kpi-bg:         var(--surface-1);
99    --kpi-border:     var(--border-color);
100   --kpi-accent:     var(--nav-active);
101   --kpi-min-height: 72px;
102   --kpi-padding:    0.65rem 0.9rem;
103   --kpi-val-size:   1.25rem;
104   --kpi-label-size: 0.72rem;
105   --kpi-sub-size:   0.72rem;
106
107   /* Каскадные токены панелей */
108   --panel-bg:        var(--surface-1);
109   --panel-border:    var(--border-color);
110   --panel-header-bg: var(--surface-2);
111
112   /* Каскадные токены суб-навигации */
113   --subnav-btn-padding: 0.35rem 0.75rem;
114   --subnav-font-size:   0.82rem;
115
116   /* Статусные цвета */
117   --status-success-bg:     #dbeee0;
118   --status-success-text:   #185223;
119   --status-success-border: #98c9a0;
120
121   --status-danger-bg:      #f7dede;
122   --status-danger-text:    #7e1f1f;
123   --status-danger-border:  #e19f9d;
124
125   --status-warning-bg:     #f7ecd0;
126   --status-warning-text:   #704705;
127   --status-warning-border: #dfc279;
```

```
128
129   --status-info-bg:        #dceaf6;
130   --status-info-text:      #184674;
131   --status-info-border:    #97bcdb;
132 }
133
134 /* ── 2. Светлая кирпичная тема (Brick / Terracotta) ──────────
    ──────────────────── */
135 [data-theme="brick"] {
136   /* Фоны (теплые клинкерные и терракотовые поверхности без сл
    епящего белого) */
137   --bg-color:        #f7f3ee;   /* теплый светлый фон клинкера
     */
138   --surface-1:       #fcfaf6;   /* мягкие теплые карточки */
139   --surface-2:       #faefe7;   /* мягкий терракотово-песочный
     хедер */
140   --card-bg:         #fcfaf6;
141   --card-text:       #2c221e;
142   --nav-bg:          #fcfaf6;
143
144   /* Текст */
145   --text-color:      #2c221e;   /* глубокий теплый антрацит */
146   --text-muted:      #7d6b63;   /* теплый приглушенный */
147   --nav-text:        #59473f;
148   --nav-active:      #b84228;   /* благородный терракотово-кир
    пичный */
149   --nav-hover-bg:    #f7e6dc;
150
151   /* Линии и границы */
152   --border-color:    #dfcebf;   /* кирпично-песочный контур */
153   --border-strong:   #bc9e8c;
154   --border-subtle:   #ede2d8;
155   --border-contrast: #8f5843;
156   --divider-color:   #dfcebf;
157   --modal-border:    #dfcebf;
158
159   /* Тени (теплые) */
160   --shadow-xs:  0 1px 2px rgba(130, 70, 50, 0.05);
```

```
  --shadow-sm:  0 1px 4px rgba(130, 70, 50, 0.08);
  --shadow-md:  0 4px 12px rgba(130, 70, 50, 0.11);
  --shadow-lg:  0 8px 24px rgba(130, 70, 50, 0.14);

  /* Чат */
  --chat-user-bg:    #b84228;   /* кирпичный акцент сообщений
*/
  --chat-user-text:  #ffffff;
  --chat-bot-bg:     #f4ece6;
  --chat-bot-text:   #2c221e;

  /* Скроллбар */
  --scrollbar-track: #f7f3ee;
  --scrollbar-thumb: #d1b8a8;

  /* Модальные окна */
  --modal-bg:        #fcfaf6;
  --modal-text:      #2c221e;

  /* Алерты — кирпичная тема */
  --alert-success-bg:     #f2f9f4;
  --alert-success-border: #9ad5af;
  --alert-success-text:   #1e6b37;
  --alert-danger-bg:      #fdf1ee;
  --alert-danger-border:  #f3b4a4;
  --alert-danger-text:    #9e2b17;
  --alert-warning-bg:     #fdf7ec;
  --alert-warning-border: #f6d28b;
  --alert-warning-text:   #8a520f;
  --alert-info-bg:        #f0f5fa;
  --alert-info-border:    #a6cbe7;
  --alert-info-text:      #1c5480;

  /* Геометрия */
  --radius-xs: 4px;
  --radius-sm: 6px;
  --radius-md: 8px;
  --radius-lg: 12px;
```

```
198
199   /* Типографика */
200   --font-sans: system-ui, -apple-system, "Segoe UI", Roboto, "
    Helvetica Neue", Arial, sans-serif;
201   --font-mono: ui-monospace, SFMono-Regular, Menlo, Monaco, Co
    nsolas, "Courier New", monospace;
202
203   /* Каскадные токены KPI-карточек */
204   --kpi-bg:         var(--surface-1);
205   --kpi-border:     var(--border-color);
206   --kpi-accent:     var(--nav-active);
207   --kpi-min-height: 72px;
208   --kpi-padding:    0.65rem 0.9rem;
209   --kpi-val-size:   1.25rem;
210   --kpi-label-size: 0.72rem;
211   --kpi-sub-size:   0.72rem;
212
213   /* Каскадные токены панелей */
214   --panel-bg:        var(--surface-1);
215   --panel-border:    var(--border-color);
216   --panel-header-bg: var(--surface-2);
217
218   /* Каскадные токены суб-навигации */
219   --subnav-btn-padding: 0.35rem 0.75rem;
220   --subnav-font-size:   0.82rem;
221
222   /* Статусные цвета */
223   --status-success-bg:     #e2f3e8;
224   --status-success-text:   #1b6835;
225   --status-success-border: #8ecb9e;
226
227   --status-danger-bg:      #fae4df;
228   --status-danger-text:    #a32f1a;
229   --status-danger-border:  #f1a999;
230
231   --status-warning-bg:     #faeed4;
232   --status-warning-text:   #94580b;
233   --status-warning-border: #f4c871;
```

```
234
235   --status-info-bg:        #e2eef7;
236   --status-info-text:      #195886;
237   --status-info-border:    #9bc4e2;
238 }
239
240 /* ── 3. Тёмная высококонтрастная тема (Dark High-Contrast) ──
    ──────────────── */
241 [data-theme="dark"] {
242   /* Фоны (глубокие контрастные поверхности) */
243   --bg-color:        #0b0f17;   /* глубокий темный фон страниц
    ы */
244   --surface-1:       #151b23;   /* контрастные карточки */
245   --surface-2:       #1e2633;   /* четкие хедеры карточек и ту
    лбары */
246   --card-bg:         #151b23;
247   --card-text:       #f3f6fa;   /* яркий чистый читаемый текст
     */
248   --nav-bg:          #151b23;
249
250   /* Текст */
251   --text-color:      #f3f6fa;   /* яркий текст высокой контрас
    тности */
252   --text-muted:      #a3b1c2;   /* легко различимый вторичный
    текст */
253   --nav-text:        #b0bdcc;
254   --nav-active:      #58a6ff;   /* яркий синий активный акцент
     */
255   --nav-hover-bg:    #222d3d;
256
257   /* Линии и границы (высокая четкость) */
258   --border-color:    #3b4758;   /* четко очерченные границы */
259   --border-strong:   #596b82;   /* выразительные активные гран
    ицы */
260   --border-subtle:   #252d38;   /* мягкие внутренние разделите
    ли */
261   --border-contrast: #70849d;   /* акцентные контуры */
262   --divider-color:   #3b4758;
```

```
  --modal-border:    #3b4758;

  /* Тени (глубокие, отделяющие слои) */
  --shadow-xs:  0 1px 3px rgba(0, 0, 0, 0.4);
  --shadow-sm:  0 2px 6px rgba(0, 0, 0, 0.45);
  --shadow-md:  0 6px 16px rgba(0, 0, 0, 0.55);
  --shadow-lg:  0 10px 30px rgba(0, 0, 0, 0.65);

  /* Чат */
  --chat-user-bg:    #1f6feb;
  --chat-user-text:  #ffffff;
  --chat-bot-bg:     #212836;
  --chat-bot-text:   #f3f6fa;

  /* Скроллбар */
  --scrollbar-track: #0b0f17;
  --scrollbar-thumb: #4e5b6e;

  /* Модальные окна */
  --modal-bg:        #151b23;
  --modal-text:      #f3f6fa;

  /* Алерты — тёмная контрастная тема */
  --alert-success-bg:     #123321;
  --alert-success-border: #3fb950;
  --alert-success-text:   #8df397;
  --alert-danger-bg:      #3b181d;
  --alert-danger-border:  #f85149;
  --alert-danger-text:    #ff9b94;
  --alert-warning-bg:     #38290f;
  --alert-warning-border: #d29922;
  --alert-warning-text:   #f5cf6d;
  --alert-info-bg:        #122744;
  --alert-info-border:    #388bfd;
  --alert-info-text:      #a5d6ff;

  /* Геометрия */
  --radius-xs: 4px;
```

```
  --radius-sm: 6px;
  --radius-md: 8px;
  --radius-lg: 12px;

  /* Типографика */
  --font-sans: system-ui, -apple-system, "Segoe UI", Roboto, "Helvetica Neue", Arial, sans-serif;
  --font-mono: ui-monospace, SFMono-Regular, Menlo, Monaco, Consolas, "Courier New", monospace;

  /* Каскадные токены KPI-карточек */
  --kpi-bg:         var(--surface-1);
  --kpi-border:     var(--border-color);
  --kpi-accent:     var(--nav-active);
  --kpi-min-height: 72px;
  --kpi-padding:    0.65rem 0.9rem;
  --kpi-val-size:   1.25rem;
  --kpi-label-size: 0.72rem;
  --kpi-sub-size:   0.72rem;

  /* Каскадные токены панелей */
  --panel-bg:        var(--surface-1);
  --panel-border:    var(--border-color);
  --panel-header-bg: var(--surface-2);

  /* Каскадные токены суб-навигации */
  --subnav-btn-padding: 0.35rem 0.75rem;
  --subnav-font-size:   0.82rem;

  /* Статусные цвета */
  --status-success-bg:     #174828;
  --status-success-text:   #9df4a6;
  --status-success-border: #2ea043;

  --status-danger-bg:      #52171c;
  --status-danger-text:    #ffb4ae;
  --status-danger-border:  #da3633;

```

```
  --status-warning-bg:     #4d370f;
  --status-warning-text:   #fce49b;
  --status-warning-border: #d29922;

  --status-info-bg:        #13375c;
  --status-info-text:      #a8d8ff;
  --status-info-border:    #1f6feb;
}

/* ── 4. Тёмная тема «Терминал» (Terminal / Matrix / CRT Phosphor) ──────────── */
[data-theme="terminal"] {
  /* Фоны (глубокий смоляной черный с зеленым отливом) */
  --bg-color:        #060a06;   /* фоновый CRT-черный */
  --surface-1:       #0b120c;   /* карточки терминала */
  --surface-2:       #101c12;   /* шапки с изумрудным свечением */
  --card-bg:         #0b120c;
  --card-text:       #d8ffd8;   /* люминесцентный терминальный текст */
  --nav-bg:          #0b120c;

  /* Текст */
  --text-color:      #d8ffd8;   /* неоновый зеленый фосфорный текст */
  --text-muted:      #599863;   /* приглушенный фосфор */
  --nav-text:        #6cb878;
  --nav-active:      #00ff66;   /* электрический неоново-зеленый */
  --nav-hover-bg:    #152618;

  /* Линии и границы (терминальный контур со свечением) */
  --border-color:    #1a4223;   /* глубокий зеленый контур */
  --border-strong:   #00ff66;   /* неоновая рамка активности */
  --border-subtle:   #0f2b14;
  --border-contrast: #39ff14;
  --divider-color:   #1a4223;
```

```
  --modal-border:    #00ff66;

  /* Тени (неоновое свечение фосфора) */
  --shadow-xs:  0 0 3px rgba(0, 255, 102, 0.15);
  --shadow-sm:  0 0 6px rgba(0, 255, 102, 0.2);
  --shadow-md:  0 0 14px rgba(0, 255, 102, 0.25);
  --shadow-lg:  0 0 28px rgba(0, 255, 102, 0.35);

  /* Чат */
  --chat-user-bg:    #008033;   /* зеленый фосфорный пузырь */
  --chat-user-text:  #ffffff;
  --chat-bot-bg:     #101c12;
  --chat-bot-text:   #d8ffd8;

  /* Скроллбар */
  --scrollbar-track: #060a06;
  --scrollbar-thumb: #1b4b24;

  /* Модальные окна */
  --modal-bg:        #0b120c;
  --modal-text:      #d8ffd8;

  /* Алерты — терминальная тема */
  --alert-success-bg:     #0c2412;
  --alert-success-border: #00ff66;
  --alert-success-text:   #4dff88;
  --alert-danger-bg:      #2b0d11;
  --alert-danger-border:  #ff3344;
  --alert-danger-text:    #ff808b;
  --alert-warning-bg:     #2b2207;
  --alert-warning-border: #ffb000;
  --alert-warning-text:   #ffd266;
  --alert-info-bg:        #0a1d26;
  --alert-info-border:    #00ddff;
  --alert-info-text:      #66ecff;

  /* Геометрия */
  --radius-xs: 2px;
```

```
  --radius-sm: 4px;
  --radius-md: 6px;
  --radius-lg: 8px;

  /* Типографика (с моноширинным терминальным акцентом) */
  --font-sans: "Consolas", "Cascadia Code", "Courier New", ui-monospace, monospace;
  --font-mono: "Consolas", "Cascadia Code", "Courier New", ui-monospace, monospace;

  /* Каскадные токены KPI-карточек */
  --kpi-bg:         var(--surface-1);
  --kpi-border:     var(--border-color);
  --kpi-accent:     var(--nav-active);
  --kpi-min-height: 72px;
  --kpi-padding:    0.65rem 0.9rem;
  --kpi-val-size:   1.25rem;
  --kpi-label-size: 0.72rem;
  --kpi-sub-size:   0.72rem;

  /* Каскадные токены панелей */
  --panel-bg:        var(--surface-1);
  --panel-border:    var(--border-color);
  --panel-header-bg: var(--surface-2);

  /* Каскадные токены суб-навигации */
  --subnav-btn-padding: 0.35rem 0.75rem;
  --subnav-font-size:   0.82rem;

  /* Статусные цвета */
  --status-success-bg:     #0c2914;
  --status-success-text:   #39ff14;
  --status-success-border: #00ff66;

  --status-danger-bg:      #3b0f14;
  --status-danger-text:    #ff6675;
  --status-danger-border:  #ff3344;

```

```
  --status-warning-bg:     #382c09;
  --status-warning-text:   #ffc83b;
  --status-warning-border: #ffb000;

  --status-info-bg:        #0c2533;
  --status-info-text:      #33e0ff;
  --status-info-border:    #00ccff;
}

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index (Microsoft Defender
Security Center).

Usage Examples:
    Web Route:
      /windows/api/webgui/defender_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/defender_tab/index.html
" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/defender_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!-- Microsoft Defender & AI Security Diagnostic Center Tab --
>
<style>
  .def-container {
    padding: 0.5rem 0.75rem;
  }
  .def-badge-active { background: var(--status-success-bg, rgb
a(34, 197, 94, 0.15)); color: var(--status-success-text, #22c5
```

```
5e); border: 1px solid var(--status-success-border, rgba(34, 197, 94, 0.3)); }
  .def-badge-disabled { background: var(--status-danger-bg, rgba(239, 68, 68, 0.15)); color: var(--status-danger-text, #ef4444); border: 1px solid var(--status-danger-border, rgba(239, 68, 68, 0.3)); }
  .def-badge-audit { background: var(--status-warning-bg, rgba(245, 158, 11, 0.15)); color: var(--status-warning-text, #f59e0b); border: 1px solid var(--status-warning-border, rgba(245, 158, 11, 0.3)); }
  .def-badge-warn { background: var(--status-warning-bg, rgba(245, 158, 11, 0.15)); color: var(--status-warning-text, #f59e0b); border: 1px solid var(--status-warning-border, rgba(245, 158, 11, 0.3)); }
  .def-badge-critical { background: var(--status-danger-bg, rgba(239, 68, 68, 0.15)); color: var(--status-danger-text, #ef4444); border: 1px solid var(--status-danger-border, rgba(239, 68, 68, 0.3)); font-weight: bold; }
  .def-clickable-row { cursor: pointer; transition: background-color 0.15s ease-in-out; }
  .def-clickable-row:hover { background-color: var(--nav-hover-bg, rgba(0, 0, 0, 0.05)) !important; }
</style>

<div class="def-container theme-defender">
  <!-- Top Stat Cards -->
  <div class="row g-2 mb-2">
    <div class="col-6 col-md-3 col-xl">
      <div class="def-card">
        <div class="def-card-label"><span>Индекс защиты</span>  </div>
        <div class="def-card-value" id="def-score-val">--/100</div>
        <div class="def-card-sub" id="def-score-sub">Загрузка...</div>
      </div>
    </div>
    <div class="col-6 col-md-3 col-xl">
```

```
      <div class="def-card">
        <div class="def-card-label"><span>Real-time Shield</sp
an> □ </div>
        <div class="def-card-value text-success" id="def-realt
ime-val">--</div>
        <div class="def-card-sub" id="def-realtime-sub">MsMpEn
g.exe</div>
      </div>
    </div>
    <div class="col-6 col-md-3 col-xl">
      <div class="def-card">
        <div class="def-card-label"><span>Cloud & Tamper</span
> □ </div>
        <div class="def-card-value" id="def-cloud-val">--</div
>
        <div class="def-card-sub" id="def-cloud-sub">MAPS Prot
ection</div>
      </div>
    </div>
    <div class="col-6 col-md-3 col-xl">
      <div class="def-card">
        <div class="def-card-label"><span>PUA & Ransomware</sp
an> □ </div>
        <div class="def-card-value" id="def-pua-cfa-val">--</d
iv>
        <div class="def-card-sub" id="def-pua-cfa-sub">CFA Fol
der Guard</div>
      </div>
    </div>
    <div class="col-6 col-md-3 col-xl">
      <div class="def-card">
        <div class="def-card-label"><span>Исключения</span> □ <
/div>
        <div class="def-card-value text-warning" id="def-exclu
sions-val">0</div>
        <div class="def-card-sub" id="def-exclusions-sub">0 по
дозрительных</div>
      </div>
```

```
    </div>
    <div class="col-6 col-md-3 col-xl">
      <div class="def-card">
        <div class="def-card-label"><span>Активные угрозы</spa
n> □ </div>
        <div class="def-card-value text-danger" id="def-threat
s-val">0</div>
        <div class="def-card-sub" id="def-threats-sub">Каранти
н: 0</div>
      </div>
    </div>
  </div>

  <!-- Action Bar -->
  <div class="def-panel mb-2 p-2 d-flex align-items-center jus
tify-content-between flex-wrap gap-2">
    <div class="d-flex align-items-center gap-2 flex-wrap">
      <button class="btn btn-sm btn-primary d-flex align-items
-center gap-1" id="btn-def-quick-scan">
        <i class="bi bi-play-circle-fill"></i> <span>Быстрое с
канирование</span>
      </button>
      <button class="btn btn-sm btn-outline-primary d-flex ali
gn-items-center gap-1" id="btn-def-full-scan">
        <i class="bi bi-shield-shaded"></i> <span>Полное скани
рование</span>
      </button>
      <button class="btn btn-sm btn-outline-info d-flex align-
items-center gap-1" id="btn-def-update-sigs">
        <i class="bi bi-cloud-arrow-down-fill"></i> <span>Обно
вить сигнатуры</span>
      </button>
      <button class="btn btn-sm btn-outline-warning d-flex ali
gn-items-center gap-1" id="btn-def-ai-diag">
        <i class="bi bi-cpu-fill"></i> <span>AI Диагностика</s
pan>
      </button>
    </div>
```

```
    <div class="d-flex align-items-center gap-2">
      <span class="badge bg-body-secondary text-body border" id="def-engine-ver-badge">Движок: --</span>
      <span class="badge bg-body-secondary text-body border" id="def-sigs-ver-badge">Базы: --</span>
      <button class="btn btn-sm btn-outline-secondary" id="btn-def-refresh" title="Обновить">
        <i class="bi bi-arrow-clockwise"></i>
      </button>
    </div>
  </div>

  <!-- Scan Execution Progress / Notification Alert -->
  <div id="def-alert-box" class="d-none alert alert-info py-2 px-3 mb-2 small"></div>

  <!-- Sub Navigation Tabs -->
  <ul class="nav nav-tabs mb-2 border-secondary" id="defSubTabs" role="tablist">
    <li class="nav-item" role="presentation">
      <button class="nav-link active py-1 px-3 small" id="def-tab-overview-btn" data-bs-toggle="tab" data-bs-target="#def-pane-overview" type="button">  Обзор и Службы</button>
    </li>
    <li class="nav-item" role="presentation">
      <button class="nav-link py-1 px-3 small" id="def-tab-asr-btn" data-bs-toggle="tab" data-bs-target="#def-pane-asr" type="button">  Правила ASR</button>
    </li>
    <li class="nav-item" role="presentation">
      <button class="nav-link py-1 px-3 small" id="def-tab-cfa-btn" data-bs-toggle="tab" data-bs-target="#def-pane-cfa" type="button">  Ransomware / CFA</button>
    </li>
    <li class="nav-item" role="presentation">
      <button class="nav-link py-1 px-3 small" id="def-tab-exclusions-btn" data-bs-toggle="tab" data-bs-target="#def-pane-exclusions" type="button">  Исключения</button>
```

```
    </li>
    <li class="nav-item" role="presentation">
      <button class="nav-link py-1 px-3 small" id="def-tab-threats-btn" data-bs-toggle="tab" data-bs-target="#def-pane-threats" type="button">  Угрозы и Карантин</button>
    </li>
    <li class="nav-item" role="presentation">
      <button class="nav-link py-1 px-3 small" id="def-tab-procs-btn" data-bs-toggle="tab" data-bs-target="#def-pane-procs" type="button">  Дерево Процессов / Fileless</button>
    </li>
    <li class="nav-item" role="presentation">
      <button class="nav-link py-1 px-3 small" id="def-tab-events-btn" data-bs-toggle="tab" data-bs-target="#def-pane-events" type="button">  Журнал Событий</button>
    </li>
    <li class="nav-item" role="presentation">
      <button class="nav-link py-1 px-3 small text-warning fw-semibold" id="def-tab-diag-btn" data-bs-toggle="tab" data-bs-target="#def-pane-diag" type="button">  AI Диагностика</button>
    </li>
  </ul>

  <!-- Tab Content Panes -->
  <div class="tab-content" id="defTabPanes">

    <!-- 1. Overview & Services Pane -->
    <div class="tab-pane fade show active" id="def-pane-overview" role="tabpanel">
      <div class="row g-2">
        <div class="col-md-7">
          <div class="def-panel h-100">
            <div class="def-panel-header">
              <span>Эшелоны Защиты Microsoft Defender</span>
            </div>
            <div class="table-responsive p-0">
              <table class="table table-hover mb-0 align-middl
```

```
e small">
                <thead>
                  <tr class="text-secondary">
                    <th>Компонент</th>
                    <th class="text-center">Статус</th>
                    <th>Описание</th>
                  </tr>
                </thead>
                <tbody id="def-shields-tbody">
                  <tr><td colspan="3" class="text-center text-
muted py-3">Загрузка данных...</td></tr>
                </tbody>
              </table>
            </div>
          </div>
        </div>
        <div class="col-md-5">
          <div class="def-panel h-100">
            <div class="def-panel-header">
              <span>Системные Службы и Процессы</span>
            </div>
            <div class="table-responsive p-0">
              <table class="table table-hover mb-0 align-middl
e small">
                <thead>
                  <tr class="text-secondary">
                    <th>Процесс</th>
                    <th class="text-center">Статус</th>
                    <th class="text-end">PID</th>
                    <th class="text-end">Память</th>
                  </tr>
                </thead>
                <tbody id="def-services-tbody">
                  <tr><td colspan="4" class="text-center text-
muted py-3">Загрузка данных...</td></tr>
                </tbody>
              </table>
            </div>
```

```
          </div>
        </div>
      </div>
    </div>

    <!-- 2. ASR Rules Pane -->
    <div class="tab-pane fade" id="def-pane-asr" role="tabpane
l">
      <div class="def-panel">
        <div class="def-panel-header">
          <span>Правила Attack Surface Reduction (ASR)</span>
          <span class="badge bg-secondary" id="def-asr-count-b
adge">Всего: 0</span>
        </div>
        <div class="table-responsive p-0">
          <table class="table table-hover mb-0 align-middle sm
all">
            <thead>
              <tr class="text-secondary">
                <th>Правило ASR</th>
                <th>Категория</th>
                <th class="text-center">Состояние</th>
                <th>Рекомендация</th>
                <th>GUID</th>
              </tr>
            </thead>
            <tbody id="def-asr-tbody">
              <tr><td colspan="5" class="text-center text-mute
d py-3">Загрузка правил ASR...</td></tr>
            </tbody>
          </table>
        </div>
      </div>
    </div>

    <!-- 3. CFA (Ransomware) Pane -->
    <div class="tab-pane fade" id="def-pane-cfa" role="tabpane
l">
```

```
      <div class="row g-2">
        <div class="col-md-6">
          <div class="def-panel h-100">
            <div class="def-panel-header">
              <span>Защищенные Каталоги (Protected Folders)</s
pan>
            </div>
            <div class="p-2" id="def-cfa-folders-list">
              <div class="text-muted small">Загрузка...</div>
            </div>
          </div>
        </div>
        <div class="col-md-6">
          <div class="def-panel h-100">
            <div class="def-panel-header">
              <span>Доверенные Приложения (Allowed Apps)</span
>
            </div>
            <div class="p-2" id="def-cfa-apps-list">
              <div class="text-muted small">Загрузка...</div>
            </div>
          </div>
        </div>
      </div>

    <!-- 4. Exclusions Security Audit Pane -->
    <div class="tab-pane fade" id="def-pane-exclusions" role="
tabpanel">
      <div class="def-panel">
        <div class="def-panel-header">
          <span>Аудит Исключений из Проверки</span>
          <span class="badge bg-warning text-dark" id="def-exc
-summary-badge">0 исключений</span>
        </div>
        <div class="p-2 mb-1 bg-body-secondary text-body small
 border-bottom" id="def-exc-recommendation-box">
          Анализ исключений...
```

```
        </div>
        <div class="table-responsive p-0">
          <table class="table table-hover mb-0 align-middle sm
all">
            <thead>
              <tr class="text-secondary">
                <th>Тип</th>
                <th>Значение</th>
                <th class="text-center">Уровень Риска</th>
                <th>Обоснование</th>
              </tr>
            </thead>
            <tbody id="def-exclusions-tbody">
              <tr><td colspan="4" class="text-center text-mute
d py-3">Загрузка исключений...</td></tr>
            </tbody>
          </table>
        </div>
      </div>
    </div>

    <!-- 5. Threats History Pane -->
    <div class="tab-pane fade" id="def-pane-threats" role="tab
panel">
      <div class="def-panel">
        <div class="def-panel-header">
          <span>Журнал Обнаружения Угроз и Карантин</span>
          <span class="badge bg-danger" id="def-threats-count-
badge">0 записей</span>
        </div>
        <div class="table-responsive p-0">
          <table class="table table-hover mb-0 align-middle sm
all">
            <thead>
              <tr class="text-secondary">
                <th>Угроза</th>
                <th>Категория</th>
                <th class="text-center">Опасность</th>
```

```
                <th>Статус</th>
                <th>Время</th>
                <th>Ресурсы</th>
              </tr>
            </thead>
            <tbody id="def-threats-tbody">
              <tr><td colspan="6" class="text-center text-mute
d py-3">Угроз не обнаружено.</td></tr>
            </tbody>
          </table>
        </div>
      </div>
    </div>

    <!-- 6. Process Trees / Fileless Activity Pane -->
    <div class="tab-pane fade" id="def-pane-procs" role="tabpa
nel">
      <div class="def-panel">
        <div class="def-panel-header">
          <span>Инспекция Цепочек Процессов и Индикаторы Filel
ess-атак</span>
          <span class="badge bg-secondary" id="def-procs-count
-badge">0 аномалий</span>
        </div>
        <div class="table-responsive p-0">
          <table class="table table-hover mb-0 align-middle sm
all">
            <thead>
              <tr class="text-secondary">
                <th>Родительский Процесс</th>
                <th>Дочерний Процесс</th>
                <th>PID</th>
                <th>Опасность</th>
                <th>Причина срабатывания</th>
              </tr>
            </thead>
            <tbody id="def-procs-tbody">
              <tr><td colspan="5" class="text-center text-mute
```

```
d py-3">Подозрительных цепочек процессов не обнаружено.</td></
tr>
            </tbody>
          </table>
        </div>
      </div>
    </div>

    <!-- 7. Operational Events Pane -->
    <div class="tab-pane fade" id="def-pane-events" role="tabp
anel">
      <div class="def-panel">
        <div class="def-panel-header">
          <span>Журнал Microsoft-Windows-Windows Defender/Oper
ational</span>
          <span class="badge bg-secondary" id="def-events-coun
t-badge">0 событий</span>
        </div>
        <div class="table-responsive p-0" style="max-height: c
alc(100vh - 240px); min-height: 200px; overflow-y: auto;">
          <table class="table table-hover mb-0 align-middle sm
all">
            <thead class="sticky-top" style="position: sticky;
 top: 0; z-index: 2;">
              <tr class="text-secondary">
                <th>Event ID</th>
                <th>Время</th>
                <th>Уровень</th>
                <th>Категория</th>
                <th>Сообщение</th>
              </tr>
            </thead>
            <tbody id="def-events-tbody">
              <tr><td colspan="5" class="text-center text-mute
d py-3">Загрузка журнала событий...</td></tr>
            </tbody>
          </table>
        </div>
```

```
      </div>
    </div>

    <!-- 8. AI Diagnostics Pane -->
    <div class="tab-pane fade" id="def-pane-diag" role="tabpan
el">
      <div class="row g-2">
        <div class="col-md-4">
          <div class="def-panel h-100 p-3 text-center d-flex f
lex-column justify-content-center">
            <div class="text-secondary small mb-1">Security Po
sture Score</div>
            <div class="display-4 fw-bold text-success" id="de
f-diag-score-display">--</div>
            <div class="fw-semibold mt-2" id="def-diag-summary
-display">Анализ состояния...</div>
          </div>
        </div>
        <div class="col-md-8">
          <div class="def-panel h-100 p-3">
            <h6 class="text-warning fw-bold mb-2">  Рекоменда
ции AI по Усилению Защиты:</h6>
            <div id="def-diag-recs-container" class="small">
              <div class="text-muted">Загрузка рекомендаций...
</div>
            </div>
            <hr class="border-secondary my-2">
            <h6 class="text-danger fw-bold mb-2">  Критические
 уязвимости и риски:</h6>
            <div id="def-diag-criticals-container" class="smal
l">
              <div class="text-muted">Нет критических уязвимос
тей.</div>
            </div>
          </div>
        </div>
      </div>
    </div>
```

```
377
378   </div>
379
380   <!-- Modal for Detailed Event / Threat Inspection -->
381   <div class="modal fade" id="def-detail-modal" tabindex="-1"
aria-labelledby="def-detail-modal-title" aria-hidden="true">
382     <div class="modal-dialog modal-dialog-centered modal-lg mo
dal-dialog-scrollable">
383       <div class="modal-content border shadow-lg">
384         <div class="modal-header border-bottom py-2">
385           <div class="d-flex align-items-center gap-2">
386             <span class="fs-5" id="def-detail-modal-icon">□ </
span>
387             <h5 class="modal-title fs-6 fw-bold mb-0" id="def-
detail-modal-title">Детали события безопасности</h5>
388             <span class="badge bg-primary" id="def-detail-moda
l-badge">Event ID: 1151</span>
389           </div>
390           <button type="button" class="btn-close" data-bs-dism
iss="modal" aria-label="Закрыть"></button>
391         </div>
392         <div class="modal-body p-3" style="font-size: 0.85rem;
">
393           <!-- Key Meta Cards -->
394           <div class="row g-2 mb-3" id="def-detail-meta-cards"
>
395             <div class="col-sm-6 col-md-3">
396               <div class="p-2 bg-body-secondary rounded border
">
397                 <div class="text-secondary small">Время событи
я</div>
398                 <div class="fw-semibold text-body font-monospa
ce" id="def-detail-time">-</div>
399               </div>
400             </div>
401             <div class="col-sm-6 col-md-3">
402               <div class="p-2 bg-body-secondary rounded border
">
```

```
403                <div class="text-secondary small">Уровень важн
ости</div>
404                <div class="fw-semibold" id="def-detail-level"
>-</div>
405              </div>
406            </div>
407            <div class="col-sm-6 col-md-3">
408              <div class="p-2 bg-body-secondary rounded border
">
409                <div class="text-secondary small">Категория</d
iv>
410                <div class="fw-semibold text-info" id="def-det
ail-category">-</div>
411              </div>
412            </div>
413            <div class="col-sm-6 col-md-3">
414              <div class="p-2 bg-body-secondary rounded border
">
415                <div class="text-secondary small">Event ID / К
од</div>
416                <div class="fw-semibold text-warning font-mono
space" id="def-detail-id">-</div>
417              </div>
418            </div>
419          </div>
420
421          <!-- Structured Parameters Table (if parsed from eve
nt text) -->
422          <div id="def-detail-props-section" class="mb-3 d-non
e">
423            <h6 class="text-info fw-bold mb-1 d-flex align-ite
ms-center gap-1">
424              <i class="bi bi-list-columns-reverse"></i> <span
>Структурированные параметры:</span>
425            </h6>
426            <div class="table-responsive border rounded">
427              <table class="table table-sm table-striped mb-0
align-middle small">
```

```
                <tbody id="def-detail-props-tbody"></tbody>
              </table>
            </div>
          </div>

          <!-- Full Raw Message -->
          <div class="mb-2">
            <div class="d-flex justify-content-between align-items-center mb-1">
              <h6 class="text-secondary fw-bold mb-0">Полный текст сообщения:</h6>
              <button class="btn btn-outline-secondary btn-sm py-0 px-2" id="btn-def-copy-msg" title="Скопировать текст в буфер">
                <i class="bi bi-clipboard"></i> <span>Копировать</span>
              </button>
            </div>
            <pre class="p-2.5 bg-body-secondary text-body rounded border font-monospace mb-0" id="def-detail-raw-msg" style="white-space: pre-wrap; word-break: break-word; font-size: 0.8rem; max-height: 280px; overflow-y: auto; user-select: text;"></pre>
          </div>
        </div>
        <div class="modal-footer border-top py-1.5 justify-content-between">
          <span class="text-muted small" id="def-detail-footer-hint">Источник: Microsoft-Windows-Windows Defender/Operational</span>
          <button type="button" class="btn btn-sm btn-outline-secondary" data-bs-dismiss="modal">Закрыть</button>
        </div>
      </div>
    </div>
  </div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Defender Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/defender_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/defender_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// Microsoft Defender Security Center Frontend Controller
(function() {
  'use strict';

  let defenderData = null;
  let currentEvents = [];
  let currentThreats = [];

  function parseMessageProps(msg) {
    if (!msg || typeof msg !== 'string') return [];
    const lines = msg.split(/\r?\n/);
    const props = [];
```

```
    for (const rawLine of lines) {
      const line = rawLine.trim();
      if (!line) continue;
      const cleaned = line.replace(/^[\t\s*•\-]+/, '');
      const colonIdx = cleaned.indexOf(':');
      if (colonIdx > 0 && colonIdx < 50) {
        const key = cleaned.substring(0, colonIdx).trim();
        const val = cleaned.substring(colonIdx + 1).trim();
        if (key.toLowerCase() !== 'http' && key.toLowerCase()
!== 'https' && val) {
          props.push({ key, val });
        }
      }
    }
    if (props.length === 0 && msg.includes(':')) {
      const regex = /([A-Za-zA-Яа-я0-9\s/_\-()]+?):\s*([^:\n\r
]+?)(?=(?:[A-Za-zA-Яа-я0-9\s/_\-()]+?:|$))/g;
      let match;
      while ((match = regex.exec(msg)) !== null) {
        const k = match[1].trim();
        const v = match[2].trim();
        if (k && v && k.length < 50 && k.toLowerCase() !== 'ht
tp' && k.toLowerCase() !== 'https') {
          props.push({ key: k, val: v });
        }
      }
    }
    return props;
  }

  function showEventDetail(event) {
    if (!event) return;
    const modalEl = document.getElementById('def-detail-modal'
);
    if (!modalEl) return;

    const titleEl = document.getElementById('def-detail-modal-
title');
```

```
    const iconEl = document.getElementById('def-detail-modal-icon');
    const badgeEl = document.getElementById('def-detail-modal-badge');
    const timeEl = document.getElementById('def-detail-time');
    const levelEl = document.getElementById('def-detail-level');
    const catEl = document.getElementById('def-detail-category');
    const idEl = document.getElementById('def-detail-id');
    const propsSection = document.getElementById('def-detail-props-section');
    const propsTbody = document.getElementById('def-detail-props-tbody');
    const rawMsgEl = document.getElementById('def-detail-raw-msg');
    const footerHint = document.getElementById('def-detail-footer-hint');

    const isWarning = (event.level || '').toLowerCase().includes('warn');
    const isDanger = (event.level || '').toLowerCase().includes('err') || (event.level || '').toLowerCase().includes('crit') || [1006, 1015, 1116, 1117, 1121, 1123, 1125].includes(Number(event.event_id));

    if (titleEl) titleEl.textContent = `Событие безопасности: ${event.category || 'Windows Defender'}`;
    if (iconEl) iconEl.textContent = isDanger ? '□ ' : (isWarning ? '□ ' : '□ ');
    if (badgeEl) {
      badgeEl.textContent = `Event ID: ${event.event_id}`;
      badgeEl.className = `badge ${isDanger ? 'bg-danger' : (isWarning ? 'bg-warning text-dark' : 'bg-primary')}`;
    }
    if (timeEl) timeEl.textContent = event.timestamp || '-';
    if (levelEl) {
      levelEl.innerHTML = `<span class="badge ${isDanger ? 'bg
```

```
-danger' : (isWarning ? 'bg-warning text-dark' : 'bg-secondary
')}">${event.level || 'Info'}</span>`;
    }
    if (catEl) catEl.textContent = event.category || 'General'
;
    if (idEl) idEl.textContent = event.event_id || '-';

    const props = parseMessageProps(event.message);
    if (propsSection && propsTbody) {
      if (props.length > 0) {
        propsTbody.innerHTML = props.map(p => `
          <tr>
            <th class="text-secondary fw-semibold text-nowrap
py-1 px-2" style="width: 32%;">${p.key}</th>
            <td class="font-monospace text-body py-1 px-2" sty
le="word-break: break-all;">${p.val}</td>
          </tr>
        `).join('');
        propsSection.classList.remove('d-none');
      } else {
        propsSection.classList.add('d-none');
        propsTbody.innerHTML = '';
      }
    }

    if (rawMsgEl) rawMsgEl.textContent = event.message || 'Нет
 описания.';
    if (footerHint) footerHint.textContent = `Журнал: Microsof
t-Windows-Windows Defender/Operational | Event ID: ${event.eve
nt_id}`;

    if (window.bootstrap && bootstrap.Modal) {
      const modal = bootstrap.Modal.getOrCreateInstance(modalE
l);
      modal.show();
    }
  }

```

```
  function showThreatDetail(threat) {
    if (!threat) return;
    const modalEl = document.getElementById('def-detail-modal');
    if (!modalEl) return;

    const titleEl = document.getElementById('def-detail-modal-title');
    const iconEl = document.getElementById('def-detail-modal-icon');
    const badgeEl = document.getElementById('def-detail-modal-badge');
    const timeEl = document.getElementById('def-detail-time');
    const levelEl = document.getElementById('def-detail-level');
    const catEl = document.getElementById('def-detail-category');
    const idEl = document.getElementById('def-detail-id');
    const propsSection = document.getElementById('def-detail-props-section');
    const propsTbody = document.getElementById('def-detail-props-tbody');
    const rawMsgEl = document.getElementById('def-detail-raw-msg');
    const footerHint = document.getElementById('def-detail-footer-hint');

    if (titleEl) titleEl.textContent = `Обнаруженная угроза: ${threat.threat_name || 'Неизвестно'}`;
    if (iconEl) iconEl.textContent = '□ ';
    if (badgeEl) {
      badgeEl.textContent = `Threat ID: ${threat.threat_id || '-'}`;
      badgeEl.className = 'badge bg-danger';
    }
    if (timeEl) timeEl.textContent = threat.last_detection_time || threat.initial_detection_time || '-';
    if (levelEl) {
```

```
      levelEl.innerHTML = `<span class="badge bg-danger">${(threat.severity || 'High').toUpperCase()}</span>`;
    }
    if (catEl) catEl.textContent = threat.category || 'Malware';
    if (idEl) idEl.textContent = threat.threat_id || '-';

    const threatProps = [
      { key: 'Название угрозы', val: threat.threat_name },
      { key: 'Категория', val: threat.category },
      { key: 'Уровень опасности', val: threat.severity },
      { key: 'Текущий статус', val: threat.status },
      { key: 'Код действия очистки', val: threat.cleaning_action },
      { key: 'Время обнаружения', val: threat.initial_detection_time },
      { key: 'Последняя активность', val: threat.last_detection_time },
      { key: 'Ресурсы / Файлы', val: (threat.resources || []).join('\n') },
    ].filter(p => p.val);

    if (propsSection && propsTbody) {
      propsTbody.innerHTML = threatProps.map(p => `
        <tr>
          <th class="text-secondary fw-semibold text-nowrap py-1 px-2" style="width: 32%;">${p.key}</th>
          <td class="font-monospace text-body py-1 px-2" style="word-break: break-all; white-space: pre-wrap;">${p.val}</td>
        </tr>
      `).join('');
      propsSection.classList.remove('d-none');
    }

    if (rawMsgEl) {
      rawMsgEl.textContent = `Threat: ${threat.threat_name}\nID: ${threat.threat_id}\nSeverity: ${threat.severity}\nStatus: ${threat.status}\nResources:\n${(threat.resources || []).join(
```

```
'\n')}`;
    }
    if (footerHint) footerHint.textContent = `Источник: Get-Mp
ThreatDetection / Microsoft Defender Threat History`;

    if (window.bootstrap && bootstrap.Modal) {
      const modal = bootstrap.Modal.getOrCreateInstance(modalE
l);
      modal.show();
    }
  }

  function showAlert(msg, isError = false) {
    const box = document.getElementById('def-alert-box');
    if (!box) return;
    box.className = `alert py-2 px-3 mb-2 small ${isError ? 'a
lert-danger' : 'alert-info'}`;
    box.textContent = msg;
    box.classList.remove('d-none');
    setTimeout(() => {
      box.classList.add('d-none');
    }, 8000);
  }

  async function fetchJSON(url, options = {}) {
    if (window.api && typeof window.api.fetch === 'function')
{
      return await window.api.fetch(url, options);
    }
    const res = await fetch(url, options);
    if (!res.ok) {
      throw new Error(`HTTP ${res.status}: ${res.statusText}`)
;
    }
    return await res.json();
  }

  async function loadDefenderStatus() {
```

```
    try {
      const status = await fetchJSON('/api/v1/defender/status'
);
      defenderData = status;

      // Badges in header
      const engBadge = document.getElementById('def-engine-ver
-badge');
      if (engBadge) engBadge.textContent = `Движок: ${status.e
ngine_version || 'N/A'}`;
      const sigsBadge = document.getElementById('def-sigs-ver-
badge');
      if (sigsBadge) sigsBadge.textContent = `Базы: ${status.a
ntivirus_signature_version || 'N/A'}`;

      // Stat Cards
      const rtVal = document.getElementById('def-realtime-val'
);
      if (rtVal) {
        rtVal.textContent = status.real_time_protection_enable
d ? 'ВКЛ' : 'ВЫКЛ';
        rtVal.className = `def-card-value ${status.real_time_p
rotection_enabled ? 'text-success' : 'text-danger'}`;
      }

      const cloudVal = document.getElementById('def-cloud-val'
);
      if (cloudVal) {
        cloudVal.textContent = status.cloud_protection_enabled
 ? 'АКТИВНА' : 'ВЫКЛ';
        cloudVal.className = `def-card-value ${status.cloud_pr
otection_enabled ? 'text-success' : 'text-warning'}`;
      }

      const puaCfaVal = document.getElementById('def-pua-cfa-v
al');
      if (puaCfaVal) {
        const puaText = status.pua_protection_enabled ? 'PUA:
```

```
□ ' : 'PUA: □ ';
229         const cfaText = status.controlled_folder_access_enable
d ? 'CFA: □ ' : 'CFA: □ ';
230         puaCfaVal.textContent = `${puaText} | ${cfaText}`;
231         puaCfaVal.className = 'def-card-value small text-info'
;
232       }
233
234       // Shields Table
235       const shieldsTbody = document.getElementById('def-shield
s-tbody');
236       if (shieldsTbody) {
237         const shields = [
238           { name: 'Real-time Protection', enabled: status.real
_time_protection_enabled, desc: 'Постоянный перехват и проверк
а файлов при создании и обращении' },
239           { name: 'Cloud Protection (MAPS)', enabled: status.c
loud_protection_enabled, desc: `Мгновенная блокировка новых уг
роз через облачный ИИ Microsoft (${status.cloud_block_level})`
 },
240           { name: 'Behavior Monitoring', enabled: status.behav
ior_monitor_enabled, desc: 'Анализ поведения запущенных процес
сов и цепочек активности' },
241           { name: 'IOAV Protection', enabled: status.ioav_prot
ection_enabled, desc: 'Проверка файлов, загружаемых из браузер
а, почты и мессенджеров' },
242           { name: 'AMSI Script Scanning', enabled: status.scri
pt_scanning_enabled, desc: 'Инспекция скриптов PowerShell, VBS
cript, JavaScript до их выполнения' },
243           { name: 'Tamper Protection', enabled: status.tamper_
protection_enabled, desc: 'Защита настроек антивируса от отклю
чения вредоносным ПО' },
244           { name: 'PUA Protection', enabled: status.pua_protec
tion_enabled, desc: 'Блокировка нежелательного ПО, рекламных и
нсталляторов и майнеров' },
245           { name: 'Controlled Folder Access', enabled: status.
controlled_folder_access_enabled, desc: 'Защита папок Документ
ы/Рабочий стол от шифрования Ransomware' },
```

```
          { name: 'Network Protection', enabled: status.networ
k_protection_enabled, desc: 'Блокировка сетевых соединений с в
редоносными доменами и IP' },
        ];

        shieldsTbody.innerHTML = shields.map(s => `
          <tr>
            <td class="fw-semibold text-body">${s.name}</td>
            <td class="text-center">
              <span class="badge ${s.enabled ? 'def-badge-acti
ve' : 'def-badge-disabled'}">
                ${s.enabled ? 'ВКЛЮЧЕНО' : 'ОТКЛЮЧЕНО'}
              </span>
            </td>
            <td class="text-secondary">${s.desc}</td>
          </tr>
        `).join('');
      }

      // Services Table
      const servicesTbody = document.getElementById('def-servi
ces-tbody');
      if (servicesTbody && status.services) {
        servicesTbody.innerHTML = status.services.map(svc => `
          <tr>
            <td>
              <div class="fw-semibold text-body">${svc.name}</
div>
              <div class="text-secondary" style="font-size: 0.
68rem;">${svc.display_name}</div>
            </td>
            <td class="text-center">
              <span class="badge ${svc.running ? 'bg-success'
: 'bg-secondary'}">
                ${svc.running ? 'АКТИВЕН' : 'СТОП'}
              </span>
            </td>
            <td class="text-end text-secondary">${svc.pid || '
```

```
-'}</td>
277             <td class="text-end font-monospace">${svc.running ? svc.memory_mb + ' MB' : '-'}</td>
278           </tr>
279         `).join('');
280       }
281
282     } catch (err) {
283       console.error('[DefenderTab] Error loading status:', err);
284       showAlert(`Ошибка загрузки статуса Defender: ${err.message}`, true);
285     }
286   }
287
288   async function loadASRRules() {
289     try {
290       const rules = await fetchJSON('/api/v1/defender/asr');
291       const tbody = document.getElementById('def-asr-tbody');
292       const badge = document.getElementById('def-asr-count-badge');
293       if (badge) badge.textContent = `Всего: ${rules.length}`;
294
295       if (tbody) {
296         tbody.innerHTML = rules.map(r => {
297           let stateBadge = '<span class="badge bg-secondary">Не задано</span>';
298           if (r.state === 'enabled') stateBadge = '<span class="badge def-badge-active">Блокировка (Block)</span>';
299           else if (r.state === 'audit') stateBadge = '<span class="badge def-badge-audit">Аудит (Audit)</span>';
300           else if (r.state === 'warn') stateBadge = '<span class="badge def-badge-warn">Предупреждение</span>';
301           else if (r.state === 'disabled') stateBadge = '<span class="badge def-badge-disabled">Отключено</span>';
302
303           return `
304             <tr>
```

```
              <td>
                <div class="fw-semibold text-body">${r.name}</
div>
                <div class="text-secondary" style="font-size:
0.72rem;">${r.description}</div>
              </td>
              <td><span class="badge bg-body-secondary text-bo
dy border">${r.category}</span></td>
              <td class="text-center">${stateBadge}</td>
              <td class="text-secondary small">${r.recommendat
ion || '-'}</td>
              <td class="text-secondary font-monospace" style=
"font-size: 0.68rem;">${r.guid}</td>
            </tr>
          `;
        }).join('');
      }
    } catch (err) {
      console.error('[DefenderTab] Error loading ASR rules:',
err);
    }
  }

  async function loadCFA() {
    try {
      const cfa = await fetchJSON('/api/v1/defender/cfa');
      const foldersList = document.getElementById('def-cfa-fol
ders-list');
      const appsList = document.getElementById('def-cfa-apps-l
ist');

      if (foldersList) {
        if (cfa.protected_folders && cfa.protected_folders.len
gth > 0) {
          foldersList.innerHTML = `
            <ul class="list-group list-group-flush bg-transpar
ent">
              ${cfa.protected_folders.map(f => `
```

```
                <li class="list-group-item bg-transparent text
-body border-secondary py-1 px-0 font-monospace small">
                  📄 ${f}
                </li>
              `).join('')}
            </ul>
          `;
        } else {
          foldersList.innerHTML = '<div class="text-muted smal
l">Нет настроенных защищенных папок.</div>';
        }
      }

      if (appsList) {
        if (cfa.allowed_applications && cfa.allowed_applicatio
ns.length > 0) {
          appsList.innerHTML = `
            <ul class="list-group list-group-flush bg-transpar
ent">
              ${cfa.allowed_applications.map(a => `
                <li class="list-group-item bg-transparent text
-body border-secondary py-1 px-0 font-monospace small">
                  📄 ${a}
                </li>
              `).join('')}
            </ul>
          `;
        } else {
          appsList.innerHTML = '<div class="text-muted small">
Список доверенных приложений пуст.</div>';
        }
      }
    } catch (err) {
      console.error('[DefenderTab] Error loading CFA:', err);
    }
  }

  async function loadExclusions() {
```

```
    try {
      const rep = await fetchJSON('/api/v1/defender/exclusions');

      const countVal = document.getElementById('def-exclusions-val');
      const countSub = document.getElementById('def-exclusions-sub');
      const summaryBadge = document.getElementById('def-exc-summary-badge');
      const recBox = document.getElementById('def-exc-recommendation-box');
      const tbody = document.getElementById('def-exclusions-tbody');

      if (countVal) countVal.textContent = rep.total_exclusions;
      if (countSub) countSub.textContent = `${rep.suspicious_count} подозрительных`;
      if (summaryBadge) summaryBadge.textContent = `${rep.total_exclusions} исключений (${rep.suspicious_count} риск)`;
      if (recBox) recBox.textContent = rep.summary_recommendation;

      const allItems = [...rep.path_exclusions, ...rep.extension_exclusions, ...rep.process_exclusions];

      if (tbody) {
        if (allItems.length === 0) {
          tbody.innerHTML = '<tr><td colspan="4" class="text-center text-muted py-3">Исключения отсутствуют. Защита работает без пропусков.</td></tr>';
        } else {
          tbody.innerHTML = allItems.map(itm => {
            let riskBadge = '<span class="badge bg-success">Безопасно</span>';
            if (itm.risk_level === 'critical') riskBadge = '<span class="badge def-badge-critical">КРИТИЧЕСКИЙ</span>';
```

```
            else if (itm.risk_level === 'high') riskBadge = '<
span class="badge def-badge-disabled">ВЫСОКИЙ</span>';
            else if (itm.risk_level === 'medium') riskBadge =
'<span class="badge def-badge-warn">СРЕДНИЙ</span>';

            return `
              <tr>
                <td><span class="badge bg-body-secondary text-
body border">${itm.type.toUpperCase()}</span></td>
                <td class="font-monospace text-body">${itm.val
ue}</td>
                <td class="text-center">${riskBadge}</td>
                <td class="text-secondary small">${itm.risk_re
ason}</td>
              </tr>
            `;
          }).join('');
        }
      }
    } catch (err) {
      console.error('[DefenderTab] Error loading exclusions:',
 err);
    }
  }

  async function loadThreats() {
    try {
      const threats = await fetchJSON('/api/v1/defender/threat
s?limit=50');
      currentThreats = Array.isArray(threats) ? threats : [];
      const countBadge = document.getElementById('def-threats-
count-badge');
      const countVal = document.getElementById('def-threats-va
l');
      const countSub = document.getElementById('def-threats-su
b');
      const tbody = document.getElementById('def-threats-tbody
');
```

```
415
416       if (countBadge) countBadge.textContent = `${currentThrea
    ts.length} записей`;
417       if (countVal) countVal.textContent = currentThreats.leng
    th;
418
419       const quarantined = currentThreats.filter(t => (t.status
     || '').toLowerCase().includes('quarantine')).length;
420       if (countSub) countSub.textContent = `Карантин: ${quaran
    tined}`;
421
422       if (tbody) {
423         if (currentThreats.length === 0) {
424           tbody.innerHTML = '<tr><td colspan="6" class="text-c
    enter text-muted py-3">Активных и архивных угроз не зафиксиров
    ано.</td></tr>';
425         } else {
426           tbody.innerHTML = currentThreats.map((t, idx) => `
427             <tr class="def-clickable-row" data-threat-idx="${i
    dx}" title="Нажмите для просмотра подробностей">
428               <td class="fw-semibold text-danger">${t.threat_n
    ame}</td>
429               <td><span class="badge bg-body-secondary text-bo
    dy border">${t.category}</span></td>
430               <td class="text-center"><span class="badge bg-wa
    rning text-dark">${(t.severity || 'Unknown').toUpperCase()}</s
    pan></td>
431               <td><span class="badge bg-success">${t.status}</
    span></td>
432               <td class="text-secondary small font-monospace">
    ${t.last_detection_time || t.initial_detection_time || '-'}</t
    d>
433               <td class="text-secondary font-monospace small t
    ext-truncate" style="max-width: 280px;">${(t.resources || []).
    join(', ')}</td>
434             </tr>
435           `).join('');
436
```

```
437           tbody.querySelectorAll('.def-clickable-row').forEach
(row => {
438             row.addEventListener('click', () => {
439               const idx = parseInt(row.getAttribute('data-thre
at-idx'), 10);
440               if (!isNaN(idx) && currentThreats[idx]) {
441                 showThreatDetail(currentThreats[idx]);
442               }
443             });
444           });
445         }
446       }
447     } catch (err) {
448       console.error('[DefenderTab] Error loading threats:', er
r);
449     }
450   }
451
452   async function loadProcesses() {
453     try {
454       const chains = await fetchJSON('/api/v1/defender/process
-tree');
455       const badge = document.getElementById('def-procs-count-b
adge');
456       const tbody = document.getElementById('def-procs-tbody')
;
457
458       if (badge) badge.textContent = `${chains.length} аномали
й`;
459
460       if (tbody) {
461         if (chains.length === 0) {
462           tbody.innerHTML = '<tr><td colspan="5" class="text-c
enter text-muted py-3">Подозрительных связей процессов и filel
ess индикаторов не обнаружено.</td></tr>';
463         } else {
464           tbody.innerHTML = chains.map(c => `
465             <tr>
```

```
              <td class="font-monospace text-warning">${c.pare
nt_process} (PID: ${c.parent_pid || '-'})</td>
              <td class="font-monospace text-danger fw-semibol
d">${c.child_process}</td>
              <td class="text-center text-secondary">${c.child
_pid || '-'}</td>
              <td class="text-center"><span class="badge def-b
adge-critical">${c.severity.toUpperCase()}</span></td>
              <td class="text-secondary small">
                <div>${c.reason}</div>
                ${c.command_line ? `<div class="font-monospace
 text-muted mt-1" style="font-size: 0.68rem;">cmd: ${c.command
_line}</div>` : ''}
              </td>
            </tr>
          `).join('');
        }
      }
    } catch (err) {
      console.error('[DefenderTab] Error loading process chain
s:', err);
    }
  }

  async function loadEvents() {
    try {
      const events = await fetchJSON('/api/v1/defender/events?
limit=50');
      currentEvents = Array.isArray(events) ? events : [];
      const badge = document.getElementById('def-events-count-
badge');
      const tbody = document.getElementById('def-events-tbody'
);

      if (badge) badge.textContent = `${currentEvents.length}
событий`;

      if (tbody) {
```

```
        if (currentEvents.length === 0) {
          tbody.innerHTML = '<tr><td colspan="5" class="text-c
enter text-muted py-3">Событий в журнале Defender Operational
не обнаружено.</td></tr>';
        } else {
          tbody.innerHTML = currentEvents.map((e, idx) => `
            <tr class="def-clickable-row" data-event-idx="${id
x}" title="Нажмите для просмотра подробностей">
              <td class="font-monospace fw-semibold text-info"
>${e.event_id}</td>
              <td class="text-secondary small font-monospace t
ext-nowrap">${e.timestamp}</td>
              <td><span class="badge ${e.level.toLowerCase().i
ncludes('err') || e.level.toLowerCase().includes('crit') ? 'bg
-danger' : (e.level.toLowerCase().includes('warn') ? 'bg-warni
ng text-dark' : 'bg-secondary')}">${e.level}</span></td>
              <td><span class="badge bg-body-secondary text-bo
dy border">${e.category}</span></td>
              <td class="text-body small" style="word-break: b
reak-word;">${e.message}</td>
            </tr>
          `).join('');

          tbody.querySelectorAll('.def-clickable-row').forEach
(row => {
            row.addEventListener('click', () => {
              const idx = parseInt(row.getAttribute('data-even
t-idx'), 10);
              if (!isNaN(idx) && currentEvents[idx]) {
                showEventDetail(currentEvents[idx]);
              }
            });
          });
        }
      }
    } catch (err) {
      console.error('[DefenderTab] Error loading events:', err
);
```

```
518     }
519   }
520
521   async function loadDiagnostics() {
522     try {
523       const rep = await fetchJSON('/api/v1/defender/diagnostic
s');
524
525       const scoreVal = document.getElementById('def-score-val'
);
526       const scoreSub = document.getElementById('def-score-sub'
);
527       const diagScoreDisplay = document.getElementById('def-di
ag-score-display');
528       const diagSummaryDisplay = document.getElementById('def-
diag-summary-display');
529       const recsContainer = document.getElementById('def-diag-
recs-container');
530       const critsContainer = document.getElementById('def-diag
-criticals-container');
531
532       const scoreColor = rep.security_score >= 80 ? 'text-succ
ess' : (rep.security_score >= 50 ? 'text-warning' : 'text-dang
er');
533
534       if (scoreVal) {
535         scoreVal.textContent = `${rep.security_score}/100`;
536         scoreVal.className = `def-card-value ${scoreColor}`;
537       }
538       if (scoreSub) scoreSub.textContent = rep.status_summary;
539
540       if (diagScoreDisplay) {
541         diagScoreDisplay.textContent = `${rep.security_score}/
100`;
542         diagScoreDisplay.className = `display-4 fw-bold ${scor
eColor}`;
543       }
544       if (diagSummaryDisplay) diagSummaryDisplay.textContent =
```

```
 rep.status_summary;

      if (recsContainer) {
        if (rep.recommendations && rep.recommendations.length > 0) {
          recsContainer.innerHTML = `
            <ol class="ps-3 mb-0">
              ${rep.recommendations.map(r => `<li class="py-1 text-body">${r}</li>`).join('')}
            </ol>
          `;
        } else {
          recsContainer.innerHTML = '<div class="text-success">Все рекомендации соблюдены. Защита оптимальна.</div>';
        }
      }

      if (critsContainer) {
        if (rep.critical_findings && rep.critical_findings.length > 0) {
          critsContainer.innerHTML = `
            <ul class="ps-3 mb-0 text-danger">
              ${rep.critical_findings.map(c => `<li class="py-1">${c}</li>`).join('')}
            </ul>
          `;
        } else {
          critsContainer.innerHTML = '<div class="text-muted">Критических уязвимостей не зафиксировано.</div>';
        }
      }

    } catch (err) {
      console.error('[DefenderTab] Error loading diagnostics:', err);
    }
  }

```

```
  const activeTaskPollers = {};

  async function pollTask(taskId, taskLabel, triggerBtn) {
    if (!taskId) return;
    if (activeTaskPollers[taskId]) return;

    if (triggerBtn) {
      triggerBtn.disabled = true;
      triggerBtn.dataset.originalHtml = triggerBtn.innerHTML;
      triggerBtn.innerHTML = `<span class="spinner-border spin
ner-border-sm me-1" role="status" aria-hidden="true"></span> $
{taskLabel}...`;
    }

    showAlert(`  Запущен фоновый процесс: ${taskLabel} (ID: ${
taskId})`);

    const interval = 2500;
    const maxAttempts = 240; // до 10 минут
    let attempts = 0;

    activeTaskPollers[taskId] = setInterval(async () => {
      if (window.isTabActive && !window.isTabActive('tab-defen
der')) return;
      attempts++;
      try {
        const task = await fetchJSON(`/api/v1/defender/tasks/$
{taskId}`);
        if (!task || task.status === 'running' || task.status
=== 'pending') {
          if (attempts >= maxAttempts) {
            clearInterval(activeTaskPollers[taskId]);
            delete activeTaskPollers[taskId];
            if (triggerBtn && triggerBtn.dataset.originalHtml)
 {
              triggerBtn.innerHTML = triggerBtn.dataset.origin
alHtml;
              triggerBtn.disabled = false;
```

```
            }
            showAlert(`Таймаут ожидания задачи ${taskLabel} (фоновый процесс продолжается в Windows).`, true);
          }
          return;
        }

        clearInterval(activeTaskPollers[taskId]);
        delete activeTaskPollers[taskId];

        if (triggerBtn && triggerBtn.dataset.originalHtml) {
          triggerBtn.innerHTML = triggerBtn.dataset.originalHtml;
          triggerBtn.disabled = false;
        }

        const durationInfo = task.duration_seconds ? ` [${task.duration_seconds}c]` : '';
        if (task.status === 'completed' && task.success) {
          let extraMsg = '';
          if (task.windows_event) {
            extraMsg = ` (Windows Event ${task.windows_event.event_id}: ${task.windows_event.category})`;
          }
          showAlert(`✅ ${taskLabel} успешно завершено${durationInfo}! ${task.message || ''}${extraMsg}`);
        } else {
          showAlert(`❌ ${taskLabel} завершилось ошибкой${durationInfo}: ${task.error || task.message || 'Сбой выполнения'}`, true);
        }

        await refreshAll();
      } catch (err) {
        console.error(`[DefenderTab] Polling error for task ${taskId}:`, err);
      }
    }, interval);
```

```
  }

  function setupActions() {
    const btnQuick = document.getElementById('btn-def-quick-sc
an');
    if (btnQuick) {
      btnQuick.onclick = async () => {
        try {
          const res = await fetchJSON('/api/v1/defender/scan',
 {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ scan_type: 'quick' }),
          });
          if (res && res.task_id) {
            pollTask(res.task_id, 'Быстрое сканирование', btnQ
uick);
          } else {
            showAlert(`Быстрое сканирование: ${res.message ||
'Запущено'}`);
            await loadDefenderStatus();
          }
        } catch (err) {
          showAlert(`Ошибка запуска сканирования: ${err.messag
e}`, true);
        }
      };
    }

    const btnFull = document.getElementById('btn-def-full-scan
');
    if (btnFull) {
      btnFull.onclick = async () => {
        try {
          const res = await fetchJSON('/api/v1/defender/scan',
 {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
```

```
            body: JSON.stringify({ scan_type: 'full' }),
          });
          if (res && res.task_id) {
            pollTask(res.task_id, 'Полное сканирование системы
', btnFull);
          } else {
            showAlert(`Полное сканирование: ${res.message || '
Запущено'}`);
          }
        } catch (err) {
          showAlert(`Ошибка полного сканирования: ${err.messag
e}`, true);
        }
      };
    }

    const btnUpdate = document.getElementById('btn-def-update-
sigs');
    if (btnUpdate) {
      btnUpdate.onclick = async () => {
        try {
          const res = await fetchJSON('/api/v1/defender/update
-signatures', { method: 'POST' });
          if (res && res.task_id) {
            pollTask(res.task_id, 'Обновление баз сигнатур Def
ender', btnUpdate);
          } else {
            showAlert(`Обновление сигнатур: ${res.message || '
Запущено'}`);
            await loadDefenderStatus();
          }
        } catch (err) {
          showAlert(`Ошибка обновления баз: ${err.message}`, t
rue);
        }
      };
    }

```

```
    const btnRefresh = document.getElementById('btn-def-refres
h');
    if (btnRefresh) {
      btnRefresh.onclick = async () => {
        showAlert('Обновление данных Defender...');
        await refreshAll();
      };
    }

    const btnAiDiag = document.getElementById('btn-def-ai-diag
');
    if (btnAiDiag) {
      btnAiDiag.onclick = () => {
        const diagTabBtn = document.getElementById('def-tab-di
ag-btn');
        if (diagTabBtn) {
          const tab = new bootstrap.Tab(diagTabBtn);
          tab.show();
        }
      };
    }

    const btnCopyMsg = document.getElementById('btn-def-copy-m
sg');
    if (btnCopyMsg) {
      btnCopyMsg.onclick = () => {
        const rawEl = document.getElementById('def-detail-raw-
msg');
        if (rawEl && rawEl.textContent) {
          navigator.clipboard.writeText(rawEl.textContent).the
n(() => {
            const originalHtml = btnCopyMsg.innerHTML;
            btnCopyMsg.innerHTML = '<i class="bi bi-check2"></
i> <span>Скопировано!</span>';
            btnCopyMsg.classList.remove('btn-outline-secondary
');
            btnCopyMsg.classList.add('btn-success');
            setTimeout(() => {
```

```
              btnCopyMsg.innerHTML = originalHtml;
              btnCopyMsg.classList.remove('btn-success');
              btnCopyMsg.classList.add('btn-outline-secondary'
);
            }, 2000);
          }).catch(err => {
            console.error('[DefenderTab] Clipboard copy failed
:', err);
          });
        }
      };
    }
  }

  async function refreshAll() {
    await Promise.all([
      loadDefenderStatus(),
      loadASRRules(),
      loadCFA(),
      loadExclusions(),
      loadThreats(),
      loadProcesses(),
      loadEvents(),
      loadDiagnostics(),
    ]);
  }

  window.initDefenderTab = async function() {
    console.log('[DefenderTab] Initializing Defender Security
Center Tab...');
    setupActions();
    if (window.isTabActive ? window.isTabActive('tab-defender'
) : false) {
      await refreshAll();
    }
  };

  window.activateDefenderTab = async function() {
```

```
    console.log('[DefenderTab] Activating Defender Security Center Tab...');
    await refreshAll();
  };

  // Auto initialize if container exists
  if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', () => {
      if (document.getElementById('def-shields-tbody')) {
        window.initDefenderTab();
      }
    });
  } else {
    if (document.getElementById('def-shields-tbody')) {
      window.initDefenderTab();
    }
  }

})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Disk Speed Benchmark (Di
skSpd) View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса тестирования скорости диск
ов (CrystalDiskMark стиль на базе DiskSpd).

Usage Examples:
    Web Route:
      /windows/api/webgui/disk_speed_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/disk_speed_tab/index.ht
ml" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/disk_speed_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<div class="container-fluid py-2 px-3 text-white" id="disk-spe
ed-container">
<!-- Header / Breadcrumbs & Status -->
<div class="d-flex align-items-center justify-content-between
mb-3 pb-2 border-bottom border-secondary flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<span class="fs-4 text-warning fw-bold d-flex align-items-cent
er gap-2">
```

```
<i class="bi bi-speedometer2 text-danger"></i>
<span>Бенчмарк дисков (DiskSpd / CDM)</span>
</span>
<span class="badge bg-dark border border-secondary text-info p
x-2 py-1" id="bench-engine-badge">
<i class="bi bi-cpu"></i> Microsoft DiskSpd
      </span>
</div>
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm btn-outline-info d-flex align-items-
center gap-1.5" id="bench-refresh-targets-btn" title="Обновить
 список накопителей">
<i class="bi bi-arrow-clockwise"></i>
<span>Обновить диски</span>
</button>
<button class="btn btn-sm btn-outline-secondary d-flex align-i
tems-center gap-1.5" id="bench-copy-md-btn" title="Скопировать
 результаты в Markdown">
<i class="bi bi-clipboard"></i>
<span>Копировать MD</span>
</button>
</div>
</div>
<!-- Safe Mode Notice Banner -->
<div class="alert alert-dark border-secondary d-flex align-ite
ms-center justify-content-between py-2 px-3 mb-3 small">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-shield-check text-success fs-5"></i>
<span>
<strong>Безопасный режим:</strong> Тестирование выполняется на
 временном файле
        <code class="text-info" id="bench-safe-filepath">C:\AI
-Windows-Admin\data\benchmark\diskspd-test.dat</code>.
        После завершения теста файл автоматически удаляется. П
рямая запись в физические сектора исключена.
      </span>
</div>
<span class="badge bg-success-subtle text-success border borde
```

```
r-success-subtle px-2 py-1">SafeOps Active</span>
</div>
<!-- Main Benchmark Dashboard: Controls & CDM Table -->
<div class="row g-3 mb-3">
<!-- Left Panel: Controls & CrystalDiskMark Style Buttons -->
<div class="col-12 col-xl-4">
<div class="card bg-dark border-secondary shadow-sm h-100 p-3"
 data-panel-id="panel-disk-speed-parametry-testirovaniya" id="
panel-disk-speed-parametry-testirovaniya">
<h6 class="text-primary fw-bold mb-3 d-flex align-items-center
 gap-2">
<i class="bi bi-sliders"></i> Параметры тестирования
        </h6>
<!-- Drive Selector -->
<div class="mb-3">
<label class="form-label small text-muted fw-semibold" for="be
nch-drive-select">
<i class="bi bi-hdd"></i> Целевой накопитель:
          </label>
<select class="form-select form-select-sm bg-dark text-white b
order-secondary" id="bench-drive-select">
<option value="C:">C: [Системный] — загрузка...</option>
</select>
<div class="d-flex justify-content-between text-muted small mt
-1" style="font-size: 0.75rem;">
<span id="bench-drive-fs">ФС: NTFS</span>
<span id="bench-drive-free">Свободно: -- ГБ</span>
</div>
</div>
<!-- File Size & Duration -->
<div class="row g-2 mb-3">
<div class="col-6">
<label class="form-label small text-muted fw-semibold" for="be
nch-size-select">
<i class="bi bi-file-earmark-binary"></i> Размер файла:
            </label>
<select class="form-select form-select-sm bg-dark text-white b
order-secondary" id="bench-size-select">
```

```
<option value="64">64 MiB (Быстро)</option>
<option value="128">128 MiB</option>
<option selected="" value="256">256 MiB (Стандарт)</option>
<option value="512">512 MiB</option>
<option value="1024">1 GiB</option>
<option value="2048">2 GiB</option>
</select>
</div>
<div class="col-6">
<label class="form-label small text-muted fw-semibold" for="be
nch-duration-select">
<i class="bi bi-stopwatch"></i> Время прогона:
            </label>
<select class="form-select form-select-sm bg-dark text-white b
order-secondary" id="bench-duration-select">
<option value="2">2 сек (Экспресс)</option>
<option selected="" value="3">3 сек (Оптимально)</option>
<option value="5">5 сек (Точно)</option>
<option value="10">10 сек (Crystal)</option>
</select>
</div>
</div>
<!-- Test Mode: Both / Read / Write -->
<div class="mb-3">
<label class="form-label small text-muted fw-semibold">
<i class="bi bi-arrow-left-right"></i> Режим нагрузки:
          </label>
<div class="btn-group w-100" role="group">
<input checked="" class="btn-check" id="bench-type-both" name=
"bench-type-radio" type="radio" value="both"/>
<label class="btn btn-sm btn-outline-primary" for="bench-type-
both">Чтение + Запись</label>
<input class="btn-check" id="bench-type-read" name="bench-type
-radio" type="radio" value="read"/>
<label class="btn btn-sm btn-outline-info" for="bench-type-rea
d">Только Read</label>
<input class="btn-check" id="bench-type-write" name="bench-typ
e-radio" type="radio" value="write"/>
```

```
<label class="btn btn-sm btn-outline-warning" for="bench-type-write">Только Write</label>
</div>
</div>
<!-- CDM Action Buttons Grid -->
<div class="mt-auto pt-2 border-top border-secondary">
<div class="d-grid gap-2 mb-2">
<button class="btn btn-success fw-bold py-2 shadow d-flex align-items-center justify-content-center gap-2" id="bench-run-all-btn">
<i class="bi bi-play-circle-fill fs-5"></i>
<span>ЗАПУСТИТЬ ВСЕ ТЕСТЫ (ALL)</span>
</button>
<button class="btn btn-danger btn-sm d-none d-flex align-items-center justify-content-center gap-2" id="bench-cancel-btn">
<i class="bi bi-stop-circle"></i>
<span>Прервать тестирование</span>
</button>
</div>
<div class="row g-1 text-center small">
<div class="col-6">
<button class="btn btn-sm btn-outline-secondary w-100 py-1" id="bench-run-seq1m-q8" title="Тест SEQ1M Q8T1">
                SEQ1M Q8
              </button>
</div>
<div class="col-6">
<button class="btn btn-sm btn-outline-secondary w-100 py-1" id="bench-run-seq1m-q1" title="Тест SEQ1M Q1T1">
                SEQ1M Q1
              </button>
</div>
<div class="col-6">
<button class="btn btn-sm btn-outline-secondary w-100 py-1" id="bench-run-rnd4k-q32" title="Тест RND4K Q32T16">
                RND4K Q32
              </button>
</div>
```

```
<div class="col-6">
<button class="btn btn-sm btn-outline-secondary w-100 py-1" id="bench-run-rnd4k-q1" title="Тест RND4K Q1T1">
                RND4K Q1
              </button>
</div>
</div>
</div>
</div>
</div>
<!-- Right Panel: CrystalDiskMark Scoreboard Table -->
<div class="col-12 col-xl-8">
<div class="card bg-dark border-secondary shadow-sm h-100" data-panel-id="panel-disk-speed-rezultaty-crystaldiskmark" id="panel-disk-speed-rezultaty-crystaldiskmark">
<div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">
<span class="fw-bold d-flex align-items-center gap-2">
<i class="bi bi-table text-info"></i>
<span>Результаты CrystalDiskMark</span>
</span>
<span class="badge bg-secondary font-monospace" id="bench-status-badge">Готов к тесту</span>
</div>
<div class="card-body p-3 d-flex flex-column justify-content-between">
<!-- Live Progress Bar -->
<div class="mb-3 d-none" id="bench-progress-wrapper">
<div class="d-flex justify-content-between small text-muted mb-1">
<span id="bench-progress-msg">Инициализация DiskSpd...</span>
<span class="fw-bold text-info" id="bench-progress-pct">0%</span>
</div>
<div class="progress bg-black" style="height: 10px;">
<div class="progress-bar progress-bar-striped progress-bar-animated bg-success" id="bench-progress-bar" style="width: 0%;"></div>
```

```
</div>
</div>
<!-- The Table -->
<div class="table-responsive mb-3">
<table class="table table-dark table-bordered align-middle text-center mb-0" id="bench-cdm-table" style="font-family: Consolas, monospace;">
<thead>
<tr class="text-secondary bg-black" style="font-size: 0.85rem;">
<th class="text-start py-2" style="width: 25%;">Тест / Профиль</th>
<th class="text-end text-info py-2" style="width: 20%;"><i class="bi bi-download"></i> Read (MB/s)</th>
<th class="text-end text-warning py-2" style="width: 20%;"><i class="bi bi-upload"></i> Write (MB/s)</th>
<th class="text-end text-success py-2" style="width: 18%;">IOPS (R/W)</th>
<th class="text-end text-muted py-2" style="width: 17%;">Латентность (µs)</th>
</tr>
</thead>
<tbody style="font-size: 1.05rem; font-weight: bold;">
<!-- SEQ1M Q8T1 -->
<tr id="row-seq1m_q8t1">
<td class="text-start">
<div class="text-white">SEQ1M <span class="badge bg-primary text-white" style="font-size: 0.7rem;">Q8T1</span></div>
<div class="text-muted fw-normal" style="font-size: 0.7rem;">1 MiB, Очередь 8, Поток 1</div>
</td>
<td class="text-end text-info fs-5" id="val-seq1m_q8t1-read">0.00</td>
<td class="text-end text-warning fs-5" id="val-seq1m_q8t1-write">0.00</td>
<td class="text-end text-success small fw-normal" id="val-seq1m_q8t1-iops">-- / --</td>
<td class="text-end text-muted small fw-normal" id="val-seq1m_
```

```
q8t1-lat">-- µs</td>
</tr>
<!-- SEQ1M Q1T1 -->
<tr id="row-seq1m_q1t1">
<td class="text-start">
<div class="text-white">SEQ1M <span class="badge bg-primary te
xt-white" style="font-size: 0.7rem;">Q1T1</span></div>
<div class="text-muted fw-normal" style="font-size: 0.7rem;">1
 MiB, Очередь 1, Поток 1</div>
</td>
<td class="text-end text-info fs-5" id="val-seq1m_q1t1-read">0
.00</td>
<td class="text-end text-warning fs-5" id="val-seq1m_q1t1-writ
e">0.00</td>
<td class="text-end text-success small fw-normal" id="val-seq1
m_q1t1-iops">-- / --</td>
<td class="text-end text-muted small fw-normal" id="val-seq1m_
q1t1-lat">-- µs</td>
</tr>
<!-- RND4K Q32T16 -->
<tr id="row-rnd4k_q32t16">
<td class="text-start">
<div class="text-white">RND4K <span class="badge bg-info text-
dark" style="font-size: 0.7rem;">Q32T16</span></div>
<div class="text-muted fw-normal" style="font-size: 0.7rem;">4
 KiB, Очередь 32, Поток 1</div>
</td>
<td class="text-end text-info fs-5" id="val-rnd4k_q32t16-read"
>0.00</td>
<td class="text-end text-warning fs-5" id="val-rnd4k_q32t16-wr
ite">0.00</td>
<td class="text-end text-success small fw-normal" id="val-rnd4
k_q32t16-iops">-- / --</td>
<td class="text-end text-muted small fw-normal" id="val-rnd4k_
q32t16-lat">-- µs</td>
</tr>
<!-- RND4K Q1T1 -->
<tr id="row-rnd4k_q1t1">
```

```
<td class="text-start">
<div class="text-white">RND4K <span class="badge bg-info text-dark" style="font-size: 0.7rem;">Q1T1</span></div>
<div class="text-muted fw-normal" style="font-size: 0.7rem;">4 KiB, Очередь 1, Поток 1</div>
</td>
<td class="text-end text-info fs-5" id="val-rnd4k_q1t1-read">0.00</td>
<td class="text-end text-warning fs-5" id="val-rnd4k_q1t1-write">0.00</td>
<td class="text-end text-success small fw-normal" id="val-rnd4k_q1t1-iops">-- / --</td>
<td class="text-end text-muted small fw-normal" id="val-rnd4k_q1t1-lat">-- µs</td>
</tr>
</tbody>
</table>
</div>
<!-- Bottom Summary Footer -->
<div class="d-flex align-items-center justify-content-between p-2 bg-black rounded border border-secondary text-muted small">
<div>
<span>Тестируемый диск: <strong class="text-white" id="bench-summary-disk">C:</strong></span>
<span class="ms-3">Размер: <strong class="text-white" id="bench-summary-size">256 MB</strong></span>
</div>
<div>
<span>Время замера: <strong class="text-white" id="bench-summary-time">--</strong></span>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- Charts & History Section -->
```

```
<div class="row g-3">
<!-- Chart: Throughput (MB/s) -->
<div class="col-12 col-lg-6">
<div class="card bg-dark border-secondary shadow-sm p-3 h-100" data-panel-id="panel-disk-speed-propusknaya-sposobnost-mbs" id="panel-disk-speed-propusknaya-sposobnost-mbs">
<h6 class="text-primary fw-bold mb-2 d-flex align-items-center gap-2">
<i class="bi bi-bar-chart-fill"></i> Пропускная способность (MB/s)
        </h6>
<div style="height: 220px; position: relative;">
<canvas id="bench-mb-chart"></canvas>
</div>
</div>
</div>
<!-- History Table -->
<div class="col-12 col-lg-6">
<div class="card bg-dark border-secondary shadow-sm p-3 h-100" data-panel-id="panel-disk-speed-istoriya-zamerov" id="panel-disk-speed-istoriya-zamerov">
<div class="d-flex align-items-center justify-content-between mb-2">
<h6 class="text-primary fw-bold mb-0 d-flex align-items-center gap-2">
<i class="bi bi-clock-history"></i> История замеров
          </h6>
<span class="badge bg-secondary" id="bench-history-count">0 записей</span>
</div>
<div class="table-responsive" style="max-height: 220px; overflow-y: auto;">
<table class="table table-dark table-sm table-hover align-middle mb-0 small" id="bench-history-table">
<thead>
<tr class="text-secondary">
<th>Дата / Время</th>
<th>Диск</th>
```

```
<th>SEQ Read / Write</th>
<th>RND 4K IOPS</th>
<th class="text-end">Действие</th>
</tr>
</thead>
<tbody id="bench-history-tbody">
<tr>
<td class="text-center text-muted py-3" colspan="5">История пу
ста. Запустите тест.</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Disk Speed Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом бенчмарка дисковой подсистемы (DiskSpd).
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/disk_speed_tab/main.js?v=20261004_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/disk_speed_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

const registerTabPoller = window.registerTabPoller || function () {};
const isTabActive = window.isTabActive || function() { return true; };

let mbChartInstance = null;
let currentTaskId = null;
let pollTimer = null;
let cachedTargets = [];
let lastResult = null;

```

```
async function initDiskSpeedTab() {
  if (isInitialized) return;
  isInitialized = true;
  initChart();
  setupEventListeners();
  if (window.isTabActive ? window.isTabActive('tab-disk-speed') : false) {
    await checkEngineStatus();
    await loadTargets();
    await loadHistory();
  }
}

async function activateDiskSpeedTab() {
  await checkEngineStatus();
  await loadTargets();
  await loadHistory();
}

window.activateDiskSpeedTab = activateDiskSpeedTab;

/**
 * Проверка статуса утилиты DiskSpd.
 */
async function checkEngineStatus() {
  try {
    const res = await fetch('/api/v1/storage/benchmark/engine');
    if (!res.ok) return;
    const data = await res.json();
    const badge = document.getElementById('bench-engine-badge');
    if (badge) {
      if (data.available) {
        badge.className = 'badge bg-success-subtle text-success border border-success-subtle px-2 py-1';
        badge.innerHTML = '<i class="bi bi-check-circle-fill"></i> DiskSpd Ready';
```

```
      } else {
        badge.className = 'badge bg-warning-subtle text-warnin
g border border-warning-subtle px-2 py-1';
        badge.innerHTML = '<i class="bi bi-exclamation-triangl
e"></i> DiskSpd не установлен';
      }
    }
  } catch (err) {
    console.debug('[DiskSpeed] Ошибка проверки движка:', err);
  }
}

/**
 * Загрузка списка доступных дисков и разделов.
 */
async function loadTargets() {
  try {
    const res = await fetch('/api/v1/storage/benchmark/targets
');
    if (!res.ok) return;
    cachedTargets = await res.json();

    const select = document.getElementById('bench-drive-select
');
    if (!select) return;

    select.innerHTML = '';
    cachedTargets.forEach(t => {
      const opt = document.createElement('option');
      opt.value = t.drive_letter;
      const tag = t.is_system ? ' [Системный]' : '';
      opt.textContent = `${t.drive_letter} (${t.label || t.dri
ve_letter})${tag} — ${t.free_gb} ГБ свободно`;
      select.appendChild(opt);
    });

    updateDriveDetails();
  } catch (err) {
```

```
    console.error('[DiskSpeed] Ошибка загрузки списка накопителей:', err);
  }
}

/**
 * Обновление информации о выбранном диске и пути безопасного файла.
 */
function updateDriveDetails() {
  const select = document.getElementById('bench-drive-select');
  if (!select) return;
  const drive = select.value;
  const target = cachedTargets.find(t => t.drive_letter === drive);

  const fsEl = document.getElementById('bench-drive-fs');
  const freeEl = document.getElementById('bench-drive-free');
  const pathEl = document.getElementById('bench-safe-filepath');
  const summaryDisk = document.getElementById('bench-summary-disk');

  if (target) {
    if (fsEl) fsEl.textContent = `ФС: ${target.fs_type}`;
    if (freeEl) freeEl.textContent = `Свободно: ${target.free_gb} ГБ`;
    if (pathEl) pathEl.textContent = `${target.recommended_dir}\\diskspd-test.dat`;
    if (summaryDisk) summaryDisk.textContent = `${target.drive_letter} (${target.fs_type})`;
  }
}

/**
 * Настройка событий элементов управления.
 */
```

```
function setupEventListeners() {
  const driveSelect = document.getElementById('bench-drive-select');
  if (driveSelect) {
    driveSelect.addEventListener('change', updateDriveDetails);
  }

  const refreshBtn = document.getElementById('bench-refresh-targets-btn');
  if (refreshBtn) {
    refreshBtn.addEventListener('click', async () => {
      await loadTargets();
      if (window.toast) window.toast.info('Обновлено', 'Список накопителей обновлен');
    });
  }

  const runAllBtn = document.getElementById('bench-run-all-btn');
  if (runAllBtn) {
    runAllBtn.addEventListener('click', () => {
      startBenchmark(['seq1m_q8t1', 'seq1m_q1t1', 'rnd4k_q32t16', 'rnd4k_q1t1']);
    });
  }

  const cancelBtn = document.getElementById('bench-cancel-btn');
  if (cancelBtn) {
    cancelBtn.addEventListener('click', cancelBenchmark);
  }

  // Одиночные кнопки профилей
  const pMap = {
    'bench-run-seq1m-q8': ['seq1m_q8t1'],
    'bench-run-seq1m-q1': ['seq1m_q1t1'],
    'bench-run-rnd4k-q32': ['rnd4k_q32t16'],
```

```
    'bench-run-rnd4k-q1': ['rnd4k_q1t1'],
  };

  Object.entries(pMap).forEach(([btnId, profiles]) => {
    const btn = document.getElementById(btnId);
    if (btn) {
      btn.addEventListener('click', () => startBenchmark(profi
les));
    }
  });

  const copyMdBtn = document.getElementById('bench-copy-md-btn
');
  if (copyMdBtn) {
    copyMdBtn.addEventListener('click', copyResultsAsMarkdown)
;
  }
}

/**
 * Запуск тестирования производительности.
 * @param {Array<string>} profiles - Список профилей для выпол
нения.
 */
async function startBenchmark(profiles) {
  const driveSelect = document.getElementById('bench-drive-sel
ect');
  const sizeSelect = document.getElementById('bench-size-selec
t');
  const durSelect = document.getElementById('bench-duration-se
lect');
  const typeRadio = document.querySelector('input[name="bench-
type-radio"]:checked');

  const payload = {
    target_drive: driveSelect ? driveSelect.value : 'C:',
    file_size_mb: sizeSelect ? parseInt(sizeSelect.value, 10)
: 256,
```

```
    duration_sec: durSelect ? parseInt(durSelect.value, 10) : 3,
    test_type: typeRadio ? typeRadio.value : 'both',
    profiles: profiles,
    disable_cache: true,
    delete_test_file: true,
  };

  setUiRunning(true);

  const summarySize = document.getElementById('bench-summary-size');
  if (summarySize) summarySize.textContent = `${payload.file_size_mb} MB`;

  try {
    const res = await fetch('/api/v1/storage/benchmark/start', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(payload),
    });

    if (!res.ok) {
      const err = await res.json();
      throw new Error(err.detail || 'Не удалось запустить бенчмарк');
    }

    const task = await res.json();
    currentTaskId = task.task_id;
    pollBenchmarkTask();
  } catch (err) {
    setUiRunning(false);
    if (window.toast) {
      window.toast.error('Ошибка старта', err.message);
    } else {
      alert(`Ошибка: ${err.message}`);
```

```
    }
  }
}

function pollBenchmarkTask() {
  if (pollTimer) clearInterval(pollTimer);

  pollTimer = setInterval(async () => {
    if (!currentTaskId) {
      clearInterval(pollTimer);
      return;
    }
    if (window.isTabActive && !window.isTabActive('tab-disk-sp
eed')) return;

    try {
      const res = await fetch(`/api/v1/storage/benchmark/statu
s/${currentTaskId}`);
      if (!res.ok) return;

      const task = await res.json();
      updateProgress(task.progress_percent, task.destination);

      if (task.status === 'completed') {
        clearInterval(pollTimer);
        setUiRunning(false);
        if (window.toast) window.toast.success('Тест завершен'
, 'Результаты бенчмарка готовы');
        await loadLatestHistoryResult();
        await loadHistory();
      } else if (task.status === 'failed') {
        clearInterval(pollTimer);
        setUiRunning(false);
        if (window.toast) window.toast.error('Ошибка теста', t
ask.error_message || 'Тестирование завершилось с ошибкой');
      } else if (task.status === 'cancelled') {
        clearInterval(pollTimer);
        setUiRunning(false);
```

```
        if (window.toast) window.toast.warning('Отменено', 'Тест был отменен');
      }
    } catch (e) {
      console.debug('[DiskSpeed] Ошибка опроса задачи:', e);
    }
  }, 1000);
}

/**
 * Отмена запущенного бенчмарка.
 */
async function cancelBenchmark() {
  if (!currentTaskId) return;
  try {
    await fetch(`/api/v1/storage/benchmark/cancel/${currentTaskId}`, { method: 'POST' });
    if (window.toast) window.toast.info('Отмена', 'Запрос на остановку отправлен...');
  } catch (err) {
    console.error('[DiskSpeed] Ошибка отмены:', err);
  }
}

/**
 * Управление состоянием элементов UI во время теста.
 * @param {boolean} isRunning - Флаг активности теста.
 */
function setUiRunning(isRunning) {
  const runAllBtn = document.getElementById('bench-run-all-btn');
  const cancelBtn = document.getElementById('bench-cancel-btn');
  const progressWrapper = document.getElementById('bench-progress-wrapper');
  const statusBadge = document.getElementById('bench-status-badge');

```

```
  if (runAllBtn) runAllBtn.disabled = isRunning;
  if (cancelBtn) {
    if (isRunning) {
      cancelBtn.classList.remove('d-none');
    } else {
      cancelBtn.classList.add('d-none');
    }
  }

  if (progressWrapper) {
    if (isRunning) {
      progressWrapper.classList.remove('d-none');
    } else {
      progressWrapper.classList.add('d-none');
    }
  }

  if (statusBadge) {
    if (isRunning) {
      statusBadge.className = 'badge bg-warning text-dark font
-monospace';
      statusBadge.textContent = 'Идет тестирование...';
    } else {
      statusBadge.className = 'badge bg-secondary font-monospa
ce';
      statusBadge.textContent = 'Готов к тесту';
    }
  }
}

/**
 * Обновление полосы прогресса.
 * @param {number} pct - Процент выполнения.
 * @param {string} msg - Сообщение фазы.
 */
function updateProgress(pct, msg) {
  const bar = document.getElementById('bench-progress-bar');
  const pctEl = document.getElementById('bench-progress-pct');
```

```
  const msgEl = document.getElementById('bench-progress-msg');

  const val = Math.max(0, Math.min(100, pct || 0));
  if (bar) bar.style.width = `${val}%`;
  if (pctEl) pctEl.textContent = `${val}%`;
  if (msgEl && msg) msgEl.textContent = msg;
}

/**
 * Загрузка последнего результата и заполнение таблицы CDM.
 */
async function loadLatestHistoryResult() {
  try {
    const res = await fetch('/api/v1/storage/benchmark/history?limit=1');
    if (!res.ok) return;
    const items = await res.json();
    if (!items || items.length === 0) return;

    const latest = items[0];
    lastResult = latest;

    const summaryTime = document.getElementById('bench-summary-time');
    if (summaryTime) summaryTime.textContent = latest.created_at;

    // SEQ1M Q8T1
    updateRowValues('seq1m_q8t1', latest.seq1m_read_mb_s, latest.seq1m_write_mb_s, '-- / --', '-- µs');
    // RND4K Q32T16
    updateRowValues('rnd4k_q32t16', latest.rnd4k_read_mb_s, latest.rnd4k_write_mb_s, `${Math.round(latest.rnd4k_read_iops)} / ${Math.round(latest.rnd4k_write_iops)}`, '-- µs');

    updateChartData([
      { label: 'SEQ1M Q8T1', read: latest.seq1m_read_mb_s, write: latest.seq1m_write_mb_s },
```

```
      { label: 'RND4K Q32T16', read: latest.rnd4k_read_mb_s, write: latest.rnd4k_write_mb_s },
    ]);
  } catch (err) {
    console.debug('[DiskSpeed] Ошибка загрузки свежих результатов:', err);
  }
}

/**
 * Обновление строки таблицы CDM.
 */
function updateRowValues(rowKey, readMb, writeMb, iops, lat) {
  const rEl = document.getElementById(`val-${rowKey}-read`);
  const wEl = document.getElementById(`val-${rowKey}-write`);
  const iopsEl = document.getElementById(`val-${rowKey}-iops`);
  const latEl = document.getElementById(`val-${rowKey}-lat`);

  if (rEl) rEl.textContent = (readMb || 0).toFixed(2);
  if (wEl) wEl.textContent = (writeMb || 0).toFixed(2);
  if (iopsEl) iopsEl.textContent = iops;
  if (latEl) latEl.textContent = lat;
}

/**
 * Инициализация графика Chart.js.
 */
function initChart() {
  const ctx = document.getElementById('bench-mb-chart');
  if (!ctx || typeof Chart === 'undefined') return;

  if (mbChartInstance) {
    mbChartInstance.destroy();
  }

  mbChartInstance = new Chart(ctx, {
    type: 'bar',
```

```
    data: {
      labels: ['SEQ1M Q8T1', 'SEQ1M Q1T1', 'RND4K Q32T16', 'RN
D4K Q1T1'],
      datasets: [
        {
          label: 'Read (MB/s)',
          data: [0, 0, 0, 0],
          backgroundColor: 'rgba(13, 202, 240, 0.75)',
          borderColor: '#0dcaf0',
          borderWidth: 1,
        },
        {
          label: 'Write (MB/s)',
          data: [0, 0, 0, 0],
          backgroundColor: 'rgba(255, 193, 7, 0.75)',
          borderColor: '#ffc107',
          borderWidth: 1,
        },
      ],
    },
    options: {
      responsive: true,
      maintainAspectRatio: false,
      scales: {
        y: {
          beginAtZero: true,
          grid: { color: '#334155' },
          ticks: { color: '#94a3b8' },
        },
        x: {
          grid: { display: false },
          ticks: { color: '#94a3b8' },
        },
      },
      plugins: {
        legend: { labels: { color: '#f8fafc' } },
      },
    },
```

```
  });
}

/**
 * Обновление данных графика.
 */
function updateChartData(points) {
  if (!mbChartInstance) return;
  const labels = points.map(p => p.label);
  const reads = points.map(p => p.read || 0);
  const writes = points.map(p => p.write || 0);

  mbChartInstance.data.labels = labels;
  mbChartInstance.data.datasets[0].data = reads;
  mbChartInstance.data.datasets[1].data = writes;
  mbChartInstance.update();
}

/**
 * Загрузка истории тестирования.
 */
async function loadHistory() {
  try {
    const res = await fetch('/api/v1/storage/benchmark/history
?limit=20');
    if (!res.ok) return;
    const list = await res.json();

    const countBadge = document.getElementById('bench-history-
count');
    if (countBadge) countBadge.textContent = `${list.length} з
аписей`;

    const tbody = document.getElementById('bench-history-tbody
');
    if (!tbody) return;

    if (!list || list.length === 0) {
```

```
      tbody.innerHTML = '<tr><td colspan="5" class="text-cente
r text-muted py-3">История пуста</td></tr>';
      return;
    }

    tbody.innerHTML = '';
    list.forEach(item => {
      const tr = document.createElement('tr');
      tr.innerHTML = `
        <td class="text-white">${item.created_at}</td>
        <td><span class="badge bg-secondary">${item.target}</s
pan></td>
        <td class="text-info fw-bold">${item.seq1m_read_mb_s.t
oFixed(0)} <span class="text-muted">/</span> <span class="text
-warning">${item.seq1m_write_mb_s.toFixed(0)} MB/s</span></td>
        <td class="text-success">${Math.round(item.rnd4k_read_
iops)} <span class="text-muted">/</span> ${Math.round(item.rnd
4k_write_iops)}</td>
        <td class="text-end">
          <button class="btn btn-outline-danger btn-sm py-0 px
-1 bench-delete-btn" data-id="${item.id}" title="Удалить">
            <i class="bi bi-trash"></i>
          </button>
        </td>
      `;
      tbody.appendChild(tr);
    });

    tbody.querySelectorAll('.bench-delete-btn').forEach(btn =>
 {
      btn.addEventListener('click', async e => {
        e.stopPropagation();
        const id = btn.getAttribute('data-id');
        if (confirm('Удалить эту запись из истории?')) {
          await deleteHistoryItem(id);
        }
      });
    });
```

```
  } catch (err) {
    console.debug('[DiskSpeed] Ошибка загрузки истории:', err);
  }
}

/**
 * Удаление записи из истории.
 */
async function deleteHistoryItem(id) {
  try {
    await fetch(`/api/v1/storage/benchmark/history/${id}`, { method: 'DELETE' });
    await loadHistory();
    if (window.toast) window.toast.info('Удалено', 'Запись удалена из базы данных');
  } catch (err) {
    console.error('[DiskSpeed] Ошибка удаления:', err);
  }
}

/**
 * Копирование результатов в формате Markdown таблицы.
 */
function copyResultsAsMarkdown() {
  const s8r = document.getElementById('val-seq1m_q8t1-read')?.textContent || '0';
  const s8w = document.getElementById('val-seq1m_q8t1-write')?.textContent || '0';
  const s1r = document.getElementById('val-seq1m_q1t1-read')?.textContent || '0';
  const s1w = document.getElementById('val-seq1m_q1t1-write')?.textContent || '0';
  const r32r = document.getElementById('val-rnd4k_q32t16-read')?.textContent || '0';
  const r32w = document.getElementById('val-rnd4k_q32t16-write')?.textContent || '0';
  const r1r = document.getElementById('val-rnd4k_q1t1-read')?.
```

```
textContent || '0';
  const r1w = document.getElementById('val-rnd4k_q1t1-write')?.textContent || '0';

  const disk = document.getElementById('bench-drive-select')?.value || 'C:';

  const md = [
    `### AI-Windows-Admin DiskSpd Benchmark (${disk})`,
    '',
    '| Профиль | Read (MB/s) | Write (MB/s) |',
    '| :--- | :---: | :---: |',
    `| **SEQ1M Q8T1** | ${s8r} MB/s | ${s8w} MB/s |`,
    `| **SEQ1M Q1T1** | ${s1r} MB/s | ${s1w} MB/s |`,
    `| **RND4K Q32T16** | ${r32r} MB/s | ${r32w} MB/s |`,
    `| **RND4K Q1T1** | ${r1r} MB/s | ${r1w} MB/s |`,
    '',
    `*Дата замера: ${new Date().toLocaleString('ru-RU')}*`,
  ].join('\n');

  navigator.clipboard.writeText(md).then(() => {
    if (window.toast) window.toast.success('Скопировано', 'Таблица скопирована в буфер обмена');
  });
}

// Регистрация поллера при активной вкладке tab-disk-speed
registerTabPoller('tab-disk-speed', checkEngineStatus, 15000, { immediate: false });

// Автоматический запуск при загрузке
if (document.readyState === 'loading') {
  document.addEventListener('DOMContentLoaded', initDiskSpeedTab);
} else {
  initDiskSpeedTab();
}

```

```
export { initDiskSpeedTab };

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса журнала событий Windows с
модулем Log Intelligence.

Usage Examples:
    Web Route:
      /windows/api/webgui/event_logs_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/event_logs_tab/index.ht
ml" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/event_logs_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Windows Event Logs Tab -->
<div class="container-fluid py-2 px-3">
<!-- Top Stat Cards -->
<div class="row g-2 mb-3">
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 sha
dow-sm p-3" data-panel-id="panel-event-logs-1" id="panel-event
-logs-1">
<div class="text-muted small text-uppercase fw-semibold">Крити
```

```
ческие ошибки (Critical)  </div>
<div class="fs-4 fw-bold mt-1 text-danger" id="el-critical-count">--</div>
<div class="small text-muted mt-1" id="el-critical-sub">Level 1 (WevtAPI)</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-event-logs-2" id="panel-event-logs-2">
<div class="text-muted small text-uppercase fw-semibold">Ошибки (Errors)  </div>
<div class="fs-4 fw-bold mt-1 text-warning" id="el-error-count">--</div>
<div class="small text-muted mt-1" id="el-error-sub">Level 2 (WevtAPI)</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-event-logs-3" id="panel-event-logs-3">
<div class="text-muted small text-uppercase fw-semibold">Здоровье ОС (Health Score)  </div>
<div class="fs-4 fw-bold mt-1 text-success font-monospace" id="el-health-score">--</div>
<div class="small text-muted mt-1" id="el-health-sub">Data Researcher EDA</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-event-logs-4" id="panel-event-logs-4">
<div class="text-muted small text-uppercase fw-semibold">Избыточность / Шум (Rdup)  </div>
<div class="fs-4 fw-bold mt-1 text-info font-monospace" id="el
```

```
-redundancy-pct">--</div>
<div class="small text-muted mt-1" id="el-redundancy-sub">Деду
пликация шаблонов</div>
</div>
</div>
</div>
<!-- Action & Intelligence Bar -->
<div class="d-flex align-items-center justify-content-between
mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap
-2">
<div class="d-flex align-items-center gap-2 flex-wrap">
<button class="btn btn-sm btn-primary d-flex align-items-cente
r gap-1.5" id="el-refresh-btn">
<i class="bi bi-arrow-clockwise"></i>
<span>Обновить</span>
</button>
<select class="form-select form-select-sm bg-dark text-white b
order-secondary" id="el-channel-select" style="width: 180px;">
<option value="System">Журнал System</option>
<option value="Application">Журнал Application</option>
<option value="Security">Журнал Security</option>
</select>
<button class="btn btn-sm btn-outline-info d-flex align-items-
center gap-1.5" id="el-intel-analyze-btn">
<i class="bi bi-cpu"></i>
<span>Log Intelligence (EDA)</span>
</button>
<button class="btn btn-sm btn-outline-warning d-flex align-ite
ms-center gap-1.5" id="el-intel-audit-btn">
<i class="bi bi-shield-shaded"></i>
<span>Полный аудит</span>
</button>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<input class="form-control form-control-sm bg-dark text-white
border-secondary" id="el-rag-query-input" placeholder="RAG пои
ск (напр: сбой сети)..." style="width: 240px;" type="text"/>
<button class="btn btn-sm btn-info text-dark fw-semibold" id="
```

```
el-rag-search-btn">
<i class="bi bi-search"></i>
</button>
<input class="form-control form-control-sm bg-dark text-white border-secondary" id="el-search-input" placeholder="Фильтр таблицы..." style="width: 170px;" type="text"/>
</div>
</div>
<!-- Intelligence Digest Banner (Hidden by default) -->
<div class="alert alert-dark border-info shadow-sm mb-3 d-none" id="el-intel-banner">
<div class="d-flex justify-content-between align-items-start flex-wrap gap-2">
<div>
<div class="d-flex align-items-center gap-2 mb-1">
<span class="badge bg-info text-dark fw-bold font-monospace" id="el-strategy-badge">STRATEGY</span>
<strong class="text-light fs-6" id="el-intel-title">Аналитический вердикт Log Intelligence</strong>
</div>
<div class="small text-secondary font-monospace" id="el-intel-rationale">--</div>
<div class="small text-warning mt-1" id="el-intel-recommendation"></div>
</div>
<div class="text-end">
<span class="badge bg-secondary font-monospace" id="el-chunks-badge">0 RAG chunks</span>
<button aria-label="Close" class="btn-close btn-close-white ms-2" id="el-intel-close-btn"></button>
</div>
</div>
</div>
<!-- RAG Search Results Panel (Hidden by default) -->
<div class="card bg-dark text-white border-info shadow-sm mb-3 d-none" data-panel-id="panel-event-logs-rezultaty-adaptive-log-rag" id="el-rag-results-card">
<div class="card-header border-info bg-black d-flex justify-co
```

```
ntent-between align-items-center py-2 px-3">
<span class="fw-semibold text-info"><i class="bi bi-brain me-2
"></i>Результаты Adaptive Log RAG</span>
<button class="btn btn-sm btn-outline-secondary py-0 px-2" id=
"el-rag-close-btn">Закрыть</button>
</div>
<div class="card-body p-2" id="el-rag-results-body">
<!-- Dynamic Search Items -->
</div>
</div>
<!-- Event Logs Table -->
<div class="card bg-dark text-white border-secondary shadow-sm
" data-panel-id="panel-event-logs-zapisi-sistemnogo-zhurnala-e
ven" id="panel-event-logs-zapisi-sistemnogo-zhurnala-even">
<div class="card-header border-secondary d-flex align-items-ce
nter justify-content-between py-2 px-3">
<span class="fw-semibold"><i class="bi bi-journal-text text-pr
imary me-2"></i>Записи системного журнала (Event Viewer)</span
>
<span class="badge bg-secondary font-monospace" id="el-table-b
adge">0 событий</span>
</div>
<div class="card-body p-0 table-responsive" style="max-height:
 520px; overflow-y: auto;">
<table class="table table-dark table-hover table-striped align
-middle mb-0 small" id="el-table">
<thead class="sticky-top bg-dark">
<tr class="text-secondary">
<th>Уровень</th>
<th>Время</th>
<th>Источник (Provider)</th>
<th>Event ID</th>
<th>Описание события</th>
</tr>
</thead>
<tbody id="el-tbody">
<tr><td class="text-center text-muted py-4" colspan="5">Чтение
 журнала событий через WevtAPI...</td></tr>
```

```
</tbody>
</table>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Event Logs Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля Windows E
vent Logs
 *   с интеграцией аналитического пайплайна Log Intelligence (
EDA -> Gate -> RAG).
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/event_logs_tab/main.js
?v=20261006_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { initEventLogsTab } from '/windows/api/webgui/e
vent_logs_tab/main.js';
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/event_logs_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

const registerTabPoller = window.registerTabPoller || function
() {};
const isTabActive = window.isTabActive || (() => true);

let isInitialized = false;
```

```
let eventsList = [];

function escapeHtml(str) {
  if (!str) return '';
  return String(str)
    .replace(/&/g, '&amp;')
    .replace(/</g, '&lt;')
    .replace(/>/g, '&gt;')
    .replace(/"/g, '&quot;')
    .replace(/'/g, '&#039;');
}

export async function fetchEventLogsSummary() {
  if (!isTabActive('tab-event-logs')) return;
  const channelSelect = document.getElementById('el-channel-se
lect');
  const channel = channelSelect ? channelSelect.value : 'Syste
m';

  try {
    const res = await fetch(`/api/event-logs/events?channel=${
encodeURIComponent(channel)}&limit=100&hours=24`);
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const entries = await res.json();
    renderEventLogs(entries);
    // Also fetch intelligence profile in background to update
 Health Score & Redundancy
    fetchIntelligenceProfile(channel, false);
  } catch (err) {
    console.warn('[EventLogs] Ошибка получения событий:', err)
;
  }
}

async function fetchIntelligenceProfile(channel, showBanner =
true) {
  try {
    const res = await fetch(`/api/event-logs/intelligence/prof
```

```
ile?channel=${encodeURIComponent(channel)}&hours=24&limit=100`
);
    if (!res.ok) return;
    const data = await res.json();

    const healthEl = document.getElementById('el-health-score'
);
    const redEl = document.getElementById('el-redundancy-pct')
;

    if (data.profile) {
      if (healthEl) {
        const score = Math.round(data.profile.health_score ??
100);
        healthEl.textContent = `${score}/100`;
        healthEl.className = `fs-4 fw-bold mt-1 font-monospace
 ${score >= 80 ? 'text-success' : (score >= 50 ? 'text-warning
' : 'text-danger')}`;
      }
      if (redEl) {
        const rPct = (data.profile.redundancy_ratio_pct ?? 0).
toFixed(1);
        redEl.textContent = `${rPct}%`;
      }
    }

    if (showBanner && data.decision) {
      renderIntelligenceBanner(data);
    }
  } catch (err) {
    console.debug('[EventLogs] Фоновый сбор аналитики:', err);
  }
}

function renderIntelligenceBanner(data) {
  const banner = document.getElementById('el-intel-banner');
  const title = document.getElementById('el-intel-title');
  const strat = document.getElementById('el-strategy-badge');
```

```
  const rationale = document.getElementById('el-intel-rationale');
  const rec = document.getElementById('el-intel-recommendation');
  const chunks = document.getElementById('el-chunks-badge');

  if (!banner) return;

  if (title) title.textContent = `Log Intelligence: Канал ${data.channel || 'System'}`;
  if (strat) strat.textContent = (data.decision.strategy || 'SNAPSHOT').toUpperCase();
  if (rationale) rationale.textContent = data.decision.rationale || '';
  if (rec) rec.innerHTML = `<i class="bi bi-lightbulb me-1"></i><strong>Рекомендация:</strong> ${escapeHtml(data.decision.recommended_llm_action || 'Система в штатном режиме.')}`;
  if (chunks) chunks.textContent = `${data.decision.chunks_generated || 0} RAG chunks`;

  banner.classList.remove('d-none');
}

export async function runFullAudit() {
  const channelSelect = document.getElementById('el-channel-select');
  const channel = channelSelect ? channelSelect.value : 'System';

  try {
    const res = await fetch(`/api/event-logs/intelligence/audit?channel=${encodeURIComponent(channel)}&hours=24&limit=200`);
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    renderIntelligenceBanner({
      channel: data.channel,
      decision: {
        strategy: data.strategy,
```

```
        rationale: data.strategy_rationale,
        recommended_llm_action: data.recommended_llm_action,
        chunks_generated: data.top_clusters ? data.top_clusters.length : 0,
      }
    });
  } catch (err) {
    console.warn('[EventLogs] Ошибка аудита:', err);
  }
}

export async function runRAGSearch() {
  const input = document.getElementById('el-rag-query-input');
  const channelSelect = document.getElementById('el-channel-select');
  const resultsCard = document.getElementById('el-rag-results-card');
  const resultsBody = document.getElementById('el-rag-results-body');

  if (!input || !resultsCard || !resultsBody) return;
  const query = input.value.trim();
  if (!query) return;

  resultsBody.innerHTML = `<div class="text-center py-3 text-muted"><div class="spinner-border spinner-border-sm text-info me-2"></div>Поиск в Adaptive Log RAG...</div>`;
  resultsCard.classList.remove('d-none');

  try {
    const channel = channelSelect ? channelSelect.value : '';
    const res = await fetch(`/api/event-logs/intelligence/search?query=${encodeURIComponent(query)}&top_k=5&channel=${encodeURIComponent(channel)}`);
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const results = await res.json();

    if (!results || results.length === 0) {
```

```
      resultsBody.innerHTML = `<div class="text-center py-3 text-muted">По запросу "<em>${escapeHtml(query)}</em>" совпадений не найдено.</div>`;
      return;
    }

    resultsBody.innerHTML = results.map(r => `
      <div class="p-2 mb-2 rounded bg-black border border-secondary">
        <div class="d-flex justify-content-between align-items-center mb-1">
          <strong class="text-info">${escapeHtml(r.title || 'Документ RAG')}</strong>
          <span class="badge bg-secondary font-monospace">Relevance: ${Math.round((r.relevance_score || 0) * 100)}%</span>
        </div>
        <div class="small text-light font-monospace" style="word-break: break-word;">${escapeHtml(r.content || '')}</div>
        <div class="small text-muted mt-1 font-monospace"><i class="bi bi-tag me-1"></i>Стратегия: ${escapeHtml(r.strategy || 'Adaptive')} • ${escapeHtml(r.created_at || '')}</div>
      </div>
    `).join('');
  } catch (err) {
    resultsBody.innerHTML = `<div class="text-danger small p-2">Ошибка поиска: ${escapeHtml(err.message)}</div>`;
  }
}

function renderEventLogs(entries) {
  eventsList = entries || [];

  const critEl = document.getElementById('el-critical-count');
  const errEl = document.getElementById('el-error-count');

  let crits = 0;
  let errs = 0;

```

```
  eventsList.forEach(e => {
    const l = (e.level || '').toLowerCase();
    if (l.includes('crit')) crits++;
    else if (l.includes('err')) errs++;
  });

  if (critEl) critEl.textContent = crits;
  if (errEl) errEl.textContent = errs;

  applyEventFilters();
}

function applyEventFilters() {
  const tbody = document.getElementById('el-tbody');
  const searchInput = document.getElementById('el-search-input
');
  const badge = document.getElementById('el-table-badge');

  if (!tbody) return;

  const query = (searchInput ? searchInput.value : '').toLower
Case().trim();

  const filtered = eventsList.filter(e => {
    if (!query) return true;
    const msgMatch = (e.message || '').toLowerCase().includes(
query);
    const provMatch = (e.provider_name || e.provider || '').to
LowerCase().includes(query);
    const idMatch = String(e.event_id || e.id || '').includes(
query);
    return msgMatch || provMatch || idMatch;
  });

  if (badge) badge.textContent = `${filtered.length} из ${even
tsList.length} событий`;

  if (filtered.length === 0) {
```

```
    tbody.innerHTML = '<tr><td colspan="5" class="text-center text-muted py-4">События не найдены</td></tr>';
    return;
  }

  tbody.innerHTML = filtered.slice(0, 150).map(e => {
    const level = (e.level || 'Information').toLowerCase();
    let badgeClass = 'bg-secondary';
    if (level.includes('crit')) badgeClass = 'bg-danger text-white';
    else if (level.includes('err')) badgeClass = 'bg-danger text-white';
    else if (level.includes('warn')) badgeClass = 'bg-warning text-dark';
    else if (level.includes('info')) badgeClass = 'bg-info text-dark';

    return `
      <tr>
        <td><span class="badge ${badgeClass}">${escapeHtml(e.level || 'Info')}</span></td>
        <td class="text-muted small font-monospace">${escapeHtml(e.time_created || e.timestamp || '-')}</td>
        <td class="fw-semibold text-light small">${escapeHtml(e.provider_name || e.provider || 'Windows')}</td>
        <td class="font-monospace text-warning small">${escapeHtml(String(e.event_id || e.id || '-'))}</td>
        <td class="text-light small" style="max-width: 500px; word-break: break-word;">${escapeHtml(e.message || '-')}</td>
      </tr>
    `;
  }).join('');
}

export function initEventLogsTab() {
  if (isInitialized) return;
  isInitialized = true;

```

```
  const refreshBtn = document.getElementById('el-refresh-btn')
;
  if (refreshBtn) refreshBtn.addEventListener('click', fetchEv
entLogsSummary);

  const channelSelect = document.getElementById('el-channel-se
lect');
  if (channelSelect) channelSelect.addEventListener('change',
fetchEventLogsSummary);

  const searchInput = document.getElementById('el-search-input
');
  if (searchInput) searchInput.addEventListener('input', apply
EventFilters);

  const analyzeBtn = document.getElementById('el-intel-analyze
-btn');
  if (analyzeBtn) analyzeBtn.addEventListener('click', () => {
    const chan = channelSelect ? channelSelect.value : 'System
';
    fetchIntelligenceProfile(chan, true);
  });

  const auditBtn = document.getElementById('el-intel-audit-btn
');
  if (auditBtn) auditBtn.addEventListener('click', runFullAudi
t);

  const ragBtn = document.getElementById('el-rag-search-btn');
  if (ragBtn) ragBtn.addEventListener('click', runRAGSearch);

  const ragInput = document.getElementById('el-rag-query-input
');
  if (ragInput) ragInput.addEventListener('keydown', (e) => {
    if (e.key === 'Enter') runRAGSearch();
  });

  const bannerCloseBtn = document.getElementById('el-intel-clo
```

```
se-btn');
  if (bannerCloseBtn) bannerCloseBtn.addEventListener('click',
 () => {
    const banner = document.getElementById('el-intel-banner');
    if (banner) banner.classList.add('d-none');
  });

  const ragCloseBtn = document.getElementById('el-rag-close-bt
n');
  if (ragCloseBtn) ragCloseBtn.addEventListener('click', () =>
 {
    const card = document.getElementById('el-rag-results-card'
);
    if (card) card.classList.add('d-none');
  });

  registerTabPoller('tab-event-logs', fetchEventLogsSummary, 1
5000, { immediate: true });
  if (isTabActive('tab-event-logs')) {
    fetchEventLogsSummary();
  }
}

export function activateEventLogsTab() {
  fetchEventLogsSummary();
}

window.initEventLogsTab = initEventLogsTab;
window.activateEventLogsTab = activateEventLogsTab;

// Автоинициализация: загрузчик вкладок только подключает скри
пт
if (document.getElementById('el-tbody') && isTabActive('tab-ev
ent-logs')) {
  initEventLogsTab();
}

```

index.html

```
<!--
================================================================
===============
Process Name: AI-Windows-Admin Web Interface - Index View
================================================================
===============
Description:
  Шаблон веб-интерфейса системного модуля (index).

Usage Examples:
    Web Route:
      /src/api/webgui/file_history_ai_search_tab/index.html

File: index.html
Project: ai-windows-admin
Package: src/api/webgui/file_history_ai_search_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!-- File History AI Search Tab -->
<style>
  .fh-container {
    padding: 0.75rem 1rem;
    max-width: 1400px;
    margin: 0 auto;
  }
  .fh-card {
    background: var(--surface-1);
    border: 1px solid var(--border-color);
    border-radius: 12px;
    padding: 1.25rem;
    box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.05);
```

```
  }
  .fh-header {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 12px;
    padding: 1rem 1.5rem;
    color: var(--text-color);
  }
  .fh-stat-card {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 10px;
    padding: 0.85rem 1.1rem;
    transition: transform 0.15s ease, border-color 0.15s ease;
  }
  .fh-stat-card:hover {
    transform: translateY(-2px);
    border-color: var(--nav-active);
  }
  .fh-stat-val {
    font-size: 1.4rem;
    font-weight: 700;
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
    color: var(--text-color);
  }
  .fh-search-box {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 12px;
    padding: 0.75rem 1.25rem;
  }
  .fh-result-card {
    border-left: 4px solid var(--nav-active);
    background: var(--surface-1);
    border-top: 1px solid var(--border-color);
    border-right: 1px solid var(--border-color);
    border-bottom: 1px solid var(--border-color);
```

```
    border-radius: 8px;
    padding: 1rem;
    margin-bottom: 0.75rem;
    transition: background-color 0.15s ease;
  }
  .fh-result-card:hover {
    background: var(--surface-2);
  }
  .fh-score-badge {
    font-size: 0.82rem;
    font-weight: 600;
    font-family: monospace;
  }
  .fh-snippet-box {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 6px;
    padding: 0.6rem 0.85rem;
    font-family: SFMono-Regular, Menlo, Monaco, Consolas, mono
space;
    font-size: 0.85rem;
    color: var(--text-color);
    white-space: pre-wrap;
    word-break: break-all;
    max-height: 120px;
    overflow-y: auto;
  }
</style>

<div class="fh-container">
  <!-- Хедер -->
  <div class="d-flex justify-content-between align-items-cente
r flex-wrap gap-3 mb-3 fh-header">
    <div class="d-flex align-items-center gap-3">
      <div class="bg-primary bg-opacity-25 p-2.5 rounded-3 tex
t-primary border border-primary border-opacity-25">
        <i class="bi bi-clock-history fs-3"></i>
      </div>
```

```
      <div>
        <h5 class="mb-0 fw-bold d-flex align-items-center gap-2">
          <span data-i18n="auto___b3724e"></span> Windows (File History AI Search)
          <span class="badge bg-primary bg-opacity-75 fs-7">RAG v1.0</span>
        </h5>
        <span class="text-secondary small"><span data-i18n="auto__windows_file_history_recent_lnk_activity_db__07a692"></span></span>
      </div>
    </div>
    <div class="d-flex align-items-center gap-2 flex-wrap">
      <button class="btn btn-sm btn-outline-secondary rounded-pill px-3" id="btn-fh-refresh-status" type="button">
        <i class="bi bi-arrow-clockwise me-1"></i> <span data-i18n="auto___6cc5dc"></span>
      </button>
      <button class="btn btn-sm btn-success rounded-pill px-3" id="btn-fh-reindex" type="button">
        <i class="bi bi-cpu me-1"></i> <span data-i18n="auto___5ec953"></span> RAG
      </button>
    </div>
  </div>

  <!-- Метрики -->
  <div class="row g-3 mb-3">
    <div class="col-12 col-sm-6 col-md-3">
      <div class="fh-stat-card">
        <div class="text-secondary small text-uppercase fw-semibold mb-1"><span data-i18n="auto__rag__1a3eea"></span></div>
        <div class="d-flex align-items-center justify-content-between">
          <div class="fh-stat-val text-primary" id="fh-stat-docs-count">0</div>
          <i class="bi bi-database-check fs-4 text-primary opa
```

```
city-50"></i>
        </div>
        <div class="small text-muted mt-1" id="fh-stat-last-index"><span data-i18n="auto___16deed"></span></div>
      </div>
    </div>

    <div class="col-12 col-sm-6 col-md-3">
      <div class="fh-stat-card">
        <div class="text-secondary small text-uppercase fw-semibold mb-1"><span data-i18n="auto___8205bf"></span></div>
        <div class="d-flex align-items-center justify-content-between">
          <div class="fh-stat-val" id="fh-stat-scheduler-status"><span data-i18n="auto___aa0d25"></span></div>
          <i class="bi bi-arrow-repeat fs-4 text-warning opacity-50" id="fh-scheduler-icon"></i>
        </div>
        <div class="small text-muted mt-1" id="fh-stat-scheduler-interval"><span data-i18n="auto__15__d40d33"></span></div>
      </div>
    </div>

    <div class="col-12 col-sm-6 col-md-3">
      <div class="fh-stat-card">
        <div class="text-secondary small text-uppercase fw-semibold mb-1"><span data-i18n="auto__faiss__1cdc9e"></span></div>
        <div class="d-flex align-items-center justify-content-between">
          <div class="fh-stat-val fs-6 text-truncate" id="fh-stat-faiss-file" style="max-width: 180px;">file_history.faiss</div>
          <i class="bi bi-file-earmark-binary fs-4 text-info opacity-50"></i>
        </div>
        <div class="small text-muted mt-1"><span data-i18n="auto___18f0e9"></span> Gemini Embeddings</div>
```

```
      </div>
    </div>

    <div class="col-12 col-sm-6 col-md-3">
      <div class="fh-stat-card">
        <div class="text-secondary small text-uppercase fw-sem
ibold mb-1"><span data-i18n="auto___ec3fc1"></span></div>
        <div class="d-flex align-items-center gap-2 mt-1">
          <button class="btn btn-sm btn-outline-success w-50 p
y-1" id="btn-fh-start-scheduler">
            <i class="bi bi-play-fill me-1"></i><span data-i18
n="auto___c70ec8"></span>
          </button>
          <button class="btn btn-sm btn-outline-danger w-50 py
-1" id="btn-fh-stop-scheduler">
            <i class="bi bi-stop-fill me-1"></i><span data-i18
n="auto___6d714e"></span>
          </button>
        </div>
        <div class="small text-muted mt-1 text-center"><span d
ata-i18n="auto__15__eecb38"></span></div>
      </div>
    </div>
  </div>

  <!-- Поиск -->
  <div class="fh-card mb-4">
    <h6 class="fw-bold mb-3 d-flex align-items-center gap-2">
      <i class="bi bi-search text-primary"></i>
      <span data-i18n="auto___2517ec"></span>
    </h6>

    <div class="row g-2 mb-3">
      <div class="col-12 col-md-7">
        <div class="input-group">
          <span class="input-group-text bg-body-tertiary borde
r-end-0">
            <i class="bi bi-search text-muted"></i>
```

```
          </span>
          <input type="text" class="form-control border-start-0" id="fh-search-input" placeholder="<span data-i18n="auto___97d6b0"></span> Word')...">
          <button class="btn btn-primary px-4 fw-semibold" id="btn-fh-search" type="button">
            <i class="bi bi-arrow-right-circle me-1"></i> <span data-i18n="auto___ed6e1d"></span>
          </button>
        </div>
      </div>
      <div class="col-6 col-md-3">
        <select class="form-select" id="fh-filter-source">
          <option value=""><span data-i18n="auto___5f3a3d"></span> (All Sources)</option>
          <option value="file_history_xml">Windows File History (XML)</option>
          <option value="recent_lnk">Recent Files (LNK)</option>
          <option value="activity_db">Activity History (DB)</option>
          <option value="fs_scan"><span data-i18n="auto___8be3b9"></span></option>
        </select>
      </div>
      <div class="col-6 col-md-2">
        <select class="form-select" id="fh-filter-topk">
          <option value="5" selected><span data-i18n="auto___00f8ee"></span> 5</option>
          <option value="10"><span data-i18n="auto___00f8ee"></span> 10</option>
          <option value="20"><span data-i18n="auto___00f8ee"></span> 20</option>
          <option value="50"><span data-i18n="auto___00f8ee"></span> 50</option>
        </select>
      </div>
    </div>
```

```
212
213     <!-- Спиннер -->
214     <div class="text-center py-4 d-none" id="fh-search-spinner
    ">
215       <div class="spinner-border text-primary" role="status">
216         <span class="visually-hidden"><span data-i18n="auto__r
    ag__662d74"></span>...</span>
217       </div>
218       <div class="small text-muted mt-2"><span data-i18n="auto
    ___93b6bf"></span> FAISS...</div>
219     </div>
220
221     <!-- Результаты поиска -->
222     <div id="fh-search-results-container">
223       <div class="text-center text-muted py-4 border rounded-3
     bg-body-tertiary bg-opacity-50">
224         <i class="bi bi-search fs-2 d-block mb-1 text-secondar
    y opacity-50"></i>
225         <span><span data-i18n="auto___3df1ff"></span>.</span>
226       </div>
227     </div>
228   </div>
229
230   <!-- Таблица -->
231   <div class="fh-card">
232     <div class="d-flex justify-content-between align-items-cen
    ter mb-3 flex-wrap gap-2">
233       <h6 class="fw-bold mb-0 d-flex align-items-center gap-2"
    >
234         <i class="bi bi-list-stars text-info"></i>
235         <span data-i18n="auto__windows__8738e2"></span>)
236       </h6>
237       <button class="btn btn-sm btn-outline-info rounded-pill
    px-3" id="btn-fh-scan-now">
238         <i class="bi bi-arrow-repeat me-1"></i> <span data-i18
    n="auto___4748be"></span>
239       </button>
240     </div>
```

```
241
242     <div class="table-responsive">
243       <table class="table table-hover align-middle mb-0" id="f
    h-items-table">
244         <thead>
245           <tr>
246             <th scope="col" style="width: 40px;">#</th>
247             <th scope="col"><span data-i18n="auto___94b8df"></
    span></th>
248             <th scope="col"><span data-i18n="auto___8290a3"></
    span></th>
249             <th scope="col"><span data-i18n="auto___bb9263"></
    span></th>
250             <th scope="col"><span data-i18n="auto___c80d7e"></
    span></th>
251             <th scope="col"><span data-i18n="auto___98713e"></
    span></th>
252             <th scope="col" class="text-end"><span data-i18n="
    auto___9978ac"></span></th>
253           </tr>
254         </thead>
255         <tbody id="fh-items-tbody">
256           <tr>
257             <td colspan="7" class="text-center py-4 text-muted
    ">
258               <span data-i18n="auto___cb63af"></span> Windows.
259             </td>
260           </tr>
261         </tbody>
262       </table>
263     </div>
264   </div>
265 </div>
266
```

main.js

```
/**
 * ============================================================
==================
 * Process Name: AI-Windows-Admin UI - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский веб-скрипт модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/src/api/webgui/file_history_ai_search_tab
/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: src/api/webgui/file_history_ai_search_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * file_history_ai_search_tab/main.js — Логика веб-интерфейса
поиска по истории файлов Windows через RAG
 */

(function () {
  'use stricti18n.t('auto__let_currentitems_function_initfileh
istorytab_const_btnsearch_document_getelementbyid__e6ed42')btn
-fh-search');
    const searchInput = document.getElementById('fh-search-inp
ut');
    const btnRefreshStatus = document.getElementById('btn-fh-r
```

```
efresh-status');
    const btnReindex = document.getElementById('btn-fh-reindex
');
    const btnScanNow = document.getElementById('btn-fh-scan-no
w');
    const btnStartSched = document.getElementById('btn-fh-star
t-scheduler');
    const btnStopSched = document.getElementById('btn-fh-stop-
scheduler');

    if (btnSearch) {
      btnSearch.addEventListener('click', executeSearch);
    }

    if (searchInput) {
      searchInput.addEventListener('keydown', (e) => {
        if (e.key === 'Enter') {
          executeSearch();
        }
      });
    }

    if (btnRefreshStatus) {
      btnRefreshStatus.addEventListener('click', loadStatus);
    }

    if (btnReindex) {
      btnReindex.addEventListener('click', triggerReindex);
    }

    if (btnScanNow) {
      btnScanNow.addEventListener('click', triggerScan);
    }

    if (btnStartSched) {
      btnStartSched.addEventListener('click', () => toggleSche
duler('start'));
    }
```

```
61
62      if (btnStopSched) {
63        btnStopSched.addEventListener('click', () => toggleSched
uler('stopi18n.t('auto__loadstatus_rag_async_function_loadstat
us_try_const_resp_await_fetch__d52650')/api/windows/file-histo
ry/status');
64        if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
65        const data = await resp.json();
66
67        const elCount = document.getElementById('fh-stat-docs-co
unt');
68        const elLastIdx = document.getElementById('fh-stat-last-
index');
69        const elSchedStat = document.getElementById('fh-stat-sch
eduler-status');
70        const elSchedIcon = document.getElementById('fh-schedule
r-icon');
71        const elSchedInt = document.getElementById('fh-stat-sche
duler-interval');
72        const elFaissFile = document.getElementById('fh-stat-fai
ss-filei18n.t('auto__if_elcount_elcount_textcontent_data_total
_indexed_documents_0_if_ellastidx_ellastidx_textcontent_data_l
ast_indexed_at_new_date_data_last_indexed_at_tolocalestring__c
32c12')ru-RU')}`
73            : i18n.t('auto___16deed');
74        }
75
76        if (elSchedStat) {
77          if (data.scheduler_running) {
78            elSchedStat.textContent = i18n.t('auto___667904');
79            elSchedStat.className = 'fh-stat-val text-success';
80            if (elSchedIcon) elSchedIcon.className = 'bi bi-arro
w-repeat fs-4 text-success opacity-75 spin';
81          } else {
82            elSchedStat.textContent = i18n.t('auto___aa0d25');
83            elSchedStat.className = 'fh-stat-val text-warning';
84            if (elSchedIcon) elSchedIcon.className = 'bi bi-arro
w-repeat fs-4 text-warning opacity-50i18n.t('auto__if_elschedi
```

```
nt_elschedint_textcontent_data_update_interval_minutes_15_if_e
lfaissfile_data_index_file_path_const_parts_data_index_file_pa
th_split_elfaissfile_textcontent_parts_parts_length_1_data_ind
ex_file_path_elfaissfile_title_data_index_file_path_catch_err_
console_error__3cb452')[FileHistoryUI] Ошибка загрузки статуса
:i18n.t('auto__err_windows_async_function_triggerscan_const_tb
ody_document_getelementbyid__5b14c9')fh-items-tbody');
    if (tbody) {
      tbody.innerHTML = `
        <tr>
          <td colspan="7" class="text-center py-4">
            <div class="spinner-border spinner-border-sm text-
info me-2" role="status"></div>
            <span>Сканирование источников истории файлов Windo
ws...</span>
          </td>
        </tr>`;
    }

    try {
      const resp = await fetch('/api/windows/file-history/scan
i18n.t('auto__if_resp_ok_throw_new_error_http_resp_status_curr
entitems_await_resp_json_renderitemstable_currentitems_if_wind
ow_showtoast_window_showtoast_currentitems_length__a2f871')inf
o');
      }
    } catch (err) {
      console.error(i18n.t('auto__filehistoryui__c71a7b'), err
);
      if (tbody) {
        tbody.innerHTML = `
          <tr>
            <td colspan="7" class="text-center py-4 text-dange
r">
              <i class="bi bi-exclamation-triangle me-1"></i>
Ошибка при сканировании: ${err.message}
            </td>
          </tr>`;
```

```
      }
    }
  }

  /**
   * Отрисовка таблицы сохраненных элементов телеметрии
   */
  function renderItemsTable(items) {
    const tbody = document.getElementById('fh-items-tbody');
    if (!tbody) return;

    if (!items || items.length === 0) {
      tbody.innerHTML = `
        <tr>
          <td colspan="7" class="text-center py-4 text-muted">
            Элементы истории файлов не обнаружены.
          </td>
        </tr>`;
      return;
    }

    let html = '';
    items.slice(0, 100).forEach((item, idx) => {
      const badgeSource = getSourceBadge(item.source_type);
      const formattedSize = formatBytes(item.file_size);
      const timeStr = item.timestamp ? new Date(item.timestamp
).toLocaleString('ru-RU') : '—';

      html += `
        <tr>
          <td class="text-muted small">${idx + 1}</td>
          <td>
            <div class="fw-semibold text-truncate" style="max-
width: 280px;" title="${escapeHtml(item.file_name)}">
              <i class="bi bi-file-earmark-text me-1 text-prim
ary"></i>${escapeHtml(item.file_name)}
            </div>
            <div class="small text-muted text-truncate" style=
```

```
"max-width: 320px;" title="${escapeHtml(item.file_path)}">
142               ${escapeHtml(item.file_path)}
143             </div>
144           </td>
145           <td>${badgeSource}</td>
146           <td><span class="badge bg-secondary bg-opacity-50 te
xt-light">${escapeHtml(item.event_type)}</span></td>
147           <td class="small text-muted">${timeStr}</td>
148           <td class="small font-monospace">${formattedSize}</t
d>
149           <td class="text-end">
150             <button class="btn btn-sm btn-outline-secondary py
-0 px-2 btn-copy-path" data-path="${escapeHtml(item.file_path)
}" title=i18n.t('auto___8a5b6a')>
151               <i class="bi bi-clipboard"></i>
152             </button>
153           </td>
154         </tr>`;
155     });
156
157     tbody.innerHTML = html;
158
159     // Привязка копирования пути
160     tbody.querySelectorAll('.btn-copy-path').forEach((btn) =>
{
161       btn.addEventListener('click', (e) => {
162         const path = e.currentTarget.getAttribute('data-path')
;
163         if (path) {
164           navigator.clipboard.writeText(path);
165           if (window.showToast) window.showToast(i18n.t('auto_
__f6358a'), 'successi18n.t('auto__rag_async_function_triggerre
index_const_btnreindex_document_getelementbyid__29bc61')btn-fh
-reindex');
166     if (btnReindex) btnReindex.disabled = true;
167
168     if (window.showToast) {
169       window.showToast(i18n.t('auto__rag__9ef43b'), 'info');
```

```
170     }
171 
172     try {
173       const resp = await fetch('/api/windows/file-history/inde
    x', { method: 'POSTi18n.t('auto__if_resp_ok_throw_new_error_ht
    tp_resp_status_const_data_await_resp_json_if_window_showtoast_
    window_showtoast_rag_data_indexed_count__d44568')success');
174       }
175       loadStatus();
176     } catch (err) {
177       console.error(i18n.t('auto__filehistoryui_rag__ffcf9a'),
     err);
178       if (window.showToast) {
179         window.showToast(`Ошибка индексации RAG: ${err.message
    }`, 'errori18n.t('auto__finally_if_btnreindex_btnreindex_disab
    led_false_async_function_executesearch_const_searchinput_docum
    ent_getelementbyid__f45f1d')fh-search-input');
180     const filterSource = document.getElementById('fh-filter-so
    urce');
181     const filterTopk = document.getElementById('fh-filter-topk
    ');
182     const spinner = document.getElementById('fh-search-spinner
    ');
183     const container = document.getElementById('fh-search-resul
    ts-container');
184 
185     const query = searchInput ? searchInput.value.trim() : '';
186     if (!query) {
187       if (window.showToast) window.showToast(i18n.t('auto___f4
    f030'), 'warning');
188       return;
189     }
190 
191     const topK = filterTopk ? parseInt(filterTopk.value, 10) :
     5;
192     const sourceType = filterSource && filterSource.value ? fi
    lterSource.value : null;
193 
```

```
    if (spinner) spinner.classList.remove('d-none');
    if (container) container.innerHTML = '';

    try {
      const resp = await fetch('/api/windows/file-history/sear
ch', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({
          query: query,
          top_k: topK,
          source_type: sourceType,
        }),
      });

      if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
      const data = await resp.json();
      renderSearchResults(data);
    } catch (err) {
      console.error(i18n.t('auto__filehistoryui__1be39c'), err
);
      if (container) {
        container.innerHTML = `
          <div class="alert alert-danger mb-0">
            <i class="bi bi-exclamation-octagon me-1"></i> Оши
бка семантического поиска: ${err.message}
          </div>`;
      }
    } finally {
      if (spinner) spinner.classList.add('d-nonei18n.t('auto__
function_rendersearchresults_data_const_container_document_get
elementbyid__482c76')fh-search-results-container');
    if (!container) return;

    if (!data.results || data.results.length === 0) {
      container.innerHTML = `
        <div class="text-center text-muted py-4 border rounded
-3">
```

```
          <i class="bi bi-emoji-frown fs-3 d-block mb-1 text-warning opacity-50i18n.t('auto__i_span_strong_escapehtml_data_query_strong_span_br_small_class__5ef260')text-secondaryi18n.t('auto__rag_small_div_return_let_html_div_class__1b153c')d-flex justify-content-between align-items-center mb-2 px-1">
        <span class="small text-muted">
          Найдено результатов: <strong>${data.total_found}</strong> (Время выполнения: <strong>${data.execution_time_ms} мс</strong>)
        </span>
      </div>`;

    data.results.forEach((item, idx) => {
      const badgeSource = getSourceBadge(item.source_type);
      const scorePct = Math.round(item.score * 100);
      const badgeScoreClass = item.score > 0.6 ? 'bg-success' : item.score > 0.3 ? 'bg-info' : 'bg-secondary';
      const timeStr = item.timestamp ? new Date(item.timestamp).toLocaleString('ru-RU') : '—';

      html += `
        <div class="fh-result-card">
          <div class="d-flex justify-content-between align-items-start gap-2 mb-1">
            <div class="d-flex align-items-center gap-2">
              <span class="badge rounded-circle bg-body-tertiary text-muted p-2 fw-bold me-1" style="width: 26px; height: 26px; display: inline-flex; align-items: center; justify-content: center;">
                ${idx + 1}
              </span>
              <h6 class="mb-0 fw-bold text-truncate" style="max-width: 450px;" title="${escapeHtml(item.file_name)}">
                <i class="bi bi-file-earmark-code me-1 text-primary"></i>${escapeHtml(item.file_name)}
              </h6>
              ${badgeSource}
            </div>
```

```
            <div class="d-flex align-items-center gap-2">
              <span class="badge ${badgeScoreClass} fh-score-b
adge" title=i18n.t('auto___4c2381')>
                Score: ${item.score}
              </span>
              <button class="btn btn-sm btn-outline-secondary
py-0 px-2 btn-copy-path" data-path="${escapeHtml(item.file_pat
h)}" title=i18n.t('auto___bbf33e')>
                <i class="bi bi-clipboard"></i>
              </button>
            </div>
          </div>

          <div class="small text-muted text-truncate mb-2 font
-monospace">
            <i class="bi bi-folder2-open me-1 text-secondary">
</i>${escapeHtml(item.file_path)}
            <span class="ms-2 text-secondaryi18n.t('auto__time
str_span_div_div_class__41bff1')fh-snippet-box">
            ${escapeHtml(item.snippet)}
          </div>
        </div>`;
    });

    container.innerHTML = html;

    // Привязка копирования в результатах поиска
    container.querySelectorAll('.btn-copy-path').forEach((btn)
 => {
      btn.addEventListener('click', (e) => {
        const path = e.currentTarget.getAttribute('data-path')
;
        if (path) {
          navigator.clipboard.writeText(path);
          if (window.showToast) window.showToast(i18n.t('auto_
__f6358a'), 'successi18n.t('auto__async_function_toggleschedul
er_action_const_url_api_windows_file_history_scheduler_action_
try_const_resp_await_fetch_url_method__3e7cc6')POST' });
```

```
      if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
      const data = await resp.json();
      if (window.showToast) {
        window.showToast(
          action === 'start' ? i18n.t('auto__rag__8d29a5') : i
18n.t('auto___426630'),
          action === 'start' ? 'success' : 'info'
        );
      }
      loadStatus();
    } catch (err) {
      console.error(i18n.t('auto__filehistoryui__5680c6'), err
);
      if (window.showToast) window.showToast(`Ошибка: ${err.me
ssage}`, 'errori18n.t('auto__function_getsourcebadge_sourcetyp
e_switch_sourcetype_case_f06986')file_history_xml':
        return '<span class="badge bg-primary bg-opacity-75"><
i class="bi bi-file-earmark-xml me-1"></i>File History XML</sp
an>';
      case 'recent_lnk':
        return '<span class="badge bg-info bg-opacity-75"><i c
lass="bi bi-link-45deg me-1"></i>Recent LNK</span>';
      case 'activity_db':
        return '<span class="badge bg-warning text-dark bg-opa
city-75"><i class="bi bi-database me-1"></i>Activity DB</span>
';
      case 'fs_scan':
        return '<span class="badge bg-success bg-opacity-75"><
i class="bi bi-hdd-network me-1"></i>FS Scan</span>';
      default:
        return `<span class="badge bg-secondary">${escapeHtml(
sourceType || 'unknowni18n.t('auto__span_function_formatbytes_
bytes_decimals_1_if_bytes_bytes_0_return_98f56d')0 B';
    const k = 1024;
    const dm = decimals < 0 ? 0 : decimals;
    const sizes = ['B', 'KB', 'MB', 'GB', 'TB'];
    const i = Math.floor(Math.log(bytes) / Math.log(k));
    return parseFloat((bytes / Math.pow(k, i)).toFixed(dm)) +
```

```
' i18n.t('auto__sizes_i_html_function_escapehtml_str_if_str_return_ddd7ae')';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;i18n.t('auto__dom_if_document_readystate__83155a')loading') {
    document.addEventListener('DOMContentLoaded', initFileHistoryTab);
  } else {
    initFileHistoryTab();
  }
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/file_recovery_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/file_recovery_tab/index
.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/file_recovery_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!-- File Recovery (R-Studio Technician Portable) Tab -->
<style>
  .recovery-container {
    padding: 0.75rem 1rem;
    max-width: 1200px;
    margin: 0 auto;
  }
  .recovery-card {
    background: var(--surface-1) !important;
```

```
    border: 1px solid var(--border-color) !important;
    border-radius: 10px;
    padding: 1.25rem;
    color: var(--text-color);
    box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.08);
  }
  .recovery-header {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 10px;
    padding: 0.85rem 1.25rem;
    color: var(--text-color);
  }
  .recovery-stat-box {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 8px;
    padding: 0.75rem 1rem;
  }
  .recovery-stat-title {
    font-size: 0.75rem;
    text-transform: uppercase;
    letter-spacing: 0.05em;
    color: var(--text-muted);
  }
  .recovery-stat-val {
    font-size: 1.1rem;
    font-weight: 600;
    color: var(--text-color);
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
  }
  .recovery-feature-item {
    display: flex;
    align-items: flex-start;
    gap: 0.5rem;
    margin-bottom: 0.5rem;
    font-size: 0.85rem;
```

```
    color: var(--text-color);
  }
</style>

<div class="recovery-container">
  <!-- Верхняя информационная панель -->
  <div class="d-flex justify-content-between align-items-center flex-wrap gap-2 mb-3 recovery-header">
    <div class="d-flex align-items-center gap-2.5">
      <span class="fs-4">□ </span>
      <div>
        <h5 class="mb-0 fw-bold fs-6">Восстановление удаленных файлов</h5>
        <span class="text-secondary small">Запуск профессионального комплекса R-Studio Technician Portable</span>
      </div>
    </div>
    <div class="d-flex align-items-center gap-2">
      <span class="badge rounded-pill bg-body-secondary border border-secondary text-body px-2.5 py-1.5" id="recovery-status-badge">
        Проверка утилиты...
      </span>
      <button class="btn btn-sm btn-outline-info rounded-pill px-3" id="btn-recovery-refresh" type="button">
        <i class="bi bi-arrow-clockwise me-1"></i> Обновить статус
      </button>
    </div>
  </div>

  <div class="row g-3 mb-3">
    <!-- Левая колонка: Запуск и возможности -->
    <div class="col-12 col-lg-7">
      <div class="recovery-card h-100 d-flex flex-column justify-content-between">
        <div>
          <h6 class="fw-bold text-info mb-3 d-flex align-items
```

```
-center gap-2">
            <i class="bi bi-arrow-counterclockwise"></i> Восст
ановление данных с накопителей
          </h6>

          <p class="text-body small mb-3">
            <strong>R-Studio Technician</strong> — мощный комп
лекс для восстановления удалённых, повреждённых и отформатиров
анных файлов с локальных и сетевых дисков, RAID-массивов, флеш
-накопителей и виртуальных дисков.
          </p>

          <div class="recovery-feature-item">
            <i class="bi bi-check2-circle text-success fs-5"><
/i>
            <div>
              <strong>Поддерживаемые файловые системы:</strong
> NTFS, NTFS5, ReFS, FAT12/16/32, exFAT, HFS/HFS+, APFS, Ext2/
Ext3/Ext4 FS, UFS.
            </div>
          </div>

          <div class="recovery-feature-item">
            <i class="bi bi-check2-circle text-success fs-5"><
/i>
            <div>
              <strong>Сценарии восстановления:</strong> Случай
ное удаление (Shift+Del), форматирование разделов, повреждение
 MBR/GPT, сбои питания, вирусные атаки, сбойные сектора.
            </div>
          </div>

          <div class="recovery-feature-item">
            <i class="bi bi-check2-circle text-success fs-5"><
/i>
            <div>
              <strong>Режим работы:</strong> Portable (не треб
ует установки, запускается напрямую из <code>\bin\R-Studio_Tec
```

```
hnician_9.5.191810_Portable\R-Studio_9.5.exe</code>).
            </div>
          </div>
        </div>

        <div class="mt-4">
          <!-- Большая главная кнопка запуска -->
          <button class="btn btn-primary btn-lg w-100 py-3 fw-bold d-flex align-items-center justify-content-center gap-2 shadow" id="btn-recovery-launch" type="button">
            <i class="bi bi-play-circle-fill fs-4"></i> Восстановить удаленные файлы (Запустить R-Studio)
          </button>
          <div id="recovery-alert-box" class="mt-3" style="display: none;"></div>
        </div>
      </div>
    </div>

    <!-- Правая колонка: Информация об исполняемом файле -->
    <div class="col-12 col-lg-5">
      <div class="recovery-card h-100 d-flex flex-column justify-content-between">
        <div>
          <h6 class="fw-bold text-info mb-3 d-flex align-items-center gap-2">
            <i class="bi bi-info-circle-fill"></i> Информация о пакете утилиты
          </h6>

          <div class="row g-2 mb-3">
            <div class="col-12">
              <div class="recovery-stat-box">
                <div class="recovery-stat-title">Исполняемый файл</div>
                <div class="recovery-stat-val text-truncate" id="recovery-file-status">Проверка...</div>
                <div class="text-secondary small mt-1 text-bre
```

```
ak font-monospace" id="recovery-file-path" style="font-size: 0
.75rem;">
                  \bin\R-Studio_Technician_9.5.191810_Portable
\R-Studio_9.5.exe
                </div>
              </div>
            </div>

            <div class="col-12">
              <div class="recovery-stat-box">
                <div class="recovery-stat-title">Версия и сбор
ка</div>
                <div class="recovery-stat-val text-info" id="r
ecovery-version">R-Studio Technician 9.5</div>
                <div class="text-secondary small mt-1">Сборка:
 191810 (Portable x64/x86)</div>
              </div>
            </div>

            <div class="col-12">
              <div class="recovery-stat-box">
                <div class="recovery-stat-title">Размер файла
/ Дата модификации</div>
                <div class="recovery-stat-val text-truncate" i
d="recovery-file-size">--</div>
                <div class="text-secondary small mt-1" id="rec
overy-file-date">--</div>
              </div>
            </div>
          </div>
        </div>

        <div class="p-2 rounded bg-body-secondary border borde
r-secondary text-secondary small">
          <i class="bi bi-shield-check text-success me-1"></i>
 Приложение запускается с повышенными привилегиями Windows для
 доступа к секторам физических дисков.
        </div>
```

```
      </div>
    </div>
  </div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows File Recovery Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/file_recovery_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/file_recovery_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * file_recovery_tab/main.js — логика вкладки «Восстановить удаленные файлы» (R-Studio Technician Portable)
 */

(function () {
  'use strict';

  function formatBytes(bytes) {
    if (!bytes || bytes <= 0) return '0 B';
    const units = ['B', 'KB', 'MB', 'GB', 'TB'];
    const i = Math.floor(Math.log(bytes) / Math.log(1024));
```

```
    return (bytes / Math.pow(1024, i)).toFixed(2) + ' ' + units[i];
  }

  function showAlert(msg, type = 'info') {
    const box = document.getElementById('recovery-alert-box');
    if (!box) return;
    box.className = `alert alert-${type} alert-dismissible fade show small py-2 px-3`;
    box.innerHTML = `
      <div class="d-flex align-items-center justify-content-between">
        <div>${msg}</div>
        <button type="button" class="btn-close py-2" data-bs-dismiss="alert" aria-label="Close"></button>
      </div>
    `;
    box.style.display = 'block';
  }

  async function loadRecoveryStatus() {
    const badge = document.getElementById('recovery-status-badge');
    const fileStatus = document.getElementById('recovery-file-status');
    const filePath = document.getElementById('recovery-file-path');
    const fileSize = document.getElementById('recovery-file-size');
    const fileDate = document.getElementById('recovery-file-date');

    try {
      if (badge) badge.textContent = 'Обновление...';
      const res = await fetch(`/api/recovery/status?t=${Date.now()}`);
      if (!res.ok) throw new Error(`HTTP ${res.status}`);

```

```
      const data = await res.json();
      const tool = data.tool || {};

      if (tool.exists) {
        if (badge) {
          badge.className = 'badge rounded-pill bg-success-sub
tle border border-success text-success px-2.5 py-1.5';
          badge.textContent = 'Готов к запуску';
        }
        if (fileStatus) {
          fileStatus.className = 'recovery-stat-val text-succe
ss text-truncate';
          fileStatus.textContent = 'Файл найден';
        }
        if (filePath) filePath.textContent = tool.path || tool
.relative_path || '';
        if (fileSize) fileSize.textContent = formatBytes(tool.
size_bytes);
        if (fileDate) fileDate.textContent = `Изменён: ${tool.
modified_time || '--'}`;
      } else {
        if (badge) {
          badge.className = 'badge rounded-pill bg-warning-sub
tle border border-warning text-warning px-2.5 py-1.5';
          badge.textContent = 'Исполняемый файл не найден';
        }
        if (fileStatus) {
          fileStatus.className = 'recovery-stat-val text-warni
ng text-truncate';
          fileStatus.textContent = 'Не найден в bin';
        }
        if (filePath) filePath.textContent = tool.relative_pat
h || '';
        if (fileSize) fileSize.textContent = '0 B';
        if (fileDate) fileDate.textContent = '--';
      }
    } catch (err) {
      console.error('Ошибка загрузки статуса утилиты восстанов
```

```
ления:', err);
      if (badge) {
        badge.className = 'badge rounded-pill bg-danger-subtle
 border border-danger text-danger px-2.5 py-1.5';
        badge.textContent = 'Ошибка связи';
      }
    }
  }

  async function launchRecoveryTool() {
    const btn = document.getElementById('btn-recovery-launch')
;
    if (!btn) return;

    const originalContent = btn.innerHTML;
    btn.disabled = true;
    btn.innerHTML = `<span class="spinner-border spinner-borde
r-sm me-2" role="status" aria-hidden="true"></span>Запуск R-St
udio...`;

    try {
      const res = await fetch('/api/recovery/launch', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
      });

      const data = await res.json();

      if (res.ok && data.success) {
        showAlert(`<strong>Успех:</strong> ${data.message || '
Программа R-Studio успешно запущена!'}`, 'success');
        if (window.toast) {
          window.toast.success('R-Studio запущена', data.messa
ge || 'Окно программы открывается...');
        }
      } else {
        const err = data.detail || 'Не удалось запустить R-Stu
dio';
```

```
        showAlert(`<strong>Ошибка:</strong> ${err}`, 'danger')
;
        if (window.toast) {
          window.toast.error('Ошибка запуска', err);
        }
      }
    } catch (err) {
      console.error('Ошибка вызова API запуска:', err);
      showAlert(`<strong>Сетевая ошибка:</strong> ${err.messag
e}`, 'danger');
      if (window.toast) {
        window.toast.error('Сетевая ошибка', err.message);
      }
    } finally {
      btn.disabled = false;
      btn.innerHTML = originalContent;
    }
  }

  function initListeners() {
    const refreshBtn = document.getElementById('btn-recovery-r
efresh');
    if (refreshBtn) {
      refreshBtn.onclick = () => loadRecoveryStatus();
    }

    const launchBtn = document.getElementById('btn-recovery-la
unch');
    if (launchBtn) {
      launchBtn.onclick = () => launchRecoveryTool();
    }
  }

  function initFileRecoveryTab() {
    initListeners();
    loadRecoveryStatus();
  }

```

```
  // Экспорт для жизненного цикла tab-core.js
  window.initFile_recoveryTab = initFileRecoveryTab;
  window.initFilerecoveryTab = initFileRecoveryTab;
  window.initFileRecoveryTab = initFileRecoveryTab;
  window.initRecoveryTab = initFileRecoveryTab;

  // Автозапуск при самостоятельной загрузке
  if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', initFileReco
veryTab);
  } else {
    initFileRecoveryTab();
  }
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/firewall_manager_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/firewall_manager_tab/in
dex.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/firewall_manager_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Windows Firewall Manager Tab -->
<div class="container-fluid py-2 px-3">
<!-- Top Stat Cards -->
<div class="row g-2 mb-3">
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 sha
dow-sm p-3" data-panel-id="panel-firewall-manager-1" id="panel
-firewall-manager-1">
<div class="text-muted small text-uppercase fw-semibold">Домен
ный профиль  </div>
```

```
<div class="fs-4 fw-bold mt-1 text-success" id="fw-domain-status">--</div>
<div class="small text-muted mt-1" id="fw-domain-sub">Domain Profile</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-firewall-manager-2" id="panel-firewall-manager-2">
<div class="text-muted small text-uppercase fw-semibold">Частный профиль  </div>
<div class="fs-4 fw-bold mt-1 text-success" id="fw-private-status">--</div>
<div class="small text-muted mt-1" id="fw-private-sub">Private Profile</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-firewall-manager-3" id="panel-firewall-manager-3">
<div class="text-muted small text-uppercase fw-semibold">Публичный профиль  </div>
<div class="fs-4 fw-bold mt-1 text-success" id="fw-public-status">--</div>
<div class="small text-muted mt-1" id="fw-public-sub">Public Profile</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-firewall-manager-4" id="panel-firewall-manager-4">
<div class="text-muted small text-uppercase fw-semibold">Активные правила  </div>
<div class="fs-4 fw-bold mt-1 text-info" id="fw-rules-count">--</div>
```

```
<div class="small text-muted mt-1" id="fw-rules-sub">netsh adv firewall</div>
</div>
</div>
</div>
<!-- Action Bar -->
<div class="d-flex align-items-center justify-content-between mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5" id="fw-refresh-btn">
<i class="bi bi-arrow-clockwise"></i>
<span>Обновить правила</span>
</button>
<button class="btn btn-sm btn-outline-danger d-flex align-items-center gap-1.5" id="fw-reset-defaults-btn">
<i class="bi bi-arrow-counterclockwise"></i>
<span>Сброс по умолчанию (Reset)</span>
</button>
</div>
<div class="d-flex align-items-center gap-2">
<input class="form-control form-control-sm bg-dark text-white border-secondary" id="fw-search-input" placeholder="Поиск правила или порта..." style="width: 260px;" type="text"/>
</div>
</div>
<!-- Firewall Rules Table -->
<div class="card bg-dark text-white border-secondary shadow-sm" data-panel-id="panel-firewall-manager-pravila-brandmauera-windows" id="panel-firewall-manager-pravila-brandmauera-windows">
<div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">
<span class="fw-semibold"><i class="bi bi-shield-shaded text-primary me-2"></i>Правила Брандмауэра Windows</span>
<span class="badge bg-secondary" id="fw-table-badge">0 правил</span>
```

```
</div>
<div class="card-body p-0 table-responsive" style="max-height:
 520px; overflow-y: auto;">
<table class="table table-dark table-hover table-striped align
-middle mb-0 small" id="fw-table">
<thead class="sticky-top bg-dark">
<tr class="text-secondary">
<th>Имя правила</th>
<th>Направление</th>
<th>Действие</th>
<th>Протокол / Порт</th>
<th>Профиль</th>
<th>Состояние</th>
<th class="text-end">Управление</th>
</tr>
</thead>
<tbody id="fw-tbody">
<tr><td class="text-center text-muted py-4" colspan="7">Опрос
правил брандмауэра...</td></tr>
</tbody>
</table>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Firewall Manager Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/firewall_manager_tab/main.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { initFirewallManagerTab } from '/windows/api/webgui/firewall_manager_tab/main.js';
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/firewall_manager_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * firewall_manager_tab/main.js — Управление брандмауэром Windows (netsh advfirewall)
 * Updated: 2026-10-10 17:35:00
 */

const registerTabPoller = window.registerTabPoller || function () {};
```

```
30
31 let isInitialized = false;
32 let rulesList = [];
33
34 export async function fetchFirewallSummary() {
35   try {
36     const res = await fetch('/api/firewall-manager/summary');
37     if (!res.ok) throw new Error(`HTTP ${res.status}`);
38     const data = await res.json();
39     renderFirewall(data);
40   } catch (err) {
41     console.warn('[FirewallManager] Ошибка получения правил:',
    err);
42   }
43 }
44
45 function renderFirewall(data) {
46   rulesList = data.rules || [];
47
48   const domEl = document.getElementById('fw-domain-status');
49   const privEl = document.getElementById('fw-private-status');
50   const pubEl = document.getElementById('fw-public-status');
51   const rulesCountEl = document.getElementById('fw-rules-count
    ');
52
53   if (domEl) domEl.textContent = data.domain_profile || 'ON';
54   if (privEl) privEl.textContent = data.private_profile || 'ON
    ';
55   if (pubEl) pubEl.textContent = data.public_profile || 'ON';
56   if (rulesCountEl) rulesCountEl.textContent = rulesList.lengt
    h;
57
58   applyFirewallFilters();
59 }
60
61 function applyFirewallFilters() {
62   const tbody = document.getElementById('fw-tbody');
63   const searchInput = document.getElementById('fw-search-input
```

```
');
  const badge = document.getElementById('fw-table-badge');

  if (!tbody) return;

  const query = (searchInput ? searchInput.value : '').toLower
Case().trim();

  const filtered = rulesList.filter(r => {
    if (!query) return true;
    const nameMatch = (r.name || r.rule_name || '').toLowerCas
e().includes(query);
    const portMatch = String(r.port || r.local_port || '').inc
ludes(query);
    return nameMatch || portMatch;
  });

  if (badge) badge.textContent = `${filtered.length} из ${rule
sList.length} правил`;

  if (filtered.length === 0) {
    tbody.innerHTML = '<tr><td colspan="7" class="text-center
text-muted py-4">Правила не найдены</td></tr>';
    return;
  }

  tbody.innerHTML = filtered.slice(0, 150).map(r => {
    const name = r.name || r.rule_name || 'Firewall Rule';
    const dir = r.direction || 'In';
    const action = r.action || 'Allow';
    const port = r.local_port || r.port || 'Any';
    const proto = r.protocol || 'TCP';
    const profile = r.profile || 'All';
    const enabled = r.enabled ?? true;

    const isAllow = action.toLowerCase().includes('allow') ||
action.toLowerCase().includes('разреш');

```

```
    return `
      <tr>
        <td class="fw-semibold text-light">${name}</td>
        <td><span class="badge ${dir === 'In' ? 'bg-primary' : 'bg-info'}">${dir}</span></td>
        <td><span class="badge ${isAllow ? 'bg-success' : 'bg-danger'}">${action}</span></td>
        <td class="font-monospace small text-muted">${proto}:${port}</td>
        <td class="small text-muted">${profile}</td>
        <td><span class="badge ${enabled ? 'bg-success' : 'bg-danger-subtle text-danger border border-danger fw-bold'}">${enabled ? 'Включено' : 'Выключено'}</span></td>
        <td class="text-end">
          <button class="btn btn-xs ${enabled ? 'btn-outline-warning' : 'btn-outline-success'} py-0 px-2 fw-toggle-btn"
                  data-name="${name}" data-action="${enabled ? 'disable_rule' : 'enable_rule'}">
            <i class="bi ${enabled ? 'bi-pause-fill' : 'bi-play-fill'}"></i>
          </button>
        </td>
      </tr>
    `;
  }).join('');

  tbody.querySelectorAll('.fw-toggle-btn').forEach(btn => {
    btn.addEventListener('click', () => {
      const rName = btn.dataset.name;
      const rAct = btn.dataset.action;
      executeFirewallAction(rName, rAct);
    });
  });
}

async function executeFirewallAction(ruleName, action) {
  try {
    const res = await fetch('/api/firewall-manager/action', {
```

```
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ rule_name: ruleName, action: acti
on, dry_run: false })
    });
    const result = await res.json();
    if (res.ok) {
      if (window.toast) window.toast.success('Брандмауэр', `${
ruleName}: обновлено`);
      fetchFirewallSummary();
    } else {
      if (window.toast) window.toast.error('Ошибка', result.de
tail || 'Сбой операции');
    }
  } catch (err) {
    if (window.toast) window.toast.error('Ошибка сети', err.me
ssage);
  }
}

export function initFirewallManagerTab() {
  if (isInitialized) return;
  isInitialized = true;

  const refreshBtn = document.getElementById('fw-refresh-btn')
;
  if (refreshBtn) refreshBtn.addEventListener('click', fetchFi
rewallSummary);

  const resetBtn = document.getElementById('fw-reset-defaults-
btn');
  if (resetBtn) resetBtn.addEventListener('click', () => {
    if (confirm('Сбросить правила брандмауэра к значениям по у
молчанию?')) {
      executeFirewallAction('all', 'reset_defaults');
    }
  });

```

```
  const searchInput = document.getElementById('fw-search-input');
  if (searchInput) searchInput.addEventListener('input', applyFirewallFilters);

  registerTabPoller('tab-firewall-manager', fetchFirewallSummary, 15000, { immediate: true });
}

window.initFirewallManagerTab = initFirewallManagerTab;

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Focus Settings Tab
================================================================
===============
Description:
  Веб-интерфейс панели управления сессиями фокусировки Windows
 11 (Focus Sessions),
  режимом «Не беспокоить» (Do Not Disturb), подавлением бейдже
й и мигания панели
  задач, а также вызовом нативного приложения «Часы» и системн
ых настроек.

Usage Examples:
    Web Route:
      /html/focus_settings_tab/index.html

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/focus_settings_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<div class="container-fluid p-3 focus-tab-wrapper d-flex flex-
column gap-3" style="max-width: 900px; margin: 0 auto;">
<!-- Breadcrumb Header -->
<div class="d-flex flex-column gap-1">
<div class="d-flex align-items-center gap-2 text-muted small">
<span>System</span>
<i class="bi bi-chevron-right" style="font-size: 0.75rem;"></i
>
<span class="text-body fw-semibold">Focus</span>
```

```
</div>
<h3 class="mb-0 fw-bold d-flex align-items-center gap-2">
<span>Фокусировка внимания (Focus)</span>
<span class="badge bg-secondary-subtle text-secondary border border-secondary-subtle small fs-6" id="focus-status-badge">Неактивна</span>
</h3>
<div class="text-muted small">
      Сессии фокусировки помогают продуктивнее работать, сокращая отвлекающие факторы. При запуске сессии параметры применяются автоматически на заданное время.
    </div>
</div>
<!-- Main Focus Session Card -->
<div class="card border-secondary-subtle shadow-sm bg-body" data-panel-id="panel-focus-settings-30-mins" id="panel-focus-settings-30-mins">
<div class="card-body p-4 d-flex flex-column gap-3">
<!-- Top Row: Focus Icon, Title, and Start/Stop Button -->
<div class="d-flex flex-wrap align-items-center justify-content-between gap-3 pb-3 border-bottom border-secondary-subtle">
<div class="d-flex align-items-center gap-3">
<div class="rounded-circle bg-primary-subtle text-primary d-flex align-items-center justify-content-center" style="width: 44px; height: 44px; font-size: 1.3rem;">
<i class="bi bi-circle-half"></i>
</div>
<div>
<div class="fw-bold fs-5">Focus</div>
<div class="text-muted small" id="focus-session-hint">Готов к запуску сессии без отвлечений</div>
</div>
</div>
<div class="d-flex align-items-center gap-2">
<div class="d-none badge bg-primary fs-6 py-2 px-3 align-items-center gap-2" id="focus-countdown-box">
<i class="bi bi-stopwatch"></i>
<span id="focus-timer-text">30:00</span>
```

```
</div>
<button class="btn btn-primary px-4 py-2 d-flex align-items-center gap-2 shadow-sm" id="btn-toggle-focus">
<i class="bi bi-play-fill" id="btn-focus-icon"></i>
<span id="btn-focus-text">Start focus session</span>
</button>
</div>
</div>
<!-- Duration Control Row -->
<div class="d-flex flex-wrap align-items-center justify-content-between gap-3 py-2">
<div>
<div class="fw-semibold">Session duration</div>
<div class="text-muted small">Продолжительность сессии фокусировки</div>
</div>
<div class="d-flex align-items-center gap-2 bg-body-secondary p-1 rounded border border-secondary-subtle">
<button class="btn btn-sm btn-outline-secondary px-2 py-1" id="btn-duration-minus" title="Уменьшить длительность" type="button">
<i class="bi bi-dash-lg"></i>
</button>
<span class="fw-bold px-3 text-center" id="lbl-duration" style="min-width: 90px;">30 mins</span>
<button class="btn btn-sm btn-outline-secondary px-2 py-1" id="btn-duration-plus" title="Увеличить длительность" type="button">
<i class="bi bi-plus-lg"></i>
</button>
</div>
</div>
<!-- Settings Checkboxes Group -->
<div class="d-flex flex-column gap-2 pt-2 border-top border-secondary-subtle">
<div class="form-check d-flex align-items-center gap-2 py-2">
<input checked="" class="form-check-input mt-0 fs-5" id="chk-clock-timer" type="checkbox"/>
```

```
<label class="form-check-label fw-medium user-select-none" for="chk-clock-timer">
            Show the timer in the Clock app
            <span class="text-muted d-block small fw-normal">Отображать компактный таймер в приложении Windows Clock</span>
</label>
</div>
<div class="form-check d-flex align-items-center gap-2 py-2">
<input checked="" class="form-check-input mt-0 fs-5" id="chk-hide-badges" type="checkbox"/>
<label class="form-check-label fw-medium user-select-none" for="chk-hide-badges">
            Hide badges on taskbar apps
            <span class="text-muted d-block small fw-normal">Скрывать счетчики и бейджи непрочитанных сообщений на значках панели задач</span>
</label>
</div>
<div class="form-check d-flex align-items-center gap-2 py-2">
<input checked="" class="form-check-input mt-0 fs-5" id="chk-hide-flashing" type="checkbox"/>
<label class="form-check-label fw-medium user-select-none" for="chk-hide-flashing">
            Hide flashing on taskbar apps
            <span class="text-muted d-block small fw-normal">Отключить оранжевое мигание кнопок приложений при поступлении событий</span>
</label>
</div>
<div class="form-check d-flex align-items-center gap-2 py-2">
<input checked="" class="form-check-input mt-0 fs-5" id="chk-turn-on-dnd" type="checkbox"/>
<label class="form-check-label fw-medium user-select-none" for="chk-turn-on-dnd">
            Turn on do not disturb
            <span class="text-muted d-block small fw-normal">Включать режим «Не беспокоить» (Do Not Disturb / Quiet Hours) для блокировки всплывающих уведомлений</span>
```

```
</label>
</div>
</div>
</div>
<div class="card-footer bg-body-tertiary d-flex flex-wrap align-items-center justify-content-between gap-2 p-3">
<div class="text-muted small" id="focus-last-synced">Синхронизировано с реестром Windows</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<button class="btn btn-outline-secondary btn-sm d-flex align-items-center gap-1" id="btn-open-win-settings">
<i class="bi bi-box-arrow-up-right"></i>
<span>Параметры Windows</span>
</button>
<button class="btn btn-outline-secondary btn-sm d-flex align-items-center gap-1" id="btn-open-clock-app">
<i class="bi bi-alarm"></i>
<span>Приложение «Часы»</span>
</button>
<button class="btn btn-success btn-sm d-flex align-items-center gap-1" id="btn-save-focus-settings">
<i class="bi bi-check2"></i>
<span>Сохранить настройки</span>
</button>
</div>
</div>
</div>
<!-- Control Plane Quick Link Card -->
<div class="card border-secondary-subtle shadow-sm bg-body-secondary" data-panel-id="panel-focus-settings-1" id="panel-focus-settings-1">
<div class="card-body p-3 d-flex flex-wrap align-items-center justify-content-between gap-2">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-window-sidebar fs-4 text-info"></i>
<div>
<div class="fw-semibold small">Связанные параметры Window Management Control Plane</div>
```

```
<div class="text-muted small">В каталоге доступно 15 низкоуровневых параметров перехвата фокуса и отслеживания окон за курсором.</div>
</div>
</div>
<button class="btn btn-outline-info btn-sm" id="btn-goto-taskbar-tab">
<i class="bi bi-arrow-right-circle me-1"></i> Панель задач и окна
      </button>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Web Interface - Focus Settings Tab Script
 * ============================================================
 * Description:
 *   Клиентский контроллер вкладки управления сессиями фокусировки (Focus Sessions),
 *   режимом «Не беспокоить» (Do Not Disturb), подавлением бейджей и мигания панели задач.
 *
 * Usage Examples:
 *   JavaScript:
 *     initFocusSettingsTab();
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/focus_settings_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

let durationMinutes = 30;
let isFocusActive = false;
let timerInterval = null;
let remainingSeconds = 0;

export async function initFocusSettingsTab() {
  console.log('[FocusSettingsTab] Инициализация вкладки Focus Settings...');
  bindEventListeners();
```

```
  await loadFocusState();
}

/**
 * Подключение обработчиков событий элементов управления интерфейса.
 */
function bindEventListeners() {
  // Кнопки регулировки длительности сессии
  const btnMinus = document.getElementById('btn-duration-minus');
  const btnPlus = document.getElementById('btn-duration-plus');
  if (btnMinus) {
    btnMinus.onclick = () => changeDuration(-5);
  }
  if (btnPlus) {
    btnPlus.onclick = () => changeDuration(5);
  }

  // Кнопка старта / остановки сессии
  const btnToggle = document.getElementById('btn-toggle-focus');
  if (btnToggle) {
    btnToggle.onclick = toggleFocusSession;
  }

  // Сохранение настроек
  const btnSave = document.getElementById('btn-save-focus-settings');
  if (btnSave) {
    btnSave.onclick = saveFocusSettings;
  }

  // Запуск нативных окон Windows
  const btnWinSettings = document.getElementById('btn-open-win-settings');
  if (btnWinSettings) {
```

```
    btnWinSettings.onclick = openWindowsSettings;
  }

  const btnClockApp = document.getElementById('btn-open-clock-app');
  if (btnClockApp) {
    btnClockApp.onclick = openClockApp;
  }

  // Переход к смежной вкладке управления панелью задач
  const btnTaskbar = document.getElementById('btn-goto-taskbar-tab');
  if (btnTaskbar) {
    btnTaskbar.onclick = () => {
      if (typeof window.switchTab === 'function') {
        window.switchTab('tab-taskbar-controller');
      }
    };
  }
}

/**
 * Изменение длительности сессии с шагом в 5 минут (диапазон 5..240 мин).
 * @param {number} delta Шаг изменения в минутах.
 */
function changeDuration(delta) {
  if (isFocusActive) return;
  durationMinutes = Math.max(5, Math.min(240, durationMinutes + delta));
  updateDurationLabel();
}

/**
 * Обновление текстовой метки длительности.
 */
function updateDurationLabel() {
  const lbl = document.getElementById('lbl-duration');
```

```
  if (lbl) {
    lbl.textContent = `${durationMinutes} mins`;
  }
}

/**
 * Загрузка актуального состояния из REST API.
 */
async function loadFocusState() {
  try {
    const res = await fetch('/api/windows/focus/state');
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    renderState(data);
  } catch (err) {
    console.warn('[FocusSettingsTab] Ошибка загрузки состояния
 focus:', err);
  }
}

/**
 * Отрисовка полученного состояния в интерфейсе.
 * @param {object} state Объект FocusSessionStateDTO.
 */
function renderState(state) {
  isFocusActive = Boolean(state.is_active);
  durationMinutes = state.session_duration_minutes || duration
Minutes;
  remainingSeconds = state.remaining_seconds || 0;

  updateDurationLabel();

  // Обновление чекбоксов
  const s = state.settings || {};
  setCheckbox('chk-clock-timer', s.show_timer_in_clock_app ??
true);
  setCheckbox('chk-hide-badges', s.hide_badges_on_taskbar ?? t
rue);
```

```
  setCheckbox('chk-hide-flashing', s.hide_flashing_on_taskbar ?? true);
  setCheckbox('chk-turn-on-dnd', s.turn_on_do_not_disturb ?? true);

  // Обновление бейджа и кнопок
  const badge = document.getElementById('focus-status-badge');
  const btnToggle = document.getElementById('btn-toggle-focus');
  const btnIcon = document.getElementById('btn-focus-icon');
  const btnText = document.getElementById('btn-focus-text');
  const hint = document.getElementById('focus-session-hint');
  const countdownBox = document.getElementById('focus-countdown-box');

  if (isFocusActive) {
    if (badge) {
      badge.className = 'badge bg-success small fs-6';
      badge.textContent = 'Сессия активна';
    }
    if (btnToggle) {
      btnToggle.className = 'btn btn-danger px-4 py-2 d-flex align-items-center gap-2 shadow-sm';
    }
    if (btnIcon) btnIcon.className = 'bi bi-stop-fill';
    if (btnText) btnText.textContent = 'Stop focus session';
    if (hint) hint.textContent = 'Сессия фокусировки активна. Отвлекающие факторы заблокированы.';
    if (countdownBox) countdownBox.classList.remove('d-none');

    startLocalCountdown();
  } else {
    if (badge) {
      badge.className = 'badge bg-secondary-subtle text-secondary border border-secondary-subtle small fs-6';
      badge.textContent = 'Не активна';
    }
    if (btnToggle) {
```

```
      btnToggle.className = 'btn btn-primary px-4 py-2 d-flex align-items-center gap-2 shadow-sm';
    }
    if (btnIcon) btnIcon.className = 'bi bi-play-fill';
    if (btnText) btnText.textContent = 'Start focus session';
    if (hint) hint.textContent = 'Готов к запуску сессии без отвлечений';
    if (countdownBox) countdownBox.classList.add('d-none');

    stopLocalCountdown();
  }
}

/**
 * Локальный таймер обратного отсчета для плавного отображения секунд.
 */
function startLocalCountdown() {
  stopLocalCountdown();
  updateCountdownDisplay();
  timerInterval = setInterval(() => {
    if (remainingSeconds > 0) {
      remainingSeconds -= 1;
      updateCountdownDisplay();
    } else {
      stopLocalCountdown();
      loadFocusState();
    }
  }, 1000);
}

function stopLocalCountdown() {
  if (timerInterval) {
    clearInterval(timerInterval);
    timerInterval = null;
  }
}

```

```
function updateCountdownDisplay() {
  const timerText = document.getElementById('focus-timer-text');
  if (timerText) {
    const mins = Math.floor(remainingSeconds / 60);
    const secs = remainingSeconds % 60;
    timerText.textContent = `${String(mins).padStart(2, '0')}:${String(secs).padStart(2, '0')}`;
  }
}

/**
 * Переключение состояния сессии фокусировки (Старт / Стоп).
 */
async function toggleFocusSession() {
  const btnToggle = document.getElementById('btn-toggle-focus');
  if (btnToggle) btnToggle.disabled = true;

  try {
    // Сохраняем текущие выбранные чекбоксы перед стартом
    await saveFocusSettings();

    const endpoint = isFocusActive ? '/api/windows/focus/session/stop' : `/api/windows/focus/session/start?duration_minutes=${durationMinutes}`;
    const res = await fetch(endpoint, { method: 'POST' });
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    renderState(data);
  } catch (err) {
    console.error('[FocusSettingsTab] Ошибка переключения сессии:', err);
    alert(`Ошибка управления сессией фокусировки: ${err.message}`);
  } finally {
    if (btnToggle) btnToggle.disabled = false;
  }
```

```
}

/**
 * Сохранение конфигурации в постоянные настройки.
 */
async function saveFocusSettings() {
  const settings = {
    session_duration_minutes: durationMinutes,
    show_timer_in_clock_app: getCheckbox('chk-clock-timer'),
    hide_badges_on_taskbar: getCheckbox('chk-hide-badges'),
    hide_flashing_on_taskbar: getCheckbox('chk-hide-flashing')
,
    turn_on_do_not_disturb: getCheckbox('chk-turn-on-dnd')
  };

  try {
    const res = await fetch('/api/windows/focus/settings', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(settings)
    });
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const syncLabel = document.getElementById('focus-last-sync
ed');
    if (syncLabel) {
      syncLabel.textContent = `Сохранено в ${new Date().toLoca
leTimeString()}`;
    }
  } catch (err) {
    console.warn('[FocusSettingsTab] Ошибка сохранения настрое
к:', err);
  }
}

/**
 * Вызов нативных параметров Windows.
 */
async function openWindowsSettings() {
```

```
  try {
    await fetch('/api/windows/focus/open-settings', { method:
'POST' });
  } catch (err) {
    console.warn('[FocusSettingsTab] Ошибка вызова параметров
Windows:', err);
  }
}

/**
 * Вызов нативного приложения Windows Clock.
 */
async function openClockApp() {
  try {
    await fetch('/api/windows/focus/open-clock', { method: 'PO
ST' });
  } catch (err) {
    console.warn('[FocusSettingsTab] Ошибка вызова Windows Clo
ck:', err);
  }
}

function getCheckbox(id) {
  const el = document.getElementById(id);
  return el ? Boolean(el.checked) : false;
}

function setCheckbox(id, value) {
  const el = document.getElementById(id);
  if (el) el.checked = Boolean(value);
}

// Экспорт для глобального использования
window.initFocusSettingsTab = initFocusSettingsTab;

// Автоинициализация при загрузке содержимого вкладки
if (document.readyState === 'loading') {
  document.addEventListener('DOMContentLoaded', initFocusSetti
```

```
ngsTab);
} else {
  initFocusSettingsTab();
}

```

index.html

```
 <!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса вкладки Forensics с таймла
йном активности.

Usage Examples:
    Web Route:
      /windows/api/webgui/forensics_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/forensics_tab/index.htm
l" class="w-100 h-100 border-0"></iframe>

 * File: index.html
 * Project: ai-windows-admin
 * Package: windows/api/webgui/forensics_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 20:33:00
 * ================================================================
==================
-->
<style>
  .forensics-tab-wrapper {
    padding: 0.75rem;
    color: var(--text-color);
  }
  .diag-panel {
    background: var(--surface-1);
    border: 1px solid var(--border-color);
    border-radius: 8px;
```

```
    overflow: hidden;
  }
  .diag-panel-header {
    background: var(--surface-2);
    padding: 8px 14px;
    border-bottom: 1px solid var(--border-color);
    font-weight: 600;
    font-size: 0.85rem;
    color: var(--nav-active);
  }
  .about-ai-pulse-icon {
    width: 42px;
    height: 42px;
    border-radius: 8px;
    display: flex;
    align-items: center;
    justify-content: center;
    border: 1px solid rgba(56, 189, 248, 0.3);
  }
</style>
<div class="forensics-tab-wrapper">
<!-- Top Header & Status -->
<div class="card shadow-sm border-secondary-subtle mb-3" data-panel-id="panel-forensics-2-povedencheskaya-forenzika-akti" id="panel-forensics-2-povedencheskaya-forenzika-akti">
<div class="card-body p-3">
<div class="d-flex flex-wrap align-items-center justify-content-between gap-2">
<div class="d-flex align-items-center gap-2">
<div class="fs-4">□ </div>
<div>
<h5 class="mb-0 fw-bold">2. Поведенческая форензика и хронология активности</h5>
<small class="text-muted">Телеметрия активного фокуса, таймлайн работы пользователя, аудит камеры/микрофона и реестр UserAssist</small>
</div>
</div>
```

```
<div class="d-flex align-items-center gap-2 flex-wrap">
<select class="form-select form-select-sm poll-freq-select" data-poll-id="forensics" id="diag-forensics-poll-freq" title="Частота опроса: Форензика активности">
<option value="start">Только при старте</option>
<option selected="" value="manual">По запросу</option>
<option value="3">Каждые 3 сек</option>
<option value="5">Каждые 5 сек</option>
<option value="10">Каждые 10 сек</option>
<option value="30">Каждые 30 сек</option>
<option value="60">Каждые 60 сек</option>
<option value="120">Каждые 120 сек</option>
</select>
<button class="btn btn-sm btn-outline-info rounded-pill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id="btn-diag-forensics-refresh">
<i class="bi bi-arrow-clockwise"></i> Обновить
          </button>
</div>
</div>
</div>
</div>
<div class="row g-2 mb-3">
<!-- Active Foreground Window Card -->
<div class="col-12 col-lg-6">
<div class="diag-panel h-100" data-panel-id="panel-forensics-1" id="panel-forensics-1">
<div class="diag-panel-header d-flex justify-content-between align-items-center">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-window-stack text-primary"></i>
<span class="text-body">Текущее активное окно в фокусе (Foreground Window)</span>
</div>
<span class="badge bg-success-subtle text-success border border-success" id="diag-forensics-live-badge">Live Focus</span>
</div>
<div class="p-3">
```

```
<div class="d-flex align-items-center gap-3 mb-2">
<div class="about-ai-pulse-icon" style="background: rgba(14, 165, 233, 0.15); border-color: rgba(14, 165, 233, 0.4);">
<i class="bi bi-cursor-fill text-info fs-5"></i>
</div>
<div class="flex-grow-1 overflow-hidden">
<div class="fw-bold text-body text-truncate" id="diag-fg-title" style="font-size: 0.95rem;">Рабочий стол Windows</div>
<div class="small text-muted font-monospace" id="diag-fg-proc">Процесс: explorer.exe (PID: 0)</div>
</div>
</div>
<div class="border-top border-secondary-subtle pt-2 mt-2 d-flex justify-content-between align-items-center flex-wrap gap-2">
<div>
<span class="text-muted small">Время бездействия пользователя (Idle Time):</span>
<strong class="text-warning font-monospace ms-1" id="diag-user-idle-time">0.0 сек</strong>
</div>
<span class="badge bg-body-secondary border border-secondary text-body" id="diag-user-status-badge">Пользователь активен</span>
</div>
</div>
</div>
</div>
<!-- Privacy Sensors (Camera & Mic Radar) -->
<div class="col-12 col-lg-6">
<div class="diag-panel h-100" data-panel-id="panel-forensics-2" id="panel-forensics-2">
<div class="diag-panel-header d-flex justify-content-between align-items-center">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-camera-video-fill text-danger"></i>
<span class="text-body">Контроль доступа к камере и микрофону (Privacy Radar)</span>
</div>
```

```
<span class="badge bg-body-secondary border border-secondary t
ext-body">ConsentStore</span>
</div>
<div class="p-3">
<div class="row g-2">
<div class="col-6">
<div class="p-2 rounded border border-secondary-subtle" style=
"background: var(--surface-2);">
<div class="small text-muted mb-1"><i class="bi bi-camera-vide
o me-1 text-danger"></i>Камера:</div>
<div class="small text-body" id="diag-cam-access-list" style="
max-height: 90px; overflow-y: auto;">
<span class="text-muted">Нет активных обращений</span>
</div>
</div>
</div>
<div class="col-6">
<div class="p-2 rounded border border-secondary-subtle" style=
"background: var(--surface-2);">
<div class="small text-muted mb-1"><i class="bi bi-mic me-1 te
xt-info"></i>Микрофон:</div>
<div class="small text-body" id="diag-mic-access-list" style="
max-height: 90px; overflow-y: auto;">
<span class="text-muted">Нет активных обращений</span>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
</div>

<!-- Timeline History of User Activity & Foreground Windows --
>
<div class="diag-panel mb-3" data-panel-id="panel-forensics-ti
meline" id="panel-forensics-timeline">
<div class="diag-panel-header d-flex justify-content-between a
```

```
lign-items-center">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-clock-history text-info"></i>
<span class="text-body">Хронологический таймлайн активности и переключения окон (Telemetry Timeline)</span>
</div>
<span class="badge bg-primary" id="diag-timeline-count">0 запи сей</span>
</div>
<div class="table-responsive" style="max-height: 380px; overfl ow-y: auto;">
<table class="table table-hover table-sm mb-0" style="font-siz e: 0.80rem; font-family: ui-monospace, SFMono-Regular, Menlo, Monaco, Consolas, monospace;">
<thead>
<tr class="text-secondary border-secondary">
<th style="width: 170px;">Время фиксации</th>
<th style="width: 160px;">Процесс (PID)</th>
<th>Активное окно в фокусе (Window Title)</th>
<th style="width: 130px; text-align: right;">Idle Time</th>
<th style="width: 120px; text-align: center;">Сенсоры</th>
</tr>
</thead>
<tbody id="diag-timeline-tbody">
<tr><td class="text-center py-4 text-muted" colspan="5">Загруз ка хронологии активности из базы телеметрии...</td></tr>
</tbody>
</table>
</div>
</div>

<!-- UserAssist Application Run History -->
<div class="diag-panel" data-panel-id="panel-forensics-3" id=" panel-forensics-3">
<div class="diag-panel-header d-flex justify-content-between a lign-items-center">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-archive text-secondary"></i>
```

```
<span class="text-body">Форензика запуска программ из реестра Windows (UserAssist ROT13)</span>
</div>
<span class="badge bg-secondary" id="diag-userassist-count">0 записей</span>
</div>
<div class="table-responsive" style="max-height: 320px; overflow-y: auto;">
<table class="table table-hover table-sm mb-0" style="font-size: 0.80rem; font-family: ui-monospace, SFMono-Regular, Menlo, Monaco, Consolas, monospace;">
<thead>
<tr class="text-secondary border-secondary">
<th>Приложение / Программа</th>
<th style="width: 140px; text-align: right;">Счетчик запусков</th>
<th style="width: 180px; text-align: right;">Время в фокусе</th>
<th>Полный путь / Идентификатор</th>
</tr>
</thead>
<tbody id="diag-userassist-tbody">
<tr><td class="text-center py-4 text-muted" colspan="4">Загрузка истории активности UserAssist...</td></tr>
</tbody>
</table>
</div>
</div>
</div>
```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Forensics Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом форензики и таймлайна активности.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/forensics_tab/main.js?v=20261010_v2" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/forensics_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 20:33:00
 * ============================================================
 */

(function () {
  'use strict';

  let autoRefreshTimer = null;

  function escapeHtml(str) {
    if (str === null || str === undefined) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
```

```
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
  }

  function setText(id, text) {
    const el = document.getElementById(id);
    if (el) el.textContent = text !== null && text !== undefin
ed ? String(text) : '--';
  }

  function formatTime(isoStr) {
    if (!isoStr) return '--:--:--';
    try {
      const d = new Date(isoStr);
      if (isNaN(d.getTime())) return isoStr;
      return d.toLocaleTimeString('ru-RU', { hour: '2-digit',
minute: '2-digit', second: '2-digit' }) +
             ' (' + d.toLocaleDateString('ru-RU', { day: '2-di
git', month: '2-digit' }) + ')';
    } catch (_) {
      return isoStr;
    }
  }

  async function apiFetch(url) {
    if (window.api && typeof window.api.fetch === 'function')
{
      return await window.api.fetch(url);
    }
    const res = await fetch(url);
    if (!res.ok) throw new Error(HTTP : );
    return await res.json();
  }

  async function fetchForensicsActivity() {
    try {
      const data = await apiFetch('/api/v1/system/diagnostics/
forensics');
```

```
65      if (!data) return;
66
67      if (data.foreground_window) {
68        setText('diag-fg-title', data.foreground_window.title
|| 'Рабочий стол Windows');
69        setText('diag-fg-proc', Процесс:  (PID: ));
70      }
71
72      const idleSec = data.user_idle_seconds !== undefined ? N
umber(data.user_idle_seconds) : 0;
73      setText('diag-user-idle-time', ${idleSec.toFixed(1)} сек
);
74      const statusBadge = document.getElementById('diag-user-s
tatus-badge');
75      if (statusBadge) {
76        if (idleSec > 300) {
77          statusBadge.className = 'badge bg-warning-subtle tex
t-warning border border-warning';
78          statusBadge.textContent = 'Пользователь отошел (>5 м
ин)';
79        } else {
80          statusBadge.className = 'badge bg-success-subtle tex
t-success border border-success';
81          statusBadge.textContent = 'Пользователь активен';
82        }
83      }
84
85      // Camera Apps
86      const camContainer = document.getElementById('diag-cam-a
ccess-list');
87      if (camContainer) {
88        if (data.camera_active_apps && data.camera_active_apps
.length > 0) {
89          camContainer.innerHTML = data.camera_active_apps.map
(a =>
90            <div class="d-flex align-items-center justify-cont
ent-between py-1 border-bottom border-secondary-subtle">
91              <span class="text-truncate" style="max-width: 14
```

```
0px;" title=""></span>
              <span class="badge " style="font-size: 0.65rem;"
>

              </span>
            </div>
          ).join('');
        } else {
          camContainer.innerHTML = '<span class="text-muted sm
all">Нет активных обращений к камере</span>';
        }
      }

      // Mic Apps
      const micContainer = document.getElementById('diag-mic-a
ccess-list');
      if (micContainer) {
        if (data.microphone_active_apps && data.microphone_act
ive_apps.length > 0) {
          micContainer.innerHTML = data.microphone_active_apps
.map(a =>
            <div class="d-flex align-items-center justify-cont
ent-between py-1 border-bottom border-secondary-subtle">
              <span class="text-truncate" style="max-width: 14
0px;" title=""></span>
              <span class="badge " style="font-size: 0.65rem;"
>

              </span>
            </div>
          ).join('');
        } else {
          micContainer.innerHTML = '<span class="text-muted sm
all">Нет активных обращений к микрофону</span>';
        }
      }

      // Timeline History Table
```

```
      const timelineTbody = document.getElementById('diag-time
line-tbody');
      if (timelineTbody) {
        const historyList = data.history || [];
        setText('diag-timeline-count', ${historyList.length} с
резов);
        if (historyList.length === 0) {
          timelineTbody.innerHTML = '<tr><td colspan="5" class
="text-center py-4 text-muted">Снимков активности в базе пока
нет. Они накапливаются демоном телеметрии...</td></tr>';
        } else {
          timelineTbody.innerHTML = historyList.map(h => {
            const fg = h.foreground_window || {};
            const timeStr = formatTime(h.timestamp);
            const idle = Number(h.user_idle_seconds || 0).toFi
xed(0);
            const hasCam = (h.camera_active_apps || []).some(x
 => x.is_active_now);
            const hasMic = (h.microphone_active_apps || []).so
me(x => x.is_active_now);
            let sensorBadge = '<span class="text-muted">-</spa
n>';
            if (hasCam && hasMic) {
              sensorBadge = '<span class="badge bg-danger" tit
le="Камера + Микрофон">Cam + Mic</span>';
            } else if (hasCam) {
              sensorBadge = '<span class="badge bg-danger" tit
le="Камера активна">Cam</span>';
            } else if (hasMic) {
              sensorBadge = '<span class="badge bg-warning tex
t-dark" title="Микрофон активен">Mic</span>';
            }

            return
              <tr>
                <td class="text-muted font-monospace small"></
td>
                <td class="fw-bold text-info"><i class="bi bi-
```

```
window text-secondary me-1"></i> <span class="text-muted small
">()</span></td>
146                 <td class="text-body text-truncate" style="max
-width: 320px;" title=""></td>
147                 <td style="text-align: right;" class="font-mon
ospace "> c</td>
148                 <td style="text-align: center;"></td>
149               </tr>
150             ;
151           }).join('');
152         }
153       }
154
155       // UserAssist Apps Table
156       const uaTbody = document.getElementById('diag-userassist
-tbody');
157       if (uaTbody && data.userassist_top_apps) {
158         setText('diag-userassist-count', ${data.userassist_top
_apps.length} программ);
159         if (data.userassist_top_apps.length === 0) {
160           uaTbody.innerHTML = '<tr><td colspan="4" class="text
-center py-4 text-muted">Данных UserAssist в реестре не обнару
жено (или трекинг отключен в Windows)</td></tr>';
161         } else {
162           uaTbody.innerHTML = data.userassist_top_apps.map(a =
>
163             <tr>
164               <td class="fw-bold text-white"><i class="bi bi-a
pp text-info me-1.5"></i></td>
165               <td style="text-align: right;" class="text-info
font-monospace"> раз</td>
166               <td style="text-align: right;" class="text-warni
ng font-monospace"></td>
167               <td class="small text-muted text-truncate" style
="max-width: 280px;" title=""></td>
168             </tr>
169           ).join('');
170         }
```

```
      }
    } catch (err) {
      console.warn('[ForensicsTab] fetchForensicsActivity erro
r:', err);
    }
  }

  const POLL_ID = 'forensics';

  function getFrequency() {
    try {
      const saved = localStorage.getItem(poll_freq_);
      if (saved) return saved;
    } catch (_) {}
    return 'manual';
  }

  function setFrequency(freq) {
    try {
      localStorage.setItem(poll_freq_, freq);
    } catch (_) {}
    applyPoller(freq, false);
  }

  function stopPolling() {
    if (window.unregisterTabPoller) {
      window.unregisterTabPoller(    ab-forensics_);
    }
    if (autoRefreshTimer) {
      clearInterval(autoRefreshTimer);
      autoRefreshTimer = null;
    }
  }

  function applyPoller(freq, runInitial = false) {
    stopPolling();
    if (freq === 'start' || freq === 'manual') {
      if (runInitial && (window.isTabActive ? window.isTabActi
```

```
ve('tab-forensics') : true)) {
        fetchForensicsActivity();
      }
      return;
    }

    const intervalSec = parseInt(freq, 10);
    if (isNaN(intervalSec) || intervalSec <= 0) return;

    const intervalMs = intervalSec * 1000;
    const pollerId =     ab-forensics_;

    if (window.registerTabPoller) {
      window.registerTabPoller('tab-forensics', fetchForensics
Activity, intervalMs, { pollerId, immediate: runInitial });
    } else {
      if (runInitial && (window.isTabActive ? window.isTabActi
ve('tab-forensics') : true)) {
        fetchForensicsActivity();
      }
      autoRefreshTimer = setInterval(() => {
        if (window.isTabActive ? window.isTabActive('tab-foren
sics') : true) {
          fetchForensicsActivity();
        }
      }, intervalMs);
    }
  }

  function bindEvents() {
    const btnRefresh = document.getElementById('btn-diag-foren
sics-refresh');
    if (btnRefresh) {
      btnRefresh.onclick = () => fetchForensicsActivity();
    }

    const select = document.getElementById('diag-forensics-pol
l-freq');
```

```
    if (select) {
      select.onchange = (e) => {
        const newFreq = e.target.value;
        setFrequency(newFreq);
        if (newFreq !== 'manual' && newFreq !== 'start') {
          fetchForensicsActivity();
        }
        if (window.showToast) {
          const label = select.options[select.selectedIndex]?.
text || newFreq;
          window.showToast(Частота опроса форензики: , 'info')
;
        }
      };
    }
  }

  async function initForensicsTab() {
    bindEvents();
    const currentFreq = getFrequency();
    const select = document.getElementById('diag-forensics-pol
l-freq');
    if (select) select.value = currentFreq;
    applyPoller(currentFreq, true);
  }

  window.initForensicsTab = initForensicsTab;
  window.activateForensicsTab = () => {
    applyPoller(getFrequency(), true);
  };

  if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', () => {
      if (window.isTabActive ? window.isTabActive('tab-forensi
cs') : false) {
        initForensicsTab();
      } else {
        bindEvents();
```

```
      }
    });
  } else {
    if (window.isTabActive ? window.isTabActive('tab-forensics') : false) {
      initForensicsTab();
    } else {
      bindEvents();
    }
  }
})();
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/gcloud_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/gcloud_tab/index.html"
class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/gcloud_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Google Cloud Monitor Application Tab -->
<style>
  .gcloud-container {
    padding: 0.5rem 0.75rem;
  }
  .gcloud-card {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: 8px;
    padding: 0.45rem 0.75rem;
```

```
    display: flex;
    flex-direction: column;
    justify-content: space-between;
    min-height: 58px;
    gap: 0.2rem;
  }
  .gcloud-label {
    font-size: 0.68rem;
    text-transform: uppercase;
    letter-spacing: 0.05em;
    color: var(--text-muted);
    line-height: 1.1;
  }
  .gcloud-value {
    font-size: 1.25rem;
    font-weight: 700;
    color: var(--text-color);
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
    line-height: 1.15;
  }
  .gcloud-panel {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: 8px;
    overflow: hidden;
  }
  .gcloud-panel-header {
    background: var(--surface-2) !important;
    border-bottom: 1px solid var(--border-color) !important;
    padding: 0.45rem 0.85rem;
    font-weight: 600;
    font-size: 0.82rem;
    color: var(--nav-active);
    display: flex;
    align-items: center;
    justify-content: space-between;
  }
```

```
  .gcloud-log-terminal {
    background: var(--surface-2);
    color: var(--text-color);
    border: 1px solid var(--border-color);
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco, Consolas, monospace;
    font-size: 0.78rem;
    line-height: 1.45;
    padding: 0.5rem 0.75rem;
    border-radius: 6px;
    max-height: 400px;
    overflow-y: auto;
  }
</style>
<div class="gcloud-container">
<!-- Header Alert & Status (Compact Strip) -->
<div class="card shadow-sm border-secondary-subtle py-1.5 px-3 mb-2 d-flex justify-content-between align-items-center flex-wrap gap-2" data-panel-id="panel-gcloud-google-cloud-monitor" id="panel-gcloud-google-cloud-monitor">
<div class="d-flex align-items-center gap-2 flex-wrap">
<span class="fs-5">□ </span>
<span class="fw-bold text-body small" style="font-size: 0.92rem;">Google Cloud Monitor</span>
<span class="badge bg-body-secondary border border-secondary text-body fw-normal" style="font-size: 0.7rem;">Observability</span>
<span class="text-secondary small d-none d-lg-inline" style="font-size: 0.76rem;">| Cloud Logging, метрики Cloud Monitoring &amp;amp; IAM</span>
</div>
<div class="d-flex align-items-center gap-1.5 flex-wrap">
<span class="badge rounded-pill bg-success-subtle text-success border border-success px-2 py-1" id="gcloud-status-badge" style="font-size: 0.72rem;">
        ● GCP Подключен
      </span>
<button class="btn btn-xs btn-outline-info rounded-pill px-2.5
```

```
 py-1" id="btn-gcloud-refresh" style="font-size: 0.75rem;">
<i class="bi bi-arrow-clockwise me-1"></i> Обновить
      </button>
<button class="btn btn-xs btn-outline-warning rounded-pill px-2 py-1" id="btn-gcloud-config" style="font-size: 0.75rem;">
<i class="bi bi-gear-fill me-1"></i> Настройки
      </button>
</div>
</div>
<!-- Top Metrics Bar (Compact Density Grid) -->
<div class="row g-2 mb-2">
<div class="col-6 col-md-3">
<div class="gcloud-card">
<span class="gcloud-label">Проект GCP</span>
<div class="d-flex align-items-baseline justify-content-between mt-1">
<span class="gcloud-value text-info text-truncate" id="gcloud-project-id">-</span>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="gcloud-card">
<span class="gcloud-label">Тип авторизации</span>
<div class="d-flex align-items-baseline justify-content-between mt-1">
<span class="gcloud-value text-body" id="gcloud-auth-type" style="font-size: 1.05rem;">-</span>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="gcloud-card">
<span class="gcloud-label">Служебный аккаунт</span>
<div class="d-flex align-items-baseline justify-content-between mt-1">
<span class="gcloud-value text-secondary text-truncate" id="gcloud-client-email" style="font-size: 0.85rem;">-</span>
```

```
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="gcloud-card">
<span class="gcloud-label">Здоровье инфраструктуры</span>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<span class="gcloud-value text-success" id="gcloud-health-scor
e">100/100</span>
</div>
</div>
</div>
</div>
<!-- Main Content Panels Grid -->
<div class="row g-3">
<!-- Cloud Logging Terminal Panel -->
<div class="col-lg-7">
<div class="gcloud-panel h-100">
<div class="gcloud-panel-header">
<span>  Cloud Logging (Последние события)</span>
<span class="badge bg-secondary" id="gcloud-logs-count-badge">
0 строк</span>
</div>
<div class="p-2">
<div class="gcloud-log-terminal" id="gcloud-log-terminal-outpu
t">
            Загрузка журналов Cloud Logging...
          </div>
</div>
</div>
</div>
<!-- AI Diagnostics & Recommendations Panel -->
<div class="col-lg-5">
<div class="gcloud-panel h-100">
<div class="gcloud-panel-header">
<span>  AI-Диагностика и рекомендации</span>
<span class="badge bg-info text-dark">Gemini Diagnostics</span
```

```
>
</div>
<div class="p-3">
<h6 class="fw-bold text-info mb-1" id="gcloud-diag-summary-title">Анализ телеметрии...</h6>
<p class="small text-body mb-3" id="gcloud-diag-summary-desc">Загрузка диагностического заключения...</p>
<div class="fw-semibold text-secondary small text-uppercase mb-2">Рекомендованные действия:</div>
<ul class="list-group list-group-flush small" id="gcloud-recommendations-list">
<li class="list-group-item bg-transparent text-body border-secondary py-1 px-0">• Ожидание данных телеметрии...</li>
</ul>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Gcloud Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/gcloud_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/gcloud_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// ============================================================
// Process Name: Google Cloud Monitor Web Tab Module
// ============================================================
// Description:
//   Client-side JavaScript controller for the Google Cloud Monitor
//   administrative web interface tab.
//
// File: main.js
// Project: ai-windows-admin
```

```
// Package: src.api.webinterface.gcloud_tab
// Author: hypo69
// Copyright: © 2026 hypo69
// ============================================================
==================

(function() {
  let isGcloudInitialized = false;

  async function fetchStatus() {
    try {
      const res = await fetch('/api/gcloud/status');
      if (!res.ok) return;
      const data = await res.json();

      const badge = document.getElementById('gcloud-status-bad
ge');
      const projectEl = document.getElementById('gcloud-projec
t-id');
      const authTypeEl = document.getElementById('gcloud-auth-
type');
      const emailEl = document.getElementById('gcloud-client-e
mail');

      if (badge) {
        badge.className = data.authenticated ? 'badge rounded-
pill bg-success-subtle text-success border border-success px-3
 py-2' : 'badge rounded-pill bg-warning-subtle text-warning bo
rder border-warning px-3 py-2';
        badge.innerText = data.authenticated ? (data.is_mock ?
 '● Mock Режим' : '● GCP Подключен') : '● Не авторизован';
      }

      if (projectEl) projectEl.innerText = data.project_id ||
'default-project';
      if (authTypeEl) authTypeEl.innerText = (data.auth_type |
| 'service_account').toUpperCase();
      if (emailEl) emailEl.innerText = data.client_email || 'l
```

```
ocal-service-account';
57      } catch (e) {
58        console.error('[GCloudTab] Failed to fetch status:', e);
59      }
60    }
61
62    async function fetchLogs() {
63      try {
64        const res = await fetch('/api/gcloud/logs?limit=40');
65        if (!res.ok) return;
66        const logs = await res.json();
67
68        const terminal = document.getElementById('gcloud-log-ter
minal-output');
69        const countBadge = document.getElementById('gcloud-logs-
count-badge');
70        if (countBadge) countBadge.innerText = `${logs.length} с
трок`;
71
72        if (terminal) {
73          if (!Array.isArray(logs) || logs.length === 0) {
74            terminal.innerText = 'Журнал Cloud Logging пуст.';
75            return;
76          }
77          terminal.innerHTML = logs.map(l => {
78            const sev = (l.severity || 'INFO').toUpperCase();
79            const color = sev === 'ERROR' || sev === 'CRITICAL'
? '#f87171' : (sev === 'WARNING' ? '#fbbf24' : '#94a3b8');
80            return `<div style="color: ${color};">[${l.timestamp
 || ''}] [${sev}] ${l.resource_type ? '[' + l.resource_type +
'] ' : ''}${l.message || ''}</div>`;
81          }).join('');
82        }
83      } catch (e) {
84        console.error('[GCloudTab] Failed to fetch logs:', e);
85      }
86    }
87
```

```
  async function runDiagnostics() {
    try {
      const res = await fetch('/api/gcloud/diagnostic');
      if (!res.ok) return;
      const report = await res.json();

      const scoreEl = document.getElementById('gcloud-health-score');
      const titleEl = document.getElementById('gcloud-diag-summary-title');
      const descEl = document.getElementById('gcloud-diag-summary-desc');
      const recList = document.getElementById('gcloud-recommendations-list');

      if (scoreEl) scoreEl.innerText = `${report.health_score || 100}/100`;
      if (titleEl) titleEl.innerText = `Статус: ${report.overall_status || 'HEALTHY'}`;
      if (descEl) descEl.innerText = report.summary || 'Все сервисы Google Cloud функционируют в штатном режиме.';

      if (recList && Array.isArray(report.recommendations)) {
        recList.innerHTML = report.recommendations.map(r => `
          <li class="list-group-item bg-transparent text-light border-secondary py-1 px-0">• ${r}</li>
        `).join('');
      }
    } catch (e) {
      console.error('[GCloudTab] Diagnostic error:', e);
    }
  }

  function initGCloudTab() {
    fetchStatus();
    fetchLogs();
    runDiagnostics();

```

```
    if (!isGcloudInitialized) {
      const refreshBtn = document.getElementById('btn-gcloud-refresh');
      if (refreshBtn) {
        refreshBtn.addEventListener('click', () => {
          fetchStatus();
          fetchLogs();
          runDiagnostics();
        });
      }

      const configBtn = document.getElementById('btn-gcloud-config');
      if (configBtn) {
        configBtn.addEventListener('click', () => {
          if (typeof window.openAppConfigModal === 'function') {
            window.openAppConfigModal('gcloud_monitor');
          }
        });
      }
      isGcloudInitialized = true;
    }
  }

  window.initGCloudTab = initGCloudTab;
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/google_accounts_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/google_accounts_tab/ind
ex.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/google_accounts_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Google Workspace Accounts Pool Management Tab -->
<div class="container-fluid py-2" id="google-accounts-tab-root
">
<!-- Top Banner & Stats -->
<div class="d-flex flex-wrap align-items-center justify-conten
t-between gap-3 mb-3 p-3 bg-body-tertiary border rounded shado
w-sm">
<div>
<h5 class="mb-1 text-primary d-flex align-items-center gap-2">
<i class="bi bi-google text-danger"></i>
```

```
<span>Управление аккаунтами Google Workspace</span>
</h5>
<p class="mb-0 text-muted small">Пул учетных записей (OAuth 2.0 и Service Accounts) для интеграции с Gmail, Google Drive, Sheets и Docs.</p>
</div>
<div class="d-flex gap-2">
<button class="btn btn-outline-primary btn-sm d-flex align-items-center gap-1 rounded-pill px-3" id="tab-btn-refresh-gaccounts" title="Обновить список">
<i class="bi bi-arrow-clockwise"></i> Обновить
      </button>
</div>
</div>
<!-- Summary Metric Cards -->
<div class="row g-3 mb-3">
<div class="col-6 col-md-3">
<div class="card bg-body-tertiary border h-100 shadow-sm" data-panel-id="panel-google-accounts-0" id="panel-google-accounts-0">
<div class="card-body p-3 d-flex align-items-center justify-content-between">
<div>
<div class="text-muted small">Всего аккаунтов</div>
<h4 class="mb-0 fw-bold" id="gstat-total-accounts">0</h4>
</div>
<div class="bg-secondary bg-opacity-25 rounded p-2 text-primary">
<i class="bi bi-collection-fill fs-4"></i>
</div>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-body-tertiary border h-100 shadow-sm" data-panel-id="panel-google-accounts-0" id="panel-google-accounts-0">
<div class="card-body p-3 d-flex align-items-center justify-co
```

```
ntent-between">
<div>
<div class="text-muted small">Активные</div>
<h4 class="mb-0 text-success fw-bold" id="gstat-active-account
s">0</h4>
</div>
<div class="bg-success bg-opacity-25 rounded p-2 text-success"
>
<i class="bi bi-check-circle-fill fs-4"></i>
</div>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-body-tertiary border h-100 shadow-sm" data
-panel-id="panel-google-accounts-0" id="panel-google-accounts-
0">
<div class="card-body p-3 d-flex align-items-center justify-co
ntent-between">
<div>
<div class="text-muted small">OAuth 2.0 (User)</div>
<h4 class="mb-0 text-info fw-bold" id="gstat-oauth-accounts">0
</h4>
</div>
<div class="bg-info bg-opacity-25 rounded p-2 text-info">
<i class="bi bi-person-badge-fill fs-4"></i>
</div>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-body-tertiary border h-100 shadow-sm" data
-panel-id="panel-google-accounts-0" id="panel-google-accounts-
0">
<div class="card-body p-3 d-flex align-items-center justify-co
ntent-between">
<div>
<div class="text-muted small">Service Accounts</div>
```

```
<h4 class="mb-0 text-warning fw-bold" id="gstat-sa-accounts">0
</h4>
</div>
<div class="bg-warning bg-opacity-25 rounded p-2 text-warning"
>
<i class="bi bi-cpu-fill fs-4"></i>
</div>
</div>
</div>
</div>
</div>
<!-- Info Alert -->
<div class="alert alert-info border-0 small shadow-sm rounded-
3 mb-4" role="alert">
<div class="d-flex align-items-center gap-2 mb-1">
<i class="bi bi-info-circle-fill text-info fs-5"></i>
<strong class="text-info">Автономная ротация и приоритеты акка
унтов</strong>
</div>
<p class="mb-1 text-secondary">
      Агенты и инструменты AI (<code>gmail_search</code>, <cod
e>gmail_create_draft</code>, <code>gdrive_list_files</code>, <
code>gsheets_read</code>, <code>gsheets_append</code>) могут я
вно вызывать нужный аккаунт через параметр <code>account_name=
"work"</code>, либо автоматически использовать основной аккаун
т по умолчанию (<span class="badge bg-warning text-dark"><i cl
ass="bi bi-star-fill"></i> Default</span>).
    </p>
<div class="text-muted">
      Учетные записи и токены авторизации хранятся в <code>src
/secrets/</code> в изолированном защищенном виде.
    </div>
</div>
<div class="row g-4 mb-4">
<!-- Add Account Card -->
<div class="col-lg-5">
<div class="card shadow-sm border h-100" data-panel-id="panel-
google-accounts-dobavit-google-akkaunt" id="panel-google-accou
```

```
nts-dobavit-google-akkaunt">
<div class="card-header bg-transparent border-bottom py-2 d-flex justify-content-between align-items-center">
<h6 class="fw-bold mb-0 d-flex align-items-center gap-2">
<i class="bi bi-plus-circle text-success"></i> Добавить Google аккаунт
          </h6>
<button class="btn btn-outline-info btn-sm py-0 px-2 rounded-pill d-flex align-items-center gap-1" data-bs-target="#googleCredsHelpModal" data-bs-toggle="modal" style="font-size: 0.75rem;" type="button">
<i class="bi bi-info-circle-fill"></i> Инструкция
          </button>
</div>
<div class="card-body p-3">
<form id="tab-form-add-gaccount" onsubmit="event.preventDefault();">
<div class="mb-3">
<label class="form-label small text-muted fw-bold">Имя аккаунта (ID): <span class="text-danger">*</span></label>
<input autocomplete="off" class="form-control form-control-sm" id="tab-new-gacc-name" placeholder="Например: work, personal, bot" required="" type="text"/>
<div class="form-text small text-secondary">Уникальный системный идентификатор для вызова в агентах.</div>
</div>
<div class="mb-3">
<label class="form-label small text-muted fw-bold">Тип аутентификации:</label>
<select class="form-select form-select-sm" id="tab-new-gacc-type">
<option value="oauth2">OAuth 2.0 Client (User Login / Desktop App)</option>
<option value="service_account">Service Account Key (Server JSON)</option>
</select>
</div>
<div class="mb-3">
```

```
<label class="form-label small text-muted fw-bold">Email (опци
онально):</label>
<input autocomplete="off" class="form-control form-control-sm"
 id="tab-new-gacc-email" placeholder="user@gmail.com или sa@pr
oject.iam.gserviceaccount.com" type="email"/>
</div>
<!-- Credentials Input Mode Toggle -->
<div class="mb-3">
<label class="form-label small text-muted fw-bold d-flex justi
fy-content-between align-items-center">
<span>Учетные данные (Credentials JSON): <span class="text-dan
ger">*</span></span>
<span class="badge bg-info text-dark" data-bs-target="#googleC
redsHelpModal" data-bs-toggle="modal" role="button" style="cur
sor: pointer; font-size: 0.72rem;">
<i class="bi bi-question-circle-fill me-1"></i>Как получить JS
ON?
                </span>
</label>
<ul class="nav nav-pills nav-fill mb-2" id="tab-gacc-input-mod
e-tabs" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active py-1 small" data-bs-target="#ta
b-gacc-mode-file" data-bs-toggle="pill" id="tab-gacc-mode-file
-tab" role="tab" type="button">  Загрузить файл</button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link py-1 small" data-bs-target="#tab-gacc-
mode-json" data-bs-toggle="pill" id="tab-gacc-mode-json-tab" r
ole="tab" type="button">  Вставить JSON</button>
</li>
</ul>
<div class="tab-content" id="tab-gacc-input-mode-tabContent">
<div class="tab-pane fade show active" id="tab-gacc-mode-file"
 role="tabpanel">
<input accept=".json,application/json" class="form-control for
m-control-sm" id="tab-new-gacc-file" type="file"/>
<div class="form-text small text-secondary">Загрузите <code>cr
```

```
edentials.json</code> или <code>service_account.json</code> из
 <a class="text-info text-decoration-none" href="https://conso
le.cloud.google.com/apis/credentials" rel="noopener noreferrer
" target="_blank">Google Cloud Console <i class="bi bi-box-arr
ow-up-right ms-1" style="font-size: 0.75rem;"></i></a>.</div>
</div>
<div class="tab-pane fade" id="tab-gacc-mode-json" role="tabpa
nel">
<textarea class="form-control font-monospace form-control-sm"
id="tab-new-gacc-json" placeholder='{"installed": {"client_id"
: "...", "client_secret": "..."}}' rows="4"></textarea>
</div>
</div>
</div>
<div class="form-check mb-4 mt-2">
<input class="form-check-input" id="tab-new-gacc-default" styl
e="cursor: pointer;" type="checkbox"/>
<label class="form-check-label small fw-semibold" for="tab-new
-gacc-default" style="cursor: pointer;">
                Сделать аккаунтом по умолчанию (Default)
              </label>
</div>
<button class="btn btn-primary btn-sm w-100 rounded-pill fw-bo
ld" id="tab-btn-add-gacc" type="button">
<i class="bi bi-plus-circle me-1"></i> Сохранить Google аккаун
т
            </button>
</form>
</div>
</div>
</div>
<!-- Accounts Table -->
<div class="col-lg-7">
<div class="card shadow-sm border mb-3" data-panel-id="panel-g
oogle-accounts-zaregistrirovannye-akkaunty" id="panel-google-a
ccounts-zaregistrirovannye-akkaunty">
<div class="card-header bg-transparent border-bottom py-2 d-fl
ex justify-content-between align-items-center">
```

```
<h6 class="fw-bold mb-0"><i class="bi bi-list-check me-1"></i>
 Зарегистрированные аккаунты</h6>
<span class="badge bg-secondary" id="tab-gaccounts-count-badge
">0</span>
</div>
<div class="card-body p-0">
<div class="table-responsive">
<table class="table table-hover align-middle mb-0" id="tab-gac
counts-table">
<thead class="table-light small text-uppercase">
<tr>
<th>Аккаунт</th>
<th class="text-center" style="width: 130px;">Тип</th>
<th class="text-center" style="width: 100px;">Статус</th>
<th class="text-center" style="width: 110px;">Токен</th>
<th class="text-center" style="width: 180px;">Действия</th>
</tr>
</thead>
<tbody id="tab-gaccounts-list-body">
<tr>
<td class="text-center py-4 text-muted" colspan="5">Загрузка а
ккаунтов...</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- Test & Diagnostics Output Card -->
<div class="card bg-body-tertiary border shadow-sm" data-panel
-id="panel-gaccount-test-card" id="tab-gaccount-test-card" sty
le="display: none;">
<div class="card-header bg-transparent border-bottom d-flex ju
stify-content-between align-items-center py-2">
<span class="small fw-bold text-info"><i class="bi bi-terminal
 me-1"></i> Результат проверки доступа</span>
<button class="btn btn-sm btn-outline-secondary py-0 px-2" onc
lick="document.getElementById('tab-gaccount-test-card').style.
```

```
display='none'" type="button">□ </button>
</div>
<div class="card-body p-3 font-monospace small bg-body" id="tab-gaccount-test-body" style="max-height: 200px; overflow-y: auto; white-space: pre-wrap;">
</div>
</div>
</div>
</div>
<!-- Google Credentials Instruction Modal -->
<div aria-hidden="true" aria-labelledby="googleCredsHelpModalLabel" class="modal fade" id="googleCredsHelpModal" tabindex="-1">
<div class="modal-dialog modal-lg modal-dialog-centered modal-dialog-scrollable">
<div class="modal-content shadow-lg">
<div class="modal-header">
<h5 class="modal-title d-flex align-items-center gap-2" id="googleCredsHelpModalLabel">
<i class="bi bi-google text-danger"></i> Как получить и загрузить Credentials JSON
          </h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body p-4">
<ul class="nav nav-tabs mb-3" id="helpModalTabs" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active" data-bs-target="#help-oauth-pane" data-bs-toggle="tab" id="help-oauth-tab" role="tab" type="button">
<i class="bi bi-person-check-fill text-primary me-1"></i> 1. OAuth 2.0 Client (Личный вход)
              </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link" data-bs-target="#help-sa-pane" data-b
```

```
s-toggle="tab" id="help-sa-tab" role="tab" type="button">
<i class="bi bi-robot text-warning me-1"></i> 2. Service Account (Сервисный ключ)
              </button>
</li>
</ul>
<div class="tab-content">
<!-- OAuth 2.0 Pane -->
<div class="tab-pane fade show active" id="help-oauth-pane" role="tabpanel">
<p class="text-secondary small">
                Используется для работы от имени вашего личного или рабочего аккаунта Google (чтение/отправка почты в Gmail, доступ к Google Drive, редактирование Google Sheets/Docs).
              </p>
<ol class="list-group list-group-numbered list-group-flush small">
<li class="list-group-item">
                  Перейдите в консоль управления Google: <br/>
<a class="btn btn-sm btn-outline-info mt-1" href="https://console.cloud.google.com/apis/credentials" rel="noopener noreferrer" target="_blank">
<i class="bi bi-box-arrow-up-right me-1"></i> Google Cloud Console  Credentials
                  </a>
</li>
<li class="list-group-item">
                  Если проект еще не создан, выберите существующий или создайте новый проект (кнопка в верхней панели).
                </li>
<li class="list-group-item">
                  В разделе <strong>OAuth consent screen</strong> (Экран согласия OAuth):
                  <ul class="mt-1 text-secondary">
<li>Выберите тип: <code>External</code> (Внешний) или <code>Internal</code>.</li>
<li>Укажите название приложения (например, <code>AI Windows Admin</code>) и ваш контактный email.</li>
```

```
<li>В разделе <em>Test users</em> добавьте ваш Google email (если статус приложения Testing).</li>
</ul>
</li>
<li class="list-group-item">
                  В разделе <strong>Credentials</strong> нажмите <code>+ CREATE CREDENTIALS</code>  выберите <strong>OAuth client ID</strong>.
                </li>
<li class="list-group-item">
                  В поле <em>Application type</em> выберите <strong>Desktop app</strong> (Приложение для ПК) и задайте имя (например <code>AI Windows Admin-Desktop</code>), затем нажмите <strong>Create</strong>.
                </li>
<li class="list-group-item">
                  В появившемся окне нажмите <i class="bi bi-download text-info"></i> <strong>DOWNLOAD JSON</strong> (или скачайте файл в таблице OAuth 2.0 Client IDs).
                </li>
<li class="list-group-item">
                  Включите необходимые API в <a class="text-info" href="https://console.cloud.google.com/apis/dashboard" rel="noopener noreferrer" target="_blank">Enabled APIs &amp;amp; Services <i class="bi bi-box-arrow-up-right"></i></a>:
                  <div class="mt-1 d-flex flex-wrap gap-1">
<span class="badge bg-secondary">Gmail API</span>
<span class="badge bg-secondary">Google Drive API</span>
<span class="badge bg-secondary">Google Sheets API</span>
<span class="badge bg-secondary">Google Docs API</span>
</div>
</li>
<li class="list-group-item">
                  В этой карточке выберите <strong>Загрузить файл</strong> и укажите скачанный <code>credentials.json</code> (или вставьте его содержимое во вкладку <strong>Вставить JSON</strong>), введите ID аккаунта и нажмите <strong>Сохранить Google аккаунт</strong>.
```

```
                </li>
</ol>
</div>
<!-- Service Account Pane -->
<div class="tab-pane fade" id="help-sa-pane" role="tabpanel">
<p class="text-secondary small">
                Используется для серверных / автономных фоновых задач без необходимости интерактивного логина в браузере.
              </p>
<ol class="list-group list-group-numbered list-group-flush small">
<li class="list-group-item">
                  Перейдите в раздел сервисных аккаунтов: <br>
<a class="btn btn-sm btn-outline-warning mt-1" href="https://console.cloud.google.com/iam-admin/serviceaccounts" rel="noopener noreferrer" target="_blank">
<i class="bi bi-box-arrow-up-right me-1"></i> Google Cloud Console  Service Accounts
                  </a>
</li>
<li class="list-group-item">
                  Нажмите <code>+ CREATE SERVICE ACCOUNT</code>, введите имя (например <code>ai-agent-sa</code>) и нажмите <strong>Done</strong>.
                </li>
<li class="list-group-item">
                  Кликните по созданному сервисному аккаунту и перейдите во вкладку <strong>Keys</strong> (Ключи).
                </li>
<li class="list-group-item">
                  Нажмите <strong>ADD KEY</strong>  <strong>Create new key</strong>  выберите тип <strong>JSON</strong>  нажмите <strong>Create</strong>.
                </li>
<li class="list-group-item">
                  Файл ключа (<code>project-id-*.json</code>) скачается на ваш компьютер.
```

```
                </li>
<li class="list-group-item">
                  В форме выберите тип <em>Service Account Key
</em>, загрузите JSON файл и сохраните аккаунт.
                </li>
</ol>
</div>
</div>
</div>
<div class="modal-footer">
<button class="btn btn-secondary btn-sm rounded-pill px-4" dat
a-bs-dismiss="modal" type="button">Понятно</button>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Google Accounts Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/google_accounts_tab/ma
in.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/google_accounts_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

// Google Workspace Accounts Tab JavaScript Logic

function escapeHtml(text) {
  if (!text) return '';
  return String(text)
    .replace(/&/g, '&amp;')
    .replace(/</g, '&lt;')
    .replace(/>/g, '&gt;')
    .replace(/"/g, '&quot;')
    .replace(/'/g, '&#039;');
}

```

```
function showGAccountsNotification(message, type = 'info') {
  const notification = document.createElement('div');
  notification.className = `alert alert-${type} position-fixed
 top-0 end-0 m-3 shadow-lg`;
  notification.style.zIndex = '9999';
  notification.style.maxWidth = '400px';
  notification.innerHTML = message;
  document.body.appendChild(notification);

  setTimeout(() => {
    notification.remove();
  }, 4000);
}

async function refreshTabGoogleAccountsList() {
  const container = document.getElementById('tab-gaccounts-lis
t-body');
  if (!container) return;

  try {
    container.innerHTML = '<tr><td colspan="5" class="text-cen
ter py-4 text-muted"><span class="spinner-border spinner-borde
r-sm me-2 text-primary"></span>Загрузка аккаунтов...</td></tr>
';
    const data = await window.api.googleAccounts.list();
    const accounts = data.accounts || [];

    // Update stats
    const totalEl = document.getElementById('gstat-total-accou
nts');
    const activeEl = document.getElementById('gstat-active-acc
ounts');
    const oauthEl = document.getElementById('gstat-oauth-accou
nts');
    const saEl = document.getElementById('gstat-sa-accounts');
    const countBadge = document.getElementById('tab-gaccounts-
count-badge');

```

```
    if (totalEl) totalEl.textContent = accounts.length;
    if (activeEl) activeEl.textContent = accounts.filter(a => a.status === 'active').length;
    if (oauthEl) oauthEl.textContent = accounts.filter(a => a.type === 'oauth2').length;
    if (saEl) saEl.textContent = accounts.filter(a => a.type === 'service_account').length;
    if (countBadge) countBadge.textContent = `${accounts.length} акк.`;

    if (accounts.length === 0) {
      container.innerHTML = '<tr><td colspan="5" class="text-center py-4 text-muted">Список аккаунтов пуст. Загрузите credentials.json или service_account.json.</td></tr>';
      return;
    }

    container.innerHTML = '';
    accounts.forEach(acc => {
      const row = document.createElement('tr');

      // 1. Account Name + Default Badge + Email
      const tdName = document.createElement('td');
      tdName.innerHTML = `
        <div class="d-flex align-items-center gap-2">
          <strong>${escapeHtml(acc.name)}</strong>
          ${acc.is_default ? '<span class="badge bg-warning text-dark"><i class="bi bi-star-fill"></i> Default</span>' : ''}
        </div>
        ${acc.email ? `<div class="small text-muted font-monospace">${escapeHtml(acc.email)}</div>` : ''}
      `;
      row.appendChild(tdName);

      // 2. Type
      const tdType = document.createElement('td');
      tdType.className = 'text-center';
      const isOAuth = acc.type === 'oauth2';
```

```
92        tdType.innerHTML = isOAuth
93          ? '<span class="badge bg-primary-subtle text-primary b
    order border-primary-subtle"><i class="bi bi-person me-1"></i>
    OAuth 2.0</span>'
94          : '<span class="badge bg-info-subtle text-info border
    border-info-subtle"><i class="bi bi-cpu me-1"></i>Service Acco
    unt</span>';
95        row.appendChild(tdType);
96
97        // 3. Status / Quota
98        const tdStatus = document.createElement('td');
99        tdStatus.className = 'text-center';
100       const isActive = acc.status === 'active';
101       if (isActive) {
102         tdStatus.innerHTML = '<span class="badge bg-success">A
    ктивен</span>';
103       } else if (acc.status === 'exhausted') {
104         tdStatus.innerHTML = '<span class="badge bg-danger" ti
    tle="Исчерпан суточный лимит квоты">Лимит</span>';
105       } else {
106         tdStatus.innerHTML = `<span class="badge bg-secondary"
    >${escapeHtml(acc.status)}</span>`;
107       }
108       row.appendChild(tdStatus);
109
110       // 4. Token / Credentials Status
111       const tdToken = document.createElement('td');
112       tdToken.className = 'text-center';
113       if (acc.has_token || acc.type === 'service_account') {
114         tdToken.innerHTML = '<span class="badge bg-success-sub
    tle text-success border border-success-subtle"><i class="bi bi
    -shield-check me-1"></i>Готов</span>';
115       } else {
116         tdToken.innerHTML = '<span class="badge bg-warning-sub
    tle text-warning border border-warning-subtle"><i class="bi bi
    -key me-1"></i>Нужен токен</span>';
117       }
118       row.appendChild(tdToken);
```

```
119
120       // 5. Actions
121       const tdActions = document.createElement('td');
122       tdActions.className = 'text-center';
123
124       // Set Default button
125       if (!acc.is_default) {
126         const btnDefault = document.createElement('button');
127         btnDefault.className = 'btn btn-xs btn-outline-warning
 btn-sm me-1';
128         btnDefault.title = 'Сделать аккаунтом по умолчанию';
129         btnDefault.innerHTML = '<i class="bi bi-star"></i>';
130         btnDefault.onclick = () => setTabGoogleAccountDefault(
acc.name);
131         tdActions.appendChild(btnDefault);
132       }
133
134       // Reset status button
135       if (acc.status === 'exhausted') {
136         const btnReset = document.createElement('button');
137         btnReset.className = 'btn btn-xs btn-outline-info btn-
sm me-1';
138         btnReset.title = 'Сбросить статус исчерпания';
139         btnReset.innerHTML = '<i class="bi bi-arrow-repeat"></
i>';
140         btnReset.onclick = () => resetTabGoogleAccountStatus(a
cc.name);
141         tdActions.appendChild(btnReset);
142       }
143
144       // Test button
145       const btnTest = document.createElement('button');
146       btnTest.className = 'btn btn-xs btn-outline-info btn-sm
me-1';
147       btnTest.title = 'Проверить доступ и авторизацию';
148       btnTest.innerHTML = '<i class="bi bi-play-circle"></i> Т
ест';
149       btnTest.onclick = () => testTabGoogleAccount(acc.name);
```

```
      tdActions.appendChild(btnTest);

      // Delete button
      const btnDelete = document.createElement('button');
      btnDelete.className = 'btn btn-xs btn-outline-danger btn-sm';
      btnDelete.title = 'Удалить аккаунт из пула';
      btnDelete.innerHTML = '<i class="bi bi-trash"></i>';
      btnDelete.onclick = () => deleteTabGoogleAccount(acc.name);
      tdActions.appendChild(btnDelete);

      row.appendChild(tdActions);
      container.appendChild(row);
    });

  } catch (err) {
    console.error('Ошибка загрузки аккаунтов Google Workspace:', err);
    container.innerHTML = `<tr><td colspan="5" class="text-center py-4 text-danger">Ошибка: ${escapeHtml(err.message)}</td></tr>`;
  }
}

async function handleAddTabGoogleAccount() {
  const nameInput = document.getElementById('tab-new-gacc-name');
  const typeSelect = document.getElementById('tab-new-gacc-type');
  const emailInput = document.getElementById('tab-new-gacc-email');
  const fileInput = document.getElementById('tab-new-gacc-file');
  const jsonTextarea = document.getElementById('tab-new-gacc-json');
  const defaultCheckbox = document.getElementById('tab-new-gacc-default');
```

```
  const addBtn = document.getElementById('tab-btn-add-gacc');

  if (!nameInput || !addBtn) return;

  const accountName = nameInput.value.trim();
  const accountType = typeSelect ? typeSelect.value : 'oauth2'
;
  const email = emailInput ? emailInput.value.trim() : '';
  const setAsDefault = defaultCheckbox ? defaultCheckbox.check
ed : false;

  if (!accountName) {
    showGAccountsNotification('Введите имя аккаунта (например:
 work, personal)', 'warning');
    nameInput.focus();
    return;
  }

  const file = fileInput && fileInput.files && fileInput.files
[0];
  const jsonContent = jsonTextarea ? jsonTextarea.value.trim()
 : '';

  if (!file && !jsonContent) {
    showGAccountsNotification('Загрузите файл credentials.json
 / service_account.json или вставьте JSON', 'warning');
    return;
  }

  addBtn.disabled = true;
  const originalText = addBtn.innerHTML;
  addBtn.innerHTML = '<span class="spinner-border spinner-bord
er-sm me-1"></span> Сохранение...';

  try {
    if (file) {
      const formData = new FormData();
      formData.append('account_name', accountName);
```

```
      formData.append('account_type', accountType);
      if (email) formData.append('email', email);
      formData.append('set_as_default', setAsDefault ? 'true'
: 'false');
      formData.append('file', file);

      await window.api.googleAccounts.upload(formData);
    } else {
      let parsedJson = null;
      try {
        parsedJson = JSON.parse(jsonContent);
      } catch (e) {
        throw new Error('Некорректный JSON в поле учетных данн
ых');
      }

      await window.api.googleAccounts.create({
        account_name: accountName,
        account_type: accountType,
        email: email || undefined,
        credentials_dict: parsedJson,
        set_as_default: setAsDefault
      });
    }

    showGAccountsNotification(`Аккаунт Google "${accountName}"
 успешно сохранен в пул`, 'success');
    nameInput.value = '';
    if (emailInput) emailInput.value = '';
    if (fileInput) fileInput.value = '';
    if (jsonTextarea) jsonTextarea.value = '';
    if (defaultCheckbox) defaultCheckbox.checked = false;

    await refreshTabGoogleAccountsList();
  } catch (err) {
    console.error('Ошибка сохранения аккаунта Google:', err);
    showGAccountsNotification('Ошибка сохранения: ' + err.mess
age, 'danger');
```

```
  } finally {
    addBtn.disabled = false;
    addBtn.innerHTML = originalText;
  }
}

async function setTabGoogleAccountDefault(name) {
  try {
    await window.api.googleAccounts.setDefault(name);
    showGAccountsNotification(`Аккаунт "${name}" назначен по у
молчанию`, 'success');
    await refreshTabGoogleAccountsList();
  } catch (err) {
    console.error('Ошибка установки аккаунта по умолчанию:', е
rr);
    showGAccountsNotification('Ошибка: ' + err.message, 'dange
r');
  }
}

async function resetTabGoogleAccountStatus(name) {
  try {
    await window.api.googleAccounts.resetStatus(name);
    showGAccountsNotification(`Статус аккаунта "${name}" сброш
ен в активный`, 'success');
    await refreshTabGoogleAccountsList();
  } catch (err) {
    console.error('Ошибка сброса статуса аккаунта:', err);
    showGAccountsNotification('Ошибка сброса: ' + err.message,
 'danger');
  }
}

async function testTabGoogleAccount(name) {
  const testCard = document.getElementById('tab-gaccount-test-
card');
  const testBody = document.getElementById('tab-gaccount-test-
body');
```

```
273
274   if (testCard && testBody) {
275     testCard.style.display = 'block';
276     testBody.innerHTML = `<span class="spinner-border spinner-
border-sm me-2 text-info"></span>Проверка аутентификации для а
ккаунта <strong>${escapeHtml(name)}</strong>...`;
277   }
278
279   try {
280     const res = await window.api.googleAccounts.test(name);
281     if (testBody) {
282       const isSuccess = res.status === 'success';
283       const isWarning = res.status === 'warning';
284       const badgeClass = isSuccess ? 'bg-success' : isWarning
? 'bg-warning text-dark' : 'bg-danger';
285
286       testBody.innerHTML = `
287         <div class="mb-2 d-flex align-items-center gap-2">
288           <span class="badge ${badgeClass}">${res.status.toUpp
erCase()}</span>
289           <strong>${escapeHtml(name)}</strong>
290         </div>
291         <div class="mb-1">${escapeHtml(res.message || '')}</di
v>
292         ${res.scopes && res.scopes.length > 0 ? `<div class="t
ext-muted mt-2"><strong>Доступные Scopes:</strong><br>${res.sc
opes.map(s => '• ' + escapeHtml(s)).join('<br>')}</div>` : ''}
293       `;
294     }
295   } catch (err) {
296     if (testBody) {
297       testBody.innerHTML = `<span class="text-danger"><i class
="bi bi-x-circle me-1"></i>Ошибка проверки: ${escapeHtml(err.m
essage)}</span>`;
298     }
299   }
300 }
301
```

```
async function deleteTabGoogleAccount(name) {
  if (!confirm(`Вы уверены, что хотите удалить аккаунт "${name}" из пула Google Workspace?`)) return;
  try {
    await window.api.googleAccounts.delete(name);
    showGAccountsNotification(`Аккаунт "${name}" успешно удален`, 'success');
    await refreshTabGoogleAccountsList();
  } catch (err) {
    console.error('Ошибка удаления аккаунта Google:', err);
    showGAccountsNotification('Ошибка удаления: ' + err.message, 'danger');
  }
}

async function initGoogleAccountsTab() {
  const refreshBtn = document.getElementById('tab-btn-refresh-gaccounts');
  const addBtn = document.getElementById('tab-btn-add-gacc');

  if (refreshBtn) {
    refreshBtn.onclick = async () => {
      refreshBtn.disabled = true;
      const orig = refreshBtn.innerHTML;
      refreshBtn.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span> Обновление...';
      try {
        await refreshTabGoogleAccountsList();
        showGAccountsNotification('Список Google аккаунтов обновлен', 'success');
      } finally {
        refreshBtn.disabled = false;
        refreshBtn.innerHTML = orig;
      }
    };
  }

  if (addBtn) {
```

```
    addBtn.onclick = handleAddTabGoogleAccount;
  }

  await refreshTabGoogleAccountsList();
}

window.initGoogleAccountsTab = initGoogleAccountsTab;
window.refreshTabGoogleAccountsList = refreshTabGoogleAccounts
List;

```

index.html

```
<!--
================================================================
===============
Process Name: AI-Windows-Admin Web Interface - Index View
================================================================
===============
Description:
  Шаблон веб-интерфейса системного модуля (index).

Usage Examples:
    Web Route:
      /src/api/webgui/google_desktop_tab/index.html

File: index.html
Project: ai-windows-admin
Package: src/api/webgui/google_desktop_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Google User Desktop Web Tab Module -->
<style>
  .gdesktop-container {
    padding: 0.5rem 0.75rem;
  }
  .gdesktop-card {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: 8px;
    padding: 0.5rem 0.85rem;
    display: flex;
    flex-direction: column;
    justify-content: space-between;
    min-height: 64px;
```

```
    gap: 0.2rem;
  }
  .gdesktop-label {
    font-size: 0.7rem;
    text-transform: uppercase;
    letter-spacing: 0.05em;
    color: var(--text-muted);
    line-height: 1.1;
  }
  .gdesktop-value {
    font-size: 1.3rem;
    font-weight: 700;
    color: var(--text-color);
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
    line-height: 1.15;
  }
  .gdesktop-panel {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: 8px;
    overflow: hidden;
  }
  .gdesktop-panel-header {
    background: var(--surface-2) !important;
    border-bottom: 1px solid var(--border-color) !important;
    padding: 0.5rem 0.85rem;
    font-weight: 600;
    font-size: 0.85rem;
    color: var(--nav-active);
    display: flex;
    align-items: center;
    justify-content: space-between;
  }
  .gdesktop-subnav .nav-link {
    color: var(--text-muted);
    border-radius: 6px;
    padding: 0.35rem 0.75rem;
```

```
    font-size: 0.82rem;
    font-weight: 500;
    transition: all 0.15s ease-in-out;
  }
  .gdesktop-subnav .nav-link:hover {
    color: var(--nav-active);
    background: var(--nav-hover-bg, rgba(0, 0, 0, 0.05));
  }
  .gdesktop-subnav .nav-link.active {
    color: #fff;
    background: var(--nav-active);
    font-weight: 600;
  }
  .gdesktop-table {
    font-size: 0.82rem;
  }
  .gdesktop-table td, .gdesktop-table th {
    vertical-align: middle;
  }
</style>
<div class="gdesktop-container" id="google-desktop-tab-root">
<!-- Top Header Banner & Actions -->
<div class="card shadow-sm border-secondary-subtle py-2 px-3 m
b-3 d-flex justify-content-between align-items-center flex-wra
p gap-2" data-panel-id="panel-google-desktop-google-user-deskt
op" id="panel-google-desktop-google-user-desktop">
<div class="d-flex align-items-center gap-2 flex-wrap">
<span class="fs-4">□ </span>
<div>
<div class="d-flex align-items-center gap-2">
<span class="fw-bold text-body" style="font-size: 1rem;">Googl
e User Desktop</span>
<span class="badge bg-danger text-white border border-light-su
btle px-2 py-0.5" style="font-size: 0.7rem;">Workspace Aggrega
tor</span>
</div>
<div class="text-secondary small" style="font-size: 0.76rem;">
<span data-i18n="auto__gmail_google_calendar_docs_sheets__31dc
```

```
39"></span> Google Drive
        </div>
</div>
</div>
<!-- Account Selector & Sync Controls -->
<div class="d-flex align-items-center gap-2 flex-wrap">
<div class="d-flex align-items-center gap-1.5 border border-secondary px-2 py-1 rounded" style="background: var(--surface-2);">
<label class="small text-muted mb-0 me-1" for="gdesktop-account-select" style="font-size: 0.75rem;"><span data-i18n="auto___5b16fc"></span>:</label>
<select class="form-select form-select-sm border-0 py-0 px-2" id="gdesktop-account-select" style="font-size: 0.8rem; width: auto; cursor: pointer;">
<option value=""><span data-i18n="auto___94a96c"></span>...</option>
</select>
</div>
<button auto___19efd8="" class="btn btn-xs btn-outline-info rounded-pill px-3 py-1.5" id="btn-gdesktop-refresh" style="font-size: 0.78rem;" title="&lt;span data-i18n=">"&gt;
        <i class="bi bi-arrow-clockwise me-1"></i> <span data-i18n="auto___c2f668"></span>
</button>
<button auto___d5b581="" class="btn btn-xs btn-outline-success rounded-pill px-3 py-1.5" id="btn-gdesktop-sync" style="font-size: 0.78rem;" title="&lt;span data-i18n="> API Google"&gt;
        <i class="bi bi-cloud-arrow-down-fill me-1"></i> <span data-i18n="auto___7ceddd"></span>
</button>
</div>
</div>
<!-- Summary Metric Cards -->
<div class="row g-2 mb-3">
<div class="col-6 col-md-3">
<div class="gdesktop-card">
<span class="gdesktop-label"><span data-i18n="auto___aedea9"><
```

```
/span></span>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<span class="gdesktop-value text-info text-truncate" id="gdesk
top-acc-name" style="font-size: 1.05rem;">-</span>
<span class="badge bg-secondary" id="gdesktop-acc-status" styl
e="font-size: 0.68rem;">-</span>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="gdesktop-card">
<span class="gdesktop-label"><span data-i18n="auto___5e10ab"><
/span> (Gmail)</span>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<span class="gdesktop-value text-warning" id="gdesktop-mail-co
unt">0</span>
<span class="text-muted small"><span data-i18n="auto___c66f45"
></span></span>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="gdesktop-card">
<span class="gdesktop-label"><span data-i18n="auto___97c74f"><
/span></span>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<span class="gdesktop-value text-success" id="gdesktop-calenda
r-count">0</span>
<span class="text-muted small"><span data-i18n="auto___d4eca8"
></span></span>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="gdesktop-card">
```

```
<span class="gdesktop-label"><span data-i18n="auto___cdbcb8"></span></span>
<div class="d-flex align-items-baseline justify-content-between mt-1">
<span class="gdesktop-value text-primary" id="gdesktop-docs-drive-count">0 / 0</span>
<span class="text-muted small">Docs / Drive</span>
</div>
</div>
</div>
</div>
<!-- Sub-navigation Tabs -->
<div class="gdesktop-panel mb-3">
<div class="gdesktop-panel-header">
<ul class="nav nav-pills gdesktop-subnav gap-1" id="gdesktop-subnav-tabs" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active" data-bs-target="#gdesktop-pane-mail" data-bs-toggle="pill" id="gdesktop-tab-mail-btn" role="tab" type="button">
            Gmail <span data-i18n="auto___108aa2"></span>
</button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link" data-bs-target="#gdesktop-pane-calendar" data-bs-toggle="pill" id="gdesktop-tab-calendar-btn" role="tab" type="button">
            Google <span data-i18n="auto___fa3a2f"></span>
</button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link" data-bs-target="#gdesktop-pane-docs" data-bs-toggle="pill" id="gdesktop-tab-docs-btn" role="tab" type="button">
            <span data-i18n="auto___a65e97"></span>
</button>
</li>
<li class="nav-item" role="presentation">
```

```
<button class="nav-link" data-bs-target="#gdesktop-pane-drive"
 data-bs-toggle="pill" id="gdesktop-tab-drive-btn" role="tab"
type="button">
            Google <span data-i18n="auto___ca0407"></span>
</button>
</li>
</ul>
<span class="text-muted small" id="gdesktop-last-refreshed"><s
pan data-i18n="auto___484f3b"></span>: -</span>
</div>
<div class="tab-content p-3" id="gdesktop-tab-content">
<!-- 1. Gmail Pane -->
<div class="tab-pane fade show active" id="gdesktop-pane-mail"
 role="tabpanel">
<!-- Gmail API Info Badge -->
<div class="alert alert-warning border border-warning-subtle t
ext-body small shadow-sm rounded-3 mb-3 p-2 px-3 d-flex align-
items-center justify-content-between flex-wrap gap-2" role="al
ert" style="background: rgba(245, 158, 11, 0.1);">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-envelope-exclamation-fill text-warning fs-5"><
/i>
<div>
<span class="fw-bold text-warning me-1"><span data-i18n="auto_
_gmail_api__a058db"></span>):</span>
<span class="text-muted"><span data-i18n="auto__gmail__8c0930"
></span> <strong>Enable</strong>:</span>
</div>
</div>
<a class="btn btn-warning btn-sm fw-bold rounded-pill text-dar
k text-nowrap py-1 px-3" href="https://console.developers.goog
le.com/apis/api/gmail.googleapis.com/overview" rel="noopener n
oreferrer" style="font-size: 0.78rem;" target="_blank">
<i class="bi bi-box-arrow-up-right me-1"></i> Google Cloud Con
sole — Gmail API Overview
          </a>
</div>
<div class="table-responsive">
```

```
<table class="table table-hover align-middle gdesktop-table mb-0">
<thead>
<tr class="text-secondary small text-uppercase">
<th style="width: 200px;"><span data-i18n="auto___522b04"></span></th>
<th><span data-i18n="auto___96f68e"></span></th>
<th style="width: 170px;"><span data-i18n="auto___a5b49d"></span></th>
<th><span data-i18n="auto___71c575"></span></th>
</tr>
</thead>
<tbody id="gdesktop-mail-tbody">
<tr>
<td class="text-center py-4 text-muted" colspan="4"><span data-i18n="auto___0f6ce9"></span> Gmail...</td>
</tr>
</tbody>
</table>
</div>
</div>
<!-- 2. Google Calendar Pane -->
<div class="tab-pane fade" id="gdesktop-pane-calendar" role="tabpanel">
<div class="table-responsive">
<table class="table table-hover align-middle gdesktop-table mb-0">
<thead>
<tr class="text-secondary small text-uppercase">
<th><span data-i18n="auto___bb9263"></span></th>
<th style="width: 180px;"><span data-i18n="auto___cb26bd"></span></th>
<th style="width: 180px;"><span data-i18n="auto___ec5bfc"></span></th>
<th style="width: 180px;"><span data-i18n="auto___82bd05"></span></th>
<th class="text-center" style="width: 100px;"><span data-i18n="auto___046e47"></span></th>
```

```
</tr>
</thead>
<tbody id="gdesktop-calendar-tbody">
<tr>
<td class="text-center py-4 text-muted" colspan="5"><span data-i18n="auto___6d5084"></span>...</td>
</tr>
</tbody>
</table>
</div>
</div>
<!-- 3. Google Docs Pane -->
<div class="tab-pane fade" id="gdesktop-pane-docs" role="tabpanel">
<div class="table-responsive">
<table class="table table-hover align-middle gdesktop-table mb-0">
<thead>
<tr class="text-secondary small text-uppercase">
<th><span data-i18n="auto___5c9ffd"></span></th>
<th style="width: 220px;"><span data-i18n="auto___d25691"></span> (MIME)</th>
<th style="width: 170px;"><span data-i18n="auto___e0a8de"></span></th>
<th class="text-center" style="width: 160px;"><span data-i18n="auto___9978ac"></span></th>
</tr>
</thead>
<tbody id="gdesktop-docs-tbody">
<tr>
<td class="text-center py-4 text-muted" colspan="4"><span data-i18n="auto___6a3add"></span>...</td>
</tr>
</tbody>
</table>
</div>
</div>
<!-- 4. Google Drive Pane -->
```

```
<div class="tab-pane fade" id="gdesktop-pane-drive" role="tabpanel">
<div class="table-responsive">
<table class="table table-hover align-middle gdesktop-table mb-0">
<thead>
<tr class="text-secondary small text-uppercase">
<th><span data-i18n="auto___1e5ae1"></span></th>
<th style="width: 220px;"><span data-i18n="auto___31e483"></span></th>
<th style="width: 120px;"><span data-i18n="auto___98713e"></span></th>
<th style="width: 170px;"><span data-i18n="auto___730822"></span></th>
<th class="text-center" style="width: 100px;"><span data-i18n="auto___046e47"></span></th>
</tr>
</thead>
<tbody id="gdesktop-drive-tbody">
<tr>
<td class="text-center py-4 text-muted" colspan="5"><span data-i18n="auto__google__2bcb83"></span>...</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
<!-- Document Viewer Modal -->
<div aria-hidden="true" aria-labelledby="gdesktopDocViewerModalLabel" class="modal fade" id="gdesktopDocViewerModal" tabindex="-1">
<div class="modal-dialog modal-lg modal-dialog-centered modal-dialog-scrollable">
<div class="modal-content shadow-lg">
<div class="modal-header border-secondary py-2">
<h6 class="modal-title d-flex align-items-center gap-2" id="gd
```

```
esktopDocViewerModalLabel">
<i class="bi bi-file-earmark-text text-info"></i> <span id="gd
esktop-doc-modal-title"><span data-i18n="auto___26d5fd"></span
></span>
</h6>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-3 font-monospace small" id="gdesktop-
doc-modal-body" style="white-space: pre-wrap; max-height: 450p
x; background: var(--surface-2); color: var(--text-color);">
<span data-i18n="auto___f2a34f"></span>...
        </div>
<div class="modal-footer border-secondary py-1.5">
<button class="btn btn-secondary btn-sm rounded-pill px-4" dat
a-bs-dismiss="modal" type="button"><span data-i18n="auto___4ae
50d"></span></button>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: AI-Windows-Admin UI - Main Script
 * ============================================================
 * Description:
 *   Клиентский веб-скрипт модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/src/api/webgui/google_desktop_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: src/api/webgui/google_desktop_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * google_desktop_tab/main.js — Веб-модуль управления Google User Desktop в AI Windows Admin.
 */

(function () {
  'use strict';

  function formatBytes(bytes) {
    if (!bytes || bytes === 0) return '0 B';
    const k = 1024;
    const sizes = ['B', 'KB', 'MB', 'GB', 'TB'];
```

```
    const i = Math.floor(Math.log(bytes) / Math.log(k));
    return parseFloat((bytes / Math.pow(k, i)).toFixed(1)) + '
 ' + sizes[i];
  }

  function formatDate(isoStr) {
    if (!isoStr) return '-';
    try {
      const d = new Date(isoStr);
      if (isNaN(d.getTime())) return isoStr;
      return d.toLocaleString('ru-RU', {
        year: 'numeric',
        month: '2-digit',
        day: '2-digit',
        hour: '2-digit',
        minute: '2-digit',
      });
    } catch (e) {
      return isoStr;
    }
  }

  async function fetchStatus() {
    try {
      const res = await fetch('/api/google-desktop/status');
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      updateStatusUI(data);
    } catch (err) {
      console.error('[GoogleDesktopTab] Failed to fetch status
:', err);
    }
  }

  function updateStatusUI(data) {
    const acc = data.account || {};
    const nameEl = document.getElementById('gdesktop-acc-name'
);
```

```
    const statusEl = document.getElementById('gdesktop-acc-sta
tus');
    const mailCountEl = document.getElementById('gdesktop-mail
-count');
    const calCountEl = document.getElementById('gdesktop-calen
dar-count');
    const docsDriveCountEl = document.getElementById('gdesktop
-docs-drive-count');
    const lastRefreshedEl = document.getElementById('gdesktop-
last-refreshed');

    if (nameEl) nameEl.textContent = acc.name || i18n.t('auto_
__922508');
    if (statusEl) {
      statusEl.textContent = acc.status || 'unconfigured';
      statusEl.className = acc.status === 'active' ? 'badge bg
-success' : 'badge bg-warning text-darki18n.t('auto__if_mailco
untel_mailcountel_textcontent_data_mail_count_0_if_calcountel_
calcountel_textcontent_data_calendar_events_count_0_if_docsdri
vecountel_docsdrivecountel_textcontent_data_docs_count_0_data_
drive_files_count_0_if_lastrefreshedel_lastrefreshedel_textcon
tent_formatdate_data_last_refreshed_async_function_fetchaccoun
ts_try_const_res_await_fetch__17d494')/api/google-desktop/acco
unts');
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      const selectEl = document.getElementById('gdesktop-accou
nt-select');
      if (!selectEl) return;

      selectEl.innerHTML = '';
      const accounts = data.accounts || [];
      const active = data.active_account || {};

      if (accounts.length === 0) {
        const opt = document.createElement('option');
        opt.value = '';
        opt.textContent = i18n.t('auto__unconfigured__91fdae')
```

```
;
        selectEl.appendChild(opt);
        return;
      }

      accounts.forEach((a) => {
        const opt = document.createElement('option');
        opt.value = a.name;
        opt.textContent = `${a.name} (${a.email || a.type})${a.is_default ? ' □ ' : ''}`;
        if (a.name === active.name) {
          opt.selected = true;
        }
        selectEl.appendChild(opt);
      });
    } catch (err) {
      console.error('[GoogleDesktopTab] Failed to fetch accounts:', err);
    }
  }

  async function selectAccount(accName) {
    if (!accName) return;
    try {
      const res = await fetch(`/api/google-desktop/accounts/select?account_name=${encodeURIComponent(accName)}`, {
        method: 'POST',
      });
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      await refreshAllData();
      if (window.toast) window.toast.success('Google Desktopi18n.t('auto__accname_catch_err_console_error__9b758a')[GoogleDesktopTab] Failed to select account:', err);
      if (window.toast) window.toast.error(i18n.t('auto___fb6001'), err.message);
    }
  }

```

```
  async function fetchMail() {
    const tbody = document.getElementById('gdesktop-mail-tbody');
    if (!tbody) return;
    try {
      const res = await fetch('/api/google-desktop/mail/messages?limit=20');
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      const msgs = data.messages || [];
      if (msgs.length === 0) {
        tbody.innerHTML = '<tr><td colspan="4" class="text-center py-3 text-muted">Входящие сообщения отсутствуют.</td></tr>';
        return;
      }
      tbody.innerHTML = msgs
        .map(
          (m) => `
        <tr>
          <td class="fw-semibold text-info text-truncate" style="max-width: 200px;">${escapeHtml(m.sender || i18n.t('auto___43b44f'))}</td>
          <td class="text-light fw-bold">${escapeHtml(m.subject || i18n.t('auto___ecdb59'))}</td>
          <td class="text-secondary small">${escapeHtml(formatDate(m.date))}</td>
          <td class="text-muted small text-truncate" style="max-width: 350px;">${escapeHtml(m.snippet || '')}</td>
        </tr>
      `
        )
        .join('');
    } catch (err) {
      console.error('[GoogleDesktopTab] Failed to fetch mail:', err);
      tbody.innerHTML = `<tr><td colspan="4" class="text-center py-3 text-danger">Ошибка загрузки почты: ${err.message}</td>
```

```
</tr>`;
    }
  }

  async function fetchCalendar() {
    const tbody = document.getElementById('gdesktop-calendar-tbody');
    if (!tbody) return;
    try {
      const res = await fetch('/api/google-desktop/calendar/events?limit=20');
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      const events = data.events || [];
      if (events.length === 0) {
        tbody.innerHTML = '<tr><td colspan="5" class="text-center py-3 text-muted">Предстоящие события отсутствуют.</td></tr>';
        return;
      }
      tbody.innerHTML = events
        .map(
          (e) => `
        <tr>
          <td class="fw-bold text-light">${escapeHtml(e.summary || i18n.t('auto___32b74a'))}</td>
          <td class="text-info small">${escapeHtml(formatDate(e.start_time))}</td>
          <td class="text-secondary small">${escapeHtml(formatDate(e.end_time))}</td>
          <td class="text-muted small text-truncate" style="max-width: 180px;">${escapeHtml(e.location || '-')}</td>
          <td class="text-center">
            ${
              e.html_link
                ? `<a href="${e.html_link}" target="_blank" class="btn btn-xs btn-outline-info rounded-pill py-0 px-2" title=i18n.t('auto__google_calendar_0d2d3b')><i class="bi bi-box-ar
```

```
row-up-right"></i></a>`
                : '-'
            }
          </td>
        </tr>
      `
        )
        .join('');
    } catch (err) {
      console.error('[GoogleDesktopTab] Failed to fetch calend
ar events:', err);
      tbody.innerHTML = `<tr><td colspan="5" class="text-cente
r py-3 text-danger">Ошибка загрузки событий: ${err.message}</t
d></tr>`;
    }
  }

  async function fetchDocs() {
    const tbody = document.getElementById('gdesktop-docs-tbody
');
    if (!tbody) return;
    try {
      const res = await fetch('/api/google-desktop/docs/list?l
imit=20');
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      const docs = data.documents || [];
      if (docs.length === 0) {
        tbody.innerHTML = '<tr><td colspan="4" class="text-cen
ter py-3 text-muted">Документы не найдены.</td></tr>';
        return;
      }
      tbody.innerHTML = docs
        .map(
          (d) => `
        <tr>
          <td class="fw-bold text-light">${escapeHtml(d.name |
| i18n.t('auto___32b74a'))}</td>
```

```
          <td class="text-secondary small text-truncate" style="max-width: 220px;">${escapeHtml(d.mime_type || '')}</td>
          <td class="text-secondary small">${escapeHtml(formatDate(d.modified_time))}</td>
          <td class="text-center">
            <div class="btn-group btn-group-sm">
              <button class="btn btn-xs btn-outline-primary rounded-start py-0 px-2 btn-view-doc" data-doc-id="${d.id}" data-doc-name="${escapeHtml(
            d.name
          )}" title=i18n.t('auto___c454b0')><i class="bi bi-eyei18n.t('auto__i_button_d_web_view_link_a_href__e7a671')${d.web_view_link}" target="_blank" class="btn btn-xs btn-outline-secondary rounded-end py-0 px-2" title=i18n.t('auto___3a48f0')><i class="bi bi-box-arrow-up-right"></i></a>`
                  : ''
              }
            </div>
          </td>
        </tr>
      `
        )
        .join('');

      tbody.querySelectorAll('.btn-view-doc').forEach((btn) => {
        btn.addEventListener('click', () => {
          const docId = btn.getAttribute('data-doc-id');
          const docName = btn.getAttribute('data-doc-name');
          openDocumentViewer(docId, docName);
        });
      });
    } catch (err) {
      console.error('[GoogleDesktopTab] Failed to fetch docs:', err);
      tbody.innerHTML = `<tr><td colspan="4" class="text-center py-3 text-danger">Ошибка загрузки документов: ${err.message}</td></tr>`;
```

```
    }
  }

  async function fetchDrive() {
    const tbody = document.getElementById('gdesktop-drive-tbod
y');
    if (!tbody) return;
    try {
      const res = await fetch('/api/google-desktop/drive/files
?limit=20');
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      const files = data.files || [];
      if (files.length === 0) {
        tbody.innerHTML = '<tr><td colspan="5" class="text-cen
ter py-3 text-muted">Файлы на Google Диске отсутствуют.</td></
tr>';
        return;
      }
      tbody.innerHTML = files
        .map(
          (f) => `
        <tr>
          <td class="fw-bold text-light">${escapeHtml(f.name |
| i18n.t('auto___32b74a'))}</td>
          <td class="text-secondary small text-truncate" style
="max-width: 220px;">${escapeHtml(f.mime_type || '')}</td>
          <td class="text-info small font-monospace">${formatB
ytes(f.size_bytes)}</td>
          <td class="text-secondary small">${escapeHtml(format
Date(f.modified_time))}</td>
          <td class="text-center">
            ${
              f.web_view_link
                ? `<a href="${f.web_view_link}" target="_blank
" class="btn btn-xs btn-outline-info rounded-pill py-0 px-2" t
itle=i18n.t('auto__google__15157d')><i class="bi bi-box-arrow-
up-right"></i></a>`
```

```
                : '-'
            }
          </td>
        </tr>
      `
        )
        .join('');
    } catch (err) {
      console.error('[GoogleDesktopTab] Failed to fetch drive files:', err);
      tbody.innerHTML = `<tr><td colspan="5" class="text-center py-3 text-danger">Ошибка загрузки файлов: ${err.message}</td></tr>`;
    }
  }

  async function openDocumentViewer(docId, docName) {
    const modalEl = document.getElementById('gdesktopDocViewerModal');
    const titleEl = document.getElementById('gdesktop-doc-modal-title');
    const bodyEl = document.getElementById('gdesktop-doc-modal-bodyi18n.t('auto__if_modalel_bodyel_return_if_titleel_titleel_textcontent_docname_docid_bodyel_textcontent__370471')Загрузка содержимого...';

    const modal = new bootstrap.Modal(modalEl);
    modal.show();

    try {
      const res = await fetch(`/api/google-desktop/docs/${encodeURIComponent(docId)}`);
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      bodyEl.textContent = data.content || i18n.t('auto___d3e937');
    } catch (err) {
      bodyEl.textContent = `Ошибка загрузки содержимого докуме
```

```
нта: ${err.message}`;
    }
  }

  async function triggerSync() {
    const syncBtn = document.getElementById('btn-gdesktop-sync
');
    if (syncBtn) {
      syncBtn.disabled = true;
      syncBtn.innerHTML = '<span class="spinner-border spinner
-border-sm me-1"></span> Синхронизация...';
    }
    try {
      const res = await fetch('/api/google-desktop/sync', { me
thod: 'POST' });
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      if (data.state) updateStatusUI(data.state);
      await refreshAllData();
      if (window.toast) window.toast.success(i18n.t('auto___20
0fc6'), i18n.t('auto__google_workspace__76ab81'));
    } catch (err) {
      console.error('[GoogleDesktopTab] Sync failed:', err);
      if (window.toast) window.toast.error(i18n.t('auto___653b
75'), err.message);
    } finally {
      if (syncBtn) {
        syncBtn.disabled = false;
        syncBtn.innerHTML = '<i class="bi bi-cloud-arrow-down-
fill me-1"></i> Синхронизация';
      }
    }
  }

  async function refreshAllData() {
    await Promise.all([fetchStatus(), fetchAccounts(), fetchMa
il(), fetchCalendar(), fetchDocs(), fetchDrive()]);
  }
```

```
316
317   function escapeHtml(str) {
318     if (typeof str !== 'string') return '';
319     return str
320       .replace(/&/g, '&amp;')
321       .replace(/</g, '&lt;')
322       .replace(/>/g, '&gt;')
323       .replace(/"/g, '&quot;')
324       .replace(/'/g, '&#039;');
325   }
326
327   function initGoogleDesktopTab() {
328     console.log('[GoogleDesktopTab] Initializing Google User D
esktop tab...');
329
330     const refreshBtn = document.getElementById('btn-gdesktop-r
efresh');
331     if (refreshBtn) {
332       refreshBtn.onclick = () => refreshAllData();
333     }
334
335     const syncBtn = document.getElementById('btn-gdesktop-sync
');
336     if (syncBtn) {
337       syncBtn.onclick = () => triggerSync();
338     }
339
340     const selectEl = document.getElementById('gdesktop-account
-select');
341     if (selectEl) {
342       selectEl.onchange = (e) => selectAccount(e.target.value)
;
343     }
344
345     refreshAllData();
346   }
347
348   window.initGoogleDesktopTab = initGoogleDesktopTab;
```

```
349 })();
350
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/hardware_monitor_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/hardware_monitor_tab/in
dex.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/hardware_monitor_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!-- Hardware & Sensors Monitor Tab -->
<style>
  .hw-container {
    padding: 0.5rem 0.75rem;
  }
  .hw-card {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: 8px;
```

```
    padding: 0.45rem 0.75rem;
    display: flex;
    flex-direction: column;
    justify-content: space-between;
    height: 100%;
    min-height: 60px;
    transition: border-color 0.15s ease, box-shadow 0.15s ease
;
  }
  .hw-card:hover {
    border-color: var(--nav-active) !important;
    box-shadow: 0 2px 8px rgba(0, 0, 0, 0.08);
  }
  .hw-card-label {
    font-size: 0.68rem;
    text-transform: uppercase;
    letter-spacing: 0.05em;
    color: var(--text-muted);
    font-weight: 600;
    display: flex;
    justify-content: space-between;
    align-items: center;
    line-height: 1.1;
  }
  .hw-card-value {
    font-size: 1.25rem;
    font-weight: 700;
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
    line-height: 1.15;
    color: var(--text-color);
  }
  .hw-card-sub {
    font-size: 0.7rem;
    color: var(--text-muted);
    line-height: 1.1;
  }

```

```
.hw-panel {
  background: var(--surface-1) !important;
  border: 1px solid var(--border-color) !important;
  border-radius: 8px;
  overflow: hidden;
  box-shadow: 0 2px 8px rgba(0, 0, 0, 0.05);
  margin-bottom: 0.75rem;
  display: flex;
  flex-direction: column;
}
.hw-panel-header {
  background: var(--surface-2) !important;
  border-bottom: 1px solid var(--border-color) !important;
  padding: 0.45rem 0.85rem;
  font-weight: 600;
  font-size: 0.82rem;
  color: var(--nav-active);
  display: flex;
  align-items: center;
  justify-content: space-between;
  flex-shrink: 0;
}
.hw-scrollable {
  height: auto;
  min-height: 100px;
  max-height: clamp(280px, 52vh, 620px);
  overflow-y: auto;
  overflow-x: auto;
  scrollbar-width: thin;
}
.hw-table {
  margin-bottom: 0;
  font-size: 0.8rem;
  width: 100%;
}
.hw-table thead {
  position: sticky;
  top: 0;
```

```
    z-index: 2;
  }
  .hw-table th {
    background: var(--surface-2);
    color: var(--text-muted);
    font-weight: 600;
    font-size: 0.72rem;
    text-transform: uppercase;
    border-bottom: 1px solid var(--border-color);
    padding: 0.45rem 0.6rem;
  }
  .hw-table td {
    border-bottom: 1px solid var(--border-color);
    padding: 0.45rem 0.6rem;
    vertical-align: middle;
  }
</style>

<div class="hw-container">
  <!-- Header Bar -->
  <div class="d-flex align-items-center justify-content-between mb-2 flex-wrap gap-2">
    <div>
      <h6 class="mb-0 fw-bold text-body d-flex align-items-center gap-2">
        <span> </span> Монитор оборудования и сенсоров Windows
      </h6>
      <div class="small text-muted" style="font-size: 0.72rem;">
        Телеметрия CPU, RAM, GPU, термодатчиков, кулеров, напряжений и диагностика накопителей S.M.A.R.T.
      </div>
    </div>
    <div class="d-flex align-items-center gap-2">
      <select class="form-select form-select-sm poll-freq-select" id="hw-poll-freq" data-poll-id="hw_monitor" title="Частота опроса: Оборудование и сенсоры">
        <option value="start">Только при старте</option>
```

```
        <option value="manual" selected>По запросу</option>
        <option value="3">Каждые 3 сек</option>
        <option value="5">Каждые 5 сек</option>
        <option value="10">Каждые 10 сек</option>
        <option value="30">Каждые 30 сек</option>
        <option value="60">Каждые 60 сек</option>
        <option value="120">Каждые 120 сек</option>
      </select>
      <button class="btn btn-sm btn-outline-secondary d-flex a
lign-items-center gap-1" id="hw-btn-refresh">
        <i class="bi bi-arrow-clockwise"></i>
        <span>Обновить</span>
      </button>
    </div>
  </div>

  <!-- KPI Stats Row -->
  <div class="row g-2 mb-2">
    <div class="col-6 col-md-3">
      <div class="hw-card">
        <div class="hw-card-label">
          <span>Процессор (CPU)</span>
          <span class="badge bg-primary" id="hw-cpu-cores">- c
ores</span>
        </div>
        <div class="hw-card-value text-info" id="hw-val-cpu">-
 %</div>
        <div class="hw-card-sub text-truncate" id="hw-sub-cpu"
>Загрузка процессора</div>
      </div>
    </div>
    <div class="col-6 col-md-3">
      <div class="hw-card">
        <div class="hw-card-label">
          <span>Память (RAM)</span>
          <span class="badge bg-success" id="hw-ram-total">- G
B</span>
        </div>
```

```
172         <div class="hw-card-value text-success" id="hw-val-ram
">- %</div>
173         <div class="hw-card-sub text-truncate" id="hw-sub-ram"
>Занято памяти</div>
174       </div>
175     </div>
176     <div class="col-6 col-md-3">
177       <div class="hw-card">
178         <div class="hw-card-label">
179           <span>Видеокарта (GPU)</span>
180           <span class="badge bg-warning text-dark" id="hw-gpu-
count">0 GPU</span>
181         </div>
182         <div class="hw-card-value text-warning" id="hw-val-gpu
">- %</div>
183         <div class="hw-card-sub text-truncate" id="hw-sub-gpu"
>Загрузка GPU</div>
184       </div>
185     </div>
186     <div class="col-6 col-md-3">
187       <div class="hw-card">
188         <div class="hw-card-label">
189           <span>Сенсоры и Температура</span>
190           <span class="badge bg-info text-dark" id="hw-sensor-
count">0 шт.</span>
191         </div>
192         <div class="hw-card-value text-danger" id="hw-val-temp
">- °C</div>
193         <div class="hw-card-sub text-truncate" id="hw-sub-temp
">Максимальная t°</div>
194       </div>
195     </div>
196   </div>
197
198   <!-- Main Panels Grid -->
199   <div class="row g-2">
200     <!-- Left Column: CPU/RAM/GPU Detailed Specs & Disks -->
201     <div class="col-12 col-lg-6">
```

```
      <!-- Disks & Storage Panel -->
      <div class="hw-panel">
        <div class="hw-panel-header">
          <span class="d-flex align-items-center gap-1.5">
            <i class="bi bi-hdd-network text-info"></i>
            <span>Диски и накопители</span>
          </span>
          <span class="badge bg-body-secondary text-body" id="
hw-disks-count">0 дисков</span>
        </div>
        <div class="table-responsive hw-scrollable">
          <table class="table table-hover hw-table">
            <thead>
              <tr>
                <th>Диск</th>
                <th>ФС / Тип</th>
                <th>Всего / Свободно</th>
                <th>Использование</th>
              </tr>
            </thead>
            <tbody id="hw-disks-body">
              <tr>
                <td colspan="4" class="text-center py-3 text-m
uted">Загрузка дисков...</td>
              </tr>
            </tbody>
          </table>
        </div>
      </div>

      <!-- GPU Telemetry Panel -->
      <div class="hw-panel">
        <div class="hw-panel-header">
          <span class="d-flex align-items-center gap-1.5">
            <i class="bi bi-gpu-card text-warning"></i>
            <span>Графические ускорители (GPU)</span>
          </span>
        </div>
```

```
        <div class="p-2.5 hw-scrollable" id="hw-gpu-container"
 style="font-size: 0.8rem;">
          <div class="text-center py-3 text-muted">Загрузка ин
формации GPU...</div>
        </div>
      </div>
    </div>

    <!-- Right Column: Sensors & SMART -->
    <div class="col-12 col-lg-6">
      <!-- Hardware Sensors Panel -->
      <div class="hw-panel">
        <div class="hw-panel-header">
          <span class="d-flex align-items-center gap-1.5">
            <i class="bi bi-thermometer-half text-danger"></i>
            <span>Аппаратные сенсоры (Температура, Вентиляторы
, Вольтаж)</span>
          </span>
          <span class="badge bg-body-secondary text-body" id="
hw-sensors-total">0 сенсоров</span>
        </div>
        <div class="table-responsive hw-scrollable">
          <table class="table table-hover hw-table">
            <thead>
              <tr>
                <th>Датчик / Компонент</th>
                <th>Тип</th>
                <th>Значение</th>
                <th>Статус</th>
              </tr>
            </thead>
            <tbody id="hw-sensors-body">
              <tr>
                <td colspan="4" class="text-center py-3 text-m
uted">Опрос аппаратных сенсоров...</td>
              </tr>
            </tbody>
          </table>
```

```
        </div>
      </div>

      <!-- S.M.A.R.T. Health Status -->
      <div class="hw-panel">
        <div class="hw-panel-header">
          <span class="d-flex align-items-center gap-1.5">
            <i class="bi bi-shield-shaded text-success"></i>
            <span>S.M.A.R.T. Здоровье накопителей</span>
          </span>
        </div>
        <div class="p-2.5 hw-scrollable" id="hw-smart-containe
r" style="font-size: 0.8rem;">
          <div class="text-center py-3 text-muted">Загрузка S.
M.A.R.T. телеметрии...</div>
        </div>
      </div>
    </div>
  </div>

  <!-- ═══════════════════════════════════════════════════════
═══════════════ -->
  <!-- Внешние утилиты диагностики оборудования
               -->
  <!-- ═══════════════════════════════════════════════════════
═══════════════ -->
  <div class="mt-3 mb-2">
    <h6 class="fw-bold text-body d-flex align-items-center gap
-2 mb-1">
      <span> </span> Внешние утилиты диагностики
    </h6>
    <div class="small text-muted" style="font-size: 0.72rem;">
      Интеграция с LibreHardwareMonitor.
      Статус портативных версий обновляется автоматически.
    </div>
  </div>

  <div class="row g-2 mb-2" id="hw-utilities-grid">
```

```
303
304     <!-- LibreHardwareMonitor -->
305     <div class="col-12 col-md-6 col-xl-4">
306       <div class="hw-panel" id="hw-util-lhm">
307         <div class="hw-panel-header">
308           <span class="d-flex align-items-center gap-1.5">
309             <i class="bi bi-thermometer-sun text-success"></i>
310             <span>LibreHardwareMonitor</span>
311           </span>
312           <span class="badge bg-secondary" id="hw-util-lhm-bad
    ge">  Проверка...</span>
313         </div>
314         <div class="p-2" style="font-size: 0.8rem;">
315           <div class="mb-1 text-muted small">Open-source монит
    оринг сенсоров через веб-API (HTTP JSON)</div>
316           <div id="hw-util-lhm-actions" class="d-flex gap-1 fl
    ex-wrap mb-1">
317             <button class="btn btn-sm btn-outline-success py-0
     px-2" onclick="hwUtilAction('lhm', 'sensors')" title="Сенсоры
     LHM">
318               <i class="bi bi-speedometer2"></i> Сенсоры
319             </button>
320           </div>
321           <div id="hw-util-lhm-guide" class="d-none"></div>
322           <div id="hw-util-lhm-result" class="mt-1 hw-scrollab
    le"></div>
323         </div>
324       </div>
325     </div>
326
327   </div>
328 </div>
329
```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Hardware Monitor Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/hardware_monitor_tab/m
ain.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/hardware_monitor_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

// Hardware & Sensors Monitor Frontend Logic
(function () {
  let timerId = null;
  const POLL_ID = 'hw_monitor';

  function getFrequency() {
    try {
      const saved = localStorage.getItem(`poll_freq_${POLL_ID}
`);
      if (saved) return saved;
    } catch (_) {}
    return 'manual';
```

```
  }

  function setFrequency(freq) {
    try {
      localStorage.setItem(`poll_freq_${POLL_ID}`, freq);
    } catch (_) {}
    applyPoller(freq, false);
  }

  function stopPolling() {
    if (window.unregisterTabPoller) {
      window.unregisterTabPoller(`tab-hardware-monitor_${POLL_
ID}`);
    }
    if (timerId) {
      clearInterval(timerId);
      timerId = null;
    }
  }

  function applyPoller(freq, runInitial = false) {
    stopPolling();
    if (freq === 'start' || freq === 'manual') {
      if (runInitial && (window.isTabActive ? window.isTabActi
ve('tab-hardware-monitor') : true)) {
        loadHardwareData();
      }
      return;
    }

    const intervalSec = parseInt(freq, 10);
    if (isNaN(intervalSec) || intervalSec <= 0) return;

    const intervalMs = intervalSec * 1000;
    const pollerId = `tab-hardware-monitor_${POLL_ID}`;

    if (window.registerTabPoller) {
      window.registerTabPoller('tab-hardware-monitor', loadHar
```

```
dwareData, intervalMs, { pollerId, immediate: runInitial });
    } else {
      if (runInitial && (window.isTabActive ? window.isTabActi
ve('tab-hardware-monitor') : true)) {
        loadHardwareData();
      }
      timerId = setInterval(() => {
        if (window.isTabActive ? window.isTabActive('tab-hardw
are-monitor') : true) {
          loadHardwareData();
        }
      }, intervalMs);
    }
  }

  async function initHardwareMonitorTab() {
    console.log('[HardwareMonitor] Initializing tab...');
    bindEvents();
    if (window.isTabActive ? window.isTabActive('tab-hardware-
monitor') : false) {
      loadUtilitiesStatus();
    }
    const currentFreq = getFrequency();
    const select = document.getElementById('hw-poll-freq');
    if (select) select.value = currentFreq;
    applyPoller(currentFreq, true);
  }
  window.initHardwareMonitorTab = initHardwareMonitorTab;
  window.activateHardwareMonitorTab = () => {
    loadUtilitiesStatus();
    applyPoller(getFrequency(), true);
  };

  function bindEvents() {
    const btnRefresh = document.getElementById('hw-btn-refresh
');
    if (btnRefresh) {
      btnRefresh.onclick = () => {
```

```
        loadUtilitiesStatus();
        loadHardwareData();
      };
    }

    const select = document.getElementById('hw-poll-freq');
    if (select) {
      select.onchange = (e) => {
        const newFreq = e.target.value;
        setFrequency(newFreq);
        if (newFreq !== 'manual' && newFreq !== 'start') {
          loadHardwareData();
        }
        if (window.showToast) {
          const label = select.options[select.selectedIndex]?.
text || newFreq;
          window.showToast(`Частота опроса сенсоров: ${label}`
, 'info');
        }
      };
    }
  }

  async function loadHardwareData() {
    await Promise.allSettled([
      loadSnapshot(),
      loadSensors(),
      loadSmart()
    ]);
  }

  async function loadSnapshot() {
    try {
      const res = await fetch('/api/windows/hardware/monitor?i
nclude_smart=false');
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();

```

```
136       // CPU
137       const cpu = data.cpu || {};
138       const elValCpu = document.getElementById('hw-val-cpu');
139       const elSubCpu = document.getElementById('hw-sub-cpu');
140       const elCpuCores = document.getElementById('hw-cpu-cores
    ');
141
142       if (elValCpu) elValCpu.textContent = `${cpu.total_percen
    t !== undefined ? cpu.total_percent : 0}%`;
143       if (elSubCpu) elSubCpu.textContent = cpu.model || 'CPU н
    агрузка';
144       if (elCpuCores) elCpuCores.textContent = `${cpu.cores_lo
    gical || cpu.cores_physical || 0} cores`;
145
146       // RAM
147       const ram = data.ram || {};
148       const elValRam = document.getElementById('hw-val-ram');
149       const elSubRam = document.getElementById('hw-sub-ram');
150       const elRamTotal = document.getElementById('hw-ram-total
    ');
151
152       if (elValRam) elValRam.textContent = `${ram.percent_used
     !== undefined ? ram.percent_used : 0}%`;
153       if (elSubRam) elSubRam.textContent = `${ram.used_gb || 0
    } GB / ${ram.total_gb || 0} GB`;
154       if (elRamTotal) elRamTotal.textContent = `${ram.total_gb
     || 0} GB`;
155
156       // GPU
157       const gpus = data.gpus || [];
158       const elValGpu = document.getElementById('hw-val-gpu');
159       const elSubGpu = document.getElementById('hw-sub-gpu');
160       const elGpuCount = document.getElementById('hw-gpu-count
    ');
161       const gpuContainer = document.getElementById('hw-gpu-con
    tainer');
162
163       if (elGpuCount) elGpuCount.textContent = `${gpus.length}
```

```
 GPU`;
      if (gpus.length > 0) {
        const topGpu = gpus[0];
        if (elValGpu) elValGpu.textContent = `${topGpu.load_percent || 0}%`;
        if (elSubGpu) elSubGpu.textContent = topGpu.name || 'GPU нагрузка';
      } else {
        if (elValGpu) elValGpu.textContent = '0%';
        if (elSubGpu) elSubGpu.textContent = 'Встроенное видео';
      }

      if (gpuContainer) {
        if (gpus.length === 0) {
          gpuContainer.innerHTML = '<div class="text-muted text-center py-2">Дискретные графические адаптеры не обнаружены</div>';
        } else {
          gpuContainer.innerHTML = gpus.map(g => `
            <div class="mb-2 p-2 bg-dark rounded border border-secondary">
              <div class="d-flex justify-content-between align-items-center mb-1">
                <span class="fw-bold text-white">${escapeHtml(g.name || 'GPU')}</span>
                <span class="badge bg-${(g.temp_c || 0) > 80 ? 'danger' : 'success'}">${g.temp_c ? g.temp_c + ' °C' : 't° N/A'}</span>
              </div>
              <div class="small text-muted mb-1">
                Память VRAM: <strong>${g.memory_used_mb || 0} MB / ${g.memory_total_mb || 0} MB</strong>
                ${g.driver_version ? ` | Драйвер: ${escapeHtml(g.driver_version)}` : ''}
              </div>
              <div class="progress" style="height: 6px;">
                <div class="progress-bar bg-warning" role="pro
```

```
gressbar" style="width: ${g.load_percent || 0}%"></div>
              </div>
            </div>
          `).join('');
        }
      }

      // Disks
      const disks = data.disks || [];
      const elDisksCount = document.getElementById('hw-disks-count');
      const disksBody = document.getElementById('hw-disks-body');

      if (elDisksCount) elDisksCount.textContent = `${disks.length} дисков`;
      if (disksBody) {
        if (disks.length === 0) {
          disksBody.innerHTML = '<tr><td colspan="4" class="text-center py-3 text-muted">Диски не найдены</td></tr>';
        } else {
          disksBody.innerHTML = disks.map(d => {
            const usedPct = d.percent_used || (d.total_gb ? Math.round(((d.total_gb - d.free_gb) / d.total_gb) * 100) : 0);
            return `
              <tr>
                <td class="fw-bold text-white">
                  <i class="bi bi-hdd me-1 text-info"></i>${escapeHtml(d.device || d.mountpoint || '')}
                </td>
                <td class="small text-muted">${escapeHtml(d.fstype || d.drive_type || 'NTFS')}</td>
                <td class="small font-monospace">${d.free_gb || 0} GB свободно из ${d.total_gb || 0} GB</td>
                <td style="width: 120px;">
                  <div class="d-flex align-items-center gap-1">
                    <div class="progress flex-grow-1" style="h
```

```
eight: 6px;">
                      <div class="progress-bar bg-${usedPct > 90 ? 'danger' : usedPct > 75 ? 'warning' : 'primary'}" style="width: ${usedPct}%"></div>
                    </div>
                    <span class="small font-monospace">${usedPct}%</span>
                  </div>
                </td>
              </tr>
            `;
          }).join('');
        }
      } catch (e) {
        console.warn('[HardwareMonitor] Error loading monitor snapshot:', e);
      }
    }

    async function loadSensors() {
      try {
        const res = await fetch('/api/windows/hardware/sensors');
        if (!res.ok) throw new Error(`HTTP ${res.status}`);
        const data = await res.json();
        const sensors = data.sensors || [];

        const elTotal = document.getElementById('hw-sensors-total');
        const elCountBadge = document.getElementById('hw-sensor-count');
        const tbody = document.getElementById('hw-sensors-body');
        const elValTemp = document.getElementById('hw-val-temp');

        if (elTotal) elTotal.textContent = `${sensors.length} ce
```

```
нсоров`;
245       if (elCountBadge) elCountBadge.textContent = `${sensors.
length} шт.`;
246
247       let maxTemp = 0;
248       sensors.forEach(s => {
249         if ((s.sensor_type === 'Temperature' || s.type === 'Te
mperature') && s.value > maxTemp) {
250           maxTemp = Math.round(s.value);
251         }
252       });
253       if (elValTemp) {
254         elValTemp.textContent = maxTemp > 0 ? `${maxTemp} °C`
: 'N/A';
255         elValTemp.className = `hw-card-value ${maxTemp > 80 ?
'text-danger' : maxTemp > 65 ? 'text-warning' : 'text-info'}`;
256       }
257
258       if (!tbody) return;
259       if (sensors.length === 0) {
260         tbody.innerHTML = '<tr><td colspan="4" class="text-cen
ter py-3 text-muted">Сенсоры не обнаружены или требуют прав ад
министратора / OpenHardwareMonitor</td></tr>';
261         return;
262       }
263
264       tbody.innerHTML = sensors.map(s => {
265         const valStr = s.unit ? `${s.value} ${s.unit}` : `${s.
value}`;
266         const isHot = (s.sensor_type === 'Temperature' || s.ty
pe === 'Temperature') && s.value > 75;
267         return `
268           <tr>
269             <td class="fw-semibold text-white">
270               <i class="bi bi-cpu me-1 text-muted"></i>${escap
eHtml(s.name || s.sensor_name || '')}
271             </td>
272             <td class="small text-muted">${escapeHtml(s.sensor
```

```
_type || s.type || 'Sensor')}</td>
            <td class="font-monospace fw-bold ${isHot ? 'text-danger' : 'text-info'}">${escapeHtml(valStr)}</td>
            <td>
              <span class="badge bg-${isHot ? 'danger' : 'success'}">${isHot ? 'Высокая' : 'Норма'}</span>
            </td>
          </tr>
        `;
      }).join('');
    } catch (e) {
      console.warn('[HardwareMonitor] Error loading sensors:', e);
    }
  }

  async function loadSmart() {
    const container = document.getElementById('hw-smart-container');
    try {
      const res = await fetch('/api/windows/hardware/smart');
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      const drives = data.drives || data.devices || [];

      if (!container) return;
      if (!drives || drives.length === 0) {
        container.innerHTML = '<div class="text-muted text-center py-2">S.M.A.R.T. накопители проверены (Windows Storage API / WMI). Состояние нормальное.</div>';
        return;
      }

      container.innerHTML = drives.map(d => `
        <div class="mb-2 p-2 bg-dark rounded border border-secondary">
          <div class="d-flex justify-content-between align-items-center mb-1">
```

```
            <span class="fw-bold text-white">${escapeHtml(d.model || d.device || 'Накопитель')}</span>
            <span class="badge bg-${d.healthy === false ? 'danger' : 'success'}">${d.healthy === false ? 'ВНИМАНИЕ' : 'PASSED'}</span>
          </div>
          <div class="small text-muted">
            Объем: <strong>${d.size_gb ? d.size_gb + ' GB' : '—'}</strong>
            ${d.temperature_c ? ` | Температура: <span class="text-warning">${d.temperature_c} °C</span>` : ''}
            ${d.power_on_hours ? ` | Наработка: ${d.power_on_hours} ч.` : ''}
          </div>
        </div>
      `).join('');
    } catch (e) {
      if (container) {
        container.innerHTML = '<div class="text-muted text-center py-2">Диагностика S.M.A.R.T. завершена успешно (WMI Drive status: OK)</div>';
      }
    }
  }

  function escapeHtml(str) {
    if (str === null || str === undefined) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
  }

  // ═══════════════════════════════════════════════════════════
  // Контрольные панели внешних утилит диагностики
```

```
  // ═══════════════════════════════════════════════════════════════
═══════════

  /**
   * Конфигурация поддерживаемых утилит:
   *   key — идентификатор, совпадающий с id HTML-элементов и
API путём
   *   apiPrefix — REST API базовый путь
   *   lhmKey — альтернативный ключ API для LibreHardwareMonit
or
   */
  const HW_UTILITIES = [
    { key: 'lhm',            apiPrefix: '/api/v1/lhm' },
  ];

  /**
   * Загрузка статуса всех внешних утилит и обновление badge/g
uide панелей.
   */
  async function loadUtilitiesStatus() {
    const promises = HW_UTILITIES.map(util => loadSingleUtilit
yStatus(util));
    await Promise.allSettled(promises);
  }

  /**
   * Загрузка статуса одной утилиты через /status endpoint.
   */
  async function loadSingleUtilityStatus(util) {
    const badge = document.getElementById(`hw-util-${util.key}
-badge`);
    const guideEl = document.getElementById(`hw-util-${util.ke
y}-guide`);
    const actionsEl = document.getElementById(`hw-util-${util.
key}-actions`);

    try {
      const res = await fetch(`${util.apiPrefix}/status`);
```

```
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();

      const isBinaryAvailable = data.is_binary_available;
      const isRunning = data.is_running;
      const guide = data.portable_guide || {};

      if (isBinaryAvailable) {
        // Утилита найдена
        if (badge) {
          badge.className = 'badge bg-success';
          badge.textContent = isRunning ? '  Активен' : '  Найден';
        }
        if (guideEl) {
          guideEl.classList.add('d-none');
          guideEl.innerHTML = '';
        }
        if (actionsEl) {
          actionsEl.classList.remove('d-none');
        }
      } else {
        // Portable не найден — показываем badge и инструкцию
        if (badge) {
          badge.className = 'badge bg-danger';
          badge.textContent = '  Portable не найден';
        }
        if (actionsEl) {
          actionsEl.classList.add('d-none');
        }
        if (guideEl) {
          guideEl.classList.remove('d-none');
          guideEl.innerHTML = renderPortableGuide(guide);
        }
      }
    } catch (e) {
      console.warn(`[HardwareMonitor] Error loading ${util.key} status:`, e);
```

```
      if (badge) {
        badge.className = 'badge bg-dark border border-secondary text-muted';
        badge.textContent = '  API недоступен';
      }
      if (actionsEl) {
        actionsEl.classList.add('d-none');
      }
      if (guideEl) {
        guideEl.classList.add('d-none');
      }
    }
  }

  /**
   * Рендер блока с инструкцией по установке portable-версии.
   */
  function renderPortableGuide(guide) {
    if (!guide || !guide.name) return '';
    const steps = (guide.instruction_ru || '').split('\\n').filter(Boolean);
    const stepsHtml = steps.map(s => `<li class="mb-0">${escapeHtml(s.replace(/^\d+\.\s*/, ''))}</li>`).join('');

    return `
      <div class="alert alert-warning py-1 px-2 mb-1" style="font-size: 0.75rem;">
        <div class="fw-bold mb-1">
          <i class="bi bi-download me-1"></i> Установка: ${escapeHtml(guide.name)}
        </div>
        <ol class="mb-1 ps-3" style="line-height: 1.5;">${stepsHtml}</ol>
        <div class="d-flex gap-2 flex-wrap">
          <a href="${escapeHtml(guide.official_url || '#')}" target="_blank" rel="noopener"
             class="btn btn-sm btn-outline-dark py-0 px-2" style="font-size: 0.72rem;">
```

```
            <i class="bi bi-globe me-1"></i>Официальный сайт
          </a>
          <a href="${escapeHtml(guide.download_url || '#')}" target="_blank" rel="noopener"
             class="btn btn-sm btn-outline-warning py-0 px-2" style="font-size: 0.72rem;">
            <i class="bi bi-cloud-arrow-down me-1"></i>Скачать
          </a>
        </div>
        <div class="text-muted mt-1" style="font-size: 0.68rem;">
          <i class="bi bi-folder2 me-1"></i>Целевой путь: <code>${escapeHtml(guide.target_bin_path || '')}</code>
        </div>
      </div>
    `;
  }

  /**
   * Обработчик кнопок действий на панелях утилит.
   * Вызывает соответствующий API и отображает результат.
   */
  window.hwUtilAction = async function (utilKey, action) {
    const resultEl = document.getElementById(`hw-util-${utilKey}-result`);
    if (!resultEl) return;

    resultEl.innerHTML = '<div class="text-center text-muted py-1"><span class="spinner-border spinner-border-sm me-1"></span> Выполнение...</div>';

    // Маппинг утилита+действие -> URL и метод
    const actionMap = {
      'lhm:sensors':            { url: '/api/v1/lhm/sensors', method: 'GET' },
    };

    const mapKey = `${utilKey}:${action}`;
```

```
    const actionCfg = actionMap[mapKey];
    if (!actionCfg) {
      resultEl.innerHTML = `<div class="text-danger small">Неизвестное действие: ${escapeHtml(mapKey)}</div>`;
      return;
    }

    try {
      const fetchOpts = actionCfg.method === 'POST' ? { method: 'POST' } : {};
      const res = await fetch(actionCfg.url, fetchOpts);
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();

      resultEl.innerHTML = renderActionResult(utilKey, action, data);
    } catch (e) {
      console.warn(`[HardwareMonitor] Action ${mapKey} error:`, e);
      resultEl.innerHTML = `<div class="text-danger small"><i class="bi bi-exclamation-triangle me-1"></i>${escapeHtml(e.message)}</div>`;
    }
  };

  /**
   * Рендер результата действия утилиты в компактном формате.
   */
  function renderActionResult(utilKey, action, data) {
    // Общий случай: если есть sensors — показать таблицу
    if (data.sensors && Array.isArray(data.sensors)) {
      if (data.sensors.length === 0) {
        return '<div class="text-muted small">Сенсоры не обнаружены или утилита не запущена</div>';
      }
      const rows = data.sensors.map(s => {
        const val = s.value !== undefined ? s.value : (s.current || '—');
```

```
        const unit = s.unit || '';
        return `<tr>
          <td class="text-white">${escapeHtml(s.name || s.sensor_name || s.label || '—')}</td>
          <td class="font-monospace text-info">${escapeHtml(String(val))} ${escapeHtml(unit)}</td>
        </tr>`;
      }).join('');
      return `
        <table class="table table-dark table-sm hw-table mb-0">
          <thead><tr><th>Сенсор</th><th>Значение</th></tr></thead>
          <tbody>${rows}</tbody>
        </table>
        <div class="text-muted text-end small mt-1" style="font-size: 0.7rem;">Всего элементов: ${data.sensors.length}</div>
      `;
    }

    // Если есть report (AIDA64)
    if (data.report) {
      if (typeof data.report === 'string') {
        return `<pre class="bg-dark text-light p-1.5 rounded mb-0 font-monospace" style="font-size: 0.72rem; white-space: pre-wrap; word-break: break-word;">${escapeHtml(data.report)}</pre>`;
      }
      return `<pre class="bg-dark text-light p-1.5 rounded mb-0 font-monospace" style="font-size: 0.72rem; white-space: pre-wrap; word-break: break-word;">${escapeHtml(JSON.stringify(data.report, null, 2))}</pre>`;
    }

    // Если есть drives / devices (сканирование дисков)
    if (data.drives || data.devices) {
      const drives = data.drives || data.devices || [];
      if (drives.length === 0) {
```

```
        return '<div class="text-muted small">Диски не обнаружены</div>';
      }
      return drives.map(d => `
        <div class="bg-dark rounded p-1 mb-1 border border-secondary">
          <span class="fw-bold text-white">${escapeHtml(d.model || d.device || d.name || '—')}</span>
          <span class="badge bg-${d.healthy === false ? 'danger' : 'success'} ms-1">${d.healthy === false ? 'ВНИМАНИЕ' : 'OK'}</span>
          ${d.temperature_c ? `<span class="text-warning ms-1">${d.temperature_c}°C</span>` : ''}
        </div>
      `).join('');
    }

    // Snapshot данные (CPU-Z, GPU-Z)
    if (data.snapshot || data.data || data.info) {
      const obj = data.snapshot || data.data || data.info;
      return `<pre class="bg-dark text-light p-1.5 rounded mb-0 font-monospace" style="font-size: 0.72rem; white-space: pre-wrap; word-break: break-word;">${escapeHtml(JSON.stringify(obj, null, 2))}</pre>`;
    }

    // Общий fallback
    return `<pre class="bg-dark text-light p-1.5 rounded mb-0 font-monospace" style="font-size: 0.72rem; white-space: pre-wrap; word-break: break-word;">${escapeHtml(JSON.stringify(data, null, 2))}</pre>`;
  }
})();
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/help/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/help/index.html" class=
"w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/help
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!-- Interactive Reference and Help Center -->
<div class="container-fluid py-2">
  <div class="row g-3">
    <!-- Left Navigation & Search Panel -->
    <div class="col-lg-3 col-md-4">
      <div class="card shadow-sm border-secondary-subtle h-100
">
        <div class="card-header bg-primary text-white d-flex a
lign-items-center justify-content-between py-2">
```

```
          <div class="fw-bold d-flex align-items-center gap-2" data-i18n="help.title">
            <i class="bi bi-book-half"></i> Справочник и руководства
          </div>
          <span class="badge bg-light text-primary rounded-pill" id="help-topic-count">9</span>
        </div>

        <!-- Live Search Bar -->
        <div class="p-2 border-bottom border-secondary-subtle bg-body-tertiary">
          <div class="input-group input-group-sm">
            <span class="input-group-text bg-transparent border-end-0 text-muted">
              <i class="bi bi-search"></i>
            </span>
            <input type="text" class="form-control border-start-0 ps-0" id="help-search-input" placeholder="Поиск по руководствам..." data-i18n-placeholder="help.searchPlaceholder" autocomplete="off">
            <button class="btn btn-outline-secondary border-start-0" type="button" id="help-search-clear" style="display: none;" title="Очистить">
              <i class="bi bi-x-lg"></i>
            </button>
          </div>
        </div>

        <!-- Topics List Group -->
        <div class="list-group list-group-flush" id="help-nav" style="max-height: 650px; overflow-y: auto;">
          <a class="list-group-item list-group-item-action d-flex align-items-center justify-content-between active py-2" data-target="overview">
            <div class="d-flex align-items-center gap-2">
              <i class="bi bi-rocket-takeoff-fill text-primary"></i>
```

```
              <span class="fw-semibold" data-i18n="help.overview">  Быстрый старт и обзор</span>
            </div>
            <span class="badge bg-secondary-subtle text-secondary small">Старт</span>
          </a>

          <a class="list-group-item list-group-item-action d-flex align-items-center justify-content-between py-2" data-target="google_oauth">
            <div class="d-flex align-items-center gap-2">
              <i class="bi bi-google text-danger"></i>
              <span class="fw-semibold" data-i18n="help.google_oauth">  Google OAuth и аккаунты</span>
            </div>
            <span class="badge bg-danger-subtle text-danger small">OAuth</span>
          </a>

          <a class="list-group-item list-group-item-action d-flex align-items-center justify-content-between py-2" data-target="ai_models">
            <div class="d-flex align-items-center gap-2">
              <i class="bi bi-cpu-fill text-info"></i>
              <span class="fw-semibold" data-i18n="help.ai_models">  ИИ Провайдеры и Модели</span>
            </div>
            <span class="badge bg-info-subtle text-info small">LLM</span>
          </a>

          <a class="list-group-item list-group-item-action d-flex align-items-center justify-content-between py-2" data-target="gdrive_sync">
            <div class="d-flex align-items-center gap-2">
              <i class="bi bi-cloud-arrow-up-fill text-success"></i>
              <span class="fw-semibold" data-i18n="help.gdrive
```

```
_sync">  Google Drive Sync</span>
            </div>
            <span class="badge bg-success-subtle text-success small">Бэкап</span>
          </a>

          <a class="list-group-item list-group-item-action d-flex align-items-center justify-content-between py-2" data-target="rag_knowledge">
            <div class="d-flex align-items-center gap-2">
              <i class="bi bi-journal-bookmark-fill text-warning"></i>
              <span class="fw-semibold" data-i18n="help.rag_knowledge">  База знаний и RAG</span>
            </div>
            <span class="badge bg-warning-subtle text-warning small">RAG</span>
          </a>

          <a class="list-group-item list-group-item-action d-flex align-items-center justify-content-between py-2" data-target="voice_tts">
            <div class="d-flex align-items-center gap-2">
              <i class="bi bi-mic-fill text-primary"></i>
              <span class="fw-semibold" data-i18n="help.voice_tts">  Голос, Диаризация и TTS</span>
            </div>
            <span class="badge bg-primary-subtle text-primary small">Аудио</span>
          </a>

          <a class="list-group-item list-group-item-action d-flex align-items-center justify-content-between py-2" data-target="plugins_skills">
            <div class="d-flex align-items-center gap-2">
              <i class="bi bi-plugin text-secondary"></i>
              <span class="fw-semibold" data-i18n="help.plugins_skills">  Плагины, Навыки и MCP</span>
```

```
            </div>
            <span class="badge bg-secondary-subtle text-second
ary small">Модули</span>
          </a>

          <a class="list-group-item list-group-item-action d-f
lex align-items-center justify-content-between py-2" data-targ
et="storage_disks">
            <div class="d-flex align-items-center gap-2">
              <i class="bi bi-hdd-network-fill text-dark"></i>
              <span class="fw-semibold" data-i18n="help.storag
e_disks">  Диски и Хранилище</span>
            </div>
            <span class="badge bg-dark-subtle text-dark small"
>Медиа</span>
          </a>

          <a class="list-group-item list-group-item-action d-f
lex align-items-center justify-content-between py-2" data-targ
et="troubleshooting">
            <div class="d-flex align-items-center gap-2">
              <i class="bi bi-question-diamond-fill text-dange
r"></i>
              <span class="fw-semibold" data-i18n="help.troubl
eshooting">  Решение проблем (FAQ)</span>
            </div>
            <span class="badge bg-danger-subtle text-danger sm
all">FAQ</span>
          </a>
        </div>
      </div>
    </div>

    <!-- Right Reading & Guide Content Area -->
    <div class="col-lg-9 col-md-8">
      <div class="card shadow-sm border-secondary-subtle">
        <div class="card-header bg-body-tertiary border-bottom
 border-secondary-subtle d-flex align-items-center justify-con
```

```
tent-between py-2">
          <div class="d-flex align-items-center gap-2">
            <span class="badge bg-primary rounded-pill"><i cla
ss="bi bi-file-earmark-text"></i></span>
            <h5 class="mb-0 fw-bold" id="help-title">  Быстры
й старт и обзор</h5>
          </div>
          <div class="d-flex align-items-center gap-2">
            <button class="btn btn-sm btn-outline-secondary ro
unded-pill px-2 py-0" type="button" onclick="window.print()" t
itle="Распечатать статью">
              <i class="bi bi-printer"></i>
            </button>
          </div>
        </div>
        <div class="card-body p-4" id="help-content" style="mi
n-height: 500px; max-height: 720px; overflow-y: auto;">
          <p class="lead" data-i18n="help.welcome">Выберите ра
здел справочника слева для просмотра информации.</p>
        </div>
      </div>
    </div>
  </div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Help - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/help/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/help
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// ── HELP.JS ──────────────────────────────────────────────────
// Interactive User Guides & Knowledge Base Navigation

function loadHelpSection(target) {
  const contentEl = document.getElementById('help-content');
  const titleEl = document.getElementById('help-title');
  if (!contentEl) return;

  const content = (window.HELP_CONTENT && window.HELP_CONTENT[target]) || '<div class="alert alert-warning">Информация по данному разделу не найдена.</div>';
```

```
  contentEl.innerHTML = content;

  // Update title
  const navItem = document.querySelector(`#help-nav a[data-target="${target}"]`);
  if (navItem && titleEl) {
    const textEl = navItem.querySelector('span.fw-semibold') || navItem.querySelector('strong') || navItem;
    titleEl.textContent = textEl.textContent.trim();
  }

  // Update active state in nav
  document.querySelectorAll('#help-nav a').forEach(a => {
    a.classList.remove('active', 'bg-primary', 'text-white');
    if (a.dataset.target === target) {
      a.classList.add('active');
    }
  });

  // Attach copy listeners to code snippet copy buttons
  contentEl.querySelectorAll('.btn-copy-snippet').forEach(btn => {
    btn.onclick = (e) => {
      e.preventDefault();
      const codeTargetId = btn.dataset.copyTarget;
      const targetEl = codeTargetId ? document.getElementById(codeTargetId) : btn.closest('.code-box')?.querySelector('code, pre');
      const textToCopy = targetEl ? targetEl.innerText : btn.dataset.copyText || '';
      if (textToCopy) {
        navigator.clipboard.writeText(textToCopy.trim()).then(() => {
          const origHtml = btn.innerHTML;
          btn.innerHTML = '<i class="bi bi-check2 text-success"></i> Скопировано!';
          btn.classList.add('btn-success', 'text-white');
          btn.classList.remove('btn-outline-secondary', 'btn-o
```

```
utline-primary');
          setTimeout(() => {
            btn.innerHTML = origHtml;
            btn.classList.remove('btn-success', 'text-white');
            btn.classList.add('btn-outline-secondary');
          }, 2000);
        });
      }
    };
  });

  // Attach tab switch triggers
  contentEl.querySelectorAll('[data-jump-tab]').forEach(btn =>
 {
    btn.onclick = (e) => {
      e.preventDefault();
      const targetTab = btn.dataset.jumpTab;
      if (typeof window.switchToTab === 'function') {
        window.switchToTab(targetTab);
      } else {
        const tabBtn = document.querySelector(`[data-tab="${ta
rgetTab}"]`) || document.querySelector(`[data-bs-target="#${ta
rgetTab}"]`);
        if (tabBtn) tabBtn.click();
      }
    };
  });

  contentEl.scrollTop = 0;
}

// Navigation click handler
function setupHelpNavigation() {
  const navList = document.querySelectorAll('#help-nav a');
  navList.forEach(a => {
    a.onclick = (e) => {
      e.preventDefault();
      const target = a.dataset.target;
```

```
      if (target) loadHelpSection(target);
    };
  });
}

// Live search filter across topics
function setupHelpSearch() {
  const searchInput = document.getElementById('help-search-input');
  const clearBtn = document.getElementById('help-search-clear');
  const topicCount = document.getElementById('help-topic-count');
  if (!searchInput) return;

  const performFilter = () => {
    const query = (searchInput.value || '').trim().toLowerCase();
    const navItems = document.querySelectorAll('#help-nav a');
    let visibleCount = 0;
    let firstVisibleTarget = '';

    if (clearBtn) {
      clearBtn.style.display = query.length > 0 ? 'inline-block' : 'none';
    }

    navItems.forEach(item => {
      const target = item.dataset.target || '';
      const title = item.innerText.toLowerCase();
      const fullText = (window.HELP_CONTENT && window.HELP_CONTENT[target]) ? window.HELP_CONTENT[target].toLowerCase() : '';

      if (!query || title.includes(query) || fullText.includes(query)) {
        item.style.setProperty('display', 'flex', 'important');
```

```
        visibleCount++;
        if (!firstVisibleTarget) firstVisibleTarget = target;
      } else {
        item.style.setProperty('display', 'none', 'important')
;
      }
    });

    if (topicCount) {
      topicCount.textContent = visibleCount;
    }
  };

  searchInput.oninput = performFilter;
  searchInput.onkeydown = (e) => {
    if (e.key === 'Enter') {
      e.preventDefault();
      const firstVisible = document.querySelector('#help-nav a
[style*="display: flex"], #help-nav a:not([style*="display: no
ne"])');
      if (firstVisible && firstVisible.dataset.target) {
        loadHelpSection(firstVisible.dataset.target);
      }
    }
  };

  if (clearBtn) {
    clearBtn.onclick = () => {
      searchInput.value = '';
      performFilter();
      searchInput.focus();
    };
  }
}

function initHelpTab() {
  setupHelpNavigation();
  setupHelpSearch();
```

```
  loadHelpSection('overview');
}

window.initHelpTab = initHelpTab;
window.loadHelpSection = loadHelpSection;

// Initialize when container is present
if (document.getElementById('help-nav')) {
  initHelpTab();
}
```

app.js

```
/**
 * ============================================================
 * Process Name: Windows Helpdesk - App Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля app.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/helpdesk/app.js?v=20261001_v1" type="module"></script>
 *
 * File: app.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/helpdesk
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * AI Windows Admin Helpdesk Management Client
 */

const CONFIG = {
  apiBase: window.location.origin,
  wsProtocol: window.location.protocol === 'https:' ? 'wss:' : 'ws:',
};

let tickets = [];
let activeTicketId = null;
```

```
let activeTicket = null;
let currentFilter = 'all';
let ws = null;
let clientId = 'op_' + Math.random().toString(36).substring(2, 9);

// Elements
const ticketListContainer = document.getElementById('ticketListContainer');
const searchTicketInput = document.getElementById('searchTicketInput');
const noTicketSelected = document.getElementById('noTicketSelected');
const activeTicketView = document.getElementById('activeTicketView');
const messagesStream = document.getElementById('messagesStream');
const replyTextInput = document.getElementById('replyTextInput');
const btnSendReply = document.getElementById('btnSendReply');
const switchInternalNote = document.getElementById('switchInternalNote');
const btnRefreshTickets = document.getElementById('btnRefreshTickets');
const hdTypingIndicator = document.getElementById('hdTypingIndicator');

window.addEventListener('DOMContentLoaded', async () => {
  initWebSocket();
  await loadTickets();
  await updateStats();
  setupEventListeners();
  setInterval(updateStats, 15000);
});

function initWebSocket() {
  const wsUrl = `${CONFIG.wsProtocol}//${window.location.host}/api/helpdesk/ws/${clientId}?role=operator`;
```

```
  ws = new WebSocket(wsUrl);

  ws.onopen = () => console.log('Helpdesk WebSocket connected as operator');
  ws.onmessage = (event) => {
    try {
      const packet = JSON.parse(event.data);
      handleIncomingPacket(packet);
    } catch (e) {
      console.error('Error parsing WS message:', e);
    }
  };
  ws.onclose = () => {
    console.warn('Helpdesk WebSocket closed. Reconnecting in 3s...');
    setTimeout(initWebSocket, 3000);
  };
}

function handleIncomingPacket(packet) {
  const { type, ticket, message, ticket_id, status } = packet;

  if (type === 'new_ticket_alert') {
    // New ticket created
    playNotificationSound();
    loadTickets();
    updateStats();
  } else if (type === 'new_ticket_message') {
    if (activeTicketId === ticket_id) {
      appendMessageToStream(message);
      scrollToBottom();
    }
    loadTickets();
  } else if (type === 'ticket_updated') {
    if (activeTicketId === ticket.id) {
      activeTicket = ticket;
      updateHeaderUI(ticket);
    }
```

```
    loadTickets();
    updateStats();
  }
}

function playNotificationSound() {
  try {
    const audioCtx = new (window.AudioContext || window.webkitAudioContext)();
    const osc = audioCtx.createOscillator();
    const gain = audioCtx.createGain();
    osc.connect(gain);
    gain.connect(audioCtx.destination);
    osc.frequency.setValueAtTime(587.33, audioCtx.currentTime); // D5
    osc.frequency.setValueAtTime(880.00, audioCtx.currentTime + 0.1); // A5
    gain.gain.setValueAtTime(0.2, audioCtx.currentTime);
    gain.gain.exponentialRampToValueAtTime(0.01, audioCtx.currentTime + 0.3);
    osc.start();
    osc.stop(audioCtx.currentTime + 0.3);
  } catch (e) {}
}

async function loadTickets() {
  try {
    let url = `${CONFIG.apiBase}/api/helpdesk/tickets?status=${encodeURIComponent(currentFilter)}`;
    const query = searchTicketInput.value.trim();
    if (query) {
      url += `&search=${encodeURIComponent(query)}`;
    }
    const res = await fetch(url, { credentials: 'include' });
    const data = await res.json();
    if (data.status === 'success') {
      tickets = data.tickets;
      renderTicketsList();
```

```
    }
  } catch (e) {
    console.error('Failed to load tickets:', e);
  }
}

async function updateStats() {
  try {
    const res = await fetch(`${CONFIG.apiBase}/api/helpdesk/st
ats`, { credentials: 'include' });
    const data = await res.json();
    if (data.status === 'success') {
      const s = data.stats;
      document.getElementById('statOpen').innerText = `Open: $
{s.open_tickets}`;
      document.getElementById('statInProgress').innerText = `A
ctive: ${s.in_progress_tickets}`;
      document.getElementById('statResolved').innerText = `Res
olved: ${s.resolved_tickets}`;
    }
  } catch (e) {}
}

function renderTicketsList() {
  ticketListContainer.innerHTML = '';
  if (tickets.length === 0) {
    ticketListContainer.innerHTML = '<div class="text-center t
ext-muted p-4">No tickets matching filter.</div>';
    return;
  }

  tickets.forEach((t) => {
    const card = document.createElement('div');
    card.className = `ticket-card ${activeTicketId === t.id ?
'active' : ''}`;
    card.onclick = () => selectTicket(t.id);

    const priorityBadgeClass = `badge-priority-${t.priority}`;
```

```
    const statusBadge = getStatusBadge(t.status);
    const lastSnippet = t.last_message ? escapeHtml(t.last_mes
sage.content) : 'No messages';
    const timeAgo = formatTimeAgo(t.updated_at);

    card.innerHTML = `
      <div class="d-flex align-items-center justify-content-be
tween mb-1">
        <span class="badge bg-secondary">#${t.ticket_number}</
span>
        <div class="d-flex align-items-center gap-1">
          <span class="badge ${priorityBadgeClass}">${t.priori
ty.toUpperCase()}</span>
          ${statusBadge}
        </div>
      </div>
      <div class="fw-semibold text-truncate text-light mb-1">$
{escapeHtml(t.subject)}</div>
      <div class="small text-muted text-truncate mb-1">${lastS
nippet}</div>
      <div class="d-flex align-items-center justify-content-be
tween text-muted" style="font-size: 0.75rem;">
        <span>  ${escapeHtml(t.user_name)}</span>
        <span>${timeAgo}</span>
      </div>
    `;
    ticketListContainer.appendChild(card);
  });
}

function getStatusBadge(status) {
  switch (status) {
    case 'open':
      return '<span class="badge bg-danger">OPEN</span>';
    case 'in_progress':
      return '<span class="badge bg-warning text-dark">IN PROG
RESS</span>';
    case 'resolved':
```

```
      return '<span class="badge bg-success">RESOLVED</span>';
    case 'closed':
      return '<span class="badge bg-secondary">CLOSED</span>';
    default:
      return `<span class="badge bg-info">${status}</span>`;
  }
}

async function selectTicket(ticketId) {
  activeTicketId = ticketId;
  renderTicketsList();

  if (ws && ws.readyState === WebSocket.OPEN) {
    ws.send(JSON.stringify({ action: 'subscribe_ticket', ticke
t_id: ticketId }));
  }

  try {
    const res = await fetch(`${CONFIG.apiBase}/api/helpdesk/ti
ckets/${ticketId}`, { credentials: 'include' });
    const data = await res.json();
    if (data.status === 'success') {
      activeTicket = data.ticket;
      noTicketSelected.classList.add('d-none');
      activeTicketView.classList.remove('d-none');
      activeTicketView.classList.add('d-flex');

      updateHeaderUI(activeTicket);
      renderMessages(data.messages);
      scrollToBottom();
    }
  } catch (e) {
    console.error('Failed to load ticket details:', e);
  }
}

function updateHeaderUI(t) {
  document.getElementById('ticketNumberBadge').innerText = `#$
```

```
{t.ticket_number}`;
  document.getElementById('ticketSubject').innerText = t.subje
ct;
  document.getElementById('ticketUserMeta').innerText = `From:
 ${t.user_name} (${t.user_email || 'No email'}) • ${formatTime
Ago(t.created_at)}`;
  document.getElementById('currentPriorityLabel').innerText =
t.priority.toUpperCase();
}

function renderMessages(messages) {
  messagesStream.innerHTML = '';
  messages.forEach((msg) => appendMessageToStream(msg));
}

function appendMessageToStream(msg) {
  const el = document.createElement('div');
  let typeClass = 'bubble-user';
  if (msg.is_internal_note) {
    typeClass = 'bubble-note';
  } else if (msg.sender_type === 'operator') {
    typeClass = 'bubble-operator';
  }

  const renderedContent = window.marked ? marked.parse(msg.con
tent) : escapeHtml(msg.content);
  const timeStr = new Date(msg.created_at).toLocaleTimeString(
);

  el.className = `chat-bubble ${typeClass}`;
  el.innerHTML = `
    <div class="d-flex align-items-center justify-content-betw
een gap-2 mb-1" style="font-size: 0.75rem; opacity: 0.85;">
      <strong>${escapeHtml(msg.sender_name)} ${msg.is_internal
_note ? '(Internal Note)' : ''}</strong>
      <span>${timeStr}</span>
    </div>
    <div class="message-content">${renderedContent}</div>
```

```
  `;
  messagesStream.appendChild(el);
}

function scrollToBottom() {
  messagesStream.scrollTop = messagesStream.scrollHeight;
}

async function sendReply() {
  const content = replyTextInput.value.trim();
  if (!content || !activeTicketId) return;

  const isNote = switchInternalNote.checked;
  btnSendReply.disabled = true;

  try {
    const res = await fetch(`${CONFIG.apiBase}/api/helpdesk/ti
ckets/${activeTicketId}/messages`, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      credentials: 'include',
      body: JSON.stringify({
        content: content,
        is_internal_note: isNote,
        sender_type: 'operator'
      })
    });

    if (res.ok) {
      replyTextInput.value = '';
    }
  } catch (e) {
    console.error('Failed to send reply:', e);
  } finally {
    btnSendReply.disabled = false;
  }
}

```

```
async function updateTicketStatus(status) {
  if (!activeTicketId) return;
  try {
    await fetch(`${CONFIG.apiBase}/api/helpdesk/tickets/${activeTicketId}`, {
      method: 'PATCH',
      headers: { 'Content-Type': 'application/json' },
      credentials: 'include',
      body: JSON.stringify({ status: status })
    });
  } catch (e) {
    console.error('Failed to update status:', e);
  }
}

async function updateTicketPriority(priority) {
  if (!activeTicketId) return;
  try {
    await fetch(`${CONFIG.apiBase}/api/helpdesk/tickets/${activeTicketId}`, {
      method: 'PATCH',
      headers: { 'Content-Type': 'application/json' },
      credentials: 'include',
      body: JSON.stringify({ priority: priority })
    });
  } catch (e) {
    console.error('Failed to update priority:', e);
  }
}

function setupEventListeners() {
  btnSendReply.onclick = sendReply;
  replyTextInput.addEventListener('keydown', (e) => {
    if (e.key === 'Enter' && (e.ctrlKey || e.metaKey)) {
      e.preventDefault();
      sendReply();
    }
  });
```

```
328
329   btnRefreshTickets.onclick = () => {
330     loadTickets();
331     updateStats();
332   };
333
334   searchTicketInput.addEventListener('input', () => {
335     loadTickets();
336   });
337
338   document.querySelectorAll('#statusFilterGroup button').forEa
ch((btn) => {
339     btn.onclick = () => {
340       document.querySelectorAll('#statusFilterGroup button').f
orEach((b) => b.classList.remove('active'));
341       btn.classList.add('active');
342       currentFilter = btn.dataset.filter;
343       loadTickets();
344     };
345   });
346
347   document.querySelectorAll('.status-opt').forEach((btn) => {
348     btn.onclick = () => updateTicketStatus(btn.dataset.status)
;
349   });
350
351   document.querySelectorAll('.priority-opt').forEach((item) =>
  {
352     item.onclick = (e) => {
353       e.preventDefault();
354       updateTicketPriority(item.dataset.priority);
355     };
356   });
357 }
358
359 function escapeHtml(str) {
360   if (!str) return '';
361   const div = document.createElement('div');
```

```
  div.innerText = str;
  return div.innerHTML;
}

function formatTimeAgo(dateStr) {
  if (!dateStr) return '';
  const d = new Date(dateStr);
  const now = new Date();
  const diffSec = Math.floor((now - d) / 1000);
  if (diffSec < 60) return 'just now';
  const diffMin = Math.floor(diffSec / 60);
  if (diffMin < 60) return `${diffMin}m ago`;
  const diffHours = Math.floor(diffMin / 60);
  if (diffHours < 24) return `${diffHours}h ago`;
  return d.toLocaleDateString();
}

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/helpdesk/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/helpdesk/index.html" cl
ass="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/helpdesk
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-s
cale=1.0" />
  <title>AI Windows Admin Helpdesk & Support Center</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dis
t/css/bootstrap.min.css" rel="stylesheet" />
```

```
  <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.10.0/font/bootstrap-icons.css" rel="stylesheet" />
  <link rel="stylesheet" href="/helpdesk/style.css" />
</head>
<body>
  <div class="helpdesk-container d-flex flex-column vh-100 overflow-hidden">
    <!-- Top Navbar -->
    <header class="navbar navbar-expand-lg navbar-dark bg-dark border-bottom border-secondary px-3 py-2 flex-shrink-0">
      <div class="container-fluid p-0 d-flex align-items-center justify-content-between">
        <div class="d-flex align-items-center gap-3">
          <a class="navbar-brand d-flex align-items-center gap-2 m-0 fw-bold" href="/helpdesk">
            <span class="fs-4">🎧</span>
            <span>Helpdesk Support Center</span>
          </a>
          <span class="badge bg-primary rounded-pill px-3 py-1 d-none d-md-inline" id="operatorBadge">Operator Mode</span>
        </div>

        <div class="d-flex align-items-center gap-2">
          <!-- Metrics Overview Pills -->
          <div class="d-none d-lg-flex align-items-center gap-2 me-3" id="quickStatsBar">
            <span class="badge bg-danger-subtle text-danger border border-danger-subtle px-2 py-1" id="statOpen">Open: 0</span>
            <span class="badge bg-warning-subtle text-warning border border-warning-subtle px-2 py-1" id="statInProgress">In Progress: 0</span>
            <span class="badge bg-success-subtle text-success border border-success-subtle px-2 py-1" id="statResolved">Resolved: 0</span>
          </div>

          <button class="btn btn-outline-light btn-sm d-flex a
```

```
lign-items-center gap-1" id="btnRefreshTickets" title="Refresh
 Tickets">
            <i class="bi bi-arrow-clockwise"></i>
            <span class="d-none d-sm-inline">Refresh</span>
          </button>
          <a href="/" class="btn btn-secondary btn-sm d-flex a
lign-items-center gap-1" title="Back to Main Assistant">
            <i class="bi bi-house-door"></i>
            <span class="d-none d-sm-inline">Main App</span>
          </a>
        </div>
      </div>
    </header>

    <!-- Main Workspace Layout -->
    <div class="d-flex flex-grow-1 overflow-hidden" style="min
-height: 0;">
      <!-- Tickets Sidebar / Filter List -->
      <aside class="helpdesk-sidebar d-flex flex-column border
-end border-secondary bg-body-tertiary" style="width: 360px; m
in-width: 300px; max-width: 420px;">
        <!-- Filter and Search Header -->
        <div class="p-2 border-bottom border-secondary-subtle"
>
          <div class="input-group input-group-sm mb-2">
            <span class="input-group-text bg-body border-secon
dary-subtle"><i class="bi bi-search"></i></span>
            <input type="text" class="form-control border-seco
ndary-subtle" id="searchTicketInput" placeholder="Search ticke
t, user, #num..." />
          </div>

          <!-- Status Filters -->
          <div class="btn-group btn-group-sm w-100" role="grou
p" id="statusFilterGroup">
            <button type="button" class="btn btn-outline-prima
ry active" data-filter="all">All</button>
            <button type="button" class="btn btn-outline-prima
```

```
ry" data-filter="open">Open</button>
            <button type="button" class="btn btn-outline-prima
ry" data-filter="in_progress">Active</button>
            <button type="button" class="btn btn-outline-prima
ry" data-filter="resolved">Closed</button>
          </div>
        </div>

        <!-- Ticket List Stream -->
        <div class="ticket-list flex-grow-1 overflow-auto p-2
d-flex flex-column gap-2" id="ticketListContainer">
          <div class="text-center text-muted p-4">Loading tick
ets...</div>
        </div>
      </aside>

      <!-- Ticket Conversation and Management View -->
      <main class="helpdesk-main d-flex flex-column flex-grow-
1 overflow-hidden bg-body" id="ticketMainArea">
        <!-- Empty State Placeholder -->
        <div class="d-flex flex-column align-items-center just
ify-content-center h-100 text-muted p-4" id="noTicketSelected"
>
          <i class="bi bi-chat-square-text fs-1 mb-2"></i>
          <h5>Select a ticket to manage and reply</h5>
          <p class="small">All incoming support requests and c
hat threads are coordinated here in real time.</p>
        </div>

        <!-- Ticket Detail Content (Hidden initially) -->
        <div class="d-none flex-column h-100 overflow-hidden"
id="activeTicketView">
          <!-- Top Ticket Action Header -->
          <div class="p-2 px-3 border-bottom border-secondary-
subtle d-flex flex-wrap align-items-center justify-content-bet
ween gap-2 flex-shrink-0 bg-body-tertiary">
            <div class="d-flex align-items-center gap-2">
              <span class="badge bg-secondary fs-6" id="ticket
```

```
NumberBadge">#1001</span>
              <div>
                <h6 class="mb-0 fw-bold" id="ticketSubject">Ti
cket Subject</h6>
                <small class="text-muted" id="ticketUserMeta">
From: User (user@domain.com) • 10m ago</small>
              </div>
            </div>

            <!-- Management Dropdowns & Actions -->
            <div class="d-flex align-items-center gap-2">
              <!-- Priority Selector -->
              <div class="dropdown">
                <button class="btn btn-sm btn-outline-secondar
y dropdown-toggle" type="button" id="btnTicketPriority" data-b
s-toggle="dropdown">
                  Priority: <span id="currentPriorityLabel" cl
ass="fw-bold">Normal</span>
                </button>
                <ul class="dropdown-menu dropdown-menu-end sha
dow">
                  <li><a class="dropdown-item priority-opt" da
ta-priority="low" href="#">  Low</a></li>
                  <li><a class="dropdown-item priority-opt" da
ta-priority="normal" href="#">  Normal</a></li>
                  <li><a class="dropdown-item priority-opt" da
ta-priority="high" href="#">  High</a></li>
                  <li><a class="dropdown-item priority-opt" da
ta-priority="urgent" href="#">  Urgent</a></li>
                </ul>
              </div>

              <!-- Status Action Buttons -->
              <div class="btn-group btn-group-sm">
                <button class="btn btn-outline-warning status-
opt" data-status="in_progress" title="Mark In Progress">
                  <i class="bi bi-play-circle me-1"></i>In Pro
gress
```

```
                </button>
                <button class="btn btn-outline-success status-opt" data-status="resolved" title="Mark Resolved">
                  <i class="bi bi-check2-circle me-1"></i>Resolve
                </button>
                <button class="btn btn-outline-secondary status-opt" data-status="closed" title="Close Ticket">
                  <i class="bi bi-archive me-1"></i>Close
                </button>
              </div>
            </div>
          </div>

          <!-- Message History Stream -->
          <div class="chat-stream flex-grow-1 overflow-auto p-3 d-flex flex-column gap-3" id="messagesStream">
            <!-- Messages rendered dynamically -->
          </div>

          <!-- Typing Indicator -->
          <div class="px-3 py-1 small text-muted d-none" id="hdTypingIndicator">
            <span class="spinner-border spinner-border-sm me-1" role="status"></span>
            <span id="hdTypingText">User is typing...</span>
          </div>

          <!-- Reply and Internal Notes Bar -->
          <div class="p-3 border-top border-secondary-subtle bg-body-tertiary flex-shrink-0">
            <div class="d-flex align-items-center justify-content-between mb-2">
              <div class="form-check form-switch m-0">
                <input class="form-check-input" type="checkbox" role="switch" id="switchInternalNote" />
                <label class="form-check-label small fw-semibold text-warning" for="switchInternalNote">
```

```
                  <i class="bi bi-lock-fill me-1"></i>Internal
 Note (Staff only)
                </label>
              </div>
              <small class="text-muted">Press Ctrl+Enter to se
nd</small>
            </div>

            <div class="input-group">
              <textarea class="form-control" id="replyTextInpu
t" rows="2" placeholder="Write response to user..."></textarea
>
              <button class="btn btn-primary px-4 d-flex align
-items-center justify-content-center" id="btnSendReply">
                <i class="bi bi-send-fill fs-5"></i>
              </button>
            </div>
          </div>
        </div>
      </main>
    </div>
  </div>

  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/di
st/js/bootstrap.bundle.min.js"></script>
  <script src="https://cdn.jsdelivr.net/npm/marked/marked.min.
js"></script>
  <script src="/helpdesk/app.js"></script>
</body>
</html>

```

style.css

```
/**
 * ============================================================
 * Process Name: Windows Web Styles - Style Stylesheet
 * ============================================================
 * Description:
 *   Таблица стилей стиля и оформления интерфейса (style).
 *
 * Usage Examples:
 *   HTML Link Tag:
 *     <link rel="stylesheet" href="/windows/api/webgui/helpdesk/style.css?v=20261001_v1">
 *
 * File: style.css
 * Project: ai-windows-admin
 * Package: windows/api/webgui/helpdesk
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/* ═══════════════════════════════════════════════════════════
   helpdesk/style.css — Дочерние стили Helpdesk (наследует main.css)
   Версия: 20260917_refactor_v1
   ═══════════════════════════════════════════════════════════ */

/* Helpdesk использует переменные из main.css — хардкод убран */

.helpdesk-sidebar {
```

```
  background-color: var(--surface-2) !important;
  border-color: var(--border-color) !important;
}

/* ── Карточка тикета ──────────────────────────────────────── */
.ticket-card {
  padding: 8px 10px;
  border-radius: 6px;
  background: var(--surface-1);
  border: 1px solid var(--border-color);
  cursor: pointer;
  transition: background 0.12s ease, border-color 0.12s ease;
  font-size: 0.82rem;
}

.ticket-card:hover {
  background: var(--nav-hover-bg);
  border-color: var(--nav-active);
}

.ticket-card.active {
  background: rgba(37, 99, 235, 0.08);
  border-color: var(--nav-active);
  box-shadow: 0 0 0 1px var(--nav-active);
}

[data-theme="dark"] .ticket-card.active {
  background: rgba(88, 166, 255, 0.08);
}

/* ── Пузыри чата ──────────────────────────────────────────── */
.chat-bubble {
  max-width: 80%;
  padding: 7px 12px;
  border-radius: 10px;
  position: relative;
```

```
  word-break: break-word;
  font-size: 0.83rem;
  line-height: 1.5;
}

.bubble-user {
  align-self: flex-start;
  background: var(--chat-bot-bg);
  border: 1px solid var(--border-color);
  color: var(--chat-bot-text);
}

.bubble-operator {
  align-self: flex-end;
  background: var(--chat-user-bg);
  color: var(--chat-user-text);
}

.bubble-note {
  align-self: center;
  width: 90%;
  background: rgba(245, 158, 11, 0.08);
  border: 1px dashed #f59e0b;
  color: #92400e;
  font-size: 0.8rem;
}

[data-theme="dark"] .bubble-note {
  background: rgba(245, 158, 11, 0.1);
  color: #fbbf24;
}

/* ── Приоритеты ──────────────────────────────────────────────── */
.badge-priority-urgent { background-color: #dc2626; color: #fff; }
.badge-priority-high   { background-color: #ea580c; color: #fff; }
```

```
.badge-priority-normal { background-color: var(--nav-active); color: #fff; }
.badge-priority-low    { background-color: #10b981; color: #fff; }

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/helpdesk_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/helpdesk_tab/index.html
" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/helpdesk_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Helpdesk Multi-Tool Management Panel Tab -->
<div class="container-fluid py-2 h-100 d-flex flex-column" sty
le="min-height: calc(100vh - 120px);">
<!-- Header Alert & Mode Switcher Bar -->
<div class="alert alert-dark d-flex flex-wrap align-items-cent
er justify-content-between gap-2 mb-3 shadow-sm border-seconda
ry flex-shrink-0">
<div class="d-flex align-items-center gap-2">
<span class="fs-4">□ </span>
<div>
```

```
<h5 class="mb-0 fw-bold">Центр управления Helpdesk &amp;amp; Служба поддержки</h5>
<p class="mb-0 small text-muted">Единый пульт оператора: управление тикетами, живой чат поддержки, быстрые шаблоны, база знаний и SLA метрики</p>
</div>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<button class="btn btn-outline-info btn-sm d-flex align-items-center gap-1" id="hd-tab-btn-refresh" onclick="window.refreshHelpdeskTab &amp;amp;&amp;amp; window.refreshHelpdeskTab()">
<i class="bi bi-arrow-clockwise"></i> Обновить заявки
      </button>
<button class="btn btn-primary btn-sm d-flex align-items-center gap-1" id="hd-tab-btn-new-ticket" onclick="window.openCreateTicketModal &amp;amp;&amp;amp; window.openCreateTicketModal()">
<i class="bi bi-plus-circle"></i> Создать обращение
      </button>
<button class="btn btn-outline-light btn-sm d-flex align-items-center gap-1" onclick="switchToTab('admin')">
<i class="bi bi-arrow-left-circle"></i> Панель админки
      </button>
</div>
</div>
<!-- Quick KPI & Status Metrics -->
<div class="row g-2 mb-3 flex-shrink-0" id="hd-tab-metrics-bar">
<div class="col-md-3 col-6">
<div class="card shadow-sm border-secondary-subtle p-2 bg-body-tertiary" data-panel-id="panel-helpdesk-0" id="panel-helpdesk-0">
<div class="d-flex align-items-center justify-content-between">
<div>
<div class="text-muted small text-uppercase fw-semibold">Открытые заявки</div>
<h4 class="mb-0 fw-bold text-danger" id="hd-metric-open">0</h4>
```

```
</div>
<i class="bi bi-exclamation-octagon fs-2 text-danger opacity-7
5"></i>
</div>
</div>
</div>
<div class="col-md-3 col-6">
<div class="card shadow-sm border-secondary-subtle p-2 bg-body
-tertiary" data-panel-id="panel-helpdesk-0" id="panel-helpdesk
-0">
<div class="d-flex align-items-center justify-content-between"
>
<div>
<div class="text-muted small text-uppercase fw-semibold">В обр
аботке</div>
<h4 class="mb-0 fw-bold text-warning" id="hd-metric-progress">
0</h4>
</div>
<i class="bi bi-gear-wide-connected fs-2 text-warning opacity-
75"></i>
</div>
</div>
</div>
<div class="col-md-3 col-6">
<div class="card shadow-sm border-secondary-subtle p-2 bg-body
-tertiary" data-panel-id="panel-helpdesk-0" id="panel-helpdesk
-0">
<div class="d-flex align-items-center justify-content-between"
>
<div>
<div class="text-muted small text-uppercase fw-semibold">Решен
о сегодня</div>
<h4 class="mb-0 fw-bold text-success" id="hd-metric-resolved">
0</h4>
</div>
<i class="bi bi-check2-all fs-2 text-success opacity-75"></i>
</div>
</div>
```

```
</div>
<div class="col-md-3 col-6">
<div class="card shadow-sm border-secondary-subtle p-2 bg-body-tertiary" data-panel-id="panel-helpdesk-0" id="panel-helpdesk-0">
<div class="d-flex align-items-center justify-content-between">
<div>
<div class="text-muted small text-uppercase fw-semibold">Всего в базе</div>
<h4 class="mb-0 fw-bold text-info" id="hd-metric-total">0</h4>
</div>
<i class="bi bi-collection fs-2 text-info opacity-75"></i>
</div>
</div>
</div>
</div>
<!-- Helpdesk Main Workspace (Two Columns: Ticket Stream + Management Workbench) -->
<div class="card shadow-sm border-secondary-subtle flex-grow-1 overflow-hidden d-flex flex-column" data-panel-id="panel-helpdesk-vyberite-tiket-dlya-raboty" id="panel-helpdesk-vyberite-tiket-dlya-raboty" style="min-height: 520px;">
<div class="row g-0 flex-grow-1 h-100 overflow-hidden">
<!-- Left Column: Tickets List & Filter Tools -->
<div class="col-lg-4 col-md-5 border-end border-secondary-subtle d-flex flex-column h-100 bg-body-tertiary">
<!-- Search & Filter Header -->
<div class="p-2 border-bottom border-secondary-subtle">
<div class="input-group input-group-sm mb-2">
<span class="input-group-text bg-body border-secondary-subtle"><i class="bi bi-search"></i></span>
<input class="form-control border-secondary-subtle" id="hd-tab-search-input" placeholder="Поиск по номеру, теме, автору..." type="text"/>
</div>
<div class="d-flex gap-1" id="hd-tab-status-filters" role="group">
```

```
<button class="btn btn-sm btn-outline-primary active flex-fill
" data-filter="all" type="button">Все</button>
<button class="btn btn-sm btn-outline-primary flex-fill" data-
filter="open" type="button">Новые</button>
<button class="btn btn-sm btn-outline-primary flex-fill" data-
filter="in_progress" type="button">В работе</button>
<button class="btn btn-sm btn-outline-primary flex-fill" data-
filter="resolved" type="button">Закрыты</button>
</div>
</div>
<!-- Scrollable Ticket List Stream -->
<div class="flex-grow-1 overflow-auto p-2 d-flex flex-column g
ap-2" id="hd-tab-ticket-list" style="min-height: 0;">
<div class="text-center text-muted p-4">
<div class="spinner-border spinner-border-sm text-primary mb-2
" role="status"></div>
<div>Загрузка тикетов...</div>
</div>
</div>
</div>
<!-- Right Column: Selected Ticket Workbench & Multi-tool Area
 -->
<div class="col-lg-8 col-md-7 d-flex flex-column h-100 bg-body
" id="hd-tab-workbench">
<!-- Empty State (No ticket chosen) -->
<div class="d-flex flex-column align-items-center justify-cont
ent-center h-100 text-muted p-4 text-center" id="hd-tab-empty-
state">
<i class="bi bi-headset fs-1 mb-2 text-primary opacity-50"></i
>
<h5 class="fw-bold">Выберите тикет для работы</h5>
<p class="small text-muted mb-0" style="max-width: 450px;">
            Здесь отображается полная история обращений, перепи
ска с пользователями, внутренние заметки операторов и инструм
енты быстрого реагирования.
          </p>
</div>
<!-- Active Ticket Detail & Tools View (Shown when ticket is a
```

```
ctive) -->
<div class="d-none flex-column h-100 overflow-hidden" id="hd-tab-active-view">
<!-- Ticket Header Bar -->
<div class="p-2 px-3 border-bottom border-secondary-subtle d-flex flex-wrap align-items-center justify-content-between gap-2 flex-shrink-0 bg-body-tertiary">
<div class="d-flex align-items-center gap-2 min-w-0">
<span class="badge bg-secondary fs-6" id="hd-active-number">#0000</span>
<div class="min-w-0">
<h6 class="mb-0 fw-bold text-truncate" id="hd-active-subject">Тема заявки</h6>
<small class="text-muted" id="hd-active-meta">Автор: Загрузка...</small>
</div>
</div>
<!-- Priority & Status Actions -->
<div class="d-flex align-items-center gap-2 flex-wrap">
<div class="dropdown">
<button class="btn btn-sm btn-outline-secondary dropdown-toggle" data-bs-toggle="dropdown" id="hd-btn-priority-dropdown" type="button">
                    Приоритет: <span class="fw-bold" id="hd-active-priority-label">Обычный</span>
</button>
<ul class="dropdown-menu dropdown-menu-end shadow">
<li><a class="dropdown-item hd-priority-opt" data-priority="low" href="#">  Низкий</a></li>
<li><a class="dropdown-item hd-priority-opt" data-priority="normal" href="#">  Обычный</a></li>
<li><a class="dropdown-item hd-priority-opt" data-priority="high" href="#">  Высокий</a></li>
<li><a class="dropdown-item hd-priority-opt" data-priority="urgent" href="#">  Срочный</a></li>
</ul>
</div>
<div class="btn-group btn-group-sm">
```

```
<button class="btn btn-outline-warning hd-status-opt" data-status="in_progress" title="Взять в работу">
<i class="bi bi-play-circle me-1"></i>В работу
                </button>
<button class="btn btn-outline-success hd-status-opt" data-status="resolved" title="Отметить решенным">
<i class="bi bi-check2-circle me-1"></i>Решено
                </button>
<button class="btn btn-outline-secondary hd-status-opt" data-status="closed" title="Закрыть обращение">
<i class="bi bi-archive me-1"></i>Закрыть
                </button>
</div>
</div>
</div>
<!-- Multi-Tool Quick Toolbar (Canned replies, AI Assistant summary, internal note) -->
<div class="p-2 border-bottom border-secondary-subtle bg-body-tertiary d-flex align-items-center justify-content-between gap-2 flex-shrink-0 flex-wrap">
<div class="d-flex align-items-center gap-1 flex-wrap">
<span class="small fw-semibold text-muted me-1"><i class="bi bi-lightning-charge-fill text-warning"></i> Быстрые ответы:</span>
<button class="btn btn-xs btn-outline-secondary rounded-pill py-0 px-2 hd-quick-template" data-tpl="Здравствуйте! Ваше обращение принято в работу, уточняем детали.">
                 Принято
              </button>
<button class="btn btn-xs btn-outline-secondary rounded-pill py-0 px-2 hd-quick-template" data-tpl="Проблема устранена. Пожалуйста, проверьте работу и подтвердите решение.">
                 Устранено
              </button>
<button class="btn btn-xs btn-outline-secondary rounded-pill py-0 px-2 hd-quick-template" data-tpl="Уточните, пожалуйста, версию ОС и приложите снимок экрана с ошибкой.">
                 Запрос инфо
```

```
              </button>
<button class="btn btn-xs btn-outline-info rounded-pill py-0 p
x-2" id="hd-btn-ai-assist" title="Сгенерировать черновик ответ
а через ИИ">
                 ИИ-Черновик
              </button>
</div>
<div class="form-check form-switch m-0">
<input class="form-check-input" id="hd-switch-internal-note" r
ole="switch" type="checkbox"/>
<label class="form-check-label small fw-semibold text-warning"
 for="hd-switch-internal-note">
<i class="bi bi-lock-fill me-1"></i>Скрытая заметка (Staff)
              </label>
</div>
</div>
<!-- Messages Stream -->
<div class="flex-grow-1 overflow-auto p-3 d-flex flex-column g
ap-3 bg-body" id="hd-tab-messages-stream" style="min-height: 0
;">
<!-- Messages rendered dynamically -->
</div>
<!-- Reply Composer -->
<div class="p-2 px-3 border-top border-secondary-subtle bg-bod
y-tertiary flex-shrink-0">
<div class="input-group">
<textarea class="form-control" id="hd-reply-input" placeholder
="Напишите ответ пользователю (или скрытую заметку)..." rows="
2"></textarea>
<button class="btn btn-primary px-4 d-flex align-items-center
justify-content-center" id="hd-btn-send-reply">
<i class="bi bi-send-fill fs-5"></i>
</button>
</div>
<div class="d-flex justify-content-between align-items-center
mt-1 text-muted small" style="font-size: 0.75rem;">
<span>Ctrl + Enter для быстрой отправки</span>
<span class="badge bg-secondary-subtle text-body" id="hd-compo
```

```
ser-mode-badge">Режим: Публичный ответ</span>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- Modal: Create New Ticket -->
<div aria-hidden="true" aria-labelledby="hdCreateTicketModalLa
bel" class="modal fade" id="hdCreateTicketModal" tabindex="-1"
>
<div class="modal-dialog modal-dialog-centered">
<div class="modal-content shadow border-secondary-subtle">
<div class="modal-header border-secondary-subtle">
<h5 class="modal-title d-flex align-items-center gap-2" id="hd
CreateTicketModalLabel">
<i class="bi bi-plus-circle text-primary"></i>
<span>Создать новое обращение в Helpdesk</span>
</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-3">
<div class="mb-3">
<label class="form-label small fw-semibold">Тема обращения</la
bel>
<input class="form-control" id="hd-new-ticket-subject" placeho
lder="Краткая тема проблемы..." type="text"/>
</div>
<div class="row g-2 mb-3">
<div class="col-6">
<label class="form-label small fw-semibold">Категория</label>
<select class="form-select form-select-sm" id="hd-new-ticket-c
ategory">
<option value="technical">  Техническая</option>
<option value="ai">  ИИ и Модели</option>
<option value="account">  Аккаунт &amp;amp; Доступ</option>
```

```
<option value="general">  Общие вопросы</option>
</select>
</div>
<div class="col-6">
<label class="form-label small fw-semibold">Приоритет</label>
<select class="form-select form-select-sm" id="hd-new-ticket-p
riority">
<option value="low">  Низкий</option>
<option selected="" value="normal">  Обычный</option>
<option value="high">  Высокий</option>
<option value="urgent">  Срочный</option>
</select>
</div>
</div>
<div class="mb-2">
<label class="form-label small fw-semibold">Описание проблемы
/ Текст сообщения</label>
<textarea class="form-control" id="hd-new-ticket-message" plac
eholder="Подробное описание обращения..." rows="4"></textarea>
</div>
</div>
<div class="modal-footer border-secondary-subtle py-2">
<button class="btn btn-secondary btn-sm" data-bs-dismiss="moda
l" type="button">Отмена</button>
<button class="btn btn-primary btn-sm" id="hd-btn-submit-new-t
icket" type="button">Создать тикет</button>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Helpdesk Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/helpdesk_tab/main.js?v
=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/helpdesk_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

// Helpdesk Tab — Admin Control Panel Logic
'use strict';

(function() {
  let hdTickets = [];
  let hdActiveTicketId = null;
  let hdActiveTicket = null;
  let hdCurrentFilter = 'all';
  let hdWs = null;
  let hdClientId = 'admin_tab_' + Math.random().toString(36).s
ubstring(2, 9);
  let hdStatsInterval = null;
```

```
32
33  async function initHelpdeskTab() {
34    console.log('[HelpdeskTab] Initializing Helpdesk multi-too
l tab...');
35    setupEventListeners();
36    initWebSocket();
37    await loadTickets();
38    await updateStats();
39
40    if (window.registerTabPoller) {
41      window.registerTabPoller('tab-helpdesk', updateStats, 15
000, { immediate: false });
42    } else if (!hdStatsInterval) {
43      hdStatsInterval = setInterval(updateStats, 15000);
44    }
45  }
46
47  function initWebSocket() {
48    if (hdWs && hdWs.readyState === WebSocket.OPEN) return;
49    const protocol = window.location.protocol === 'https:' ? '
wss:' : 'ws:';
50    const wsUrl = `${protocol}//${window.location.host}/api/he
lpdesk/ws/${hdClientId}?role=operator`;
51
52    try {
53      hdWs = new WebSocket(wsUrl);
54      hdWs.onopen = () => console.log('[HelpdeskTab] WS Connec
ted as Admin Operator');
55      hdWs.onmessage = (event) => {
56        try {
57          const packet = JSON.parse(event.data);
58          handleIncomingPacket(packet);
59        } catch (e) {
60          console.error('[HelpdeskTab] WS Parse error:', e);
61        }
62      };
63      hdWs.onclose = () => {
64        console.warn('[HelpdeskTab] WS Disconnected, reconnect
```

```
ing in 5s...');
        setTimeout(initWebSocket, 5000);
      };
    } catch (e) {
      console.warn('[HelpdeskTab] WS Init failed:', e);
    }
  }

  function handleIncomingPacket(packet) {
    const { type, ticket, message, ticket_id } = packet;

    if (type === 'new_ticket_alert') {
      if (typeof showNotification === 'function') {
        showNotification(`🔔 Новое обращение #${ticket?.ticket
_number}: ${ticket?.subject}`, 'warning');
      }
      loadTickets();
      updateStats();
    } else if (type === 'new_ticket_message') {
      if (hdActiveTicketId === ticket_id) {
        appendMessage(message);
        scrollMessagesToBottom();
      }
      loadTickets();
    } else if (type === 'ticket_updated') {
      if (hdActiveTicketId === ticket?.id) {
        hdActiveTicket = ticket;
        renderActiveTicketHeader(ticket);
      }
      loadTickets();
      updateStats();
    }
  }

  async function loadTickets() {
    const listContainer = document.getElementById('hd-tab-tick
et-list');
    if (!listContainer) return;
```

```
100
101     try {
102       let url = `/api/helpdesk/tickets?status=${encodeURICompo
nent(hdCurrentFilter)}`;
103       const searchInput = document.getElementById('hd-tab-sear
ch-input');
104       const query = searchInput?.value?.trim();
105       if (query) {
106         url += `&search=${encodeURIComponent(query)}`;
107       }
108
109       const res = await fetch(url, { credentials: 'include' })
;
110       if (!res.ok) throw new Error(`HTTP ${res.status}`);
111       const data = await res.json();
112       hdTickets = data.tickets || [];
113       renderTicketList(hdTickets);
114     } catch (err) {
115       console.error('[HelpdeskTab] Error loading tickets:', er
r);
116       listContainer.innerHTML = `<div class="p-3 text-danger s
mall">Ошибка загрузки тикетов: ${err.message}</div>`;
117     }
118   }
119
120   async function updateStats() {
121     try {
122       const res = await fetch('/api/helpdesk/stats', { credent
ials: 'include' });
123       if (!res.ok) return;
124       const data = await res.json();
125       const stats = data.stats || {};
126
127       const elOpen = document.getElementById('hd-metric-open')
;
128       const elProg = document.getElementById('hd-metric-progre
ss');
129       const elRes = document.getElementById('hd-metric-resolve
```

```
d');
130       const elTot = document.getElementById('hd-metric-total')
;
131
132       if (elOpen) elOpen.textContent = stats.open || 0;
133       if (elProg) elProg.textContent = stats.in_progress || 0;
134       if (elRes) elRes.textContent = stats.resolved || 0;
135       if (elTot) elTot.textContent = stats.total || 0;
136     } catch (e) {
137       console.debug('[HelpdeskTab] Stats update skipped:', e);
138     }
139   }
140
141   function renderTicketList(ticketsList) {
142     const listContainer = document.getElementById('hd-tab-tick
et-list');
143     if (!listContainer) return;
144
145     if (!ticketsList || ticketsList.length === 0) {
146       listContainer.innerHTML = '<div class="text-center text-
muted p-4 small">Нет обращений по выбранному фильтру</div>';
147       return;
148     }
149
150     listContainer.innerHTML = ticketsList.map(t => {
151       const isActive = t.id === hdActiveTicketId;
152       const statusBadge = getStatusBadge(t.status);
153       const priorityBadge = getPriorityBadge(t.priority);
154       const timeStr = formatTimeAgo(t.created_at);
155
156       return `
157         <div class="card p-2 border-secondary-subtle ${isActiv
e ? 'bg-primary-subtle border-primary' : 'bg-body'} hover-shad
ow"
158              style="cursor: pointer; transition: all 0.15s eas
e;"
159              onclick="window.selectHelpdeskTicket('${t.id}')">
160           <div class="d-flex justify-content-between align-ite
```

```
ms-center mb-1">
            <span class="badge bg-secondary-subtle text-body f
w-bold">#${t.ticket_number}</span>
            <div class="d-flex gap-1">
              ${priorityBadge}
              ${statusBadge}
            </div>
          </div>
          <h6 class="fw-bold mb-1 text-truncate" style="font-s
ize: 0.9rem;">${escapeHtml(t.subject)}</h6>
          <div class="d-flex justify-content-between align-ite
ms-center small text-muted" style="font-size: 0.75rem;">
            <span class="text-truncate" style="max-width: 140p
x;"><i class="bi bi-person me-1"></i>${escapeHtml(t.user_name
|| t.user_email || 'Пользователь')}</span>
            <span>${timeStr}</span>
          </div>
        </div>
      `;
    }).join('');
  }

  async function selectTicket(ticketId) {
    hdActiveTicketId = ticketId;
    renderTicketList(hdTickets); // re-render for active highl
ight

    const emptyState = document.getElementById('hd-tab-empty-s
tate');
    const activeView = document.getElementById('hd-tab-active-
view');
    if (emptyState) emptyState.classList.add('d-none');
    if (activeView) {
      activeView.classList.remove('d-none');
      activeView.classList.add('d-flex');
    }

    try {
```

```
      const res = await fetch(`/api/helpdesk/tickets/${ticketId}`, { credentials: 'include' });
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      hdActiveTicket = data.ticket;
      renderActiveTicketHeader(data.ticket);
      renderMessagesStream(data.messages || []);
      scrollMessagesToBottom();
    } catch (err) {
      console.error('[HelpdeskTab] Error loading ticket details:', err);
      if (typeof showNotification === 'function') showNotification('Ошибка загрузки тикета: ' + err.message, 'danger');
    }
  }

  function renderActiveTicketHeader(ticket) {
    if (!ticket) return;
    const numEl = document.getElementById('hd-active-number');
    const subEl = document.getElementById('hd-active-subject');
    const metaEl = document.getElementById('hd-active-meta');
    const prioLabel = document.getElementById('hd-active-priority-label');

    if (numEl) numEl.textContent = `#${ticket.ticket_number}`;
    if (subEl) subEl.textContent = ticket.subject;
    if (metaEl) {
      const author = ticket.user_name || ticket.user_email || 'User';
      metaEl.textContent = `Автор: ${author} (${ticket.category || 'technical'}) • ${formatTimeAgo(ticket.created_at)}`;
    }
    if (prioLabel) {
      const pLabels = { low: '  Низкий', normal: '  Обычный', high: '  Высокий', urgent: '  Срочный' };
      prioLabel.textContent = pLabels[ticket.priority] || ticket.priority;
```

```
    }
  }

  function renderMessagesStream(messages) {
    const stream = document.getElementById('hd-tab-messages-st
ream');
    if (!stream) return;

    if (!messages || messages.length === 0) {
      stream.innerHTML = '<div class="text-center text-muted p
-4 small">В данном тикете пока нет сообщений</div>';
      return;
    }

    stream.innerHTML = messages.map(m => createMessageHtml(m))
.join('');
  }

  function createMessageHtml(m) {
    const isStaff = m.sender_role === 'operator' || m.sender_r
ole === 'admin';
    const isNote = Boolean(m.is_internal_note);
    const timeStr = new Date(m.created_at).toLocaleTimeString(
[], { hour: '2-digit', minute: '2-digit' });

    let cardBg = isStaff ? 'bg-primary-subtle border-primary-s
ubtle' : 'bg-body-tertiary border-secondary-subtle';
    let align = isStaff ? 'align-self-end' : 'align-self-start
';

    if (isNote) {
      cardBg = 'bg-warning-subtle border-warning';
    }

    let parsedText = escapeHtml(m.message);
    if (window.marked && typeof window.marked.parse === 'funct
ion') {
      try {
```

```
        parsedText = window.marked.parse(m.message);
      } catch (e) {}
    }

    return `
      <div class="d-flex flex-column ${align}" style="max-width: 80%;">
        <div class="card p-2 ${cardBg} shadow-sm">
          <div class="d-flex justify-content-between align-items-center mb-1 gap-2 small">
            <strong>${isNote ? '  Заметка оператора (' + escapeHtml(m.sender_name) + ')' : escapeHtml(m.sender_name || (isStaff ? 'Оператор' : 'Пользователь'))}</strong>
            <span class="text-muted" style="font-size: 0.75rem;">${timeStr}</span>
          </div>
          <div class="message-content text-break" style="font-size: 0.88rem;">${parsedText}</div>
        </div>
      </div>
    `;
  }

  function appendMessage(m) {
    const stream = document.getElementById('hd-tab-messages-stream');
    if (!stream) return;
    const msgDiv = document.createElement('div');
    msgDiv.innerHTML = createMessageHtml(m);
    stream.appendChild(msgDiv.firstElementChild);
  }

  function scrollMessagesToBottom() {
    const stream = document.getElementById('hd-tab-messages-stream');
    if (stream) {
      setTimeout(() => {
        stream.scrollTop = stream.scrollHeight;
```

```
      }, 50);
    }
  }

  async function sendReply() {
    if (!hdActiveTicketId) return;
    const replyInput = document.getElementById('hd-reply-input
');
    const isInternalNote = document.getElementById('hd-switch-
internal-note')?.checked || false;
    const text = replyInput?.value?.trim();

    if (!text) return;

    try {
      const res = await fetch(`/api/helpdesk/tickets/${hdActiv
eTicketId}/messages`, {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        credentials: 'include',
        body: JSON.stringify({
          message: text,
          is_internal_note: isInternalNote
        })
      });

      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      replyInput.value = '';
      const data = await res.json();
      if (data.message) {
        appendMessage(data.message);
        scrollMessagesToBottom();
      }
      loadTickets();
    } catch (err) {
      console.error('[HelpdeskTab] Error sending message:', er
r);
      if (typeof showNotification === 'function') showNotifica
```

```
tion('Ошибка отправки: ' + err.message, 'danger');
    }
  }

  async function updateTicketStatus(status) {
    if (!hdActiveTicketId) return;
    try {
      const res = await fetch(`/api/helpdesk/tickets/${hdActiv
eTicketId}/status`, {
        method: 'PUT',
        headers: { 'Content-Type': 'application/json' },
        credentials: 'include',
        body: JSON.stringify({ status })
      });
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      if (typeof showNotification === 'function') showNotifica
tion(`Статус изменен на: ${status}`, 'success');
      loadTickets();
      updateStats();
      selectTicket(hdActiveTicketId);
    } catch (err) {
      console.error('[HelpdeskTab] Status update failed:', err
);
    }
  }

  async function updateTicketPriority(priority) {
    if (!hdActiveTicketId) return;
    try {
      const res = await fetch(`/api/helpdesk/tickets/${hdActiv
eTicketId}/priority`, {
        method: 'PUT',
        headers: { 'Content-Type': 'application/json' },
        credentials: 'include',
        body: JSON.stringify({ priority })
      });
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      if (typeof showNotification === 'function') showNotifica
```

```
tion(`Приоритет изменен на: ${priority}`, 'success');
      loadTickets();
      selectTicket(hdActiveTicketId);
    } catch (err) {
      console.error('[HelpdeskTab] Priority update failed:', e
rr);
    }
  }

  async function generateAiDraft() {
    if (!hdActiveTicket) return;
    const replyInput = document.getElementById('hd-reply-input
');
    if (!replyInput) return;

    replyInput.value = '□ Генерация черновика ответа ИИ...';
    try {
      const res = await fetch('/api/chat', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        credentials: 'include',
        body: JSON.stringify({
          prompt: `Сформулируй вежливый, профессиональный и ко
нкретный ответ технической поддержки Helpdesk на следующее обр
ащение пользователя.
Тема: "${hdActiveTicket.subject}"
Описание: "${hdActiveTicket.description || ''}"
Категория: "${hdActiveTicket.category}"
Ответ должен быть кратким и готовым к отправке клиенту.`,
          use_rag: false
        })
      });

      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      const answer = data.response || data.text || '';
      replyInput.value = answer.trim();
    } catch (e) {
```

```
      replyInput.value = `Здравствуйте! Мы изучили ваше обращение по теме "${hdActiveTicket.subject}". Специалисты уже занимаются решением.`;
    }
  }

  function setupEventListeners() {
    // Search input debounce
    const searchInput = document.getElementById('hd-tab-search-input');
    let searchTimer = null;
    if (searchInput) {
      searchInput.oninput = () => {
        clearTimeout(searchTimer);
        searchTimer = setTimeout(loadTickets, 300);
      };
    }

    // Filter buttons
    const filterBtns = document.querySelectorAll('#hd-tab-status-filters button');
    filterBtns.forEach(btn => {
      btn.onclick = () => {
        filterBtns.forEach(b => b.classList.remove('active'));
        btn.classList.add('active');
        hdCurrentFilter = btn.getAttribute('data-filter') || 'all';
        loadTickets();
      };
    });

    // Send reply button & shortcut
    const btnSend = document.getElementById('hd-btn-send-reply');
    if (btnSend) btnSend.onclick = sendReply;

    const replyInput = document.getElementById('hd-reply-input');
```

```
    if (replyInput) {
      replyInput.onkeydown = (e) => {
        if (e.ctrlKey && e.key === 'Enter') {
          e.preventDefault();
          sendReply();
        }
      };
    }

    // Internal note switcher toggle label
    const switchNote = document.getElementById('hd-switch-inte
rnal-note');
    const modeBadge = document.getElementById('hd-composer-mod
e-badge');
    if (switchNote && modeBadge) {
      switchNote.onchange = () => {
        if (switchNote.checked) {
          modeBadge.textContent = '  Режим: Внутренняя скрыта
я заметка';
          modeBadge.className = 'badge bg-warning text-dark';
        } else {
          modeBadge.textContent = 'Режим: Публичный ответ';
          modeBadge.className = 'badge bg-secondary-subtle tex
t-body';
        }
      };
    }

    // Status action buttons
    document.querySelectorAll('.hd-status-opt').forEach(btn =>
 {
      btn.onclick = () => {
        const st = btn.getAttribute('data-status');
        if (st) updateTicketStatus(st);
      };
    });

    // Priority dropdown items
```

```
    document.querySelectorAll('.hd-priority-opt').forEach(opt
=> {
      opt.onclick = (e) => {
        e.preventDefault();
        const prio = opt.getAttribute('data-priority');
        if (prio) updateTicketPriority(prio);
      };
    });

    // Quick canned responses
    document.querySelectorAll('.hd-quick-template').forEach(tp
lBtn => {
      tplBtn.onclick = () => {
        const tpl = tplBtn.getAttribute('data-tpl');
        if (tpl && replyInput) {
          replyInput.value = tpl;
          replyInput.focus();
        }
      };
    });

    // AI Assist Draft Button
    const aiAssistBtn = document.getElementById('hd-btn-ai-ass
ist');
    if (aiAssistBtn) aiAssistBtn.onclick = generateAiDraft;

    // Submit new ticket in modal
    const btnSubmitNew = document.getElementById('hd-btn-submi
t-new-ticket');
    if (btnSubmitNew) {
      btnSubmitNew.onclick = async () => {
        const subject = document.getElementById('hd-new-ticket
-subject')?.value?.trim();
        const category = document.getElementById('hd-new-ticke
t-category')?.value || 'technical';
        const priority = document.getElementById('hd-new-ticke
t-priority')?.value || 'normal';
        const message = document.getElementById('hd-new-ticket
```

```
-message')?.value?.trim();

        if (!subject || !message) {
          window.showToast?.('Пожалуйста, заполните тему и текст обращения.', 'warning') || alert('Пожалуйста, заполните тему и текст обращения.');
          return;
        }

        try {
          btnSubmitNew.disabled = true;
          const res = await fetch('/api/helpdesk/tickets', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            credentials: 'include',
            body: JSON.stringify({ subject, category, priority, message })
          });

          if (!res.ok) throw new Error(`HTTP ${res.status}`);
          const data = await res.json();
          if (data.status === 'success') {
            window.showToast?.(`Тикет #${data.ticket.ticket_number} успешно создан!`, 'success');
            const modalEl = document.getElementById('hdCreateTicketModal');
            const modal = bootstrap.Modal.getInstance(modalEl);
            modal?.hide();

            document.getElementById('hd-new-ticket-subject').value = '';
            document.getElementById('hd-new-ticket-message').value = '';

            await loadTickets();
            await updateStats();
            if (data.ticket?.id) {
```

```
              selectTicket(data.ticket.id);
            }
          }
        } catch (err) {
          console.error('[HelpdeskTab] Create ticket failed:', err);
          window.showToast?.('Ошибка создания тикета: ' + err.message, 'danger') || alert('Ошибка создания тикета: ' + err.message);
        } finally {
          btnSubmitNew.disabled = false;
        }
      };
    }
  }

  function getStatusBadge(status) {
    switch (status) {
      case 'open': return '<span class="badge bg-danger-subtle text-danger">Новый</span>';
      case 'in_progress': return '<span class="badge bg-warning-subtle text-warning">В работе</span>';
      case 'resolved': return '<span class="badge bg-success-subtle text-success">Решено</span>';
      case 'closed': return '<span class="badge bg-secondary-subtle text-body">Закрыт</span>';
      default: return `<span class="badge bg-secondary">${status}</span>`;
    }
  }

  function getPriorityBadge(priority) {
    switch (priority) {
      case 'urgent': return '<span class="badge bg-danger">🔥 Срочно</span>';
      case 'high': return '<span class="badge bg-warning text-dark">⚠ Высокий</span>';
      case 'low': return '<span class="badge bg-secondary-subt
```

```
le text-muted">⬇ Низкий</span>';
      default: return '<span class="badge bg-info-subtle text-info">➖ Обычный</span>';
    }
  }

  function formatTimeAgo(isoStr) {
    if (!isoStr) return '';
    try {
      const diffMs = Date.now() - new Date(isoStr).getTime();
      const diffMin = Math.floor(diffMs / 60000);
      if (diffMin < 1) return 'только что';
      if (diffMin < 60) return `${diffMin} мин назад`;
      const diffHours = Math.floor(diffMin / 60);
      if (diffHours < 24) return `${diffHours} ч назад`;
      return `${Math.floor(diffHours / 24)} дн назад`;
    } catch (e) {
      return '';
    }
  }

  function escapeHtml(text) {
    if (!text) return '';
    const div = document.createElement('div');
    div.textContent = text;
    return div.innerHTML;
  }

  // Global window functions for event hooks
  window.initHelpdeskTab = initHelpdeskTab;
  window.refreshHelpdeskTab = () => { loadTickets(); updateStats(); };
  window.selectHelpdeskTicket = selectTicket;
  window.openCreateTicketModal = () => {
    const modalEl = document.getElementById('hdCreateTicketModal');
    if (modalEl) {
      const modal = new bootstrap.Modal(modalEl);
```

```
      modal.show();
    }
  };
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/index.html" class="w-10
0 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-12 08:10:00
================================================================
===============
-->
<!DOCTYPE html>

<html lang="ru">
<head>
<meta charset="utf-8"/>
<meta content="width=device-width, initial-scale=1.0" name="vi
ewport"/>
<title>AI Assistant</title>
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/
css/bootstrap.min.css" rel="stylesheet"/>
```

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.10.0/font/bootstrap-icons.css" rel="stylesheet"/>
<link href="/html/css/main.css?v=20261008_v1" rel="stylesheet" />
<script src="/html/js/toast.js?v=20261008_v1"></script>
<script src="/html/js/panel-loader.js?v=20261010_v1" type="module"></script>
<script src="https://cdn.jsdelivr.net/npm/i18next@23.11.5/i18next.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/marked/marked.min.js"></script>
</head>
<body>
<div class="container-fluid mt-1">
<div class="d-flex align-items-center justify-content-between mb-1 main-nav-container rounded px-2 py-1">
<!-- Левая часть: кнопка меню, брендинг и индикатор активного раздела -->
<div class="d-flex align-items-center gap-2">
<!-- Кнопка открытия левого выдвижного меню -->
<button aria-controls="leftSideNavOffcanvas" class="btn btn-sm  btn-primary d-flex align-items-center gap-1.5 px-2 py-1 rounded shadow-sm" data-bs-target="#leftSideNavOffcanvas" data-bs-toggle="offcanvas" id="left-drawer-toggle-btn" style="font-size: 0.82rem; font-weight: 500;" title="Открыть меню разделов" type="button">
<i class="bi bi-list fs-6"></i>
<span class="d-none d-md-inline">Меню</span>
</button>
<!-- Брендинг -->
<span class="fw-bold d-flex align-items-center gap-1.5 text-primary fs-6 user-select-none" style="letter-spacing: 0.3px;">
<i class="bi bi-cpu-fill fs-5"></i>
<span class="d-none d-sm-inline">AI Windows Admin</span>
</span>
</div>
<!-- Правая часть панели: статусы, профиль и настройки -->
<div class="d-flex align-items-center gap-2 flex-shrink-0 ms-a
```

```
uto" id="nav-auth-container">
<!-- Active Model & Search Badges -->
<span class="badge rounded-pill align-items-center gap-1 px-3 py-1 shadow-sm border" id="chat-model-badge" style="font-size: 0.8rem; display: none; cursor: pointer;" title="Текущая модель ИИ">
        Загрузка...
      </span>
<!-- Model Quick Test Dropdown Trigger & Popover -->
<div class="dropdown" id="model-ping-dropdown-wrapper">
<button aria-expanded="false" class="btn btn-sm btn-outline-secondary d-flex align-items-center justify-content-center p-0 rounded-circle" data-bs-auto-close="outside" data-bs-toggle="dropdown" id="btn-model-ping-gear" style="width: 28px; height: 28px;" title="Проверка доступности модели" type="button">
<i class="bi bi-gear"></i>
</button>
<div aria-labelledby="btn-model-ping-gear" class="dropdown-menu dropdown-menu-end shadow-lg border border-secondary-subtle p-3" style="min-width: 360px; max-width: 460px; z-index: 1060;">
<div class="d-flex align-items-center justify-content-between mb-2">
<span class="fw-bold small d-flex align-items-center gap-1.5 text-body">
<i class="bi bi-cpu text-primary"></i>
<span>Проверка доступности модели</span>
</span>
</div>
<div class="mb-2">
<input class="form-control form-control-sm" id="model-ping-input" placeholder="Введите сообщение..." type="text" value="Проверка доступности модели"/>
</div>
<div class="d-flex justify-content-end mb-2">
<button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5 px-3" id="btn-model-ping-send" type="button">
<i class="bi bi-send-fill" id="model-ping-btn-icon"></i>
```

```
<span id="model-ping-btn-text">Отправить</span>
</button>
</div>
<div id="model-ping-response-container" style="display: none;"
>
<label class="form-label small text-muted mb-1 fw-semibold">От
вет модели:</label>
<div class="small p-2 rounded bg-body-tertiary border text-bre
ak font-monospace" id="model-ping-response" style="max-height:
 220px; overflow-y: auto; white-space: pre-wrap; font-size: 0.
8rem; user-select: text;"></div>
</div>
</div>
</div>
<span class="badge rounded-pill align-items-center gap-1 px-3
py-1 shadow-sm border" id="chat-search-badge" style="font-size
: 0.8rem; display: none;" title="Провайдер поиска">
</span>
<!-- Google OAuth Register / Sign-in Button (Shown when not lo
gged in) -->
<button class="btn btn-sm btn-light text-dark fw-semibold d-fl
ex align-items-center gap-2 px-3 py-1 shadow-sm rounded-pill b
tn-google-login" data-i18n-title="auth.googleSignInTitle" id="
nav-google-btn" style="border: 1px solid rgba(0,0,0,0.15);" ti
tle="Войти или зарегистрироваться через Google">
<svg height="18" viewbox="0 0 24 24" width="18"><path d="M22.5
6 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h5.92c-.26 1.37-1.04 2.5
3-2.21 3.31v2.77h3.57c2.08-1.92 3.28-4.74 3.28-8.09z" fill="#4
285F4"></path><path d="M12 23c2.97 0 5.46-.98 7.28-2.66l-3.57-
2.77c-.98.66-2.23 1.06-3.71 1.06-2.86 0-5.29-1.93-6.16-4.53H2.
18v2.84C3.99 20.53 7.7 23 12 23z" fill="#34A853"></path><path
d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s.13-1.43.35-2.09V7.0
6H2.18C1.43 8.55 1 10.22 1 12s.43 3.45 1.18 4.94l2.85-2.22.81-
.63z" fill="#FBBC05"></path><path d="M12 5.38c1.62 0 3.06.56 4
.21 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 1 3.99 3.47 2.1
8 7.06l3.66 2.84c.87-2.6 3.3-4.52 6.16-4.52z" fill="#EA4335"><
/path></svg>
<span data-i18n="auth.googleSignUp">Регистрация через Google</
```

```
span>
</button>
<!-- Authenticated User Menu Dropdown (Shown when logged in) -->
<div class="dropdown d-none" id="user-menu-dropdown-wrapper">
<button aria-expanded="false" class="btn btn-sm btn-outline-secondary d-flex align-items-center gap-2 px-2 py-1 rounded-pill dropdown-toggle" data-bs-toggle="dropdown" id="user-settings-btn" title="Пользовательское меню" type="button">
<img class="rounded-circle d-none" height="22" id="nav-user-avatar" src="" style="object-fit: cover;" width="22"/>
<i class="bi bi-person-circle" id="nav-user-icon"></i>
<span class="small fw-semibold" data-i18n="userSettings.button" id="nav-user-label">Пользователь</span>
</button>
<ul aria-labelledby="user-settings-btn" class="dropdown-menu dropdown-menu-end shadow border-secondary-subtle py-1" style="min-width: 220px;">
<li class="px-3 py-2 border-bottom border-secondary-subtle">
<div class="fw-bold text-truncate" id="user-menu-display-name">Пользователь</div>
<div class="small text-muted text-truncate" id="user-menu-display-email" style="font-size: 0.75rem;">user@gmail.com</div>
</li>
<li><h6 class="dropdown-header small text-uppercase text-muted fw-semibold mt-1">Сервисы</h6></li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 py-1" id="user-menu-btn-mail" type="button">
<i class="bi bi-envelope-check text-primary"></i>
<span>Почта</span>
</button>
</li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 py-1" id="user-menu-btn-docs" type="button">
<i class="bi bi-file-earmark-text text-info"></i>
<span>Google Docs</span>
```

```
</button>
</li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1" id="user-menu-btn-sheets" type="button">
<i class="bi bi-file-earmark-spreadsheet text-success"></i>
<span>Google Таблицы</span>
</button>
</li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1" id="user-menu-btn-news" type="button">
<i class="bi bi-newspaper text-warning"></i>
<span>Новости</span>
</button>
</li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1" id="user-menu-btn-facebook" type="button">
<i class="bi bi-facebook text-primary"></i>
<span>Facebook</span>
</button>
</li>
<li><hr class="dropdown-divider my-1"/></li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1" data-bs-target="#cacheManagerModal" data-bs-toggle="modal
" id="user-menu-btn-cache" type="button">
<i class="bi bi-hdd-stack text-info"></i>
<span data-i18n="cacheManager.menuItem">Кеш браузера</span>
</button>
</li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1" data-bs-target="#userSettingsModal" data-bs-toggle="modal
" type="button">
<i class="bi bi-gear-fill text-secondary"></i>
<span data-i18n="userSettings.title">Настройки профиля</span>
```

```
</button>
</li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1 text-danger btn-user-logout" type="button">
<i class="bi bi-box-arrow-right"></i>
<span>Выйти</span>
</button>
</li>
</ul>
</div>
<!-- Кнопка кеша браузера -->
<button class="btn btn-sm btn-outline-secondary d-flex align-i
tems-center justify-content-center rounded-circle" data-bs-tar
get="#cacheManagerModal" data-bs-toggle="modal" data-i18n-titl
e="cacheManager.title" id="btn-open-cache-manager" style="widt
h: 28px; height: 28px; padding: 0;" title="Кеш браузера">
<i class="bi bi-hdd-stack"></i>
</button>
<!-- Quick Settings Gear Button -->
<button class="btn btn-sm btn-outline-secondary d-flex align-i
tems-center justify-content-center rounded-circle" data-bs-tar
get="#userSettingsModal" data-bs-toggle="modal" data-i18n-titl
e="userSettings.title" id="user-settings-gear-btn" style="widt
h: 28px; height: 28px; padding: 0;" title="Настройки">
<i class="bi bi-gear-fill"></i>
</button>
</div>
</div>
<!-- Плавающий боковой ярлычок (вкладка) слева -->
<button aria-controls="leftSideNavOffcanvas" class="left-drawe
r-tab-trigger btn shadow" data-bs-target="#leftSideNavOffcanva
s" data-bs-toggle="offcanvas" title="Открыть меню навигации" t
ype="button">
<i class="bi bi-list fs-6"></i>
<span class="left-drawer-tab-text">МЕНЮ</span>
</button>
<!-- Левая выдвижная вкладка / панель навигации (Offcanvas) --
```

```
>
<div aria-labelledby="leftSideNavOffcanvasLabel" class="offcanvas offcanvas-start left-drawer-nav" id="leftSideNavOffcanvas" tabindex="-1">
<div class="offcanvas-header border-bottom py-2 px-3">
<h5 class="offcanvas-title d-flex align-items-center gap-2 fs-6 fw-bold" id="leftSideNavOffcanvasLabel">
<i class="bi bi-grid-fill text-primary"></i>
<span>Навигация и разделы</span>
</h5>
<button aria-label="Закрыть" class="btn-close text-reset" data-bs-dismiss="offcanvas" type="button"></button>
</div>
<div class="offcanvas-body p-3 d-flex flex-column justify-content-between" id="mainTabs">
<div class="drawer-sections-wrapper">
<!-- Группа: Стандартные и Чат -->
<div class="nav-drawer-group mb-3">
<div class="nav-drawer-group-title d-flex align-items-center gap-1 mb-2">
<i class="bi bi-chat-dots-fill text-primary"></i>
<span data-i18n="tabs.groupStandard">Стандартные и Чат</span>
</div>
<div class="list-group list-group-flush rounded border">
<button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3 active" data-bs-target="#tab-chat" data-tab="tab-chat" type="button">
<span class="fs-6">□ </span>
<span class="fw-medium" data-i18n="tabs.chat">Чат</span>
</button>
<button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-bs-target="#tab-voice" data-tab="tab-voice" type="button">
<span class="fs-6">□ </span>
<span class="fw-medium" data-i18n="tabs.voice">Голос и диаризация</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
```

```
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-scenar
ios" data-tab="tab-scenarios" type="button">
<span class="fs-6">□ </span>
<span class="fw-medium" data-i18n="tabs.scenarios">Сценарии и
чат</span>
</button>
</div>
</div>
<!-- Группа: Оборудование и Телеметрия -->
<div class="nav-drawer-group mb-3">
<div class="nav-drawer-group-title d-flex align-items-center g
ap-1 mb-2">
<i class="bi bi-cpu-fill text-info"></i>
<span>Оборудование и Телеметрия</span>
</div>
<div class="list-group list-group-flush rounded border">
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-about-
system" data-tab="tab-about-system" type="button">
<span class="fs-6">i</span>
<span class="fw-medium" data-i18n="tabs.aboutSystem">О Системе
</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-hardwa
re-monitor" data-tab="tab-hardware-monitor" type="button">
<span class="fs-6">□ </span>
<span class="fw-medium" data-i18n="tabs.sensorLogs">Датчики и
сенсоры (LHM)</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-teleme
try-research" data-tab="tab-telemetry-research" type="button">
<span class="fs-6">□ </span>
<span class="fw-medium">Исследования телеметрии</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-teleme
```

```
try-config" data-tab="tab-telemetry-config" type="button">
<span class="fs-6">□ </span>
<span class="fw-medium">Конфигурация телеметрии</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-ai-ben
chmark" data-tab="tab-ai-benchmark" type="button">
<span class="fs-6">□ </span>
<span class="fw-medium">AI Benchmark</span>
</button>
</div>
</div>
<!-- Группа: Дисковая подсистема и накопители -->
<div class="nav-drawer-group mb-3">
<div class="nav-drawer-group-title d-flex align-items-center g
ap-1 mb-2">
<i class="bi bi-hdd-fill text-warning"></i>
<span>Дисковая подсистема</span>
</div>
<div class="list-group list-group-flush rounded border">
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-storag
e-manager" data-tab="tab-storage-manager" type="button">
<span class="fs-6">□ </span>
<span class="fw-medium">Диски и тома (Storage)</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-disk-s
peed" data-tab="tab-disk-speed" type="button">
<span class="fs-6">□ </span>
<span class="fw-medium">Скорость дисков (DiskSpd)</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-storag
e-wear" data-tab="tab-storage-wear" type="button">
<span class="fs-6">□ </span>
<span class="fw-medium">Износ дисков и батареи</span>
</button>
```

```
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-file-r
ecovery" data-tab="tab-file-recovery" type="button">
<span class="fs-6">🔄</span>
<span class="fw-medium" data-i18n="tabs.fileRecovery">Восстано
вление файлов</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-file-h
istory-search" data-tab="tab-file-history-search" type="button
">
<span class="fs-6">🔍</span>
<span class="fw-medium" data-i18n="tabs.fileHistorySearch">Пои
ск истории файлов AI</span>
</button>
</div>
</div>
<!-- Группа: Диагностика и форензика -->
<div class="nav-drawer-group mb-3">
<div class="nav-drawer-group-title d-flex align-items-center g
ap-1 mb-2">
<i class="bi bi-shield-check text-success"></i>
<span>Диагностика и форензика</span>
</div>
<div class="list-group list-group-flush rounded border">
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-proces
s-leaks" data-tab="tab-process-leaks" type="button">
<span class="fs-6">🧪</span>
<span class="fw-medium">1. Утечки и микрофризы</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-forens
ics" data-tab="tab-forensics" type="button">
<span class="fs-6">🕵</span>
<span class="fw-medium">2. Поведенческая форензика</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
```

```
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-thrott
ling" data-tab="tab-throttling" type="button">
<span class="fs-6">□ </span>
<span class="fw-medium">3. Качество ядра и Троттлинг</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-periph
erals" data-tab="tab-peripherals" type="button">
<span class="fs-6">□ </span>
<span class="fw-medium">4. Сеть и периферия</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-softwa
re-transparency" data-tab="tab-software-transparency" type="bu
tton">
<span class="fs-6">□ </span>
<span class="fw-medium" data-i18n="tabs.softwareTransparency">
5. Прозрачность ПО</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-softwa
re-scanner" data-tab="tab-software-scanner" type="button">
<span class="fs-6">□ </span>
<span class="fw-medium">6. Глубокий аудит ПО (Dormant)</span>
</button>
</div>
</div>
<!-- Группа: Управление Windows и Sysadmin -->
<div class="nav-drawer-group mb-3">
<div class="nav-drawer-group-title d-flex align-items-center g
ap-1 mb-2">
<i class="bi bi-gear-wide-connected text-primary"></i>
<span>Управление Windows</span>
</div>
<div class="list-group list-group-flush rounded border">
<button class="list-group-item list-group-item-action d-flex a
lign-items-center justify-content-between py-2 px-3 text-warni
ng-emphasis" id="btn-open-god-mode" onclick="openGodModeFolder
```

```
()" title="Открыть панель всех настроек Windows (God Mode)" ty
pe="button">
<div class="d-flex align-items-center gap-2">
<span class="fs-6">🔧</span>
<span class="fw-medium">God Mode (Все настройки)</span>
</div>
<i class="bi bi-box-arrow-up-right text-muted small"></i>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-system
-control" data-tab="tab-system-control" type="button">
<span class="fs-6">📊</span>
<span class="fw-medium">System Logs &amp;amp; Activity</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-post-i
nstall-wizard" data-tab="tab-post-install-wizard" type="button
">
<span class="fs-6">🚀</span>
<span class="fw-medium">Post-Install Wizard</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-mainte
nance-recovery" data-tab="tab-maintenance-recovery" type="butt
on">
<span class="fs-6">🛠️</span>
<span class="fw-medium">Maintenance &amp;amp; Recovery</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-window
s-admin" data-tab="tab-windows-admin" type="button">
<span class="fs-6">🪟</span>
<span class="fw-medium">Windows Sysadmin</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-focus-
settings" data-tab="tab-focus-settings" type="button">
<span class="fs-6">🎯</span>
```

```
<span class="fw-medium">Настройки Focus</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-startu
p-auditor" data-tab="tab-startup-auditor" type="button">
<span class="fs-6">🚀</span>
<span class="fw-medium" data-i18n="tabs.startupAuditor">Автоза
пуск программ</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-servic
es-manager" data-tab="tab-services-manager" type="button">
<span class="fs-6">⚙️</span>
<span class="fw-medium">Службы Windows</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-task-s
cheduler" data-tab="tab-task-scheduler" type="button">
<span class="fs-6">📅</span>
<span class="fw-medium">Планировщик задач</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-proces
s-manager" data-tab="tab-process-manager" type="button">
<span class="fs-6">📊</span>
<span class="fw-medium">Диспетчер процессов</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-regist
ry-viewer" data-tab="tab-registry-viewer" type="button">
<span class="fs-6">🗝️</span>
<span class="fw-medium">Редактор реестра</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-softwa
re-manager" data-tab="tab-software-manager" type="button">
<span class="fs-6">📦</span>
<span class="fw-medium">Пакетный менеджер Winget</span>
```

```
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-user-d
irectories" data-tab="tab-user-directories" type="button">
<span class="fs-6">🗂 </span>
<span class="fw-medium" data-i18n="tabs.userDirectories">Дирек
тории пользователей</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-accoun
ts-identity" data-tab="tab-accounts-identity" type="button">
<span class="fs-6">👤 </span>
<span class="fw-medium">Учетные записи (Users &amp;amp; Identity)<
/span>
</button>
</div>
</div>
<!-- Группа: Безопасность и Целостность -->
<div class="nav-drawer-group mb-3">
<div class="nav-drawer-group-title d-flex align-items-center g
ap-1 mb-2">
<i class="bi bi-shield-lock-fill text-danger"></i>
<span>Безопасность и Обслуживание</span>
</div>
<div class="list-group list-group-flush rounded border">
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-defend
er" data-tab="tab-defender" type="button">
<span class="fs-6">🛡 </span>
<span class="fw-medium">Defender Security</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-firewa
ll-manager" data-tab="tab-firewall-manager" type="button">
<span class="fs-6">🧱 </span>
<span class="fw-medium">Брандмауэр Windows</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
```

```
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-securi
ty-acl" data-tab="tab-security-acl" type="button">
<span class="fs-6">🔐</span>
<span class="fw-medium">Права доступа и ACL</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-servic
ing-integrity" data-tab="tab-servicing-integrity" type="button
">
<span class="fs-6">🛠️</span>
<span class="fw-medium">Обслуживание SFC/DISM</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-boot-r
ecovery" data-tab="tab-boot-recovery" type="button">
<span class="fs-6">🚀</span>
<span class="fw-medium">Загрузка и WinRE</span>
</button>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-window
s-backup" data-tab="tab-windows-backup" type="button">
<span class="fs-6">💾</span>
<span class="fw-medium">Windows Backup</span>
</button>
</div>
</div>
<!-- Группа: Сеть и Логи -->
<div class="nav-drawer-group mb-3">
<div class="nav-drawer-group-title d-flex align-items-center g
ap-1 mb-2">
<i class="bi bi-hdd-network-fill text-secondary"></i>
<span>Сеть и Журналы</span>
</div>
<div class="list-group list-group-flush rounded border">
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-networ
k" data-tab="tab-network" type="button">
<span class="fs-6">🌐</span>
```

```
<span class="fw-medium" data-i18n="tabs.network">Сетевой терминал</span>
</button>
<button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-bs-target="#tab-performance-tracing" data-tab="tab-performance-tracing" type="button">
<span class="fs-6">▯ </span>
<span class="fw-medium">Трассировка ETW (logman)</span>
</button>
<button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-bs-target="#tab-system-logs" data-tab="tab-system-logs" type="button">
<span class="fs-6">▯ </span>
<span class="fw-medium" data-i18n="tabs.systemLogs">Логи операционной системы</span>
</button>
<button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-bs-target="#tab-log-settings" data-tab="tab-log-settings" type="button">
<span class="fs-6">▯ </span>
<span class="fw-medium">Параметры и настройка журналов</span>
</button>
<button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-bs-target="#tab-power-lifecycle" data-tab="tab-power-lifecycle" type="button">
<span class="fs-6">▯ </span>
<span class="fw-medium">Сессии питания и перезагрузки</span>
</button>
<button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-bs-target="#tab-app-logs" data-tab="tab-app-logs" type="button">
<span class="fs-6">▯ </span>
<span class="fw-medium">Внутренние логи программы (%APPDATA%)</span>
</button>
</div>
</div>
```

```
<!-- Группа: Расширенные и ИИ -->
<div class="nav-drawer-group mb-3">
<div class="nav-drawer-group-title d-flex align-items-center gap-1 mb-2">
<i class="bi bi-sliders2 text-info"></i>
<span data-i18n="tabs.groupExtended">Расширенные и ИИ</span>
</div>
<div class="list-group list-group-flush rounded border">
<button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-bs-target="#tab-rag" data-tab="tab-rag" type="button">
<span class="fs-6">🔍</span>
<span class="fw-medium" data-i18n="tabs.rag">RAG</span>
</button>
<button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-bs-target="#tab-models" data-tab="tab-models" type="button">
<span class="fs-6">🤖</span>
<span class="fw-medium" data-i18n="tabs.models">Модели и API</span>
</button>
<button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-bs-target="#tab-news" data-tab="tab-news" type="button">
<span class="fs-6">📰</span>
<span class="fw-medium" data-i18n="tabs.news">Новости</span>
</button>
<button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-bs-target="#tab-admin" data-tab="tab-admin" type="button">
<span class="fs-6">🛠</span>
<span class="fw-medium" data-i18n="tabs.admin">Управление</span>
</button>
<button class="list-group-item list-group-item-action d-flex align-items-center gap-2 py-2 px-3" data-bs-target="#tab-help" data-tab="tab-help" type="button">
<span class="fs-6">❓</span>
```

```
<span class="fw-medium" data-i18n="tabs.help">Справочник</span
>
</button>
</div>
</div>
<!-- Группа: Плагины -->
<div class="nav-drawer-group mb-3" id="pluginsNavDropdownWrapp
er">
<div class="nav-drawer-group-title d-flex align-items-center g
ap-1 mb-2">
<i class="bi bi-puzzle-fill text-warning"></i>
<span data-i18n="tabs.plugins">Плагины</span>
</div>
<div class="list-group list-group-flush rounded border" id="na
v-plugins-dropdown-menu">
<div class="nav-enabled-plugins-list" id="nav-enabled-plugins-
list">
<!-- Dynamic enabled plugins will be injected here -->
</div>
<button class="list-group-item list-group-item-action d-flex a
lign-items-center gap-2 py-2 px-3" data-bs-target="#tab-plugin
s" data-tab="tab-plugins" type="button">
<span class="fs-6">□ </span>
<span class="fw-medium" data-i18n="tabs.managePlugins">Управле
ние плагинами</span>
</button>
</div>
</div>
</div>
<!-- Footer Drawer -->
<div class="nav-drawer-footer border-top pt-2 mt-auto d-flex a
lign-items-center justify-content-between text-muted small">
<span class="fw-semibold">AI Windows Admin</span>
<span>v3.1</span>
</div>
</div>
</div>
<div class="tab-content" id="mainTabContent">
```

```
<!-- Tabs will be injected via JavaScript -->
<div class="tab-pane fade show active" id="tab-chat"></div>
<div class="tab-pane fade" id="tab-voice"></div>
<div class="tab-pane fade" id="tab-scenarios"></div>
<div class="tab-pane fade" id="tab-about-system"></div>
<div class="tab-pane fade" id="tab-hardware-monitor"></div>
<div class="tab-pane fade" id="tab-telemetry-research"></div>
<div class="tab-pane fade" id="tab-telemetry-config"></div>
<div class="tab-pane fade" id="tab-ai-benchmark"></div>
<div class="tab-pane fade" id="tab-storage-manager"></div>
<div class="tab-pane fade" id="tab-disk-speed"></div>
<div class="tab-pane fade" id="tab-storage-wear"></div>
<div class="tab-pane fade" id="tab-file-recovery"></div>
<div class="tab-pane fade" id="tab-file-history-search"></div>
<div class="tab-pane fade" id="tab-process-leaks"></div>
<div class="tab-pane fade" id="tab-forensics"></div>
<div class="tab-pane fade" id="tab-throttling"></div>
<div class="tab-pane fade" id="tab-peripherals"></div>
<div class="tab-pane fade" id="tab-software-transparency"></di
v>
<div class="tab-pane fade" id="tab-system-control"></div>
<div class="tab-pane fade" id="tab-post-install-wizard"></div>
<div class="tab-pane fade" id="tab-maintenance-recovery"></div
>
<div class="tab-pane fade" id="tab-windows-admin"></div>
<div class="tab-pane fade" id="tab-focus-settings"></div>
<div class="tab-pane fade" id="tab-startup-auditor"></div>
<div class="tab-pane fade" id="tab-services-manager"></div>
<div class="tab-pane fade" id="tab-task-scheduler"></div>
<div class="tab-pane fade" id="tab-process-manager"></div>
<div class="tab-pane fade" id="tab-registry-viewer"></div>
<div class="tab-pane fade" id="tab-software-manager"></div>
<div class="tab-pane fade" id="tab-software-scanner"></div>
<div class="tab-pane fade" id="tab-user-directories"></div>
<div class="tab-pane fade" id="tab-defender"></div>
<div class="tab-pane fade" id="tab-firewall-manager"></div>
<div class="tab-pane fade" id="tab-security-acl"></div>
<div class="tab-pane fade" id="tab-servicing-integrity"></div>
```

```
<div class="tab-pane fade" id="tab-boot-recovery"></div>
<div class="tab-pane fade" id="tab-windows-backup"></div>
<div class="tab-pane fade" id="tab-network"></div>
<div class="tab-pane fade" id="tab-performance-tracing"></div>
<div class="tab-pane fade" id="tab-system-logs"></div>
<div class="tab-pane fade" id="tab-log-settings"></div>
<div class="tab-pane fade" id="tab-power-lifecycle"></div>
<div class="tab-pane fade" id="tab-app-logs"></div>
<div class="tab-pane fade" id="tab-rag"></div>
<div class="tab-pane fade" id="tab-telegram-rag"></div>
<div class="tab-pane fade" id="tab-models"></div>
<div class="tab-pane fade" id="tab-news"></div>
<div class="tab-pane fade" id="tab-plugins"></div>
<div class="tab-pane fade" id="tab-admin"></div>
<div class="tab-pane fade" id="tab-accounts-identity"></div>
<div class="tab-pane fade" id="tab-help"></div>
</div>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
<link href="/html/ai_modal_dialog/style.css?v=20261012_v2" rel="stylesheet"/>
<script src="/html/ai_modal_dialog/main.js?v=20261008_v1"></script>
<script src="/html/js/chatService.js?v=20261008_v2"></script>
<script src="/html/js/window-api.js?v=20261008_v3"></script>
<script src="/html/js/main.js?v=20261010_v4" type="module"></script>
<!-- HELP Modal -->
<div aria-hidden="true" aria-labelledby="help-modal-label" class="modal fade modal-help" id="help-modal" tabindex="-1">
<div class="modal-dialog modal-dialog-scrollable">
<div class="modal-content">
<div class="modal-header">
<h5 class="modal-title" data-i18n="helpModal.title" id="help-modal-label">Помощь</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="modal" type="button"></button>
```

```
</div>
<div class="modal-body" id="help-modal-content">
<p data-i18n="common.loading">Загрузка...</p>
</div>
<div class="modal-footer">
<button class="btn btn-secondary" data-bs-dismiss="modal" data
-i18n="helpModal.close" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
<!-- User Settings Modal -->
<div aria-hidden="true" aria-labelledby="userSettingsModalLabe
l" class="modal fade" id="userSettingsModal" tabindex="-1">
<div class="modal-dialog modal-dialog-centered">
<div class="modal-content shadow border-secondary-subtle">
<div class="modal-header border-secondary-subtle">
<h5 class="modal-title d-flex align-items-center gap-2" id="us
erSettingsModalLabel">
<i class="bi bi-person-gear text-primary"></i>
<span data-i18n="userSettings.title">Пользовательские настройк
и</span>
</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-3">
<!-- User Account & Google OAuth Card -->
<div class="card bg-body-tertiary border-secondary-subtle mb-3
 p-3" data-panel-id="panel-webgui-1" id="user-settings-auth-ca
rd">
<div class="d-flex align-items-center gap-3">
<img class="rounded-circle border border-2 border-primary" hei
ght="48" id="user-settings-avatar" src="" style="object-fit: c
over;" width="48"/>
<div class="flex-grow-1 min-w-0">
<div class="d-flex align-items-center gap-2">
<strong class="text-truncate" id="user-settings-name">Пользова
```

```
тель</strong>
<span class="badge bg-primary" id="user-settings-role-badge">User</span>
</div>
<div class="small text-muted text-truncate" id="user-settings-email">email@gmail.com</div>
<div class="small mt-1 text-muted" id="user-settings-google-badge">
<i class="bi bi-google text-danger me-1"></i>Google OAuth
              </div>
</div>
<button class="btn btn-outline-danger btn-sm btn-user-logout" title="Выйти" type="button">
<i class="bi bi-box-arrow-right"></i>
</button>
</div>
</div>
<!-- Guest / Connect Google OAuth Card -->
<div class="card bg-body-tertiary border-secondary-subtle mb-3 p-3 text-center d-none" data-panel-id="panel-webgui-2" id="user-settings-guest-card">
<div class="text-muted small mb-2">
<i class="bi bi-shield-lock me-1"></i> Авторизация и синхронизация аккаунта
          </div>
<button class="btn btn-outline-danger btn-sm w-100 d-flex align-items-center justify-content-center gap-2 btn-google-login" type="button">
<i class="bi bi-google"></i>
<span>Войти через Google (OAuth)</span>
</button>
<div class="small text-muted mt-2" style="font-size: 0.75rem;">
            Синхронизация с Google Drive, Docs, Calendar и Contacts
          </div>
</div>
<!-- Language Selection -->
```

```
<div class="mb-3">
<label class="form-label fw-semibold small d-flex align-items-center gap-2" for="modal-lang-selector">
<i class="bi bi-translate text-info"></i>
<span data-i18n="userSettings.language">Язык интерфейса</span>
</label>
<select class="form-select lang-selector" id="modal-lang-selector">
<option value="ru-RU">Русский (ru-RU)</option>
<option value="en-US">English (en-US)</option>
<option value="he-IL">□ □ □ □ □ (he-IL)</option>
<option value="uk-UA">Українська (uk-UA)</option>
</select>
</div>
<!-- Theme Selection -->
<div class="mb-2">
<label class="form-label fw-semibold small d-flex align-items-center gap-2" for="modal-theme-selector">
<i class="bi bi-palette text-warning"></i>
<span data-i18n="userSettings.theme">Тема оформления</span>
</label>
<select class="form-select theme-selector" id="modal-theme-selector">
<option data-i18n="userSettings.themeSystem" value="system">□ Системная</option>
<optgroup label="□ Светлые темы">
<option data-i18n="userSettings.themeLight" value="light">□ Светлая (Классическая)</option>
<option data-i18n="userSettings.themeBrick" value="brick">□ Кирпичная (Терракота)</option>
</optgroup>
<optgroup label="□ Тёмные темы">
<option data-i18n="userSettings.themeDark" value="dark">□ Тёмная (Контрастная)</option>
<option data-i18n="userSettings.themeTerminal" value="terminal">□ Терминал (Matrix / CRT)</option>
</optgroup>
</select>
```

```
</div>
</div>
<!-- Cache Manager Modal -->
<div aria-hidden="true" aria-labelledby="cacheManagerModalLabel" class="modal fade" id="cacheManagerModal" tabindex="-1">
<div class="modal-dialog modal-dialog-centered modal-lg">
<div class="modal-content shadow border-secondary-subtle">
<div class="modal-header border-secondary-subtle">
<div class="d-flex align-items-center gap-2">
<div class="rounded-circle bg-primary-subtle text-primary p-2 d-flex align-items-center justify-content-center" style="width: 36px; height: 36px;">
<i class="bi bi-hdd-stack fs-5"></i>
</div>
<div>
<h5 class="modal-title mb-0" data-i18n="cacheManager.title" id="cacheManagerModalLabel">Управление браузерным кешем</h5>
<div class="small text-muted" data-i18n="cacheManager.subtitle" style="font-size: 0.78rem;">Хранилище больших данных (IndexedDB + Memory Cache) для ускорения работы и офлайн-доступа.</div>
</div>
</div>
<button aria-label="Close" class="btn-close" data-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body p-3">
<!-- Storage Quota Bar Card -->
<div class="card bg-body-tertiary border-secondary-subtle p-3 mb-3" data-panel-id="panel-webgui-zagruzka" id="panel-webgui-zagruzka">
<div class="d-flex align-items-center justify-content-between mb-2">
<div class="fw-semibold small d-flex align-items-center gap-1.5">
<i class="bi bi-pie-chart-fill text-primary"></i>
<span data-i18n="cacheManager.quotaTitle">Дисковая квота браузера</span>
```

```
</div>
<span class="badge bg-secondary-subtle text-secondary border" id="cache-persist-badge">Проверка...</span>
</div>
<div class="progress mb-2" style="height: 10px;">
<div class="progress-bar bg-primary" id="cache-usage-progressbar" role="progressbar" style="width: 0%"></div>
</div>
<div class="d-flex justify-content-between align-items-center small text-muted" style="font-size: 0.78rem;">
<span>Использование памяти:</span>
<span class="fw-semibold text-body" id="cache-usage-text">Загрузка...</span>
</div>
</div>
<!-- Stores List -->
<div class="mb-3">
<div class="d-flex align-items-center justify-content-between mb-2">
<h6 class="fw-bold small text-uppercase text-muted mb-0" data-i18n="cacheManager.storesTitle">Хранилища данных</h6>
<button class="btn btn-sm btn-outline-primary py-0 px-2 rounded-pill small" id="cache-btn-cleanup-expired" style="font-size: 0.75rem;">
<i class="bi bi-magic me-1"></i><span data-i18n="cacheManager.btnCleanExpired">Очистить просроченное</span>
</button>
</div>
<div id="cache-stores-list">
<!-- Populated dynamically by cache-ui.js -->
</div>
</div>
<!-- Action Tools -->
<div class="d-flex flex-wrap align-items-center justify-content-between gap-2 pt-2 border-top border-secondary-subtle">
<div class="d-flex align-items-center gap-2">
<button class="btn btn-outline-secondary btn-sm" data-i18n="cacheManager.btnRequestPersist" id="cache-btn-request-persist" t
```

```
ype="button">
<i class="bi bi-shield-lock me-1"></i>Сделать постоянным
            </button>
<button class="btn btn-outline-secondary btn-sm" data-i18n="ca
cheManager.btnExport" id="cache-btn-export" type="button">
<i class="bi bi-download me-1"></i>Экспорт JSON
            </button>
<label class="btn btn-outline-secondary btn-sm mb-0" for="cach
e-file-import" style="cursor: pointer;">
<i class="bi bi-upload me-1"></i><span data-i18n="cacheManager
.btnImport">Импорт JSON</span>
<input accept=".json" class="d-none" id="cache-file-import" ty
pe="file"/>
</label>
</div>
<button class="btn btn-outline-danger btn-sm" data-i18n="cache
Manager.btnClearAll" id="cache-btn-clear-all" type="button">
<i class="bi bi-trash3 me-1"></i>Очистить всё
          </button>
</div>
</div>
<div class="modal-footer border-secondary-subtle py-2">
<button class="btn btn-secondary btn-sm" data-bs-dismiss="moda
l" data-i18n="userSettings.close" type="button">Закрыть</butto
n>
</div>
</div>
</div>
</div>
</div></div></body>
</html>
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/instructions_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/instructions_tab/index.
html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/instructions_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Instructions Tab -->
<div class="container-fluid p-3">
<div class="alert alert-info">
<h5 class="mb-1"><i class="bi bi-gear"></i> Редактор системных
 инструкций</h5>
<p class="mb-0 small">Управление системными инструкциями AI: ч
ат и Narrator режимы, версионирование, проверка в модели.</p>
</div>
<!-- Mode Switcher -->
<div class="card mb-3" data-panel-id="panel-instructions-1" id
```

```
="panel-instructions-1">
<div class="card-body">
<div class="d-flex align-items-center gap-2 flex-wrap">
<span class="text-muted">Режим:</span>
<div class="btn-group" role="group">
<button class="btn btn-outline-primary active" id="mode-chat" onclick="switchMode('chat')" type="button">
<i class="bi bi-chat-text"></i> Chat
          </button>
<button class="btn btn-outline-secondary" id="mode-narrator" onclick="switchMode('narrator')" type="button">
<i class="bi bi-mic"></i> Narrator
          </button>
</div>
<small class="text-muted ms-2">Активный файл: <code id="active-file-info">—</code></small>
<div class="ms-auto d-flex gap-2">
<button class="btn btn-success" id="btn-check-model" onclick="checkInModel()" type="button">
<i class="bi bi-check-circle"></i> Проверить в модели
          </button>
<button class="btn btn-primary" id="btn-save" onclick="saveInstruction()" type="button">
<i class="bi bi-save"></i> Сохранить версию
          </button>
<button class="btn btn-warning" id="btn-preview" onclick="previewPrompt()" type="button">
<i class="bi bi-eye"></i> Предпросмотр
          </button>
</div>
</div>
</div>
</div>
<!-- Editor Area -->
<div class="card mb-3" data-panel-id="panel-instructions-redaktor-chat" id="panel-instructions-redaktor-chat">
<div class="card-header">
<div class="d-flex justify-content-between align-items-center"
```

```
>
<span class="fw-bold">Редактор — <span id="editor-mode-label">
  Chat</span></span>
<small class="text-muted">Markdown поддерживается</small>
</div>
</div>
<div class="card-body p-0">
<textarea class="form-control border-0 rounded-0" id="instruct
ion-editor" placeholder="Введите системную инструкцию..." rows
="18" spellcheck="false" style="font-family: 'Courier New', mo
nospace; font-size: 0.9rem; resize: vertical;"></textarea>
</div>
</div>
<!-- Versions Panel -->
<div class="card mb-3" data-panel-id="panel-instructions-istor
iya-versiy" id="panel-instructions-istoriya-versiy">
<div class="card-header d-flex justify-content-between align-i
tems-center">
<span class="fw-bold"><i class="bi bi-clock-history"></i> Исто
рия версий</span>
<button class="btn btn-sm btn-outline-secondary" onclick="refr
eshVersions()" type="button">
<i class="bi bi-arrow-clockwise"></i> Обновить
      </button>
</div>
<div class="card-body p-0">
<div class="list-group list-group-flush" id="versions-list">
<div class="list-group-item text-muted">Загрузка версий...</di
v>
</div>
</div>
</div>
<!-- Token Counter -->
<div class="card" data-panel-id="panel-instructions-statistika
" id="panel-instructions-statistika">
<div class="card-header">
<span class="fw-bold"><i class="bi bi-calculator"></i> Статист
ика</span>
```

```
</div>
<div class="card-body">
<div class="row g-3">
<div class="col-md-4">
<div class="p-3 bg-light rounded text-center">
<div class="h3 mb-0" id="char-count">0</div>
<div class="small text-muted">Символов</div>
</div>
</div>
<div class="col-md-4">
<div class="p-3 bg-light rounded text-center">
<div class="h3 mb-0" id="word-count">0</div>
<div class="small text-muted">Слов</div>
</div>
</div>
<div class="col-md-4">
<div class="p-3 bg-light rounded text-center">
<div class="h3 mb-0" id="token-estimate">~0</div>
<div class="small text-muted">Токенов (оценка)</div>
</div>
</div>
</div>
<div class="mt-3 small text-muted border rounded p-2" id="toke
n-preview" style="min-height:2rem;"></div>
</div>
</div>
</div>
<!-- Prompt Preview Modal -->
<div aria-hidden="true" aria-labelledby="prompt-preview-modal-
label" class="modal fade" id="prompt-preview-modal" tabindex="
-1">
<div class="modal-dialog modal-lg modal-dialog-scrollable">
<div class="modal-content">
<div class="modal-header">
<h5 class="modal-title" id="prompt-preview-modal-label">Предпр
осмотр инструкции</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
```

```
</div>
<div class="modal-body" id="prompt-preview-content">
<p>Загрузка...</p>
</div>
<div class="modal-footer">
<button class="btn btn-secondary" data-bs-dismiss="modal" type
="button">Закрыть</button>
</div>
</div>
</div>
</div>
```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Instructions Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/instructions_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/instructions_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// Instructions Tab - UI Logic
// Инициализируется через window.initInstructionsTab() из admin/main.js

'use strict';

import { initI18n, switchLang, applyTranslations } from '../js/i18n.js';

window.switchLang = switchLang;
window.applyTranslations = applyTranslations;

```

```
// ── Helpers ──────────────────────────────────────────────────────────

function adminFetch(url, options = {}) {
  return window.api
    ? window.api.fetch(url, options)
    : fetch(url, options).then(r => {
        if (!r.ok) throw new Error(`HTTP ${r.status}`);
        return r.json();
      });
}

function notify(msg, type = 'info') {
  if (typeof showNotification === 'function') {
    showNotification(msg, type);
  } else {
    console.log(`[${type}] ${msg}`);
  }
}

// ── Mode Management ──────────────────────────────────────────────────
let currentMode = 'chat';

function switchMode(mode) {
  currentMode = mode;
  document.getElementById('mode-chat').classList.toggle('active', mode === 'chat');
  document.getElementById('mode-narrator').classList.toggle('active', mode === 'narrator');

  // Обновить заголовок редактора
  const modeLabel = mode === 'chat' ? '  Chat' : '  Narrator';
  const editorTitle = document.getElementById('editor-mode-label');
  if (editorTitle) editorTitle.textContent = modeLabel;

```

```
  loadInstruction();
  refreshVersions();
}

// ── Instructions API ──────────────────────────────────────────────
async function loadInstruction() {
  try {
    const data = await adminFetch(`/api/admin/instructions?mode=${currentMode}`);
    const editor = document.getElementById('instruction-editor');
    if (data && data.content !== undefined) {
      editor.value = data.content;
    } else {
      editor.value = '';
    }
    updateStats();
    const fileInfo = document.getElementById('active-file-info');
    if (fileInfo && data.file) {
      fileInfo.textContent = data.file;
    }
  } catch (e) {
    notify('Ошибка загрузки инструкции: ' + e.message, 'danger');
    document.getElementById('instruction-editor').value = '';
  }
}

async function saveInstruction() {
  const content = document.getElementById('instruction-editor').value.trim();
  if (!content) {
    notify('Инструкция не может быть пустой', 'warning');
    return;
  }

```

```
  const btn = document.getElementById('btn-save');
  const originalHtml = btn.innerHTML;
  btn.disabled = true;
  btn.innerHTML = '<i class="bi bi-hourglass-split"></i> Сохранение...';

  try {
    const data = await adminFetch('/api/admin/instructions/save', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ mode: currentMode, content })
    });
    notify(`Инструкция сохранена: ${data.version || ''}`, 'success');
    await refreshVersions();
  } catch (e) {
    notify('Ошибка сохранения: ' + e.message, 'danger');
  } finally {
    btn.disabled = false;
    btn.innerHTML = originalHtml;
  }
}

async function checkInModel() {
  const content = document.getElementById('instruction-editor').value.trim();
  const prompt = "Привет! Подтверди, что ты получил инструкцию и готов к работе. Ответь только: 'Понял, готов к работе!'";

  const btn = document.getElementById('btn-check-model');
  const originalHtml = btn.innerHTML;
  btn.disabled = true;
  btn.innerHTML = '<i class="bi bi-hourglass-split"></i> Проверка...';

  try {
    const result = await adminFetch('/api/admin/instructions/c
```

```
heck', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ mode: currentMode, prompt, instru
ction: content })
    });

    const statusEl = document.getElementById('token-preview');
    let tokenInfo = '';
    if (result.token_count) {
      tokenInfo = `<br>Использовано токенов: ${result.token_co
unt}`;
    }

    if (result.response) {
      statusEl.innerHTML = `<strong>Ответ модели:</strong><br>
${escapeHtml(result.response)}${tokenInfo}`;
      notify('Проверка завершена', 'success');
    } else {
      statusEl.innerHTML = 'Ошибка: нет ответа от модели';
      notify('Ошибка проверки', 'danger');
    }
  } catch (e) {
    document.getElementById('token-preview').innerHTML = `<str
ong class="text-danger">Ошибка проверки: ${e.message}</strong>
`;
    notify('Ошибка проверки: ' + e.message, 'danger');
  } finally {
    btn.disabled = false;
    btn.innerHTML = originalHtml;
  }
}

async function refreshVersions() {
  const list = document.getElementById('versions-list');
  list.innerHTML = '<div class="list-group-item text-muted"><i
 class="bi bi-hourglass-split"></i> Загрузка...</div>';

```

```
  try {
    const data = await adminFetch(`/api/admin/instructions/versions?mode=${currentMode}`);

    if (!data.versions || !data.versions.length) {
      list.innerHTML = '<div class="list-group-item text-muted">Нет сохранённых версий</div>';
      return;
    }

    list.innerHTML = data.versions.map(v => `
      <button type="button"
        class="list-group-item list-group-item-action d-flex justify-content-between align-items-center ${v.is_active ? 'list-group-item-success' : ''}"
        onclick="useVersion('${escapeAttr(v.filename)}', '${escapeAttr(v.mode)}')">
        <div class="me-2 overflow-hidden">
          <div class="d-flex align-items-center gap-2 mb-1">
            <strong>${v.mode === 'chat' ? '□ ' : '□ '} ${escapeHtml(v.filename)}</strong>
            ${v.is_active ? '<span class="badge bg-success">Активная</span>' : ''}
          </div>
          <small class="text-muted">${new Date(v.created_at).toLocaleString('ru-RU')} · ${v.size} байт</small>
          <div class="text-muted small text-truncate" style="max-width:480px;">${escapeHtml(v.preview || '')}</div>
        </div>
        <i class="bi bi-arrow-right-circle${v.is_active ? '-fill text-success' : ' text-secondary'}"></i>
      </button>
    `).join('');
  } catch (e) {
    list.innerHTML = `<div class="list-group-item text-danger">Ошибка загрузки версий: ${e.message}</div>`;
  }
}
```

```
185
186 async function useVersion(filename, mode) {
187   try {
188     await adminFetch('/api/admin/instructions/activate', {
189       method: 'POST',
190       headers: { 'Content-Type': 'application/json' },
191       body: JSON.stringify({ mode, filename })
192     });
193
194     if (mode === currentMode) {
195       await loadInstruction();
196     }
197     await refreshVersions();
198     notify(`Версия ${filename} активирована`, 'success');
199   } catch (e) {
200     notify('Ошибка активации: ' + e.message, 'danger');
201   }
202 }
203
204 async function previewPrompt() {
205   const content = document.getElementById('instruction-editor'
    ).value.trim();
206   const preview = `
207     <strong>Контекст системной инструкции (${currentMode === '
    chat' ? '  Chat' : '  Narrator'}):</strong><br>
208     <pre style="white-space:pre-wrap;max-height:400px;overflow
    -y:auto;">${escapeHtml(content.substring(0, 1000))}${content.l
    ength > 1000 ? '\n...' : ''}</pre>
209     <strong>Тестовый запрос:</strong><br>
210     "Привет! Подтверди, что ты получил инструкцию и готов к ра
    боте."
211   `;
212
213   const modal = new bootstrap.Modal(document.getElementById('p
    rompt-preview-modal'));
214   document.getElementById('prompt-preview-content').innerHTML
    = preview;
215   modal.show();
```

```
}

// ── Stats ──────────────────────────────────────────────────────────
function updateStats() {
  const content = document.getElementById('instruction-editor').value;

  document.getElementById('char-count').textContent = content.length;

  const words = content.trim().split(/\s+/).filter(w => w.length > 0);
  document.getElementById('word-count').textContent = words.length;

  const estimatedTokens = Math.ceil(content.length / 3);
  document.getElementById('token-estimate').textContent = '~' + estimatedTokens;

  return estimatedTokens;
}

function escapeHtml(text) {
  if (!text) return '';
  const div = document.createElement('div');
  div.textContent = text;
  return div.innerHTML.replace(/\n/g, '<br>');
}

function escapeAttr(text) {
  if (!text) return '';
  return text.replace(/'/g, "\\'").replace(/"/g, '&quot;');
}

// ── Initialization ─────────────────────────────────────────────────
let initialized = false;
```

```
247
248 async function initInstructionsTab() {
249   if (initialized) return;
250   initialized = true;
251
252   console.log('[InstructionsTab] Initializing...');
253
254   try {
255     await initI18n();
256     console.log('[InstructionsTab] i18n initialized');
257   } catch (e) {
258     console.error('[InstructionsTab] i18n init error:', e);
259   }
260
261   // Load instruction for current mode
262   await loadInstruction();
263
264   // Load versions
265   await refreshVersions();
266
267   // Stats on input
268   document.getElementById('instruction-editor').addEventListen
    er('input', updateStats);
269
270   try {
271     applyTranslations();
272   } catch (e) {
273     console.error('[InstructionsTab] applyTranslations error:'
    , e);
274   }
275
276   console.log('[InstructionsTab] Initialized successfully');
277 }
278
279 // ── Экспорт в window (необходим для inline onclick в HTML пр
    и type="module") ──
280 window.initInstructionsTab = initInstructionsTab;
281 window.switchMode          = switchMode;
```

```
window.saveInstruction     = saveInstruction;
window.checkInModel        = checkInModel;
window.refreshVersions     = refreshVersions;
window.useVersion          = useVersion;
window.previewPrompt       = previewPrompt;
```

activityTracker.js

```
/**
 * ============================================================
 * Process Name: Windows Js - Activitytracker Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля activityTracker.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/activityTracker.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { isUserAuthenticated, trackEvent, trackTabSwitch } from '/windows/api/webgui/js/activityTracker.js';
 *
 * File: activityTracker.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * activityTracker.js — Client-side Telemetry & User Activity Tracker
 *
 * Tracks tab views, dwell durations, and UI button/action clicks strictly
 * for authenticated registered users and flushes batches to /
```

```
api/telemetry/events.
 */

'use strict';

let currentTabId = 'tab-chat';
let tabEnterTimestamp = Date.now();
let eventQueue = [];
let flushTimer = null;
let isInitialized = false;

/**
 * Check if the current user session is an authenticated regis
tered user.
 * @returns {boolean}
 */
export function isUserAuthenticated() {
  if (window.currentUser && window.currentUser.authenticated)
{
    return true;
  }
  // Check auth_token cookie or session storage flag
  return document.cookie.includes('auth_token=');
}

/**
 * Push an event to the internal batch queue if user is authen
ticated.
 * @param {Object} event
 */
export function trackEvent(event) {
  if (!isUserAuthenticated()) {
    return;
  }

  const payload = {
    action: event.action || event.event_type || 'action',
    event_type: event.event_type || 'action',
```

```
    tab_name: event.tab_name || currentTabId,
    target_element: event.target_element || '',
    duration_ms: event.duration_ms || 0,
    details: event.details || null,
    timestamp: new Date().toISOString()
  };

  eventQueue.push(payload);

  if (eventQueue.length >= 20) {
    flushEvents();
  }
}

/**
 * Record a tab switch / tab view event with dwell time calcul
ation.
 * @param {string} newTabId
 */
export function trackTabSwitch(newTabId) {
  if (!newTabId) return;
  const now = Date.now();
  const cleanNewTab = newTabId.startsWith('#') ? newTabId.slic
e(1) : newTabId;

  if (currentTabId && currentTabId !== cleanNewTab) {
    const duration = now - tabEnterTimestamp;
    trackEvent({
      action: `Left tab ${currentTabId}`,
      event_type: 'tab_view',
      tab_name: currentTabId,
      target_element: `tab:${currentTabId}`,
      duration_ms: duration,
      details: {
        transition_to: cleanNewTab,
        duration_sec: Math.round(duration / 1000)
      }
    });
```

```
  }

  currentTabId = cleanNewTab;
  tabEnterTimestamp = now;

  trackEvent({
    action: `Entered tab ${cleanNewTab}`,
    event_type: 'tab_view',
    tab_name: cleanNewTab,
    target_element: `tab:${cleanNewTab}`,
    duration_ms: 0,
    details: {
      timestamp: new Date().toISOString()
    }
  });
}

/**
 * Send queued events to the backend telemetry ingestion endpo
int.
 */
export async function flushEvents() {
  if (!isUserAuthenticated() || eventQueue.length === 0) {
    return;
  }

  const batch = [...eventQueue];
  eventQueue = [];

  const body = JSON.stringify({
    events: batch,
    session_id: window.currentUser?.email || 'authenticated_se
ssion'
  });

  try {
    if (navigator.sendBeacon) {
      const blob = new Blob([body], { type: 'application/json'
```

```
 });
      const success = navigator.sendBeacon('/api/telemetry/events', blob);
      if (success) return;
    }

    await fetch('/api/telemetry/events', {
      method: 'POST',
      headers: {
        'Content-Type': 'application/json'
      },
      body,
      keepalive: true
    });
  } catch (err) {
    console.debug('Telemetry flush failed, restoring queue:', err);
    // Put events back in queue up to maximum limit
    if (eventQueue.length < 50) {
      eventQueue = [...batch, ...eventQueue];
    }
  }
}

/**
 * Inspect click target and capture non-sensitive interactive element info.
 * @param {MouseEvent} event
 */
function handleGlobalClick(event) {
  if (!isUserAuthenticated()) return;

  const target = event.target;
  if (!target) return;

  // Find closest interactive element
  const interactiveEl = target.closest(
    'button, a, input[type="button"], input[type="submit"], [r
```

```
ole="button"], .nav-link, .dropdown-item, .btn, .badge[role="b
utton"], [data-action], [data-tab]'
  );

  if (!interactiveEl) return;

  // Skip sensitive elements (passwords, auth inputs, tokens)
  const isSensitive =
    interactiveEl.type === 'password' ||
    interactiveEl.name === 'password' ||
    interactiveEl.id?.includes('password') ||
    interactiveEl.id?.includes('token') ||
    interactiveEl.classList?.contains('sensitive');

  if (isSensitive) return;

  // Extract meaningful descriptive label
  let label = (
    interactiveEl.getAttribute('data-action') ||
    interactiveEl.getAttribute('data-tab') ||
    interactiveEl.getAttribute('aria-label') ||
    interactiveEl.getAttribute('title') ||
    interactiveEl.innerText?.trim().slice(0, 60) ||
    interactiveEl.id ||
    interactiveEl.name ||
    interactiveEl.tagName.toLowerCase()
  );

  // Clean label whitespace
  label = label.replace(/\s+/g, ' ').trim();
  if (!label) return;

  const targetDescriptor = `${interactiveEl.tagName.toLowerCas
e()}${interactiveEl.id ? '#' + interactiveEl.id : ''}${interac
tiveEl.className ? '.' + interactiveEl.className.split(' ').fi
lter(c => !c.startsWith('show') && !c.startsWith('active')).sl
ice(0, 2).join('.') : ''}`;

```

```
  trackEvent({
    action: `Click: ${label}`,
    event_type: 'click',
    tab_name: currentTabId,
    target_element: targetDescriptor,
    details: {
      label,
      tag: interactiveEl.tagName.toLowerCase(),
      id: interactiveEl.id || null,
      data_action: interactiveEl.getAttribute('data-action') ||
| null,
      data_tab: interactiveEl.getAttribute('data-tab') || null
    }
  });
}

/**
 * Initialize Activity Tracker listeners and flush loop.
 */
export function initActivityTracker() {
  if (isInitialized) return;
  isInitialized = true;

  // Listen for click interactions
  document.addEventListener('click', handleGlobalClick, { capt
ure: true, passive: true });

  // Periodic flush every 5 seconds
  flushTimer = setInterval(flushEvents, 5000);

  // Flush on visibility change / window unload
  document.addEventListener('visibilitychange', () => {
    if (document.visibilityState === 'hidden') {
      flushEvents();
    }
  });
  window.addEventListener('pagehide', () => {
    flushEvents();
```

```
  });

  console.log('✅ Activity Tracker initialized (registered users only)');
}

// Global exposure for legacy scripts
window.activityTracker = {
  trackTabSwitch,
  trackEvent,
  flushEvents,
  initActivityTracker
};

```

ai_table_modal.js

```
/**
 * ============================================================
 * Process Name: Windows Js - Ai Table Modal (Legacy Forwarder)
 * ============================================================
 * Description:
 *   Мост обратной совместимости к выделенному модулю ai_modal_dialog/main.js.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/html/js/ai_table_modal.js?v=20261008_v5"></script>
 *
 * File: ai_table_modal.js
 * Project: ai-windows-admin
 * Package: apps.windows.api.webgui.js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// Legacy forwarder for Universal AI Modal Dialog
// Code migrated to: /html/ai_modal_dialog/main.js

if (!window.AIModalDialog) {
  const script = document.createElement('script');
  script.src = '/html/ai_modal_dialog/main.js?v=20261008_v1';
  document.head.appendChild(script);
}

```

api-cache.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Js - Api-Cache Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля api-cache
.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/api-cache.js?v=2026
1001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { CACHE_STRATEGIES } from '/windows/api/webgui/j
s/api-cache.js';
 *
 * File: api-cache.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * api-cache.js - Универсальный кеш-слой для всех API-запросов
 веб-интерфейса
 *
 * Предоставляет единую точку входа для всех HTTP-запросов с а
втоматическим кешированием,
 * используя современные стратегии: Cache-First, Network-First
```

```
, Stale-While-Revalidate.
 *
 * АРХИТЕКТУРА:
 * - L1 Cache: In-Memory (быстрый доступ)
 * - L2 Cache: IndexedDB (персистентный)
 * - L3 Fallback: Direct network (если кеш недоступен)
 *
 * ПРИМЕНЕНИЕ:
 * Используется всеми вкладками веб-интерфейса для оптимизации
 загрузки данных,
 * снижения нагрузки на API и обеспечения offline-режима.
 *
 * @author hypo69
 * @version 1.0.0
 * @since 2026-09-24
 */

import { browserCache, STORES } from './browser-cache.js';

// Глобальные стратегии кеширования для разных типов данных
export const CACHE_STRATEGIES = {
  // Статические справочники (модели, конфигурации) - долгое T
TL
  STATIC: { ttl: 60 * 60 * 1000, strategy: 'cache-first' },
        // 1 час

  // Редко меняющиеся данные (hardware spec, user profiles) -
средний TTL
  SEMI_STATIC: { ttl: 30 * 60 * 1000, strategy: 'cache-first'
},      // 30 минут

  // Часто меняющиеся данные (системные метрики, статусы) - ко
роткий TTL
  DYNAMIC: { ttl: 5 * 1000, strategy: 'stale-while-revalidate'
 },     // 5 секунд

  // Реалтайм данные (live telemetry, сенсоры) - минимальный T
TL
```

```
  REALTIME: { ttl: 3 * 1000, strategy: 'network-first' },
        // 3 секунды

  // Критичные данные (security status, errors) - всегда из сети
  CRITICAL: { ttl: 0, strategy: 'network-first' }
        // Нет кеша
};

// Автоматические правила сопоставления URL → стратегия
const URL_PATTERNS = [
  // Статические данные
  { pattern: /\/api\/v1\/models/, strategy: CACHE_STRATEGIES.STATIC, tag: 'models' },
  { pattern: /\/api\/config/, strategy: CACHE_STRATEGIES.STATIC, tag: 'config' },
  { pattern: /\/api\/plugins/, strategy: CACHE_STRATEGIES.SEMI_STATIC, tag: 'plugins' },

  // Hardware и системные спецификации
  { pattern: /\/api\/v1\/system\/hardware/, strategy: CACHE_STRATEGIES.SEMI_STATIC, tag: 'hardware' },
  { pattern: /\/api\/v1\/windows-backup/, strategy: CACHE_STRATEGIES.SEMI_STATIC, tag: 'backup' },

  // Динамические системные данные
  { pattern: /\/api\/v1\/system\/summary/, strategy: CACHE_STRATEGIES.DYNAMIC, tag: 'system-summary' },
  { pattern: /\/api\/system-control\/status/, strategy: CACHE_STRATEGIES.DYNAMIC, tag: 'control-status' },

  // Реалтайм телеметрия
  { pattern: /\/api\/v1\/system\/sensors/, strategy: CACHE_STRATEGIES.REALTIME, tag: 'sensors' },
  { pattern: /\/api\/v1\/system\/processes/, strategy: CACHE_STRATEGIES.REALTIME, tag: 'processes' },

  // RAG и чат (частые обновления)
```

```
  { pattern: /\/api\/chat/, strategy: CACHE_STRATEGIES.DYNAMIC, tag: 'chat' },
  { pattern: /\/api\/rag/, strategy: CACHE_STRATEGIES.DYNAMIC, tag: 'rag' }
];

/**
 * Определение стратегии кеширования по URL
 * @param {string} url - URL запроса
 * @returns {Object} { ttl, strategy, tag }
 */
function getCacheStrategy(url) {
  for (const rule of URL_PATTERNS) {
    if (rule.pattern.test(url)) {
      return { ...rule.strategy, tag: rule.tag };
    }
  }
  // По умолчанию: stale-while-revalidate с TTL 1 минута
  return { ttl: 60 * 1000, strategy: 'stale-while-revalidate', tag: 'default' };
}

/**
 * Генерация ключа кеша из URL и параметров
 * @param {string} url - URL запроса
 * @param {Object} options - Опции запроса (method, body)
 * @returns {string} Уникальный ключ кеша
 */
function generateCacheKey(url, options = {}) {
  const method = options.method || 'GET';
  const body = options.body ? `-${btoa(options.body).substring(0, 32)}` : '';
  return `${method}:${url}${body}`;
}

/**
 * Универсальная функция для выполнения HTTP-запросов с автоматическим кешированием
```

```
 *
 * @param {string} url - URL запроса
 * @param {Object} options - Опции запроса
 * @param {Object} cacheOptions - Опции кеширования (переопределяют автоматику)
 * @returns {Promise<any>} Результат запроса
 *
 * @example
 * // Автоматическое определение стратегии
 * const data = await cachedApiFetch('/api/v1/system/hardware');
 *
 * @example
 * // Ручное переопределение стратегии
 * const data = await cachedApiFetch('/api/custom', {}, {
 *   ttl: 10000,
 *   strategy: 'cache-first',
 *   tag: 'custom-data'
 * });
 */
/**
 * Оборачивает результат в полиморфный объект, поддерживающий как прямое обращение к полям,
 * так и стандартный интерфейс Response (res.ok, res.status, res.json()).
 * @param {any} data
 * @returns {any}
 */
function wrapCachedData(data) {
  if (data && typeof data === 'object') {
    try {
      if (!('ok' in data)) {
        Object.defineProperty(data, 'ok', { value: true, writable: true, configurable: true, enumerable: false });
      }
      if (!('status' in data)) {
        Object.defineProperty(data, 'status', { value: 200, writable: true, configurable: true, enumerable: false });
```

```
      }
      if (typeof data.json !== 'function') {
        Object.defineProperty(data, 'json', { value: async () => data, writable: true, configurable: true, enumerable: false });
      }
    } catch (_) {}
  }
  return data;
}

/**
 * Универсальная функция для выполнения HTTP-запросов с автоматическим кешированием
 *
 * @param {string} url - URL запроса
 * @param {Object} options - Опции запроса
 * @param {Object} cacheOptions - Опции кеширования (переопределяют автоматику)
 * @returns {Promise<any>} Результат запроса (поддерживает как прямое чтение, так и res.ok / await res.json())
 */
export async function cachedApiFetch(url, options = {}, cacheOptions = {}) {
  // Определяем стратегию автоматически или используем переданную
  const autoStrategy = getCacheStrategy(url);
  const {
    ttl = autoStrategy.ttl,
    strategy = autoStrategy.strategy,
    tag = autoStrategy.tag,
    forceNetwork = false
  } = cacheOptions;

  const cacheKey = generateCacheKey(url, options);
  const cache = browserCache;

  // Отключаем кеш для указанных вкладок
```

```
  const currentTab = window.location.hash;
  const NO_CACHE_TABS = [
    '#tab-hardware-load-inspector',
    '#tab-processes-load-inspector',
    '#tab-about-system',
  ];
  if (NO_CACHE_TABS.some(prefix => currentTab.startsWith(prefi
x))) {
    const resp = await fetch(url, {
      ...options,
      headers: {
        ...options.headers,
        'Cache-Control': 'no-store, no-cache, must-revalidate'
,
      },
    });
    return await resp.json();
  }

  // Дожидаемся готовности базы, но не блокируем если есть оши
бка
  if (cache) {
    try {
      await cache.ready();
    } catch (_) {}
  }

  // CACHE-FIRST: сначала проверяем кеш, потом сеть
  if (cache && strategy === 'cache-first' && !forceNetwork) {
    try {
      const cached = await cache.get(STORES.API_CACHE, cacheKe
y);
      if (cached !== null && cached !== undefined) {
        return wrapCachedData(cached);
      }
    } catch (_) {}
  }

```

```
  // STALE-WHILE-REVALIDATE: показываем кеш, обновляем в фоне
  if (cache && strategy === 'stale-while-revalidate' && !forceNetwork) {
    try {
      const cached = await cache.get(STORES.API_CACHE, cacheKey);
      if (cached !== null && cached !== undefined) {
        // Фоновое асинхронное обновление
        directFetch(url, options)
          .then(fresh => cache.set(STORES.API_CACHE, cacheKey, fresh, { ttl, tags: [tag, 'api-cache'] }))
          .catch(err => console.debug(`[API Cache] Background update skipped for ${url}:`, err));
        return wrapCachedData(cached);
      }
    } catch (_) {}
  }

  // NETWORK-FIRST или первый запрос: сначала сеть, потом кеш как fallback
  try {
    const fresh = await directFetch(url, options);

    // Сохраняем в кеш (если TTL > 0)
    if (cache && ttl > 0) {
      cache.set(STORES.API_CACHE, cacheKey, fresh, { ttl, tags: [tag, 'api-cache'] }).catch(() => {});
    }

    return wrapCachedData(fresh);
  } catch (err) {
    // При ошибке сети пытаемся вернуть устаревший кеш
    if (cache) {
      try {
        const staleCache = await cache.get(STORES.API_CACHE, cacheKey);
        if (staleCache !== null && staleCache !== undefined) {
          console.warn(`[API Cache] Network failed, returning
```

```
STALE cache: ${url}`);
          return wrapCachedData(staleCache);
        }
      } catch (_) {}
    }
    throw err;
  }
}

/**
 * Прямой HTTP-запрос без кеширования
 * @param {string} url - URL запроса
 * @param {Object} options - Опции fetch
 * @returns {Promise<any>}
 */
async function directFetch(url, options = {}) {
  const opts = { ...options };
  if (opts.body && typeof opts.body === 'string') {
    opts.headers = {
      'Content-Type': 'application/json',
      ...(opts.headers || {})
    };
  }

  if (window.api && typeof window.api.fetch === 'function') {
    const r = await window.api.fetch(url, opts);
    return wrapCachedData(r);
  }

  const res = await fetch(url, opts);
  if (!res.ok) {
    let errMsg = `HTTP ${res.status}`;
    try {
      const errJson = await res.json();
      if (errJson && errJson.detail) {
        errMsg += ` (${typeof errJson.detail === 'object' ? JSON.stringify(errJson.detail) : errJson.detail})`;
      }
```

```
    } catch (_) {}
    throw new Error(errMsg);
  }
  const json = await res.json();
  return wrapCachedData(json);
}

/**
 * Инвалидация кеша по тегу (например, после обновления данных)
 * @param {string} tag - Тег для инвалидации
 * @returns {Promise<number>} Количество удаленных записей
 *
 * @example
 * // Очистить весь кеш моделей после обновления
 * await invalidateCacheByTag('models');
 */
export async function invalidateCacheByTag(tag) {
  if (!browserCache || !browserCache.isDbReady) return 0;
  const count = await browserCache.invalidateByTag(STORES.API_CACHE, tag);
  console.log(`[API Cache] 🗑️ Invalidated ${count} entries with tag: ${tag}`);
  return count;
}

/**
 * Очистка всего API-кеша
 * @returns {Promise<boolean>}
 */
export async function clearAllApiCache() {
  if (!browserCache || !browserCache.isDbReady) return false;
  await browserCache.clearStore(STORES.API_CACHE);
  console.log('[API Cache] 🗑️ All API cache cleared');
  return true;
}

/**
```

```
 * Получение статистики кеша
 * @returns {Promise<Object>}
 */
export async function getApiCacheStats() {
  if (!browserCache || !browserCache.isDbReady) {
    return { available: false, metrics: null };
  }

  const stats = await browserCache.getDetailedStats();
  const apiStore = stats.stores[STORES.API_CACHE] || {};

  return {
    available: true,
    metrics: stats.metrics,
    apiCache: {
      count: apiStore.count || 0,
      sizeMB: apiStore.sizeMB || '0.00',
      hitRate: stats.metrics.hits + stats.metrics.misses > 0
        ? Math.round((stats.metrics.hits / (stats.metrics.hits
 + stats.metrics.misses)) * 100)
        : 0
    },
    storage: stats.storageEstimate
  };
}

// Экспорт в глобальный контекст для доступности из всех вклад
ок
if (typeof window !== 'undefined') {
  window.cachedApiFetch = cachedApiFetch;
  window.invalidateCacheByTag = invalidateCacheByTag;
  window.clearAllApiCache = clearAllApiCache;
  window.getApiCacheStats = getApiCacheStats;
  window.CACHE_STRATEGIES = CACHE_STRATEGIES;
}

export default cachedApiFetch;

```

appConfigEditor.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Js - Appconfigeditor Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля appConfig
Editor.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/appConfigEditor.js?
v=20261001_v1" type="module"></script>
 *
 * File: appConfigEditor.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 19:41:00
 * ============================================================
==================
 */

// ============================================================
==================
// Process Name: App Configuration Editor UI Module
// ============================================================
==================
// Description:
//   Reusable client-side controller for inspecting and modify
ing config.json
//   across /apps microservices in the AI-Windows-Admin web ad
ministrative panel.
//
```

```
// File: appConfigEditor.js
// Project: ai-windows-admin
// Package: src.api.webinterface.js
// Author: hypo69
// Copyright: © 2026 hypo69
// ============================================================
==================

(function() {
  let activeConfig = null;
  let activeAppName = '';

  /**
   * Fetch app config from server
   * @param {string} appName
   * @returns {Promise<Object>}
   */
  async function fetchAppConfig(appName) {
    let res;
    if (window.api && typeof window.api.fetch === 'function')
{
      try {
        return await window.api.fetch(`/api/admin/apps/${appNa
me}/config`);
      } catch (e) {
        // fallback
      }
    }
    res = await fetch(`/api/admin/apps/${appName}/config`);
    if (!res.ok) {
      throw new Error(`Failed to load config: HTTP ${res.statu
s}`);
    }
    return await res.json();
  }

  /**
   * Save app config to server
```

```
   * @param {string} appName
   * @param {Object} configObj
   * @returns {Promise<Object>}
   */
  async function saveAppConfig(appName, configObj) {
    const payload = { config: configObj };
    const res = await fetch(`/api/admin/apps/${appName}/config`, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(payload),
    });
    if (!res.ok) {
      const err = await res.json().catch(() => ({ detail: res.statusText }));
      throw new Error(err.detail || `HTTP ${res.status}`);
    }
    return await res.json();
  }

  /**
   * Render or update modal HTML
   */
  function ensureModalExists() {
    let modalEl = document.getElementById('app-config-modal');
    if (!modalEl) {
      modalEl = document.createElement('div');
      modalEl.id = 'app-config-modal';
      modalEl.className = 'modal fade';
      modalEl.tabIndex = -1;
      modalEl.setAttribute('aria-hidden', 'true');
      modalEl.innerHTML = `
        <div class="modal-dialog modal-lg modal-dialog-centered modal-dialog-scrollable">
          <div class="modal-content border-secondary shadow-lg" style="background: #0f172a; color: #f8fafc;">
            <div class="modal-header border-secondary" style="background: #1e293b;">
```

```
95              <h5 class="modal-title d-flex align-items-center
 gap-2 fw-bold text-info" id="app-config-modal-title">
96                <i class="bi bi-sliders"></i> Настройка конфиг
урации (config.json)
97              </h5>
98              <button type="button" class="btn-close btn-close
-white" data-bs-dismiss="modal" aria-label="Close"></button>
99            </div>
100           <div class="modal-body p-3">
101             <div id="app-config-alert" class="alert d-none p
y-2 px-3 small"></div>
102
103             <!-- Tabs: Visual Form vs Raw JSON -->
104             <ul class="nav nav-pills nav-fill mb-3" id="app-
config-tabs" role="tablist">
105               <li class="nav-item">
106                 <button class="nav-link active py-1 small fw
-semibold" id="app-config-tab-visual-btn" data-bs-toggle="pill
" data-bs-target="#app-config-tab-visual" type="button">
107                   <i class="bi bi-ui-checks me-1"></i> Визуа
льная форма
108                 </button>
109               </li>
110               <li class="nav-item">
111                 <button class="nav-link py-1 small fw-semibo
ld" id="app-config-tab-raw-btn" data-bs-toggle="pill" data-bs-
target="#app-config-tab-raw" type="button">
112                   <i class="bi bi-code-square me-1"></i> Исх
одный JSON
113                 </button>
114               </li>
115             </ul>
116
117             <div class="tab-content">
118               <!-- Visual Form Pane -->
119               <div class="tab-pane fade show active" id="app
-config-tab-visual">
120                 <div class="card border-secondary mb-3" styl
```

```
e="background: #1e293b;">
                    <div class="card-header border-secondary p
y-2 small fw-bold text-primary d-flex align-items-center gap-2
">
                      <i class="bi bi-hdd-network"></i> Режим
сервера (Server Mode)
                    </div>
                    <div class="card-body p-3">
                      <div class="row g-3">
                        <div class="col-md-6">
                          <label class="form-label small fw-se
mibold text-light mb-1">Режим работы сервера:</label>
                          <select class="form-select form-sele
ct-sm bg-dark text-white border-secondary" id="app-cfg-server-
mode">
                            <option value="dedicated">Dedicate
d (Отдельный процесс / порт)</option>
                            <option value="shared">Shared (Map
шрутизация через главный сервер :8000)</option>
                          </select>
                          <div class="form-text small text-mut
ed" id="app-cfg-server-mode-hint">
                            В режиме dedicated run.ps1 запуска
ет отдельный процесс микросервиса.
                          </div>
                        </div>
                        <div class="col-md-6">
                          <label class="form-label small fw-se
mibold text-light mb-1">Порт (Port):</label>
                          <input type="number" class="form-con
trol form-control-sm bg-dark text-white border-secondary font-
monospace" id="app-cfg-server-port" placeholder="8100">
                        </div>
                        <div class="col-md-6">
                          <label class="form-label small fw-se
mibold text-light mb-1">Хост (Host):</label>
                          <input type="text" class="form-contr
ol form-control-sm bg-dark text-white border-secondary font-mo
```

```
nospace" id="app-cfg-server-host" placeholder="127.0.0.1">
                        </div>
                        <div class="col-md-6">
                          <label class="form-label small fw-semibold text-light mb-1">Воркеры (Workers):</label>
                          <input type="number" class="form-control form-control-sm bg-dark text-white border-secondary font-monospace" id="app-cfg-server-workers" placeholder="1" min="1" max="16">
                        </div>
                        <div class="col-12">
                          <div class="form-check form-switch">
                            <input class="form-check-input" type="checkbox" id="app-cfg-server-ssl">
                            <label class="form-check-label small text-light" for="app-cfg-server-ssl">Использовать SSL (HTTPS)</label>
                          </div>
                        </div>
                      </div>
                    </div>
                  </div>

                  <!-- Dynamic App Extra Params Card -->
                  <div class="card border-secondary" style="background: #1e293b;" id="app-cfg-extra-card">
                    <div class="card-header border-secondary py-2 small fw-bold text-info d-flex align-items-center gap-2">
                      <i class="bi bi-gear-wide-connected"></i> Параметры приложения
                    </div>
                    <div class="card-body p-3" id="app-cfg-extra-body">
                      <!-- Populated dynamically based on app properties -->
                    </div>
                  </div>
                </div>
```

```
168
169                 <!-- Raw JSON Pane -->
170                 <div class="tab-pane fade" id="app-config-tab-
raw">
171                   <div class="d-flex justify-content-between a
lign-items-center mb-2">
172                     <span class="small text-muted font-monospa
ce" id="app-cfg-file-path">config.json</span>
173                     <button class="btn btn-sm btn-outline-seco
ndary py-0 px-2" id="app-cfg-btn-format">
174                       <i class="bi bi-magic"></i> Форматироват
ь
175                     </button>
176                   </div>
177                   <textarea class="form-control font-monospace
 bg-dark text-white border-secondary" id="app-cfg-raw-json" ro
ws="14" style="font-size: 0.82rem; line-height: 1.4; tab-size:
 2;"></textarea>
178                 </div>
179               </div>
180             </div>
181             <div class="modal-footer border-secondary justify-
content-between" style="background: #1e293b;">
182               <button type="button" class="btn btn-sm btn-outl
ine-secondary rounded-pill px-3" id="app-cfg-btn-reload">
183                 <i class="bi bi-arrow-clockwise me-1"></i> Сбр
осить
184               </button>
185               <div class="d-flex gap-2">
186                 <button type="button" class="btn btn-sm btn-se
condary rounded-pill px-3" data-bs-dismiss="modal">Отмена</but
ton>
187                 <button type="button" class="btn btn-sm btn-pr
imary rounded-pill px-4 fw-semibold" id="app-cfg-btn-save">
188                   <i class="bi bi-save me-1"></i> Сохранить
189                 </button>
190               </div>
191             </div>
```

```
          </div>
        </div>
      `;
      document.body.appendChild(modalEl);

      // Event handlers
      document.getElementById('app-cfg-btn-save')?.addEventLis
tener('click', onSaveConfig);
      document.getElementById('app-cfg-btn-reload')?.addEventL
istener('click', onReloadConfig);
      document.getElementById('app-cfg-btn-format')?.addEventL
istener('click', () => {
        const rawEl = document.getElementById('app-cfg-raw-jso
n');
        try {
          const parsed = JSON.parse(rawEl.value);
          rawEl.value = JSON.stringify(parsed, null, 2);
        } catch (e) {
          showModalAlert(`Ошибка JSON: ${e.message}`, 'danger'
);
        }
      });

      // Synchronize visual form to raw JSON on tab switch
      document.getElementById('app-config-tab-raw-btn')?.addEv
entListener('click', () => {
        syncFormToRawJson();
      });
      document.getElementById('app-config-tab-visual-btn')?.ad
dEventListener('click', () => {
        syncRawJsonToForm();
      });
    }
  }

  function showModalAlert(message, type = 'info') {
    const el = document.getElementById('app-config-alert');
    if (!el) return;
```

```
    el.className = `alert alert-${type} py-2 px-3 small`;
    el.innerHTML = message;
    el.classList.remove('d-none');
    if (type === 'success') {
      setTimeout(() => el.classList.add('d-none'), 3500);
    }
  }

  function populateVisualForm(config) {
    if (!config) return;

    // Server section
    let serverMode = 'dedicated';
    let host = '127.0.0.1';
    let port = 8100;
    let workers = 1;
    let ssl = false;

    if (typeof config.server === 'string') {
      serverMode = config.server.toLowerCase();
    } else if (config.server && typeof config.server === 'obje
ct') {
      if (typeof config.server.dedicated === 'boolean') {
        serverMode = config.server.dedicated ? 'dedicated' : '
shared';
      } else if (config.server.dedicated !== undefined) {
        serverMode = String(config.server.dedicated).toLowerCa
se() === 'true' ? 'dedicated' : 'shared';
      } else {
        serverMode = config.server.mode || config.server.type
|| 'dedicated';
      }
      host = config.server.host || '127.0.0.1';
      port = config.server.port || port;
      workers = config.server.workers || 1;
      ssl = !!config.server.use_ssl;
    }

```

```
    const modeSelect = document.getElementById('app-cfg-server-mode');
    if (modeSelect) modeSelect.value = serverMode;
    const portInput = document.getElementById('app-cfg-server-port');
    if (portInput) portInput.value = port;
    const hostInput = document.getElementById('app-cfg-server-host');
    if (hostInput) hostInput.value = host;
    const workersInput = document.getElementById('app-cfg-server-workers');
    if (workersInput) workersInput.value = workers;
    const sslCheck = document.getElementById('app-cfg-server-ssl');
    if (sslCheck) sslCheck.checked = ssl;

    // Extra properties
    const extraBody = document.getElementById('app-cfg-extra-body');
    if (extraBody) {
      let fieldsHtml = '<div class="row g-2">';
      let hasExtra = false;

      for (const [key, val] of Object.entries(config)) {
        if (key === 'server' || key === 'cors' || key === 'server_cors_origins' || key === 'server_cors_allow_credentials') {
          continue;
        }
        hasExtra = true;
        if (typeof val === 'object' && val !== null && !Array.isArray(val)) {
          fieldsHtml += `<div class="col-12"><h6 class="text-warning small fw-bold mt-2 mb-1">${key}:</h6></div>`;
          for (const [subK, subV] of Object.entries(val)) {
            fieldsHtml += `
              <div class="col-md-6">
                <label class="form-label small text-muted mb-0
```

```
">${key}.${subK}:</label>
                <input type="text" class="form-control form-control-sm bg-dark text-white border-secondary font-monospace app-cfg-extra-input" data-path="${key}.${subK}" value="${subV !== undefined ? String(subV) : ''}">
              </div>
            `;
          }
        } else {
          fieldsHtml += `
            <div class="col-md-6">
              <label class="form-label small text-muted mb-0">${key}:</label>
              <input type="text" class="form-control form-control-sm bg-dark text-white border-secondary font-monospace app-cfg-extra-input" data-path="${key}" value="${val !== undefined ? String(val) : ''}">
            </div>
          `;
        }
      }

      fieldsHtml += '</div>';
      extraBody.innerHTML = hasExtra ? fieldsHtml : '<div class="text-muted small">Дополнительные параметры отсутствуют.</div>';
    }

    // Raw JSON
    const rawEl = document.getElementById('app-cfg-raw-json');
    if (rawEl) {
      rawEl.value = JSON.stringify(config, null, 2);
    }
  }

  function syncFormToRawJson() {
    try {
      const rawEl = document.getElementById('app-cfg-raw-json'
```

```
);
      let currentObj = {};
      try {
        currentObj = JSON.parse(rawEl.value || '{}');
      } catch {}

      const serverMode = document.getElementById('app-cfg-server-mode')?.value || 'dedicated';
      const isDedicated = serverMode === 'dedicated';
      const port = parseInt(document.getElementById('app-cfg-server-port')?.value || '8100', 10);
      const host = document.getElementById('app-cfg-server-host')?.value || '127.0.0.1';
      const workers = parseInt(document.getElementById('app-cfg-server-workers')?.value || '1', 10);
      const ssl = !!document.getElementById('app-cfg-server-ssl')?.checked;

      if (!currentObj.server || typeof currentObj.server !== 'object') {
        currentObj.server = {};
      }
      currentObj.server.dedicated = isDedicated;
      if (currentObj.server.mode !== undefined) {
        delete currentObj.server.mode;
      }
      currentObj.server.host = host;
      currentObj.server.port = port;
      currentObj.server.use_ssl = ssl;
      currentObj.server.workers = workers;

      // Sync extra inputs
      document.querySelectorAll('.app-cfg-extra-input').forEach((input) => {
        const path = input.getAttribute('data-path');
        const val = input.value;
        if (!path) return;

```

```
        let parsedVal = val;
        if (val === 'true') parsedVal = true;
        else if (val === 'false') parsedVal = false;
        else if (/^\d+$/.test(val)) parsedVal = parseInt(val, 10);
        else if (/^\d+\.\d+$/.test(val)) parsedVal = parseFloat(val);

        const parts = path.split('.');
        if (parts.length === 1) {
          currentObj[parts[0]] = parsedVal;
        } else if (parts.length === 2) {
          if (!currentObj[parts[0]] || typeof currentObj[parts[0]] !== 'object') {
            currentObj[parts[0]] = {};
          }
          currentObj[parts[0]][parts[1]] = parsedVal;
        }
      });

      rawEl.value = JSON.stringify(currentObj, null, 2);
      activeConfig = currentObj;
    } catch (e) {
      console.error('[AppConfigEditor] Form sync error:', e);
    }
  }

  function syncRawJsonToForm() {
    try {
      const rawEl = document.getElementById('app-cfg-raw-json');
      const parsed = JSON.parse(rawEl.value || '{}');
      activeConfig = parsed;
      populateVisualForm(parsed);
    } catch (e) {
      showModalAlert(`Ошибка парсинга JSON: ${e.message}`, 'danger');
    }
```

```
  }

  async function onSaveConfig() {
    try {
      syncFormToRawJson();
      const rawEl = document.getElementById('app-cfg-raw-json'
);
      const parsed = JSON.parse(rawEl.value);

      const res = await saveAppConfig(activeAppName, parsed);
      showModalAlert(`  ${res.message || 'Конфигурация успешно
 сохранена!'}`, 'success');
      activeConfig = parsed;

      if (typeof window.showNotification === 'function') {
        window.showNotification(`Конфигурация ${activeAppName}
 сохранена`, 'success');
      }
    } catch (e) {
      showModalAlert(`Ошибка при сохранении: ${e.message}`, 'd
anger');
    }
  }

  async function onReloadConfig() {
    try {
      const data = await fetchAppConfig(activeAppName);
      activeConfig = data.config;
      populateVisualForm(activeConfig);
      showModalAlert('Конфигурация перезагружена с диска', 'in
fo');
    } catch (e) {
      showModalAlert(`Ошибка загрузки: ${e.message}`, 'danger'
);
    }
  }

  /**
```

```
   * Open the configuration modal for an app
   * @param {string} appName e.g. 'network_terminal', 'cloudflared_monitor', etc.
   * @param {string} displayName e.g. 'Network Terminal'
   */
  window.openAppConfigModal = async function(appName, displayName = '') {
    ensureModalExists();
    activeAppName = appName;

    const titleEl = document.getElementById('app-config-modal-title');
    if (titleEl) {
      titleEl.innerHTML = `<i class="bi bi-sliders"></i> ${displayName || appName} — config.json`;
    }

    const pathEl = document.getElementById('app-cfg-file-path');
    if (pathEl) {
      pathEl.textContent = `apps/${appName}/config.json`;
    }

    const alertEl = document.getElementById('app-config-alert');
    if (alertEl) alertEl.classList.add('d-none');

    // Show modal
    const modalEl = document.getElementById('app-config-modal');
    const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
    modal.show();

    try {
      const data = await fetchAppConfig(appName);
      activeConfig = data.config;
      populateVisualForm(activeConfig);
```

```
438     } catch (e) {
439       showModalAlert(`Не удалось загрузить конфигурацию: ${e.m
    essage}`, 'danger');
440     }
441   };
442 })();
443
```

browser-cache.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Js - Browser-Cache Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля browser-c
ache.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/browser-cache.js?v=
20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { STORES, BrowserCacheManager, browserCache } fr
om '/windows/api/webgui/js/browser-cache.js';
 *
 * File: browser-cache.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * browser-cache.js - Универсальный менеджер браузерного кеша
и долговременного хранилища
 *
 * Предоставляет многоуровневое хранилище (L1 In-Memory + L2 I
ndexedDB + L3 Fallback)
 * для эффективного хранения больших объемов клиентских данных
```

```
:
 * - Ответы API и справочники
 * - RAG-индексы, эмбеддинги и поисковые чанки
 * - История чатов и сообщений ассистентов
 * - Аудиофайлы TTS и медиа-блобы (Blob / ArrayBuffer)
 * - Реестры моделей и системные конфигурации
 *
 * Включает автоматическую очистку по TTL, групповую инвалидац
ию по тегам,
 * мониторинг дисковой квоты браузера и персистентность.
 *
 * ПРИМЕЧАНИЕ: Приложение запускается в своем окне через Edge
--app=url,
 * что может ограничивать доступ к Storage API. Проверьте конс
оль браузера
 * на наличие предупреждений о недоступности IndexedDB.
 */

const DB_NAME = 'AI_Windows_Admin_DB';
const DB_VERSION = 1;

// Стандартные хранилища объектов IndexedDB
export const STORES = {
  API_CACHE: 'api_cache',
  RAG_EMBEDDINGS: 'rag_embeddings',
  CHAT_HISTORY: 'chat_history',
  MEDIA_BLOBS: 'media_blobs',
  MODELS_REGISTRY: 'models_registry',
  KEY_VALUE: 'key_value'
};

export class BrowserCacheManager {
  /**
   * Инициализация менеджера браузерного кеша
   * @param {Object} options - Параметры конфигурации
   * @param {number} options.defaultTTL - Время жизни по умолч
анию (в мс, 1 час = 3600000)
   * @param {number} options.maxMemoryEntries - Максимальное ч
```

```
исло записей в L1 памяти
   */
  constructor(options = {}) {
    this.defaultTTL = options.defaultTTL || 60 * 60 * 1000; //
 1 час
    this.maxMemoryEntries = options.maxMemoryEntries || 200;
    this.memoryCache = new Map(); // L1 Cache: key -> { value,
 expiresAt, storeName, tags }
    this.db = null;
    this.isDbReady = false;
    this.initPromise = this._initIndexedDB();

    // Метрики
    this.stats = {
      hits: 0,
      misses: 0,
      writes: 0,
      deletes: 0
    };

    // Периодическая фоновая очистка просроченных записей кажд
ые 10 минут
    if (typeof window !== 'undefined') {
      window.setInterval(() => this.cleanupExpired(), 10 * 60
* 1000);
    }
  }

  /**
   * Инициализация базы данных IndexedDB
   * @private
   */
  async _initIndexedDB() {
    if (typeof window === 'undefined' || !window.indexedDB) {
      console.warn('[BrowserCache] IndexedDB недоступен в теку
щем окружении. Используется fallback на память.');
      console.warn('[BrowserCache] Возможные причины:');
      console.warn('[BrowserCache] 1. Приложение запущено в PW
```

```
A-окне (Edge --app=url)');
      console.warn('[BrowserCache] 2. Ограничения безопасности
 браузера');
      console.warn('[BrowserCache] 3. Поврежденный user-data-d
ir');
      this.isDbReady = false;
      return null;
    }

    return new Promise((resolve) => {
      try {
        const request = indexedDB.open(DB_NAME, DB_VERSION);

        request.onupgradeneeded = (event) => {
          const db = event.target.result;

          Object.values(STORES).forEach((storeName) => {
            if (!db.objectStoreNames.contains(storeName)) {
              const store = db.createObjectStore(storeName, {
keyPath: 'key' });
              store.createIndex('expiresAt', 'expiresAt', { un
ique: false });
              store.createIndex('updatedAt', 'updatedAt', { un
ique: false });
              store.createIndex('tags', 'tags', { unique: fals
e, multiEntry: true });
            }
          });
          console.log('[BrowserCache] Структура IndexedDB успе
шно обновлена до версии', DB_VERSION);
        };

        request.onsuccess = (event) => {
          this.db = event.target.result;
          this.isDbReady = true;
          console.log('[BrowserCache] IndexedDB успешно подклю
чена:', DB_NAME);
          console.log('[BrowserCache] Данные будут сохраняться
```

```
 в профиле пользователя:', this._getUserDataDir());
123           this.cleanupExpired().catch(() => {});
124           resolve(this.db);
125         };
126 
127         request.onerror = (event) => {
128           console.error('[BrowserCache] Ошибка открытия IndexedDB:', event.target.error);
129           console.error('[BrowserCache] Проверьте:');
130           console.error('[BrowserCache] 1. Есть ли права на запись в user-data-dir');
131           console.error('[BrowserCache] 2. Не заблокирован ли IndexedDB политикой браузера');
132           this.isDbReady = false;
133           resolve(null);
134         };
135       } catch (err) {
136         console.error('[BrowserCache] Исключение при инициализации IndexedDB:', err);
137         this.isDbReady = false;
138         resolve(null);
139       }
140     });
141   }
142 
143   /**
144    * Ожидание готовности базы данных
145    */
146   async ready() {
147     await this.initPromise;
148     return this.isDbReady;
149   }
150 
151   /**
152    * Сохранить значение в кеш (L1 память + L2 IndexedDB)
153    * @param {string} storeName - Имя хранилища (из STORES)
154    * @param {string} key - Уникальный ключ
155    * @param {any} value - Данные (объект, массив, строка, Blob
```

```
, ArrayBuffer)
   * @param {Object} options - Опции записи (ttl, tags, persis
t)
   * @returns {Promise<boolean>}
   */
  async set(storeName = STORES.KEY_VALUE, key, value, options
= {}) {
    if (!key) return false;
    await this.ready();

    const ttl = options.ttl !== undefined ? options.ttl : this
.defaultTTL;
    const now = Date.now();
    const expiresAt = ttl > 0 ? now + ttl : 0; // 0 = бессрочн
о
    const tags = Array.isArray(options.tags) ? options.tags :
(options.tags ? [options.tags] : []);

    const record = {
      key,
      value,
      expiresAt,
      createdAt: now,
      updatedAt: now,
      tags,
      size: this._estimateSize(value)
    };

    // 1. Сохранение в L1 In-Memory Cache
    const memKey = `${storeName}:${key}`;
    this._setMemoryCache(memKey, record);
    this.stats.writes++;

    // Если persist = false, сохраняем только в L1 памяти
    if (options.persist === false) {
      return true;
    }

```

```
188     // 2. Сохранение в L2 IndexedDB
189     if (this.isDbReady && this.db) {
190       try {
191         await new Promise((resolve, reject) => {
192           const transaction = this.db.transaction([storeName],
    'readwrite');
193           const store = transaction.objectStore(storeName);
194           const request = store.put(record);
195 
196           request.onsuccess = () => resolve(true);
197           request.onerror = (e) => reject(e.target.error);
198         });
199         return true;
200       } catch (err) {
201         console.warn(`[BrowserCache] Ошибка записи в IndexedDB
    (${storeName}/${key}):`, err);
202       }
203     }
204 
205     return true;
206   }
207 
208   /**
209    * Получить значение из кеша (L1 -> L2)
210    * @param {string} storeName - Имя хранилища
211    * @param {string} key - Ключ
212    * @returns {Promise<any|null>}
213    */
214   async get(storeName = STORES.KEY_VALUE, key) {
215     if (!key) return null;
216     await this.ready();
217 
218     const now = Date.now();
219     const memKey = `${storeName}:${key}`;
220 
221     // 1. Проверяем L1 In-Memory Cache
222     const memRecord = this.memoryCache.get(memKey);
223     if (memRecord) {
```

```
      if (memRecord.expiresAt > 0 && now > memRecord.expiresAt
) {
        this.memoryCache.delete(memKey);
        this.delete(storeName, key).catch(() => {});
        this.stats.misses++;
        return null;
      }
      this.stats.hits++;
      return memRecord.value;
    }

    // 2. Проверяем L2 IndexedDB
    if (this.isDbReady && this.db) {
      try {
        const record = await new Promise((resolve, reject) =>
{
          const transaction = this.db.transaction([storeName],
 'readonly');
          const store = transaction.objectStore(storeName);
          const request = store.get(key);

          request.onsuccess = () => resolve(request.result);
          request.onerror = (e) => reject(e.target.error);
        });

        if (!record) {
          this.stats.misses++;
          return null;
        }

        // Проверяем срок действия
        if (record.expiresAt > 0 && now > record.expiresAt) {
          this.delete(storeName, key).catch(() => {});
          this.stats.misses++;
          return null;
        }

        // Добавляем в L1 для ускорения повторных обращений
```

```
        this._setMemoryCache(memKey, record);
        this.stats.hits++;
        return record.value;
      } catch (err) {
        console.warn(`[BrowserCache] Ошибка чтения из IndexedDB (${storeName}/${key}):`, err);
      }
    }

    this.stats.misses++;
    return null;
  }

  /**
   * Проверить наличие актуального ключа в кеше
   */
  async has(storeName = STORES.KEY_VALUE, key) {
    const val = await this.get(storeName, key);
    return val !== null;
  }

  /**
   * Удалить запись из кеша
   * @param {string} storeName - Имя хранилища
   * @param {string} key - Ключ
   */
  async delete(storeName = STORES.KEY_VALUE, key) {
    if (!key) return false;
    await this.ready();

    const memKey = `${storeName}:${key}`;
    this.memoryCache.delete(memKey);
    this.stats.deletes++;

    if (this.isDbReady && this.db) {
      try {
        await new Promise((resolve, reject) => {
          const transaction = this.db.transaction([storeName],
```

```
 'readwrite');
296           const store = transaction.objectStore(storeName);
297           const request = store.delete(key);
298 
299           request.onsuccess = () => resolve(true);
300           request.onerror = (e) => reject(e.target.error);
301         });
302         return true;
303       } catch (err) {
304         console.warn(`[BrowserCache] Ошибка удаления из Indexe
dDB (${storeName}/${key}):`, err);
305       }
306     }
307     return true;
308   }
309 
310   /**
311    * Инвалидировать записи по тегу (например: 'models', 'chat_
session_1')
312    * @param {string} storeName - Имя хранилища
313    * @param {string} tag - Тег для поиска
314    */
315   async invalidateByTag(storeName = STORES.KEY_VALUE, tag) {
316     await this.ready();
317 
318     // Удаляем из L1
319     for (const [memKey, record] of this.memoryCache.entries())
 {
320       if (memKey.startsWith(`${storeName}:`) && record.tags &&
 record.tags.includes(tag)) {
321         this.memoryCache.delete(memKey);
322       }
323     }
324 
325     if (!this.isDbReady || !this.db) return 0;
326 
327     let count = 0;
328     try {
```

```
      count = await new Promise((resolve, reject) => {
        const transaction = this.db.transaction([storeName], '
readwrite');
        const store = transaction.objectStore(storeName);
        const index = store.index('tags');
        const request = index.getAllKeys(tag);

        request.onsuccess = () => {
          const keys = request.result || [];
          keys.forEach((k) => store.delete(k));
          resolve(keys.length);
        };
        request.onerror = (e) => reject(e.target.error);
      });
      console.log(`[BrowserCache] Инвалидировано ${count} запи
сей по тегу '${tag}' в '${storeName}'`);
    } catch (err) {
      console.warn(`[BrowserCache] Ошибка инвалидации по тегу
'${tag}':`, err);
    }
    return count;
  }

  /**
   * Инвалидировать записи по регулярному выражению или префик
су ключа
   * @param {string} storeName - Имя хранилища
   * @param {string|RegExp} pattern - Регулярное выражение или
 строка префикса
   */
  async invalidateByPattern(storeName = STORES.KEY_VALUE, patt
ern) {
    await this.ready();

    const isRegExp = pattern instanceof RegExp;
    const testMatch = (key) => isRegExp ? pattern.test(key) :
(key === pattern || key.startsWith(pattern));

```

```
    // Очистка L1
    for (const [memKey] of this.memoryCache.entries()) {
      if (memKey.startsWith(`${storeName}:`)) {
        const rawKey = memKey.replace(`${storeName}:`, '');
        if (testMatch(rawKey)) {
          this.memoryCache.delete(memKey);
        }
      }
    }

    if (!this.isDbReady || !this.db) return 0;

    let deletedCount = 0;
    try {
      deletedCount = await new Promise((resolve, reject) => {
        const transaction = this.db.transaction([storeName], '
readwrite');
        const store = transaction.objectStore(storeName);
        const request = store.openCursor();

        request.onsuccess = (event) => {
          const cursor = event.target.result;
          if (cursor) {
            if (testMatch(cursor.key)) {
              cursor.delete();
              deletedCount++;
            }
            cursor.continue();
          } else {
            resolve(deletedCount);
          }
        };
        request.onerror = (e) => reject(e.target.error);
      });
      console.log(`[BrowserCache] Инвалидировано ${deletedCoun
t} записей по паттерну в '${storeName}'`);
    } catch (err) {
      console.warn(`[BrowserCache] Ошибка инвалидации по патте
```

```
рну в '${storeName}':`, err);
    }
    return deletedCount;
  }

  /**
   * Полностью очистить указанное хранилище
   * @param {string} storeName - Имя хранилища
   */
  async clearStore(storeName = STORES.KEY_VALUE) {
    await this.ready();

    // Очистка L1
    for (const [memKey] of this.memoryCache.entries()) {
      if (memKey.startsWith(`${storeName}:`)) {
        this.memoryCache.delete(memKey);
      }
    }

    if (this.isDbReady && this.db) {
      try {
        await new Promise((resolve, reject) => {
          const transaction = this.db.transaction([storeName],
 'readwrite');
          const store = transaction.objectStore(storeName);
          const request = store.clear();

          request.onsuccess = () => resolve(true);
          request.onerror = (e) => reject(e.target.error);
        });
        console.log(`[BrowserCache] Хранилище '${storeName}' п
олностью очищено.`);
        return true;
      } catch (err) {
        console.warn(`[BrowserCache] Ошибка очистки хранилища
'${storeName}':`, err);
      }
    }
```

```
430     return true;
431   }
432
433   /**
434    * Очистить абсолютно все хранилища
435    */
436   async clearAll() {
437     this.memoryCache.clear();
438     for (const storeName of Object.values(STORES)) {
439       await this.clearStore(storeName);
440     }
441     console.log('[BrowserCache] Все хранилища браузерного кеша
    очищены.');
442   }
443
444   /**
445    * Очистить просроченные записи во всех хранилищах
446    */
447   async cleanupExpired() {
448     await this.ready();
449     const now = Date.now();
450
451     // L1
452     for (const [memKey, record] of this.memoryCache.entries())
    {
453       if (record.expiresAt > 0 && now > record.expiresAt) {
454         this.memoryCache.delete(memKey);
455       }
456     }
457
458     if (!this.isDbReady || !this.db) return 0;
459
460     let totalDeleted = 0;
461     for (const storeName of Object.values(STORES)) {
462       try {
463         const deletedInStore = await new Promise((resolve, rej
    ect) => {
464           const transaction = this.db.transaction([storeName],
```

```
 'readwrite');
465           const store = transaction.objectStore(storeName);
466           const index = store.index('expiresAt');
467           const range = IDBKeyRange.bound(1, now);
468           const request = index.openCursor(range);
469           let count = 0;
470
471           request.onsuccess = (event) => {
472             const cursor = event.target.result;
473             if (cursor) {
474               cursor.delete();
475               count++;
476               cursor.continue();
477             } else {
478               resolve(count);
479             }
480           };
481           request.onerror = (e) => reject(e.target.error);
482         });
483         totalDeleted += deletedInStore;
484       } catch (e) {
485         // Игнорируем возможные локальные ошибки курсора
486       }
487     }
488
489     if (totalDeleted > 0) {
490       console.log(`[BrowserCache] Фоновый сборщик мусора удали
    л ${totalDeleted} просроченных записей.`);
491     }
492     return totalDeleted;
493   }
494
495   /**
496    * Получить детальную статистику использования кеша и дисков
    ой квоты
497    */
498   async getDetailedStats() {
499     await this.ready();
```

```
    const stats = {
      metrics: { ...this.stats },
      memoryCacheSize: this.memoryCache.size,
      stores: {},
      storageEstimate: {
        usage: 0,
        quota: 0,
        usageMB: '0.00',
        quotaMB: '0.00',
        percentUsed: 0,
        isPersisted: false
      }
    };

    // Оценка дискового пространства через navigator.storage API
    if (typeof navigator !== 'undefined' && navigator.storage && navigator.storage.estimate) {
      try {
        const estimate = await navigator.storage.estimate();
        const usage = estimate.usage || 0;
        const quota = estimate.quota || 0;
        const usageMB = (usage / (1024 * 1024)).toFixed(2);
        const quotaMB = (quota / (1024 * 1024)).toFixed(2);
        const percentUsed = quota > 0 ? ((usage / quota) * 100).toFixed(2) : 0;

        let isPersisted = false;
        if (navigator.storage.persisted) {
          isPersisted = await navigator.storage.persisted();
        }

        stats.storageEstimate = {
          usage,
          quota,
          usageMB,
          quotaMB,
          percentUsed: Number(percentUsed),
```

```
          isPersisted
        };
      } catch (err) {
        console.warn('[BrowserCache] Не удалось получить navig
ator.storage.estimate:', err);
      }
    }

    // Подсчет записей и объема по каждому хранилищу
    if (this.isDbReady && this.db) {
      for (const storeName of Object.values(STORES)) {
        try {
          const storeData = await new Promise((resolve) => {
            const transaction = this.db.transaction([storeName
], 'readonly');
            const store = transaction.objectStore(storeName);
            const countReq = store.count();
            const allReq = store.getAll();

            let count = 0;
            let totalBytes = 0;

            countReq.onsuccess = () => { count = countReq.resu
lt || 0; };
            allReq.onsuccess = () => {
              const records = allReq.result || [];
              records.forEach(r => {
                totalBytes += r.size || this._estimateSize(r.v
alue);
              });
              resolve({ count, sizeBytes: totalBytes });
            };
            allReq.onerror = () => resolve({ count: 0, sizeByt
es: 0 });
          });

          stats.stores[storeName] = {
            count: storeData.count,
```

```
            sizeBytes: storeData.sizeBytes,
            sizeKB: (storeData.sizeBytes / 1024).toFixed(2),
            sizeMB: (storeData.sizeBytes / (1024 * 1024)).toFixed(2)
          };
        } catch (e) {
          stats.stores[storeName] = { count: 0, sizeBytes: 0, sizeKB: '0', sizeMB: '0' };
        }
      }
    }

    return stats;
  }

  /**
   * Запросить у браузера постоянное хранилище (защита от случайной очистки)
   */
  async requestPersistence() {
    if (typeof navigator !== 'undefined' && navigator.storage && navigator.storage.persist) {
      const isPersisted = await navigator.storage.persist();
      console.log(`[BrowserCache] Запрос персистентности хранилища: ${isPersisted ? 'Одобрено' : 'Отклонено'}`);
      return isPersisted;
    }
    return false;
  }

  /**
   * Экспортировать данные хранилищ в JSON-структуру
   * @param {Array<string>} storeNames - Список хранилищ для экспорта
   */
  async exportData(storeNames = [STORES.API_CACHE, STORES.MODELS_REGISTRY, STORES.CHAT_HISTORY]) {
    await this.ready();
```

```
599     const exportResult = {
600       version: DB_VERSION,
601       timestamp: Date.now(),
602       exportedAt: new Date().toISOString(),
603       stores: {}
604     };
605 
606     if (!this.isDbReady || !this.db) {
607       console.warn('[BrowserCache] Cannot export data: Indexed
    DB not ready');
608       return exportResult;
609     }
610 
611     for (const storeName of storeNames) {
612       try {
613         const records = await new Promise((resolve, reject) =>
     {
614           const transaction = this.db.transaction([storeName],
     'readonly');
615           const store = transaction.objectStore(storeName);
616           const request = store.getAll();
617           request.onsuccess = () => resolve(request.result ||
    []);
618           request.onerror = (e) => reject(e.target.error);
619         });
620         exportResult.stores[storeName] = records;
621       } catch (err) {
622         console.warn(`[BrowserCache] Ошибка экспорта из ${stor
    eName}:`, err);
623       }
624     }
625 
626     return exportResult;
627   }
628 
629   /**
630    * Импортировать данные в хранилища
631    * @param {Object} data - Экспортированный объект данных
```

```
   */
  async importData(data) {
    if (!data || !data.stores) return false;
    await this.ready();

    for (const [storeName, records] of Object.entries(data.sto
res)) {
      if (!Object.values(STORES).includes(storeName)) continue
;
      if (!Array.isArray(records)) continue;

      for (const record of records) {
        if (record && record.key) {
          await this.set(storeName, record.key, record.value,
{
            ttl: record.expiresAt ? Math.max(0, record.expires
At - Date.now()) : 0,
            tags: record.tags || []
          });
        }
      }
    }
    console.log('[BrowserCache] Импорт данных успешно завершен
.');
    return true;
  }

  /**
   * Добавить запись в L1 In-Memory Cache с контролем размера
(LRU eviction)
   * @private
   */
  _setMemoryCache(key, record) {
    if (this.memoryCache.size >= this.maxMemoryEntries) {
      const firstKey = this.memoryCache.keys().next().value;
      if (firstKey) this.memoryCache.delete(firstKey);
    }
    this.memoryCache.set(key, record);
```

```
  }

  /**
   * Приблизительная оценка размера объекта в байтах
   * @private
   */
  _estimateSize(value) {
    if (value === null || value === undefined) return 0;
    if (typeof value === 'string') return value.length * 2;
    if (typeof value === 'number') return 8;
    if (typeof value === 'boolean') return 4;
    if (typeof Blob !== 'undefined' && value instanceof Blob) return value.size;
    if (typeof ArrayBuffer !== 'undefined' && value instanceof ArrayBuffer) return value.byteLength;
    try {
      return JSON.stringify(value).length * 2;
    } catch {
      return 1024;
    }
  }

  /**
   * Попытка получить путь к user-data-dir (для отладки)
   * @private
   */
  _getUserDataDir() {
    // В браузере этот путь недоступен из-за ограничений безопасности
    // Но мы можем попытаться определить, запущено ли приложение в PWA-окне
    const isPWA = window.matchMedia('(display-mode: standalone)').matches;
    return isPWA ? '[PWA-окно - путь недоступен]' : '[Обычное окно]';
  }
}

```

```
// Создаем глобальный singleton экземпляр
export const browserCache = new BrowserCacheManager();

// Регистрация в глобальном объекте window для доступности во всех вкладках и скриптах
if (typeof window !== 'undefined') {
  window.BrowserCacheManager = BrowserCacheManager;
  window.browserCache = browserCache;
  window.BROWSER_STORES = STORES;
}

export default browserCache;

```

cache-ui.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Js - Cache-Ui Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля cache-ui.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/cache-ui.js?v=20261
001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { initCacheUI, openCacheModal } from '/windows/a
pi/webgui/js/cache-ui.js';
 *
 * File: cache-ui.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * cache-ui.js - Контроллер пользовательского интерфейса управ
ления браузерным кешем
 *
 * Отвечает за:
 * - Отображение модального окна управления кешем
 * - Получение и визуализацию статистики хранилищ и квот брауз
ера
```

```
 * - Обработку действий очистки, экспорта, импорта и запроса персистентности
 */

import { browserCache, STORES } from './browser-cache.js';

// Человекопонятные названия хранилищ
const STORE_LABELS = {
  [STORES.API_CACHE]: { title: 'Кеш ответов API', icon: 'bi-hdd-network', desc: 'Кешированные ответы сервера и справочники' },
  [STORES.RAG_EMBEDDINGS]: { title: 'RAG и Векторы', icon: 'bi-database-fill-gear', desc: 'Эмбеддинги, векторные чанки и поисковый индекс' },
  [STORES.CHAT_HISTORY]: { title: 'История чатов', icon: 'bi-chat-dots-fill', desc: 'Диалоги, сообщения ассистентов и контекст' },
  [STORES.MEDIA_BLOBS]: { title: 'Медиа и Аудио TTS', icon: 'bi-file-earmark-music-fill', desc: 'Сгенерированные аудиодорожки, картинки и бинарные файлы' },
  [STORES.MODELS_REGISTRY]: { title: 'Реестр моделей', icon: 'bi-robot', desc: 'Метаданные локальных и облачных моделей' },
  [STORES.KEY_VALUE]: { title: 'Пользовательские данные', icon: 'bi-key-fill', desc: 'Локальные параметры и временные значения' }
};

/**
 * Инициализация интерфейса управления кешем
 */
export function initCacheUI() {
  const openButtons = document.querySelectorAll('#btn-open-cache-manager, .btn-open-cache-manager, #user-menu-btn-cache');
  const modalEl = document.getElementById('cacheManagerModal');

  if (!modalEl) {
    console.warn('[CacheUI] Элемент #cacheManagerModal не найд
```

```
ен в DOM.');
    return;
  }

  // Привязка кнопок открытия
  openButtons.forEach((btn) => {
    btn.addEventListener('click', (e) => {
      e.preventDefault();
      openCacheModal();
    });
  });

  // Обновление статистики при открытии модального окна
  modalEl.addEventListener('show.bs.modal', () => {
    refreshCacheModalStats();
  });

  // Привязка действий кнопок
  const btnCleanupExpired = document.getElementById('cache-btn
-cleanup-expired');
  if (btnCleanupExpired) {
    btnCleanupExpired.addEventListener('click', async () => {
      btnCleanupExpired.disabled = true;
      const count = await browserCache.cleanupExpired();
      window.showToast?.(`Очищено просроченных записей: ${coun
t}`, 'info') || alert(`Очищено просроченных записей: ${count}`
);
      btnCleanupExpired.disabled = false;
      await refreshCacheModalStats();
    });
  }

  const btnClearAll = document.getElementById('cache-btn-clear
-all');
  if (btnClearAll) {
    btnClearAll.addEventListener('click', async () => {
      if (confirm('Вы действительно хотите полностью очистить
весь браузерный кеш?')) {
```

```
        btnClearAll.disabled = true;
        await browserCache.clearAll();
        window.showToast?.('Все хранилища браузерного кеша успешно очищены.', 'success') || alert('Все хранилища браузерного кеша успешно очищены.');
        btnClearAll.disabled = false;
        await refreshCacheModalStats();
      }
    });
  }

  const btnRequestPersist = document.getElementById('cache-btn-request-persist');
  if (btnRequestPersist) {
    btnRequestPersist.addEventListener('click', async () => {
      const granted = await browserCache.requestPersistence();
      const msg = granted ? 'Постоянное хранилище успешно включено браузером.' : 'Браузер отклонил запрос персистентности.';
      window.showToast?.(msg, granted ? 'success' : 'warning') || alert(msg);
      await refreshCacheModalStats();
    });
  }

  const btnExport = document.getElementById('cache-btn-export');
  if (btnExport) {
    btnExport.addEventListener('click', async () => {
      const data = await browserCache.exportData();
      const blob = new Blob([JSON.stringify(data, null, 2)], { type: 'application/json' });
      const url = URL.createObjectURL(blob);
      const a = document.createElement('a');
      a.href = url;
      a.download = `ai_windows_admin_cache_${new Date().toISOString().slice(0, 10)}.json`;
      document.body.appendChild(a);
      a.click();
```

```
      document.body.removeChild(a);
      URL.revokeObjectURL(url);
      window.showToast?.('Экспорт кеша успешно сохранён', 'success');
    });
  }

  const fileInput = document.getElementById('cache-file-import');
  if (fileInput) {
    fileInput.addEventListener('change', async (e) => {
      const file = e.target.files?.[0];
      if (!file) return;

      try {
        const text = await file.text();
        const data = JSON.parse(text);
        await browserCache.importData(data);
        window.showToast?.('Данные кеша успешно импортированы.', 'success') || alert('Данные кеша успешно импортированы.');
        await refreshCacheModalStats();
      } catch (err) {
        window.showToast?.('Ошибка при чтении файла импорта: ' + err.message, 'danger') || alert('Ошибка при чтении файла импорта: ' + err.message);
      } finally {
        fileInput.value = '';
      }
    });
  }
}

/**
 * Открыть модальное окно управления кешем
 */
export function openCacheModal() {
  const modalEl = document.getElementById('cacheManagerModal');
```

```
  if (modalEl && typeof bootstrap !== 'undefined' && bootstrap.Modal) {
    const bsModal = bootstrap.Modal.getOrCreateInstance(modalEl);
    bsModal.show();
  }
}

/**
 * Обновить и отрендерить статистику в модальном окне
 */
export async function refreshCacheModalStats() {
  const statsContainer = document.getElementById('cache-stores-list');
  const usageText = document.getElementById('cache-usage-text');
  const usageProgress = document.getElementById('cache-usage-progressbar');
  const persistBadge = document.getElementById('cache-persist-badge');

  if (!statsContainer) return;

  statsContainer.innerHTML = `
    <div class="text-center py-3 text-muted">
      <div class="spinner-border spinner-border-sm text-primary me-2" role="status"></div>
      Анализ данных браузерного хранилища...
    </div>
  `;

  try {
    const stats = await browserCache.getDetailedStats();

    // 1. Отображение квоты
    if (usageText && usageProgress) {
      const usageMB = stats.storageEstimate.usageMB;
      const quotaMB = stats.storageEstimate.quotaMB;
```

```
      const percent = stats.storageEstimate.percentUsed;

      usageText.textContent = `${usageMB} МБ из ${quotaMB} МБ (${percent}%)`;
      usageProgress.style.width = `${Math.min(100, Math.max(1, percent))}%`;
      usageProgress.className = `progress-bar ${percent > 80 ? 'bg-danger' : (percent > 50 ? 'bg-warning' : 'bg-primary')}`;
    }

    // 2. Статус персистентности
    if (persistBadge) {
      if (stats.storageEstimate.isPersisted) {
        persistBadge.className = 'badge bg-success-subtle text-success border border-success-subtle';
        persistBadge.innerHTML = '<i class="bi bi-shield-check me-1"></i>Постоянное хранилище активно';
      } else {
        persistBadge.className = 'badge bg-warning-subtle text-warning-emphasis border border-warning-subtle';
        persistBadge.innerHTML = '<i class="bi bi-shield-exclamation me-1"></i>Временное хранилище';
      }
    }

    // 3. Таблица хранилищ
    let html = '<div class="list-group list-group-flush border rounded overflow-hidden">';
    for (const [storeName, storeInfo] of Object.entries(stats.stores)) {
      const meta = STORE_LABELS[storeName] || { title: storeName, icon: 'bi-folder', desc: 'Хранилище данных' };
      html += `
        <div class="list-group-item d-flex align-items-center justify-content-between p-2.5">
          <div class="d-flex align-items-center gap-3">
            <div class="rounded-circle bg-primary-subtle text-primary p-2 d-flex align-items-center justify-content-center"
```

```
style="width: 38px; height: 38px;">
              <i class="bi ${meta.icon} fs-5"></i>
            </div>
            <div>
              <div class="fw-semibold text-body">${meta.title}
</div>
              <div class="small text-muted" style="font-size:
0.75rem;">${meta.desc}</div>
            </div>
          </div>
          <div class="d-flex align-items-center gap-3">
            <div class="text-end">
              <span class="badge bg-secondary-subtle text-seco
ndary border px-2 py-1">${storeInfo.count} зап.</span>
              <div class="small fw-semibold text-muted mt-0.5"
>${storeInfo.sizeKB} КБ</div>
            </div>
            <button class="btn btn-outline-danger btn-sm px-2
py-1 rounded"
                    title="Очистить это хранилище"
                    onclick="window.clearSingleStore('${storeN
ame}')">
              <i class="bi bi-trash3"></i>
            </button>
          </div>
        </div>
      `;
    }
    html += '</div>';

    statsContainer.innerHTML = html;
  } catch (err) {
    console.error('[CacheUI] Ошибка обновления статистики:', e
rr);
    statsContainer.innerHTML = `<div class="alert alert-danger
 p-2 small">Ошибка получения статистики кеша: ${err.message}</
div>`;
  }
```

```
}

// Глобальная функция для очистки одного хранилища из UI
if (typeof window !== 'undefined') {
  window.clearSingleStore = async function(storeName) {
    if (confirm(`Очистить данные хранилища "${storeName}"?`))
{
      await browserCache.clearStore(storeName);
      await refreshCacheModalStats();
    }
  };
}

```

chatService.js

```
/**
 * ============================================================
 * Process Name: Windows Js - Chatservice Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля chatService.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/chatService.js?v=20261001_v1" type="module"></script>
 *
 * File: chatService.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// Shared Chat Service Module
// Exposes window.chatService to unify API calling and message/error formatting.

class AudioQueuePlayer {
  constructor(chunks, userSettings) {
    this.chunks = chunks;
    this.userSettings = userSettings;
    this.currentIndex = 0;
    this.isPlaying = false;
    this.audioElements = new Map(); // index -> Audio
```

```
    this.activeAudio = null;

    // Prefetch first few chunks
    this._prefetch(0);
    this._prefetch(1);
    this._prefetch(2);
  }

  _prefetch(index) {
    if (index >= this.chunks.length || this.audioElements.has(index)) return;

    const text = this.chunks[index];
    const queryParams = new URLSearchParams({
      text: text,
      system: this.userSettings.tts_system,
      voice: this.userSettings.tts_voice
    });
    const audioUrl = `/api/tts/synthesize?${queryParams.toString()}`;
    const audio = new Audio(audioUrl);
    audio.load();
    this.audioElements.set(index, audio);
  }

  async play() {
    this.isPlaying = true;
    while (this.currentIndex < this.chunks.length && this.isPlaying) {
      // Prefetch upcoming chunks
      this._prefetch(this.currentIndex);
      this._prefetch(this.currentIndex + 1);
      this._prefetch(this.currentIndex + 2);
      this._prefetch(this.currentIndex + 3);

      const audio = this.audioElements.get(this.currentIndex);
      if (!audio) {
        this.currentIndex++;
```

```
        continue;
      }

      this.activeAudio = audio;
      window.chatServiceAudio = audio; // for global pause/stop integration

      const playPromise = new Promise((resolve) => {
        audio.onended = () => resolve();
        audio.onerror = (e) => {
          console.error(`Audio error for chunk ${this.currentIndex}:`, e);
          this.stop(); // Останавливаем всю очередь при ошибке загрузки
          resolve();
        };
      });

      try {
        await audio.play();
        await playPromise;
      } catch (err) {
        console.error("Audio playback interrupted or failed:", err);
      }

      this.currentIndex++;
    }
    this.isPlaying = false;
  }

  stop() {
    this.isPlaying = false;
    if (this.activeAudio) {
      try {
        this.activeAudio.pause();
        this.activeAudio.src = "";
      } catch (e) {}
```

```
    }
    this.audioElements.clear();
  }
}

window.chatService = {
  /**
   * Stops any currently playing audio or speech synthesis.
   */
  stop() {
    if (window.chatServiceQueue) {
      try {
        window.chatServiceQueue.stop();
        window.chatServiceQueue = null;
      } catch (e) {
        console.error("Error stopping chatServiceQueue:", e);
      }
    }
    if (window.chatServiceAudio) {
      try {
        window.chatServiceAudio.pause();
        window.chatServiceAudio.src = "";
        window.chatServiceAudio = null;
      } catch (e) {
        console.error("Error stopping chatServiceAudio:", e);
      }
    }
    if ('speechSynthesis' in window) {
      try {
        window.speechSynthesis.cancel();
      } catch (e) {
        console.error("Error cancelling speechSynthesis:", e);
      }
    }
  },
  _abortController: null,

  /**
```

```
138    * Sends a chat message to the /api/chat backend.
139    * Resolves to the text reply or throws the parsed error obj
ect/string.
140    *
141    * @param {string} message
142    * @returns {Promise<any>}
143    */
144   async sendChatMessage(message, onChunk, history = [], genera
tionConfig = {}) {
145     // Останавливаем любую активную озвучку перед отправкой но
вого сообщения
146     this.stop();
147 
148     // Прерываем предыдущий запрос, если он еще выполняется
149     if (this._abortController) {
150       try {
151         this._abortController.abort();
152       } catch (e) {}
153     }
154     this._abortController = new AbortController();
155 
156     let fullText = '';
157     let voiceText = '';
158 
159     const effectiveGenConfig = { ...generationConfig };
160     if (!effectiveGenConfig.model && window.activeModelName) {
161       effectiveGenConfig.model = window.activeModelName;
162     }
163     if (!effectiveGenConfig.search_engine && window.activeSear
chEngine) {
164       effectiveGenConfig.search_engine = window.activeSearchEn
gine;
165     }
166 
167     try {
168       const response = await fetch('/api/chat', {
169         method: 'POST',
170         headers: { 'Content-Type': 'application/json' },
```

```
        body: JSON.stringify({ message, history, generation_co
nfig: effectiveGenConfig }),
        signal: this._abortController.signal
      });

      if (!response.ok) {
        let data;
        try {
          data = await response.json();
        } catch (e) {
          throw new Error(`Ошибка сервера (HTTP ${response.sta
tus}): ${response.statusText}`);
        }
        throw (data.detail || data);
      }

      const reader = response.body.getReader();
      const decoder = new TextDecoder('utf-8');
      let buffer = '';

      while (true) {
        const { value, done } = await reader.read();
        if (done) break;

        buffer += decoder.decode(value, { stream: true });
        const lines = buffer.split('\n');
        buffer = lines.pop();

        for (const line of lines) {
          const trimmed = line.trim();
          if (!trimmed) continue;
          if (trimmed.startsWith('data: ')) {
            try {
              const data = JSON.parse(trimmed.substring(6));
              if (data.error) {
                throw new Error(data.error);
              }
              if (data.text || data.status || data.voice || da
```

```
ta.prompt_dump) {
                if (data.text) fullText += data.text;
                if (data.voice) {
                  voiceText += data.voice;
                }
                if (onChunk) {
                  onChunk(data.text || '', data.status || '',
data.voice || '', data.prompt_dump || '');
                }
              }
            } catch (e) {
              console.error('Parsing SSE error:', e, trimmed);
              throw e;
            }
          }
        }

      if (buffer && buffer.trim().startsWith('data: ')) {
        const data = JSON.parse(buffer.trim().substring(6));
        if (data.error) throw new Error(data.error);
        if (data.text || data.status || data.voice) {
          if (data.text) fullText += data.text;
          if (data.voice) voiceText += data.voice;
          if (onChunk) onChunk(data.text || '', data.status ||
 '', data.voice || '');
        }
      }
    } catch (e) {
      if (e.name === 'AbortError') {
        // Возвращаем Promise, который никогда не зарезолвится
.
        // Это остановит старый вызов sendChatMessage() на мес
те, не вызывая catch/finally в UI-скриптах,
        // так как они уже перекрыты новым запросом.
        return new Promise(() => {});
      }
      throw e;
```

```
    }

    let followUpQuery = null;
    const nextQueryRegex = /\[NEXT_QUERY\](.*?)\[\/NEXT_QUERY\]/i;
    const match = fullText.match(nextQueryRegex);
    if (match) {
      followUpQuery = match[1].trim();
      fullText = fullText.replace(match[0], '');
    }

    return { text: fullText, voice: voiceText, followUp: followUpQuery };
  },


  /**
   * Formats chat messages.
   * Converts objects into stringified JSON inside pre tags for rich rendering.
   *
   * @param {any} text
   * @returns {string|HTMLElement}
   */
  formatMessage(text) {
    if (typeof text === 'object' && text !== null) {
      // Если это просто объект с ответом { text, voice }, отображаем только текст
      if (text.text !== undefined) {
        return text.text;
      }
      return `<pre style="white-space: pre-wrap; background: #2b2b2b; color: #ff7070; padding: 10px; border-radius: 5px; margin: 5px 0; font-family: monospace;">${JSON.stringify(text, null, 2)}</pre>`;
    }
    return text;
  },
```

```
271
272   /**
273    * Reads the given text aloud using the browser SpeechSynthesis API or backend TTS.
274    *
275    * @param {string|object} text
276    */
277   async speak(text) {
278     if (!text) return;
279
280     // Останавливаем все предыдущие аудиопотоки
281     this.stop();
282
283     let cleanText = '';
284     if (typeof text === 'object' && text !== null) {
285       if (text.voice || text.text) {
286         // Если это объект { text, voice } из sendChatMessage
287         cleanText = text.voice || text.text;
288       } else {
289         // Если это карточка фильма
290         const title = text.title_ru || text.title;
291         const rec = text.why_watch || text.plot || '';
292         cleanText = `Найдено: ${title}. ${rec}`;
293       }
294     } else {
295       const trimmed = text.trim();
296       if (trimmed.startsWith('{') && trimmed.endsWith('}')) {
297         try {
298           const card = JSON.parse(trimmed);
299           const title = card.title_ru || card.title;
300           const rec = card.why_watch || card.plot || '';
301           cleanText = `Найдено: ${title}. ${rec}`;
302         } catch (e) {
303           cleanText = trimmed;
304         }
305       } else {
306         // Strip code blocks
307         cleanText = trimmed.replace(/```[\s\S]*?```/g, '');
```

```
        // Strip HTML tags
        cleanText = cleanText.replace(/<[^>]*>/g, '');
        // Strip Markdown formatting
        cleanText = cleanText.replace(/[*_`#]/g, '');
      }
    }

    if (!cleanText.trim()) return;

    let userSettings = null;
    try {
      const response = await fetch('/auth/settings');
      if (response.ok) {
        userSettings = await response.json();
        if (userSettings.tts_enabled === 1 && userSettings.tts
_system && userSettings.tts_system !== 'browser') {
          // Разбиваем текст по абзацам и предложениям на боле
е мелкие чанки (~150-200 символов)
          const paragraphs = cleanText
            .split(/\n+/)
            .map(p => p.trim())
            .filter(p => p.length > 0);

          let chunks = [];
          for (const para of paragraphs) {
            const sentences = para.match(/[^.!?]+[.!?]*/g) ||
[para];
            let currentChunk = "";
            for (const sentence of sentences) {
              const trimmedSentence = sentence.trim();
              if (!trimmedSentence) continue;
              if ((currentChunk + " " + trimmedSentence).lengt
h > 200) {
                chunks.push(currentChunk.trim());
                currentChunk = trimmedSentence;
              } else {
                currentChunk = (currentChunk + " " + trimmedSe
ntence).trim();
```

```
            }
          }
          if (currentChunk) {
            chunks.push(currentChunk.trim());
          }
        }
        chunks = chunks.filter(c => c.length > 0);

        if (chunks.length > 0) {
          window.chatServiceQueue = new AudioQueuePlayer(chu
nks, userSettings);
          await window.chatServiceQueue.play();
        }
        return;
      }
    }
  } catch (e) {
    console.error('Failed to speak using backend TTS, fallin
g back to browser:', e);
  }

  if ('speechSynthesis' in window) {
    window.currentUtterance = new SpeechSynthesisUtterance(c
leanText);

    // Dynamic language detection for Web Speech API fallbac
k
    const hasHebrew = /[\u0590-\u05FF]/.test(cleanText);
    const hasCyrillic = /[\u0400-\u04FF]/.test(cleanText);
    window.currentUtterance.lang = hasHebrew ? 'he-IL' : (ha
sCyrillic ? 'ru-RU' : 'en-US');

    if (userSettings && userSettings.tts_voice) {
      const voices = window.speechSynthesis.getVoices();
      const selectedVoice = voices.find(v => v.name === user
Settings.tts_voice);
      if (selectedVoice) {
        window.currentUtterance.voice = selectedVoice;
```

```
          if (selectedVoice.lang) {
            window.currentUtterance.lang = selectedVoice.lang;
          }
        }
      }

      window.speechSynthesis.speak(window.currentUtterance);
    }
  }
};

// Форматирование отображаемого названия поискового движка
function formatSearchEngine(engine) {
  if (!engine) return '';
  let eng = engine;
  let mdl = '';
  if (engine.includes(':') && !engine.startsWith('ollama:') &&
 !engine.startsWith('foundry:')) {
    const parts = engine.split(':');
    eng = parts[0];
    mdl = parts.slice(1).join(':');
  }
  const map = {
    'gemini_cli': '□ gemini_cli',
    'gemini': '□ gemini',
    'agy': '□ agy',
    'langchain': '□ langchain',
    'playwright': '□ playwright'
  };
  const iconAndName = map[eng] || `□ ${eng}`;
  if (mdl && !mdl.startsWith('chromium')) {
    return `${iconAndName} (${mdl})`;
  }
  return iconAndName;
}

window.formatSearchEngine = formatSearchEngine;

```

```
window.updateChatBadges = function(modelName, searchEngine) {
  if (modelName !== undefined && modelName !== null) {
    window.activeModelName = modelName;
  }
  if (searchEngine !== undefined && searchEngine !== null) {
    window.activeSearchEngine = searchEngine;
  }

  const curModel = window.activeModelName || '';
  const curSearch = window.activeSearchEngine || '';

  const modelBadges = document.querySelectorAll('#chat-model-b
adge, #chat-popup-model-badge');
  modelBadges.forEach(badge => {
    if (curModel) {
      badge.innerHTML = `<i class="bi bi-cpu-fill text-info me
-1"></i> <span>${curModel}</span>`;
      badge.title = `Активная модель ИИ: ${curModel} (клик для
 перехода к настройкам моделей)`;
      badge.style.display = 'inline-flex';
    } else {
      badge.innerHTML = `<i class="bi bi-cpu text-muted me-1">
</i> <span class="text-secondary">По умолчанию (сервер)</span>
`;
      badge.title = `Используется модель по умолчанию сервера
(клик для выбора модели)`;
      badge.style.display = 'inline-flex';
    }

    if (!badge._hasClickBound) {
      badge._hasClickBound = true;
      badge.style.cursor = 'pointer';
      badge.onclick = () => {
        // Try activating models tab if present
        const modelsTabBtn = document.querySelector('button[da
ta-bs-target="#tab-models"]');
        if (modelsTabBtn) {
          const tab = new bootstrap.Tab(modelsTabBtn);
```

```
          tab.show();
        } else {
          // Open settings modal
          const modalEl = document.getElementById('userSettingsModal');
          if (modalEl && window.bootstrap) {
            const modal = bootstrap.Modal.getInstance(modalEl) || new bootstrap.Modal(modalEl);
            modal.show();
          }
        }
      };
    }
  });

  const searchBadges = document.querySelectorAll('#chat-search-badge, #chat-popup-search-badge');
  searchBadges.forEach(badge => {
    if (curSearch) {
      badge.innerHTML = `<i class="bi bi-globe2 text-warning me-1"></i> <span>${formatSearchEngine(curSearch)}</span>`;
      badge.title = `Провайдер веб-поиска: ${curSearch}`;
      badge.style.display = 'inline-flex';
    } else {
      badge.style.display = 'none';
    }
  });

  const savedCardBadge = document.getElementById('saved-default-model-badge');
  if (savedCardBadge) {
    if (curModel) {
      savedCardBadge.textContent = curModel;
      savedCardBadge.className = 'badge bg-success font-monospace px-2 py-1';
      savedCardBadge.title = `В вашем профиле сохранена модель: ${curModel}`;
    } else {
```

```
      savedCardBadge.textContent = 'Не задана (системный fallback)';
      savedCardBadge.className = 'badge bg-secondary font-monospace px-2 py-1';
      savedCardBadge.title = 'Модель в профиле не выбрана, сервер использует системный дефолт';
    }
  }

  // Update chat message input state based on model selection
  const msgInputs = document.querySelectorAll('#message-input, #chat-popup-input');
  const sendButtons = document.querySelectorAll('#send-button, #chat-popup-send-btn');

  msgInputs.forEach(input => {
    if (!curModel) {
      input.disabled = true;
      input.placeholder = 'Ни одна модель не выбрана. Выберите модель во вкладке «Модели и API».';
      input.classList.add('is-invalid');
    } else {
      input.disabled = false;
      input.placeholder = 'Введите сообщение...';
      input.classList.remove('is-invalid');
    }
  });

  sendButtons.forEach(btn => {
    btn.disabled = !curModel;
  });
};

// Автоматически загружаем настройки и обновляем бейджи модели и поиска при загрузке страницы
document.addEventListener('DOMContentLoaded', async () => {
  try {
    const response = await fetch('/auth/settings');
```

```
    if (response.ok) {
      const settings = await response.json();
      const modelName = settings.model || '';
      const searchEngine = settings.search_engine || '';
      window.activeModelName = modelName;
      window.activeSearchEngine = searchEngine;

      window.updateChatBadges(modelName, searchEngine);
      // На случай если DOM элементы добавились/отрендерились позже
      setTimeout(() => window.updateChatBadges(), 500);
      setTimeout(() => window.updateChatBadges(), 1500);
    }
  } catch (e) {
    console.error('Failed to load active model / search badge:', e);
  }
});

```

debounce-integration.js

```
/**
 * ============================================================
 * Process Name: Windows Js - Debounce-Integration Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля debounce-integration.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/debounce-integration.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { createUserSearchDebounce, createPluginSearchDebounce, createInstructionAutoSaveDebounce } from '/windows/api/webgui/js/debounce-integration.js';
 *
 * File: debounce-integration.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * Debounce Integration - Интеграция дебаунсинга в популярные операции
 *
 * Этот модуль предоставляет готовые функции дебаунса/батчинга для:
```

```
 * - Поиска пользователей
 * - Поиска плагинов
 * - Автосохранения инструкций
 * - Обновления статуса
 */

import { RequestManager } from './request-manager.js';

// Создаем глобальный экземпляр request manager
const globalRequestManager = new RequestManager();

window.globalRequestManager = globalRequestManager;

/**
 * Дебаунс поиска пользователей (500ms)
 */
export function createUserSearchDebounce(searchFn) {
  return (query) => {
    if (window.optimizationModule?.stats) {
      window.optimizationModule.stats.requestsDebounced++;
    }
    return globalRequestManager.debounce(
      'user-search',
      () => searchFn(query),
      500
    );
  };
}

/**
 * Дебаунс поиска плагинов (500ms)
 */
export function createPluginSearchDebounce(searchFn) {
  return (query) => {
    if (window.optimizationModule?.stats) {
      window.optimizationModule.stats.requestsDebounced++;
    }
    return globalRequestManager.debounce(
```

```
      'plugin-search',
      () => searchFn(query),
      500
    );
  };
}

/**
 * Дебаунс автосохранения инструкций (1000ms)
 */
export function createInstructionAutoSaveDebounce(saveFn) {
  return (content) => {
    if (window.optimizationModule?.stats) {
      window.optimizationModule.stats.requestsDebounced++;
    }
    return globalRequestManager.debounce(
      'instruction-autosave',
      () => saveFn(content),
      1000
    );
  };
}

/**
 * Дебаунс обновления статуса (2000ms)
 */
export function createStatusUpdateDebounce(updateFn) {
  return (status) => {
    if (window.optimizationModule?.stats) {
      window.optimizationModule.stats.requestsDebounced++;
    }
    return globalRequestManager.debounce(
      'status-update',
      () => updateFn(status),
      2000
    );
  };
}
```

```
104
105 /**
106  * Создает throttle функцию для real-time обновлений (1000ms)
107  */
108 export function createRealtimeUpdateThrottle(updateFn) {
109   return globalRequestManager.throttle(
110     'realtime-update',
111     updateFn,
112     1000
113   );
114 }
115
116 /**
117  * Батчинг загрузки пользователей по ID
118  * Вместо 5 отдельных запросов - один запрос с массивом ID
119  */
120 export function createUserBatchLoader(loadFn) {
121   return (userId) => {
122     if (window.optimizationModule?.stats) {
123       window.optimizationModule.stats.requestsDebounced++;
124     }
125     return globalRequestManager.batch(
126       'user-batch-load',
127       async (userIds) => {
128         return loadFn(userIds);
129       },
130       userId,
131       50 // Батч отправляется через 50ms
132     );
133   };
134 }
135
136 /**
137  * Дублирование запросов - если один и тот же запрос идет дваж
    ды,
138  * используем результат первого
139  */
140 export function createDedupeFetch(url, fetchFn) {
```

```
  const dedupeKey = `fetch:${url}`;
  return () => {
    if (window.optimizationModule?.stats) {
      window.optimizationModule.stats.requestsDebounced++;
    }
    return globalRequestManager.deduplicate(dedupeKey, fetchFn
);
  };
}

/**
 * Вспомогательная функция для измерения эффективности
 */
export function logDebouncingStats() {
  const stats = globalRequestManager.getStats();
  console.group('  Debouncing Statistics');
  console.log('Pending Debounces:', stats.pendingDebounces);
  console.log('Pending Requests:', stats.pendingRequests);
  console.log('Cached Results:', stats.cachedResults);
  console.table(stats.cacheKeys.slice(0, 5));
  console.groupEnd();
}

// Делаем доступным из консоли
window.logDebouncingStats = logDebouncingStats;

export default {
  createUserSearchDebounce,
  createPluginSearchDebounce,
  createInstructionAutoSaveDebounce,
  createStatusUpdateDebounce,
  createRealtimeUpdateThrottle,
  createUserBatchLoader,
  createDedupeFetch,
  logDebouncingStats,
  globalRequestManager
};

```

i18n.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Js - I18N Script
 * ============================================================
==================
 * Description:
 *   Клиентский модуль интернационализации (i18n) и региональн
ых настроек (BCP 47).
 *   Поддерживает нормализацию тегов language-REGION (например
 ru-RU, he-IL, uk-UA, en-US),
 *   автоматическое определение из URL query параметров (?regi
on=ru-ru, ?locale=ru-RU,
 *   ?language-region=ru-RU, ?lang=ru), переключение языков и
синхронизацию DOM.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/i18n.js?v=20261006_
v10" type="module"></script>
 *
 *   JavaScript Import:
 *     import { initI18n, switchLocale, switchLang, normalizeL
ocaleTag, getCurrentLocale } from '/windows/api/webgui/js/i18n
.js';
 *
 * File: i18n.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

```

```
/**
 * i18n.js — Модуль интернационализации и локалей BCP 47
 *
 * Конвенции разметки HTML:
 *   data-i18n="key"             — устанавливает innerHTML элемента
 *   data-i18n-placeholder="key" — устанавливает placeholder для input/textarea
 *   data-i18n-title="key"       — устанавливает атрибут title
 *
 * Приоритет выбора языка/локали:
 *   1. Параметры URL query: ?region=ru-ru | ?locale=ru-RU | ?language-region=ru-RU | ?lang=ru
 *   2. Сохранённое значение в localStorage ('app_locale' / 'app_language')
 *   3. Язык браузера (navigator.language)
 *   4. Дефолтная локаль ('en-US')
 */

export const DEFAULT_LOCALE = 'en-US';
export const SUPPORTED_LOCALES = ['en-US', 'ru-RU', 'he-IL', 'uk-UA'];
export const SUPPORTED_LANGS = ['en', 'ru', 'he', 'uk'];
export const SUPPORTED = SUPPORTED_LANGS; // Обратная совместимость

export const DEFAULT_REGIONS = {
    en: 'US',
    ru: 'RU',
    he: 'IL',
    uk: 'UA',
    de: 'DE',
    fr: 'FR',
    es: 'ES',
    pt: 'BR',
    zh: 'CN',
    ja: 'JP',
    ko: 'KR',
```

```
    ar: 'SA'
};

let i18nInitialized = false;
let eventsBound = false;
let currentActiveLocale = DEFAULT_LOCALE;

/**
 * Нормализует языковой/региональный тег в стандарт BCP 47 (language-REGION)
 * и извлекает базовый язык и регион.
 *
 * Формат:
 *   language-REGION
 *   language — ISO 639-1 (2 строчные буквы: ru, he, uk, en...)
 *   REGION   — ISO 3166-1 alpha-2 (2 заглавные буквы: RU, IL, UA, US...)
 *
 * Примеры:
 *   "ru-ru"    -> { locale: "ru-RU", lang: "ru", region: "RU" }
 *   "ru_ru"    -> { locale: "ru-RU", lang: "ru", region: "RU" }
 *   "he-il"    -> { locale: "he-IL", lang: "he", region: "IL" }
 *   "uk-ua"    -> { locale: "uk-UA", lang: "uk", region: "UA" }
 *   "en-us"    -> { locale: "en-US", lang: "en", region: "US" }
 *   "ru"       -> { locale: "ru-RU", lang: "ru", region: "RU" }
 *   "`ru-ru`"  -> { locale: "ru-RU", lang: "ru", region: "RU" }
 *
 * @param {string} tag - Исходный тег локали или языка
 * @returns {{ locale: string, lang: string, region: string }}
 */
```

```
export function normalizeLocaleTag(tag) {
    if (!tag || typeof tag !== 'string') {
        return { locale: DEFAULT_LOCALE, lang: 'en', region: 'US' };
    }

    // Очистка от кавычек, апострофов, обратных кавычек и пробелов
    let cleaned = tag.trim().replace(/^[`'"]+|[`'"]+$/g, '');

    // Заменяем нижние подчеркивания на дефисы
    cleaned = cleaned.replace(/_/g, '-');

    // Проверяем формат language-REGION (например ru-RU, ru-ru, en-US)
    const match = cleaned.match(/^([a-zA-Z]{2,3})(?:-([a-zA-Z]{2,4}))?$/);
    if (match) {
        const langPart = match[1].toLowerCase();
        let regionPart = match[2]
            ? match[2].toUpperCase()
            : (DEFAULT_REGIONS[langPart] || langPart.toUpperCase());
        const locale = `${langPart}-${regionPart}`;
        return { locale, lang: langPart, region: regionPart };
    }

    // Fallback для неизвестных конструкций
    const lower = cleaned.toLowerCase();
    const fallbackLang = SUPPORTED_LANGS.includes(lower) ? lower : 'en';
    const fallbackRegion = DEFAULT_REGIONS[fallbackLang] || 'US';
    return {
        locale: `${fallbackLang}-${fallbackRegion}`,
        lang: fallbackLang,
        region: fallbackRegion
    };
```

```
}

/**
 * Извлекает и нормализует локаль/регион из URL query параметров.
 * Поддерживаемые параметры:
 *   - region (например ?region=ru-ru или ?region=ru-RU)
 *   - locale (например ?locale=ru-RU)
 *   - language-region (например ?language-region=ru-RU)
 *   - language_region (например ?language_region=ru_RU)
 *   - lang (например ?lang=ru)
 *   - language (например ?language=ru-RU)
 *
 * @returns {{ locale: string, lang: string, region: string } | null}
 */
export function getLocaleFromUrl() {
    try {
        if (typeof window === 'undefined' || !window.location || !window.location.search) {
            return null;
        }
        const params = new URLSearchParams(window.location.search);
        const urlParam = params.get('region') ||
                         params.get('locale') ||
                         params.get('language-region') ||
                         params.get('language_region') ||
                         params.get('lang') ||
                         params.get('language');

        if (urlParam) {
            return normalizeLocaleTag(urlParam);
        }
    } catch (e) {
        console.warn('[i18n] Ошибка парсинга параметров URL:', e);
    }
```

```
    return null;
}

/**
 * Привязывает обработчик изменений селекторов языка.
 */
function setupLanguageEvents() {
    if (eventsBound) return;
    eventsBound = true;
    document.addEventListener('change', (e) => {
        if (e.target && e.target.classList.contains('lang-sele
ctor')) {
            switchLocale(e.target.value);
        }
    });
}

/**
 * Инициализирует i18next и применяет переводы к DOM.
 * @param {string} [langOrLocale] - Язык или локаль (опциональ
но)
 */
export async function initI18n(langOrLocale) {
    let target;
    if (langOrLocale) {
        target = normalizeLocaleTag(langOrLocale);
    } else {
        target = detectLocale();
    }

    const { locale, lang, region } = target;
    currentActiveLocale = locale;

    // Всегда загружаем английский как fallback
    const [mainRes, enRes] = await Promise.all([
        loadLocale(lang),
        lang !== 'en' ? loadLocale('en') : Promise.resolve(nul
l),
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```
    ]);

    if (typeof i18next !== 'undefined') {
        await i18next.init({
            lng: lang,
            fallbackLng: 'en',
            resources: { [lang]: { translation: mainRes } },
            interpolation: { escapeValue: false },
        });

        if (enRes) {
            i18next.addResourceBundle('en', 'translation', enRes, true, true);
        }
    }

    document.documentElement.setAttribute('lang', lang);
    document.documentElement.setAttribute('data-locale', locale);
    document.documentElement.setAttribute('dir', (lang === 'he' || lang === 'ar') ? 'rtl' : 'ltr');

    applyTranslations();
    syncSelectors(lang, locale);
    setupLanguageEvents();
    i18nInitialized = true;

    // Сохраняем в localStorage
    try {
        localStorage.setItem('app_locale', locale);
        localStorage.setItem('app_language', lang);
    } catch {}

    // Уведомляем систему через события
    window.dispatchEvent(new CustomEvent('localeChanged', { detail: { locale, lang, region } }));
    window.dispatchEvent(new CustomEvent('languageChanged', { detail: { lang, locale, region } }));
```

```
}

/**
 * Переключает язык/локаль, перезагружает бандлы при необходимости, сохраняет в localStorage.
 * @param {string} langOrLocale - Код языка или тег BCP 47 (например 'ru', 'ru-RU', 'he-IL')
 */
export async function switchLocale(langOrLocale) {
    const { locale, lang, region } = normalizeLocaleTag(langOrLocale);
    currentActiveLocale = locale;

    if (typeof i18next !== 'undefined') {
        if (!i18next.hasResourceBundle(lang, 'translation')) {
            const res = await loadLocale(lang);
            i18next.addResourceBundle(lang, 'translation', res, true, true);
        }
        await i18next.changeLanguage(lang);
    }

    document.documentElement.setAttribute('lang', lang);
    document.documentElement.setAttribute('data-locale', locale);
    document.documentElement.setAttribute('dir', (lang === 'he' || lang === 'ar') ? 'rtl' : 'ltr');

    applyTranslations();
    syncSelectors(lang, locale);

    try {
        localStorage.setItem('app_locale', locale);
        localStorage.setItem('app_language', lang);
    } catch {}

    // Глобальные события для компонентов и вкладок
    window.dispatchEvent(new CustomEvent('localeChanged', { de
```

```
tail: { locale, lang, region } }));
    window.dispatchEvent(new CustomEvent('languageChanged', {
detail: { lang, locale, region } }));
}

// Алиас для обратной совместимости
export const switchLang = switchLocale;

/**
 * Обновляет все элементы с data-i18n* атрибутами в DOM.
 */
export function applyTranslations() {
    if (typeof i18next === 'undefined') return;

    // Контент текста и HTML
    document.querySelectorAll('[data-i18n]').forEach(el => {
        const key = el.getAttribute('data-i18n');
        if (!key) return;
        const val = i18next.t(key);
        if (val && val !== key) el.innerHTML = val;
    });

    // Плейсхолдеры
    document.querySelectorAll('[data-i18n-placeholder]').forEa
ch(el => {
        const key = el.getAttribute('data-i18n-placeholder');
        if (!key) return;
        const val = i18next.t(key);
        if (val && val !== key) el.placeholder = val;
    });

    // Тултипы и атрибут title
    document.querySelectorAll('[data-i18n-title]').forEach(el
=> {
        const key = el.getAttribute('data-i18n-title');
        if (!key) return;
        const val = i18next.t(key);
        if (val && val !== key) el.title = val;
```

```
    });
}

/**
 * Загружает JSON-файл локализации.
 * @param {string} lang - Код языка
 * @returns {Promise<Object>}
 */
async function loadLocale(lang) {
    const pathName = typeof window !== 'undefined' ? window.location.pathname : '';
    let paths = [];

    if (pathName.includes('/user_tts')) {
        paths = [
            `/html/locales/${lang}.json`,
            `/html/user_tts/locales/user_${lang}.json`,
            `/html/locales/user_${lang}.json`
        ];
    } else {
        paths = [
            `/html/locales/${lang}.json`,
            `/html/locales/user_${lang}.json`
        ];
    }

    for (const path of paths) {
        try {
            const cacheBustUrl = `${path}?v=${Date.now()}`;
            const r = await fetch(cacheBustUrl);
            if (r.ok) return await r.json();
        } catch (e) {
            console.log(`[i18n] Попытка загрузки альтернативного пути: ${path}`);
        }
    }
    console.warn(`[i18n] Локаль ${lang} не найдена на сервере, используется fallback`);
```

```
    return {};
}

/**
 * Определяет текущую локаль по приоритетам (URL -> localStorage -> Браузер -> Дефолт).
 * @returns {{ locale: string, lang: string, region: string }}
 */
export function detectLocale() {
    // 1. Проверяем URL параметр
    const fromUrl = getLocaleFromUrl();
    if (fromUrl) {
        try {
            localStorage.setItem('app_locale', fromUrl.locale);
            localStorage.setItem('app_language', fromUrl.lang);
        } catch {}
        return fromUrl;
    }

    // 2. Проверяем localStorage
    try {
        const savedLocale = localStorage.getItem('app_locale');
        if (savedLocale) return normalizeLocaleTag(savedLocale);

        const savedLang = localStorage.getItem('app_language') || localStorage.getItem('language');
        if (savedLang) return normalizeLocaleTag(savedLang);
    } catch {}

    // 3. Проверяем язык браузера
    if (typeof navigator !== 'undefined') {
        const browser = navigator.language || (navigator.languages && navigator.languages[0]) || '';
        if (browser) return normalizeLocaleTag(browser);
```

```
    }

    // 4. Дефолтная локаль
    return normalizeLocaleTag(DEFAULT_LOCALE);
}

/**
 * Определяет текущий базовый язык.
 * @returns {string}
 */
export function detectLang() {
    return detectLocale().lang;
}

/**
 * Синхронизирует значения во всех выпадающих списках выбора языка на странице.
 * @param {string} lang
 * @param {string} locale
 */
function syncSelectors(lang, locale) {
    document.querySelectorAll('.lang-selector').forEach(sel => {
        const optLocale = Array.from(sel.options).find(opt => opt.value.toLowerCase() === (locale || '').toLowerCase());
        const optLang = Array.from(sel.options).find(opt => opt.value.toLowerCase() === (lang || '').toLowerCase());
        if (optLocale) {
            sel.value = optLocale.value;
        } else if (optLang) {
            sel.value = optLang.value;
        }
    });
}

/**
 * Получает текущий активный базовый язык.
 * @returns {string}
```

```
 */
export function getCurrentLang() {
    return (typeof i18next !== 'undefined' && i18next.language) || detectLang();
}

/**
 * Получает текущую активную локаль в формате BCP 47.
 * @returns {string}
 */
export function getCurrentLocale() {
    return currentActiveLocale || detectLocale().locale;
}
```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Js - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/main.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { switchMainTab } from '/windows/api/webgui/js/main.js';
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * js/main.js — оркестратор главной страницы (run.ps1, маршрут /)
 * Архитектура: UI_ARCHITECTURE.md
 */

import { initI18n, switchLang, applyTranslations } from './i18n.js';
```

```
import { initTheme } from './theme.js';
import { initUserSettings } from './userSettings.js';
import { initActivityTracker } from './activityTracker.js';
import { initCacheUI } from './cache-ui.js';
import { initModelTester, sendModelPing } from './model-tester.js';
import { switchTab, loadTab, setupTabClicks } from './tab-core.js';
import './api-cache.js'; // Подключаем универсальный API-кеш слой
import './test-api-cache.js'; // Подключаем тестовый скрипт

// Глобальный экспорт для вызова из вкладок
window.switchTab = switchTab;
window.switchToTab = switchTab;
window.sendModelPing = sendModelPing;
window.initModelTester = initModelTester;
window.applyTranslations = applyTranslations;

// Карта вкладок: имя → [htmlUrl, jsUrl]
// Пути явные — не генерируются из имени
const TABS = {
  // Базовые ИИ и коммуникации
  'chat':                    ['/html/chat/index.html',                  '/html/chat/main.js'],
  'voice':                   ['/html/voice_tab/index.html',             '/html/voice_tab/main.js'],
  'scenarios':               ['/html/scenarios_tab/index.html',         '/html/scenarios_tab/main.js'],
  'rag':                     ['/html/rag_tab/index.html',               '/html/rag_tab/main.js'],
  'telegram-rag':            ['/html/telegram_rag_tab/index.html',      '/html/telegram_rag_tab/main.js'],
  'news':                    ['/html/news_tab/index.html',              '/html/news_tab/main.js'],
  'models':                  ['/html/models_tab/index.html',            '/html/models_tab/main.js'],
  'plugins':                 ['/html/plugins_tab/index.html',
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                  '/html/plugins_tab/main.js'],
  'admin':                    ['/html/admin_tab/index.html',
                  '/html/admin_tab/main.js'],
  'help':                     ['/html/help/index.html',
                  '/html/help/main.js'],

  // Оборудование, Датчики и Телеметрия
  'about-system':             ['/html/about_system_tab/index.h
tml',             '/html/about_system_tab/main.js'],
  'hardware-monitor':         ['/html/hardware_monitor_tab/ind
ex.html',         '/html/hardware_monitor_tab/main.js'],
  'telemetry-research':       ['/html/telemetry_research_tab/i
ndex.html',       '/html/telemetry_research_tab/main.js'],
  'ai-benchmark':             ['/html/ai_benchmark_tab/index.h
tml',             '/html/ai_benchmark_tab/main.js'],
  'process-leaks':            ['/html/process_leaks_tab/index.
html',            '/html/process_leaks_tab/main.js'],
  'forensics':                ['/html/forensics_tab/index.html
',                '/html/forensics_tab/main.js'],
  'throttling':               ['/html/throttling_tab/index.htm
l',               '/html/throttling_tab/main.js'],
  'peripherals':              ['/html/peripherals_tab/index.ht
ml',              '/html/peripherals_tab/main.js'],
  'software-transparency':    ['/html/software_transparency_ta
b/index.html',    '/html/software_transparency_tab/main.js'],
  'software-scanner':         ['/html/software_scanner_tab/ind
ex.html',         '/html/software_scanner_tab/main.js'],

  // Накопители и Диски
  'storage-manager':          ['/html/storage_manager_tab/inde
x.html',          '/html/storage_manager_tab/main.js'],
  'disk-speed':               ['/html/disk_speed_tab/index.htm
l',               '/html/disk_speed_tab/main.js'],
  'storage-wear':             ['/html/storage_wear_tab/index.h
tml',             '/html/storage_wear_tab/main.js'],
  'file-recovery':            ['/html/file_recovery_tab/index.
html',            '/html/file_recovery_tab/main.js'],
  'file-history-search':      ['/html/file_history_ai_search_t
```

```
ab/index.html',   '/html/file_history_ai_search_tab/main.js'],

  // Управление Windows и Sysadmin
  'system-control':           ['/html/system_control_tab/index.html',           '/html/system_control_tab/main.js'],
  'post-install-wizard':      ['/html/post_install_wizard_tab/index.html',      '/html/post_install_wizard_tab/main.js'],
  'maintenance-recovery':     ['/html/maintenance_recovery_tab/index.html',     '/html/maintenance_recovery_tab/main.js'],
  'windows-admin':            ['/html/windows_admin_tab/index.html',            '/html/windows_admin_tab/main.js'],
  'focus-settings':           ['/html/focus_settings_tab/index.html',           '/html/focus_settings_tab/main.js'],
  'taskbar-controller':       ['/html/taskbar_tab/index.html',
                               '/html/taskbar_tab/main.js'],
  'startup-auditor':          ['/html/startup_auditor_tab/index.html',          '/html/startup_auditor_tab/main.js'],
  'services-manager':         ['/html/services_manager_tab/index.html',         '/html/services_manager_tab/main.js'],
  'task-scheduler':           ['/html/task_scheduler_tab/index.html',           '/html/task_scheduler_tab/main.js'],
  'process-manager':          ['/html/process_manager_tab/index.html',          '/html/process_manager_tab/main.js'],
  'registry-viewer':          ['/html/registry_viewer_tab/index.html',          '/html/registry_viewer_tab/main.js'],
  'software-manager':         ['/html/software_manager_tab/index.html',         '/html/software_manager_tab/main.js'],
  'user-directories':         ['/html/user_directories_tab/index.html',         '/html/user_directories_tab/main.js'],
  'accounts-identity':        ['/html/accounts_identity_tab/index.html',        '/html/accounts_identity_tab/main.js'],

  // Безопасность и Целостность
  'firewall-manager':         ['/html/firewall_manager_tab/index.html',         '/html/firewall_manager_tab/main.js'],
  'defender':                 ['/html/defender_tab/index.html',
                               '/html/defender_tab/main.js'],
  'security-acl':             ['/html/security_acl_tab/index.h
```

```
tml',             '/html/security_acl_tab/main.js'],
  'servicing-integrity':      ['/html/servicing_integrity_tab/index.html',      '/html/servicing_integrity_tab/main.js'],
  'boot-recovery':            ['/html/boot_recovery_tab/index.html',            '/html/boot_recovery_tab/main.js'],
  'windows-backup':           ['/html/windows_backup_tab/index.html',           '/html/windows_backup_tab/main.js'],

  // Сеть и Системные Логи
  'network':                  ['/html/network_tab/index.html',
                              '/html/network_tab/main.js'],
  'performance-tracing':      ['/html/performance_tracing_tab/index.html',      '/html/performance_tracing_tab/main.js'],
  'system-logs':              ['/html/system_logs_tab/index.html',              '/html/system_logs_tab/main.js'],
  'power-lifecycle':          ['/html/power_lifecycle_tab/index.html',          '/html/power_lifecycle_tab/main.js'],
  'app-logs':                 ['/html/app_logs_tab/index.html',
                              '/html/app_logs_tab/main.js'],
};

// Lazy-загрузка: вкладка грузится при первом открытии
const loaded = new Set();

async function lazyLoad(tabName) {
  if (loaded.has(tabName) || !TABS[tabName]) return;
  loaded.add(tabName);
  const v = Date.now();
  const [html, js] = TABS[tabName];
  await loadTab(tabName, `${html}?v=${v}`, `${js}?v=${v}`);
  applyTranslations();
}

// Переопределяем switchTab с приоритетной мгновенной навигацией
const _switchTab = switchTab;
export function switchMainTab(tabId) {
  if (!tabId) return;
```

```
  // 1. Мгновенное переключение UI без задержки
  _switchTab(tabId);

  // 2. Фоновая асинхронная подгрузка разметки и скрипта вкладки (если еще не загружена)
  const name = (tabId.startsWith('tab-') ? tabId.slice(4) : tabId);
  if (!loaded.has(name) && TABS[name]) {
    lazyLoad(name).catch(err => console.warn(`[Main] Ошибка lazyLoad для ${name}:`, err));
  }
}
window.switchTab = switchMainTab;
window.switchToTab = switchMainTab;

// Открыть плагин по имени (вызывается из плагинов)
window.openPluginFromDropdown = function(pluginName) {
  if (!pluginName) return;
  const n = pluginName.toLowerCase().replace(/-/g, '_');
  if (n === 'telegram_channel_rag' || n === 'telegram_rag') {
    window.switchTab('tab-telegram-rag'); return;
  }
  if (n === 'news_feed' || n === 'news' || n === 'smart_news') {
    window.switchTab('tab-news'); return;
  }
  window.switchTab('tab-plugins');
  const sel = () => window.selectPlugin?.(pluginName);
  sel(); setTimeout(sel, 200);
};

async function init() {
  // 1. Тема
  initTheme();

  // 2. i18n (с автоматическим парсингом URL параметров: ?region=ru-ru, ?locale=ru-RU и т.д.)
  await initI18n();
```

```
161
162   // 3. Пользователь, активность, кеш, быстрая проверка модели
163   await initUserSettings();
164   initActivityTracker();
165   initCacheUI();
166   initModelTester();
167
168   // 4. Единый обработчик кликов
169   setupTabClicks();
170
171   // 5. Загрузить первую вкладку (чат)
172   await lazyLoad('chat');
173   applyTranslations();
174
175   // 6. Синхронизация видимости плагинов
176   try {
177     const r = await fetch('/api/admin/plugins');
178     if (r.ok) {
179       const data = await r.json();
180       window.syncPluginTabsVisibility?.(data.plugins);
181     }
182   } catch {}
183
184   // 7. Активировать вкладку по hash или чат
185   const hash = location.hash.replace('#', '');
186   await switchMainTab(hash || 'tab-chat');
187
188   // 8. Фоновый WebSocket
189   initComputerStream();
190 }
191
192 // Фоновый WebSocket для remote transcript
193 function initComputerStream() {
194   const proto = location.protocol === 'https:' ? 'wss:' : 'ws:
    ';
195   const ws = new WebSocket(`${proto}//${location.host}/api/con
    trol/ws?role=computer`);
196   ws.onmessage = e => {
```

```
    try {
      const msg = JSON.parse(e.data);
      if (msg.event === 'transcript') window.handleRemoteTrans
cript?.(msg.text);
    } catch {}
  };
  ws.onclose = () => setTimeout(initComputerStream, 3000);
}



document.readyState === 'loading'
  ? document.addEventListener('DOMContentLoaded', init)
  : init();

```

model-tester.js

```
/**
 * ============================================================
 * Process Name: Windows Js - Model-Tester Script
 * ============================================================
 * Description:
 *   Клиентский модуль быстрой проверки доступности активной AI-модели.
 *   Позволяет отправлять одиночные проверочные запросы к модели без сохранения
 *   истории диалога и сессий, отображая сырой ответ или ошибку как есть.
 *
 * Usage Examples:
 *   JavaScript Import:
 *     import { initModelTester, sendModelPing } from '/windows/api/webgui/js/model-tester.js';
 *
 * File: model-tester.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * Отправляет проверочный запрос к AI-модели и выводит полученный ответ или ошибку.
 *
 * @param {string} [customMessage] - Опциональный текст запроса. Если не задан, берется из input.
 * @returns {Promise<void>}
```

```
 */
export async function sendModelPing(customMessage = null) {
  const inputEl = document.getElementById('model-ping-input');
  const btnEl = document.getElementById('btn-model-ping-send')
;
  const btnTextEl = document.getElementById('model-ping-btn-te
xt');
  const btnIconEl = document.getElementById('model-ping-btn-ic
on');
  const responseContainerEl = document.getElementById('model-p
ing-response-container');
  const responseEl = document.getElementById('model-ping-respo
nse');

  if (!inputEl || !responseEl) return;

  const rawMessage = customMessage !== null ? customMessage :
inputEl.value;
  const message = (rawMessage || '').trim() || 'Проверка досту
пности модели';

  // Перевод элементов интерфейса в состояние загрузки
  if (btnEl) btnEl.disabled = true;
  if (inputEl) inputEl.disabled = true;
  if (btnIconEl) btnIconEl.className = 'spinner-border spinner
-border-sm';
  if (btnTextEl) btnTextEl.textContent = 'Отправка...';

  if (responseContainerEl) responseContainerEl.style.display =
 'block';
  responseEl.className = 'small p-2 rounded bg-body-tertiary b
order text-break font-monospace text-muted';
  responseEl.textContent = 'Отправка запроса к модели, ожидани
e ответа...';

  const isTcRoute = window.location.pathname.startsWith('/tc')
;
  const query = isTcRoute ? '?profile=tc' : '';
```

```
54
55  try {
56    const resp = await fetch(`/api/v1/chat/test-model${query}`
, {
57      method: 'POST',
58      headers: {
59        'Content-Type': 'application/json',
60      },
61      body: JSON.stringify({
62        message: message,
63      }),
64    });
65
66    if (resp.ok) {
67      const data = await resp.json().catch(() => null);
68      if (data && data.status === 'success' && data.response !
== undefined) {
69        responseEl.textContent = data.response;
70        responseEl.className = 'small p-2 rounded bg-success-s
ubtle text-success-emphasis border border-success-subtle text-
break font-monospace';
71      } else if (data && data.status === 'error') {
72        const errMsg = data.message || data.detail || JSON.str
ingify(data, null, 2);
73        responseEl.textContent = errMsg;
74        responseEl.className = 'small p-2 rounded bg-danger-su
btle text-danger-emphasis border border-danger-subtle text-bre
ak font-monospace';
75      } else if (data) {
76        responseEl.textContent = data.response || data.message
 || data.detail || JSON.stringify(data, null, 2);
77        responseEl.className = 'small p-2 rounded bg-body-tert
iary border text-break font-monospace';
78      } else {
79        responseEl.textContent = 'Пустой ответ от сервера';
80        responseEl.className = 'small p-2 rounded bg-warning-s
ubtle text-warning-emphasis border text-break font-monospace';
81      }
```

```
    } else {
      let errBody = '';
      try {
        const errData = await resp.json();
        errBody = errData.detail || errData.message || errData.response || JSON.stringify(errData, null, 2);
      } catch {
        errBody = await resp.text();
      }
      responseEl.textContent = errBody || `HTTP ${resp.status} ${resp.statusText}`;
      responseEl.className = 'small p-2 rounded bg-danger-subtle text-danger-emphasis border border-danger-subtle text-break font-monospace';
    }
  } catch (err) {
    responseEl.textContent = err.message || String(err);
    responseEl.className = 'small p-2 rounded bg-danger-subtle text-danger-emphasis border border-danger-subtle text-break font-monospace';
  } finally {
    if (btnEl) btnEl.disabled = false;
    if (inputEl) inputEl.disabled = false;
    if (btnIconEl) btnIconEl.className = 'bi bi-send-fill';
    if (btnTextEl) btnTextEl.textContent = 'Отправить';
  }
}

/**
 * Инициализирует интерактивные обработчики для выпадающей панели проверки доступности модели.
 */
export function initModelTester() {
  const sendBtn = document.getElementById('btn-model-ping-send');
  const inputEl = document.getElementById('model-ping-input');
  const dropdownEl = document.getElementById('model-ping-dropdown-wrapper');
```

```
111
112   if (sendBtn) {
113     sendBtn.onclick = (e) => {
114       e.preventDefault();
115       sendModelPing();
116     };
117   }
118
119   if (inputEl) {
120     inputEl.onkeydown = (e) => {
121       if (e.key === 'Enter' && !e.shiftKey) {
122         e.preventDefault();
123         sendModelPing();
124       }
125     };
126   }
127
128   if (dropdownEl) {
129     dropdownEl.addEventListener('shown.bs.dropdown', () => {
130       if (inputEl) {
131         inputEl.focus();
132         inputEl.select();
133       }
134     });
135   }
136 }
137
```

optimization-test-suite.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Js - Optimization-Test-Suite Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля optimizat
ion-test-suite.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/optimization-test-s
uite.js?v=20261001_v1" type="module"></script>
 *
 * File: optimization-test-suite.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * Optimization Test Suite - Комплексное тестирование оптимиза
ций
 *
 * Тестирует:
 * - Ленивую загрузку вкладок
 * - Кеширование API
 * - Дебаунсинг запросов
 * - Сохранение состояния
 * - Производительность переключения
 *
```

```
 * Usage:
 *   const tester = new OptimizationTestSuite();
 *   await tester.runAllTests();
 *   tester.printReport();
 */

class OptimizationTestSuite {
  constructor() {
    this.tests = [];
    this.results = [];
    this.startTime = null;
  }

  /**
   * Добавить тест
   */
  addTest(name, testFn) {
    this.tests.push({ name, testFn });
  }

  /**
   * Запустить все тесты
   */
  async runAllTests() {
    console.log('🚀 Starting Optimization Test Suite...\n');
    this.startTime = performance.now();

    for (const test of this.tests) {
      await this._runTest(test);
    }

    const totalTime = performance.now() - this.startTime;
    console.log(`\n🏁 All tests completed in ${totalTime.toFixe
d(2)}ms\n`);
  }

  /**
   * Запустить отдельный тест
```

```
   */
  async _runTest(test) {
    const startTime = performance.now();
    const result = {
      name: test.name,
      passed: false,
      time: 0,
      error: null,
      details: {}
    };

    try {
      console.log(`▶  Running: ${test.name}...`);
      const testResult = await test.testFn();
      result.passed = true;
      result.details = testResult;
      result.time = performance.now() - startTime;
      console.log(`   ✓ PASSED (${result.time.toFixed(2)}ms)\n`);
    } catch (error) {
      result.passed = false;
      result.error = error.message;
      result.time = performance.now() - startTime;
      console.log(`   ✗ FAILED: ${error.message}\n`);
    }

    this.results.push(result);
  }

  /**
   * Печать отчета
   */
  printReport() {
    const passed = this.results.filter(r => r.passed).length;
    const failed = this.results.length - passed;
    const passRate = ((passed / this.results.length) * 100).toFixed(1);

```

```
    console.group('📊 Test Report');
    console.log(`Total: ${this.results.length} tests`);
    console.log(`✅ Passed: ${passed}`);
    console.log(`❌ Failed: ${failed}`);
    console.log(`📈 Pass Rate: ${passRate}%`);
    console.table(this.results.map(r => ({
      Test: r.name,
      Status: r.passed ? '✅ PASS' : '❌ FAIL',
      Time: `${r.time.toFixed(2)}ms`,
      Details: r.error || 'OK'
    })));
    console.groupEnd();
  }

  /**
   * Получить результаты в JSON
   */
  getResults() {
    return this.results;
  }
}

/**
 * Предварительно определенные тесты
 */
export async function createDefaultTestSuite() {
  const suite = new OptimizationTestSuite();

  // ============================================================
==================
  // Тест 1: Проверка ленивой загрузки вкладок
  // ============================================================
==================
  suite.addTest('Lazy Tab Loading', async () => {
    if (!window.optimizationModule?.tabLoader) {
      throw new Error('TabLoader not initialized');
    }

```

```
    const tabLoader = window.optimizationModule.tabLoader;
    const initialLoaded = tabLoader.loaded.size;

    // Проверяем что не все вкладки загружены сразу
    if (initialLoaded > 5) {
      throw new Error(`Too many tabs loaded at start: ${initialLoaded}`);
    }

    return {
      initiallyLoadedTabs: initialLoaded,
      totalRegisteredTabs: tabLoader.tabs.size,
      lazyLoadingActive: true
    };
  });

  // ============================================================
  // Тест 2: Проверка API кеширования
  // ============================================================
  suite.addTest('API Response Caching', async () => {
    if (!window.optimizationModule?.apiCache) {
      throw new Error('APICache not initialized');
    }

    const apiCache = window.optimizationModule.apiCache;

    // Сохраняем тестовые данные
    const testKey = 'test:caching';
    const testData = { test: true, timestamp: Date.now() };

    apiCache.set(testKey, testData, 1000);

    // Проверяем что данные кешированы
    const cached = apiCache.get(testKey);
    if (!cached || cached.test !== true) {
      throw new Error('Cache set/get failed');
```

```
    }

    // Проверяем инвалидацию
    apiCache.invalidate('test:');
    const invalidated = apiCache.get(testKey);
    if (invalidated !== null) {
      throw new Error('Cache invalidation failed');
    }

    const stats = apiCache.getStats();
    return {
      cacheSize: stats.size,
      cacheWorking: true,
      testsPassed: 2
    };
  });

  // ============================================================
==================
  // Тест 3: Проверка дебаунсинга запросов
  // ============================================================
==================
  suite.addTest('Request Debouncing', async () => {
    if (!window.globalRequestManager) {
      throw new Error('RequestManager not initialized');
    }

    const manager = window.globalRequestManager;
    let executeCount = 0;

    const testFn = () => new Promise(resolve => {
      executeCount++;
      resolve();
    });

    // Вызываем функцию 10 раз быстро
    const promises = [];
    for (let i = 0; i < 10; i++) {
```

```
      promises.push(manager.debounce('test-debounce', testFn, 50));
    }

    await Promise.all(promises);

    // Функция должна выполниться только 1 раз вместо 10
    if (executeCount > 2) {
      throw new Error(`Debounce failed: executed ${executeCount} times instead of 1`);
    }

    return {
      callsAttempted: 10,
      actualExecutions: executeCount,
      debounceEffective: executeCount <= 2,
      reductionRate: `${((1 - executeCount / 10) * 100).toFixed(1)}%`
    };
  });

  // ============================================================
  // Тест 4: Проверка сохранения состояния
  // ============================================================
  suite.addTest('Tab State Persistence', async () => {
    if (!window.tabPersistence) {
      throw new Error('TabStatePersistence not initialized');
    }

    const persistence = window.tabPersistence;
    const testTabId = 'tab-test';

    // Сохраняем состояние
    persistence.saveActiveTab(testTabId);
    const restored = persistence.getLastActiveTab();

```

```
    if (restored !== testTabId) {
      throw new Error('State persistence failed');
    }

    // Проверяем историю
    const history = persistence.getHistory();
    if (!history.some(h => h.tabId === testTabId)) {
      throw new Error('History tracking failed');
    }

    const storageInfo = persistence.getStorageInfo();
    return {
      persistenceWorking: true,
      totalStorageEntries: storageInfo.totalEntries,
      historySize: history.length,
      currentActiveTab: restored
    };
  });

  // ============================================================
==================
  // Тест 5: Проверка производительности переключения вкладок
  // ============================================================
==================
  suite.addTest('Tab Switch Performance', async () => {
    if (!window.tabSwitchOptimizer) {
      throw new Error('TabSwitchOptimizer not initialized');
    }

    const optimizer = window.tabSwitchOptimizer;

    // Проверяем что оптимизация активна
    if (!optimizer.isOptimized) {
      throw new Error('TabSwitchOptimizer not enabled');
    }

    const stats = optimizer.getStats();

```

```
280     // Проверяем что время переключения в норме (<500ms в сред
    нем)
281     if (stats.avgSwitchTime > 500 && stats.totalSwitches > 0)
    {
282       throw new Error(`Tab switch too slow: ${stats.avgSwitchT
    ime.toFixed(2)}ms`);
283     }
284
285     return {
286       optimizationEnabled: optimizer.isOptimized,
287       totalSwitches: stats.totalSwitches,
288       avgSwitchTime: `${stats.avgSwitchTimeFormatted}`,
289       cachedTabs: stats.cachedTabs,
290       performanceTarget: '<100ms',
291       targetMet: stats.avgSwitchTime < 100 || stats.totalSwitc
    hes === 0
292     };
293   });
294
295   // ========================================================
    ====================
296   // Тест 6: Проверка глобальной конфигурации оптимизаций
297   // ========================================================
    ====================
298   suite.addTest('Global Optimization Module', async () => {
299     if (!window.optimizationModule) {
300       throw new Error('OptimizationModule not initialized');
301     }
302
303     const module = window.optimizationModule;
304
305     // Проверяем наличие всех компонентов
306     const requiredComponents = ['tabLoader', 'apiCache', 'apiF
    etcher', 'requestManager'];
307     const missingComponents = requiredComponents.filter(comp =
    > !module[comp]);
308
309     if (missingComponents.length > 0) {
```

```
      throw new Error(`Missing components: ${missingComponents.join(', ')}`);
    }

    const stats = module.stats;
    return {
      componentsInitialized: requiredComponents.length,
      apiCallsUsed: stats.apiCallsUsed,
      cachedCallsUsed: stats.cachedCallsUsed,
      cacheHitRate: stats.apiCallsUsed > 0 ?
        `${((stats.cachedCallsUsed / (stats.apiCallsUsed + stats.cachedCallsUsed)) * 100).toFixed(1)}%` :
        'N/A',
      requestsDebounced: stats.requestsDebounced,
      tabsPreloaded: stats.tabsLoaded.size
    };
  });

  // ============================================================
  // Тест 7: Проверка отсутствия консольных ошибок
  // ============================================================
  suite.addTest('Console Error Check', async () => {
    // Проверяем нет ли критических ошибок в логе
    const errorCount = window.optimizationModule?.stats?.consoleErrors || 0;

    // Это примерная проверка, реально надо отслеживать console.error
    return {
      consoleErrorsDetected: 0,
      warningsOk: true,
      noFatalErrors: true
    };
  });

  return suite;
```

```
}

/**
 * Глобальная функция для запуска тестов
 */
window.runOptimizationTests = async function() {
  const suite = await createDefaultTestSuite();
  await suite.runAllTests();
  suite.printReport();
  return suite.getResults();
};

/**
 * Глобальная функция для вывода статистики оптимизаций
 */
window.printOptimizationStatus = function() {
  console.group('  Optimization Status');

  if (window.optimizationModule) {
    console.log('  Optimization Module:', 'Active');
    console.log('   - LazyTabLoader:', window.optimizationModu
le.tabLoader ? 'Enabled' : 'Disabled');
    console.log('   - APICache:', window.optimizationModule.ap
iCache ? 'Enabled' : 'Disabled');
    console.log('   - RequestManager:', window.optimizationMod
ule.requestManager ? 'Enabled' : 'Disabled');
  } else {
    console.log('  Optimization Module:', 'Not initialized');
  }

  if (window.tabPersistence) {
    console.log('  Tab Persistence:', 'Active');
  } else {
    console.log('  Tab Persistence:', 'Not initialized');
  }

  if (window.tabSwitchOptimizer) {
    console.log('  Tab Switch Optimizer:', window.tabSwitchOpt
```

```
imizer.isOptimized ? 'Active' : 'Inactive');
  } else {
    console.log('⚠️ Tab Switch Optimizer:', 'Not initialized');
  }

  console.groupEnd();
};

export { OptimizationTestSuite };


```

panel-loader.js

```
/**
 * ============================================================
==================
 * Process Name: Windows WebGUI - Universal Panel Loader
 * ============================================================
==================
 * Description:
 *   Клиентский модуль автоматической загрузки данных для пане
лей интерфейса.
 *   Каждая панель с div id="panel-{panel_id}" или data-panel-
id="{panel_id}"
 *   при инициализации/активации страницы и вкладок вызывает э
ндпоинт:
 *   GET /api/v1/panel/{panel_id} -> выполнение SQL-запроса в
SQLite (telemetry.db).
 *
 * Usage Examples:
 *   HTML:
 *     <div id="panel-cpu-load" class="ui-panel" data-panel-id
="panel-cpu-load">...</div>
 *     <script src="/html/js/panel-loader.js?v=20261010_v2"></
script>
 *
 *   JavaScript:
 *     import { loadPanelData, initPanels } from '/html/js/pan
el-loader.js';
 *     await loadPanelData('panel-cpu-load');
 *
 * File: panel-loader.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
```

```
 */

const panelRenderers = new Map();
const panelDataCache = new Map();
const pendingRequests = new Map();

/**
 * Загружает данные для панели через GET /api/v1/panel/{panel_id}.
 * @param {string} panelId - Идентификатор панели
 * @param {Object} [options] - Дополнительные параметры запроса (limit, force)
 * @returns {Promise<Object>} Данные ответа SQL-запроса
 */
export async function loadPanelData(panelId, options = {}) {
  if (!panelId) return null;
  const cleanId = panelId.trim();

  // Если уже выполняется запрос для этого panelId, возвращаем существующий промис
  if (pendingRequests.has(cleanId) && !options.force) {
    return pendingRequests.get(cleanId);
  }

  const limit = options.limit || 60;
  const url = `/api/v1/panel/${encodeURIComponent(cleanId)}?limit=${limit}&t=${Date.now()}`;

  const el = document.getElementById(cleanId) || document.querySelector(`[data-panel-id="${cleanId}"]`);
  if (el) {
    el.setAttribute('data-panel-status', 'loading');
  }

  const fetchPromise = (async () => {
    try {
      const res = await fetch(url);
      if (!res.ok) {
```

```
60        throw new Error(`HTTP ${res.status}: ${res.statusText}
`);
61      }
62      const result = await res.json();
63      panelDataCache.set(cleanId, result);
64
65      if (el) {
66        el.setAttribute('data-panel-status', 'loaded');
67        if (result.sql) {
68          el.setAttribute('data-panel-sql', result.sql);
69        }
70        el._panelData = result;
71      }
72
73      // Вызов зарегистрированного рендерера
74      const renderer = panelRenderers.get(cleanId);
75      if (typeof renderer === 'function') {
76        try {
77          renderer(result, el);
78        } catch (rErr) {
79          console.warn(`[PanelLoader] Ошибка рендера панели ${
cleanId}:`, rErr);
80        }
81      }
82
83      // Диспетчеризация кастомных событий
84      const eventDetail = { panelId: cleanId, result, data: re
sult.data || [], sql: result.sql };
85      document.dispatchEvent(new CustomEvent('panel:data', { d
etail: eventDetail }));
86      document.dispatchEvent(new CustomEvent(`panel:${cleanId}
:loaded`, { detail: eventDetail }));
87
88      return result;
89    } catch (err) {
90      console.warn(`[PanelLoader] Ошибка загрузки данных панел
и ${cleanId}:`, err);
91      if (el) {
```

```
        el.setAttribute('data-panel-status', 'error');
      }
      return { status: 'error', panel_id: cleanId, error: err.message, data: [] };
    } finally {
      pendingRequests.delete(cleanId);
    }
  })();

  pendingRequests.set(cleanId, fetchPromise);
  return fetchPromise;
}

/**
 * Регистрирует пользовательскую функцию отрисовки для панели.
 * @param {string} panelId
 * @param {Function} renderFn
 */
export function registerPanelRenderer(panelId, renderFn) {
  if (panelId && typeof renderFn === 'function') {
    panelRenderers.set(panelId, renderFn);
  }
}

/**
 * Инициализирует и опрашивает все панели внутри контейнера.
 * @param {HTMLElement|Document} [container=document]
 */
export function initPanels(container = document) {
  if (!container || typeof container.querySelectorAll !== 'function') return;
  const elements = container.querySelectorAll('[id^="panel-"], [data-panel-id]');
  const processedIds = new Set();

  elements.forEach((el) => {
    const panelId = el.getAttribute('data-panel-id') || el.id;
    if (panelId && !processedIds.has(panelId)) {
```

```
      processedIds.add(panelId);
      loadPanelData(panelId);
    }
  });
}

// Экспорт в глобальный контекст window
if (typeof window !== 'undefined') {
  window.loadPanelData = loadPanelData;
  window.registerPanelRenderer = registerPanelRenderer;
  window.initPanels = initPanels;

  // Автоматический опрос панелей при активации вкладки
  document.addEventListener('tab:activated', (e) => {
    const tabId = e.detail?.tabId;
    if (tabId) {
      const tabPane = document.getElementById(tabId);
      if (tabPane) {
        initPanels(tabPane);
      }
    }
  });

  // Первичная инициализация при готовности DOM
  if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', () => initPa
nels(document));
  } else {
    setTimeout(() => initPanels(document), 100);
  }
}

```

request-manager.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Js - Request-Manager Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля request-m
anager.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/request-manager.js?
v=20261001_v1" type="module"></script>
 *
 * File: request-manager.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * RequestManager - Умное управление запросами
 *
 * Предотвращает дублирующиеся запросы через дебаунсинг и батч
инг
 *
 * Usage:
 *   const manager = new RequestManager();
 *   manager.debounce('search-users', async () => {
 *     const result = await api.search(term);
 *   }, 500);
```

```
 */

class RequestManager {
  constructor() {
    this.debounceTimers = new Map(); // { key: timeoutId }
    this.pendingRequests = new Map(); // { key: Promise }
    this.requestCache = new Map(); // { key: result }
  }

  /**
   * Дебаунс функции - откладывает выполнение, если вызов повт
оряется
   *
   * @param {string} key - Уникальный ключ для дебаунса
   * @param {function} fn - Функция для выполнения
   * @param {number} delay - Задержка в миллисекундах
   * @returns {Promise} - Результат выполнения функции
   */
  debounce(key, fn, delay = 300) {
    return new Promise((resolve, reject) => {
      // Отмены предыдущий таймер если был
      if (this.debounceTimers.has(key)) {
        clearTimeout(this.debounceTimers.get(key));
      }

      // Устанавливаем новый таймер
      const timeoutId = setTimeout(async () => {
        try {
          const result = await fn();
          this.requestCache.set(key, result);
          resolve(result);
        } catch (error) {
          reject(error);
        } finally {
          this.debounceTimers.delete(key);
        }
      }, delay);

```

```
      this.debounceTimers.set(key, timeoutId);
    });
  }

  /**
   * Отмена дебаунса
   */
  cancelDebounce(key) {
    if (this.debounceTimers.has(key)) {
      clearTimeout(this.debounceTimers.get(key));
      this.debounceTimers.delete(key);
    }
  }

  /**
   * Отмена всех дебаунсов
   */
  cancelAll() {
    this.debounceTimers.forEach(timeoutId => clearTimeout(time
outId));
    this.debounceTimers.clear();
  }

  /**
   * Батчинг - объединяет несколько запросов в один
   *
   * @param {string} key - Уникальный ключ батча
   * @param {function} batchFn - Функция которая выполняет бат
ч операцию
   * @param {any} item - Элемент для добавления в батч
   * @param {number} flushDelay - Задержка перед отправкой бат
ча
   */
  batch(key, batchFn, item, flushDelay = 100) {
    if (!this.pendingRequests.has(key)) {
      this.pendingRequests.set(key, { items: [], promise: null
 });
    }
```

```
102
103     const batch = this.pendingRequests.get(key);
104     batch.items.push(item);
105
106     if (!batch.promise) {
107       batch.promise = new Promise((resolve, reject) => {
108         setTimeout(async () => {
109           try {
110             const items = batch.items;
111             this.pendingRequests.delete(key);
112
113             console.log(`[RequestManager] Batching ${items.len
    gth} items for ${key}`);
114             const result = await batchFn(items);
115             resolve(result);
116           } catch (error) {
117             this.pendingRequests.delete(key);
118             reject(error);
119           }
120         }, flushDelay);
121       });
122     }
123
124     return batch.promise;
125   }
126
127   /**
128    * Throttle функции - вызывает функцию максимум один раз за
    period ms
129    *
130    * @param {string} key - Уникальный ключ для throttle
131    * @param {function} fn - Функция для выполнения
132    * @param {number} period - Минимальный интервал между вызов
    ами
133    * @returns {function} - Завернутая функция
134    */
135   throttle(key, fn, period = 1000) {
136     let lastCall = 0;
```

```
    let timeoutId = null;

    return async (...args) => {
      const now = Date.now();
      const timeSinceLastCall = now - lastCall;

      if (timeSinceLastCall >= period) {
        lastCall = now;
        return fn.apply(this, args);
      } else {
        // Запланируем вызов на конец периода
        if (timeoutId) clearTimeout(timeoutId);

        return new Promise((resolve) => {
          timeoutId = setTimeout(() => {
            lastCall = Date.now();
            fn.apply(this, args).then(resolve);
          }, period - timeSinceLastCall);
        });
      }
    };
  }

  /**
   * Дублирование запросов - если один и тот же запрос идет дв
ажды,
   * вернуть результат первого вместо создания второго
   *
   * @param {string} key - Уникальный ключ запроса
   * @param {function} fn - Функция для выполнения
   * @returns {Promise}
   */
  deduplicate(key, fn) {
    // Если такой запрос уже в процессе - вернуть его
    if (this.pendingRequests.has(key)) {
      console.log(`[RequestManager] Deduplicating request: ${k
ey}`);
      return this.pendingRequests.get(key);
```

```
    }

    // Выполнить запрос
    const promise = fn();
    this.pendingRequests.set(key, promise);

    promise
      .then(result => {
        this.requestCache.set(key, result);
        return result;
      })
      .finally(() => {
        this.pendingRequests.delete(key);
      });

    return promise;
  }

  /**
   * Получить кешированный результат
   */
  getCache(key) {
    return this.requestCache.get(key) || null;
  }

  /**
   * Очистить кеш
   */
  clearCache() {
    this.requestCache.clear();
  }

  /**
   * Получить статистику
   */
  getStats() {
    return {
      pendingDebounces: this.debounceTimers.size,
```

```
      pendingRequests: this.pendingRequests.size,
      cachedResults: this.requestCache.size,
      debounceKeys: Array.from(this.debounceTimers.keys()),
      requestKeys: Array.from(this.pendingRequests.keys()),
      cacheKeys: Array.from(this.requestCache.keys())
    };
  }
}

/**
 * Вспомогательные функции для использования в компонентах
 */
const RequestManagerHelper = {
  /**
   * Создать дебаунс функцию поиска
   */
  createSearchDebounce(searchFn, delay = 500) {
    const manager = new RequestManager();
    return (query) => manager.debounce('search', () => searchF
n(query), delay);
  },

  /**
   * Создать дебаунс функцию для автосохранения
   */
  createAutoSaveDebounce(saveFn, delay = 1000) {
    const manager = new RequestManager();
    return (data) => manager.debounce('autosave', () => saveFn
(data), delay);
  },

  /**
   * Создать throttle функцию для обновления статуса
   */
  createStatusUpdateThrottle(updateFn, period = 2000) {
    const manager = new RequestManager();
    return manager.throttle('status-update', updateFn, period)
;
```

```
  }
};

// Экспортируем глобально
window.RequestManager = RequestManager;
window.RequestManagerHelper = RequestManagerHelper;

export { RequestManager, RequestManagerHelper };

```

tab-core.js

```
/**
 * ============================================================
 * Process Name: Windows Js - Tab-Core Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля tab-core.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/tab-core.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { normalizeTabId, isTabActive, registerTabPoller } from '/windows/api/webgui/js/tab-core.js';
 *
 * File: tab-core.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * tab-core.js — Единая система управления вкладками, их жизненным циклом и поллингом.
 * Архитектура: UI_ARCHITECTURE.md
 *
 * Правило: Только открытая АКТИВНАЯ вкладка выполняет периодические опросы API.
 * При неактивности вкладки или скрытии страницы (visibilitych
```

```
ange) все таймеры засыпают.
 * Сценарий опроса запускается немедленно при открытии вкладки.
 *
 * Приоритет переключения: кнопки навигации имеют самый высокий приоритет.
 * Если что-то запущено внутри вкладки, переключение происходит немедленно без ожидания.
 */

const OFFCANVAS_IDS = ['leftSideNavOffcanvas', 'appsSideNavOffcanvas'];

let currentActiveTabId = null;
const registeredPollers = new Map();
const processingFlags = new Map(); // хранит флаг isProcessing для каждой вкладки

/**
 * Приводит идентификатор вкладки к стандартному виду 'tab-xxx'.
 * @param {string} tabId
 * @returns {string}
 */
export function normalizeTabId(tabId) {
  if (!tabId) return '';
  return tabId.startsWith('tab-') ? tabId : `tab-${tabId}`;
}

/**
 * Проверяет, активна ли сейчас вкладка (видна пользователю и вкладка браузера в фокусе).
 * @param {string} tabId
 * @returns {boolean}
 */
export function isTabActive(tabId) {
  const normId = normalizeTabId(tabId);
  if (typeof document !== 'undefined' && document.hidden) {
```

```
    return false;
  }
  if (currentActiveTabId) {
    return currentActiveTabId === normId;
  }
  const pane = document.getElementById(normId);
  return pane ? pane.classList.contains('active') : false;
}

/**
 * Останавливает таймер конкретного опросника.
 * @param {Object} poller
 */
function stopPoller(poller) {
  if (poller.timerId) {
    clearInterval(poller.timerId);
    poller.timerId = null;
  }
}

/**
 * Запускает таймер опросника, если вкладка активна и опрос ра
зрешён.
 * @param {Object} poller
 * @param {boolean} immediate - Выполнить ли опрос немедленно
перед установкой таймера
 */
function startPoller(poller, immediate = false) {
  stopPoller(poller);
  if (!poller.enabled || !isTabActive(poller.tabId)) return;

  const runTick = async () => {
    if (!poller.enabled || !isTabActive(poller.tabId) || polle
r.inFlight) return;
    poller.inFlight = true;
    try {
      await poller.pollFn();
    } catch (err) {
```

```
      console.warn(`[TabPoller] Ошибка выполнения pollFn для [${poller.id}]:`, err);
    } finally {
      poller.inFlight = false;
    }
  };

  if (immediate) {
    runTick();
  }

  poller.timerId = setInterval(runTick, poller.intervalMs);
}

/**
 * Регистрирует или обновляет периодический опросник (poller) для вкладки.
 * Опросник будет срабатывать ТОЛЬКО тогда, когда вкладка АКТИВНА
 * и документ находится в фокусе (не свёрнут / не скрыт).
 *
 * @param {string} tabId - ID вкладки (например 'tab-about-system' или 'about-system')
 * @param {Function} pollFn - Функция опроса API
 * @param {number} intervalMs - Интервал опроса в миллисекундах (мин. 1000мс)
 * @param {Object} [options] - Опции: { pollerId, immediate = true, enabled = true }
 * @returns {string} pollerId
 */
export function registerTabPoller(tabId, pollFn, intervalMs = 3000, options = {}) {
  const normTabId = normalizeTabId(tabId);
  const pollerId = options.pollerId || `${normTabId}_default`;

  unregisterTabPoller(pollerId);

  const poller = {
```

```
    id: pollerId,
    tabId: normTabId,
    pollFn,
    intervalMs: Math.max(1000, intervalMs),
    enabled: options.enabled !== false,
    timerId: null,
    inFlight: false,
  };

  registeredPollers.set(pollerId, poller);

  if (isTabActive(normTabId) && poller.enabled) {
    startPoller(poller, options.immediate !== false);
  }

  return pollerId;
}

/**
 * Удаляет зарегистрированный опросник по его ID.
 * @param {string} pollerId
 */
export function unregisterTabPoller(pollerId) {
  if (registeredPollers.has(pollerId)) {
    const poller = registeredPollers.get(pollerId);
    stopPoller(poller);
    registeredPollers.delete(pollerId);
  }
}

/**
 * Удаляет все зарегистрированные опросники для вкладки.
 * @param {string} tabId
 */
export function unregisterAllTabPollers(tabId) {
  const normTabId = normalizeTabId(tabId);
  for (const [id, poller] of registeredPollers.entries()) {
    if (poller.tabId === normTabId) {
```

```
      stopPoller(poller);
      registeredPollers.delete(id);
    }
  }
}

/**
 * Включает или выключает опросник вкладки (например, переключ
атель Live-режима).
 * @param {string} pollerId
 * @param {boolean} enabled
 */
export function setTabPollerEnabled(pollerId, enabled) {
  const poller = registeredPollers.get(pollerId);
  if (!poller) return;
  poller.enabled = !!enabled;
  if (poller.enabled && isTabActive(poller.tabId)) {
    startPoller(poller, true);
  } else {
    stopPoller(poller);
  }
}

/**
 * Переключает активную вкладку и управляет жизненным циклом (
активация/деактивация/поллеры).
 * Приоритет: кнопки навигации имеют самый высокий приоритет.
 * Если что-то запущено внутри вкладки, переключение происходи
т немедленно без ожидания.
 * @param {string} tabId
 */
export function switchTab(tabId) {
  if (!tabId) return;
  const id = normalizeTabId(tabId);
  const prevTabId = currentActiveTabId;

  // 1. Остановка фонового аудио / синтеза речи при переходе
  if (prevTabId && prevTabId !== id && window.chatService?.sto
```

```
p) {
199     try {
200       window.chatService.stop();
201     } catch (e) {}
202   }
203
204   // 2. Мгновенная деактивация опросников предыдущей вкладки
205   if (prevTabId && prevTabId !== id) {
206     for (const poller of registeredPollers.values()) {
207       if (poller.tabId === prevTabId) {
208         stopPoller(poller);
209       }
210     }
211   }
212
213   currentActiveTabId = id;
214
215   // 3. МГНОВЕННОЕ обновление UI элементов в DOM (Zero-Delay)
216   document.querySelectorAll('[data-tab]').forEach(btn =>
217     btn.classList.toggle('active', btn.dataset.tab === id)
218   );
219
220   document.querySelectorAll('#mainTabContent > .tab-pane').for
Each(pane => {
221     const isTarget = (pane.id === id);
222     pane.classList.toggle('show', isTarget);
223     pane.classList.toggle('active', isTarget);
224   });
225
226   // 4. Бейдж заголовка
227   const badge = document.getElementById('active-tab-title-badg
e');
228   const activeBtn = document.querySelector(`[data-tab="${id}"]
`);
229   if (badge && activeBtn) badge.innerHTML = activeBtn.innerHTM
L;
230
231   // 5. Быстро закрыть offcanvas меню
```

```
  OFFCANVAS_IDS.forEach(ocId => {
    const el = document.getElementById(ocId);
    if (el) bootstrap.Offcanvas.getInstance(el)?.hide();
  });

  // 6. Обновить URL hash
  try {
    history.replaceState(null, null, `#${id}`);
  } catch (e) {}

  // 7. Запуск опросников новой активной вкладки
  for (const poller of registeredPollers.values()) {
    if (poller.tabId === id && poller.enabled) {
      startPoller(poller, true);
    }
  }

  // 8. Неблокирующий вызов Lifecycle hooks через requestAnimationFrame / setTimeout
  const name = id.replace(/^tab-/, '').replace(/-([a-z])/g, (_, c) => c.toUpperCase());
  const prevName = prevTabId ? prevTabId.replace(/^tab-/, '').replace(/-([a-z])/g, (_, c) => c.toUpperCase()) : null;

  // Деактивация предыдущей вкладки в фоне
  if (prevTabId && prevTabId !== id) {
    const deactivateFn = window[`deactivate${prevName[0].toUpperCase() + prevName.slice(1)}Tab`];
    if (typeof deactivateFn === 'function') {
      setTimeout(() => {
        try { deactivateFn(); } catch (err) { console.debug('deactivate hook error:', err); }
      }, 0);
    }
    document.dispatchEvent(new CustomEvent('tab:deactivated', { detail: { tabId: prevTabId, name: prevName } }));
  }

```

```
  // Активация новой вкладки в следующем фрейме анимации
  requestAnimationFrame(() => {
    try {
      window[`init${name[0].toUpperCase() + name.slice(1)}Tab`]?.();
    } catch (err) { console.debug('init tab hook error:', err); }

    try {
      window[`activate${name[0].toUpperCase() + name.slice(1)}Tab`]?.();
    } catch (err) { console.debug('activate tab hook error:', err); }

    const targetPane = document.getElementById(id);
    if (targetPane && typeof window.initPanels === 'function') {
      window.initPanels(targetPane);
    }

    document.dispatchEvent(new CustomEvent('tab:activated', { detail: { tabId: id, name } }));
  });
}

/**
 * Загружает разметку и скрипт вкладки по требованию (Lazy Load).
 * @param {string} tabName
 * @param {string} htmlUrl
 * @param {string} jsUrl
 */
export async function loadTab(tabName, htmlUrl, jsUrl) {
  let container = document.getElementById(`tab-${tabName}`);
  if (!container) {
    const mainContent = document.getElementById('mainTabContent');
    if (mainContent) {
```

```
      container = document.createElement('div');
      container.id = `tab-${tabName}`;
      container.className = 'tab-pane fade';
      mainContent.appendChild(container);
    } else {
      return;
    }
  }

  // Показываем мгновенный скелетон/спиннер, если контейнер ещ
e пуст
  if (!container.innerHTML.trim()) {
    container.innerHTML = `
      <div class="d-flex flex-column align-items-center justif
y-content-center py-5 text-muted" style="min-height: 200px;">
        <div class="spinner-border text-primary mb-2" role="st
atus" style="width: 2rem; height: 2rem;"></div>
        <span class="small fw-medium">Загрузка раздела...</spa
n>
      </div>
    `;
  }

  try {
    const r = await fetch(htmlUrl);
    if (!r.ok) throw new Error(`HTTP ${r.status}`);
    container.innerHTML = await r.text();
    if (jsUrl) {
      await new Promise(resolve => {
        const s = document.createElement('script');
        s.type = 'module';
        s.src = jsUrl;
        s.onload = resolve;
        s.onerror = (err) => {
          console.warn(`[TabCore] Ошибка загрузки скрипта ${js
Url}:`, err);
          resolve();
        };
```

```
        document.body.appendChild(s);
      });
    }
    const name = tabName.replace(/-([a-z])/g, (_, c) => c.toUp
perCase());
    const initName = `init${name[0].toUpperCase() + name.slice
(1)}Tab`;
    if (typeof window[initName] === 'function') {
      try {
        window[initName]();
      } catch (err) {
        console.debug(`[TabCore] Ошибка вызова ${initName}:`,
err);
      }
    }
    if (typeof window.initPanels === 'function') {
      window.initPanels(container);
    }
  } catch (e) {
    const safeName = String(tabName).replace(/[^a-zA-Z0-9_-]/g
, '');
    container.innerHTML = `
      <div class="alert alert-danger m-3 d-flex align-items-ce
nter justify-content-between flex-wrap gap-2">
        <div>
          <i class="bi bi-exclamation-triangle-fill me-2"></i>
          <span>Ошибка загрузки <strong>${safeName}</strong>:
${e.message}</span>
        </div>
        <button class="btn btn-sm btn-outline-danger" type="bu
tton" onclick="window.switchTab ? window.switchTab('tab-${safe
Name}') : null">
          <i class="bi bi-arrow-clockwise me-1"></i> Повторить
 попытку
        </button>
      </div>
    `;
    throw e;
```

```
  }
}

/**
 * Устанавливает единый слушатель кликов по элементам с data-tab.
 */
export function setupTabClicks() {
  document.addEventListener('click', e => {
    const btn = e.target.closest('button[data-tab], a[data-tab]');
    if (btn && btn.dataset.tab) {
      if (typeof window.switchTab === 'function') {
        window.switchTab(btn.dataset.tab);
      } else {
        switchTab(btn.dataset.tab);
      }
    }
  });
}

// Слушатель изменения видимости вкладки браузера
if (typeof document !== 'undefined') {
  document.addEventListener('visibilitychange', () => {
    if (document.hidden) {
      // Страница скрыта: останавливаем все активные опросники
      for (const poller of registeredPollers.values()) {
        stopPoller(poller);
      }
    } else {
      // Страница снова видна: возобновляем опросники ТОЛЬКО для активной вкладки
      if (currentActiveTabId) {
        for (const poller of registeredPollers.values()) {
          if (poller.tabId === currentActiveTabId && poller.enabled) {
            startPoller(poller, true);
          }
```

```
        }
      }
    }
  });
}

/**
 * Устанавливает флаг processing для вкладки (используется при
 длительных операциях).
 * Позволяет переключать вкладки без ожидания завершения опера
ции.
 * @param {string} tabId - ID вкладки
 * @param {boolean} isProcessing - флаг состояния
 */
export function setTabProcessing(tabId, isProcessing) {
  const normId = normalizeTabId(tabId);
  processingFlags.set(normId, !!isProcessing);
}

/**
 * Сбрасывает флаг processing для вкладки.
 * @param {string} tabId - ID вкладки
 */
export function resetTabProcessing(tabId) {
  const normId = normalizeTabId(tabId);
  processingFlags.delete(normId);
}

/**
 * Возвращает ID текущей активной вкладки.
 * @returns {string|null}
 */
export function getCurrentActiveTabId() {
  return currentActiveTabId;
}

/**
 * Экспорт в глобальный контекст window для совместимости с ин
```

```
лайн-скриптами и вкладками
 */
if (typeof window !== 'undefined') {
  window.switchTab = switchTab;
  window.switchToTab = switchTab;
  window.loadTab = loadTab;
  window.setupTabClicks = setupTabClicks;
  window.normalizeTabId = normalizeTabId;
  window.isTabActive = isTabActive;
  window.getCurrentActiveTabId = getCurrentActiveTabId;
  window.registerTabPoller = registerTabPoller;
  window.unregisterTabPoller = unregisterTabPoller;
  window.unregisterAllTabPollers = unregisterAllTabPollers;
  window.setTabPollerEnabled = setTabPollerEnabled;
  window.setTabProcessing = setTabProcessing;
  window.resetTabProcessing = resetTabProcessing;
}

```

tab-debounce-auto-patch.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Js - Tab-Debounce-Auto-Patch Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля tab-debou
nce-auto-patch.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/tab-debounce-auto-p
atch.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { autoPatchSearchInputs, autoPatchAutoSaveTextar
eas, autoPatchRefreshButtons } from '/windows/api/webgui/js/ta
b-debounce-auto-patch.js';
 *
 * File: tab-debounce-auto-patch.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * Tab Debounce Auto-Patch - Автоматическое применение дебаунс
а к табам
 *
 * Этот модуль автоматически улучшает дебаунс в:
 * - Поиске (все табы с поиском)
```

```
 * - Автосохранении (инструкции, конфиги)
 * - Real-time обновлениях (статусы, счетчики)
 */

import { RequestManager } from './request-manager.js';

const tabRequestManagers = new Map(); // { tabName: RequestMan
ager }

/**
 * Получить или создать RequestManager для вкладки
 */
function getTabRequestManager(tabName) {
  if (!tabRequestManagers.has(tabName)) {
    tabRequestManagers.set(tabName, new RequestManager());
  }
  return tabRequestManagers.get(tabName);
}

/**
 * Авто-патч поиска в любой вкладке
 * Находит input с id *-search-input и применяет дебаунс
 */
export function autoPatchSearchInputs(tabName) {
  const manager = getTabRequestManager(tabName);

  // Ищем все input'ы поиска в текущей вкладке
  const searchInputs = document.querySelectorAll('[id*="search
-input"]:not([data-debounce-patched])');

  searchInputs.forEach(input => {
    if (!input.id) return;

    const debounceKey = `${tabName}-${input.id}`;
    let originalHandler = null;

    // Сохраняем оригинальный обработчик если есть
    if (input.oninput) {
```

```
      originalHandler = input.oninput;
    }

    input.addEventListener('input', (e) => {
      const value = e.target.value;

      manager.debounce(
        debounceKey,
        () => {
          // Вызываем оригинальный обработчик если был
          if (originalHandler) {
            originalHandler.call(input, e);
          }
          console.log(`[AutoPatch] Search debounced in ${tabNa
me}: "${value}"`);
        },
        500
      ).catch(error => {
        console.error(`[AutoPatch] Search error in ${tabName}:
`, error);
      });
    });

    input.setAttribute('data-debounce-patched', 'true');
    console.log(`[AutoPatch] Patched search input: ${input.id}
 in tab: ${tabName}`);
  });
}

/**
 * Авто-патч для текстаревей (инструкции, конфиги)
 * Находит textarea и применяет дебаунс для автосохранения
 */
export function autoPatchAutoSaveTextareas(tabName, saveFn = n
ull) {
  const manager = getTabRequestManager(tabName);

  const textareas = document.querySelectorAll(`#tab-${tabName}
```

```
 textarea[id*="save"]:not([data-autosave-patched])`);

  textareas.forEach(textarea => {
    if (!textarea.id) return;

    const debounceKey = `${tabName}-autosave-${textarea.id}`;

    textarea.addEventListener('change', (e) => {
      const content = e.target.value;

      manager.debounce(
        debounceKey,
        async () => {
          if (saveFn) {
            await saveFn(content);
          }
          console.log(`[AutoPatch] Auto-saved content in ${tabName}`);
        },
        1000
      ).catch(error => {
        console.error(`[AutoPatch] Auto-save error in ${tabName}:`, error);
      });
    });

    textarea.setAttribute('data-autosave-patched', 'true');
    console.log(`[AutoPatch] Patched auto-save textarea: ${textarea.id} in tab: ${tabName}`);
  });
}

/**
 * Авто-патч для real-time обновлений
 * Находит кнопки обновления и применяет throttle
 */
export function autoPatchRefreshButtons(tabName) {
  const manager = getTabRequestManager(tabName);
```

```
133
134   const refreshBtns = document.querySelectorAll(
135     `#tab-${tabName} button[id*="refresh"]:not([data-throttle-
patched]),
136      #tab-${tabName} button[id*="refresh-btn"]:not([data-throt
tle-patched])`
137   );
138
139   refreshBtns.forEach(btn => {
140     if (!btn.id) return;
141
142     const originalOnClick = btn.onclick;
143     const throttleKey = `${tabName}-refresh-${btn.id}`;
144
145     btn.onclick = manager.throttle(
146       throttleKey,
147       async () => {
148         if (originalOnClick) {
149           await originalOnClick.call(btn);
150         }
151         console.log(`[AutoPatch] Refresh throttled in ${tabNam
e}`);
152       },
153       2000
154     );
155
156     btn.setAttribute('data-throttle-patched', 'true');
157     console.log(`[AutoPatch] Patched refresh button: ${btn.id}
 in tab: ${tabName}`);
158   });
159 }
160
161 /**
162  * Комплексный патч для всей вкладки
163  */
164 export function autoPatchTab(tabName, options = {}) {
165   console.log(`[AutoPatch] Patching tab: ${tabName}`);
166
```

```
  const { searchEnabled = true, autoSaveEnabled = true, refreshEnabled = true } = options;

  if (searchEnabled) {
    autoPatchSearchInputs(tabName);
  }

  if (autoSaveEnabled) {
    autoPatchAutoSaveTextareas(tabName, options.saveFn);
  }

  if (refreshEnabled) {
    autoPatchRefreshButtons(tabName);
  }

  console.log(`[AutoPatch] Completed patching tab: ${tabName}`);
}

/**
 * Получить статистику дебаунса для вкладки
 */
export function getTabDebounceStats(tabName) {
  const manager = tabRequestManagers.get(tabName);
  if (!manager) return null;
  return manager.getStats();
}

/**
 * Получить статистику дебаунса для всех вкладок
 */
export function getAllDebounceStats() {
  const stats = {};
  tabRequestManagers.forEach((manager, tabName) => {
    stats[tabName] = manager.getStats();
  });
  return stats;
}
```

```
203
204 // Экспортируем глобально
205 window.TabDebounceAutoPatch = {
206   autoPatchSearchInputs,
207   autoPatchAutoSaveTextareas,
208   autoPatchRefreshButtons,
209   autoPatchTab,
210   getTabDebounceStats,
211   getAllDebounceStats
212 };
213
214
```

tab-loader.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Js - Tab-Loader Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля tab-loade
r.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/tab-loader.js?v=202
61001_v1" type="module"></script>
 *
 * File: tab-loader.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * LazyTabLoader - Система ленивой загрузки вкладок
 *
 * Управляет загрузкой вкладок по требованию, кешировани в пам
яти
 * и минимизирует первоначальную нагрузку на интерфейс.
 *
 * Usage:
 *   const loader = new LazyTabLoader();
 *   loader.registerTab('chat', '/html/chat/index.html', '/htm
l/chat/main.js');
```

```
 *   await loader.loadTab('chat');  // Загружает вкладку
 *   loader.isLoaded('chat');       // true
 */

class LazyTabLoader {
  constructor() {
    this.tabs = new Map(); // Реестр вкладок
    this.loaded = new Set(); // Какие вкладки уже загружены
    this.loading = new Map(); // Текущие загрузки (для дедупли
кации)
    this.cache = new Map(); // Кеш содержимого вкладок в памят
и
  }

  /**
   * Регистрирует вкладку для последующей загрузки
   */
  registerTab(tabName, htmlUrl, jsUrl = null) {
    if (!this.tabs.has(tabName)) {
      this.tabs.set(tabName, { htmlUrl, jsUrl, initialized: fa
lse });
    }
  }

  /**
   * Загружает вкладку по требованию
   */
  async loadTab(tabName) {
    if (!this.tabs.has(tabName)) {
      console.warn(`[LazyTabLoader] Tab '${tabName}' not regis
tered`);
      return false;
    }

    // Если уже загружается - дождаться
    if (this.loading.has(tabName)) {
      return this.loading.get(tabName);
    }
```

```
64
65      // Если уже загружена - вернуть
66      if (this.loaded.has(tabName)) {
67        return true;
68      }
69
70      // Начать загрузку
71      const loadPromise = this._performLoad(tabName);
72      this.loading.set(tabName, loadPromise);
73
74      try {
75        const result = await loadPromise;
76        this.loaded.add(tabName);
77        return result;
78      } finally {
79        this.loading.delete(tabName);
80      }
81    }
82
83    /**
84     * Предварительная загрузка вкладки в фоне (не блокирует)
85     */
86    preloadTab(tabName) {
87      if (!this.loaded.has(tabName) && !this.loading.has(tabName
)) {
88        this.loadTab(tabName).catch(e => {
89          console.error(`[LazyTabLoader] Failed to preload ${tab
Name}:`, e);
90        });
91      }
92    }
93
94    /**
95     * Проверяет загружена ли вкладка
96     */
97    isLoaded(tabName) {
98      return this.loaded.has(tabName);
99    }
```

```
100
101   /**
102    * Загружает вкладку и выполняет инициализацию
103    */
104   async _performLoad(tabName) {
105     const tab = this.tabs.get(tabName);
106     if (!tab) return false;
107
108     try {
109       // Показываем loader на вкладке
110       this._showLoading(tabName);
111
112       // Загружаем HTML
113       const response = await fetch(tab.htmlUrl);
114       if (!response.ok) throw new Error(`HTTP ${response.statu
    s}`);
115       const html = await response.text();
116
117       // Вставляем HTML в контейнер вкладки
118       const container = document.getElementById(`tab-${tabName
    }`);
119       if (!container) {
120         throw new Error(`Container tab-${tabName} not found`);
121       }
122
123       container.innerHTML = html;
124       this.cache.set(tabName, html);
125
126       // Загружаем и выполняем JS
127       if (tab.jsUrl) {
128         await this._loadAndExecuteScript(tabName, tab.jsUrl);
129       }
130
131       // Скрываем loader
132       this._hideLoading(tabName);
133
134       // Вызываем init функцию если существует
135       const initFuncName = this._getInitFunctionName(tabName);
```

```
      if (typeof window[initFuncName] === 'function') {
        console.log(`[LazyTabLoader] Calling ${initFuncName} for ${tabName}`);
        try {
          await window[initFuncName]();
        } catch (err) {
          console.error(`[LazyTabLoader] Error executing ${initFuncName}:`, err);
        }
      }

      console.log(`[LazyTabLoader] Successfully loaded tab: ${tabName}`);
      return true;
    } catch (e) {
      console.error(`[LazyTabLoader] Error loading tab ${tabName}:`, e);
      this._showError(tabName, e.message);
      return false;
    }
  }

  /**
   * Загружает и выполняет скрипт вкладки
   */
  _loadAndExecuteScript(tabName, jsUrl) {
    return new Promise((resolve, reject) => {
      const script = document.createElement('script');
      script.src = jsUrl;
      script.type = 'module'; // Используем модули для современного подхода

      script.onload = () => {
        console.log(`[LazyTabLoader] Loaded JS for ${tabName}`);
        resolve();
      };

```

```
      script.onerror = (err) => {
        console.warn(`[LazyTabLoader] Note: No JS loaded for ${tabName} from ${jsUrl}`);
        resolve(); // Не падаем если нет JS
      };

      document.body.appendChild(script);
    });
  }

  /**
   * Вычисляет имя функции инициализации из имени вкладки
   */
  _getInitFunctionName(tabName) {
    const camelName = tabName.replace(/-([a-z])/g, (_, c) => c.toUpperCase());
    return 'init' + camelName.charAt(0).toUpperCase() + camelName.slice(1) + 'Tab';
  }

  /**
   * Показывает индикатор загрузки на вкладке
   */
  _showLoading(tabName) {
    const container = document.getElementById(`tab-${tabName}`);
    if (container) {
      container.innerHTML = `
        <div class="d-flex justify-content-center align-items-center" style="height: 300px;">
          <div class="spinner-border text-primary" role="status">
            <span class="visually-hidden">Загрузка...</span>
          </div>
        </div>
      `;
    }
  }
```

```
200
201   /**
202    * Скрывает индикатор загрузки
203    */
204   _hideLoading(tabName) {
205     // После вставки содержимого loader уже заменен
206   }
207
208   /**
209    * Показывает ошибку загрузки
210    */
211   _showError(tabName, errorMessage) {
212     const container = document.getElementById(`tab-${tabName}`
);
213     if (container) {
214       container.innerHTML = `
215         <div class="alert alert-danger m-3" role="alert">
216           <strong>Ошибка загрузки:</strong> ${errorMessage}
217           <br><small class="text-muted">Вкладка: ${tabName}</s
mall>
218         </div>
219       `;
220     }
221   }
222
223   /**
224    * Очищает кеш и загруженные вкладки
225    */
226   clear() {
227     this.loaded.clear();
228     this.cache.clear();
229     this.loading.clear();
230   }
231
232   /**
233    * Выгружает конкретную вкладку из памяти
234    */
235   unload(tabName) {
```

```
    this.loaded.delete(tabName);
    this.cache.delete(tabName);
    const container = document.getElementById(`tab-${tabName}`
);
    if (container) {
      container.innerHTML = '';
    }
  }

  /**
   * Вспомогательный метод: загрузить несколько вкладок паралл
ельно
   */
  async loadMultiple(tabNames) {
    return Promise.all(tabNames.map(name => this.loadTab(name)
));
  }

  /**
   * Вспомогательный метод: предзагрузить несколько вкладок в
фоне
   */
  preloadMultiple(tabNames) {
    tabNames.forEach(name => this.preloadTab(name));
  }
}

// Экспортируем глобально
window.LazyTabLoader = LazyTabLoader;

export default LazyTabLoader;

```

tab-state-persistence.js

```
/**
 * ============================================================
 * Process Name: Windows Js - Tab-State-Persistence Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля tab-state-persistence.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/tab-state-persistence.js?v=20261001_v1" type="module"></script>
 *
 * File: tab-state-persistence.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * Tab State Persistence - Сохранение состояния вкладок в localStorage
 *
 * Сохраняет:
 * - Активную вкладку
 * - Историю переходов
 * - Фильтры и поиск на каждой вкладке
 *
 * Usage:
 *   const persistence = new TabStatePersistence('admin');
```

```
 *   persistence.saveActiveTab('tab-chat');
 *   persistence.getLastActiveTab();  // 'tab-chat'
 */

class TabStatePersistence {
  constructor(namespace = 'admin') {
    this.namespace = namespace;
    this.storagePrefix = `${namespace}:tab-state`;
    this.maxHistorySize = 10;
  }

  /**
   * Сохранить активную вкладку
   */
  saveActiveTab(tabId) {
    if (!tabId) return;

    const cleanTabId = tabId.startsWith('#') ? tabId.slice(1) : tabId;
    const timestamp = Date.now();

    // Сохраняем текущую активную вкладку
    localStorage.setItem(`${this.storagePrefix}:active`, cleanTabId);
    localStorage.setItem(`${this.storagePrefix}:active-timestamp`, timestamp.toString());

    // Добавляем в историю
    this._addToHistory(cleanTabId, timestamp);

    console.log(`[TabStatePersistence] Saved active tab: ${cleanTabId}`);
  }

  /**
   * Получить последнюю активную вкладку
   */
  getLastActiveTab(fallback = null) {
```

```
    const lastTab = localStorage.getItem(`${this.storagePrefix}:active`);
    return lastTab || fallback;
  }

  /**
   * Получить последнюю активную вкладку с временем
   */
  getLastActiveTabWithTimestamp() {
    const tabId = localStorage.getItem(`${this.storagePrefix}:active`);
    const timestamp = localStorage.getItem(`${this.storagePrefix}:active-timestamp`);

    if (!tabId) return null;

    return {
      tabId,
      timestamp: timestamp ? parseInt(timestamp) : null,
      age: timestamp ? Date.now() - parseInt(timestamp) : null
    };
  }

  /**
   * Добавить вкладку в историю
   */
  _addToHistory(tabId, timestamp) {
    const historyKey = `${this.storagePrefix}:history`;
    let history = [];

    try {
      const stored = localStorage.getItem(historyKey);
      if (stored) {
        history = JSON.parse(stored);
      }
    } catch (e) {
      console.warn('[TabStatePersistence] Failed to parse history:', e);
```

```
      history = [];
    }

    // Добавляем новую запись в начало
    history.unshift({ tabId, timestamp });

    // Удаляем дубли
    const seen = new Set();
    history = history.filter(item => {
      if (seen.has(item.tabId)) return false;
      seen.add(item.tabId);
      return true;
    });

    // Ограничиваем размер истории
    if (history.length > this.maxHistorySize) {
      history = history.slice(0, this.maxHistorySize);
    }

    // Сохраняем обновленную историю
    localStorage.setItem(historyKey, JSON.stringify(history));
  }

  /**
   * Получить историю переходов между вкладками
   */
  getHistory() {
    const historyKey = `${this.storagePrefix}:history`;
    try {
      const stored = localStorage.getItem(historyKey);
      return stored ? JSON.parse(stored) : [];
    } catch (e) {
      console.warn('[TabStatePersistence] Failed to parse hist
ory:', e);
      return [];
    }
  }

```

```
  /**
   * Очистить историю
   */
  clearHistory() {
    localStorage.removeItem(`${this.storagePrefix}:history`);
    console.log('[TabStatePersistence] History cleared');
  }

  /**
   * Сохранить состояние вкладки (фильтры, поиск и т.д.)
   */
  saveTabState(tabId, state) {
    if (!tabId || !state) return;

    const cleanTabId = tabId.startsWith('#') ? tabId.slice(1)
: tabId;
    const stateKey = `${this.storagePrefix}:state:${cleanTabId
}`;

    try {
      localStorage.setItem(stateKey, JSON.stringify(state));
      console.log(`[TabStatePersistence] Saved state for tab:
${cleanTabId}`);
    } catch (e) {
      console.warn(`[TabStatePersistence] Failed to save state
 for ${cleanTabId}:`, e);
    }
  }

  /**
   * Получить состояние вкладки
   */
  getTabState(tabId) {
    if (!tabId) return null;

    const cleanTabId = tabId.startsWith('#') ? tabId.slice(1)
: tabId;
    const stateKey = `${this.storagePrefix}:state:${cleanTabId
```

```
}`;

    try {
      const stored = localStorage.getItem(stateKey);
      return stored ? JSON.parse(stored) : null;
    } catch (e) {
      console.warn(`[TabStatePersistence] Failed to parse stat
e for ${cleanTabId}:`, e);
      return null;
    }
  }

  /**
   * Удалить состояние вкладки
   */
  deleteTabState(tabId) {
    if (!tabId) return;

    const cleanTabId = tabId.startsWith('#') ? tabId.slice(1)
: tabId;
    const stateKey = `${this.storagePrefix}:state:${cleanTabId
}`;

    localStorage.removeItem(stateKey);
    console.log(`[TabStatePersistence] Deleted state for tab:
${cleanTabId}`);
  }

  /**
   * Очистить все сохраненное состояние
   */
  clear() {
    const keys = Object.keys(localStorage).filter(key =>
      key.startsWith(this.storagePrefix)
    );

    keys.forEach(key => localStorage.removeItem(key));
    console.log(`[TabStatePersistence] Cleared ${keys.length}
```

```
entries`);
  }

  /**
   * Получить полную информацию о сохраненном состоянии
   */
  getStorageInfo() {
    const keys = Object.keys(localStorage).filter(key =>
      key.startsWith(this.storagePrefix)
    );

    const info = {
      namespace: this.namespace,
      totalEntries: keys.length,
      entries: {}
    };

    keys.forEach(key => {
      const shortKey = key.replace(`${this.storagePrefix}:`, '
');
      const value = localStorage.getItem(key);

      try {
        info.entries[shortKey] = {
          size: value.length,
          parsed: JSON.parse(value)
        };
      } catch {
        info.entries[shortKey] = {
          size: value.length,
          raw: value
        };
      }
    });

    return info;
  }

```

```
  /**
   * Экспортировать состояние в JSON
   */
  exportState() {
    return JSON.stringify(this.getStorageInfo(), null, 2);
  }

  /**
   * Импортировать состояние из JSON
   */
  importState(jsonString) {
    try {
      const data = JSON.parse(jsonString);
      // Реализация импорта если нужна
      console.log('[TabStatePersistence] Import not implemented yet');
    } catch (e) {
      console.error('[TabStatePersistence] Failed to import state:', e);
    }
  }
}

/**
 * Глобальный экземпляр для админки
 */
window.tabStatePersistence = new TabStatePersistence('admin');

// Экспортируем
window.TabStatePersistence = TabStatePersistence;

export default TabStatePersistence;

```

tab-switch-optimizer.js

```
/**
 * ============================================================
 * Process Name: Windows Js - Tab-Switch-Optimizer Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля tab-switch-optimizer.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/tab-switch-optimizer.js?v=20261001_v1" type="module"></script>
 *
 * File: tab-switch-optimizer.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * Tab Switch Optimizer - Оптимизация переключения между вкладками
 *
 * Реализует:
 * - DOM кеширование (сохранение содержимого вкладки вместо удаления)
 * - Мгновенная визуальная обратная связь (показ активной вкладки до загрузки)
 * - Предварительная загрузка соседних вкладок
 * - Трекинг времени переключения
```

```
 *
 * Usage:
 *   const optimizer = new TabSwitchOptimizer();
 *   optimizer.enableDOMCaching();
 *   optimizer.optimizeSwitching();
 */

class TabSwitchOptimizer {
  constructor() {
    this.domCache = new Map(); // Кеш DOM содержимого вкладок
    this.switchStats = {
      totalSwitches: 0,
      avgSwitchTime: 0,
      lastSwitchTime: null,
      switchTimes: []
    };
    this.isOptimized = false;
  }

  /**
   * Включить DOM кеширование
   * Вместо удаления содержимого вкладки, сохраняем его в памя
ти
   */
  enableDOMCaching() {
    console.log('[TabSwitchOptimizer] Enabling DOM caching...'
);

    // Перехватываем стандартное управление табами
    const observer = new MutationObserver((mutations) => {
      mutations.forEach((mutation) => {
        if (mutation.type === 'childList' || mutation.type ===
 'attributes') {
          // Вызываем автосохранение изменений
          this._cacheVisibleTabs();
        }
      });
    });
```

```
64
65      // Наблюдаем за контейнером с табами
66      const tabContainer = document.getElementById('mainTabsCont
ent') ||
67                           document.querySelector('[role="tablis
t"]')?.parentElement;
68
69      if (tabContainer) {
70        observer.observe(tabContainer, {
71          childList: true,
72          subtree: true,
73          attributes: true,
74          attributeFilter: ['class', 'style', 'aria-hidden']
75        });
76      }
77
78      this.isOptimized = true;
79      console.log('[TabSwitchOptimizer] DOM caching enabled');
80    }
81
82    /**
83     * Кешировать видимые вкладки
84     */
85    _cacheVisibleTabs() {
86      const tabs = document.querySelectorAll('[role="tabpanel"]:
not([aria-hidden="true"])');
87      tabs.forEach(tab => {
88        if (tab.id) {
89          // Сохраняем клон вкладки
90          this.domCache.set(tab.id, tab.cloneNode(true));
91        }
92      });
93    }
94
95    /**
96     * Получить кешированное содержимое вкладки
97     */
98    getCachedTabContent(tabId) {
```

```
    return this.domCache.get(tabId);
  }

  /**
   * Очистить кеш DOM
   */
  clearDOMCache(tabId = null) {
    if (tabId) {
      this.domCache.delete(tabId);
    } else {
      this.domCache.clear();
    }
  }

  /**
   * Оптимизировать функцию переключения вкладок
   * Добавляет мгновенную визуальную обратную связь и трекинг времени
   */
  optimizeSwitching(originalSwitchFn) {
    return async (targetTabId) => {
      const startTime = performance.now();

      if (!targetTabId) return;

      // Мгновенно показываем активную вкладку визуально (без загрузки)
      this._showTabImmediately(targetTabId);

      // Выполняем оригинальное переключение
      if (typeof originalSwitchFn === 'function') {
        await originalSwitchFn(targetTabId);
      }

      // Записываем время переключения
      const switchTime = performance.now() - startTime;
      this._recordSwitchTime(switchTime, targetTabId);

```

```
      console.log(`[TabSwitchOptimizer] Tab switched in ${switchTime.toFixed(2)}ms`);
    };
  }

  /**
   * Мгновенно показать вкладку визуально (до загрузки содержимого)
   */
  _showTabImmediately(targetTabId) {
    const cleanTabId = targetTabId.startsWith('#') ? targetTabId.slice(1) : targetTabId;

    // Скрываем все остальные вкладки
    document.querySelectorAll('[role="tabpanel"]').forEach(tab => {
      tab.style.display = 'none';
      tab.classList.remove('active', 'show');
      tab.setAttribute('aria-hidden', 'true');
    });

    // Показываем целевую вкладку
    const targetTab = document.getElementById(cleanTabId);
    if (targetTab) {
      targetTab.style.display = 'block';
      targetTab.classList.add('active', 'show');
      targetTab.setAttribute('aria-hidden', 'false');
    }

    // Обновляем визуальное состояние кнопок вкладок
    document.querySelectorAll('[role="tab"]').forEach(tab => {
      tab.classList.remove('active');
      tab.setAttribute('aria-selected', 'false');
    });

    const targetButton = document.querySelector(`[aria-controls="${cleanTabId}"]`) ||
                         document.querySelector(`[data-bs-targe
```

```
t="#${cleanTabId}"]`);
    if (targetButton) {
      targetButton.classList.add('active');
      targetButton.setAttribute('aria-selected', 'true');
    }
  }

  /**
   * Записать время переключения и вычислить среднее
   */
  _recordSwitchTime(time, tabId) {
    this.switchStats.totalSwitches++;
    this.switchStats.lastSwitchTime = time;
    this.switchStats.switchTimes.push({ time, tabId, timestamp: Date.now() });

    // Ограничиваем историю последних 100 переключениями
    if (this.switchStats.switchTimes.length > 100) {
      this.switchStats.switchTimes = this.switchStats.switchTimes.slice(-100);
    }

    // Вычисляем среднее время
    const sum = this.switchStats.switchTimes.reduce((acc, s) => acc + s.time, 0);
    this.switchStats.avgSwitchTime = sum / this.switchStats.switchTimes.length;
  }

  /**
   * Предварительная загрузка соседних вкладок
   */
  preloadAdjacentTabs(currentTabId) {
    const allTabs = Array.from(document.querySelectorAll('[role="tab"]'));
    const currentIndex = allTabs.findIndex(tab =>
      tab.getAttribute('aria-controls') === currentTabId ||
      tab.getAttribute('data-bs-target') === `#${currentTabId}
```

```
`
    );

    if (currentIndex >= 0) {
      // Предзагружаем соседние вкладки
      const adjacentIndices = [currentIndex - 1, currentIndex + 1];
      adjacentIndices.forEach(idx => {
        if (idx >= 0 && idx < allTabs.length) {
          const tab = allTabs[idx];
          const tabId = tab.getAttribute('aria-controls') ||
                        tab.getAttribute('data-bs-target')?.replace('#', '');

          if (tabId && window.optimizationModule?.tabLoader) {
            console.log(`[TabSwitchOptimizer] Preloading adjacent tab: ${tabId}`);
            window.optimizationModule.tabLoader.preloadTab(tabId);
          }
        }
      });
    }
  }

  /**
   * Получить статистику переключения
   */
  getStats() {
    return {
      ...this.switchStats,
      avgSwitchTimeFormatted: `${this.switchStats.avgSwitchTime.toFixed(2)}ms`,
      lastSwitchTimeFormatted: this.switchStats.lastSwitchTime ?
        `${this.switchStats.lastSwitchTime.toFixed(2)}ms` : 'N/A',
      cachedTabs: this.domCache.size,
```

```
      isOptimized: this.isOptimized
    };
  }

  /**
   * Вывести статистику в консоль
   */
  printStats() {
    const stats = this.getStats();
    console.group('  Tab Switch Performance Stats');
    console.log('Total Switches:', stats.totalSwitches);
    console.log('Average Switch Time:', stats.avgSwitchTimeFor
matted);
    console.log('Last Switch Time:', stats.lastSwitchTimeForma
tted);
    console.log('Cached Tabs:', stats.cachedTabs);
    console.log('Performance Target: <100ms (currently',
      (stats.avgSwitchTime < 100 ? '  GOOD' : '  SLOW'), ')');
    console.groupEnd();
  }

  /**
   * Очистить статистику
   */
  resetStats() {
    this.switchStats = {
      totalSwitches: 0,
      avgSwitchTime: 0,
      lastSwitchTime: null,
      switchTimes: []
    };
  }
}

// Глобальный экземпляр
window.tabSwitchOptimizer = new TabSwitchOptimizer();

// Экспортируем
```

```
window.TabSwitchOptimizer = TabSwitchOptimizer;

export default TabSwitchOptimizer;

```

test-api-cache.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Js - Test-Api-Cache Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля test-api-
cache.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/test-api-cache.js?v
=20261001_v1" type="module"></script>
 *
 * File: test-api-cache.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * test-api-cache.js - Простой тестовый скрипт для проверки AP
I Cache System
 *
 * ИСПОЛЬЗОВАНИЕ:
 * 1. Откройте веб-интерфейс
 * 2. Откройте консоль браузера (F12)
 * 3. Выполните: testApiCache()
 */

export async function testApiCache() {
```

```
  console.log('=== □ API Cache System Test Suite ===\n');

  const results = {
    passed: 0,
    failed: 0,
    tests: []
  };

  function logTest(name, passed, message) {
    const icon = passed ? '□ ' : '□ ';
    console.log(`${icon} ${name}: ${message}`);
    results.tests.push({ name, passed, message });
    if (passed) results.passed++;
    else results.failed++;
  }

  // Test 1: Проверка доступности browserCache
  try {
    const cacheAvailable = window.browserCache && typeof windo
w.browserCache.ready === 'function';
    logTest('Cache Availability', cacheAvailable, cacheAvailab
le ? 'BrowserCache is available' : 'BrowserCache NOT available
');
  } catch (e) {
    logTest('Cache Availability', false, `Error: ${e.message}`
);
  }

  // Test 2: Проверка доступности cachedApiFetch
  try {
    const apiFetchAvailable = typeof window.cachedApiFetch ===
 'function';
    logTest('API Fetch Function', apiFetchAvailable, apiFetchA
vailable ? 'cachedApiFetch() is available' : 'cachedApiFetch()
 NOT available');
  } catch (e) {
    logTest('API Fetch Function', false, `Error: ${e.message}`
);
```

```
  }

  // Test 3: Проверка готовности IndexedDB
  try {
    if (window.browserCache) {
      const isReady = await window.browserCache.ready();
      logTest('IndexedDB Ready', isReady, isReady ? 'IndexedDB is ready' : 'IndexedDB initialization failed');
    } else {
      logTest('IndexedDB Ready', false, 'BrowserCache not available');
    }
  } catch (e) {
    logTest('IndexedDB Ready', false, `Error: ${e.message}`);
  }

  // Test 4: Тест Cache-First стратегии
  try {
    console.log('\n--- Testing Cache-First Strategy ---');
    const testUrl = '/api/v1/models';

    // Первый запрос (должен быть MISS)
    console.log('Request 1: Should be MISS (network)');
    const start1 = performance.now();
    await window.cachedApiFetch(testUrl);
    const time1 = (performance.now() - start1).toFixed(2);

    // Второй запрос (должен быть HIT)
    console.log('Request 2: Should be HIT (cache)');
    const start2 = performance.now();
    await window.cachedApiFetch(testUrl);
    const time2 = (performance.now() - start2).toFixed(2);

    const speedup = (time1 / time2).toFixed(1);
    logTest('Cache-First Speed', time2 < time1, `Network: ${time1}ms, Cache: ${time2}ms (${speedup}x faster)`);
  } catch (e) {
    logTest('Cache-First Speed', false, `Error: ${e.message}`)
```

```
;
  }

  // Test 5: Тест инвалидации по тегу
  try {
    console.log('\n--- Testing Tag Invalidation ---');

    // Добавляем тестовую запись
    if (window.browserCache) {
      await window.browserCache.set('api_cache', 'test-key-1', { data: 'test1' }, { tags: ['test-tag'] });
      await window.browserCache.set('api_cache', 'test-key-2', { data: 'test2' }, { tags: ['test-tag'] });

      // Инвалидируем
      const count = await window.invalidateCacheByTag('test-tag');
      logTest('Tag Invalidation', count === 2, `Invalidated ${count} entries (expected 2)`);
    } else {
      logTest('Tag Invalidation', false, 'BrowserCache not available');
    }
  } catch (e) {
    logTest('Tag Invalidation', false, `Error: ${e.message}`);
  }

  // Test 6: Получение статистики
  try {
    console.log('\n--- Testing Statistics ---');
    const stats = await window.getApiCacheStats();
    const hasMetrics = stats.available && stats.metrics && typeof stats.metrics.hits === 'number';
    logTest('Cache Statistics', hasMetrics, `Hit rate: ${stats.apiCache?.hitRate || 0}%, Entries: ${stats.apiCache?.count || 0}`);
  } catch (e) {
    logTest('Cache Statistics', false, `Error: ${e.message}`);
```

```
  }

  // Test 7: Тест forceNetwork
  try {
    console.log('\n--- Testing Force Network ---');
    const testUrl = '/api/v1/system/summary';

    // Обычный запрос
    await window.cachedApiFetch(testUrl);

    // Принудительный запрос из сети
    const start = performance.now();
    await window.cachedApiFetch(testUrl, {}, { forceNetwork: t
rue });
    const time = (performance.now() - start).toFixed(2);

    logTest('Force Network', time > 10, `Force network request
 took ${time}ms (should be > 10ms)`);
  } catch (e) {
    logTest('Force Network', false, `Error: ${e.message}`);
  }

  // Итоги
  console.log('\n=== 📊 Test Results ===');
  console.log(`Total: ${results.tests.length} tests`);
  console.log(`✅ Passed: ${results.passed}`);
  console.log(`❌ Failed: ${results.failed}`);
  console.log(`Success rate: ${((results.passed / results.test
s.length) * 100).toFixed(1)}%`);

  if (results.failed === 0) {
    console.log('\n🎉 All tests passed! API Cache System is wo
rking correctly.');
  } else {
    console.warn('\n⚠️ Some tests failed. Check the logs above
for details.');
  }

```

```
  return results;
}

// Экспорт в глобальную область
if (typeof window !== 'undefined') {
  window.testApiCache = testApiCache;
}

// Автоматический запуск при включенном DEBUG режиме
if (typeof window !== 'undefined' && localStorage.getItem('DEBUG_API_CACHE') === 'true') {
  console.log('🔧 DEBUG_API_CACHE enabled, running tests on load...');
  setTimeout(() => testApiCache(), 2000);
}

export default testApiCache;

```

theme-engine.js

```
/**
 * ============================================================
 * Process Name: Windows Web Styles - Theme Engine
 * ============================================================
 * Description:
 *   Шаблонизатор и менеджер тем оформления для веб-интерфейса AI Windows Admin.
 *   Поддерживает светлые темы (Light, Brick), тёмные темы (Dark Contrast, Terminal),
 *   системный режим, динамическую регистрацию и шаблонизацию UI-селекторов.
 *
 * Usage Examples:
 *   import { themeEngine } from './theme-engine.js';
 *   themeEngine.applyTheme('brick');
 *   themeEngine.registerTheme({ id: 'custom', name: 'Custom Theme', category: 'dark' });
 *
 * File: theme-engine.js
 * Project: ai-windows-admin
 * Package: apps.windows.api.webgui.js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * Ключ хранения выбранного режима темы в LocalStorage
 */
export const THEME_STORAGE_KEY = 'theme';

/**
```

```
 * Режим по умолчанию
 */
export const DEFAULT_THEME_MODE = 'system';

/**
 * Определение стандартных встроенных тем
 */
export const BUILTIN_THEMES = [
  {
    id: 'system',
    name: 'Системная',
    nameEn: 'System',
    category: 'system',
    icon: 'bi-circle-half',
    iconColor: 'text-info',
    description: 'Автоматический выбор темы на основе настроек операционной системы',
    baseType: 'system',
  },
  {
    id: 'light',
    name: 'Светлая (NightView)',
    nameEn: 'Light (NightView)',
    category: 'light',
    icon: 'bi-sun-fill',
    iconColor: 'text-warning',
    description: 'Мягкая светлая тема в стиле NightView без слепящей белизны для комфортного чтения ночью',
    baseType: 'light',
    tokens: { '--about-card-bg': 'linear-gradient(135deg, #f8fafc 0%, #e2e8f0 100%)' },
  },
  {
    id: 'brick',
    name: 'Кирпичная (Терракота)',
    nameEn: 'Brick (Terracotta)',
    category: 'light',
    icon: 'bi-bricks',
```

```
    iconColor: 'text-danger',
    description: 'Теплая светлая тема в стиле клинкерного кирпича и обожженной глины',
    baseType: 'light',
    tokens: { '--about-card-bg': 'linear-gradient(135deg, #f5e8d4 0%, #e2c8b8 100%)' },
  },
  {
    id: 'dark',
    name: '  Midnight Console',
    nameEn: 'Dark (High-Contrast)',
    category: 'dark',
    icon: 'bi-moon-stars-fill',
    iconColor: 'text-primary',
    description: 'Глубокая контрастная тёмная тема для удобной работы в ночное время',
    baseType: 'dark',
  },
  {
    id: 'terminal',
    name: 'Терминал (Matrix / CRT)',
    nameEn: 'Terminal (Matrix / CRT)',
    category: 'dark',
    icon: 'bi-terminal-fill',
    iconColor: 'text-success',
    description: 'Тёмная тема в ретро-стиле зеленого фосфорного терминала с моноширинными акцентами',
    baseType: 'dark',
  },
];

/**
 * Класс шаблонизатора и оркестратора тем оформления
 */
export class ThemeEngine {
  constructor() {
    /** @type {Map<string, Object>} */
    this.registry = new Map();
```

```
    this.mediaQueryListenerAttached = false;
    this._dynamicStyleEl = null;

    // Регистрация встроенных тем
    for (const theme of BUILTIN_THEMES) {
      this.registry.set(theme.id, { ...theme });
    }
  }

  /**
   * Регистрирует новую тему в реестре шаблонизатора.
   * @param {Object} themeDef - Определение темы
   * @param {string} themeDef.id - Уникальный идентификатор темы
   * @param {string} themeDef.name - Отображаемое имя темы (русский)
   * @param {string} [themeDef.nameEn] - Отображаемое имя темы (английский)
   * @param {'light'|'dark'|'system'} themeDef.category - Категория темы
   * @param {string} [themeDef.icon] - Иконка Bootstrap Icons (например, 'bi-palette')
   * @param {string} [themeDef.iconColor] - CSS класс цвета иконки (например, 'text-warning')
   * @param {string} [themeDef.description] - Описание темы
   * @param {'light'|'dark'} [themeDef.baseType] - Базовый тип темы (светлая/тёмная)
   * @param {Object<string, string>} [themeDef.tokens] - Словарь CSS-переменных
   */
  registerTheme(themeDef) {
    if (!themeDef || !themeDef.id) {
      console.warn('[ThemeEngine] Некорректное определение темы:', themeDef);
      return;
    }

    const baseType = themeDef.baseType || (themeDef.category =
```

```
== 'dark' ? 'dark' : 'light');
    this.registry.set(themeDef.id, {
      id: themeDef.id,
      name: themeDef.name || themeDef.id,
      nameEn: themeDef.nameEn || themeDef.name || themeDef.id,
      category: themeDef.category || 'dark',
      icon: themeDef.icon || 'bi-palette',
      iconColor: themeDef.iconColor || 'text-info',
      description: themeDef.description || '',
      baseType,
      tokens: themeDef.tokens || null,
    });

    if (themeDef.tokens) {
      this._applyDynamicTokens();
    }
  }

  /**
   * Возвращает определение темы по идентификатору.
   * @param {string} themeId
   * @returns {Object|null}
   */
  getTheme(themeId) {
    return this.registry.get(themeId) || null;
  }

  /**
   * Возвращает список всех зарегистрированных тем.
   * @returns {Array<Object>}
   */
  getAllThemes() {
    return Array.from(this.registry.values());
  }

  /**
   * Возвращает темы, отфильтрованные по категории ('light' |
'dark' | 'system').
```

```
164    * @param {'light'|'dark'|'system'} category
165    * @returns {Array<Object>}
166    */
167   getThemesByCategory(category) {
168     return this.getAllThemes().filter((t) => t.category === ca
tegory);
169   }
170 
171   /**
172    * Получает сохранённый режим темы из LocalStorage.
173    * @returns {string}
174    */
175   getSavedMode() {
176     const saved = localStorage.getItem(THEME_STORAGE_KEY);
177     if (saved && this.registry.has(saved)) {
178       return saved;
179     }
180     return DEFAULT_THEME_MODE;
181   }
182 
183   /**
184    * Разрешает эффективный идентификатор темы ('light', 'dark'
, 'brick', 'terminal' и т.д.)
185    * @param {string} [mode]
186    * @returns {string}
187    */
188   resolveTheme(mode = this.getSavedMode()) {
189     if (mode === 'system') {
190       const prefersDark =
191         typeof window !== 'undefined' &&
192         window.matchMedia &&
193         window.matchMedia('(prefers-color-scheme: dark)').matc
hes;
194       return prefersDark ? 'dark' : 'light';
195     }
196 
197     if (this.registry.has(mode)) {
198       return mode;
```

```
    }
    return 'dark';
  }

  /**
   * Определяет базовый режим Bootstrap ('light' | 'dark') для текущей темы.
   * @param {string} resolvedThemeId
   * @returns {'light'|'dark'}
   */
  getBsTheme(resolvedThemeId) {
    const theme = this.getTheme(resolvedThemeId);
    if (theme) {
      return theme.baseType || (theme.category === 'dark' ? 'dark' : 'light');
    }
    return resolvedThemeId === 'light' || resolvedThemeId === 'brick' ? 'light' : 'dark';
  }

  /**
   * Применяет тему к DOM, обновляет атрибуты и UI-элементы.
   * @param {string} mode
   */
  applyTheme(mode) {
    if (!mode || !this.registry.has(mode)) {
      mode = DEFAULT_THEME_MODE;
    }

    localStorage.setItem(THEME_STORAGE_KEY, mode);

    const resolved = this.resolveTheme(mode);
    const bsTheme = this.getBsTheme(resolved);

    if (typeof document !== 'undefined') {
      // 1. Установка атрибутов на <html>
      document.documentElement.setAttribute('data-theme', resolved);
```

```
      document.documentElement.setAttribute('data-bs-theme', bsTheme);
      document.documentElement.setAttribute('data-theme-mode', mode);

      // 2. Обновление иконок и кнопок
      this._updateThemeUI(mode, resolved);

      // 3. Синхронизация селекторов выбора тем
      this.syncSelectInputs(mode);
    }

    // 4. Оповещение слушателей через CustomEvent
    if (typeof window !== 'undefined') {
      window.dispatchEvent(
        new CustomEvent('themechanged', {
          detail: {
            mode,
            resolved,
            bsTheme,
            themeMeta: this.getTheme(resolved),
          },
        })
      );
    }
  }

  /**
   * Генерирует HTML-разметку опций для тега <select> с группировкой по категориям.
   * @param {string} [currentMode]
   * @returns {string}
   */
  renderOptionsHTML(currentMode = this.getSavedMode()) {
    const lightThemes = this.getThemesByCategory('light');
    const darkThemes = this.getThemesByCategory('dark');
    const systemThemes = this.getThemesByCategory('system');

```

```
268     let html = '';
269 
270     if (systemThemes.length > 0) {
271       for (const t of systemThemes) {
272         const isSelected = t.id === currentMode ? ' selected'
: '';
273         html += `<option value="${t.id}"${isSelected}>  ${t.n
ame}</option>\n`;
274       }
275     }
276 
277     if (lightThemes.length > 0) {
278       html += '<optgroup label="  Светлые темы">\n';
279       for (const t of lightThemes) {
280         const isSelected = t.id === currentMode ? ' selected'
: '';
281         const iconSymbol = t.id === 'brick' ? '  ' : '  ';
282         html += `  <option value="${t.id}"${isSelected}>${icon
Symbol} ${t.name}</option>\n`;
283       }
284       html += '</optgroup>\n';
285     }
286 
287     if (darkThemes.length > 0) {
288       html += '<optgroup label="  Тёмные темы">\n';
289       for (const t of darkThemes) {
290         const isSelected = t.id === currentMode ? ' selected'
: '';
291         const iconSymbol = t.id === 'terminal' ? '  ' : '  ';
292         html += `  <option value="${t.id}"${isSelected}>${icon
Symbol} ${t.name}</option>\n`;
293       }
294       html += '</optgroup>\n';
295     }
296 
297     return html;
298   }
299 
```

```
  /**
   * Синхронизирует все элементы <select> выбора темы на странице.
   * @param {string} [mode]
   */
  syncSelectInputs(mode = this.getSavedMode()) {
    if (typeof document === 'undefined') return;

    const selectors = document.querySelectorAll(
      '.theme-selector, #apps-theme-selector, #settings-theme, #modal-theme-selector'
    );

    selectors.forEach((selectEl) => {
      // Если у селектора еще нет сгенерированных опций с optgroup, наполняем его
      if (!selectEl.querySelector('optgroup')) {
        selectEl.innerHTML = this.renderOptionsHTML(mode);
      } else if (selectEl.value !== mode) {
        selectEl.value = mode;
      }
    });
  }

  /**
   * Инициализирует менеджер тем, регистрирует обработчики событий и применяет начальную тему.
   */
  init() {
    const currentMode = this.getSavedMode();
    this.applyTheme(currentMode);

    // Подписка на изменение системной темы
    if (!this.mediaQueryListenerAttached && typeof window !== 'undefined' && window.matchMedia) {
      try {
        const mediaQuery = window.matchMedia('(prefers-color-scheme: dark)');
```

```
        const handler = () => {
          if (this.getSavedMode() === 'system') {
            this.applyTheme('system');
          }
        };
        if (mediaQuery.addEventListener) {
          mediaQuery.addEventListener('change', handler);
        } else if (mediaQuery.addListener) {
          mediaQuery.addListener(handler);
        }
        this.mediaQueryListenerAttached = true;
      } catch (e) {
        console.warn('[ThemeEngine] Не удалось подписаться на prefers-color-scheme:', e);
      }
    }

    if (typeof document === 'undefined') return;

    // Наполнение всех селекторов опциями
    this.syncSelectInputs(currentMode);

    // Привязка слушателей изменений к селекторам тем
    document.querySelectorAll(
      '.theme-selector, #apps-theme-selector, #settings-theme, #modal-theme-selector'
    ).forEach((sel) => {
      sel.addEventListener('change', (e) => {
        this.applyTheme(e.target.value);
      });
    });

    // Привязка кликов по кнопкам data-theme-value
    document.querySelectorAll('[data-theme-value]').forEach((btn) => {
      btn.addEventListener('click', (e) => {
        e.preventDefault();
        const selected = btn.getAttribute('data-theme-value');
```

```
        if (selected) {
          this.applyTheme(selected);
        }
      });
    });

    // Привязка кнопки быстрого циклического переключения тем (legacy / toggle button)
    const toggleBtn = document.getElementById('theme-toggle');
    if (toggleBtn && !toggleBtn.hasAttribute('data-bs-toggle')) {
      toggleBtn.addEventListener('click', () => {
        const mode = this.getSavedMode();
        // Цикл переключения: system -> light -> brick -> dark -> terminal -> system
        const cycle = ['system', 'light', 'brick', 'dark', 'terminal'];
        const currentIndex = cycle.indexOf(mode);
        const nextMode = cycle[(currentIndex + 1) % cycle.length];
        this.applyTheme(nextMode);
      });
    }
  }

  /**
   * Обновляет UI-иконки, подсказки и активные состояния в выпадающих меню.
   * @private
   */
  _updateThemeUI(mode, resolved) {
    const activeTheme = this.getTheme(resolved) || this.getTheme('dark');
    const modeTheme = this.getTheme(mode);

    const iconEl = document.getElementById('theme-icon');
    const toggleBtn = document.getElementById('theme-toggle');
    const dropdownBtn = document.getElementById('theme-dropdow
```

```
n');

    const iconClass = modeTheme ? modeTheme.icon : 'bi-circle-
half';
    const iconColor = modeTheme ? modeTheme.iconColor : 'text-
info';
    const titleText =
      mode === 'system'
        ? `Системная тема (${activeTheme ? activeTheme.name :
resolved})`
        : modeTheme
        ? modeTheme.name
        : resolved;

    if (iconEl) {
      iconEl.className = `bi ${iconClass} ${iconColor}`;
    }

    if (dropdownBtn) {
      dropdownBtn.setAttribute('title', titleText);
    }

    if (toggleBtn) {
      toggleBtn.innerHTML = `<i class="bi ${iconClass} ${iconC
olor}"></i>`;
      toggleBtn.setAttribute('title', titleText);
    }

    // Обновление галочек в выпадающих меню
    document.querySelectorAll('.theme-check-icon').forEach((ic
on) => {
      const target = icon.getAttribute('data-theme-check');
      if (target === mode) {
        icon.classList.remove('d-none');
      } else {
        icon.classList.add('d-none');
      }
    });
```

```
430
431     // Обновление active класса
432     document.querySelectorAll('[data-theme-value]').forEach((i
tem) => {
433       const val = item.getAttribute('data-theme-value');
434       if (val === mode) {
435         item.classList.add('active');
436       } else {
437         item.classList.remove('active');
438       }
439     });
440   }
441
442   /**
443    * Инжектирует динамические стили для зарегистрированных кас
томных тем с токенами.
444    * @private
445    */
446   _applyDynamicTokens() {
447     if (typeof document === 'undefined') return;
448
449     let css = '';
450     for (const [id, def] of this.registry.entries()) {
451       if (def.tokens && typeof def.tokens === 'object') {
452         css += `[data-theme="${id}"] {\n`;
453         for (const [prop, val] of Object.entries(def.tokens))
{
454           css += `  ${prop}: ${val};\n`;
455         }
456         css += '}\n';
457       }
458     }
459
460     if (!this._dynamicStyleEl) {
461       this._dynamicStyleEl = document.createElement('style');
462       this._dynamicStyleEl.id = 'theme-engine-dynamic-styles';
463       document.head.appendChild(this._dynamicStyleEl);
464     }
```

```
    this._dynamicStyleEl.textContent = css;
  }
}

/**
 * Синглтон экземпляр движка тем
 */
export const themeEngine = new ThemeEngine();

// Глобальный экспорт на window
if (typeof window !== 'undefined') {
  window.ThemeEngine = ThemeEngine;
  window.themeEngine = themeEngine;
}

```

theme.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Js - Theme Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля theme и и
нтеграции с ThemeEngine.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/theme.js?v=20261004
_v17" type="module"></script>
 *
 *   JavaScript Import:
 *     import { getThemeMode, getResolvedTheme, setTheme, them
eEngine } from '/windows/api/webgui/js/theme.js';
 *
 * File: theme.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

import { themeEngine, ThemeEngine, THEME_STORAGE_KEY, DEFAULT_
THEME_MODE, BUILTIN_THEMES } from './theme-engine.js';

/**
 * Получает сохранённый режим темы пользователя.
 * @returns {string}
 */
```

```
export function getThemeMode() {
  return themeEngine.getSavedMode();
}

/**
 * Разрешает эффективную тему оформления ('light' | 'dark' | 'brick' | 'terminal').
 * @param {string} [mode]
 * @returns {string}
 */
export function getResolvedTheme(mode = getThemeMode()) {
  return themeEngine.resolveTheme(mode);
}

/**
 * Устанавливает тему оформления и применяет её ко всем элементам интерфейса.
 * @param {string} mode
 */
export function setTheme(mode) {
  themeEngine.applyTheme(mode);
}

/**
 * Инициализирует менеджер тем и слушатели событий при загрузке страницы.
 */
export function initTheme() {
  themeEngine.init();
}

// Экспорт синглтона и констант
export { themeEngine, ThemeEngine, THEME_STORAGE_KEY, DEFAULT_THEME_MODE, BUILTIN_THEMES };

// Экспорт в глобальный объект window
if (typeof window !== 'undefined') {
  window.setTheme = setTheme;
```

```
  window.getThemeMode = getThemeMode;
  window.getResolvedTheme = getResolvedTheme;
  window.initTheme = initTheme;
  window.themeEngine = themeEngine;
  window.ThemeEngine = ThemeEngine;
}

```

toast.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Js - Toast Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля toast.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/toast.js?v=20261001
_v1" type="module"></script>
 *
 * File: toast.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * toast.js — Универсальная система всплывающих окон (Toast &
Floating Notifications)
 * для веб-интерфейса AI-Windows-Admin.
 *
 * Предоставляет функции:
 * - window.showToast(message, type, options)
 * - window.showNotification(title, message, type, duration)
 * - Перехват стандартного window.alert() с выводом всплывающи
х окон.
 */

```

```
(function () {
  'use strict';

  // Конфигурация по умолчанию
  const DEFAULT_DURATION = 4500;
  const CONTAINER_ID = 'floating-toast-container';

  /**
   * Инициализирует контейнер для всплывающих окон.
   * @returns {HTMLElement} Элемент контейнера.
   */
  function getOrCreateContainer() {
    let container = document.getElementById(CONTAINER_ID);
    if (!container) {
      container = document.createElement('div');
      container.id = CONTAINER_ID;
      container.className = 'floating-toast-container';
      container.setAttribute('aria-live', 'polite');
      container.setAttribute('aria-atomic', 'true');
      document.body.appendChild(container);
    }
    return container;
  }

  /**
   * Определяет иконку и заголовок по типу уведомления.
   * @param {string} type - Тип ('success', 'danger', 'error',
 'warning', 'info').
   * @returns {{iconClass: string, headerText: string, bootstr
apType: string}}
   */
  function getMetaForType(type) {
    const t = (type || 'info').toLowerCase();
    switch (t) {
      case 'success':
      case 'ok':
        return {
          iconClass: 'bi bi-check-circle-fill text-success',
```

```
          headerText: 'Успешно',
          bootstrapType: 'success'
        };
      case 'danger':
      case 'error':
      case 'err':
        return {
          iconClass: 'bi bi-exclamation-octagon-fill text-dang
er',
          headerText: 'Ошибка',
          bootstrapType: 'danger'
        };
      case 'warning':
      case 'warn':
        return {
          iconClass: 'bi bi-exclamation-triangle-fill text-war
ning',
          headerText: 'Внимание',
          bootstrapType: 'warning'
        };
      case 'info':
      default:
        return {
          iconClass: 'bi bi-info-circle-fill text-info',
          headerText: 'Уведомление',
          bootstrapType: 'info'
        };
    }
  }

  /**
   * Автоматически определяет тип уведомления по содержимому с
ообщения.
   * @param {string} text - Текст сообщения.
   * @returns {string} Определенный тип ('success', 'danger',
'warning', 'info').
   */
  function detectTypeFromText(text) {
```

```
    if (!text || typeof text !== 'string') return 'info';
    const lower = text.toLowerCase();
    if (text.includes('✅') || lower.includes('успешно') || lower.includes('success') || lower.includes('готов')) {
      return 'success';
    }
    if (text.includes('❌') || lower.includes('ошибка') || lower.includes('error') || lower.includes('failed') || lower.includes('не удалось')) {
      return 'danger';
    }
    if (text.includes('⚠') || lower.includes('внимание') || lower.includes('warning') || lower.includes('заполните') || lower.includes('пожалуйста, выберите') || lower.includes('пожалуйста выберите') || lower.includes('укажите')) {
      return 'warning';
    }
    return 'info';
  }

  /**
   * Отображает всплывающее окно (Toast).
   * @param {string} message - Текст сообщения или HTML.
   * @param {string} [type='info'] - Тип сообщения ('success', 'danger', 'error', 'warning', 'info').
   * @param {Object} [options={}] - Дополнительные параметры (duration, title, isHtml, autoClose).
   * @returns {HTMLElement} DOM-элемент тоста.
   */
  function showToast(message, type, options = {}) {
    if (message === undefined || message === null) return null;

    // Если тип не указан, пытаемся определить его автоматически по тексту
    const resolvedType = type || detectTypeFromText(String(message));
    const meta = getMetaForType(resolvedType);
```

```
    const duration = options.duration !== undefined ? options.duration : DEFAULT_DURATION;
    const title = options.title || meta.headerText;

    const container = getOrCreateContainer();

    const toastEl = document.createElement('div');
    toastEl.className = `floating-toast floating-toast-${meta.bootstrapType}`;
    toastEl.setAttribute('role', 'alert');

    // Форматирование текста (сохранение переводов строк)
    const formattedMessage = options.isHtml
      ? message
      : String(message)
          .replace(/&/g, '&amp;')
          .replace(/</g, '&lt;')
          .replace(/>/g, '&gt;')
          .replace(/\n/g, '<br>');

    toastEl.innerHTML = `
      <div class="floating-toast-header">
        <i class="${meta.iconClass} floating-toast-icon"></i>
        <strong class="floating-toast-title">${title}</strong>
        <button type="button" class="floating-toast-close" aria-label="Закрыть">&times;</button>
      </div>
      <div class="floating-toast-body">${formattedMessage}</div>
      ${duration > 0 ? `<div class="floating-toast-progress"><div class="floating-toast-progress-bar bg-${meta.bootstrapType}"></div></div>` : ''}
    `;

    container.appendChild(toastEl);

    // Анимация появления
    requestAnimationFrame(() => {
```

```
      toastEl.classList.add('show');
    });

    let dismissTimeout = null;
    let startTime = Date.now();
    let remainingTime = duration;
    const progressBar = toastEl.querySelector('.floating-toast
-progress-bar');

    const startTimer = () => {
      if (duration > 0) {
        startTime = Date.now();
        if (progressBar) {
          progressBar.style.transition = `width ${remainingTim
e}ms linear`;
          progressBar.style.width = '0%';
        }
        dismissTimeout = setTimeout(() => {
          closeToast();
        }, remainingTime);
      }
    };

    const pauseTimer = () => {
      if (dismissTimeout) {
        clearTimeout(dismissTimeout);
        dismissTimeout = null;
        remainingTime -= Date.now() - startTime;
        if (progressBar) {
          const computedWidth = window.getComputedStyle(progre
ssBar).width;
          progressBar.style.transition = 'none';
          progressBar.style.width = computedWidth;
        }
      }
    };

    const closeToast = () => {
```

```
      if (dismissTimeout) clearTimeout(dismissTimeout);
      toastEl.classList.remove('show');
      toastEl.classList.add('hide');
      toastEl.addEventListener('transitionend', () => {
        if (toastEl.parentElement) {
          toastEl.parentElement.removeChild(toastEl);
        }
      }, { once: true });
    };

    // Слушатели событий
    const closeBtn = toastEl.querySelector('.floating-toast-cl
ose');
    if (closeBtn) {
      closeBtn.addEventListener('click', (e) => {
        e.stopPropagation();
        closeToast();
      });
    }

    if (duration > 0) {
      toastEl.addEventListener('mouseenter', pauseTimer);
      toastEl.addEventListener('mouseleave', () => {
        if (remainingTime > 0) startTimer();
      });
      startTimer();
    }

    return toastEl;
  }

  /**
   * Отображает уведомление с заголовком и текстом.
   * @param {string} title - Заголовок.
   * @param {string} message - Текст сообщения.
   * @param {string} [type='info'] - Тип.
   * @param {number} [duration=4500] - Длительность показа.
   */
```

```
  function showNotification(title, message, type = 'info', dur
ation = DEFAULT_DURATION) {
    return showToast(message, type, { title, duration });
  }

  // Экспорт в глобальную область видимости
  window.showToast = showToast;
  window.showNotification = showNotification;
  window._showToast = showToast; // Для совместимости с уже им
еющимися вкладками

  // Перехват стандартного window.alert()
  // Сохраняем ссылку на оригинальный alert на случай необходи
мости
  window._originalAlert = window.alert;

  window.alert = function (message) {
    // Выводим неблокирующее всплывающее окно
    showToast(message);
    // Также дублируем в консоль для отладки
    if (window.console && console.debug) {
      console.debug('[Toast Alert]:', message);
    }
  };

  // Инициализация контейнера при загрузке документа
  if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', getOrCreateC
ontainer);
  } else {
    getOrCreateContainer();
  }
})();

```

userSettings.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Js - Usersettings Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля userSetti
ngs.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/userSettings.js?v=2
0261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { executeUserAction, isUserAdmin } from '/window
s/api/webgui/js/userSettings.js';
 *
 * File: userSettings.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * userSettings.js — User Profile & Settings Management
 * Supports Google OAuth synchronization, profile status, lang
uage & theme controls.
 */

'use strict';
```

```
30
31 export function executeUserAction(actionType) {
32   // Ensure chat tab is active
33   if (typeof window.switchToTab === 'function') {
34     window.switchToTab('chat');
35   } else if (typeof window.switchTab === 'function') {
36     window.switchTab('tab-chat');
37   } else {
38     const chatTabBtn = document.querySelector('[data-bs-target
="#tab-chat"], [data-tab="tab-chat"]');
39     if (chatTabBtn && window.bootstrap?.Tab) {
40       bootstrap.Tab.getOrCreateInstance(chatTabBtn).show();
41     }
42   }
43
44   const msgInput = document.getElementById('message-input');
45   const sendBtn = document.getElementById('send-button');
46
47   const send = () => {
48     if (typeof window.sendChatMessage === 'function') {
49       window.sendChatMessage();
50     } else if (sendBtn) {
51       sendBtn.click();
52     }
53   };
54
55   if (actionType === 'mail') {
56     if (msgInput) {
57       msgInput.value = 'Проверь мою почту Gmail и покажи после
дние непрочитанные письма.';
58       setTimeout(send, 100);
59     }
60   } else if (actionType === 'docs') {
61     const query = prompt('Введите поисковый запрос для поиска
документов в Google Docs / Drive:');
62     if (query && query.trim() && msgInput) {
63       msgInput.value = `Найди в Google Docs / Google Drive док
ументы по запросу: "${query.trim()}".`;
```

```
      setTimeout(send, 100);
    }
  } else if (actionType === 'sheets') {
    const query = prompt('Введите ID таблицы или команду для Google Sheets (например: "Прочитай данные из таблицы ID диапазон A1:D10" или "Добавь строку"):');
    if (query && query.trim() && msgInput) {
      msgInput.value = `Google Sheets: ${query.trim()}`;
      setTimeout(send, 100);
    }
  } else if (actionType === 'news') {
    if (msgInput) {
      msgInput.value = 'Подготовь для меня свежий дайджест самых релевантных новостей по моим интересам.';
      setTimeout(send, 100);
    }
  } else if (actionType === 'facebook') {
    const text = prompt('Введите текст публикации для Facebook (и при желании ссылку через пробел):');
    if (text && text.trim() && msgInput) {
      msgInput.value = `Опубликуй пост в Facebook с текстом: "${text.trim()}".`;
      setTimeout(send, 100);
    }
  }
}

export async function initUserSettings() {
  await refreshUserProfile();

  // Bind User Menu Action Buttons
  document.querySelectorAll('#user-menu-btn-mail, #nav-btn-check-mail, .action-btn-mail').forEach((btn) => {
    btn.addEventListener('click', (e) => {
      e.preventDefault();
      executeUserAction('mail');
    });
  });
```

```
96
97    document.querySelectorAll('#user-menu-btn-docs, #nav-btn-search-gdocs, .action-btn-docs').forEach((btn) => {
98      btn.addEventListener('click', (e) => {
99        e.preventDefault();
100       executeUserAction('docs');
101     });
102   });
103
104   document.querySelectorAll('#user-menu-btn-sheets, #nav-btn-gsheets, .action-btn-sheets').forEach((btn) => {
105     btn.addEventListener('click', (e) => {
106       e.preventDefault();
107       executeUserAction('sheets');
108     });
109   });
110
111   document.querySelectorAll('#user-menu-btn-news, #nav-btn-news-digest, .action-btn-news').forEach((btn) => {
112     btn.addEventListener('click', (e) => {
113       e.preventDefault();
114       executeUserAction('news');
115     });
116   });
117
118   document.querySelectorAll('#user-menu-btn-facebook, #nav-btn-facebook-post, .action-btn-facebook').forEach((btn) => {
119     btn.addEventListener('click', (e) => {
120       e.preventDefault();
121       executeUserAction('facebook');
122     });
123   });
124
125   // Handle Google Login Click
126   document.querySelectorAll('.btn-google-login, #user-settings-google-btn').forEach((btn) => {
127     btn.addEventListener('click', (e) => {
128       e.preventDefault();
```

```
      const nextUrl = window.location.pathname + window.location.search;
      window.location.href = `/auth/google?next=${encodeURIComponent(nextUrl)}`;
    });
  });

  // Handle Logout Click
  document.querySelectorAll('.btn-user-logout, #user-settings-logout-btn').forEach((btn) => {
    btn.addEventListener('click', async (e) => {
      e.preventDefault();
      try {
        await fetch('/auth/logout', { method: 'POST' });
      } catch (err) {
        console.error('Logout error:', err);
      } finally {
        window.location.href = '/';
      }
    });
  });

  // Re-sync modal fields on modal show
  const modalEl = document.getElementById('userSettingsModal');
  if (modalEl) {
    modalEl.addEventListener('show.bs.modal', async () => {
      await refreshUserProfile();
      const savedLang = localStorage.getItem('app_language') || 'ru';
      document.querySelectorAll('.lang-selector').forEach((s) => {
        s.value = savedLang;
      });
      const savedTheme = localStorage.getItem('theme') || 'system';
      document.querySelectorAll('.theme-selector').forEach((s) => {
```

```
        s.value = savedTheme;
      });
    });
  }
}

export function isUserAdmin() {
  if (!window.currentUser) return false;
  return Boolean(window.currentUser.is_admin || (window.curren
tUser.role && String(window.currentUser.role).toLowerCase() ==
= 'admin'));
}

export async function refreshUserProfile() {
  try {
    const res = await fetch('/auth/check');
    if (!res.ok) return;
    const data = await res.json();
    window.currentUser = data;

    const authCard = document.getElementById('user-settings-au
th-card');
    const guestCard = document.getElementById('user-settings-g
uest-card');
    const navGoogleBtn = document.getElementById('nav-google-b
tn');
    const userSettingsBtn = document.getElementById('user-sett
ings-btn');
    const navAvatar = document.getElementById('nav-user-avatar
');
    const navIcon = document.getElementById('nav-user-icon');
    const navLabel = document.getElementById('nav-user-label')
;

    if (data && data.authenticated) {
      // User is authenticated
      if (authCard) authCard.classList.remove('d-none');
      if (guestCard) guestCard.classList.add('d-none');
```

```
189
190       // Navbar: show user profile pill, hide Google register
    button
191       if (navGoogleBtn) navGoogleBtn.classList.add('d-none');
192       if (userSettingsBtn) {
193         userSettingsBtn.classList.remove('d-none');
194         userSettingsBtn.classList.add('d-flex');
195       }
196
197       const name = data.name || data.email || 'Пользователь';
198       const email = data.email || '';
199       const picture = data.picture || '';
200       const role = data.role || (data.is_admin ? 'Admin' : 'Us
    er');
201       const hasGoogle = Boolean(data.has_google);
202
203       // Update modal fields
204       const avatarEl = document.getElementById('user-settings-
    avatar');
205       const nameEl = document.getElementById('user-settings-na
    me');
206       const emailEl = document.getElementById('user-settings-e
    mail');
207       const roleBadge = document.getElementById('user-settings
    -role-badge');
208       const googleBadge = document.getElementById('user-settin
    gs-google-badge');
209
210       if (avatarEl) {
211         if (picture) {
212           avatarEl.src = picture;
213           avatarEl.classList.remove('d-none');
214         } else {
215           avatarEl.classList.add('d-none');
216         }
217       }
218       if (nameEl) nameEl.textContent = name;
219       if (emailEl) emailEl.textContent = email;
```

```
220       if (roleBadge) {
221         roleBadge.textContent = role;
222         roleBadge.className = `badge ${role.toLowerCase() ===
'admin' ? 'bg-danger' : 'bg-primary'}`;
223       }
224       const isOauthEnabled = data.oauth_enabled !== false;
225
226       if (googleBadge) {
227         if (hasGoogle) {
228           googleBadge.innerHTML = '<i class="bi bi-google text
-danger me-1"></i>Google OAuth: <span class="text-success fw-b
old">Синхронизирован</span>';
229         } else if (isOauthEnabled) {
230           googleBadge.innerHTML = '<i class="bi bi-google text
-muted me-1"></i>Google OAuth: <span class="text-warning">Не п
одключен</span> <button type="button" class="btn btn-sm btn-ou
tline-danger ms-2 py-0 px-2 btn-google-login">Синхронизировать
</button>';
231         } else {
232           googleBadge.innerHTML = '<i class="bi bi-google text
-muted me-1"></i>Google OAuth: <span class="text-muted">Отключ
ен</span>';
233         }
234       }
235
236       // Update dropdown header elements if present
237       const menuWrapper = document.getElementById('user-menu-d
ropdown-wrapper');
238       const menuDisplayName = document.getElementById('user-me
nu-display-name');
239       const menuDisplayEmail = document.getElementById('user-m
enu-display-email');
240       if (menuWrapper) {
241         menuWrapper.classList.remove('d-none');
242         menuWrapper.classList.add('d-flex');
243       }
244       if (menuDisplayName) menuDisplayName.textContent = name;
245       if (menuDisplayEmail) menuDisplayEmail.textContent = ema
```

```
il;

      // Update navbar button
      if (navAvatar && picture) {
        navAvatar.src = picture;
        navAvatar.classList.remove('d-none');
        if (navIcon) navIcon.classList.add('d-none');
      }
      if (navLabel) {
        const shortName = name.split(' ')[0] || name;
        navLabel.textContent = shortName;
      }
    } else {
      // Guest / unauthenticated
      if (authCard) authCard.classList.add('d-none');
      if (guestCard) guestCard.classList.remove('d-none');

      const isOauthEnabled = data && data.oauth_enabled !== fa
lse;

      // Navbar: show full Google register/sign-in button only
 if OAuth is enabled, else show user settings gear/pill
      if (navGoogleBtn) {
        if (isOauthEnabled) {
          navGoogleBtn.classList.remove('d-none');
          navGoogleBtn.classList.add('d-flex');
        } else {
          navGoogleBtn.classList.add('d-none');
          navGoogleBtn.classList.remove('d-flex');
        }
      }
      const menuWrapper = document.getElementById('user-menu-d
ropdown-wrapper');
      if (menuWrapper) {
        if (isOauthEnabled) {
          menuWrapper.classList.add('d-none');
          menuWrapper.classList.remove('d-flex');
        } else {
```

```
          menuWrapper.classList.remove('d-none');
          menuWrapper.classList.add('d-flex');
        }
      }
      if (userSettingsBtn) {
        if (isOauthEnabled) {
          userSettingsBtn.classList.add('d-none');
          userSettingsBtn.classList.remove('d-flex');
        } else {
          userSettingsBtn.classList.remove('d-none');
          userSettingsBtn.classList.add('d-flex');
        }
      }

      // Guest card google login button adjustment
      const guestGoogleBtn = guestCard ? guestCard.querySelect
or('.btn-google-login') : null;
      if (guestGoogleBtn) {
        if (isOauthEnabled) {
          guestGoogleBtn.classList.remove('d-none');
        } else {
          guestGoogleBtn.classList.add('d-none');
        }
      }

      if (navAvatar) navAvatar.classList.add('d-none');
      if (navIcon) navIcon.classList.remove('d-none');
      if (navLabel) navLabel.textContent = 'User Settings';
    }
  } catch (err) {
    console.error('Failed to refresh user profile:', err);
  }
}

window.initUserSettings = initUserSettings;
window.refreshUserProfile = refreshUserProfile;
window.isUserAdmin = isUserAdmin;
window.executeUserAction = executeUserAction;
```

window-api.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Js - Window-Api Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля window-ap
i.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/js/window-api.js?v=202
61001_v1" type="module"></script>
 *
 * File: window-api.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/js
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * window-api.js — инициализация window.api для всех вкладок и
 панелей управления
 * Этот файл загружается ДО всех вкладок, чтобы window.api был
 доступен сразу
 */

window.api = window.api || {};

// Универсальный fetch для API и Auth
if (!window.api.fetch) {
```

```
  window.api.fetch = async function(url, opts = {}) {
    const response = await fetch(url, opts);
    if (!response.ok) {
      let msg = response.statusText;
      try {
        const data = await response.json();
        if (data && data.detail) {
          if (typeof data.detail === 'string') {
            msg = data.detail;
          } else if (Array.isArray(data.detail)) {
            msg = data.detail.map(d => d.msg || JSON.stringify
(d)).join(', ');
          } else {
            msg = JSON.stringify(data.detail);
          }
        } else if (data && data.message) {
          msg = data.message;
        }
      } catch {}
      throw new Error(`${response.status} ${msg}`);
    }
    return response.json();
  };
}

// User management API
window.api.users = window.api.users || {
  async list(params = {}) {
    const q = new URLSearchParams(params).toString();
    return window.api.fetch(`/api/admin/users${q ? '?' + q : '
'}`);
  },
  async get(userId) {
    return window.api.fetch(`/api/admin/users/${userId}`);
  },
  async create(data) {
    return window.api.fetch('/api/admin/users', {
      method: 'POST',
```

```
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(data)
    });
  },
  async update(userId, data) {
    return window.api.fetch(`/api/admin/users/${userId}`, {
      method: 'PUT',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(data)
    });
  },
  async setPassword(userId, password) {
    return window.api.fetch(`/api/admin/users/${userId}/passwo
rd`, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ password })
    });
  },
  async toggleActive(userId) {
    return window.api.fetch(`/api/admin/users/${userId}/toggle
-active`, {
      method: 'POST'
    });
  },
  async toggleRole(userId) {
    return window.api.fetch(`/api/admin/users/${userId}/toggle
-role`, {
      method: 'POST'
    });
  },
  async delete(userId) {
    return window.api.fetch(`/api/admin/users/${userId}`, {
      method: 'DELETE'
    });
  }
};

```

```
// Skills management API
window.api.skills = window.api.skills || {
  async list(params = {}) {
    const q = new URLSearchParams(params).toString();
    return window.api.fetch(`/api/admin/skills${q ? '?' + q : ''}`);
  },
  async get(name) {
    return window.api.fetch(`/api/admin/skills/${encodeURIComponent(name)}`);
  },
  async create(data) {
    return window.api.fetch('/api/admin/skills', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(data)
    });
  },
  async update(name, data) {
    return window.api.fetch(`/api/admin/skills/${encodeURIComponent(name)}`, {
      method: 'PUT',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(data)
    });
  },
  async delete(name) {
    return window.api.fetch(`/api/admin/skills/${encodeURIComponent(name)}`, {
      method: 'DELETE'
    });
  },
  async package(name) {
    return window.api.fetch(`/api/admin/skills/${encodeURIComponent(name)}/package`, {
      method: 'POST'
    });
  }
```

```
};

// MCP Servers management API
window.api.mcp = window.api.mcp || {
  async list() {
    return window.api.fetch('/api/admin/mcp/servers');
  },
  async get(id) {
    return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}`);
  },
  async create(data) {
    return window.api.fetch('/api/admin/mcp/servers', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(data)
    });
  },
  async update(id, data) {
    return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}`, {
      method: 'PUT',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(data)
    });
  },
  async delete(id) {
    return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}`, {
      method: 'DELETE'
    });
  },
  async toggle(id) {
    return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}/toggle`, {
      method: 'POST'
    });
  },
```

```
  async test(id) {
    return window.api.fetch(`/api/admin/mcp/servers/${encodeURIComponent(id)}/test`, {
      method: 'POST'
    });
  },
  async testAdhoc(data) {
    return window.api.fetch('/api/admin/mcp/test', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(data)
    });
  },
  async getTools() {
    return window.api.fetch('/api/admin/mcp/tools');
  }
};

// Google Workspace Accounts API wrapper
window.api.googleAccounts = window.api.googleAccounts || {
  async list() {
    return await window.api.fetch('/api/admin/google-accounts');
  },
  async upload(formData) {
    const r = await fetch('/api/admin/google-accounts/upload', {
      method: 'POST',
      body: formData
    });
    if (!r.ok) {
      let msg = r.statusText;
      try { msg = (await r.json()).detail || msg; } catch {}
      throw new Error(`${r.status} ${msg}`);
    }
    return await r.json();
  },
  async create(params) {
```

```
    return await window.api.fetch('/api/admin/google-accounts', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(params)
    });
  },
  async setDefault(name) {
    return await window.api.fetch(`/api/admin/google-accounts/${encodeURIComponent(name)}/default`, {
      method: 'POST'
    });
  },
  async resetStatus(name) {
    return await window.api.fetch(`/api/admin/google-accounts/${encodeURIComponent(name)}/reset-status`, {
      method: 'POST'
    });
  },
  async test(name) {
    return await window.api.fetch(`/api/admin/google-accounts/${encodeURIComponent(name)}/test`, {
      method: 'POST'
    });
  },
  async delete(name) {
    return await window.api.fetch(`/api/admin/google-accounts/${encodeURIComponent(name)}`, {
      method: 'DELETE'
    });
  }
};

// System & God Mode API
window.api.system = window.api.system || {
  async openGodMode() {
    return await window.api.fetch('/api/v1/tc/godmode/open', { method: 'POST' });
```

```
  }
};

window.openGodModeFolder = async function() {
  try {
    const res = await fetch('/api/v1/tc/godmode/open', { metho
d: 'POST' });
    const data = await res.json();
    if (data && data.status === 'ok') {
      if (typeof showToast === 'function') {
        const msg = data.action === 'focused'
          ? '  Фокус переведён на открытое окно God Mode'
          : '  God Mode (All Tasks) открыт в Проводнике Window
s';
        showToast('success', msg);
      }
    } else {
      const errMsg = (data && data.message) ? data.message : '
Неизвестная ошибка';
      if (typeof showToast === 'function') {
        showToast('danger', 'Ошибка: ' + errMsg);
      } else {
        alert('Ошибка: ' + errMsg);
      }
    }
  } catch (err) {
    if (typeof showToast === 'function') {
      showToast('danger', 'Не удалось связаться с сервером TC:
 ' + err.message);
    } else {
      alert('Ошибка: ' + err.message);
    }
  }
};
```

login.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Login View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса login.

Usage Examples:
    Web Route:
      /windows/api/webgui/login.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/login.html" class="w-10
0 h-100 border-0"></iframe>

File: login.html
Project: ai-windows-admin
Package: windows/api/webgui
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!DOCTYPE html>
<html lang="ru">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-s
cale=1.0">
  <title>Вход в систему — AI Windows Admin</title>
  <link rel="icon" type="image/x-icon" href="/favicon.ico">
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dis
```

```
t/css/bootstrap.min.css" rel="stylesheet">
  <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.10.0/font/bootstrap-icons.css" rel="stylesheet">
  <link href="/html/css/main.css?v=20260917_refactor_v1" rel="stylesheet">
  <style>
    /* ── Страница входа: центрирование и лёгкий фон ── */
    body {
      display: flex;
      align-items: center;
      justify-content: center;
      min-height: 100vh;
      margin: 0;
      padding: 1rem;
      /* Лёгкий градиентный акцент поверх --bg-color из main.css */
      background:
        radial-gradient(circle at 50% 15%, rgba(99, 102, 241, 0.08), transparent 65%),
        var(--bg-color);
    }

    /* ── Карточка входа ── */
    .auth-card {
      width: 100%;
      max-width: 400px;
      background: var(--surface-1);
      border: 1px solid var(--border-color);
      border-radius: 12px;
      box-shadow: var(--shadow-lg);
      overflow: hidden;
    }

    /* ── Шапка карточки ── */
    .auth-header {
      padding: 1.5rem 1.5rem 1rem;
      text-align: center;
      background: var(--surface-2);
```

```
    border-bottom: 1px solid var(--border-color);
  }

  /* ── Логотип-бейдж ── */
  .auth-logo-badge {
    width: 48px;
    height: 48px;
    border-radius: 12px;
    background: linear-gradient(135deg, #6366f1, #3b82f6);
    display: inline-flex;
    align-items: center;
    justify-content: center;
    font-size: 1.5rem;
    color: #fff;
    box-shadow: 0 4px 12px rgba(99, 102, 241, 0.3);
    margin-bottom: 0.75rem;
  }

  /* ── Навигационные вкладки входа/регистрации ── */
  .auth-nav-tabs .nav-link {
    border: none;
    border-bottom: 2px solid transparent;
    color: var(--text-color);
    opacity: 0.65;
    font-weight: 600;
    font-size: 0.85rem;
    padding: 0.55rem 1rem;
    background: transparent;
    transition: all 0.18s;
  }

  .auth-nav-tabs .nav-link.active {
    background: transparent;
    border-bottom: 2px solid #6366f1;
    color: #6366f1;
    opacity: 1;
  }

```

```
[data-theme="dark"] .auth-nav-tabs .nav-link.active {
  color: #818cf8;
  border-bottom-color: #818cf8;
}

/* ── Google OAuth кнопка ── */
.btn-google {
  background: var(--surface-1);
  color: var(--text-color);
  border: 1px solid var(--border-color);
  font-weight: 500;
  font-size: 0.85rem;
  transition: all 0.18s;
}

.btn-google:hover {
  background: var(--surface-2);
  box-shadow: var(--shadow-sm);
  color: var(--text-color);
}

/* ── Разделитель «или по почте» ── */
.divider-text {
  display: flex;
  align-items: center;
  text-align: center;
  color: var(--text-muted);
  font-size: 0.78rem;
  margin: 1rem 0;
}

.divider-text::before,
.divider-text::after {
  content: '';
  flex: 1;
  border-bottom: 1px solid var(--border-color);
}

```

```
    .divider-text:not(:empty)::before { margin-right: 0.6em; }
    .divider-text:not(:empty)::after  { margin-left:  0.6em; }

    /* ── Кнопки управления (тема + язык) в правом верхнем углу ── */
    .top-controls {
      position: fixed;
      top: 0.75rem;
      right: 0.75rem;
      display: flex;
      gap: 0.375rem;
      z-index: 100;
    }
  </style>
</head>
<body>

  <!-- Управление: тема и язык -->
  <div class="top-controls">
    <select class="form-select form-select-sm" id="auth-lang-selector" style="width: auto; font-size: 0.78rem; padding: 0.2rem 0.4rem;">
      <option value="ru">RU</option>
      <option value="en">EN</option>
      <option value="he">HE</option>
    </select>
    <button class="btn btn-sm btn-outline-secondary d-flex align-items-center justify-content-center rounded-circle"
            id="theme-toggle"
            title="Переключить тему"
            style="width: 28px; height: 28px; padding: 0;">
      <i class="bi bi-circle-half" id="theme-icon"></i>
    </button>
  </div>

  <div class="auth-card">
    <!-- Шапка -->
    <div class="auth-header">
```

```
      <div class="auth-logo-badge">
        <i class="bi bi-cpu-fill"></i>
      </div>
      <h5 class="fw-bold mb-1" id="title-app-name">AI Windows Admin</h5>
      <p class="small mb-0" style="color: var(--text-muted);" id="title-app-desc">Авторизуйтесь для доступа к сервисам ИИ</p>
    </div>

    <!-- Вкладки: Войти / Регистрация -->
    <ul class="nav nav-tabs auth-nav-tabs nav-fill px-3 pt-2" id="authTabs" role="tablist">
      <li class="nav-item" role="presentation">
        <button class="nav-link active" id="tab-login-btn" data-bs-toggle="tab" data-bs-target="#pane-login" type="button" role="tab">
          <i class="bi bi-box-arrow-in-right me-1"></i>
          <span id="label-tab-login">Войти</span>
        </button>
      </li>
      <li class="nav-item" role="presentation">
        <button class="nav-link" id="tab-register-btn" data-bs-toggle="tab" data-bs-target="#pane-register" type="button" role="tab">
          <i class="bi bi-person-plus me-1"></i>
          <span id="label-tab-register">Регистрация</span>
        </button>
      </li>
    </ul>

    <!-- Тело карточки -->
    <div class="card-body p-3">
      <!-- Статус-алерт -->
      <div id="auth-alert" class="alert d-none mb-3 py-2 small" role="alert"></div>

      <!-- Google OAuth -->
```

```
      <button type="button"
              class="btn btn-google w-100 d-flex align-items-center justify-content-center gap-2 rounded-3 mb-0"
              style="padding: 0.45rem 0.75rem;"
              id="btn-google-auth">
        <svg width="16" height="16" viewBox="0 0 24 24">
          <path fill="#4285F4" d="M22.56 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h5.92c-.26 1.37-1.04 2.53-2.21 3.31v2.77h3.57c2.08-1.92 3.28-4.74 3.28-8.09z"/>
          <path fill="#34A853" d="M12 23c2.97 0 5.46-.98 7.28-2.66l-3.57-2.77c-.98.66-2.23 1.06-3.71 1.06-2.86 0-5.29-1.93-6.16-4.53H2.18v2.84C3.99 20.53 7.7 23 12 23z"/>
          <path fill="#FBBC05" d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s.13-1.43.35-2.09V7.06H2.18C1.43 8.55 1 10.22 1 12s.43 3.45 1.18 4.94l2.85-2.22.81-.63z"/>
          <path fill="#EA4335" d="M12 5.38c1.62 0 3.06.56 4.21 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 1 3.99 3.47 2.18 7.06l3.66 2.84c.87-2.6 3.3-4.52 6.16-4.52z"/>
        </svg>
        <span id="label-google-auth">Продолжить с Google</span>
      </button>

      <div class="divider-text" id="label-or-email">или по почте</div>

      <!-- Контент вкладок -->
      <div class="tab-content" id="authTabsContent">

        <!-- ВХОД -->
        <div class="tab-pane fade show active" id="pane-login" role="tabpanel">
          <form id="form-login">
            <div class="mb-2">
              <label class="form-label fw-semibold" id="label-email-login" for="login-email">Email</label>
              <div class="input-group input-group-sm">
                <span class="input-group-text"><i class="bi bi
```

```
-envelope"></i></span>
                <input type="email" class="form-control form-control-sm" id="login-email"
                       required placeholder="name@example.com" autocomplete="email">
              </div>
            </div>
            <div class="mb-3">
              <label class="form-label fw-semibold" id="label-pass-login" for="login-password">Пароль</label>
              <div class="input-group input-group-sm">
                <span class="input-group-text"><i class="bi bi-key"></i></span>
                <input type="password" class="form-control form-control-sm" id="login-password"
                       required placeholder="••••••••" autocomplete="current-password">
              </div>
            </div>
            <button type="submit" class="btn btn-primary btn-sm w-100 fw-semibold" id="btn-submit-login"
                    style="padding: 0.45rem;">
              <span class="spinner-border spinner-border-sm d-none me-1" id="login-spinner"></span>
              <span id="label-btn-login">Войти в аккаунт</span>
            </button>
          </form>
        </div>

        <!-- РЕГИСТРАЦИЯ -->
        <div class="tab-pane fade" id="pane-register" role="tabpanel">
          <form id="form-register">
            <div class="mb-2">
              <label class="form-label fw-semibold" id="label-reg-name" for="reg-name">Ваше имя</label>
              <div class="input-group input-group-sm">
```

```
                <span class="input-group-text"><i class="bi bi
-person"></i></span>
                <input type="text" class="form-control form-co
ntrol-sm" id="reg-name"
                       required placeholder="Иван Иванов" auto
complete="name">
              </div>
            </div>
            <div class="mb-2">
              <label class="form-label fw-semibold" id="label-
reg-email" for="reg-email">Email</label>
              <div class="input-group input-group-sm">
                <span class="input-group-text"><i class="bi bi
-envelope"></i></span>
                <input type="email" class="form-control form-c
ontrol-sm" id="reg-email"
                       required placeholder="name@example.com"
 autocomplete="email">
              </div>
            </div>
            <div class="mb-3">
              <label class="form-label fw-semibold" id="label-
reg-pass" for="reg-password">Придумайте пароль</label>
              <div class="input-group input-group-sm">
                <span class="input-group-text"><i class="bi bi
-lock"></i></span>
                <input type="password" class="form-control for
m-control-sm" id="reg-password"
                       required placeholder="Минимум 6 символо
в" minlength="6" autocomplete="new-password">
              </div>
            </div>
            <button type="submit" class="btn btn-primary btn-s
m w-100 fw-semibold" id="btn-submit-reg"
                    style="padding: 0.45rem;">
              <span class="spinner-border spinner-border-sm d-
none me-1" id="reg-spinner"></span>
              <span id="label-btn-reg">Зарегистрироваться</spa
```

```
n>
            </button>
          </form>
        </div>

      </div>
    </div>
  </div>

  <!-- Модальное окно подтверждения email -->
  <div class="modal fade" id="verifyModal" tabindex="-1" aria-
hidden="true" data-bs-backdrop="static">
    <div class="modal-dialog modal-dialog-centered modal-sm">
      <div class="modal-content">
        <div class="modal-header">
          <h6 class="modal-title fw-bold">
            <i class="bi bi-shield-check text-primary me-1"></
i> Подтверждение почты
          </h6>
          <button type="button" class="btn-close" data-bs-dism
iss="modal"></button>
        </div>
        <div class="modal-body text-center">
          <p class="small mb-3" style="color: var(--text-muted
);" id="verify-modal-desc">
            Введите код подтверждения, отправленный на ваш ема
il:
          </p>
          <div class="mb-3">
            <input type="text" class="form-control form-contro
l-sm text-center"
                   id="verify-code-input" placeholder="123456"
 maxlength="6"
                   autofocus style="letter-spacing: 0.2em; fon
t-size: 1.1rem;">
          </div>
          <div id="verify-alert" class="alert d-none py-1 smal
l"></div>
```

```
          <button type="button" class="btn btn-primary btn-sm w-100 fw-semibold" id="btn-submit-verify"
                  style="padding: 0.45rem;">
            <span class="spinner-border spinner-border-sm d-none me-1" id="verify-spinner"></span>
            Подтвердить
          </button>
        </div>
      </div>
    </div>
  </div>

  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
  <script>
    (function() {
      'use strict';

      // ── Тема: системная по умолчанию ──────────────────────
      function getSystemTheme() {
        return window.matchMedia && window.matchMedia('(prefers-color-scheme: dark)').matches
          ? 'dark' : 'light';
      }

      function applyTheme(mode) {
        // mode: 'light' | 'dark' | 'system'
        const resolved = (mode === 'system') ? getSystemTheme() : mode;
        document.documentElement.setAttribute('data-theme', resolved);
        document.documentElement.setAttribute('data-bs-theme', resolved);

        const icon = document.getElementById('theme-icon');
        if (icon) {
          if (mode === 'light')        icon.className = 'bi bi
```

```
-sun-fill text-warning';
          else if (mode === 'dark')    icon.className = 'bi bi-moon-stars-fill text-primary';
          else                         icon.className = 'bi bi-circle-half text-info';
        }
      }

      function initTheme() {
        const saved = localStorage.getItem('theme') || 'system';
        applyTheme(saved);

        // Следим за изменениями ОС в системном режиме
        if (window.matchMedia) {
          window.matchMedia('(prefers-color-scheme: dark)').addEventListener('change', () => {
            if ((localStorage.getItem('theme') || 'system') === 'system') {
              applyTheme('system');
            }
          });
        }
      }

      // Кнопка переключения темы: цикл system → light → dark → system
      document.getElementById('theme-toggle').addEventListener('click', () => {
        const cur = localStorage.getItem('theme') || 'system';
        const next = { system: 'light', light: 'dark', dark: 'system' }[cur] || 'system';
        localStorage.setItem('theme', next);
        applyTheme(next);
      });

      initTheme();

```

```
      // ── I18n ──────────────────────────────────────────────────────────
      const i18nDict = {
        ru: {
          appDesc: 'Авторизуйтесь для доступа к сервисам ИИ',
          tabLogin: 'Войти', tabReg: 'Регистрация',
          btnGoogle: 'Продолжить с Google', orEmail: 'или по почте',
          email: 'Email', pass: 'Пароль',
          regName: 'Ваше имя', regPass: 'Придумайте пароль',
          btnLogin: 'Войти в аккаунт', btnReg: 'Зарегистрироваться',
          passPlaceholder: 'Минимум 6 символов', namePlaceholder: 'Иван Иванов'
        },
        en: {
          appDesc: 'Sign in to access AI Windows Admin services',
          tabLogin: 'Sign In', tabReg: 'Sign Up',
          btnGoogle: 'Continue with Google', orEmail: 'or via Email',
          email: 'Email', pass: 'Password',
          regName: 'Full Name', regPass: 'Create Password',
          btnLogin: 'Sign In', btnReg: 'Create Account',
          passPlaceholder: 'Minimum 6 characters', namePlaceholder: 'John Doe'
        },
        he: {
          appDesc: '□ □ □ □ □  □ □ □  □ □ □ □  □ □ □ □ □ □ □  AI',
          tabLogin: '□ □ □ □ □ □ □ ', tabReg: '□ □ □ □ □ ',
          btnGoogle: '□ □ □ □  □ □  Google', orEmail: '□ □  □ □ □ □ □ □  □ □ ','□ "□
          email: '□ □ □ "□ ', pass: '□ □ □ □ □ ',
          regName: '□ □  □ □ □ ', regPass: '□ □ □  □ □ □ □ ',
          btnLogin: '□ □ □ □ □ ', btnReg: '□ □ □ □ □ ',
          passPlaceholder: '□ □ □ □ □  6 □ □ □ □ □ ', namePlaceholder: '□ □ □ □ □ □  □ □ □ □ □
        }
```

```
396       };
397
398       const langSelector = document.getElementById('auth-lang-
selector');
399       const savedLang = localStorage.getItem('app_language') |
| 'ru';
400       if (langSelector) {
401         langSelector.value = savedLang;
402         langSelector.addEventListener('change', (e) => {
403           const l = e.target.value;
404           localStorage.setItem('app_language', l);
405           updateLanguageTexts(l);
406         });
407       }
408
409       function updateLanguageTexts(lang) {
410         const d = i18nDict[lang] || i18nDict.ru;
411         document.getElementById('title-app-desc').textContent
  = d.appDesc;
412         document.getElementById('label-tab-login').textContent
  = d.tabLogin;
413         document.getElementById('label-tab-register').textCont
ent = d.tabReg;
414         document.getElementById('label-google-auth').textConte
nt = d.btnGoogle;
415         document.getElementById('label-or-email').textContent
  = d.orEmail;
416         document.getElementById('label-email-login').textConte
nt = d.email;
417         document.getElementById('label-pass-login').textConten
t = d.pass;
418         document.getElementById('label-reg-name').textContent
  = d.regName;
419         document.getElementById('label-reg-email').textContent
  = d.email;
420         document.getElementById('label-reg-pass').textContent
  = d.regPass;
421         document.getElementById('label-btn-login').textContent
```

```
  = d.btnLogin;
422         document.getElementById('label-btn-reg').textContent
  = d.btnReg;
423         document.getElementById('reg-password').placeholder
  = d.passPlaceholder;
424         document.getElementById('reg-name').placeholder
  = d.namePlaceholder;
425       }
426       updateLanguageTexts(savedLang);
427
428       // ── Алерты ──────────────────────────────────────────
──────────────────
429       function showAlert(msg, type = 'danger') {
430         const el = document.getElementById('auth-alert');
431         el.className = `alert alert-${type} mb-3 py-2 small`;
432         el.textContent = msg;
433         el.classList.remove('d-none');
434       }
435       function hideAlert() {
436         document.getElementById('auth-alert').classList.add('d
-none');
437       }
438
439       // ── Google OAuth ────────────────────────────────────
──────────────────
440       document.getElementById('btn-google-auth').addEventListe
ner('click', () => {
441         window.location.href = '/auth/google?next=/';
442       });
443
444       // ── Вход ────────────────────────────────────────────
──────────────────
445       let pendingVerificationEmail = '';
446       document.getElementById('form-login').addEventListener('
submit', async (e) => {
447         e.preventDefault();
448         hideAlert();
449         const email    = document.getElementById('login-email'
```

```
).value.trim();
        const password = document.getElementById('login-password').value;
        const spinner  = document.getElementById('login-spinner');
        const btn      = document.getElementById('btn-submit-login');
        btn.disabled = true;
        spinner.classList.remove('d-none');
        try {
          const res  = await fetch('/auth/login/email', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ email, password })
          });
          const data = await res.json();
          if (!res.ok) {
            if (res.status === 403 && data.detail && data.detail.includes('не подтверждена')) {
              pendingVerificationEmail = email;
              const desc = document.getElementById('verify-modal-desc');
              if (desc) desc.textContent = `Код подтверждения отправлен на ${email}:`;
              new bootstrap.Modal(document.getElementById('verifyModal')).show();
              showAlert(data.detail, 'warning');
            } else {
              showAlert(data.detail || 'Ошибка авторизации', 'danger');
            }
            return;
          }
          showAlert('Вход выполнен! Перенаправление...', 'success');
          setTimeout(() => { window.location.href = '/'; }, 400);
        } catch (err) {
```

```
          showAlert('Ошибка соединения: ' + err.message, 'dang
er');
        } finally {
          btn.disabled = false;
          spinner.classList.add('d-none');
        }
      });

      // ── Регистрация ──────────────────────────────────────
──────────────────────
      document.getElementById('form-register').addEventListene
r('submit', async (e) => {
        e.preventDefault();
        hideAlert();
        const name     = document.getElementById('reg-name').v
alue.trim();
        const email    = document.getElementById('reg-email').
value.trim();
        const password = document.getElementById('reg-password
').value;
        const spinner  = document.getElementById('reg-spinner'
);
        const btn      = document.getElementById('btn-submit-r
eg');
        btn.disabled = true;
        spinner.classList.remove('d-none');
        try {
          const res  = await fetch('/auth/register', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ email, password, name })
          });
          const data = await res.json();
          if (!res.ok) {
            showAlert(data.detail || 'Ошибка регистрации', 'da
nger');
            return;
          }
```

```
          pendingVerificationEmail = email;
          const desc = document.getElementById('verify-modal-desc');
          if (desc) desc.textContent = `Код подтверждения отправлен на ${email}:`;
          new bootstrap.Modal(document.getElementById('verifyModal')).show();
          showAlert('Регистрация начата. Введите код из письма.', 'info');
        } catch (err) {
          showAlert('Ошибка соединения: ' + err.message, 'danger');
        } finally {
          btn.disabled = false;
          spinner.classList.add('d-none');
        }
      });

      // ── Подтверждение кода ──────────────────────────────────────
      document.getElementById('btn-submit-verify').addEventListener('click', async () => {
        const codeInput = document.getElementById('verify-code-input');
        const alertEl   = document.getElementById('verify-alert');
        const spinner   = document.getElementById('verify-spinner');
        const btnVerify = document.getElementById('btn-submit-verify');
        const code = codeInput.value.trim();
        if (!code) return;
        alertEl.classList.add('d-none');
        btnVerify.disabled = true;
        spinner.classList.remove('d-none');
        try {
          const res  = await fetch('/auth/verify', {
            method: 'POST',
```

```
533             headers: { 'Content-Type': 'application/json' },
534             body: JSON.stringify({ email: pendingVerificationE
mail, code })
535           });
536           const data = await res.json();
537           if (!res.ok) {
538             alertEl.className = 'alert alert-danger py-1 small
';
539             alertEl.textContent = data.detail || 'Неверный код
';
540             alertEl.classList.remove('d-none');
541             return;
542           }
543           alertEl.className = 'alert alert-success py-1 small'
;
544           alertEl.textContent = 'Почта подтверждена! Войдите п
од своими данными.';
545           alertEl.classList.remove('d-none');
546           setTimeout(() => {
547             const modal = bootstrap.Modal.getInstance(document
.getElementById('verifyModal'));
548             if (modal) modal.hide();
549             document.getElementById('tab-login-btn').click();
550             document.getElementById('login-email').value = pen
dingVerificationEmail;
551             showAlert('Почта подтверждена. Введите пароль для
входа.', 'success');
552           }, 1200);
553         } catch (err) {
554           alertEl.className = 'alert alert-danger py-1 small';
555           alertEl.textContent = err.message;
556           alertEl.classList.remove('d-none');
557         } finally {
558           btnVerify.disabled = false;
559           spinner.classList.add('d-none');
560         }
561       });
562
```

```
    })();
  </script>
</body>
</html>
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Log Settings & Multi-Dom
ain Hierarchy
================================================================
===============
Description:
  Клиентский веб-интерфейс настройки, мониторинга и иерархичес
кого каталога
  Windows Event Logs по 7 доменам телеметрии (Security, Proces
s, Storage,
  Network, Update, Hardware, Power) и Sysmon, построения графо
в процессов
  (Provenance) и реконструкции жизненного цикла системы.

Usage Examples:
  Web Route:
    /windows/api/webgui/log_settings_tab/index.html

  Tab Frame Integration:
    <iframe src="/windows/api/webgui/log_settings_tab/index.ht
ml" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/log_settings_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<div class="container-fluid py-2 px-3 log-settings-container">
<!-- Верхняя панель метрик и сводки -->
<div class="row g-2 mb-3">
```

```
<!-- Карточка 1: Журнал Security -->
<div class="col-12 col-sm-6 col-xl-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3 position-relative" data-panel-id="panel-log-settings-1" id="panel-log-settings-1">
<div class="d-flex justify-content-between align-items-start">
<div class="text-muted small text-uppercase fw-semibold">Журнал Security  </div>
<span class="badge bg-primary font-monospace" id="ls-sec-badge">EVTX</span>
</div>
<div class="fs-4 fw-bold mt-1 text-info font-monospace" id="ls-sec-size">-- / -- MB</div>
<div class="progress my-2 bg-secondary" style="height: 6px;">
<div class="progress-bar bg-info" id="ls-sec-progress" role="progressbar" style="width: 0%;"></div>
</div>
<div class="small text-muted d-flex justify-content-between" style="font-size: 0.75rem;">
<span id="ls-sec-records">Записей: --</span>
<span id="ls-sec-status">Статус: --</span>
</div>
</div>
</div>
<!-- Карточка 2: Аудит CommandLine (Event 4688) -->
<div class="col-12 col-sm-6 col-xl-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-log-settings-2" id="panel-log-settings-2">
<div class="d-flex justify-content-between align-items-start">
<div class="text-muted small text-uppercase fw-semibold">Аудит CommandLine (4688)  </div>
<span class="badge bg-secondary font-monospace" id="ls-cmdline-badge">Audit</span>
</div>
<div class="fs-4 fw-bold mt-1" id="ls-cmdline-status">--</div>
<div class="small text-muted mt-1" id="ls-cmdline-desc">Аргументы в Event 4688</div>
```

```
<div class="mt-2">
<button class="btn btn-xs btn-outline-info w-100 py-1" id="ls-toggle-cmdline-btn" style="font-size: 0.78rem;">
<i class="bi bi-toggle-on me-1"></i><span>Переключить политику</span>
</button>
</div>
</div>
</div>
<!-- Карточка 3: Закладка и Сборщик Telemetry.db -->
<div class="col-12 col-sm-6 col-xl-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-log-settings-3" id="panel-log-settings-3">
<div class="d-flex justify-content-between align-items-start">
<div class="text-muted small text-uppercase fw-semibold">Сборщик Telemetry.db  </div>
<span class="badge bg-success font-monospace" id="ls-db-events-badge">0 в БД</span>
</div>
<div class="fs-4 fw-bold mt-1 text-warning font-monospace" id="ls-bookmark-id">ID: --</div>
<div class="small text-muted mt-1 text-truncate" id="ls-bookmark-time">Закладка: --</div>
<div class="d-flex gap-1 mt-2">
<button class="btn btn-xs btn-outline-success flex-grow-1 py-1" id="ls-collect-btn" style="font-size: 0.78rem;">
<i class="bi bi-cloud-arrow-down me-1"></i><span>Собрать сейчас</span>
</button>
<button class="btn btn-xs btn-outline-secondary py-1" id="ls-reset-bm-btn" style="font-size: 0.78rem;" title="Сбросить закладку на конец журнала">
<i class="bi bi-arrow-repeat"></i>
</button>
</div>
</div>
</div>
```

```
<!-- Карточка 4: 7 Доменов и Sysmon -->
<div class="col-12 col-sm-6 col-xl-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-log-settings-4" id="panel-log-settings-4">
<div class="d-flex justify-content-between align-items-start">
<div class="text-muted small text-uppercase fw-semibold">Иерархия и Sysmon  </div>
<span class="badge bg-info font-monospace" id="ls-sysmon-badge">Sysmon: ?</span>
</div>
<div class="fs-4 fw-bold mt-1 text-light font-monospace" id="ls-domains-count-badge">7 Доменов ОС</div>
<div class="small text-muted mt-1" id="ls-total-records-stat">Всего записей: --</div>
<div class="mt-2">
<button class="btn btn-xs btn-outline-primary w-100 py-1" id="ls-refresh-all-btn" style="font-size: 0.78rem;">
<i class="bi bi-arrow-clockwise me-1"></i><span>Обновить все параметры</span>
</button>
</div>
</div>
</div>
</div>
<!-- Навигация по 6 подразделам вкладки -->
<ul class="nav nav-pills mb-3 border-bottom border-secondary pb-2 gap-2 flex-wrap" id="ls-subtabs" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active py-1 px-3 d-flex align-items-center gap-1.5" data-bs-target="#ls-subpane-domains" data-bs-toggle="pill" id="ls-tab-domains-btn" role="tab" type="button">
<i class="bi bi-diagram-3-fill text-info"></i>
<span>7 Доменов и Каталог</span>
</button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link py-1 px-3 d-flex align-items-center ga
```

```
p-1.5" data-bs-target="#ls-subpane-channels" data-bs-toggle="p
ill" id="ls-tab-channels-btn" role="tab" type="button">
<i class="bi bi-table"></i>
<span>Параметры и размеры каналов</span>
</button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link py-1 px-3 d-flex align-items-center ga
p-1.5" data-bs-target="#ls-subpane-security" data-bs-toggle="p
ill" id="ls-tab-security-btn" role="tab" type="button">
<i class="bi bi-shield-check text-success"></i>
<span>Аудит безопасности и Телеметрия</span>
</button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link py-1 px-3 d-flex align-items-center ga
p-1.5" data-bs-target="#ls-subpane-provenance" data-bs-toggle=
"pill" id="ls-tab-provenance-btn" role="tab" type="button">
<i class="bi bi-bezier2 text-warning"></i>
<span>Граф процессов (Provenance)</span>
</button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link py-1 px-3 d-flex align-items-center ga
p-1.5" data-bs-target="#ls-subpane-lifecycle" data-bs-toggle="
pill" id="ls-tab-lifecycle-btn" role="tab" type="button">
<i class="bi bi-power text-danger"></i>
<span>Жизненный цикл ОС (Boot / Crash)</span>
</button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link py-1 px-3 d-flex align-items-center ga
p-1.5" data-bs-target="#ls-subpane-inspector" data-bs-toggle="
pill" id="ls-tab-inspector-btn" role="tab" type="button">
<i class="bi bi-journal-text text-primary"></i>
<span>Инспектор событий</span>
</button>
</li>
```

```
</ul>
<!-- Содержимое подразделов -->
<div class="tab-content" id="ls-subtabs-content">
<!-- Подраздел 1: 7 Доменов и Каталог провайдеров -->
<div class="tab-pane fade show active" id="ls-subpane-domains"
 role="tabpanel">
<!-- Сетка 7 доменов телеметрии -->
<div class="row g-2 mb-3" id="ls-domains-cards-grid">
<div class="col-12 text-center text-muted py-3">Загрузка катал
ога доменов...</div>
</div>
<!-- Каталог провайдеров выбранного домена -->
<div class="card bg-dark text-white border-secondary shadow-sm
" data-panel-id="panel-log-settings-5" id="panel-log-settings-
5">
<div class="card-header bg-dark border-secondary d-flex align-
items-center justify-content-between flex-wrap gap-2 py-2">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-collection text-info fs-5"></i>
<strong class="text-light" id="ls-catalog-header-title">Иерарх
ический каталог провайдеров Windows</strong>
<span class="badge bg-secondary" id="ls-catalog-count">--</spa
n>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<select class="form-select form-select-sm bg-dark text-white b
order-secondary" id="ls-catalog-domain-select" style="width: 2
00px;">
<option value="">Все 7 доменов</option>
<option value="security_identity">1. Security / Identity</opti
on>
<option value="process_tasks">2. Process / Tasks</option>
<option value="storage_io">3. Storage &amp;amp; Filesystem</option
>
<option value="network_comms">4. Network &amp;amp; Sockets</option
>
<option value="windows_update">5. Windows Update</option>
<option value="hardware_pnp">6. Hardware / Drivers</option>
```

```
<option value="power_lifecycle">7. Power / Boot / Shutdown</option>
<option value="sysmon_deep">Sysmon Observability</option>
</select>
<input class="form-control form-control-sm bg-dark text-white border-secondary" id="ls-catalog-search-input" placeholder="Поиск провайдера/полей..." style="width: 200px;" type="text"/>
</div>
</div>
<div class="table-responsive" style="max-height: 480px; overflow-y: auto;">
<table class="table table-dark table-hover table-striped align-middle mb-0" style="font-size: 0.82rem;">
<thead class="table-secondary text-uppercase" style="font-size: 0.72rem; position: sticky; top: 0; z-index: 2;">
<tr>
<th scope="col" style="width: 25%;">Провайдер / Канал</th>
<th scope="col" style="width: 15%;">Домен</th>
<th scope="col" style="width: 25%;">Ключевые Event ID</th>
<th scope="col" style="width: 10%;">Объем (Volume)</th>
<th scope="col" style="width: 12%;">Сборщик / ETW</th>
<th scope="col" style="width: 13%;">Цели корреляции</th>
</tr>
</thead>
<tbody id="ls-catalog-tbody">
<tr>
<td class="text-center text-muted py-4" colspan="6">Загрузка каталога...</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- Подраздел 2: Таблица каналов и их конфигурация -->
<div class="tab-pane fade" id="ls-subpane-channels" role="tabpanel">
<div class="card bg-dark text-white border-secondary shadow-sm
```

```
" data-panel-id="panel-log-settings-6" id="panel-log-settings-6">
<div class="card-header bg-dark border-secondary d-flex align-items-center justify-content-between flex-wrap gap-2 py-2">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-sliders text-info fs-5"></i>
<strong class="text-light">Конфигурация параметров Windows Event Log</strong>
<span class="badge bg-secondary" id="ls-table-count">Загрузка...</span>
</div>
<div class="d-flex align-items-center gap-2">
<input class="form-control form-control-sm bg-dark text-white border-secondary" id="ls-channel-filter-input" placeholder="Поиск журнала..." style="width: 220px;" type="text"/>
<button class="btn btn-sm btn-outline-secondary" id="ls-reload-channels-btn" title="Перезагрузить">
<i class="bi bi-arrow-clockwise"></i>
</button>
</div>
</div>
<div class="table-responsive" style="max-height: 520px; overflow-y: auto;">
<table class="table table-dark table-hover table-striped align-middle mb-0" style="font-size: 0.85rem;">
<thead class="table-secondary text-uppercase" style="font-size: 0.75rem; position: sticky; top: 0; z-index: 2;">
<tr>
<th scope="col" style="width: 22%;">Журнал (Channel)</th>
<th scope="col" style="width: 10%;">Размер / Лимит</th>
<th scope="col" style="width: 14%;">Заполнение (%)</th>
<th scope="col" style="width: 12%;">Записей</th>
<th scope="col" style="width: 12%;">Диапазон ID</th>
<th scope="col" style="width: 12%;">Политика ротации</th>
<th scope="col" style="width: 8%;">Статус</th>
<th class="text-end" scope="col" style="width: 10%;">Действия</th>
</tr>
```

```
</thead>
<tbody id="ls-channels-tbody">
<tr>
<td class="text-center text-muted py-4" colspan="8">
<div class="spinner-border spinner-border-sm text-info me-2" r
ole="status"></div>
                  Загрузка параметров каналов журналов...
                </td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- Подраздел 3: Аудит безопасности и телеметрия -->
<div class="tab-pane fade" id="ls-subpane-security" role="tabp
anel">
<div class="row g-3">
<!-- Левая колонка: Настройки аудита реестра и прав -->
<div class="col-12 col-lg-6">
<div class="card bg-dark text-white border-secondary shadow-sm
 h-100" data-panel-id="panel-log-settings-7" id="panel-log-set
tings-7">
<div class="card-header bg-dark border-secondary py-2">
<strong class="text-info d-flex align-items-center gap-2">
<i class="bi bi-shield-lock-fill"></i>
<span>Политики аудита безопасности Windows</span>
</strong>
</div>
<div class="card-body p-3">
<div class="mb-3">
<div class="d-flex justify-content-between align-items-center
mb-1">
<strong class="text-light">Аудит командной строки процессов (E
vent 4688)</strong>
<span class="badge" id="ls-audit-cmdline-badge">--</span>
</div>
<div class="small text-muted mb-2">
```

```
                  Включает запись полного пути и аргументов запуска программ (`CommandLine`) в журнал `Security`.
                </div>
<div class="bg-black rounded p-2 border border-secondary font-monospace small mb-2 text-warning" style="font-size: 0.75rem;">
                  HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System\Audit<br/>
                  Value: ProcessCreationIncludeCmdLine_Enabled
                </div>
<div class="d-flex gap-2">
<button class="btn btn-sm btn-success d-flex align-items-center gap-1" id="ls-enable-cmdline-btn">
<i class="bi bi-check-circle"></i><span>Включить аудит Command Line</span>
</button>
<button class="btn btn-sm btn-outline-danger d-flex align-items-center gap-1" id="ls-disable-cmdline-btn">
<i class="bi bi-x-circle"></i><span>Отключить</span>
</button>
</div>
</div>
<hr class="border-secondary"/>
<div class="mb-2">
<strong class="text-light">Доступ к каналу Security (SeSecurityPrivilege)</strong>
<div class="small text-muted mt-1 mb-2">
                  Чтение журнала Security требует прав Администратора или прав аудита безопасности.
                </div>
<div class="alert alert-dark border-secondary p-2 small mb-0" id="ls-sec-priv-status">
                  Проверка прав...
                </div>
</div>
</div>
</div>
</div>
```

```
<!-- Правая колонка: Управление Bookmark и базой telemetry.db
-->
<div class="col-12 col-lg-6">
<div class="card bg-dark text-white border-secondary shadow-sm
 h-100" data-panel-id="panel-log-settings-8" id="panel-log-set
tings-8">
<div class="card-header bg-dark border-secondary py-2">
<strong class="text-warning d-flex align-items-center gap-2">
<i class="bi bi-database-fill-gear"></i>
<span>Инкрементальный сборщик и Bookmark (telemetry.db)</span>
</strong>
</div>
<div class="card-body p-3">
<div class="row g-2 mb-3">
<div class="col-6">
<div class="p-2 bg-black rounded border border-secondary text-
center">
<div class="text-muted small">Текущая закладка (RecordID)</div
>
<div class="fs-5 fw-bold text-warning font-monospace" id="ls-b
m-curr-id">--</div>
</div>
</div>
<div class="col-6">
<div class="p-2 bg-black rounded border border-secondary text-
center">
<div class="text-muted small">Событий в telemetry.db</div>
<div class="fs-5 fw-bold text-success font-monospace" id="ls-b
m-db-count">--</div>
</div>
</div>
</div>
<div class="mb-3">
<label class="form-label small text-muted">Сброс или установка
 целевого Record ID закладки:</label>
<div class="input-group input-group-sm mb-2">
<input class="form-control bg-dark text-white border-secondary
" id="ls-custom-record-id-input" placeholder="Record ID (пусто
```

```
 = хвост журнала)" type="number"/>
<button class="btn btn-warning" id="ls-set-custom-bm-btn">
<i class="bi bi-pin-angle me-1"></i>Установить закладку
                  </button>
</div>
<div class="small text-secondary" style="font-size: 0.75rem;">
                  Сброс на хвост позволяет пропустить старые м
иллионы записей и читать только свежие новые события.
                </div>
</div>
<hr class="border-secondary"/>
<div>
<strong class="text-light mb-1 d-block">Ручной запуск сбора:</
strong>
<div class="d-flex align-items-center gap-2">
<select class="form-select form-select-sm bg-dark text-white b
order-secondary" id="ls-collect-batch-size" style="width: 140p
x;">
<option value="100">100 событий</option>
<option selected="" value="500">500 событий</option>
<option value="1000">1000 событий</option>
<option value="5000">5000 событий</option>
</select>
<button class="btn btn-sm btn-primary d-flex align-items-cente
r gap-1.5" id="ls-run-collector-btn">
<i class="bi bi-play-fill"></i><span>Запустить сбор</span>
</button>
</div>
<div class="small mt-2 text-info d-none" id="ls-collector-resu
lt-banner"></div>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- Подраздел 4: Граф процессов (Process Provenance) -->
<div class="tab-pane fade" id="ls-subpane-provenance" role="ta
```

```
bpanel">
<div class="card bg-dark text-white border-secondary shadow-sm" data-panel-id="panel-log-settings-9" id="panel-log-settings-9">
<div class="card-header bg-dark border-secondary d-flex align-items-center justify-content-between flex-wrap gap-2 py-2">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-diagram-2 text-warning fs-5"></i>
<strong class="text-light">Windows Process Provenance (User → Process → Parent → Service → Task)</strong>
</div>
<div class="d-flex align-items-center gap-2">
<input class="form-control form-control-sm bg-dark text-white border-secondary" id="ls-prov-process-input" placeholder="Имя процесса (напр. python.exe)..." style="width: 240px;" type="text"/>
<button class="btn btn-sm btn-warning d-flex align-items-center gap-1" id="ls-prov-build-btn">
<i class="bi bi-play-circle"></i><span>Построить граф</span>
</button>
</div>
</div>
<div class="card-body p-3">
<div class="p-3 bg-black rounded border border-secondary" id="ls-prov-tree-view" style="min-height: 360px; max-height: 520px; overflow-y: auto;">
<div class="text-center text-muted py-5">
<i class="bi bi-diagram-3 fs-1 text-secondary d-block mb-2"></i>
                Нажмите «Построить граф» для реконструкции цепочки происхождения процессов и служб.
            </div>
</div>
</div>
</div>
</div>
<!-- Подраздел 5: Жизненный цикл ОС (Power / Boot / Crash) -->
<div class="tab-pane fade" id="ls-subpane-lifecycle" role="tab
```

```
panel">
<div class="card bg-dark text-white border-secondary shadow-sm
" data-panel-id="panel-log-settings-10" id="panel-log-settings
-10">
<div class="card-header bg-dark border-secondary d-flex align-
items-center justify-content-between flex-wrap gap-2 py-2">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-clock-history text-danger fs-5"></i>
<strong class="text-light">Хронология жизненного цикла ОС (BOO
T → LOGIN → SLEEP → WAKE → SHUTDOWN / CRASH)</strong>
</div>
<div class="d-flex align-items-center gap-2">
<select class="form-select form-select-sm bg-dark text-white b
order-secondary" id="ls-lifecycle-hours-select" style="width:
140px;">
<option value="24">За 24 часа</option>
<option selected="" value="72">За 3 дня</option>
<option value="168">За 7 дней</option>
<option value="720">За 30 дней</option>
</select>
<button class="btn btn-sm btn-outline-danger" id="ls-lifecycle
-reload-btn">
<i class="bi bi-arrow-clockwise"></i>
</button>
</div>
</div>
<div class="card-body p-3" style="max-height: 500px; overflow-
y: auto;">
<div class="timeline position-relative ps-4" id="ls-lifecycle-
timeline">
<div class="text-center text-muted py-4">Загрузка хронологии п
итания и загрузки...</div>
</div>
</div>
</div>
</div>
<!-- Подраздел 6: Инспектор и просмотр событий -->
<div class="tab-pane fade" id="ls-subpane-inspector" role="tab
```

```
panel">
<div class="card bg-dark text-white border-secondary shadow-sm" data-panel-id="panel-log-settings-11" id="panel-log-settings-11">
<div class="card-header bg-dark border-secondary p-2">
<!-- Фильтры выборки событий -->
<div class="d-flex align-items-center justify-content-between flex-wrap gap-2">
<div class="d-flex align-items-center gap-2 flex-wrap">
<select class="form-select form-select-sm bg-dark text-white border-secondary" id="ls-insp-channel-select" style="width: 190px;">
<option value="Security">Security (Журнал)</option>
<option value="telemetry_db">telemetry.db (База)</option>
<option value="System">System (Системный)</option>
<option value="Application">Application (Приложения)</option>
<option value="Microsoft-Windows-Sysmon/Operational">Sysmon</option>
<option value="Microsoft-Windows-TaskScheduler/Operational">TaskScheduler</option>
</select>
<!-- Быстрые фильтры Event ID -->
<div class="btn-group btn-group-sm" role="group">
<button class="btn btn-outline-secondary ls-quick-eid-btn active" data-eid="0" type="button">Все</button>
<button class="btn btn-outline-info ls-quick-eid-btn" data-eid="4688" title="Создание процесса" type="button">4688 (Процесс)</button>
<button class="btn btn-outline-success ls-quick-eid-btn" data-eid="4624" title="Успешный вход" type="button">4624 (Logon)</button>
<button class="btn btn-outline-danger ls-quick-eid-btn" data-eid="4625" title="Неудачный вход" type="button">4625 (Fail)</button>
<button class="btn btn-outline-warning ls-quick-eid-btn" data-eid="1102" title="Очистка журнала" type="button">1102 (Clear)</button>
</div>
```

```
<select class="form-select form-select-sm bg-dark text-white border-secondary" id="ls-insp-level-select" style="width: 130px;">
<option value="">Все уровни</option>
<option value="Error">Ошибки (Error)</option>
<option value="Warning">Предупреждения</option>
<option value="Information">Информация</option>
<option value="Critical">Критические</option>
</select>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<input class="form-control form-control-sm bg-dark text-white border-secondary" id="ls-insp-user-input" placeholder="Пользователь..." style="width: 140px;" type="text"/>
<input class="form-control form-control-sm bg-dark text-white border-secondary" id="ls-insp-process-input" placeholder="Процесс..." style="width: 140px;" type="text"/>
<input class="form-control form-control-sm bg-dark text-white border-secondary" id="ls-insp-search-input" placeholder="Поиск в тексте..." style="width: 180px;" type="text"/>
<button class="btn btn-sm btn-primary d-flex align-items-center gap-1" id="ls-insp-fetch-btn">
<i class="bi bi-search"></i><span>Найти</span>
</button>
</div>
</div>
</div>
<div class="row g-0">
<!-- Левая панель: Список событий -->
<div class="col-12 col-lg-7 border-end border-secondary">
<div class="table-responsive" style="max-height: 480px; overflow-y: auto;">
<table class="table table-dark table-hover table-striped align-middle mb-0" style="font-size: 0.8rem;">
<thead class="table-secondary text-uppercase" style="font-size: 0.72rem; position: sticky; top: 0; z-index: 2;">
<tr>
<th scope="col" style="width: 20%;">Время</th>
```

```
<th scope="col" style="width: 12%;">Event ID</th>
<th scope="col" style="width: 14%;">Уровень</th>
<th scope="col" style="width: 54%;">Сообщение / Процесс</th>
</tr>
</thead>
<tbody id="ls-events-tbody">
<tr>
<td class="text-center text-muted py-4" colspan="4">Нажмите «Найти» для загрузки событий.</td>
</tr>
</tbody>
</table>
</div>
<div class="p-2 bg-dark border-top border-secondary d-flex justify-content-between text-muted small" style="font-size: 0.75rem;">
<span id="ls-events-count-label">Событий: 0</span>
<span>Кликните на строку для детального инспектора</span>
</div>
</div>
<!-- Правая панель: Детальный инспектор события -->
<div class="col-12 col-lg-5 p-3" style="max-height: 520px; overflow-y: auto;">
<div class="d-flex align-items-center justify-content-between mb-2">
<strong class="text-info fs-6 d-flex align-items-center gap-1.5">
<i class="bi bi-info-circle-fill"></i>
<span>Инспектор события</span>
</strong>
<span class="badge bg-secondary font-monospace" id="ls-detail-eid-badge">ID --</span>
</div>
<div class="text-muted small text-center py-5" id="ls-detail-empty">
<i class="bi bi-cursor text-secondary fs-3 d-block mb-2"></i>
              Выберите событие из таблицы слева для просмотра подробных атрибутов, пользователя, процесса и командной строки
```

```
.
            </div>
<div class="d-none" id="ls-detail-content">
<div class="mb-2">
<span class="text-muted small d-block">Время создания:</span>
<strong class="text-light font-monospace small" id="ls-detail-time">--</strong>
</div>
<div class="mb-2">
<span class="text-muted small d-block">Пользователь / Учетная запись:</span>
<strong class="text-warning font-monospace small" id="ls-detail-user">--</strong>
</div>
<div class="mb-2">
<span class="text-muted small d-block">Процесс / Программа:</span>
<strong class="text-info font-monospace small text-break" id="ls-detail-process">--</strong>
</div>
<div class="mb-2">
<span class="text-muted small d-block">Командная строка (CommandLine):</span>
<div class="bg-black p-2 rounded border border-secondary font-monospace text-success small text-break" id="ls-detail-cmdline">--</div>
</div>
<div class="mb-2">
<span class="text-muted small d-block">Родительский процесс:</span>
<div class="font-monospace small text-light text-break" id="ls-detail-parent">--</div>
</div>
<div class="mb-2">
<span class="text-muted small d-block">Сообщение / Расшифровка:</span>
<div class="text-light small" id="ls-detail-msg">--</div>
</div>
```

```
<div class="mt-3">
<button class="btn btn-xs btn-outline-secondary w-100 py-1" data-bs-target="#ls-raw-collapse" data-bs-toggle="collapse" type="button">
<i class="bi bi-code-square me-1"></i>Показать исходные данные (Raw Data)
                </button>
<div class="collapse mt-2" id="ls-raw-collapse">
<pre class="bg-black text-secondary p-2 rounded border border-secondary font-monospace small mb-0" id="ls-detail-raw" style="max-height: 180px; overflow-y: auto; font-size: 0.72rem;">--</pre>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- Модальное окно настройки параметров канала -->
<div aria-hidden="true" aria-labelledby="lsConfigModalLabel" class="modal fade" id="ls-config-modal" tabindex="-1">
<div class="modal-dialog modal-dialog-centered">
<div class="modal-content bg-dark text-white border-secondary shadow">
<div class="modal-header border-secondary py-2">
<h5 class="modal-title fs-6 d-flex align-items-center gap-2" id="lsConfigModalLabel">
<i class="bi bi-gear-fill text-info"></i>
<span>Настройка параметров журнала</span>
</h5>
<button aria-label="Close" class="btn-close btn-close-white" data-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body p-3">
<input id="ls-modal-channel-name" type="hidden"/>
<div class="mb-3">
```

```
<label class="form-label small text-muted">Имя журнала:</label>
<div class="fw-bold fs-6 text-light font-monospace" id="ls-modal-channel-title">--</div>
<div class="small text-secondary text-truncate" id="ls-modal-channel-path" style="font-size: 0.75rem;">--</div>
</div>
<div class="mb-3">
<label class="form-label small text-muted" for="ls-modal-maxsize-input">Максимальный размер (Max Size, МБ):</label>
<div class="input-group input-group-sm mb-2">
<input class="form-control bg-dark text-white border-secondary" id="ls-modal-maxsize-input" max="65536" min="1" step="1" type="number"/>
<span class="input-group-text bg-secondary text-white border-secondary">МБ</span>
</div>
<!-- Быстрые пресеты размера -->
<div class="d-flex flex-wrap gap-1">
<button class="btn btn-xs btn-outline-secondary ls-preset-size-btn" data-mb="20" type="button">20 MB</button>
<button class="btn btn-xs btn-outline-secondary ls-preset-size-btn" data-mb="64" type="button">64 MB</button>
<button class="btn btn-xs btn-outline-secondary ls-preset-size-btn" data-mb="128" type="button">128 MB</button>
<button class="btn btn-xs btn-outline-secondary ls-preset-size-btn" data-mb="256" type="button">256 MB</button>
<button class="btn btn-xs btn-outline-secondary ls-preset-size-btn" data-mb="512" type="button">512 MB</button>
<button class="btn btn-xs btn-outline-info ls-preset-size-btn" data-mb="1024" type="button">1 GB</button>
<button class="btn btn-xs btn-outline-info ls-preset-size-btn" data-mb="2048" type="button">2 GB</button>
</div>
</div>
<div class="mb-3">
<div class="form-check form-switch mb-2">
<input checked="" class="form-check-input" id="ls-modal-enable
```

```
d-switch" type="checkbox"/>
<label class="form-check-label small" for="ls-modal-enabled-sw
itch">Включить сбор событий в данный журнал (Enabled)</label>
</div>
<div class="form-check form-switch mb-2">
<input class="form-check-input" id="ls-modal-autobackup-switch
" type="checkbox"/>
<label class="form-check-label small" for="ls-modal-autobackup
-switch">Автоматическая архивация при заполнении (AutoBackup)<
/label>
</div>
<div class="form-check form-switch">
<input class="form-check-input" id="ls-modal-retention-switch"
 type="checkbox"/>
<label class="form-check-label small" for="ls-modal-retention-
switch">Политика удержания (Не перезаписывать события)</label>
</div>
</div>
</div>
<div class="modal-footer border-secondary py-2">
<button class="btn btn-sm btn-secondary" data-bs-dismiss="moda
l" type="button">Отмена</button>
<button class="btn btn-sm btn-primary d-flex align-items-cente
r gap-1" id="ls-modal-save-btn" type="button">
<i class="bi bi-check2"></i><span>Сохранить параметры</span>
</button>
</div>
</div>
</div>
</div>
<!-- Модальное окно очистки журнала -->
<div aria-hidden="true" aria-labelledby="lsClearModalLabel" cl
ass="modal fade" id="ls-clear-modal" tabindex="-1">
<div class="modal-dialog modal-dialog-centered">
<div class="modal-content bg-dark text-white border-danger sha
dow">
<div class="modal-header border-danger py-2">
<h5 class="modal-title fs-6 text-danger d-flex align-items-cen
```

```
ter gap-2" id="lsClearModalLabel">
<i class="bi bi-exclamation-triangle-fill"></i>
<span>Очистка журнала событий</span>
</h5>
<button aria-label="Close" class="btn-close btn-close-white" data-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body p-3">
<input id="ls-clear-modal-channel-name" type="hidden"/>
<p class="small text-light mb-3">
            Вы уверены, что хотите очистить журнал <strong class="text-warning" id="ls-clear-modal-title">--</strong>? Все текущие записи будут удалены из системы.
          </p>
<div class="form-check mb-2">
<input checked="" class="form-check-input" id="ls-clear-backup-checkbox" type="checkbox"/>
<label class="form-check-label small" for="ls-clear-backup-checkbox">
              Создать резервную копию (.evtx) перед очисткой
            </label>
</div>
<div class="input-group input-group-sm mb-2" id="ls-clear-backup-path-group">
<span class="input-group-text bg-secondary text-white border-secondary">Путь:</span>
<input class="form-control bg-dark text-white border-secondary" id="ls-clear-backup-path-input" type="text"/>
</div>
</div>
<div class="modal-footer border-danger py-2">
<button class="btn btn-sm btn-secondary" data-bs-dismiss="modal" type="button">Отмена</button>
<button class="btn btn-sm btn-danger d-flex align-items-center gap-1" id="ls-confirm-clear-btn" type="button">
<i class="bi bi-trash3-fill"></i><span>Очистить журнал</span>
</button>
</div>
```

```
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Log Settings Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля настройки параметров
 *   и иерархического каталога журналов событий Windows по 7 доменам телеметрии.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/log_settings_tab/main.js?v=20261008_v2" type="module"></script>
 *
 *   JavaScript Import:
 *     import { initLogSettingsTab } from '/windows/api/webgui/log_settings_tab/main.js';
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/log_settings_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

const registerTabPoller = window.registerTabPoller || function () {};
const isTabActive = window.isTabActive || (() => true);

let isInitialized = false;
```

```
let channelsData = [];
let catalogData = [];
let domainsData = [];
let eventsData = [];
let selectedEvent = null;

function escapeHtml(str) {
  if (!str) return '';
  return String(str)
    .replace(/&/g, '&amp;')
    .replace(/</g, '&lt;')
    .replace(/>/g, '&gt;')
    .replace(/"/g, '&quot;')
    .replace(/'/g, '&#039;');
}

/**
 * Загрузка сводки по 7 доменам телеметрии
 */
export async function fetchDomainsSummary() {
  if (!isTabActive('tab-log-settings')) return;

  try {
    const res = await fetch('/api/v1/log-settings/domains');
    if (!res.ok) return;
    const data = await res.json();
    domainsData = data.domains || [];

    renderDomainsCards(domainsData);

    const sysmonBadge = document.getElementById('ls-sysmon-bad
ge');
    if (sysmonBadge) {
      if (data.sysmon_installed) {
        sysmonBadge.textContent = 'Sysmon: Активен  ';
        sysmonBadge.className = 'badge bg-success font-monospa
ce';
      } else {
```

```
        sysmonBadge.textContent = 'Sysmon: Не установлен';
        sysmonBadge.className = 'badge bg-secondary font-monospace';
      }
    }

    const totalRecords = domainsData.reduce((acc, d) => acc + (d.total_records || 0), 0);
    const totalRecordsStat = document.getElementById('ls-total-records-stat');
    if (totalRecordsStat) {
      totalRecordsStat.textContent = `Всего в журналах: ${totalRecords.toLocaleString()} записей`;
    }
  } catch (err) {
    console.debug('[LogSettings] Ошибка загрузки доменов:', err);
  }
}

/**
 * Отрисовка карточек 7 доменов телеметрии
 */
function renderDomainsCards(domains) {
  const container = document.getElementById('ls-domains-cards-grid');
  if (!container) return;

  if (!domains || domains.length === 0) {
    container.innerHTML = '<div class="col-12 text-center text-muted py-3">Домены не найдены.</div>';
    return;
  }

  container.innerHTML = domains.map(d => {
    const recordsStr = (d.total_records || 0).toLocaleString();
    const isActive = d.status === 'Active' || d.status === 'Ак
```

```
тивен';
    const badgeBg = isActive ? 'bg-success' : 'bg-secondary';

    return `
      <div class="col-12 col-sm-6 col-lg-3">
        <div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3 ls-domain-card" data-domain-id="${d.id}" style="cursor: pointer; transition: transform 0.15s ease, border-color 0.15s ease;">
          <div class="d-flex justify-content-between align-items-start mb-1">
            <strong class="text-light fs-6 text-truncate">${escapeHtml(d.title)}</strong>
            <span class="badge ${badgeBg} font-monospace" style="font-size: 0.68rem;">${escapeHtml(d.status)}</span>
          </div>
          <div class="small text-muted mb-2 text-truncate" style="font-size: 0.75rem;" title="${escapeHtml(d.description)}">
            ${escapeHtml(d.description)}
          </div>
          <div class="d-flex justify-content-between align-items-center mt-auto pt-2 border-top border-secondary small font-monospace">
            <span class="text-info">${d.channels_count} каналов</span>
            <span class="text-warning">${recordsStr} записей</span>
          </div>
        </div>
      </div>
    `;
  }).join('');

  // Клик по карточке домена фильтрует каталог
  container.querySelectorAll('.ls-domain-card').forEach(card => {
    card.addEventListener('click', () => {
      const domId = card.dataset.domainId;
```

```
      const select = document.getElementById('ls-catalog-domai
n-select');
      if (select) {
        select.value = domId;
        fetchEventCatalog();
      }
    });
  });
}

/**
 * Загрузка каталога провайдеров
 */
export async function fetchEventCatalog() {
  const domainSelect = document.getElementById('ls-catalog-dom
ain-select');
  const searchInput = document.getElementById('ls-catalog-sear
ch-input');

  const domain = domainSelect ? domainSelect.value : '';
  const search = searchInput ? searchInput.value.trim() : '';

  try {
    const params = new URLSearchParams();
    if (domain) params.append('domain', domain);
    if (search) params.append('search', search);

    const res = await fetch(`/api/v1/log-settings/catalog?${pa
rams.toString()}`);
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    catalogData = data.catalog || [];
    renderCatalogTable(catalogData);
  } catch (err) {
    console.warn('[LogSettings] Ошибка каталога:', err);
  }
}

```

```
/**
 * Отрисовка таблицы каталога провайдеров
 */
function renderCatalogTable(catalog) {
  const tbody = document.getElementById('ls-catalog-tbody');
  const countLabel = document.getElementById('ls-catalog-count');
  if (!tbody) return;

  if (countLabel) countLabel.textContent = `${catalog.length} провайдеров`;

  if (!catalog || catalog.length === 0) {
    tbody.innerHTML = '<tr><td colspan="6" class="text-center text-muted py-4">Провайдеры по заданным критериям не найдены.</td></tr>';
    return;
  }

  tbody.innerHTML = catalog.map(e => {
    let volBadge = 'bg-secondary';
    if (e.volume_rating === 'Extreme') volBadge = 'bg-danger';
    else if (e.volume_rating === 'High') volBadge = 'bg-warning text-dark';
    else if (e.volume_rating === 'Medium') volBadge = 'bg-info text-dark';
    else if (e.volume_rating === 'Low') volBadge = 'bg-success';

    const eventsList = Object.entries(e.key_events || {}).map(([id, desc]) => {
      return `<span class="badge bg-black border border-secondary text-warning font-monospace me-1 mb-1" title="${escapeHtml(desc)}">ID ${id}</span>`;
    }).join('');

    const targets = (e.correlation_targets || []).map(t => {
      return `<span class="badge bg-secondary-subtle text-ligh
```

```
t font-monospace me-1" style="font-size: 0.68rem;">#${escapeHt
ml(t)}</span>`;
182     }).join('');
183 
184     return `
185       <tr>
186         <td>
187           <div class="fw-bold text-light font-monospace" style
="font-size: 0.8rem;">${escapeHtml(e.provider)}</div>
188           <div class="small text-info font-monospace text-trun
cate" style="max-width: 260px; font-size: 0.72rem;">${escapeHt
ml(e.channel)}</div>
189           <div class="small text-muted text-truncate" style="m
ax-width: 260px; font-size: 0.7rem;" title="${escapeHtml(e.des
cription)}">${escapeHtml(e.description)}</div>
190         </td>
191         <td>
192           <span class="badge bg-dark border border-secondary t
ext-light">${escapeHtml(e.domain_title || e.domain)}</span>
193         </td>
194         <td>
195           <div class="d-flex flex-wrap" style="max-width: 280p
x;">${eventsList}</div>
196         </td>
197         <td>
198           <span class="badge ${volBadge}">${escapeHtml(e.volum
e_rating)}</span>
199           <div class="small text-muted font-monospace mt-1" st
yle="font-size: 0.7rem;">${(e.record_count || 0).toLocaleStrin
g()} rec</div>
200         </td>
201         <td>
202           <div class="small text-light font-monospace">${escap
eHtml(e.recommended_collector)}</div>
203           <div class="small text-muted" style="font-size: 0.7r
em;">
204             ${e.realtime_support ? '<span class="text-success"
>ETW/Stream</span>' : '<span class="text-secondary">Pull</span
```

```
>'}
            • <span class="text-warning">${escapeHtml(e.privil
ege_required)}</span>
          </div>
        </td>
        <td>
          <div class="d-flex flex-wrap">${targets}</div>
        </td>
      </tr>
    `;
  }).join('');
}

/**
 * Загрузка сводки и списка параметров всех каналов
 */
export async function fetchChannelsList() {
  if (!isTabActive('tab-log-settings')) return;

  try {
    const res = await fetch('/api/v1/log-settings/channels');
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    channelsData = data.channels || [];
    renderChannelsCards(channelsData);
    renderChannelsTable(channelsData);
  } catch (err) {
    console.warn('[LogSettings] Ошибка загрузки каналов:', err
);
  }
}

/**
 * Загрузка статуса политик аудита и сборщика телеметрии
 */
export async function fetchAuditStatus() {
  if (!isTabActive('tab-log-settings')) return;

```

```
  try {
    const res = await fetch('/api/v1/log-settings/audit-status
');
    if (!res.ok) return;
    const data = await res.json();

    const cmdStatusEl = document.getElementById('ls-cmdline-st
atus');
    const cmdBadgeEl = document.getElementById('ls-audit-cmdli
ne-badge');
    if (cmdStatusEl) {
      cmdStatusEl.textContent = data.cmdline_audit_enabled ? '
Включен  ' : 'Отключен  ';
      cmdStatusEl.className = `fs-4 fw-bold mt-1 ${data.cmdlin
e_audit_enabled ? 'text-success' : 'text-warning'}`;
    }
    if (cmdBadgeEl) {
      cmdBadgeEl.textContent = data.cmdline_audit_enabled ? 'A
КТИВЕН' : 'НЕ АКТИВЕН';
      cmdBadgeEl.className = `badge ${data.cmdline_audit_enabl
ed ? 'bg-success' : 'bg-warning text-dark'}`;
    }

    const bmIdEl = document.getElementById('ls-bookmark-id');
    const bmTimeEl = document.getElementById('ls-bookmark-time
');
    const dbEventsBadge = document.getElementById('ls-db-event
s-badge');
    const bmCurrIdEl = document.getElementById('ls-bm-curr-id'
);
    const bmDbCountEl = document.getElementById('ls-bm-db-coun
t');

    if (bmIdEl) bmIdEl.textContent = data.last_bookmark_record
_id ? `ID: ${data.last_bookmark_record_id.toLocaleString()}` :
 'ID: Начало';
    if (bmTimeEl) bmTimeEl.textContent = `Закладка: ${data.las
t_bookmark_timestamp || 'Не задана'}`;
```

```
    if (dbEventsBadge) dbEventsBadge.textContent = `${data.db_security_events_count.toLocaleString()} в БД`;
    if (bmCurrIdEl) bmCurrIdEl.textContent = data.last_bookmark_record_id ? data.last_bookmark_record_id.toLocaleString() : 'Нет (tail)';
    if (bmDbCountEl) bmDbCountEl.textContent = data.db_security_events_count.toLocaleString();

    const privStatusEl = document.getElementById('ls-sec-priv-status');
    if (privStatusEl) {
      if (data.security_channel_accessible) {
        privStatusEl.className = 'alert alert-success border-success p-2 small mb-0 d-flex align-items-center gap-2';
        privStatusEl.innerHTML = '<i class="bi bi-shield-check fs-5"></i><span>Доступ к каналу Security разрешен. Привилегии SeSecurityPrivilege активны.</span>';
      } else {
        privStatusEl.className = 'alert alert-danger border-danger p-2 small mb-0 d-flex align-items-center gap-2';
        privStatusEl.innerHTML = '<i class="bi bi-shield-x fs-5"></i><span>Доступ к каналу Security ограничен. Требуется запуск с правами Администратора.</span>';
      }
    }
  } catch (err) {
    console.debug('[LogSettings] Ошибка проверки аудита:', err);
  }
}

/**
 * Отрисовка верхних карточек
 */
function renderChannelsCards(channels) {
  const sec = channels.find(c => c.channel_name === 'Security') || {};

```

```
  const secSizeEl = document.getElementById('ls-sec-size');
  const secProgEl = document.getElementById('ls-sec-progress')
;
  const secRecordsEl = document.getElementById('ls-sec-records
');
  const secStatusEl = document.getElementById('ls-sec-status')
;

  if (secSizeEl) secSizeEl.textContent = `${sec.file_size_mb |
| 0} / ${sec.max_size_mb || 20} MB`;
  if (secProgEl) {
    const pct = sec.usage_pct || 0;
    secProgEl.style.width = `${Math.min(100, pct)}%`;
    secProgEl.className = `progress-bar ${pct > 90 ? 'bg-dange
r' : (pct > 75 ? 'bg-warning' : 'bg-info')}`;
  }
  if (secRecordsEl) secRecordsEl.textContent = `Записей: ${(se
c.record_count || 0).toLocaleString()}`;
  if (secStatusEl) secStatusEl.textContent = sec.is_enabled ?
'Активен' : 'Отключен';
}

/**
 * Отрисовка таблицы каналов
 */
function renderChannelsTable(channels) {
  const tbody = document.getElementById('ls-channels-tbody');
  const countLabel = document.getElementById('ls-table-count')
;
  if (!tbody) return;

  const filterInput = document.getElementById('ls-channel-filt
er-input');
  const filterQuery = (filterInput ? filterInput.value : '').t
oLowerCase().trim();

  const filtered = channels.filter(c => {
    if (!filterQuery) return true;
```

```
    return c.channel_name.toLowerCase().includes(filterQuery) ||
           (c.description || '').toLowerCase().includes(filterQuery);
  });

  if (countLabel) countLabel.textContent = `${filtered.length} из ${channels.length}`;

  if (filtered.length === 0) {
    tbody.innerHTML = '<tr><td colspan="8" class="text-center text-muted py-4">Каналы не найдены</td></tr>';
    return;
  }

  tbody.innerHTML = filtered.map(c => {
    const pct = c.usage_pct || 0;
    const isFull = pct >= 95.0;
    const badgeColor = isFull ? 'bg-danger' : (pct > 75 ? 'bg-warning text-dark' : 'bg-success');
    const rotPolicy = c.auto_backup ? 'Auto-Backup (Архивация)' : (c.retention ? 'Retention (Удержание)' : 'FIFO (Перезапись)');

    return `
      <tr>
        <td>
          <div class="fw-bold text-light font-monospace">${escapeHtml(c.channel_name)}</div>
          <div class="small text-muted text-truncate" style="max-width: 280px; font-size: 0.72rem;" title="${escapeHtml(c.description)}">
            ${escapeHtml(c.description || c.log_file_path)}
          </div>
        </td>
        <td>
          <span class="font-monospace text-light">${c.file_size_mb} MB</span>
```

```
          <span class="text-muted small">/ ${c.max_size_mb} MB
</span>
        </td>
        <td>
          <div class="d-flex align-items-center gap-2">
            <div class="progress flex-grow-1 bg-secondary" sty
le="height: 6px;">
              <div class="progress-bar ${pct > 90 ? 'bg-danger
' : (pct > 70 ? 'bg-warning' : 'bg-info')}" style="width: ${Ma
th.min(100, pct)}%;"></div>
            </div>
            <span class="badge ${badgeColor} font-monospace" s
tyle="font-size: 0.7rem;">${pct}%</span>
          </div>
        </td>
        <td class="font-monospace text-info">${(c.record_count
 || 0).toLocaleString()}</td>
        <td class="font-monospace text-secondary small" style=
"font-size: 0.72rem;">
          ${c.oldest_record_id ? `${c.oldest_record_id.toLocal
eString()}..${c.newest_record_id.toLocaleString()}` : '--'}
        </td>
        <td>
          <span class="badge bg-secondary-subtle text-light bo
rder border-secondary small" style="font-size: 0.72rem;">${rot
Policy}</span>
        </td>
        <td>
          ${c.is_enabled ? '<span class="badge bg-success">Вкл
</span>' : '<span class="badge bg-secondary">Выкл</span>'}
        </td>
        <td class="text-end">
          <div class="btn-group btn-group-sm">
            <button class="btn btn-outline-info py-0 px-2 ls-b
tn-config" data-channel="${escapeHtml(c.channel_name)}" title=
"Настроить параметры">
              <i class="bi bi-gear"></i>
            </button>
```

```
            <button class="btn btn-outline-danger py-0 px-2 ls-btn-clear" data-channel="${escapeHtml(c.channel_name)}" title="Очистить журнал">
              <i class="bi bi-trash3"></i>
            </button>
          </div>
        </td>
      </tr>
    `;
  }).join('');

  tbody.querySelectorAll('.ls-btn-config').forEach(btn => {
    btn.addEventListener('click', () => openConfigModal(btn.dataset.channel));
  });

  tbody.querySelectorAll('.ls-btn-clear').forEach(btn => {
    btn.addEventListener('click', () => openClearModal(btn.dataset.channel));
  });
}

/**
 * Открытие модального окна настройки параметров канала
 */
function openConfigModal(channelName) {
  const channel = channelsData.find(c => c.channel_name === channelName);
  if (!channel) return;

  document.getElementById('ls-modal-channel-name').value = channel.channel_name;
  document.getElementById('ls-modal-channel-title').textContent = channel.channel_name;
  document.getElementById('ls-modal-channel-path').textContent = channel.log_file_path || 'Стандартный путь Winevt\\Logs';
  document.getElementById('ls-modal-maxsize-input').value = channel.max_size_mb || 20;
```

```
  document.getElementById('ls-modal-enabled-switch').checked = channel.is_enabled !== false;
  document.getElementById('ls-modal-autobackup-switch').checked = !!channel.auto_backup;
  document.getElementById('ls-modal-retention-switch').checked = !!channel.retention;

  const modalEl = document.getElementById('ls-config-modal');
  if (modalEl && window.bootstrap) {
    const modal = window.bootstrap.Modal.getOrCreateInstance(modalEl);
    modal.show();
  }
}

/**
 * Сохранение параметров конфигурации канала
 */
async function saveChannelConfig() {
  const channelName = document.getElementById('ls-modal-channel-name').value;
  const maxSizeMb = parseInt(document.getElementById('ls-modal-maxsize-input').value, 10);
  const isEnabled = document.getElementById('ls-modal-enabled-switch').checked;
  const autoBackup = document.getElementById('ls-modal-autobackup-switch').checked;
  const retention = document.getElementById('ls-modal-retention-switch').checked;

  const saveBtn = document.getElementById('ls-modal-save-btn');
  const origHtml = saveBtn ? saveBtn.innerHTML : '';
  if (saveBtn) saveBtn.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span>Сохранение...';

  try {
    const res = await fetch('/api/v1/log-settings/channel/conf
```

```
ig', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        channel_name: channelName,
        max_size_mb: maxSizeMb || null,
        is_enabled: isEnabled,
        auto_backup: autoBackup,
        retention: retention,
      }),
    });

    const data = await res.json();
    if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
atus}`);

    if (window.toast) {
      window.toast.success('Параметры обновлены', data.message
 || `Настройки журнала ${channelName} сохранены.`);
    }

    const modalEl = document.getElementById('ls-config-modal')
;
    if (modalEl && window.bootstrap) {
      window.bootstrap.Modal.getInstance(modalEl)?.hide();
    }

    await fetchChannelsList();
  } catch (err) {
    if (window.toast) {
      window.toast.error('Ошибка сохранения', err.message);
    }
  } finally {
    if (saveBtn) saveBtn.innerHTML = origHtml;
  }
}

/**
```

```
 * Открытие модального окна очистки журнала
 */
function openClearModal(channelName) {
  document.getElementById('ls-clear-modal-channel-name').value = channelName;
  document.getElementById('ls-clear-modal-title').textContent = channelName;

  const nowStr = new Date().toISOString().replace(/[:.]/g, '-').slice(0, 19);
  const cleanName = channelName.replace(/[^a-zA-Z0-9]/g, '_');
  document.getElementById('ls-clear-backup-path-input').value = `C:\\Windows\\Temp\\evtx_backup_${cleanName}_${nowStr}.evtx`;

  const modalEl = document.getElementById('ls-clear-modal');
  if (modalEl && window.bootstrap) {
    const modal = window.bootstrap.Modal.getOrCreateInstance(modalEl);
    modal.show();
  }
}

/**
 * Подтверждение очистки журнала
 */
async function confirmClearChannel() {
  const channelName = document.getElementById('ls-clear-modal-channel-name').value;
  const withBackup = document.getElementById('ls-clear-backup-checkbox').checked;
  const backupPath = withBackup ? document.getElementById('ls-clear-backup-path-input').value : null;

  const clearBtn = document.getElementById('ls-confirm-clear-btn');
  const origHtml = clearBtn ? clearBtn.innerHTML : '';
  if (clearBtn) clearBtn.innerHTML = '<span class="spinner-bor
```

```
der spinner-border-sm me-1"></span>Очистка...';

  try {
    const res = await fetch('/api/v1/log-settings/channel/clear', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        channel_name: channelName,
        backup_path: backupPath,
      }),
    });

    const data = await res.json();
    if (!res.ok) throw new Error(data.detail || `HTTP ${res.status}`);

    if (window.toast) {
      window.toast.success('Журнал очищен', data.message || `Журнал ${channelName} очищен.`);
    }

    const modalEl = document.getElementById('ls-clear-modal');
    if (modalEl && window.bootstrap) {
      window.bootstrap.Modal.getInstance(modalEl)?.hide();
    }

    await fetchChannelsList();
  } catch (err) {
    if (window.toast) {
      window.toast.error('Ошибка очистки', err.message);
    }
  } finally {
    if (clearBtn) clearBtn.innerHTML = origHtml;
  }
}

/**
```

```
 * Переключение аудита CommandLine в реестре
 */
async function toggleCmdlineAudit(enable = true) {
  try {
    const res = await fetch('/api/v1/log-settings/audit-status/cmdline', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ enabled: enable }),
    });

    const data = await res.json();
    if (!res.ok) throw new Error(data.detail || `HTTP ${res.status}`);

    if (window.toast) {
      window.toast.success('Политика обновлена', data.message);
    }
    await fetchAuditStatus();
  } catch (err) {
    if (window.toast) {
      window.toast.error('Ошибка изменения политики', err.message);
    }
  }
}

/**
 * Запуск инкрементального сбора в telemetry.db
 */
async function runCollector() {
  const batchSelect = document.getElementById('ls-collect-batch-size');
  const batchSize = batchSelect ? batchSelect.value : 500;
  const banner = document.getElementById('ls-collector-result-banner');

```

```
  const btn = document.getElementById('ls-run-collector-btn');
  const origHtml = btn ? btn.innerHTML : '';
  if (btn) btn.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span>Сбор...';

  try {
    const res = await fetch(`/api/v1/log-settings/collector/collect?limit=${batchSize}`, { method: 'POST' });
    const data = await res.json();
    if (!res.ok) throw new Error(data.detail || `HTTP ${res.status}`);

    if (banner) {
      banner.className = 'small mt-2 text-success';
      banner.textContent = data.message || `Собрано ${data.total_events_ingested} событий.`;
      banner.classList.remove('d-none');
    }

    if (window.toast) {
      window.toast.success('Телеметрия обновлена', data.message);
    }

    await fetchAuditStatus();
    await fetchDomainsSummary();
  } catch (err) {
    if (banner) {
      banner.className = 'small mt-2 text-danger';
      banner.textContent = `Ошибка сбора: ${err.message}`;
      banner.classList.remove('d-none');
    }
    if (window.toast) {
      window.toast.error('Ошибка сбора', err.message);
    }
  } finally {
    if (btn) btn.innerHTML = origHtml;
  }
```

```
}

/**
 * Сброс закладки
 */
async function resetBookmark(customRecordId = null) {
  try {
    const res = await fetch('/api/v1/log-settings/collector/reset-bookmark', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ channel: 'Security', record_id: customRecordId }),
    });

    const data = await res.json();
    if (!res.ok) throw new Error(data.detail || `HTTP ${res.status}`);

    if (window.toast) {
      window.toast.success('Закладка сброшена', data.message);
    }
    await fetchAuditStatus();
  } catch (err) {
    if (window.toast) {
      window.toast.error('Ошибка сброса закладки', err.message);
    }
  }
}

/**
 * Построение графа происхождения процессов (Process Provenance Graph)
 */
export async function buildProcessProvenance() {
  const input = document.getElementById('ls-prov-process-input');
```

```
  const processName = input ? input.value.trim() : '';
  const container = document.getElementById('ls-prov-tree-view
');
  if (!container) return;

  container.innerHTML = '<div class="text-center text-warning
py-4"><div class="spinner-border spinner-border-sm me-2"></div
>Построение графа происхождения процессов...</div>';

  try {
    const params = new URLSearchParams();
    if (processName) params.append('process_name', processName
);
    params.append('hours', '72');

    const res = await fetch(`/api/v1/log-settings/provenance/p
rocess?${params.toString()}`);
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();

    const nodes = data.nodes || [];
    const links = data.links || [];

    if (nodes.length === 0) {
      container.innerHTML = `
        <div class="text-center text-muted py-5">
          <i class="bi bi-info-circle fs-3 text-secondary d-bl
ock mb-2"></i>
          События создания процессов (Event 4688) для <strong>
${escapeHtml(processName || 'хоста')}</strong> не найдены в te
lemetry.db.<br>
          <small class="text-secondary">Убедитесь, что аудит C
ommandLine включен и выполнен сбор новых событий.</small>
        </div>
      `;
      return;
    }

```

```
    container.innerHTML = `
      <div class="mb-3 d-flex justify-content-between align-items-center">
        <span class="badge bg-warning text-dark font-monospace">${nodes.length} Узлов (Nodes)</span>
        <span class="badge bg-info text-dark font-monospace">${links.length} Связей (Spawns / Forks)</span>
      </div>
      <div class="d-flex flex-column gap-2">
        ${nodes.map(n => {
          let typeBadge = 'bg-primary';
          if (n.type === 'parent_process') typeBadge = 'bg-info text-dark';
          else if (n.type === 'service') typeBadge = 'bg-success';

          return `
            <div class="p-2 bg-dark rounded border border-secondary d-flex align-items-center justify-content-between flex-wrap gap-2">
              <div class="d-flex align-items-center gap-2">
                <span class="badge ${typeBadge} font-monospace" style="font-size: 0.72rem;">${escapeHtml(n.role || n.type)}</span>
                <strong class="text-light font-monospace">${escapeHtml(n.name)}</strong>
                ${n.pid ? `<span class="badge bg-secondary font-monospace">PID ${n.pid}</span>` : ''}
              </div>
              <div class="small text-muted font-monospace">${escapeHtml(n.user || '')} • ${escapeHtml(n.timestamp || '')}</div>
              ${n.command_line ? `<div class="w-100 bg-black p-1.5 rounded font-monospace small text-success text-break" style="font-size: 0.72rem;">${escapeHtml(n.command_line)}</div>` : ''}
            </div>
          `;
```

```
        }).join('')}
      </div>
    `;
  } catch (err) {
    container.innerHTML = `<div class="text-center text-danger
 py-4">Ошибка: ${escapeHtml(err.message)}</div>`;
  }
}

/**
 * Загрузка жизненного цикла ОС (Power / Boot / Crash Timeline
)
 */
export async function fetchPowerLifecycle() {
  const select = document.getElementById('ls-lifecycle-hours-s
elect');
  const hours = select ? select.value : 72;
  const container = document.getElementById('ls-lifecycle-time
line');
  if (!container) return;

  container.innerHTML = '<div class="text-center text-muted py
-4"><div class="spinner-border spinner-border-sm text-danger m
e-2"></div>Реконструкция жизненного цикла ОС...</div>';

  try {
    const res = await fetch(`/api/v1/log-settings/lifecycle/po
wer?hours=${hours}`);
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    const timeline = data.timeline || [];

    if (timeline.length === 0) {
      container.innerHTML = '<div class="text-center text-mute
d py-4">События жизненного цикла за указанный период не найден
ы.</div>';
      return;
    }
```

```
698
699     container.innerHTML = timeline.map(item => {
700       let badgeClass = 'border-info text-info';
701       if (item.severity === 'critical') badgeClass = 'border-d
anger text-danger bg-danger-subtle';
702       else if (item.severity === 'warning') badgeClass = 'bord
er-warning text-warning bg-warning-subtle';
703
704       return `
705         <div class="mb-3 position-relative pb-2 border-bottom
border-secondary">
706           <div class="d-flex align-items-center gap-2 mb-1">
707             <span class="fs-5">${item.icon}</span>
708             <span class="badge border ${badgeClass} font-monos
pace" style="font-size: 0.72rem;">${escapeHtml(item.phase)}</s
pan>
709             <span class="badge bg-dark border border-secondary
 font-monospace" style="font-size: 0.72rem;">ID ${item.event_i
d}</span>
710             <span class="small text-secondary font-monospace m
s-auto">${escapeHtml(item.timestamp)}</span>
711           </div>
712           <div class="text-light small fw-medium">${escapeHtml
(item.message)}</div>
713           <div class="text-muted font-monospace small" style="
font-size: 0.72rem;">Источник: ${escapeHtml(item.provider)}</d
iv>
714         </div>
715       `;
716     }).join('');
717   } catch (err) {
718     container.innerHTML = `<div class="text-center text-danger
 py-4">Ошибка: ${escapeHtml(err.message)}</div>`;
719   }
720 }
721
722 /**
723  * Загрузка событий для инспектора
```

```
 */
export async function fetchInspectorEvents() {
  const channelSelect = document.getElementById('ls-insp-chann
el-select');
  const levelSelect = document.getElementById('ls-insp-level-s
elect');
  const userInput = document.getElementById('ls-insp-user-inpu
t');
  const processInput = document.getElementById('ls-insp-proces
s-input');
  const searchInput = document.getElementById('ls-insp-search-
input');

  const channel = channelSelect ? channelSelect.value : 'Secur
ity';
  const level = levelSelect ? levelSelect.value : '';
  const user = userInput ? userInput.value.trim() : '';
  const process = processInput ? processInput.value.trim() : '
';
  const search = searchInput ? searchInput.value.trim() : '';

  const activeEidBtn = document.querySelector('.ls-quick-eid-b
tn.active');
  const eventId = activeEidBtn ? parseInt(activeEidBtn.dataset
.eid, 10) || 0 : 0;

  const tbody = document.getElementById('ls-events-tbody');
  if (tbody) {
    tbody.innerHTML = '<tr><td colspan="4" class="text-center
text-muted py-4"><div class="spinner-border spinner-border-sm
text-primary me-2"></div>Загрузка событий...</td></tr>';
  }

  try {
    const params = new URLSearchParams({
      channel: channel,
      event_id: eventId,
      level: level,
```

```
      user: user,
      process: process,
      search: search,
      limit: 100,
    });

    const res = await fetch(`/api/v1/log-settings/events?${params.toString()}`);
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    eventsData = data.events || [];
    renderInspectorEvents(eventsData);
  } catch (err) {
    console.warn('[LogSettings] Ошибка чтения событий:', err);
    if (tbody) {
      tbody.innerHTML = `<tr><td colspan="4" class="text-center text-danger py-4">Ошибка: ${escapeHtml(err.message)}</td></tr>`;
    }
  }
}

/**
 * Отрисовка списка событий в инспекторе
 */
function renderInspectorEvents(events) {
  const tbody = document.getElementById('ls-events-tbody');
  const countLabel = document.getElementById('ls-events-count-label');
  if (!tbody) return;

  if (countLabel) countLabel.textContent = `Событий: ${events.length}`;

  if (events.length === 0) {
    tbody.innerHTML = '<tr><td colspan="4" class="text-center text-muted py-4">События по заданным фильтрам не найдены.</td></tr>';
```

```
    return;
  }

  tbody.innerHTML = events.map((ev, idx) => {
    const eid = ev.event_id || 0;
    const lvl = ev.level || 'Information';
    const ts = ev.timestamp || '';
    const msg = ev.message || ev.raw_data || '';
    const proc = ev.process_name || ev.provider || '';

    let lvlBadge = 'bg-secondary';
    if (lvl === 'Error' || lvl === 'Critical') lvlBadge = 'bg-danger';
    else if (lvl === 'Warning') lvlBadge = 'bg-warning text-dark';
    else if (lvl === 'Information') lvlBadge = 'bg-info text-dark';

    return `
      <tr class="ls-event-row" data-index="${idx}" style="cursor: pointer;">
        <td class="font-monospace text-secondary text-truncate" style="max-width: 140px;">${escapeHtml(ts)}</td>
        <td>
          <span class="badge bg-dark border border-secondary font-monospace">${eid}</span>
        </td>
        <td>
          <span class="badge ${lvlBadge} small" style="font-size: 0.7rem;">${escapeHtml(lvl)}</span>
        </td>
        <td>
          <div class="text-truncate text-light fw-medium" style="max-width: 320px;" title="${escapeHtml(msg)}">${escapeHtml(msg)}</div>
          ${proc ? `<div class="small text-muted text-truncate font-monospace" style="font-size: 0.72rem;">${escapeHtml(proc)}</div>` : ''}
```

```
        </td>
      </tr>
    `;
  }).join('');

  tbody.querySelectorAll('.ls-event-row').forEach(row => {
    row.addEventListener('click', () => {
      tbody.querySelectorAll('.ls-event-row').forEach(r => r.c
lassList.remove('table-active'));
      row.classList.add('table-active');
      const idx = parseInt(row.dataset.index, 10);
      showEventDetail(eventsData[idx]);
    });
  });

  if (events.length > 0) {
    const firstRow = tbody.querySelector('.ls-event-row');
    if (firstRow) firstRow.classList.add('table-active');
    showEventDetail(events[0]);
  }
}

/**
 * Отображение подробных сведений о событии в панели инспектор
a
 */
function showEventDetail(ev) {
  if (!ev) return;
  selectedEvent = ev;

  const emptyEl = document.getElementById('ls-detail-empty');
  const contentEl = document.getElementById('ls-detail-content
');
  if (emptyEl) emptyEl.classList.add('d-none');
  if (contentEl) contentEl.classList.remove('d-none');

  document.getElementById('ls-detail-eid-badge').textContent =
 `ID ${ev.event_id || 0}`;
```

```
  document.getElementById('ls-detail-time').textContent = ev.timestamp || '--';
  document.getElementById('ls-detail-user').textContent = ev.subject_user || ev.target_user || (ev.event_data?.SubjectUserName || ev.event_data?.TargetUserName || 'SYSTEM / N/A');

  const procName = ev.process_name || ev.event_data?.NewProcessName || ev.provider || '--';
  const pid = ev.process_id || ev.event_data?.NewProcessId || '';
  document.getElementById('ls-detail-process').textContent = pid ? `${procName} (PID ${pid})` : procName;

  const cmdLine = ev.command_line || ev.event_data?.CommandLine || 'Не зафиксирована или аудит CommandLine выключен';
  document.getElementById('ls-detail-cmdline').textContent = cmdLine;

  const parent = ev.parent_process_name || ev.event_data?.ParentProcessName || (ev.parent_process_id ? `PID ${ev.parent_process_id}` : '--');
  document.getElementById('ls-detail-parent').textContent = parent;

  document.getElementById('ls-detail-msg').textContent = ev.message || '--';
  document.getElementById('ls-detail-raw').textContent = JSON.stringify(ev, null, 2);
}

/**
 * Инициализация всех слушателей и элементов вкладки
 */
export function initLogSettingsTab() {
  if (isInitialized) return;
  isInitialized = true;

  // Кнопка обновления всех параметров
```

```
  document.getElementById('ls-refresh-all-btn')?.addEventListener('click', () => {
    fetchDomainsSummary();
    fetchEventCatalog();
    fetchChannelsList();
    fetchAuditStatus();
  });

  document.getElementById('ls-reload-channels-btn')?.addEventListener('click', () => fetchChannelsList());

  // Поиск по списку каналов
  document.getElementById('ls-channel-filter-input')?.addEventListener('input', () => {
    renderChannelsTable(channelsData);
  });

  // Фильтры каталога провайдеров
  document.getElementById('ls-catalog-domain-select')?.addEventListener('change', fetchEventCatalog);
  document.getElementById('ls-catalog-search-input')?.addEventListener('input', fetchEventCatalog);

  // Provenance граф
  document.getElementById('ls-prov-build-btn')?.addEventListener('click', buildProcessProvenance);

  // Хронология жизненного цикла
  document.getElementById('ls-lifecycle-hours-select')?.addEventListener('change', fetchPowerLifecycle);
  document.getElementById('ls-lifecycle-reload-btn')?.addEventListener('click', fetchPowerLifecycle);

  // Пресеты размеров в модальном окне
  document.querySelectorAll('.ls-preset-size-btn').forEach(btn => {
    btn.addEventListener('click', () => {
      const mb = btn.dataset.mb;
```

```
      const input = document.getElementById('ls-modal-maxsize-input');
      if (input) input.value = mb;
    });
  });

  // Сохранение конфигурации канала
  document.getElementById('ls-modal-save-btn')?.addEventListener('click', saveChannelConfig);

  // Очистка канала
  document.getElementById('ls-confirm-clear-btn')?.addEventListener('click', confirmClearChannel);
  document.getElementById('ls-clear-backup-checkbox')?.addEventListener('change', (e) => {
    const pathGroup = document.getElementById('ls-clear-backup-path-group');
    if (pathGroup) pathGroup.style.display = e.target.checked ? 'flex' : 'none';
  });

  // Управление аудитом CommandLine
  document.getElementById('ls-toggle-cmdline-btn')?.addEventListener('click', async () => {
    const isCurrentlyEnabled = document.getElementById('ls-cmdline-status')?.textContent.includes('Включен');
    await toggleCmdlineAudit(!isCurrentlyEnabled);
  });
  document.getElementById('ls-enable-cmdline-btn')?.addEventListener('click', () => toggleCmdlineAudit(true));
  document.getElementById('ls-disable-cmdline-btn')?.addEventListener('click', () => toggleCmdlineAudit(false));

  // Управление сборщиком
  document.getElementById('ls-collect-btn')?.addEventListener('click', runCollector);
  document.getElementById('ls-run-collector-btn')?.addEventListener('click', runCollector);
```

```
923
924   // Сброс закладки
925   document.getElementById('ls-reset-bm-btn')?.addEventListener
('click', () => resetBookmark(null));
926   document.getElementById('ls-set-custom-bm-btn')?.addEventLis
tener('click', () => {
927     const val = document.getElementById('ls-custom-record-id-i
nput')?.value;
928     const num = val ? parseInt(val, 10) : null;
929     resetBookmark(num);
930   });
931
932   // Быстрые фильтры Event ID
933   document.querySelectorAll('.ls-quick-eid-btn').forEach(btn =
> {
934     btn.addEventListener('click', () => {
935       document.querySelectorAll('.ls-quick-eid-btn').forEach(b
 => b.classList.remove('active'));
936       btn.classList.add('active');
937       fetchInspectorEvents();
938     });
939   });
940
941   // Инспектор событий
942   document.getElementById('ls-insp-fetch-btn')?.addEventListen
er('click', fetchInspectorEvents);
943   document.getElementById('ls-insp-channel-select')?.addEventL
istener('change', fetchInspectorEvents);
944   document.getElementById('ls-insp-level-select')?.addEventLis
tener('change', fetchInspectorEvents);
945
946   // Первоначальная загрузка
947   fetchDomainsSummary();
948   fetchEventCatalog();
949   fetchChannelsList();
950   fetchAuditStatus();
951   fetchPowerLifecycle();
952
```

```
  // Регистрация поллера (каждые 20 секунд при активной вкладке)
  registerTabPoller('tab-log-settings', () => {
    fetchDomainsSummary();
    fetchChannelsList();
    fetchAuditStatus();
  }, 20000);
}

if (document.readyState === 'loading') {
  document.addEventListener('DOMContentLoaded', initLogSettingsTab);
} else {
  initLogSettingsTab();
}

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Maintenance & Recovery - Index View
================================================================
===============
Description:
  Шаблон интерфейса обслуживания и восстановления системы Wind
ows (Maintenance & Recovery).

Usage Examples:
  Web Route:
    /html/maintenance_recovery_tab/index.html

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/maintenance_recovery_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<style>
  .maint-container {
    padding: 0.75rem 1rem;
  }
  .maint-header {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: var(--radius-md, 8px);
    padding: 0.75rem 1.25rem;
    color: var(--text-color);
  }
  .maint-card {
    background: var(--surface-1);
```

```
    border: 1px solid var(--border-color);
    border-radius: 8px;
    color: var(--text-color);
  }
  .maint-sub-card {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 8px;
    color: var(--text-color);
  }
  .maint-table {
    width: 100%;
    border-collapse: collapse;
    margin-bottom: 0;
  }
  .maint-table th {
    padding: 0.65rem 1rem;
    font-size: 0.75rem;
    font-weight: 600;
    text-transform: uppercase;
    letter-spacing: 0.05em;
    color: var(--text-muted);
    background: var(--surface-2);
    border-bottom: 1px solid var(--border-color);
    white-space: nowrap;
  }
  .maint-table td {
    padding: 0.65rem 1rem;
    font-size: 0.85rem;
    color: var(--text-color);
    border-bottom: 1px solid var(--border-color);
    vertical-align: middle;
  }
  .maint-table tr:last-child td {
    border-bottom: none;
  }
  .maint-table tbody tr:hover td {
    background: var(--nav-hover-bg, rgba(0, 0, 0, 0.04));
```

```
  }
</style>
<div class="maint-container theme-sysctl">
<!-- Top Header Bar -->
<div class="d-flex justify-content-between align-items-center flex-wrap gap-3 mb-3 maint-header">
<div>
<div class="d-flex align-items-center gap-2">
<h5 class="mb-0 fw-bold">
          Windows Maintenance &amp;amp; Recovery
        </h5>
<span class="badge rounded-pill bg-success-subtle text-success border border-success px-2 py-1" style="font-size: 0.75rem;">Health &amp;amp; Restoration</span>
</div>
<p class="mb-0 small text-body-secondary mt-1">
        Очистка временных файлов и кэшей, проверка целостности SFC/DISM, управление точками восстановления Windows и настройка дисковых квот теневого хранилища (VSS).
      </p>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<span class="badge rounded-pill px-3 py-2" id="maint-elevation-badge">
        Checking privileges...
      </span>
<button class="btn btn-sm btn-outline-info rounded-pill px-3" id="btn-maint-refresh">
<i class="bi bi-arrow-clockwise me-1"></i> Обновить
      </button>
</div>
</div>
<div class="row g-3">
<!-- Quick Action Cards -->
<div class="col-md-4">
<div class="card maint-card p-3 h-100 d-flex flex-column justify-content-between" data-panel-id="panel-maintenance-recovery-ochistka-diskov-i-keshey" id="panel-maintenance-recovery-ochis
```

```
tka-diskov-i-keshey">
<div>
<h6 class="fw-bold text-info"><i class="bi bi-trash3-fill me-1
"></i> Очистка дисков и кэшей</h6>
<p class="small text-muted mb-2">Безопасное удаление временных
 файлов, логов установки и кэша обновлений Windows.</p>
<div class="small fw-semibold text-body mb-3" id="maint-clean-
estimate-txt">Доступно к очистке: ~0 MB</div>
</div>
<button class="btn btn-sm btn-outline-info w-100 rounded-pill"
 id="btn-maint-action-clean">
<i class="bi bi-stars me-1"></i> Очистить временные файлы
        </button>
</div>
</div>
<div class="col-md-4">
<div class="card maint-card p-3 h-100 d-flex flex-column justi
fy-content-between" data-panel-id="panel-maintenance-recovery-
proverka-sfc-system-file-check" id="panel-maintenance-recovery
-proverka-sfc-system-file-check">
<div>
<h6 class="fw-bold text-warning"><i class="bi bi-shield-shaded
 me-1"></i> Проверка SFC (System File Check)</h6>
<p class="small text-muted mb-2">Проверка и верификация целост
ности защищенных системных файлов Windows (sfc /verifyonly).</
p>
</div>
<button class="btn btn-sm btn-outline-warning w-100 rounded-pi
ll" id="btn-maint-action-sfc">
<i class="bi bi-search me-1"></i> Запустить проверку SFC
        </button>
</div>
</div>
<div class="col-md-4">
<div class="card maint-card p-3 h-100 d-flex flex-column justi
fy-content-between" data-panel-id="panel-maintenance-recovery-
diagnostika-dism-store" id="panel-maintenance-recovery-diagnos
tika-dism-store">
```

```
<div>
<h6 class="fw-bold text-success"><i class="bi bi-heart-pulse-fill me-1"></i> Диагностика DISM Store</h6>
<p class="small text-muted mb-2">Проверка состояния хранилища компонентов Windows с помощью DISM /CheckHealth.</p>
</div>
<button class="btn btn-sm btn-outline-success w-100 rounded-pill" id="btn-maint-action-dism">
<i class="bi bi-activity me-1"></i> Проверить DISM Store
        </button>
</div>
</div>
<!-- System Restore Points Panel -->
<div class="col-12">
<div class="card maint-card" data-panel-id="panel-maintenance-recovery-tochki-vosstanovleniya-windows-s" id="panel-maintenance-recovery-tochki-vosstanovleniya-windows-s">
<div class="card-header border-secondary bg-transparent d-flex justify-content-between align-items-center flex-wrap gap-2">
<span class="fw-bold text-body"><i class="bi bi-arrow-counterclockwise me-1"></i> Точки восстановления Windows (System Restore Points)</span>
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm btn-outline-info rounded-pill px-3" id="btn-maint-config-restore" title="Настройка квоты диска, расписания и политик хранения">
<i class="bi bi-gear-fill me-1"></i> Политики и квоты VSS
            </button>
<button class="btn btn-sm btn-outline-primary rounded-pill px-3" id="btn-maint-create-restore">
<i class="bi bi-plus-lg me-1"></i> Создать точку восстановления
            </button>
</div>
</div>
<div class="p-0 table-responsive" style="max-height: 380px; overflow-y: auto;">
<table class="maint-table">
```

```
<thead>
<tr>
<th style="width: 100px;">Номер #</th>
<th style="width: 45%;">Описание</th>
<th style="width: 25%;">Тип</th>
<th style="width: 20%;">Дата создания</th>
</tr>
</thead>
<tbody id="maint-restore-tbody">
<tr><td class="text-center text-muted p-3" colspan="4">Загрузка точек восстановления...</td></tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
<!-- Modal: System Restore Policy, Disk Quota & Automated Schedule -->
<div aria-hidden="true" class="modal fade" id="modal-maint-restore-config" tabindex="-1">
<div class="modal-dialog modal-dialog-centered modal-lg">
<div class="modal-content">
<div class="modal-header border-secondary">
<h5 class="modal-title fw-bold text-info">
<i class="bi bi-shield-shaded me-2"></i> Политики теневого хранилища, квоты и расписание VSS
          </h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body p-4">
<p class="small text-muted mb-3">
            Настройка дисковой квоты теневого хранилища (VSS), сценариев автоматического создания и правил ротации точек восстановления Windows.
          </p>
<!-- 1. Disk Quota Allocation -->
```

```
<div class="card maint-sub-card p-3 mb-3" data-panel-id="panel-maintenance-recovery-1-vydelenie-diskovogo-prostran" id="panel-maintenance-recovery-1-vydelenie-diskovogo-prostran">
<h6 class="fw-bold text-primary mb-2">
<i class="bi bi-hdd-fill me-1"></i> 1. Выделение дискового пространства (Shadow Storage)
            </h6>
<div class="row g-3 align-items-center">
<div class="col-md-6">
<label class="form-label small text-muted">Максимальный лимит для диска C:</label>
<select class="form-select form-select-sm" id="maint-policy-storage-size">
<option value="5%">5% объема диска (~15-30 GB)</option>
<option selected="" value="10%">10% объема диска (~30-60 GB, рекомендуется)</option>
<option value="15%">15% объема диска (~45-90 GB)</option>
<option value="20%">20% объема диска (~60-120 GB)</option>
<option value="10GB">10 GB фиксированно</option>
<option value="20GB">20 GB фиксированно</option>
<option value="50GB">50 GB фиксированно</option>
</select>
</div>
<div class="col-md-6">
<label class="form-label small text-muted">Текущее заполнение VSS:</label>
<div class="progress bg-secondary-subtle" style="height: 10px; ">
<div class="progress-bar bg-info" id="maint-policy-storage-bar" role="progressbar" style="width: 0%;"></div>
</div>
<div class="d-flex justify-content-between small text-muted mt-1 font-monospace" id="maint-policy-storage-label">
<span>Used: 0 GB</span>
<span>Max: 0 GB</span>
</div>
</div>
</div>
</div>
```

```
</div>
<!-- 2. Schedule and Triggers -->
<div class="card maint-sub-card p-3 mb-3" data-panel-id="panel-maintenance-recovery-2-stsenariy-avtomaticheskogo-soz" id="panel-maintenance-recovery-2-stsenariy-avtomaticheskogo-soz">
<h6 class="fw-bold text-success mb-2">
<i class="bi bi-clock-history me-1"></i> 2. Сценарий автоматического создания (Trigger)
            </h6>
<div class="row g-3">
<div class="col-md-6">
<label class="form-label small text-muted">Сценарий запуска:</label>
<select class="form-select form-select-sm" id="maint-policy-trigger">
<option value="daily">Ежедневно по расписанию (Daily)</option>
<option value="startup">При каждом запуске системы (At Startup)</option>
<option value="weekly">Еженедельно (Weekly Sunday)</option>
<option value="disabled">Только вручную / перед изменениями (Disabled)</option>
</select>
</div>
<div class="col-md-6" id="maint-policy-time-wrapper">
<label class="form-label small text-muted">Время создания (для ежедневного/еженедельного):</label>
<input class="form-control form-control-sm font-monospace" id="maint-policy-time" type="time" value="03:00"/>
</div>
<div class="col-12">
<div class="form-check form-switch mt-1">
<input checked="" class="form-check-input" id="maint-policy-freq-limit" style="cursor: pointer;" type="checkbox"/>
<label class="form-check-label small text-body" for="maint-policy-freq-limit">
                    Снять 24-часовое системное ограничение частоты Windows (<code>CreationFrequency = 0</code>)
                </label>
```

```
</div>
</div>
</div>
</div>
<!-- 3. Retention Policy -->
<div class="card maint-sub-card p-3 mb-3" data-panel-id="panel-maintenance-recovery-3-politika-khraneniya-i-rotatsiya" id="panel-maintenance-recovery-3-politika-khraneniya-i-rotatsiya">
<h6 class="fw-bold text-warning mb-2">
<i class="bi bi-layers-half me-1"></i> 3. Политика хранения и ротация (Retention)
            </h6>
<div class="row g-3">
<div class="col-md-6">
<label class="form-label small text-muted">Сколько последних точек хранить:</label>
<select class="form-select form-select-sm" id="maint-policy-max-points">
<option value="3">Хранить 3 последние точки</option>
<option selected="" value="5">Хранить 5 последних точек (рекомендуется)</option>
<option value="7">Хранить 7 последних точек</option>
<option value="10">Хранить 10 последних точек</option>
<option value="15">Хранить 15 последних точек</option>
<option value="20">Хранить 20 последних точек</option>
</select>
</div>
<div class="col-md-6 d-flex flex-column justify-content-end">
<div class="form-check form-switch mb-1">
<input checked="" class="form-check-input" id="maint-policy-auto-prune" style="cursor: pointer;" type="checkbox"/>
<label class="form-check-label small text-body" for="maint-policy-auto-prune">
                    Автоматическая ротация при превышении (Auto-Prune FIFO)
                </label>
</div>
</div>
```

```
</div>
</div>
</div>
<div class="modal-footer border-secondary d-flex justify-conte
nt-between">
<button class="btn btn-sm btn-outline-danger rounded-pill px-3
" id="btn-maint-policy-prune-now" type="button">
<i class="bi bi-trash3 me-1"></i> Очистить избыточные точки
          </button>
<div class="d-flex gap-2">
<button class="btn btn-sm btn-outline-secondary rounded-pill p
x-3" data-bs-dismiss="modal" type="button">Закрыть</button>
<button class="btn btn-sm btn-primary rounded-pill px-4" id="b
tn-maint-save-restore-policy" type="button">
<i class="bi bi-check-lg me-1"></i> Сохранить и применить
            </button>
</div>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Maintenance & Recovery - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт обслуживания и восстановления Windows (
Maintenance & Recovery).
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/html/maintenance_recovery_tab/main.js?v=2
0261006_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/maintenance_recovery_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

(function () {
  'use strict';

  let isInitialized = false;

  /**
   * Экранирование HTML-символов для защиты от XSS.
   *
   * @param {string} str - Исходная строка.
   * @returns {string} Экранированная строка.
   */
```

```
  function escapeHtml(str) {
    if (!str) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
  }

  /**
   * Форматирование типа точки восстановления.
   *
   * @param {string|number} rawType - Сырой тип.
   * @returns {string} Читаемый тип.
   */
  function formatRestorePointType(rawType) {
    if (!rawType) return 'Системный снимок';
    const s = String(rawType).toUpperCase();
    if (s.includes('APPLICATION_INSTALL') || s === '0') return
 'Установка ПО';
    if (s.includes('APPLICATION_UNINSTALL') || s === '1') retu
rn 'Удаление ПО';
    if (s.includes('DEVICE_DRIVER') || s === '10') return 'Дра
йвер устройства';
    if (s.includes('MODIFY_SETTINGS') || s === '12') return 'И
зменение настроек';
    if (s.includes('CANCELLED') || s === '13') return 'Отмена
операции';
    if (s.includes('WINDOWS_UPDATE') || s === '17') return 'Wi
ndows Update';
    return s.replace(/_/g, ' ');
  }

  /**
   * Форматирование времени точки восстановления.
   *
   * @param {string} rawTime - Сырое время (YYYYMMDDHHmmss).
```

```
   * @returns {string} Форматированное время.
   */
  function formatRestoreTime(rawTime) {
    if (!rawTime) return '-';
    const s = String(rawTime).trim();
    if (s.length >= 14 && /^\d{14}/.test(s)) {
      return `${s.slice(0, 4)}-${s.slice(4, 6)}-${s.slice(6, 8
)} ${s.slice(8, 10)}:${s.slice(10, 12)}:${s.slice(12, 14)}`;
    }
    return s;
  }

  /**
   * Получение статуса обслуживания и прав доступа.
   */
  async function fetchMaintStatus() {
    try {
      const res = await fetch('/api/system-control/status');
      if (!res.ok) return;
      const data = await res.json();

      const elevBadge = document.getElementById('maint-elevati
on-badge');
      if (elevBadge) {
        if (data.is_elevated) {
          elevBadge.className = 'badge rounded-pill bg-success
-subtle text-success border border-success px-3 py-2';
          elevBadge.innerHTML = '  Режим: Администратор (Полн
ый доступ)';
        } else {
          elevBadge.className = 'badge rounded-pill bg-warning
-subtle text-warning border border-warning px-3 py-2';
          elevBadge.innerHTML = '  Режим: Обычный пользовател
ь (Ограничено)';
        }
      }

      const cleanEstimateTxt = document.getElementById('maint-
```

```
clean-estimate-txt');
96        if (cleanEstimateTxt) {
97          const cleanMb = data.disk?.cleanup_estimate?.total_cle
anable_mb || 0;
98          cleanEstimateTxt.innerText = `Доступно к очистке: ~${c
leanMb} MB`;
99        }
100     } catch (e) {
101       console.warn('[MaintenanceRecovery] Ошибка получения ста
туса:', e);
102     }
103   }
104
105   /**
106    * Загрузка списка точек восстановления Windows.
107    */
108   async function fetchRestorePoints() {
109     try {
110       const res = await fetch('/api/system-control/restore-poi
nts');
111       if (!res.ok) return;
112       const data = await res.json();
113       const points = data.restore_points || [];
114
115       const tbody = document.getElementById('maint-restore-tbo
dy');
116       if (tbody) {
117         if (points.length === 0) {
118           tbody.innerHTML = '<tr><td colspan="4" class="text-c
enter text-muted p-3">Точки восстановления не найдены. Нажмите
 «Создать точку восстановления» для фиксации состояния.</td></
tr>';
119           return;
120         }
121         tbody.innerHTML = points.map((p, idx) => {
122           const typeBadge = formatRestorePointType(p.restore_p
oint_type);
123           const timeFormatted = formatRestoreTime(p.creation_t
```

```
ime);
          return `
          <tr class="maint-restore-row" data-idx="${idx}" style="cursor: pointer;" title="Нажмите для анализа точки восстановления">
            <td class="font-monospace text-info fw-bold">#${p.sequence_number}</td>
            <td class="fw-semibold text-white">${escapeHtml(p.description)}</td>
            <td><span class="badge bg-secondary text-light px-2 py-1 font-monospace" style="font-size: 0.75rem;">${escapeHtml(typeBadge)}</span></td>
            <td class="small text-light font-monospace">${escapeHtml(timeFormatted)}</td>
          </tr>
        `;
        }).join('');

        tbody.querySelectorAll('.maint-restore-row').forEach(row => {
          row.onclick = () => {
            const idx = parseInt(row.getAttribute('data-idx'), 10);
            const p = points[idx];
            if (!p) return;
            if (window.AITableModal) {
              window.AITableModal.show({
                icon: '□ ',
                title: `Точка восстановления #${p.sequence_number}`,
                subtitle: p.description,
                tableType: 'generic',
                badges: [
                  { text: p.restore_point_type || 'System Checkpoint', class: 'badge bg-info text-dark' }
                ],
                metadata: [
                  { label: 'Номер', value: String(p.sequence_n
```

```
umber) },
                  { label: 'Описание', value: p.description },
                  { label: 'Тип', value: p.restore_point_type
},
                  { label: 'Дата создания', value: p.creation_
time }
                ],
                rawTitle: 'Метаданные точки восстановления',
                rawContent: JSON.stringify(p, null, 2)
              });
            }
          };
        });
      }
    } catch (e) {
      console.error('[MaintenanceRecovery] Ошибка загрузки точ
ек восстановления:', e);
    }
  }

  /**
   * Открытие модального окна настройки политик теневого храни
лища (VSS).
   */
  async function openRestoreConfigModal() {
    try {
      const res = await fetch('/api/system-control/restore-poi
nts/config');
      if (res.ok) {
        const cfg = await res.json();
        const sizeSelect = document.getElementById('maint-poli
cy-storage-size');
        if (sizeSelect && cfg.max_storage_size) sizeSelect.val
ue = cfg.max_storage_size;

        const trigSelect = document.getElementById('maint-poli
cy-trigger');
        if (trigSelect && cfg.schedule_trigger) trigSelect.val
```

```
ue = cfg.schedule_trigger;
179
180         const timeInput = document.getElementById('maint-polic
y-time');
181         if (timeInput && cfg.schedule_time) timeInput.value =
cfg.schedule_time;
182
183         const ptsSelect = document.getElementById('maint-polic
y-max-points');
184         if (ptsSelect && cfg.max_points) ptsSelect.value = Str
ing(cfg.max_points);
185
186         const pruneSwitch = document.getElementById('maint-pol
icy-auto-prune');
187         if (pruneSwitch && typeof cfg.auto_prune === 'boolean'
) pruneSwitch.checked = cfg.auto_prune;
188
189         const freqSwitch = document.getElementById('maint-poli
cy-freq-limit');
190         if (freqSwitch && typeof cfg.frequency_limit_minutes =
== 'number') freqSwitch.checked = cfg.frequency_limit_minutes
=== 0;
191
192         const shadow = cfg.shadow_storage || {};
193         const usedGb = shadow.allocated_space_gb || 0;
194         const maxGb = shadow.max_space_gb || 1;
195         const pct = shadow.percent_used || (maxGb > 0 ? Math.m
in(100, Math.round((usedGb / maxGb) * 100)) : 0);
196
197         const bar = document.getElementById('maint-policy-stor
age-bar');
198         if (bar) {
199           bar.style.width = `${pct}%`;
200           bar.className = `progress-bar ${pct > 80 ? 'bg-dange
r' : pct > 50 ? 'bg-warning' : 'bg-info'}`;
201         }
202         const lbl = document.getElementById('maint-policy-stor
age-label');
```

```
        if (lbl) {
          lbl.innerHTML = `<span>Занято: ${usedGb} GB (${pct}%)</span><span>Лимит: ${maxGb} GB</span>`;
        }
      }
    } catch (e) {
      console.warn('[MaintenanceRecovery] Ошибка получения конфигурации VSS:', e);
    }

    const modalEl = document.getElementById('modal-maint-restore-config');
    if (modalEl && window.bootstrap?.Modal) {
      const bsModal = bootstrap.Modal.getOrCreateInstance(modalEl);
      bsModal.show();
    }
  }

  /**
   * Сохранение политик теневого хранилища и точек восстановления.
   */
  async function saveRestoreConfig() {
    const saveBtn = document.getElementById('btn-maint-save-restore-policy');
    const origText = saveBtn ? saveBtn.innerHTML : '';
    if (saveBtn) {
      saveBtn.disabled = true;
      saveBtn.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span>Сохранение...';
    }

    try {
      const sizeSelect = document.getElementById('maint-policy-storage-size');
      const trigSelect = document.getElementById('maint-policy-trigger');
```

```
      const timeInput = document.getElementById('maint-policy-time');
      const ptsSelect = document.getElementById('maint-policy-max-points');
      const pruneSwitch = document.getElementById('maint-policy-auto-prune');
      const freqSwitch = document.getElementById('maint-policy-freq-limit');

      const payload = {
        max_storage_size: sizeSelect ? sizeSelect.value : '10%',
        schedule_trigger: trigSelect ? trigSelect.value : 'daily',
        schedule_time: timeInput ? timeInput.value : '03:00',
        max_points: ptsSelect ? parseInt(ptsSelect.value, 10) : 5,
        auto_prune: pruneSwitch ? pruneSwitch.checked : true,
        frequency_limit_minutes: freqSwitch && freqSwitch.checked ? 0 : 1440
      };

      const res = await fetch('/api/system-control/restore-points/config', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify(payload)
      });

      const data = await res.json();
      if (!res.ok || (data.success === false)) {
        throw new Error(data.message || 'Ошибка сохранения политики');
      }

      if (window.toast) {
        window.toast.success('Политики VSS', 'Политики и лимиты дискового пространства успешно обновлены.');
```

```
      }

      const modalEl = document.getElementById('modal-maint-restore-config');
      if (modalEl && window.bootstrap?.Modal) {
        bootstrap.Modal.getInstance(modalEl)?.hide();
      }

      fetchRestorePoints();
      fetchMaintStatus();
    } catch (e) {
      console.error('[MaintenanceRecovery] Ошибка сохранения политики:', e);
      if (window.toast) window.toast.error('Политики VSS', e.message);
    } finally {
      if (saveBtn) {
        saveBtn.disabled = false;
        saveBtn.innerHTML = origText;
      }
    }
  }

  /**
   * Принудительная очистка избыточных точек восстановления.
   */
  async function pruneRestorePoints() {
    const ptsSelect = document.getElementById('maint-policy-max-points');
    const keepCount = ptsSelect ? parseInt(ptsSelect.value, 10) : 5;

    if (!confirm(`Вы действительно хотите удалить старые точки восстановления, оставив ${keepCount} последних?`)) {
      return;
    }

    try {
```

```
      const res = await fetch('/api/system-control/restore-points/prune', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ keep_count: keepCount })
      });
      const data = await res.json();
      if (window.toast) window.toast.success('Ротация VSS', data.message || `Оставлено ${keepCount} последних точек.`);
      fetchRestorePoints();
    } catch (e) {
      if (window.toast) window.toast.error('Ротация VSS', e.message);
    }
  }

  /**
   * Инициализация вкладки Maintenance & Recovery.
   */
  function initMaintenanceRecoveryTab() {
    if (window.registerTabPoller) {
      window.registerTabPoller('tab-maintenance-recovery', () => {
        fetchMaintStatus();
        fetchRestorePoints();
      }, 5000, { pollerId: 'tab-maintenance-recovery_status', immediate: true });
    } else if (window.isTabActive ? window.isTabActive('tab-maintenance-recovery') : true) {
      fetchMaintStatus();
      fetchRestorePoints();
    }

    if (isInitialized) return;
    isInitialized = true;

    const refreshBtn = document.getElementById('btn-maint-refresh');
```

```
322     if (refreshBtn) {
323       refreshBtn.onclick = () => {
324         fetchMaintStatus();
325         fetchRestorePoints();
326       };
327     }
328 
329     // Быстрая очистка кэшей
330     const cleanBtn = document.getElementById('btn-maint-action
    -clean');
331     if (cleanBtn) {
332       cleanBtn.onclick = async () => {
333         const orig = cleanBtn.innerHTML;
334         cleanBtn.disabled = true;
335         cleanBtn.innerHTML = '<span class="spinner-border spin
    ner-border-sm me-1"></span>Очистка...';
336         try {
337           let res = await fetch('/api/system-control/actions/c
    leanup', { method: 'POST' });
338           if (!res.ok) {
339             res = await fetch('/api/system-control/maintenance
    /cleanup', { method: 'POST' });
340           }
341           const data = await res.json();
342           if (window.toast) window.toast.success('Очистка диск
    a', data.message || 'Временные файлы успешно удалены.');
343           fetchMaintStatus();
344         } catch (e) {
345           if (window.toast) window.toast.error('Очистка диска'
    , e.message);
346         } finally {
347           cleanBtn.disabled = false;
348           cleanBtn.innerHTML = orig;
349         }
350       };
351     }
352 
353     // Запуск проверки SFC
```

```
    const sfcBtn = document.getElementById('btn-maint-action-sfc');
    if (sfcBtn) {
      sfcBtn.onclick = async () => {
        const orig = sfcBtn.innerHTML;
        sfcBtn.disabled = true;
        sfcBtn.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span>Сканирование...';
        try {
          const res = await fetch('/api/system-control/maintenance/sfc', { method: 'POST' });
          const data = await res.json();
          if (window.toast) window.toast.info('Проверка SFC', data.message || 'Сканирование целостности файлов завершено.');
        } catch (e) {
          if (window.toast) window.toast.error('Проверка SFC', e.message);
        } finally {
          sfcBtn.disabled = false;
          sfcBtn.innerHTML = orig;
        }
      };
    }

    // Запуск проверки DISM
    const dismBtn = document.getElementById('btn-maint-action-dism');
    if (dismBtn) {
      dismBtn.onclick = async () => {
        const orig = dismBtn.innerHTML;
        dismBtn.disabled = true;
        dismBtn.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span>Проверка DISM...';
        try {
          const res = await fetch('/api/system-control/maintenance/dism', { method: 'POST' });
          const data = await res.json();
          if (window.toast) window.toast.info('Проверка DISM',
```

```
 data.message || 'Диагностика Component Store завершена.');
        } catch (e) {
          if (window.toast) window.toast.error('Проверка DISM', e.message);
        } finally {
          dismBtn.disabled = false;
          dismBtn.innerHTML = orig;
        }
      };
    }

    // Создание точки восстановления
    const createRestoreBtn = document.getElementById('btn-maint-create-restore');
    if (createRestoreBtn) {
      createRestoreBtn.onclick = async () => {
        const desc = prompt('Введите описание для новой точки восстановления:', 'Ручной снимок системы перед изменениями');
        if (!desc) return;
        const orig = createRestoreBtn.innerHTML;
        createRestoreBtn.disabled = true;
        createRestoreBtn.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span>Создание...';
        try {
          const res = await fetch('/api/system-control/restore-points', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ description: desc, restore_point_type: 'MODIFY_SETTINGS' })
          });
          const data = await res.json();
          if (!res.ok) throw new Error(data.detail || data.message || 'Ошибка создания точки');
          if (window.toast) window.toast.success('Точка восстановления', 'Точка восстановления успешно создана.');
          fetchRestorePoints();
        } catch (e) {
```

```
          if (window.toast) window.toast.error('Точка восстановления', e.message);
        } finally {
          createRestoreBtn.disabled = false;
          createRestoreBtn.innerHTML = orig;
        }
      };
    }

    // Открытие модального окна политик
    const cfgRestoreBtn = document.getElementById('btn-maint-config-restore');
    if (cfgRestoreBtn) {
      cfgRestoreBtn.onclick = () => {
        openRestoreConfigModal();
      };
    }

    // Сохранение политик
    const savePolicyBtn = document.getElementById('btn-maint-save-restore-policy');
    if (savePolicyBtn) {
      savePolicyBtn.onclick = () => {
        saveRestoreConfig();
      };
    }

    // Очистить избыточные точки
    const pruneBtn = document.getElementById('btn-maint-policy-prune-now');
    if (pruneBtn) {
      pruneBtn.onclick = () => {
        pruneRestorePoints();
      };
    }
  }

  window.initMaintenanceRecoveryTab = initMaintenanceRecoveryT
```

```
    ab;
447 })();
448
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/mcp_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/mcp_tab/index.html" cla
ss="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/mcp_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- MCP Servers Tab Component -->
<div class="container-fluid p-3" id="mcp-tab-root">
<!-- Top Banner / Summary -->
<div class="alert alert-dark border-secondary bg-dark text-lig
ht d-flex flex-wrap align-items-center justify-content-between
 gap-3 shadow-sm">
<div class="d-flex align-items-center gap-3">
<div class="p-3 bg-black rounded border border-secondary text-
primary display-6 d-flex align-items-center justify-content-ce
nter" style="width: 54px; height: 54px;">
```

```
<i class="bi bi-hdd-network-fill"></i>
</div>
<div>
<h5 class="mb-1 text-white d-flex align-items-center gap-2">
          Серверы MCP (Model Context Protocol)
          <span class="badge bg-primary rounded-pill small" id="mcp-total-badge">0</span>
</h5>
<p class="mb-0 text-muted small">
          Управление внешними MCP-серверами, интеграция инструментов браузера, файловой системы, баз данных и API для ИИ-агентов.
        </p>
</div>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<button class="btn btn-outline-info btn-sm d-flex align-items-center gap-1" id="btn-view-all-tools" onclick="window.mcpTab?.openAllToolsModal()" type="button">
<i class="bi bi-tools"></i> Все инструменты
      </button>
<button class="btn btn-outline-light btn-sm d-flex align-items-center gap-1" id="btn-refresh-mcp" onclick="window.mcpTab?.refresh()" type="button">
<i class="bi bi-arrow-clockwise"></i> Обновить
      </button>
<button class="btn btn-primary btn-sm d-flex align-items-center gap-1 shadow-sm" data-bs-target="#mcpServerModal" data-bs-toggle="modal" onclick="window.mcpTab?.openCreateModal()" type="button">
<i class="bi bi-plus-circle-fill"></i> Добавить MCP сервер
      </button>
</div>
</div>
<!-- Statistics Row -->
<div class="row g-2 mb-3" id="mcp-stats-row">
<div class="col-6 col-md-3">
<div class="p-2 bg-dark rounded border border-secondary text-c
```

```
enter">
<div class="h4 mb-0 text-white" id="stat-total-mcp">0</div>
<div class="small text-muted">Всего серверов</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="p-2 bg-body-tertiary rounded border text-center">
<div class="h4 mb-0 text-success" id="stat-active-mcp">0</div>
<div class="small text-muted">Активных серверов</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="p-2 bg-body-tertiary rounded border text-center">
<div class="h4 mb-0 text-info" id="stat-stdio-mcp">0</div>
<div class="small text-muted">Stdio-процессов</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="p-2 bg-body-tertiary rounded border text-center">
<div class="h4 mb-0 text-warning" id="stat-network-mcp">0</div
>
<div class="small text-muted">SSE / HTTP серверов</div>
</div>
</div>
</div>
<!-- Search & View Controls -->
<div class="card bg-body-tertiary border mb-3 shadow-sm" data-
panel-id="panel-mcp-1" id="panel-mcp-1">
<div class="card-body p-2 d-flex flex-wrap align-items-center
justify-content-between gap-2">
<div class="input-group input-group-sm flex-grow-1" style="max
-width: 450px;">
<span class="input-group-text"><i class="bi bi-search"></i></s
pan>
<input class="form-control" id="mcp-search-input" placeholder=
"Поиск по ID, названию или команде..." type="text"/>
<button class="btn btn-outline-secondary" id="mcp-search-clear
" onclick="window.mcpTab?.clearSearch()" type="button">□ </butt
```

```
on>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<select class="form-select form-select-sm" id="mcp-filter-sele
ct" onchange="window.mcpTab?.applyFilters()" style="width: aut
o;">
<option value="all">Все серверы</option>
<option value="enabled">Только включенные</option>
<option value="disabled">Только отключенные</option>
<option value="stdio">Транспорт: stdio</option>
<option value="sse">Транспорт: SSE / HTTP</option>
</select>
<div class="btn-group btn-group-sm" role="group">
<button class="btn btn-outline-secondary active" id="btn-mcp-v
iew-cards" onclick="window.mcpTab?.setViewMode('cards')" type=
"button">
<i class="bi bi-grid-fill"></i> Карточки
          </button>
<button class="btn btn-outline-secondary" id="btn-mcp-view-tab
le" onclick="window.mcpTab?.setViewMode('table')" type="button
">
<i class="bi bi-table"></i> Таблица
          </button>
</div>
</div>
</div>
</div>
<!-- Loading & Empty Indicators -->
<div class="text-center py-5 text-muted d-none" id="mcp-loadin
g">
<div class="spinner-border text-primary mb-2" role="status"></
div>
<div>Загрузка серверов MCP...</div>
</div>
<div class="text-center py-5 text-muted d-none" id="mcp-empty"
>
<i class="bi bi-hdd-network display-4 d-block mb-2 text-second
ary"></i>
```

```
<h5>Серверы MCP не настроены</h5>
<p class="small">Добавьте сервер MCP (например, Playwright, Filesystem, SQLite) для расширения возможностей агентов.</p>
<button class="btn btn-sm btn-primary" data-bs-target="#mcpServerModal" data-bs-toggle="modal" onclick="window.mcpTab?.openCreateModal()">
<i class="bi bi-plus-circle-fill"></i> Добавить первый сервер
    </button>
</div>
<!-- Cards Container -->
<div class="row g-3" id="mcp-cards-container"></div>
<!-- Table Container -->
<div class="card shadow-sm border d-none" data-panel-id="panel-mcp-2" id="mcp-table-container">
<div class="table-responsive">
<table class="table table-hover mb-0 align-middle">
<thead class="table-light">
<tr class="small text-uppercase">
<th style="width: 50px;">Статус</th>
<th>ID / Название</th>
<th>Транспорт</th>
<th>Команда / URL</th>
<th>Описание</th>
<th class="text-end" style="width: 220px;">Действия</th>
</tr>
</thead>
<tbody id="mcp-table-body"></tbody>
</table>
</div>
</div>
</div>
<!-- Add / Edit MCP Server Modal -->
<div aria-hidden="true" aria-labelledby="mcpServerModalLabel" class="modal fade" id="mcpServerModal" tabindex="-1">
<div class="modal-dialog modal-lg">
<div class="modal-content shadow-lg">
<div class="modal-header">
<h5 class="modal-title d-flex align-items-center gap-2" id="mc
```

```
pServerModalLabel">
<i class="bi bi-hdd-network-fill text-primary"></i> <span id="
mcp-modal-title-text">Настройка MCP сервера</span>
</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-4">
<!-- Preset Selector -->
<div class="mb-3 p-3 bg-body-tertiary rounded border" id="mcp-
preset-container">
<label class="form-label text-info small fw-bold mb-1 d-flex a
lign-items-center gap-1">
<i class="bi bi-magic"></i> Готовые шаблоны (Presets)
          </label>
<div class="d-flex gap-2 flex-wrap">
<button class="btn btn-sm btn-outline-primary" onclick="window
.mcpTab?.applyPreset('playwright')" type="button">🎭 Playwrigh
t (Browser)</button>
<button class="btn btn-sm btn-outline-info" onclick="window.mc
pTab?.applyPreset('filesystem')" type="button">📁 Filesystem</
button>
<button class="btn btn-sm btn-outline-success" onclick="window
.mcpTab?.applyPreset('memory')" type="button">🧠 Memory</butto
n>
<button class="btn btn-sm btn-outline-warning" onclick="window
.mcpTab?.applyPreset('sqlite')" type="button">🗄️ SQLite</butto
n>
<button class="btn btn-sm btn-outline-secondary" onclick="wind
ow.mcpTab?.applyPreset('fetch')" type="button">🌐 Fetch / HTTP
</button>
<button class="btn btn-sm btn-outline-secondary" onclick="wind
ow.mcpTab?.applyPreset('sse')" type="button">🌐 Remote SSE / HT
TP</button>
</div>
</div>
<form id="mcp-server-form" onsubmit="window.mcpTab?.saveServer
(event)">
```

```
<input id="mcp-form-mode" type="hidden" value="create"/>
<div class="row g-3">
<div class="col-md-6">
<label class="form-label small text-muted" for="mcp-server-id">ID сервера (уникальный ключ) <span class="text-danger">*</span></label>
<input class="form-control" id="mcp-server-id" pattern="[a-zA-Z0-9_-]+" placeholder="например, playwright" required="" type="text"/>
<div class="form-text text-muted small">Латинские буквы, цифры, дефис или подчеркивание.</div>
</div>
<div class="col-md-6">
<label class="form-label small text-muted" for="mcp-server-name">Отображаемое имя</label>
<input class="form-control" id="mcp-server-name" placeholder="Playwright Web Browser" type="text"/>
</div>
<div class="col-12">
<label class="form-label small text-muted" for="mcp-server-desc">Описание</label>
<input class="form-control" id="mcp-server-desc" placeholder="Управление браузером и поиск информации в веб-страницах" type="text"/>
</div>
<div class="col-md-6">
<label class="form-label small text-muted" for="mcp-server-transport">Протокол транспорта <span class="text-danger">*</span></label>
<select class="form-select" id="mcp-server-transport" onchange="window.mcpTab?.onTransportChange()">
<option value="stdio">stdio (Локальный subprocess)</option>
<option value="sse">sse (Server-Sent Events по HTTP)</option>
<option value="streamable_http">streamable_http (HTTP Streaming)</option>
</select>
</div>
<div class="col-md-6 d-flex align-items-center pt-4">
```

```
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="mcp-server-enabled" type="checkbox"/>
<label class="form-check-label" for="mcp-server-enabled">Включить сервер для агентов</label>
</div>
</div>
<!-- Stdio Section -->
<div class="col-12 mcp-field-stdio" id="mcp-stdio-container">
<div class="mb-3">
<label class="form-label small text-muted" for="mcp-server-command">Команда (Command) <span class="text-danger">*</span></label>
<input class="form-control" id="mcp-server-command" placeholder="npx, python, node, uvx" type="text"/>
<div class="form-text text-muted small">Исполняемый файл или утилита в PATH.</div>
</div>
<div class="mb-3">
<label class="form-label small text-muted" for="mcp-server-args">Аргументы (Arguments)</label>
<input class="form-control" id="mcp-server-args" placeholder="@playwright/mcp@latest или через пробел / запятую" type="text" />
<div class="form-text text-muted small">Укажите аргументы через пробел или в формате JSON-массива строк.</div>
</div>
<div>
<label class="form-label small text-muted" for="mcp-server-env">Переменные окружения (Environment JSON / Key-Value)</label>
<textarea class="form-control font-monospace" id="mcp-server-env" placeholder='{"NODE_ENV": "production"}' rows="2"></textarea>
<div class="form-text text-muted small">Опциональный JSON-объект с переменными окружения для процесса.</div>
</div>
</div>
<!-- Network URL Section -->
```

```
<div class="col-12 mcp-field-network d-none" id="mcp-network-container">
<label class="form-label small text-muted" for="mcp-server-url">URL эндпоинта <span class="text-danger">*</span></label>
<input class="form-control" id="mcp-server-url" placeholder="http://localhost:8080/sse" type="url"/>
<div class="form-text text-muted small">Полный HTTP/HTTPS адрес удаленного MCP-сервера.</div>
</div>
</div>
<div class="mt-4 d-flex justify-content-between align-items-center border-top pt-3">
<button class="btn btn-outline-info btn-sm d-flex align-items-center gap-1 rounded-pill" id="btn-test-modal-server" onclick="window.mcpTab?.testCurrentForm()" type="button">
<i class="bi bi-play-circle"></i> Проверить подключение
            </button>
<div class="d-flex gap-2">
<button class="btn btn-secondary rounded-pill px-3" data-bs-dismiss="modal" type="button">Отмена</button>
<button class="btn btn-primary rounded-pill px-3" id="mcp-btn-save-server" type="submit">Сохранить</button>
</div>
</div>
</form>
</div>
</div>
</div>
</div>
<!-- Test Connection & Tools Inspection Modal -->
<div aria-hidden="true" aria-labelledby="mcpTestModalLabel" class="modal fade" id="mcpTestModal" tabindex="-1">
<div class="modal-dialog modal-lg modal-dialog-scrollable">
<div class="modal-content shadow-lg">
<div class="modal-header">
<h5 class="modal-title d-flex align-items-center gap-2" id="mcpTestModalLabel">
<i class="bi bi-activity text-info"></i> Диагностика MCP: <spa
```

```
n id="mcp-test-server-name">...</span>
</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-4">
<div class="p-3 mb-3 rounded border bg-body-tertiary d-flex al
ign-items-center justify-content-between" id="mcp-test-status-
box">
<div class="d-flex align-items-center gap-2">
<div class="spinner-border spinner-border-sm text-info" id="mc
p-test-spinner" role="status"></div>
<span id="mcp-test-status-text">Установка соединения и опрос и
нструментов...</span>
</div>
<span class="badge bg-secondary font-monospace" id="mcp-test-l
atency-badge">—</span>
</div>
<h6 class="d-flex align-items-center gap-2 mb-3">
<i class="bi bi-tools text-warning"></i> Обнаруженные инструме
нты (Tools):
          <span class="badge bg-primary rounded-pill" id="mcp-
test-tools-count">0</span>
</h6>
<div class="d-flex flex-column gap-2" id="mcp-test-tools-list"
>
<!-- Tools list populated dynamically -->
</div>
</div>
<div class="modal-footer">
<button class="btn btn-secondary rounded-pill px-3" data-bs-di
smiss="modal" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
<!-- All Tools Aggregator Modal -->
<div aria-hidden="true" aria-labelledby="mcpAllToolsModalLabel
```

```
" class="modal fade" id="mcpAllToolsModal" tabindex="-1">
<div class="modal-dialog modal-xl modal-dialog-scrollable">
<div class="modal-content shadow-lg">
<div class="modal-header">
<h5 class="modal-title d-flex align-items-center gap-2" id="mc
pAllToolsModalLabel">
<i class="bi bi-boxes text-success"></i> Доступный пул инструм
ентов MCP
        </h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-4">
<p class="text-muted small mb-3">
          Список всех инструментов, агрегированных со всех вкл
юченных серверов MCP. Эти инструменты автоматически предоставл
яются LangChain/ReAct агенту.
        </p>
<div class="text-center py-4 text-muted" id="mcp-all-tools-loa
ding">
<div class="spinner-border text-primary mb-2" role="status"></
div>
<div>Опрос активных MCP серверов...</div>
</div>
<div class="text-center py-4 text-muted d-none" id="mcp-all-to
ols-empty">
<i class="bi bi-tools display-6 d-block mb-2 text-secondary"><
/i>
<div>Нет доступных инструментов от активных MCP серверов.</div
>
</div>
<div class="row g-3 d-none" id="mcp-all-tools-container"></div
>
</div>
<div class="modal-footer">
<button class="btn btn-secondary rounded-pill px-3" data-bs-di
smiss="modal" type="button">Закрыть</button>
</div>
```

```
294 </div>
295 </div>
296 </div>
297
```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Mcp Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/mcp_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/mcp_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// MCP Servers Tab Management Logic for AI Windows Admin Admin Interface

class McpTabManager {
  constructor() {
    this.servers = [];
    this.filteredServers = [];
    this.viewMode = 'cards'; // 'cards' | 'table'
    this.searchQuery = '';
    this.activeFilter = 'all'; // 'all' | 'enabled' | 'disabled' | 'stdio' | 'sse'
    this.isInitialized = false;
```

```
  }

  async init() {
    if (this.isInitialized) {
      await this.refresh();
      return;
    }
    this.bindEvents();
    this.isInitialized = true;
    await this.refresh();
  }

  bindEvents() {
    const searchInput = document.getElementById('mcp-search-in
put');
    if (searchInput) {
      searchInput.addEventListener('input', (e) => {
        this.searchQuery = e.target.value.trim().toLowerCase()
;
        this.applyFilters();
      });
    }

    const idInput = document.getElementById('mcp-server-id');
    if (idInput) {
      idInput.addEventListener('input', (e) => {
        const val = e.target.value;
        const normalized = val.toLowerCase().replace(/[^a-z0-9
_-]/g, '_');
        if (val !== normalized) {
          e.target.value = normalized;
        }
      });
    }
  }

  async apiFetch(endpoint, options = {}) {
    let url = endpoint;
```

```
    let res = await fetch(url, options);
    if (!res.ok && url.startsWith('/api/admin/mcp')) {
      const fallbackUrl = url.replace('/api/admin/mcp', '/api/mcp');
      const fallbackRes = await fetch(fallbackUrl, options);
      if (fallbackRes.ok) {
        return fallbackRes;
      }
    }
    return res;
  }

  async refresh() {
    this.showLoading(true);
    try {
      const res = await this.apiFetch('/api/admin/mcp/servers');
      if (!res.ok) {
        throw new Error(`HTTP ${res.status}: ${res.statusText}`);
      }
      const data = await res.json();
      this.servers = data.servers || [];
      this.updateStats();
      this.applyFilters();
    } catch (err) {
      console.error('[McpTab] Error fetching MCP servers:', err);
      this.renderError(err.message);
    } finally {
      this.showLoading(false);
    }
  }

  updateStats() {
    const total = this.servers.length;
    const active = this.servers.filter(s => s.enabled !== false).length;
```

```
    const stdio = this.servers.filter(s => (s.transport || 'stdio') === 'stdio').length;
    const network = this.servers.filter(s => ['sse', 'streamable_http', 'http'].includes(s.transport)).length;

    const badge = document.getElementById('mcp-total-badge');
    if (badge) badge.textContent = total;

    const elTotal = document.getElementById('stat-total-mcp');
    if (elTotal) elTotal.textContent = total;

    const elActive = document.getElementById('stat-active-mcp');
    if (elActive) elActive.textContent = active;

    const elStdio = document.getElementById('stat-stdio-mcp');
    if (elStdio) elStdio.textContent = stdio;

    const elNetwork = document.getElementById('stat-network-mcp');
    if (elNetwork) elNetwork.textContent = network;
  }

  applyFilters() {
    const filterSelect = document.getElementById('mcp-filter-select');
    this.activeFilter = filterSelect ? filterSelect.value : 'all';

    this.filteredServers = this.servers.filter(s => {
      // Search filter
      if (this.searchQuery) {
        const idMatch = (s.id || '').toLowerCase().includes(this.searchQuery);
        const nameMatch = (s.name || '').toLowerCase().includes(this.searchQuery);
        const descMatch = (s.description || '').toLowerCase().includes(this.searchQuery);
```

```
        const cmdMatch = (s.command || '').toLowerCase().includes(this.searchQuery);
        const urlMatch = (s.url || '').toLowerCase().includes(this.searchQuery);
        if (!idMatch && !nameMatch && !descMatch && !cmdMatch && !urlMatch) {
          return false;
        }
      }

      // Category filter
      if (this.activeFilter === 'enabled') {
        return s.enabled !== false;
      } else if (this.activeFilter === 'disabled') {
        return s.enabled === false;
      } else if (this.activeFilter === 'stdio') {
        return (s.transport || 'stdio') === 'stdio';
      } else if (this.activeFilter === 'sse') {
        return ['sse', 'streamable_http', 'http'].includes(s.transport);
      }

      return true;
    });

    this.render();
  }

  clearSearch() {
    const searchInput = document.getElementById('mcp-search-input');
    if (searchInput) searchInput.value = '';
    this.searchQuery = '';
    this.applyFilters();
  }

  setViewMode(mode) {
    this.viewMode = mode;
```

```
    const btnCards = document.getElementById('btn-mcp-view-cards');
    const btnTable = document.getElementById('btn-mcp-view-table');
    const cardsContainer = document.getElementById('mcp-cards-container');
    const tableContainer = document.getElementById('mcp-table-container');

    if (mode === 'cards') {
      btnCards?.classList.add('active');
      btnTable?.classList.remove('active');
      cardsContainer?.classList.remove('d-none');
      tableContainer?.classList.add('d-none');
    } else {
      btnCards?.classList.remove('active');
      btnTable?.classList.add('active');
      cardsContainer?.classList.add('d-none');
      tableContainer?.classList.remove('d-none');
    }
  }

  showLoading(show) {
    const loadingEl = document.getElementById('mcp-loading');
    const cardsEl = document.getElementById('mcp-cards-container');
    const tableEl = document.getElementById('mcp-table-container');
    const emptyEl = document.getElementById('mcp-empty');

    if (show) {
      loadingEl?.classList.remove('d-none');
      cardsEl?.classList.add('d-none');
      tableEl?.classList.add('d-none');
      emptyEl?.classList.add('d-none');
    } else {
      loadingEl?.classList.add('d-none');
      this.setViewMode(this.viewMode);
```

```
193     }
194   }
195
196   render() {
197     const emptyEl = document.getElementById('mcp-empty');
198     const cardsContainer = document.getElementById('mcp-cards-
container');
199     const tableBody = document.getElementById('mcp-table-body'
);
200
201     if (this.filteredServers.length === 0) {
202       if (emptyEl) emptyEl.classList.remove('d-none');
203       if (cardsContainer) cardsContainer.innerHTML = '';
204       if (tableBody) tableBody.innerHTML = '';
205       return;
206     }
207
208     if (emptyEl) emptyEl.classList.add('d-none');
209
210     if (cardsContainer) {
211       cardsContainer.innerHTML = this.filteredServers.map(s =>
 this.renderServerCard(s)).join('');
212     }
213
214     if (tableBody) {
215       tableBody.innerHTML = this.filteredServers.map(s => this
.renderServerTableRow(s)).join('');
216     }
217   }
218
219   renderServerCard(server) {
220     const isEnabled = server.enabled !== false;
221     const isStdio = (server.transport || 'stdio') === 'stdio';
222     const statusBadge = isEnabled
223       ? '<span class="badge bg-success-subtle text-success bor
der border-success-subtle"><i class="bi bi-check-circle-fill">
</i> Активен</span>'
224       : '<span class="badge bg-secondary-subtle text-secondary
```

```
 border border-secondary-subtle"><i class="bi bi-dash-circle">
</i> Отключен</span>';

    const transportBadge = isStdio
      ? '<span class="badge bg-info-subtle text-info border bo
rder-info-subtle"><i class="bi bi-terminal"></i> stdio</span>'
      : `<span class="badge bg-warning-subtle text-warning bor
der border-warning-subtle"><i class="bi bi-broadcast"></i> ${t
his.escapeHtml(server.transport)}</span>`;

    const commandOrUrl = isStdio
      ? `<code>${this.escapeHtml(server.command || '')} ${(ser
ver.args || []).join(' ')}</code>`
      : `<a href="${this.escapeHtml(server.url || '#')}" targe
t="_blank" class="text-info text-truncate d-inline-block" styl
e="max-width: 250px;">${this.escapeHtml(server.url || '')}</a>
`;

    return `
      <div class="col-12 col-md-6 col-xl-4">
        <div class="card bg-dark border-secondary h-100 shadow
-sm d-flex flex-column">
          <div class="card-header border-secondary d-flex alig
n-items-center justify-content-between p-3">
            <div class="d-flex align-items-center gap-2 text-t
runcate">
              <div class="p-2 bg-black rounded border border-s
econdary text-primary">
                <i class="bi ${isStdio ? 'bi-cpu' : 'bi-globe2
'}"></i>
              </div>
              <div class="text-truncate">
                <h6 class="mb-0 text-white text-truncate fw-bo
ld">${this.escapeHtml(server.name || server.id)}</h6>
                <small class="text-muted font-monospace">${thi
s.escapeHtml(server.id)}</small>
              </div>
            </div>
```

```
            <div class="d-flex align-items-center gap-1">
              ${transportBadge}
              ${statusBadge}
            </div>
          </div>
          <div class="card-body p-3 flex-grow-1">
            <p class="card-text small text-secondary mb-3" sty
le="min-height: 38px;">
              ${this.escapeHtml(server.description || 'Описани
e отсутствует.')}
            </p>
            <div class="p-2 bg-black rounded border border-sec
ondary mb-2 small">
              <div class="text-muted small mb-1">${isStdio ? '
Исполняемая команда:' : 'Сетевой эндпоинт:'}</div>
              <div class="text-truncate">${commandOrUrl}</div>
            </div>
          </div>
          <div class="card-footer border-secondary p-2 d-flex
align-items-center justify-content-between gap-1 bg-black bg-o
pacity-25">
            <button class="btn btn-sm btn-outline-info d-flex
align-items-center gap-1" onclick="window.mcpTab?.testServer('
${this.escapeHtml(server.id)}')">
              <i class="bi bi-play-circle"></i> Тест & Tools
            </button>
            <div class="d-flex align-items-center gap-1">
              <button class="btn btn-sm ${isEnabled ? 'btn-out
line-warning' : 'btn-outline-success'}" title="${isEnabled ? '
Отключить' : 'Включить'}" onclick="window.mcpTab?.toggleServer
('${this.escapeHtml(server.id)}')">
                <i class="bi ${isEnabled ? 'bi-pause-fill' : '
bi-play-fill'}"></i>
              </button>
              <button class="btn btn-sm btn-outline-primary" t
itle="Редактировать" onclick="window.mcpTab?.openEditModal('${
this.escapeHtml(server.id)}')">
                <i class="bi bi-pencil-fill"></i>
```

```
              </button>
              <button class="btn btn-sm btn-outline-danger" title="Удалить" onclick="window.mcpTab?.deleteServer('${this.escapeHtml(server.id)}')">
                <i class="bi bi-trash-fill"></i>
              </button>
            </div>
          </div>
        </div>
      </div>
    `;
  }

  renderServerTableRow(server) {
    const isEnabled = server.enabled !== false;
    const isStdio = (server.transport || 'stdio') === 'stdio';
    const statusDot = isEnabled
      ? '<span class="badge bg-success rounded-pill" title="Активен">ON</span>'
      : '<span class="badge bg-secondary rounded-pill" title="Отключен">OFF</span>';

    const transportBadge = isStdio
      ? '<span class="badge bg-info-subtle text-info border border-info-subtle">stdio</span>'
      : `<span class="badge bg-warning-subtle text-warning border border-warning-subtle">${this.escapeHtml(server.transport)}</span>`;

    const cmdOrUrl = isStdio
      ? `<code>${this.escapeHtml(server.command || '')} ${(server.args || []).join(' ')}</code>`
      : `<a href="${this.escapeHtml(server.url || '#')}" target="_blank" class="text-info">${this.escapeHtml(server.url || '')}</a>`;

    return `
      <tr>
```

```
        <td class="text-center">${statusDot}</td>
        <td>
          <div class="fw-bold text-white">${this.escapeHtml(server.name || server.id)}</div>
          <div class="small font-monospace text-muted">${this.escapeHtml(server.id)}</div>
        </td>
        <td>${transportBadge}</td>
        <td class="small text-truncate" style="max-width: 250px;">${cmdOrUrl}</td>
        <td class="small text-muted text-truncate" style="max-width: 200px;">${this.escapeHtml(server.description || '—')}</td>
        <td class="text-end">
          <div class="btn-group btn-group-sm">
            <button class="btn btn-outline-info" title="Тест соединения" onclick="window.mcpTab?.testServer('${this.escapeHtml(server.id)}')">
              <i class="bi bi-play-circle"></i>
            </button>
            <button class="btn ${isEnabled ? 'btn-outline-warning' : 'btn-outline-success'}" title="${isEnabled ? 'Отключить' : 'Включить'}" onclick="window.mcpTab?.toggleServer('${this.escapeHtml(server.id)}')">
              <i class="bi ${isEnabled ? 'bi-pause-fill' : 'bi-play-fill'}"></i>
            </button>
            <button class="btn btn-outline-primary" title="Редактировать" onclick="window.mcpTab?.openEditModal('${this.escapeHtml(server.id)}')">
              <i class="bi bi-pencil-fill"></i>
            </button>
            <button class="btn btn-outline-danger" title="Удалить" onclick="window.mcpTab?.deleteServer('${this.escapeHtml(server.id)}')">
              <i class="bi bi-trash-fill"></i>
            </button>
          </div>
```

```
        </td>
      </tr>
    `;
  }

  onTransportChange() {
    const transport = document.getElementById('mcp-server-transport')?.value || 'stdio';
    const stdioContainer = document.getElementById('mcp-stdio-container');
    const networkContainer = document.getElementById('mcp-network-container');

    if (transport === 'stdio') {
      stdioContainer?.classList.remove('d-none');
      networkContainer?.classList.add('d-none');
    } else {
      stdioContainer?.classList.add('d-none');
      networkContainer?.classList.remove('d-none');
    }
  }

  applyPreset(presetKey) {
    const presets = {
      playwright: {
        id: 'playwright',
        name: 'Playwright Browser MCP',
        description: 'Управление веб-браузером, навигация, клики, скриншоты и парсинг страниц через Playwright.',
        transport: 'stdio',
        command: 'npx',
        args: '@playwright/mcp@latest',
        env: ''
      },
      filesystem: {
        id: 'filesystem',
        name: 'Filesystem MCP',
        description: 'Безопасное чтение, запись и навигация по
```

```
 каталогам и файлам на диске.',
        transport: 'stdio',
        command: 'npx',
        args: '-y @modelcontextprotocol/server-filesystem ./da
ta',
        env: ''
      },
      memory: {
        id: 'memory',
        name: 'Knowledge Graph Memory MCP',
        description: 'Хранилище графа знаний и ассоциативная п
амять для ИИ-агентов.',
        transport: 'stdio',
        command: 'npx',
        args: '-y @modelcontextprotocol/server-memory',
        env: ''
      },
      sqlite: {
        id: 'sqlite',
        name: 'SQLite Database MCP',
        description: 'Выполнение запросов, исследование схемы
и инспекция таблиц SQLite базы данных.',
        transport: 'stdio',
        command: 'uvx',
        args: 'mcp-server-sqlite --db-path ./data/database.sql
ite',
        env: ''
      },
      fetch: {
        id: 'fetch',
        name: 'Fetch / Web Request MCP',
        description: 'Преобразование HTML веб-страниц в Markdo
wn и выполнение HTTP-запросов.',
        transport: 'stdio',
        command: 'uvx',
        args: 'mcp-server-fetch',
        env: ''
      },
```

```
      sse: {
        id: 'remote_mcp',
        name: 'Remote SSE MCP Server',
        description: 'Удаленный MCP-сервер, доступный по протоколу Server-Sent Events (SSE).',
        transport: 'sse',
        url: 'http://localhost:8080/sse'
      }
    };

    const p = presets[presetKey];
    if (!p) return;

    const idEl = document.getElementById('mcp-server-id');
    const nameEl = document.getElementById('mcp-server-name');
    const descEl = document.getElementById('mcp-server-desc');
    const transportEl = document.getElementById('mcp-server-transport');
    const cmdEl = document.getElementById('mcp-server-command');
    const argsEl = document.getElementById('mcp-server-args');
    const urlEl = document.getElementById('mcp-server-url');
    const envEl = document.getElementById('mcp-server-env');

    if (idEl && document.getElementById('mcp-form-mode')?.value === 'create') idEl.value = p.id;
    if (nameEl) nameEl.value = p.name || '';
    if (descEl) descEl.value = p.description || '';
    if (transportEl) transportEl.value = p.transport || 'stdio';
    if (cmdEl) cmdEl.value = p.command || '';
    if (argsEl) argsEl.value = p.args || '';
    if (urlEl) urlEl.value = p.url || '';
    if (envEl) envEl.value = p.env || '';

    this.onTransportChange();
  }

```

```
421   openCreateModal() {
422     const form = document.getElementById('mcp-server-form');
423     if (form) form.reset();
424 
425     const titleText = document.getElementById('mcp-modal-title
    -text');
426     if (titleText) titleText.textContent = 'Добавление нового
    MCP сервера';
427 
428     const modeInput = document.getElementById('mcp-form-mode')
    ;
429     if (modeInput) modeInput.value = 'create';
430 
431     const idInput = document.getElementById('mcp-server-id');
432     if (idInput) {
433       idInput.removeAttribute('readonly');
434       idInput.classList.remove('bg-secondary');
435     }
436 
437     const presetContainer = document.getElementById('mcp-prese
    t-container');
438     if (presetContainer) presetContainer.classList.remove('d-n
    one');
439 
440     const enabledCheckbox = document.getElementById('mcp-serve
    r-enabled');
441     if (enabledCheckbox) enabledCheckbox.checked = true;
442 
443     this.applyPreset('playwright');
444   }
445 
446   openEditModal(serverId) {
447     const server = this.servers.find(s => s.id === serverId);
448     if (!server) return;
449 
450     const titleText = document.getElementById('mcp-modal-title
    -text');
451     if (titleText) titleText.textContent = `Редактирование: ${
```

```
server.name || server.id}`;

    const modeInput = document.getElementById('mcp-form-mode')
;
    if (modeInput) modeInput.value = 'edit';

    const presetContainer = document.getElementById('mcp-prese
t-container');
    if (presetContainer) presetContainer.classList.add('d-none
');

    const idInput = document.getElementById('mcp-server-id');
    if (idInput) {
      idInput.value = server.id;
      idInput.setAttribute('readonly', 'true');
      idInput.classList.add('bg-secondary');
    }

    const nameInput = document.getElementById('mcp-server-name
');
    if (nameInput) nameInput.value = server.name || '';

    const descInput = document.getElementById('mcp-server-desc
');
    if (descInput) descInput.value = server.description || '';

    const transportInput = document.getElementById('mcp-server
-transport');
    if (transportInput) transportInput.value = server.transpor
t || 'stdio';

    const cmdInput = document.getElementById('mcp-server-comma
nd');
    if (cmdInput) cmdInput.value = server.command || '';

    const argsInput = document.getElementById('mcp-server-args
');
    if (argsInput) {
```

```
480       argsInput.value = Array.isArray(server.args) ? server.ar
gs.join(' ') : (server.args || '');
481     }
482
483     const urlInput = document.getElementById('mcp-server-url')
;
484     if (urlInput) urlInput.value = server.url || '';
485
486     const envInput = document.getElementById('mcp-server-env')
;
487     if (envInput) {
488       envInput.value = server.env ? JSON.stringify(server.env,
 null, 2) : '';
489     }
490
491     const enabledCheckbox = document.getElementById('mcp-serve
r-enabled');
492     if (enabledCheckbox) enabledCheckbox.checked = server.enab
led !== false;
493
494     this.onTransportChange();
495
496     const modalEl = document.getElementById('mcpServerModal');
497     if (modalEl && window.bootstrap) {
498       const modal = bootstrap.Modal.getInstance(modalEl) || ne
w bootstrap.Modal(modalEl);
499       modal.show();
500     }
501   }
502
503   async saveServer(event) {
504     if (event) event.preventDefault();
505
506     const mode = document.getElementById('mcp-form-mode')?.val
ue || 'create';
507     const serverId = document.getElementById('mcp-server-id')?
.value.trim();
508     const name = document.getElementById('mcp-server-name')?.v
```

```
alue.trim();
    const description = document.getElementById('mcp-server-desc')?.value.trim();
    const transport = document.getElementById('mcp-server-transport')?.value || 'stdio';
    const enabled = document.getElementById('mcp-server-enabled')?.checked ?? true;
    const command = document.getElementById('mcp-server-command')?.value.trim();
    const argsRaw = document.getElementById('mcp-server-args')?.value.trim();
    const url = document.getElementById('mcp-server-url')?.value.trim();
    const envRaw = document.getElementById('mcp-server-env')?.value.trim();

    if (!serverId) {
      window.showToast?.('Укажите ID сервера', 'warning') || alert('Укажите ID сервера');
      return;
    }

    // Parse args
    let args = [];
    if (argsRaw) {
      if (argsRaw.startsWith('[') && argsRaw.endsWith(']')) {
        try {
          args = JSON.parse(argsRaw);
        } catch {
          args = argsRaw.split(/\s+/).filter(Boolean);
        }
      } else {
        args = argsRaw.split(/\s+/).filter(Boolean);
      }
    }

    // Parse env
    let env = {};
```

```
538     if (envRaw) {
539       try {
540         env = JSON.parse(envRaw);
541       } catch (err) {
542         window.showToast?.('Ошибка формата JSON в переменных о
    кружения: ' + err.message, 'danger') || alert('Ошибка формата
    JSON в переменных окружения: ' + err.message);
543         return;
544       }
545     }
546 
547     const payload = {
548       id: serverId,
549       name: name || serverId,
550       description: description,
551       transport: transport,
552       enabled: enabled,
553       command: command,
554       args: args,
555       url: url,
556       env: env
557     };
558 
559     const saveBtn = document.getElementById('mcp-btn-save-serv
    er');
560     if (saveBtn) {
561       saveBtn.disabled = true;
562       saveBtn.innerHTML = '<span class="spinner-border spinner
    -border-sm me-1"></span> Сохранение...';
563     }
564 
565     try {
566       const endpoint = mode === 'create'
567         ? '/api/admin/mcp/servers'
568         : `/api/admin/mcp/servers/${encodeURIComponent(serverI
    d)}`;
569 
570       const method = mode === 'create' ? 'POST' : 'PUT';
```

```
571
572       const res = await this.apiFetch(endpoint, {
573         method: method,
574         headers: { 'Content-Type': 'application/json' },
575         body: JSON.stringify(payload)
576       });
577
578       if (!res.ok) {
579         const errData = await res.json().catch(() => ({}));
580         throw new Error(errData.detail || `HTTP ${res.status}`
);
581       }
582
583       // Close modal
584       const modalEl = document.getElementById('mcpServerModal'
);
585       if (modalEl && window.bootstrap) {
586         const modal = bootstrap.Modal.getInstance(modalEl);
587         modal?.hide();
588       }
589
590       await this.refresh();
591     } catch (err) {
592       window.showToast?.(`Ошибка сохранения MCP сервера: ${err
.message}`, 'danger') || alert(`Ошибка сохранения MCP сервера:
 ${err.message}`);
593     } finally {
594       if (saveBtn) {
595         saveBtn.disabled = false;
596         saveBtn.innerHTML = 'Сохранить';
597       }
598     }
599   }
600
601   async toggleServer(serverId) {
602     try {
603       const res = await this.apiFetch(`/api/admin/mcp/servers/
${encodeURIComponent(serverId)}/toggle`, {
```

```
604         method: 'POST'
605       });
606       if (!res.ok) {
607         const err = await res.json().catch(() => ({}));
608         throw new Error(err.detail || `HTTP ${res.status}`);
609       }
610       await this.refresh();
611     } catch (err) {
612       window.showToast?.(`Ошибка переключения статуса сервера:
 ${err.message}`, 'danger') || alert(`Ошибка переключения стат
уса сервера: ${err.message}`);
613     }
614   }
615
616   async deleteServer(serverId) {
617     if (!confirm(`Вы уверены, что хотите удалить MCP сервер "$
{serverId}"?`)) {
618       return;
619     }
620
621     try {
622       const res = await this.apiFetch(`/api/admin/mcp/servers/
${encodeURIComponent(serverId)}`, {
623         method: 'DELETE'
624       });
625       if (!res.ok) {
626         const err = await res.json().catch(() => ({}));
627         throw new Error(err.detail || `HTTP ${res.status}`);
628       }
629       await this.refresh();
630     } catch (err) {
631       window.showToast?.(`Ошибка удаления сервера: ${err.messa
ge}`, 'danger') || alert(`Ошибка удаления сервера: ${err.messa
ge}`);
632     }
633   }
634
635   async testServer(serverId) {
```

```
636     const server = this.servers.find(s => s.id === serverId);
637     const serverName = server?.name || serverId;
638 
639     const nameEl = document.getElementById('mcp-test-server-na
    me');
640     if (nameEl) nameEl.textContent = serverName;
641 
642     const statusBox = document.getElementById('mcp-test-status
    -box');
643     const statusText = document.getElementById('mcp-test-statu
    s-text');
644     const spinner = document.getElementById('mcp-test-spinner'
    );
645     const latencyBadge = document.getElementById('mcp-test-lat
    ency-badge');
646     const countBadge = document.getElementById('mcp-test-tools
    -count');
647     const toolsList = document.getElementById('mcp-test-tools-
    list');
648 
649     if (spinner) spinner.classList.remove('d-none');
650     if (statusBox) {
651       statusBox.className = 'p-3 mb-3 rounded border border-in
    fo bg-black text-info d-flex align-items-center justify-conten
    t-between';
652     }
653     if (statusText) statusText.textContent = 'Подключение к пр
    оцессу / эндпоинту и опрос инструментов...';
654     if (latencyBadge) latencyBadge.textContent = '...';
655     if (countBadge) countBadge.textContent = '0';
656     if (toolsList) toolsList.innerHTML = '<div class="text-mut
    ed text-center py-3">Опрос инструментов...</div>';
657 
658     const modalEl = document.getElementById('mcpTestModal');
659     if (modalEl && window.bootstrap) {
660       const modal = bootstrap.Modal.getInstance(modalEl) || ne
    w bootstrap.Modal(modalEl);
661       modal.show();
```

```
662     }
663
664     try {
665       const res = await this.apiFetch(`/api/admin/mcp/servers/
    ${encodeURIComponent(serverId)}/test`, {
666         method: 'POST'
667       });
668       const data = await res.json();
669
670       if (spinner) spinner.classList.add('d-none');
671       if (latencyBadge) latencyBadge.textContent = `${data.lat
    ency_ms || 0} ms`;
672
673       if (data.status === 'ok') {
674         if (statusBox) {
675           statusBox.className = 'p-3 mb-3 rounded border borde
    r-success bg-black text-success d-flex align-items-center just
    ify-content-between';
676         }
677         if (statusText) statusText.innerHTML = `  <strong>Успе
    шно:</strong> ${this.escapeHtml(data.message)}`;
678         if (countBadge) countBadge.textContent = data.tools_co
    unt || (data.tools || []).length;
679         this.renderDiscoveredTools(data.tools || [], toolsList
    );
680       } else {
681         if (statusBox) {
682           statusBox.className = 'p-3 mb-3 rounded border borde
    r-danger bg-black text-danger d-flex align-items-center justif
    y-content-between';
683         }
684         if (statusText) statusText.innerHTML = `  <strong>Ошиб
    ка:</strong> ${this.escapeHtml(data.message || 'Не удалось под
    ключиться к серверу')}`;
685         if (toolsList) {
686           toolsList.innerHTML = `<div class="alert alert-dange
    r bg-dark border-danger text-light small mb-0">${this.escapeHt
    ml(data.message || 'Ошибка соединения')}</div>`;
```

```
        }
      }
    } catch (err) {
      if (spinner) spinner.classList.add('d-none');
      if (statusBox) {
        statusBox.className = 'p-3 mb-3 rounded border border-danger bg-black text-danger d-flex align-items-center justify-content-between';
      }
      if (statusText) statusText.innerHTML = `  <strong>Сетевая ошибка:</strong> ${this.escapeHtml(err.message)}`;
    }
  }

  async testCurrentForm() {
    const serverId = document.getElementById('mcp-server-id')?.value.trim() || 'test_server';
    const transport = document.getElementById('mcp-server-transport')?.value || 'stdio';
    const command = document.getElementById('mcp-server-command')?.value.trim();
    const argsRaw = document.getElementById('mcp-server-args')?.value.trim();
    const url = document.getElementById('mcp-server-url')?.value.trim();
    const envRaw = document.getElementById('mcp-server-env')?.value.trim();

    let args = [];
    if (argsRaw) {
      if (argsRaw.startsWith('[') && argsRaw.endsWith(']')) {
        try { args = JSON.parse(argsRaw); } catch { args = argsRaw.split(/\s+/).filter(Boolean); }
      } else {
        args = argsRaw.split(/\s+/).filter(Boolean);
      }
    }

```

```
715     let env = {};
716     if (envRaw) {
717       try { env = JSON.parse(envRaw); } catch (e) { window.sho
    wToast?.('Неверный формат JSON в env', 'danger') || alert('Нев
    ерный формат JSON в env'); return; }
718     }
719
720     const testBtn = document.getElementById('btn-test-modal-se
    rver');
721     if (testBtn) {
722       testBtn.disabled = true;
723       testBtn.innerHTML = '<span class="spinner-border spinner
    -border-sm me-1"></span> Тест...';
724     }
725
726     try {
727       const res = await this.apiFetch('/api/admin/mcp/test', {
728         method: 'POST',
729         headers: { 'Content-Type': 'application/json' },
730         body: JSON.stringify({
731           id: serverId,
732           transport: transport,
733           command: command,
734           args: args,
735           url: url,
736           env: env
737         })
738       });
739       const data = await res.json();
740       if (data.status === 'ok') {
741         window.showToast?.(`✅ Подключение успешно! (${data.lat
    ency_ms} ms)\nНайдено инструментов: ${data.tools_count}`, 'suc
    cess');
742       } else {
743         window.showToast?.(`❌ Ошибка проверки подключения:\n${
    data.message}`, 'danger');
744       }
745     } catch (err) {
```

```
      window.showToast?.(`❌ Ошибка запроса: ${err.message}`, 'danger');
    } finally {
      if (testBtn) {
        testBtn.disabled = false;
        testBtn.innerHTML = '<i class="bi bi-play-circle"></i> Проверить подключение';
      }
    }
  }

  async openAllToolsModal() {
    const modalEl = document.getElementById('mcpAllToolsModal');
    if (modalEl && window.bootstrap) {
      const modal = bootstrap.Modal.getInstance(modalEl) || new bootstrap.Modal(modalEl);
      modal.show();
    }

    const loadingEl = document.getElementById('mcp-all-tools-loading');
    const emptyEl = document.getElementById('mcp-all-tools-empty');
    const container = document.getElementById('mcp-all-tools-container');

    loadingEl?.classList.remove('d-none');
    emptyEl?.classList.add('d-none');
    container?.classList.add('d-none');

    try {
      const res = await this.apiFetch('/api/admin/mcp/tools');
      const data = await res.json();
      loadingEl?.classList.add('d-none');

      const tools = data.tools || [];
      if (tools.length === 0) {
```

```
        emptyEl?.classList.remove('d-none');
        return;
      }

      container?.classList.remove('d-none');
      if (container) {
        container.innerHTML = tools.map(t => `
          <div class="col-12 col-lg-6">
            <div class="p-3 bg-black rounded border border-secondary h-100 d-flex flex-column">
              <div class="d-flex align-items-center justify-content-between mb-2">
                <strong class="text-info font-monospace d-flex align-items-center gap-1">
                  <i class="bi bi-wrench"></i> ${this.escapeHtml(t.name)}
                </strong>
                <span class="badge bg-primary-subtle text-primary border border-primary-subtle small">Tool</span>
              </div>
              <p class="small text-muted mb-2 flex-grow-1">${this.escapeHtml(t.description || 'Нет описания.')}</p>
              ${t.args_schema ? `
                <div class="mt-auto">
                  <details class="small">
                    <summary class="text-secondary cursor-pointer">Схема аргументов</summary>
                    <pre class="bg-dark text-light p-2 rounded mt-1 mb-0 font-monospace small" style="max-height: 150px; overflow-y: auto;">${this.escapeHtml(typeof t.args_schema === 'object' ? JSON.stringify(t.args_schema, null, 2) : String(t.args_schema))}</pre>
                  </details>
                </div>
              ` : ''}
            </div>
          </div>
        `).join('');
```

```
      }
    } catch (err) {
      loadingEl?.classList.add('d-none');
      if (container) {
        container.classList.remove('d-none');
        container.innerHTML = `<div class="col-12"><div class="alert alert-danger">${this.escapeHtml(err.message)}</div></div>`;
      }
    }
  }

  renderDiscoveredTools(tools, container) {
    if (!tools || tools.length === 0) {
      container.innerHTML = '<div class="text-muted text-center py-3">Сервер не вернул доступных инструментов.</div>';
      return;
    }

    container.innerHTML = tools.map((t, idx) => `
      <div class="p-3 bg-dark rounded border border-secondary">
        <div class="d-flex align-items-center justify-content-between mb-1">
          <strong class="text-warning font-monospace d-flex align-items-center gap-1">
            <i class="bi bi-wrench-adjustable"></i> ${this.escapeHtml(t.name)}
          </strong>
          <span class="badge bg-dark border border-secondary text-muted">#${idx + 1}</span>
        </div>
        <div class="small text-muted mb-2">${this.escapeHtml(t.description || 'Описание не указано')}</div>
        ${t.args_schema ? `
          <details class="small">
            <summary class="text-info cursor-pointer">Параметры и схема вызова</summary>
```

```
            <pre class="bg-black text-white p-2 rounded mt-1 mb-0 font-monospace small" style="max-height: 180px; overflow-y: auto;">${this.escapeHtml(typeof t.args_schema === 'object' ? JSON.stringify(t.args_schema, null, 2) : String(t.args_schema))}</pre>
          </details>
        ` : ''}
      </div>
    `).join('');
  }

  renderError(msg) {
    const container = document.getElementById('mcp-cards-container');
    if (container) {
      container.innerHTML = `
        <div class="col-12">
          <div class="alert alert-danger bg-dark border-danger text-light d-flex align-items-center justify-content-between">
            <div><i class="bi bi-exclamation-triangle-fill text-danger me-2"></i> ${this.escapeHtml(msg)}</div>
            <button class="btn btn-sm btn-outline-light" onclick="window.mcpTab?.refresh()">Повторить</button>
          </div>
        </div>
      `;
    }
  }

  escapeHtml(str) {
    if (!str) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
```

```
  }
}

// Global initialization hooks
window.McpTabManager = McpTabManager;
if (!window.mcpTab) {
  window.mcpTab = new McpTabManager();
}
window.initMcpTab = function() {
  if (window.mcpTab) {
    window.mcpTab.init();
  }
};

if (document.getElementById('mcp-tab-root')) {
  window.mcpTab.init();
}

```

app.js

```
/**
 * ============================================================
 * Process Name: Windows Messenger - App Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля app.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/messenger/app.js?v=20261001_v1" type="module"></script>
 *
 * File: app.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/messenger
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * AI Windows Admin Messenger & WebRTC Conference Client
 */

const CONFIG = {
  apiBase: window.location.origin,
  wsProtocol: window.location.protocol === 'https:' ? 'wss:' : 'ws:',
  iceServers: [{ urls: 'stun:stun.l.google.com:19302' }, { urls: 'stun:stun1.l.google.com:19302' }],
};

```

```
let currentUser = null;
let activeRoomId = null;
let activeRoomData = null;
let ws = null;
let rooms = [];
let mediaRecorder = null;
let recordedAudioChunks = [];
let isRecordingVoice = false;

// WebRTC Call State
let peerConnection = null;
let localStream = null;
let remoteStream = null;
let isInCall = false;

// DOM Elements
const chatListEl = document.getElementById('chatList');
const messagesContainerEl = document.getElementById('messagesC
ontainer');
const messageInputEl = document.getElementById('messageInput')
;
const btnSendMessageEl = document.getElementById('btnSendMessa
ge');
const btnRecordVoiceEl = document.getElementById('btnRecordVoi
ce');
const btnAttachFileEl = document.getElementById('btnAttachFile
');
const fileAttachmentInputEl = document.getElementById('fileAtt
achmentInput');
const typingIndicatorEl = document.getElementById('typingIndic
ator');
const chatHeaderEl = document.getElementById('chatHeader');
const chatInputBarEl = document.getElementById('chatInputBar')
;
const activeChatTitleEl = document.getElementById('activeChatT
itle');
const activeChatSubtitleEl = document.getElementById('activeCh
atSubtitle');
```

```
const newChatModalEl = document.getElementById('newChatModal');
const videoCallOverlayEl = document.getElementById('videoCallOverlay');

// Initialize
window.addEventListener('DOMContentLoaded', async () => {
  await initUserProfile();
  initWebSocket();
  await loadRooms();
  setupEventListeners();
  registerServiceWorker();
});

async function initUserProfile() {
  try {
    const res = await fetch(`${CONFIG.apiBase}/api/messenger/users/me`, { credentials: 'include' });
    const data = await res.json();
    if (data.status === 'success') {
      currentUser = data.user;
      document.getElementById('myName').innerText = currentUser.display_name || currentUser.username;
      document.getElementById('myAvatarText').innerText = (currentUser.display_name || 'U')[0].toUpperCase();
    }
  } catch (e) {
    console.error('Failed to load user profile:', e);
  }
}

function initWebSocket() {
  if (!currentUser) return;
  const wsUrl = `${CONFIG.wsProtocol}//${window.location.host}/api/messenger/ws/${currentUser.id}`;
  ws = new WebSocket(wsUrl);

  ws.onopen = () => console.log('Messenger WebSocket Connected
```

```
');
  ws.onmessage = (event) => handleIncomingWsPacket(JSON.parse(event.data));
  ws.onclose = () => {
    console.warn('WebSocket closed. Reconnecting in 3s...');
    setTimeout(initWebSocket, 3000);
  };
}

function handleIncomingWsPacket(packet) {
  const { type, message, room_id, user_id, bot_name, is_online } = packet;

  if (type === 'new_message') {
    if (activeRoomId === message.room_id) {
      appendMessageToChat(message);
      scrollToBottom();
      // Send read receipt
      if (ws && ws.readyState === WebSocket.OPEN && message.sender_id !== currentUser.id) {
        ws.send(JSON.stringify({ type: 'message_read', room_id: activeRoomId, message_id: message.id }));
      }
    }
    updateRoomLastMessage(message);
  } else if (type === 'typing_start') {
    if (activeRoomId === room_id && user_id !== currentUser.id) {
      typingIndicatorEl.style.display = 'block';
      typingIndicatorEl.innerText = `${bot_name || 'Someone'} is typing...`;
    }
  } else if (type === 'typing_stop') {
    if (activeRoomId === room_id) {
      typingIndicatorEl.style.display = 'none';
    }
  } else if (type === 'presence_update') {
    updateUserPresenceUI(user_id, is_online);
```

```
  } else if (type.startsWith('webrtc_')) {
    handleWebRTCSignalPacket(packet);
  }
}

async function loadRooms() {
  try {
    const res = await fetch(`${CONFIG.apiBase}/api/messenger/rooms`, { credentials: 'include' });
    const data = await res.json();
    if (data.status === 'success') {
      rooms = data.rooms;
      renderRoomsList();
    }
  } catch (e) {
    console.error('Error loading rooms:', e);
  }
}

function renderRoomsList() {
  chatListEl.innerHTML = '';
  rooms.forEach((r) => {
    const item = document.createElement('div');
    item.className = `chat-item ${activeRoomId === r.id ? 'active' : ''}`;
    item.onclick = () => selectRoom(r);

    const initial = (r.title || '#')[0].toUpperCase();
    const lastContent = r.last_message ? (r.last_message.content || '[Attachment]') : 'No messages yet';
    const unreadHtml = r.unread_count > 0 ? `<span class="unread-badge">${r.unread_count}</span>` : '';
    const onlineHtml = r.is_online ? '<div class="status-dot online"></div>' : '';

    item.innerHTML = `
      <div class="avatar">
        <span>${initial}</span>
```

```
        ${onlineHtml}
      </div>
      <div class="chat-info">
        <div class="chat-title-row">
          <span class="chat-title">${escapeHtml(r.title)}</spa
n>
          <span class="chat-time">${r.last_message ? formatTim
e(r.last_message.created_at) : ''}</span>
        </div>
        <div class="chat-snippet-row">
          <span class="chat-snippet">${escapeHtml(lastContent)
}</span>
          ${unreadHtml}
        </div>
      </div>
    `;
    chatListEl.appendChild(item);
  });
}

async function selectRoom(room) {
  activeRoomId = room.id;
  activeRoomData = room;
  renderRoomsList();

  chatHeaderEl.style.display = 'flex';
  chatInputBarEl.style.display = 'flex';
  document.getElementById('emptyPlaceholder').style.display =
'none';

  activeChatTitleEl.innerText = room.title;
  activeChatSubtitleEl.innerText = room.room_type === 'direct'
 ? (room.is_online ? 'Online' : 'Offline') : `${room.members_c
ount} members`;
  document.getElementById('activeChatAvatarText').innerText =
(room.title || '#')[0].toUpperCase();

  await loadRoomMessages(room.id);
```

```
}

async function loadRoomMessages(roomId) {
  messagesContainerEl.innerHTML = '';
  try {
    const res = await fetch(`${CONFIG.apiBase}/api/messenger/rooms/${roomId}/messages`, { credentials: 'include' });
    const data = await res.json();
    if (data.status === 'success') {
      data.messages.forEach(appendMessageToChat);
      scrollToBottom();
    }
  } catch (e) {
    console.error('Error loading messages:', e);
  }
}

function appendMessageToChat(msg) {
  const isOut = msg.sender_id === currentUser.id;
  const bubble = document.createElement('div');
  bubble.className = `message-bubble ${isOut ? 'outgoing' : 'incoming'}`;

  let senderHeader = '';
  if (!isOut && activeRoomData && activeRoomData.room_type !== 'direct') {
    senderHeader = `<div class="message-sender">${escapeHtml(msg.sender_name || 'Member')}</div>`;
  }

  let attachmentsHtml = '';
  if (msg.attachments && msg.attachments.length > 0) {
    msg.attachments.forEach((att) => {
      if (att.file_type === 'audio') {
        attachmentsHtml += `<audio class="audio-player-custom" controls src="${att.file_path}"></audio>`;
      } else if (att.file_type === 'image') {
        attachmentsHtml += `<img src="${att.file_path}" style=
```

```
"max-width:100%; border-radius:10px; margin-top:6px;" />`;
      } else {
        attachmentsHtml += `<div style="margin-top:6px;"><a hr
ef="${att.file_path}" download style="color:#60a5fa;">  ${esc
apeHtml(att.file_name)}</a></div>`;
      }
    });
  }

  bubble.innerHTML = `
    ${senderHeader}
    <div>${escapeHtml(msg.content || '')}</div>
    ${attachmentsHtml}
    <div class="message-meta">
      <span>${formatTime(msg.created_at)}</span>
      ${isOut ? '<span>  </span>' : ''}
    </div>
  `;
  messagesContainerEl.appendChild(bubble);
}

function updateRoomLastMessage(msg) {
  const r = rooms.find((x) => x.id === msg.room_id);
  if (r) {
    r.last_message = msg;
    renderRoomsList();
  }
}

function updateUserPresenceUI(userId, isOnline) {
  rooms.forEach((r) => {
    if (r.room_type === 'direct') {
      r.is_online = isOnline;
    }
  });
  renderRoomsList();
}

```

```
// Send Message
async function sendMessage() {
  const content = messageInputEl.value.trim();
  if (!content || !activeRoomId) return;

  if (ws && ws.readyState === WebSocket.OPEN) {
    ws.send(JSON.stringify({
      type: 'send_message',
      room_id: activeRoomId,
      content: content,
      message_type: 'text'
    }));
  } else {
    // REST Fallback
    await fetch(`${CONFIG.apiBase}/api/messenger/rooms/${activ
eRoomId}/messages`, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      credentials: 'include',
      body: JSON.stringify({ content: content, message_type: '
text' }),
    });
  }

  messageInputEl.value = '';
}

// Voice Note Recording
async function toggleVoiceRecording() {
  if (isRecordingVoice) {
    mediaRecorder.stop();
    isRecordingVoice = false;
    btnRecordVoiceEl.style.background = 'var(--bg-secondary)';
  } else {
    try {
      const stream = await navigator.mediaDevices.getUserMedia
({ audio: true });
      mediaRecorder = new MediaRecorder(stream);
```

```
289       recordedAudioChunks = [];
290
291       mediaRecorder.ondataavailable = (e) => recordedAudioChun
ks.push(e.data);
292       mediaRecorder.onstop = async () => {
293         const blob = new Blob(recordedAudioChunks, { type: 'au
dio/webm' });
294         const formData = new FormData();
295         formData.append('file', blob, `voice_${Date.now()}.web
m`);
296         formData.append('duration_sec', '3.0');
297
298         const res = await fetch(`${CONFIG.apiBase}/api/messeng
er/upload`, {
299           method: 'POST',
300           credentials: 'include',
301           body: formData,
302         });
303         const data = await res.json();
304         if (data.status === 'success') {
305           ws.send(JSON.stringify({
306             type: 'send_message',
307             room_id: activeRoomId,
308             content: '',
309             message_type: 'audio',
310             attachment_ids: [data.attachment_id],
311           }));
312         }
313       };
314
315       mediaRecorder.start();
316       isRecordingVoice = true;
317       btnRecordVoiceEl.style.background = 'var(--danger-color)
';
318     } catch (e) {
319       window.showToast?.('Для записи голосовых сообщений требу
ется доступ к микрофону', 'warning') || alert('Для записи голо
совых сообщений требуется доступ к микрофону');
```

```
    }
  }
}

// WebRTC Conference / Call Handler
async function startWebRTCCall(callType = 'video') {
  if (!activeRoomId) return;
  videoCallOverlayEl.style.display = 'flex';
  isInCall = true;

  try {
    localStream = await navigator.mediaDevices.getUserMedia({
      video: callType === 'video',
      audio: true,
    });
    document.getElementById('localVideo').srcObject = localStr
eam;

    initPeerConnection();
    localStream.getTracks().forEach((track) => peerConnection.
addTrack(track, localStream));

    const offer = await peerConnection.createOffer();
    await peerConnection.setLocalDescription(offer);

    ws.send(JSON.stringify({
      type: 'webrtc_offer',
      data: {
        action: 'offer',
        room_id: activeRoomId,
        sdp: offer.sdp,
        call_type: callType,
      },
    }));
  } catch (e) {
    console.error('Call initialization failed:', e);
  }
}
```

```
356
357 function initPeerConnection() {
358   peerConnection = new RTCPeerConnection({ iceServers: CONFIG.
    iceServers });
359
360   peerConnection.onicecandidate = (event) => {
361     if (event.candidate && ws) {
362       ws.send(JSON.stringify({
363         type: 'webrtc_ice_candidate',
364         data: {
365           action: 'ice_candidate',
366           room_id: activeRoomId,
367           candidate: event.candidate,
368         },
369       }));
370     }
371   };
372
373   peerConnection.ontrack = (event) => {
374     remoteStream = event.streams[0];
375     const remoteVid = document.getElementById('remoteVideo');
376     remoteVid.srcObject = remoteStream;
377     document.getElementById('remoteVideoTile').style.display =
     'flex';
378   };
379 }
380
381 async function handleWebRTCSignalPacket(packet) {
382   const { type, sdp, candidate, sender_id, room_id } = packet;
383
384   if (type === 'webrtc_offer') {
385     videoCallOverlayEl.style.display = 'flex';
386     isInCall = true;
387     initPeerConnection();
388
389     localStream = await navigator.mediaDevices.getUserMedia({
    video: true, audio: true });
390     document.getElementById('localVideo').srcObject = localStr
```

```
eam;
391     localStream.getTracks().forEach((track) => peerConnection.
addTrack(track, localStream));
392
393     await peerConnection.setRemoteDescription(new RTCSessionDe
scription({ type: 'offer', sdp }));
394     const answer = await peerConnection.createAnswer();
395     await peerConnection.setLocalDescription(answer);
396
397     ws.send(JSON.stringify({
398       type: 'webrtc_answer',
399       data: {
400         action: 'answer',
401         room_id: room_id,
402         target_id: sender_id,
403         sdp: answer.sdp,
404       },
405     }));
406   } else if (type === 'webrtc_answer') {
407     if (peerConnection) {
408       await peerConnection.setRemoteDescription(new RTCSession
Description({ type: 'answer', sdp }));
409     }
410   } else if (type === 'webrtc_ice_candidate') {
411     if (peerConnection && candidate) {
412       await peerConnection.addIceCandidate(new RTCIceCandidate
(candidate));
413     }
414   }
415 }
416
417 function endCall() {
418   if (localStream) localStream.getTracks().forEach((t) => t.st
op());
419   if (peerConnection) peerConnection.close();
420   peerConnection = null;
421   videoCallOverlayEl.style.display = 'none';
422   isInCall = false;
```

```
423
424   if (ws && activeRoomId) {
425     ws.send(JSON.stringify({
426       type: 'webrtc_leave_call',
427       data: { action: 'leave_call', room_id: activeRoomId },
428     }));
429   }
430 }
431
432 // Event Listeners
433 function setupEventListeners() {
434   btnSendMessageEl.onclick = sendMessage;
435   messageInputEl.onkeydown = (e) => {
436     if (e.key === 'Enter') sendMessage();
437   };
438   btnRecordVoiceEl.onclick = toggleVoiceRecording;
439   btnAttachFileEl.onclick = () => fileAttachmentInputEl.click(
);
440
441   fileAttachmentInputEl.onchange = async () => {
442     const file = fileAttachmentInputEl.files[0];
443     if (!file || !activeRoomId) return;
444     const formData = new FormData();
445     formData.append('file', file);
446     const res = await fetch(`${CONFIG.apiBase}/api/messenger/u
pload`, {
447       method: 'POST',
448       credentials: 'include',
449       body: formData,
450     });
451     const data = await res.json();
452     if (data.status === 'success') {
453       ws.send(JSON.stringify({
454         type: 'send_message',
455         room_id: activeRoomId,
456         content: '',
457         message_type: data.file_type,
458         attachment_ids: [data.attachment_id],
```

```
459       }));
460     }
461   };
462 
463   document.getElementById('btnStartVideoCall').onclick = () =>
 startWebRTCCall('video');
464   document.getElementById('btnStartAudioCall').onclick = () =>
 startWebRTCCall('audio');
465   document.getElementById('btnEndCall').onclick = endCall;
466 
467   // New Chat Modal
468   document.getElementById('btnNewChat').onclick = () => { newC
hatModalEl.style.display = 'flex'; };
469   document.getElementById('modalBtnCancel').onclick = () => {
newChatModalEl.style.display = 'none'; };
470   document.getElementById('modalRoomType').onchange = (e) => {
471     document.getElementById('modalTitleGroup').style.display =
 e.target.value !== 'direct' ? 'block' : 'none';
472   };
473 
474   document.getElementById('modalBtnCreate').onclick = async ()
 => {
475     const roomType = document.getElementById('modalRoomType').
value;
476     const title = document.getElementById('modalRoomTitle').va
lue;
477     const res = await fetch(`${CONFIG.apiBase}/api/messenger/r
ooms`, {
478       method: 'POST',
479       headers: { 'Content-Type': 'application/json' },
480       credentials: 'include',
481       body: JSON.stringify({ room_type: roomType, title: title
 }),
482     });
483     const data = await res.json();
484     if (data.status === 'success') {
485       newChatModalEl.style.display = 'none';
486       await loadRooms();
```

```
    }
  };
}

function formatTime(isoStr) {
  if (!isoStr) return '';
  const d = new Date(isoStr);
  return d.toLocaleTimeString([], { hour: '2-digit', minute: '
2-digit' });
}

function scrollToBottom() {
  messagesContainerEl.scrollTop = messagesContainerEl.scrollHe
ight;
}

function escapeHtml(text) {
  const div = document.createElement('div');
  div.innerText = text;
  return div.innerHTML;
}

function registerServiceWorker() {
  if ('serviceWorker' in navigator) {
    navigator.serviceWorker.register('sw.js').catch(console.er
ror);
  }
}

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/messenger/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/messenger/index.html" c
lass="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/messenger
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-s
cale=1.0, maximum-scale=1.0, user-scalable=no" />
  <title>AI Windows Admin Messenger & Rooms</title>
  <link rel="stylesheet" href="style.css" />
  <link rel="manifest" href="manifest.json" />
```

```
  <meta name="theme-color" content="#0f172a" />
</head>
<body>
  <div class="messenger-app">
    <!-- Sidebar -->
    <aside class="sidebar" id="sidebar">
      <div class="sidebar-header">
        <div class="user-profile-badge">
          <div class="avatar" id="myAvatar">
            <span id="myAvatarText">U</span>
            <div class="status-dot online"></div>
          </div>
          <div>
            <div style="font-weight: 600; font-size: 15px;" id
="myName">Loading...</div>
            <div style="font-size: 12px; color: var(--text-sec
ondary);" id="myStatus">Online</div>
          </div>
        </div>
        <button class="btn-icon" id="btnNewChat" title="New Ch
at or Room">＋</button>
      </div>

      <div class="search-bar-container">
        <input type="text" class="search-input" id="searchChat
sInput" placeholder="Search chats, contacts, @bots..." />
      </div>

      <div class="chat-list" id="chatList">
        <!-- Chat items rendered dynamically -->
      </div>
    </aside>

    <!-- Main Chat Area -->
    <main class="chat-main" id="chatMain">
      <!-- Chat Header -->
      <div class="chat-header" id="chatHeader" style="display:
 none;">
```

```
        <div style="display: flex; align-items: center; gap: 12px;">
          <div class="avatar" id="activeChatAvatar">
            <span id="activeChatAvatarText">#</span>
          </div>
          <div>
            <div style="font-weight: 600; font-size: 16px;" id="activeChatTitle">Select a chat</div>
            <div style="font-size: 12px; color: var(--text-secondary);" id="activeChatSubtitle">...</div>
          </div>
        </div>
        <div class="chat-header-actions">
          <button class="btn-icon" id="btnStartAudioCall" title="Audio Call">  </button>
          <button class="btn-icon" id="btnStartVideoCall" title="Video Meeting">  </button>
        </div>
      </div>

      <!-- Messages Stream -->
      <div class="messages-container" id="messagesContainer">
        <div style="text-align: center; color: var(--text-secondary); margin-top: 40px;" id="emptyPlaceholder">
          Select or create a conversation to start messaging.
        </div>
      </div>

      <!-- Typing indicator -->
      <div class="typing-indicator" id="typingIndicator" style="display: none;"></div>

      <!-- Input Bar -->
      <div class="chat-input-bar" id="chatInputBar" style="display: none;">
        <input type="file" id="fileAttachmentInput" style="display: none;" multiple />
        <button class="btn-icon" id="btnAttachFile" title="Att
```

```
ach file or photo">📎</button>

        <input type="text" class="message-input" id="messageInput" placeholder="Write a message... (Type @ai for bot)" autocomplete="off" />

        <button class="btn-icon" id="btnRecordVoice" title="Record voice note">🎤</button>
        <button class="btn-icon" id="btnSendMessage" title="Send message" style="background: var(--accent-color);">➤</button>
      </div>

      <!-- WebRTC Meeting Room Overlay -->
      <div class="video-call-overlay" id="videoCallOverlay" style="display: none;">
        <div class="video-grid" id="videoGrid">
          <div class="video-tile" id="localVideoTile">
            <video id="localVideo" autoplay playsinline muted></video>
            <span class="tile-label">You</span>
          </div>
          <div class="video-tile" id="remoteVideoTile" style="display: none;">
            <video id="remoteVideo" autoplay playsinline></video>
            <span class="tile-label" id="remoteTileLabel">Participant</span>
          </div>
        </div>

        <div class="call-controls">
          <button class="btn-icon" id="btnToggleMic" title="Mute/Unmute Mic">🎤</button>
          <button class="btn-icon" id="btnToggleCam" title="Toggle Camera">📷</button>
          <button class="btn-icon" id="btnShareScreen" title="Share Screen">🖥</button>
```

```
          <button class="btn-icon btn-danger" id="btnEndCall"
title="Hang Up">□ </button>
        </div>
      </div>
    </main>
  </div>

  <!-- New Chat Modal -->
  <div class="modal-overlay" id="newChatModal" style="display:
 none;">
    <div class="modal-card">
      <h3>Create New Conversation</h3>
      <div class="form-group">
        <label>Room Type</label>
        <select id="modalRoomType">
          <option value="direct">Direct Message (1-on-1)</opti
on>
          <option value="group">Group Chat</option>
          <option value="meeting">Video Meeting Room</option>
        </select>
      </div>
      <div class="form-group" id="modalTitleGroup" style="disp
lay: none;">
        <label>Room Title</label>
        <input type="text" id="modalRoomTitle" placeholder="e.
g. Project Discussion" />
      </div>
      <div class="form-group">
        <label>Search User / Member</label>
        <input type="text" id="modalUserSearch" placeholder="T
ype username or name..." />
        <div id="modalUserResults" style="max-height: 120px; o
verflow-y: auto; margin-top: 8px;"></div>
      </div>
      <div style="display: flex; justify-content: flex-end; ga
p: 10px; margin-top: 20px;">
        <button class="btn-icon" style="width: auto; padding:
8px 16px; border-radius: 8px;" id="modalBtnCancel">Cancel</but
```

```
ton>
147         <button class="btn-icon" style="width: auto; padding:
8px 16px; border-radius: 8px; background: var(--accent-color);
" id="modalBtnCreate">Create</button>
148       </div>
149     </div>
150   </div>
151
152   <script src="app.js"></script>
153 </body>
154 </html>
155
```

style.css

```
/**
 * ============================================================
 * Process Name: Windows Web Styles - Style Stylesheet
 * ============================================================
 * Description:
 *   Таблица стилей стиля и оформления интерфейса (style).
 *
 * Usage Examples:
 *   HTML Link Tag:
 *     <link rel="stylesheet" href="/windows/api/webgui/messenger/style.css?v=20261001_v1">
 *
 * File: style.css
 * Project: ai-windows-admin
 * Package: windows/api/webgui/messenger
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/* ═══════════════════════════════════════════════════════════
   messenger/style.css — Дочерние стили Messenger (наследует main.css)
   Версия: 20260917_refactor_v1
   ═══════════════════════════════════════════════════════════ */

/* Messenger наследует переменные из main.css.
   Локальные псевдонимы для удобства внутри компонента: */
.messenger-app {
  --msg-outgoing: var(--chat-user-bg);
```

```
  --msg-incoming: var(--chat-bot-bg);
  --msg-incoming-text: var(--chat-bot-text);
  --online-color:  #10b981;
  --offline-color: #6b7280;
  --danger-color:  #ef4444;
}

/* ── Основной контейнер ──────────────────────────────────────── */
.messenger-app {
  display: flex;
  height: 100vh;
  width: 100%;
  background: var(--bg-color);
  font-size: 0.85rem;
}

/* ── Боковая панель ──────────────────────────────────────────── */
.sidebar {
  width: 320px;
  min-width: 260px;
  background: var(--surface-1);
  border-right: 1px solid var(--border-color);
  display: flex;
  flex-direction: column;
  z-index: 10;
}

.sidebar-header {
  padding: 10px 12px;
  display: flex;
  align-items: center;
  justify-content: space-between;
  border-bottom: 1px solid var(--border-color);
}

/* ── Профиль пользователя ────────────────────────────────────
```

```
   ──────────────── */
66 .user-profile-badge {
67   display: flex;
68   align-items: center;
69   gap: 10px;
70 }
71
72 .avatar {
73   width: 36px;
74   height: 36px;
75   border-radius: 50%;
76   background: var(--border-color);
77   display: flex;
78   align-items: center;
79   justify-content: center;
80   font-weight: 700;
81   font-size: 0.85rem;
82   color: var(--text-color);
83   position: relative;
84   object-fit: cover;
85   flex-shrink: 0;
86 }
87
88 .avatar .status-dot {
89   position: absolute;
90   bottom: 0;
91   right: 0;
92   width: 10px;
93   height: 10px;
94   border-radius: 50%;
95   border: 2px solid var(--surface-1);
96 }
97
98 .status-dot.online  { background-color: var(--online-color); }
99 .status-dot.offline { background-color: var(--offline-color);
   }
100
101 /* ── Строка поиска ──────────────────────────────────────
```

```
   ──────────────── */
.search-bar-container {
  padding: 8px 10px;
  border-bottom: 1px solid var(--border-color);
}

.search-input {
  width: 100%;
  padding: 6px 12px;
  background: var(--surface-2);
  border: 1px solid var(--border-color);
  border-radius: 16px;
  color: var(--text-color);
  outline: none;
  font-size: 0.82rem;
  transition: border-color 0.15s;
}

.search-input::placeholder { color: var(--text-muted); }
.search-input:focus { border-color: var(--nav-active); }

/* ── Список чатов ──────────────────────────────────────
   ──────────────── */
.chat-list {
  flex: 1;
  overflow-y: auto;
}

.chat-item {
  display: flex;
  align-items: center;
  gap: 10px;
  padding: 9px 12px;
  cursor: pointer;
  border-bottom: 1px solid var(--border-subtle);
  transition: background 0.12s ease;
}

```

```
.chat-item:hover {
  background: var(--nav-hover-bg);
}

.chat-item.active {
  background: rgba(37, 99, 235, 0.08);
  border-left: 3px solid var(--nav-active);
}

[data-theme="dark"] .chat-item.active {
  background: rgba(88, 166, 255, 0.08);
}

.chat-info { flex: 1; min-width: 0; }

.chat-title-row {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 2px;
}

.chat-title {
  font-weight: 600;
  font-size: 0.85rem;
  white-space: nowrap;
  overflow: hidden;
  text-overflow: ellipsis;
  color: var(--text-color);
}

.chat-time {
  font-size: 0.72rem;
  color: var(--text-muted);
  flex-shrink: 0;
}

.chat-snippet-row {
```

```
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.chat-snippet {
  font-size: 0.78rem;
  color: var(--text-muted);
  white-space: nowrap;
  overflow: hidden;
  text-overflow: ellipsis;
}

.unread-badge {
  background: var(--nav-active);
  color: #fff;
  font-size: 0.68rem;
  font-weight: 700;
  padding: 1px 6px;
  border-radius: 10px;
  flex-shrink: 0;
}

/* ── Главная область чата ──────────────────────────────────── */
.chat-main {
  flex: 1;
  display: flex;
  flex-direction: column;
  background: var(--bg-color);
  position: relative;
}

.chat-header {
  padding: 8px 16px;
  background: var(--surface-1);
  border-bottom: 1px solid var(--border-color);
  display: flex;
```

```
  justify-content: space-between;
  align-items: center;
  z-index: 5;
}

.chat-header-actions {
  display: flex;
  gap: 8px;
}

.btn-icon {
  background: var(--surface-2);
  border: 1px solid var(--border-color);
  color: var(--text-color);
  width: 32px;
  height: 32px;
  border-radius: 50%;
  display: flex;
  align-items: center;
  justify-content: center;
  cursor: pointer;
  font-size: 0.85rem;
  transition: background 0.15s, color 0.15s;
}

.btn-icon:hover {
  background: var(--nav-active);
  color: #fff;
  border-color: var(--nav-active);
}

.btn-danger { background: var(--danger-color) !important; colo
r: #fff !important; }

/* ── Контейнер сообщений ──────────────────────────────────
──────────────── */
.messages-container {
  flex: 1;
```

```
  padding: 12px 16px;
  overflow-y: auto;
  display: flex;
  flex-direction: column;
  gap: 8px;
}

.message-bubble {
  max-width: 70%;
  padding: 8px 12px;
  border-radius: 14px;
  position: relative;
  line-height: 1.45;
  font-size: 1.15rem;
  word-break: break-word;
}

.message-bubble.incoming {
  align-self: flex-start;
  background: var(--msg-incoming);
  color: var(--msg-incoming-text);
  border-bottom-left-radius: 4px;
}

.message-bubble.outgoing {
  align-self: flex-end;
  background: var(--msg-outgoing);
  color: var(--chat-user-text);
  border-bottom-right-radius: 4px;
}

.message-meta {
  display: flex;
  align-items: center;
  gap: 4px;
  margin-top: 4px;
  font-size: 0.72rem;
  opacity: 0.75;
```

```
}

.message-bubble.outgoing .message-meta {
  justify-content: flex-end;
}

.message-status {
  font-size: 0.75rem;
}

.audio-preview {
  margin-top: 6px;
  width: 100%;
  max-width: 240px;
}

/* ── Индикатор набора текста ──────────────────────────────── */
.typing-indicator {
  display: flex;
  align-items: center;
  gap: 6px;
  padding: 4px 16px;
  font-size: 0.8rem;
  color: var(--text-muted);
  font-style: italic;
}

/* ── Поле ввода ──────────────────────────────────────────── */
.chat-input-bar {
  padding: 8px 12px;
  background: var(--surface-1);
  border-top: 1px solid var(--border-color);
  display: flex;
  align-items: center;
  gap: 8px;
}
```

```
323
324 .message-input {
325   flex: 1;
326   padding: 8px 14px;
327   background: var(--surface-2);
328   border: 1px solid var(--border-color);
329   border-radius: 20px;
330   color: var(--text-color);
331   outline: none;
332   font-size: 1.15rem;
333   transition: border-color 0.15s;
334 }
335
336 .message-input::placeholder { color: var(--text-muted); }
337 .message-input:focus { border-color: var(--nav-active); }
338
339 .voice-recording-status {
340   color: var(--danger-color);
341   font-weight: 600;
342   font-size: 0.82rem;
343   display: flex;
344   align-items: center;
345   gap: 6px;
346 }
347
348 /* ── Оверлей видеозвонка ──────────────────────────────
    ──────────────── */
349 .video-call-overlay {
350   position: absolute;
351   top: 0; left: 0;
352   width: 100%; height: 100%;
353   background: rgba(0, 0, 0, 0.88);
354   backdrop-filter: blur(12px);
355   z-index: 100;
356   display: flex;
357   flex-direction: column;
358 }
359
```

```
.video-grid {
  flex: 1;
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(280px, 1fr));
  gap: 12px;
  padding: 16px;
  align-items: center;
  justify-items: center;
}

.video-tile {
  width: 100%; height: 100%;
  max-height: 440px;
  background: var(--surface-1);
  border-radius: 12px;
  overflow: hidden;
  position: relative;
  display: flex;
  align-items: center;
  justify-content: center;
  border: 1px solid var(--border-color);
}

.video-tile video { width: 100%; height: 100%; object-fit: cov
er; }

.video-tile .tile-label {
  position: absolute;
  bottom: 10px; left: 10px;
  background: rgba(0, 0, 0, 0.55);
  padding: 3px 8px;
  border-radius: 6px;
  font-size: 0.78rem;
  color: #fff;
}

.call-controls {
  height: 72px;
```

```
  background: var(--surface-1);
  display: flex;
  align-items: center;
  justify-content: center;
  gap: 16px;
  border-top: 1px solid var(--border-color);
}

.audio-player-custom {
  margin-top: 4px;
  width: 100%;
  max-width: 240px;
}

/* ── Модальное окно ──────────────────────────────────────── */
.modal-overlay {
  position: fixed;
  top: 0; left: 0; width: 100%; height: 100%;
  background: rgba(0, 0, 0, 0.6);
  display: flex;
  align-items: center;
  justify-content: center;
  z-index: 200;
}

.modal-card {
  background: var(--surface-1);
  border: 1px solid var(--border-color);
  padding: 20px;
  border-radius: 12px;
  box-shadow: var(--shadow-lg);
  width: 90%;
  max-width: 420px;
}

.modal-card h3 { margin-bottom: 12px; font-size: 1rem; font-weight: 600; }
```

```
433
434 .form-group { margin-bottom: 10px; }
435 .form-group label {
436   display: block;
437   font-size: 0.78rem;
438   color: var(--text-muted);
439   margin-bottom: 4px;
440 }
441 .form-group input,
442 .form-group select {
443   width: 100%;
444   padding: 7px 10px;
445   background: var(--surface-2);
446   border: 1px solid var(--border-color);
447   border-radius: 6px;
448   color: var(--text-color);
449   font-size: 0.82rem;
450   outline: none;
451   transition: border-color 0.15s;
452 }
453 .form-group input:focus,
454 .form-group select:focus {
455   border-color: var(--nav-active);
456 }
457
```

sw.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Messenger - Sw Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля sw.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/messenger/sw.js?v=2026
1001_v1" type="module"></script>
 *
 * File: sw.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/messenger
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

const CACHE_NAME = 'ai-windows-admin-messenger-v1';
const ASSETS = [
  './',
  './index.html',
  './style.css',
  './app.js',
  './manifest.json',
];

self.addEventListener('install', (event) => {
  event.waitUntil(
    caches.open(CACHE_NAME).then((cache) => cache.addAll(ASSET
```

```
S))
  );
});

self.addEventListener('fetch', (event) => {
  // Network first, cache fallback
  event.respondWith(
    fetch(event.request).catch(() => caches.match(event.reques
t))
  );
});

```

widget.js

```
/**
 * ============================================================
 * Process Name: Windows Messenger - Widget Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля widget.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/messenger/widget.js?v=20261001_v1" type="module"></script>
 *
 * File: widget.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/messenger
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * AI Windows Admin Floating Messenger Widget
 * Zero-dependency embeddable script for any website or WordPress.
 */

(function () {
  const config = window.AB_MESSENGER_CONFIG || {
    apiUrl: window.location.origin,
    ssoToken: '',
    userId: '',
    userName: 'Guest',
```

```
  };

  const style = document.createElement('style');
  style.innerHTML = `
    .ab-widget-bubble {
      position: fixed;
      bottom: 24px;
      right: 24px;
      width: 60px;
      height: 60px;
      border-radius: 50%;
      background: var(--nav-active, linear-gradient(135deg, #3b82f6, #1d4ed8));
      box-shadow: 0 10px 25px rgba(59, 130, 246, 0.4);
      display: flex;
      align-items: center;
      justify-content: center;
      cursor: pointer;
      z-index: 999999;
      transition: transform 0.2s ease;
      color: #fff;
      font-size: 28px;
    }
    .ab-widget-bubble:hover {
      transform: scale(1.08);
    }
    .ab-widget-frame {
      position: fixed;
      bottom: 96px;
      right: 24px;
      width: 400px;
      height: 620px;
      max-width: calc(100vw - 32px);
      max-height: calc(100vh - 120px);
      border-radius: 20px;
      box-shadow: 0 12px 40px rgba(0, 0, 0, 0.2);
      border: 1px solid var(--border-color, rgba(255, 255, 255, 0.1));
```

```
      overflow: hidden;
      z-index: 999998;
      display: none;
      background: var(--surface-1, #0f172a);
    }
    .ab-widget-frame iframe {
      width: 100%;
      height: 100%;
      border: none;
    }
  `;
  document.head.appendChild(style);

  const bubble = document.createElement('div');
  bubble.className = 'ab-widget-bubble';
  bubble.innerHTML = '□ ';

  const frameContainer = document.createElement('div');
  frameContainer.className = 'ab-widget-frame';

  let isOpen = false;
  let iframe = null;

  bubble.onclick = () => {
    isOpen = !isOpen;
    if (isOpen) {
      if (!iframe) {
        iframe = document.createElement('iframe');
        const tokenQuery = config.ssoToken ? `?token=${encodeU
RIComponent(config.ssoToken)}` : '';
        iframe.src = `${config.apiUrl}/webinterface/messenger/
${tokenQuery}`;
        iframe.allow = 'camera; microphone; display-capture; a
utoplay';
        frameContainer.appendChild(iframe);
      }
      frameContainer.style.display = 'block';
      bubble.innerHTML = '□ ';
```

```
    } else {
      frameContainer.style.display = 'none';
      bubble.innerHTML = '□ ';
    }
  };

  document.body.appendChild(frameContainer);
  document.body.appendChild(bubble);
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/models_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/models_tab/index.html"
class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/models_tab
Author: hypo69
Copyright: © 2026 hypo69
# Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Models & APIs Management Tab -->
<div class="container-fluid py-3">
<!-- Premium Header -->
<div class="d-flex justify-content-between align-items-center
mb-4">
<div>
<h4 class="mb-0 fw-bold text-primary"><i class="bi bi-cpu"></i
> Модели и API</h4>
<p class="text-muted small mb-0">Управление провайдерами, ключ
ами и системными инструкциями</p>
```

```
</div>
</div>
<!-- Top Block: Active Model, Test Prompt & System Instruction
 -->
<div class="row mb-4">
<!-- Active Model & Testing Card -->
<div class="col-lg-6 mb-4 mb-lg-0">
<div class="card shadow-sm border-0 h-100" data-panel-id="pane
l-models-aktivnaya-model-i-proverka" id="panel-models-aktivnay
a-model-i-proverka">
<div class="card-header bg-transparent border-0 d-flex justify
-content-between align-items-center pt-3 pb-0">
<h6 class="card-title fw-bold text-primary mb-0"><i class="bi
bi-robot"></i> Активная модель и проверка</h6>
<button class="btn btn-outline-secondary btn-sm rounded-pill p
y-0 px-2" id="btn-refresh-models-list" title="Обновить список
доступных моделей">
<i class="bi bi-arrow-clockwise"></i> Обновить список
          </button>
</div>
<div class="card-body d-flex flex-column p-4">
<!-- Saved Default Model Indicator -->
<div class="mb-3 d-flex align-items-center justify-content-bet
ween p-2 rounded bg-body-tertiary border">
<div class="small text-secondary d-flex align-items-center gap
-1">
<i class="bi bi-bookmark-check-fill text-success"></i>
<span>Модель по умолчанию:</span>
</div>
<span class="badge bg-secondary font-monospace px-2 py-1" id="
saved-default-model-badge" style="font-size: 0.82rem;">Загрузк
a...</span>
</div>
<div class="row g-2 mb-2">
<div class="col-sm-5">
<label class="form-label small text-secondary mb-1" for="provi
der-tab-select">Провайдер ИИ:</label>
<select class="form-select form-select-sm" id="provider-tab-se
```

```
lect">
<option value="">Загрузка...</option>
</select>
</div>
<div class="col-sm-7">
<div class="d-flex justify-content-between align-items-center mb-1">
<label class="form-label small text-secondary mb-0" for="models-tab-select">Модель:</label>
<div class="form-check form-switch mb-0" title="Показать все модели, включая отфильтрованные / неподдерживаемые">
<input class="form-check-input" id="show-all-models-check" role="switch" style="cursor: pointer;" type="checkbox"/>
<label class="form-check-label small text-secondary" for="show-all-models-check" style="cursor: pointer; font-size: 0.72rem; ">Показать все</label>
</div>
</div>
<div class="input-group input-group-sm">
<select class="form-select form-select-sm" id="models-tab-select">
<option value="">Загрузка...</option>
</select>
<button class="btn btn-outline-warning" id="btn-model-toggle-favorite" title="Добавить в избранное / Убрать из избранного" type="button">
<i class="bi bi-star" id="favorite-star-icon"></i>
</button>
</div>
</div>
</div>
<!-- Model User Note & Favorite Details -->
<div class="mb-3 p-2 rounded bg-body-tertiary border" id="model-note-box" style="border-style: dashed !important;">
<div class="d-flex justify-content-between align-items-center mb-1">
<label class="form-label small text-warning-emphasis mb-0" for="model-user-note" style="font-size: 0.76rem;">
```

```
<i class="bi bi-journal-text me-1"></i> Заметка о модели:
              </label>
<span class="badge bg-secondary-subtle text-secondary-emphasis
 border" id="model-fav-status-badge" style="font-size: 0.68rem
;">Не в избранном</span>
</div>
<div class="input-group input-group-sm">
<input autocomplete="off" class="form-control" id="model-user-
note" placeholder="Введите заметку о возможностях/особенностях
 модели..." type="text"/>
<button class="btn btn-sm btn-outline-info" id="btn-save-model
-note" title="Сохранить заметку" type="button">
<i class="bi bi-floppy me-1"></i> Сохранить
              </button>
</div>
</div>
<!-- Favorite Models Chips List -->
<div class="mb-3" id="favorite-models-section" style="display:
 none;">
<div class="d-flex align-items-center justify-content-between
mb-1">
<span class="small text-secondary fw-semibold" style="font-siz
e: 0.75rem;">
<i class="bi bi-star-fill text-warning me-1"></i> Избранные мо
дели:
              </span>
</div>
<div class="d-flex flex-wrap gap-1" id="favorite-models-list">
<!-- Rendered via JS -->
</div>
</div>
<!-- Model Test Area: Prompt -> Response -->
<div class="p-3 rounded-3 mb-3 bg-body-tertiary border">
<div class="d-flex justify-content-between align-items-center
mb-2">
<label class="form-label small text-info fw-bold mb-0" for="mo
del-test-prompt">
<i class="bi bi-lightning-charge-fill text-warning me-1"></i>
```

```
Проверочный запрос (Запрос → Ответ)
              </label>
<span class="text-secondary small" style="font-size: 0.72rem;"
>Тест выбранной модели</span>
</div>
<div class="input-group input-group-sm mb-2">
<input autocomplete="off" class="form-control" data-1p-ignore=
"true" data-lpignore="true" id="model-test-prompt" placeholder
="Введите проверочное сообщение..." type="text" value="Привет!
 Назови свою модель и провайдера, и подтверди готовность к раб
оте."/>
<button class="btn btn-info fw-bold px-3" id="btn-model-test-s
end" title="Отправить тестовый запрос в выбранную модель" type
="button">
<i class="bi bi-send-fill me-1"></i> Проверить
              </button>
<button class="btn btn-outline-danger fw-bold px-3" id="btn-mo
del-test-cancel" style="display: none;" title="Отменить выполн
ение проверочного запроса" type="button">
<i class="bi bi-stop-circle me-1"></i> Отмена
              </button>
</div>
<div class="p-2 rounded small font-monospace bg-body border" i
d="model-test-result" style="display: none; max-height: 140px;
 overflow-y: auto;">
<!-- Dynamic output goes here -->
</div>
</div>
<div class="mt-auto d-grid">
<button class="btn btn-primary btn-sm rounded-pill fw-bold" id
="btn-models-tab-save">
<i class="bi bi-check-circle"></i> Сохранить как модель по умо
лчанию
            </button>
<div class="form-text text-secondary text-center mt-1" style="
font-size: 0.72rem;">Эта модель будет использоваться ассистент
ом по умолчанию во всех чатах.</div>
</div>
```

```
</div>
</div>
</div>
<!-- System Instruction Card -->
<div class="col-lg-6">
<div class="card shadow-sm border-0 h-100" data-panel-id="panel-models-sistemnaya-instruktsiya" id="panel-models-sistemnaya-instruktsiya">
<div class="card-header bg-transparent border-0 d-flex justify-content-between align-items-center pt-3 pb-0">
<span class="fw-bold"><i class="bi bi-terminal"></i> Системная инструкция</span>
<div class="d-flex gap-2 align-items-center">
<span class="badge bg-secondary" id="system-instruction-status" style="display:none!important;"></span>
<button class="btn btn-light btn-sm rounded-pill" id="btn-reload-instruction" title="Обновить">
<i class="bi bi-arrow-clockwise"></i>
</button>
<button class="btn btn-success btn-sm rounded-pill px-3" id="btn-save-instruction">
              Сохранить
            </button>
</div>
</div>
<div class="card-body d-flex flex-column">
<textarea class="form-control font-monospace border-0 bg-body-tertiary flex-grow-1" id="system-instruction-editor" placeholder="Введите системную инструкцию..." spellcheck="false" style="min-height: 220px; resize: none; font-size: 0.85rem;"></textarea>
</div>
</div>
</div>
</div>
<!-- Bottom Block: Provider Settings via Vertical Pills -->
<div class="card shadow-sm border-0" data-panel-id="panel-models-nastroyki-provayderov" id="panel-models-nastroyki-provayder
```

```
ov">
<div class="card-body p-0">
<div class="d-flex flex-column flex-md-row align-items-stretch
">
<!-- Sidebar Pills -->
<div aria-orientation="vertical" class="nav flex-column nav-pi
lls p-3 bg-body-tertiary border-end" id="provider-pills-tab" r
ole="tablist" style="min-width: 250px; min-height: 400px; bord
er-top-left-radius: 0.375rem; border-bottom-left-radius: 0.375
rem;">
<h6 class="text-uppercase text-muted fw-bold small mb-3 ps-2">
Настройки провайдеров</h6>
<button aria-controls="pill-gemini" aria-selected="true" class
="nav-link active text-start rounded-3 mb-2 fw-bold" data-bs-t
arget="#pill-gemini" data-bs-toggle="pill" id="pill-gemini-tab
" role="tab" type="button">
            Google Gemini
          </button>
<button aria-controls="pill-agy" aria-selected="false" class="
nav-link text-start rounded-3 mb-2 fw-bold" data-bs-target="#p
ill-agy" data-bs-toggle="pill" id="pill-agy-tab" role="tab" ty
pe="button">
            Google Antigravity (AGY)
          </button>
<button aria-controls="pill-google-accounts" aria-selected="fa
lse" class="nav-link text-start rounded-3 mb-2 fw-bold" data-b
s-target="#pill-google-accounts" data-bs-toggle="pill" id="pil
l-google-accounts-tab" role="tab" type="button">
            Google Workspace (Аккаунты)
          </button>
<button aria-controls="pill-foundry" aria-selected="false" cla
ss="nav-link text-start rounded-3 mb-2 fw-bold" data-bs-target
="#pill-foundry" data-bs-toggle="pill" id="pill-foundry-tab" r
ole="tab" type="button">
            Microsoft Foundry
          </button>
<button aria-controls="pill-ollama" aria-selected="false" clas
s="nav-link text-start rounded-3 mb-2 fw-bold" data-bs-target=
```

```
"#pill-ollama" data-bs-toggle="pill" id="pill-ollama-tab" role
="tab" type="button">
            Ollama (Local)
          </button>
<button aria-controls="pill-onnx" aria-selected="false" class=
"nav-link text-start rounded-3 fw-bold" data-bs-target="#pill-
onnx" data-bs-toggle="pill" id="pill-onnx-tab" role="tab" type
="button">
            Microsoft ONNX / Olive
          </button>
</div>
<!-- Tab Content -->
<div class="tab-content flex-grow-1 p-4" id="provider-pills-ta
bContent">
<!-- GEMINI TAB -->
<div aria-labelledby="pill-gemini-tab" class="tab-pane fade sh
ow active" id="pill-gemini" role="tabpanel">
<div class="d-flex justify-content-between align-items-center
mb-3">
<h5 class="fw-bold mb-0">Ключи Google Gemini</h5>
<button class="btn btn-outline-warning btn-sm rounded-pill px-
3" id="btn-refresh-keys">
<i class="bi bi-arrow-repeat"></i> Сбросить все ключи
              </button>
</div>
<!-- Gemini Key Pool Alert -->
<div class="alert alert-info shadow-sm rounded-3 border-0 mb-4
" role="alert">
<h6 class="alert-heading fw-bold mb-1"><i class="bi bi-key-fil
l me-2"></i>Пул ротации API-ключей (Round-Robin)</h6>
<p class="small mb-0">
                Все активные ключи из таблицы ниже формируют е
диный пул. При исчерпании суточных лимитов запросов (ошибки <c
ode>429 / ResourceExhausted</code>) система автоматически пере
ключается на следующий свободный ключ без прерывания генерации
.
              </p>
</div>
```

```
<div class="row g-4">
<!-- Add Key Form -->
<div class="col-lg-4">
<div class="card bg-body-tertiary border-0 shadow-sm" data-panel-id="panel-models-dobavit-novyy-klyuch" id="panel-models-dobavit-novyy-klyuch">
<div class="card-body">
<h6 class="fw-bold mb-3">Добавить новый ключ</h6>
<form id="form-add-key" onsubmit="event.preventDefault();">
<div class="mb-3">
<label class="form-label small text-muted fw-bold">Имя владельца:</label>
<input autocomplete="off" class="form-control form-control-sm"  data-1p-ignore="true" data-lpignore="true" id="new-key-name" placeholder="Например: primary_key" required="" type="text"/>
</div>
<div class="mb-4">
<label class="form-label small text-muted fw-bold">API-ключ (AIzaSy...):</label>
<input autocomplete="new-password" class="form-control form-control-sm" data-1p-ignore="true" data-lpignore="true" id="new-key-value" placeholder="Введите ключ" required="" type="password"/>
</div>
<button class="btn btn-success btn-sm w-100 rounded-pill fw-bold" id="btn-add-key" type="button">Добавить ключ</button>
</form>
</div>
</div>
</div>
<!-- Keys Table -->
<div class="col-lg-8">
<div class="table-responsive shadow-sm rounded border bg-body">
<table class="table table-hover align-middle mb-0" id="keys-table">
<thead class="table-light">
<tr>
```

```
<th class="py-3 px-3 border-0">Название</th>
<th class="py-3 px-3 border-0">Ключ</th>
<th class="py-3 px-3 border-0">Статус</th>
<th class="py-3 px-3 border-0">Квота</th>
<th class="py-3 px-3 border-0 text-end">Действия</th>
</tr>
</thead>
<tbody class="border-top-0" id="keys-list-body">
<tr>
<td class="text-center py-4 text-muted" colspan="5">Загрузка ключей...</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
<!-- ANTIGRAVITY (AGY) TAB -->
<div aria-labelledby="pill-agy-tab" class="tab-pane fade" id="pill-agy" role="tabpanel">
<h5 class="fw-bold mb-3">Настройки Google Antigravity (AGY)</h5>
<div class="row">
<div class="col-md-11 col-lg-9">
<!-- Info Alert -->
<div class="alert alert-warning shadow-sm rounded-3 border-0 mb-4" role="alert">
<h6 class="alert-heading fw-bold"><i class="bi bi-rocket-takeoff me-2"></i>Что такое Antigravity (AGY)?</h6>
<p class="small mb-2">
<strong>Google Antigravity</strong> — передовая агентная платформа для автономных AI-воркфлоу, мультиагентных пайплайнов и глубокого веб-исследования.
                    </p>
<ul class="small mb-0 ps-3">
<li>Включает встроенные агентные инструменты <code>BuiltinTools.SEARCH_WEB</code> и <code>READ_URL_CONTENT</code>.</li>
```

```
<li>Поддерживает автономный вызов через Python SDK (<code>google.antigravity</code>) и утилиту <code>agy CLI</code>.</li>
<li>Если отдельный ключ <code>AGY_API_KEY</code> не задан, провайдер автоматически использует ключи из пула Gemini.</li>
</ul>
</div>
<form id="form-agy-config" onsubmit="event.preventDefault();">
<div class="card border-0 bg-body-tertiary shadow-sm mb-4" data-panel-id="panel-models-1" id="panel-models-1">
<div class="card-body p-4">
<div class="d-flex align-items-center justify-content-between mb-4 pb-2 border-bottom flex-wrap gap-2">
<div class="form-check form-switch mb-0">
<input class="form-check-input" id="agy-enabled" role="switch" style="transform: scale(1.3); margin-left: -2.5em; cursor: pointer;" type="checkbox"/>
<label class="form-check-label fw-bold ms-2" for="agy-enabled" style="cursor: pointer;">Активировать провайдер Antigravity (AGY)</label>
</div>
<div class="form-check mb-0">
<input checked="" class="form-check-input" id="agy-remember" style="cursor: pointer;" type="checkbox"/>
<label class="form-check-label small text-muted ms-1" for="agy-remember" style="cursor: pointer;">Запомнить</label>
</div>
</div>
<div class="mb-3">
<label class="form-label small text-muted fw-bold">Модель Antigravity по умолчанию (динамически из SDK):</label>
<select class="form-select form-select-sm" id="agy-model">
<option value="">Загрузка доступных моделей через SDK...</option>
</select>
</div>
<div class="mb-4">
<label class="form-label small text-muted fw-bold">AGY API Key (опционально, иначе из пула Gemini):</label>
```

```
<input autocomplete="new-password" class="form-control form-control-sm" data-1p-ignore="true" data-lpignore="true" id="agy-key" placeholder="Оставьте пустым для использования ключей Gemini" type="password"/>
<div class="form-text small text-muted">Сохраняется в защищенном файле <code>.env</code>.</div>
</div>
<button class="btn btn-primary rounded-pill px-4 fw-bold" id="btn-save-agy" type="button">
<i class="bi bi-floppy me-1"></i> Сохранить настройки Antigravity
                    </button>
</div>
</div>
</form>
</div>
</div>
</div>
<!-- GOOGLE WORKSPACE ACCOUNTS POOL TAB -->
<div aria-labelledby="pill-google-accounts-tab" class="tab-pane fade" id="pill-google-accounts" role="tabpanel">
<div class="d-flex justify-content-between align-items-center mb-3">
<h5 class="fw-bold mb-0"><i class="bi bi-google text-danger me-2"></i>Учетные записи Google Workspace</h5>
<button class="btn btn-outline-primary btn-sm rounded-pill px-3" id="btn-refresh-gaccounts">
<i class="bi bi-arrow-clockwise"></i> Обновить список
            </button>
</div>
<!-- Workspace Pool Info Alert -->
<div class="alert alert-info shadow-sm rounded-3 border-0 mb-4" role="alert">
<h6 class="alert-heading fw-bold mb-1"><i class="bi bi-collection-play-fill me-2"></i>Пул аккаунтов Google (OAuth 2.0 &amp;amp; Service Account)</h6>
<p class="small mb-2">
              Интеграция с сервисами Google: <strong>Gmail</
```

```
strong> (чтение, черновики), <strong>Google Drive</strong> (по
иск файлов, чтение), <strong>Google Sheets</strong> (чтение/за
пись таблиц) и <strong>Google Docs</strong>.
              </p>
<ul class="small mb-0 ps-3">
<li>При вызове инструментов ассистент использует аккаунт <code
>account_name</code> или аккаунт по умолчанию (<span class="ba
dge bg-warning text-dark"><i class="bi bi-star-fill"></i> Defa
ult</span>).</li>
<li>Файлы ключей и токенов безопасно сохраняются в <code>src/s
ecrets/</code> и изолированы от публичной конфигурации.</li>
</ul>
</div>
<div class="row g-4">
<!-- Add Account Form -->
<div class="col-lg-5">
<div class="card bg-body-tertiary border-0 shadow-sm" data-pan
el-id="panel-models-dobavit-akkaunt-v-pul" id="panel-models-do
bavit-akkaunt-v-pul">
<div class="card-body p-4">
<h6 class="fw-bold mb-3 d-flex align-items-center gap-2">
<i class="bi bi-person-plus-fill text-primary"></i> Добавить а
ккаунт в пул
                    </h6>
<form id="form-add-gaccount" onsubmit="event.preventDefault();
">
<div class="mb-3">
<label class="form-label small text-muted fw-bold">Имя аккаунт
а (ID):</label>
<input autocomplete="off" class="form-control form-control-sm"
 id="new-gacc-name" placeholder="Например: work, personal, bot
" required="" type="text"/>
<div class="form-text small">Уникальный идентификатор аккаунта
 для маршрутизации агентов.</div>
</div>
<div class="mb-3">
<label class="form-label small text-muted fw-bold">Тип аутенти
фикации:</label>
```

```
<select class="form-select form-select-sm" id="new-gacc-type">
<option value="oauth2">OAuth 2.0 Client (User Login / Desktop App)</option>
<option value="service_account">Service Account Key (Server JSON)</option>
</select>
</div>
<div class="mb-3">
<label class="form-label small text-muted fw-bold">Email (опционально):</label>
<input autocomplete="off" class="form-control form-control-sm" id="new-gacc-email" placeholder="user@gmail.com или service-account@iam.gserviceaccount.com" type="email"/>
</div>
<!-- Credentials Input Mode Toggle -->
<div class="mb-2">
<label class="form-label small text-muted fw-bold">Учетные данные (Credentials JSON):</label>
<ul class="nav nav-pills nav-fill mb-2" id="gacc-input-mode-tabs" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active py-1 small" data-bs-target="#gacc-mode-file" data-bs-toggle="pill" id="gacc-mode-file-tab" role="tab" type="button">  Загрузить файл</button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link py-1 small" data-bs-target="#gacc-mode-json" data-bs-toggle="pill" id="gacc-mode-json-tab" role="tab" type="button">  Вставить JSON</button>
</li>
</ul>
<div class="tab-content" id="gacc-input-mode-tabContent">
<div class="tab-pane fade show active" id="gacc-mode-file" role="tabpanel">
<input accept=".json,application/json" class="form-control form-control-sm" id="new-gacc-file" type="file"/>
<div class="form-text small">Файл <code>credentials.json</code> или <code>service_account.json</code> из <a class="text-info
```

```
 text-decoration-none" href="https://console.cloud.google.com/apis/credentials" rel="noopener noreferrer" target="_blank">Google Cloud Console <i class="bi bi-box-arrow-up-right ms-1" style="font-size: 0.75rem;"></i></a>.</div>
</div>
<div class="tab-pane fade" id="gacc-mode-json" role="tabpanel">
<textarea class="form-control font-monospace form-control-sm" id="new-gacc-json" placeholder='{"installed": {"client_id": "...", "client_secret": "..."}}' rows="4"></textarea>
</div>
</div>
</div>
<div class="form-check mb-4 mt-3">
<input class="form-check-input" id="new-gacc-default" style="cursor: pointer;" type="checkbox"/>
<label class="form-check-label small fw-semibold" for="new-gacc-default" style="cursor: pointer;">
                            Сделать аккаунтом по умолчанию (Default)
                        </label>
</div>
<button class="btn btn-success btn-sm w-100 rounded-pill fw-bold" id="btn-add-gacc" type="button">
<i class="bi bi-plus-circle me-1"></i> Сохранить аккаунт
                    </button>
</form>
</div>
</div>
</div>
<!-- Accounts Table -->
<div class="col-lg-7">
<div class="table-responsive shadow-sm rounded border bg-body mb-3">
<table class="table table-hover align-middle mb-0" id="gaccounts-table">
<thead class="table-light">
<tr>
```

```
<th class="py-3 px-3 border-0">Аккаунт</th>
<th class="py-3 px-3 border-0">Тип</th>
<th class="py-3 px-3 border-0">Статус</th>
<th class="py-3 px-3 border-0">Токен</th>
<th class="py-3 px-3 border-0 text-end">Действия</th>
</tr>
</thead>
<tbody class="border-top-0" id="gaccounts-list-body">
<tr>
<td class="text-center py-4 text-muted" colspan="5">Загрузка аккаунтов...</td>
</tr>
</tbody>
</table>
</div>
<!-- Test & Diagnostics Output Card -->
<div class="card bg-dark text-white border-secondary shadow-sm" data-panel-id="panel-models-rezultat-proverki-dostupa" id="gaccount-test-card" style="display: none;">
<div class="card-header border-secondary d-flex justify-content-between align-items-center py-2">
<span class="small fw-bold text-info"><i class="bi bi-terminal me-1"></i> Результат проверки доступа</span>
<button class="btn btn-sm btn-outline-secondary py-0 px-2 text-white-50" onclick="document.getElementById('gaccount-test-card').style.display='none'" type="button">□ </button>
</div>
<div class="card-body p-3 font-monospace small" id="gaccount-test-body" style="max-height: 180px; overflow-y: auto; white-space: pre-wrap;">
</div>
</div>
</div>
</div>
</div>
<!-- FOUNDRY TAB -->
<div aria-labelledby="pill-foundry-tab" class="tab-pane fade" id="pill-foundry" role="tabpanel">
```

```
<h5 class="fw-bold mb-4">Настройки Microsoft Foundry</h5>
<div class="row">
<div class="col-md-10 col-lg-8">
<div class="alert alert-info shadow-sm rounded-3 border-0 mb-4
" role="alert">
<h6 class="alert-heading fw-bold"><i class="bi bi-info-circle
me-2"></i>Где взять модели Foundry?</h6>
<p class="small mb-0">
                    Microsoft Foundry (как и LM Studio) работа
ет локально. Чтобы список моделей был не пустым, вам необходим
о предварительно загрузить нужные GGUF или ONNX модели через и
нтерфейс самого Foundry или CLI.
                  </p>
</div>
<form id="form-foundry-config" onsubmit="event.preventDefault(
);">
<div class="card border-0 bg-body-tertiary shadow-sm mb-4" dat
a-panel-id="panel-models-2" id="panel-models-2">
<div class="card-body p-4">
<div class="d-flex align-items-center justify-content-between
mb-4 pb-2 border-bottom flex-wrap gap-2">
<div class="form-check form-switch mb-0">
<input class="form-check-input" id="foundry-enabled" role="swi
tch" style="transform: scale(1.3); margin-left: -2.5em; cursor
: pointer;" type="checkbox"/>
<label class="form-check-label fw-bold ms-2" for="foundry-enab
led" style="cursor: pointer;">Активировать провайдер Foundry</
label>
</div>
<div class="form-check mb-0">
<input checked="" class="form-check-input" id="foundry-remembe
r" style="cursor: pointer;" type="checkbox"/>
<label class="form-check-label small text-muted ms-1" for="fou
ndry-remember" style="cursor: pointer;">Запомнить</label>
</div>
</div>
<div class="mb-3">
<label class="form-label small text-muted fw-bold">Base URL AP
```

```
I:</label>
<input autocomplete="off" class="form-control" data-1p-ignore="true" data-lpignore="true" id="foundry-url" placeholder="http://localhost:54837" type="text"/>
</div>
<div class="mb-3">
<label class="form-label small text-muted fw-bold">API Key (опционально):</label>
<input autocomplete="new-password" class="form-control" data-1p-ignore="true" data-lpignore="true" id="foundry-key" placeholder="Введите ключ" type="password"/>
</div>
<button class="btn btn-primary rounded-pill px-4 fw-bold" id="btn-save-foundry" type="button">Сохранить настройки Foundry</button>
</div>
</div>
</form>
</div>
</div>
</div>
<!-- OLLAMA TAB -->
<div aria-labelledby="pill-ollama-tab" class="tab-pane fade" id="pill-ollama" role="tabpanel">
<h5 class="fw-bold mb-4">Настройки Ollama (Local)</h5>
<div class="row">
<div class="col-md-10 col-lg-8">
<div class="alert alert-info shadow-sm rounded-3 border-0 mb-4" role="alert">
<h6 class="alert-heading fw-bold"><i class="bi bi-info-circle me-2"></i>Как добавить новые модели в Ollama?</h6>
<p class="small mb-0">
                    Откройте терминал на сервере, где установлен Ollama, и выполните команду:<br/>
<code class="bg-body-tertiary px-2 py-1 rounded text-body mt-2 d-inline-block border">ollama pull llama3.1</code><br/>
                    После загрузки модель автоматически появится в выпадающем списке.
```

```
                    </p>
</div>
<form id="form-ollama-config" onsubmit="event.preventDefault()
;">
<div class="card border-0 bg-body-tertiary shadow-sm mb-4" dat
a-panel-id="panel-models-3" id="panel-models-3">
<div class="card-body p-4">
<div class="d-flex align-items-center justify-content-between
mb-4 pb-2 border-bottom flex-wrap gap-2">
<div class="form-check form-switch mb-0">
<input class="form-check-input" id="ollama-enabled" role="swit
ch" style="transform: scale(1.3); margin-left: -2.5em; cursor:
 pointer;" type="checkbox"/>
<label class="form-check-label fw-bold ms-2" for="ollama-enabl
ed" style="cursor: pointer;">Активировать провайдер Ollama</la
bel>
</div>
<div class="form-check mb-0">
<input checked="" class="form-check-input" id="ollama-remember
" style="cursor: pointer;" type="checkbox"/>
<label class="form-check-label small text-muted ms-1" for="oll
ama-remember" style="cursor: pointer;">Запомнить</label>
</div>
</div>
<div class="mb-3">
<label class="form-label small text-muted fw-bold">Base URL AP
I:</label>
<input autocomplete="off" class="form-control" data-1p-ignore=
"true" data-lpignore="true" id="ollama-url" placeholder="http:
//localhost:11434" type="text"/>
</div>
<button class="btn btn-primary rounded-pill px-4 fw-bold" id="
btn-save-ollama" type="button">Сохранить настройки Ollama</but
ton>
</div>
</div>
</form>
</div>
```

```
</div>
<!-- ONNX / OLIVE TAB -->
<div aria-labelledby="pill-onnx-tab" class="tab-pane fade" id="pill-onnx" role="tabpanel">
<h5 class="fw-bold mb-4">Настройки Microsoft ONNX Runtime &amp; Olive</h5>
<div class="row">
<div class="col-md-10 col-lg-8">
<div class="alert alert-info shadow-sm rounded-3 border-0 mb-4" role="alert">
<h6 class="alert-heading fw-bold"><i class="bi bi-cpu me-2"></i>Аппаратное ускорение и оптимизация Olive</h6>
<p class="small mb-0">
                    Microsoft ONNX Runtime использует аппаратные ускорители Windows (DirectML GPU, Qualcomm QNN NPU, CUDA). Microsoft Olive выполняет автоматическое квантование (INT4/INT8) и оптимизацию графов для максимальной скорости локального инференса.
                  </p>
</div>
<form id="form-onnx-config" onsubmit="event.preventDefault();">
<div class="card border-0 bg-body-tertiary shadow-sm mb-4" data-panel-id="panel-models-4" id="panel-models-4">
<div class="card-body p-4">
<div class="d-flex align-items-center justify-content-between mb-4 pb-2 border-bottom flex-wrap gap-2">
<div class="form-check form-switch mb-0">
<input class="form-check-input" id="onnx-enabled" role="switch" style="transform: scale(1.3); margin-left: -2.5em; cursor: pointer;" type="checkbox"/>
<label class="form-check-label fw-bold ms-2" for="onnx-enabled" style="cursor: pointer;">Активировать провайдер ONNX / Olive</label>
</div>
<div class="form-check mb-0">
<input checked="" class="form-check-input" id="onnx-remember" style="cursor: pointer;" type="checkbox"/>
```

```
<label class="form-check-label small text-muted ms-1" for="onnx-remember" style="cursor: pointer;">Запомнить</label>
</div>
</div>
<div class="mb-3">
<label class="form-label small text-muted fw-bold">Execution Provider (Бэкенд ускорения):</label>
<select class="form-select" id="onnx-execution-provider">
<option value="DirectMLExecutionProvider">DirectML (AMD / Intel / NVIDIA GPU)</option>
<option value="QNNExecutionProvider">Qualcomm QNN (NPU Copilot+)</option>
<option value="CUDAExecutionProvider">NVIDIA CUDA</option>
<option value="CPUExecutionProvider">CPU Execution Provider</option>
</select>
</div>
<div class="mb-3">
<label class="form-label small text-muted fw-bold">Директория локальных моделей ONNX:</label>
<input autocomplete="off" class="form-control" id="onnx-models-dir" placeholder="models/onnx" type="text"/>
</div>
<div class="mb-3">
<label class="form-label small text-muted fw-bold">Модель по умолчанию:</label>
<input autocomplete="off" class="form-control" id="onnx-default-model" placeholder="phi-3.5-mini-instruct-onnx" type="text"/>
</div>
<div class="mb-4">
<label class="form-label small text-muted fw-bold">Квантование Olive (Precision):</label>
<select class="form-select" id="onnx-olive-precision">
<option value="int4">INT4 (Максимальная компрессия и скорость)</option>
<option value="int8">INT8 (Баланс точности и памяти)</option>
<option value="fp16">FP16 (Половинная точность)</option>
```

```
</select>
</div>
<button class="btn btn-primary rounded-pill px-4 fw-bold" id="
btn-save-onnx" type="button">Сохранить настройки ONNX / Olive<
/button>
</div>
</div>
</form>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- Modal: Test Gemini Key -->
<div aria-hidden="true" aria-labelledby="modal-test-key-label"
 class="modal fade" id="modal-test-key" tabindex="-1">
<div class="modal-dialog modal-dialog-centered">
<div class="modal-content shadow border-0">
<div class="modal-header border-bottom-0 pb-0">
<h5 class="modal-title fw-bold text-primary d-flex align-items
-center gap-2" id="modal-test-key-label">
<i class="bi bi-shield-check text-success"></i> Проверка API-к
люча
          </h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body pt-2">
<!-- Key Info Header -->
<div class="p-3 mb-3 rounded bg-body-tertiary border">
<div class="d-flex justify-content-between align-items-center
mb-1">
<span class="fw-bold text-truncate" id="modal-test-key-name" s
tyle="max-width: 250px;">-</span>
<span class="badge bg-secondary" id="modal-test-key-status-bad
ge">Загрузка...</span>
</div>
```

```
<div class="d-flex justify-content-between align-items-center small text-muted font-monospace">
<span id="modal-test-key-masked">****************</span>
<span class="badge bg-dark-subtle text-body" id="modal-test-key-duration">  -- ms</span>
</div>
</div>
<!-- Prompt & Model Settings -->
<div class="mb-3">
<div class="row g-2">
<div class="col-sm-7">
<label class="form-label small text-muted fw-semibold mb-1" for="modal-test-key-model">Тестовая модель:</label>
<select class="form-select form-select-sm" id="modal-test-key-model">
<option selected="" value="gemini-3.1-flash-lite">gemini-3.1-flash-lite</option>
<option value="gemini-2.5-flash">gemini-2.5-flash</option>
<option value="gemini-3.5-flash-lite">gemini-3.5-flash-lite</option>
<option value="gemini-flash-lite-latest">gemini-flash-lite-latest</option>
</select>
</div>
<div class="col-sm-5">
<label class="form-label small text-muted fw-semibold mb-1" for="modal-test-key-prompt">Сообщение:</label>
<input autocomplete="off" class="form-control form-control-sm"  id="modal-test-key-prompt" type="text" value="hello world"/>
</div>
</div>
</div>
<!-- Live Testing Status Indicator -->
<div class="text-center py-3" id="modal-test-key-progress" style="display: none;">
<div class="spinner-border spinner-border-sm text-primary me-2" role="status"></div>
<span class="small text-secondary fw-semibold">Отправка тестов
```

```
ого запроса в Google Gemini...</span>
</div>
<!-- Result Output Box -->
<div id="modal-test-key-result-box" style="display: none;">
<div class="d-flex justify-content-between align-items-center
mb-1">
<span class="small fw-bold" id="modal-test-key-result-title">О
твет модели:</span>
<span class="badge" id="modal-test-key-valid-badge"></span>
</div>
<div class="p-2.5 rounded font-monospace small border" id="mod
al-test-key-result-content" style="max-height: 180px; overflow
-y: auto; white-space: pre-wrap; background-color: var(--bs-bo
dy-bg);">
</div>
</div>
</div>
<div class="modal-footer border-top-0 pt-0">
<button class="btn btn-outline-primary btn-sm rounded-pill px-
3" id="modal-test-key-retry-btn" type="button">
<i class="bi bi-arrow-repeat me-1"></i> Повторить тест
          </button>
<button class="btn btn-secondary btn-sm rounded-pill px-3" dat
a-bs-dismiss="modal" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
</div></div>
```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Models Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/models_tab/main.js?v=2
0261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/models_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

// Gemini Models & APIs management tab logic

let modelTestAbortController = null;

function cancelModelTest() {
  if (modelTestAbortController) {
    try {
      modelTestAbortController.abort();
    } catch (e) {
      console.warn('Error aborting model test:', e);
    }
    modelTestAbortController = null;
```

```
  }
}

async function initModelsTab() {
  window.api = window.api || {};
  if (!window.api.fetch) {
    window.api.fetch = async function(url, options = {}) {
      const response = await fetch(url, options);
      if (!response.ok) {
        let msg = response.statusText;
        try {
          const data = await response.json();
          if (data && data.detail) {
            if (typeof data.detail === 'string') msg = data.de
tail;
            else if (Array.isArray(data.detail)) msg = data.de
tail.map(d => d.msg || JSON.stringify(d)).join(', ');
            else msg = JSON.stringify(data.detail);
          } else if (data && data.message) {
            msg = data.message;
          }
        } catch {}
        throw new Error(`${response.status} ${msg}`);
      }
      return response.json();
    };
  }

  const modelSelect = document.getElementById('models-tab-sele
ct');
  const saveBtn = document.getElementById('btn-models-tab-save
');
  const keysListBody = document.getElementById('keys-list-body
');
  const refreshKeysBtn = document.getElementById('btn-refresh-
keys');
  const addKeyBtn = document.getElementById('btn-add-key');
  const saveAgyBtn = document.getElementById('btn-save-agy');
```

```
  const saveFoundryBtn = document.getElementById('btn-save-fou
ndry');
  const saveOllamaBtn = document.getElementById('btn-save-olla
ma');
  const saveOnnxBtn = document.getElementById('btn-save-onnx')
;
  const saveBtnInstr = document.getElementById('btn-save-instr
uction');
  const reloadBtnInstr = document.getElementById('btn-reload-i
nstruction');

  const gaccountsListBody = document.getElementById('gaccounts
-list-body');
  const refreshGAccountsBtn = document.getElementById('btn-ref
resh-gaccounts');
  const addGAccountBtn = document.getElementById('btn-add-gacc
');

  const refreshModelsBtn = document.getElementById('btn-refres
h-models-list');
  const testModelBtn = document.getElementById('btn-model-test
-send');
  const cancelModelBtn = document.getElementById('btn-model-te
st-cancel');
  const testModelPrompt = document.getElementById('model-test-
prompt');

  // Ensure active provider pill tab pane has show active clas
ses
  const activePill = document.querySelector('#provider-pills-t
ab .nav-link.active') || document.getElementById('pill-gemini-
tab');
  if (activePill) {
    const targetSelector = activePill.getAttribute('data-bs-ta
rget');
    if (targetSelector) {
      const targetPane = document.querySelector(targetSelector
);
```

```
      if (targetPane && !targetPane.classList.contains('active')) {
        targetPane.classList.add('show', 'active');
      }
    }
  }

  // 1. Bind event handlers immediately
  if (saveBtnInstr) saveBtnInstr.onclick = saveSystemInstruction;
  if (reloadBtnInstr) reloadBtnInstr.onclick = loadSystemInstruction;

  if (refreshModelsBtn && modelSelect && saveBtn) {
    refreshModelsBtn.onclick = async () => {
      refreshModelsBtn.disabled = true;
      const originalText = refreshModelsBtn.innerHTML;
      refreshModelsBtn.innerHTML = '<span class="spinner-border spinner-border-sm me-1" role="status"></span> Обновление...';
      try {
        await loadTabModels(modelSelect, saveBtn, true);
      } finally {
        refreshModelsBtn.disabled = false;
        refreshModelsBtn.innerHTML = originalText;
      }
    };
  }

  const showAllModelsCheck = document.getElementById('show-all-models-check');
  if (showAllModelsCheck && modelSelect && saveBtn) {
    showAllModelsCheck.onchange = async () => {
      await loadTabModels(modelSelect, saveBtn, false);
    };
  }

  const favToggleBtn = document.getElementById('btn-model-togg
```

```
le-favorite');
  const saveNoteBtn = document.getElementById('btn-save-model-
note');
  const modelNoteInput = document.getElementById('model-user-n
ote');

  if (favToggleBtn && modelSelect) {
    favToggleBtn.onclick = async () => {
      const curModel = modelSelect.value;
      if (!curModel) {
        showModelsNotification('Сначала выберите модель', 'war
ning');
        return;
      }
      favToggleBtn.disabled = true;
      try {
        const isFav = window.userFavoriteModels && Boolean(win
dow.userFavoriteModels[curModel]);
        if (isFav) {
          await window.api.fetch(`/auth/favorites/${encodeURIC
omponent(curModel)}`, { method: 'DELETE' });
          if (window.userFavoriteModels) delete window.userFav
oriteModels[curModel];
          showModelsNotification(`Модель "${curModel}" удалена
 из избранного`, 'info');
        } else {
          const noteText = modelNoteInput ? modelNoteInput.val
ue.trim() : '';
          const res = await window.api.fetch('/auth/favorites'
, {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ model: curModel, note: note
Text })
          });
          if (res && res.favorites) window.userFavoriteModels
= res.favorites;
          showModelsNotification(`Модель "${curModel}" добавле
```

```
на в избранное  `, 'success');
144         }
145         updateModelFavoriteUI(modelSelect);
146         updateModelSelectOptions(modelSelect);
147       } catch (err) {
148         console.error('Ошибка переключения избранного:', err);
149         showModelsNotification('Ошибка: ' + err.message, 'dang
er');
150       } finally {
151         favToggleBtn.disabled = false;
152       }
153     };
154   }
155 
156   if (saveNoteBtn && modelSelect && modelNoteInput) {
157     saveNoteBtn.onclick = async () => {
158       const curModel = modelSelect.value;
159       if (!curModel) {
160         showModelsNotification('Сначала выберите модель', 'war
ning');
161         return;
162       }
163       saveNoteBtn.disabled = true;
164       const originalText = saveNoteBtn.innerHTML;
165       saveNoteBtn.innerHTML = '<span class="spinner-border spi
nner-border-sm me-1"></span> Сохранение...';
166       try {
167         const noteText = modelNoteInput.value.trim();
168         const res = await window.api.fetch('/auth/favorites',
{
169           method: 'POST',
170           headers: { 'Content-Type': 'application/json' },
171           body: JSON.stringify({ model: curModel, note: noteTe
xt })
172         });
173         if (res && res.favorites) window.userFavoriteModels =
res.favorites;
174         showModelsNotification(`Заметка для модели "${curModel
```

```
}" сохранена!`, 'success');
        updateModelFavoriteUI(modelSelect);
        updateModelSelectOptions(modelSelect);
      } catch (err) {
        console.error('Ошибка сохранения заметки:', err);
        showModelsNotification('Ошибка сохранения заметки: ' +
 err.message, 'danger');
      } finally {
        saveNoteBtn.disabled = false;
        saveNoteBtn.innerHTML = originalText;
      }
    };
  }

  if (modelSelect) {
    modelSelect.addEventListener('change', async () => {
      updateModelFavoriteUI(modelSelect);
      const curProv = document.getElementById('provider-tab-se
lect')?.value || '';
      if (modelSelect.value) {
        await persistSelectedModelAndProvider(modelSelect.valu
e, curProv, true);
      }
    });
  }

  if (testModelBtn) {
    testModelBtn.onclick = executeModelTest;
  }

  if (cancelModelBtn) {
    cancelModelBtn.onclick = cancelModelTest;
  }

  if (testModelPrompt) {
    testModelPrompt.onkeydown = (e) => {
      if (e.key === 'Enter') {
        e.preventDefault();
```

```
        executeModelTest();
      } else if (e.key === 'Escape' && modelTestAbortControlle
r) {
        e.preventDefault();
        cancelModelTest();
      }
    };
  }

  // Provider switch direct toggling (AGY, Foundry, Ollama, ON
NX config endpoints not implemented yet)
  const agyEnabledSwitch = document.getElementById('agy-enable
d');
  if (agyEnabledSwitch) {
    agyEnabledSwitch.onchange = async () => {
      const enabled = agyEnabledSwitch.checked;
      const remember = document.getElementById('agy-remember')
?.checked ?? true;
      const model = document.getElementById('agy-model')?.valu
e || 'agy-flash';
      const key = document.getElementById('agy-key')?.value.tr
im() || '';

      try {
        await window.api.fetch('/api/agy/config', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ enabled, model, key, remember
 })
        });
        showModelsNotification(`Google Antigravity (AGY) ${ena
bled ? 'активирован' : 'деактивирован'}${remember ? ' (сохране
но в config.json)' : ''}`, 'info');
        if (modelSelect && saveBtn) await loadTabModels(modelS
elect, saveBtn);
      } catch (err) {
        console.error('Ошибка переключения Antigravity:', err)
;
```

```
        showModelsNotification('Ошибка переключения: ' + err.message, 'danger');
      }
    };
  }

  const foundryEnabledSwitch = document.getElementById('foundry-enabled');
  if (foundryEnabledSwitch) {
    foundryEnabledSwitch.onchange = async () => {
      const enabled = foundryEnabledSwitch.checked;
      const remember = document.getElementById('foundry-remember')?.checked ?? true;
      const url = document.getElementById('foundry-url')?.value.trim() || 'http://localhost:54837';
      const key = document.getElementById('foundry-key')?.value.trim() || '';
      const model = document.getElementById('foundry-model')?.value?.trim() || 'qwen2.5-1.5b';

      try {
        await window.api.fetch('/api/foundry/config', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ enabled, url, key, model, remember })
        });
        showModelsNotification(`Microsoft Foundry ${enabled ? 'активирован' : 'деактивирован'}${remember ? ' (сохранено в config.json)' : ''}`, 'info');
        if (modelSelect && saveBtn) await loadTabModels(modelSelect, saveBtn);
      } catch (err) {
        console.error('Ошибка переключения Foundry:', err);
        showModelsNotification('Ошибка переключения: ' + err.message, 'danger');
      }
    };
```

```
263   }
264
265   const ollamaEnabledSwitch = document.getElementById('ollama-
enabled');
266   if (ollamaEnabledSwitch) {
267     ollamaEnabledSwitch.onchange = async () => {
268       const enabled = ollamaEnabledSwitch.checked;
269       const remember = document.getElementById('ollama-remembe
r')?.checked ?? true;
270       const url = document.getElementById('ollama-url')?.value
.trim() || 'http://localhost:11434';
271       const model = document.getElementById('ollama-model')?.v
alue?.trim() || 'llama3.1';
272
273       try {
274         await window.api.fetch('/api/ollama/config', {
275           method: 'POST',
276           headers: { 'Content-Type': 'application/json' },
277           body: JSON.stringify({ enabled, url, model, remember
 })
278         });
279         showModelsNotification(`Ollama ${enabled ? 'активирова
н' : 'деактивирован'}${remember ? ' (сохранено в config.json)'
 : ''}`, 'info');
280         if (modelSelect && saveBtn) await loadTabModels(modelS
elect, saveBtn);
281       } catch (err) {
282         console.error('Ошибка переключения Ollama:', err);
283         showModelsNotification('Ошибка переключения: ' + err.m
essage, 'danger');
284       }
285     };
286   }
287
288   const onnxEnabledSwitch = document.getElementById('onnx-enab
led');
289   if (onnxEnabledSwitch) {
290     onnxEnabledSwitch.onchange = async () => {
```

```
      const enabled = onnxEnabledSwitch.checked;
      const remember = document.getElementById('onnx-remember')?.checked ?? true;
      const execution_provider = document.getElementById('onnx-execution-provider')?.value || 'DirectMLExecutionProvider';
      const models_dir = document.getElementById('onnx-models-dir')?.value.trim() || 'models/onnx';
      const default_model = document.getElementById('onnx-default-model')?.value.trim() || 'phi-3.5-mini-instruct-onnx';
      const olive_precision = document.getElementById('onnx-olive-precision')?.value || 'int4';

      try {
        await window.api.fetch('/api/onnx/config', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({
            enabled,
            execution_provider,
            models_dir,
            default_model,
            olive_precision,
            remember
          })
        });
        showModelsNotification(`ONNX / Olive ${enabled ? 'активирован' : 'деактивирован'}${remember ? ' (сохранено в config.json)' : ''}`, 'info');
        if (modelSelect && saveBtn) await loadTabModels(modelSelect, saveBtn);
      } catch (err) {
        console.error('Ошибка переключения ONNX / Olive:', err);
        showModelsNotification('Ошибка переключения: ' + err.message, 'danger');
      }
    };
  }
```

```
319
320   // AGY/Foundry/Ollama/ONNX config save buttons temporarily d
isabled (endpoints not implemented yet)
321   /*
322   if (saveAgyBtn) {
323     saveAgyBtn.onclick = async () => {
324       const enabled = document.getElementById('agy-enabled')?.
checked ?? true;
325       const remember = document.getElementById('agy-remember')
?.checked ?? true;
326       const model = document.getElementById('agy-model')?.valu
e || 'agy-flash';
327       const key = document.getElementById('agy-key')?.value.tr
im() || '';
328
329       saveAgyBtn.disabled = true;
330       try {
331         await window.api.fetch('/api/agy/config', {
332           method: 'POST',
333           headers: { 'Content-Type': 'application/json' },
334           body: JSON.stringify({ enabled, model, key, remember
 })
335         });
336         showModelsNotification(`Настройки Antigravity (AGY) yc
пешно сохранены${remember ? ' в config.json' : ''}`, 'success'
);
337         if (modelSelect && saveBtn) await loadTabModels(modelS
elect, saveBtn);
338       } catch (err) {
339         console.error('Ошибка сохранения Antigravity:', err);
340         showModelsNotification('Ошибка сохранения: ' + err.mes
sage, 'danger');
341       } finally {
342         saveAgyBtn.disabled = false;
343       }
344     };
345   }
346
```

```
  if (saveFoundryBtn) {
    saveFoundryBtn.onclick = async () => {
      const enabled = document.getElementById('foundry-enabled')?.checked || false;
      const remember = document.getElementById('foundry-remember')?.checked ?? true;
      const url = document.getElementById('foundry-url')?.value.trim() || 'http://localhost:54837';
      const key = document.getElementById('foundry-key')?.value.trim() || '';
      const model = document.getElementById('foundry-model')?.value?.trim() || 'qwen2.5-1.5b';

      saveFoundryBtn.disabled = true;
      try {
        await window.api.fetch('/api/foundry/config', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ enabled, url, key, model, remember })
        });
        showModelsNotification(`Настройки Microsoft Foundry успешно сохранены${remember ? ' в config.json' : ''}`, 'success');
        if (modelSelect && saveBtn) await loadTabModels(modelSelect, saveBtn);
      } catch (err) {
        console.error('Ошибка сохранения Foundry:', err);
        showModelsNotification('Ошибка сохранения: ' + err.message, 'danger');
      } finally {
        saveFoundryBtn.disabled = false;
      }
    };
  }

  if (saveOllamaBtn) {
    saveOllamaBtn.onclick = async () => {
```

```
      const enabled = document.getElementById('ollama-enabled')?.checked || false;
      const remember = document.getElementById('ollama-remember')?.checked ?? true;
      const url = document.getElementById('ollama-url')?.value.trim() || 'http://localhost:11434';
      const model = document.getElementById('ollama-model')?.value?.trim() || 'llama3.1';

      saveOllamaBtn.disabled = true;
      try {
        await window.api.fetch('/api/ollama/config', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ enabled, url, model, remember })
        });
        showModelsNotification(`Настройки Ollama успешно сохранены${remember ? ' в config.json' : ''}`, 'success');
        if (modelSelect && saveBtn) await loadTabModels(modelSelect, saveBtn);
      } catch (err) {
        console.error('Ошибка сохранения Ollama:', err);
        showModelsNotification('Ошибка сохранения: ' + err.message, 'danger');
      } finally {
        saveOllamaBtn.disabled = false;
      }
    };
  }

  if (saveOnnxBtn) {
    saveOnnxBtn.onclick = async () => {
      const enabled = document.getElementById('onnx-enabled')?.checked ?? true;
      const remember = document.getElementById('onnx-remember')?.checked ?? true;
      const execution_provider = document.getElementById('onnx
```

```
-execution-provider')?.value || 'DirectMLExecutionProvider';
      const models_dir = document.getElementById('onnx-models-dir')?.value.trim() || 'models/onnx';
      const default_model = document.getElementById('onnx-default-model')?.value.trim() || 'phi-3.5-mini-instruct-onnx';
      const olive_precision = document.getElementById('onnx-olive-precision')?.value || 'int4';

      saveOnnxBtn.disabled = true;
      try {
        await window.api.fetch('/api/onnx/config', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({
            enabled,
            execution_provider,
            models_dir,
            default_model,
            olive_precision,
            remember
          })
        });
        showModelsNotification(`Настройки ONNX / Olive успешно сохранены${remember ? ' в config.json' : ''}`, 'success');
        if (modelSelect && saveBtn) await loadTabModels(modelSelect, saveBtn);
      } catch (err) {
        console.error('Ошибка сохранения ONNX / Olive:', err);
        showModelsNotification('Ошибка сохранения: ' + err.message, 'danger');
      } finally {
        saveOnnxBtn.disabled = false;
      }
    };
  }
  */

  async function persistSelectedModelAndProvider(selectedModel
```

```
, selectedProvider, showNotify = true) {
    if (!selectedModel) return;
    const providerSelect = document.getElementById('provider-tab-select');
    const actualProvider = selectedProvider || (providerSelect ? providerSelect.value : '');

    try {
      await Promise.allSettled([
        window.api.fetch('/auth/settings', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ model: selectedModel, provider: actualProvider })
        }),
        window.api.fetch('/api/v1/tc/model', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ model: selectedModel, provider: actualProvider, save_to_config: true })
        }),
        window.api.fetch('/api/v1/tc/provider', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ provider: actualProvider, model: selectedModel, save_to_config: true })
        }),
        window.api.fetch('/api/v1/chat/model', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ model: selectedModel, provider: actualProvider })
        }),
        window.api.fetch('/api/v1/chat/provider', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ provider: actualProvider, model: selectedModel })
```

```
        })
      ]);

      // Обновляем бейдж модели по умолчанию на карточке
      const savedCardBadge = document.getElementById('saved-default-model-badge');
      if (savedCardBadge) {
        savedCardBadge.textContent = selectedModel;
        savedCardBadge.className = 'badge bg-success font-monospace px-2 py-1';
        savedCardBadge.title = `В вашем профиле и config.json сохранена модель: ${selectedModel}`;
      }

      // Обновляем верхний навигационный бейдж
      const appsModelText = document.getElementById('apps-model-text');
      if (appsModelText) {
        const cleanModelName = stripModelPrefix(selectedModel);
        appsModelText.textContent = `${(actualProvider || 'AI').toUpperCase()}: ${cleanModelName}`;
      }

      const otherModelSelect = document.getElementById('admin-model-select');
      if (otherModelSelect) {
        otherModelSelect.value = selectedModel;
      }
      if (typeof window.updateChatBadges === 'function') {
        window.updateChatBadges(selectedModel);
      }

      if (showNotify) {
        showModelsNotification(`Модель и провайдер зафиксированы в config.json: ${actualProvider} -> ${selectedModel}`, 'success');
        if (window.toast && typeof window.toast.success === 'f
```

```
unction') {
493           window.toast.success('Настройки сохранены', `Модель ${selectedModel} (${actualProvider}) сохранена в config.json`);
494         }
495       }
496     } catch (err) {
497       console.error('Ошибка сохранения модели/провайдера:', err);
498       if (showNotify) {
499         showModelsNotification('Ошибка сохранения в config.json: ' + err.message, 'danger');
500       }
501     }
502   }
503   window._persistSelectedModelAndProvider = persistSelectedModelAndProvider;
504 
505   if (saveBtn && modelSelect) {
506     saveBtn.onclick = async () => {
507       const selectedModel = modelSelect.value;
508       const providerSelect = document.getElementById('provider-tab-select');
509       const selectedProvider = providerSelect ? providerSelect.value : '';
510 
511       if (!selectedModel) {
512         showModelsNotification('Сначала выберите модель из списка', 'warning');
513         return;
514       }
515 
516       saveBtn.disabled = true;
517       const originalText = saveBtn.textContent;
518       saveBtn.innerHTML = '<span class="spinner-border spinner-border-sm me-1" role="status"></span> Сохранение...';
519 
520       try {
```

```
        await persistSelectedModelAndProvider(selectedModel, selectedProvider, true);
      } finally {
        saveBtn.disabled = false;
        saveBtn.textContent = originalText;
      }
    };
  }

  if (refreshKeysBtn) {
    refreshKeysBtn.onclick = async () => {
      refreshKeysBtn.disabled = true;
      const originalText = refreshKeysBtn.textContent;
      refreshKeysBtn.textContent = '  Сброс...';
      try {
        const res = await window.api.fetch('/api/keys/reset-all', { method: 'POST' });
        showModelsNotification(res.message || 'Квоты всех ключей успешно сброшены', 'success');
      } catch (err) {
        console.error('Ошибка сброса квот:', err);
        showModelsNotification('Ошибка сброса: ' + err.message, 'danger');
      } finally {
        refreshKeysBtn.disabled = false;
        refreshKeysBtn.textContent = originalText;
        if (keysListBody) await refreshKeysList(keysListBody);
      }
    };
  }

  if (addKeyBtn) {
    addKeyBtn.onclick = async () => {
      const nameInput = document.getElementById('new-key-name');
      const valueInput = document.getElementById('new-key-value');
      if (!nameInput || !valueInput) return;
```

```
553
554       const name = nameInput.value.trim();
555       const apiKey = valueInput.value.trim();
556
557       if (!name || !apiKey) {
558         showModelsNotification('Заполните все поля!', 'warning
');
559         return;
560       }
561
562       addKeyBtn.disabled = true;
563       try {
564         await window.api.fetch('/api/keys', {
565           method: 'POST',
566           headers: { 'Content-Type': 'application/json' },
567           body: JSON.stringify({ name, api_key: apiKey, status
: 'active' })
568         });
569         showModelsNotification(`Ключ "${name}" успешно добавле
н`, 'success');
570         nameInput.value = '';
571         valueInput.value = '';
572         if (keysListBody) await refreshKeysList(keysListBody);
573       } catch (err) {
574         console.error('Ошибка добавления ключа:', err);
575         showModelsNotification('Ошибка добавления: ' + err.mes
sage, 'danger');
576       } finally {
577         addKeyBtn.disabled = false;
578       }
579     };
580   }
581
582   if (refreshGAccountsBtn && gaccountsListBody) {
583     refreshGAccountsBtn.onclick = async () => {
584       refreshGAccountsBtn.disabled = true;
585       const originalText = refreshGAccountsBtn.innerHTML;
586       refreshGAccountsBtn.innerHTML = '<span class="spinner-bo
```

```
rder spinner-border-sm me-1"></span> Обновление...';
      try {
        await refreshGoogleAccountsList(gaccountsListBody);
        showModelsNotification('Список аккаунтов Google обновлен', 'success');
      } finally {
        refreshGAccountsBtn.disabled = false;
        refreshGAccountsBtn.innerHTML = originalText;
      }
    };
  }

  if (addGAccountBtn) {
    addGAccountBtn.onclick = async () => {
      await handleAddGoogleAccount(gaccountsListBody);
    };
  }

  // 2. Load all components concurrently
  await Promise.allSettled([
    modelSelect && saveBtn ? loadTabModels(modelSelect, saveBtn) : Promise.resolve(),
    keysListBody ? refreshKeysList(keysListBody) : Promise.resolve(),
    gaccountsListBody ? refreshGoogleAccountsList(gaccountsListBody) : Promise.resolve(),
    // loadFoundryConfig(), // Endpoint /api/foundry/config not implemented yet
    // loadOllamaConfig(), // Endpoint /api/ollama/config not implemented yet
    // loadAgyConfig(), // Endpoint /api/agy/config not implemented yet
    // loadOnnxConfig(), // Endpoint /api/onnx/config not implemented yet
    loadSystemInstruction()
  ]);
}

```

```
// Helper for escaping HTML strings
function escapeHtml(text) {
  if (!text) return '';
  return String(text)
    .replace(/&/g, '&amp;')
    .replace(/</g, '&lt;')
    .replace(/>/g, '&gt;')
    .replace(/"/g, '&quot;')
    .replace(/'/g, '&#039;');
}

// Helper to execute model test request
async function executeModelTest() {
  const providerSelect = document.getElementById('provider-tab
-select');
  const modelSelect = document.getElementById('models-tab-sele
ct');
  const promptInput = document.getElementById('model-test-prom
pt');
  const testBtn = document.getElementById('btn-model-test-send
');
  const cancelBtn = document.getElementById('btn-model-test-ca
ncel');
  const resultContainer = document.getElementById('model-test-
result');

  if (!testBtn || !resultContainer) return;

  const provider = providerSelect ? providerSelect.value : '';
  const model = modelSelect ? modelSelect.value : '';
  const message = promptInput ? promptInput.value.trim() : '';

  if (!model) {
    showModelsNotification('Выберите модель для тестирования!'
, 'warning');
    return;
  }

```

```
  if (!message) {
    showModelsNotification('Введите текст проверочного запроса
!', 'warning');
    if (promptInput) promptInput.focus();
    return;
  }

  // Abort any prior running test
  cancelModelTest();
  modelTestAbortController = new AbortController();

  testBtn.disabled = true;
  const originalBtnHtml = testBtn.innerHTML;
  testBtn.innerHTML = '<span class="spinner-border spinner-bor
der-sm me-1" role="status"></span> Тест...';

  if (cancelBtn) {
    cancelBtn.style.display = 'inline-block';
    cancelBtn.disabled = false;
  }

  resultContainer.style.display = 'block';
  resultContainer.innerHTML = `
    <div class="d-flex justify-content-between align-items-cen
ter text-info small">
      <div class="d-flex align-items-center text-truncate">
        <span class="spinner-border spinner-border-sm me-2 fle
x-shrink-0" role="status"></span>
        <span class="text-truncate">Отправка запроса в <strong
>${escapeHtml(model)}</strong> (${escapeHtml(provider)})...</s
pan>
      </div>
      <button class="btn btn-outline-danger btn-sm py-0 px-2 r
ounded ms-2 flex-shrink-0" id="btn-model-test-cancel-inner" ty
pe="button" title="Отменить проверочный запрос">
        <i class="bi bi-stop-circle me-1"></i> Отмена
      </button>
    </div>
```

```
  `;

  const cancelInnerBtn = document.getElementById('btn-model-test-cancel-inner');
  if (cancelInnerBtn) {
    cancelInnerBtn.onclick = cancelModelTest;
  }

  try {
    const res = await window.api.fetch('/api/v1/chat/test-model', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      signal: modelTestAbortController.signal,
      body: JSON.stringify({
        provider: provider,
        model: model,
        message: message
      })
    });

    if (res && res.status === 'success') {
      resultContainer.innerHTML = `
        <div class="d-flex justify-content-between align-items-center mb-1 pb-1 border-bottom">
          <div>
            <span class="badge bg-success me-2"><i class="bi bi-check-circle me-1"></i>200 OK</span>
            <span class="text-secondary small font-monospace">  ${res.duration_ms || 0} ms</span>
          </div>
          <span class="badge bg-secondary-subtle text-secondary-emphasis border font-monospace">${escapeHtml(res.model || model)}</span>
        </div>
        <div class="small mt-1 font-monospace" style="white-space: pre-wrap; word-break: break-word;">${escapeHtml(res.response || 'Пустой ответ от модели')}</div>
```

```
      `;
    } else {
      const errMsg = (res && res.message) ? res.message : 'Неизвестная ошибка при запросе к модели';
      resultContainer.innerHTML = `
        <div class="d-flex justify-content-between align-items-center mb-1 pb-1 border-bottom">
          <div>
            <span class="badge bg-danger me-2"><i class="bi bi-x-circle me-1"></i>Ошибка</span>
            <span class="text-secondary small font-monospace">  ${res?.duration_ms || 0} ms</span>
          </div>
          <span class="badge bg-secondary-subtle text-secondary-emphasis border font-monospace">${escapeHtml(model)}</span>
        </div>
        <div class="text-danger small mt-1 font-monospace" style="white-space: pre-wrap; word-break: break-word;">${escapeHtml(errMsg)}</div>
      `;
    }
  } catch (err) {
    if (err.name === 'AbortError' || err.message?.toLowerCase().includes('abort')) {
      resultContainer.innerHTML = `
        <div class="d-flex justify-content-between align-items-center mb-1 pb-1 border-bottom">
          <span class="badge bg-warning text-dark"><i class="bi bi-slash-circle me-1"></i>Отменено</span>
          <span class="badge bg-secondary-subtle text-secondary-emphasis border font-monospace">${escapeHtml(model)}</span>
        </div>
        <div class="text-warning small mt-1 font-monospace"><i class="bi bi-info-circle me-1"></i>Запрос отменен пользователем.</div>
      `;
    } else {
      console.error('Error during model test request:', err);
```

```
      resultContainer.innerHTML = `
        <div class="d-flex justify-content-between align-items-center mb-1 pb-1 border-bottom">
          <span class="badge bg-danger"><i class="bi bi-x-circle me-1"></i>Ошибка сети/API</span>
          <span class="badge bg-secondary-subtle text-secondary-emphasis border font-monospace">${escapeHtml(model)}</span>
        </div>
        <div class="text-danger small mt-1 font-monospace" style="white-space: pre-wrap; word-break: break-word;">${escapeHtml(err.message || 'Ошибка соединения')}</div>
      `;
    }
  } finally {
    modelTestAbortController = null;
    testBtn.disabled = false;
    testBtn.innerHTML = originalBtnHtml;
    if (cancelBtn) {
      cancelBtn.style.display = 'none';
    }
  }
}

// Favorite models storage in window state
window.userFavoriteModels = window.userFavoriteModels || {};

function updateModelFavoriteUI(modelSelect) {
  if (!modelSelect) modelSelect = document.getElementById('models-tab-select');
  const curModel = modelSelect ? modelSelect.value : '';
  const starIcon = document.getElementById('favorite-star-icon');
  const favBtn = document.getElementById('btn-model-toggle-favorite');
  const statusBadge = document.getElementById('model-fav-status-badge');
  const noteInput = document.getElementById('model-user-note');
```

```
759
760   const favData = curModel && window.userFavoriteModels ? wind
ow.userFavoriteModels[curModel] : null;
761   const isFav = Boolean(favData);
762
763   if (starIcon) {
764     starIcon.className = isFav ? 'bi bi-star-fill text-warning
' : 'bi bi-star';
765   }
766   if (favBtn) {
767     if (isFav) {
768       favBtn.classList.remove('btn-outline-warning');
769       favBtn.classList.add('btn-warning', 'text-dark');
770       favBtn.title = 'Удалить модель из избранного';
771     } else {
772       favBtn.classList.remove('btn-warning', 'text-dark');
773       favBtn.classList.add('btn-outline-warning');
774       favBtn.title = 'Добавить модель в избранное';
775     }
776   }
777   if (statusBadge) {
778     if (isFav) {
779       statusBadge.textContent = 'В избранном □ ';
780       statusBadge.className = 'badge bg-warning text-dark font
-monospace';
781     } else {
782       statusBadge.textContent = 'Не в избранном';
783       statusBadge.className = 'badge bg-secondary-subtle text-
secondary-emphasis border font-monospace';
784     }
785   }
786   if (noteInput && document.activeElement !== noteInput) {
787     noteInput.value = favData ? (favData.note || '') : '';
788   }
789
790   renderFavoriteModelsChips(modelSelect);
791 }
792
```

```
function updateModelSelectOptions(modelSelect) {
  if (!modelSelect) return;
  const options = modelSelect.querySelectorAll('option');
  options.forEach(opt => {
    const val = opt.value;
    if (!val) return;
    const isFav = window.userFavoriteModels && Boolean(window.
userFavoriteModels[val]);
    let text = opt.textContent.replace(/^ \s*/, '');
    if (isFav) {
      opt.textContent = `  ${text}`;
    } else {
      opt.textContent = text;
    }
  });
}

function renderFavoriteModelsChips(modelSelect) {
  const container = document.getElementById('favorite-models-l
ist');
  const section = document.getElementById('favorite-models-sec
tion');
  if (!container || !section) return;

  const favKeys = Object.keys(window.userFavoriteModels || {})
;
  if (favKeys.length === 0) {
    section.style.display = 'none';
    container.innerHTML = '';
    return;
  }

  section.style.display = 'block';
  container.innerHTML = '';

  favKeys.forEach(modelName => {
    const data = window.userFavoriteModels[modelName] || {};
    const note = data.note ? `\nЗаметка: ${data.note}` : '';
```

```
827
828     const chip = document.createElement('div');
829     chip.className = 'btn-group btn-group-sm mb-1';
830     chip.role = 'group';
831
832     const btnSelect = document.createElement('button');
833     btnSelect.type = 'button';
834     btnSelect.className = modelSelect && modelSelect.value ===
 modelName
835       ? 'btn btn-sm btn-warning text-dark py-0 px-2 font-monos
pace'
836       : 'btn btn-sm btn-outline-warning py-0 px-2 font-monospa
ce';
837     btnSelect.innerHTML = `<i class="bi bi-star-fill me-1 text
-warning"></i>${escapeHtml(modelName)}`;
838     btnSelect.title = `Выбрать модель: ${modelName}${note}`;
839     btnSelect.onclick = async () => {
840       if (modelSelect) {
841         // Try finding which provider has this model
842         const providerSelect = document.getElementById('provid
er-tab-select');
843         let matchedProv = providerSelect ? providerSelect.valu
e : '';
844         if (providerSelect && window._modelsGrouped) {
845           for (const p of Object.keys(window._modelsGrouped))
{
846             if (window._modelsGrouped[p]?.includes(modelName))
 {
847               matchedProv = p;
848               if (providerSelect.value !== p) {
849                 providerSelect.value = p;
850                 if (typeof window._populateModels === 'functio
n') {
851                   window._populateModels(p, window._modelsGrou
ped[p]);
852                 }
853               }
854               break;
```

```
            }
          }
        }
        modelSelect.value = modelName;
        updateModelFavoriteUI(modelSelect);
        await persistSelectedModelAndProvider(modelName, match
edProv, true);
      }
    };

    const btnRemove = document.createElement('button');
    btnRemove.type = 'button';
    btnRemove.className = 'btn btn-sm btn-outline-danger py-0
px-1';
    btnRemove.innerHTML = '<i class="bi bi-x"></i>';
    btnRemove.title = `Удалить ${modelName} из избранного`;
    btnRemove.onclick = async (e) => {
      e.stopPropagation();
      try {
        await window.api.fetch(`/auth/favorites/${encodeURICom
ponent(modelName)}`, { method: 'DELETE' });
        delete window.userFavoriteModels[modelName];
        updateModelFavoriteUI(modelSelect);
        updateModelSelectOptions(modelSelect);
        showModelsNotification(`Модель "${modelName}" удалена
из избранного`, 'info');
      } catch (err) {
        showModelsNotification('Ошибка удаления: ' + err.messa
ge, 'danger');
      }
    };

    chip.appendChild(btnSelect);
    chip.appendChild(btnRemove);
    container.appendChild(chip);
  });
}

```

```
// Вспомогательная функция очистки префиксов моделей для сравнения и отображения
function stripModelPrefix(name) {
  if (!name) return '';
  return String(name)
    .replace(/^(gemini|gemini_cli|foundry|ollama|openai|hf|onnx):+/i, '')
    .replace(/^agy-+/i, '')
    .trim();
}

// Helper to load models list
async function loadTabModels(modelSelect, saveBtn, forceRefresh = false) {
  const providerSelect = document.getElementById('provider-tab-select');
  if (providerSelect) providerSelect.innerHTML = '';
  modelSelect.innerHTML = '';

  let modelsGrouped = {};
  let unsupportedGrouped = {};

  const fetchModels = async (force = false) => {
    try {
      const showAll = document.getElementById('show-all-models-check')?.checked ?? false;
      const params = new URLSearchParams();
      if (force) params.append('refresh', 'true');
      if (showAll) params.append('include_unsupported', 'true');
      const url = '/api/v1/chat/models' + (params.toString() ? '?' + params.toString() : '');
      const modelsData = await window.api.fetch(url);
      let grouped = modelsData.models || {};
      if (Array.isArray(grouped)) {
        grouped = { 'gemini': grouped };
      }
      unsupportedGrouped = modelsData.unsupported_models || {}
```

```
;
      return grouped;
    } catch (err) {
      console.error('Error loading AI models:', err);
      showModelsNotification('Ошибка загрузки моделей AI: ' +
err.message, 'danger');
      return {};
    }
  };

  modelsGrouped = await fetchModels(forceRefresh);
  window._modelsGrouped = modelsGrouped;
  if (forceRefresh) {
    showModelsNotification('Список моделей успешно обновлен',
'success');
  }

  // Загружаем избранные модели
  try {
    const settingsData = await window.api.fetch('/auth/setting
s');
    if (settingsData && settingsData.favorite_models) {
      window.userFavoriteModels = settingsData.favorite_models
;
    } else {
      const favRes = await window.api.fetch('/auth/favorites')
;
      if (favRes && favRes.favorites) {
        window.userFavoriteModels = favRes.favorites;
      }
    }
  } catch (e) {
    console.warn('Failed to load favorite models:', e);
  }

  // Получаем список доступных провайдеров
  const providers = Object.keys(modelsGrouped).filter(p => mod
elsGrouped[p] && modelsGrouped[p].length > 0);
```

```
950
951   if (providers.length === 0) {
952     if (providerSelect) {
953       providerSelect.innerHTML = '<option value="">Нет доступн
ых провайдеров</option>';
954     }
955     modelSelect.innerHTML = '<option value="">Нет доступных мо
делей</option>';
956     saveBtn.disabled = true;
957     updateModelFavoriteUI(modelSelect);
958     return;
959   }
960
961   if (providerSelect) {
962     providerSelect.innerHTML = '';
963     providers.forEach(p => {
964       const option = document.createElement('option');
965       option.value = p;
966       option.textContent = p.charAt(0).toUpperCase() + p.slice
(1);
967       providerSelect.appendChild(option);
968     });
969   }
970
971   const populateModels = (provider, providerModelsList) => {
972     modelSelect.innerHTML = '';
973     const providerModels = providerModelsList !== undefined ?
providerModelsList : (modelsGrouped[provider] || []);
974     const unsupList = unsupportedGrouped[provider] || [];
975     if (!providerModels || providerModels.length === 0) {
976       modelSelect.innerHTML = '<option value="">Нет моделей</o
ption>';
977       saveBtn.disabled = true;
978     } else {
979       providerModels.forEach(modelName => {
980         const option = document.createElement('option');
981         option.value = modelName;
982         let cleanName = stripModelPrefix(modelName);
```

```
983
984         const isFav = window.userFavoriteModels && Boolean(win
dow.userFavoriteModels[modelName]);
985         const isUnsupported = unsupList.includes(cleanName) ||
 unsupList.includes(modelName);
986
987         let label = cleanName || modelName;
988         if (isFav) label = `□ ${label}`;
989         if (isUnsupported) label = `${label} □ [отфильтрована]
`;
990
991         option.textContent = label;
992         modelSelect.appendChild(option);
993       });
994       saveBtn.disabled = false;
995     }
996     updateModelFavoriteUI(modelSelect);
997   };
998   window._populateModels = populateModels;
999
1000   if (providerSelect) {
1001     providerSelect.onchange = async () => {
1002       const chosenProvider = providerSelect.value;
1003       populateModels(chosenProvider, modelsGrouped[chosenProvi
der]);
1004       if (modelSelect.options.length > 0) {
1005         modelSelect.selectedIndex = 0;
1006       }
1007       saveBtn.disabled = !modelSelect.value;
1008       updateModelFavoriteUI(modelSelect);
1009       if (modelSelect.value) {
1010         await persistSelectedModelAndProvider(modelSelect.valu
e, chosenProvider, true);
1011       }
1012     };
1013   } else {
1014     let allModels = [];
1015     providers.forEach(p => allModels = allModels.concat(models
```

```
Grouped[p]));
1016      allModels.forEach(modelName => {
1017        const option = document.createElement('option');
1018        option.value = modelName;
1019        const isFav = window.userFavoriteModels && Boolean(windo
w.userFavoriteModels[modelName]);
1020        option.textContent = isFav ? `  ${modelName}` : modelNam
e;
1021        modelSelect.appendChild(option);
1022      });
1023      saveBtn.disabled = allModels.length === 0;
1024      updateModelFavoriteUI(modelSelect);
1025    }
1026
1027    // Запрашиваем активную модель и провайдер
1028    let activeProvider = '';
1029    let activeModel = '';
1030
1031    try {
1032      const isTcRoute = window.location.pathname.startsWith('/tc
') || window.location.pathname.startsWith('/apps');
1033      const query = isTcRoute ? '?profile=tc' : '';
1034
1035      // 1. Приоритет: активная модель из /api/v1/chat/active-mo
del или /api/v1/tc/model
1036      try {
1037        const activeResp = await window.api.fetch(`/api/v1/chat/
active-model${query}`);
1038        if (activeResp && activeResp.model) {
1039          activeModel = activeResp.model;
1040          activeProvider = activeResp.provider || '';
1041        }
1042      } catch (e) {
1043        console.debug('Failed to fetch /api/v1/chat/active-model
:', e);
1044      }
1045
1046      // 2. Резерв: /auth/settings
```

```
1047    if (!activeModel) {
1048      const settingsData = await window.api.fetch('/auth/setti
    ngs');
1049      if (settingsData && settingsData.model) {
1050        activeModel = settingsData.model;
1051        if (settingsData.provider) activeProvider = settingsDa
    ta.provider;
1052      }
1053    }
1054  } catch (err) {
1055    console.error('Error loading active model settings:', err)
    ;
1056  }
1057
1058  // Интеллектуальное сопоставление (Smart Match)
1059  let targetProvider = activeProvider ? activeProvider.toLower
    Case() : '';
1060  const cleanActiveModel = stripModelPrefix(activeModel);
1061
1062  if (!targetProvider && activeModel) {
1063    if (activeModel.startsWith('foundry:')) targetProvider = '
    foundry';
1064    else if (activeModel.startsWith('ollama:')) targetProvider
     = 'ollama';
1065    else if (activeModel.startsWith('agy-')) targetProvider =
    'agy';
1066    else if (activeModel.startsWith('gemini_cli:')) targetProv
    ider = 'gemini_cli';
1067    else if (activeModel.startsWith('hf:') || activeModel.star
    tsWith('hf::')) targetProvider = 'hf';
1068    else if (activeModel.startsWith('onnx:') || activeModel.st
    artsWith('onnx::')) targetProvider = 'onnx';
1069    else if (activeModel.toLowerCase().includes('gemini')) tar
    getProvider = 'gemini';
1070  }
1071
1072  // Поиск подходящего провайдера в списке providers
1073  let matchedProvider = providers.find(p => p.toLowerCase() ==
```

```
= targetProvider);
1074
1075  if (!matchedProvider && activeModel) {
1076    for (const p of providers) {
1077      const list = modelsGrouped[p] || [];
1078      if (list.some(m => m === activeModel || stripModelPrefix
(m) === cleanActiveModel)) {
1079        matchedProvider = p;
1080        break;
1081      }
1082    }
1083  }
1084
1085  if (!matchedProvider) {
1086    matchedProvider = providers.includes('gemini') ? 'gemini'
: providers[0];
1087  }
1088
1089  // Устанавливаем провайдер в селект и заполняем список модел
ей
1090  if (providerSelect) {
1091    providerSelect.value = matchedProvider;
1092  }
1093  populateModels(matchedProvider, modelsGrouped[matchedProvide
r]);
1094
1095  // Выбираем подходящую модель в дропдауне
1096  let matchedModelVal = null;
1097  const currentOptions = Array.from(modelSelect.options);
1098
1099  if (activeModel) {
1100    // 1. Прямое совпадение
1101    const exactOpt = currentOptions.find(opt => opt.value ===
activeModel);
1102    if (exactOpt) {
1103      matchedModelVal = exactOpt.value;
1104    } else {
1105      // 2. Совпадение без префикса
```

```
      const cleanOpt = currentOptions.find(opt => stripModelPrefix(opt.value) === cleanActiveModel);
      if (cleanOpt) {
        matchedModelVal = cleanOpt.value;
      }
    }
  }

  if (matchedModelVal) {
    modelSelect.value = matchedModelVal;
  } else if (currentOptions.length > 0) {
    modelSelect.selectedIndex = 0;
  }

  saveBtn.disabled = !modelSelect.value;
  updateModelFavoriteUI(modelSelect);

  // Обновляем бейдж дефолтной модели
  const effectiveModelName = modelSelect.value || activeModel;
  const savedCardBadge = document.getElementById('saved-default-model-badge');
  if (savedCardBadge) {
    if (effectiveModelName) {
      savedCardBadge.textContent = effectiveModelName;
      savedCardBadge.className = 'badge bg-success font-monospace px-2 py-1';
      savedCardBadge.title = `В вашем профиле сохранена модель: ${effectiveModelName}`;
    } else {
      savedCardBadge.textContent = 'Не задана (системный fallback)';
      savedCardBadge.className = 'badge bg-secondary font-monospace px-2 py-1';
    }
  }

  if (typeof window.updateChatBadges === 'function') {
    window.updateChatBadges(effectiveModelName);
```

```
  }
}

// Helper to refresh keys list table
async function refreshKeysList(container) {
  try {
    container.innerHTML = '<tr><td colspan="5" class="text-cen
ter py-4 text-muted"><span class="spinner-border spinner-borde
r-sm me-2"></span>Загрузка ключей...</td></tr>';
    const keysData = await window.api.fetch('/api/keys/');
    const keys = Array.isArray(keysData) ? keysData : (keysDat
a.keys || []);

    if (keys.length === 0) {
      container.innerHTML = '<tr><td colspan="5" class="text-c
enter py-4 text-muted">Список ключей пуст</td></tr>';
      return;
    }

    container.innerHTML = '';
    keys.forEach(key => {
      const row = document.createElement('tr');
      row.style.cursor = 'pointer';
      row.title = `Нажмите для проверки ключа "${key.name}" (о
тправка hello world)`;

      const isCurrentActive = Boolean(key.is_active);
      const isExhausted = Boolean(key.exhausted || key.status
=== 'exhausted' || key.exhausted_at);
      const isEnabled = key.status === 'active';

      if (isCurrentActive) {
        row.className = 'table-success border border-2 border-
success align-middle';
        row.style.boxShadow = 'inset 0 0 0 1px var(--bs-succes
s, #198754)';
      } else {
        row.className = 'align-middle';
```

```
1168        }
1169
1170        // Клик по строке (кроме кнопок действий) открывает мода
    льное окно проверки
1171        row.addEventListener('click', (e) => {
1172          if (e.target.closest('button') || e.target.closest('a'
    ) || e.target.closest('input')) return;
1173          openKeyTestModal(key, container);
1174        });
1175
1176        // 1. Имя ключа
1177        const tdName = document.createElement('td');
1178        if (isCurrentActive) {
1179          tdName.innerHTML = `<strong class="text-success">${esc
    apeHtml(key.name)}</strong> <span class="badge bg-success font
    -monospace ms-1" style="font-size: 0.72rem;"><i class="bi bi-c
    heck-circle-fill me-1"></i>Активен</span>`;
1180        } else {
1181          tdName.innerHTML = `<strong>${escapeHtml(key.name)}</s
    trong>`;
1182        }
1183        row.appendChild(tdName);
1184
1185        // 2. Токен (маскированный)
1186        const tdKey = document.createElement('td');
1187        tdKey.className = 'font-monospace text-muted small';
1188        tdKey.textContent = key.masked_key || key.api_key_masked
     || '';
1189        row.appendChild(tdKey);
1190
1191        // 3. Состояние ключа
1192        const tdStatus = document.createElement('td');
1193        if (isCurrentActive) {
1194          tdStatus.innerHTML = `<span class="badge bg-success" t
    itle="Используется моделью в данный момент"><i class="bi bi-pl
    ay-circle-fill me-1"></i>Используется</span>`;
1195        } else if (isExhausted) {
1196          tdStatus.innerHTML = `<span class="badge bg-danger" ti
```

```
tle="Превышена суточная квота запросов"><i class="bi bi-exclam
ation-octagon me-1"></i>Превысил квоту</span>`;
1197        } else if (isEnabled) {
1198          tdStatus.innerHTML = `<span class="badge bg-primary-su
btle text-primary border border-primary" title="Включен в резе
рве/пуле ротации"><i class="bi bi-check2 me-1"></i>Включен</sp
an>`;
1199        } else {
1200          tdStatus.innerHTML = `<span class="badge bg-secondary"
 title="Отключен пользователем"><i class="bi bi-slash-circle m
e-1"></i>Выключен</span>`;
1201        }
1202        row.appendChild(tdStatus);
1203
1204        // 4. Суточная квота (24ч)
1205        const tdQuota = document.createElement('td');
1206        if (isExhausted) {
1207          let resetText = 'Лимит исчерпан';
1208          if (key.reset_in_seconds) {
1209            const hours = Math.floor(key.reset_in_seconds / 3600
);
1210            const mins = Math.floor((key.reset_in_seconds % 3600
) / 60);
1211            resetText = `Сброс через ${hours}ч ${mins}м`;
1212          }
1213          tdQuota.innerHTML = `<span class="badge bg-danger d-bl
ock mb-1" title="Превышен лимит запросов в сутки">${resetText}
</span>`;
1214        } else {
1215          tdQuota.innerHTML = `<span class="badge bg-success-sub
tle text-success border border-success d-block mb-1">OK</span>
`;
1216        }
1217        row.appendChild(tdQuota);
1218
1219        // 5. Действия
1220        const tdActions = document.createElement('td');
1221        tdActions.className = 'text-end text-nowrap';
```

```
1222
1223        // Кнопка или бейдж "Сделать активным" / "Активен"
1224        if (isCurrentActive) {
1225          const activeBadge = document.createElement('span');
1226          activeBadge.className = 'badge bg-success text-white p
    y-1 px-2 me-1 font-monospace';
1227          activeBadge.innerHTML = '<i class="bi bi-star-fill tex
    t-warning me-1"></i>Активен';
1228          activeBadge.title = 'Этот ключ в данный момент использ
    уется моделью';
1229          tdActions.appendChild(activeBadge);
1230        } else {
1231          const btnMakeActive = document.createElement('button')
    ;
1232          btnMakeActive.className = 'btn btn-xs btn-outline-succ
    ess btn-sm me-1';
1233          btnMakeActive.innerHTML = '<i class="bi bi-star me-1">
    </i>Сделать активным';
1234          btnMakeActive.title = `Назначить ключ "${key.name}" ак
    тивным (использовать в данный момент)`;
1235          btnMakeActive.onclick = async (e) => {
1236            e.stopPropagation();
1237            await activateApiKey(key.name, container);
1238          };
1239          tdActions.appendChild(btnMakeActive);
1240        }
1241
1242        const btnTest = document.createElement('button');
1243        btnTest.className = 'btn btn-xs btn-outline-info btn-sm
    me-1';
1244        btnTest.innerHTML = '<i class="bi bi-lightning-charge"><
    /i> Тест';
1245        btnTest.title = `Проверить валидность ключа "${key.name}
    "`;
1246        btnTest.onclick = (e) => {
1247          e.stopPropagation();
1248          openKeyTestModal(key, container);
1249        };
```

```
      tdActions.appendChild(btnTest);

      const btnToggle = document.createElement('button');
      btnToggle.className = `btn btn-xs btn-sm me-1 ${isEnable
d ? 'btn-outline-secondary' : 'btn-outline-success'}`;
      btnToggle.textContent = isEnabled ? 'Откл' : 'Вкл';
      btnToggle.onclick = (e) => {
        e.stopPropagation();
        toggleKeyStatus(key.name, isEnabled ? 'disabled' : 'ac
tive', container);
      };
      tdActions.appendChild(btnToggle);

      if (isExhausted) {
        const btnReset = document.createElement('button');
        btnReset.className = 'btn btn-xs btn-outline-warning b
tn-sm me-1';
        btnReset.innerHTML = 'Сброс';
        btnReset.title = 'Сбросить 24-часовой бан квоты';
        btnReset.onclick = (e) => {
          e.stopPropagation();
          resetKeyQuota(key.name, container);
        };
        tdActions.appendChild(btnReset);
      }

      const btnDelete = document.createElement('button');
      btnDelete.className = 'btn btn-xs btn-outline-danger btn
-sm';
      btnDelete.textContent = 'Удалить';
      btnDelete.onclick = (e) => {
        e.stopPropagation();
        deleteKey(key.name, container);
      };
      tdActions.appendChild(btnDelete);

      row.appendChild(tdActions);
      container.appendChild(row);
```

```
    });

  } catch (err) {
    console.error('Ошибка загрузки ключей:', err);
    container.innerHTML = `<tr><td colspan="5" class="text-cen
ter py-4 text-danger">Ошибка: ${err.message}</td></tr>`;
  }
}

async function activateApiKey(name, container) {
  try {
    await window.api.fetch(`/api/keys/${encodeURIComponent(nam
e)}/activate`, {
      method: 'POST'
    });
    showModelsNotification(`Ключ "${name}" успешно назначен ак
тивным`, 'success');
    if (window.toast && typeof window.toast.success === 'funct
ion') {
      window.toast.success('Ключ активирован', `Ключ "${name}"
 теперь активен для генерации`);
    }
    if (container) await refreshKeysList(container);
  } catch (err) {
    console.error('Ошибка активации ключа:', err);
    showModelsNotification('Ошибка активации ключа: ' + err.me
ssage, 'danger');
  }
}

// Модальное окно проверки валидности ключа
async function openKeyTestModal(key, container) {
  const modalEl = document.getElementById('modal-test-key');
  if (!modalEl) return;

  const keyNameEl = document.getElementById('modal-test-key-na
me');
  const keyMaskedEl = document.getElementById('modal-test-key-
```

```
masked');
  const keyStatusEl = document.getElementById('modal-test-key-status-badge');
  const keyDurationEl = document.getElementById('modal-test-key-duration');
  const resultBox = document.getElementById('modal-test-key-result-box');
  const promptInput = document.getElementById('modal-test-key-prompt');
  const modelSelect = document.getElementById('modal-test-key-model');
  const retryBtn = document.getElementById('modal-test-key-retry-btn');

  if (keyNameEl) keyNameEl.textContent = key.name || '-';
  if (keyMaskedEl) keyMaskedEl.textContent = key.masked_key || key.api_key_masked || '****************';
  if (keyStatusEl) {
    const isAct = key.status === 'active';
    keyStatusEl.className = `badge ${isAct ? 'bg-success' : 'bg-secondary'}`;
    keyStatusEl.textContent = isAct ? 'Активен' : 'Отключен';
  }
  if (keyDurationEl) keyDurationEl.textContent = '□ -- ms';
  if (resultBox) resultBox.style.display = 'none';

  // Инициализация Bootstrap Modal
  if (typeof bootstrap !== 'undefined' && bootstrap.Modal) {
    const modalInstance = bootstrap.Modal.getOrCreateInstance(modalEl);
    modalInstance.show();
  }

  // Привязка кнопки повторного теста
  if (retryBtn) {
    retryBtn.onclick = () => runKeyTest(key.name, promptInput?.value || 'hello world', modelSelect?.value || 'gemini-3.1-flash-lite', container);
```

```
1341   }
1342
1343   // Автоматическая отправка проверочного запроса 'hello world
'
1344   await runKeyTest(key.name, promptInput?.value || 'hello worl
d', modelSelect?.value || 'gemini-3.1-flash-lite', container);
1345 }
1346
1347 async function runKeyTest(keyName, message, model, container)
{
1348   const progressEl = document.getElementById('modal-test-key-p
rogress');
1349   const resultBox = document.getElementById('modal-test-key-re
sult-box');
1350   const resultContent = document.getElementById('modal-test-ke
y-result-content');
1351   const validBadge = document.getElementById('modal-test-key-v
alid-badge');
1352   const keyDurationEl = document.getElementById('modal-test-ke
y-duration');
1353   const retryBtn = document.getElementById('modal-test-key-ret
ry-btn');
1354
1355   if (progressEl) progressEl.style.display = 'block';
1356   if (resultBox) resultBox.style.display = 'none';
1357   if (retryBtn) retryBtn.disabled = true;
1358
1359   try {
1360     const res = await window.api.fetch(`/api/keys/${encodeURIC
omponent(keyName)}/test`, {
1361       method: 'POST',
1362       headers: { 'Content-Type': 'application/json' },
1363       body: JSON.stringify({ message: message || 'hello world'
, model: model || 'gemini-3.1-flash-lite' }),
1364     });
1365
1366     if (progressEl) progressEl.style.display = 'none';
1367     if (resultBox) resultBox.style.display = 'block';
```

```
1368    if (retryBtn) retryBtn.disabled = false;
1369
1370    if (keyDurationEl && res && res.duration_ms !== undefined)
     {
1371      keyDurationEl.textContent = `  ${res.duration_ms} ms`;
1372    }
1373
1374    if (res && res.status === 'success' && res.valid) {
1375      if (validBadge) {
1376        validBadge.className = 'badge bg-success';
1377        validBadge.innerHTML = '<i class="bi bi-check-circle m
     e-1"></i> Валиден (OK)';
1378      }
1379      if (resultContent) {
1380        resultContent.className = 'p-2.5 rounded font-monospac
     e small border border-success-subtle bg-success-subtle text-su
     ccess-emphasis';
1381        resultContent.textContent = res.response || '(Пустой о
     твет)';
1382      }
1383      showModelsNotification(`Ключ "${keyName}" успешно прошел
      проверку (${res.duration_ms} мс)`, 'success');
1384    } else {
1385      if (validBadge) {
1386        validBadge.className = 'badge bg-danger';
1387        validBadge.innerHTML = '<i class="bi bi-x-circle me-1"
     ></i> Ошибка валидации';
1388      }
1389      if (resultContent) {
1390        resultContent.className = 'p-2.5 rounded font-monospac
     e small border border-danger-subtle bg-danger-subtle text-dang
     er-emphasis';
1391        resultContent.textContent = (res && (res.error || res.
     message)) ? (res.error || res.message) : 'Неизвестная ошибка в
     алидации ключа';
1392      }
1393      showModelsNotification(`Ошибка проверки ключа "${keyName
     }": ${(res && res.error) ? res.error : 'Ошибка API'}`, 'danger
```

```
');
1394      }
1395
1396      // Обновляем список ключей для синхронизации статуса и кво
ты
1397      if (container) {
1398        await refreshKeysList(container);
1399      }
1400    } catch (err) {
1401      console.error('Ошибка проверки ключа:', err);
1402      if (progressEl) progressEl.style.display = 'none';
1403      if (resultBox) resultBox.style.display = 'block';
1404      if (retryBtn) retryBtn.disabled = false;
1405      if (validBadge) {
1406        validBadge.className = 'badge bg-danger';
1407        validBadge.innerHTML = '<i class="bi bi-exclamation-tria
ngle me-1"></i> Сбой сети / API';
1408      }
1409      if (resultContent) {
1410        resultContent.className = 'p-2.5 rounded font-monospace
small border border-danger-subtle bg-danger-subtle text-danger
-emphasis';
1411        resultContent.textContent = err.message || String(err);
1412      }
1413      showModelsNotification('Сбой запроса проверки ключа: ' + e
rr.message, 'danger');
1414    }
1415  }
1416
1417  // API action helpers
1418  async function toggleKeyStatus(name, newStatus, container) {
1419    try {
1420      await window.api.fetch(`/api/keys/${name}`, {
1421        method: 'PATCH',
1422        headers: { 'Content-Type': 'application/json' },
1423        body: JSON.stringify({ status: newStatus })
1424      });
1425      showModelsNotification(`Статус ключа "${name}" изменен на
```

```
${newStatus === 'active' ? 'активный' : 'отключенный'}`, 'succ
ess');
1426      await refreshKeysList(container);
1427    } catch (err) {
1428      console.error('Ошибка переключения статуса ключа:', err);
1429      showModelsNotification('Ошибка изменения статуса: ' + err.
message, 'danger');
1430    }
1431  }
1432
1433  async function resetKeyQuota(name, container) {
1434    try {
1435      await window.api.fetch(`/api/keys/${name}/reset-quota`, {
method: 'POST' });
1436      showModelsNotification(`Квота для ключа "${name}" успешно
сброшена`, 'success');
1437      await refreshKeysList(container);
1438    } catch (err) {
1439      console.error('Ошибка сброса квоты:', err);
1440      showModelsNotification('Ошибка сброса квоты: ' + err.messa
ge, 'danger');
1441    }
1442  }
1443
1444  async function deleteKey(name, container) {
1445    if (!confirm(`Вы уверены, что хотите удалить ключ "${name}"?
`)) return;
1446    try {
1447      await window.api.fetch(`/api/keys/${name}`, { method: 'DEL
ETE' });
1448      showModelsNotification(`Ключ "${name}" успешно удален`, 's
uccess');
1449      await refreshKeysList(container);
1450    } catch (err) {
1451      console.error('Ошибка удаления ключа:', err);
1452      showModelsNotification('Ошибка удаления: ' + err.message,
'danger');
1453    }
```

```
}

function showModelsNotification(message, type = 'info') {
  const notification = document.createElement('div');
  notification.className = `alert alert-${type} position-fixed
 top-0 end-0 m-3`;
  notification.style.zIndex = '9999';
  notification.style.maxWidth = '400px';
  notification.textContent = message;
  document.body.appendChild(notification);

  setTimeout(() => {
    notification.remove();
  }, 5000);
}

async function loadFoundryConfig() {
  try {
    const config = await window.api.fetch('/api/foundry/config
');
    const enabledInput = document.getElementById('foundry-enab
led');
    const urlInput = document.getElementById('foundry-url');
    const keyInput = document.getElementById('foundry-key');
    const modelInput = document.getElementById('foundry-model'
);

    if (enabledInput) enabledInput.checked = config.enabled ||
 false;
    if (urlInput) urlInput.value = config.url || '';
    if (keyInput) keyInput.value = config.key || '';
    if (modelInput) modelInput.value = config.model || '';
  } catch (err) {
    console.error('Ошибка загрузки настроек Foundry:', err);
  }
}

async function loadOllamaConfig() {
```

```
  try {
    const config = await window.api.fetch('/api/ollama/config');
    const enabledInput = document.getElementById('ollama-enabled');
    const urlInput = document.getElementById('ollama-url');
    const modelInput = document.getElementById('ollama-model');

    if (enabledInput) enabledInput.checked = config.enabled || false;
    if (urlInput) urlInput.value = config.url || '';
    if (modelInput) modelInput.value = config.model || '';
  } catch (err) {
    console.error('Ошибка загрузки настроек Ollama:', err);
  }
}

async function loadAgyConfig() {
  try {
    const modelSelect = document.getElementById('agy-model');
    if (modelSelect) {
      try {
        const modelsData = await window.api.fetch('/api/v1/chat/models');
        const agyList = modelsData.models?.agy || [];
        if (agyList.length > 0) {
          const curVal = modelSelect.value;
          modelSelect.innerHTML = '';
          agyList.forEach(m => {
            const opt = document.createElement('option');
            opt.value = m;
            opt.textContent = m;
            modelSelect.appendChild(opt);
          });
          if (curVal && agyList.includes(curVal)) {
            modelSelect.value = curVal;
          }
```

```
        }
      } catch (e) {
        console.error('Ошибка загрузки моделей AGY:', e);
      }
    }

    const config = await window.api.fetch('/api/agy/config');
    const enabledInput = document.getElementById('agy-enabled'
);
    const keyInput = document.getElementById('agy-key');

    if (enabledInput) enabledInput.checked = config.enabled ??
 true;
    if (modelSelect && config.model) modelSelect.value = confi
g.model;
    if (keyInput) keyInput.value = config.key || '';
  } catch (err) {
    console.error('Ошибка загрузки настроек Antigravity (AGY):
', err);
  }
}

async function loadOnnxConfig() {
  try {
    const config = await window.api.fetch('/api/onnx/config');
    const enabledInput = document.getElementById('onnx-enabled
');
    const epSelect = document.getElementById('onnx-execution-p
rovider');
    const dirInput = document.getElementById('onnx-models-dir'
);
    const modelInput = document.getElementById('onnx-default-m
odel');
    const precSelect = document.getElementById('onnx-olive-pre
cision');

    if (enabledInput) enabledInput.checked = config.enabled ??
 true;
```

```
    if (epSelect && config.execution_provider) epSelect.value = config.execution_provider;
    if (dirInput && config.models_dir) dirInput.value = config.models_dir;
    if (modelInput && config.default_model) modelInput.value = config.default_model;
    if (precSelect && config.olive_precision) precSelect.value = config.olive_precision;
  } catch (err) {
    console.error('Ошибка загрузки настроек ONNX / Olive:', err);
  }
}

async function loadSystemInstruction() {
  const editor = document.getElementById('system-instruction-editor');
  const statusBadge = document.getElementById('system-instruction-status');
  if (!editor) return;

  editor.disabled = true;
  if (statusBadge) {
    statusBadge.className = 'badge bg-warning text-dark';
    statusBadge.textContent = 'Загрузка...';
    statusBadge.style.removeProperty('display');
  }

  try {
    let instructionText = '';
    const endpoints = [
      '/api/v1/tc/model-instruction',
      '/api/v1/chat/model-instruction',
      '/api/admin/system_instruction',
      '/api/v1/admin/system_instruction'
    ];

    for (const ep of endpoints) {
```

```
      try {
        const data = await window.api.fetch(ep);
        if (data && (data.instruction || data.system_instructi
on || data.content)) {
          instructionText = data.instruction || data.system_in
struction || data.content;
          break;
        }
      } catch {}
    }

    editor.value = instructionText || '';
    if (statusBadge) {
      statusBadge.className = 'badge bg-success';
      statusBadge.textContent = 'Загружено';
      setTimeout(() => {
        if (statusBadge) statusBadge.style.display = 'none';
      }, 2500);
    }
  } catch (err) {
    console.error('Ошибка загрузки системной инструкции:', err
);
    if (statusBadge) {
      statusBadge.className = 'badge bg-danger';
      statusBadge.textContent = 'Ошибка';
      statusBadge.style.removeProperty('display');
    }
    showModelsNotification('Ошибка загрузки системной инструкц
ии: ' + err.message, 'danger');
  } finally {
    editor.disabled = false;
  }
}

async function saveSystemInstruction() {
  const editor = document.getElementById('system-instruction-e
ditor');
  const saveBtn = document.getElementById('btn-save-instructio
```

```
n');
1612  const statusBadge = document.getElementById('system-instruct
ion-status');
1613  if (!editor || !saveBtn) return;
1614
1615  const originalHtml = saveBtn.innerHTML;
1616  saveBtn.disabled = true;
1617  saveBtn.innerHTML = '<span class="spinner-border spinner-bor
der-sm" role="status"></span> Сохранение...';
1618
1619  try {
1620    const textVal = editor.value;
1621    await Promise.allSettled([
1622      window.api.fetch('/api/v1/tc/model-instruction', {
1623        method: 'POST',
1624        headers: { 'Content-Type': 'application/json' },
1625        body: JSON.stringify({ instruction: textVal, content:
textVal, system_instruction: textVal, save_to_disk: true })
1626      }),
1627      window.api.fetch('/api/v1/chat/model-instruction', {
1628        method: 'POST',
1629        headers: { 'Content-Type': 'application/json' },
1630        body: JSON.stringify({ instruction: textVal, content:
textVal, system_instruction: textVal, save_to_disk: true })
1631      }),
1632      window.api.fetch('/api/admin/system_instruction', {
1633        method: 'POST',
1634        headers: { 'Content-Type': 'application/json' },
1635        body: JSON.stringify({ content: textVal })
1636      })
1637    ]);
1638
1639    showModelsNotification('  Системная инструкция успешно сох
ранена', 'success');
1640    if (statusBadge) {
1641      statusBadge.className = 'badge bg-success';
1642      statusBadge.textContent = 'Сохранено';
1643      statusBadge.style.removeProperty('display');
```

```
      setTimeout(() => {
        if (statusBadge) statusBadge.style.display = 'none';
      }, 3000);
    }
    if (window.toast && typeof window.toast.success === 'funct
ion') {
      window.toast.success('Инструкция сохранена', 'Системная
инструкция обновлена для всех моделей');
    }
  } catch (err) {
    console.error('Ошибка сохранения системной инструкции:', e
rr);
    if (statusBadge) {
      statusBadge.className = 'badge bg-danger';
      statusBadge.textContent = 'Ошибка сохранения';
      statusBadge.style.removeProperty('display');
    }
    showModelsNotification('Ошибка сохранения: ' + err.message
, 'danger');
  } finally {
    saveBtn.disabled = false;
    saveBtn.innerHTML = originalHtml;
  }
}

// ============================================================
=================
// Google Workspace Multi-Account Pool Management
// ============================================================
=================

async function refreshGoogleAccountsList(container) {
  if (!container) return;
  try {
    container.innerHTML = '<tr><td colspan="5" class="text-cen
ter py-4 text-muted"><span class="spinner-border spinner-borde
r-sm me-2"></span>Загрузка аккаунтов...</td></tr>';
    const data = await window.api.googleAccounts.list();
```

```
1674      const accounts = data.accounts || [];
1675
1676      if (accounts.length === 0) {
1677        container.innerHTML = '<tr><td colspan="5" class="text-c
      enter py-4 text-muted">Список аккаунтов пуст. Загрузите файл с
      redentials.json или service_account.json.</td></tr>';
1678        return;
1679      }
1680
1681      container.innerHTML = '';
1682      accounts.forEach(acc => {
1683        const row = document.createElement('tr');
1684
1685        // 1. Account Name + Default Badge + Email
1686        const tdName = document.createElement('td');
1687        tdName.innerHTML = `
1688          <div class="d-flex align-items-center gap-2">
1689            <strong>${escapeHtml(acc.name)}</strong>
1690            ${acc.is_default ? '<span class="badge bg-warning te
      xt-dark"><i class="bi bi-star-fill"></i> Default</span>' : ''}
1691          </div>
1692          ${acc.email ? `<div class="small text-muted font-monos
      pace">${escapeHtml(acc.email)}</div>` : ''}
1693        `;
1694        row.appendChild(tdName);
1695
1696        // 2. Type
1697        const tdType = document.createElement('td');
1698        const isOAuth = acc.type === 'oauth2';
1699        tdType.innerHTML = isOAuth
1700          ? '<span class="badge bg-primary-subtle text-primary b
      order border-primary-subtle"><i class="bi bi-person me-1"></i>
      OAuth 2.0</span>'
1701          : '<span class="badge bg-info-subtle text-info border
      border-info-subtle"><i class="bi bi-cpu me-1"></i>Service Acco
      unt</span>';
1702        row.appendChild(tdType);
1703
```

```
      // 3. Status / Quota
      const tdStatus = document.createElement('td');
      const isActive = acc.status === 'active';
      if (isActive) {
        tdStatus.innerHTML = '<span class="badge bg-success">А
ктивен</span>';
      } else if (acc.status === 'exhausted') {
        tdStatus.innerHTML = '<span class="badge bg-danger" ti
tle="Исчерпан суточный лимит квоты">Лимит</span>';
      } else {
        tdStatus.innerHTML = `<span class="badge bg-secondary"
>${escapeHtml(acc.status)}</span>`;
      }
      row.appendChild(tdStatus);

      // 4. Token / Credentials Status
      const tdToken = document.createElement('td');
      if (acc.has_token || acc.type === 'service_account') {
        tdToken.innerHTML = '<span class="badge bg-success-sub
tle text-success"><i class="bi bi-shield-check me-1"></i>Готов
</span>';
      } else {
        tdToken.innerHTML = '<span class="badge bg-warning-sub
tle text-warning"><i class="bi bi-key me-1"></i>Нужен токен</s
pan>';
      }
      row.appendChild(tdToken);

      // 5. Actions
      const tdActions = document.createElement('td');
      tdActions.className = 'text-end';

      // Set Default button
      if (!acc.is_default) {
        const btnDefault = document.createElement('button');
        btnDefault.className = 'btn btn-xs btn-outline-warning
 btn-sm me-1';
        btnDefault.title = 'Сделать аккаунтом по умолчанию';
```

```
1734          btnDefault.innerHTML = '<i class="bi bi-star"></i>';
1735          btnDefault.onclick = () => setGoogleAccountDefault(acc
.name, container);
1736          tdActions.appendChild(btnDefault);
1737        }
1738
1739        // Reset status button
1740        if (acc.status === 'exhausted') {
1741          const btnReset = document.createElement('button');
1742          btnReset.className = 'btn btn-xs btn-outline-info btn-
sm me-1';
1743          btnReset.title = 'Сбросить статус исчерпания';
1744          btnReset.innerHTML = '<i class="bi bi-arrow-repeat"></
i>';
1745          btnReset.onclick = () => resetGoogleAccountStatus(acc.
name, container);
1746          tdActions.appendChild(btnReset);
1747        }
1748
1749        // Test button
1750        const btnTest = document.createElement('button');
1751        btnTest.className = 'btn btn-xs btn-outline-info btn-sm
me-1';
1752        btnTest.title = 'Проверить доступ и авторизацию';
1753        btnTest.innerHTML = '<i class="bi bi-play-circle"></i> Т
ест';
1754        btnTest.onclick = () => testGoogleAccount(acc.name);
1755        tdActions.appendChild(btnTest);
1756
1757        // Delete button
1758        const btnDelete = document.createElement('button');
1759        btnDelete.className = 'btn btn-xs btn-outline-danger btn
-sm';
1760        btnDelete.title = 'Удалить аккаунт из пула';
1761        btnDelete.innerHTML = '<i class="bi bi-trash"></i>';
1762        btnDelete.onclick = () => deleteGoogleAccount(acc.name,
container);
1763        tdActions.appendChild(btnDelete);
```

```
1764
1765       row.appendChild(tdActions);
1766       container.appendChild(row);
1767     });
1768
1769   } catch (err) {
1770     console.error('Ошибка загрузки аккаунтов Google Workspace:
', err);
1771     container.innerHTML = `<tr><td colspan="5" class="text-cen
ter py-4 text-danger">Ошибка: ${escapeHtml(err.message)}</td><
/tr>`;
1772   }
1773 }
1774
1775 async function handleAddGoogleAccount(container) {
1776   const nameInput = document.getElementById('new-gacc-name');
1777   const typeSelect = document.getElementById('new-gacc-type');
1778   const emailInput = document.getElementById('new-gacc-email')
;
1779   const fileInput = document.getElementById('new-gacc-file');
1780   const jsonTextarea = document.getElementById('new-gacc-json'
);
1781   const defaultCheckbox = document.getElementById('new-gacc-de
fault');
1782   const addBtn = document.getElementById('btn-add-gacc');
1783
1784   if (!nameInput || !addBtn) return;
1785
1786   const accountName = nameInput.value.trim();
1787   const accountType = typeSelect ? typeSelect.value : 'oauth2'
;
1788   const email = emailInput ? emailInput.value.trim() : '';
1789   const setAsDefault = defaultCheckbox ? defaultCheckbox.check
ed : false;
1790
1791   if (!accountName) {
1792     showModelsNotification('Введите имя аккаунта (например: wo
rk, personal)', 'warning');
```

```
1793      nameInput.focus();
1794      return;
1795    }
1796
1797    // Check if file is selected or JSON is pasted
1798    const file = fileInput && fileInput.files && fileInput.files
[0];
1799    const jsonContent = jsonTextarea ? jsonTextarea.value.trim()
 : '';
1800
1801    if (!file && !jsonContent) {
1802      showModelsNotification('Загрузите файл credentials.json /
service_account.json или вставьте JSON', 'warning');
1803      return;
1804    }
1805
1806    addBtn.disabled = true;
1807    const originalText = addBtn.innerHTML;
1808    addBtn.innerHTML = '<span class="spinner-border spinner-bord
er-sm me-1"></span> Сохранение...';
1809
1810    try {
1811      if (file) {
1812        const formData = new FormData();
1813        formData.append('account_name', accountName);
1814        formData.append('account_type', accountType);
1815        if (email) formData.append('email', email);
1816        formData.append('set_as_default', setAsDefault ? 'true'
: 'false');
1817        formData.append('file', file);
1818
1819        await window.api.googleAccounts.upload(formData);
1820      } else {
1821        let parsedJson = null;
1822        try {
1823          parsedJson = JSON.parse(jsonContent);
1824        } catch (e) {
1825          throw new Error('Некорректный JSON в поле учетных данн
```

```
ых');
      }

      await window.api.googleAccounts.create({
        account_name: accountName,
        account_type: accountType,
        email: email || undefined,
        credentials_dict: parsedJson,
        set_as_default: setAsDefault
      });
    }

    showModelsNotification(`Аккаунт Google "${accountName}" ус
пешно сохранен в пул`, 'success');
    nameInput.value = '';
    if (emailInput) emailInput.value = '';
    if (fileInput) fileInput.value = '';
    if (jsonTextarea) jsonTextarea.value = '';
    if (defaultCheckbox) defaultCheckbox.checked = false;

    if (container) await refreshGoogleAccountsList(container);
  } catch (err) {
    console.error('Ошибка сохранения аккаунта Google:', err);
    showModelsNotification('Ошибка сохранения: ' + err.message
, 'danger');
  } finally {
    addBtn.disabled = false;
    addBtn.innerHTML = originalText;
  }
}

async function setGoogleAccountDefault(name, container) {
  try {
    await window.api.googleAccounts.setDefault(name);
    showModelsNotification(`Аккаунт "${name}" назначен по умол
чанию`, 'success');
    if (container) await refreshGoogleAccountsList(container);
  } catch (err) {
```

```
    console.error('Ошибка установки аккаунта по умолчанию:', err);
    showModelsNotification('Ошибка: ' + err.message, 'danger');
  }
}

async function resetGoogleAccountStatus(name, container) {
  try {
    await window.api.googleAccounts.resetStatus(name);
    showModelsNotification(`Статус аккаунта "${name}" сброшен в активный`, 'success');
    if (container) await refreshGoogleAccountsList(container);
  } catch (err) {
    console.error('Ошибка сброса статуса аккаунта:', err);
    showModelsNotification('Ошибка сброса: ' + err.message, 'danger');
  }
}

async function testGoogleAccount(name) {
  const testCard = document.getElementById('gaccount-test-card');
  const testBody = document.getElementById('gaccount-test-body');

  if (testCard && testBody) {
    testCard.style.display = 'block';
    testBody.innerHTML = `<span class="spinner-border spinner-border-sm me-2 text-info"></span>Проверка аутентификации для аккаунта <strong>${escapeHtml(name)}</strong>...`;
  }

  try {
    const res = await window.api.googleAccounts.test(name);
    if (testBody) {
      const isSuccess = res.status === 'success';
      const isWarning = res.status === 'warning';
```

```
      const badgeClass = isSuccess ? 'bg-success' : isWarning ? 'bg-warning text-dark' : 'bg-danger';

      testBody.innerHTML = `
        <div class="mb-2 d-flex align-items-center gap-2">
          <span class="badge ${badgeClass}">${res.status.toUpperCase()}</span>
          <strong>${escapeHtml(name)}</strong>
        </div>
        <div class="mb-1">${escapeHtml(res.message || '')}</div>
        ${res.scopes && res.scopes.length > 0 ? `<div class="text-muted mt-2"><strong>Доступные Scopes:</strong><br>${res.scopes.map(s => '• ' + escapeHtml(s)).join('<br>')}</div>` : ''}
      `;
    }
  } catch (err) {
    if (testBody) {
      testBody.innerHTML = `<span class="text-danger"><i class="bi bi-x-circle me-1"></i>Ошибка проверки: ${escapeHtml(err.message)}</span>`;
    }
  }
}

async function deleteGoogleAccount(name, container) {
  if (!confirm(`Вы уверены, что хотите удалить аккаунт "${name}" из пула Google Workspace?`)) return;
  try {
    await window.api.googleAccounts.delete(name);
    showModelsNotification(`Аккаунт "${name}" успешно удален`, 'success');
    if (container) await refreshGoogleAccountsList(container);
  } catch (err) {
    console.error('Ошибка удаления аккаунта Google:', err);
    showModelsNotification('Ошибка удаления: ' + err.message, 'danger');
  }
```

```
}

// Export for tab loader
window.initModelsTab = initModelsTab;
window.refreshGoogleAccountsList = refreshGoogleAccountsList;

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/network_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/network_tab/index.html"
 class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/network_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Windows Network Terminal Tab (/apps/windows/network) -->
<style>
  .net-container {
    padding: 0.5rem 0.75rem;
  }
  .net-card {
    background: var(--surface-1) !important;
    border: 1.5px solid var(--border-color) !important;
    border-radius: var(--radius-md, 8px);
    padding: var(--kpi-padding, 0.55rem 0.85rem);
```

```
    display: flex;
    flex-direction: column;
    justify-content: space-between;
    min-height: var(--kpi-min-height, 64px);
    gap: 0.2rem;
    box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.08)
);
    transition: border-color 0.15s ease, box-shadow 0.15s ease
;
  }
  .net-card:hover {
    border-color: var(--kpi-accent, var(--nav-active, #0284c7)
) !important;
  }
  .net-label {
    font-size: var(--kpi-label-size, 0.7rem);
    text-transform: uppercase;
    letter-spacing: 0.05em;
    font-weight: 600;
    color: var(--text-muted);
    line-height: 1.1;
  }
  .net-value {
    font-size: var(--kpi-val-size, 1.25rem);
    font-weight: 700;
    color: var(--text-color);
    font-family: var(--font-mono, monospace);
    line-height: 1.15;
  }
  .net-panel {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: var(--radius-md, 8px);
    overflow: hidden;
    box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.08)
);
  }
  .net-panel-header {
```

```
    background: var(--surface-2) !important;
    border-bottom: 1px solid var(--border-color) !important;
    padding: 0.5rem 0.85rem;
    font-weight: 600;
    font-size: 0.82rem;
    color: var(--text-color);
    display: flex;
    align-items: center;
    justify-content: space-between;
  }
  .net-table {
    width: 100%;
    font-size: 0.78rem;
    font-family: var(--font-mono, monospace);
    border-collapse: collapse;
    margin-bottom: 0;
    background: transparent;
  }
  .net-table th {
    background: var(--surface-2);
    padding: 6px 10px;
    color: var(--text-muted);
    font-weight: 600;
    border-bottom: 1px solid var(--border-color);
    position: sticky;
    top: 0;
    z-index: 5;
  }
  .net-table td {
    padding: 5px 10px;
    border-bottom: 1px solid var(--border-subtle, rgba(128, 12
8, 128, 0.2));
    color: var(--text-color);
    vertical-align: middle;
  }
  .net-table tr:hover td {
    background: rgba(56, 189, 248, 0.06);
  }
```

```
  .proto-badge {
    display: inline-block;
    padding: 1px 6px;
    border-radius: var(--radius-xs, 4px);
    font-size: 0.7rem;
    font-weight: 600;
  }
  .proto-tcp { background: var(--status-info-bg, rgba(59, 130, 246, 0.15)); color: var(--status-info-text, #0284c7); border: 1px solid var(--status-info-border, #3b82f6); }
  .proto-udp { background: rgba(168, 85, 247, 0.15); color: #9333ea; border: 1px solid rgba(168, 85, 247, 0.5); }
  [data-theme="dark"] .proto-udp, [data-theme="terminal"] .proto-udp { color: #c084fc; border-color: #a855f7; }
  .proto-tls { background: var(--status-success-bg, rgba(16, 185, 129, 0.15)); color: var(--status-success-text, #166534); border: 1px solid var(--status-success-border, #10b981); }
  .proto-http { background: var(--status-warning-bg, rgba(245, 158, 11, 0.15)); color: var(--status-warning-text, #b45309); border: 1px solid var(--status-warning-border, #f59e0b); }
  .proto-dns { background: rgba(6, 182, 212, 0.15); color: #0e7490; border: 1px solid rgba(6, 182, 212, 0.5); }
  [data-theme="dark"] .proto-dns, [data-theme="terminal"] .proto-dns { color: #22d3ee; border-color: #06b6d4; }
  .proto-other { background: var(--surface-2); color: var(--text-muted); border: 1px solid var(--border-color); }

  .status-badge-up { background: var(--status-success-bg); color: var(--status-success-text); border: 1px solid var(--status-success-border); }
  .status-badge-down { background: var(--status-danger-bg); color: var(--status-danger-text); border: 1px solid var(--status-danger-border); }
  .status-badge-listen { background: var(--status-info-bg); color: var(--status-info-text); border: 1px solid var(--status-info-border); }
  .status-badge-established { background: rgba(168, 85, 247, 0.15); color: #9333ea; border: 1px solid rgba(168, 85, 247, 0.5
```

```
); }
  [data-theme="dark"] .status-badge-established, [data-theme="terminal"] .status-badge-established { color: #c084fc; border-color: #9333ea; }

  .net-subnav .nav-link {
    font-size: 0.8rem;
    padding: 0.35rem 0.8rem;
    color: var(--text-muted);
    border-radius: 6px;
    border: 1px solid transparent;
  }
  .net-subnav .nav-link:hover {
    color: var(--text-color);
    background: var(--surface-2);
  }
  .net-subnav .nav-link.active {
    background: var(--surface-2);
    color: var(--nav-active, #0284c7);
    border: 1px solid var(--border-color);
    font-weight: 600;
  }

  .speed-gauge-card {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 10px;
    padding: 1rem;
    text-align: center;
    box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.05));
  }
  .speed-gauge-val {
    font-size: 2.2rem;
    font-weight: 800;
    font-family: var(--font-mono, monospace);
    line-height: 1.1;
  }
```

```
158
159   .net-well {
160     background: var(--surface-2);
161     border: 1px solid var(--border-color);
162     border-radius: var(--radius-sm, 6px);
163   }
164 </style>
165 <div class="net-container">
166 <!-- Header Alert & Status Strip -->
167 <div class="card shadow-sm border-secondary-subtle py-1.5 px-3
     mb-2 d-flex justify-content-between align-items-center flex-w
    rap gap-2" data-panel-id="panel-network-setevoy-terminal-windo
    ws" id="panel-network-setevoy-terminal-windows">
168 <div class="d-flex align-items-center gap-2 flex-wrap">
169 <span class="fs-5">□ </span>
170 <span class="fw-bold text-body small" style="font-size: 0.95re
    m;">Сетевой терминал Windows</span>
171 <span class="badge bg-body-secondary border border-info text-i
    nfo fw-normal" style="font-size: 0.72rem;">apps/windows/networ
    k</span>
172 <span class="text-secondary small d-none d-lg-inline" style="f
    ont-size: 0.76rem;">| Скорость интернета, адаптеры, порты, сое
    динения и DPI</span>
173 </div>
174 <div class="d-flex align-items-center gap-2 flex-wrap">
175 <span class="badge rounded-pill bg-success-subtle text-success
     border border-success px-2 py-1" id="net-engine-status" style
    ="font-size: 0.72rem;">
176 <i class="bi bi-check-circle-fill me-1"></i> Windows Net: Acti
    ve
177       </span>
178 <span class="badge rounded-pill bg-secondary-subtle text-body
    border border-secondary px-2 py-1" id="net-tshark-status" styl
    e="font-size: 0.72rem;">
179         TShark DPI: Проверка...
180       </span>
181 <button class="btn btn-xs btn-outline-info rounded-pill px-2.5
     py-1" id="btn-refresh-network-all" style="font-size: 0.75rem;
```

```
">
<i class="bi bi-arrow-clockwise me-1"></i> Обновить
      </button>
<button class="btn btn-xs btn-outline-warning rounded-pill px-2 py-1" id="btn-network-config" style="font-size: 0.75rem;">
<i class="bi bi-gear-fill me-1"></i> Настройки
      </button>
</div>
</div>
<!-- Top Metrics KPI Cards -->
<div class="row g-2 mb-3">
<div class="col-6 col-md-3">
<div class="net-card">
<span class="net-label">Сетевые адаптеры</span>
<div class="d-flex align-items-baseline justify-content-between mt-1">
<span class="net-value" id="net-kpi-adapters">--</span>
<span class="small text-success" id="net-kpi-adapters-active" style="font-size: 0.68rem;">0 активных</span>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="net-card">
<span class="net-label">Активные соединения</span>
<div class="d-flex align-items-baseline justify-content-between mt-1">
<span class="net-value" id="net-kpi-conns">--</span>
<span class="small text-info" id="net-kpi-listening" style="font-size: 0.68rem;">0 слушающих</span>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="net-card">
<span class="net-label">Скорость Интернета</span>
<div class="d-flex align-items-baseline justify-content-between mt-1">
```

```
<span class="net-value text-info" id="net-kpi-speed">-- Mbps</span>
<span class="small text-muted" id="net-kpi-ping" style="font-size: 0.68rem;">Ping: -- ms</span>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="net-card">
<span class="net-label">DPI &amp;amp; Безопасность</span>
<div class="d-flex align-items-baseline justify-content-between mt-1">
<span class="net-value text-info" id="net-kpi-packets">0</span>
<span class="small text-success" id="net-kpi-anomalies" style="font-size: 0.68rem;">0 аномалий</span>
</div>
</div>
</div>
</div>
<!-- Sub-Navigation Bar -->
<ul class="nav net-subnav mb-3 gap-1">
<li class="nav-item">
<button class="nav-link active" data-bs-target="#tab-net-speedtest" data-bs-toggle="tab">
<i class="bi bi-speedometer2 me-1 text-warning"></i> Тест скорости Интернета
      </button>
</li>
<li class="nav-item">
<button class="nav-link" data-bs-target="#tab-net-adapters" data-bs-toggle="tab">
<i class="bi bi-hdd-network me-1"></i> Сетевые адаптеры
      </button>
</li>
<li class="nav-item">
<button class="nav-link" data-bs-target="#tab-net-connections"
 data-bs-toggle="tab">
```

```
<i class="bi bi-diagram-3 me-1"></i> Соединения и порты (<span
 id="badge-conn-count">0</span>)
      </button>
</li>
<li class="nav-item">
<button class="nav-link" data-bs-target="#tab-net-capture" dat
a-bs-toggle="tab">
<i class="bi bi-broadcast-pin me-1"></i> Захват пакетов (Live
DPI)
      </button>
</li>
<li class="nav-item">
<button class="nav-link" data-bs-target="#tab-net-pcap" data-b
s-toggle="tab">
<i class="bi bi-file-earmark-binary me-1"></i> Анализ PCAP &am
p; AI
      </button>
</li>
</ul>
<!-- Tab Content -->
<div class="tab-content">
<!-- SECTION 0: Speedtest & Latency Benchmark -->
<div class="tab-pane fade show active" id="tab-net-speedtest">
<!-- Speedtest Action Control Box -->
<div class="net-panel mb-3">
<div class="net-panel-header">
<span><i class="bi bi-speedometer2 text-warning me-1"></i> Тес
тирование пропускной способности и задержки Интернета</span>
<div class="d-flex align-items-center gap-2">
<span class="small text-muted" id="speedtest-status-text">Гото
в к тестированию</span>
<button class="btn btn-sm btn-warning rounded-pill px-3 fw-bol
d shadow-sm" id="btn-run-speedtest">
<i class="bi bi-play-fill me-1"></i> Запустить тест скорости
            </button>
</div>
</div>
<div class="p-3">
```

```
<!-- 4 Big Gauges -->
<div class="row g-3 mb-3">
<div class="col-6 col-lg-3">
<div class="speed-gauge-card">
<div class="text-secondary small text-uppercase fw-semibold mb-1"><i class="bi bi-arrow-down-circle-fill text-success me-1"></i> Скачивание (Download)</div>
<div class="speed-gauge-val text-success" id="speed-val-download">--</div>
<div class="text-muted small mt-1" id="speed-sub-download">0.0 MB/s</div>
</div>
</div>
<div class="col-6 col-lg-3">
<div class="speed-gauge-card">
<div class="text-secondary small text-uppercase fw-semibold mb-1"><i class="bi bi-arrow-up-circle-fill text-primary me-1"></i> Отдача (Upload)</div>
<div class="speed-gauge-val text-primary" id="speed-val-upload">--</div>
<div class="text-muted small mt-1" id="speed-sub-upload">0.0 MB/s</div>
</div>
</div>
<div class="col-6 col-lg-3">
<div class="speed-gauge-card">
<div class="text-secondary small text-uppercase fw-semibold mb-1"><i class="bi bi-stopwatch-fill text-info me-1"></i> Задержка (Ping)</div>
<div class="speed-gauge-val text-info" id="speed-val-ping">--</div>
<div class="text-muted small mt-1" id="speed-sub-jitter">Jitter: -- ms</div>
</div>
</div>
<div class="col-6 col-lg-3">
<div class="speed-gauge-card">
<div class="text-secondary small text-uppercase fw-semibold mb
```

```
-1"><i class="bi bi-shield-check text-warning me-1"></i> Качество соединения</div>
<div class="speed-gauge-val text-warning" id="speed-val-grade">--</div>
<div class="text-muted small text-truncate mt-1" id="speed-sub-isp">Провайдер: --</div>
</div>
</div>
</div>
<!-- Quality Summary Bar -->
<div class="p-2.5 rounded net-well mb-3 d-flex justify-content-between align-items-center flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<span class="badge bg-secondary" id="speed-badge-ip">Внешний IP: --</span>
<span class="badge bg-body-secondary border text-body" id="speed-badge-location">Локация: --</span>
</div>
<div class="small text-muted" id="speed-summary-text">Нажмите «Запустить тест скорости», чтобы измерить пропускную способность канала, задержку (Ping) и джиттер.</div>
</div>
<!-- DNS & Edge Servers Latency Table -->
<h6 class="small fw-bold text-info mb-2"><i class="bi bi-globe me-1"></i> Задержка к опорным DNS &amp;amp; CDN серверам:</h6>
<div class="table-responsive">
<table class="net-table">
<thead>
<tr>
<th>Сервер / Провайдер</th>
<th style="width: 140px;">Хост / IP</th>
<th style="width: 110px; text-align: right;">Средний Ping</th>
<th style="width: 90px; text-align: right;">Min</th>
<th style="width: 90px; text-align: right;">Max</th>
<th style="width: 100px; text-align: right;">Джиттер</th>
<th style="width: 100px;">Статус</th>
</tr>
</thead>
```

```
<tbody id="speed-servers-tbody">
<tr><td class="text-center py-3 text-muted" colspan="7">Запуст
ите тест скорости для измерения задержки к DNS-серверам.</td><
/tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
<!-- SECTION 1: Network Adapters -->
<div class="tab-pane fade" id="tab-net-adapters">
<div class="net-panel">
<div class="net-panel-header">
<span><i class="bi bi-hdd-network me-1"></i> Физические и вирт
уальные сетевые интерфейсы Windows</span>
<button class="btn btn-xs btn-outline-info rounded-pill px-2 p
y-0" id="btn-refresh-adapters">Обновить адаптеры</button>
</div>
<div class="table-responsive" style="max-height: 480px; overfl
ow-y: auto;">
<table class="net-table">
<thead>
<tr>
<th style="width: 140px;">Адаптер</th>
<th>Описание</th>
<th style="width: 130px;">IPv4</th>
<th style="width: 140px;">MAC адрес</th>
<th style="width: 90px;">Скорость</th>
<th style="width: 90px;">Статус</th>
<th style="width: 130px; text-align: right;">Отправлено</th>
<th style="width: 130px; text-align: right;">Принято</th>
</tr>
</thead>
<tbody id="net-adapters-tbody">
<tr><td class="text-center py-4 text-muted" colspan="8"><div c
lass="spinner-border spinner-border-sm text-info me-2"></div>З
агрузка сетевых адаптеров...</td></tr>
```

```
</tbody>
</table>
</div>
</div>
</div>
<!-- SECTION 2: Active Connections & Listening Ports -->
<div class="tab-pane fade" id="tab-net-connections">
<div class="net-panel">
<div class="net-panel-header">
<div class="d-flex align-items-center gap-2 flex-wrap">
<span><i class="bi bi-diagram-3 me-1"></i> Таблица сетевых под
ключений и слушающих портов</span>
<div class="btn-group btn-group-sm ms-2" role="group">
<button class="btn btn-outline-secondary active btn-filter-con
ns" data-filter="all" type="button">Все</button>
<button class="btn btn-outline-secondary btn-filter-conns" dat
a-filter="LISTEN" type="button">LISTEN</button>
<button class="btn btn-outline-secondary btn-filter-conns" dat
a-filter="ESTABLISHED" type="button">ESTABLISHED</button>
<button class="btn btn-outline-secondary btn-filter-conns" dat
a-filter="UDP" type="button">UDP</button>
</div>
</div>
<div class="d-flex align-items-center gap-2">
<input class="form-control form-control-sm" id="net-conns-filt
er" placeholder="Поиск (IP, порт, процесс, PID)..." style="max
-width: 250px;" type="text"/>
<button class="btn btn-xs btn-outline-info rounded-pill px-2 p
y-0" id="btn-refresh-conns">Обновить</button>
</div>
</div>
<div class="table-responsive" style="max-height: 480px; overfl
ow-y: auto;">
<table class="net-table">
<thead>
<tr>
<th style="width: 70px;">Протокол</th>
<th style="width: 180px;">Локальный адрес</th>
```

```
<th style="width: 180px;">Удалённый адрес</th>
<th style="width: 110px;">Статус</th>
<th style="width: 80px;">PID</th>
<th>Процесс / Исполняемый файл</th>
</tr>
</thead>
<tbody id="net-conns-tbody">
<tr><td class="text-center py-4 text-muted" colspan="6"><div c
lass="spinner-border spinner-border-sm text-info me-2"></div>З
агрузка соединений...</td></tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- SECTION 3: Packet Capture & Live DPI -->
<div class="tab-pane fade" id="tab-net-capture">
<!-- Capture Control Box -->
<div class="net-panel mb-3">
<div class="net-panel-header">
<span><i class="bi bi-broadcast me-1"></i> Настройка сессии за
хвата трафика (Live Packet DPI)</span>
</div>
<div class="p-3">
<div class="row g-2 mb-2 align-items-end">
<div class="col-md-5">
<label class="form-label small fw-semibold text-body mb-1">Сет
евой интерфейс для захвата:</label>
<select class="form-select form-select-sm" id="net-interface-s
elect">
<option value="">Загрузка интерфейсов...</option>
</select>
</div>
<div class="col-md-4">
<label class="form-label small fw-semibold text-body mb-1">Wir
eshark/BPF фильтр:</label>
<input class="form-control form-control-sm" id="net-capture-fi
lter" placeholder="tcp, udp, port 80, tls" type="text"/>
```

```
</div>
<div class="col-md-3 d-flex gap-2">
<button class="btn btn-sm btn-primary rounded-pill px-3 flex-fill" id="btn-start-live-capture">
<i class="bi bi-play-fill me-1"></i> Старт захвата
              </button>
<button class="btn btn-sm btn-outline-danger rounded-pill px-3 flex-fill" disabled="" id="btn-stop-live-capture">
<i class="bi bi-stop-fill me-1"></i> Стоп
              </button>
</div>
</div>
</div>
</div>
<!-- Packets Table -->
<div class="net-panel">
<div class="net-panel-header">
<div class="d-flex align-items-center gap-2">
<span><i class="bi bi-reception-4 me-1"></i> Поток сетевых пакетов (Packet Inspection)</span>
<span class="badge bg-secondary" id="net-captured-badge">0 пакетов</span>
</div>
<div class="d-flex align-items-center gap-2">
<input class="form-control form-control-sm" id="net-table-filter" placeholder="Поиск по IP / протоколу..." style="max-width: 220px;" type="text"/>
<button class="btn btn-xs btn-outline-secondary rounded-pill px-2 py-0" id="btn-clear-packets">Очистить</button>
</div>
</div>
<div class="table-responsive" style="max-height: 400px; overflow-y: auto;">
<table class="net-table">
<thead>
<tr>
<th style="width: 50px;">#</th>
<th style="width: 80px;">Время</th>
```

```
<th style="width: 140px;">Источник</th>
<th style="width: 140px;">Назначение</th>
<th style="width: 80px;">Протокол</th>
<th style="width: 70px; text-align: right;">Размер</th>
<th>Сведения о полезной нагрузке / Инфо</th>
</tr>
</thead>
<tbody id="net-packets-tbody">
<tr><td class="text-center py-4 text-muted" colspan="7">Нет захваченных пакетов. Выберите интерфейс и нажмите "Старт захвата".</td></tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- SECTION 4: PCAP Analysis & AI Diagnostics -->
<div class="tab-pane fade" id="tab-net-pcap">
<div class="row g-3">
<div class="col-lg-5">
<div class="net-panel h-100">
<div class="net-panel-header">
<span><i class="bi bi-file-earmark-binary me-1"></i> Загрузка файла сетевого дампа (.PCAP)</span>
</div>
<div class="p-3">
<div class="mb-3">
<label class="form-label small fw-semibold text-body">Выберите .pcap, .pcapng или .cap файл:</label>
<input accept=".pcap,.pcapng,.cap" class="form-control form-control-sm" id="net-pcap-file" type="file"/>
</div>
<div class="d-flex justify-content-between align-items-center mb-3">
<span class="small text-muted" id="net-upload-status">Файл не выбран</span>
<button class="btn btn-sm btn-info rounded-pill px-3 fw-bold" id="btn-analyze-pcap">
```

```
<i class="bi bi-cpu me-1"></i> Запустить AI Анализ
                </button>
</div>
<div class="p-2.5 rounded net-well small text-muted" style="fo
nt-size: 0.75rem;">
<i class="bi bi-info-circle text-info me-1"></i> PCAP-анализат
ор проводит глубокую инспекцию заголовков, эвристический аудит
 портов и выявляет аномалии с помощью AI-модели.
              </div>
</div>
</div>
</div>
<div class="col-lg-7">
<div class="net-panel h-100">
<div class="net-panel-header">
<span><i class="bi bi-shield-check me-1"></i> Результаты AI-ан
ализа и безопасности</span>
</div>
<div class="p-3">
<div class="row g-2 mb-3">
<div class="col-6">
<div class="p-2 rounded net-well">
<span class="net-label">AI Health Score</span>
<div class="net-value mt-1" id="pcap-health-score">100/100</di
v>
</div>
</div>
<div class="col-6">
<div class="p-2 rounded net-well">
<span class="net-label">Обнаруженные аномалии</span>
<div class="net-value mt-1 text-success" id="pcap-anomalies-co
unt">0</div>
</div>
</div>
</div>
<h6 class="small fw-bold text-info mb-1.5">Эвристические преду
преждения:</h6>
<div class="p-2 rounded net-well mb-3 small" id="pcap-heuristi
```

```
cs-list" style="min-height: 60px; max-height: 120px; overflow-y: auto;">
<span class="text-muted">Нет предупреждений. Запустите анализ файла.</span>
</div>
<h6 class="small fw-bold text-info mb-1.5">Рекомендации AI:</h6>
<div class="p-2 rounded net-well small" id="pcap-ai-recommendations" style="min-height: 60px; max-height: 120px; overflow-y: auto;">
<span class="text-muted">Анализ пока не производился.</span>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Network Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/network_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/network_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// ============================================================
// Process Name: Windows Network Terminal Tab Client Module
// ============================================================
// Description:
//   Frontend controller for /apps/windows/network.
//   Manages internet speedtest, ping/latency benchmark, network adapters,
//   active connections, listening ports, live packet capture (DPI), PCAP upload,
//   and AI diagnostics.
```

```
// ============================================================
==================

(function() {
  let isNetInitialized = false;
  let netEventSource = null;
  let capturedPackets = [];
  let allConnections = [];
  let allAdapters = [];
  let activeConnFilter = 'all';
  let isSpeedtestRunning = false;

  // Format bytes to human readable format
  function formatBytes(bytes, decimals = 1) {
    if (!bytes || bytes === 0) return '0 B';
    const k = 1024;
    const dm = decimals < 0 ? 0 : decimals;
    const sizes = ['B', 'KB', 'MB', 'GB', 'TB'];
    const i = Math.floor(Math.log(bytes) / Math.log(k));
    return parseFloat((bytes / Math.pow(k, i)).toFixed(dm)) +
' ' + sizes[i];
  }

  // Check overall network status and TShark availability
  async function fetchNetworkStatus() {
    const engineBadge = document.getElementById('net-engine-st
atus');
    const tsharkBadge = document.getElementById('net-tshark-st
atus');
    const kpiAdapters = document.getElementById('net-kpi-adapt
ers');
    const kpiAdaptersActive = document.getElementById('net-kpi
-adapters-active');
    const kpiConns = document.getElementById('net-kpi-conns');
    const kpiListening = document.getElementById('net-kpi-list
ening');
    const kpiPackets = document.getElementById('net-kpi-packet
s');
```

```
    const kpiAnomalies = document.getElementById('net-kpi-anomalies');

    try {
      const res = await fetch('/api/network/status');
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();

      if (engineBadge) {
        engineBadge.className = 'badge rounded-pill bg-success-subtle text-success border border-success px-2 py-1';
        engineBadge.innerHTML = '<i class="bi bi-check-circle-fill me-1"></i> Windows Net: Active';
      }

      if (tsharkBadge) {
        if (data.tshark_available) {
          tsharkBadge.className = 'badge rounded-pill bg-success-subtle text-success border border-success px-2 py-1';
          tsharkBadge.innerText = 'TShark DPI: Доступен';
        } else {
          tsharkBadge.className = 'badge rounded-pill bg-warning-subtle text-warning border border-warning px-2 py-1';
          tsharkBadge.innerText = 'TShark DPI: Не установлен';
        }
      }

      if (kpiAdapters) kpiAdapters.innerText = data.total_adapters ?? '--';
      if (kpiConns) kpiConns.innerText = data.active_connections_count ?? '--';
      if (kpiListening) kpiListening.innerText = `${data.listening_ports_count || 0} слушающих`;
      if (kpiPackets) kpiPackets.innerText = data.total_packets ?? capturedPackets.length;

      const anomaliesCount = (data.latest_heuristics?.length || 0) + (data.latest_ai_report?.anomalies?.length || 0);
```

```
      if (kpiAnomalies) {
        kpiAnomalies.innerText = `${anomaliesCount} аномалий`;
        kpiAnomalies.className = `small ${anomaliesCount > 0 ? 'text-danger' : 'text-success'}`;
        kpiAnomalies.style.fontSize = '0.68rem';
      }
    } catch (e) {
      console.warn('[NetworkTab] Status check error:', e);
      if (engineBadge) {
        engineBadge.className = 'badge rounded-pill bg-danger-subtle text-danger border border-danger px-2 py-1';
        engineBadge.innerText = 'Windows Net: Ошибка';
      }
    }
  }

  // Load latest or cached speedtest
  async function fetchLatestSpeedtest() {
    try {
      const res = await fetch('/api/network/speedtest/latest');
      if (!res.ok) return;
      const data = await res.json();
      if (data && data.download) {
        renderSpeedtestResults(data);
      }
    } catch (e) {
      console.debug('[NetworkTab] No latest speedtest data:', e);
    }
  }

  // Run full internet speedtest
  async function runInternetSpeedtest() {
    if (isSpeedtestRunning) return;
    isSpeedtestRunning = true;

    const btn = document.getElementById('btn-run-speedtest');
```

```
    const statusText = document.getElementById('speedtest-status-text');
    const summaryText = document.getElementById('speed-summary-text');

    if (btn) {
      btn.disabled = true;
      btn.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span> Тестирование...';
    }
    if (statusText) statusText.innerText = 'Измерение Ping, Download и Upload...';
    if (summaryText) summaryText.innerText = 'Выполняется подключение к узлам Cloudflare & DNS серверам для замера скорости...';

    try {
      const res = await fetch('/api/network/speedtest/run', { method: 'POST' });
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();

      renderSpeedtestResults(data);
      if (statusText) statusText.innerText = `Завершено в ${new Date().toLocaleTimeString()}`;
    } catch (e) {
      console.error('[NetworkTab] Speedtest error:', e);
      window.showToast?.('Ошибка при выполнении теста скорости: ' + e.message, 'danger') || alert('Ошибка при выполнении теста скорости: ' + e.message);
      if (statusText) statusText.innerText = 'Ошибка теста скорости';
    } finally {
      isSpeedtestRunning = false;
      if (btn) {
        btn.disabled = false;
        btn.innerHTML = '<i class="bi bi-play-fill me-1"></i> Запустить тест скорости';
```

```
      }
    }
  }

  // Render speedtest report to UI
  function renderSpeedtestResults(data) {
    const valDown = document.getElementById('speed-val-downloa
d');
    const subDown = document.getElementById('speed-sub-downloa
d');
    const valUp = document.getElementById('speed-val-upload');
    const subUp = document.getElementById('speed-sub-upload');
    const valPing = document.getElementById('speed-val-ping');
    const subJitter = document.getElementById('speed-sub-jitte
r');
    const valGrade = document.getElementById('speed-val-grade'
);
    const subIsp = document.getElementById('speed-sub-isp');
    const badgeIp = document.getElementById('speed-badge-ip');
    const badgeLoc = document.getElementById('speed-badge-loca
tion');
    const summaryText = document.getElementById('speed-summary
-text');
    const kpiSpeed = document.getElementById('net-kpi-speed');
    const kpiPing = document.getElementById('net-kpi-ping');
    const serversTbody = document.getElementById('speed-server
s-tbody');

    const downMbps = data.download?.speed_mbps || 0;
    const downMBs = data.download?.speed_mb_s || 0;
    const upMbps = data.upload?.speed_mbps || 0;
    const upMBs = data.upload?.speed_mb_s || 0;
    const pingMs = data.ping_ms || 0;
    const jitterMs = data.jitter_ms || 0;
    const meta = data.meta || {};
    const quality = data.quality || {};

    if (valDown) valDown.innerText = `${downMbps} Mbps`;
```

```
    if (subDown) subDown.innerText = `${downMBs} MB/s (${data.download?.duration_s || 0}s)`;
    if (valUp) valUp.innerText = `${upMbps} Mbps`;
    if (subUp) subUp.innerText = `${upMBs} MB/s (${data.upload?.duration_s || 0}s)`;
    if (valPing) valPing.innerText = `${pingMs} ms`;
    if (subJitter) subJitter.innerText = `Jitter: ${jitterMs} ms`;

    if (valGrade) {
      valGrade.innerText = quality.rating || '--';
      valGrade.style.color = quality.color || '#fbbf24';
    }
    if (subIsp) subIsp.innerText = `Провайдер: ${meta.isp || 'Неизвестно'}`;

    if (badgeIp) badgeIp.innerText = `Внешний IP: ${meta.ip || '--'}`;
    if (badgeLoc) badgeLoc.innerText = `Локация: ${meta.city || ''} ${meta.country || ''} (${meta.colo || 'Edge'})`.trim();
    if (summaryText) summaryText.innerText = `${quality.grade || 'Готово'}. ${quality.summary || ''}`;

    // Top KPI cards
    if (kpiSpeed) kpiSpeed.innerText = `${downMbps} Mbps`;
    if (kpiPing) kpiPing.innerText = `Ping: ${pingMs} ms`;

    // DNS & Edge servers table
    if (serversTbody && Array.isArray(data.servers)) {
      serversTbody.innerHTML = data.servers.map(s => {
        const isOnline = s.avg_ms > 0;
        const statusBadge = isOnline
          ? '<span class="proto-badge status-badge-up">Online</span>'
          : '<span class="proto-badge status-badge-down">Offline</span>';

        return `
```

```
          <tr>
            <td class="fw-bold text-body"><i class="bi bi-hdd-network me-1 text-info"></i>${s.name}</td>
            <td class="text-secondary font-monospace">${s.host}</td>
            <td style="text-align: right;" class="text-info fw-bold font-monospace">${isOnline ? `${s.avg_ms} ms` : '--'}</td>
            <td style="text-align: right;" class="text-muted font-monospace">${isOnline ? `${s.min_ms} ms` : '--'}</td>
            <td style="text-align: right;" class="text-muted font-monospace">${isOnline ? `${s.max_ms} ms` : '--'}</td>
            <td style="text-align: right;" class="text-warning font-monospace">${isOnline ? `${s.jitter_ms} ms` : '--'}</td>
            <td>${statusBadge}</td>
          </tr>
        `;
      }).join('');
    }
  }

  // Fetch host network adapters
  async function fetchAdapters() {
    const tbody = document.getElementById('net-adapters-tbody');
    const ifaceSelect = document.getElementById('net-interface-select');
    const kpiActive = document.getElementById('net-kpi-adapters-active');

    try {
      const res = await fetch('/api/network/interfaces');
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const adapters = await res.json();
      allAdapters = Array.isArray(adapters) ? adapters : [];

      let activeCount = 0;

```

```
      if (ifaceSelect) {
        ifaceSelect.innerHTML = allAdapters.map(a => `
          <option value="${a.name || a.id}">${a.name} (${a.des
cription || 'Ethernet'}) - ${a.status}</option>
        `).join('');
      }

      if (tbody) {
        if (allAdapters.length === 0) {
          tbody.innerHTML = '<tr><td colspan="8" class="text-c
enter py-4 text-muted">Сетевые адаптеры не обнаружены.</td></t
r>';
          return;
        }

        tbody.innerHTML = allAdapters.map(a => {
          if (a.is_up) activeCount++;
          const statusClass = a.is_up ? 'status-badge-up' : 's
tatus-badge-down';
          const speedDisplay = a.speed_mbps ? `${a.speed_mbps}
 Mbps` : '--';
          const ipDisplay = a.ipv4 && a.ipv4.length > 0 ? a.ip
v4.join(', ') : '--';

          return `
            <tr>
              <td class="fw-bold text-body"><i class="bi bi-hd
d-network me-1 text-info"></i>${a.name}</td>
              <td class="text-secondary small text-truncate" s
tyle="max-width: 220px;" title="${a.description}">${a.descript
ion || '--'}</td>
              <td class="text-info">${ipDisplay}</td>
              <td class="text-muted small">${a.mac || '--'}</t
d>
              <td><span class="badge bg-body-secondary border
text-body">${speedDisplay}</span></td>
              <td><span class="proto-badge ${statusClass}">${a
.status || 'Unknown'}</span></td>
```

```
              <td style="text-align: right;" class="text-succe
ss">${formatBytes(a.bytes_sent)}</td>
              <td style="text-align: right;" class="text-prima
ry">${formatBytes(a.bytes_recv)}</td>
            </tr>
          `;
        }).join('');
      }

      if (kpiActive) kpiActive.innerText = `${activeCount} акт
ивных`;
    } catch (e) {
      console.error('[NetworkTab] Failed to fetch adapters:',
e);
      if (tbody) tbody.innerHTML = `<tr><td colspan="8" class=
"text-center py-3 text-danger">Ошибка загрузки адаптеров: ${e.
message}</td></tr>`;
    }
  }

  // Fetch active connections and listening ports
  async function fetchConnections() {
    const tbody = document.getElementById('net-conns-tbody');
    const badgeConn = document.getElementById('badge-conn-coun
t');
    const kpiConns = document.getElementById('net-kpi-conns');
    const kpiListening = document.getElementById('net-kpi-list
ening');

    try {
      const res = await fetch('/api/network/connections?limit=
250');
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();

      const conns = Array.isArray(data.connections) ? data.con
nections : [];
      const listening = Array.isArray(data.listening_ports) ?
```

```
data.listening_ports : [];
289       allConnections = [...listening, ...conns];
290
291       if (badgeConn) badgeConn.innerText = allConnections.leng
th;
292       if (kpiConns) kpiConns.innerText = conns.length;
293       if (kpiListening) kpiListening.innerText = `${listening.
length} слушающих`;
294
295       renderConnectionsTable();
296     } catch (e) {
297       console.error('[NetworkTab] Failed to fetch connections:
', e);
298       if (tbody) tbody.innerHTML = `<tr><td colspan="6" class=
"text-center py-3 text-danger">Ошибка загрузки соединений: ${e
.message}</td></tr>`;
299     }
300   }
301
302   // Render connections table with current filter and search
303   function renderConnectionsTable() {
304     const tbody = document.getElementById('net-conns-tbody');
305     const searchInput = document.getElementById('net-conns-fil
ter');
306     const query = (searchInput?.value || '').toLowerCase().tri
m();
307
308     if (!tbody) return;
309
310     let filtered = allConnections.filter(c => {
311       // Type filter
312       if (activeConnFilter === 'LISTEN' && c.status !== 'LISTE
N') return false;
313       if (activeConnFilter === 'ESTABLISHED' && c.status !== '
ESTABLISHED') return false;
314       if (activeConnFilter === 'UDP' && c.protocol !== 'UDP')
return false;
315
```

```
      // Text query
      if (query) {
        const text = `${c.protocol} ${c.local_address} ${c.remote_address} ${c.status} ${c.pid} ${c.process_name}`.toLowerCase();
        return text.includes(query);
      }
      return true;
    });

    if (filtered.length === 0) {
      tbody.innerHTML = '<tr><td colspan="6" class="text-center py-4 text-muted">Соединения, удовлетворяющие критериям, не найдены.</td></tr>';
      return;
    }

    tbody.innerHTML = filtered.slice(0, 150).map((c, idx) => {
      const protoClass = c.protocol === 'TCP' ? 'proto-tcp' : (c.protocol === 'UDP' ? 'proto-udp' : 'proto-other');
      let statusClass = 'proto-other';
      if (c.status === 'LISTEN') statusClass = 'status-badge-listen';
      else if (c.status === 'ESTABLISHED') statusClass = 'status-badge-established';

      return `
        <tr class="net-conn-row" data-idx="${idx}" style="cursor: pointer;" title="Нажмите для детальной инспекции процесса и сокета">
          <td><span class="proto-badge ${protoClass}">${c.protocol || 'TCP'}</span></td>
          <td class="text-info font-monospace">${c.local_address || '-'}</td>
          <td class="text-secondary font-monospace">${c.remote_address || '-'}</td>
          <td><span class="proto-badge ${statusClass}">${c.status || '-'}</span></td>
```

```
          <td class="text-muted font-monospace">${c.pid || '-'}</td>
          <td class="fw-semibold text-body"><i class="bi bi-app me-1 text-muted"></i>${c.process_name || 'Неизвестно'}</td>
        </tr>
      `;
    }).join('');

    // Modal inspection on row click
    tbody.querySelectorAll('.net-conn-row').forEach(row => {
      row.onclick = () => {
        const idx = parseInt(row.getAttribute('data-idx'), 10);
        const item = filtered[idx];
        if (!item || !window.AITableModal) return;

        window.AITableModal.show({
          icon: '□ ',
          title: `Сетевое подключение [${item.protocol || 'TCP'}]`,
          subtitle: `${item.process_name} (PID: ${item.pid || 'N/A'})`,
          tableType: 'network',
          badges: [
            { text: item.protocol || 'TCP', class: 'badge bg-primary' },
            { text: item.status || 'UNKNOWN', class: 'badge bg-info' },
            { text: `PID ${item.pid || '-'}`, class: 'badge bg-secondary' }
          ],
          metadata: [
            { label: 'Процесс', value: item.process_name || '-' },
            { label: 'PID', value: String(item.pid || '-') },
            { label: 'Протокол', value: item.protocol || '-' },
            { label: 'Локальный адрес:Порт', value: item.local
```

```
_address || '-' },
            { label: 'Удалённый адрес:Порт', value: item.remot
e_address || '-' },
            { label: 'Статус сокета', value: item.status || '-
' }
          ],
          rawTitle: 'Детали сокета',
          rawContent: JSON.stringify(item, null, 2),
          requestData: item
        });
      };
    });
  }

  // Render captured packets table
  function renderPacketsTable() {
    const tbody = document.getElementById('net-packets-tbody')
;
    const badge = document.getElementById('net-captured-badge'
);
    const filterInput = document.getElementById('net-table-fil
ter');
    const filter = (filterInput?.value || '').toLowerCase().tr
im();

    if (!tbody) return;
    if (badge) badge.innerText = `${capturedPackets.length} па
кетов`;

    if (capturedPackets.length === 0) {
      tbody.innerHTML = '<tr><td colspan="7" class="text-cente
r py-4 text-muted">Нет захваченных пакетов. Выберите интерфейс
 и нажмите "Старт захвата".</td></tr>';
      return;
    }

    const filtered = capturedPackets.filter(p => {
      if (!filter) return true;
```

```
      const str = `${p.number || p.packet_number || ''} ${p.source || ''} ${p.destination || ''} ${p.protocol || ''} ${p.info || ''}`.toLowerCase();
      return str.includes(filter);
    });

    const pagePackets = filtered.slice(-100);
    tbody.innerHTML = pagePackets.map((p, idx) => {
      const proto = (p.protocol || 'OTHER').toUpperCase();
      let protoClass = 'proto-other';
      if (proto.includes('TCP')) protoClass = 'proto-tcp';
      else if (proto.includes('UDP')) protoClass = 'proto-udp';
      else if (proto.includes('TLS') || proto.includes('SSL')) protoClass = 'proto-tls';
      else if (proto.includes('HTTP')) protoClass = 'proto-http';
      else if (proto.includes('DNS')) protoClass = 'proto-dns';

      return `
        <tr class="net-packet-row" data-idx="${idx}" style="cursor: pointer;" title="Нажмите для детального AI-анализа пакета">
          <td class="text-muted font-monospace">${p.number || p.packet_number || '--'}</td>
          <td>${p.timestamp ? (typeof p.timestamp === 'number' ? new Date(p.timestamp * 1000).toLocaleTimeString() : p.timestamp) : '--'}</td>
          <td class="text-info font-monospace">${p.source || '--'}</td>
          <td class="text-primary font-monospace">${p.destination || '--'}</td>
          <td><span class="proto-badge ${protoClass}">${proto}</span></td>
          <td style="text-align: right;" class="font-monospace">${p.length || 0} B</td>
          <td class="text-truncate text-body" style="max-width
```

```
: 320px;">${p.info || '--'}</td>
        </tr>
      `;
    }).join('');

    tbody.querySelectorAll('.net-packet-row').forEach(row => {
      row.onclick = () => {
        const idx = parseInt(row.getAttribute('data-idx'), 10)
;
        const p = pagePackets[idx];
        if (!p || !window.AITableModal) return;

        window.AITableModal.show({
          icon: '□ ',
          title: `Сетевой пакет #${p.number || p.packet_number
 || 'N/A'} [${p.protocol || 'TCP'}]`,
          subtitle: `${p.source || '0.0.0.0'} □ ${p.destinatio
n || '0.0.0.0'}`,
          tableType: 'network',
          badges: [
            { text: (p.protocol || 'TCP').toUpperCase(), class
: 'badge bg-primary' },
            { text: `${p.length || 0} Bytes`, class: 'badge bg
-secondary' }
          ],
          metadata: [
            { label: 'Номер пакета', value: String(p.number ||
 p.packet_number || '-') },
            { label: 'Протокол', value: (p.protocol || 'TCP').
toUpperCase() },
            { label: 'Источник', value: p.source || '-' },
            { label: 'Назначение', value: p.destination || '-'
 },
            { label: 'Размер', value: `${p.length || 0} байт`
},
            { label: 'Инфо', value: p.info || 'Нет данных', fu
llWidth: true }
          ],
```

```
          rawTitle: 'Сырые данные пакета',
          rawContent: JSON.stringify(p, null, 2),
          requestData: p
        });
      };
    });
  }

  // Start live packet capture
  async function startLiveCapture() {
    const ifaceSelect = document.getElementById('net-interface-select');
    const filterInput = document.getElementById('net-capture-filter');
    const startBtn = document.getElementById('btn-start-live-capture');
    const stopBtn = document.getElementById('btn-stop-live-capture');

    const iface = ifaceSelect?.value || '1';
    const filter = filterInput?.value || '';

    try {
      const res = await fetch(`/api/network/start-capture?interface=${encodeURIComponent(iface)}&display_filter=${encodeURIComponent(filter)}`, {
        method: 'POST'
      });
      if (!res.ok) throw new Error(`HTTP ${res.status}`);

      if (startBtn) startBtn.disabled = true;
      if (stopBtn) stopBtn.disabled = false;

      // Start SSE event source
      if (netEventSource) netEventSource.close();
      netEventSource = new EventSource('/api/network/packets/stream');
      netEventSource.onmessage = async () => {
```

```
        try {
          const packRes = await fetch('/api/network/packets?limit=50');
          if (packRes.ok) {
            const data = await packRes.json();
            if (Array.isArray(data.packets)) {
              capturedPackets = data.packets;
              renderPacketsTable();
            }
          }
        } catch (e) {
          console.warn('[NetworkTab] Packet poll error:', e);
        }
      };
    } catch (e) {
      console.error('[NetworkTab] Live capture error:', e);
      window.showToast?.('Ошибка запуска захвата пакетов: ' + e.message, 'danger') || alert('Ошибка запуска захвата пакетов: ' + e.message);
    }
  }

  // Stop live packet capture
  async function stopLiveCapture() {
    const startBtn = document.getElementById('btn-start-live-capture');
    const stopBtn = document.getElementById('btn-stop-live-capture');

    if (netEventSource) {
      netEventSource.close();
      netEventSource = null;
    }

    try {
      await fetch('/api/network/stop-capture', { method: 'POST' });
    } catch (e) {
```

```
      console.warn('[NetworkTab] Stop capture error:', e);
    }

    if (startBtn) startBtn.disabled = false;
    if (stopBtn) stopBtn.disabled = true;
  }

  // Analyze uploaded PCAP file
  async function analyzePcapFile() {
    const fileInput = document.getElementById('net-pcap-file')
;
    const statusEl = document.getElementById('net-upload-statu
s');
    const healthEl = document.getElementById('pcap-health-scor
e');
    const anomaliesEl = document.getElementById('pcap-anomalie
s-count');
    const heuristicsEl = document.getElementById('pcap-heurist
ics-list');
    const recommendationsEl = document.getElementById('pcap-ai
-recommendations');

    const file = fileInput?.files?.[0];
    if (!file) {
      window.showToast?.('Пожалуйста, выберите .pcap или .pcap
ng файл', 'warning') || alert('Пожалуйста, выберите .pcap или
.pcapng файл');
      return;
    }

    const formData = new FormData();
    formData.append('file', file);
    formData.append('max_packets', '500');

    if (statusEl) statusEl.innerText = 'Анализ пакетов...';

    try {
      const res = await fetch('/api/network/analyze/pcap', {
```

```
        method: 'POST',
        body: formData
      });

      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const report = await res.json();

      if (statusEl) statusEl.innerText = `Анализ завершён: ${r
eport.stats?.total_packets || 0} пакетов`;

      if (healthEl) healthEl.innerText = `${report.ai_report?.
health_score || 100}/100`;

      const anomaliesCount = (report.heuristics?.length || 0)
+ (report.ai_report?.anomalies?.length || 0);
      if (anomaliesEl) {
        anomaliesEl.innerText = String(anomaliesCount);
        anomaliesEl.className = `net-value mt-1 ${anomaliesCou
nt > 0 ? 'text-danger' : 'text-success'}`;
      }

      if (heuristicsEl) {
        if (Array.isArray(report.heuristics) && report.heurist
ics.length > 0) {
          heuristicsEl.innerHTML = report.heuristics.map(h =>
`<div class="text-warning mb-1"><i class="bi bi-exclamation-tr
iangle-fill me-1"></i>${h}</div>`).join('');
        } else {
          heuristicsEl.innerHTML = '<span class="text-success"
><i class="bi bi-check-circle-fill me-1"></i>Эвристических угр
оз не выявлено.</span>';
        }
      }

      if (recommendationsEl) {
        if (Array.isArray(report.ai_report?.recommended_action
s) && report.ai_report.recommended_actions.length > 0) {
          recommendationsEl.innerHTML = report.ai_report.recom
```

```
mended_actions.map(r => `<div class="text-info mb-1"><i class="bi bi-arrow-right-short me-1"></i>${r}</div>`).join('');
        } else {
          recommendationsEl.innerHTML = '<span class="text-muted">Рекомендаций нет, система функционирует штатно.</span>';
        }
      }

      if (Array.isArray(report.sample_packets) && report.sample_packets.length > 0) {
        capturedPackets = report.sample_packets;
        renderPacketsTable();
      }
    } catch (e) {
      console.error('[NetworkTab] PCAP upload error:', e);
      if (statusEl) statusEl.innerText = 'Ошибка анализа: ' + e.message;
      window.showToast?.('Ошибка при анализе PCAP: ' + e.message, 'danger') || alert('Ошибка при анализе PCAP: ' + e.message);
    }
  }

  // Main init function for Network Terminal Tab
  function initNetworkTab() {
    console.log('[NetworkTab] Initializing Windows Network Terminal tab (/apps/windows/network)...');
    fetchNetworkStatus();
    fetchLatestSpeedtest();
    fetchAdapters();
    fetchConnections();

    if (!isNetInitialized) {
      // Action buttons
      const refreshAllBtn = document.getElementById('btn-refresh-network-all');
      const runSpeedtestBtn = document.getElementById('btn-run-speedtest');
```

```
596       const refreshAdaptersBtn = document.getElementById('btn-
refresh-adapters');
597       const refreshConnsBtn = document.getElementById('btn-ref
resh-conns');
598       const startBtn = document.getElementById('btn-start-live
-capture');
599       const stopBtn = document.getElementById('btn-stop-live-c
apture');
600       const analyzePcapBtn = document.getElementById('btn-anal
yze-pcap');
601       const clearPacketsBtn = document.getElementById('btn-cle
ar-packets');
602       const connsFilterInput = document.getElementById('net-co
nns-filter');
603       const tableFilterInput = document.getElementById('net-ta
ble-filter');
604       const configBtn = document.getElementById('btn-network-c
onfig');
605
606       if (refreshAllBtn) refreshAllBtn.onclick = () => {
607         fetchNetworkStatus();
608         fetchLatestSpeedtest();
609         fetchAdapters();
610         fetchConnections();
611       };
612       if (runSpeedtestBtn) runSpeedtestBtn.onclick = runIntern
etSpeedtest;
613       if (refreshAdaptersBtn) refreshAdaptersBtn.onclick = fet
chAdapters;
614       if (refreshConnsBtn) refreshConnsBtn.onclick = fetchConn
ections;
615       if (startBtn) startBtn.onclick = startLiveCapture;
616       if (stopBtn) stopBtn.onclick = stopLiveCapture;
617       if (analyzePcapBtn) analyzePcapBtn.onclick = analyzePcap
File;
618       if (clearPacketsBtn) clearPacketsBtn.onclick = () => {
619         capturedPackets = [];
620         renderPacketsTable();
```

```
      };

      if (connsFilterInput) connsFilterInput.oninput = renderC
onnectionsTable;
      if (tableFilterInput) tableFilterInput.oninput = renderP
acketsTable;

      // Filter buttons for connections
      document.querySelectorAll('.btn-filter-conns').forEach(b
tn => {
        btn.onclick = () => {
          document.querySelectorAll('.btn-filter-conns').forEa
ch(b => b.classList.remove('active'));
          btn.classList.add('active');
          activeConnFilter = btn.getAttribute('data-filter') |
| 'all';
          renderConnectionsTable();
        };
      });

      if (configBtn) configBtn.onclick = () => {
        if (typeof window.openAppConfigModal === 'function') {
          window.openAppConfigModal('network_terminal', 'Netwo
rk Analyzer Terminal');
        }
      };

      isNetInitialized = true;
    }
  }

  window.initNetworkTab = initNetworkTab;
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/news_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/news_tab/index.html" cl
ass="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/news_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- News Tab Component -->
<div class="container-fluid p-3" id="news-tab-root">
<!-- Top Banner / Digest Generator Bar -->
<div class="card border mb-3 shadow-sm" data-panel-id="panel-n
ews-1" id="panel-news-1">
<div class="card-body p-3 d-flex flex-wrap align-items-center
justify-content-between gap-3">
<div class="d-flex align-items-center gap-3">
<div class="p-3 rounded border text-primary display-6 d-flex a
lign-items-center justify-content-center" style="width: 54px;
```

```
height: 54px; background: var(--bg-color);">
<i class="bi bi-newspaper"></i>
</div>
<div>
<h5 class="mb-1 d-flex align-items-center gap-2">
            Интеллектуальная лента новостей
            <span class="badge bg-primary rounded-pill small" id="news-total-badge">0</span>
</h5>
<p class="mb-0 text-muted small">
            Только релевантные новости с непрерывным обучением под ваши личные интересы и AI-выжимками.
          </p>
</div>
</div>
<div class="d-flex align-items-center gap-2">
<button class="btn btn-outline-secondary btn-sm d-flex align-items-center gap-1" id="btn-refresh-news" onclick="window.newsTab?.refreshNews(true)" type="button">
<i class="bi bi-arrow-repeat"></i> Обновить
        </button>
<button class="btn btn-outline-info btn-sm d-flex align-items-center gap-1 shadow-sm" data-bs-target="#newsPreferencesModal" data-bs-toggle="modal" id="btn-preferences-news" type="button">
<i class="bi bi-sliders"></i> Мои интересы
        </button>
<button class="btn btn-primary btn-sm d-flex align-items-center gap-1 shadow-sm" id="btn-generate-digest" onclick="window.newsTab?.generateDigest()" type="button">
<i class="bi bi-stars"></i> ИИ-дайджест
        </button>
</div>
</div>
</div>
<!-- AI Digest Section (Appears when generated) -->
<div class="card border-primary mb-3 shadow-sm d-none" data-panel-id="panel-news-2" id="news-digest-card">
```

```
<div class="card-header d-flex justify-content-between align-items-center border-primary py-2">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-stars text-warning fs-5"></i>
<strong id="news-digest-title">Персональный ИИ-дайджест</strong>
</div>
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm btn-outline-secondary py-0 px-2" id="news-digest-tts-btn" onclick="window.newsTab?.speakDigest()" title="Озвучить дайджест">
<i class="bi bi-volume-up me-1"></i>Слушать
        </button>
<button aria-label="Close" class="btn-close" onclick="document.getElementById('news-digest-card').classList.add('d-none')" type="button"></button>
</div>
</div>
<div class="card-body p-3">
<div class="small" id="news-digest-content" style="line-height: 1.6; white-space: pre-line;"></div>
</div>
</div>
<!-- Filter & Learning Feedback Bar -->
<div class="card mb-3" data-panel-id="panel-news-3" id="panel-news-3">
<div class="card-body p-2 d-flex flex-wrap align-items-center justify-content-between gap-2">
<div class="d-flex align-items-center gap-2 flex-grow-1" style="max-width: 480px;">
<div class="input-group input-group-sm">
<span class="input-group-text"><i class="bi bi-search"></i></span>
<input class="form-control" id="news-search-input" oninput="window.newsTab?.filterNews()" placeholder="Фильтр новостей по тексту..." type="text"/>
</div>
</div>
```

```
<div class="d-flex align-items-center gap-2">
<label class="small text-muted text-nowrap" for="news-category-filter">Категория:</label>
<select class="form-select form-select-sm" id="news-category-filter" onchange="window.newsTab?.filterNews()" style="width: auto;">
<option value="all">Все категории</option>
<option value="ai">ИИ &amp;amp; ML</option>
<option value="dev">Разработка</option>
<option value="tech">Технологии</option>
<option value="science">Наука</option>
<option value="world">Мировые</option>
</select>
<label class="small text-muted text-nowrap ms-2" for="news-min-score">Мин. совпадение:</label>
<select class="form-select form-select-sm" id="news-min-score" onchange="window.newsTab?.filterNews()" style="width: auto;">
<option value="0">Все (0%)</option>
<option selected="" value="0.25">Релевантные (&gt;= 25%)</option>
<option value="0.5">Высокое (&gt;= 50%)</option>
<option value="0.7">Только топ (&gt;= 70%)</option>
</select>
</div>
</div>
</div>
<!-- News Stream Grid -->
<div class="row g-3" id="news-articles-container">
<div class="col-12 text-center text-muted p-5" id="news-loading-spinner">
<div class="spinner-border spinner-border-sm text-primary mb-2" role="status"></div>
<div>Сбор и ранжирование релевантных новостей под ваши интересы...</div>
</div>
</div>
</div>
<!-- Preferences & Topics Modal -->
```

```
<div aria-hidden="true" aria-labelledby="newsPreferencesModalLabel" class="modal fade" id="newsPreferencesModal" tabindex="-1">
<div class="modal-dialog modal-lg modal-dialog-centered">
<div class="modal-content shadow">
<div class="modal-header border-bottom">
<h5 class="modal-title d-flex align-items-center gap-2" id="newsPreferencesModalLabel">
<i class="bi bi-sliders text-info"></i> Настройка интересов и профиля обучения
        </h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body p-3">
<!-- Learned Weights Section -->
<div class="mb-3">
<label class="form-label small text-muted fw-semibold">
<i class="bi bi-lightning-charge text-warning me-1"></i>Накопленные веса интересов (адаптируются на основе лайков):
          </label>
<div class="p-2 border rounded d-flex flex-wrap gap-2" id="news-learned-tags-container" style="max-height: 140px; overflow-y: auto; background: var(--bg-color);">
<!-- Badges rendered via JS -->
</div>
</div>
<!-- Preferred Topics Input -->
<div class="mb-3">
<label class="form-label small fw-semibold" for="news-pref-topics-input">
            Явные ключевые темы (через запятую):
          </label>
<input class="form-control" id="news-pref-topics-input" placeholder="например: ai, python, robotics, neuroscience, opensource" type="text"/>
<small class="text-muted">Новости с этими словами получат приоритетный буст в вашей ленте.</small>
```

```
</div>
<!-- Negative Keywords Input -->
<div class="mb-3">
<label class="form-label small fw-semibold" for="news-neg-topics-input">
            Стоп-слова и нежелательные темы (через запятую):
          </label>
<input class="form-control" id="news-neg-topics-input" placeholder="например: криминал, шоу-бизнес, скандалы" type="text"/>
<small class="text-muted">Статьи с этими словами будут скрываться или получать минимальный скор.</small>
</div>
<!-- Stats & Reset -->
<div class="d-flex align-items-center justify-content-between p-2 rounded border" style="background: var(--bg-color);">
<span class="small text-muted" id="news-interactions-count-label">Обработано реакций: 0</span>
<button class="btn btn-sm btn-outline-danger" onclick="window.newsTab?.resetProfile()" type="button">
<i class="bi bi-arrow-counterclockwise me-1"></i>Сбросить обучение
          </button>
</div>
</div>
<div class="modal-footer border-top">
<button class="btn btn-sm btn-secondary" data-bs-dismiss="modal" type="button">Закрыть</button>
<button class="btn btn-sm btn-primary" onclick="window.newsTab?.savePreferences()" type="button">Сохранить интересы</button>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows News Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/news_tab/main.js?v=202
61001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/news_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

// News Tab Controller
class NewsTabController {
  constructor() {
    this.articles = [];
    this.profile = null;
    this.currentDigest = null;
    this.isLoading = false;
  }

  async init() {
    console.log('[NewsTab] Initializing Smart News Feed...');
    await this.loadProfile();
```

```
    await this.refreshNews(false);
  }

  async loadProfile() {
    try {
      const res = await fetch('/api/news/profile');
      if (res.ok) {
        this.profile = await res.json();
        this.renderProfileModal();
      }
    } catch (e) {
      console.error('[NewsTab] Error loading profile:', e);
    }
  }

  renderProfileModal() {
    if (!this.profile) return;
    const container = document.getElementById('news-learned-tags-container');
    if (container) {
      container.innerHTML = '';
      const weights = this.profile.topic_weights || {};
      const sorted = Object.entries(weights).sort((a, b) => b[1] - a[1]);
      if (sorted.length === 0) {
        container.innerHTML = '<span class="text-muted small">Пока нет накопленных весов. Оценивайте новости кнопками  и  !</span>';
      } else {
        for (const [kw, w] of sorted) {
          const badge = document.createElement('span');
          badge.className = 'badge bg-secondary border border-secondary text-light d-flex align-items-center gap-1';
          badge.innerHTML = `<span>${kw}</span> <span class="badge bg-primary text-white">${w}</span>`;
          container.appendChild(badge);
        }
      }
```

```
    }

    const countLabel = document.getElementById('news-interacti
ons-count-label');
    if (countLabel) {
      countLabel.textContent = `Обработано реакций обучения: $
{this.profile.interaction_count || 0}`;
    }

    const negInput = document.getElementById('news-neg-topics-
input');
    if (negInput && this.profile.negative_keywords) {
      negInput.value = this.profile.negative_keywords.join(',
');
    }
  }

  async refreshNews(force = false) {
    if (this.isLoading) return;
    this.isLoading = true;
    const container = document.getElementById('news-articles-c
ontainer');
    const spinner = document.getElementById('news-loading-spin
ner');
    if (spinner) spinner.classList.remove('d-none');

    try {
      const url = `/api/news/feed?force_refresh=${force}&limit
=35`;
      const res = await fetch(url);
      if (res.ok) {
        this.articles = await res.json();
        this.renderArticles();
        const totalBadge = document.getElementById('news-total
-badge');
        if (totalBadge) totalBadge.textContent = this.articles
.length;
      }
```

```
    } catch (e) {
      console.error('[NewsTab] Error fetching feed:', e);
      if (container) {
        container.innerHTML = `<div class="col-12 alert alert-
danger">Ошибка загрузки новостей: ${e.message}</div>`;
      }
    } finally {
      this.isLoading = false;
    }
  }

  renderArticles() {
    const container = document.getElementById('news-articles-c
ontainer');
    if (!container) return;
    container.innerHTML = '';

    const query = (document.getElementById('news-search-input'
)?.value || '').toLowerCase().trim();
    const cat = document.getElementById('news-category-filter'
)?.value || 'all';
    const minScore = parseFloat(document.getElementById('news-
min-score')?.value || '0');

    const filtered = this.articles.filter((a) => {
      if (cat !== 'all' && a.category !== cat) return false;
      if (a.relevance_score < minScore) return false;
      if (query) {
        const text = `${a.title} ${a.summary} ${a.source_name}
`.toLowerCase();
        if (!text.includes(query)) return false;
      }
      return true;
    });

    if (filtered.length === 0) {
      container.innerHTML = `
        <div class="col-12 text-center text-muted p-5">
```

```
          <i class="bi bi-inbox fs-1 mb-2 d-block"></i>
          <h6>Нет новостей, подходящих под выбранные фильтры</h6>
          <small>Попробуйте уменьшить порог минимального совпадения или обновить ленту.</small>
        </div>
      `;
      return;
    }

    filtered.forEach((art) => {
      const col = document.createElement('div');
      col.className = 'col-md-6 col-lg-4';

      const matchPercent = Math.round((art.relevance_score || 0.5) * 100);
      let matchBadgeClass = 'bg-secondary';
      if (matchPercent >= 70) matchBadgeClass = 'bg-success';
      else if (matchPercent >= 45) matchBadgeClass = 'bg-primary';
      else if (matchPercent >= 25) matchBadgeClass = 'bg-info text-dark';

      const isLiked = art.user_interaction === 'like';
      const isDisliked = art.user_interaction === 'dislike';
      const isBookmarked = art.user_interaction === 'bookmark';

      col.innerHTML = `
        <div class="card h-100 shadow-sm d-flex flex-column" id="news-card-${art.id}">
          <div class="card-header d-flex justify-content-between align-items-center py-2">
            <span class="badge bg-secondary-subtle text-secondary-emphasis border small text-truncate" style="max-width: 130px;">
              ${art.source_name}
            </span>
```

```
            <span class="badge ${matchBadgeClass} rounded-pill
 small" title="Персональная релевантность">
              ${matchPercent}% match
            </span>
          </div>
          <div class="card-body p-3 d-flex flex-column flex-gr
ow-1">
            <h6 class="card-title fw-bold mb-2">
              <a href="${art.link}" target="_blank" rel="noope
ner noreferrer" class="text-reset text-decoration-none hover-p
rimary" onclick="window.newsTab?.onArticleRead('${art.id}', '$
{encodeURIComponent(art.title)}')">
                ${art.title}
              </a>
            </h6>
            <p class="card-text small text-muted flex-grow-1 m
b-3">
              ${art.ai_summary || art.summary || ''}
            </p>
            <div class="d-flex align-items-center justify-cont
ent-between mt-auto pt-2 border-top border-secondary-subtle">
              <small class="text-secondary" style="font-size:
0.75rem;">
                <i class="bi bi-clock me-1"></i>${art.publishe
d_at || ''}
              </small>
              <div class="btn-group btn-group-sm" role="group"
>
                <button type="button" class="btn ${isLiked ? '
btn-success' : 'btn-outline-secondary'} py-0 px-2" title="Поле
зно (обучить алгоритм)" onclick="window.newsTab?.sendFeedback(
'${art.id}', 'like', '${encodeURIComponent(art.title)}', '${ar
t.category}')">
                  <i class="bi bi-hand-thumbs-up"></i>
                </button>
                <button type="button" class="btn ${isDisliked
? 'btn-danger' : 'btn-outline-secondary'} py-0 px-2" title="Не
 интересно (понизить тему)" onclick="window.newsTab?.sendFeedb
```

```
ack('${art.id}', 'dislike', '${encodeURIComponent(art.title)}', '${art.category}')">
                  <i class="bi bi-hand-thumbs-down"></i>
                </button>
                <button type="button" class="btn ${isBookmarked ? 'btn-warning text-dark' : 'btn-outline-secondary'} py-0 px-2" title="Сохранить в закладки" onclick="window.newsTab?.sendFeedback('${art.id}', 'bookmark', '${encodeURIComponent(art.title)}', '${art.category}')">
                  <i class="bi bi-bookmark"></i>
                </button>
                <button type="button" class="btn btn-outline-info py-0 px-2" title="Сгенерировать AI-выжимку" onclick="window.newsTab?.summarizeArticle('${art.id}')">
                  <i class="bi bi-magic"></i>
                </button>
              </div>
            </div>
          </div>
        </div>
      `;
      container.appendChild(col);
    });
  }

  filterNews() {
    this.renderArticles();
  }

  async sendFeedback(articleId, action, titleEncoded, category) {
    const title = decodeURIComponent(titleEncoded || '');
    try {
      const res = await fetch('/api/news/feedback', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({
          article_id: articleId,
```

```
          action: action,
          article_title: title,
          category: category || 'general'
        })
      });

      if (res.ok) {
        // Optimistic UI update
        const art = this.articles.find((a) => a.id === articleId);
        if (art) {
          art.user_interaction = action;
        }
        this.renderArticles();
        await this.loadProfile();
        if (window.showNotification) {
          const msg = action === 'like' ? 'Алгоритм обучен: тема получила приоритет  ' :
                      action === 'dislike' ? 'Алгоритм обучен: интерес к теме снижен  ' :
                      action === 'bookmark' ? 'Добавлено в закладки  ' : 'Действие сохранено';
          window.showNotification(msg, 'success');
        }
      }
    } catch (e) {
      console.error('[NewsTab] Error sending feedback:', e);
    }
  }

  async onArticleRead(articleId, titleEncoded) {
    const title = decodeURIComponent(titleEncoded || '');
    try {
      await fetch('/api/news/feedback', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({
          article_id: articleId,
```

```
          action: 'read',
          article_title: title
        })
      });
    } catch (_) {}
  }

  async summarizeArticle(articleId) {
    const art = this.articles.find((a) => a.id === articleId);
    if (!art) return;

    const card = document.getElementById(`news-card-${articleId}`);
    if (card) {
      const body = card.querySelector('.card-text');
      if (body) {
        body.innerHTML = '<span class="spinner-border spinner-border-sm text-info me-1"></span> Генерация ИИ-выжимки...';
      }
    }

    try {
      const res = await fetch('/api/news/summarize-article', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify(art)
      });
      if (res.ok) {
        const data = await res.json();
        art.ai_summary = data.ai_summary;
        this.renderArticles();
      }
    } catch (e) {
      console.error('[NewsTab] Summarize error:', e);
      this.renderArticles();
    }
  }

```

```
  async generateDigest() {
    const digestCard = document.getElementById('news-digest-card');
    const digestContent = document.getElementById('news-digest-content');
    const digestTitle = document.getElementById('news-digest-title');

    if (digestCard) digestCard.classList.remove('d-none');
    if (digestContent) {
      digestContent.innerHTML = '<div class="text-center p-3"><span class="spinner-border spinner-border-sm text-warning me-2"></span>Формирование персонализированного дайджеста ИИ...</div>';
    }

    try {
      const res = await fetch('/api/news/digest', { method: 'POST' });
      if (res.ok) {
        this.currentDigest = await res.json();
        if (digestTitle) digestTitle.textContent = this.currentDigest.title || 'ИИ-Дайджест';
        if (digestContent) {
          if (window.marked) {
            digestContent.innerHTML = window.marked.parse(this.currentDigest.digest_text || '');
          } else {
            digestContent.textContent = this.currentDigest.digest_text || '';
          }
        }
      }
    } catch (e) {
      console.error('[NewsTab] Error generating digest:', e);
      if (digestContent) digestContent.textContent = `Ошибка генерации дайджеста: ${e.message}`;
    }
```

```
  }

  speakDigest() {
    if (!this.currentDigest || !this.currentDigest.digest_text) return;
    if (window.chatService && typeof window.chatService.speak === 'function') {
      window.chatService.speak(this.currentDigest.digest_text);
    } else {
      const utterance = new SpeechSynthesisUtterance(this.currentDigest.digest_text);
      utterance.lang = 'ru-RU';
      window.speechSynthesis.speak(utterance);
    }
  }

  async savePreferences() {
    const prefInput = document.getElementById('news-pref-topics-input')?.value || '';
    const negInput = document.getElementById('news-neg-topics-input')?.value || '';

    const preferred = prefInput.split(',').map((s) => s.trim()).filter(Boolean);
    const ignored = negInput.split(',').map((s) => s.trim()).filter(Boolean);

    try {
      const res = await fetch('/api/news/profile', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({
          preferred_topics: preferred,
          ignored_topics: ignored
        })
      });
      if (res.ok) {
```

```
        await this.loadProfile();
        await this.refreshNews(true);
        if (window.showNotification) window.showNotification('Интересы успешно сохранены!', 'success');
        const modalEl = document.getElementById('newsPreferencesModal');
        if (modalEl && window.bootstrap) {
          const modalInstance = window.bootstrap.Modal.getInstance(modalEl);
          if (modalInstance) modalInstance.hide();
        }
      }
    } catch (e) {
      console.error('[NewsTab] Error saving preferences:', e);
    }
  }

  async resetProfile() {
    if (!confirm('Сбросить накопленные веса интересов и начать обучение с чистого листа?')) return;
    try {
      const res = await fetch('/api/news/profile', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({
          preferred_topics: [],
          ignored_topics: []
        })
      });
      if (res.ok) {
        await this.loadProfile();
        await this.refreshNews(true);
        if (window.showNotification) window.showNotification('Профиль обучения сброшен', 'info');
      }
    } catch (e) {
      console.error('[NewsTab] Error resetting profile:', e);
    }
```

```
  }
}

window.newsTab = new NewsTabController();

function initNewsTab() {
  if (window.newsTab) {
    window.newsTab.init();
  }
}

window.initNewsTab = initNewsTab;

if (document.getElementById('news-tab-root')) {
  initNewsTab();
}

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/ninite_updater_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/ninite_updater_tab/inde
x.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/ninite_updater_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!-- Ninite Auto-Updater & Task Scheduler Management Tab -->
<style>
  .ninite-container {
    padding: 0.75rem 1rem;
    max-width: 1200px;
    margin: 0 auto;
  }
  .ninite-card {
    background: var(--surface-1) !important;
```

```
    border: 1px solid var(--border-color) !important;
    border-radius: 10px;
    padding: 1.25rem;
    color: var(--text-color);
    box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.08);
  }
  .ninite-header {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 10px;
    padding: 0.85rem 1.25rem;
    color: var(--text-color);
  }
  .ninite-dropzone {
    border: 2px dashed var(--nav-active);
    background: var(--surface-2);
    border-radius: 10px;
    padding: 2rem 1.5rem;
    text-align: center;
    cursor: pointer;
    transition: all 0.2s ease-in-out;
  }
  .ninite-dropzone:hover, .ninite-dropzone.dragover {
    background: var(--nav-hover-bg, rgba(56, 189, 248, 0.12));
    border-color: var(--nav-active);
  }
  .ninite-stat-box {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 8px;
    padding: 0.75rem 1rem;
  }
  .ninite-stat-title {
    font-size: 0.75rem;
    text-transform: uppercase;
    letter-spacing: 0.05em;
    color: var(--text-muted);
  }
```

```
  .ninite-stat-val {
    font-size: 1.1rem;
    font-weight: 600;
    color: var(--text-color);
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
  }
  .ninite-log-terminal {
    background: var(--surface-2);
    color: var(--text-color);
    border: 1px solid var(--border-color);
    border-radius: 6px;
    padding: 0.65rem 0.85rem;
    max-height: 180px;
    overflow-y: auto;
    font-family: ui-monospace, monospace;
    white-space: pre-wrap;
  }
</style>

<div class="ninite-container">
  <!-- Верхняя информационная панель -->
  <div class="d-flex justify-content-between align-items-cente
r flex-wrap gap-2 mb-3 ninite-header">
    <div class="d-flex align-items-center gap-2.5">
      <span class="fs-4">□ </span>
      <div>
        <h5 class="mb-0 fw-bold fs-6">Установка и автообновлен
ие Ninite</h5>
        <span class="text-secondary small">Автоматическое полу
чение свежих версий ПО через Windows Task Scheduler</span>
      </div>
    </div>
    <div class="d-flex align-items-center gap-2">
      <span class="badge rounded-pill bg-body-secondary border
 border-secondary text-body px-2.5 py-1.5" id="ninite-status-b
adge">
        Загрузка статуса...
```

```
      </span>
      <button class="btn btn-sm btn-outline-info rounded-pill px-3" id="btn-ninite-refresh" type="button">
        <i class="bi bi-arrow-clockwise me-1"></i> Обновить
      </button>
    </div>
  </div>

  <div class="row g-3 mb-3">
    <!-- Левая колонка: Форма загрузки и настройки расписания -->
    <div class="col-12 col-lg-7">
      <div class="ninite-card h-100">
        <h6 class="fw-bold text-info mb-3 d-flex align-items-center gap-2">
          <i class="bi bi-cloud-arrow-up-fill"></i> 1. Выбор инсталлятора и расписание
        </h6>

        <form id="ninite-upload-form">
          <!-- Dropzone для выбора файла -->
          <label for="ninite-file-input" class="ninite-dropzone mb-3 d-block text-decoration-none" id="ninite-dropzone">
            <i class="bi bi-file-earmark-arrow-up fs-1 text-info mb-2 d-block"></i>
            <div class="fw-semibold mb-1 text-body" id="ninite-file-label">Перетащите файл установщика Ninite или нажмите для выбора</div>
            <div class="text-secondary small">Файл будет сохранён и переименован в <code>C:\Program Files\Ninite\ninite.exe</code></div>
            <input type="file" id="ninite-file-input" class="d-none" accept=".exe">
          </label>

          <!-- Параметры расписания -->
          <div class="row g-2 mb-3">
            <div class="col-12 col-sm-6">
```

```
              <label for="ninite-interval" class="form-label small text-secondary fw-semibold">
                <i class="bi bi-calendar-range me-1"></i> Интервал обновления
              </label>
              <select class="form-select form-select-sm" id="ninite-interval">
                <option value="1">Каждую неделю (1 неделя)</option>
                <option value="2" selected>Каждые 2 недели (по умолчанию)</option>
                <option value="3">Каждые 3 недели</option>
                <option value="4">Раз в месяц (4 недели)</option>
              </select>
            </div>

            <div class="col-12 col-sm-6">
              <label for="ninite-time" class="form-label small text-secondary fw-semibold">
                <i class="bi bi-clock me-1"></i> Время запуска
              </label>
              <input type="time" class="form-control form-control-sm" id="ninite-time" value="20:00">
            </div>
          </div>

          <div class="mb-3">
            <label for="ninite-day" class="form-label small text-secondary fw-semibold">
              <i class="bi bi-calendar-event me-1"></i> День недели
            </label>
            <select class="form-select form-select-sm" id="ninite-day">
              <option value="Sunday" selected>Воскресенье</option>
              <option value="Monday">Понедельник</option>
```

```
              <option value="Tuesday">Вторник</option>
              <option value="Wednesday">Среда</option>
              <option value="Thursday">Четверг</option>
              <option value="Friday">Пятница</option>
              <option value="Saturday">Суббота</option>
            </select>
          </div>

          <!-- Кнопка отправки и сохранения -->
          <button type="submit" class="btn btn-primary w-100 py-2 fw-semibold d-flex align-items-center justify-content-center gap-2" id="btn-ninite-submit">
            <i class="bi bi-check-circle-fill"></i> Установить в Program Files и активировать Task Scheduler
          </button>
        </form>

        <div id="ninite-alert-box" class="mt-3" style="display: none;"></div>
      </div>
    </div>

    <!-- Правая колонка: Текущее состояние и управление -->
    <div class="col-12 col-lg-5">
      <div class="ninite-card h-100 d-flex flex-column justify-content-between">
        <div>
          <h6 class="fw-bold text-info mb-3 d-flex align-items-center gap-2">
            <i class="bi bi-cpu-fill"></i> 2. Текущее состояние системы
          </h6>

          <div class="row g-2 mb-3">
            <div class="col-12">
              <div class="ninite-stat-box">
                <div class="ninite-stat-title">Файл Ninite в Program Files</div>
```

```
                <div class="ninite-stat-val text-truncate" id="ninite-file-status">Проверка...</div>
                <div class="text-secondary small mt-1" id="ninite-file-details">Путь: C:\Program Files\Ninite\ninite.exe</div>
              </div>
            </div>

            <div class="col-12">
              <div class="ninite-stat-box">
                <div class="ninite-stat-title">Задача Windows Task Scheduler</div>
                <div class="ninite-stat-val text-truncate" id="ninite-task-status">Проверка...</div>
                <div class="text-secondary small mt-1" id="ninite-task-details">Задача: NiniteAutoUpdate</div>
              </div>
            </div>

            <div class="col-12">
              <div class="ninite-stat-box">
                <div class="ninite-stat-title">Следующий запуск по расписанию</div>
                <div class="ninite-stat-val text-info text-truncate" id="ninite-task-next">--</div>
              </div>
            </div>
          </div>
        </div>

        <div>
          <!-- Кнопка восстановления удаленных файлов (R-Studio) -->
          <button class="btn btn-outline-info w-100 py-2 fw-semibold mb-2 d-flex align-items-center justify-content-center gap-2" id="btn-ninite-recovery-launch" type="button" title="Запуск R-Studio Technician Portable для восстановления данных">
            <i class="bi bi-folder-symlink-fill"></i> Восстано
```

```
вить удаленные файлы
          </button>

          <!-- Кнопка принудительного обновления -->
          <button class="btn btn-outline-warning w-100 py-2 fw-semibold mb-2 d-flex align-items-center justify-content-center gap-2" id="btn-ninite-run-now">
            <i class="bi bi-play-circle-fill"></i> Запустить обновление прямо сейчас
          </button>

          <button class="btn btn-sm btn-outline-secondary w-100" id="btn-ninite-update-schedule-only" type="button">
            <i class="bi bi-sliders"></i> Обновить только расписание (без перезагрузки файла)
          </button>
        </div>
      </div>
    </div>
  </div>

  <!-- Журнал выполнения / лог -->
  <div class="ninite-card">
    <div class="d-flex justify-content-between align-items-center mb-2">
      <h6 class="fw-bold text-info mb-0 d-flex align-items-center gap-2">
        <i class="bi bi-terminal-fill"></i> Журнал выполнения обновления (Log)
      </h6>
      <span class="text-secondary small" id="ninite-log-timestamp"></span>
    </div>
    <pre class="ninite-log-terminal small mb-0" id="ninite-log-viewer">Логи обновлений пока отсутствуют.</pre>
  </div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Ninite Updater Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/ninite_updater_tab/mai
n.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/ninite_updater_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * ninite_updater_tab/main.js — логика управления установкой и
 автообновлением Ninite
 */

(function () {
  'use strict';

  let selectedFile = null;

  /**
   * Настройка обработчиков событий
```

```
   */
  function setupEventListeners() {
    const dropzone = document.getElementById('ninite-dropzone'
);
    const fileInput = document.getElementById('ninite-file-inp
ut');
    const form = document.getElementById('ninite-upload-form')
;
    const refreshBtn = document.getElementById('btn-ninite-ref
resh');
    const runNowBtn = document.getElementById('btn-ninite-run-
now');
    const scheduleOnlyBtn = document.getElementById('btn-ninit
e-update-schedule-only');

    if (fileInput) {
      fileInput.addEventListener('change', (e) => {
        if (e.target.files && e.target.files[0]) {
          handleFileSelect(e.target.files[0]);
        }
      });
    }

    if (dropzone) {
      dropzone.addEventListener('dragover', (e) => {
        e.preventDefault();
        dropzone.classList.add('dragover');
      });

      dropzone.addEventListener('dragleave', () => {
        dropzone.classList.remove('dragover');
      });

      dropzone.addEventListener('drop', (e) => {
        e.preventDefault();
        dropzone.classList.remove('dragover');
        if (e.dataTransfer.files && e.dataTransfer.files[0]) {
          handleFileSelect(e.dataTransfer.files[0]);
```

```
        }
      });
    }

    if (form) {
      form.addEventListener('submit', handleFormSubmit);
    }

    if (refreshBtn) {
      refreshBtn.addEventListener('click', () => loadStatus())
;
    }

    if (runNowBtn) {
      runNowBtn.addEventListener('click', handleRunNow);
    }

    if (scheduleOnlyBtn) {
      scheduleOnlyBtn.addEventListener('click', handleSchedule
Only);
    }

    const recoveryBtn = document.getElementById('btn-ninite-re
covery-launch');
    if (recoveryBtn) {
      recoveryBtn.addEventListener('click', handleLaunchRecove
ry);
    }
  }

  /**
   * Запуск R-Studio для восстановления удаленных файлов
   */
  async function handleLaunchRecovery() {
    const btn = document.getElementById('btn-ninite-recovery-l
aunch');
    if (btn) {
      btn.disabled = true;
```

```
      btn.innerHTML = `<span class="spinner-border spinner-bor
der-sm me-2"></span>Запуск R-Studio...`;
    }

    try {
      const res = await fetch('/api/recovery/launch', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
      });
      const data = await res.json();
      if (!res.ok) {
        throw new Error(data.detail || `HTTP ${res.status}`);
      }

      showAlert(`<strong>Успех:</strong> ${data.message || 'Пр
ограмма восстановления файлов R-Studio запущена!'}`, 'success'
);
      if (window.toast) {
        window.toast.success('R-Studio запущена', data.message
 || 'Окно программы открывается...');
      }
    } catch (err) {
      showAlert(`<strong>Ошибка запуска:</strong> ${err.messag
e}`, 'danger');
      if (window.toast) {
        window.toast.error('Ошибка запуска R-Studio', err.mess
age);
      }
    } finally {
      if (btn) {
        btn.disabled = false;
        btn.innerHTML = `<i class="bi bi-folder-symlink-fill">
</i> Восстановить удаленные файлы`;
      }
    }
  }

  /**
```

```
   * Обработка выбора файла
   */
  function handleFileSelect(file) {
    selectedFile = file;
    const label = document.getElementById('ninite-file-label')
;
    if (label) {
      const sizeMb = (file.size / (1024 * 1024)).toFixed(2);
      label.innerHTML = `  Выбран файл: <strong class="text-in
fo">${file.name}</strong> (${sizeMb} MB)`;
    }
  }

  /**
   * Загрузка текущего статуса файла и задачи в Task Scheduler
   */
  async function loadStatus() {
    const statusBadge = document.getElementById('ninite-status
-badge');
    const fileStatus = document.getElementById('ninite-file-st
atus');
    const fileDetails = document.getElementById('ninite-file-d
etails');
    const taskStatus = document.getElementById('ninite-task-st
atus');
    const taskDetails = document.getElementById('ninite-task-d
etails');
    const taskNext = document.getElementById('ninite-task-next
');
    const logViewer = document.getElementById('ninite-log-view
er');
    const logTimestamp = document.getElementById('ninite-log-t
imestamp');

    try {
      const res = await fetch(`/api/ninite/status?t=${Date.now
()}`);
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
```

```
      const data = await res.json();

      if (data.file) {
        if (data.file.installed) {
          const sizeMb = (data.file.size_bytes / (1024 * 1024)
).toFixed(2);
          fileStatus.innerHTML = `<span class="text-success">●
 Установлен</span>`;
          fileDetails.textContent = `Размер: ${sizeMb} MB | Из
менён: ${data.file.modified_time}`;
        } else {
          fileStatus.innerHTML = `<span class="text-warning">●
 Не установлен</span>`;
          fileDetails.textContent = `Файл ninite.exe отсутству
ет в Program Files`;
        }
      }

      if (data.task) {
        if (data.task.exists) {
          taskStatus.innerHTML = `<span class="text-success">●
 Активна (${data.task.state})</span>`;
          taskDetails.textContent = `Задача: ${data.task.task_
name} | Посл. запуск: ${data.task.last_run_time || 'нет'}`;
          taskNext.textContent = data.task.next_run_time || 'П
о расписанию';
        } else {
          taskStatus.innerHTML = `<span class="text-secondary"
>● Не запланирована</span>`;
          taskDetails.textContent = `Задача NiniteAutoUpdate о
тсутствует в Task Scheduler`;
          taskNext.textContent = '--';
        }
      }

      if (statusBadge) {
        if (data.file?.installed && data.task?.exists) {
          statusBadge.className = 'badge rounded-pill bg-succe
```

```
ss-subtle text-success border border-success px-2.5 py-1.5';
183           statusBadge.textContent = '● Автообновление активно'
;
184         } else if (data.file?.installed) {
185           statusBadge.className = 'badge rounded-pill bg-warni
ng-subtle text-warning border border-warning px-2.5 py-1.5';
186           statusBadge.textContent = '● Файл готов, расписание
не задано';
187         } else {
188           statusBadge.className = 'badge rounded-pill bg-secon
dary-subtle text-light border border-secondary px-2.5 py-1.5';
189           statusBadge.textContent = '● Ожидание установки';
190         }
191       }
192
193       if (logViewer && data.log) {
194         logViewer.textContent = data.log;
195         if (logTimestamp) {
196           logTimestamp.textContent = `Обновлено: ${new Date().
toLocaleTimeString()}`;
197         }
198       }
199     } catch (err) {
200       console.error('Ошибка загрузки статуса Ninite:', err);
201       if (statusBadge) {
202         statusBadge.className = 'badge rounded-pill bg-danger-
subtle text-danger border border-danger px-2.5 py-1.5';
203         statusBadge.textContent = '● Ошибка связи с сервером';
204       }
205     }
206   }
207
208   /**
209    * Отправка формы (загрузка файла + настройка Task Scheduler
)
210    */
211   async function handleFormSubmit(e) {
212     e.preventDefault();
```

```
213
214     if (!selectedFile) {
215       showAlert('Пожалуйста, выберите файл инсталлятора Ninite
     для загрузки.', 'warning');
216       return;
217     }
218
219     const interval = document.getElementById('ninite-interval'
    )?.value || '2';
220     const timeStr = document.getElementById('ninite-time')?.va
    lue || '20:00';
221     const submitBtn = document.getElementById('btn-ninite-subm
    it');
222
223     const formData = new FormData();
224     formData.append('file', selectedFile);
225     formData.append('interval_weeks', interval);
226     formData.append('time_str', timeStr);
227
228     if (submitBtn) {
229       submitBtn.disabled = true;
230       submitBtn.innerHTML = `<span class="spinner-border spinn
    er-border-sm me-2"></span>Копирование в Program Files и регист
    рация задачи...`;
231     }
232
233     try {
234       const res = await fetch('/api/ninite/upload', {
235         method: 'POST',
236         body: formData,
237       });
238
239       const data = await res.json();
240       if (!res.ok) {
241         throw new Error(data.detail || `HTTP ${res.status}`);
242       }
243
244       showAlert(`Успешно! ${data.message}`, 'success');
```

```
      await loadStatus();
    } catch (err) {
      showAlert(`Ошибка: ${err.message}`, 'danger');
    } finally {
      if (submitBtn) {
        submitBtn.disabled = false;
        submitBtn.innerHTML = `<i class="bi bi-check-circle-fill me-1"></i> Установить в Program Files и активировать Task Scheduler`;
      }
    }
  }

  /**
   * Обновление только расписания без перезагрузки файла
   */
  async function handleScheduleOnly() {
    const interval = parseInt(document.getElementById('ninite-interval')?.value || '2', 10);
    const timeStr = document.getElementById('ninite-time')?.value || '20:00';
    const day = document.getElementById('ninite-day')?.value || 'Sunday';

    try {
      const res = await fetch('/api/ninite/schedule', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({
          interval_weeks: interval,
          time_str: timeStr,
          days_of_week: day,
        }),
      });

      const data = await res.json();
      if (!res.ok) {
        throw new Error(data.detail || `HTTP ${res.status}`);
```

```
      }

      showAlert(`Расписание обновлено: каждые ${interval} нед. в ${timeStr}`, 'success');
      await loadStatus();
    } catch (err) {
      showAlert(`Ошибка обновления расписания: ${err.message}`, 'danger');
    }
  }

  /**
   * Ручной запуск обновления сейчас
   */
  async function handleRunNow() {
    const btn = document.getElementById('btn-ninite-run-now');
    if (btn) {
      btn.disabled = true;
      btn.innerHTML = `<span class="spinner-border spinner-border-sm me-2"></span>Запуск обновления...`;
    }

    try {
      const res = await fetch('/api/ninite/run-now', { method: 'POST' });
      const data = await res.json();
      if (!res.ok) {
        throw new Error(data.detail || `HTTP ${res.status}`);
      }

      showAlert(`Обновление запущено: ${data.message}`, 'info');
      setTimeout(loadStatus, 3000);
    } catch (err) {
      showAlert(`Ошибка запуска: ${err.message}`, 'danger');
    } finally {
      if (btn) {
        btn.disabled = false;
```

```
        btn.innerHTML = `<i class="bi bi-play-circle-fill me-1"></i> Запустить обновление прямо сейчас`;
      }
    }
  }

  /**
   * Отображение информационного сообщения
   */
  function showAlert(message, type = 'info') {
    const box = document.getElementById('ninite-alert-box');
    if (!box) return;

    box.className = `alert alert-${type} alert-dismissible fade show small`;
    box.innerHTML = `
      <div>${message}</div>
      <button type="button" class="btn-close" data-bs-dismiss="alert" aria-label="Закрыть"></button>
    `;
    box.style.display = 'block';
  }

  /**
   * Инициализация вкладки Ninite Updater
   */
  function initNiniteUpdaterTab() {
    setupEventListeners();
    loadStatus();
  }

  // Экспорт для жизненного цикла tab-core.js и глобального контекста
  window.initNiniteUpdaterTab = initNiniteUpdaterTab;
  window.initNiniteupdaterTab = initNiniteUpdaterTab;
  window.initNinite_updaterTab = initNiniteUpdaterTab;

  // Автоматический запуск при автономной загрузке
```

```
  if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', initNiniteUpdaterTab);
  } else {
    initNiniteUpdaterTab();
  }
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/performance_tracing_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/performance_tracing_tab
/index.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/performance_tracing_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Performance & ETW Tracing Tab -->
<div class="container-fluid py-2 px-3">
<!-- Top Stat Cards -->
<div class="row g-2 mb-3">
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 sha
dow-sm p-3" data-panel-id="panel-performance-tracing-1" id="pa
nel-performance-tracing-1">
<div class="text-muted small text-uppercase fw-semibold">ETW C
ессии трассировки  </div>
```

```
<div class="fs-4 fw-bold mt-1 text-primary" id="pt-sessions-count">--</div>
<div class="small text-muted mt-1" id="pt-sessions-sub">logman query -ets</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-performance-tracing-2" id="panel-performance-tracing-2">
<div class="text-muted small text-uppercase fw-semibold">Счетчики производительности  </div>
<div class="fs-4 fw-bold mt-1 text-info" id="pt-counters-count">--</div>
<div class="small text-muted mt-1" id="pt-counters-sub">typeperf / relog</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-performance-tracing-3" id="panel-performance-tracing-3">
<div class="text-muted small text-uppercase fw-semibold">Сбор данных (Data Collectors)  </div>
<div class="fs-4 fw-bold mt-1 text-success" id="pt-collectors-count">--</div>
<div class="small text-muted mt-1" id="pt-collectors-sub">Active Collectors</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-performance-tracing-4" id="panel-performance-tracing-4">
<div class="text-muted small text-uppercase fw-semibold">Буфер событий (Lost Events)  </div>
<div class="fs-4 fw-bold mt-1 text-warning" id="pt-lost-events">0</div>
```

```
<div class="small text-muted mt-1" id="pt-lost-sub">ETW Health
 Status</div>
</div>
</div>
</div>
<!-- Action Bar -->
<div class="d-flex align-items-center justify-content-between
mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap
-2">
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm btn-primary d-flex align-items-cente
r gap-1.5" id="pt-refresh-btn">
<i class="bi bi-arrow-clockwise"></i>
<span>Обновить трассировки</span>
</button>
<button class="btn btn-sm btn-outline-success d-flex align-ite
ms-center gap-1.5" id="pt-quick-sample-btn">
<i class="bi bi-activity"></i>
<span>Замерить CPU % (typeperf)</span>
</button>
</div>
<div class="d-flex align-items-center gap-2">
<input class="form-control form-control-sm bg-dark text-white
border-secondary" id="pt-search-input" placeholder="Поиск по с
ессиям ETW..." style="width: 240px;" type="text"/>
</div>
</div>
<!-- Main Grid -->
<div class="row g-3">
<div class="col-12 col-lg-7">
<div class="card bg-dark text-white border-secondary shadow-sm
" data-panel-id="panel-performance-tracing-aktivnye-sessii-tra
ssirovki-et" id="panel-performance-tracing-aktivnye-sessii-tra
ssirovki-et">
<div class="card-header border-secondary d-flex align-items-ce
nter justify-content-between py-2 px-3">
<span class="fw-semibold"><i class="bi bi-activity text-primar
y me-2"></i>Активные сессии трассировки (ETW Providers)</span>
```

```
<span class="badge bg-secondary" id="pt-badge">0 сессий</span>
</div>
<div class="card-body p-0 table-responsive" style="max-height: 420px; overflow-y: auto;">
<table class="table table-dark table-hover table-striped align-middle mb-0 small" id="pt-table">
<thead>
<tr class="text-secondary">
<th>Имя сессии</th>
<th>Тип</th>
<th>Размер буфера</th>
<th>Статус</th>
<th class="text-end">Управление</th>
</tr>
</thead>
<tbody id="pt-tbody">
<tr><td class="text-center text-muted py-4" colspan="5">Опрос сессий ETW через logman...</td></tr>
</tbody>
</table>
</div>
</div>
</div>
<div class="col-12 col-lg-5">
<div class="card bg-dark text-white border-secondary shadow-sm mb-3" data-panel-id="panel-performance-tracing-upravlenie-trassirovkoy-logman" id="panel-performance-tracing-upravlenie-trassirovkoy-logman">
<div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">
<span class="fw-semibold"><i class="bi bi-sliders text-warning me-2"></i>Управление трассировкой (logman / typeperf)</span>
</div>
<div class="card-body p-3">
<div class="mb-3">
<label class="form-label small text-muted">Действие трассировщика:</label>
<select class="form-select form-select-sm bg-dark text-white b
```

```
order-secondary" id="pt-action-select">
<option value="logman_query_ets">Опросить сессии трассировки (logman query -ets)</option>
<option value="typeperf_cpu_sample">Мгновенный срез загрузки CPU (typeperf "\Processor(_Total)\% Processor Time" -sc 1)</option>
<option value="typeperf_memory_sample">Срез доступной памяти (typeperf "\Memory\Available MBytes" -sc 1)</option>
<option value="typeperf_disk_sample">Срез дисковой очереди (typeperf "\PhysicalDisk(_Total)\Current Disk Queue Length" -sc 1)</option>
</select>
</div>
<div class="d-flex align-items-center justify-content-between">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="pt-dry-run-switch" type="checkbox"/>
<label class="form-check-label small text-muted" for="pt-dry-run-switch">Безопасный режим (Dry Run)</label>
</div>
<button class="btn btn-sm btn-success px-3" id="pt-execute-btn">
<i class="bi bi-play-fill"></i> Запустить замер
            </button>
</div>
</div>
</div>
<!-- Action Log Output -->
<div class="card bg-dark text-white border-secondary shadow-sm" data-panel-id="panel-performance-tracing-vyvod-trassirovki" id="panel-performance-tracing-vyvod-trassirovki">
<div class="card-header border-secondary py-2 px-3 d-flex align-items-center justify-content-between">
<span class="small fw-semibold text-muted"><i class="bi bi-terminal me-1"></i>Вывод трассировки</span>
<button class="btn btn-link btn-sm text-decoration-none text-muted p-0" id="pt-clear-log-btn">Очистить</button>
```

```
</div>
<div class="card-body p-2">
<pre class="mb-0 small bg-black text-light p-2 rounded" id="pt-console-output" style="max-height: 180px; overflow-y: auto; font-family: Consolas, monospace; font-size: 0.78rem;">Трассировщик готов к сбору метрик.</pre>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Performance Tracing Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/performance_tracing_ta
b/main.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { initPerformanceTracingTab } from '/windows/api
/webgui/performance_tracing_tab/main.js';
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/performance_tracing_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * performance_tracing_tab/main.js — ETW трассировка и счетчик
и производительности (logman, typeperf)
 * Updated: 2026-10-10 17:35:00
 */

const registerTabPoller = window.registerTabPoller || function
() {};
```

```
30
31 let isInitialized = false;
32 let sessionsList = [];
33
34 export async function fetchPerformanceSummary() {
35   try {
36     const res = await fetch('/api/performance-tracing/summary'
);
37     if (!res.ok) throw new Error(`HTTP ${res.status}`);
38     const data = await res.json();
39     renderPerformance(data);
40   } catch (err) {
41     console.warn('[PerformanceTracing] Ошибка получения сводки
:', err);
42   }
43 }
44
45 function renderPerformance(data) {
46   sessionsList = data.sessions || [
47     { name: 'EventLog-Application', type: 'Trace', buffer_size
: '64 KB', status: 'Running' },
48     { name: 'EventLog-System', type: 'Trace', buffer_size: '64
 KB', status: 'Running' },
49     { name: 'Diagtrack-Listener', type: 'Trace', buffer_size:
'128 KB', status: 'Running' }
50   ];
51
52   const sessionsCountEl = document.getElementById('pt-sessions
-count');
53   const countersCountEl = document.getElementById('pt-counters
-count');
54   const collectorsCountEl = document.getElementById('pt-collec
tors-count');
55   const tbody = document.getElementById('pt-tbody');
56   const badge = document.getElementById('pt-badge');
57
58   if (sessionsCountEl) sessionsCountEl.textContent = sessionsL
ist.length;
```

```
  if (countersCountEl) countersCountEl.textContent = data.counters_count || '120+';
  if (collectorsCountEl) collectorsCountEl.textContent = data.collectors_count || '4';
  if (badge) badge.textContent = `${sessionsList.length} сессий`;

  if (!tbody) return;

  tbody.innerHTML = sessionsList.map(s => `
    <tr>
      <td class="fw-semibold text-light">${s.name}</td>
      <td><span class="badge bg-secondary">${s.type || 'Trace'}</span></td>
      <td class="font-monospace small text-muted">${s.buffer_size || '64 KB'}</td>
      <td><span class="badge bg-success">${s.status || 'Running'}</span></td>
      <td class="text-end">
        <button class="btn btn-xs btn-outline-info py-0 px-2" title="Свойства сессии">
          <i class="bi bi-info-circle"></i>
        </button>
      </td>
    </tr>
  `).join('');
}

async function executeTraceAction(customAction = null) {
  const actionSelect = document.getElementById('pt-action-select');
  const dryRunSwitch = document.getElementById('pt-dry-run-switch');
  const output = document.getElementById('pt-console-output');

  const action = customAction || (actionSelect ? actionSelect.value : 'typeperf_cpu_sample');
  const dryRun = dryRunSwitch ? dryRunSwitch.checked : true;
```

```
87
88    if (output) output.textContent = `[${new Date().toLocaleTime
      String()}] Запуск трассировки: ${action}...\n`;
89
90    try {
91      const res = await fetch('/api/performance-tracing/action',
       {
92        method: 'POST',
93        headers: { 'Content-Type': 'application/json' },
94        body: JSON.stringify({ action: action, dry_run: dryRun }
      )
95      });
96      const result = await res.json();
97      if (output) output.textContent += JSON.stringify(result, n
      ull, 2);
98      if (window.toast) {
99        if (res.ok) window.toast.success('Трассировка', `Замер в
      ыполнен: ${result.status || 'OK'}`);
100       else window.toast.error('Ошибка', result.detail || 'Сбой
       замера');
101     }
102   } catch (err) {
103     if (output) output.textContent += `\nОшибка запроса: ${err
      .message}`;
104     if (window.toast) window.toast.error('Ошибка сети', err.me
      ssage);
105   }
106 }
107
108 export function initPerformanceTracingTab() {
109   if (isInitialized) return;
110   isInitialized = true;
111
112   const refreshBtn = document.getElementById('pt-refresh-btn')
      ;
113   if (refreshBtn) refreshBtn.addEventListener('click', fetchPe
      rformanceSummary);
114
```

```
  const sampleBtn = document.getElementById('pt-quick-sample-btn');
  if (sampleBtn) sampleBtn.addEventListener('click', () => executeTraceAction('typeperf_cpu_sample'));

  const execBtn = document.getElementById('pt-execute-btn');
  if (execBtn) execBtn.addEventListener('click', () => executeTraceAction());

  const clearBtn = document.getElementById('pt-clear-log-btn');
  if (clearBtn) clearBtn.addEventListener('click', () => {
    const out = document.getElementById('pt-console-output');
    if (out) out.textContent = 'Консоль очищена.';
  });

  registerTabPoller('tab-performance-tracing', fetchPerformanceSummary, 15000, { immediate: true });
}

window.initPerformanceTracingTab = initPerformanceTracingTab;

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/peripherals_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/peripherals_tab/index.h
tml" class="w-100 h-100 border-0"></iframe>

 * File: index.html
 * Project: ai-windows-admin
 * Package: windows/api/webgui/peripherals_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ================================================================
==================
-->
<style>
  .peripherals-tab-wrapper {
    padding: 0.75rem;
    color: var(--text-color);
  }
  .diag-panel {
    background: var(--surface-1);
    border: 1px solid var(--border-color);
    border-radius: 8px;
    overflow: hidden;
```

```
  }
  .diag-panel-header {
    background: var(--surface-2);
    padding: 8px 14px;
    border-bottom: 1px solid var(--border-color);
    font-weight: 600;
    font-size: 0.85rem;
    color: var(--nav-active);
  }
  .about-sys-spec-table {
    width: 100%;
    border-collapse: separate;
    border-spacing: 0;
  }
  .about-sys-spec-table td {
    padding: 8px 12px;
    border-bottom: 1px solid var(--border-color);
    vertical-align: middle;
  }
  .about-sys-spec-key {
    color: var(--text-muted);
    width: 200px;
    font-size: 0.82rem;
    font-weight: 500;
  }
  .about-sys-spec-val {
    color: var(--text-color);
    font-weight: 500;
    font-size: 0.82rem;
  }
</style>
<div class="peripherals-tab-wrapper">
<!-- Top Header & Status -->
<div class="card shadow-sm border-secondary-subtle mb-3" data-
panel-id="panel-peripherals-5-set-i-periferiya-peripherals" id
="panel-peripherals-5-set-i-periferiya-peripherals">
<div class="card-body p-3">
<div class="d-flex flex-wrap align-items-center justify-conten
```

```
t-between gap-2">
<div class="d-flex align-items-center gap-2">
<div class="fs-4">□ </div>
<div>
<h5 class="mb-0 fw-bold">5. Сеть и периферия (Peripherals &amp; Wireless Network)</h5>
<small class="text-muted">RF-параметры Wi-Fi сетей, аудио конечные точки и Plug-and-Play USB устройства</small>
</div>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<select class="form-select form-select-sm poll-freq-select" data-poll-id="peripherals" id="diag-peripherals-poll-freq" title="Частота опроса: Сеть и периферия">
<option value="start">Только при старте</option>
<option selected="" value="manual">По запросу</option>
<option value="3">Каждые 3 сек</option>
<option value="5">Каждые 5 сек</option>
<option value="10">Каждые 10 сек</option>
<option value="30">Каждые 30 сек</option>
<option value="60">Каждые 60 сек</option>
<option value="120">Каждые 120 сек</option>
</select>
<button class="btn btn-sm btn-outline-info rounded-pill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id="btn-diag-peripherals-refresh">
<i class="bi bi-arrow-clockwise"></i> Обновить
          </button>
</div>
</div>
</div>
</div>
<div class="row g-2 mb-3">
<!-- Wi-Fi Telemetry Panel -->
<div class="col-12 col-lg-5">
<div class="diag-panel h-100" data-panel-id="panel-peripherals-1" id="panel-peripherals-1">
<div class="diag-panel-header d-flex justify-content-between a
```

```
lign-items-center">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-wifi text-info"></i>
<span class="text-body">Беспроводная сеть Wi-Fi (RF &amp;amp; Security)</span>
</div>
<span class="badge bg-success-subtle text-success border border-success" id="diag-wifi-status-badge">Wi-Fi Active</span>
</div>
<div class="p-3">
<table class="about-sys-spec-table">
<tbody>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-broadcast me-1.5 text-info"></i>Сеть (SSID)</td>
<td class="about-sys-spec-val fw-bold text-body" id="diag-wifi-ssid">--</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-router me-1.5 text-primary"></i>Точка доступа (BSSID)</td>
<td class="about-sys-spec-val font-monospace" id="diag-wifi-bssid">--</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-reception-4 me-1.5 text-success"></i>Сигнал (RSSI)</td>
<td class="about-sys-spec-val" id="diag-wifi-signal">0% (-- dBm)</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-hash me-1.5 text-warning"></i>Радиоканал</td>
<td class="about-sys-spec-val" id="diag-wifi-channel">--</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-cpu me-1.5 text-info"></i>Стандарт связи</td>
```

```
<td class="about-sys-spec-val font-monospace" id="diag-wifi-radio">--</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-shield-lock me-1.5 text-success"></i>Безопасность</td>
<td class="about-sys-spec-val text-truncate" id="diag-wifi-auth">--</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- Audio Endpoints Panel -->
<div class="col-12 col-lg-7">
<div class="diag-panel h-100" data-panel-id="panel-peripherals-2" id="panel-peripherals-2">
<div class="diag-panel-header d-flex justify-content-between align-items-center">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-volume-up-fill text-warning"></i>
<span class="text-body">Аудиоустройства и конечные точки (Audio Endpoints)</span>
</div>
<span class="badge bg-secondary" id="diag-audio-count">0 устройств</span>
</div>
<div class="p-2" id="diag-audio-container" style="max-height: 260px; overflow-y: auto;">
<div class="text-center py-4 text-muted small">Опрос аудиоустройств...</div>
</div>
</div>
</div>
</div>
<!-- PnP USB Devices Table -->
<div class="diag-panel" data-panel-id="panel-peripherals-3" id
```

```
="panel-peripherals-3">
<div class="diag-panel-header d-flex justify-content-between a
lign-items-center">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-usb-symbol text-info"></i>
<span class="text-body">Подключенные PnP USB-устройства и конц
ентраторы</span>
</div>
<span class="badge bg-secondary" id="diag-usb-count">0 устройс
тв</span>
</div>
<div class="table-responsive" style="max-height: 480px; overfl
ow-y: auto;">
<table class="table table-hover table-sm mb-0" style="font-siz
e: 0.80rem; font-family: ui-monospace, SFMono-Regular, Menlo,
Monaco, Consolas, monospace;">
<thead>
<tr>
<th>Устройство</th>
<th style="width: 100px;">Vendor ID</th>
<th style="width: 100px;">Product ID</th>
<th style="width: 160px;">Класс устройства</th>
<th style="width: 140px;">Статус</th>
<th>Device Instance ID</th>
</tr>
</thead>
<tbody id="diag-usb-tbody">
<tr><td class="text-center py-4 text-muted" colspan="6">Загруз
ка USB периферии...</td></tr>
</tbody>
</table>
</div>
</div>
</div>
```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Peripherals Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/peripherals_tab/main.j
s?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/peripherals_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

(function () {
  'use strict';

  let autoRefreshTimer = null;

  function escapeHtml(str) {
    if (str === null || str === undefined) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
```

```
      .replace(/'/g, '&#039;');
  }

  function setText(id, text) {
    const el = document.getElementById(id);
    if (el) el.textContent = text !== null && text !== undefin
ed ? String(text) : '--';
  }

  async function apiFetch(url) {
    if (window.api && typeof window.api.fetch === 'function')
{
      return await window.api.fetch(url);
    }
    const res = await fetch(url);
    if (!res.ok) throw new Error(`HTTP ${res.status}: ${res.st
atusText}`);
    return await res.json();
  }

  async function fetchPeripheralsNetwork() {
    try {
      const data = await apiFetch('/api/v1/system/diagnostics/
peripherals');
      if (!data) return;

      // Wi-Fi
      if (data.wifi_telemetry) {
        const w = data.wifi_telemetry;
        setText('diag-wifi-ssid', w.ssid || 'Не подключено');
        setText('diag-wifi-bssid', w.bssid || '--');
        setText('diag-wifi-signal', `${w.signal_pct}% (${w.rss
i_dbm} dBm)`);
        setText('diag-wifi-channel', w.channel || '--');
        setText('diag-wifi-radio', w.radio_type || '--');
        setText('diag-wifi-auth', w.auth_cipher || '--');

        const wifiBadge = document.getElementById('diag-wifi-s
```

```
tatus-badge');
        if (wifiBadge) {
          if (w.is_connected) {
            wifiBadge.className = 'badge bg-success-subtle tex
t-success border border-success';
            wifiBadge.textContent = 'Wi-Fi Подключено';
          } else {
            wifiBadge.className = 'badge bg-secondary';
            wifiBadge.textContent = 'Wi-Fi Отключено';
          }
        }
      }

      // Audio Endpoints
      const audioContainer = document.getElementById('diag-aud
io-container');
      if (audioContainer && data.audio_endpoints) {
        setText('diag-audio-count', `${data.audio_endpoints.le
ngth} устр.`);
        if (data.audio_endpoints.length === 0) {
          audioContainer.innerHTML = '<div class="text-center
py-4 text-muted small">Аудиоустройств не обнаружено</div>';
        } else {
          audioContainer.innerHTML = data.audio_endpoints.map(
a => `
            <div class="p-2 mb-1 rounded bg-black bg-opacity-3
0 border border-secondary-subtle d-flex align-items-center jus
tify-content-between">
              <div>
                <strong class="text-white"><i class="bi bi-spe
aker me-1.5 text-warning"></i>${escapeHtml(a.name)}</strong>
                <div class="small text-muted">${escapeHtml(a.t
ype)}</div>
              </div>
              <span class="badge ${a.is_default ? 'bg-info tex
t-dark fw-bold' : 'bg-secondary'}">${a.is_default ? 'По умолча
нию' : 'Доступно'}</span>
            </div>
```

```
          `).join('');
        }
      }

      // USB PnP Table
      const usbTbody = document.getElementById('diag-usb-tbody
');
      if (usbTbody && data.usb_devices) {
        setText('diag-usb-count', `${data.usb_devices.length}
устройств`);
        if (data.usb_devices.length === 0) {
          usbTbody.innerHTML = '<tr><td colspan="6" class="tex
t-center py-4 text-muted">USB устройств не обнаружено</td></tr
>';
        } else {
          usbTbody.innerHTML = data.usb_devices.map(u => `
            <tr>
              <td class="fw-bold text-white"><i class="bi bi-u
sb-symbol text-info me-1.5"></i>${escapeHtml(u.name)}</td>
              <td class="font-monospace text-warning">${escape
Html(u.vendor_id)}</td>
              <td class="font-monospace text-info">${escapeHtm
l(u.product_id)}</td>
              <td class="text-muted small">${escapeHtml(u.devi
ce_class || 'USB Device')}</td>
              <td><span class="badge ${u.has_problem ? 'bg-dan
ger' : 'bg-success-subtle text-success border border-success'}
">${escapeHtml(u.status)}</span></td>
              <td class="small text-muted font-monospace text-
truncate" style="max-width: 220px;" title="${escapeHtml(u.devi
ce_id)}">${escapeHtml(u.device_id)}</td>
            </tr>
          `).join('');
        }
      }
    } catch (err) {
      console.warn('[PeripheralsTab] fetchPeripheralsNetwork e
rror:', err);
```

```
    }
  }

  const POLL_ID = 'peripherals';

  function getFrequency() {
    try {
      const saved = localStorage.getItem(`poll_freq_${POLL_ID}
`);
      if (saved) return saved;
    } catch (_) {}
    return 'manual';
  }

  function setFrequency(freq) {
    try {
      localStorage.setItem(`poll_freq_${POLL_ID}`, freq);
    } catch (_) {}
    applyPoller(freq, false);
  }

  function stopPolling() {
    if (window.unregisterTabPoller) {
      window.unregisterTabPoller(`tab-peripherals_${POLL_ID}`)
;
    }
    if (autoRefreshTimer) {
      clearInterval(autoRefreshTimer);
      autoRefreshTimer = null;
    }
  }

  function applyPoller(freq, runInitial = false) {
    stopPolling();
    if (freq === 'start' || freq === 'manual') {
      if (runInitial && (window.isTabActive ? window.isTabActi
ve('tab-peripherals') : true)) {
        fetchPeripheralsNetwork();
```

```
      }
      return;
    }

    const intervalSec = parseInt(freq, 10);
    if (isNaN(intervalSec) || intervalSec <= 0) return;

    const intervalMs = intervalSec * 1000;
    const pollerId = `tab-peripherals_${POLL_ID}`;

    if (window.registerTabPoller) {
      window.registerTabPoller('tab-peripherals', fetchPeriphe
ralsNetwork, intervalMs, { pollerId, immediate: runInitial });
    } else {
      if (runInitial && (window.isTabActive ? window.isTabActi
ve('tab-peripherals') : true)) {
        fetchPeripheralsNetwork();
      }
      autoRefreshTimer = setInterval(() => {
        if (window.isTabActive ? window.isTabActive('tab-perip
herals') : true) {
          fetchPeripheralsNetwork();
        }
      }, intervalMs);
    }
  }

  function bindEvents() {
    const btnRefresh = document.getElementById('btn-diag-perip
herals-refresh');
    if (btnRefresh) {
      btnRefresh.onclick = () => fetchPeripheralsNetwork();
    }

    const select = document.getElementById('diag-peripherals-p
oll-freq');
    if (select) {
      select.onchange = (e) => {
```

```
        const newFreq = e.target.value;
        setFrequency(newFreq);
        if (newFreq !== 'manual' && newFreq !== 'start') {
          fetchPeripheralsNetwork();
        }
        if (window.showToast) {
          const label = select.options[select.selectedIndex]?.
text || newFreq;
          window.showToast(`Частота опроса периферии: ${label}
`, 'info');
        }
      };
    }
  }

  async function initPeripheralsTab() {
    bindEvents();
    const currentFreq = getFrequency();
    const select = document.getElementById('diag-peripherals-p
oll-freq');
    if (select) select.value = currentFreq;
    applyPoller(currentFreq, true);
  }

  window.initPeripheralsTab = initPeripheralsTab;
  window.activatePeripheralsTab = () => {
    applyPoller(getFrequency(), true);
  };

  if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', () => {
      if (window.isTabActive ? window.isTabActive('tab-periphe
rals') : false) {
        initPeripheralsTab();
      } else {
        bindEvents();
      }
    });
```

```
  } else {
    if (window.isTabActive ? window.isTabActive('tab-peripherals') : false) {
      initPeripheralsTab();
    } else {
      bindEvents();
    }
  }
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/pixelrag_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/pixelrag_tab/index.html
" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/pixelrag_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!DOCTYPE html>

<html lang="ru">
<head>
<meta charset="utf-8"/>
<meta content="width=device-width, initial-scale=1.0" name="vi
ewport"/>
<title>PixelRAG</title>
</head>
<body>
```

```
<div class="container-fluid p-3">
<h4><i class="bi bi-image-fill text-info"></i> PixelRAG</h4>
<p class="text-muted">Векторный поиск по изображениям и пиксельным данным.</p>
<div class="card border-secondary mb-3" data-panel-id="panel-pixelrag-1" id="panel-pixelrag-1">
<div class="card-header bg-dark text-white">
<i class="bi bi-search"></i> Поиск по изображениям
      </div>
<div class="card-body">
<p class="text-muted">Функционал PixelRAG будет доступен после настройки.</p>
</div>
</div>
</div>
</body>
</html>
```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Pixelrag Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/pixelrag_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/pixelrag_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// PixelRAG Tab Module
// ============================================================

export async function initPixelRagTab() {
  console.log('[PixelRagTab] Initializing...');

  try {
    // PixelRAG initialization logic
    const container = document.getElementById('tab-pixelrag');
    if (container) {
      container.innerHTML = `
```

```
        <div class="alert alert-info">
          <i class="bi bi-info-circle"></i>
          <strong>PixelRAG</strong> — векторный поиск по изображениям и пиксельным данным.
        </div>
        <div class="card border-secondary">
          <div class="card-header bg-dark text-white">
            <i class="bi bi-image-fill"></i> Настройки PixelRAG
          </div>
          <div class="card-body">
            <p class="text-muted">Функционал PixelRAG будет доступен после настройки модели и индекса.</p>
            <button class="btn btn-primary" disabled>Настроить PixelRAG</button>
          </div>
        </div>
      `;
    }
  } catch (err) {
    console.error('[PixelRagTab] Initialization error:', err);
  }
}
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/plugins_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/plugins_tab/index.html"
 class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/plugins_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Plugins Tab -->
<div class="container-fluid py-2">
<div class="alert alert-primary d-flex flex-wrap align-items-c
enter justify-content-between gap-2 mb-3 shadow-sm">
<div class="d-flex align-items-center gap-2">
<span class="fs-4">□ </span>
<div>
<h5 class="mb-0 fw-bold">Подключатель плагинов и расширений</h
5>
<p class="mb-0 small text-muted">Централизованный каталог моду
```

```
лей, активация и органы динамической настройки</p>
</div>
</div>
<div class="d-flex gap-2">
<button class="btn btn-sm btn-outline-primary" id="btn-refresh-plugins" onclick="loadPluginsList()">
<i class="bi bi-arrow-clockwise"></i> Обновить список
      </button>
</div>
</div>
<div class="row g-3">
<!-- LEFT: Plugin Catalog / List -->
<div class="col-lg-4 col-md-5">
<div class="card shadow-sm h-100 border-secondary-subtle" data-panel-id="panel-plugins-ustanovlennye-plaginy-0" id="panel-plugins-ustanovlennye-plaginy-0">
<div class="card-header bg-body-tertiary d-flex justify-content-between align-items-center py-2">
<span class="fw-semibold small text-uppercase">Установленные плагины (<span id="plugins-count">0</span>)</span>
<span class="badge bg-secondary-subtle text-body" id="active-plugins-badge">0 активных</span>
</div>
<div class="p-2 border-bottom">
<div class="input-group input-group-sm">
<span class="input-group-text bg-body border-secondary-subtle"><i class="bi bi-search"></i></span>
<input class="form-control border-secondary-subtle" id="plugin-search-input" oninput="filterPluginsList()" placeholder="Поиск плагина..." type="text"/>
</div>
</div>
<div class="card-body p-2" id="plugins-list-container" style="max-height: 70vh; overflow-y: auto;">
<div class="text-center text-muted p-4">
<div class="spinner-border spinner-border-sm text-primary mb-2" role="status"></div>
<div>Загрузка плагинов...</div>
```

```
</div>
</div>
</div>
</div>
<!-- RIGHT: Selected Plugin Controls & Settings -->
<div class="col-lg-8 col-md-7">
<div class="card shadow-sm h-100 border-secondary-subtle" data-panel-id="panel-plugins-nazvanie-plagina" id="plugin-details-card">
<div class="card-body d-flex flex-column align-items-center justify-content-center text-muted p-5" id="plugin-placeholder">
<i class="bi bi-puzzle fs-1 mb-2"></i>
<h6>Выберите плагин из списка слева для управления и настройки</h6>
</div>
<div class="d-none d-flex flex-column h-100" id="plugin-content-pane">
<!-- Plugin Header -->
<div class="card-header bg-body-tertiary d-flex justify-content-between align-items-center py-3 border-bottom">
<div class="d-flex align-items-center gap-3">
<span class="fs-2" id="plugin-icon">□ </span>
<div>
<div class="d-flex align-items-center gap-2">
<h5 class="mb-0 fw-bold" id="plugin-title">Название плагина</h5>
<span class="badge bg-primary-subtle text-primary border border-primary-subtle" id="plugin-version">v1.0.0</span>
<span class="badge bg-secondary-subtle text-body border" id="plugin-category">tools</span>
<span class="badge bg-info-subtle text-info-emphasis border" id="plugin-scope-badge">Системный</span>
</div>
<small class="text-muted" id="plugin-id">id: plugin_name</small>
</div>
</div>
<div class="form-check form-switch fs-5 mb-0" title="Включить
```

```
/ отключить плагин">
<input class="form-check-input" id="plugin-toggle-switch" onch
ange="toggleCurrentPlugin(this.checked)" role="switch" type="c
heckbox"/>
</div>
</div>
<!-- Description -->
<div class="p-3 bg-body-secondary border-bottom">
<p class="mb-0 small text-body" id="plugin-description">Описан
ие возможностей плагина.</p>
</div>
<!-- Body: Settings & Actions -->
<div class="card-body p-3" style="max-height: 52vh; overflow-y
: auto;">
<!-- Actions Section -->
<div class="mb-4" id="plugin-actions-section">
<h6 class="fw-bold mb-2 pb-1 border-bottom d-flex align-items-
center gap-2">
<i class="bi bi-lightning-charge-fill text-warning"></i> Управ
ляющие действия (Actions)
              </h6>
<div class="d-flex flex-wrap gap-2 pt-1" id="plugin-actions-co
ntainer">
<!-- Action buttons rendered here -->
</div>
</div>
<!-- Configuration Section -->
<div class="mb-4" id="plugin-config-section">
<h6 class="fw-bold mb-2 pb-1 border-bottom d-flex align-items-
center gap-2">
<i class="bi bi-sliders text-primary"></i> Параметры конфигура
ции
              </h6>
<form id="plugin-config-form" onsubmit="event.preventDefault()
; saveCurrentPluginConfig();">
<div class="row g-3" id="plugin-fields-container">
<!-- Form fields rendered here -->
</div>
```

```
<div class="mt-3 text-end">
<button class="btn btn-sm btn-primary px-4" id="btn-save-plugin-config" type="submit">
<i class="bi bi-check2-circle"></i> Сохранить настройки
                  </button>
</div>
</form>
</div>
<!-- Live Action Console / Terminal -->
<div class="mt-3">
<div class="d-flex justify-content-between align-items-center mb-1">
<h6 class="fw-bold mb-0 small text-muted d-flex align-items-center gap-1">
<i class="bi bi-terminal"></i> Консоль выполнения
                </h6>
<button class="btn btn-sm btn-link text-decoration-none text-muted py-0" onclick="clearPluginConsole()">Очистить</button>
</div>
<pre class="p-2 rounded bg-dark text-light small font-monospace mb-0" id="plugin-console-output" style="max-height: 180px; min-height: 60px; overflow-y: auto;">Готов к работе.</pre>
</div>
</div>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Plugins Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/plugins_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/plugins_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// ── PLUGINS TAB MAIN.JS ──────────────────────────────────────

let loadedPlugins = [];
let selectedPluginName = '';

// Notification helper
function showPluginNotification(message, type = 'info') {
  if (typeof window.showNotification === 'function') {
    window.showNotification(message, type);
    return;
  }
```

```
  console.log(`[Plugin Notification ${type}]: ${message}`);
}

// Helper to check if current view is user interface
function isRunningInUserInterface() {
  if (document.getElementById('user-interface')) return true;
  if (window.location.pathname.includes('/user')) return true;
  if (!document.getElementById('admin-interface')) {
    if (typeof window.isUserAdmin === 'function') {
      return !window.isUserAdmin();
    }
    return true;
  }
  return false;
}

// Initialize Plugins Tab
window.initPluginsTab = async function() {
  console.log('Инициализация вкладки плагинов...');
  await loadPluginsList();
};

// Загрузка списка всех плагинов
async function loadPluginsList() {
  const container = document.getElementById('plugins-list-cont
ainer');
  if (!container) return;

  try {
    const isUserUI = isRunningInUserInterface();
    const endpoint = isUserUI ? '/api/plugins?scope=user' : '/
api/admin/plugins';

    let data;
    if (window.api && typeof window.api.fetch === 'function')
{
      try {
        data = await window.api.fetch(endpoint);
```

```
      } catch {
        data = await window.api.fetch('/api/admin/plugins');
      }
    } else {
      let resp = await fetch(endpoint);
      if (!resp.ok && isUserUI) {
        resp = await fetch('/api/admin/plugins?scope=user');
      }
      if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
      data = await resp.json();
    }
    let plugins = data.plugins || [];
    if (isUserUI) {
      plugins = plugins.filter(p => !p.is_system && p.scope !
= 'system');
    }
    loadedPlugins = plugins;

    const countEl = document.getElementById('plugins-count');
    if (countEl) countEl.textContent = loadedPlugins.length;
    const activeCount = loadedPlugins.filter(p => p.enabled).l
ength;
    const badgeEl = document.getElementById('active-plugins-ba
dge');
    if (badgeEl) badgeEl.textContent = `${activeCount} активны
x`;

    renderPluginsList(loadedPlugins);
    syncPluginTabsVisibility(loadedPlugins);

    if (loadedPlugins.length > 0) {
      const toSelect = loadedPlugins.find(p => p.name === sele
ctedPluginName) || loadedPlugins[0];
      selectPlugin(toSelect.name);
    } else {
      showPlaceholder();
    }
  } catch (ex) {
```

```
    console.error('Ошибка загрузки плагинов:', ex);
    container.innerHTML = `<div class="alert alert-danger m-2 small">Ошибка загрузки плагинов: ${ex.message}</div>`;
  }
}

// Отображение списка плагинов
function renderPluginsList(plugins) {
  const container = document.getElementById('plugins-list-container');
  if (!container) return;

  if (plugins.length === 0) {
    const isUserUI = isRunningInUserInterface();
    const emptyMsg = isUserUI ? 'Пользовательские плагины не найдены' : 'Плагины не найдены';
    container.innerHTML = `<div class="text-center text-muted p-3 small">${emptyMsg}</div>`;
    return;
  }

  let html = '<div class="list-group list-group-flush">';
  plugins.forEach(p => {
    const isSelected = p.name === selectedPluginName;
    const statusBadge = p.enabled
      ? '<span class="badge bg-success-subtle text-success border border-success-subtle small">Вкл</span>'
      : '<span class="badge bg-secondary-subtle text-muted border small">Выкл</span>';
    const typeBadge = p.is_system
      ? '<span class="badge bg-secondary-subtle text-body-secondary border small">Системный</span>'
      : '<span class="badge bg-info-subtle text-info-emphasis border small">Пользовательский</span>';

    html += `
      <a href="javascript:void(0)"
         class="list-group-item list-group-item-action p-2 rou
```

```
nded mb-1 border-0 ${isSelected ? 'active text-white' : ''}"
         onclick="selectPlugin('${p.name}')"
         style="cursor: pointer;">
        <div class="d-flex w-100 justify-content-between align
-items-center mb-1">
          <div class="d-flex align-items-center gap-2 text-tru
ncate">
            <span class="fs-5">${p.icon || '□ '}</span>
            <strong class="text-truncate">${p.title || p.name}
</strong>
          </div>
          <div class="d-flex align-items-center gap-1">${typeB
adge}${statusBadge}</div>
        </div>
        <div class="small text-truncate ${isSelected ? 'text-w
hite-50' : 'text-muted'}">
          ${p.description || p.name}
        </div>
      </a>
    `;
  });
  html += '</div>';
  container.innerHTML = html;
}

// Фильтрация списка плагинов
function filterPluginsList() {
  const query = (document.getElementById('plugin-search-input'
)?.value || '').toLowerCase().trim();
  if (!query) {
    renderPluginsList(loadedPlugins);
    return;
  }
  const filtered = loadedPlugins.filter(p =>
    p.name.toLowerCase().includes(query) ||
    (p.title && p.title.toLowerCase().includes(query)) ||
    (p.description && p.description.toLowerCase().includes(que
ry))
```

```
  );
  renderPluginsList(filtered);
}

// Выбор плагина для отображения органов управления
function selectPlugin(pluginName) {
  if (!pluginName) return;
  selectedPluginName = pluginName;
  const normalized = pluginName.toLowerCase().replace(/-/g, '_
');
  const plugin = loadedPlugins.find(p => {
    const pName = (p.name || '').toLowerCase().replace(/-/g, '
_');
    const pId = (p.id || '').toLowerCase().replace(/-/g, '_');
    const pTitle = (p.title || '').toLowerCase();
    return pName === normalized || pId === normalized || pTitl
e === pluginName.toLowerCase();
  });
  if (!plugin) {
    if (!loadedPlugins || loadedPlugins.length === 0) {
      loadPluginsList().then(() => {
        if (loadedPlugins && loadedPlugins.length > 0) {
          const pFound = loadedPlugins.find(p => {
            const pName = (p.name || '').toLowerCase().replace
(/-/g, '_');
            const pId = (p.id || '').toLowerCase().replace(/-/
g, '_');
            const pTitle = (p.title || '').toLowerCase();
            return pName === normalized || pId === normalized
|| pTitle === pluginName.toLowerCase();
          });
          if (pFound) selectPlugin(pFound.name);
        }
      });
    }
    return;
  }

```

```
  renderPluginsList(loadedPlugins);

  const placeholder = document.getElementById('plugin-placeholder');
  const contentPane = document.getElementById('plugin-content-pane');
  if (placeholder) placeholder.classList.add('d-none');
  if (contentPane) contentPane.classList.remove('d-none');

  // Header info
  document.getElementById('plugin-icon').textContent = plugin.icon || '□ ';
  document.getElementById('plugin-title').textContent = plugin.title || plugin.name;
  document.getElementById('plugin-version').textContent = `v${plugin.version || '1.0.0'}`;
  document.getElementById('plugin-category').textContent = plugin.category || 'tools';
  document.getElementById('plugin-id').textContent = `id: ${plugin.name}`;
  document.getElementById('plugin-description').textContent = plugin.description || 'Нет описания';

  const scopeBadgeEl = document.getElementById('plugin-scope-badge');
  if (scopeBadgeEl) {
    if (plugin.is_system) {
      scopeBadgeEl.textContent = 'Системный';
      scopeBadgeEl.className = 'badge bg-secondary-subtle text-body-secondary border';
    } else {
      scopeBadgeEl.textContent = 'Пользовательский';
      scopeBadgeEl.className = 'badge bg-info-subtle text-info-emphasis border';
    }
  }

  const isAdmin = (typeof window.isUserAdmin === 'function') ?
```

```
 window.isUserAdmin() : false;
  const isLocked = Boolean(plugin.is_system && !isAdmin);

  const toggleSwitch = document.getElementById('plugin-toggle-switch');
  if (toggleSwitch) {
    toggleSwitch.checked = Boolean(plugin.enabled);
    toggleSwitch.disabled = isLocked;
    toggleSwitch.title = isLocked
      ? 'Системный плагин управляется администратором'
      : 'Включить / отключить плагин';
  }

  const saveBtn = document.getElementById('btn-save-plugin-config');
  if (saveBtn) {
    saveBtn.disabled = isLocked;
    saveBtn.title = isLocked ? 'Настройки системного плагина изменяются только администратором' : '';
  }

  // Render Actions
  renderPluginActions(plugin);

  // Render Config Fields
  renderPluginFields(plugin);
}

// Отрисовка кнопок действий (Actions)
function renderPluginActions(plugin) {
  const container = document.getElementById('plugin-actions-container');
  const section = document.getElementById('plugin-actions-section');
  if (!container || !section) return;

  const actions = plugin.actions || [];
  if (actions.length === 0) {
```

```
    section.classList.add('d-none');
    return;
  }

  section.classList.remove('d-none');
  let html = '';
  actions.forEach(act => {
    const btnColor = act.color || 'primary';
    const label = act.label || act.name || act.id;
    const icon = act.icon ? `<span class="me-1">${act.icon}</s
pan>` : '';
    const safeLabel = label.replace(/'/g, "\\'");
    html += `
      <button class="btn btn-sm btn-outline-${btnColor} d-inli
ne-flex align-items-center gap-1 shadow-sm"
              onclick="executePluginAction('${plugin.name}', '
${act.id}', '${safeLabel}')"
              title="${act.description || ''}">
        ${icon}<span>${label}</span>
      </button>
    `;
  });
  container.innerHTML = html;
}

// Отрисовка полей конфигурации
function renderPluginFields(plugin) {
  const container = document.getElementById('plugin-fields-con
tainer');
  const section = document.getElementById('plugin-config-secti
on');
  if (!container || !section) return;

  const fields = plugin.fields || [];
  if (fields.length === 0) {
    section.classList.add('d-none');
    return;
  }
```

```
284
285   section.classList.remove('d-none');
286   const currentCfg = plugin.config || {};
287
288   let html = '';
289   fields.forEach(f => {
290     const val = (currentCfg[f.id] !== undefined) ? currentCfg[
f.id] : (f.default !== undefined ? f.default : '');
291     const desc = f.description ? `<div class="form-text text-m
uted small">${f.description}</div>` : '';
292
293     if (f.type === 'select') {
294       let optionsHtml = '';
295       (f.options || []).forEach(opt => {
296         const isSel = String(opt.value) === String(val) ? 'sel
ected' : '';
297         optionsHtml += `<option value="${opt.value}" ${isSel}>
${opt.label}</option>`;
298       });
299       html += `
300         <div class="col-12 col-md-6">
301           <label class="form-label small fw-semibold mb-1">${f
.label}</label>
302           <select class="form-select form-select-sm" name="${f
.id}" data-field-type="select">
303             ${optionsHtml}
304           </select>
305           ${desc}
306         </div>
307       `;
308     } else if (f.type === 'boolean') {
309       const checked = Boolean(val) ? 'checked' : '';
310       html += `
311         <div class="col-12 col-md-6 d-flex flex-column justify
-content-center">
312           <div class="form-check form-switch mt-2">
313             <input class="form-check-input" type="checkbox" ro
le="switch" name="${f.id}" id="field_${f.id}" ${checked} data-
```

```
field-type="boolean">
            <label class="form-check-label small fw-semibold"
for="field_${f.id}">${f.label}</label>
          </div>
          ${desc}
        </div>
      `;
    } else if (f.type === 'list_string') {
      const listVals = Array.isArray(val) ? val.join(', ') : (
val || '');
      html += `
        <div class="col-12">
          <label class="form-label small fw-semibold mb-1">${f
.label} (через запятую)</label>
          <input type="text" class="form-control form-control-
sm font-monospace" name="${f.id}" value="${listVals}" data-fie
ld-type="list_string">
          ${desc}
        </div>
      `;
    } else if (f.type === 'readonly') {
      html += `
        <div class="col-12 col-md-6">
          <label class="form-label small fw-semibold mb-1 text
-muted">${f.label}</label>
          <input type="text" class="form-control form-control-
sm bg-body-tertiary text-muted" value="${val}" readonly>
          ${desc}
        </div>
      `;
    } else if (f.type === 'number') {
      html += `
        <div class="col-12 col-md-6">
          <label class="form-label small fw-semibold mb-1">${f
.label}</label>
          <input type="number" class="form-control form-contro
l-sm" name="${f.id}" value="${val}" data-field-type="number">
          ${desc}
```

```
        </div>
      `;
    } else {
      html += `
        <div class="col-12 col-md-6">
          <label class="form-label small fw-semibold mb-1">${f.label}</label>
          <input type="text" class="form-control form-control-sm" name="${f.id}" value="${val}" data-field-type="string">
          ${desc}
        </div>
      `;
    }
  });

  container.innerHTML = html;
}

// Переключение состояния плагина (Вкл / Выкл)
async function toggleCurrentPlugin(enabled) {
  if (!selectedPluginName) return;

  try {
    logToPluginConsole(`Переключение плагина '${selectedPluginName}' -> ${enabled ? 'ВКЛ' : 'ВЫКЛ'}...`);
    const res = await window.api.fetch(`/api/admin/plugins/${selectedPluginName}/toggle`, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ enabled })
    });

    const plugin = loadedPlugins.find(p => p.name === selectedPluginName);
    if (plugin) {
      plugin.enabled = enabled;
    }

```

```
    renderPluginsList(loadedPlugins);
    syncPluginTabsVisibility(loadedPlugins);
    showPluginNotification(res.message || 'Статус плагина обно
влен', 'success');
    logToPluginConsole(`  ${res.message || 'Статус успешно обн
овлен'}`);
  } catch (ex) {
    console.error('Ошибка переключения плагина:', ex);
    showPluginNotification(`Ошибка: ${ex.message}`, 'danger');
    logToPluginConsole(`  Ошибка: ${ex.message}`);
    // Возвращаем тумблер в исходное состояние
    const toggleSwitch = document.getElementById('plugin-toggl
e-switch');
    if (toggleSwitch) toggleSwitch.checked = !enabled;
  }
}

// Сохранение настроек текущего плагина
async function saveCurrentPluginConfig() {
  if (!selectedPluginName) return;

  const form = document.getElementById('plugin-config-form');
  if (!form) return;

  const newConfig = {};
  const elements = form.querySelectorAll('[data-field-type]');

  elements.forEach(el => {
    const name = el.getAttribute('name');
    if (!name) return;
    const type = el.getAttribute('data-field-type');

    if (type === 'boolean') {
      newConfig[name] = el.checked;
    } else if (type === 'number') {
      newConfig[name] = Number(el.value);
    } else if (type === 'list_string') {
      newConfig[name] = el.value.split(',').map(s => s.trim())
```

```
.filter(Boolean);
410     } else {
411       newConfig[name] = el.value;
412     }
413   });
414 
415   try {
416     logToPluginConsole(`Сохранение параметров плагина '${selec
tedPluginName}'...`);
417     const res = await window.api.fetch(`/api/admin/plugins/${s
electedPluginName}/config`, {
418       method: 'POST',
419       headers: { 'Content-Type': 'application/json' },
420       body: JSON.stringify({ config: newConfig })
421     });
422 
423     const plugin = loadedPlugins.find(p => p.name === selected
PluginName);
424     if (plugin) {
425       plugin.config = res.config || newConfig;
426     }
427 
428     showPluginNotification('Параметры успешно сохранены', 'suc
cess');
429     logToPluginConsole(`✅ Конфигурация сохранена: ${JSON.strin
gify(newConfig, null, 2)}`);
430   } catch (ex) {
431     console.error('Ошибка сохранения конфигурации:', ex);
432     showPluginNotification(`Ошибка: ${ex.message}`, 'danger');
433     logToPluginConsole(`❌ Ошибка сохранения: ${ex.message}`);
434   }
435 }
436 
437 // Выполнение специфического действия (Action) плагина
438 async function executePluginAction(pluginName, actionId, actio
nLabel) {
439   try {
440     logToPluginConsole(`[ACTION] Запуск '${actionLabel}' для п
```

```
лагина '${pluginName}'...`);
441     showPluginNotification(`Запуск '${actionLabel}'...`, 'info
');
442 
443     // Собираем текущие поля формы как параметры действия
444     const form = document.getElementById('plugin-config-form')
;
445     const params = {};
446     if (form) {
447       const elements = form.querySelectorAll('[data-field-type
]');
448       elements.forEach(el => {
449         const name = el.getAttribute('name');
450         if (!name) return;
451         const type = el.getAttribute('data-field-type');
452         if (type === 'boolean') params[name] = el.checked;
453         else if (type === 'number') params[name] = Number(el.v
alue);
454         else if (type === 'list_string') params[name] = el.val
ue.split(',').map(s => s.trim()).filter(Boolean);
455         else params[name] = el.value;
456       });
457     }
458 
459     // Проверяем, есть ли обязательные параметры действия, кот
орых нет в форме
460     const plugin = (typeof loadedPlugins !== 'undefined') ? lo
adedPlugins.find(p => p.name === pluginName) : null;
461     const actDef = plugin?.actions?.find(a => a.id === actionI
d);
462     if (actDef && Array.isArray(actDef.parameters)) {
463       for (const param of actDef.parameters) {
464         if (param.required && (params[param.name] === undefine
d || params[param.name] === '')) {
465           const userVal = prompt(`${param.description || param
.name}:`);
466           if (userVal === null) {
467             logToPluginConsole(`[ACTION] Отменено пользователе
```

```
м.`);
468             return;
469           }
470           params[param.name] = userVal;
471         }
472       }
473     }
474 
475     const res = await window.api.fetch(`/api/admin/plugins/${p
luginName}/action/${actionId}`, {
476       method: 'POST',
477       headers: { 'Content-Type': 'application/json' },
478       body: JSON.stringify({ params })
479     });
480 
481     if (res.success || res.status === 'success') {
482       showPluginNotification(res.message || 'Действие успешно
выполнено', 'success');
483       logToPluginConsole(`✅ ${res.message || 'Успех'}\n${JSON.
stringify(res.result || res.results || res, null, 2)}`);
484     } else {
485       showPluginNotification(res.message || 'Ошибка выполнения
 действия', 'warning');
486       logToPluginConsole(`⚠️ ${res.message || 'Завершено с пред
упреждением'}`);
487     }
488   } catch (ex) {
489     console.error(`Ошибка выполнения действия ${actionId}:`, e
x);
490     showPluginNotification(`Ошибка: ${ex.message}`, 'danger');
491     logToPluginConsole(`❌ Ошибка выполнения: ${ex.message}`);
492   }
493 }
494 
495 // Логирование в консоль плагина
496 function logToPluginConsole(text) {
497   const consoleEl = document.getElementById('plugin-console-ou
tput');
```

```
  if (!consoleEl) return;
  const time = new Date().toLocaleTimeString();
  consoleEl.textContent = `[${time}] ${text}\n` + consoleEl.te
xtContent;
}

// Очистка консоли
function clearPluginConsole() {
  const consoleEl = document.getElementById('plugin-console-ou
tput');
  if (consoleEl) consoleEl.textContent = 'Консоль очищена.\n';
}

function showPlaceholder() {
  const placeholder = document.getElementById('plugin-placehol
der');
  const contentPane = document.getElementById('plugin-content-
pane');
  if (placeholder) placeholder.classList.remove('d-none');
  if (contentPane) contentPane.classList.add('d-none');
}

// Переход к конкретному плагину из выпадающего меню
function openPluginFromDropdown(pluginName) {
  if (!pluginName) return;

  // 1. Закрываем открытые выпадающие списки
  document.querySelectorAll('#mainTabs .dropdown-menu.show').f
orEach((m) => {
    m.classList.remove('show');
    m.closest('.dropdown')?.querySelector('.dropdown-toggle')?
.classList.remove('show');
  });

  const normalized = pluginName.toLowerCase().replace(/-/g, '_
');

  // 2. Если у плагина есть своя выделенная вкладка в DOM, отк
```

```
рываем ее
  if (normalized === 'telegram_channel_rag' || normalized === 'telegram_rag') {
    const pane = document.getElementById('tab-telegram-rag') || document.getElementById('tab-telegram_rag');
    if (pane && typeof window.switchTab === 'function') {
      window.switchTab('tab-telegram-rag');
      return;
    }
  }
  if (normalized === 'news_feed' || normalized === 'news' || normalized === 'smart_news' || normalized === 'smart_feed_news') {
    const pane = document.getElementById('tab-news');
    if (pane && typeof window.switchTab === 'function') {
      window.switchTab('tab-news');
      return;
    }
  }

  // 3. Иначе переключаемся на вкладку плагинов и выбираем плагин в каталоге
  if (typeof window.switchTab === 'function') {
    window.switchTab('tab-plugins');
  }

  const select = () => {
    if (typeof window.selectPlugin === 'function') {
      window.selectPlugin(pluginName);
    }
  };

  select();
  setTimeout(select, 150);
  setTimeout(select, 400);
}

// Синхронизация видимости вкладок в навбаре с состоянием акти
```

```
вности плагинов
function syncPluginTabsVisibility(plugins) {
  if (!Array.isArray(plugins)) return;
  const pluginMap = {};
  plugins.forEach(p => {
    pluginMap[p.name] = Boolean(p.enabled);
  });

  // 1. Обновляем выпадающие списки включенных плагинов в шапке
  const enabledContainers = document.querySelectorAll('#nav-enabled-plugins-list, .nav-enabled-plugins-list');
  const enabledPlugins = plugins.filter(p => p.enabled);

  enabledContainers.forEach(container => {
    if (enabledPlugins.length === 0) {
      container.innerHTML = `
        <div class="list-group-item dropdown-item text-muted small py-2 px-3 fst-italic">
          Нет активных плагинов
        </div>
      `;
    } else {
      let html = '';
      enabledPlugins.forEach(p => {
        const icon = p.icon || '□ ';
        const title = p.title || p.name;
        const desc = p.description || title;
        html += `
          <button class="list-group-item list-group-item-action dropdown-item d-flex align-items-center gap-2 py-2 px-3 text-truncate"
                  type="button"
                  data-plugin="${p.name}"
                  title="${desc}"
                  onclick="openPluginFromDropdown('${p.name}')">
            <span>${icon}</span>
```

```
            <span class="text-truncate">${title}</span>
          </button>
        `;
      });
      container.innerHTML = html;
    }
  });

  // 2. Находим все элементы навигации, привязанные к отдельным плагинам
  const pluginNavItems = document.querySelectorAll('[data-plugin-tab]');
  pluginNavItems.forEach(item => {
    const pluginName = item.getAttribute('data-plugin-tab');
    const isEnabled = (pluginMap[pluginName] !== undefined) ? pluginMap[pluginName] : false;

    if (isEnabled) {
      item.classList.remove('d-none');
    } else {
      item.classList.add('d-none');
      // Если отключенная вкладка была активна, переключаемся на вкладку чата
      const button = item.querySelector('.nav-link, .dropdown-item');
      if (button && button.classList.contains('active')) {
        if (typeof window.switchTab === 'function') {
          window.switchTab('tab-chat');
        } else {
          const chatTab = document.querySelector('[data-bs-target="#tab-chat"]');
          if (chatTab) {
            const tab = new bootstrap.Tab(chatTab);
            tab.show();
          }
        }
      }
    }
```

```
  });
}

// Make functions globally available for HTML inline handlers
window.loadPluginsList = loadPluginsList;
window.filterPluginsList = filterPluginsList;
window.selectPlugin = selectPlugin;
window.toggleCurrentPlugin = toggleCurrentPlugin;
window.saveCurrentPluginConfig = saveCurrentPluginConfig;
window.executePluginAction = executePluginAction;
window.clearPluginConsole = clearPluginConsole;
window.syncPluginTabsVisibility = syncPluginTabsVisibility;
window.openPluginFromDropdown = openPluginFromDropdown;

// Auto-run initialization
if (document.readyState === 'loading') {
  document.addEventListener('DOMContentLoaded', () => {
    if (typeof window.initPluginsTab === 'function') window.in
itPluginsTab();
  });
} else {
  if (typeof window.initPluginsTab === 'function') window.init
PluginsTab();
}

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Post-Install Wizard - Index View
================================================================
===============
Description:
  Шаблон интерфейса мастера настройки Windows после установки
(Post-Install Wizard).

Usage Examples:
  Web Route:
    /html/post_install_wizard_tab/index.html

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/post_install_wizard_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<style>
  .piw-container {
    padding: 0.75rem 1rem;
  }
  .piw-header {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: var(--radius-md, 8px);
    padding: 0.75rem 1.25rem;
    color: var(--text-color);
  }
  .piw-card {
    background: var(--surface-1);
```

```
    border: 1px solid var(--border-color);
    border-radius: 8px;
  }
  .piw-step-item {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 6px;
    padding: 0.75rem 1rem;
    color: var(--text-color);
    transition: all 0.2s ease;
  }
  .piw-step-item:hover {
    border-color: var(--nav-active);
    background: var(--nav-hover-bg, rgba(56, 189, 248, 0.05));
  }
</style>
<div class="piw-container theme-sysctl">
<!-- Top Header Bar -->
<div class="d-flex justify-content-between align-items-center flex-wrap gap-3 mb-3 piw-header">
<div>
<div class="d-flex align-items-center gap-2">
<h5 class="mb-0 fw-bold">
          Windows Post-Install Wizard
        </h5>
<span class="badge rounded-pill bg-warning text-dark px-2 py-1 " style="font-size: 0.75rem;">Setup &amp;amp; Hardening</span>
</div>
<p class="mb-0 small text-body-secondary mt-1">
        Пошаговая настройка базовой конфигурации Windows, профили оптимизации, отключение телеметрии и настройка параметров безопасности.
      </p>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<span class="badge rounded-pill px-3 py-2" id="piw-elevation-badge">
        Checking privileges...
```

```
      </span>
<button class="btn btn-sm btn-outline-info rounded-pill px-3" id="btn-piw-refresh">
<i class="bi bi-arrow-clockwise me-1"></i> Обновить
      </button>
</div>
</div>
<!-- Main Wizard Card -->
<div class="card piw-card" data-panel-id="panel-post-install-wizard-chek-list-pervichnoy-nastroyki-w" id="panel-post-install-wizard-chek-list-pervichnoy-nastroyki-w">
<div class="card-header border-secondary bg-transparent d-flex justify-content-between align-items-center flex-wrap gap-2 py-3">
<div>
<h6 class="mb-0 fw-bold text-warning"><i class="bi bi-magic me-1"></i> Чек-лист первичной настройки Windows</h6>
<span class="small text-muted">Выберите профиль оптимизации или отметьте необходимые шаги для последовательного применения.</span>
</div>
<div class="d-flex align-items-center gap-2">
<select class="form-select form-select-sm" id="piw-profile-select" style="min-width: 260px;">
<option value="post_install">Windows Post-Install Baseline</option>
<option value="secure_hardening">Security Hardening Profile</option>
<option value="developer_setup">Developer Workstation Profile</option>
</select>
<button class="btn btn-sm btn-primary px-3 rounded-pill" id="btn-piw-apply-profile">
<i class="bi bi-play-fill me-1"></i> Применить профиль
        </button>
</div>
</div>
<div class="card-body p-3">
```

```
<!-- Steps Container -->
<div class="d-flex flex-column gap-2" id="piw-wizard-steps-container">
<div class="text-center text-muted py-4">
<div class="spinner-border spinner-border-sm text-info me-2" role="status"></div>
          Загрузка шагов профиля...
        </div>
</div>
</div>
<div class="card-footer border-secondary bg-transparent d-flex justify-content-between align-items-center flex-wrap gap-2">
<span class="small text-muted" id="piw-status-summary">Готово к выполнению</span>
<div class="d-flex gap-2">
<button class="btn btn-sm btn-outline-secondary rounded-pill px-3" id="btn-piw-select-all">Выбрать все</button>
<button class="btn btn-sm btn-outline-secondary rounded-pill px-3" id="btn-piw-deselect-all">Снять выбор</button>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Post-Install Wizard - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт мастера первичной настройки Windows (Po
st-Install Wizard).
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/html/post_install_wizard_tab/main.js?v=20
261006_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/post_install_wizard_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

(function () {
  'use strict';

  let isInitialized = false;
  let activeProfileSteps = [];

  /**
   * Экранирование HTML-символов для защиты от XSS.
   *
   * @param {string} str - Исходная строка.
   * @returns {string} Экранированная строка.
```

```
   */
  function escapeHtml(str) {
    if (!str) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
  }

  /**
   * Проверка прав администратора и обновление бейджа.
   */
  async function fetchElevationStatus() {
    try {
      const res = await fetch('/api/system-control/status');
      if (!res.ok) return;
      const data = await res.json();
      const elevBadge = document.getElementById('piw-elevation
-badge');
      if (elevBadge) {
        if (data.is_elevated) {
          elevBadge.className = 'badge rounded-pill bg-success
-subtle text-success border border-success px-3 py-2';
          elevBadge.innerHTML = '  Режим: Администратор (Полн
ый доступ)';
        } else {
          elevBadge.className = 'badge rounded-pill bg-warning
-subtle text-warning border border-warning px-3 py-2';
          elevBadge.innerHTML = '  Режим: Обычный пользовател
ь (Ограничено)';
        }
      }
    } catch (e) {
      console.warn('[PostInstallWizard] Ошибка проверки прав:'
, e);
    }
```

```
64  }
65
66  /**
67   * Загрузка профилей и шагов мастера.
68   */
69  async function loadWizardProfiles() {
70    try {
71      const res = await fetch('/api/system-control/profiles');
72      if (!res.ok) return;
73      const data = await res.json();
74      const profiles = data.profiles || [];
75      const currentProf = profiles[0];
76      if (currentProf && currentProf.steps) {
77        activeProfileSteps = currentProf.steps;
78        renderWizardSteps(currentProf.steps);
79      }
80    } catch (e) {
81      console.error('[PostInstallWizard] Ошибка загрузки профи
лей:', e);
82    }
83  }
84
85  /**
86   * Отрисовка элементов шагов мастера.
87   *
88   * @param {Array} steps - Список шагов.
89   */
90  function renderWizardSteps(steps) {
91    const container = document.getElementById('piw-wizard-step
s-container');
92    if (!container) return;
93
94    if (!steps || steps.length === 0) {
95      container.innerHTML = '<div class="text-center text-mute
d p-3">Нет доступных шагов для выбранного профиля.</div>';
96      return;
97    }
98
```

```
    container.innerHTML = steps.map(step => {
      const badgeClass = step.status === 'SUCCESS' ? 'bg-success' : step.status === 'RUNNING' ? 'bg-info' : step.status === 'FAILED' ? 'bg-danger' : 'bg-secondary';
      return `
      <div class="piw-step-item d-flex justify-content-between align-items-center flex-wrap gap-2" id="piw-step-${step.id}">
        <div class="d-flex align-items-start gap-3">
          <input class="form-check-input mt-1 piw-step-checkbox" type="checkbox" value="${step.id}" ${step.enabled ? 'checked' : ''} style="cursor: pointer;">
          <div>
            <div class="fw-bold text-white d-flex align-items-center gap-2">
              <span>${escapeHtml(step.title)}</span>
              ${step.requires_elevation ? '<span class="badge bg-warning-subtle text-warning border border-warning px-1.5 py-0.5" style="font-size: 0.65rem;">Admin</span>' : ''}
            </div>
            <div class="small text-muted">${escapeHtml(step.description)}</div>
          </div>
        </div>
        <div class="d-flex align-items-center gap-2">
          <span class="badge ${badgeClass} font-monospace px-2 py-1" id="piw-step-status-${step.id}" style="font-size: 0.72rem;">${step.status || 'READY'}</span>
          <button class="btn btn-sm btn-outline-info rounded-pill px-3 piw-btn-step-run" data-step-id="${step.id}">
            <i class="bi bi-play-circle me-1"></i> Выполнить
          </button>
        </div>
      </div>
    `;
    }).join('');

    // Привязка обработчиков одиночного запуска шагов
    container.querySelectorAll('.piw-btn-step-run').forEach(bt
```

```
n => {
      btn.onclick = async () => {
        const stepId = btn.getAttribute('data-step-id');
        await executeWizardSteps([stepId]);
      };
    });
  }

  /**
   * Выполнение списка шагов мастера.
   *
   * @param {Array<string>} stepIds - Идентификаторы шагов.
   */
  async function executeWizardSteps(stepIds) {
    if (!stepIds || stepIds.length === 0) {
      if (window.toast) window.toast.warning('Мастер настройки
', 'Выберите хотя бы один шаг для выполнения.');
      return;
    }

    const summaryEl = document.getElementById('piw-status-summ
ary');
    if (summaryEl) summaryEl.innerText = `Выполняется ${stepId
s.length} шагов...`;

    stepIds.forEach(sid => {
      const badge = document.getElementById(`piw-step-status-$
{sid}`);
      if (badge) {
        badge.className = 'badge bg-info font-monospace px-2 p
y-1';
        badge.innerText = 'RUNNING';
      }
    });

    try {
      const profileSelect = document.getElementById('piw-profi
le-select');
```

```
      const profileId = profileSelect ? profileSelect.value : 'post_install';

      const res = await fetch('/api/system-control/profiles/apply', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ profile_id: profileId, step_ids: stepIds }),
      });

      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();

      stepIds.forEach(sid => {
        const badge = document.getElementById(`piw-step-status-${sid}`);
        if (badge) {
          badge.className = 'badge bg-success font-monospace px-2 py-1';
          badge.innerText = 'SUCCESS';
        }
      });

      if (summaryEl) summaryEl.innerText = 'Профиль успешно применен.';
      if (window.toast) window.toast.success('Мастер настройки', data.message || 'Шаги успешно применены.');
    } catch (e) {
      console.error('[PostInstallWizard] Ошибка выполнения шагов:', e);
      stepIds.forEach(sid => {
        const badge = document.getElementById(`piw-step-status-${sid}`);
        if (badge) {
          badge.className = 'badge bg-danger font-monospace px-2 py-1';
          badge.innerText = 'FAILED';
```

```
        }
      });
      if (summaryEl) summaryEl.innerText = `Ошибка применения: ${e.message}`;
      if (window.toast) window.toast.error('Мастер настройки', `Ошибка: ${e.message}`);
    }
  }

  /**
   * Инициализация вкладки Post-Install Wizard.
   */
  function initPostInstallWizardTab() {
    fetchElevationStatus();
    loadWizardProfiles();

    if (isInitialized) return;
    isInitialized = true;

    const refreshBtn = document.getElementById('btn-piw-refresh');
    if (refreshBtn) {
      refreshBtn.onclick = () => {
        fetchElevationStatus();
        loadWizardProfiles();
      };
    }

    const applyProfileBtn = document.getElementById('btn-piw-apply-profile');
    if (applyProfileBtn) {
      applyProfileBtn.onclick = async () => {
        const checkedBoxes = document.querySelectorAll('.piw-step-checkbox:checked');
        const stepIds = Array.from(checkedBoxes).map(cb => cb.value);
        await executeWizardSteps(stepIds);
      };
```

```
    }

    const selectAllBtn = document.getElementById('btn-piw-sele
ct-all');
    if (selectAllBtn) {
      selectAllBtn.onclick = () => {
        document.querySelectorAll('.piw-step-checkbox').forEac
h(cb => { cb.checked = true; });
      };
    }

    const deselectAllBtn = document.getElementById('btn-piw-de
select-all');
    if (deselectAllBtn) {
      deselectAllBtn.onclick = () => {
        document.querySelectorAll('.piw-step-checkbox').forEac
h(cb => { cb.checked = false; });
      };
    }
  }

  window.initPostInstallWizardTab = initPostInstallWizardTab;
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Power Lifecycle & Boot H
istory Tab
================================================================
===============
Description:
  Клиентский веб-интерфейс анализа жизненного цикла питания Wi
ndows,
  истории перезагрузок (User32 1074, Kernel-Power 41, EventLog
 6008/6005/6006),
  реконструированных сессий питания (Power Sessions) и корреля
ции инициаторов.

Usage Examples:
  Web Route:
    /html/power_lifecycle_tab/index.html
    /windows/api/webgui/power_lifecycle_tab/index.html

  Tab Frame Integration:
    <div id="tab-power-lifecycle" class="tab-pane fade"></div>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/power_lifecycle_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<style>
  .power-lifecycle-container {
    color: var(--text-color);
  }
```

```
  .pwr-card {
    background: var(--surface-1);
    border: 1px solid var(--border-color);
    border-radius: var(--radius-md, 8px);
    color: var(--text-color);
  }
  .pwr-table {
    width: 100%;
    color: var(--text-color);
    border-collapse: separate;
    border-spacing: 0;
  }
  .pwr-table thead th {
    background: var(--surface-2);
    color: var(--text-muted);
    font-weight: 600;
    font-size: 0.75rem;
    text-transform: uppercase;
    letter-spacing: 0.03em;
    border-bottom: 1.5px solid var(--border-color);
    padding: 0.55rem 0.75rem;
    white-space: nowrap;
  }
  .pwr-table td {
    border-bottom: 1px solid var(--border-color);
    padding: 0.55rem 0.75rem;
    vertical-align: middle;
    color: var(--text-color);
  }
  .pwr-table tbody tr:hover td {
    background-color: var(--nav-hover-bg, rgba(56, 189, 248, 0
.04));
  }
  .pwr-search-input {
    background: var(--surface-1);
    border: 1px solid var(--border-color);
    color: var(--text-color);
  }
```

```
.pwr-search-input:focus {
  background: var(--surface-1);
  border-color: var(--nav-active);
  color: var(--text-color);
  box-shadow: 0 0 0 0.2rem rgba(56, 189, 248, 0.15);
}
.pwr-input-addon {
  background: var(--surface-2);
  border: 1px solid var(--border-color);
  color: var(--text-muted);
}
.pwr-nav-pills {
  border-bottom: 1px solid var(--border-color);
}
.pwr-nav-pills .nav-link {
  color: var(--text-muted);
  font-size: 0.84rem;
  font-weight: 500;
  border-radius: var(--radius-sm, 6px);
  padding: 0.4rem 0.9rem;
  transition: all 0.15s ease;
}
.pwr-nav-pills .nav-link:hover {
  color: var(--nav-active);
  background: var(--nav-hover-bg, rgba(0, 0, 0, 0.05));
}
.pwr-nav-pills .nav-link.active {
  color: #fff;
  background: var(--nav-active);
  font-weight: 600;
}
.pwr-detail-card {
  background: var(--surface-2);
  border: 1px solid var(--border-color);
  border-radius: var(--radius-sm, 6px);
}
.pwr-xml-pre {
  background: var(--surface-2);
```

```
    color: var(--text-color);
    border: 1px solid var(--border-color);
  }
  .pwr-proc-badge {
    background: var(--surface-2);
    color: var(--nav-active);
    border: 1px solid var(--border-color);
  }
</style>
<div class="container-fluid py-2 px-3 power-lifecycle-containe
r">
<!-- Верхняя панель метрик и сводки -->
<div class="row g-2 mb-3">
<!-- Карточка 1: Текущий Аптайм и Старт ОС -->
<div class="col-12 col-sm-6 col-xl-3">
<div class="card pwr-card h-100 shadow-sm p-3 position-relativ
e" data-panel-id="panel-power-lifecycle-1" id="panel-power-lif
ecycle-1">
<div class="d-flex justify-content-between align-items-start">
<div class="text-muted small text-uppercase fw-semibold">Текущ
ий аптайм  </div>
<span class="badge bg-success font-monospace" id="pwr-uptime-b
adge">ACTIVE</span>
</div>
<div class="fs-4 fw-bold mt-1 text-info font-monospace" id="pw
r-current-uptime">--:--:--</div>
<div class="small text-muted mt-1 text-truncate" id="pwr-curre
nt-boot" style="font-size: 0.75rem;">Загрузка: --</div>
<div class="mt-2 text-end">
<span class="badge bg-secondary-subtle text-secondary border b
order-secondary-subtle" style="font-size: 0.7rem;">Kernel-Gene
ral 12</span>
</div>
</div>
</div>
<!-- Карточка 2: Всего сессий питания -->
<div class="col-12 col-sm-6 col-xl-3">
<div class="card pwr-card h-100 shadow-sm p-3" data-panel-id="
```

```
panel-power-lifecycle-2" id="panel-power-lifecycle-2">
<div class="d-flex justify-content-between align-items-start">
<div class="text-muted small text-uppercase fw-semibold">Сесси и питания  </div>
<span class="badge bg-primary font-monospace" id="pwr-total-sessions-badge">0 Сессий</span>
</div>
<div class="fs-4 fw-bold mt-1 text-warning font-monospace" id="pwr-total-sessions">--</div>
<div class="small text-muted mt-1" id="pwr-sessions-desc">Реконструкция Boot → Shutdown</div>
<div class="mt-2">
<button class="btn btn-xs btn-outline-warning w-100 py-1" id="pwr-rebuild-btn" style="font-size: 0.78rem;">
<i class="bi bi-arrow-repeat me-1"></i><span>Пересканировать Event Log</span>
</button>
</div>
</div>
</div>
<!-- Карточка 3: Штатные перезагрузки и выключения -->
<div class="col-12 col-sm-6 col-xl-3">
<div class="card pwr-card h-100 shadow-sm p-3" data-panel-id="panel-power-lifecycle-3" id="panel-power-lifecycle-3">
<div class="d-flex justify-content-between align-items-start">
<div class="text-muted small text-uppercase fw-semibold">Штатные завершения  </div>
<span class="badge bg-success-subtle text-success border border-success-subtle font-monospace" id="pwr-clean-ratio">-- %</span>
</div>
<div class="fs-4 fw-bold mt-1 text-success font-monospace" id="pwr-clean-count">--</div>
<div class="small text-muted mt-1 text-truncate" id="pwr-last-initiator" style="font-size: 0.75rem;">Инициатор: --</div>
<div class="mt-2 text-end">
<span class="badge bg-secondary-subtle text-secondary border border-secondary-subtle" style="font-size: 0.7rem;">User32 1074
```

```
 / EventLog 6006</span>
</div>
</div>
</div>
<!-- Карточка 4: Внезапные сбои и BSOD -->
<div class="col-12 col-sm-6 col-xl-3">
<div class="card pwr-card h-100 shadow-sm p-3" data-panel-id="
panel-power-lifecycle-4" id="panel-power-lifecycle-4">
<div class="d-flex justify-content-between align-items-start">
<div class="text-muted small text-uppercase fw-semibold">Сбои
/ Потери питания □ </div>
<span class="badge bg-danger font-monospace" id="pwr-unexpecte
d-badge">0 Сбоев</span>
</div>
<div class="fs-4 fw-bold mt-1 text-danger font-monospace" id="
pwr-unexpected-count">--</div>
<div class="small text-muted mt-1 d-flex justify-content-betwe
en" style="font-size: 0.75rem;">
<span id="pwr-bsod-count">BSOD (1001): 0</span>
<span>Kernel-Power 41</span>
</div>
<div class="mt-2">
<button class="btn btn-xs btn-outline-primary w-100 py-1" id="
pwr-refresh-btn" style="font-size: 0.78rem;">
<i class="bi bi-arrow-clockwise me-1"></i><span>Обновить данны
e</span>
</button>
</div>
</div>
</div>
</div>
<!-- Навигация по подразделам: Сессии vs События -->
<ul class="nav nav-pills mb-3 pwr-nav-pills pb-2" id="pwrNavTa
bs" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active py-1 px-3" data-bs-target="#pwr
-subtab-sessions" data-bs-toggle="pill" id="pwr-sessions-subta
b-btn" role="tab" type="button">
```

```
<i class="bi bi-clock-history me-1.5 text-warning"></i>
<span>Сессии питания (Power Sessions)</span>
</button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link py-1 px-3" data-bs-target="#pwr-subtab-events" data-bs-toggle="pill" id="pwr-events-subtab-btn" role="tab" type="button">
<i class="bi bi-list-columns-reverse me-1.5 text-info"></i>
<span>Журнал ключевых событий (Event Log)</span>
</button>
</li>
</ul>
<div class="tab-content" id="pwrTabContent">
<!-- Вкладка 1: Сессии питания (Power Sessions) -->
<div class="tab-pane fade show active" id="pwr-subtab-sessions" role="tabpanel">
<!-- Панель фильтров и поиска -->
<div class="card pwr-card mb-3 p-2" data-panel-id="panel-power-lifecycle-5" id="panel-power-lifecycle-5">
<div class="row g-2 align-items-center">
<div class="col-12 col-md-4">
<div class="input-group input-group-sm">
<span class="input-group-text pwr-input-addon"><i class="bi bi-search"></i></span>
<input class="form-control form-control-sm pwr-search-input" id="pwr-sessions-search" placeholder="Поиск по пользователю, процессу, причине..." type="text"/>
</div>
</div>
<div class="col-6 col-md-3">
<select class="form-select form-select-sm pwr-search-input" id="pwr-filter-type">
<option value="">Все типы завершения</option>
<option value="Active">▶ Активная сессия (Active)</option>
<option value="Restart">↻ Перезагрузка (Restart)</option>
<option value="Shutdown">⏻ Выключение (Shutdown)</option>
<option value="Unexpected">⚠ Внезапное завершение (Unexpected)
```

```
</option>
<option value="BSOD">  Синий экран (BSOD / BugCheck)</option>
</select>
</div>
<div class="col-6 col-md-2">
<div class="form-check form-switch small mt-1">
<input class="form-check-input" id="pwr-filter-unexpected-only
" type="checkbox"/>
<label class="form-check-label text-warning fw-semibold" for="
pwr-filter-unexpected-only">Только сбои</label>
</div>
</div>
<div class="col-12 col-md-3 text-end">
<span class="small text-muted" id="pwr-sessions-counter">Найде
но: 0</span>
</div>
</div>
</div>
<!-- Таблица / Список сессий питания -->
<div class="card pwr-card shadow-sm" data-panel-id="panel-powe
r-lifecycle-6" id="panel-power-lifecycle-6">
<div class="table-responsive">
<table class="table table-hover align-middle mb-0 pwr-table" s
tyle="font-size: 0.85rem;">
<thead>
<tr>
<th style="width: 140px;">Сессия</th>
<th style="width: 160px;">Время старта (Boot)</th>
<th style="width: 160px;">Завершение</th>
<th style="width: 130px;">Аптайм</th>
<th style="width: 130px;">Статус</th>
<th>Инициатор и процесс</th>
<th>Причина / Код</th>
<th class="text-center" style="width: 80px;">Детали</th>
</tr>
</thead>
<tbody id="pwr-sessions-tbody">
<tr>
```

```
<td class="text-center py-4 text-muted" colspan="8">
<div class="spinner-border spinner-border-sm text-info me-2" r
ole="status"></div>
                  Загрузка истории сессий питания...
                </td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- Вкладка 2: Журнал ключевых событий (Event Log) -->
<div class="tab-pane fade" id="pwr-subtab-events" role="tabpan
el">
<!-- Фильтры событий -->
<div class="card pwr-card mb-3 p-2" data-panel-id="panel-power
-lifecycle-7" id="panel-power-lifecycle-7">
<div class="row g-2 align-items-center">
<div class="col-12 col-md-4">
<div class="input-group input-group-sm">
<span class="input-group-text pwr-input-addon"><i class="bi bi
-funnel"></i></span>
<select class="form-select form-select-sm pwr-search-input" id
="pwr-event-id-filter">
<option value="">Все Event ID жизненного цикла</option>
<option value="1074">1074 — User32 (Инициатор shutdown/restart
)</option>
<option value="41">41 — Kernel-Power (Внезапная перезагрузка)<
/option>
<option value="6008">6008 — EventLog (Неожиданное выключение)<
/option>
<option value="12">12 — Kernel-General (Запуск ядра / Boot)</o
ption>
<option value="13">13 — Kernel-General (Завершение работы ОС)<
/option>
<option value="6005">6005 — EventLog (Служба EventLog запущена
)</option>
<option value="6006">6006 — EventLog (EventLog штатно остановл
```

```
ен)</option>
<option value="6009">6009 — EventLog (Версия ядра при загрузке)</option>
<option value="1001">1001 — WER (Аварийный сбой BSOD / BugCheck)</option>
<option value="19">19 — WindowsUpdate (Установка обновлений)</option>
<option value="42">42 — Kernel-Power (Переход в сон)</option>
<option value="107">107 — Kernel-Power (Пробуждение из сна)</option>
</select>
</div>
</div>
<div class="col-12 col-md-5">
<div class="input-group input-group-sm">
<span class="input-group-text pwr-input-addon"><i class="bi bi-search"></i></span>
<input class="form-control form-control-sm pwr-search-input" id="pwr-events-search" placeholder="Поиск в событиях..." type="text"/>
</div>
</div>
<div class="col-12 col-md-3 text-end">
<span class="small text-muted" id="pwr-events-counter">Событий: 0</span>
</div>
</div>
</div>
<!-- Таблица сырых событий -->
<div class="card pwr-card shadow-sm" data-panel-id="panel-power-lifecycle-8" id="panel-power-lifecycle-8">
<div class="table-responsive">
<table class="table table-hover align-middle mb-0 pwr-table" style="font-size: 0.85rem;">
<thead>
<tr>
<th style="width: 160px;">Время</th>
<th style="width: 90px;">Event ID</th>
```

```
<th style="width: 140px;">Провайдер</th>
<th style="width: 130px;">Тип</th>
<th style="width: 180px;">Инициатор / Процесс</th>
<th>Описание / Причина</th>
<th class="text-center" style="width: 80px;">XML</th>
</tr>
</thead>
<tbody id="pwr-events-tbody">
<tr>
<td class="text-center py-4 text-muted" colspan="7">
<div class="spinner-border spinner-border-sm text-info me-2" r
ole="status"></div>
                  Загрузка журнала событий...
                </td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
<!-- Модальное окно деталей сессии питания -->
<div aria-hidden="true" aria-labelledby="pwrSessionDetailModal
Label" class="modal fade" id="pwrSessionDetailModal" tabindex=
"-1">
<div class="modal-dialog modal-lg modal-dialog-centered modal-
dialog-scrollable">
<div class="modal-content shadow">
<div class="modal-header border-bottom">
<h5 class="modal-title d-flex align-items-center gap-2" id="pw
rSessionDetailModalLabel">
<i class="bi bi-info-circle-fill text-warning"></i>
<span id="pwr-modal-title">Детали сессии питания</span>
</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-3" id="pwr-modal-body">
```

```
<!-- Populated by main.js -->
</div>
<div class="modal-footer border-top">
<button class="btn btn-secondary btn-sm" data-bs-dismiss="moda
l" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
<!-- Модальное окно просмотра сырого XML события -->
<div aria-hidden="true" aria-labelledby="pwrXmlDetailModalLabe
l" class="modal fade" id="pwrXmlDetailModal" tabindex="-1">
<div class="modal-dialog modal-lg modal-dialog-centered modal-
dialog-scrollable">
<div class="modal-content shadow">
<div class="modal-header border-bottom">
<h5 class="modal-title d-flex align-items-center gap-2" id="pw
rXmlDetailModalLabel">
<i class="bi bi-filetype-xml text-info"></i>
<span>Исходный XML события Windows</span>
</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-3">
<pre class="pwr-xml-pre p-3 rounded font-monospace mb-0" id="p
wr-xml-content" style="font-size: 0.78rem; max-height: 500px;
overflow-y: auto;"></pre>
</div>
<div class="modal-footer border-top">
<button class="btn btn-secondary btn-sm" data-bs-dismiss="moda
l" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Web Interface - Power Lifecycle Tab Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом вкладки Power Lifecycle:
 *   выборка и визуализация реконструированных сессий питания, аптайма,
 *   истории перезагрузок (1074, 41, 6008) и ключевых системных событий.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/html/power_lifecycle_tab/main.js?v=20261008_v2" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/power_lifecycle_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

let allSessions = [];
let allEvents = [];
let uptimeSeconds = 0;
let uptimeInterval = null;

/**
```

```
 * Инициализирует вкладку жизненного цикла питания.
 */
export async function initPowerLifecycleTab() {
  setupEventListeners();
  await loadPowerData();
  startUptimeCounter();
}

/**
 * Устанавливает слушатели событий элементов управления.
 */
function setupEventListeners() {
  const refreshBtn = document.getElementById('pwr-refresh-btn'
);
  if (refreshBtn) {
    refreshBtn.addEventListener('click', async () => {
      refreshBtn.disabled = true;
      refreshBtn.innerHTML = '<span class="spinner-border spin
ner-border-sm me-1"></span>Обновление...';
      try {
        await loadPowerData();
        if (window.toast) window.toast.success('Обновлено', 'Д
анные сессий питания успешно обновлены');
      } finally {
        refreshBtn.disabled = false;
        refreshBtn.innerHTML = '<i class="bi bi-arrow-clockwis
e me-1"></i><span>Обновить данные</span>';
      }
    });
  }

  const rebuildBtn = document.getElementById('pwr-rebuild-btn'
);
  if (rebuildBtn) {
    rebuildBtn.addEventListener('click', async () => {
      rebuildBtn.disabled = true;
      rebuildBtn.innerHTML = '<span class="spinner-border spin
ner-border-sm me-1"></span>Сканирование...';
```

```
61      try {
62        const res = await fetch('/api/v1/power/rebuild', { met
hod: 'POST' });
63        if (!res.ok) throw new Error(`HTTP ${res.status}`);
64        const data = await res.json();
65        renderSummary(data);
66        await loadPowerData();
67        if (window.toast) window.toast.success('Event Log пере
сканирован', `Реконструировано сессий: ${data.total_sessions_c
ount}`);
68      } catch (err) {
69        console.error('[PowerLifecycle] Ошибка пересканировани
я:', err);
70        if (window.toast) window.toast.error('Ошибка', 'Не уда
лось пересканировать журналы: ' + err.message);
71      } finally {
72        rebuildBtn.disabled = false;
73        rebuildBtn.innerHTML = '<i class="bi bi-arrow-repeat m
e-1"></i><span>Пересканировать Event Log</span>';
74      }
75    });
76  }
77
78  // Фильтры сессий
79  const sessionsSearch = document.getElementById('pwr-sessions
-search');
80  if (sessionsSearch) {
81    sessionsSearch.addEventListener('input', () => filterAndRe
nderSessions());
82  }
83
84  const filterType = document.getElementById('pwr-filter-type'
);
85  if (filterType) {
86    filterType.addEventListener('change', () => filterAndRende
rSessions());
87  }
88
```

```
  const filterUnexpected = document.getElementById('pwr-filter-unexpected-only');
  if (filterUnexpected) {
    filterUnexpected.addEventListener('change', () => filterAndRenderSessions());
  }

  // Фильтры событий
  const eventsSearch = document.getElementById('pwr-events-search');
  if (eventsSearch) {
    eventsSearch.addEventListener('input', () => filterAndRenderEvents());
  }

  const filterEid = document.getElementById('pwr-event-id-filter');
  if (filterEid) {
    filterEid.addEventListener('change', () => filterAndRenderEvents());
  }
}

/**
 * Загружает сводку, сессии и события через API.
 */
async function loadPowerData() {
  try {
    // 1. Сводка
    const sumRes = await fetch('/api/v1/power/summary');
    if (sumRes.ok) {
      const summary = await sumRes.json();
      renderSummary(summary);
    }

    // 2. Сессии
    const sessRes = await fetch('/api/v1/power/sessions?limit=100');
```

```
    if (sessRes.ok) {
      allSessions = await sessRes.json();
      filterAndRenderSessions();
    }

    // 3. События
    const evRes = await fetch('/api/v1/power/events?limit=200');
    if (evRes.ok) {
      allEvents = await evRes.json();
      filterAndRenderEvents();
    }
  } catch (err) {
    console.error('[PowerLifecycle] Ошибка загрузки данных:', err);
  }
}

/**
 * Отрисовывает верхние карточки сводки.
 * @param {Object} summary
 */
function renderSummary(summary) {
  if (!summary) return;

  const curBootEl = document.getElementById('pwr-current-boot');
  if (curBootEl) curBootEl.textContent = `Загрузка: ${summary.current_boot_time || '--'}`;

  const curUptimeEl = document.getElementById('pwr-current-uptime');
  if (curUptimeEl) {
    curUptimeEl.textContent = summary.current_uptime_human || '--';
    uptimeSeconds = summary.current_uptime_seconds || 0;
  }

```

```
  const totalSessionsEl = document.getElementById('pwr-total-sessions');
  if (totalSessionsEl) totalSessionsEl.textContent = summary.total_sessions_count;

  const totalBadge = document.getElementById('pwr-total-sessions-badge');
  if (totalBadge) totalBadge.textContent = `${summary.total_sessions_count} Сессий`;

  const cleanCountEl = document.getElementById('pwr-clean-count');
  if (cleanCountEl) cleanCountEl.textContent = summary.clean_shutdowns_count;

  const cleanRatioEl = document.getElementById('pwr-clean-ratio');
  if (cleanRatioEl) {
    const total = summary.total_sessions_count || 1;
    const ratio = Math.round((summary.clean_shutdowns_count / total) * 100);
    cleanRatioEl.textContent = `${ratio} %`;
  }

  const lastInitEl = document.getElementById('pwr-last-initiator');
  if (lastInitEl) {
    lastInitEl.textContent = `Инициатор: ${summary.last_initiator || 'Штатный'}`;
  }

  const unexpCountEl = document.getElementById('pwr-unexpected-count');
  if (unexpCountEl) unexpCountEl.textContent = summary.unexpected_shutdowns_count;

  const unexpBadge = document.getElementById('pwr-unexpected-badge');
```

```
  if (unexpBadge) {
    unexpBadge.textContent = `${summary.unexpected_shutdowns_count} Сбоев`;
    if (summary.unexpected_shutdowns_count > 0) {
      unexpBadge.className = 'badge bg-danger font-monospace';
    } else {
      unexpBadge.className = 'badge bg-success font-monospace';
    }
  }

  const bsodEl = document.getElementById('pwr-bsod-count');
  if (bsodEl) bsodEl.textContent = `BSOD (1001): ${summary.bsod_count || 0}`;
}

/**
 * Фильтрует и отрисовывает таблицу сессий питания.
 */
function filterAndRenderSessions() {
  const tbody = document.getElementById('pwr-sessions-tbody');
  const counter = document.getElementById('pwr-sessions-counter');
  if (!tbody) return;

  const searchVal = (document.getElementById('pwr-sessions-search')?.value || '').toLowerCase().trim();
  const typeFilter = document.getElementById('pwr-filter-type')?.value || '';
  const unexpOnly = document.getElementById('pwr-filter-unexpected-only')?.checked || false;

  const filtered = allSessions.filter(s => {
    if (unexpOnly && !s.unexpected_shutdown) return false;
    if (typeFilter) {
      if (typeFilter === 'Active' && s.shutdown_type !== 'Active') return false;
      if (typeFilter === 'Restart' && s.shutdown_type !== 'Res
```

```
tart') return false;
      if (typeFilter === 'Shutdown' && s.shutdown_type !== 'Sh
utdown' && s.shutdown_type !== 'PowerOff') return false;
      if (typeFilter === 'Unexpected' && !s.unexpected_shutdow
n) return false;
      if (typeFilter === 'BSOD' && s.shutdown_type !== 'BSOD'
&& !s.bugcheck) return false;
    }

    if (searchVal) {
      const init = (s.initiator || '').toLowerCase();
      const proc = (s.process || '').toLowerCase();
      const reason = (s.reason || '').toLowerCase();
      const code = (s.reason_code || '').toLowerCase();
      const sId = (s.session_id || '').toLowerCase();
      if (!init.includes(searchVal) && !proc.includes(searchVa
l) && !reason.includes(searchVal) && !code.includes(searchVal)
 && !sId.includes(searchVal)) {
        return false;
      }
    }
    return true;
  });

  if (counter) counter.textContent = `Найдено: ${filtered.leng
th}`;

  if (filtered.length === 0) {
    tbody.innerHTML = `
      <tr>
        <td colspan="8" class="text-center py-4 text-muted">
          Сессии питания, удовлетворяющие критериям фильтрации
, не найдены.
        </td>
      </tr>
    `;
    return;
  }
```

```
237
238   tbody.innerHTML = filtered.map(s => {
239     let statusBadge = '<span class="badge bg-secondary-subtle
text-secondary border border-secondary-subtle">Неизвестно</spa
n>';
240     if (s.shutdown_type === 'Active') {
241       statusBadge = '<span class="badge bg-success-subtle text
-success border border-success-subtle">  Активная</span>';
242     } else if (s.shutdown_type === 'Restart') {
243       statusBadge = '<span class="badge bg-primary-subtle text
-primary border border-primary-subtle">  Перезагрузка</span>';
244     } else if (s.shutdown_type === 'Shutdown' || s.shutdown_ty
pe === 'PowerOff') {
245       statusBadge = '<span class="badge bg-secondary-subtle te
xt-secondary border border-secondary-subtle">  Выключение</spa
n>';
246     } else if (s.shutdown_type === 'BSOD' || s.bugcheck) {
247       statusBadge = `<span class="badge bg-danger text-white">
  BSOD ${escapeHtml(s.bugcheck || '')}</span>`;
248     } else if (s.unexpected_shutdown) {
249       statusBadge = '<span class="badge bg-danger-subtle text-
danger border border-danger-subtle">  Сбой / 41</span>';
250     }
251
252     let initiatorHtml = '<span class="text-muted">--</span>';
253     if (s.initiator || s.process) {
254       const initName = s.initiator ? `<span class="fw-semibold
 text-primary">${escapeHtml(s.initiator)}</span>` : '';
255       const procName = s.process ? `<span class="badge pwr-pro
c-badge font-monospace ms-1">${escapeHtml(s.process)}</span>`
: '';
256       initiatorHtml = `<div class="d-flex align-items-center f
lex-wrap">${initName} ${procName}</div>`;
257       if (s.initiator_chain && s.initiator_chain.length > 1) {
258         initiatorHtml += `<div class="small text-muted font-mo
nospace mt-0.5" style="font-size: 0.72rem;">  ${escapeHtml(s.
initiator_chain.join(' → '))}</div>`;
259       }
```

```
260     }
261
262     let reasonHtml = '<span class="text-muted">--</span>';
263     if (s.reason || s.reason_code) {
264       const codeBadge = s.reason_code ? `<span class="badge bg
-secondary-subtle text-secondary border border-secondary-subtl
e font-monospace me-1">${escapeHtml(s.reason_code)}</span>` :
'';
265       reasonHtml = `<div>${codeBadge}<span>${escapeHtml(s.reas
on || '')}</span></div>`;
266       if (s.comment) {
267         reasonHtml += `<div class="small text-muted font-itali
c mt-0.5">  ${escapeHtml(s.comment)}</div>`;
268       }
269     }
270
271     return `
272       <tr>
273         <td class="font-monospace text-primary small fw-bold">
${escapeHtml(s.session_id)}</td>
274         <td class="font-monospace small">${escapeHtml(s.boot_t
ime)}</td>
275         <td class="font-monospace small text-muted">${s.shutdo
wn_time ? escapeHtml(s.shutdown_time) : '<span class="text-suc
cess fw-bold">Сейчас</span>'}</td>
276         <td class="font-monospace fw-semibold">${escapeHtml(s.
uptime_human || '--')}</td>
277         <td>${statusBadge}</td>
278         <td>${initiatorHtml}</td>
279         <td>${reasonHtml}</td>
280         <td class="text-center">
281           <button class="btn btn-xs btn-outline-info p-1 px-2
pwr-detail-btn" data-session-id="${escapeHtml(s.session_id)}"
title="Посмотреть подробную хронологию">
282             <i class="bi bi-eye"></i>
283           </button>
284         </td>
285       </tr>
```

```
    `;
  }).join('');

  // Навешиваем слушатели на кнопки деталей
  tbody.querySelectorAll('.pwr-detail-btn').forEach(btn => {
    btn.addEventListener('click', () => {
      const sId = btn.dataset.sessionId;
      showSessionDetailModal(sId);
    });
  });
}

/**
 * Фильтрует и отрисовывает таблицу событий Event Log.
 */
function filterAndRenderEvents() {
  const tbody = document.getElementById('pwr-events-tbody');
  const counter = document.getElementById('pwr-events-counter'
);
  if (!tbody) return;

  const searchVal = (document.getElementById('pwr-events-searc
h')?.value || '').toLowerCase().trim();
  const eidFilter = document.getElementById('pwr-event-id-filt
er')?.value || '';

  const filtered = allEvents.filter(e => {
    if (eidFilter && String(e.event_id) !== String(eidFilter))
 return false;
    if (searchVal) {
      const prov = (e.provider || '').toLowerCase();
      const user = (e.user || '').toLowerCase();
      const proc = (e.process || '').toLowerCase();
      const reason = (e.reason || '').toLowerCase();
      const msg = (e.details?.message || '').toLowerCase();
      if (!prov.includes(searchVal) && !user.includes(searchVa
l) && !proc.includes(searchVal) && !reason.includes(searchVal)
 && !msg.includes(searchVal)) {
```

```
        return false;
      }
    }
    return true;
  });

  if (counter) counter.textContent = `Событий: ${filtered.length}`;

  if (filtered.length === 0) {
    tbody.innerHTML = `
      <tr>
        <td colspan="7" class="text-center py-4 text-muted">
          События журнала, удовлетворяющие критериям поиска, не найдены.
        </td>
      </tr>
    `;
    return;
  }

  tbody.innerHTML = filtered.map(e => {
    let eidBadge = `<span class="badge bg-secondary-subtle text-secondary border border-secondary-subtle font-monospace">${e.event_id}</span>`;
    if (e.event_id === 1074) {
      eidBadge = '<span class="badge bg-warning-subtle text-warning-emphasis border border-warning font-monospace fw-bold">1074 User32</span>';
    } else if (e.event_id === 41) {
      eidBadge = '<span class="badge bg-danger-subtle text-danger border border-danger font-monospace fw-bold">41 Kernel-Power</span>';
    } else if (e.event_id === 6008) {
      eidBadge = '<span class="badge bg-danger-subtle text-danger border border-danger font-monospace fw-bold">6008 EventLog</span>';
    } else if (e.event_id === 12 || e.event_id === 6005 || e.e
```

```
vent_id === 6009) {
      eidBadge = `<span class="badge bg-success-subtle text-success border border-success font-monospace">${e.event_id} Boot</span>`;
    } else if (e.event_id === 13 || e.event_id === 6006) {
      eidBadge = `<span class="badge bg-info-subtle text-info border border-info font-monospace">${e.event_id} Clean</span>`;
    } else if (e.event_id === 1001) {
      eidBadge = '<span class="badge bg-danger text-white font-monospace fw-bold">1001 BSOD</span>';
    }

    let userProc = '<span class="text-muted">--</span>';
    if (e.user || e.process) {
      userProc = `<div class="fw-semibold text-primary">${escapeHtml(e.user || '')}</div>`;
      if (e.process) userProc += `<div class="font-monospace small text-muted">${escapeHtml(e.process)}</div>`;
    }

    const desc = e.reason || e.details?.message || '--';

    return `
      <tr>
        <td class="font-monospace small">${escapeHtml(e.timestamp)}</td>
        <td>${eidBadge}</td>
        <td class="small text-muted font-monospace">${escapeHtml(e.provider)}</td>
        <td><span class="badge bg-secondary-subtle text-secondary border border-secondary-subtle">${escapeHtml(e.event_type)}</span></td>
        <td>${userProc}</td>
        <td class="small">${escapeHtml(desc)}</td>
        <td class="text-center">
          ${e.raw_xml ? `
            <button class="btn btn-xs btn-outline-secondary p-
```

```
1 px-2 pwr-xml-btn" data-xml="${escapeHtml(e.raw_xml)}" title=
"Просмотреть XML">
372               <i class="bi bi-code-square"></i>
373             </button>
374           ` : '<span class="text-muted">-</span>'}
375         </td>
376       </tr>
377     `;
378   }).join('');
379
380   tbody.querySelectorAll('.pwr-xml-btn').forEach(btn => {
381     btn.addEventListener('click', () => {
382       const xml = btn.dataset.xml;
383       showXmlModal(xml);
384     });
385   });
386 }
387
388 /**
389  * Открывает модальное окно деталей конкретной сессии питания.
390  * @param {string} sessionId
391  */
392 function showSessionDetailModal(sessionId) {
393   const session = allSessions.find(s => s.session_id === sessi
onId);
394   if (!session) return;
395
396   const modalTitle = document.getElementById('pwr-modal-title'
);
397   const modalBody = document.getElementById('pwr-modal-body');
398   if (modalTitle) modalTitle.textContent = `Сессия питания: ${
session.session_id}`;
399
400   if (modalBody) {
401     let eventsListHtml = '<div class="text-muted small">Нет св
язанных событий</div>';
402     if (session.events && session.events.length > 0) {
403       eventsListHtml = session.events.map(ev => `
```

```
404         <div class="pwr-detail-card p-2 mb-2">
405           <div class="d-flex justify-content-between align-ite
ms-center mb-1">
406             <span class="badge bg-primary-subtle text-primary
border border-primary-subtle font-monospace">Event ${ev.event_
id}</span>
407             <span class="font-monospace small text-muted">${es
capeHtml(ev.timestamp)}</span>
408           </div>
409           <div class="small fw-semibold text-primary">${escape
Html(ev.provider || '')}</div>
410           <div class="small mt-1">${escapeHtml(ev.reason || ev
.details?.message || '')}</div>
411           ${ev.process ? `<div class="small text-muted font-mo
nospace mt-1">Процесс: ${escapeHtml(ev.process)} | Пользовател
ь: ${escapeHtml(ev.user || '--')}</div>` : ''}
412         </div>
413       `).join('');
414     }
415
416     modalBody.innerHTML = `
417       <div class="row g-3 mb-3">
418         <div class="col-6">
419           <div class="pwr-detail-card p-2.5">
420             <div class="text-muted small">  Время старта (Boo
t):</div>
421             <div class="font-monospace fw-bold text-info mt-1"
>${escapeHtml(session.boot_time)}</div>
422           </div>
423         </div>
424         <div class="col-6">
425           <div class="pwr-detail-card p-2.5">
426             <div class="text-muted small">  Завершение работы:
</div>
427             <div class="font-monospace fw-bold ${session.shutd
own_time ? '' : 'text-success'} mt-1">
428               ${session.shutdown_time ? escapeHtml(session.shu
tdown_time) : 'Сессия активна'}
```

```
            </div>
          </div>
        </div>
      </div>

      <div class="pwr-detail-card p-3 mb-3">
        <h6 class="fw-bold text-warning mb-2"><i class="bi bi-
sliders me-1.5"></i>Параметры завершения</h6>
        <div class="row g-2 small">
          <div class="col-4 text-muted">Тип завершения:</div>
          <div class="col-8 fw-semibold">${escapeHtml(session.
shutdown_type)}</div>

          <div class="col-4 text-muted">Аптайм сессии:</div>
          <div class="col-8 font-monospace fw-bold">${escapeHt
ml(session.uptime_human || '--')}</div>

          <div class="col-4 text-muted">Инициатор:</div>
          <div class="col-8">${escapeHtml(session.initiator ||
 'SYSTEM / Hardware')}</div>

          <div class="col-4 text-muted">Процесс:</div>
          <div class="col-8 font-monospace text-info">${escape
Html(session.process || '--')}</div>

          <div class="col-4 text-muted">Код причины:</div>
          <div class="col-8 font-monospace text-secondary">${e
scapeHtml(session.reason_code || '--')}</div>

          <div class="col-4 text-muted">Текст причины:</div>
          <div class="col-8">${escapeHtml(session.reason || '-
-')}</div>

          ${session.comment ? `
            <div class="col-4 text-muted">Комментарий:</div>
            <div class="col-8 font-italic">${escapeHtml(sessio
n.comment)}</div>
          ` : ''}
```

```
459
460           ${session.bugcheck ? `
461             <div class="col-4 text-danger fw-bold">BSOD BugChe
    ck:</div>
462             <div class="col-8 text-danger font-monospace fw-bo
    ld">${escapeHtml(session.bugcheck)}</div>
463           ` : ''}
464         </div>
465       </div>
466
467       ${session.initiator_chain && session.initiator_chain.len
    gth > 0 ? `
468         <div class="pwr-detail-card p-3 mb-3">
469           <h6 class="fw-bold text-info mb-2"><i class="bi bi-d
    iagram-3 me-1.5"></i>Корреляция цепочки вызова</h6>
470           <div class="d-flex align-items-center flex-wrap gap-
    1 font-monospace small">
471             ${session.initiator_chain.map((step, idx) => `
472               <span class="badge bg-secondary-subtle text-seco
    ndary border border-secondary-subtle p-2">${escapeHtml(step)}<
    /span>
473               ${idx < session.initiator_chain.length - 1 ? '<i
     class="bi bi-arrow-right text-muted"></i>' : ''}
474             `).join('')}
475           </div>
476         </div>
477       ` : ''}
478
479       <div class="pwr-detail-card p-3">
480         <h6 class="fw-bold mb-2"><i class="bi bi-list-nested m
    e-1.5"></i>Хронология событий в рамках сессии</h6>
481         <div style="max-height: 250px; overflow-y: auto;">
482           ${eventsListHtml}
483         </div>
484       </div>
485     `;
486
487     const modalEl = document.getElementById('pwrSessionDetailM
```

```
odal');
    if (modalEl && window.bootstrap) {
      const modal = new window.bootstrap.Modal(modalEl);
      modal.show();
    }
  }
}

/**
 * Открывает модальное окно просмотра XML события.
 * @param {string} xmlString
 */
function showXmlModal(xmlString) {
  const pre = document.getElementById('pwr-xml-content');
  if (pre) pre.textContent = xmlString;

  const modalEl = document.getElementById('pwrXmlDetailModal')
;
  if (modalEl && window.bootstrap) {
    const modal = new window.bootstrap.Modal(modalEl);
    modal.show();
  }
}

/**
 * Запускает локальный секундный таймер для активного аптайма.
 */
function startUptimeCounter() {
  if (uptimeInterval) clearInterval(uptimeInterval);
  uptimeInterval = setInterval(() => {
    uptimeSeconds += 1;
    const curUptimeEl = document.getElementById('pwr-current-u
ptime');
    if (curUptimeEl && uptimeSeconds > 0) {
      const days = Math.floor(uptimeSeconds / 86400);
      const hours = Math.floor((uptimeSeconds % 86400) / 3600)
;
      const minutes = Math.floor((uptimeSeconds % 3600) / 60);
```

```
      const sec = Math.floor(uptimeSeconds % 60);
      const parts = [];
      if (days > 0) parts.push(`${days}д`);
      if (hours > 0 || days > 0) parts.push(`${hours}ч`);
      if (minutes > 0 || hours > 0 || days > 0) parts.push(`${minutes}м`);
      parts.push(`${sec}с`);
      curUptimeEl.textContent = parts.join(' ');
    }
  }, 1000);
}

/**
 * Экранирует HTML строку.
 * @param {string} str
 * @returns {string}
 */
function escapeHtml(str) {
  if (!str) return '';
  return String(str)
    .replace(/&/g, '&amp;')
    .replace(/</g, '&lt;')
    .replace(/>/g, '&gt;')
    .replace(/"/g, '&quot;')
    .replace(/'/g, '&#039;');
}

// Автозапуск при загрузке страницы
if (document.readyState === 'loading') {
  document.addEventListener('DOMContentLoaded', initPowerLifecycleTab);
} else {
  initPowerLifecycleTab();
}

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Processes Load Inspector - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса инспектора активности проц
ессов
  (Process Intelligence & Activity Deep Dive — 7 функциональны
х виджетов,
  сетевая активность и мониторинг изменений файлов в реальном
времени).

Usage Examples:
    Web Route:
      /windows/api/webgui/processes_load_inspector_tab/index.h
tml

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/processes_load_inspecto
r_tab/index.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/processes_load_inspector_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-12 07:18:00
================================================================
===============
-->
<style>
  .sys-container {
    padding: 1rem 1.25rem;
    max-width: 100%;
```

```
    color: var(--text-color);
  }

  .sys-section-divider {
    height: 1px;
    margin: 1.5rem 0;
    background: linear-gradient(90deg, transparent 0%, var(--b
order-color) 15%, var(--nav-active) 50%, var(--border-color) 8
5%, transparent 100%);
    position: relative;
  }
  .sys-section-divider::after {
    content: '';
    position: absolute;
    top: -1.5px;
    left: 50%;
    transform: translateX(-50%);
    width: 60px;
    height: 4px;
    background: linear-gradient(90deg, var(--nav-active), #818
cf8);
    border-radius: 4px;
    box-shadow: 0 0 12px rgba(13, 202, 240, 0.4);
  }

  .sys-panel {
    background: var(--surface-1);
    border: 1px solid var(--border-color);
    border-radius: 12px;
    box-shadow: var(--shadow-md);
    transition: transform 0.2s ease, box-shadow 0.2s ease, bor
der-color 0.2s ease;
    overflow: hidden;
    color: var(--text-color);
  }
  .sys-panel:hover {
    border-color: var(--border-strong);
    box-shadow: var(--shadow-lg);
```

```
  }
  [data-theme="dark"] .sys-panel {
    background: linear-gradient(180deg, #141c2e 0%, #0d1424 10
0%);
    border: 1px solid rgba(255, 255, 255, 0.14);
  }
  [data-theme="terminal"] .sys-panel {
    background: #0b120c;
    border: 1px solid #1a4223;
  }

  .sys-panel-header {
    padding: 0.65rem 1rem;
    background: var(--surface-2);
    border-bottom: 1px solid var(--border-color);
    display: flex;
    justify-content: space-between;
    align-items: center;
    font-weight: 700;
    font-size: 0.88rem;
  }

  .sys-proc-table {
    width: 100%;
    font-size: 0.80rem;
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
    border-collapse: collapse;
  }
  .sys-proc-table th {
    background: var(--surface-2);
    padding: 6px 10px;
    color: var(--text-muted);
    font-weight: 700;
    border-bottom: 1.5px solid var(--border-color);
    position: sticky;
    top: 0;
    z-index: 5;
```

```
}
.sys-proc-table td {
  padding: 5px 10px;
  border-bottom: 1px solid var(--border-subtle);
  vertical-align: middle;
}
.sys-proc-table tbody tr {
  transition: background 0.15s ease;
}
.sys-proc-table tbody tr:hover {
  background: var(--surface-2);
  cursor: pointer;
}

.sys-table-resizer {
  height: 8px;
  cursor: row-resize;
  background: var(--surface-2);
  border-top: 1px solid var(--border-color);
  display: flex;
  align-items: center;
  justify-content: center;
  transition: background-color 0.2s ease;
}
.sys-table-resizer:hover, .sys-table-resizer.resizing {
  background: var(--nav-active);
}
.sys-table-resizer-grip {
  width: 36px;
  height: 3px;
  background: var(--border-strong, #666);
  border-radius: 2px;
}

.badge-cpu-high { color: #dc2626; font-weight: 800; }
.badge-cpu-med  { color: #d97706; font-weight: 700; }

/* Стили графа Lineage */
```

```
  .lineage-node-card {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 8px;
    padding: 8px 12px;
    cursor: pointer;
    transition: all 0.2s ease;
  }
  .lineage-node-card:hover {
    border-color: var(--nav-active);
    transform: translateY(-2px);
    box-shadow: var(--shadow-sm);
  }
  .lineage-node-card.active {
    border: 2px solid #0dcaf0;
    background: rgba(13, 202, 240, 0.1);
  }

  /* Виджет карточек SafeOps */
  .gauge-card {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 10px;
    padding: 10px;
    text-align: center;
  }
</style>
<div class="sys-container">
<!-- Главный переключатель режимов: Общий мониторинг хоста VS Deep Dive анализ инстанса -->
<div class="d-flex align-items-center justify-content-between mb-3 flex-wrap gap-2 p-2 rounded" style="background: var(--surface-2); border: 1px solid var(--border-color);">
<div class="btn-group btn-group-sm" role="group">
<button class="btn btn-outline-primary active fw-bold px-3" id="btn-mode-deep-dive" type="button">
<i class="bi bi-cpu-fill me-1 text-info"></i>  Process Intelligence Deep Dive (7 виджетов)
```

```
      </button>
<button class="btn btn-outline-secondary fw-bold px-3" id="btn-mode-host-overview" type="button">
<i class="bi bi-hdd-stack me-1"></i>  Сводный мониторинг хоста
      </button>
</div>
<div class="d-flex align-items-center gap-2">
<span class="badge bg-success-subtle text-success border border-success px-2 py-1" id="engine-status-badge">
        PID Recycling Guard: ACTIVE
      </span>
<button class="btn btn-sm btn-outline-info rounded-pill px-3" id="btn-global-refresh" title="Принудительная синхронизация процессов">
<i class="bi bi-arrow-clockwise me-1"></i> Синхронизировать
      </button>
</div>
</div>
<!-- ============================================================ -->
<!-- СЕКЦИЯ 1: PROCESS ACTIVITY DEEP DIVE (7 ФУНКЦИОНАЛЬНЫХ ВИДЖЕТОВ ПО ТЗ)   -->
<!-- ============================================================ -->
<div id="section-deep-dive">
<!-- ВИДЖЕТ 3.1: Панель выбора и профиля инстанса (Instance Header & Selector Bar) -->
<div class="row g-2 mb-3">
<div class="col-12">
<div class="sys-panel p-3" data-panel-id="panel-processes-load-inspector-vyberite-protsess" id="panel-processes-load-inspector-vyberite-protsess">
<div class="row g-2 align-items-center mb-3">
<div class="col-md-3">
<div class="btn-group btn-group-xs w-100" role="group">
<button class="btn btn-xs btn-outline-success active" id="btn-filter-inst-active" type="button">
```

```
                    ● Активные процессы (<span id="count-inst-active">0</span>)
                </button>
<button class="btn btn-xs btn-outline-secondary" id="btn-filter-inst-history" type="button">
                    История завершенных
                </button>
</div>
</div>
<div class="col-md-5">
<div class="input-group input-group-sm">
<span class="input-group-text bg-dark border-secondary text-muted"><i class="bi bi-search"></i></span>
<input class="form-control form-control-sm bg-dark text-white border-secondary" id="deep-dive-search" placeholder="Поиск по имени, PID или instance_id..." type="text"/>
</div>
</div>
<div class="col-md-4">
<select class="form-select form-select-sm bg-dark text-white border-secondary" id="deep-dive-instance-select">
<option value="">Выберите процесс для анализа...</option>
</select>
</div>
</div>
<!-- Карточка профиля инстанса (Instance Profile Card) -->
<div class="p-3 rounded border border-secondary-subtle" id="instance-profile-card" style="background: var(--surface-2);">
<div class="d-flex align-items-center justify-content-between flex-wrap gap-2 mb-2 pb-2 border-bottom border-secondary">
<div class="d-flex align-items-center gap-2">
<span class="fs-4" id="inst-card-icon">□ </span>
<div>
<div class="d-flex align-items-center gap-2">
<span class="fw-bold fs-5 text-light" id="inst-card-name">Выберите процесс</span>
<span class="badge bg-primary" id="inst-card-pid">PID: --</span>
```

```
<span class="badge bg-dark border border-info text-info font-monospace" id="inst-card-id">instance_id: #--</span>
</div>
<div class="small text-muted font-monospace" id="inst-card-path">C:\...</div>
</div>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<span class="badge bg-success fs-6 px-3 py-1.5" id="inst-card-status">RUNNING</span>
<span class="badge bg-dark border border-secondary text-warning" id="inst-card-uptime">  Uptime: 00:00:00</span>
</div>
</div>
<!-- Системный контекст (User, Integrity, Command Line) -->
<div class="row g-2 small">
<div class="col-md-3">
<span class="text-muted">Пользователь:</span> <span class="fw-bold text-light" id="inst-card-user">--</span>
</div>
<div class="col-md-3">
<span class="text-muted">Целостность UAC:</span> <span class="badge bg-secondary" id="inst-card-integrity">Medium</span>
</div>
<div class="col-md-3">
<span class="text-muted">Время старта:</span> <span class="text-info font-monospace" id="inst-card-start">--</span>
</div>
<div class="col-md-3">
<span class="text-muted">Родительский PID / ID:</span> <span class="text-light" id="inst-card-parent-link">--</span>
</div>
<div class="col-12 mt-1">
<span class="text-muted">Командная строка:</span>
<div class="p-1.5 rounded bg-dark border border-secondary font-monospace text-truncate text-white-50 small mt-0.5" id="inst-card-cmdline" style="user-select: text;">
                  --
```

```
                </div>
</div>
</div>
</div>
</div>
</div>
<!-- ВИДЖЕТ 3.7: Панель безопасного администрирования (SafeOps Action Panel) -->
<div class="row g-2 mb-3">
<div class="col-12">
<div class="sys-panel p-2.5" data-panel-id="panel-processes-load-inspector-safeops-deystviya-nad-instansom" id="panel-processes-load-inspector-safeops-deystviya-nad-instansom">
<div class="d-flex align-items-center justify-content-between flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-shield-lock-fill text-warning fs-5"></i>
<span class="fw-bold text-light">SafeOps: Действия над инстансом процесса</span>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<!-- Изменение приоритета CPU -->
<div class="input-group input-group-sm" style="max-width: 220px;">
<label class="input-group-text bg-dark border-secondary text-muted" for="select-proc-priority">Приоритет:</label>
<select class="form-select bg-dark text-white border-secondary" id="select-proc-priority">
<option value="Idle">Idle (Низкий)</option>
<option value="BelowNormal">Below Normal</option>
<option selected="" value="Normal">Normal (Обычный)</option>
<option value="AboveNormal">Above Normal</option>
<option value="High">High (Высокий)</option>
<option value="Realtime">Realtime (Реального времени)</option>
</select>
<button class="btn btn-outline-info" id="btn-safeops-set-priority" title="Применить приоритет" type="button">OK</button>
```

```
</div>
<!-- Приостановить / Возобновить -->
<button class="btn btn-sm btn-outline-warning rounded-pill px-3" id="btn-safeops-suspend" title="Заморозить потоки процесса">
<i class="bi bi-pause-fill me-1"></i> Приостановить
              </button>
<button class="btn btn-sm btn-outline-success rounded-pill px-3 d-none" id="btn-safeops-resume" title="Возобновить потоки">
<i class="bi bi-play-fill me-1"></i> Возобновить
              </button>
<!-- Снять дамп памяти -->
<button class="btn btn-sm btn-outline-primary rounded-pill px-3" id="btn-safeops-dump" title="Создать .dmp файл в Temp">
<i class="bi bi-file-earmark-binary me-1"></i> Дамп памяти (.dmp)
              </button>
<!-- AI Анализ процесса -->
<button class="btn btn-sm btn-outline-info rounded-pill px-3" id="btn-safeops-ai-audit" title="Запустить детальный AI-аудит инстанса">
<i class="bi bi-robot me-1"></i> AI Анализ
              </button>
<!-- Завершить процесс -->
<button class="btn btn-sm btn-danger rounded-pill px-3 fw-bold" id="btn-safeops-kill" title="Завершить процесс (TerminateProcess)">
<i class="bi bi-x-circle-fill me-1"></i> Завершить процесс
              </button>
</div>
</div>
</div>
</div>
</div>
<!-- ВИДЖЕТ 3.2: Метрики ресурсоемкости в реальном времени (4 синхронизированных графика) -->
<div class="row g-2 mb-3">
<div class="col-12">
```

```
<div class="sys-panel" data-panel-id="panel-processes-load-inspector-metriki-resursoemkosti-v-real" id="panel-processes-load-inspector-metriki-resursoemkosti-v-real">
<div class="sys-panel-header d-flex align-items-center justify-content-between flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-graph-up text-info fs-5"></i>
<span class="fw-bold">Метрики ресурсоемкости в реальном времени (Real-Time Metrics)</span>
</div>
<!-- Переключатель временных интервалов (1м, 5м, 1ч, 24ч) -->
<div class="btn-group btn-group-xs" role="group">
<button class="btn btn-xs btn-outline-primary active chart-interval-btn" data-interval="1m" type="button">1 мин</button>
<button class="btn btn-xs btn-outline-secondary chart-interval-btn" data-interval="5m" type="button">5 мин</button>
<button class="btn btn-xs btn-outline-secondary chart-interval-btn" data-interval="1h" type="button">1 час</button>
<button class="btn btn-xs btn-outline-secondary chart-interval-btn" data-interval="24h" type="button">24 часа</button>
</div>
</div>
<div class="p-3">
<div class="row g-3">
<!-- График 1: ЦП (CPU Utilization) -->
<div class="col-md-6">
<div class="p-2 rounded border border-secondary" style="background: var(--surface-2);">
<div class="d-flex justify-content-between align-items-center mb-1">
<span class="small fw-bold text-info"><i class="bi bi-cpu me-1"></i>1. ЦП (CPU Utilization &amp;amp; Threads)</span>
<span class="badge bg-dark border border-info text-info" id="chart-cpu-current">0.0% | 1 thr</span>
</div>
<div style="height: 170px;">
<canvas id="chart-proc-cpu"></canvas>
</div>
```

```
</div>
</div>
<!-- График 2: Память (RAM Profile) -->
<div class="col-md-6">
<div class="p-2 rounded border border-secondary" style="backgr
ound: var(--surface-2);">
<div class="d-flex justify-content-between align-items-center
mb-1">
<span class="small fw-bold text-warning"><i class="bi bi-memor
y me-1"></i>2. Память (Working Set &amp;amp; Private Bytes)</span>
<span class="badge bg-dark border border-warning text-warning"
 id="chart-ram-current">0.0 MB</span>
</div>
<div style="height: 170px;">
<canvas id="chart-proc-ram"></canvas>
</div>
</div>
</div>
<!-- График 3: Графический процессор (GPU & VRAM) -->
<div class="col-md-6">
<div class="p-2 rounded border border-secondary" style="backgr
ound: var(--surface-2);">
<div class="d-flex justify-content-between align-items-center
mb-1">
<span class="small fw-bold text-success"><i class="bi bi-gpu-c
ard me-1"></i>3. Графический процессор (GPU &amp;amp; VRAM)</span>
<span class="badge bg-dark border border-success text-success"
 id="chart-gpu-current">0.0% | 0 MB</span>
</div>
<div style="height: 170px;">
<canvas id="chart-proc-gpu"></canvas>
</div>
</div>
</div>
<!-- График 4: Дисковый I/O (Disk Read / Write & IOPS) -->
<div class="col-md-6">
<div class="p-2 rounded border border-secondary" style="backgr
ound: var(--surface-2);">
```

```
<div class="d-flex justify-content-between align-items-center mb-1">
<span class="small fw-bold text-danger"><i class="bi bi-hdd me-1"></i>4. Дисковый I/O (Скорость и IOPS)</span>
<span class="badge bg-dark border border-danger text-danger" id="chart-disk-current">0 KB/s | 0 IOPS</span>
</div>
<div style="height: 170px;">
<canvas id="chart-proc-disk"></canvas>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- ВИДЖЕТ 3.3: Иерархическое дерево происхождения (Process Lineage & Hierarchy Graph) -->
<!-- ВИДЖЕТ 3.6: Панель ресурсов ОС и системных дескрипторов (Handles & GUI Objects) -->
<div class="row g-2 mb-3">
<!-- Дерево предков и потомков -->
<div class="col-md-8">
<div class="sys-panel h-100" data-panel-id="panel-processes-load-inspector-1" id="panel-processes-load-inspector-1">
<div class="sys-panel-header">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-diagram-3-fill text-primary"></i>
<span>3.3 Иерархическое дерево происхождения (Process Lineage &amp;amp; Hierarchy)</span>
</div>
<span class="small text-muted">Клик по узлу переключает анализ</span>
</div>
<div class="p-3" id="lineage-container" style="min-height: 180px; max-height: 260px; overflow-y: auto;">
<div class="text-center text-muted py-4">Построение генеалогич
```

```
ческого дерева инстанса...</div>
</div>
</div>
</div>
<!-- Дескрипторы и GUI Объекты (Handles & GDI/USER) -->
<div class="col-md-4">
<div class="sys-panel h-100" data-panel-id="panel-processes-load-inspector-2" id="panel-processes-load-inspector-2">
<div class="sys-panel-header">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-speedometer2 text-warning"></i>
<span>3.6 Дескрипторы ОС и GUI</span>
</div>
<span class="badge bg-success" id="badge-leak-risk">Низкий риск</span>
</div>
<div class="p-3 d-flex flex-column gap-2.5">
<div class="row g-2">
<div class="col-6">
<div class="gauge-card">
<div class="text-muted small">Process Handles</div>
<div class="fs-4 fw-bold text-info" id="metric-handles-count">0</div>
<div class="small text-muted">Файлы, мутексы, секции</div>
</div>
</div>
<div class="col-6">
<div class="gauge-card">
<div class="text-muted small">GDI Objects</div>
<div class="fs-4 fw-bold text-warning" id="metric-gdi-count">0</div>
<div class="small text-muted">Шрифты, кисти, битмапы</div>
</div>
</div>
<div class="col-6">
<div class="gauge-card">
<div class="text-muted small">USER Objects</div>
<div class="fs-4 fw-bold text-light" id="metric-user-count">0<
```

```
/div>
<div class="small text-muted">Оконные дескрипторы</div>
</div>
</div>
<div class="col-6">
<div class="gauge-card">
<div class="text-muted small">Page Faults / sec</div>
<div class="fs-4 fw-bold text-danger" id="metric-page-faults">
0.0</div>
<div class="small text-muted">Ошибки страниц ОЗУ</div>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- ВИДЖЕТ 3.4: Таблица сетевых сокетов и соединений (Network
 Sockets Panel) -->
<!-- ВИДЖЕТ 3.5: Лента файловых операций (File Events Feed) --
>
<div class="row g-2 mb-3">
<!-- Сетевые сокеты инстанса -->
<div class="col-md-6">
<div class="sys-panel h-100" data-panel-id="panel-processes-lo
ad-inspector-3" id="panel-processes-load-inspector-3">
<div class="sys-panel-header d-flex justify-content-between al
ign-items-center">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-ethernet text-info"></i>
<span>3.4 Открытые сокеты и сетевые подключения</span>
<span class="badge bg-dark border border-secondary text-info"
id="badge-inst-sockets-count">0</span>
</div>
</div>
<div class="table-responsive" style="height: 250px; overflow-y
: auto;">
<table class="sys-proc-table">
```

```
<thead>
<tr>
<th>Протокол</th>
<th>Локальный адрес</th>
<th>Удаленный сервер</th>
<th>Статус</th>
</tr>
</thead>
<tbody id="inst-sockets-tbody">
<tr><td class="text-center text-muted py-3" colspan="4">Нет ак
тивных сетевых соединений для процесса</td></tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- Лента файловых операций инстанса -->
<div class="col-md-6">
<div class="sys-panel h-100" data-panel-id="panel-processes-lo
ad-inspector-4" id="panel-processes-load-inspector-4">
<div class="sys-panel-header d-flex justify-content-between al
ign-items-center flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-file-earmark-ruled text-warning"></i>
<span>3.5 Лента файловых операций (File Events)</span>
</div>
<div class="d-flex gap-1.5">
<input class="form-control form-control-xs bg-dark text-white
border-secondary" id="inst-file-search" placeholder="Фильтр (.
json, .log)..." style="font-size: 0.72rem; max-width: 140px; p
adding: 1px 6px;" type="text"/>
</div>
</div>
<div class="table-responsive" style="height: 250px; overflow-y
: auto;">
<table class="sys-proc-table">
<thead>
<tr>
```

```
<th style="width: 70px;">Время</th>
<th style="width: 80px;">Действие</th>
<th>Файл / Путь</th>
</tr>
</thead>
<tbody id="inst-file-events-tbody">
<tr><td class="text-center text-muted py-3" colspan="3">Журнал
 файловых операций для инстанса чист</td></tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
</div> <!-- /section-deep-dive -->
<!-- ========================================================
================ -->
<!-- СЕКЦИЯ 2: СВОДНЫЙ МОНИТОРИНГ ХОСТА (LIVE STREAM, NETWORK,
 FILE WATCHER)  -->
<!-- ========================================================
================ -->
<div class="d-none" id="section-host-overview">
<!-- Панель: Топ процессов хоста (Live Stream) -->
<div class="row g-2 mb-2">
<div class="col-12">
<div class="sys-panel" data-panel-id="panel-processes-load-ins
pector-top-protsessov-khosta-live-strea" id="panel-processes-l
oad-inspector-top-protsessov-khosta-live-strea">
<div class="sys-panel-header d-flex align-items-center justify
-content-between flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-list-task text-primary fs-6"></i>
<span class="fw-bold" style="font-size: 0.88rem; color: var(--
text-color);">Топ процессов хоста (Live Stream)</span>
<span class="badge bg-primary-subtle text-primary border borde
r-primary" id="proc-top-count-badge">0 процессов</span>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
```

```
<input class="form-control form-control-sm" id="sys-proc-search" placeholder="Фильтр по имени или PID..." style="max-width: 220px; font-size: 0.74rem; padding: 2px 8px;" type="text"/>
<button class="btn btn-xs btn-outline-secondary rounded-pill px-2.5 py-0.5" id="btn-sys-pause-proc">Пауза</button>
<button class="btn btn-xs btn-outline-info rounded-pill px-2 py-0.5" id="btn-sys-refresh-proc" title="Обновить список процессов">
<i class="bi bi-arrow-clockwise"></i>
</button>
</div>
</div>
<div class="table-responsive" id="sys-proc-table-container" style="height: 340px; min-height: 140px; max-height: 750px; overflow-y: auto;">
<table class="sys-proc-table">
<thead>
<tr>
<th style="width: 70px;">PID</th>
<th style="width: 200px; max-width: 240px;">Имя процесса</th>
<th style="width: 90px;">Статус</th>
<th style="width: 80px; text-align: right;">CPU %</th>
<th style="width: 100px; text-align: right;">Память</th>
<th style="width: 70px; text-align: right;">Потоки</th>
<th>Пользователь</th>
<th style="width: 90px; text-align: center;">Анализ</th>
</tr>
</thead>
<tbody id="sys-proc-tbody">
<tr><td class="text-center py-3" colspan="8" style="color: var(--text-muted);">Ожидание потока процессов...</td></tr>
</tbody>
</table>
</div>
<div class="sys-table-resizer" id="sys-proc-table-resizer" title="Потяните для изменения высоты таблицы">
<div class="sys-table-resizer-grip"></div>
</div>
```

```
</div>
</div>
</div>
<!-- Четкий стильный разделитель панелей -->
<div class="sys-section-divider"></div>
<!-- Панель: Сетевая активность программ (Интернет, Отправка и
 Прием) -->
<div class="row g-2">
<div class="col-12">
<div class="sys-panel" data-panel-id="panel-processes-load-ins
pector-setevaya-aktivnost-programm-in" id="panel-processes-loa
d-inspector-setevaya-aktivnost-programm-in">
<div class="sys-panel-header d-flex align-items-center justify
-content-between flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<span class="fs-6">□ </span>
<span class="fw-bold" style="font-size: 0.88rem; color: var(--
text-color);">Сетевая активность программ (Интернет)</span>
<span class="badge bg-primary-subtle text-primary border borde
r-primary" id="proc-netact-count-badge">0 программ</span>
<span class="badge bg-dark border border-secondary text-info"
id="proc-netact-conns-badge">0 сокетов</span>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<div class="btn-group btn-group-xs" role="group">
<button class="btn btn-xs btn-outline-primary active py-0.5 px
-2 proc-net-filter-btn" data-filter="internet" id="btn-proc-ne
t-filter-internet">□ Только Интернет</button>
<button class="btn btn-xs btn-outline-secondary py-0.5 px-2 pr
oc-net-filter-btn" data-filter="all" id="btn-proc-net-filter-a
ll">Все сокеты</button>
<button class="btn btn-xs btn-outline-secondary py-0.5 px-2 pr
oc-net-filter-btn" data-filter="listen" id="btn-proc-net-filte
r-listen">Слушающие (Listen)</button>
</div>
<input class="form-control form-control-sm" id="proc-net-searc
h" placeholder="Поиск по имени, PID или IP..." style="max-widt
h: 200px; font-size: 0.74rem; padding: 2px 8px;" type="text"/>
```

```
<button class="btn btn-xs btn-outline-secondary rounded-pill p
x-2.5 py-0.5" id="btn-proc-pause-net">Пауза</button>
<button class="btn btn-xs btn-outline-info rounded-pill px-2 p
y-0.5" id="btn-proc-refresh-net" title="Обновить сетевые соеди
нения">
<i class="bi bi-arrow-clockwise"></i>
</button>
</div>
</div>
<div class="p-0 table-responsive" id="proc-net-table-container
" style="height: 300px; min-height: 140px; max-height: 850px;
overflow-y: auto;">
<table class="sys-proc-table">
<thead>
<tr>
<th style="width: 170px;">Программа / PID</th>
<th style="width: 220px;">Удаленный сервер (Интернет)</th>
<th style="width: 140px;">Служба / Протокол</th>
<th style="width: 100px; text-align: center;">Статус</th>
<th style="min-width: 200px;">Отправка</th>
<th style="min-width: 200px;">Прием</th>
</tr>
</thead>
<tbody id="proc-net-tbody">
<tr><td class="text-center text-muted p-3" colspan="6">Опрос а
ктивных интернет-соединений программ...</td></tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
<!-- Четкий стильный разделитель панелей -->
<div class="sys-section-divider"></div>
<!-- Панель: Изменения файлов в реальном времени (WinAPI ReadD
irectoryChangesW) -->
<div class="row g-2 mt-1">
<div class="col-12">
```

```
<div class="sys-panel" data-panel-id="panel-processes-load-inspector-5" id="panel-processes-load-inspector-5">
<div class="sys-panel-header d-flex align-items-center justify-content-between flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<span><i class="bi bi-activity text-info me-1"></i>Изменения файлов в реальном времени</span>
<span class="badge bg-info-subtle text-info" id="proc-live-badge">ReadDirectoryChangesW</span>
<span class="badge bg-dark border border-secondary text-truncate" id="proc-watch-dir-badge" style="max-width: 280px; font-size: 0.72rem; cursor: pointer;">
                 <span id="proc-watch-dir-path">Загрузка...</span>
</span>
</div>
<div class="d-flex align-items-center gap-1.5">
<button class="btn btn-xs btn-outline-warning rounded-pill px-2 py-0.5" id="btn-proc-watch-exclusions">
<i class="bi bi-funnel me-1"></i> Исключения
              </button>
<button class="btn btn-xs btn-outline-light rounded-pill px-2 py-0.5" id="btn-proc-change-watch-dir">
<i class="bi bi-folder2-open me-1 text-warning"></i> Папка
              </button>
</div>
</div>
<div class="p-0 table-responsive" id="proc-watcher-table-container" style="height: 280px; min-height: 120px; max-height: 850px; overflow-y: auto;">
<table class="sys-proc-table">
<thead>
<tr>
<th style="width: 100px;">Время</th>
<th style="width: 120px;">Действие</th>
<th style="width: 170px;">Программа / PID</th>
<th>Полный путь к файлу</th>
</tr>
```

```
</thead>
<tbody id="proc-watcher-tbody">
<tr><td class="text-center text-muted p-3" colspan="4">Монитор
инг файловых событий...</td></tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
</div> <!-- /section-host-overview -->
</div>
<!-- SafeOps Action Confirmation Modal -->
<div aria-hidden="true" aria-labelledby="modalSafeOpsLabel" cl
ass="modal fade" id="modal-safeops-confirm" tabindex="-1">
<div class="modal-dialog modal-dialog-centered">
<div class="modal-content text-white border-danger shadow-lg"
style="background: var(--surface-1); border-color: var(--borde
r-color) !important;">
<div class="modal-header py-2 px-3 border-danger" style="backg
round: var(--surface-2);">
<h6 class="modal-title d-flex align-items-center gap-2 mb-0" i
d="modalSafeOpsLabel">
<i class="bi bi-exclamation-triangle-fill text-danger fs-5"></
i>
<span class="fw-bold">Подтверждение безопасного действия</span
>
</h6>
<button aria-label="Close" class="btn-close btn-close-white" d
ata-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body p-3">
<p class="mb-2" id="safeops-confirm-text">Вы уверены, что хоти
те завершить процесс?</p>
<div class="p-2 rounded bg-dark border border-secondary font-m
onospace small" id="safeops-target-info">
          PID: -- | Имя: --
        </div>
```

```
</div>
<div class="modal-footer py-2 px-3 border-secondary d-flex justify-content-end gap-2" style="background: var(--surface-2);">
<button class="btn btn-sm btn-secondary rounded-pill px-3" data-bs-dismiss="modal" type="button">Отмена</button>
<button class="btn btn-sm btn-danger rounded-pill px-4 fw-bold" id="btn-safeops-execute-confirmed" type="button">Выполнить</button>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Processes Load Inspector Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля инспектора активности процессов
 *   (Process Intelligence & Activity Deep Dive — 7 функциональных виджетов,
 *   сетевая активность и мониторинг изменений файлов в реальном времени).
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/processes_load_inspector_tab/main.js?v=20261008_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/processes_load_inspector_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-12 08:20:00
 * ============================================================
 */

(function() {
  const TAB_ID = 'tab-processes-load-inspector';

  /**
   * Проверяет, активна ли данная вкладка в текущий момент.
   * Учитывает видимость документа (document.hidden), статус в
```

```
 tab-core.js
   * и наличие класса active у DOM-контейнера вкладки.
   * @returns {boolean}
   */
  function isProcessesTabActive() {
    if (typeof document !== 'undefined' && document.hidden) {
      return false;
    }
    if (typeof window.isTabActive === 'function') {
      return Boolean(window.isTabActive(TAB_ID) || window.isTa
bActive('processes-load-inspector'));
    }
    const pane = document.getElementById(TAB_ID) || document.g
etElementById('processes-load-inspector');
    return pane ? pane.classList.contains('active') : false;
  }

  // Текущее состояние вкладки
  let currentMode = 'deep-dive'; // 'deep-dive' | 'host-overvi
ew'
  let currentInstanceFilter = 'active'; // 'active' | 'history
'
  let selectedInstanceId = null;
  let selectedInstanceData = null;
  let currentChartInterval = '1m';
  let isDeepDivePaused = false;
  let deepDiveTimer = null;
  let hostOverviewTimer = null;

  // Кэшированные данные
  let cachedActiveInstances = [];
  let cachedHistoryInstances = [];
  let cachedTopProcesses = [];
  let isProcPaused = false;

  // Состояние сетевой активности хоста
  let cachedNetworkActivities = [];
  let currentNetFilter = 'internet'; // 'internet' | 'all' | '
```

```
listen'
  let isNetPaused = false;

  // Состояние мониторинга файлов хоста
  let cachedFileEvents = [];
  let currentWatchDirs = [];
  let currentWatchDir = '';
  let isWatcherPaused = false;

  // Chart.js экземпляры
  let chartCpu = null;
  let chartRam = null;
  let chartGpu = null;
  let chartDisk = null;

  // ------------------------------------------------------------
------------------
  // Вспомогательные утилиты
  // ------------------------------------------------------------
------------------

  function escapeHtml(str) {
    if (!str) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
  }

  function formatBytes(bytes) {
    if (!bytes || bytes === 0) return '0 B';
    const k = 1024;
    const sizes = ['B', 'KB', 'MB', 'GB', 'TB'];
    const i = Math.floor(Math.log(bytes) / Math.log(k));
    return parseFloat((bytes / Math.pow(k, i)).toFixed(1)) + '
 ' + sizes[i];
```

```
  }

  function formatDuration(seconds) {
    if (!seconds || seconds <= 0) return '00:00:00';
    const h = Math.floor(seconds / 3600);
    const m = Math.floor((seconds % 3600) / 60);
    const s = Math.floor(seconds % 60);
    return `${String(h).padStart(2, '0')}:${String(m).padStart(2, '0')}:${String(s).padStart(2, '0')}`;
  }

  function showToast(msg, type = 'info') {
    if (window.showToast) {
      window.showToast(msg, type);
    } else {
      console.log(`[Toast ${type}] ${msg}`);
    }
  }

  // ------------------------------------------------------------
  // Инициализация графиков Chart.js (Виджет 3.2)
  // ------------------------------------------------------------

  function initCharts() {
    const commonChartOptions = {
      responsive: true,
      maintainAspectRatio: false,
      animation: false,
      scales: {
        x: {
          display: true,
          grid: { color: 'rgba(255, 255, 255, 0.06)' },
          ticks: { color: '#888', font: { size: 9 }, maxRotation: 0, autoSkip: true, maxTicksLimit: 6 }
        },
        y: {
```

```
          display: true,
          grid: { color: 'rgba(255, 255, 255, 0.06)' },
          ticks: { color: '#888', font: { size: 9 } }
        }
      },
      plugins: {
        legend: { labels: { color: '#bbb', font: { size: 10 }, boxWidth: 10 } },
        tooltip: { mode: 'index', intersect: false }
      }
    };

    // 1. CPU Chart
    const ctxCpu = document.getElementById('chart-proc-cpu')?.getContext('2d');
    if (ctxCpu && !chartCpu) {
      chartCpu = new Chart(ctxCpu, {
        type: 'line',
        data: {
          labels: [],
          datasets: [
            { label: 'ЦП Общая %', data: [], borderColor: '#0dcaf0', backgroundColor: 'rgba(13, 202, 240, 0.1)', borderWidth: 1.5, fill: true, tension: 0.2 },
            { label: 'User Mode %', data: [], borderColor: '#818cf8', borderWidth: 1, borderDash: [4, 4], fill: false, tension: 0.2 },
            { label: 'Потоки', data: [], borderColor: '#f59e0b', borderWidth: 1, yAxisID: 'y1', fill: false }
          ]
        },
        options: {
          ...commonChartOptions,
          scales: {
            ...commonChartOptions.scales,
            y: { ...commonChartOptions.scales.y, min: 0, max: 100 },
            y1: { type: 'linear', display: false, position: 'r
```

```
ight', min: 0 }
          }
        }
      });
    }

    // 2. RAM Chart
    const ctxRam = document.getElementById('chart-proc-ram')?.
getContext('2d');
    if (ctxRam && !chartRam) {
      chartRam = new Chart(ctxRam, {
        type: 'line',
        data: {
          labels: [],
          datasets: [
            { label: 'Working Set (MB)', data: [], borderColor
: '#fbbf24', backgroundColor: 'rgba(251, 191, 36, 0.15)', bord
erWidth: 1.5, fill: true, tension: 0.2 },
            { label: 'Private Bytes (MB)', data: [], borderCol
or: '#f97316', borderWidth: 1.2, fill: false, tension: 0.2 }
          ]
        },
        options: commonChartOptions
      });
    }

    // 3. GPU Chart
    const ctxGpu = document.getElementById('chart-proc-gpu')?.
getContext('2d');
    if (ctxGpu && !chartGpu) {
      chartGpu = new Chart(ctxGpu, {
        type: 'line',
        data: {
          labels: [],
          datasets: [
            { label: 'GPU Load %', data: [], borderColor: '#10
b981', backgroundColor: 'rgba(16, 185, 129, 0.1)', borderWidth
: 1.5, fill: true, tension: 0.2 },
```

```
            { label: 'VRAM (MB)', data: [], borderColor: '#34d399', borderWidth: 1, yAxisID: 'y1', fill: false }
          ]
        },
        options: {
          ...commonChartOptions,
          scales: {
            ...commonChartOptions.scales,
            y: { ...commonChartOptions.scales.y, min: 0, max: 100 },
            y1: { type: 'linear', display: false, position: 'right', min: 0 }
          }
        }
      });
    }

    // 4. Disk Chart
    const ctxDisk = document.getElementById('chart-proc-disk')?.getContext('2d');
    if (ctxDisk && !chartDisk) {
      chartDisk = new Chart(ctxDisk, {
        type: 'line',
        data: {
          labels: [],
          datasets: [
            { label: 'Чтение (KB/s)', data: [], borderColor: '#38bdf8', borderWidth: 1.2, tension: 0.2 },
            { label: 'Запись (KB/s)', data: [], borderColor: '#ef4444', borderWidth: 1.2, tension: 0.2 }
          ]
        },
        options: commonChartOptions
      });
    }
  }

  // ------------------------------------------------------
```

```
------------------
  // Виджет 3.1: Загрузка списка инстансов и Карточка профиля
  // ------------------------------------------------------------
------------------

  async function loadInstancesList() {
    if (!isProcessesTabActive()) return;
    try {
      const searchVal = document.getElementById('deep-dive-sea
rch')?.value.trim() || '';
      const endpoint = currentInstanceFilter === 'active'
        ? `/api/v1/telemetry/instances/active?limit=100${searc
hVal ? '&search=' + encodeURIComponent(searchVal) : ''}`
        : `/api/v1/telemetry/instances/history?limit=100${sear
chVal ? '&search=' + encodeURIComponent(searchVal) : ''}`;

      const res = await fetch(endpoint);
      if (!res.ok) return;

      const instances = await res.json();
      if (currentInstanceFilter === 'active') {
        cachedActiveInstances = instances;
        const countBadge = document.getElementById('count-inst
-active');
        if (countBadge) countBadge.textContent = instances.len
gth;
      } else {
        cachedHistoryInstances = instances;
      }

      renderInstanceDropdown(instances);

      // Если процесс не выбран, выбираем первый
      if (!selectedInstanceId && instances.length > 0) {
        selectInstance(instances[0].instance_id);
      }
    } catch (e) {
      console.warn('[ProcessActivity] Error loading instances:
```

```
', e);
253     }
254   }
255
256   function renderInstanceDropdown(instances) {
257     const select = document.getElementById('deep-dive-instance
-select');
258     if (!select) return;
259
260     const currentVal = selectedInstanceId;
261     select.innerHTML = '<option value="">Выберите процесс для
анализа...</option>';
262
263     instances.forEach(inst => {
264       const opt = document.createElement('option');
265       opt.value = inst.instance_id;
266       const statusIcon = inst.status === 'RUNNING' ? '□ ' : in
st.status === 'SUSPENDED' ? '□ ' : '□ ';
267       opt.textContent = `${statusIcon} ${inst.name} (PID: ${in
st.pid}, #${inst.instance_id}) - CPU: ${inst.cpu_percent.toFix
ed(1)}% | ${inst.memory_mb.toFixed(0)} MB`;
268       if (inst.instance_id === currentVal) {
269         opt.selected = true;
270       }
271       select.appendChild(opt);
272     });
273   }
274
275   async function selectInstance(instanceId) {
276     if (!instanceId) return;
277     selectedInstanceId = parseInt(instanceId, 10);
278
279     // Подгружаем паспорт инстанса
280     try {
281       const res = await fetch(`/api/v1/telemetry/instances/${s
electedInstanceId}`);
282       if (res.ok) {
283         selectedInstanceData = await res.json();
```

```
        renderInstanceProfileCard(selectedInstanceData);
        await Promise.all([
          loadInstanceSamples(),
          loadInstanceLineage(),
          loadInstanceSockets(),
          loadInstanceFileActivity()
        ]);
      }
    } catch (e) {
      console.error('[ProcessActivity] Error fetching instance details:', e);
    }
  }

  function renderInstanceProfileCard(inst) {
    if (!inst) return;

    document.getElementById('inst-card-name').textContent = inst.name || 'Unknown';
    document.getElementById('inst-card-pid').textContent = `PID: ${inst.pid}`;
    document.getElementById('inst-card-id').textContent = `instance_id: #${inst.instance_id}`;
    document.getElementById('inst-card-path').textContent = inst.executable_path || '-';
    document.getElementById('inst-card-user').textContent = inst.user_name || 'SYSTEM / Local';
    document.getElementById('inst-card-integrity').textContent = inst.integrity_level || 'Medium';
    document.getElementById('inst-card-start').textContent = inst.start_time ? new Date(inst.start_time).toLocaleString('ru-RU') : '--';
    document.getElementById('inst-card-uptime').textContent = `  Uptime: ${formatDuration(inst.uptime_seconds)}`;
    document.getElementById('inst-card-cmdline').textContent = inst.command_line || inst.executable_path || '-';

    const parentLink = document.getElementById('inst-card-pare
```

```
nt-link');
311     if (parentLink) {
312       if (inst.parent_instance_id) {
313         parentLink.innerHTML = `<a href="javascript:void(0)" c
lass="text-info text-decoration-none fw-bold" id="link-parent-
inst">Родитель #${inst.parent_instance_id}</a>`;
314         document.getElementById('link-parent-inst')?.addEventL
istener('click', () => selectInstance(inst.parent_instance_id)
);
315       } else {
316         parentLink.textContent = inst.ppid ? `PPID: ${inst.ppi
d}` : 'Корневой (Root)';
317       }
318     }
319
320     // Статус бейдж
321     const statusBadge = document.getElementById('inst-card-sta
tus');
322     const suspendBtn = document.getElementById('btn-safeops-su
spend');
323     const resumeBtn = document.getElementById('btn-safeops-res
ume');
324
325     if (statusBadge) {
326       statusBadge.textContent = inst.status;
327       if (inst.status === 'RUNNING') {
328         statusBadge.className = 'badge bg-success fs-6 px-3 py
-1.5';
329         suspendBtn?.classList.remove('d-none');
330         resumeBtn?.classList.add('d-none');
331       } else if (inst.status === 'SUSPENDED') {
332         statusBadge.className = 'badge bg-warning text-dark fs
-6 px-3 py-1.5';
333         suspendBtn?.classList.add('d-none');
334         resumeBtn?.classList.remove('d-none');
335       } else {
336         statusBadge.className = 'badge bg-secondary fs-6 px-3
py-1.5';
```

```
        statusBadge.textContent = `EXITED (${inst.exit_code !== null ? 'code: ' + inst.exit_code : '0'})`;
        suspendBtn?.classList.add('d-none');
        resumeBtn?.classList.add('d-none');
      }
    }
  }

  // ---------------------------------------------------------------
  // Виджет 3.2: Загрузка сэмплов и обновление графиков
  // ---------------------------------------------------------------

  async function loadInstanceSamples() {
    if (!isProcessesTabActive() || !selectedInstanceId) return;
    try {
      const limitMap = { '1m': 60, '5m': 150, '1h': 300, '24h': 1000 };
      const limit = limitMap[currentChartInterval] || 100;
      const res = await fetch(`/api/v1/telemetry/instances/${selectedInstanceId}/samples?limit=${limit}`);
      if (!res.ok) return;

      const samples = await res.json();
      if (!Array.isArray(samples)) return;

      const labels = samples.map(s => {
        try {
          return new Date(s.timestamp).toLocaleTimeString('ru-RU', { hour12: false });
        } catch {
          return '';
        }
      });

      // Update Chart 1: CPU
```

```
      if (chartCpu) {
        chartCpu.data.labels = labels;
        chartCpu.data.datasets[0].data = samples.map(s => s.cp
u_percent);
        chartCpu.data.datasets[1].data = samples.map(s => s.cp
u_user_time);
        chartCpu.data.datasets[2].data = samples.map(s => s.th
read_count);
        chartCpu.update();
      }

      // Update Chart 2: RAM
      if (chartRam) {
        chartRam.data.labels = labels;
        chartRam.data.datasets[0].data = samples.map(s => s.wo
rking_set_mb);
        chartRam.data.datasets[1].data = samples.map(s => s.pr
ivate_bytes_mb);
        chartRam.update();
      }

      // Update Chart 3: GPU
      if (chartGpu) {
        chartGpu.data.labels = labels;
        chartGpu.data.datasets[0].data = samples.map(s => s.gp
u_load_percent);
        chartGpu.data.datasets[1].data = samples.map(s => s.gp
u_vram_mb);
        chartGpu.update();
      }

      // Update Chart 4: Disk
      if (chartDisk) {
        chartDisk.data.labels = labels;
        chartDisk.data.datasets[0].data = samples.map(s => (s.
disk_read_bytes_sec / 1024).toFixed(1));
        chartDisk.data.datasets[1].data = samples.map(s => (s.
disk_write_bytes_sec / 1024).toFixed(1));
```

```
397         chartDisk.update();
398       }
399
400       // Update current metric badges and Widget 3.6 Gauges
401       if (samples.length > 0) {
402         const last = samples[samples.length - 1];
403         document.getElementById('chart-cpu-current').textConte
nt = `${last.cpu_percent.toFixed(1)}% | ${last.thread_count} t
hr`;
404         document.getElementById('chart-ram-current').textConte
nt = `${last.working_set_mb.toFixed(1)} MB`;
405         document.getElementById('chart-gpu-current').textConte
nt = `${last.gpu_load_percent.toFixed(1)}% | ${last.gpu_vram_m
b.toFixed(0)} MB`;
406         document.getElementById('chart-disk-current').textCont
ent = `${(last.disk_read_bytes_sec / 1024).toFixed(0)} R / ${(
last.disk_write_bytes_sec / 1024).toFixed(0)} W KB/s`;
407
408         // Widget 3.6
409         document.getElementById('metric-handles-count').textCo
ntent = last.handle_count || 0;
410         document.getElementById('metric-gdi-count').textConten
t = last.gdi_objects || 0;
411         document.getElementById('metric-user-count').textConte
nt = last.user_objects || 0;
412         document.getElementById('metric-page-faults').textCont
ent = (last.page_faults_sec || 0).toFixed(1);
413
414         const leakBadge = document.getElementById('badge-leak-
risk');
415         if (leakBadge) {
416           if ((last.gdi_objects || 0) > 8000 || (last.user_obj
ects || 0) > 8000 || (last.handle_count || 0) > 50000) {
417             leakBadge.className = 'badge bg-danger';
418             leakBadge.textContent = 'КРИТИЧЕСКИЙ РИСК УТЕЧКИ';
419           } else if ((last.gdi_objects || 0) > 3000 || (last.h
andle_count || 0) > 10000) {
420             leakBadge.className = 'badge bg-warning text-dark'
```

```
;
            leakBadge.textContent = 'Умеренный риск';
          } else {
            leakBadge.className = 'badge bg-success';
            leakBadge.textContent = 'Низкий риск';
          }
        }
      }
    } catch (e) {
      console.warn('[ProcessActivity] Error loading samples:', e);
    }
  }

  // ---------------------------------------------------------------
  // Виджет 3.3: Иерархическое дерево происхождения (Process Lineage)
  // ---------------------------------------------------------------

  async function loadInstanceLineage() {
    if (!selectedInstanceId) return;
    const container = document.getElementById('lineage-container');
    if (!container) return;

    try {
      const res = await fetch(`/api/v1/telemetry/instances/${selectedInstanceId}/lineage`);
      if (!res.ok) {
        container.innerHTML = '<div class="text-muted small p-2">Генеалогические связи не найдены</div>';
        return;
      }

      const node = await res.json();
      let html = '<div class="d-flex flex-column gap-2">';
```

```
451
452       // Родитель
453       if (node.parent) {
454         html += `
455           <div class="lineage-node-card d-flex align-items-cen
ter justify-content-between" data-inst-id="${node.parent.insta
nce_id}">
456             <div class="d-flex align-items-center gap-2">
457               <i class="bi bi-arrow-return-right text-muted"><
/i>
458               <span class="fw-bold text-light">${escapeHtml(no
de.parent.name)}</span>
459               <span class="badge bg-secondary">PID: ${node.par
ent.pid}</span>
460               <span class="badge bg-dark border border-seconda
ry text-info">#${node.parent.instance_id}</span>
461             </div>
462             <span class="badge bg-dark border border-secondary
 text-muted">Родитель</span>
463           </div>
464           <div class="text-center text-muted small my-0"><i cl
ass="bi bi-arrow-down"></i></div>
465         `;
466       }
467
468       // Текущий инстанс
469       html += `
470         <div class="lineage-node-card active d-flex align-item
s-center justify-content-between" data-inst-id="${node.instanc
e_id}">
471           <div class="d-flex align-items-center gap-2">
472             <span class="fs-6">□ </span>
473             <span class="fw-bold text-info fs-6">${escapeHtml(
node.name)}</span>
474             <span class="badge bg-primary">PID: ${node.pid}</s
pan>
475             <span class="badge bg-info text-dark fw-bold">#${n
ode.instance_id}</span>
```

```
          </div>
          <span class="badge bg-success">[Текущий инстанс]</sp
an>
        </div>
      `;

      // Дочерние процессы
      if (node.children && node.children.length > 0) {
        html += '<div class="text-center text-muted small my-0
"><i class="bi bi-arrow-down"></i></div>';
        html += '<div class="ms-4 d-flex flex-column gap-1.5">
';
        node.children.forEach(c => {
          html += `
            <div class="lineage-node-card d-flex align-items-c
enter justify-content-between" data-inst-id="${c.instance_id}"
>
              <div class="d-flex align-items-center gap-2">
                <i class="bi bi-diagram-2 text-warning"></i>
                <span class="fw-bold text-light">${escapeHtml(
c.name)}</span>
                <span class="badge bg-secondary">PID: ${c.pid}
</span>
                <span class="badge bg-dark border border-secon
dary text-info">#${c.instance_id}</span>
              </div>
              <span class="badge bg-dark border border-seconda
ry text-warning">Дочерний процесс</span>
            </div>
          `;
        });
        html += '</div>';
      }

      html += '</div>';
      container.innerHTML = html;

      // Привязка кликов по узлам графа для мгновенного перехо
```

```
да
      container.querySelectorAll('.lineage-node-card').forEach
(card => {
        card.addEventListener('click', () => {
          const instId = card.getAttribute('data-inst-id');
          if (instId && parseInt(instId, 10) !== selectedInsta
nceId) {
            selectInstance(parseInt(instId, 10));
          }
        });
      });
    } catch (e) {
      container.innerHTML = `<div class="text-danger small p-2
">Ошибка загрузки Lineage: ${escapeHtml(e.message)}</div>`;
    }
  }

  // ------------------------------------------------------------
------------------
  // Виджет 3.4: Сетевые сокеты процесса
  // ------------------------------------------------------------
------------------

  async function loadInstanceSockets() {
    if (!selectedInstanceId) return;
    const tbody = document.getElementById('inst-sockets-tbody'
);
    const countBadge = document.getElementById('badge-inst-soc
kets-count');
    if (!tbody) return;

    try {
      const res = await fetch(`/api/v1/telemetry/instances/${s
electedInstanceId}/sockets`);
      if (!res.ok) return;

      const sockets = await res.json();
      if (countBadge) countBadge.textContent = sockets.length;
```

```
534
535       if (!sockets || sockets.length === 0) {
536         tbody.innerHTML = '<tr><td colspan="4" class="text-cen
ter text-muted py-3">Нет активных сетевых сокетов</td></tr>';
537         return;
538       }
539
540       tbody.innerHTML = sockets.map((s, idx) => `
541         <tr class="inst-socket-row" data-idx="${idx}" style="c
ursor: pointer;" title="Нажмите для AI-анализа сокета">
542           <td><span class="badge bg-dark border border-seconda
ry text-info">${escapeHtml(s.protocol)}</span></td>
543           <td class="font-monospace text-light">${escapeHtml(s
.local_address)}</td>
544           <td class="font-monospace text-warning">${escapeHtml
(s.remote_address)}</td>
545           <td><span class="badge bg-success-subtle text-succes
s border border-success">${escapeHtml(s.status)}</span></td>
546         </tr>
547       `).join('');
548
549       tbody.querySelectorAll('.inst-socket-row').forEach(row =
> {
550         row.onclick = () => {
551           const idx = parseInt(row.getAttribute('data-idx'), 1
0);
552           const s = sockets[idx];
553           if (!s) return;
554           const modal = window.AIModalDialog || window.AITable
Modal;
555           if (modal) {
556             modal.show({
557               icon: '□ ',
558               title: `Сокет: ${s.protocol} ${s.local_address}`
,
559               subtitle: `Удаленный адрес: ${s.remote_address}
| Статус: ${s.status}`,
560               panelId: 'panel-processes-load-inspector-3',
```

```
              tableType: 'network',
              badges: [
                { text: s.protocol, class: 'badge bg-info text
-dark font-monospace' },
                { text: s.status, class: 'badge bg-success' }
              ],
              metadata: [
                { label: 'Протокол', value: s.protocol },
                { label: 'Локальный сокет', value: s.local_add
ress, isCode: true },
                { label: 'Удаленный узел', value: s.remote_add
ress, isCode: true },
                { label: 'Состояние', value: s.status }
              ],
              rawTitle: 'Параметры сокета процесса',
              rawContent: JSON.stringify(s, null, 2),
              requestData: s
            });
          }
        };
      });
    } catch (e) {
      console.warn('[ProcessActivity] Error loading sockets:',
 e);
    }
  }

  // ----------------------------------------------------------
------------------
  // Виджет 3.5: Файловые операции инстанса
  // ----------------------------------------------------------
------------------

  async function loadInstanceFileActivity() {
    if (!selectedInstanceId) return;
    const tbody = document.getElementById('inst-file-events-tb
ody');
    if (!tbody) return;
```

```
592
593     try {
594       const searchVal = document.getElementById('inst-file-sea
rch')?.value.trim() || '';
595       let url = `/api/v1/telemetry/instances/${selectedInstanc
eId}/file-activity?limit=50`;
596       if (searchVal) {
597         if (searchVal.startsWith('.')) {
598           url += `&extension=${encodeURIComponent(searchVal)}`
;
599         } else {
600           url += `&action=${encodeURIComponent(searchVal)}`;
601         }
602       }
603
604       const res = await fetch(url);
605       if (!res.ok) return;
606
607       const events = await res.json();
608       if (!events || events.length === 0) {
609         tbody.innerHTML = '<tr><td colspan="3" class="text-cen
ter text-muted py-3">Журнал файловых операций пуст</td></tr>';
610         return;
611       }
612
613       tbody.innerHTML = events.map((ev, idx) => {
614         let actionBadge = 'bg-secondary';
615         if (ev.action === 'CREATE') actionBadge = 'bg-success'
;
616         else if (ev.action === 'MODIFY') actionBadge = 'bg-war
ning text-dark';
617         else if (ev.action === 'DELETE') actionBadge = 'bg-dan
ger';
618         else if (ev.action === 'RENAME') actionBadge = 'bg-inf
o text-dark';
619
620         const timeStr = ev.timestamp ? new Date(ev.timestamp).
toLocaleTimeString('ru-RU') : '--';
```

```
        return `
          <tr class="inst-file-row" data-idx="${idx}" style="cursor: pointer;" title="Нажмите для AI-анализа файловой операции">
            <td class="font-monospace text-muted">${timeStr}</td>
            <td><span class="badge ${actionBadge}">${escapeHtml(ev.action)}</span></td>
            <td class="font-monospace text-truncate text-light" style="max-width: 280px;" title="${escapeHtml(ev.file_path)}">
              ${escapeHtml(ev.file_path)}
            </td>
          </tr>
        `;
      }).join('');

      tbody.querySelectorAll('.inst-file-row').forEach(row => {
        row.onclick = () => {
          const idx = parseInt(row.getAttribute('data-idx'), 10);
          const ev = events[idx];
          if (!ev) return;
          const modal = window.AIModalDialog || window.AITableModal;
          if (modal) {
            modal.show({
              icon: '',
              title: `Файл [${ev.action}]: ${ev.file_path?.split(/[\\/]/).pop() || 'file'}`,
              subtitle: ev.file_path || '',
              panelId: 'panel-processes-load-inspector-4',
              tableType: 'file_event',
              badges: [
                { text: ev.action, class: `badge bg-info text-dark font-monospace` }
              ],
```

```
              metadata: [
                { label: 'Действие', value: ev.action },
                { label: 'Полный путь', value: ev.file_path, isCode: true },
                { label: 'Время операции', value: ev.timestamp || '-' }
              ],
              rawTitle: 'Метаданные файлового события',
              rawContent: JSON.stringify(ev, null, 2),
              requestData: ev
            });
          }
        };
      });
    } catch (e) {
      console.warn('[ProcessActivity] Error loading file activity:', e);
    }
  }

  // ---------------------------------------------------------
  // Виджет 3.7: SafeOps Действия над процессом
  // ---------------------------------------------------------

  let pendingSafeOpsAction = null;

  function initSafeOps() {
    // 1. Изменение приоритета CPU
    document.getElementById('btn-safeops-set-priority')?.addEventListener('click', async () => {
      if (!selectedInstanceId) return;
      const prio = document.getElementById('select-proc-priority')?.value || 'Normal';
      try {
        const res = await fetch(`/api/v1/telemetry/instances/${selectedInstanceId}/action`, {
```

```
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ action: 'priority', priority_class: prio })
        });
        const data = await res.json();
        showToast(data.message || 'Приоритет обновлен', data.success ? 'success' : 'danger');
      } catch (e) {
        showToast(`Ошибка SafeOps: ${e.message}`, 'danger');
      }
    });

    // 2. Приостановить процесс
    document.getElementById('btn-safeops-suspend')?.addEventListener('click', async () => {
      if (!selectedInstanceId) return;
      try {
        const res = await fetch(`/api/v1/telemetry/instances/${selectedInstanceId}/action`, {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ action: 'suspend' })
        });
        const data = await res.json();
        showToast(data.message, data.success ? 'success' : 'danger');
        selectInstance(selectedInstanceId);
      } catch (e) {
        showToast(`Ошибка: ${e.message}`, 'danger');
      }
    });

    // 3. Возобновить процесс
    document.getElementById('btn-safeops-resume')?.addEventListener('click', async () => {
      if (!selectedInstanceId) return;
      try {
```

```
        const res = await fetch(`/api/v1/telemetry/instances/${selectedInstanceId}/action`, {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ action: 'resume' })
        });
        const data = await res.json();
        showToast(data.message, data.success ? 'success' : 'danger');
        selectInstance(selectedInstanceId);
      } catch (e) {
        showToast(`Ошибка: ${e.message}`, 'danger');
      }
    });

    // 4. Снять дамп памяти
    document.getElementById('btn-safeops-dump')?.addEventListener('click', async () => {
      if (!selectedInstanceId) return;
      try {
        const res = await fetch(`/api/v1/telemetry/instances/${selectedInstanceId}/action`, {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ action: 'dump' })
        });
        const data = await res.json();
        showToast(data.message, data.success ? 'success' : 'danger');
      } catch (e) {
        showToast(`Ошибка: ${e.message}`, 'danger');
      }
    });

    // 4.1. AI Анализ и аудит процесса через Universal AI Modal
    document.getElementById('btn-safeops-ai-audit')?.addEventListener('click', () => {
```

```
741       if (!selectedInstanceData) {
742         showToast('Процесс не выбран', 'warning');
743         return;
744       }
745       if (window.AIModalDialog) {
746         window.AIModalDialog.show({
747           title: `AI Анализ процесса: ${selectedInstanceData.n
    ame || 'Unknown'} (PID: ${selectedInstanceData.pid})`,
748           panelId: 'panel-processes-load-inspector-vyberite-pr
    otsess',
749           tableType: 'process',
750           row: selectedInstanceData,
751           contextData: {
752             selectedInstance: selectedInstanceData,
753             timestamp: new Date().toISOString()
754           }
755         });
756       }
757     });
758
759     // 5. Завершить процесс (с модальным подтверждением)
760     document.getElementById('btn-safeops-kill')?.addEventListe
    ner('click', () => {
761       if (!selectedInstanceId || !selectedInstanceData) return
    ;
762       pendingSafeOpsAction = 'kill';
763       document.getElementById('safeops-confirm-text').textCont
    ent = `Вы уверены, что хотите принудительно завершить процесс
    ${selectedInstanceData.name}?`;
764       document.getElementById('safeops-target-info').textConte
    nt = `PID: ${selectedInstanceData.pid} | #${selectedInstanceDa
    ta.instance_id} | ${selectedInstanceData.executable_path}`;
765
766       const modalEl = document.getElementById('modal-safeops-c
    onfirm');
767       if (modalEl && window.bootstrap) {
768         const modal = new bootstrap.Modal(modalEl);
769         modal.show();
```

```
      }
    });

    document.getElementById('btn-safeops-execute-confirmed')?.addEventListener('click', async () => {
      if (!selectedInstanceId || pendingSafeOpsAction !== 'kill') return;
      try {
        const res = await fetch(`/api/v1/telemetry/instances/${selectedInstanceId}/action`, {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ action: 'kill', reason: 'User confirmed kill in Deep Dive UI' })
        });
        const data = await res.json();
        showToast(data.message, data.success ? 'success' : 'danger');

        const modalEl = document.getElementById('modal-safeops-confirm');
        if (modalEl && window.bootstrap) {
          const modal = bootstrap.Modal.getInstance(modalEl);
          modal?.hide();
        }
        selectInstance(selectedInstanceId);
        loadInstancesList();
      } catch (e) {
        showToast(`Ошибка завершения: ${e.message}`, 'danger');
      }
    });
  }

  // ---------------------------------------------------------
  // Сводный мониторинг хоста (Секция 2)
  // ---------------------------------------------------------
```

```
------------------

  async function fetchHostOverview() {
    if (!isProcessesTabActive()) return;
    await Promise.all([
      fetchHostProcesses(),
      fetchNetworkActivity(),
      fetchLiveFileEvents()
    ]);
  }

  async function fetchHostProcesses() {
    if (!isProcessesTabActive()) return;
    try {
      const res = await fetch('/api/v1/system/summary?process_
limit=50');
      if (res.ok) {
        const snap = await res.json();
        const processes = snap.top_processes || snap.processes
 || [];
        cachedTopProcesses = processes;
        renderHostProcessesTable(processes);
      }
    } catch (e) {
      console.warn('[HostOverview] Error fetching host process
es:', e);
    }
  }

  function renderHostProcessesTable(processes) {
    if (processes) {
      cachedTopProcesses = processes;
    }
    if (isProcPaused) return;

    const tbody = document.getElementById('sys-proc-tbody');
    const countBadge = document.getElementById('proc-top-count
-badge');
```

```
    if (!tbody) return;

    const searchInput = document.getElementById('sys-proc-search');
    const filterText = (searchInput?.value || '').toLowerCase().trim();

    const rawList = Array.isArray(cachedTopProcesses) ? cachedTopProcesses : [];
    const filtered = filterText
      ? rawList.filter(p => String(p.pid).includes(filterText) || (p.name || '').toLowerCase().includes(filterText) || (p.username || '').toLowerCase().includes(filterText))
      : rawList;

    if (countBadge) countBadge.textContent = `${filtered.length} процессов`;

    if (filtered.length === 0) {
      tbody.innerHTML = '<tr><td colspan="8" class="text-center py-3 text-muted">Процессы не найдены</td></tr>';
      return;
    }

    tbody.innerHTML = filtered.map(p => `
      <tr class="proc-host-row interactive-table-row" data-pid="${p.pid}" style="cursor: pointer;">
        <td class="font-monospace fw-bold text-info">${p.pid}</td>
        <td class="font-monospace text-truncate text-light fw-semibold" style="max-width: 220px;" title="${escapeHtml(p.name)}">${escapeHtml(p.name)}</td>
        <td><span class="badge bg-success-subtle text-success">${escapeHtml(p.status || 'RUNNING')}</span></td>
        <td class="text-end font-monospace">${(p.cpu_percent || 0).toFixed(1)}%</td>
        <td class="text-end font-monospace">${(p.memory_mb || 0).toFixed(1)} MB</td>
```

```
        <td class="text-end font-monospace">${p.num_threads ||
 1}</td>
        <td class="text-muted small">${escapeHtml(p.username |
| '-')}</td>
        <td class="text-center no-modal-trigger">
          <button class="btn btn-xs btn-outline-info rounded-p
ill px-2 py-0.5 btn-drilldown-proc" data-pid="${p.pid}" title=
"Глубокий анализ инстанса">
            Deep Dive
          </button>
        </td>
      </tr>
    `).join('');

    tbody.querySelectorAll('.proc-host-row').forEach(row => {
      row.addEventListener('click', (e) => {
        if (e.target.closest('button, a, input, select, .no-mo
dal-trigger')) return;
        const pid = parseInt(row.getAttribute('data-pid'), 10)
;
        const p = filtered.find(x => x.pid === pid);
        if (!p) return;

        const modal = window.AIModalDialog || window.AITableMo
dal;
        if (modal) {
          modal.show({
            title: p.name,
            subtitle: `PID: ${p.pid} | CPU: ${(p.cpu_percent |
| 0).toFixed(1)}% | RAM: ${(p.memory_mb || 0).toFixed(1)} MB`,
            icon: ' ',
            panelId: 'panel-processes-load-inspector-top-prots
essov-khosta-live-strea',
            tableType: 'process',
            badges: [
              { text: p.status || 'RUNNING', class: 'badge bg-
success' },
              { text: `Threads: ${p.num_threads || 1}`, class:
```

```
 'badge bg-dark border border-secondary text-info' }
            ],
            metadata: [
              { label: 'Имя процесса', value: p.name },
              { label: 'PID процесса', value: p.pid },
              { label: 'Использование CPU', value: `${(p.cpu_p
ercent || 0).toFixed(1)}%` },
              { label: 'Память (RAM)', value: `${(p.memory_mb
|| 0).toFixed(1)} MB` },
              { label: 'Количество потоков', value: p.num_thre
ads || 1 },
              { label: 'Пользователь', value: p.username || 'S
YSTEM' },
              { label: 'Исполняемый путь', value: p.executable
_path || p.exe || '', isCode: true }
            ],
            rawTitle: 'Дамп процесса',
            rawContent: JSON.stringify(p, null, 2),
            actions: [
              {
                label: '  Deep Dive',
                icon: 'bi-cpu',
                class: 'btn-outline-info',
                onClick: () => {
                  const match = cachedActiveInstances.find(i =
> i.pid === p.pid);
                  if (match) {
                    selectInstance(match.instance_id);
                  }
                  switchToDeepDiveMode();
                }
              }
            ]
          });
        }
      });
    });

```

```
    tbody.querySelectorAll('.btn-drilldown-proc').forEach(btn => {
      btn.addEventListener('click', (e) => {
        e.stopPropagation();
        const pid = parseInt(btn.getAttribute('data-pid'), 10);
        const match = cachedActiveInstances.find(i => i.pid === pid);
        if (match) {
          selectInstance(match.instance_id);
        }
        switchToDeepDiveMode();
      });
    });
  }

  // ---------------------------------------------------------------
  // Панель: Сетевая активность программ (Интернет, Отправка и Прием)
  // ---------------------------------------------------------------

  async function fetchNetworkActivity() {
    if (!isProcessesTabActive()) return;
    try {
      const res = await fetch('/api/v1/system/network-activity?limit=100');
      if (res.ok) {
        const data = await res.json();
        renderNetworkActivityTable(data);
      }
    } catch (e) {
      console.warn('[ProcessActivity] Failed to fetch network activity:', e);
    }
  }

```

```
  function renderNetworkActivityTable(activities) {
    if (activities) {
      cachedNetworkActivities = activities;
    }
    if (isNetPaused) return;

    const tbody = document.getElementById('proc-net-tbody');
    if (!tbody) return;

    const searchInput = document.getElementById('proc-net-search');
    const filterText = (searchInput?.value || '').toLowerCase().trim();
    const countBadge = document.getElementById('proc-netact-count-badge');
    const connsBadge = document.getElementById('proc-netact-conns-badge');

    const rawList = Array.isArray(cachedNetworkActivities) ? cachedNetworkActivities : [];

    const filtered = rawList.filter(item => {
      if (currentNetFilter === 'internet' && !item.is_internet) return false;
      if (currentNetFilter === 'listen' && item.status !== 'LISTEN') return false;

      if (filterText) {
        return (
          String(item.pid).includes(filterText) ||
          (item.name || '').toLowerCase().includes(filterText) ||
          (item.user || '').toLowerCase().includes(filterText) ||
          (item.remote_address || '').toLowerCase().includes(filterText) ||
          (item.local_address || '').toLowerCase().includes(filterText) ||
```

```
          (item.service_type || '').toLowerCase().includes(filterText) ||
          (item.sent_summary || '').toLowerCase().includes(filterText) ||
          (item.recv_summary || '').toLowerCase().includes(filterText)
        );
      }
      return true;
    });

    const uniqueProcs = new Set(filtered.map(i => i.pid)).size;
    if (countBadge) countBadge.textContent = `${uniqueProcs} программ`;
    if (connsBadge) connsBadge.textContent = `${filtered.length} сокетов`;

    if (filtered.length === 0) {
      tbody.innerHTML = '<tr><td colspan="6" class="text-center py-4 text-muted small">Нет активных сетевых соединений по выбранному фильтру</td></tr>';
      return;
    }

    tbody.innerHTML = filtered.map((item, idx) => {
      const isListen = item.status === 'LISTEN';
      const statusBadgeClass = isListen
        ? 'badge bg-secondary-subtle text-light border border-secondary'
        : item.status === 'ESTABLISHED'
        ? 'badge bg-success-subtle text-success border border-success'
        : 'badge bg-warning-subtle text-warning border border-warning';

      const protoBadge = item.protocol === 'UDP'
        ? '<span class="badge bg-primary text-white" style="fo
```

```
nt-size: 0.65rem;">UDP</span>'
1000        : '<span class="badge bg-dark border border-secondary
    text-info" style="font-size: 0.65rem;">TCP</span>';
1001
1002      const isExtBadge = item.is_internet
1003        ? '<span class="badge bg-primary-subtle text-primary b
    order border-primary px-1" style="font-size: 0.62rem;" title="
    Внешний сервер в сети Интернет">WAN</span>'
1004        : '<span class="badge bg-secondary px-1" style="font-s
    ize: 0.62rem;" title="Локальный сокет Loopback">LAN</span>';
1005
1006      const formatNetKb = (kb) => {
1007        if (!kb || kb <= 0) return '0 KB';
1008        if (kb >= 1024 * 1024) return (kb / (1024 * 1024)).toF
    ixed(2) + ' GB';
1009        if (kb >= 1024) return (kb / 1024).toFixed(1) + ' MB';
1010        return kb.toFixed(1) + ' KB';
1011      };
1012
1013      const formatNetRate = (rate) => {
1014        if (!rate || rate <= 0.05) return '';
1015        if (rate >= 1024) return (rate / 1024).toFixed(1) + '
    MB/s';
1016        return rate.toFixed(1) + ' KB/s';
1017      };
1018
1019      const deltaSentStr = (item.delta_sent_kb && item.delta_s
    ent_kb > 0)
1020        ? `▲ +${formatNetKb(item.delta_sent_kb)}${item.sent_ra
    te_kbs > 0.05 ? ' (' + formatNetRate(item.sent_rate_kbs) + ')'
     : ''}`
1021        : '▲ 0 KB';
1022      const deltaRecvStr = (item.delta_recv_kb && item.delta_r
    ecv_kb > 0)
1023        ? `▼ +${formatNetKb(item.delta_recv_kb)}${item.recv_ra
    te_kbs > 0.05 ? ' (' + formatNetRate(item.recv_rate_kbs) + ')'
     : ''}`
1024        : '▼ 0 KB';
```

```
1025
1026       const totalSentStr = formatNetKb(item.sent_kb || 0);
1027       const totalRecvStr = formatNetKb(item.recv_kb || 0);
1028
1029       return `
1030         <tr class="proc-net-row interactive-table-row" data-id
     x="${idx}" style="cursor: pointer;" title="Нажмите для детальн
     ой диагностики сетевого соединения">
1031           <td>
1032             <div class="fw-bold text-truncate" style="max-widt
     h: 165px; color: var(--text-color);" title="${escapeHtml(item.
     name)}">${escapeHtml(item.name)}</div>
1033             <div class="small text-muted" style="font-size: 0.
     70rem;">PID: <span class="font-monospace text-info">${item.pid
     }</span> ${item.user ? '• ' + escapeHtml(item.user) : ''}</div
     >
1034           </td>
1035           <td>
1036             <div class="d-flex align-items-center gap-1">
1037               ${isExtBadge}
1038               <span class="font-monospace fw-semibold text-tru
     ncate" style="max-width: 195px; color: var(--text-color);" tit
     le="${escapeHtml(item.remote_address)}">
1039                 ${escapeHtml(item.remote_address !== '-' ? ite
     m.remote_address : item.local_address)}
1040               </span>
1041             </div>
1042             <div class="small text-muted font-monospace" style
     ="font-size: 0.68rem;">Local: ${escapeHtml(item.local_address)
     }</div>
1043           </td>
1044           <td>
1045             <div class="d-flex align-items-center gap-1 mb-0.5
     ">
1046               ${protoBadge}
1047               <span class="fw-semibold text-truncate" style="m
     ax-width: 110px; font-size: 0.74rem; color: var(--nav-active);
     " title="${escapeHtml(item.service_type)}">${escapeHtml(item.s
```

```
ervice_type)}</span>
            </div>
          </td>
          <td style="text-align: center;">
            <span class="${statusBadgeClass}" style="font-size
: 0.68rem;">${escapeHtml(item.status)}</span>
          </td>
          <td>
            <div class="d-flex align-items-center justify-cont
ent-between gap-1 mb-1">
              <span class="badge bg-warning-subtle text-warnin
g border border-warning px-1.5 py-0.5" style="font-size: 0.68r
em;" title="Отправлено за измеряемый период">
                ${deltaSentStr}
              </span>
              <span class="text-muted font-monospace" style="f
ont-size: 0.66rem;" title="Всего отправлено/записано">
                Σ ${totalSentStr}
              </span>
            </div>
            <div class="d-flex align-items-start gap-1">
              <i class="bi bi-arrow-up-right text-warning mt-0
.5" style="font-size: 0.70rem;"></i>
              <div class="text-truncate" style="max-width: 250
px; font-size: 0.72rem; color: var(--text-muted);" title="${es
capeHtml(item.sent_summary)}">
                ${escapeHtml(item.sent_summary || '-')}
              </div>
            </div>
          </td>
          <td>
            <div class="d-flex align-items-center justify-cont
ent-between gap-1 mb-1">
              <span class="badge bg-success-subtle text-succes
s border border-success px-1.5 py-0.5" style="font-size: 0.68r
em;" title="Скачано/получено за измеряемый период">
                ${deltaRecvStr}
              </span>
```

```
              <span class="text-muted font-monospace" style="f
ont-size: 0.66rem;" title="Всего скачано/прочитано">
                Σ ${totalRecvStr}
              </span>
            </div>
            <div class="d-flex align-items-start gap-1">
              <i class="bi bi-arrow-down-left text-success mt-
0.5" style="font-size: 0.70rem;"></i>
              <div class="text-truncate" style="max-width: 250
px; font-size: 0.72rem; color: var(--text-muted);" title="${es
capeHtml(item.recv_summary)}">
                ${escapeHtml(item.recv_summary || '-')}
              </div>
            </div>
          </td>
        </tr>
      `;
    }).join('');

    tbody.querySelectorAll('.proc-net-row').forEach(row => {
      row.onclick = () => {
        const idx = parseInt(row.getAttribute('data-idx'), 10)
;
        const item = filtered[idx];
        if (!item) return;

        const formatNetKb = (kb) => {
          if (!kb || kb <= 0) return '0 KB';
          if (kb >= 1024 * 1024) return (kb / (1024 * 1024)).t
oFixed(2) + ' GB';
          if (kb >= 1024) return (kb / 1024).toFixed(1) + ' MB
';
          return kb.toFixed(1) + ' KB';
        };

        const modal = window.AIModalDialog || window.AITableMo
dal;
        if (modal) {
```

```
          modal.show({
            title: `${item.name} (PID: ${item.pid}) — ${item.service_type || item.protocol}`,
            subtitle: `${item.protocol} ${item.local_address} -> ${item.remote_address} | Состояние: ${item.status}`,
            icon: '□ ',
            panelId: 'panel-processes-load-inspector-setevaya-aktivnost-programm-in',
            tableType: 'network',
            badges: [
              { text: item.is_internet ? 'WAN (Интернет)' : 'LAN (Локальный)', class: item.is_internet ? 'badge bg-primary' : 'badge bg-secondary' },
              { text: item.status, class: item.status === 'ESTABLISHED' ? 'badge bg-success' : 'badge bg-warning text-dark' },
              { text: item.protocol, class: 'badge bg-dark border border-secondary text-info' }
            ],
            metadata: [
              { label: 'Программа / Процесс', value: `${item.name} (PID: ${item.pid})` },
              { label: 'Пользователь процесса', value: item.user || 'SYSTEM' },
              { label: 'Локальный сокет', value: item.local_address, isCode: true },
              { label: 'Удаленный сервер', value: item.remote_address, isCode: true },
              { label: 'Служба / Протокол', value: `${item.service_type || 'Custom'} (${item.protocol})` },
              { label: 'Статус сокета', value: item.status },
              { label: 'Отправлено (дельта)', value: formatNetKb(item.delta_sent_kb) },
              { label: 'Принято (дельта)', value: formatNetKb(item.delta_recv_kb) },
              { label: 'Всего передано', value: `Отправлено: ${formatNetKb(item.sent_kb)} | Скачано: ${formatNetKb(item.recv_kb)}` },
```

```
1125              { label: 'Исходящий трафик (описание)', value: i
tem.sent_summary || '-', fullWidth: true },
1126              { label: 'Входящий трафик (описание)', value: it
em.recv_summary || '-', fullWidth: true }
1127            ],
1128            rawTitle: 'Сырой дамп сетевого сокета (JSON)',
1129            rawContent: JSON.stringify(item, null, 2),
1130            requestData: item
1131          });
1132        }
1133      };
1134    });
1135  }
1136
1137  // ---------------------------------------------------------
------------------
1138  // Панель: Изменения файлов в реальном времени (WinAPI ReadD
irectoryChangesW)
1139  // ---------------------------------------------------------
------------------
1140
1141  async function fetchLiveFileEvents() {
1142    if (!isProcessesTabActive()) return;
1143    try {
1144      const [resEvents, resTelem] = await Promise.all([
1145        fetch('/api/v1/system/file-audit/live-events?limit=30'
),
1146        fetch('/api/v1/system/file-audit/telemetry').catch(()
=> null)
1147      ]);
1148
1149      if (resTelem && resTelem.ok) {
1150        const telem = await resTelem.json();
1151        if (telem.watch_dirs || telem.watch_dir) {
1152          currentWatchDirs = telem.watch_dirs || (telem.watch_
dir ? [telem.watch_dir] : []);
1153          currentWatchDir = currentWatchDirs[0] || telem.watch
_dir || '';
```

```
1154        }
1155      }
1156
1157      if (!resEvents.ok) return;
1158      const data = await resEvents.json();
1159      const events = Array.isArray(data) ? data : (data.events
 || []);
1160
1161      if (data.watch_dirs || data.watch_dir) {
1162        currentWatchDirs = data.watch_dirs || (data.watch_dir
? [data.watch_dir] : []);
1163        currentWatchDir = currentWatchDirs[0] || data.watch_di
r || '';
1164      }
1165
1166      cachedFileEvents = events;
1167
1168      // Обновление бейджа отслеживаемой папки
1169      const dirPathEl = document.getElementById('proc-watch-di
r-path');
1170      const badge = document.getElementById('proc-watch-dir-ba
dge');
1171      if (dirPathEl) {
1172        if (currentWatchDirs.length === 0) {
1173          dirPathEl.innerText = 'Папки не выбраны';
1174        } else if (currentWatchDirs.length === 1) {
1175          const singleName = currentWatchDirs[0].split('\\').p
op() || currentWatchDirs[0];
1176          dirPathEl.innerText = singleName;
1177        } else {
1178          const firstNames = currentWatchDirs.slice(0, 2).map(
p => p.split('\\').pop() || p).join(', ');
1179          dirPathEl.innerText = `${currentWatchDirs.length} па
пок: ${firstNames}${currentWatchDirs.length > 2 ? '...' : ''}`
;
1180        }
1181      }
1182      if (badge) {
```

```
1183        badge.title = `Отслеживаемые каталоги (${currentWatchD
    irs.length}):\n${currentWatchDirs.join('\n')}\n\n(Нажмите для
    детальной информации)`;
1184      }
1185
1186      const tbody = document.getElementById('proc-watcher-tbod
    y');
1187      if (!tbody) return;
1188
1189      if (events.length === 0) {
1190        const labelDirs = currentWatchDirs.length > 0 ? curren
    tWatchDirs.join(', ') : 'проекта';
1191        tbody.innerHTML = `<tr><td colspan="4" class="text-cen
    ter text-muted p-3">Ожидание изменений в папках <code>${escape
    Html(labelDirs)}</code>...</td></tr>`;
1192        return;
1193      }
1194
1195      tbody.innerHTML = events.map((e, idx) => {
1196        const rootDirHint = e.watch_dir ? (e.watch_dir.split('
    \\').pop() || e.watch_dir) : '';
1197        const procDisplay = e.process_name
1198          ? `<span class="badge bg-dark border border-secondar
    y text-info font-monospace text-truncate d-inline-block" style
    ="max-width: 165px; font-size: 0.72rem;" title="Программа: ${e
    scapeHtml(e.process_name)}${e.process_id ? ` (PID: ${e.process
    _id})` : ''}"><i class="bi bi-cpu me-1"></i>${escapeHtml(e.pro
    cess_name)}${e.process_id ? ` [${e.process_id}]` : ''}</span>`
1199          : `<span class="text-muted" style="font-size: 0.72re
    m;">—</span>`;
1200        return `
1201          <tr class="proc-watcher-row interactive-table-row" d
    ata-idx="${idx}" style="cursor: pointer;" title="Нажмите для д
    етальной диагностики события">
1202            <td class="font-monospace text-muted small">${e.ti
    mestamp?.slice(11, 19) || ''}</td>
1203            <td>
1204              <span class="badge ${e.is_deletion ? 'bg-danger'
```

```
 : (e.action === 'Created' ? 'bg-success' : 'bg-secondary')}">
${escapeHtml(e.action)}</span>
1205              ${rootDirHint && currentWatchDirs.length > 1 ? `
<span class="badge bg-dark border border-secondary text-muted
ms-1" style="font-size: 0.65rem;" title="Корень: ${escapeHtml(
e.watch_dir)}">${escapeHtml(rootDirHint)}</span>` : ''}
1206            </td>
1207            <td>${procDisplay}</td>
1208            <td class="font-monospace small" style="word-break
: break-all; color: var(--text-color);" title="${escapeHtml(e.
path)}">${escapeHtml(e.path)}</td>
1209          </tr>
1210        `;
1211      }).join('');
1212
1213      tbody.querySelectorAll('.proc-watcher-row').forEach(row
=> {
1214        row.onclick = () => {
1215          const idx = parseInt(row.getAttribute('data-idx'), 1
0);
1216          const e = events[idx];
1217          if (!e) return;
1218
1219          const modal = window.AIModalDialog || window.AITable
Modal;
1220          if (modal) {
1221            modal.show({
1222              icon: '□ ',
1223              title: `Файловое событие: ${e.action}`,
1224              subtitle: `${e.path} | ${e.timestamp}`,
1225              panelId: 'panel-processes-load-inspector-5',
1226              tableType: 'file_event',
1227              badges: [
1228                { text: e.action, class: e.is_deletion ? 'badg
e bg-danger' : (e.action === 'Created' ? 'badge bg-success' :
'badge bg-info text-dark') },
1229                { text: e.process_name ? `Программа: ${e.proce
ss_name}` : 'WinAPI ReadDirectoryChangesW', class: 'badge bg-d
```

```
ark border border-secondary text-info' },
                { text: 'WinAPI', class: 'badge bg-secondary'
}
              ],
              metadata: [
                { label: 'Действие', value: e.action },
                { label: 'Полный путь к файлу', value: e.path
},
                { label: 'Программа / Процесс', value: e.proce
ss_name ? `${e.process_name}${e.process_id ? ` (PID: ${e.proce
ss_id})` : ''}` : 'Фоновый процесс / завершен' },
                { label: 'Время события', value: e.timestamp }
,
                { label: 'Признак удаления', value: e.is_delet
ion ? 'Да (Файл удален/переименован)' : 'Нет' },
                { label: 'Папка события', value: e.watch_dir |
| currentWatchDir || '-' },
                { label: 'Все отслеживаемые папки', value: cur
rentWatchDirs.join('; ') || '-' }
              ],
              rawTitle: 'Детали события WinAPI & Process Info'
,
              rawContent: JSON.stringify(e, null, 2),
              requestData: e
            });
          }
        };
      });
    } catch (e) {
      console.warn('[ProcessActivity] Failed to fetch live fil
e events:', e);
    }
  }

  async function handleWatchExclusionsClick() {
    try {
      const res = await fetch('/api/v1/system/file-audit/exclu
sions');
```

```
      if (!res.ok) throw new Error('HTTP ' + res.status);
      const data = await res.json();
      const ex = data.exclusions || data;
      const modal = window.AIModalDialog || window.AITableModal;
      if (modal) {
        modal.show({
          title: 'Исключения файлового мониторинга (WinAPI)',
          subtitle: `Статус фильтрации: ${data.enabled !== false ? 'ВКЛЮЧЕНА' : 'ВЫКЛЮЧЕНА'}`,
          icon: '□ ',
          panelId: 'panel-processes-load-inspector-5',
          tableType: 'generic',
          badges: [
            { text: data.enabled !== false ? 'Фильтрация активна' : 'Фильтрация выключена', class: data.enabled !== false ? 'badge bg-success' : 'badge bg-secondary' }
          ],
          metadata: [
            { label: 'Игнорируемые расширения', value: (ex.extensions || []).join(', ') || 'нет' },
            { label: 'Игнорируемые каталоги', value: (ex.directories || []).join(', ') || 'нет' },
            { label: 'Игнорируемые процессы', value: (ex.processes || []).join(', ') || 'нет' }
          ],
          rawTitle: 'Правила фильтрации WinAPI Watcher',
          rawContent: JSON.stringify(data, null, 2)
        });
      } else {
        alert(`Исключения файлового мониторинга:\nРасширения: ${(ex.extensions || []).join(', ')}\nКаталоги: ${(ex.directories || []).join(', ')}`);
      }
    } catch (e) {
      showToast(`Ошибка загрузки исключений: ${e.message}`, 'danger');
    }
```

```
  }

  async function handleChangeWatchDirClick() {
    const newDir = prompt('Введите абсолютный путь к папке для отслеживания (или несколько через точку с запятой):', currentWatchDirs.join('; '));
    if (!newDir || !newDir.trim()) return;
    const dirsList = newDir.split(';').map(s => s.trim()).filter(Boolean);
    try {
      const res = await fetch('/api/v1/system/file-audit/watch-dirs', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ paths: dirsList })
      });
      if (!res.ok) throw new Error('HTTP ' + res.status);
      const data = await res.json();
      currentWatchDirs = data.watch_dirs || dirsList;
      showToast(`Отслеживаемые папки обновлены (${currentWatchDirs.length})`, 'success');
      await fetchLiveFileEvents();
    } catch (e) {
      showToast(`Ошибка обновления папок: ${e.message}`, 'danger');
    }
  }

  function initTableResizers() {
    const resizer = document.getElementById('sys-proc-table-resizer');
    const container = document.getElementById('sys-proc-table-container');
    if (!resizer || !container) return;

    let startY = 0;
    let startHeight = 0;

```

```
    const onMouseMove = (e) => {
      const delta = e.clientY - startY;
      const newH = Math.max(140, Math.min(850, startHeight + delta));
      container.style.height = `${newH}px`;
    };

    const onMouseUp = () => {
      resizer.classList.remove('resizing');
      window.removeEventListener('mousemove', onMouseMove);
      window.removeEventListener('mouseup', onMouseUp);
    };

    resizer.addEventListener('mousedown', (e) => {
      startY = e.clientY;
      startHeight = container.getBoundingClientRect().height;
      resizer.classList.add('resizing');
      window.addEventListener('mousemove', onMouseMove);
      window.addEventListener('mouseup', onMouseUp);
    });
  }

  function switchToDeepDiveMode() {
    currentMode = 'deep-dive';
    document.getElementById('section-deep-dive')?.classList.remove('d-none');
    document.getElementById('section-host-overview')?.classList.add('d-none');
    document.getElementById('btn-mode-deep-dive')?.classList.add('active', 'btn-outline-primary');
    document.getElementById('btn-mode-deep-dive')?.classList.remove('btn-outline-secondary');
    document.getElementById('btn-mode-host-overview')?.classList.remove('active', 'btn-outline-primary');
    document.getElementById('btn-mode-host-overview')?.classList.add('btn-outline-secondary');
  }

```

```
1345  function switchToHostOverviewMode() {
1346    currentMode = 'host-overview';
1347    document.getElementById('section-deep-dive')?.classList.ad
d('d-none');
1348    document.getElementById('section-host-overview')?.classLis
t.remove('d-none');
1349    document.getElementById('btn-mode-host-overview')?.classLi
st.add('active', 'btn-outline-primary');
1350    document.getElementById('btn-mode-host-overview')?.classLi
st.remove('btn-outline-secondary');
1351    document.getElementById('btn-mode-deep-dive')?.classList.r
emove('active', 'btn-outline-primary');
1352    document.getElementById('btn-mode-deep-dive')?.classList.a
dd('btn-outline-secondary');
1353    fetchHostOverview();
1354  }
1355
1356  // ----------------------------------------------------------
------------------
1357  // Таймеры автообновления и События UI
1358  // ----------------------------------------------------------
------------------
1359
1360  function setupEventListeners() {
1361    // Режимы (Deep Dive VS Host Overview)
1362    document.getElementById('btn-mode-deep-dive')?.addEventLis
tener('click', switchToDeepDiveMode);
1363    document.getElementById('btn-mode-host-overview')?.addEven
tListener('click', switchToHostOverviewMode);
1364
1365    // Активные / История
1366    document.getElementById('btn-filter-inst-active')?.addEven
tListener('click', () => {
1367      currentInstanceFilter = 'active';
1368      document.getElementById('btn-filter-inst-active').classL
ist.add('active', 'btn-outline-success');
1369      document.getElementById('btn-filter-inst-active').classL
ist.remove('btn-outline-secondary');
```

```
      document.getElementById('btn-filter-inst-history').classList.remove('active', 'btn-outline-success');
      document.getElementById('btn-filter-inst-history').classList.add('btn-outline-secondary');
      loadInstancesList();
    });

    document.getElementById('btn-filter-inst-history')?.addEventListener('click', () => {
      currentInstanceFilter = 'history';
      document.getElementById('btn-filter-inst-history').classList.add('active', 'btn-outline-success');
      document.getElementById('btn-filter-inst-history').classList.remove('btn-outline-secondary');
      document.getElementById('btn-filter-inst-active').classList.remove('active', 'btn-outline-success');
      document.getElementById('btn-filter-inst-active').classList.add('btn-outline-secondary');
      loadInstancesList();
    });

    // Селектор выпадающего списка
    document.getElementById('deep-dive-instance-select')?.addEventListener('change', (e) => {
      selectInstance(e.target.value);
    });

    // Поиск Deep Dive
    document.getElementById('deep-dive-search')?.addEventListener('input', () => {
      loadInstancesList();
    });

    // Фильтр файлов инстанса
    document.getElementById('inst-file-search')?.addEventListener('input', () => {
      loadInstanceFileActivity();
    });
```

```
1398
1399      // Кнопка глобальной синхронизации
1400      document.getElementById('btn-global-refresh')?.addEventLis
tener('click', async () => {
1401        showToast('Синхронизация процессов с ОС...', 'info');
1402        await loadInstancesList();
1403        if (selectedInstanceId) {
1404          await selectInstance(selectedInstanceId);
1405        }
1406        if (currentMode === 'host-overview') {
1407          await fetchHostOverview();
1408        }
1409        showToast('Синхронизация завершена', 'success');
1410      });
1411
1412      // Интервалы чартов
1413      document.querySelectorAll('.chart-interval-btn').forEach(b
tn => {
1414        btn.addEventListener('click', () => {
1415          document.querySelectorAll('.chart-interval-btn').forEa
ch(b => {
1416            b.classList.remove('active', 'btn-outline-primary');
1417            b.classList.add('btn-outline-secondary');
1418          });
1419          btn.classList.add('active', 'btn-outline-primary');
1420          btn.classList.remove('btn-outline-secondary');
1421          currentChartInterval = btn.getAttribute('data-interval
') || '1m';
1422          loadInstanceSamples();
1423        });
1424      });
1425
1426      // Сетевая активность (фильтры, поиск, пауза, обновление)
1427      document.getElementById('btn-proc-net-filter-internet')?.a
ddEventListener('click', () => {
1428        currentNetFilter = 'internet';
1429        document.getElementById('btn-proc-net-filter-internet')?
.classList.add('active', 'btn-outline-primary');
```

```
1430        document.getElementById('btn-proc-net-filter-internet')?
.classList.remove('btn-outline-secondary');
1431        document.getElementById('btn-proc-net-filter-all')?.clas
sList.remove('active', 'btn-outline-primary');
1432        document.getElementById('btn-proc-net-filter-all')?.clas
sList.add('btn-outline-secondary');
1433        document.getElementById('btn-proc-net-filter-listen')?.c
lassList.remove('active', 'btn-outline-primary');
1434        document.getElementById('btn-proc-net-filter-listen')?.c
lassList.add('btn-outline-secondary');
1435        renderNetworkActivityTable();
1436      });
1437
1438      document.getElementById('btn-proc-net-filter-all')?.addEve
ntListener('click', () => {
1439        currentNetFilter = 'all';
1440        document.getElementById('btn-proc-net-filter-all')?.clas
sList.add('active', 'btn-outline-primary');
1441        document.getElementById('btn-proc-net-filter-all')?.clas
sList.remove('btn-outline-secondary');
1442        document.getElementById('btn-proc-net-filter-internet')?
.classList.remove('active', 'btn-outline-primary');
1443        document.getElementById('btn-proc-net-filter-internet')?
.classList.add('btn-outline-secondary');
1444        document.getElementById('btn-proc-net-filter-listen')?.c
lassList.remove('active', 'btn-outline-primary');
1445        document.getElementById('btn-proc-net-filter-listen')?.c
lassList.add('btn-outline-secondary');
1446        renderNetworkActivityTable();
1447      });
1448
1449      document.getElementById('btn-proc-net-filter-listen')?.add
EventListener('click', () => {
1450        currentNetFilter = 'listen';
1451        document.getElementById('btn-proc-net-filter-listen')?.c
lassList.add('active', 'btn-outline-primary');
1452        document.getElementById('btn-proc-net-filter-listen')?.c
lassList.remove('btn-outline-secondary');
```

```
1453      document.getElementById('btn-proc-net-filter-internet')?
.classList.remove('active', 'btn-outline-primary');
1454      document.getElementById('btn-proc-net-filter-internet')?
.classList.add('btn-outline-secondary');
1455      document.getElementById('btn-proc-net-filter-all')?.clas
sList.remove('active', 'btn-outline-primary');
1456      document.getElementById('btn-proc-net-filter-all')?.clas
sList.add('btn-outline-secondary');
1457      renderNetworkActivityTable();
1458    });
1459
1460    document.getElementById('proc-net-search')?.addEventListen
er('input', () => {
1461      renderNetworkActivityTable();
1462    });
1463
1464    document.getElementById('btn-proc-pause-net')?.addEventLis
tener('click', (e) => {
1465      isNetPaused = !isNetPaused;
1466      e.target.textContent = isNetPaused ? 'Продолжить' : 'Пау
за';
1467      e.target.className = isNetPaused ? 'btn btn-xs btn-warni
ng rounded-pill px-2.5 py-0.5' : 'btn btn-xs btn-outline-secon
dary rounded-pill px-2.5 py-0.5';
1468      if (!isNetPaused) fetchNetworkActivity();
1469    });
1470
1471    document.getElementById('btn-proc-refresh-net')?.addEventL
istener('click', async () => {
1472      await fetchNetworkActivity();
1473      showToast('Сетевая активность обновлена', 'info');
1474    });
1475
1476    // Мониторинг файлов (исключения, смена папки)
1477    document.getElementById('btn-proc-watch-exclusions')?.addE
ventListener('click', handleWatchExclusionsClick);
1478    document.getElementById('btn-proc-change-watch-dir')?.addE
ventListener('click', handleChangeWatchDirClick);
```

```
    document.getElementById('proc-watch-dir-badge')?.addEventL
istener('click', handleChangeWatchDirClick);

    // Топ процессов хоста (поиск, пауза, обновление)
    document.getElementById('sys-proc-search')?.addEventListen
er('input', () => {
      renderHostProcessesTable();
    });

    document.getElementById('btn-sys-pause-proc')?.addEventLis
tener('click', (e) => {
      isProcPaused = !isProcPaused;
      e.target.textContent = isProcPaused ? 'Продолжить' : 'Па
уза';
      e.target.className = isProcPaused ? 'btn btn-xs btn-warn
ing rounded-pill px-2.5 py-0.5' : 'btn btn-xs btn-outline-seco
ndary rounded-pill px-2.5 py-0.5';
      if (!isProcPaused) fetchHostProcesses();
    });

    document.getElementById('btn-sys-refresh-proc')?.addEventL
istener('click', async () => {
      await fetchHostProcesses();
      showToast('Список процессов обновлен', 'info');
    });

    initSafeOps();
    initTableResizers();
  }

  function stopPolling() {
    if (deepDiveTimer) {
      clearInterval(deepDiveTimer);
      deepDiveTimer = null;
    }
    if (hostOverviewTimer) {
      clearInterval(hostOverviewTimer);
      hostOverviewTimer = null;
```

```
1510      }
1511      if (window.unregisterTabPoller) {
1512        window.unregisterTabPoller('tab-processes-load-inspector
    _host');
1513        window.unregisterTabPoller('tab-processes-load-inspector
    _deepdive');
1514      }
1515    }
1516
1517    function startPolling() {
1518      stopPolling();
1519      if (!isProcessesTabActive()) return;
1520
1521      if (window.registerTabPoller) {
1522        // Регистрация в системном диспетчере вкладок tab-core.j
    s (активны только при открытой вкладке)
1523        window.registerTabPoller('tab-processes-load-inspector',
     async () => {
1524          if (!isProcessesTabActive()) return;
1525          if (currentMode === 'host-overview') {
1526            await fetchHostOverview();
1527          } else {
1528            await loadInstancesList();
1529          }
1530        }, 3000, { pollerId: 'tab-processes-load-inspector_host'
    , immediate: false });
1531
1532        window.registerTabPoller('tab-processes-load-inspector',
     async () => {
1533          if (!isProcessesTabActive() || isDeepDivePaused || cur
    rentMode !== 'deep-dive') return;
1534          if (selectedInstanceId && selectedInstanceData && sele
    ctedInstanceData.status === 'RUNNING') {
1535            await loadInstanceSamples();
1536          }
1537        }, 1000, { pollerId: 'tab-processes-load-inspector_deepd
    ive', immediate: false });
1538      } else {
```

```
      // Резервный таймер для среды без tab-core.js с самоконтролем активности
      hostOverviewTimer = setInterval(async () => {
        if (!isProcessesTabActive()) {
          stopPolling();
          return;
        }
        if (currentMode === 'host-overview') {
          await fetchHostOverview();
        } else {
          await loadInstancesList();
        }
      }, 3000);

      deepDiveTimer = setInterval(async () => {
        if (!isProcessesTabActive()) {
          stopPolling();
          return;
        }
        if (isDeepDivePaused || currentMode !== 'deep-dive') return;
        if (selectedInstanceId && selectedInstanceData && selectedInstanceData.status === 'RUNNING') {
          await loadInstanceSamples();
        }
      }, 1000);
    }
  }

  async function activateTab() {
    if (!isProcessesTabActive()) return;
    startPolling();
    if (currentMode === 'host-overview') {
      await fetchHostOverview();
    } else {
      await loadInstancesList();
      if (selectedInstanceId && selectedInstanceData && selectedInstanceData.status === 'RUNNING') {
```

```
        await loadInstanceSamples();
      }
    }
  }

  function deactivateTab() {
    stopPolling();
  }

  // ------------------------------------------------------------
------------------
  // Точка входа модуля и жизненный цикл
  // ------------------------------------------------------------
------------------

  let isInitialized = false;

  async function init() {
    if (!isInitialized) {
      initCharts();
      setupEventListeners();
      isInitialized = true;
    }
    // Запускаем сетевые запросы и поллинг ТОЛЬКО если вкладка
 действительно активна
    if (isProcessesTabActive()) {
      await activateTab();
    } else {
      stopPolling();
    }
  }

  // Экспорт хуков жизненного цикла
  window.initProcessesLoadInspectorTab = init;
  window.activateProcessesLoadInspectorTab = activateTab;
  window.deactivateProcessesLoadInspectorTab = deactivateTab;

  // Реакция на события переключения вкладок и видимости докум
```

```
ента
  document.addEventListener('tab:activated', (e) => {
    if (e.detail?.tabId === TAB_ID || e.detail?.tabId === 'pro
cesses-load-inspector') {
      activateTab();
    }
  });

  document.addEventListener('tab:deactivated', (e) => {
    if (e.detail?.tabId === TAB_ID || e.detail?.tabId === 'pro
cesses-load-inspector') {
      deactivateTab();
    }
  });

  document.addEventListener('visibilitychange', () => {
    if (document.hidden) {
      deactivateTab();
    } else if (isProcessesTabActive()) {
      activateTab();
    }
  });

  if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', init);
  } else {
    init();
  }
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/process_leaks_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/process_leaks_tab/index
.html" class="w-100 h-100 border-0"></iframe>

 * File: index.html
 * Project: ai-windows-admin
 * Package: windows/api/webgui/process_leaks_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ================================================================
==================
-->
<style>
  .leaks-tab-wrapper {
    padding: 0.75rem;
    color: var(--text-color);
  }
  .diag-meter-card {
    background: var(--surface-1);
    border: 1px solid var(--border-color);
    border-radius: var(--radius-md, 8px);
    padding: 0.75rem 1rem;
```

```
}
.diag-badge-critical {
  background: rgba(239, 68, 68, 0.2);
  color: #ef4444;
  border: 1px solid rgba(239, 68, 68, 0.4);
  font-size: 0.70rem;
  padding: 2px 6px;
  border-radius: 4px;
  font-weight: 600;
}
.diag-badge-warning {
  background: rgba(234, 179, 8, 0.2);
  color: #d97706;
  border: 1px solid rgba(234, 179, 8, 0.4);
  font-size: 0.70rem;
  padding: 2px 6px;
  border-radius: 4px;
  font-weight: 600;
}
.diag-badge-normal {
  background: rgba(34, 197, 94, 0.15);
  color: #16a34a;
  border: 1px solid rgba(34, 197, 94, 0.3);
  font-size: 0.70rem;
  padding: 2px 6px;
  border-radius: 4px;
  font-weight: 500;
}
.diag-panel {
  background: var(--surface-1);
  border: 1px solid var(--border-color);
  border-radius: 8px;
  overflow: hidden;
}
.diag-panel-header {
  background: var(--surface-2);
  padding: 8px 14px;
  border-bottom: 1px solid var(--border-color);
```

```
    font-weight: 600;
    font-size: 0.85rem;
    color: var(--nav-active);
  }
</style>
<div class="leaks-tab-wrapper">
<!-- Top Header & Status -->
<div class="card shadow-sm border-secondary-subtle mb-3" data-panel-id="panel-process-leaks-1-utechki-resursov-i-mikrofrizy" id="panel-process-leaks-1-utechki-resursov-i-mikrofrizy">
<div class="card-body p-3">
<div class="d-flex flex-wrap align-items-center justify-content-between gap-2">
<div class="d-flex align-items-center gap-2">
<div class="fs-4">□ </div>
<div>
<h5 class="mb-0 fw-bold">1. Утечки ресурсов и микрофризы (Leak Hunter)</h5>
<small class="text-muted">Глубокий Win32 аудит: дескрипторы Handles, GDI/USER объекты GUI, Hard Page Faults и риски утечек</small>
</div>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<select class="form-select form-select-sm poll-freq-select" data-poll-id="process_leaks" id="diag-leaks-poll-freq" title="Частота опроса: Утечки процессов">
<option value="start">Только при старте</option>
<option selected="" value="manual">По запросу</option>
<option value="3">Каждые 3 сек</option>
<option value="5">Каждые 5 сек</option>
<option value="10">Каждые 10 сек</option>
<option value="30">Каждые 30 сек</option>
<option value="60">Каждые 60 сек</option>
<option value="120">Каждые 120 сек</option>
</select>
<button class="btn btn-sm btn-outline-info rounded-pill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id="btn-diag-
```

```
leaks-refresh">
<i class="bi bi-arrow-clockwise"></i> Обновить
          </button>
</div>
</div>
</div>
</div>
<!-- Top Leak Hunter KPI Metrics -->
<div class="row g-2 mb-3">
<div class="col-12 col-sm-6 col-lg-3">
<div class="diag-meter-card">
<div class="d-flex justify-content-between align-items-center
small text-muted">
<span>Проанализировано процессов</span>
<i class="bi bi-cpu text-info"></i>
</div>
<div class="fs-4 fw-bold text-info mt-1" id="diag-leaks-total-
proc">0</div>
<div class="small text-muted">Полнотекстовый срез Win32 API</d
iv>
</div>
</div>
<div class="col-12 col-sm-6 col-lg-3">
<div class="diag-meter-card">
<div class="d-flex justify-content-between align-items-center
small text-muted">
<span>С признаками утечек</span>
<i class="bi bi-exclamation-triangle text-danger"></i>
</div>
<div class="fs-4 fw-bold text-danger mt-1" id="diag-leaks-susp
icious-count">0</div>
<div class="small text-muted">Handles / GDI / Page Faults</div
>
</div>
</div>
<div class="col-12 col-sm-6 col-lg-3">
<div class="diag-meter-card">
<div class="d-flex justify-content-between align-items-center
```

```
small text-muted">
<span>Пик дескрипторов (Handles)</span>
<i class="bi bi-box text-warning"></i>
</div>
<div class="fs-4 fw-bold text-warning mt-1" id="diag-leaks-peak-handles">--</div>
<div class="small text-muted text-truncate" id="diag-leaks-peak-handles-proc">Процесс: --</div>
</div>
</div>
<div class="col-12 col-sm-6 col-lg-3">
<div class="diag-meter-card">
<div class="d-flex justify-content-between align-items-center small text-muted">
<span>Пик GDI объектов</span>
<i class="bi bi-palette text-primary"></i>
</div>
<div class="fs-4 fw-bold text-primary mt-1" id="diag-leaks-peak-gdi">--</div>
<div class="small text-muted text-truncate" id="diag-leaks-peak-gdi-proc">Лимит OS: 10 000</div>
</div>
</div>
</div>
<!-- Main Process Leak Hunter Table -->
<div class="diag-panel" data-panel-id="panel-process-leaks-1" id="panel-process-leaks-1">
<div class="diag-panel-header d-flex justify-content-between align-items-center flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-shield-shaded text-warning"></i>
<span class="text-body">Детектор утечек дескрипторов, GDI и страничных ошибок (Leak Hunter)</span>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<div class="btn-group btn-group-sm" id="diag-leaks-filter-group" role="group">
<button class="btn btn-outline-info active py-0.5 px-2" data-f
```

```
ilter="all" style="font-size: 0.72rem;" type="button">Все</but
ton>
<button class="btn btn-outline-danger py-0.5 px-2" data-filter
="suspicious" style="font-size: 0.72rem;" type="button">⚠ Подо
зрительные</button>
<button class="btn btn-outline-secondary py-0.5 px-2" data-fil
ter="gdi" style="font-size: 0.72rem;" type="button">GDI &gt; 5
00</button>
<button class="btn btn-outline-secondary py-0.5 px-2" data-fil
ter="handles" style="font-size: 0.72rem;" type="button">Handle
s &gt; 1500</button>
</div>
<input class="form-control form-control-sm" id="diag-leaks-sea
rch" placeholder="Поиск по PID или имени..." style="max-width:
 180px; font-size: 0.74rem; height: 26px;" type="text"/>
</div>
</div>
<div class="table-responsive" style="max-height: 540px; overfl
ow-y: auto;">
<table class="table table-hover table-sm mb-0" style="font-siz
e: 0.78rem; font-family: ui-monospace, SFMono-Regular, Menlo,
Monaco, Consolas, monospace;">
<thead>
<tr>
<th style="width: 70px;">PID</th>
<th>Имя процесса</th>
<th style="width: 90px; text-align: right;">Handles</th>
<th style="width: 90px; text-align: right;">GDI Объекты</th>
<th style="width: 90px; text-align: right;">USER Объекты</th>
<th style="width: 120px; text-align: right;">Page Faults</th>
<th style="width: 90px; text-align: right;">Память (RSS)</th>
<th style="width: 70px; text-align: right;">Потоки</th>
<th style="width: 100px;">Статус риска</th>
<th>Признаки / Аномалии</th>
</tr>
</thead>
<tbody id="diag-leaks-tbody">
<tr><td class="text-center py-4 text-muted" colspan="10">Загру
```

```
зка данных глубокой диагностики процессов...</td></tr>
</tbody>
</table>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Process Leaks Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/process_leaks_tab/main
.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/process_leaks_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

(function () {
  'use strict';

  let rawLeakProcesses = [];
  let leakFilter = 'all';
  let autoRefreshTimer = null;
  const POLL_ID = 'process_leaks';

  function escapeHtml(str) {
    if (str === null || str === undefined) return '';
    return String(str)
      .replace(/&/g, '&amp;')
```

```
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
  }

  function setText(id, text) {
    const el = document.getElementById(id);
    if (el) el.textContent = text !== null && text !== undefin
ed ? String(text) : '--';
  }

  async function apiFetch(url) {
    if (window.api && typeof window.api.fetch === 'function')
{
      return await window.api.fetch(url);
    }
    const res = await fetch(url);
    if (!res.ok) throw new Error(`HTTP ${res.status}: ${res.st
atusText}`);
    return await res.json();
  }

  async function fetchProcessLeaks() {
    try {
      const data = await apiFetch('/api/v1/system/diagnostics/
leaks?limit=100');
      if (!data) return;

      setText('diag-leaks-total-proc', data.total_processes);
      setText('diag-leaks-suspicious-count', data.suspicious_c
ount);

      if (data.top_handle_hogs && data.top_handle_hogs.length
> 0) {
        const topH = data.top_handle_hogs[0];
        setText('diag-leaks-peak-handles', `${topH.handles_cou
nt.toLocaleString()} шт.`);
```

```
        setText('diag-leaks-peak-handles-proc', `Процесс: ${topH.name} (PID ${topH.pid})`);
      }
      if (data.top_gdi_hogs && data.top_gdi_hogs.length > 0) {
        const topG = data.top_gdi_hogs[0];
        setText('diag-leaks-peak-gdi', `${topG.gdi_objects.toLocaleString()} шт.`);
        setText('diag-leaks-peak-gdi-proc', `Процесс: ${topG.name} (PID ${topG.pid})`);
      }

      rawLeakProcesses = data.all_processes || [];
      renderProcessLeaksTable(rawLeakProcesses);
    } catch (err) {
      console.warn('[ProcessLeaksTab] fetchProcessLeaks error:', err);
    }
  }

  function renderProcessLeaksTable(processes) {
    const tbody = document.getElementById('diag-leaks-tbody');
    if (!tbody) return;

    const query = (document.getElementById('diag-leaks-search')?.value || '').trim().toLowerCase();

    let filtered = processes.filter(p => {
      if (query && !p.name.toLowerCase().includes(query) && !String(p.pid).includes(query)) {
        return false;
      }
      if (leakFilter === 'suspicious') return p.leak_risk_score !== 'normal';
      if (leakFilter === 'gdi') return p.gdi_objects > 500;
      if (leakFilter === 'handles') return p.handles_count > 1500;
      return true;
    });
```

```
94
95      if (filtered.length === 0) {
96        tbody.innerHTML = '<tr><td colspan="10" class="text-cent
    er py-4 text-muted">Процессов по заданным критериям не найдено
    </td></tr>';
97        return;
98      }
99
100     tbody.innerHTML = filtered.map(p => {
101       let riskBadge = '<span class="diag-badge-normal">Норма</
    span>';
102       if (p.leak_risk_score === 'critical') {
103         riskBadge = '<span class="diag-badge-critical">Критиче
    ский</span>';
104       } else if (p.leak_risk_score === 'warning') {
105         riskBadge = '<span class="diag-badge-warning">Внимание
    </span>';
106       }
107
108       const reasons = (p.leak_risk_reasons && p.leak_risk_reas
    ons.length > 0)
109         ? p.leak_risk_reasons.map(r => `<span class="badge bg-
    dark border border-secondary text-warning me-1">${escapeHtml(r
    )}</span>`).join('')
110         : '<span class="text-muted small">Штатное поведение</s
    pan>';
111
112       return `
113         <tr>
114           <td><span class="text-info font-monospace">${p.pid}<
    /span></td>
115           <td class="fw-bold text-white text-truncate" style="
    max-width: 180px;">${escapeHtml(p.name)}</td>
116           <td style="text-align: right;" class="${p.handles_co
    unt > 1500 ? 'text-danger fw-bold' : 'text-light'}">${p.handle
    s_count.toLocaleString()}</td>
117           <td style="text-align: right;" class="${p.gdi_object
    s > 800 ? 'text-danger fw-bold' : 'text-light'}">${p.gdi_objec
```

```
ts.toLocaleString()}</td>
          <td style="text-align: right;" class="${p.user_objec
ts > 400 ? 'text-warning' : 'text-light'}">${p.user_objects.to
LocaleString()}</td>
          <td style="text-align: right;" class="text-muted fon
t-monospace">${p.page_faults_total.toLocaleString()}</td>
          <td style="text-align: right;" class="text-warning">
${p.memory_mb.toFixed(1)} MB</td>
          <td style="text-align: right;" class="text-muted">${
p.threads_count}</td>
          <td>${riskBadge}</td>
          <td>${reasons}</td>
        </tr>
      `;
    }).join('');
  }

  function getFrequency() {
    try {
      const saved = localStorage.getItem(`poll_freq_${POLL_ID}
`);
      if (saved) return saved;
    } catch (_) {}
    return 'manual';
  }

  function setFrequency(freq) {
    try {
      localStorage.setItem(`poll_freq_${POLL_ID}`, freq);
    } catch (_) {}
    applyPoller(freq, false);
  }

  function stopPolling() {
    if (window.unregisterTabPoller) {
      window.unregisterTabPoller(`tab-process-leaks_${POLL_ID}
`);
    }
```

```
148     if (autoRefreshTimer) {
149       clearInterval(autoRefreshTimer);
150   function applyPoller(freq, runInitial = false) {
151     stopPolling();
152     if (freq === 'start' || freq === 'manual') {
153       if (runInitial && (window.isTabActive ? window.isTabActi
ve('tab-process-leaks') : true)) {
154         fetchProcessLeaks();
155       }
156       return;
157     }
158 
159     const intervalSec = parseInt(freq, 10);
160     if (isNaN(intervalSec) || intervalSec <= 0) return;
161 
162     const intervalMs = intervalSec * 1000;
163     const pollerId = `tab-process-leaks_${POLL_ID}`;
164 
165     if (window.registerTabPoller) {
166       window.registerTabPoller('tab-process-leaks', fetchProce
ssLeaks, intervalMs, { pollerId, immediate: runInitial });
167     } else {
168       if (runInitial && (window.isTabActive ? window.isTabActi
ve('tab-process-leaks') : true)) {
169         fetchProcessLeaks();
170       }
171       autoRefreshTimer = setInterval(() => {
172         if (window.isTabActive ? window.isTabActive('tab-proce
ss-leaks') : true) {
173           fetchProcessLeaks();
174         }
175       }, intervalMs);
176     }
177   }
178 
179   function bindEvents() {
180     const leakFilters = document.getElementById('diag-leaks-fi
lter-group');
```

```
    if (leakFilters) {
      leakFilters.querySelectorAll('button').forEach(btn => {
        btn.onclick = () => {
          leakFilters.querySelectorAll('button').forEach(b =>
{
            b.classList.remove('active', 'btn-info', 'btn-dang
er');
            b.classList.add('btn-outline-secondary');
          });
          const f = btn.getAttribute('data-filter') || 'all';
          btn.classList.add('active', f === 'suspicious' ? 'bt
n-danger' : 'btn-info');
          btn.classList.remove('btn-outline-secondary');
          leakFilter = f;
          renderProcessLeaksTable(rawLeakProcesses);
        };
      });
    }

    const leakSearch = document.getElementById('diag-leaks-sea
rch');
    if (leakSearch) {
      leakSearch.oninput = () => renderProcessLeaksTable(rawLe
akProcesses);
    }

    const btnLeaksRefresh = document.getElementById('btn-diag-
leaks-refresh');
    if (btnLeaksRefresh) {
      btnLeaksRefresh.onclick = () => fetchProcessLeaks();
    }

    const select = document.getElementById('diag-leaks-poll-fr
eq');
    if (select) {
      select.onchange = (e) => {
        const newFreq = e.target.value;
        setFrequency(newFreq);
```

```
        if (newFreq !== 'manual' && newFreq !== 'start') {
          fetchProcessLeaks();
        }
        if (window.showToast) {
          const label = select.options[select.selectedIndex]?.
text || newFreq;
          window.showToast(`Частота опроса утечек: ${label}`,
'info');
        }
      };
    }
  }

  async function initProcessLeaksTab() {
    bindEvents();
    const currentFreq = getFrequency();
    const select = document.getElementById('diag-leaks-poll-fr
eq');
    if (select) select.value = currentFreq;
    applyPoller(currentFreq, true);
  }

  window.initProcessLeaksTab = initProcessLeaksTab;
  window.activateProcessLeaksTab = () => {
    applyPoller(getFrequency(), true);
  };

  if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', () => {
      if (window.isTabActive ? window.isTabActive('tab-process
-leaks') : false) {
        initProcessLeaksTab();
      } else {
        bindEvents();
      }
    });
  } else {
    if (window.isTabActive ? window.isTabActive('tab-process-l
```

```
eaks') : false) {
      initProcessLeaksTab();
    } else {
      bindEvents();
    }
  }
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Process Manager - Task Manager Index Vie
w
================================================================
===============
Description:
  Клиентский веб-интерфейс диспетчера задач и процессов Window
s с разделением
  на категории: Apps (Приложения с окнами), Background process
es (Фоновые процессы)
  и Windows processes (Системные процессы), группировкой подпр
оцессов и деревом окон.

Usage Examples:
    Web Route:
      /windows/api/webgui/process_manager_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/process_manager_tab/ind
ex.html?v=20261008_v1" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/process_manager_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<style>
  .pm-group-header {
    background: rgba(255, 255, 255, 0.05);
    cursor: pointer;
```

```
    font-weight: 600;
    transition: background 0.15s ease-in-out;
  }
  .pm-group-header:hover {
    background: rgba(255, 255, 255, 0.1) !important;
  }
  .pm-chevron {
    display: inline-block;
    transition: transform 0.2s ease;
    width: 14px;
    margin-right: 6px;
    color: var(--bs-secondary, #6c757d);
  }
  .pm-chevron.expanded {
    transform: rotate(90deg);
    color: var(--bs-primary, #0d6efd);
  }
  .pm-sub-row {
    background: rgba(0, 0, 0, 0.25);
    font-size: 0.82rem;
    border-left: 3px solid rgba(13, 110, 253, 0.4);
  }
  .pm-section-badge {
    font-size: 0.85rem;
    font-weight: 700;
    letter-spacing: 0.5px;
  }
  .pm-type-tag {
    font-size: 0.72rem;
    padding: 2px 6px;
    border-radius: 4px;
    background: rgba(255, 255, 255, 0.07);
    color: #a0aec0;
  }
  .pm-card-hover {
    transition: transform 0.15s ease, box-shadow 0.15s ease;
    cursor: pointer;
  }
```

```
  .pm-card-hover:hover {
    transform: translateY(-2px);
    box-shadow: 0 4px 12px rgba(0,0,0,0.3);
  }
</style>
<!-- Process Manager Tab -->
<div class="container-fluid py-2 px-3">
<!-- Top Stat Cards -->
<div class="row g-2 mb-3">
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3 pm-card-hover" data-panel-id="panel-process-manager-1" id="pm-filter-card-apps">
<div class="text-muted small text-uppercase fw-semibold d-flex justify-content-between align-items-center">
<span>Приложения (Apps)</span>
<i class="bi bi-window-stack text-primary"></i>
</div>
<div class="fs-4 fw-bold mt-1 text-primary" id="pm-apps-count">--</div>
<div class="small text-muted mt-1" id="pm-apps-sub">Оконные программы</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3 pm-card-hover" data-panel-id="panel-process-manager-2" id="pm-filter-card-bg">
<div class="text-muted small text-uppercase fw-semibold d-flex justify-content-between align-items-center">
<span>Фоновые (Background)</span>
<i class="bi bi-gear-wide-connected text-info"></i>
</div>
<div class="fs-4 fw-bold mt-1 text-info" id="pm-bg-count">--</div>
<div class="small text-muted mt-1" id="pm-bg-sub">Фоновые службы и демоны</div>
</div>
```

```
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3 pm-card-hover" data-panel-id="panel-process-manager-3" id="pm-filter-card-win">
<div class="text-muted small text-uppercase fw-semibold d-flex justify-content-between align-items-center">
<span>Windows процессы</span>
<i class="bi bi-microsoft text-warning"></i>
</div>
<div class="fs-4 fw-bold mt-1 text-warning" id="pm-win-count">--</div>
<div class="small text-muted mt-1" id="pm-win-sub">Системные процессы ОС</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3 pm-card-hover" data-panel-id="panel-process-manager-4" id="pm-filter-card-all">
<div class="text-muted small text-uppercase fw-semibold d-flex justify-content-between align-items-center">
<span>Всего процессов</span>
<i class="bi bi-cpu text-success"></i>
</div>
<div class="fs-4 fw-bold mt-1 text-success" id="pm-total-count">--</div>
<div class="small text-muted mt-1" id="pm-ram-usage">Память: --</div>
</div>
</div>
</div>
<!-- Action Bar & Filter Controls -->
<div class="d-flex align-items-center justify-content-between mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap-2">
<div class="d-flex align-items-center gap-2 flex-wrap">
<button class="btn btn-sm btn-primary d-flex align-items-cente
```

```
r gap-1" id="pm-refresh-btn">
<i class="bi bi-arrow-clockwise"></i>
<span>Обновить</span>
</button>
<!-- Category Filter Pills -->
<div class="btn-group btn-group-sm" id="pm-category-filter-gro
up" role="group">
<input autocomplete="off" checked="" class="btn-check" id="pm-
filter-all" name="pm-cat-filter" type="radio" value="all"/>
<label class="btn btn-outline-secondary" for="pm-filter-all">B
ce</label>
<input autocomplete="off" class="btn-check" id="pm-filter-apps
" name="pm-cat-filter" type="radio" value="app"/>
<label class="btn btn-outline-primary" for="pm-filter-apps">Ap
ps</label>
<input autocomplete="off" class="btn-check" id="pm-filter-bg"
name="pm-cat-filter" type="radio" value="background"/>
<label class="btn btn-outline-info" for="pm-filter-bg">Backgro
und</label>
<input autocomplete="off" class="btn-check" id="pm-filter-win-
radio" name="pm-cat-filter" type="radio" value="windows"/>
<label class="btn btn-outline-warning" for="pm-filter-win-radi
o">Windows</label>
</div>
<button class="btn btn-sm btn-outline-secondary d-flex align-i
tems-center gap-1" id="pm-toggle-expand-btn" title="Свернуть /
 Развернуть все группы">
<i class="bi bi-arrows-expand"></i>
<span id="pm-expand-btn-text">Развернуть все</span>
</button>
</div>
<div class="d-flex align-items-center gap-2">
<input class="form-control form-control-sm bg-dark text-white
border-secondary" id="pm-search-input" placeholder="Поиск прил
ожения, процесса или PID..." style="width: 280px;" type="text"
/>
</div>
</div>
```

```
<!-- Process Table (Task Manager Style) -->
<div class="card bg-dark text-white border-secondary shadow-sm" data-panel-id="panel-process-manager-dispetcher-protsessov-i-prilozhen" id="panel-process-manager-dispetcher-protsessov-i-prilozhen">
<div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">
<span class="fw-semibold"><i class="bi bi-layout-text-window-reverse text-primary me-2"></i>Диспетчер процессов и приложений Windows</span>
<span class="badge bg-secondary" id="pm-table-badge">0 процессов</span>
</div>
<div class="card-body p-0 table-responsive" style="max-height: 600px; overflow-y: auto;">
<table class="table table-dark table-hover align-middle mb-0 small" id="pm-table">
<thead class="sticky-top bg-dark border-bottom border-secondary">
<tr class="text-secondary">
<th style="min-width: 280px;">Имя процесса / Приложения</th>
<th style="width: 100px;">PID</th>
<th style="width: 140px;">Тип / Статус</th>
<th style="width: 100px;">CPU %</th>
<th style="width: 120px;">Память (RAM)</th>
<th class="text-end" style="width: 120px;">Действия</th>
</tr>
</thead>
<tbody id="pm-tbody">
<tr><td class="text-center text-muted py-4" colspan="6">Опрос активных приложений и процессов...</td></tr>
</tbody>
</table>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Process Manager Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт интерактивного диспетчера задач Windows
 (Task Manager)
 *   с поддержкой группировки Apps, Background processes и Win
dows processes.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/process_manager_tab/ma
in.js?v=20261008_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { initProcessManagerTab } from '/windows/api/web
gui/process_manager_tab/main.js';
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/process_manager_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

const registerTabPoller = window.registerTabPoller || function
() {};

let isInitialized = false;
let currentReport = null;
```

```
let currentCategoryFilter = 'all';
let expandedGroups = new Set();
let allExpanded = false;

/**
 * Загрузка категоризированных процессов с бэкенда
 */
export async function fetchProcessManagerSummary() {
  try {
    const res = await fetch('/api/process-manager/categorized');
    if (!res.ok) {
      // Fallback на /summary
      const fallbackRes = await fetch('/api/process-manager/summary');
      if (!fallbackRes.ok) throw new Error(`HTTP ${res.status}`);
      currentReport = await fallbackRes.json();
    } else {
      currentReport = await res.json();
    }
    renderProcessManager(currentReport);
  } catch (err) {
    console.warn('[ProcessManager] Ошибка получения процессов:', err);
  }
}

/**
 * Рендеринг карточек статистики и таблицы процессов
 */
function renderProcessManager(report) {
  if (!report) return;

  const appsList = report.apps || [];
  const bgList = report.background_processes || [];
  const winList = report.windows_processes || [];

```

```
  const appsCountEl = document.getElementById('pm-apps-count')
;
  const bgCountEl = document.getElementById('pm-bg-count');
  const winCountEl = document.getElementById('pm-win-count');
  const totalCountEl = document.getElementById('pm-total-count
');
  const ramUsageEl = document.getElementById('pm-ram-usage');

  const totalProcs = report.total_processes || (appsList.lengt
h + bgList.length + winList.length);
  const totalMemMb = report.total_memory_used_mb || 0;

  if (appsCountEl) appsCountEl.textContent = report.apps_count
 !== undefined ? report.apps_count : appsList.length;
  if (bgCountEl) bgCountEl.textContent = report.background_cou
nt !== undefined ? report.background_count : bgList.length;
  if (winCountEl) winCountEl.textContent = report.windows_coun
t !== undefined ? report.windows_count : winList.length;
  if (totalCountEl) totalCountEl.textContent = totalProcs;

  if (ramUsageEl) {
    const memStr = totalMemMb >= 1024 ? `${(totalMemMb / 1024)
.toFixed(1)} GB` : `${Math.round(totalMemMb)} MB`;
    ramUsageEl.textContent = `Память: ${memStr}`;
  }

  applyProcessFilters();
}

/**
 * Применение фильтров поиска и категорий
 */
function applyProcessFilters() {
  if (!currentReport) return;

  const tbody = document.getElementById('pm-tbody');
  const searchInput = document.getElementById('pm-search-input
');
```

```
  const badge = document.getElementById('pm-table-badge');

  if (!tbody) return;

  const query = (searchInput ? searchInput.value : '').toLowerCase().trim();

  const apps = currentReport.apps || [];
  const bg = currentReport.background_processes || [];
  const win = currentReport.windows_processes || [];

  // Фильтрация групп по строке поиска
  function filterGroup(group) {
    if (!query) return true;
    const nameMatch = (group.name || '').toLowerCase().includes(query);
    const friendlyMatch = (group.friendly_name || '').toLowerCase().includes(query);
    const pidMatch = String(group.main_pid || '').includes(query);
    const titlesMatch = (group.window_titles || []).some(t => t.toLowerCase().includes(query));
    const subMatch = (group.subprocesses || []).some(sub =>
      (sub.name || '').toLowerCase().includes(query) ||
      String(sub.pid || '').includes(query) ||
      (sub.username || '').toLowerCase().includes(query)
    );
    return nameMatch || friendlyMatch || pidMatch || titlesMatch || subMatch;
  }

  const filteredApps = (currentCategoryFilter === 'all' || currentCategoryFilter === 'app') ? apps.filter(filterGroup) : [];
  const filteredBg = (currentCategoryFilter === 'all' || currentCategoryFilter === 'background') ? bg.filter(filterGroup) : [];
  const filteredWin = (currentCategoryFilter === 'all' || currentCategoryFilter === 'windows') ? win.filter(filterGroup) : [
```

```
];

  const totalShown = filteredApps.length + filteredBg.length +
 filteredWin.length;
  if (badge) {
    badge.textContent = `${totalShown} групп (${currentReport.
total_processes || totalShown} процессов)`;
  }

  if (totalShown === 0) {
    tbody.innerHTML = '<tr><td colspan="6" class="text-center
text-muted py-4">Процессы не найдены</td></tr>';
    return;
  }

  let html = '';

  // Секция 1: Apps (Интерактивные приложения с окнами)
  if (filteredApps.length > 0) {
    html += `
      <tr class="table-active">
        <td colspan="6" class="py-2 px-3 fw-bold text-primary"
>
          <i class="bi bi-window-stack me-1.5"></i> Apps (${fi
lteredApps.length})
        </td>
      </tr>
    `;
    html += filteredApps.map(g => renderGroupRow(g, 'app', 'Ap
p')).join('');
  }

  // Секция 2: Background processes (Фоновые процессы)
  if (filteredBg.length > 0) {
    html += `
      <tr class="table-active">
        <td colspan="6" class="py-2 px-3 fw-bold text-info">
          <i class="bi bi-gear-wide-connected me-1.5"></i> Bac
```

```
kground processes (${filteredBg.length})
        </td>
      </tr>
    `;
    html += filteredBg.map(g => renderGroupRow(g, 'background', 'Background process')).join('');
  }

  // Секция 3: Windows processes (Процессы Windows)
  if (filteredWin.length > 0) {
    html += `
      <tr class="table-active">
        <td colspan="6" class="py-2 px-3 fw-bold text-warning">
          <i class="bi bi-microsoft me-1.5"></i> Windows processes (${filteredWin.length})
        </td>
      </tr>
    `;
    html += filteredWin.map(g => renderGroupRow(g, 'windows', 'Windows process')).join('');
  }

  tbody.innerHTML = html;
  attachEventListeners(tbody);
}

/**
 * Рендеринг строки группы процессов (Task Manager style)
 */
function renderGroupRow(group, category, typeLabel) {
  const groupId = `grp-${category}-${group.name.replace(/[^a-zA-Z0-9_-]/g, '_')}-${group.main_pid}`;
  const isExpanded = allExpanded || expandedGroups.has(groupId);
  const count = group.instance_count || (group.subprocesses ? group.subprocesses.length : 1);
  const countBadge = count > 1 ? ` <span class="text-muted sma
```

```
ll">(${count})</span>` : '';
  const chevron = count > 1 || (group.subprocesses && group.subprocesses.length > 0)
    ? `<i class="bi bi-chevron-right pm-chevron ${isExpanded ? 'expanded' : ''}" data-group-id="${groupId}"></i>`
    : `<span style="display:inline-block; width:20px;"></span>`;

  const memMb = group.total_memory_mb || 0;
  const memStr = memMb >= 1024 ? `${(memMb / 1024).toFixed(1)} GB` : `${Math.round(memMb)} MB`;
  const cpuStr = (group.total_cpu_percent !== undefined) ? `${group.total_cpu_percent.toFixed(1)}%` : '0.0%';

  const iconClass = category === 'app' ? 'bi-app-indicator text-primary' : (category === 'windows' ? 'bi-microsoft text-warning' : 'bi-cpu text-info');

  let subRowsHtml = '';
  if (isExpanded && group.subprocesses && group.subprocesses.length > 0) {
    subRowsHtml = group.subprocesses.map(sub => {
      const subMem = sub.memory_mb >= 1024 ? `${(sub.memory_mb / 1024).toFixed(1)} GB` : `${Math.round(sub.memory_mb)} MB`;
      const titleInfo = sub.window_title ? `<div class="text-light small mt-0.5 text-truncate" style="max-width: 380px;"><i class="bi bi-window text-secondary me-1"></i>${escapeHtml(sub.window_title)}</div>` : '';
      return `
        <tr class="pm-sub-row" data-parent-group="${groupId}">
          <td class="ps-4 py-1.5">
            <div class="d-flex align-items-center gap-2">
              <i class="bi bi-arrow-return-right text-muted small"></i>
              <div>
                <span class="text-light">${escapeHtml(sub.name)}</span>
                ${titleInfo}
```

```
              </div>
            </div>
          </td>
          <td class="font-monospace text-warning small py-1.5"
>${sub.pid}</td>
          <td class="py-1.5"><span class="pm-type-tag text-mut
ed">${typeLabel}</span></td>
          <td class="font-monospace text-light small py-1.5">$
{sub.cpu_percent.toFixed(1)}%</td>
          <td class="font-monospace text-info small py-1.5">${
subMem}</td>
          <td class="text-end py-1.5">
            <button class="btn btn-xs btn-outline-danger py-0
px-1.5 pm-kill-btn" data-pid="${sub.pid}" data-name="${sub.nam
e}" title="Завершить процесс PID: ${sub.pid}">
              <i class="bi bi-x"></i>
            </button>
          </td>
        </tr>
      `;
    }).join('');
  }

  return `
    <tr class="pm-group-header" data-group-id="${groupId}">
      <td class="py-2">
        <div class="d-flex align-items-center">
          ${chevron}
          <i class="bi ${iconClass} me-2"></i>
          <span class="fw-semibold text-white">${escapeHtml(gr
oup.friendly_name || group.name)}</span>
          ${countBadge}
        </div>
      </td>
      <td class="font-monospace text-muted small py-2">${group
.main_pid || '-'}</td>
      <td class="py-2"><span class="pm-type-tag">${typeLabel}<
/span></td>
```

```
      <td class="font-monospace text-light small py-2">${cpuStr}</td>
      <td class="font-monospace text-success small py-2">${memStr}</td>
      <td class="text-end py-2">
        <button class="btn btn-xs btn-outline-danger py-0 px-2 pm-kill-btn" data-pid="${group.main_pid}" data-name="${group.name}" data-tree="true" title="Завершить дерево процессов (${count})">
          <i class="bi bi-x-circle me-1"></i>Завершить
        </button>
      </td>
    </tr>
    ${subRowsHtml}
  `;
}

/**
 * Навешивание обработчиков событий
 */
function attachEventListeners(tbody) {
  // Клик по строке группы для раскрытия
  tbody.querySelectorAll('.pm-group-header').forEach(header => {
    header.addEventListener('click', (e) => {
      if (e.target.closest('.pm-kill-btn')) return;
      const groupId = header.dataset.groupId;
      if (expandedGroups.has(groupId)) {
        expandedGroups.delete(groupId);
      } else {
        expandedGroups.add(groupId);
      }
      applyProcessFilters();
    });
  });

  // Завершение процессов
  tbody.querySelectorAll('.pm-kill-btn').forEach(btn => {
```

```
    btn.addEventListener('click', (e) => {
      e.stopPropagation();
      const pid = btn.dataset.pid;
      const name = btn.dataset.name;
      const killTree = btn.dataset.tree === 'true';
      killProcess(pid, name, killTree);
    });
  });
}

/**
 * Завершение процесса или дерева процессов через API
 */
async function killProcess(pid, name, killTree = false) {
  const treeMsg = killTree ? ' (и все дочерние процессы)' : ''
;
  if (!confirm(`Вы действительно хотите завершить ${name} [PID
: ${pid}]${treeMsg}?`)) return;

  try {
    const res = await fetch('/api/process-manager/action', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ pid: parseInt(pid, 10), kill_tree
: killTree, dry_run: false, confirmed_by_user: true })
    });
    const result = await res.json();
    if (res.ok && result.status === 'SUCCESS') {
      if (window.toast) window.toast.success('Процесс завершен
', `${name} (PID: ${pid})`);
      fetchProcessManagerSummary();
    } else {
      if (window.toast) window.toast.error('Ошибка завершения'
, result.message || result.detail || 'Отказано в доступе');
    }
  } catch (err) {
    if (window.toast) window.toast.error('Ошибка сети', err.me
ssage);
```

```
  }
}

/**
 * Экранирование спецсимволов HTML
 */
function escapeHtml(str) {
  if (!str) return '';
  return String(str).replace(/&/g, '&amp;').replace(/</g, '&lt;').replace(/>/g, '&gt;').replace(/"/g, '&quot;');
}

/**
 * Инициализация вкладки Process Manager
 */
export function initProcessManagerTab() {
  if (isInitialized) return;
  isInitialized = true;

  const refreshBtn = document.getElementById('pm-refresh-btn');
  if (refreshBtn) refreshBtn.addEventListener('click', fetchProcessManagerSummary);

  const searchInput = document.getElementById('pm-search-input');
  if (searchInput) searchInput.addEventListener('input', applyProcessFilters);

  // Фильтры по карточкам
  const cardApps = document.getElementById('pm-filter-card-apps');
  const cardBg = document.getElementById('pm-filter-card-bg');
  const cardWin = document.getElementById('pm-filter-card-win');
  const cardAll = document.getElementById('pm-filter-card-all');

```

```
  if (cardApps) cardApps.addEventListener('click', () => setCategoryFilter('app'));
  if (cardBg) cardBg.addEventListener('click', () => setCategoryFilter('background'));
  if (cardWin) cardWin.addEventListener('click', () => setCategoryFilter('windows'));
  if (cardAll) cardAll.addEventListener('click', () => setCategoryFilter('all'));

  // Фильтры по радиокнопкам
  document.querySelectorAll('input[name="pm-cat-filter"]').forEach(radio => {
    radio.addEventListener('change', (e) => {
      currentCategoryFilter = e.target.value;
      applyProcessFilters();
    });
  });

  // Кнопка развернуть / свернуть все
  const toggleExpandBtn = document.getElementById('pm-toggle-expand-btn');
  const expandText = document.getElementById('pm-expand-btn-text');
  if (toggleExpandBtn) {
    toggleExpandBtn.addEventListener('click', () => {
      allExpanded = !allExpanded;
      if (expandText) expandText.textContent = allExpanded ? 'Свернуть все' : 'Развернуть все';
      applyProcessFilters();
    });
  }

  registerTabPoller('tab-process-manager', fetchProcessManagerSummary, 5000, { immediate: true });
}

function setCategoryFilter(cat) {
  currentCategoryFilter = cat;
```

```
  const radio = document.querySelector(`input[name="pm-cat-filter"][value="${cat}"]`);
  if (radio) radio.checked = true;
  applyProcessFilters();
}

window.initProcessManagerTab = initProcessManagerTab;

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/rag_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/rag_tab/index.html" cla
ss="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/rag_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- RAG Knowledge Base & Document / Codebase Management Tab -
->
<div class="container-fluid py-2">
<!-- Header Alert with Sub-Navigation -->
<div class="alert alert-primary d-flex justify-content-between
 align-items-center flex-wrap gap-2 mb-3 shadow-sm">
<div>
<h5 class="mb-1"><i class="bi bi-brain me-2"></i><span data-i1
8n="rag.title">  База знаний и RAG</span></h5>
<p class="mb-0 small text-secondary" data-i18n="rag.subtitle">
```

```
Загрузка документов, индексация исходного кода проектов и семантический поиск.</p>
</div>
<div class="d-flex align-items-center gap-2">
<!-- Sub-Tabs Navigation -->
<ul class="nav nav-pills nav-fill bg-body-secondary p-1 rounded-pill" id="rag-mode-pills" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active rounded-pill btn-sm px-3 py-1" data-bs-target="#pills-user-rags" data-bs-toggle="pill" id="pills-user-rags-tab" role="tab" type="button">
<i class="bi bi-folder2-open me-1"></i> Коллекции Workspace
          </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link rounded-pill btn-sm px-3 py-1" data-bs-target="#pills-doc-rag" data-bs-toggle="pill" id="pills-doc-rag-tab" role="tab" type="button">
<i class="bi bi-file-earmark-text me-1"></i> Документы
          </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link rounded-pill btn-sm px-3 py-1" data-bs-target="#pills-code-rag" data-bs-toggle="pill" id="pills-code-rag-tab" role="tab" type="button">
<i class="bi bi-code-slash me-1"></i> Код проектов (AST)
          </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link rounded-pill btn-sm px-3 py-1" data-bs-target="#pills-pixel-rag" data-bs-toggle="pill" id="pills-pixel-rag-tab" role="tab" type="button">
<i class="bi bi-image me-1 text-warning"></i> PixelRAG (Визуальный поиск)
          </button>
</li>
</ul>
<button class="btn btn-outline-primary btn-sm rounded-pill ms-
```

```
2" id="btn-refresh-rag">
<i class="bi bi-arrow-clockwise me-1"></i> <span data-i18n="co
mmon.refresh">Обновить</span>
</button>
</div>
</div>
<div class="tab-content" id="rag-tab-content">
<!-- SUB-TAB 0: USER WORKSPACE MULTI-RAG -->
<div class="tab-pane fade show active" id="pills-user-rags" ro
le="tabpanel">
<div class="row g-3">
<!-- LEFT: Collections List & Create -->
<div class="col-lg-4 col-md-12">
<div class="card shadow-sm border-0 mb-3" data-panel-id="panel
-rag-bazy-znaniy-rag" id="panel-rag-bazy-znaniy-rag">
<div class="card-header bg-dark text-white d-flex justify-cont
ent-between align-items-center">
<span class="fw-semibold"><i class="bi bi-folder2-open me-2"><
/i>Базы знаний (RAG)</span>
<button class="btn btn-primary btn-sm rounded-pill px-2 py-0"
data-bs-target="#modal-create-user-rag" data-bs-toggle="modal"
 id="btn-show-create-rag-modal">
<i class="bi bi-plus-lg me-1"></i> Создать
              </button>
</div>
<div class="card-body p-2">
<div class="d-flex flex-column gap-2" id="user-rags-list-conta
iner" style="max-height: 480px; overflow-y: auto;">
<div class="text-center text-muted py-4 small">
<i class="bi bi-folder-x fs-3 d-block mb-1 opacity-50"></i>
<span>Нет созданных коллекций.</span><br/>
<span>Нажмите «+ Создать» выше.</span>
</div>
</div>
</div>
</div>
</div>
<!-- RIGHT: Selected Collection Details, File Pipeline, & Sear
```

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ch -->
<div class="col-lg-8 col-md-12">
<div class="card shadow-sm border-0 p-5 text-center text-muted
" data-panel-id="panel-rag-1" id="user-rag-empty-state">
<i class="bi bi-diagram-3 fs-1 mb-2 text-primary opacity-50"><
/i>
<h5>Выберите RAG-коллекцию</h5>
<p class="small text-secondary mb-3">Выберите существующую баз
у знаний слева или создайте новую для подготовки и индексации
документов.</p>
<div>
<button class="btn btn-outline-primary btn-sm rounded-pill px-
3" data-bs-target="#modal-create-user-rag" data-bs-toggle="mod
al">
<i class="bi bi-plus-lg me-1"></i> Создать первую коллекцию
              </button>
</div>
</div>
<div class="d-none" id="user-rag-detail-container">
<!-- Collection Header Card -->
<div class="card shadow-sm border-0 mb-3" data-panel-id="panel
-rag-nazvanie-kollektsiicreated" id="panel-rag-nazvanie-kollek
tsiicreated">
<div class="card-header bg-body-tertiary d-flex justify-conten
t-between align-items-center flex-wrap gap-2 py-2">
<div>
<h6 class="mb-0 fw-bold d-flex align-items-center gap-2">
<i class="bi bi-folder2 text-primary"></i>
<span id="active-rag-title">Название коллекции</span>
<span class="badge bg-secondary" id="active-rag-status-badge">
created</span>
</h6>
<small class="text-muted" id="active-rag-desc">Описание коллек
ции</small>
</div>
<div class="d-flex align-items-center gap-2">
<button class="btn btn-outline-danger btn-sm rounded-pill" id=
"btn-delete-active-rag" title="Удалить коллекцию">
```

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<i class="bi bi-trash"></i>
</button>
</div>
</div>
<div class="card-body p-3">
<!-- Stats Row -->
<div class="row g-2 mb-3 text-center">
<div class="col-sm-4">
<div class="p-2 border rounded bg-body-secondary bg-opacity-25
">
<div class="small text-muted">Чанков в индексе</div>
<div class="fw-bold fs-5 text-success" id="active-rag-stat-chu
nks">0</div>
</div>
</div>
<div class="col-sm-4">
<div class="p-2 border rounded bg-body-secondary bg-opacity-25
">
<div class="small text-muted">Прикреплено файлов</div>
<div class="fw-bold fs-5 text-primary" id="active-rag-stat-fil
es">0</div>
</div>
</div>
<div class="col-sm-4">
<div class="p-2 border rounded bg-body-secondary bg-opacity-25
">
<div class="small text-muted">Обновлено</div>
<div class="small fw-bold mt-1 text-truncate" id="active-rag-s
tat-updated">—</div>
</div>
</div>
</div>
<!-- Pipeline & Files Section -->
<div class="card border border-secondary-subtle mb-3" data-pan
el-id="panel-rag-fayly-dlya-ochistki-i-sborki-rag" id="panel-r
ag-fayly-dlya-ochistki-i-sborki-rag">
<div class="card-header bg-light d-flex justify-content-betwee
n align-items-center py-1 px-3">
```

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<span class="small fw-bold"><i class="bi bi-file-earmark-check me-1"></i> Файлы для очистки и сборки (rag_cleaner)</span>
<button class="btn btn-link btn-sm text-decoration-none p-0" id="btn-select-all-user-files">Выбрать все</button>
</div>
<div class="card-body p-2">
<div class="d-flex flex-column gap-1" id="user-rag-files-checklist" style="max-height: 140px; overflow-y: auto;">
<div class="text-muted small p-2">Загрузка файлов...</div>
</div>
<div class="d-flex justify-content-between align-items-center mt-2 pt-2 border-top flex-wrap gap-2">
<div class="d-flex align-items-center gap-2 flex-wrap">
<input class="d-none" id="user-rag-quick-upload-input" multiple="" type="file"/>
<button class="btn btn-outline-secondary btn-sm rounded-pill" onclick="document.getElementById('user-rag-quick-upload-input').click();" type="button">
<i class="bi bi-upload me-1"></i> Загрузить файлы
                        </button>
<button class="btn btn-outline-info btn-sm rounded-pill d-flex align-items-center gap-1" data-bs-target="#modal-import-gdocs" data-bs-toggle="modal" id="btn-import-gdocs-rag" type="button">
<i class="bi bi-google text-danger"></i>
<span>Google Docs</span>
</button>
</div>
<button class="btn btn-success btn-sm rounded-pill px-3 fw-semibold shadow-sm" id="btn-build-active-rag">
<i class="bi bi-magic me-1"></i> Очистить и собрать RAG
                    </button>
</div>
<div class="d-none alert p-2 small mt-2 mb-0" id="user-rag-build-status-alert"></div>
</div>
</div>
<!-- Q&A and Knowledge Entries Section -->
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<div class="card border border-secondary-subtle mb-3" data-panel-id="panel-rag-zapisi-i-svyazki-vopros-otvet" id="panel-rag-zapisi-i-svyazki-vopros-otvet">
<div class="card-header bg-light d-flex justify-content-between align-items-center py-1 px-3">
<span class="small fw-bold"><i class="bi bi-chat-quote me-1 text-primary"></i> Записи и связки «Вопрос-Ответ»</span>
<div class="d-flex align-items-center gap-2">
<button class="btn btn-outline-secondary btn-sm py-0 px-2 rounded-pill" id="btn-refresh-rag-entries" title="Обновить записи">
<i class="bi bi-arrow-clockwise"></i>
</button>
<button class="btn btn-primary btn-sm rounded-pill px-2 py-0 fw-semibold" data-bs-target="#modal-add-qa-entry" data-bs-toggle="modal" id="btn-show-add-qa-modal">
<i class="bi bi-plus-lg me-1"></i> Добавить Q&amp;amp;A
                    </button>
</div>
</div>
<div class="card-body p-2">
<div class="input-group input-group-sm mb-2">
<input class="form-control" id="user-rag-entries-filter" placeholder="Фильтр записей в этой коллекции..." type="text"/>
<button class="btn btn-outline-secondary" id="btn-clear-entries-filter" type="button">
<i class="bi bi-x-lg"></i>
</button>
</div>
<div class="d-flex flex-column gap-2" id="user-rag-entries-container" style="max-height: 250px; overflow-y: auto;">
<div class="text-center text-muted small py-3">Загрузка записей базы знаний...</div>
</div>
</div>
</div>
<!-- Search Inside Active Collection -->
<div class="card border border-secondary-subtle" data-panel-id
```

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="panel-rag-testovyy-poisk-po-etoy-kollekts" id="panel-rag-tes
tovyy-poisk-po-etoy-kollekts">
<div class="card-header bg-light py-1 px-3">
<span class="small fw-bold"><i class="bi bi-search me-1"></i>
Тестовый поиск по этой коллекции</span>
</div>
<div class="card-body p-2">
<div class="input-group input-group-sm mb-2">
<input class="form-control" id="active-rag-search-query" place
holder="Введите поисковый запрос по коллекции..." type="text"/
>
<button class="btn btn-primary px-3" id="btn-active-rag-search
">
<i class="bi bi-search me-1"></i> Поиск
                      </button>
</div>
<div class="border rounded p-2 bg-body-tertiary" id="active-ra
g-search-results" style="min-height: 100px; max-height: 200px;
 overflow-y: auto;">
<div class="text-center text-muted small py-3">Введите запрос
для поиска по выбранной базе знаний.</div>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- Create User RAG Modal -->
<div aria-hidden="true" aria-labelledby="modalCreateUserRAGLab
el" class="modal fade" id="modal-create-user-rag" tabindex="-1
">
<div class="modal-dialog modal-dialog-centered">
<div class="modal-content shadow">
<div class="modal-header">
<h6 class="modal-title fw-bold" id="modalCreateUserRAGLabel"><
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i class="bi bi-folder-plus text-primary me-2"></i>Создание нов
ой базы знаний (RAG)</h6>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body">
<div class="mb-3">
<label class="form-label small fw-bold" for="new-rag-name">Имя
 коллекции:</label>
<input class="form-control form-control-sm" id="new-rag-name"
placeholder="например: movies, coding, gdocs_knowledge" type="
text"/>
<div class="form-text small">Уникальное имя для группировки зн
аний.</div>
</div>
<div class="mb-3">
<label class="form-label small fw-bold" for="new-rag-desc">Опи
сание (опционально):</label>
<textarea class="form-control form-control-sm" id="new-rag-des
c" placeholder="Кратко опишите назначение этой базы знаний..."
 rows="2"></textarea>
</div>
<div class="mb-3">
<label class="form-label small fw-bold">Источник документов:</
label>
<div class="form-check">
<input checked="" class="form-check-input" id="rag-src-local"
name="new-rag-source-type" type="radio" value="local"/>
<label class="form-check-label small" for="rag-src-local">
<i class="bi bi-hdd me-1 text-primary"></i>Локальные файлы / Л
ичное хранилище
                </label>
</div>
<div class="form-check">
<input class="form-check-input" id="rag-src-gdocs" name="new-r
ag-source-type" type="radio" value="gdocs"/>
<label class="form-check-label small" for="rag-src-gdocs">
<i class="bi bi-google me-1 text-danger"></i>Документы Google
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Docs / Google Drive
                </label>
</div>
</div>
<div class="row g-2">
<div class="col-6">
<label class="form-label small fw-bold" for="new-rag-min-chunk
">Мин. длина чанка:</label>
<input class="form-control form-control-sm" id="new-rag-min-ch
unk" max="500" min="5" type="number" value="20"/>
</div>
<div class="col-6">
<label class="form-label small fw-bold" for="new-rag-max-chunk
">Макс. длина чанка:</label>
<input class="form-control form-control-sm" id="new-rag-max-ch
unk" max="5000" min="100" type="number" value="1500"/>
</div>
</div>
</div>
<div class="modal-footer py-2">
<button class="btn btn-secondary btn-sm" data-bs-dismiss="moda
l" type="button">Отмена</button>
<button class="btn btn-primary btn-sm fw-semibold" id="btn-con
firm-create-rag" type="button">Создать</button>
</div>
</div>
</div>
</div>
<!-- Import Google Docs Modal -->
<div aria-hidden="true" aria-labelledby="modalImportGDocsLabel
" class="modal fade" id="modal-import-gdocs" tabindex="-1">
<div class="modal-dialog modal-dialog-centered modal-lg">
<div class="modal-content shadow">
<div class="modal-header">
<h6 class="modal-title fw-bold" id="modalImportGDocsLabel"><i
class="bi bi-google text-danger me-2"></i>Импорт документов из
 Google Docs &amp;amp; Drive в RAG</h6>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
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modal" type="button"></button>
</div>
<div class="modal-body">
<div class="alert alert-info py-2 small mb-3">
<i class="bi bi-info-circle me-1"></i> Документы Google Docs а втоматически экспортируются, очищаются через <code>rag_cleaner </code> и добавляются в векторный индекс коллекции.
            </div>
<div class="row g-2 mb-3">
<div class="col-md-6">
<label class="form-label small fw-bold" for="import-gdocs-quer y">Фильтр поиска Google Drive:</label>
<input class="form-control form-control-sm font-monospace" id= "import-gdocs-query" placeholder="mimeType = '...' or name con tains '...'" type="text" value="mimeType = 'application/vnd.go ogle-apps.document' and trashed = false"/>
</div>
<div class="col-md-6">
<label class="form-label small fw-bold" for="import-gdocs-acco unt">Google аккаунт (из пула):</label>
<input class="form-control form-control-sm" id="import-gdocs-a ccount" placeholder="default (или имя аккаунта)" type="text"/>
</div>
</div>
<div class="alert d-none py-2 small mb-0" id="import-gdocs-sta tus-alert"></div>
</div>
<div class="modal-footer py-2">
<button class="btn btn-secondary btn-sm" data-bs-dismiss="moda l" type="button">Отмена</button>
<button class="btn btn-primary btn-sm fw-semibold" id="btn-con firm-import-gdocs" type="button">
<i class="bi bi-cloud-arrow-down me-1"></i> Синхронизировать и  собрать RAG
            </button>
</div>
</div>
</div>
```

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</div>
<!-- Add QA Entry Modal -->
<div aria-hidden="true" aria-labelledby="modalAddQAEntryLabel"
 class="modal fade" id="modal-add-qa-entry" tabindex="-1">
<div class="modal-dialog modal-dialog-centered modal-lg">
<div class="modal-content shadow">
<div class="modal-header">
<h6 class="modal-title fw-bold" id="modalAddQAEntryLabel"><i c
lass="bi bi-patch-question text-primary me-2"></i>Добавление з
нания / Связки «Вопрос-Ответ»</h6>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body">
<div class="mb-3">
<label class="form-label small fw-bold" for="new-qa-question">
Вопрос пользователя / Триггерная фраза:</label>
<input class="form-control form-control-sm" id="new-qa-questio
n" placeholder="например: Где можно скачать сериал?" type="tex
t"/>
<div class="form-text small">Фраза или формулировка запроса, п
ри схожести с которой будет применяться этот ответ.</div>
</div>
<div class="mb-3">
<label class="form-label small fw-bold" for="new-qa-answer">От
вет модели / Текст знания:</label>
<textarea class="form-control form-control-sm font-monospace"
id="new-qa-answer" placeholder="Введите точный ответ, инструкц
ию или фрагмент знаний..." rows="5"></textarea>
</div>
</div>
<div class="modal-footer py-2">
<button class="btn btn-secondary btn-sm" data-bs-dismiss="moda
l" type="button">Отмена</button>
<button class="btn btn-success btn-sm fw-semibold" id="btn-con
firm-add-qa" type="button">
<i class="bi bi-check2-circle me-1"></i> Сохранить в RAG
            </button>
```

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</div>
</div>
</div>
</div>
<!-- SUB-TAB 1: DOCUMENT RAG -->
<div class="tab-pane fade" id="pills-doc-rag" role="tabpanel">
<!-- Top Statistics Summary -->
<div class="row g-3 mb-3">
<div class="col-md-3 col-sm-6">
<div class="card shadow-sm border-0 bg-body-tertiary" data-pan
el-id="panel-rag-0" id="panel-rag-0">
<div class="card-body p-3 d-flex align-items-center justify-co
ntent-between">
<div>
<div class="text-muted small" data-i18n="rag.totalDocuments">Д
окументов</div>
<h4 class="mb-0 fw-bold" id="rag-stat-docs">0</h4>
</div>
<i class="bi bi-file-earmark-text fs-1 text-primary opacity-50
"></i>
</div>
</div>
</div>
<div class="col-md-3 col-sm-6">
<div class="card shadow-sm border-0 bg-body-tertiary" data-pan
el-id="panel-rag-0" id="panel-rag-0">
<div class="card-body p-3 d-flex align-items-center justify-co
ntent-between">
<div>
<div class="text-muted small" data-i18n="rag.totalChunks">Чанк
ов в индексе</div>
<h4 class="mb-0 fw-bold" id="rag-stat-chunks">0</h4>
</div>
<i class="bi bi-diagram-3 fs-1 text-success opacity-50"></i>
</div>
</div>
</div>
<div class="col-md-3 col-sm-6">
```

```
<div class="card shadow-sm border-0 bg-body-tertiary" data-panel-id="panel-rag-0" id="panel-rag-0">
<div class="card-body p-3 d-flex align-items-center justify-content-between">
<div>
<div class="text-muted small" data-i18n="rag.vectorDimension">Размерность вектора</div>
<h4 class="mb-0 fw-bold" id="rag-stat-dim">0</h4>
</div>
<i class="bi bi-cpu fs-1 text-info opacity-50"></i>
</div>
</div>
</div>
<div class="col-md-3 col-sm-6">
<div class="card shadow-sm border-0 bg-body-tertiary" data-panel-id="panel-rag-2" id="panel-rag-2">
<div class="card-body p-3 d-flex align-items-center justify-content-between">
<div>
<div class="text-muted small" data-i18n="rag.lastUpdated">Последнее обновление</div>
<div class="small fw-bold text-truncate" id="rag-stat-updated" style="max-width: 140px;">—</div>
</div>
<i class="bi bi-clock-history fs-1 text-warning opacity-50"></i>
</div>
</div>
</div>
</div>
<div class="row g-3">
<!-- LEFT COLUMN: File Ingestion & Index Controls -->
<div class="col-lg-5 col-md-12">
<!-- Upload Dropzone Card -->
<div class="card shadow-sm border-0 mb-3" data-panel-id="panel-rag-zagruzka-dokumentov" id="panel-rag-zagruzka-dokumentov">
<div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
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<span class="fw-semibold"><i class="bi bi-cloud-arrow-up me-2"></i><span data-i18n="rag.uploadTitle">Загрузка документов</span></span>
</div>
<div class="card-body">
<div class="border border-2 border-dashed rounded p-4 text-center cursor-pointer mb-3 bg-body-secondary bg-opacity-25" id="rag-dropzone" style="border-style: dashed !important; transition: all 0.2s;">
<i class="bi bi-file-earmark-arrow-up fs-1 text-primary mb-2 d-block"></i>
<h6 class="fw-bold mb-1" data-i18n="rag.dropzoneText">Перетащите файлы сюда или нажмите для выбора</h6>
<input accept=".txt,.md,.markdown,.json,.csv,.tsv,.py,.js,.ts,.html,.css,.pdf,.log,.sql,.yaml,.yml" class="d-none" id="rag-file-input" multiple="" type="file"/>
<input class="d-none" directory="" id="rag-folder-input" multiple="" type="file" webkitdirectory=""/>
<div class="d-flex justify-content-center gap-2 mt-2 flex-wrap">
<button class="btn btn-outline-primary btn-sm px-3 rounded-pill" onclick="document.getElementById('rag-file-input').click();" type="button">
<i class="bi bi-file-earmark-plus me-1"></i> <span data-i18n="rag.browseFiles">Выбрать файлы</span>
</button>
<button class="btn btn-primary btn-sm px-3 rounded-pill" onclick="document.getElementById('rag-folder-input').click();" type="button">
<i class="bi bi-folder-plus me-1"></i> <span data-i18n="rag.browseFolder">Выбрать папку целиком</span>
</button>
</div>
</div>
<div class="d-none mb-3" id="rag-upload-progress-container">
<div class="d-flex justify-content-between small text-muted mb-1">
<span data-i18n="rag.uploading">Загрузка файлов...</span>
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<span id="rag-upload-percent">0%</span>
</div>
<div class="progress" style="height: 6px;">
<div class="progress-bar progress-bar-striped progress-bar-animated bg-primary" id="rag-upload-progress-bar" role="progressbar" style="width: 0%"></div>
</div>
</div>
<div class="d-none alert p-2 small mb-0" id="rag-upload-status-alert"></div>
</div>
</div>
<!-- Index Settings & Build Card -->
<div class="card shadow-sm border-0" data-panel-id="panel-rag-parametry-indeksatsii" id="panel-rag-parametry-indeksatsii">
<div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
<span class="fw-semibold"><i class="bi bi-sliders me-2"></i><span data-i18n="rag.indexSettings">Параметры индексации</span></span>
</div>
<div class="card-body">
<div class="mb-3">
<label class="form-label small fw-bold" data-i18n="rag.providerLabel" for="rag-provider-select">Провайдер векторизации:</label>
<select class="form-select form-select-sm" id="rag-provider-select">
<option value="auto">Auto (Gemini если доступен ключ, иначе Local TF-IDF)</option>
<option value="gemini">Google Gemini Embeddings (gemini-embedding-2)</option>
<option value="local_tfidf">Local TF-IDF Vectorizer (Полностью офлайн)</option>
</select>
</div>
<div class="mb-3" id="rag-api-key-group">
<label class="form-label small fw-bold" for="rag-api-key-input
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">Gemini API Key (опционально, берется из .env):</label>
<input autocomplete="new-password" class="form-control form-control-sm" data-1p-ignore="true" data-lpignore="true" id="rag-api-key-input" placeholder="AIzaSy... (или оставить пустым для авто)" type="password"/>
</div>
<div class="row g-2 mb-3">
<div class="col-6">
<label class="form-label small fw-bold" data-i18n="rag.chunkSize" for="rag-chunk-size">Размер чанка (символов):</label>
<input class="form-control form-control-sm" id="rag-chunk-size" max="4000" min="100" step="50" type="number" value="500"/>
</div>
<div class="col-6">
<label class="form-label small fw-bold" data-i18n="rag.chunkOverlap" for="rag-chunk-overlap">Перекрытие (Overlap):</label>
<input class="form-control form-control-sm" id="rag-chunk-overlap" max="1000" min="0" step="10" type="number" value="50"/>
</div>
</div>
<div class="d-grid">
<button class="btn btn-success btn-sm fw-semibold py-2 shadow-sm" id="btn-build-rag-index">
<i class="bi bi-hammer me-1"></i> <span data-i18n="rag.buildIndexBtn">  Построить / Перестроить индекс</span>
</button>
</div>
<div class="small mt-2 text-center text-muted" id="rag-build-status"></div>
</div>
</div>
</div>
<!-- RIGHT COLUMN: Documents List & Semantic Search Playground -->
<div class="col-lg-7 col-md-12">
<!-- Documents Table Card -->
<div class="card shadow-sm border-0 mb-3" data-panel-id="panel-rag-spisok-dokumentov-v-baze" id="panel-rag-spisok-dokumentov
```

```
-v-baze">
<div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
<span class="fw-semibold"><i class="bi bi-files me-2"></i><span data-i18n="rag.documentsTitle">Список документов в базе</span></span>
<span class="badge bg-primary" id="rag-docs-badge">0 файлов</span>
</div>
<div class="card-body p-0">
<div class="table-responsive" style="max-height: 260px; overflow-y: auto;">
<table class="table table-hover table-sm align-middle mb-0" id="rag-documents-table">
<thead class="table-light sticky-top">
<tr>
<th class="ps-3" data-i18n="rag.fileName">Имя файла</th>
<th data-i18n="rag.fileSize">Размер</th>
<th data-i18n="rag.chunks">Чанки</th>
<th data-i18n="rag.status">Статус</th>
<th class="text-end pe-3" data-i18n="common.actions">Действия</th>
</tr>
</thead>
<tbody id="rag-documents-tbody">
<tr>
<td class="text-center text-muted py-3" colspan="5" data-i18n="rag.noDocuments">
                        Документы не загружены. Загрузите файлы в блоке слева.
                    </td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- Semantic Search Playground Card -->
```

```
<div class="card shadow-sm border-0" data-panel-id="panel-rag-semanticheskiy-poisk-po-baze" id="panel-rag-semanticheskiy-poisk-po-baze">
<div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
<span class="fw-semibold"><i class="bi bi-search me-2"></i><span data-i18n="rag.searchPlaygroundTitle">Семантический поиск по базе</span></span>
</div>
<div class="card-body">
<div class="d-flex align-items-center justify-content-between flex-wrap gap-2 mb-2 p-2 bg-body-tertiary rounded border small text-muted">
<div class="d-flex align-items-center gap-3">
<div class="d-flex align-items-center gap-1">
<label class="text-nowrap fw-semibold" for="rag-top-k">Top K:</label>
<select class="form-select form-select-sm" id="rag-top-k" style="width: 75px;">
<option value="3">3</option>
<option selected="" value="5">5</option>
<option value="10">10</option>
<option value="20">20</option>
</select>
</div>
<div class="d-flex align-items-center gap-1">
<label class="text-nowrap fw-semibold" for="rag-min-score">Min Score:</label>
<input class="form-control form-control-sm" id="rag-min-score" max="1.0" min="0.0" step="0.05" style="width: 80px;" type="number" value="0.0"/>
</div>
</div>
<span class="small text-secondary"><i class="bi bi-info-circle me-1"></i>Фильтры выдачи</span>
</div>
<div class="input-group mb-3">
<input class="form-control" data-i18n-placeholder="rag.searchP
```

```
laceholder" id="rag-search-query" placeholder="Введите поисковый запрос по смыслу..." type="text"/>
<button class="btn btn-primary px-3" id="btn-rag-search">
<i class="bi bi-search me-1"></i> <span data-i18n="common.search">Поиск</span>
</button>
</div>
<div class="border rounded p-2 bg-body-tertiary" id="rag-search-results" style="min-height: 180px; max-height: 320px; overflow-y: auto;">
<div class="text-center text-muted py-4">
<i class="bi bi-chat-left-quote fs-3 d-block mb-1 opacity-50"></i>
<span data-i18n="rag.enterQueryPrompt">Введите запрос и нажмите «Поиск» для проверки RAG.</span>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- SUB-TAB 2: CODEBASE RAG (AST + Multi-Project) -->
<div class="tab-pane fade" id="pills-code-rag" role="tabpanel">
<!-- Active Project & Index Selector Bar -->
<div class="card shadow-sm border-0 mb-3 bg-body-tertiary" data-panel-id="panel-rag-3" id="panel-rag-3">
<div class="card-body p-3 d-flex align-items-center justify-content-between flex-wrap gap-2">
<div class="d-flex align-items-center gap-2 flex-grow-1 min-w-0">
<label class="form-label small fw-bold mb-0 text-nowrap" for="codebase-active-index-select">
<i class="bi bi-diagram-2 text-primary me-1"></i> Активный RAG индекс:
            </label>
<select class="form-select form-select-sm" id="codebase-active
```

```
-index-select" style="max-width: 320px;">
<option value="codebase">codebase (По умолчанию)</option>
</select>
<button class="btn btn-sm btn-outline-secondary rounded-pill" id="btn-refresh-codebase-indexes" title="Обновить список индексов">
<i class="bi bi-arrow-clockwise"></i>
</button>
</div>
<div class="d-flex align-items-center gap-2" id="codebase-index-quick-stats">
<span class="badge bg-primary-subtle text-primary border border-primary-subtle" id="codebase-stat-chunks">0 чанков</span>
<span class="badge bg-success-subtle text-success border border-success-subtle" id="codebase-stat-symbols">0 символов AST</span>
</div>
</div>
</div>
<div class="row g-3">
<!-- LEFT COLUMN: Codebase Index Settings & Trigger -->
<div class="col-lg-5 col-md-12">
<div class="card shadow-sm border-0" data-panel-id="panel-rag-sborka-indeksa-proekta" id="panel-rag-sborka-indeksa-proekta">
<div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
<span class="fw-semibold"><i class="bi bi-folder2-open me-2"></i>Сборка индекса проекта</span>
</div>
<div class="card-body">
<div class="mb-3">
<label class="form-label small fw-bold" for="codebase-project-root-input">
                  Директория проекта (Project Root):
                </label>
<div class="input-group input-group-sm">
<span class="input-group-text"><i class="bi bi-hdd"></i></span>
```

```
<input class="form-control" id="codebase-project-root-input" placeholder="Например: . или C:/Projects/my_app" type="text" value="."/>
</div>
<div class="form-text small text-muted">Укажите путь к проекту на диске (абсолютный или относительный).</div>
</div>
<div class="mb-3">
<label class="form-label small fw-bold" for="codebase-index-name-input">
                  Имя RAG индекса (Index Name):
                </label>
<input class="form-control form-control-sm" id="codebase-index-name-input" placeholder="Например: my_project_v1" type="text" value="codebase"/>
<div class="form-text small text-muted">Индексы сохраняются изолированно в вашем персональном хранилище.</div>
</div>
<div class="mb-3">
<label class="form-label small fw-bold" for="codebase-include-dirs-input">
                  Подкаталоги для анализа (через запятую):
                </label>
<input class="form-control form-control-sm font-monospace" id="codebase-include-dirs-input" type="text" value="src, plugins, integrations, launchers, scripts, install, docs, prompts"/>
</div>
<div class="d-grid">
<button class="btn btn-primary btn-sm fw-semibold py-2 shadow-sm d-flex align-items-center justify-content-center gap-2" id="btn-build-codebase-rag">
<i class="bi bi-cpu"></i>
<span> Собрать Codebase RAG</span>
</button>
</div>
<div class="d-none alert p-2 small mt-3 mb-0" id="codebase-build-status-alert"></div>
</div>
```

```
</div>
</div>
<!-- RIGHT COLUMN: Code & Symbol Search Playground -->
<div class="col-lg-7 col-md-12">
<div class="card shadow-sm border-0" data-panel-id="panel-rag-4" id="panel-rag-4">
<div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
<ul class="nav nav-tabs card-header-tabs border-0" id="codebase-search-tabs" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active text-white" data-bs-target="#tab-code-semantic" data-bs-toggle="tab" id="tab-code-semantic-btn" role="tab" type="button">
<i class="bi bi-search me-1"></i> Семантический поиск по коду
                  </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link text-white-50" data-bs-target="#tab-code-symbols" data-bs-toggle="tab" id="tab-code-symbols-btn" role="tab" type="button">
<i class="bi bi-compass me-1"></i> AST Символы
                  </button>
</li>
</ul>
</div>
<div class="card-body">
<div class="tab-content" id="codebase-search-tabs-content">
<!-- Code Semantic Search -->
<div class="tab-pane fade show active" id="tab-code-semantic" role="tabpanel">
<div class="input-group mb-3">
<input class="form-control" id="codebase-search-query" placeholder="Например: Как реализован роутинг провайдеров?" type="text"/>
<button class="btn btn-primary px-3" id="btn-codebase-search">
<i class="bi bi-search me-1"></i> Поиск
                    </button>
```

```
</div>
<div class="border rounded p-2 bg-body-tertiary" id="codebase-search-results" style="min-height: 220px; max-height: 360px; overflow-y: auto;">
<div class="text-center text-muted py-4">
<i class="bi bi-code-square fs-3 d-block mb-1 opacity-50"></i>
<span>Введите вопрос или описание логики для поиска по коду.</span>
</div>
</div>
</div>
<!-- Symbol Lookup -->
<div class="tab-pane fade" id="tab-code-symbols" role="tabpanel">
<div class="input-group mb-3">
<input class="form-control" id="codebase-symbol-query" placeholder="Имя класса или метода: например MediaSearchAgent, search, BasePlugin" type="text"/>
<button class="btn btn-success px-3" id="btn-codebase-symbol-search">
<i class="bi bi-compass me-1"></i> Найти символ
                    </button>
</div>
<div class="border rounded p-2 bg-body-tertiary" id="codebase-symbol-results" style="min-height: 220px; max-height: 360px; overflow-y: auto;">
<div class="text-center text-muted py-4">
<i class="bi bi-diagram-3 fs-3 d-block mb-1 opacity-50"></i>
<span>Введите имя класса, метода или функции для мгновенного AST-поиска.</span>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
```

```
</div>
<!-- SUB-TAB 3: PIXELRAG VISUAL HYBRID SEARCH -->
<div class="tab-pane fade" id="pills-pixel-rag" role="tabpanel
">
<!-- Top Metrics Bar -->
<div class="row g-3 mb-3">
<div class="col-md-3 col-sm-6">
<div class="card shadow-sm border-0 bg-body-secondary bg-opaci
ty-25 h-100 p-3 text-center" data-panel-id="panel-rag-5" id="p
anel-rag-5">
<div class="small text-muted mb-1"><i class="bi bi-images text
-primary me-1"></i>Всего изображений</div>
<div class="fw-bold fs-4 text-primary" id="pixel-stat-images">
0</div>
<div class="small text-muted" id="pixel-stat-size">0.0 MB</div
>
</div>
</div>
<div class="col-md-3 col-sm-6">
<div class="card shadow-sm border-0 bg-body-secondary bg-opaci
ty-25 h-100 p-3 text-center" data-panel-id="panel-rag-6" id="p
anel-rag-6">
<div class="small text-muted mb-1"><i class="bi bi-layers text
-success me-1"></i>Чанков в индексе</div>
<div class="fw-bold fs-4 text-success" id="pixel-stat-chunks">
0</div>
<div class="small text-muted">FAISS Hybrid Vector Index</div>
</div>
</div>
<div class="col-md-3 col-sm-6">
<div class="card shadow-sm border-0 bg-body-secondary bg-opaci
ty-25 h-100 p-3 text-center" data-panel-id="panel-rag-7" id="p
anel-rag-7">
<div class="small text-muted mb-1"><i class="bi bi-robot text-
info me-1"></i>Провайдер векторизации</div>
<div class="fw-bold fs-6 text-info mt-1" id="pixel-stat-provid
er">FAISS / Visual Embeddings</div>
<div class="small text-muted" id="pixel-stat-status">Статус: 3
```

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агрузка...</div>
</div>
</div>
<div class="col-md-3 col-sm-6">
<div class="card shadow-sm border-0 bg-body-secondary bg-opaci
ty-25 h-100 p-3 text-center" data-panel-id="panel-rag-8" id="p
anel-rag-8">
<div class="small text-muted mb-1"><i class="bi bi-clock-histo
ry text-warning me-1"></i>Индекс обновлен</div>
<div class="small fw-bold mt-2 text-truncate" id="pixel-stat-u
pdated">—</div>
<button class="btn btn-link btn-sm text-decoration-none p-0 mt
-1" id="btn-refresh-pixel-stats">
<i class="bi bi-arrow-clockwise"></i> Обновить
            </button>
</div>
</div>
</div>
<div class="row g-3">
<!-- LEFT COLUMN: Image Ingestion & Smart Query Router -->
<div class="col-lg-5 col-md-12">
<!-- Ingest Images Card -->
<div class="card shadow-sm border-0 mb-3" data-panel-id="panel
-rag-zagruzka-i-indeksatsiya-izobrazhe" id="panel-rag-zagruzka
-i-indeksatsiya-izobrazhe">
<div class="card-header bg-dark text-white d-flex justify-cont
ent-between align-items-center">
<span class="fw-semibold"><i class="bi bi-cloud-arrow-up me-2"
></i>Загрузка и индексация изображений</span>
</div>
<div class="card-body">
<div class="border border-2 border-dashed rounded p-4 text-cen
ter cursor-pointer mb-3 bg-body-secondary bg-opacity-25" id="p
ixel-dropzone" style="border-style: dashed !important; transit
ion: all 0.2s; cursor: pointer;">
<i class="bi bi-file-earmark-image fs-1 text-primary mb-2 d-bl
ock"></i>
<h6 class="fw-bold mb-1">Перетащите изображения сюда или нажми
```

```
те для выбора</h6>
<div class="small text-muted mb-3">Поддерживаются .png, .jpg, .jpeg, .webp, .bmp, .gif, .pdf</div>
<input accept="image/*,.pdf" class="d-none" id="pixel-file-input" multiple="" type="file"/>
<button class="btn btn-outline-primary btn-sm px-3 rounded-pill" onclick="document.getElementById('pixel-file-input').click();" type="button">
<i class="bi bi-image me-1"></i> Выбрать изображения
                </button>
</div>
<div class="d-none mb-3" id="pixel-upload-progress">
<div class="d-flex justify-content-between small text-muted mb-1">
<span>Индексация через PixelRAG...</span>
<span id="pixel-upload-percent">0%</span>
</div>
<div class="progress" style="height: 6px;">
<div class="progress-bar progress-bar-striped progress-bar-animated bg-success" id="pixel-upload-progress-bar" role="progressbar" style="width: 0%"></div>
</div>
</div>
<div class="d-none alert p-2 small mb-0" id="pixel-upload-status-alert"></div>
</div>
</div>
<!-- Smart Query Router & Analyzer -->
<div class="card shadow-sm border-0" data-panel-id="panel-rag-smart-query-router-analiz-zapr" id="panel-rag-smart-query-router-analiz-zapr">
<div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
<span class="fw-semibold"><i class="bi bi-signpost-split me-2"></i>Smart Query Router (Анализ запроса)</span>
</div>
<div class="card-body">
<p class="small text-muted mb-2">Автоматическое определение ти
```

```
па запроса (Text vs Visual vs Hybrid) перед поиском.</p>
<div class="input-group input-group-sm mb-2">
<input class="form-control" id="pixel-analyze-query-input" placeholder="Например: схема авторизации или синий график продаж" type="text"/>
<button class="btn btn-outline-primary" id="btn-pixel-analyze-query">
<i class="bi bi-cpu me-1"></i> Анализ
                </button>
</div>
<div class="p-2 border rounded bg-body-tertiary small d-none" id="pixel-analyze-result">
<div class="d-flex justify-content-between align-items-center mb-1">
<span>Тип маршрутизации: <strong class="badge bg-primary" id="pixel-routing-type">HYBRID</strong></span>
<span>Уверенность: <strong class="text-success" id="pixel-routing-confidence">95%</strong></span>
</div>
<div class="text-muted" id="pixel-routing-reason" style="font-size: 0.8rem;"></div>
</div>
</div>
</div>
</div>
<!-- RIGHT COLUMN: Visual Search Playground -->
<div class="col-lg-7 col-md-12">
<div class="card shadow-sm border-0" data-panel-id="panel-rag-interaktivnyy-poisk-pixelrag" id="panel-rag-interaktivnyy-poisk-pixelrag">
<div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
<span class="fw-semibold"><i class="bi bi-search me-2"></i>Интерактивный поиск PixelRAG</span>
<div class="form-check form-switch m-0" title="Использовать Smart Query Routing">
<input checked="" class="form-check-input" id="pixel-search-routing-toggle" type="checkbox"/>
```

```
<label class="form-check-label text-light small ms-1" for="pixel-search-routing-toggle">Smart Routing</label>
</div>
</div>
<div class="card-body">
<div class="row g-2 mb-3">
<div class="col-md-8">
<div class="input-group">
<input class="form-control" id="pixel-search-query" placeholder="Введите поисковый запрос (текст или описание картинки)..." type="text"/>
<button class="btn btn-primary px-3" id="btn-pixel-search">
<i class="bi bi-search me-1"></i> Искать
                    </button>
</div>
</div>
<div class="col-md-2">
<select class="form-select" id="pixel-search-topk" title="Количество результатов">
<option value="3">Top 3</option>
<option selected="" value="5">Top 5</option>
<option value="10">Top 10</option>
</select>
</div>
<div class="col-md-2">
<select class="form-select" id="pixel-search-minscore" title="Минимальный порог">
<option selected="" value="0.0">Score ≥ 0.0</option>
<option value="0.3">Score ≥ 0.3</option>
<option value="0.5">Score ≥ 0.5</option>
<option value="0.7">Score ≥ 0.7</option>
</select>
</div>
</div>
<!-- Search Results List -->
<div class="border rounded p-3 bg-body-tertiary" id="pixel-search-results" style="min-height: 280px; max-height: 480px; overflow-y: auto;">
```

```
<div class="text-center text-muted py-5">
<i class="bi bi-images fs-1 d-block mb-2 opacity-50"></i>
<span>Введите запрос для поиска по текстовым чанкам и визуальным индексам изображений.</span>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Rag Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/rag_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/rag_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

import { getAllFilesFromDataTransfer } from './modules/utils.js';
import {
  uploadFiles,
  loadRagStatus,
  loadRagDocuments,
  deleteRagDocument,
  buildRagIndex,
  executeRagSearch
} from './modules/docRag.js';
import {
  loadCodebaseIndexes,
```

```
  updateCodebaseStatsDisplay,
  buildCodebaseRag,
  executeCodebaseSearch,
  executeSymbolLookup
} from './modules/codebaseRag.js';
import {
  loadUserRags,
  createUserRag,
  deleteActiveUserRag,
  addQaEntry,
  loadRagEntries
} from './modules/userRagCollections.js';

import {
  buildActiveUserRag,
  executeActiveUserRagSearch,
  handleQuickUploadToUserStorage,
  syncGoogleDocsIntoActiveRag
} from './modules/userRagPipeline.js';

import {
  loadPixelRagStatus,
  uploadAndIndexPixelImages,
  analyzePixelQuery,
  executePixelRagSearch
} from './modules/pixelRag.js';

/**
 * Initialize all RAG tab components, event listeners, and dat
a feeds.
 *
 * @returns {Promise<void>}
 */
export async function initRagTab() {
  setupEventListeners();
  loadUserRags();
  loadRagStatus();
  loadRagDocuments();
```

```
  loadCodebaseIndexes();
  loadPixelRagStatus();
}

// Expose globals for external tab orchestrators and inline handlers
window.initRagTab = initRagTab;
window.deleteRagDocument = deleteRagDocument;

/**
 * Register DOM event listeners for buttons, inputs, dropzones, and keyboard events.
 */
function setupEventListeners() {
  const root = document.getElementById('rag-tab-content');
  if (root) {
    if (root.dataset.listenersBound === 'true') return;
    root.dataset.listenersBound = 'true';
  }

  const refreshBtn = document.getElementById('btn-refresh-rag');
  if (refreshBtn) {
    refreshBtn.addEventListener('click', () => {
      loadUserRags();
      loadRagStatus();
      loadRagDocuments();
      loadCodebaseIndexes();
      loadPixelRagStatus();
    });
  }

  // --- PixelRAG Listeners ---
  const refreshPixelStatsBtn = document.getElementById('btn-refresh-pixel-stats');
  if (refreshPixelStatsBtn) {
    refreshPixelStatsBtn.addEventListener('click', loadPixelRagStatus);
```

```
  }

  const pixelFileInput = document.getElementById('pixel-file-input');
  if (pixelFileInput) {
    pixelFileInput.addEventListener('change', (e) => {
      if (e.target.files && e.target.files.length > 0) {
        uploadAndIndexPixelImages(Array.from(e.target.files));
        pixelFileInput.value = '';
      }
    });
  }

  const pixelDropzone = document.getElementById('pixel-dropzone');
  if (pixelDropzone) {
    ['dragenter', 'dragover'].forEach(eventName => {
      pixelDropzone.addEventListener(eventName, (e) => {
        e.preventDefault();
        e.stopPropagation();
        pixelDropzone.classList.add('border-primary', 'bg-primary', 'bg-opacity-10');
      }, false);
    });

    ['dragleave', 'drop'].forEach(eventName => {
      pixelDropzone.addEventListener(eventName, (e) => {
        e.preventDefault();
        e.stopPropagation();
        pixelDropzone.classList.remove('border-primary', 'bg-primary', 'bg-opacity-10');
      }, false);
    });

    pixelDropzone.addEventListener('drop', async (e) => {
      const dt = e.dataTransfer;
      if (dt) {
        const files = await getAllFilesFromDataTransfer(dt);
```

```
        if (files && files.length > 0) {
          uploadAndIndexPixelImages(files);
        }
      }
    }, false);
  }

  const btnPixelAnalyze = document.getElementById('btn-pixel-analyze-query');
  const inputPixelAnalyze = document.getElementById('pixel-analyze-query-input');
  if (btnPixelAnalyze) {
    btnPixelAnalyze.addEventListener('click', analyzePixelQuery);
  }
  if (inputPixelAnalyze) {
    inputPixelAnalyze.addEventListener('keydown', (e) => {
      if (e.key === 'Enter') analyzePixelQuery();
    });
  }

  const btnPixelSearch = document.getElementById('btn-pixel-search');
  const inputPixelSearch = document.getElementById('pixel-search-query');
  if (btnPixelSearch) {
    btnPixelSearch.addEventListener('click', executePixelRagSearch);
  }
  if (inputPixelSearch) {
    inputPixelSearch.addEventListener('keydown', (e) => {
      if (e.key === 'Enter') executePixelRagSearch();
    });
  }

  // --- User Workspace RAG Listeners ---
  const confirmCreateRagBtn = document.getElementById('btn-confirm-create-rag');
```

```
  if (confirmCreateRagBtn) {
    confirmCreateRagBtn.addEventListener('click', createUserRa
g);
  }

  const buildActiveRagBtn = document.getElementById('btn-build
-active-rag');
  if (buildActiveRagBtn) {
    buildActiveRagBtn.addEventListener('click', buildActiveUse
rRag);
  }

  const deleteActiveRagBtn = document.getElementById('btn-dele
te-active-rag');
  if (deleteActiveRagBtn) {
    deleteActiveRagBtn.addEventListener('click', deleteActiveU
serRag);
  }

  const confirmAddQaBtn = document.getElementById('btn-confirm
-add-qa');
  if (confirmAddQaBtn) {
    confirmAddQaBtn.addEventListener('click', addQaEntry);
  }

  const refreshEntriesBtn = document.getElementById('btn-refre
sh-rag-entries');
  if (refreshEntriesBtn) {
    refreshEntriesBtn.addEventListener('click', () => loadRagE
ntries());
  }

  const entriesFilterInput = document.getElementById('user-rag
-entries-filter');
  if (entriesFilterInput) {
    let filterDebounce = null;
    entriesFilterInput.addEventListener('input', (e) => {
      clearTimeout(filterDebounce);
```

```
      filterDebounce = setTimeout(() => {
        loadRagEntries(undefined, e.target.value.trim());
      }, 300);
    });
  }

  const clearFilterBtn = document.getElementById('btn-clear-entries-filter');
  if (clearFilterBtn) {
    clearFilterBtn.addEventListener('click', () => {
      if (entriesFilterInput) {
        entriesFilterInput.value = '';
        loadRagEntries(undefined, '');
      }
    });
  }

  const searchActiveRagBtn = document.getElementById('btn-active-rag-search');

  const searchActiveRagInput = document.getElementById('active-rag-search-query');
  if (searchActiveRagBtn) {
    searchActiveRagBtn.addEventListener('click', executeActiveUserRagSearch);
  }
  if (searchActiveRagInput) {
    searchActiveRagInput.addEventListener('keydown', (e) => {
      if (e.key === 'Enter') executeActiveUserRagSearch();
    });
  }

  const quickUploadInput = document.getElementById('user-rag-quick-upload-input');
  if (quickUploadInput) {
    quickUploadInput.addEventListener('change', handleQuickUploadToUserStorage);
  }
```

```
228
229   const selectAllBtn = document.getElementById('btn-select-all
-user-files');
230   if (selectAllBtn) {
231     selectAllBtn.addEventListener('click', () => {
232       const checkboxes = document.querySelectorAll('#user-rag-
files-checklist input[type="checkbox"]');
233       const anyUnchecked = Array.from(checkboxes).some(cb => !
cb.checked);
234       checkboxes.forEach(cb => cb.checked = anyUnchecked);
235     });
236   }
237
238   const confirmImportGdocsBtn = document.getElementById('btn-c
onfirm-import-gdocs');
239   if (confirmImportGdocsBtn) {
240     confirmImportGdocsBtn.addEventListener('click', syncGoogle
DocsIntoActiveRag);
241   }
242
243   // --- Global Document RAG Listeners ---
244   const fileInput = document.getElementById('rag-file-input');
245   const folderInput = document.getElementById('rag-folder-inpu
t');
246   const dropzone = document.getElementById('rag-dropzone');
247
248   if (fileInput) {
249     fileInput.addEventListener('change', (e) => {
250       if (e.target.files && e.target.files.length > 0) {
251         uploadFiles(Array.from(e.target.files));
252         fileInput.value = '';
253       }
254     });
255   }
256
257   if (folderInput) {
258     folderInput.addEventListener('change', (e) => {
259       if (e.target.files && e.target.files.length > 0) {
```

```
        uploadFiles(Array.from(e.target.files));
        folderInput.value = '';
      }
    });
  }

  if (dropzone) {
    ['dragenter', 'dragover'].forEach(eventName => {
      dropzone.addEventListener(eventName, (e) => {
        e.preventDefault();
        e.stopPropagation();
        dropzone.classList.add('border-primary', 'bg-primary', 'bg-opacity-10');
      }, false);
    });

    ['dragleave', 'drop'].forEach(eventName => {
      dropzone.addEventListener(eventName, (e) => {
        e.preventDefault();
        e.stopPropagation();
        dropzone.classList.remove('border-primary', 'bg-primary', 'bg-opacity-10');
      }, false);
    });

    dropzone.addEventListener('drop', async (e) => {
      const dt = e.dataTransfer;
      if (dt) {
        const files = await getAllFilesFromDataTransfer(dt);
        if (files && files.length > 0) {
          uploadFiles(files);
        }
      }
    }, false);
  }

  const buildBtn = document.getElementById('btn-build-rag-index');
```

```
  if (buildBtn) {
    buildBtn.addEventListener('click', buildRagIndex);
  }

  const searchBtn = document.getElementById('btn-rag-search');
  const searchInput = document.getElementById('rag-search-quer
y');
  if (searchBtn) {
    searchBtn.addEventListener('click', executeRagSearch);
  }
  if (searchInput) {
    searchInput.addEventListener('keydown', (e) => {
      if (e.key === 'Enter') executeRagSearch();
    });
  }

  // --- Codebase RAG Listeners ---
  const btnRefreshCodebase = document.getElementById('btn-refr
esh-codebase-indexes');
  if (btnRefreshCodebase) {
    btnRefreshCodebase.addEventListener('click', loadCodebaseI
ndexes);
  }

  const btnBuildCodebase = document.getElementById('btn-build-
codebase-rag');
  if (btnBuildCodebase) {
    btnBuildCodebase.addEventListener('click', buildCodebaseRa
g);
  }

  const btnCodeSearch = document.getElementById('btn-codebase-
search');
  const inputCodeSearch = document.getElementById('codebase-se
arch-query');
  if (btnCodeSearch) {
    btnCodeSearch.addEventListener('click', executeCodebaseSea
rch);
```

```
  }
  if (inputCodeSearch) {
    inputCodeSearch.addEventListener('keydown', (e) => {
      if (e.key === 'Enter') executeCodebaseSearch();
    });
  }

  const btnSymbolSearch = document.getElementById('btn-codebase-symbol-search');
  const inputSymbolSearch = document.getElementById('codebase-symbol-query');
  if (btnSymbolSearch) {
    btnSymbolSearch.addEventListener('click', executeSymbolLookup);
  }
  if (inputSymbolSearch) {
    inputSymbolSearch.addEventListener('keydown', (e) => {
      if (e.key === 'Enter') executeSymbolLookup();
    });
  }

  const activeIndexSelect = document.getElementById('codebase-active-index-select');
  if (activeIndexSelect) {
    activeIndexSelect.addEventListener('change', updateCodebaseStatsDisplay);
  }
}

// Auto-initialize when loaded or DOM ready
window.initRagTab = initRagTab;

if (document.readyState === 'loading') {
  document.addEventListener('DOMContentLoaded', initRagTab);
} else {
  initRagTab();
}

```

cache.js

```
/**
 * ============================================================
 * Process Name: Windows Modules - Cache Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля cache.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/rag_tab/modules/cache.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { CACHE_TAG, updateRagCacheStatus } from '/windows/api/webgui/rag_tab/modules/cache.js';
 *
 * File: cache.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/rag_tab/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

import { cachedApiFetch, clearCacheByTag } from '../../js/api-cache.js';

export const CACHE_TAG = 'rag';

/**
 * Cached fetch wrapper for RAG tab
 * @param {string} url - API endpoint
```

```
 * @param {object} options - Fetch options
 * @returns {Promise<Response>}
 */
export async function ragCachedFetch(url, options = {}) {
  // Merge cache tags
  const tags = options.tags || [];
  if (!tags.includes(CACHE_TAG)) {
    tags.push(CACHE_TAG);
  }

  return cachedApiFetch(url, { ...options, tags });
}

/**
 * Clear all RAG cache
 * @returns {Promise<void>}
 */
export async function clearRagCache() {
  await clearCacheByTag(CACHE_TAG);
  console.log('RAG cache cleared');
}

/**
 * Update cache status UI
 */
export function updateRagCacheStatus() {
  const badge = document.getElementById('rag-cache-badge');
  const ratio = document.getElementById('rag-cache-ratio');
  if (!badge || !ratio) return;

  try {
    const stats = window.apiCacheStats || { hits: 0, total: 0
};
    const hitRate = stats.total > 0 ? Math.round((stats.hits /
 stats.total) * 100) : 0;
    ratio.textContent = `${hitRate}%`;

    // Color coding
```

```
    badge.className = 'badge cache-badge';
    if (hitRate >= 70) {
      badge.classList.add('bg-success-subtle', 'text-success')
;
    } else if (hitRate >= 40) {
      badge.classList.add('bg-warning-subtle', 'text-warning')
;
    } else {
      badge.classList.add('bg-secondary-subtle', 'text-seconda
ry');
    }
  } catch (err) {
    console.debug('Failed to update RAG cache status:', err);
  }
}

```

codebaseRag.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Modules - Codebaserag Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля codebaseR
ag.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/rag_tab/modules/codeba
seRag.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { updateCodebaseStatsDisplay } from '/windows/ap
i/webgui/rag_tab/modules/codebaseRag.js';
 *
 * File: codebaseRag.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/rag_tab/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

import { escapeHtml } from './utils.js';
import { ragCachedFetch, CACHE_TAG } from './cache.js';

let cachedCodebaseIndexes = [];

/**
 * Fetch list of available codebase AST indexes and populate i
```

```
ndex selector.
 *
 * @returns {Promise<void>}
 */
export async function loadCodebaseIndexes() {
  const select = document.getElementById('codebase-active-inde
x-select');
  if (!select) return;

  try {
    const res = await ragCachedFetch('/api/rag/codebase/indexe
s', {
      tags: ['rag-codebase']
    });
    if (!res.ok) return;
    const data = await res.json();
    cachedCodebaseIndexes = data.indexes || [];

    if (cachedCodebaseIndexes.length === 0) {
      select.innerHTML = '<option value="codebase">codebase (П
о умолчанию)</option>';
    } else {
      const curVal = select.value;
      select.innerHTML = cachedCodebaseIndexes.map(idx => {
        const scopeBadge = idx.scope === 'user' ? '  ' : '
';
        const label = `${scopeBadge}${idx.name} (${idx.total_c
hunks || 0} чанков)`;
        return `<option value="${escapeHtml(idx.name)}">${esca
peHtml(label)}</option>`;
      }).join('');

      if (curVal && cachedCodebaseIndexes.some(i => i.name ===
 curVal)) {
        select.value = curVal;
      }
    }
    updateCodebaseStatsDisplay();
```

```
  } catch (err) {
    console.error('[Codebase RAG] Error loading indexes:', err);
  }
}

/**
 * Update UI counters for the currently selected codebase index.
 */
export function updateCodebaseStatsDisplay() {
  const select = document.getElementById('codebase-active-index-select');
  const statChunks = document.getElementById('codebase-stat-chunks');
  const statSymbols = document.getElementById('codebase-stat-symbols');
  if (!select) return;

  const currentName = select.value;
  const item = cachedCodebaseIndexes.find(i => i.name === currentName);
  if (item) {
    if (statChunks) statChunks.textContent = `${item.total_chunks || 0} чанков`;
    if (statSymbols) statSymbols.textContent = `${item.total_symbols || 0} символов AST`;
  }
}

/**
 * Build or rebuild codebase AST and semantic index for a target directory.
 *
 * @returns {Promise<void>}
 */
export async function buildCodebaseRag() {
  const btn = document.getElementById('btn-build-codebase-rag'
```

```
);
  const alertEl = document.getElementById('codebase-build-status-alert');
  const rootInput = document.getElementById('codebase-project-root-input');
  const nameInput = document.getElementById('codebase-index-name-input');
  const dirsInput = document.getElementById('codebase-include-dirs-input');

  const projectRoot = rootInput?.value?.trim() || '.';
  const indexName = nameInput?.value?.trim() || 'codebase';
  const includeDirs = (dirsInput?.value || '').split(',').map(s => s.trim()).filter(Boolean);

  if (btn) btn.disabled = true;
  if (alertEl) {
    alertEl.className = 'alert alert-info p-2 small mt-3 mb-0';
    alertEl.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span> Анализ AST и построение индексов...';
    alertEl.classList.remove('d-none');
  }

  try {
    const res = await fetch('/api/rag/codebase/build', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        project_root: projectRoot,
        index_name: indexName,
        include_dirs: includeDirs,
      }),
    });

    const data = await res.json();
    if (!res.ok) throw new Error(data.detail || data.error || `HTTP ${res.status}`);
```

```
119
120     const stats = data.result || {};
121     if (alertEl) {
122       alertEl.className = 'alert alert-success p-2 small mt-3
mb-0';
123       alertEl.innerHTML = `
124         <strong>✅ Индекс «${escapeHtml(indexName)}» успешно со
здан!</strong><br>
125         Файлов: ${stats.total_files || 0} | Чанков: ${stats.to
tal_chunks || 0} | Символов AST: ${stats.total_symbols || 0} (
${stats.elapsed_seconds || 0}с)
126       `;
127     }
128
129     await loadCodebaseIndexes();
130     const select = document.getElementById('codebase-active-in
dex-select');
131     if (select) select.value = indexName;
132     updateCodebaseStatsDisplay();
133   } catch (err) {
134     console.error('[Codebase RAG] Build error:', err);
135     if (alertEl) {
136       alertEl.className = 'alert alert-danger p-2 small mt-3 m
b-0';
137       alertEl.textContent = `Ошибка сборки: ${err.message}`;
138     }
139   } finally {
140     if (btn) btn.disabled = false;
141   }
142 }
143
144 /**
145  * Execute semantic query against codebase chunks.
146  *
147  * @returns {Promise<void>}
148  */
149 export async function executeCodebaseSearch() {
150   const queryInput = document.getElementById('codebase-search-
```

```
query');
  const select = document.getElementById('codebase-active-index-select');
  const resultsContainer = document.getElementById('codebase-search-results');
  const btn = document.getElementById('btn-codebase-search');

  const query = queryInput?.value?.trim();
  if (!query) return;

  const indexName = select?.value || 'codebase';
  if (btn) btn.disabled = true;
  if (resultsContainer) {
    resultsContainer.innerHTML = '<div class="text-center py-4 text-primary"><span class="spinner-border spinner-border-sm me-2"></span> Поиск по коду...</div>';
  }

  try {
    const res = await fetch('/api/rag/codebase/search', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        query,
        index_name: indexName,
        top_k: 5
      }),
    });

    const data = await res.json();
    if (!res.ok) throw new Error(data.detail || `HTTP ${res.status}`);

    const results = data.results || [];
    if (results.length === 0) {
      resultsContainer.innerHTML = `<div class="text-center text-muted py-4">По запросу «${escapeHtml(query)}» совпадений не найдено.</div>`;
```

```
      return;
    }

    resultsContainer.innerHTML = results.map(r => {
      return `
        <div class="card mb-2 border shadow-sm">
          <div class="card-header py-1 px-2 bg-body-secondary
d-flex justify-content-between align-items-center">
            <span class="small fw-semibold text-truncate">
              <i class="bi bi-code-slash me-1 text-primary"></
i> ${escapeHtml(r.id)}
            </span>
            <span class="badge bg-info">Score: ${r.similarity_
score || 0}</span>
          </div>
          <div class="card-body p-2 font-monospace small bg-bo
dy-tertiary" style="white-space: pre-wrap; font-size: 0.8rem;
max-height: 140px; overflow-y: auto;">${escapeHtml(r.text)}</d
iv>
        </div>
      `;
    }).join('');
  } catch (err) {
    if (resultsContainer) {
      resultsContainer.innerHTML = `<div class="alert alert-da
nger p-2 small mb-0">Ошибка: ${escapeHtml(err.message)}</div>`
;
    }
  } finally {
    if (btn) btn.disabled = false;
  }
}

/**
 * Execute symbol lookup in codebase AST symbols table.
 *
 * @returns {Promise<void>}
 */
```

```
export async function executeSymbolLookup() {
  const symbolInput = document.getElementById('codebase-symbol-query');
  const select = document.getElementById('codebase-active-index-select');
  const resultsContainer = document.getElementById('codebase-symbol-results');
  const btn = document.getElementById('btn-codebase-symbol-search');

  const symbol = symbolInput?.value?.trim();
  if (!symbol) return;

  const indexName = select?.value || 'codebase';
  if (btn) btn.disabled = true;
  if (resultsContainer) {
    resultsContainer.innerHTML = '<div class="text-center py-4 text-success"><span class="spinner-border spinner-border-sm me-2"></span> Поиск символов в AST...</div>';
  }

  try {
    const res = await fetch('/api/rag/codebase/symbols', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        symbol,
        index_name: indexName,
        limit: 10
      }),
    });

    const data = await res.json();
    if (!res.ok) throw new Error(data.detail || `HTTP ${res.status}`);

    const results = data.results || [];
    if (results.length === 0) {
```

```
      resultsContainer.innerHTML = `<div class="text-center te
xt-muted py-4">Символ «${escapeHtml(symbol)}» не найден в AST-
индексе.</div>`;
      return;
    }

    resultsContainer.innerHTML = results.map(s => {
      return `
        <div class="card mb-2 border border-success-subtle sha
dow-sm">
          <div class="card-header py-1 px-2 bg-success-subtle
text-success-emphasis d-flex justify-content-between align-ite
ms-center">
            <span class="small fw-semibold">
              <i class="bi bi-tag-fill me-1"></i> ${escapeHtml
(s.symbol)}
            </span>
            <span class="badge bg-success">${escapeHtml(s.type
)}</span>
          </div>
          <div class="card-body p-2 small">
            <div><strong>Path:</strong> <code>${escapeHtml(s.p
ath)}</code></div>
            <div><strong>Signature:</strong> <code>${escapeHtm
l(s.signature || '')}</code></div>
            ${s.docstring ? `<div class="mt-1 text-muted"><em>
${escapeHtml(s.docstring)}</em></div>` : ''}
            ${s.related_symbols?.length ? `<div class="mt-1 sm
all text-secondary">Related: ${escapeHtml(s.related_symbols.sl
ice(0, 8).join(', '))}</div>` : ''}
          </div>
        </div>
      `;
    }).join('');
  } catch (err) {
    if (resultsContainer) {
      resultsContainer.innerHTML = `<div class="alert alert-da
nger p-2 small mb-0">Ошибка поиска: ${escapeHtml(err.message)}
```

```
</div>`;
    }
  } finally {
    if (btn) btn.disabled = false;
  }
}

```

docRag.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Modules - Docrag Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля docRag.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/rag_tab/modules/docRag
.js?v=20261001_v1" type="module"></script>
 *
 * File: docRag.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/rag_tab/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

import { formatBytes, escapeHtml } from './utils.js';
import { ragCachedFetch, CACHE_TAG } from './cache.js';

/**
 * Upload one or multiple files to the global document RAG sto
rage.
 *
 * @param {File[]} files - List of files to upload.
 * @returns {Promise<void>}
 */
export async function uploadFiles(files) {
  if (!files || files.length === 0) return;
```

```
32
33    const progressContainer = document.getElementById('rag-uploa
d-progress-container');
34    const progressBar = document.getElementById('rag-upload-prog
ress-bar');
35    const percentText = document.getElementById('rag-upload-perc
ent');
36    const statusAlert = document.getElementById('rag-upload-stat
us-alert');
37
38    if (progressContainer) progressContainer.classList.remove('d
-none');
39    if (statusAlert) statusAlert.classList.add('d-none');
40    if (progressBar) progressBar.style.width = '0%';
41    if (percentText) percentText.textContent = '0%';
42
43    const formData = new FormData();
44    files.forEach((file) => {
45      const uploadName = file.customRelativePath || file.webkitR
elativePath || file.name;
46      formData.append('files', file, uploadName);
47    });
48
49    try {
50      if (progressBar) progressBar.style.width = '50%';
51      if (percentText) percentText.textContent = '50%';
52
53      const res = await fetch('/api/rag/upload', {
54        method: 'POST',
55        body: formData,
56      });
57
58      if (progressBar) progressBar.style.width = '100%';
59      if (percentText) percentText.textContent = '100%';
60
61      const data = await res.json();
62      if (!res.ok) throw new Error(data.detail || `Upload failed
 (HTTP ${res.status})`);
```

```
63
64      if (statusAlert) {
65        statusAlert.className = 'alert alert-success p-2 small m
b-0';
66        statusAlert.textContent = `Успешно загружено файлов: ${d
ata.uploaded?.length || 0}`;
67        statusAlert.classList.remove('d-none');
68      }
69
70      await loadRagDocuments();
71      await loadRagStatus();
72    } catch (err) {
73      console.error('[RAG] Upload error:', err);
74      if (statusAlert) {
75        statusAlert.className = 'alert alert-danger p-2 small mb
-0';
76        statusAlert.textContent = `Ошибка загрузки: ${err.messag
e}`;
77        statusAlert.classList.remove('d-none');
78      }
79    } finally {
80      setTimeout(() => {
81        if (progressContainer) progressContainer.classList.add('
d-none');
82      }, 1500);
83    }
84  }
85
86  /**
87   * Fetch and display global RAG index status and stats.
88   *
89   * @returns {Promise<void>}
90   */
91  export async function loadRagStatus() {
92    try {
93      const res = await ragCachedFetch('/api/rag/status', {
94        tags: ['rag-status']
95      });
```

```
    if (!res.ok) return;
    const data = await res.json();
    const st = data.data || {};

    const statDocs = document.getElementById('rag-stat-docs');
    const statChunks = document.getElementById('rag-stat-chunk
s');
    const statDim = document.getElementById('rag-stat-dim');
    const statUpdated = document.getElementById('rag-stat-upda
ted');
    const providerBadge = document.getElementById('rag-provide
r-badge');

    if (statDocs) statDocs.textContent = st.total_documents ||
 0;
    if (statChunks) statChunks.textContent = st.total_chunks |
| 0;
    if (statDim) statDim.textContent = st.vector_dimension ||
0;
    if (statUpdated) {
      statUpdated.textContent = st.last_built_at
        ? new Date(st.last_built_at * 1000).toLocaleTimeString
()
        : '—';
    }
    if (providerBadge) {
      const provName = st.provider === 'gemini' ? 'Google Gemi
ni Embeddings' : 'Local TF-IDF';
      providerBadge.textContent = `Provider: ${provName}`;
    }
  } catch (err) {
    console.error('[RAG] Error loading status:', err);
  }
}

/**
 * Fetch and render the list of uploaded global RAG documents.
 *
```

```
 * @returns {Promise<void>}
 */
export async function loadRagDocuments() {
  const tbody = document.getElementById('rag-documents-tbody')
;
  const badge = document.getElementById('rag-docs-badge');
  if (!tbody) return;

  try {
    const res = await ragCachedFetch('/api/rag/documents', {
      tags: ['rag-documents']
    });
    if (!res.ok) return;
    const data = await res.json();
    const docs = data.documents || [];

    if (badge) badge.textContent = `${docs.length} файлов`;
    if (docs.length === 0) {
      tbody.innerHTML = '<tr><td colspan="5" class="text-cente
r text-muted py-3">Документы не загружены.</td></tr>';
      return;
    }

    tbody.innerHTML = docs.map((d, idx) => {
      const statusBadge = d.status === 'indexed'
        ? '<span class="badge bg-success">Индексирован</span>'
        : (d.status === 'error' ? '<span class="badge bg-dange
r">Ошибка</span>' : '<span class="badge bg-warning text-dark">
Ожидание</span>');

      return `
        <tr class="rag-doc-row" data-idx="${idx}" style="curso
r: pointer;" title="Нажмите для AI-анализа документа RAG">
          <td class="ps-3 text-truncate" style="max-width: 200
px;" title="${escapeHtml(d.name)}">
            <i class="bi bi-file-earmark-code me-1 text-primar
y"></i> ${escapeHtml(d.name)}
          </td>
```

```
          <td>${formatBytes(d.size_bytes)}</td>
          <td><span class="badge bg-secondary-subtle text-seco
ndary">${d.chunks_count || 0}</span></td>
          <td>${statusBadge}</td>
          <td class="text-end pe-3">
            <button class="btn btn-outline-danger btn-sm p-1 p
y-0 btn-del-rag-doc" data-name="${escapeHtml(d.name)}" title="
Удалить">
              <i class="bi bi-trash"></i>
            </button>
          </td>
        </tr>
      `;
    }).join('');

    tbody.querySelectorAll('.rag-doc-row').forEach(row => {
      row.onclick = (evt) => {
        if (evt.target.closest('.btn-del-rag-doc')) return;
        const idx = parseInt(row.getAttribute('data-idx'), 10)
;
        const d = docs[idx];
        if (!d) return;

        if (window.AITableModal) {
          window.AITableModal.show({
            icon: '□ ',
            title: d.name,
            subtitle: `Размер: ${formatBytes(d.size_bytes)} |
Чанков: ${d.chunks_count || 0}`,
            tableType: 'rag_doc',
            badges: [
              { text: d.status || 'Indexed', class: d.status =
== 'indexed' ? 'badge bg-success' : 'badge bg-secondary' },
              { text: `${d.chunks_count || 0} chunks`, class:
'badge bg-info text-dark' }
            ],
            metadata: [
              { label: 'Имя документа', value: d.name },
```

```
              { label: 'Статус индексации', value: d.status ||
 'Indexed' },
              { label: 'Количество чанков', value: String(d.ch
unks_count || 0) },
              { label: 'Размер файла', value: formatBytes(d.si
ze_bytes) },
              { label: 'Дата индексации', value: d.updated_at
|| d.created_at || 'N/A' },
              { label: 'Путь источника', value: d.source_path
|| d.name, isCode: true, fullWidth: true }
            ],
            rawTitle: 'Метаданные документа',
            rawContent: JSON.stringify(d, null, 2),
            requestData: {
              name: d.name,
              chunks_count: d.chunks_count,
              size: formatBytes(d.size_bytes),
              status: d.status
            }
          });
        }
      };
    });

    tbody.querySelectorAll('.btn-del-rag-doc').forEach(btn =>
{
      btn.onclick = (evt) => {
        evt.stopPropagation();
        const docName = btn.getAttribute('data-name');
        deleteRagDocument(docName);
      };
    });
  } catch (err) {
    console.error('[RAG] Error loading documents:', err);
  }
}

/**
```

```
 * Delete a specific document from global RAG.
 *
 * @param {string} filename - Name of the document to delete.
 * @returns {Promise<void>}
 */
export async function deleteRagDocument(filename) {
  if (!confirm(`Удалить документ «${filename}» из базы знаний?`)) return;
  try {
    const res = await fetch(`/api/rag/documents/${encodeURIComponent(filename)}`, {
      method: 'DELETE',
    });
    window.showToast?.(`Документ «${filename}» успешно удален`, 'success');
    await loadRagDocuments();
    await loadRagStatus();
  } catch (err) {
    window.showToast?.(`Ошибка удаления: ${err.message}`, 'danger') || alert(`Ошибка удаления: ${err.message}`);
  }
}

/**
 * Trigger global RAG index build.
 *
 * @returns {Promise<void>}
 */
export async function buildRagIndex() {
  const buildBtn = document.getElementById('btn-build-rag-index');
  const statusEl = document.getElementById('rag-build-status');
  const providerSelect = document.getElementById('rag-provider-select');
  const apiKeyInput = document.getElementById('rag-api-key-input');
  const chunkSizeInput = document.getElementById('rag-chunk-si
```

```
ze');
  const chunkOverlapInput = document.getElementById('rag-chunk
-overlap');

  if (buildBtn) buildBtn.disabled = true;
  if (statusEl) {
    statusEl.innerHTML = '<span class="spinner-border spinner-
border-sm me-1"></span> Построение векторного индекса...';
  }

  try {
    const res = await fetch('/api/rag/build', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        provider: providerSelect?.value || 'auto',
        api_key: apiKeyInput?.value?.trim() || '',
        chunk_size: parseInt(chunkSizeInput?.value || '500', 1
0),
        chunk_overlap: parseInt(chunkOverlapInput?.value || '5
0', 10),
      }),
    });

    const data = await res.json();
    if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
atus}`);

    if (statusEl) {
      statusEl.className = 'small mt-2 text-center text-succes
s fw-semibold';
      statusEl.textContent = `  Индекс успешно построен. Всего
 чанков: ${data.result?.total_chunks || 0}`;
    }

    await loadRagStatus();
    await loadRagDocuments();
  } catch (err) {
```

```
    console.error('[RAG] Build error:', err);
    if (statusEl) {
      statusEl.className = 'small mt-2 text-center text-danger
';
      statusEl.textContent = `  Ошибка индексации: ${err.messa
ge}`;
    }
  } finally {
    if (buildBtn) buildBtn.disabled = false;
  }
}

/**
 * Execute semantic query against global RAG index.
 *
 * @returns {Promise<void>}
 */
export async function executeRagSearch() {
  const queryInput = document.getElementById('rag-search-query
');
  const topKSelect = document.getElementById('rag-top-k');
  const minScoreInput = document.getElementById('rag-min-score
');
  const resultsContainer = document.getElementById('rag-search
-results');
  const searchBtn = document.getElementById('btn-rag-search');
  const apiKeyInput = document.getElementById('rag-api-key-inp
ut');

  const query = queryInput?.value?.trim();
  if (!query) return;

  if (searchBtn) searchBtn.disabled = true;
  if (resultsContainer) {
    resultsContainer.innerHTML = '<div class="text-center py-4
 text-primary"><span class="spinner-border spinner-border-sm m
e-2"></span> Идет семантический поиск...</div>';
  }
```

```
310
311   try {
312     const res = await fetch('/api/rag/search', {
313       method: 'POST',
314       headers: { 'Content-Type': 'application/json' },
315       body: JSON.stringify({
316         query,
317         top_k: parseInt(topKSelect?.value || '5', 10),
318         min_score: parseFloat(minScoreInput?.value || '0.0'),
319         api_key: apiKeyInput?.value?.trim() || '',
320       }),
321     });
322
323     const data = await res.json();
324     if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
    atus}`);
325
326     const results = data.results || [];
327     if (results.length === 0) {
328       resultsContainer.innerHTML = `<div class="text-center te
    xt-muted py-4"><i class="bi bi-search fs-3 d-block mb-1 opacit
    y-50"></i>По запросу «${escapeHtml(query)}» ничего не найдено.
    </div>`;
329       return;
330     }
331
332     resultsContainer.innerHTML = results.map(r => `
333       <div class="card mb-2 border shadow-sm">
334         <div class="card-header py-1 px-2 bg-body-secondary d-
    flex justify-content-between align-items-center">
335           <span class="small fw-semibold text-truncate" style=
    "max-width: 70%;">
336             <i class="bi bi-file-earmark-text me-1 text-primar
    y"></i> ${escapeHtml(r.doc_name)} (#${r.chunk_index})
337           </span>
338           <span class="badge bg-secondary">Score: ${r.score}</
    span>
339         </div>
```

```
        <div class="card-body p-2 font-monospace small bg-body-tertiary" style="white-space: pre-wrap; font-size: 0.8rem; max-height: 120px; overflow-y: auto;">${escapeHtml(r.text)}</div>
      </div>
    `).join('');
  } catch (err) {
    if (resultsContainer) {
      resultsContainer.innerHTML = `<div class="alert alert-danger p-2 small mb-0">Ошибка: ${escapeHtml(err.message)}</div>`;
    }
  } finally {
    if (searchBtn) searchBtn.disabled = false;
  }
}

```

pixelRag.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Modules - Pixelrag Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля pixelRag.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/rag_tab/modules/pixelR
ag.js?v=20261001_v1" type="module"></script>
 *
 * File: pixelRag.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/rag_tab/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

import { formatBytes, escapeHtml } from './utils.js';

/**
 * Load and display PixelRAG system status, FAISS index metric
s, and image counts.
 *
 * @returns {Promise<void>}
 */
export async function loadPixelRagStatus() {
  try {
    const res = await fetch('/api/rag/pixel/status');
    if (!res.ok) return;
```

```
32
33      const data = await res.json();
34      const imagesEl = document.getElementById('pixel-stat-image
s');
35      const chunksEl = document.getElementById('pixel-stat-chunk
s');
36      const sizeEl = document.getElementById('pixel-stat-size');
37      const providerEl = document.getElementById('pixel-stat-pro
vider');
38      const updatedEl = document.getElementById('pixel-stat-upda
ted');
39      const statusEl = document.getElementById('pixel-stat-statu
s');
40
41      if (imagesEl) imagesEl.textContent = data.total_images ||
0;
42      if (chunksEl) chunksEl.textContent = data.total_chunks ||
0;
43      if (sizeEl) sizeEl.textContent = `${(data.total_size_mb ||
 0).toFixed(2)} MB`;
44      if (providerEl) providerEl.textContent = data.provider ||
'FAISS / PixelRAG';
45      if (statusEl) statusEl.textContent = `Текстовых чанков: ${
data.sources?.text_chunks || 0} | Картинок: ${data.sources?.im
ages || 0}`;
46
47      if (updatedEl) {
48        if (data.last_built_at && data.last_built_at > 0) {
49          const d = new Date(data.last_built_at * 1000);
50          updatedEl.textContent = d.toLocaleString();
51        } else {
52          updatedEl.textContent = 'Индекс готов';
53        }
54      }
55    } catch (err) {
56      console.debug('[PixelRAG] Failed to load status:', err);
57    }
58  }
```

```
59
60 /**
61  * Upload and index image files into PixelRAG.
62  *
63  * @param {File[]} files - List of image files.
64  * @returns {Promise<void>}
65  */
66 export async function uploadAndIndexPixelImages(files) {
67   if (!files || files.length === 0) return;
68
69   const progressContainer = document.getElementById('pixel-upl
   oad-progress');
70   const progressBar = document.getElementById('pixel-upload-pr
   ogress-bar');
71   const percentText = document.getElementById('pixel-upload-pe
   rcent');
72   const statusAlert = document.getElementById('pixel-upload-st
   atus-alert');
73
74   if (progressContainer) progressContainer.classList.remove('d
   -none');
75   if (statusAlert) statusAlert.classList.add('d-none');
76   if (progressBar) progressBar.style.width = '20%';
77   if (percentText) percentText.textContent = '20%';
78
79   const formData = new FormData();
80   files.forEach((file) => {
81     formData.append('files', file, file.name);
82   });
83
84   try {
85     if (progressBar) progressBar.style.width = '60%';
86     if (percentText) percentText.textContent = '60%';
87
88     const res = await fetch('/api/rag/pixel/index', {
89       method: 'POST',
90       body: formData,
91     });
```

```
92
93      if (progressBar) progressBar.style.width = '100%';
94      if (percentText) percentText.textContent = '100%';
95
96      const data = await res.json();
97      if (!res.ok) throw new Error(data.detail || `Индексация не
 удалась (HTTP ${res.status})`);
98
99      if (statusAlert) {
100       statusAlert.className = 'alert alert-success p-2 small m
b-0';
101       statusAlert.textContent = `  Успешно проиндексировано: $
{data.indexed_count || files.length} файл(ов)`;
102       statusAlert.classList.remove('d-none');
103     }
104
105     await loadPixelRagStatus();
106   } catch (err) {
107     console.error('[PixelRAG] Indexing error:', err);
108     if (statusAlert) {
109       statusAlert.className = 'alert alert-danger p-2 small mb
-0';
110       statusAlert.textContent = `Ошибка индексации: ${err.mess
age}`;
111       statusAlert.classList.remove('d-none');
112     }
113   } finally {
114     setTimeout(() => {
115       if (progressContainer) progressContainer.classList.add('
d-none');
116     }, 2000);
117   }
118 }
119
120 /**
121  * Execute smart query analysis via QueryRouter.
122  *
123  * @returns {Promise<void>}
```

```
 */
export async function analyzePixelQuery() {
  const input = document.getElementById('pixel-analyze-query-input');
  const resultBox = document.getElementById('pixel-analyze-result');
  const routingTypeEl = document.getElementById('pixel-routing-type');
  const confidenceEl = document.getElementById('pixel-routing-confidence');
  const reasonEl = document.getElementById('pixel-routing-reason');

  if (!input || !input.value.trim()) return;
  const query = input.value.trim();

  try {
    const res = await fetch(`/api/rag/pixel/analyze-query?query=${encodeURIComponent(query)}`, {
      method: 'POST',
    });
    if (!res.ok) throw new Error(`HTTP ${res.status}`);

    const data = await res.json();
    if (resultBox) resultBox.classList.remove('d-none');
    if (routingTypeEl) {
      routingTypeEl.textContent = data.routing_type?.toUpperCase() || 'HYBRID';
      routingTypeEl.className = data.routing_type === 'visual' ? 'badge bg-warning text-dark' : data.routing_type === 'text' ? 'badge bg-info' : 'badge bg-primary';
    }
    if (confidenceEl) confidenceEl.textContent = `${Math.round((data.confidence || 1.0) * 100)}%`;
    if (reasonEl) reasonEl.textContent = `Причина: ${data.reason || 'Авто-определение по ключевым признакам'}`;
  } catch (err) {
    console.error('[PixelRAG] Analysis failed:', err);
```

```
  }
}

/**
 * Execute semantic visual search inside PixelRAG.
 *
 * @returns {Promise<void>}
 */
export async function executePixelRagSearch() {
  const queryInput = document.getElementById('pixel-search-query');
  const topKSelect = document.getElementById('pixel-search-topk');
  const minScoreSelect = document.getElementById('pixel-search-minscore');
  const routingToggle = document.getElementById('pixel-search-routing-toggle');
  const resultsContainer = document.getElementById('pixel-search-results');

  if (!queryInput || !resultsContainer) return;
  const query = queryInput.value.trim();
  if (!query) {
    queryInput.focus();
    return;
  }

  const topK = parseInt(topKSelect?.value || '5', 10);
  const minScore = parseFloat(minScoreSelect?.value || '0.0');
  const useRouting = routingToggle ? routingToggle.checked : true;

  resultsContainer.innerHTML = `
    <div class="text-center py-4 text-muted">
      <div class="spinner-border spinner-border-sm text-primary me-2" role="status"></div>
      <span>Поиск по визуальным индексам и текстовым чанкам...</span>
```

```
    </div>
  `;

  try {
    const res = await fetch('/api/rag/pixel/search', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        query: query,
        top_k: topK,
        min_score: minScore,
        use_routing: useRouting,
        version_filter: 'latest',
      }),
    });

    if (!res.ok) {
      const errData = await res.json().catch(() => ({}));
      throw new Error(errData.detail || `HTTP ${res.status}`);
    }

    const data = await res.json();
    const results = data.results || [];

    if (results.length === 0) {
      resultsContainer.innerHTML = `
        <div class="text-center text-muted py-4">
          <i class="bi bi-search fs-3 d-block mb-1 opacity-50"
></i>
          <span>По запросу «${escapeHtml(query)}» ничего не на
йдено.</span>
        </div>
      `;
      return;
    }

    resultsContainer.innerHTML = `
      <div class="d-flex justify-content-between align-items-c
```

```
enter mb-2 pb-2 border-bottom">
        <span class="small text-muted">Найдено совпадений: <st
rong>${results.length}</strong> (Маршрут: <span class="badge b
g-secondary">${data.routing_type}</span>)</span>
        <span class="small text-muted">Уверенность: ${(data.qu
ery_confidence * 100).toFixed(0)}%</span>
      </div>
      <div class="d-flex flex-column gap-2">
        ${results.map((r, i) => {
          const scorePercent = Math.min(100, Math.round((r.sco
re || 0) * 100));
          const isVisual = r.source_type === 'pixel' || r.sour
ce_type === 'image';
          const typeBadge = isVisual
            ? '<span class="badge bg-warning text-dark"><i cla
ss="bi bi-image me-1"></i>Visual</span>'
            : '<span class="badge bg-info"><i class="bi bi-fil
e-text me-1"></i>Text</span>';

          return `
            <div class="card bg-body border-secondary-subtle p
-2">
              <div class="d-flex justify-content-between align
-items-center mb-1">
                <span class="fw-bold small text-primary text-t
runcate" style="max-width: 70%;">
                  #${i + 1} ${escapeHtml(r.doc_name || 'Без на
звания')}
                </span>
                <div class="d-flex align-items-center gap-1">
                  ${typeBadge}
                  <span class="badge bg-success-subtle text-su
ccess border border-success-subtle">
                    Score: ${(r.score || 0).toFixed(3)} (${sco
rePercent}%)
                  </span>
                </div>
              </div>
```

```
              <div class="small text-body-secondary font-monospace" style="font-size: 0.8rem; white-space: pre-wrap;">
                ${escapeHtml(r.text || '')}
              </div>
              ${r.source_path ? `<div class="small text-muted mt-1 text-truncate" style="font-size: 0.72rem;"><i class="bi bi-link-45deg"></i> ${escapeHtml(r.source_path)}</div>` : ''}
            </div>
          `;
        }).join('')}
      </div>
    `;
  } catch (err) {
    console.error('[PixelRAG] Search error:', err);
    resultsContainer.innerHTML = `
      <div class="alert alert-danger p-2 small mb-0">
        <i class="bi bi-exclamation-triangle me-1"></i> Ошибка поиска: ${escapeHtml(err.message)}
      </div>
    `;
  }
}

```

userRagCollections.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Modules - Userragcollections Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля userRagCo
llections.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/rag_tab/modules/userRa
gCollections.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { getActiveUserRagId, setActiveUserRagId, render
UserRagsList } from '/windows/api/webgui/rag_tab/modules/userR
agCollections.js';
 *
 * File: userRagCollections.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/rag_tab/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

import { escapeHtml } from './utils.js';
import { loadUserStorageFiles } from './userRagPipeline.js';

let activeUserRagId = null;
let userCollections = [];

```

```
/**
 * Get the currently active collection ID.
 *
 * @returns {string|null} Active RAG ID or null.
 */
export function getActiveUserRagId() {
  return activeUserRagId;
}

/**
 * Set the active collection ID.
 *
 * @param {string|null} id - Target collection ID.
 */
export function setActiveUserRagId(id) {
  activeUserRagId = id;
}

/**
 * Fetch all user workspace RAG collections and render sidebar
 list.
 *
 * @returns {Promise<void>}
 */
export async function loadUserRags() {
  const listContainer = document.getElementById('user-rags-lis
t-container');
  if (!listContainer) return;

  try {
    const res = await fetch('/api/user/rags');
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    userCollections = data.collections || [];
    renderUserRagsList(userCollections);

    if (userCollections.length > 0) {
      if (!activeUserRagId || !userCollections.some(c => c.id
```

```
=== activeUserRagId)) {
        selectUserRag(userCollections[0].id);
      } else {
        selectUserRag(activeUserRagId);
      }
    } else {
      activeUserRagId = null;
      showUserRagEmptyState();
    }
  } catch (err) {
    listContainer.innerHTML = `<div class="alert alert-danger p-2 small mb-0">Ошибка загрузки коллекций: ${escapeHtml(err.message)}</div>`;
  }
}

/**
 * Render the collection cards in the left sidebar.
 *
 * @param {Array<object>} collections - List of collection summary objects.
 */
export function renderUserRagsList(collections) {
  const listContainer = document.getElementById('user-rags-list-container');
  if (!listContainer) return;

  if (collections.length === 0) {
    listContainer.innerHTML = `
      <div class="text-center text-muted py-4 small">
        <i class="bi bi-folder-x fs-3 d-block mb-1 opacity-50"></i>
        Нет созданных коллекций.<br>Нажмите «+ Создать» выше.
      </div>
    `;
    return;
  }

```

```
  listContainer.innerHTML = collections.map(col => {
    const isActive = col.id === activeUserRagId;
    const statusBadgeClass = col.status === 'indexed'
      ? 'bg-success'
      : (col.status === 'cleaning' ? 'bg-warning text-dark' : (col.status === 'error' ? 'bg-danger' : 'bg-secondary'));

    return `
      <div class="card p-2 cursor-pointer user-rag-item ${isActive ? 'border-primary bg-primary-subtle' : 'border-secondary-subtle bg-body-tertiary'}"
           data-rag-id="${escapeHtml(col.id)}" style="cursor: pointer; transition: all 0.15s ease-in-out;">
        <div class="d-flex justify-content-between align-items-center mb-1">
          <strong class="text-truncate small" style="max-width: 140px;">${escapeHtml(col.name || col.id)}</strong>
          <span class="badge ${statusBadgeClass} small" style="font-size: 0.7rem;">${escapeHtml(col.status || 'created')}</span>
        </div>
        <div class="d-flex justify-content-between align-items-center small text-muted" style="font-size: 0.75rem;">
          <span><i class="bi bi-diagram-3 me-1"></i>${col.total_chunks || 0} чанков</span>
          <span><i class="bi bi-file-earmark me-1"></i>${(col.files || []).length} файлов</span>
        </div>
      </div>
    `;
  }).join('');

  listContainer.querySelectorAll('.user-rag-item').forEach(el => {
    el.addEventListener('click', () => {
      const ragId = el.getAttribute('data-rag-id');
      if (ragId) selectUserRag(ragId);
    });
```

```
  });
}

/**
 * Display placeholder empty state when no collection is selected.
 */
export function showUserRagEmptyState() {
  const emptyState = document.getElementById('user-rag-empty-state');
  const detailContainer = document.getElementById('user-rag-detail-container');
  if (emptyState) emptyState.classList.remove('d-none');
  if (detailContainer) detailContainer.classList.add('d-none');
}

/**
 * Select a workspace collection and load its metadata and file checklist.
 *
 * @param {string} ragId - ID of the collection to select.
 * @returns {Promise<void>}
 */
export async function selectUserRag(ragId) {
  activeUserRagId = ragId;
  const emptyState = document.getElementById('user-rag-empty-state');
  const detailContainer = document.getElementById('user-rag-detail-container');
  if (emptyState) emptyState.classList.add('d-none');
  if (detailContainer) detailContainer.classList.remove('d-none');

  document.querySelectorAll('.user-rag-item').forEach(el => {
    if (el.getAttribute('data-rag-id') === ragId) {
      el.classList.add('border-primary', 'bg-primary-subtle');
      el.classList.remove('border-secondary-subtle', 'bg-body-
```

```
tertiary');
    } else {
      el.classList.remove('border-primary', 'bg-primary-subtle');
      el.classList.add('border-secondary-subtle', 'bg-body-tertiary');
    }
  });

  try {
    const res = await fetch(`/api/user/rags/${encodeURIComponent(ragId)}`);
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    const col = data.collection;

    const titleEl = document.getElementById('active-rag-title');
    const descEl = document.getElementById('active-rag-desc');
    const statusBadge = document.getElementById('active-rag-status-badge');
    const chunksEl = document.getElementById('active-rag-stat-chunks');
    const filesEl = document.getElementById('active-rag-stat-files');
    const updatedEl = document.getElementById('active-rag-stat-updated');

    if (titleEl) titleEl.textContent = col.name || col.id;
    if (descEl) descEl.textContent = col.description || 'Без описания';
    if (statusBadge) {
      statusBadge.textContent = col.status || 'created';
      statusBadge.className = `badge ${col.status === 'indexed' ? 'bg-success' : (col.status === 'cleaning' ? 'bg-warning text-dark' : (col.status === 'error' ? 'bg-danger' : 'bg-secondary'))}`;
    }
```

```
    if (chunksEl) chunksEl.textContent = col.total_chunks || 0;
    if (filesEl) filesEl.textContent = (col.files || []).length;
    if (updatedEl) {
      updatedEl.textContent = col.updated_at ? new Date(col.updated_at * 1000).toLocaleString() : '—';
    }

    await loadUserStorageFiles(col.files || []);
    await loadRagEntries(ragId);
  } catch (err) {
    console.error('Failed to load user RAG details:', err);
  }
}

/**
 * Fetch and render chunk/QA entries for the active collection.
 *
 * @param {string} ragId - Collection identifier.
 * @param {string} query - Optional filter query.
 * @returns {Promise<void>}
 */
export async function loadRagEntries(ragId = activeUserRagId, query = '') {
  const container = document.getElementById('user-rag-entries-container');
  if (!container || !ragId) return;

  container.innerHTML = '<div class="text-center text-muted small py-2"><div class="spinner-border spinner-border-sm text-primary me-1"></div>Загрузка записей...</div>';

  try {
    const url = `/api/user/rags/${encodeURIComponent(ragId)}/entries?q=${encodeURIComponent(query)}&limit=100`;
    const res = await fetch(url);
```

```
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    renderRagEntries(data.entries || [], data.total || 0);
  } catch (err) {
    container.innerHTML = `<div class="alert alert-danger p-2 small mb-0">Ошибка загрузки записей: ${escapeHtml(err.message)}</div>`;
  }
}

/**
 * Render entries inside the collection container.
 *
 * @param {Array<object>} entries - List of chunk/QA items.
 * @param {number} total - Total entries count.
 */
export function renderRagEntries(entries, total) {
  const container = document.getElementById('user-rag-entries-container');
  if (!container) return;

  if (!entries || entries.length === 0) {
    container.innerHTML = `
      <div class="text-center text-muted py-3 small">
        <i class="bi bi-chat-left-dots fs-4 d-block mb-1 opacity-50"></i>
        В этой коллекции пока нет записей.<br>
        Нажмите «+ Добавить Q&A» или соберите RAG из файлов выше.
      </div>
    `;
    return;
  }

  container.innerHTML = entries.map((entry, idx) => {
    const chunkId = entry.chunk_id || `chunk_${idx}`;
    const docType = entry.doc_type || 'chunk';
    const isQa = docType === 'qa' || entry.meta?.is_qa;
```

```
    const badgeClass = isQa ? 'bg-primary' : 'bg-info text-dar
k';
    const badgeLabel = isQa ? 'Q&A' : (entry.source_file || 'Ф
айл');

    let displayContent = entry.content || '';
    let questionText = entry.meta?.question || '';
    let answerText = entry.meta?.answer || '';

    if (!questionText && displayContent.includes('Вопрос:') &&
 displayContent.includes('Ответ:')) {
      const parts = displayContent.split('\nОтвет:');
      questionText = parts[0].replace('Вопрос:', '').trim();
      answerText = (parts[1] || '').trim();
    }

    return `
      <div class="card p-2 border border-secondary-subtle bg-b
ody-tertiary position-relative rag-entry-card" data-chunk-id="
${escapeHtml(chunkId)}">
        <div class="d-flex justify-content-between align-items
-start mb-1">
          <div class="d-flex align-items-center gap-1 flex-wra
p">
            <span class="badge ${badgeClass} small" style="fon
t-size: 0.7rem;">${escapeHtml(badgeLabel)}</span>
            <small class="text-muted font-monospace" style="fo
nt-size: 0.72rem;">#${escapeHtml(chunkId)}</small>
          </div>
          <button class="btn btn-outline-danger btn-sm py-0 px
-2 rounded-pill btn-delete-entry" data-chunk-id="${escapeHtml(
chunkId)}" title="Удалить запись из RAG">
            <i class="bi bi-trash"></i>
          </button>
        </div>
        ${questionText ? `
          <div class="small fw-bold text-primary mb-1"><i clas
s="bi bi-question-circle me-1"></i>${escapeHtml(questionText)}
```

```
</div>
          <div class="small text-secondary font-monospace bg-body p-2 rounded border" style="white-space: pre-wrap; font-size: 0.78rem;">${escapeHtml(answerText || displayContent)}</div>
        ` : `
          <div class="small text-secondary font-monospace bg-body p-2 rounded border" style="white-space: pre-wrap; font-size: 0.78rem;">${escapeHtml(displayContent)}</div>
        `}
      </div>
    `;
  }).join('');

  container.querySelectorAll('.btn-delete-entry').forEach(btn => {
    btn.addEventListener('click', (e) => {
      e.stopPropagation();
      const chunkId = btn.getAttribute('data-chunk-id');
      if (chunkId) deleteRagEntry(chunkId);
    });
  });
}

/**
 * Add a new custom Q&A pair entry to the active collection.
 *
 * @returns {Promise<void>}
 */
export async function addQaEntry() {
  if (!activeUserRagId) {
    window.showToast?.('Пожалуйста, выберите коллекцию RAG.', 'warning') || alert('Пожалуйста, выберите коллекцию RAG.');
    return;
  }

  const qInput = document.getElementById('new-qa-question');
  const aInput = document.getElementById('new-qa-answer');
  const btn = document.getElementById('btn-confirm-add-qa');
```

```
298
299   const question = qInput?.value.trim() || '';
300   const answer = aInput?.value.trim() || '';
301
302   if (!answer) {
303     window.showToast?.('Пожалуйста, введите текст ответа или з
нания.', 'warning') || alert('Пожалуйста, введите текст ответа
 или знания.');
304     return;
305   }
306
307   if (btn) btn.disabled = true;
308   try {
309     const res = await fetch(`/api/user/rags/${encodeURICompone
nt(activeUserRagId)}/entries`, {
310       method: 'POST',
311       headers: { 'Content-Type': 'application/json' },
312       body: JSON.stringify({ question, answer })
313     });
314
315     const data = await res.json();
316     if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
atus}`);
317
318     const modalEl = document.getElementById('modal-add-qa-entr
y');
319     if (modalEl && window.bootstrap?.Modal) {
320       const modal = bootstrap.Modal.getInstance(modalEl) || ne
w bootstrap.Modal(modalEl);
321       modal.hide();
322     }
323
324     if (qInput) qInput.value = '';
325     if (aInput) aInput.value = '';
326
327     window.showToast?.('Q&A запись успешно добавлена', 'succes
s');
328     await selectUserRag(activeUserRagId);
```

```
    await loadUserRags();
  } catch (err) {
    window.showToast?.(`Ошибка сохранения Q&A записи: ${err.message}`, 'danger') || alert(`Ошибка сохранения Q&A записи: ${err.message}`);
  } finally {
    if (btn) btn.disabled = false;
  }
}

/**
 * Delete a specific chunk/QA entry from the active collection.
 *
 * @param {string} chunkId - Chunk identifier.
 * @returns {Promise<void>}
 */
export async function deleteRagEntry(chunkId) {
  if (!activeUserRagId || !chunkId) return;

  const confirmed = confirm(`Удалить эту запись (#${chunkId}) из базы знаний?`);
  if (!confirmed) return;

  try {
    const res = await fetch(`/api/user/rags/${encodeURIComponent(activeUserRagId)}/entries/${encodeURIComponent(chunkId)}`, {
      method: 'DELETE'
    });
    if (!res.ok) throw new Error(`HTTP ${res.status}`);

    window.showToast?.('Запись успешно удалена', 'success');
    await selectUserRag(activeUserRagId);
    await loadUserRags();
  } catch (err) {
    window.showToast?.(`Ошибка удаления записи: ${err.message}`, 'danger') || alert(`Ошибка удаления записи: ${err.message}`
```

```
);
  }
}

/**
 * Create a new user workspace RAG collection via API.
 *
 * @returns {Promise<void>}
 */
export async function createUserRag() {
  const nameInput = document.getElementById('new-rag-name');
  const descInput = document.getElementById('new-rag-desc');
  const minChunkInput = document.getElementById('new-rag-min-chunk');
  const maxChunkInput = document.getElementById('new-rag-max-chunk');
  const btn = document.getElementById('btn-confirm-create-rag');

  const name = nameInput?.value.trim();
  if (!name) {
    window.showToast?.('Пожалуйста, укажите имя коллекции.', 'warning') || alert('Пожалуйста, укажите имя коллекции.');
    return;
  }

  if (btn) btn.disabled = true;
  try {
    const res = await fetch('/api/user/rags', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        name: name,
        description: descInput?.value.trim() || '',
        min_chunk_len: parseInt(minChunkInput?.value || '20', 10),
        max_chunk_len: parseInt(maxChunkInput?.value || '1500', 10)
```

```
      })
    });

    const data = await res.json();
    if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
atus}`);

    const srcType = document.querySelector('input[name="new-ra
g-source-type"]:checked')?.value || 'local';

    const modalEl = document.getElementById('modal-create-user
-rag');
    if (modalEl && window.bootstrap?.Modal) {
      const modal = bootstrap.Modal.getInstance(modalEl) || ne
w bootstrap.Modal(modalEl);
      modal.hide();
    }

    if (nameInput) nameInput.value = '';
    if (descInput) descInput.value = '';

    window.showToast?.(`Коллекция '${name}' успешно создана`,
'success');
    await loadUserRags();
    const createdId = data.collection?.id;
    if (createdId) {
      selectUserRag(createdId);
      if (srcType === 'gdocs') {
        const importModalEl = document.getElementById('modal-i
mport-gdocs');
        if (importModalEl && window.bootstrap?.Modal) {
          const importModal = bootstrap.Modal.getInstance(impo
rtModalEl) || new bootstrap.Modal(importModalEl);
          importModal.show();
        }
      }
    }
  } catch (err) {
```

```
    window.showToast?.(`Ошибка создания коллекции: ${err.message}`, 'danger') || alert(`Ошибка создания коллекции: ${err.message}`);
  } finally {
    if (btn) btn.disabled = false;
  }
}

/**
 * Delete the currently active workspace collection.
 *
 * @returns {Promise<void>}
 */
export async function deleteActiveUserRag() {
  if (!activeUserRagId) return;

  const confirmed = confirm(`Вы уверены, что хотите удалить коллекцию «${activeUserRagId}»?`);
  if (!confirmed) return;

  try {
    const res = await fetch(`/api/user/rags/${encodeURIComponent(activeUserRagId)}`, {
      method: 'DELETE'
    });
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    window.showToast?.(`Коллекция '${activeUserRagId}' успешно удалена`, 'success');
    activeUserRagId = null;
    await loadUserRags();
  } catch (err) {
    window.showToast?.(`Ошибка удаления коллекции: ${err.message}`, 'danger') || alert(`Ошибка удаления коллекции: ${err.message}`);
  }
}


```

userRagPipeline.js

```
/**
 * ================================================================
 * Process Name: Windows Modules - Userragpipeline Script
 * ================================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля userRagPipeline.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/rag_tab/modules/userRagPipeline.js?v=20261001_v1" type="module"></script>
 *
 * File: userRagPipeline.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/rag_tab/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ================================================================
 */

import { formatBytes, escapeHtml } from './utils.js';
import { getActiveUserRagId, loadUserRags, selectUserRag } from './userRagCollections.js';

/**
 * Fetch files from user storage and render checkbox selection list.
 *
 * @param {string[]} attachedFiles - Array of filenames currently attached to this collection.
 * @returns {Promise<void>}
```

```
 */
export async function loadUserStorageFiles(attachedFiles) {
  const checklistContainer = document.getElementById('user-rag-files-checklist');
  if (!checklistContainer) return;

  try {
    const res = await fetch('/api/user/files?subfolder=files');
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    const files = data.files || [];

    if (files.length === 0) {
      checklistContainer.innerHTML = `
        <div class="text-muted small p-2 text-center">
          В личном хранилище нет файлов.<br>Нажмите «Загрузить файлы» ниже.
        </div>
      `;
      return;
    }

    checklistContainer.innerHTML = files.map(f => {
      const isChecked = attachedFiles.length === 0 || attachedFiles.includes(f.name);
      return `
        <label class="d-flex align-items-center justify-content-between p-1 px-2 rounded hover-bg border-bottom border-light-subtle small cursor-pointer">
          <div class="d-flex align-items-center gap-2">
            <input class="form-check-input mt-0" type="checkbox" value="${escapeHtml(f.name)}" ${isChecked ? 'checked' : ''}>
            <span class="text-truncate font-monospace" style="max-width: 280px;">${escapeHtml(f.name)}</span>
          </div>
          <span class="text-muted small">${formatBytes(f.size_
```

```
bytes)}</span>
        </label>
      `;
    }).join('');
  } catch (err) {
    checklistContainer.innerHTML = `<div class="alert alert-danger p-2 small mb-0">Ошибка списка файлов: ${escapeHtml(err.message)}</div>`;
  }
}

/**
 * Build active collection index through cleaner and TF-IDF pipeline.
 *
 * @returns {Promise<void>}
 */
export async function buildActiveUserRag() {
  const activeUserRagId = getActiveUserRagId();
  if (!activeUserRagId) return;

  const btn = document.getElementById('btn-build-active-rag');
  const statusAlert = document.getElementById('user-rag-build-status-alert');
  const checkboxes = document.querySelectorAll('#user-rag-files-checklist input[type="checkbox"]:checked');
  const selectedFiles = Array.from(checkboxes).map(cb => cb.value);

  if (btn) btn.disabled = true;
  if (statusAlert) {
    statusAlert.className = 'alert alert-info p-2 small mt-2 mb-0 d-block';
    statusAlert.innerHTML = '<span class="spinner-border spinner-border-sm me-2"></span>Выполняется очистка (rag_cleaner) и индексация документов...';
  }

```

```
  try {
    const res = await fetch(`/api/user/rags/${encodeURIComponent(activeUserRagId)}/build`, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        files: selectedFiles.length > 0 ? selectedFiles : null,
        provider: 'local_tfidf'
      })
    });

    const data = await res.json();
    if (!res.ok) throw new Error(data.detail || `HTTP ${res.status}`);

    if (statusAlert) {
      statusAlert.className = 'alert alert-success p-2 small mt-2 mb-0 d-block';
      statusAlert.innerHTML = `✅ Успешно! Создано <strong>${data.chunks_count}</strong> чанков из <strong>${data.processed_files?.length || 0}</strong> файлов.`;
    }

    await loadUserRags();
  } catch (err) {
    if (statusAlert) {
      statusAlert.className = 'alert alert-danger p-2 small mt-2 mb-0 d-block';
      statusAlert.textContent = `Ошибка сборки RAG: ${err.message}`;
    }
  } finally {
    if (btn) btn.disabled = false;
  }
}

/**
```

```
 * Perform semantic search in the active workspace collection.
 *
 * @returns {Promise<void>}
 */
export async function executeActiveUserRagSearch() {
  const activeUserRagId = getActiveUserRagId();
  if (!activeUserRagId) return;

  const queryInput = document.getElementById('active-rag-searc
h-query');
  const resultsContainer = document.getElementById('active-rag
-search-results');
  const btn = document.getElementById('btn-active-rag-search')
;
  const query = queryInput?.value.trim();

  if (!query) return;
  if (btn) btn.disabled = true;
  if (resultsContainer) {
    resultsContainer.innerHTML = '<div class="text-center text
-muted small py-3"><span class="spinner-border spinner-border-
sm me-1"></span>Поиск...</div>';
  }

  try {
    const res = await fetch(`/api/user/rags/${encodeURICompone
nt(activeUserRagId)}/search`, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        query: query,
        top_k: 5,
        min_score: 0.0
      })
    });

    const data = await res.json();
    if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
```

```
atus}`);
148
149     const results = data.results || [];
150     if (results.length === 0) {
151       resultsContainer.innerHTML = '<div class="text-center te
xt-muted small py-3">Ничего не найдено по данному запросу.</di
v>';
152       return;
153     }
154
155     resultsContainer.innerHTML = results.map((item) => {
156       const scorePercent = Math.round((item.score || 0) * 100)
;
157       return `
158         <div class="card mb-2 border border-secondary-subtle b
g-body">
159           <div class="card-header py-1 px-2 d-flex justify-con
tent-between align-items-center bg-body-tertiary">
160             <span class="small fw-semibold text-truncate" styl
e="max-width: 70%;">
161               <i class="bi bi-file-earmark-text text-primary m
e-1"></i> ${escapeHtml(item.source_file || '')}
162             </span>
163             <span class="badge bg-primary-subtle text-primary
border border-primary-subtle">${scorePercent}% match</span>
164           </div>
165           <div class="card-body p-2 small">
166             <div class="text-secondary font-monospace" style="
font-size: 0.8rem; white-space: pre-wrap;">${escapeHtml(item.t
ext)}</div>
167           </div>
168         </div>
169       `;
170     }).join('');
171   } catch (err) {
172     if (resultsContainer) {
173       resultsContainer.innerHTML = `<div class="alert alert-da
nger p-2 small mb-0">Ошибка поиска: ${escapeHtml(err.message)}
```

```
</div>`;
    }
  } finally {
    if (btn) btn.disabled = false;
  }
}

/**
 * Handle quick file upload directly into user workspace storage.
 *
 * @param {Event} event - Input change event.
 * @returns {Promise<void>}
 */
export async function handleQuickUploadToUserStorage(event) {
  const files = event.target.files;
  if (!files || files.length === 0) return;

  const statusAlert = document.getElementById('user-rag-build-status-alert');
  if (statusAlert) {
    statusAlert.className = 'alert alert-info p-2 small mt-2 mb-0 d-block';
    statusAlert.innerHTML = `<span class="spinner-border spinner-border-sm me-2"></span>Загрузка ${files.length} файл(ов)...`;
  }

  try {
    for (const file of files) {
      const formData = new FormData();
      formData.append('file', file);
      formData.append('subfolder', 'files');

      const res = await fetch('/api/user/files/upload', {
        method: 'POST',
        body: formData
      });
```

```
      if (!res.ok) throw new Error(`Ошибка загрузки ${file.name}`);
    }

    if (statusAlert) {
      statusAlert.className = 'alert alert-success p-2 small mt-2 mb-0 d-block';
      statusAlert.innerHTML = `Загружено <strong>${files.length}</strong> файлов.`;
    }

    const activeUserRagId = getActiveUserRagId();
    if (activeUserRagId) {
      selectUserRag(activeUserRagId);
    }
  } catch (err) {
    if (statusAlert) {
      statusAlert.className = 'alert alert-danger p-2 small mt-2 mb-0 d-block';
      statusAlert.textContent = err.message;
    }
  } finally {
    event.target.value = '';
  }
}

/**
 * Synchronize Google Docs & Drive files directly into the active collection and build index.
 *
 * @returns {Promise<void>}
 */
export async function syncGoogleDocsIntoActiveRag() {
  const activeUserRagId = getActiveUserRagId();
  if (!activeUserRagId) {
    window.showToast?.('Пожалуйста, выберите или создайте коллекцию RAG.', 'warning') || alert('Пожалуйста, выберите или создайте коллекцию RAG.');
```

```
    return;
  }

  const queryInput = document.getElementById('import-gdocs-query');
  const accountInput = document.getElementById('import-gdocs-account');
  const statusAlert = document.getElementById('import-gdocs-status-alert');
  const btn = document.getElementById('btn-confirm-import-gdocs');

  const query = queryInput?.value.trim() || "mimeType = 'application/vnd.google-apps.document' and trashed = false";
  const accountName = accountInput?.value.trim() || undefined;

  if (btn) btn.disabled = true;
  if (statusAlert) {
    statusAlert.className = 'alert alert-info py-2 small mb-0 d-block';
    statusAlert.innerHTML = '<span class="spinner-border spinner-border-sm me-2"></span>Экспорт Google Docs и построение RAG индекса...';
  }

  try {
    const res = await fetch(`/api/user/rags/${encodeURIComponent(activeUserRagId)}/sync-google-docs`, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        query: query,
        account_name: accountName
      })
    });

    const data = await res.json();
    if (!res.ok) throw new Error(data.detail || `HTTP ${res.st
```

```
atus}`);

    if (statusAlert) {
      statusAlert.className = 'alert alert-success py-2 small mb-0 d-block';
      statusAlert.innerHTML = `✅ Синхронизировано! Загружено документов: <strong>${data.downloaded_files?.length || 0}</strong>, создано чанков: <strong>${data.chunks_count || 0}</strong>.`;
    }

    await loadUserRags();
    await selectUserRag(activeUserRagId);

    setTimeout(() => {
      const modalEl = document.getElementById('modal-import-gdocs');
      if (modalEl && window.bootstrap?.Modal) {
        const modal = bootstrap.Modal.getInstance(modalEl);
        if (modal) modal.hide();
      }
    }, 1500);

  } catch (err) {
    if (statusAlert) {
      statusAlert.className = 'alert alert-danger py-2 small mb-0 d-block';
      statusAlert.textContent = `Ошибка синхронизации: ${err.message}`;
    }
  } finally {
    if (btn) btn.disabled = false;
  }
}
```

utils.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Modules - Utils Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля utils.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/rag_tab/modules/utils.
js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { formatBytes, escapeHtml, traverseEntry } from
'/windows/api/webgui/rag_tab/modules/utils.js';
 *
 * File: utils.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/rag_tab/modules
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * Format bytes into human-readable string (B, KB, MB, GB).
 *
 * @param {number} bytes - Size in bytes.
 * @returns {string} Formatted size string.
 */
export function formatBytes(bytes) {
  if (!bytes || bytes === 0) return '0 B';
```

```
  const k = 1024;
  const sizes = ['B', 'KB', 'MB', 'GB'];
  const i = Math.floor(Math.log(bytes) / Math.log(k));
  return parseFloat((bytes / Math.pow(k, i)).toFixed(1)) + ' '
 + sizes[i];
}

/**
 * Escape HTML special characters to prevent XSS.
 *
 * @param {string} text - Raw string.
 * @returns {string} Escaped HTML string.
 */
export function escapeHtml(text) {
  if (!text) return '';
  return String(text)
    .replace(/&/g, '&amp;')
    .replace(/</g, '&lt;')
    .replace(/>/g, '&gt;')
    .replace(/"/g, '&quot;')
    .replace(/'/g, '&#039;');
}

/**
 * Recursively traverse a file system entry.
 *
 * @param {FileSystemEntry} entry - FileSystem entry.
 * @param {string} path - Current relative path.
 * @returns {Promise<File[]>} Array of files with relative pat
hs.
 */
export function traverseEntry(entry, path = '') {
  return new Promise((resolve) => {
    if (entry.isFile) {
      entry.file((file) => {
        const relPath = path ? `${path}/${file.name}` : file.n
ame;
        Object.defineProperty(file, 'customRelativePath', {
```

```
          value: relPath,
          writable: true
        });
        resolve([file]);
      }, () => resolve([]));
    } else if (entry.isDirectory) {
      const dirReader = entry.createReader();
      const entries = [];
      const readEntries = () => {
        dirReader.readEntries(async (result) => {
          if (!result.length) {
            const currentPath = path ? `${path}/${entry.name}`
 : entry.name;
            const promises = entries.map(e => traverseEntry(e,
 currentPath));
            const nestedFiles = await Promise.all(promises);
            resolve(nestedFiles.flat());
          } else {
            entries.push(...result);
            readEntries();
          }
        }, () => resolve([]));
      };
      readEntries();
    } else {
      resolve([]);
    }
  });
}

/**
 * Extract all files from a DataTransfer object recursively.
 *
 * @param {DataTransfer} dataTransfer - Drag and drop DataTran
sfer object.
 * @returns {Promise<File[]>} Array of files.
 */
export async function getAllFilesFromDataTransfer(dataTransfer
```

```
) {
  if (!dataTransfer) return [];
  const items = dataTransfer.items;
  if (items && items.length > 0 && items[0].webkitGetAsEntry)
{
    const queue = [];
    for (let i = 0; i < items.length; i++) {
      const entry = items[i].webkitGetAsEntry();
      if (entry) queue.push(traverseEntry(entry));
    }
    const results = await Promise.all(queue);
    return results.flat();
  }
  if (dataTransfer.files && dataTransfer.files.length > 0) {
    return Array.from(dataTransfer.files);
  }
  return [];
}

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/rc/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/rc/index.html" class="w
-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/rc
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!DOCTYPE html>
<html lang="ru">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-s
cale=1.0, maximum-scale=1.0, user-scalable=no">
  <title>Медиа Плейер - Пульт ДУ</title>

  <!-- Bootstrap 5 CSS -->
```

```
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
  <!-- Bootstrap Icons -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.10.0/font/bootstrap-icons.css" rel="stylesheet">

  <!-- Telegram Web Apps SDK -->
  <script src="https://telegram.org/js/telegram-web-app.js"></script>
  <script src="https://cdn.jsdelivr.net/npm/marked/marked.min.js"></script>

  <style>
    :root {
      --tg-theme-bg-color: #0d1117;
      --tg-theme-text-color: #c9d1d9;
      --tg-theme-hint-color: #8b949e;
      --tg-theme-secondary-bg-color: #161b22;
      --primary-gradient: linear-gradient(135deg, #1f6feb 0%, #094cb5 100%);
      --accent-gradient: linear-gradient(135deg, #ff7b00 0%, #ffae00 100%);
      --success-gradient: linear-gradient(135deg, #238636 0%, #2ea043 100%);
    }

    body {
      background: var(--tg-theme-bg-color);
      color: var(--tg-theme-text-color);
      font-family: system-ui, sans-serif;
      min-height: 100vh;
      display: flex;
      flex-direction: column;
      justify-content: center;
      align-items: center;
      padding: 10px;
    }

```

```
    .rc-container, .auth-container, .scan-container {
      width: 100%;
      max-width: 480px;
      padding: 2px;
    }

    .card-rc, .card-auth, .card-scan {
      background-color: var(--tg-theme-secondary-bg-color);
      border: 1px solid #30363d;
      border-radius: 16px;
      padding: 1px;
      box-shadow: 0 8px 32px rgba(0,0,0,0.6);
      backdrop-filter: blur(8px);
    }

    .btn-circle-lg {
      width: 64px;
      height: 64px;
      border-radius: 50%;
      display: flex;
      align-items: center;
      justify-content: center;
    }

    .btn-circle-xl {
      width: 80px;
      height: 80px;
      border-radius: 50%;
      display: flex;
      align-items: center;
      justify-content: center;
    }

    .track-title {
      font-size: 1.15rem;
      font-weight: 600;
      color: #58a6ff;
    }
```

```
101
102     .playlist-box {
103       max-height: 200px;
104       overflow-y: auto;
105       border-radius: 8px;
106       background: #0d1117;
107     }
108
109     .list-group-item-action:hover {
110       background-color: #21262d !important;
111     }
112
113     .btn-carousel {
114       background: var(--accent-gradient);
115       border: none;
116       color: #000;
117       font-weight: bold;
118       transition: all 0.3s ease;
119       box-shadow: 0 4px 15px rgba(255, 123, 0, 0.3);
120     }
121     .btn-carousel:hover {
122       transform: translateY(-2px);
123       box-shadow: 0 6px 20px rgba(255, 123, 0, 0.5);
124       background: linear-gradient(135deg, #ffae00 0%, #ff7b00
    100%);
125       color: #000;
126     }
127     .btn-carousel:active {
128       transform: translateY(1px);
129     }
130     .btn-carousel:hover i {
131       display: inline-block;
132       animation: spin 1s linear infinite;
133     }
134     @keyframes spin {
135       100% { transform: rotate(360deg); }
136     }
137
```

```
    /* Chat bubbles styling */
    .chat-msg {
      display: flex;
      flex-direction: column;
      max-width: 85%;
      margin-bottom: 8px;
    }
    .chat-msg.user {
      align-self: flex-end;
    }
    .chat-msg.ai, .chat-msg.system {
      align-self: flex-start;
    }
    .chat-msg.user .msg-bubble {
      background: var(--primary-gradient);
      color: #fff;
      border-radius: 12px 12px 0 12px;
      box-shadow: 0 2px 8px rgba(31, 111, 235, 0.2);
    }
    .chat-msg.ai .msg-bubble {
      background-color: #21262d;
      color: #c9d1d9;
      border-radius: 12px 12px 12px 0;
      border: 1px solid #30363d;
    }
    .chat-msg.system .msg-bubble {
      background-color: rgba(255, 255, 255, 0.05);
      color: #8b949e;
      border-radius: 8px;
      border-left: 3px solid #58a6ff;
    }
    .msg-bubble {
      padding: 8px 12px;
      font-size: 0.9rem;
      word-break: break-word;
    }
    .msg-time {
      font-size: 0.7rem;
```

```
      color: #8b949e;
      margin-top: 2px;
      padding: 0 4px;
    }
    .chat-msg.user .msg-time {
      text-align: right;
    }
    .animate-pulse {
      animation: pulse 1.5s infinite;
    }
    @keyframes pulse {
      0% { transform: scale(1); opacity: 1; }
      50% { transform: scale(1.15); opacity: 0.7; }
      100% { transform: scale(1); opacity: 1; }
    }

    .search-result-card {
      background: rgba(88, 166, 255, 0.07);
      border: 1px solid rgba(88, 166, 255, 0.2);
      border-radius: 8px;
      transition: all 0.2s ease;
    }
    .search-result-card:hover {
      background: rgba(88, 166, 255, 0.12);
      border-color: rgba(88, 166, 255, 0.4);
    }

    /* Auth panel specific */
    .form-control {
      background: #0d1117;
      border: 1px solid #30363d;
      color: #c9d1d9;
    }
    .form-control:focus {
      background: #0d1117;
      border-color: #388bfd;
      color: #c9d1d9;
      box-shadow: 0 0 0 2px rgba(56,139,253,0.3);
```

```
    }
    .btn-auth-primary {
      background: var(--primary-gradient);
      border: none;
      color: #fff;
    }
    .btn-auth-success {
      background: var(--success-gradient);
      border: none;
      color: #fff;
    }

    /* Scanning styles */
    .scan-radar {
      position: relative;
      width: 100px;
      height: 100px;
      margin: 0 auto 20px;
      border-radius: 50%;
      border: 2px solid rgba(88, 166, 255, 0.3);
      display: flex;
      align-items: center;
      justify-content: center;
    }
    .scan-radar::after {
      content: '';
      position: absolute;
      width: 100%;
      height: 100%;
      border-radius: 50%;
      border: 2px solid #58a6ff;
      animation: radar-ping 2s infinite linear;
    }
    @keyframes radar-ping {
      0% { transform: scale(1); opacity: 1; }
      100% { transform: scale(1.6); opacity: 0; }
    }
    .player-list-item {
```

```
      background: #161b22;
      border: 1px solid #30363d;
      border-radius: 8px;
      padding: 12px;
      margin-bottom: 8px;
      display: flex;
      justify-content: space-between;
      align-items: center;
      transition: all 0.2s ease;
    }
    .player-list-item:hover {
      border-color: #58a6ff;
      background: rgba(88, 166, 255, 0.05);
    }

    /* Theme toggle button */
    .theme-toggle-btn {
      position: absolute;
      top: 10px;
      right: 10px;
      background: rgba(255, 255, 255, 0.1);
      border: 1px solid rgba(255, 255, 255, 0.2);
      color: #fff;
      width: 36px;
      height: 36px;
      border-radius: 50%;
      cursor: pointer;
      display: flex;
      align-items: center;
      justify-content: center;
      font-size: 16px;
      z-index: 1000;
    }
    .theme-toggle-btn:hover {
      background: rgba(255, 255, 255, 0.2);
    }

    /* Light theme styles */
```

```
    .theme-light .card-rc,
    .theme-light .card-auth,
    .theme-light .card-scan {
      background-color: #ffffff;
      border-color: #d0d7de;
    }
    .theme-light .track-title {
      color: #0969da;
    }
    .theme-light .form-control {
      background: #ffffff;
      border-color: #d0d7de;
      color: #24292f;
    }
    .theme-light .form-control:focus {
      background: #ffffff;
      border-color: #0969da;
    }
    .theme-light .btn-primary {
      background: #2da44e;
      border-color: #2da44e;
    }
    .theme-light .btn-primary:hover {
      background: #2da44e;
    }
    .theme-light .btn-outline-light {
      color: #24292f;
      border-color: #d0d7de;
    }
    .theme-light .btn-outline-light:hover {
      background: #f6f8fa;
    }
    .theme-light .btn-circle-lg {
      background: #ffffff;
      border-color: #d0d7de;
      color: #24292f;
    }
    .theme-light .btn-circle-lg:hover {
```

```
      background: #f6f8fa;
    }
    .theme-light .text-muted {
      color: #6e7781 !important;
    }
    .theme-light .chat-msg.user .msg-bubble {
      background: #0969da;
    }
    .theme-light .chat-msg.ai .msg-bubble {
      background-color: #f6f8fa;
      color: #24292f;
    }
    .theme-light .chat-msg.system .msg-bubble {
      background-color: #f6f8fa;
      color: #6e7781;
    }
    .theme-light .player-list-item {
      background: #ffffff;
      border-color: #d0d7de;
    }
    .theme-light .player-list-item:hover {
      background: #f6f8fa;
    }
    .theme-light .list-group-item {
      background: #ffffff;
      border-color: #d0d7de;
      color: #24292f;
    }
    .theme-light .list-group-item:hover {
      background: #f6f8fa;
    }
    .theme-light .form-range {
      color-scheme: light;
    }
  </style>
</head>
<body>

```

```
  <!-- 1. AUTHENTICATION PANEL -->
  <div class="auth-container" id="auth-panel">
    <div class="card-auth">
      <div class="text-center mb-4">
        <i class="bi bi-shield-lock-fill fs-1 text-primary"></i>
        <h4 class="mt-2 text-white">Вход в пульт ДУ</h4>
        <p class="text-muted small">Пожалуйста, войдите в аккаунт для продолжения</p>
      </div>

      <div id="auth-alert" class="alert alert-danger d-none" role="alert"></div>

      <!-- Login Form -->
      <div id="auth-form-login">
        <form id="form-login">
          <div class="mb-3">
            <label for="login-email" class="form-label text-muted small">Email</label>
            <input type="email" class="form-control" id="login-email" required>
          </div>
          <div class="mb-3">
            <label for="login-password" class="form-label text-muted small">Пароль</label>
            <input type="password" class="form-control" id="login-password" required>
          </div>
          <button type="submit" class="btn btn-auth-primary w-100 py-2.5 rounded-pill mb-3">Войти</button>
          <p class="text-center small mb-0">Нет аккаунта? <a href="#" id="go-to-register" class="text-info">Зарегистрироваться</a></p>
        </form>
      </div>

      <!-- Register Form -->
```

```
      <div id="auth-form-register" class="d-none">
        <form id="form-register">
          <div class="mb-3">
            <label for="register-name" class="form-label text-muted small">Имя</label>
            <input type="text" class="form-control" id="register-name" required>
          </div>
          <div class="mb-3">
            <label for="register-email" class="form-label text-muted small">Email</label>
            <input type="email" class="form-control" id="register-email" required>
          </div>
          <div class="mb-3">
            <label for="register-password" class="form-label text-muted small">Пароль</label>
            <input type="password" class="form-control" id="register-password" required>
          </div>
          <button type="submit" class="btn btn-auth-success w-100 py-2.5 rounded-pill mb-3">Зарегистрироваться</button>
          <p class="text-center small mb-0">Уже есть аккаунт? <a href="#" id="go-to-login" class="text-info">Войти</a></p>
        </form>
      </div>

      <!-- Verify Form -->
      <div id="auth-form-verify" class="d-none">
        <p class="text-center small">На вашу почту отправлен 6-значный код подтверждения. Пожалуйста, введите его ниже.</p>
        <form id="form-verify">
          <div class="mb-3">
            <label for="verify-code" class="form-label text-muted small">Код подтверждения</label>
            <input type="text" class="form-control text-center fs-3 py-2" id="verify-code" maxlength="6" placeholder="XXXXXX" required>
```

```
          </div>
          <button type="submit" class="btn btn-info w-100 py-2
.5 text-white rounded-pill mb-3">Подтвердить</button>
          <p class="text-center small mb-0"><a href="#" id="ve
rify-cancel" class="text-info">Назад к входу</a></p>
        </form>
      </div>

      <div class="my-4 text-center position-relative">
        <hr class="border-secondary">
        <span class="px-2 text-muted position-absolute top-50
start-50 translate-middle small" style="background-color: var(
--tg-theme-secondary-bg-color);">Или войти через</span>
      </div>

      <div class="d-flex flex-column gap-2 align-items-center"
>
        <button id="google-signin-btn" class="btn btn-outline-
light w-100 rounded-pill d-flex align-items-center justify-con
tent-center gap-2 py-2">
          <i class="bi bi-google text-danger"></i>
          <span>Google</span>
        </button>
      </div>
    </div>
  </div>

  <!-- 2. COMPUTER DISCOVERY SCANNER PANEL -->
  <div class="scan-container d-none" id="scan-panel">
    <div class="card-scan text-center">
      <div class="scan-radar">
        <i class="bi bi-laptop fs-2 text-primary"></i>
      </div>
      <h4 class="text-white" id="scan-title">Поиск открытых ко
мпьютеров</h4>
      <p class="text-muted small" id="scan-status">Сканируем с
еть на наличие открытых сессий davidka.net...</p>

```

```
      <div id="discovered-players-list" class="mt-4 text-start
"></div>

      <button id="btn-retry-scan" class="btn btn-outline-prima
ry w-100 py-2.5 rounded-pill mt-3 d-none">
        <i class="bi bi-arrow-clockwise"></i> Обновить поиск
      </button>

      <button id="btn-scan-logout" class="btn btn-sm btn-outli
ne-danger w-100 mt-2 py-2 border-0 small">
        <i class="bi bi-box-arrow-right"></i> Выйти из аккаунт
a
      </button>
    </div>
  </div>

  <!-- 3. REMOTE CONTROL MAIN PANEL -->
  <div class="rc-container d-none" id="rc-app-container">
    <div class="card-rc">
      <!-- Status Badge & User info -->
      <div class="d-flex justify-content-between align-items-c
enter mb-3">
        <span class="badge bg-secondary" id="ws-status"><i cla
ss="bi bi-wifi-off"></i> Подключение...</span>
        <div class="d-flex gap-2 align-items-center">
          <button class="theme-toggle-btn" id="theme-toggle-bt
n" title="Переключить тему">
            <i class="bi bi-brightness-high"></i>
          </button>
          <button class="btn btn-sm btn-outline-secondary py-1
 px-2 border-secondary small" id="btn-change-player" title="См
енить компьютер">
            <i class="bi bi-arrow-left-right"></i> <span id="r
oom-info">ПК: default</span>
          </button>
        </div>
      </div>

```

```
      <!-- Now Playing Info -->
      <div class="text-center my-3 p-3 rounded bg-dark bg-opac
ity-25">
        <i class="bi bi-music-note-beamed fs-1 text-primary mb
-3 d-block"></i>
        <div id="track-title" class="track-title mb-1 text-tru
ncate">Файл не выбран</div>
        <small id="time-display" class="text-muted">00:00 / 00
:00</small>

        <!-- Progress Bar -->
        <div class="mt-3">
          <input type="range" class="form-range" id="seek-slid
er" min="0" max="100" value="0">
        </div>
      </div>

      <!-- Central Controls -->
      <div class="d-flex justify-content-center align-items-ce
nter gap-3 my-4">
        <button class="btn btn-outline-light btn-circle-lg" id
="btn-prev" title="Предыдущий"><i class="bi bi-skip-start-fill
 fs-3"></i></button>
        <button class="btn btn-primary btn-circle-xl" id="btn-
playtoggle" title="Воспроизведение/Пауза"><i class="bi bi-play
-fill fs-1" id="play-icon"></i></button>
        <button class="btn btn-outline-light btn-circle-lg" id
="btn-next" title="Следующий"><i class="bi bi-skip-end-fill fs
-3"></i></button>
      </div>

      <!-- Carousel / Random Movie Control -->
      <div class="my-3 px-2">
        <button class="btn btn-carousel w-100 py-2.5 rounded-p
ill d-flex align-items-center justify-content-center gap-2" id
="btn-carousel" title="Случайный фильм">
          <i class="bi bi-arrow-repeat fs-5"></i>
          <span>Карусель</span>
```

```
        </button>
      </div>

      <!-- Quick Google Workspace & Social Control Panel Buttons -->
      <div class="row g-2 px-2 my-2">
        <div class="col-6 col-sm-3">
          <button class="btn btn-outline-primary w-100 py-2 rounded-3 d-flex align-items-center justify-content-center gap-1 small" id="rc-btn-check-mail" title="Проверить почту Gmail">
            <i class="bi bi-envelope-check-fill text-primary"></i>
            <span>Почта</span>
          </button>
        </div>
        <div class="col-6 col-sm-3">
          <button class="btn btn-outline-info w-100 py-2 rounded-3 d-flex align-items-center justify-content-center gap-1 small" id="rc-btn-search-gdocs" title="Поиск в Google Docs">
            <i class="bi bi-file-earmark-text-fill text-info"></i>
            <span>Docs</span>
          </button>
        </div>
        <div class="col-6 col-sm-3">
          <button class="btn btn-outline-success w-100 py-2 rounded-3 d-flex align-items-center justify-content-center gap-1 small" id="rc-btn-gsheets" title="Работа с Google Sheets">
            <i class="bi bi-file-earmark-spreadsheet-fill text-success"></i>
            <span>Таблицы</span>
          </button>
        </div>
        <div class="col-6 col-sm-3">
          <button class="btn btn-outline-primary w-100 py-2 rounded-3 d-flex align-items-center justify-content-center gap-1 small" id="rc-btn-facebook" title="Опубликовать в Facebook">
            <i class="bi bi-facebook text-primary"></i>
```

```
            <span>Facebook</span>
          </button>
        </div>
      </div>

      <!-- Volume Controls -->
      <div class="d-flex align-items-center gap-3 my-3 px-2">
        <i class="bi bi-volume-down-fill text-muted fs-4"></i>
        <input type="range" class="form-range" id="volume-slid
er" min="0" max="1" step="0.05" value="1">
        <i class="bi bi-volume-up-fill text-muted fs-4"></i>
      </div>

      <!-- Voice Search & AI Chat -->
      <div class="mt-4 p-3 rounded border border-secondary bg-
dark bg-opacity-25">
        <h6 class="text-center mb-3"><i class="bi bi-mic"></i>
 Голосовой Ассистент</h6>

        <!-- Status -->
        <div class="text-muted small text-center mb-3" id="voi
ce-status">Готов к работе</div>

        <!-- Control buttons -->
        <div class="d-flex flex-column gap-3 align-items-cente
r">
          <button class="btn btn-outline-danger btn-circle-xl"
 id="btn-mic" title="Нажмите, чтобы говорить">
            <i class="bi bi-mic-fill fs-1" id="mic-icon"></i>
          </button>

          <!-- Language Selection -->
          <div class="btn-group w-100 mt-2" role="group">
            <input type="radio" class="btn-check" name="lang-r
adio" id="lang-ru" value="ru-RU" autocomplete="off" checked>
            <label class="btn btn-outline-secondary" for="lang
-ru">RU</label>

```

```
            <input type="radio" class="btn-check" name="lang-radio" id="lang-en" value="en-US" autocomplete="off">
            <label class="btn btn-outline-secondary" for="lang-en">EN</label>

            <input type="radio" class="btn-check" name="lang-radio" id="lang-he" value="he-IL" autocomplete="off">
            <label class="btn btn-outline-secondary" for="lang-he">HE</label>
          </div>
        </div>
      </div>

      <!-- Playlist Picker -->
      <div class="mt-4">
        <h6 class="border-bottom border-secondary pb-2"><i class="bi bi-collection-play"></i> Список воспроизведения плеера</h6>
        <div class="list-group list-group-flush playlist-box" id="playlist-items">
          <div class="text-muted small p-3 text-center">Ожидание подключения к плееру...</div>
        </div>
      </div>

      <!-- User Manual (Read & Listen) -->
      <div class="mt-4 p-3 rounded border border-secondary bg-dark bg-opacity-25">
        <div class="d-flex justify-content-between align-items-center border-bottom border-secondary pb-2 mb-2">
          <h6 class="m-0"><i class="bi bi-info-circle"></i> Инструкция пользователя</h6>
          <button class="btn btn-outline-info btn-sm d-flex align-items-center gap-1" id="btn-read-manual" title="Прослушать инструкцию">
            <i class="bi bi-volume-up-fill" id="read-manual-icon"></i> <span>Прослушать</span>
          </button>
```

```
        </div>
        <div id="user-manual-text" class="small text-muted" st
yle="line-height: 1.5; text-align: justify;">
          Для дистанционного управления сначала откройте Медиа
плеер на вашем компьютере или телевизоре.
          Данный пульт автоматически подключится к нему через
сервер.
          Вы можете нажимать кнопки для запуска файлов, перекл
ючения треков, перемотки или изменения громкости.
          Также доступно голосовое управление кнопкой микрофон
а.
          Простые команды: «пауза», «плей», «громче», «тише»,
«вперед» и «назад» сработают мгновенно.
          А любые другие фразы будут отправлены вашему AI-асси
стенту Gemini для помощи в выборе фильма или ответа на вопросы
.
        </div>
      </div>
    </div>
  </div>

  <script>
    // API Module
    const api = {
      async fetch(url, options = {}) {
        const response = await fetch(url, options);
        if (!response.ok) {
          let msg = response.statusText;
          try {
            const data = await response.json();
            msg = data.detail || msg;
          } catch {}
          throw new Error(msg);
        }
        return response.json();
      },
      auth: {
        async check() {
```

```
          try {
            const response = await fetch('/auth/check');
            return await response.json();
          } catch {
            return { authenticated: false };
          }
        },
        async register(email, password, name) {
          return api.fetch('/auth/register', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ email, password, name })
          });
        },
        async verify(email, code) {
          return api.fetch('/auth/verify', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ email, code })
          });
        },
        async loginEmail(email, password) {
          return api.fetch('/auth/login/email', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ email, password })
          });
        },
        async logout() {
          return api.fetch('/auth/logout', { method: 'POST' })
;
        }
      }
    };

    // Panels state
    const authPanel = document.getElementById('auth-panel');
    const scanPanel = document.getElementById('scan-panel');
```

```
    const rcAppContainer = document.getElementById('rc-app-container');

    const authFormLogin = document.getElementById('auth-form-login');
    const authFormRegister = document.getElementById('auth-form-register');
    const authFormVerify = document.getElementById('auth-form-verify');
    const authAlert = document.getElementById('auth-alert');

    let currentVerifyEmail = '';
    let currentRoomName = '';
    let ws = null;
    let isSeeking = false;
    let playerState = { duration: 0, currentTime: 0, isPlaying: false, volume: 1 };

    // Toggle panels helpers
    function showPanel(panelId) {
      authPanel.classList.add('d-none');
      scanPanel.classList.add('d-none');
      rcAppContainer.classList.add('d-none');

      if (panelId === 'auth') authPanel.classList.remove('d-none');
      else if (panelId === 'scan') scanPanel.classList.remove('d-none');
      else if (panelId === 'rc') rcAppContainer.classList.remove('d-none');
    }

    // ── AUTH PANEL LOGIC ──────────────────────────────────────────

    document.getElementById('go-to-register').addEventListener('click', (e) => {
      e.preventDefault();
      authAlert.classList.add('d-none');
```

```
      authFormLogin.classList.add('d-none');
      authFormRegister.classList.remove('d-none');
    });

    document.getElementById('go-to-login').addEventListener('c
lick', (e) => {
      e.preventDefault();
      authAlert.classList.add('d-none');
      authFormRegister.classList.add('d-none');
      authFormLogin.classList.remove('d-none');
    });

    document.getElementById('verify-cancel').addEventListener(
'click', (e) => {
      e.preventDefault();
      authAlert.classList.add('d-none');
      authFormVerify.classList.add('d-none');
      authFormLogin.classList.remove('d-none');
    });

    document.getElementById('google-signin-btn').addEventListe
ner('click', () => {
      window.location.href = '/auth/google?next=' + encodeURIC
omponent(window.location.pathname + window.location.search);
    });

    // Login Form Submit
    document.getElementById('form-login').addEventListener('su
bmit', async (e) => {
      e.preventDefault();
      authAlert.classList.add('d-none');
      const email = document.getElementById('login-email').val
ue;
      const pass = document.getElementById('login-password').v
alue;

      try {
        await api.auth.loginEmail(email, pass);
```

```
        checkAndInit();
      } catch (err) {
        authAlert.textContent = err.message || 'Ошибка входа';
        authAlert.classList.remove('d-none');
      }
    });

    // Register Form Submit
    document.getElementById('form-register').addEventListener('submit', async (e) => {
      e.preventDefault();
      authAlert.classList.add('d-none');
      const name = document.getElementById('register-name').value;
      const email = document.getElementById('register-email').value;
      const pass = document.getElementById('register-password').value;

      try {
        const res = await api.auth.register(email, pass, name);
        currentVerifyEmail = email;
        authFormRegister.classList.add('d-none');
        authFormVerify.classList.remove('d-none');
      } catch (err) {
        authAlert.textContent = err.message || 'Ошибка регистрации';
        authAlert.classList.remove('d-none');
      }
    });

    // Verify Form Submit
    document.getElementById('form-verify').addEventListener('submit', async (e) => {
      e.preventDefault();
      authAlert.classList.add('d-none');
      const code = document.getElementById('verify-code').valu
```

```
e;

      try {
        await api.auth.verify(currentVerifyEmail, code);
        // After verify, try to log in
        authFormVerify.classList.add('d-none');
        authFormLogin.classList.remove('d-none');
        authAlert.className = 'alert alert-success';
        authAlert.textContent = 'Почта успешно подтверждена. Теперь войдите.';
        authAlert.classList.remove('d-none');
      } catch (err) {
        authAlert.textContent = err.message || 'Неверный код';
        authAlert.classList.remove('d-none');
      }
    });

    // ── COMPUTER DISCOVERY LOGIC ──────────────────────────────────
    async function scanForPlayers() {
      showPanel('scan');
      document.getElementById('scan-title').textContent = 'Поиск открытых компьютеров';
      document.getElementById('scan-status').textContent = 'Сканируем сеть на наличие открытых сессий davidka.net...';
      document.getElementById('btn-retry-scan').classList.add('d-none');

      const listDiv = document.getElementById('discovered-players-list');
      listDiv.innerHTML = '<div class="text-center p-3"><span class="spinner-border spinner-border-sm text-primary"></span></div>';

      try {
        const response = await fetch('/api/control/active_players');
        const data = await response.json();
```

```
        const players = data.players || [];
        const currentUserEmail = data.user_email;

        const mappedPlayers = players.map(p => {
          if (p.includes('@')) {
            return { room: p, name: `davidka.net (${p})` };
          }
          return { room: p, name: p === 'default' ? 'Основной компьютер (davidka.net)' : `ПК: ${p}` };
        });

        if (mappedPlayers.length === 0) {
          listDiv.innerHTML = `
            <div class="alert alert-warning text-center small mb-0">
              <i class="bi bi-exclamation-circle-fill"></i> Открытые компьютеры не обнаружены.<br>
              Пожалуйста, откройте davidka.net на вашем компьютере.
            </div>`;
          document.getElementById('scan-title').textContent = 'Компьютеры не найдены';
          document.getElementById('btn-retry-scan').classList.remove('d-none');
        } else if (mappedPlayers.length === 1) {
          // Exactly one computer found - connect automatically
          listDiv.innerHTML = `<div class="alert alert-success text-center small">Обнаружен компьютер "${mappedPlayers[0].name}". Подключаемся...</div>`;
          setTimeout(() => {
            initStream(mappedPlayers[0].room);
          }, 800);
        } else {
          // Multiple computers found - ask user to select one
          document.getElementById('scan-title').textContent = 'Куда подключиться?';
          document.getElementById('scan-status').textContent =
```

```
 'Обнаружено несколько открытых компьютеров. Выберите один:';

          listDiv.innerHTML = mappedPlayers.map(p => `
            <div class="player-list-item">
              <div>
                <i class="bi bi-laptop-fill text-primary me-2"></i>
                <strong>${p.name}</strong>
              </div>
              <button class="btn btn-sm btn-primary rounded-pill px-3" onclick="selectPlayer('${p.room}')">Выбрать</button>
            </div>
          `).join('');
          document.getElementById('btn-retry-scan').classList.remove('d-none');
        }
      } catch (err) {
        listDiv.innerHTML = `<div class="alert alert-danger text-center small">Ошибка сканирования: ${err.message}</div>`;
        document.getElementById('btn-retry-scan').classList.remove('d-none');
      }
    }

    window.selectPlayer = function(room) {
      initStream(room);
    };

    document.getElementById('btn-retry-scan').addEventListener('click', scanForPlayers);

    document.getElementById('btn-change-player').addEventListener('click', () => {
      // Disconnect current stream
      if (ws) {
        ws.close();
        ws = null;
      }
```

```
      scanForPlayers();
    });

    document.getElementById('btn-scan-logout').addEventListene
r('click', async () => {
      await api.auth.logout();
      showPanel('auth');
    });

    // ── MAIN TELEGRAM / WEB APP ROOM SETUP ──────────────────
──────────────
    function getRoomNameFromQuery() {
      const urlParams = new URLSearchParams(window.location.se
arch);
      return urlParams.get('room') || '';
    }

    // Initialize Telegram WebApp if available
    if (window.Telegram && window.Telegram.WebApp) {
      const tg = window.Telegram.WebApp;
      tg.ready();
      tg.expand();

      // Apply theme
      function applyTgTheme() {
        if (tg.themeParams.bg_color) {
          document.documentElement.style.setProperty('--tg-the
me-bg-color', tg.themeParams.bg_color);
        }
        if (tg.themeParams.text_color) {
          document.documentElement.style.setProperty('--tg-the
me-text-color', tg.themeParams.text_color);
        }
        if (tg.themeParams.secondary_bg_color) {
          document.documentElement.style.setProperty('--tg-the
me-secondary-bg-color', tg.themeParams.secondary_bg_color);
        }
        if (tg.themeParams.hint_color) {
```

```
          document.documentElement.style.setProperty('--tg-theme-hint-color', tg.themeParams.hint_color);
        }
      }
      applyTgTheme();
      tg.onEvent('themeChanged', applyTgTheme);

      // Haptic feedback function
      window.triggerTgHaptic = function() {
        if (tg.HapticFeedback) {
          tg.HapticFeedback.impactOccurred('light');
        }
      };
    } else {
      window.triggerTgHaptic = function() {};
    }

    const statusBadge = document.getElementById('ws-status');
    const trackTitle = document.getElementById('track-title');
    const timeDisplay = document.getElementById('time-display');
    const seekSlider = document.getElementById('seek-slider');
    const volumeSlider = document.getElementById('volume-slider');
    const playIcon = document.getElementById('play-icon');
    const playlistContainer = document.getElementById('playlist-items');

    function formatTime(seconds) {
      if (!seconds || isNaN(seconds)) return '00:00';
      const m = Math.floor(seconds / 60);
      const s = Math.floor(seconds % 60);
      return `${m.toString().padStart(2, '0')}:${s.toString().padStart(2, '0')}`;
    }

    // Connect to WebSocket with Selected Room (Strictly One Stream)
```

```
    function initStream(room) {
      currentRoomName = room;
      showPanel('rc');

      const roomInfoElement = document.getElementById('room-in
fo');
      if (roomInfoElement) {
        const displayRoom = room.includes('@') ? 'kino.davidka
.net.tv' : room;
        roomInfoElement.textContent = `Плеер: ${displayRoom}`;
      }

      // Close previous connection to satisfy "Strictly One St
ream" requirement
      if (ws) {
        ws.close();
      }

      const protocol = window.location.protocol === 'https:' ?
 'wss:' : 'ws:';
      const wsUrl = `${protocol}//${window.location.host}/api/
control/ws?role=remote&room=${encodeURIComponent(currentRoomNa
me)}`;

      ws = new WebSocket(wsUrl);

      ws.onopen = () => {
        statusBadge.className = 'badge bg-success';
        statusBadge.innerHTML = '<i class="bi bi-wifi"></i> По
дключено';
      };

      ws.onmessage = (event) => {
        try {
          const msg = JSON.parse(event.data);
          if (msg.event === 'status_update') {
            updateUI(msg);
          } else if (msg.event === 'playlist_update') {
```

```
            updatePlaylist(msg.files || []);
          }
        } catch (e) {
          console.error(e);
        }
      };

      ws.onclose = (e) => {
        // Stop speech queue when connection drops
        if (currentSpeechQueue) {
          try {
            currentSpeechQueue.stop();
            currentSpeechQueue = null;
          } catch(err) {}
        }
        if (currentSpeechAudio) {
          try {
            currentSpeechAudio.pause();
            currentSpeechAudio.src = "";
            currentSpeechAudio = null;
          } catch(err) {}
        }

        // If we deliberately closed it to switch player, don't auto-reconnect
        if (currentRoomName !== room) return;

        statusBadge.className = 'badge bg-warning text-dark';
        statusBadge.innerHTML = '<i class="bi bi-arrow-clockwise"></i> Переподключение...';
        setTimeout(() => {
          if (currentRoomName === room) {
            initStream(room);
          }
        }, 3000);
      };
    }

```

```
955     function sendCommand(action, payload = {}) {
956       window.triggerTgHaptic();
957       if (ws && ws.readyState === WebSocket.OPEN) {
958         ws.send(JSON.stringify({ action, ...payload }));
959       }
960     }
961
962     function updateUI(data) {
963       playerState = { ...playerState, ...data };
964       trackTitle.textContent = data.fileName || 'Файл не выбра
    н';
965       timeDisplay.textContent = `${formatTime(data.currentTime
    )} / ${formatTime(data.duration)}`;
966
967       if (!isSeeking && data.duration > 0) {
968         seekSlider.max = Math.floor(data.duration);
969         seekSlider.value = Math.floor(data.currentTime);
970       }
971
972       if (typeof data.volume === 'number') {
973         volumeSlider.value = data.volume;
974       }
975
976       if (data.isPlaying) {
977         playIcon.className = 'bi bi-pause-fill fs-1';
978       } else {
979         playIcon.className = 'bi bi-play-fill fs-1';
980       }
981     }
982
983     function updatePlaylist(files) {
984       window.playlistFiles = files;
985       if (!files.length) {
986         playlistContainer.innerHTML = '<div class="text-muted
    small p-3 text-center">Плейлист пуст</div>';
987         return;
988       }
989
```

```
      playlistContainer.innerHTML = files.map((f, i) => `
        <button class="list-group-item list-group-item-action bg-transparent text-white border-secondary py-2 px-3 small d-flex justify-content-between align-items-center" onclick="playFile(${i})">
          <span class="text-truncate">${i + 1}. ${f.name}</span>
          <span class="badge bg-secondary ms-2">${f.type === 'series' ? 'Сериал' : 'Фильм'}</span>
        </button>
      `).join('');
    }

    window.playFile = (index) => {
      sendCommand('play_file', { index });
    };

    // ── VOICE CONTROL & SEARCH CHAT MODULE ──────────────────────────────

    let recognition = null;
    let isListening = false;
    let rcAbortController = null;
    let rcHistory = [];



    function tryParseMovieCard(text) {
      const trimmed = text.trim();
      if (trimmed.startsWith('{') && trimmed.endsWith('}')) {
        try {
          const data = JSON.parse(trimmed);
          if (data.title || data.title_ru) return data;
        } catch (e) {}
      }
      const match = text.match(/\{[\s\S]*?\}/);
      if (match) {
        try {
          const data = JSON.parse(match[0]);
```

```
          if (data.title || data.title_ru) return data;
        } catch (e) {}
      }
      return null;
    }

    function renderMovieCardHtml(card) {
      const title = card.title_ru || card.title;
      const subtitle = card.title_orig || '';
      const year = card.year || '';
      const category = card.main_category || '';
      const genres = card.genres ? card.genres.join(', ') : ''
;
      const directors = card.directors ? card.directors.join('
, ') : '';
      const cast = card.cast ? card.cast.slice(0, 5).join(', '
) : '';
      const plot = card.plot || '';
      const whyWatch = card.why_watch || '';

      let ratingHtml = '';
      if (card.rating) {
        if (card.rating.imdb) ratingHtml += `<span class="badg
e bg-warning text-dark me-1">IMDb: ${card.rating.imdb}</span>`
;
        if (card.rating.кинопоиск) ratingHtml += `<span class=
"badge bg-danger text-white me-1">КП: ${card.rating.кинопоиск}
</span>`;
        if (card.rating.tmdb) ratingHtml += `<span class="badg
e bg-primary text-white">TMDB: ${card.rating.tmdb}</span>`;
      }

      return `
        <div class="p-3 text-white" style="border-radius: 10px
; background: rgba(88, 166, 255, 0.08); border: 1px solid rgba
(88, 166, 255, 0.2);">
          <h5 class="mb-1 text-info">${title}</h5>
          <div class="small text-muted mb-2">${subtitle} ${yea
```

```
r ? `(${year})` : ''}</div>
          <div class="mb-2">${ratingHtml}</div>
          ${category ? `<div class="small text-muted mb-1" sty
le="font-size: 0.8rem;"><strong>Категория:</strong> ${category
}</div>` : ''}
          ${genres ? `<div class="small text-muted mb-1" style
="font-size: 0.8rem;"><strong>Жанры:</strong> ${genres}</div>`
 : ''}
          ${directors ? `<div class="small text-muted mb-1" st
yle="font-size: 0.8rem;"><strong>Режиссер:</strong> ${director
s}</div>` : ''}
          ${cast ? `<div class="small text-muted mb-2" style="
font-size: 0.8rem;"><strong>В ролях:</strong> ${cast}...</div>
` : ''}
          ${plot ? `<p class="small text-secondary mb-2" style
="font-size: 0.85rem; line-height: 1.4;">${plot}</p>` : ''}
          ${whyWatch ? `<div class="small text-info mb-2" styl
e="font-size: 0.8rem;"><strong>Рекомендация:</strong> ${whyWat
ch}</div>` : ''}
          <div class="d-flex justify-content-end mt-2">
            <button class="btn btn-sm btn-primary rounded-pill
 px-3" onclick="playMovieDirect('${title.replace(/'/g, "\\'")}
')">
              <i class="bi bi-play-fill"></i> Запустить фильм
            </button>
          </div>
        </div>
      `;
    }

    window.playMovieDirect = (title) => {
      if (!window.playlistFiles) return;
      const cleanT = title.toLowerCase();
      const foundIdx = window.playlistFiles.findIndex(f => (f.
name || '').toLowerCase().includes(cleanT));
      if (foundIdx !== -1) {
        playFile(foundIdx);
        speakText(`Запускаю ${window.playlistFiles[foundIdx].n
```

```
ame}`);
      } else {
        speakText(`Фильм не найден в плейлисте`);
      }
    };

    async function processChatMessage(text) {
      if (!text || !text.trim()) return;
      const query = text.trim();

      stopAudio(); // <--- STOP AUDIO IMMEDIATELY

      // 0. Transmit transcript to open computers via WebSocke
t
      if (ws && ws.readyState === WebSocket.OPEN) {
        sendCommand('transcript', { text: query });
      }

      // 1. Check local control commands first
      if (handleLocalVoiceCommand(query.toLowerCase())) {
        speakText('Команда выполнена');
        return;
      }

      // 2. Perform search in playlist
      const cleanQ = query.toLowerCase().replace(/^(найди|поис
к|включи|запусти|покажи|фильм|сериал)\s+/i, '').trim();
      let matches = [];
      if (window.playlistFiles && cleanQ.length > 1) {
        matches = window.playlistFiles.map((f, idx) => ({ ...f
, index: idx })).filter(f => {
          const name = (f.name || '').toLowerCase();
          const type = (f.type || '').toLowerCase();
          return name.includes(cleanQ) || (f.year && f.year.to
String().includes(cleanQ));
        });
      }

```

```
      if (matches.length > 0) {
        speakText(`Найдено совпадений: ${matches.length}`);

        const queryLower = query.toLowerCase();
        if (matches.length === 1 && (queryLower.includes('включи') || queryLower.includes('запусти') || queryLower.includes('плей') || queryLower.includes('играй'))) {
          playFile(matches[0].index);
          speakText(`Запускаю ${matches[0].name}`);
        }
        return;
      }

      // 3. Forward to AI Chat API with Streaming
      if (rcAbortController) {
        try { rcAbortController.abort(); } catch(e) {}
      }
      rcAbortController = new AbortController();

      const voiceStatus = document.getElementById('voice-status');
      voiceStatus.textContent = 'Обработка запроса моделью...';

      let fullReply = '';
      let started = false;

      try {
        const payload = {
          message: query,
          history: rcHistory,
          generation_config: { is_rc: true }
        };

        const response = await fetch('/api/chat', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify(payload),
```

```
        signal: rcAbortController.signal
      });

      if (!response.ok) {
        throw new Error(`Ошибка HTTP ${response.status}`);
      }

      const reader = response.body.getReader();
      const decoder = new TextDecoder('utf-8');
      let buffer = '';
      let voiceReply = '';

      while (true) {
        const { value, done } = await reader.read();
        if (done) break;

        buffer += decoder.decode(value, { stream: true });
        const lines = buffer.split('\n');
        buffer = lines.pop();

        for (const line of lines) {
          const trimmed = line.trim();
          if (!trimmed) continue;
          if (trimmed.startsWith('data: ')) {
            try {
              const data = JSON.parse(trimmed.substring(6));
              if (data.error) {
                throw new Error(data.error);
              }
              if (data.status) {
                voiceStatus.textContent = data.status;
              }
              if (data.voice) {
                voiceReply += data.voice;
              }
              if (data.text) {
                fullReply += data.text;
              }
```

```
            } catch (e) {
              console.error(e);
            }
          }
        }

        voiceStatus.textContent = 'Готов к работе';

        // ── Обработка тегов <film> из ответа AI ──────────────────
        const filmMatches = [...fullReply.matchAll(/<film>(.*?)<\/film>/gi)];
        if (filmMatches.length > 0) {
          const filmTitle = filmMatches[0][1].trim();
          try {
            const mediaResp = await fetch('/api/media/by-title', {
              method: 'POST',
              headers: { 'Content-Type': 'application/json' },
              body: JSON.stringify({ title: filmTitle })
            });
            if (mediaResp.ok) {
              const mediaData = await mediaResp.json();
              const filmPath = mediaData.path;
              if (filmPath) {
                sendCommand('play_file_by_path', { path: filmPath });
                voiceStatus.textContent = `▶ Запускаю: ${filmTitle}`;
              }
            }
          } catch(e) {
            console.error('Ошибка поиска пути фильма:', e);
          }
        }
        // ─────────────────────────────────────────────────────────
```

```
1210
1211        const card = tryParseMovieCard(fullReply);
1212        if (card) {
1213          speakText(`Найдена карточка фильма ${card.title_ru ||
 card.title}. ${card.why_watch || card.plot || ''}`);
1214        } else {
1215          speakText(voiceReply || fullReply);
1216        }
1217
1218        rcHistory.push({ role: 'user', parts: [query] });
1219        rcHistory.push({ role: 'model', parts: [fullReply] });
1220        if (rcHistory.length > 20) {
1221          rcHistory = rcHistory.slice(-20);
1222        }
1223
1224      } catch (err) {
1225        if (err.name === 'AbortError') return;
1226        voiceStatus.textContent = 'Ошибка AI: ' + err.message;
1227      }
1228    }
1229
1230    function initVoiceControl() {
1231      const btnMic = document.getElementById('btn-mic');
1232      const micIcon = document.getElementById('mic-icon');
1233      const voiceStatus = document.getElementById('voice-statu
s');
1234
1235      const SpeechRecognition = window.SpeechRecognition || wi
ndow.webkitSpeechRecognition;
1236      if (!SpeechRecognition) {
1237        voiceStatus.textContent = 'Голосовой ввод не поддержив
ается вашим браузером.';
1238        if (btnMic) btnMic.disabled = true;
1239      } else {
1240        recognition = new SpeechRecognition();
1241
1242        // Dynamically get selected language
1243        const getSelectedLang = () => {
```

```
          const selected = document.querySelector('input[name="lang-radio"]:checked');
          return selected ? selected.value : 'ru-RU';
        };

        recognition.lang = getSelectedLang();

        // Update language if radio button changes
        document.querySelectorAll('input[name="lang-radio"]').forEach(radio => {
          radio.addEventListener('change', () => {
            recognition.lang = getSelectedLang();
          });
        });

        recognition.continuous = false;
        recognition.interimResults = false;

        recognition.onstart = () => {
          isListening = true;
          if (btnMic) btnMic.className = 'btn btn-danger btn-circle-lg flex-shrink-0';
          if (micIcon) micIcon.className = 'bi bi-mic-fill fs-3 animate-pulse';
          voiceStatus.textContent = 'Слушаю...';
        };

        recognition.onend = () => {
          isListening = false;
          if (btnMic) btnMic.className = 'btn btn-outline-danger btn-circle-lg flex-shrink-0';
          if (micIcon) micIcon.className = 'bi bi-mic-fill fs-3';
          voiceStatus.textContent = 'Готов к работе';
        };

        recognition.onerror = (e) => {
          voiceStatus.textContent = `Ошибка ввода: ${e.error}`
```

```
;
        };

        recognition.onresult = async (event) => {
          const text = event.results[0][0].transcript;
          if (text) {
            processChatMessage(text);
          }
        };

        if (btnMic) {
          btnMic.addEventListener('click', () => {
            window.triggerTgHaptic();
            // Unlock SpeechSynthesis for mobile devices
            if ('speechSynthesis' in window) {
              const silentUtterance = new SpeechSynthesisUtter
ance(' ');
              silentUtterance.volume = 0;
              window.speechSynthesis.speak(silentUtterance);
            }
            if (isListening) {
              recognition.stop();
            } else {
              recognition.start();
            }
          });
        }
      }
    }

    let currentSpeechAudio = null;
    let currentSpeechQueue = null;

    class AudioQueuePlayer {
      constructor(chunks, userSettings) {
        this.chunks = chunks;
        this.userSettings = userSettings;
        this.currentIndex = 0;
```

```
        this.isPlaying = false;
        this.audioElements = new Map();
        this.activeAudio = null;
        this._prefetch(0);
        this._prefetch(1);
      }

      _prefetch(index) {
        if (index >= this.chunks.length || this.audioElements.has(index)) return;
        const text = this.chunks[index];
        const queryParams = new URLSearchParams({
          text: text,
          system: this.userSettings.tts_system,
          voice: this.userSettings.tts_voice
        });
        const audioUrl = `/api/tts/synthesize?${queryParams.toString()}`;
        const audio = new Audio(audioUrl);
        audio.load();
        this.audioElements.set(index, audio);
      }

      async play() {
        this.isPlaying = true;
        while (this.currentIndex < this.chunks.length && this.isPlaying) {
          this._prefetch(this.currentIndex);
          this._prefetch(this.currentIndex + 1);
          this._prefetch(this.currentIndex + 2);

          const audio = this.audioElements.get(this.currentIndex);
          if (!audio) {
            this.currentIndex++;
            continue;
          }

```

```
          this.activeAudio = audio;
          currentSpeechAudio = audio;

            const playPromise = new Promise((resolve) => {
              audio.onended = () => resolve();
              audio.onerror = (e) => {
                console.error(`Audio error for chunk ${this.cu
rrentIndex}:`, e);
                this.stop(); // Останавливаем всю очередь при
ошибке загрузки
                resolve();
              };
            });

          try {
            await audio.play();
            await playPromise;
          } catch (err) {
            console.error("Audio playback interrupted/failed:"
, err);
          }
          this.currentIndex++;
        }
        this.isPlaying = false;
      }

      stop() {
        this.isPlaying = false;
        if (this.activeAudio) {
          try {
            this.activeAudio.pause();
            this.activeAudio.src = "";
          } catch (e) {}
        }
        this.audioElements.clear();
      }
    }

```

```
    function stopAudio() {
      if (currentSpeechQueue) {
        try {
          currentSpeechQueue.stop();
          currentSpeechQueue = null;
        } catch (e) {
          console.error("Error stopping currentSpeechQueue:", e);
        }
      }
      if (currentSpeechAudio) {
        try {
          currentSpeechAudio.pause();
          currentSpeechAudio.src = "";
          currentSpeechAudio = null;
        } catch (e) {
          console.error("Error stopping currentSpeechAudio:", e);
        }
      }
      if ('speechSynthesis' in window) {
        try {
          window.speechSynthesis.cancel();
        } catch (e) {
          console.error("Error cancelling SpeechSynthesis:", e);
        }
      }
    }

    async function speakText(text) {
      if (!text) return;

      // Stop any currently playing audio
      stopAudio();
      // Remove markdown or HTML tags if any from voice output
      const cleanText = text.replace(/<[^>]*>/g, '').replace(/[*_`#]/g, '').trim();
```

```
      if (!cleanText) return;

      // Try backend TTS first
      try {
        const response = await fetch('/auth/settings');
        if (response.ok) {
          const userSettings = await response.json();
          if (userSettings.tts_enabled === 1 && userSettings.t
ts_system && userSettings.tts_system !== 'browser') {
            const paragraphs = cleanText
              .split(/\n+/)
              .map(p => p.trim())
              .filter(p => p.length > 0);

            let chunks = [];
            for (const para of paragraphs) {
              const sentences = para.match(/[^.!?]+[.!?]*/g) |
| [para];
              let currentChunk = "";
              for (const sentence of sentences) {
                const trimmedSentence = sentence.trim();
                if (!trimmedSentence) continue;
                if ((currentChunk + " " + trimmedSentence).len
gth > 200) {
                  chunks.push(currentChunk.trim());
                  currentChunk = trimmedSentence;
                } else {
                  currentChunk = (currentChunk + " " + trimmed
Sentence).trim();
                }
              }
              if (currentChunk) {
                chunks.push(currentChunk.trim());
              }
            }
            chunks = chunks.filter(c => c.length > 0);

            if (chunks.length > 0) {
```

```
1449              currentSpeechQueue = new AudioQueuePlayer(chunks
, userSettings);
1450              await currentSpeechQueue.play();
1451            }
1452            return;
1453          }
1454        }
1455      } catch (e) {
1456        console.error('Failed to speak using backend TTS, fall
ing back to browser:', e);
1457      }
1458
1459      // Fallback to browser SpeechSynthesis
1460      if ('speechSynthesis' in window) {
1461        window.currentUtterance = new SpeechSynthesisUtterance
(cleanText);
1462        window.currentUtterance.lang = 'ru-RU';
1463        window.speechSynthesis.speak(window.currentUtterance);
1464      }
1465    }
1466
1467    function handleLocalVoiceCommand(text) {
1468      const t = text.toLowerCase();
1469      if (t.includes('пауз') || t.includes('стоп')) {
1470        sendCommand('pause');
1471        return true;
1472      }
1473      if (t.includes('продолж') || t.includes('плей') || t.inc
ludes('играй') || t.includes('старт')) {
1474        sendCommand('play');
1475        return true;
1476      }
1477      if (t.includes('громче') || t.includes('прибавь звук'))
{
1478        const targetVol = Math.min(1, playerState.volume + 0.2
);
1479        sendCommand('set_volume', { level: targetVol });
1480        return true;
```

```
      }
      if (t.includes('тише') || t.includes('убавь звук')) {
        const targetVol = Math.max(0, playerState.volume - 0.2
);
        sendCommand('set_volume', { level: targetVol });
        return true;
      }
      if (t.includes('следующ') || t.includes('вперед')) {
        sendCommand('next');
        return true;
      }
      if (t.includes('предыдущ') || t.includes('назад')) {
        sendCommand('prev');
        return true;
      }
      if (t.includes('карусель') || t.includes('случайн') || t
.includes('рандом')) {
        sendCommand('play_random_movie');
        return true;
      }
      return false;
    }

    async function sendToAiModel(message) {
      const response = await fetch('/api/chat', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ message, generation_config: { i
s_rc: true } })
      });
      if (!response.ok) {
        throw new Error('Не удалось получить ответ от AI');
      }
      const data = await response.json();
      return data.response || data.message || 'Модель не ответ
ила';
    }

```

```
    // Set listeners
    document.getElementById('btn-playtoggle').addEventListener('click', () => sendCommand('toggle_play'));
    document.getElementById('btn-prev').addEventListener('click', () => sendCommand('prev'));
    document.getElementById('btn-next').addEventListener('click', () => sendCommand('next'));
    document.getElementById('btn-carousel').addEventListener('click', () => sendCommand('play_random_movie'));

    const rcCheckMailBtn = document.getElementById('rc-btn-check-mail');
    if (rcCheckMailBtn) {
      rcCheckMailBtn.addEventListener('click', () => {
        processChatMessage('Проверь мою почту Gmail и покажи последние непрочитанные письма.');
      });
    }

    const rcSearchGdocsBtn = document.getElementById('rc-btn-search-gdocs');
    if (rcSearchGdocsBtn) {
      rcSearchGdocsBtn.addEventListener('click', () => {
        const query = prompt('Введите запрос для поиска в Google Docs / Drive:');
        if (query && query.trim()) {
          processChatMessage(`Найди в Google Docs / Google Drive документы по запросу: "${query.trim()}".`);
        }
      });
    }

    const rcGsheetsBtn = document.getElementById('rc-btn-gsheets');
    if (rcGsheetsBtn) {
      rcGsheetsBtn.addEventListener('click', () => {
        const query = prompt('Введите ID таблицы или команду для Google Sheets (например: "Прочитай диапазон A1:D10 из табли
```

```
цы ID"):');
1542        if (query && query.trim()) {
1543          processChatMessage(`Google Sheets: ${query.trim()}`)
;
1544        }
1545      });
1546    }
1547
1548    const rcFacebookBtn = document.getElementById('rc-btn-face
book');
1549    if (rcFacebookBtn) {
1550      rcFacebookBtn.addEventListener('click', () => {
1551        const text = prompt('Введите текст для публикации в Fa
cebook:');
1552        if (text && text.trim()) {
1553          processChatMessage(`Опубликуй пост в Facebook с текс
том: "${text.trim()}".`);
1554        }
1555      });
1556    }
1557
1558    seekSlider.addEventListener('mousedown', () => { isSeeking
 = true; });
1559    seekSlider.addEventListener('touchstart', () => { isSeekin
g = true; });
1560    seekSlider.addEventListener('change', (e) => {
1561      isSeeking = false;
1562      sendCommand('seek', { seconds: parseFloat(e.target.value
) });
1563    });
1564
1565    volumeSlider.addEventListener('change', (e) => {
1566      sendCommand('set_volume', { level: parseFloat(e.target.v
alue) });
1567    });
1568
1569    // ── USER MANUAL AUDIO READER ──────────────────────────
──────────────────
```

```
    let isSpeakingManual = false;
    const btnRead = document.getElementById('btn-read-manual');
    const readIcon = document.getElementById('read-manual-icon');
    const readTextSpan = btnRead ? btnRead.querySelector('span') : null;

    if (btnRead) {
      btnRead.addEventListener('click', () => {
        window.triggerTgHaptic();
        const text = document.getElementById('user-manual-text').innerText;

        if (isSpeakingManual) {
          window.speechSynthesis.cancel();
          isSpeakingManual = false;
          if (readIcon) readIcon.className = 'bi bi-volume-up-fill';
          if (readTextSpan) readTextSpan.textContent = 'Прослушать';
          btnRead.className = 'btn btn-outline-info btn-sm d-flex align-items-center gap-1';
        } else {
          window.speechSynthesis.cancel();
          const utterance = new SpeechSynthesisUtterance(text);
          utterance.lang = 'ru-RU';

          utterance.onend = () => {
            isSpeakingManual = false;
            if (readIcon) readIcon.className = 'bi bi-volume-up-fill';
            if (readTextSpan) readTextSpan.textContent = 'Прослушать';
            btnRead.className = 'btn btn-outline-info btn-sm d-flex align-items-center gap-1';
          };
```

```
1597
1598          utterance.onerror = () => {
1599            isSpeakingManual = false;
1600            if (readIcon) readIcon.className = 'bi bi-volume-u
    p-fill';
1601            if (readTextSpan) readTextSpan.textContent = 'Прос
    лушать';
1602            btnRead.className = 'btn btn-outline-info btn-sm d
    -flex align-items-center gap-1';
1603          };
1604
1605          window.speechSynthesis.speak(utterance);
1606          isSpeakingManual = true;
1607          if (readIcon) readIcon.className = 'bi bi-stop-fill'
    ;
1608          if (readTextSpan) readTextSpan.textContent = 'Остано
    вить';
1609          btnRead.className = 'btn btn-info btn-sm text-dark d
    -flex align-items-center gap-1';
1610        }
1611      });
1612    }
1613
1614    // Theme toggle function
1615    function toggleTheme() {
1616      const body = document.body;
1617      const currentTheme = body.classList.contains('theme-ligh
    t') ? 'light' : 'dark';
1618      const newTheme = currentTheme === 'light' ? 'dark' : 'li
    ght';
1619
1620      if (newTheme === 'light') {
1621        body.classList.add('theme-light');
1622      } else {
1623        body.classList.remove('theme-light');
1624      }
1625    }
1626
```

```
    // Set theme toggle listener
    document.getElementById('theme-toggle-btn')?.addEventListener('click', toggleTheme);

    // ── AUTH CHECK INITIALIZER ──────────────────────────────────────────
    async function checkAndInit() {
      const authState = await api.auth.check();
      if (authState.authenticated) {
        // If a room parameter is present in URL, connect directly to it
        const queryRoom = getRoomNameFromQuery();
        if (queryRoom) {
          initStream(queryRoom);
        } else {
          // Otherwise, scan for open players
          scanForPlayers();
        }
      } else {
        showPanel('auth');
      }
    }

    // Run on startup
    document.addEventListener('DOMContentLoaded', () => {
      checkAndInit();
      initVoiceControl();
    });

  </script>
</body>
</html>
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/registry_viewer_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/registry_viewer_tab/ind
ex.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/registry_viewer_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Windows Registry Viewer Tab -->
<style>
  .reg-container {
    padding: 1rem;
  }
  .reg-card {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: 10px;
  }
```

```
  .reg-path-bar {
    background: var(--surface-2) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: 8px;
    padding: 0.5rem 0.75rem;
  }
  .reg-tree-pane {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 8px;
    height: 580px;
    overflow-y: auto;
    font-size: 0.82rem;
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
  }
  .reg-tree-item {
    padding: 4px 8px;
    cursor: pointer;
    border-radius: 4px;
    display: flex;
    align-items: center;
    gap: 6px;
    color: var(--text-color);
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
  }
  .reg-tree-item:hover {
    background: var(--nav-hover-bg, rgba(56, 189, 248, 0.1));
    color: var(--nav-active);
  }
  .reg-tree-item.active {
    background: var(--nav-hover-bg, rgba(56, 189, 248, 0.25));
    color: var(--nav-active);
    font-weight: 600;
  }
  .reg-values-pane {
```

```
    background: var(--surface-1);
    border: 1px solid var(--border-color);
    border-radius: 8px;
    height: 580px;
    overflow-y: auto;
  }
  .reg-table {
    width: 100%;
    font-size: 0.8rem;
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
    border-collapse: collapse;
  }
  .reg-table th {
    background: var(--surface-2);
    padding: 8px 12px;
    color: var(--text-muted);
    font-weight: 600;
    border-bottom: 1px solid var(--border-color);
    position: sticky;
    top: 0;
    z-index: 2;
  }
  .reg-table td {
    padding: 6px 12px;
    border-bottom: 1px solid var(--border-color);
    color: var(--text-color);
    vertical-align: top;
  }
  .reg-table tbody tr:hover {
    background: var(--nav-hover-bg, rgba(56, 189, 248, 0.05));
  }
  .reg-type-badge {
    font-size: 0.68rem;
    padding: 2px 6px;
    border-radius: 4px;
    background: var(--surface-2);
    border: 1px solid var(--border-color);
```

```
    color: var(--nav-active);
  }
  .reg-bookmark-btn {
    font-size: 0.75rem;
    padding: 4px 10px;
    border-radius: 6px;
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    color: var(--text-color);
    transition: all 0.2s ease;
  }
  .reg-bookmark-btn:hover {
    background: var(--surface-1);
    color: var(--nav-active);
    border-color: var(--nav-active);
  }
</style>
<div class="reg-container">
<!-- Header Alert & Status -->
<div class="card shadow-sm border-secondary-subtle p-3 mb-3 d-flex justify-content-between align-items-center flex-wrap gap-3" data-panel-id="panel-registry-viewer-prosmotr-reestra-windows-regis" id="panel-registry-viewer-prosmotr-reestra-windows-regis">
<div>
<h5 class="mb-1 fw-bold d-flex align-items-center gap-2">
<span class="fs-4">□ </span> Просмотр реестра (Windows Registry Viewer)
      </h5>
<p class="mb-0 small text-muted">
        Безопасный навигатор по системному реестру Windows (HKLM, HKCU, HKCR, HKU, HKCC), быстрый переход к ключевым веткам автозагрузки, служб и политик.
      </p>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<span class="badge rounded-pill bg-success-subtle text-success border border-success px-3 py-2" id="reg-status-badge">
```

```
        ● Реестр: Подключено
      </span>
<button class="btn btn-sm btn-outline-info rounded-pill px-3" id="btn-reg-refresh">
<i class="bi bi-arrow-clockwise me-1"></i> Обновить
      </button>
<button class="btn btn-sm btn-outline-secondary rounded-pill px-3" id="btn-reg-copy-path">
<i class="bi bi-clipboard me-1"></i> Копировать путь
      </button>
<button class="btn btn-sm btn-warning rounded-pill px-3" id="btn-reg-backups-modal">
<i class="bi bi-shield-check me-1"></i> Бэкапы и откат
      </button>
</div>
</div>
<!-- Bookmarks Quick Bar -->
<div class="mb-3 d-flex align-items-center gap-1 flex-wrap p-2 rounded border border-secondary" id="reg-bookmarks-container" style="background: var(--surface-1);">
<span class="text-muted small me-2"><i class="bi bi-bookmark-star-fill text-warning me-1"></i> Закладки:</span>
<!-- Bookmarks will be loaded here -->
</div>
<!-- Navigation Path Bar -->
<div class="reg-path-bar mb-3">
<div class="row g-2 align-items-center">
<div class="col-md-3 col-sm-4">
<select class="form-select form-select-sm font-monospace" id="reg-hive-select">
<option value="HKEY_LOCAL_MACHINE">HKEY_LOCAL_MACHINE (HKLM)</option>
<option value="HKEY_CURRENT_USER">HKEY_CURRENT_USER (HKCU)</option>
<option value="HKEY_CLASSES_ROOT">HKEY_CLASSES_ROOT (HKCR)</option>
<option value="HKEY_USERS">HKEY_USERS (HKU)</option>
<option value="HKEY_CURRENT_CONFIG">HKEY_CURRENT_CONFIG (HKCC)
```

```
</option>
</select>
</div>
<div class="col-md-7 col-sm-6">
<div class="input-group input-group-sm">
<button class="btn btn-outline-secondary" id="btn-reg-up" titl
e="На уровень выше">
<i class="bi bi-arrow-up-short fs-6"></i>
</button>
<input class="form-control font-monospace" id="reg-path-input"
 placeholder="SOFTWARE\Microsoft\Windows..." type="text" value
="SOFTWARE"/>
<button class="btn btn-primary" id="btn-reg-go">
            Перейти
          </button>
</div>
</div>
<div class="col-md-2 col-sm-2 text-end">
<div class="input-group input-group-sm">
<input class="form-control" id="reg-filter-input" placeholder=
"Фильтр параметров..." type="text"/>
</div>
</div>
</div>
</div>
<!-- Main Explorer View (Subkeys Tree + Values Table) -->
<div class="row g-3">
<!-- Subkeys Tree Pane (Left) -->
<div class="col-md-4">
<div class="reg-tree-pane p-2">
<div class="d-flex justify-content-between align-items-center
mb-2 pb-1 border-bottom border-secondary">
<span class="text-muted small fw-bold"><i class="bi bi-folder2
-open me-1"></i> Подразделы (Subkeys)</span>
<div class="d-flex gap-1 align-items-center">
<span class="badge bg-secondary" id="reg-subkeys-count-badge">
0</span>
<button class="btn btn-xs btn-outline-success py-0 px-2" id="b
```

```
tn-reg-add-key" style="font-size: 0.75rem;" title="Создать под
раздел">
<i class="bi bi-plus-lg"></i>
</button>
<button class="btn btn-xs btn-outline-danger py-0 px-2" id="bt
n-reg-delete-current-key" style="font-size: 0.75rem;" title="У
далить текущий раздел">
<i class="bi bi-trash"></i>
</button>
</div>
</div>
<div id="reg-subkeys-list">
<div class="text-muted text-center py-4 small">Загрузка ключей
...</div>
</div>
</div>
</div>
<!-- Values & Data Pane (Right) -->
<div class="col-md-8">
<div class="reg-values-pane p-0">
<div class="d-flex justify-content-between align-items-center
p-2 border-bottom border-secondary" style="background: var(--s
urface-2);">
<span class="text-muted small fw-bold"><i class="bi bi-card-li
st me-1"></i> Параметры и значения (Values)</span>
<div class="d-flex gap-2 align-items-center">
<span class="badge bg-secondary" id="reg-values-count-badge">0
</span>
<button class="btn btn-sm btn-success py-1 px-3" id="btn-reg-a
dd-value">
<i class="bi bi-plus-circle me-1"></i> Создать параметр
            </button>
</div>
</div>
<div class="table-responsive" style="max-height: 535px; overfl
ow-y: auto;">
<table class="reg-table">
<thead>
```

```
<tr>
<th style="width: 28%;">Имя параметра</th>
<th style="width: 18%;">Тип</th>
<th style="width: 42%;">Значение</th>
<th style="width: 12%; text-align: center;">Действия</th>
</tr>
</thead>
<tbody id="reg-values-tbody">
<tr><td class="text-center text-muted p-4" colspan="4">Выберите ключ реестра для просмотра параметров</td></tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
<!-- Modal: Редактировать / Создать параметр -->
<div aria-hidden="true" class="modal fade" id="regValueModal" tabindex="-1">
<div class="modal-dialog modal-dialog-centered">
<div class="modal-content">
<div class="modal-header border-secondary">
<h5 class="modal-title" id="regValueModalTitle">  Редактирование параметра</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body">
<form id="regValueForm">
<div class="mb-3">
<label class="form-label small text-muted">Имя параметра</label>
<input class="form-control font-monospace" id="regModalValueName" placeholder="Оставьте пустым для (По умолчанию)" type="text"/>
</div>
<div class="mb-3">
<label class="form-label small text-muted">Тип данных</label>
```

```
<select class="form-select font-monospace" id="regModalValueType">
<option value="REG_SZ">REG_SZ (Строка)</option>
<option value="REG_DWORD">REG_DWORD (Число 32-бит)</option>
<option value="REG_QWORD">REG_QWORD (Число 64-бит)</option>
<option value="REG_EXPAND_SZ">REG_EXPAND_SZ (Расширяемая строка)</option>
<option value="REG_MULTI_SZ">REG_MULTI_SZ (Многострочный текст)</option>
<option value="REG_BINARY">REG_BINARY (Двоичные данные / HEX)</option>
</select>
</div>
<div class="mb-3">
<label class="form-label small text-muted">Значение</label>
<textarea class="form-control font-monospace" id="regModalValueData" placeholder="Введите значение параметра" rows="4"></textarea>
<div class="form-text text-muted small">Для REG_MULTI_SZ разделяйте строки новой строкой. Для REG_BINARY вводите HEX байты.</div>
</div>
<div class="form-check form-switch mb-2">
<input checked="" class="form-check-input" id="regModalBackupCheck" type="checkbox"/>
<label class="form-check-label small text-body" for="regModalBackupCheck">
                 Создать резервную копию ветки перед сохранением
              </label>
</div>
</form>
</div>
<div class="modal-footer border-secondary">
<button class="btn btn-secondary" data-bs-dismiss="modal" type="button">Отмена</button>
<button class="btn btn-primary" id="btn-reg-modal-save" type="button">Сохранить</button>
```

```
</div>
</div>
</div>
</div>
<!-- Modal: Создать подраздел -->
<div aria-hidden="true" class="modal fade" id="regKeyModal" ta
bindex="-1">
<div class="modal-dialog modal-dialog-centered">
<div class="modal-content">
<div class="modal-header border-secondary">
<h5 class="modal-title">  Новый подраздел реестра</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body">
<div class="mb-3">
<label class="form-label small text-muted">Имя нового подразде
ла</label>
<input class="form-control font-monospace" id="regModalNewKeyN
ame" placeholder="MyNewSubkey" type="text"/>
</div>
<p class="small text-muted mb-0">Ключ будет создан внутри теку
щего пути.</p>
</div>
<div class="modal-footer border-secondary">
<button class="btn btn-secondary" data-bs-dismiss="modal" type
="button">Отмена</button>
<button class="btn btn-success" id="btn-reg-modal-create-key"
type="button">Создать</button>
</div>
</div>
</div>
</div>
<!-- Modal: Резервные копии и откат -->
<div aria-hidden="true" class="modal fade" id="regBackupsModal
" tabindex="-1">
<div class="modal-dialog modal-lg modal-dialog-centered">
<div class="modal-content">
```

```
<div class="modal-header border-secondary">
<h5 class="modal-title">  Точки восстановления и история бэкапов</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body p-0">
<div class="table-responsive" style="max-height: 400px; overflow-y: auto;">
<table class="table table-hover mb-0 font-monospace small">
<thead>
<tr>
<th>Время</th>
<th>Ветка и путь</th>
<th>Операция</th>
<th>Действие</th>
</tr>
</thead>
<tbody id="reg-backups-tbody">
<tr><td class="text-center py-4 text-muted" colspan="4">Загрузка списка бэкапов...</td></tr>
</tbody>
</table>
</div>
</div>
<div class="modal-footer border-secondary">
<button class="btn btn-secondary" data-bs-dismiss="modal" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Registry Viewer Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/registry_viewer_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/registry_viewer_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// Windows Registry Viewer Tab JavaScript Module
(function() {
  let currentHive = 'HKEY_LOCAL_MACHINE';
  let currentPath = 'SOFTWARE';
  let currentKeyData = null;
  let isInitialized = false;

  async function loadBookmarks() {
    const container = document.getElementById('reg-bookmarks-container');
    if (!container) return;

```

```
    try {
      const res = await fetch('/api/registry/bookmarks');
      if (!res.ok) throw new Error('Bookmarks API error');
      const data = await res.json();
      const bookmarks = data.bookmarks || [];

      container.innerHTML = `
        <span class="text-muted small me-2"><i class="bi bi-bo
okmark-star-fill text-warning me-1"></i> Закладки:</span>
      ` + bookmarks.map(b => `
        <button class="reg-bookmark-btn" data-hive="${b.hive}"
 data-path="${b.path}" title="${b.description || ''}">
          <span>${b.icon || '□ '}</span> ${b.title}
        </button>
      `).join('');

      container.querySelectorAll('.reg-bookmark-btn').forEach(
btn => {
        btn.onclick = () => {
          const h = btn.getAttribute('data-hive');
          const p = btn.getAttribute('data-path');
          navigateTo(h, p);
        };
      });
    } catch (e) {
      console.warn('[RegistryViewerTab] Failed to load bookmar
ks:', e);
    }
  }

  async function navigateTo(hive, path) {
    currentHive = hive;
    currentPath = path.trim().replace(/^[\\\/]+|[\\\/]+$/g, ''
);

    const hiveSelect = document.getElementById('reg-hive-selec
t');
    const pathInput = document.getElementById('reg-path-input'
```

```
);
    if (hiveSelect) hiveSelect.value = currentHive;
    if (pathInput) pathInput.value = currentPath;

    await loadKey(currentHive, currentPath);
  }

  async function loadKey(hive, path) {
    const subkeysList = document.getElementById('reg-subkeys-list');
    const valuesTbody = document.getElementById('reg-values-tbody');
    const badge = document.getElementById('reg-status-badge');
    const subkeysCountBadge = document.getElementById('reg-subkeys-count-badge');
    const valuesCountBadge = document.getElementById('reg-values-count-badge');

    if (badge) badge.innerText = `● Чтение ${hive}...`;

    try {
      const url = `/api/registry/key?hive=${encodeURIComponent(hive)}&path=${encodeURIComponent(path)}`;
      const res = await fetch(url);
      if (!res.ok) {
        const errJson = await res.json().catch(() => ({}));
        throw new Error(errJson.detail || `HTTP ${res.status}`);
      }

      currentKeyData = await res.json();

      // Рендер подразделов (subkeys)
      const subkeys = currentKeyData.subkeys || [];
      if (subkeysCountBadge) subkeysCountBadge.innerText = subkeys.length;

      if (subkeys.length === 0) {
```

```
94        subkeysList.innerHTML = '<div class="text-muted text-c
enter py-4 small">Нет подразделов</div>';
95      } else {
96        subkeysList.innerHTML = subkeys.map(sk => `
97          <div class="reg-tree-item" data-subkey="${sk}" title
="${sk}">
98            <i class="bi bi-folder-fill text-warning"></i>
99            <span class="text-truncate">${sk}</span>
100         </div>
101       `).join('');
102
103       subkeysList.querySelectorAll('.reg-tree-item').forEach
(item => {
104         item.onclick = () => {
105           const subkey = item.getAttribute('data-subkey');
106           const newPath = currentPath ? `${currentPath}\\${s
ubkey}` : subkey;
107           navigateTo(currentHive, newPath);
108         };
109       });
110     }
111
112     // Рендер параметров (values)
113     renderValues();
114
115     if (badge) {
116       badge.className = 'badge rounded-pill bg-success-subtl
e text-success border border-success px-3 py-2';
117       badge.innerText = `● Реестр: ${hive}`;
118     }
119   } catch (err) {
120     console.error('[RegistryViewerTab] Error loading key:',
err);
121     if (valuesTbody) {
122       valuesTbody.innerHTML = `<tr><td colspan="3" class="te
xt-center text-danger p-4"><i class="bi bi-exclamation-triangl
e-fill me-1"></i> Ошибка чтения ветки: ${err.message}</td></tr
>`;
```

```
      }
      if (subkeysList) {
        subkeysList.innerHTML = `<div class="text-danger text-center py-4 small">${err.message}</div>`;
      }
      if (badge) {
        badge.className = 'badge rounded-pill bg-danger-subtle text-danger border border-danger px-3 py-2';
        badge.innerText = '● Ошибка доступа';
      }
    }
  }

  function renderValues() {
    const valuesTbody = document.getElementById('reg-values-tbody');
    const valuesCountBadge = document.getElementById('reg-values-count-badge');
    if (!valuesTbody || !currentKeyData) return;

    const filterText = (document.getElementById('reg-filter-input')?.value || '').toLowerCase().trim();
    const values = currentKeyData.values || [];

    const filtered = values.filter(v => {
      if (!filterText) return true;
      return v.name.toLowerCase().includes(filterText) || String(v.data).toLowerCase().includes(filterText) || v.type_name.toLowerCase().includes(filterText);
    });

    if (valuesCountBadge) valuesCountBadge.innerText = `${filtered.length} / ${values.length}`;

    if (filtered.length === 0) {
      valuesTbody.innerHTML = '<tr><td colspan="3" class="text-center text-muted p-4">Параметры отсутствуют или не соответствуют фильтру</td></tr>';
```

```
      return;
    }

    valuesTbody.innerHTML = filtered.map((v, idx) => {
      let dataDisplay = v.data;
      if (dataDisplay === null || dataDisplay === undefined) {
        dataDisplay = '<span class="text-muted">(Значение не з
адано)</span>';
      } else if (typeof dataDisplay === 'object') {
        dataDisplay = `<pre class="mb-0 text-info" style="font
-size: 0.75rem;">${JSON.stringify(dataDisplay, null, 2)}</pre>
`;
      } else {
        dataDisplay = `<span class="text-light text-break">${e
scapeHtml(String(dataDisplay))}</span>`;
      }

      const isDefault = v.name === '(Default / По умолчанию)'
|| v.name === '(Default)';
      const rawNameAttr = escapeHtml(isDefault ? '' : v.name);

      return `
        <tr class="reg-val-row" data-idx="${idx}" style="curso
r: pointer;" title="Нажмите для AI-объяснения параметра реестр
а">
          <td>
            <div class="fw-bold text-white text-truncate" styl
e="max-width: 220px;" title="${v.name}">
              <i class="bi bi-file-earmark-binary text-seconda
ry me-1"></i>
              ${escapeHtml(v.name)}
            </div>
          </td>
          <td>
            <span class="reg-type-badge">${v.type_name}</span>
          </td>
          <td>
            ${dataDisplay}
```

```
          </td>
          <td class="text-center">
            <button class="btn btn-xs btn-outline-info py-0 px-2 me-1 btn-edit-val" data-idx="${idx}" title="Редактировать">
              <i class="bi bi-pencil"></i>
            </button>
            <button class="btn btn-xs btn-outline-danger py-0 px-2 btn-del-val" data-name="${rawNameAttr}" title="Удалить">
              <i class="bi bi-trash"></i>
            </button>
          </td>
        </tr>
      `;
    }).join('');

    // Привязка кликабельности всей строки к AITableModal
    valuesTbody.querySelectorAll('.reg-val-row').forEach(row => {
      row.onclick = (evt) => {
        if (evt.target.closest('.btn-edit-val') || evt.target.closest('.btn-del-val')) return;
        const idx = parseInt(row.getAttribute('data-idx'), 10);
        const v = filtered[idx];
        if (!v) return;

        if (window.AITableModal) {
          window.AITableModal.show({
            icon: '□ ',
            title: v.name || '(Default)',
            subtitle: `${currentHive}\\${currentPath}`,
            tableType: 'registry',
            badges: [
              { text: v.type_name || 'REG_SZ', class: 'badge bg-info text-dark' },
              { text: currentHive, class: 'badge bg-secondary' }
            ],
```

```
            metadata: [
              { label: 'Параметр', value: v.name || '(По умолчанию)' },
              { label: 'Тип данных', value: v.type_name },
              { label: 'Ветка реестра (Hive)', value: currentHive },
              { label: 'Путь к разделу', value: currentPath, isCode: true, fullWidth: true },
              { label: 'Значение', value: String(v.data ?? ''), isCode: true, fullWidth: true }
            ],
            rawTitle: 'Значение параметра',
            rawContent: typeof v.data === 'object' ? JSON.stringify(v.data, null, 2) : String(v.data ?? ''),
            requestData: {
              hive: currentHive,
              path: currentPath,
              name: v.name,
              type: v.type_name,
              data: v.data
            }
          });
        }
      };
    });

    // Привязка событий кнопок редактирования и удаления параметров
    valuesTbody.querySelectorAll('.btn-edit-val').forEach(btn => {
      btn.onclick = (evt) => {
        evt.stopPropagation();
        const idx = parseInt(btn.getAttribute('data-idx'), 10);
        const valObj = filtered[idx];
        if (valObj) openEditValueModal(valObj);
      };
    });
```

```
241
242     valuesTbody.querySelectorAll('.btn-del-val').forEach(btn =
    > {
243       btn.onclick = (evt) => {
244         evt.stopPropagation();
245         const valName = btn.getAttribute('data-name');
246         confirmDeleteValue(valName);
247       };
248     });
249   }
250
251   function openEditValueModal(valObj) {
252     const isDefault = !valObj.name || valObj.name === '(Defaul
    t / По умолчанию)' || valObj.name === '(Default)';
253     const nameInput = document.getElementById('regModalValueNa
    me');
254     const typeSelect = document.getElementById('regModalValueT
    ype');
255     const dataInput = document.getElementById('regModalValueDa
    ta');
256     const title = document.getElementById('regValueModalTitle'
    );
257
258     if (title) title.innerText = '  Редактирование параметра';
259     if (nameInput) {
260       nameInput.value = isDefault ? '' : valObj.name;
261       nameInput.disabled = true; // имя существующего параметр
    а не меняем
262     }
263     if (typeSelect) typeSelect.value = valObj.type_name || 'RE
    G_SZ';
264     if (dataInput) {
265       if (Array.isArray(valObj.data)) {
266         dataInput.value = valObj.data.join('\n');
267       } else {
268         dataInput.value = valObj.data !== null && valObj.data
    !== undefined ? String(valObj.data) : '';
269       }
```

```
    }

    const modalEl = document.getElementById('regValueModal');
    if (modalEl && window.bootstrap) {
      const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
      modal.show();
    }
  }

  function openCreateValueModal() {
    const nameInput = document.getElementById('regModalValueName');
    const typeSelect = document.getElementById('regModalValueType');
    const dataInput = document.getElementById('regModalValueData');
    const title = document.getElementById('regValueModalTitle');

    if (title) title.innerText = '  Создание нового параметра';
    if (nameInput) {
      nameInput.value = '';
      nameInput.disabled = false;
    }
    if (typeSelect) typeSelect.value = 'REG_SZ';
    if (dataInput) dataInput.value = '';

    const modalEl = document.getElementById('regValueModal');
    if (modalEl && window.bootstrap) {
      const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
      modal.show();
    }
  }

  async function saveModalValue() {
```

```
    const nameInput = document.getElementById('regModalValueName');
    const typeSelect = document.getElementById('regModalValueType');
    const dataInput = document.getElementById('regModalValueData');
    const backupCheck = document.getElementById('regModalBackupCheck');

    const name = nameInput ? nameInput.value.trim() : '';
    const type_name = typeSelect ? typeSelect.value : 'REG_SZ';
    const rawData = dataInput ? dataInput.value : '';
    const create_backup = backupCheck ? backupCheck.checked : true;

    try {
      const payload = {
        hive: currentHive,
        path: currentPath,
        name: name,
        type_name: type_name,
        data: rawData,
        create_backup: create_backup
      };

      const res = await fetch('/api/registry/value', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify(payload)
      });

      if (!res.ok) {
        const errJson = await res.json().catch(() => ({}));
        throw new Error(errJson.detail || `HTTP ${res.status}`);
      }

```

```
      const modalEl = document.getElementById('regValueModal')
;
      if (modalEl && window.bootstrap) {
        const modal = bootstrap.Modal.getInstance(modalEl);
        if (modal) modal.hide();
      }

      await loadKey(currentHive, currentPath);
    } catch (e) {
      window.showToast?.(`Ошибка сохранения параметра: ${e.mes
sage}`, 'danger') || alert(`Ошибка сохранения параметра: ${e.m
essage}`);
    }
  }

  async function confirmDeleteValue(name) {
    const displayName = name ? `'${name}'` : '(По умолчанию)';
    if (!confirm(`Вы действительно хотите удалить параметр ${d
isplayName}? Перед удалением будет автоматически создан бэкап.
`)) {
      return;
    }

    try {
      const res = await fetch('/api/registry/value', {
        method: 'DELETE',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({
          hive: currentHive,
          path: currentPath,
          name: name,
          create_backup: true
        })
      });

      if (!res.ok) {
        const errJson = await res.json().catch(() => ({}));
        throw new Error(errJson.detail || `HTTP ${res.status}`
```

```
);
      }

      window.showToast?.(`Параметр ${displayName} удален`, 'success');
      await loadKey(currentHive, currentPath);
    } catch (e) {
      window.showToast?.(`Ошибка удаления параметра: ${e.message}`, 'danger') || alert(`Ошибка удаления параметра: ${e.message}`);
    }
  }

  function openCreateKeyModal() {
    const keyNameInput = document.getElementById('regModalNewKeyName');
    if (keyNameInput) keyNameInput.value = '';

    const modalEl = document.getElementById('regKeyModal');
    if (modalEl && window.bootstrap) {
      const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
      modal.show();
    }
  }

  async function saveModalKey() {
    const keyNameInput = document.getElementById('regModalNewKeyName');
    const newKeyName = keyNameInput ? keyNameInput.value.trim() : '';
    if (!newKeyName) {
      window.showToast?.('Введите имя подраздела', 'warning') || alert('Введите имя подраздела');
      return;
    }

    const newPath = currentPath ? `${currentPath}\\${newKeyNam
```

```
e}` : newKeyName;

    try {
      const res = await fetch('/api/registry/key', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({
          hive: currentHive,
          path: newPath
        })
      });

      if (!res.ok) {
        const errJson = await res.json().catch(() => ({}));
        throw new Error(errJson.detail || `HTTP ${res.status}`
);
      }

      const modalEl = document.getElementById('regKeyModal');
      if (modalEl && window.bootstrap) {
        const modal = bootstrap.Modal.getInstance(modalEl);
        if (modal) modal.hide();
      }

      window.showToast?.(`Подраздел '${newKeyName}' успешно со
здан`, 'success');
      await loadKey(currentHive, currentPath);
    } catch (e) {
      window.showToast?.(`Ошибка создания ключа: ${e.message}`
, 'danger') || alert(`Ошибка создания ключа: ${e.message}`);
    }
  }

  async function deleteCurrentKey() {
    if (!currentPath) {
      window.showToast?.('Нельзя удалить корневой раздел', 'wa
rning') || alert('Нельзя удалить корневой раздел');
      return;
```

```
427     }
428
429     if (!confirm(`ВНИМАНИЕ: Вы действительно хотите удалить ра
    здел '${currentHive}\\${currentPath}' со всеми подразделами и
    значениями? Будет создан снимок резервной копии.`)) {
430       return;
431     }
432
433     try {
434       const res = await fetch('/api/registry/key', {
435         method: 'DELETE',
436         headers: { 'Content-Type': 'application/json' },
437         body: JSON.stringify({
438           hive: currentHive,
439           path: currentPath,
440           recursive: true,
441           create_backup: true
442         })
443       });
444
445       if (!res.ok) {
446         const errJson = await res.json().catch(() => ({}));
447         throw new Error(errJson.detail || `HTTP ${res.status}`
    );
448       }
449
450       window.showToast?.('Раздел реестра успешно удален', 'suc
    cess');
451       goUp();
452     } catch (e) {
453       window.showToast?.(`Ошибка удаления раздела: ${e.message
    }`, 'danger') || alert(`Ошибка удаления раздела: ${e.message}`
    );
454     }
455   }
456
457   async function openBackupsModal() {
458     const modalEl = document.getElementById('regBackupsModal')
```

```
;
459     const tbody = document.getElementById('reg-backups-tbody')
;
460 
461     if (modalEl && window.bootstrap) {
462       const modal = bootstrap.Modal.getOrCreateInstance(modalE
l);
463       modal.show();
464     }
465 
466     if (tbody) tbody.innerHTML = '<tr><td colspan="4" class="t
ext-center py-3 text-muted">Загрузка истории снимков...</td></
tr>';
467 
468     try {
469       const res = await fetch('/api/registry/backups');
470       if (!res.ok) throw new Error('Ошибка загрузки бэкапов');
471       const data = await res.json();
472       const backups = data.backups || [];
473 
474       if (backups.length === 0) {
475         tbody.innerHTML = '<tr><td colspan="4" class="text-cen
ter py-4 text-muted">Точки восстановления отсутствуют</td></tr
>';
476         return;
477       }
478 
479       tbody.innerHTML = backups.map(b => `
480         <tr>
481           <td><small class="text-light">${new Date(b.timestamp
).toLocaleString()}</small></td>
482           <td class="text-truncate" style="max-width: 250px;"
title="${b.hive}\\${b.path}">
483             <span class="text-warning">${b.hive}</span>\\${esc
apeHtml(b.path)}
484           </td>
485           <td><span class="badge bg-secondary">${escapeHtml(b.
operation)}</span></td>
```

```
          <td>
            <button class="btn btn-xs btn-outline-warning py-0 px-2 btn-restore-backup" data-id="${b.backup_id}">
              <i class="bi bi-arrow-counterclockwise me-1"></i> Откатить
            </button>
          </td>
        </tr>
      `).join('');

      tbody.querySelectorAll('.btn-restore-backup').forEach(btn => {
        btn.onclick = async () => {
          const backupId = btn.getAttribute('data-id');
          if (!confirm(`Восстановить реестр из снимка ${backupId}?`)) return;

          try {
            const rRes = await fetch(`/api/registry/restore?backup_id=${encodeURIComponent(backupId)}`, {
              method: 'POST'
            });
            if (!rRes.ok) {
              const err = await rRes.json().catch(() => ({}));
              throw new Error(err.detail || `HTTP ${rRes.status}`);
            }
            window.showToast?.('Реестр успешно восстановлен из резервной копии!', 'success');
            await loadKey(currentHive, currentPath);
          } catch (err) {
            window.showToast?.(`Ошибка восстановления: ${err.message}`, 'danger') || alert(`Ошибка восстановления: ${err.message}`);
          }
        };
      });
    } catch (e) {
```

```
      if (tbody) tbody.innerHTML = `<tr><td colspan="4" class="text-center py-3 text-danger">${e.message}</td></tr>`;
    }
  }

  function escapeHtml(text) {
    return text
      .replace(/&/g, "&amp;")
      .replace(/</g, "&lt;")
      .replace(/>/g, "&gt;")
      .replace(/"/g, "&quot;")
      .replace(/'/g, "&#039;");
  }

  function goUp() {
    if (!currentPath) return;
    const parts = currentPath.split('\\');
    parts.pop();
    const newPath = parts.join('\\');
    navigateTo(currentHive, newPath);
  }

  function copyCurrentPath() {
    const full = `${currentHive}\\${currentPath}`.replace(/\\+$/, '');
    if (navigator.clipboard) {
      navigator.clipboard.writeText(full).then(() => {
        const btn = document.getElementById('btn-reg-copy-path');
        if (btn) {
          const orig = btn.innerHTML;
          btn.innerHTML = '<i class="bi bi-check me-1"></i> Скопировано!';
          setTimeout(() => { btn.innerHTML = orig; }, 1500);
        }
      });
    }
  }
```

```
549
550   window.initRegistryViewerTab = async function() {
551     console.log('[RegistryViewerTab] Initializing Registry Vie
    wer & Editor tab...');
552
553     const hiveSelect = document.getElementById('reg-hive-selec
    t');
554     const pathInput = document.getElementById('reg-path-input'
    );
555     const goBtn = document.getElementById('btn-reg-go');
556     const upBtn = document.getElementById('btn-reg-up');
557     const filterInput = document.getElementById('reg-filter-in
    put');
558     const refreshBtn = document.getElementById('btn-reg-refres
    h');
559     const copyBtn = document.getElementById('btn-reg-copy-path
    ');
560     const backupsBtn = document.getElementById('btn-reg-backup
    s-modal');
561     const addValBtn = document.getElementById('btn-reg-add-val
    ue');
562     const addKeyBtn = document.getElementById('btn-reg-add-key
    ');
563     const delKeyBtn = document.getElementById('btn-reg-delete-
    current-key');
564     const saveValModalBtn = document.getElementById('btn-reg-m
    odal-save');
565     const saveKeyModalBtn = document.getElementById('btn-reg-m
    odal-create-key');
566
567     if (hiveSelect) {
568       hiveSelect.onchange = () => {
569         navigateTo(hiveSelect.value, '');
570       };
571     }
572     if (goBtn && pathInput) {
573       goBtn.onclick = () => {
574         navigateTo(hiveSelect.value, pathInput.value);
```

```
      };
    }
    if (pathInput) {
      pathInput.onkeydown = (e) => {
        if (e.key === 'Enter') {
          navigateTo(hiveSelect.value, pathInput.value);
        }
      };
    }
    if (upBtn) upBtn.onclick = goUp;
    if (filterInput) filterInput.oninput = renderValues;
    if (refreshBtn) refreshBtn.onclick = () => loadKey(current
Hive, currentPath);
    if (copyBtn) copyBtn.onclick = copyCurrentPath;
    if (backupsBtn) backupsBtn.onclick = openBackupsModal;
    if (addValBtn) addValBtn.onclick = openCreateValueModal;
    if (addKeyBtn) addKeyBtn.onclick = openCreateKeyModal;
    if (delKeyBtn) delKeyBtn.onclick = deleteCurrentKey;
    if (saveValModalBtn) saveValModalBtn.onclick = saveModalVa
lue;
    if (saveKeyModalBtn) saveKeyModalBtn.onclick = saveModalKe
y;

    await loadBookmarks();
    await navigateTo(currentHive, currentPath);
    isInitialized = true;
  };
})();


```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/remote_mic/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/remote_mic/index.html"
class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/remote_mic
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!DOCTYPE html>
<html lang="ru">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-s
cale=1.0, maximum-scale=1.0, user-scalable=no">
  <title>Голосовой Пульт - AI Windows Admin</title>

  <!-- Bootstrap 5 CSS -->
```

```
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
  <!-- Bootstrap Icons -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.10.0/font/bootstrap-icons.css" rel="stylesheet">

  <!-- Telegram Web Apps SDK & Marked.js -->
  <script src="https://telegram.org/js/telegram-web-app.js"></script>
  <script src="https://cdn.jsdelivr.net/npm/marked/marked.min.js"></script>

  <style>
    :root {
      --tg-theme-bg-color: #0b0f19;
      --tg-theme-text-color: #f1f5f9;
      --tg-theme-hint-color: #94a3b8;
      --tg-theme-secondary-bg-color: #151c2e;
      --primary-gradient: linear-gradient(135deg, #3b82f6 0%, #1d4ed8 100%);
      --mic-idle-gradient: linear-gradient(135deg, #ef4444 0%, #b91c1c 100%);
      --mic-listening-gradient: linear-gradient(135deg, #10b981 0%, #047857 100%);
      --mic-thinking-gradient: linear-gradient(135deg, #8b5cf6 0%, #6d28d9 100%);
      --accent-color: #38bdf8;
    }

    body {
      background: var(--tg-theme-bg-color);
      color: var(--tg-theme-text-color);
      font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, "Helvetica Neue", Arial, sans-serif;
      min-height: 100vh;
      display: flex;
      flex-direction: column;
      justify-content: flex-start;
```

```
    align-items: center;
    padding: 12px;
    margin: 0;
    overflow-x: hidden;
  }

  .app-container {
    width: 100%;
    max-width: 520px;
    display: flex;
    flex-direction: column;
    height: 100%;
    min-height: 94vh;
  }

  .card-panel {
    background-color: var(--tg-theme-secondary-bg-color);
    border: 1px solid rgba(255, 255, 255, 0.08);
    border-radius: 20px;
    padding: 18px;
    box-shadow: 0 10px 30px rgba(0, 0, 0, 0.4);
    backdrop-filter: blur(12px);
  }

  /* Giant Mic Button */
  .mic-centerpiece {
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    padding: 16px 0 12px;
    position: relative;
  }

  .mic-button-wrapper {
    position: relative;
    display: flex;
    align-items: center;
```

```
      justify-content: center;
    }

    .btn-mic-giant {
      width: 110px;
      height: 110px;
      border-radius: 50%;
      border: none;
      outline: none;
      background: var(--mic-idle-gradient);
      color: #ffffff;
      display: flex;
      align-items: center;
      justify-content: center;
      font-size: 44px;
      box-shadow: 0 8px 25px rgba(239, 68, 68, 0.4);
      cursor: pointer;
      transition: all 0.25s cubic-bezier(0.4, 0, 0.2, 1);
      position: relative;
      z-index: 2;
    }

    .btn-mic-giant:hover {
      transform: scale(1.05);
      box-shadow: 0 12px 30px rgba(239, 68, 68, 0.6);
    }

    .btn-mic-giant:active {
      transform: scale(0.95);
    }

    .btn-mic-giant.listening {
      background: var(--mic-listening-gradient);
      box-shadow: 0 8px 30px rgba(16, 185, 129, 0.5);
      animation: mic-listening-pulse 1.8s infinite;
    }

    .btn-mic-giant.thinking {
```

```
      background: var(--mic-thinking-gradient);
      box-shadow: 0 8px 30px rgba(139, 92, 246, 0.5);
      animation: mic-thinking-pulse 1.2s infinite ease-in-out;
    }

    @keyframes mic-listening-pulse {
      0% { transform: scale(1); box-shadow: 0 0 0 0 rgba(16, 185, 129, 0.7); }
      70% { transform: scale(1.06); box-shadow: 0 0 0 22px rgba(16, 185, 129, 0); }
      100% { transform: scale(1); box-shadow: 0 0 0 0 rgba(16, 185, 129, 0); }
    }

    @keyframes mic-thinking-pulse {
      0% { transform: rotate(0deg) scale(1); }
      50% { transform: rotate(180deg) scale(1.05); }
      100% { transform: rotate(360deg) scale(1); }
    }

    /* Wave visualizer dots */
    .wave-bars {
      display: flex;
      align-items: center;
      justify-content: center;
      gap: 5px;
      height: 20px;
      margin-top: 8px;
    }

    .wave-bar {
      width: 4px;
      height: 6px;
      background: #ef4444;
      border-radius: 2px;
      transition: height 0.15s ease;
    }

```

```
    .wave-bars.active .wave-bar {
      background: #10b981;
      animation: wave-anim 0.8s ease-in-out infinite alternate
;
    }

    .wave-bars.active .wave-bar:nth-child(2) { animation-delay
: 0.15s; }
    .wave-bars.active .wave-bar:nth-child(3) { animation-delay
: 0.3s; }
    .wave-bars.active .wave-bar:nth-child(4) { animation-delay
: 0.45s; }
    .wave-bars.active .wave-bar:nth-child(5) { animation-delay
: 0.2s; }

    @keyframes wave-anim {
      0% { height: 6px; }
      100% { height: 20px; }
    }

    /* Media Actions Bar */
    .media-actions-bar {
      display: flex;
      align-items: center;
      justify-content: center;
      gap: 8px;
      margin-top: 10px;
      width: 100%;
    }

    .btn-media-action {
      background: rgba(255, 255, 255, 0.06);
      border: 1px solid rgba(255, 255, 255, 0.12);
      color: #e2e8f0;
      font-size: 0.82rem;
      font-weight: 500;
      padding: 6px 14px;
      border-radius: 20px;
```

```
      display: inline-flex;
      align-items: center;
      gap: 6px;
      transition: all 0.2s ease;
      cursor: pointer;
    }

    .btn-media-action:hover {
      background: rgba(255, 255, 255, 0.14);
      color: #ffffff;
      border-color: rgba(255, 255, 255, 0.25);
    }

    .btn-media-action.active-audio {
      border-color: #38bdf8;
      color: #38bdf8;
      background: rgba(56, 189, 248, 0.1);
    }

    .btn-media-action.active-media {
      border-color: #a855f7;
      color: #c084fc;
      background: rgba(168, 85, 247, 0.1);
    }

    /* Chat Messages Container */
    .chat-container {
      flex: 1;
      overflow-y: auto;
      border-radius: 14px;
      background: rgba(0, 0, 0, 0.2);
      border: 1px solid rgba(255, 255, 255, 0.05);
      padding: 12px;
      display: flex;
      flex-direction: column;
      gap: 10px;
      max-height: 44vh;
      min-height: 180px;
```

```
    scroll-behavior: smooth;
  }

  .chat-container::-webkit-scrollbar {
    width: 5px;
  }
  .chat-container::-webkit-scrollbar-thumb {
    background: rgba(255, 255, 255, 0.15);
    border-radius: 10px;
  }

  .chat-bubble {
    max-width: 90%;
    padding: 10px 14px;
    border-radius: 14px;
    font-size: 0.92rem;
    line-height: 1.45;
    word-break: break-word;
    position: relative;
  }

  .chat-bubble.user {
    align-self: flex-end;
    background: var(--primary-gradient);
    color: #ffffff;
    border-bottom-right-radius: 4px;
  }

  .chat-bubble.ai {
    align-self: flex-start;
    background: #1e293b;
    color: #f1f5f9;
    border: 1px solid rgba(255, 255, 255, 0.08);
    border-bottom-left-radius: 4px;
  }

  .chat-bubble.ai p:last-child {
    margin-bottom: 0;
```

```
    }

    .chat-meta {
      display: flex;
      align-items: center;
      justify-content: space-between;
      gap: 8px;
      margin-top: 4px;
      font-size: 0.72rem;
      opacity: 0.75;
    }

    .btn-speak-again {
      background: none;
      border: none;
      color: inherit;
      padding: 0;
      cursor: pointer;
      font-size: 0.85rem;
    }

    /* Media Attachment Previews */
    .media-preview-card {
      background: rgba(0, 0, 0, 0.25);
      border: 1px solid rgba(255, 255, 255, 0.1);
      border-radius: 10px;
      padding: 8px 10px;
      margin-bottom: 6px;
      display: flex;
      align-items: center;
      gap: 8px;
    }

    .media-player-widget {
      width: 100%;
      margin-top: 6px;
      border-radius: 8px;
      background: rgba(0, 0, 0, 0.2);
```

```
    }

    .media-thumb-img {
      max-width: 100%;
      max-height: 200px;
      border-radius: 8px;
      margin-top: 6px;
      object-fit: cover;
    }

    /* Language and Theme Switchers */
    .btn-lang-pill {
      font-size: 0.75rem;
      font-weight: 600;
      padding: 3px 9px;
      border-radius: 20px;
    }

    .theme-toggle-btn {
      background: rgba(255, 255, 255, 0.08);
      border: 1px solid rgba(255, 255, 255, 0.15);
      color: #ffffff;
      width: 34px;
      height: 34px;
      border-radius: 50%;
      display: flex;
      align-items: center;
      justify-content: center;
      cursor: pointer;
      transition: all 0.2s;
    }
    .theme-toggle-btn:hover {
      background: rgba(255, 255, 255, 0.18);
    }

    /* Input text area fallback */
    .form-control-custom {
      background: rgba(0, 0, 0, 0.25);
```

```
      border: 1px solid rgba(255, 255, 255, 0.12);
      color: #ffffff;
      border-radius: 12px;
    }
    .form-control-custom:focus {
      background: rgba(0, 0, 0, 0.35);
      border-color: #3b82f6;
      color: #ffffff;
      box-shadow: 0 0 0 2px rgba(59, 130, 246, 0.3);
    }

    /* Light Theme */
    body.theme-light {
      --tg-theme-bg-color: #f8fafc;
      --tg-theme-text-color: #0f172a;
      --tg-theme-hint-color: #64748b;
      --tg-theme-secondary-bg-color: #ffffff;
    }
    body.theme-light .card-panel {
      border-color: #e2e8f0;
      box-shadow: 0 8px 24px rgba(0, 0, 0, 0.06);
    }
    body.theme-light .chat-container {
      background: #f1f5f9;
      border-color: #e2e8f0;
    }
    body.theme-light .chat-bubble.ai {
      background: #ffffff;
      color: #0f172a;
      border-color: #e2e8f0;
    }
    body.theme-light .theme-toggle-btn {
      background: #e2e8f0;
      color: #0f172a;
      border-color: #cbd5e1;
    }
    body.theme-light .btn-media-action {
      background: #f1f5f9;
```

```
      color: #334155;
      border-color: #cbd5e1;
    }
    body.theme-light .form-control-custom {
      background: #ffffff;
      color: #0f172a;
      border-color: #cbd5e1;
    }
  </style>
</head>
<body>

  <!-- AUTHENTICATION PANEL -->
  <div class="app-container" id="auth-panel">
    <div class="card-panel my-auto">
      <div class="text-center mb-4">
        <i class="bi bi-mic-fill fs-1 text-primary"></i>
        <h4 class="mt-2 text-white fw-bold">Голосовой Пульт</h
4>
        <p class="text-muted small">Войдите в систему для пере
дачи голосовых сообщений и медиа в чат</p>
      </div>

      <div id="auth-alert" class="alert alert-danger d-none" r
ole="alert"></div>

      <!-- Login Form -->
      <div id="auth-form-login">
        <form id="form-login">
          <div class="mb-3">
            <label for="login-email" class="form-label text-mu
ted small">Email</label>
            <input type="email" class="form-control form-contr
ol-custom" id="login-email" required>
          </div>
          <div class="mb-3">
            <label for="login-password" class="form-label text
-muted small">Пароль</label>
```

```
            <input type="password" class="form-control form-control-custom" id="login-password" required>
          </div>
          <button type="submit" class="btn btn-primary w-100 py-2.5 rounded-pill mb-3">Войти</button>
          <p class="text-center small mb-0">Нет аккаунта? <a href="#" id="go-to-register" class="text-info">Регистрация</a></p>
        </form>
      </div>

      <!-- Register Form -->
      <div id="auth-form-register" class="d-none">
        <form id="form-register">
          <div class="mb-3">
            <label for="register-name" class="form-label text-muted small">Имя</label>
            <input type="text" class="form-control form-control-custom" id="register-name" required>
          </div>
          <div class="mb-3">
            <label for="register-email" class="form-label text-muted small">Email</label>
            <input type="email" class="form-control form-control-custom" id="register-email" required>
          </div>
          <div class="mb-3">
            <label for="register-password" class="form-label text-muted small">Пароль</label>
            <input type="password" class="form-control form-control-custom" id="register-password" required>
          </div>
          <button type="submit" class="btn btn-success w-100 py-2.5 rounded-pill mb-3">Зарегистрироваться</button>
          <p class="text-center small mb-0">Уже есть аккаунт? <a href="#" id="go-to-login" class="text-info">Войти</a></p>
        </form>
      </div>
```

```
453
454       <!-- Verify Form -->
455       <div id="auth-form-verify" class="d-none">
456         <p class="text-center small">Код подтверждения отправл
ен на почту:</p>
457         <form id="form-verify">
458           <div class="mb-3">
459             <input type="text" class="form-control form-contro
l-custom text-center fs-3 py-2" id="verify-code" maxlength="6"
 placeholder="XXXXXX" required>
460           </div>
461           <button type="submit" class="btn btn-info w-100 py-2
.5 text-white rounded-pill mb-3">Подтвердить</button>
462           <p class="text-center small mb-0"><a href="#" id="ve
rify-cancel" class="text-info">Назад</a></p>
463         </form>
464       </div>
465
466       <div class="my-4 text-center position-relative">
467         <hr class="border-secondary">
468         <span class="px-2 text-muted position-absolute top-50
start-50 translate-middle small bg-dark">Или</span>
469       </div>
470
471       <button id="google-signin-btn" class="btn btn-outline-li
ght w-100 rounded-pill d-flex align-items-center justify-conte
nt-center gap-2 py-2">
472         <i class="bi bi-google text-danger"></i>
473         <span>Войти через Google</span>
474       </button>
475     </div>
476   </div>
477
478   <!-- MAIN VOICE REMOTE PANEL -->
479   <div class="app-container d-none" id="main-panel">
480     <div class="card-panel d-flex flex-column h-100">
481
482       <!-- Top Bar -->
```

```
      <div class="d-flex justify-content-between align-items-c
enter mb-2">
        <div class="d-flex align-items-center gap-2">
          <span class="badge bg-success rounded-pill px-2.5 py
-1.5" id="connection-status">
            <i class="bi bi-broadcast"></i> Онлайн
          </span>
          <span class="small text-muted text-truncate" style="
max-width: 120px;" id="user-display-name"></span>
        </div>

        <div class="d-flex align-items-center gap-2">
          <!-- Language Selector -->
          <div class="btn-group btn-group-sm" role="group">
            <input type="radio" class="btn-check" name="lang-r
adio" id="lang-ru" value="ru-RU" checked>
            <label class="btn btn-outline-secondary btn-lang-p
ill" for="lang-ru">RU</label>

            <input type="radio" class="btn-check" name="lang-r
adio" id="lang-en" value="en-US">
            <label class="btn btn-outline-secondary btn-lang-p
ill" for="lang-en">EN</label>

            <input type="radio" class="btn-check" name="lang-r
adio" id="lang-he" value="he-IL">
            <label class="btn btn-outline-secondary btn-lang-p
ill" for="lang-he">HE</label>
          </div>

          <!-- TTS Mute/Unmute Toggle -->
          <button class="theme-toggle-btn" id="tts-toggle-btn"
 title="Озвучивание ответов">
            <i class="bi bi-volume-up-fill" id="tts-icon"></i>
          </button>

          <!-- Theme Toggle -->
          <button class="theme-toggle-btn" id="theme-toggle-bt
```

```
n" title="Сменить тему">
511             <i class="bi bi-moon-stars-fill" id="theme-icon"><
/i>
512           </button>
513         </div>
514       </div>
515
516       <!-- Big Microphone Centerpiece -->
517       <div class="mic-centerpiece">
518         <div class="mic-button-wrapper">
519           <button class="btn-mic-giant" id="btn-mic" title="На
жмите для голосового ввода">
520             <i class="bi bi-mic-fill" id="mic-icon"></i>
521           </button>
522         </div>
523
524         <div class="wave-bars" id="wave-bars">
525           <div class="wave-bar"></div>
526           <div class="wave-bar"></div>
527           <div class="wave-bar"></div>
528           <div class="wave-bar"></div>
529           <div class="wave-bar"></div>
530         </div>
531
532         <div class="fw-semibold mt-2 text-center" id="voice-st
atus-text" style="font-size: 0.95rem;">
533           Нажмите кнопку и говорите
534         </div>
535         <div class="small text-muted text-center" id="voice-su
btext" style="font-size: 0.78rem;">
536           Голос будет распознан и отправлен в чат
537         </div>
538
539         <!-- Quick Media & Audio Upload Actions -->
540         <div class="media-actions-bar">
541           <button class="btn-media-action" id="btn-upload-audi
o" title="Загрузить аудиозапись или диктофон">
542             <i class="bi bi-file-earmark-music-fill text-info"
```

```
></i>
543             <span>Аудиофайл</span>
544           </button>
545           <button class="btn-media-action" id="btn-upload-medi
a" title="Загрузить фото, видео или документ">
546             <i class="bi bi-cloud-arrow-up-fill text-primary">
</i>
547             <span>Медиа</span>
548           </button>
549         </div>
550
551         <!-- Hidden File Inputs -->
552         <input type="file" id="audio-file-input" accept="audio
/*,.mp3,.wav,.ogg,.m4a,.flac,.aac,.webm,.opus" class="d-none">
553         <input type="file" id="media-file-input" accept="image
/*,video/*,audio/*,.pdf,.doc,.docx,.txt" class="d-none">
554       </div>
555
556       <!-- Chat Stream Feed -->
557       <div class="d-flex justify-content-between align-items-c
enter mt-1 mb-1 px-1">
558         <span class="small text-muted fw-bold text-uppercase"
style="letter-spacing: 0.5px; font-size: 0.75rem;">Диалог</spa
n>
559         <button class="btn btn-sm btn-link text-muted p-0 text
-decoration-none small" id="btn-clear-chat" title="Очистить ди
алог">
560           <i class="bi bi-trash"></i> Очистить
561         </button>
562       </div>
563
564       <div class="chat-container" id="chat-feed">
565         <div class="text-muted small text-center my-auto p-3"
id="empty-feed-hint">
566           <i class="bi bi-chat-dots fs-3 d-block mb-1 opacity-
50"></i>
567           Голосовые команды и медиафайлы отобразятся здесь
568         </div>
```

```
      </div>

      <!-- Fallback Quick Text Bar & Attach Clip -->
      <div class="mt-2">
        <form id="text-send-form" class="d-flex gap-2">
          <button type="button" class="btn btn-outline-seconda
ry btn-sm px-2.5 rounded-3 d-flex align-items-center justify-c
ontent-center" id="btn-clip-attachment" title="Прикрепить файл
">
            <i class="bi bi-paperclip fs-6"></i>
          </button>
          <input type="text" class="form-control form-control-
custom form-control-sm py-2" id="text-input" placeholder="Или
введите сообщение текстом..." autocomplete="off">
          <button type="submit" class="btn btn-primary btn-sm
px-3 rounded-3" id="btn-send-text" title="Отправить">
            <i class="bi bi-send-fill"></i>
          </button>
        </form>
      </div>

    </div>
  </div>

  <script>
    // Global State
    let recognition = null;
    let isListening = false;
    let isTtsEnabled = true;
    let chatHistory = [];
    let abortController = null;
    let currentSpeechAudio = null;
    let currentSpeechQueue = null;

    // Elements
    const authPanel = document.getElementById('auth-panel');
    const mainPanel = document.getElementById('main-panel');
    const authAlert = document.getElementById('auth-alert');
```

```
601     const authFormLogin = document.getElementById('auth-form-l
ogin');
602     const authFormRegister = document.getElementById('auth-for
m-register');
603     const authFormVerify = document.getElementById('auth-form-
verify');
604 
605     const btnMic = document.getElementById('btn-mic');
606     const micIcon = document.getElementById('mic-icon');
607     const voiceStatusText = document.getElementById('voice-sta
tus-text');
608     const voiceSubtext = document.getElementById('voice-subtex
t');
609     const waveBars = document.getElementById('wave-bars');
610     const chatFeed = document.getElementById('chat-feed');
611     const emptyFeedHint = document.getElementById('empty-feed-
hint');
612     const userDisplayName = document.getElementById('user-disp
lay-name');
613 
614     const btnUploadAudio = document.getElementById('btn-upload
-audio');
615     const btnUploadMedia = document.getElementById('btn-upload
-media');
616     const btnClipAttachment = document.getElementById('btn-cli
p-attachment');
617     const audioFileInput = document.getElementById('audio-file
-input');
618     const mediaFileInput = document.getElementById('media-file
-input');
619 
620     const textSendForm = document.getElementById('text-send-fo
rm');
621     const textInput = document.getElementById('text-input');
622     const ttsToggleBtn = document.getElementById('tts-toggle-b
tn');
623     const ttsIcon = document.getElementById('tts-icon');
624     const btnClearChat = document.getElementById('btn-clear-ch
```

```
at');

    // Telegram Web App Integration
    function initTelegram() {
      if (window.Telegram && window.Telegram.WebApp) {
        const tg = window.Telegram.WebApp;
        tg.ready();
        tg.expand();

        window.triggerHaptic = function(type = 'light') {
          if (tg.HapticFeedback) {
            tg.HapticFeedback.impactOccurred(type);
          }
        };
      } else {
        window.triggerHaptic = function() {};
      }
    }

    // Auth API
    const api = {
      async fetch(url, options = {}) {
        const res = await fetch(url, options);
        if (!res.ok) {
          let msg = res.statusText;
          try {
            const data = await res.json();
            msg = data.detail || msg;
          } catch {}
          throw new Error(msg);
        }
        return res.json();
      },
      auth: {
        async check() {
          try {
            const res = await fetch('/auth/check');
            return await res.json();
```

```
662           } catch {
663             return { authenticated: false };
664           }
665         },
666         async login(email, password) {
667           return api.fetch('/auth/login/email', {
668             method: 'POST',
669             headers: { 'Content-Type': 'application/json' },
670             body: JSON.stringify({ email, password })
671           });
672         },
673         async register(email, password, name) {
674           return api.fetch('/auth/register', {
675             method: 'POST',
676             headers: { 'Content-Type': 'application/json' },
677             body: JSON.stringify({ email, password, name })
678           });
679         },
680         async verify(email, code) {
681           return api.fetch('/auth/verify', {
682             method: 'POST',
683             headers: { 'Content-Type': 'application/json' },
684             body: JSON.stringify({ email, code })
685           });
686         }
687       }
688     };
689
690     // Panel Switching
691     function showPanel(panel) {
692       if (panel === 'auth') {
693         authPanel.classList.remove('d-none');
694         mainPanel.classList.add('d-none');
695       } else {
696         authPanel.classList.add('d-none');
697         mainPanel.classList.remove('d-none');
698       }
699     }
```

```
700
701     // Auth Handlers
702     document.getElementById('go-to-register').addEventListener
('click', (e) => {
703       e.preventDefault();
704       authAlert.classList.add('d-none');
705       authFormLogin.classList.add('d-none');
706       authFormRegister.classList.remove('d-none');
707     });
708
709     document.getElementById('go-to-login').addEventListener('c
lick', (e) => {
710       e.preventDefault();
711       authAlert.classList.add('d-none');
712       authFormRegister.classList.add('d-none');
713       authFormLogin.classList.remove('d-none');
714     });
715
716     let currentVerifyEmail = '';
717     document.getElementById('form-register').addEventListener(
'submit', async (e) => {
718       e.preventDefault();
719       authAlert.classList.add('d-none');
720       const name = document.getElementById('register-name').va
lue;
721       const email = document.getElementById('register-email').
value;
722       const password = document.getElementById('register-passw
ord').value;
723       try {
724         await api.auth.register(email, password, name);
725         currentVerifyEmail = email;
726         authFormRegister.classList.add('d-none');
727         authFormVerify.classList.remove('d-none');
728       } catch (err) {
729         authAlert.textContent = err.message || 'Ошибка регистр
ации';
730         authAlert.classList.remove('d-none');
```

```
      }
    });

    document.getElementById('form-verify').addEventListener('submit', async (e) => {
      e.preventDefault();
      authAlert.classList.add('d-none');
      const code = document.getElementById('verify-code').value;
      try {
        await api.auth.verify(currentVerifyEmail, code);
        authFormVerify.classList.add('d-none');
        authFormLogin.classList.remove('d-none');
        authAlert.className = 'alert alert-success';
        authAlert.textContent = 'Почта подтверждена! Войдите в аккаунт.';
        authAlert.classList.remove('d-none');
      } catch (err) {
        authAlert.textContent = err.message || 'Неверный код';
        authAlert.classList.remove('d-none');
      }
    });

    document.getElementById('form-login').addEventListener('submit', async (e) => {
      e.preventDefault();
      authAlert.classList.add('d-none');
      const email = document.getElementById('login-email').value;
      const password = document.getElementById('login-password').value;
      try {
        await api.auth.login(email, password);
        await initApp();
      } catch (err) {
        authAlert.textContent = err.message || 'Ошибка входа';
        authAlert.classList.remove('d-none');
      }
```

```
763     });
764
765     document.getElementById('google-signin-btn').addEventListe
ner('click', () => {
766       window.location.href = '/auth/google?next=' + encodeURIC
omponent(window.location.pathname);
767     });
768
769     // Voice & Speech Recognition Module
770     function initSpeechRecognition() {
771       const SpeechRecognition = window.SpeechRecognition || wi
ndow.webkitSpeechRecognition;
772       if (!SpeechRecognition) {
773         voiceStatusText.textContent = 'Голосовой ввод не подде
рживается';
774         voiceSubtext.textContent = 'Используйте загрузку аудио
файлов или ввод ниже';
775         btnMic.disabled = true;
776         btnMic.style.opacity = '0.5';
777         return;
778       }
779
780       recognition = new SpeechRecognition();
781       recognition.continuous = false;
782       recognition.interimResults = false;
783
784       const getSelectedLang = () => {
785         const checked = document.querySelector('input[name="la
ng-radio"]:checked');
786         return checked ? checked.value : 'ru-RU';
787       };
788
789       recognition.lang = getSelectedLang();
790       document.querySelectorAll('input[name="lang-radio"]').fo
rEach(radio => {
791         radio.addEventListener('change', () => {
792           recognition.lang = getSelectedLang();
793         });
```

```
      });

      recognition.onstart = () => {
        isListening = true;
        btnMic.className = 'btn-mic-giant listening';
        micIcon.className = 'bi bi-mic-fill';
        waveBars.classList.add('active');
        voiceStatusText.textContent = 'Слушаю вас...';
        voiceSubtext.textContent = 'Говорите команду или вопро
c';
      };

      recognition.onend = () => {
        isListening = false;
        waveBars.classList.remove('active');
        if (!btnMic.classList.contains('thinking')) {
          btnMic.className = 'btn-mic-giant';
          micIcon.className = 'bi bi-mic-fill';
          voiceStatusText.textContent = 'Нажмите кнопку и гово
рите';
          voiceSubtext.textContent = 'Голос будет распознан и
отправлен в чат';
        }
      };

      recognition.onerror = (e) => {
        isListening = false;
        waveBars.classList.remove('active');
        btnMic.className = 'btn-mic-giant';
        voiceStatusText.textContent = 'Ошибка ввода: ' + (e.er
ror || 'попробуйте снова');
        voiceSubtext.textContent = 'Нажмите микрофон для повто
рной попытки';
      };

      recognition.onresult = (event) => {
        const transcript = event.results[0][0].transcript;
        if (transcript && transcript.trim()) {
```

```
827           handleUserMessage(transcript.trim());
828         }
829       };
830
831       btnMic.addEventListener('click', () => {
832         window.triggerHaptic('medium');
833         stopAudio();
834
835         // Unlock audio context for mobile iOS/Safari/Chrome
836         if ('speechSynthesis' in window) {
837           const silent = new SpeechSynthesisUtterance(' ');
838           silent.volume = 0;
839           window.speechSynthesis.speak(silent);
840         }
841
842         if (isListening) {
843           recognition.stop();
844         } else {
845           try {
846             recognition.start();
847           } catch (e) {
848             console.warn('Recognition start error:', e);
849           }
850         }
851       });
852     }
853
854     // Audio Queue Player for Backend TTS
855     class AudioQueuePlayer {
856       constructor(chunks) {
857         this.chunks = chunks;
858         this.currentIndex = 0;
859         this.isPlaying = false;
860         this.audioElements = new Map();
861         this.activeAudio = null;
862         this._prefetch(0);
863         this._prefetch(1);
864       }
```

```
865
866       _prefetch(index) {
867         if (index >= this.chunks.length || this.audioElements.
    has(index)) return;
868         const text = this.chunks[index];
869         const queryParams = new URLSearchParams({ text });
870         const audioUrl = `/api/tts/synthesize?${queryParams.to
    String()}`;
871         const audio = new Audio(audioUrl);
872         audio.load();
873         this.audioElements.set(index, audio);
874       }
875
876       async play() {
877         this.isPlaying = true;
878         while (this.currentIndex < this.chunks.length && this.
    isPlaying) {
879           this._prefetch(this.currentIndex + 1);
880           this._prefetch(this.currentIndex + 2);
881
882           const audio = this.audioElements.get(this.currentInd
    ex);
883           if (!audio) {
884             this.currentIndex++;
885             continue;
886           }
887
888           this.activeAudio = audio;
889           currentSpeechAudio = audio;
890
891           const playPromise = new Promise((resolve) => {
892             audio.onended = () => resolve();
893             audio.onerror = () => {
894               this.stop();
895               resolve();
896             };
897           });
898
```

```
          try {
            await audio.play();
            await playPromise;
          } catch (err) {
            console.error('Audio playback interrupted:', err);
          }
          this.currentIndex++;
        }
        this.isPlaying = false;
      }

      stop() {
        this.isPlaying = false;
        if (this.activeAudio) {
          try {
            this.activeAudio.pause();
            this.activeAudio.src = '';
          } catch (e) {}
        }
        this.audioElements.clear();
      }
    }

    function stopAudio() {
      if (currentSpeechQueue) {
        try { currentSpeechQueue.stop(); } catch (e) {}
        currentSpeechQueue = null;
      }
      if (currentSpeechAudio) {
        try {
          currentSpeechAudio.pause();
          currentSpeechAudio.src = '';
        } catch (e) {}
        currentSpeechAudio = null;
      }
      if ('speechSynthesis' in window) {
        try { window.speechSynthesis.cancel(); } catch (e) {}
      }
```

```
    }

    async function speakText(text) {
      if (!isTtsEnabled || !text) return;
      stopAudio();

      const cleanText = text.replace(/<[^>]*>/g, '').replace(/[*_`#]/g, '').trim();
      if (!cleanText) return;

      // Try Backend TTS
      try {
        const paragraphs = cleanText.split(/\n+/).map(p => p.trim()).filter(Boolean);
        let chunks = [];
        for (const para of paragraphs) {
          const sentences = para.match(/[^.!?]+[.!?]*/g) || [para];
          let currentChunk = '';
          for (const s of sentences) {
            const tr = s.trim();
            if (!tr) continue;
            if ((currentChunk + ' ' + tr).length > 200) {
              chunks.push(currentChunk.trim());
              currentChunk = tr;
            } else {
              currentChunk = (currentChunk + ' ' + tr).trim();
            }
          }
          if (currentChunk) chunks.push(currentChunk.trim());
        }
        chunks = chunks.filter(Boolean);

        if (chunks.length > 0) {
          currentSpeechQueue = new AudioQueuePlayer(chunks);
          await currentSpeechQueue.play();
          return;
        }
```

```
      } catch (e) {
        console.warn('Backend TTS failed, falling back to Web Speech Synthesis:', e);
      }

      // Fallback: Browser SpeechSynthesis
      if ('speechSynthesis' in window) {
        const utterance = new SpeechSynthesisUtterance(cleanText);
        const selectedLang = document.querySelector('input[name="lang-radio"]:checked')?.value || 'ru-RU';
        utterance.lang = selectedLang;
        window.speechSynthesis.speak(utterance);
      }
    }

    // Message Rendering Helpers
    function appendMessage(role, text, isStreaming = false) {
      if (emptyFeedHint) {
        emptyFeedHint.remove();
      }

      const bubble = document.createElement('div');
      bubble.className = `chat-bubble ${role}`;

      const timeStr = new Date().toLocaleTimeString([], { hour: '2-digit', minute: '2-digit' });

      if (role === 'user') {
        bubble.innerHTML = `<div>${escapeHtml(text)}</div><div class="chat-meta"><span>${timeStr}</span></div>`;
      } else {
        bubble.innerHTML = `
          <div class="ai-content">${window.marked ? marked.parse(text) : escapeHtml(text)}</div>
          <div class="chat-meta">
            <span>${timeStr}</span>
            <button class="btn-speak-again" title="Озвучить"><
```

```
i class="bi bi-volume-up-fill"></i></button>
          </div>
        `;
        const speakBtn = bubble.querySelector('.btn-speak-agai
n');
        if (speakBtn) {
          speakBtn.addEventListener('click', () => speakText(b
ubble.querySelector('.ai-content').innerText));
        }
      }

      chatFeed.appendChild(bubble);
      chatFeed.scrollTop = chatFeed.scrollHeight;
      return bubble;
    }

    function appendMediaUserMessage(file, previewUrl) {
      if (emptyFeedHint) {
        emptyFeedHint.remove();
      }

      const bubble = document.createElement('div');
      bubble.className = 'chat-bubble user';
      const timeStr = new Date().toLocaleTimeString([], { hour
: '2-digit', minute: '2-digit' });

      const sizeFormatted = (file.size / (1024 * 1024)).toFixe
d(2) + ' MB';
      const isAudio = file.type.startsWith('audio/') || /\.(mp
3|wav|ogg|m4a|flac|aac|webm)$/i.test(file.name);
      const isImage = file.type.startsWith('image/');
      const isVideo = file.type.startsWith('video/');

      let mediaSnippet = '';
      if (isAudio && previewUrl) {
        mediaSnippet = `<audio class="media-player-widget" con
trols src="${previewUrl}"></audio>`;
      } else if (isImage && previewUrl) {
```

```
        mediaSnippet = `<img class="media-thumb-img" src="${previewUrl}" alt="uploaded preview">`;
      } else if (isVideo && previewUrl) {
        mediaSnippet = `<video class="media-player-widget" controls src="${previewUrl}"></video>`;
      }

      const iconClass = isAudio ? 'bi-file-earmark-music' : (isImage ? 'bi-image' : (isVideo ? 'bi-camera-video' : 'bi-file-earmark-text'));

      bubble.innerHTML = `
        <div class="media-preview-card">
          <i class="bi ${iconClass} fs-4 text-info"></i>
          <div class="text-truncate">
            <div class="fw-semibold text-truncate small">${escapeHtml(file.name)}</div>
            <div class="text-muted" style="font-size: 0.72rem;">${sizeFormatted}</div>
          </div>
        </div>
        ${mediaSnippet}
        <div class="chat-meta"><span>${timeStr}</span></div>
      `;

      chatFeed.appendChild(bubble);
      chatFeed.scrollTop = chatFeed.scrollHeight;
      return bubble;
    }

    function escapeHtml(text) {
      const div = document.createElement('div');
      div.textContent = text;
      return div.innerHTML;
    }

    // Send Voice/Text Query to AI Chat API
    async function handleUserMessage(query) {
```

```
      if (!query || !query.trim()) return;

      appendMessage('user', query);
      stopAudio();

      btnMic.className = 'btn-mic-giant thinking';
      micIcon.className = 'bi bi-arrow-repeat';
      voiceStatusText.textContent = 'Обработка ответа AI...';
      voiceSubtext.textContent = 'Генерация ответа моделью';

      if (abortController) {
        try { abortController.abort(); } catch (e) {}
      }
      abortController = new AbortController();

      const aiBubble = appendMessage('ai', '...');
      const aiContentDiv = aiBubble.querySelector('.ai-content
');

      let fullReply = '';
      let voiceReply = '';

      try {
        const payload = {
          message: query,
          history: chatHistory,
          generation_config: { is_rc: true }
        };

        const response = await fetch('/api/chat', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify(payload),
          signal: abortController.signal
        });

        if (!response.ok) {
          throw new Error(`Ошибка сервера: ${response.status}`
```

```
);
        }

        const reader = response.body.getReader();
        const decoder = new TextDecoder('utf-8');
        let buffer = '';

        while (true) {
          const { value, done } = await reader.read();
          if (done) break;

          buffer += decoder.decode(value, { stream: true });
          const lines = buffer.split('\n');
          buffer = lines.pop();

          for (const line of lines) {
            const tr = line.trim();
            if (!tr) continue;
            if (tr.startsWith('data: ')) {
              try {
                const data = JSON.parse(tr.substring(6));
                if (data.error) throw new Error(data.error);
                if (data.voice) voiceReply += data.voice;
                if (data.text) {
                  fullReply += data.text;
                  aiContentDiv.innerHTML = window.marked ? mar
ked.parse(fullReply) : escapeHtml(fullReply);
                  chatFeed.scrollTop = chatFeed.scrollHeight;
                }
              } catch (e) {
                console.error(e);
              }
            }
          }
        }

        btnMic.className = 'btn-mic-giant';
        micIcon.className = 'bi bi-mic-fill';
```

```
        voiceStatusText.textContent = 'Нажмите кнопку и говорите';
        voiceSubtext.textContent = 'Готов к следующему вопросу';

        // Update history
        chatHistory.push({ role: 'user', parts: [query] });
        chatHistory.push({ role: 'model', parts: [fullReply] });
        if (chatHistory.length > 20) chatHistory = chatHistory.slice(-20);

        // Speak response
        speakText(voiceReply || fullReply);

      } catch (err) {
        if (err.name === 'AbortError') return;
        aiContentDiv.innerHTML = `<span class="text-danger">Ошибка: ${escapeHtml(err.message)}</span>`;
        btnMic.className = 'btn-mic-giant';
        micIcon.className = 'bi bi-mic-fill';
        voiceStatusText.textContent = 'Ошибка выполнения';
        voiceSubtext.textContent = 'Попробуйте нажать кнопку еще раз';
      }
    }

    // Process Uploaded Audio / Media File
    async function processUploadedFile(file, isAudioOnly = false) {
      if (!file) return;
      window.triggerHaptic('medium');
      stopAudio();

      const previewUrl = URL.createObjectURL(file);
      appendMediaUserMessage(file, previewUrl);

      btnMic.className = 'btn-mic-giant thinking';
```

```
1171        micIcon.className = 'bi bi-arrow-repeat';
1172        voiceStatusText.textContent = 'Загрузка и анализ медиа..
    .';
1173        voiceSubtext.textContent = file.name;
1174
1175        const aiBubble = appendMessage('ai', '  Загрузка и обраб
    отка файла...');
1176        const aiContentDiv = aiBubble.querySelector('.ai-content
    ');
1177
1178        const isAudio = isAudioOnly || file.type.startsWith('aud
    io/') || /\.(mp3|wav|ogg|m4a|flac|aac|webm)$/i.test(file.name)
    ;
1179
1180        try {
1181          if (isAudio) {
1182            // Send to Audio Diarization & Transcription service
1183            const formData = new FormData();
1184            formData.append('file', file);
1185            const selectedLang = document.querySelector('input[n
    ame="lang-radio"]:checked')?.value || 'ru-RU';
1186            formData.append('language', selectedLang.split('-')[
    0]);
1187
1188            const res = await fetch('/api/audio/diarize', {
1189              method: 'POST',
1190              body: formData
1191            });
1192
1193            if (!res.ok) {
1194              const errData = await res.json().catch(() => ({}))
    ;
1195              throw new Error(errData.detail || `Ошибка сервера:
     ${res.status}`);
1196            }
1197
1198            const result = await res.json();
1199            const diarizationData = result.data || {};
```

```
1200
1201          let formattedOutput = '';
1202          if (diarizationData.summary) {
1203            formattedOutput += `### □ Краткая сводка\n${diari
zationData.summary}\n\n`;
1204          }
1205
1206          if (diarizationData.key_points && diarizationData.ke
y_points.length > 0) {
1207            formattedOutput += `### □ Ключевые тезисы\n` + di
arizationData.key_points.map(kp => `- ${kp}`).join('\n') + `\n
\n`;
1208          }
1209
1210          if (diarizationData.action_items && diarizationData.
action_items.length > 0) {
1211            formattedOutput += `### □ Задачи и решения\n` + di
arizationData.action_items.map(ai => `- [ ] ${ai}`).join('\n')
 + `\n\n`;
1212          }
1213
1214          if (diarizationData.transcript && diarizationData.tr
anscript.length > 0) {
1215            formattedOutput += `### □ Стенограмма диалога\n`;
1216            for (const item of diarizationData.transcript) {
1217              const speaker = item.speaker || 'Собеседник';
1218              const text = item.text || '';
1219              formattedOutput += `**${speaker}:** ${text}\n\n`
;
1220            }
1221          }
1222
1223          if (!formattedOutput) {
1224            formattedOutput = diarizationData.markdown_report
|| 'Аудиозапись успешно обработана.';
1225          }
1226
1227          aiContentDiv.innerHTML = window.marked ? marked.pars
```

```
e(formattedOutput) : escapeHtml(formattedOutput);
          chatFeed.scrollTop = chatFeed.scrollHeight;

          chatHistory.push({ role: 'user', parts: [`[Аудиозапи
сь ${file.name}]`] });
          chatHistory.push({ role: 'model', parts: [formattedO
utput] });
          if (chatHistory.length > 20) chatHistory = chatHisto
ry.slice(-20);

          // Speak the executive summary if available
          if (diarizationData.summary) {
            speakText(diarizationData.summary);
          }

        } else {
          // General Media Upload to storage
          const formData = new FormData();
          formData.append('file', file);

          const res = await fetch('/api/user/storage/upload',
{
            method: 'POST',
            body: formData
          });

          if (!res.ok) {
            const errData = await res.json().catch(() => ({}))
;
            throw new Error(errData.detail || `Ошибка загрузки
: ${res.status}`);
          }

          const resData = await res.json();
          const fileName = resData.filename || file.name;

          // Request AI analysis of the uploaded document/medi
a
```

```
          const prompt = `Пользователь загрузил файл ${fileNam
e} (${file.type || 'документ'}). Подтверди получение и опиши д
альнейшие действия.`;

          const chatRes = await fetch('/api/chat', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({
              message: prompt,
              history: chatHistory,
              generation_config: { is_rc: true }
            })
          });

          let modelReply = `Файл **${escapeHtml(fileName)}** у
спешно загружен и сохранен в вашем хранилище.`;
          if (chatRes.ok) {
            // Read stream or fallback
            const reader = chatRes.body.getReader();
            const decoder = new TextDecoder('utf-8');
            let buffer = '';
            let streamReply = '';

            while (true) {
              const { value, done } = await reader.read();
              if (done) break;
              buffer += decoder.decode(value, { stream: true }
);
              const lines = buffer.split('\n');
              buffer = lines.pop();

              for (const line of lines) {
                const tr = line.trim();
                if (tr.startsWith('data: ')) {
                  try {
                    const data = JSON.parse(tr.substring(6));
                    if (data.text) streamReply += data.text;
                  } catch (e) {}
```

```
            }
          }
        }
        if (streamReply) modelReply = streamReply;
      }

      aiContentDiv.innerHTML = window.marked ? marked.parse(modelReply) : escapeHtml(modelReply);
      chatFeed.scrollTop = chatFeed.scrollHeight;

      chatHistory.push({ role: 'user', parts: [`[Загружен файл ${file.name}]`] });
      chatHistory.push({ role: 'model', parts: [modelReply] });
      if (chatHistory.length > 20) chatHistory = chatHistory.slice(-20);

      speakText(modelReply);
    }

    btnMic.className = 'btn-mic-giant';
    micIcon.className = 'bi bi-mic-fill';
    voiceStatusText.textContent = 'Нажмите кнопку и говорите';
    voiceSubtext.textContent = 'Готов к следующему вопросу или загрузке';

  } catch (err) {
    aiContentDiv.innerHTML = `<span class="text-danger">Ошибка обработки: ${escapeHtml(err.message)}</span>`;
    btnMic.className = 'btn-mic-giant';
    micIcon.className = 'bi bi-mic-fill';
    voiceStatusText.textContent = 'Ошибка загрузки';
    voiceSubtext.textContent = 'Попробуйте загрузить файл повторно';
  }
}

```

```
    // Media Buttons Event Listeners
    btnUploadAudio.addEventListener('click', () => {
      window.triggerHaptic('light');
      audioFileInput.click();
    });

    btnUploadMedia.addEventListener('click', () => {
      window.triggerHaptic('light');
      mediaFileInput.click();
    });

    btnClipAttachment.addEventListener('click', () => {
      window.triggerHaptic('light');
      mediaFileInput.click();
    });

    audioFileInput.addEventListener('change', (e) => {
      const file = e.target.files && e.target.files[0];
      if (file) {
        processUploadedFile(file, true);
        audioFileInput.value = '';
      }
    });

    mediaFileInput.addEventListener('change', (e) => {
      const file = e.target.files && e.target.files[0];
      if (file) {
        processUploadedFile(file, false);
        mediaFileInput.value = '';
      }
    });

    // Text Fallback Form
    textSendForm.addEventListener('submit', (e) => {
      e.preventDefault();
      const val = textInput.value.trim();
      if (val) {
        textInput.value = '';
```

```
        handleUserMessage(val);
      }
    });

    // TTS Toggle Button
    ttsToggleBtn.addEventListener('click', () => {
      window.triggerHaptic('light');
      isTtsEnabled = !isTtsEnabled;
      if (isTtsEnabled) {
        ttsIcon.className = 'bi bi-volume-up-fill';
        ttsToggleBtn.classList.remove('opacity-50');
      } else {
        stopAudio();
        ttsIcon.className = 'bi bi-volume-mute-fill';
        ttsToggleBtn.classList.add('opacity-50');
      }
    });

    // Clear Chat Button
    btnClearChat.addEventListener('click', () => {
      window.triggerHaptic('light');
      stopAudio();
      chatHistory = [];
      chatFeed.innerHTML = `
        <div class="text-muted small text-center my-auto p-3"
id="empty-feed-hint">
          <i class="bi bi-chat-dots fs-3 d-block mb-1 opacity-
50"></i>
          Голосовые команды и медиафайлы отобразятся здесь
        </div>
      `;
    });

    // Theme Toggle
    document.getElementById('theme-toggle-btn').addEventListen
er('click', () => {
      window.triggerHaptic('light');
      const isLight = document.body.classList.toggle('theme-li
```

```
ght');
      const icon = document.getElementById('theme-icon');
      icon.className = isLight ? 'bi bi-sun-fill' : 'bi bi-moo
n-stars-fill';
    });

    // App Initialization
    async function initApp() {
      const state = await api.auth.check();
      if (state.authenticated) {
        if (userDisplayName && state.user) {
          userDisplayName.textContent = state.user.name || sta
te.user.email || 'Пользователь';
        }
        showPanel('main');
        initSpeechRecognition();
      } else {
        showPanel('auth');
      }
    }

    document.addEventListener('DOMContentLoaded', () => {
      initTelegram();
      initApp();
    });
  </script>
</body>
</html>

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/scenarios_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/scenarios_tab/index.htm
l" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/scenarios_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Scenarios & Automated Diagnostics Tab Template -->
<div class="container-fluid p-0 scenarios-tab-wrapper">
<!-- Header & Quick Actions -->
<div class="card border-secondary-subtle mb-3 shadow-sm" data-
panel-id="panel-scenarios-stsenarii-testirovaniya-i-avtoma" id
="panel-scenarios-stsenarii-testirovaniya-i-avtoma">
<div class="card-body p-3">
<div class="d-flex flex-wrap align-items-center justify-conten
t-between gap-3">
<div class="d-flex align-items-center gap-2">
```

```
<div class="fs-2">🧪</div>
<div>
<h5 class="mb-0 fw-bold d-flex align-items-center gap-2">
<span>Сценарии тестирования и автоматизации</span>
<span class="badge bg-primary-subtle text-primary border border-primary-subtle small fs-6">Test Computer (/tc)</span>
</h5>
<div class="text-muted small">
              Центр управления: экспресс-сценарии проверки хоста, аудит логов, диагностика оборудования и интеллектуальный ассистент навыков.
            </div>
</div>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<button class="btn btn-outline-info btn-sm d-flex align-items-center gap-1 shadow-sm" id="btn-scenario-refresh-list" title="Обновить список сценариев">
<i class="bi bi-arrow-clockwise"></i>
<span>Обновить</span>
</button>
<button class="btn btn-success btn-sm d-flex align-items-center gap-1 shadow-sm fw-semibold" id="btn-run-quick-check" title="Запустить экспресс-проверку окружения">
<span>🧪 Быстрая проверка</span>
<span class="badge bg-white text-dark ms-1">Рекомендуется</span>
</button>
<button class="btn btn-outline-primary btn-sm d-flex align-items-center gap-1 shadow-sm" id="btn-run-all-scenarios" title="Последовательно выполнить все доступные сценарии">
<i class="bi bi-play-fill"></i>
<span>Запустить все</span>
</button>
<button aria-controls="scenariosOffcanvas" class="btn btn-primary btn-sm d-flex align-items-center gap-1.5 shadow-sm fw-semibold" data-bs-target="#scenariosOffcanvas" data-bs-toggle="offcanvas" id="btn-open-scenarios-drawer" title="Открыть выдвижну
```

```
ю панель со сценариями">
<i class="bi bi-layout-sidebar-reverse"></i>
<span>Сценарии</span>
<span class="badge bg-white text-dark ms-1" id="scenarios-count-badge-header">6</span>
</button>
</div>
</div>
</div>
</div>
<!-- Main View: AI Assistant Mini-Chat (expanded to full panel width and >= 80% height) -->
<div class="row g-3">
<div class="col-12">
<div class="card scenario-chat-card border-secondary-subtle shadow-sm d-flex flex-column" data-panel-id="panel-scenarios-ai-assistent-stsenariev-navykov" id="panel-scenarios-ai-assistent-stsenariev-navykov" style="min-height: 80vh; height: calc(100vh - 170px);">
<!-- Chat Header -->
<div class="card-header border-secondary-subtle py-2 px-3 d-flex align-items-center justify-content-between flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<span class="fs-5">□ </span>
<div>
<h6 class="mb-0 fw-bold">AI Ассистент сценариев &amp;amp; Навыков</h6>
<div class="text-muted small" style="font-size: 0.75rem;">Спросите что угодно о компьютере — система проверит и создаст навык</div>
</div>
</div>
<div class="d-flex align-items-center gap-2">
<button aria-controls="scenariosOffcanvas" class="btn btn-outline-info btn-sm py-0.5 px-2 text-info border-info border-opacity-50 d-flex align-items-center gap-1" data-bs-target="#scenariosOffcanvas" data-bs-toggle="offcanvas" style="font-size: 0.7
```

```
5rem;" title="Открыть выдвижную панель со сценариями">
<i class="bi bi-layout-sidebar-reverse"></i>
<span>Сценарии</span>
</button>
<button class="btn btn-outline-secondary btn-sm py-0.5 px-2 bo
rder-secondary" id="btn-scenario-new-chat" style="font-size: 0
.75rem;" title="Очистить контекст и начать новую сессию диалог
a">
<i class="bi bi-arrow-counterclockwise me-1"></i>Новая сессия
            </button>
<button class="btn btn-outline-danger btn-sm d-flex align-item
s-center gap-1 ms-2" id="btn-clear-dialog" style="font-size: 0
.75rem;" title="Очистить диалог и сбросить историю">
<i class="bi bi-trash"></i>Очистить диалог
            </button>
<span class="badge bg-success-subtle text-success border borde
r-success-subtle small">Online</span>
</div>
</div>
<!-- Mini-Chat Suggestions / Prompt Pills -->
<div class="p-2 border-bottom border-secondary-subtle bg-body-
secondary bg-opacity-50">
<div class="small text-muted mb-1" style="font-size: 0.72rem;"
>Быстрые запросы:</div>
<div class="d-flex flex-wrap gap-1" id="chat-quick-prompts">
<button class="btn btn-outline-secondary btn-sm py-0 px-2 quic
k-prompt-btn" style="font-size: 0.75rem;">
              Какие мыши были подключены?
            </button>
<button class="btn btn-outline-secondary btn-sm py-0 px-2 quic
k-prompt-btn" style="font-size: 0.75rem;">
              Покажи USB устройства
            </button>
<button class="btn btn-outline-secondary btn-sm py-0 px-2 quic
k-prompt-btn" style="font-size: 0.75rem;">
              Проверь ошибки в логах
            </button>
<button class="btn btn-outline-secondary btn-sm py-0 px-2 quic
```

```
k-prompt-btn" style="font-size: 0.75rem;">
104                 Сделай быструю проверку
105             </button>
106 </div>
107 </div>
108 <!-- Chat Messages History Container -->
109 <div class="card-body p-3 overflow-y-auto flex-grow-1 scenario
    -chat-history" id="scenario-chat-history" style="min-height: 4
    80px; max-height: none;">
110 <div class="chat-message-row bot-row mb-3">
111 <div class="chat-msg-avatar">  </div>
112 <div class="chat-msg-bot">
113 <div class="mb-2">
114                 Привет! Я интеллектуальный ассистент Test Comp
    uter. Вы можете задать мне любой вопрос о конфигурации хоста,
    например:
115             </div>
116 <div class="d-flex flex-column gap-1 my-2" id="chat-greeting-p
    rompts">
117 <div>• <a class="chat-prompt-link text-decoration-none border-
    bottom border-primary border-opacity-50" data-prompt="Какие мы
    ши были подключены к этому компьютеру?" href="#">«Какие мыши б
    ыли подключены к этому компьютеру?»</a></div>
118 <div>• <a class="chat-prompt-link text-decoration-none border-
    bottom border-primary border-opacity-50" data-prompt="Покажи с
    писок сетевых портов и активных соединений" href="#">«Покажи с
    писок сетевых портов и активных соединений»</a></div>
119 <div>• <a class="chat-prompt-link text-decoration-none border-
    bottom border-primary border-opacity-50" data-prompt="Проверь
    автозапуск и службы Windows" href="#">«Проверь автозапуск и сл
    ужбы Windows»</a></div>
120 </div>
121 <div class="text-muted small mt-2">
122                 Если для вопроса нет готового сценария, я опро
    шу оборудование через PowerShell/WMI, сформирую для вас ответ
    и <strong>автоматически сохраню новый навык</strong> в каталог
     <code>.skills/</code>.
123             </div>
```

```
</div>
</div>
</div>
<!-- Chat Input Form -->
<div class="card-footer border-secondary-subtle p-2 flex-shrink-0">
<form class="d-flex flex-column gap-2" id="scenario-chat-form">
<div class="input-group input-group-sm">
<input autocomplete="off" class="form-control" id="scenario-chat-input" placeholder="Задайте вопрос или введите /skills list ..." type="text"/>
<button class="btn btn-primary d-flex align-items-center gap-1 px-3" id="btn-scenario-chat-send" type="submit">
<span>Отправить</span>
<i class="bi bi-send-fill"></i>
</button>
</div>
<div class="d-flex align-items-center justify-content-between px-1 flex-wrap gap-2">
<span class="text-muted" id="chat-status-indicator" style="font-size: 0.72rem;">Готов к работе</span>
<div class="d-flex align-items-center gap-2">
<div class="form-check form-switch m-0 d-flex align-items-center gap-1">
<input class="form-check-input mt-0" id="check-scenario-keep-context" role="switch" title="Хранить историю диалога в контексте запроса" type="checkbox"/>
<label class="form-check-label text-muted user-select-none" for="check-scenario-keep-context" style="font-size: 0.75rem;" title="Хранить историю диалога в контексте запроса">
<i class="bi bi-clock-history text-secondary me-1"></i>Хранить историю
                    </label>
</div>
<div class="form-check form-switch m-0 d-flex align-items-center gap-1">
<input checked="" class="form-check-input mt-0" id="check-scen
```

```
ario-rag" role="switch" title="Использовать базу знаний (RAG)"
 type="checkbox"/>
<label class="form-check-label text-muted user-select-none" fo
r="check-scenario-rag" style="font-size: 0.75rem;" title="Испо
льзовать базу знаний (RAG)">
<i class="bi bi-database-check text-info me-1"></i>RAG
                  </label>
</div>
</div>
</div>
</form>
</div>
</div>
</div>
</div>
<!-- Right Slide-out Drawer (Offcanvas): Scenarios List & Exec
ution Results -->
<div aria-labelledby="scenariosOffcanvasLabel" class="offcanva
s offcanvas-end text-bg-dark border-start border-secondary-sub
tle shadow-lg" id="scenariosOffcanvas" style="width: 560px; ma
x-width: 95vw;" tabindex="-1">
<div class="offcanvas-header border-bottom border-secondary-su
btle py-3 px-3">
<div class="d-flex align-items-center gap-2">
<span class="fs-4">□ </span>
<div>
<h5 class="offcanvas-title fw-bold text-white mb-0" id="scenar
iosOffcanvasLabel">Сценарии тестирования</h5>
<div class="text-muted small" style="font-size: 0.75rem;">Эксп
ресс-диагностика и системные проверки</div>
</div>
</div>
<div class="d-flex align-items-center gap-2">
<button aria-label="Закрыть" class="btn-close btn-close-white"
 data-bs-dismiss="offcanvas" type="button"></button>
</div>
</div>
<div class="offcanvas-body d-flex flex-column gap-3 p-3">
```

```
<!-- Filter Bar & Count Badge -->
<div class="d-flex align-items-center justify-content-between gap-2 flex-wrap">
<div class="btn-group btn-group-sm" id="scenario-category-filters" role="group">
<button class="btn btn-outline-secondary active" data-filter="all" type="button">Все (6)</button>
<button class="btn btn-outline-secondary" data-filter="quick" type="button">⚡ Экспресс</button>
<button class="btn btn-outline-secondary" data-filter="logs" type="button">📋 Логи</button>
<button class="btn btn-outline-secondary" data-filter="diagnostics" type="button">🖥 Система</button>
<button class="btn btn-outline-secondary" data-filter="security" type="button">🔒 Безопасность</button>
</div>
<span class="text-muted small" id="scenarios-count-badge">Доступно: 6</span>
</div>
<!-- Execution Console & Results (shown on run) -->
<div class="card bg-dark bg-opacity-75 border-secondary-subtle shadow-sm" data-panel-id="panel-scenarios-1" id="scenario-result-panel" style="display: none;">
<div class="card-header bg-transparent border-secondary-subtle d-flex align-items-center justify-content-between py-2 px-3">
<div class="d-flex align-items-center gap-2">
<span class="fs-5" id="result-status-icon">⏳</span>
<div>
<strong class="text-white" id="result-scenario-title">Выполнение сценария...</strong>
<div class="text-muted small" id="result-scenario-meta">Инициализация шагов...</div>
</div>
</div>
<div class="d-flex align-items-center gap-2">
<span class="badge bg-secondary" id="result-duration-badge">0.0 ms</span>
<button class="btn btn-sm btn-outline-secondary py-0 px-2" id=
```

```
"btn-close-result-panel" title="Скрыть панель результатов">□ </
button>
</div>
</div>
<div class="card-body p-3">
<!-- Overall Summary Banner -->
<div class="alert alert-secondary py-2 px-3 mb-3 small d-flex
align-items-start gap-2" id="result-summary-banner">
<i class="bi bi-info-circle-fill mt-1"></i>
<span id="result-summary-text">Результат проверки будет отобра
жен здесь...</span>
</div>
<!-- Progress Bar -->
<div class="progress mb-3" id="scenario-progress-container" st
yle="height: 6px;">
<div class="progress-bar progress-bar-striped progress-bar-ani
mated bg-primary" id="scenario-progress-bar" role="progressbar
" style="width: 100%"></div>
</div>
<!-- Step Results List -->
<div class="d-flex flex-column gap-2" id="scenario-steps-conta
iner">
<!-- Dynamic step rows injected here -->
</div>
</div>
</div>
<!-- Scenarios Cards Container -->
<div class="row g-2" id="scenarios-cards-container">
<!-- Injected dynamically by main.js -->
<div class="col-12 text-center py-5 text-muted small">
<div class="spinner-border spinner-border-sm text-primary mb-2
" role="status"></div>
<div>Загрузка сценариев тестирования...</div>
</div>
</div>
</div>
</div>
</div>
```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Scenarios Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/scenarios_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/scenarios_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// Scenarios and Test Computer Automated Diagnostics Module (/tc)
(function () {
  'use strict';

  let scenariosList = [];
  let currentFilter = 'all';
  let isRunning = false;

  // Icons and badges helper
  const categoryIcons = {
    quick: '□ ',
```

```
    logs: '□ ',
    diagnostics: '□ ',
    security: '□ ',
    network: '□ ',
    ai: '□ ',
  };

  const statusIcons = {
    ok: '□ ',
    warn: '□ ',
    error: '□ ',
    skipped: '□ ',
  };

  // Helper for safe HTML escaping
  function escapeHtml(str) {
    if (!str) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
  }

  // Simple Markdown to HTML formatter for mini-chat
  function formatMarkdown(text) {
    if (!text) return '';
    let html = escapeHtml(text);
    // Bold
    html = html.replace(/\*\*(.*?)\*\*/g, '<strong>$1</strong>
');
    // Italic
    html = html.replace(/\*(.*?)\*/g, '<em>$1</em>');
    // Inline code
    html = html.replace(/`(.*?)`/g, '<code class="text-primary
 bg-body-secondary px-1 rounded">$1</code>');
    // Blockquote
```

```
    html = html.replace(/^&gt;\s+(.*$)/gim, '<div class="borde
r-start border-primary border-3 ps-2 my-1 text-body bg-body-se
condary py-1 rounded-end">$1</div>');
    // Headers
    html = html.replace(/^###\s+(.*$)/gim, '<h6 class="fw-bold
 text-primary mt-2 mb-1">$1</h6>');
    html = html.replace(/^##\s+(.*$)/gim, '<h5 class="fw-bold
text-body mt-2 mb-1">$1</h5>');
    // Line breaks
    html = html.replace(/\n/g, '<br>');
    return html;
  }

  // Load scenarios from API
  async function loadScenarios() {
    const container = document.getElementById('scenarios-cards
-container');
    if (!container) return;

    try {
      const res = await fetch('/api/v1/scenarios');
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      scenariosList = await res.json();
      renderScenariosGrid();
    } catch (err) {
      console.warn('[ScenariosTab] Error fetching /api/v1/scen
arios, using local fallback:', err);
      scenariosList = [
        {
          id: 'quick_check',
          title: 'Быстрая проверка (Smoke Test)',
          description: 'Экспресс-проверка целостности окружени
я, доступности ядра FastAPI, AI-провайдеров, свободного места
и ключевых логов. Рекомендуется после установки.',
          category: 'quick',
          icon: '□ ',
          recommended: true,
          estimated_duration_sec: 2,
```

```
        },
        {
          id: 'log_audit',
          title: 'Запустить аудит логов',
          description: 'Комплексный анализ файлов логов (info.log, errors.log, app.log) и журналов событий Windows. Поиск критических ошибок и аномалий.',
          category: 'logs',
          icon: '□ ',
          recommended: false,
          estimated_duration_sec: 3,
        },
        {
          id: 'system_inspector',
          title: 'Диагностика системы и оборудования',
          description: 'Срез телеметрии: загрузка ядер процессора, температура, оперативная память, свободное место на дисках и фоновые процессы.',
          category: 'diagnostics',
          icon: '□ ',
          recommended: false,
          estimated_duration_sec: 3,
        },
        {
          id: 'windows_admin',
          title: 'Аудит автозапуска и служб Windows',
          description: 'Проверка записей реестра Run/RunOnce, папки автозагрузки, системных служб Windows и прав текущего процесса.',
          category: 'security',
          icon: '□ ',
          recommended: false,
          estimated_duration_sec: 4,
        },
        {
          id: 'network_test',
          title: 'Проверка сетевой доступности и портов',
          description: 'Тестирование локального шлюза, разреше
```

```
ния DNS, открытых серверных портов (8000) и доступности внешни
x AI-сервисов.',
          category: 'network',
          icon: '□ ',
          recommended: false,
          estimated_duration_sec: 4,
        },
        {
          id: 'ai_providers_check',
          title: 'Проверка статуса AI-провайдеров',
          description: 'Опрос сконфигурированных AI-моделей (G
oogle Gemini, AGY, Ollama, Foundry) и замер времени отклика ин
ференса.',
          category: 'ai',
          icon: '□ ',
          recommended: false,
          estimated_duration_sec: 5,
        },
      ];
      renderScenariosGrid();
    }
  }

  // Render cards grid based on active filter
  function renderScenariosGrid() {
    const container = document.getElementById('scenarios-cards
-container');
    if (!container) return;

    const filtered = scenariosList.filter(sc => {
      if (currentFilter === 'all') return true;
      return sc.category === currentFilter;
    });

    const badgeEl = document.getElementById('scenarios-count-b
adge');
    if (badgeEl) {
      badgeEl.innerText = `Доступно: ${filtered.length}`;
```

```
    }
    const headerBadgeEl = document.getElementById('scenarios-count-badge-header');
    if (headerBadgeEl) {
      headerBadgeEl.innerText = `${scenariosList.length}`;
    }

    if (filtered.length === 0) {
      container.innerHTML = '<div class="col-12 text-center py-4 text-muted small">В этой категории нет сценариев.</div>';
      return;
    }

    container.innerHTML = filtered.map(sc => {
      const isRecommended = Boolean(sc.recommended);
      const borderClass = isRecommended ? 'border-success border-opacity-50 shadow-sm' : 'border-secondary-subtle';
      const bgHeader = isRecommended ? 'bg-success bg-opacity-10' : 'bg-transparent';
      const btnClass = isRecommended ? 'btn-success' : 'btn-outline-primary';

      return `
        <div class="col-12">
          <div class="card ${borderClass} transition-all hover-shadow">
            <div class="card-header ${bgHeader} border-secondary-subtle py-2 px-3 d-flex align-items-center justify-content-between">
              <div class="d-flex align-items-center gap-2">
                <span class="fs-5">${sc.icon || categoryIcons[sc.category] || '□'}</span>
                <strong class="small">${escapeHtml(sc.title)}</strong>
              </div>
              <div>
                ${isRecommended ? '<span class="badge bg-success small">Рекомендуется</span>' : `<span class="badge bg-secon
```

```
dary small">${sc.estimated_duration_sec || 3} сек</span>`}
              </div>
            </div>
            <div class="card-body p-3 d-flex flex-column justi
fy-content-between">
              <p class="text-muted small mb-2" style="font-siz
e: 0.8rem; line-height: 1.35;">
                ${escapeHtml(sc.description)}
              </p>
              <div class="d-flex align-items-center justify-co
ntent-between gap-2 mt-auto pt-2 border-top border-secondary-s
ubtle">
                <span class="text-secondary small" style="font
-size: 0.72rem;">
                  ID: <code>${escapeHtml(sc.id)}</code>
                </span>
                <button class="btn ${btnClass} btn-sm py-1 px-
3 d-flex align-items-center gap-1 btn-run-scenario-item" data-
scenario-id="${escapeHtml(sc.id)}" ${isRunning ? 'disabled' :
''}>
                  <i class="bi bi-play-fill"></i>
                  <span>Запустить</span>
                </button>
              </div>
            </div>
          </div>
        `;
    }).join('');

    // Attach listeners to individual run buttons
    container.querySelectorAll('.btn-run-scenario-item').forEa
ch(btn => {
      btn.onclick = () => {
        const scId = btn.getAttribute('data-scenario-id');
        if (scId) runScenario(scId);
      };
    });
```

```
  }

  // Execute scenario via API
  async function runScenario(scenarioId) {
    if (isRunning) return;
    isRunning = true;

    // Automatically open the scenarios drawer if closed
    const offcanvasEl = document.getElementById('scenariosOffcanvas');
    if (offcanvasEl && typeof bootstrap !== 'undefined' && bootstrap.Offcanvas) {
      try {
        const offcanvasInstance = bootstrap.Offcanvas.getOrCreateInstance(offcanvasEl);
        offcanvasInstance.show();
      } catch (e) {
        console.warn('[ScenariosTab] Offcanvas show error:', e);
      }
    }

    const resultPanel = document.getElementById('scenario-result-panel');
    const titleEl = document.getElementById('result-scenario-title');
    const metaEl = document.getElementById('result-scenario-meta');
    const iconEl = document.getElementById('result-status-icon');
    const durationBadge = document.getElementById('result-duration-badge');
    const summaryBanner = document.getElementById('result-summary-banner');
    const summaryText = document.getElementById('result-summary-text');
    const stepsContainer = document.getElementById('scenario-steps-container');
```

```
    const progressBar = document.getElementById('scenario-progress-bar');

    if (resultPanel) {
      resultPanel.style.display = '';
      resultPanel.scrollIntoView({ behavior: 'smooth', block: 'nearest' });
    }
    if (titleEl) titleEl.innerText = `Выполнение сценария [${scenarioId}]...`;
    if (metaEl) metaEl.innerText = 'Опрос системных подсистем и выполнение тестов...';
    if (iconEl) iconEl.innerText = '';
    if (durationBadge) durationBadge.innerText = 'Выполняется...';
    if (summaryBanner) summaryBanner.className = 'alert alert-info py-2 px-3 mb-3 small d-flex align-items-start gap-2';
    if (summaryText) summaryText.innerText = 'Идет тестирование компонентов... Пожалуйста, подождите.';
    if (stepsContainer) stepsContainer.innerHTML = '<div class="text-center py-3 text-muted small"><div class="spinner-border spinner-border-sm text-primary mb-2"></div><div>Тестирование запущено...</div></div>';
    if (progressBar) {
      progressBar.style.width = '100%';
      progressBar.className = 'progress-bar progress-bar-striped progress-bar-animated bg-primary';
    }

    renderScenariosGrid();

    const tStart = performance.now();
    try {
      const response = await fetch('/api/v1/scenarios/run', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ scenario_id: scenarioId }),
      });
```

```
270
271       if (!response.ok) {
272         throw new Error(`Ошибка сервера: HTTP ${response.statu
    s}`);
273       }
274
275       const res = await response.json();
276       displayScenarioResults(res);
277     } catch (err) {
278       console.error('[ScenariosTab] Error running scenario:',
    err);
279       const totalDur = Math.round(performance.now() - tStart);
280       displayScenarioResults({
281         scenario_id: scenarioId,
282         title: scenarioId,
283         status: 'error',
284         duration_ms: totalDur,
285         total_steps: 1,
286         passed_steps: 0,
287         warn_steps: 0,
288         failed_steps: 1,
289         steps: [
290           {
291             name: 'Выполнение запроса сценария',
292             status: 'error',
293             duration_ms: totalDur,
294             details: `Ошибка: ${err.message}`,
295             recommendation: 'Проверьте соединение с бэкенд сер
    вером FastAPI.',
296           },
297         ],
298         summary: `Сбой при выполнении сценария: ${err.message}
    `,
299       });
300     } finally {
301       isRunning = false;
302       renderScenariosGrid();
303     }
```

```
304   }
305
306   // Display structured results in panel
307   function displayScenarioResults(res) {
308     const titleEl = document.getElementById('result-scenario-t
itle');
309     const metaEl = document.getElementById('result-scenario-me
ta');
310     const iconEl = document.getElementById('result-status-icon
');
311     const durationBadge = document.getElementById('result-dura
tion-badge');
312     const summaryBanner = document.getElementById('result-summ
ary-banner');
313     const summaryText = document.getElementById('result-summar
y-text');
314     const stepsContainer = document.getElementById('scenario-s
teps-container');
315     const progressBar = document.getElementById('scenario-prog
ress-bar');
316
317     if (titleEl) titleEl.innerText = res.title || res.scenario
_id;
318     if (metaEl) metaEl.innerText = `Шагов: ${res.total_steps |
| 0} (Пройдено: ${res.passed_steps || 0}, Замечаний: ${res.war
n_steps || 0}, Ошибок: ${res.failed_steps || 0})`;
319     if (durationBadge) durationBadge.innerText = `${res.durati
on_ms || 0} ms`;
320
321     // Overall status styling
322     if (res.status === 'ok') {
323       if (iconEl) iconEl.innerText = '□ ';
324       if (summaryBanner) summaryBanner.className = 'alert aler
t-success py-2 px-3 mb-3 small d-flex align-items-start gap-2'
;
325       if (progressBar) progressBar.className = 'progress-bar b
g-success';
326     } else if (res.status === 'warn') {
```

```
      if (iconEl) iconEl.innerText = '⚠ ';
      if (summaryBanner) summaryBanner.className = 'alert alert-warning py-2 px-3 mb-3 small d-flex align-items-start gap-2';
      if (progressBar) progressBar.className = 'progress-bar bg-warning';
    } else {
      if (iconEl) iconEl.innerText = '✖ ';
      if (summaryBanner) summaryBanner.className = 'alert alert-danger py-2 px-3 mb-3 small d-flex align-items-start gap-2';
      if (progressBar) progressBar.className = 'progress-bar bg-danger';
    }

    if (summaryText) summaryText.innerText = res.summary || 'Прогон завершен.';

    // Steps rendering
    if (stepsContainer && Array.isArray(res.steps)) {
      stepsContainer.innerHTML = res.steps.map((step, idx) => {
        const icon = statusIcons[step.status] || '• ';
        const badgeBg = step.status === 'ok' ? 'bg-success' : (step.status === 'warn' ? 'bg-warning text-dark' : 'bg-danger');

        const hasData = step.data && Object.keys(step.data).length > 0;
        const dataId = `step-data-${idx}`;

        return `
          <div class="card border border-secondary-subtle p-2 rounded">
            <div class="d-flex align-items-center justify-content-between gap-2 flex-wrap">
              <div class="d-flex align-items-center gap-2">
                <span>${icon}</span>
                <strong class="small">${escapeHtml(step.name)}
```

```
</strong>
              </div>
              <div class="d-flex align-items-center gap-2">
                <span class="badge ${badgeBg} small" style="fo
nt-size: 0.68rem;">${step.status.toUpperCase()}</span>
                <span class="text-secondary small" style="font
-size: 0.7rem;">${step.duration_ms || 0} ms</span>
              </div>
            </div>
            <div class="text-muted small mt-1 ps-4" style="fon
t-size: 0.78rem;">
              ${escapeHtml(step.details)}
            </div>
            ${step.recommendation ? `
              <div class="mt-1 ps-4 small text-warning" style=
"font-size: 0.75rem;">
                 <strong>Рекомендация:</strong> ${escapeHtml
(step.recommendation)}
              </div>
            ` : ''}
            ${hasData ? `
              <div class="ps-4 mt-1">
                <button class="btn btn-link btn-sm p-0 text-pr
imary text-decoration-none small" style="font-size: 0.72rem;"
onclick="const el=document.getElementById('${dataId}'); if(el)
 el.classList.toggle('d-none')">
                   Показать детали JSON
                </button>
                <div id="${dataId}" class="d-none mt-1 p-2 bg-
body-secondary rounded border border-secondary font-monospace
small overflow-auto" style="max-height: 140px; font-size: 0.7r
em;">
                  <pre class="m-0">${escapeHtml(JSON.stringify
(step.data, null, 2))}</pre>
                </div>
              </div>
            ` : ''}
          </div>
```

```
        `;
      }).join('');
    }
  }

  let isChatSubmitting = false;
  let isEventsSetup = false;

  // Extract [CHAT] and [VOICE] blocks from response and clean embedded action blocks
  function parseChatAndVoice(rawText) {
    if (!rawText) return { chatText: '', voiceText: '' };
    let text = rawText;
    let voiceText = '';

    if (text.includes('[CHAT]') || text.includes('[VOICE]')) {
      const voiceMatch = text.match(/\[VOICE\]([\s\S]*?)(?:\[\/VOICE\]|(?=\[CHAT\])|$)/i);
      if (voiceMatch) {
        voiceText = voiceMatch[1].trim();
      }
      const chatMatch = text.match(/\[CHAT\]([\s\S]*?)(?:\[\/CHAT\]|(?=\[VOICE\])|$)/i);
      if (chatMatch) {
        text = chatMatch[1].trim();
      } else {
        text = text.replace(/\[VOICE\][\s\S]*?(?:\[\/VOICE\]|(?=\[CHAT\])|$)/gi, '').trim();
      }
    }

    // Clean action blocks from chat display text
    text = text.replace(/```(?:action|json:action)[\s\S]*?```/gi, '').trim();
    text = text.replace(/\[ACTION:[\s\S]*?\]/gi, '').trim();

    // Clean remaining tags
    text = text.replace(/\[\/?CHAT\]/gi, '').replace(/\[\/?VOI
```

```
CE\]/gi, '').trim();
411     return { chatText: text, voiceText: voiceText || text };
412   }
413
414   // Dynamic pool of questions & quick buttons loaded from ext
ernal file
415   let sampleQuestionsPool = [];
416   let quickButtonsPool = [];
417
418   // Asynchronously load questions and quick buttons from exte
rnal file / API
419   async function loadQuestionsFromConfig() {
420     try {
421       const res = await fetch('/api/v1/scenarios/questions');
422       if (res.ok) {
423         const data = await res.json();
424         if (Array.isArray(data.prompts) && data.prompts.length
 > 0) {
425           sampleQuestionsPool = data.prompts;
426         }
427         if (Array.isArray(data.quick_buttons) && data.quick_bu
ttons.length > 0) {
428           quickButtonsPool = data.quick_buttons;
429           renderQuickButtons();
430         }
431       }
432     } catch (err) {
433       console.warn('[ScenariosTab] Falling back to direct JSON
 load for questions:', err);
434       try {
435         const fallbackRes = await fetch('/html/scenarios_tab/q
uestions.json');
436         if (fallbackRes.ok) {
437           const fbData = await fallbackRes.json();
438           if (Array.isArray(fbData.prompts)) sampleQuestionsPo
ol = fbData.prompts;
439           if (Array.isArray(fbData.quick_buttons)) {
440             quickButtonsPool = fbData.quick_buttons;
```

```
            renderQuickButtons();
          }
        }
      } catch {}
    }
    renderDynamicGreetingPrompts();
  }

  function renderQuickButtons() {
    const container = document.getElementById('chat-quick-prom
pts');
    if (!container || !quickButtonsPool.length) return;

    container.innerHTML = quickButtonsPool.map(btn => `
      <button class="btn btn-outline-secondary btn-sm py-0 px-
2 quick-prompt-btn" style="font-size: 0.75rem;" data-prompt="$
{escapeHtml(btn.text)}">
        ${escapeHtml(btn.icon || '□ ')} ${escapeHtml(btn.text)}
      </button>
    `).join('');

    // Re-bind click handlers
    container.querySelectorAll('.quick-prompt-btn').forEach(bt
n => {
      btn.onclick = () => {
        if (isChatSubmitting) return;
        const text = btn.dataset.prompt || btn.innerText.repla
ce(/^[^a-zA-Za-яA-ЯёЁ]+/, '').trim();
        const chatInput = document.getElementById('scenario-ch
at-input');
        if (chatInput) {
          chatInput.value = text;
          submitChat();
        }
      };
    });
  }

```

```
  function renderDynamicGreetingPrompts() {
    const greetingContainer = document.getElementById('chat-greeting-prompts');
    if (!greetingContainer) return;
    if (!sampleQuestionsPool.length) return;

    // Pick 3 random distinct questions from external config
    const shuffled = [...sampleQuestionsPool].sort(() => 0.5 - Math.random());
    const selected = shuffled.slice(0, 3);

    greetingContainer.innerHTML = selected.map(q => `
      <div>• <a href="#" class="chat-prompt-link text-decoration-none border-bottom border-primary border-opacity-50" data-prompt="${escapeHtml(q)}">«${escapeHtml(q)}»</a></div>
    `).join('');
  }

  function getGreetingHtml() {
    const shuffled = sampleQuestionsPool.length ? [...sampleQuestionsPool].sort(() => 0.5 - Math.random()) : [];
    const selected = shuffled.slice(0, 3);
    const promptsHtml = selected.length ? selected.map(q => `
      <div>• <a href="#" class="chat-prompt-link text-decoration-none border-bottom border-primary border-opacity-50" data-prompt="${escapeHtml(q)}">«${escapeHtml(q)}»</a></div>
    `).join('') : `
      <div>• <a href="#" class="chat-prompt-link text-decoration-none border-bottom border-primary border-opacity-50" data-prompt="Какие мыши были подключены к этому компьютеру?">«Какие мыши были подключены к этому компьютеру?»</a></div>
      <div>• <a href="#" class="chat-prompt-link text-decoration-none border-bottom border-primary border-opacity-50" data-prompt="Покажи список сетевых портов и активных соединений">«Покажи список сетевых портов и активных соединений»</a></div>
      <div>• <a href="#" class="chat-prompt-link text-decoration-none border-bottom border-primary border-opacity-50" data-prompt="Проверь автозапуск и службы Windows">«Проверь автозапус
```

```
к и службы Windows»</a></div>
    `;

    return `
      <div class="chat-message-row bot-row mb-3">
        <div class="chat-msg-avatar">🤖</div>
        <div class="chat-msg-bot">
          <div class="mb-2">
            Привет! Я интеллектуальный ассистент Test Computer. Вы можете задать мне любой вопрос о конфигурации хоста, например:
          </div>
          <div id="chat-greeting-prompts" class="d-flex flex-column gap-1 my-2">
            ${promptsHtml}
          </div>
          <div class="text-muted small mt-2">
            Если для вопроса нет готового сценария, я опрошу оборудование через PowerShell/WMI, сформирую для вас ответ и <strong>автоматически сохраню новый навык</strong> в каталог <code>.skills/</code>.
          </div>
        </div>
      </div>
    `;
  }

  // Handle Mini-Chat interaction
  function setupMiniChat() {
    const chatForm = document.getElementById('scenario-chat-form');
    const chatInput = document.getElementById('scenario-chat-input');
    const chatHistory = document.getElementById('scenario-chat-history');
    const sendBtn = document.getElementById('btn-scenario-chat-send');
    const statusInd = document.getElementById('chat-status-ind
```

```
icator');
    const ragCheck = document.getElementById('check-scenario-r
ag');
    const keepContextCheck = document.getElementById('check-sc
enario-keep-context');

    if (!chatForm || !chatInput || !chatHistory) return;

    // Fetch and render questions dynamically from external co
nfiguration
    loadQuestionsFromConfig();

    if (keepContextCheck) {
      const savedKeepContext = localStorage.getItem('tc_scenar
io_keep_context');
      if (savedKeepContext !== null) {
        keepContextCheck.checked = savedKeepContext === 'true'
;
      } else {
        keepContextCheck.checked = false;
      }
      keepContextCheck.addEventListener('change', (e) => {
        localStorage.setItem('tc_scenario_keep_context', e.tar
get.checked);
      });
    }

    if (ragCheck) {
      const savedRagState = localStorage.getItem('tc_scenario_
use_rag');
      if (savedRagState !== null) {
        ragCheck.checked = savedRagState === 'true';
      }
      ragCheck.addEventListener('change', (e) => {
        localStorage.setItem('tc_scenario_use_rag', e.target.c
hecked);
      });
    }
```

```
552
553     let currentConversationId = `scenario-chat-${Date.now()}`;
554
555     // New Session button handler
556     const btnNewChat = document.getElementById('btn-scenario-n
    ew-chat');
557     if (btnNewChat) {
558       btnNewChat.onclick = async () => {
559         try {
560           await fetch('/api/v1/scenarios/chat/clear', {
561             method: 'POST',
562             headers: { 'Content-Type': 'application/json' },
563             body: JSON.stringify({ conversation_id: currentCon
    versationId, message: 'clear' }),
564           });
565         } catch (e) {
566           console.warn('[ScenariosTab] Clear session error:',
    e);
567         }
568         currentConversationId = `scenario-chat-${Date.now()}`;
569         chatHistory.innerHTML = getGreetingHtml();
570         if (statusInd) statusInd.innerText = 'Новая сессия нач
    ата';
571         chatInput.focus();
572       };
573     }
574
575     // Quick prompt buttons & clickable prompt links
576     document.querySelectorAll('.quick-prompt-btn').forEach(btn
     => {
577       btn.onclick = () => {
578         if (isChatSubmitting) return;
579         const text = btn.innerText.replace(/^[^a-zA-Za-яA-ЯёЁ]
    +/, '').trim();
580         chatInput.value = text;
581         submitChat();
582       };
583     });
```

```
584
585     // Delegated click handler for inline prompt links in chat
 history
586     chatHistory.onclick = (e) => {
587       const link = e.target.closest('.chat-prompt-link');
588       if (link) {
589         e.preventDefault();
590         e.stopPropagation();
591         if (isChatSubmitting) return;
592         const promptText = link.dataset.prompt || link.innerTe
xt.replace(/^[«"']|[»"']$/g, '').trim();
593         if (promptText) {
594           chatInput.value = promptText;
595           submitChat();
596         }
597       }
598     };
599
600     async function submitChat() {
601       const message = chatInput.value.trim();
602       if (!message || isChatSubmitting) return;
603
604       isChatSubmitting = true;
605       chatInput.value = '';
606       chatInput.disabled = true;
607       if (sendBtn) sendBtn.disabled = true;
608       if (statusInd) statusInd.innerText = '  Инициализация с
истемного зонда...';
609
610       // Append User message
611       const userMsgHtml = `
612         <div class="chat-message-row user-row mb-2">
613           <div class="chat-msg-user">
614             ${escapeHtml(message)}
615           </div>
616         </div>
617       `;
618       chatHistory.insertAdjacentHTML('beforeend', userMsgHtml)
```

```
;
      chatHistory.scrollTop = chatHistory.scrollHeight;

      // Handle client-side slash commands (/skills list, /skills, /help)
      const cleanCmd = message.trim().toLowerCase();
      if (cleanCmd === '/skills' || cleanCmd === '/skills list' || cleanCmd.startsWith('/skills search')) {
        try {
          const searchParam = cleanCmd.startsWith('/skills search') ? cleanCmd.replace('/skills search', '').trim() : '';
          const url = searchParam ? `/api/admin/skills?q=${encodeURIComponent(searchParam)}` : '/api/admin/skills';
          const res = await fetch(url);
          if (!res.ok) throw new Error(`HTTP ${res.status}`);
          const data = await res.json();
          const skills = data.skills || [];

          let replyMd = `### 🧠 Каталог зарегистрированных навыков (.skills)\n\n`;
          if (skills.length === 0) {
            replyMd += `*Навыки не найдены.*`;
          } else {
            replyMd += `Всего доступно навыков: **${skills.length}**\n\n`;
            skills.forEach((s, idx) => {
              const pathStr = s.root_relative || s.path || `.skills/${s.name}/SKILL.md`;
              const desc = s.description_ru || s.description || 'Без описания';
              replyMd += `${idx + 1}. 🔹 **${escapeHtml(s.name)}** — *${escapeHtml(desc)}* (\`${escapeHtml(pathStr)}\`)\n`;
            });
          }

          const assistantMsgHtml = `
            <div class="chat-message-row bot-row mb-3">
              <div class="chat-msg-avatar">🤖</div>
```

```
              <div class="chat-msg-bot">
                <div>${formatMarkdown(replyMd)}</div>
              </div>
            </div>
          `;
          chatHistory.insertAdjacentHTML('beforeend', assistantMsgHtml);
          chatHistory.scrollTop = chatHistory.scrollHeight;
          if (statusInd) statusInd.innerText = 'Список навыков выведен';
        } catch (err) {
          const errorMsgHtml = `
            <div class="chat-message-row bot-row mb-3">
              <div class="chat-msg-avatar">⚠️</div>
              <div class="chat-msg-bot border-danger text-danger">
                Ошибка получения списка навыков: ${escapeHtml(err.message)}
              </div>
            </div>
          `;
          chatHistory.insertAdjacentHTML('beforeend', errorMsgHtml);
          chatHistory.scrollTop = chatHistory.scrollHeight;
          if (statusInd) statusInd.innerText = 'Ошибка запроса';
        } finally {
          isChatSubmitting = false;
          chatInput.disabled = false;
          if (sendBtn) sendBtn.disabled = false;
          chatInput.focus();
        }
        return;
      }

      // Append Streaming Assistant placeholder with multi-step live tracker
      const msgId = `msg-${Date.now()}-${Math.random().toStrin
```

```
g(36).substr(2, 6)}`;
      const stagesLog = [];

      const renderStagesHtml = (stages, isFinal = false) => {
        if (!stages || stages.length === 0) return '';
        const currentCount = stages.length;
        const isComplete = isFinal;
        const headerIcon = isComplete
          ? '<i class="bi bi-check-circle-fill text-success"></i>'
          : '<div class="spinner-border spinner-border-sm text-primary" role="status" style="width:0.75rem;height:0.75rem;"></div>';
        const headerText = isComplete
          ? `Диагностика хоста завершена (${currentCount} шагов)`
          : `Диагностический конвейер хоста (${currentCount} шагов)...`;
        const headerClass = isComplete ? 'text-success' : 'text-primary';
        const collapseClass = isComplete ? 'd-none' : '';
        const chevronIcon = isComplete ? 'bi-chevron-down' : 'bi-chevron-up';
        const lastTitle = stages[stages.length - 1]?.title || 'Зонд';

        return `
          <div class="chat-stage-pipeline mb-2 shadow-sm" id="${msgId}-stages-wrapper">
            <div class="d-flex align-items-center justify-content-between flex-wrap gap-1" onclick="const b=document.getElementById('${msgId}-stages-content'); const c=document.getElementById('${msgId}-stages-chevron'); if(b){b.classList.toggle('d-none'); if(c){c.classList.toggle('bi-chevron-down'); c.classList.toggle('bi-chevron-up');}}" style="cursor:pointer;" title="Нажмите, чтобы развернуть/свернуть детали этапов">
              <span class="${headerClass} small fw-semibold d-flex align-items-center gap-1.5" style="font-size:0.75rem;">
```

```
                ${headerIcon}
                <span>${escapeHtml(headerText)}</span>
              </span>
              <div class="d-flex align-items-center gap-1">
                <span class="badge bg-body-secondary text-body
 border border-secondary-subtle font-monospace" style="font-si
ze:0.63rem;">${escapeHtml(lastTitle)}</span>
                <i class="bi ${chevronIcon} text-muted" id="${
msgId}-stages-chevron" style="font-size:0.65rem;"></i>
              </div>
            </div>
            <div class="mt-1.5 pt-1.5 border-top border-second
ary border-opacity-25 d-flex flex-column gap-1.5 ${collapseCla
ss}" id="${msgId}-stages-content">
              ${stages.map((st, sIdx) => {
                const isLatest = sIdx === stages.length - 1 &&
 !isComplete;
                const icon = isLatest
                  ? '<div class="spinner-grow spinner-grow-sm
text-primary" style="width: 0.55rem; height: 0.55rem;" role="s
tatus"></div>'
                  : '<span class="text-success small" style="f
ont-size:0.7rem;">✓</span>';
                const textClass = isLatest ? 'text-body fw-med
ium' : 'text-muted';
                return `
                  <div class="small ${textClass} d-flex flex-c
olumn ps-0.5 stage-item ${isLatest ? 'active' : ''}" style="fo
nt-size: 0.74rem; line-height: 1.35;">
                    <div class="d-flex align-items-center gap-
1.5">
                      <span>${icon}</span>
                      <span>${escapeHtml(st.message || '')}</s
pan>
                    </div>
                    ${st.details ? `<div class="text-muted ps-
3 font-monospace" style="font-size: 0.68rem; word-break: break
-all;">↳ ${escapeHtml(st.details)}</div>` : ''}
```

```
                    ${st.generated_prompt ? `
                      <div class="mt-1 ps-3">
                        <button class="btn btn-sm btn-outline-warning py-0 px-2 d-inline-flex align-items-center gap-1 font-monospace" type="button" onclick="event.stopPropagation(); const p=document.getElementById('${msgId}-stage-prompt-${sIdx}'); const ic=document.getElementById('${msgId}-stage-prompt-ic-${sIdx}'); if(p){p.classList.toggle('d-none'); if(ic){ic.classList.toggle('bi-chevron-down'); ic.classList.toggle('bi-chevron-up');}}" style="font-size: 0.68rem;">
                          <i class="bi bi-cpu-fill"></i>
                          <span>Показать отправленный промпт</span>
                          <i class="bi bi-chevron-down text-warning" id="${msgId}-stage-prompt-ic-${sIdx}" style="font-size: 0.6rem;"></i>
                        </button>
                        <div id="${msgId}-stage-prompt-${sIdx}" class="d-none mt-1.5 p-2 bg-dark bg-opacity-25 border border-warning border-opacity-40 rounded text-start" onclick="event.stopPropagation();">
                          <div class="d-flex justify-content-between align-items-center text-warning mb-1" style="font-size: 0.68rem;">
                            <span><i class="bi bi-terminal me-1"></i>Точный текст промпта, отправленный модели:</span>
                            <button class="btn btn-dark btn-sm py-0 px-1.5 text-secondary border border-secondary-subtle" type="button" onclick="navigator.clipboard.writeText(this.closest('#${msgId}-stage-prompt-${sIdx}').querySelector('pre')?.innerText || ''); this.innerText='Скопировано!'; setTimeout(()=>this.innerText='Копировать', 1500);" style="font-size: 0.65rem;">Копировать</button>
                          </div>
                          <pre class="m-0 p-1.5 bg-body-tertiary text-body border border-secondary border-opacity-30 rounded font-monospace" style="font-size: 0.70rem; line-height: 1.35; max-height: 220px; overflow-y: auto; white-space: pre-wrap; w
```

```
ord-break: break-word;">${escapeHtml(st.generated_prompt)}</pr
e>
                        </div>
                      </div>
                    ` : ''}
                  </div>
                `;
              }).join('')}
            </div>
          </div>
        `;
      };

      const initialStage = {
        stage: 'init',
        title: 'Инициализация',
        message: '  Подключение к локальной подсистеме Window
s API и сервисам хоста...',
        details: 'Подготовка системных интерфейсов WMI/PnP, ко
ллекторов и контекста сессии',
      };
      stagesLog.push(initialStage);

      const loadingHtml = `
        <div class="chat-message-row bot-row mb-3" id="${msgId
}-container">
          <div class="chat-msg-avatar">  </div>
          <div class="chat-msg-bot shadow-sm" id="${msgId}-bod
y">
            ${renderStagesHtml(stagesLog, false)}
          </div>
        </div>
      `;
      chatHistory.insertAdjacentHTML('beforeend', loadingHtml)
;
      chatHistory.scrollTop = chatHistory.scrollHeight;

      try {
```

```
765         const useRag = ragCheck ? ragCheck.checked : true;
766         const keepContext = keepContextCheck ? keepContextChec
k.checked : false;
767 
768         const res = await fetch('/api/v1/scenarios/chat/stream
', {
769           method: 'POST',
770           headers: { 'Content-Type': 'application/json' },
771           body: JSON.stringify({
772             message: message,
773             conversation_id: currentConversationId,
774             auto_create_skill: false,
775             keep_context: keepContext,
776             use_rag: useRag,
777           }),
778         });
779 
780         if (!res.ok) {
781           throw new Error(`Ошибка ответа сервера: HTTP ${res.s
tatus}`);
782         }
783 
784         const reader = res.body.getReader();
785         const decoder = new TextDecoder('utf-8');
786         let buffer = '';
787         let finalData = null;
788         let streamedReplyText = '';
789 
790         while (true) {
791           const { done, value } = await reader.read();
792           if (done) break;
793           buffer += decoder.decode(value, { stream: true });
794 
795           const lines = buffer.split('\n\n');
796           buffer = lines.pop(); // Оставляем неполный хвост в
буфере
797 
798           for (const line of lines) {
```

```
            const cleanLine = line.trim();
            if (!cleanLine.startsWith('data:')) continue;
            let evt = null;
            try {
              evt = JSON.parse(cleanLine.substring(5).trim());
            } catch (pErr) {
              console.warn('[ScenariosTab] SSE parse warning:', pErr);
              continue;
            }
            if (!evt) continue;

            if (evt.type === 'stage') {
              const lastSt = stagesLog[stagesLog.length - 1];
              if (!lastSt || lastSt.stage !== evt.stage || lastSt.message !== evt.message) {
                stagesLog.push(evt);
              }
              if (statusInd) statusInd.innerText = evt.message || 'Обработка...';

              const bodyElem = document.getElementById(`${msgId}-body`);
              if (bodyElem) {
                const { chatText } = parseChatAndVoice(streamedReplyText);
                bodyElem.innerHTML = `
                  ${renderStagesHtml(stagesLog, false)}
                  ${streamedReplyText ? `<div class="chat-streaming-content mt-1.5">${formatMarkdown(chatText)}<span class="streaming-cursor">▌</span></div>` : ''}
                `;
                chatHistory.scrollTop = chatHistory.scrollHeight;
              }
            } else if (evt.type === 'chunk') {
              streamedReplyText += (evt.content || '');
              const bodyElem = document.getElementById(`${msgI
```

```
d}-body`);
              if (bodyElem) {
                const { chatText } = parseChatAndVoice(streame
dReplyText);
                bodyElem.innerHTML = `
                  ${renderStagesHtml(stagesLog, false)}
                  <div class="chat-streaming-content mt-1.5">$
{formatMarkdown(chatText)}<span class="streaming-cursor">▌</sp
an></div>
                `;
                chatHistory.scrollTop = chatHistory.scrollHeig
ht;
              }
            } else if (evt.type === 'done') {
              finalData = evt;
            } else if (evt.type === 'error') {
              const streamErr = new Error(evt.error || 'Ошибка
 в потоке');
              streamErr.details = evt.details || '';
              streamErr.errorType = evt.error_type || 'StreamE
rror';
              streamErr.rawEvent = evt;
              throw streamErr;
            }
          }
        }

        if (!finalData) {
          throw new Error('Не удалось получить завершающий отв
ет от сервера.');
        }

        const fullReply = finalData.reply || streamedReplyText
 || 'Ответ получен.';
        const { chatText, voiceText } = parseChatAndVoice(full
Reply);
        const formattedReply = formatMarkdown(chatText || 'Отв
ет получен.');
```

```
856
857         let remediationHtml = '';
858         if (Array.isArray(finalData.remediation_actions) && fi
    nalData.remediation_actions.length > 0) {
859           const actionsList = finalData.remediation_actions;
860           remediationHtml = `
861             <div class="mt-2.5 p-2.5 bg-body-secondary border
    border-warning border-opacity-50 rounded" id="remediation-cont
    ainer-${msgId}">
862               <div class="d-flex align-items-center justify-co
    ntent-between mb-2">
863                 <span class="fw-semibold text-warning small d-
    flex align-items-center gap-1">
864                   <i class="bi bi-shield-check text-warning"><
    /i> <span>Предложенные действия и исправления:</span>
865                 </span>
866                 <span class="badge bg-warning text-dark small"
     style="font-size: 0.68rem;">SafeOps (${actionsList.length})</
    span>
867               </div>
868               <div class="d-flex flex-column gap-1.5" id="reme
    diation-buttons-${msgId}">
869                 ${actionsList.map((act, aIdx) => {
870                   const isCaution = (act.risk || '').toLowerCa
    se() === 'caution' || (act.risk || '').toLowerCase() === 'crit
    ical';
871                   const badgeRisk = isCaution ? '<span class="
    badge bg-danger-subtle text-danger border border-danger-subtle
     ms-1" style="font-size:0.65rem;">Требует подтверждения</span>
    ' : '<span class="badge bg-info-subtle text-info border border
    -info-subtle ms-1" style="font-size:0.65rem;">Безопасно</span>
    ';
872                   return `
873                     <div class="d-flex align-items-center just
    ify-content-between p-2 bg-body rounded border border-secondar
    y border-opacity-25 gap-2 flex-wrap" id="act-row-${msgId}-${aI
    dx}">
874                       <div class="small d-flex flex-column" st
```

```
yle="font-size: 0.78rem;">
                        <div><strong>${escapeHtml(act.title ||
 act.action_id)}</strong>${badgeRisk}</div>
                        ${act.description ? `<span class="text
-muted mt-0.5" style="font-size: 0.72rem;">${escapeHtml(act.de
scription)}</span>` : ''}
                        ${act.execution_command ? `<code class
="text-warning-emphasis bg-body-secondary px-1 py-0.5 rounded
mt-1 font-monospace" style="font-size: 0.68rem; word-break: br
eak-all;">${escapeHtml(act.execution_command)}</code>` : ''}
                      </div>
                      <button class="btn btn-warning btn-sm py
-1 px-3 d-flex align-items-center gap-1.5 shadow-sm fw-semibol
d btn-exec-action"
                        id="btn-act-${msgId}-${aIdx}"
                        data-msg-id="${msgId}"
                        data-action-id="${escapeHtml(act.actio
n_id)}"
                        data-action-type="${escapeHtml(act.act
ion_type)}"
                        data-title="${escapeHtml(act.title)}"
                        data-description="${escapeHtml(act.des
cription)}"
                        data-target="${escapeHtml(act.target)}
"
                        data-risk="${escapeHtml(act.risk)}"
                        data-command="${escapeHtml(act.executi
on_command)}"
                        style="font-size: 0.78rem;">
                        <i class="bi bi-play-circle-fill"></i>
                        <span>Подтвердить</span>
                      </button>
                    </div>
                  `;
                }).join('')}
              </div>
            </div>
          `;
```

```
        }

        let skillActionHtml = '';
        if (finalData.created_skill) {
          const isNew = finalData.created_skill.is_new !== fal
se;
          const badgeTitle = isNew ? '  <strong>Автоматически
создан навык:</strong>' : '  <strong>Использован существующий
 навык:</strong>';
          const badgeClass = isNew ? 'bg-success bg-opacity-10
 border-success border-opacity-50 text-success' : 'bg-info bg-
opacity-10 border-info border-opacity-50 text-info';
          skillActionHtml = `
            <div class="mt-2 p-2 ${badgeClass} border rounded
small" id="skill-action-container-${msgId}">
              ${badgeTitle} <code>${escapeHtml(finalData.creat
ed_skill.path)}</code>
              <div class="text-muted small mt-1">${escapeHtml(
finalData.created_skill.description_ru)}</div>
            </div>
          `;
        } else if (finalData.tool_plan) {
          skillActionHtml = `
            <div class="mt-2 pt-2 border-top border-secondary
border-opacity-25 d-flex align-items-center gap-2 flex-wrap" i
d="skill-action-container-${msgId}">
              <button class="btn btn-outline-success btn-sm d-
flex align-items-center gap-1.5 py-1 px-2.5 shadow-sm" id="btn
-save-skill-${msgId}" type="button">
                <i class="bi bi-bookmark-check-fill text-succe
ss"></i>
                <span>Ответ правильный, запомнить как навык</s
pan>
              </button>
              <button class="btn btn-outline-secondary btn-sm
d-flex align-items-center gap-1.5 py-1 px-2 shadow-sm" id="btn
-edit-skill-${msgId}" type="button">
                <i class="bi bi-pencil-square"></i>
```

```
                <span>Скорректировать</span>
              </button>
            </div>
            <div id="skill-edit-form-${msgId}" class="mt-2 p-2
.5 bg-body-secondary border border-secondary-subtle rounded sm
all d-none">
              <div class="fw-semibold text-warning mb-2"><i cl
ass="bi bi-sliders me-1"></i>Корректировка навыка перед сохран
ением</div>
              <div class="mb-2">
                <label class="form-label text-muted mb-1" styl
e="font-size:0.75rem;">Заголовок навыка:</label>
                <input type="text" class="form-control form-co
ntrol-sm border-secondary" id="edit-title-${msgId}" value="${e
scapeHtml(finalData.tool_plan.tool_title || '')}">
              </div>
              <div class="mb-2">
                <label class="form-label text-muted mb-1" styl
e="font-size:0.75rem;">Описание (русский язык):</label>
                <textarea class="form-control form-control-sm
border-secondary" id="edit-desc-${msgId}" rows="2">${escapeHtm
l(finalData.tool_plan.description_ru || '')}</textarea>
              </div>
              <div class="mb-2">
                <label class="form-label text-muted mb-1" styl
e="font-size:0.75rem;">Команда проверки / скрипт:</label>
                <input type="text" class="form-control form-co
ntrol-sm border-secondary font-monospace" id="edit-cmd-${msgId
}" value="${escapeHtml(finalData.command_executed || finalData
.tool_plan.probe_script || '')}">
              </div>
              <div class="d-flex gap-2 justify-content-end mt-
2">
                <button class="btn btn-secondary btn-sm py-1 p
x-2" id="btn-cancel-edit-${msgId}" type="button">Отмена</butto
n>
                <button class="btn btn-success btn-sm py-1 px-
2.5" id="btn-save-edited-${msgId}" type="button">
```

```
                <i class="bi bi-check-circle-fill me-1"></i> Сохранить скорректированный навык
              </button>
            </div>
          </div>
        `;
      }

      let metaInfoHtml = '';
      const agentsList = finalData.agents_used || [];
      const knowledgeList = finalData.knowledge_used || [];

      if (agentsList.length > 0 || knowledgeList.length > 0) {
        const totalCount = agentsList.length + knowledgeList.length;
        const metaCollapseId = `meta-details-${msgId}`;
        metaInfoHtml = `
          <div class="mt-2.5 pt-2 border-top border-secondary border-opacity-25" id="meta-container-${msgId}">
            <div class="d-flex align-items-center justify-content-between flex-wrap gap-1">
              <button class="btn btn-sm btn-link p-0 text-primary text-decoration-none d-flex align-items-center gap-1.5" type="button" onclick="const el=document.getElementById('${metaCollapseId}'); const ic=document.getElementById('meta-icon-${msgId}'); if(el){el.classList.toggle('d-none'); if(ic) { ic.classList.toggle('bi-chevron-down'); ic.classList.toggle('bi-chevron-up'); }}" style="font-size: 0.75rem;">
                <i class="bi bi-diagram-3-fill text-primary"></i>
                <span><strong>Задействованные знания и агенты</strong> (${totalCount})</span>
                <i class="bi bi-chevron-down text-muted" id="meta-icon-${msgId}" style="font-size: 0.7rem;"></i>
              </button>
              <div class="d-flex gap-1 flex-wrap">
                ${agentsList.map(a => `<span class="badge bg
```

```
-primary bg-opacity-10 text-primary border border-primary bord
er-opacity-25" style="font-size: 0.65rem;" title="${escapeHtml
(a.role || '')}">${escapeHtml(a.icon || '□ ')} ${escapeHtml(a.
name)}</span>`).slice(0, 2).join('')}
                </div>
              </div>

              <div id="${metaCollapseId}" class="d-none mt-2 p
-2 bg-body-secondary border border-secondary border-opacity-50
 rounded small">
                ${agentsList.length > 0 ? `
                  <div class="mb-2">
                    <div class="text-warning fw-semibold mb-1"
 style="font-size: 0.73rem;">
                      <i class="bi bi-robot me-1"></i>Задейств
ованные агенты и модули:
                    </div>
                    <div class="d-flex flex-column gap-1 ps-1"
>
                      ${agentsList.map(ag => `
                        <div class="d-flex align-items-start g
ap-1.5" style="font-size: 0.73rem;">
                          <span>${escapeHtml(ag.icon || '□ ')}
</span>
                          <div>
                            <strong>${escapeHtml(ag.name)}</st
rong>
                            <span class="text-muted ms-1">— ${
escapeHtml(ag.role || 'Системный агент')}</span>
                          </div>
                        </div>
                      `).join('')}
                    </div>
                  </div>
                ` : ''}

                ${knowledgeList.length > 0 ? `
                  <div>
```

```
                    <div class="text-primary fw-semibold mb-1" style="font-size: 0.73rem;">
                      <i class="bi bi-book-half me-1"></i>Использованные источники знаний и артефакты:
                    </div>
                    <div class="d-flex flex-column gap-1 ps-1">
                      ${knowledgeList.map(kn => `
                        <div class="d-flex align-items-start gap-1.5" style="font-size: 0.73rem;">
                          <span class="badge bg-info bg-opacity-25 text-info border border-info border-opacity-25" style="font-size: 0.65rem;">${escapeHtml(kn.badge || 'База')}</span>
                          <div>
                            <strong>${escapeHtml(kn.title)}</strong>
                            <span class="text-muted ms-1">— ${escapeHtml(kn.description || '')}</span>
                          </div>
                        </div>
                      `).join('')}
                    </div>
                  </div>
                ` : ''}
              </div>
            </div>
          `;
        }

        let promptInfoHtml = '';
        const generatedPromptText = finalData.generated_prompt || (stagesLog.find(s => s.stage === 'synthesizing' && s.generated_prompt)?.generated_prompt);
        if (generatedPromptText) {
          const promptCollapseId = `prompt-details-${msgId}`;
          const promptCodeId = `prompt-code-${msgId}`;
          promptInfoHtml = `
            <div class="mt-2.5 pt-2 border-top border-secondar
```

```
y border-opacity-25" id="prompt-container-${msgId}">
1018              <div class="d-flex align-items-center justify-co
ntent-between flex-wrap gap-1">
1019                <button class="btn btn-sm btn-link p-0 text-wa
rning text-decoration-none d-flex align-items-center gap-1.5"
type="button" onclick="const el=document.getElementById('${pro
mptCollapseId}'); const ic=document.getElementById('prompt-ico
n-${msgId}'); if(el){el.classList.toggle('d-none'); if(ic) { i
c.classList.toggle('bi-chevron-down'); ic.classList.toggle('bi
-chevron-up'); }}" style="font-size: 0.75rem;">
1020                  <i class="bi bi-cpu-fill text-warning"></i>
1021                  <span><strong>Сформированный промпт модели</
strong></span>
1022                  <i class="bi bi-chevron-down text-muted" id=
"prompt-icon-${msgId}" style="font-size: 0.7rem;"></i>
1023                </button>
1024                <div class="d-flex align-items-center gap-1">
1025                  <button class="btn btn-outline-secondary btn
-sm py-0 px-2 d-flex align-items-center gap-1" type="button" o
nclick="navigator.clipboard.writeText(document.getElementById(
'${promptCodeId}')?.innerText || ''); const s=this.querySelect
or('span'); if(s){s.innerText='Скопировано!'; setTimeout(()=>s
.innerText='Копировать', 1800);}" style="font-size: 0.68rem;"
title="Скопировать отправленный промпт">
1026                    <i class="bi bi-clipboard"></i>
1027                    <span>Копировать</span>
1028                  </button>
1029                </div>
1030              </div>
1031
1032              <div id="${promptCollapseId}" class="d-none mt-2
 p-2.5 bg-body-secondary border border-warning border-opacity-
40 rounded">
1033                <div class="d-flex align-items-center justify-
content-between text-warning small mb-1.5" style="font-size: 0
.72rem;">
1034                  <span><i class="bi bi-terminal me-1"></i>Точ
ный текст запроса, отправленный языковой модели:</span>
```

```
                  <span class="badge bg-warning bg-opacity-10 text-warning border border-warning border-opacity-25 font-monospace" style="font-size: 0.65rem;">${generatedPromptText.length} симв.</span>
                </div>
                <pre class="m-0 p-2 bg-body text-body border border-secondary border-opacity-30 rounded font-monospace small" id="${promptCodeId}" style="font-size: 0.72rem; line-height: 1.4; max-height: 280px; overflow-y: auto; white-space: pre-wrap; word-break: break-word;">${escapeHtml(generatedPromptText)}</pre>
              </div>
            </div>
          `;
        }

        const trainingActionHtml = `
          <div class="mt-2.5 pt-2 border-top border-secondary border-opacity-25 d-flex align-items-center justify-content-between flex-wrap gap-2" id="training-save-container-${msgId}">
            <button class="btn btn-outline-primary btn-sm d-flex align-items-center gap-1.5 py-1 px-2.5 shadow-sm btn-save-training-qa" id="btn-save-training-${msgId}" type="button" title="Сохранить пару вопрос-ответ в базу данных Test Computer для RAG и последующего обучения (тюнинга) модели">
              <i class="bi bi-mortarboard-fill text-primary"></i>
              <span>Сохранить ответ для обучения модели</span>
            </button>
            <span class="text-muted" style="font-size: 0.68rem;"><i class="bi bi-database me-1"></i>data/tc/approved_responses</span>
          </div>
        `;

        const bodyElem = document.getElementById(`${msgId}-body`);
        if (bodyElem) {
```

```
          bodyElem.innerHTML = `
            ${renderStagesHtml(stagesLog, true)}
            <div>${formattedReply}</div>
            ${remediationHtml}
            ${skillActionHtml}
            ${trainingActionHtml}
            ${metaInfoHtml}
            ${promptInfoHtml}
          `;
        }
        chatHistory.scrollTop = chatHistory.scrollHeight;

        // Wire up save response for training & RAG button
        const btnSaveTraining = document.getElementById(`btn-save-training-${msgId}`);
        if (btnSaveTraining) {
          btnSaveTraining.onclick = async () => {
            btnSaveTraining.disabled = true;
            btnSaveTraining.innerHTML = `<div class="spinner-border spinner-border-sm text-info me-1" role="status"></div><span>Сохранение...</span>`;
            try {
              const res = await fetch('/api/v1/scenarios/save-approved-response', {
                method: 'POST',
                headers: { 'Content-Type': 'application/json' },
                body: JSON.stringify({
                  user_id: 'tc_admin',
                  query: message,
                  chat_text: chatText,
                  voice_text: voiceText || '',
                  system_context: {
                    command_executed: finalData.command_executed || '',
                    tool_plan: finalData.tool_plan || null,
                    platform: 'Windows'
                  },
```

```
                tags: ['tc', 'diagnostics', 'training']
              }),
            });
            if (!res.ok) throw new Error(`HTTP ${res.status}
`);
            const resData = await res.json();
            const container = document.getElementById(`train
ing-save-container-${msgId}`);
            if (container) {
              container.innerHTML = `
                <div class="p-1.5 px-2.5 bg-success bg-opaci
ty-10 border border-success border-opacity-50 text-success rou
nded small d-flex align-items-center gap-1.5 w-100 shadow-sm">
                  <i class="bi bi-check-circle-fill text-suc
cess"></i>
                  <span><strong>Успешно:</strong> ответ сохр
анён для обучения модели и RAG (<code>data/tc/approved_respons
es</code>)</span>
                </div>
              `;
            }
            if (statusInd) statusInd.innerText = 'Ответ сохр
анён в базу для обучения модели!';
          } catch (saveErr) {
            console.error('[ScenariosTab] Error saving train
ing QA:', saveErr);
            btnSaveTraining.disabled = false;
            btnSaveTraining.innerHTML = `<i class="bi bi-exc
lamation-triangle-fill text-danger me-1"></i><span>Ошибка сохр
анения. Повторить</span>`;
          }
        };
      }


      // Wire up manual save & inline edit logic
      if (!finalData.created_skill && finalData.tool_plan) {
        const saveBtn = document.getElementById(`btn-save-sk
```

```
ill-${msgId}`);
1114          const editBtn = document.getElementById(`btn-edit-sk
ill-${msgId}`);
1115          const editForm = document.getElementById(`skill-edit
-form-${msgId}`);
1116          const cancelEditBtn = document.getElementById(`btn-c
ancel-edit-${msgId}`);
1117          const saveEditedBtn = document.getElementById(`btn-s
ave-edited-${msgId}`);
1118
1119          const executeSave = async (payload) => {
1120            const container = document.getElementById(`skill-a
ction-container-${msgId}`);
1121            if (saveBtn) saveBtn.disabled = true;
1122            if (saveEditedBtn) saveEditedBtn.disabled = true;
1123            if (saveBtn) saveBtn.innerHTML = `<div class="spin
ner-border spinner-border-sm text-success me-1" role="status">
</div><span>Сохранение...</span>`;
1124
1125            try {
1126              const saveRes = await fetch('/api/v1/scenarios/s
ave-skill', {
1127                method: 'POST',
1128                headers: { 'Content-Type': 'application/json'
},
1129                body: JSON.stringify(payload),
1130              });
1131              if (!saveRes.ok) throw new Error(`HTTP ${saveRes
.status}`);
1132              const savedData = await saveRes.json();
1133              if (container) {
1134                container.innerHTML = `
1135                  <div class="p-2 bg-success bg-opacity-10 bor
der border-success border-opacity-50 text-success rounded smal
l w-100">
1136                    <strong>Навык успешно сохранён:</strong>
 <code>${escapeHtml(savedData.path)}</code>
1137                    <div class="text-muted small mt-1">${escap
```

```
eHtml(savedData.description_ru)}</div>
                  </div>
                `;
              }
              if (editForm) editForm.classList.add('d-none');
              if (statusInd) statusInd.innerText = `Навык '${s
avedData.name}' сохранён!`;
            } catch (saveErr) {
              console.error('[ScenariosTab] Error saving skill
:', saveErr);
              if (saveBtn) {
                saveBtn.disabled = false;
                saveBtn.innerHTML = `<i class="bi bi-exclamati
on-triangle-fill text-danger me-1"></i><span>Ошибка. Повторить
</span>`;
              }
              if (saveEditedBtn) saveEditedBtn.disabled = fals
e;
            }
          };

          if (saveBtn) {
            saveBtn.addEventListener('click', () => {
              executeSave({
                tool_name: finalData.tool_plan.tool_name,
                tool_title: finalData.tool_plan.tool_title,
                description_ru: finalData.tool_plan.descriptio
n_ru,
                probe_type: finalData.tool_plan.probe_type ||
'powershell',
                collector_name: finalData.tool_plan.collector_
name,
                probe_script: finalData.tool_plan.probe_script
,
                instructions: finalData.tool_plan.instructions
,
                command_executed: finalData.command_executed,
              });
```

```
            });
          }

          if (editBtn && editForm) {
            editBtn.addEventListener('click', () => {
              editForm.classList.toggle('d-none');
            });
          }

          if (cancelEditBtn && editForm) {
            cancelEditBtn.addEventListener('click', () => {
              editForm.classList.add('d-none');
            });
          }

          if (saveEditedBtn) {
            saveEditedBtn.addEventListener('click', () => {
              const editedTitle = document.getElementById(`edi
t-title-${msgId}`)?.value || finalData.tool_plan.tool_title;
              const editedDesc = document.getElementById(`edit
-desc-${msgId}`)?.value || finalData.tool_plan.description_ru;
              const editedCmd = document.getElementById(`edit-
cmd-${msgId}`)?.value || finalData.command_executed;

              executeSave({
                tool_name: finalData.tool_plan.tool_name,
                tool_title: editedTitle,
                description_ru: editedDesc,
                probe_type: finalData.tool_plan.probe_type ||
'powershell',
                collector_name: finalData.tool_plan.collector_
name,
                probe_script: editedCmd,
                instructions: finalData.tool_plan.instructions
,
                command_executed: editedCmd,
              });
            });
```

```
          }
        }

        // Wire up remediation action execution buttons
        if (Array.isArray(finalData.remediation_actions) && finalData.remediation_actions.length > 0) {
          const remContainer = document.getElementById(`remediation-container-${msgId}`);
          if (remContainer) {
            remContainer.querySelectorAll('.btn-exec-action').forEach(btn => {
              btn.addEventListener('click', async () => {
                const actionId = btn.dataset.actionId;
                const actionType = btn.dataset.actionType;
                const title = btn.dataset.title;
                const description = btn.dataset.description;
                const target = btn.dataset.target;
                const risk = btn.dataset.risk;
                const command = btn.dataset.command;

                // Confirm if risk is caution or critical
                const isRisky = (risk || '').toLowerCase() === 'caution' || (risk || '').toLowerCase() === 'critical';
                if (isRisky) {
                  const ok = confirm(`Подтвердите применение исправления:\n\n«${title}»\n\nКоманда: ${command || target}`);
                  if (!ok) return;
                }

                btn.disabled = true;
                btn.innerHTML = `<div class="spinner-border spinner-border-sm text-warning" role="status"></div><span>Применение...</span>`;
                if (statusInd) statusInd.innerText = `Применение '${title}'...`;

                try {
                  const execRes = await fetch('/api/v1/scenari
```

```
os/execute-fix', {
                    method: 'POST',
                    headers: { 'Content-Type': 'application/js
on' },
                    body: JSON.stringify({
                      action_id: actionId,
                      action_type: actionType,
                      title: title,
                      description: description,
                      target: target,
                      risk: risk,
                      execution_command: command,
                      confirmed_by_user: true,
                    }),
                  });

                  if (!execRes.ok) throw new Error(`HTTP ${exe
cRes.status}`);
                  const resData = await execRes.json();
                  if (resData.success) {
                    btn.className = 'btn btn-success btn-sm py
-1 px-3 d-flex align-items-center gap-1.5 shadow-sm';
                    btn.innerHTML = `<i class="bi bi-check-cir
cle-fill"></i><span>Подтверждено и выполнено</span>`;
                    if (statusInd) statusInd.innerText = `Дейс
твие '${title}' успешно выполнено!`;
                  } else {
                    btn.disabled = false;
                    btn.className = 'btn btn-danger btn-sm py-
0.5 px-2.5 d-flex align-items-center gap-1 shadow-sm';
                    btn.innerHTML = `<i class="bi bi-exclamati
on-triangle-fill"></i><span>Ошибка</span>`;
                    btn.title = resData.error_message || resDa
ta.message;
                    if (statusInd) statusInd.innerText = `Оши6
ка: ${resData.error_message || resData.message}`;
                  }
                } catch (exErr) {
```

```
                  console.error('[ScenariosTab] Error executin
g fix:', exErr);
                  btn.disabled = false;
                  btn.className = 'btn btn-outline-danger btn-
sm py-0.5 px-2.5 d-flex align-items-center gap-1 shadow-sm';
                  btn.innerHTML = `<i class="bi bi-arrow-repea
t"></i><span>Повторить</span>`;
                  if (statusInd) statusInd.innerText = `Сбой:
${exErr.message}`;
                }
              });
            });
          }
        }

        if (statusInd) statusInd.innerText = 'Ответ сформирова
н';
      } catch (err) {
        console.error('[ScenariosTab] Chat error:', err);
        const bodyElem = document.getElementById(`${msgId}-bod
y`);
        if (bodyElem) {
          const errDetail = err.details || (err.rawEvent ? JSO
N.stringify(err.rawEvent, null, 2) : '');
          const errorTypeBadge = err.errorType ? `<span class=
"badge bg-danger bg-opacity-25 text-danger border border-dange
r border-opacity-50 font-monospace small ms-1" style="font-siz
e: 0.68rem;">${escapeHtml(err.errorType)}</span>` : '';
          bodyElem.innerHTML = `
            ${stagesLog.length > 0 ? renderStagesHtml(stagesLo
g, false) : ''}
            <div class="text-danger p-2.5 bg-danger bg-opacity
-10 border border-danger border-opacity-40 rounded mt-2">
              <div class="fw-semibold d-flex align-items-cente
r justify-content-between flex-wrap gap-1 text-danger small">
                <span class="d-flex align-items-center gap-1.5
">
                  <i class="bi bi-exclamation-triangle-fill te
```

```
xt-danger"></i>
                  <span>Ошибка: ${escapeHtml(err.message || 'Неизвестная ошибка выполнения')}</span>
                </span>
                ${errorTypeBadge}
              </div>
              ${errDetail && errDetail.trim() !== (err.message || '').trim() ? `
                <div class="mt-2 pt-1.5 border-top border-danger border-opacity-25">
                  <div class="text-muted small mb-1" style="font-size: 0.72rem;"><i class="bi bi-code-slash me-1"></i>Параметры и стек ошибки:</div>
                  <pre class="m-0 p-2 bg-body-secondary text-danger border border-danger border-opacity-25 rounded font-monospace small" style="font-size: 0.72rem; line-height: 1.35; max-height: 220px; overflow-y: auto; white-space: pre-wrap; word-break: break-all;">${escapeHtml(errDetail)}</pre>
                </div>
              ` : ''}
            </div>
          `;
        }
        chatHistory.scrollTop = chatHistory.scrollHeight;
        if (statusInd) statusInd.innerText = `Ошибка: ${err.message || 'Сбой'}`;
      } finally {
        isChatSubmitting = false;
        chatInput.disabled = false;
        if (sendBtn) sendBtn.disabled = false;
        chatInput.focus();
      }
    }

    chatForm.onsubmit = (e) => {
      e.preventDefault();
      submitChat();
    };
```

```
  }

  // Setup UI event handlers
  function setupEvents() {
    if (isEventsSetup) return;
    isEventsSetup = true;

    // Quick Check Button
    const btnQuick = document.getElementById('btn-run-quick-ch
eck');
    if (btnQuick) {
      btnQuick.onclick = () => runScenario('quick_check');
    }

    // Run All Button
    const btnAll = document.getElementById('btn-run-all-scenar
ios');
    if (btnAll) {
      btnAll.onclick = () => runScenario('all');
    }

    // Refresh List Button
    const btnRefresh = document.getElementById('btn-scenario-r
efresh-list');
    if (btnRefresh) {
      btnRefresh.onclick = () => loadScenarios();
    }

    // Close Results Button
    const btnCloseRes = document.getElementById('btn-close-res
ult-panel');
    if (btnCloseRes) {
      btnCloseRes.onclick = () => {
        const panel = document.getElementById('scenario-result
-panel');
        if (panel) panel.style.display = 'none';
      };
    }
```

```
1339
1340    // Category Filter Buttons
1341    const filterContainer = document.getElementById('scenario-
category-filters');
1342    if (filterContainer) {
1343      filterContainer.querySelectorAll('button').forEach(btn =
> {
1344        btn.onclick = () => {
1345          filterContainer.querySelectorAll('button').forEach(b
 => b.classList.remove('active'));
1346          btn.classList.add('active');
1347          currentFilter = btn.getAttribute('data-filter') || '
all';
1348          renderScenariosGrid();
1349        };
1350      });
1351    }
1352
1353    setupMiniChat();
1354  }
1355
1356  // Public initialization entrypoint
1357  window.initScenariosTab = async function () {
1358    console.log('[ScenariosTab] Initializing Scenarios & Test
Computer Tab...');
1359    setupEvents();
1360    loadQuestionsFromConfig();
1361    await loadScenarios();
1362    console.log('[ScenariosTab] Scenarios tab ready.');
1363  };
1364  window.initChatTab = window.initScenariosTab;
1365
1366  // Auto-init if container exists
1367  if (document.getElementById('scenarios-cards-container') ||
document.getElementById('scenario-chat-history')) {
1368    window.initScenariosTab();
1369  }
1370 })();
```

1371

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/search_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/search_tab/index.html"
class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/search_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Web Search MCP Providers Management Tab -->
<div class="container-fluid py-3">
<!-- Header -->
<div class="alert alert-info d-flex justify-content-between al
ign-items-center flex-wrap gap-2 mb-3 shadow-sm">
<div>
<h5 class="mb-1"><i class="bi bi-globe2 me-2"></i> Провайдер
ы веб-поиска (MCP)</h5>
<p class="mb-0 small text-secondary">Управление серверами инте
рнет-поиска, алгоритмами извлечения и тестирование выдачи.</p>
```

```
</div>
<div class="input-group input-group-sm" style="max-width: 440px;">
<span class="input-group-text fw-bold">Активный движок поиска:</span>
<select class="form-select" id="search-tab-engine-selector">
<option value="playwright">🎭 Playwright MCP</option>
<option value="langchain">🦜 LangChain MCP</option>
<option value="gemini">✨ Google Gemini (Grounding)</option>
<option value="gemini_cli">✨ Google Gemini CLI</option>
<option value="agy">🚀 Antigravity (AGY)</option>
</select>
<button class="btn btn-primary" id="btn-save-search-tab-engine">
<i class="bi bi-check-circle me-1"></i> Сохранить
      </button>
</div>
</div>
<!-- Reference Info Block (Collapse, matching RAG tab style) -->
<div class="card mb-4 border-0 shadow-sm bg-light" data-panel-id="panel-search-spravka-arkhitektura-i-printsipy" id="panel-search-spravka-arkhitektura-i-printsipy">
<div class="card-body p-3">
<div class="d-flex align-items-center justify-content-between">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-info-circle-fill text-primary fs-5"></i>
<span class="fw-bold">Справка: архитектура и принципы работы 5 серверов веб-поиска</span>
</div>
<button aria-controls="searchArchCollapse" aria-expanded="true" class="btn btn-sm btn-link text-decoration-none p-0 text-secondary" data-bs-target="#searchArchCollapse" data-bs-toggle="collapse" type="button">
<i class="bi bi-chevron-down"></i> Развернуть / Скрыть
        </button>
</div>
```

```
<div class="collapse show" id="searchArchCollapse">
<div class="row g-3 mt-2 pt-2 border-top">
<!-- 1. Playwright -->
<div class="col-xl col-md-6">
<div class="p-3 border rounded bg-white h-100 shadow-sm">
<div class="fw-bold text-primary mb-1">
                 Playwright MCP
              </div>
<div class="small text-muted mb-2">
                 Прямой headless-браузер (Chromium).
              </div>
<ul class="small text-secondary mb-0 ps-3">
<li>Поиск в DuckDuckGo, Bing и Google.</li>
<li>Глубокий сбор DOM-структуры и текста сайтов.</li>
<li><strong>Плюс:</strong> не расходует токены ИИ на сбор данн
ых.</li>
<li><strong>Особенность:</strong> требует запущенного Playwrig
ht.</li>
</ul>
</div>
</div>
<!-- 2. LangChain -->
<div class="col-xl col-md-6">
<div class="p-3 border rounded bg-white h-100 shadow-sm">
<div class="fw-bold text-success mb-1">
                 LangChain MCP
              </div>
<div class="small text-muted mb-2">
                 Автономный ReAct-агент AI Windows Admin.
              </div>
<ul class="small text-secondary mb-0 ps-3">
<li>Инструменты: веб-поиск, RAG, Python-вычисления.</li>
<li>Агент сам принимает решение о вызове инструментов.</li>
<li><strong>Плюс:</strong> гибкое решение комплексных задач.</
li>
<li><strong>Особенность:</strong> многошаговый reasoning-цикл.
</li>
</ul>
```

```
</div>
</div>
<!-- 3. Gemini Grounding -->
<div class="col-xl col-md-6">
<div class="p-3 border rounded bg-white h-100 shadow-sm">
<div class="fw-bold text-info mb-1">
                 Google Gemini Grounding
              </div>
<div class="small text-muted mb-2">
                 Официальный Google Search Grounding.
              </div>
<ul class="small text-secondary mb-0 ps-3">
<li>Прямой поиск в реальном индексе Google Search.</li>
<li>Автоматическая ротация пула ключей при ошибках 429.</li>
<li><strong>Плюс:</strong> проверенная точность фактов и цитаты.</li>
<li><strong>Особенность:</strong> требует валидных API-ключей.</li>
</ul>
</div>
</div>
<!-- 4. Gemini CLI -->
<div class="col-xl col-md-6">
<div class="p-3 border rounded bg-white h-100 shadow-sm">
<div class="fw-bold text-primary mb-1">
                 Google Gemini CLI
              </div>
<div class="small text-muted mb-2">
                 Локальный терминальный агент Google.
              </div>
<ul class="small text-secondary mb-0 ps-3">
<li>Прямое выполнение через CLI подпроцесс.</li>
<li>Встроенные агентные инструменты веб-поиска.</li>
<li><strong>Плюс:</strong> автономный агент, модель по умолчанию flash-lite.</li>
<li><strong>Особенность:</strong> требует установленной утилиты gemini CLI.</li>
</ul>
```

```
</div>
</div>
<!-- 5. Antigravity AGY -->
<div class="col-xl col-md-6">
<div class="p-3 border rounded bg-white h-100 shadow-sm">
<div class="fw-bold text-warning mb-1">
                 Antigravity (AGY)
              </div>
<div class="small text-muted mb-2">
                 Мультиагентный поиск платформы Google.
              </div>
<ul class="small text-secondary mb-0 ps-3">
<li>Инструменты <code>SEARCH_WEB</code> и <code>READ_URL_CONTE
NT</code>.</li>
<li>Глубокий ресерч и синтез сложных многоходовок.</li>
<li><strong>Плюс:</strong> автономный агентный цикл.</li>
<li><strong>Особенность:</strong> использует Antigravity SDK /
 CLI.</li>
</ul>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- Detailed Provider Settings & Live Search Tester -->
<div class="row g-4 mb-4">
<!-- LEFT: Provider Detailed Settings -->
<div class="col-lg-6">
<div class="card shadow-sm border-0 h-100" data-panel-id="pane
l-search-gemini-grounding" id="panel-search-gemini-grounding">
<div class="card-header bg-dark text-white fw-bold">
<i class="bi bi-sliders me-1"></i> Параметры поисковых модулей
        </div>
<div class="card-body">
<!-- Gemini Grounding Settings -->
<div class="mb-4 p-3 border rounded bg-light">
<h6 class="fw-bold text-info mb-2"><i class="bi bi-stars me-1"
```

```
></i> Gemini Grounding</h6>
<div class="mb-2">
<label class="form-label small text-muted fw-bold">Модель для поиска Grounding (динамически из SDK):</label>
<select class="form-select form-select-sm" id="search-gemini-model">
<option value="">Загрузка доступных моделей через Google GenAI SDK...</option>
</select>
</div>
<div class="small text-muted">
              Ключи автоматически считываются из вкладки «Модели и API» и ротируются по кругу.
            </div>
</div>
<!-- Gemini CLI Settings -->
<div class="mb-4 p-3 border rounded bg-light">
<h6 class="fw-bold text-primary mb-2"><i class="bi bi-terminal-fill me-1"></i> Google Gemini CLI</h6>
<div class="mb-2">
<label class="form-label small text-muted fw-bold">Модель Gemini CLI (динамически из инструмента):</label>
<select class="form-select form-select-sm" id="search-gemini-cli-model">
<option value="">Загрузка доступных моделей через Gemini CLI...</option>
</select>
</div>
<div class="small text-muted">
              Использует локальную утилиту <code>gemini</code> с автоматическим вызовом поисковых инструментов.
            </div>
</div>
<!-- Antigravity AGY Settings -->
<div class="mb-4 p-3 border rounded bg-light">
<h6 class="fw-bold text-warning mb-2"><i class="bi bi-rocket-takeoff me-1"></i> Antigravity (AGY) Search</h6>
<div class="mb-2">
```

```
<label class="form-label small text-muted fw-bold">Модель агента Antigravity (динамически из SDK):</label>
<select class="form-select form-select-sm" id="search-agy-model">
<option value="">Загрузка доступных моделей через Antigravity SDK...</option>
</select>
</div>
<div class="small text-muted">
              Использует встроенные агентные инструменты <code>SEARCH_WEB</code> и <code>READ_URL_CONTENT</code>.
            </div>
</div>
<div class="d-grid">
<button class="btn btn-success fw-bold" id="btn-save-search-params">
<i class="bi bi-floppy me-1"></i> Сохранить настройки модулей
            </button>
</div>
</div>
</div>
</div>
<!-- RIGHT: Live Interactive Search Tester -->
<div class="col-lg-6">
<div class="card shadow-sm border-0 h-100" data-panel-id="panel-search-interaktivnyy-test-poiska" id="panel-search-interaktivnyy-test-poiska">
<div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
<span class="fw-bold"><i class="bi bi-terminal me-1"></i> Интерактивный тест поиска</span>
<span class="badge bg-primary" id="test-search-engine-badge">playwright</span>
</div>
<div class="card-body d-flex flex-column">
<div class="mb-3">
<label class="form-label small text-muted fw-bold">Тестовый поисковый запрос:</label>
```

```
<div class="input-group">
<input class="form-control form-control-sm" id="test-search-query-input" placeholder="Например: последние новости марсохода 2026" type="text"/>
<button class="btn btn-primary btn-sm px-3" id="btn-run-test-search">
<i class="bi bi-search me-1"></i> Искать
              </button>
</div>
</div>
<div class="flex-grow-1 d-flex flex-column">
<label class="form-label small text-muted fw-bold">Результат работы провайдера:</label>
<div class="border rounded p-3 bg-light flex-grow-1 font-monospace" id="test-search-output-container" style="min-height: 230px; max-height: 350px; overflow-y: auto; font-size: 0.82rem; white-space: pre-wrap;">
<span class="text-muted">Введите запрос и нажмите «Искать» для проверки выбранного провайдера.</span>
</div>
</div>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Search Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/search_tab/main.js?v=2
0261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/search_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

// Search Tab Logic - Web Search MCP Providers Management

async function initSearchTab() {
  const engineSelect = document.getElementById('search-tab-eng
ine-selector');
  const btnSaveEngine = document.getElementById('btn-save-sear
ch-tab-engine');
  const btnSaveParams = document.getElementById('btn-save-sear
ch-params');
  const geminiModelSelect = document.getElementById('search-ge
mini-model');
  const geminiCliModelSelect = document.getElementById('search
```

```
-gemini-cli-model');
  const agyModelSelect = document.getElementById('search-agy-model');
  const engineBadge = document.getElementById('test-search-engine-badge');
  const btnRunTest = document.getElementById('btn-run-test-search');
  const testQueryInput = document.getElementById('test-search-query-input');
  const testOutputContainer = document.getElementById('test-search-output-container');

  if (!engineSelect) return;

  // --- 1. Load Dynamic Models via SDK / Tools ---
  async function loadModels() {
    try {
      const modelsData = await window.api.fetch('/api/chat/models');
      const modelsGrouped = modelsData.models || {};
      const geminiList = modelsGrouped.gemini || [];
      const geminiCliList = modelsGrouped.gemini_cli || ['gemini-3.1-flash-lite', 'gemini-3.1-flash', 'gemini-2.5-pro'];
      const agyList = modelsGrouped.agy || [];

      if (geminiModelSelect && geminiList.length > 0) {
        const curVal = geminiModelSelect.value;
        geminiModelSelect.innerHTML = '';
        geminiList.forEach(m => {
          const opt = document.createElement('option');
          opt.value = m;
          opt.textContent = m;
          geminiModelSelect.appendChild(opt);
        });
        if (curVal && geminiList.includes(curVal)) {
          geminiModelSelect.value = curVal;
        }
      }
```

```
59
60        if (geminiCliModelSelect && geminiCliList.length > 0) {
61          const curVal = geminiCliModelSelect.value;
62          geminiCliModelSelect.innerHTML = '';
63          geminiCliList.forEach(m => {
64            const opt = document.createElement('option');
65            opt.value = m;
66            opt.textContent = m;
67            geminiCliModelSelect.appendChild(opt);
68          });
69          if (curVal && geminiCliList.includes(curVal)) {
70            geminiCliModelSelect.value = curVal;
71          } else if (geminiCliList.includes('gemini-3.1-flash-li
te')) {
72            geminiCliModelSelect.value = 'gemini-3.1-flash-lite'
;
73          }
74        }
75
76        if (agyModelSelect && agyList.length > 0) {
77          const curVal = agyModelSelect.value;
78          agyModelSelect.innerHTML = '';
79          agyList.forEach(m => {
80            const opt = document.createElement('option');
81            opt.value = m;
82            opt.textContent = m;
83            agyModelSelect.appendChild(opt);
84          });
85          if (curVal && agyList.includes(curVal)) {
86            agyModelSelect.value = curVal;
87          }
88        }
89      } catch (err) {
90        console.error('[SearchTab] Ошибка загрузки динамических
моделей SDK/CLI:', err);
91      }
92    }
93
```

```
  // --- 2. Load Current Config ---
  async function loadConfig() {
    try {
      await loadModels();
      const data = await window.api.fetch('/api/admin/web-search/config');
      if (data) {
        if (data.engine && engineSelect) {
          engineSelect.value = data.engine;
          if (engineBadge) engineBadge.textContent = data.engine;
        }
        if (data.gemini_model && geminiModelSelect) {
          geminiModelSelect.value = data.gemini_model;
        }
        if (data.gemini_cli_model && geminiCliModelSelect) {
          geminiCliModelSelect.value = data.gemini_cli_model;
        }
        if (data.agy_model && agyModelSelect) {
          agyModelSelect.value = data.agy_model;
        }
      }
    } catch (e) {
      console.error('[SearchTab] Ошибка загрузки конфигурации поиска:', e);
      notifySearch('Ошибка загрузки настроек поиска: ' + e.message, 'danger');
    }
  }

  // --- 3. Save Config Helper ---
  async function saveConfig() {
    const engine = engineSelect ? engineSelect.value : 'playwright';
    const gemini_model = geminiModelSelect ? geminiModelSelect.value : 'gemini-flash-lite-latest';
    const gemini_cli_model = geminiCliModelSelect ? geminiCliModelSelect.value : 'gemini-3.1-flash-lite';
```

```
    const agy_model = agyModelSelect ? agyModelSelect.value : 'agy-flash';

    try {
      await window.api.fetch('/api/admin/web-search/config', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ engine, gemini_model, gemini_cli_model, agy_model })
      });
      if (engineBadge) engineBadge.textContent = engine;
      notifySearch(`  Настройки веб-поиска сохранены (активен: ${engine})`, 'success');

      let compoundSearch = engine;
      if (engine === 'gemini') compoundSearch = `${engine}:${gemini_model}`;
      if (engine === 'gemini_cli') compoundSearch = `${engine}:${gemini_cli_model}`;
      if (engine === 'agy') compoundSearch = `${engine}:${agy_model}`;

      if (typeof window.updateChatBadges === 'function') {
        window.updateChatBadges(undefined, compoundSearch);
      }
    } catch (e) {
      console.error('[SearchTab] Ошибка сохранения:', e);
      notifySearch('  Ошибка сохранения настроек: ' + e.message, 'danger');
    }
  }

  if (btnSaveEngine) btnSaveEngine.onclick = saveConfig;
  if (btnSaveParams) btnSaveParams.onclick = saveConfig;

  if (engineSelect) {
    engineSelect.onchange = () => {
      if (engineBadge) engineBadge.textContent = engineSelect.
```

```
value;
156     };
157   }
158
159   // --- 3. Interactive Search Tester ---
160   if (btnRunTest && testQueryInput && testOutputContainer) {
161     const runTest = async () => {
162       const query = testQueryInput.value.trim();
163       if (!query) {
164         notifySearch('Введите поисковый запрос для теста!', 'w
arning');
165         testQueryInput.focus();
166         return;
167       }
168
169       const engine = engineSelect ? engineSelect.value : 'play
wright';
170       btnRunTest.disabled = true;
171       const originalText = btnRunTest.innerHTML;
172       btnRunTest.innerHTML = '<span class="spinner-border spin
ner-border-sm me-1" role="status"></span> Поиск...';
173       testOutputContainer.innerHTML = `<span class="text-info"
>  Выполнение поиска через <strong>${engine}</strong> по запро
су: "${query}"...</span>`;
174
175       try {
176         const response = await window.api.fetch('/api/admin/we
b-search/test', {
177           method: 'POST',
178           headers: { 'Content-Type': 'application/json' },
179           body: JSON.stringify({ query, engine })
180         });
181
182         if (response.status === 'ok') {
183           testOutputContainer.textContent = response.result ||
 'Пустой ответ.';
184         } else {
185           testOutputContainer.innerHTML = `<span class="text-d
```

```
anger">❌ Ошибка: ${response.message || 'Неизвестная ошибка'}</
span>`;
        }
      } catch (err) {
        console.error('[SearchTab] Ошибка теста:', err);
        testOutputContainer.innerHTML = `<span class="text-dan
ger">❌ Ошибка запроса: ${err.message}</span>`;
      } finally {
        btnRunTest.disabled = false;
        btnRunTest.innerHTML = originalText;
      }
    };

    btnRunTest.onclick = runTest;
    testQueryInput.onkeydown = (e) => {
      if (e.key === 'Enter') {
        e.preventDefault();
        runTest();
      }
    };
  }

  // Helper notification
  function notifySearch(msg, type = 'info') {
    if (typeof showNotification === 'function') {
      showNotification(msg, type);
    } else if (typeof showModelsNotification === 'function') {
      showModelsNotification(msg, type);
    } else {
      console.log(`[${type}] ${msg}`);
    }
  }

  await loadConfig();
}

window.initSearchTab = initSearchTab;

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/security_acl_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/security_acl_tab/index.
html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/security_acl_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Security & ACL Management Tab -->
<div class="container-fluid py-2 px-3">
<!-- Top Stat Cards -->
<div class="row g-2 mb-3">
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 sha
dow-sm p-3" data-panel-id="panel-security-acl-1" id="panel-sec
urity-acl-1">
<div class="text-muted small text-uppercase fw-semibold">Владе
лец объекта  </div>
```

```
<div class="fs-5 fw-bold mt-1 text-primary text-truncate" id="acl-owner-val">--</div>
<div class="small text-muted mt-1" id="acl-owner-sub">takeown.exe</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-security-acl-2" id="panel-security-acl-2">
<div class="text-muted small text-uppercase fw-semibold">Наследование прав □ </div>
<div class="fs-4 fw-bold mt-1 text-info" id="acl-inherit-val">Включено</div>
<div class="small text-muted mt-1" id="acl-inherit-sub">Inheritance Mode</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-security-acl-3" id="panel-security-acl-3">
<div class="text-muted small text-uppercase fw-semibold">Количество ACE □ </div>
<div class="fs-4 fw-bold mt-1 text-success" id="acl-ace-count">--</div>
<div class="small text-muted mt-1" id="acl-ace-sub">icacls entries</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-security-acl-4" id="panel-security-acl-4">
<div class="text-muted small text-uppercase fw-semibold">Целостность ACL □ </div>
<div class="fs-4 fw-bold mt-1 text-warning" id="acl-integrity-val">Normal</div>
```

```
<div class="small text-muted mt-1" id="acl-integrity-sub">ACL Health</div>
</div>
</div>
</div>
<!-- Path & Inspect Bar -->
<div class="d-flex align-items-center justify-content-between mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap-2">
<div class="d-flex align-items-center gap-2 flex-grow-1">
<input class="form-control form-control-sm bg-dark text-white border-secondary" id="acl-path-input" placeholder="Путь к файлу или папке (например: C:\Windows\System32 или C:\Data)" type="text" value="C:\Windows\System32"/>
<button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5 flex-shrink-0" id="acl-inspect-btn">
<i class="bi bi-search"></i>
<span>Инспектировать ACL</span>
</button>
<button class="btn btn-sm btn-outline-warning d-flex align-items-center gap-1.5 flex-shrink-0" id="acl-takeown-btn">
<i class="bi bi-person-check"></i>
<span>Взять владение (takeown)</span>
</button>
</div>
</div>
<!-- Main Grid -->
<div class="row g-3">
<div class="col-12 col-lg-7">
<div class="card bg-dark text-white border-secondary shadow-sm" data-panel-id="panel-security-acl-zapisi-kontrolya-dostupa-ace-li" id="panel-security-acl-zapisi-kontrolya-dostupa-ace-li">
<div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">
<span class="fw-semibold"><i class="bi bi-shield-lock text-primary me-2"></i>Записи контроля доступа (ACE List)</span>
<span class="badge bg-secondary" id="acl-badge">0 записей</spa
```

```
n>
</div>
<div class="card-body p-0 table-responsive" style="max-height:
 420px; overflow-y: auto;">
<table class="table table-dark table-hover table-striped align
-middle mb-0 small" id="acl-table">
<thead>
<tr class="text-secondary">
<th>Пользователь / Группа (Principal)</th>
<th>Права доступа</th>
<th>Тип</th>
<th>Наследование</th>
</tr>
</thead>
<tbody id="acl-tbody">
<tr><td class="text-center text-muted py-4" colspan="4">Введит
е путь и нажмите «Инспектировать ACL»</td></tr>
</tbody>
</table>
</div>
</div>
</div>
<div class="col-12 col-lg-5">
<div class="card bg-dark text-white border-secondary shadow-sm
 mb-3" data-panel-id="panel-security-acl-izmenenie-razresheniy
-icacls" id="panel-security-acl-izmenenie-razresheniy-icacls">
<div class="card-header border-secondary d-flex align-items-ce
nter justify-content-between py-2 px-3">
<span class="fw-semibold"><i class="bi bi-person-gear text-war
ning me-2"></i>Изменение разрешений (icacls)</span>
</div>
<div class="card-body p-3">
<div class="mb-2">
<label class="form-label small text-muted">Пользователь / Субъ
ект:</label>
<input class="form-control form-control-sm bg-dark text-white
border-secondary" id="acl-user-input" placeholder="Administrat
ors или Users" type="text" value="Administrators"/>
```

```
</div>
<div class="mb-3">
<label class="form-label small text-muted">Уровень разрешений:
</label>
<select class="form-select form-select-sm bg-dark text-white b
order-secondary" id="acl-perm-select">
<option value="F">Полный доступ (Full Control - :F)</option>
<option value="M">Изменение (Modify - :M)</option>
<option value="RX">Чтение и выполнение (Read/Execute - :RX)</o
ption>
<option value="R">Только чтение (Read - :R)</option>
<option value="W">Только запись (Write - :W)</option>
</select>
</div>
<div class="d-flex align-items-center justify-content-between"
>
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="acl-dry-run-swi
tch" type="checkbox"/>
<label class="form-check-label small text-muted" for="acl-dry-
run-switch">Безопасный режим (Dry Run)</label>
</div>
<button class="btn btn-sm btn-success px-3" id="acl-grant-btn"
>
<i class="bi bi-check-lg"></i> Применить права
            </button>
</div>
</div>
</div>
<!-- Action Log Output -->
<div class="card bg-dark text-white border-secondary shadow-sm
" data-panel-id="panel-security-acl-konsolnyy-vyvod-icacls" id
="panel-security-acl-konsolnyy-vyvod-icacls">
<div class="card-header border-secondary py-2 px-3 d-flex alig
n-items-center justify-content-between">
<span class="small fw-semibold text-muted"><i class="bi bi-ter
minal me-1"></i>Консольный вывод icacls</span>
<button class="btn btn-link btn-sm text-decoration-none text-m
```

```
uted p-0" id="acl-clear-log-btn">Очистить</button>
</div>
<div class="card-body p-2">
<pre class="mb-0 small bg-black text-light p-2 rounded" id="acl-console-output" style="max-height: 180px; overflow-y: auto; font-family: Consolas, monospace; font-size: 0.78rem;">Готов к инспекции прав доступа.</pre>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Security Acl Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/security_acl_tab/main.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { initSecurityAclTab } from '/windows/api/webgui/security_acl_tab/main.js';
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/security_acl_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * security_acl_tab/main.js — Управление правами доступа и дескрипторами безопасности (icacls, takeown)
 * Updated: 2026-10-10 17:35:00
 */

const registerTabPoller = window.registerTabPoller || function () {};
```

```
30
31 let isInitialized = false;
32
33 export async function inspectPathAcl() {
34   const pathInput = document.getElementById('acl-path-input');
35   const path = pathInput ? pathInput.value.trim() : 'C:\\Windo
ws\\System32';
36
37   try {
38     const res = await fetch(`/api/security-acl/summary?path=${
encodeURIComponent(path)}`);
39     if (!res.ok) throw new Error(`HTTP ${res.status}`);
40     const data = await res.json();
41     renderAclSummary(data);
42   } catch (err) {
43     console.warn('[SecurityACL] Ошибка получения ACL:', err);
44   }
45 }
46
47 function renderAclSummary(data) {
48   const ownerEl = document.getElementById('acl-owner-val');
49   const inheritEl = document.getElementById('acl-inherit-val')
;
50   const aceCountEl = document.getElementById('acl-ace-count');
51   const tbody = document.getElementById('acl-tbody');
52   const badge = document.getElementById('acl-badge');
53
54   if (ownerEl) ownerEl.textContent = data.owner || 'NT AUTHORI
TY\\SYSTEM';
55   if (inheritEl) inheritEl.textContent = data.inheritance_enab
led ? 'Включено' : 'Отключено';
56
57   const aces = data.aces || [
58     { principal: 'NT AUTHORITY\\SYSTEM', permissions: 'Full Co
ntrol', type: 'Allow', inheritance: 'Inherited' },
59     { principal: 'BUILTIN\\Administrators', permissions: 'Full
  Control', type: 'Allow', inheritance: 'Inherited' },
60     { principal: 'BUILTIN\\Users', permissions: 'Read & Execut
```

```
e', type: 'Allow', inheritance: 'Inherited' }
  ];

  if (aceCountEl) aceCountEl.textContent = aces.length;
  if (badge) badge.textContent = `${aces.length} записей`;

  if (!tbody) return;

  tbody.innerHTML = aces.map(a => `
    <tr>
      <td class="fw-semibold text-light">${a.principal || '-'}
</td>
      <td class="text-info font-monospace small">${a.permissio
ns || 'Read'}</td>
      <td><span class="badge ${a.type === 'Allow' ? 'bg-succes
s' : 'bg-danger'}">${a.type || 'Allow'}</span></td>
      <td class="small text-muted">${a.inheritance || 'Direct'
}</td>
    </tr>
  `).join('');
}

async function executeAclAction(customAction = null) {
  const pathInput = document.getElementById('acl-path-input');
  const userInput = document.getElementById('acl-user-input');
  const permSelect = document.getElementById('acl-perm-select'
);
  const dryRunSwitch = document.getElementById('acl-dry-run-sw
itch');
  const output = document.getElementById('acl-console-output')
;

  const path = pathInput ? pathInput.value.trim() : 'C:\\Windo
ws\\System32';
  const user = userInput ? userInput.value.trim() : 'Administr
ators';
  const perm = permSelect ? permSelect.value : 'F';
  const action = customAction || 'icacls_grant';
```

```
  const dryRun = dryRunSwitch ? dryRunSwitch.checked : true;

  if (output) output.textContent = `[${new Date().toLocaleTime
String()}] Выполнение ${action} для "${path}"...\n`;

  try {
    const res = await fetch('/api/security-acl/action', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ path: path, user: user, permissio
n: perm, action: action, dry_run: dryRun })
    });
    const result = await res.json();
    if (output) output.textContent += JSON.stringify(result, n
ull, 2);
    if (window.toast) {
      if (res.ok) window.toast.success('Права обновлены', `Ста
тус: ${result.status || 'OK'}`);
      else window.toast.error('Ошибка', result.detail || 'Сбой
 выполнения icacls');
    }
  } catch (err) {
    if (output) output.textContent += `\nОшибка запроса: ${err
.message}`;
    if (window.toast) window.toast.error('Ошибка сети', err.me
ssage);
  }
}

export function initSecurityAclTab() {
  if (isInitialized) return;
  isInitialized = true;

  const inspectBtn = document.getElementById('acl-inspect-btn'
);
  if (inspectBtn) inspectBtn.addEventListener('click', inspect
PathAcl);

```

```
  const takeownBtn = document.getElementById('acl-takeown-btn');
  if (takeownBtn) takeownBtn.addEventListener('click', () => executeAclAction('takeown'));

  const grantBtn = document.getElementById('acl-grant-btn');
  if (grantBtn) grantBtn.addEventListener('click', () => executeAclAction('icacls_grant'));

  const clearBtn = document.getElementById('acl-clear-log-btn');
  if (clearBtn) clearBtn.addEventListener('click', () => {
    const out = document.getElementById('acl-console-output');
    if (out) out.textContent = 'Консоль очищена.';
  });

  registerTabPoller('tab-security-acl', inspectPathAcl, 20000, { immediate: true });
}

window.initSecurityAclTab = initSecurityAclTab;

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Services Manager View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса диспетчера служб Windows.

Usage Examples:
    Web Route:
      /windows/api/webgui/services_manager_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/services_manager_tab/in
dex.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/services_manager_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<style>
  .svc-manager-wrapper {
    color: var(--text-color);
  }
  .svc-card {
    background: var(--surface-1);
    border: 1px solid var(--border-color);
    color: var(--text-color);
  }
  .svc-card-header {
```

```
    background: var(--surface-2);
    border-bottom: 1px solid var(--border-color);
    color: var(--text-color);
  }
  #svc-table tbody tr {
    border-bottom: 1px solid var(--border-color);
  }
  #svc-table tbody tr:hover {
    background-color: var(--surface-2, rgba(56, 189, 248, 0.06
)) !important;
  }
  #svc-table td {
    background: transparent;
    color: var(--text-color);
  }
</style>
<!-- Services Manager Tab -->
<div class="container-fluid py-2 px-3 svc-manager-wrapper">
<!-- Top Stat Cards -->
<div class="row g-2 mb-3">
<div class="col-6 col-md-3">
<div class="card svc-card h-100 shadow-sm p-3" data-panel-id="
panel-services-manager-1" id="panel-services-manager-1">
<div class="text-muted small text-uppercase fw-semibold">Всего
 служб  </div>
<div class="fs-4 fw-bold mt-1 text-primary" id="svc-total-coun
t">--</div>
<div class="small text-muted mt-1" id="svc-total-sub">sc query
 state= all</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card svc-card h-100 shadow-sm p-3" data-panel-id="
panel-services-manager-2" id="panel-services-manager-2">
<div class="text-muted small text-uppercase fw-semibold">Работ
ают (Running)  </div>
<div class="fs-4 fw-bold mt-1 text-success" id="svc-running-co
unt">--</div>
```

```
<div class="small text-muted mt-1" id="svc-running-sub">Активные процессы</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card svc-card h-100 shadow-sm p-3" data-panel-id="panel-services-manager-3" id="panel-services-manager-3">
<div class="text-muted small text-uppercase fw-semibold">Остановлены (Stopped)  </div>
<div class="fs-4 fw-bold mt-1 text-secondary" id="svc-stopped-count">--</div>
<div class="small text-muted mt-1" id="svc-stopped-sub">В режиме ожидания</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card svc-card h-100 shadow-sm p-3" data-panel-id="panel-services-manager-4" id="panel-services-manager-4">
<div class="text-muted small text-uppercase fw-semibold">С автозапуском (Auto)  </div>
<div class="fs-4 fw-bold mt-1 text-info" id="svc-auto-count">--</div>
<div class="small text-muted mt-1" id="svc-auto-sub">Системный старт</div>
</div>
</div>
</div>
<!-- Action Bar -->
<div class="d-flex align-items-center justify-content-between mb-3 p-2 rounded flex-wrap gap-2 svc-card">
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5" id="svc-refresh-btn">
<i class="bi bi-arrow-clockwise"></i>
<span>Обновить список</span>
</button>
<div class="btn-group btn-group-sm">
<button class="btn btn-sm btn-outline-secondary active" id="sv
```

```
c-filter-all">Все</button>
<button class="btn btn-sm btn-outline-secondary" id="svc-filte
r-running">Работающие</button>
<button class="btn btn-sm btn-outline-secondary" id="svc-filte
r-stopped">Остановленные</button>
</div>
</div>
<div class="d-flex align-items-center gap-2">
<input class="form-control form-control-sm" id="svc-search-inp
ut" placeholder="Поиск службы (имя или отображение)..." style=
"width: 260px; background: var(--surface-2); color: var(--text
-color); border-color: var(--border-color);" type="text"/>
</div>
</div>
<!-- Services Table -->
<div class="card svc-card shadow-sm" data-panel-id="panel-serv
ices-manager-sluzhby-windows-service-control" id="panel-servic
es-manager-sluzhby-windows-service-control">
<div class="card-header svc-card-header d-flex align-items-cen
ter justify-content-between py-2 px-3">
<span class="fw-semibold"><i class="bi bi-gear-wide-connected
text-primary me-2"></i>Службы Windows (Service Control Manager
)</span>
<span class="badge bg-secondary-subtle text-secondary border b
order-secondary" id="svc-table-badge">0 служб</span>
</div>
<div class="card-body p-0 table-responsive" style="max-height:
 560px; overflow-y: auto;">
<table class="table table-hover align-middle mb-0 small" id="s
vc-table">
<thead class="sticky-top" style="background: var(--surface-2);
 border-bottom: 2px solid var(--border-color);">
<tr>
<th style="background: var(--surface-2); color: var(--text-col
or);">Имя службы</th>
<th style="background: var(--surface-2); color: var(--text-col
or);">Отображаемое имя</th>
<th style="background: var(--surface-2); color: var(--text-col
```

```
or);">Состояние</th>
<th style="background: var(--surface-2); color: var(--text-col
or);">Тип запуска</th>
<th style="background: var(--surface-2); color: var(--text-col
or);">PID</th>
<th class="text-end" style="background: var(--surface-2); colo
r: var(--text-color);">Управление</th>
</tr>
</thead>
<tbody id="svc-tbody">
<tr><td class="text-center text-muted py-4" colspan="6">Опрос
диспетчера служб Windows...</td></tr>
</tbody>
</table>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Services Manager Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом диспетчера служб
 Windows.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/services_manager_tab/m
ain.js?v=20261006_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { initServicesManagerTab } from '/windows/api/we
bgui/services_manager_tab/main.js';
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/services_manager_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

const registerTabPoller = window.registerTabPoller || function
() {};

let isInitialized = false;
let servicesList = [];
let currentFilter = 'all';

```

```
function escapeHtml(str) {
  if (str === null || str === undefined) return '';
  return String(str)
    .replace(/&/g, '&amp;')
    .replace(/</g, '&lt;')
    .replace(/>/g, '&gt;')
    .replace(/"/g, '&quot;')
    .replace(/'/g, '&#039;');
}

export async function fetchServicesSummary() {
  const refreshBtn = document.getElementById('svc-refresh-btn'
);
  if (refreshBtn) {
    const icon = refreshBtn.querySelector('i');
    if (icon) icon.classList.add('spin-animation');
  }

  try {
    const res = await fetch('/api/services-manager/summary');
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    renderServices(data);
  } catch (err) {
    console.warn('[ServicesManager] Ошибка получения списка сл
ужб:', err);
  } finally {
    if (refreshBtn) {
      const icon = refreshBtn.querySelector('i');
      if (icon) icon.classList.remove('spin-animation');
    }
  }
}

function renderServices(data) {
  servicesList = data.services || [];

  const totalEl = document.getElementById('svc-total-count');
```

```
  const runningEl = document.getElementById('svc-running-count');
  const stoppedEl = document.getElementById('svc-stopped-count');
  const autoEl = document.getElementById('svc-auto-count');

  const runningCount = servicesList.filter(s => (s.state || '').toLowerCase().includes('running') || s.state === '4').length;
  const stoppedCount = servicesList.filter(s => (s.state || '').toLowerCase().includes('stopped') || s.state === '1').length;
  const autoCount = servicesList.filter(s => (s.start_type || '').toLowerCase().includes('auto')).length;

  if (totalEl) totalEl.textContent = servicesList.length;
  if (runningEl) runningEl.textContent = runningCount;
  if (stoppedEl) stoppedEl.textContent = stoppedCount;
  if (autoEl) autoEl.textContent = autoCount;

  applyServiceFilters();
}

function applyServiceFilters() {
  const tbody = document.getElementById('svc-tbody');
  const searchInput = document.getElementById('svc-search-input');
  const badge = document.getElementById('svc-table-badge');

  if (!tbody) return;

  const query = (searchInput ? searchInput.value : '').toLowerCase().trim();

  const filtered = servicesList.filter(s => {
    const isRunning = (s.state || '').toLowerCase().includes('running') || s.state === '4';
    if (currentFilter === 'running' && !isRunning) return fals
```

```
e;
94      if (currentFilter === 'stopped' && isRunning) return false
;
95
96      if (query) {
97        const matchName = (s.name || '').toLowerCase().includes(
query);
98        const matchDisp = (s.display_name || '').toLowerCase().i
ncludes(query);
99        return matchName || matchDisp;
100     }
101     return true;
102   });
103
104   if (badge) badge.textContent = `${filtered.length} из ${serv
icesList.length} служб`;
105
106   if (filtered.length === 0) {
107     tbody.innerHTML = '<tr><td colspan="6" class="text-center
text-muted py-4">Службы не найдены</td></tr>';
108     return;
109   }
110
111   tbody.innerHTML = filtered.slice(0, 150).map(s => {
112     const isRunning = (s.state || '').toLowerCase().includes('
running') || s.state === '4';
113     const badgeClass = isRunning
114       ? 'badge bg-success-subtle text-success border border-su
ccess'
115       : 'badge bg-secondary-subtle text-secondary border borde
r-secondary';
116     const statusText = isRunning ? 'Running' : 'Stopped';
117
118     return `
119       <tr>
120         <td class="fw-bold font-monospace" style="color: var(-
-accent-color, #0ea5e9);">${escapeHtml(s.name)}</td>
121         <td style="color: var(--text-color);">${escapeHtml(s.d
```

```
isplay_name || s.name)}</td>
        <td>
          <span class="${badgeClass}">
            ${statusText}
          </span>
        </td>
        <td class="small text-muted">${escapeHtml(s.start_type
 || 'Manual')}</td>
        <td class="font-monospace small text-muted">${escapeHt
ml(String(s.pid || '-'))}</td>
        <td class="text-end">
          <button class="btn btn-xs ${isRunning ? 'btn-outline
-danger' : 'btn-outline-success'} py-0 px-2 svc-action-btn"
                  data-service="${escapeHtml(s.name)}" data-ac
tion="${isRunning ? 'sc_stop' : 'sc_start'}">
            <i class="bi ${isRunning ? 'bi-stop-fill' : 'bi-pl
ay-fill'}"></i> ${isRunning ? 'Стоп' : 'Старт'}
          </button>
        </td>
      </tr>
    `;
  }).join('');

  tbody.querySelectorAll('.svc-action-btn').forEach(btn => {
    btn.addEventListener('click', () => {
      const sName = btn.dataset.service;
      const sAct = btn.dataset.action;
      executeServiceCmd(sName, sAct);
    });
  });
}

async function executeServiceCmd(serviceName, action) {
  if (window.showToast) window.showToast(`Команда отправлена:
${action} ${serviceName}`, 'info');
  else if (window.toast) window.toast.info('Команда отправлена
', `${action}: ${serviceName}`);

```

```
  try {
    const res = await fetch('/api/services-manager/action', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ service_name: serviceName, action
: action, dry_run: false })
    });
    const result = await res.json();
    if (res.ok) {
      if (window.showToast) window.showToast(`  ${serviceName}
: команда выполнена`, 'success');
      else if (window.toast) window.toast.success('Успех', `${
serviceName}: команда выполнена`);
      fetchServicesSummary();
    } else {
      const errDetail = result.detail || result.message || 'Сб
ой операции';
      if (window.showToast) window.showToast(`Ошибка: ${errDet
ail}`, 'danger');
      else if (window.toast) window.toast.error('Ошибка', errD
etail);
    }
  } catch (err) {
    const errMsg = `Ошибка сети: ${err.message}`;
    if (window.showToast) window.showToast(errMsg, 'danger');
    else if (window.toast) window.toast.error('Ошибка сети', e
rr.message);
  }
}

export function initServicesManagerTab() {
  if (isInitialized) return;
  isInitialized = true;

  const refreshBtn = document.getElementById('svc-refresh-btn'
);
  if (refreshBtn) refreshBtn.addEventListener('click', fetchSe
rvicesSummary);
```

```
181
182   const searchInput = document.getElementById('svc-search-inpu
    t');
183   if (searchInput) searchInput.addEventListener('input', apply
    ServiceFilters);
184
185   const filterAll = document.getElementById('svc-filter-all');
186   const filterRunning = document.getElementById('svc-filter-ru
    nning');
187   const filterStopped = document.getElementById('svc-filter-st
    opped');
188
189   [filterAll, filterRunning, filterStopped].forEach(btn => {
190     if (!btn) return;
191     btn.addEventListener('click', () => {
192       [filterAll, filterRunning, filterStopped].forEach(b => {
193         if (!b) return;
194         b.classList.remove('active', 'btn-primary');
195         b.classList.add('btn-outline-secondary');
196       });
197       btn.classList.add('active', 'btn-primary');
198       btn.classList.remove('btn-outline-secondary');
199       currentFilter = btn.id.replace('svc-filter-', '');
200       applyServiceFilters();
201     });
202   });
203
204   registerTabPoller('tab-services-manager', fetchServicesSumma
    ry, 10000, { immediate: true });
205 }
206
207 window.initServicesManagerTab = initServicesManagerTab;
208
209 // Автозапуск если вкладка уже в DOM
210 if (document.getElementById('svc-table') || document.getElemen
    tById('tab-services-manager')) {
211   if (document.readyState === 'loading') {
212     document.addEventListener('DOMContentLoaded', () => initSe
```

```
rvicesManagerTab());
  } else {
    setTimeout(() => initServicesManagerTab(), 10);
  }
}


```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/servicing_integrity_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/servicing_integrity_tab
/index.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/servicing_integrity_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Servicing & Component Integrity Tab -->
<div class="container-fluid py-2 px-3">
<!-- Top Stat Cards -->
<div class="row g-2 mb-3">
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 sha
dow-sm p-3" data-panel-id="panel-servicing-integrity-1" id="pa
nel-servicing-integrity-1">
<div class="text-muted small text-uppercase fw-semibold">Храни
лище компонентов WinSxS  </div>
```

```
<div class="fs-4 fw-bold mt-1 text-info" id="si-winsxs-size">--</div>
<div class="small text-muted mt-1" id="si-winsxs-sub">DISM AnalyzeComponentStore</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-servicing-integrity-2" id="panel-servicing-integrity-2">
<div class="text-muted small text-uppercase fw-semibold">Рекомендуется очистка  </div>
<div class="fs-4 fw-bold mt-1 text-warning" id="si-cleanup-recommended">--</div>
<div class="small text-muted mt-1" id="si-cleanup-sub">StartComponentCleanup</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-servicing-integrity-3" id="panel-servicing-integrity-3">
<div class="text-muted small text-uppercase fw-semibold">Целостность файлов SFC  </div>
<div class="fs-4 fw-bold mt-1 text-success" id="si-sfc-status">OK</div>
<div class="small text-muted mt-1" id="si-sfc-sub">sfc /scannow</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-servicing-integrity-4" id="panel-servicing-integrity-4">
<div class="text-muted small text-uppercase fw-semibold">Состояние образа DISM  </div>
<div class="fs-4 fw-bold mt-1 text-primary" id="si-dism-status">Healthy</div>
```

```
<div class="small text-muted mt-1" id="si-dism-sub">CheckHealth / ScanHealth</div>
</div>
</div>
</div>
<!-- Action Bar -->
<div class="d-flex align-items-center justify-content-between mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5" id="si-refresh-btn">
<i class="bi bi-arrow-clockwise"></i>
<span>Обновить состояние</span>
</button>
<button class="btn btn-sm btn-outline-info d-flex align-items-center gap-1.5" id="si-dism-check-btn">
<i class="bi bi-heart-pulse"></i>
<span>DISM CheckHealth</span>
</button>
<button class="btn btn-sm btn-outline-warning d-flex align-items-center gap-1.5" id="si-sfc-scan-btn">
<i class="bi bi-shield-check"></i>
<span>SFC Scannow</span>
</button>
<button class="btn btn-sm btn-outline-danger d-flex align-items-center gap-1.5" id="si-winsxs-clean-btn">
<i class="bi bi-trash"></i>
<span>Очистить WinSxS</span>
</button>
</div>
</div>
<!-- Main Grid -->
<div class="row g-3">
<div class="col-12 col-lg-6">
<div class="card bg-dark text-white border-secondary shadow-sm mb-3" data-panel-id="panel-servicing-integrity-parametry-obsluzhivaniya-obraza" id="panel-servicing-integrity-parametry-obs
```

```
luzhivaniya-obraza">
<div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">
<span class="fw-semibold"><i class="bi bi-tools text-primary me-2"></i>Параметры обслуживания образа</span>
</div>
<div class="card-body p-3">
<div class="mb-3">
<label class="form-label small text-muted">Выберите операцию обслуживания:</label>
<select class="form-select form-select-sm bg-dark text-white border-secondary" id="si-action-select">
<option value="dism_check_health">DISM: Быстрая проверка повреждений (/CheckHealth)</option>
<option value="dism_scan_health">DISM: Полное сканирование целостности (/ScanHealth)</option>
<option value="dism_restore_health">DISM: Восстановление поврежденного образа (/RestoreHealth)</option>
<option value="dism_analyze_component_store">DISM: Анализ размера хранилища WinSxS (/AnalyzeComponentStore)</option>
<option value="dism_start_component_cleanup">DISM: Очистка старых версий компонентов (/StartComponentCleanup)</option>
<option value="dism_reset_base">DISM: Фиксация базы обновлений (/ResetBase)</option>
<option value="sfc_scannow">SFC: Проверка и восстановление системных файлов (sfc /scannow)</option>
<option value="sfc_verifyonly">SFC: Только проверка без восстановления (sfc /verifyonly)</option>
</select>
</div>
<div class="d-flex align-items-center justify-content-between">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="si-dry-run-switch" type="checkbox"/>
<label class="form-check-label small text-muted" for="si-dry-run-switch">Безопасный режим (Dry Run)</label>
</div>
```

```
<button class="btn btn-sm btn-success px-3" id="si-execute-btn
">
<i class="bi bi-play-fill"></i> Запустить операцию
            </button>
</div>
</div>
</div>
<!-- Quick Info Card -->
<div class="card bg-dark text-white border-secondary shadow-sm
" data-panel-id="panel-servicing-integrity-spravochnaya-inform
atsiya" id="panel-servicing-integrity-spravochnaya-informatsiy
a">
<div class="card-header border-secondary py-2 px-3">
<span class="small fw-semibold text-info"><i class="bi bi-info
-circle me-1"></i>Справочная информация</span>
</div>
<div class="card-body p-3 small text-muted">
<p class="mb-2"><strong class="text-light">DISM</strong> (Depl
oyment Image Servicing and Management) обслуживает компоненты
Windows и исправит ошибки хранилища манифестов.</p>
<p class="mb-0"><strong class="text-light">SFC</strong> (Syste
m File Checker) сверяет контрольные суммы системных dll и exe
файлов с резервными копиями в WinSxS.</p>
</div>
</div>
</div>
<div class="col-12 col-lg-6">
<div class="card bg-dark text-white border-secondary shadow-sm
 h-100" data-panel-id="panel-servicing-integrity-zhurnal-vypol
neniya-obsluzhivaniya" id="panel-servicing-integrity-zhurnal-v
ypolneniya-obsluzhivaniya">
<div class="card-header border-secondary py-2 px-3 d-flex alig
n-items-center justify-content-between">
<span class="small fw-semibold text-muted"><i class="bi bi-ter
minal me-1"></i>Журнал выполнения обслуживания</span>
<button class="btn btn-link btn-sm text-decoration-none text-m
uted p-0" id="si-clear-log-btn">Очистить</button>
</div>
```

```
<div class="card-body p-2 d-flex flex-column">
<pre class="mb-0 small bg-black text-light p-2 rounded flex-grow-1" id="si-console-output" style="min-height: 280px; max-height: 480px; overflow-y: auto; font-family: Consolas, monospace; font-size: 0.78rem;">Готов к диагностике компонентов.</pre>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Servicing Integrity Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/servicing_integrity_ta
b/main.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { initServicingIntegrityTab } from '/windows/api
/webgui/servicing_integrity_tab/main.js';
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/servicing_integrity_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * servicing_integrity_tab/main.js — Обслуживание образов DISM
 и целостность SFC
 * Updated: 2026-10-10 17:35:00
 */

const registerTabPoller = window.registerTabPoller || function
() {};
```

```
30
31 let isInitialized = false;
32
33 export async function fetchServicingSummary() {
34   try {
35     const res = await fetch('/api/servicing-integrity/summary'
   );
36     if (!res.ok) throw new Error(`HTTP ${res.status}`);
37     const data = await res.json();
38     renderServicingSummary(data);
39   } catch (err) {
40     console.warn('[ServicingIntegrity] Ошибка получения сводки
   :', err);
41   }
42 }
43
44 function renderServicingSummary(data) {
45   const winsxsSizeEl = document.getElementById('si-winsxs-size
   ');
46   const cleanupRecommendedEl = document.getElementById('si-cle
   anup-recommended');
47   const sfcStatusEl = document.getElementById('si-sfc-status')
   ;
48   const dismStatusEl = document.getElementById('si-dism-status
   ');
49
50   if (winsxsSizeEl) winsxsSizeEl.textContent = data.winsxs_siz
   e || '8.4 GB';
51   if (cleanupRecommendedEl) cleanupRecommendedEl.textContent =
    data.cleanup_recommended ? 'Да' : 'Нет';
52   if (sfcStatusEl) sfcStatusEl.textContent = data.sfc_status |
   | 'OK';
53   if (dismStatusEl) dismStatusEl.textContent = data.dism_healt
   h || 'Healthy';
54 }
55
56 async function executeServicingAction(customAction = null) {
57   const actionSelect = document.getElementById('si-action-sele
```

```
ct');
  const dryRunSwitch = document.getElementById('si-dry-run-swi
tch');
  const output = document.getElementById('si-console-output');

  const op = customAction || (actionSelect ? actionSelect.valu
e : 'dism_check_health');
  const dryRun = dryRunSwitch ? dryRunSwitch.checked : true;

  if (output) output.textContent = `[${new Date().toLocaleTime
String()}] Запуск операции обслуживания: ${op} (DryRun: ${dryR
un})...\nПожалуйста, подождите...\n`;

  try {
    const res = await fetch('/api/servicing-integrity/action',
 {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ action: op, dry_run: dryRun })
    });
    const result = await res.json();
    if (output) output.textContent += JSON.stringify(result, n
ull, 2);
    if (window.toast) {
      if (res.ok) window.toast.success('Операция завершена', `
Статус: ${result.status || 'OK'}`);
      else window.toast.error('Ошибка', result.detail || 'Сбой
 обслуживания');
    }
  } catch (err) {
    if (output) output.textContent += `\nОшибка запроса: ${err
.message}`;
    if (window.toast) window.toast.error('Ошибка сети', err.me
ssage);
  }
}

export function initServicingIntegrityTab() {
```

```
  if (isInitialized) return;
  isInitialized = true;

  const refreshBtn = document.getElementById('si-refresh-btn')
;
  if (refreshBtn) refreshBtn.addEventListener('click', fetchSe
rvicingSummary);

  const dismCheckBtn = document.getElementById('si-dism-check-
btn');
  if (dismCheckBtn) dismCheckBtn.addEventListener('click', ()
=> executeServicingAction('dism_check_health'));

  const sfcScanBtn = document.getElementById('si-sfc-scan-btn'
);
  if (sfcScanBtn) sfcScanBtn.addEventListener('click', () => e
xecuteServicingAction('sfc_scannow'));

  const winsxsCleanBtn = document.getElementById('si-winsxs-cl
ean-btn');
  if (winsxsCleanBtn) winsxsCleanBtn.addEventListener('click',
 () => executeServicingAction('dism_start_component_cleanup'))
;

  const execBtn = document.getElementById('si-execute-btn');
  if (execBtn) execBtn.addEventListener('click', () => execute
ServicingAction());

  const clearBtn = document.getElementById('si-clear-log-btn')
;
  if (clearBtn) clearBtn.addEventListener('click', () => {
    const out = document.getElementById('si-console-output');
    if (out) out.textContent = 'Журнал очищен.';
  });

  registerTabPoller('tab-servicing-integrity', fetchServicingS
ummary, 20000, { immediate: true });
}
```

```
111
112 window.initServicingIntegrityTab = initServicingIntegrityTab;
113
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/skills_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/skills_tab/index.html"
class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/skills_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Skills Tab Component -->
<div class="container-fluid p-3" id="skills-tab-root">
<!-- Top Banner / Summary -->
<div class="alert alert-dark border-secondary bg-dark text-lig
ht d-flex flex-wrap align-items-center justify-content-between
 gap-3 shadow-sm">
<div class="d-flex align-items-center gap-3">
<div class="p-3 bg-black rounded border border-secondary text-
warning display-6 d-flex align-items-center justify-content-ce
nter" style="width: 54px; height: 54px;">
```

```
<i class="bi bi-lightning-charge-fill"></i>
</div>
<div>
<h5 class="mb-1 text-white d-flex align-items-center gap-2">
          Управление навыками (Agent Skills)
          <span class="badge bg-primary rounded-pill small" id="skills-total-badge">0</span>
</h5>
<p class="mb-0 text-muted small">
          Просмотр, создание, настройка и упаковка переиспользуемых навыков для AI-агентов (Antigravity, Gemini CLI, AI Windows Admin).
        </p>
</div>
</div>
<div class="d-flex align-items-center gap-2">
<button class="btn btn-outline-light btn-sm d-flex align-items-center gap-1" id="btn-refresh-skills" onclick="window.skillsTab?.refresh()" type="button">
<i class="bi bi-arrow-clockwise"></i> Обновить
      </button>
<button class="btn btn-primary btn-sm d-flex align-items-center gap-1 shadow-sm" data-bs-target="#createSkillModal" data-bs-toggle="modal" type="button">
<i class="bi bi-plus-circle-fill"></i> Новый навык
      </button>
</div>
</div>
<!-- Statistics Row -->
<div class="row g-2 mb-3" id="skills-stats-row">
<div class="col-6 col-md-3">
<div class="p-2 bg-dark rounded border border-secondary text-center">
<div class="h4 mb-0 text-white" id="stat-total-skills">0</div>
<div class="small text-muted">Всего навыков</div>
</div>
</div>
<div class="col-6 col-md-3">
```

```
<div class="p-2 bg-dark rounded border border-secondary text-center">
<div class="h4 mb-0 text-info" id="stat-scripts-skills">0</div>
<div class="small text-muted">С исполняемыми скриптами</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="p-2 bg-body-tertiary rounded border text-center">
<div class="h4 mb-0 text-warning" id="stat-refs-skills">0</div>
<div class="small text-muted">Со справочниками</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="p-2 bg-body-tertiary rounded border text-center">
<div class="h4 mb-0 text-success" id="stat-packaged-skills">0</div>
<div class="small text-muted">Скомпилировано (.skill)</div>
</div>
</div>
</div>
<!-- Search & View Controls -->
<div class="card bg-body-tertiary border mb-3 shadow-sm" data-panel-id="panel-skills-1" id="panel-skills-1">
<div class="card-body p-2 d-flex flex-wrap align-items-center justify-content-between gap-2">
<div class="input-group input-group-sm flex-grow-1" style="max-width: 450px;">
<span class="input-group-text"><i class="bi bi-search"></i></span>
<input class="form-control" id="skills-search-input" placeholder="Поиск по названию или описанию навыка..." type="text"/>
<button class="btn btn-outline-secondary" id="skills-search-clear" onclick="window.skillsTab?.clearSearch()" type="button">
</button>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
```

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<select class="form-select form-select-sm" id="skills-filter-select" onchange="window.skillsTab?.applyFilters()" style="width: auto;">
<option value="all">Все навыки</option>
<option value="scripts">Только со скриптами</option>
<option value="references">Только со справочниками</option>
<option value="packaged">Собранные (.skill)</option>
</select>
<div class="btn-group btn-group-sm" role="group">
<button class="btn btn-outline-secondary active" id="btn-view-cards" onclick="window.skillsTab?.setViewMode('cards')" type="button">
<i class="bi bi-grid-fill"></i> Карточки
          </button>
<button class="btn btn-outline-secondary" id="btn-view-table" onclick="window.skillsTab?.setViewMode('table')" type="button">
<i class="bi bi-table"></i> Таблица
          </button>
</div>
</div>
</div>
</div>
<!-- Loading & Empty Indicators -->
<div class="text-center py-5 text-muted d-none" id="skills-loading">
<div class="spinner-border text-primary mb-2" role="status"></div>
<div>Загрузка навыков...</div>
</div>
<div class="text-center py-5 text-muted d-none" id="skills-empty">
<i class="bi bi-folder-x display-4 d-block mb-2 text-secondary"></i>
<h5>Навыки не найдены</h5>
<p class="small">Попробуйте изменить поисковый запрос или создайте новый навык.</p>
</div>
```

```
<!-- Cards Container -->
<div class="row g-3" id="skills-cards-container"></div>
<!-- Table Container -->
<div class="card shadow-sm border d-none" data-panel-id="panel-skills-2" id="skills-table-card">
<div class="card-body p-0">
<div class="table-responsive">
<table class="table table-hover align-middle mb-0">
<thead class="table-light text-uppercase small">
<tr>
<th scope="col" style="width: 25%;">Название навыка</th>
<th scope="col" style="width: 35%;">Описание</th>
<th scope="col" style="width: 15%;">Расположение</th>
<th scope="col" style="width: 10%;">Компоненты</th>
<th scope="col" style="width: 15%; text-align: right;">Действия</th>
</tr>
</thead>
<tbody id="skills-table-body"></tbody>
</table>
</div>
</div>
</div>
</div>
<!-- Modal: Create New Skill Wizard -->
<div aria-hidden="true" aria-labelledby="createSkillModalLabel" class="modal fade" id="createSkillModal" tabindex="-1">
<div class="modal-dialog modal-xl modal-dialog-centered modal-dialog-scrollable">
<div class="modal-content bg-dark text-white border-secondary shadow-lg">
<div class="modal-header border-secondary py-2 px-3 bg-black bg-opacity-40">
<div class="d-flex align-items-center gap-2">
<span class="fs-5 text-warning">□ </span>
<div>
<h5 class="modal-title mb-0 fs-6 fw-bold" id="createSkillModalLabel">Мастер создания навыков (Skill Wizard)</h5>
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<small class="text-muted">Создавайте переиспользуемые навыки для AI Windows Admin, Gemini CLI и Antigravity</small>
</div>
</div>
<button aria-label="Close" class="btn-close btn-close-white" data-bs-dismiss="modal" type="button"></button>
</div>
<!-- Mode Selector Pills -->
<div class="px-3 pt-3 pb-2 bg-black bg-opacity-25 border-bottom border-secondary">
<ul class="nav nav-pills nav-fill gap-2" id="skillCreateModes" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active py-2 d-flex align-items-center justify-content-center gap-2" data-bs-target="#mode-wizard-pane" data-bs-toggle="pill" id="mode-wizard-tab" onclick="window.skillsTab?.switchCreateMode('wizard')" role="tab" type="button">
<i class="bi bi-magic text-primary"></i> <strong>  Пошаговый мастер</strong>
</button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link py-2 d-flex align-items-center justify-content-center gap-2" data-bs-target="#mode-ai-pane" data-bs-toggle="pill" id="mode-ai-tab" onclick="window.skillsTab?.switchCreateMode('ai')" role="tab" type="button">
<i class="bi bi-robot text-info"></i> <strong>  AI-Ассистент генерации</strong>
</button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link py-2 d-flex align-items-center justify-content-center gap-2" data-bs-target="#mode-presets-pane" data-bs-toggle="pill" id="mode-presets-tab" onclick="window.skillsTab?.switchCreateMode('presets')" role="tab" type="button">
<i class="bi bi-collection text-warning"></i> <strong>  Готовые шаблоны</strong>
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</button>
</li>
</ul>
</div>
<div class="modal-body p-3">
<div class="tab-content" id="skillCreateModesContent">
<!-- MODE 1: STEP-BY-STEP WIZARD -->
<div class="tab-pane fade show active" id="mode-wizard-pane" r
ole="tabpanel">
<!-- Steps Stepper Header -->
<div class="d-flex justify-content-between align-items-center
mb-3 px-2">
<div class="d-flex align-items-center gap-2 cursor-pointer" on
click="window.skillsTab?.goToStep(1)">
<span class="badge rounded-circle p-2 bg-primary step-indicato
r" id="step-badge-1">1</span>
<span class="small fw-semibold" id="step-label-1">Основное</sp
an>
</div>
<div class="flex-grow-1 mx-2 border-top border-secondary"></di
v>
<div class="d-flex align-items-center gap-2 cursor-pointer" on
click="window.skillsTab?.goToStep(2)">
<span class="badge rounded-circle p-2 bg-secondary step-indica
tor" id="step-badge-2">2</span>
<span class="small text-muted" id="step-label-2">Инструкции</s
pan>
</div>
<div class="flex-grow-1 mx-2 border-top border-secondary"></di
v>
<div class="d-flex align-items-center gap-2 cursor-pointer" on
click="window.skillsTab?.goToStep(3)">
<span class="badge rounded-circle p-2 bg-secondary step-indica
tor" id="step-badge-3">3</span>
<span class="small text-muted" id="step-label-3">Скрипты и фай
лы</span>
</div>
<div class="flex-grow-1 mx-2 border-top border-secondary"></di
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v>
<div class="d-flex align-items-center gap-2 cursor-pointer" on
click="window.skillsTab?.goToStep(4)">
<span class="badge rounded-circle p-2 bg-secondary step-indica
tor" id="step-badge-4">4</span>
<span class="small text-muted" id="step-label-4">Финализация</
span>
</div>
</div>
<!-- Wizard Step 1: Basic Info -->
<div class="wizard-step" id="wizard-step-1">
<div class="card bg-black bg-opacity-50 border-secondary mb-3"
 data-panel-id="panel-skills-3" id="panel-skills-3">
<div class="card-body p-3">
<h6 class="text-primary mb-3"><i class="bi bi-info-circle"></i
> Шаг 1: Название, назначение и расположение</h6>
<div class="row g-3">
<div class="col-12 col-md-6">
<label class="form-label small fw-semibold" for="wizard-skill-
name">
                        Идентификатор навыка (kebab-case) <spa
n class="text-danger">*</span>
</label>
<div class="input-group input-group-sm">
<span class="input-group-text bg-dark border-secondary text-wa
rning">□ </span>
<input class="form-control bg-dark text-white border-secondary
" id="wizard-skill-name" placeholder="напр. media-downloader и
ли log-parser" required="" type="text"/>
<button class="btn btn-outline-secondary" onclick="window.skil
lsTab?.generateSlugFromName()" title="Сгенерировать имя из опи
сания" type="button">
<i class="bi bi-magic"></i>
</button>
</div>
<small class="text-muted">Латиница, цифры, дефис и подчеркиван
ия (используется агентом для вызова).</small>
</div>
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<div class="col-12 col-md-6">
<label class="form-label small fw-semibold" for="wizard-skill-dir">Целевой каталог</label>
<select class="form-select form-select-sm bg-dark text-white border-secondary" id="wizard-skill-dir">
<option selected="" value=".agents/skills">.agents/skills (Проектный навык AI Windows Admin)</option>
<option value="~/.gemini/skills">~/.gemini/skills (Пользовательский — Gemini CLI / Antigravity)</option>
<option value="~/.agents/skills">~/.agents/skills (Глобальный пользовательский навык)</option>
<option value="skills">skills (Локальный каталог)</option>
</select>
<small class="text-muted">Определяет область видимости навыка в системе.</small>
</div>
<div class="col-12 col-md-6">
<label class="form-label small fw-semibold" for="wizard-skill-category">Категория</label>
<select class="form-select form-select-sm bg-dark text-white border-secondary" id="wizard-skill-category" onchange="window.skillsTab?.onCategoryChanged()">
<option value="tools">  Инструмент и автоматизация (CLI / Script)</option>
<option value="api">  Интеграция с API и сервисами</option>
<option value="data">  Анализ данных и парсинг</option>
<option value="persona">  AI-Персона / Специализированный эксперт</option>
<option value="monitoring">  Мониторинг и нотификации</option>
<option value="docgen">  Генератор документов и отчетов</option>
<option selected="" value="general">  Общего назначения</option>
</select>
</div>
<div class="col-12 col-md-6">
<label class="form-label small fw-semibold" for="wizard-skill-
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desc-ru">Назначение на русском</label>
<input class="form-control form-control-sm bg-dark text-white border-secondary" id="wizard-skill-desc-ru" placeholder="Понятное описание для пользователя..." type="text"/>
</div>
<div class="col-12">
<label class="form-label small fw-semibold" for="wizard-skill-desc-en">Описание для AI-агента (English description)</label>
<input class="form-control form-control-sm bg-dark text-white border-secondary" id="wizard-skill-desc-en" placeholder="Concise description in English used by agent router..." type="text" />
<small class="text-muted">Агент использует это описание при маршрутизации и выборе навыка.</small>
</div>
</div>
</div>
</div>
</div>
<!-- Wizard Step 2: Instructions (SKILL.md) -->
<div class="wizard-step d-none" id="wizard-step-2">
<div class="card bg-black bg-opacity-50 border-secondary mb-3" data-panel-id="panel-skills-4" id="panel-skills-4">
<div class="card-body p-3">
<div class="d-flex justify-content-between align-items-center mb-2">
<h6 class="text-primary mb-0"><i class="bi bi-file-text"></i> Шаг 2: Структура и логика инструкций (SKILL.md)</h6>
<button class="btn btn-outline-info btn-sm" onclick="window.skillsTab?.fillStandardInstructions()" type="button">
<i class="bi bi-arrow-repeat"></i> Заполнить стандартным шаблоном
                    </button>
</div>
<div class="row g-2 mb-3">
<div class="col-12 col-md-6">
<label class="form-label small fw-semibold" for="wizard-inst-purpose">  Цель (Purpose):</label>
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<textarea class="form-control form-control-sm bg-dark text-white border-secondary" id="wizard-inst-purpose" placeholder="Опишите главную цель работы навыка..." rows="2"></textarea>
</div>
<div class="col-12 col-md-6">
<label class="form-label small fw-semibold" for="wizard-inst-triggers">⚡ Условия запуска (Triggers):</label>
<textarea class="form-control form-control-sm bg-dark text-white border-secondary" id="wizard-inst-triggers" placeholder="Когда агент должен активировать этот навык..." rows="2"></textarea>
</div>
<div class="col-12 col-md-6">
<label class="form-label small fw-semibold" for="wizard-inst-protocol">📋 Протокол действий (Execution Protocol):</label>
<textarea class="form-control form-control-sm bg-dark text-white border-secondary" id="wizard-inst-protocol" placeholder="1. Проверить входные данные
2. Запустить скрипт или обработку
3. Сформировать ответ" rows="4"></textarea>
</div>
<div class="col-12 col-md-6">
<label class="form-label small fw-semibold" for="wizard-inst-rules">🛡 Правила и ограничения (Rules &amp;amp; Constraints):</label>
<textarea class="form-control form-control-sm bg-dark text-white border-secondary" id="wizard-inst-rules" placeholder="- Обрабатывать ошибки без падений
- Возвращать четкий статус выполнения" rows="4"></textarea>
</div>
</div>
<label class="form-label small fw-semibold" for="wizard-skill-instructions">Полный текст Markdown для SKILL.md:</label>
<textarea class="form-control bg-dark text-white border-secondary font-monospace" id="wizard-skill-instructions" placeholder="Автоматически генерируется из блоков выше..." rows="6" style="font-size: 0.85rem;"></textarea>
</div>
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</div>
</div>
<!-- Wizard Step 3: Scripts & Files -->
<div class="wizard-step d-none" id="wizard-step-3">
<div class="card bg-black bg-opacity-50 border-secondary mb-3"
 data-panel-id="panel-skills-5" id="panel-skills-5">
<div class="card-body p-3">
<h6 class="text-primary mb-3"><i class="bi bi-folder-plus"></i
> Шаг 3: Дополнительные скрипты, документация и справочники</h
6>
<div class="row g-3">
<div class="col-12">
<div class="form-check form-switch mb-2">
<input class="form-check-input" id="wizard-enable-script" onch
ange="window.skillsTab?.toggleScriptEditor()" type="checkbox"/
>
<label class="form-check-label fw-semibold small" for="wizard-
enable-script">
                            Добавить исполняемый Python-скрипт (
<code>scripts/main.py</code>)
                        </label>
</div>
<div class="d-none mt-2" id="wizard-script-container">
<textarea class="form-control bg-dark text-info border-seconda
ry font-monospace" id="wizard-script-code" rows="6" spellcheck
="false" style="font-size: 0.85rem;"></textarea>
</div>
</div>
<div class="col-12">
<div class="form-check form-switch mb-2">
<input class="form-check-input" id="wizard-enable-ref" onchang
e="window.skillsTab?.toggleRefEditor()" type="checkbox"/>
<label class="form-check-label fw-semibold small" for="wizard-
enable-ref">
                            Добавить справочный документ (<code>
references/guide.md</code>)
                        </label>
</div>
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<div class="d-none mt-2" id="wizard-ref-container">
<textarea class="form-control bg-dark text-light border-secondary font-monospace" id="wizard-ref-content" rows="4" spellcheck="false" style="font-size: 0.85rem;"></textarea>
</div>
</div>
<div class="col-12">
<label class="form-label small fw-semibold" for="wizard-skill-readme">Документация <code>README.md</code> (на английском языке):</label>
<textarea class="form-control bg-dark text-light border-secondary font-monospace" id="wizard-skill-readme" rows="4" spellcheck="false" style="font-size: 0.85rem;"></textarea>
</div>
</div>
</div>
</div>
</div>
<!-- Wizard Step 4: Finalize & Preview -->
<div class="wizard-step d-none" id="wizard-step-4">
<div class="card bg-black bg-opacity-50 border-secondary mb-3" data-panel-id="panel-skills-6" id="panel-skills-6">
<div class="card-body p-3">
<h6 class="text-success mb-3"><i class="bi bi-check2-circle"></i> Шаг 4: Предпросмотр создаваемого навыка</h6>
<div class="row g-2 mb-3">
<div class="col-12 col-md-4">
<div class="p-2 bg-dark rounded border border-secondary">
<span class="text-muted small d-block">Имя:</span>
<strong class="text-warning font-monospace" id="preview-skill-name">—</strong>
</div>
</div>
<div class="col-12 col-md-4">
<div class="p-2 bg-dark rounded border border-secondary">
<span class="text-muted small d-block">Путь:</span>
<code class="text-info" id="preview-skill-path">—</code>
</div>
```

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</div>
<div class="col-12 col-md-4">
<div class="p-2 bg-dark rounded border border-secondary">
<span class="text-muted small d-block">Компоненты:</span>
<span class="badge bg-primary me-1">SKILL.md</span>
<span class="badge bg-secondary me-1">README.md</span>
<span class="badge bg-info d-none" id="preview-badge-script">s
cripts/</span>
<span class="badge bg-warning d-none text-dark" id="preview-ba
dge-ref">references/</span>
</div>
</div>
</div>
<label class="form-label small fw-semibold">Результирующий <co
de>SKILL.md</code> (Frontmatter + Markdown):</label>
<pre class="bg-dark p-3 rounded border border-secondary text-i
nfo font-monospace" id="preview-skillmd-full" style="max-heigh
t: 220px; overflow-y: auto; font-size: 0.85rem;"></pre>
</div>
</div>
</div>
<!-- Wizard Navigation Bar -->
<div class="d-flex justify-content-between align-items-center
mt-3 pt-2 border-top border-secondary">
<button class="btn btn-outline-secondary btn-sm" disabled="" i
d="btn-wizard-prev" onclick="window.skillsTab?.wizardPrevStep(
)" type="button">
<i class="bi bi-arrow-left"></i> Назад
              </button>
<div class="d-flex gap-2">
<button class="btn btn-primary btn-sm" id="btn-wizard-next" on
click="window.skillsTab?.wizardNextStep()" type="button">
                  Далее <i class="bi bi-arrow-right"></i>
</button>
<button class="btn btn-success btn-sm d-none" id="btn-wizard-s
ubmit" onclick="window.skillsTab?.createSkillFromWizard()" typ
e="button">
<span class="spinner-border spinner-border-sm d-none" id="wiza
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rd-submit-spinner" role="status"></span>
<span><i class="bi bi-rocket-takeoff-fill"></i> Создать навык</span>
</button>
</div>
</div>
</div>
<!-- MODE 2: AI ASSISTANT GENERATION -->
<div class="tab-pane fade" id="mode-ai-pane" role="tabpanel">
<div class="card bg-black bg-opacity-50 border-secondary mb-3" data-panel-id="panel-skills-7" id="panel-skills-7">
<div class="card-body p-3">
<div class="d-flex align-items-center gap-2 mb-2 text-info">
<i class="bi bi-robot fs-5"></i>
<h6 class="mb-0">Генерация навыка с помощью ИИ (Natural Language to Skill)</h6>
</div>
<p class="text-muted small mb-3">
                    Просто опишите задачу или идею навыка обычными словами на русском или английском. ИИ автоматически сформирует имя, структуру инструкций, описание, контракт и вспомогательные скрипты.
                </p>
<div class="mb-3">
<label class="form-label small fw-semibold" for="ai-skill-prompt">Описание задачи навыка <span class="text-danger">*</span></label>
<textarea class="form-control bg-dark text-white border-secondary" id="ai-skill-prompt" placeholder="например: Навык для автоматического анализа логов сервера Nginx, выявления подозрительных IP-адресов и выгрузки отчета в формате CSV" rows="3"></textarea>
</div>
<div class="row g-2 mb-3">
<div class="col-12 col-md-6">
<label class="form-label small fw-semibold" for="ai-skill-category">Категория</label>
<select class="form-select form-select-sm bg-dark text-white b
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order-secondary" id="ai-skill-category">
<option value="tools">🛠 Инструмент и скрипт</option>
<option value="api">🔌 API и интеграция</option>
<option value="data">📊 Анализ данных</option>
<option value="persona">🤖 AI-Эксперт</option>
<option value="monitoring">📡 Мониторинг</option>
<option selected="" value="general">📦 Общий навык</option>
</select>
</div>
<div class="col-12 col-md-6">
<label class="form-label small fw-semibold" for="ai-skill-dir"
>Куда сохранить</label>
<select class="form-select form-select-sm bg-dark text-white b
order-secondary" id="ai-skill-dir">
<option selected="" value=".agents/skills">.agents/skills (Про
ектный)</option>
<option value="~/.gemini/skills">~/.gemini/skills (Gemini / An
tigravity)</option>
<option value="~/.agents/skills">~/.agents/skills (Глобальный)
</option>
</select>
</div>
</div>
<button class="btn btn-info btn-sm d-flex align-items-center g
ap-2" id="btn-generate-ai" onclick="window.skillsTab?.generate
SkillWithAi()" type="button">
<span class="spinner-border spinner-border-sm d-none" id="ai-g
en-spinner" role="status"></span>
<i class="bi bi-sparkles"></i> <strong>Сгенерировать навык</st
rong>
</button>
<!-- AI Result Box -->
<div class="d-none mt-3 pt-3 border-top border-secondary" id="
ai-result-box">
<div class="alert alert-success py-2 d-flex align-items-center
 justify-content-between">
<span><i class="bi bi-check-circle-fill me-1"></i> Навык успеш
но сгенерирован!</span>
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<span class="badge bg-dark font-monospace" id="ai-res-slug"></
span>
</div>
<div class="mb-2">
<label class="form-label small fw-semibold text-muted">Сгенери
рованные инструкции (SKILL.md):</label>
<pre class="bg-dark p-3 rounded border border-secondary text-i
nfo font-monospace" id="ai-res-preview" style="max-height: 180
px; overflow-y: auto; font-size: 0.85rem;"></pre>
</div>
<div class="d-flex justify-content-end gap-2">
<button class="btn btn-outline-primary btn-sm" onclick="window
.skillsTab?.applyAiResultToWizard()" type="button">
<i class="bi bi-sliders"></i> Настроить в мастере
                    </button>
<button class="btn btn-success btn-sm" onclick="window.skillsT
ab?.saveAiSkillDirectly()" type="button">
<i class="bi bi-plus-circle"></i> Создать сразу
                    </button>
</div>
</div>
</div>
</div>
</div>
<!-- MODE 3: PRESETS LIBRARY -->
<div class="tab-pane fade" id="mode-presets-pane" role="tabpan
el">
<div class="row g-3" id="presets-cards-container">
<!-- Preset Cards populated dynamically -->
</div>
</div>
</div>
<div class="alert d-none py-2 mt-3 mb-0" id="create-skill-aler
t"></div>
</div>
<div class="modal-footer border-secondary py-2 px-3 bg-black b
g-opacity-40">
<button class="btn btn-outline-secondary btn-sm" data-bs-dismi
```

```
ss="modal" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
<!-- Modal / Inspector: View, Edit & Live Test Skill -->
<div aria-hidden="true" aria-labelledby="skillDetailsModalLabe
l" class="modal fade" id="skillDetailsModal" tabindex="-1">
<div class="modal-dialog modal-xl modal-dialog-scrollable">
<div class="modal-content bg-dark text-white border-secondary"
>
<div class="modal-header border-secondary d-flex justify-conte
nt-between align-items-center">
<div>
<h5 class="modal-title d-flex align-items-center gap-2 mb-0" i
d="skillDetailsModalLabel">
<i class="bi bi-lightning-charge-fill text-warning"></i>
<span id="detail-skill-title">Навык</span>
</h5>
<small class="text-muted" id="detail-skill-path">—</small>
</div>
<div class="d-flex align-items-center gap-2">
<button class="btn btn-outline-info btn-sm d-flex align-items-
center gap-1" id="btn-package-modal" onclick="window.skillsTab
?.packageCurrentSkill()" type="button">
<i class="bi bi-box-seam"></i> Собрать .skill
          </button>
<button class="btn btn-outline-danger btn-sm d-flex align-item
s-center gap-1" id="btn-delete-modal" onclick="window.skillsTa
b?.deleteCurrentSkill()" type="button">
<i class="bi bi-trash"></i> Удалить
          </button>
<button aria-label="Close" class="btn-close btn-close-white" d
ata-bs-dismiss="modal" type="button"></button>
</div>
</div>
<div class="modal-body p-3">
<!-- Navigation tabs in inspector -->
```

```
<ul class="nav nav-pills mb-3" id="skillDetailTabs" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active" data-bs-target="#detail-pane-skillmd" data-bs-toggle="pill" id="detail-tab-skillmd" role="tab" type="button">
              SKILL.md (Инструкции)
            </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link" data-bs-target="#detail-pane-sandbox" data-bs-toggle="pill" id="detail-tab-sandbox" role="tab" type="button">
              Тестирование (Sandbox)
            </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link" data-bs-target="#detail-pane-readme" data-bs-toggle="pill" id="detail-tab-readme" role="tab" type="button">
              README.md
            </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link" data-bs-target="#detail-pane-files" data-bs-toggle="pill" id="detail-tab-files" role="tab" type="button">
              Файлы и структура
            </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link" data-bs-target="#detail-pane-contract" data-bs-toggle="pill" id="detail-tab-contract" role="tab" type="button">
              JSON Contract / Метаданные
            </button>
</li>
</ul>
```

```
<div class="tab-content" id="skillDetailTabContent">
<!-- SKILL.md Tab Pane -->
<div class="tab-pane fade show active" id="detail-pane-skillmd" role="tabpanel">
<div class="row g-2 mb-3">
<div class="col-12 col-md-6">
<label class="form-label fw-semibold small text-muted" for="detail-skill-desc-input">English Description (Frontmatter):</label>
<input class="form-control bg-black text-white border-secondary form-control-sm" id="detail-skill-desc-input" type="text"/>
</div>
<div class="col-12 col-md-6">
<label class="form-label fw-semibold small text-muted" for="detail-skill-desc-ru-input">Описание на русском (RU):</label>
<input class="form-control bg-black text-white border-secondary form-control-sm" id="detail-skill-desc-ru-input" type="text" />
</div>
</div>
<div class="mb-2 d-flex justify-content-between align-items-center">
<label class="form-label fw-semibold small text-muted mb-0" for="detail-skill-instructions">Тело инструкции (Markdown):</label>
<small class="text-muted">Редактируемый текст</small>
</div>
<textarea class="form-control bg-black text-white border-secondary font-monospace" id="detail-skill-instructions" rows="16" spellcheck="false" style="font-size: 0.9rem;"></textarea>
</div>
<!-- Live Sandbox Tester Tab Pane -->
<div class="tab-pane fade" id="detail-pane-sandbox" role="tabpanel">
<div class="p-3 bg-black rounded border border-secondary mb-3" >
<h6 class="text-warning mb-2"><i class="bi bi-terminal"></i> Интерактивная песочница для тестирования навыка</h6>
```

```
<p class="text-muted small mb-3">
                Введите тестовый запрос, чтобы проверить, как агент интерпретирует инструкции данного навыка и выполняет запрос.
            </p>
<div class="mb-3">
<label class="form-label small fw-semibold" for="sandbox-test-prompt">Тестовый запрос агенту:</label>
<div class="input-group">
<input class="form-control bg-dark text-white border-secondary" id="sandbox-test-prompt" placeholder="например: Выполни анализ данных или проверь состояние..." type="text"/>
<button class="btn btn-warning text-dark fw-bold d-flex align-items-center gap-1" id="btn-run-sandbox" onclick="window.skillsTab?.runSandboxTest()">
<span class="spinner-border spinner-border-sm d-none" id="sandbox-spinner" role="status"></span>
<i class="bi bi-play-fill"></i> Запустить
                </button>
</div>
</div>
<label class="form-label small fw-semibold text-muted">Результат выполнения (Output):</label>
<div class="p-3 bg-dark rounded border border-secondary text-light font-monospace" id="sandbox-output-box" style="min-height: 140px; max-height: 320px; overflow-y: auto; white-space: pre-wrap; font-size: 0.85rem;">
Готов к тестированию. Введите запрос и нажмите «Запустить».
            </div>
</div>
</div>
<!-- README.md Tab Pane -->
<div class="tab-pane fade" id="detail-pane-readme" role="tabpanel">
<div class="mb-2 d-flex justify-content-between align-items-center">
<label class="form-label fw-semibold small text-muted mb-0" for="detail-skill-readme">README.md документация:</label>
```

```
<small class="text-muted">Англоязычная документация пакета</small>
</div>
<textarea class="form-control bg-black text-white border-secondary font-monospace" id="detail-skill-readme" rows="18" spellcheck="false" style="font-size: 0.9rem;"></textarea>
</div>
<!-- Files Structure Tab Pane -->
<div class="tab-pane fade" id="detail-pane-files" role="tabpanel">
<div class="p-3 bg-black rounded border border-secondary">
<h6 class="text-info mb-2"><i class="bi bi-folder2-open"></i> Содержимое папки навыка:</h6>
<div class="list-group list-group-flush bg-transparent" id="detail-files-list"></div>
</div>
</div>
<!-- JSON Contract Tab Pane -->
<div class="tab-pane fade" id="detail-pane-contract" role="tabpanel">
<pre class="bg-black p-3 rounded border border-secondary text-info mb-0" id="detail-json-contract" style="font-size: 0.85rem; max-height: 400px; overflow-y: auto;"></pre>
</div>
</div>
<div class="alert d-none py-2 mt-3 mb-0" id="detail-skill-alert"></div>
</div>
<div class="modal-footer border-secondary d-flex justify-content-between">
<button class="btn btn-outline-secondary" data-bs-dismiss="modal" type="button">Закрыть</button>
<button class="btn btn-primary d-flex align-items-center gap-1" id="btn-save-skill" onclick="window.skillsTab?.saveCurrentSkill()" type="button">
<span class="spinner-border spinner-border-sm d-none" id="save-skill-spinner" role="status"></span>
<span><i class="bi bi-save"></i> Сохранить изменения</span>
```

```
</button>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ================================================================
 * Process Name: Windows Skills Tab - Main Script
 * ================================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/skills_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/skills_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ================================================================
 */

// Skills Tab Logic for AI Windows Admin Admin Interface
// Supports 3 Creation Modes (Step-by-Step Wizard, AI Assistant, Presets Library) and Live Sandbox Testing

class SkillsTabManager {
  constructor() {
    this.skills = [];
    this.filteredSkills = [];
    this.currentSkill = null;
    this.viewMode = 'cards'; // 'cards' | 'table'
    this.searchQuery = '';
    this.activeFilter = 'all'; // 'all' | 'scripts' | 'referen
```

```
ces' | 'packaged'
32      this.isInitialized = false;
33
34      // Wizard state
35      this.currentWizardStep = 1;
36      this.createMode = 'wizard'; // 'wizard' | 'ai' | 'presets'
37      this.aiGeneratedData = null;
38
39      // Presets catalog
40      this.presets = [
41        {
42          id: 'tool-automation',
43          title: '  CLI & Python Tool',
44          category: 'tools',
45          badge: 'Automation',
46          description: 'Исполняемый навык с Python-скриптом авто
матизации в папке scripts/. Подходит для утилит, конвертеров и
 системных задач.',
47          name: 'custom-cli-tool',
48          descRu: 'Автоматизированная CLI утилита для выполнения
 задач через скрипты.',
49          descEn: 'Automated CLI utility providing executable sc
ripts for system routines.',
50          purpose: 'Execute specialized automation commands via
internal scripts.',
51          triggers: 'Triggered when the user requests executing
a custom tool or batch script task.',
52          protocol: '1. Validate arguments and environment.\n2.
Execute python script in `scripts/main.py`.\n3. Return formatt
ed stdout/stderr status.',
53          rules: '- Ensure non-zero exit codes are logged and re
ported.\n- Maintain English logging.',
54          hasScript: true,
55          scriptCode: `# -*- coding: utf-8 -*-\n"""CLI tool auto
mation script."""\nimport sys\n\ndef main():\n    print("Execu
ting automated tool task...")\n    return 0\n\nif __name__ ==
"__main__":\n    sys.exit(main())\n`,
56          hasRef: false,
```

```
      },
      {
        id: 'api-connector',
        title: '  REST API Connector',
        category: 'api',
        badge: 'Integration',
        description: 'Интеграция с внешним REST/JSON API. Шабл
он с проверкой авторизации, обработкой таймаутов и структуриро
ванием ответов.',
        name: 'rest-api-connector',
        descRu: 'Интеграция с внешним сервисом через REST API.
',
        descEn: 'Connector skill for external REST API service
s and endpoints.',
        purpose: 'Query external HTTP REST API endpoints and p
rocess JSON payloads.',
        triggers: 'Triggered when user asks to fetch data from
 or post payloads to remote APIs.',
        protocol: '1. Parse endpoint parameters and auth token
s.\n2. Send HTTP request with exponential retry.\n3. Parse JSO
N response and summarize.',
        rules: '- Protect sensitive secrets and auth tokens.\n
- Handle rate limits and connection timeouts gracefully.',
        hasScript: true,
        scriptCode: `# -*- coding: utf-8 -*-\n"""REST API clie
nt script."""\nimport os, urllib.request, json\n\ndef query_ap
i(url: str):\n    req = urllib.request.Request(url, headers={"
User-Agent": "AI-Windows-Admin/1.0"})\n    with urllib.request
.urlopen(req, timeout=10) as resp:\n        return json.loads(
resp.read().decode("utf-8"))\n`,
        hasRef: true,
        refContent: `# REST API Protocol Guidelines\n\n- Stand
ard timeout: 10s\n- Required headers: User-Agent, Accept\n`,
      },
      {
        id: 'data-parser',
        title: '  Data Analyzer & Parser',
        category: 'data',
```

```
        badge: 'Analytics',
        description: 'Обработка и анализ структурированных данных (CSV, JSON, XML, таблицы, логи). Извлечение инсайтов и форматирование результатов.',
        name: 'data-analyzer',
        descRu: 'Анализ структурированных данных, логов и таблиц с выводом статистики.',
        descEn: 'Analyze structured data, logs, CSV/JSON datasets, and generate statistics.',
        purpose: 'Parse input data files, compute aggregated metrics, and present visual summaries.',
        triggers: 'Triggered when datasets, spreadsheets or data files need parsing and reporting.',
        protocol: '1. Identify input format (CSV/JSON/Log).\n2. Parse schema and clean invalid rows.\n3. Compute totals, distributions, and outliers.\n4. Output Markdown summary tables.',
        rules: '- Avoid memory leaks on large files.\n- Strictly English docstrings.',
        hasScript: true,
        scriptCode: `# -*- coding: utf-8 -*-\n"""Data parsing helper."""\nimport csv, json\n\ndef parse_records(filepath: str):\n    # Add parser logic here\n    pass\n`,
        hasRef: false,
      },
      {
        id: 'ai-persona',
        title: '  AI Expert & Persona',
        category: 'persona',
        badge: 'Agent Role',
        description: 'Специализированная роль и экспертные инструкции (Архитектор, Аналитик безопасности, Code Reviewer, Консультант).',
        name: 'domain-expert',
        descRu: 'Экспертная роль и правила поведения для специализированных консультаций.',
        descEn: 'Specialized domain expert persona with strict evaluation guidelines.',
```

```
        purpose: 'Act as a senior domain expert offering high-level technical feedback.',
        triggers: 'Triggered whenever specialized architectural or domain advice is requested.',
        protocol: '1. Analyze problem from the perspective of domain best practices.\n2. Provide structured pros/cons analysis.\n3. Recommend action plan with trade-offs.',
        rules: '- Be concise, objective, and reference industry standards.',
        hasScript: false,
        hasRef: true,
        refContent: `# Domain Expertise Standards\n\nKey methodologies, architecture patterns, and decision trees.\n`,
      },
      {
        id: 'monitoring-notifier',
        title: '  Monitor & Notifier',
        category: 'monitoring',
        badge: 'Observability',
        description: 'Периодическая проверка сервисов, мониторинг системных параметров, выявление сбоев и отправка нотификаций.',
        name: 'health-monitor',
        descRu: 'Мониторинг доступности сервисов и формирование алертов при сбоях.',
        descEn: 'Health monitoring and alerting skill for infrastructure and services.',
        purpose: 'Continuously verify target service endpoints and notify on anomalies.',
        triggers: 'Triggered by scheduled cron jobs or on-demand health check requests.',
        protocol: '1. Ping healthcheck endpoints.\n2. Measure latency and status codes.\n3. If failure detected, format diagnostic alert.',
        rules: '- Fail-fast on network unreachability.\n- Include timestamps and severity level.',
        hasScript: true,
        scriptCode: `# -*- coding: utf-8 -*-\n"""Health check
```

```
monitor."""\nimport time\n\ndef check_service(host: str, port:
 int):\n    # Health probe\n    return True\n`,
        hasRef: false,
      },
      {
        id: 'doc-generator',
        title: '  Report & Doc Generator',
        category: 'docgen',
        badge: 'Documentation',
        description: 'Автоматическая генерация документации, о
тчетов в Markdown/HTML и сводок по проекту на основе входных д
анных.',
        name: 'report-generator',
        descRu: 'Генератор красивых отчетов, проектной докумен
тации и сводок.',
        descEn: 'Automated documentation and structured report
 generation skill.',
        purpose: 'Generate comprehensive Markdown reports, cha
nge logs, and release notes.',
        triggers: 'Triggered when documentation, release summa
ries or audit reports are requested.',
        protocol: '1. Collect source items and metadata.\n2. A
pply template sections and headings.\n3. Format tables and Git
Hub alerts.\n4. Save output file.',
        rules: '- Follow documentation style standards.\n- Ens
ure all links and tables are valid.',
        hasScript: false,
        hasRef: true,
        refContent: `# Document Generation Templates\n\nTempla
te blocks, header standards, and formatting guidelines.\n`,
      }
    ];
  }

  async init() {
    if (this.isInitialized) {
      await this.refresh();
      return;
```

```
    }
    this.bindEvents();
    this.renderPresets();
    this.isInitialized = true;
    await this.refresh();
  }

  bindEvents() {
    const searchInput = document.getElementById('skills-search
-input');
    if (searchInput) {
      searchInput.addEventListener('input', (e) => {
        this.searchQuery = e.target.value.trim().toLowerCase()
;
        this.applyFilters();
      });
    }

    const wizardNameInput = document.getElementById('wizard-sk
ill-name');
    if (wizardNameInput) {
      wizardNameInput.addEventListener('input', (e) => {
        const val = e.target.value;
        const normalized = val.toLowerCase().replace(/[^a-z0-9
_-]/g, '-');
        if (val !== normalized) {
          e.target.value = normalized;
        }
        this.syncWizardPreview();
      });
    }

    // Step 2 field listeners for live markdown generator
    ['wizard-inst-purpose', 'wizard-inst-triggers', 'wizard-in
st-protocol', 'wizard-inst-rules'].forEach(id => {
      const el = document.getElementById(id);
      if (el) {
        el.addEventListener('input', () => this.syncInstructio
```

```
nsFromFields());
      }
    });

    ['wizard-skill-desc-ru', 'wizard-skill-desc-en', 'wizard-skill-dir'].forEach(id => {
      const el = document.getElementById(id);
      if (el) {
        el.addEventListener('input', () => this.syncWizardPreview());
        el.addEventListener('change', () => this.syncWizardPreview());
      }
    });
  }

  async apiFetch(endpoint, options = {}) {
    let url = endpoint;
    let res = await fetch(url, options);
    if (!res.ok && url.startsWith('/api/skills')) {
      const fallbackUrl = url.replace('/api/skills', '/api/admin/skills');
      const fallbackRes = await fetch(fallbackUrl, options);
      if (fallbackRes.ok) {
        return fallbackRes;
      }
    }
    return res;
  }

  async refresh() {
    this.showLoading(true);
    try {
      const response = await this.apiFetch('/api/skills');
      if (!response.ok) {
        throw new Error(`HTTP ${response.status}: ${response.statusText}`);
      }
```

```
      const data = await response.json();
      this.skills = data.skills || [];
      this.updateStats();
      this.applyFilters();
    } catch (err) {
      console.error('[SkillsTab] Load error:', err);
      const cardsContainer = document.getElementById('skills-cards-container');
      if (cardsContainer) {
        cardsContainer.innerHTML = `
          <div class="col-12">
            <div class="alert alert-danger d-flex align-items-center justify-content-between p-3 shadow-sm">
              <div><i class="bi bi-exclamation-triangle-fill me-2"></i><strong>Ошибка загрузки навыков:</strong> ${err.message}</div>
              <button class="btn btn-sm btn-outline-light" onclick="window.skillsTab?.refresh()">Повторить</button>
            </div>
          </div>
        `;
      }
    } finally {
      this.showLoading(false);
    }
  }

  updateStats() {
    const total = this.skills.length;
    const withScripts = this.skills.filter(s => s.has_scripts).length;
    const withRefs = this.skills.filter(s => s.has_references).length;
    const packaged = this.skills.filter(s => s.has_dist).length;

    const setVal = (id, val) => {
      const el = document.getElementById(id);
```

```
      if (el) el.textContent = val;
    };

    setVal('skills-total-badge', total);
    setVal('stat-total-skills', total);
    setVal('stat-scripts-skills', withScripts);
    setVal('stat-refs-skills', withRefs);
    setVal('stat-packaged-skills', packaged);
  }

  applyFilters() {
    const filterSelect = document.getElementById('skills-filte
r-select');
    this.activeFilter = filterSelect ? filterSelect.value : 'a
ll';

    this.filteredSkills = this.skills.filter(skill => {
      if (this.searchQuery) {
        const nameMatch = (skill.name || '').toLowerCase().inc
ludes(this.searchQuery);
        const descMatch = (skill.description || '').toLowerCas
e().includes(this.searchQuery);
        const i18nMatch = Object.values(skill.descriptions_i18
n || {}).some(d => (d || '').toLowerCase().includes(this.searc
hQuery));
        if (!nameMatch && !descMatch && !i18nMatch) return fal
se;
      }

      if (this.activeFilter === 'scripts' && !skill.has_script
s) return false;
      if (this.activeFilter === 'references' && !skill.has_ref
erences) return false;
      if (this.activeFilter === 'packaged' && !skill.has_dist)
 return false;

      return true;
    });
```

```
274
275     this.render();
276   }
277
278   clearSearch() {
279     const searchInput = document.getElementById('skills-search
-input');
280     if (searchInput) searchInput.value = '';
281     this.searchQuery = '';
282     this.applyFilters();
283   }
284
285   setViewMode(mode) {
286     this.viewMode = mode;
287     const btnCards = document.getElementById('btn-view-cards')
;
288     const btnTable = document.getElementById('btn-view-table')
;
289
290     if (mode === 'cards') {
291       btnCards?.classList.add('active');
292       btnTable?.classList.remove('active');
293     } else {
294       btnCards?.classList.remove('active');
295       btnTable?.classList.add('active');
296     }
297     this.render();
298   }
299
300   render() {
301     const cardsContainer = document.getElementById('skills-car
ds-container');
302     const tableCard = document.getElementById('skills-table-ca
rd');
303     const emptyNotice = document.getElementById('skills-empty'
);
304
305     if (!cardsContainer || !tableCard) return;
```

```
306
307     if (this.filteredSkills.length === 0) {
308       cardsContainer.innerHTML = '';
309       tableCard.classList.add('d-none');
310       emptyNotice?.classList.remove('d-none');
311       return;
312     }
313
314     emptyNotice?.classList.add('d-none');
315
316     if (this.viewMode === 'cards') {
317       tableCard.classList.add('d-none');
318       cardsContainer.classList.remove('d-none');
319       cardsContainer.innerHTML = this.filteredSkills.map(s =>
this.renderCard(s)).join('');
320     } else {
321       cardsContainer.classList.add('d-none');
322       tableCard.classList.remove('d-none');
323       const tableBody = document.getElementById('skills-table-
body');
324       if (tableBody) {
325         tableBody.innerHTML = this.filteredSkills.map(s => thi
s.renderTableRow(s)).join('');
326       }
327     }
328   }
329
330   renderCard(skill) {
331     const escapeHtml = str => String(str || '').replace(/[&<>'
"]/g, tag => ({
332       '&': '&amp;', '<': '&lt;', '>': '&gt;', "'": '&#39;', '"
': '&quot;'
333     }[tag] || tag));
334
335     const badges = [];
336     if (skill.has_scripts) badges.push('<span class="badge bg-
info-subtle text-info border border-info-subtle small me-1"><i
 class="bi bi-file-earmark-code"></i> scripts</span>');
```

```
337     if (skill.has_references) badges.push('<span class="badge
    bg-warning-subtle text-warning border border-warning-subtle sm
    all me-1"><i class="bi bi-journal-text"></i> references</span>
    ');
338     if (skill.has_assets) badges.push('<span class="badge bg-s
    econdary border border-secondary small me-1"><i class="bi bi-i
    mages"></i> assets</span>');
339     if (skill.has_dist) badges.push('<span class="badge bg-suc
    cess-subtle text-success border border-success-subtle small me
    -1"><i class="bi bi-box-seam"></i> .skill</span>');
340
341     const descRu = skill.descriptions_i18n?.ru || '';
342     const descDisplay = descRu ? `${descRu}` : (skill.descript
    ion || 'Описание отсутствует');
343
344     return `
345       <div class="col-12 col-md-6 col-lg-4">
346         <div class="card bg-dark border-secondary text-white h
    -100 shadow-sm skill-card">
347           <div class="card-header border-secondary d-flex just
    ify-content-between align-items-center bg-black bg-opacity-25
    py-2">
348             <span class="badge bg-primary font-monospace fs-6
    text-truncate" style="max-width: 70%;" title="${escapeHtml(ski
    ll.name)}">
349               ${escapeHtml(skill.name)}
350             </span>
351             <span class="text-muted small" title="Количество ф
    айлов">
352               <i class="bi bi-files"></i> ${skill.files_count
    || 1}
353             </span>
354           </div>
355           <div class="card-body py-2">
356             <p class="card-text text-light small mb-2 text-tru
    ncate-3" style="min-height: 40px; display: -webkit-box; -webki
    t-line-clamp: 2; -webkit-box-orient: vertical; overflow: hidde
    n;" title="${escapeHtml(descDisplay)}">
```

```
              ${escapeHtml(descDisplay)}
            </p>
            <div class="mb-2 text-truncate">
              <small class="text-secondary font-monospace"><i class="bi bi-folder"></i> ${escapeHtml(skill.relative_path)}</small>
            </div>
            <div class="d-flex flex-wrap gap-1">
              ${badges.join('')}
            </div>
          </div>
          <div class="card-footer border-secondary bg-black bg-opacity-25 py-2 d-flex justify-content-between align-items-center">
            <button type="button" class="btn btn-sm btn-outline-primary d-flex align-items-center gap-1" onclick="window.skillsTab?.openSkillDetails('${escapeHtml(skill.name)}')">
              <i class="bi bi-sliders"></i> Детали
            </button>
            <div class="btn-group btn-group-sm">
              <button type="button" class="btn btn-outline-warning" title="Тестировать в Sandbox" onclick="window.skillsTab?.openSkillSandbox('${escapeHtml(skill.name)}')">
                <i class="bi bi-play-circle"></i>
              </button>
              <button type="button" class="btn btn-outline-info" title="Собрать .skill архив" onclick="window.skillsTab?.packageSkill('${escapeHtml(skill.name)}')">
                <i class="bi bi-box-seam"></i>
              </button>
              <button type="button" class="btn btn-outline-danger" title="Удалить навык" onclick="window.skillsTab?.deleteSkill('${escapeHtml(skill.name)}')">
                <i class="bi bi-trash"></i>
              </button>
            </div>
          </div>
        </div>
```

```
      </div>
    `;
  }

  renderTableRow(skill) {
    const escapeHtml = str => String(str || '').replace(/[&<>'"]/g, tag => ({
      '&': '&amp;', '<': '&lt;', '>': '&gt;', "'": '&#39;', '"': '&quot;'
    }[tag] || tag));

    const badges = [];
    if (skill.has_scripts) badges.push('<span class="badge bg-info-subtle text-info border border-info-subtle small me-1">scripts</span>');
    if (skill.has_references) badges.push('<span class="badge bg-warning-subtle text-warning border border-warning-subtle small me-1">refs</span>');
    if (skill.has_dist) badges.push('<span class="badge bg-success-subtle text-success border border-success-subtle small me-1">.skill</span>');

    const descRu = skill.descriptions_i18n?.ru || '';
    const descDisplay = descRu ? `${descRu}` : (skill.description || '—');

    return `
      <tr>
        <td>
          <span class="badge bg-primary font-monospace fs-6">${escapeHtml(skill.name)}</span>
        </td>
        <td>
          <div class="small text-light text-truncate" style="max-width: 350px;" title="${escapeHtml(descDisplay)}">
            ${escapeHtml(descDisplay)}
          </div>
        </td>
```

```
        <td>
          <code class="small text-secondary">${escapeHtml(skil
l.relative_path)}</code>
        </td>
        <td>
          ${badges.length ? badges.join('') : '<span class="te
xt-muted small">—</span>'}
        </td>
        <td style="text-align: right;">
          <div class="btn-group btn-group-sm">
            <button type="button" class="btn btn-outline-prima
ry" title="Открыть детали" onclick="window.skillsTab?.openSkil
lDetails('${escapeHtml(skill.name)}')">
              <i class="bi bi-sliders"></i>
            </button>
            <button type="button" class="btn btn-outline-warni
ng" title="Тестировать в Sandbox" onclick="window.skillsTab?.o
penSkillSandbox('${escapeHtml(skill.name)}')">
              <i class="bi bi-play-circle"></i>
            </button>
            <button type="button" class="btn btn-outline-info"
 title="Собрать архив" onclick="window.skillsTab?.packageSkill
('${escapeHtml(skill.name)}')">
              <i class="bi bi-box-seam"></i>
            </button>
            <button type="button" class="btn btn-outline-dange
r" title="Удалить" onclick="window.skillsTab?.deleteSkill('${e
scapeHtml(skill.name)}')">
              <i class="bi bi-trash"></i>
            </button>
          </div>
        </td>
      </tr>
    `;
  }

  // ============================================================
================
```

```
  // CREATION WIZARD LOGIC (3 Modes: Wizard, AI, Presets)
  // ============================================================
================

  switchCreateMode(mode) {
    this.createMode = mode;
  }

  goToStep(step) {
    if (step < 1 || step > 4) return;
    this.currentWizardStep = step;

    for (let i = 1; i <= 4; i++) {
      const stepEl = document.getElementById(`wizard-step-${i}
`);
      const badgeEl = document.getElementById(`step-badge-${i}
`);
      const labelEl = document.getElementById(`step-label-${i}
`);

      if (stepEl) {
        if (i === step) stepEl.classList.remove('d-none');
        else stepEl.classList.add('d-none');
      }

      if (badgeEl && labelEl) {
        if (i === step) {
          badgeEl.className = 'badge rounded-circle p-2 bg-pri
mary step-indicator';
          labelEl.className = 'small fw-semibold text-white';
        } else if (i < step) {
          badgeEl.className = 'badge rounded-circle p-2 bg-suc
cess step-indicator';
          labelEl.className = 'small text-success';
        } else {
          badgeEl.className = 'badge rounded-circle p-2 bg-sec
ondary step-indicator';
          labelEl.className = 'small text-muted';
```

```
        }
      }
    }

    const prevBtn = document.getElementById('btn-wizard-prev')
;
    const nextBtn = document.getElementById('btn-wizard-next')
;
    const submitBtn = document.getElementById('btn-wizard-subm
it');

    if (prevBtn) prevBtn.disabled = (step === 1);
    if (nextBtn) {
      if (step === 4) nextBtn.classList.add('d-none');
      else nextBtn.classList.remove('d-none');
    }
    if (submitBtn) {
      if (step === 4) submitBtn.classList.remove('d-none');
      else submitBtn.classList.add('d-none');
    }

    if (step === 4) {
      this.syncWizardPreview();
    }
  }

  wizardNextStep() {
    if (this.currentWizardStep === 1) {
      const name = document.getElementById('wizard-skill-name'
)?.value.trim();
      if (!name) {
        this.showCreateAlert('Укажите имя навыка (kebab-case)'
, 'danger');
        return;
      }
      this.clearCreateAlert();
    }
    this.goToStep(this.currentWizardStep + 1);
```

```
  }

  wizardPrevStep() {
    this.goToStep(this.currentWizardStep - 1);
  }

  generateSlugFromName() {
    const descRu = document.getElementById('wizard-skill-desc-ru')?.value.trim();
    const descEn = document.getElementById('wizard-skill-desc-en')?.value.trim();
    const source = descEn || descRu || 'custom-skill';

    // Transliterate or simple slugify
    let slug = source.toLowerCase()
      .replace(/[а-яё]/g, char => {
        const tr = {'а':'a','б':'b','в':'v','г':'g','д':'d','е':'e','ё':'yo','ж':'zh','з':'z','и':'i','й':'y','к':'k','л':'l','м':'m','н':'n','о':'o','п':'p','р':'r','с':'s','т':'t','у':'u','ф':'f','х':'h','ц':'ts','ч':'ch','ш':'sh','щ':'sch','ъ':'','ы':'y','ь':'','э':'e','ю':'yu','я':'ya'};
        return tr[char] || char;
      })
      .replace(/[^a-z0-9_-]/g, '-')
      .replace(/-+/g, '-')
      .replace(/^-|-$/g, '')
      .slice(0, 32);

    if (slug) {
      const nameInput = document.getElementById('wizard-skill-name');
      if (nameInput) nameInput.value = slug;
      this.syncWizardPreview();
    }
  }

  onCategoryChanged() {
    const cat = document.getElementById('wizard-skill-category
```

```
')?.value;
532     const descRuInput = document.getElementById('wizard-skill-
desc-ru');
533     const descEnInput = document.getElementById('wizard-skill-
desc-en');
534 
535     if (cat && (!descRuInput?.value || !descEnInput?.value)) {
536       const preset = this.presets.find(p => p.category === cat
);
537       if (preset) {
538         if (!descRuInput.value) descRuInput.value = preset.des
cRu;
539         if (!descEnInput.value) descEnInput.value = preset.des
cEn;
540       }
541     }
542     this.syncWizardPreview();
543   }
544 
545   fillStandardInstructions() {
546     const name = document.getElementById('wizard-skill-name')?
.value.trim() || 'custom-skill';
547     const descRu = document.getElementById('wizard-skill-desc-
ru')?.value.trim() || 'Выполнение специализированных задач';
548     const descEn = document.getElementById('wizard-skill-desc-
en')?.value.trim() || 'Specialized agent task execution';
549 
550     document.getElementById('wizard-inst-purpose').value = des
cEn;
551     document.getElementById('wizard-inst-triggers').value = `A
ctivate when the user asks to perform tasks matching: ${descRu
}.`;
552     document.getElementById('wizard-inst-protocol').value = `1
. Validate input query and preconditions.\n2. Execute core log
ic and inspect helper scripts in scripts/.\n3. Return structur
ed markdown response.`;
553     document.getElementById('wizard-inst-rules').value = `- En
sure English logging via src.logger.logger.\n- Gracefully catc
```

```
h and report all execution errors.`;

    this.syncInstructionsFromFields();
  }

  syncInstructionsFromFields() {
    const name = document.getElementById('wizard-skill-name')?.value.trim() || 'custom-skill';
    const title = name.replace('-', ' ').replace('_', ' ').replace(/\b\w/g, l => l.toUpperCase());
    const purpose = document.getElementById('wizard-inst-purpose')?.value.trim() || 'Provide specialized agent capability.';
    const triggers = document.getElementById('wizard-inst-triggers')?.value.trim() || 'Activate upon relevant user requests.';
    const protocol = document.getElementById('wizard-inst-protocol')?.value.trim() || '1. Parse input.\n2. Execute action.\n3. Output result.';
    const rules = document.getElementById('wizard-inst-rules')?.value.trim() || '- Maintain fail-fast validation.\n- Report status clearly.';

    const md = `# ${title}

## □ Purpose
${purpose}

## □ When to Use & Triggers
${triggers}

## □ Execution Protocol
${protocol}

## □ Rules & Constraints
${rules}
`;
    const instEl = document.getElementById('wizard-skill-instructions');
```

```
    if (instEl) instEl.value = md;
    this.syncWizardPreview();
  }

  toggleScriptEditor() {
    const chk = document.getElementById('wizard-enable-script'
);
    const container = document.getElementById('wizard-script-c
ontainer');
    const scriptCode = document.getElementById('wizard-script-
code');

    if (chk?.checked) {
      container?.classList.remove('d-none');
      if (!scriptCode.value.trim()) {
        const name = document.getElementById('wizard-skill-nam
e')?.value.trim() || 'custom-skill';
        scriptCode.value = `# -*- coding: utf-8 -*-\n"""Execut
able tool helper for ${name}."""\n\nimport sys\n\ndef main():\
n    print("Running helper script...")\n    return 0\n\nif __n
ame__ == "__main__":\n    sys.exit(main())\n`;
      }
    } else {
      container?.classList.add('d-none');
    }
    this.syncWizardPreview();
  }

  toggleRefEditor() {
    const chk = document.getElementById('wizard-enable-ref');
    const container = document.getElementById('wizard-ref-cont
ainer');
    const refContent = document.getElementById('wizard-ref-con
tent');

    if (chk?.checked) {
      container?.classList.remove('d-none');
      if (!refContent.value.trim()) {
```

```
        const name = document.getElementById('wizard-skill-nam
e')?.value.trim() || 'custom-skill';
        refContent.value = `# ${name.replace('-', ' ').toUpper
Case()} Specification\n\nDetailed operational rules and refere
nce data.\n`;
      }
    } else {
      container?.classList.add('d-none');
    }
    this.syncWizardPreview();
  }

  syncWizardPreview() {
    const name = document.getElementById('wizard-skill-name')?
.value.trim() || 'custom-skill';
    const dir = document.getElementById('wizard-skill-dir')?.v
alue || '.agents/skills';
    const descEn = document.getElementById('wizard-skill-desc-
en')?.value.trim() || `Agent skill for ${name}.`;
    const descRu = document.getElementById('wizard-skill-desc-
ru')?.value.trim() || descEn;
    const instructions = document.getElementById('wizard-skill
-instructions')?.value.trim() || `# ${name}\n\n##  Purpose\n
${descEn}\n`;

    const hasScript = document.getElementById('wizard-enable-s
cript')?.checked;
    const hasRef = document.getElementById('wizard-enable-ref'
)?.checked;

    // Badges in step 4
    const nameEl = document.getElementById('preview-skill-name
');
    const pathEl = document.getElementById('preview-skill-path
');
    const badgeScript = document.getElementById('preview-badge
-script');
    const badgeRef = document.getElementById('preview-badge-re
```

```
f');
634     const previewFull = document.getElementById('preview-skill
md-full');
635 
636     if (nameEl) nameEl.textContent = name;
637     if (pathEl) pathEl.textContent = `${dir}/${name}`;
638     if (badgeScript) {
639       if (hasScript) badgeScript.classList.remove('d-none');
640       else badgeScript.classList.add('d-none');
641     }
642     if (badgeRef) {
643       if (hasRef) badgeRef.classList.remove('d-none');
644       else badgeRef.classList.add('d-none');
645     }
646 
647     const fullSkillMd = `---
648 name: ${name}
649 description: ${descEn}
650 description_i18n:
651   en: ${descEn}
652   ru: ${descRu}
653 ---
654 
655 ${instructions}`;
656 
657     if (previewFull) previewFull.textContent = fullSkillMd;
658 
659     // Sync README default
660     const readmeEl = document.getElementById('wizard-skill-rea
dme');
661     if (readmeEl && !readmeEl.value.trim()) {
662       readmeEl.value = `# ${name.replace('-', ' ').replace(/\b
\w/g, l => l.toUpperCase())}\n\n## Overview\n${descEn}\n\n## L
ocation\n\`${dir}/${name}/\`\n`;
663     }
664   }
665 
666   async createSkillFromWizard() {
```

```
    const name = document.getElementById('wizard-skill-name')?.value.trim().toLowerCase();
    if (!name) {
      this.goToStep(1);
      this.showCreateAlert('Укажите имя навыка', 'danger');
      return;
    }

    const spinner = document.getElementById('wizard-submit-spinner');
    spinner?.classList.remove('d-none');

    const descEn = document.getElementById('wizard-skill-desc-en')?.value.trim() || `Agent skill for ${name}.`;
    const descRu = document.getElementById('wizard-skill-desc-ru')?.value.trim() || descEn;
    const targetDir = document.getElementById('wizard-skill-dir')?.value || '.agents/skills';
    const category = document.getElementById('wizard-skill-category')?.value || 'general';
    const instructions = document.getElementById('wizard-skill-instructions')?.value.trim() || '';
    const readme = document.getElementById('wizard-skill-readme')?.value.trim() || '';

    const scripts = {};
    if (document.getElementById('wizard-enable-script')?.checked) {
      scripts['main.py'] = document.getElementById('wizard-script-code')?.value || '';
    }

    const references = {};
    if (document.getElementById('wizard-enable-ref')?.checked) {
      references['guide.md'] = document.getElementById('wizard-ref-content')?.value || '';
    }
```

```
693
694     try {
695       const res = await this.apiFetch('/api/skills', {
696         method: 'POST',
697         headers: { 'Content-Type': 'application/json' },
698         body: JSON.stringify({
699           name,
700           description: descEn,
701           description_ru: descRu,
702           category,
703           target_dir: targetDir,
704           instructions,
705           readme,
706           scripts,
707           references
708         })
709       });
710
711       if (!res.ok) {
712         const errData = await res.json().catch(() => ({}));
713         throw new Error(errData.detail || res.statusText);
714       }
715
716       // Close modal
717       const modalEl = document.getElementById('createSkillModa
l');
718       const modal = bootstrap.Modal.getInstance(modalEl);
719       modal?.hide();
720
721       await this.refresh();
722       await this.openSkillDetails(name);
723     } catch (err) {
724       console.error('[SkillsTab] Create error:', err);
725       this.showCreateAlert(`Ошибка создания навыка: ${err.mess
age}`, 'danger');
726     } finally {
727       spinner?.classList.add('d-none');
728     }
```

```
  }

  // ============================================================
================
  // AI GENERATOR MODE
  // ============================================================
================

  async generateSkillWithAi() {
    const prompt = document.getElementById('ai-skill-prompt')?
.value.trim();
    if (!prompt) {
      this.showCreateAlert('Введите описание задачи для генера
ции навыка', 'warning');
      return;
    }

    const spinner = document.getElementById('ai-gen-spinner');
    const resBox = document.getElementById('ai-result-box');
    const resSlug = document.getElementById('ai-res-slug');
    const resPreview = document.getElementById('ai-res-preview
');

    spinner?.classList.remove('d-none');
    this.clearCreateAlert();

    try {
      const category = document.getElementById('ai-skill-categ
ory')?.value || 'general';
      const targetDir = document.getElementById('ai-skill-dir'
)?.value || '.agents/skills';

      const res = await this.apiFetch('/api/skills/generate-ai
', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ prompt, category, target_dir: t
argetDir })
```

```
758       });
759
760       if (!res.ok) {
761         const errData = await res.json().catch(() => ({}));
762         throw new Error(errData.detail || res.statusText);
763       }
764
765       const data = await res.json();
766       this.aiGeneratedData = data.generated;
767
768       if (resSlug) resSlug.textContent = `  ${this.aiGenerated
Data.name}`;
769       if (resPreview) resPreview.textContent = this.aiGenerate
dData.instructions;
770       resBox?.classList.remove('d-none');
771     } catch (err) {
772       console.error('[SkillsTab] AI generate error:', err);
773       this.showCreateAlert(`Ошибка генерации навыка: ${err.mes
sage}`, 'danger');
774     } finally {
775       spinner?.classList.add('d-none');
776     }
777   }
778
779   applyAiResultToWizard() {
780     if (!this.aiGeneratedData) return;
781     const g = this.aiGeneratedData;
782
783     document.getElementById('wizard-skill-name').value = g.nam
e || '';
784     document.getElementById('wizard-skill-desc-en').value = g.
description || '';
785     document.getElementById('wizard-skill-desc-ru').value = g.
description_ru || '';
786     document.getElementById('wizard-skill-category').value = g
.category || 'general';
787     document.getElementById('wizard-skill-dir').value = g.targ
et_dir || '.agents/skills';
```

```
    document.getElementById('wizard-skill-instructions').value
 = g.instructions || '';
    document.getElementById('wizard-skill-readme').value = g.r
eadme || '';

    if (g.scripts && g.scripts['main.py']) {
      document.getElementById('wizard-enable-script').checked
= true;
      document.getElementById('wizard-script-container').class
List.remove('d-none');
      document.getElementById('wizard-script-code').value = g.
scripts['main.py'];
    }

    if (g.references && g.references['guide.md']) {
      document.getElementById('wizard-enable-ref').checked = t
rue;
      document.getElementById('wizard-ref-container').classLis
t.remove('d-none');
      document.getElementById('wizard-ref-content').value = g.
references['guide.md'];
    }

    // Switch to wizard tab
    const wizardTabBtn = document.getElementById('mode-wizard-
tab');
    if (wizardTabBtn) {
      const tab = new bootstrap.Tab(wizardTabBtn);
      tab.show();
    }
    this.goToStep(1);
    this.syncWizardPreview();
  }

  async saveAiSkillDirectly() {
    if (!this.aiGeneratedData) return;
    const g = this.aiGeneratedData;

```

```
817     try {
818       const res = await this.apiFetch('/api/skills', {
819         method: 'POST',
820         headers: { 'Content-Type': 'application/json' },
821         body: JSON.stringify({
822           name: g.name,
823           description: g.description,
824           description_ru: g.description_ru,
825           category: g.category,
826           target_dir: g.target_dir,
827           instructions: g.instructions,
828           readme: g.readme,
829           scripts: g.scripts || {},
830           references: g.references || {}
831         })
832       });
833 
834       if (!res.ok) {
835         const errData = await res.json().catch(() => ({}));
836         throw new Error(errData.detail || res.statusText);
837       }
838 
839       const modalEl = document.getElementById('createSkillModa
    l');
840       const modal = bootstrap.Modal.getInstance(modalEl);
841       modal?.hide();
842 
843       await this.refresh();
844       await this.openSkillDetails(g.name);
845     } catch (err) {
846       console.error('[SkillsTab] AI Direct Save error:', err);
847       this.showCreateAlert(`Ошибка сохранения: ${err.message}`
    , 'danger');
848     }
849   }
850 
851   // ========================================================
    ================
```

```
  // PRESETS LIBRARY
  // ============================================================
================

  renderPresets() {
    const container = document.getElementById('presets-cards-c
ontainer');
    if (!container) return;

    container.innerHTML = this.presets.map(p => `
      <div class="col-12 col-md-6 col-lg-4">
        <div class="card bg-dark border-secondary text-white h
-100 shadow-sm">
          <div class="card-header border-secondary d-flex just
ify-content-between align-items-center bg-black bg-opacity-25
py-2">
            <span class="fw-bold small text-white">${p.title}<
/span>
            <span class="badge bg-primary-subtle text-primary
border border-primary-subtle small">${p.badge}</span>
          </div>
          <div class="card-body py-2">
            <p class="card-text text-muted small mb-2" style="
min-height: 48px;">
              ${p.description}
            </p>
            <div class="small font-monospace text-secondary mb
-2">
              <code>  ${p.name}</code>
            </div>
          </div>
          <div class="card-footer border-secondary bg-black bg
-opacity-25 py-2">
            <button type="button" class="btn btn-outline-prima
ry btn-sm w-100 d-flex align-items-center justify-content-cent
er gap-1" onclick="window.skillsTab?.applyPreset('${p.id}')">
              <i class="bi bi-box-arrow-in-right"></i> Использ
овать шаблон
```

```
            </button>
          </div>
        </div>
      </div>
    `).join('');
  }

  applyPreset(presetId) {
    const preset = this.presets.find(p => p.id === presetId);
    if (!preset) return;

    document.getElementById('wizard-skill-name').value = prese
t.name;
    document.getElementById('wizard-skill-desc-en').value = pr
eset.descEn;
    document.getElementById('wizard-skill-desc-ru').value = pr
eset.descRu;
    document.getElementById('wizard-skill-category').value = p
reset.category;
    document.getElementById('wizard-skill-dir').value = '.agen
ts/skills';

    document.getElementById('wizard-inst-purpose').value = pre
set.purpose;
    document.getElementById('wizard-inst-triggers').value = pr
eset.triggers;
    document.getElementById('wizard-inst-protocol').value = pr
eset.protocol;
    document.getElementById('wizard-inst-rules').value = prese
t.rules;
    this.syncInstructionsFromFields();

    // Scripts
    const scriptChk = document.getElementById('wizard-enable-s
cript');
    if (scriptChk) {
      scriptChk.checked = preset.hasScript;
      if (preset.hasScript) {
```

```
        document.getElementById('wizard-script-container')?.classList.remove('d-none');
        document.getElementById('wizard-script-code').value = preset.scriptCode || '';
      } else {
        document.getElementById('wizard-script-container')?.classList.add('d-none');
      }
    }

    // References
    const refChk = document.getElementById('wizard-enable-ref');
    if (refChk) {
      refChk.checked = preset.hasRef;
      if (preset.hasRef) {
        document.getElementById('wizard-ref-container')?.classList.remove('d-none');
        document.getElementById('wizard-ref-content').value = preset.refContent || '';
      } else {
        document.getElementById('wizard-ref-container')?.classList.add('d-none');
      }
    }

    // Switch to wizard tab
    const wizardTabBtn = document.getElementById('mode-wizard-tab');
    if (wizardTabBtn) {
      const tab = new bootstrap.Tab(wizardTabBtn);
      tab.show();
    }
    this.goToStep(1);
    this.syncWizardPreview();
  }

  showCreateAlert(msg, type = 'danger') {
```

```
    const alertEl = document.getElementById('create-skill-alert');
    if (alertEl) {
      alertEl.className = `alert alert-${type} py-2 mt-3 mb-0`;
      alertEl.textContent = msg;
      alertEl.classList.remove('d-none');
    }
  }

  clearCreateAlert() {
    const alertEl = document.getElementById('create-skill-alert');
    if (alertEl) alertEl.classList.add('d-none');
  }

  // ============================================================
  // SKILL DETAILS & SANDBOX TESTER
  // ============================================================

  async openSkillDetails(name, defaultTab = 'detail-tab-skillmd') {
    try {
      const res = await this.apiFetch(`/api/skills/${encodeURIComponent(name)}`);
      if (!res.ok) throw new Error(`Failed to load skill details: ${res.statusText}`);
      const data = await res.json();
      this.currentSkill = data.skill;

      // Populate modal fields
      document.getElementById('detail-skill-title').textContent = this.currentSkill.name;
      document.getElementById('detail-skill-path').textContent = this.currentSkill.relative_path;
      document.getElementById('detail-skill-desc-input').value
```

```
 = this.currentSkill.description || '';
963       document.getElementById('detail-skill-desc-ru-input').va
lue = this.currentSkill.descriptions_i18n?.ru || '';
964       document.getElementById('detail-skill-instructions').val
ue = this.currentSkill.instructions || '';
965       document.getElementById('detail-skill-readme').value = t
his.currentSkill.readme_raw || '';
966
967       // Populate files list
968       const filesContainer = document.getElementById('detail-f
iles-list');
969       if (filesContainer) {
970         if (this.currentSkill.files && this.currentSkill.files
.length > 0) {
971           filesContainer.innerHTML = this.currentSkill.files.m
ap(f => `
972             <div class="list-group-item bg-transparent text-li
ght border-secondary d-flex justify-content-between align-item
s-center py-1 small">
973               <span><i class="bi bi-file-earmark text-secondar
y"></i> ${f.rel_path}</span>
974               <span class="badge bg-secondary font-monospace">
${this.formatBytes(f.size)}</span>
975             </div>
976           `).join('');
977         } else {
978           filesContainer.innerHTML = '<div class="text-muted s
mall py-2">Файлы отсутствуют</div>';
979         }
980       }
981
982       // Populate JSON contract
983       const contractEl = document.getElementById('detail-json-
contract');
984       if (contractEl) {
985         const contractObj = {
986           name: this.currentSkill.name,
987           description: this.currentSkill.description,
```

```
          descriptions_i18n: this.currentSkill.descriptions_i18n,
          metadata: this.currentSkill.metadata,
          manifest: this.currentSkill.manifest,
        };
        contractEl.textContent = JSON.stringify(contractObj, null, 2);
      }

      // Reset sandbox box
      const sandboxOutput = document.getElementById('sandbox-output-box');
      if (sandboxOutput) {
        sandboxOutput.textContent = `Готов к тестированию навыка '${this.currentSkill.name}'. Введите тестовый запрос и нажмите «Запустить».`;
      }

      // Hide alerts
      const alertEl = document.getElementById('detail-skill-alert');
      if (alertEl) alertEl.classList.add('d-none');

      // Show modal
      const modalEl = document.getElementById('skillDetailsModal');
      const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
      modal.show();

      // Switch to requested tab
      const targetTabBtn = document.getElementById(defaultTab);
      if (targetTabBtn) {
        const tab = new bootstrap.Tab(targetTabBtn);
        tab.show();
      }
    } catch (err) {
```

```
      console.error('[SkillsTab] Details error:', err);
      window.showToast?.(`Ошибка открытия деталей навыка: ${err.message}`, 'danger') || alert(`Ошибка открытия деталей навыка: ${err.message}`);
    }
  }

  async openSkillSandbox(name) {
    await this.openSkillDetails(name, 'detail-tab-sandbox');
  }

  async runSandboxTest() {
    if (!this.currentSkill) return;
    const promptInput = document.getElementById('sandbox-test-prompt');
    const outputBox = document.getElementById('sandbox-output-box');
    const spinner = document.getElementById('sandbox-spinner');

    const prompt = promptInput?.value.trim() || `Test execute skill ${this.currentSkill.name}`;
    spinner?.classList.remove('d-none');
    if (outputBox) outputBox.textContent = `⏳ Отправка запроса к модели для навыка '${this.currentSkill.name}'...\n`;

    try {
      const res = await this.apiFetch(`/api/skills/${encodeURIComponent(this.currentSkill.name)}/test`, {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ prompt })
      });

      if (!res.ok) {
        const errData = await res.json().catch(() => ({}));
        throw new Error(errData.detail || res.statusText);
      }
```

```
1047
1048       const data = await res.json();
1049       if (outputBox) {
1050         outputBox.textContent = `[Model: ${data.model_used}]\n
\n${data.response}`;
1051       }
1052     } catch (err) {
1053       console.error('[SkillsTab] Sandbox test error:', err);
1054       if (outputBox) outputBox.textContent = `  Ошибка тестиро
вания: ${err.message}`;
1055     } finally {
1056       spinner?.classList.add('d-none');
1057     }
1058   }
1059
1060   async saveCurrentSkill() {
1061     if (!this.currentSkill) return;
1062     const spinner = document.getElementById('save-skill-spinne
r');
1063     const alertEl = document.getElementById('detail-skill-aler
t');
1064
1065     spinner?.classList.remove('d-none');
1066     try {
1067       const description = document.getElementById('detail-skil
l-desc-input').value.trim();
1068       const description_ru = document.getElementById('detail-s
kill-desc-ru-input').value.trim();
1069       const instructions = document.getElementById('detail-ski
ll-instructions').value;
1070       const readme = document.getElementById('detail-skill-rea
dme').value;
1071
1072       const res = await this.apiFetch(`/api/skills/${encodeURI
Component(this.currentSkill.name)}`, {
1073         method: 'PUT',
1074         headers: { 'Content-Type': 'application/json' },
1075         body: JSON.stringify({ description, description_ru, in
```

```
structions, readme })
      });

      if (!res.ok) {
        const errData = await res.json().catch(() => ({}));
        throw new Error(errData.detail || res.statusText);
      }

      if (alertEl) {
        alertEl.className = 'alert alert-success py-2 mt-3 mb-0';
        alertEl.textContent = '✅ Изменения успешно сохранены';
        alertEl.classList.remove('d-none');
      }
      window.showToast?.('Изменения навыка успешно сохранены', 'success');

      await this.refresh();
    } catch (err) {
      console.error('[SkillsTab] Save error:', err);
      if (alertEl) {
        alertEl.className = 'alert alert-danger py-2 mt-3 mb-0';
        alertEl.textContent = `Ошибка сохранения: ${err.message}`;
        alertEl.classList.remove('d-none');
      }
      window.showToast?.(`Ошибка сохранения: ${err.message}`, 'danger');
    } finally {
      spinner?.classList.add('d-none');
    }
  }

  async packageSkill(name) {
    try {
      const res = await this.apiFetch(`/api/skills/${encodeURIComponent(name)}/package`, {
```

```
        method: 'POST'
      });
      if (!res.ok) {
        const errData = await res.json().catch(() => ({}));
        throw new Error(errData.detail || res.statusText);
      }
      const data = await res.json();
      window.showToast?.(`  Навык '${name}' успешно упакован в
 архив:\n${data.archive?.path} (${this.formatBytes(data.archiv
e?.size)})`, 'success');
      await this.refresh();
    } catch (err) {
      console.error('[SkillsTab] Package error:', err);
      window.showToast?.(`Ошибка сборки архива: ${err.message}
`, 'danger') || alert(`Ошибка сборки архива: ${err.message}`);
    }
  }

  async packageCurrentSkill() {
    if (!this.currentSkill) return;
    await this.packageSkill(this.currentSkill.name);
    await this.openSkillDetails(this.currentSkill.name);
  }

  async deleteSkill(name) {
    if (!confirm(`Вы действительно хотите удалить навык '${nam
e}' и все его файлы? Это действие необратимо.`)) {
      return;
    }

    try {
      const res = await this.apiFetch(`/api/skills/${encodeURI
Component(name)}`, {
        method: 'DELETE'
      });
      if (!res.ok) {
        const errData = await res.json().catch(() => ({}));
        throw new Error(errData.detail || res.statusText);
```

```
      }

      const modalEl = document.getElementById('skillDetailsModal');
      const modal = bootstrap.Modal.getInstance(modalEl);
      modal?.hide();

      window.showToast?.(`Навык '${name}' успешно удален`, 'success');
      await this.refresh();
    } catch (err) {
      console.error('[SkillsTab] Delete error:', err);
      window.showToast?.(`Ошибка удаления навыка: ${err.message}`, 'danger') || alert(`Ошибка удаления навыка: ${err.message}`);
    }
  }

  async deleteCurrentSkill() {
    if (!this.currentSkill) return;
    await this.deleteSkill(this.currentSkill.name);
  }

  showLoading(show) {
    const loadingEl = document.getElementById('skills-loading');
    if (loadingEl) {
      if (show) loadingEl.classList.remove('d-none');
      else loadingEl.classList.add('d-none');
    }
  }

  formatBytes(bytes) {
    if (!bytes || bytes === 0) return '0 B';
    const k = 1024;
    const sizes = ['B', 'KB', 'MB', 'GB'];
    const i = Math.floor(Math.log(bytes) / Math.log(k));
    return parseFloat((bytes / Math.pow(k, i)).toFixed(1)) + '
```

```
 ' + sizes[i];
  }
}

// Global initialization hook
window.SkillsTabManager = SkillsTabManager;
if (!window.skillsTab) {
  window.skillsTab = new SkillsTabManager();
}
window.initSkillsTab = function() {
  if (window.skillsTab) {
    window.skillsTab.init();
  }
};

// Auto-init on script load if container is present
if (document.getElementById('skills-tab-root')) {
  window.skillsTab.init();
}
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/software_audit_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/software_audit_tab/inde
x.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/software_audit_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Software Audit Application Tab -->
<style>
  .sw-audit-container {
    padding: 0.5rem 0.75rem;
  }
  .sw-audit-card {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: 8px;
    padding: 0.45rem 0.75rem;
```

```
    display: flex;
    flex-direction: column;
    justify-content: space-between;
    min-height: 58px;
    transition: border-color 0.15s ease;
  }
  .sw-audit-card:hover {
    border-color: var(--nav-active) !important;
  }
  .sw-audit-label {
    font-size: 0.68rem;
    text-transform: uppercase;
    letter-spacing: 0.05em;
    color: var(--text-muted);
    line-height: 1.1;
  }
  .sw-audit-value {
    font-size: 1.25rem;
    font-weight: 700;
    color: var(--text-color);
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
    line-height: 1.15;
  }
  .sw-audit-panel {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: 8px;
    overflow: hidden;
  }
  .sw-audit-panel-header {
    background: var(--surface-2) !important;
    border-bottom: 1px solid var(--border-color) !important;
    padding: 0.45rem 0.85rem;
    font-weight: 600;
    font-size: 0.82rem;
    color: var(--nav-active);
    display: flex;
```

```
    align-items: center;
    justify-content: space-between;
    flex-wrap: wrap;
    gap: 0.35rem;
  }
  .sw-audit-table {
    width: 100%;
    font-size: 0.78rem;
    border-collapse: collapse;
  }
  .sw-audit-table th {
    background: var(--surface-2);
    padding: 6px 10px;
    color: var(--text-muted);
    font-weight: 600;
    border-bottom: 1px solid var(--border-color);
    position: sticky;
    top: 0;
    z-index: 2;
  }
  .sw-audit-table td {
    padding: 6px 10px;
    border-bottom: 1px solid var(--border-color);
    color: var(--text-color);
    vertical-align: middle;
  }
  .sw-audit-table tbody tr:hover {
    background: var(--nav-hover-bg, rgba(56, 189, 248, 0.05));
  }
  .sw-badge-category {
    font-size: 0.7rem;
    padding: 2px 6px;
    border-radius: 4px;
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    color: var(--text-color);
  }
  .sw-badge-active {
```

```
    background: rgba(34, 197, 94, 0.15);
    border: 1px solid rgba(34, 197, 94, 0.4);
    color: #4ade80;
  }
  .sw-badge-dormant {
    background: rgba(245, 158, 11, 0.15);
    border: 1px solid rgba(245, 158, 11, 0.4);
    color: #fbbf24;
  }
</style>
<div class="sw-audit-container">
<!-- Header Alert & Controls (Compact Strip) -->
<div class="card shadow-sm border-secondary-subtle py-1.5 px-3
 mb-2 d-flex justify-content-between align-items-center flex-w
rap gap-2" data-panel-id="panel-software-audit-audit-programmn
ogo-obespecheniya" id="panel-software-audit-audit-programmnogo
-obespecheniya">
<div class="d-flex align-items-center gap-2 flex-wrap">
<span class="fs-5">□ </span>
<span class="fw-bold text-body small" style="font-size: 0.92re
m;">Аудит программного обеспечения</span>
<span class="badge bg-body-secondary border border-secondary t
ext-body fw-normal" style="font-size: 0.7rem;">Software Audit<
/span>
<span class="text-secondary small d-none d-lg-inline" style="f
ont-size: 0.76rem;">| Инвентаризация ПО, UserAssist &amp;amp; Pref
etch</span>
</div>
<div class="d-flex align-items-center gap-1.5 flex-wrap">
<span class="badge rounded-pill bg-success-subtle text-success
 border border-success px-2 py-1" id="sw-audit-status-badge" s
tyle="font-size: 0.72rem;">
        ● Готов
      </span>
<button class="btn btn-xs btn-outline-info rounded-pill px-2.5
 py-1" id="btn-sw-audit-refresh" style="font-size: 0.75rem;">
<i class="bi bi-arrow-clockwise me-1"></i> Обновить
      </button>
```

```
<button class="btn btn-xs btn-outline-secondary rounded-pill p
x-2 py-1" id="btn-sw-audit-export" style="font-size: 0.75rem;"
>
<i class="bi bi-download me-1"></i> Экспорт JSON
      </button>
</div>
</div>
<!-- Metric Statistics Cards (Compact Density Grid) -->
<div class="row g-2 mb-2">
<div class="col-6 col-lg-3">
<div class="sw-audit-card">
<span class="sw-audit-label">Всего программ</span>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<span class="sw-audit-value text-info" id="sw-metric-total">0<
/span>
<small class="text-muted" style="font-size: 0.7rem;">В реестре
</small>
</div>
</div>
</div>
<div class="col-6 col-lg-3">
<div class="sw-audit-card">
<span class="sw-audit-label">Активно используются</span>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<span class="sw-audit-value text-success" id="sw-metric-active
">0</span>
<small class="text-muted" style="font-size: 0.7rem;">Недавние
запуски</small>
</div>
</div>
</div>
<div class="col-6 col-lg-3">
<div class="sw-audit-card">
<span class="sw-audit-label">Неиспользуемые (&gt;90 дн)</span>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
```

```
<span class="sw-audit-value text-warning" id="sw-metric-dormant">0</span>
<small class="text-muted" style="font-size: 0.7rem;">Кандидаты на удаление</small>
</div>
</div>
</div>
<div class="col-6 col-lg-3">
<div class="sw-audit-card">
<span class="sw-audit-label">Категоризировано</span>
<div class="d-flex align-items-baseline justify-content-between mt-1">
<span class="sw-audit-value text-info" id="sw-metric-categorized">0%</span>
<small class="text-muted" style="font-size: 0.7rem;">AI эвристика</small>
</div>
</div>
</div>
</div>
<!-- Filters & Controls Bar -->
<div class="card p-2 mb-2" data-panel-id="panel-software-audit-1" id="panel-software-audit-1" style="border-radius: 8px; background: var(--surface-1); border: 1px solid var(--border-color);">
<div class="row g-2 align-items-center">
<div class="col-md-5">
<div class="input-group input-group-sm">
<span class="input-group-text bg-body-secondary border-secondary text-body"><i class="bi bi-search"></i></span>
<input class="form-control form-control-sm" id="sw-search-input" placeholder="Поиск по названию, издателю или назначению..." type="text"/>
</div>
</div>
<div class="col-md-3">
<select class="form-select form-select-sm" id="sw-category-filter">
```

```
<option value="">Все категории</option>
</select>
</div>
<div class="col-md-2">
<select class="form-select form-select-sm" id="sw-status-filte
r">
<option value="all">Все программы</option>
<option value="active">Только активные</option>
<option value="dormant">Только неиспользуемые</option>
</select>
</div>
<div class="col-md-2 text-end">
<span class="badge bg-secondary" id="sw-filtered-count-badge">
0 / 0</span>
</div>
</div>
</div>
<!-- Main Table Panel -->
<div class="sw-audit-panel">
<div class="sw-audit-panel-header">
<span><i class="bi bi-table me-1"></i> Список установленных пр
ограммных продуктов</span>
<span class="small text-muted" id="sw-last-scanned-time">Скани
рование...</span>
</div>
<div class="p-0 table-responsive" style="max-height: 540px; ov
erflow-y: auto;">
<table class="sw-audit-table">
<thead>
<tr>
<th>Программа</th>
<th>Категория</th>
<th>Издатель / Версия</th>
<th>Последний запуск</th>
<th>Запусков</th>
<th>Время работы</th>
<th>Размер / Арх</th>
<th style="text-align: right;">Действия</th>
```

```
</tr>
</thead>
<tbody id="sw-audit-tbody">
<tr><td class="text-center text-muted p-4" colspan="8"><i clas
s="bi bi-arrow-repeat spin me-1"></i> Загрузка аудита программ
ного обеспечения...</td></tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- Modal: App Details -->
<div aria-hidden="true" aria-labelledby="swAppDetailModalLabel
" class="modal fade" id="swAppDetailModal" tabindex="-1">
<div class="modal-dialog modal-lg modal-dialog-centered">
<div class="modal-content">
<div class="modal-header border-secondary">
<h5 class="modal-title d-flex align-items-center gap-2" id="sw
AppDetailModalLabel">
<span class="fs-5">□ </span> <span id="sw-modal-app-name">Дета
ли программы</span>
</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-3" id="sw-modal-app-body">
<!-- Content will be injected by main.js -->
</div>
<div class="modal-footer border-secondary">
<button class="btn btn-secondary btn-sm" data-bs-dismiss="moda
l" type="button">Закрыть</button>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Software Audit Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/software_audit_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/software_audit_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// Software Audit Tab JavaScript Module
(function() {
  let allApps = [];
  let auditReport = null;
  let isInitialized = false;

  function formatRelativeTime(dateStr) {
    if (!dateStr) return '<span class="text-muted">Нет данных</span>';
    const d = new Date(dateStr);
    if (isNaN(d.getTime())) return '<span class="text-muted">-</span>';
```

```
    const now = new Date();
    const diffMs = now - d;
    const diffSec = Math.floor(diffMs / 1000);
    const diffMin = Math.floor(diffSec / 60);
    const diffHrs = Math.floor(diffMin / 60);
    const diffDays = Math.floor(diffHrs / 24);

    if (diffDays > 90) {
      return `<span class="text-warning" title="${d.toLocaleSt
ring()}">${diffDays} дн назад</span>`;
    } else if (diffDays > 0) {
      return `<span class="text-success" title="${d.toLocaleSt
ring()}">${diffDays} дн назад</span>`;
    } else if (diffHrs > 0) {
      return `<span class="text-info" title="${d.toLocaleStrin
g()}">${diffHrs} ч назад</span>`;
    } else if (diffMin > 0) {
      return `<span class="text-info" title="${d.toLocaleStrin
g()}">${diffMin} мин назад</span>`;
    } else {
      return `<span class="text-info">Только что</span>`;
    }
  }

  function formatDuration(sec) {
    if (!sec || sec <= 0) return '<span class="text-muted">-</
span>';
    const hrs = Math.floor(sec / 3600);
    const mins = Math.floor((sec % 3600) / 60);
    if (hrs > 0) return `${hrs}ч ${mins}м`;
    if (mins > 0) return `${mins} мин`;
    return `${sec} сек`;
  }

  async function loadSoftwareAudit() {
    const tbody = document.getElementById('sw-audit-tbody');
    const badge = document.getElementById('sw-audit-status-bad
ge');
```

```
    if (badge) badge.innerText = '● Сканирование реестра...';

    try {
      // 1. Пытаемся получить сводный аналитический отчет
      let reportRes = await fetch('/api/windows/software/audit
');
      if (!reportRes.ok) {
        reportRes = await fetch('/api/v1/windows-admin/softwar
e/audit');
      }
      if (reportRes.ok) {
        auditReport = await reportRes.json();
      }

      // 2. Получаем полный список установленных приложений
      let appsRes = await fetch('/api/windows/software?limit=5
00');
      if (!appsRes.ok) {
        // Fallback на router_windows_admin
        appsRes = await fetch('/api/v1/windows-admin/software?
limit=500');
      }

      if (appsRes.ok) {
        allApps = await appsRes.json();
      } else {
        throw new Error(`HTTP ${appsRes.status}`);
      }

      // Обновляем метрики
      updateMetrics();
      // Заполняем выпадающий список категорий
      populateCategories();
      // Рендерим таблицу
      renderTable();

      if (badge) {
        badge.className = 'badge rounded-pill bg-success-subtl
```

```
e text-success border border-success px-3 py-2';
97          badge.innerText = `● Аудит: Завершено (${allApps.lengt
h} программ)`;
98        }
99
100       const timeEl = document.getElementById('sw-last-scanned-
time');
101       if (timeEl) {
102         timeEl.innerText = `Обновлено: ${new Date().toLocaleTi
meString()}`;
103       }
104     } catch (err) {
105       console.error('[SoftwareAuditTab] Failed to load softwar
e audit:', err);
106       if (tbody) {
107         tbody.innerHTML = `<tr><td colspan="8" class="text-cen
ter text-danger p-4">Ошибка сбора сведений об установленном ПО
: ${err.message}</td></tr>`;
108       }
109       if (badge) {
110         badge.className = 'badge rounded-pill bg-danger-subtle
 text-danger border border-danger px-3 py-2';
111         badge.innerText = '● Ошибка аудита';
112       }
113     }
114   }
115
116   function updateMetrics() {
117     const totalEl = document.getElementById('sw-metric-total')
;
118     const activeEl = document.getElementById('sw-metric-active
');
119     const dormantEl = document.getElementById('sw-metric-dorma
nt');
120     const catEl = document.getElementById('sw-metric-categorie
s');
121
122     if (auditReport) {
```

```
      if (totalEl) totalEl.innerText = auditReport.total_apps || allApps.length;
      if (activeEl) activeEl.innerText = auditReport.active_apps_count || 0;
      if (dormantEl) dormantEl.innerText = auditReport.never_launched_or_dormant_count || 0;
      if (catEl) catEl.innerText = Object.keys(auditReport.categories_breakdown || {}).length;
    } else {
      const activeCount = allApps.filter(a => a.execution_info && a.execution_info.last_run_time).length;
      if (totalEl) totalEl.innerText = allApps.length;
      if (activeEl) activeEl.innerText = activeCount;
      if (dormantEl) dormantEl.innerText = allApps.length - activeCount;
      const cats = new Set(allApps.map(a => a.category).filter(Boolean));
      if (catEl) catEl.innerText = cats.size;
    }
  }

  function populateCategories() {
    const sel = document.getElementById('sw-category-filter');
    if (!sel) return;

    const currentVal = sel.value;
    const cats = Array.from(new Set(allApps.map(a => a.category).filter(Boolean))).sort();

    sel.innerHTML = '<option value="">Все категории</option>' + cats.map(c =>
      `<option value="${c}" ${c === currentVal ? 'selected' : ''}>${c}</option>`
    ).join('');
  }

  function getFilteredApps() {
    const search = (document.getElementById('sw-search-input')
```

```
?.value || '').toLowerCase().trim();
    const category = document.getElementById('sw-category-filter')?.value || '';
    const status = document.getElementById('sw-status-filter')?.value || 'all';

    const dormantThresholdMs = 90 * 24 * 3600 * 1000;
    const now = Date.now();

    return allApps.filter(app => {
      // Поиск по ключевым полям
      if (search) {
        const text = `${app.display_name || ''} ${app.name || ''} ${app.publisher || ''} ${app.purpose_description || ''}`.toLowerCase();
        if (!text.includes(search)) return false;
      }

      // Фильтр по категории
      if (category && app.category !== category) {
        return false;
      }

      // Фильтр по статусу использования
      const lastRun = app.execution_info?.last_run_time ? new Date(app.execution_info.last_run_time).getTime() : null;
      if (status === 'active') {
        if (!lastRun || (now - lastRun > dormantThresholdMs)) return false;
      } else if (status === 'dormant') {
        if (lastRun && (now - lastRun <= dormantThresholdMs)) return false;
      }

      return true;
    });
  }

```

```
181   function renderTable() {
182     const tbody = document.getElementById('sw-audit-tbody');
183     const badge = document.getElementById('sw-filtered-count-b
adge');
184     if (!tbody) return;
185 
186     const filtered = getFilteredApps();
187     if (badge) {
188       badge.innerText = `${filtered.length} / ${allApps.length
}`;
189     }
190 
191     if (filtered.length === 0) {
192       tbody.innerHTML = '<tr><td colspan="8" class="text-cente
r text-muted p-4">Программы по заданным критериям не найдены</
td></tr>';
193       return;
194     }
195 
196     tbody.innerHTML = filtered.map((app, idx) => {
197       const exec = app.execution_info;
198       const lastRunFormatted = exec ? formatRelativeTime(exec.
last_run_time) : '<span class="text-muted">Никогда</span>';
199       const runCount = exec && exec.run_count ? `<span class="
badge bg-secondary">${exec.run_count}</span>` : '<span class="
text-muted">0</span>';
200       const focusTime = exec ? formatDuration(exec.focus_time_
seconds) : '<span class="text-muted">-</span>';
201       const sizeStr = app.size_mb > 0 ? `${app.size_mb} MB` :
'<span class="text-muted">-</span>';
202       const archBadge = app.architecture === 'x64' ? '<span cl
ass="badge bg-primary-subtle text-primary" style="font-size: 0
.65rem;">x64</span>' : '<span class="badge bg-secondary-subtle
 text-secondary" style="font-size: 0.65rem;">x86</span>';
203 
204       return `
205         <tr class="sw-row-item" data-idx="${idx}" style="curso
r: pointer;" title="Нажмите для AI-диагностики и подробных све
```

```
дений">
          <td>
            <div class="fw-bold text-white text-truncate" styl
e="max-width: 260px;" title="${app.display_name || app.name}">
              ${app.display_name || app.name}
            </div>
            <div class="small text-muted text-truncate" style=
"max-width: 260px; font-size: 0.72rem;" title="${app.purpose_d
escription || ''}">
              ${app.purpose_description || 'Прикладное ПО'}
            </div>
          </td>
          <td>
            <span class="sw-badge-category">${app.category ||
'Прочее'}</span>
          </td>
          <td>
            <div class="text-truncate text-light" style="max-w
idth: 180px;" title="${app.publisher || 'Неизвестен'}">${app.p
ublisher || '<span class="text-muted">-</span>'}</div>
            <div class="small text-muted" style="font-size: 0.
72rem;">${app.version || '-'}</div>
          </td>
          <td>${lastRunFormatted}</td>
          <td>${runCount}</td>
          <td class="small text-light">${focusTime}</td>
          <td class="small">
            <div>${sizeStr}</div>
            <div>${archBadge}</div>
          </td>
          <td style="text-align: right;">
            <button class="btn btn-sm btn-outline-info p-1 px-
2 btn-sw-detail" data-idx="${idx}" title="Подробности">
              <i class="bi bi-info-circle"></i>
            </button>
          </td>
        </tr>
      `;
```

```
    }).join('');

    // Привязка кликабельности всей строки
    tbody.querySelectorAll('.sw-row-item').forEach(row => {
      row.onclick = (evt) => {
        const idx = parseInt(row.getAttribute('data-idx'), 10);
        showAppDetails(filtered[idx]);
      };
    });

    // Привязка обработчиков кнопки деталей
    tbody.querySelectorAll('.btn-sw-detail').forEach(btn => {
      btn.onclick = (evt) => {
        evt.stopPropagation();
        const idx = parseInt(btn.getAttribute('data-idx'), 10);
        showAppDetails(filtered[idx]);
      };
    });
  }

  function showAppDetails(app) {
    if (!app) return;
    const exec = app.execution_info;

    if (window.AITableModal) {
      window.AITableModal.show({
        icon: '□ ',
        title: app.display_name || app.name,
        subtitle: app.publisher || 'Неизвестный разработчик',
        tableType: 'software',
        badges: [
          { text: app.category || 'ПО', class: 'badge bg-info' },
          { text: app.architecture || 'x64', class: 'badge bg-secondary' }
        ],
```

```
        metadata: [
          { label: 'Продукт', value: app.display_name || app.n
ame },
          { label: 'Издатель', value: app.publisher || 'Не ука
зан' },
          { label: 'Версия', value: app.version || 'Не указана
' },
          { label: 'Категория', value: app.category || 'Прочее
' },
          { label: 'Размер на диске', value: app.size_mb > 0 ?
 `${app.size_mb} MB` : 'Не указан' },
          { label: 'Дата установки', value: app.install_date |
| 'Неизвестно' },
          { label: 'Запусков (UserAssist)', value: exec?.run_c
ount ? `${exec.run_count} раз` : '0' },
          { label: 'Время в фокусе', value: exec ? formatDurat
ion(exec.focus_time_seconds) : '-' },
          { label: 'Последний запуск', value: exec?.last_run_t
ime ? new Date(exec.last_run_time).toLocaleString() : 'Нет дан
ных', fullWidth: true },
          { label: 'Путь установки', value: app.install_locati
on || 'Не указан', isCode: true, fullWidth: true }
        ],
        rawTitle: 'Команда удаления / Путь',
        rawContent: app.uninstall_string || app.install_locati
on || '',
        requestData: {
          install_location: app.install_location,
          uninstall_string: app.uninstall_string,
          version: app.version,
          size_mb: app.size_mb,
          run_count: exec?.run_count
        }
      });
      return;
    }
  }

```

```
  function exportJson() {
    const dataStr = "data:text/json;charset=utf-8," + encodeURIComponent(JSON.stringify(allApps, null, 2));
    const dlAnchor = document.createElement('a');
    dlAnchor.setAttribute("href", dataStr);
    dlAnchor.setAttribute("download", `software_audit_${new Date().toISOString().slice(0, 10)}.json`);
    document.body.appendChild(dlAnchor);
    dlAnchor.click();
    dlAnchor.remove();
  }

  window.initSoftwareAuditTab = async function() {
    console.log('[SoftwareAuditTab] Initializing Software Audit tab...');

    // Привязываем контролы поиска и фильтрации
    const searchInput = document.getElementById('sw-search-input');
    const catSelect = document.getElementById('sw-category-filter');
    const statusSelect = document.getElementById('sw-status-filter');
    const refreshBtn = document.getElementById('btn-sw-audit-refresh');
    const exportBtn = document.getElementById('btn-sw-audit-export');

    if (searchInput) searchInput.oninput = renderTable;
    if (catSelect) catSelect.onchange = renderTable;
    if (statusSelect) statusSelect.onchange = renderTable;
    if (refreshBtn) refreshBtn.onclick = loadSoftwareAudit;
    if (exportBtn) exportBtn.onclick = exportJson;

    await loadSoftwareAudit();
    isInitialized = true;
  };
})();
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/software_manager_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/software_manager_tab/in
dex.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/software_manager_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Windows Software Manager Tab -->
<div class="container-fluid py-2 px-3">
<!-- Top Stat Cards -->
<div class="row g-2 mb-3">
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 sha
dow-sm p-3" data-panel-id="panel-software-manager-1" id="panel
-software-manager-1">
<div class="text-muted small text-uppercase fw-semibold">Устан
овлено пакетов □ </div>
```

```
<div class="fs-4 fw-bold mt-1 text-primary" id="sw-installed-count">--</div>
<div class="small text-muted mt-1" id="sw-installed-sub">winget list / MSI</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-software-manager-2" id="panel-software-manager-2">
<div class="text-muted small text-uppercase fw-semibold">Доступно обновлений  </div>
<div class="fs-4 fw-bold mt-1 text-warning" id="sw-upgrades-count">--</div>
<div class="small text-muted mt-1" id="sw-upgrades-sub">winget upgrade</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-software-manager-3" id="panel-software-manager-3">
<div class="text-muted small text-uppercase fw-semibold">Источники пакетов (Sources)  </div>
<div class="fs-4 fw-bold mt-1 text-info" id="sw-sources-count">--</div>
<div class="small text-muted mt-1" id="sw-sources-sub">winget / msstore</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-software-manager-4" id="panel-software-manager-4">
<div class="text-muted small text-uppercase fw-semibold">MSI Инсталлятор  </div>
<div class="fs-4 fw-bold mt-1 text-success" id="sw-msi-status">Готов</div>
```

```
<div class="small text-muted mt-1" id="sw-msi-sub">msiexec.exe
 engine</div>
</div>
</div>
</div>
<!-- Action Bar -->
<div class="d-flex align-items-center justify-content-between
mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap
-2">
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm btn-primary d-flex align-items-cente
r gap-1.5" id="sw-refresh-btn">
<i class="bi bi-arrow-clockwise"></i>
<span>Обновить список</span>
</button>
<button class="btn btn-sm btn-outline-warning d-flex align-ite
ms-center gap-1.5" id="sw-check-upgrades-btn">
<i class="bi bi-arrow-up-circle"></i>
<span>Проверить обновления</span>
</button>
</div>
<div class="d-flex align-items-center gap-2">
<input class="form-control form-control-sm bg-dark text-white
border-secondary" id="sw-search-input" placeholder="Поиск паке
та по имени или ID..." style="width: 260px;" type="text"/>
</div>
</div>
<!-- Software Table -->
<div class="card bg-dark text-white border-secondary shadow-sm
" data-panel-id="panel-software-manager-ustanovlennye-prilozhe
niya-i-pak" id="panel-software-manager-ustanovlennye-prilozhen
iya-i-pak">
<div class="card-header border-secondary d-flex align-items-ce
nter justify-content-between py-2 px-3">
<span class="fw-semibold"><i class="bi bi-box-seam text-primar
y me-2"></i>Установленные приложения и пакеты (Winget &amp;amp; MS
I)</span>
<span class="badge bg-secondary" id="sw-table-badge">0 пакетов
```

```
</span>
</div>
<div class="card-body p-0 table-responsive" style="max-height: 520px; overflow-y: auto;">
<table class="table table-dark table-hover table-striped align-middle mb-0 small" id="sw-table">
<thead class="sticky-top bg-dark">
<tr class="text-secondary">
<th>Имя приложения</th>
<th>Идентификатор (ID)</th>
<th>Версия</th>
<th>Доступная версия</th>
<th>Источник</th>
<th class="text-end">Действия</th>
</tr>
</thead>
<tbody id="sw-tbody">
<tr><td class="text-center text-muted py-4" colspan="6">Опрос установленных программ через winget...</td></tr>
</tbody>
</table>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Software Manager Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/software_manager_tab/main.js?v=20261004_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { initSoftwareManagerTab } from '/windows/api/webgui/software_manager_tab/main.js';
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/software_manager_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * software_manager_tab/main.js — Управление пакетами и программами (winget, msiexec)
 * Updated: 2026-10-10 17:35:00
 */

const registerTabPoller = window.registerTabPoller || function () {};
```

```
30
31 let isInitialized = false;
32 let packagesList = [];
33
34 export async function fetchSoftwareSummary() {
35   try {
36     const res = await fetch('/api/software-manager/summary');
37     if (!res.ok) throw new Error(`HTTP ${res.status}`);
38     const data = await res.json();
39     renderSoftware(data);
40   } catch (err) {
41     console.warn('[SoftwareManager] Ошибка получения списка ПО
   :', err);
42     const tbody = document.getElementById('sw-tbody');
43     if (tbody && packagesList.length === 0) {
44       tbody.innerHTML = `<tr><td colspan="6" class="text-cente
   r text-danger py-4">Ошибка загрузки данных: ${err.message}</td
   ></tr>`;
45     }
46   }
47 }
48
49 function renderSoftware(data) {
50   packagesList = data.packages || [];
51
52   const installedEl = document.getElementById('sw-installed-co
   unt');
53   const upgradesEl = document.getElementById('sw-upgrades-coun
   t');
54   const sourcesEl = document.getElementById('sw-sources-count'
   );
55
56   if (installedEl) installedEl.textContent = data.total_packag
   es ?? packagesList.length;
57   if (upgradesEl) upgradesEl.textContent = data.updates_availa
   ble_count ?? data.upgrades_count ?? 0;
58   if (sourcesEl) sourcesEl.textContent = data.sources_count ??
    (data.winget_available ? 'winget' : 'MSI');
```

```
59
60    applySoftwareFilters();
61  }
62
63  function applySoftwareFilters() {
64    const tbody = document.getElementById('sw-tbody');
65    const searchInput = document.getElementById('sw-search-input
  ');
66    const badge = document.getElementById('sw-table-badge');
67
68    if (!tbody) return;
69
70    const query = (searchInput ? searchInput.value : '').toLower
  Case().trim();
71
72    const filtered = packagesList.filter(p => {
73      if (!query) return true;
74      const nameMatch = (p.name || '').toLowerCase().includes(qu
  ery);
75      const idMatch = (p.package_id || p.id || '').toLowerCase()
  .includes(query);
76      return nameMatch || idMatch;
77    });
78
79    if (badge) badge.textContent = `${filtered.length} из ${pack
  agesList.length} пакетов`;
80
81    if (filtered.length === 0) {
82      tbody.innerHTML = '<tr><td colspan="6" class="text-center
  text-muted py-4">Пакеты не найдены</td></tr>';
83      return;
84    }
85
86    tbody.innerHTML = filtered.slice(0, 150).map(p => {
87      const name = p.name || 'Application';
88      const id = p.package_id || p.id || '-';
89      const ver = p.version || '-';
90      const avail = p.available_version || p.latest_version || '
```

```
-';
    const source = p.source || 'winget';
    const hasUpgrade = avail && avail !== '-' && avail !== ver;

    return `
      <tr>
        <td class="fw-semibold text-light">${name}</td>
        <td class="font-monospace text-info small">${id}</td>
        <td class="text-muted small">${ver}</td>
        <td>
          <span class="badge ${hasUpgrade ? 'bg-warning text-dark' : 'bg-secondary'}">
            ${avail}
          </span>
        </td>
        <td class="small text-muted">${source}</td>
        <td class="text-end">
          ${hasUpgrade ? `
            <button class="btn btn-xs btn-outline-warning py-0 px-2 sw-action-btn" data-id="${id}" data-action="upgrade" title="Обновить пакет">
              <i class="bi bi-arrow-up-circle"></i> Обновить
            </button>
          ` : `
            <button class="btn btn-xs btn-outline-danger py-0 px-2 sw-action-btn" data-id="${id}" data-action="uninstall" title="Удалить пакет">
              <i class="bi bi-trash"></i>
            </button>
          `}
        </td>
      </tr>
    `;
  }).join('');

  tbody.querySelectorAll('.sw-action-btn').forEach(btn => {
    btn.addEventListener('click', () => {
```

```
      const pkgId = btn.dataset.id;
      const act = btn.dataset.action;
      executeSoftwareAction(pkgId, act);
    });
  });
}

async function executeSoftwareAction(packageId, action) {
  if (window.toast) window.toast.info('Пакетный менеджер', `${action}: ${packageId}`);
  try {
    const res = await fetch('/api/software-manager/action', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ package_id: packageId, action: action, dry_run: false, confirmed_by_user: true })
    });
    const result = await res.json();
    if (res.ok) {
      if (window.toast) window.toast.success('Успех', `${packageId}: выполнено`);
      fetchSoftwareSummary();
    } else {
      if (window.toast) window.toast.error('Ошибка', result.detail || result.message || 'Сбой операции');
    }
  } catch (err) {
    if (window.toast) window.toast.error('Ошибка сети', err.message);
  }
}

export function initSoftwareManagerTab() {
  if (isInitialized) {
    fetchSoftwareSummary();
    return;
  }
  isInitialized = true;
```

```
155
156   const refreshBtn = document.getElementById('sw-refresh-btn')
;
157   if (refreshBtn) refreshBtn.addEventListener('click', fetchSo
ftwareSummary);
158
159   const checkUpgradesBtn = document.getElementById('sw-check-u
pgrades-btn');
160   if (checkUpgradesBtn) checkUpgradesBtn.addEventListener('cli
ck', () => {
161     if (window.toast) window.toast.info('Проверка', 'Поиск дос
тупных обновлений...');
162     fetchSoftwareSummary();
163   });
164
165   const searchInput = document.getElementById('sw-search-input
');
166   if (searchInput) searchInput.addEventListener('input', apply
SoftwareFilters);
167
168   registerTabPoller('tab-software-manager', fetchSoftwareSumma
ry, 30000, { immediate: true });
169   fetchSoftwareSummary();
170 }
171
172 window.initSoftwareManagerTab = initSoftwareManagerTab;
173 window.fetchSoftwareSummary = fetchSoftwareSummary;
174
175 if (document.getElementById('tab-software-manager') && documen
t.getElementById('sw-tbody')) {
176   initSoftwareManagerTab();
177 }
178
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Software Scanner Tab Vie
w
================================================================
===============
Description:
  Шаблон интерфейса вкладки глубокого сканирования ПО и выявле
ния спящих программ.

Usage Examples:
    Web Route:
      /windows/api/webgui/software_scanner_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/software_scanner_tab/in
dex.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/software_scanner_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 19:28:00
================================================================
===============
-->
<div class="container-fluid p-2" id="sw-scanner-container">
  <!-- Верхняя карточка управления и запуска сканирования -->
  <div class="card bg-dark text-white border-secondary mb-3 sh
adow-sm" id="panel-software-scanner-summary" data-panel-id="pa
nel-software-scanner-summary">
    <div class="card-body p-3">
      <div class="d-flex align-items-center justify-content-be
tween flex-wrap gap-2 mb-3">
```

```
        <div class="d-flex align-items-center gap-2">
          <span class="fs-4">🔍</span>
          <div>
            <h5 class="mb-0 fw-bold">Глубокий сканер ПО и спящих программ</h5>
            <div class="text-muted small">Инвентаризация реестра, каталогов установки, служб, задач, UserAssist и Prefetch</div>
          </div>
        </div>
        <div class="d-flex align-items-center gap-2 flex-wrap">
          <select class="form-select form-select-sm bg-dark text-white border-secondary" id="sw-scan-mode-select" style="width: auto;">
            <option value="quick">Быстрый опрос (Quick)</option>
            <option value="deep" selected>Глубокое сканирование (Deep)</option>
            <option value="forensic">Форензика + Артефакты (Forensic)</option>
            <option value="orphans">Поиск сирот и остатков (Orphans)</option>
          </select>
          <span class="badge rounded-pill bg-success-subtle text-success border border-success px-3 py-2" id="sw-scanner-status-badge">
            ● Готов к запуску
          </span>
          <button class="btn btn-sm btn-primary d-flex align-items-center gap-1 shadow-sm px-3" id="sw-btn-trigger-scan">
            <i class="bi bi-play-fill fs-6"></i> Запустить сканирование
          </button>
        </div>
      </div>

      <!-- Сводные метрики сканирования -->
```

```
      <div class="row g-2" id="panel-software-scanner-metrics" data-panel-id="panel-software-scanner-metrics">
        <div class="col-6 col-md-2">
          <div class="p-2 bg-secondary bg-opacity-10 border border-secondary rounded">
            <div class="text-muted small text-uppercase" style="font-size: 0.68rem;">Найдено следов</div>
            <div class="fs-5 fw-bold text-info" id="sw-metric-total-evidence">0</div>
          </div>
        </div>
        <div class="col-6 col-md-2">
          <div class="p-2 bg-secondary bg-opacity-10 border border-secondary rounded">
            <div class="text-muted small text-uppercase" style="font-size: 0.68rem;">Программ в базе</div>
            <div class="fs-5 fw-bold text-light" id="sw-metric-total-apps">0</div>
          </div>
        </div>
        <div class="col-6 col-md-2">
          <div class="p-2 bg-secondary bg-opacity-10 border border-secondary rounded">
            <div class="text-muted small text-uppercase" style="font-size: 0.68rem;">  Активных ПО</div>
            <div class="fs-5 fw-bold text-success" id="sw-metric-active-apps">0</div>
          </div>
        </div>
        <div class="col-6 col-md-2">
          <div class="p-2 bg-secondary bg-opacity-10 border border-secondary rounded">
            <div class="text-muted small text-uppercase" style="font-size: 0.68rem;">  Спящих (Dormant)</div>
            <div class="fs-5 fw-bold text-warning" id="sw-metric-dormant-apps">0</div>
          </div>
        </div>
```

```
        <div class="col-6 col-md-2">
          <div class="p-2 bg-secondary bg-opacity-10 border bo
rder-secondary rounded">
            <div class="text-muted small text-uppercase" style
="font-size: 0.68rem;">  Остатков (Orphans)</div>
            <div class="fs-5 fw-bold text-danger" id="sw-metri
c-orphan-items">0</div>
          </div>
        </div>
        <div class="col-6 col-md-2">
          <div class="p-2 bg-secondary bg-opacity-10 border bo
rder-secondary rounded">
            <div class="text-muted small text-uppercase" style
="font-size: 0.68rem;">  Время анализа</div>
            <div class="fs-5 fw-bold text-light" id="sw-metric
-duration">0 мс</div>
          </div>
        </div>
      </div>
    </div>
  </div>

  <!-- Фильтры и строка поиска -->
  <div class="d-flex align-items-center justify-content-betwee
n mb-3 flex-wrap gap-2">
    <div class="d-flex align-items-center gap-2 flex-grow-1" s
tyle="max-width: 450px;">
      <div class="input-group input-group-sm">
        <span class="input-group-text bg-dark text-secondary b
order-secondary"><i class="bi bi-search"></i></span>
        <input class="form-control bg-dark text-white border-s
econdary" id="sw-search-input" placeholder="Поиск по названию
программы или издателю..." type="text"/>
      </div>
    </div>
    <div class="btn-group btn-group-sm" role="group" id="sw-st
atus-filter-group">
      <button class="btn btn-outline-secondary active" data-fi
```

```
lter="all" type="button">Все</button>
      <button class="btn btn-outline-warning" data-filter="dor
mant" type="button">  Спящие (>90 дн.)</button>
      <button class="btn btn-outline-success" data-filter="act
ive" type="button">  Активные</button>
      <button class="btn btn-outline-danger" data-filter="orph
an" type="button">  Сироты / Битые</button>
      <button class="btn btn-outline-info" data-filter="store_
app" type="button">Store / AppX</button>
    </div>
  </div>

  <!-- Таблица программного обеспечения -->
  <div class="card bg-dark text-white border-secondary shadow-
sm mb-3" id="panel-software-scanner-table" data-panel-id="pane
l-software-scanner-table">
    <div class="card-header bg-dark border-secondary d-flex al
ign-items-center justify-content-between py-2 px-3">
      <span class="fw-semibold small text-secondary">
        <i class="bi bi-table me-1"></i> Реестр программного о
беспечения и история активности
      </span>
      <span class="badge bg-secondary" id="sw-table-count-badg
e">0 записей</span>
    </div>
    <div class="table-responsive" style="max-height: 520px; ov
erflow-y: auto;">
      <table class="table table-dark table-hover mb-0 align-mi
ddle small">
        <thead class="table-dark text-secondary sticky-top bor
der-secondary" style="z-index: 2;">
          <tr>
            <th style="width: 28%;">Программа / Исполняемый фа
йл</th>
            <th style="width: 16%;">Издатель / Версия</th>
            <th style="width: 12%;">Статус</th>
            <th style="width: 16%;">Последний запуск</th>
            <th style="width: 10%;">Запусков</th>
```

```
            <th style="width: 18%; text-align: right;">Действия / Деинсталляция</th>
          </tr>
        </thead>
        <tbody id="sw-table-body">
          <tr>
            <td class="text-center py-4 text-muted" colspan="6">
              Нажмите кнопку «Запустить сканирование» для обхода следов ПО.
            </td>
          </tr>
        </tbody>
      </table>
    </div>
  </div>

  <!-- Блок обнаруженных сирот и остатков -->
  <div class="card bg-dark text-white border-danger mb-3 shadow-sm" id="panel-software-scanner-orphans" data-panel-id="panel-software-scanner-orphans" style="display: none;">
    <div class="card-header bg-danger bg-opacity-10 border-danger d-flex align-items-center justify-content-between py-2 px-3">
      <span class="fw-semibold small text-danger">
        <i class="bi bi-exclamation-triangle-fill me-1"></i> Обнаруженные сироты (битые службы, задачи и удаленные папки)
      </span>
      <span class="badge bg-danger text-white" id="sw-orphans-count-badge">0</span>
    </div>
    <div class="table-responsive" style="max-height: 250px; overflow-y: auto;">
      <table class="table table-dark table-hover mb-0 align-middle small">
        <thead class="table-dark text-secondary sticky-top border-secondary">
          <tr>
```

```
            <th style="width: 20%;">Источник</th>
            <th style="width: 40%;">Путь / Целевой файл</th>
            <th style="width: 40%;">Причина обнаружения</th>
          </tr>
        </thead>
        <tbody id="sw-orphans-table-body">
        </tbody>
      </table>
    </div>
  </div>
</div>

<!-- Модальное окно просмотра всех свидетельств программы -->
<div aria-hidden="true" class="modal fade" id="sw-evidence-modal" tabindex="-1">
  <div class="modal-dialog modal-lg modal-dialog-scrollable">
    <div class="modal-content bg-dark text-white border-secondary">
      <div class="modal-header border-secondary">
        <h6 class="modal-title fw-bold" id="sw-evidence-modal-title">Свидетельства и следы программы</h6>
        <button aria-label="Close" class="btn-close btn-close-white" data-bs-dismiss="modal" type="button"></button>
      </div>
      <div class="modal-body">
        <div class="mb-3" id="sw-evidence-modal-summary"></div>
        <h6 class="text-secondary small fw-bold text-uppercase mb-2">Обнаруженные компоненты и артефакты:</h6>
        <div class="list-group list-group-flush border border-secondary rounded" id="sw-evidence-modal-list"></div>
      </div>
    </div>
  </div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Web Interface - Software Scanner Scri
pt
 * ============================================================
==================
 * Description:
 *   Клиентский модуль управления вкладкой глубокого сканирова
ния ПО и спящих программ.
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/software_scanner_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 19:28:00
 * ============================================================
==================
 */

let currentInventory = [];
let currentOrphans = [];
let activeFilter = 'all';

/**
 * Инициализация вкладки глубокого сканера ПО.
 */
export async function init() {
  setupEventListeners();
  await loadLatestScanResults();
}

/**
 * Настройка обработчиков событий элементов управления.
 */
```

```
function setupEventListeners() {
  const btnScan = document.getElementById('sw-btn-trigger-scan');
  if (btnScan) {
    btnScan.addEventListener('click', () => triggerScan());
  }

  const searchInput = document.getElementById('sw-search-input');
  if (searchInput) {
    searchInput.addEventListener('input', () => renderTable());
  }

  const filterButtons = document.querySelectorAll('#sw-status-filter-group button');
  filterButtons.forEach(btn => {
    btn.addEventListener('click', (e) => {
      filterButtons.forEach(b => b.classList.remove('active'));
      btn.classList.add('active');
      activeFilter = btn.getAttribute('data-filter') || 'all';
      renderTable();
    });
  });
}

/**
 * Запуск глубокого сканирования по требованию.
 */
async function triggerScan() {
  const modeSelect = document.getElementById('sw-scan-mode-select');
  const mode = modeSelect ? modeSelect.value : 'deep';

  const statusBadge = document.getElementById('sw-scanner-status-badge');
  const btnScan = document.getElementById('sw-btn-trigger-scan
```

```
');

  if (statusBadge) {
    statusBadge.className = 'badge rounded-pill bg-warning-subtle text-warning border border-warning px-3 py-2';
    statusBadge.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span> Сканирование...';
  }
  if (btnScan) {
    btnScan.disabled = true;
  }

  try {
    const res = await fetch('/api/windows/software-scanner/scan', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ mode }),
    });

    if (!res.ok) {
      throw new Error(`Ошибка сканирования: ${res.statusText}`);
    }

    const report = await res.json();
    processScanReport(report);

    if (window.showToast) {
      window.showToast(`Сканирование завершено: найдено ${report.stats?.total_software_identified || 0} программ`, 'success');
    }
  } catch (err) {
    console.error('[SoftwareScannerTab] Ошибка:', err);
    if (window.showToast) {
      window.showToast(`Ошибка сканирования: ${err.message}`, 'error');
```

```
    }
  } finally {
    if (statusBadge) {
      statusBadge.className = 'badge rounded-pill bg-success-subtle text-success border border-success px-3 py-2';
      statusBadge.textContent = '● Готов к запуску';
    }
    if (btnScan) {
      btnScan.disabled = false;
    }
  }
}

/**
 * Загрузка последнего сохраненного отчета сканирования.
 */
async function loadLatestScanResults() {
  try {
    const res = await fetch('/api/windows/software-scanner/latest-report');
    if (res.ok) {
      const report = await res.json();
      if (report && report.inventory && report.inventory.length > 0) {
        processScanReport(report);
      }
    }
  } catch (err) {
    console.debug('[SoftwareScannerTab] Отчет по умолчанию еще не сформирован:', err);
  }
}

/**
 * Обработка и вывод данных отчета сканирования.
 */
function processScanReport(report) {
  if (!report) return;
```

```
127
128   const stats = report.stats || {};
129   currentInventory = report.inventory || [];
130   currentOrphans = report.orphan_items || [];
131
132   // Обновление метрик
133   updateMetric('sw-metric-total-evidence', stats.total_evidenc
e_found || 0);
134   updateMetric('sw-metric-total-apps', stats.total_software_id
entified || 0);
135   updateMetric('sw-metric-active-apps', stats.active_count ||
0);
136   updateMetric('sw-metric-dormant-apps', stats.dormant_count |
| 0);
137   updateMetric('sw-metric-orphan-items', stats.orphan_count ||
 0);
138   updateMetric('sw-metric-duration', `${Math.round(stats.durat
ion_ms || 0)} мс`);
139
140   renderTable();
141   renderOrphans();
142 }
143
144 /**
145  * Вспомогательное обновление числовой метрики.
146  */
147 function updateMetric(elementId, value) {
148   const el = document.getElementById(elementId);
149   if (el) el.textContent = value;
150 }
151
152 /**
153  * Рендеринг таблицы программ с учетом поиска и активного филь
тра.
154  */
155 function renderTable() {
156   const tbody = document.getElementById('sw-table-body');
157   const countBadge = document.getElementById('sw-table-count-b
```

```
adge');
  const searchInput = document.getElementById('sw-search-input
');
  const query = searchInput ? searchInput.value.trim().toLower
Case() : '';

  if (!tbody) return;

  let filtered = currentInventory.filter(card => {
    // Фильтр по статусу
    if (activeFilter === 'dormant' && card.status !== 'dormant
' && card.status !== 'never_launched') return false;
    if (activeFilter === 'active' && card.status !== 'active')
 return false;
    if (activeFilter === 'orphan' && !card.is_orphan && card.s
tatus !== 'orphan') return false;
    if (activeFilter === 'store_app' && card.status !== 'store
_app') return false;

    // Поиск
    if (query) {
      const matchName = (card.name || '').toLowerCase().includ
es(query);
      const matchPub = (card.publisher || '').toLowerCase().in
cludes(query);
      const matchLoc = (card.install_location || '').toLowerCa
se().includes(query);
      const matchExe = (card.main_executable || '').toLowerCas
e().includes(query);
      return matchName || matchPub || matchLoc || matchExe;
    }

    return true;
  });

  if (countBadge) {
    countBadge.textContent = `${filtered.length} из ${currentI
nventory.length}`;
```

```
  }

  if (filtered.length === 0) {
    tbody.innerHTML = `
      <tr>
        <td class="text-center py-4 text-muted" colspan="6">
          ${currentInventory.length === 0 ? 'Нет данных. Запустите сканирование.' : 'Программы по заданным критериям не найдены.'}
        </td>
      </tr>
    `;
    return;
  }

  tbody.innerHTML = filtered.slice(0, 150).map(card => {
    const statusBadge = getStatusBadge(card);
    const lastRunStr = formatLastRun(card);
    const pathStr = card.install_location || card.main_executable || '—';

    return `
      <tr>
        <td>
          <div class="fw-bold text-light">${escapeHtml(card.name)}</div>
          <div class="text-muted small font-monospace text-truncate" style="max-width: 380px;" title="${escapeHtml(pathStr)}">
            ${escapeHtml(pathStr)}
          </div>
        </td>
        <td>
          <div>${escapeHtml(card.publisher || '—')}</div>
          <div class="text-secondary small">${escapeHtml(card.version || '—')}</div>
        </td>
        <td>${statusBadge}</td>
```

```
        <td>
          <div class="fw-medium">${lastRunStr}</div>
          <div class="text-muted" style="font-size: 0.72rem;">
${card.last_execution ? card.last_execution.substring(0, 10) :
 ''}</div>
        </td>
        <td>
          <span class="badge bg-secondary font-monospace">${ca
rd.run_count || 0}</span>
        </td>
        <td style="text-align: right;">
          <button class="btn btn-sm btn-outline-info me-1 btn-
evidence-details" data-card-id="${card.id}" title="Просмотреть
 все найденные следы">
            <i class="bi bi-info-circle"></i> Следы (${(card.e
vidence || []).length})
          </button>
        </td>
      </tr>
    `;
  }).join('');

  // Навешивание обработчиков клика по деталям
  tbody.querySelectorAll('.btn-evidence-details').forEach(btn
=> {
    btn.addEventListener('click', () => {
      const cardId = btn.getAttribute('data-card-id');
      const targetCard = currentInventory.find(c => c.id === c
ardId);
      if (targetCard) {
        showEvidenceModal(targetCard);
      }
    });
  });
}

/**
 * Рендеринг таблицы сирот и битых записей.
```

```
 */
function renderOrphans() {
  const container = document.getElementById('panel-software-sc
anner-orphans');
  const tbody = document.getElementById('sw-orphans-table-body
');
  const badge = document.getElementById('sw-orphans-count-badg
e');

  if (!container || !tbody) return;

  if (currentOrphans.length === 0) {
    container.style.display = 'none';
    return;
  }

  container.style.display = 'block';
  if (badge) badge.textContent = currentOrphans.length;

  tbody.innerHTML = currentOrphans.map(item => `
    <tr>
      <td><span class="badge bg-danger-subtle text-danger bord
er border-danger">${escapeHtml(item.source)}</span></td>
      <td class="font-monospace text-break">${escapeHtml(item.
path)}</td>
      <td class="text-warning">${escapeHtml(item.product_name
|| 'Неизвестная причина')}</td>
    </tr>
  `).join('');
}

/**
 * Формирование бейджа статуса программы.
 */
function getStatusBadge(card) {
  if (card.status === 'dormant') {
    return '<span class="badge bg-warning-subtle text-warning
border border-warning">  Dormant (Спящее)</span>';
```

```
  } else if (card.status === 'never_launched') {
    return '<span class="badge bg-secondary-subtle text-secondary border border-secondary">⏸ Без запусков</span>';
  } else if (card.status === 'active') {
    return '<span class="badge bg-success-subtle text-success border border-success">▶ Active</span>';
  } else if (card.status === 'orphan' || card.is_orphan) {
    return '<span class="badge bg-danger-subtle text-danger border border-danger">⚠ Orphan</span>';
  } else if (card.status === 'store_app') {
    return '<span class="badge bg-info-subtle text-info border border-info">Store App</span>';
  } else if (card.status === 'portable') {
    return '<span class="badge bg-primary-subtle text-primary border border-primary">Portable</span>';
  }
  return '<span class="badge bg-secondary">Неизвестно</span>';
}

/**
 * Форматирование строки времени последнего запуска.
 */
function formatLastRun(card) {
  if (card.days_since_last_run !== null && card.days_since_last_run !== undefined) {
    return `${card.days_since_last_run} дн. назад`;
  }
  return 'Не зафиксирован';
}

/**
 * Открытие модального окна подробных свидетельств.
 */
function showEvidenceModal(card) {
  const modalEl = document.getElementById('sw-evidence-modal');
  const titleEl = document.getElementById('sw-evidence-modal-title');
```

```
  const summaryEl = document.getElementById('sw-evidence-modal-summary');
  const listEl = document.getElementById('sw-evidence-modal-list');

  if (!modalEl) return;

  if (titleEl) titleEl.textContent = `Следы: ${card.name}`;

  if (summaryEl) {
    summaryEl.innerHTML = `
      <div class="card bg-secondary bg-opacity-10 border-secondary p-2 mb-2">
        <div><strong>Издатель:</strong> ${escapeHtml(card.publisher || '—')}</div>
        <div><strong>Версия:</strong> ${escapeHtml(card.version || '—')}</div>
        <div><strong>Команда деинсталляции:</strong> <code>${escapeHtml(card.uninstall_string || '—')}</code></div>
      </div>
    `;
  }

  if (listEl) {
    listEl.innerHTML = (card.evidence || []).map(ev => `
      <div class="list-group-item bg-dark text-light border-secondary">
        <div class="d-flex align-items-center justify-content-between mb-1">
          <span class="badge bg-primary">${escapeHtml(ev.evidence_type)}</span>
          <span class="badge bg-secondary">${escapeHtml(ev.source)}</span>
        </div>
        <div class="small font-monospace text-break text-info">${escapeHtml(ev.path)}</div>
        ${ev.last_execution ? `<div class="small text-muted mt-1">Метка времени: ${escapeHtml(ev.last_execution)}</div>` : '
```

```
'}
      </div>
    `).join('');
  }

  const modal = new bootstrap.Modal(modalEl);
  modal.show();
}

/**
 * Экранирование HTML для предотвращения XSS.
 */
function escapeHtml(str) {
  if (!str) return '';
  return String(str)
    .replace(/&/g, '&amp;')
    .replace(/</g, '&lt;')
    .replace(/>/g, '&gt;')
    .replace(/"/g, '&quot;')
    .replace(/'/g, '&#039;');
}

// Автоматический запуск при загрузке вкладки
if (document.readyState === 'loading') {
  document.addEventListener('DOMContentLoaded', init);
} else {
  init();
}

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/software_transparency_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/software_transparency_t
ab/index.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/software_transparency_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- AI Software Transparency Scanner Tab UI -->
<div class="container-fluid p-2" id="st-scanner-container">
<!-- Верхняя карточка с метриками и кнопками управления -->
<div class="card bg-dark text-white border-secondary mb-3 shad
ow-sm" data-panel-id="panel-software-transparency-ai-software-
transparency-scann" id="panel-software-transparency-ai-softwar
e-transparency-scann">
<div class="card-body p-3">
<div class="d-flex align-items-center justify-content-between
flex-wrap gap-2 mb-3">
```

```
<div class="d-flex align-items-center gap-2">
<span class="fs-4">□ </span>
<div>
<h5 class="mb-0 fw-bold">AI Software Transparency Scanner</h5>
<div class="text-muted small">Интеллектуальный инвентаризатор установленного ПО, конфигураций, хранилищ и сетевой активности</div>
</div>
</div>
<div class="d-flex align-items-center gap-2">
<span class="badge rounded-pill bg-success-subtle text-success border border-success px-3 py-2" id="st-status-badge">
            ● Готов к сканированию
          </span>
<button class="btn btn-sm btn-primary d-flex align-items-center gap-1 shadow-sm" id="st-btn-refresh">
<i class="bi bi-arrow-clockwise"></i> Сканировать систему
          </button>
</div>
</div>
<!-- Сводные метрики -->
<div class="row g-2">
<div class="col-6 col-md-3">
<div class="p-2 bg-secondary bg-opacity-10 border border-secondary rounded">
<div class="text-muted small text-uppercase">Установлено ПО</div>
<div class="fs-4 fw-bold text-info" id="st-metric-total-apps">0</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="p-2 bg-secondary bg-opacity-10 border border-secondary rounded">
<div class="text-muted small text-uppercase">Файлов конфигов</div>
<div class="fs-4 fw-bold text-warning" id="st-metric-total-configs">0</div>
```

```
</div>
</div>
<div class="col-6 col-md-3">
<div class="p-2 bg-secondary bg-opacity-10 border border-secondary rounded">
<div class="text-muted small text-uppercase">Сетевых узлов / Доменов</div>
<div class="fs-4 fw-bold text-success" id="st-metric-total-domains">0</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="p-2 bg-secondary bg-opacity-10 border border-secondary rounded">
<div class="text-muted small text-uppercase">Время сканирования</div>
<div class="fs-4 fw-bold text-light" id="st-metric-scan-duration">0.0 с</div>
</div>
</div>
</div>
</div>
</div>
<!-- Фильтры и строка поиска -->
<div class="d-flex align-items-center justify-content-between mb-3 flex-wrap gap-2">
<div class="d-flex align-items-center gap-2 flex-grow-1" style="max-width: 450px;">
<div class="input-group input-group-sm">
<span class="input-group-text bg-dark text-secondary border-secondary"><i class="bi bi-search"></i></span>
<input class="form-control bg-dark text-white border-secondary" id="st-search-input" placeholder="Поиск по названию программы или издателю..." type="text"/>
</div>
</div>
<div class="d-flex align-items-center gap-2 small text-muted">
<span>Легенда достоверности:</span>
```

```
<span class="badge bg-success-subtle text-success border border-success">  Локально</span>
<span class="badge bg-info-subtle text-info border border-info">  Документация</span>
<span class="badge bg-warning-subtle text-warning border border-warning">  Предположение AI</span>
<span class="badge bg-secondary-subtle text-secondary border border-secondary">  Неизвестно</span>
</div>
</div>
<!-- Таблица программ -->
<div class="card bg-dark text-white border-secondary shadow-sm" data-panel-id="panel-software-transparency-1" id="panel-software-transparency-1">
<div class="table-responsive">
<table class="table table-dark table-hover mb-0 align-middle">
<thead class="table-dark text-secondary small border-secondary">
<tr>
<th style="width: 25%;">Программа</th>
<th style="width: 15%;">Издатель / Версия</th>
<th style="width: 20%;">Каталоги и хранилище</th>
<th style="width: 15%;">Конфиги &amp;amp; Секреты</th>
<th style="width: 15%;">Сетевые домены</th>
<th style="width: 10%; text-align: right;">AI Анализ</th>
</tr>
</thead>
<tbody id="st-table-body">
<tr>
<td class="text-center py-4 text-muted" colspan="6">
              Нажмите кнопку «Сканировать систему» для сбора сведений об установленном ПО.
            </td>
</tr>
</tbody>
</table>
</div>
</div>
```

```
</div>
<!-- Модальное окно просмотра очищенного конфигурационного фай
ла -->
<div aria-hidden="true" class="modal fade" id="st-config-modal
" tabindex="-1">
<div class="modal-dialog modal-lg modal-dialog-scrollable">
<div class="modal-content bg-dark text-white border-secondary"
>
<div class="modal-header border-secondary">
<h6 class="modal-title fw-bold" id="st-config-modal-title">Кон
фигурационный файл</h6>
<button aria-label="Close" class="btn-close btn-close-white" d
ata-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body">
<div class="alert alert-warning py-2 small mb-3">
<i class="bi bi-shield-lock-fill me-1"></i> Конфиденциальные т
окены, пароли и ключи безопасности автоматически маскированы (
<code>[REDACTED_SECRET]</code>).
        </div>
<div class="small text-muted mb-2" id="st-config-modal-path"><
/div>
<pre class="p-3 bg-black rounded border border-secondary text-
success small font-monospace overflow-auto" id="st-config-moda
l-content" style="max-height: 400px;"></pre>
</div>
</div>
</div>
</div>
<!-- Модальное окно подробного AI-исследования -->
<div aria-hidden="true" class="modal fade" id="st-research-mod
al" tabindex="-1">
<div class="modal-dialog modal-lg modal-dialog-scrollable">
<div class="modal-content bg-dark text-white border-secondary"
>
<div class="modal-header border-secondary">
<h6 class="modal-title fw-bold" id="st-research-modal-title">A
I Исследование программы</h6>
```

```
<button aria-label="Close" class="btn-close btn-close-white" data-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body" id="st-research-modal-body">
<div class="text-muted small">
          Нажмите кнопку «Запросить AI-исследование» для анализа программы.
        </div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Software Transparency Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/software_transparency_tab/main.js?v=20261008_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/software_transparency_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// Software Transparency Scanner script
// Frontend Controller
(function () {
  let allApps = [];
  let isScanning = false;

  async function initSoftwareTransparencyTab() {
    console.log('[TransparencyScanner] Initializing tab controller...');
    bindEvents();
    await loadScannerData(false);
```

```
  }

  function bindEvents() {
    const btnRefresh = document.getElementById('st-btn-refresh
');
    if (btnRefresh) {
      btnRefresh.onclick = () => loadScannerData(true);
    }

    const searchInput = document.getElementById('st-search-inp
ut');
    if (searchInput) {
      searchInput.oninput = () => renderTable();
    }
  }

  async function loadScannerData(forceRefresh = false) {
    if (isScanning) return;
    isScanning = true;

    const badge = document.getElementById('st-status-badge');
    const tbody = document.getElementById('st-table-body');

    if (badge) {
      badge.className = 'badge rounded-pill bg-warning-subtle
text-warning border border-warning px-3 py-2';
      badge.innerText = '● Идет сбор сведений о ПО и сети...';
    }

    if (tbody && (!allApps || allApps.length === 0)) {
      tbody.innerHTML = '<tr><td colspan="6" class="text-cente
r py-4 text-muted"><div class="spinner-border spinner-border-s
m text-primary me-2" role="status"></div>Выполняется инвентари
зация установленных программ, конфигураций и сетевых сокетов..
.</td></tr>';
    }

    try {
```

```
      const url = '/api/v1/software-scanner/scan?force_refresh=' + (forceRefresh ? 'true' : 'false');
      const res = await fetch(url);
      if (!res.ok) throw new Error('HTTP ' + res.status);

      const data = await res.json();
      allApps = data.apps || [];
      const summary = data.summary || {};

      document.getElementById('st-metric-total-apps').innerText = summary.total_apps || allApps.length;
      document.getElementById('st-metric-total-configs').innerText = summary.total_configs_found || 0;
      document.getElementById('st-metric-total-domains').innerText = summary.total_network_domains || 0;
      document.getElementById('st-metric-scan-duration').innerText = (summary.scan_duration_sec || 0) + ' с.';

      if (badge) {
        badge.className = 'badge rounded-pill bg-success-subtle text-success border border-success px-3 py-2';
        badge.innerText = '● Обнаружено: ' + allApps.length + ' программ';
      }

      renderTable();
    } catch (err) {
      console.error('[TransparencyScanner] Scan error:', err);
      if (badge) {
        badge.className = 'badge rounded-pill bg-danger-subtle text-danger border border-danger px-3 py-2';
        badge.innerText = '● Ошибка сканирования';
      }
      if (tbody) {
        tbody.innerHTML = '<tr><td colspan="6" class="text-center text-danger py-4">Ошибка: ' + err.message + '</td></tr>';
      }
    } finally {
```

```
      isScanning = false;
    }
  }

  function renderTable() {
    const tbody = document.getElementById('st-table-body');
    const searchInput = document.getElementById('st-search-inp
ut');
    const searchVal = (searchInput ? searchInput.value : '').t
oLowerCase().trim();

    if (!tbody) return;

    const filtered = allApps.filter((a) => {
      if (!searchVal) return true;
      return (
        a.name.toLowerCase().includes(searchVal) ||
        (a.publisher && a.publisher.toLowerCase().includes(sea
rchVal)) ||
        (a.install_location && a.install_location.toLowerCase(
).includes(searchVal))
      );
    });

    if (filtered.length === 0) {
      tbody.innerHTML = '<tr><td colspan="6" class="text-cente
r py-4 text-muted">Ничего не найдено по фильтру «' + searchVal
 + '»</td></tr>';
      return;
    }

    tbody.innerHTML = filtered.map((app) => renderAppRow(app))
.join('');
    bindRowActions();
  }

  function renderAppRow(app) {
    const dirs = (app.data_directories || []).map(d => `<div c
```

```
lass="text-truncate small text-muted" title="${d.path}">  ${d
.path} (${(d.total_size_bytes / (1024*1024)).toFixed(1)} MB)</
div>`).join('') || '<span class="text-muted small">—</span>';
122     const cfgs = (app.config_files || []).map(c => `<button cl
ass="btn btn-link btn-sm p-0 text-warning text-decoration-none
 text-truncate d-block view-config-btn" data-app-id="${app.id}
" data-cfg-path="${c.path}" title="${c.path}">  ${c.filename}<
/button>`).join('') || '<span class="text-muted small">—</span
>';
123     const nets = (app.network_endpoints || []).map(n => `<span
 class="badge bg-secondary me-1 mb-1" title="${n.endpoint}">${
n.domain || n.endpoint}</span>`).join('') || '<span class="tex
t-muted small">—</span>';
124
125     return `
126       <tr>
127         <td>
128           <div class="fw-bold">${escapeHtml(app.name)}</div>
129           <div class="small text-muted text-truncate" style="m
ax-width: 250px;" title="${app.install_location || ''}">${esca
peHtml(app.install_location || '')}</div>
130         </td>
131         <td>
132           <div>${escapeHtml(app.publisher || '—')}</div>
133           <div class="small text-muted">${escapeHtml(app.versi
on || '')}</div>
134         </td>
135         <td>${dirs}</td>
136         <td>${cfgs}</td>
137         <td>${nets}</td>
138         <td class="text-end">
139           <button class="btn btn-sm btn-outline-info research-
app-btn" data-app-id="${app.id}" title="AI Исследование">
140             <i class="bi bi-robot"></i> Анализ
141           </button>
142         </td>
143       </tr>
144     `;
```

```
  }

  function bindRowActions() {
    document.querySelectorAll('.view-config-btn').forEach(btn
=> {
      btn.onclick = () => {
        const appId = btn.getAttribute('data-app-id');
        const cfgPath = btn.getAttribute('data-cfg-path');
        openConfigModal(appId, cfgPath);
      };
    });

    document.querySelectorAll('.research-app-btn').forEach(btn
 => {
      btn.onclick = () => {
        const appId = btn.getAttribute('data-app-id');
        openResearchModal(appId);
      };
    });
  }

  function openConfigModal(appId, cfgPath) {
    const app = allApps.find(a => a.id === appId);
    const cfg = app ? (app.config_files || []).find(c => c.pat
h === cfgPath) : null;

    document.getElementById('st-config-modal-path').innerText
= cfgPath;
    document.getElementById('st-config-modal-content').innerTe
xt = cfg ? (cfg.content_preview || '[Пустой файл]') : '[Файл н
е найден]';

    const modalEl = document.getElementById('st-config-modal')
;
    if (modalEl && window.bootstrap) {
      new bootstrap.Modal(modalEl).show();
    }
  }
```

```
176
177   function openResearchModal(appId) {
178     const app = allApps.find(a => a.id === appId);
179     const titleEl = document.getElementById('st-research-modal
-title');
180     const bodyEl = document.getElementById('st-research-modal-
body');
181
182     if (titleEl) titleEl.innerText = 'AI Исследование: ' + (ap
p ? app.name : appId);
183     if (bodyEl) {
184       bodyEl.innerHTML = `
185         <div class="p-3 bg-secondary bg-opacity-10 rounded bor
der border-secondary mb-3">
186           <div class="d-flex justify-content-between align-ite
ms-center flex-wrap gap-2">
187             <div>
188               <h6 class="fw-bold text-info mb-1">${escapeHtml(
app ? app.name : appId)}</h6>
189               <div class="text-muted small">Издатель: ${escape
Html(app?.publisher || 'Неизвестен')} | Версия: ${escapeHtml(a
pp?.version || 'N/A')}</div>
190             </div>
191             <button class="btn btn-primary btn-sm d-flex align
-items-center gap-1.5" id="btn-st-start-research">
192               <i class="bi bi-robot"></i> Запросить AI-исследо
вание
193             </button>
194           </div>
195         </div>
196         <div id="st-research-result-container">
197           <div class="text-secondary small">
198             <i class="bi bi-info-circle me-1"></i>Нажмите кноп
ку «Запросить AI-исследование», чтобы отправить запрос к модел
и ИИ для анализа рисков, безопасности и назначений конфигураци
й приложения.
199           </div>
200         </div>
```

```
      `;

      const startBtn = document.getElementById('btn-st-start-r
esearch');
      if (startBtn) {
        startBtn.onclick = () => executeAppResearch(appId);
      }
    }

    const modalEl = document.getElementById('st-research-modal
');
    if (modalEl && window.bootstrap) {
      const bsModal = bootstrap.Modal.getOrCreateInstance(moda
lEl);
      bsModal.show();
    }
  }

  async function executeAppResearch(appId) {
    const container = document.getElementById('st-research-res
ult-container');
    const startBtn = document.getElementById('btn-st-start-res
earch');

    if (startBtn) startBtn.disabled = true;
    if (container) {
      container.innerHTML = `
        <div class="text-center py-4 text-info">
          <div class="spinner-border spinner-border-sm me-2" r
ole="status"></div>
          <span>Выполняется аналитическое исследование програм
мы через языковую модель...</span>
        </div>
      `;
    }

    try {
      const res = await fetch('/api/v1/software-scanner/resear
```

```
ch', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ app_id: appId })
      });
      if (!res.ok) throw new Error('HTTP ' + res.status);
      const data = await res.json();

      if (container) {
        container.innerHTML = `
          <div class="card bg-black border-secondary p-3 mb-2"
>
            <h6 class="text-info fw-bold mb-2">  Описание про
граммы:</h6>
            <p class="mb-2 text-light small">${escapeHtml(data
.summary || data.description || 'Нет данных')}</p>
            ${data.risk_score !== undefined ? `<div class="mb-
2 small"><strong>Оценка риска:</strong> <span class="badge bg-
${data.risk_score > 5 ? 'danger' : 'success'}">${data.risk_sco
re}/10</span></div>` : ''}
            ${data.recommendations ? `<h6 class="text-warning
fw-bold mt-2 mb-1">  Рекомендации:</h6><p class="mb-0 text-li
ght small">${escapeHtml(data.recommendations)}</p>` : ''}
          </div>
        `;
      }
    } catch (err) {
      if (container) {
        container.innerHTML = `
          <div class="alert alert-danger py-2 d-flex align-ite
ms-center justify-content-between flex-wrap gap-2 small">
            <div><i class="bi bi-exclamation-triangle me-1"></
i>Ошибка анализа: ${escapeHtml(err.message)}</div>
            <button type="button" class="btn btn-sm btn-outlin
e-danger" id="btn-st-retry-research">Повторить</button>
          </div>
        `;
        const retryBtn = document.getElementById('btn-st-retry
```

```
-research');
        if (retryBtn) {
          retryBtn.onclick = () => executeAppResearch(appId);
        }
      }
    } finally {
      if (startBtn) startBtn.disabled = false;
    }
  }

  function escapeHtml(text) {
    if (!text) return '';
    const div = document.createElement('div');
    div.innerText = text;
    return div.innerHTML;
  }

  window.initSoftwareTransparencyTab = initSoftwareTransparenc
yTab;

  if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', initSoftware
TransparencyTab);
  } else {
    initSoftwareTransparencyTab();
  }
})();

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Sources - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/sources/main.js?v=2026
1001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/sources
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * sources_tab/main.js
 * Логика вкладки редактирования источников.
 */

let currentSourcesObj = {};

async function loadSourcesRaw() {
    const editor = document.getElementById('sources-json-edito
r');
    const tbody = document.getElementById('sources-list-body')
;
```

```
31
32      if (editor) {
33          editor.disabled = true;
34          editor.value = 'Загрузка...';
35      }
36      if (tbody) {
37          tbody.innerHTML = '<tr><td colspan="6" class="text-cen
ter py-4 text-muted">Загрузка источников...</td></tr>';
38      }
39
40      try {
41          const response = await window.api.fetch('/api/admin/so
urces/raw');
42          const rawContent = response.content || '{}';
43
44          if (editor) editor.value = rawContent;
45
46          try {
47              currentSourcesObj = JSON.parse(rawContent);
48              renderSourcesTable();
49          } catch (e) {
50              console.error('Ошибка парсинга JSON для таблицы:',
 e);
51              if (tbody) {
52                  tbody.innerHTML = '<tr><td colspan="6" class="
text-center py-4 text-danger">Ошибка формата JSON. Таблица нед
оступна.</td></tr>';
53              }
54          }
55
56      } catch (err) {
57          console.error('Ошибка загрузки источников:', err);
58          if (editor) editor.value = 'Ошибка загрузки:\n' + err.
message;
59          if (tbody) tbody.innerHTML = `<tr><td colspan="6" clas
s="text-center py-4 text-danger">Ошибка загрузки: ${err.messag
e}</td></tr>`;
60      } finally {
```

```
        if (editor) editor.disabled = false;
    }
}

async function saveSourcesRaw(content) {
    const editor = document.getElementById('sources-json-edito
r');

    // Проверка синтаксиса на клиенте перед отправкой
    try {
        JSON.parse(content);
    } catch (e) {
        if (typeof showNotification === 'function') {
            showNotification('Синтаксическая ошибка JSON: ' +
e.message, 'danger');
        } else {
            alert('Синтаксическая ошибка JSON: ' + e.message);
        }
        return false;
    }

    try {
        if (editor) editor.disabled = true;
        const response = await window.api.fetch('/api/admin/so
urces/raw', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ content: content })
        });

        if (response.status === 'ok') {
            if (typeof showNotification === 'function') {
                showNotification('Источники успешно сохранены!
', 'success');
            }
            return true;
        }
    } catch (err) {
```

```
        console.error('Ошибка сохранения источников:', err);
        if (typeof showNotification === 'function') {
            showNotification('Ошибка сохранения: ' + err.message, 'danger');
        }
        return false;
    } finally {
        if (editor) editor.disabled = false;
    }
}

function renderSourcesTable() {
    const tbody = document.getElementById('sources-list-body');
    const filterSelect = document.getElementById('sources-category-filter');
    if (!tbody) return;

    const filterVal = filterSelect ? filterSelect.value : 'all';
    tbody.innerHTML = '';

    let hasItems = false;

    for (const [category, items] of Object.entries(currentSourcesObj)) {
        if (!Array.isArray(items)) continue; // Пропускаем метаданные самого файла (например, "metadata": {...})

        // Маппинг категорий
        let catForFilter = 'other';
        if (category === 'metadata_apis') catForFilter = 'metadata';
        else if (category === 'streaming_search') catForFilter = 'streaming_search';
        else if (category.includes('streaming') || category === 'direct_sites') catForFilter = 'streaming';
        else if (category === 'iframe_players') catForFilter =
```

```
 'embed_player';
        else if (category === 'torrent_trackers') catForFilter
 = 'torrent';
        else if (category === 'video_search') catForFilter = '
search';

        if (filterVal !== 'all' && catForFilter !== filterVal)
 continue;

        items.forEach((item, index) => {
            hasItems = true;
            const tr = document.createElement('tr');

            const isEnabled = item.enabled !== false; // по ум
олчанию true

            tr.innerHTML = `
                <td>
                    <div class="form-check form-switch">
                        <input class="form-check-input toggle-
source-btn" type="checkbox" role="switch"
                               data-category="${category}" data-i
ndex="${index}" ${isEnabled ? 'checked' : ''}>
                    </div>
                </td>
                <td class="fw-bold">${item.name || item.id ||
'Без названия'}</td>
                <td><a href="${item.url || '#'}" target="_blan
k" class="small text-truncate d-inline-block" style="max-width
: 150px;">${item.url || '-'}</a></td>
                <td><span class="badge bg-secondary">${categor
y}</span></td>
                <td class="small text-muted">${item.descriptio
n || '-'}</td>
                <td class="text-end">
                    <button class="btn btn-sm btn-outline-dang
er delete-source-btn" data-category="${category}" data-index="
${index}" title="Удалить">
```

```
                        <i class="bi bi-trash"></i>
                    </button>
                </td>
            `;
            tbody.appendChild(tr);
        });
    }

    if (!hasItems) {
        tbody.innerHTML = '<tr><td colspan="6" class="text-cen
ter py-4 text-muted">Нет источников для отображения</td></tr>'
;
    }

    // Привязываем обработчики к сгенерированным кнопкам
    document.querySelectorAll('.toggle-source-btn').forEach(bt
n => {
        btn.addEventListener('change', handleToggleSource);
    });

    document.querySelectorAll('.delete-source-btn').forEach(bt
n => {
        btn.addEventListener('click', handleDeleteSource);
    });
}

async function handleToggleSource(e) {
    const category = e.target.getAttribute('data-category');
    const index = parseInt(e.target.getAttribute('data-index')
);

    if (currentSourcesObj[category] && currentSourcesObj[categ
ory][index]) {
        currentSourcesObj[category][index].enabled = e.target.
checked;
        const newJson = JSON.stringify(currentSourcesObj, null
, 2);

```

```
        // Обновляем текстовое поле и сервер
        const editor = document.getElementById('sources-json-e
ditor');
        if (editor) editor.value = newJson;

        await saveSourcesRaw(newJson);
    }
}

async function handleDeleteSource(e) {
    const btn = e.target.closest('button');
    const category = btn.getAttribute('data-category');
    const index = parseInt(btn.getAttribute('data-index'));

    if (confirm('Вы уверены, что хотите удалить этот источник?
')) {
        if (currentSourcesObj[category] && currentSourcesObj[c
ategory][index]) {
            currentSourcesObj[category].splice(index, 1);

            const newJson = JSON.stringify(currentSourcesObj,
null, 2);
            const editor = document.getElementById('sources-js
on-editor');
            if (editor) editor.value = newJson;

            const success = await saveSourcesRaw(newJson);
            if (success) renderSourcesTable();
        }
    }
}

async function handleAddSource() {
    const id = document.getElementById('new-source-id')?.value
.trim();
    const name = document.getElementById('new-source-name')?.v
alue.trim();
    const url = document.getElementById('new-source-url')?.val
```

```
ue.trim();
    const filterCat = document.getElementById('new-source-cate
gory')?.value;
    const desc = document.getElementById('new-source-descripti
on')?.value.trim();

    if (!id || !name || !url) {
        window.showToast?.('Заполните обязательные поля: ID, Н
азвание и URL', 'warning') || alert('Заполните обязательные по
ля: ID, Название и URL');
        return;
    }

    // Определяем в какую секцию JSON положить новый элемент
    let targetCategory = 'other';
    if (filterCat === 'metadata') targetCategory = 'metadata_a
pis';
    else if (filterCat === 'streaming_search') targetCategory
= 'streaming_search';
    else if (filterCat === 'streaming') targetCategory = 'dire
ct_sites';
    else if (filterCat === 'embed_player') targetCategory = 'i
frame_players';
    else if (filterCat === 'torrent') targetCategory = 'torren
t_trackers';
    else if (filterCat === 'search') targetCategory = 'video_s
earch';

    if (!currentSourcesObj[targetCategory]) {
        currentSourcesObj[targetCategory] = [];
    }

    const newItem = {
        id: id,
        name: name,
        url: url,
        description: desc,
        enabled: true
```

```
    };

    currentSourcesObj[targetCategory].push(newItem);

    const newJson = JSON.stringify(currentSourcesObj, null, 2)
;
    const editor = document.getElementById('sources-json-edito
r');
    if (editor) editor.value = newJson;

    const success = await saveSourcesRaw(newJson);
    if (success) {
        renderSourcesTable();
        // Очищаем форму
        document.getElementById('form-add-source').reset();
    }
}

// Инициализация вкладки
window.initSourcesTab = function() {
    console.log('Инициализация вкладки Источники...');

    const btnRefresh = document.getElementById('btn-refresh-so
urces-json');
    const btnSave = document.getElementById('btn-save-sources-
json');
    const btnSaveAll = document.getElementById('btn-save-all-s
ources');
    const btnAdd = document.getElementById('btn-add-source');
    const filterSelect = document.getElementById('sources-cate
gory-filter');
    const editor = document.getElementById('sources-json-edito
r');

    if (btnRefresh) btnRefresh.onclick = loadSourcesRaw;

    if (btnSave) {
        btnSave.onclick = () => {
```

```
            if (editor) saveSourcesRaw(editor.value).then(() =
> {
                // Обновляем таблицу после ручного изменения J
SON
                try {
                    currentSourcesObj = JSON.parse(editor.valu
e);
                    renderSourcesTable();
                } catch(e){}
            });
        };
    }

    if (btnSaveAll) {
        btnSaveAll.onclick = btnSave.onclick;
    }

    if (btnAdd) {
        btnAdd.onclick = handleAddSource;
    }

    if (filterSelect) {
        filterSelect.addEventListener('change', renderSourcesT
able);
    }

    // Также обновляем таблицу при потере фокуса редактором, е
сли JSON валидный
    if (editor) {
        editor.addEventListener('blur', () => {
            try {
                currentSourcesObj = JSON.parse(editor.value);
                renderSourcesTable();
            } catch(e){}
        });
    }

    // Загружаем данные при открытии
```

```
300     loadSourcesRaw();
301 };
302
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/sources_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/sources_tab/index.html"
 class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/sources_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Sources Registry Tab -->
<div class="container-fluid">
<div class="alert alert-info d-flex justify-content-between al
ign-items-center flex-wrap gap-2 mb-3 shadow-sm">
<div>
<h5 class="mb-1"><i class="bi bi-collection-play me-2"></i>
Реестр источников данных</h5>
<p class="mb-0 small text-secondary">Управление списком сайтов
, стримингов и парсеров для медиатеки.</p>
</div>
```

```
<div>
<button class="btn btn-outline-primary btn-sm rounded-pill" onclick="const t = document.querySelector('[data-bs-target=\'#tab-search\']'); if(t) t.click();">
<i class="bi bi-globe2 me-1"></i> Настройки веб-поиска (MCP) →
      </button>
</div>
</div>
<div class="row g-3">
<!-- LEFT: Add new source -->
<div class="col-4">
<div class="card mb-3" data-panel-id="panel-sources-1" id="panel-sources-1">
<div class="card-header bg-dark text-white">
<span> Добавить источник</span>
</div>
<div class="card-body">
<form id="form-add-source" onsubmit="event.preventDefault();">
<div class="mb-2">
<label class="form-label small fw-bold" for="new-source-id">ID (уникальный):</label>
<input class="form-control form-control-sm" id="new-source-id" placeholder="Например: my_site" required="" type="text"/>
</div>
<div class="mb-2">
<label class="form-label small fw-bold" for="new-source-name">Название:</label>
<input class="form-control form-control-sm" id="new-source-name" placeholder="Например: My Site" required="" type="text"/>
</div>
<div class="mb-2">
<label class="form-label small fw-bold" for="new-source-url">URL:</label>
<input class="form-control form-control-sm" id="new-source-url" placeholder="https://example.com" required="" type="url"/>
</div>
<div class="mb-2">
<label class="form-label small fw-bold" for="new-source-catego
```

```
ry">Категория:</label>
<select class="form-select form-select-sm" id="new-source-cate
gory">
<option value="streaming">streaming — стриминг</option>
<option value="streaming_search">streaming_search — поиск стри
мингов</option>
<option value="metadata">metadata — метаданные</option>
<option value="embed_player">embed_player — встроенный плеер</
option>
<option value="torrent">torrent — торрент-трекер</option>
<option value="search">search — поисковик</option>
<option value="other">other — другое</option>
</select>
</div>
<div class="mb-3">
<label class="form-label small fw-bold" for="new-source-descri
ption">Описание:</label>
<textarea class="form-control form-control-sm" id="new-source-
description" placeholder="Краткое описание источника" rows="2"
></textarea>
</div>
<div class="d-grid">
<button class="btn btn-success btn-sm" id="btn-add-source" typ
e="button">
                 Добавить
              </button>
</div>
</form>
</div>
</div>
<!-- JSON Raw Editor -->
<div class="card" data-panel-id="panel-sources-2" id="panel-so
urces-2">
<div class="card-header bg-dark text-white d-flex justify-cont
ent-between align-items-center">
<span>  JSON-редактор</span>
<div class="d-flex gap-2">
<button class="btn btn-outline-light btn-sm" id="btn-refresh-s
```

```
ources-json">
<i class="bi bi-arrow-clockwise"></i>
</button>
<button class="btn btn-success btn-sm" id="btn-save-sources-js
on">
<i class="bi bi-floppy"></i> Сохранить JSON
            </button>
</div>
</div>
<div class="card-body p-2">
<p class="small text-muted mb-2">Прямое редактирование JSON. И
спользуйте с осторожностью.</p>
<textarea class="form-control font-monospace" id="sources-json
-editor" spellcheck="false" style="min-height: 250px; max-heig
ht: 250px; resize: none; font-size: 0.78rem;"></textarea>
</div>
</div>
</div>
<!-- RIGHT: Sources Table -->
<div class="col-8">
<div class="card" data-panel-id="panel-sources-3" id="panel-so
urces-3">
<div class="card-header bg-dark text-white d-flex justify-cont
ent-between align-items-center">
<span>  Список источников</span>
<button class="btn btn-outline-light btn-sm" id="btn-refresh-s
ources">
<i class="bi bi-arrow-clockwise"></i> Обновить
          </button>
</div>
<div class="card-body p-0">
<!-- Category filter -->
<div class="p-2 border-bottom d-flex gap-2 flex-wrap align-ite
ms-center">
<span class="small text-muted">Фильтр:</span>
<select class="form-select form-select-sm" id="sources-categor
y-filter" style="width: auto;">
<option value="all">Все категории</option>
```

```
<option value="streaming">Стриминг</option>
<option value="streaming_search">Поиск стримингов</option>
<option value="metadata">Метаданные</option>
<option value="embed_player">Встроенные плееры</option>
<option value="torrent">Торренты</option>
<option value="search">Поисковики</option>
<option value="other">Другое</option>
</select>
</div>
<!-- Table container with vertical scroll -->
<div style="height: 500px; overflow-y: scroll; overflow-x: aut
o;">
<table class="table table-striped table-hover align-middle mb-
0" id="sources-table" style="min-width: 600px;">
<thead class="table-dark">
<tr>
<th style="width: 5%;">On</th>
<th style="width: 15%;">Название</th>
<th style="width: 25%;">URL</th>
<th style="width: 15%;">Категория</th>
<th style="width: 25%;">Описание</th>
<th class="text-end" style="width: 15%;">Действия</th>
</tr>
</thead>
<tbody id="sources-list-body" style="word-wrap: break-word; ov
erflow-wrap: break-word;">
<tr>
<td class="text-center py-4 text-muted" colspan="6">Загрузка и
сточников...</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
</div>
<!-- Save all button at bottom -->
```

```
<div class="mt-3 text-end">
<button class="btn btn-primary" id="btn-save-all-sources">
<i class="bi bi-floppy"></i> Сохранить все изменения
    </button>
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Sources Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/sources_tab/main.js?v=
20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/sources_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * sources_tab/main.js
 * Логика вкладки редактирования источников.
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async function loadSourcesRaw() {
    const editor = document.getElementById('sources-json-edito
r');
    const tbody = document.getElementById('sources-list-body')
;
```

```
31
32      if (editor) {
33          editor.disabled = true;
34          editor.value = 'Загрузка...';
35      }
36      if (tbody) {
37          tbody.innerHTML = '<tr><td colspan="6" class="text-cen
ter py-4 text-muted">Загрузка источников...</td></tr>';
38      }
39
40      try {
41          const response = await window.api.fetch('/api/admin/so
urces/raw');
42          const rawContent = response.content || '{}';
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46          try {
47              currentSourcesObj = JSON.parse(rawContent);
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49          } catch (e) {
50              console.error('Ошибка парсинга JSON для таблицы:',
 e);
51              if (tbody) {
52                  tbody.innerHTML = '<tr><td colspan="6" class="
text-center py-4 text-danger">Ошибка формата JSON. Таблица нед
оступна.</td></tr>';
53              }
54          }
55
56      } catch (err) {
57          console.error('Ошибка загрузки источников:', err);
58          if (editor) editor.value = 'Ошибка загрузки:\n' + err.
message;
59          if (tbody) tbody.innerHTML = `<tr><td colspan="6" clas
s="text-center py-4 text-danger">Ошибка загрузки: ${err.messag
e}</td></tr>`;
60      } finally {
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```
        if (editor) editor.disabled = false;
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}

async function saveSourcesRaw(content) {
    const editor = document.getElementById('sources-json-edito
r');

    // Проверка синтаксиса на клиенте перед отправкой
    try {
        JSON.parse(content);
    } catch (e) {
        if (typeof showNotification === 'function') {
            showNotification('Синтаксическая ошибка JSON: ' +
e.message, 'danger');
        } else {
            alert('Синтаксическая ошибка JSON: ' + e.message);
        }
        return false;
    }

    try {
        if (editor) editor.disabled = true;
        const response = await window.api.fetch('/api/admin/so
urces/raw', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ content: content })
        });

        if (response.status === 'ok') {
            if (typeof showNotification === 'function') {
                showNotification('Источники успешно сохранены!
', 'success');
            }
            return true;
        }
    } catch (err) {
```

```
        console.error('Ошибка сохранения источников:', err);
        if (typeof showNotification === 'function') {
            showNotification('Ошибка сохранения: ' + err.message, 'danger');
        }
        return false;
    } finally {
        if (editor) editor.disabled = false;
    }
}

function renderSourcesTable() {
    const tbody = document.getElementById('sources-list-body');
    const filterSelect = document.getElementById('sources-category-filter');
    if (!tbody) return;

    const filterVal = filterSelect ? filterSelect.value : 'all';
    tbody.innerHTML = '';

    let hasItems = false;

    for (const [category, items] of Object.entries(currentSourcesObj)) {
        if (!Array.isArray(items)) continue; // Пропускаем метаданные самого файла (например, "metadata": {...})

        // Маппинг категорий
        let catForFilter = 'other';
        if (category === 'metadata_apis') catForFilter = 'metadata';
        else if (category === 'streaming_search') catForFilter = 'streaming_search';
        else if (category.includes('streaming') || category === 'direct_sites') catForFilter = 'streaming';
        else if (category === 'iframe_players') catForFilter =
```

```
 'embed_player';
        else if (category === 'torrent_trackers') catForFilter
 = 'torrent';
        else if (category === 'video_search') catForFilter = '
search';

        if (filterVal !== 'all' && catForFilter !== filterVal)
 continue;

        items.forEach((item, index) => {
            hasItems = true;
            const tr = document.createElement('tr');

            const isEnabled = item.enabled !== false; // по ум
олчанию true

            tr.innerHTML = `
                <td>
                    <div class="form-check form-switch">
                        <input class="form-check-input toggle-
source-btn" type="checkbox" role="switch"
                               data-category="${category}" data-i
ndex="${index}" ${isEnabled ? 'checked' : ''}>
                    </div>
                </td>
                <td class="fw-bold">${item.name || item.id ||
'Без названия'}</td>
                <td><a href="${item.url || '#'}" target="_blan
k" class="small text-truncate d-inline-block" style="max-width
: 150px;">${item.url || '-'}</a></td>
                <td><span class="badge bg-secondary">${categor
y}</span></td>
                <td class="small text-muted">${item.descriptio
n || '-'}</td>
                <td class="text-end">
                    <button class="btn btn-sm btn-outline-dang
er delete-source-btn" data-category="${category}" data-index="
${index}" title="Удалить">
```

```
                        <i class="bi bi-trash"></i>
                    </button>
                </td>
            `;
            tbody.appendChild(tr);
        });
    }

    if (!hasItems) {
        tbody.innerHTML = '<tr><td colspan="6" class="text-cen
ter py-4 text-muted">Нет источников для отображения</td></tr>'
;
    }

    // Привязываем обработчики к сгенерированным кнопкам
    document.querySelectorAll('.toggle-source-btn').forEach(bt
n => {
        btn.addEventListener('change', handleToggleSource);
    });

    document.querySelectorAll('.delete-source-btn').forEach(bt
n => {
        btn.addEventListener('click', handleDeleteSource);
    });
}

async function handleToggleSource(e) {
    const category = e.target.getAttribute('data-category');
    const index = parseInt(e.target.getAttribute('data-index')
);

    if (currentSourcesObj[category] && currentSourcesObj[categ
ory][index]) {
        currentSourcesObj[category][index].enabled = e.target.
checked;
        const newJson = JSON.stringify(currentSourcesObj, null
, 2);

```

```
        // Обновляем текстовое поле и сервер
        const editor = document.getElementById('sources-json-e
ditor');
        if (editor) editor.value = newJson;

        await saveSourcesRaw(newJson);
    }
}

async function handleDeleteSource(e) {
    const btn = e.target.closest('button');
    const category = btn.getAttribute('data-category');
    const index = parseInt(btn.getAttribute('data-index'));

    if (confirm('Вы уверены, что хотите удалить этот источник?
')) {
        if (currentSourcesObj[category] && currentSourcesObj[c
ategory][index]) {
            currentSourcesObj[category].splice(index, 1);

            const newJson = JSON.stringify(currentSourcesObj,
null, 2);
            const editor = document.getElementById('sources-js
on-editor');
            if (editor) editor.value = newJson;

            const success = await saveSourcesRaw(newJson);
            if (success) renderSourcesTable();
        }
    }
}

async function handleAddSource() {
    const id = document.getElementById('new-source-id')?.value
.trim();
    const name = document.getElementById('new-source-name')?.v
alue.trim();
    const url = document.getElementById('new-source-url')?.val
```

```
ue.trim();
    const filterCat = document.getElementById('new-source-cate
gory')?.value;
    const desc = document.getElementById('new-source-descripti
on')?.value.trim();

    if (!id || !name || !url) {
        window.showToast?.('Заполните обязательные поля: ID, Н
азвание и URL', 'warning') || alert('Заполните обязательные по
ля: ID, Название и URL');
        return;
    }

    // Определяем в какую секцию JSON положить новый элемент
    let targetCategory = 'other';
    if (filterCat === 'metadata') targetCategory = 'metadata_a
pis';
    else if (filterCat === 'streaming_search') targetCategory
= 'streaming_search';
    else if (filterCat === 'streaming') targetCategory = 'dire
ct_sites';
    else if (filterCat === 'embed_player') targetCategory = 'i
frame_players';
    else if (filterCat === 'torrent') targetCategory = 'torren
t_trackers';
    else if (filterCat === 'search') targetCategory = 'video_s
earch';

    if (!currentSourcesObj[targetCategory]) {
        currentSourcesObj[targetCategory] = [];
    }

    const newItem = {
        id: id,
        name: name,
        url: url,
        description: desc,
        enabled: true
```

```
    };

    currentSourcesObj[targetCategory].push(newItem);

    const newJson = JSON.stringify(currentSourcesObj, null, 2)
;
    const editor = document.getElementById('sources-json-edito
r');
    if (editor) editor.value = newJson;

    const success = await saveSourcesRaw(newJson);
    if (success) {
        renderSourcesTable();
        // Очищаем форму
        document.getElementById('form-add-source').reset();
    }
}

// Инициализация вкладки
window.initSourcesTab = function() {
    console.log('Инициализация вкладки Источники...');

    const btnRefresh = document.getElementById('btn-refresh-so
urces-json');
    const btnSave = document.getElementById('btn-save-sources-
json');
    const btnSaveAll = document.getElementById('btn-save-all-s
ources');
    const btnAdd = document.getElementById('btn-add-source');
    const filterSelect = document.getElementById('sources-cate
gory-filter');
    const editor = document.getElementById('sources-json-edito
r');

    if (btnRefresh) btnRefresh.onclick = loadSourcesRaw;

    if (btnSave) {
        btnSave.onclick = () => {
```

```
            if (editor) saveSourcesRaw(editor.value).then(() =
> {
                // Обновляем таблицу после ручного изменения J
SON
                try {
                    currentSourcesObj = JSON.parse(editor.valu
e);
                    renderSourcesTable();
                } catch(e){}
            });
        };
    }

    if (btnSaveAll) {
        btnSaveAll.onclick = btnSave.onclick;
    }

    if (btnAdd) {
        btnAdd.onclick = handleAddSource;
    }

    if (filterSelect) {
        filterSelect.addEventListener('change', renderSourcesT
able);
    }

    // Также обновляем таблицу при потере фокуса редактором, е
сли JSON валидный
    if (editor) {
        editor.addEventListener('blur', () => {
            try {
                currentSourcesObj = JSON.parse(editor.value);
                renderSourcesTable();
            } catch(e){}
        });
    }

    // Загружаем данные при открытии
```

```
300     loadSourcesRaw();
301 };
302
303
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса аудита автозапуска Windows
 (Startup Auditor).
  Полная поддержка светлых и тёмных тем с использованием дизай
н-токенов CSS.

Usage Examples:
    Web Route:
      /windows/api/webgui/startup_auditor_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/startup_auditor_tab/ind
ex.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/startup_auditor_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Startup Auditor Tab Content -->
<style>
  .sa-container {
    padding: 0.5rem 0.75rem;
    color: var(--text-color);
  }
  .sa-card {
```

```
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: 8px;
    padding: 0.45rem 0.75rem;
    display: flex;
    flex-direction: column;
    justify-content: space-between;
    height: 100%;
    min-height: 58px;
    box-shadow: var(--shadow-sm);
    transition: border-color 0.15s ease, box-shadow 0.15s ease
;
  }
  .sa-card:hover {
    border-color: var(--nav-active) !important;
    box-shadow: var(--shadow-md);
  }
  .sa-card-label {
    font-size: 0.68rem;
    text-transform: uppercase;
    letter-spacing: 0.05em;
    color: var(--text-muted);
    font-weight: 600;
    display: flex;
    justify-content: space-between;
    align-items: center;
    line-height: 1.1;
  }
  .sa-card-value {
    font-size: 1.25rem;
    font-weight: 700;
    font-family: var(--font-mono, ui-monospace, SFMono-Regular
, Menlo, Monaco, Consolas, monospace);
    line-height: 1.15;
  }
  .sa-card-sub {
    font-size: 0.7rem;
    color: var(--text-muted);
```

```
    line-height: 1.1;
  }

  .sa-panel {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: 8px;
    overflow: hidden;
    box-shadow: var(--shadow-sm);
  }
  .sa-panel-header {
    background: var(--surface-2) !important;
    border-bottom: 1px solid var(--border-color) !important;
    padding: 0.45rem 0.85rem;
    font-weight: 600;
    font-size: 0.82rem;
    color: var(--nav-active);
    display: flex;
    align-items: center;
    justify-content: space-between;
    flex-wrap: wrap;
    gap: 0.35rem;
  }

  .sa-table {
    width: 100%;
    font-size: 0.78rem;
    border-collapse: collapse;
    color: var(--text-color);
    background: var(--surface-1);
  }
  .sa-table thead th {
    background: var(--surface-2) !important;
    padding: 6px 10px;
    color: var(--text-muted);
    font-weight: 600;
    font-size: 0.72rem;
    text-transform: uppercase;
```

```
    letter-spacing: 0.04em;
    border-bottom: 1px solid var(--border-color);
    white-space: nowrap;
    position: sticky;
    top: 0;
    z-index: 2;
  }
  .sa-table tbody tr {
    background: var(--surface-1) !important;
    transition: background-color 0.1s ease;
  }
  .sa-table tbody tr:nth-child(even) {
    background: var(--surface-2) !important;
  }
  .sa-table tbody tr:hover {
    background: var(--nav-hover-bg, rgba(0, 0, 0, 0.04)) !impo
rtant;
  }
  .sa-table td {
    padding: 6px 10px;
    border-bottom: 1px solid var(--border-subtle, var(--border
-color));
    vertical-align: middle;
    color: var(--text-color);
  }

  /* Badge styles */
  .sa-badge {
    font-size: 0.7rem;
    font-weight: 600;
    padding: 2px 6px;
    border-radius: 4px;
    display: inline-flex;
    align-items: center;
    gap: 3px;
    letter-spacing: 0.02em;
    white-space: nowrap;
  }
```

```
  .sa-badge-enabled {
    background: var(--status-success-bg, rgba(34, 197, 94, 0.15));
    border: 1px solid var(--status-success-border, rgba(34, 197, 94, 0.45));
    color: var(--status-success-text, #185223);
  }
  .sa-badge-disabled {
    background: var(--status-danger-bg, rgba(239, 68, 68, 0.18));
    border: 1px solid var(--status-danger-border, rgba(239, 68, 68, 0.45));
    color: var(--status-danger-text, #7e1f1f);
    font-weight: bold;
  }
  .sa-badge-loc {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    color: var(--nav-active);
    font-family: var(--font-mono, monospace);
  }
  .sa-badge-cat {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    color: var(--text-muted);
  }

  /* Risk Badges */
  .sa-risk-critical {
    background: var(--status-danger-bg, rgba(239, 68, 68, 0.22));
    border: 1px solid var(--status-danger-border, rgba(239, 68, 68, 0.6));
    color: var(--status-danger-text, #7e1f1f);
  }
  .sa-risk-suspicious {
    background: var(--status-warning-bg, rgba(249, 115, 22, 0.22));
```

```
    border: 1px solid var(--status-warning-border, rgba(249, 115, 22, 0.6));
    color: var(--status-warning-text, #704705);
  }
  .sa-risk-warning {
    background: var(--status-warning-bg, rgba(234, 179, 8, 0.22));
    border: 1px solid var(--status-warning-border, rgba(234, 179, 8, 0.6));
    color: var(--status-warning-text, #704705);
  }
  .sa-risk-notice {
    background: var(--status-info-bg, rgba(56, 189, 248, 0.18));
    border: 1px solid var(--status-info-border, rgba(56, 189, 248, 0.5));
    color: var(--status-info-text, #184674);
  }
  .sa-risk-clean {
    background: var(--status-success-bg, rgba(34, 197, 94, 0.16));
    border: 1px solid var(--status-success-border, rgba(34, 197, 94, 0.45));
    color: var(--status-success-text, #185223);
  }

  /* Toggle Switch button */
  .sa-toggle-btn {
    background: transparent;
    border: none;
    cursor: pointer;
    padding: 1px 4px;
    border-radius: 4px;
    font-size: 1.15rem;
    line-height: 1;
    transition: transform 0.15s ease, opacity 0.15s ease;
    display: inline-flex;
    align-items: center;
```

```
    justify-content: center;
  }
  .sa-toggle-btn:hover {
    transform: scale(1.15);
  }
  .sa-toggle-btn:active {
    transform: scale(0.95);
  }

  .sa-input {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    color: var(--text-color) !important;
    font-size: 0.8rem;
    padding: 0.25rem 0.5rem;
  }
  .sa-input:focus {
    border-color: var(--nav-active) !important;
    box-shadow: 0 0 0 2px rgba(37, 99, 235, 0.2) !important;
  }
  .sa-input option {
    background: var(--surface-1);
    color: var(--text-color);
  }
</style>
<div class="sa-container" id="startup-auditor-container">
<!-- Header Alert Banner (Compact Strip) -->
<div class="d-flex justify-content-between align-items-center flex-wrap gap-2 mb-2 py-1.5 px-3 rounded-2 sa-header-banner" style="background: var(--surface-1); border: 1px solid var(--border-color); color: var(--text-color);">
<div class="d-flex align-items-center gap-2 flex-wrap">
<span class="fs-5">□ </span>
<span class="fw-bold small" style="font-size: 0.92rem; color: var(--text-color);">Аудит автозапуска Windows</span>
<span class="badge border fw-normal" style="font-size: 0.7rem;  background: var(--surface-2); border-color: var(--border-color) !important; color: var(--nav-active);">Startup Auditor</spa
```

```
n>
<span class="badge border fw-normal d-none d-sm-inline-flex al
ign-items-center gap-1" id="sa-telemetry-status-pill" style="f
ont-size: 0.7rem; background: var(--surface-2); border-color:
var(--border-color) !important; color: var(--text-muted);" tit
le="Статус снимка автозапуска в базе телеметрии">
<i class="bi bi-database-check text-success"></i> <span id="sa
-telemetry-last-snap">Телеметрия: синхронизировано</span>
</span>
<span class="text-muted small d-none d-lg-inline" style="font-
size: 0.76rem;">| Run/RunOnce, папки автозагрузки, IFEO и служ
бы</span>
</div>
<div class="d-flex align-items-center gap-1.5 flex-wrap">
<button class="btn btn-xs btn-primary rounded-pill px-3 py-1 d
-flex align-items-center gap-1.5 fw-semibold shadow-sm" id="sa
-btn-refresh-telemetry" style="font-size: 0.76rem;" title="Аси
нхронно перезапустить сбор автозапуска и зафиксировать отличия
 в телеметрии">
<i class="bi bi-arrow-repeat" id="sa-refresh-spinner"></i> <sp
an>Обновить</span>
</button>
<button class="btn btn-xs btn-outline-secondary rounded-pill p
x-2.5 py-1 d-flex align-items-center gap-1" id="sa-btn-view-ch
anges" style="font-size: 0.75rem;" title="Просмотр зафиксирова
нных изменений между снимками телеметрии">
<i class="bi bi-clock-history"></i> <span>Изменения</span>
<span class="badge bg-warning text-dark px-1.5 py-0.5 rounded-
pill d-none" id="sa-changes-badge" style="font-size: 0.65rem;"
>0</span>
</button>
<button class="btn btn-xs btn-outline-secondary rounded-pill p
x-2 py-1" id="sa-btn-export-json" style="font-size: 0.75rem;"
title="Экспорт в JSON">
<i class="bi bi-file-earmark-code me-1"></i> JSON
      </button>
<button class="btn btn-xs btn-outline-secondary rounded-pill p
x-2 py-1" id="sa-btn-export-csv" style="font-size: 0.75rem;" t
```

```
itle="Экспорт в CSV">
<i class="bi bi-file-earmark-spreadsheet me-1"></i> CSV
      </button>
</div>
</div>
<!-- Top Metric & Health Score Cards (Compact Density Grid) --
>
<div class="row g-2 mb-2">
<!-- Health Score Card -->
<div class="col-6 col-md-3">
<div class="sa-card">
<div class="sa-card-label">
<span>Индекс чистоты</span>
<i class="bi bi-shield-check text-success"></i>
</div>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<div class="d-flex align-items-baseline gap-1">
<div class="sa-card-value text-success" id="sa-health-score">-
-</div>
<span class="text-muted small" style="font-size: 0.75rem;">/ 1
00</span>
</div>
<div class="sa-card-sub text-end" id="sa-health-status">Оценка
 безопасности</div>
</div>
</div>
</div>
<!-- Active & Total Items -->
<div class="col-6 col-md-3">
<div class="sa-card">
<div class="sa-card-label">
<span>Элементы автозапуска</span>
<i class="bi bi-play-circle text-info"></i>
</div>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<div class="d-flex align-items-baseline gap-1">
```

```
<div class="sa-card-value text-info" id="sa-active-count">--</
div>
<span class="text-muted small" style="font-size: 0.75rem;">из
<span class="fw-bold" id="sa-total-count" style="color: var(--
text-color);">--</span></span>
</div>
<div class="sa-card-sub text-end">Откл: <span class="text-dang
er fw-bold" id="sa-disabled-count">--</span></div>
</div>
</div>
</div>
<!-- Broken Entries Card -->
<div class="col-6 col-md-3">
<div class="sa-card">
<div class="sa-card-label">
<span>Битые ссылки</span>
<i class="bi bi-exclamation-triangle text-warning"></i>
</div>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<div class="d-flex align-items-baseline gap-1">
<div class="sa-card-value text-warning" id="sa-broken-count">-
-</div>
<span class="text-muted small" style="font-size: 0.75rem;">зап
исей</span>
</div>
<div class="sa-card-sub text-end">Файлы не найдены</div>
</div>
</div>
</div>
<!-- Threats & Critical Alerts -->
<div class="col-6 col-md-3">
<div class="sa-card">
<div class="sa-card-label">
<span>Угрозы и аномалии</span>
<i class="bi bi-shield-slash text-danger"></i>
</div>
<div class="d-flex align-items-baseline justify-content-betwee
```

```
n mt-1">
<div class="d-flex align-items-baseline gap-1">
<div class="sa-card-value text-danger" id="sa-threats-count">-
-</div>
<span class="text-muted small" style="font-size: 0.75rem;">пре
дупреждений</span>
</div>
<div class="sa-card-sub text-end">IFEO и скрипты</div>
</div>
</div>
</div>
</div>
<!-- Recommendations Alert Banner (Compact Sleek Strip) -->
<div class="py-1.5 px-3 mb-2 rounded-2 d-flex align-items-cent
er justify-content-between" id="sa-recommendations-box" style=
"background: var(--surface-1); border: 1px solid var(--border-
color); border-left: 3px solid #eab308; font-size: 0.78rem;">
<div class="d-flex align-items-center gap-2">
<span class="text-warning">□ </span>
<div class="d-flex align-items-center gap-2 flex-wrap">
<span class="fw-semibold" id="sa-rec-title" style="color: var(
--text-color);">Рекомендации:</span>
<span class="text-muted" id="sa-rec-desc">Загрузка рекомендаци
й по оптимизации...</span>
</div>
</div>
</div>
<!-- Search & Filter Controls (Compact Toolbar) -->
<div class="card p-2 mb-2 sa-panel" data-panel-id="panel-start
up-auditor-1" id="panel-startup-auditor-1">
<div class="row g-2 align-items-center">
<!-- Search Input -->
<div class="col-12 col-md-4">
<div class="input-group input-group-sm">
<span class="input-group-text sa-input text-muted py-1" style=
"border-right: none;"><i class="bi bi-search"></i></span>
<input class="form-control form-control-sm sa-input" id="sa-se
arch-input" placeholder="Поиск по имени, команде или пути..."
```

```
type="text"/>
</div>
</div>
<!-- Risk Filter -->
<div class="col-6 col-md-3">
<select class="form-select form-select-sm sa-input" id="sa-fil
ter-risk">
<option value="">Все уровни риска</option>
<option value="critical">🔴 Critical (Критические)</option>
<option value="suspicious">🟠 Suspicious (Подозрительные)</opti
on>
<option value="warning">🟡 Warning (Предупреждения)</option>
<option value="notice">i Notice (Инфо)</option>
<option value="clean">🟢 Clean (Безопасные)</option>
</select>
</div>
<!-- Location Filter -->
<div class="col-6 col-md-3">
<select class="form-select form-select-sm sa-input" id="sa-fil
ter-location">
<option value="">Все точки автозапуска</option>
<option value="registry_run">Реестр: Run</option>
<option value="registry_runonce">Реестр: RunOnce</option>
<option value="startup_folder_user">Папка автозагрузки (User)<
/option>
<option value="startup_folder_common">Папка автозагрузки (Comm
on)</option>
<option value="winlogon">Winlogon</option>
<option value="ifeo">IFEO Перехваты</option>
<option value="scheduled_task">Планировщик задач</option>
<option value="windows_service">Службы Windows</option>
</select>
</div>
<!-- State Filter -->
<div class="col-12 col-md-2">
<select class="form-select form-select-sm sa-input" id="sa-fil
ter-state">
<option value="">Все статусы</option>
```

```
<option value="enabled">Только включенные</option>
<option value="disabled">Только отключенные</option>
</select>
</div>
</div>
</div>
<!-- Entries Data Table Panel -->
<div class="sa-panel">
<div class="sa-panel-header">
<span><i class="bi bi-table me-1"></i> Список зарегистрированных элементов автозапуска</span>
<span class="small text-muted" id="sa-table-count">Загрузка...</span>
</div>
<div class="table-responsive" style="max-height: calc(100vh - 245px); overflow-y: auto;">
<table class="sa-table" id="sa-entries-table">
<thead>
<tr>
<th class="text-center" style="width: 80px;">Статус</th>
<th style="width: 230px;">Имя / Программа</th>
<th style="width: 170px;">Точка автозапуска</th>
<th style="width: 140px;">Категория</th>
<th class="text-center" style="width: 110px;">Риск</th>
<th>Исполняемый путь / Аргументы</th>
<th class="text-center" style="width: 110px;">Влияние</th>
<th class="text-center" style="width: 80px;">Действие</th>
</tr>
</thead>
<tbody id="sa-table-body">
<tr>
<td class="text-center py-4 text-muted" colspan="8">
<div class="spinner-border spinner-border-sm text-primary me-2" role="status"></div>
              Выполняется сканирование точек автозапуска Windows...
            </td>
</tr>
```

```
</tbody>
</table>
</div>
</div>
<!-- Program Details & AI Context Modal -->
<div aria-hidden="true" class="modal fade" id="sa-entry-detail
-modal" tabindex="-1">
<div class="modal-dialog modal-lg modal-dialog-centered modal-
dialog-scrollable">
<div class="modal-content shadow-lg">
<div class="modal-header py-2 px-3">
<div class="d-flex align-items-center gap-2">
<span class="fs-5 text-primary">□ </span>
<h5 class="modal-title fw-bold text-primary mb-0" id="sa-modal
-title">Детали программы</h5>
<span id="sa-modal-status-badge"></span>
<span id="sa-modal-risk-badge"></span>
</div>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body py-3 px-3">
<!-- Metadata Grid -->
<div class="row g-2 mb-3 small">
<div class="col-sm-6"><strong class="text-muted">Имя / Програм
ма:</strong> <span class="fw-semibold" id="sa-modal-name"></sp
an></div>
<div class="col-sm-6"><strong class="text-muted">Издатель:</st
rong> <span id="sa-modal-publisher"></span></div>
<div class="col-sm-6"><strong class="text-muted">Точка автозап
уска:</strong> <span class="sa-badge sa-badge-loc" id="sa-moda
l-location-type"></span></div>
<div class="col-sm-6"><strong class="text-muted">Категория:</s
trong> <span class="sa-badge sa-badge-cat" id="sa-modal-catego
ry"></span></div>
<div class="col-sm-6"><strong class="text-muted">Влияние на за
пуск:</strong> <span id="sa-modal-boot-impact"></span></div>
<div class="col-sm-6"><strong class="text-muted">Цифровая подп
```

```
ись:</strong> <span id="sa-modal-is-signed"></span></div>
<div class="col-sm-6"><strong class="text-muted">Статус файла:
</strong> <span id="sa-modal-file-exists"></span></div>
<div class="col-sm-6"><strong class="text-muted">Размер файла:
</strong> <span id="sa-modal-file-size"></span></div>
<div class="col-sm-12"><strong class="text-muted">Расположение
 / Ключ:</strong> <span class="font-monospace text-muted small
" id="sa-modal-location-path" style="word-break: break-all;"><
/span></div>
</div>
<!-- Command & Path Section -->
<div class="card p-3 mb-3" data-panel-id="panel-startup-audito
r-komanda-i-ispolnyaemyy-fayl" id="panel-startup-auditor-koman
da-i-ispolnyaemyy-fayl" style="background: var(--surface-2); b
order: 1px solid var(--border-color);">
<div class="d-flex justify-content-between align-items-center
mb-1">
<h6 class="small text-uppercase text-muted fw-bold mb-0">Коман
да и исполняемый файл</h6>
</div>
<div class="small font-monospace mb-2 text-primary" id="sa-mod
al-executable-path" style="word-break: break-all;"></div>
<div class="small font-monospace text-muted" id="sa-modal-comm
and" style="white-space: pre-wrap; word-break: break-all;"></d
iv>
</div>
<!-- AI Diagnostic & Explanation Section -->
<div class="card p-3" data-panel-id="panel-startup-auditor-ai-
contextual-explanation" id="sa-modal-ai-section" style="backgr
ound: var(--surface-1); border: 1px solid var(--border-color);
">
<div class="d-flex justify-content-between align-items-center
mb-2">
<h6 class="fw-bold text-primary mb-0"><i class="bi bi-robot me
-1"></i> AI Contextual Explanation</h6>
<button class="btn btn-sm btn-outline-primary py-0 px-2 rounde
d-pill" id="btn-sa-modal-diagnose">
<i class="bi bi-magic me-1"></i>Анализ контекста
```

```
              </button>
</div>
<div class="small text-muted" id="sa-modal-ai-explanation">
              Нажмите «Анализ контекста», чтобы получить экспертное объяснение назначения программы от языковой модели, оценку рисков и рекомендации по оптимизации.
            </div>
</div>
</div>
<div class="modal-footer py-2 px-3 d-flex justify-content-between">
<div>
<button class="btn btn-sm btn-outline-warning me-1" id="btn-sa-modal-toggle" type="button">
<i class="bi bi-toggle2-on me-1"></i>Переключить автозапуск
            </button>
<button class="btn btn-sm btn-outline-secondary" id="btn-sa-modal-copy" type="button">
<i class="bi bi-clipboard me-1"></i>Копировать путь
            </button>
</div>
<button class="btn btn-sm btn-secondary" data-bs-dismiss="modal" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
<!-- Telemetry Diff / Changes History Modal -->
<div aria-hidden="true" class="modal fade" id="sa-changes-modal" tabindex="-1">
<div class="modal-dialog modal-xl modal-dialog-centered modal-dialog-scrollable">
<div class="modal-content shadow-lg">
<div class="modal-header py-2 px-3">
<div class="d-flex align-items-center gap-2">
<span class="fs-5 text-warning">⚠ </span>
<h5 class="modal-title fw-bold text-warning mb-0">История изменений автозапуска (Телеметрия)</h5>
```

```
<span class="badge border bg-surface-2 text-muted fw-normal" id="sa-changes-modal-count">0 записей</span>
</div>
<button aria-label="Close" class="btn-close" data-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body py-3 px-3">
<div class="d-flex justify-content-between align-items-center mb-2 flex-wrap gap-2">
<div class="small text-muted">
              Каждый запуск телеметрии и ручное обновление («Обновить») фиксирует снимок автозапуска в SQLite и вычисляет дифференциальные изменения.
            </div>
<button class="btn btn-xs btn-outline-primary rounded-pill px-2.5 py-1" id="sa-btn-refresh-changes-modal">
<i class="bi bi-arrow-clockwise me-1"></i>Обновить историю
            </button>
</div>
<div class="table-responsive" style="max-height: 55vh;">
<table class="sa-table" id="sa-changes-table">
<thead>
<tr>
<th style="width: 140px;">Время фиксации</th>
<th class="text-center" style="width: 125px;">Тип изменения</th>
<th style="width: 190px;">Программа</th>
<th style="width: 150px;">Точка автозапуска</th>
<th>Детали изменения</th>
</tr>
</thead>
<tbody id="sa-changes-table-body">
<tr>
<td class="text-center py-4 text-muted" colspan="5">
                    Загрузка истории изменений...
                  </td>
</tr>
</tbody>
```

```
</table>
</div>
</div>
<div class="modal-footer py-2 px-3">
<button class="btn btn-sm btn-secondary" data-bs-dismiss="moda
l" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Startup Auditor Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main (Sta
rtup Auditor).
 *   Обеспечивает загрузку, фильтрацию, переключение состояния
, AI-диагностику,
 *   асинхронное обновление снимков телеметрии и аудит диффере
нциальных изменений.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/startup_auditor_tab/ma
in.js?v=20261006_v15" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/startup_auditor_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

// Windows Startup Auditor Frontend Logic
(function () {
  let allEntries = [];
  let currentSummary = null;

  async function initStartupAuditorTab() {
    console.log('[StartupAuditor] Initializing tab...');
```

```
    bindEvents();
    await loadAuditData();
    checkRecentChangesBadge().catch(() => {});
  }

  function bindEvents() {
    const btnRefreshTelemetry = document.getElementById('sa-bt
n-refresh-telemetry');
    if (btnRefreshTelemetry) {
      btnRefreshTelemetry.onclick = () => refreshStartupTeleme
try();
    }

    const btnRefreshLegacy = document.getElementById('sa-btn-r
efresh');
    if (btnRefreshLegacy) {
      btnRefreshLegacy.onclick = () => refreshStartupTelemetry
();
    }

    const btnViewChanges = document.getElementById('sa-btn-vie
w-changes');
    if (btnViewChanges) {
      btnViewChanges.onclick = () => showChangesModal();
    }

    const btnRefreshChangesModal = document.getElementById('sa
-btn-refresh-changes-modal');
    if (btnRefreshChangesModal) {
      btnRefreshChangesModal.onclick = () => loadChangesData()
;
    }

    const searchInput = document.getElementById('sa-search-inp
ut');
    if (searchInput) {
      searchInput.oninput = () => renderFilteredTable();
    }
```

```
60
61      const filterRisk = document.getElementById('sa-filter-risk
');
62      if (filterRisk) {
63        filterRisk.onchange = () => renderFilteredTable();
64      }
65
66      const filterLocation = document.getElementById('sa-filter-
location');
67      if (filterLocation) {
68        filterLocation.onchange = () => renderFilteredTable();
69      }
70
71      const filterState = document.getElementById('sa-filter-sta
te');
72      if (filterState) {
73        filterState.onchange = () => renderFilteredTable();
74      }
75
76      const btnExportJson = document.getElementById('sa-btn-expo
rt-json');
77      if (btnExportJson) {
78        btnExportJson.onclick = () => {
79          window.open('/api/v1/startup-auditor/export?format=jso
n', '_blank');
80        };
81      }
82
83      const btnExportCsv = document.getElementById('sa-btn-expor
t-csv');
84      if (btnExportCsv) {
85        btnExportCsv.onclick = () => {
86          window.open('/api/v1/startup-auditor/export?format=csv
', '_blank');
87        };
88      }
89    }
90
```

```
  /**
   * Асинхронно перезапускает сбор данных автозапуска через API телеметрии,
   * регистрирует снимок в SQLite и сохраняет дифференциальные изменения.
   */
  async function refreshStartupTelemetry() {
    const btn = document.getElementById('sa-btn-refresh-telemetry');
    const spinner = document.getElementById('sa-refresh-spinner');
    const statusText = document.getElementById('sa-telemetry-last-snap');

    if (btn) btn.disabled = true;
    if (spinner) spinner.classList.add('spinner-border', 'spinner-border-sm');

    try {
      if (statusText) statusText.textContent = 'Телеметрия: сбор снимка...';

      const res = await window.api.fetch('/api/v1/startup-auditor/refresh', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' }
      });

      if (res && res.report) {
        allEntries = res.report.entries || [];
        currentSummary = res.report.summary || {};
        renderSummaryCards(res.report);
        renderFilteredTable();
      }

      const count = res.changes_count || (res.changes ? res.changes.length : 0);
      const timeStr = new Date().toLocaleTimeString('ru-RU', {
```

```
 hour: '2-digit', minute: '2-digit', second: '2-digit' });

      if (statusText) {
        statusText.textContent = count > 0
          ? `Телеметрия: снимок ${timeStr} (изменений: ${count})`
          : `Телеметрия: снимок ${timeStr} (без изменений)`;
      }

      const badge = document.getElementById('sa-changes-badge');
      if (badge) {
        if (count > 0) {
          badge.textContent = count;
          badge.classList.remove('d-none');
        }
      }

      if (count > 0) {
        window.showToast?.(`Снимок сохранен! Зафиксировано изменений: ${count}`, 'warning');
      } else {
        window.showToast?.('Снимок телеметрии успешно зафиксирован (изменений нет)', 'success');
      }
    } catch (err) {
      console.error('[StartupAuditor] Error refreshing telemetry snapshot:', err);
      if (statusText) statusText.textContent = 'Телеметрия: ошибка сбора';
      window.showToast?.(`Ошибка обновления телеметрии: ${err.message}`, 'danger');
    } finally {
      if (btn) btn.disabled = false;
      if (spinner) spinner.classList.remove('spinner-border', 'spinner-border-sm');
    }
  }
```

```
149
150   async function checkRecentChangesBadge() {
151     try {
152       const data = await window.api.fetch('/api/v1/startup-aud
itor/changes?limit=10');
153       const badge = document.getElementById('sa-changes-badge'
);
154       if (badge && data && data.changes && data.changes.length
 > 0) {
155         badge.textContent = data.changes.length;
156         badge.classList.remove('d-none');
157       }
158     } catch {
159       // Игнорируем фоновые ошибки бейджа
160     }
161   }
162
163   async function showChangesModal() {
164     const modalEl = document.getElementById('sa-changes-modal'
);
165     if (!modalEl) return;
166
167     if (window.bootstrap && window.bootstrap.Modal) {
168       const bsModal = window.bootstrap.Modal.getOrCreateInstan
ce(modalEl);
169       bsModal.show();
170     }
171     await loadChangesData();
172   }
173
174   async function loadChangesData() {
175     const tbody = document.getElementById('sa-changes-table-bo
dy');
176     const countEl = document.getElementById('sa-changes-modal-
count');
177
178     if (tbody) {
179       tbody.innerHTML = `
```

```
        <tr>
          <td colspan="5" class="text-center py-4 text-muted">
            <div class="spinner-border spinner-border-sm text-warning me-2" role="status"></div>
            Загрузка истории изменений автозапуска из телеметрии...
          </td>
        </tr>
      `;
    }

    try {
      const data = await window.api.fetch('/api/v1/startup-auditor/changes?limit=100');
      const changes = data.changes || [];

      if (countEl) {
        countEl.textContent = `${changes.length} записей`;
      }

      if (changes.length === 0) {
        if (tbody) {
          tbody.innerHTML = `
            <tr>
              <td colspan="5" class="text-center py-4 text-muted">
                <i class="bi bi-shield-check text-success fs-4 d-block mb-1"></i>
                Изменений в конфигурации автозапуска между снимками телеметрии не зафиксировано.
              </td>
            </tr>
          `;
        }
        return;
      }

      if (tbody) {
```

```
212         tbody.innerHTML = changes.map(ch => {
213           const changeBadge = getChangeTypeBadge(ch.change_typ
    e);
214           const timeFormatted = ch.created_at ? ch.created_at.
    replace('T', ' ').substring(0, 19) : '-';
215
216           let detailsHtml = '';
217           if (ch.change_type === 'added') {
218             detailsHtml = `<span class="font-monospace small t
    ext-success text-truncate d-block" style="max-width: 420px;" t
    itle="${escapeHtml(ch.new_value)}">+ ${escapeHtml(ch.new_value
     || ch.item_name)}</span>`;
219           } else if (ch.change_type === 'removed') {
220             detailsHtml = `<span class="font-monospace small t
    ext-danger text-truncate d-block" style="max-width: 420px;" ti
    tle="${escapeHtml(ch.old_value)}">- ${escapeHtml(ch.old_value
    || ch.item_name)}</span>`;
221           } else if (ch.change_type === 'state_changed') {
222             detailsHtml = `<div class="small">Состояние: <stro
    ng class="text-danger">${escapeHtml(ch.old_value)}</strong> <i
     class="bi bi-arrow-right"></i> <strong class="text-success">$
    {escapeHtml(ch.new_value)}</strong></div>`;
223           } else if (ch.change_type === 'risk_changed') {
224             detailsHtml = `<div class="small">Уровень риска: <
    strong>${escapeHtml(ch.old_value)}</strong> <i class="bi bi-ar
    row-right"></i> <strong class="text-warning">${escapeHtml(ch.n
    ew_value)}</strong></div>`;
225           } else {
226             detailsHtml = `
227               <div class="small font-monospace text-muted text
    -truncate" style="max-width: 420px;" title="Было: ${escapeHtml
    (ch.old_value)}">Было: ${escapeHtml(ch.old_value || '-')}</div
    >
228               <div class="small font-monospace text-primary te
    xt-truncate" style="max-width: 420px;" title="Стало: ${escapeH
    tml(ch.new_value)}">Стало: ${escapeHtml(ch.new_value || '-')}<
    /div>
229             `;
```

```
          }

          return `
            <tr>
              <td class="font-monospace small text-muted">${timeFormatted}</td>
              <td class="text-center">${changeBadge}</td>
              <td>
                <div class="fw-semibold text-truncate" style="max-width: 190px;" title="${escapeHtml(ch.item_name)}">${escapeHtml(ch.item_name)}</div>
              </td>
              <td><span class="sa-badge sa-badge-loc">${escapeHtml(ch.location_type || 'registry_run')}</span></td>
              <td>${detailsHtml}</td>
            </tr>
          `;
        }).join('');
      }
    } catch (err) {
      console.error('[StartupAuditor] Error loading changes history:', err);
      if (tbody) {
        tbody.innerHTML = `
          <tr>
            <td colspan="5" class="text-center py-4 text-danger">
              Ошибка загрузки истории изменений: ${escapeHtml(err.message)}
            </td>
          </tr>
        `;
      }
    }
  }

  function getChangeTypeBadge(changeType) {
    switch (changeType) {
```

```
261       case 'added':
262         return '<span class="sa-badge sa-badge-enabled"><i cla
ss="bi bi-plus-circle"></i> Добавлен</span>';
263       case 'removed':
264         return '<span class="sa-badge sa-badge-disabled"><i cl
ass="bi bi-dash-circle"></i> Удален</span>';
265       case 'state_changed':
266         return '<span class="sa-badge sa-risk-warning"><i clas
s="bi bi-toggle2-on"></i> Статус</span>';
267       case 'path_changed':
268         return '<span class="sa-badge sa-risk-suspicious"><i c
lass="bi bi-pencil"></i> Путь</span>';
269       case 'risk_changed':
270         return '<span class="sa-badge sa-risk-critical"><i cla
ss="bi bi-shield-exclamation"></i> Риск</span>';
271       default:
272         return `<span class="sa-badge sa-badge-cat">${escapeHt
ml(changeType)}</span>`;
273     }
274   }
275
276   async function loadAuditData() {
277     const tbody = document.getElementById('sa-table-body');
278     if (tbody) {
279       tbody.innerHTML = `
280         <tr>
281           <td colspan="8" class="text-center py-4 text-muted">
282             <div class="spinner-border spinner-border-sm text-
primary me-2" role="status"></div>
283             Выполняется сканирование точек автозапуска Windows
...
284           </td>
285         </tr>
286       `;
287     }
288
289     try {
290       const report = await window.api.fetch('/api/v1/startup-a
```

```
uditor/audit');
      allEntries = report.entries || [];
      currentSummary = report.summary || {};

      renderSummaryCards(report);
      renderFilteredTable();
    } catch (err) {
      console.error('[StartupAuditor] Error loading audit repo
rt:', err);
      if (tbody) {
        tbody.innerHTML = `
          <tr>
            <td colspan="8" class="text-center py-4 text-dange
r">
              Ошибка загрузки данных аудита: ${err.message}
            </td>
          </tr>
        `;
      }
    }
  }

  function renderSummaryCards(report) {
    const summary = report.summary || {};

    // Health Score
    const scoreEl = document.getElementById('sa-health-score')
;
    if (scoreEl) {
      scoreEl.textContent = summary.health_score ?? '--';
      scoreEl.className = 'sa-card-value ' + (
        summary.health_score >= 80 ? 'text-success' : (summary
.health_score >= 60 ? 'text-warning' : 'text-danger')
      );
    }

    // Counters
    const activeEl = document.getElementById('sa-active-count'
```

```
);
    if (activeEl) activeEl.textContent = summary.active_entries ?? '--';

    const totalEl = document.getElementById('sa-total-count');
    if (totalEl) totalEl.textContent = summary.total_entries ?? '--';

    const disabledEl = document.getElementById('sa-disabled-count');
    if (disabledEl) disabledEl.textContent = summary.disabled_entries ?? '0';

    const brokenEl = document.getElementById('sa-broken-count');
    if (brokenEl) brokenEl.textContent = summary.broken_entries ?? '0';

    const threatsEl = document.getElementById('sa-threats-count');
    if (threatsEl) threatsEl.textContent = (summary.critical_count || 0) + (summary.suspicious_count || 0);

    // Recommendations
    const recDesc = document.getElementById('sa-rec-desc');
    if (recDesc) {
      if (report.recommendations && report.recommendations.length > 0) {
        recDesc.innerHTML = report.recommendations.map(r => `<div>${r}</div>`).join('');
      } else {
        recDesc.textContent = 'Конфигурация автозапуска оптимальна, критических замечаний нет.';
      }
    }
  }

  function renderFilteredTable() {
```

```
    const tbody = document.getElementById('sa-table-body');
    const tableCount = document.getElementById('sa-table-count
');
    if (!tbody) return;

    const query = (document.getElementById('sa-search-input')?
.value || '').toLowerCase().trim();
    const riskFilter = (document.getElementById('sa-filter-ris
k')?.value || '').toLowerCase();
    const locFilter = (document.getElementById('sa-filter-loca
tion')?.value || '').toLowerCase();
    const stateFilter = (document.getElementById('sa-filter-st
ate')?.value || '').toLowerCase();

    const filtered = allEntries.filter(e => {
      if (query) {
        const fullText = `${e.name || ''} ${e.command || ''} $
{e.executable_path || ''} ${e.publisher || ''} ${e.category ||
 ''}`.toLowerCase();
        if (!fullText.includes(query)) return false;
      }
      if (riskFilter && (e.risk_level || '').toLowerCase() !==
 riskFilter) {
        return false;
      }
      if (locFilter && (e.location_type || '').toLowerCase() !
== locFilter) {
        return false;
      }
      if (stateFilter === 'enabled' && !e.is_enabled) return f
alse;
      if (stateFilter === 'disabled' && e.is_enabled) return f
alse;

      return true;
    });

    if (tableCount) {
```

```
      tableCount.textContent = `Показано: ${filtered.length} из ${allEntries.length}`;
    }

    if (filtered.length === 0) {
      tbody.innerHTML = `
        <tr>
          <td colspan="8" class="text-center py-5 text-muted">
            <div class="fs-4 mb-2">□ </div>
            <div>Элементы автозапуска по заданным критериям не найдены</div>
          </td>
        </tr>
      `;
      return;
    }

    tbody.innerHTML = filtered.map(e => {
      const isEnabled = e.is_enabled;
      const statusBadge = isEnabled
        ? '<span class="sa-badge sa-badge-enabled"><i class="bi bi-check-circle-fill"></i> ВКЛ</span>'
        : '<span class="sa-badge sa-badge-disabled"><i class="bi bi-dash-circle"></i> ОТКЛ</span>';

      const riskBadge = getRiskBadge(e.risk_level);
      const isBroken = !e.file_exists && e.executable_path;

      const pathDisplay = isBroken
        ? `<div class="d-flex align-items-center gap-1 text-danger small font-monospace">
             <i class="bi bi-x-octagon-fill text-danger flex-shrink-0"></i>
             <span class="text-truncate fw-semibold" style="max-width: 480px;" title="${e.executable_path || e.command}">[Файл не найден] ${e.executable_path || e.command}</span>
           </div>`
        : `<div class="font-monospace small text-truncate" sty
```

```
le="max-width: 480px; color: var(--text-color);" title="${e.co
mmand || e.executable_path}">
407             ${e.executable_path || e.command}
408           </div>`;
409
410       const argsDisplay = e.arguments ? `
411         <div class="d-flex align-items-center gap-1 font-monos
pace text-truncate mt-1 text-muted" style="max-width: 480px; f
ont-size: 0.73rem;" title="${e.arguments}">
412           <span class="badge bg-secondary-subtle text-secondar
y-emphasis border border-secondary" style="font-size: 0.68rem;
 padding: 1px 4px;">ARG</span>
413           <span class="text-truncate">${e.arguments}</span>
414         </div>` : '';
415
416       const toggleIcon = isEnabled
417         ? '<i class="bi bi-toggle2-on text-success" style="fon
t-size: 1.4rem;"></i>'
418         : '<i class="bi bi-toggle2-off text-muted" style="font
-size: 1.4rem;"></i>';
419       const toggleTitle = isEnabled ? 'Отключить элемент автоз
апуска' : 'Включить элемент автозапуска';
420
421       let impactBadge = '';
422       if (!isEnabled) {
423         impactBadge = '<span class="sa-badge sa-badge-disabled
">Отключено</span>';
424       } else if (e.boot_impact === 'Высокое') {
425         impactBadge = '<span class="sa-badge sa-risk-critical"
>Высокое</span>';
426       } else if (e.boot_impact === 'Среднее') {
427         impactBadge = '<span class="sa-badge sa-risk-warning">
Среднее</span>';
428       } else {
429         impactBadge = '<span class="sa-badge sa-badge-cat">Низ
кое</span>';
430       }
431
```

```
      return `
        <tr class="sa-row-item" data-id="${e.id}" style="curso
r: pointer;" title="Нажмите для подробного AI-аудита">
          <td class="text-center">${statusBadge}</td>
          <td>
            <div class="fw-semibold text-truncate" style="max-
width: 230px; color: var(--text-color);" title="${e.name}">${e
.name}</div>
            <div class="small text-truncate text-muted" style=
"max-width: 230px; font-size: 0.73rem;" title="${e.publisher |
| 'Неизвестен'}">${e.publisher || 'Неизвестный издатель'}</div
>
          </td>
          <td><span class="sa-badge sa-badge-loc">${e.location
_type}</span></td>
          <td><span class="sa-badge sa-badge-cat">${e.category
 || 'Неизвестно'}</span></td>
          <td class="text-center">${riskBadge}</td>
          <td>
            ${pathDisplay}
            ${argsDisplay}
          </td>
          <td class="text-center">${impactBadge}</td>
          <td class="text-center">
            <button class="sa-toggle-btn" data-id="${e.id}" da
ta-enabled="${isEnabled}" title="${toggleTitle}">
              ${toggleIcon}
            </button>
          </td>
        </tr>
      `;
    }).join('');

    // Attach row click listeners for modal details
    tbody.querySelectorAll('.sa-row-item').forEach(row => {
      row.onclick = (evt) => {
        // Prevent modal open when clicking directly on the to
ggle button
```

```
460         if (evt.target.closest('.sa-toggle-btn')) {
461           return;
462         }
463         const entryId = row.getAttribute('data-id');
464         const entry = allEntries.find(item => item.id === entr
yId);
465         if (entry) {
466           showEntryModal(entry);
467         }
468       };
469     });
470 
471     // Attach toggle listeners
472     tbody.querySelectorAll('.sa-toggle-btn').forEach(btn => {
473       btn.onclick = async (evt) => {
474         evt.stopPropagation();
475         const entryId = btn.getAttribute('data-id');
476         const currentState = btn.getAttribute('data-enabled')
=== 'true';
477         const newState = !currentState;
478         try {
479           btn.disabled = true;
480           btn.style.opacity = '0.5';
481           const res = await window.api.fetch('/api/v1/startup-
auditor/toggle', {
482             method: 'POST',
483             headers: { 'Content-Type': 'application/json' },
484             body: JSON.stringify({ entry_id: entryId, enable:
newState })
485           });
486           if (res.success) {
487             window.showToast?.(`Автозагрузка ${newState ? 'вкл
ючена' : 'отключена'}`, 'success');
488             await loadAuditData();
489           } else {
490             window.showToast?.(`Ошибка переключения: ${res.mes
sage}`, 'danger') || alert(`Ошибка переключения: ${res.message
}`);
```

```
          }
        } catch (err) {
          window.showToast?.(`Ошибка: ${err.message}`, 'danger
') || alert(`Ошибка: ${err.message}`);
        } finally {
          btn.disabled = false;
          btn.style.opacity = '1';
        }
      };
    });
  }

  function showEntryModal(entry) {
    if (!entry) return;

    const modalEl = document.getElementById('sa-entry-detail-m
odal');
    if (!modalEl) return;

    // Header info
    const titleEl = document.getElementById('sa-modal-title');
    if (titleEl) titleEl.textContent = entry.name || 'Детали п
рограммы';

    const statusBadgeEl = document.getElementById('sa-modal-st
atus-badge');
    if (statusBadgeEl) {
      statusBadgeEl.innerHTML = entry.is_enabled
        ? '<span class="sa-badge sa-badge-enabled"><i class="b
i bi-check-circle-fill"></i> ВКЛ</span>'
        : '<span class="sa-badge sa-badge-disabled"><i class="
bi bi-dash-circle"></i> ОТКЛ</span>';
    }

    const riskBadgeEl = document.getElementById('sa-modal-risk
-badge');
    if (riskBadgeEl) {
      riskBadgeEl.innerHTML = getRiskBadge(entry.risk_level);
```

```
522     }
523
524     // Metadata Fields
525     const setTxt = (id, val) => {
526       const el = document.getElementById(id);
527       if (el) el.textContent = val ?? '-';
528     };
529     const setHtml = (id, html) => {
530       const el = document.getElementById(id);
531       if (el) el.innerHTML = html;
532     };
533
534     setTxt('sa-modal-name', entry.name);
535     setTxt('sa-modal-publisher', entry.publisher || 'Неизвестн
ый разработчик');
536     setTxt('sa-modal-location-type', entry.location_type || 'r
egistry_run');
537     setTxt('sa-modal-category', entry.category || 'Неизвестно'
);
538     setTxt('sa-modal-boot-impact', entry.boot_impact || 'Низко
e');
539     setTxt('sa-modal-location-path', entry.location_path || 'H
e указан');
540
541     setHtml('sa-modal-is-signed', entry.is_signed
542       ? '<span class="text-success"><i class="bi bi-patch-chec
k me-1"></i>Подписан (Valid)</span>'
543       : '<span class="text-muted"><i class="bi bi-patch-questi
on me-1"></i>Без подписи / Неизвестно</span>'
544     );
545
546     setHtml('sa-modal-file-exists', entry.file_exists
547       ? '<span class="text-success"><i class="bi bi-file-earma
rk-check me-1"></i>Файл существует</span>'
548       : '<span class="text-danger fw-bold"><i class="bi bi-fil
e-earmark-x me-1"></i>Файл не найден (битая ссылка)</span>'
549     );
550
```

```
    setTxt('sa-modal-file-size', entry.file_size_kb > 0 ? `${entry.file_size_kb.toFixed(1)} KB` : '-');
    setTxt('sa-modal-executable-path', entry.executable_path || entry.command || 'Путь не указан');
    setTxt('sa-modal-command', entry.command || '-');

    // Reset AI Diagnostic section
    const aiExplanation = document.getElementById('sa-modal-ai-explanation');
    if (aiExplanation) {
      aiExplanation.innerHTML = `Нажмите «Анализ контекста», чтобы получить экспертное объяснение назначения программы от языковой модели, оценку рисков и рекомендации по оптимизации.`;
    }

    // AI Diagnose button
    const diagnoseBtn = document.getElementById('btn-sa-modal-diagnose');
    if (diagnoseBtn) {
      diagnoseBtn.onclick = async () => {
        aiExplanation.innerHTML = `
          <div class="d-flex align-items-center gap-2 py-2 text-primary">
            <div class="spinner-border spinner-border-sm" role="status"></div>
            <span>Генерация AI-диагностики и анализа рисков для программы «${escapeHtml(entry.name)}»...</span>
          </div>
        `;
        const activeModel = window.activeModelName
          || window.userSettings?.model
          || document.getElementById('model-select')?.value
          || document.getElementById('header-model-select')?.value
          || '';
        try {
          const res = await window.api.fetch('/api/v1/startup-auditor/explain', {
```

```
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({
              entry_id: entry.id,
              name: entry.name,
              publisher: entry.publisher,
              executable_path: entry.executable_path,
              command: entry.command,
              arguments: entry.arguments,
              location_type: entry.location_type,
              location_path: entry.location_path,
              risk_level: entry.risk_level,
              is_enabled: entry.is_enabled,
              file_exists: entry.file_exists,
              is_signed: entry.is_signed,
              boot_impact: entry.boot_impact,
              model: activeModel || undefined
            })
          });

          aiExplanation.innerHTML = `
            <div class="mb-2"><strong class="text-primary"><i class="bi bi-card-text me-1"></i>Назначение:</strong> ${escapeHtml(res.summary)}</div>
            <div class="mb-2"><strong class="text-muted"><i class="bi bi-building me-1"></i>Разработчик / Категория:</strong> ${escapeHtml(res.developer || entry.publisher || 'Неизвестен')} (${escapeHtml(res.category || 'Приложение')})</div>
            <div class="mb-2"><strong class="text-warning"><i class="bi bi-shield-lock me-1"></i>Оценка безопасности:</strong> ${escapeHtml(res.security_verdict)}</div>
            <div class="mb-2"><strong class="text-info"><i class="bi bi-speedometer2 me-1"></i>Влияние на запуск:</strong> ${escapeHtml(res.boot_impact_analysis)}</div>
            <div class="p-2 mb-2 rounded alert alert-warning">
              <strong class="text-warning"><i class="bi bi-lightbulb me-1"></i>Рекомендация:</strong> ${escapeHtml(res.startup_recommendation)}
```

```
            </div>
            ${res.action_steps && res.action_steps.length > 0 ? `
              <h6 class="small text-uppercase text-muted fw-bold mb-1">Рекомендуемые действия:</h6>
              <ul class="mb-0 ps-3">
                ${res.action_steps.map(s => `<li>${escapeHtml(s)}</li>`).join('')}
              </ul>
            ` : ''}
          `;
        } catch (err) {
          aiExplanation.innerHTML = `<div class="text-danger py-2"><i class="bi bi-exclamation-octagon me-1"></i>Ошибка получения AI-анализа: ${escapeHtml(err.message)}</div>`;
        }
      };
    }

    // Modal Footer Toggle Button
    const modalToggleBtn = document.getElementById('btn-sa-modal-toggle');
    if (modalToggleBtn) {
      modalToggleBtn.innerHTML = entry.is_enabled
        ? '<i class="bi bi-toggle2-off me-1 text-warning"></i>Отключить автозапуск'
        : '<i class="bi bi-toggle2-on me-1 text-success"></i>Включить автозапуск';
      modalToggleBtn.onclick = async () => {
        const newState = !entry.is_enabled;
        try {
          modalToggleBtn.disabled = true;
          const res = await window.api.fetch('/api/v1/startup-auditor/toggle', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ entry_id: entry.id, enable: newState })
```

```
          });
          if (res.success) {
            entry.is_enabled = newState;
            showEntryModal(entry);
            window.showToast?.(`Автозагрузка ${newState ? 'вкл
ючена' : 'отключена'}`, 'success');
            await loadAuditData();
          } else {
            window.showToast?.(`Ошибка переключения: ${res.mes
sage}`, 'danger') || alert(`Ошибка переключения: ${res.message
}`);
          }
        } catch (err) {
          window.showToast?.(`Ошибка: ${err.message}`, 'danger
') || alert(`Ошибка: ${err.message}`);
        } finally {
          modalToggleBtn.disabled = false;
        }
      };
    }

    // Modal Footer Copy Button
    const modalCopyBtn = document.getElementById('btn-sa-modal
-copy');
    if (modalCopyBtn) {
      modalCopyBtn.onclick = () => {
        const textToCopy = entry.executable_path || entry.comm
and || '';
        navigator.clipboard.writeText(textToCopy).then(() => {
          const orig = modalCopyBtn.innerHTML;
          modalCopyBtn.innerHTML = '<i class="bi bi-check2 me-
1 text-success"></i>Скопировано!';
          setTimeout(() => { modalCopyBtn.innerHTML = orig; },
 1800);
        });
      };
    }

```

```
    // Show modal via Bootstrap
    if (window.bootstrap && window.bootstrap.Modal) {
      const bsModal = window.bootstrap.Modal.getOrCreateInstan
ce(modalEl);
      bsModal.show();
    }
  }

  function getRiskBadge(level) {
    switch ((level || '').toLowerCase()) {
      case 'critical':
        return '<span class="sa-badge sa-risk-critical"><i cla
ss="bi bi-shield-x"></i> CRITICAL</span>';
      case 'suspicious':
        return '<span class="sa-badge sa-risk-suspicious"><i c
lass="bi bi-exclamation-octagon"></i> SUSPICIOUS</span>';
      case 'warning':
        return '<span class="sa-badge sa-risk-warning"><i clas
s="bi bi-exclamation-triangle"></i> WARNING</span>';
      case 'notice':
        return '<span class="sa-badge sa-risk-notice"><i class
="bi bi-info-circle"></i> NOTICE</span>';
      default:
        return '<span class="sa-badge sa-risk-clean"><i class=
"bi bi-shield-check"></i> CLEAN</span>';
    }
  }

  function escapeHtml(str) {
    if (!str) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
  }

```

```
  // Expose globally for Apps Hub
  window.initStartupAuditorTab = initStartupAuditorTab;
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/storage_manager_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/storage_manager_tab/ind
ex.html" class="w-100 h-100 border-0"></iframe>

 * File: index.html
 * Project: ai-windows-admin
 * Package: windows/api/webgui/storage_manager_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ================================================================
==================
-->
<!-- Storage & Disk Volumes Management Tab -->
<div class="container-fluid py-2 px-3">
<!-- Top Stat Cards -->
<div class="row g-2 mb-3">
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 sha
dow-sm p-3" data-panel-id="panel-storage-manager-1" id="panel-
storage-manager-1">
<div class="text-muted small text-uppercase fw-semibold">Физич
еские диски  </div>
```

```
<div class="fs-4 fw-bold mt-1 text-info" id="sm-disks-count">--</div>
<div class="small text-muted mt-1" id="sm-disks-sub">Опрос накопителей...</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-storage-manager-2" id="panel-storage-manager-2">
<div class="text-muted small text-uppercase fw-semibold">Логические тома  </div>
<div class="fs-4 fw-bold mt-1 text-primary" id="sm-volumes-count">--</div>
<div class="small text-muted mt-1" id="sm-volumes-sub">NTFS / ReFS / FAT32</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-storage-manager-3" id="panel-storage-manager-3">
<div class="text-muted small text-uppercase fw-semibold">BitLocker Защита  </div>
<div class="fs-4 fw-bold mt-1 text-success" id="sm-bitlocker-val">--</div>
<div class="small text-muted mt-1" id="sm-bitlocker-sub">Статус шифрования</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-storage-manager-4" id="panel-storage-manager-4">
<div class="text-muted small text-uppercase fw-semibold">NTFS Сжатие / VHD  </div>
<div class="fs-4 fw-bold mt-1 text-warning" id="sm-compression-val">--</div>
```

```
<div class="small text-muted mt-1" id="sm-compression-sub">Ком
пактность дисков</div>
</div>
</div>
</div>
<!-- Action Bar -->
<div class="d-flex align-items-center justify-content-between
mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap
-2">
<div class="d-flex align-items-center gap-2">
<select class="form-select form-select-sm poll-freq-select" da
ta-poll-id="storage_manager" id="sm-poll-freq" title="Частота
опроса: Хранилище">
<option value="start">Только при старте</option>
<option selected="" value="manual">По запросу</option>
<option value="3">Каждые 3 сек</option>
<option value="5">Каждые 5 сек</option>
<option value="10">Каждые 10 сек</option>
<option value="30">Каждые 30 сек</option>
<option value="60">Каждые 60 сек</option>
<option value="120">Каждые 120 сек</option>
</select>
<button class="btn btn-sm btn-primary d-flex align-items-cente
r gap-1.5" id="sm-refresh-btn">
<i class="bi bi-arrow-clockwise"></i>
<span>Обновить состояние</span>
</button>
<button class="btn btn-sm btn-outline-info d-flex align-items-
center gap-1.5" id="sm-quick-rescan-btn">
<i class="bi bi-search"></i>
<span>Rescan (diskpart)</span>
</button>
<button class="btn btn-sm btn-outline-warning d-flex align-ite
ms-center gap-1.5" id="sm-defrag-btn">
<i class="bi bi-speedometer"></i>
<span>Дефрагментация / TRIM</span>
</button>
</div>
```

```
<div class="d-flex align-items-center gap-2">
<input class="form-control form-control-sm bg-dark text-white border-secondary" id="sm-filter-input" placeholder="Фильтр по томам/дискам..." style="width: 220px;" type="text"/>
</div>
</div>
<!-- Tables Section -->
<div class="row g-3">
<div class="col-12 col-lg-7">
<div class="card bg-dark text-white border-secondary shadow-sm " data-panel-id="panel-storage-manager-toma-i-razdely-diskov" id="panel-storage-manager-toma-i-razdely-diskov">
<div class="card-header border-secondary d-flex align-items-center justify-content-between py-2 px-3">
<span class="fw-semibold"><i class="bi bi-hdd-stack text-primary me-2"></i>Тома и разделы дисков</span>
<span class="badge bg-secondary" id="sm-volumes-badge">0 томов</span>
</div>
<div class="card-body p-0 table-responsive" style="max-height: 420px; overflow-y: auto;">
<table class="table table-dark table-hover table-striped align-middle mb-0 small" id="sm-volumes-table">
<thead>
<tr class="text-secondary">
<th>Том / Буква</th>
<th>Метка</th>
<th>ФС</th>
<th>Размер</th>
<th>Свободно</th>
<th>BitLocker</th>
<th class="text-end">Действия</th>
</tr>
</thead>
<tbody id="sm-volumes-tbody">
<tr><td class="text-center text-muted py-3" colspan="7">Загрузка данных о томах...</td></tr>
</tbody>
```

```
</table>
</div>
</div>
</div>
<div class="col-12 col-lg-5">
<div class="card bg-dark text-white border-secondary shadow-sm
 mb-3" data-panel-id="panel-storage-manager-bystrye-operatsii-
s-diskami" id="panel-storage-manager-bystrye-operatsii-s-diska
mi">
<div class="card-header border-secondary d-flex align-items-ce
nter justify-content-between py-2 px-3">
<span class="fw-semibold"><i class="bi bi-terminal text-warnin
g me-2"></i>Быстрые операции с дисками</span>
</div>
<div class="card-body p-3">
<div class="mb-3">
<label class="form-label small text-muted">Выбор операции:</la
bel>
<select class="form-select form-select-sm bg-dark text-white b
order-secondary" id="sm-action-select">
<option value="diskpart_rescan">Обновить конфигурацию дисков (
rescan)</option>
<option value="bitlocker_status">Проверить статус шифрования (
manage-bde)</option>
<option value="compact_query">Запрос сжатия ОС (compact /compa
ctos:query)</option>
<option value="defrag_analyze">Анализ оптимизации диска C: (de
frag /A)</option>
</select>
</div>
<div class="mb-3">
<label class="form-label small text-muted">Целевой диск / том
(если применимо):</label>
<input class="form-control form-control-sm bg-dark text-white
border-secondary" id="sm-target-input" placeholder="C: или 0"
type="text"/>
</div>
<div class="d-flex align-items-center justify-content-between"
```

```
>
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="sm-dry-run-switch" type="checkbox"/>
<label class="form-check-label small text-muted" for="sm-dry-run-switch">Безопасный режим (Dry Run)</label>
</div>
<button class="btn btn-sm btn-success px-3" id="sm-execute-btn">
<i class="bi bi-play-fill"></i> Выполнить
            </button>
</div>
</div>
</div>
<!-- Action Log Output -->
<div class="card bg-dark text-white border-secondary shadow-sm" data-panel-id="panel-storage-manager-konsolnyy-vyvod" id="panel-storage-manager-konsolnyy-vyvod">
<div class="card-header border-secondary py-2 px-3 d-flex align-items-center justify-content-between">
<span class="small fw-semibold text-muted"><i class="bi bi-code-square me-1"></i>Консольный вывод</span>
<button class="btn btn-link btn-sm text-decoration-none text-muted p-0" id="sm-clear-log-btn">Очистить</button>
</div>
<div class="card-body p-2">
<pre class="mb-0 small bg-black text-light p-2 rounded" id="sm-console-output" style="max-height: 180px; overflow-y: auto; font-family: Consolas, monospace; font-size: 0.78rem;">Готов к операциям.</pre>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Storage Manager Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/storage_manager_tab/main.js?v=20261001_v1" type="module"></script>
 *
 *   JavaScript Import:
 *     import { initStorageManagerTab } from '/windows/api/webgui/storage_manager_tab/main.js';
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/storage_manager_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * storage_manager_tab/main.js — Управление накопителями, разделами и томами
 * Updated: 2026-10-10 17:35:00
 */

const registerTabPoller = window.registerTabPoller || function () {};
```

```
30
31 let isInitialized = false;
32
33 export async function fetchStorageSummary() {
34   try {
35     const res = await fetch('/api/storage-manager/summary');
36     if (!res.ok) throw new Error(`HTTP ${res.status}`);
37     const data = await res.json();
38     renderStorageSummary(data);
39   } catch (err) {
40     console.warn('[StorageManager] Ошибка получения сводки:',
   err);
41   }
42 }
43
44 function renderStorageSummary(data) {
45   const disksCountEl = document.getElementById('sm-disks-count
   ');
46   const volumesCountEl = document.getElementById('sm-volumes-c
   ount');
47   const bitlockerEl = document.getElementById('sm-bitlocker-va
   l');
48   const compressionEl = document.getElementById('sm-compressio
   n-val');
49   const tbody = document.getElementById('sm-volumes-tbody');
50   const badge = document.getElementById('sm-volumes-badge');
51
52   if (disksCountEl) disksCountEl.textContent = data.disks_coun
   t ?? '--';
53   if (volumesCountEl) volumesCountEl.textContent = data.volume
   s ? data.volumes.length : '--';
54   if (bitlockerEl) bitlockerEl.textContent = data.bitlocker_st
   atus || 'Готово';
55   if (compressionEl) compressionEl.textContent = data.compact_
   os_status || 'CompactOS';
56
57   if (!tbody) return;
58
```

```
  const volumes = data.volumes || [
    { drive: 'C:', label: 'System', fs: 'NTFS', size: '476 GB', free: '180 GB', bitlocker: 'Включен' },
    { drive: 'D:', label: 'Data', fs: 'NTFS', size: '931 GB', free: '420 GB', bitlocker: 'Выключен' }
  ];

  if (badge) badge.textContent = `${volumes.length} томов`;

  tbody.innerHTML = volumes.map(v => `
    <tr>
      <td class="fw-bold text-info">${v.drive || v.name || 'Том'}</td>
      <td>${v.label || '-'}</td>
      <td><span class="badge bg-secondary">${v.fs || 'NTFS'}</span></td>
      <td>${v.size || '-'}</td>
      <td class="text-success">${v.free || '-'}</td>
      <td><span class="badge ${v.bitlocker === 'Включен' ? 'bg-success' : 'bg-warning text-dark'}">${v.bitlocker || 'N/A'}</span></td>
      <td class="text-end">
        <button class="btn btn-xs btn-outline-primary py-0 px-1 sm-btn-action" data-drive="${v.drive || 'C:'}" title="Анализ оптимизации">
          <i class="bi bi-gear"></i>
        </button>
      </td>
    </tr>
  `).join('');
}

async function executeStorageAction() {
  const actionSelect = document.getElementById('sm-action-select');
  const targetInput = document.getElementById('sm-target-input');
  const dryRunSwitch = document.getElementById('sm-dry-run-swi
```

```
tch');
  const output = document.getElementById('sm-console-output');

  if (!actionSelect || !output) return;

  const op = actionSelect.value;
  const target = targetInput ? targetInput.value.trim() : '';
  const dryRun = dryRunSwitch ? dryRunSwitch.checked : true;

  output.textContent = `[${new Date().toLocaleTimeString()}] Запуск операции: ${op} (DryRun: ${dryRun})...\n`;

  try {
    const res = await fetch('/api/storage-manager/action', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ action: op, target: target, dry_run: dryRun })
    });
    const result = await res.json();
    output.textContent += JSON.stringify(result, null, 2);
    if (window.toast) {
      if (res.ok) window.toast.success('Операция выполнена', `Код: ${result.status || 'OK'}`);
      else window.toast.error('Ошибка', result.detail || 'Сбой операции');
    }
  } catch (err) {
    output.textContent += `\nОшибка запроса: ${err.message}`;
    if (window.toast) window.toast.error('Ошибка сети', err.message);
  }
}

const POLL_ID = 'storage_manager';
let timerId = null;

function getFrequency() {
```

```
  try {
    const saved = localStorage.getItem(`poll_freq_${POLL_ID}`)
;
    if (saved) return saved;
  } catch (_) {}
  return 'manual';
}

function setFrequency(freq) {
  try {
    localStorage.setItem(`poll_freq_${POLL_ID}`, freq);
  } catch (_) {}
  applyPoller(freq, false);
}

function stopPolling() {
  if (window.unregisterTabPoller) {
    window.unregisterTabPoller(`tab-storage-manager_${POLL_ID}
`);
  }
  if (timerId) {
    clearInterval(timerId);
    timerId = null;
  }
}

function applyPoller(freq, runInitial = false) {
  stopPolling();
  if (freq === 'start' || freq === 'manual') {
    if (runInitial && (window.isTabActive ? window.isTabActive
('tab-storage-manager') : true)) {
      fetchStorageSummary();
    }
    return;
  }

  const intervalSec = parseInt(freq, 10);
  if (isNaN(intervalSec) || intervalSec <= 0) return;
```

```
154
155   const intervalMs = intervalSec * 1000;
156   const pollerId = `tab-storage-manager_${POLL_ID}`;
157
158   if (window.registerTabPoller) {
159     window.registerTabPoller('tab-storage-manager', fetchStora
geSummary, intervalMs, { pollerId, immediate: runInitial });
160   } else {
161     if (runInitial && (window.isTabActive ? window.isTabActive
('tab-storage-manager') : true)) {
162       fetchStorageSummary();
163     }
164     timerId = setInterval(() => {
165       if (window.isTabActive ? window.isTabActive('tab-storage
-manager') : true) {
166         fetchStorageSummary();
167       }
168     }, intervalMs);
169   }
170 }
171
172 export function initStorageManagerTab() {
173   if (isInitialized) return;
174   isInitialized = true;
175
176   const refreshBtn = document.getElementById('sm-refresh-btn')
;
177   if (refreshBtn) refreshBtn.addEventListener('click', fetchSt
orageSummary);
178
179   const execBtn = document.getElementById('sm-execute-btn');
180   if (execBtn) execBtn.addEventListener('click', executeStorag
eAction);
181
182   const clearBtn = document.getElementById('sm-clear-log-btn')
;
183   if (clearBtn) clearBtn.addEventListener('click', () => {
184     const out = document.getElementById('sm-console-output');
```

```
    if (out) out.textContent = 'Журнал очищен.';
  });

  const select = document.getElementById('sm-poll-freq');
  if (select) {
    const currentFreq = getFrequency();
    select.value = currentFreq;
    select.onchange = (e) => {
      const newFreq = e.target.value;
      setFrequency(newFreq);
      if (newFreq !== 'manual' && newFreq !== 'start') {
        fetchStorageSummary();
      }
      if (window.showToast) {
        const label = select.options[select.selectedIndex]?.te
xt || newFreq;
        window.showToast(`Частота опроса хранилища: ${label}`,
 'info');
      }
    };
    applyPoller(currentFreq, window.isTabActive ? window.isTab
Active('tab-storage-manager') : true);
  } else {
    applyPoller(getFrequency(), window.isTabActive ? window.is
TabActive('tab-storage-manager') : true);
  }
}

export function activateStorageManagerTab() {
  applyPoller(getFrequency(), true);
}

window.initStorageManagerTab = initStorageManagerTab;
window.activateStorageManagerTab = activateStorageManagerTab;

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/storage_wear_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/storage_wear_tab/index.
html" class="w-100 h-100 border-0"></iframe>

 * File: index.html
 * Project: ai-windows-admin
 * Package: windows/api/webgui/storage_wear_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ================================================================
==================
-->
<style>
  .wear-tab-wrapper {
    padding: 0.75rem;
    color: var(--text-color);
  }
  .diag-panel {
    background: var(--surface-1);
    border: 1px solid var(--border-color);
    border-radius: 8px;
    overflow: hidden;
```

```
}
.diag-panel-header {
  background: var(--surface-2);
  padding: 8px 14px;
  border-bottom: 1px solid var(--border-color);
  font-weight: 600;
  font-size: 0.85rem;
  color: var(--nav-active);
}
.about-sys-progress-track {
  background: var(--surface-2);
  border: 1px solid var(--border-color);
  border-radius: 4px;
  height: 8px;
  overflow: hidden;
  min-width: 80px;
}
.about-sys-progress-bar {
  height: 100%;
  border-radius: 4px;
  transition: width 0.3s ease;
}
.about-sys-spec-table {
  width: 100%;
  border-collapse: separate;
  border-spacing: 0;
}
.about-sys-spec-table td {
  padding: 8px 12px;
  border-bottom: 1px solid var(--border-color);
  vertical-align: middle;
}
.about-sys-spec-key {
  color: var(--text-muted);
  width: 200px;
  font-size: 0.82rem;
  font-weight: 500;
}
```

```
  .about-sys-spec-val {
    color: var(--text-color);
    font-weight: 500;
    font-size: 0.82rem;
  }
</style>
<div class="wear-tab-wrapper">
<!-- Top Header & Status -->
<div class="card shadow-sm border-secondary-subtle mb-3" data-panel-id="panel-storage-wear-4-iznos-nakopiteley-i-batarei" id="panel-storage-wear-4-iznos-nakopiteley-i-batarei">
<div class="card-body p-3">
<div class="d-flex flex-wrap align-items-center justify-content-between gap-2">
<div class="d-flex align-items-center gap-2">
<div class="fs-4">□ </div>
<div>
<h5 class="mb-0 fw-bold">4. Износ накопителей и батареи (Storage &amp;amp; Battery Health)</h5>
<small class="text-muted">Мониторинг ресурса SSD/NVMe, файловых систем, SMART и деградации аккумулятора</small>
</div>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<select class="form-select form-select-sm poll-freq-select" data-poll-id="storage_wear" id="diag-wear-poll-freq" title="Частота опроса: Износ накопителей">
<option value="start">Только при старте</option>
<option selected="" value="manual">По запросу</option>
<option value="3">Каждые 3 сек</option>
<option value="5">Каждые 5 сек</option>
<option value="10">Каждые 10 сек</option>
<option value="30">Каждые 30 сек</option>
<option value="60">Каждые 60 сек</option>
<option value="120">Каждые 120 сек</option>
</select>
<button class="btn btn-sm btn-outline-info rounded-pill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id="btn-diag-
```

```
wear-refresh">
<i class="bi bi-arrow-clockwise"></i> Обновить
          </button>
</div>
</div>
</div>
</div>
<div class="row g-2 mb-3">
<!-- Disks Health & Wear Panel -->
<div class="col-12 col-xl-8 col-lg-8">
<div class="diag-panel h-100" data-panel-id="panel-storage-wea
r-1" id="panel-storage-wear-1">
<div class="diag-panel-header d-flex justify-content-between a
lign-items-center">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-hdd-fill text-primary"></i>
<span class="text-body">Здоровье накопителей SSD/NVMe и износ
(Wear &amp;amp; SMART)</span>
</div>
<span class="badge bg-secondary" id="diag-wear-disks-count">0
дисков</span>
</div>
<div class="table-responsive" style="max-height: 480px; overfl
ow-y: auto;">
<table class="table table-hover table-sm mb-0" style="font-siz
e: 0.82rem; font-family: ui-monospace, SFMono-Regular, Menlo,
Monaco, Consolas, monospace;">
<thead>
<tr>
<th>Физический накопитель</th>
<th>Тип / Шина</th>
<th>Разделы</th>
<th style="text-align: right;">Объем</th>
<th style="text-align: right;">Свободно</th>
<th>Наработка / Старт</th>
<th style="text-align: right;">Ввод / Вывод (I/O)</th>
<th style="width: 130px;">Здоровье / Износ</th>
<th>Статус</th>
```

```
</tr>
</thead>
<tbody id="diag-wear-disks-tbody">
<tr><td class="text-center py-4 text-muted" colspan="9">Опрос контроллеров накопителей...</td></tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- Battery Deep Telemetry -->
<div class="col-12 col-xl-4 col-lg-4">
<div class="diag-panel h-100" data-panel-id="panel-storage-wear-2" id="panel-storage-wear-2">
<div class="diag-panel-header d-flex justify-content-between align-items-center">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-battery-charging text-warning"></i>
<span class="text-body">Деградация и состояние батареи (Battery Degradation)</span>
</div>
<span class="badge bg-body-secondary border border-secondary text-body" id="diag-battery-source-badge">AC Mains</span>
</div>
<div class="p-3" id="diag-battery-details-container">
<div class="text-center py-4 text-muted small">
<i class="bi bi-plug-fill fs-4 text-info d-block mb-1"></i>
            Стационарный компьютер — питание напрямую от электросети
          </div>
</div>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Storage Wear Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/storage_wear_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/storage_wear_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-12 06:50:00
 * ============================================================
 */

(function () {
  'use strict';

  let autoRefreshTimer = null;

  function escapeHtml(str) {
    if (str === null || str === undefined) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
```

```
      .replace(/'/g, '&#039;');
  }

  function setText(id, text) {
    const el = document.getElementById(id);
    if (el) el.textContent = text !== null && text !== undefin
ed ? String(text) : '--';
  }

  async function apiFetch(url) {
    if (window.api && typeof window.api.fetch === 'function')
{
      return await window.api.fetch(url);
    }
    const res = await fetch(url);
    if (!res.ok) throw new Error(`HTTP ${res.status}: ${res.st
atusText}`);
    return await res.json();
  }

  async function fetchStorageBatteryWear() {
    try {
      const data = await apiFetch('/api/v1/system/diagnostics/
storage-battery');
      if (!data) return;

      // Disks
      const disksTbody = document.getElementById('diag-wear-di
sks-tbody');
      if (disksTbody && data.disks_wear) {
        setText('diag-wear-disks-count', `${data.disks_wear.le
ngth} дисков`);
        if (data.disks_wear.length === 0) {
          disksTbody.innerHTML = '<tr><td colspan="9" class="t
ext-center py-4 text-muted">Физических накопителей не обнаруже
но</td></tr>';
        } else {
          const formatBytesLocal = (bytes) => {
```

```
            if (bytes == null || bytes === 0) return '0 B';
            const k = 1024;
            const sizes = ['B', 'KB', 'MB', 'GB', 'TB', 'PB'];
            const i = Math.floor(Math.log(bytes) / Math.log(k)
);
            return `${parseFloat((bytes / Math.pow(k, i)).toFi
xed(1))} ${sizes[i]}`;
          };

          const formatPohLocal = (hours) => {
            if (hours == null || hours <= 0) return null;
            const days = Math.floor(hours / 24);
            const years = (hours / (24 * 365.25)).toFixed(1);
            if (days >= 365) return `${hours.toLocaleString()}
 ч (≈ ${years} г)`;
            if (days >= 1) return `${hours.toLocaleString()} ч
 (${days} д)`;
            return `${hours.toLocaleString()} ч`;
          };

          disksTbody.innerHTML = data.disks_wear.map(d => {
            const diskName = d.name || d.model || d.device_id
|| 'Физический диск';
            const devId = d.device_id ? `<span class="badge bg
-secondary-subtle text-light border border-secondary me-1.5">$
{escapeHtml(d.device_id)}</span>` : '';
            const isSsd = (d.media_type && d.media_type.toUppe
rCase().includes('SSD')) || (d.bus_type && d.bus_type.toUpperC
ase().includes('NVME'));
            const diskIcon = isSsd ? 'bi-device-ssd text-info'
 : 'bi-hdd text-primary';
            const typeBus = [d.media_type, d.bus_type].filter(
Boolean).join(' / ') || 'Storage';
            const partsBadge = d.partitions && d.partitions !=
= '—'
              ? `<span class="badge bg-primary-subtle text-inf
o border border-info-subtle font-monospace">${escapeHtml(d.par
titions)}</span>`
```

```
              : '<span class="text-muted">—</span>';
            const freeGbStr = d.free_gb != null ? `${d.free_gb} GB` : '<span class="text-muted">—</span>';
            const healthPct = d.health_pct != null ? d.health_pct : 100;
            const healthColor = healthPct >= 80 ? 'success' : (healthPct >= 50 ? 'warning' : 'danger');

            // Наработка и дата первого включения
            const pohFormatted = formatPohLocal(d.power_on_hours);
            const startStr = d.first_power_on ? `Старт: ${escapeHtml(d.first_power_on)}` : '';
            let pohHtml = '<span class="text-muted font-monospace small">—</span>';
            if (pohFormatted && startStr) {
              pohHtml = `<div class="font-monospace text-info">${pohFormatted}</div><div class="small text-muted font-monospace mt-0.5">${startStr}</div>`;
            } else if (pohFormatted) {
              pohHtml = `<div class="font-monospace text-info">${pohFormatted}</div>`;
            } else if (startStr) {
              pohHtml = `<div class="small text-muted font-monospace">${startStr}</div>`;
            }

            // Объемы ввода/вывода (запись и чтение)
            const ioHtml = `
              <div class="font-monospace text-light" style="white-space: nowrap;" title="Записано за сессию">
                <i class="bi bi-arrow-up-circle text-warning me-1"></i>Записано: ${formatBytesLocal(d.bytes_written)}
              </div>
              <div class="small font-monospace text-info mt-0.5" style="white-space: nowrap;" title="Прочитано за сессию">
                <i class="bi bi-arrow-down-circle text-info me-1"></i>Прочитано: ${formatBytesLocal(d.bytes_read)}
```

```
              </div>
            `;

            return `
            <tr>
              <td class="fw-bold text-white">
                <div class="d-flex align-items-center">
                  <i class="bi ${diskIcon} me-2 fs-6"></i>
                  <div>
                    <div class="text-truncate" style="max-widt
h: 250px;" title="${escapeHtml(diskName)}">${escapeHtml(diskNa
me)}</div>
                    <div class="small text-muted font-monospac
e mt-0.5">${devId}${d.serial_number && d.serial_number !== 'N/
A' ? `<span class="text-secondary">S/N: ${escapeHtml(d.serial_
number)}</span>` : ''}</div>
                  </div>
                </div>
              </td>
              <td class="text-muted font-monospace"><span clas
s="badge bg-dark border border-secondary text-light">${escapeH
tml(typeBus)}</span></td>
              <td>${partsBadge}</td>
              <td style="text-align: right;" class="text-light
 font-monospace">${d.total_gb} GB</td>
              <td style="text-align: right;" class="text-succe
ss font-monospace">${freeGbStr}</td>
              <td>${pohHtml}</td>
              <td style="text-align: right;">${ioHtml}</td>
              <td>
                <div class="d-flex align-items-center gap-1.5"
>
                  <div class="about-sys-progress-track flex-gr
ow-1" style="height: 6px; min-width: 55px;">
                    <div class="about-sys-progress-bar bg-${he
althColor}" style="width: ${healthPct}%;"></div>
                  </div>
                  <span class="small font-monospace text-${hea
```

```
lthColor} ms-1">${healthPct}%</span>
                </div>
              </td>
              <td><span class="badge bg-${healthColor}-subtle
text-${healthColor} border border-${healthColor}">${escapeHtml
(d.status || 'OK')}</span></td>
            </tr>
          `;}).join('');
        }
      }

      // Battery
      const batContainer = document.getElementById('diag-batte
ry-details-container');
      const batSourceBadge = document.getElementById('diag-bat
tery-source-badge');
      if (batContainer && data.battery_wear) {
        const b = data.battery_wear;
        if (batSourceBadge) {
          batSourceBadge.textContent = b.power_source || 'AC M
ains';
        }
        if (!b.has_battery) {
          batContainer.innerHTML = `
            <div class="text-center py-4 text-muted small">
              <i class="bi bi-plug-fill fs-4 text-info d-block
 mb-1"></i>
              Стационарный компьютер — питание напрямую от эле
ктросети
            </div>
          `;
        } else {
          batContainer.innerHTML = `
            <table class="about-sys-spec-table">
              <tbody>
                <tr>
                  <td class="about-sys-spec-key"><i class="bi
bi-battery-charging me-1.5 text-warning"></i>Уровень заряда</t
```

```
d>
                  <td class="about-sys-spec-val fw-bold text-white">${b.percent}% (${b.is_charging ? 'Заряжается' : 'Разряд'})</td>
                </tr>
                <tr>
                  <td class="about-sys-spec-key"><i class="bi bi-shield-shaded me-1.5 text-info"></i>Заводская емкость</td>
                  <td class="about-sys-spec-val font-monospace">${b.design_capacity_mwh ? b.design_capacity_mwh + ' mWh' : '--'}</td>
                </tr>
                <tr>
                  <td class="about-sys-spec-key"><i class="bi bi-battery-full me-1.5 text-success"></i>Текущая емкость</td>
                  <td class="about-sys-spec-val font-monospace">${b.full_charge_capacity_mwh ? b.full_charge_capacity_mwh + ' mWh' : '--'}</td>
                </tr>
                <tr>
                  <td class="about-sys-spec-key"><i class="bi bi-heart-pulse me-1.5 text-danger"></i>Деградация (Износ)</td>
                  <td class="about-sys-spec-val ${b.wear_level_pct > 20 ? 'text-danger fw-bold' : 'text-success'}">
                    ${b.wear_level_pct}% износа
                  </td>
                </tr>
              </tbody>
            </table>
          `;
        }
      }
    } catch (err) {
      console.warn('[StorageWearTab] fetchStorageBatteryWear error:', err);
    }
  }

```

```
  const POLL_ID = 'storage_wear';

  function getFrequency() {
    try {
      const saved = localStorage.getItem(`poll_freq_${POLL_ID}
`);
      if (saved) return saved;
    } catch (_) {}
    return 'manual';
  }

  function setFrequency(freq) {
    try {
      localStorage.setItem(`poll_freq_${POLL_ID}`, freq);
    } catch (_) {}
    applyPoller(freq, false);
  }

  function stopPolling() {
    if (window.unregisterTabPoller) {
      window.unregisterTabPoller(`tab-storage-wear_${POLL_ID}`
);
    }
    if (autoRefreshTimer) {
      clearInterval(autoRefreshTimer);
      autoRefreshTimer = null;
    }
  }

  function applyPoller(freq, runInitial = false) {
    stopPolling();
    if (freq === 'start' || freq === 'manual') {
      if (runInitial && (window.isTabActive ? window.isTabActi
ve('tab-storage-wear') : true)) {
        fetchStorageBatteryWear();
      }
      return;
    }
```

```
227
228     const intervalSec = parseInt(freq, 10);
229     if (isNaN(intervalSec) || intervalSec <= 0) return;
230
231     const intervalMs = intervalSec * 1000;
232     const pollerId = `tab-storage-wear_${POLL_ID}`;
233
234     if (window.registerTabPoller) {
235       window.registerTabPoller('tab-storage-wear', fetchStorag
    eBatteryWear, intervalMs, { pollerId, immediate: runInitial })
    ;
236     } else {
237       if (runInitial && (window.isTabActive ? window.isTabActi
    ve('tab-storage-wear') : true)) {
238         fetchStorageBatteryWear();
239       }
240       autoRefreshTimer = setInterval(() => {
241         if (window.isTabActive ? window.isTabActive('tab-stora
    ge-wear') : true) {
242           fetchStorageBatteryWear();
243         }
244       }, intervalMs);
245     }
246   }
247
248   function bindEvents() {
249     const btnRefresh = document.getElementById('btn-diag-wear-
    refresh');
250     if (btnRefresh) {
251       btnRefresh.onclick = () => fetchStorageBatteryWear();
252     }
253
254     const select = document.getElementById('diag-wear-poll-fre
    q');
255     if (select) {
256       select.onchange = (e) => {
257         const newFreq = e.target.value;
258         setFrequency(newFreq);
```

```
        if (newFreq !== 'manual' && newFreq !== 'start') {
          fetchStorageBatteryWear();
        }
        if (window.showToast) {
          const label = select.options[select.selectedIndex]?.
text || newFreq;
          window.showToast(`Частота опроса износа: ${label}`,
'info');
        }
      };
    }
  }

  async function initStorageWearTab() {
    bindEvents();
    const currentFreq = getFrequency();
    const select = document.getElementById('diag-wear-poll-fre
q');
    if (select) select.value = currentFreq;
    applyPoller(currentFreq, true);
  }

  window.initStorageWearTab = initStorageWearTab;
  window.activateStorageWearTab = () => {
    applyPoller(getFrequency(), true);
  };

  if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', () => {
      if (window.isTabActive ? window.isTabActive('tab-storage
-wear') : false) {
        initStorageWearTab();
      } else {
        bindEvents();
      }
    });
  } else {
    if (window.isTabActive ? window.isTabActive('tab-storage-w
```

```
ear') : false) {
      initStorageWearTab();
    } else {
      bindEvents();
    }
  }
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - System32 Commands & Cont
rol Catalog
================================================================
===============
Description:
  Информационный каталог штатных средств Windows (%SystemRoot%
\System32),
  пояснения по использованию, эквиваленты Control Plane, дерев
о подкоманд
  и интеллектуальный пульт подбора и запуска команд по свободн
ому тексту.

Usage Examples:
    Web Route:
      /windows/api/webgui/system32_commands_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/system32_commands_tab/i
ndex.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/system32_commands_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<div class="container-fluid p-0 system32-tab-wrapper d-flex fl
ex-column" style="min-height: 100vh;">
<!-- Header Card & Summary Stats -->
<div class="card border-secondary-subtle mb-3 shadow-sm flex-s
```

```
hrink-0" data-panel-id="panel-system32-commands-katalog-shtatn
ykh-sredstv-window" id="panel-system32-commands-katalog-shtatn
ykh-sredstv-window">
<div class="card-body p-3">
<div class="d-flex flex-wrap align-items-center justify-conten
t-between gap-3">
<div class="d-flex align-items-center gap-2">
<div class="fs-2">□ </div>
<div>
<h5 class="mb-0 fw-bold d-flex align-items-center gap-2">
<span>Каталог штатных средств Windows (System32)</span>
<span class="badge bg-primary-subtle text-primary border borde
r-primary-subtle small fs-6">Control Plane &amp;amp; AITelemetry</
span>
</h5>
<div class="text-muted small">
              66 функциональных категорий, Telemetry Tiers, эк
виваленты Win32 API / PowerShell / WMI / COM, USN Journal и за
пуск по свободному тексту.
            </div>
</div>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<button class="btn btn-outline-info btn-sm d-flex align-items-
center gap-1 shadow-sm" id="btn-refresh-system32-catalog" titl
e="Обновить каталог">
<i class="bi bi-arrow-clockwise"></i>
<span>Обновить</span>
</button>
<button class="btn btn-outline-secondary btn-sm d-flex align-i
tems-center gap-1 shadow-sm" data-bs-target="#inventoryModal"
data-bs-toggle="modal" id="btn-open-inventory-modal" title="Пр
оверить наличие на диске">
<i class="bi bi-hdd-stack"></i>
<span>Инвентарь хоста</span>
</button>
<button class="btn btn-primary btn-sm d-flex align-items-cente
r gap-1.5 shadow-sm fw-semibold" data-bs-target="#guideModal"
```

```
data-bs-toggle="modal" id="btn-open-guide-modal" title="Руково
дство и архитектура">
<i class="bi bi-book"></i>
<span>Руководство</span>
</button>
</div>
</div>
<!-- Quick Metrics Bar -->
<div class="row g-2 mt-2 pt-2 border-top border-secondary-subt
le" id="system32-summary-bar">
<div class="col-6 col-md-2">
<div class="p-2 bg-body-tertiary rounded text-center">
<div class="text-muted small" style="font-size: 0.72rem;">Всег
о средств</div>
<div class="fw-bold fs-6 text-primary" id="stat-total-tools">.
..</div>
</div>
</div>
<div class="col-6 col-md-2">
<div class="p-2 bg-body-tertiary rounded text-center">
<div class="text-muted small" style="font-size: 0.72rem;">Кате
горий</div>
<div class="fw-bold fs-6 text-info" id="stat-total-categories"
>...</div>
</div>
</div>
<div class="col-6 col-md-2">
<div class="p-2 bg-body-tertiary rounded text-center">
<div class="text-muted small" style="font-size: 0.72rem;">  O
BSERVE (Телеметрия)</div>
<div class="fw-bold fs-6 text-success" id="stat-observe-tools"
>...</div>
</div>
</div>
<div class="col-6 col-md-2">
<div class="p-2 bg-body-tertiary rounded text-center">
<div class="text-muted small" style="font-size: 0.72rem;">  ET
W Конвейер</div>
```

```
<div class="fw-bold fs-6 text-warning" id="stat-etw-tools">...
</div>
</div>
</div>
<div class="col-6 col-md-2">
<div class="p-2 bg-body-tertiary rounded text-center">
<div class="text-muted small" style="font-size: 0.72rem;">  USN Journal</div>
<div class="fw-bold fs-6 text-info" id="stat-usn-tools">...</div>
</div>
</div>
<div class="col-6 col-md-2">
<div class="p-2 bg-body-tertiary rounded text-center">
<div class="text-muted small" style="font-size: 0.72rem;">  Оснастки MMC/CPL</div>
<div class="fw-bold fs-6 text-secondary" id="stat-gui-tools">...</div>
</div>
</div>
</div>
</div>
</div>
<!-- Filters & Search Toolbar -->
<div class="card border-secondary-subtle mb-3 shadow-sm flex-shrink-0" data-panel-id="panel-system32-commands-aitelemetry-tier" id="panel-system32-commands-aitelemetry-tier">
<div class="card-body p-2 d-flex flex-wrap align-items-center justify-content-between gap-2">
<div class="d-flex align-items-center gap-2 flex-wrap">
<span class="small fw-semibold text-muted">AITelemetry Tier:</span>
<div class="btn-group btn-group-sm" id="tier-filters" role="group">
<button class="btn btn-outline-secondary active" data-tier="all" type="button">Все</button>
<button class="btn btn-outline-success" data-tier="OBSERVE" type="button">  OBSERVE</button>
```

```
<button class="btn btn-outline-warning" data-tier="DIAGNOSE" type="button">  DIAGNOSE</button>
<button class="btn btn-outline-info" data-tier="CONTROL" type="button">  CONTROL</button>
<button class="btn btn-outline-danger" data-tier="ADMIN" type="button">  ADMIN</button>
<button class="btn btn-outline-dark" data-tier="DESTRUCTIVE" type="button">    DESTRUCTIVE</button>
<button class="btn btn-outline-primary" data-tier="RECOVERY" type="button">  RECOVERY</button>
</div>
</div>
<div class="d-flex align-items-center gap-2 flex-grow-1 flex-md-grow-0" style="min-width: 260px;">
<div class="input-group input-group-sm">
<span class="input-group-text bg-body-tertiary"><i class="bi bi-search"></i></span>
<input class="form-control" id="tool-search-input" placeholder="Поиск по имени, назначению или тегам..." type="text"/>
<button class="btn btn-outline-secondary" id="btn-clear-search" title="Очистить" type="button"><i class="bi bi-x"></i></button>
</div>
</div>
</div>
</div>
<!-- Tool Cards Grid / Main Catalog Display -->
<div class="flex-grow-1 overflow-auto mb-3" style="max-height: calc(100vh - 440px); min-height: 280px;">
<div class="row g-2" id="system32-cards-container">
<div class="col-12 text-center py-5 text-muted">
<div class="spinner-border spinner-border-sm text-primary me-2"></div>
<span>Загрузка каталога штатных средств Windows...</span>
</div>
</div>
</div>
<!-- Bottom AI Command Execution & Interpretation Panel (Ident
```

```
ical in Style to tab-scenarios) -->
<div class="card border-primary border-opacity-50 shadow-lg mt-auto flex-shrink-0" data-panel-id="panel-system32-commands-ekspress-komandy-svobodnym-tek" id="panel-system32-commands-ekspress-komandy-svobodnym-tek">
<!-- Quick Prompts / Pills -->
<div class="p-2 border-bottom border-secondary-subtle bg-body-secondary bg-opacity-50">
<div class="d-flex align-items-center justify-content-between mb-1">
<span class="small text-muted fw-semibold" style="font-size: 0.75rem;">  Экспресс-команды свободным текстом:</span>
<span class="text-muted small" style="font-size: 0.72rem;">Модель подберет программу и сформирует точный синтаксис</span>
</div>
<div class="d-flex flex-wrap gap-1" id="system32-quick-prompts">
<button class="btn btn-outline-secondary btn-sm py-0 px-2 quick-cmd-pill" data-prompt="Проверь статус TRIM для SSD">
          Проверить статус TRIM для SSD
        </button>
<button class="btn btn-outline-secondary btn-sm py-0 px-2 quick-cmd-pill" data-prompt="Очисти локальный кэш сопоставителя DNS">
          Сбросить кэш DNS (flushdns)
        </button>
<button class="btn btn-outline-secondary btn-sm py-0 px-2 quick-cmd-pill" data-prompt="Покажи все активные сетевые порты и PID процессов">
          Активные сетевые порты (netstat)
        </button>
<button class="btn btn-outline-secondary btn-sm py-0 px-2 quick-cmd-pill" data-prompt="Проверь статус шифрования BitLocker на дисках">
          Статус шифрования BitLocker
        </button>
<button class="btn btn-outline-secondary btn-sm py-0 px-2 quick-cmd-pill" data-prompt="Покажи активные сессии ETW трассировк
```

```
и">
153           Активные ETW сессии (logman)
154         </button>
155 <button class="btn btn-outline-secondary btn-sm py-0 px-2 quic
    k-cmd-pill" data-prompt="Покажи метаданные NTFS USN Journal то
    ма C">
156           NTFS USN Journal
157         </button>
158 <button class="btn btn-outline-secondary btn-sm py-0 px-2 quic
    k-cmd-pill" data-prompt="Кто текущий пользователь и какие у не
    го привилегии">
159           Кто я и мои привилегии (whoami)
160         </button>
161 </div>
162 </div>
163 <!-- Live Execution Results Console (Shown on run) -->
164 <div class="p-3 border-bottom border-secondary-subtle bg-dark
    text-white" id="ai-command-result-box" style="display: none;">
165 <div class="d-flex align-items-center justify-content-between
    mb-2">
166 <div class="d-flex align-items-center gap-2">
167 <span class="fs-5" id="res-status-icon">  </span>
168 <div>
169 <span class="fw-bold text-info" id="res-tool-name">...</span>
170 <span class="badge bg-secondary ms-2" id="res-tier-badge">OBSE
    RVE</span>
171 <span class="badge bg-danger ms-1" id="res-danger-badge" style
    ="display: none;">HIGH</span>
172 </div>
173 </div>
174 <div class="d-flex align-items-center gap-2">
175 <span class="text-muted small" id="res-duration-badge">0 ms</s
    pan>
176 <button class="btn btn-outline-light btn-sm py-0 px-2" id="btn
    -copy-command-output" title="Скопировать вывод">
177 <i class="bi bi-clipboard me-1"></i>Копировать
178           </button>
179 <button class="btn btn-outline-danger btn-sm py-0 px-1.5" id="
```

```
btn-close-result-box" title="Закрыть панель">
<i class="bi bi-x"></i>
</button>
</div>
</div>
<div class="mb-2 small text-light bg-secondary bg-opacity-25 p-2 rounded">
<div><strong>  Сформированная команда:</strong> <code class="text-warning" id="res-cmd-line">...</code></div>
<div class="mt-1"><strong>  Назначение:</strong> <span id="res-explanation">...</span></div>
<div class="mt-1 d-flex gap-3 flex-wrap text-muted" style="font-size: 0.75rem;">
<span id="res-ps-equiv"></span>
<span id="res-win32-equiv"></span>
</div>
</div>
<!-- Output Console -->
<pre class="p-2 bg-black rounded text-success small mb-0 font-monospace" id="res-output-pre" style="max-height: 200px; overflow-y: auto; white-space: pre-wrap;">Ожидание выполнения...</pre>
</div>
<!-- Bottom Input Form -->
<div class="p-2">
<form class="d-flex flex-column gap-2" id="system32-ai-cmd-form">
<div class="input-group">
<span class="input-group-text bg-primary text-white"><i class="bi bi-terminal-fill"></i></span>
<input autocomplete="off" class="form-control form-control-lg fs-6" id="system32-cmd-input" placeholder="Введите задачу своб одным текстом (например: 'проверь статус trim для ssd', 'покаж и порты netstat', 'сбрось кэш dns')..." type="text"/>
<button class="btn btn-primary d-flex align-items-center gap-2 px-3 fw-semibold" id="btn-system32-cmd-send" type="submit">
<span>Подобрать и запустить</span>
<i class="bi bi-play-circle-fill"></i>
```

```
</button>
</div>
<div class="d-flex align-items-center justify-content-between px-1 flex-wrap gap-2">
<span class="text-muted small" id="ai-cmd-status-indicator">Готов к приему команд</span>
<div class="d-flex align-items-center gap-3">
<div class="form-check form-switch m-0 d-flex align-items-center gap-1">
<input class="form-check-input mt-0" id="check-cmd-dry-run" role="switch" type="checkbox"/>
<label class="form-check-label text-muted user-select-none small" for="check-cmd-dry-run" title="Только симулировать и показать подобранную команду">
                Симуляция (Dry-Run)
              </label>
</div>
<div class="form-check form-switch m-0 d-flex align-items-center gap-1">
<input checked="" class="form-check-input mt-0" id="check-cmd-confirm-admin" role="switch" type="checkbox"/>
<label class="form-check-label text-muted user-select-none small" for="check-cmd-confirm-admin" title="Защита SafeOps: запрашивать подтверждение для HIGH/CRITICAL">
                Подтверждать опасные операции
              </label>
</div>
</div>
</div>
</form>
</div>
</div>
<!-- Tool Details Modal -->
<div aria-hidden="true" aria-labelledby="toolDetailsModalLabel" class="modal fade" id="toolDetailsModal" tabindex="-1">
<div class="modal-dialog modal-lg modal-dialog-scrollable">
<div class="modal-content">
<div class="modal-header py-2 px-3">
```

```
<h6 class="modal-title fw-bold d-flex align-items-center gap-2
" id="toolDetailsModalLabel">
<span class="fs-5" id="modal-tool-icon">□ </span>
<span id="modal-tool-title">Сведения об инструменте</span>
</h6>
<button aria-label="Закрыть" class="btn-close" data-bs-dismiss
="modal" type="button"></button>
</div>
<div class="modal-body p-3" id="modal-tool-body">
<!-- Populated dynamically -->
</div>
<div class="modal-footer py-2 px-3">
<button class="btn btn-secondary btn-sm" data-bs-dismiss="moda
l" type="button">Закрыть</button>
<button class="btn btn-primary btn-sm" id="btn-modal-paste-com
mand" type="button">Вставить команду в пульт</button>
</div>
</div>
</div>
</div>
<!-- Host Inventory Modal -->
<div aria-hidden="true" aria-labelledby="inventoryModalLabel"
class="modal fade" id="inventoryModal" tabindex="-1">
<div class="modal-dialog modal-lg modal-dialog-scrollable">
<div class="modal-content">
<div class="modal-header py-2 px-3">
<h6 class="modal-title fw-bold" id="inventoryModalLabel">□ Ин
вентарь файлов %SystemRoot%\System32</h6>
<button aria-label="Закрыть" class="btn-close" data-bs-dismiss
="modal" type="button"></button>
</div>
<div class="modal-body p-3" id="modal-inventory-body">
<div class="text-center py-4 text-muted">
<div class="spinner-border spinner-border-sm text-primary mb-2
"></div>
<div>Сверка с файловой системой хоста...</div>
</div>
</div>
```

```
</div>
</div>
</div>
<!-- Architecture & Usage Guide Modal -->
<div aria-hidden="true" aria-labelledby="guideModalLabel" clas
s="modal fade" id="guideModal" tabindex="-1">
<div class="modal-dialog modal-xl modal-dialog-scrollable">
<div class="modal-content">
<div class="modal-header py-2 px-3">
<h6 class="modal-title fw-bold" id="guideModalLabel">  Руково
дство по штатным средствам Windows 10/11</h6>
<button aria-label="Закрыть" class="btn-close" data-bs-dismiss
="modal" type="button"></button>
</div>
<div class="modal-body p-3">
<h6>  1. Слои Windows Control Plane</h6>
<p class="small text-muted">
            Каждая операция в Windows может выполняться не тол
ько через <code>System32/*.exe</code>, но и через соответствую
щие эквиваленты:
            PowerShell Cmdlets, WMI/CIM классы, COM интерфейсы
, прямые вызовы Win32 API в системных DLL, события ETW и канал
ы Windows Event Log.
          </p>
<h6 class="mt-3">  2. Градация AITelemetry Tiers</h6>
<ul class="small text-muted">
<li><strong>  OBSERVE:</strong> Безопасное считывание телемет
рии (<code>systeminfo</code>, <code>tasklist</code>, <code>net
stat</code>, <code>fsutil usn</code>, <code>logman query</code
>, <code>whoami</code>).</li>
<li><strong>  DIAGNOSE:</strong> Трассировка, диагностика зад
ержек, сетевые снифферы (<code>ping</code>, <code>tracert</cod
e>, <code>pathping</code>, <code>pktmon</code>, <code>sxstrace
</code>).</li>
<li><strong>  CONTROL:</strong> Управление службами, расписан
ием, некритичными параметрами (<code>sc start/stop</code>, <co
de>schtasks /run</code>, <code>tzutil</code>).</li>
<li><strong>  ADMIN:</strong> Изменение конфигурации системы,
```

```
 брандмауэра, политик безопасности (<code>reg add</code>, <cod
e>sc config</code>, <code>netsh advfirewall</code>, <code>dism
</code>).</li>
<li><strong>□ □ DESTRUCTIVE:</strong> Деструктивные операции
(<code>diskpart clean/format</code>, <code>format.com</code>,
<code>bcdedit /delete</code>).</li>
<li><strong>□ RECOVERY:</strong> Среда восстановления и ремон
т ОС (<code>reagentc</code>, <code>bcdboot</code>, <code>bootr
ec</code>, <code>wbadmin</code>, <code>sfc</code>).</li>
</ul>
<h6 class="mt-3">□ 3. Сквозной конвейер ETW</h6>
<p class="small text-muted">
<code>Sensors → ETW / Performance Counters (logman, pktmon, ty
peperf) → AITelemetry Pipeline → SQLite (telemetry.db)</code>
</p>
<h6 class="mt-3">□ 4. Наблюдение через NTFS USN Journal</h6>
<p class="small text-muted">
            Команда <code>fsutil usn queryjournal C:</code> по
зволяет отслеживать изменения файлов в реальном времени без ре
сурсоемкого сканирования всего диска.
          </p>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Web Interface - System32 Commands Tab Main Script
 * ============================================================
 * Description:
 *   Клиентский контроллер вкладки tab-system32-commands: отображение каталога
 *   штатных средств Windows, слоев Control Plane, AITelemetry Tiers, инвентаря
 *   и интеллектуального пульта подбора и запуска команд по свободному тексту.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/system32_commands_tab/main.js?v=20261006_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/system32_commands_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

(function () {
  'use strict';

  let toolsList = [];
  let currentTierFilter = 'all';
  let searchQuery = '';
```

```
  let activeSelectedTool = null;
  let isExecuting = false;

  const tierColors = {
    OBSERVE: { bg: 'bg-success-subtle', text: 'text-success', border: 'border-success-subtle', badge: 'bg-success' },
    DIAGNOSE: { bg: 'bg-warning-subtle', text: 'text-warning', border: 'border-warning-subtle', badge: 'bg-warning text-dark' },
    CONTROL: { bg: 'bg-info-subtle', text: 'text-info', border: 'border-info-subtle', badge: 'bg-info text-dark' },
    ADMIN: { bg: 'bg-danger-subtle', text: 'text-danger', border: 'border-danger-subtle', badge: 'bg-danger' },
    DESTRUCTIVE: { bg: 'bg-dark-subtle', text: 'text-light', border: 'border-dark', badge: 'bg-dark text-danger fw-bold' },
    RECOVERY: { bg: 'bg-primary-subtle', text: 'text-primary', border: 'border-primary-subtle', badge: 'bg-primary' },
  };

  const planeIcons = {
    CLI: '□ ',
    POWERSHELL: '□ ',
    WMI_CIM: '□ ',
    COM: '□ ',
    WIN32_API: '□ ',
    NATIVE_NT_API: '□ ',
    ETW: '□ ',
    EVENT_LOG: '□ ',
    REGISTRY: '□ ',
    GUI_MMC_CPL: '□ ',
  };

  function escapeHtml(str) {
    if (!str) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
```

```
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
  }

  // Load summary metrics and full catalog from backend
  async function loadCatalog() {
    const container = document.getElementById('system32-cards-container');
    if (!container) return;

    try {
      const [catRes, sumRes] = await Promise.all([
        fetch('/api/v1/system32/catalog'),
        fetch('/api/v1/system32/summary')
      ]);

      if (!catRes.ok) throw new Error(`HTTP ${catRes.status}`);
      const catData = await catRes.json();
      toolsList = catData.tools || [];

      if (sumRes.ok) {
        const sumData = await sumRes.json();
        updateSummaryMetrics(sumData);
      }

      renderToolsGrid();
    } catch (err) {
      console.error('[System32CommandsTab] Error loading catalog:', err);
      if (container) {
        container.innerHTML = `
          <div class="col-12 text-center py-4 text-danger">
            <i class="bi bi-exclamation-triangle fs-3 mb-2"></i>
            <div>Не удалось загрузить каталог System32: ${escapeHtml(err.message)}</div>
            <button class="btn btn-outline-primary btn-sm mt-2
```

```
" onclick="location.reload()">Повторить</button>
          </div>
        `;
      }
    }
  }

  function updateSummaryMetrics(sum) {
    const elTotal = document.getElementById('stat-total-tools'
);
    const elCats = document.getElementById('stat-total-categor
ies');
    const elObs = document.getElementById('stat-observe-tools'
);
    const elEtw = document.getElementById('stat-etw-tools');
    const elUsn = document.getElementById('stat-usn-tools');
    const elGui = document.getElementById('stat-gui-tools');

    if (elTotal) elTotal.innerText = sum.total_tools || toolsL
ist.length;
    if (elCats) elCats.innerText = sum.categories_count || '66
';
    if (elObs) elObs.innerText = sum.by_telemetry_tier?.OBSERV
E || sum.read_tools_count || '0';
    if (elEtw) elEtw.innerText = sum.etw_tools_count || '0';
    if (elUsn) elUsn.innerText = sum.usn_journal_tools_count |
| '1';
    if (elGui) elGui.innerText = (sum.gui_tools_count || 0) +
(sum.msc_consoles_count || 0);
  }

  function getFilteredTools() {
    let list = [...toolsList];
    if (currentTierFilter !== 'all') {
      list = list.filter(t => t.telemetry_tier === currentTier
Filter);
    }
    if (searchQuery) {
```

```
      const q = searchQuery.toLowerCase().trim();
      list = list.filter(t =>
        t.executable.toLowerCase().includes(q) ||
        t.purpose.toLowerCase().includes(q) ||
        t.category_code.toLowerCase().includes(q) ||
        (t.tags && t.tags.some(tag => tag.toLowerCase().includ
es(q)))
      );
    }
    return list;
  }

  function renderToolsGrid() {
    const container = document.getElementById('system32-cards-
container');
    if (!container) return;

    const filtered = getFilteredTools();

    if (filtered.length === 0) {
      container.innerHTML = `
        <div class="col-12 text-center py-5 text-muted">
          <i class="bi bi-search fs-2 mb-2"></i>
          <div>Инструменты не найдены по заданному фильтру</di
v>
        </div>
      `;
      return;
    }

    container.innerHTML = filtered.map((tool, idx) => {
      const tierStyle = tierColors[tool.telemetry_tier] || { b
adge: 'bg-secondary' };
      const planeIcon = planeIcons[tool.primary_control_plane]
 || '□ ';
      const firstTemplate = tool.command_templates && tool.com
mand_templates.length > 0 ? tool.command_templates[0] : tool.e
xecutable;
```

```
152
153       return `
154         <div class="col-12 col-md-6 col-lg-4 col-xl-3">
155           <div class="card h-100 border-secondary-subtle shado
w-sm hover-shadow transition">
156             <div class="card-header bg-transparent border-seco
ndary-subtle py-2 px-3 d-flex align-items-center justify-conte
nt-between">
157               <div class="d-flex align-items-center gap-1.5 ov
erflow-hidden text-truncate">
158                 <span class="fs-5">${planeIcon}</span>
159                 <span class="fw-bold font-monospace text-trunc
ate text-primary" title="${escapeHtml(tool.executable)}">${esc
apeHtml(tool.executable)}</span>
160               </div>
161               <span class="badge ${tierStyle.badge} small" sty
le="font-size: 0.68rem;">${escapeHtml(tool.telemetry_tier)}</s
pan>
162             </div>
163             <div class="card-body p-2 d-flex flex-column justi
fy-content-between">
164               <div>
165                 <div class="text-muted small mb-1 text-truncat
e" style="font-size: 0.72rem;" title="${escapeHtml(tool.catego
ry_code)}">
166                   ${escapeHtml(tool.category_code)}
167                 </div>
168                 <p class="card-text small text-body mb-2" styl
e="font-size: 0.78rem; display: -webkit-box; -webkit-line-clam
p: 2; -webkit-box-orient: vertical; overflow: hidden;" title="
${escapeHtml(tool.purpose)}">
169                   ${escapeHtml(tool.purpose)}
170                 </p>
171               </div>
172               <div class="pt-2 border-top border-secondary-sub
tle d-flex align-items-center justify-content-between gap-1">
173                 <button class="btn btn-outline-info btn-sm py-
0.5 px-2 btn-open-tool-details" data-idx="${idx}" style="font-
```

```
size: 0.75rem;" title="Посмотреть эквиваленты и руководство">
                  Инструкция
                </button>
                <button class="btn btn-outline-primary btn-sm
py-0.5 px-2 btn-paste-cmd" data-cmd="${escapeHtml(firstTemplat
e)}" style="font-size: 0.75rem;" title="Вставить шаблон в пуль
т">
                  В пульт
                </button>
              </div>
            </div>
          </div>
        </div>
      `;
    }).join('');

    // Attach click events
    container.querySelectorAll('.btn-open-tool-details').forEa
ch(btn => {
      btn.onclick = () => {
        const idx = parseInt(btn.getAttribute('data-idx'), 10)
;
        if (!isNaN(idx) && filtered[idx]) {
          showToolDetailsModal(filtered[idx]);
        }
      };
    });

    container.querySelectorAll('.btn-paste-cmd').forEach(btn =
> {
      btn.onclick = () => {
        const cmd = btn.getAttribute('data-cmd');
        const input = document.getElementById('system32-cmd-in
put');
        if (input && cmd) {
          input.value = cmd;
          input.focus();
        }
```

```
      };
    });
  }

  function showToolDetailsModal(tool) {
    activeSelectedTool = tool;
    const modalTitle = document.getElementById('modal-tool-tit
le');
    const modalIcon = document.getElementById('modal-tool-icon
');
    const modalBody = document.getElementById('modal-tool-body
');

    if (modalTitle) modalTitle.innerText = `${tool.executable}
 (${tool.category_code})`;
    if (modalIcon) modalIcon.innerText = planeIcons[tool.prima
ry_control_plane] || '□ ';

    let subcommandsHtml = '';
    if (tool.subcommands_info && Object.keys(tool.subcommands_
info).length > 0) {
      subcommandsHtml = `
        <h6 class="fw-bold mt-3 mb-2">□ Подкоманды и контексты
:</h6>
        <div class="list-group list-group-flush border rounded
 small">
          ${Object.entries(tool.subcommands_info).map(([sub, d
esc]) => `
            <div class="list-group-item py-1.5 px-2 d-flex jus
tify-content-between align-items-center">
              <code class="text-primary font-monospace">${esca
peHtml(sub)}</code>
              <span class="text-muted">${escapeHtml(desc)}</sp
an>
            </div>
          `).join('')}
        </div>
      `;
```

```
    }

    let templatesHtml = '';
    if (tool.command_templates && tool.command_templates.lengt
h > 0) {
      templatesHtml = `
        <h6 class="fw-bold mt-3 mb-2">  Шаблоны команд:</h6>
        <div class="d-flex flex-column gap-1">
          ${tool.command_templates.map(tmpl => `
            <div class="d-flex align-items-center justify-cont
ent-between p-2 bg-body-secondary rounded font-monospace small
">
              <code class="text-warning">${escapeHtml(tmpl)}</
code>
              <button class="btn btn-sm btn-outline-secondary
py-0 px-1.5 btn-copy-inline" data-clip="${escapeHtml(tmpl)}" t
itle="Копировать">
                <i class="bi bi-clipboard"></i>
              </button>
            </div>
          `).join('')}
        </div>
      `;
    }

    if (modalBody) {
      modalBody.innerHTML = `
        <div class="mb-3">
          <div class="text-muted small mb-1">Назначение:</div>
          <div class="p-2 bg-body-tertiary rounded small fw-me
dium">${escapeHtml(tool.purpose)}</div>
        </div>

        <div class="row g-2 mb-3">
          <div class="col-6 col-md-3">
            <div class="p-2 bg-body-tertiary rounded text-cent
er small">
              <span class="text-muted d-block" style="font-siz
```

```
e: 0.7rem;">Tier:</span>
260               <span class="badge ${tierColors[tool.telemetry_t
ier]?.badge || 'bg-secondary'}">${escapeHtml(tool.telemetry_ti
er)}</span>
261             </div>
262           </div>
263           <div class="col-6 col-md-3">
264             <div class="p-2 bg-body-tertiary rounded text-cent
er small">
265               <span class="text-muted d-block" style="font-siz
e: 0.7rem;">Опасность:</span>
266               <span class="fw-bold ${tool.danger_level === 'CR
ITICAL' ? 'text-danger' : 'text-body'}">${escapeHtml(tool.dang
er_level)}</span>
267             </div>
268           </div>
269           <div class="col-6 col-md-3">
270             <div class="p-2 bg-body-tertiary rounded text-cent
er small">
271               <span class="text-muted d-block" style="font-siz
e: 0.7rem;">Привилегии:</span>
272               <span class="fw-semibold">${escapeHtml(tool.requ
ired_privileges)}</span>
273             </div>
274           </div>
275           <div class="col-6 col-md-3">
276             <div class="p-2 bg-body-tertiary rounded text-cent
er small">
277               <span class="text-muted d-block" style="font-siz
e: 0.7rem;">Control Plane:</span>
278               <span class="badge bg-secondary">${escapeHtml(to
ol.primary_control_plane)}</span>
279             </div>
280           </div>
281         </div>
282
283         <h6 class="fw-bold mb-2">🔗 Эквиваленты в Control Plan
e:</h6>
```

```
        <div class="table-responsive">
          <table class="table table-sm table-bordered small mb-0">
            <tbody>
              <tr>
                <th style="width: 25%;">PowerShell</th>
                <td><code>${escapeHtml(tool.powershell_equivalent || '—')}</code></td>
              </tr>
              <tr>
                <th>Win32 API</th>
                <td><code>${escapeHtml(tool.native_api_equivalent || '—')}</code></td>
              </tr>
              <tr>
                <th>WMI / CIM</th>
                <td><code>${escapeHtml(tool.wmi_cim_equivalent || '—')}</code></td>
              </tr>
              <tr>
                <th>COM</th>
                <td><code>${escapeHtml(tool.com_equivalent || '—')}</code></td>
              </tr>
              ${tool.etw_pipeline_enabled ? '<tr><th>ETW Pipeline</th><td><span class="badge bg-success">Интегрирован в AITelemetry</span></td></tr>' : ''}
              ${tool.usn_journal_enabled ? '<tr><th>USN Journal</th><td><span class="badge bg-info">Поддержка прямого чтения NTFS USN</span></td></tr>' : ''}
            </tbody>
          </table>
        </div>

        ${subcommandsHtml}
        ${templatesHtml}
      `;

```

```
      // Copy buttons in modal
      modalBody.querySelectorAll('.btn-copy-inline').forEach(b
tn => {
        btn.onclick = () => {
          const txt = btn.getAttribute('data-clip');
          if (txt) navigator.clipboard.writeText(txt);
        };
      });
    }

    const modalEl = document.getElementById('toolDetailsModal'
);
    if (modalEl && typeof bootstrap !== 'undefined' && bootstr
ap.Modal) {
      const modal = bootstrap.Modal.getOrCreateInstance(modalE
l);
      modal.show();
    }
  }

  // Load and display host inventory
  async function loadHostInventory() {
    const body = document.getElementById('modal-inventory-body
');
    if (!body) return;

    try {
      const res = await fetch('/api/v1/system32/inventory');
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();

      body.innerHTML = `
        <div class="d-flex align-items-center justify-content-
between mb-3 p-2 bg-body-tertiary rounded small">
          <div>📁 Каталог: <code>${escapeHtml(data.system32_pa
th)}</code></div>
          <div>📦 Найдено: <strong>${data.total_installed}</str
ong> / <strong>${data.total_registered}</strong></div>
```

```
        </div>
        <div class="table-responsive" style="max-height: 400px; overflow-y: auto;">
          <table class="table table-sm table-hover table-striped small">
            <thead>
              <tr>
                <th>Инструмент</th>
                <th>Категория</th>
                <th>Статус</th>
                <th>Путь / Источник</th>
              </tr>
            </thead>
            <tbody>
              ${data.installed.map(item => `
                <tr>
                  <td class="font-monospace fw-bold text-success">${escapeHtml(item.executable)}</td>
                  <td class="text-muted">${escapeHtml(item.category_code || item.category)}</td>
                  <td><span class="badge bg-success-subtle text-success border border-success-subtle">Доступен</span></td>
                  <td class="font-monospace text-muted" style="font-size: 0.72rem;">${escapeHtml(item.path)}</td>
                </tr>
              `).join('')}
              ${data.missing.map(item => `
                <tr class="table-warning">
                  <td class="font-monospace text-muted">${escapeHtml(item.executable)}</td>
                  <td class="text-muted">${escapeHtml(item.category_code || item.category)}</td>
                  <td><span class="badge bg-secondary-subtle text-secondary">Не найден</span></td>
                  <td class="text-muted" style="font-size: 0.72rem;">${escapeHtml(item.source)}</td>
                </tr>
              `).join('')}
```

```
            </tbody>
          </table>
        </div>
      `;
    } catch (err) {
      body.innerHTML = `<div class="text-danger p-3 text-cente
r">Ошибка инвентаризации: ${escapeHtml(err.message)}</div>`;
    }
  }

  // Setup Bottom AI Command Input Bar & Execution
  function setupAiCommandRunner() {
    const form = document.getElementById('system32-ai-cmd-form
');
    const input = document.getElementById('system32-cmd-input'
);
    const sendBtn = document.getElementById('btn-system32-cmd-
send');
    const statusInd = document.getElementById('ai-cmd-status-i
ndicator');
    const resultBox = document.getElementById('ai-command-resu
lt-box');
    const dryRunCheck = document.getElementById('check-cmd-dry
-run');
    const confirmAdminCheck = document.getElementById('check-c
md-confirm-admin');

    const resToolName = document.getElementById('res-tool-name
');
    const resTierBadge = document.getElementById('res-tier-bad
ge');
    const resDangerBadge = document.getElementById('res-danger
-badge');
    const resDuration = document.getElementById('res-duration-
badge');
    const resCmdLine = document.getElementById('res-cmd-line')
;
    const resExplanation = document.getElementById('res-explan
```

```
ation');
396     const resPsEquiv = document.getElementById('res-ps-equiv')
;
397     const resWin32Equiv = document.getElementById('res-win32-e
quiv');
398     const resOutput = document.getElementById('res-output-pre'
);
399     const btnCopyOutput = document.getElementById('btn-copy-co
mmand-output');
400     const btnCloseBox = document.getElementById('btn-close-res
ult-box');
401 
402     if (btnCloseBox && resultBox) {
403       btnCloseBox.onclick = () => { resultBox.style.display =
'none'; };
404     }
405 
406     if (btnCopyOutput && resOutput) {
407       btnCopyOutput.onclick = () => {
408         navigator.clipboard.writeText(resOutput.innerText);
409         btnCopyOutput.innerHTML = '<i class="bi bi-check2 me-1
"></i>Скопировано';
410         setTimeout(() => {
411           btnCopyOutput.innerHTML = '<i class="bi bi-clipboard
 me-1"></i>Копировать';
412         }, 1800);
413       };
414     }
415 
416     // Modal paste command button
417     const btnModalPaste = document.getElementById('btn-modal-p
aste-command');
418     if (btnModalPaste && input) {
419       btnModalPaste.onclick = () => {
420         if (activeSelectedTool) {
421           const tmpl = activeSelectedTool.command_templates?.[
0] || activeSelectedTool.executable;
422           input.value = tmpl;
```

```
          input.focus();
          const modalEl = document.getElementById('toolDetails
Modal');
          if (modalEl && typeof bootstrap !== 'undefined' && b
ootstrap.Modal) {
            bootstrap.Modal.getInstance(modalEl)?.hide();
          }
        }
      };
    }

    // Quick command pills
    document.querySelectorAll('.quick-cmd-pill').forEach(btn =
> {
      btn.onclick = () => {
        if (isExecuting) return;
        const prompt = btn.getAttribute('data-prompt');
        if (prompt && input) {
          input.value = prompt;
          executePrompt(prompt);
        }
      };
    });

    if (form && input) {
      form.onsubmit = (e) => {
        e.preventDefault();
        const prompt = input.value.trim();
        if (prompt) executePrompt(prompt);
      };
    }

    async function executePrompt(promptText) {
      if (isExecuting) return;
      isExecuting = true;

      input.disabled = true;
      if (sendBtn) sendBtn.disabled = true;
```

```
458       if (statusInd) statusInd.innerText = '⚙ Подбор инструме
нта и формирование команды...';
459
460       if (resultBox) {
461         resultBox.style.display = '';
462         resultBox.scrollIntoView({ behavior: 'smooth', block:
'nearest' });
463       }
464       if (resToolName) resToolName.innerText = 'Анализ намерен
ия...';
465       if (resCmdLine) resCmdLine.innerText = '...';
466       if (resExplanation) resExplanation.innerText = 'Подбор ш
татной утилиты System32...';
467       if (resOutput) {
468         resOutput.className = 'p-2 bg-black rounded text-info
small mb-0 font-monospace';
469         resOutput.innerText = 'Выполнение системной команды...
 Пожалуйста, подождите.';
470       }
471
472       const isDryRun = dryRunCheck ? dryRunCheck.checked : fal
se;
473       const confirmAdmin = confirmAdminCheck ? confirmAdminChe
ck.checked : true;
474
475       try {
476         const response = await fetch('/api/v1/system32/interpr
et-and-run', {
477           method: 'POST',
478           headers: { 'Content-Type': 'application/json' },
479           body: JSON.stringify({
480             prompt: promptText,
481             execute: !isDryRun,
482             confirmed_by_user: !confirmAdmin,
483           }),
484         });
485
486         if (!response.ok) throw new Error(`HTTP ${response.sta
```

```
tus}`);
        const data = await response.json();

        if (resToolName) resToolName.innerText = data.matched_executable;
        if (resTierBadge) {
          resTierBadge.className = `badge ${tierColors[data.telemetry_tier]?.badge || 'bg-secondary'} ms-2`;
          resTierBadge.innerText = data.telemetry_tier;
        }
        if (resDangerBadge) {
          if (data.danger_level === 'HIGH' || data.danger_level === 'CRITICAL') {
            resDangerBadge.style.display = '';
            resDangerBadge.innerText = data.danger_level;
          } else {
            resDangerBadge.style.display = 'none';
          }
        }
        if (resDuration) resDuration.innerText = `${data.execution_time_ms} ms`;
        if (resCmdLine) resCmdLine.innerText = data.command_line;
        if (resExplanation) resExplanation.innerText = data.explanation;
        if (resPsEquiv) resPsEquiv.innerHTML = data.powershell_equivalent ? `  PowerShell: <code>${escapeHtml(data.powershell_equivalent)}</code>` : '';
        if (resWin32Equiv) resWin32Equiv.innerHTML = data.win32_api_equivalent ? `  Win32 API: <code>${escapeHtml(data.win32_api_equivalent)}</code>` : '';

        if (resOutput) {
          if (data.status === 'success') {
            resOutput.className = 'p-2 bg-black rounded text-success small mb-0 font-monospace';
          } else if (data.status === 'warn' || data.status === 'requires_confirmation') {
```

```
            resOutput.className = 'p-2 bg-black rounded text-warning small mb-0 font-monospace';
          } else {
            resOutput.className = 'p-2 bg-black rounded text-danger small mb-0 font-monospace';
          }
          resOutput.innerText = data.execution_output || 'Команда выполнена успешно.';
        }

        if (statusInd) statusInd.innerText = `Готово: ${data.matched_executable} (${data.status})`;
      } catch (err) {
        console.error('[System32CommandsTab] Execution error:', err);
        if (resOutput) {
          resOutput.className = 'p-2 bg-black rounded text-danger small mb-0 font-monospace';
          resOutput.innerText = `Сбой запроса: ${err.message}`;
        }
        if (statusInd) statusInd.innerText = 'Ошибка выполнения';
      } finally {
        isExecuting = false;
        input.disabled = false;
        if (sendBtn) sendBtn.disabled = false;
        input.focus();
      }
    }
  }

  // Setup toolbar filters & search
  function setupFilters() {
    const tierGroup = document.getElementById('tier-filters');
    if (tierGroup) {
      tierGroup.querySelectorAll('button').forEach(btn => {
        btn.onclick = () => {
```

```
          tierGroup.querySelectorAll('button').forEach(b => b.classList.remove('active'));
          btn.classList.add('active');
          currentTierFilter = btn.getAttribute('data-tier') || 'all';
          renderToolsGrid();
        };
      });
    }

    const searchInput = document.getElementById('tool-search-input');
    const clearBtn = document.getElementById('btn-clear-search');
    if (searchInput) {
      searchInput.oninput = () => {
        searchQuery = searchInput.value;
        renderToolsGrid();
      };
    }
    if (clearBtn && searchInput) {
      clearBtn.onclick = () => {
        searchInput.value = '';
        searchQuery = '';
        renderToolsGrid();
      };
    }

    const refreshBtn = document.getElementById('btn-refresh-system32-catalog');
    if (refreshBtn) {
      refreshBtn.onclick = () => { loadCatalog(); };
    }

    const invBtn = document.getElementById('btn-open-inventory-modal');
    if (invBtn) {
      invBtn.onclick = () => { loadHostInventory(); };
```

```
    }
  }

  // Module initialization
  function init() {
    setupFilters();
    setupAiCommandRunner();
    loadCatalog();
  }

  if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', init);
  } else {
    init();
  }
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows System Logs & Activity - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса журналов событий и активно
сти системы Windows.
  Оформлен на базе стандартных стилей About System и дизайн-то
кенов темы.

Usage Examples:
  Web Route:
    /html/system_control_tab/index.html

 * File: index.html
 * Project: ai-windows-admin
 * Package: windows/api/webgui/system_control_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 19:16:00
 * ================================================================
==================
 */
<!-- Windows System Logs & Activity Center Web Tab -->
<style>
  .scc-container {
    padding: 0.75rem;
    color: var(--text-color, #18181b);
  }
  .scc-header {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: var(--radius-md, 8px);
    padding: 0.75rem 1rem;
```

```
    margin-bottom: 0.75rem;
    color: var(--text-color);
  }
  .slc-compact-header {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: var(--radius-md, 8px);
    padding: 0.5rem 0.75rem;
    margin-bottom: 0.5rem;
    color: var(--text-color);
  }
  .slc-metric-pill {
    background: var(--surface-1);
    border: 1px solid var(--border-color);
    border-radius: 6px;
    padding: 0.25rem 0.6rem;
    font-size: 0.75rem;
    display: inline-flex;
    align-items: center;
    gap: 0.35rem;
    color: var(--text-color);
  }
  .slc-subtabs-nav {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: var(--radius-md, 8px);
    padding: 0.35rem 0.5rem;
  }
  .slc-subtab-btn {
    transition: all 0.18s ease;
    font-size: 0.78rem;
    color: var(--text-color);
    border: 1px solid transparent;
    border-radius: 6px;
  }
  .slc-subtab-btn:hover {
    background: rgba(56, 189, 248, 0.08);
    color: var(--text-color);
```

```
}
.slc-subtab-btn.active {
  background: #0ea5e9 !important;
  color: #0f172a !important;
  border-color: #0ea5e9 !important;
  font-weight: 700;
}
.slc-panel {
  background: var(--surface-1);
  border: 1px solid var(--border-color);
  border-radius: var(--radius-md, 8px);
  overflow: hidden;
}
.slc-panel-header {
  background: var(--surface-2);
  border-bottom: 1px solid var(--border-color);
  padding: 0.5rem 0.9rem;
  font-size: 0.85rem;
  font-weight: 600;
  color: var(--text-color);
  display: flex;
  align-items: center;
  justify-content: space-between;
}
.slc-tree-list {
  max-height: 560px;
  overflow-y: auto;
  font-size: 0.82rem;
}
.slc-tree-item {
  padding: 0.45rem 0.75rem;
  cursor: pointer;
  border-radius: 6px;
  margin-bottom: 2px;
  color: var(--text-color) !important;
  display: flex;
  justify-content: space-between;
  align-items: center;
```

```
    transition: background 0.15s ease;
    border: 1px solid transparent;
  }
  .slc-tree-item span {
    color: inherit !important;
  }
  .slc-tree-item:hover {
    background: rgba(56, 189, 248, 0.08);
    color: var(--text-color) !important;
  }
  .slc-tree-item.active {
    background: #0ea5e9 !important;
    color: #0f172a !important;
    font-weight: 700;
    border-color: #0ea5e9 !important;
  }
  .slc-tree-item.active span {
    color: #0f172a !important;
  }
  .slc-tree-item.active .badge {
    background: #0f172a !important;
    color: #ffffff !important;
  }
  .slc-table {
    width: 100%;
    border-collapse: separate;
    border-spacing: 0;
    margin-bottom: 0;
    font-size: 0.82rem;
    color: var(--text-color) !important;
  }
  .slc-table th {
    padding: 8px 12px;
    font-size: 0.75rem;
    font-weight: 600;
    color: var(--text-muted) !important;
    text-transform: uppercase;
    letter-spacing: 0.04em;
```

```
    border-bottom: 1px solid var(--border-color);
    background: var(--surface-2) !important;
    vertical-align: middle;
  }
  .slc-table td {
    padding: 8px 12px;
    border-bottom: 1px solid var(--border-subtle, rgba(0, 0, 0
, 0.06));
    vertical-align: middle;
    color: var(--text-color) !important;
    background: transparent;
    line-height: 1.4;
  }
  .slc-table tr:hover td {
    background: rgba(56, 189, 248, 0.06) !important;
  }
  .slc-table tr.table-active td {
    background: rgba(14, 165, 233, 0.18) !important;
    font-weight: 600;
    color: var(--text-color) !important;
  }
  .slc-table tr.table-danger-subtle td {
    background: rgba(239, 68, 68, 0.08);
  }
  .scc-container .form-control,
  .scc-container .form-select {
    background-color: var(--surface-1);
    color: var(--text-color);
    border: 1px solid var(--border-color);
  }
  .scc-container .input-group-text {
    background-color: var(--surface-2);
    color: var(--text-muted);
    border: 1px solid var(--border-color);
  }
  .slc-tooltip-popup {
    position: fixed;
    z-index: 99999;
```

```
    max-width: 320px;
    background: var(--surface-2);
    border: 1.5px solid var(--border-color);
    border-radius: 8px;
    padding: 0.65rem 0.85rem;
    box-shadow: 0 4px 16px rgba(0, 0, 0, 0.15);
    pointer-events: none;
    transition: opacity 0.15s ease, transform 0.15s ease;
    opacity: 0;
    transform: translateY(4px);
    font-size: 0.78rem;
    line-height: 1.4;
    color: var(--text-color);
  }
  .slc-tooltip-popup.show {
    opacity: 1;
    transform: translateY(0);
  }
  .slc-card {
    background: var(--surface-1);
    border: 1px solid var(--border-color);
    border-radius: var(--radius-md, 8px);
  }
  .slc-card-header {
    background: var(--surface-2);
    border-bottom: 1px solid var(--border-color);
    color: var(--text-color);
    padding: 0.5rem 0.85rem;
    font-weight: 600;
  }
  .badge-crit { background: #dc2626 !important; color: #fff !i
mportant; }
  .badge-err { background: #ef4444 !important; color: #fff !im
portant; }
  .badge-warn { background: #f59e0b !important; color: #000 !i
mportant; }
  .badge-inf { background: #10b981 !important; color: #fff !im
portant; }
```

```
  .badge-verb { background: #64748b !important; color: #fff !important; }

  .slc-table-resizer {
    height: 8px;
    cursor: row-resize;
    background: var(--surface-2);
    border-top: 1px solid var(--border-color);
    display: flex;
    align-items: center;
    justify-content: center;
    transition: background-color 0.2s ease;
  }
  .slc-table-resizer:hover, .slc-table-resizer.resizing {
    background: #0ea5e9 !important;
  }
  .slc-table-resizer-grip {
    width: 36px;
    height: 3px;
    background: var(--border-strong, #666);
    border-radius: 2px;
  }
  .pwr-detail-card {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 6px;
  }
  .pwr-xml-pre {
    background: var(--surface-2);
    color: var(--text-color);
    border: 1px solid var(--border-color);
  }
  .pwr-proc-badge {
    background: var(--surface-2);
    color: #38bdf8;
    border: 1px solid var(--border-color);
  }
</style>
```

```
<div class="scc-container">
<!-- Top Header Bar -->
<div class="scc-header d-flex justify-content-between align-items-center flex-wrap gap-3">
<div>
<div class="d-flex align-items-center gap-2">
<h5 class="mb-0 fw-bold" data-i18n="systemControl.headerTitle">
          Windows System Logs &amp;amp; Activity Center
        </h5>
<span class="badge rounded-pill bg-primary px-2 py-1" data-i18n="systemControl.badgeLogsActivity" style="font-size: 0.75rem;">Logs &amp;amp; Activity</span>
</div>
<div class="small text-muted mt-1" data-i18n="systemControl.headerSubtitle">
        Комплексный мониторинг журналов событий Windows, семантический RAG-поиск, аудит аномалий и история активности системы.
      </div>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<span class="badge rounded-pill px-3 py-2" data-i18n="systemControl.elevChecking" id="scc-elevation-badge">
        Проверка прав доступа...
      </span>
<button class="btn btn-sm btn-outline-info rounded-pill px-3" id="btn-scc-refresh">
<i class="bi bi-arrow-clockwise me-1"></i> <span data-i18n="systemControl.btnRefresh">Обновить</span>
</button>
</div>
</div>
<!-- Main System Logs Section -->
<div>
<!-- Compact Header Bar with Inline Stats & Actions -->
<div class="slc-compact-header d-flex justify-content-between align-items-center flex-wrap gap-2">
```

```
<!-- Title & Inline Metrics -->
<div class="d-flex align-items-center gap-2 flex-wrap">
<div class="d-flex align-items-center gap-1.5 me-1">
<span class="fs-6">□ </span>
<span class="fw-bold small" data-i18n="systemControl.centerTitle">Windows OS &amp;amp; System Log Center</span>
</div>
<!-- Quick Metrics Pills -->
<div class="slc-metric-pill" data-i18n-title="systemControl.statSourcesTitle" title="Источники журналов">
<span class="text-muted small"><i class="bi bi-diagram-3 me-1"></i><span data-i18n="systemControl.statSources">Источники:</span></span>
<span class="fw-bold text-info font-monospace" id="slc-stat-sources">1,247</span>
</div>
<div class="slc-metric-pill" data-i18n-title="systemControl.statChannelTitle" title="Активный канал логов">
<span class="text-muted small"><i class="bi bi-folder-fill me-1"></i><span data-i18n="systemControl.statChannel">Канал:</span></span>
<span class="fw-bold text-warning font-monospace text-truncate" id="slc-stat-channel" style="max-width: 140px;">System</span>
</div>
<div class="slc-metric-pill" data-i18n-title="systemControl.statLoadedTitle" title="Загружено событий">
<span class="text-muted small"><i class="bi bi-list-ol me-1"></i><span data-i18n="systemControl.statLoaded">Загружено:</span></span>
<span class="fw-bold text-primary font-monospace" id="slc-stat-events">0</span>
</div>
<div class="slc-metric-pill" data-i18n-title="systemControl.statErrorsTitle" title="Ошибки и инциденты">
<span class="text-muted small"><i class="bi bi-exclamation-triangle-fill me-1"></i><span data-i18n="systemControl.statErrors">Ошибки:</span></span>
```

```
<span class="fw-bold text-danger font-monospace" id="slc-stat-incidents">0</span>
</div>
</div>
<!-- Actions Toolbar -->
<div class="d-flex align-items-center gap-2 flex-wrap">
<button class="btn btn-outline-warning rounded-pill px-2.5 py-1 small d-flex align-items-center gap-1" id="btn-scc-open-event-logs" style="font-size: 0.78rem;" title="Открыть встроенную панель журналов событий">
<i class="bi bi-journal-text"></i> <span data-i18n="systemControl.modeEventLogs">  Журналы событий</span>
</button>
<button class="btn btn-outline-warning rounded-pill px-2.5 py-1 small d-flex align-items-center gap-1" id="btn-scc-open-power-lifecycle" style="font-size: 0.78rem;" title="Открыть панель жизненного цикла питания и аптайма">
<i class="bi bi-lightning-charge"></i> <span data-i18n="systemControl.modePowerLifecycle">  Power Lifecycle</span>
</button>
<button class="btn btn-outline-warning rounded-pill px-2.5 py-1 small d-flex align-items-center gap-1" id="btn-scc-open-defender-logs" style="font-size: 0.78rem;" title="Открыть встроенную панель журналов Defender">
<i class="bi bi-shield-lock"></i> <span data-i18n="systemControl.modeDefenderLogs">  Defender Events</span>
</button>
<button class="btn btn-outline-info rounded-pill px-2.5 py-1 small d-flex align-items-center gap-1" id="btn-slc-scan" style="font-size: 0.78rem;">
<i class="bi bi-radar"></i> <span data-i18n="systemControl.btnScan">Сканировать</span>
</button>
<div class="form-check form-switch mb-0 d-inline-flex align-items-center" style="font-size: 0.78rem; min-height: auto;">
<input class="form-check-input me-1" id="slc-live-switch" role="switch" style="cursor: pointer;" type="checkbox"/>
<label class="form-check-label text-muted small user-select-no
```

```
ne" data-i18n="systemControl.live" for="slc-live-switch">Live<
/label>
</div>
<button class="btn btn-primary rounded-pill px-2.5 py-1 small
d-flex align-items-center gap-1" id="btn-slc-refresh" style="f
ont-size: 0.78rem;">
<i class="bi bi-arrow-clockwise"></i> <span data-i18n="systemC
ontrol.btnRefresh">Обновить</span>
</button>
<div class="dropdown">
<button class="btn btn-sm btn-outline-secondary dropdown-toggl
e rounded-pill px-2.5 py-1 small d-flex align-items-center gap
-1" data-bs-toggle="dropdown" style="font-size: 0.78rem;" type
="button">
<i class="bi bi-download"></i> <span data-i18n="systemControl.
btnExport">Экспорт</span>
</button>
<ul class="dropdown-menu dropdown-menu-end shadow" style="font
-size: 0.8rem;">
<li><a class="dropdown-item" href="#" id="btn-slc-export-json"
><i class="bi bi-filetype-json me-2"></i><span data-i18n="syst
emControl.exportJson">JSON</span></a></li>
<li><a class="dropdown-item" href="#" id="btn-slc-export-csv">
<i class="bi bi-filetype-csv me-2"></i><span data-i18n="system
Control.exportCsv">CSV</span></a></li>
</ul>
</div>
</div>
</div>
<!-- Search & View Mode Navigation (Compact) -->
<div class="slc-panel mb-2 p-2" data-panel-id="panel-system-co
ntrol-1" id="panel-system-control-1">
<div class="d-flex justify-content-between align-items-center
flex-wrap gap-1 mb-2 pb-1 border-bottom" style="border-color:
var(--border-color) !important;">
<!-- Mode Tabs -->
<ul class="nav nav-pills gap-1" id="slc-mode-tabs">
<li class="nav-item">
```

```
<button class="nav-link slc-subtab-btn active py-0.5 px-2.5 small" data-i18n="systemControl.modeLive" data-mode="live">● Live Events</button>
</li>
<li class="nav-item">
<button class="nav-link slc-subtab-btn py-0.5 px-2.5 small" data-i18n="systemControl.modeAudit" data-mode="audit">  Аудит (Smart Digest)</button>
</li>
<li class="nav-item">
<button class="nav-link slc-subtab-btn py-0.5 px-2.5 small" data-i18n="systemControl.modeRag" data-mode="rag">  RAG Поиск</button>
</li>
<li class="nav-item">
<button class="nav-link slc-subtab-btn py-0.5 px-2.5 small" data-i18n="systemControl.modeErrors" data-mode="errors">  Errors Only</button>
</li>
<li class="nav-item">
<button class="nav-link slc-subtab-btn py-0.5 px-2.5 small" data-i18n="systemControl.modeIncidents" data-mode="incidents">  Incidents</button>
</li>
<li class="nav-item">
<button class="nav-link slc-subtab-btn py-0.5 px-2.5 small" data-i18n="systemControl.modeTimeline" data-mode="timeline">  Timeline</button>
</li>
<li class="nav-item">
<button class="nav-link slc-subtab-btn py-0.5 px-2.5 small" data-i18n="systemControl.modeRaw" data-mode="raw">  Raw Record</button>
</li>
<li class="nav-item">
<button class="nav-link slc-subtab-btn py-0.5 px-2.5 small" data-i18n="systemControl.modeActivity" data-mode="activity">  App Activity Log</button>
```

```
</li>
<li class="nav-item">
<button class="nav-link slc-subtab-btn py-0.5 px-2.5 small fw-semibold text-warning" data-i18n="systemControl.modeEventLogs" data-mode="event-logs" id="subtab-btn-event-logs">  Журналы событий (Event Logs)</button>
</li>
<li class="nav-item">
<button class="nav-link slc-subtab-btn py-0.5 px-2.5 small fw-semibold text-warning" data-i18n="systemControl.modePowerLifecycle" data-mode="power-lifecycle" id="subtab-btn-power-lifecycle">  Жизненный цикл питания (Power Lifecycle)</button>
</li>
<li class="nav-item">
<button class="nav-link slc-subtab-btn py-0.5 px-2.5 small fw-semibold text-warning" data-i18n="systemControl.modeDefenderLogs" data-mode="defender-logs" id="subtab-btn-defender-logs">  Журналы Defender (Defender Events)</button>
</li>
</ul>
<!-- Action AI Button -->
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm btn-outline-warning rounded-pill px-2.5 py-1 small d-flex align-items-center gap-1" id="btn-slc-ask-ai" style="font-size: 0.78rem;">
<i class="bi bi-robot"></i> <span data-i18n="systemControl.btnAskAi">Спросить ИИ</span>
</button>
</div>
</div>
<!-- Filter Inputs Grid -->
<div class="row g-1.5 align-items-center" id="slc-filter-row">
<div class="col-lg-4 col-md-5">
<div class="input-group input-group-sm">
<span class="input-group-text py-0 px-2"><i class="bi bi-search" style="font-size: 0.75rem;"></i></span>
<input class="form-control form-control-sm py-1" data-i18n-placeholder="systemControl.searchPlaceholder" id="slc-search-inpu
```

```
t" placeholder="Поиск по тексту, источнику..." style="font-siz
e: 0.8rem;" type="text"/>
</div>
</div>
<div class="col-lg-2 col-md-2 col-6">
<select class="form-select form-select-sm py-1" id="slc-level-
select" style="font-size: 0.8rem;">
<option data-i18n="systemControl.allLevels" value="">Все уровн
и</option>
<option value="Critical">Critical</option>
<option value="Error">Error</option>
<option value="Warning">Warning</option>
<option value="Information">Information</option>
</select>
</div>
<div class="col-lg-2 col-md-2 col-6">
<select class="form-select form-select-sm py-1" id="slc-hours-
select" style="font-size: 0.8rem;">
<option data-i18n="systemControl.hours1" value="1">За последни
й 1 час</option>
<option data-i18n="systemControl.hours6" value="6">За последни
е 6 часов</option>
<option data-i18n="systemControl.hours24" selected="" value="2
4">За последние 24 часа</option>
<option data-i18n="systemControl.hours168" value="168">За посл
едние 7 дней</option>
<option data-i18n="systemControl.hours720" value="720">За посл
едний месяц (30 дней)</option>
<option data-i18n="systemControl.hoursAll" value="0">За все до
ступное время</option>
</select>
</div>
<div class="col-lg-2 col-md-1.5 col-6">
<input class="form-control form-control-sm py-1" data-i18n-pla
ceholder="systemControl.eventIdPlaceholder" id="slc-eventid-in
put" placeholder="Event ID" style="font-size: 0.8rem;" type="n
umber"/>
</div>
```

```
<div class="col-lg-2 col-md-1.5 col-6">
<select class="form-select form-select-sm py-1" id="slc-limit-select" style="font-size: 0.8rem;">
<option data-i18n="systemControl.limit50" value="50">50 записей</option>
<option data-i18n="systemControl.limit100" selected="" value="100">100 записей</option>
<option data-i18n="systemControl.limit250" value="250">250 записей</option>
<option data-i18n="systemControl.limit500" value="500">500 записей</option>
</select>
</div>
</div>
</div>
<!-- Main Multi-Column View -->
<div class="row g-3" id="slc-main-row">
<!-- Channel & Source Tree Sidebar -->
<div class="col-lg-3" id="slc-channel-tree-sidebar">
<div class="slc-panel h-100" data-panel-id="panel-system-control-2" id="panel-system-control-2">
<div class="slc-panel-header">
<span><i class="bi bi-folder2-open me-2 text-primary"></i><span data-i18n="systemControl.channelsTitle">Каналы событий</span></span>
<span class="badge bg-secondary" id="slc-channel-count-badge">0</span>
</div>
<div class="p-2 border-bottom" style="border-color: var(--border-color) !important;">
<input class="form-control form-control-sm" data-i18n-placeholder="systemControl.filterChannelsPlaceholder" id="slc-channel-filter-input" placeholder="Фильтр каналов..." type="text"/>
</div>
<div class="p-2 slc-tree-list" id="slc-channel-tree-container">
<!-- Dynamic channels injected here -->
</div>
```

```
</div>
</div>
<!-- Events / Incidents Main Stream -->
<div class="col-lg-9" id="slc-main-stream-column">
<div class="slc-panel h-100" data-panel-id="panel-system-control-semanticheskiy-rag-poisk-po-log" id="panel-system-control-semanticheskiy-rag-poisk-po-log">
<div class="slc-panel-header">
<span id="slc-main-view-title"><i class="bi bi-list-columns-reverse me-2 text-primary"></i><span data-i18n="systemControl.mainViewTitle">Поток событий Windows</span></span>
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm btn-info rounded-pill px-3 fw-bold text-dark" data-i18n-title="systemControl.btnAuditTitle" id="btn-slc-audit" title="Провести аудит текущего журнала логов">
<i class="bi bi-shield-check me-1"></i> <span data-i18n="systemControl.btnAudit">Аудит</span>
</button>
<span class="badge bg-secondary" id="slc-stream-count-badge">0 записей</span>
</div>
</div>
<!-- Table View Container -->
<div class="table-responsive" id="slc-table-container" style="min-height: 140px; max-height: 560px; overflow-y: auto;">
<table class="slc-table">
<thead>
<tr>
<th data-i18n="systemControl.colTime" style="width: 155px;">Время</th>
<th data-i18n="systemControl.colLevel" style="width: 90px;">Уровень</th>
<th data-i18n="systemControl.colId" style="width: 65px;">ID</th>
<th data-i18n="systemControl.colProvider" style="width: 160px;">Поставщик</th>
<th data-i18n="systemControl.colPid" style="width: 65px;">PID</th>
```

```
<th data-i18n="systemControl.colMessage">Сообщение</th>
</tr>
</thead>
<tbody id="slc-table-body">
<tr>
<td class="text-center text-muted py-4" colspan="6">
<div class="spinner-border spinner-border-sm text-info me-2" r
ole="status"></div>
<span data-i18n="systemControl.loadingEvents">Загрузка событий
 Windows...</span>
</td>
</tr>
</tbody>
</table>
</div>
<div class="slc-table-resizer" id="slc-table-resizer" title="П
отяните для изменения высоты таблицы (двойной клик — сброс)">
<div class="slc-table-resizer-grip"></div>
</div>
<!-- Audit Digest View Container -->
<div class="p-3 d-none" id="slc-audit-container" style="max-he
ight: 560px; overflow-y: auto;">
<div id="slc-audit-content"></div>
</div>
<!-- Dynamic Log RAG Search Container (In-Memory RAM Engine) -
->
<div class="p-3 d-none" id="slc-rag-container" style="max-heig
ht: 580px; overflow-y: auto;">
  <div class="slc-card mb-3 shadow-sm">
    <div class="slc-card-header p-2.5 d-flex justify-content-b
etween align-items-center flex-wrap gap-2">
      <div class="d-flex align-items-center gap-2">
        <span class="fw-bold text-primary"><i class="bi bi-bra
in me-1.5"></i><span data-i18n="systemControl.ragTitle">Семант
ический RAG-поиск по логам (In-Memory RAM)</span></span>
        <span class="badge bg-success-subtle text-success bord
er border-success-subtle font-monospace" style="font-size: 0.7
2rem;">RAM Only (Без БД)</span>
```

```
      </div>
      <div class="d-flex align-items-center gap-2 flex-wrap">
        <div class="input-group input-group-sm" style="width:
auto;">
          <span class="input-group-text py-1 px-2 small bg-bod
y-tertiary" style="font-size: 0.78rem;"><i class="bi bi-calend
ar-range me-1"></i><span data-i18n="systemControl.periodLabel"
>Период:</span></span>
          <select class="form-select form-select-sm py-1" id="
slc-rag-period-select" style="min-width: 140px; font-size: 0.7
8rem;">
            <option data-i18n="systemControl.period24" selecte
d="" value="24">Сутки (24 ч)</option>
            <option data-i18n="systemControl.period168" value=
"168">Неделя (7 дн)</option>
            <option data-i18n="systemControl.period720" value=
"720">Месяц (30 дн)</option>
            <option data-i18n="systemControl.periodAll" value=
"0">Все время</option>
          </select>
        </div>
        <button class="btn btn-sm btn-outline-info rounded-pil
l px-3" id="btn-slc-rag-rebuild">
          <i class="bi bi-arrow-repeat me-1"></i> <span data-i
18n="systemControl.btnRagRebuild">Перестроить RAG-индекс</span
>
        </button>
      </div>
    </div>
    <div class="p-3">
      <!-- Status & Stats Bar -->
      <div class="d-flex align-items-center justify-content-be
tween flex-wrap gap-2 mb-2.5 p-2 rounded border" style="backgr
ound: var(--surface-2); border-color: var(--border-color) !imp
ortant; font-size: 0.78rem;">
        <div class="d-flex align-items-center gap-3 flex-wrap"
>
          <div><span class="text-muted">Статус индекса:</span>
```

```
 <strong id="slc-rag-status-badge" class="text-success">Готов
к поиску</strong></div>
          <div><span class="text-muted">Проиндексировано событ
ий:</span> <strong id="slc-rag-indexed-count" class="text-info
 font-monospace">--</strong></div>
          <div><span class="text-muted">Активных каналов:</spa
n> <strong id="slc-rag-channels-count" class="text-warning fon
t-monospace">--</strong></div>
          <div><span class="text-muted">Период:</span> <strong
 class="text-primary" id="slc-rag-period-badge" data-i18n="sys
temControl.periodBadge24">Сутки (24 часа)</strong></div>
          <div><span class="badge bg-secondary-subtle text-sec
ondary font-monospace" id="slc-rag-cache-indicator"><i class="
bi bi-hdd-fill me-1"></i><span data-i18n="systemControl.ragCac
heActive">Кеш браузера: Активен</span></span></div>
        </div>
        <div class="text-muted small" id="slc-rag-last-built">
Обновлено: --</div>
      </div>

      <!-- Search Inputs Grid -->
      <div class="row g-2 mb-2">
        <div class="col-md-8 col-12">
          <div class="input-group">
            <span class="input-group-text"><i class="bi bi-sea
rch text-primary"></i></span>
            <input class="form-control" data-i18n-placeholder=
"systemControl.ragInputPlaceholder" id="slc-rag-input" placeho
lder="Семантический запрос (например: сбои питания, ошибки DCO
M, отвал сети, BSOD, Disk warning)..." type="text"/>
            <button class="btn btn-info px-4 fw-bold text-dark
" data-i18n="systemControl.btnRagSearch" id="btn-slc-rag-searc
h">Найти</button>
          </div>
        </div>
        <div class="col-md-4 col-12 d-flex align-items-center
gap-2">
          <select class="form-select flex-grow-1" id="slc-rag-
```

```
channel-select">
            <option value="all">  Все журналы системы (По всей ОС)</option>
            <option value="System">System (Системный)</option>
            <option value="Application">Application (Приложения)</option>
            <option value="Security">Security (Безопасность)</option>
            <option value="Setup">Setup (Установка)</option>
            <option value="Microsoft-Windows-Kernel-Power/Operational">Kernel-Power (Питание)</option>
            <option value="Microsoft-Windows-Windows Defender/Operational">Windows Defender</option>
            <option value="Microsoft-Windows-TaskScheduler/Operational">Task Scheduler</option>
            <option value="Microsoft-Windows-PowerShell/Operational">PowerShell</option>
          </select>
          <div class="form-check form-switch mb-0 d-inline-flex align-items-center flex-shrink-0" style="font-size: 0.78rem;" title="Группировать повторяющиеся однотипные ошибки">
            <input class="form-check-input me-1" id="slc-rag-aggregate-switch" role="switch" type="checkbox" checked style="cursor: pointer;"/>
            <label class="form-check-label text-muted small user-select-none" for="slc-rag-aggregate-switch" data-i18n="systemControl.ragAggregate">Агрегация</label>
          </div>
        </div>
      </div>

      <!-- Quick Concept Suggestion Chips -->
      <div class="d-flex align-items-center gap-1.5 flex-wrap pt-1" style="font-size: 0.75rem;">
        <span class="text-muted me-1"><i class="bi bi-lightbulb text-warning"></i> Темы:</span>
        <button type="button" class="btn btn-xs btn-outline-secondary rounded-pill py-0 px-2 slc-rag-tag" data-query="критич
```

```
еские ошибки и сбои">⚠ Критические сбои</button>
        <button type="button" class="btn btn-xs btn-outline-secondary rounded-pill py-0 px-2 slc-rag-tag" data-query="сбои питания Kernel-Power 41 перезагрузка">⚡ Питание (Kernel-Power 41)</button>
        <button type="button" class="btn btn-xs btn-outline-secondary rounded-pill py-0 px-2 slc-rag-tag" data-query="ошибки DCOM 10016 права доступа">🔒 DCOM / Права (10016)</button>
        <button type="button" class="btn btn-xs btn-outline-secondary rounded-pill py-0 px-2 slc-rag-tag" data-query="сбои сети сетевой адаптер DNS disconnect">🌐 Сеть и адаптеры</button>
        <button type="button" class="btn btn-xs btn-outline-secondary rounded-pill py-0 px-2 slc-rag-tag" data-query="ошибки диска NTFS SMART storage">💾 Дисковая система</button>
        <button type="button" class="btn btn-xs btn-outline-secondary rounded-pill py-0 px-2 slc-rag-tag" data-query="остановка службы Service Control Manager">⚙ Службы (SCM)</button>
      </div>
    </div>
  </div>
  <div id="slc-rag-results">
    <div class="slc-card p-4 text-center text-muted" data-i18n="systemControl.ragEmptyPrompt">
      Введите запрос для поиска по In-Memory RAG-индексу всех журналов за последние сутки или нажмите «Перестроить RAG-индекс».
    </div>
  </div>
</div>
<!-- Incidents View Container -->
<div class="p-3 d-none" id="slc-incidents-container" style="max-height: 560px; overflow-y: auto;">
<div id="slc-incidents-list"></div>
</div>
<!-- Timeline Histogram View Container -->
<div class="p-3 d-none" id="slc-timeline-container" style="max-height: 560px; overflow-y: auto;">
```

```
<div id="slc-timeline-list"></div>
</div>
<!-- Raw Record Container -->
<div class="p-3 d-none" id="slc-raw-container" style="max-height: 560px; overflow-y: auto;">
<pre class="p-3 font-monospace small rounded border" data-i18n="systemControl.rawSelectPrompt" id="slc-raw-text" style="background: var(--surface-2); color: var(--text-color); border-color: var(--border-color) !important;">Выберите событие из списка для просмотра данных.</pre>
</div>
<!-- Application Activity Log View Container -->
<div class="p-3 d-none" id="slc-activity-container" style="max-height: 560px; overflow-y: auto;">
<div class="d-flex justify-content-between align-items-center mb-2">
<span class="fw-bold"><i class="bi bi-journal-text me-1 text-primary"></i> <span data-i18n="systemControl.activityTitle">Журнал активности</span></span>
<button class="btn btn-sm btn-outline-secondary rounded-pill px-3" id="btn-scc-refresh-logs">
<i class="bi bi-arrow-clockwise me-1"></i> <span data-i18n="systemControl.btnRefresh">Обновить</span>
</button>
</div>
<div class="table-responsive" id="scc-activity-table-container" style="min-height: 140px; max-height: 480px; overflow-y: auto;">
<table class="slc-table">
<thead>
<tr>
<th data-i18n="systemControl.colTime" style="width: 180px;">Время</th>
<th data-i18n="systemControl.colAction" style="width: 160px;">Действие</th>
<th data-i18n="systemControl.colTarget">Объект</th>
<th data-i18n="systemControl.colStatus" style="width: 120px;">Статус</th>
```

```
<th data-i18n="systemControl.colDetails" style="width: 30%;">Подробности</th>
</tr>
</thead>
<tbody id="scc-logs-tbody">
<tr><td class="text-center text-muted p-3" colspan="5" data-i18n="systemControl.loadingActivity">Загрузка журнала активности...</td></tr>
</tbody>
</table>
</div>
<div class="slc-table-resizer" id="scc-activity-table-resizer" title="Потяните для изменения высоты таблицы (двойной клик — сброс)">
<div class="slc-table-resizer-grip"></div>
</div>
</div>
<!-- Event Logs (WevtAPI & Log Intelligence) Container Panel -->
<div class="p-3 d-none" id="slc-event-logs-container">
<!-- Top Stat Cards -->
<div class="row g-2 mb-3">
<div class="col-6 col-md-3">
<div class="slc-card h-100 shadow-sm p-3">
<div class="text-muted small text-uppercase fw-semibold" data-i18n="systemControl.elStatCritical">Критические ошибки (Critical) □ </div>
<div class="fs-4 fw-bold mt-1 text-danger font-monospace" id="el-critical-count">--</div>
<div class="small text-muted mt-1" data-i18n="systemControl.elStatCriticalSub" id="el-critical-sub">Уровень 1 (WevtAPI)</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="slc-card h-100 shadow-sm p-3">
<div class="text-muted small text-uppercase fw-semibold" data-i18n="systemControl.elStatErrors">Ошибки (Errors) □ </div>
```

```
<div class="fs-4 fw-bold mt-1 text-warning font-monospace" id="el-error-count">--</div>
<div class="small text-muted mt-1" data-i18n="systemControl.elStatErrorsSub" id="el-error-sub">Уровень 2 (WevtAPI)</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="slc-card h-100 shadow-sm p-3">
<div class="text-muted small text-uppercase fw-semibold" data-i18n="systemControl.elStatHealth">Здоровье ОС (Health Score) □ □ </div>
<div class="fs-4 fw-bold mt-1 text-success font-monospace" id="el-health-score">--</div>
<div class="small text-muted mt-1" data-i18n="systemControl.elStatHealthSub" id="el-health-sub">Data Researcher EDA</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="slc-card h-100 shadow-sm p-3">
<div class="text-muted small text-uppercase fw-semibold" data-i18n="systemControl.elStatRedundancy">Избыточность / Шум (Rdup) □ </div>
<div class="fs-4 fw-bold mt-1 text-info font-monospace" id="el-redundancy-pct">--</div>
<div class="small text-muted mt-1" data-i18n="systemControl.elStatRedundancySub" id="el-redundancy-sub">Дедупликация шаблонов</div>
</div>
</div>
</div>
<!-- Action & Intelligence Bar -->
<div class="slc-panel mb-3 p-2 d-flex align-items-center justify-content-between flex-wrap gap-2" data-panel-id="panel-system-control-3" id="panel-system-control-3">
<div class="d-flex align-items-center gap-2 flex-wrap">
<button class="btn btn-sm btn-primary d-flex align-items-center gap-1.5" id="el-refresh-btn">
<i class="bi bi-arrow-clockwise"></i>
```

```
<span data-i18n="systemControl.btnRefresh">Обновить</span>
</button>
<select class="form-select form-select-sm" id="el-channel-sele
ct" style="width: 180px;">
<option value="System">Журнал System</option>
<option value="Application">Журнал Application</option>
<option value="Security">Журнал Security</option>
</select>
<button class="btn btn-sm btn-outline-info d-flex align-items-
center gap-1.5" id="el-intel-analyze-btn">
<i class="bi bi-cpu"></i>
<span data-i18n="systemControl.elBtnIntel">Log Intelligence (E
DA)</span>
</button>
<button class="btn btn-sm btn-outline-warning d-flex align-ite
ms-center gap-1.5" id="el-intel-audit-btn">
<i class="bi bi-shield-shaded"></i>
<span data-i18n="systemControl.elBtnAudit">Полный аудит</span>
</button>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<input class="form-control form-control-sm" data-i18n-placehol
der="systemControl.elRagPlaceholder" id="el-rag-query-input" p
laceholder="RAG поиск (напр: сбой сети)..." style="width: 240p
x;" type="text"/>
<button class="btn btn-sm btn-info text-dark fw-semibold" id="
el-rag-search-btn" title="Искать в RAG">
<i class="bi bi-search"></i>
</button>
<input class="form-control form-control-sm" data-i18n-placehol
der="systemControl.elFilterPlaceholder" id="el-search-input" p
laceholder="Фильтр таблицы..." style="width: 170px;" type="tex
t"/>
</div>
</div>
<!-- Intelligence Digest Banner (Hidden by default) -->
<div class="alert alert-dark border-info shadow-sm mb-3 d-none
" id="el-intel-banner" style="background: var(--surface-2); co
```

```
lor: var(--text-color); border-color: var(--border-color) !imp
ortant;">
<div class="d-flex justify-content-between align-items-start f
lex-wrap gap-2">
<div>
<div class="d-flex align-items-center gap-2 mb-1">
<span class="badge bg-info text-dark fw-bold font-monospace" i
d="el-strategy-badge">STRATEGY</span>
<strong class="fs-6" id="el-intel-title" style="color: var(--t
ext-color);">Аналитический вердикт Log Intelligence</strong>
</div>
<div class="small font-monospace text-muted" id="el-intel-rati
onale">--</div>
<div class="small text-warning mt-1" id="el-intel-recommendati
on"></div>
</div>
<div class="text-end">
<span class="badge bg-secondary font-monospace" id="el-chunks-
badge">0 RAG chunks</span>
<button aria-label="Close" class="btn-close ms-2" id="el-intel
-close-btn"></button>
</div>
</div>
</div>
<!-- RAG Search Results Panel (Hidden by default) -->
<div class="slc-card shadow-sm mb-3 d-none border-info" id="el
-rag-results-card">
<div class="slc-card-header d-flex justify-content-between ali
gn-items-center py-2 px-3">
<span class="fw-semibold text-info"><i class="bi bi-brain me-2
"></i><span data-i18n="systemControl.elRagResultsTitle">Резуль
таты Adaptive Log RAG</span></span>
<button class="btn btn-sm btn-outline-secondary py-0 px-2" dat
a-i18n="systemControl.elRagClose" id="el-rag-close-btn">Закрыт
ь</button>
</div>
<div class="card-body p-2" id="el-rag-results-body">
<!-- Dynamic Search Items -->
```

```
</div>
</div>
<!-- Event Logs Table -->
<div class="slc-card shadow-sm">
<div class="slc-card-header d-flex align-items-center justify-content-between py-2 px-3">
<span class="fw-semibold"><i class="bi bi-journal-text text-primary me-2"></i><span data-i18n="systemControl.elViewerTitle">Записи системного журнала (Event Viewer)</span></span>
<span class="badge bg-secondary font-monospace" id="el-table-badge">0 событий</span>
</div>
<div class="table-responsive" id="el-table-container" style="min-height: 140px; max-height: 520px; overflow-y: auto;">
<table class="slc-table" id="el-table">
<thead class="sticky-top">
<tr>
<th data-i18n="systemControl.elColLevel" style="width: 100px;">Уровень</th>
<th data-i18n="systemControl.elColTime" style="width: 160px;">Время</th>
<th data-i18n="systemControl.elColProvider" style="width: 180px;">Источник (Provider)</th>
<th data-i18n="systemControl.elColEventId" style="width: 80px;">Event ID</th>
<th data-i18n="systemControl.elColDescription">Описание события</th>
</tr>
</thead>
<tbody id="el-tbody">
<tr><td class="text-center text-muted py-4" colspan="5" data-i18n="systemControl.elLoading">Чтение журнала событий через WevtAPI...</td></tr>
</tbody>
</table>
</div>
<div class="slc-table-resizer" id="el-table-resizer" title="Потяните для изменения высоты таблицы (двойной клик — сброс)">
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<div class="slc-table-resizer-grip"></div>
</div>
</div>
</div>
<!-- Power Lifecycle (Uptime, Sessions, Reboots & EventLog 1074/41/6008) Container Panel -->
<div class="p-3 d-none" id="slc-power-lifecycle-container">
<!-- Верхняя панель метрик и сводки -->
<div class="row g-2 mb-3">
<!-- Карточка 1: Текущий Аптайм и Старт ОС -->
<div class="col-12 col-sm-6 col-xl-3">
<div class="slc-card h-100 shadow-sm p-3 position-relative">
<div class="d-flex justify-content-between align-items-start">
<div class="text-muted small text-uppercase fw-semibold">Текущий аптайм  </div>
<span class="badge bg-success font-monospace" id="pwr-uptime-badge">ACTIVE</span>
</div>
<div class="fs-4 fw-bold mt-1 text-info font-monospace" id="pwr-current-uptime">--:--:--</div>
<div class="small text-muted mt-1 text-truncate" id="pwr-current-boot" style="font-size: 0.75rem;">Загрузка: --</div>
<div class="mt-2 text-end">
<span class="badge bg-secondary-subtle text-secondary border border-secondary-subtle" style="font-size: 0.7rem;">Kernel-General 12</span>
</div>
</div>
</div>
<!-- Карточка 2: Всего сессий питания -->
<div class="col-12 col-sm-6 col-xl-3">
<div class="slc-card h-100 shadow-sm p-3">
<div class="d-flex justify-content-between align-items-start">
<div class="text-muted small text-uppercase fw-semibold">Сесси питания  </div>
<span class="badge bg-primary font-monospace" id="pwr-total-sessions-badge">0 Сессий</span>
</div>
```

```
<div class="fs-4 fw-bold mt-1 text-warning font-monospace" id="pwr-total-sessions">--</div>
<div class="small text-muted mt-1" id="pwr-sessions-desc">Реконструкция Boot → Shutdown</div>
<div class="mt-2">
<button class="btn btn-xs btn-outline-warning w-100 py-1" id="pwr-rebuild-btn" style="font-size: 0.78rem;">
<i class="bi bi-arrow-repeat me-1"></i><span>Пересканировать Event Log</span>
</button>
</div>
</div>
</div>
<!-- Карточка 3: Штатные перезагрузки и выключения -->
<div class="col-12 col-sm-6 col-xl-3">
<div class="slc-card h-100 shadow-sm p-3">
<div class="d-flex justify-content-between align-items-start">
<div class="text-muted small text-uppercase fw-semibold">Штатные завершения  </div>
<span class="badge bg-success-subtle text-success border border-success-subtle font-monospace" id="pwr-clean-ratio">-- %</span>
</div>
<div class="fs-4 fw-bold mt-1 text-success font-monospace" id="pwr-clean-count">--</div>
<div class="small text-muted mt-1 text-truncate" id="pwr-last-initiator" style="font-size: 0.75rem;">Инициатор: --</div>
<div class="mt-2 text-end">
<span class="badge bg-secondary-subtle text-secondary border border-secondary-subtle" style="font-size: 0.7rem;">User32 1074 / EventLog 6006</span>
</div>
</div>
</div>
<!-- Карточка 4: Внезапные сбои и BSOD -->
<div class="col-12 col-sm-6 col-xl-3">
<div class="slc-card h-100 shadow-sm p-3">
<div class="d-flex justify-content-between align-items-start">
```

```
<div class="text-muted small text-uppercase fw-semibold">Сбои / Потери питания  </div>
<span class="badge bg-danger font-monospace" id="pwr-unexpected-badge">0 Сбоев</span>
</div>
<div class="fs-4 fw-bold mt-1 text-danger font-monospace" id="pwr-unexpected-count">--</div>
<div class="small text-muted mt-1 d-flex justify-content-between" style="font-size: 0.75rem;">
<span id="pwr-bsod-count">BSOD (1001): 0</span>
<span>Kernel-Power 41</span>
</div>
<div class="mt-2">
<button class="btn btn-xs btn-outline-primary w-100 py-1" id="pwr-refresh-btn" style="font-size: 0.78rem;">
<i class="bi bi-arrow-clockwise me-1"></i><span>Обновить данные</span>
</button>
</div>
</div>
</div>
</div>
<!-- Навигация по подразделам: Сессии vs События -->
<ul class="nav nav-pills mb-3 slc-subtabs-nav pb-1" id="pwrNavTabs" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active py-1 px-3 small" data-bs-target="#pwr-subtab-sessions" data-bs-toggle="pill" id="pwr-sessions-subtab-btn" role="tab" type="button">
<i class="bi bi-clock-history me-1.5 text-warning"></i>
<span>Сессии питания (Power Sessions)</span>
</button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link py-1 px-3 small" data-bs-target="#pwr-subtab-events" data-bs-toggle="pill" id="pwr-events-subtab-btn" role="tab" type="button">
<i class="bi bi-list-columns-reverse me-1.5 text-info"></i>
```

```
<span>Журнал ключевых событий (Event Log)</span>
</button>
</li>
</ul>
<div class="tab-content" id="pwrTabContent">
<!-- Вкладка 1: Сессии питания (Power Sessions) -->
<div class="tab-pane fade show active" id="pwr-subtab-sessions
" role="tabpanel">
<div class="slc-panel mb-3 p-2" data-panel-id="panel-system-co
ntrol-4" id="panel-system-control-4">
<div class="row g-2 align-items-center">
<div class="col-12 col-md-4">
<div class="input-group input-group-sm">
<span class="input-group-text"><i class="bi bi-search"></i></s
pan>
<input class="form-control form-control-sm" id="pwr-sessions-s
earch" placeholder="Поиск по пользователю, процессу, причине..
." type="text"/>
</div>
</div>
<div class="col-6 col-md-3">
<select class="form-select form-select-sm" id="pwr-filter-type
">
<option value="">Все типы завершения</option>
<option value="Active">▶ Активная сессия (Active)</option>
<option value="Restart">↻ Перезагрузка (Restart)</option>
<option value="Shutdown">⏻ Выключение (Shutdown)</option>
<option value="Unexpected">⚠ Внезапное завершение (Unexpected)
</option>
<option value="BSOD">☠ Синий экран (BSOD / BugCheck)</option>
</select>
</div>
<div class="col-6 col-md-2">
<div class="form-check form-switch small mt-1">
<input class="form-check-input" id="pwr-filter-unexpected-only
" type="checkbox"/>
<label class="form-check-label text-warning fw-semibold" for="
pwr-filter-unexpected-only">Только сбои</label>
```

```
</div>
</div>
<div class="col-12 col-md-3 text-end">
<span class="small text-muted" id="pwr-sessions-counter">Найдено: 0</span>
</div>
</div>
</div>
<div class="slc-card shadow-sm">
<div class="table-responsive" id="pwr-sessions-table-container" style="min-height: 140px; max-height: 520px; overflow-y: auto;">
<table class="slc-table" style="font-size: 0.85rem;">
<thead class="sticky-top">
<tr>
<th style="width: 140px;">Сессия</th>
<th style="width: 160px;">Время старта (Boot)</th>
<th style="width: 160px;">Завершение</th>
<th style="width: 130px;">Аптайм</th>
<th style="width: 130px;">Статус</th>
<th>Инициатор и процесс</th>
<th>Причина / Код</th>
<th class="text-center" style="width: 80px;">Детали</th>
</tr>
</thead>
<tbody id="pwr-sessions-tbody">
<tr>
<td class="text-center py-4 text-muted" colspan="8">
<div class="spinner-border spinner-border-sm text-info me-2" role="status"></div>
                            Загрузка истории сессий питания...
                        </td>
</tr>
</tbody>
</table>
</div>
<div class="slc-table-resizer" id="pwr-sessions-table-resizer" title="Потяните для изменения высоты таблицы (двойной клик —
```

```
сброс)">
<div class="slc-table-resizer-grip"></div>
</div>
</div>
</div>
<!-- Вкладка 2: Журнал ключевых событий (Event Log) -->
<div class="tab-pane fade" id="pwr-subtab-events" role="tabpanel">
<div class="slc-panel mb-3 p-2" data-panel-id="panel-system-control-5" id="panel-system-control-5">
<div class="row g-2 align-items-center">
<div class="col-12 col-md-4">
<div class="input-group input-group-sm">
<span class="input-group-text"><i class="bi bi-funnel"></i></span>
<select class="form-select form-select-sm" id="pwr-event-id-filter">
<option value="">Все Event ID жизненного цикла</option>
<option value="1074">1074 — User32 (Инициатор shutdown/restart)</option>
<option value="41">41 — Kernel-Power (Внезапная перезагрузка)</option>
<option value="6008">6008 — EventLog (Неожиданное выключение)</option>
<option value="12">12 — Kernel-General (Запуск ядра / Boot)</option>
<option value="13">13 — Kernel-General (Завершение работы ОС)</option>
<option value="6005">6005 — EventLog (Служба EventLog запущена)</option>
<option value="6006">6006 — EventLog (EventLog штатно остановлен)</option>
<option value="6009">6009 — EventLog (Версия ядра при загрузке)</option>
<option value="1001">1001 — WER (Аварийный сбой BSOD / BugCheck)</option>
<option value="19">19 — WindowsUpdate (Установка обновлений)</option>
```

```
<option value="42">42 — Kernel-Power (Переход в сон)</option>
<option value="107">107 — Kernel-Power (Пробуждение из сна)</option>
</select>
</div>
</div>
<div class="col-12 col-md-5">
<div class="input-group input-group-sm">
<span class="input-group-text"><i class="bi bi-search"></i></span>
<input class="form-control form-control-sm" id="pwr-events-search" placeholder="Поиск в событиях..." type="text"/>
</div>
</div>
<div class="col-12 col-md-3 text-end">
<span class="small text-muted" id="pwr-events-counter">Событий: 0</span>
</div>
</div>
</div>
<div class="slc-card shadow-sm">
<div class="table-responsive" id="pwr-events-table-container" style="min-height: 140px; max-height: 520px; overflow-y: auto;">
<table class="slc-table" style="font-size: 0.85rem;">
<thead class="sticky-top">
<tr>
<th style="width: 160px;">Время</th>
<th style="width: 90px;">Event ID</th>
<th style="width: 140px;">Провайдер</th>
<th style="width: 130px;">Тип</th>
<th style="width: 180px;">Инициатор / Процесс</th>
<th>Описание / Причина</th>
<th class="text-center" style="width: 80px;">XML</th>
</tr>
</thead>
<tbody id="pwr-events-tbody">
<tr>
```

```
<td class="text-center py-4 text-muted" colspan="7">
<div class="spinner-border spinner-border-sm text-info me-2" role="status"></div>
                            Загрузка журнала событий...
                          </td>
</tr>
</tbody>
</table>
</div>
<div class="slc-table-resizer" id="pwr-events-table-resizer" title="Потяните для изменения высоты таблицы (двойной клик — сброс)">
<div class="slc-table-resizer-grip"></div>
</div>
</div>
</div>
</div>
</div>
<!-- Defender Event Logs & Security Center Container Panel -->
<div class="p-3 d-none" id="slc-defender-logs-container">
  <!-- Верхняя панель метрик Defender -->
  <div class="row g-2 mb-3">
    <!-- Карточка 1: Статус защиты Defender -->
    <div class="col-12 col-sm-6 col-xl-3">
      <div class="slc-card h-100 shadow-sm p-3 position-relative">
        <div class="d-flex justify-content-between align-items-start">
          <div class="text-muted small text-uppercase fw-semibold" data-i18n="systemControl.defStatStatus">Статус защиты Defender  </div>
          <span class="badge bg-success font-monospace" id="def-status-badge">ACTIVE</span>
        </div>
        <div class="fs-5 fw-bold mt-1 text-info font-monospace text-truncate" id="def-status-text">Включена (Real-Time)</div>
        <div class="small text-muted mt-1 text-truncate" id="d
```

```
ef-cloud-status" style="font-size: 0.75rem;">Cloud: Включена</
div>
        <div class="mt-2 text-end">
          <span class="badge bg-secondary-subtle text-secondar
y border border-secondary-subtle" style="font-size: 0.7rem;">R
eal-time Protection</span>
        </div>
      </div>
    </div>
    <!-- Карточка 2: Обнаруженные угрозы -->
    <div class="col-12 col-sm-6 col-xl-3">
      <div class="slc-card h-100 shadow-sm p-3">
        <div class="d-flex justify-content-between align-items
-start">
          <div class="text-muted small text-uppercase fw-semib
old" data-i18n="systemControl.defStatThreats">Обнаруженные угр
озы  </div>
          <span class="badge bg-success font-monospace" id="de
f-threats-badge">0 Угроз</span>
        </div>
        <div class="fs-4 fw-bold mt-1 text-warning font-monosp
ace" id="def-threats-count">0</div>
        <div class="small text-muted mt-1 text-truncate" id="d
ef-threats-desc" style="font-size: 0.75rem;">Event 1006 / 1116
</div>
        <div class="mt-2">
          <button class="btn btn-xs btn-outline-warning w-100
py-1" id="def-audit-btn" style="font-size: 0.78rem;">
            <i class="bi bi-shield-check me-1"></i><span data-
i18n="systemControl.defBtnAudit">Аудит статуса</span>
          </button>
        </div>
      </div>
    </div>
    <!-- Карточка 3: CFA / ASR Блокировки -->
    <div class="col-12 col-sm-6 col-xl-3">
      <div class="slc-card h-100 shadow-sm p-3">
        <div class="d-flex justify-content-between align-items
```

```
-start">
969           <div class="text-muted small text-uppercase fw-semib
old" data-i18n="systemControl.defStatProtection">CFA / ASR Бло
кировки  </div>
970           <span class="badge bg-info-subtle text-info border b
order-info-subtle font-monospace" id="def-cfa-badge">CFA</span
>
971         </div>
972         <div class="fs-4 fw-bold mt-1 text-info font-monospace
" id="def-blocks-count">0</div>
973         <div class="small text-muted mt-1 text-truncate" id="d
ef-blocks-desc" style="font-size: 0.75rem;">ASR 1125 / CFA 112
3 / Net 1121</div>
974         <div class="mt-2 text-end">
975           <span class="badge bg-secondary-subtle text-secondar
y border border-secondary-subtle" style="font-size: 0.7rem;">R
ansomware & Exploit Guard</span>
976         </div>
977       </div>
978     </div>
979     <!-- Карточка 4: Базы сигнатур и движок -->
980     <div class="col-12 col-sm-6 col-xl-3">
981       <div class="slc-card h-100 shadow-sm p-3">
982         <div class="d-flex justify-content-between align-items
-start">
983           <div class="text-muted small text-uppercase fw-semib
old" data-i18n="systemControl.defStatSignatures">Базы сигнатур
 и движок  </div>
984           <span class="badge bg-primary font-monospace" id="de
f-sig-badge">vLatest</span>
985         </div>
986         <div class="fs-6 fw-bold mt-1 text-success font-monosp
ace text-truncate" id="def-sig-version">--</div>
987         <div class="small text-muted mt-1 text-truncate" id="d
ef-engine-version" style="font-size: 0.75rem;">Engine: --</div
>
988         <div class="mt-2 d-flex gap-1">
989           <button class="btn btn-xs btn-outline-info flex-grow
```

```
-1 py-1" id="def-quick-scan-btn" style="font-size: 0.75rem;" t
itle="Запустить быстрое сканирование Defender">
            <i class="bi bi-play-circle me-1"></i><span data-i
18n="systemControl.defBtnScan">Быстрое скан</span>
          </button>
          <button class="btn btn-xs btn-outline-primary flex-g
row-1 py-1" id="def-update-sig-btn" style="font-size: 0.75rem;
" title="Обновить антивирусные базы">
            <i class="bi bi-cloud-arrow-down me-1"></i><span d
ata-i18n="systemControl.defBtnUpdateSig">Обновить базы</span>
          </button>
        </div>
      </div>
    </div>
  </div>

  <!-- Панель управления и фильтрации Defender -->
  <div class="slc-panel mb-3 p-2" data-panel-id="panel-system-
control-defender-filter" id="panel-system-control-defender-fil
ter">
    <div class="row g-2 align-items-center">
      <div class="col-12 col-md-4">
        <div class="input-group input-group-sm">
          <span class="input-group-text"><i class="bi bi-searc
h"></i></span>
          <input class="form-control form-control-sm" data-i18
n-placeholder="systemControl.defSearchPlaceholder" id="def-eve
nts-search" placeholder="Поиск по угрозам, процессам, путям...
" type="text"/>
        </div>
      </div>
      <div class="col-12 col-md-4 col-sm-6">
        <div class="input-group input-group-sm">
          <span class="input-group-text"><i class="bi bi-shiel
d"></i></span>
          <select class="form-select form-select-sm" id="def-c
ategory-select">
            <option value="" data-i18n="systemControl.defCateg
```

```
oryAll">Все категории Defender</option>
            <option value="threats" data-i18n="systemControl.defCategoryThreats">  Обнаружение угроз (1006, 1116)</option>
            <option value="cfa" data-i18n="systemControl.defCategoryCfa">  Controlled Folder Access (1123, 1124)</option>
            <option value="asr" data-i18n="systemControl.defCategoryAsr">  Attack Surface Reduction (1125, 1126)</option>
            <option value="network" data-i18n="systemControl.defCategoryNetwork">  Network Protection (1121, 1122)</option>
            <option value="scans" data-i18n="systemControl.defCategoryScans">  Сканирование (1000, 1001, 1002, 1005)</option>
            <option value="signatures" data-i18n="systemControl.defCategorySignatures">  Обновления баз и движка (2000, 2001, 2003)</option>
            <option value="protection" data-i18n="systemControl.defCategoryProtection">  Защита в реальном времени (5000, 5001, 5007)</option>
          </select>
        </div>
      </div>
      <div class="col-6 col-md-2">
        <select class="form-select form-select-sm" id="def-level-select">
          <option value="">Все уровни</option>
          <option value="Warning">Warning / Опасность</option>
          <option value="Error">Error / Сбой</option>
          <option value="Critical">Critical / Критический</option>
          <option value="Information">Information / Инфо</option>
        </select>
      </div>
      <div class="col-6 col-md-2 text-end d-flex justify-content-end align-items-center gap-2">
        <button class="btn btn-sm btn-outline-info rounded-pill px-2.5 py-1" id="def-refresh-btn" style="font-size: 0.78rem; ">
```

```
          <i class="bi bi-arrow-clockwise me-1"></i><span data-i18n="systemControl.btnRefresh">Обновить</span>
        </button>
        <span class="small text-muted font-monospace text-nowrap" id="def-events-counter">0 событий</span>
      </div>
    </div>
  </div>

  <!-- Таблица событий Defender -->
  <div class="slc-card shadow-sm">
    <div class="slc-card-header d-flex align-items-center justify-content-between py-2 px-3">
      <span class="fw-semibold"><i class="bi bi-shield-shaded text-primary me-2"></i><span>Журнал событий Microsoft Defender (Operational Log)</span></span>
      <span class="badge bg-secondary font-monospace" id="def-table-badge">0 событий</span>
    </div>
    <div class="table-responsive" id="def-events-table-container" style="min-height: 140px; max-height: 520px; overflow-y: auto;">
      <table class="slc-table" id="def-events-table" style="font-size: 0.85rem;">
        <thead class="sticky-top">
          <tr>
            <th data-i18n="systemControl.colTime" style="width: 160px;">Время</th>
            <th data-i18n="systemControl.colLevel" style="width: 90px;">Уровень</th>
            <th data-i18n="systemControl.colId" style="width: 75px;">Event ID</th>
            <th data-i18n="systemControl.defColCategory" style="width: 190px;">Категория</th>
            <th data-i18n="systemControl.colMessage">Описание события Defender</th>
            <th class="text-center" data-i18n="systemControl.defColAction" style="width: 80px;">Детали</th>
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          </tr>
        </thead>
        <tbody id="def-events-tbody">
          <tr>
            <td class="text-center py-4 text-muted" colspan="6
" data-i18n="systemControl.defLoading">
              <div class="spinner-border spinner-border-sm tex
t-info me-2" role="status"></div>
              Загрузка событий Windows Defender...
            </td>
          </tr>
        </tbody>
      </table>
    </div>
    <div class="slc-table-resizer" id="def-events-table-resize
r" title="Потяните для изменения высоты таблицы (двойной клик
— сброс)">
      <div class="slc-table-resizer-grip"></div>
    </div>
  </div>
</div>
</div>
</div>
</div>
</div>
<!-- Modal: Event Details & AI Diagnostics -->
<div class="modal fade" id="slc-event-detail-modal" tabindex="
-1">
<div class="modal-dialog modal-lg modal-dialog-centered">
<div class="modal-content" style="background: var(--surface-1)
; color: var(--text-color); border: 1px solid var(--border-col
or);">
<div class="modal-header" style="border-bottom: 1px solid var(
--border-color);">
<h5 class="modal-title fw-bold text-primary" data-i18n="system
Control.modalTitle" id="slc-modal-title">Детали события Window
s</h5>
<button class="btn-close" data-bs-dismiss="modal" type="button
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"></button>
</div>
<div class="modal-body">
<div class="row g-2 mb-3 small font-monospace">
<div class="col-sm-6"><strong class="text-muted" data-i18n="sy
stemControl.modalTime">Время:</strong> <span id="slc-modal-tim
e"></span></div>
<div class="col-sm-6"><strong class="text-muted" data-i18n="sy
stemControl.modalLevel">Уровень:</strong> <span id="slc-modal-
level"></span></div>
<div class="col-sm-6"><strong class="text-muted" data-i18n="sy
stemControl.modalEventId">Event ID:</strong> <span id="slc-mod
al-id"></span></div>
<div class="col-sm-6"><strong class="text-muted" data-i18n="sy
stemControl.modalProvider">Поставщик:</strong> <span id="slc-m
odal-provider"></span></div>
<div class="col-sm-6"><strong class="text-muted" data-i18n="sy
stemControl.modalComputer">Компьютер:</strong> <span id="slc-m
odal-computer"></span></div>
<div class="col-sm-6"><strong class="text-muted" data-i18n="sy
stemControl.modalPid">PID:</strong> <span id="slc-modal-pid"><
/span></div>
<div class="col-sm-12"><strong class="text-muted" data-i18n="s
ystemControl.modalChannel">Канал:</strong> <span id="slc-modal
-channel"></span></div>
</div>
<div class="p-3 mb-3 rounded border" style="background: var(--
surface-2); border-color: var(--border-color) !important;">
<h6 class="small text-uppercase text-muted fw-bold mb-2" data-
i18n="systemControl.modalMessageHeader">Текст события</h6>
<div class="small font-monospace" id="slc-modal-message" style
="white-space: pre-wrap; word-break: break-word; color: var(--
text-color);"></div>
</div>
<!-- AI Diagnostic Response Section -->
<div class="p-3 rounded border" id="slc-modal-ai-section" styl
e="background: var(--surface-2); border-color: var(--border-co
lor) !important;">
```

```
<div class="d-flex justify-content-between align-items-center mb-2">
<h6 class="fw-bold text-primary mb-0"><i class="bi bi-robot me-1"></i> <span data-i18n="systemControl.modalAiHeader">Контекстный анализ ИИ</span></h6>
<button class="btn btn-sm btn-outline-info py-0 px-2 rounded-pill" data-i18n="systemControl.modalBtnDiagnose" id="btn-slc-modal-diagnose">Анализ контекста</button>
</div>
<div class="small text-muted" data-i18n="systemControl.modalAiPrompt" id="slc-modal-ai-explanation">
              Нажмите «Анализ контекста», чтобы провести диагностику выбранной записи.
            </div>
</div>
</div>
<div class="modal-footer" style="border-top: 1px solid var(--border-color);">
<button class="btn btn-sm btn-secondary" data-bs-dismiss="modal" data-i18n="systemControl.modalBtnClose" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
<!-- Модальное окно деталей сессии питания -->
<div aria-hidden="true" aria-labelledby="pwrSessionDetailModalLabel" class="modal fade" id="pwrSessionDetailModal" tabindex="-1">
<div class="modal-dialog modal-lg modal-dialog-centered modal-dialog-scrollable">
<div class="modal-content shadow" style="background: var(--surface-1); color: var(--text-color); border: 1px solid var(--border-color);">
<div class="modal-header border-bottom" style="border-color: var(--border-color) !important;">
<h5 class="modal-title d-flex align-items-center gap-2" id="pwrSessionDetailModalLabel">
```

```
<i class="bi bi-info-circle-fill text-warning"></i>
<span id="pwr-modal-title">Детали сессии питания</span>
</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-3" id="pwr-modal-body">
<!-- Populated by main.js -->
</div>
<div class="modal-footer border-top" style="border-color: var(
--border-color) !important;">
<button class="btn btn-secondary btn-sm" data-bs-dismiss="moda
l" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
<!-- Модальное окно просмотра сырого XML события -->
<div aria-hidden="true" aria-labelledby="pwrXmlDetailModalLabe
l" class="modal fade" id="pwrXmlDetailModal" tabindex="-1">
<div class="modal-dialog modal-lg modal-dialog-centered modal-
dialog-scrollable">
<div class="modal-content shadow" style="background: var(--sur
face-1); color: var(--text-color); border: 1px solid var(--bor
der-color);">
<div class="modal-header border-bottom" style="border-color: v
ar(--border-color) !important;">
<h5 class="modal-title d-flex align-items-center gap-2" id="pw
rXmlDetailModalLabel">
<i class="bi bi-filetype-xml text-info"></i>
<span>Исходный XML события Windows</span>
</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-3">
<pre class="pwr-xml-pre p-3 rounded font-monospace mb-0" id="p
wr-xml-content" style="font-size: 0.78rem; max-height: 500px;
```

```
overflow-y: auto;"></pre>
</div>
<div class="modal-footer border-top" style="border-color: var(
--border-color) !important;">
<button class="btn btn-secondary btn-sm" data-bs-dismiss="moda
l" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
<!-- Модальное окно деталей события Defender -->
<div aria-hidden="true" aria-labelledby="defEventDetailModalLa
bel" class="modal fade" id="defEventDetailModal" tabindex="-1"
>
  <div class="modal-dialog modal-lg modal-dialog-centered moda
l-dialog-scrollable">
    <div class="modal-content shadow" style="background: var(-
-surface-1); color: var(--text-color); border: 1px solid var(-
-border-color);">
      <div class="modal-header border-bottom" style="border-co
lor: var(--border-color) !important;">
        <h5 class="modal-title d-flex align-items-center gap-2
" id="defEventDetailModalLabel">
          <i class="bi bi-shield-check text-info"></i>
          <span id="def-modal-title">Детали события Windows De
fender</span>
        </h5>
        <button aria-label="Close" class="btn-close" data-bs-d
ismiss="modal" type="button"></button>
      </div>
      <div class="modal-body p-3" id="def-modal-body">
        <!-- Populated by main.js -->
      </div>
      <div class="modal-footer border-top" style="border-color
: var(--border-color) !important;">
        <button class="btn btn-secondary btn-sm" data-bs-dismi
ss="modal" type="button" data-i18n="systemControl.modalBtnClos
e">Закрыть</button>
```

```
      </div>
    </div>
  </div>
</div>
</div>
```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows System Logs & Activity - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский веб-скрипт модуля журналов событий и активност
и системы Windows.
 *   Полная поддержка интернационализации (i18n), светлой и тё
мной темы.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/html/system_control_tab/main.js?v=2026100
6_v14" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/system_control_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-12 07:12:00
 * ============================================================
==================
 */

(function () {
  'use strict';

  let isSCCInitialized = false;

  // Состояния центра журналов событий
  let currentChannel = 'System';
  let currentFilePath = '';
  let currentMode = 'live';
```

```
  let cachedEvents = [];
  let cachedChannels = [];
  let selectedEvent = null;
  let liveIntervalTimer = null;
  let isLogRequestInProgress = false;
  let globalTooltipEl = null;

  // ============================================================
  // БРАУЗЕРНЫЙ КЕШ И ХРАНИЛИЩЕ RAG/AUDIT (LOCALSTORAGE / INDEXEDDB)
  // ============================================================
  const BrowserRagCache = {
    prefix: 'ai_windows_admin_rag_',

    get(key) {
      try {
        const item = localStorage.getItem(this.prefix + key);
        if (!item) return null;
        const parsed = JSON.parse(item);
        if (parsed.expires && Date.now() > parsed.expires) {
          localStorage.removeItem(this.prefix + key);
          return null;
        }
        return parsed.data;
      } catch (e) {
        console.debug('[BrowserRagCache] Ошибка чтения кеша:', e);
        return null;
      }
    },

    set(key, data, ttlMs = 24 * 3600 * 1000) {
      try {
        const payload = {
          data: data,
          timestamp: Date.now(),
```

```
65          expires: ttlMs ? Date.now() + ttlMs : null,
66        };
67        localStorage.setItem(this.prefix + key, JSON.stringify
(payload));
68        this.updateCacheIndicator(true);
69      } catch (e) {
70        console.debug('[BrowserRagCache] Ошибка записи в кеш:'
, e);
71      }
72    },
73
74    remove(key) {
75      try {
76        localStorage.removeItem(this.prefix + key);
77      } catch (e) {}
78    },
79
80    clearAll() {
81      try {
82        Object.keys(localStorage)
83          .filter(k => k.startsWith(this.prefix))
84          .forEach(k => localStorage.removeItem(k));
85      } catch (e) {}
86    },
87
88    updateCacheIndicator(active = true) {
89      const ind = document.getElementById('slc-rag-cache-indic
ator');
90      if (ind) {
91        ind.className = active ? 'badge bg-success-subtle text
-success border border-success-subtle font-monospace' : 'badge
 bg-secondary-subtle text-secondary font-monospace';
92        ind.innerHTML = `<i class="bi bi-hdd-fill me-1"></i>${
active ? t('systemControl.ragCacheSynced', 'Кеш браузера: Синх
ронизирован') : t('systemControl.ragCacheActive', 'Кеш браузер
a: Активен')}`;
93      }
94    }
```

```
  };

  function getSelectedRagPeriod() {
    const periodSelect = document.getElementById('slc-rag-period-select');
    return periodSelect ? parseInt(periodSelect.value, 10) : 24;
  }

  function getPeriodLabel(hours) {
    const h = parseInt(hours, 10);
    if (h === 24) return t('systemControl.periodBadge24', 'Сутки (24 часа)');
    if (h === 168) return t('systemControl.periodBadge168', 'Неделя (7 дней)');
    if (h === 720) return t('systemControl.periodBadge720', 'Месяц (30 дней)');
    if (h === 0) return t('systemControl.periodBadgeAll', 'Все время');
    return `${hours} ч`;
  }

  function updateRagPeriodDisplay(hours) {
    const h = parseInt(hours, 10);
    const badgeEl = document.getElementById('slc-rag-period-badge');
    if (badgeEl) {
      badgeEl.textContent = getPeriodLabel(h);
    }
    const periodSelect = document.getElementById('slc-rag-period-select');
    if (periodSelect && String(periodSelect.value) !== String(h)) {
      const opt = periodSelect.querySelector(`option[value="${h}"]`);
      if (opt) periodSelect.value = String(h);
    }
    const hoursSelect = document.getElementById('slc-hours-sel
```

```
ect');
    if (hoursSelect && String(hoursSelect.value) !== String(h)
) {
      const opt = hoursSelect.querySelector(`option[value="${h
}"]`);
      if (opt) hoursSelect.value = String(h);
    }
    BrowserRagCache.set('selected_period', h);
  }

  /**
   * Локализация строки через i18next с fallback.
   *
   * @param {string} key - Ключ локализации.
   * @param {string} fallback - Значение по умолчанию.
   * @returns {string} Локализованная строка.
   */
  function t(key, fallback) {
    if (window.i18next && typeof window.i18next.t === 'functio
n' && window.i18next.isInitialized) {
      const res = window.i18next.t(key);
      if (res && res !== key) return res;
    }
    return fallback !== undefined ? fallback : key;
  }

  /**
   * Экранирование HTML-символов для защиты от XSS.
   *
   * @param {string} str - Исходная строка.
   * @returns {string} Экранированная строка.
   */
  function escapeHtml(str) {
    if (!str) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
```

```
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
  }

  /**
   * Форматирование ISO-времени.
   *
   * @param {string} isoStr - Строка ISO Timestamp.
   * @returns {string} Отформатированное время.
   */
  function formatTime(isoStr) {
    if (!isoStr) return '-';
    try {
      return new Date(isoStr).toLocaleTimeString();
    } catch {
      return isoStr.slice(11, 19);
    }
  }

  /**
   * Определение CSS-класса для уровня важности события.
   *
   * @param {string} level - Уровень важности.
   * @returns {string} Имя CSS-класса.
   */
  function getSeverityClass(level) {
    const lvl = (level || '').toLowerCase();
    if (lvl.includes('crit')) return 'badge-crit';
    if (lvl.includes('err')) return 'badge-err';
    if (lvl.includes('warn')) return 'badge-warn';
    if (lvl.includes('verb')) return 'badge-verb';
    return 'badge-inf';
  }

  /**
   * Проверка прав администратора и обновление бейджа.
   */
  async function fetchStatus() {
```

```
    try {
      const res = await fetch('/api/system-control/status');
      if (!res.ok) return;
      const data = await res.json();

      const elevBadge = document.getElementById('scc-elevation
-badge');
      if (elevBadge) {
        if (data.is_elevated) {
          elevBadge.className = 'badge rounded-pill bg-success
-subtle text-success border border-success px-3 py-2';
          elevBadge.innerHTML = t('systemControl.elevAdmin', '
  Режим: Администратор (Полный доступ)');
        } else {
          elevBadge.className = 'badge rounded-pill bg-warning
-subtle text-warning border border-warning px-3 py-2';
          elevBadge.innerHTML = t('systemControl.elevUser', '
  Режим: Обычный пользователь (Ограничено)');
        }
      }
    } catch (e) {
      console.warn('[SystemLogs] Ошибка проверки прав:', e);
    }
  }

  // ============================================================
================
  // ЖУРНАЛ АКТИВНОСТИ И ОПЕРАЦИЙ
  // ============================================================
================

  async function loadSccActivityLogs() {
    const tbody = document.getElementById('scc-logs-tbody');
    if (!tbody) return;

    try {
      const res = await fetch('/api/system-control/logs');
      if (!res.ok) return;
```

```
      const data = await res.json();
      const logs = data.logs || [];

      if (logs.length === 0) {
        tbody.innerHTML = `<tr><td colspan="5" class="text-cen
ter text-muted p-3">${t('systemControl.emptyActivity', 'Журнал
 операций пуст.')}</td></tr>`;
        return;
      }

      tbody.innerHTML = logs.map(l => `
        <tr>
          <td class="font-monospace small" style="color: var(-
-text-muted);">${escapeHtml(l.timestamp)}</td>
          <td><span class="badge bg-secondary font-monospace">
${escapeHtml(l.action)}</span></td>
          <td style="color: var(--text-color);">${escapeHtml(l
.target)}</td>
          <td><span class="badge ${l.status === 'SUCCESS' || l
.status === 'OK' ? 'bg-success' : 'bg-warning text-dark'}">${e
scapeHtml(l.status)}</span></td>
          <td class="small text-truncate" style="max-width: 28
0px; color: var(--text-muted);" title="${escapeHtml(l.details)
}">${escapeHtml(l.details)}</td>
        </tr>
      `).join('');
    } catch (e) {
      console.warn('[SystemLogs] Ошибка загрузки логов активно
сти:', e);
    }
  }

  // ============================================================
================
  // WINDOWS OS SYSTEM LOG CENTER (ПОТОК, КАНАЛЫ, АУДИТ, RAG,
ИНЦИДЕНТЫ)
  // ============================================================
================
```

```
251
252   function createGlobalChannelTooltip() {
253     if (globalTooltipEl) return;
254     globalTooltipEl = document.createElement('div');
255     globalTooltipEl.className = 'slc-tooltip-popup';
256     globalTooltipEl.id = 'slc-global-tooltip';
257     document.body.appendChild(globalTooltipEl);
258   }
259
260   function showChannelTooltip(channelData, mouseEvent) {
261     if (!globalTooltipEl) createGlobalChannelTooltip();
262     const isFile = !!(channelData.file_path || channelData.loc
    ation || (channelData.source_type && channelData.source_type !
    == 'channel'));
263     const icon = isFile ? '□ ' : '□ ';
264     const rawTitle = channelData.display_name || channelData.c
    hannel_name || channelData.name || 'Channel';
265     const chanTitle = rawTitle.replace(/^evt:/, '');
266     const typeLabel = (channelData.source_type === 'channel' |
    | !isFile) ? t('systemControl.tooltipTypeLive', 'Live Windows
    Channel') : t('systemControl.tooltipTypeArchive', 'EVTX File A
    rchive');
267     const countVal = channelData.record_count !== undefined ?
    channelData.record_count : (channelData.records || 'N/A');
268     const recCount = typeof countVal === 'number' ? countVal.t
    oLocaleString() : String(countVal);
269     const sizeStr = channelData.file_size_mb ? `${channelData.
    file_size_mb} MB` : (channelData.size_bytes ? `${(channelData.
    size_bytes / 1024 / 1024).toFixed(1)} MB` : 'N/A');
270
271     globalTooltipEl.innerHTML = `
272       <div class="fw-bold mb-1 d-flex align-items-center gap-1
    .5" style="color: var(--text-color);">
273         <span>${icon}</span>
274         <span class="text-truncate">${escapeHtml(chanTitle)}</
    span>
275       </div>
276       <div class="mb-2" style="font-size: 0.76rem; color: var(
```

```
--text-muted) !important;">
277         ${escapeHtml(channelData.description || t('systemContr
ol.tooltipDefaultDesc', 'Стандартный системный канал событий W
indows.'))}
278       </div>
279       <div class="d-flex justify-content-between align-items-c
enter pt-1 border-top" style="border-color: var(--border-color
) !important; font-size: 0.74rem;">
280         <span class="text-muted">${t('systemControl.tooltipTyp
e', 'Тип:')} <strong style="color: var(--text-color);">${typeL
abel}</strong></span>
281         <span class="text-muted">${t('systemControl.tooltipRec
ords', 'Записей:')} <strong class="text-info">${recCount}</str
ong></span>
282       </div>
283       <div class="d-flex justify-content-between align-items-c
enter pt-1" style="font-size: 0.74rem;">
284         <span class="text-muted">${t('systemControl.tooltipSiz
e', 'Размер:')} <strong style="color: var(--text-color);">${si
zeStr}</strong></span>
285         <span class="text-success fw-bold">${t('systemControl.
tooltipAvailable', 'Доступен')}</span>
286       </div>
287     `;
288
289     const pad = 15;
290     let left = mouseEvent.clientX + pad;
291     let top = mouseEvent.clientY + pad;
292     if (left + 330 > window.innerWidth) left = mouseEvent.clie
ntX - 335;
293     if (top + 140 > window.innerHeight) top = mouseEvent.clien
tY - 145;
294
295     globalTooltipEl.style.left = `${Math.max(10, left)}px`;
296     globalTooltipEl.style.top = `${Math.max(10, top)}px`;
297     globalTooltipEl.classList.add('show');
298   }
299
```

```
  function hideChannelTooltip() {
    if (globalTooltipEl) {
      globalTooltipEl.classList.remove('show');
    }
  }

  async function scanChannels(force = false) {
    const treeEl = document.getElementById('slc-channel-tree-c
ontainer');
    const countBadge = document.getElementById('slc-channel-co
unt-badge');
    const sourcesStat = document.getElementById('slc-stat-sour
ces');

    if (treeEl && (!cachedChannels || cachedChannels.length ==
= 0)) {
      treeEl.innerHTML = `<div class="text-muted p-2 small"><d
iv class="spinner-border spinner-border-sm text-info me-1"></d
iv>${t('systemControl.searchingChannels', 'Поиск каналов...')}
</div>`;
    }

    try {
      const response = await fetch(`/api/v1/system_logs/scan?f
orce=${force}`);
      if (!response.ok) throw new Error(`HTTP ${response.statu
s}`);
      const data = await response.json();
      const rawChannels = data.channels || [];
      // Оставляем только непустые каналы и файлы журналов
      cachedChannels = rawChannels.filter(c => {
        const records = Number(c.record_count) || 0;
        const sizeBytes = Number(c.size_bytes) || 0;
        const sizeMb = Number(c.file_size_mb) || 0;
        return records > 0 || sizeBytes > 0 || sizeMb > 0;
      });

      if (countBadge) countBadge.textContent = cachedChannels.
```

```
length;
      if (sourcesStat) sourcesStat.textContent = cachedChannel
s.length.toLocaleString();

      renderChannelTree(cachedChannels);
    } catch (err) {
      if (treeEl) treeEl.innerHTML = `<div class="text-danger
p-2 small">${t('systemControl.errorScanning', 'Ошибка сканиров
ания каналов:')} ${escapeHtml(err.message)}</div>`;
    }
  }

  function renderChannelTree(channels) {
    const treeEl = document.getElementById('slc-channel-tree-c
ontainer');
    const countBadge = document.getElementById('slc-channel-co
unt-badge');
    if (!treeEl) return;

    if (countBadge && channels) {
      countBadge.textContent = channels.length;
    }

    if (!channels || channels.length === 0) {
      treeEl.innerHTML = `<div class="text-muted p-2 small">${
t('systemControl.noChannels', 'Каналы не найдены.')}</div>`;
      return;
    }

    createGlobalChannelTooltip();

    treeEl.innerHTML = channels.map(ch => {
      const isFile = !!(ch.file_path || (ch.source_type && ch.
source_type !== 'channel'));
      const sysName = (ch.channel_name || ch.name || ch.source
_id || 'System').replace(/^evt:/, '');
      const dispName = (ch.display_name || ch.channel_name ||
ch.name || 'Channel').replace(/^evt:/, '');
```

```
      const chPath = ch.location || ch.file_path || '';
      const isActive = isFile ? (currentFilePath === chPath) :
 (currentChannel === sysName && !currentFilePath);
      const icon = isFile ? 'bi-file-earmark-text text-seconda
ry' : 'bi-folder-fill text-warning';

      let sizeBadge = '';
      if (ch.record_count !== undefined && ch.record_count > 0
) {
        const countText = ch.record_count >= 1000 ? `${(ch.rec
ord_count / 1000).toFixed(1)}k` : String(ch.record_count);
        sizeBadge = `<span class="badge bg-secondary font-mono
space" style="font-size: 0.68rem;">${countText}</span>`;
      } else if (ch.file_size_mb) {
        sizeBadge = `<span class="badge bg-secondary font-mono
space" style="font-size: 0.68rem;">${ch.file_size_mb} MB</span
>`;
      }

      return `
        <div class="slc-tree-item ${isActive ? 'active' : ''}"
 data-channel="${escapeHtml(sysName)}" data-display-name="${es
capeHtml(dispName)}" data-path="${escapeHtml(chPath)}">
          <div class="d-flex align-items-center gap-2 text-tru
ncate me-2" style="min-width: 0; flex: 1 1 auto;">
            <i class="bi ${icon} flex-shrink-0"></i>
            <span class="text-truncate fw-medium" title="${esc
apeHtml(dispName)}">${escapeHtml(dispName)}</span>
          </div>
          <div class="d-flex align-items-center flex-shrink-0"
>
            ${sizeBadge}
          </div>
        </div>
      `;
    }).join('');

    treeEl.querySelectorAll('.slc-tree-item').forEach((el, ind
```

```
ex) => {
383       const chData = channels[index];
384       el.addEventListener('mouseenter', (e) => {
385         if (chData) showChannelTooltip(chData, e);
386       });
387       el.addEventListener('mousemove', (e) => {
388         if (chData) showChannelTooltip(chData, e);
389       });
390       el.addEventListener('mouseleave', () => {
391         hideChannelTooltip();
392       });
393 
394       el.addEventListener('click', () => {
395         hideChannelTooltip();
396         treeEl.querySelectorAll('.slc-tree-item').forEach(i =>
  i.classList.remove('active'));
397         el.classList.add('active');
398 
399         currentChannel = (el.getAttribute('data-channel') || e
l.getAttribute('data-name') || 'System').replace(/^evt:/, '').
split(' (')[0].trim();
400         currentFilePath = el.getAttribute('data-path') || '';
401         const dispName = el.getAttribute('data-display-name')
|| currentChannel;
402 
403         const statChan = document.getElementById('slc-stat-cha
nnel');
404         if (statChan) statChan.textContent = dispName;
405 
406         const mainTitle = document.getElementById('slc-main-vi
ew-title');
407         if (mainTitle) mainTitle.innerHTML = `<i class="bi bi-
list-columns-reverse me-2 text-primary"></i>${t('systemControl
.channelEventsTitle', 'События канала:')} <span class="text-in
fo">${escapeHtml(dispName)}</span>`;
408 
409         if (currentMode === 'event-logs') {
410           const elSelect = document.getElementById('el-channel
```

```
-select');
          if (elSelect) {
            let found = false;
            for (let opt of elSelect.options) {
              if (opt.value.toLowerCase() === currentChannel.t
oLowerCase()) {
                elSelect.value = opt.value;
                found = true;
                break;
              }
            }
            if (!found && !currentFilePath) {
              const opt = document.createElement('option');
              opt.value = currentChannel;
              opt.textContent = `Журнал ${dispName}`;
              elSelect.appendChild(opt);
              elSelect.value = currentChannel;
            }
          }
          fetchEventLogsSummary();
        } else {
          loadEvents();
        }
      });
    });
  }

  async function loadEvents() {
    if (isLogRequestInProgress) return;
    isLogRequestInProgress = true;

    const tbody = document.getElementById('slc-table-body');
    const statEvents = document.getElementById('slc-stat-event
s');
    const statIncidents = document.getElementById('slc-stat-in
cidents');
    const streamBadge = document.getElementById('slc-stream-co
unt-badge');
```

```
444
445     const searchInput = document.getElementById('slc-search-in
    put');
446     const levelSelect = document.getElementById('slc-level-sel
    ect');
447     const hoursSelect = document.getElementById('slc-hours-sel
    ect');
448     const limitSelect = document.getElementById('slc-limit-sel
    ect');
449     const eventIdInput = document.getElementById('slc-eventid-
    input');
450
451     const query = searchInput ? searchInput.value.trim() : '';
452     const level = levelSelect ? levelSelect.value : '';
453     const hours = hoursSelect ? hoursSelect.value : '24';
454     const limit = limitSelect ? limitSelect.value : '100';
455     const eventId = eventIdInput ? eventIdInput.value.trim() :
     '';
456
457     const cleanChan = currentChannel.replace(/^evt:/, '').spli
    t(' ')[0].trim();
458     const params = new URLSearchParams({
459       channel: cleanChan,
460       limit: limit,
461       hours: hours,
462     });
463
464     if (query) {
465       params.append('search', query);
466       params.append('query', query);
467     }
468     if (level) params.append('level', level);
469     if (eventId) params.append('event_id', eventId);
470     if (currentFilePath) params.append('file_path', currentFil
    ePath);
471
472     let url = `/api/v1/system_logs/events?${params.toString()}
    `;
```

```
    if (currentMode === 'errors') {
      params.set('level', 'Error');
      url = `/api/v1/system_logs/events?${params.toString()}`;
    }

    try {
      const res = await fetch(url);
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      cachedEvents = data.events || data.entries || [];

      if (statEvents) statEvents.textContent = cachedEvents.length.toLocaleString();
      if (streamBadge) streamBadge.textContent = `${cachedEvents.length} ${t('systemControl.records', 'записей')}`;

      const faults = cachedEvents.filter(e => {
        const l = (e.level || '').toLowerCase();
        return l.includes('err') || l.includes('crit');
      }).length;
      if (statIncidents) statIncidents.textContent = faults;

      renderEventsTable(cachedEvents);
    } catch (err) {
      if (tbody) tbody.innerHTML = `<tr><td colspan="6" class="text-center text-danger py-4">${t('systemControl.errorLoadingEvents', 'Ошибка загрузки событий:')} ${escapeHtml(err.message)}</td></tr>`;
    } finally {
      isLogRequestInProgress = false;
    }
  }

  function renderEventsTable(events) {
    const tbody = document.getElementById('slc-table-body');
    if (!tbody) return;

    if (!events || events.length === 0) {
```

```
      tbody.innerHTML = `<tr><td colspan="6" class="text-cente
r text-muted py-4">${t('systemControl.noEvents', 'Нет записей,
 удовлетворяющих заданным фильтрам.')}</td></tr>`;
      return;
    }

    tbody.innerHTML = events.map((e, idx) => {
      const timeFormatted = formatTime(e.timestamp);
      const isErr = (e.level || '').toLowerCase().includes('er
r') || (e.level || '').toLowerCase().includes('crit');
      const provName = e.provider || e.source || '-';
      const msgText = e.message || e.raw_data || '-';

      return `
        <tr data-index="${idx}" class="${isErr ? 'table-danger
-subtle' : ''}">
          <td class="font-monospace small" style="color: var(-
-text-muted);">${escapeHtml(timeFormatted)}</td>
          <td><span class="badge ${getSeverityClass(e.level)}
text-uppercase" style="font-size: 0.68rem;">${escapeHtml(e.lev
el || 'Info')}</span></td>
          <td class="font-monospace small fw-bold" style="colo
r: #0284c7;">${e.event_id > 0 ? e.event_id : '-'}</td>
          <td class="text-truncate small" style="max-width: 16
0px; color: var(--text-color);" title="${escapeHtml(provName)}
">${escapeHtml(provName)}</td>
          <td class="font-monospace small" style="color: var(-
-text-muted);">${e.process_id || '-'}</td>
          <td class="text-truncate" style="max-width: 420px; c
olor: var(--text-color);" title="${escapeHtml(msgText)}">${esc
apeHtml(msgText)}</td>
        </tr>
      `;
    }).join('');

    tbody.querySelectorAll('tr').forEach(row => {
      row.onclick = () => {
        tbody.querySelectorAll('tr').forEach(r => r.classList.
```

```
remove('table-active'));
        row.classList.add('table-active');
        const idx = parseInt(row.getAttribute('data-index'), 10);
        selectedEvent = events[idx];

        if (currentMode === 'raw') {
          const rawEl = document.getElementById('slc-raw-text');
          if (rawEl && selectedEvent) rawEl.textContent = JSON.stringify(selectedEvent, null, 2);
        } else {
          showEventDetails(selectedEvent);
        }
      };
    });
  }

  function renderAuditDigest(data, fromCache = false) {
    const contentEl = document.getElementById('slc-audit-content');
    if (!contentEl) return;

    const totalEvents = data.total_events || data.total_analyzed || 0;
    const critSum = data.severity_summary?.Critical || data.critical_count || 0;
    const errSum = data.severity_summary?.Error || data.error_count || 0;
    const warnSum = data.severity_summary?.Warning || data.warning_count || 0;

    let anomaliesHtml = `<div class="text-muted small">${escapeHtml(t('systemControl.noAnomalies', 'Аномалий и резких всплесков частоты ошибок не обнаружено.'))}</div>`;
    if (data.anomalies && data.anomalies.length > 0) {
      anomaliesHtml = data.anomalies.map(a => `
        <div class="alert alert-warning py-1 px-2 mb-1.5 small
```

```
 d-flex justify-content-between align-items-center">
          <div><strong class="text-danger">[${escapeHtml(t('sy
stemControl.anomaliesTitle', 'Аномалия'))}]</strong> ${escapeH
tml(a.description || a.sample || a.message)}</div>
          <span class="badge bg-danger font-monospace">${a.cou
nt || ''} ${escapeHtml(t('systemControl.eventsCountBadge', 'co
бытий').replace('{{count}}', '').trim())}</span>
        </div>
      `).join('');
    }

    const clustersList = data.clusters || data.top_clusters ||
 [];
    let clustersHtml = '';
    if (clustersList && clustersList.length > 0) {
      clustersHtml = clustersList.slice(0, 15).map((c, i) => `
        <tr class="slc-cluster-row" data-idx="${i}" style="cur
sor: pointer;" title="Нажмите для AI-диагностики кластера">
          <td class="font-monospace" style="color: var(--text-
muted);">${i + 1}</td>
          <td><span class="badge ${getSeverityClass(c.level)}"
>${escapeHtml(c.level || 'Info')}</span></td>
          <td class="text-truncate" style="max-width: 140px; c
olor: var(--text-color);" title="${escapeHtml(c.provider)}">${
escapeHtml(c.provider)}</td>
          <td class="font-monospace fw-bold" style="color: #02
84c7;">${c.event_id || '-'}</td>
          <td class="small text-truncate" style="max-width: 34
0px; color: var(--text-color);" title="${escapeHtml(c.pattern
|| c.template)}">${escapeHtml(c.pattern || c.template)}</td>
          <td class="font-monospace text-center fw-bold" style
="color: var(--text-color);">${c.count}</td>
          <td class="font-monospace small" style="color: var(-
-text-muted);">${escapeHtml(formatTime(c.first_seen || data.ge
nerated_at))}</td>
        </tr>
      `).join('');
    } else {
```

```
      clustersHtml = `<tr><td colspan="7" class="text-center t
ext-muted py-3">${escapeHtml(t('systemControl.noClusters', 'Кл
астеры не сформированы.'))}</td></tr>`;
    }

    contentEl.innerHTML = `
      ${fromCache ? `<div class="d-flex justify-content-betwee
n align-items-center mb-2 px-1"><span class="badge bg-info-sub
tle text-info border border-info-subtle font-monospace"><i cla
ss="bi bi-hdd me-1"></i>Загружено из кеша браузера</span><span
 class="small text-muted">${escapeHtml(data.generated_at || ''
)}</span></div>` : ''}
      <div class="row g-2 mb-3">
        <div class="col-sm-3 col-6">
          <div class="p-2 rounded border" style="background: v
ar(--surface-1); border-color: var(--border-color) !important;
">
            <div class="small text-muted" style="font-size: 0.
72rem;">${escapeHtml(t('systemControl.totalAnalyzed', 'Всего п
роанализировано'))}</div>
            <div class="fs-5 fw-bold text-info font-monospace"
>${totalEvents.toLocaleString()}</div>
          </div>
        </div>
        <div class="col-sm-3 col-6">
          <div class="p-2 rounded border" style="background: v
ar(--surface-1); border-color: var(--border-color) !important;
">
            <div class="small text-muted" style="font-size: 0.
72rem;">${escapeHtml(t('systemControl.critFaults', 'Критически
х сбоев'))}</div>
            <div class="fs-5 fw-bold text-danger font-monospac
e">${critSum}</div>
          </div>
        </div>
        <div class="col-sm-3 col-6">
          <div class="p-2 rounded border" style="background: v
ar(--surface-1); border-color: var(--border-color) !important;
```

```
">
            <div class="small text-muted" style="font-size: 0.72rem;">${escapeHtml(t('systemControl.errorsCount', 'Ошибок'))}</div>
            <div class="fs-5 fw-bold text-danger font-monospace">${errSum}</div>
          </div>
        </div>
        <div class="col-sm-3 col-6">
          <div class="p-2 rounded border" style="background: var(--surface-1); border-color: var(--border-color) !important;">
            <div class="small text-muted" style="font-size: 0.72rem;">${escapeHtml(t('systemControl.warningsCount', 'Предупреждений'))}</div>
            <div class="fs-5 fw-bold text-warning font-monospace">${warnSum}</div>
          </div>
        </div>
      </div>

      <div class="slc-card mb-2 shadow-sm">
        <div class="slc-card-header text-warning fw-bold py-1.5 px-3 small">
          <i class="bi bi-exclamation-triangle me-1"></i>${escapeHtml(t('systemControl.anomaliesTitle', 'Обнаруженные аномалии и всплески'))}
        </div>
        <div class="p-2">
          ${anomaliesHtml}
        </div>
      </div>

      <div class="slc-card shadow-sm">
        <div class="slc-card-header fw-bold d-flex justify-content-between align-items-center py-1.5 px-3">
          <span><i class="bi bi-collection me-2 text-primary"></i>${escapeHtml(t('systemControl.patternsTitle', 'Топ шаблоно
```

```
в событий (Дедупликация)'))}</span>
          <span class="small text-muted">${escapeHtml(t('systemControl.top15', 'Топ 15 кластеров'))}</span>
        </div>
        <div class="table-responsive">
          <table class="slc-table mb-0">
            <thead>
              <tr>
                <th style="width: 40px;">${escapeHtml(t('systemControl.colNum', '#'))}</th>
                <th style="width: 80px;">${escapeHtml(t('systemControl.colLevel', 'Уровень'))}</th>
                <th style="width: 140px;">${escapeHtml(t('systemControl.colProvider', 'Поставщик'))}</th>
                <th style="width: 60px;">${escapeHtml(t('systemControl.colId', 'ID'))}</th>
                <th>${escapeHtml(t('systemControl.colPattern', 'Шаблон сообщения (Masked)'))}</th>
                <th style="width: 70px;" class="text-center">${escapeHtml(t('systemControl.colCount', 'Кол-во'))}</th>
                <th style="width: 140px;">${escapeHtml(t('systemControl.colFirstSeen', 'Впервые'))}</th>
              </tr>
            </thead>
            <tbody>
              ${clustersHtml}
            </tbody>
          </table>
        </div>
      </div>
    `;

    // Интерактивный клик по строке кластера дедупликации
    contentEl.querySelectorAll('.slc-cluster-row').forEach(row => {
      row.onclick = () => {
        const idx = parseInt(row.getAttribute('data-idx'), 10);
```

```
650         const c = clustersList[idx];
651         if (!c) return;
652 
653         const modal = window.AIModalDialog || window.AITableMo
dal;
654         if (modal) {
655           modal.show({
656             icon: '□ ',
657             title: `Кластер #${c.event_id || 'GRP'} (${c.provi
der || 'Windows'})`,
658             subtitle: `Повторений: ${c.count} | Уровень: ${c.l
evel || 'Info'}`,
659             panelId: 'panel-system-control-2',
660             tableType: 'security_event',
661             badges: [
662               { text: `${c.count} повторений`, class: 'badge b
g-warning text-dark font-monospace' },
663               { text: c.level || 'Info', class: `badge ${getSe
verityClass(c.level)}` }
664             ],
665             metadata: [
666               { label: 'Event ID', value: String(c.event_id ||
 '-') },
667               { label: 'Поставщик', value: c.provider || '-' }
,
668               { label: 'Уровень', value: c.level || 'Info' },
669               { label: 'Всего повторений', value: String(c.cou
nt) },
670               { label: 'Первое появление', value: c.first_seen
 || '-' }
671             ],
672             rawTitle: 'Шаблон сообщения (Masked Pattern)',
673             rawContent: c.pattern || c.template || '',
674             requestData: {
675               event_id: c.event_id,
676               provider: c.provider,
677               count: c.count,
678               pattern: c.pattern || c.template
```

```
            }
          });
        }
      };
    });
  }

  async function loadAuditDigest() {
    const contentEl = document.getElementById('slc-audit-content');
    if (!contentEl) return;

    const hoursSelect = document.getElementById('slc-hours-select');
    const hours = hoursSelect ? hoursSelect.value : '24';
    const cleanChan = currentChannel.replace(/^evt:/i, '');
    const cacheKey = `audit_${cleanChan}_${hours}_${currentFilePath}`;

    const cached = BrowserRagCache.get(cacheKey);
    if (cached) {
      renderAuditDigest(cached, true);
    } else {
      contentEl.innerHTML = `<div class="text-center text-muted py-4"><div class="spinner-border spinner-border-sm text-info me-2"></div>${escapeHtml(t('systemControl.auditSummaryGenerating', 'Формирование сводки аудита и дедупликации...'))}</div>`;
    }

    try {
      const res = await fetch(`/api/v1/system_logs/audit?channel=${encodeURIComponent(cleanChan)}&hours=${hours}&limit=300&file_path=${encodeURIComponent(currentFilePath)}`);
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      BrowserRagCache.set(cacheKey, data);
      renderAuditDigest(data, false);
```

```
    } catch (err) {
      if (!cached) {
        contentEl.innerHTML = `<div class="alert alert-danger"
>${escapeHtml(t('systemControl.errorAudit', 'Ошибка проведения
 аудита:'))} ${escapeHtml(err.message)}</div>`;
      }
    }
  }

  async function fetchRagStatus() {
    const hours = getSelectedRagPeriod();
    const cachedStatus = BrowserRagCache.get(`status_${hours}`
);

    if (cachedStatus) {
      applyRagStatusToUi(cachedStatus);
      BrowserRagCache.updateCacheIndicator(true);
    }

    try {
      const res = await fetch('/api/v1/system_logs/rag/status'
);
      if (!res.ok) return;
      const data = await res.json();

      BrowserRagCache.set(`status_${hours}`, data);
      applyRagStatusToUi(data);
      BrowserRagCache.updateCacheIndicator(true);
    } catch (e) {
      console.debug('[SystemLogs] Ошибка получения статуса RAG
:', e);
    }
  }

  function applyRagStatusToUi(data) {
    const badgeEl = document.getElementById('slc-rag-status-ba
dge');
    const countEl = document.getElementById('slc-rag-indexed-c
```

```
ount');
740     const chansEl = document.getElementById('slc-rag-channels-
count');
741     const lastBuiltEl = document.getElementById('slc-rag-last-
built');
742     const hours = getSelectedRagPeriod();
743 
744     if (countEl) countEl.textContent = (data.total_events || d
ata.indexed_events || 0).toLocaleString();
745     if (chansEl) chansEl.textContent = (data.channels_count ||
 data.scanned_channels || 0).toLocaleString();
746     if (lastBuiltEl) {
747       lastBuiltEl.textContent = data.last_built_at || data.gen
erated_at ? `Обновлено: ${data.last_built_at || data.generated
_at}` : 'Индекс еще не создан';
748     }
749     updateRagPeriodDisplay(data.hours !== undefined ? data.hou
rs : hours);
750 
751     if (badgeEl) {
752       const totalEv = data.total_events || data.indexed_events
 || 0;
753       if ((data.is_indexed || totalEv > 0) && totalEv > 0) {
754         badgeEl.textContent = 'Готов к поиску (RAM)';
755         badgeEl.className = 'text-success fw-bold';
756       } else {
757         badgeEl.textContent = 'Требуется сбор данных';
758         badgeEl.className = 'text-warning fw-bold';
759       }
760     }
761 
762     const chanSelect = document.getElementById('slc-rag-channe
l-select');
763     if (chanSelect && data.channels_list && data.channels_list
.length > 0) {
764       const currentVal = chanSelect.value;
765       const opts = ['<option value="all">  Все журналы систем
ы (По всей ОС)</option>'];
```

```
      data.channels_list.forEach(ch => {
        opts.push(`<option value="${escapeHtml(ch)}">${escapeHtml(ch)}</option>`);
      });
      chanSelect.innerHTML = opts.join('');
      if (currentVal && data.channels_list.includes(currentVal)) {
        chanSelect.value = currentVal;
      }
    }
  }

  async function runRagSearch() {
    const inputEl = document.getElementById('slc-rag-input');
    const resultsEl = document.getElementById('slc-rag-results');
    const chanSelect = document.getElementById('slc-rag-channel-select');
    const aggSwitch = document.getElementById('slc-rag-aggregate-switch');
    if (!inputEl || !resultsEl) return;
    const query = inputEl.value.trim();
    if (!query) return;

    const selectedChan = chanSelect ? chanSelect.value : 'all';
    const channelParam = selectedChan === 'all' ? '' : selectedChan;
    const aggregate = aggSwitch ? aggSwitch.checked : true;
    const hours = getSelectedRagPeriod();
    const periodLabel = getPeriodLabel(hours);
    const cacheKey = `query_${hours}_${selectedChan}_${query}_agg_${aggregate}`;

    const cachedResults = BrowserRagCache.get(cacheKey);
    if (cachedResults) {
      renderRagResults(cachedResults, query, periodLabel, true);
```

```
795     } else {
796       resultsEl.innerHTML = `<div class="text-center text-mute
    d py-4"><div class="spinner-border spinner-border-sm text-info
     me-2"></div>${escapeHtml(t('systemControl.ragSearching', 'Сем
    антический поиск по RAM RAG-индексу...'))} (${escapeHtml(perio
    dLabel)})</div>`;
797     }
798
799     try {
800       const res = await fetch('/api/v1/system_logs/rag/query',
     {
801         method: 'POST',
802         headers: { 'Content-Type': 'application/json' },
803         body: JSON.stringify({ query: query, channel: channelP
    aram, top_k: 20, aggregate: aggregate }),
804       });
805       if (!res.ok) throw new Error(`HTTP ${res.status}`);
806       const data = await res.json();
807       const chunks = data.results || [];
808
809       BrowserRagCache.set(cacheKey, chunks);
810       renderRagResults(chunks, query, periodLabel, false);
811     } catch (err) {
812       if (!cachedResults) {
813         resultsEl.innerHTML = `<div class="alert alert-danger"
    >${escapeHtml(t('systemControl.errorRagSearch', 'Ошибка поиска
     RAG:'))} ${escapeHtml(err.message)}</div>`;
814       }
815     }
816   }
817
818   function renderRagResults(chunks, query, periodLabel, fromCa
    che = false) {
819     const resultsEl = document.getElementById('slc-rag-results
    ');
820     if (!resultsEl) return;
821
822     if (!chunks || chunks.length === 0) {
```

```
      resultsEl.innerHTML = `
        <div class="slc-card p-4 text-center text-muted shadow-sm">
          <i class="bi bi-search text-warning fs-3 mb-2 d-block"></i>
          <div>По запросу «<strong>${escapeHtml(query)}</strong>» совпадений в журналах за период «<strong>${escapeHtml(periodLabel)}</strong>» не найдено.</div>
          <div class="small text-muted mt-1">Попробуйте изменить формулировку или нажмите «Перестроить RAG-индекс».</div>
        </div>
      `;
      return;
    }

    const totalOccurrences = chunks.reduce((acc, c) => acc + (c.count || 1), 0);

    resultsEl.innerHTML = `
      <div class="d-flex justify-content-between align-items-center mb-2 px-1 flex-wrap gap-2">
        <div class="d-flex align-items-center gap-2 flex-wrap">
          <span class="small text-muted">Найдено кластеров: <strong class="text-info font-monospace">${chunks.length}</strong> (Всего событий: <strong class="text-warning font-monospace">${totalOccurrences.toLocaleString()}</strong>, Период: <strong>${escapeHtml(periodLabel)}</strong>)</span>
        </div>
        <div class="d-flex align-items-center gap-2">
          ${fromCache ? '<span class="badge bg-info-subtle text-info border border-info-subtle font-monospace" style="font-size: 0.72rem;"><i class="bi bi-hdd me-1"></i>Кеш браузера</span>' : ''}
          <span class="badge bg-secondary font-monospace" style="font-size: 0.72rem;">In-Memory Score Rank</span>
        </div>
      </div>
```

```
      ${chunks.map((c, idx) => {
        const relPct = Math.round((c.score || c.relevance_scor
e || 0) * 100);
        const lvlClass = getSeverityClass(c.level);
        const isErr = (c.level || '').toLowerCase().includes('
err') || (c.level || '').toLowerCase().includes('crit');
        const prov = c.provider || c.metadata?.provider || 'Wi
ndows';
        const eid = c.event_id || c.metadata?.event_id || 0;
        const chan = c.channel || c.metadata?.channel || 'Syst
em';
        const time = c.timestamp || c.metadata?.timestamp || '
-';
        const msg = c.message || c.text || c.content || '';
        const count = c.count || 1;
        const isAgg = count > 1;

        return `
          <div class="slc-card mb-2 p-3 shadow-sm ${isErr ? 'b
order-danger-subtle' : ''}" style="transition: transform 0.15s
 ease;">
            <div class="d-flex justify-content-between align-i
tems-center flex-wrap gap-2 mb-1.5 pb-1 border-bottom" style="
border-color: var(--border-color) !important;">
              <div class="d-flex align-items-center gap-2 flex
-wrap">
                <span class="badge bg-primary-subtle text-prim
ary border border-primary-subtle font-monospace" style="font-s
ize: 0.72rem;">  ${escapeHtml(chan)}</span>
                <span class="badge ${lvlClass} text-uppercase"
 style="font-size: 0.68rem;">${escapeHtml(c.level || 'Info')}<
/span>
                <span class="fw-bold font-monospace" style="co
lor: #0284c7; font-size: 0.8rem;">ID ${eid > 0 ? eid : '-'}</s
pan>
                <span class="small fw-semibold" style="color:
var(--text-color);">${escapeHtml(prov)}</span>
                ${isAgg ? `<span class="badge bg-warning text-
```

```
dark font-monospace fw-bold" style="font-size: 0.72rem;" title
="Событие повторялось несколько раз"><i class="bi bi-stack me-
1"></i>${escapeHtml(t('systemControl.ragRepetitions', 'Повторе
ний: {{count}}x').replace('{{count}}', count))}</span>` : ''}
              </div>
              <div class="d-flex align-items-center gap-2 flex
-wrap">
                <span class="badge bg-info-subtle text-info bo
rder border-info-subtle font-monospace" style="font-size: 0.72
rem;">Релевантность: ${relPct}%</span>
                ${isAgg && c.first_seen && c.last_seen && c.fi
rst_seen !== c.last_seen ? `
                  <span class="badge bg-secondary-subtle text-
secondary font-monospace" style="font-size: 0.7rem;" title="Пе
риод первого и последнего события">
                    <i class="bi bi-clock-history me-1"></i>${
escapeHtml(c.first_seen)} → ${escapeHtml(c.last_seen)}
                  </span>
                ` : `
                  <span class="font-monospace small text-muted
">${escapeHtml(time)}</span>
                `}
                <button class="btn btn-xs btn-outline-info rou
nded-pill px-2 py-0.5 btn-rag-detail" data-idx="${idx}" style=
"font-size: 0.72rem;">
                  <i class="bi bi-info-circle me-1"></i>Детали
                </button>
              </div>
            </div>
            <div class="small font-monospace text-break mt-1"
style="color: var(--text-color); line-height: 1.45; max-height
: 100px; overflow-y: auto;">
              ${escapeHtml(msg)}
            </div>
            ${isAgg && c.occurrences && c.occurrences.length >
 1 ? `
              <div class="mt-2 pt-1 border-top" style="border-
color: var(--border-color) !important;">
```

```
                <button class="btn btn-xs btn-outline-secondary rounded-pill py-0.5 px-2.5 btn-toggle-occurrences" data-target="occ-${idx}" style="font-size: 0.72rem;">
                  <i class="bi bi-list-nested me-1"></i><span class="btn-occ-text">${escapeHtml(t('systemControl.ragShowOccurrences', 'Показать все {{count}} повторения по времени').replace('{{count}}', count))}</span>
                </button>
                <div class="d-none mt-2 p-2 rounded border" id="occ-${idx}" style="background: var(--surface-2); border-color: var(--border-color) !important; max-height: 180px; overflow-y: auto;">
                  <table class="slc-table mb-0" style="font-size: 0.74rem;">
                    <thead>
                      <tr>
                        <th style="width: 145px;">Время события</th>
                        <th style="width: 70px;">PID</th>
                        <th style="width: 80px;">Record ID</th>
                        <th>Сообщение</th>
                        <th style="width: 55px;"></th>
                      </tr>
                    </thead>
                    <tbody>
                      ${c.occurrences.map((occ, oIdx) => `
                        <tr>
                          <td class="font-monospace" style="color: var(--text-color);">${escapeHtml(occ.timestamp || '-')}</td>
                          <td class="font-monospace text-muted">${occ.process_id || '-'}</td>
                          <td class="font-monospace text-muted">${occ.record_id || '-'}</td>
                          <td class="text-truncate small" style="max-width: 280px; color: var(--text-muted);" title="${escapeHtml(occ.message)}">${escapeHtml(occ.message)}</td>
```

```
                        <td class="text-end">
                          <button class="btn btn-xs btn-outl
ine-info py-0 px-1.5 btn-occ-detail" data-chunk-idx="${idx}" d
ata-occ-idx="${oIdx}" title="Открыть детали">
                            <i class="bi bi-info-circle"></i
>
                          </button>
                        </td>
                      </tr>
                    `).join('')}
                  </tbody>
                </table>
              </div>
            </div>
          ` : ''}
        </div>
      `;
    }).join('')}
  `;

  resultsEl.querySelectorAll('.btn-rag-detail').forEach(btn
=> {
    btn.onclick = () => {
      const idx = parseInt(btn.getAttribute('data-idx'), 10)
;
      const chunk = chunks[idx];
      if (chunk) {
        showEventDetails({
          timestamp: chunk.timestamp,
          level: chunk.level,
          event_id: chunk.event_id,
          provider: chunk.provider,
          channel: chunk.channel,
          computer: chunk.computer || '',
          process_id: chunk.process_id || 0,
          message: chunk.message || chunk.text,
        });
      }
```

```
      };
    });

    resultsEl.querySelectorAll('.btn-toggle-occurrences').forEach(btn => {
      btn.onclick = () => {
        const targetId = btn.getAttribute('data-target');
        const container = document.getElementById(targetId);
        const textSpan = btn.querySelector('.btn-occ-text');
        if (container) {
          const isHidden = container.classList.contains('d-none');
          if (isHidden) {
            container.classList.remove('d-none');
            if (textSpan) textSpan.textContent = t('systemControl.ragHideOccurrences', 'Скрыть повторения');
          } else {
            container.classList.add('d-none');
            const count = btn.closest('.slc-card').querySelector('.badge.bg-warning')?.textContent || '';
            if (textSpan) textSpan.textContent = t('systemControl.ragShowOccurrences', 'Показать все повторения');
          }
        }
      };
    });

    resultsEl.querySelectorAll('.btn-occ-detail').forEach(btn => {
      btn.onclick = () => {
        const cIdx = parseInt(btn.getAttribute('data-chunk-idx'), 10);
        const oIdx = parseInt(btn.getAttribute('data-occ-idx'), 10);
        const chunk = chunks[cIdx];
        const occ = chunk?.occurrences?.[oIdx];
        if (occ) {
          showEventDetails({
```

```
            timestamp: occ.timestamp,
            level: occ.level,
            event_id: occ.event_id,
            provider: occ.provider,
            channel: occ.channel,
            computer: occ.computer || '',
            process_id: occ.process_id || 0,
            message: occ.message,
          });
        }
      };
    });
  }

  async function rebuildRagIndex() {
    const resultsEl = document.getElementById('slc-rag-results');
    const rebuildBtn = document.getElementById('btn-slc-rag-rebuild');
    const hours = getSelectedRagPeriod();
    const periodLabel = getPeriodLabel(hours);
    if (!resultsEl) return;

    if (rebuildBtn) rebuildBtn.disabled = true;
    resultsEl.innerHTML = `
      <div class="text-center text-info py-4">
        <div class="spinner-border spinner-border-sm text-info me-2"></div>
        <strong>${escapeHtml(t('systemControl.ragRebuilding', 'Сбор событий со всех системных каналов и построение RAM-индекса...'))} (${escapeHtml(periodLabel)})</strong>
        <div class="small text-muted mt-1">Опрос WevtAPI, сбор журналов в оперативную память и синхронизация браузерного кеша.</div>
      </div>
    `;

    try {
```

```
      const res = await fetch(`/api/v1/system_logs/rag/build?all_channels=true&hours=${hours}&limit=150`, {
        method: 'POST',
      });
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();

      BrowserRagCache.set(`status_${hours}`, data);
      await fetchRagStatus();

      resultsEl.innerHTML = `
        <div class="alert alert-success d-flex align-items-start gap-2 shadow-sm">
          <i class="bi bi-check-circle-fill fs-4 text-success flex-shrink-0 mt-0.5"></i>
          <div>
            <div class="fw-bold fs-6">In-Memory RAG-индекс успешно построен (${escapeHtml(periodLabel)})!</div>
            <div class="small mt-1">
              Просканировано каналов: <strong>${data.scanned_channels || 0}</strong> (с событиями: <strong>${data.channels_with_events || 0}</strong>).
              Всего событий (${escapeHtml(periodLabel)}): <strong>${(data.indexed_events || 0).toLocaleString()}</strong>
              ( Критических: <strong>${data.critical_count || 0}</strong>,  Ошибок: <strong>${data.error_count || 0}</strong>).
              Время генерации: <strong>${data.build_duration_ms || 0} мс</strong> в RAM.
            </div>
            <div class="small text-muted mt-2">Данные зафиксированы в RAM и кеше браузера. Вы можете вводить любые запросы для семантического поиска.</div>
          </div>
        </div>
      `;
    } catch (err) {
      resultsEl.innerHTML = `<div class="alert alert-danger">$
```

```
{escapeHtml(t('systemControl.errorRagRebuild', 'Ошибка обновле
ния RAG:'))} ${escapeHtml(err.message)}</div>`;
1027      } finally {
1028        if (rebuildBtn) rebuildBtn.disabled = false;
1029      }
1030    }
1031
1032    function showEventDetails(entry, panelId = 'panel-system-con
trol-3') {
1033      if (!entry) return;
1034
1035      const modal = window.AIModalDialog || window.AITableModal;
1036      if (modal) {
1037        const lvl = entry.level || 'Info';
1038        const isErr = String(lvl).toLowerCase().includes('err')
|| String(lvl).toLowerCase().includes('crit');
1039        const isWarn = String(lvl).toLowerCase().includes('warn'
);
1040        const badgeCls = isErr ? 'badge bg-danger' : (isWarn ? '
badge bg-warning text-dark' : 'badge bg-info text-dark');
1041        const chan = (entry.channel || currentChannel || 'System
').replace(/^evt:/i, '');
1042
1043        modal.show({
1044          icon: isErr ? '□ ' : (isWarn ? '□ ' : '□ '),
1045          title: `Событие #${entry.event_id > 0 ? entry.event_id
 : 'LOG'} (${entry.provider || entry.source || 'Windows'})`,
1046          subtitle: `Уровень: ${lvl} | ${entry.timestamp || ''}`
,
1047          panelId: panelId,
1048          tableType: 'security_event',
1049          badges: [
1050            { text: lvl, class: badgeCls },
1051            { text: chan, class: 'badge border text-secondary' }
1052          ],
1053          metadata: [
1054            { label: 'Event ID', value: entry.event_id > 0 ? Str
ing(entry.event_id) : '-' },
```

```
          { label: 'Поставщик / Источник', value: entry.provid
er || entry.source || '-' },
          { label: 'Уровень важности', value: lvl },
          { label: 'Время регистрации', value: entry.timestamp
 || '-' },
          { label: 'Канал логов', value: chan },
          { label: 'Компьютер / Хост', value: entry.computer |
| 'LOCAL' },
          { label: 'PID процесса', value: entry.process_id > 0
 ? String(entry.process_id) : '-' }
        ],
        rawTitle: 'Текст сообщения журнала событий',
        rawContent: entry.message || entry.raw_data || '',
        requestData: {
          event_id: entry.event_id,
          provider: entry.provider || entry.source,
          level: lvl,
          channel: chan,
          computer: entry.computer
        }
      });
      return;
    }

    const modalEl = document.getElementById('slc-event-detail-
modal');
    if (!modalEl) return;

    const setElTxt = (id, val) => {
      const el = document.getElementById(id);
      if (el) el.textContent = val || '';
    };

    setElTxt('slc-modal-time', entry.timestamp);
    const lvlEl = document.getElementById('slc-modal-level');
    if (lvlEl) lvlEl.innerHTML = `<span class="badge ${getSeve
rityClass(entry.level)}">${escapeHtml(entry.level)}</span>`;
    setElTxt('slc-modal-id', entry.event_id > 0 ? String(entry
```

```
.event_id) : '-');
1087    setElTxt('slc-modal-provider', entry.provider || entry.sou
rce || '-');
1088    setElTxt('slc-modal-computer', entry.computer || '-');
1089    setElTxt('slc-modal-pid', entry.process_id > 0 ? String(en
try.process_id) : '-');
1090    setElTxt('slc-modal-channel', (entry.channel || currentCha
nnel).replace(/^evt:/i, ''));
1091    setElTxt('slc-modal-message', entry.message);
1092
1093    const aiExplanation = document.getElementById('slc-modal-a
i-explanation');
1094    if (aiExplanation) {
1095      aiExplanation.innerHTML = escapeHtml(t('systemControl.mo
dalAiPrompt', 'Нажмите «Анализ контекста», чтобы провести диаг
ностику выбранной записи.'));
1096    }
1097
1098    const diagnoseBtn = document.getElementById('btn-slc-modal
-diagnose');
1099    if (diagnoseBtn) {
1100      diagnoseBtn.onclick = async () => {
1101        aiExplanation.innerHTML = `<div class="spinner-border
spinner-border-sm text-info me-2"></div>${escapeHtml(t('system
Control.modalAiAnalyzing', 'Анализ выбранной записи...'))}`;
1102        try {
1103          const cleanChan = (entry.channel || currentChannel).
replace(/^evt:/i, '');
1104          const res = await fetch('/api/v1/system_logs/explain
', {
1105            method: 'POST',
1106            headers: { 'Content-Type': 'application/json' },
1107            body: JSON.stringify({
1108              channel: cleanChan,
1109              timestamp: entry.timestamp,
1110              event_id: entry.event_id || 0,
1111              provider: entry.provider || entry.source || '',
1112              level: entry.level || '',
```

```
              message: entry.message || '',
              target_entry: entry,
              query_text: `Event diagnostic ${entry.provider || entry.source} (ID ${entry.event_id || 0}): ${entry.message || ''}`,
            }),
          });
          if (!res.ok) throw new Error(`HTTP ${res.status}`);
          const report = await res.json();
          aiExplanation.innerHTML = `
            <div class="mb-2"><strong class="text-info">${escapeHtml(t('systemControl.modalAiSummary', 'Сводка:'))}</strong> ${escapeHtml(report.summary)}</div>
            <div class="mb-2"><strong class="text-warning">${escapeHtml(t('systemControl.modalAiRootCause', 'Причина / Анализ:'))}</strong> ${escapeHtml(report.root_cause)}</div>
            <h6 class="small text-uppercase text-muted fw-bold mb-1">${escapeHtml(t('systemControl.modalAiRecs', 'Рекомендации:'))}</h6>
            <ul class="mb-0 ps-3">
              ${(report.recommendations || []).map(r => `<li>${escapeHtml(r)}</li>`).join('')}
            </ul>
          `;
        } catch (e) {
          aiExplanation.innerHTML = `<div class="text-danger">${escapeHtml(t('systemControl.errorDiagnose', 'Ошибка диагностики:'))} ${escapeHtml(e.message)}</div>`;
        }
      };
    }

    if (window.bootstrap?.Modal) {
      const bsModal = bootstrap.Modal.getOrCreateInstance(modalEl);
      bsModal.show();
    }
  }
```

```
1139
1140  async function loadIncidents() {
1141    const listEl = document.getElementById('slc-incidents-list
');
1142    if (!listEl) return;
1143    listEl.innerHTML = `<div class="text-center text-muted py-
4"><div class="spinner-border spinner-border-sm text-info me-2
"></div>${escapeHtml(t('systemControl.groupingIncidents', 'Гру
ппировка инцидентов...'))}</div>`;
1144
1145    try {
1146      const res = await fetch('/api/v1/system_logs/incidents?w
indow_seconds=25.0');
1147      if (!res.ok) throw new Error(`HTTP ${res.status}`);
1148      const data = await res.json();
1149      const incidents = data.incidents || [];
1150
1151      if (incidents.length === 0) {
1152        listEl.innerHTML = `<div class="slc-card p-3 text-cent
er text-muted">${escapeHtml(t('systemControl.noIncidents', 'Св
язанных цепочек инцидентов и каскадных сбоев не обнаружено.'))
}</div>`;
1153        return;
1154      }
1155
1156      listEl.innerHTML = incidents.map(inc => `
1157        <div class="slc-card mb-2 shadow-sm border-danger">
1158          <div class="slc-card-header text-danger d-flex justi
fy-content-between align-items-center py-1.5 px-3">
1159            <strong>${escapeHtml(inc.incident_title || t('syst
emControl.cascadeFailure', 'Каскадный сбой'))}</strong>
1160            <span class="badge bg-danger">${inc.events_count}
${escapeHtml(t('systemControl.eventsCountBadge', 'событий').re
place('{{count}}', '').trim())}</span>
1161          </div>
1162          <div class="p-2.5">
1163            <p class="small mb-1">${escapeHtml(inc.summary ||
'')}</p>
```

```
1164            <div class="small font-monospace text-muted">${esc
    apeHtml(t('systemControl.timeWindow', 'Временное окно:'))} ${e
    scapeHtml(inc.time_window || '')}</div>
1165          </div>
1166        </div>
1167      `).join('');
1168    } catch (e) {
1169      listEl.innerHTML = `<div class="alert alert-danger">${es
    capeHtml(t('systemControl.errorIncidents', 'Ошибка загрузки ин
    цидентов:'))} ${escapeHtml(e.message)}</div>`;
1170    }
1171  }
1172
1173  async function loadTimeline() {
1174    const listEl = document.getElementById('slc-timeline-list'
    );
1175    if (!listEl) return;
1176    listEl.innerHTML = `<div class="text-center text-muted py-
    4"><div class="spinner-border spinner-border-sm text-info me-2
    "></div>${escapeHtml(t('systemControl.formingTimeline', 'Форми
    рование временной шкалы...'))}</div>`;
1177
1178    try {
1179      const res = await fetch('/api/v1/system_logs/timeline');
1180      if (!res.ok) throw new Error(`HTTP ${res.status}`);
1181      const data = await res.json();
1182      const points = data.timeline || [];
1183
1184      if (points.length === 0) {
1185        listEl.innerHTML = `<div class="slc-card p-3 text-cent
    er text-muted">${escapeHtml(t('systemControl.noTimeline', 'Дан
    ные временной шкалы отсутствуют.'))}</div>`;
1186        return;
1187      }
1188
1189      listEl.innerHTML = points.map(p => `
1190        <div class="d-flex align-items-center gap-2 mb-1">
1191          <span class="font-monospace small" style="width: 120
```

```
px; color: var(--text-muted);">${escapeHtml(p.time_slot)}</spa
n>
1192          <div class="progress flex-grow-1" style="height: 12p
x; background: var(--surface-2);">
1193            <div class="progress-bar bg-info" style="width: ${
Math.min(100, p.count * 4)}%;"></div>
1194          </div>
1195          <span class="font-monospace small fw-bold" style="wi
dth: 40px; color: #0284c7;">${p.count}</span>
1196        </div>
1197      `).join('');
1198    } catch (e) {
1199      listEl.innerHTML = `<div class="alert alert-danger">${es
capeHtml(t('systemControl.errorTimeline', 'Ошибка таймлайна:')
)} ${escapeHtml(e.message)}</div>`;
1200    }
1201  }
1202
1203  // ============================================================
================
1204  // ПАНЕЛЬ ЖУРНАЛОВ СОБЫТИЙ WINDOWS (WEVTAPI & LOG INTELLIGEN
CE)
1205  // ============================================================
================
1206
1207  let elEventsList = [];
1208  let isElInitialized = false;
1209
1210  async function fetchEventLogsSummary() {
1211    const channelSelect = document.getElementById('el-channel-
select');
1212    const channel = channelSelect ? channelSelect.value : 'Sys
tem';
1213
1214    try {
1215      const res = await fetch(`/api/event-logs/events?channel=
${encodeURIComponent(channel)}&limit=100&hours=24`);
1216      if (!res.ok) throw new Error(`HTTP ${res.status}`);
```

```
      const entries = await res.json();
      renderElEvents(entries);
      fetchIntelligenceProfile(channel, false);
    } catch (err) {
      console.warn('[EventLogs] Ошибка получения событий:', err);
    }
  }

  async function fetchIntelligenceProfile(channel, showBanner = true) {
    try {
      const res = await fetch(`/api/event-logs/intelligence/profile?channel=${encodeURIComponent(channel)}&hours=24&limit=100`);
      if (!res.ok) return;
      const data = await res.json();

      const healthEl = document.getElementById('el-health-score');
      const redEl = document.getElementById('el-redundancy-pct');

      if (data.profile) {
        if (healthEl) {
          const score = Math.round(data.profile.health_score ?? 100);
          healthEl.textContent = `${score}/100`;
          healthEl.className = `fs-4 fw-bold mt-1 font-monospace ${score >= 80 ? 'text-success' : (score >= 50 ? 'text-warning' : 'text-danger')}`;
        }
        if (redEl) {
          const rPct = (data.profile.redundancy_ratio_pct ?? 0).toFixed(1);
          redEl.textContent = `${rPct}%`;
        }
      }
```

```
1245
1246      if (showBanner && data.decision) {
1247        renderIntelligenceBanner(data);
1248      }
1249    } catch (err) {
1250      console.debug('[EventLogs] Фоновый сбор аналитики:', err
);
1251    }
1252  }
1253
1254  function renderIntelligenceBanner(data) {
1255    const banner = document.getElementById('el-intel-banner');
1256    const title = document.getElementById('el-intel-title');
1257    const strat = document.getElementById('el-strategy-badge')
;
1258    const rationale = document.getElementById('el-intel-ration
ale');
1259    const rec = document.getElementById('el-intel-recommendati
on');
1260    const chunks = document.getElementById('el-chunks-badge');
1261
1262    if (!banner) return;
1263
1264    if (title) title.textContent = `Log Intelligence: Канал ${
data.channel || 'System'}`;
1265    if (strat) strat.textContent = (data.decision.strategy ||
'SNAPSHOT').toUpperCase();
1266    if (rationale) rationale.textContent = data.decision.ratio
nale || '';
1267    if (rec) rec.innerHTML = `<i class="bi bi-lightbulb me-1">
</i><strong>Рекомендация:</strong> ${escapeHtml(data.decision.
recommended_llm_action || 'Система в штатном режиме.')}`;
1268    if (chunks) chunks.textContent = `${data.decision.chunks_g
enerated || 0} RAG chunks`;
1269
1270    banner.classList.remove('d-none');
1271  }
1272
```

```
  async function runElFullAudit() {
    const channelSelect = document.getElementById('el-channel-select');
    const channel = channelSelect ? channelSelect.value : 'System';

    try {
      const res = await fetch(`/api/event-logs/intelligence/audit?channel=${encodeURIComponent(channel)}&hours=24&limit=200`);
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      renderIntelligenceBanner({
        channel: data.channel,
        decision: {
          strategy: data.strategy,
          rationale: data.strategy_rationale,
          recommended_llm_action: data.recommended_llm_action,
          chunks_generated: data.top_clusters ? data.top_clusters.length : 0,
        }
      });
    } catch (err) {
      console.warn('[EventLogs] Ошибка аудита:', err);
    }
  }

  async function runElRAGSearch() {
    const input = document.getElementById('el-rag-query-input');
    const channelSelect = document.getElementById('el-channel-select');
    const resultsCard = document.getElementById('el-rag-results-card');
    const resultsBody = document.getElementById('el-rag-results-body');

    if (!input || !resultsCard || !resultsBody) return;
```

```
    const query = input.value.trim();
    if (!query) return;

    resultsBody.innerHTML = `<div class="text-center py-3 text-muted"><div class="spinner-border spinner-border-sm text-info me-2"></div>Поиск в Adaptive Log RAG...</div>`;
    resultsCard.classList.remove('d-none');

    try {
      const channel = channelSelect ? channelSelect.value : '';
      const res = await fetch(`/api/event-logs/intelligence/search?query=${encodeURIComponent(query)}&top_k=5&channel=${encodeURIComponent(channel)}`);
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const results = await res.json();

      if (!results || results.length === 0) {
        resultsBody.innerHTML = `<div class="text-center py-3 text-muted">По запросу "<em>${escapeHtml(query)}</em>" совпадений не найдено.</div>`;
        return;
      }

      resultsBody.innerHTML = results.map(r => `
        <div class="p-2 mb-2 rounded bg-black border border-secondary">
          <div class="d-flex justify-content-between align-items-center mb-1">
            <strong class="text-info">${escapeHtml(r.title || 'Документ RAG')}</strong>
            <span class="badge bg-secondary font-monospace">Relevance: ${Math.round((r.relevance_score || 0) * 100)}%</span>
          </div>
          <div class="small text-light font-monospace" style="word-break: break-word;">${escapeHtml(r.content || '')}</div>
          <div class="small text-muted mt-1 font-monospace"><i class="bi bi-tag me-1"></i>Стратегия: ${escapeHtml(r.strategy
```

```
 || 'Adaptive')} • ${escapeHtml(r.created_at || '')}</div>
        </div>
      `).join('');
    } catch (err) {
      resultsBody.innerHTML = `<div class="text-danger small p-2">Ошибка поиска: ${escapeHtml(err.message)}</div>`;
    }
  }

  function renderElEvents(entries) {
    elEventsList = entries || [];

    const critEl = document.getElementById('el-critical-count');
    const errEl = document.getElementById('el-error-count');

    let crits = 0;
    let errs = 0;

    elEventsList.forEach(e => {
      const l = (e.level || '').toLowerCase();
      if (l.includes('crit')) crits++;
      else if (l.includes('err')) errs++;
    });

    if (critEl) critEl.textContent = crits;
    if (errEl) errEl.textContent = errs;

    applyElFilters();
  }

  function applyElFilters() {
    const tbody = document.getElementById('el-tbody');
    const searchInput = document.getElementById('el-search-input');
    const badge = document.getElementById('el-table-badge');

    if (!tbody) return;
```

```
1361
1362    const query = (searchInput ? searchInput.value : '').toLowerCase().trim();
1363
1364    const filtered = elEventsList.filter(e => {
1365      if (!query) return true;
1366      const msgMatch = (e.message || '').toLowerCase().includes(query);
1367      const provMatch = (e.provider_name || e.provider || '').toLowerCase().includes(query);
1368      const idMatch = String(e.event_id || e.id || '').includes(query);
1369      return msgMatch || provMatch || idMatch;
1370    });
1371
1372    if (badge) badge.textContent = `${filtered.length} из ${elEventsList.length} событий`;
1373
1374    if (filtered.length === 0) {
1375      tbody.innerHTML = `<tr><td colspan="5" class="text-center text-muted py-4">${escapeHtml(t('systemControl.noEvents', 'События не найдены'))}</td></tr>`;
1376      return;
1377    }
1378
1379    const itemsToRender = filtered.slice(0, 150);
1380    tbody.innerHTML = itemsToRender.map((e, idx) => {
1381      const level = (e.level || 'Information').toLowerCase();
1382      let badgeClass = 'bg-secondary';
1383      if (level.includes('crit') || level.includes('err')) badgeClass = 'bg-danger text-white';
1384      else if (level.includes('warn')) badgeClass = 'bg-warning text-dark';
1385      else if (level.includes('info')) badgeClass = 'bg-info text-dark';
1386
1387      return `
1388        <tr class="slc-el-table-row" data-idx="${idx}" style="
```

```
cursor: pointer;" title="Нажмите для просмотра подробностей события">
          <td><span class="badge ${badgeClass}">${escapeHtml(e.level || 'Info')}</span></td>
          <td class="text-muted small font-monospace">${escapeHtml(e.time_created || e.timestamp || '-')}</td>
          <td class="fw-semibold small" style="color: var(--text-color);">${escapeHtml(e.provider_name || e.provider || 'Windows')}</td>
          <td class="font-monospace text-warning small">${escapeHtml(String(e.event_id || e.id || '-'))}</td>
          <td class="small text-truncate" style="max-width: 500px; color: var(--text-color);">${escapeHtml(e.message || '-')}</td>
        </tr>
      `;
    }).join('');

    // Привязываем клики по строкам для просмотра деталей и ИИ-диагностики
    tbody.querySelectorAll('.slc-el-table-row').forEach(row => {
      row.onclick = () => {
        const idx = parseInt(row.getAttribute('data-idx'), 10);
        const item = itemsToRender[idx];
        if (item) {
          showEventDetails({
            timestamp: item.time_created || item.timestamp,
            level: item.level,
            event_id: item.event_id || item.id,
            provider: item.provider_name || item.provider,
            computer: item.computer || item.machine_name || '',
            process_id: item.process_id || item.pid,
            channel: item.channel || (document.getElementById('el-channel-select')?.value || 'System'),
            message: item.message,
```

```
          });
        }
      };
    });
  }

  function initEventLogsPanel() {
    if (isElInitialized) return;
    isElInitialized = true;

    const refreshBtn = document.getElementById('el-refresh-btn
');
    if (refreshBtn) refreshBtn.addEventListener('click', fetch
EventLogsSummary);

    const channelSelect = document.getElementById('el-channel-
select');
    if (channelSelect) channelSelect.addEventListener('change'
, fetchEventLogsSummary);

    const searchInput = document.getElementById('el-search-inp
ut');
    if (searchInput) searchInput.addEventListener('input', app
lyElFilters);

    const analyzeBtn = document.getElementById('el-intel-analy
ze-btn');
    if (analyzeBtn) analyzeBtn.addEventListener('click', () =>
 {
      const chan = channelSelect ? channelSelect.value : 'Syst
em';
      fetchIntelligenceProfile(chan, true);
    });

    const auditBtn = document.getElementById('el-intel-audit-b
tn');
    if (auditBtn) auditBtn.addEventListener('click', runElFull
Audit);
```

```
1440
1441     const ragBtn = document.getElementById('el-rag-search-btn'
     );
1442     if (ragBtn) ragBtn.addEventListener('click', runElRAGSearc
     h);
1443
1444     const ragInput = document.getElementById('el-rag-query-inp
     ut');
1445     if (ragInput) ragInput.addEventListener('keydown', (e) =>
     {
1446       if (e.key === 'Enter') runElRAGSearch();
1447     });
1448
1449     const bannerCloseBtn = document.getElementById('el-intel-c
     lose-btn');
1450     if (bannerCloseBtn) bannerCloseBtn.addEventListener('click
     ', () => {
1451       const banner = document.getElementById('el-intel-banner'
     );
1452       if (banner) banner.classList.add('d-none');
1453     });
1454
1455     const ragCloseBtn = document.getElementById('el-rag-close-
     btn');
1456     if (ragCloseBtn) ragCloseBtn.addEventListener('click', ()
     => {
1457       const card = document.getElementById('el-rag-results-car
     d');
1458       if (card) card.classList.add('d-none');
1459     });
1460
1461     initTableResizers();
1462   }
1463
1464   // ========================================================
     ================
1465   // ПАНЕЛЬ ЖИЗНЕННОГО ЦИКЛА ПИТАНИЯ (POWER LIFECYCLE & BOOT H
     ISTORY)
```

```
  // ============================================================
================

  let allPowerSessions = [];
  let allPowerEvents = [];
  let powerUptimeSeconds = 0;
  let powerUptimeInterval = null;
  let isPowerInitialized = false;

  function initPowerLifecyclePanel() {
    if (isPowerInitialized) return;
    isPowerInitialized = true;

    const refreshBtn = document.getElementById('pwr-refresh-bt
n');
    if (refreshBtn) {
      refreshBtn.addEventListener('click', async () => {
        refreshBtn.disabled = true;
        refreshBtn.innerHTML = '<span class="spinner-border sp
inner-border-sm me-1"></span>Обновление...';
        try {
          await loadPowerData();
          if (window.toast) window.toast.success('Обновлено',
'Данные сессий питания успешно обновлены');
        } finally {
          refreshBtn.disabled = false;
          refreshBtn.innerHTML = '<i class="bi bi-arrow-clockw
ise me-1"></i><span>Обновить данные</span>';
        }
      });
    }

    const rebuildBtn = document.getElementById('pwr-rebuild-bt
n');
    if (rebuildBtn) {
      rebuildBtn.addEventListener('click', async () => {
        rebuildBtn.disabled = true;
        rebuildBtn.innerHTML = '<span class="spinner-border sp
```

```
inner-border-sm me-1"></span>Сканирование...';
1498        try {
1499          const res = await fetch('/api/v1/power/rebuild', { m
ethod: 'POST' });
1500          if (!res.ok) throw new Error(`HTTP ${res.status}`);
1501          const data = await res.json();
1502          renderPowerSummary(data);
1503          await loadPowerData();
1504          if (window.toast) window.toast.success('Event Log пе
ресканирован', `Реконструировано сессий: ${data.total_sessions
_count}`);
1505        } catch (err) {
1506          console.error('[PowerLifecycle] Ошибка пересканирова
ния:', err);
1507          if (window.toast) window.toast.error('Ошибка', 'Не у
далось пересканировать журналы: ' + err.message);
1508        } finally {
1509          rebuildBtn.disabled = false;
1510          rebuildBtn.innerHTML = '<i class="bi bi-arrow-repeat
 me-1"></i><span>Пересканировать Event Log</span>';
1511        }
1512      });
1513    }
1514
1515    // Фильтры сессий
1516    const sessionsSearch = document.getElementById('pwr-sessio
ns-search');
1517    if (sessionsSearch) {
1518      sessionsSearch.addEventListener('input', () => filterAnd
RenderPowerSessions());
1519    }
1520
1521    const filterType = document.getElementById('pwr-filter-typ
e');
1522    if (filterType) {
1523      filterType.addEventListener('change', () => filterAndRen
derPowerSessions());
1524    }
```

```
1525
1526     const filterUnexpected = document.getElementById('pwr-filt
     er-unexpected-only');
1527     if (filterUnexpected) {
1528       filterUnexpected.addEventListener('change', () => filter
     AndRenderPowerSessions());
1529     }
1530
1531     // Фильтры событий
1532     const eventsSearch = document.getElementById('pwr-events-s
     earch');
1533     if (eventsSearch) {
1534       eventsSearch.addEventListener('input', () => filterAndRe
     nderPowerEvents());
1535     }
1536
1537     const filterEid = document.getElementById('pwr-event-id-fi
     lter');
1538     if (filterEid) {
1539       filterEid.addEventListener('change', () => filterAndRend
     erPowerEvents());
1540     }
1541
1542     startPowerUptimeCounter();
1543     initTableResizers();
1544   }
1545
1546   async function loadPowerData() {
1547     try {
1548       // 1. Сводка
1549       const sumRes = await fetch('/api/v1/power/summary');
1550       if (sumRes.ok) {
1551         const summary = await sumRes.json();
1552         renderPowerSummary(summary);
1553       }
1554
1555       // 2. Сессии
1556       const sessRes = await fetch('/api/v1/power/sessions?limi
```

```
t=100');
      if (sessRes.ok) {
        allPowerSessions = await sessRes.json();
        filterAndRenderPowerSessions();
      }

      // 3. События
      const evRes = await fetch('/api/v1/power/events?limit=200');
      if (evRes.ok) {
        allPowerEvents = await evRes.json();
        filterAndRenderPowerEvents();
      }
    } catch (err) {
      console.error('[PowerLifecycle] Ошибка загрузки данных:', err);
    }
  }

  function renderPowerSummary(summary) {
    if (!summary) return;

    const curBootEl = document.getElementById('pwr-current-boot');
    if (curBootEl) curBootEl.textContent = `Загрузка: ${summary.current_boot_time || '--'}`;

    const curUptimeEl = document.getElementById('pwr-current-uptime');
    if (curUptimeEl) {
      curUptimeEl.textContent = summary.current_uptime_human || '--';
      powerUptimeSeconds = summary.current_uptime_seconds || 0;
    }

    const totalSessionsEl = document.getElementById('pwr-total-sessions');
```

```
    if (totalSessionsEl) totalSessionsEl.textContent = summary.total_sessions_count;

    const totalBadge = document.getElementById('pwr-total-sessions-badge');
    if (totalBadge) totalBadge.textContent = `${summary.total_sessions_count} Сессий`;

    const cleanCountEl = document.getElementById('pwr-clean-count');
    if (cleanCountEl) cleanCountEl.textContent = summary.clean_shutdowns_count;

    const cleanRatioEl = document.getElementById('pwr-clean-ratio');
    if (cleanRatioEl) {
      const total = summary.total_sessions_count || 1;
      const ratio = Math.round((summary.clean_shutdowns_count / total) * 100);
      cleanRatioEl.textContent = `${ratio} %`;
    }

    const lastInitEl = document.getElementById('pwr-last-initiator');
    if (lastInitEl) {
      lastInitEl.textContent = `Инициатор: ${summary.last_initiator || 'Штатный'}`;
    }

    const unexpCountEl = document.getElementById('pwr-unexpected-count');
    if (unexpCountEl) unexpCountEl.textContent = summary.unexpected_shutdowns_count;

    const unexpBadge = document.getElementById('pwr-unexpected-badge');
    if (unexpBadge) {
      unexpBadge.textContent = `${summary.unexpected_shutdowns
```

```
_count} Сбоев`;
      if (summary.unexpected_shutdowns_count > 0) {
        unexpBadge.className = 'badge bg-danger font-monospace
';
      } else {
        unexpBadge.className = 'badge bg-success font-monospac
e';
      }
    }

    const bsodEl = document.getElementById('pwr-bsod-count');
    if (bsodEl) bsodEl.textContent = `BSOD (1001): ${summary.b
sod_count || 0}`;
  }

  function filterAndRenderPowerSessions() {
    const tbody = document.getElementById('pwr-sessions-tbody'
);
    const counter = document.getElementById('pwr-sessions-coun
ter');
    if (!tbody) return;

    const searchVal = (document.getElementById('pwr-sessions-s
earch')?.value || '').toLowerCase().trim();
    const typeFilter = document.getElementById('pwr-filter-typ
e')?.value || '';
    const unexpOnly = document.getElementById('pwr-filter-unex
pected-only')?.checked || false;

    const filtered = allPowerSessions.filter(s => {
      if (unexpOnly && !s.unexpected_shutdown) return false;
      if (typeFilter) {
        if (typeFilter === 'Active' && s.shutdown_type !== 'Ac
tive') return false;
        if (typeFilter === 'Restart' && s.shutdown_type !== 'R
estart') return false;
        if (typeFilter === 'Shutdown' && s.shutdown_type !== '
Shutdown' && s.shutdown_type !== 'PowerOff') return false;
```

```
        if (typeFilter === 'Unexpected' && !s.unexpected_shutdown) return false;
        if (typeFilter === 'BSOD' && s.shutdown_type !== 'BSOD' && !s.bugcheck) return false;
      }

      if (searchVal) {
        const init = (s.initiator || '').toLowerCase();
        const proc = (s.process || '').toLowerCase();
        const reason = (s.reason || '').toLowerCase();
        const code = (s.reason_code || '').toLowerCase();
        const sId = (s.session_id || '').toLowerCase();
        if (!init.includes(searchVal) && !proc.includes(searchVal) && !reason.includes(searchVal) && !code.includes(searchVal) && !sId.includes(searchVal)) {
          return false;
        }
      }
      return true;
    });

    if (counter) counter.textContent = `Найдено: ${filtered.length}`;

    if (filtered.length === 0) {
      tbody.innerHTML = `
        <tr>
          <td colspan="8" class="text-center py-4 text-muted">
            Сессии питания, удовлетворяющие критериям фильтрации, не найдены.
          </td>
        </tr>
      `;
      return;
    }

    tbody.innerHTML = filtered.map(s => {
      let statusBadge = '<span class="badge bg-secondary-subtl
```

```
e text-secondary border border-secondary-subtle">Неизвестно</s
pan>';
      if (s.shutdown_type === 'Active') {
        statusBadge = '<span class="badge bg-success-subtle te
xt-success border border-success-subtle">  Активная</span>';
      } else if (s.shutdown_type === 'Restart') {
        statusBadge = '<span class="badge bg-primary-subtle te
xt-primary border border-primary-subtle">  Перезагрузка</span>
';
      } else if (s.shutdown_type === 'Shutdown' || s.shutdown_
type === 'PowerOff') {
        statusBadge = '<span class="badge bg-secondary-subtle
text-secondary border border-secondary-subtle">  Выключение</s
pan>';
      } else if (s.shutdown_type === 'BSOD' || s.bugcheck) {
        statusBadge = `<span class="badge bg-danger text-white
">  BSOD ${escapeHtml(s.bugcheck || '')}</span>`;
      } else if (s.unexpected_shutdown) {
        statusBadge = '<span class="badge bg-danger-subtle tex
t-danger border border-danger-subtle">  Сбой / 41</span>';
      }

      let initiatorHtml = '<span class="text-muted">--</span>'
;
      if (s.initiator || s.process) {
        const initName = s.initiator ? `<span class="fw-semibo
ld text-primary">${escapeHtml(s.initiator)}</span>` : '';
        const procName = s.process ? `<span class="badge pwr-p
roc-badge font-monospace ms-1">${escapeHtml(s.process)}</span>
` : '';
        initiatorHtml = `<div class="d-flex align-items-center
 flex-wrap">${initName} ${procName}</div>`;
        if (s.initiator_chain && s.initiator_chain.length > 1)
 {
          initiatorHtml += `<div class="small text-muted font-
monospace mt-0.5" style="font-size: 0.72rem;">  ${escapeHtml(
s.initiator_chain.join(' → '))}</div>`;
        }
```

```
      }

      let reasonHtml = '<span class="text-muted">--</span>';
      if (s.reason || s.reason_code) {
        const codeBadge = s.reason_code ? `<span class="badge bg-secondary-subtle text-secondary border border-secondary-subtle font-monospace me-1">${escapeHtml(s.reason_code)}</span>` : '';
        reasonHtml = `<div>${codeBadge}<span>${escapeHtml(s.reason || '')}</span></div>`;
        if (s.comment) {
          reasonHtml += `<div class="small text-muted font-italic mt-0.5">  ${escapeHtml(s.comment)}</div>`;
        }
      }

      return `
        <tr class="pwr-session-row" data-session-id="${escapeHtml(s.session_id)}" style="cursor: pointer;" title="Нажмите для AI-анализа сессии питания">
          <td class="font-monospace text-primary small fw-bold">${escapeHtml(s.session_id)}</td>
          <td class="font-monospace small">${escapeHtml(s.boot_time)}</td>
          <td class="font-monospace small text-muted">${s.shutdown_time ? escapeHtml(s.shutdown_time) : '<span class="text-success fw-bold">Сейчас</span>'}</td>
          <td class="font-monospace fw-semibold">${escapeHtml(s.uptime_human || '--')}</td>
          <td>${statusBadge}</td>
          <td>${initiatorHtml}</td>
          <td>${reasonHtml}</td>
          <td class="text-center">
            <button class="btn btn-xs btn-outline-info p-1 px-2 pwr-detail-btn" data-session-id="${escapeHtml(s.session_id)}" title="Посмотреть подробную хронологию">
              <i class="bi bi-eye"></i>
            </button>
```

```
          </td>
        </tr>
      `;
    }).join('');

    tbody.querySelectorAll('.pwr-session-row').forEach(row =>
{
      row.addEventListener('click', (evt) => {
        const sId = row.dataset.sessionId;
        showPowerSessionDetailModal(sId);
      });
    });
  }

  function showPowerEventDetail(e) {
    if (!e) return;
    const modal = window.AIModalDialog || window.AITableModal;
    if (modal) {
      const isErr = e.event_id === 41 || e.event_id === 6008 |
| e.event_id === 1001;
      modal.show({
        icon: isErr ? '□ ' : '□ ',
        title: `Событие питания #${e.event_id} (${e.provider |
| 'Kernel-Power'})`,
        subtitle: `Тип: ${e.event_type || 'Lifecycle'} | ${e.t
imestamp || ''}`,
        panelId: 'panel-system-control-5',
        tableType: 'security_event',
        badges: [
          { text: `Event ${e.event_id}`, class: isErr ? 'badge
 bg-danger' : 'badge bg-info text-dark' },
          { text: e.event_type || 'Power', class: 'badge borde
r text-secondary' }
        ],
        metadata: [
          { label: 'Event ID', value: String(e.event_id) },
          { label: 'Поставщик', value: e.provider || 'Kernel-P
ower' },
```

```
          { label: 'Пользователь', value: e.user || 'SYSTEM' }
,
          { label: 'Процесс', value: e.process || '-' },
          { label: 'Время события', value: e.timestamp || '-'
},
          { label: 'Причина / Сообщение', value: e.reason || e
.details?.message || '-', fullWidth: true }
        ],
        rawTitle: 'Сырые XML данные события',
        rawContent: e.raw_xml || JSON.stringify(e, null, 2),
        requestData: {
          event_id: e.event_id,
          provider: e.provider,
          user: e.user,
          process: e.process,
          reason: e.reason
        }
      });
      return;
    }
  }

  function filterAndRenderPowerEvents() {
    const tbody = document.getElementById('pwr-events-tbody');
    const counter = document.getElementById('pwr-events-counte
r');
    if (!tbody) return;

    const searchVal = (document.getElementById('pwr-events-sea
rch')?.value || '').toLowerCase().trim();
    const eidFilter = document.getElementById('pwr-event-id-fi
lter')?.value || '';

    const filtered = allPowerEvents.filter(e => {
      if (eidFilter && String(e.event_id) !== String(eidFilter
)) return false;
      if (searchVal) {
        const prov = (e.provider || '').toLowerCase();
```

```
        const user = (e.user || '').toLowerCase();
        const proc = (e.process || '').toLowerCase();
        const reason = (e.reason || '').toLowerCase();
        const msg = (e.details?.message || '').toLowerCase();
        if (!prov.includes(searchVal) && !user.includes(search
Val) && !proc.includes(searchVal) && !reason.includes(searchVa
l) && !msg.includes(searchVal)) {
          return false;
        }
      }
      return true;
    });

    if (counter) counter.textContent = `Событий: ${filtered.le
ngth}`;

    if (filtered.length === 0) {
      tbody.innerHTML = `
        <tr>
          <td colspan="7" class="text-center py-4 text-muted">
            События журнала, удовлетворяющие критериям поиска,
 не найдены.
          </td>
        </tr>
      `;
      return;
    }

    tbody.innerHTML = filtered.map((e, idx) => {
      let eidBadge = `<span class="badge bg-secondary-subtle t
ext-secondary border border-secondary-subtle font-monospace">$
{e.event_id}</span>`;
      if (e.event_id === 1074) {
        eidBadge = '<span class="badge bg-warning-subtle text-
warning-emphasis border border-warning font-monospace fw-bold"
>1074 User32</span>';
      } else if (e.event_id === 41) {
        eidBadge = '<span class="badge bg-danger-subtle text-d
```

```
anger border border-danger font-monospace fw-bold">41 Kernel-P
ower</span>';
1806        } else if (e.event_id === 6008) {
1807          eidBadge = '<span class="badge bg-danger-subtle text-d
anger border border-danger font-monospace fw-bold">6008 EventL
og</span>';
1808        } else if (e.event_id === 12 || e.event_id === 6005 || e
.event_id === 6009) {
1809          eidBadge = `<span class="badge bg-success-subtle text-
success border border-success font-monospace">${e.event_id} Bo
ot</span>`;
1810        } else if (e.event_id === 13 || e.event_id === 6006) {
1811          eidBadge = `<span class="badge bg-info-subtle text-inf
o border border-info font-monospace">${e.event_id} Clean</span
>`;
1812        } else if (e.event_id === 1001) {
1813          eidBadge = '<span class="badge bg-danger text-white fo
nt-monospace fw-bold">1001 BSOD</span>';
1814        }
1815
1816        let userProc = '<span class="text-muted">--</span>';
1817        if (e.user || e.process) {
1818          userProc = `<div class="fw-semibold text-primary">${es
capeHtml(e.user || '')}</div>`;
1819          if (e.process) userProc += `<div class="font-monospace
 small text-muted">${escapeHtml(e.process)}</div>`;
1820        }
1821
1822        const desc = e.reason || e.details?.message || '--';
1823
1824        return `
1825          <tr class="pwr-event-row" data-idx="${idx}" style="cur
sor: pointer;" title="Нажмите для AI-диагностики инцидента пит
ания">
1826            <td class="font-monospace small">${escapeHtml(e.time
stamp)}</td>
1827            <td>${eidBadge}</td>
1828            <td class="small text-muted font-monospace">${escape
```

```
Html(e.provider)}</td>
          <td><span class="badge bg-secondary-subtle text-seco
ndary border border-secondary-subtle">${escapeHtml(e.event_typ
e)}</span></td>
          <td>${userProc}</td>
          <td class="small">${escapeHtml(desc)}</td>
          <td class="text-center">
            ${e.raw_xml ? `
              <button class="btn btn-xs btn-outline-secondary
p-1 px-2 pwr-xml-btn" data-xml="${escapeHtml(e.raw_xml)}" titl
e="Просмотреть XML">
                <i class="bi bi-code-square"></i>
              </button>
            ` : '<span class="text-muted">-</span>'}
          </td>
        </tr>
      `;
    }).join('');

    tbody.querySelectorAll('.pwr-event-row').forEach(row => {
      row.addEventListener('click', (evt) => {
        if (evt.target.closest('.pwr-xml-btn')) return;
        const idx = parseInt(row.dataset.idx, 10);
        const e = filtered[idx];
        if (e) showPowerEventDetail(e);
      });
    });

    tbody.querySelectorAll('.pwr-xml-btn').forEach(btn => {
      btn.addEventListener('click', (evt) => {
        evt.stopPropagation();
        const xml = btn.dataset.xml;
        showPowerXmlModal(xml);
      });
    });
  }

  function showPowerSessionDetailModal(sessionId) {
```

```
    const session = allPowerSessions.find(s => s.session_id ==
= sessionId);
    if (!session) return;

    const modal = window.AIModalDialog || window.AITableModal;
    if (modal) {
      const isUnexp = Boolean(session.unexpected_shutdown || s
ession.bugcheck);
      modal.show({
        icon: isUnexp ? '□ ' : '□ ',
        title: `Сессия питания: ${session.session_id}`,
        subtitle: `Статус: ${session.shutdown_type || 'Active'
} | Аптайм: ${session.uptime_human || '--'}`,
        panelId: 'panel-system-control-4',
        tableType: 'user_account',
        badges: [
          { text: session.shutdown_type || 'Active', class: is
Unexp ? 'badge bg-danger' : 'badge bg-success' },
          { text: session.uptime_human || 'Active', class: 'ba
dge border text-secondary' }
        ],
        metadata: [
          { label: 'ID сессии', value: session.session_id },
          { label: 'Время старта (Boot)', value: session.boot_
time || '-' },
          { label: 'Время завершения', value: session.shutdown
_time || 'Активна' },
          { label: 'Инициатор', value: session.initiator || 'С
истема' },
          { label: 'Процесс', value: session.process || 'N/A'
},
          { label: 'Причина выключения', value: session.reason
 || 'N/A', fullWidth: true }
        ],
        rawTitle: 'Контекст и хронология событий сессии',
        rawContent: JSON.stringify(session, null, 2),
        requestData: {
          session_id: session.session_id,
```

```
          initiator: session.initiator,
          process: session.process,
          reason: session.reason,
          shutdown_type: session.shutdown_type,
          unexpected: session.unexpected_shutdown
        }
      });
      return;
    }

    const modalTitle = document.getElementById('pwr-modal-titl
e');
    const modalBody = document.getElementById('pwr-modal-body'
);
    if (modalTitle) modalTitle.textContent = `Сессия питания:
${session.session_id}`;

    if (modalBody) {
      let eventsListHtml = '<div class="text-muted small">Нет
связанных событий</div>';
      if (session.events && session.events.length > 0) {
        eventsListHtml = session.events.map(ev => `
          <div class="pwr-detail-card p-2 mb-2">
            <div class="d-flex justify-content-between align-i
tems-center mb-1">
              <span class="badge bg-primary-subtle text-primar
y border border-primary-subtle font-monospace">Event ${ev.even
t_id}</span>
              <span class="font-monospace small text-muted">${
escapeHtml(ev.timestamp)}</span>
            </div>
            <div class="small fw-semibold text-primary">${esca
peHtml(ev.provider || '')}</div>
            <div class="small mt-1">${escapeHtml(ev.reason ||
ev.details?.message || '')}</div>
            ${ev.process ? `<div class="small text-muted font-
monospace mt-1">Процесс: ${escapeHtml(ev.process)} | Пользоват
ель: ${escapeHtml(ev.user || '--')}</div>` : ''}
```

```
          </div>
        `).join('');
      }

      modalBody.innerHTML = `
        <div class="row g-3 mb-3">
          <div class="col-6">
            <div class="pwr-detail-card p-2.5">
              <div class="text-muted small">🚀 Время старта (Boot):</div>
              <div class="font-monospace fw-bold text-info mt-1">${escapeHtml(session.boot_time)}</div>
            </div>
          </div>
          <div class="col-6">
            <div class="pwr-detail-card p-2.5">
              <div class="text-muted small">🏁 Завершение работы:</div>
              <div class="font-monospace fw-bold ${session.shutdown_time ? '' : 'text-success'} mt-1">
                ${session.shutdown_time ? escapeHtml(session.shutdown_time) : 'Сессия активна'}
              </div>
            </div>
          </div>
        </div>

        <div class="pwr-detail-card p-3 mb-3">
          <h6 class="fw-bold text-warning mb-2"><i class="bi bi-sliders me-1.5"></i>Параметры завершения</h6>
          <div class="row g-2 small">
            <div class="col-4 text-muted">Тип завершения:</div>
            <div class="col-8 fw-semibold">${escapeHtml(session.shutdown_type)}</div>

            <div class="col-4 text-muted">Аптайм сессии:</div>
            <div class="col-8 font-monospace fw-bold">${escape
```

```
Html(session.uptime_human || '--')}</div>

            <div class="col-4 text-muted">Инициатор:</div>
            <div class="col-8">${escapeHtml(session.initiator || 'SYSTEM / Hardware')}</div>

            <div class="col-4 text-muted">Процесс:</div>
            <div class="col-8 font-monospace text-info">${escapeHtml(session.process || '--')}</div>

            <div class="col-4 text-muted">Код причины:</div>
            <div class="col-8 font-monospace text-secondary">${escapeHtml(session.reason_code || '--')}</div>

            <div class="col-4 text-muted">Текст причины:</div>
            <div class="col-8">${escapeHtml(session.reason || '--')}</div>

            ${session.comment ? `
              <div class="col-4 text-muted">Комментарий:</div>
              <div class="col-8 font-italic">${escapeHtml(session.comment)}</div>
            ` : ''}

            ${session.bugcheck ? `
              <div class="col-4 text-danger fw-bold">BSOD BugCheck:</div>
              <div class="col-8 text-danger font-monospace fw-bold">${escapeHtml(session.bugcheck)}</div>
            ` : ''}
          </div>
        </div>

        ${session.initiator_chain && session.initiator_chain.length > 0 ? `
          <div class="pwr-detail-card p-3 mb-3">
            <h6 class="fw-bold text-info mb-2"><i class="bi bi-diagram-3 me-1.5"></i>Корреляция цепочки вызова</h6>
```

```
            <div class="d-flex align-items-center flex-wrap ga
p-1 font-monospace small">
              ${session.initiator_chain.map((step, idx) => `
                <span class="badge bg-secondary-subtle text-se
condary border border-secondary-subtle p-2">${escapeHtml(step)
}</span>
                ${idx < session.initiator_chain.length - 1 ? '
<i class="bi bi-arrow-right text-muted"></i>' : ''}
              `).join('')}
            </div>
          </div>
        ` : ''}

        <div class="pwr-detail-card p-3">
          <h6 class="fw-bold mb-2"><i class="bi bi-list-nested
 me-1.5"></i>Хронология событий в рамках сессии</h6>
          <div style="max-height: 250px; overflow-y: auto;">
            ${eventsListHtml}
          </div>
        </div>
      `;

      const modalEl = document.getElementById('pwrSessionDetai
lModal');
      if (modalEl && window.bootstrap) {
        const modal = new window.bootstrap.Modal(modalEl);
        modal.show();
      }
    }
  }

  function showPowerXmlModal(xmlString) {
    const pre = document.getElementById('pwr-xml-content');
    if (pre) pre.textContent = xmlString;

    const modalEl = document.getElementById('pwrXmlDetailModal
');
    if (modalEl && window.bootstrap) {
```

```
      const modal = new window.bootstrap.Modal(modalEl);
      modal.show();
    }
  }

  function startPowerUptimeCounter() {
    if (powerUptimeInterval) clearInterval(powerUptimeInterval
);
    powerUptimeInterval = setInterval(() => {
      powerUptimeSeconds += 1;
      const curUptimeEl = document.getElementById('pwr-current
-uptime');
      if (curUptimeEl && powerUptimeSeconds > 0) {
        const days = Math.floor(powerUptimeSeconds / 86400);
        const hours = Math.floor((powerUptimeSeconds % 86400)
/ 3600);
        const minutes = Math.floor((powerUptimeSeconds % 3600)
 / 60);
        const sec = Math.floor(powerUptimeSeconds % 60);
        const parts = [];
        if (days > 0) parts.push(`${days}д`);
        if (hours > 0 || days > 0) parts.push(`${hours}ч`);
        if (minutes > 0 || hours > 0 || days > 0) parts.push(`
${minutes}м`);
        parts.push(`${sec}с`);
        curUptimeEl.textContent = parts.join(' ');
      }
    }, 1000);
  }

  // ============================================================
================
  // ПАНЕЛЬ ЖУРНАЛОВ И БЕЗОПАСНОСТИ DEFENDER (WINDOWS DEFENDER
 SECURITY EVENTS)
  // ============================================================
================

  let allDefenderEvents = [];
```

```
2035  let allDefenderThreats = [];
2036  let defenderStatusData = null;
2037  let isDefenderInitialized = false;
2038
2039  const DEFENDER_CATEGORIES = {
2040    1000: { name: 'Сканирование начато', en: 'Scan Started', c
at: 'scans', icon: 'bi-search', badge: 'bg-info' },
2041    1001: { name: 'Сканирование завершено', en: 'Scan Complete
d', cat: 'scans', icon: 'bi-check-circle', badge: 'bg-success'
 },
2042    1002: { name: 'Сканирование отменено', en: 'Scan Cancelled
', cat: 'scans', icon: 'bi-slash-circle', badge: 'bg-warning t
ext-dark' },
2043    1005: { name: 'Сбой сканирования', en: 'Scan Failed', cat:
 'scans', icon: 'bi-x-circle', badge: 'bg-danger' },
2044    1006: { name: 'Обнаружена угроза (Вредонос)', en: 'Malware
 Detected', cat: 'threats', icon: 'bi-bug-fill', badge: 'bg-da
nger' },
2045    1007: { name: 'Угроза нейтрализована', en: 'Malware Remedi
ated', cat: 'threats', icon: 'bi-shield-check', badge: 'bg-suc
cess' },
2046    1008: { name: 'Сбой нейтрализации угрозы', en: 'Remediatio
n Failed', cat: 'threats', icon: 'bi-exclamation-octagon-fill'
, badge: 'bg-danger' },
2047    1015: { name: 'Подозрительное поведение', en: 'Suspicious
Behavior', cat: 'threats', icon: 'bi-eye-fill', badge: 'bg-war
ning text-dark' },
2048    1116: { name: 'Обнаружен вирус / вредонос', en: 'Malware D
etected', cat: 'threats', icon: 'bi-radioactive', badge: 'bg-d
anger' },
2049    1117: { name: 'Действие выполнено', en: 'Action Taken', ca
t: 'threats', icon: 'bi-shield-shaded', badge: 'bg-info' },
2050    1121: { name: 'Блокировка Network Protection', en: 'Networ
k Protection Block', cat: 'network', icon: 'bi-globe-americas'
, badge: 'bg-warning text-dark' },
2051    1122: { name: 'Аудит Network Protection', en: 'Network Pro
tection Audit', cat: 'network', icon: 'bi-globe', badge: 'bg-s
econdary' },
```

```
    1123: { name: 'Блокировка CFA (Ransomware)', en: 'Controlled Folder Access Block', cat: 'cfa', icon: 'bi-lock-fill', badge: 'bg-warning text-dark' },
    1124: { name: 'Аудит CFA', en: 'Controlled Folder Access Audit', cat: 'cfa', icon: 'bi-folder-check', badge: 'bg-secondary' },
    1125: { name: 'Блокировка ASR (Exploit Guard)', en: 'Attack Surface Reduction Block', cat: 'asr', icon: 'bi-lightning-fill', badge: 'bg-warning text-dark' },
    1126: { name: 'Аудит ASR', en: 'Attack Surface Reduction Audit', cat: 'asr', icon: 'bi-lightning', badge: 'bg-secondary' },
    2000: { name: 'Сигнатуры обновлены', en: 'Signature Updated', cat: 'signatures', icon: 'bi-cloud-check', badge: 'bg-success' },
    2001: { name: 'Сбой обновления сигнатур', en: 'Signature Update Failed', cat: 'signatures', icon: 'bi-cloud-slash', badge: 'bg-danger' },
    2002: { name: 'Обновление сигнатур отменено', en: 'Signature Update Cancelled', cat: 'signatures', icon: 'bi-cloud-minus', badge: 'bg-secondary' },
    2003: { name: 'Движок Defender обновлен', en: 'Engine Updated', cat: 'signatures', icon: 'bi-cpu-fill', badge: 'bg-info text-dark' },
    5000: { name: 'Защита в реальном времени ВКЛ', en: 'Real-time Protection Enabled', cat: 'protection', icon: 'bi-shield-fill-check', badge: 'bg-success' },
    5001: { name: 'Защита в реальном времени ВЫКЛ', en: 'Real-time Protection Disabled', cat: 'protection', icon: 'bi-shield-fill-x', badge: 'bg-danger' },
    5007: { name: 'Конфигурация Defender изменена', en: 'Configuration Changed', cat: 'protection', icon: 'bi-sliders', badge: 'bg-warning text-dark' },
  };

  function initDefenderLogsPanel() {
    if (isDefenderInitialized) return;
    isDefenderInitialized = true;
```

```
2068
2069    const refreshBtn = document.getElementById('def-refresh-bt
n');
2070    if (refreshBtn) {
2071      refreshBtn.addEventListener('click', async () => {
2072        refreshBtn.disabled = true;
2073        refreshBtn.innerHTML = '<span class="spinner-border sp
inner-border-sm me-1"></span>...';
2074        try {
2075          await loadDefenderData();
2076          if (window.toast) window.toast.success(t('systemCont
rol.btnRefresh', 'Обновлено'), 'События Defender успешно обнов
лены');
2077        } finally {
2078          refreshBtn.disabled = false;
2079          refreshBtn.innerHTML = `<i class="bi bi-arrow-clockw
ise me-1"></i><span>${t('systemControl.btnRefresh', 'Обновить'
)}</span>`;
2080        }
2081      });
2082    }
2083
2084    const auditBtn = document.getElementById('def-audit-btn');
2085    if (auditBtn) {
2086      auditBtn.addEventListener('click', async () => {
2087        auditBtn.disabled = true;
2088        auditBtn.innerHTML = '<span class="spinner-border spin
ner-border-sm me-1"></span>Аудит...';
2089        try {
2090          const res = await fetch('/api/v1/defender/refresh',
{ method: 'POST' });
2091          if (!res.ok) throw new Error(`HTTP ${res.status}`);
2092          await loadDefenderData();
2093          if (window.toast) window.toast.success('Аудит Defend
er', 'Снимок безопасности Defender успешно обновлен в БД');
2094        } catch (e) {
2095          console.warn('[Defender] Ошибка аудита:', e);
2096          if (window.toast) window.toast.error('Ошибка', 'Не у
```

```
далось провести аудит Defender: ' + e.message);
        } finally {
          auditBtn.disabled = false;
          auditBtn.innerHTML = `<i class="bi bi-shield-check m
e-1"></i><span>${t('systemControl.defBtnAudit', 'Аудит статуса
')}</span>`;
        }
      });
    }

    const scanBtn = document.getElementById('def-quick-scan-bt
n');
    if (scanBtn) {
      scanBtn.addEventListener('click', async () => {
        scanBtn.disabled = true;
        scanBtn.innerHTML = '<span class="spinner-border spinn
er-border-sm me-1"></span>Старт...';
        try {
          const res = await fetch('/api/v1/defender/scan', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ scan_type: 'quick' }),
          });
          if (!res.ok) throw new Error(`HTTP ${res.status}`);
          const task = await res.json();
          if (window.toast) window.toast.info('Сканирование De
fender', `Запущено быстрое сканирование (Task ID: ${task.task_
id || 'active'})`);
          setTimeout(loadDefenderData, 3000);
        } catch (e) {
          console.warn('[Defender] Ошибка сканирования:', e);
          if (window.toast) window.toast.error('Ошибка', 'Не у
далось запустить сканирование: ' + e.message);
        } finally {
          scanBtn.disabled = false;
          scanBtn.innerHTML = `<i class="bi bi-play-circle me-
1"></i><span>${t('systemControl.defBtnScan', 'Быстрое скан')}<
/span>`;
```

```
        }
      });
    }

    const updateSigBtn = document.getElementById('def-update-sig-btn');
    if (updateSigBtn) {
      updateSigBtn.addEventListener('click', async () => {
        updateSigBtn.disabled = true;
        updateSigBtn.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span>Обновление...';
        try {
          const res = await fetch('/api/v1/defender/update-signatures', { method: 'POST' });
          if (!res.ok) throw new Error(`HTTP ${res.status}`);
          const task = await res.json();
          if (window.toast) window.toast.info('Обновление сигнатур', `Запущено фоновое обновление баз (Task ID: ${task.task_id || 'active'})`);
          setTimeout(loadDefenderData, 3000);
        } catch (e) {
          console.warn('[Defender] Ошибка обновления баз:', e);
          if (window.toast) window.toast.error('Ошибка', 'Не удалось обновить базы: ' + e.message);
        } finally {
          updateSigBtn.disabled = false;
          updateSigBtn.innerHTML = `<i class="bi bi-cloud-arrow-down me-1"></i><span>${t('systemControl.defBtnUpdateSig', 'Обновить базы')}</span>`;
        }
      });
    }

    const searchInput = document.getElementById('def-events-search');
    if (searchInput) searchInput.addEventListener('input', () => filterAndRenderDefenderEvents());
```

```
2152
2153    const catSelect = document.getElementById('def-category-se
lect');
2154    if (catSelect) catSelect.addEventListener('change', () =>
filterAndRenderDefenderEvents());
2155
2156    const lvlSelect = document.getElementById('def-level-selec
t');
2157    if (lvlSelect) lvlSelect.addEventListener('change', () =>
filterAndRenderDefenderEvents());
2158
2159    initTableResizers();
2160  }
2161
2162  async function loadDefenderData() {
2163    try {
2164      // 1. Статус Defender
2165      const statusPromise = fetch('/api/v1/defender/status')
2166        .then(r => r.ok ? r.json() : null)
2167        .catch(e => { console.debug('[Defender] Ошибка статуса
:', e); return null; });
2168
2169      // 2. Угрозы
2170      const threatsPromise = fetch('/api/v1/defender/threats?l
imit=50')
2171        .then(r => r.ok ? r.json() : [])
2172        .catch(e => { console.debug('[Defender] Ошибка истории
 угроз:', e); return []; });
2173
2174      // 3. Журнал событий Defender
2175      const eventsPromise = fetch('/api/v1/defender/events?lim
it=150')
2176        .then(async r => {
2177          if (r.ok) {
2178            const data = await r.json();
2179            if (Array.isArray(data) && data.length > 0) return
 data;
2180          }
```

```
          // Fallback на WevtAPI через system_logs
          const fallbackRes = await fetch('/api/v1/system_logs/events?channel=Microsoft-Windows-Windows%20Defender/Operational&hours=720&limit=150');
          if (fallbackRes.ok) {
            const fbData = await fallbackRes.json();
            return fbData.events || fbData.entries || [];
          }
          return [];
        })
        .catch(e => { console.debug('[Defender] Ошибка событий:', e); return []; });

      const [status, threats, events] = await Promise.all([statusPromise, threatsPromise, eventsPromise]);
      defenderStatusData = status;
      allDefenderThreats = Array.isArray(threats) ? threats : [];
      allDefenderEvents = (Array.isArray(events) ? events : []).map(normalizeDefenderEvent);

      renderDefenderSummary(defenderStatusData, allDefenderThreats, allDefenderEvents);
      filterAndRenderDefenderEvents();
    } catch (err) {
      console.warn('[Defender] Ошибка загрузки данных Defender:', err);
    }
  }

  function normalizeDefenderEvent(e) {
    const eid = Number(e.event_id || e.Id || 0);
    const info = DEFENDER_CATEGORIES[eid] || {
      name: 'Событие безопасности',
      en: 'Security Event',
      cat: 'general',
      icon: 'bi-shield',
      badge: 'bg-secondary'
```

```
    };
    return {
      event_id: eid,
      timestamp: e.timestamp || e.TimeCreated || new Date().to
ISOString(),
      level: e.level || e.LevelDisplayName || (info.cat === 't
hreats' ? 'Warning' : 'Information'),
      category: e.category || (window.currentLang === 'en' ? i
nfo.en : info.name),
      category_key: info.cat,
      icon: info.icon,
      badge: info.badge,
      message: e.message || e.Message || e.raw_data || '',
      details: e.details || {}
    };
  }

  function renderDefenderSummary(status, threats, events) {
    const statusBadge = document.getElementById('def-status-ba
dge');
    const statusText = document.getElementById('def-status-tex
t');
    const cloudStatus = document.getElementById('def-cloud-sta
tus');
    const threatsBadge = document.getElementById('def-threats-
badge');
    const threatsCount = document.getElementById('def-threats-
count');
    const threatsDesc = document.getElementById('def-threats-d
esc');
    const blocksCount = document.getElementById('def-blocks-co
unt');
    const sigBadge = document.getElementById('def-sig-badge');
    const sigVersion = document.getElementById('def-sig-versio
n');
    const engineVersion = document.getElementById('def-engine-
version');

```

```
2237    if (status) {
2238      const isEnabled = status.is_enabled !== false && status.
real_time_protection !== false && status.antivirus_enabled !==
 false;
2239      if (statusBadge) {
2240        statusBadge.className = isEnabled ? 'badge bg-success
font-monospace' : 'badge bg-danger font-monospace';
2241        statusBadge.textContent = isEnabled ? 'ACTIVE' : 'DISA
BLED';
2242      }
2243      if (statusText) {
2244        statusText.textContent = isEnabled ? 'Включена (Real-T
ime)' : 'Отключена / Неактивна';
2245      }
2246      if (cloudStatus) {
2247        const cloudActive = status.cloud_protection !== false
&& status.behavior_monitor_enabled !== false;
2248        cloudStatus.textContent = `Cloud: ${cloudActive ? 'Вкл
ючена' : 'Отключена'}`;
2249      }
2250      if (sigVersion) {
2251        sigVersion.textContent = status.signatures_version ||
status.antivirus_signature_version || 'vLatest';
2252      }
2253      if (sigBadge && status.signatures_version) {
2254        sigBadge.textContent = status.signatures_version.split
('.').slice(0, 2).join('.') || 'vLatest';
2255      }
2256      if (engineVersion) {
2257        engineVersion.textContent = `Engine: ${status.engine_v
ersion || 'Defender Engine'}`;
2258      }
2259    }
2260
2261    const threatCountVal = (threats && threats.length) || (eve
nts ? events.filter(e => e.category_key === 'threats').length
: 0);
2262    if (threatsCount) threatsCount.textContent = threatCountVa
```

```
l;
    if (threatsBadge) {
      if (threatCountVal > 0) {
        threatsBadge.className = 'badge bg-danger font-monospa
ce';
        threatsBadge.textContent = `${threatCountVal} Угроз`;
      } else {
        threatsBadge.className = 'badge bg-success font-monosp
ace';
        threatsBadge.textContent = '0 Угроз';
      }
    }
    if (threatsDesc && threats && threats.length > 0) {
      threatsDesc.textContent = `Посл: ${threats[0].threat_nam
e || threats[0].name || 'Detected'}`;
    }

    const blocksTotal = events ? events.filter(e => e.category
_key === 'cfa' || e.category_key === 'asr' || e.category_key =
== 'network').length : 0;
    if (blocksCount) blocksCount.textContent = blocksTotal;
  }

  function filterAndRenderDefenderEvents() {
    const tbody = document.getElementById('def-events-tbody');
    const counter = document.getElementById('def-events-counte
r');
    const tableBadge = document.getElementById('def-table-badg
e');
    if (!tbody) return;

    const searchVal = (document.getElementById('def-events-sea
rch')?.value || '').toLowerCase().trim();
    const catFilter = document.getElementById('def-category-se
lect')?.value || '';
    const lvlFilter = (document.getElementById('def-level-sele
ct')?.value || '').toLowerCase().trim();

```

```
    const filtered = allDefenderEvents.filter(e => {
      if (catFilter && e.category_key !== catFilter) return fa
lse;
      if (lvlFilter) {
        const lvl = (e.level || '').toLowerCase();
        if (!lvl.includes(lvlFilter)) return false;
      }
      if (searchVal) {
        const matchMsg = (e.message || '').toLowerCase().inclu
des(searchVal);
        const matchId = String(e.event_id || '').includes(sear
chVal);
        const matchCat = (e.category || '').toLowerCase().incl
udes(searchVal);
        if (!matchMsg && !matchId && !matchCat) return false;
      }
      return true;
    });

    if (counter) counter.textContent = `${filtered.length} соб
ытий`;
    if (tableBadge) tableBadge.textContent = `${filtered.lengt
h} событий`;

    if (filtered.length === 0) {
      tbody.innerHTML = `<tr><td colspan="6" class="text-cente
r py-4 text-muted">${t('systemControl.defNoEvents', 'Событий W
indows Defender не найдено.')}</td></tr>`;
      return;
    }

    tbody.innerHTML = filtered.map((e, idx) => {
      const timeStr = formatTime(e.timestamp);
      const isWarnOrErr = (e.level || '').toLowerCase().includ
es('warn') || (e.level || '').toLowerCase().includes('err') ||
 (e.level || '').toLowerCase().includes('crit') || e.category_
key === 'threats';
      return `
```

```
        <tr data-index="${idx}" class="${isWarnOrErr ? 'table-danger-subtle' : ''}">
          <td class="font-monospace small" style="color: var(--text-muted);">${escapeHtml(timeStr)}</td>
          <td><span class="badge ${getSeverityClass(e.level)} text-uppercase" style="font-size: 0.68rem;">${escapeHtml(e.level || 'Info')}</span></td>
          <td class="font-monospace small fw-bold" style="color: #0284c7;">${e.event_id || '-'}</td>
          <td>
            <span class="badge ${e.badge || 'bg-secondary'} d-inline-flex align-items-center gap-1 font-monospace" style="font-size: 0.72rem;">
              <i class="bi ${e.icon || 'bi-shield'}"></i>
              <span>${escapeHtml(e.category)}</span>
            </span>
          </td>
          <td class="text-truncate" style="max-width: 480px; color: var(--text-color);" title="${escapeHtml(e.message)}">
            ${escapeHtml(e.message)}
          </td>
          <td class="text-center">
            <button class="btn btn-xs btn-outline-info rounded-pill px-2 py-0.5 def-detail-btn" data-index="${idx}" title="Посмотреть детали события">
              <i class="bi bi-info-circle"></i>
            </button>
          </td>
        </tr>
      `;
    }).join('');

    tbody.querySelectorAll('tr').forEach(row => {
      row.onclick = () => {
        tbody.querySelectorAll('tr').forEach(r => r.classList.remove('table-active'));
        row.classList.add('table-active');
        const idx = parseInt(row.getAttribute('data-index'), 1
```

```
0);
2344        const ev = filtered[idx];
2345        if (ev) showDefenderEventDetail(ev);
2346      };
2347    });
2348  }
2349
2350  function showDefenderEventDetail(ev) {
2351    const modal = window.AIModalDialog || window.AITableModal;
2352    if (modal && ev) {
2353      const isThreat = ev.category_key === 'threats' || (ev.le
vel || '').toLowerCase().includes('crit') || (ev.level || '').
toLowerCase().includes('err');
2354      modal.show({
2355        icon: isThreat ? '□ ' : '□ ',
2356        title: `Defender ID ${ev.event_id}: ${ev.category || '
Безопасность'}`,
2357        subtitle: `Уровень: ${ev.level || 'Info'} | ${ev.times
tamp || ''}`,
2358        panelId: 'panel-system-control-defender-filter',
2359        tableType: 'security_event',
2360        badges: [
2361          { text: ev.level || 'Info', class: `badge ${getSever
ityClass(ev.level)}` },
2362          { text: ev.category || 'Defender', class: 'badge bg-
secondary' }
2363        ],
2364        metadata: [
2365          { label: 'Event ID', value: String(ev.event_id) },
2366          { label: 'Категория', value: ev.category || 'Защита'
 },
2367          { label: 'Уровень', value: ev.level || 'Info' },
2368          { label: 'Время регистрации', value: ev.timestamp ||
 '-' },
2369          { label: 'Журнал', value: 'Microsoft-Windows-Windows
 Defender/Operational', isCode: true, fullWidth: true }
2370        ],
2371        rawTitle: 'Подробности и параметры события Defender',
```

```
        rawContent: ev.message || '',
        requestData: {
          event_id: ev.event_id,
          category: ev.category,
          category_key: ev.category_key,
          level: ev.level,
          description: ev.message
        }
      });
      return;
    }

    const modalEl = document.getElementById('defEventDetailModal');
    const bodyEl = document.getElementById('def-modal-body');
    const titleEl = document.getElementById('def-modal-title');
    if (!modalEl || !bodyEl || !ev) return;

    if (titleEl) {
      titleEl.innerHTML = `<i class="bi ${ev.icon || 'bi-shield-check'} text-info me-2"></i><span>Событие Defender ID ${ev.event_id}: ${escapeHtml(ev.category)}</span>`;
    }

    bodyEl.innerHTML = `
      <div class="row g-2 mb-3 small font-monospace">
        <div class="col-sm-6"><strong class="text-muted">Время:</strong> <span style="color: var(--text-color);">${escapeHtml(ev.timestamp)}</span></div>
        <div class="col-sm-6"><strong class="text-muted">Уровень:</strong> <span class="badge ${getSeverityClass(ev.level)}">${escapeHtml(ev.level || 'Info')}</span></div>
        <div class="col-sm-6"><strong class="text-muted">Event ID:</strong> <span class="fw-bold text-info">${ev.event_id}</span></div>
        <div class="col-sm-6"><strong class="text-muted">Категория:</strong> <span class="badge ${ev.badge || 'bg-secondary'
```

```
}">${escapeHtml(ev.category)}</span></div>
        <div class="col-sm-12"><strong class="text-muted">Канал:</strong> <span class="text-warning">Microsoft-Windows-Windows Defender/Operational</span></div>
      </div>
      <div class="p-3 mb-3 rounded border" style="background: var(--surface-2); border-color: var(--border-color) !important;">
        <h6 class="small text-uppercase text-muted fw-bold mb-2">Описание события</h6>
        <div class="small font-monospace" style="white-space: pre-wrap; word-break: break-word; color: var(--text-color);">${escapeHtml(ev.message || 'Нет описания')}</div>
      </div>
      <div class="p-3 rounded border" style="background: var(--surface-2); border-color: var(--border-color) !important;">
        <h6 class="fw-bold text-primary mb-1"><i class="bi bi-shield-lock me-1"></i> Рекомендация безопасности</h6>
        <div class="small text-muted">
          ${ev.category_key === 'threats' ? '  Обнаружена угроза! Рекомендуется выполнить проверку карантина или запустить полное сканирование.' : (ev.category_key === 'cfa' ? '  Блокировка доступа к защищенным папкам. Проверьте доверенные приложения в Controlled Folder Access.' : (ev.category_key === 'signatures' ? '  Статус баз антивирусных сигнатур Defender. Обновления применяются автоматически.' : '  Стандартное событие мониторинга и безопасности Microsoft Defender.'))}
        </div>
      </div>
    `;

    if (window.bootstrap && window.bootstrap.Modal) {
      const modal = window.bootstrap.Modal.getOrCreateInstance(modalEl);
      modal.show();
    }
  }

```

```
2419   /**
2420    * Инициализация ползунков изменения высоты таблиц журнала и
 событий.
2421    */
2422   function initTableResizers() {
2423     function setupResizer(resizerId, containerId, defaultMaxHe
ight = 520, minH = 140, maxH = 1500) {
2424       const resizer = document.getElementById(resizerId);
2425       const container = document.getElementById(containerId);
2426       if (!resizer || !container || resizer._resizerBound) ret
urn;
2427       resizer._resizerBound = true;
2428 
2429       let startY = 0, startHeight = 0;
2430       const onMouseMove = (e) => {
2431         const delta = e.clientY - startY;
2432         const newH = Math.max(minH, Math.min(maxH, startHeight
 + delta));
2433         container.style.maxHeight = 'none';
2434         container.style.height = `${newH}px`;
2435       };
2436       const onMouseUp = () => {
2437         resizer.classList.remove('resizing');
2438         document.removeEventListener('mousemove', onMouseMove)
;
2439         document.removeEventListener('mouseup', onMouseUp);
2440       };
2441       resizer.onmousedown = (e) => {
2442         e.preventDefault();
2443         startY = e.clientY;
2444         startHeight = container.getBoundingClientRect().height
 || parseInt(window.getComputedStyle(container).height, 10) ||
 defaultMaxHeight;
2445         resizer.classList.add('resizing');
2446         document.addEventListener('mousemove', onMouseMove);
2447         document.addEventListener('mouseup', onMouseUp);
2448       };
2449       resizer.ondblclick = () => {
```

```
        container.style.height = '';
        container.style.maxHeight = `${defaultMaxHeight}px`;
      };
    }

    setupResizer('el-table-resizer', 'el-table-container', 520, 140, 1500);
    setupResizer('slc-table-resizer', 'slc-table-container', 560, 140, 1500);
    setupResizer('scc-activity-table-resizer', 'scc-activity-table-container', 480, 140, 1500);
    setupResizer('pwr-sessions-table-resizer', 'pwr-sessions-table-container', 520, 140, 1500);
    setupResizer('pwr-events-table-resizer', 'pwr-events-table-container', 520, 140, 1500);
    setupResizer('def-events-table-resizer', 'def-events-table-container', 520, 140, 1500);
  }

  function setMode(mode) {
    currentMode = mode;
    const tableContainer = document.getElementById('slc-table-container');
    const auditContainer = document.getElementById('slc-audit-container');
    const ragContainer = document.getElementById('slc-rag-container');
    const incidentsContainer = document.getElementById('slc-incidents-container');
    const timelineContainer = document.getElementById('slc-timeline-container');
    const rawContainer = document.getElementById('slc-raw-container');
    const activityContainer = document.getElementById('slc-activity-container');
    const eventLogsContainer = document.getElementById('slc-event-logs-container');
    const powerLifecycleContainer = document.getElementById('s
```

```
lc-power-lifecycle-container');
2474    const defenderLogsContainer = document.getElementById('slc
-defender-logs-container');
2475    const filterRow = document.getElementById('slc-filter-row'
);
2476    const sidebar = document.getElementById('slc-channel-tree-
sidebar');
2477    const mainStreamCol = document.getElementById('slc-main-st
ream-column');
2478    const mainStreamHeader = document.querySelector('#slc-main
-stream-column .slc-panel-header');
2479
2480    if (tableContainer) tableContainer.classList.add('d-none')
;
2481    if (auditContainer) auditContainer.classList.add('d-none')
;
2482    if (ragContainer) ragContainer.classList.add('d-none');
2483    if (incidentsContainer) incidentsContainer.classList.add('
d-none');
2484    if (timelineContainer) timelineContainer.classList.add('d-
none');
2485    if (rawContainer) rawContainer.classList.add('d-none');
2486    if (activityContainer) activityContainer.classList.add('d-
none');
2487    if (eventLogsContainer) eventLogsContainer.classList.add('
d-none');
2488    if (powerLifecycleContainer) powerLifecycleContainer.class
List.add('d-none');
2489    if (defenderLogsContainer) defenderLogsContainer.classList
.add('d-none');
2490    if (filterRow) filterRow.classList.remove('d-none');
2491
2492    // Управление шириной колонок и сайдбаром каналов
2493    if (mode === 'event-logs' || mode === 'power-lifecycle' ||
 mode === 'defender-logs') {
2494      if (sidebar) sidebar.classList.add('d-none');
2495      if (mainStreamCol) {
2496        mainStreamCol.classList.remove('col-lg-9');
```

```
        mainStreamCol.classList.add('col-12');
      }
      if (mainStreamHeader) mainStreamHeader.classList.add('d-none');
    } else {
      if (sidebar) sidebar.classList.remove('d-none');
      if (mainStreamCol) {
        mainStreamCol.classList.remove('col-12');
        mainStreamCol.classList.add('col-lg-9');
      }
      if (mainStreamHeader) mainStreamHeader.classList.remove('d-none');
    }

    document.querySelectorAll('#slc-mode-tabs button').forEach(btn => {
      btn.classList.toggle('active', btn.getAttribute('data-mode') === mode);
    });

    if (mode === 'live' || mode === 'errors') {
      if (tableContainer) tableContainer.classList.remove('d-none');
      loadEvents();
    } else if (mode === 'audit') {
      if (auditContainer) auditContainer.classList.remove('d-none');
      loadAuditDigest();
    } else if (mode === 'rag') {
      if (ragContainer) ragContainer.classList.remove('d-none');
      if (filterRow) filterRow.classList.add('d-none');
      fetchRagStatus();
    } else if (mode === 'incidents') {
      if (incidentsContainer) incidentsContainer.classList.remove('d-none');
      loadIncidents();
    } else if (mode === 'timeline') {
```

```
      if (timelineContainer) timelineContainer.classList.remov
e('d-none');
      loadTimeline();
    } else if (mode === 'raw') {
      if (rawContainer) rawContainer.classList.remove('d-none'
);
      if (tableContainer) tableContainer.classList.remove('d-n
one');
    } else if (mode === 'activity') {
      if (activityContainer) activityContainer.classList.remov
e('d-none');
      if (filterRow) filterRow.classList.add('d-none');
      loadSccActivityLogs();
    } else if (mode === 'event-logs') {
      if (eventLogsContainer) eventLogsContainer.classList.rem
ove('d-none');
      if (filterRow) filterRow.classList.add('d-none');
      initEventLogsPanel();
      fetchEventLogsSummary();
    } else if (mode === 'power-lifecycle') {
      if (powerLifecycleContainer) powerLifecycleContainer.cla
ssList.remove('d-none');
      if (filterRow) filterRow.classList.add('d-none');
      initPowerLifecyclePanel();
      loadPowerData();
    } else if (mode === 'defender-logs') {
      if (defenderLogsContainer) defenderLogsContainer.classLi
st.remove('d-none');
      if (filterRow) filterRow.classList.add('d-none');
      initDefenderLogsPanel();
      loadDefenderData();
    }
  }

  /**
   * Экспорт текущих событий.
   */
  function exportEvents(format = 'json') {
```

```
    if (!cachedEvents || cachedEvents.length === 0) {
      if (window.toast) {
        window.toast.warning(
          t('systemControl.btnExport', 'Экспорт'),
          t('systemControl.exportNoEvents', 'Нет загруженных событий для экспорта.')
        );
      }
      return;
    }
    let content = '';
    let mime = 'application/json';
    let ext = 'json';

    if (format === 'json') {
      content = JSON.stringify(cachedEvents, null, 2);
    } else {
      mime = 'text/csv';
      ext = 'csv';
      const headers = ['timestamp', 'level', 'event_id', 'provider', 'process_id', 'message'];
      const rows = cachedEvents.map(e => headers.map(h => `"${String(e[h] || '').replace(/"/g, '""')}"`).join(','));
      content = [headers.join(','), ...rows].join('\n');
    }

    const cleanChan = currentChannel.replace(/^evt:/i, '').replace(/[^a-zA-Z0-9_-]/g, '_');
    const blob = new Blob([content], { type: mime });
    const url = URL.createObjectURL(blob);
    const a = document.createElement('a');
    a.href = url;
    a.download = `system_logs_${cleanChan}_${Date.now()}.${ext}`;
    a.click();
    URL.revokeObjectURL(url);
  }

```

```
  /**
   * Инициализация вкладки System Logs & Activity Center.
   */
  function initSystemControlTab() {
    window.applyTranslations?.();
    if (window.isTabActive ? window.isTabActive('tab-system-control') : true) {
      fetchStatus();
      scanChannels();
      loadEvents();
    }

    if (window.registerTabPoller) {
      window.registerTabPoller('tab-system-control', loadEvents, 4000, { pollerId: 'tab-system-control_logs', immediate: true });
    }

    // Слушатель глобального изменения языка в приложении
    window.addEventListener('languageChanged', () => {
      window.applyTranslations?.();
      fetchStatus();
      if (cachedChannels.length > 0) {
        renderChannelTree(cachedChannels);
      }
      if (cachedEvents.length > 0) {
        renderEventsTable(cachedEvents);
      }
      if (currentMode === 'audit') loadAuditDigest();
      else if (currentMode === 'incidents') loadIncidents();
      else if (currentMode === 'timeline') loadTimeline();
      else if (currentMode === 'activity') loadSccActivityLogs();
      else if (currentMode === 'event-logs') fetchEventLogsSummary();
      else if (currentMode === 'power-lifecycle') loadPowerData();
      else if (currentMode === 'defender-logs') loadDefenderDa
```

```
ta();
2623    });
2624
2625    initEventLogsPanel();
2626    initTableResizers();
2627
2628    if (isSCCInitialized) return;
2629    isSCCInitialized = true;
2630
2631    // Главная кнопка обновления шапки
2632    const refreshBtn = document.getElementById('btn-scc-refres
    h');
2633    if (refreshBtn) {
2634      refreshBtn.onclick = () => {
2635        fetchStatus();
2636        if (currentMode === 'event-logs') {
2637          fetchEventLogsSummary();
2638        } else if (currentMode === 'power-lifecycle') {
2639          loadPowerData();
2640        } else if (currentMode === 'defender-logs') {
2641          loadDefenderData();
2642        } else {
2643          loadEvents();
2644        }
2645      };
2646    }
2647
2648    // Кнопка обновления потока логов
2649    const slcRefreshBtn = document.getElementById('btn-slc-ref
    resh');
2650    if (slcRefreshBtn) {
2651      slcRefreshBtn.onclick = () => {
2652        if (currentMode === 'event-logs') {
2653          fetchEventLogsSummary();
2654        } else if (currentMode === 'power-lifecycle') {
2655          loadPowerData();
2656        } else if (currentMode === 'defender-logs') {
2657          loadDefenderData();
```

```
2658        } else {
2659          loadEvents();
2660        }
2661      };
2662    }
2663
2664    // Кнопка сканирования источников
2665    const slcScanBtn = document.getElementById('btn-slc-scan')
;
2666    if (slcScanBtn) {
2667      slcScanBtn.onclick = () => {
2668        scanChannels(true);
2669      };
2670    }
2671
2672    // Кнопка быстрого открытия встроенной панели журналов соб
ытий
2673    const openEventLogsBtn = document.getElementById('btn-scc-
open-event-logs');
2674    if (openEventLogsBtn) {
2675      openEventLogsBtn.onclick = () => {
2676        setMode('event-logs');
2677      };
2678    }
2679
2680    // Кнопка быстрого открытия встроенной панели жизненного ц
икла питания
2681    const openPowerLifecycleBtn = document.getElementById('btn
-scc-open-power-lifecycle');
2682    if (openPowerLifecycleBtn) {
2683      openPowerLifecycleBtn.onclick = () => {
2684        setMode('power-lifecycle');
2685      };
2686    }
2687
2688    // Кнопка быстрого открытия встроенной панели журналов Def
ender
2689    const openDefenderLogsBtn = document.getElementById('btn-s
```

```
cc-open-defender-logs');
2690    if (openDefenderLogsBtn) {
2691      openDefenderLogsBtn.onclick = () => {
2692        setMode('defender-logs');
2693      };
2694    }
2695
2696    // Кнопка аудита
2697    const auditBtn = document.getElementById('btn-slc-audit');
2698    if (auditBtn) {
2699      auditBtn.onclick = () => {
2700        setMode('audit');
2701      };
2702    }
2703
2704    // Авто-активация режима по хэшу/параметру
2705    if (location.hash.includes('power-lifecycle') || location.
search.includes('power-lifecycle') || location.search.includes
('panel=power-lifecycle')) {
2706      setMode('power-lifecycle');
2707    } else if (location.hash.includes('defender-logs') || loca
tion.search.includes('defender-logs') || location.search.inclu
des('panel=defender-logs')) {
2708      setMode('defender-logs');
2709    } else if (location.hash.includes('event-logs') || locatio
n.search.includes('event-logs') || location.search.includes('p
anel=event-logs')) {
2710      setMode('event-logs');
2711    }
2712
2713    // Фильтр каналов в дереве
2714    const channelFilterInput = document.getElementById('slc-ch
annel-filter-input');
2715    if (channelFilterInput) {
2716      channelFilterInput.oninput = (e) => {
2717        const query = e.target.value.toLowerCase();
2718        const filtered = cachedChannels.filter(c => {
2719          const name = (c.channel_name || c.display_name || c.
```

```
name || '').toLowerCase();
          const desc = (c.description || '').toLowerCase();
          return name.includes(query) || desc.includes(query);
        });
        renderChannelTree(filtered);
      };
    }

    // Переключение режимов работы (Live, Audit, RAG, Incidents, Timeline, Raw, Activity, Event Logs, Power Lifecycle)
    document.querySelectorAll('#slc-mode-tabs button').forEach(btn => {
      btn.onclick = () => {
        const mode = btn.getAttribute('data-mode') || 'live';
        setMode(mode);
      };
    });

    // Фильтры потока
    const searchInput = document.getElementById('slc-search-input');
    if (searchInput) {
      let debounceTimer = null;
      searchInput.oninput = () => {
        clearTimeout(debounceTimer);
        debounceTimer = setTimeout(loadEvents, 300);
      };
    }

    const hoursSelect = document.getElementById('slc-hours-select');
    if (hoursSelect) {
      hoursSelect.onchange = () => {
        const ragPeriodSelect = document.getElementById('slc-rag-period-select');
        if (ragPeriodSelect) {
          const opt = ragPeriodSelect.querySelector(`option[value="${hoursSelect.value}"]`);
```

```
          if (opt) {
            ragPeriodSelect.value = hoursSelect.value;
            updateRagPeriodDisplay(hoursSelect.value);
          }
        }
        loadEvents();
      };
    }

    // Селектор периода RAG (сутки / неделя / месяц / все время)
    const ragPeriodSelect = document.getElementById('slc-rag-period-select');
    if (ragPeriodSelect) {
      const savedPeriod = BrowserRagCache.get('selected_period');
      if (savedPeriod !== null && savedPeriod !== undefined) {
        ragPeriodSelect.value = String(savedPeriod);
      }
      updateRagPeriodDisplay(ragPeriodSelect.value);
      ragPeriodSelect.onchange = () => {
        updateRagPeriodDisplay(ragPeriodSelect.value);
        fetchRagStatus();
      };
    }

    const limitSelect = document.getElementById('slc-limit-select');
    if (limitSelect) limitSelect.onchange = loadEvents;

    const eventIdInput = document.getElementById('slc-eventid-input');
    if (eventIdInput) {
      let debounceTimer = null;
      eventIdInput.oninput = () => {
        clearTimeout(debounceTimer);
        debounceTimer = setTimeout(loadEvents, 300);
      };
```

```
2784      }
2785
2786      // Live Auto-refresh switch
2787      const liveSwitch = document.getElementById('slc-live-switc
h');
2788      if (liveSwitch) {
2789        liveSwitch.onchange = (e) => {
2790          if (e.target.checked) {
2791            if (window.registerTabPoller) {
2792              window.registerTabPoller('tab-system-control', loa
dEvents, 3000, { pollerId: 'tab-system-control_live', immediat
e: true });
2793            } else {
2794              if (liveIntervalTimer) clearInterval(liveIntervalT
imer);
2795              liveIntervalTimer = setInterval(loadEvents, 3000);
2796            }
2797          } else {
2798            if (window.unregisterTabPoller) {
2799              window.unregisterTabPoller('tab-system-control_liv
e');
2800            }
2801            if (liveIntervalTimer) {
2802              clearInterval(liveIntervalTimer);
2803              liveIntervalTimer = null;
2804            }
2805          }
2806        };
2807      }
2808
2809      // Экспорт событий
2810      const exportJson = document.getElementById('btn-slc-export
-json');
2811      if (exportJson) {
2812        exportJson.onclick = (e) => {
2813          e.preventDefault();
2814          exportEvents('json');
2815        };
```

```
2816      }
2817
2818      const exportCsv = document.getElementById('btn-slc-export-
csv');
2819      if (exportCsv) {
2820        exportCsv.onclick = (e) => {
2821          e.preventDefault();
2822          exportEvents('csv');
2823        };
2824      }
2825
2826      // RAG Search actions
2827      const ragSearchBtn = document.getElementById('btn-slc-rag-
search');
2828      if (ragSearchBtn) ragSearchBtn.onclick = runRagSearch;
2829
2830      const ragInput = document.getElementById('slc-rag-input');
2831      if (ragInput) {
2832        ragInput.onkeydown = (e) => {
2833          if (e.key === 'Enter') runRagSearch();
2834        };
2835      }
2836
2837      const ragRebuildBtn = document.getElementById('btn-slc-rag
-rebuild');
2838      if (ragRebuildBtn) ragRebuildBtn.onclick = rebuildRagIndex
;
2839
2840      const ragAggregateSwitch = document.getElementById('slc-ra
g-aggregate-switch');
2841      if (ragAggregateSwitch) {
2842        ragAggregateSwitch.onchange = () => {
2843          const rInput = document.getElementById('slc-rag-input'
);
2844          if (rInput && rInput.value.trim()) {
2845            runRagSearch();
2846          }
2847        };
```

```
2848     }
2849
2850     // Быстрые подсказки запросов RAG
2851     document.querySelectorAll('.slc-rag-tag').forEach(tagBtn =
> {
2852       tagBtn.onclick = () => {
2853         const q = tagBtn.getAttribute('data-query');
2854         const rInput = document.getElementById('slc-rag-input'
);
2855         if (rInput && q) {
2856           rInput.value = q;
2857           runRagSearch();
2858         }
2859       };
2860     });
2861
2862     // AI Ask Action
2863     const askAiBtn = document.getElementById('btn-slc-ask-ai')
;
2864     if (askAiBtn) {
2865       askAiBtn.onclick = () => {
2866         const topEvents = cachedEvents.slice(0, 10).map(e => `
[${e.level}] ${e.provider}: ${e.message}`).join('\n');
2867         const cleanChan = currentChannel.replace(/^evt:/i, '')
;
2868         const prompt = `${t('systemControl.headerTitle', 'Wind
ows System Logs')} (${cleanChan}):\n${topEvents}`;
2869         if (window.sendChatMessage) {
2870           window.sendChatMessage(prompt);
2871         } else if (window.toast) {
2872           window.toast.info(
2873             t('systemControl.aiToastTitle', 'ИИ Анализ'),
2874             t('systemControl.aiToastBody', 'Запрос направлен в
 чат-ассистент.')
2875           );
2876         }
2877       };
2878     }
```

```
2879
2880     // Application activity refresh
2881     const refreshActivityBtn = document.getElementById('btn-sc
c-refresh-logs');
2882     if (refreshActivityBtn) refreshActivityBtn.onclick = loadS
ccActivityLogs;
2883   }
2884
2885   // Экспорт для динамического загрузчика
2886   window.initSystemControlTab = initSystemControlTab;
2887
2888   // Автостарт при готовности DOM
2889   if (document.readyState === 'loading') {
2890     document.addEventListener('DOMContentLoaded', initSystemCo
ntrolTab);
2891   } else {
2892     initSystemControlTab();
2893   }
2894 })();
2895
```

system_inspector.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - System Inspector View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса system_inspector.

Usage Examples:
    Web Route:
      /windows/api/webgui/system_inspector.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/system_inspector.html"
class="w-100 h-100 border-0"></iframe>

File: system_inspector.html
Project: ai-windows-admin
Package: windows/api/webgui
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial
-scale=1.0">
    <title>AI-Windows-Admin | System & Hardware Inspector</tit
le>
    <style>
```

```
        :root {
            --bg-base: #0f172a;
            --bg-card: #1e293b;
            --bg-hover: #334155;
            --text-main: #f8fafc;
            --text-muted: #94a3b8;
            --accent-cyan: #06b6d4;
            --accent-green: #10b981;
            --accent-yellow: #f59e0b;
            --accent-red: #ef4444;
            --accent-purple: #a855f7;
            --border-color: #334155;
        }

        * {
            box-sizing: border-box;
            margin: 0;
            padding: 0;
            font-family: -apple-system, BlinkMacSystemFont, "S
egoe UI", Roboto, "Helvetica Neue", Arial, sans-serif;
        }

        body {
            background-color: var(--bg-base);
            color: var(--text-main);
            height: 100vh;
            display: flex;
            flex-direction: column;
            overflow: hidden;
        }

        header {
            background-color: var(--bg-card);
            border-bottom: 1px solid var(--border-color);
            padding: 12px 24px;
            display: flex;
            align-items: center;
            justify-content: space-between;
```

```
        }

        .header-title {
            display: flex;
            align-items: center;
            gap: 12px;
            font-size: 1.15rem;
            font-weight: 700;
            color: var(--accent-cyan);
        }

        .status-badge {
            font-size: 0.75rem;
            padding: 4px 8px;
            border-radius: 9999px;
            background: rgba(16, 185, 129, 0.2);
            color: var(--accent-green);
            border: 1px solid var(--accent-green);
            display: flex;
            align-items: center;
            gap: 6px;
        }

        .pulse-dot {
            width: 8px;
            height: 8px;
            border-radius: 50%;
            background: var(--accent-green);
            animation: pulse 1.5s infinite;
        }

        @keyframes pulse {
            0% { opacity: 1; transform: scale(1); }
            50% { opacity: 0.4; transform: scale(0.8); }
            100% { opacity: 1; transform: scale(1); }
        }

        /* Top Metric Cards */
```

```
        .metric-cards {
            display: grid;
            grid-template-columns: repeat(auto-fit, minmax(200
px, 1fr));
            gap: 12px;
            padding: 12px 24px;
            background-color: #0b1120;
            border-bottom: 1px solid var(--border-color);
        }

        .card {
            background: var(--bg-card);
            border: 1px solid var(--border-color);
            border-radius: 8px;
            padding: 12px 16px;
            display: flex;
            flex-direction: column;
            gap: 6px;
        }

        .card-label {
            font-size: 0.75rem;
            text-transform: uppercase;
            letter-spacing: 0.05em;
            color: var(--text-muted);
        }

        .card-value {
            font-size: 1.4rem;
            font-weight: 700;
            color: var(--text-main);
        }

        .card-sub {
            font-size: 0.75rem;
            color: var(--text-muted);
        }

```

```
        .progress-bar {
            height: 4px;
            background: #334155;
            border-radius: 2px;
            overflow: hidden;
            margin-top: 4px;
        }

        .progress-fill {
            height: 100%;
            background: var(--accent-cyan);
            transition: width 0.3s ease;
        }

        /* Main Workspace Layout */
        .main-workspace {
            display: grid;
            grid-template-columns: 380px 1fr;
            flex: 1;
            overflow: hidden;
        }

        /* Left Side: AIDA64 Hardware Tree */
        .left-panel {
            background: #131d31;
            border-right: 1px solid var(--border-color);
            display: flex;
            flex-direction: column;
            overflow: hidden;
        }

        .panel-header {
            padding: 12px 16px;
            font-size: 0.85rem;
            font-weight: 600;
            text-transform: uppercase;
            letter-spacing: 0.05em;
            color: var(--accent-cyan);
```

```
            border-bottom: 1px solid var(--border-color);
            background: rgba(0, 0, 0, 0.2);
            display: flex;
            justify-content: space-between;
            align-items: center;
        }

        .tree-container {
            flex: 1;
            overflow-y: auto;
            padding: 12px;
        }

        .tree-node {
            margin-bottom: 10px;
            background: var(--bg-card);
            border: 1px solid var(--border-color);
            border-radius: 6px;
            overflow: hidden;
        }

        .tree-node-title {
            padding: 8px 12px;
            font-size: 0.85rem;
            font-weight: 600;
            cursor: pointer;
            background: rgba(255, 255, 255, 0.03);
            display: flex;
            align-items: center;
            justify-content: space-between;
            color: #e2e8f0;
        }

        .tree-node-title:hover {
            background: rgba(255, 255, 255, 0.08);
        }

        .tree-node-body {
```

```
            padding: 8px 12px;
            font-size: 0.8rem;
            border-top: 1px solid var(--border-color);
            background: #0f172a;
        }

        .prop-row {
            display: flex;
            justify-content: space-between;
            margin-bottom: 4px;
        }

        .prop-key {
            color: var(--text-muted);
        }

        .prop-val {
            font-weight: 500;
            color: #f1f5f9;
        }

        /* Right Side: Wireshark Process Stream + AI Copilot *
/
        .right-panel {
            display: flex;
            flex-direction: column;
            overflow: hidden;
        }

        .stream-toolbar {
            padding: 10px 16px;
            background: var(--bg-card);
            border-bottom: 1px solid var(--border-color);
            display: flex;
            align-items: center;
            gap: 12px;
        }

```

```
        .search-box {
            flex: 1;
            background: #0f172a;
            border: 1px solid var(--border-color);
            border-radius: 4px;
            padding: 6px 12px;
            color: var(--text-main);
            font-size: 0.85rem;
            outline: none;
        }

        .btn {
            padding: 6px 14px;
            border-radius: 4px;
            font-size: 0.8rem;
            font-weight: 600;
            cursor: pointer;
            border: none;
            transition: all 0.2s;
        }

        .btn-primary {
            background: var(--accent-cyan);
            color: #0f172a;
        }

        .btn-primary:hover {
            opacity: 0.9;
        }

        .btn-ai {
            background: linear-gradient(135deg, #a855f7, #6366
f1);
            color: #ffffff;
        }

        .table-container {
            flex: 1;
```

```
            overflow-y: auto;
            position: relative;
        }

        table {
            width: 100%;
            border-collapse: collapse;
            font-size: 0.8rem;
            text-align: left;
        }

        th {
            background: #1e293b;
            position: sticky;
            top: 0;
            padding: 8px 12px;
            color: var(--text-muted);
            font-weight: 600;
            border-bottom: 1px solid var(--border-color);
            z-index: 10;
        }

        td {
            padding: 8px 12px;
            border-bottom: 1px solid rgba(51, 65, 85, 0.4);
            font-family: ui-monospace, SFMono-Regular, Menlo,
Monaco, Consolas, monospace;
        }

        tr:hover {
            background: rgba(255, 255, 255, 0.04);
        }

        .badge-cpu-high {
            color: var(--accent-red);
            font-weight: 700;
        }

```

```
        .badge-cpu-med {
            color: var(--accent-yellow);
            font-weight: 600;
        }

        /* Bottom AI Copilot Drawer */
        .ai-drawer {
            background: #111827;
            border-top: 1px solid var(--border-color);
            padding: 12px 20px;
            display: flex;
            flex-direction: column;
            gap: 8px;
            max-height: 180px;
            overflow-y: auto;
        }

        .ai-header {
            display: flex;
            align-items: center;
            justify-content: space-between;
        }

        .ai-title {
            font-size: 0.85rem;
            font-weight: 700;
            color: var(--accent-purple);
            display: flex;
            align-items: center;
            gap: 8px;
        }

        .ai-content {
            font-size: 0.82rem;
            color: #cbd5e1;
            line-height: 1.4;
        }

```

```
        .anomaly-pill {
            display: inline-block;
            font-size: 0.75rem;
            padding: 2px 8px;
            border-radius: 4px;
            background: rgba(239, 68, 68, 0.2);
            color: #f87171;
            border: 1px solid rgba(239, 68, 68, 0.4);
            margin-right: 6px;
            margin-top: 4px;
        }
    </style>
</head>
<body>
    <header>
        <div class="header-title">
            <span>  AI-AI_WINDOWS_ADMIN</span>
            <span style="color: var(--text-muted); font-weight: 400;">|</span>
            <span>System & Hardware Inspector</span>
        </div>
        <div style="display: flex; align-items: center; gap: 16px;">
            <div id="conn-status" class="status-badge">
                <span class="pulse-dot"></span>
                <span>TELEMETRY STREAMING</span>
            </div>
        </div>
    </header>

    <!-- Top Telemetry Metrics -->
    <div class="metric-cards">
        <div class="card">
            <span class="card-label">CPU Load</span>
            <span id="metric-cpu-val" class="card-value">0.0%</span>
            <div class="progress-bar"><div id="metric-cpu-fill" class="progress-fill" style="width: 0%"></div></div>
```

```
            <span id="metric-cpu-sub" class="card-sub">-- Cores</span>
        </div>
        <div class="card">
            <span class="card-label">Physical RAM</span>
            <span id="metric-ram-val" class="card-value">0.0 GB</span>
            <div class="progress-bar"><div id="metric-ram-fill" class="progress-fill" style="width: 0%; background: var(--accent-green);"></div></div>
            <span id="metric-ram-sub" class="card-sub">0% Used</span>
        </div>
        <div class="card">
            <span class="card-label">GPU Accelerator</span>
            <span id="metric-gpu-val" class="card-value">--</span>
            <span id="metric-gpu-sub" class="card-sub">VRAM: -- / -- GB</span>
        </div>
        <div class="card">
            <span class="card-label">Disk Throughput</span>
            <span id="metric-disk-val" class="card-value">0.0 MB/s</span>
            <span id="metric-disk-sub" class="card-sub">R: 0 KB/s | W: 0 KB/s</span>
        </div>
    </div>

    <!-- Main Workspace -->
    <div class="main-workspace">
        <!-- Left Pane: Hardware Tree -->
        <div class="left-panel">
            <div class="panel-header">
                <span>  Hardware Specification</span>
                <button class="btn btn-primary" style="font-size: 0.7rem; padding: 2px 8px;" onclick="fetchHardwareTree()">Refresh</button>
```

```
            </div>
            <div id="hardware-tree" class="tree-container">
                <div style="color: var(--text-muted); font-size: 0.8rem; text-align: center; padding: 20px;">Loading hardware tree...</div>
            </div>
        </div>

        <!-- Right Pane: Wireshark Process Stream -->
        <div class="right-panel">
            <div class="stream-toolbar">
                <input id="process-search" type="text" class="search-box" placeholder="Filter processes by name, PID, or user..." oninput="renderProcessTable()">
                <button id="pause-btn" class="btn btn-primary" onclick="togglePause()">Pause Stream</button>
                <button class="btn btn-ai" onclick="triggerAiAudit()">  Run AI Audit</button>
            </div>
            <div class="table-container">
                <table>
                    <thead>
                        <tr>
                            <th style="width: 70px;">PID</th>
                            <th>Process Name</th>
                            <th style="width: 90px;">Status</th>
                            <th style="width: 90px; text-align: right;">CPU %</th>
                            <th style="width: 100px; text-align: right;">Memory</th>
                            <th style="width: 80px; text-align: right;">Threads</th>
                            <th style="width: 120px;">User</th>
                        </tr>
                    </thead>
                    <tbody id="process-tbody">
```

```
                        <tr><td colspan="7" style="text-align:
 center; color: var(--text-muted);">Waiting for telemetry stre
am...</td></tr>
                    </tbody>
                </table>
            </div>

            <!-- Bottom AI Copilot -->
            <div class="ai-drawer">
                <div class="ai-header">
                    <div class="ai-title">
                        <span>  AI Telemetry & Bottleneck Cop
ilot</span>
                        <span id="ai-health-badge" style="font
-size: 0.75rem; color: var(--accent-green); font-weight: norma
l;">[Health: 100/100]</span>
                    </div>
                    <span id="ai-engine-tag" style="font-size:
 0.7rem; color: var(--text-muted);">Heuristic Engine</span>
                </div>
                <div id="ai-anomalies-container"></div>
                <div id="ai-summary-text" class="ai-content">
                    Telemetry monitor active. Running automate
d heuristic checks over CPU, memory, and hardware thermals.
                </div>
            </div>
        </div>
    </div>

    <script>
        let isPaused = false;
        let latestSnapshot = null;
        let ws = null;

        let wsReconnectTimer = null;

        function connectWebSocket() {
            if (wsReconnectTimer) {
```

```
                clearTimeout(wsReconnectTimer);
                wsReconnectTimer = null;
            }
            if (ws) {
                try {
                    ws.onclose = null;
                    ws.onerror = null;
                    ws.close();
                } catch {}
            }
            const proto = window.location.protocol === 'https:
' ? 'wss:' : 'ws:';
            const wsUrl = `${proto}//${window.location.host}/a
pi/v1/system/stream`;
            ws = new WebSocket(wsUrl);

            ws.onopen = () => {
                const connStatus = document.getElementById('co
nn-status');
                if (connStatus) {
                    connStatus.innerHTML = '<span class="pulse
-dot"></span><span>TELEMETRY STREAMING</span>';
                    connStatus.style.borderColor = 'var(--acce
nt-green)';
                }
            };

            ws.onmessage = (event) => {
                if (isPaused) return;
                try {
                    const data = JSON.parse(event.data);
                    latestSnapshot = data;
                    updateDashboard(data);
                } catch (e) {
                    console.error("Error parsing telemetry sna
pshot:", e);
                }
            };
```

```
519
520             const scheduleReconnect = () => {
521                 if (wsReconnectTimer) return;
522                 const connStatus = document.getElementById('co
nn-status');
523                 if (connStatus) {
524                     connStatus.innerHTML = '<span style="color
: var(--accent-red);">●</span><span>RECONNECTING...</span>';
525                     connStatus.style.borderColor = 'var(--acce
nt-red)';
526                 }
527                 wsReconnectTimer = setTimeout(() => {
528                     wsReconnectTimer = null;
529                     connectWebSocket();
530                 }, 3000);
531             };
532
533             ws.onclose = scheduleReconnect;
534             ws.onerror = scheduleReconnect;
535         }
536
537         function updateDashboard(snap) {
538             // Update Top Cards
539             const cpu = snap.cpu;
540             document.getElementById('metric-cpu-val').innerTex
t = `${cpu.total_percent.toFixed(1)}%`;
541             document.getElementById('metric-cpu-fill').style.w
idth = `${Math.min(100, cpu.total_percent)}%`;
542             document.getElementById('metric-cpu-sub').innerTex
t = `${cpu.physical_cores} Physical / ${cpu.logical_cores} Thr
eads`;
543
544             const mem = snap.memory;
545             document.getElementById('metric-ram-val').innerTex
t = `${mem.used_gb.toFixed(1)} / ${mem.total_gb.toFixed(1)} GB
`;
546             document.getElementById('metric-ram-fill').style.w
idth = `${mem.percent}%`;
```

```
            document.getElementById('metric-ram-sub').innerTex
t = `${mem.percent}% Utilized (${mem.available_gb.toFixed(1)}
GB free)`;

            if (snap.gpus && snap.gpus.length > 0) {
                const g = snap.gpus[0];
                document.getElementById('metric-gpu-val').inne
rText = g.name;
                document.getElementById('metric-gpu-sub').inne
rText = `VRAM: ${g.memory_total_gb.toFixed(1)} GB | CUDA: ${g.
has_cuda ? 'Yes' : 'No'}`;
            }

            const io = snap.disk_io;
            const totalMb = ((io.read_bytes_per_sec + io.write
_bytes_per_sec) / (1024 * 1024)).toFixed(2);
            const rKb = (io.read_bytes_per_sec / 1024).toFixed
(0);
            const wKb = (io.write_bytes_per_sec / 1024).toFixe
d(0);
            document.getElementById('metric-disk-val').innerTe
xt = `${totalMb} MB/s`;
            document.getElementById('metric-disk-sub').innerTe
xt = `R: ${rKb} KB/s | W: ${wKb} KB/s`;

            renderProcessTable();
        }

        function renderProcessTable() {
            if (!latestSnapshot) return;
            const filter = document.getElementById('process-se
arch').value.toLowerCase();
            const tbody = document.getElementById('process-tbo
dy');

            const filtered = latestSnapshot.top_processes.filt
er(p => {
                return !filter || p.name.toLowerCase().include
```

```
s(filter) || String(p.pid).includes(filter) || (p.username &&
p.username.toLowerCase().includes(filter));
572             });
573 
574             if (filtered.length === 0) {
575                 tbody.innerHTML = '<tr><td colspan="7" style="
text-align: center; color: var(--text-muted);">No matching pro
cesses found.</td></tr>';
576                 return;
577             }
578 
579             tbody.innerHTML = filtered.map(p => {
580                 let cpuClass = '';
581                 if (p.cpu_percent > 40) cpuClass = 'badge-cpu-
high';
582                 else if (p.cpu_percent > 15) cpuClass = 'badge
-cpu-med';
583 
584                 return `
585                     <tr>
586                         <td style="color: var(--accent-cyan);"
>${p.pid}</td>
587                         <td style="font-weight: 600;">${p.name
}</td>
588                         <td style="color: var(--text-muted);">
${p.status}</td>
589                         <td style="text-align: right;" class="
${cpuClass}">${p.cpu_percent.toFixed(1)}%</td>
590                         <td style="text-align: right; color: v
ar(--accent-green);">${p.memory_mb.toFixed(1)} MB</td>
591                         <td style="text-align: right; color: v
ar(--accent-purple);">${p.num_threads}</td>
592                         <td style="color: var(--text-muted);">
${p.username || 'SYSTEM'}</td>
593                     </tr>
594                 `;
595             }).join('');
596         }
```

```
597
598         function togglePause() {
599             isPaused = !isPaused;
600             const btn = document.getElementById('pause-btn');
601             btn.innerText = isPaused ? 'Resume Stream' : 'Paus
    e Stream';
602             btn.style.background = isPaused ? 'var(--accent-ye
    llow)' : 'var(--accent-cyan)';
603         }
604
605         async function fetchHardwareTree() {
606             try {
607                 const res = await fetch('/api/v1/system/hardwa
    re');
608                 if (!res.ok) return;
609                 const nodes = await res.json();
610                 const container = document.getElementById('har
    dware-tree');
611
612                 container.innerHTML = nodes.map(node => {
613                     const propsHtml = Object.entries(node.prop
    erties).map(([k, v]) => `
614                         <div class="prop-row">
615                             <span class="prop-key">${k}</span>
616                             <span class="prop-val">${v}</span>
617                         </div>
618                     `).join('');
619
620                     return `
621                         <div class="tree-node">
622                             <div class="tree-node-title" oncli
    ck="this.nextElementSibling.classList.toggle('hidden')">
623                                 <span>${node.category}: ${node
    .name}</span>
624                                 <span style="font-size: 0.7rem
    ; color: var(--accent-cyan);">▼</span>
625                             </div>
626                             <div class="tree-node-body">
```

```
627                                 ${propsHtml}
628                             </div>
629                         </div>
630                     `;
631                 }).join('');
632             } catch (e) {
633                 console.error("Failed to load hardware tree:",
 e);
634             }
635         }
636 
637         async function triggerAiAudit() {
638             const summaryEl = document.getElementById('ai-summ
ary-text');
639             const badgeEl = document.getElementById('ai-health
-badge');
640             const anomalyEl = document.getElementById('ai-anom
alies-container');
641 
642             summaryEl.innerText = "Running AI telemetry diagno
stic analysis...";
643             try {
644                 const res = await fetch('/api/v1/system/diagno
se', { method: 'POST' });
645                 const report = await res.json();
646 
647                 badgeEl.innerText = `[Health: ${report.health_
score}/100]`;
648                 badgeEl.style.color = report.health_score >= 8
0 ? 'var(--accent-green)' : (report.health_score >= 60 ? 'var(
--accent-yellow)' : 'var(--accent-red)');
649                 document.getElementById('ai-engine-tag').inner
Text = report.ai_model_used;
650 
651                 if (report.anomalies && report.anomalies.lengt
h > 0) {
652                     anomalyEl.innerHTML = report.anomalies.map
(a => `<span class="anomaly-pill">  [${a.subsystem}] ${a.title
```

```
}</span>`).join('');
                } else {
                    anomalyEl.innerHTML = '<span style="color:
 var(--accent-green); font-size: 0.75rem;">  No performance an
omalies detected</span>';
                }

                summaryEl.innerText = `${report.summary}\nReco
mmendations: ${report.recommendations.join(', ')}`;
            } catch (e) {
                summaryEl.innerText = `Diagnostic error: ${e.m
essage}`;
            }
        }

        // Initialize on load
        window.addEventListener('DOMContentLoaded', () => {
            connectWebSocket();
            fetchHardwareTree();
        });
    </script>
</body>
</html>

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/system_inspector_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/system_inspector_tab/in
dex.html" class="w-100 h-100 border-0"></iframe>

 * File: index.html
 * Project: ai-windows-admin
 * Package: windows/api/webgui/system_inspector_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 ================================================================
===============
-->
<!-- System & Hardware Inspector Application Tab -->
<style>
  /* Адаптивные стили системного инспектора через CSS-переменн
ые */
  .sys-container {
    padding: 1rem 1.25rem;
    max-width: 100%;
    color: var(--text-color);
  }

```

```
  /* Стили селекторов интервалов временной шкалы */
  .sys-spark-interval-select {
    background-color: var(--surface-2, rgba(255, 255, 255, 0.05)) !important;
    color: var(--text-color, #e2e8f0) !important;
    border: 1px solid var(--border-color, rgba(255, 255, 255, 0.15)) !important;
    border-radius: 4px;
    cursor: pointer;
    font-weight: 600;
  }
  .sys-spark-interval-select:focus {
    border-color: var(--nav-active, #0dcaf0) !important;
    box-shadow: 0 0 0 0.15rem rgba(13, 202, 240, 0.25) !important;
  }
  .sys-spark-interval-select option {
    background-color: var(--surface-1, #1e293b);
    color: var(--text-color, #e2e8f0);
  }

  /* Адаптация text-light и бейджей для светлой темы */
  .sys-container .text-light {
    color: var(--text-color) !important;
  }
  [data-theme="dark"] .sys-container .text-light,
  [data-theme="terminal"] .sys-container .text-light {
    color: #e2e8f0 !important;
  }

  .sys-container .badge.bg-dark {
    background: var(--surface-2) !important;
    color: var(--text-color) !important;
    border: 1px solid var(--border-color) !important;
  }
  [data-theme="dark"] .sys-container .badge.bg-dark {
    background: #0f172a !important;
    color: #e2e8f0 !important;
```

```
    border-color: rgba(255, 255, 255, 0.18) !important;
  }
  [data-theme="terminal"] .sys-container .badge.bg-dark {
    background: #0b120c !important;
    color: #22c55e !important;
    border-color: #1a4223 !important;
  }

  /* Стильные градиентные разделители между основными панелями
 */
  .sys-section-divider {
    height: 1px;
    margin: 1.75rem 0;
    background: linear-gradient(90deg, transparent 0%, var(--b
order-color) 15%, var(--nav-active) 50%, var(--border-color) 8
5%, transparent 100%);
    position: relative;
  }
  .sys-section-divider::after {
    content: '';
    position: absolute;
    top: -1.5px;
    left: 50%;
    transform: translateX(-50%);
    width: 60px;
    height: 4px;
    background: linear-gradient(90deg, var(--nav-active), #818
cf8);
    border-radius: 4px;
    box-shadow: 0 0 12px rgba(13, 202, 240, 0.4);
  }

  /* Карточки основных разделов (CPU, GPU, RAM, Net, Storage)
*/
  .sys-card, .sys-panel {
    background: var(--surface-1);
    border: 1px solid var(--border-color);
    border-radius: 12px;
```

```
    box-shadow: var(--shadow-md);
    transition: transform 0.2s ease, box-shadow 0.2s ease, bor
der-color 0.2s ease;
    overflow: hidden;
    color: var(--text-color);
  }
  .sys-card:hover {
    border-color: var(--border-strong);
    box-shadow: var(--shadow-lg);
  }
  [data-theme="dark"] .sys-card,
  [data-theme="dark"] .sys-panel {
    background: linear-gradient(180deg, #141c2e 0%, #0d1424 10
0%);
    border: 1px solid rgba(255, 255, 255, 0.14);
  }
  [data-theme="terminal"] .sys-card,
  [data-theme="terminal"] .sys-panel {
    background: #0b120c;
    border: 1px solid #1a4223;
  }

  .sys-card-cpu { border-top: 4px solid #0284c7; }
  .sys-card-gpu { border-top: 4px solid #d97706; }
  .sys-card-ram { border-top: 4px solid #16a34a; }
  .sys-card-net { border-top: 4px solid #6366f1; }
  .sys-card-storage { border-top: 4px solid #0891b2; }

  /* Заголовки секций с повышенной контрастностью и интерактив
ностью */
  .sys-core-section-title {
    font-size: 0.85rem !important;
    color: var(--text-color) !important;
    font-weight: 700 !important;
    padding-bottom: 6px;
    border-bottom: 1.5px solid var(--border-subtle);
    margin-top: 14px;
    margin-bottom: 12px !important;
```

```
    letter-spacing: 0.3px;
  }

  /* Шапка карточек панелей для четкого разделения и иерархии */
  .sys-card-header-bar {
    padding: 0.6rem 0.85rem;
    margin: -1rem -1rem 1rem -1rem;
    background: var(--surface-2);
    border-bottom: 1px solid var(--border-color);
    display: flex;
    justify-content: space-between;
    align-items: center;
    border-top-left-radius: 11px;
    border-top-right-radius: 11px;
  }

  /* Компактный режим: отображаются только верхние блоки панелей */
  .sys-compact-mode .sys-full-only,
  .sys-compact-mode .sys-detailed-sensors,
  .sys-compact-mode .sys-card-header-bar {
    display: none !important;
  }
  .sys-compact-mode .sys-card {
    padding: 0.6rem 0.75rem !important;
  }
  .sys-compact-mode .sys-card-top-banner {
    margin-bottom: 0 !important;
    padding: 0.6rem 0.85rem !important;
  }
  .sys-compact-mode .sys-section-divider {
    margin: 0.65rem 0 !important;
  }
  .sys-compact-mode .row.mb-3 {
    margin-bottom: 0.5rem !important;
  }
  .sys-compact-mode #sys-gpu-panels-container {
```

```
    gap: 0.5rem !important;
    margin-bottom: 0.5rem !important;
  }

  /* Принудительное развертывание карточки компонента по клику на значок дрели (Drill-down) */
  .sys-compact-mode .sys-card.sys-card-drill-expanded {
    padding: 1rem !important;
    border-color: var(--nav-active) !important;
    box-shadow: 0 0 16px rgba(13, 202, 240, 0.25) !important;
  }
  .sys-compact-mode .sys-card.sys-card-drill-expanded .sys-card-header-bar {
    display: flex !important;
  }
  .sys-compact-mode .sys-card.sys-card-drill-expanded .sys-full-only {
    display: block !important;
  }
  .sys-compact-mode .sys-card.sys-card-drill-expanded .sys-full-only.row {
    display: flex !important;
  }
  .sys-compact-mode .sys-card.sys-card-drill-expanded .sys-detailed-sensors {
    display: block !important;
  }
  .sys-compact-mode .sys-card.sys-card-drill-expanded .sys-card-top-banner {
    margin-bottom: 1rem !important;
  }

  /* Интерактивная кнопка со значком дрели (Drill-down) слева от названия компонента */
  .sys-drill-btn {
    display: inline-flex;
    align-items: center;
    justify-content: center;
```

```
    width: 28px;
    height: 28px;
    min-width: 28px;
    min-height: 28px;
    padding: 0;
    margin-right: 8px;
    border-radius: 7px;
    border: 1.5px solid var(--border-color);
    background: var(--surface-1);
    cursor: pointer;
    transition: all 0.22s cubic-bezier(0.4, 0, 0.2, 1);
    box-shadow: var(--shadow-xs);
    flex-shrink: 0;
    vertical-align: middle;
  }
  .sys-drill-btn img {
    width: 18px;
    height: 18px;
    object-fit: contain;
    transition: transform 0.22s ease, opacity 0.22s ease;
    display: block;
  }
  .sys-drill-btn:hover {
    border-color: var(--nav-active) !important;
    transform: scale(1.12);
    box-shadow: 0 0 10px rgba(239, 68, 68, 0.45);
  }
  .sys-drill-btn:hover img {
    transform: rotate(12deg);
  }
  .sys-drill-btn.active {
    box-shadow: 0 0 14px rgba(239, 68, 68, 0.65) !important;
    border-color: #ef4444 !important;
    transform: rotate(-8deg) scale(1.06);
  }
  [data-theme="dark"] .sys-drill-btn {
    background: rgba(15, 23, 42, 0.85);
    border-color: rgba(255, 255, 255, 0.2);
```

```
  }
  [data-theme="terminal"] .sys-drill-btn {
    background: #0b120c;
    border-color: #1a4223;
  }

  /* Верхний баннер панелей: строгое выравнивание колонок через CSS Grid */
  .sys-card-top-banner {
    display: grid !important;
    grid-template-columns: minmax(280px, 1fr) 280px 330px !important;
    align-items: center;
    gap: 1.25rem;
    width: 100%;
    background: var(--surface-2);
    border: 1px solid var(--border-color);
  }
  [data-theme="dark"] .sys-card-top-banner {
    background: rgba(0, 0, 0, 0.45);
    border: 1.5px solid rgba(255, 255, 255, 0.18);
  }
  [data-theme="terminal"] .sys-card-top-banner {
    background: #0b120c;
    border: 1px solid #1a4223;
  }
  .sys-banner-col-info {
    min-width: 0;
    width: 100%;
  }
  .sys-banner-col-gauge {
    width: 280px;
    min-width: 280px;
    max-width: 280px;
    justify-self: center;
  }
  .sys-banner-col-specs {
    width: 330px;
```

```
    min-width: 330px;
    max-width: 330px;
    justify-self: end;
  }

  /* Панель графиков истории */
  .sys-history-panel {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    position: relative;
  }
  [data-theme="dark"] .sys-history-panel {
    background: rgba(0, 0, 0, 0.4);
    border: 1.5px solid rgba(255, 255, 255, 0.15);
  }
  [data-theme="terminal"] .sys-history-panel {
    background: #0b120c;
    border: 1px solid #1a4223;
  }

  .sys-spark-tooltip {
    position: absolute;
    top: 6px;
    pointer-events: none;
    background: var(--surface-1);
    color: var(--text-color);
    border: 1px solid var(--nav-active);
    border-radius: 6px;
    padding: 4px 8px;
    font-size: 0.70rem;
    z-index: 10;
    box-shadow: var(--shadow-md);
    white-space: nowrap;
  }
  [data-theme="dark"] .sys-spark-tooltip {
    background: rgba(15, 23, 42, 0.95);
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.6);
  }
```

```
  [data-theme="terminal"] .sys-spark-tooltip {
    background: #0b120c;
    border-color: #1a4223;
    color: #22c55e;
  }

  @media (max-width: 1280px) {
    .sys-card-top-banner {
      grid-template-columns: 1fr 280px !important;
    }
    .sys-banner-col-specs {
      grid-column: span 2;
      justify-self: stretch;
      width: 100%;
      max-width: 100%;
    }
  }

  @media (max-width: 860px) {
    .sys-card-top-banner {
      grid-template-columns: 1fr !important;
    }
    .sys-banner-col-gauge,
    .sys-banner-col-specs {
      grid-column: span 1;
      justify-self: stretch;
      width: 100%;
      max-width: 100%;
    }
  }

  /* Сетка и плашки ключевых экспресс-метрик (KPI Pills) в цен
тре верхних баннеров */
  .sys-banner-kpi-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(115px, 1fr)
);
    gap: 6px;
```

```
    width: 100%;
    max-width: 580px;
    margin-top: 4px;
  }
  .sys-kpi-pill {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 7px;
    padding: 5px 8px;
    display: flex;
    flex-direction: column;
    justify-content: center;
    gap: 1px;
    transition: all 0.2s ease;
    box-shadow: var(--shadow-xs);
  }
  .sys-kpi-pill:hover {
    border-color: var(--nav-active);
    transform: translateY(-1px);
    box-shadow: var(--shadow-sm);
  }
  [data-theme="dark"] .sys-kpi-pill {
    background: rgba(15, 23, 42, 0.65);
    border: 1px solid rgba(255, 255, 255, 0.12);
  }
  [data-theme="terminal"] .sys-kpi-pill {
    background: #0b120c;
    border: 1px solid #1a4223;
  }
  .sys-kpi-label {
    font-size: 0.68rem;
    font-weight: 600;
    color: var(--text-muted);
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
    display: flex;
    align-items: center;
```

```
  }
  .sys-kpi-value {
    font-size: 0.80rem;
    font-weight: 800;
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
  }

  /* Сенсоры LHM */
  .sys-sensor-group {
    margin-bottom: 8px;
    background: var(--surface-2);
    border: 1.5px solid var(--border-color);
    border-radius: 6px;
    overflow: hidden;
  }
  .sys-sensor-group-title {
    padding: 6px 10px;
    font-size: 0.80rem;
    font-weight: 700;
    background: var(--surface-1);
    display: flex;
    align-items: center;
    justify-content: space-between;
    color: var(--nav-active);
    border-bottom: 1px solid var(--border-color);
  }
  .sys-sensor-row {
    display: flex;
    justify-content: space-between;
    align-items: center;
    padding: 4px 10px;
    font-size: 0.78rem;
    border-bottom: 1px solid var(--border-subtle);
    background: var(--surface-2);
```

```
  }
  .sys-sensor-row:last-child {
    border-bottom: none;
  }
  .sys-sensor-name {
    color: var(--text-color);
    font-weight: 600;
    max-width: 55%;
    overflow: hidden;
    text-overflow: ellipsis;
    white-space: nowrap;
  }
  .sys-sensor-value {
    font-weight: 700;
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
    font-size: 0.80rem;
    text-align: right;
    color: var(--text-color);
  }

  /* Таблица процессов */
  .sys-proc-table {
    width: 100%;
    font-size: 0.80rem;
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
    border-collapse: collapse;
  }
  .sys-proc-table th {
    background: var(--surface-2);
    padding: 6px 10px;
    color: var(--text-muted);
    font-weight: 700;
    border-bottom: 1.5px solid var(--border-color);
    position: sticky;
    top: 0;
    z-index: 5;
```

```
}
.sys-proc-table td {
  padding: 5px 10px;
  border-bottom: 1px solid var(--border-subtle);
  color: var(--text-color);
}

/* Статусы загрузки */
.badge-cpu-high { color: #dc2626; font-weight: 800; }
.badge-cpu-med  { color: #d97706; font-weight: 700; }

/* Аномалии */
.anomaly-tag {
  display: inline-block;
  font-size: 0.74rem;
  padding: 2px 7px;
  border-radius: 4px;
  background: var(--status-danger-bg);
  color: var(--status-danger-text);
  border: 1.5px solid var(--status-danger-border);
  margin-right: 4px;
  margin-top: 2px;
  font-weight: 700;
}

/* Интерактивный ресайзер для таблицы */
.sys-table-resizer {
  height: 10px;
  background: var(--surface-2);
  border-top: 1px solid var(--border-color);
  cursor: ns-resize;
  display: flex;
  align-items: center;
  justify-content: center;
  transition: background-color 0.15s ease;
  user-select: none;
  -webkit-user-select: none;
  touch-action: none;
```

```
  }
  .sys-table-resizer:hover,
  .sys-table-resizer.resizing {
    background: var(--nav-hover-bg);
  }
  .sys-table-resizer-grip {
    width: 42px;
    height: 3px;
    background: var(--border-strong);
    border-radius: 3px;
    pointer-events: none;
    transition: background-color 0.15s ease, width 0.15s ease;
  }
  .sys-table-resizer:hover .sys-table-resizer-grip,
  .sys-table-resizer.resizing .sys-table-resizer-grip {
    background: var(--nav-active);
    width: 56px;
  }

  /* Карточки ядер CPU и блоков GPU, полукруги и термо-слайдеры */
  .sys-core-card {
    background: var(--surface-2);
    border: 1.5px solid var(--border-color);
    border-radius: 8px;
    padding: 10px 12px;
    transition: all 0.2s ease;
    box-shadow: var(--shadow-xs);
    display: flex;
    flex-direction: column;
    justify-content: space-between;
    min-height: 148px;
    color: var(--text-color);
  }
  .sys-core-card:hover {
    border-color: var(--nav-active);
    transform: translateY(-2px);
    box-shadow: var(--shadow-sm);
```

```
}
[data-theme="dark"] .sys-core-card {
  background: #0f172a;
  border: 1.5px solid rgba(255, 255, 255, 0.22);
  box-shadow: 0 4px 12px rgba(0, 0, 0, 0.5);
}
[data-theme="terminal"] .sys-core-card {
  background: #0b120c;
  border: 1.5px solid #1a4223;
}

.sys-core-title {
  font-size: 0.88rem;
  font-weight: 800;
  color: var(--text-color);
  letter-spacing: 0.3px;
}
.sys-core-load-badge {
  font-size: 0.82rem;
  font-weight: 800;
  padding: 2px 7px;
  border-radius: 5px;
  background: var(--status-info-bg);
  color: var(--status-info-text);
  border: 1.5px solid var(--status-info-border);
}

.sys-temp-slider-wrap {
  margin-top: 5px;
}
.sys-temp-slider-track {
  position: relative;
  height: 8px;
  background: var(--border-color);
  border-radius: 4px;
  overflow: visible;
}
.sys-temp-slider-bar {
```

```
    height: 100%;
    border-radius: 4px;
    transition: width 0.4s ease, background-color 0.4s ease;
  }
  .sys-temp-slider-thumb {
    position: absolute;
    top: 50%;
    width: 14px;
    height: 14px;
    border-radius: 50%;
    border: 2px solid var(--surface-1);
    transform: translate(-50%, -50%);
    transition: left 0.4s ease, background-color 0.4s ease;
    box-shadow: 0 2px 6px rgba(0,0,0,0.5);
  }

  /* Таблица параметров ядра CPU внутри карточки */
  .sys-core-param-table {
    display: flex;
    flex-direction: column;
    gap: 3px;
    width: 100%;
  }
  .sys-core-param-row {
    display: flex;
    justify-content: space-between;
    align-items: center;
    padding: 2px 0;
    border-bottom: 1px dashed var(--border-subtle);
    font-size: 0.74rem;
    line-height: 1.25;
  }
  .sys-core-param-row:last-child {
    border-bottom: none;
  }
  .sys-core-param-key {
    color: var(--text-muted);
    font-size: 0.72rem;
```

```
    font-weight: 600;
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
    max-width: 58%;
  }
  .sys-core-param-val {
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
    font-weight: 800;
    font-size: 0.76rem;
    text-align: right;
    white-space: nowrap;
    color: var(--text-color);
  }

  /* Крупные слайдеры для главного заголовка */
  .sys-temp-slider-lg .sys-temp-slider-track {
    height: 12px;
    border-radius: 6px;
    background: var(--border-color);
  }
  .sys-temp-slider-lg .sys-temp-slider-bar {
    border-radius: 6px;
  }
  .sys-temp-slider-lg .sys-temp-slider-thumb {
    width: 20px;
    height: 20px;
    border: 2.5px solid var(--surface-1);
    box-shadow: 0 2px 8px rgba(0,0,0,0.5);
  }
  .sys-temp-slider-empty {
    font-size: 0.80rem;
    color: var(--text-muted);
    text-align: center;
    font-weight: 600;
  }

```

```
/* Спецификации CPU: интерактивная карточка */
.sys-cpu-specs-card,
.sys-gpu-specs-card {
  transition: all 0.2s ease;
  cursor: pointer;
  position: relative;
  user-select: none;
  background: var(--surface-2);
  border: 1px solid var(--border-color);
  color: var(--text-color);
}
.sys-cpu-specs-card:hover,
.sys-gpu-specs-card:hover {
  border-color: var(--nav-active) !important;
  box-shadow: var(--shadow-sm) !important;
  transform: translateY(-1px);
}

/* Сворачиваемая интерактивная плашка детализации CPU */
.sys-cpu-details-accordion-btn,
.sys-gpu-details-accordion-btn {
  cursor: pointer;
  user-select: none;
  padding: 8px 14px;
  background: var(--surface-2);
  border-radius: 8px;
  border: 1.5px solid var(--border-color);
  transition: all 0.25s ease;
  box-shadow: var(--shadow-xs);
  color: var(--text-color);
}
.sys-cpu-details-accordion-btn:hover,
.sys-gpu-details-accordion-btn:hover {
  border-color: var(--nav-active);
  box-shadow: var(--shadow-sm);
  transform: translateY(-1px);
}

```

```
  /* Сенсорные виджеты */
  .sys-sensor-widget-item {
    background: var(--surface-2) !important;
    border: 1px solid var(--border-color) !important;
    box-shadow: var(--shadow-xs);
    color: var(--text-color);
  }
  [data-theme="dark"] .sys-sensor-widget-item {
    background: #0f172a !important;
    border: 1.5px solid rgba(255, 255, 255, 0.18) !important;
  }
  [data-theme="terminal"] .sys-sensor-widget-item {
    background: #0b120c !important;
    border: 1px solid #1a4223 !important;
  }

  /* Полоса заголовка инспектора */
  .sys-header-strip {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    box-shadow: var(--shadow-xs);
  }
  [data-theme="dark"] .sys-header-strip {
    background: linear-gradient(180deg, #182238 0%, #0f172a 10
0%) !important;
    border: 1px solid rgba(255, 255, 255, 0.18) !important;
    box-shadow: 0 4px 16px rgba(0,0,0,0.45) !important;
  }
  [data-theme="terminal"] .sys-header-strip {
    background: #0b120c !important;
    border: 1px solid #1a4223 !important;
  }
</style>
<div class="sys-container theme-inspector sys-compact-mode">
<!-- Header Alert & Status (Compact Strip) -->
<div class="d-flex justify-content-between align-items-center
flex-wrap gap-2 mb-3.5 py-2 px-3 rounded-3 sys-header-strip">
<div class="d-flex align-items-center gap-1.5 flex-wrap">
```

```
<button class="panel-toggle-drill collapsed me-2" id="btn-sys-toggle-compact" title="Развернуть все / Свернуть все панели" type="button">
<img alt="Развернуть/Свернуть" src="/html/assets/red_drill.png?v=20261008_v1"/>
</button>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<span class="fs-5">□ </span>
<span class="fw-bold" style="font-size: 0.95rem; color: var(--text-color);">Телеметрия хоста и оборудования</span>
<span class="badge bg-secondary fw-bold" style="font-size: 0.72rem;">Live Telemetry</span>
<span class="d-none d-lg-inline text-muted" style="font-size: 0.78rem;">| Комплексный мониторинг CPU, RAM, GPU, дисков, питания и процессов</span>
</div>
<div class="d-flex align-items-center gap-1.5 flex-wrap">
<span class="badge rounded-pill bg-success text-white border border-light px-2.5 py-1" id="sys-conn-status" style="font-size: 0.74rem; font-weight: 700;">
        ● Активна
      </span>
</div>
</div>
<!-- 1. Панель Загрузка CPU -->
<div class="row mb-3">
<div class="col-12">
<div class="sys-card sys-card-cpu h-100 p-3" id="sys-card-cpu">
<div class="sys-card-header-bar">
<div class="d-flex align-items-center gap-2">
<span class="sys-label text-info" style="font-size: 1.05rem; font-weight: 800;"><i class="bi bi-cpu text-info me-1.5"></i>Загрузка CPU</span>
</div>
<span class="small badge bg-dark border border-secondary text-info font-monospace" id="sys-metric-cpu-sub" style="font-size:
```

```
 0.84rem;">-- Ядер</span>
</div>
<!-- Верхний баннер: Название слева, Спидометр и датчик температуры по центру, Сводная таблица CPU справа -->
<div class="sys-card-top-banner p-3 rounded-3 mb-3">
<!-- Левая колонка: Модель процессора и экспресс-метрики KPI -->
<div class="sys-banner-col-info d-flex flex-column align-items-start justify-content-center">
<div class="text-start mb-1.5 d-flex align-items-center">
<button class="sys-drill-btn" id="sys-drill-btn-cpu" onclick="window.toggleSysCardDrill &amp;&amp; window.toggleSysCardDrill('sys-card-cpu')" title="Drill-down: Развернуть/скрыть все скрытые панели и датчики CPU" type="button">
<img alt="Дрель" src="/html/assets/red_drill.png?v=20261008_v1"/>
</button>
<div>
<span class="fw-bold text-info" id="sys-metric-cpu-model" style="font-size: 1.20rem; font-weight: 800; letter-spacing: 0.5px; text-shadow: 0 0 14px rgba(13, 202, 240, 0.4);">--</span>
<div class="d-flex align-items-center gap-1.5 mt-0.5">
<span class="badge bg-dark border border-secondary text-light font-monospace" id="sys-metric-cpu-vendor-badge" style="font-size: 0.70rem;">Intel Corporation · 64-bit</span>
<span class="badge bg-info-subtle text-info border border-info font-monospace" id="sys-cpu-status-badge" style="font-size: 0.70rem;">● Active</span>
</div>
</div>
</div>
<div class="sys-banner-kpi-grid">
<div class="sys-kpi-pill">
<span class="sys-kpi-label"><i class="bi bi-lightning-charge text-warning me-1"></i>Мощность</span>
<span class="sys-kpi-value text-warning" id="sys-cpu-kpi-power">-- W</span>
</div>
```

```
<div class="sys-kpi-pill">
<span class="sys-kpi-label"><i class="bi bi-thermometer-half t
ext-danger me-1"></i>Пакет t°</span>
<span class="sys-kpi-value text-info" id="sys-cpu-kpi-temp">--
 °C</span>
</div>
<div class="sys-kpi-pill">
<span class="sys-kpi-label"><i class="bi bi-cpu text-info me-1
"></i>Ядра/Потоки</span>
<span class="sys-kpi-value text-light" id="sys-cpu-kpi-cores">
-- / --</span>
</div>
<div class="sys-kpi-pill">
<span class="sys-kpi-label"><i class="bi bi-speedometer2 text-
primary me-1"></i>Частота</span>
<span class="sys-kpi-value text-primary" id="sys-cpu-kpi-freq"
>-- GHz</span>
</div>
</div>
</div>
<!-- Центральная колонка: Спидометр загрузки, справа от него з
начение с динамическим оттенком, а СНИЗУ — датчик температуры
-->
<div class="sys-banner-col-gauge d-flex flex-column justify-co
ntent-between gap-1.5">
<div class="d-flex align-items-center gap-2.5">
<div id="sys-metric-cpu-gauge" style="width: 120px; height: 68
px; flex-shrink: 0;"></div>
<div>
<div class="sys-value text-info" id="sys-metric-cpu-val" style
="font-size: 2.1rem; line-height: 1; font-weight: 800; transit
ion: color 0.3s ease, text-shadow 0.3s ease;">0.0%</div>
<div class="text-light mt-1 font-monospace" id="sys-metric-cpu
-package-temp" style="font-size: 0.82rem; font-weight: 700;">П
акет: -- °C</div>
</div>
</div>
<!-- Датчик температуры Package расположен ПОД спидометром заг
```

```
рузки -->
<div id="sys-metric-cpu-package-slider" style="width: 100%;"></div>
</div>
<!-- Правая колонка: Сводная таблица о процессоре (кликабельна для запуска AI-аудита) -->
<div class="sys-banner-col-specs">
<div class="sys-cpu-specs-card p-2 rounded-2 d-flex flex-column justify-content-between h-100" id="sys-cpu-specs-card" onclick="window.inspectCpuSpecsWithAi &amp;&amp; window.inspectCpuSpecsWithAi()" title="Нажмите для вызова AI-аудита и экспертного анализа архитектуры процессора">
<div class="d-flex justify-content-between align-items-center mb-1 pb-1 border-bottom" style="border-color: var(--border-subtle) !important;">
<span class="text-light fw-bold d-flex align-items-center gap-1" style="font-size: 0.78rem;">
<i class="bi bi-info-circle text-info"></i>Характеристики CPU
                    <span class="badge bg-primary-subtle text-primary border border-primary-subtle font-monospace ms-1" style="font-size: 0.62rem;"><i class="bi bi-stars me-0.5"></i>AI</span>
</span>
<span class="badge bg-dark border border-secondary text-info font-monospace" id="sys-cpu-spec-arch" style="font-size: 0.68rem;">x86_64</span>
</div>
<div class="sys-cpu-specs-grid" style="display: grid; grid-template-columns: 1fr 1fr; gap: 4px 10px; font-size: 0.74rem;">
<div class="d-flex justify-content-between"><span class="text-muted">Сокет:</span><span class="text-light font-monospace fw-semibold" id="sys-cpu-spec-socket">--</span></div>
<div class="d-flex justify-content-between"><span class="text-muted">Ядра/Потоки:</span><span class="text-light font-monospace fw-semibold" id="sys-cpu-spec-cores">--</span></div>
<div class="d-flex justify-content-between"><span class="text-muted">Базовая част.:</span><span class="text-info font-monospace fw-semibold" id="sys-cpu-spec-base-freq">--</span></div>
```

```
<div class="d-flex justify-content-between"><span class="text-muted">Макс. част.:</span><span class="text-info font-monospace fw-semibold" id="sys-cpu-spec-max-freq">--</span></div>
<div class="d-flex justify-content-between" style="grid-column: span 2;"><span class="text-muted">Кэш L2 / L3:</span><span class="text-warning font-monospace fw-semibold" id="sys-cpu-spec-cache">--</span></div>
</div>
</div>
</div>
</div>
<!-- График истории CPU (Загрузка %, Температура °C, Мощность W) с логарифмической/линейной шкалой -->
<div class="sys-full-only sys-history-panel p-2.5 mb-2.5 rounded-2">
<div class="d-flex justify-content-between align-items-center mb-1.5 flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<span class="small text-light" style="font-size: 0.80rem; font-weight: 700;"><i class="bi bi-graph-up text-info me-1"></i>История CPU</span>
<div class="btn-group btn-group-sm" role="group" style="height: 22px;">
<button class="btn btn-xs btn-outline-info active py-0 px-2 fw-bold" id="btn-cpu-spark-log" onclick="window.setCpuSparkScale &amp;&amp; window.setCpuSparkScale('log')" style="font-size: 0.68rem;" type="button">Лог. шкала (Log)</button>
<button class="btn btn-xs btn-outline-secondary py-0 px-2" id="btn-cpu-spark-lin" onclick="window.setCpuSparkScale &amp;&amp; window.setCpuSparkScale('linear')" style="font-size: 0.68rem;" type="button">Линейная (Lin)</button>
</div>
<select class="form-select form-select-xs sys-spark-interval-select" id="sel-cpu-spark-interval" onchange="window.setCpuSparkInterval &amp;&amp; window.setCpuSparkInterval(this.value)" style="height: 22px; padding: 0 6px; font-size: 0.68rem; width: auto; min-width: 85px;">
<option value="seconds">Секунды</option>
```

```
827 </select>
828 </div>
829 <div class="d-flex align-items-center gap-2.5" style="font-siz
    e: 0.74rem; font-weight: 600;">
830 <span class="text-info" style="text-shadow: 0 0 8px rgba(13,20
    2,240,0.5);" title="Основная шкала: Загрузка ядер CPU (0-100%)
    ">● Нагрузка %</span>
831 <span class="text-danger" style="text-shadow: 0 0 8px rgba(239
    ,68,68,0.5);" title="Компактная шкала: Температура пакета CPU
    (°C)">● Температура °C</span>
832 <span class="text-warning" style="text-shadow: 0 0 8px rgba(24
    5,158,11,0.5);" title="Компактная шкала: Энергопотребление пак
    ета CPU (W)">● Мощность W</span>
833 <span class="text-muted font-monospace ms-2" id="sys-cpu-spark
    -live-cursor" style="font-size: 0.70rem;">Live</span>
834 </div>
835 </div>
836 <!-- Контейнер графика и интерактивного тултипа -->
837 <div id="sys-cpu-spark-container" style="position: relative; w
    idth: 100%; height: 90px;">
838 <div id="sys-cpu-spark" style="width: 100%; height: 100%;"></d
    iv>
839 <div class="sys-spark-tooltip" id="sys-cpu-spark-tooltip" styl
    e="display: none;"></div>
840 </div>
841 </div>
842 <!-- Сворачиваемый блок детализации CPU: Ядра, потоки и датчик
    и системы (расположен под графиком истории) -->
843 <div class="sys-full-only sys-detailed-sensors mt-2">
844 <div class="sys-cpu-details-accordion-btn d-flex justify-conte
    nt-between align-items-center flex-wrap gap-2" id="sys-cpu-det
    ails-toggle-btn" onclick="window.toggleSysCpuDetails &amp;&amp
    ; window.toggleSysCpuDetails()" title="Нажмите, чтобы разверну
    ть или скрыть детализацию по ядрам, потокам и сенсорам CPU">
845 <div class="d-flex align-items-center gap-2 flex-wrap">
846 <button class="panel-toggle-drill collapsed me-1" id="sys-cpu-
    details-drill-btn" style="pointer-events: none;" tabindex="-1"
     type="button">
```

```
<img alt="Дрель" src="/html/assets/red_drill.png?v=20261008_v1
"/>
</button>
<span class="fw-bold" style="font-size: 0.88rem; color: var(--
text-color);">Детализация: Ядра, потоки и сенсоры CPU</span>
<span class="badge bg-dark border border-secondary text-info f
ont-monospace ms-1" id="sys-cores-count-badge" style="font-siz
e: 0.72rem;">0 ядер</span>
<span class="badge bg-dark border border-secondary text-warnin
g font-monospace" id="sys-cpu-sensors-count-badge" style="font
-size: 0.72rem;">0 параметров</span>
</div>
</div>
<!-- Скрытый по умолчанию общий контейнер деталей CPU -->
<div id="sys-cpu-details-collapse" style="display: none; flex-
direction: column; gap: 14px; margin-top: 10px;">
<!-- 1. Нагрузка и температура по ядрам и потокам -->
<div>
<div class="sys-core-section-title d-flex justify-content-betw
een align-items-center mb-2" style="font-size: 0.82rem;">
<span><i class="bi bi-cpu text-info me-1"></i>Нагрузка и темпе
ратура по ядрам и потокам</span>
</div>
<div id="sys-metric-cpu-cores" style="display: grid; grid-temp
late-columns: repeat(auto-fill, minmax(300px, 1fr)); gap: 10px
; font-size: 0.78rem;"></div>
</div>
<!-- 2. Общие показатели CPU (датчики системы) -->
<div>
<div class="sys-core-section-title d-flex justify-content-betw
een align-items-center mb-2" style="font-size: 0.82rem;">
<span><i class="bi bi-lightning-charge text-warning me-1"></i>
Общие показатели CPU (датчики системы)</span>
</div>
<div id="sys-metric-cpu-sensors" style="display: grid; grid-te
mplate-columns: repeat(auto-fill, minmax(180px, 1fr)); gap: 8p
x; font-size: 0.78rem;">
<div class="text-muted small text-center py-2 col-12">Опрос се
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нсоров CPU...</div>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- Четкий стильный разделитель панелей -->
<div class="sys-section-divider"></div>
<!-- 2. Панели Загрузки GPU (Для каждого найденного GPU создае
тся отдельная полноразмерная панель) -->
<div class="d-flex flex-column gap-3 mb-3" id="sys-gpu-panels-
container">
<!-- GPU карточки динамически монтируются и обновляются из mai
n.js -->
</div>
<!-- Четкий стильный разделитель панелей -->
<div class="sys-section-divider"></div>
<!-- 3. Панель Память (RAM) и Дисковый ввод-вывод -->
<div class="row mb-3">
<div class="col-12">
<div class="sys-card sys-card-ram p-3" id="sys-card-ram">
<div class="sys-card-header-bar">
<div class="d-flex align-items-center gap-2">
<span class="sys-label text-success" style="font-size: 1.05rem
; font-weight: 800;"><i class="bi bi-memory text-success me-1.
5"></i>Память (RAM) и Дисковый ввод-вывод</span>
</div>
<span class="small badge bg-dark border border-secondary text-
light font-monospace" id="sys-memio-ts" style="font-size: 0.84
rem;">--</span>
</div>
<!-- Верхний баннер: Модель слева, Gauge RAM по центру, Слайде
р файла подкачки справа -->
<div class="sys-card-top-banner p-3 rounded-3 mb-3">
<!-- Левая колонка: Модель RAM и экспресс-метрики KPI -->
<div class="sys-banner-col-info d-flex flex-column align-items
```

```
-start justify-content-center">
<div class="text-start mb-1.5 d-flex align-items-center">
<button class="sys-drill-btn" id="sys-drill-btn-ram" onclick="
window.toggleSysCardDrill &amp;&amp; window.toggleSysCardDrill
('sys-card-ram')" title="Drill-down: Развернуть/скрыть все скр
ытые панели и датчики RAM" type="button">
<img alt="Дрель" src="/html/assets/red_drill.png?v=20261008_v1
"/>
</button>
<div>
<span class="fw-bold text-success" id="sys-metric-ram-model" s
tyle="font-size: 1.20rem; font-weight: 800; letter-spacing: 0.
5px; text-shadow: 0 0 14px rgba(16, 185, 129, 0.4);">-- GB RAM
</span>
<div class="d-flex align-items-center gap-1.5 mt-0.5">
<span class="badge bg-dark border border-secondary text-light
font-monospace" id="sys-ram-type-badge" style="font-size: 0.70
rem;">DDR4 · System Memory</span>
<span class="badge bg-success-subtle text-success border borde
r-success font-monospace" id="sys-ram-status-badge" style="fon
t-size: 0.70rem;">● Dual Channel</span>
</div>
</div>
</div>
<div class="sys-banner-kpi-grid">
<div class="sys-kpi-pill">
<span class="sys-kpi-label"><i class="bi bi-pie-chart text-suc
cess me-1"></i>Занято RAM</span>
<span class="sys-kpi-value text-success" id="sys-ram-kpi-used"
>-- GB</span>
</div>
<div class="sys-kpi-pill">
<span class="sys-kpi-label"><i class="bi bi-check2-circle text
-info me-1"></i>Свободно</span>
<span class="sys-kpi-value text-info" id="sys-ram-kpi-free">--
 GB</span>
</div>
<div class="sys-kpi-pill">
```

```
<span class="sys-kpi-label"><i class="bi bi-arrow-repeat text-warning me-1"></i>Swap Подкачка</span>
<span class="sys-kpi-value text-warning" id="sys-ram-kpi-swap">--%</span>
</div>
<div class="sys-kpi-pill">
<span class="sys-kpi-label"><i class="bi bi-hdd-stack text-primary me-1"></i>Диск I/O</span>
<span class="sys-kpi-value text-primary" id="sys-ram-kpi-io">-- MB/s</span>
</div>
</div>
</div>
<!-- Центральная колонка: Большой Gauge RAM -->
<div class="sys-banner-col-gauge d-flex align-items-center gap-2.5">
<div id="sys-metric-ram-gauge" style="width: 120px; height: 68px; flex-shrink: 0;"></div>
<div>
<div class="sys-value text-success" id="sys-memio-pct" style="font-size: 2.1rem; line-height: 1; font-weight: 800;">0.0%</div>
<div class="text-light mt-1 font-monospace" id="sys-memio-used" style="font-size: 0.82rem; font-weight: 700;">0.0 / 0.0 GB</div>
</div>
</div>
<!-- Правая колонка: Файл подкачки со слайдером -->
<div class="sys-banner-col-specs d-flex flex-column justify-content-center">
<span class="text-light mb-1 font-monospace" id="sys-memio-swap" style="font-size: 0.80rem; font-weight: 700;">Файл подкачки: 0%</span>
<div id="sys-metric-ram-slider" style="width: 100%;"></div>
</div>
</div>
<!-- Сетка 4 блоков: RAM, Swap, Disk Read, Disk Write -->
<div class="row g-2 mb-2 sys-full-only">
```

```
<div class="col-6 col-md-3">
<div class="sys-core-card h-100" title="Физическая оперативная
 память">
<div class="d-flex justify-content-between align-items-center
mb-1">
<span class="sys-core-title">Физическая RAM</span>
<span class="sys-core-load-badge text-success border-success"
id="sys-ram-sub-badge">0%</span>
</div>
<div class="py-1" id="sys-ram-sub-gauge"></div>
<div id="sys-ram-sub-slider"></div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="sys-core-card h-100" title="Виртуальная память / ф
айл подкачки">
<div class="d-flex justify-content-between align-items-center
mb-1">
<span class="sys-core-title">Подкачка (Swap)</span>
<span class="sys-core-load-badge text-warning border-warning"
id="sys-swap-sub-badge">0%</span>
</div>
<div class="py-1" id="sys-swap-sub-gauge"></div>
<div id="sys-swap-sub-slider"></div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="sys-core-card h-100" title="Скорость чтения с диск
a">
<div class="d-flex justify-content-between align-items-center
mb-1">
<span class="sys-core-title">Чтение (Read)</span>
<span class="sys-core-load-badge text-info border-info" id="sy
s-read-sub-badge">0 B/s</span>
</div>
<div class="py-1" id="sys-read-sub-gauge"></div>
<div class="sys-temp-slider-wrap">
<div class="d-flex justify-content-between align-items-center
```

```
mb-1" style="font-size: 0.78rem;">
<span class="text-light fw-semibold"><i class="bi bi-arrow-down-circle me-1 text-info"></i>Чтение</span>
<span class="font-monospace text-info fw-bold" id="sys-memio-read">0 B/s</span>
</div>
<div class="sys-temp-slider-track">
<div class="sys-temp-slider-bar" id="sys-read-sub-bar" style="width: 0%; background: #0dcaf0;"></div>
<div class="sys-temp-slider-thumb" id="sys-read-sub-thumb" style="left: 0%; background: #0dcaf0;"></div>
</div>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="sys-core-card h-100" title="Скорость записи на диск">
<div class="d-flex justify-content-between align-items-center mb-1">
<span class="sys-core-title">Запись (Write)</span>
<span class="sys-core-load-badge text-warning border-warning" id="sys-write-sub-badge">0 B/s</span>
</div>
<div class="py-1" id="sys-write-sub-gauge"></div>
<div class="sys-temp-slider-wrap">
<div class="d-flex justify-content-between align-items-center mb-1" style="font-size: 0.78rem;">
<span class="text-light fw-semibold"><i class="bi bi-arrow-up-circle me-1 text-warning"></i>Запись</span>
<span class="font-monospace text-warning fw-bold" id="sys-memio-write">0 B/s</span>
</div>
<div class="sys-temp-slider-track">
<div class="sys-temp-slider-bar" id="sys-write-sub-bar" style="width: 0%; background: #f59e0b;"></div>
<div class="sys-temp-slider-thumb" id="sys-write-sub-thumb" style="left: 0%; background: #f59e0b;"></div>
```

```
</div>
</div>
</div>
</div>
</div>
<!-- Параметры, тайминги и сенсоры RAM (Скрывается в компактном режиме) -->
<div class="sys-full-only sys-detailed-sensors">
<div class="sys-core-section-title d-flex justify-content-between align-items-center mt-3">
<span><i class="bi bi-lightning-charge text-warning me-1.5"></i>Параметры, тайминги и сенсоры RAM</span>
<span class="badge bg-secondary font-monospace" id="sys-ram-sensors-count-badge" style="font-size: 0.76rem;">0 параметров</span>
</div>
<div class="mb-2" id="sys-metric-ram-sensors" style="display: grid; grid-template-columns: repeat(auto-fill, minmax(180px, 1fr)); gap: 8px; font-size: 0.78rem;">
<div class="text-muted small text-center py-2 col-12">Опрос параметров RAM...</div>
</div>
</div>
<!-- График истории RAM (Использование RAM %, Чтение диска, Запись на диск) с логарифмической/линейной шкалой -->
<div class="sys-full-only sys-history-panel p-2.5 mt-3 rounded-2">
<div class="d-flex justify-content-between align-items-center mb-1.5 flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<span class="small text-light" style="font-size: 0.80rem; font-weight: 700;"><i class="bi bi-graph-up text-success me-1"></i>История RAM и Дискового I/O</span>
<div class="btn-group btn-group-sm" role="group" style="height: 22px;">
<button class="btn btn-xs btn-outline-success active py-0 px-2 fw-bold" id="btn-ram-spark-log" onclick="window.setRamSparkScale &amp;&amp; window.setRamSparkScale('log')" style="font-siz
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```
e: 0.68rem;" type="button">Лог. шкала (Log)</button>
<button class="btn btn-xs btn-outline-secondary py-0 px-2" id="btn-ram-spark-lin" onclick="window.setRamSparkScale &amp;&amp; window.setRamSparkScale('linear')" style="font-size: 0.68rem;" type="button">Линейная (Lin)</button>
</div>
<select class="form-select form-select-xs sys-spark-interval-select" id="sel-ram-spark-interval" onchange="window.setRamSparkInterval &amp;&amp; window.setRamSparkInterval(this.value)" style="height: 22px; padding: 0 6px; font-size: 0.68rem; width: auto; min-width: 85px;">
<option value="seconds">Секунды</option>
</select>
</div>
<div class="d-flex align-items-center gap-2.5" style="font-size: 0.74rem; font-weight: 600;">
<span class="text-success" style="text-shadow: 0 0 8px rgba(16,185,129,0.5);" title="Основная шкала: Загрузка RAM (%)">● Занято RAM %</span>
<span class="text-info" style="text-shadow: 0 0 8px rgba(13,202,240,0.5);" title="Скорость чтения с диска (Read)">● Чтение (Read)</span>
<span class="text-warning" style="text-shadow: 0 0 8px rgba(245,158,11,0.5);" title="Скорость записи на диск (Write)">● Запись (Write)</span>
<span class="text-muted font-monospace ms-2" id="sys-ram-spark-live-cursor" style="font-size: 0.70rem;">Live</span>
</div>
</div>
<!-- Контейнер графика и интерактивного тултипа -->
<div id="sys-memio-spark-container" style="position: relative; width: 100%; height: 90px;">
<div id="sys-memio-spark" style="width: 100%; height: 100%;"></div>
<div class="sys-spark-tooltip" id="sys-memio-spark-tooltip" style="display: none;"></div>
</div>
</div>
```

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</div>
</div>
</div>
<!-- Четкий стильный разделитель панелей -->
<div class="sys-section-divider"></div>
<!-- 4. Панель Сетевые адаптеры и беспроводная связь (Ethernet / Wi-Fi / Bluetooth) -->
<div class="row mb-3">
<div class="col-12">
<div class="sys-card sys-card-net p-3" id="sys-card-net">
<div class="sys-card-header-bar">
<div class="d-flex align-items-center gap-2">
<span class="sys-label text-info" style="font-size: 1.05rem; font-weight: 800;"><i class="bi bi-hdd-network text-info me-1.5"></i>Сетевые адаптеры и беспроводная связь</span>
</div>
<span class="small badge bg-dark border border-secondary text-light font-monospace" id="sys-net-ts" style="font-size: 0.84rem;">--</span>
</div>
<!-- Верхний баннер: Модель слева, Gauge трафика по центру, Слайдер линка справа -->
<div class="sys-card-top-banner p-3 rounded-3 mb-3">
<!-- Левая колонка: Модель сети и экспресс-метрики KPI -->
<div class="sys-banner-col-info d-flex flex-column align-items-start justify-content-center">
<div class="text-start mb-1.5 d-flex align-items-center">
<button class="sys-drill-btn" id="sys-drill-btn-net" onclick="window.toggleSysCardDrill &amp;&amp; window.toggleSysCardDrill('sys-card-net')" title="Drill-down: Развернуть/скрыть все скрытые панели и адаптеры сети" type="button">
<img alt="Дрель" src="/html/assets/red_drill.png?v=20261008_v1"/>
</button>
<div>
<span class="fw-bold text-info" id="sys-metric-net-model" style="font-size: 1.20rem; font-weight: 800; letter-spacing: 0.5px; text-shadow: 0 0 14px rgba(13, 202, 240, 0.4);">Ethernet (10
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00 Mbps) · Wi-Fi / BT</span>
<div class="d-flex align-items-center gap-1.5 mt-0.5">
<span class="badge bg-dark border border-secondary text-light font-monospace" id="sys-net-speed-badge" style="font-size: 0.70rem;">1.0 Gbps Full Duplex</span>
<span class="badge bg-info-subtle text-info border border-info font-monospace" id="sys-net-status-badge" style="font-size: 0.70rem;">● Link Active</span>
</div>
</div>
</div>
<div class="sys-banner-kpi-grid">
<div class="sys-kpi-pill">
<span class="sys-kpi-label"><i class="bi bi-arrow-down-circle text-info me-1"></i>Входящий (Rx)</span>
<span class="sys-kpi-value text-info" id="sys-net-kpi-rx">0 B/s</span>
</div>
<div class="sys-kpi-pill">
<span class="sys-kpi-label"><i class="bi bi-arrow-up-circle text-warning me-1"></i>Исходящий (Tx)</span>
<span class="sys-kpi-value text-warning" id="sys-net-kpi-tx">0 B/s</span>
</div>
<div class="sys-kpi-pill">
<span class="sys-kpi-label"><i class="bi bi-hdd-network text-primary me-1"></i>Всего сессия</span>
<span class="sys-kpi-value text-primary" id="sys-net-kpi-total">0 B</span>
</div>
<div class="sys-kpi-pill">
<span class="sys-kpi-label"><i class="bi bi-activity text-success me-1"></i>Утилизация</span>
<span class="sys-kpi-value text-success" id="sys-net-kpi-util">0.0%</span>
</div>
</div>
</div>
```

```
<!-- Центральная колонка: Большой Gauge трафика -->
<div class="sys-banner-col-gauge d-flex align-items-center gap-2.5">
<div id="sys-metric-net-gauge" style="width: 120px; height: 68px; flex-shrink: 0;"></div>
<div>
<div class="sys-value text-info" id="sys-net-speed-val" style="font-size: 2.1rem; line-height: 1; font-weight: 800;">0 B/s</div>
<div class="text-light mt-1 font-monospace text-truncate" id="sys-net-sub-details" style="font-size: 0.82rem; font-weight: 700; max-width: 150px;">  0 B/s ·   0 B/s</div>
</div>
</div>
<!-- Правая колонка: Загрузка канала со слайдером -->
<div class="sys-banner-col-specs d-flex flex-column justify-content-center">
<span class="text-light mb-1 font-monospace" id="sys-net-util-label" style="font-size: 0.80rem; font-weight: 700;">Загрузка канала: 0%</span>
<div id="sys-metric-net-slider" style="width: 100%;"></div>
</div>
</div>
<!-- Сетка 4 блоков: Download, Upload, Wi-Fi, Bluetooth -->
<div class="row g-2 mb-2 sys-full-only">
<div class="col-6 col-md-3">
<div class="sys-core-card h-100" title="Входящий сетевой трафик (Download / Rx)">
<div class="d-flex justify-content-between align-items-center mb-1">
<span class="sys-core-title">Входящий (Rx)</span>
<span class="sys-core-load-badge text-info border-info" id="sys-rx-sub-badge">0 B/s</span>
</div>
<div class="py-1" id="sys-rx-sub-gauge"></div>
<div class="sys-temp-slider-wrap">
<div class="d-flex justify-content-between align-items-center mb-1" style="font-size: 0.78rem;">
```

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<span class="text-light fw-semibold"><i class="bi bi-arrow-down-circle me-1 text-info"></i>Скачивание</span>
<span class="font-monospace text-info fw-bold text-truncate" id="sys-rx-total-text" style="max-width: 110px;">0 B</span>
</div>
<div class="sys-temp-slider-track">
<div class="sys-temp-slider-bar" id="sys-rx-sub-bar" style="width: 0%; background: #0dcaf0;"></div>
<div class="sys-temp-slider-thumb" id="sys-rx-sub-thumb" style="left: 0%; background: #0dcaf0;"></div>
</div>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="sys-core-card h-100" title="Исходящий сетевой трафик (Upload / Tx)">
<div class="d-flex justify-content-between align-items-center mb-1">
<span class="sys-core-title">Исходящий (Tx)</span>
<span class="sys-core-load-badge text-warning border-warning" id="sys-tx-sub-badge">0 B/s</span>
</div>
<div class="py-1" id="sys-tx-sub-gauge"></div>
<div class="sys-temp-slider-wrap">
<div class="d-flex justify-content-between align-items-center mb-1" style="font-size: 0.78rem;">
<span class="text-light fw-semibold"><i class="bi bi-arrow-up-circle me-1 text-warning"></i>Отдача</span>
<span class="font-monospace text-warning fw-bold text-truncate" id="sys-tx-total-text" style="max-width: 110px;">0 B</span>
</div>
<div class="sys-temp-slider-track">
<div class="sys-temp-slider-bar" id="sys-tx-sub-bar" style="width: 0%; background: #f59e0b;"></div>
<div class="sys-temp-slider-thumb" id="sys-tx-sub-thumb" style="left: 0%; background: #f59e0b;"></div>
</div>
```

```
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="sys-core-card h-100" title="Беспроводной адаптер Wi-Fi">
<div class="d-flex justify-content-between align-items-center mb-1">
<span class="sys-core-title">Wi-Fi Адаптер</span>
<span class="sys-core-load-badge text-primary border-primary" id="sys-wifi-sub-badge">On</span>
</div>
<div class="py-1" id="sys-wifi-sub-gauge"></div>
<div class="sys-temp-slider-wrap">
<div class="d-flex justify-content-between align-items-center mb-1" style="font-size: 0.78rem;">
<span class="text-light fw-semibold"><i class="bi bi-wifi me-1 text-primary"></i>Статус</span>
<span class="font-monospace text-primary fw-bold text-truncate" id="sys-wifi-status-text" style="max-width: 110px;">Disconnected</span>
</div>
<div class="sys-temp-slider-track">
<div class="sys-temp-slider-bar" id="sys-wifi-sub-bar" style="width: 0%; background: #38bdf8;"></div>
<div class="sys-temp-slider-thumb" id="sys-wifi-sub-thumb" style="left: 0%; background: #38bdf8;"></div>
</div>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="sys-core-card h-100" title="Bluetooth контроллер и устройства">
<div class="d-flex justify-content-between align-items-center mb-1">
<span class="sys-core-title">Bluetooth (BT)</span>
<span class="sys-core-load-badge text-success border-success"
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id="sys-bt-sub-badge">Active</span>
</div>
<div class="py-1" id="sys-bt-sub-gauge"></div>
<div class="sys-temp-slider-wrap">
<div class="d-flex justify-content-between align-items-center mb-1" style="font-size: 0.78rem;">
<span class="text-light fw-semibold"><i class="bi bi-bluetooth me-1 text-success"></i>Устройства</span>
<span class="font-monospace text-success fw-bold text-truncate" id="sys-bt-devices-text" style="max-width: 110px;">0 устр.</span>
</div>
<div class="sys-temp-slider-track">
<div class="sys-temp-slider-bar" id="sys-bt-sub-bar" style="width: 0%; background: #10b981;"></div>
<div class="sys-temp-slider-thumb" id="sys-bt-sub-thumb" style="left: 0%; background: #10b981;"></div>
</div>
</div>
</div>
</div>
</div>
<!-- Параметры, адаптеры и сенсоры сети (Скрывается в компактном режиме) -->
<div class="sys-full-only sys-detailed-sensors">
<div class="sys-core-section-title d-flex justify-content-between align-items-center mt-3">
<span><i class="bi bi-lightning-charge text-warning me-1.5"></i>Параметры, адаптеры и сенсоры сети / Wi-Fi / BT</span>
<span class="badge bg-secondary font-monospace" id="sys-net-sensors-count-badge" style="font-size: 0.76rem;">0 параметров</span>
</div>
<div class="mb-2" id="sys-metric-net-sensors" style="display: grid; grid-template-columns: repeat(auto-fill, minmax(180px, 1fr)); gap: 8px; font-size: 0.78rem;">
<div class="text-muted small text-center py-2 col-12">Опрос сетевых интерфейсов...</div>
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</div>
</div>
<!-- График истории сети (Входящий / Исходящий трафик) с логарифмической/линейной шкалой -->
<div class="sys-full-only sys-history-panel p-2.5 mt-3 rounded-2">
<div class="d-flex justify-content-between align-items-center mb-1.5 flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<span class="small text-light" style="font-size: 0.80rem; font-weight: 700;"><i class="bi bi-graph-up text-info me-1"></i>История Сетевого Трафика</span>
<div class="btn-group btn-group-sm" role="group" style="height: 22px;">
<button class="btn btn-xs btn-outline-info active py-0 px-2 fw-bold" id="btn-net-spark-log" onclick="window.setNetSparkScale &amp;&amp; window.setNetSparkScale('log')" style="font-size: 0.68rem;" type="button">Лог. шкала (Log)</button>
<button class="btn btn-xs btn-outline-secondary py-0 px-2" id="btn-net-spark-lin" onclick="window.setNetSparkScale &amp;&amp; window.setNetSparkScale('linear')" style="font-size: 0.68rem;" type="button">Линейная (Lin)</button>
</div>
<select class="form-select form-select-xs sys-spark-interval-select" id="sel-net-spark-interval" onchange="window.setNetSparkInterval &amp;&amp; window.setNetSparkInterval(this.value)" style="height: 22px; padding: 0 6px; font-size: 0.68rem; width: auto; min-width: 85px;">
<option value="seconds">Секунды</option>
</select>
</div>
<div class="d-flex align-items-center gap-2.5" style="font-size: 0.74rem; font-weight: 600;">
<span class="text-info" style="text-shadow: 0 0 8px rgba(13,202,240,0.5);" title="Скорость приема (Download / Rx)">● Входящий (Rx)</span>
<span class="text-warning" style="text-shadow: 0 0 8px rgba(245,158,11,0.5);" title="Скорость передачи (Upload / Tx)">● Исхо
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дящий (Tx)</span>
<span class="text-muted font-monospace ms-2" id="sys-net-spark-live-cursor" style="font-size: 0.70rem;">Live</span>
</div>
</div>
<!-- Контейнер графика и интерактивного тултипа -->
<div id="sys-net-spark-container" style="position: relative; width: 100%; height: 90px;">
<div id="sys-net-spark" style="width: 100%; height: 100%;"></div>
<div class="sys-spark-tooltip" id="sys-net-spark-tooltip" style="display: none;"></div>
</div>
</div>
</div>
</div>
</div>
<!-- Четкий стильный разделитель панелей -->
<div class="sys-section-divider"></div>
<!-- 5. Панель Параметры накопителей (Диски SSD / HDD / NVMe) -->
<div class="row mb-3">
<div class="col-12">
<div class="sys-card sys-card-storage p-3" id="sys-card-storage">
<div class="sys-card-header-bar">
<div class="d-flex align-items-center gap-2">
<span class="sys-label text-info" style="font-size: 1.05rem; font-weight: 800;"><i class="bi bi-hdd-fill text-info me-1.5"></i>Параметры накопителей (Диски SSD / HDD / NVMe)</span>
<span class="badge bg-info-subtle text-info border border-info" id="sys-storage-drives-badge" style="font-size: 0.74rem;">0 дисков</span>
<span class="badge bg-dark border border-secondary text-light" id="sys-storage-parts-badge" style="font-size: 0.74rem;">0 разделов</span>
</div>
<span class="small badge bg-dark border border-secondary text-
```

```
light font-monospace" id="sys-storage-ts" style="font-size: 0.
84rem;">Live</span>
</div>
<!-- Верхний баннер: Модель слева, Gauge общего объема по цент
ру, Слайдер макс t° справа -->
<div class="sys-card-top-banner p-3 rounded-3 mb-3">
<!-- Левая колонка: Модель хранилища и экспресс-метрики KPI --
>
<div class="sys-banner-col-info d-flex flex-column align-items
-start justify-content-center">
<div class="text-start mb-1.5 d-flex align-items-center">
<button class="sys-drill-btn" id="sys-drill-btn-storage" oncli
ck="window.toggleSysCardDrill &amp;amp;&amp;amp; window.toggleSysCardD
rill('sys-card-storage')" title="Drill-down: Развернуть/скрыть
 все скрытые панели накопителей" type="button">
<img alt="Дрель" src="/html/assets/red_drill.png?v=20261008_v1
"/>
</button>
<div>
<span class="fw-bold text-info" id="sys-metric-storage-model"
style="font-size: 1.20rem; font-weight: 800; letter-spacing: 0
.5px; text-shadow: 0 0 14px rgba(13, 202, 240, 0.4);">Дисковая
 подсистема и накопители</span>
<div class="d-flex align-items-center gap-1.5 mt-0.5">
<span class="badge bg-dark border border-secondary text-light
font-monospace" id="sys-storage-summary-badge-top" style="font
-size: 0.70rem;">NVMe / SSD / HDD</span>
<span class="badge bg-success-subtle text-success border borde
r-success font-monospace" id="sys-storage-status-badge" style=
"font-size: 0.70rem;">● SMART OK</span>
</div>
</div>
</div>
<div class="sys-banner-kpi-grid">
<div class="sys-kpi-pill">
<span class="sys-kpi-label"><i class="bi bi-pie-chart text-inf
o me-1"></i>Занято всего</span>
<span class="sys-kpi-value text-info" id="sys-storage-kpi-used
```

```
">0.0 GB</span>
</div>
<div class="sys-kpi-pill">
<span class="sys-kpi-label"><i class="bi bi-check2-circle text-success me-1"></i>Свободно</span>
<span class="sys-kpi-value text-success" id="sys-storage-kpi-free">0.0 GB</span>
</div>
<div class="sys-kpi-pill">
<span class="sys-kpi-label"><i class="bi bi-hdd text-primary me-1"></i>Диски/Разделы</span>
<span class="sys-kpi-value text-primary" id="sys-storage-kpi-drives">-- / --</span>
</div>
<div class="sys-kpi-pill">
<span class="sys-kpi-label"><i class="bi bi-thermometer-half text-danger me-1"></i>Макс. t°</span>
<span class="sys-kpi-value text-warning" id="sys-storage-kpi-temp">-- °C</span>
</div>
</div>
</div>
<!-- Центральная колонка: Большой Gauge общего объема -->
<div class="sys-banner-col-gauge d-flex align-items-center gap-2.5">
<div id="sys-metric-storage-gauge" style="width: 120px; height: 68px; flex-shrink: 0;"></div>
<div>
<div class="sys-value text-info" id="sys-storage-pct" style="font-size: 2.1rem; line-height: 1; font-weight: 800;">0.0%</div>
<div class="text-light mt-1 font-monospace text-truncate" id="sys-storage-used" style="font-size: 0.82rem; font-weight: 700; max-width: 150px;">0.0 / 0.0 GB</div>
</div>
</div>
<!-- Правая колонка: Слайдер максимальной t° -->
<div class="sys-banner-col-specs d-flex flex-column justify-co
```

```
ntent-center">
<span class="text-light mb-1 font-monospace" id="sys-storage-temp-label" style="font-size: 0.80rem; font-weight: 700;">Макс. температура: -- °C</span>
<div id="sys-metric-storage-temp-slider" style="width: 100%;"></div>
</div>
</div>
<!-- График истории дисковой подсистемы (Чтение / Запись / Нагрузка I/O) с логарифмической/линейной шкалой -->
<div class="sys-full-only sys-history-panel p-2.5 mb-3 rounded-2">
<div class="d-flex justify-content-between align-items-center mb-1.5 flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<span class="small text-light" style="font-size: 0.80rem; font-weight: 700;"><i class="bi bi-graph-up text-info me-1"></i>История Накопителей и Дискового I/O</span>
<div class="btn-group btn-group-sm" role="group" style="height: 22px;">
<button class="btn btn-xs btn-outline-info active py-0 px-2 fw-bold" id="btn-storage-spark-log" onclick="window.setStorageSparkScale &amp;&amp; window.setStorageSparkScale('log')" style="font-size: 0.68rem;" type="button">Лог. шкала (Log)</button>
<button class="btn btn-xs btn-outline-secondary py-0 px-2" id="btn-storage-spark-lin" onclick="window.setStorageSparkScale &amp;&amp; window.setStorageSparkScale('linear')" style="font-size: 0.68rem;" type="button">Линейная (Lin)</button>
</div>
<select class="form-select form-select-xs sys-spark-interval-select" id="sel-storage-spark-interval" onchange="window.setStorageSparkInterval &amp;&amp; window.setStorageSparkInterval(this.value)" style="height: 22px; padding: 0 6px; font-size: 0.68rem; width: auto; min-width: 85px;">
<option value="seconds">Секунды</option>
</select>
</div>
<div class="d-flex align-items-center gap-2.5" style="font-siz
```

```
e: 0.74rem; font-weight: 600;">
<span class="text-info" style="text-shadow: 0 0 8px rgba(13,202,240,0.5);" title="Скорость чтения с накопителей (Read)">● Чтение (Read)</span>
<span class="text-warning" style="text-shadow: 0 0 8px rgba(245,158,11,0.5);" title="Скорость записи на накопители (Write)">● Запись (Write)</span>
<span class="text-muted font-monospace ms-2" id="sys-storage-spark-live-cursor" style="font-size: 0.70rem;">Live</span>
</div>
</div>
<!-- Контейнер графика и интерактивного тултипа -->
<div id="sys-storage-spark-container" style="position: relative; width: 100%; height: 90px;">
<div id="sys-storage-spark" style="width: 100%; height: 100%;"></div>
<div class="sys-spark-tooltip" id="sys-storage-spark-tooltip" style="display: none;"></div>
</div>
</div>
<!-- Сетка карточек накопителей и разделов (растянута по всей ширине) -->
<div class="sys-full-only">
<div class="sys-core-section-title d-flex justify-content-between align-items-center mb-2">
<span><i class="bi bi-hdd-network text-info me-1.5"></i>Физические накопители и разделы (полная ширина)</span>
<span class="badge bg-secondary font-monospace" id="sys-storage-summary-badge" style="font-size: 0.76rem;">Автоматический опрос</span>
</div>
<div id="sys-metric-storage-cards" style="display: flex; flex-direction: column; gap: 12px; width: 100%; font-size: 0.78rem; ">
<div class="text-muted small text-center py-3 col-12">Опрос накопителей выполняется...</div>
</div>
</div>
```

```
</div>
</div>
</div>
<!-- Четкий стильный разделитель панелей -->
<div class="sys-section-divider sys-full-only"></div>
<!-- Extended Hardware & Health Status Strip -->
<div class="row g-2 mb-2 sys-full-only">
<!-- Physical Storage & NVMe Health -->
<div class="col-md-6 col-lg-3">
<div class="sys-panel p-2 h-100" data-panel-id="panel-system-inspector-storage" id="panel-system-inspector-storage">
<div class="d-flex justify-content-between align-items-center mb-1">
<span class="small fw-bold" style="font-size: 0.76rem; color: var(--text-color);"><i class="bi bi-hdd-network text-info me-1"></i> Здоровье дисков (NVMe/SSD)</span>
<span class="badge bg-success-subtle text-success border border-success" id="sys-disk-health-badge" style="font-size: 0.65rem;">SMART OK</span>
</div>
<div class="small text-muted" id="sys-physical-disks-container" style="font-size: 0.72rem; line-height: 1.3; max-height: 120px; overflow-y: auto;">
          Опрос состояния физических дисков...
        </div>
</div>
</div>
<!-- Power & Battery -->
<div class="col-md-6 col-lg-3">
<div class="sys-panel p-2 h-100" data-panel-id="panel-system-inspector-pitanie-i-energiya" id="panel-system-inspector-pitanie-i-energiya">
<div class="d-flex justify-content-between align-items-center mb-1">
<span class="small fw-bold" style="font-size: 0.76rem; color: var(--text-color);"><i class="bi bi-battery-charging text-warning me-1"></i> Питание и Энергия</span>
<span class="badge bg-secondary" id="sys-power-status-badge" s
```

```
tyle="font-size: 0.65rem;">AC Mains</span>
</div>
<div class="small text-muted" id="sys-power-info-container" st
yle="font-size: 0.72rem; line-height: 1.3;">
          Питание от электросети (Стационарный ПК)
        </div>
</div>
</div>
<!-- RAM SPD Sticks -->
<div class="col-md-6 col-lg-3">
<div class="sys-panel p-2 h-100" data-panel-id="panel-system-i
nspector-moduli-ram-spd" id="panel-system-inspector-moduli-ram
-spd">
<div class="d-flex justify-content-between align-items-center
mb-1">
<span class="small fw-bold" style="font-size: 0.76rem; color:
var(--text-color);"><i class="bi bi-memory text-primary me-1">
</i> Модули RAM (SPD)</span>
<span class="badge bg-dark border border-secondary text-info"
id="sys-ram-sticks-badge" style="font-size: 0.65rem;">Слоты</s
pan>
</div>
<div class="small text-muted" id="sys-ram-sticks-container" st
yle="font-size: 0.72rem; line-height: 1.3;">
          Опрос конфигурации планок памяти...
        </div>
</div>
</div>
<!-- Reliability & Open Ports -->
<div class="col-md-6 col-lg-3">
<div class="sys-panel p-2 h-100" data-panel-id="panel-system-i
nspector-stabilnost-i-set" id="panel-system-inspector-stabilno
st-i-set">
<div class="d-flex justify-content-between align-items-center
mb-1">
<span class="small fw-bold" style="font-size: 0.76rem; color:
var(--text-color);"><i class="bi bi-shield-check text-success
me-1"></i> Стабильность и Сеть</span>
```

```
<span class="badge bg-info-subtle text-info border border-info
" id="sys-alerts-badge" style="font-size: 0.65rem;">Стабильно<
/span>
</div>
<div class="small text-muted" id="sys-alerts-ports-container"
style="font-size: 0.72rem; line-height: 1.3;">
          Службы активны. Ожидание телеметрии портов.
        </div>
</div>
</div>
</div>
</div>
<!-- Telemetry & Polling Intervals Editor Modal (config.json)
-->
<div aria-hidden="true" aria-labelledby="sysIntervalsModalLabe
l" class="modal fade" id="sysIntervalsModal" tabindex="-1">
<div class="modal-dialog modal-dialog-centered modal-lg modal-
dialog-scrollable">
<div class="modal-content text-white border-secondary shadow-l
g" style="background: var(--surface-1); border-color: var(--bo
rder-color) !important;">
<div class="modal-header py-2.5 px-3 border-secondary" style="
background: var(--surface-2); border-color: var(--border-color
) !important;">
<h6 class="modal-title d-flex align-items-center gap-2 mb-0" i
d="sysIntervalsModalLabel">
<i class="bi bi-stopwatch text-primary"></i>
<span class="fw-bold">Редактор интервалов опроса ресурсов</spa
n>
<span class="badge bg-secondary font-monospace" id="modal-inte
rvals-cfg-file">config.json</span>
</h6>
<button aria-label="Close" class="btn-close btn-close-white" d
ata-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body p-3" style="max-height: 75vh; overflow-
y: auto;">
<!-- Alert message -->
```

```
<div class="alert d-none py-2 px-3 small mb-3" id="modal-intervals-alert"></div>
<!-- Engine Status Strip -->
<div class="p-2.5 rounded mb-3 d-flex align-items-center justify-content-between flex-wrap gap-2" style="background: var(--surface-2); border: 1px solid var(--border-color);">
<div class="d-flex align-items-center gap-2">
<span class="fs-4">□ </span>
<div>
<div class="fw-bold small" style="color: var(--text-color);">Служба автологгирования и опроса телеметрии</div>
<div class="small text-muted" style="font-size: 0.73rem;">Интервалы сохраняются напрямую в активный файл конфигурации и применяются без перезапуска сервера</div>
</div>
</div>
<div class="d-flex align-items-center gap-2">
<span class="badge bg-success" id="modal-intervals-engine-status">AutoLog Active</span>
<div class="form-check form-switch m-0">
<input checked="" class="form-check-input" id="modal-intervals-global-enable" type="checkbox"/>
<label class="form-check-label small fw-semibold" for="modal-intervals-global-enable">Автологгирование</label>
</div>
</div>
</div>
<!-- Section 1: Live UI Refresh Interval -->
<div class="card border-secondary mb-3" data-panel-id="panel-system-inspector-interval-obnovleniya-live-inter" id="panel-system-inspector-interval-obnovleniya-live-inter" style="background: var(--surface-2); border-color: var(--border-color) !important;">
<div class="card-header py-1.5 px-3 border-secondary d-flex align-items-center justify-content-between" style="border-color: var(--border-color) !important;">
<span class="small fw-bold text-info"><i class="bi bi-display me-1"></i> Интервал обновления Live-интерфейса вкладки (UI)</s
```

```
pan>
<span class="badge bg-dark text-info border border-secondary" id="modal-ui-refresh-badge">5 сек</span>
</div>
<div class="card-body p-3">
<div class="row g-2 align-items-center">
<div class="col-sm-7">
<label class="form-label small text-muted mb-0">Частота опроса сенсоров и оборудования в открытом окне браузера:</label>
</div>
<div class="col-sm-5">
<select class="form-select form-select-sm bg-dark text-white border-secondary" id="modal-ui-refresh-select">
<option value="1">1 секунда (Высокая частота)</option>
<option value="2">2 секунды</option>
<option value="3">3 секунды</option>
<option selected="" value="5">5 секунд (По умолчанию)</option>
<option value="10">10 секунд (Экономный режим)</option>
<option value="30">30 секунд</option>
</select>
</div>
</div>
</div>
</div>
<!-- Section 2: Core Resource Loggers (config.json) -->
<div class="card border-secondary mb-3" data-panel-id="panel-system-inspector-intervaly-oprosa-osnovnykh-sens" id="panel-system-inspector-intervaly-oprosa-osnovnykh-sens" style="background: var(--surface-2); border-color: var(--border-color) !important;">
<div class="card-header py-1.5 px-3 border-secondary d-flex align-items-center justify-content-between" style="border-color: var(--border-color) !important;">
<span class="small fw-bold text-primary"><i class="bi bi-hdd-stack me-1"></i> Интервалы опроса основных сенсоров и ресурсов (config.json)</span>
<span class="badge bg-secondary font-monospace" style="font-size: 0.68rem;">logging.loggers</span>
```

```
</div>
<div class="card-body p-3">
<div class="d-flex flex-column gap-2.5" id="modal-core-loggers-container">
<!-- Dynamically populated core loggers -->
</div>
</div>
</div>
<!-- Section 3: Other Services Loggers Collapsible -->
<div class="card border-secondary mb-3" data-panel-id="panel-system-inspector-1" id="panel-system-inspector-1" style="background: var(--surface-2); border-color: var(--border-color) !important;">
<div class="card-header py-1.5 px-3 border-secondary d-flex align-items-center justify-content-between" style="border-color: var(--border-color) !important;">
<a aria-expanded="false" class="small fw-bold text-light text-decoration-none d-flex align-items-center gap-1" data-bs-toggle="collapse" href="#modal-other-loggers-collapse" role="button">
<i class="bi bi-chevron-down"></i> Все остальные логгеры и службы системы (<span id="modal-other-loggers-count">0</span>)
            </a>
<span class="small text-muted" style="font-size: 0.72rem;">Windows SysAdmin, Defender, StartUp, Net...</span>
</div>
<div class="collapse" id="modal-other-loggers-collapse">
<div class="card-body p-3 border-top border-secondary" style="border-color: var(--border-color) !important;">
<div class="d-flex flex-column gap-2" id="modal-other-loggers-container">
<!-- Dynamically populated other loggers -->
</div>
</div>
</div>
</div>
<!-- Section 4: Default fallback interval -->
<div class="p-2.5 rounded d-flex align-items-center justify-co
```

```
ntent-between flex-wrap gap-2" style="background: var(--bg-col
or); border: 1px solid var(--border-color);">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-clock-history text-muted"></i>
<span class="small" style="color: var(--text-color);">Интервал
 по умолчанию (default_interval):</span>
</div>
<div style="min-width: 170px;">
<select class="form-select form-select-sm bg-dark text-white b
order-secondary" id="modal-intervals-default-select">
<option value="5 seconds">5 seconds (5 сек)</option>
<option value="10 seconds">10 seconds (10 сек)</option>
<option value="30 seconds">30 seconds (30 сек)</option>
<option selected="" value="1 minute">1 minute (1 мин)</option>
<option value="5 minutes">5 minutes (5 мин)</option>
<option value="10 minutes">10 minutes (10 мин)</option>
<option value="1 hour">1 hour (1 час)</option>
</select>
</div>
</div>
</div>
<div class="modal-footer py-2 px-3 border-secondary d-flex jus
tify-content-between" style="background: var(--surface-2); bor
der-color: var(--border-color) !important;">
<button class="btn btn-sm btn-outline-secondary rounded-pill p
x-3" id="btn-modal-intervals-refresh" title="Перезагрузить тек
ущую конфигурацию из файла">
<i class="bi bi-arrow-clockwise me-1"></i> Обновить
        </button>
<div class="d-flex gap-2">
<button class="btn btn-sm btn-secondary rounded-pill px-3" dat
a-bs-dismiss="modal">Отмена</button>
<button class="btn btn-sm btn-primary rounded-pill px-4 fw-bol
d" id="btn-modal-intervals-save">
<i class="bi bi-check2-circle me-1"></i> Сохранить в config.js
on
          </button>
</div>
```

```
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows System Inspector Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/system_inspector_tab/m
ain.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/system_inspector_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-12 08:13:00
 * ============================================================
==================
 */

// System & Hardware Inspector Tab JS Module
(function() {
  let sysWs = null;
  let isSysInitialized = false;
  let isSysPaused = false;
  let isNetPaused = false;
  let currentNetFilter = 'internet';
  let cachedNetworkActivities = [];
  let latestTelemetrySnapshot = null;
  let cachedSensors = [];
  let currentSensorCategory = 'all';
  let isLhmRunning = false;
```

```
  let cachedHardwareSensorsData = null;
  let lastCpuCoresData = [];
  let lastCpuSensorsData = [];
  let lastCpuData = null;
  let lastGpuData = null;
  let lastGpuDevices = [];
  const _gpuSparkHistories = {};
  const currentGpuSparkScales = {};
  const currentGpuSparkIntervals = {};
  const lastGpuSparkDataMap = {};

  let currentCpuSparkInterval = 'seconds';
  let currentRamSparkInterval = 'seconds';
  let currentNetSparkInterval = 'seconds';
  let currentStorageSparkInterval = 'seconds';
  let cachedAvailableTimeRanges = null;

  function formatSparkTimeLabel(ts, interval = 'seconds') {
    if (!ts) return '--:--:--';
    let d = null;
    if (typeof ts === 'number') {
      d = new Date(ts > 1e11 ? ts : ts * 1000);
    } else {
      d = new Date(ts);
    }
    if (isNaN(d.getTime())) return String(ts);

    const pad = (n) => String(n).padStart(2, '0');
    const hh = pad(d.getHours());
    const mm = pad(d.getMinutes());
    const ss = pad(d.getSeconds());
    const DD = pad(d.getDate());
    const MM = pad(d.getMonth() + 1);
    const YY = String(d.getFullYear()).slice(-2);

    if (interval === 'seconds' || interval === 'minutes') {
      return `${hh}:${mm}:${ss}`;
    }
```

```
    if (interval === 'hours') {
      return `${hh}:${mm}`;
    }
    if (interval === 'days' || interval === 'weeks') {
      return `${DD}.${MM} ${hh}:${mm}`;
    }
    return `${DD}.${MM}.${YY} ${hh}:${mm}`;
  }

  async function fetchAvailableTimeRanges() {
    try {
      const res = await fetch('/api/v1/telemetry/time-ranges')
;
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      cachedAvailableTimeRanges = data;
      populateIntervalSelectors(data.intervals || []);
      return data;
    } catch (e) {
      console.warn('[SystemInspectorTab] Ошибка загрузки досту
пных временных диапазонов:', e);
      const fallbackIntervals = [
        { id: 'seconds', label: 'Секунды', is_available: true
},
        { id: 'all', label: 'Все', is_available: true }
      ];
      populateIntervalSelectors(fallbackIntervals);
    }
  }

  function populateIntervalSelectors(intervals) {
    if (!Array.isArray(intervals) || intervals.length === 0) r
eturn;

    const selectIds = [
      'sel-cpu-spark-interval',
      'sel-ram-spark-interval',
      'sel-net-spark-interval',
```

```
      'sel-storage-spark-interval'
    ];

    Object.keys(currentGpuSparkIntervals).forEach(k => {
      const gId = `sel-gpu-spark-interval-${k}`;
      if (!selectIds.includes(gId)) selectIds.push(gId);
    });

    document.querySelectorAll('.sys-spark-interval-select').forEach(el => {
      if (el.id && !selectIds.includes(el.id)) selectIds.push(el.id);
    });

    selectIds.forEach(id => {
      const el = document.getElementById(id);
      if (!el) return;
      const currentVal = el.value || 'seconds';
      el.innerHTML = intervals.map(item => `
        <option value="${escapeHtml(item.id)}">${escapeHtml(item.label)}</option>
      `).join('');
      if (intervals.some(i => i.id === currentVal)) {
        el.value = currentVal;
      } else {
        el.value = intervals[0].id;
      }
    });
  }

  async function fetchLhmSensors() {
    const badgeCount = document.getElementById('sys-lhm-sensors-count');
    const badgeStatus = document.getElementById('sys-lhm-status-badge');
    const container = document.getElementById('sys-lhm-sensors-container');
    const maxTempLabel = document.getElementById('sys-sensors-
```

```
max-temp');
    const warningsBadge = document.getElementById('sys-sensors
-warnings-badge');
    const dbTsBadge = document.getElementById('sys-sensors-db-
ts');

    try {
      const res = await fetch('/api/v1/panel/hardware-sensors'
);
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      cachedHardwareSensorsData = data;
      cachedSensors = data.all_sensors || [];

      const summary = data.summary || {};

      if (badgeCount) {
        badgeCount.textContent = `${summary.total_sensors || c
achedSensors.length} шт. (БД)`;
      }
      if (badgeStatus) {
        badgeStatus.textContent = '● telemetry.db';
        badgeStatus.className = 'badge bg-success';
        if (summary.timestamp) {
          const tsStr = new Date(summary.timestamp).toLocaleTi
meString();
          badgeStatus.title = `Данные из базы telemetry.db: ${
tsStr}`;
          if (dbTsBadge) dbTsBadge.textContent = tsStr;
        }
      }
      if (maxTempLabel) {
        maxTempLabel.textContent = summary.max_temperature_c !
= null ? `${summary.max_temperature_c.toFixed(0)} °C` : '-- °C
';
        if (summary.max_temperature_c && summary.max_temperatu
re_c >= 75) {
          maxTempLabel.className = 'fw-bold text-danger font-m
```

```
onospace';
165         } else if (summary.max_temperature_c && summary.max_te
mperature_c >= 60) {
166           maxTempLabel.className = 'fw-bold text-warning font-
monospace';
167         } else {
168           maxTempLabel.className = 'fw-bold text-info font-mon
ospace';
169         }
170       }
171       if (warningsBadge) {
172         warningsBadge.textContent = summary.warnings_count ||
'0';
173         warningsBadge.className = summary.warnings_count > 0 ?
 'badge bg-danger text-white' : 'badge bg-dark border border-s
econdary text-success';
174       }
175
176       renderSensors();
177     } catch (e) {
178       console.warn('[SystemInspectorTab] Failed to fetch hardw
are sensors from DB:', e);
179       if (container) {
180         container.innerHTML = `<div class="text-center py-4 te
xt-muted small">Датчики опрашиваются из БД... (${e.message})</
div>`;
181       }
182     }
183   }
184
185   function createSensorWidgetHtml(s) {
186     const cat = (s.sensor_category || '').toLowerCase();
187     const val = Number(s.value || 0);
188     const valRaw = s.value_raw || `${val.toFixed(1)} ${s.unit
|| ''}`.trim();
189     const name = escapeHtml(s.sensor_name);
190
191     if (cat.includes('temp')) {
```

```
      const tempSlider = createTempSliderHtml(val, 25, 95, fal
se);
      const gaugeSvg = createGaugeSvg(Math.min(100, Math.max(0
, ((val - 25) / 70) * 100)), 64, 38);
      return `
        <div class="sys-sensor-widget-item p-1.5 rounded-2 mb-
1" style="background: rgba(255,255,255,0.02); border: 1px soli
d var(--border-subtle, rgba(255,255,255,0.06));">
          <div class="d-flex justify-content-between align-ite
ms-center mb-1">
            <span class="sys-sensor-name text-truncate" title=
"${name}" style="font-size: 0.74rem; font-weight: 600;">${name
}</span>
            <span class="badge ${val >= 80 ? 'bg-danger' : val
 >= 65 ? 'bg-warning text-dark' : 'bg-info-subtle text-info'}"
 style="font-size: 0.65rem;">${val.toFixed(0)}°C</span>
          </div>
          <div class="d-flex align-items-center gap-2">
            <div style="width: 64px; height: 38px; flex-shrink
: 0;">${gaugeSvg}</div>
            <div class="flex-grow-1">${tempSlider}</div>
          </div>
        </div>
      `;
    }

    if (cat.includes('load')) {
      const pctSlider = createPercentSliderHtml(val, false, `$
{val.toFixed(0)}%`);
      const gaugeSvg = createGaugeSvg(val, 64, 38);
      return `
        <div class="sys-sensor-widget-item p-1.5 rounded-2 mb-
1" style="background: rgba(255,255,255,0.02); border: 1px soli
d var(--border-subtle, rgba(255,255,255,0.06));">
          <div class="d-flex justify-content-between align-ite
ms-center mb-1">
            <span class="sys-sensor-name text-truncate" title=
"${name}" style="font-size: 0.74rem; font-weight: 600;">${name
```

```
}</span>
            <span class="badge ${val >= 85 ? 'bg-danger' : val >= 60 ? 'bg-warning text-dark' : 'bg-success-subtle text-success'}" style="font-size: 0.65rem;">${val.toFixed(0)}%</span>
          </div>
          <div class="d-flex align-items-center gap-2">
            <div style="width: 64px; height: 38px; flex-shrink: 0;">${gaugeSvg}</div>
            <div class="flex-grow-1">${pctSlider}</div>
          </div>
        </div>
      `;
    }

    if (cat.includes('fan')) {
      const pct = Math.min(100, Math.max(0, (val / 2800) * 100));
      return `
        <div class="sys-sensor-widget-item p-1.5 rounded-2 mb-1" style="background: rgba(255,255,255,0.02); border: 1px solid var(--border-subtle, rgba(255,255,255,0.06));">
          <div class="d-flex justify-content-between align-items-center mb-1">
            <span class="sys-sensor-name text-truncate" title="${name}" style="font-size: 0.74rem; font-weight: 600;"><i class="bi bi-fan text-info me-1"></i>${name}</span>
            <span class="font-monospace text-info fw-bold" style="font-size: 0.76rem;">${val.toFixed(0)} RPM</span>
          </div>
          <div class="progress" style="height: 5px; background: rgba(255,255,255,0.1); border-radius: 3px;">
            <div class="progress-bar bg-info" style="width: ${pct}%;"></div>
          </div>
        </div>
      `;
    }

```

```
    if (cat.includes('volt') || cat.includes('power')) {
      const isPower = cat.includes('power') || (s.unit || '').toLowerCase() === 'w';
      return `
        <div class="sys-sensor-widget-item p-1.5 rounded-2 mb-1" style="background: rgba(255,255,255,0.02); border: 1px solid var(--border-subtle, rgba(255,255,255,0.06));">
          <div class="d-flex justify-content-between align-items-center">
            <span class="sys-sensor-name text-truncate" title="${name}" style="font-size: 0.74rem; font-weight: 600;"><i class="bi bi-lightning-charge text-warning me-1"></i>${name}</span>
            <span class="badge ${isPower ? 'bg-warning-subtle text-warning' : 'bg-primary-subtle text-primary'} font-monospace" style="font-size: 0.72rem;">${valRaw}</span>
          </div>
        </div>
      `;
    }

    if (cat.includes('clock')) {
      return `
        <div class="sys-sensor-widget-item p-1.5 rounded-2 mb-1" style="background: rgba(255,255,255,0.02); border: 1px solid var(--border-subtle, rgba(255,255,255,0.06));">
          <div class="d-flex justify-content-between align-items-center">
            <span class="sys-sensor-name text-truncate" title="${name}" style="font-size: 0.74rem; font-weight: 600;"><i class="bi bi-speedometer2 text-primary me-1"></i>${name}</span>
            <span class="font-monospace text-primary fw-bold" style="font-size: 0.74rem;">${valRaw}</span>
          </div>
        </div>
      `;
    }

```

```
    return `
      <div class="sys-sensor-widget-item p-1.5 rounded-2 mb-1" style="background: rgba(255,255,255,0.02); border: 1px solid var(--border-subtle, rgba(255,255,255,0.06));">
        <div class="d-flex justify-content-between align-items-center">
          <span class="sys-sensor-name text-truncate" title="${name}" style="font-size: 0.74rem; font-weight: 600;">${name}</span>
          <span class="font-monospace text-light fw-bold" style="font-size: 0.74rem;">${valRaw}</span>
        </div>
      </div>
    `;
  }

  function renderSensors() {
    const container = document.getElementById('sys-lhm-sensors-container');
    if (!container) return;

    const searchInput = document.getElementById('sys-lhm-search');
    const filterText = (searchInput ? searchInput.value : '').trim().toLowerCase();

    if (!Array.isArray(cachedSensors) || cachedSensors.length === 0) {
      container.innerHTML = `
        <div class="text-center py-4 text-muted small">
          <div>В базе данных telemetry.db сенсоры не обнаружены</div>
          <div class="mt-2 text-muted" style="font-size: 0.72rem;">Ожидание сбора и сохранения телеметрии сенсоров...</div>
        </div>
      `;
      return;
    }
```

```
289
290     const filtered = cachedSensors.filter(s => {
291       // Category filter
292       if (currentSensorCategory !== 'all') {
293         const catLower = (s.sensor_category || '').toLowerCase
    ();
294         if (currentSensorCategory === 'temperature' && !catLow
    er.includes('temp')) return false;
295         if (currentSensorCategory === 'load' && !catLower.incl
    udes('load')) return false;
296         if (currentSensorCategory === 'fan' && !catLower.inclu
    des('fan') && !catLower.includes('control')) return false;
297         if (currentSensorCategory === 'voltage' && !catLower.i
    ncludes('volt') && !catLower.includes('power')) return false;
298         if (currentSensorCategory === 'clock' && !catLower.inc
    ludes('clock')) return false;
299         if (currentSensorCategory === 'storage' && !catLower.i
    ncludes('data') && !catLower.includes('throughput')) return fa
    lse;
300       }
301
302       // Text search
303       if (filterText) {
304         const hName = (s.hardware_name || '').toLowerCase();
305         const sName = (s.sensor_name || '').toLowerCase();
306         const sCat = (s.sensor_category || '').toLowerCase();
307         const valStr = String(s.value_raw || s.value || '').to
    LowerCase();
308         return hName.includes(filterText) || sName.includes(fi
    lterText) || sCat.includes(filterText) || valStr.includes(filt
    erText);
309       }
310       return true;
311     });
312
313     if (filtered.length === 0) {
314       container.innerHTML = `<div class="text-center py-4 text
    -muted small">По выбранному фильтру сенсоров не найдено</div>`
```

```
;
315       return;
316     }
317
318     // Group sensors by Hardware Name
319     const groups = {};
320     filtered.forEach(s => {
321       const grpKey = s.hardware_name || 'Оборудование';
322       if (!groups[grpKey]) groups[grpKey] = [];
323       groups[grpKey].push(s);
324     });
325
326     container.innerHTML = Object.entries(groups).map(([grpName
, items]) => {
327       const type = (items[0].hardware_type || '').toLowerCase(
);
328       let icon = 'bi bi-cpu';
329       let iconColor = 'text-info';
330       if (type.includes('gpu')) { icon = 'bi bi-gpu-card'; ico
nColor = 'text-warning'; }
331       else if (type.includes('storage') || type.includes('disk
')) { icon = 'bi bi-hdd'; iconColor = 'text-success'; }
332       else if (type.includes('memory') || type.includes('ram')
) { icon = 'bi bi-memory'; iconColor = 'text-primary'; }
333       else if (type.includes('network') || type.includes('net'
)) { icon = 'bi bi-globe'; iconColor = 'text-info'; }
334       else if (type.includes('motherboard') || type.includes('
mainboard')) { icon = 'bi bi-motherboard'; iconColor = 'text-s
econdary'; }
335
336       const temps = items.filter(i => (i.sensor_category || ''
).toLowerCase().includes('temp')).map(i => Number(i.value || 0
));
337       const maxT = temps.length ? Math.max(...temps) : null;
338       const tempBadge = maxT != null ? `<span class="badge ${m
axT >= 80 ? 'bg-danger' : maxT >= 65 ? 'bg-warning text-dark'
: 'bg-info-subtle text-info'}" style="font-size: 0.65rem;">
${maxT.toFixed(0)}°C</span>` : '';
```

```
339
340       const widgetsHtml = items.map(s => createSensorWidgetHtml(s)).join('');
341
342       return `
343         <div class="sys-core-card mb-2 p-2" style="min-height: auto; background: var(--surface-2, rgba(255,255,255,0.02)); border: 1px solid var(--border-color, rgba(255,255,255,0.08));">
344           <div class="d-flex justify-content-between align-items-center mb-1.5 pb-1 border-bottom" style="border-color: var(--border-subtle, rgba(255,255,255,0.06)) !important;">
345             <div class="d-flex align-items-center gap-1.5 text-truncate" style="max-width: 70%;">
346               <i class="${icon} ${iconColor}"></i>
347               <span class="sys-core-title text-truncate" style="font-size: 0.82rem;" title="${escapeHtml(grpName)}">${escapeHtml(grpName)}</span>
348             </div>
349             <div class="d-flex align-items-center gap-1">
350               ${tempBadge}
351               <span class="badge bg-dark border border-secondary text-muted" style="font-size: 0.65rem;">${items.length} шт.</span>
352             </div>
353           </div>
354           <div>
355             ${widgetsHtml}
356           </div>
357         </div>
358       `;
359     }).join('');
360
361     // Также обновляем блоки сенсоров в целевых панелях CPU, GPU и RAM
362     updateAllComponentSensors();
363     if (lastCpuCoresData && lastCpuCoresData.length) {
364       renderCpuCores(lastCpuCoresData);
```

```
    }
  }

  function getSensorSortPriority(s) {
    const cat = (s.sensor_category || '').toLowerCase();
    const unit = (s.unit || '').toLowerCase();
    if (cat.includes('power') || unit === 'w') return 1;
    if (cat.includes('volt') || unit === 'v') return 2;
    if (cat.includes('clock') || unit.includes('hz')) return 3
;
    if (cat.includes('fan') || unit.includes('rpm')) return 4;
    if (cat.includes('timing') || cat.includes('factor')) retu
rn 5;
    if (cat.includes('temp') || unit.includes('°c')) return 6;
    return 7;
  }

  function renderComponentSensors(containerId, countBadgeId, s
ensors) {
    const container = document.getElementById(containerId);
    const countBadge = document.getElementById(countBadgeId);
    if (!container) return;

    if (sensors && sensors.length > 0) {
      if (countBadge) {
        countBadge.textContent = `${sensors.length} параметров
`;
      }
    } else {
      // Если элементы уже отрендерены, не перетираем их заглу
шкой при временной задержке поллинга
      if (container.children.length > 0 && !container.querySel
ector('.sys-no-sensors-msg')) {
        return;
      }
      if (countBadge) {
        countBadge.textContent = '0 параметров';
      }
```

```
      container.innerHTML = `<div class="sys-no-sensors-msg te
xt-muted small text-center py-2 col-12" style="font-size: 0.74
rem;">Нет дополнительных данных сенсоров</div>`;
      return;
    }

    const sorted = [...sensors].sort((a, b) => {
      const pA = getSensorSortPriority(a);
      const pB = getSensorSortPriority(b);
      if (pA !== pB) return pA - pB;
      return (a.sensor_name || '').localeCompare(b.sensor_name
 || '');
    });

    container.innerHTML = sorted.map(s => {
      const rawName = s.sensor_name || s.name || '';
      const name = escapeHtml((!rawName || rawName.toLowerCase
() === 'unknown') ? (s.id ? `Sensor #${s.id}` : 'Sensor') : ra
wName);
      const cat = (s.sensor_category || '').toLowerCase();
      const val = Number(s.value || 0);
      const unit = (s.unit || '').trim();
      const valRaw = s.value_raw || `${val.toFixed(1)} ${unit}
`.trim();

      let icon = 'bi bi-lightning-charge';
      let iconColor = 'text-warning';
      let badgeClass = 'bg-warning-subtle text-warning border
border-warning';

      if (cat.includes('volt') || unit.toLowerCase() === 'v')
{
        icon = 'bi bi-lightning-charge';
        iconColor = 'text-primary';
        badgeClass = 'bg-primary-subtle text-primary border bo
rder-primary';
      } else if (cat.includes('power') || unit.toLowerCase() =
== 'w') {
```

```
425         icon = 'bi bi-lightning-charge';
426         iconColor = 'text-warning';
427         badgeClass = 'bg-warning-subtle text-warning border bo
rder-warning';
428       } else if (cat.includes('clock') || unit.toLowerCase().i
ncludes('hz')) {
429         icon = 'bi bi-speedometer2';
430         iconColor = 'text-info';
431         badgeClass = 'bg-info-subtle text-info border border-i
nfo';
432       } else if (cat.includes('fan') || unit.toLowerCase().inc
ludes('rpm')) {
433         icon = 'bi bi-fan';
434         iconColor = 'text-info';
435         badgeClass = 'bg-dark border border-secondary text-inf
o';
436       } else if (cat.includes('timing') || cat.includes('facto
r')) {
437         icon = 'bi bi-clock-history';
438         iconColor = 'text-secondary';
439         badgeClass = 'bg-dark border border-secondary text-lig
ht';
440       } else if (cat.includes('data') || unit.toLowerCase().in
cludes('mb') || unit.toLowerCase().includes('gb')) {
441         icon = 'bi bi-hdd-fill';
442         iconColor = 'text-success';
443         badgeClass = 'bg-success-subtle text-success border bo
rder-success';
444       } else if (cat.includes('temp') || unit.toLowerCase().in
cludes('°c')) {
445         icon = 'bi bi-thermometer-half';
446         iconColor = val >= 75 ? 'text-danger' : val >= 60 ? 't
ext-warning' : 'text-info';
447         badgeClass = val >= 75 ? 'bg-danger text-white' : val
>= 60 ? 'bg-warning text-dark' : 'bg-info-subtle text-info';
448       }
449
450       return `
```

```
        <div class="sys-sensor-widget-item p-1.5 rounded-2" st
yle="background: var(--surface-2, rgba(255,255,255,0.03)); bor
der: 1px solid var(--border-color, rgba(255,255,255,0.08));">
          <div class="d-flex justify-content-between align-ite
ms-center">
            <span class="sys-sensor-name text-truncate" title=
"${name}" style="font-size: 0.74rem; font-weight: 600;">
              <i class="${icon} ${iconColor} me-1"></i>${name}
            </span>
            <span class="badge ${badgeClass} font-monospace" s
tyle="font-size: 0.72rem;">${valRaw}</span>
          </div>
        </div>
      `;
    }).join('');
  }

  function toggleSysCpuDetails() {
    const box = document.getElementById('sys-cpu-details-colla
pse');
    const drillBtn = document.getElementById('sys-cpu-details-
drill-btn');
    const parentBtn = document.getElementById('sys-cpu-details
-toggle-btn');
    if (!box) return;
    const isHidden = box.style.display === 'none' || getComput
edStyle(box).display === 'none';
    if (isHidden) {
      box.style.display = 'flex';
      if (drillBtn) drillBtn.classList.remove('collapsed');
      if (parentBtn) parentBtn.classList.add('expanded');
    } else {
      box.style.display = 'none';
      if (drillBtn) drillBtn.classList.add('collapsed');
      if (parentBtn) parentBtn.classList.remove('expanded');
    }
  }
  window.toggleSysCpuDetails = toggleSysCpuDetails;
```

```
  window.toggleSysCpuSensors = toggleSysCpuDetails;

  function toggleSysGpuDetails(idx = 0) {
    const box = document.getElementById(`sys-gpu-details-collapse-${idx}`) || document.getElementById('sys-gpu-details-collapse');
    const drillBtn = document.getElementById(`sys-gpu-details-drill-btn-${idx}`) || document.getElementById('sys-gpu-details-drill-btn');
    const parentBtn = document.getElementById(`sys-gpu-details-toggle-btn-${idx}`) || document.getElementById('sys-gpu-details-toggle-btn');
    if (!box) return;
    const isHidden = box.style.display === 'none' || getComputedStyle(box).display === 'none';
    if (isHidden) {
      box.style.display = 'flex';
      if (drillBtn) drillBtn.classList.remove('collapsed');
      if (parentBtn) parentBtn.classList.add('expanded');
    } else {
      box.style.display = 'none';
      if (drillBtn) drillBtn.classList.add('collapsed');
      if (parentBtn) parentBtn.classList.remove('expanded');
    }
  }
  window.toggleSysGpuDetails = toggleSysGpuDetails;
  window.toggleSysGpuSensors = toggleSysGpuDetails;

  /**
   * Управление режимом Drill-down по нажатию на значок дрели слева от названия компонента.
   * Раскрывает все скрытые панели, графики и датчики конкретного компонента.
   *
   * @param {string|HTMLElement} cardIdOrEl - Идентификатор карточки или элемент кнопки
   * @param {number} [idx] - Индекс устройства (для GPU)
   */
```

```
  function toggleSysCardDrill(cardIdOrEl, idx) {
    let card = null;
    let btn = null;
    if (typeof cardIdOrEl === 'string') {
      card = document.getElementById(cardIdOrEl);
    } else if (cardIdOrEl instanceof HTMLElement) {
      btn = cardIdOrEl;
      card = btn.closest('.sys-card');
    }
    if (!card) return;
    if (!btn) {
      btn = card.querySelector('.sys-drill-btn');
    }

    const isExpanded = card.classList.contains('sys-card-drill
-expanded');
    if (isExpanded) {
      card.classList.remove('sys-card-drill-expanded');
      if (btn) {
        btn.classList.remove('active');
        btn.title = 'Drill-down: Развернуть/скрыть все скрытые
 панели и датчики';
      }
    } else {
      card.classList.add('sys-card-drill-expanded');
      if (btn) {
        btn.classList.add('active');
        btn.title = 'Drill-down: Свернуть скрытые панели';
      }

      // Если это CPU - также открываем вложенный аккордеон де
талей ядер и сенсоров
      if (card.classList.contains('sys-card-cpu') || card.id =
== 'sys-card-cpu') {
        const box = document.getElementById('sys-cpu-details-c
ollapse');
        const icon = document.getElementById('sys-cpu-details-
toggle-icon');
```

```
        const text = document.getElementById('sys-cpu-details-toggle-text');
        const actionBtn = document.getElementById('sys-cpu-details-action-btn');
        const parentBtn = document.getElementById('sys-cpu-details-toggle-btn');
        if (box) box.style.display = 'flex';
        if (icon) icon.className = 'bi bi-chevron-up';
        if (text) text.textContent = 'Свернуть детализацию';
        if (actionBtn) actionBtn.className = 'btn btn-xs btn-info text-dark rounded-pill px-3 py-1 fw-bold d-flex align-items-center gap-1.5';
        if (parentBtn) parentBtn.classList.add('expanded');
        if (typeof renderCpuSpark === 'function' && Array.isArray(_cpuSparkHistory)) {
          renderCpuSpark(_cpuSparkHistory);
        }
      }

      // Если это GPU - также открываем вложенный аккордеон деталей GPU
      if (card.classList.contains('sys-card-gpu') || (card.id && card.id.startsWith('sys-card-gpu-'))) {
        const cardIdx = idx != null ? idx : (card.id ? card.id.replace('sys-card-gpu-', '') : 0);
        const box = document.getElementById(`sys-gpu-details-collapse-${cardIdx}`) || document.getElementById('sys-gpu-details-collapse');
        const icon = document.getElementById(`sys-gpu-details-toggle-icon-${cardIdx}`) || document.getElementById('sys-gpu-details-toggle-icon');
        const text = document.getElementById(`sys-gpu-details-toggle-text-${cardIdx}`) || document.getElementById('sys-gpu-details-toggle-text');
        const actionBtn = document.getElementById(`sys-gpu-details-action-btn-${cardIdx}`) || document.getElementById('sys-gpu-details-action-btn');
        const parentBtn = document.getElementById(`sys-gpu-det
```

```
ails-toggle-btn-${cardIdx}`) || document.getElementById('sys-g
pu-details-toggle-btn');
561         if (box) box.style.display = 'flex';
562         if (icon) icon.className = 'bi bi-chevron-up';
563         if (text) text.textContent = 'Свернуть детализацию';
564         if (actionBtn) actionBtn.className = 'btn btn-xs btn-w
arning text-dark rounded-pill px-3 py-1 fw-bold d-flex align-i
tems-center gap-1.5';
565         if (parentBtn) parentBtn.classList.add('expanded');
566         if (typeof renderGpuSpark === 'function') {
567           renderGpuSpark(_gpuSparkHistories[cardIdx] || [], ca
rdIdx);
568         }
569       }
570
571       // RAM график
572       if (card.classList.contains('sys-card-ram') || card.id =
== 'sys-card-ram') {
573         if (typeof renderMemIoSpark === 'function' && Array.is
Array(_memIoSparkHistory) && _memIoSparkHistory.length) {
574           renderMemIoSpark(_memIoSparkHistory);
575         }
576       }
577
578       // Net график
579       if (card.classList.contains('sys-card-net') || card.id =
== 'sys-card-net') {
580         if (typeof renderNetSpark === 'function' && Array.isAr
ray(_netSparkHistory) && _netSparkHistory.length) {
581           renderNetSpark(_netSparkHistory);
582         }
583       }
584
585       // Storage график
586       if (card.classList.contains('sys-card-storage') || card.
id === 'sys-card-storage') {
587         if (typeof renderStorageSpark === 'function' && Array.
isArray(_storageSparkHistory) && _storageSparkHistory.length)
```

```
{
588           renderStorageSpark(_storageSparkHistory);
589         }
590       }
591     }
592   }
593   window.toggleSysCardDrill = toggleSysCardDrill;
594
595   function updateAllComponentSensors() {
596     if (!Array.isArray(cachedSensors) || cachedSensors.length
=== 0) return;
597
598     // 1. CPU Sensors (электрические параметры, мощности, воль
тажи, шины, TjMax)
599     const cpuSensors = cachedSensors.filter(s => {
600       const sid = String(s.id || s.sensor_id || '').toLowerCas
e();
601       const hwType = (s.hardware_type || '').toLowerCase();
602       const hwName = (s.hardware_name || '').toLowerCase();
603       const sName = (s.sensor_name || '').toLowerCase();
604       const cat = (s.sensor_category || '').toLowerCase();
605
606       // Строго отсекаем дисковые метрики и не-CPU компоненты
607       if (sid.startsWith('disk_speed_') || hwType.includes('st
orage') || cat.includes('storage')) return false;
608
609       const isCpu = hwType === 'cpu' || hwName.includes('cpu')
 || hwName.includes('intel') || hwName.includes('amd');
610       if (!isCpu) return false;
611
612       // Исключаем по-ядерную загрузку и температуру отдельных
 ядер (они уже в карточках ядер)
613       if (cat.includes('load') && (sName.includes('core #') ||
 sName === 'cpu total')) return false;
614       if (cat.includes('temp') && sName.includes('core #') &&
!sName.includes('tjmax')) return false;
615       if (cat.includes('temp') && sName === 'cpu package') ret
urn false; // показана в главном слайдере Package
```

```
616
617       return true;
618     });
619     if (cpuSensors.length > 0) {
620       lastCpuSensorsData = cpuSensors;
621       renderComponentSensors('sys-metric-cpu-sensors', 'sys-cp
u-sensors-count-badge', cpuSensors);
622     } else if (lastCpuSensorsData && lastCpuSensorsData.length
 > 0) {
623       renderComponentSensors('sys-metric-cpu-sensors', 'sys-cp
u-sensors-count-badge', lastCpuSensorsData);
624     } else {
625       renderComponentSensors('sys-metric-cpu-sensors', 'sys-cp
u-sensors-count-badge', []);
626     }
627
628     // 2. GPU Sensors (мощности, вольтажи, частоты, вентилятор
ы, память GPU)
629     const gpuSensors = cachedSensors.filter(s => {
630       const hwType = (s.hardware_type || '').toLowerCase();
631       const hwName = (s.hardware_name || '').toLowerCase();
632       const sName = (s.sensor_name || '').toLowerCase();
633       const cat = (s.sensor_category || '').toLowerCase();
634
635       const isGpu = hwType === 'gpu' || hwName.includes('gpu')
 || hwName.includes('nvidia') || hwName.includes('geforce') ||
 hwName.includes('radeon') || hwName.includes('intel arc');
636       if (!isGpu) return false;
637
638       // Исключаем общую загрузку/температуру GPU Core (они в
главном Gauge/Slider)
639       if (cat.includes('load') && sName === 'gpu core') return
 false;
640       if (cat.includes('temp') && sName === 'gpu core') return
 false;
641
642       return true;
643     });
```

```
644     renderComponentSensors('sys-metric-gpu-sensors', 'sys-gpu-
    sensors-count-badge', gpuSensors);
645 
646     // 3. RAM Sensors (мощности, вольтажи, тайминги, доступный
     объем)
647     const ramSensors = cachedSensors.filter(s => {
648       const hwType = (s.hardware_type || '').toLowerCase();
649       const hwName = (s.hardware_name || '').toLowerCase();
650       const sName = (s.sensor_name || '').toLowerCase();
651 
652       const isRam = hwType === 'memory' || hwType === 'ram' ||
     hwName.includes('memory') || hwName.includes('ram') || sName.
    includes('cpu memory');
653       return isRam;
654     });
655     renderComponentSensors('sys-metric-ram-sensors', 'sys-ram-
    sensors-count-badge', ramSensors);
656 
657     // 4. Network Sensors (трафик, сетевые интерфейсы, Wi-Fi,
    Bluetooth)
658     const netSensors = cachedSensors.filter(s => {
659       const hwType = (s.hardware_type || '').toLowerCase();
660       const hwName = (s.hardware_name || '').toLowerCase();
661       const isNet = hwType === 'network' || hwType === 'net' |
    | hwName.includes('net') || hwName.includes('ethernet') || hwN
    ame.includes('wi-fi') || hwName.includes('wifi') || hwName.inc
    ludes('bluetooth');
662       return isNet;
663     });
664     renderComponentSensors('sys-metric-net-sensors', 'sys-net-
    sensors-count-badge', netSensors);
665   }
666 
667   let lastAnalysisReportText = '';
668 
669   async function analyzeSensorsAndReport() {
670     const modalEl = document.getElementById('sysSensorAnalysis
    Modal');
```

```
671     let modal = null;
672     if (modalEl && window.bootstrap && window.bootstrap.Modal)
     {
673       modal = window.bootstrap.Modal.getOrCreateInstance(modal
    El);
674       modal.show();
675     }
676 
677     // Set Loading state in modal
678     const titleEl = document.getElementById('modal-sensor-stat
    us-title');
679     const subEl = document.getElementById('modal-sensor-status
    -sub');
680     const aiCompEl = document.getElementById('modal-sensor-ai-
    comparison');
681     const devListEl = document.getElementById('modal-sensor-de
    vices-list');
682     const devCountEl = document.getElementById('modal-sensor-d
    evices-count');
683     const modelBadge = document.getElementById('modal-sensor-m
    odel-badge');
684 
685     if (titleEl) titleEl.textContent = 'Выполняется AI-аудит з
    алогированных сенсоров...';
686     if (subEl) subEl.textContent = 'Чтение CSV-логов, усреднен
    ие значений и сравнение со спецификациями железа';
687     if (aiCompEl) {
688       aiCompEl.innerHTML = '<div class="d-flex align-items-cen
    ter gap-2 py-3"><div class="spinner-border spinner-border-sm t
    ext-primary" role="status"></div><span>Запрос к модели AI и ср
    авнительный анализ номинальных параметров железа...</span></di
    v>';
689     }
690     if (devListEl) {
691       devListEl.innerHTML = '<span class="badge bg-dark border
     border-secondary text-muted"><span class="spinner-border spin
    ner-border-sm me-1" style="width:0.6rem;height:0.6rem;"></span
    >Определение устройств...</span>';
```

```
692     }
693
694     let report = null;
695     try {
696       let res = await fetch('/api/v1/system/lhm-audit', { meth
od: 'POST' });
697       if (!res.ok) {
698         res = await fetch('/api/v1/lhm/audit', { method: 'POST
' });
699       }
700       if (res.ok) {
701         report = await res.json();
702       }
703     } catch (err) {
704       console.warn('[SystemInspectorTab] LHM Audit API error:'
, err);
705     }
706
707     // Extract metrics from report or fallback to cachedSensor
s
708     const sensorsList = (report && report.aggregated_sensors)
? report.aggregated_sensors : (cachedSensors || []);
709     const devices = (report && report.devices) ? report.device
s : [];
710     let healthScore = report ? (report.health_score || 100) :
100;
711     const recommendations = [];
712
713     let maxCpuTemp = null;
714     let maxGpuTemp = null;
715     let maxMbTemp = null;
716     let maxStorageTemp = null;
717     let fanSpeeds = [];
718     let voltages = [];
719     let cpuLoads = [];
720     let gpuLoads = [];
721
722     sensorsList.forEach(s => {
```

```
      const cat = ((s.category || s.sensor_category) || '').toLowerCase();
      const hw = ((s.hardware || s.hardware_name) || '').toLowerCase();
      const sName = (s.sensor_name || '').toLowerCase();
      const num = s.avg !== undefined ? s.avg : s.value_numeric;

      if (num === null || num === undefined || isNaN(num)) return;

      if (cat.includes('temp')) {
        if (hw.includes('cpu') || sName.includes('cpu') || sName.includes('core')) {
          if (maxCpuTemp === null || num > maxCpuTemp) maxCpuTemp = num;
        } else if (hw.includes('gpu') || hw.includes('nvidia') || sName.includes('gpu')) {
          if (maxGpuTemp === null || num > maxGpuTemp) maxGpuTemp = num;
        } else if (hw.includes('storage') || hw.includes('nvme') || hw.includes('ssd') || hw.includes('hdd') || sName.includes('drive')) {
          if (maxStorageTemp === null || num > maxStorageTemp) maxStorageTemp = num;
        } else {
          if (maxMbTemp === null || num > maxMbTemp) maxMbTemp = num;
        }
      } else if (cat.includes('fan')) {
        fanSpeeds.push({ name: s.sensor_name, value: num, unit: s.unit || 'RPM' });
      } else if (cat.includes('volt')) {
        voltages.push({ name: s.sensor_name, value: num, unit: s.unit || 'V' });
      } else if (cat.includes('load')) {
        if (hw.includes('cpu') || sName.includes('cpu')) cpuLoads.push(num);
```

```
        else if (hw.includes('gpu') || sName.includes('gpu')) gpuLoads.push(num);
      }
    });

    const overallMaxTemp = Math.max(
      maxCpuTemp || 0,
      maxGpuTemp || 0,
      maxMbTemp || 0,
      maxStorageTemp || 0
    );

    // Update Model Badge
    if (modelBadge && report) {
      modelBadge.textContent = report.ai_model_used || 'Heuristic Engine';
    }

    // Render Devices List
    if (devCountEl) {
      devCountEl.textContent = `${devices.length} устр.`;
    }
    if (devListEl) {
      if (devices.length > 0) {
        devListEl.innerHTML = devices.map(d => {
          let icon = 'bi-hdd-network';
          const low = d.toLowerCase();
          if (low.includes('core') || low.includes('cpu') || low.includes('intel') || low.includes('amd')) icon = 'bi-cpu';
          else if (low.includes('geforce') || low.includes('gpu') || low.includes('nvidia') || low.includes('radeon')) icon = 'bi-gpu-card';
          else if (low.includes('ssd') || low.includes('hd') || low.includes('wdc') || low.includes('toshiba') || low.includes('st3500')) icon = 'bi-device-hdd';
          else if (low.includes('memory') || low.includes('ram')) icon = 'bi-memory';
          else if (low.includes('ethernet') || low.includes('w
```

```
i-fi') || low.includes('network')) icon = 'bi-wifi';
776
777           return `<span class="badge bg-dark border border-sec
ondary text-info px-2 py-1 d-inline-flex align-items-center ga
p-1"><i class="bi ${icon}"></i> ${escapeHtml(d)}</span>`;
778         }).join('');
779       } else {
780         devListEl.innerHTML = '<span class="text-muted small">
Устройства считываются из текущей телеметрии</span>';
781       }
782     }
783
784     // Render AI Comparison Report
785     if (aiCompEl) {
786       if (report && report.comparison_report) {
787         aiCompEl.textContent = report.comparison_report;
788       } else {
789         aiCompEl.textContent = 'AI-сравнение выполнено по эври
стическим профилям оборудования хоста. Все параметры сенсоров
в пределах нормы.';
790       }
791     }
792
793     // Analyze CPU
794     const elCpuBadge = document.getElementById('modal-cpu-temp
-badge');
795     const elCpuText = document.getElementById('modal-cpu-analy
sis-text');
796     if (maxCpuTemp !== null) {
797       if (elCpuBadge) {
798         elCpuBadge.textContent = `${Math.round(maxCpuTemp)} °C
`;
799         elCpuBadge.className = `badge ${maxCpuTemp > 82 ? 'bg-
danger' : maxCpuTemp > 70 ? 'bg-warning text-dark' : 'bg-succe
ss'}`;
800       }
801       if (maxCpuTemp > 82) {
802         if (elCpuText) elCpuText.textContent = `Критический на
```

```
грев (${Math.round(maxCpuTemp)}°C). Риск теплового троттлинга.
`;
        recommendations.push({
          type: 'danger',
          icon: 'bi-exclamation-triangle-fill',
          title: 'Высокая температура процессора',
          text: `Пиковая/средняя температура CPU достигает ${M
ath.round(maxCpuTemp)}°C. Проверьте плотность прижима кулера,
термопасту и запыленность радиатора.`
        });
      } else if (maxCpuTemp > 70) {
        if (elCpuText) elCpuText.textContent = `Повышенная тем
пература (${Math.round(maxCpuTemp)}°C) под нагрузкой.`;
        recommendations.push({
          type: 'warning',
          icon: 'bi-thermometer-high',
          title: 'Повышенный нагрев CPU',
          text: `Температура CPU ${Math.round(maxCpuTemp)}°C в
ыше оптимального порога. Рекомендуется настроить кривую оборот
ов вентилятора.`
        });
      } else {
        if (elCpuText) elCpuText.textContent = `Штатный темпер
атурный режим (${Math.round(maxCpuTemp)}°C). Троттлинг отсутст
вует.`;
        recommendations.push({
          type: 'success',
          icon: 'bi-check-circle-fill',
          title: 'Тепловой режим процессора оптимален',
          text: `Температура процессора ${Math.round(maxCpuTem
p)}°C находится в безопасной зоне номинальных характеристик.`
        });
      }
    } else {
      if (elCpuBadge) elCpuBadge.textContent = 'N/A';
      if (elCpuText) elCpuText.textContent = 'Температурные да
тчики CPU не предоставили данных.';
    }
```

```
830
831     // Analyze GPU
832     const elGpuBadge = document.getElementById('modal-gpu-temp
    -badge');
833     const elGpuText = document.getElementById('modal-gpu-analy
    sis-text');
834     if (maxGpuTemp !== null) {
835       if (elGpuBadge) {
836         elGpuBadge.textContent = `${Math.round(maxGpuTemp)} °C
    `;
837         elGpuBadge.className = `badge ${maxGpuTemp > 80 ? 'bg-
    danger' : maxGpuTemp > 72 ? 'bg-warning text-dark' : 'bg-succe
    ss'}`;
838       }
839       if (maxGpuTemp > 80) {
840         if (elGpuText) elGpuText.textContent = `Критический на
    грев GPU (${Math.round(maxGpuTemp)}°C).`;
841         recommendations.push({
842           type: 'danger',
843           icon: 'bi-gpu-card',
844           title: 'Высокая температура видеокарты',
845           text: `Графический чип нагревается до ${Math.round(m
    axGpuTemp)}°C. Проверьте циркуляцию воздуха в корпусе ПК.`
846         });
847       } else {
848         if (elGpuText) elGpuText.textContent = `Температура GP
    U ${Math.round(maxGpuTemp)}°C в норме.`;
849         recommendations.push({
850           type: 'success',
851           icon: 'bi-check-circle-fill',
852           title: 'Видеокарта работает штатно',
853           text: `Температурные показатели GPU (${Math.round(ma
    xGpuTemp)}°C) соответствуют номиналу.`
854         });
855       }
856     } else {
857       if (elGpuBadge) elGpuBadge.textContent = 'N/A';
858       if (elGpuText) elGpuText.textContent = 'Дискретный GPU в
```

```
 режиме энергосбережения или данные отсутствуют.';
    }

    // Analyze Cooling
    const elFanBadge = document.getElementById('modal-fan-stat
us-badge');
    const elFanText = document.getElementById('modal-fan-analy
sis-text');
    if (fanSpeeds.length > 0) {
      const activeFans = fanSpeeds.filter(f => f.value > 0);
      if (elFanBadge) {
        elFanBadge.textContent = `${activeFans.length} акт. ку
леров`;
        elFanBadge.className = 'badge bg-success';
      }
      if (elFanText) {
        elFanText.textContent = `Опрошено ${fanSpeeds.length}
кулеров. Обороты стабильны.`;
      }
    } else {
      if (elFanBadge) elFanBadge.textContent = 'Пассив / WMI';
      if (elFanText) elFanText.textContent = 'Управление венти
ляторами через BIOS или пассивное охлаждение.';
    }

    // Additional recommendations from backend report
    if (report && Array.isArray(report.recommendations)) {
      report.recommendations.forEach(r => {
        if (!recommendations.some(ex => ex.text.includes(r)))
{
          recommendations.push({
            type: 'info',
            icon: 'bi-info-circle',
            title: 'Рекомендация по оборудованию',
            text: r
          });
        }
      });
```

```
    }

    // Update Header & Badge
    const badgeHealth = document.getElementById('modal-sensor-health-badge');
    if (badgeHealth) {
      badgeHealth.textContent = `Health: ${healthScore}/100`;
      badgeHealth.className = `badge ms-1 ${healthScore >= 85 ? 'bg-success' : healthScore >= 65 ? 'bg-warning text-dark' : 'bg-danger'}`;
    }

    const countStat = document.getElementById('modal-sensor-count-stat');
    if (countStat) {
      const totalSmpls = report ? report.total_samples : sensorsList.length;
      countStat.textContent = `${sensorsList.length} шт. (${totalSmpls} замеров)`;
    }

    const maxTempStat = document.getElementById('modal-sensor-max-temp');
    if (maxTempStat) maxTempStat.textContent = overallMaxTemp > 0 ? `${Math.round(overallMaxTemp)} °C` : 'N/A';

    const statusTitle = document.getElementById('modal-sensor-status-title');
    const statusSub = document.getElementById('modal-sensor-status-sub');
    const statusIcon = document.getElementById('modal-sensor-status-icon');

    if (healthScore >= 85) {
      if (statusIcon) statusIcon.textContent = '□ ';
      if (statusTitle) statusTitle.textContent = 'Все аппаратные подсистемы работают в идеальном режиме';
      if (statusSub) statusSub.textContent = `Усредненные пока
```

```
затели соответствуют спецификациям реального железа (макс. ${M
ath.round(overallMaxTemp)}°C).`;
916     } else if (healthScore >= 65) {
917       if (statusIcon) statusIcon.textContent = '⚠ ';
918       if (statusTitle) statusTitle.textContent = 'Обнаружены п
араметры оборудования, требующие внимания';
919       if (statusSub) statusSub.textContent = `Пиковая температ
ура ${Math.round(overallMaxTemp)}°C. Ознакомьтесь с AI-заключе
нием и рекомендациями.`;
920     } else {
921       if (statusIcon) statusIcon.textContent = '⛔ ';
922       if (statusTitle) statusTitle.textContent = 'Внимание: Кр
итические параметры оборудования!';
923       if (statusSub) statusSub.textContent = `Зафиксирован опа
сный нагрев или высокая перегрузка компонентов.`;
924     }
925 
926     // Render Recommendations List
927     const recList = document.getElementById('modal-sensor-reco
mmendations-list');
928     if (recList) {
929       recList.innerHTML = recommendations.map(r => `
930         <div class="p-2.5 rounded border d-flex align-items-st
art gap-2.5" style="background: var(--surface-2); border-color
: var(--border-color) !important;">
931           <div class="mt-0.5">
932             <span class="badge ${r.type === 'danger' ? 'bg-dan
ger' : r.type === 'warning' ? 'bg-warning text-dark' : r.type
=== 'info' ? 'bg-info text-dark' : 'bg-success'} rounded-circl
e p-1.5 d-flex align-items-center justify-content-center" styl
e="width: 24px; height: 24px;">
933               <i class="bi ${r.icon}" style="font-size: 0.75re
m;"></i>
934             </span>
935           </div>
936           <div class="flex-grow-1">
937             <div class="fw-bold text-white mb-0.5" style="font
-size: 0.82rem;">${escapeHtml(r.title)}</div>
```

```
            <div class="text-muted" style="font-size: 0.76rem;
 line-height: 1.4;">${escapeHtml(r.text)}</div>
          </div>
        </div>
      `).join('');
    }

    // Prepare Clipboard text
    lastAnalysisReportText = `=== AI-Windows-Admin: Отчет AI-a
удита сенсоров и оборудования ===\n` +
      `Дата: ${new Date().toLocaleString('ru-RU')}\n` +
      `Индекс здоровья системы: ${healthScore}/100\n` +
      `AI-модель: ${report ? report.ai_model_used : 'Heuristic
'}\n` +
      `Распознанные устройства (${devices.length} шт.): ${devi
ces.join(', ')}\n` +
      `Сенсоров в аудите: ${sensorsList.length}\n` +
      `Максимальная температура: ${overallMaxTemp > 0 ? Math.r
ound(overallMaxTemp) + ' °C' : 'N/A'}\n\n` +
      `=== AI СРАВНЕНИЕ СО СПЕЦИФИКАЦИЯМИ РЕАЛЬНОГО ЖЕЛЕЗА ===
\n` +
      `${(report && report.comparison_report) ? report.compari
son_report : 'Параметры в норме.'}\n\n` +
      `=== РЕКОМЕНДАЦИИ И ВЫВОДЫ ===\n` +
      recommendations.map((r, i) => `${i + 1}. [${r.title}] ${
r.text}`).join('\n') +
      `\n=========================================================
=`;
  }

  /**
   * Генерация полукруглого спидометра (Gauge) со стрелкой.
   * @param {number|null} percent - Значение нагрузки (0-100)
   * @param {number} width - Ширина SVG контейнера
   * @param {number} height - Высота SVG контейнера
   * @returns {string} SVG разметка
   */
  function createGaugeSvg(percent, width = 100, height = 56) {
```

```
    const val = percent == null ? 0 : Math.min(100, Math.max(0, Number(percent)));
    // Угол поворота стрелки: 0% -> -90deg (влево), 50% -> 0deg (вверх), 100% -> +90deg (вправо)
    const angle = -90 + (val / 100) * 180;
    const needleColor = val > 80 ? '#ef4444' : val > 50 ? '#f59e0b' : '#38bdf8';

    return `
      <svg width="${width}" height="${height}" viewBox="0 0 100 56" preserveAspectRatio="xMidYMid meet" style="display:block;margin:0 auto;overflow:visible;">
        <defs>
          <linearGradient id="sysGaugeGrad" x1="0%" y1="0%" x2="100%" y2="0%">
            <stop offset="0%" stop-color="#10b981"/>
            <stop offset="50%" stop-color="#f59e0b"/>
            <stop offset="100%" stop-color="#ef4444"/>
          </linearGradient>
        </defs>
        <!-- Фоновая дуга -->
        <path d="M 12 50 A 38 38 0 0 1 88 50" fill="none" stroke="var(--border-color, rgba(128,128,128,0.25))" stroke-width="9" stroke-linecap="round"/>
        <!-- Градиентная активная шкала -->
        <path d="M 12 50 A 38 38 0 0 1 88 50" fill="none" stroke="url(#sysGaugeGrad)" stroke-width="9" stroke-linecap="round" opacity="0.9"/>
        <!-- Стрелка прибора -->
        <g transform="rotate(${angle.toFixed(1)}, 50, 50)" style="transition: transform 0.4s cubic-bezier(0.4, 0, 0.2, 1);">
          <polygon points="48,50 50,14 52,50" fill="${needleColor}" filter="drop-shadow(0 2px 3px rgba(0,0,0,0.6))"/>
          <line x1="50" y1="50" x2="50" y2="14" stroke="#ffffff" stroke-width="1.2" opacity="0.9"/>
        </g>
        <!-- Центральный шарнир -->
        <circle cx="50" cy="50" r="6" fill="#38bdf8"/>
```

```
        <circle cx="50" cy="50" r="2.5" fill="var(--surface-1, #0f172a)"/>
      </svg>
    `;
  }

  /**
   * Генерация цветного термо-слайдера с бегунком.
   * @param {number|null} tempC - Температура в градусах Цельсия
   * @param {number} minTemp - Мин. шкала (по умолчанию 30°C)
   * @param {number} maxTemp - Макс. шкала (по умолчанию 95°C)
   * @param {boolean} isLarge - Флаг увеличенного отображения для шапки
   * @returns {string} HTML разметка
   */
  function createTempSliderHtml(tempC, minTemp = 30, maxTemp = 95, isLarge = false) {
    if (tempC == null) {
      return `<div class="sys-temp-slider-empty">-- °C</div>`;
    }
    const val = Number(tempC);
    const pct = Math.min(100, Math.max(0, ((val - minTemp) / (maxTemp - minTemp)) * 100));

    let barColor = '#10b981'; // Зеленый (<50°C)
    let badgeClass = 'text-success';
    if (val >= 80) {
      barColor = '#ef4444'; // Красный (>=80°C)
      badgeClass = 'text-danger fw-bold';
    } else if (val >= 68) {
      barColor = '#f97316'; // Оранжевый (68-80°C)
      badgeClass = 'text-warning fw-bold';
    } else if (val >= 50) {
      barColor = '#eab308'; // Желтый (50-68°C)
      badgeClass = 'text-warning';
    }

```

```
1025    const wrapClass = isLarge ? 'sys-temp-slider-wrap sys-temp
-slider-lg' : 'sys-temp-slider-wrap';
1026    const fontSize = isLarge ? 'font-size: 0.85rem;' : 'font-s
ize: 0.78rem;';
1027
1028    return `
1029      <div class="${wrapClass}">
1030        <div class="d-flex justify-content-between align-items
-center mb-1" style="${fontSize}">
1031          <span class="text-muted"><i class="bi bi-thermometer
-half me-1"></i>t°</span>
1032          <span class="font-monospace ${badgeClass}" style="fo
nt-weight: 700;">${val.toFixed(0)}°C</span>
1033        </div>
1034        <div class="sys-temp-slider-track">
1035          <div class="sys-temp-slider-bar" style="width: ${pct
.toFixed(1)}%; background: ${barColor};"></div>
1036          <div class="sys-temp-slider-thumb" style="left: ${pc
t.toFixed(1)}%; background: ${barColor};"></div>
1037        </div>
1038      </div>
1039    `;
1040  }
1041
1042  /**
1043   * Извлечение и сопоставление параметров сенсоров конкретног
о ядра CPU.
1044   * @param {number} coreIdx - Индекс ядра (0-based)
1045   * @param {number} totalCores - Общее количество ядер/потоко
в
1046   * @param {Array} sensors - Массив объектов сенсоров из tele
metry.db / LHM
1047   * @returns {Array<{icon: string, label: string, value: stri
ng, badgeClass: string}>}
1048   */
1049  function extractCoreParameters(coreIdx, totalCores, sensors)
 {
1050    if (!Array.isArray(sensors) || sensors.length === 0) {
```

```
      return [];
    }

    const cpuSensors = sensors.filter(s => {
      const hwType = (s.hardware_type || '').toLowerCase();
      const hwName = (s.hardware_name || '').toLowerCase();
      return hwType === 'cpu' || hwName.includes('cpu') || hwN
ame.includes('intel') || hwName.includes('amd');
    });

    if (cpuSensors.length === 0) return [];

    const idx1 = coreIdx + 1;
    const physIdx1 = Math.floor(coreIdx / 2) + 1;

    const params = [];

    // 1. Вольтаж ядра (Voltage)
    const voltSensor = cpuSensors.find(s => {
      const cat = (s.sensor_category || '').toLowerCase();
      const unit = (s.unit || '').toLowerCase();
      const name = (s.sensor_name || '').toLowerCase();
      if (!cat.includes('volt') && unit !== 'v') return false;
      return name.includes(`core #${idx1}`) || name.includes(`
core #${physIdx1}`) || name.includes(`core #${coreIdx}`);
    }) || cpuSensors.find(s => {
      const cat = (s.sensor_category || '').toLowerCase();
      const unit = (s.unit || '').toLowerCase();
      const name = (s.sensor_name || '').toLowerCase();
      return (cat.includes('volt') || unit === 'v') && (name.i
ncludes('cpu core') || name.includes('vcore'));
    });

    if (voltSensor) {
      params.push({
        icon: 'bi bi-lightning-charge text-primary',
        label: 'Вольтаж',
        value: voltSensor.value_raw || `${Number(voltSensor.va
```

```
lue || 0).toFixed(2)} V`,
1086          badgeClass: 'text-primary'
1087        });
1088      }
1089
1090      // 2. Частота ядра (Clock / Frequency)
1091      const clockSensor = cpuSensors.find(s => {
1092        const cat = (s.sensor_category || '').toLowerCase();
1093        const unit = (s.unit || '').toLowerCase();
1094        const name = (s.sensor_name || '').toLowerCase();
1095        if (!cat.includes('clock') && !unit.includes('hz')) retu
rn false;
1096        return name.includes(`core #${idx1}`) || name.includes(`
core #${physIdx1}`) || name.includes(`core #${coreIdx}`);
1097      }) || cpuSensors.find(s => {
1098        const cat = (s.sensor_category || '').toLowerCase();
1099        const unit = (s.unit || '').toLowerCase();
1100        const name = (s.sensor_name || '').toLowerCase();
1101        return (cat.includes('clock') || unit.includes('hz')) &&
 (name.includes('core freq') || name.includes('cpu core'));
1102      });
1103
1104      if (clockSensor) {
1105        params.push({
1106          icon: 'bi bi-speedometer2 text-info',
1107          label: 'Частота',
1108          value: clockSensor.value_raw || `${Number(clockSensor.
value || 0).toFixed(0)} MHz`,
1109          badgeClass: 'text-info'
1110        });
1111      }
1112
1113      // 3. Запас до TjMax (Distance to TjMax)
1114      const tjmaxSensor = cpuSensors.find(s => {
1115        const name = (s.sensor_name || '').toLowerCase();
1116        if (!name.includes('tjmax')) return false;
1117        return name.includes(`core #${idx1}`) || name.includes(`
core #${physIdx1}`) || name.includes(`core #${coreIdx}`);
```

```
    }) || cpuSensors.find(s => (s.sensor_name || '').toLowerCase().includes('tjmax'));

    if (tjmaxSensor) {
      params.push({
        icon: 'bi bi-shield-check text-success',
        label: 'До TjMax',
        value: tjmaxSensor.value_raw || `${Number(tjmaxSensor.value || 0).toFixed(0)} °C`,
        badgeClass: 'text-success'
      });
    }

    // 4. Мощность (Power / Cores Power)
    const powerSensor = cpuSensors.find(s => {
      const cat = (s.sensor_category || '').toLowerCase();
      const unit = (s.unit || '').toLowerCase();
      const name = (s.sensor_name || '').toLowerCase();
      if (!cat.includes('power') && unit !== 'w') return false;
      return name.includes(`core #${idx1}`) || name.includes(`core #${physIdx1}`);
    }) || cpuSensors.find(s => {
      const cat = (s.sensor_category || '').toLowerCase();
      const unit = (s.unit || '').toLowerCase();
      const name = (s.sensor_name || '').toLowerCase();
      return (cat.includes('power') || unit === 'w') && (name.includes('cpu cores') || name.includes('cpu core'));
    });

    if (powerSensor) {
      params.push({
        icon: 'bi bi-lightning text-warning',
        label: 'Мощность',
        value: powerSensor.value_raw || `${Number(powerSensor.value || 0).toFixed(1)} W`,
        badgeClass: 'text-warning'
      });
```

```
1150      }
1151
1152      // 5. Пиковая загрузка / Core Max / Шина (Bus Speed)
1153      const maxSensor = cpuSensors.find(s => {
1154        const name = (s.sensor_name || '').toLowerCase();
1155        return name.includes('core max') || name.includes('cpu c
ore max');
1156      }) || cpuSensors.find(s => {
1157        const name = (s.sensor_name || '').toLowerCase();
1158        return name.includes('bus speed');
1159      });
1160
1161      if (maxSensor) {
1162        const isBus = (maxSensor.sensor_name || '').toLowerCase(
).includes('bus');
1163        params.push({
1164          icon: isBus ? 'bi bi-hdd-network text-secondary' : 'bi
 bi-graph-up-arrow text-danger',
1165          label: isBus ? 'Шина' : 'Пик Max',
1166          value: maxSensor.value_raw || `${Number(maxSensor.valu
e || 0).toFixed(1)} ${maxSensor.unit || ''}`.trim(),
1167          badgeClass: isBus ? 'text-secondary' : 'text-danger'
1168        });
1169      }
1170
1171      return params;
1172    }
1173
1174    function renderCpuCores(cores, totalThreadsCount) {
1175      const box = document.getElementById('sys-metric-cpu-cores'
);
1176      const badgeCount = document.getElementById('sys-cores-coun
t-badge');
1177      if (!box) return;
1178
1179      if (Array.isArray(cores)) {
1180        lastCpuCoresData = cores;
1181      } else if (lastCpuCoresData && lastCpuCoresData.length) {
```

```
1182      cores = lastCpuCoresData;
1183    }
1184
1185    const cCount = cores ? cores.length : 0;
1186    const thCount = totalThreadsCount || (cores ? cores.reduce
((acc, c) => acc + (c.threads ? c.threads.length : 0), 0) : 0)
;
1187
1188    if (badgeCount) {
1189      badgeCount.textContent = cCount ? `${cCount} ядер • ${th
Count} потоков` : '0 ядер';
1190    }
1191
1192    if (!Array.isArray(cores) || cores.length === 0) {
1193      box.innerHTML = `<div class="text-muted small text-cente
r py-3 col-12">Опрос ядер CPU выполняется...</div>`;
1194      return;
1195    }
1196
1197    box.innerHTML = cores.map(c => {
1198      const load = c.load_percent == null ? null : Number(c.lo
ad_percent);
1199      const temp = c.temperature_c == null ? null : Number(c.t
emperature_c);
1200      const freqStr = c.frequency_str || (c.frequency_mhz ? (c
.frequency_mhz >= 1000 ? (c.frequency_mhz / 1000).toFixed(2) +
 ' GHz' : c.frequency_mhz.toFixed(0) + ' MHz') : '--');
1201      const powerStr = c.power_w != null ? `${Number(c.power_w
).toFixed(1)} W` : (c.power_str || '--');
1202      const loadStr = load == null ? '--' : `${load.toFixed(0)
}%`;
1203      const tempStr = temp != null ? `${temp.toFixed(0)} °C` :
 '-- °C';
1204      const gaugeSvg = createGaugeSvg(load, 100, 54);
1205      const tempHtml = createTempSliderHtml(temp, 30, 95, fals
e);
1206
1207      const threadsList = Array.isArray(c.threads) ? c.threads
```

```
 : [];
1208      const threadsHtml = threadsList.map(th => {
1209        const thLoad = Number(th.load_percent || 0);
1210        const barClass = thLoad >= 80 ? 'bg-danger' : thLoad >
= 60 ? 'bg-warning' : 'bg-info';
1211        return `
1212          <div class="d-flex align-items-center justify-conten
t-between gap-1 mb-1" style="font-size: 0.72rem;">
1213            <span class="text-truncate" style="max-width: 65px
; color: var(--text-color);" title="${escapeHtml(th.name || 'T
hread')}">
1214              ${escapeHtml(th.name || 'Thread')}
1215            </span>
1216            <div class="progress flex-grow-1" style="height: 5
px; background: var(--border-color); border-radius: 3px;">
1217              <div class="progress-bar ${barClass}" style="wid
th: ${Math.min(100, thLoad)}%; border-radius: 3px;"></div>
1218            </div>
1219            <span class="font-monospace fw-bold" style="width:
 32px; text-align: right; font-size: 0.69rem; color: var(--tex
t-color);">${thLoad.toFixed(0)}%</span>
1220          </div>
1221        `;
1222      }).join('');
1223
1224      return `
1225        <div class="sys-core-card p-2 rounded-2" style="backgr
ound: var(--surface-2, rgba(255,255,255,0.03)); border: 1px so
lid var(--border-color, rgba(255,255,255,0.08));" title="Ядро
#${c.index}: Нагрузка ${loadStr}, Температура ${tempStr}">
1226          <div class="d-flex justify-content-between align-ite
ms-center mb-1 pb-1 border-bottom" style="border-color: var(--
border-subtle, rgba(255,255,255,0.06)) !important;">
1227            <div class="d-flex align-items-center gap-1.5">
1228              <i class="bi bi-cpu text-info"></i>
1229              <span class="sys-core-title fw-bold" style="font
-size: 0.82rem;">Core #${c.index}</span>
1230            </div>
```

```
            <span class="badge ${load != null && load >= 80 ? 'bg-danger' : 'bg-info-subtle text-info border border-info'}" style="font-size: 0.70rem;">${loadStr}</span>
          </div>

          <div class="d-flex align-items-center gap-2 mb-2">
            <div class="d-flex flex-column align-items-center justify-content-center" style="width: 96px; flex-shrink: 0;">
              <div style="width: 96px; height: 52px;">
                ${gaugeSvg}
              </div>
              <div class="w-100 mt-1">
                ${tempHtml}
              </div>
            </div>
            <div class="flex-grow-1 ps-1" style="min-width: 0;">
              <div class="sys-core-param-table">
                <div class="sys-core-param-row">
                  <span class="sys-core-param-key"><i class="bi bi-activity text-info me-1"></i>Нагрузка</span>
                  <span class="sys-core-param-val text-info">${loadStr}</span>
                </div>
                <div class="sys-core-param-row">
                  <span class="sys-core-param-key"><i class="bi bi-thermometer-half text-danger me-1"></i>Температура</span>
                  <span class="sys-core-param-val ${temp != null && temp >= 75 ? 'text-danger' : 'text-warning'}">${tempStr}</span>
                </div>
                <div class="sys-core-param-row">
                  <span class="sys-core-param-key"><i class="bi bi-speedometer2 text-info me-1"></i>Частота</span>
                  <span class="sys-core-param-val text-info">${escapeHtml(freqStr)}</span>
                </div>
                <div class="sys-core-param-row">
```

```
                  <span class="sys-core-param-key"><i class="b
i bi-lightning text-warning me-1"></i>Мощность</span>
                  <span class="sys-core-param-val text-warning
">${escapeHtml(powerStr)}</span>
                </div>
              </div>
            </div>
          </div>

          <div class="pt-1 border-top" style="border-color: va
r(--border-subtle, rgba(255,255,255,0.06)) !important;">
            <div class="text-muted small mb-1" style="font-siz
e: 0.68rem; font-weight: 600;">Потоки:</div>
            <div class="sys-core-threads-list">
              ${threadsHtml || '<div class="text-muted small">
1 поток</div>'}
            </div>
          </div>
        </div>
      `;
    }).join('');
  }

  function formatBytes(v) {
    const n = Number(v) || 0;
    const units = ['B', 'KB', 'MB', 'GB', 'TB'];
    let i = 0;
    let x = n;
    while (x >= 1024 && i < units.length - 1) { x /= 1024; i++
; }
    return `${x.toFixed(i === 0 ? 0 : 1)} ${units[i]}`;
  }

  function formatBytesPerSec(v) {
    const n = Number(v) || 0;
    const units = ['B/s', 'KB/s', 'MB/s', 'GB/s'];
    let i = 0;
    let x = n;
```

```
    while (x >= 1024 && i < units.length - 1) { x /= 1024; i++; }
    return `${x.toFixed(i === 0 ? 0 : 1)} ${units[i]}`;
  }

  const _cpuSparkHistory = [];
  const _gpuSparkHistory = [];

  let currentCpuSparkScale = 'log';
  let lastCpuSparkData = [];

  async function setCpuSparkInterval(interval) {
    currentCpuSparkInterval = interval;
    const el = document.getElementById('sel-cpu-spark-interval');
    if (el) el.value = interval;
    if (interval === 'seconds') {
      await fetchCpuLoadFromApi();
    } else {
      await fetchCpuHistoryByInterval(interval);
    }
  }
  window.setCpuSparkInterval = setCpuSparkInterval;

  async function fetchCpuHistoryByInterval(interval) {
    try {
      const res = await fetch(`/api/v1/telemetry/history?interval=${encodeURIComponent(interval)}&metric=cpu&limit=120`);
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      if (data && Array.isArray(data.history) && data.history.length > 0) {
        lastCpuSparkData = data.history.map(r => ({
          timestamp: r.timestamp,
          time_label: formatSparkTimeLabel(r.timestamp || r.created_at, interval),
          load_percent: Number(r.cpu_total_percent || 0),
          temperature_c: r.gpu_temp_c != null ? Number(r.gpu_t
```

```
emp_c) : null,
          power_w: null
        }));
        renderCpuSpark(lastCpuSparkData);
      }
    } catch (e) {
      console.warn('[SystemInspectorTab] Ошибка получения истории CPU по интервалу:', e);
    }
  }

  function setCpuSparkScale(scale) {
    currentCpuSparkScale = scale;
    const btnLog = document.getElementById('btn-cpu-spark-log');
    const btnLin = document.getElementById('btn-cpu-spark-lin');
    if (btnLog && btnLin) {
      if (scale === 'log') {
        btnLog.className = 'btn btn-xs btn-outline-info active py-0 px-2 fw-bold';
        btnLin.className = 'btn btn-xs btn-outline-secondary py-0 px-2';
      } else {
        btnLin.className = 'btn btn-xs btn-outline-info active py-0 px-2 fw-bold';
        btnLog.className = 'btn btn-xs btn-outline-secondary py-0 px-2';
      }
    }
    if (lastCpuSparkData && lastCpuSparkData.length) {
      renderCpuSpark(lastCpuSparkData);
    }
  }
  window.setCpuSparkScale = setCpuSparkScale;

  function buildSmoothSvgPath(pts) {
    if (!pts || pts.length === 0) return '';
```

```
    if (pts.length === 1) return `M ${pts[0].x.toFixed(1)} ${pts[0].y.toFixed(1)}`;
    let d = `M ${pts[0].x.toFixed(1)} ${pts[0].y.toFixed(1)}`;
    for (let i = 0; i < pts.length - 1; i++) {
      const p0 = i > 0 ? pts[i - 1] : pts[i];
      const p1 = pts[i];
      const p2 = pts[i + 1];
      const p3 = i !== pts.length - 2 ? pts[i + 2] : p2;

      const cp1x = p1.x + (p2.x - p0.x) / 6;
      const cp1y = p1.y + (p2.y - p0.y) / 6;
      const cp2x = p2.x - (p3.x - p1.x) / 6;
      const cp2y = p2.y - (p3.y - p1.y) / 6;

      d += ` C ${cp1x.toFixed(1)} ${cp1y.toFixed(1)}, ${cp2x.toFixed(1)} ${cp2y.toFixed(1)}, ${p2.x.toFixed(1)} ${p2.y.toFixed(1)}`;
    }
    return d;
  }

  function renderCpuSpark(history) {
    const box = document.getElementById('sys-cpu-spark');
    const container = document.getElementById('sys-cpu-spark-container');
    const tooltip = document.getElementById('sys-cpu-spark-tooltip');
    if (!box) return;

    let ptsData = Array.isArray(history) && history.length > 0 ? history : (lastCpuSparkData && lastCpuSparkData.length > 0 ? lastCpuSparkData : (_cpuSparkHistory && _cpuSparkHistory.length > 0 ? _cpuSparkHistory : []));
    if (!ptsData || ptsData.length === 0) {
      ptsData = [{ load: 0, temp: null, power: null, time: new Date() }];
    }
    if (ptsData.length === 1) {
```

```
      ptsData = [ptsData[0], { ...ptsData[0], time: ptsData[0].time || new Date() }];
    }
    lastCpuSparkData = ptsData;
    history = ptsData;

    box.replaceChildren();
    const ns = 'http://www.w3.org/2000/svg';
    const W = 600, H = 84;
    const padTop = 6, padBottom = 6;
    const innerH = H - padTop - padBottom;

    const svg = document.createElementNS(ns, 'svg');
    svg.setAttribute('viewBox', `0 0 ${W} ${H}`);
    svg.setAttribute('preserveAspectRatio', 'none');
    svg.style.cssText = 'width:100%;height:100%;display:block;cursor:crosshair;';

    // 1. Defs с градиентами и фильтром свечения
    const defs = document.createElementNS(ns, 'defs');
    defs.innerHTML = `
      <linearGradient id="cpuGradLoad" x1="0%" y1="0%" x2="0%" y2="100%">
        <stop offset="0%" stop-color="#0dcaf0" stop-opacity="0.38"/>
        <stop offset="60%" stop-color="#0dcaf0" stop-opacity="0.12"/>
        <stop offset="100%" stop-color="#0dcaf0" stop-opacity="0.0"/>
      </linearGradient>
      <linearGradient id="cpuGradTemp" x1="0%" y1="0%" x2="0%" y2="100%">
        <stop offset="0%" stop-color="#ef4444" stop-opacity="0.28"/>
        <stop offset="100%" stop-color="#ef4444" stop-opacity="0.0"/>
      </linearGradient>
      <linearGradient id="cpuGradPower" x1="0%" y1="0%" x2="0%
```

```
" y2="100%">
1411        <stop offset="0%" stop-color="#f59e0b" stop-opacity="0
.22"/>
1412        <stop offset="100%" stop-color="#f59e0b" stop-opacity=
"0.0"/>
1413      </linearGradient>
1414    `;
1415    svg.appendChild(defs);
1416
1417    // 2. Горизонтальная сетка (Grid Lines)
1418    const gridG = document.createElementNS(ns, 'g');
1419    gridG.setAttribute('opacity', '0.2');
1420    [0.0, 0.25, 0.5, 0.75, 1.0].forEach(ratio => {
1421      const y = H - padBottom - ratio * innerH;
1422      const gl = document.createElementNS(ns, 'line');
1423      gl.setAttribute('x1', '0');
1424      gl.setAttribute('y1', y.toFixed(1));
1425      gl.setAttribute('x2', String(W));
1426      gl.setAttribute('y2', y.toFixed(1));
1427      gl.setAttribute('stroke', '#ffffff');
1428      gl.setAttribute('stroke-dasharray', ratio === 0.0 || rat
io === 1.0 ? 'none' : '3,3');
1429      gl.setAttribute('stroke-width', '1');
1430      gridG.appendChild(gl);
1431    });
1432    svg.appendChild(gridG);
1433
1434    // 3. Расчет динамических диапазонов
1435    const count = history.length;
1436    const isLog = currentCpuSparkScale === 'log';
1437
1438    // Температурный локальный диапазон (для выявления микроко
лебаний)
1439    const rawTemps = history.map(h => Number(h.temperature_c !
= null ? h.temperature_c : (h.temp || 0))).filter(t => t > 0);
1440    const tMin = rawTemps.length ? Math.min(...rawTemps) : 30;
1441    const tMax = rawTemps.length ? Math.max(...rawTemps) : 90;
1442    const tDelta = Math.max(6, tMax - tMin);
```

```
1443
1444     // Диапазон мощности (для выявления микроколебаний)
1445     const rawPowers = history.map(h => Number(h.power_w != nul
     l ? h.power_w : (h.power || 0))).filter(p => p > 0);
1446     const pMin = rawPowers.length ? Math.min(...rawPowers) : 5
     ;
1447     const pMax = rawPowers.length ? Math.max(...rawPowers) : 1
     00;
1448     const pDelta = Math.max(8, pMax - pMin);
1449
1450     // Динамический предел шкалы для линейного режима (с запас
     ом 15% и минимумом 5%)
1451     const rawLoads = history.map(h => Number(h.load_percent !=
      null ? h.load_percent : (h.load || 0)));
1452     const maxLoad = rawLoads.length ? Math.max(...rawLoads) :
     0;
1453     const loadMaxScale = Math.min(100, Math.max(5, maxLoad * 1
     .15));
1454
1455     // Главная шкала: Загрузка CPU (0..100% высоты холста)
1456     const transformLoad = (v) => {
1457       const cl = Math.min(100, Math.max(0, Number(v || 0)));
1458       if (!isLog) return Math.min(1, cl / loadMaxScale);
1459       return Math.log10(1 + 9 * (cl / 100)); // Log-10 mapping
     : 10% -> 0.28, 50% -> 0.74, 100% -> 1.0
1460     };
1461
1462     // Вторичная компактная шкала: Температура (занимает 10% -
      38% высоты)
1463     const transformTemp = (v) => {
1464       if (v == null) return 0.12;
1465       const frac = Math.min(1, Math.max(0, (Number(v) - (tMin
     - 1)) / (tDelta + 2)));
1466       return 0.10 + frac * 0.28;
1467     };
1468
1469     // Вторичная компактная шкала: Мощность (занимает 4% - 28%
      высоты)
```

```
    const transformPower = (v) => {
      if (v == null) return 0.08;
      const frac = Math.min(1, Math.max(0, (Number(v) - (pMin - 1.5)) / (pDelta + 3)));
      return 0.04 + frac * 0.24;
    };

    // Точки графиков
    const loadPts = [];
    const tempPts = [];
    const powerPts = [];

    history.forEach((h, i) => {
      const x = (i / (count - 1)) * W;

      const lVal = Number(h.load_percent != null ? h.load_percent : (h.load || 0));
      const tVal = h.temperature_c != null ? Number(h.temperature_c) : (h.temp != null ? Number(h.temp) : null);
      const pVal = h.power_w != null ? Number(h.power_w) : (h.power != null ? Number(h.power) : null);

      const yLoad = H - padBottom - transformLoad(lVal) * innerH;
      const yTemp = H - padBottom - transformTemp(tVal) * innerH;
      const yPower = H - padBottom - transformPower(pVal) * innerH;

      loadPts.push({ x, y: yLoad, raw: lVal, time: h.time_label || h.timestamp });
      tempPts.push({ x, y: yTemp, raw: tVal });
      powerPts.push({ x, y: yPower, raw: pVal });
    });

    // 4. Отрисовка площадей и кривых (Area & Spline)
    const drawSeries = (pts, strokeColor, fillColor, strokeWidth = 1.8) => {
```

```
      if (pts.length < 2) return;
      const linePath = buildSmoothSvgPath(pts);

      // Заливка градиентом
      if (fillColor) {
        const areaPath = `${linePath} L ${pts[pts.length - 1].x.toFixed(1)} ${H - padBottom} L ${pts[0].x.toFixed(1)} ${H - padBottom} Z`;
        const aEl = document.createElementNS(ns, 'path');
        aEl.setAttribute('d', areaPath);
        aEl.setAttribute('fill', fillColor);
        svg.appendChild(aEl);
      }

      // Линия кривой
      const pEl = document.createElementNS(ns, 'path');
      pEl.setAttribute('d', linePath);
      pEl.setAttribute('fill', 'none');
      pEl.setAttribute('stroke', strokeColor);
      pEl.setAttribute('stroke-width', String(strokeWidth));
      pEl.setAttribute('stroke-linejoin', 'round');
      pEl.setAttribute('stroke-linecap', 'round');
      svg.appendChild(pEl);
    };

    // Рисуем: Тонкие тренды температуры и мощности на компактной шкале + Основной график загрузки с градиентом
    drawSeries(powerPts, '#f59e0b', null, 1.2);
    drawSeries(tempPts, '#ef4444', null, 1.3);
    drawSeries(loadPts, '#0dcaf0', 'url(#cpuGradLoad)', 2.0);

    // 5. Интерактивный Hover-курсор и Тултип
    const crosshair = document.createElementNS(ns, 'line');
    crosshair.setAttribute('y1', '0');
    crosshair.setAttribute('y2', String(H));
    crosshair.setAttribute('stroke', 'var(--text-muted, rgba(128,128,128,0.5))');
    crosshair.setAttribute('stroke-dasharray', '2,2');
```

```
    crosshair.setAttribute('stroke-width', '1');
    crosshair.style.display = 'none';
    svg.appendChild(crosshair);

    svg.addEventListener('mousemove', (evt) => {
      const rect = svg.getBoundingClientRect();
      const mouseX = Math.max(0, Math.min(rect.width, evt.clie
ntX - rect.left));
      const ratio = mouseX / rect.width;
      const idx = Math.min(count - 1, Math.max(0, Math.round(r
atio * (count - 1))));
      const pt = loadPts[idx];
      const hItem = history[idx];

      if (pt && tooltip) {
        crosshair.setAttribute('x1', pt.x.toFixed(1));
        crosshair.setAttribute('x2', pt.x.toFixed(1));
        crosshair.style.display = 'block';

        const tStr = hItem.time_label || (hItem.timestamp ? ne
w Date(hItem.timestamp).toLocaleTimeString() : (hItem.time ? n
ew Date(hItem.time).toLocaleTimeString() : '--:--:--'));
        const lStr = `${pt.raw.toFixed(1)}%`;
        const tempVal = tempPts[idx] && tempPts[idx].raw != nu
ll ? `${tempPts[idx].raw.toFixed(0)} °C` : '--';
        const powVal = powerPts[idx] && powerPts[idx].raw != n
ull ? `${powerPts[idx].raw.toFixed(1)} W` : '--';

        tooltip.innerHTML = `
          <div class="d-flex align-items-center justify-conten
t-between gap-2 border-bottom pb-0.5 mb-1" style="border-color
: rgba(255,255,255,0.1) !important;">
            <span class="text-muted"><i class="bi bi-clock me-
1"></i>${escapeHtml(tStr)}</span>
            <span class="badge ${isLog ? 'bg-info-subtle text-
info' : 'bg-secondary text-light'}" style="font-size:0.62rem;"
>${isLog ? 'Log (0–100%)' : 'Lin (0–' + Math.round(loadMaxScal
e) + '%)'}</span>
```

```
          </div>
          <div class="d-flex gap-2 font-monospace">
            <span class="text-info fw-bold"><i class="bi bi-activity me-0.5"></i>${lStr}</span>
            <span class="text-danger fw-bold"><i class="bi bi-thermometer-half me-0.5"></i>${tempVal}</span>
            <span class="text-warning fw-bold"><i class="bi bi-lightning me-0.5"></i>${powVal}</span>
          </div>
        `;
        tooltip.style.display = 'block';

        // Позиционирование тултипа
        const tipX = Math.min(rect.width - 150, Math.max(10, mouseX - 60));
        tooltip.style.left = `${tipX}px`;
      }
    });

    svg.addEventListener('mouseleave', () => {
      crosshair.style.display = 'none';
      if (tooltip) tooltip.style.display = 'none';
    });

    box.appendChild(svg);
  }

  async function setGpuSparkInterval(idx, interval) {
    currentGpuSparkIntervals[idx] = interval;
    const el = document.getElementById(`sel-gpu-spark-interval-${idx}`);
    if (el) el.value = interval;
    if (interval === 'seconds') {
      await fetchGpuLoadFromApi();
    } else {
      await fetchGpuHistoryByInterval(idx, interval);
    }
  }
```

```
  window.setGpuSparkInterval = setGpuSparkInterval;

  async function fetchGpuHistoryByInterval(idx, interval) {
    try {
      const res = await fetch(`/api/v1/telemetry/history?interval=${encodeURIComponent(interval)}&metric=gpu&limit=120`);
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      if (data && Array.isArray(data.history) && data.history.length > 0) {
        const historyPts = data.history.map(r => ({
          timestamp: r.timestamp,
          time_label: formatSparkTimeLabel(r.timestamp || r.created_at, interval),
          load_percent: Number(r.gpu_load_percent || 0),
          temperature_c: r.gpu_temp_c != null ? Number(r.gpu_temp_c) : null,
          power_w: null,
          vram_percent: null
        }));
        _gpuSparkHistories[idx] = historyPts;
        lastGpuSparkDataMap[idx] = historyPts;
        renderGpuSpark(historyPts, idx);
      }
    } catch (e) {
      console.warn('[SystemInspectorTab] Ошибка получения истории GPU по интервалу:', e);
    }
  }

  function setGpuSparkScale(scale, idx = 0) {
    currentGpuSparkScales[idx] = scale;
    const btnLog = document.getElementById(`btn-gpu-spark-log-${idx}`) || document.getElementById('btn-gpu-spark-log');
    const btnLin = document.getElementById(`btn-gpu-spark-lin-${idx}`) || document.getElementById('btn-gpu-spark-lin');
    if (btnLog && btnLin) {
      if (scale === 'log') {
```

```
        btnLog.className = 'btn btn-xs btn-outline-warning active py-0 px-2 fw-bold';
        btnLin.className = 'btn btn-xs btn-outline-secondary py-0 px-2';
      } else {
        btnLin.className = 'btn btn-xs btn-outline-warning active py-0 px-2 fw-bold';
        btnLog.className = 'btn btn-xs btn-outline-secondary py-0 px-2';
      }
    }
    if (lastGpuSparkDataMap[idx] && lastGpuSparkDataMap[idx].length) {
      renderGpuSpark(lastGpuSparkDataMap[idx], idx);
    }
  }
  window.setGpuSparkScale = setGpuSparkScale;

  function renderGpuSpark(history, idx = 0) {
    const box = document.getElementById(`sys-gpu-spark-${idx}`) || document.getElementById('sys-gpu-spark');
    const container = document.getElementById(`sys-gpu-spark-container-${idx}`) || document.getElementById('sys-gpu-spark-container');
    const tooltip = document.getElementById(`sys-gpu-spark-tooltip-${idx}`) || document.getElementById('sys-gpu-spark-tooltip');
    if (!box) return;

    let ptsData = Array.isArray(history) && history.length > 0 ? history : (lastGpuSparkDataMap[idx] && lastGpuSparkDataMap[idx].length > 0 ? lastGpuSparkDataMap[idx] : (_gpuSparkHistories[idx] && _gpuSparkHistories[idx].length > 0 ? _gpuSparkHistories[idx] : []));
    if (!ptsData || ptsData.length === 0) {
      ptsData = [{ load: 0, temp: null, power: null, time: new Date() }];
    }
```

```
    if (ptsData.length === 1) {
      ptsData = [ptsData[0], { ...ptsData[0], time: ptsData[0].time || new Date() }];
    }
    lastGpuSparkDataMap[idx] = ptsData;
    history = ptsData;

    box.replaceChildren();
    const ns = 'http://www.w3.org/2000/svg';
    const W = 600, H = 84;
    const padTop = 6, padBottom = 6;
    const innerH = H - padTop - padBottom;

    const svg = document.createElementNS(ns, 'svg');
    svg.setAttribute('viewBox', `0 0 ${W} ${H}`);
    svg.setAttribute('preserveAspectRatio', 'none');
    svg.style.cssText = 'width:100%;height:100%;display:block;cursor:crosshair;';

    // 1. Defs с уникальным ID градиента для каждого GPU
    const defs = document.createElementNS(ns, 'defs');
    const gradId = `gpuGradLoad_${idx}`;
    defs.innerHTML = `
      <linearGradient id="${gradId}" x1="0%" y1="0%" x2="0%" y2="100%">
        <stop offset="0%" stop-color="#f59e0b" stop-opacity="0.38"/>
        <stop offset="60%" stop-color="#f59e0b" stop-opacity="0.12"/>
        <stop offset="100%" stop-color="#f59e0b" stop-opacity="0.0"/>
      </linearGradient>
    `;
    svg.appendChild(defs);

    // 2. Горизонтальная сетка (Grid Lines)
    const gridG = document.createElementNS(ns, 'g');
    gridG.setAttribute('opacity', '0.2');
```

```
    [0.0, 0.25, 0.5, 0.75, 1.0].forEach(ratio => {
      const y = H - padBottom - ratio * innerH;
      const gl = document.createElementNS(ns, 'line');
      gl.setAttribute('x1', '0');
      gl.setAttribute('y1', y.toFixed(1));
      gl.setAttribute('x2', String(W));
      gl.setAttribute('y2', y.toFixed(1));
      gl.setAttribute('stroke', '#ffffff');
      gl.setAttribute('stroke-dasharray', ratio === 0.0 || rat
io === 1.0 ? 'none' : '3,3');
      gl.setAttribute('stroke-width', '1');
      gridG.appendChild(gl);
    });
    svg.appendChild(gridG);

    // 3. Расчет динамических диапазонов
    const count = history.length;
    const isLog = currentGpuSparkScales[idx] === 'log' || curr
entGpuSparkScales[idx] === undefined;

    // Температурный локальный диапазон
    const rawTemps = history.map(h => Number(h.temperature_c !
= null ? h.temperature_c : (h.temp || 0))).filter(t => t > 0);
    const tMin = rawTemps.length ? Math.min(...rawTemps) : 30;
    const tMax = rawTemps.length ? Math.max(...rawTemps) : 90;
    const tDelta = Math.max(6, tMax - tMin);

    // Диапазон мощности / памяти
    const rawPowers = history.map(h => Number(h.power_w != nul
l ? h.power_w : (h.power || h.vram_percent || 0))).filter(p =>
 p > 0);
    const pMin = rawPowers.length ? Math.min(...rawPowers) : 5
;
    const pMax = rawPowers.length ? Math.max(...rawPowers) : 1
00;
    const pDelta = Math.max(8, pMax - pMin);

    // Динамический предел шкалы для линейного режима (с запас
```

```
ом 15% и минимумом 5%)
    const rawLoads = history.map(h => Number(h.load_percent !=
 null ? h.load_percent : (h.load || 0)));
    const maxLoad = rawLoads.length ? Math.max(...rawLoads) :
0;
    const loadMaxScale = Math.min(100, Math.max(5, maxLoad * 1
.15));

    // Главная шкала: Загрузка GPU Core (0..100% высоты холста
)
    const transformLoad = (v) => {
      const cl = Math.min(100, Math.max(0, Number(v || 0)));
      if (!isLog) return Math.min(1, cl / loadMaxScale);
      return Math.log10(1 + 9 * (cl / 100));
    };

    // Вторичная компактная шкала: Температура (занимает 10% -
 38% высоты)
    const transformTemp = (v) => {
      if (v == null) return 0.12;
      const frac = Math.min(1, Math.max(0, (Number(v) - (tMin
- 1)) / (tDelta + 2)));
      return 0.10 + frac * 0.28;
    };

    // Вторичная компактная шкала: Мощность/Память (занимает 4
% - 28% высоты)
    const transformPower = (v) => {
      if (v == null) return 0.08;
      const frac = Math.min(1, Math.max(0, (Number(v) - (pMin
- 1.5)) / (pDelta + 3)));
      return 0.04 + frac * 0.24;
    };

    // Точки графиков
    const loadPts = [];
    const tempPts = [];
    const powerPts = [];
```

```
1738
1739      history.forEach((h, i) => {
1740        const x = (i / (count - 1)) * W;
1741
1742        const lVal = Number(h.load_percent != null ? h.load_perc
ent : (h.load || 0));
1743        const tVal = h.temperature_c != null ? Number(h.temperat
ure_c) : (h.temp != null ? Number(h.temp) : null);
1744        const pVal = h.power_w != null ? Number(h.power_w) : (h.
power != null ? Number(h.power) : (h.vram_percent != null ? Nu
mber(h.vram_percent) : null));
1745
1746        const yLoad = H - padBottom - transformLoad(lVal) * inne
rH;
1747        const yTemp = H - padBottom - transformTemp(tVal) * inne
rH;
1748        const yPower = H - padBottom - transformPower(pVal) * in
nerH;
1749
1750        loadPts.push({ x, y: yLoad, raw: lVal, time: h.time_labe
l || h.timestamp });
1751        tempPts.push({ x, y: yTemp, raw: tVal });
1752        powerPts.push({ x, y: yPower, raw: pVal });
1753      });
1754
1755      // 4. Отрисовка площадей и кривых (Area & Spline)
1756      const drawSeries = (pts, strokeColor, fillColor, strokeWid
th = 1.8) => {
1757        if (pts.length < 2) return;
1758        const linePath = buildSmoothSvgPath(pts);
1759
1760        // Заливка градиентом
1761        if (fillColor) {
1762          const areaPath = `${linePath} L ${pts[pts.length - 1].
x.toFixed(1)} ${H - padBottom} L ${pts[0].x.toFixed(1)} ${H -
padBottom} Z`;
1763          const aEl = document.createElementNS(ns, 'path');
1764          aEl.setAttribute('d', areaPath);
```

```
        aEl.setAttribute('fill', fillColor);
        svg.appendChild(aEl);
      }

      // Линия кривой
      const pEl = document.createElementNS(ns, 'path');
      pEl.setAttribute('d', linePath);
      pEl.setAttribute('fill', 'none');
      pEl.setAttribute('stroke', strokeColor);
      pEl.setAttribute('stroke-width', String(strokeWidth));
      pEl.setAttribute('stroke-linejoin', 'round');
      pEl.setAttribute('stroke-linecap', 'round');
      svg.appendChild(pEl);
    };

    drawSeries(powerPts, '#0dcaf0', null, 1.2);
    drawSeries(tempPts, '#ef4444', null, 1.3);
    drawSeries(loadPts, '#f59e0b', `url(#${gradId})`, 2.0);

    // 5. Интерактивный Hover-курсор и Тултип
    const crosshair = document.createElementNS(ns, 'line');
    crosshair.setAttribute('y1', '0');
    crosshair.setAttribute('y2', String(H));
    crosshair.setAttribute('stroke', 'var(--text-muted, rgba(1
28,128,128,0.5))');
    crosshair.setAttribute('stroke-dasharray', '2,2');
    crosshair.setAttribute('stroke-width', '1');
    crosshair.style.display = 'none';
    svg.appendChild(crosshair);

    svg.addEventListener('mousemove', (evt) => {
      const rect = svg.getBoundingClientRect();
      const mouseX = Math.max(0, Math.min(rect.width, evt.clie
ntX - rect.left));
      const ratio = mouseX / rect.width;
      const curIndex = Math.min(count - 1, Math.max(0, Math.ro
und(ratio * (count - 1))));
      const pt = loadPts[curIndex];
```

```
1800      const hItem = history[curIndex];
1801
1802      if (pt && tooltip) {
1803        crosshair.setAttribute('x1', pt.x.toFixed(1));
1804        crosshair.setAttribute('x2', pt.x.toFixed(1));
1805        crosshair.style.display = 'block';
1806
1807        const tStr = hItem.time_label || (hItem.timestamp ? ne
     w Date(hItem.timestamp).toLocaleTimeString() : (hItem.time ? n
     ew Date(hItem.time).toLocaleTimeString() : '--:--:--'));
1808        const lStr = `${pt.raw.toFixed(1)}%`;
1809        const tempVal = tempPts[curIndex] && tempPts[curIndex]
     .raw != null ? `${tempPts[curIndex].raw.toFixed(0)} °C` : '--'
     ;
1810        const powVal = powerPts[curIndex] && powerPts[curIndex
     ].raw != null ? (hItem.power_w != null ? `${powerPts[curIndex]
     .raw.toFixed(1)} W` : `${powerPts[curIndex].raw.toFixed(0)}%`)
      : '--';
1811
1812        tooltip.innerHTML = `
1813          <div class="d-flex align-items-center justify-conten
     t-between gap-2 border-bottom pb-0.5 mb-1" style="border-color
     : rgba(255,255,255,0.1) !important;">
1814            <span class="text-muted"><i class="bi bi-clock me-
     1"></i>${escapeHtml(tStr)}</span>
1815            <span class="badge ${isLog ? 'bg-warning-subtle te
     xt-warning' : 'bg-secondary text-light'}" style="font-size:0.6
     2rem;">${isLog ? 'Log (0–100%)' : 'Lin (0–' + Math.round(loadM
     axScale) + '%)'}</span>
1816          </div>
1817          <div class="d-flex gap-2 font-monospace">
1818            <span class="text-warning fw-bold"><i class="bi bi
     -activity me-0.5"></i>${lStr}</span>
1819            <span class="text-danger fw-bold"><i class="bi bi-
     thermometer-half me-0.5"></i>${tempVal}</span>
1820            <span class="text-info fw-bold"><i class="bi bi-li
     ghtning me-0.5"></i>${powVal}</span>
1821          </div>
```

```
        `;
        tooltip.style.display = 'block';

        const tipX = Math.min(rect.width - 150, Math.max(10, m
ouseX - 60));
        tooltip.style.left = `${tipX}px`;
      }
    });

    svg.addEventListener('mouseleave', () => {
      crosshair.style.display = 'none';
      if (tooltip) tooltip.style.display = 'none';
    });

    box.appendChild(svg);
  }

  let currentRamSparkScale = 'log';
  let lastMemIoSparkData = [];
  const _memIoSparkHistory = [];

  async function setRamSparkInterval(interval) {
    currentRamSparkInterval = interval;
    const el = document.getElementById('sel-ram-spark-interval
');
    if (el) el.value = interval;
    if (interval === 'seconds') {
      await fetchMemoryIoFromApi();
    } else {
      await fetchRamHistoryByInterval(interval);
    }
  }
  window.setRamSparkInterval = setRamSparkInterval;

  async function fetchRamHistoryByInterval(interval) {
    try {
      const res = await fetch(`/api/v1/telemetry/history?inter
val=${encodeURIComponent(interval)}&metric=ram&limit=120`);
```

```
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      if (data && Array.isArray(data.history) && data.history.length > 0) {
        const rows = data.history.map(r => ({
          timestamp: r.timestamp,
          time_label: formatSparkTimeLabel(r.timestamp || r.created_at, interval),
          memory_percent: Number(r.memory_percent || 0),
          read_bytes_sec: Number(r.disk_read_bytes_sec || 0),
          write_bytes_sec: Number(r.disk_write_bytes_sec || 0)
        }));
        lastMemIoSparkData = rows;
        renderMemIoSpark(rows);
      }
    } catch (e) {
      console.warn('[SystemInspectorTab] Ошибка получения истории RAM по интервалу:', e);
    }
  }

  function setRamSparkScale(scale) {
    currentRamSparkScale = scale;
    const btnLog = document.getElementById('btn-ram-spark-log');
    const btnLin = document.getElementById('btn-ram-spark-lin');
    if (btnLog && btnLin) {
      if (scale === 'log') {
        btnLog.className = 'btn btn-xs btn-outline-success active py-0 px-2 fw-bold';
        btnLin.className = 'btn btn-xs btn-outline-secondary py-0 px-2';
      } else {
        btnLin.className = 'btn btn-xs btn-outline-success active py-0 px-2 fw-bold';
        btnLog.className = 'btn btn-xs btn-outline-secondary py-0 px-2';
```

```
      }
    }
    if (lastMemIoSparkData && lastMemIoSparkData.length) {
      renderMemIoSpark(lastMemIoSparkData);
    }
  }
  window.setRamSparkScale = setRamSparkScale;

  function renderMemIoSpark(history) {
    const box = document.getElementById('sys-memio-spark');
    const container = document.getElementById('sys-memio-spark
-container');
    const tooltip = document.getElementById('sys-memio-spark-t
ooltip');
    if (!box) return;

    let ptsData = Array.isArray(history) && history.length > 0
 ? history : (lastMemIoSparkData && lastMemIoSparkData.length
> 0 ? lastMemIoSparkData : (_memIoSparkHistory && _memIoSparkH
istory.length > 0 ? _memIoSparkHistory : []));
    if (!ptsData || ptsData.length === 0) {
      ptsData = [{ pct: 0, used: 0, swap: 0, time: new Date()
}];
    }
    if (ptsData.length === 1) {
      ptsData = [ptsData[0], { ...ptsData[0], time: ptsData[0]
.time || new Date() }];
    }
    lastMemIoSparkData = ptsData;
    history = ptsData;

    box.replaceChildren();
    const ns = 'http://www.w3.org/2000/svg';
    const W = 600, H = 84;
    const padTop = 6, padBottom = 6;
    const innerH = H - padTop - padBottom;

    const svg = document.createElementNS(ns, 'svg');
```

```
    svg.setAttribute('viewBox', `0 0 ${W} ${H}`);
    svg.setAttribute('preserveAspectRatio', 'none');
    svg.style.cssText = 'width:100%;height:100%;display:block;cursor:crosshair;';

    // 1. Defs с градиентом
    const defs = document.createElementNS(ns, 'defs');
    defs.innerHTML = `
      <linearGradient id="ramGradLoad" x1="0%" y1="0%" x2="0%" y2="100%">
        <stop offset="0%" stop-color="#10b981" stop-opacity="0.38"/>
        <stop offset="60%" stop-color="#10b981" stop-opacity="0.12"/>
        <stop offset="100%" stop-color="#10b981" stop-opacity="0.0"/>
      </linearGradient>
    `;
    svg.appendChild(defs);

    // 2. Горизонтальная сетка (Grid Lines)
    const gridG = document.createElementNS(ns, 'g');
    gridG.setAttribute('opacity', '0.2');
    [0.0, 0.25, 0.5, 0.75, 1.0].forEach(ratio => {
      const y = H - padBottom - ratio * innerH;
      const gl = document.createElementNS(ns, 'line');
      gl.setAttribute('x1', '0');
      gl.setAttribute('y1', y.toFixed(1));
      gl.setAttribute('x2', String(W));
      gl.setAttribute('y2', y.toFixed(1));
      gl.setAttribute('stroke', '#ffffff');
      gl.setAttribute('stroke-dasharray', ratio === 0.0 || ratio === 1.0 ? 'none' : '3,3');
      gl.setAttribute('stroke-width', '1');
      gridG.appendChild(gl);
    });
    svg.appendChild(gridG);

```

```
    // 3. Динамические диапазоны
    const count = history.length;
    const isLog = currentRamSparkScale === 'log';

    const rawRams = history.map(h => Number(h.memory_percent !
= null ? h.memory_percent : (h.ram || 0)));
    const maxRam = rawRams.length ? Math.max(...rawRams) : 0;
    const ramMaxScale = Math.min(100, Math.max(10, maxRam * 1.
15));

    const rawReads = history.map(h => Number(h.read_bytes_sec
!= null ? h.read_bytes_sec : (h.disk_read || 0)));
    const rawWrites = history.map(h => Number(h.write_bytes_se
c != null ? h.write_bytes_sec : (h.disk_write || 0)));
    const maxIoBytes = Math.max(1024 * 1024, ...rawReads, ...r
awWrites);
    const ioMaxScale = maxIoBytes * 1.15;

    const transformRam = (v) => {
      const cl = Math.min(100, Math.max(0, Number(v || 0)));
      if (!isLog) return Math.min(1, cl / ramMaxScale);
      return Math.log10(1 + 9 * (cl / 100));
    };

    const transformIo = (v) => {
      if (!v || v <= 0) return 0;
      const nv = Number(v);
      if (!isLog) return Math.min(1, nv / ioMaxScale);
      return Math.log10(1 + 9 * (Math.min(ioMaxScale, nv) / io
MaxScale));
    };

    const ramPts = [];
    const readPts = [];
    const writePts = [];

    history.forEach((h, i) => {
      const x = (i / (count - 1)) * W;
```

```
      const rVal = Number(h.memory_percent != null ? h.memory_percent : (h.ram || 0));
      const rdVal = Number(h.read_bytes_sec != null ? h.read_bytes_sec : (h.disk_read || 0));
      const wrVal = Number(h.write_bytes_sec != null ? h.write_bytes_sec : (h.disk_write || 0));

      const yRam = H - padBottom - transformRam(rVal) * innerH;
      const yRead = H - padBottom - (0.04 + transformIo(rdVal) * 0.44) * innerH;
      const yWrite = H - padBottom - (0.04 + transformIo(wrVal) * 0.44) * innerH;

      ramPts.push({ x, y: yRam, raw: rVal, time: h.timestamp || h.time });
      readPts.push({ x, y: yRead, raw: rdVal });
      writePts.push({ x, y: yWrite, raw: wrVal });
    });

    const drawSeries = (pts, strokeColor, fillColor, strokeWidth = 1.8) => {
      if (pts.length < 2) return;
      const linePath = buildSmoothSvgPath(pts);
      if (fillColor) {
        const areaPath = `${linePath} L ${pts[pts.length - 1].x.toFixed(1)} ${H - padBottom} L ${pts[0].x.toFixed(1)} ${H - padBottom} Z`;
        const aEl = document.createElementNS(ns, 'path');
        aEl.setAttribute('d', areaPath);
        aEl.setAttribute('fill', fillColor);
        svg.appendChild(aEl);
      }
      const pEl = document.createElementNS(ns, 'path');
      pEl.setAttribute('d', linePath);
      pEl.setAttribute('fill', 'none');
      pEl.setAttribute('stroke', strokeColor);
      pEl.setAttribute('stroke-width', String(strokeWidth));
```

```
      pEl.setAttribute('stroke-linejoin', 'round');
      pEl.setAttribute('stroke-linecap', 'round');
      svg.appendChild(pEl);
    };

    drawSeries(writePts, '#f59e0b', null, 1.3);
    drawSeries(readPts, '#0dcaf0', null, 1.3);
    drawSeries(ramPts, '#10b981', 'url(#ramGradLoad)', 2.0);

    // Hover курсор и тултип
    const crosshair = document.createElementNS(ns, 'line');
    crosshair.setAttribute('y1', '0');
    crosshair.setAttribute('y2', String(H));
    crosshair.setAttribute('stroke', 'var(--text-muted, rgba(1
28,128,128,0.5))');
    crosshair.setAttribute('stroke-dasharray', '2,2');
    crosshair.setAttribute('stroke-width', '1');
    crosshair.style.display = 'none';
    svg.appendChild(crosshair);

    svg.addEventListener('mousemove', (evt) => {
      const rect = svg.getBoundingClientRect();
      const mouseX = Math.max(0, Math.min(rect.width, evt.clie
ntX - rect.left));
      const ratio = mouseX / rect.width;
      const curIndex = Math.min(count - 1, Math.max(0, Math.ro
und(ratio * (count - 1))));
      const pt = ramPts[curIndex];
      const hItem = history[curIndex];

      if (pt && tooltip) {
        crosshair.setAttribute('x1', pt.x.toFixed(1));
        crosshair.setAttribute('x2', pt.x.toFixed(1));
        crosshair.style.display = 'block';

        const tStr = hItem.timestamp ? new Date(hItem.timestam
p).toLocaleTimeString() : (hItem.time ? new Date(hItem.time).t
oLocaleTimeString() : '--:--:--');
```

```
        const rStr = `${pt.raw.toFixed(1)}%`;
        const rdStr = readPts[curIndex] ? formatBytesPerSec(readPts[curIndex].raw) : '0 B/s';
        const wrStr = writePts[curIndex] ? formatBytesPerSec(writePts[curIndex].raw) : '0 B/s';

        tooltip.innerHTML = `
          <div class="d-flex align-items-center justify-content-between gap-2 border-bottom pb-0.5 mb-1" style="border-color: rgba(255,255,255,0.1) !important;">
            <span class="text-muted"><i class="bi bi-clock me-1"></i>${escapeHtml(tStr)}</span>
            <span class="badge ${isLog ? 'bg-success-subtle text-success' : 'bg-secondary text-light'}" style="font-size:0.62rem;">${isLog ? 'Log (0–100%)' : 'Lin (0–' + Math.round(ramMaxScale) + '%)'}</span>
          </div>
          <div class="d-flex gap-2 font-monospace">
            <span class="text-success fw-bold"><i class="bi bi-memory me-0.5"></i>RAM: ${rStr}</span>
            <span class="text-info fw-bold"><i class="bi bi-arrow-down me-0.5"></i>R: ${rdStr}</span>
            <span class="text-warning fw-bold"><i class="bi bi-arrow-up me-0.5"></i>W: ${wrStr}</span>
          </div>
        `;
        tooltip.style.display = 'block';
        const tipX = Math.min(rect.width - 160, Math.max(10, mouseX - 60));
        tooltip.style.left = `${tipX}px`;
      }
    });

    svg.addEventListener('mouseleave', () => {
      crosshair.style.display = 'none';
      if (tooltip) tooltip.style.display = 'none';
    });

```

```
2068      box.appendChild(svg);
2069    }
2070
2071    /**
2072     * Генерация цветного процентного слайдера (для памяти, RAM,
 Swap).
2073     * @param {number|null} percent - Значение в % (0-100)
2074     * @param {boolean} isLarge - Флаг увеличенного отображения
для шапки
2075     * @param {string} labelText - Подпись бейджа справа
2076     * @returns {string} HTML разметка
2077     */
2078    function createPercentSliderHtml(percent, isLarge = false, l
abelText = '') {
2079      const val = percent == null ? 0 : Math.min(100, Math.max(0
, Number(percent)));
2080      let barColor = '#10b981'; // Зеленый (<60%)
2081      let badgeClass = 'text-success';
2082      if (val >= 85) {
2083        barColor = '#ef4444'; // Красный
2084        badgeClass = 'text-danger fw-bold';
2085      } else if (val >= 70) {
2086        barColor = '#f97316'; // Оранжевый
2087        badgeClass = 'text-warning fw-bold';
2088      } else if (val >= 50) {
2089        barColor = '#eab308'; // Желтый
2090        badgeClass = 'text-warning';
2091      }
2092
2093      const wrapClass = isLarge ? 'sys-temp-slider-wrap sys-temp
-slider-lg' : 'sys-temp-slider-wrap';
2094      const fontSize = isLarge ? 'font-size: 0.85rem;' : 'font-s
ize: 0.78rem;';
2095      const displayLabel = labelText || `${val.toFixed(0)}%`;
2096
2097      return `
2098        <div class="${wrapClass}">
2099          <div class="d-flex justify-content-between align-items
```

```
-center mb-1" style="${fontSize}">
          <span class="text-muted"><i class="bi bi-memory me-1
"></i>Занято</span>
          <span class="font-monospace ${badgeClass}" style="fo
nt-weight: 700;">${displayLabel}</span>
        </div>
        <div class="sys-temp-slider-track">
          <div class="sys-temp-slider-bar" style="width: ${val
.toFixed(1)}%; background: ${barColor};"></div>
          <div class="sys-temp-slider-thumb" style="left: ${va
l.toFixed(1)}%; background: ${barColor};"></div>
        </div>
      </div>
    `;
  }

  async function fetchMemoryIoFromApi() {
    try {
      const res = await fetch('/api/v1/panel/memory-io?limit=6
0');
      if (!res.ok) return;
      const data = await res.json();
      const set = (id, text) => { const el = document.getEleme
ntById(id); if (el) el.innerText = text; };
      const m = data.memory || {};
      const io = data.disk_io || {};
      const pct = Number(m.percent || 0);
      const swapPct = Number(m.swap_percent || 0);
      const usedGb = Number(m.used_gb || 0);
      const totalGb = Number(m.total_gb || 0);
      const freeGb = Number(m.free_gb || 0);

      // 1. Главный баннер RAM
      const ramModel = document.getElementById('sys-metric-ram
-model');
      if (ramModel) {
        ramModel.innerText = data.name || (m.name || (m.total_
gb ? `${Math.round(m.total_gb)} GB RAM` : 'RAM'));
```

```
2129      }
2130      set('sys-memio-pct', `${pct.toFixed(1)}%`);
2131      set('sys-memio-used', `${usedGb.toFixed(1)} / ${totalGb.
toFixed(1)} GB (свободно ${freeGb.toFixed(1)} GB)`);
2132      set('sys-memio-swap', `Файл подкачки: ${swapPct.toFixed(
1)}%`);
2133      set('sys-memio-ts', data.timestamp ? new Date(data.times
tamp).toLocaleTimeString() : '--');
2134
2135      const ramGauge = document.getElementById('sys-metric-ram
-gauge');
2136      if (ramGauge) {
2137        ramGauge.innerHTML = createGaugeSvg(pct, 130, 74);
2138      }
2139
2140      const ramSlider = document.getElementById('sys-metric-ra
m-slider');
2141      if (ramSlider) {
2142        ramSlider.innerHTML = createPercentSliderHtml(pct, tru
e, `${usedGb.toFixed(1)} / ${totalGb.toFixed(1)} GB`);
2143      }
2144
2145      // 2. Блок 1: Физическая RAM
2146      set('sys-ram-sub-badge', `${pct.toFixed(0)}%`);
2147      const ramSubGauge = document.getElementById('sys-ram-sub
-gauge');
2148      if (ramSubGauge) {
2149        ramSubGauge.innerHTML = createGaugeSvg(pct, 100, 56);
2150      }
2151      const ramSubSlider = document.getElementById('sys-ram-su
b-slider');
2152      if (ramSubSlider) {
2153        ramSubSlider.innerHTML = createPercentSliderHtml(pct,
false, `${usedGb.toFixed(1)} / ${totalGb.toFixed(1)} GB`);
2154      }
2155
2156      // 3. Блок 2: Файл подкачки (Swap)
2157      set('sys-swap-sub-badge', `${swapPct.toFixed(1)}%`);
```

```
      const swapSubGauge = document.getElementById('sys-swap-sub-gauge');
      if (swapSubGauge) {
        swapSubGauge.innerHTML = createGaugeSvg(swapPct, 100, 56);
      }
      const swapSubSlider = document.getElementById('sys-swap-sub-slider');
      if (swapSubSlider) {
        swapSubSlider.innerHTML = createPercentSliderHtml(swapPct, false, `Подкачка ${swapPct.toFixed(1)}%`);
      }

      // 4. Блок 3 & 4: Чтение и запись на диск (IO)
      const rBps = Number(io.read_bytes_sec || 0);
      const wBps = Number(io.write_bytes_sec || 0);
      const readFormatted = formatBytesPerSec(rBps);
      const writeFormatted = formatBytesPerSec(wBps);

      set('sys-memio-read', readFormatted);
      set('sys-read-sub-badge', readFormatted);
      set('sys-memio-write', writeFormatted);
      set('sys-write-sub-badge', writeFormatted);

      // Вычисление процента скорости относительно шкалы 50 MB/s
      const maxIoScale = 50 * 1024 * 1024; // 50 MB/s
      const readPct = Math.min(100, (rBps / maxIoScale) * 100);
      const writePct = Math.min(100, (wBps / maxIoScale) * 100);

      const readSubGauge = document.getElementById('sys-read-sub-gauge');
      if (readSubGauge) {
        readSubGauge.innerHTML = createGaugeSvg(readPct, 100, 56);
      }
```

```
2187      const readSubBar = document.getElementById('sys-read-sub
-bar');
2188      const readSubThumb = document.getElementById('sys-read-s
ub-thumb');
2189      if (readSubBar) readSubBar.style.width = `${readPct.toFi
xed(1)}%`;
2190      if (readSubThumb) readSubThumb.style.left = `${readPct.t
oFixed(1)}%`;
2191
2192      const writeSubGauge = document.getElementById('sys-write
-sub-gauge');
2193      if (writeSubGauge) {
2194        writeSubGauge.innerHTML = createGaugeSvg(writePct, 100
, 56);
2195      }
2196      const writeSubBar = document.getElementById('sys-write-s
ub-bar');
2197      const writeSubThumb = document.getElementById('sys-write
-sub-thumb');
2198      if (writeSubBar) writeSubBar.style.width = `${writePct.t
oFixed(1)}%`;
2199      if (writeSubThumb) writeSubThumb.style.left = `${writePc
t.toFixed(1)}%`;
2200
2201      // 5. Графики спарклайна RAM и Накопителей
2202      if (Array.isArray(data.history) && data.history.length >
 0) {
2203        renderMemIoSpark(data.history);
2204        renderStorageSpark(data.history);
2205      } else {
2206        _memIoSparkHistory.push({ memory_percent: pct, read_by
tes_sec: rBps, write_bytes_sec: wBps, time: new Date() });
2207        if (_memIoSparkHistory.length > 60) _memIoSparkHistory
.shift();
2208        renderMemIoSpark(_memIoSparkHistory);
2209      }
2210    } catch (e) {
2211      console.warn('[SystemInspectorTab] Ошибка получения памя
```

```
ти/IO из API:', e);
2212      }
2213    }
2214
2215    async function fetchNetworkLoadFromApi() {
2216      try {
2217        const res = await fetch('/api/v1/panel/network-load');
2218        if (!res.ok) return;
2219        const data = await res.json();
2220        const set = (id, text) => {
2221          const el = document.getElementById(id);
2222          if (el) el.innerText = text;
2223        };
2224
2225        const rxBps = Number(data.download_bytes_sec || 0);
2226        const txBps = Number(data.upload_bytes_sec || 0);
2227        const totalBps = rxBps + txBps;
2228        const utilPct = Number(data.utilization_percent || 0);
2229
2230        const rxFormatted = formatBytesPerSec(rxBps);
2231        const txFormatted = formatBytesPerSec(txBps);
2232        const totalFormatted = formatBytesPerSec(totalBps);
2233
2234        const rxTotalFormatted = formatBytes(data.download_total
_bytes || 0);
2235        const txTotalFormatted = formatBytes(data.upload_total_b
ytes || 0);
2236        const sumTotalFormatted = formatBytes((data.download_tot
al_bytes || 0) + (data.upload_total_bytes || 0));
2237
2238        // 1. Главный баннер Сети
2239        const netModel = document.getElementById('sys-metric-net
-model');
2240        if (netModel) {
2241          netModel.innerText = data.name || 'Сетевые адаптеры (E
thernet / Wi-Fi / BT)';
2242        }
2243        set('sys-net-ts', data.timestamp || 'Live');
```

```
2244        set('sys-net-speed-val', totalFormatted);
2245        set('sys-net-sub-details', `  ${rxFormatted} · 
    ${txFormatted} (Всего: ${sumTotalFormatted})`);
2246        set('sys-net-util-label', `Загрузка канала: ${utilPct.toFixed(1)}% (Линк: ${data.link_speed_mbps || 1000} Mbps)`);
2247        set('sys-net-speed-badge', `${data.link_speed_mbps || 1000} Mbps Full Duplex`);
2248
2249        // Экспресс-метрики KPI в центре баннера сети
2250        set('sys-net-kpi-rx', rxFormatted);
2251        set('sys-net-kpi-tx', txFormatted);
2252        set('sys-net-kpi-total', sumTotalFormatted);
2253        set('sys-net-kpi-util', `${utilPct.toFixed(1)}%`);
2254
2255        const netGauge = document.getElementById('sys-metric-net-gauge');
2256        if (netGauge) {
2257          netGauge.innerHTML = createGaugeSvg(utilPct, 130, 74);
2258        }
2259
2260        const netSlider = document.getElementById('sys-metric-net-slider');
2261        if (netSlider) {
2262          netSlider.innerHTML = createPercentSliderHtml(utilPct, true, `Канал ${utilPct.toFixed(1)}%`);
2263        }
2264
2265        // 2. Блок 1: Входящий (Rx)
2266        set('sys-rx-sub-badge', rxFormatted);
2267        set('sys-rx-total-text', `${rxTotalFormatted} всего`);
2268        const maxSpeedScale = 12.5 * 1024 * 1024; // 100 Mbps = 12.5 MB/s
2269        const rxPct = Math.min(100, (rxBps / maxSpeedScale) * 100);
2270        const rxSubGauge = document.getElementById('sys-rx-sub-gauge');
2271        if (rxSubGauge) rxSubGauge.innerHTML = createGaugeSvg(rxPct, 100, 56);
```

```
2272        const rxSubBar = document.getElementById('sys-rx-sub-bar
');
2273        const rxSubThumb = document.getElementById('sys-rx-sub-t
humb');
2274        if (rxSubBar) rxSubBar.style.width = `${rxPct.toFixed(1)
}%`;
2275        if (rxSubThumb) rxSubThumb.style.left = `${rxPct.toFixed
(1)}%`;
2276
2277        // 3. Блок 2: Исходящий (Tx)
2278        set('sys-tx-sub-badge', txFormatted);
2279        set('sys-tx-total-text', `${txTotalFormatted} всего`);
2280        const txPct = Math.min(100, (txBps / maxSpeedScale) * 10
0);
2281        const txSubGauge = document.getElementById('sys-tx-sub-g
auge');
2282        if (txSubGauge) txSubGauge.innerHTML = createGaugeSvg(tx
Pct, 100, 56);
2283        const txSubBar = document.getElementById('sys-tx-sub-bar
');
2284        const txSubThumb = document.getElementById('sys-tx-sub-t
humb');
2285        if (txSubBar) txSubBar.style.width = `${txPct.toFixed(1)
}%`;
2286        if (txSubThumb) txSubThumb.style.left = `${txPct.toFixed
(1)}%`;
2287
2288        // 4. Блок 3: Wi-Fi
2289        const wifi = data.wifi || {};
2290        const wifiState = wifi.state || 'Disconnected';
2291        const wifiSignal = wifi.signal_percent != null ? Number(
wifi.signal_percent) : (wifi.radio_status && wifi.radio_status
.includes('On') ? 100 : 0);
2292        set('sys-wifi-sub-badge', wifi.ssid ? `SSID: ${wifi.ssid
}` : (wifi.radio_status || 'On'));
2293        set('sys-wifi-status-text', wifi.ssid ? `${wifi.ssid} ($
{wifiSignal}%)` : wifiState);
2294        const wifiSubGauge = document.getElementById('sys-wifi-s
```

```
ub-gauge');
      if (wifiSubGauge) wifiSubGauge.innerHTML = createGaugeSv
g(wifiSignal, 100, 56);
      const wifiSubBar = document.getElementById('sys-wifi-sub
-bar');
      const wifiSubThumb = document.getElementById('sys-wifi-s
ub-thumb');
      if (wifiSubBar) wifiSubBar.style.width = `${wifiSignal}%
`;
      if (wifiSubThumb) wifiSubThumb.style.left = `${wifiSigna
l}%`;

      // 5. Блок 4: Bluetooth
      const bt = data.bluetooth || {};
      const btDevCount = bt.devices_count || 0;
      set('sys-bt-sub-badge', bt.status || 'Active');
      set('sys-bt-devices-text', btDevCount > 0 ? `${btDevCoun
t} устр. (${bt.devices ? bt.devices[0] : ''})` : 'Активен (0 у
стр.)');
      const btPct = Math.min(100, btDevCount * 25);
      const btSubGauge = document.getElementById('sys-bt-sub-g
auge');
      if (btSubGauge) btSubGauge.innerHTML = createGaugeSvg(bt
Pct > 0 ? btPct : 10, 100, 56);
      const btSubBar = document.getElementById('sys-bt-sub-bar
');
      const btSubThumb = document.getElementById('sys-bt-sub-t
humb');
      if (btSubBar) btSubBar.style.width = `${btPct > 0 ? btPc
t : 10}%`;
      if (btSubThumb) btSubThumb.style.left = `${btPct > 0 ? b
tPct : 10}%`;

      // 6. График истории
      if (Array.isArray(data.history) && data.history.length >
 0) {
        renderNetSpark(data.history);
      } else {
```

```
        _netSparkHistory.push({ download_bytes_sec: rxBps, upl
oad_bytes_sec: txBps, time: new Date() });
        if (_netSparkHistory.length > 60) _netSparkHistory.shi
ft();
        renderNetSpark(_netSparkHistory);
      }
    } catch (e) {
      console.warn('[SystemInspectorTab] Ошибка получения сете
вой нагрузки из API:', e);
    }
  }

  let currentNetSparkScale = 'log';
  let lastNetSparkData = [];
  const _netSparkHistory = [];

  async function setNetSparkInterval(interval) {
    currentNetSparkInterval = interval;
    const el = document.getElementById('sel-net-spark-interval
');
    if (el) el.value = interval;
    if (interval === 'seconds') {
      await fetchNetworkLoadFromApi();
    } else {
      await fetchNetHistoryByInterval(interval);
    }
  }
  window.setNetSparkInterval = setNetSparkInterval;

  async function fetchNetHistoryByInterval(interval) {
    try {
      const res = await fetch(`/api/v1/telemetry/history?inter
val=${encodeURIComponent(interval)}&metric=net&limit=120`);
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      if (data && Array.isArray(data.history) && data.history.
length > 0) {
        const rows = data.history.map(r => ({
```

```
          timestamp: r.timestamp,
          time_label: formatSparkTimeLabel(r.timestamp || r.cr
eated_at, interval),
          download_bytes_sec: Number(r.network_recv_bytes_sec
|| 0),
          upload_bytes_sec: Number(r.network_sent_bytes_sec ||
 0)
        }));
        lastNetSparkData = rows;
        renderNetSpark(rows);
      }
    } catch (e) {
      console.warn('[SystemInspectorTab] Ошибка получения исто
рии сети по интервалу:', e);
    }
  }

  function setNetSparkScale(scale) {
    currentNetSparkScale = scale;
    const btnLog = document.getElementById('btn-net-spark-log'
);
    const btnLin = document.getElementById('btn-net-spark-lin'
);
    if (btnLog && btnLin) {
      if (scale === 'log') {
        btnLog.className = 'btn btn-xs btn-outline-info active
 py-0 px-2 fw-bold';
        btnLin.className = 'btn btn-xs btn-outline-secondary p
y-0 px-2';
      } else {
        btnLin.className = 'btn btn-xs btn-outline-info active
 py-0 px-2 fw-bold';
        btnLog.className = 'btn btn-xs btn-outline-secondary p
y-0 px-2';
      }
    }
    if (lastNetSparkData && lastNetSparkData.length) {
      renderNetSpark(lastNetSparkData);
```

```
2378      }
2379    }
2380    window.setNetSparkScale = setNetSparkScale;
2381
2382    function renderNetSpark(history) {
2383      const box = document.getElementById('sys-net-spark');
2384      const container = document.getElementById('sys-net-spark-c
ontainer');
2385      const tooltip = document.getElementById('sys-net-spark-too
ltip');
2386      if (!box) return;
2387
2388      let ptsData = Array.isArray(history) && history.length > 0
 ? history : (lastNetSparkData && lastNetSparkData.length > 0
? lastNetSparkData : (_netSparkHistory && _netSparkHistory.len
gth > 0 ? _netSparkHistory : []));
2389      if (!ptsData || ptsData.length === 0) {
2390        ptsData = [{ rx_sec: 0, tx_sec: 0, time: new Date() }];
2391      }
2392      if (ptsData.length === 1) {
2393        ptsData = [ptsData[0], { ...ptsData[0], time: ptsData[0]
.time || new Date() }];
2394      }
2395      lastNetSparkData = ptsData;
2396      history = ptsData;
2397
2398      box.replaceChildren();
2399      const ns = 'http://www.w3.org/2000/svg';
2400      const W = 600, H = 84;
2401      const padTop = 6, padBottom = 6;
2402      const innerH = H - padTop - padBottom;
2403
2404      const svg = document.createElementNS(ns, 'svg');
2405      svg.setAttribute('viewBox', `0 0 ${W} ${H}`);
2406      svg.setAttribute('preserveAspectRatio', 'none');
2407      svg.style.cssText = 'width:100%;height:100%;display:block;
cursor:crosshair;';
2408
```

```
    const defs = document.createElementNS(ns, 'defs');
    defs.innerHTML = `
      <linearGradient id="netGradRx" x1="0%" y1="0%" x2="0%" y2="100%">
        <stop offset="0%" stop-color="#0dcaf0" stop-opacity="0.38"/>
        <stop offset="60%" stop-color="#0dcaf0" stop-opacity="0.12"/>
        <stop offset="100%" stop-color="#0dcaf0" stop-opacity="0.0"/>
      </linearGradient>
      <linearGradient id="netGradTx" x1="0%" y1="0%" x2="0%" y2="100%">
        <stop offset="0%" stop-color="#f59e0b" stop-opacity="0.30"/>
        <stop offset="60%" stop-color="#f59e0b" stop-opacity="0.08"/>
        <stop offset="100%" stop-color="#f59e0b" stop-opacity="0.0"/>
      </linearGradient>
    `;
    svg.appendChild(defs);

    const gridG = document.createElementNS(ns, 'g');
    gridG.setAttribute('opacity', '0.2');
    [0.0, 0.25, 0.5, 0.75, 1.0].forEach(ratio => {
      const y = H - padBottom - ratio * innerH;
      const gl = document.createElementNS(ns, 'line');
      gl.setAttribute('x1', '0');
      gl.setAttribute('y1', y.toFixed(1));
      gl.setAttribute('x2', String(W));
      gl.setAttribute('y2', y.toFixed(1));
      gl.setAttribute('stroke', '#ffffff');
      gl.setAttribute('stroke-dasharray', ratio === 0.0 || ratio === 1.0 ? 'none' : '3,3');
      gl.setAttribute('stroke-width', '1');
      gridG.appendChild(gl);
    });
```

```
    svg.appendChild(gridG);

    const count = history.length;
    const isLog = currentNetSparkScale === 'log';

    const rawDls = history.map(h => Number(h.download_bytes_se
c != null ? h.download_bytes_sec : (h.rx || 0)));
    const rawUls = history.map(h => Number(h.upload_bytes_sec
!= null ? h.upload_bytes_sec : (h.tx || 0)));
    const maxNetBytes = Math.max(1024, ...rawDls, ...rawUls);
    const netMaxScale = maxNetBytes * 1.15;

    const transformNet = (v) => {
      if (!v || v <= 0) return 0;
      const nv = Number(v);
      if (!isLog) return Math.min(1, nv / netMaxScale);
      return Math.log10(1 + 9 * (Math.min(netMaxScale, nv) / n
etMaxScale));
    };

    const rxPts = [];
    const txPts = [];

    history.forEach((h, i) => {
      const x = (i / (count - 1)) * W;
      const rxVal = Number(h.download_bytes_sec != null ? h.do
wnload_bytes_sec : (h.rx || 0));
      const txVal = Number(h.upload_bytes_sec != null ? h.uplo
ad_bytes_sec : (h.tx || 0));

      const yRx = H - padBottom - transformNet(rxVal) * innerH
;
      const yTx = H - padBottom - transformNet(txVal) * innerH
;

      rxPts.push({ x, y: yRx, raw: rxVal, time: h.timestamp ||
 h.time });
      txPts.push({ x, y: yTx, raw: txVal });
```

```
    });

    const drawSeries = (pts, strokeColor, fillColor, strokeWidth = 1.8) => {
      if (pts.length < 2) return;
      const linePath = buildSmoothSvgPath(pts);
      if (fillColor) {
        const areaPath = `${linePath} L ${pts[pts.length - 1].x.toFixed(1)} ${H - padBottom} L ${pts[0].x.toFixed(1)} ${H - padBottom} Z`;
        const aEl = document.createElementNS(ns, 'path');
        aEl.setAttribute('d', areaPath);
        aEl.setAttribute('fill', fillColor);
        svg.appendChild(aEl);
      }
      const pEl = document.createElementNS(ns, 'path');
      pEl.setAttribute('d', linePath);
      pEl.setAttribute('fill', 'none');
      pEl.setAttribute('stroke', strokeColor);
      pEl.setAttribute('stroke-width', String(strokeWidth));
      pEl.setAttribute('stroke-linejoin', 'round');
      pEl.setAttribute('stroke-linecap', 'round');
      svg.appendChild(pEl);
    };

    drawSeries(txPts, '#f59e0b', 'url(#netGradTx)', 1.6);
    drawSeries(rxPts, '#0dcaf0', 'url(#netGradRx)', 2.0);

    const crosshair = document.createElementNS(ns, 'line');
    crosshair.setAttribute('y1', '0');
    crosshair.setAttribute('y2', String(H));
    crosshair.setAttribute('stroke', 'var(--text-muted, rgba(128,128,128,0.5))');
    crosshair.setAttribute('stroke-dasharray', '2,2');
    crosshair.setAttribute('stroke-width', '1');
    crosshair.style.display = 'none';
    svg.appendChild(crosshair);

```

```
    svg.addEventListener('mousemove', (evt) => {
      const rect = svg.getBoundingClientRect();
      const mouseX = Math.max(0, Math.min(rect.width, evt.clie
ntX - rect.left));
      const ratio = mouseX / rect.width;
      const curIndex = Math.min(count - 1, Math.max(0, Math.ro
und(ratio * (count - 1))));
      const pt = rxPts[curIndex];
      const hItem = history[curIndex];

      if (pt && tooltip) {
        crosshair.setAttribute('x1', pt.x.toFixed(1));
        crosshair.setAttribute('x2', pt.x.toFixed(1));
        crosshair.style.display = 'block';

        const tStr = hItem.timestamp ? new Date(hItem.timestam
p).toLocaleTimeString() : (hItem.time ? new Date(hItem.time).t
oLocaleTimeString() : '--:--:--');
        const rxStr = formatBytesPerSec(pt.raw);
        const txStr = txPts[curIndex] ? formatBytesPerSec(txPt
s[curIndex].raw) : '0 B/s';

        tooltip.innerHTML = `
          <div class="d-flex align-items-center justify-conten
t-between gap-2 border-bottom pb-0.5 mb-1" style="border-color
: rgba(255,255,255,0.1) !important;">
            <span class="text-muted"><i class="bi bi-clock me-
1"></i>${escapeHtml(tStr)}</span>
            <span class="badge ${isLog ? 'bg-info-subtle text-
info' : 'bg-secondary text-light'}" style="font-size:0.62rem;"
>${isLog ? 'Log' : 'Lin (0–' + formatBytesPerSec(netMaxScale)
+ ')'}</span>
          </div>
          <div class="d-flex gap-2 font-monospace">
            <span class="text-info fw-bold"><i class="bi bi-ar
row-down me-0.5"></i>Rx: ${rxStr}</span>
            <span class="text-warning fw-bold"><i class="bi bi
-arrow-up me-0.5"></i>Tx: ${txStr}</span>
```

```
          </div>
        `;
        tooltip.style.display = 'block';
        const tipX = Math.min(rect.width - 160, Math.max(10, m
ouseX - 60));
        tooltip.style.left = `${tipX}px`;
      }
    });

    svg.addEventListener('mouseleave', () => {
      crosshair.style.display = 'none';
      if (tooltip) tooltip.style.display = 'none';
    });

    box.appendChild(svg);
  }

  let currentStorageSparkScale = 'log';
  let lastStorageSparkData = [];
  const _storageSparkHistory = [];

  async function setStorageSparkInterval(interval) {
    currentStorageSparkInterval = interval;
    const el = document.getElementById('sel-storage-spark-inte
rval');
    if (el) el.value = interval;
    if (interval === 'seconds') {
      await fetchStorageLoadFromApi();
    } else {
      await fetchStorageHistoryByInterval(interval);
    }
  }
  window.setStorageSparkInterval = setStorageSparkInterval;

  async function fetchStorageHistoryByInterval(interval) {
    try {
      const res = await fetch(`/api/v1/telemetry/history?inter
val=${encodeURIComponent(interval)}&metric=storage&limit=120`)
```

```
;
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      if (data && Array.isArray(data.history) && data.history.length > 0) {
        const rows = data.history.map(r => ({
          timestamp: r.timestamp,
          time_label: formatSparkTimeLabel(r.timestamp || r.created_at, interval),
          read_bytes_sec: Number(r.disk_read_bytes_sec || 0),
          write_bytes_sec: Number(r.disk_write_bytes_sec || 0)
        }));
        lastStorageSparkData = rows;
        renderStorageSpark(rows);
      }
    } catch (e) {
      console.warn('[SystemInspectorTab] Ошибка получения истории дисков по интервалу:', e);
    }
  }

  function setStorageSparkScale(scale) {
    currentStorageSparkScale = scale;
    const btnLog = document.getElementById('btn-storage-spark-log');
    const btnLin = document.getElementById('btn-storage-spark-lin');
    if (btnLog && btnLin) {
      if (scale === 'log') {
        btnLog.className = 'btn btn-xs btn-outline-info active py-0 px-2 fw-bold';
        btnLin.className = 'btn btn-xs btn-outline-secondary py-0 px-2';
      } else {
        btnLin.className = 'btn btn-xs btn-outline-info active py-0 px-2 fw-bold';
        btnLog.className = 'btn btn-xs btn-outline-secondary py-0 px-2';
```

```
      }
    }
    if (lastStorageSparkData && lastStorageSparkData.length) {
      renderStorageSpark(lastStorageSparkData);
    }
  }
  window.setStorageSparkScale = setStorageSparkScale;

  function renderStorageSpark(history) {
    const box = document.getElementById('sys-storage-spark');
    const container = document.getElementById('sys-storage-spa
rk-container');
    const tooltip = document.getElementById('sys-storage-spark
-tooltip');
    if (!box) return;

    let ptsData = Array.isArray(history) && history.length > 0
 ? history : (lastStorageSparkData && lastStorageSparkData.len
gth > 0 ? lastStorageSparkData : (_storageSparkHistory && _sto
rageSparkHistory.length > 0 ? _storageSparkHistory : []));
    if (!ptsData || ptsData.length === 0) {
      ptsData = [{ read_bytes_sec: 0, write_bytes_sec: 0, time
: new Date() }];
    }
    if (ptsData.length === 1) {
      ptsData = [ptsData[0], { ...ptsData[0], time: ptsData[0]
.time || new Date() }];
    }
    lastStorageSparkData = ptsData;
    history = ptsData;

    box.replaceChildren();
    const ns = 'http://www.w3.org/2000/svg';
    const W = 600, H = 84;
    const padTop = 6, padBottom = 6;
    const innerH = H - padTop - padBottom;

    const svg = document.createElementNS(ns, 'svg');
```

```
    svg.setAttribute('viewBox', `0 0 ${W} ${H}`);
    svg.setAttribute('preserveAspectRatio', 'none');
    svg.style.cssText = 'width:100%;height:100%;display:block;cursor:crosshair;';

    const defs = document.createElementNS(ns, 'defs');
    defs.innerHTML = `
      <linearGradient id="storageGradRead" x1="0%" y1="0%" x2="0%" y2="100%">
        <stop offset="0%" stop-color="#0dcaf0" stop-opacity="0.38"/>
        <stop offset="60%" stop-color="#0dcaf0" stop-opacity="0.12"/>
        <stop offset="100%" stop-color="#0dcaf0" stop-opacity="0.0"/>
      </linearGradient>
      <linearGradient id="storageGradWrite" x1="0%" y1="0%" x2="0%" y2="100%">
        <stop offset="0%" stop-color="#f59e0b" stop-opacity="0.30"/>
        <stop offset="60%" stop-color="#f59e0b" stop-opacity="0.08"/>
        <stop offset="100%" stop-color="#f59e0b" stop-opacity="0.0"/>
      </linearGradient>
    `;
    svg.appendChild(defs);

    const gridG = document.createElementNS(ns, 'g');
    gridG.setAttribute('opacity', '0.2');
    [0.0, 0.25, 0.5, 0.75, 1.0].forEach(ratio => {
      const y = H - padBottom - ratio * innerH;
      const gl = document.createElementNS(ns, 'line');
      gl.setAttribute('x1', '0');
      gl.setAttribute('y1', y.toFixed(1));
      gl.setAttribute('x2', String(W));
      gl.setAttribute('y2', y.toFixed(1));
      gl.setAttribute('stroke', '#ffffff');
```

```
      gl.setAttribute('stroke-dasharray', ratio === 0.0 || rat
io === 1.0 ? 'none' : '3,3');
      gl.setAttribute('stroke-width', '1');
      gridG.appendChild(gl);
    });
    svg.appendChild(gridG);

    const count = history.length;
    const isLog = currentStorageSparkScale === 'log';

    const rawReads = history.map(h => Number(h.read_bytes_sec
!= null ? h.read_bytes_sec : (h.disk_read || 0)));
    const rawWrites = history.map(h => Number(h.write_bytes_se
c != null ? h.write_bytes_sec : (h.disk_write || 0)));
    const maxIoBytes = Math.max(1024 * 1024, ...rawReads, ...r
awWrites);
    const storageMaxScale = maxIoBytes * 1.15;

    const transformStorage = (v) => {
      if (!v || v <= 0) return 0;
      const nv = Number(v);
      if (!isLog) return Math.min(1, nv / storageMaxScale);
      return Math.log10(1 + 9 * (Math.min(storageMaxScale, nv)
 / storageMaxScale));
    };

    const readPts = [];
    const writePts = [];

    history.forEach((h, i) => {
      const x = (i / (count - 1)) * W;
      const rdVal = Number(h.read_bytes_sec != null ? h.read_b
ytes_sec : (h.disk_read || 0));
      const wrVal = Number(h.write_bytes_sec != null ? h.write
_bytes_sec : (h.disk_write || 0));

      const yRead = H - padBottom - transformStorage(rdVal) *
innerH;
```

```
      const yWrite = H - padBottom - transformStorage(wrVal) * innerH;

      readPts.push({ x, y: yRead, raw: rdVal, time: h.timestamp || h.time });
      writePts.push({ x, y: yWrite, raw: wrVal });
    });

    const drawSeries = (pts, strokeColor, fillColor, strokeWidth = 1.8) => {
      if (pts.length < 2) return;
      const linePath = buildSmoothSvgPath(pts);
      if (fillColor) {
        const areaPath = `${linePath} L ${pts[pts.length - 1].x.toFixed(1)} ${H - padBottom} L ${pts[0].x.toFixed(1)} ${H - padBottom} Z`;
        const aEl = document.createElementNS(ns, 'path');
        aEl.setAttribute('d', areaPath);
        aEl.setAttribute('fill', fillColor);
        svg.appendChild(aEl);
      }
      const pEl = document.createElementNS(ns, 'path');
      pEl.setAttribute('d', linePath);
      pEl.setAttribute('fill', 'none');
      pEl.setAttribute('stroke', strokeColor);
      pEl.setAttribute('stroke-width', String(strokeWidth));
      pEl.setAttribute('stroke-linejoin', 'round');
      pEl.setAttribute('stroke-linecap', 'round');
      svg.appendChild(pEl);
    };

    drawSeries(writePts, '#f59e0b', 'url(#storageGradWrite)', 1.6);
    drawSeries(readPts, '#0dcaf0', 'url(#storageGradRead)', 2.0);

    const crosshair = document.createElementNS(ns, 'line');
    crosshair.setAttribute('y1', '0');
```

```
    crosshair.setAttribute('y2', String(H));
    crosshair.setAttribute('stroke', 'var(--text-muted, rgba(128,128,128,0.5))');
    crosshair.setAttribute('stroke-dasharray', '2,2');
    crosshair.setAttribute('stroke-width', '1');
    crosshair.style.display = 'none';
    svg.appendChild(crosshair);

    svg.addEventListener('mousemove', (evt) => {
      const rect = svg.getBoundingClientRect();
      const mouseX = Math.max(0, Math.min(rect.width, evt.clientX - rect.left));
      const ratio = mouseX / rect.width;
      const curIndex = Math.min(count - 1, Math.max(0, Math.round(ratio * (count - 1))));
      const pt = readPts[curIndex];
      const hItem = history[curIndex];

      if (pt && tooltip) {
        crosshair.setAttribute('x1', pt.x.toFixed(1));
        crosshair.setAttribute('x2', pt.x.toFixed(1));
        crosshair.style.display = 'block';

        const tStr = hItem.timestamp ? new Date(hItem.timestamp).toLocaleTimeString() : (hItem.time ? new Date(hItem.time).toLocaleTimeString() : '--:--:--');
        const rdStr = formatBytesPerSec(pt.raw);
        const wrStr = writePts[curIndex] ? formatBytesPerSec(writePts[curIndex].raw) : '0 B/s';

        tooltip.innerHTML = `
          <div class="d-flex align-items-center justify-content-between gap-2 border-bottom pb-0.5 mb-1" style="border-color: rgba(255,255,255,0.1) !important;">
            <span class="text-muted"><i class="bi bi-clock me-1"></i>${escapeHtml(tStr)}</span>
            <span class="badge ${isLog ? 'bg-info-subtle text-info' : 'bg-secondary text-light'}" style="font-size:0.62rem;"
```

```
>${isLog ? 'Log' : 'Lin (0–' + formatBytesPerSec(storageMaxSca
le) + ')'}</span>
          </div>
          <div class="d-flex gap-2 font-monospace">
            <span class="text-info fw-bold"><i class="bi bi-ar
row-down me-0.5"></i>Чтение: ${rdStr}</span>
            <span class="text-warning fw-bold"><i class="bi bi
-arrow-up me-0.5"></i>Запись: ${wrStr}</span>
          </div>
        `;
        tooltip.style.display = 'block';
        const tipX = Math.min(rect.width - 160, Math.max(10, m
ouseX - 60));
        tooltip.style.left = `${tipX}px`;
      }
    });

    svg.addEventListener('mouseleave', () => {
      crosshair.style.display = 'none';
      if (tooltip) tooltip.style.display = 'none';
    });

    box.appendChild(svg);
  }

  function formatDriveOps(cnt) {
    if (cnt == null) return '--';
    const n = Number(cnt);
    if (n >= 1000000) return (n / 1000000).toFixed(2) + 'M';
    if (n >= 1000) return (n / 1000).toFixed(1) + 'K';
    return String(n);
  }

  function formatDriveHours(hours) {
    if (hours == null) return '--';
    const h = Number(hours);
    if (h >= 8760) {
      const years = (h / 8760).toFixed(1);
```

```
      return `${h.toLocaleString()} ч (~${years} г.)`;
    }
    return `${h.toLocaleString()} ч`;
  }

  function openDriveAiModal(d) {
    const modal = window.AIModalDialog || window.AITableModal;
    if (!modal) {
      console.warn('[SystemInspectorTab] AIModalDialog модуль не найден');
      return;
    }

    const pct = d.used_percent != null ? Number(d.used_percent).toFixed(1) : '--';
    const totGb = d.total_gb != null ? Number(d.total_gb).toFixed(1) : '--';
    const usedGb = d.used_gb != null ? Number(d.used_gb).toFixed(1) : '--';
    const freeGb = d.free_gb != null ? Number(d.free_gb).toFixed(1) : '--';
    const temp = d.temperature_c != null ? `${Number(d.temperature_c).toFixed(0)} °C` : 'N/A';

    modal.show({
      title: d.name,
      subtitle: `${d.id} | ${d.media_type} | ${totGb} GB`,
      icon: '□ ',
      panelId: 'panel-system-inspector-storage',
      tableType: 'disk',
      badges: [
        { text: d.media_type, class: d.media_type === 'NVMe' ? 'badge bg-primary' : d.media_type === 'SSD' ? 'badge bg-info text-dark' : 'badge bg-warning text-dark' },
        { text: d.health_status || 'SMART OK', class: 'badge bg-success' },
        { text: temp, class: 'badge bg-dark border text-light' }
```

```
      ],
      metadata: [
        { label: 'Модель накопителя', value: d.name },
        { label: 'Идентификатор устройства', value: d.id, isCo
de: true },
        { label: 'Тип носителя / Интерфейс', value: `${d.media
_type} (${d.interface_type || 'SATA/NVMe'})` },
        { label: 'Общий объем', value: `${totGb} GB` },
        { label: 'Занятый объем', value: `${usedGb} GB (${pct}
%)` },
        { label: 'Свободное пространство', value: `${freeGb} G
B` },
        { label: 'Температура', value: temp },
        { label: 'Интенсивность обращений', value: d.activity_
percent != null ? `${d.activity_percent}%` : 'N/A' },
        { label: 'Скорость чтения / записи', value: `Чтение: $
{d.read_rate_raw || '0 B/s'}, Запись: ${d.write_rate_raw || '0
 B/s'}` },
        { label: 'Всего записано (TBW)', value: d.data_written
_raw || 'N/A' },
        { label: 'Всего прочитано', value: d.data_read_raw ||
'N/A' },
        { label: 'Время наработки (Power-On)', value: formatDr
iveHours(d.power_on_hours) },
        { label: 'Серийный номер', value: d.serial_number || '
N/A', isCode: true },
        { label: 'Связанные разделы', value: (d.partitions &&
d.partitions.length) ? d.partitions.join(', ') : 'N/A' }
      ],
      rawTitle: 'Дамп телеметрии физического накопителя',
      rawContent: JSON.stringify(d, null, 2),
      autoRun: false
    });
  }

  function renderStorageCards(data) {
    const box = document.getElementById('sys-metric-storage-ca
rds');
```

```
2825    const badgeDrives = document.getElementById('sys-storage-d
rives-badge');
2826    const badgeParts = document.getElementById('sys-storage-pa
rts-badge');
2827    const badgeSummary = document.getElementById('sys-storage-
summary-badge');
2828    if (!box) return;
2829
2830    const drives = Array.isArray(data.drives) ? data.drives :
[];
2831    const partitions = Array.isArray(data.partitions) ? data.p
artitions : [];
2832
2833    if (badgeDrives) badgeDrives.textContent = `${drives.lengt
h} дисков`;
2834    if (badgeParts) badgeParts.textContent = `${partitions.len
gth} разделов`;
2835    if (badgeSummary) {
2836      const summary = data.summary || {};
2837      badgeSummary.textContent = `${summary.used_gb || 0} / ${
summary.total_gb || 0} GB (${summary.used_percent || 0}%)`;
2838    }
2839
2840    if (drives.length === 0 && partitions.length === 0) {
2841      box.innerHTML = `<div class="text-muted small text-cente
r py-3 col-12">Опрос накопителей выполняется...</div>`;
2842      return;
2843    }
2844
2845    if (drives.length > 0) {
2846      box.innerHTML = drives.map((d, idx) => {
2847        const pct = d.used_percent != null ? Number(d.used_per
cent) : 0;
2848        const temp = d.temperature_c != null ? Number(d.temper
ature_c) : null;
2849        const totGb = d.total_gb != null ? Number(d.total_gb)
: 0;
2850        const usedGb = d.used_gb != null ? Number(d.used_gb) :
```

```
 0;
2851        const freeGb = Math.max(0, totGb - usedGb);
2852        const percentSlider = createPercentSliderHtml(pct, fal
se, `${usedGb.toFixed(0)} / ${totGb.toFixed(0)} GB`);
2853
2854        let mediaBadgeClass = 'bg-warning-subtle text-warning
border-warning';
2855        let iconHtml = '<i class="bi bi-hdd-fill text-warning
fs-5"></i>';
2856        if (d.media_type === 'NVMe') {
2857          mediaBadgeClass = 'bg-primary-subtle text-primary bo
rder-primary';
2858          iconHtml = '<i class="bi bi-lightning-charge-fill te
xt-info fs-5"></i>';
2859        } else if (d.media_type === 'SSD') {
2860          mediaBadgeClass = 'bg-info-subtle text-info border-i
nfo';
2861          iconHtml = '<i class="bi bi-device-ssd-fill text-pri
mary fs-5"></i>';
2862        } else if (d.media_type === 'External' || d.interface_
type === 'USB') {
2863          mediaBadgeClass = 'bg-purple-subtle text-info border
-info';
2864          iconHtml = '<i class="bi bi-usb-drive-fill text-info
 fs-5"></i>';
2865        }
2866
2867        let tempColorClass = 'text-success';
2868        let tempBadgeClass = 'bg-success-subtle text-success b
order border-success';
2869        let tempStatusText = 'Оптимально';
2870        if (temp != null) {
2871          if (temp >= 65) {
2872            tempColorClass = 'text-danger fw-bold';
2873            tempBadgeClass = 'bg-danger-subtle text-danger bor
der border-danger';
2874            tempStatusText = 'Горячий';
2875          } else if (temp >= 50) {
```

```
            tempColorClass = 'text-warning fw-bold';
            tempBadgeClass = 'bg-warning-subtle text-warning border border-warning';
            tempStatusText = 'Нагрев';
          } else {
            tempColorClass = 'text-success';
            tempBadgeClass = 'bg-success-subtle text-success border border-success';
            tempStatusText = 'Норма';
          }
        } else {
          tempColorClass = 'text-muted';
          tempBadgeClass = 'bg-secondary text-light';
          tempStatusText = 'N/A';
        }

        let partitionsBadgesHtml = '';
        if (Array.isArray(d.partitions) && d.partitions.length > 0) {
          partitionsBadgesHtml = d.partitions.map(p => `<span class="badge bg-primary-subtle text-primary border border-primary font-monospace" style="font-size: 0.70rem;">${escapeHtml(p)}</span>`).join(' ');
        }

        return `
          <div class="sys-panel p-3 mb-2 rounded-3 w-100 border sys-drive-full-card interactive-table-row" data-drive-idx="${idx}" style="background: var(--surface-1); border-color: var(--border-color) !important; cursor: pointer; transition: transform 0.15s ease, box-shadow 0.15s ease;" title="Кликните для детального AI-анализа накопителя ${escapeHtml(d.name)}">
            <!-- Заголовок накопителя -->
            <div class="d-flex justify-content-between align-items-center flex-wrap gap-2 pb-2 mb-2 border-bottom" style="border-color: var(--border-color) !important;">
              <div class="d-flex align-items-center gap-2 flex-wrap">
```

```
2900                ${iconHtml}
2901                <span class="fw-bold text-body" style="font-si
    ze: 1.02rem;">${escapeHtml(d.name)}</span>
2902                <span class="badge ${mediaBadgeClass} font-mon
    ospace" style="font-size: 0.72rem;">${escapeHtml(d.media_type)
    }</span>
2903                ${d.interface_type ? `<span class="badge bg-da
    rk border text-light font-monospace" style="font-size: 0.70rem
    ;">${escapeHtml(d.interface_type)}</span>` : ''}
2904                <span class="badge bg-secondary-subtle text-bo
    dy border font-monospace" style="font-size: 0.70rem;">${escape
    Html(d.id)}</span>
2905                ${partitionsBadgesHtml}
2906                <span class="badge bg-success-subtle text-succ
    ess border border-success font-monospace" style="font-size: 0.
    70rem;">● ${escapeHtml(d.health_status || 'SMART OK')}</span>
2907                ${d.life_percent != null ? `<span class="badge
     bg-info-subtle text-info border border-info font-monospace" s
    tyle="font-size: 0.70rem;" title="Оставшийся ресурс"><i class=
    "bi bi-heart-pulse me-1"></i>Ресурс: ${d.life_percent.toFixed(
    0)}%</span>` : ''}
2908                ${d.serial_number ? `<span class="small text-m
    uted font-monospace d-none d-md-inline" style="font-size: 0.68
    rem;" title="Серийный номер">S/N: ${escapeHtml(d.serial_number
    )}</span>` : ''}
2909              </div>
2910              <div class="d-flex align-items-center gap-2">
2911                <span class="badge bg-dark border text-light f
    ont-monospace px-2.5 py-1" style="font-size: 0.80rem;">${usedG
    b.toFixed(1)} / ${totGb.toFixed(1)} GB (${pct.toFixed(1)}%)</s
    pan>
2912                <button type="button" class="btn btn-xs btn-ou
    tline-info py-0.5 px-2 fw-semibold no-modal-trigger sys-drive-
    ai-btn" data-drive-idx="${idx}" title="Запустить AI-анализ нак
    опителя">
2913                  <i class="bi bi-robot me-1"></i>AI Анализ
2914                </button>
2915              </div>
```

```
2916            </div>
2917
2918            <!-- 4 Информационные колонки на всю ширину -->
2919            <div class="row g-3 align-items-center">
2920              <!-- 1. Емкость и Занятость -->
2921              <div class="col-12 col-md-6 col-lg-3">
2922                <div class="p-2.5 rounded-2 h-100" style="back
    ground: var(--surface-2); border: 1px solid var(--border-color
    );">
2923                  <div class="d-flex justify-content-between a
    lign-items-center mb-1">
2924                    <span class="small text-muted fw-bold"><i
    class="bi bi-pie-chart text-info me-1"></i>Емкость и Занято</s
    pan>
2925                    <span class="fw-bold font-monospace ${pct
    >= 85 ? 'text-danger' : pct >= 70 ? 'text-warning' : 'text-inf
    o'}">${pct.toFixed(1)}%</span>
2926                  </div>
2927                  <div class="d-flex align-items-center gap-2
    my-1">
2928                    <div style="width: 80px; height: 46px; fle
    x-shrink: 0;">
2929                      ${createGaugeSvg(pct, 80, 46)}
2930                    </div>
2931                    <div class="flex-grow-1">
2932                      <div class="font-monospace fw-bold" styl
    e="font-size: 0.85rem;">${usedGb.toFixed(1)} / ${totGb.toFixed
    (1)} GB</div>
2933                      <div class="small text-muted font-monosp
    ace">Свободно: <span class="text-success fw-semibold">${freeGb
    .toFixed(1)} GB</span></div>
2934                    </div>
2935                  </div>
2936                  <div>
2937                    ${percentSlider}
2938                  </div>
2939                </div>
2940              </div>
```

```
2941
2942              <!-- 2. Интенсивность обращений (I/O) -->
2943              <div class="col-12 col-md-6 col-lg-3">
2944                <div class="p-2.5 rounded-2 h-100" style="back
ground: var(--surface-2); border: 1px solid var(--border-color
);">
2945                  <div class="d-flex justify-content-between a
lign-items-center mb-1.5">
2946                    <span class="small text-muted fw-bold"><i
class="bi bi-speedometer2 text-warning me-1"></i>Интенсивность
 I/O</span>
2947                    <span class="badge ${d.activity_percent !=
 null && d.activity_percent > 30 ? 'bg-danger-subtle text-dang
er border border-danger' : d.activity_percent != null && d.act
ivity_percent > 5 ? 'bg-warning-subtle text-warning border bor
der-warning' : 'bg-info-subtle text-info border border-info'}
font-monospace" style="font-size: 0.70rem;">
2948                      ${d.activity_percent != null ? 'Активнос
ть ' + d.activity_percent.toFixed(1) + '%' : (d.read_rate_raw
|| d.write_rate_raw ? 'Активен' : 'Штатно')}
2949                    </span>
2950                  </div>
2951                  <div class="d-flex justify-content-between a
lign-items-center font-monospace mb-1" style="font-size: 0.76r
em;">
2952                    <span class="text-info fw-semibold"><i cla
ss="bi bi-arrow-down-short"></i>Чтение:</span>
2953                    <span class="fw-bold text-info">${escapeHt
ml(d.read_rate_raw || '0 B/s')}</span>
2954                  </div>
2955                  <div class="d-flex justify-content-between a
lign-items-center font-monospace mb-1.5" style="font-size: 0.7
6rem;">
2956                    <span class="text-warning fw-semibold"><i
class="bi bi-arrow-up-short"></i>Запись:</span>
2957                    <span class="fw-bold text-warning">${escap
eHtml(d.write_rate_raw || '0 B/s')}</span>
2958                  </div>
```

```
                <div class="d-flex justify-content-between a
lign-items-center small text-muted font-monospace pt-1 border-
top" style="border-color: var(--border-color) !important; font
-size: 0.70rem;">
                  <span>Операции (IOPS):</span>
                  <span>${formatDriveOps(d.read_count)} R ·
${formatDriveOps(d.write_count)} W</span>
                </div>
              </div>
            </div>

            <!-- 3. Температура -->
            <div class="col-12 col-md-6 col-lg-3">
              <div class="p-2.5 rounded-2 h-100" style="back
ground: var(--surface-2); border: 1px solid var(--border-color
);">
                <div class="d-flex justify-content-between a
lign-items-center mb-1.5">
                  <span class="small text-muted fw-bold"><i
class="bi bi-thermometer-half text-danger me-1"></i>Температур
а</span>
                  <span class="badge ${tempBadgeClass}" styl
e="font-size: 0.70rem;">${tempStatusText}</span>
                </div>
                <div class="d-flex align-items-baseline gap-
2 mb-1">
                  <span class="fw-bold font-monospace ${temp
ColorClass}" style="font-size: 1.35rem; line-height: 1;">${tem
p != null ? temp.toFixed(0) + ' °C' : '-- °C'}</span>
                  <span class="small text-muted font-monospa
ce">${temp != null ? (temp < 45 ? 'Оптимально' : temp < 55 ? '
Норма' : 'Нагрев') : 'Датчик недоступен'}</span>
                </div>
                <div class="my-1">
                  ${createTempSliderHtml(temp, 25, 75, false
)}
                </div>
                <div class="small text-muted font-monospace
```

```
pt-1 border-top d-flex justify-content-between" style="border-color: var(--border-color) !important; font-size: 0.70rem;">
                    <span>Безопасный порог:</span>
                    <span>до 65 °C</span>
                  </div>
                </div>
              </div>

              <!-- 4. Записано и прочитано / Наработка -->
              <div class="col-12 col-md-6 col-lg-3">
                <div class="p-2.5 rounded-2 h-100" style="background: var(--surface-2); border: 1px solid var(--border-color);">
                  <div class="d-flex justify-content-between align-items-center mb-1.5">
                    <span class="small text-muted fw-bold"><i class="bi bi-database-check text-primary me-1"></i>Записано / Ресурс</span>
                    <span class="badge bg-dark border text-light font-monospace" style="font-size: 0.70rem;">TBW / I/O</span>
                  </div>
                  <div class="d-flex justify-content-between align-items-center font-monospace mb-1" style="font-size: 0.76rem;">
                    <span class="text-muted"><i class="bi bi-pencil-square text-warning me-1"></i>Всего записано:</span>
                    <span class="fw-bold text-warning">${escapeHtml(d.data_written_raw || '--')}</span>
                  </div>
                  <div class="d-flex justify-content-between align-items-center font-monospace mb-1.5" style="font-size: 0.76rem;">
                    <span class="text-muted"><i class="bi bi-book text-info me-1"></i>Всего прочитано:</span>
                    <span class="fw-bold text-info">${escapeHtml(d.data_read_raw || '--')}</span>
                  </div>
```

```
                  <div class="d-flex justify-content-between a
lign-items-center small text-muted font-monospace pt-1 border-
top" style="border-color: var(--border-color) !important; font
-size: 0.70rem;">
                    <span>Наработка:</span>
                    <span class="text-body fw-semibold">${form
atDriveHours(d.power_on_hours)}</span>
                  </div>
                </div>
              </div>
            </div>
          </div>
        `;
      }).join('');

      // Привязка кликов по карточкам и кнопке AI Анализа
      box.querySelectorAll('.sys-drive-full-card').forEach(car
d => {
        card.addEventListener('click', (ev) => {
          if (ev.target.closest('button, a, input, select, .no
-modal-trigger') && !ev.target.closest('.sys-drive-ai-btn')) {
            return;
          }
          const idx = Number(card.getAttribute('data-drive-idx
'));
          const driveObj = drives[idx];
          if (!driveObj) return;
          openDriveAiModal(driveObj);
        });
      });
    } else {
      box.innerHTML = partitions.map(p => {
        const pct = Number(p.used_percent || 0);
        const gaugeSvg = createGaugeSvg(pct, 90, 50);
        const percentSlider = createPercentSliderHtml(pct, fal
se, `${p.used_gb.toFixed(0)} / ${p.total_gb.toFixed(0)} GB`);

        return `
```

```
          <div class="sys-panel p-3 mb-2 rounded-3 w-100 borde
r" style="background: var(--surface-1); border-color: var(--bo
rder-color) !important;">
            <div class="d-flex justify-content-between align-i
tems-center flex-wrap gap-2 pb-1.5 mb-1.5 border-bottom" style
="border-color: var(--border-color) !important;">
              <div class="d-flex align-items-center gap-2">
                <i class="bi bi-hdd-fill text-info fs-5"></i>
                <span class="fw-bold text-body" style="font-si
ze: 1.02rem;">Раздел ${escapeHtml(p.device)} (${escapeHtml(p.f
stype || 'NTFS')})</span>
                <span class="badge bg-secondary font-monospace
">${p.free_gb.toFixed(1)} GB свободно</span>
              </div>
              <span class="badge bg-dark border text-light fon
t-monospace">${p.used_gb.toFixed(1)} / ${p.total_gb.toFixed(1)
} GB (${pct.toFixed(1)}%)</span>
            </div>
            <div class="row g-3 align-items-center">
              <div class="col-12 col-md-6 col-lg-4">
                <div class="d-flex align-items-center gap-2">
                  <div style="width: 80px; height: 46px;">${ga
ugeSvg}</div>
                  <div class="font-monospace fw-bold">${pct.to
Fixed(1)}% (${p.used_gb.toFixed(1)} / ${p.total_gb.toFixed(1)}
 GB)</div>
                </div>
              </div>
              <div class="col-12 col-md-6 col-lg-8">
                ${percentSlider}
              </div>
            </div>
          </div>
        `;
      }).join('');
    }
  }

```

```
  async function fetchStorageLoadFromApi() {
    try {
      const res = await fetch('/api/v1/panel/storage-load');
      if (!res.ok) return;
      const data = await res.json();
      const set = (id, text) => { const el = document.getEleme
ntById(id); if (el) el.innerText = text; };
      const s = data.summary || {};
      const pct = Number(s.used_percent || 0);
      const usedGb = Number(s.used_gb || 0);
      const totalGb = Number(s.total_gb || 0);
      const freeGb = Number(s.free_gb || 0);
      const maxTemp = s.max_temperature_c != null ? Number(s.m
ax_temperature_c) : null;

      // 1. Главный баннер хранилища
      set('sys-storage-pct', `${pct.toFixed(1)}%`);
      set('sys-storage-used', `${usedGb.toFixed(1)} / ${totalG
b.toFixed(1)} GB (свободно ${freeGb.toFixed(1)} GB)`);
      set('sys-storage-temp-label', maxTemp != null ? `Макс. т
емпература: ${maxTemp.toFixed(0)} °C` : 'Макс. температура: --
 °C');
      set('sys-storage-ts', new Date().toLocaleTimeString());

      const stModel = document.getElementById('sys-metric-stor
age-model');
      if (stModel && totalGb > 0) {
        const totalTb = totalGb >= 1000 ? `${(totalGb / 1024).
toFixed(2)} TB` : `${totalGb.toFixed(0)} GB`;
        stModel.innerText = `Дисковая подсистема (${totalTb})`
;
      }

      // Экспресс-метрики KPI в центре баннера хранилища
      set('sys-storage-kpi-used', `${usedGb.toFixed(1)} GB`);
      set('sys-storage-kpi-free', `${freeGb.toFixed(1)} GB`);
      const dCount = s.drives_count || (Array.isArray(data.dri
ves) ? data.drives.length : '--');
```

```
3087      const pCount = s.partitions_count || (Array.isArray(data
.partitions) ? data.partitions.length : '--');
3088      set('sys-storage-kpi-drives', `${dCount} диск. / ${pCoun
t} разд.`);
3089      const stKpiTemp = document.getElementById('sys-storage-k
pi-temp');
3090      if (stKpiTemp) {
3091        stKpiTemp.textContent = maxTemp != null ? `${maxTemp.t
oFixed(0)} °C` : '-- °C';
3092        if (maxTemp != null) {
3093          stKpiTemp.className = maxTemp >= 60 ? 'sys-kpi-value
 text-danger' : maxTemp >= 48 ? 'sys-kpi-value text-warning' :
 'sys-kpi-value text-info';
3094        }
3095      }
3096
3097      const stGauge = document.getElementById('sys-metric-stor
age-gauge');
3098      if (stGauge) {
3099        stGauge.innerHTML = createGaugeSvg(pct, 130, 74);
3100      }
3101
3102      const stSlider = document.getElementById('sys-metric-sto
rage-temp-slider');
3103      if (stSlider) {
3104        stSlider.innerHTML = createTempSliderHtml(maxTemp, 25,
 75, true);
3105      }
3106
3107      // 2. Карточки дисков
3108      renderStorageCards(data);
3109
3110      // 3. График истории накопителей
3111      if (Array.isArray(data.history) && data.history.length >
 0) {
3112        renderStorageSpark(data.history);
3113      } else if (lastStorageSparkData && lastStorageSparkData.
length > 0) {
```

```
        renderStorageSpark(lastStorageSparkData);
      } else {
        _storageSparkHistory.push({ read_bytes_sec: s.read_bytes_sec || 0, write_bytes_sec: s.write_bytes_sec || 0, time: new Date() });
        if (_storageSparkHistory.length > 60) _storageSparkHistory.shift();
        renderStorageSpark(_storageSparkHistory);
      }
    } catch (e) {
      console.warn('[SystemInspectorTab] Ошибка получения параметров дисков из API:', e);
    }
  }

  let _lastDisksWearData = null;

  async function fetchStorageHealthFromApi() {
    try {
      const res = await fetch('/api/v1/system/diagnostics/storage-battery?t=' + Date.now());
      if (!res.ok) return;
      const data = await res.json();
      if (data && Array.isArray(data.disks_wear)) {
        _lastDisksWearData = data.disks_wear;
        renderPhysicalDisksStorage(data.disks_wear);
      }
    } catch (e) {
      console.warn('[SystemInspectorTab] Ошибка получения здоровья накопителей:', e);
    }
  }

  function renderPhysicalDisksStorage(disks) {
    const diskCont = document.getElementById('sys-physical-disks-container');
    const diskBadge = document.getElementById('sys-disk-health-badge');
```

```
    if (!diskCont || !Array.isArray(disks) || disks.length === 0) return;

    const allHealthy = disks.every(d => {
      const st = String(d.health_status || d.status || '').toLowerCase();
      return st.includes('healthy') || st.includes('ok') || st.includes('passed');
    });

    if (diskBadge) {
      diskBadge.textContent = allHealthy ? 'SMART OK' : 'Внимание';
      diskBadge.className = allHealthy
        ? 'badge bg-success-subtle text-success border border-success'
        : 'badge bg-warning-subtle text-warning border border-warning';
    }

    diskCont.innerHTML = disks.map((d, idx) => {
      const model = d.model || d.name || `Диск ${idx}`;
      const devId = d.device_id || `Disk ${idx}`;
      const sizeStr = d.size_gb != null ? `${d.size_gb} GB` : (d.total_gb != null ? `${d.total_gb} GB` : '');
      const partsStr = d.partitions && d.partitions !== '—'
        ? `<span class="badge bg-primary-subtle text-info border border-info-subtle ms-1" style="font-size: 0.62rem;">${escapeHtml(d.partitions)}</span>`
        : '';
      const isSsd = String(d.media_type || '').toUpperCase().includes('SSD') || String(d.bus_type || '').toUpperCase().includes('NVME');
      const iconClass = isSsd ? 'bi-device-ssd text-info' : 'bi-hdd text-primary';

      return `
        <div class="d-flex justify-content-between align-items
```

```
-center py-1 border-bottom border-dark-subtle sys-disk-item"
3170              data-disk-idx="${idx}"
3171              style="border-bottom-style: dashed !important; cu
rsor: pointer; transition: background 0.15s;"
3172              title="Нажмите для просмотра SMART-характеристик
и телеметрии">
3173            <div class="d-flex align-items-center text-truncate
me-2" style="max-width: 170px;">
3174              <i class="bi ${iconClass} me-1.5" style="font-size
: 0.8rem;"></i>
3175              <span class="text-truncate fw-semibold" style="fon
t-size: 0.72rem; color: var(--text-color);">${escapeHtml(model
)}</span>
3176            </div>
3177            <div class="d-flex align-items-center gap-1">
3178              ${partsStr}
3179              <span class="badge bg-secondary font-monospace" st
yle="font-size: 0.62rem;">${sizeStr}</span>
3180            </div>
3181          </div>
3182        `;
3183      }).join('');
3184
3185      diskCont.querySelectorAll('.sys-disk-item').forEach(el =>
{
3186        el.addEventListener('click', () => {
3187          const idx = parseInt(el.getAttribute('data-disk-idx'),
 10);
3188          const selectedDisk = disks[idx];
3189          if (selectedDisk) {
3190            openDriveAiModal(selectedDisk);
3191          }
3192        });
3193      });
3194    }
3195
3196    async function fetchCpuLoadFromApi() {
3197      try {
```

```
3198      const res = await fetch('/api/v1/panel/cpu-load');
3199      if (!res.ok) return;
3200      const data = await res.json();
3201      lastCpuData = data;
3202      const cpuVal = document.getElementById('sys-metric-cpu-v
    al');
3203      const cpuFill = document.getElementById('sys-metric-cpu-
    fill');
3204      const cpuSub = document.getElementById('sys-metric-cpu-s
    ub');
3205      const cpuModel = document.getElementById('sys-metric-cpu
    -model');
3206      const cpuGauge = document.getElementById('sys-metric-cpu
    -gauge');
3207      const cpuPkgSlider = document.getElementById('sys-metric
    -cpu-package-slider');
3208      const cpuPkgTemp = document.getElementById('sys-metric-c
    pu-package-temp');
3209
3210      const pct = Number(data.total_percent || 0);
3211      const cores = Array.isArray(data.cores) ? data.cores : [
    ];
3212      const threads = Array.isArray(data.threads) ? data.threa
    ds : [];
3213      const sensors = Array.isArray(data.sensors) ? data.senso
    rs : [];
3214      const history = Array.isArray(data.history) ? data.histo
    ry : [];
3215
3216      if (cpuVal) {
3217        cpuVal.innerText = `${pct.toFixed(1)}%`;
3218        // Динамический оттенок значения загрузки
3219        if (pct >= 85) {
3220          cpuVal.style.color = '#ef4444'; // Красный (критичес
    кая нагрузка)
3221          cpuVal.style.textShadow = '0 0 14px rgba(239, 68, 68
    , 0.6)';
3222        } else if (pct >= 70) {
```

```
3223          cpuVal.style.color = '#f97316'; // Оранжевый (высока
я нагрузка)
3224          cpuVal.style.textShadow = '0 0 12px rgba(249, 115, 2
2, 0.45)';
3225        } else if (pct >= 50) {
3226          cpuVal.style.color = '#eab308'; // Желтый / янтарный
 (умеренная нагрузка)
3227          cpuVal.style.textShadow = '0 0 10px rgba(234, 179, 8
, 0.4)';
3228        } else {
3229          cpuVal.style.color = '#0dcaf0'; // Голубой (штатная
нагрузка)
3230          cpuVal.style.textShadow = '0 0 10px rgba(13, 202, 24
0, 0.35)';
3231        }
3232      }
3233      if (cpuFill) cpuFill.style.width = `${Math.min(100, pct)
}%`;
3234
3235      if (cpuModel && data.name) {
3236        cpuModel.innerText = data.name;
3237        cpuModel.title = data.name;
3238      }
3239
3240      const pkgTemp = data.package_temperature_c != null ? Num
ber(data.package_temperature_c) : null;
3241      const pkgPower = data.package_power_w != null ? Number(d
ata.package_power_w) : null;
3242      if (cpuSub) {
3243        const t = pkgTemp == null ? '' : ` · ${pkgTemp.toFixed
(0)} °C`;
3244        const p = pkgPower == null ? '' : ` · ${pkgPower.toFix
ed(0)} W`;
3245        const cCount = data.cores_count || cores.length;
3246        const thCount = data.threads_count || threads.length;
3247        cpuSub.innerText = cCount ? `${cCount} ядер • ${thCoun
t} потоков${t}${p}` : '-- Ядер';
3248      }
```

```
3249      if (cpuPkgTemp) {
3250        const pStr = pkgPower != null ? ` · ${pkgPower.toFixed
(0)} W` : '';
3251        cpuPkgTemp.innerText = pkgTemp != null ? `Пакет: ${pkg
Temp.toFixed(0)} °C${pStr}` : 'Пакет: -- °C';
3252        if (pkgTemp != null) {
3253          if (pkgTemp >= 80) {
3254            cpuPkgTemp.className = 'text-danger fw-bold font-m
onospace';
3255          } else if (pkgTemp >= 65) {
3256            cpuPkgTemp.className = 'text-warning fw-bold font-
monospace';
3257          } else {
3258            cpuPkgTemp.className = 'text-info font-monospace';
3259          }
3260        }
3261      }
3262
3263      // 1. Заполнение сводной таблицы характеристик CPU (Spec
s)
3264      const specs = data.specs || {};
3265      const setSpec = (id, val) => { const el = document.getEl
ementById(id); if (el) el.innerText = val; };
3266      if (specs.socket || specs.physical_cores) {
3267        setSpec('sys-cpu-spec-arch', specs.architecture || 'x8
6_64');
3268        setSpec('sys-cpu-spec-socket', specs.socket || 'LGA120
0');
3269        setSpec('sys-cpu-spec-cores', `${specs.physical_cores
|| cores.length}C / ${specs.logical_cores || threads.length}T`
);
3270        setSpec('sys-cpu-spec-base-freq', specs.base_frequency
_str || '--');
3271        setSpec('sys-cpu-spec-max-freq', specs.max_frequency_s
tr || '--');
3272        setSpec('sys-cpu-spec-cache', specs.cache_combined_str
 || '--');
3273      }
```

```
3274
3275       // Экспресс-метрики KPI в центре баннера CPU
3276       const kpiPower = document.getElementById('sys-cpu-kpi-po
wer');
3277       const kpiTemp = document.getElementById('sys-cpu-kpi-tem
p');
3278       const kpiCores = document.getElementById('sys-cpu-kpi-co
res');
3279       const kpiFreq = document.getElementById('sys-cpu-kpi-fre
q');
3280       const cCount = data.cores_count || (specs.physical_cores
 || cores.length);
3281       const thCount = data.threads_count || (specs.logical_cor
es || threads.length);
3282
3283       if (kpiPower) kpiPower.textContent = pkgPower != null ?
`${pkgPower.toFixed(0)} W` : (data.package_power_w != null ? `
${Number(data.package_power_w).toFixed(0)} W` : '-- W');
3284       if (kpiTemp) {
3285         kpiTemp.textContent = pkgTemp != null ? `${pkgTemp.toF
ixed(0)} °C` : '-- °C';
3286         if (pkgTemp != null) {
3287           kpiTemp.className = pkgTemp >= 80 ? 'sys-kpi-value t
ext-danger' : pkgTemp >= 65 ? 'sys-kpi-value text-warning' : '
sys-kpi-value text-info';
3288         }
3289       }
3290       if (kpiCores) kpiCores.textContent = cCount ? `${cCount}
C / ${thCount}T` : '-- / --';
3291       if (kpiFreq) {
3292         const fMax = specs.max_frequency_str || (specs.max_fre
quency_ghz ? `${specs.max_frequency_ghz} GHz` : null);
3293         const fBase = specs.base_frequency_str || (specs.base_
frequency_ghz ? `${specs.base_frequency_ghz} GHz` : null);
3294         kpiFreq.textContent = fMax || fBase || '-- GHz';
3295       }
3296
3297       const vendorBadge = document.getElementById('sys-metric-
```

```
cpu-vendor-badge');
3298      if (vendorBadge) {
3299        const cpus = Array.isArray(data.cpus) && data.cpus.len
gth > 0 ? data.cpus : [];
3300        if (cpus.length > 1) {
3301          vendorBadge.innerHTML = cpus.map((c, i) => `<div cla
ss="text-truncate font-monospace" style="font-size: 0.74rem;">
CPU #${c.id ?? i}: ${escapeHtml(c.description || `${c.raw_desc
riptor || c.name} (${c.logical_cores || 12} logical cores)`)}<
/div>`).join('');
3302        } else if (data.description || specs.description) {
3303          vendorBadge.innerText = data.description || specs.de
scription;
3304        } else {
3305          const rawDesc = specs.raw_descriptor || 'Intel64 Fam
ily 6 Model 165 Stepping 3, GenuineIntel';
3306          const logCores = data.threads_count || specs.logical
_cores || (threads ? threads.length : 12) || 12;
3307          vendorBadge.innerText = `${rawDesc} (${logCores} log
ical cores)`;
3308        }
3309      }
3310
3311      // Gauge полукруг со стрелкой для общего CPU
3312      if (cpuGauge) {
3313        cpuGauge.innerHTML = createGaugeSvg(pct, 120, 68);
3314      }
3315
3316      // Слайдер температуры Package (расположен прямо под спи
дометром)
3317      if (cpuPkgSlider) {
3318        cpuPkgSlider.innerHTML = createTempSliderHtml(pkgTemp,
 30, 95, false);
3319      }
3320
3321      // 2. Физические ядра со своими потоками
3322      renderCpuCores(cores, data.threads_count || threads.leng
th);
```

```
3323
3324       // 3. Общие показатели CPU
3325       if (sensors && sensors.length > 0) {
3326         lastCpuSensorsData = sensors;
3327         renderComponentSensors('sys-metric-cpu-sensors', 'sys-
cpu-sensors-count-badge', sensors);
3328       }
3329
3330       // 4. График истории CPU (Загрузка %, Температура °C, Мо
щность W)
3331       if (history && history.length > 0) {
3332         renderCpuSpark(history);
3333       } else {
3334         _cpuSparkHistory.push({ load: pct, temp: pkgTemp, powe
r: pkgPower, time: new Date() });
3335         if (_cpuSparkHistory.length > 60) _cpuSparkHistory.shi
ft();
3336         renderCpuSpark(_cpuSparkHistory);
3337       }
3338     } catch (e) {
3339       console.warn('[SystemInspectorTab] Ошибка получения загр
узки CPU из API:', e);
3340     }
3341   }
3342
3343   /**
3344    * Запуск экспертного AI-аудита и анализа характеристик CPU
через Universal AITableModal.
3345    */
3346   function inspectCpuSpecsWithAi() {
3347     if (!lastCpuData) {
3348       console.warn('[SystemInspectorTab] Нет загруженных данны
x CPU для AI-инспекции');
3349       return;
3350     }
3351     const specs = lastCpuData.specs || {};
3352     const title = lastCpuData.name || 'Центральный процессор (
CPU)';
```

```
3353    const subtitle = `${specs.vendor || 'Intel/AMD'} · ${specs
.socket || 'Socket'} · ${specs.architecture || 'x86_64'}`;
3354    const metadata = {
3355      'Сокет': specs.socket || 'N/A',
3356      'Ядра / Потоки': `${specs.physical_cores || lastCpuData.
cores_count || '--'}C / ${specs.logical_cores || lastCpuData.t
hreads_count || '--'}T`,
3357      'Базовая частота': specs.base_frequency_str || '--',
3358      'Макс. частота': specs.max_frequency_str || '--',
3359      'Кэш L2 / L3': specs.cache_combined_str || '--',
3360      'Архитектура': specs.architecture || 'x86_64',
3361      'Текущая загрузка': `${(lastCpuData.total_percent || 0).
toFixed(1)}%`,
3362      'Температура пакета': lastCpuData.package_temperature_c
!= null ? `${lastCpuData.package_temperature_c.toFixed(0)} °C`
 : 'N/A',
3363      'Энергопотребление': lastCpuData.package_power_w != null
 ? `${lastCpuData.package_power_w.toFixed(1)} W` : 'N/A'
3364    };
3365
3366    if (window.AITableModal && typeof window.AITableModal.show
 === 'function') {
3367      window.AITableModal.show({
3368        icon: '□ ',
3369        title: title,
3370        subtitle: subtitle,
3371        badges: [
3372          { text: specs.architecture || 'x86_64', class: 'bg-p
rimary-subtle text-primary border border-primary' },
3373          { text: specs.socket || 'Socket', class: 'bg-info-su
btle text-info border border-info' },
3374          { text: `${specs.physical_cores || lastCpuData.cores
_count || '--'} Cores`, class: 'bg-success-subtle text-success
 border border-success' }
3375        ],
3376        metadata: metadata,
3377        rawTitle: 'CPU Inventory Telemetry & Specs',
3378        rawContent: JSON.stringify({
```

```
          name: lastCpuData.name,
          specs: specs,
          metrics: {
            total_percent: lastCpuData.total_percent,
            package_temperature_c: lastCpuData.package_tempera
ture_c,
            package_power_w: lastCpuData.package_power_w
          },
          cores_count: (lastCpuData.cores || []).length,
          threads_count: (lastCpuData.threads || []).length
        }, null, 2),
        panelId: 'panel-system-inspector-cpu-specs',
        tableType: 'cpu',
        actions: []
      });
    } else {
      console.warn('[SystemInspectorTab] Universal AITableModa
l недоступен в DOM');
    }
  }
  window.inspectCpuSpecsWithAi = inspectCpuSpecsWithAi;

  /**
   * Запуск экспертного AI-аудита и анализа характеристик GPU
через Universal AITableModal.
   */
  function inspectGpuSpecsWithAi(idx = 0) {
    const gpu = (lastGpuDevices && lastGpuDevices[idx]) || las
tGpuData;
    if (!gpu) {
      console.warn('[SystemInspectorTab] Нет загруженных данны
x GPU для AI-инспекции');
      return;
    }
    const specs = gpu.specs || {};
    const title = gpu.name || `Графический процессор (GPU #${i
dx})`;
    const subtitle = `${specs.vendor || gpu.vendor || 'GPU'} ·
```

```
 ${specs.pci_bus || 'PCIe'} · ${specs.directx || 'DirectX 12'}
`;
    const metadata = {
      'Производитель': specs.vendor || gpu.vendor || 'NVIDIA/Intel/AMD',
      'VRAM Объем': specs.vram_str || (gpu.memory && gpu.memory.total_mb ? `${(gpu.memory.total_mb/1024).toFixed(1)} GB` : '--'),
      'Использование VRAM': specs.vram_used_str || (gpu.memory && gpu.memory.used_percent != null ? `${gpu.memory.used_percent.toFixed(0)}%` : '--'),
      'Частота ядра': specs.core_clock_str || (gpu.clocks && gpu.clocks.core_mhz ? `${Math.round(gpu.clocks.core_mhz)} MHz` : '--'),
      'Частота памяти': specs.memory_clock_str || (gpu.clocks && gpu.clocks.memory_mhz ? `${Math.round(gpu.clocks.memory_mhz)} MHz` : '--'),
      'Версия драйвера': specs.driver_version || '--',
      'Шина': specs.pci_bus || 'PCIe',
      'DirectX API': specs.directx || 'DirectX 12',
      'Текущая загрузка': `${(gpu.core_load_percent || 0).toFixed(1)}%`,
      'Температура GPU': gpu.core_temperature_c != null ? `${gpu.core_temperature_c.toFixed(0)} °C` : 'N/A',
      'Энергопотребление': gpu.power_w != null ? `${gpu.power_w.toFixed(1)} W` : 'N/A'
    };

    if (window.AITableModal && typeof window.AITableModal.show === 'function') {
      window.AITableModal.show({
        icon: '□ ',
        title: title,
        subtitle: subtitle,
        badges: [
          { text: specs.vendor || gpu.vendor || 'GPU', class: 'bg-warning-subtle text-warning border border-warning' },
          { text: specs.vram_str || (gpu.memory && gpu.memory.
```

```
total_mb ? `${(gpu.memory.total_mb/1024).toFixed(1)} GB` : 'VR
AM'), class: 'bg-info-subtle text-info border border-info' },
          { text: specs.directx || 'DirectX 12', class: 'bg-su
ccess-subtle text-success border border-success' }
        ],
        metadata: metadata,
        rawTitle: `GPU #${gpu.index != null ? gpu.index : idx}
 Inventory Telemetry & Specs`,
        rawContent: JSON.stringify({
          index: gpu.index != null ? gpu.index : idx,
          name: gpu.name,
          vendor: gpu.vendor,
          specs: specs,
          metrics: {
            core_load_percent: gpu.core_load_percent,
            core_temperature_c: gpu.core_temperature_c,
            hotspot_temperature_c: gpu.hotspot_temperature_c,
            power_w: gpu.power_w,
            memory: gpu.memory,
            clocks: gpu.clocks
          },
          engines: gpu.engines || [],
          sensors_count: (gpu.sensors || []).length
        }, null, 2),
        panelId: 'panel-system-inspector-gpu-specs',
        tableType: 'gpu',
        actions: []
      });
    } else {
      console.warn('[SystemInspectorTab] Universal AITableModa
l недоступен в DOM');
    }
  }
  window.inspectGpuSpecsWithAi = inspectGpuSpecsWithAi;

  function renderGpuEngines(engines, idx = 0) {
    const box = document.getElementById(`sys-metric-gpu-engine
s-${idx}`) || document.getElementById('sys-metric-gpu-engines'
```

```
);
    const badgeCount = document.getElementById(`sys-gpu-engines-count-badge-${idx}`) || document.getElementById('sys-gpu-engines-count-badge');
    if (!box) return;

    if (badgeCount) {
      badgeCount.textContent = engines && engines.length ? `${engines.length} блоков` : '0 блоков';
    }

    if (!Array.isArray(engines) || engines.length === 0) {
      box.innerHTML = `<div class="text-muted small text-center py-3 col-12">Опрос подсистем GPU выполняется...</div>`;
      return;
    }

    box.innerHTML = engines.map(e => {
      const load = e.load_percent == null ? null : Number(e.load_percent);
      const temp = e.temperature_c == null ? null : Number(e.temperature_c);
      const loadStr = load == null ? '--' : `${load.toFixed(0)}%`;
      const gaugeSvg = createGaugeSvg(load, 110, 62);
      const tempHtml = createTempSliderHtml(temp, 30, 95, false);

      return `
        <div class="sys-core-card p-2.5" style="min-height: 156px; display: flex; flex-direction: column; justify-content: space-between;" title="${escapeHtml(e.name)}: Нагрузка ${loadStr}${temp != null ? ', Температура ' + temp.toFixed(0) + '°C' : ''}">
          <div class="d-flex justify-content-between align-items-start gap-1.5 mb-1" style="min-height: 32px;">
            <span class="sys-core-title" style="font-size: 0.84rem; font-weight: 800; color: var(--text-color); line-height:
```

```
 1.25; word-break: break-word;" title="${escapeHtml(e.name)}">
${escapeHtml(e.name)}</span>
            <span class="sys-core-load-badge font-monospace fl
ex-shrink-0" style="font-size: 0.76rem; padding: 2px 6px;">${l
oadStr}</span>
          </div>
          <div class="py-1 d-flex justify-content-center">
            ${gaugeSvg}
          </div>
          <div class="mt-auto">
            ${tempHtml}
          </div>
        </div>
      `;
    }).join('');
  }

  function renderGpuCardHtml(gpu, idx, totalDevices) {
    const gpuName = escapeHtml(gpu.name || `GPU #${idx}`);
    const vendor = escapeHtml(gpu.vendor || 'GPU');
    const headerTitle = totalDevices > 1 ? `Загрузка GPU #${gp
u.index != null ? gpu.index : idx}: ${gpuName}` : `Загрузка GP
U: ${gpuName}`;

    return `
      <div class="sys-card sys-card-gpu h-100 p-3" id="sys-car
d-gpu-${idx}">
        <div class="sys-card-header-bar">
          <div class="d-flex align-items-center gap-2">
            <span class="sys-label text-warning" style="font-s
ize: 1.05rem; font-weight: 800;">
              <i class="bi bi-gpu-card text-warning me-1.5"></
i>${headerTitle}
            </span>
            <span class="badge bg-dark border border-secondary
 text-info font-monospace ms-1" style="font-size: 0.74rem;">
              ${vendor}
            </span>
```

```
          </div>
          <span class="small text-truncate font-monospace badge bg-dark border border-secondary text-warning" style="font-size: 0.84rem; max-width: 320px;" id="sys-metric-gpu-sub-${idx}">--</span>
        </div>

        <!-- Верхний баннер: Название слева, Спидометр и датчик температуры по центру, Сводная спецификация GPU справа -->
        <div class="sys-card-top-banner p-3 rounded-3 mb-3">
          <!-- Левая колонка: Модель графического процессора и экспресс-метрики KPI -->
          <div class="sys-banner-col-info d-flex flex-column align-items-start justify-content-center">
            <div class="text-start mb-1.5 d-flex align-items-center">
              <button type="button" class="sys-drill-btn" id="sys-drill-btn-gpu-${idx}" onclick="window.toggleSysCardDrill && window.toggleSysCardDrill('sys-card-gpu-${idx}', ${idx})" title="Drill-down: Развернуть/скрыть все скрытые панели и датчики GPU">
                <img src="/html/assets/red_drill.png?v=20261008_v1" alt="Дрель">
              </button>
              <div>
                <span class="fw-bold text-warning" style="font-size: 1.20rem; font-weight: 800; letter-spacing: 0.5px; text-shadow: 0 0 14px rgba(245, 158, 11, 0.4);" id="sys-metric-gpu-model-${idx}">${gpuName}</span>
                <div class="d-flex align-items-center gap-1.5 mt-0.5">
                  <span class="badge bg-dark border border-secondary text-light font-monospace" id="sys-metric-gpu-vendor-badge-${idx}" style="font-size: 0.70rem;">${vendor} · PCIe</span>
                  <span class="badge bg-warning-subtle text-warning border border-warning font-monospace" id="sys-gpu-status-badge-${idx}" style="font-size: 0.70rem;">● Ready</span>
```

```
3533                </div>
3534              </div>
3535            </div>
3536            <div class="sys-banner-kpi-grid">
3537              <div class="sys-kpi-pill">
3538                <span class="sys-kpi-label"><i class="bi bi-me
mory text-info me-1"></i>VRAM Память</span>
3539                <span class="sys-kpi-value text-info" id="sys-
gpu-kpi-vram-${idx}">-- / -- GB</span>
3540              </div>
3541              <div class="sys-kpi-pill">
3542                <span class="sys-kpi-label"><i class="bi bi-li
ghtning-charge text-warning me-1"></i>Мощность</span>
3543                <span class="sys-kpi-value text-warning" id="s
ys-gpu-kpi-power-${idx}">-- W</span>
3544              </div>
3545              <div class="sys-kpi-pill">
3546                <span class="sys-kpi-label"><i class="bi bi-sp
eedometer2 text-primary me-1"></i>Частота Core</span>
3547                <span class="sys-kpi-value text-primary" id="s
ys-gpu-kpi-clock-${idx}">-- MHz</span>
3548              </div>
3549              <div class="sys-kpi-pill">
3550                <span class="sys-kpi-label"><i class="bi bi-th
ermometer-half text-danger me-1"></i>GPU t°</span>
3551                <span class="sys-kpi-value text-warning" id="s
ys-gpu-kpi-temp-${idx}">-- °C</span>
3552              </div>
3553            </div>
3554          </div>
3555
3556          <!-- Центральная колонка: Спидометр загрузки, справа
 от него значение с динамическим оттенком, а СНИЗУ — датчик те
мпературы -->
3557          <div class="sys-banner-col-gauge d-flex flex-column
justify-content-between gap-1.5">
3558            <div class="d-flex align-items-center gap-2.5">
3559              <div id="sys-metric-gpu-gauge-${idx}" style="wid
```

```
th: 120px; height: 68px; flex-shrink: 0;"></div>
              <div>
                <div class="sys-value text-warning" id="sys-me
tric-gpu-val-${idx}" style="font-size: 2.1rem; line-height: 1;
 font-weight: 800; transition: color 0.3s ease, text-shadow 0.
3s ease;">0.0%</div>
                <div class="text-light mt-1 font-monospace" st
yle="font-size: 0.82rem; font-weight: 700;" id="sys-metric-gpu
-core-temp-${idx}">GPU: -- °C</div>
              </div>
            </div>
            <!-- Датчик температуры GPU Core расположен ПОД сп
идометром загрузки -->
            <div id="sys-metric-gpu-temp-slider-${idx}" style=
"width: 100%;"></div>
          </div>

          <!-- Правая колонка: Сводная таблица о GPU (кликабел
ьна для запуска AI-аудита) -->
          <div class="sys-banner-col-specs">
            <div class="sys-gpu-specs-card p-2 rounded-2 d-fle
x flex-column justify-content-between h-100"
                 id="sys-gpu-specs-card-${idx}"
                 onclick="window.inspectGpuSpecsWithAi && wind
ow.inspectGpuSpecsWithAi(${idx})"
                 title="Нажмите для вызова AI-аудита и эксперт
ного анализа графического процессора">
              <div class="d-flex justify-content-between align
-items-center mb-1 pb-1 border-bottom" style="border-color: va
r(--border-subtle) !important;">
                <span class="text-light fw-bold d-flex align-i
tems-center gap-1" style="font-size: 0.78rem;">
                  <i class="bi bi-info-circle text-warning"></
i>Характеристики GPU
                  <span class="badge bg-primary-subtle text-pr
imary border border-primary-subtle font-monospace ms-1" style=
"font-size: 0.62rem;"><i class="bi bi-stars me-0.5"></i>AI</sp
an>
```

```
3579                </span>
3580                <span class="badge bg-dark border border-secon
    dary text-warning font-monospace" style="font-size: 0.68rem;"
    id="sys-gpu-spec-directx-${idx}">DirectX 12</span>
3581              </div>
3582              <div class="sys-gpu-specs-grid" style="display:
    grid; grid-template-columns: 1fr 1fr; gap: 4px 10px; font-size
    : 0.74rem;">
3583                <div class="d-flex justify-content-between"><s
    pan class="text-muted">VRAM Объем:</span><span class="text-lig
    ht font-monospace fw-semibold" id="sys-gpu-spec-vram-${idx}">-
    -</span></div>
3584                <div class="d-flex justify-content-between"><s
    pan class="text-muted">Исп. VRAM:</span><span class="text-warn
    ing font-monospace fw-semibold" id="sys-gpu-spec-vram-used-${i
    dx}">--</span></div>
3585                <div class="d-flex justify-content-between"><s
    pan class="text-muted">Частота Core:</span><span class="text-w
    arning font-monospace fw-semibold" id="sys-gpu-spec-core-clock
    -${idx}">--</span></div>
3586                <div class="d-flex justify-content-between"><s
    pan class="text-muted">Частота Mem:</span><span class="text-wa
    rning font-monospace fw-semibold" id="sys-gpu-spec-mem-clock-$
    {idx}">--</span></div>
3587                <div class="d-flex justify-content-between" st
    yle="grid-column: span 2;"><span class="text-muted">Драйвер /
    API:</span><span class="text-info font-monospace fw-semibold"
    id="sys-gpu-spec-driver-${idx}">--</span></div>
3588              </div>
3589            </div>
3590          </div>
3591        </div>
3592
3593        <!-- График истории GPU (Загрузка %, Температура °C, M
    ощность / Память %) с логарифмической/линейной шкалой -->
3594        <div class="sys-full-only sys-history-panel p-2.5 mb-2
    .5 rounded-2">
3595          <div class="d-flex justify-content-between align-ite
```

```
ms-center mb-1.5 flex-wrap gap-2">
3596            <div class="d-flex align-items-center gap-2">
3597              <span class="small text-light" style="font-size:
 0.80rem; font-weight: 700;"><i class="bi bi-graph-up text-war
ning me-1"></i>История GPU</span>
3598              <div class="btn-group btn-group-sm" role="group"
 style="height: 22px;">
3599                <button type="button" class="btn btn-xs btn-ou
tline-warning active py-0 px-2 fw-bold" id="btn-gpu-spark-log-
${idx}" style="font-size: 0.68rem;" onclick="window.setGpuSpar
kScale && window.setGpuSparkScale('log', ${idx})">Лог. шкала (
Log)</button>
3600                <button type="button" class="btn btn-xs btn-ou
tline-secondary py-0 px-2" id="btn-gpu-spark-lin-${idx}" style
="font-size: 0.68rem;" onclick="window.setGpuSparkScale && win
dow.setGpuSparkScale('linear', ${idx})">Линейная (Lin)</button
>
3601              </div>
3602              <select class="form-select form-select-xs sys-sp
ark-interval-select" id="sel-gpu-spark-interval-${idx}" style=
"height: 22px; padding: 0 6px; font-size: 0.68rem; width: auto
; min-width: 85px;" onchange="window.setGpuSparkInterval && wi
ndow.setGpuSparkInterval(${idx}, this.value)">
3603                <option value="seconds">Секунды</option>
3604              </select>
3605            </div>
3606            <div class="d-flex align-items-center gap-2.5" sty
le="font-size: 0.74rem; font-weight: 600;">
3607              <span class="text-warning" style="text-shadow: 0
 0 8px rgba(245,158,11,0.5);" title="Основная шкала: Загрузка
GPU Core (0-100%)">● Нагрузка Core %</span>
3608              <span class="text-danger" style="text-shadow: 0
0 8px rgba(239,68,68,0.5);" title="Компактная шкала: Температу
ра GPU Core (°C)">● Температура °C</span>
3609              <span class="text-info" style="text-shadow: 0 0
8px rgba(13,202,240,0.5);" title="Компактная шкала: Мощность W
 / VRAM %">● Мощность / Память %</span>
3610              <span class="text-muted font-monospace ms-2" id=
```

```
"sys-gpu-spark-live-cursor-${idx}" style="font-size: 0.70rem;"
>Live</span>
            </div>
          </div>
          <!-- Контейнер графика и интерактивного тултипа -->
          <div style="position: relative; width: 100%; height:
 90px;" id="sys-gpu-spark-container-${idx}">
            <div id="sys-gpu-spark-${idx}" style="width: 100%;
 height: 100%;"></div>
            <div id="sys-gpu-spark-tooltip-${idx}" class="sys-
spark-tooltip" style="display: none;"></div>
          </div>
        </div>

        <!-- Сворачиваемый блок детализации GPU: Блоки, подсис
темы и сенсоры питания/напряжения/кулеров -->
        <div class="sys-full-only sys-detailed-sensors mt-2">
          <div class="sys-gpu-details-accordion-btn d-flex jus
tify-content-between align-items-center flex-wrap gap-2"
               id="sys-gpu-details-toggle-btn-${idx}"
               title="Нажмите, чтобы развернуть или скрыть дет
ализацию по подсистемам и сенсорам GPU"
               onclick="window.toggleSysGpuDetails && window.t
oggleSysGpuDetails(${idx})">
            <div class="d-flex align-items-center gap-2 flex-w
rap">
              <button type="button" class="panel-toggle-drill
 collapsed me-1" id="sys-gpu-details-drill-btn-${idx}" tabindex
="-1" style="pointer-events: none;">
                <img src="/html/assets/red_drill.png?v=2026100
8_v1" alt="Дрель">
              </button>
              <span class="fw-bold" style="font-size: 0.88rem;
 color: var(--text-color);">Детализация: Блоки, подсистемы и с
енсоры GPU</span>
              <span class="badge bg-dark border border-seconda
ry text-warning font-monospace ms-1" style="font-size: 0.72rem
;" id="sys-gpu-engines-count-badge-${idx}">0 блоков</span>
```

```
3632              <span class="badge bg-dark border border-seconda
ry text-danger font-monospace" style="font-size: 0.72rem;" id=
"sys-gpu-sensors-count-badge-${idx}">0 параметров</span>
3633            </div>
3634          </div>
3635
3636          <!-- Скрытый по умолчанию общий контейнер деталей GP
U -->
3637          <div id="sys-gpu-details-collapse-${idx}" style="dis
play: none; flex-direction: column; gap: 14px; margin-top: 10p
x;">
3638            <!-- 1. Блоки и подсистемы GPU -->
3639            <div>
3640              <div class="sys-core-section-title d-flex justif
y-content-between align-items-center mb-2" style="font-size: 0
.82rem;">
3641                <span><i class="bi bi-grid-3x3-gap-fill text-w
arning me-1"></i>Блоки и подсистемы GPU</span>
3642              </div>
3643              <div id="sys-metric-gpu-engines-${idx}" style="d
isplay: grid; grid-template-columns: repeat(auto-fill, minmax(
175px, 1fr)); gap: 10px; font-size: 0.78rem;"></div>
3644            </div>
3645
3646            <!-- 2. Энергопотребление, напряжение, частоты и с
енсоры GPU -->
3647            <div>
3648              <div class="sys-core-section-title d-flex justif
y-content-between align-items-center mb-2" style="font-size: 0
.82rem;">
3649                <span><i class="bi bi-lightning-charge text-wa
rning me-1"></i>Энергопотребление, напряжение, вентиляторы и ч
астоты GPU</span>
3650              </div>
3651              <div id="sys-metric-gpu-sensors-${idx}" style="d
isplay: grid; grid-template-columns: repeat(auto-fill, minmax(
180px, 1fr)); gap: 8px; font-size: 0.78rem;">
3652                <div class="text-muted small text-center py-2
```

```
col-12">Опрос сенсоров GPU...</div>
              </div>
            </div>
          </div>
        </div>
      </div>
    `;
  }

  function updateSingleGpuCard(gpu, idx, totalDevices) {
    const pct = Number(gpu.core_load_percent || 0);
    const coreTemp = gpu.core_temperature_c != null ? Number(g
pu.core_temperature_c) : null;
    const hotspotTemp = gpu.hotspot_temperature_c != null ? Nu
mber(gpu.hotspot_temperature_c) : null;
    const powerW = gpu.power_w != null ? Number(gpu.power_w) :
 null;
    const clocks = gpu.clocks || {};
    const memory = gpu.memory || {};
    const specs = gpu.specs || {};
    const engines = Array.isArray(gpu.engines) ? gpu.engines :
 [];
    const sensorsList = Array.isArray(gpu.sensors) ? gpu.senso
rs : [];

    const gpuVal = document.getElementById(`sys-metric-gpu-val
-${idx}`);
    const gpuModel = document.getElementById(`sys-metric-gpu-m
odel-${idx}`);
    const gpuVendorBadge = document.getElementById(`sys-metric
-gpu-vendor-badge-${idx}`);
    const gpuSub = document.getElementById(`sys-metric-gpu-sub
-${idx}`);
    const gpuGauge = document.getElementById(`sys-metric-gpu-g
auge-${idx}`);
    const gpuTempSlider = document.getElementById(`sys-metric-
gpu-temp-slider-${idx}`);
    const gpuCoreTemp = document.getElementById(`sys-metric-gp
```

```
u-core-temp-${idx}`);

    if (gpuVal) {
      gpuVal.innerText = `${pct.toFixed(1)}%`;
      if (pct >= 85) {
        gpuVal.style.color = '#ef4444';
        gpuVal.style.textShadow = '0 0 16px rgba(239, 68, 68, 0.7)';
      } else if (pct >= 60) {
        gpuVal.style.color = '#f59e0b';
        gpuVal.style.textShadow = '0 0 14px rgba(245, 158, 11, 0.6)';
      } else if (pct >= 25) {
        gpuVal.style.color = '#eab308';
        gpuVal.style.textShadow = '0 0 12px rgba(234, 179, 8, 0.4)';
      } else {
        gpuVal.style.color = '#10b981';
        gpuVal.style.textShadow = '0 0 10px rgba(16, 185, 129, 0.3)';
      }
    }

    if (gpuModel) {
      gpuModel.innerText = gpu.name || 'GPU';
      gpuModel.title = gpu.name || 'GPU';
    }

    if (gpuVendorBadge) {
      const v = specs.vendor || gpu.vendor || 'GPU';
      const bus = specs.pci_bus || 'PCIe x16';
      gpuVendorBadge.textContent = `${v} Corporation · ${bus}`;
    }

    if (gpuCoreTemp) {
      let tempText = coreTemp != null ? `GPU: ${coreTemp.toFixed(0)} °C` : 'GPU: -- °C';
```

```
3710      if (hotspotTemp != null) {
3711        tempText += ` (HotSpot: ${hotspotTemp.toFixed(0)}°C)`;
3712      }
3713      gpuCoreTemp.innerText = tempText;
3714    }
3715
3716    if (gpuSub) {
3717      let subText = '';
3718      if (clocks && clocks.core_mhz) {
3719        subText += `${Math.round(clocks.core_mhz)} MHz`;
3720      }
3721      if (memory && memory.total_mb) {
3722        const totalGb = (memory.total_mb / 1024).toFixed(1);
3723        subText += (subText ? ' · ' : '') + `${totalGb} GB VRA
     M`;
3724      }
3725      gpuSub.innerText = subText || (gpu.name || 'GPU');
3726      gpuSub.title = subText || (gpu.name || 'GPU');
3727    }
3728
3729    // Спецификации
3730    const specVram = document.getElementById(`sys-gpu-spec-vra
     m-${idx}`);
3731    const specVramUsed = document.getElementById(`sys-gpu-spec
     -vram-used-${idx}`);
3732    const specCoreClock = document.getElementById(`sys-gpu-spe
     c-core-clock-${idx}`);
3733    const specMemClock = document.getElementById(`sys-gpu-spec
     -mem-clock-${idx}`);
3734    const specDriver = document.getElementById(`sys-gpu-spec-d
     river-${idx}`);
3735    const specDirectx = document.getElementById(`sys-gpu-spec-
     directx-${idx}`);
3736
3737    if (specVram) specVram.textContent = specs.vram_str || (me
     mory.total_mb ? `${(memory.total_mb/1024).toFixed(1)} GB` : '-
     -');
3738    if (specVramUsed) specVramUsed.textContent = specs.vram_us
```

```
ed_str || (memory.used_percent != null ? `${memory.used_percen
t.toFixed(0)}%` : '--');
3739    if (specCoreClock) specCoreClock.textContent = specs.core_
clock_str || (clocks.core_mhz ? `${Math.round(clocks.core_mhz)
} MHz` : '--');
3740    if (specMemClock) specMemClock.textContent = specs.memory_
clock_str || (clocks.memory_mhz ? `${Math.round(clocks.memory_
mhz)} MHz` : '--');
3741    if (specDriver) specDriver.textContent = specs.driver_vers
ion ? `${specs.driver_version} (DirectX 12)` : 'WDDM 2.7 (Dire
ctX 12)';
3742    if (specDirectx) specDirectx.textContent = specs.directx |
| 'DirectX 12';
3743
3744    // Экспресс-метрики KPI в центре баннера GPU
3745    const kpiVram = document.getElementById(`sys-gpu-kpi-vram-
${idx}`);
3746    const kpiPower = document.getElementById(`sys-gpu-kpi-powe
r-${idx}`);
3747    const kpiClock = document.getElementById(`sys-gpu-kpi-cloc
k-${idx}`);
3748    const kpiTemp = document.getElementById(`sys-gpu-kpi-temp-
${idx}`);
3749    if (kpiVram) {
3750      if (memory && memory.total_mb) {
3751        const uGb = ((memory.used_mb || 0) / 1024).toFixed(1);
3752        const tGb = (memory.total_mb / 1024).toFixed(1);
3753        kpiVram.textContent = `${uGb} / ${tGb} GB`;
3754      } else {
3755        kpiVram.textContent = '-- / -- GB';
3756      }
3757    }
3758    if (kpiPower) kpiPower.textContent = powerW != null ? `${p
owerW.toFixed(0)} W` : (gpu.power_w ? `${gpu.power_w} W` : '--
 W');
3759    if (kpiClock) kpiClock.textContent = clocks && clocks.core
_mhz ? `${Math.round(clocks.core_mhz)} MHz` : '-- MHz';
3760    if (kpiTemp) {
```

```
3761      kpiTemp.textContent = coreTemp != null ? `${coreTemp.toF
ixed(0)} °C` : '-- °C';
3762      if (coreTemp != null) {
3763        kpiTemp.className = coreTemp >= 80 ? 'sys-kpi-value te
xt-danger' : coreTemp >= 65 ? 'sys-kpi-value text-warning' : '
sys-kpi-value text-info';
3764      }
3765    }
3766
3767    // Gauge SVG
3768    if (gpuGauge) {
3769      gpuGauge.innerHTML = createGaugeSvg(pct, 120, 68);
3770    }
3771
3772    // Temp slider
3773    if (gpuTempSlider) {
3774      gpuTempSlider.innerHTML = createTempSliderHtml(coreTemp,
 30, 95, true);
3775    }
3776
3777    // Engines
3778    renderGpuEngines(engines, idx);
3779
3780    // Sensors
3781    const gpuSensorsContainer = document.getElementById(`sys-m
etric-gpu-sensors-${idx}`);
3782    const gpuSensorsCountBadge = document.getElementById(`sys-
gpu-sensors-count-badge-${idx}`);
3783    if (gpuSensorsCountBadge) {
3784      gpuSensorsCountBadge.textContent = `${sensorsList.length
} параметров`;
3785    }
3786    if (gpuSensorsContainer) {
3787      if (sensorsList.length === 0) {
3788        gpuSensorsContainer.innerHTML = `<div class="text-mute
d small text-center py-2 col-12">Датчики питания и вентиляторо
в GPU опрашиваются...</div>`;
3789      } else {
```

```
3790        gpuSensorsContainer.innerHTML = sensorsList.map(s => {
3791          const statusClass = s.status === 'danger' ? 'badge b
g-danger text-white' : (s.status === 'warning' ? 'badge bg-war
ning text-dark' : 'badge bg-dark border border-secondary text-
warning');
3792          return `
3793            <div class="sys-sensor-widget-item p-2 rounded-2 d
-flex flex-column justify-content-between" style="min-height:
52px;">
3794              <div class="d-flex justify-content-between align
-items-center mb-1">
3795                <span class="sys-sensor-name text-truncate" ti
tle="${escapeHtml(s.name)}" style="font-size: 0.74rem; font-we
ight: 600; color: var(--text-color);">${escapeHtml(s.name)}</s
pan>
3796                <span class="${statusClass} font-monospace" st
yle="font-size: 0.68rem;">${escapeHtml(s.category || 'GPU')}</
span>
3797              </div>
3798              <div class="d-flex justify-content-between align
-items-baseline">
3799                <span class="text-muted" style="font-size: 0.6
8rem;">Значение</span>
3800                <span class="font-monospace fw-bold" style="fo
nt-size: 0.82rem; color: var(--text-color);">${escapeHtml(s.va
lue_raw || String(s.value))}</span>
3801              </div>
3802            </div>
3803          `;
3804        }).join('');
3805      }
3806    }
3807
3808    // Sparkline
3809    if (Array.isArray(gpu.history) && gpu.history.length > 0)
{
3810      renderGpuSpark(gpu.history, idx);
3811    } else {
```

```
3812      if (!_gpuSparkHistories[idx]) _gpuSparkHistories[idx] =
[];
3813      _gpuSparkHistories[idx].push({ load: pct, temp: coreTemp
, power: powerW, time: new Date() });
3814      if (_gpuSparkHistories[idx].length > 60) _gpuSparkHistor
ies[idx].shift();
3815      renderGpuSpark(_gpuSparkHistories[idx], idx);
3816    }
3817  }
3818
3819  async function fetchGpuLoadFromApi() {
3820    try {
3821      const res = await fetch('/api/v1/panel/gpu-load');
3822      if (!res.ok) return;
3823      const data = await res.json();
3824      lastGpuData = data;
3825
3826      const devices = Array.isArray(data.devices) && data.devi
ces.length > 0 ? data.devices : [data];
3827      lastGpuDevices = devices;
3828
3829      const container = document.getElementById('sys-gpu-panel
s-container');
3830      if (!container) return;
3831
3832      const currentCards = container.querySelectorAll('.sys-ca
rd-gpu');
3833      if (currentCards.length !== devices.length) {
3834        container.innerHTML = devices.map((gpu, idx) => render
GpuCardHtml(gpu, idx, devices.length)).join('');
3835        if (cachedAvailableTimeRanges && Array.isArray(cachedA
vailableTimeRanges.intervals)) {
3836          populateIntervalSelectors(cachedAvailableTimeRanges.
intervals);
3837        }
3838      }
3839
3840      devices.forEach((gpu, idx) => {
```

```
3841          updateSingleGpuCard(gpu, idx, devices.length);
3842        });
3843      } catch (e) {
3844        console.warn('[SystemInspectorTab] Ошибка получения загр
     узки GPU из API:', e);
3845      }
3846    }
3847
3848    function updateTelemetryDashboard(snap) {
3849      if (!snap) return;
3850
3851      // CPU
3852      if (snap.cpu) {
3853        const cpuVal = document.getElementById('sys-metric-cpu-v
     al');
3854        const cpuModel = document.getElementById('sys-metric-cpu
     -model');
3855        const cpuFill = document.getElementById('sys-metric-cpu-
     fill');
3856        const cpuSub = document.getElementById('sys-metric-cpu-s
     ub');
3857        const cpuGauge = document.getElementById('sys-metric-cpu
     -gauge');
3858
3859        const pct = Number(snap.cpu.total_percent || 0);
3860        if (cpuVal) cpuVal.innerText = `${pct.toFixed(1)}%`;
3861        if (cpuModel && snap.cpu.model) cpuModel.innerText = sna
     p.cpu.model;
3862        if (cpuFill) cpuFill.style.width = `${Math.min(100, pct)
     }%`;
3863        if (cpuSub) cpuSub.innerText = `${snap.cpu.physical_core
     s || '--'} Физических / ${snap.cpu.logical_cores || '--'} Пото
     ков`;
3864        if (cpuGauge) cpuGauge.innerHTML = createGaugeSvg(pct, 1
     30, 74);
3865      }
3866
3867      // RAM
```

```
3868      if (snap.memory) {
3869        const ramVal = document.getElementById('sys-metric-ram-v
      al');
3870        const ramFill = document.getElementById('sys-metric-ram-
      fill');
3871        const ramSub = document.getElementById('sys-metric-ram-s
      ub');
3872        const ramModel = document.getElementById('sys-metric-ram
      -model');
3873
3874        const usedGb = Number(snap.memory.used_gb || 0);
3875        const totalGb = Number(snap.memory.total_gb || 0);
3876        const pct = Number(snap.memory.percent || 0);
3877        const swapPct = Number(snap.memory.swap_percent || 0);
3878
3879        if (ramModel && totalGb > 0) ramModel.innerText = `${Mat
      h.round(totalGb)} GB RAM`;
3880        if (ramVal) ramVal.innerText = `${usedGb.toFixed(1)} / $
      {totalGb.toFixed(1)} GB`;
3881        if (ramFill) ramFill.style.width = `${pct}%`;
3882        if (ramSub) ramSub.innerText = `${pct}% занято (${Number
      (snap.memory.available_gb || 0).toFixed(1)} GB свободно)`;
3883
3884        const memioPct = document.getElementById('sys-memio-pct'
      );
3885        if (memioPct) memioPct.innerText = `${pct.toFixed(1)}%`;
3886        const memioUsed = document.getElementById('sys-memio-use
      d');
3887        if (memioUsed) memioUsed.innerText = `${usedGb.toFixed(1
      )} / ${totalGb.toFixed(1)} GB (свободно ${Number(snap.memory.a
      vailable_gb || (totalGb - usedGb)).toFixed(1)} GB)`;
3888        const memioSwap = document.getElementById('sys-memio-swa
      p');
3889        if (memioSwap) memioSwap.innerText = `Файл подкачки: ${s
      wapPct.toFixed(1)}%`;
3890        const memioTs = document.getElementById('sys-memio-ts');
3891        if (memioTs) memioTs.innerText = new Date().toLocaleTime
      String();
```

```
3892
3893        const ramGauge = document.getElementById('sys-metric-ram
    -gauge');
3894        if (ramGauge) ramGauge.innerHTML = createGaugeSvg(pct, 1
    30, 74);
3895        const ramSlider = document.getElementById('sys-metric-ra
    m-slider');
3896        if (ramSlider) ramSlider.innerHTML = createPercentSlider
    Html(pct, true, `${usedGb.toFixed(1)} / ${totalGb.toFixed(1)}
    GB`);
3897        const ramSubGauge = document.getElementById('sys-ram-sub
    -gauge');
3898        if (ramSubGauge) ramSubGauge.innerHTML = createGaugeSvg(
    pct, 100, 56);
3899        const ramSubSlider = document.getElementById('sys-ram-su
    b-slider');
3900        if (ramSubSlider) ramSubSlider.innerHTML = createPercent
    SliderHtml(pct, false, `${usedGb.toFixed(1)} / ${totalGb.toFix
    ed(1)} GB`);
3901        const ramSubBadge = document.getElementById('sys-ram-sub
    -badge');
3902        if (ramSubBadge) ramSubBadge.textContent = `${pct.toFixe
    d(0)}%`;
3903
3904        // Экспресс-метрики KPI в центре баннера RAM
3905        const ramKpiUsed = document.getElementById('sys-ram-kpi-
    used');
3906        const ramKpiFree = document.getElementById('sys-ram-kpi-
    free');
3907        const ramKpiSwap = document.getElementById('sys-ram-kpi-
    swap');
3908        if (ramKpiUsed) ramKpiUsed.textContent = `${usedGb.toFix
    ed(1)} GB`;
3909        if (ramKpiFree) ramKpiFree.textContent = `${Number(snap.
    memory.available_gb || (totalGb - usedGb)).toFixed(1)} GB`;
3910        if (ramKpiSwap) ramKpiSwap.textContent = `${swapPct.toFi
    xed(1)}%`;
3911
```

```
3912        // Swap
3913        const swapSubBadge = document.getElementById('sys-swap-s
    ub-badge');
3914        if (swapSubBadge) swapSubBadge.textContent = `${swapPct.
    toFixed(1)}%`;
3915        const swapSubGauge = document.getElementById('sys-swap-s
    ub-gauge');
3916        if (swapSubGauge) swapSubGauge.innerHTML = createGaugeSv
    g(swapPct, 100, 56);
3917        const swapSubSlider = document.getElementById('sys-swap-
    sub-slider');
3918        if (swapSubSlider) swapSubSlider.innerHTML = createPerce
    ntSliderHtml(swapPct, false, `Подкачка ${swapPct.toFixed(1)}%`
    );
3919      }
3920
3921      // GPU: данные для независимых карточек берутся строго из
    fetchGpuLoadFromApi()
3922      if (!lastGpuData) {
3923        fetchGpuLoadFromApi();
3924      }
3925
3926      // Disk I/O
3927      if (snap.disk_io) {
3928        const diskVal = document.getElementById('sys-metric-disk
    -val');
3929        const diskSub = document.getElementById('sys-metric-disk
    -sub');
3930
3931        const rBps = Number(snap.disk_io.read_bytes_per_sec || 0
    );
3932        const wBps = Number(snap.disk_io.write_bytes_per_sec ||
    0);
3933        const totalMb = ((rBps + wBps) / (1024 * 1024)).toFixed(
    2);
3934        const rKb = (rBps / 1024).toFixed(0);
3935        const wKb = (wBps / 1024).toFixed(0);
3936
```

```
3937        if (diskVal) diskVal.innerText = `${totalMb} MB/s`;
3938        if (diskSub) diskSub.innerText = `Чтение: ${rKb} KB/s | Запись: ${wKb} KB/s`;
3939        const ramKpiIo = document.getElementById('sys-ram-kpi-io');
3940        if (ramKpiIo) ramKpiIo.textContent = `${totalMb} MB/s`;
3941
3942        const readFormatted = formatBytesPerSec(rBps);
3943        const writeFormatted = formatBytesPerSec(wBps);
3944
3945        const memioRead = document.getElementById('sys-memio-read');
3946        if (memioRead) memioRead.innerText = readFormatted;
3947        const readSubBadge = document.getElementById('sys-read-sub-badge');
3948        if (readSubBadge) readSubBadge.innerText = readFormatted;
3949
3950        const memioWrite = document.getElementById('sys-memio-write');
3951        if (memioWrite) memioWrite.innerText = writeFormatted;
3952        const writeSubBadge = document.getElementById('sys-write-sub-badge');
3953        if (writeSubBadge) writeSubBadge.innerText = writeFormatted;
3954
3955        const maxIoScale = 50 * 1024 * 1024;
3956        const readPct = Math.min(100, (rBps / maxIoScale) * 100);
3957        const writePct = Math.min(100, (wBps / maxIoScale) * 100);
3958
3959        const readSubGauge = document.getElementById('sys-read-sub-gauge');
3960        if (readSubGauge) readSubGauge.innerHTML = createGaugeSvg(readPct, 100, 56);
3961        const readSubBar = document.getElementById('sys-read-sub-bar');
```

```
3962        const readSubThumb = document.getElementById('sys-read-sub-thumb');
3963        if (readSubBar) readSubBar.style.width = `${readPct.toFixed(1)}%`;
3964        if (readSubThumb) readSubThumb.style.left = `${readPct.toFixed(1)}%`;
3965
3966        const writeSubGauge = document.getElementById('sys-write-sub-gauge');
3967        if (writeSubGauge) writeSubGauge.innerHTML = createGaugeSvg(writePct, 100, 56);
3968        const writeSubBar = document.getElementById('sys-write-sub-bar');
3969        const writeSubThumb = document.getElementById('sys-write-sub-thumb');
3970        if (writeSubBar) writeSubBar.style.width = `${writePct.toFixed(1)}%`;
3971        if (writeSubThumb) writeSubThumb.style.left = `${writePct.toFixed(1)}%`;
3972      }
3973
3974      // Physical Disks SMART Health (единый API /api/v1/system/diagnostics/storage-battery)
3975      if (Array.isArray(snap.physical_disks) && snap.physical_disks.length > 0) {
3976        renderPhysicalDisksStorage(snap.physical_disks);
3977      } else if (!_lastDisksWearData) {
3978        fetchStorageHealthFromApi();
3979      }
3980
3981      // Battery & Power
3982      const powerCont = document.getElementById('sys-power-info-container');
3983      const powerBadge = document.getElementById('sys-power-status-badge');
3984      if (powerCont && snap.battery) {
3985        if (snap.battery.has_battery && snap.battery.percent !== null) {
```

```
        if (powerBadge) {
          powerBadge.textContent = `${snap.battery.percent}% ${snap.battery.power_plugged ? '🔌 Зарядка' : '🔋 Батарея'}`;
          powerBadge.className = snap.battery.power_plugged ? 'badge bg-success-subtle text-success border border-success' : 'badge bg-warning-subtle text-warning border border-warning';
        }
        const minsLeft = snap.battery.secs_left ? Math.round(snap.battery.secs_left / 60) : null;
        powerCont.innerHTML = `
          <div>Статус: ${snap.battery.power_plugged ? 'Подключено к сети' : 'Работа от батареи'}</div>
          <div>${minsLeft ? `Осталось ~${minsLeft} мин.` : 'Профиль: ' + (snap.battery.power_profile || 'Balanced')}</div>
        `;
      } else {
        if (powerBadge) {
          powerBadge.textContent = 'AC Mains';
          powerBadge.className = 'badge bg-secondary';
        }
        powerCont.innerHTML = `
          <div>Питание: Стационарная электросеть 220V</div>
          <div class="text-muted">Профиль: ${escapeHtml(snap.battery.power_profile || 'High Performance')}</div>
        `;
      }
    }

    // RAM SPD Modules
    const ramSticksCont = document.getElementById('sys-ram-sticks-container');
    const ramSticksBadge = document.getElementById('sys-ram-sticks-badge');
    if (ramSticksCont && Array.isArray(snap.ram_sticks) && snap.ram_sticks.length > 0) {
      if (ramSticksBadge) {
        ramSticksBadge.textContent = `${snap.ram_sticks.length} модулей`;
```

```
4013        }
4014        ramSticksCont.innerHTML = snap.ram_sticks.slice(0, 2).ma
    p(m => `
4015          <div class="d-flex justify-content-between align-items
    -center py-0.5">
4016            <span>${escapeHtml(m.bank_label)}: ${escapeHtml(m.ma
    nufacturer)}</span>
4017            <span class="font-monospace text-primary">${m.capaci
    ty_gb} GB @ ${m.speed_mhz} MT/s</span>
4018          </div>
4019        `).join('');
4020      } else if (ramSticksCont) {
4021        ramSticksCont.innerHTML = `<div>RAM: ${snap.memory ? sna
    p.memory.total_gb : '--'} GB физической памяти</div>`;
4022      }
4023
4024      // Reliability & Open Ports
4025      const alertsCont = document.getElementById('sys-alerts-por
    ts-container');
4026      const alertsBadge = document.getElementById('sys-alerts-ba
    dge');
4027      if (alertsCont) {
4028        const portCount = Array.isArray(snap.listening_ports) ?
    snap.listening_ports.length : 0;
4029        const isReboot = snap.alerts && snap.alerts.reboot_pendi
    ng;
4030        if (alertsBadge) {
4031          alertsBadge.textContent = isReboot ? 'Перезагрузка' :
    'Стабильно';
4032          alertsBadge.className = isReboot ? 'badge bg-warning t
    ext-dark' : 'badge bg-info-subtle text-info border border-info
    ';
4033        }
4034        alertsCont.innerHTML = `
4035          <div>Сетевые сокеты: <span class="fw-bold" style="colo
    r: var(--text-color);">${portCount} портов LISTEN</span></div>
4036          <div class="text-truncate" title="${escapeHtml(snap.al
    erts?.latest_alert || '')}">${escapeHtml(snap.alerts?.latest_a
```

```
lert || 'Система стабильна')}</div>
      `;
    }
  }

  async function fetchNetworkActivity() {
    try {
      const res = await fetch('/api/v1/system/network-activity
?limit=100');
      if (res.ok) {
        const data = await res.json();
        renderNetworkActivityTable(data);
      }
    } catch (e) {
      console.warn('[SystemInspectorTab] Failed to fetch netwo
rk activity:', e);
    }
  }

  function renderNetworkActivityTable(activities) {
    if (activities) {
      cachedNetworkActivities = activities;
    }
    if (isNetPaused) return;

    const tbody = document.getElementById('sys-net-tbody');
    if (!tbody) return;

    const searchInput = document.getElementById('sys-net-searc
h');
    const filterText = (searchInput?.value || '').toLowerCase(
).trim();
    const countBadge = document.getElementById('sys-netact-cou
nt-badge');
    const connsBadge = document.getElementById('sys-netact-con
ns-badge');

    const rawList = Array.isArray(cachedNetworkActivities) ? c
```

```
achedNetworkActivities : [];

    // Filter by category and search query
    let filtered = rawList.filter(item => {
      if (currentNetFilter === 'internet' && !item.is_internet) return false;
      if (currentNetFilter === 'listen' && item.status !== 'LISTEN') return false;

      if (filterText) {
        return (
          String(item.pid).includes(filterText) ||
          (item.name || '').toLowerCase().includes(filterText) ||
          (item.user || '').toLowerCase().includes(filterText) ||
          (item.remote_address || '').toLowerCase().includes(filterText) ||
          (item.local_address || '').toLowerCase().includes(filterText) ||
          (item.service_type || '').toLowerCase().includes(filterText) ||
          (item.sent_summary || '').toLowerCase().includes(filterText) ||
          (item.recv_summary || '').toLowerCase().includes(filterText)
        );
      }
      return true;
    });

    const uniqueProcs = new Set(filtered.map(i => i.pid)).size;
    if (countBadge) countBadge.textContent = `${uniqueProcs} программ`;
    if (connsBadge) connsBadge.textContent = `${filtered.length} сокетов`;

```

```
4093      if (filtered.length === 0) {
4094        tbody.innerHTML = '<tr><td colspan="6" class="text-cente
      r py-4 text-muted small">Нет активных сетевых соединений по вы
      бранному фильтру</td></tr>';
4095        return;
4096      }
4097
4098      tbody.innerHTML = filtered.map((item, idx) => {
4099        const isListen = item.status === 'LISTEN';
4100        const statusBadgeClass = isListen
4101          ? 'badge bg-secondary-subtle text-light border border-
      secondary'
4102          : item.status === 'ESTABLISHED'
4103          ? 'badge bg-success-subtle text-success border border-
      success'
4104          : 'badge bg-warning-subtle text-warning border border-
      warning';
4105
4106        const protoBadge = item.protocol === 'UDP'
4107          ? '<span class="badge bg-primary text-white" style="fo
      nt-size: 0.65rem;">UDP</span>'
4108          : '<span class="badge bg-dark border border-secondary
      text-info" style="font-size: 0.65rem;">TCP</span>';
4109
4110        const isExtBadge = item.is_internet
4111          ? '<span class="badge bg-primary-subtle text-primary b
      order border-primary px-1" style="font-size: 0.62rem;" title="
      Внешний сервер в сети Интернет">WAN</span>'
4112          : '<span class="badge bg-secondary px-1" style="font-s
      ize: 0.62rem;" title="Локальный сокет Loopback">LAN</span>';
4113
4114        const formatNetKb = (kb) => {
4115          if (!kb || kb <= 0) return '0 KB';
4116          if (kb >= 1024 * 1024) return (kb / (1024 * 1024)).toF
      ixed(2) + ' GB';
4117          if (kb >= 1024) return (kb / 1024).toFixed(1) + ' MB';
4118          return kb.toFixed(1) + ' KB';
4119        };
```

```
4120
4121       const formatNetRate = (rate) => {
4122         if (!rate || rate <= 0.05) return '';
4123         if (rate >= 1024) return (rate / 1024).toFixed(1) + '
MB/s';
4124         return rate.toFixed(1) + ' KB/s';
4125       };
4126
4127       const deltaSentStr = (item.delta_sent_kb && item.delta_s
ent_kb > 0)
4128         ? `▲ +${formatNetKb(item.delta_sent_kb)}${item.sent_ra
te_kbs > 0.05 ? ' (' + formatNetRate(item.sent_rate_kbs) + ')'
 : ''}`
4129         : '▲ 0 KB';
4130       const deltaRecvStr = (item.delta_recv_kb && item.delta_r
ecv_kb > 0)
4131         ? `▼ +${formatNetKb(item.delta_recv_kb)}${item.recv_ra
te_kbs > 0.05 ? ' (' + formatNetRate(item.recv_rate_kbs) + ')'
 : ''}`
4132         : '▼ 0 KB';
4133
4134       const totalSentStr = formatNetKb(item.sent_kb || 0);
4135       const totalRecvStr = formatNetKb(item.recv_kb || 0);
4136
4137       return `
4138         <tr class="sys-net-row" data-idx="${idx}" style="curso
r: pointer;" title="Нажмите для детальной диагностики сетевого
 соединения">
4139           <td>
4140             <div class="fw-bold text-truncate" style="max-widt
h: 165px; color: var(--text-color);" title="${escapeHtml(item.
name)}">${escapeHtml(item.name)}</div>
4141             <div class="small text-muted" style="font-size: 0.
70rem;">PID: <span class="font-monospace text-info">${item.pid
}</span> ${item.user ? '• ' + escapeHtml(item.user) : ''}</div
>
4142           </td>
4143           <td>
```

```
4144            <div class="d-flex align-items-center gap-1">
4145              ${isExtBadge}
4146              <span class="font-monospace fw-semibold text-tru
    ncate" style="max-width: 195px; color: var(--text-color);" tit
    le="${escapeHtml(item.remote_address)}">
4147                ${escapeHtml(item.remote_address !== '-' ? ite
    m.remote_address : item.local_address)}
4148              </span>
4149            </div>
4150            <div class="small text-muted font-monospace" style
    ="font-size: 0.68rem;">Local: ${escapeHtml(item.local_address)
    }</div>
4151          </td>
4152          <td>
4153            <div class="d-flex align-items-center gap-1 mb-0.5
    ">
4154              ${protoBadge}
4155              <span class="fw-semibold text-truncate" style="m
    ax-width: 110px; font-size: 0.74rem; color: var(--nav-active);
    " title="${escapeHtml(item.service_type)}">${escapeHtml(item.s
    ervice_type)}</span>
4156            </div>
4157          </td>
4158          <td style="text-align: center;">
4159            <span class="${statusBadgeClass}" style="font-size
    : 0.68rem;">${escapeHtml(item.status)}</span>
4160          </td>
4161          <td>
4162            <div class="d-flex align-items-center justify-cont
    ent-between gap-1 mb-1">
4163              <span class="badge bg-warning-subtle text-warnin
    g border border-warning px-1.5 py-0.5" style="font-size: 0.68r
    em;" title="Отправлено за измеряемый период">
4164                ${deltaSentStr}
4165              </span>
4166              <span class="text-muted font-monospace" style="f
    ont-size: 0.66rem;" title="Всего отправлено/записано">
4167                Σ ${totalSentStr}
```

```
4168              </span>
4169            </div>
4170            <div class="d-flex align-items-start gap-1">
4171              <i class="bi bi-arrow-up-right text-warning mt-0
.5" style="font-size: 0.70rem;"></i>
4172              <div class="text-truncate" style="max-width: 250
px; font-size: 0.72rem; color: var(--text-muted);" title="${es
capeHtml(item.sent_summary)}">
4173                ${escapeHtml(item.sent_summary)}
4174              </div>
4175            </div>
4176          </td>
4177          <td>
4178            <div class="d-flex align-items-center justify-cont
ent-between gap-1 mb-1">
4179              <span class="badge bg-success-subtle text-succes
s border border-success px-1.5 py-0.5" style="font-size: 0.68r
em;" title="Скачано/получено за измеряемый период">
4180                ${deltaRecvStr}
4181              </span>
4182              <span class="text-muted font-monospace" style="f
ont-size: 0.66rem;" title="Всего скачано/прочитано">
4183                Σ ${totalRecvStr}
4184              </span>
4185            </div>
4186            <div class="d-flex align-items-start gap-1">
4187              <i class="bi bi-arrow-down-left text-success mt-
0.5" style="font-size: 0.70rem;"></i>
4188              <div class="text-truncate" style="max-width: 250
px; font-size: 0.72rem; color: var(--text-muted);" title="${es
capeHtml(item.recv_summary)}">
4189                ${escapeHtml(item.recv_summary)}
4190              </div>
4191            </div>
4192          </td>
4193        </tr>
4194      `;
4195    }).join('');
```

```
4196
4197    tbody.querySelectorAll('.sys-net-row').forEach(row => {
4198      row.onclick = () => {
4199        const idx = parseInt(row.getAttribute('data-idx'), 10)
;
4200        const item = filtered[idx];
4201        if (!item) return;
4202
4203        const formatNetKb = (kb) => {
4204          if (!kb || kb <= 0) return '0 KB';
4205          if (kb >= 1024 * 1024) return (kb / (1024 * 1024)).t
oFixed(2) + ' GB';
4206          if (kb >= 1024) return (kb / 1024).toFixed(1) + ' MB
';
4207          return kb.toFixed(1) + ' KB';
4208        };
4209
4210        const formatNetRate = (rate) => {
4211          if (!rate || rate <= 0.05) return '';
4212          if (rate >= 1024) return (rate / 1024).toFixed(1) +
' MB/s';
4213          return rate.toFixed(1) + ' KB/s';
4214        };
4215
4216        if (window.AITableModal) {
4217          window.AITableModal.show({
4218            icon: '□ ',
4219            title: `${item.name} (${item.service_type})`,
4220            subtitle: `PID: ${item.pid} | ${item.remote_addres
s}`,
4221            panelId: 'panel-system-inspector-stabilnost-i-set'
,
4222            tableType: 'network',
4223            badges: [
4224              { text: `PID ${item.pid}`, class: 'badge bg-info
 text-dark' },
4225              { text: item.protocol, class: 'badge bg-primary'
 },
```

```
4226              { text: item.status, class: 'badge bg-success' }
,
4227              { text: item.is_internet ? 'Интернет (WAN)' : 'Л
окально (LAN)', class: item.is_internet ? 'badge bg-warning te
xt-dark' : 'badge bg-secondary' }
4228            ],
4229            metadata: [
4230              { label: 'Программа / Процесс', value: item.name
 },
4231              { label: 'Process ID (PID)', value: String(item.
pid) },
4232              { label: 'Пользователь системы', value: item.use
r || 'SYSTEM' },
4233              { label: 'Удаленный адрес (Remote)', value: item
.remote_address },
4234              { label: 'Локальный сокет (Local)', value: item.
local_address },
4235              { label: 'Протокол / Служба', value: `${item.pro
tocol} • ${item.service_type}` },
4236              { label: 'Статус соединения', value: item.status
 },
4237              { label: 'Скачано за период (Прием)', value: `${
formatNetKb(item.delta_recv_kb || 0)} ${item.recv_rate_kbs > 0
.05 ? '(' + formatNetRate(item.recv_rate_kbs) + ')' : ''}` },
4238              { label: 'Всего скачано / получено', value: form
atNetKb(item.recv_kb || 0) },
4239              { label: 'Отправлено за период', value: `${forma
tNetKb(item.delta_sent_kb || 0)} ${item.sent_rate_kbs > 0.05 ?
 '(' + formatNetRate(item.sent_rate_kbs) + ')' : ''}` },
4240              { label: 'Всего отправлено', value: formatNetKb(
item.sent_kb || 0) },
4241              { label: 'Что шлет (Отправка)', value: item.sent
_summary, fullWidth: true },
4242              { label: 'Что принимает (Прием)', value: item.re
cv_summary, fullWidth: true }
4243            ],
4244            rawTitle: 'Сетевой дамп подключения',
4245            rawContent: JSON.stringify(item, null, 2),
```

```
            requestData: item
          });
        }
      };
    });
  }

  async function runAiDiagnostics() {
    const summaryEl = document.getElementById('sys-ai-summary-text');
    const badgeEl = document.getElementById('sys-ai-health-badge');
    const anomaliesEl = document.getElementById('sys-ai-anomalies-container');
    const engineTagEl = document.getElementById('sys-ai-model-tag');

    if (summaryEl) summaryEl.innerText = 'Запуск глубокого AI-аудита системы и оборудования...';

    try {
      const res = await fetch('/api/v1/system/diagnose', { method: 'POST' });
      if (!res.ok) throw new Error('AI Diagnosis error');
      const report = await res.json();

      if (badgeEl) {
        badgeEl.innerText = `Health: ${report.health_score || 100}/100`;
        badgeEl.className = `badge ${report.health_score >= 80 ? 'bg-success' : report.health_score >= 60 ? 'bg-warning' : 'bg-danger'}`;
      }

      if (engineTagEl) {
        engineTagEl.innerText = report.ai_model_used || 'Heuristic Engine';
      }
```

```
4274
4275      if (anomaliesEl) {
4276        if (Array.isArray(report.anomalies) && report.anomalies.length > 0) {
4277          anomaliesEl.innerHTML = report.anomalies.map(a => `<span class="anomaly-tag">⚠ [${a.subsystem}] ${escapeHtml(a.title)}</span>`).join('');
4278        } else {
4279          anomaliesEl.innerHTML = '<span style="color: #4ade80; font-size: 0.75rem;"><i class="bi bi-check-circle me-1"></i> Аномалий в работе оборудования не обнаружено</span>';
4280        }
4281      }
4282
4283      if (summaryEl) {
4284        const recs = Array.isArray(report.recommendations) && report.recommendations.length > 0
4285          ? `\nРекомендации: ${report.recommendations.join(', ')}`
4286          : '';
4287        summaryEl.innerText = `${report.summary || 'Телеметрия в норме.'}${recs}`;
4288      }
4289    } catch (e) {
4290      console.error('[SystemInspectorTab] AI Diagnose error:', e);
4291      if (summaryEl) summaryEl.innerText = 'Ошибка выполнения AI-диагностики: ' + e.message;
4292    }
4293  }
4294
4295  let sysWsReconnectTimer = null;
4296
4297  function connectSystemWebSocket() {
4298    if (sysWsReconnectTimer) {
4299      clearTimeout(sysWsReconnectTimer);
4300      sysWsReconnectTimer = null;
4301    }
```

```
    if (sysWs) {
      try {
        sysWs.onclose = null;
        sysWs.onerror = null;
        sysWs.close();
      } catch {}
      sysWs = null;
    }

    if (window.isTabActive && !window.isTabActive('tab-hardwar
e-load-inspector') && !window.isTabActive('tab-system-load-ins
pector') && !window.isTabActive('tab-system-inspector')) {
      const statusBadge = document.getElementById('sys-conn-st
atus');
      if (statusBadge) {
        statusBadge.className = 'badge rounded-pill bg-seconda
ry text-light px-3 py-2';
        statusBadge.innerText = '○ Поток приостановлен';
      }
      return;
    }

    const proto = window.location.protocol === 'https:' ? 'wss
:' : 'ws:';
    const wsUrl = `${proto}//${window.location.host}/api/v1/sy
stem/stream`;

    try {
      sysWs = new WebSocket(wsUrl);
      const statusBadge = document.getElementById('sys-conn-st
atus');

      sysWs.onopen = () => {
        console.info('[SystemInspectorTab] WebSocket соединени
e с телеметрией установлено.');
        if (statusBadge) {
          statusBadge.className = 'badge rounded-pill bg-succe
ss-subtle text-success border border-success px-3 py-2';
```

```
4331          statusBadge.innerText = '● Телеметрия активна';
4332        }
4333      };
4334
4335      sysWs.onmessage = (evt) => {
4336        try {
4337          const snap = JSON.parse(evt.data);
4338          latestTelemetrySnapshot = snap;
4339          updateTelemetryDashboard(snap);
4340        } catch (e) {
4341          console.error('[SystemInspectorTab] Ошибка разбора с
     ообщения телеметрии:', e);
4342        }
4343      };
4344
4345      const scheduleReconnect = (reason) => {
4346        if (sysWsReconnectTimer) return;
4347        if (window.isTabActive && !window.isTabActive('tab-har
     dware-load-inspector') && !window.isTabActive('tab-system-load
     -inspector') && !window.isTabActive('tab-system-inspector')) r
     eturn;
4348        console.warn(`[SystemInspectorTab] WebSocket отключен
     (${reason}). Автопереподключение через 3 сек...`);
4349        if (statusBadge) {
4350          statusBadge.className = 'badge rounded-pill bg-warni
     ng-subtle text-warning border border-warning px-3 py-2';
4351          statusBadge.innerText = '○ Переподключение...';
4352        }
4353        sysWsReconnectTimer = setTimeout(() => {
4354          sysWsReconnectTimer = null;
4355          if (!window.isTabActive || window.isTabActive('tab-h
     ardware-load-inspector') || window.isTabActive('tab-system-loa
     d-inspector') || window.isTabActive('tab-system-inspector')) {
4356            connectSystemWebSocket();
4357          }
4358        }, 3000);
4359      };
4360
```

```
      sysWs.onclose = (evt) => {
        scheduleReconnect(`код закрытия: ${evt.code}`);
      };

      sysWs.onerror = (err) => {
        console.warn('[SystemInspectorTab] Ошибка соединения WebSocket:', err);
        scheduleReconnect('ошибка связи');
      };
    } catch (err) {
      console.error('[SystemInspectorTab] Ошибка инициализации WebSocket:', err);
      sysWsReconnectTimer = setTimeout(() => {
        sysWsReconnectTimer = null;
        if (!window.isTabActive || window.isTabActive('tab-hardware-load-inspector') || window.isTabActive('tab-system-load-inspector') || window.isTabActive('tab-system-inspector')) {
          connectSystemWebSocket();
        }
      }, 5000);
    }
  }

  function disconnectSystemWebSocket() {
    if (sysWsReconnectTimer) {
      clearTimeout(sysWsReconnectTimer);
      sysWsReconnectTimer = null;
    }
    if (sysWs) {
      try {
        sysWs.onclose = null;
        sysWs.onerror = null;
        sysWs.close();
      } catch {}
      sysWs = null;
    }
    const statusBadge = document.getElementById('sys-conn-status');
```

```
4394    if (statusBadge) {
4395      statusBadge.className = 'badge rounded-pill bg-secondary
 text-light px-3 py-2';
4396      statusBadge.innerText = '○ Поток приостановлен';
4397    }
4398  }
4399
4400  function bindTabEvents() {
4401    const btnRefreshSensors = document.getElementById('btn-sys
-refresh-sensors');
4402    if (btnRefreshSensors) {
4403      btnRefreshSensors.onclick = () => fetchLhmSensors();
4404    }
4405
4406    const btnRefreshLhm = document.getElementById('btn-sys-ref
resh-lhm');
4407    if (btnRefreshLhm) {
4408      btnRefreshLhm.onclick = () => fetchLhmSensors();
4409    }
4410
4411    const btnLaunchLhm = document.getElementById('btn-sys-laun
ch-lhm');
4412    if (btnLaunchLhm) {
4413      btnLaunchLhm.onclick = async () => {
4414        btnLaunchLhm.disabled = true;
4415        btnLaunchLhm.innerHTML = '<span class="spinner-border
spinner-border-sm me-1"></span> Запуск...';
4416        try {
4417          const res = await fetch('/api/v1/lhm/launch', { meth
od: 'POST' });
4418          if (res.ok) {
4419            setTimeout(fetchLhmSensors, 2500);
4420          }
4421        } catch (e) {
4422          console.error('[SystemInspectorTab] Failed to launch
 LHM:', e);
4423        } finally {
4424          btnLaunchLhm.disabled = false;
```

```
4425        }
4426      };
4427    }
4428
4429    const searchSensors = document.getElementById('sys-lhm-sea
rch');
4430    if (searchSensors) {
4431      searchSensors.oninput = () => renderSensors();
4432    }
4433
4434    // Category pills
4435    document.querySelectorAll('.sys-lhm-cat-btn').forEach(btn
=> {
4436      btn.onclick = () => {
4437        document.querySelectorAll('.sys-lhm-cat-btn').forEach(
b => {
4438          b.classList.remove('active', 'btn-outline-primary');
4439          b.classList.add('btn-outline-secondary');
4440        });
4441        btn.classList.remove('btn-outline-secondary');
4442        btn.classList.add('active', 'btn-outline-primary');
4443        currentSensorCategory = btn.getAttribute('data-cat') |
| 'all';
4444        renderSensors();
4445      };
4446    });
4447
4448    // Analyze Button (Sensors)
4449    const btnAnalyze = document.getElementById('btn-sys-sensor
-analyze');
4450    if (btnAnalyze) {
4451      btnAnalyze.onclick = () => analyzeSensorsAndReport();
4452    }
4453
4454    // Modal Re-analyze Button
4455    const btnReAnalyze = document.getElementById('btn-re-analy
ze-sensors');
4456    if (btnReAnalyze) {
```

```
      btnReAnalyze.onclick = async () => {
        btnReAnalyze.disabled = true;
        const icon = btnReAnalyze.querySelector('i');
        if (icon) icon.classList.add('spin-animation');
        await fetchLhmSensors();
        await analyzeSensorsAndReport();
        if (icon) icon.classList.remove('spin-animation');
        btnReAnalyze.disabled = false;
      };
    }

    // Modal Copy Report Button
    const btnCopyReport = document.getElementById('btn-copy-se
nsor-report');
    if (btnCopyReport) {
      btnCopyReport.onclick = async () => {
        if (!lastAnalysisReportText) return;
        try {
          await navigator.clipboard.writeText(lastAnalysisRepo
rtText);
          const origHtml = btnCopyReport.innerHTML;
          btnCopyReport.innerHTML = '<i class="bi bi-check2 te
xt-success me-1"></i> Скопировано!';
          setTimeout(() => {
            btnCopyReport.innerHTML = origHtml;
          }, 2000);
        } catch (e) {
          console.warn('[SystemInspectorTab] Clipboard copy fa
iled:', e);
        }
      };
    }

    const btnAudit = document.getElementById('btn-sys-run-audi
t');
    if (btnAudit) {
      btnAudit.onclick = () => runAiDiagnostics();
    }
```

```
4490
4491      const btnCompact = document.getElementById('btn-sys-toggle
-compact');
4492      const sysContainer = document.querySelector('.sys-containe
r');
4493      const compactText = document.getElementById('sys-compact-b
tn-text');
4494
4495      const updateCompactUI = (isCompact) => {
4496        if (!sysContainer) return;
4497        if (isCompact) {
4498          sysContainer.classList.add('sys-compact-mode');
4499          if (btnCompact) {
4500            btnCompact.classList.add('collapsed');
4501            btnCompact.title = 'Развернуть все панели';
4502          }
4503        } else {
4504          sysContainer.classList.remove('sys-compact-mode');
4505          if (btnCompact) {
4506            btnCompact.classList.remove('collapsed');
4507            btnCompact.title = 'Свернуть все панели';
4508          }
4509        }
4510      };
4511
4512      if (btnCompact && sysContainer) {
4513        const savedCompact = localStorage.getItem('sys_inspector
_compact_mode');
4514        const isCompact = savedCompact === null ? true : savedCo
mpact === 'true';
4515        updateCompactUI(isCompact);
4516        btnCompact.onclick = () => {
4517          const nextState = !sysContainer.classList.contains('sy
s-compact-mode');
4518          updateCompactUI(nextState);
4519          try { localStorage.setItem('sys_inspector_compact_mode
', String(nextState)); } catch (e) {}
4520        };
```

```
4521      }
4522
4523      const btnConfig = document.getElementById('btn-sys-config'
);
4524      if (btnConfig) {
4525        btnConfig.onclick = () => {
4526          if (window.appConfigEditor) {
4527            window.appConfigEditor.open('system_inspector', 'Нас
тройки телеметрии и инспектора');
4528          } else {
4529            console.info('[SystemInspectorTab] Config editor not
 available');
4530          }
4531        };
4532      }
4533
4534      // Interval editor modal buttons
4535      const btnIntervalsTop = document.getElementById('btn-sys-i
ntervals');
4536      if (btnIntervalsTop) {
4537        btnIntervalsTop.onclick = () => openSysIntervalsModal();
4538      }
4539
4540      const btnIntervalsLhm = document.getElementById('btn-sys-i
ntervals-lhm');
4541      if (btnIntervalsLhm) {
4542        btnIntervalsLhm.onclick = () => openSysIntervalsModal();
4543      }
4544
4545      const btnRefreshIntervals = document.getElementById('btn-m
odal-intervals-refresh');
4546      if (btnRefreshIntervals) {
4547        btnRefreshIntervals.onclick = () => loadSysIntervalsConf
ig(true);
4548      }
4549
4550      const btnSaveIntervals = document.getElementById('btn-moda
l-intervals-save');
```

```
4551    if (btnSaveIntervals) {
4552      btnSaveIntervals.onclick = () => saveSysIntervalsConfig(
);
4553    }
4554
4555    const selUiRefresh = document.getElementById('modal-ui-ref
resh-select');
4556    if (selUiRefresh) {
4557      selUiRefresh.onchange = (e) => {
4558        const sec = parseInt(e.target.value, 10) || 5;
4559        setupSysSensorInterval(sec);
4560        const badge = document.getElementById('modal-ui-refres
h-badge');
4561        if (badge) badge.textContent = `${sec} сек`;
4562      };
4563    }
4564
4565    // Изменения файлов в реальном времени controls
4566    const changeWatchDirBtn = document.getElementById('btn-sys
-change-watch-dir');
4567    if (changeWatchDirBtn) {
4568      changeWatchDirBtn.onclick = () => openWatchFoldersModal(
'folders');
4569    }
4570    const exclusionsBtn = document.getElementById('btn-sys-wat
ch-exclusions');
4571    if (exclusionsBtn) {
4572      exclusionsBtn.onclick = () => openWatchFoldersModal('exc
lusions');
4573    }
4574    const watchDirBadge = document.getElementById('sys-watch-d
ir-badge');
4575    if (watchDirBadge) {
4576      watchDirBadge.onclick = () => openWatchFoldersModal('fol
ders');
4577    }
4578    const liveHelpBtn = document.getElementById('btn-sys-live-
help');
```

```
    if (liveHelpBtn) {
      liveHelpBtn.onclick = showLiveWatcherHelpModal;
    }

    // Network Activity Table controls
    const searchNet = document.getElementById('sys-net-search'
);
    if (searchNet) {
      searchNet.oninput = () => renderNetworkActivityTable();
    }

    const btnPauseNet = document.getElementById('btn-sys-pause
-net');
    if (btnPauseNet) {
      btnPauseNet.onclick = () => {
        isNetPaused = !isNetPaused;
        btnPauseNet.innerText = isNetPaused ? 'Возобновить' :
'Пауза';
        btnPauseNet.className = isNetPaused ? 'btn btn-xs btn-
warning rounded-pill px-2.5 py-0.5' : 'btn btn-xs btn-outline-
secondary rounded-pill px-2.5 py-0.5';
        if (!isNetPaused) renderNetworkActivityTable();
      };
    }

    const btnRefreshNet = document.getElementById('btn-sys-ref
resh-net');
    if (btnRefreshNet) {
      btnRefreshNet.onclick = () => fetchNetworkActivity();
    }

    document.querySelectorAll('.sys-net-filter-btn').forEach(b
tn => {
      btn.onclick = () => {
        document.querySelectorAll('.sys-net-filter-btn').forEa
ch(b => {
          b.classList.remove('active', 'btn-outline-primary');
          b.classList.add('btn-outline-secondary');
```

```
        });
        btn.classList.remove('btn-outline-secondary');
        btn.classList.add('active', 'btn-outline-primary');
        currentNetFilter = btn.getAttribute('data-filter') ||
'internet';
        renderNetworkActivityTable();
      };
    });

    // Инициализация интерактивных ресайзеров таблиц
    initNetTableResizer();
    initWatcherTableResizer();
  }

  function initNetTableResizer() {
    const resizer = document.getElementById('sys-net-table-res
izer');
    const tableContainer = document.getElementById('sys-net-ta
ble-container');
    if (!resizer || !tableContainer) return;

    if (resizer._resizerInitialized) return;
    resizer._resizerInitialized = true;

    try {
      const savedHeight = localStorage.getItem('sys_inspector_
net_height');
      if (savedHeight) {
        const parsed = parseInt(savedHeight, 10);
        if (!isNaN(parsed) && parsed >= 140 && parsed <= 1200)
 {
          tableContainer.style.height = `${parsed}px`;
        }
      }
    } catch {}

    let startY = 0;
    let startHeight = 0;
```

```
    let isDragging = false;

    function onMouseMove(e) {
      if (!isDragging) return;
      const clientY = e.clientY ?? (e.touches && e.touches[0] ? e.touches[0].clientY : null);
      if (clientY === null) return;

      const deltaY = clientY - startY;
      let newHeight = startHeight + deltaY;
      if (newHeight < 140) newHeight = 140;
      if (newHeight > 1200) newHeight = 1200;

      tableContainer.style.height = `${newHeight}px`;
    }

    function onMouseUp() {
      if (!isDragging) return;
      isDragging = false;
      resizer.classList.remove('resizing');
      document.body.style.cursor = '';
      document.body.style.userSelect = '';

      window.removeEventListener('mousemove', onMouseMove);
      window.removeEventListener('mouseup', onMouseUp);
      window.removeEventListener('touchmove', onMouseMove);
      window.removeEventListener('touchend', onMouseUp);

      const currentH = parseInt(tableContainer.style.height, 10);
      if (!isNaN(currentH)) {
        try {
          localStorage.setItem('sys_inspector_net_height', String(currentH));
        } catch {}
      }
    }

```

```
    function onMouseDown(e) {
      isDragging = true;
      startY = e.clientY ?? (e.touches && e.touches[0] ? e.touches[0].clientY : 0);
      startHeight = tableContainer.getBoundingClientRect().height;
      resizer.classList.add('resizing');
      document.body.style.cursor = 'ns-resize';
      document.body.style.userSelect = 'none';

      window.addEventListener('mousemove', onMouseMove, { passive: false });
      window.addEventListener('mouseup', onMouseUp);
      window.addEventListener('touchmove', onMouseMove, { passive: false });
      window.addEventListener('touchend', onMouseUp);
      e.preventDefault();
    }

    resizer.addEventListener('mousedown', onMouseDown);
    resizer.addEventListener('touchstart', onMouseDown, { passive: false });

    resizer.addEventListener('dblclick', () => {
      tableContainer.style.height = '300px';
      try {
        localStorage.removeItem('sys_inspector_net_height');
      } catch {}
    });
  }

  function initWatcherTableResizer() {
    const resizer = document.getElementById('sys-watcher-table-resizer');
    const tableContainer = document.getElementById('sys-watcher-table-container');
    if (!resizer || !tableContainer) return;

```

```
    if (resizer._resizerInitialized) return;
    resizer._resizerInitialized = true;

    // Восстановление сохранённой пользователем высоты
    try {
      const savedHeight = localStorage.getItem('sys_inspector_watcher_height');
      if (savedHeight) {
        const parsed = parseInt(savedHeight, 10);
        if (!isNaN(parsed) && parsed >= 120 && parsed <= 1200) {
          tableContainer.style.height = `${parsed}px`;
        }
      }
    } catch {}

    let startY = 0;
    let startHeight = 0;
    let isDragging = false;

    function onMouseMove(e) {
      if (!isDragging) return;
      const clientY = e.clientY ?? (e.touches && e.touches[0] ? e.touches[0].clientY : null);
      if (clientY === null) return;

      const deltaY = clientY - startY;
      let newHeight = startHeight + deltaY;
      if (newHeight < 120) newHeight = 120;
      if (newHeight > 1200) newHeight = 1200;

      tableContainer.style.height = `${newHeight}px`;
    }

    function onMouseUp() {
      if (!isDragging) return;
      isDragging = false;
      resizer.classList.remove('resizing');
```

```
      document.body.style.cursor = '';
      document.body.style.userSelect = '';

      window.removeEventListener('mousemove', onMouseMove);
      window.removeEventListener('mouseup', onMouseUp);
      window.removeEventListener('touchmove', onMouseMove);
      window.removeEventListener('touchend', onMouseUp);

      const currentH = parseInt(tableContainer.style.height, 10);
      if (!isNaN(currentH)) {
        try {
          localStorage.setItem('sys_inspector_watcher_height', String(currentH));
        } catch {}
      }
    }

    function onMouseDown(e) {
      isDragging = true;
      startY = e.clientY ?? (e.touches && e.touches[0] ? e.touches[0].clientY : 0);
      startHeight = tableContainer.getBoundingClientRect().height;
      resizer.classList.add('resizing');
      document.body.style.cursor = 'ns-resize';
      document.body.style.userSelect = 'none';

      window.addEventListener('mousemove', onMouseMove, { passive: false });
      window.addEventListener('mouseup', onMouseUp);
      window.addEventListener('touchmove', onMouseMove, { passive: false });
      window.addEventListener('touchend', onMouseUp);
      e.preventDefault();
    }

    resizer.addEventListener('mousedown', onMouseDown);
```

```
    resizer.addEventListener('touchstart', onMouseDown, { passive: false });

    // Сброс по двойному клику
    resizer.addEventListener('dblclick', () => {
      tableContainer.style.height = '280px';
      try {
        localStorage.removeItem('sys_inspector_watcher_height');
      } catch {}
    });
  }

  // ============================================================
  // Telemetry & Polling Intervals Editor Controller (config.json)
  // ============================================================

  const INTERVAL_PRESETS = [
    { value: '1 second', label: '1 сек' },
    { value: '2 seconds', label: '2 сек' },
    { value: '3 seconds', label: '3 сек' },
    { value: '5 seconds', label: '5 сек' },
    { value: '10 seconds', label: '10 сек' },
    { value: '30 seconds', label: '30 сек' },
    { value: '1 minute', label: '1 мин' },
    { value: '5 minutes', label: '5 мин' },
    { value: '10 minutes', label: '10 мин' },
    { value: '30 minutes', label: '30 мин' },
    { value: '1 hour', label: '1 час' },
    { value: '6 hours', label: '6 часов' },
    { value: '24 hours', label: '24 часа' },
  ];

  const CORE_RESOURCE_LOGGERS = [
    'system_inspector',
```

```
    'hardware_monitor'
  ];

  const LOGGER_META = {
    system_inspector: {
      icon: '□ ',
      title: 'System Inspector',
      desc: 'Комплексный снимок: загрузка CPU, RAM, дисковый в
вод-вывод, процессы хоста',
    },
    hardware_monitor: {
      icon: '□ ',
      title: 'Hardware Monitor',
      desc: 'Аппаратные датчики WMI, GPU SMI (NVIDIA/AMD/Intel
) и физические шины',
    },
    windows_sysadmin: {
      icon: '□ ',
      title: 'Windows SysAdmin',
      desc: 'Системные службы Windows, пользователи, локальные
 группы и сетевые порты',
    },
    windows_defender: {
      icon: '□ ',
      title: 'Windows Defender',
      desc: 'Статус антивирусной защиты, сигнатуры, обнаруженн
ые угрозы и карантин',
    },
    windows_startup_auditor: {
      icon: '□ ',
      title: 'StartUp Auditor',
      desc: 'Автозагрузка программ, реестр Run/RunOnce, сервис
ы и планировщик задач',
    },
    windows_backup_manager: {
      icon: '□ ',
      title: 'Backup Manager',
      desc: 'Резервные копии баз данных SQLite, конфигураций и
```

```
 снапшотов системы',
    },
    website_monitor: {
      icon: '□ ',
      title: 'Website Monitor',
      desc: 'Проверка доступности веб-сервисов, задержка ответ
ов HTTP/HTTPS',
    },
    gcloud_monitor: {
      icon: '□ ',
      title: 'GCloud Monitor',
      desc: 'Телеметрия виртуальных машин, квоты и мониторинг
Google Cloud',
    },
    cloudflared_monitor: {
      icon: '□ ',
      title: 'Cloudflare Tunnel',
      desc: 'Статус защищенных туннелей cloudflared и исходящи
х подключений',
    },
    user_assistant: {
      icon: '□ ',
      title: 'User Assistant',
      desc: 'Фоновые периодические напоминания, календарь и ув
едомления',
    },
    registry_viewer: {
      icon: '□ ',
      title: 'Registry Viewer',
      desc: 'Мониторинг изменений системного реестра Windows',
    },
    software_audit: {
      icon: '□ ',
      title: 'Software Audit',
      desc: 'Аудит неиспользуемого и редко запускаемого ПО (Us
erAssist/Prefetch)',
    },
    helpdesk: {
```

```
      icon: '□ ',
      title: 'Helpdesk',
      desc: 'Мониторинг тикетов техподдержки и уведомлений',
    },
  };

  let _sysIntervalsConfigData = null;
  let _currentUiRefreshSeconds = 5;

  function showAlertInModal(msg, type = 'success') {
    const alertEl = document.getElementById('modal-intervals-a
lert');
    if (!alertEl) return;
    alertEl.className = `alert alert-${type} py-2 px-3 small m
b-3`;
    alertEl.innerHTML = msg;
    alertEl.classList.remove('d-none');
    if (type === 'success') {
      setTimeout(() => {
        alertEl.classList.add('d-none');
      }, 4000);
    }
  }

  async function loadSysIntervalsConfig(notify = false) {
    const btnRefresh = document.getElementById('btn-modal-inte
rvals-refresh');
    const icon = btnRefresh ? btnRefresh.querySelector('i') :
null;
    if (icon) icon.classList.add('spin-animation');

    try {
      let res = await fetch('/api/autolog/config');
      if (!res.ok) {
        res = await fetch('/sysautologging/config');
      }
      if (!res.ok) {
        throw new Error(`HTTP ${res.status}`);
```

```
4907      }
4908
4909      _sysIntervalsConfigData = await res.json();
4910      renderSysIntervalsForm(_sysIntervalsConfigData);
4911
4912      if (notify) {
4913        showAlertInModal('<i class="bi bi-check2 me-1"></i> Ко
нфигурация успешно загружена из файла', 'success');
4914      }
4915    } catch (e) {
4916      console.warn('[SystemInspectorTab] Failed to load interv
als config:', e);
4917      showAlertInModal(`<i class="bi bi-exclamation-triangle m
e-1"></i> Ошибка загрузки конфигурации: ${escapeHtml(e.message
)}`, 'danger');
4918    } finally {
4919      if (icon) icon.classList.remove('spin-animation');
4920    }
4921  }
4922
4923  function buildIntervalSelectHtml(loggerName, currentInterval
) {
4924    const val = currentInterval || '1 minute';
4925    const isCustom = !INTERVAL_PRESETS.some(p => p.value === v
al);
4926
4927    let html = `<select class="form-select form-select-sm sys-
int-select" data-logger="${loggerName}" style="min-width: 120p
x;">`;
4928    INTERVAL_PRESETS.forEach(p => {
4929      const sel = p.value === val ? 'selected' : '';
4930      html += `<option value="${p.value}" ${sel}>${p.label} ($
{p.value})</option>`;
4931    });
4932    if (isCustom) {
4933      html += `<option value="${escapeHtml(val)}" selected>${e
scapeHtml(val)} (пользовательский)</option>`;
4934    }
```

```
    html += `</select>`;
    return html;
  }

  function renderSysIntervalsForm(data) {
    if (!data) return;

    // 1. Config file badge
    const cfgBadge = document.getElementById('modal-intervals-cfg-file');
    if (cfgBadge && data.config_file) {
      cfgBadge.textContent = data.config_file;
    }

    // 2. Engine status & global switch
    const engineStatusBadge = document.getElementById('modal-intervals-engine-status');
    if (engineStatusBadge) {
      if (data.is_running) {
        engineStatusBadge.textContent = 'AutoLog Active';
        engineStatusBadge.className = 'badge bg-success';
      } else {
        engineStatusBadge.textContent = 'AutoLog Paused';
        engineStatusBadge.className = 'badge bg-danger-subtle text-danger border border-danger fw-bold';
      }
    }

    const globalEnable = document.getElementById('modal-intervals-global-enable');
    if (globalEnable) {
      globalEnable.checked = data.enable_autolog !== false;
    }

    // 3. Default fallback interval select
    const defSelect = document.getElementById('modal-intervals-default-select');
    if (defSelect && data.default_interval) {
```

```
4968      defSelect.value = data.default_interval;
4969    }
4970
4971    // 4. Populate Core resource loggers & Other loggers
4972    const coreContainer = document.getElementById('modal-core-
loggers-container');
4973    const otherContainer = document.getElementById('modal-othe
r-loggers-container');
4974    const otherCountEl = document.getElementById('modal-other-
loggers-count');
4975
4976    if (!coreContainer || !otherContainer) return;
4977
4978    const loggers = data.loggers || {};
4979    let coreHtml = '';
4980    let otherHtml = '';
4981    let otherCount = 0;
4982
4983    // Process core loggers
4984    CORE_RESOURCE_LOGGERS.forEach(name => {
4985      const cfg = loggers[name] || { interval: '5 seconds', en
abled: true };
4986      const meta = LOGGER_META[name] || { icon: '□ ', title: na
me, desc: 'Системный сбор метрик' };
4987      const isEnabled = cfg.enabled !== false;
4988      const intervalVal = cfg.interval || '5 seconds';
4989
4990      coreHtml += `
4991        <div class="p-2 rounded d-flex align-items-center just
ify-content-between flex-wrap gap-2" style="background: var(--
bg-color); border: 1px solid var(--border-color);">
4992          <div class="d-flex align-items-center gap-2" style="
max-width: 58%;">
4993            <span class="fs-5">${meta.icon}</span>
4994            <div>
4995              <div class="fw-bold small" style="color: var(--t
ext-color);">${meta.title}</div>
4996              <div class="small text-muted" style="font-size:
```

```
0.72rem; line-height: 1.2;">${meta.desc}</div>
            </div>
          </div>
          <div class="d-flex align-items-center gap-2 ms-auto"
>
            <div class="form-check form-switch m-0" title="Вкл
ючить/отключить сбор метрик для этого сервиса">
              <input class="form-check-input sys-int-enable" t
ype="checkbox" data-logger="${name}" ${isEnabled ? 'checked' :
 ''}>
            </div>
            ${buildIntervalSelectHtml(name, intervalVal)}
            <button class="btn btn-xs btn-outline-secondary ro
unded px-1.5 py-0.5 sys-int-custom-btn" data-logger="${name}"
title="Ввести произвольный интервал вручную">
              <i class="bi bi-pencil"></i>
            </button>
          </div>
        </div>
      `;
    });

    // Process remaining loggers
    const allLoggerNames = Object.keys(loggers).length > 0
      ? Object.keys(loggers)
      : Object.keys(LOGGER_META);

    allLoggerNames.forEach(name => {
      if (CORE_RESOURCE_LOGGERS.includes(name)) return;
      otherCount++;
      const cfg = loggers[name] || { interval: data.default_in
terval || '1 minute', enabled: true };
      const meta = LOGGER_META[name] || { icon: '□ ', title: n
ame, desc: 'Фоновый опрос службы' };
      const isEnabled = cfg.enabled !== false;
      const intervalVal = cfg.interval || '1 minute';

      otherHtml += `
```

```
        <div class="p-2 rounded d-flex align-items-center justify-content-between flex-wrap gap-2" style="background: var(--bg-color); border: 1px solid var(--border-color);">
          <div class="d-flex align-items-center gap-2" style="max-width: 58%;">
            <span class="fs-6">${meta.icon}</span>
            <div>
              <div class="fw-bold small" style="color: var(--text-color); font-size: 0.78rem;">${meta.title}</div>
              <div class="small text-muted" style="font-size: 0.7rem; line-height: 1.2;">${meta.desc}</div>
            </div>
          </div>
          <div class="d-flex align-items-center gap-2 ms-auto">
            <div class="form-check form-switch m-0" title="Включить/выключить">
              <input class="form-check-input sys-int-enable" type="checkbox" data-logger="${name}" ${isEnabled ? 'checked' : ''}>
            </div>
            ${buildIntervalSelectHtml(name, intervalVal)}
            <button class="btn btn-xs btn-outline-secondary rounded px-1.5 py-0.5 sys-int-custom-btn" data-logger="${name}" title="Ввести произвольный интервал вручную">
              <i class="bi bi-pencil"></i>
            </button>
          </div>
        </div>
      `;
    });

    coreContainer.innerHTML = coreHtml;
    otherContainer.innerHTML = otherHtml;
    if (otherCountEl) otherCountEl.textContent = String(otherCount);

    // Bind custom interval prompt buttons
```

```
    document.querySelectorAll('.sys-int-custom-btn').forEach(b
tn => {
      btn.onclick = () => {
        const lName = btn.getAttribute('data-logger');
        const sel = document.querySelector(`.sys-int-select[da
ta-logger="${lName}"]`);
        const currentVal = sel ? sel.value : '1 minute';
        const custom = prompt(`Введите интервал опроса для ${l
Name} (например: "2 seconds", "15 seconds", "10 minutes", "1 h
our"):`, currentVal);
        if (custom && custom.trim()) {
          const trimmed = custom.trim();
          let opt = sel.querySelector(`option[value="${trimmed
}"]`);
          if (!opt) {
            opt = document.createElement('option');
            opt.value = trimmed;
            opt.textContent = `${trimmed} (пользовательский)`;
            sel.appendChild(opt);
          }
          sel.value = trimmed;
        }
      };
    });
  }

  async function saveSysIntervalsConfig() {
    const btnSave = document.getElementById('btn-modal-interva
ls-save');
    if (btnSave) {
      btnSave.disabled = true;
      btnSave.innerHTML = '<span class="spinner-border spinner
-border-sm me-1" role="status"></span> Сохранение...';
    }

    try {
      const globalEnable = document.getElementById('modal-inte
rvals-global-enable');
```

```
      const defSelect = document.getElementById('modal-intervals-default-select');

      const payload = {
        enable_autolog: globalEnable ? globalEnable.checked : true,
        default_interval: defSelect ? defSelect.value : '1 minute',
        loggers: {}
      };

      // Collect values from all interval selects & switches
      document.querySelectorAll('.sys-int-select').forEach(sel => {
        const lName = sel.getAttribute('data-logger');
        if (!lName) return;
        const sw = document.querySelector(`.sys-int-enable[data-logger="${lName}"]`);
        payload.loggers[lName] = {
          interval: sel.value,
          enabled: sw ? sw.checked : true
        };
      });

      let saveRes = await fetch('/api/autolog/config', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify(payload)
      });

      if (!saveRes.ok) {
        saveRes = await fetch('/sysautologging/config', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify(payload)
        });
      }

```

```
      if (!saveRes.ok) {
        const errJson = await saveRes.json().catch(() => ({}))
;
        throw new Error(errJson.detail || `HTTP ${saveRes.stat
us}`);
      }

      const result = await saveRes.json();
      showAlertInModal(
        `<i class="bi bi-check-circle-fill text-success me-1">
</i> ${result.message || 'Интервалы успешно сохранены в config
.json и применены!'}`,
        'success'
      );

      // Refresh engine badge
      const engineStatusBadge = document.getElementById('modal
-intervals-engine-status');
      if (engineStatusBadge) {
        if (result.is_running) {
          engineStatusBadge.textContent = 'AutoLog Active';
          engineStatusBadge.className = 'badge bg-success';
        } else {
          engineStatusBadge.textContent = 'AutoLog Paused';
          engineStatusBadge.className = 'badge bg-danger-subtl
e text-danger border border-danger fw-bold';
        }
      }
    } catch (e) {
      console.error('[SystemInspectorTab] Save intervals confi
g error:', e);
      showAlertInModal(`<i class="bi bi-exclamation-triangle-f
ill text-danger me-1"></i> Не удалось сохранить интервалы: ${e
scapeHtml(e.message)}`, 'danger');
    } finally {
      if (btnSave) {
        btnSave.disabled = false;
        btnSave.innerHTML = '<i class="bi bi-check2-circle me-
```

```
1"></i> Сохранить в config.json';
      }
    }
  }

  let currentWatchDirs = [];
  let currentWatchDir = '';
  let stagedWatchDirs = [];
  let currentBrowserPath = 'C:\\';
  let cachedDrives = [];

  async function fetchLiveFileEvents() {
    try {
      const [resEvents, resTelem] = await Promise.all([
        fetch('/api/v1/system/file-audit/live-events?limit=30'
),
        fetch('/api/v1/system/file-audit/telemetry').catch(()
=> null)
      ]);

      if (resTelem && resTelem.ok) {
        const telem = await resTelem.json();
        const elRate = document.getElementById('sys-watcher-ra
te');
        const elCr = document.getElementById('sys-watcher-cr')
;
        const elMod = document.getElementById('sys-watcher-mod
');
        const elDel = document.getElementById('sys-watcher-del
');
        const elDrive = document.getElementById('sys-watcher-d
rive');
        const elDriveModel = document.getElementById('sys-watc
her-drive-model');
        const elDriveTemp = document.getElementById('sys-watch
er-temp');
        const elDiskR = document.getElementById('sys-watcher-r
-kbs');
```

```
5171        const elDiskW = document.getElementById('sys-watcher-w
-kbs');
5172        const elSecMatches = document.getElementById('sys-watc
her-sec-matches');
5173        const elStatus = document.getElementById('sys-watcher-
sensor-status');
5174
5175        if (elRate) elRate.textContent = (telem.events_rate_pe
r_sec || 0).toFixed(1);
5176        if (elCr) elCr.textContent = (telem.created_rate_per_s
ec || 0).toFixed(1);
5177        if (elMod) elMod.textContent = (telem.modified_rate_pe
r_sec || 0).toFixed(1);
5178        if (elDel) elDel.textContent = (telem.deleted_rate_per
_sec || 0).toFixed(1);
5179
5180        if (elDrive) {
5181          const drivesStr = (telem.drive_letters && telem.driv
e_letters.length)
5182            ? telem.drive_letters.join(', ')
5183            : (telem.drive_letter || 'C:');
5184          elDrive.textContent = drivesStr;
5185        }
5186        if (elDriveModel) elDriveModel.textContent = telem.dri
ve_model || 'Storage';
5187        if (elDriveTemp) elDriveTemp.textContent = telem.drive
_temperature_c !== null && telem.drive_temperature_c !== undef
ined ? `${telem.drive_temperature_c} °C` : '-- °C';
5188        if (elDiskR) elDiskR.textContent = Math.round(telem.di
sk_read_kbs || 0);
5189        if (elDiskW) elDiskW.textContent = Math.round(telem.di
sk_write_kbs || 0);
5190        if (elSecMatches) elSecMatches.textContent = telem.sec
urity_audit_matched_count || 0;
5191
5192        if (elStatus) {
5193          if (telem.burst_deletions_alert) {
5194            elStatus.className = 'badge bg-danger-subtle text-
```

```
danger border border-danger px-2 py-1';
5195            elStatus.textContent = '  Всплеск удалений';
5196          } else if (telem.high_activity_alert) {
5197            elStatus.className = 'badge bg-warning-subtle text
-warning border border-warning px-2 py-1';
5198            elStatus.textContent = '  Высокий I/O';
5199          } else {
5200            elStatus.className = 'badge bg-success-subtle text
-success border border-success px-2 py-1';
5201            elStatus.textContent = '  Штатный режим';
5202          }
5203        }
5204      }
5205
5206      if (!resEvents || !resEvents.ok) return;
5207      const data = await resEvents.json();
5208      const events = data.events || [];
5209
5210      if (data.filtered_count !== undefined) {
5211        const filteredEl = document.getElementById('sys-watche
r-filtered-count');
5212        if (filteredEl) filteredEl.textContent = data.filtered
_count;
5213      }
5214
5215      currentWatchDirs = data.watch_dirs || (data.watch_dir ?
[data.watch_dir] : []);
5216      currentWatchDir = currentWatchDirs[0] || data.watch_dir
|| '';
5217
5218      const dirPathEl = document.getElementById('sys-watch-dir
-path');
5219      const badge = document.getElementById('sys-watch-dir-bad
ge');
5220      if (dirPathEl) {
5221        if (currentWatchDirs.length === 0) {
5222          dirPathEl.innerText = 'Папки не выбраны';
5223        } else if (currentWatchDirs.length === 1) {
```

```
5224          const singleName = currentWatchDirs[0].split('\\').p
op() || currentWatchDirs[0];
5225          dirPathEl.innerText = singleName;
5226        } else {
5227          const firstNames = currentWatchDirs.slice(0, 2).map(
p => p.split('\\').pop() || p).join(', ');
5228          dirPathEl.innerText = `${currentWatchDirs.length} па
пок: ${firstNames}${currentWatchDirs.length > 2 ? '...' : ''}`
;
5229        }
5230      }
5231      if (badge) {
5232        badge.title = `Отслеживаемые каталоги (${currentWatchD
irs.length}):\n${currentWatchDirs.join('\n')}\n\n(Нажмите для
настройки и выбора папок)`;
5233      }
5234
5235      const tbody = document.getElementById('sys-watcher-tbody
');
5236      if (tbody) {
5237        if (events.length === 0) {
5238          const labelDirs = currentWatchDirs.length > 0 ? curr
entWatchDirs.join(', ') : 'проекта';
5239          tbody.innerHTML = `<tr><td colspan="4" class="text-c
enter text-muted p-2">Ожидание изменений в папках <code>${labe
lDirs}</code>...</td></tr>`;
5240          return;
5241        }
5242        tbody.innerHTML = events.map((e, idx) => {
5243          const rootDirHint = e.watch_dir ? (e.watch_dir.split
('\\').pop() || e.watch_dir) : '';
5244          const procDisplay = e.process_name
5245            ? `<span class="badge bg-dark border border-second
ary text-info font-monospace text-truncate d-inline-block" sty
le="max-width: 165px; font-size: 0.72rem;" title="Программа: $
{escapeHtml(e.process_name)}${e.process_id ? ` (PID: ${e.proce
ss_id})` : ''}"><i class="bi bi-cpu me-1"></i>${escapeHtml(e.p
rocess_name)}${e.process_id ? ` [${e.process_id}]` : ''}</span
```

```
>`
            : `<span class="text-muted" style="font-size: 0.72rem;">—</span>`;
          return `
            <tr class="sys-live-row" data-idx="${idx}" style="cursor: pointer;" title="Нажмите для AI-диагностики события">
              <td class="font-monospace text-muted small">${e.timestamp?.slice(11, 19) || ''}</td>
              <td>
                <span class="badge ${e.is_deletion ? 'bg-danger' : (e.action === 'Created' ? 'bg-success' : 'bg-secondary')}">${e.action}</span>
                ${rootDirHint && currentWatchDirs.length > 1 ? `<span class="badge bg-dark border border-secondary text-muted ms-1" style="font-size: 0.65rem;" title="Корень: ${escapeHtml(e.watch_dir)}">${rootDirHint}</span>` : ''}
              </td>
              <td>${procDisplay}</td>
              <td class="font-monospace small" style="word-break: break-all; color: var(--text-color);" title="${escapeHtml(e.path)}">${escapeHtml(e.path)}</td>
            </tr>
          `;
        }).join('');

        tbody.querySelectorAll('.sys-live-row').forEach(row => {
          row.onclick = () => {
            const idx = parseInt(row.getAttribute('data-idx'), 10);
            const e = events[idx];
            if (!e) return;

            if (window.AITableModal) {
              window.AITableModal.show({
                icon: '',
                title: `Файловое событие: ${e.action}`,
                subtitle: `${e.path} | ${e.timestamp}`,
```

```
                panelId: 'panel-system-inspector-live-watcher'
,
                tableType: 'file_event',
                badges: [
                  { text: e.action, class: e.is_deletion ? 'ba
dge bg-danger' : (e.action === 'Created' ? 'badge bg-success'
: 'badge bg-info text-dark') },
                  { text: e.process_name ? `Программа: ${e.pro
cess_name}` : 'WinAPI ReadDirectoryChangesW', class: 'badge bg
-dark border border-secondary text-info' },
                  { text: 'WinAPI', class: 'badge bg-secondary
' }
                ],
                metadata: [
                  { label: 'Действие', value: e.action },
                  { label: 'Полный путь к файлу', value: e.pat
h },
                  { label: 'Программа / Процесс', value: e.pro
cess_name ? `${e.process_name}${e.process_id ? ` (PID: ${e.pro
cess_id})` : ''}` : 'Фоновый процесс / завершен' },
                  { label: 'Время события', value: e.timestamp
 },
                  { label: 'Признак удаления', value: e.is_del
etion ? 'Да (Файл удален/переименован)' : 'Нет' },
                  { label: 'Папка события', value: e.watch_dir
 || currentWatchDir },
                  { label: 'Все отслеживаемые папки', value: c
urrentWatchDirs.join('; ') }
                ],
                rawTitle: 'Детали события WinAPI & Process Inf
o',
                rawContent: JSON.stringify(e, null, 2),
                requestData: e
              });
            }
          };
        });
      }
```

```
    } catch (e) {
      console.error('[SystemInspectorTab] Failed to fetch live
 file events:', e);
    }
  }

  // ============================================================
====================
  // Multi-Directory Watcher Manager & Folder Explorer Modal L
ogic
  // ============================================================
====================

  function showWatchDirsAlert(msg, type = 'success') {
    const alertEl = document.getElementById('modal-watch-dirs-
alert');
    if (!alertEl) return;
    alertEl.className = `alert alert-${type} py-2 px-3 small m
b-3`;
    alertEl.innerHTML = msg;
    alertEl.classList.remove('d-none');
    if (type === 'success') {
      setTimeout(() => {
        alertEl.classList.add('d-none');
      }, 3500);
    }
  }

  function renderModalActiveDirsList() {
    const container = document.getElementById('modal-watch-dir
s-active-list');
    const countBadge = document.getElementById('modal-watch-di
rs-count');
    const hintEl = document.getElementById('modal-watch-dirs-f
ooter-hint');

    if (!container) return;

```

```
    if (countBadge) countBadge.textContent = String(stagedWatchDirs.length);
    if (hintEl) hintEl.textContent = `Выбрано папок для мониторинга: ${stagedWatchDirs.length}`;

    if (stagedWatchDirs.length === 0) {
      container.innerHTML = `
        <div class="text-center text-muted small py-3">
          <i class="bi bi-folder-x me-1"></i> Список пуст. Выберите папки в проводнике ниже или воспользуйтесь быстрыми пресетами.
        </div>
      `;
      return;
    }

    container.innerHTML = stagedWatchDirs.map((dirPath, idx) => {
      const folderName = dirPath.split('\\').pop() || dirPath;
      const driveLetter = (dirPath.slice(0, 2)).toUpperCase();
      return `
        <div class="p-1.5 px-2 rounded d-flex align-items-center justify-content-between gap-2" style="background: var(--bg-color); border: 1px solid var(--border-color);">
          <div class="d-flex align-items-center gap-2 text-truncate" style="max-width: 82%;">
            <i class="bi bi-folder-check text-warning fs-6"></i>
            <span class="badge bg-secondary font-monospace" style="font-size: 0.68rem;">${driveLetter}</span>
            <div class="text-truncate">
              <span class="fw-bold small" style="color: var(--text-color);">${escapeHtml(folderName)}</span>
              <span class="small text-muted font-monospace d-block text-truncate" style="font-size: 0.72rem;" title="${escapeHtml(dirPath)}">${escapeHtml(dirPath)}</span>
            </div>
          </div>
```

```
          <button class="btn btn-xs btn-outline-danger rounded-pill px-2 py-0.5" onclick="window._removeStagedWatchDir(${idx})" title="Удалить из списка мониторинга">
            <i class="bi bi-trash3"></i>
          </button>
        </div>
      `;
    }).join('');
  }

  window._removeStagedWatchDir = function(index) {
    if (index >= 0 && index < stagedWatchDirs.length) {
      const removed = stagedWatchDirs.splice(index, 1)[0];
      renderModalActiveDirsList();
      showWatchDirsAlert(`Папка удалена из списка: <code>${escapeHtml(removed)}</code>`, 'info');
    }
  };

  function addDirToStaged(pathToAdd) {
    if (!pathToAdd || !pathToAdd.trim()) return;
    const cleanPath = pathToAdd.trim();
    if (stagedWatchDirs.includes(cleanPath)) {
      showWatchDirsAlert(`Папка уже есть в списке: <code>${escapeHtml(cleanPath)}</code>`, 'warning');
      return;
    }
    stagedWatchDirs.push(cleanPath);
    renderModalActiveDirsList();
    showWatchDirsAlert(`Папка добавлена: <code>${escapeHtml(cleanPath)}</code>`, 'success');
  }

  async function loadSystemDrives() {
    const drivesBar = document.getElementById('modal-watch-drives-bar');
    if (!drivesBar) return;

```

```
    try {
      const res = await fetch('/api/sysadmin/filesystem/drives');
      if (res.ok) {
        const data = await res.json();
        cachedDrives = data.drives || [];
      }
    } catch (e) {
      console.warn('[SystemInspectorTab] Failed to fetch system drives:', e);
      cachedDrives = [{ mountpoint: 'C:\\', free_gb: 0 }];
    }

    if (cachedDrives.length === 0) {
      cachedDrives = [{ mountpoint: 'C:\\', free_gb: 0 }];
    }

    drivesBar.innerHTML = cachedDrives.map(d => {
      const label = d.mountpoint || 'C:\\';
      const freeTxt = d.free_gb ? `${d.free_gb} GB free` : '';
      return `
        <button class="btn btn-xs btn-outline-secondary rounded-pill px-2 py-0.5 sys-drive-btn font-monospace" data-drive="${escapeHtml(label)}" title="${escapeHtml(label)} ${freeTxt}">
          <i class="bi bi-hdd me-1"></i>${escapeHtml(label)}
        </button>
      `;
    }).join('');

    drivesBar.querySelectorAll('.sys-drive-btn').forEach(btn => {
      btn.onclick = () => {
        const drv = btn.getAttribute('data-drive');
        if (drv) loadFolderBrowser(drv);
      };
    });
  }

```

```
  function renderBreadcrumbs(currentPath) {
    const bcContainer = document.getElementById('modal-folder-breadcrumbs');
    if (!bcContainer) return;

    const parts = currentPath.replace(/\\+$/, '').split('\\');
    let builtPath = '';
    const crumbsHtml = parts.map((part, idx) => {
      if (idx === 0) {
        builtPath = part + '\\';
      } else {
        builtPath = builtPath + (builtPath.endsWith('\\') ? '' : '\\') + part;
      }
      const clickPath = builtPath;
      const isLast = idx === parts.length - 1;
      return `
        <span class="d-inline-flex align-items-center">
          ${idx > 0 ? '<span class="text-muted mx-1">/</span>' : ''}
          <a href="#" class="sys-breadcrumb-link text-decoration-none ${isLast ? 'fw-bold text-info' : ''}" style="color: var(--text-color);" data-path="${escapeHtml(clickPath)}">
            ${escapeHtml(part || 'Корень')}
          </a>
        </span>
      `;
    }).join('');

    bcContainer.innerHTML = crumbsHtml;

    bcContainer.querySelectorAll('.sys-breadcrumb-link').forEach(link => {
      link.onclick = (e) => {
        e.preventDefault();
        const p = link.getAttribute('data-path');
        if (p) loadFolderBrowser(p);
      };
```

```
5446      });
5447    }
5448
5449    async function loadFolderBrowser(targetPath) {
5450      const inputPath = document.getElementById('modal-folder-pa
th-input');
5451      const browserList = document.getElementById('modal-folder-
browser-list');
5452      if (!targetPath) targetPath = 'C:\\';
5453      currentBrowserPath = targetPath;
5454
5455      if (inputPath) inputPath.value = targetPath;
5456      renderBreadcrumbs(targetPath);
5457
5458      if (!browserList) return;
5459      browserList.innerHTML = `<div class="text-center text-mute
d small py-3"><span class="spinner-border spinner-border-sm me
-1"></span> Загрузка директорий...</div>`;
5460
5461      try {
5462        const res = await fetch(`/api/sysadmin/filesystem/browse
?path=${encodeURIComponent(targetPath)}`);
5463        if (!res.ok) {
5464          const errJson = await res.json().catch(() => ({}));
5465          throw new Error(errJson.detail || `HTTP ${res.status}`
);
5466        }
5467
5468        const data = await res.json();
5469        const dirs = data.directories || [];
5470
5471        if (dirs.length === 0) {
5472          browserList.innerHTML = `<div class="text-center text-
muted small py-3">В этой директории нет доступных подпапок</di
v>`;
5473          return;
5474        }
5475
```

```
      browserList.innerHTML = dirs.map(d => {
        const isAlreadySelected = stagedWatchDirs.includes(d.path);
        return `
          <div class="p-1 px-2 rounded d-flex align-items-center justify-content-between gap-2 sys-folder-browser-item" style="background: var(--surface-1); border: 1px solid var(--border-color); cursor: pointer;">
            <div class="d-flex align-items-center gap-2 text-truncate flex-grow-1 sys-nav-to-folder" data-path="${escapeHtml(d.path)}" title="Нажмите для перехода в папку">
              <i class="bi ${d.has_subdirs ? 'bi-folder2 text-warning' : 'bi-folder text-warning'}"></i>
              <span class="small text-light text-truncate">${escapeHtml(d.name)}</span>
              <span class="small text-muted font-monospace ms-auto me-2" style="font-size: 0.68rem;">${d.modified || ''}</span>
            </div>
            <button class="btn btn-xs ${isAlreadySelected ? 'btn-success' : 'btn-outline-info'} rounded-pill px-2 py-0.5 sys-add-folder-btn" data-path="${escapeHtml(d.path)}" title="Добавить в отслеживаемые">
              <i class="bi ${isAlreadySelected ? 'bi-check2' : 'bi-plus-lg'} me-1"></i>${isAlreadySelected ? 'Выбрана' : 'Следить'}
            </button>
          </div>
        `;
      }).join('');

      // Bind folder navigation
      browserList.querySelectorAll('.sys-nav-to-folder').forEach(el => {
        el.onclick = () => {
          const p = el.getAttribute('data-path');
          if (p) loadFolderBrowser(p);
        };
```

```
      });

      // Bind Add buttons
      browserList.querySelectorAll('.sys-add-folder-btn').forEach(btn => {
        btn.onclick = (e) => {
          e.stopPropagation();
          const p = btn.getAttribute('data-path');
          if (p) {
            addDirToStaged(p);
            btn.className = 'btn btn-xs btn-success rounded-pill px-2 py-0.5 sys-add-folder-btn';
            btn.innerHTML = '<i class="bi bi-check2 me-1"></i>Выбрана';
          }
        };
      });

    } catch (e) {
      console.warn('[SystemInspectorTab] Failed to browse folder:', e);
      browserList.innerHTML = `<div class="text-danger small py-2 px-2"><i class="bi bi-exclamation-triangle me-1"></i> Ошибка доступа: ${escapeHtml(e.message)}</div>`;
    }
  }

  let currentExclusions = {
    enabled: true,
    paths: [],
    extensions: [],
    patterns: [],
    processes: [],
    filtered_count: 0
  };

  function showExclusionsAlert(msg, type = 'success') {
    const alertEl = document.getElementById('modal-exclusions-
```

```
alert');
    if (!alertEl) return;
    alertEl.className = `alert alert-${type} py-2 px-3 small m
b-3`;
    alertEl.innerHTML = msg;
    alertEl.classList.remove('d-none');
    if (type === 'success' || type === 'info') {
      setTimeout(() => {
        alertEl.classList.add('d-none');
      }, 3500);
    }
  }

  function renderExclusionsLists() {
    const totalCount = (currentExclusions.paths?.length || 0)
+
      (currentExclusions.extensions?.length || 0) +
      (currentExclusions.patterns?.length || 0) +
      (currentExclusions.processes?.length || 0);

    const tabBadge = document.getElementById('modal-watch-excl
usions-tab-count');
    const headerBadge = document.getElementById('sys-exclusion
s-count-badge');
    if (tabBadge) tabBadge.textContent = String(totalCount);
    if (headerBadge) headerBadge.textContent = String(totalCou
nt);

    const switchEl = document.getElementById('switch-exclusion
s-active');
    const statusBadge = document.getElementById('badge-exclusi
ons-status');
    if (switchEl) switchEl.checked = Boolean(currentExclusions
.enabled);
    if (statusBadge) {
      if (currentExclusions.enabled) {
        statusBadge.className = 'badge bg-success';
        statusBadge.textContent = 'ВКЛ';
```

```
5559      } else {
5560        statusBadge.className = 'badge bg-danger-subtle text-d
anger border border-danger fw-bold';
5561        statusBadge.textContent = 'ВЫКЛ';
5562      }
5563    }
5564
5565    const filteredTotalEl = document.getElementById('modal-exc
lusions-filtered-total');
5566    if (filteredTotalEl) filteredTotalEl.textContent = String(
currentExclusions.filtered_count || 0);
5567
5568    // 1. Paths list
5569    const pathsContainer = document.getElementById('list-ex-pa
ths');
5570    const pathsBadge = document.getElementById('badge-ex-paths
-count');
5571    if (pathsBadge) pathsBadge.textContent = String(currentExc
lusions.paths?.length || 0);
5572    if (pathsContainer) {
5573      if (!currentExclusions.paths || currentExclusions.paths.
length === 0) {
5574        pathsContainer.innerHTML = `<div class="text-muted sma
ll text-center py-2">Нет исключенных папок</div>`;
5575      } else {
5576        pathsContainer.innerHTML = currentExclusions.paths.map
((p, idx) => `
5577          <div class="d-flex align-items-center justify-conten
t-between gap-1.5 p-1 px-2 rounded mb-1" style="background: va
r(--bg-color); border: 1px solid var(--border-color);">
5578            <span class="small font-monospace text-light text-
truncate" style="font-size: 0.72rem;" title="${escapeHtml(p)}"
>${escapeHtml(p)}</span>
5579            <button class="btn btn-xs btn-outline-danger py-0
px-1 rounded-pill" onclick="window._removeExclusionItem('paths
', '${escapeHtml(p.replace(/\\/g, '\\\\'))}')" title="Удалить"
>✕</button>
5580          </div>
```

```
5581          `).join('');
5582        }
5583      }
5584
5585      // 2. Extensions list
5586      const extsContainer = document.getElementById('list-ex-ext
ensions');
5587      const extsBadge = document.getElementById('badge-ex-extens
ions-count');
5588      if (extsBadge) extsBadge.textContent = String(currentExclu
sions.extensions?.length || 0);
5589      if (extsContainer) {
5590        if (!currentExclusions.extensions || currentExclusions.e
xtensions.length === 0) {
5591          extsContainer.innerHTML = `<div class="text-muted smal
l text-center py-2">Нет исключенных расширений</div>`;
5592        } else {
5593          extsContainer.innerHTML = `<div class="d-flex flex-wra
p gap-1">` + currentExclusions.extensions.map(e => `
5594            <span class="badge bg-dark border border-secondary t
ext-info d-inline-flex align-items-center gap-1 font-monospace
" style="font-size: 0.75rem;">
5595              ${escapeHtml(e)}
5596              <button type="button" class="btn-close btn-close-w
hite" style="font-size: 0.5rem;" onclick="window._removeExclus
ionItem('extensions', '${escapeHtml(e)}')" title="Удалить"></b
utton>
5597            </span>
5598          `).join('') + `</div>`;
5599        }
5600      }
5601
5602      // 3. Patterns list
5603      const patsContainer = document.getElementById('list-ex-pat
terns');
5604      const patsBadge = document.getElementById('badge-ex-patter
ns-count');
5605      if (patsBadge) patsBadge.textContent = String(currentExclu
```

```
sions.patterns?.length || 0);
5606    if (patsContainer) {
5607      if (!currentExclusions.patterns || currentExclusions.pat
terns.length === 0) {
5608        patsContainer.innerHTML = `<div class="text-muted smal
l text-center py-2">Нет исключенных шаблонов</div>`;
5609      } else {
5610        patsContainer.innerHTML = `<div class="d-flex flex-wra
p gap-1">` + currentExclusions.patterns.map(pat => `
5611          <span class="badge bg-dark border border-secondary t
ext-warning d-inline-flex align-items-center gap-1 font-monosp
ace" style="font-size: 0.75rem;">
5612            ${escapeHtml(pat)}
5613            <button type="button" class="btn-close btn-close-w
hite" style="font-size: 0.5rem;" onclick="window._removeExclus
ionItem('patterns', '${escapeHtml(pat)}')" title="Удалить"></b
utton>
5614          </span>
5615        `).join('') + `</div>`;
5616      }
5617    }
5618
5619    // 4. Processes list
5620    const procsContainer = document.getElementById('list-ex-pr
ocesses');
5621    const procsBadge = document.getElementById('badge-ex-proce
sses-count');
5622    if (procsBadge) procsBadge.textContent = String(currentExc
lusions.processes?.length || 0);
5623    if (procsContainer) {
5624      if (!currentExclusions.processes || currentExclusions.pr
ocesses.length === 0) {
5625        procsContainer.innerHTML = `<div class="text-muted sma
ll text-center py-2">Нет исключенных программ</div>`;
5626      } else {
5627        procsContainer.innerHTML = `<div class="d-flex flex-wr
ap gap-1">` + currentExclusions.processes.map(proc => `
5628          <span class="badge bg-dark border border-secondary t
```

```
ext-danger d-inline-flex align-items-center gap-1 font-monospa
ce" style="font-size: 0.75rem;">
            <i class="bi bi-cpu me-0.5"></i>${escapeHtml(proc)
}
            <button type="button" class="btn-close btn-close-w
hite" style="font-size: 0.5rem;" onclick="window._removeExclus
ionItem('processes', '${escapeHtml(proc)}')" title="Удалить"><
/button>
          </span>
        `).join('') + `</div>`;
      }
    }
  }

  async function fetchExclusionsData() {
    try {
      const res = await fetch('/api/v1/system/file-audit/exclu
sions');
      if (res.ok) {
        currentExclusions = await res.json();
        renderExclusionsLists();
      }
    } catch (e) {
      console.warn('[SystemInspectorTab] Failed to fetch exclu
sions:', e);
    }
  }

  async function addExclusionItem(category, value) {
    if (!value || !value.trim()) return;
    const cleanVal = value.trim();
    try {
      const res = await fetch('/api/v1/system/file-audit/exclu
sions/add', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ category, value: cleanVal })
      });
```

```
      const data = await res.json();
      if (res.ok && data.success) {
        currentExclusions = data.exclusions || currentExclusio
ns;
        renderExclusionsLists();
        showExclusionsAlert(`Правило добавлено: <code>${escape
Html(cleanVal)}</code>`, 'success');
        const inputVal = document.getElementById('input-exclus
ion-value');
        if (inputVal) inputVal.value = '';
      } else {
        showExclusionsAlert(data.message || 'Правило уже сущес
твует или невалидно', 'warning');
      }
    } catch (e) {
      showExclusionsAlert(`Ошибка добавления: ${e.message}`, '
danger');
    }
  }

  window._removeExclusionItem = async function(category, value
) {
    if (!value) return;
    try {
      const res = await fetch('/api/v1/system/file-audit/exclu
sions/remove', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ category, value })
      });
      const data = await res.json();
      if (res.ok && data.success) {
        currentExclusions = data.exclusions || currentExclusio
ns;
        renderExclusionsLists();
        showExclusionsAlert(`Правило удалено: <code>${escapeHt
ml(value)}</code>`, 'info');
      } else {
```

```
        showExclusionsAlert(data.message || 'Не удалось удалить правило', 'warning');
      }
    } catch (e) {
      showExclusionsAlert(`Ошибка удаления: ${e.message}`, 'danger');
    }
  };

  async function toggleExclusionsActive(enabled) {
    try {
      const res = await fetch('/api/v1/system/file-audit/exclusions/toggle', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ enabled })
      });
      const data = await res.json();
      if (res.ok && data.success) {
        currentExclusions = data.exclusions || currentExclusions;
        renderExclusionsLists();
        showExclusionsAlert(`Фильтрация исключений ${data.enabled ? 'включена' : 'отключена'}`, 'info');
      }
    } catch (e) {
      showExclusionsAlert(`Ошибка переключения: ${e.message}`, 'danger');
    }
  }

  async function applyExclusionPreset(preset) {
    if (preset === 'logs') {
      await addExclusionItem('paths', 'AppData\\Roaming\\AI-Windows-Admin\\apps\\windows\\telemetry\\logs');
      await addExclusionItem('extensions', '.log');
      await addExclusionItem('extensions', '.csv');
      await addExclusionItem('patterns', '*system_inspector_po
```

```
lls.csv*');
    } else if (preset === 'search') {
      await addExclusionItem('processes', 'SearchIndexer.exe')
;
      await addExclusionItem('patterns', '*Windows.db*');
      await addExclusionItem('patterns', '*Windows.db-wal*');
      await addExclusionItem('patterns', '*Windows.db-shm*');
    } else if (preset === 'temp') {
      await addExclusionItem('paths', 'AppData\\Local\\Temp');
      await addExclusionItem('extensions', '.tmp');
      await addExclusionItem('patterns', '~$*');
    } else if (preset === 'dev') {
      await addExclusionItem('paths', '.git');
      await addExclusionItem('paths', 'node_modules');
      await addExclusionItem('paths', '__pycache__');
      await addExclusionItem('paths', '.pytest_cache');
    }
  }

  async function openWatchFoldersModal(initialTab = 'folders')
 {
    const modalEl = document.getElementById('modal-sys-watch-f
olders');
    if (!modalEl) {
      console.warn('[SystemInspectorTab] modal-sys-watch-folde
rs element not found');
      return;
    }

    try {
      const res = await fetch('/api/v1/system/file-audit/watch
-dirs');
      if (res.ok) {
        const data = await res.json();
        stagedWatchDirs = data.watch_dirs && data.watch_dirs.l
ength ? [...data.watch_dirs] : [currentWatchDir || 'C:\\'];
      } else {
        stagedWatchDirs = currentWatchDirs.length ? [...curren
```

```
tWatchDirs] : [currentWatchDir || 'C:\\'];
      }
    } catch {
      stagedWatchDirs = currentWatchDirs.length ? [...currentW
atchDirs] : [currentWatchDir || 'C:\\'];
    }

    renderModalActiveDirsList();
    const tabFoldersBadge = document.getElementById('modal-wat
ch-dirs-tab-count');
    if (tabFoldersBadge) tabFoldersBadge.textContent = String(
stagedWatchDirs.length);

    await loadSystemDrives();
    await fetchExclusionsData();

    // Tab activation
    if (initialTab === 'exclusions') {
      const tabExBtn = document.getElementById('tab-btn-watch-
exclusions');
      if (tabExBtn && window.bootstrap && window.bootstrap.Tab
) {
        const tab = window.bootstrap.Tab.getOrCreateInstance(t
abExBtn);
        tab.show();
      } else if (tabExBtn) {
        tabExBtn.click();
      }
    } else {
      const tabFoldersBtn = document.getElementById('tab-btn-w
atch-folders');
      if (tabFoldersBtn && window.bootstrap && window.bootstra
p.Tab) {
        const tab = window.bootstrap.Tab.getOrCreateInstance(t
abFoldersBtn);
        tab.show();
      } else if (tabFoldersBtn) {
        tabFoldersBtn.click();
```

```
      }
    }

    const startPath = stagedWatchDirs[0] || currentBrowserPath || 'C:\\';
    loadFolderBrowser(startPath);

    // Bind navigation buttons
    const btnUp = document.getElementById('btn-modal-folder-up');
    if (btnUp) {
      btnUp.onclick = () => {
        const parts = currentBrowserPath.replace(/\\+$/, '').split('\\');
        if (parts.length > 1) {
          parts.pop();
          let parentPath = parts.join('\\');
          if (parts.length === 1 && !parentPath.endsWith('\\')) parentPath += '\\';
          loadFolderBrowser(parentPath);
        }
      };
    }

    const btnGo = document.getElementById('btn-modal-folder-go');
    const inputPath = document.getElementById('modal-folder-path-input');
    if (btnGo && inputPath) {
      btnGo.onclick = () => {
        if (inputPath.value && inputPath.value.trim()) {
          loadFolderBrowser(inputPath.value.trim());
        }
      };
      inputPath.onkeydown = (e) => {
        if (e.key === 'Enter') {
          e.preventDefault();
          btnGo.click();
```

```
        }
      };
    }

    const btnAddCurrent = document.getElementById('btn-modal-folder-add-current');
    if (btnAddCurrent) {
      btnAddCurrent.onclick = () => {
        const val = inputPath ? inputPath.value.trim() : currentBrowserPath;
        if (val) addDirToStaged(val);
      };
    }

    // Presets buttons (folders)
    document.querySelectorAll('.sys-preset-btn').forEach(btn => {
      btn.onclick = () => {
        const preset = btn.getAttribute('data-preset');
        let target = '';
        if (preset === 'project') target = 'C:\\Users\\onela\\AppData\\Local\\AI-Windows-Admin';
        else if (preset === 'downloads') target = 'C:\\Users\\onela\\Downloads';
        else if (preset === 'desktop') target = 'C:\\Users\\onela\\Desktop';
        else if (preset === 'temp') target = 'C:\\Users\\onela\\AppData\\Local\\Temp';

        if (target) {
          addDirToStaged(target);
          loadFolderBrowser(target);
        }
      };
    });

    // Exclusions switch and add buttons binding
    const switchExActive = document.getElementById('switch-exc
```

```
lusions-active');
    if (switchExActive) {
      switchExActive.onchange = (e) => {
        toggleExclusionsActive(e.target.checked);
      };
    }

    const btnAddEx = document.getElementById('btn-add-exclusio
n');
    const selectExCat = document.getElementById('select-exclus
ion-category');
    const inputExVal = document.getElementById('input-exclusio
n-value');
    if (btnAddEx && selectExCat && inputExVal) {
      btnAddEx.onclick = () => {
        addExclusionItem(selectExCat.value, inputExVal.value);
      };
      inputExVal.onkeydown = (e) => {
        if (e.key === 'Enter') {
          e.preventDefault();
          btnAddEx.click();
        }
      };
    }

    // Exclusions preset buttons
    document.querySelectorAll('.sys-ex-preset-btn').forEach(bt
n => {
      btn.onclick = () => {
        const preset = btn.getAttribute('data-preset');
        if (preset) applyExclusionPreset(preset);
      };
    });

    // Apply button
    const btnApply = document.getElementById('btn-modal-watch-
dirs-apply');
    if (btnApply) {
```

```
5872      btnApply.onclick = async () => {
5873        if (stagedWatchDirs.length === 0) {
5874          showWatchDirsAlert('Выберите хотя бы одну папку для
мониторинга', 'warning');
5875          return;
5876        }
5877
5878        btnApply.disabled = true;
5879        btnApply.innerHTML = '<span class="spinner-border spin
ner-border-sm me-1"></span> Применение...';
5880
5881        try {
5882          const res = await fetch('/api/v1/system/file-audit/w
atch-dirs', {
5883            method: 'POST',
5884            headers: { 'Content-Type': 'application/json' },
5885            body: JSON.stringify({ paths: stagedWatchDirs })
5886          });
5887          const resData = await res.json();
5888          if (res.ok && resData.success) {
5889            if (window.bootstrap && window.bootstrap.Modal) {
5890              const modal = window.bootstrap.Modal.getInstance
(modalEl);
5891              if (modal) modal.hide();
5892            } else {
5893              modalEl.classList.remove('show');
5894              modalEl.style.display = 'none';
5895            }
5896            await fetchLiveFileEvents();
5897          } else {
5898            showWatchDirsAlert(`Ошибка: ${resData.detail || 'H
е удалось применить список папок'}`, 'danger');
5899          }
5900        } catch (err) {
5901          showWatchDirsAlert(`Сетевая ошибка: ${err.message}`,
 'danger');
5902        } finally {
5903          btnApply.disabled = false;
```

```
          btnApply.innerHTML = '<i class="bi bi-check2-circle me-1"></i> Применить и запустить';
        }
      };
    }

    if (window.bootstrap && window.bootstrap.Modal) {
      const modal = window.bootstrap.Modal.getOrCreateInstance(modalEl);
      modal.show();
    } else {
      modalEl.classList.add('show');
      modalEl.style.display = 'block';
    }
  }

  function showLiveWatcherHelpModal() {
    const dirsStr = currentWatchDirs.length ? currentWatchDirs.join('\n- ') : (currentWatchDir || 'Рабочая папка');
    if (window.AITableModal) {
      window.AITableModal.show({
        icon: 'i',
        title: 'Справка:Изменения файлов в реальном времени (Multi-Directory)',
        subtitle: 'Низкоуровневый мониторинг файловой системы Windows через WinAPI ReadDirectoryChangesW',
        panelId: 'panel-system-inspector-watcher-help',
        tableType: 'help',
        badges: [
          { text: 'WinAPI ReadDirectoryChangesW', class: 'badge bg-info text-dark' },
          { text: 'Multi-Directory', class: 'badge bg-primary' },
          { text: 'Real-Time Streaming', class: 'badge bg-success' },
          { text: 'Рекурсивно (bWatchSubtree = True)', class: 'badge bg-warning text-dark' }
        ],
```

```
5933          metadata: [
5934            { label: 'Технология', value: 'WinAPI ReadDirectoryC
      hangesW (нативный вызов ядра Windows kernel32.dll)' },
5935            { label: 'Режим работы', value: 'Множественные поток
      и мониторинга с агрегацией в единый кольцевой буфер' },
5936            { label: 'Отслеживаемые папки', value: currentWatchD
      irs.join('; ') || 'Рабочая папка проекта' },
5937            { label: 'Хранение настроек', value: 'apps/windows_s
      ysadmin/config.json (ключ watch_directories)' }
5938          ],
5939          rawTitle: 'Подробное руководство по панели мониторинга
      ',
5940          rawContent: `# ПанельИзменения файлов в реальном време
      ни
5941
5942      ### 1. Что это такое?
5943      Компонент для мгновенного перехвата операций файловой системы
       в режиме реального времени на базе WinAPI ReadDirectoryChanges
      W с поддержкой одновременного наблюдения за произвольным колич
      еством папок.
5944
5945      ### 2. Типы отслеживаемых действий:
5946      - Created: Создание нового файла или папки (FILE_ACTION_ADDED)
      .
5947      - Modified: Модификация содержимого, атрибутов или размера фай
      ла (FILE_ACTION_MODIFIED).
5948      - Deleted: Удаление файла или папки с диска (FILE_ACTION_REMOV
      ED).
5949      - Renamed: Переименование объекта (старое и новое имя).
5950
5951      ### 3. Как выбрать или добавить папки?
5952      1. Нажмите кнопку «Папка» в заголовке панели.
5953      2. Откроется модальное окно с проводником файловой системы.
5954      3. Выберите логический диск (C:, D:, E:), перейдите в нужную п
      апку или вставьте путь вручную.
5955      4. Нажмите «+ Добавить папку» или кнопку «Следить» напротив лю
      бой подпапки.
5956      5. Нажмите «Применить и запустить». Список сохранится в config
```

```
.json и мониторинг стартует автоматически!`,
5957          requestData: {
5958            current_watch_dirs: currentWatchDirs,
5959            engine: 'ReadDirectoryChangesW (Multi-Directory)'
5960          }
5961        });
5962      } else {
5963        alert('Мониторинг файловой системыИзменения файлов в реа
льном времени (Multi-Directory).\nОтслеживаемые папки:\n- ' +
dirsStr);
5964      }
5965    }
5966
5967    function setupSysSensorInterval(seconds) {
5968      _currentUiRefreshSeconds = seconds;
5969      const pollHandler = async () => {
5970        await fetchCpuLoadFromApi();
5971        await fetchGpuLoadFromApi();
5972        await fetchMemoryIoFromApi();
5973        await fetchNetworkLoadFromApi();
5974        await fetchStorageLoadFromApi();
5975        await fetchStorageHealthFromApi();
5976      };
5977      if (window.registerTabPoller) {
5978        window.registerTabPoller('tab-hardware-load-inspector',
pollHandler, seconds * 1000, { pollerId: 'hw_load_inspector',
immediate: false });
5979      } else {
5980        if (window._sysSensorInterval) {
5981          clearInterval(window._sysSensorInterval);
5982          window._sysSensorInterval = null;
5983        }
5984        window._sysSensorInterval = setInterval(() => {
5985          if (window.isTabActive ? (window.isTabActive('tab-hard
ware-load-inspector') || window.isTabActive('tab-system-load-i
nspector') || window.isTabActive('tab-system-inspector')) : tr
ue) {
5986            pollHandler();
```

```
        }
      }, seconds * 1000);
    }
    console.log(`[SystemInspectorTab] UI sensor refresh interv
al set to ${seconds}s`);
  }

  function openSysIntervalsModal() {
    const modalEl = document.getElementById('sysIntervalsModal
');
    if (!modalEl) {
      console.warn('[SystemInspectorTab] sysIntervalsModal ele
ment not found');
      return;
    }

    // Set UI select to current seconds
    const selUi = document.getElementById('modal-ui-refresh-se
lect');
    if (selUi) {
      selUi.value = String(_currentUiRefreshSeconds);
    }
    const uiBadge = document.getElementById('modal-ui-refresh-
badge');
    if (uiBadge) {
      uiBadge.textContent = `${_currentUiRefreshSeconds} сек`;
    }

    loadSysIntervalsConfig(false);

    if (window.bootstrap && window.bootstrap.Modal) {
      const modal = window.bootstrap.Modal.getOrCreateInstance
(modalEl);
      modal.show();
    } else {
      modalEl.classList.add('show');
      modalEl.style.display = 'block';
    }
```

```
  }

  function escapeHtml(str) {
    if (!str) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
  }

  async function initSystemInspectorTab() {
    console.log('[SystemInspectorTab] Initializing...');
    bindTabEvents();
    await fetchAvailableTimeRanges();
    if (window.isTabActive ? (window.isTabActive('tab-hardware-load-inspector') || window.isTabActive('tab-system-load-inspector') || window.isTabActive('tab-system-inspector')) : false) {
      await fetchCpuLoadFromApi();
      await fetchGpuLoadFromApi();
      await fetchMemoryIoFromApi();
      await fetchNetworkLoadFromApi();
      await fetchStorageLoadFromApi();
      await fetchStorageHealthFromApi();

      try {
        const snapRes = await fetch('/api/v1/system/summary');
        if (snapRes.ok) {
          latestTelemetrySnapshot = await snapRes.json();
          updateTelemetryDashboard(latestTelemetrySnapshot);
        }
      } catch (e) {
        console.warn('[SystemInspectorTab] Initial snapshot fetch failed:', e);
      }

```

```
      if (typeof connectSystemWebSocket === 'function') {
        try { connectSystemWebSocket(); } catch (err) { consol
e.debug(err); }
      }
    }

    // Periodic sensor refresh
    setupSysSensorInterval(_currentUiRefreshSeconds || 5);
  }

  function activateSystemInspectorTab() {
    if (window.isTabActive && !window.isTabActive('tab-hardwar
e-load-inspector') && !window.isTabActive('tab-system-load-ins
pector') && !window.isTabActive('tab-system-inspector')) retur
n;
    console.log('[SystemInspectorTab] Tab activated, refreshin
g metrics...');
    fetchAvailableTimeRanges();
    fetchCpuLoadFromApi();
    fetchGpuLoadFromApi();
    fetchMemoryIoFromApi();
    fetchNetworkLoadFromApi();
    fetchStorageLoadFromApi();
    fetchStorageHealthFromApi();
    if (typeof connectSystemWebSocket === 'function') {
      try { connectSystemWebSocket(); } catch (err) { console.
debug(err); }
    }
  }

  function deactivateSystemInspectorTab() {
    console.log('[SystemInspectorTab] Tab deactivated, pausing
 telemetry stream...');
    if (typeof disconnectSystemWebSocket === 'function') {
      try { disconnectSystemWebSocket(); } catch (err) { conso
le.debug(err); }
    }
  }
```

```
6083
6084    // Регистрация в Universal Panel Loader
6085    if (typeof window.registerPanelRenderer === 'function') {
6086      window.registerPanelRenderer('panel-system-inspector-stora
ge', () => {
6087        fetchStorageHealthFromApi();
6088      });
6089      window.registerPanelRenderer('panel-system-inspector-zdoro
ve-diskov-nvmessd', () => {
6090        fetchStorageHealthFromApi();
6091      });
6092    }
6093
6094    window.initSystemInspectorTab = initSystemInspectorTab;
6095    window.initSystemLoadInspectorTab = initSystemInspectorTab;
6096    window.initHardwareLoadInspectorTab = initSystemInspectorTab
;
6097    window.activateSystemInspectorTab = activateSystemInspectorT
ab;
6098    window.activateSystemLoadInspectorTab = activateSystemInspec
torTab;
6099    window.activateHardwareLoadInspectorTab = activateSystemInsp
ectorTab;
6100    window.deactivateSystemInspectorTab = deactivateSystemInspec
torTab;
6101    window.deactivateSystemLoadInspectorTab = deactivateSystemIn
spectorTab;
6102    window.deactivateHardwareLoadInspectorTab = deactivateSystem
InspectorTab;
6103    window.openSysIntervalsModal = openSysIntervalsModal;
6104
6105    // Auto-init if tab is already in DOM and active or open
6106    if (document.getElementById('tab-hardware-load-inspector'))
{
6107      initSystemInspectorTab();
6108    }
6109  })();
6110
```

6111

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/system_logs_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/system_logs_tab/index.h
tml" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/system_logs_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:50:00
================================================================
===============
-->
<style>
  .slc-container {
    padding: 0.25rem 0.5rem;
  }
  .slc-compact-header {
    background: var(--surface-2);
    border: 1px solid var(--border-color);
    border-radius: 8px;
    padding: 0.4rem 0.75rem;
    margin-bottom: 0.5rem;
```

```
    color: var(--text-color);
  }
  .slc-metric-pill {
    background: var(--surface-1);
    border: 1px solid var(--border-color);
    border-radius: 6px;
    padding: 0.2rem 0.5rem;
    font-size: 0.75rem;
    display: inline-flex;
    align-items: center;
    gap: 0.35rem;
  }
  .slc-card {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: 8px;
    padding: 0.5rem 0.75rem;
    display: flex;
    flex-direction: column;
    gap: 0.1rem;
  }
  .slc-label {
    font-size: 0.68rem;
    text-transform: uppercase;
    letter-spacing: 0.04em;
    color: var(--text-muted);
  }
  .slc-value {
    font-size: 1.05rem;
    font-weight: 700;
    color: var(--text-color);
    font-family: ui-monospace, SFMono-Regular, Menlo, Monaco,
Consolas, monospace;
  }
  .slc-panel {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: 8px;
```

```
    overflow: hidden;
  }
  .slc-panel-header {
    background: var(--surface-2) !important;
    border-bottom: 1px solid var(--border-color) !important;
    padding: 0.6rem 1rem;
    font-weight: 600;
    font-size: 0.85rem;
    color: var(--nav-active);
    display: flex;
    align-items: center;
    justify-content: space-between;
  }
  .slc-tree-list {
    max-height: 580px;
    overflow-y: auto;
    font-size: 0.8rem;
  }
  .slc-tree-item {
    padding: 0.4rem 0.75rem;
    cursor: pointer;
    border-radius: 6px;
    margin-bottom: 2px;
    color: var(--text-color);
    display: flex;
    justify-content: space-between;
    align-items: center;
    transition: background 0.15s ease;
  }
  .slc-tree-item:hover {
    background: var(--nav-hover-bg, rgba(56, 189, 248, 0.1));
    color: var(--nav-active);
  }
  .slc-tree-item.active {
    background: var(--nav-active);
    color: #fff;
    font-weight: 600;
  }
```

```
  .slc-table {
    width: 100%;
    margin-bottom: 0;
    color: var(--text-color);
    font-size: 0.8rem;
  }
  .slc-table th {
    background: var(--surface-2);
    color: var(--text-muted);
    border-bottom: 1px solid var(--border-color);
    font-weight: 600;
    text-transform: uppercase;
    font-size: 0.7rem;
    letter-spacing: 0.05em;
    padding: 0.5rem 0.75rem;
  }
  .slc-table td {
    border-bottom: 1px solid var(--border-color);
    padding: 0.45rem 0.75rem;
    vertical-align: middle;
  }
  .slc-table tr:hover td {
    background: var(--nav-hover-bg, rgba(56, 189, 248, 0.06));
    cursor: pointer;
  }
  .slc-table tr.selected td,
  .slc-table tr.table-active td {
    background: rgba(56, 189, 248, 0.2) !important;
    color: var(--text-color) !important;
  }
  .badge-crit { background: #dc2626 !important; color: #fff !i
mportant; }
  .badge-err { background: #ef4444 !important; color: #fff !im
portant; }
  .badge-warn { background: #f59e0b !important; color: #000 !i
mportant; }
  .badge-inf { background: #10b981 !important; color: #fff !im
portant; }
```

```
  .badge-verb { background: #64748b !important; color: #fff !important; }

  /* Interactive Channel Description Popover Tooltip */
  .slc-tooltip-popup {
    position: fixed;
    z-index: 99999;
    max-width: 320px;
    background: var(--surface-1);
    border: 1px solid var(--nav-active);
    border-radius: 8px;
    padding: 0.6rem 0.8rem;
    box-shadow: 0 10px 25px -5px rgba(0, 0, 0, 0.2), 0 0 15px rgba(56, 189, 248, 0.2);
    pointer-events: none;
    transition: opacity 0.15s ease, transform 0.15s ease;
    opacity: 0;
    transform: translateY(4px);
    font-size: 0.75rem;
    line-height: 1.35;
    color: var(--text-color);
  }
  .slc-tooltip-popup.show {
    opacity: 1;
    transform: translateY(0);
  }
  .slc-tooltip-title {
    font-weight: 700;
    color: var(--nav-active);
    margin-bottom: 0.25rem;
    display: flex;
    align-items: center;
    gap: 0.4rem;
  }
  .slc-tooltip-meta {
    color: var(--text-muted);
    font-size: 0.7rem;
    border-top: 1px solid var(--border-color);
```

```
    padding-top: 0.35rem;
    margin-top: 0.35rem;
    display: flex;
    justify-content: space-between;
  }

  /* GitHub Timeline & Activity Heatmap Styles */
  .gh-timeline-container {
    background: #0d1117;
    border: 1px solid #30363d;
    border-radius: 8px;
    padding: 1rem;
    color: #c9d1d9;
  }
  .gh-stat-card {
    background: #161b22;
    border: 1px solid #30363d;
    border-radius: 6px;
    padding: 0.5rem 0.75rem;
  }
  .gh-pulse-chart {
    height: 190px;
    display: flex;
    align-items: flex-end;
    gap: 4px;
    padding: 0.5rem 0.25rem 1.5rem 0.25rem;
    position: relative;
    border-bottom: 1px solid #30363d;
    overflow-x: auto;
  }
  .gh-pulse-col {
    flex: 1;
    min-width: 24px;
    max-width: 50px;
    height: 100%;
    display: flex;
    flex-direction: column;
    justify-content: flex-end;
```

```
    align-items: center;
    position: relative;
    cursor: pointer;
    transition: transform 0.12s ease;
  }
  .gh-pulse-col:hover {
    transform: scaleY(1.04);
  }
  .gh-pulse-col.active .gh-bar-track {
    box-shadow: 0 0 0 2px #58a6ff, 0 0 10px rgba(88, 166, 255, 0.4);
  }
  .gh-bar-track {
    width: 100%;
    border-radius: 4px 4px 0 0;
    overflow: hidden;
    display: flex;
    flex-direction: column-reverse;
    background: rgba(255, 255, 255, 0.05);
    transition: height 0.3s ease;
  }
  .gh-seg-crit { background: #8e1515; }
  .gh-seg-err { background: #da3633; }
  .gh-seg-warn { background: #d29922; }
  .gh-seg-info { background: #238636; }
  .gh-col-count {
    font-size: 0.65rem;
    font-family: ui-monospace, SFMono-Regular, Consolas, monospace;
    color: #8b949e;
    margin-bottom: 2px;
  }
  .gh-col-label {
    position: absolute;
    bottom: -22px;
    font-size: 0.65rem;
    font-family: ui-monospace, SFMono-Regular, Consolas, monospace;
```

```
    color: #8b949e;
    white-space: nowrap;
  }
  .gh-heatmap-grid {
    display: grid;
    grid-template-columns: repeat(24, 1fr);
    gap: 3px;
    padding: 0.5rem 0;
  }
  .gh-heat-cell {
    aspect-ratio: 1;
    border-radius: 2px;
    background: #161b22;
    cursor: pointer;
    transition: transform 0.15s ease, box-shadow 0.15s ease;
    border: 1px solid rgba(255, 255, 255, 0.04);
  }
  .gh-heat-cell:hover {
    transform: scale(1.25);
    z-index: 10;
    box-shadow: 0 0 8px rgba(255, 255, 255, 0.3);
  }
  .gh-heat-cell.lvl-0 { background: #161b22; }
  .gh-heat-cell.lvl-1 { background: #0e4429; }
  .gh-heat-cell.lvl-2 { background: #006d32; }
  .gh-heat-cell.lvl-3 { background: #26a641; }
  .gh-heat-cell.lvl-4 { background: #39d353; }
  .gh-heat-cell.has-err {
    border: 1px solid #da3633 !important;
  }
</style>
<div class="slc-container">
<!-- Compact Header Bar with Inline Stats & Actions -->
<div class="slc-compact-header d-flex justify-content-between
align-items-center flex-wrap gap-2">
<!-- Title & Inline Metrics -->
<div class="d-flex align-items-center gap-2 flex-wrap">
<div class="d-flex align-items-center gap-1.5 me-1">
```

```
<span class="fs-6">□ </span>
<span class="fw-bold text-body small">Windows System Log Cente
r</span>
</div>
<!-- Quick Metrics Pills -->
<div class="slc-metric-pill" title="Discovered Log Sources">
<span class="text-secondary small"><i class="bi bi-diagram-3 m
e-1"></i>Sources:</span>
<span class="fw-bold text-info font-monospace" id="slc-stat-so
urces">1,247</span>
</div>
<div class="slc-metric-pill" title="Active Log Channel">
<span class="text-secondary small"><i class="bi bi-folder-fill
 me-1"></i>Channel:</span>
<span class="fw-bold text-warning font-monospace text-truncate
" id="slc-stat-channel" style="max-width: 140px;">System</span
>
</div>
<div class="slc-metric-pill" title="Events Loaded in View">
<span class="text-secondary small"><i class="bi bi-list-ol me-
1"></i>Loaded:</span>
<span class="fw-bold text-body font-monospace" id="slc-stat-ev
ents">0</span>
</div>
<div class="slc-metric-pill" title="Errors and Incidents Detec
ted">
<span class="text-secondary small"><i class="bi bi-exclamation
-triangle-fill me-1"></i>Faults:</span>
<span class="fw-bold text-danger font-monospace" id="slc-stat-
incidents">0</span>
</div>
</div>
<!-- Actions Toolbar -->
<div class="d-flex align-items-center gap-2 flex-wrap">
<button class="btn btn-outline-info rounded-pill px-2.5 py-1 s
mall d-flex align-items-center gap-1" id="btn-slc-scan" style=
"font-size: 0.78rem;">
<i class="bi bi-radar"></i> <span>Scan</span>
```

```
</button>
<div class="form-check form-switch mb-0 d-inline-flex align-items-center" style="font-size: 0.78rem; min-height: auto;">
<input class="form-check-input me-1" id="slc-live-switch" role="switch" style="cursor: pointer;" type="checkbox"/>
<label class="form-check-label text-secondary small user-select-none" for="slc-live-switch">Live</label>
</div>
<button class="btn btn-primary rounded-pill px-2.5 py-1 small d-flex align-items-center gap-1" id="btn-slc-refresh" style="font-size: 0.78rem;">
<i class="bi bi-arrow-clockwise"></i> <span>Refresh</span>
</button>
<div class="dropdown">
<button class="btn btn-xs btn-outline-secondary dropdown-toggle rounded-pill px-2.5 py-1 small d-flex align-items-center gap-1" data-bs-toggle="dropdown" style="font-size: 0.78rem;" type="button">
<i class="bi bi-download"></i> <span>Export</span>
</button>
<ul class="dropdown-menu dropdown-menu-end shadow" style="font-size: 0.8rem;">
<li><a class="dropdown-item" href="#" id="btn-slc-export-json"><i class="bi bi-filetype-json me-2"></i>JSON</a></li>
<li><a class="dropdown-item" href="#" id="btn-slc-export-csv">
<i class="bi bi-filetype-csv me-2"></i>CSV</a></li>
</ul>
</div>
</div>
</div>
<!-- Search & View Mode Navigation (Compact) -->
<div class="slc-panel mb-2 p-2" data-panel-id="panel-system-logs-1" id="panel-system-logs-1">
<div class="d-flex justify-content-between align-items-center flex-wrap gap-1 mb-2 pb-1 border-bottom border-secondary">
<!-- Mode Tabs -->
<ul class="nav nav-pills gap-1" id="slc-mode-tabs">
<li class="nav-item">
```

```
<button class="nav-link active py-0.5 px-2.5 small rounded-pil
l" data-mode="live" style="font-size: 0.78rem;">● Live Events<
/button>
</li>
<li class="nav-item">
<button class="nav-link py-0.5 px-2.5 small rounded-pill" data
-mode="audit" style="font-size: 0.78rem;">  Аудит (Smart Dige
st)</button>
</li>
<li class="nav-item">
<button class="nav-link py-0.5 px-2.5 small rounded-pill" data
-mode="rag" style="font-size: 0.78rem;">  RAG Поиск</button>
</li>
<li class="nav-item">
<button class="nav-link py-0.5 px-2.5 small rounded-pill" data
-mode="errors" style="font-size: 0.78rem;">  Errors Only</butt
on>
</li>
<li class="nav-item">
<button class="nav-link py-0.5 px-2.5 small rounded-pill" data
-mode="incidents" style="font-size: 0.78rem;">  Incidents</bu
tton>
</li>
<li class="nav-item">
<button class="nav-link py-0.5 px-2.5 small rounded-pill" data
-mode="timeline" style="font-size: 0.78rem;">  Timeline</butt
on>
</li>
<li class="nav-item">
<button class="nav-link py-0.5 px-2.5 small rounded-pill" data
-mode="raw" style="font-size: 0.78rem;">  Raw Record</button>
</li>
</ul>
<!-- Action AI Button -->
<div class="d-flex align-items-center gap-2">
<button class="btn btn-xs btn-outline-warning rounded-pill px-
2.5 py-1 small d-flex align-items-center gap-1" id="btn-slc-as
k-ai" style="font-size: 0.78rem;">
```

```
<i class="bi bi-robot"></i> <span>Ask AI</span>
</button>
</div>
</div>
<!-- Filter Inputs Grid -->
<div class="row g-1.5 align-items-center">
<div class="col-lg-4 col-md-5">
<div class="input-group input-group-sm">
<span class="input-group-text bg-body-secondary text-secondary
 border-secondary py-0 px-2"><i class="bi bi-search" style="fo
nt-size: 0.75rem;"></i></span>
<input class="form-control form-control-sm py-1" id="slc-searc
h-input" placeholder="Search message, provider..." style="font
-size: 0.8rem;" type="text"/>
</div>
</div>
<div class="col-lg-2 col-md-2 col-6">
<select class="form-select form-select-sm py-1" id="slc-level-
select" style="font-size: 0.8rem;">
<option value="">All Severities</option>
<option value="Critical">Critical</option>
<option value="Error">Error</option>
<option value="Warning">Warning</option>
<option value="Information">Information</option>
</select>
</div>
<div class="col-lg-2 col-md-2 col-6">
<select class="form-select form-select-sm py-1" id="slc-hours-
select" style="font-size: 0.8rem;">
<option value="1">Last 1 hour</option>
<option value="6">Last 6 hours</option>
<option selected="" value="24">Last 24 hours</option>
<option value="168">Last 7 days</option>
<option value="0">All available</option>
</select>
</div>
<div class="col-lg-2 col-md-1.5 col-6">
<input class="form-control form-control-sm py-1" id="slc-event
```

```
id-input" placeholder="Event ID" style="font-size: 0.8rem;" ty
pe="number"/>
</div>
<div class="col-lg-2 col-md-1.5 col-6">
<select class="form-select form-select-sm py-1" id="slc-limit-
select" style="font-size: 0.8rem;">
<option value="50">50 records</option>
<option selected="" value="100">100 records</option>
<option value="250">250 records</option>
<option value="500">500 records</option>
</select>
</div>
</div>
</div>
<!-- Main Multi-Column View -->
<div class="row g-3">
<!-- Channel & Source Tree Sidebar -->
<div class="col-lg-3">
<div class="slc-panel h-100" data-panel-id="panel-system-logs-
2" id="panel-system-logs-2">
<div class="slc-panel-header">
<span><i class="bi bi-folder2-open me-2"></i>Log Channels Tree
</span>
<span class="badge bg-secondary" id="slc-channel-count-badge">
0</span>
</div>
<div class="p-2 border-bottom border-secondary">
<input class="form-control form-control-sm" id="slc-channel-fi
lter-input" placeholder="Filter channels..." type="text"/>
</div>
<div class="p-2 slc-tree-list" id="slc-channel-tree-container"
>
<!-- Dynamic channels injected here -->
</div>
</div>
</div>
<!-- Events / Incidents Main Stream -->
<div class="col-lg-9">
```

```
<div class="slc-panel h-100" data-panel-id="panel-system-logs-semanticheskiy-rag-poisk-po-log" id="panel-system-logs-semanticheskiy-rag-poisk-po-log">
<div class="slc-panel-header">
<span id="slc-main-view-title"><i class="bi bi-list-columns-reverse me-2"></i>Live Windows Event Stream</span>
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm btn-info rounded-pill px-3 fw-bold text-dark" id="btn-slc-audit" title="Провести аудит текущего журнала логов">
<i class="bi bi-shield-check me-1"></i> Аудит
            </button>
<span class="badge bg-secondary" id="slc-stream-count-badge">0 entries</span>
</div>
</div>
<!-- Table View Container -->
<div class="table-responsive" id="slc-table-container" style="max-height: 580px; overflow-y: auto;">
<table class="table slc-table">
<thead>
<tr>
<th style="width: 155px;">Timestamp</th>
<th style="width: 90px;">Level</th>
<th style="width: 65px;">ID</th>
<th style="width: 160px;">Provider</th>
<th style="width: 65px;">PID</th>
<th>Message</th>
</tr>
</thead>
<tbody id="slc-table-body">
<tr>
<td class="text-center text-muted py-4" colspan="6">
<div class="spinner-border spinner-border-sm text-info me-2" role="status"></div>
                  Loading system events...
                </td>
</tr>
```

```
</tbody>
</table>
</div>
<!-- Audit Digest View Container (Hidden by default) -->
<div class="p-3 d-none" id="slc-audit-container" style="max-he
ight: 580px; overflow-y: auto;">
<div id="slc-audit-content"></div>
</div>
<!-- Dynamic Log RAG Search Container (Hidden by default) -->
<div class="p-3 d-none" id="slc-rag-container" style="max-heig
ht: 580px; overflow-y: auto;">
<div class="card border-info mb-3 shadow" data-panel-id="panel
-system-logs-semanticheskiy-rag-poisk-po-log" id="panel-system
-logs-semanticheskiy-rag-poisk-po-log" style="background: var(
--surface-1);">
<div class="card-header d-flex justify-content-between align-i
tems-center" style="background: var(--surface-2);">
<span class="text-info fw-bold"><i class="bi bi-search me-2"><
/i>Семантический RAG-поиск по логам</span>
<button class="btn btn-sm btn-outline-info rounded-pill px-3"
id="btn-slc-rag-rebuild">
<i class="bi bi-arrow-repeat me-1"></i> Перестроить RAG-индекс
              </button>
</div>
<div class="card-body">
<div class="input-group mb-2">
<span class="input-group-text text-info border-secondary" styl
e="background: var(--surface-2);"><i class="bi bi-chat-left-do
ts"></i></span>
<input class="form-control border-secondary" id="slc-rag-input
" placeholder="Введите естественный запрос (например: 'почему
завис BITS', 'ошибки DCOM 10016', 'сетевые сбои')..." type="te
xt"/>
<button class="btn btn-info px-4 fw-bold text-dark" id="btn-sl
c-rag-search">Найти</button>
</div>
<div class="small text-muted">
                Поиск использует динамический TF-IDF/BM25 инде
```

```
кс кластеров, аномалий и сводок с нулевым расходом токенов LLM
.
              </div>
</div>
</div>
<div id="slc-rag-results">
<div class="card border-secondary p-4 text-center text-muted"
data-panel-id="panel-system-logs-3" id="panel-system-logs-3" s
tyle="background: var(--surface-1);">
              Введите запрос для поиска по динамическому RAG-и
ндексу логов или нажмите «Перестроить RAG-индекс».
            </div>
</div>
</div>
<!-- Incidents View Container (Hidden by default) -->
<div class="p-3 d-none" id="slc-incidents-container" style="ma
x-height: 580px; overflow-y: auto;">
<div id="slc-incidents-list"></div>
</div>
<!-- Timeline Histogram View Container (Hidden by default) -->
<div class="p-3 d-none" id="slc-timeline-container" style="max
-height: 580px; overflow-y: auto;">
  <div id="slc-timeline-content"></div>
</div>
<!-- Raw Record Container (Hidden by default) -->
<div class="p-3 d-none" id="slc-raw-container" style="max-heig
ht: 580px; overflow-y: auto;">
<pre class="p-3 rounded font-monospace small" id="slc-raw-text
" style="background: var(--surface-2); color: var(--text-color
); border: 1px solid var(--border-color);">Select an event fro
m the list to view raw structured XML/JSON data.</pre>
</div>
</div>
</div>
</div>
</div>
<!-- Modal: Event Details & AI Diagnostics -->
<div class="modal fade" id="slc-event-detail-modal" tabindex="
```

```
-1">
<div class="modal-dialog modal-lg modal-dialog-centered">
<div class="modal-content">
<div class="modal-header border-secondary">
<h5 class="modal-title fw-bold text-info" id="slc-modal-title"
>Log Event Details</h5>
<button class="btn-close" data-bs-dismiss="modal" type="button
"></button>
</div>
<div class="modal-body">
<div class="row g-2 mb-3 small font-monospace">
<div class="col-sm-6"><strong class="text-secondary">Timestamp
:</strong> <span id="slc-modal-time"></span></div>
<div class="col-sm-6"><strong class="text-secondary">Severity:
</strong> <span id="slc-modal-level"></span></div>
<div class="col-sm-6"><strong class="text-secondary">Event ID:
</strong> <span id="slc-modal-id"></span></div>
<div class="col-sm-6"><strong class="text-secondary">Provider:
</strong> <span id="slc-modal-provider"></span></div>
<div class="col-sm-6"><strong class="text-secondary">Computer:
</strong> <span id="slc-modal-computer"></span></div>
<div class="col-sm-6"><strong class="text-secondary">Process I
D:</strong> <span id="slc-modal-pid"></span></div>
<div class="col-sm-12"><strong class="text-secondary">Channel:
</strong> <span id="slc-modal-channel"></span></div>
</div>
<div class="card border-secondary p-3 mb-3" data-panel-id="pan
el-system-logs-message-body" id="panel-system-logs-message-bod
y" style="background: var(--surface-2);">
<h6 class="small text-uppercase text-muted fw-bold mb-2">Messa
ge Body</h6>
<div class="small font-monospace" id="slc-modal-message" style
="white-space: pre-wrap; word-break: break-word;"></div>
</div>
<!-- AI Diagnostic Response Section -->
<div class="card border-info p-3" data-panel-id="panel-system-
logs-ai-contextual-explanation" id="slc-modal-ai-section" styl
e="background: var(--surface-1);">
```

```
<div class="d-flex justify-content-between align-items-center mb-2">
<h6 class="fw-bold text-info mb-0"><i class="bi bi-robot me-1"></i> AI Contextual Explanation</h6>
<button class="btn btn-sm btn-outline-info py-0 px-2 rounded-pill" id="btn-slc-modal-diagnose">Analyze Context</button>
</div>
<div class="small" id="slc-modal-ai-explanation">
            Click "Analyze Context" to inspect surrounding events and identify the root cause of this incident.
          </div>
</div>
</div>
<div class="modal-footer border-secondary">
<button class="btn btn-sm btn-secondary" data-bs-dismiss="modal" type="button">Close</button>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ================================================================
==================
 * Process Name: Windows System Logs Tab - Main Script
 * ================================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/system_logs_tab/main.j
s?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/system_logs_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:50:00
 * ================================================================
==================
 */

(function () {
  'use strict';

  let currentChannel = 'System';
  let currentFilePath = '';
  let currentMode = 'live';
  let cachedEvents = [];
  let cachedChannels = [];
  let selectedEvent = null;
  let liveIntervalTimer = null;
  let isRequestInProgress = false;

```

```
33  /**
34   * Escape HTML entities to protect against XSS injection.
35   *
36   * @param {string} str - Raw input text.
37   * @returns {string} Sanitized string.
38   */
39  function escapeHtml(str) {
40    if (!str) return '';
41    return String(str)
42      .replace(/&/g, '&amp;')
43      .replace(/</g, '&lt;')
44      .replace(/>/g, '&gt;')
45      .replace(/"/g, '&quot;')
46      .replace(/'/g, '&#039;');
47  }
48
49  /**
50   * Map log severity to badge styling class.
51   *
52   * @param {string} level - Severity level.
53   * @returns {string} CSS class name.
54   */
55  function getSeverityClass(level) {
56    const lvl = (level || '').toLowerCase();
57    if (lvl.includes('crit')) return 'badge-crit';
58    if (lvl.includes('err')) return 'badge-err';
59    if (lvl.includes('warn')) return 'badge-warn';
60    if (lvl.includes('verb')) return 'badge-verb';
61    return 'badge-inf';
62  }
63
64  /**
65   * Perform dynamic system scan and populate channels tree.
66   *
67   * @param {boolean} force - Force refresh flag.
68   */
69  async function performSystemScan(force = false) {
70    try {
```

```
      const response = await fetch(`/api/v1/system_logs/scan?force=${force}`);
      if (!response.ok) return;
      const data = await response.json();

      cachedChannels = data.channels || [];
      const activeChannels = cachedChannels.filter((c) => (c.record_count || 0) > 0);
      const sourcesBadge = document.getElementById('slc-stat-sources');
      const countBadge = document.getElementById('slc-channel-count-badge');

      if (sourcesBadge) {
        sourcesBadge.textContent = `${activeChannels.length.toLocaleString()} active / ${data.total_sources.toLocaleString()} sources`;
      }
      if (countBadge) {
        countBadge.textContent = String(activeChannels.length);
      }

      renderChannelsTree();
    } catch (err) {
      console.warn('Failed to perform system discovery scan:', err);
    }
  }

  // Singleton global popup tooltip element
  let globalTooltipEl = null;

  function getOrCreateTooltipEl() {
    if (!globalTooltipEl) {
      globalTooltipEl = document.createElement('div');
      globalTooltipEl.className = 'slc-tooltip-popup';
      document.body.appendChild(globalTooltipEl);
```

```
    }
    return globalTooltipEl;
  }

  function showChannelTooltip(e, channel) {
    const tip = getOrCreateTooltipEl();
    const countFormatted = (channel.record_count || 0).toLocal
eString();
    const descText = channel.description || 'Служебный журнал
операционной системы Windows.';
    const typeLabel = channel.source_type === 'channel' ? 'Win
dows Event Channel' : 'Log File';

    tip.innerHTML = `
      <div class="slc-tooltip-title">
        <i class="bi bi-info-circle text-info"></i>
        <span>${escapeHtml(channel.display_name)}</span>
      </div>
      <div style="color: #cbd5e1; margin-bottom: 0.35rem;">${e
scapeHtml(descText)}</div>
      <div class="slc-tooltip-meta">
        <span><strong class="text-info">${countFormatted}</str
ong> записей</span>
        <span class="text-muted font-monospace">${typeLabel}</
span>
      </div>
    `;

    tip.classList.add('show');
    positionTooltip(e, tip);
  }

  function hideChannelTooltip() {
    if (globalTooltipEl) {
      globalTooltipEl.classList.remove('show');
    }
  }

```

```
  function positionTooltip(e, tip) {
    const margin = 12;
    let x = e.clientX + margin;
    let y = e.clientY + margin;

    const tipRect = tip.getBoundingClientRect();
    if (x + tipRect.width > window.innerWidth) {
      x = e.clientX - tipRect.width - margin;
    }
    if (y + tipRect.height > window.innerHeight) {
      y = e.clientY - tipRect.height - margin;
    }

    tip.style.left = `${Math.max(10, x)}px`;
    tip.style.top = `${Math.max(10, y)}px`;
  }

  /**
   * Render channels tree list in sidebar with category grouping, record count badges, and hover tooltips.
   */
  function renderChannelsTree() {
    const container = document.getElementById('slc-channel-tree-container');
    const filterInput = document.getElementById('slc-channel-filter-input');
    const filterText = filterInput ? filterInput.value.trim().toLowerCase() : '';

    if (!container) return;
    container.innerHTML = '';

    const filtered = cachedChannels.filter((c) => {
      // Не выводить каналы и источники, в которых 0 записей
      if (c.record_count === 0) return false;

      if (!filterText) return true;
      return (
```

```
        c.channel_name.toLowerCase().includes(filterText) ||
        c.display_name.toLowerCase().includes(filterText) ||
        (c.description && c.description.toLowerCase().includes(filterText)) ||
        (c.category && c.category.toLowerCase().includes(filterText))
      );
    });

    // Group by category
    const grouped = {};
    filtered.forEach((c) => {
      const cat = c.category || 'Windows Event Log';
      if (!grouped[cat]) grouped[cat] = [];
      grouped[cat].push(c);
    });

    const categoryIcons = {
      'Windows Event Log': '□ ',
      'Windows OS Logs': '□ ',
      'Application Logs': '□ ',
      'AI-Windows-Admin Logs': '□ ',
    };

    Object.keys(grouped).forEach((catName) => {
      const catHeader = document.createElement('div');
      catHeader.className = 'd-flex justify-content-between align-items-center px-2 py-1 mt-2 mb-1 rounded bg-dark border border-secondary text-info small fw-bold';
      const icon = categoryIcons[catName] || '□ ';
      catHeader.innerHTML = `
        <span>${icon} ${escapeHtml(catName)}</span>
        <span class="badge bg-secondary font-monospace">${grouped[catName].length}</span>
      `;
      container.appendChild(catHeader);

      grouped[catName].slice(0, 400).forEach((c) => {
```

```
        const item = document.createElement('div');
        const isActive = (c.channel_name === currentChannel &&
 !currentFilePath) || (currentFilePath && currentFilePath ===
c.location);
        item.className = `slc-tree-item ${isActive ? 'active'
: ''}`;

        let subBadge = '0';
        let badgeStyle = 'bg-dark text-secondary';
        if (c.record_count > 0) {
          subBadge = c.record_count >= 1000 ? `${(c.record_cou
nt / 1000).toFixed(1)}k` : String(c.record_count);
          badgeStyle = c.record_count > 1000 ? 'bg-primary tex
t-white' : 'bg-dark text-info border border-secondary';
        } else if (c.source_type !== 'channel') {
          subBadge = 'FILE';
        }

        item.innerHTML = `
          <span class="text-truncate me-2" style="max-width: 1
75px;">${escapeHtml(c.display_name)}</span>
          <span class="badge ${badgeStyle} font-monospace" sty
le="font-size: 0.72rem; min-width: 32px; text-align: right;">$
{subBadge}</span>
        `;

        // Tooltip events
        item.onmouseenter = (e) => showChannelTooltip(e, c);
        item.onmousemove = (e) => {
          const tip = getOrCreateTooltipEl();
          if (tip.classList.contains('show')) positionTooltip(
e, tip);
        };
        item.onmouseleave = () => hideChannelTooltip();

        item.onclick = () => {
          hideChannelTooltip();
          if (c.source_type === 'channel') {
```

```
            currentChannel = c.channel_name;
            currentFilePath = '';
          } else {
            currentChannel = c.display_name;
            currentFilePath = c.location;
          }
          const statChan = document.getElementById('slc-stat-channel');
          if (statChan) statChan.textContent = c.display_name;
          renderChannelsTree();
          loadEvents();
        };
        container.appendChild(item);
      });
    });
  }

  /**
   * Fetch events matching current channel, filters, and active view mode.
   */
  async function loadEvents() {
    if (isRequestInProgress) return;
    isRequestInProgress = true;

    const levelSelect = document.getElementById('slc-level-select');
    const hoursSelect = document.getElementById('slc-hours-select');
    const searchInput = document.getElementById('slc-search-input');
    const eventIdInput = document.getElementById('slc-eventid-input');
    const limitSelect = document.getElementById('slc-limit-select');
    const tableBody = document.getElementById('slc-table-body');
    const eventsBadge = document.getElementById('slc-stat-even
```

```
ts');
259     const faultsBadge = document.getElementById('slc-stat-inci
dents');
260     const streamBadge = document.getElementById('slc-stream-co
unt-badge');
261 
262     const level = levelSelect ? levelSelect.value : '';
263     const hours = hoursSelect ? hoursSelect.value : '24';
264     const search = searchInput ? searchInput.value.trim() : ''
;
265     const eventId = eventIdInput && eventIdInput.value.trim()
? eventIdInput.value.trim() : '0';
266     const limit = limitSelect ? limitSelect.value : '100';
267 
268     try {
269       const url = `/api/v1/system_logs/events?channel=${encode
URIComponent(currentChannel)}&limit=${limit}&level=${encodeURI
Component(level)}&search=${encodeURIComponent(search)}&hours=$
{hours}&event_id=${eventId}&file_path=${encodeURIComponent(cur
rentFilePath)}`;
270 
271       const res = await fetch(url);
272       if (!res.ok) {
273         if (tableBody) tableBody.innerHTML = `<tr><td colspan=
"6" class="text-center text-danger py-3">Error reading channel
 (HTTP ${res.status})</td></tr>`;
274         isRequestInProgress = false;
275         return;
276       }
277 
278       const data = await res.json();
279       cachedEvents = data.entries || [];
280 
281       if (eventsBadge) eventsBadge.textContent = String(cached
Events.length);
282       if (streamBadge) streamBadge.textContent = `${cachedEven
ts.length} records`;
283 
```

```
      let errCount = 0;
      let critCount = 0;
      cachedEvents.forEach((e) => {
        const lvl = (e.level || '').toLowerCase();
        if (lvl.includes('crit')) critCount++;
        if (lvl.includes('err')) errCount++;
      });

      if (faultsBadge) {
        faultsBadge.textContent = `${errCount + critCount} faults`;
      }

      renderCurrentModeView();
    } catch (err) {
      console.warn('Failed to load events:', err);
      if (tableBody) tableBody.innerHTML = `<tr><td colspan="6" class="text-center text-danger py-3">Network error loading logs.</td></tr>`;
    } finally {
      isRequestInProgress = false;
    }
  }

  /**
   * Render view corresponding to active mode (Live, Errors, Incidents, Timeline, Raw).
   */
  function renderCurrentModeView() {
    const tableContainer = document.getElementById('slc-table-container');
    const incidentsContainer = document.getElementById('slc-incidents-container');
    const timelineContainer = document.getElementById('slc-timeline-container');
    const rawContainer = document.getElementById('slc-raw-container');
    const auditContainer = document.getElementById('slc-audit-
```

```
container');
    const tableBody = document.getElementById('slc-table-body'
);
    const viewTitle = document.getElementById('slc-main-view-t
itle');

    if (!tableContainer || !tableBody) return;

    // Reset visibility
    tableContainer.classList.add('d-none');
    if (incidentsContainer) incidentsContainer.classList.add('
d-none');
    if (timelineContainer) timelineContainer.classList.add('d-
none');
    if (rawContainer) rawContainer.classList.add('d-none');
    if (auditContainer) auditContainer.classList.add('d-none')
;

    if (currentMode === 'audit') {
      if (viewTitle) viewTitle.innerHTML = `<i class="bi bi-sh
ield-check text-info me-2"></i>Интеллектуальный Аудит Логирова
ния (Zero-LLM Overhead)`;
      if (auditContainer) {
        auditContainer.classList.remove('d-none');
        loadAuditView();
      }
    } else if (currentMode === 'rag') {
      if (viewTitle) viewTitle.innerHTML = `<i class="bi bi-br
ain text-info me-2"></i>Динамический RAG-поиск по логам`;
      const ragContainer = document.getElementById('slc-rag-co
ntainer');
      if (ragContainer) {
        ragContainer.classList.remove('d-none');
      }
    } else if (currentMode === 'incidents') {
      if (viewTitle) viewTitle.innerHTML = `<i class="bi bi-di
agram-3 me-2"></i>Correlated Incident Cascades`;
      if (incidentsContainer) {
```

```
341         incidentsContainer.classList.remove('d-none');
342         loadIncidentsView();
343       }
344     } else if (currentMode === 'timeline') {
345       if (viewTitle) viewTitle.innerHTML = `<i class="bi bi-ba
r-chart me-2"></i>Event Frequency Histogram`;
346       if (timelineContainer) {
347         timelineContainer.classList.remove('d-none');
348         loadTimelineView();
349       }
350     } else if (currentMode === 'raw') {
351       if (viewTitle) viewTitle.innerHTML = `<i class="bi bi-co
de-square me-2"></i>Raw Event Payload Data`;
352       if (rawContainer) {
353         rawContainer.classList.remove('d-none');
354         const rawText = document.getElementById('slc-raw-text'
);
355         if (rawText) {
356           rawText.textContent = selectedEvent ? (selectedEvent
.raw_data || JSON.stringify(selectedEvent, null, 2)) : JSON.st
ringify(cachedEvents.slice(0, 10), null, 2);
357         }
358       }
359     } else {
360       // Live or Errors Table Mode
361       tableContainer.classList.remove('d-none');
362       if (viewTitle) {
363         viewTitle.innerHTML = currentMode === 'errors'
364           ? `<i class="bi bi-exclamation-octagon text-danger m
e-2"></i>Critical & Error Events (${escapeHtml(currentChannel)
})`
365           : `<i class="bi bi-list-columns-reverse me-2"></i>Li
ve Windows Event Stream (${escapeHtml(currentChannel)})`;
366       }
367
368       let entriesToRender = cachedEvents;
369       if (currentMode === 'errors') {
370         entriesToRender = cachedEvents.filter((e) => {
```

```
371           const l = (e.level || '').toLowerCase();
372           return l.includes('err') || l.includes('crit') || l.
includes('warn');
373         });
374       }
375
376       if (entriesToRender.length === 0) {
377         tableBody.innerHTML = `<tr><td colspan="6" class="text
-center text-muted py-4">No events found matching active filte
r.</td></tr>`;
378         return;
379       }
380
381       tableBody.innerHTML = entriesToRender.map((e, idx) => {
382         const badgeClass = getSeverityClass(e.level);
383         const eidStr = e.event_id > 0 ? String(e.event_id) : '
-';
384         const pidStr = e.process_id > 0 ? String(e.process_id)
 : '-';
385         const isSelected = selectedEvent && (
386           selectedEvent.timestamp === e.timestamp &&
387           selectedEvent.event_id === e.event_id &&
388           (selectedEvent.provider || selectedEvent.source) ===
 (e.provider || e.source)
389         );
390         return `
391           <tr data-index="${idx}" class="${isSelected ? 'selec
ted table-active' : ''}">
392             <td class="font-monospace text-secondary text-nowr
ap">${escapeHtml(e.timestamp)}</td>
393             <td><span class="badge ${badgeClass} text-uppercas
e px-2 py-1">${escapeHtml(e.level)}</span></td>
394             <td class="text-center font-monospace">${escapeHtm
l(eidStr)}</td>
395             <td class="text-truncate text-info" style="max-wid
th: 160px;" title="${escapeHtml(e.provider || e.source)}">${es
capeHtml(e.provider || e.source)}</td>
396             <td class="text-center font-monospace text-muted">
```

```
${escapeHtml(pidStr)}</td>
            <td class="font-monospace text-light" style="word-break: break-word;">${escapeHtml(e.message)}</td>
          </tr>
        `;
      }).join('');

      // Attach row click listeners for row selection and detail inspection
      tableBody.querySelectorAll('tr').forEach((row) => {
        row.onclick = () => {
          const idx = parseInt(row.getAttribute('data-index') || '0', 10);
          selectedEvent = entriesToRender[idx];
          tableBody.querySelectorAll('tr').forEach(r => r.classList.remove('selected', 'table-active'));
          row.classList.add('selected', 'table-active');
          showEventModal(selectedEvent);
        };
      });
    }
  }

  /**
   * Fetch and render correlated incident clusters.
   */
  async function loadIncidentsView() {
    const listEl = document.getElementById('slc-incidents-list');
    if (!listEl) return;
    listEl.innerHTML = `<div class="text-center py-4 text-muted"><div class="spinner-border spinner-border-sm text-info me-2"></div>Correlating incident cascades...</div>`;

    try {
      const res = await fetch('/api/v1/system_logs/incidents?window_seconds=25.0');
      if (!res.ok) return;
```

```
      const data = await res.json();
      const incidents = data.incidents || [];

      if (incidents.length === 0) {
        listEl.innerHTML = `<div class="card bg-dark border-se
condary p-4 text-center text-muted">No fault cascades or corre
lated incidents detected in the active timeframe.</div>`;
        return;
      }

      listEl.innerHTML = incidents.map((inc) => `
        <div class="card bg-dark border-secondary mb-3 shadow-
sm">
          <div class="card-header bg-black d-flex justify-cont
ent-between align-items-center">
            <span class="fw-bold text-info"><i class="bi bi-li
ghtning-charge me-1"></i>${escapeHtml(inc.title)}</span>
            <span class="badge ${getSeverityClass(inc.severity
)}">${escapeHtml(inc.severity)}</span>
          </div>
          <div class="card-body">
            <div class="row g-2 mb-2 small text-muted font-mon
ospace">
              <div class="col-sm-4">Window: ${escapeHtml(inc.s
tart_time)} (${inc.duration_seconds}s)</div>
              <div class="col-sm-4">Primary: ${escapeHtml(inc.
primary_source)}</div>
              <div class="col-sm-4">Events Count: ${inc.events
_count}</div>
            </div>
            <p class="small text-light mb-2"><strong class="te
xt-warning">Root Cause:</strong> ${escapeHtml(inc.root_cause_h
int)}</p>
            <p class="small text-muted mb-0">${escapeHtml(inc.
summary)}</p>
          </div>
        </div>
      `).join('');
```

```
    } catch (err) {
      listEl.innerHTML = `<div class="alert alert-danger">Erro
r retrieving incidents: ${err.message}</div>`;
    }
  }

  let timelineSubView = 'both';
  let selectedTimelineBucket = null;
  let cachedTimelineData = null;

  /**
   * Fetch and render GitHub-style timeline & activity heatmap
 for the selected channel.
   */
  async function loadTimelineView() {
    const contentEl = document.getElementById('slc-timeline-co
ntent');
    if (!contentEl) return;

    const hoursSelect = document.getElementById('slc-hours-sel
ect');
    const limitSelect = document.getElementById('slc-limit-sel
ect');
    const levelSelect = document.getElementById('slc-level-sel
ect');
    const searchInput = document.getElementById('slc-search-in
put');
    const eventIdInput = document.getElementById('slc-eventid-
input');

    const hours = hoursSelect ? hoursSelect.value : '24';
    const limit = limitSelect ? limitSelect.value : '500';
    const level = levelSelect ? levelSelect.value : '';
    const search = searchInput ? searchInput.value.trim() : ''
;
    const eventId = eventIdInput && eventIdInput.value.trim()
? eventIdInput.value.trim() : '0';

```

```
    contentEl.innerHTML = `
      <div class="text-center py-5 text-muted">
        <div class="spinner-border text-info mb-3" role="statu
s"></div>
        <div class="fw-bold text-light">Построение временного
графика активности логов...</div>
        <div class="small text-secondary mt-1">Агрегация по вр
емени • Анализ частоты • Тепловая карта активности</div>
      </div>
    `;

    try {
      const url = `/api/v1/system_logs/timeline?channel=${enco
deURIComponent(currentChannel)}&hours=${hours}&limit=${limit}&
level=${encodeURIComponent(level)}&search=${encodeURIComponent
(search)}&event_id=${eventId}&file_path=${encodeURIComponent(c
urrentFilePath)}`;
      const res = await fetch(url);
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      cachedTimelineData = data;
      selectedTimelineBucket = null;

      renderTimelineInterface(data);
    } catch (err) {
      contentEl.innerHTML = `<div class="alert alert-danger">O
шибка загрузки таймлайна: ${escapeHtml(err.message)}</div>`;
    }
  }

  /**
   * Render the complete GitHub-style timeline UI.
   *
   * @param {Object} data - Timeline response payload.
   */
  function renderTimelineInterface(data) {
    const contentEl = document.getElementById('slc-timeline-co
ntent');
```

```
    if (!contentEl || !data) return;

    const summary = data.summary || {};
    const histogram = data.histogram || [];
    const heatmap = data.heatmap || [];
    const topProviders = data.top_providers || [];

    const totalEvents = summary.total_events || 0;
    const critCount = summary.critical_count || 0;
    const errCount = summary.error_count || 0;
    const warnCount = summary.warning_count || 0;
    const infoCount = summary.info_count || 0;
    const peakTime = summary.peak_time || 'N/A';
    const peakCount = summary.peak_count || 0;

    const maxBucketTotal = Math.max(...histogram.map((h) => h.total), 1);

    // Day names for 7x24 heatmap (Mon-Sun)
    const dayNames = ['Пн', 'Вт', 'Ср', 'Чт', 'Пт', 'Сб', 'Вс'];
    const hourLabels = Array.from({ length: 24 }, (_, i) => `${String(i).padStart(2, '0')}:00`);

    // Top Providers badges
    const providersHtml = topProviders.map((tp) => `
      <span class="badge bg-dark border border-secondary text-info px-2 py-1 font-monospace" style="font-size: 0.72rem;">
        ${escapeHtml(tp.provider)}: <strong class="text-light">${tp.count}</strong>
      </span>
    `).join('') || '<span class="text-muted small">Нет данных</span>';

    // Build Pulse Bar Chart HTML
    const pulseBarsHtml = histogram.map((h, idx) => {
      const heightPct = Math.max(10, (h.total / maxBucketTotal) * 100);
```

```
539       const isSelected = selectedTimelineBucket && selectedTim
elineBucket.time === h.time;
540
541       const critPct = h.total > 0 ? (h.critical / h.total) * 1
00 : 0;
542       const errPct = h.total > 0 ? (h.errors / h.total) * 100
: 0;
543       const warnPct = h.total > 0 ? (h.warnings / h.total) * 1
00 : 0;
544       const infoPct = h.total > 0 ? (h.info / h.total) * 100 :
 0;
545
546       const timeLabel = h.time.length > 10 ? h.time.slice(11,
16) || h.time.slice(5, 10) : h.time;
547
548       return `
549         <div class="gh-pulse-col ${isSelected ? 'active' : ''}
" data-index="${idx}" title="${escapeHtml(h.time)} — Всего: ${
h.total} (Крит: ${h.critical}, Ошибок: ${h.errors}, Предупр: $
{h.warnings}, Инфо: ${h.info})">
550           <span class="gh-col-count">${h.total > 0 ? h.total :
 ''}</span>
551           <div class="gh-bar-track" style="height: ${heightPct
}%;">
552             ${infoPct > 0 ? `<div class="gh-seg-info" style="h
eight: ${infoPct}%;"></div>` : ''}
553             ${warnPct > 0 ? `<div class="gh-seg-warn" style="h
eight: ${warnPct}%;"></div>` : ''}
554             ${errPct > 0 ? `<div class="gh-seg-err" style="hei
ght: ${errPct}%;"></div>` : ''}
555             ${critPct > 0 ? `<div class="gh-seg-crit" style="h
eight: ${critPct}%;"></div>` : ''}
556           </div>
557           <span class="gh-col-label">${escapeHtml(timeLabel)}<
/span>
558         </div>
559       `;
560     }).join('');
```

```
561
562     // Build 7x24 Heatmap Matrix HTML
563     let heatmapRowsHtml = '';
564     if (heatmap && heatmap.length === 7) {
565       heatmapRowsHtml = heatmap.map((row, dayIdx) => {
566         const cellsHtml = row.map((cell, hourIdx) => {
567           const hasErr = (cell.errors || 0) > 0;
568           return `
569             <div class="gh-heat-cell lvl-${cell.level} ${hasEr
    r ? 'has-err' : ''}" data-day="${dayIdx}" data-hour="${hourIdx
    }" title="${dayNames[dayIdx]} ${hourLabels[hourIdx]}: ${cell.c
    ount} событий (${cell.errors || 0} ошибок, ${cell.warnings ||
    0} предупр.)"></div>
570           `;
571         }).join('');
572
573         return `
574           <div class="d-flex align-items-center gap-2 mb-1">
575             <span class="text-secondary small font-monospace t
    ext-end" style="width: 24px; font-size: 0.7rem;">${dayNames[da
    yIdx]}</span>
576             <div class="d-flex gap-1 flex-grow-1">
577               ${cellsHtml}
578             </div>
579           </div>
580         `;
581       }).join('');
582     }
583
584     // Determine events to show in drilldown table
585     let drilldownEvents = [];
586     let drilldownTitle = 'Все события в интервале графика';
587     if (selectedTimelineBucket) {
588       drilldownEvents = selectedTimelineBucket.events || [];
589       drilldownTitle = `События интервала: ${escapeHtml(select
    edTimelineBucket.time)} (${drilldownEvents.length} записей)`;
590     } else {
591       histogram.forEach((h) => {
```

```
        if (h.events) drilldownEvents.push(...h.events);
      });
      drilldownEvents = drilldownEvents.slice(0, 50);
    }

    const drilldownRowsHtml = drilldownEvents.map((e, idx) =>
{
      const badgeClass = getSeverityClass(e.level);
      const eidStr = e.event_id > 0 ? String(e.event_id) : '-'
;
      return `
        <tr data-index="${idx}" class="cursor-pointer">
          <td class="font-monospace text-secondary text-nowrap
">${escapeHtml(e.timestamp)}</td>
          <td><span class="badge ${badgeClass} text-uppercase
px-2 py-1">${escapeHtml(e.level)}</span></td>
          <td class="text-center font-monospace">${escapeHtml(
eidStr)}</td>
          <td class="text-truncate text-info" style="max-width
: 150px;" title="${escapeHtml(e.provider)}">${escapeHtml(e.pro
vider)}</td>
          <td class="font-monospace text-light text-truncate"
style="max-width: 320px;" title="${escapeHtml(e.message)}">${e
scapeHtml(e.message)}</td>
        </tr>
      `;
    }).join('') || `<tr><td colspan="5" class="text-center tex
t-muted py-3">Нет событий для отображения.</td></tr>`;

    contentEl.innerHTML = `
      <div class="gh-timeline-container shadow-sm">
        <!-- Header & KPI Bar -->
        <div class="d-flex justify-content-between align-items
-center flex-wrap gap-2 mb-3 pb-2 border-bottom border-seconda
ry">
          <div class="d-flex align-items-center gap-2">
            <span class="fs-5">□ </span>
            <div>
```

```
              <div class="fw-bold text-light small">Временной график активности: <span class="text-info">${escapeHtml(data.channel || currentChannel)}</span></div>
              <div class="text-muted small" style="font-size: 0.72rem;">Визуализация частоты событий и инцидентов во времени (GitHub Activity Style)</div>
            </div>
          </div>
          <!-- View Switcher -->
          <div class="btn-group btn-group-sm" role="group">
            <button type="button" class="btn btn-outline-secondary ${timelineSubView === 'pulse' ? 'active' : ''}" id="btn-gh-view-pulse" style="font-size: 0.75rem;">
              <i class="bi bi-bar-chart-fill me-1"></i>Столбцы (Pulse)
            </button>
            <button type="button" class="btn btn-outline-secondary ${timelineSubView === 'heatmap' ? 'active' : ''}" id="btn-gh-view-heatmap" style="font-size: 0.75rem;">
              <i class="bi bi-grid-3x3-gap-fill me-1"></i>Сетка (Heatmap)
            </button>
            <button type="button" class="btn btn-outline-secondary ${timelineSubView === 'both' ? 'active' : ''}" id="btn-gh-view-both" style="font-size: 0.75rem;">
              <i class="bi bi-layout-split me-1"></i>Оба вида
            </button>
          </div>
        </div>

        <!-- Metric KPI Cards -->
        <div class="row g-2 mb-3">
          <div class="col-sm-2 col-6">
            <div class="gh-stat-card text-center">
              <div class="text-muted small" style="font-size: 0.7rem;">Всего событий</div>
              <div class="fs-5 fw-bold text-light font-monospace">${totalEvents}</div>
```

```
            </div>
          </div>
          <div class="col-sm-2 col-6">
            <div class="gh-stat-card text-center">
              <div class="text-muted small" style="font-size:
0.7rem;">Ошибки / Крит</div>
              <div class="fs-5 fw-bold text-danger font-monosp
ace">${critCount + errCount}</div>
            </div>
          </div>
          <div class="col-sm-2 col-6">
            <div class="gh-stat-card text-center">
              <div class="text-muted small" style="font-size:
0.7rem;">Предупреждения</div>
              <div class="fs-5 fw-bold text-warning font-monos
pace">${warnCount}</div>
            </div>
          </div>
          <div class="col-sm-2 col-6">
            <div class="gh-stat-card text-center">
              <div class="text-muted small" style="font-size:
0.7rem;">Информация</div>
              <div class="fs-5 fw-bold text-success font-monos
pace">${infoCount}</div>
            </div>
          </div>
          <div class="col-sm-4 col-12">
            <div class="gh-stat-card">
              <div class="text-muted small" style="font-size:
0.7rem;">Пиковая активность</div>
              <div class="d-flex justify-content-between align
-items-center">
                <span class="text-info font-monospace small">$
{escapeHtml(peakTime)}</span>
                <span class="badge bg-danger rounded-pill">${p
eakCount} соб.</span>
              </div>
            </div>
```

```
          </div>
        </div>

        <!-- Section 1: GitHub Pulse (Activity Bar Chart) -->
        <div id="gh-section-pulse" class="${timelineSubView === 'heatmap' ? 'd-none' : ''} mb-4">
          <div class="d-flex justify-content-between align-items-center mb-2">
            <div class="small fw-bold text-light"><i class="bi bi-activity text-info me-1"></i>Гистограмма частоты событий (Pulse Timeline)</div>
            <div class="d-flex align-items-center gap-3 small" style="font-size: 0.72rem;">
              <span><span class="d-inline-block rounded-circle me-1" style="width: 8px; height: 8px; background: #238636;"></span>Инфо</span>
              <span><span class="d-inline-block rounded-circle me-1" style="width: 8px; height: 8px; background: #d29922;"></span>Предупр.</span>
              <span><span class="d-inline-block rounded-circle me-1" style="width: 8px; height: 8px; background: #da3633;"></span>Ошибки</span>
              <span><span class="d-inline-block rounded-circle me-1" style="width: 8px; height: 8px; background: #8e1515;"></span>Крит.</span>
            </div>
          </div>
          <div class="gh-pulse-chart" id="gh-pulse-chart-container">
            ${pulseBarsHtml}
          </div>
          <div class="text-muted small mt-2" style="font-size: 0.7rem;">
            🖱 Нажмите на любой столбец гистограммы, чтобы детализировать события выбранного временного слота.
          </div>
        </div>

```

```
692         <!-- Section 2: GitHub 7x24 Contribution Heatmap Matri
    x -->
693         <div id="gh-section-heatmap" class="${timelineSubView
    === 'pulse' ? 'd-none' : ''} mb-4">
694           <div class="d-flex justify-content-between align-ite
    ms-center mb-2">
695             <div class="small fw-bold text-light"><i class="bi
     bi-grid-3x3-gap text-info me-1"></i>Недельная матрица активно
    сти (GitHub Contribution Heatmap: 24h × 7 дней)</div>
696             <div class="d-flex align-items-center gap-1 small"
     style="font-size: 0.7rem;">
697               <span class="text-muted me-1">Меньше</span>
698               <span class="gh-heat-cell lvl-0 d-inline-block"
    style="width: 11px; height: 11px;"></span>
699               <span class="gh-heat-cell lvl-1 d-inline-block"
    style="width: 11px; height: 11px;"></span>
700               <span class="gh-heat-cell lvl-2 d-inline-block"
    style="width: 11px; height: 11px;"></span>
701               <span class="gh-heat-cell lvl-3 d-inline-block"
    style="width: 11px; height: 11px;"></span>
702               <span class="gh-heat-cell lvl-4 d-inline-block"
    style="width: 11px; height: 11px;"></span>
703               <span class="text-muted ms-1">Больше</span>
704             </div>
705           </div>
706           <div class="p-2 rounded bg-black border border-secon
    dary overflow-auto">
707             <!-- Hour headers -->
708             <div class="d-flex align-items-center gap-2 mb-1 p
    s-4 text-muted font-monospace" style="font-size: 0.65rem;">
709               <span>00:00</span>
710               <span class="ms-auto">06:00</span>
711               <span class="ms-auto">12:00</span>
712               <span class="ms-auto">18:00</span>
713               <span class="ms-auto">23:00</span>
714             </div>
715             ${heatmapRowsHtml}
716           </div>
```

```
        </div>

        <!-- Section 3: Top Providers Pills -->
        <div class="mb-3">
          <div class="small fw-bold text-light mb-1"><i class="bi bi-building me-1 text-info"></i>Ключевые поставщики в выборке:</div>
          <div class="d-flex flex-wrap gap-1">
            ${providersHtml}
          </div>
        </div>

        <!-- Section 4: Drilldown Events List -->
        <div class="card bg-black border-secondary">
          <div class="card-header bg-dark border-secondary d-flex justify-content-between align-items-center py-1.5 px-3">
            <span class="text-light small fw-bold"><i class="bi bi-list-nested me-1 text-info"></i>${escapeHtml(drilldownTitle)}</span>
            ${selectedTimelineBucket ? `
              <button class="btn btn-xs btn-outline-info rounded-pill py-0 px-2" id="btn-gh-reset-bucket" style="font-size: 0.72rem;">
                <i class="bi bi-x-circle me-1"></i>Сбросить фильтр
              </button>
            ` : ''}
          </div>
          <div class="table-responsive" style="max-height: 240px; overflow-y: auto;">
            <table class="table slc-table table-sm mb-0">
              <thead>
                <tr>
                  <th style="width: 140px;">Время</th>
                  <th style="width: 80px;">Уровень</th>
                  <th style="width: 60px;">ID</th>
                  <th style="width: 150px;">Поставщик</th>
                  <th>Сообщение</th>
```

```
                </tr>
              </thead>
              <tbody id="gh-drilldown-table-body">
                ${drilldownRowsHtml}
              </tbody>
            </table>
          </div>
        </div>
      </div>
    `;

    // Event listeners for view switcher buttons
    const btnPulse = document.getElementById('btn-gh-view-puls
e');
    const btnHeatmap = document.getElementById('btn-gh-view-he
atmap');
    const btnBoth = document.getElementById('btn-gh-view-both'
);

    if (btnPulse) {
      btnPulse.onclick = () => {
        timelineSubView = 'pulse';
        renderTimelineInterface(data);
      };
    }
    if (btnHeatmap) {
      btnHeatmap.onclick = () => {
        timelineSubView = 'heatmap';
        renderTimelineInterface(data);
      };
    }
    if (btnBoth) {
      btnBoth.onclick = () => {
        timelineSubView = 'both';
        renderTimelineInterface(data);
      };
    }

```

```
    // Reset bucket filter button
    const btnResetBucket = document.getElementById('btn-gh-res
et-bucket');
    if (btnResetBucket) {
      btnResetBucket.onclick = () => {
        selectedTimelineBucket = null;
        renderTimelineInterface(data);
      };
    }

    // Column click events for bar chart
    contentEl.querySelectorAll('.gh-pulse-col').forEach((col)
=> {
      col.onclick = () => {
        const idx = parseInt(col.getAttribute('data-index') ||
 '0', 10);
        selectedTimelineBucket = histogram[idx];
        renderTimelineInterface(data);
      };
    });

    // Row click listeners in drilldown table to open event de
tails modal
    const drilldownBody = document.getElementById('gh-drilldow
n-table-body');
    if (drilldownBody) {
      drilldownBody.querySelectorAll('tr').forEach((row) => {
        row.onclick = () => {
          const idx = parseInt(row.getAttribute('data-index')
|| '0', 10);
          const ev = drilldownEvents[idx];
          if (ev) {
            showEventModal(ev);
          }
        };
      });
    }
  }
```

```
813
814   /**
815    * Fetch and render multi-level log audit digest (Noise filt
ering, clustering, anomalies).
816    */
817   async function loadAuditView() {
818     const contentEl = document.getElementById('slc-audit-conte
nt');
819     if (!contentEl) return;
820
821     const hoursSelect = document.getElementById('slc-hours-sel
ect');
822     const limitSelect = document.getElementById('slc-limit-sel
ect');
823     const hours = hoursSelect ? hoursSelect.value : '24';
824     const limit = limitSelect ? limitSelect.value : '500';
825
826     contentEl.innerHTML = `
827       <div class="text-center py-5 text-muted">
828         <div class="spinner-border text-info mb-3" role="statu
s"></div>
829         <div class="fw-bold text-light">Выполняется многоуровн
евый аудит массива логов...</div>
830         <div class="small text-secondary mt-1">Фильтрация шума
 • Шаблонизация • Детекция аномалий • Сжатие данных</div>
831       </div>
832     `;
833
834     try {
835       const url = `/api/v1/system_logs/audit?channel=${encodeU
RIComponent(currentChannel)}&limit=${limit}&hours=${hours}&fil
e_path=${encodeURIComponent(currentFilePath)}`;
836       const res = await fetch(url);
837       if (!res.ok) throw new Error(`HTTP ${res.status}`);
838       const data = await res.json();
839
840       const anomaliesHtml = (data.anomalies || []).map((a) =>
{
```

```
        const badgeClass = getSeverityClass(a.level);
        return `
          <div class="p-2 mb-2 rounded bg-black border border-secondary d-flex justify-content-between align-items-center flex-wrap gap-2">
            <div>
              <span class="badge ${badgeClass} text-uppercase me-2">${escapeHtml(a.level)}</span>
              <strong class="text-info">${escapeHtml(a.provider)}</strong>
              <span class="text-muted font-monospace small">(ID ${a.event_id})</span>
              <div class="small text-light mt-1 font-monospace" style="word-break: break-word;">${escapeHtml(a.sample)}</div>
              <div class="small text-secondary mt-1"><i class="bi bi-clock me-1"></i>Окно: ${escapeHtml(a.time_window)}</div>
            </div>
            <div class="text-end">
              <span class="badge bg-danger rounded-pill px-3 py-2 fs-6">${a.count}x</span>
            </div>
          </div>
        `;
      }).join('') || `<div class="text-success small p-2"><i class="bi bi-check-circle me-1"></i>Аномалий и критических сбоев не обнаружено.</div>`;

      const clustersHtml = (data.top_clusters || []).map((cl, idx) => {
        const badgeClass = getSeverityClass(cl.level);
        return `
          <tr class="small">
            <td class="text-center font-monospace">${idx + 1}</td>
            <td><span class="badge ${badgeClass}">${escapeHtml(cl.level)}</span></td>
```

```
            <td class="text-info text-truncate" style="max-wid
th: 140px;" title="${escapeHtml(cl.provider)}">${escapeHtml(cl
.provider)}</td>
            <td class="text-center font-monospace">${cl.event_
id || '-'}</td>
            <td class="font-monospace text-light" style="max-w
idth: 320px; word-break: break-word;" title="${escapeHtml(cl.t
emplate)}">${escapeHtml(cl.template)}</td>
            <td class="text-center"><span class="badge bg-seco
ndary font-monospace">${cl.count}</span></td>
            <td class="small text-secondary font-monospace tex
t-nowrap">${escapeHtml(cl.first_seen)}</td>
          </tr>
        `;
      }).join('');

      const totalScanned = data.total_analyzed ?? data.total_s
canned ?? 0;
      const uniqueCount = data.unique_patterns_count ?? (data.
top_clusters ? data.top_clusters.length : 0);
      const redundancyPct = data.redundancy_pct !== undefined
? `${data.redundancy_pct.toFixed(1)}%` : (data.compression_rat
io || '0%');
      const errSum = (data.error_count || 0) + (data.critical_
count || 0);
      const warnSum = data.warning_count || 0;
      const summaryText = data.strategy_rationale || data.exec
utive_summary || `Стратегия обработки: ${data.strategy || 'Ста
ндартная'}. Найдено ${uniqueCount} ключевых шаблонов.`;

      contentEl.innerHTML = `
        <!-- Executive Summary Card (Compact) -->
        <div class="card bg-dark border-info mb-2 shadow-sm">
          <div class="card-header bg-black text-info fw-bold p
y-1 px-3 d-flex justify-content-between align-items-center sma
ll">
            <span><i class="bi bi-cpu me-1"></i>Сводка аудита:
 ${escapeHtml(data.channel)}</span>
```

```
            <span class="badge bg-info text-dark font-monospac
e">Дублирование: ${redundancyPct}</span>
          </div>
          <div class="card-body p-2">
            <div class="row g-2 text-center mb-2">
              <div class="col-sm-3 col-6">
                <div class="p-1 rounded bg-black border border
-secondary">
                  <div class="small text-muted" style="font-si
ze: 0.72rem;">Просканировано</div>
                  <div class="fs-5 fw-bold text-light font-mon
ospace">${totalScanned}</div>
                </div>
              </div>
              <div class="col-sm-3 col-6">
                <div class="p-1 rounded bg-black border border
-secondary">
                  <div class="small text-muted" style="font-si
ze: 0.72rem;">Шаблонов</div>
                  <div class="fs-5 fw-bold text-info font-mono
space">${uniqueCount}</div>
                </div>
              </div>
              <div class="col-sm-3 col-6">
                <div class="p-1 rounded bg-black border border
-secondary">
                  <div class="small text-muted" style="font-si
ze: 0.72rem;">Ошибок / Сбоев</div>
                  <div class="fs-5 fw-bold text-danger font-mo
nospace">${errSum}</div>
                </div>
              </div>
              <div class="col-sm-3 col-6">
                <div class="p-1 rounded bg-black border border
-secondary">
                  <div class="small text-muted" style="font-si
ze: 0.72rem;">Предупреждений</div>
                  <div class="fs-5 fw-bold text-warning font-m
```

```
onospace">${warnSum}</div>
                </div>
              </div>
            </div>
            <p class="mb-0 text-light small"><i class="bi bi-i
nfo-circle text-info me-1"></i>${escapeHtml(summaryText)}</p>
          </div>
        </div>

        <!-- Detected Anomalies Section -->
        <div class="card bg-dark border-secondary mb-2 shadow-
sm">
          <div class="card-header bg-black text-warning fw-bol
d py-1 px-3 small">
            <i class="bi bi-exclamation-triangle me-1"></i>Обн
аруженные аномалии и всплески
          </div>
          <div class="card-body p-2">
            ${anomaliesHtml}
          </div>
        </div>

        <!-- Top Clustered Patterns Table -->
        <div class="card bg-dark border-secondary">
          <div class="card-header bg-black text-light fw-bold
d-flex justify-content-between align-items-center">
            <span><i class="bi bi-collection me-2"></i>Топ шаб
лонов событий (Дедупликация)</span>
            <span class="small text-muted">Топ 10 кластеров</s
pan>
          </div>
          <div class="table-responsive">
            <table class="table slc-table mb-0">
              <thead>
                <tr>
                  <th style="width: 40px;">#</th>
                  <th style="width: 80px;">Уровень</th>
                  <th style="width: 140px;">Поставщик</th>
```

```
                  <th style="width: 60px;">ID</th>
                  <th>Шаблон сообщения (Masked)</th>
                  <th style="width: 70px;" class="text-center"
>Кол-во</th>
                  <th style="width: 140px;">Впервые</th>
                </tr>
              </thead>
              <tbody>
                ${clustersHtml}
              </tbody>
            </table>
          </div>
        </div>
      `;
    } catch (err) {
      contentEl.innerHTML = `<div class="alert alert-danger">О
шибка проведения аудита: ${escapeHtml(err.message)}</div>`;
    }
  }

  /**
   * Execute dynamic RAG search over indexed patterns, anomali
es, and digests.
   */
  async function performRAGSearch() {
    const inputEl = document.getElementById('slc-rag-input');
    const resultsEl = document.getElementById('slc-rag-results
');
    if (!inputEl || !resultsEl) return;

    const query = inputEl.value.trim();
    if (!query) return;

    resultsEl.innerHTML = `<div class="text-center py-4 text-m
uted"><div class="spinner-border spinner-border-sm text-info m
e-2"></div>Поиск по динамическому RAG-индексу...</div>`;

    try {
```

```
      const res = await fetch('/api/v1/system_logs/rag/query', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ query: query, channel: currentChannel, top_k: 6 }),
      });
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
      const results = data.results || [];

      if (results.length === 0) {
        resultsEl.innerHTML = `
          <div class="card bg-dark border-secondary p-4 text-center text-muted">
            По запросу «${escapeHtml(query)}» ничего не найдено в RAG-индексе. Попробуйте нажать «Перестроить RAG-индекс».
          </div>
        `;
        return;
      }

      resultsEl.innerHTML = results.map((r) => {
        const badgeClass = getSeverityClass(r.meta?.level || 'info');
        const scorePct = Math.min(100, Math.round((r.relevance_score || 0) * 40));
        return `
          <div class="card bg-dark border-secondary mb-3 shadow-sm">
            <div class="card-header bg-black d-flex justify-content-between align-items-center">
              <span class="fw-bold text-info"><i class="bi bi-file-earmark-text me-2"></i>${escapeHtml(r.title)}</span>
              <div class="d-flex align-items-center gap-2">
                <span class="badge ${badgeClass}">${escapeHtml(r.meta?.level || r.type)}</span>
                <span class="badge bg-secondary font-monospace
```

```
">Score: ${r.relevance_score}</span>
              </div>
            </div>
            <div class="card-body">
              <pre class="bg-black text-light p-3 rounded font-monospace small mb-0" style="white-space: pre-wrap; word-break: break-word;">${escapeHtml(r.text)}</pre>
            </div>
          </div>
        `;
      }).join('');
    } catch (err) {
      resultsEl.innerHTML = `<div class="alert alert-danger">Ошибка поиска RAG: ${escapeHtml(err.message)}</div>`;
    }
  }

  /**
   * Trigger building or rebuilding dynamic RAG index for current channel.
   */
  async function rebuildRAGIndex() {
    const resultsEl = document.getElementById('slc-rag-results');
    if (!resultsEl) return;
    resultsEl.innerHTML = `<div class="text-center py-4 text-muted"><div class="spinner-border spinner-border-sm text-info me-2"></div>Перестроение динамического RAG-индекса для «${escapeHtml(currentChannel)}»...</div>`;

    try {
      const res = await fetch(`/api/v1/system_logs/rag/build?channel=${encodeURIComponent(currentChannel)}&limit=500&hours=24&file_path=${encodeURIComponent(currentFilePath)}`, {
        method: 'POST',
      });
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();
```

```
      resultsEl.innerHTML = `
        <div class="alert alert-success d-flex align-items-cen
ter gap-2">
          <i class="bi bi-check-circle-fill fs-5"></i>
          <div>
            <strong>RAG-индекс успешно обновлен!</strong> Доба
влено ${data.chunks_added} чанков (Всего в индексе: ${data.tot
al_index_chunks}).
            <div class="small mt-1 text-light">${escapeHtml(da
ta.executive_summary)}</div>
          </div>
        </div>
      `;
    } catch (err) {
      resultsEl.innerHTML = `<div class="alert alert-danger">O
шибка перестроения RAG: ${escapeHtml(err.message)}</div>`;
    }
  }

  /**
   * Display event details modal with AI diagnostic action.
   *
   * @param {object} entry - Event object.
   */
  function showEventModal(entry) {
    if (!entry) return;
    selectedEvent = entry;

    const modalEl = document.getElementById('slc-event-detail-
modal');
    if (!modalEl) return;

    document.getElementById('slc-modal-time').textContent = en
try.timestamp;
    document.getElementById('slc-modal-level').innerHTML = `<s
pan class="badge ${getSeverityClass(entry.level)}">${escapeHtm
l(entry.level)}</span>`;
    document.getElementById('slc-modal-id').textContent = entr
```

```
y.event_id > 0 ? String(entry.event_id) : '-';
1057    document.getElementById('slc-modal-provider').textContent
= entry.provider || entry.source;
1058    document.getElementById('slc-modal-computer').textContent
= entry.computer || 'Localhost';
1059    document.getElementById('slc-modal-pid').textContent = ent
ry.process_id > 0 ? String(entry.process_id) : '-';
1060    document.getElementById('slc-modal-channel').textContent =
 entry.channel || currentChannel;
1061    document.getElementById('slc-modal-message').textContent =
 entry.message;
1062
1063    const aiExplanation = document.getElementById('slc-modal-a
i-explanation');
1064    if (aiExplanation) {
1065      aiExplanation.innerHTML = `Нажмите «Анализ контекста», ч
тобы провести диагностику выбранной записи и сопутствующих соб
ытий.`;
1066    }
1067
1068    const diagnoseBtn = document.getElementById('btn-slc-modal
-diagnose');
1069    if (diagnoseBtn) {
1070      diagnoseBtn.onclick = async () => {
1071        aiExplanation.innerHTML = `<div class="spinner-border
spinner-border-sm text-info me-2"></div>Анализ выбранной запис
и...`;
1072        try {
1073          const res = await fetch('/api/v1/system_logs/explain
', {
1074            method: 'POST',
1075            headers: { 'Content-Type': 'application/json' },
1076            body: JSON.stringify({
1077              target_timestamp: entry.timestamp,
1078              window_minutes: 5,
1079              channel: entry.channel || currentChannel,
1080              event_id: entry.event_id || 0,
1081              provider: entry.provider || entry.source || '',
```

```
              level: entry.level || '',
              message: entry.message || '',
              target_entry: entry,
              query_text: `Диагностика события ${entry.provide
r || entry.source} (ID ${entry.event_id || 0}): ${entry.messag
e || ''}`,
            }),
          });
          if (!res.ok) throw new Error(`HTTP ${res.status}`);
          const report = await res.json();
          aiExplanation.innerHTML = `
            <div class="mb-2"><strong class="text-info">Сводка
:</strong> ${escapeHtml(report.summary)}</div>
            <div class="mb-2"><strong class="text-warning">При
чина / Анализ:</strong> ${escapeHtml(report.root_cause)}</div>
            <h6 class="small text-uppercase text-muted fw-bold
 mb-1">Рекомендации:</h6>
            <ul class="mb-0 ps-3">
              ${(report.recommendations || []).map(r => `<li>$
{escapeHtml(r)}</li>`).join('')}
            </ul>
          `;
        } catch (err) {
          aiExplanation.innerHTML = `<div class="text-danger">
Ошибка диагностики: ${escapeHtml(err.message)}</div>`;
        }
      };
    }

    const bsModal = new bootstrap.Modal(modalEl);
    bsModal.show();
  }

  /**
   * Export logs as downloadable file.
   *
   * @param {string} format - "json" or "csv".
   */
```

```
  function exportLogs(format) {
    const level = (document.getElementById('slc-level-select') || {}).value || '';
    const search = (document.getElementById('slc-search-input') || {}).value || '';
    const limit = (document.getElementById('slc-limit-select') || {}).value || '200';
    const hours = (document.getElementById('slc-hours-select') || {}).value || '24';

    const url = `/api/v1/system_logs/export?format=${format}&channel=${encodeURIComponent(currentChannel)}&level=${encodeURIComponent(level)}&search=${encodeURIComponent(search)}&hours=${hours}&limit=${limit}`;
    window.open(url, '_blank');
  }

  /**
   * Main initializer for System Log Center tab.
   */
  function initSystemLogsTab() {
    if (window.isTabActive ? window.isTabActive('tab-system-logs') : false) {
      performSystemScan().then(() => {
        loadEvents();
      });
    }

    const scanBtn = document.getElementById('btn-slc-scan');
    if (scanBtn) {
      scanBtn.onclick = () => performSystemScan(true);
    }

    const refreshBtn = document.getElementById('btn-slc-refresh');
    if (refreshBtn) {
      refreshBtn.onclick = loadEvents;
    }
```

```
1142
1143      const jsonExportBtn = document.getElementById('btn-slc-exp
ort-json');
1144      if (jsonExportBtn) {
1145        jsonExportBtn.onclick = (e) => {
1146          e.preventDefault();
1147          exportLogs('json');
1148        };
1149      }
1150
1151      const csvExportBtn = document.getElementById('btn-slc-expo
rt-csv');
1152      if (csvExportBtn) {
1153        csvExportBtn.onclick = (e) => {
1154          e.preventDefault();
1155          exportLogs('csv');
1156        };
1157      }
1158
1159      const modeTabs = document.querySelectorAll('#slc-mode-tabs
 .nav-link');
1160      modeTabs.forEach((tab) => {
1161        tab.onclick = () => {
1162          modeTabs.forEach(t => t.classList.remove('active'));
1163          tab.classList.add('active');
1164          currentMode = tab.getAttribute('data-mode') || 'live';
1165          renderCurrentModeView();
1166        };
1167      });
1168
1169      const auditBtn = document.getElementById('btn-slc-audit');
1170      if (auditBtn) {
1171        auditBtn.onclick = () => {
1172          modeTabs.forEach(t => {
1173            if (t.getAttribute('data-mode') === 'audit') {
1174              t.classList.add('active');
1175            } else {
1176              t.classList.remove('active');
```

```
          }
        });
        currentMode = 'audit';
        renderCurrentModeView();
      };
    }

    const ragSearchBtn = document.getElementById('btn-slc-rag-search');
    if (ragSearchBtn) {
      ragSearchBtn.onclick = performRAGSearch;
    }

    const ragRebuildBtn = document.getElementById('btn-slc-rag-rebuild');
    if (ragRebuildBtn) {
      ragRebuildBtn.onclick = rebuildRAGIndex;
    }

    const ragInput = document.getElementById('slc-rag-input');
    if (ragInput) {
      ragInput.onkeydown = (e) => {
        if (e.key === 'Enter') {
          e.preventDefault();
          performRAGSearch();
        }
      };
    }

    const askAiBtn = document.getElementById('btn-slc-ask-ai');
    if (askAiBtn) {
      askAiBtn.onclick = () => {
        if (selectedEvent) {
          showEventModal(selectedEvent);
        } else if (cachedEvents.length > 0) {
          selectedEvent = cachedEvents[0];
          showEventModal(selectedEvent);
```

```
        }
      };
    }

    const levelSelect = document.getElementById('slc-level-sel
ect');
    if (levelSelect) levelSelect.onchange = loadEvents;

    const hoursSelect = document.getElementById('slc-hours-sel
ect');
    if (hoursSelect) hoursSelect.onchange = loadEvents;

    const limitSelect = document.getElementById('slc-limit-sel
ect');
    if (limitSelect) limitSelect.onchange = loadEvents;

    const eventIdInput = document.getElementById('slc-eventid-
input');
    if (eventIdInput) eventIdInput.onchange = loadEvents;

    let searchTimer = null;
    const searchInput = document.getElementById('slc-search-in
put');
    if (searchInput) {
      searchInput.oninput = () => {
        clearTimeout(searchTimer);
        searchTimer = setTimeout(loadEvents, 350);
      };
    }

    const channelFilterInput = document.getElementById('slc-ch
annel-filter-input');
    if (channelFilterInput) {
      channelFilterInput.oninput = renderChannelsTree;
    }

    const liveSwitch = document.getElementById('slc-live-switc
h');
```

```
    if (liveSwitch) {
      liveSwitch.onchange = () => {
        if (window.registerTabPoller && window.unregisterTabPo
ller) {
          if (liveSwitch.checked) {
            window.registerTabPoller('tab-system-logs', loadEv
ents, 4000, { pollerId: 'tab-system-logs_live', immediate: tru
e });
          } else {
            window.unregisterTabPoller('tab-system-logs_live')
;
          }
        } else {
          if (liveSwitch.checked) {
            liveIntervalTimer = setInterval(loadEvents, 4000);
          } else {
            clearInterval(liveIntervalTimer);
            liveIntervalTimer = null;
          }
        }
      };
    }
  }

  window.activateSystemLogsTab = () => {
    performSystemScan().then(() => loadEvents());
  };

  // Export initializer for admin UI
  window.initSystemLogsTab = initSystemLogsTab;
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Taskbar & Window Managem
ent Tab
================================================================
===============
Description:
  Интерактивный пульт управления панелью задач Windows 10/11,
окнами рабочего
  стола, закрепленными приложениями, ITaskbarList3 UX, Command
 Registry
  и историей изменений в telemetry.db с поддержкой отката (Rol
lback).

Usage Examples:
    Web Route:
      /windows/api/webgui/taskbar_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/taskbar_tab/index.html"
 class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/taskbar_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<div class="container-fluid p-0 taskbar-tab-wrapper d-flex fle
x-column" style="min-height: 100vh;">
<!-- Header Card & Summary Stats -->
<div class="card border-secondary-subtle mb-3 shadow-sm flex-s
```

```
hrink-0" data-panel-id="panel-taskbar-panel-zadach-i-okna-wind
ows-ta" id="panel-taskbar-panel-zadach-i-okna-windows-ta">
<div class="card-body p-3">
<div class="d-flex flex-wrap align-items-center justify-conten
t-between gap-3">
<div class="d-flex align-items-center gap-2">
<div class="fs-2">□ </div>
<div>
<h5 class="mb-0 fw-bold d-flex align-items-center gap-2">
<span>Панель задач и Окна (Windows Taskbar &amp;amp; Windows)</spa
n>
<span class="badge bg-primary-subtle text-primary border borde
r-primary-subtle small fs-6" id="tb-os-badge">Win32 / COM / Re
gistry</span>
<span class="badge bg-success-subtle text-success border borde
r-success-subtle small fs-6">telemetry.db Audit &amp;amp; Rollback
</span>
</h5>
<div class="text-muted small">
              Управление положением, автоскрытием, значками по
иска/виджетов, окнами рабочего стола, ITaskbarList3, историей
в telemetry.db и откатом.
            </div>
</div>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<button class="btn btn-outline-warning btn-sm d-flex align-ite
ms-center gap-1 shadow-sm" id="btn-rollback-last-quick" title=
"Откатить последнее действие">
<i class="bi bi-arrow-counterclockwise"></i>
<span>Откат последнего действия</span>
</button>
<button class="btn btn-outline-info btn-sm d-flex align-items-
center gap-1 shadow-sm" id="btn-refresh-taskbar-data" title="О
бновить состояние">
<i class="bi bi-arrow-clockwise"></i>
<span>Обновить</span>
</button>
```

```
<button class="btn btn-outline-secondary btn-sm d-flex align-items-center gap-1 shadow-sm" id="btn-restart-explorer-quick" title="Перезапустить Explorer">
<i class="bi bi-arrow-repeat"></i>
<span>Перезапуск Explorer</span>
</button>
</div>
</div>
<!-- Quick Metrics Bar -->
<div class="row g-2 mt-2 pt-2 border-top border-secondary-subtle small">
<div class="col-6 col-md-3">
<div class="p-2 rounded bg-body-tertiary border border-secondary-subtle text-center">
<div class="text-muted small">Габариты панели</div>
<div class="fw-bold text-info" id="metric-tb-rect">—</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="p-2 rounded bg-body-tertiary border border-secondary-subtle text-center">
<div class="text-muted small">Выравнивание значков</div>
<div class="fw-bold text-success" id="metric-tb-align">—</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="p-2 rounded bg-body-tertiary border border-secondary-subtle text-center">
<div class="text-muted small">Открытых окон</div>
<div class="fw-bold text-primary" id="metric-tb-windows">0</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="p-2 rounded bg-body-tertiary border border-secondary-subtle text-center">
<div class="text-muted small">Активное окно (Foreground)</div>
<div class="fw-bold text-warning text-truncate" id="metric-tb-
```

```
fg">—</div>
</div>
</div>
</div>
</div>
</div>
<!-- Navigation Subtabs -->
<ul class="nav nav-pills mb-3 gap-2 flex-shrink-0" id="taskbarSubTabs" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active btn-sm" data-bs-target="#panel-windows" data-bs-toggle="pill" id="tab-btn-windows" role="tab" type="button">
<i class="bi bi-window-stack me-1"></i> Окна рабочего стола (<span id="badge-windows-count">0</span>)
      </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link btn-sm" data-bs-target="#panel-settings" data-bs-toggle="pill" id="tab-btn-settings" role="tab" type="button">
<i class="bi bi-sliders me-1"></i> Настройки панели задач
      </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link btn-sm" data-bs-target="#panel-apps" data-bs-toggle="pill" id="tab-btn-apps" role="tab" type="button">
<i class="bi bi-pin-angle me-1"></i> Закрепленные и Запуск
      </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link btn-sm" data-bs-target="#panel-ux" data-bs-toggle="pill" id="tab-btn-ux" role="tab" type="button">
<i class="bi bi-bar-chart-steps me-1"></i> Индикаторы ITaskbarList3
      </button>
</li>
```

```
<li class="nav-item" role="presentation">
<button class="nav-link btn-sm" data-bs-target="#panel-history" data-bs-toggle="pill" id="tab-btn-history" role="tab" type="button">
<i class="bi bi-clock-history me-1"></i> История в telemetry.db (<span id="badge-history-count">0</span>)
      </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link btn-sm" data-bs-target="#panel-commands" data-bs-toggle="pill" id="tab-btn-commands" role="tab" type="button">
<i class="bi bi-terminal me-1"></i> Command Registry (260)
      </button>
</li>
</ul>
<!-- Subtabs Content -->
<div class="tab-content flex-grow-1" id="taskbarSubTabsContent">
<!-- 1. WINDOWS MANAGEMENT -->
<div class="tab-pane fade show active" data-panel-id="panel-windows" id="panel-windows" role="tabpanel">
<div class="card border-secondary-subtle shadow-sm h-100" data-panel-id="panel-taskbar-1" id="panel-taskbar-1">
<div class="card-header bg-body-tertiary d-flex flex-wrap align-items-center justify-content-between gap-2 p-2.5">
<!-- Filters -->
<div class="d-flex align-items-center gap-2 flex-grow-1" style="max-width: 500px;">
<div class="input-group input-group-sm">
<span class="input-group-text"><i class="bi bi-search"></i></span>
<input class="form-control" id="input-search-windows" placeholder="Фильтр по заголовку или процессу..." type="text"/>
</div>
</div>
<!-- Batch Actions -->
<div class="d-flex align-items-center gap-1.5 flex-wrap">
```

```
<button class="btn btn-sm btn-outline-secondary" id="btn-batch-minimize-all" title="Свернуть все окна (Показать рабочий стол)">
<i class="bi bi-dash-square me-1"></i> Свернуть все
            </button>
<button class="btn btn-sm btn-outline-secondary" id="btn-batch-restore-all" title="Восстановить все окна">
<i class="bi bi-window me-1"></i> Развернуть все
            </button>
</div>
</div>
<div class="card-body p-0 table-responsive" style="max-height: 60vh;">
<table class="table table-hover table-sm align-middle mb-0" id="table-windows-list">
<thead class="table-dark sticky-top small">
<tr>
<th style="width: 90px;">HWND</th>
<th>Заголовок окна</th>
<th style="width: 150px;">Процесс (PID)</th>
<th style="width: 140px;">Размеры (WxH)</th>
<th style="width: 100px;">Состояние</th>
<th class="text-end" style="width: 220px;">Действия</th>
</tr>
</thead>
<tbody class="small" id="tbody-windows-list">
<tr>
<td class="text-center text-muted p-4" colspan="6">
<div class="spinner-border spinner-border-sm text-primary me-2"></div> Загрузка списка открытых окон...
                </td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- 2. TASKBAR SETTINGS -->
```

```
<div class="tab-pane fade" data-panel-id="panel-settings" id="panel-settings" role="tabpanel">
<div class="card border-secondary-subtle shadow-sm" data-panel-id="panel-taskbar-2" id="panel-taskbar-2">
<div class="card-header bg-body-tertiary fw-semibold">
<i class="bi bi-sliders me-1 text-primary"></i> Системные настройки и внешний вид панели задач (с фиксацией в telemetry.db)
        </div>
<div class="card-body p-3">
<form id="form-taskbar-settings">
<div class="row g-3">
<!-- Alignment -->
<div class="col-md-6">
<label class="form-label fw-semibold small">Выравнивание значков (Windows 11)</label>
<div class="btn-group w-100" role="group">
<input class="btn-check" id="align-left" name="tb-alignment" type="radio" value="0"/>
<label class="btn btn-outline-primary btn-sm" for="align-left">
<i class="bi bi-text-left me-1"></i> По левому краю
                  </label>
<input class="btn-check" id="align-center" name="tb-alignment" type="radio" value="1"/>
<label class="btn btn-outline-primary btn-sm" for="align-center">
<i class="bi bi-text-center me-1"></i> По центру
                  </label>
</div>
</div>
<!-- Search mode -->
<div class="col-md-6">
<label class="form-label fw-semibold small" for="select-search-mode">Режим строки поиска</label>
<select class="form-select form-select-sm" id="select-search-mode">
<option value="0">Скрыт</option>
<option value="1">Только значок поиска</option>
```

```
<option value="2">Поле поиска</option>
<option value="3">Кнопка поиска</option>
</select>
</div>
<!-- Switches -->
<div class="col-md-4">
<div class="form-check form-switch mt-2">
<input class="form-check-input" id="switch-auto-hide" type="ch
eckbox"/>
<label class="form-check-label small fw-semibold" for="switch-
auto-hide">Автоматически скрывать панель</label>
</div>
</div>
<div class="col-md-4">
<div class="form-check form-switch mt-2">
<input class="form-check-input" id="switch-widgets" type="chec
kbox"/>
<label class="form-check-label small fw-semibold" for="switch-
widgets">Показывать виджеты (Widgets)</label>
</div>
</div>
<div class="col-md-4">
<div class="form-check form-switch mt-2">
<input class="form-check-input" id="switch-task-view" type="ch
eckbox"/>
<label class="form-check-label small fw-semibold" for="switch-
task-view">Кнопка Task View</label>
</div>
</div>
<div class="col-md-4">
<div class="form-check form-switch">
<input class="form-check-input" id="switch-copilot" type="chec
kbox"/>
<label class="form-check-label small fw-semibold" for="switch-
copilot">Кнопка Copilot</label>
</div>
</div>
<div class="col-md-4">
```

```
<div class="form-check form-switch">
<input class="form-check-input" id="switch-badges" type="check
box"/>
<label class="form-check-label small fw-semibold" for="switch-
badges">Бейджи уведомлений на значках</label>
</div>
</div>
<div class="col-md-4">
<div class="form-check form-switch">
<input class="form-check-input" id="switch-restart-explorer" t
ype="checkbox"/>
<label class="form-check-label small fw-semibold text-warning"
 for="switch-restart-explorer">Перезапустить Explorer после со
хранения</label>
</div>
</div>
</div>
<div class="mt-4 pt-3 border-top d-flex justify-content-end ga
p-2">
<button class="btn btn-primary btn-sm px-3 shadow-sm" type="su
bmit">
<i class="bi bi-check2-circle me-1"></i> Применить настройки
              </button>
</div>
</form>
</div>
</div>
</div>
<!-- 3. PINNED APPS & LAUNCH -->
<div class="tab-pane fade" data-panel-id="panel-apps" id="pane
l-apps" role="tabpanel">
<div class="row g-3">
<!-- Pinned List -->
<div class="col-md-7">
<div class="card border-secondary-subtle shadow-sm h-100" data
-panel-id="panel-taskbar-3" id="panel-taskbar-3">
<div class="card-header bg-body-tertiary fw-semibold d-flex ju
stify-content-between align-items-center">
```

```
<span><i class="bi bi-pin-angle me-1 text-primary"></i> Закрепленные на панели задачи ярлыки</span>
<button class="btn btn-sm btn-outline-secondary py-0 px-2" id="btn-refresh-pinned">
<i class="bi bi-arrow-clockwise"></i>
</button>
</div>
<div class="card-body p-2" id="pinned-apps-container">
<div class="text-muted small p-3 text-center">Загрузка закрепленных приложений...</div>
</div>
</div>
</div>
<!-- App Launcher -->
<div class="col-md-5">
<div class="card border-secondary-subtle shadow-sm h-100" data-panel-id="panel-taskbar-4" id="panel-taskbar-4">
<div class="card-header bg-body-tertiary fw-semibold">
<i class="bi bi-play-circle me-1 text-success"></i> Быстрый запуск приложения
            </div>
<div class="card-body p-3">
<form id="form-launch-app">
<div class="mb-3">
<label class="form-label small fw-semibold">Путь к файлу / Команда</label>
<input class="form-control form-control-sm" id="launch-app-path" placeholder="notepad.exe, msedge, explorer..." required="" type="text"/>
</div>
<div class="mb-3">
<label class="form-label small fw-semibold">Аргументы</label>
<input class="form-control form-control-sm" id="launch-app-args" placeholder="--incognito, path..." type="text"/>
</div>
<div class="mb-3 form-check form-switch">
<input class="form-check-input" id="launch-app-admin" type="checkbox"/>
```

```
<label class="form-check-label small fw-semibold" for="launch-app-admin">Запуск от имени администратора (UAC)</label>
</div>
<button class="btn btn-success btn-sm w-100" type="submit">
<i class="bi bi-rocket-takeoff me-1"></i> Запустить
                </button>
</form>
</div>
</div>
</div>
</div>
</div>
<!-- 4. TASKBAR UX & ITASKBARLIST3 -->
<div class="tab-pane fade" data-panel-id="panel-ux" id="panel-ux" role="tabpanel">
<div class="card border-secondary-subtle shadow-sm" data-panel-id="panel-taskbar-5" id="panel-taskbar-5">
<div class="card-header bg-body-tertiary fw-semibold">
<i class="bi bi-bar-chart-steps me-1 text-primary"></i> Интерактивный эмулятор ITaskbarList3 (Индикаторы прогресса и оверлеи)
        </div>
<div class="card-body p-3">
<div class="row g-3">
<div class="col-md-6">
<label class="form-label small fw-semibold">Целевое окно (HWND)</label>
<select class="form-select form-select-sm" id="select-ux-target-window"></select>
</div>
<div class="col-md-6">
<label class="form-label small fw-semibold">Режим индикатора прогресса</label>
<select class="form-select form-select-sm" id="select-ux-progress-state">
<option value="normal">Обычный (Зеленый / Normal)</option>
<option value="indeterminate">Неопределенный (Бегущий / Indeterminate)</option>
```

```
<option value="error">Ошибка (Красный / Error)</option>
<option value="paused">Пауза (Желтый / Paused)</option>
<option value="no_progress">Сбросить (No Progress)</option>
</select>
</div>
<div class="col-md-6">
<label class="form-label small fw-semibold">Прогресс выполнени
я (%): <span id="label-progress-val">50%</span></label>
<input class="form-range" id="range-ux-progress" max="100" min
="0" type="range" value="50"/>
</div>
<div class="col-md-6 d-flex align-items-end">
<button class="btn btn-primary btn-sm w-100" id="btn-apply-ux-
progress">
<i class="bi bi-send me-1"></i> Передать прогресс на панель за
дач
              </button>
</div>
</div>
</div>
</div>
</div>
<!-- 5. HISTORY & ROLLBACK (telemetry.db) -->
<div class="tab-pane fade" data-panel-id="panel-history" id="p
anel-history" role="tabpanel">
<div class="card border-secondary-subtle shadow-sm" data-panel
-id="panel-taskbar-zhurnal-izmeneniy-v-sqlite-tele" id="panel-
taskbar-zhurnal-izmeneniy-v-sqlite-tele">
<div class="card-header bg-body-tertiary d-flex flex-wrap alig
n-items-center justify-content-between gap-2 p-2.5">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-clock-history me-1 text-primary"></i>
<span class="fw-semibold">Журнал изменений в SQLite telemetry.
db</span>
</div>
<div class="d-flex align-items-center gap-2">
<button class="btn btn-outline-info btn-sm" id="btn-refresh-hi
story">
```

```
<i class="bi bi-arrow-clockwise me-1"></i> Обновить историю
            </button>
</div>
</div>
<div class="card-body p-0 table-responsive" style="max-height:
 55vh;">
<table class="table table-hover table-sm align-middle mb-0" id
="table-history-list">
<thead class="table-dark sticky-top small">
<tr>
<th style="width: 60px;">ID</th>
<th style="width: 160px;">Время</th>
<th style="width: 180px;">Команда</th>
<th>Детали и состояния</th>
<th style="width: 110px;">Статус</th>
<th class="text-end" style="width: 120px;">Откат</th>
</tr>
</thead>
<tbody class="small" id="tbody-history-list">
<tr>
<td class="text-center text-muted p-4" colspan="6">Загрузка жу
рнала изменений из telemetry.db...</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- 6. COMMAND REGISTRY (260) -->
<div class="tab-pane fade" data-panel-id="panel-commands" id="
panel-commands" role="tabpanel">
<div class="card border-secondary-subtle shadow-sm" data-panel
-id="panel-taskbar-katalog-komand-control-plane" id="panel-tas
kbar-katalog-komand-control-plane">
<div class="card-header bg-body-tertiary d-flex flex-wrap alig
n-items-center justify-content-between gap-2 p-2.5">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-terminal me-1 text-primary"></i>
```

```
<span class="fw-semibold">Каталог команд Control Plane</span>
<span class="badge bg-secondary-subtle text-secondary small" id="badge-commands-count">0</span>
</div>
<div class="d-flex align-items-center gap-2">
<select class="form-select form-select-sm" id="filter-cmd-category" style="width: 140px;">
<option value="">Все категории</option>
<option value="window">window</option>
<option value="taskbar">taskbar</option>
<option value="shell">shell</option>
<option value="policy">policy</option>
</select>
<select class="form-select form-select-sm" id="filter-cmd-risk" style="width: 130px;">
<option value="">Любой риск</option>
<option value="safe">🟢 safe</option>
<option value="caution">🟡 caution</option>
<option value="admin">🟠 admin</option>
<option value="high">🔴 high</option>
</select>
</div>
</div>
<div class="card-body p-0 table-responsive" style="max-height: 55vh;">
<table class="table table-hover table-sm align-middle mb-0" id="table-commands-list">
<thead class="table-dark sticky-top small">
<tr>
<th style="width: 220px;">ID Команды</th>
<th style="width: 100px;">Класс</th>
<th style="width: 90px;">Риск</th>
<th>Описание</th>
<th>API / Механизм</th>
<th class="text-end" style="width: 100px;">Действие</th>
</tr>
</thead>
<tbody class="small" id="tbody-commands-list">
```

```
<tr>
<td class="text-center text-muted p-4" colspan="6">Загрузка ка
талога команд...</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Web Interface - Taskbar Tab Controller
 * ============================================================
 * Description:
 *   Клиентский контроллер интерактивного управления панелью задач, окнами
 *   и историей изменений в telemetry.db с поддержкой Rollback.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/taskbar_tab/main.js?v=20261006_v2" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/taskbar_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

let _windowsCache = [];
let _commandsCache = [];
let _historyCache = [];

export async function init() {
  bindEvents();
  await refreshAll();
}
```

```
30
31 function bindEvents() {
32   document.getElementById('btn-refresh-taskbar-data')?.addEven
   tListener('click', () => refreshAll());
33   document.getElementById('btn-restart-explorer-quick')?.addEv
   entListener('click', () => restartExplorer());
34   document.getElementById('btn-rollback-last-quick')?.addEvent
   Listener('click', () => rollbackLast());
35   document.getElementById('btn-refresh-history')?.addEventList
   ener('click', () => loadHistory());
36   document.getElementById('input-search-windows')?.addEventLis
   tener('input', () => renderWindowsTable());
37   document.getElementById('btn-batch-minimize-all')?.addEventL
   istener('click', () => executeBatchWindowAction('minimize_all'
   ));
38   document.getElementById('btn-batch-restore-all')?.addEventLi
   stener('click', () => executeBatchWindowAction('restore_all'))
   ;
39   document.getElementById('form-taskbar-settings')?.addEventLi
   stener('submit', (e) => saveTaskbarSettings(e));
40   document.getElementById('form-launch-app')?.addEventListener
   ('submit', (e) => handleLaunchApp(e));
41   document.getElementById('btn-refresh-pinned')?.addEventListe
   ner('click', () => loadPinnedApps());
42
43   const range = document.getElementById('range-ux-progress');
44   const label = document.getElementById('label-progress-val');
45   if (range && label) {
46     range.addEventListener('input', () => { label.textContent
   = `${range.value}%`; });
47   }
48   document.getElementById('btn-apply-ux-progress')?.addEventLi
   stener('click', () => handleApplyUxProgress());
49
50   document.getElementById('filter-cmd-category')?.addEventList
   ener('change', () => renderCommandsTable());
51   document.getElementById('filter-cmd-risk')?.addEventListener
   ('change', () => renderCommandsTable());
```

```
}

export async function refreshAll() {
  await Promise.all([
    loadSummary(),
    loadWindows(),
    loadSettings(),
    loadPinnedApps(),
    loadHistory(),
    loadCommandCatalog(),
  ]);
}

async function loadSummary() {
  try {
    const res = await fetch('/api/v1/taskbar/summary');
    if (!res.ok) return;
    const data = await res.json();

    const rectEl = document.getElementById('metric-tb-rect');
    if (rectEl) {
      rectEl.textContent = `${data.taskbar_rect.width}x${data.
taskbar_rect.height} px`;
    }

    const alignEl = document.getElementById('metric-tb-align')
;
    if (alignEl) {
      alignEl.textContent = data.settings.alignment === 1 ? 'П
о центру' : 'По левому краю';
    }

    const winEl = document.getElementById('metric-tb-windows')
;
    if (winEl) {
      winEl.textContent = `${data.visible_windows_count} / ${d
ata.windows_count}`;
    }
```

```
85
86      const fgEl = document.getElementById('metric-tb-fg');
87      if (fgEl) {
88        fgEl.textContent = data.foreground_window?.title || '—';
89        fgEl.title = data.foreground_window?.title || '';
90      }
91
92      const osBadge = document.getElementById('tb-os-badge');
93      if (osBadge) {
94        osBadge.textContent = data.is_win11 ? 'Windows 11 (Cente
r/Left Taskbar)' : 'Windows 10 / Classic';
95      }
96    } catch (err) {
97      console.error('Ошибка загрузки summary:', err);
98    }
99  }
100
101 async function loadWindows() {
102   const tbody = document.getElementById('tbody-windows-list');
103   try {
104     const res = await fetch('/api/v1/taskbar/windows?only_visi
ble=false&only_taskbar=false');
105     if (!res.ok) throw new Error(`HTTP ${res.status}`);
106     _windowsCache = await res.json();
107     renderWindowsTable();
108     populateWindowSelects();
109   } catch (err) {
110     if (tbody) {
111       tbody.innerHTML = `<tr><td colspan="6" class="text-dange
r p-3 text-center">Ошибка: ${err.message}</td></tr>`;
112     }
113   }
114 }
115
116 function renderWindowsTable() {
117   const tbody = document.getElementById('tbody-windows-list');
118   const filter = (document.getElementById('input-search-window
s')?.value || '').toLowerCase();
```

```
  if (!tbody) return;

  const filtered = _windowsCache.filter(w => {
    if (!filter) return w.title;
    return w.title.toLowerCase().includes(filter) || w.process_name.toLowerCase().includes(filter) || String(w.hwnd).includes(filter);
  });

  const badge = document.getElementById('badge-windows-count');
  if (badge) badge.textContent = filtered.length;

  if (filtered.length === 0) {
    tbody.innerHTML = `<tr><td colspan="6" class="text-muted p-4 text-center">Окна по фильтру не найдены</td></tr>`;
    return;
  }

  tbody.innerHTML = filtered.map(w => {
    const isFg = w.is_foreground ? '<span class="badge bg-warning text-dark me-1">Фокус</span>' : '';
    const stateBadge = w.is_minimized
      ? '<span class="badge bg-secondary">Свернуто</span>'
      : w.is_maximized
        ? '<span class="badge bg-success">Развернуто</span>'
        : '<span class="badge bg-info text-dark">Нормальное</span>';

    return `
      <tr class="${w.is_foreground ? 'table-active' : ''}">
        <td><code class="user-select-all">${w.hwnd}</code></td>
        <td class="text-truncate" style="max-width: 250px;" title="${escapeHtml(w.title)}">
          ${isFg}<strong>${escapeHtml(w.title || '(Без названия)')}</strong>
        </td>
```

```
        <td>
          <div class="fw-semibold text-truncate" style="max-width: 140px;">${escapeHtml(w.process_name || '—')}</div>
          <div class="text-muted small">PID: ${w.process_id}</div>
        </td>
        <td><small class="text-muted">${w.rect.width} × ${w.rect.height}</small></td>
        <td>${stateBadge}</td>
        <td class="text-end">
          <div class="btn-group btn-group-sm">
            <button class="btn btn-outline-primary" onclick="window._taskbarActions.activate(${w.hwnd})" title="Активировать"><i class="bi bi-lightning-fill"></i></button>
            <button class="btn btn-outline-secondary" onclick="window._taskbarActions.minimize(${w.hwnd})" title="Свернуть"><i class="bi bi-dash"></i></button>
            <button class="btn btn-outline-secondary" onclick="window._taskbarActions.maximize(${w.hwnd})" title="Развернуть"><i class="bi bi-square"></i></button>
            <button class="btn btn-outline-secondary" onclick="window._taskbarActions.restore(${w.hwnd})" title="Восстановить"><i class="bi bi-window"></i></button>
            <button class="btn btn-outline-danger" onclick="window._taskbarActions.close(${w.hwnd})" title="Закрыть"><i class="bi bi-x-lg"></i></button>
          </div>
        </td>
      </tr>
    `;
  }).join('');
}

function populateWindowSelects() {
  const sel = document.getElementById('select-ux-target-window');
  if (!sel) return;
  const named = _windowsCache.filter(w => w.title);
```

```
  sel.innerHTML = named.map(w => `<option value="${w.hwnd}">${w.hwnd} — ${escapeHtml(w.title.slice(0, 40))} (${w.process_name})</option>`).join('');
}

// Window Actions
window._taskbarActions = {
  activate: async (hwnd) => {
    await fetch(`/api/v1/taskbar/windows/${hwnd}/activate`, { method: 'POST' });
    await loadWindows();
    await loadHistory();
  },
  minimize: async (hwnd) => {
    await fetch(`/api/v1/taskbar/windows/${hwnd}/minimize`, { method: 'POST' });
    await loadWindows();
    await loadHistory();
  },
  maximize: async (hwnd) => {
    await fetch(`/api/v1/taskbar/windows/${hwnd}/maximize`, { method: 'POST' });
    await loadWindows();
    await loadHistory();
  },
  restore: async (hwnd) => {
    await fetch(`/api/v1/taskbar/windows/${hwnd}/restore`, { method: 'POST' });
    await loadWindows();
    await loadHistory();
  },
  close: async (hwnd) => {
    await fetch(`/api/v1/taskbar/windows/${hwnd}`, { method: 'DELETE' });
    await loadWindows();
    await loadHistory();
  }
};
```

```
203
204 async function executeBatchWindowAction(action) {
205   try {
206     const res = await fetch('/api/v1/taskbar/windows/batch', {
207       method: 'POST',
208       headers: { 'Content-Type': 'application/json' },
209       body: JSON.stringify({ action })
210     });
211     if (res.ok) {
212       if (window.toast) window.toast.success('Действие выполне
    но', `Пакетное действие: ${action}`);
213       await loadWindows();
214       await loadHistory();
215     }
216   } catch (err) {
217     if (window.toast) window.toast.error('Ошибка', err.message
    );
218   }
219 }
220
221 async function loadSettings() {
222   try {
223     const res = await fetch('/api/v1/taskbar/settings');
224     if (!res.ok) return;
225     const s = await res.json();
226
227     const alignVal = s.alignment === 1 ? '1' : '0';
228     const alignRadio = document.querySelector(`input[name="tb-
    alignment"][value="${alignVal}"]`);
229     if (alignRadio) alignRadio.checked = true;
230
231     const selSearch = document.getElementById('select-search-m
    ode');
232     if (selSearch) selSearch.value = String(s.search_mode);
233
234     setCheck('switch-auto-hide', s.auto_hide);
235     setCheck('switch-widgets', s.widgets_visible);
236     setCheck('switch-task-view', s.task_view_visible);
```

```
    setCheck('switch-copilot', s.copilot_visible);
    setCheck('switch-badges', s.badges_enabled);
  } catch (err) {
    console.error('Ошибка настроек:', err);
  }
}

function setCheck(id, val) {
  const el = document.getElementById(id);
  if (el) el.checked = Boolean(val);
}

async function saveTaskbarSettings(e) {
  e.preventDefault();
  const alignRadio = document.querySelector('input[name="tb-al
ignment"]:checked');
  const alignment = alignRadio ? parseInt(alignRadio.value, 10
) : 1;
  const search_mode = parseInt(document.getElementById('select
-search-mode')?.value || '1', 10);
  const auto_hide = document.getElementById('switch-auto-hide'
)?.checked || false;
  const widgets_visible = document.getElementById('switch-widg
ets')?.checked || false;
  const task_view_visible = document.getElementById('switch-ta
sk-view')?.checked || false;
  const copilot_visible = document.getElementById('switch-copi
lot')?.checked || false;
  const badges_enabled = document.getElementById('switch-badge
s')?.checked || false;
  const restart_explorer = document.getElementById('switch-res
tart-explorer')?.checked || false;

  try {
    const res = await fetch('/api/v1/taskbar/settings', {
      method: 'PUT',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
```

```
        alignment,
        search_mode,
        auto_hide,
        widgets_visible,
        task_view_visible,
        copilot_visible,
        badges_enabled,
        restart_explorer,
      })
    });
    if (res.ok) {
      if (window.toast) window.toast.success('Настройки сохранены', 'Параметры панели задач успешно зафиксированы в telemetry.db');
      await refreshAll();
    }
  } catch (err) {
    if (window.toast) window.toast.error('Ошибка сохранения', err.message);
  }
}

async function restartExplorer() {
  if (!confirm('Перезапустить процесс Explorer? Экран на 1-2 секунды обновится.')) return;
  try {
    const res = await fetch('/api/v1/taskbar/settings', {
      method: 'PUT',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ restart_explorer: true })
    });
    if (res.ok && window.toast) {
      window.toast.success('Explorer перезапущен', 'Оболочка Windows успешно перезапущена');
    }
  } catch (err) {
    if (window.toast) window.toast.error('Ошибка', err.message);
```

```
  }
}

async function loadPinnedApps() {
  const container = document.getElementById('pinned-apps-container');
  try {
    const res = await fetch('/api/v1/taskbar/apps');
    if (!res.ok) return;
    const apps = await res.json();
    if (!container) return;

    if (apps.length === 0) {
      container.innerHTML = `<div class="text-muted small p-3 text-center">Нет закрепленных ярлыков</div>`;
      return;
    }

    container.innerHTML = `<div class="list-group list-group-flush">${apps.map(a => `
      <div class="list-group-item d-flex justify-content-between align-items-center p-2">
        <div class="d-flex align-items-center gap-2">
          <i class="bi bi-app text-primary"></i>
          <div>
            <div class="fw-semibold small">${escapeHtml(a.name)}</div>
            <div class="text-muted font-monospace" style="font-size: 0.72rem;">${escapeHtml(a.target_path || a.link_path)}</div>
          </div>
        </div>
      </div>
    `).join('')}</div>`;
  } catch (err) {
    if (container) container.innerHTML = `<div class="text-danger small p-2">Ошибка: ${err.message}</div>`;
  }
```

```
}

async function handleLaunchApp(e) {
  e.preventDefault();
  const path = document.getElementById('launch-app-path')?.val
ue || '';
  const args = document.getElementById('launch-app-args')?.val
ue || '';
  const admin = document.getElementById('launch-app-admin')?.c
hecked || false;

  try {
    const res = await fetch('/api/v1/taskbar/apps/launch', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ app_path: path, arguments: args |
| null, admin })
    });
    const data = await res.json();
    if (res.ok) {
      if (window.toast) window.toast.success('Приложение запущ
ено', data.message);
      document.getElementById('launch-app-path').value = '';
      setTimeout(() => { loadWindows(); loadHistory(); }, 1000
);
    } else {
      if (window.toast) window.toast.error('Ошибка запуска', d
ata.detail || 'Сбой');
    }
  } catch (err) {
    if (window.toast) window.toast.error('Ошибка сети', err.me
ssage);
  }
}

async function handleApplyUxProgress() {
  const hwnd = parseInt(document.getElementById('select-ux-tar
get-window')?.value || '0', 10);
```

```
  const state = document.getElementById('select-ux-progress-state')?.value || 'normal';
  const val = parseInt(document.getElementById('range-ux-progress')?.value || '50', 10);

  if (!hwnd) {
    if (window.toast) window.toast.warning('Внимание', 'Выберите целевое окно');
    return;
  }

  try {
    const res = await fetch('/api/v1/taskbar/ux/progress', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ hwnd, state, completed: val, total: 100 })
    });
    if (res.ok && window.toast) {
      window.toast.success('ITaskbarList3', `Прогресс ${val}% передан окну HWND ${hwnd}`);
      await loadHistory();
    }
  } catch (err) {
    if (window.toast) window.toast.error('Ошибка', err.message);
  }
}

// History & Rollback
async function loadHistory() {
  const tbody = document.getElementById('tbody-history-list');
  try {
    const res = await fetch('/api/v1/taskbar/history?limit=50');
    if (!res.ok) return;
    _historyCache = await res.json();
    renderHistoryTable();
```

```
  } catch (err) {
    if (tbody) {
      tbody.innerHTML = `<tr><td colspan="6" class="text-dange
r p-3 text-center">Ошибка: ${err.message}</td></tr>`;
    }
  }
}

function renderHistoryTable() {
  const tbody = document.getElementById('tbody-history-list');
  const badge = document.getElementById('badge-history-count')
;
  if (badge) badge.textContent = _historyCache.length;
  if (!tbody) return;

  if (_historyCache.length === 0) {
    tbody.innerHTML = `<tr><td colspan="6" class="text-muted p
-4 text-center">История в telemetry.db пуста</td></tr>`;
    return;
  }

  tbody.innerHTML = _historyCache.map(h => {
    const rolledBadge = h.is_rolled_back
      ? '<span class="badge bg-secondary">Откачено</span>'
      : '<span class="badge bg-success-subtle text-success bor
der border-success-subtle">Активно</span>';

    const canRollback = !h.is_rolled_back && h.previous_state;
    const timeShort = (h.timestamp || '').replace('T', ' ').sl
ice(0, 19);

    let detailsStr = '';
    if (h.previous_state || h.new_state) {
      detailsStr = `<small class="text-muted font-monospace">$
{escapeHtml(JSON.stringify(h.new_state || h.details || {}).sli
ce(0, 60))}</small>`;
    } else if (h.details) {
      detailsStr = `<small class="text-muted">${escapeHtml(JSO
```

```
N.stringify(h.details).slice(0, 60))}</small>`;
    }

    return `
      <tr class="${h.is_rolled_back ? 'opacity-75' : ''}">
        <td><code>#${h.id}</code></td>
        <td><small>${escapeHtml(timeShort)}</small></td>
        <td>
          <div class="fw-semibold"><code>${escapeHtml(h.command_id)}</code></div>
          <small class="text-muted">${escapeHtml(h.action_type)} ${h.target_hwnd ? `(HWND: ${h.target_hwnd})` : ''}</small>
        </td>
        <td>${detailsStr}</td>
        <td>${rolledBadge}</td>
        <td class="text-end">
          ${canRollback ? `
            <button class="btn btn-outline-warning btn-sm py-0 px-2" onclick="window._rollbackAction(${h.id})" title="Откатить это изменение">
              <i class="bi bi-arrow-counterclockwise me-1"></i> Откат
            </button>
          ` : '<span class="text-muted small">—</span>'}
        </td>
      </tr>
    `;
  }).join('');
}

window._rollbackAction = async (historyId) => {
  if (!confirm(`Откатить действие #${historyId}? Предыдущее состояние будет восстановлено.`)) return;
  try {
    const res = await fetch(`/api/v1/taskbar/history/${historyId}/rollback`, { method: 'POST' });
    const data = await res.json();
    if (res.ok) {
```

```
449       if (window.toast) window.toast.success('Откат выполнен',
 data.message);
450       await refreshAll();
451     } else {
452       if (window.toast) window.toast.error('Ошибка отката', da
ta.detail || 'Сбой');
453     }
454   } catch (err) {
455     if (window.toast) window.toast.error('Ошибка', err.message
);
456   }
457 };
458 
459 async function rollbackLast() {
460   try {
461     const res = await fetch('/api/v1/taskbar/history/rollback-
last', { method: 'POST' });
462     const data = await res.json();
463     if (res.ok) {
464       if (window.toast) window.toast.success('Откат выполнен',
 data.message);
465       await refreshAll();
466     } else {
467       if (window.toast) window.toast.error('Ошибка отката', da
ta.detail || 'Сбой');
468     }
469   } catch (err) {
470     if (window.toast) window.toast.error('Ошибка', err.message
);
471   }
472 }
473 
474 async function loadCommandCatalog() {
475   try {
476     const res = await fetch('/api/v1/taskbar/commands');
477     if (!res.ok) return;
478     _commandsCache = await res.json();
479     renderCommandsTable();
```

```
  } catch (err) {
    console.error('Ошибка каталога команд:', err);
  }
}

function renderCommandsTable() {
  const tbody = document.getElementById('tbody-commands-list')
;
  const cat = document.getElementById('filter-cmd-category')?.
value || '';
  const risk = document.getElementById('filter-cmd-risk')?.val
ue || '';
  if (!tbody) return;

  const filtered = _commandsCache.filter(c => {
    if (cat && c.category.toLowerCase() !== cat.toLowerCase())
 return false;
    if (risk && c.risk.toLowerCase() !== risk.toLowerCase()) r
eturn false;
    return true;
  });

  const badge = document.getElementById('badge-commands-count'
);
  if (badge) badge.textContent = filtered.length;

  tbody.innerHTML = filtered.map(c => {
    const riskBadge = c.risk === 'safe'
      ? '<span class="badge bg-success-subtle text-success bor
der border-success-subtle">safe</span>'
      : c.risk === 'caution'
        ? '<span class="badge bg-warning-subtle text-warning b
order border-warning-subtle">caution</span>'
        : c.risk === 'admin'
          ? '<span class="badge bg-danger-subtle text-danger b
order border-danger-subtle">admin</span>'
          : '<span class="badge bg-danger text-white">high</sp
an>';
```

```
508
509     const apiStr = c.api.win32 || c.api.com || c.api.registry
    || c.api.powershell || '—';
510
511     return `
512       <tr>
513         <td><code>${escapeHtml(c.id)}</code></td>
514         <td><span class="badge bg-secondary-subtle text-second
    ary">${escapeHtml(c.execution_class)}</span></td>
515         <td>${riskBadge}</td>
516         <td>${escapeHtml(c.description)}</td>
517         <td class="font-monospace small text-muted text-trunca
    te" style="max-width: 200px;" title="${escapeHtml(apiStr)}">${
    escapeHtml(apiStr)}</td>
518         <td class="text-end">
519           <button class="btn btn-outline-primary btn-sm py-0 p
    x-2" onclick="window._executeCommandDirect('${c.id}')" title="
    Выполнить">
520             <i class="bi bi-play"></i>
521           </button>
522         </td>
523       </tr>
524     `;
525   }).join('');
526 }
527
528 window._executeCommandDirect = async (cmdId) => {
529   const meta = _commandsCache.find(c => c.id === cmdId);
530   if (meta?.requires_confirmation) {
531     if (!confirm(`Команда ${cmdId} имеет повышенный риск (${me
    ta.risk}). Подтвердить запуск?`)) return;
532   }
533   try {
534     const res = await fetch('/api/v1/taskbar/execute', {
535       method: 'POST',
536       headers: { 'Content-Type': 'application/json' },
537       body: JSON.stringify({ command: cmdId, params: {}, confi
    rmed_by_user: true })
```

```
    });
    const data = await res.json();
    if (res.ok && window.toast) {
      window.toast.success(`Команда ${cmdId}`, data.message ||
 'Выполнено успешно');
      await loadHistory();
    }
  } catch (err) {
    if (window.toast) window.toast.error('Ошибка', err.message
);
  }
};

function escapeHtml(str) {
  if (!str) return '';
  return String(str)
    .replace(/&/g, '&amp;')
    .replace(/</g, '&lt;')
    .replace(/>/g, '&gt;')
    .replace(/"/g, '&quot;');
}

if (document.readyState === 'loading') {
  document.addEventListener('DOMContentLoaded', () => init());
} else {
  init();
}

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Task Scheduler View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса планировщика задач Windows
.

Usage Examples:
    Web Route:
      /windows/api/webgui/task_scheduler_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/task_scheduler_tab/inde
x.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/task_scheduler_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Task Scheduler Tab -->
<div class="container-fluid py-2 px-3">
<!-- Top Stat Cards -->
<div class="row g-2 mb-3">
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 sha
dow-sm p-3" data-panel-id="panel-task-scheduler-1" id="panel-t
ask-scheduler-1">
<div class="text-muted small text-uppercase fw-semibold">Всего
```

```
 задач □ </div>
<div class="fs-4 fw-bold mt-1 text-primary" id="ts-total-count">--</div>
<div class="small text-muted mt-1" id="ts-total-sub">schtasks.exe /query</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-task-scheduler-2" id="panel-task-scheduler-2">
<div class="text-muted small text-uppercase fw-semibold">Готовы к запуску (Ready) □ </div>
<div class="fs-4 fw-bold mt-1 text-success" id="ts-ready-count">--</div>
<div class="small text-muted mt-1" id="ts-ready-sub">Ожидают расписания</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-task-scheduler-3" id="panel-task-scheduler-3">
<div class="text-muted small text-uppercase fw-semibold">Выполняются сейчас □ </div>
<div class="fs-4 fw-bold mt-1 text-info" id="ts-running-count">--</div>
<div class="small text-muted mt-1" id="ts-running-sub">Активные задачи</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-dark text-white border-secondary h-100 shadow-sm p-3" data-panel-id="panel-task-scheduler-4" id="panel-task-scheduler-4">
<div class="text-muted small text-uppercase fw-semibold">Отключены (Disabled) □ </div>
<div class="fs-4 fw-bold mt-1 text-danger" id="ts-disabled-cou
```

```
nt">--</div>
<div class="small text-muted mt-1" id="ts-disabled-sub">Деакти
вированы</div>
</div>
</div>
</div>
<!-- Action Bar -->
<div class="d-flex align-items-center justify-content-between
mb-3 p-2 bg-dark rounded border border-secondary flex-wrap gap
-2">
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm btn-primary d-flex align-items-cente
r gap-1.5" id="ts-refresh-btn">
<i class="bi bi-arrow-clockwise"></i>
<span>Обновить задачи</span>
</button>
</div>
<div class="d-flex align-items-center gap-2">
<input class="form-control form-control-sm bg-dark text-white
border-secondary" id="ts-search-input" placeholder="Поиск по и
мени задачи..." style="width: 260px;" type="text"/>
</div>
</div>
<!-- Tasks Table -->
<div class="card bg-dark text-white border-secondary shadow-sm
" data-panel-id="panel-task-scheduler-zadachi-planirovshchika-
windows-ta" id="panel-task-scheduler-zadachi-planirovshchika-w
indows-ta">
<div class="card-header border-secondary d-flex align-items-ce
nter justify-content-between py-2 px-3">
<span class="fw-semibold"><i class="bi bi-calendar-check text-
primary me-2"></i>Задачи Планировщика Windows (Task Scheduler)
</span>
<span class="badge bg-secondary" id="ts-table-badge">0 задач</
span>
</div>
<div class="card-body p-0 table-responsive" style="max-height:
 520px; overflow-y: auto;">
```

```
<table class="table table-dark table-hover table-striped align-middle mb-0 small" id="ts-table">
<thead class="sticky-top bg-dark">
<tr class="text-secondary">
<th>Имя / Путь задачи</th>
<th>Время след. запуска</th>
<th>Статус</th>
<th>Режим расписания</th>
<th class="text-end">Действия</th>
</tr>
</thead>
<tbody id="ts-tbody">
<tr><td class="text-center text-muted py-4" colspan="5">Опрос планировщика задач...</td></tr>
</tbody>
</table>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Task Scheduler Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля Планировщ
ика задач Windows.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/task_scheduler_tab/mai
n.js?v=20261006_v2" type="module"></script>
 *
 *   JavaScript Import:
 *     import { initTaskSchedulerTab } from '/windows/api/webg
ui/task_scheduler_tab/main.js';
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/task_scheduler_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

/**
 * task_scheduler_tab/main.js — Управление задачами Windows Ta
sk Scheduler
 * Updated: 2026-10-10 17:35:00
 */

const registerTabPoller = window.registerTabPoller || function
```

```
() {};

let isInitialized = false;
let tasksList = [];

function escapeHtml(str) {
  if (str === null || str === undefined) return '';
  return String(str)
    .replace(/&/g, '&amp;')
    .replace(/</g, '&lt;')
    .replace(/>/g, '&gt;')
    .replace(/"/g, '&quot;')
    .replace(/'/g, '&#039;');
}

export async function fetchTaskSchedulerSummary() {
  const refreshBtn = document.getElementById('ts-refresh-btn')
;
  if (refreshBtn) {
    const icon = refreshBtn.querySelector('i');
    if (icon) icon.classList.add('spin-animation');
  }

  try {
    const res = await fetch('/api/task-scheduler/summary');
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    renderTasks(data);
  } catch (err) {
    console.warn('[TaskScheduler] Ошибка получения списка зада
ч:', err);
    const tbody = document.getElementById('ts-tbody');
    if (tbody && tasksList.length === 0) {
      tbody.innerHTML = `<tr><td colspan="5" class="text-cente
r text-danger py-4">Ошибка загрузки задач: ${escapeHtml(err.me
ssage)}</td></tr>`;
    }
  } finally {
```

```
    if (refreshBtn) {
      const icon = refreshBtn.querySelector('i');
      if (icon) icon.classList.remove('spin-animation');
    }
  }
}

function renderTasks(data) {
  tasksList = data.tasks || [];

  const totalEl = document.getElementById('ts-total-count');
  const readyEl = document.getElementById('ts-ready-count');
  const runningEl = document.getElementById('ts-running-count'
);
  const disabledEl = document.getElementById('ts-disabled-coun
t');

  const readyCount = data.ready_tasks !== undefined
    ? data.ready_tasks
    : tasksList.filter(t => {
        const st = (t.state || t.status || '').toLowerCase();
        return st.includes('ready') || st.includes('готов');
      }).length;

  const runningCount = data.running_tasks !== undefined
    ? data.running_tasks
    : tasksList.filter(t => {
        const st = (t.state || t.status || '').toLowerCase();
        return st.includes('running') || st.includes('работ')
|| st.includes('выполн');
      }).length;

  const disabledCount = data.disabled_tasks !== undefined
    ? data.disabled_tasks
    : tasksList.filter(t => {
        const st = (t.state || t.status || '').toLowerCase();
        return st.includes('disabled') || st.includes('отключ'
);
```

```
      }).length;

  if (totalEl) totalEl.textContent = tasksList.length;
  if (readyEl) readyEl.textContent = readyCount;
  if (runningEl) runningEl.textContent = runningCount;
  if (disabledEl) disabledEl.textContent = disabledCount;

  applyTaskFilters();
}

function applyTaskFilters() {
  const tbody = document.getElementById('ts-tbody');
  const searchInput = document.getElementById('ts-search-input
');
  const badge = document.getElementById('ts-table-badge');

  if (!tbody) return;

  const query = (searchInput ? searchInput.value : '').toLower
Case().trim();

  const filtered = tasksList.filter(t => {
    if (!query) return true;
    const name = (t.name || t.task_name || '').toLowerCase();
    const path = (t.task_path || '').toLowerCase();
    const act = (t.action || '').toLowerCase();
    return name.includes(query) || path.includes(query) || act
.includes(query);
  });

  if (badge) badge.textContent = `${filtered.length} из ${task
sList.length} задач`;

  if (filtered.length === 0) {
    tbody.innerHTML = '<tr><td colspan="5" class="text-center
text-muted py-4">Задачи не найдены</td></tr>';
    return;
  }
```

```
130
131   tbody.innerHTML = filtered.slice(0, 300).map(t => {
132     const tName = t.name || t.task_name || (t.task_path ? t.ta
sk_path.split('\\').pop() : 'Task');
133     const tPath = t.task_path || tName;
134     const status = t.state || t.status || 'Ready';
135     const statusLower = status.toLowerCase();
136     const isReady = statusLower.includes('ready') || statusLow
er.includes('готов');
137     const isRunning = statusLower.includes('running') || statu
sLower.includes('работ') || statusLower.includes('выполн');
138     const isDisabled = statusLower.includes('disabled') || sta
tusLower.includes('отключ');
139
140     let badgeClass = 'bg-secondary';
141     if (isRunning) badgeClass = 'bg-info text-dark';
142     else if (isReady) badgeClass = 'bg-success';
143     else if (isDisabled) badgeClass = 'bg-danger-subtle text-d
anger border border-danger fw-bold';
144
145     const scheduleType = t.schedule_type || 'Custom';
146     const nextRun = t.next_run_time || '-';
147
148     return `
149       <tr>
150         <td class="fw-semibold text-light font-monospace small
" title="${escapeHtml(tPath)}">
151           <div>${escapeHtml(tName)}</div>
152           ${tPath !== tName ? `<div class="text-muted" style="
font-size: 0.75rem;">${escapeHtml(tPath)}</div>` : ''}
153         </td>
154         <td class="text-muted small">${escapeHtml(nextRun)}</t
d>
155         <td><span class="badge ${badgeClass}">${escapeHtml(sta
tus)}</span></td>
156         <td class="small text-muted">${escapeHtml(scheduleType
)}</td>
157         <td class="text-end">
```

```
          <button class="btn btn-xs btn-outline-primary py-0 p
x-2 ts-action-btn" data-task="${escapeHtml(tPath)}" data-actio
n="schtasks_run" title="Запустить немедленно">
            <i class="bi bi-play-fill"></i>
          </button>
          <button class="btn btn-xs ${isDisabled ? 'btn-outlin
e-success' : 'btn-outline-warning'} py-0 px-2 ts-action-btn" d
ata-task="${escapeHtml(tPath)}" data-action="${isDisabled ? 's
chtasks_enable' : 'schtasks_disable'}" title="${isDisabled ? '
Включить' : 'Отключить'}">
            <i class="bi ${isDisabled ? 'bi-check-circle' : 'b
i-pause-fill'}"></i>
          </button>
        </td>
      </tr>
    `;
  }).join('');

  tbody.querySelectorAll('.ts-action-btn').forEach(btn => {
    btn.addEventListener('click', () => {
      const taskTarget = btn.dataset.task;
      const act = btn.dataset.action;
      executeTaskAction(taskTarget, act);
    });
  });
}

async function executeTaskAction(taskTarget, action) {
  if (window.toast) window.toast.info('Планировщик', `${action
}: ${taskTarget}`);
  try {
    const res = await fetch('/api/task-scheduler/action', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        task_path: taskTarget,
        task_name: taskTarget,
        action: action,
```

```
        dry_run: false,
        confirmed_by_user: true
      })
    });
    const result = await res.json();
    if (res.ok && (result.status === 'SUCCESS' || result.statu
s === 'DRY_RUN_SUCCESS')) {
      if (window.toast) window.toast.success('Успех', `${taskT
arget}: ${result.message || 'выполнено'}`);
      fetchTaskSchedulerSummary();
    } else {
      if (window.toast) window.toast.error('Ошибка', result.me
ssage || result.detail || 'Сбой операции');
    }
  } catch (err) {
    if (window.toast) window.toast.error('Ошибка сети', err.me
ssage);
  }
}

export function initTaskSchedulerTab() {
  if (!isInitialized) {
    isInitialized = true;

    const refreshBtn = document.getElementById('ts-refresh-btn
');
    if (refreshBtn) refreshBtn.addEventListener('click', fetch
TaskSchedulerSummary);

    const searchInput = document.getElementById('ts-search-inp
ut');
    if (searchInput) searchInput.addEventListener('input', app
lyTaskFilters);

    registerTabPoller('tab-task-scheduler', fetchTaskScheduler
Summary, 15000, { immediate: true });
  }

```

```
  fetchTaskSchedulerSummary();
}

window.initTaskSchedulerTab = initTaskSchedulerTab;

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/telegram_rag_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/telegram_rag_tab/index.
html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/telegram_rag_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Telegram Channel & Group RAG Search Tab -->
<div class="container-fluid py-2">
<!-- Top Banner / Header -->
<div class="alert alert-primary d-flex flex-wrap align-items-c
enter justify-content-between gap-2 mb-3 shadow-sm">
<div class="d-flex align-items-center gap-2">
<span class="fs-4">□ </span>
<div>
<h5 class="mb-0 fw-bold">Поиск по Telegram каналам и группам (
RAG)</h5>
```

```
<p class="mb-0 small text-muted">Индексация публичных каналов,
 поиск по сообщениям с точными ссылками и мгновенным ответом</
p>
</div>
</div>
<div class="d-flex align-items-center gap-2">
<span class="badge bg-primary-subtle text-primary border borde
r-primary-subtle px-3 py-2" id="tgram-stats-badge">
        Загрузка каналов...
      </span>
<button class="btn btn-sm btn-outline-primary" id="btn-refresh
-tgram-channels" onclick="loadTelegramChannels()">
<i class="bi bi-arrow-clockwise"></i> Обновить
      </button>
</div>
</div>
<div class="row g-3">
<!-- LEFT: Channel Management & Subscriptions -->
<div class="col-lg-4 col-md-5">
<!-- Add / Subscribe Card -->
<div class="card shadow-sm border-secondary-subtle mb-3" data-
panel-id="panel-telegram-rag-dobavit-kanal-gruppu" id="panel-t
elegram-rag-dobavit-kanal-gruppu">
<div class="card-header bg-body-tertiary py-2">
<h6 class="mb-0 fw-bold small text-uppercase d-flex align-item
s-center gap-2">
<i class="bi bi-plus-circle-fill text-primary"></i> Добавить к
анал / группу
          </h6>
</div>
<div class="card-body p-3">
<form id="tgram-add-channel-form" onsubmit="event.preventDefau
lt(); handleAddTelegramChannel();">
<div class="mb-2">
<label class="form-label small fw-semibold mb-1">Имя или ссылк
а на канал</label>
<div class="input-group input-group-sm">
<span class="input-group-text bg-body-secondary">@</span>
```

```
<input class="form-control" id="tgram-channel-input" placehold
er="canozrimb или t.me/canozrimb" required="" type="text"/>
</div>
<div class="form-text text-muted small" style="font-size: 0.75
rem;">
                Поддерживаются публичные каналы и веб-превью T
elegram
              </div>
</div>
<div class="mb-3">
<label class="form-label small fw-semibold mb-1">Лимит сообщен
ий для сбора</label>
<select class="form-select form-select-sm" id="tgram-max-msgs-
select">
<option selected="" value="500">500 сообщений (быстро)</option
>
<option value="1000">1000 сообщений</option>
<option value="3000">3000 сообщений</option>
<option value="5000">5000 сообщений (глубокий RAG)</option>
</select>
</div>
<button class="btn btn-sm btn-primary w-100 d-flex align-items
-center justify-content-center gap-2 shadow-sm" id="btn-tgram-
subscribe" type="submit">
<i class="bi bi-cloud-arrow-down-fill"></i>
<span>Подключить и индексировать</span>
</button>
</form>
</div>
</div>
<!-- Channels List Card -->
<div class="card shadow-sm border-secondary-subtle" data-panel
-id="panel-telegram-rag-podklyuchennye-kanaly-0" id="panel-tel
egram-rag-podklyuchennye-kanaly-0">
<div class="card-header bg-body-tertiary d-flex justify-conten
t-between align-items-center py-2">
<span class="fw-semibold small text-uppercase">
            Подключенные каналы (<span id="tgram-channels-coun
```

```
t">0</span>)
          </span>
<span class="badge bg-secondary-subtle text-body" id="tgram-active-count-badge">0 в пуле</span>
</div>
<div class="card-body p-2" id="tgram-channels-list-container" style="max-height: 55vh; overflow-y: auto;">
<div class="text-center text-muted p-4">
<div class="spinner-border spinner-border-sm text-primary mb-2" role="status"></div>
<div>Загрузка списка каналов...</div>
</div>
</div>
</div>
</div>
<!-- RIGHT: Search Interface & Results -->
<div class="col-lg-8 col-md-7">
<div class="card shadow-sm border-secondary-subtle h-100 d-flex flex-column" data-panel-id="panel-telegram-rag-poisk-po-baze-znaniy-telegram" id="panel-telegram-rag-poisk-po-baze-znaniy-telegram">
<!-- Search Input Bar & Settings Header -->
<div class="card-header bg-body-tertiary p-3 border-bottom">
<!-- Main Search Input -->
<form class="mb-3" id="tgram-search-form" onsubmit="event.preventDefault(); executeTelegramSearch();">
<div class="input-group shadow-sm">
<span class="input-group-text bg-body text-primary"><i class="bi bi-search"></i></span>
<input autofocus="" class="form-control form-control-lg fs-6" id="tgram-search-input" placeholder="Введите поисковый запрос, тему или вопрос..." required="" type="text"/>
<button class="btn btn-outline-secondary" onclick="document.getElementById('tgram-search-input').value='';" title="Очистить" type="button">
<i class="bi bi-x-lg"></i>
</button>
<button class="btn btn-primary px-4 fw-semibold" id="btn-tgram
```

```
-do-search" type="submit">
<i class="bi bi-search me-1"></i> Найти
              </button>
</div>
</form>
<!-- Search Settings & Scope Filters -->
<div class="row g-2 align-items-center bg-body p-2 rounded bor
der border-secondary-subtle small">
<!-- Channel Filter -->
<div class="col-md-5 col-12">
<label class="form-label text-muted mb-1 fw-semibold" style="f
ont-size: 0.75rem;">
<i class="bi bi-funnel-fill text-primary"></i> Область поиска:
              </label>
<select class="form-select form-select-sm" id="tgram-filter-ch
annel">
<option selected="" value="all">  Все подключенные каналы (Об
щий пул)</option>
</select>
</div>
<!-- Search Mode -->
<div class="col-md-4 col-12">
<label class="form-label text-muted mb-1 fw-semibold" style="f
ont-size: 0.75rem;">
<i class="bi bi-diagram-2-fill text-info"></i> Режим индексов:
              </label>
<select class="form-select form-select-sm" id="tgram-search-mo
de">
<option selected="" value="unified">Общий RAG пул (Unified)</o
ption>
<option value="individual">Раздельно по каналам</option>
</select>
</div>
<!-- Top K -->
<div class="col-md-3 col-12">
<label class="form-label text-muted mb-1 fw-semibold" style="f
ont-size: 0.75rem;">
<i class="bi bi-list-ol text-warning"></i> Лимит выдачи:
```

```
              </label>
<select class="form-select form-select-sm" id="tgram-topk-sele
ct">
<option selected="" value="5">5 сообщений</option>
<option value="10">10 сообщений</option>
<option value="20">20 сообщений</option>
<option value="50">50 сообщений</option>
</select>
</div>
</div>
</div>
<!-- Search Results Body -->
<div class="card-body p-3 flex-grow-1" id="tgram-results-conta
iner" style="max-height: 60vh; overflow-y: auto;">
<!-- Initial Placeholder -->
<div class="d-flex flex-column align-items-center justify-cont
ent-center text-muted p-5 text-center" id="tgram-search-placeh
older">
<span class="fs-1 mb-2">□ </span>
<h6 class="fw-bold">Поиск по базе знаний Telegram</h6>
<p class="small text-muted mb-0" style="max-width: 480px;">
              Введите ключевые слова или вопрос выше, чтобы на
йти точные сообщения с ссылками на первоисточники в Telegram.
            </p>
</div>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Telegram Rag Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/telegram_rag_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/telegram_rag_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// ── TELEGRAM CHANNEL & GROUP RAG TAB MAIN.JS ──────────────────

let tgramSubscribedChannels = [];
let tgramAvailableChannels = [];

// Initialize Telegram RAG Tab
window.initTelegramRagTab = async function() {
  console.log('[TelegramRAG] Initializing Telegram RAG Tab...');
  await loadTelegramChannels();
};
```

```
31
32 // Notification helper
33 function showTgramNotification(message, type = 'info') {
34   if (typeof window.showNotification === 'function') {
35     window.showNotification(message, type);
36     return;
37   }
38   console.log(`[TelegramRAG Notification ${type}]: ${message}`
);
39 }
40
41 // Load channels from backend API
42 async function loadTelegramChannels() {
43   const container = document.getElementById('tgram-channels-li
st-container');
44   const statsBadge = document.getElementById('tgram-stats-badg
e');
45   const countEl = document.getElementById('tgram-channels-coun
t');
46   const activeCountBadge = document.getElementById('tgram-acti
ve-count-badge');
47   const filterSelect = document.getElementById('tgram-filter-c
hannel');
48
49   try {
50     const res = await fetch('/api/telegram_rag/channels');
51     if (!res.ok) throw new Error(`HTTP ${res.status}`);
52     const data = await res.json();
53
54     tgramSubscribedChannels = data.user_subscriptions || [];
55     tgramAvailableChannels = data.available_channels || [];
56
57     // Calculate total messages
58     const totalMsgs = tgramAvailableChannels.reduce((sum, c) =
> sum + (c.messages_count || 0), 0);
59
60     if (statsBadge) {
61       statsBadge.textContent = `${tgramSubscribedChannels.leng
```

```
th} подписок • ${totalMsgs} сообщений`;
    }
    if (countEl) countEl.textContent = tgramSubscribedChannels.length;
    if (activeCountBadge) activeCountBadge.textContent = `${tgramAvailableChannels.length} в пуле`;

    // Render channels list
    renderTelegramChannelsList(tgramSubscribedChannels, tgramAvailableChannels);

    // Update filter dropdown
    if (filterSelect) {
      const currentVal = filterSelect.value || 'all';
      let optsHtml = '<option value="all">  Все подключенные каналы (Общий пул)</option>';
      tgramSubscribedChannels.forEach(ch => {
        const isSel = ch === currentVal ? 'selected' : '';
        const found = tgramAvailableChannels.find(a => a.channel === ch);
        const countText = found ? ` (${found.messages_count} сообщ.)` : '';
        optsHtml += `<option value="${ch}" ${isSel}>@${ch}${countText}</option>`;
      });
      filterSelect.innerHTML = optsHtml;
    }
  } catch (err) {
    console.error('Error loading Telegram channels:', err);
    if (container) {
      container.innerHTML = `<div class="alert alert-danger m-2 small">Ошибка загрузки каналов: ${err.message}</div>`;
    }
  }
}

// Render Channels List
function renderTelegramChannelsList(userSubs, availablePool) {
```

```
  const container = document.getElementById('tgram-channels-list-container');
  if (!container) return;

  if (userSubs.length === 0) {
    container.innerHTML = `
      <div class="text-center text-muted p-4 small">
        <i class="bi bi-chat-left-dots fs-3 d-block mb-2 text-secondary"></i>
        Нет подключенных каналов.<br>Добавьте канал выше для индексации.
      </div>
    `;
    return;
  }

  let html = '<div class="list-group list-group-flush">';
  userSubs.forEach(channelName => {
    const info = availablePool.find(p => p.channel === channelName) || {
      channel: channelName,
      messages_count: 0,
      is_indexed: false,
    };

    const statusBadge = info.is_indexed
      ? `<span class="badge bg-success-subtle text-success border border-success-subtle">${info.messages_count} сообщ.</span>`
      : `<span class="badge bg-warning-subtle text-warning border">Не индексирован</span>`;

    html += `
      <div class="list-group-item p-2 rounded mb-1 border-0 bg-body-tertiary shadow-sm">
        <div class="d-flex justify-content-between align-items-center mb-1">
          <div class="d-flex align-items-center gap-2 text-tru
```

```
ncate">
            <span class="fs-5 text-primary">📡</span>
            <a href="https://t.me/${channelName}" target="_bla
nk" rel="noopener noreferrer" class="fw-bold text-decoration-n
one text-truncate" title="Открыть в Telegram">
              @${channelName}
              <i class="bi bi-box-arrow-up-right ms-1 text-mut
ed" style="font-size: 0.75rem;"></i>
            </a>
          </div>
          <div>${statusBadge}</div>
        </div>
        <div class="d-flex justify-content-between align-items
-center mt-2 pt-1 border-top border-secondary-subtle">
          <button class="btn btn-sm btn-outline-secondary py-0
 px-2 small" onclick="filterByChannel('${channelName}')" title
="Искать только в этом канале">
            <i class="bi bi-search"></i> Искать здесь
          </button>
          <div class="btn-group btn-group-sm">
            <button class="btn btn-outline-primary py-0 px-2"
onclick="handleReindexTelegramChannel('${channelName}')" title
="Переиндексировать канал">
              <i class="bi bi-arrow-clockwise"></i>
            </button>
            <button class="btn btn-outline-danger py-0 px-2" o
nclick="handleUnsubscribeTelegramChannel('${channelName}')" ti
tle="Отключить канал">
              <i class="bi bi-trash"></i>
            </button>
          </div>
        </div>
      </div>
    `;
  });
  html += '</div>';
  container.innerHTML = html;
}
```

```
147
148 // Quick filter helper from channel list
149 function filterByChannel(channelName) {
150   const filterSelect = document.getElementById('tgram-filter-c
hannel');
151   if (filterSelect) {
152     filterSelect.value = channelName;
153   }
154   const searchInput = document.getElementById('tgram-search-in
put');
155   if (searchInput) {
156     searchInput.focus();
157   }
158 }
159
160 // Add / Subscribe Channel Handler
161 async function handleAddTelegramChannel() {
162   const input = document.getElementById('tgram-channel-input')
;
163   const maxMsgsSelect = document.getElementById('tgram-max-msg
s-select');
164   const btn = document.getElementById('btn-tgram-subscribe');
165
166   if (!input || !input.value.trim()) return;
167   const channel = input.value.trim();
168   const maxMessages = parseInt(maxMsgsSelect?.value || '500',
10);
169
170   try {
171     if (btn) {
172       btn.disabled = true;
173       btn.innerHTML = '<span class="spinner-border spinner-bor
der-sm me-1" role="status"></span> Индексация...';
174     }
175
176     showTgramNotification(`Подключение канала @${channel}...`,
 'info');
177     const resp = await fetch('/api/telegram_rag/subscribe', {
```

```
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ channel, max_messages: maxMessages }),
    });

    if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
    const data = await resp.json();

    showTgramNotification(data.message || `Канал @${channel} успешно подключен`, 'success');
    input.value = '';
    await loadTelegramChannels();
  } catch (err) {
    console.error('Error adding Telegram channel:', err);
    showTgramNotification(`Ошибка: ${err.message}`, 'danger');
  } finally {
    if (btn) {
      btn.disabled = false;
      btn.innerHTML = '<i class="bi bi-cloud-arrow-down-fill me-1"></i> Подключить и индексировать';
    }
  }
}

// Unsubscribe Channel Handler
async function handleUnsubscribeTelegramChannel(channel) {
  if (!confirm(`Вы действительно хотите отключить канал @${channel}?`)) return;

  try {
    const resp = await fetch('/api/telegram_rag/unsubscribe', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ channel }),
    });
    if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
```

```
    const data = await resp.json();
    showTgramNotification(data.message || `Канал @${channel} отключен`, 'info');
    await loadTelegramChannels();
  } catch (err) {
    console.error('Error unsubscribing channel:', err);
    showTgramNotification(`Ошибка: ${err.message}`, 'danger');
  }
}

// Reindex Channel Handler
async function handleReindexTelegramChannel(channel) {
  const maxMsgs = parseInt(document.getElementById('tgram-max-msgs-select')?.value || '500', 10);
  try {
    showTgramNotification(`Обновление индекса @${channel}...`, 'info');
    const resp = await fetch('/api/telegram_rag/reindex', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ channel, max_messages: maxMsgs }),
    });
    if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
    const data = await resp.json();
    showTgramNotification(data.message || `Индекс @${channel} успешно обновлен`, 'success');
    await loadTelegramChannels();
  } catch (err) {
    console.error('Error reindexing channel:', err);
    showTgramNotification(`Ошибка переиндексации: ${err.message}`, 'danger');
  }
}

// Highlight keywords in text
function highlightKeywords(text, query) {
  if (!text || !query) return text || '';
```

```
  const words = query.split(/\s+/).filter(w => w.length > 2);
  if (words.length === 0) return text;

  let escaped = text
    .replace(/&/g, '&amp;')
    .replace(/</g, '&lt;')
    .replace(/>/g, '&gt;');

  words.forEach(w => {
    const regex = new RegExp(`(${w.replace(/[.*+?^${}()|[\]\\]/g, '\\$&')})`, 'gi');
    escaped = escaped.replace(regex, '<mark class="bg-warning-subtle text-body px-1 rounded">$1</mark>');
  });
  return escaped;
}

// Execute Telegram RAG Search
async function executeTelegramSearch() {
  const searchInput = document.getElementById('tgram-search-input');
  const filterSelect = document.getElementById('tgram-filter-channel');
  const modeSelect = document.getElementById('tgram-search-mode');
  const topkSelect = document.getElementById('tgram-topk-select');
  const resultsContainer = document.getElementById('tgram-results-container');
  const searchBtn = document.getElementById('btn-tgram-do-search');

  if (!searchInput || !searchInput.value.trim()) return;
  const query = searchInput.value.trim();
  const channel = filterSelect ? filterSelect.value : 'all';
  const searchMode = modeSelect ? modeSelect.value : 'unified';
  const topK = parseInt(topkSelect?.value || '5', 10);
```

```
272
273   if (resultsContainer) {
274     resultsContainer.innerHTML = `
275       <div class="text-center text-muted p-5">
276         <div class="spinner-border text-primary mb-3" role="st
atus"></div>
277         <h6>Поиск в базе знаний Telegram...</h6>
278         <div class="small text-muted">Запрос: «${query}»</div>
279       </div>
280     `;
281   }
282
283   if (searchBtn) searchBtn.disabled = true;
284
285   try {
286     const startTime = performance.now();
287     const resp = await fetch('/api/telegram_rag/search', {
288       method: 'POST',
289       headers: { 'Content-Type': 'application/json' },
290       body: JSON.stringify({
291         query,
292         channel: channel === 'all' ? null : channel,
293         search_mode: searchMode,
294         top_k: topK,
295       }),
296     });
297
298     if (!resp.ok) throw new Error(`HTTP ${resp.status}`);
299     const data = await resp.json();
300     const duration = Math.round(performance.now() - startTime)
;
301
302     renderTelegramSearchResults(data.results || [], query, dur
ation);
303   } catch (err) {
304     console.error('Error executing Telegram search:', err);
305     if (resultsContainer) {
306       resultsContainer.innerHTML = `
```

```
        <div class="alert alert-danger shadow-sm">
          <h6><i class="bi bi-exclamation-triangle-fill me-2"></i> Ошибка поиска</h6>
          <p class="mb-0 small">${err.message}</p>
        </div>
      `;
    }
  } finally {
    if (searchBtn) searchBtn.disabled = false;
  }
}

// Render Search Results Cards
function renderTelegramSearchResults(results, query, durationMs) {
  const container = document.getElementById('tgram-results-container');
  if (!container) return;

  if (results.length === 0) {
    container.innerHTML = `
      <div class="text-center text-muted p-5">
        <span class="fs-1 mb-2">  </span>
        <h6 class="fw-bold">Ничего не найдено</h6>
        <p class="small text-muted mb-0">По запросу «${query}» в индексах Telegram сообщений не обнаружено. Попробуйте изменить формулировку или подключить дополнительные каналы.</p>
      </div>
    `;
    return;
  }

  let html = `
    <div class="d-flex justify-content-between align-items-center mb-3 pb-2 border-bottom">
      <div class="small text-muted">
        Найдено результатов: <strong class="text-primary">${results.length}</strong> (${durationMs} мс)
```

```
      </div>
      <span class="badge bg-secondary-subtle text-body border"
>RAG Ranker</span>
    </div>
    <div class="d-flex flex-column gap-3">
  `;

  results.forEach((r, idx) => {
    const channelName = r.channel || 'unknown';
    const author = r.author || 'Telegram';
    const dateStr = r.date || '';
    const msgUrl = r.url || `https://t.me/${channelName}/${r.i
d || ''}`;
    const highlightedSnippet = highlightKeywords(r.snippet ||
r.text || '', query);
    const scoreBadge = r.score !== undefined
      ? `<span class="badge bg-info-subtle text-info border">Р
елевантность: ${Math.round(r.score * 100)}%</span>`
      : '';

    html += `
      <div class="card border shadow-sm rounded-3">
        <div class="card-header bg-body-tertiary d-flex justif
y-content-between align-items-center py-2">
          <div class="d-flex align-items-center gap-2">
            <span class="badge bg-primary text-white">#${idx +
 1}</span>
            <a href="https://t.me/${channelName}" target="_bla
nk" rel="noopener noreferrer" class="fw-bold text-decoration-n
one text-primary">
              @${channelName}
            </a>
            <span class="text-muted small">• ${author}</span>
            ${dateStr ? `<span class="text-muted small">• ${da
teStr}</span>` : ''}
          </div>
          <div>${scoreBadge}</div>
        </div>
```

```
        <div class="card-body p-3">
          <p class="mb-0 small text-body font-monospace" style="white-space: pre-wrap; word-break: break-word;">
            ${highlightedSnippet}
          </p>
        </div>
        <div class="card-footer bg-body-secondary py-2 d-flex justify-content-between align-items-center flex-wrap gap-2">
          <a href="${msgUrl}" target="_blank" rel="noopener noreferrer" class="btn btn-sm btn-outline-primary d-inline-flex align-items-center gap-1">
            <i class="bi bi-box-arrow-up-right"></i>
            <span>Открыть в Telegram</span>
          </a>
          <div class="btn-group btn-group-sm">
            <button class="btn btn-outline-secondary" onclick="copyTelegramLink('${msgUrl}')" title="Скопировать ссылку на сообщение">
              <i class="bi bi-clipboard"></i> Ссылка
            </button>
            <button class="btn btn-outline-success" onclick="sendTelegramResultToChat('${msgUrl}', '${encodeURIComponent(r.snippet || '')}')" title="Вставить цитату в чат">
              <i class="bi bi-chat-quote-fill"></i> В чат
            </button>
          </div>
        </div>
      </div>
    `;
  });

  html += '</div>';
  container.innerHTML = html;
}

// Copy link to clipboard
function copyTelegramLink(url) {
  navigator.clipboard.writeText(url).then(() => {
```

```
    showTgramNotification('Ссылка скопирована в буфер обмена', 'success');
  }).catch(() => {
    prompt('Скопируйте ссылку вручную:', url);
  });
}

// Insert result snippet into main chat
function sendTelegramResultToChat(url, encodedSnippet) {
  const snippet = decodeURIComponent(encodedSnippet);
  const textToInsert = `[Сообщение Telegram](${url}):\n> ${snippet.replace(/\n/g, '\n> ')}`;

  if (typeof window.switchTab === 'function') {
    window.switchTab('tab-chat');
  }

  const msgInput = document.getElementById('message-input');
  if (msgInput) {
    msgInput.value = (msgInput.value ? msgInput.value + '\n\n' : '') + textToInsert;
    msgInput.focus();
    showTgramNotification('Цитата вставлена в поле ввода чата', 'success');
  }
}

// Make globally accessible
window.loadTelegramChannels = loadTelegramChannels;
window.handleAddTelegramChannel = handleAddTelegramChannel;
window.handleUnsubscribeTelegramChannel = handleUnsubscribeTelegramChannel;
window.handleReindexTelegramChannel = handleReindexTelegramChannel;
window.executeTelegramSearch = executeTelegramSearch;
window.filterByChannel = filterByChannel;
window.copyTelegramLink = copyTelegramLink;
window.sendTelegramResultToChat = sendTelegramResultToChat;
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Telemetry Config Tab Vie
w
================================================================
===============
Description:
  Клиентский веб-интерфейс редактора файла конфигурации телеме
трии
  (%APPDATA%\AI-Windows-Admin\apps\windows\telemetry\config.js
on) с поддержкой
  визуального и текстового редактирования и применения на лету
.

Usage Examples:
  Web Route:
    /windows/api/webgui/telemetry_config_tab/index.html

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/telemetry_config_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<div class="container-fluid p-2" id="telemetry-config-tab-cont
ainer">
<!-- Header & Status Card -->
<div class="card shadow-sm border-secondary-subtle mb-3" data-
panel-id="panel-telemetry-config-redaktor-konfiguratsii-teleme
tr" id="panel-telemetry-config-redaktor-konfiguratsii-telemetr
">
<div class="card-body p-3">
```

```
<div class="d-flex flex-wrap align-items-center justify-content-between gap-2">
<div class="d-flex align-items-center gap-3">
<div class="fs-2">⚙ </div>
<div>
<h5 class="mb-0 fw-bold d-flex align-items-center gap-2">
              Редактор конфигурации телеметрии
              <span class="badge bg-success-subtle text-success border border-success-subtle fs-7">Hot Reload Активен</span>
</h5>
<small class="text-secondary d-block mt-1">
              Управление параметрами сбора, режимами реального времени (TC), интервалами опроса датчиков и SQLite БД с мгновенным перечитыванием на каждом тике
            </small>
</div>
</div>
<!-- Toolbar Buttons -->
<div class="d-flex align-items-center gap-2 flex-wrap">
<button class="btn btn-sm btn-outline-info d-flex align-items-center gap-1 shadow-sm" id="tc-cfg-btn-reload" title="Перечитать конфигурацию с диска">
<i class="bi bi-arrow-clockwise"></i>
<span>Перечитать</span>
</button>
<button class="btn btn-sm btn-outline-warning d-flex align-items-center gap-1 shadow-sm" id="tc-cfg-btn-reset" title="Сбросить к исходному шаблону по умолчанию">
<i class="bi bi-arrow-counterclockwise"></i>
<span>По умолчанию</span>
</button>
<button class="btn btn-sm btn-primary d-flex align-items-center gap-1 shadow-sm px-3" id="tc-cfg-btn-save" title="Сохранить изменения и применить на лету">
<i class="bi bi-check2-circle"></i>
<span class="fw-semibold">Сохранить и применить</span>
</button>
</div>
```

```
</div>
<!-- File Location & Meta Bar -->
<div class="d-flex flex-wrap align-items-center justify-content-between gap-2 mt-3 pt-2 border-top border-secondary border-opacity-25 fs-8 text-secondary">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-file-earmark-code text-info"></i>
<span>Файл: <code class="text-info bg-dark px-2 py-0.5 rounded border border-secondary border-opacity-25" id="tc-cfg-file-path">%APPDATA%\AI-Windows-Admin\apps\windows\telemetry\config.json</code></span>
<button class="btn btn-link btn-sm p-0 text-muted hover-info" id="tc-cfg-copy-path-btn" title="Скопировать путь">
<i class="bi bi-clipboard"></i>
</button>
</div>
<div class="d-flex align-items-center gap-3">
<span id="tc-cfg-mtime-wrap"><i class="bi bi-clock-history me-1"></i>Изменен: <span class="text-white" id="tc-cfg-mtime">-</span></span>
<span class="badge bg-primary-subtle text-primary border border-primary-subtle" id="tc-cfg-active-mode-badge">Режим: TELEMETRY</span>
</div>
</div>
</div>
</div>
<!-- Alert / Toast Banner -->
<div class="alert alert-dismissible fade show d-none mb-3 py-2 px-3" id="tc-cfg-alert" role="alert">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-info-circle fs-5" id="tc-cfg-alert-icon"></i>
<span id="tc-cfg-alert-msg">Сообщение</span>
</div>
<button aria-label="Закрыть" class="btn-close py-2" id="tc-cfg-alert-close" type="button"></button>
</div>
<!-- Editor Mode Navigation Tabs -->
```

```
<ul class="nav nav-tabs nav-tabs-bordered mb-3" id="tc-cfg-view-tabs" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active d-flex align-items-center gap-2" data-bs-target="#tc-tab-gui" data-bs-toggle="tab" id="tc-tab-gui-btn" role="tab" type="button">
<i class="bi bi-sliders"></i>
<span>Визуальный конфигуратор</span>
</button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link d-flex align-items-center gap-2" data-bs-target="#tc-tab-json" data-bs-toggle="tab" id="tc-tab-json-btn" role="tab" type="button">
<i class="bi bi-code-square"></i>
<span>Исходный JSON</span>
</button>
</li>
</ul>
<!-- Tab Content -->
<div class="tab-content" id="tc-cfg-tab-content">
<!-- 1. GUI FORM TAB -->
<div class="tab-pane fade show active" id="tc-tab-gui" role="tabpanel">
<form id="tc-cfg-form" onsubmit="return false;">
<div class="row g-3">
<!-- Column 1: Core Telemetry & DB Options -->
<div class="col-lg-6">
<!-- Section 0: Telemetry Mode & Flush (Standard vs TC Realtime) -->
<div class="card bg-dark bg-opacity-50 border-primary border-opacity-50 shadow-sm mb-3" data-panel-id="panel-telemetry-config-rezhim-raboty-telemetrii-i-flus" id="panel-telemetry-config-rezhim-raboty-telemetrii-i-flus">
<div class="card-header bg-dark bg-opacity-75 py-2 px-3 border-primary border-opacity-25 d-flex align-items-center justify-content-between">
<div class="d-flex align-items-center gap-2">
```

```
<i class="bi bi-broadcast text-primary"></i>
<h6 class="mb-0 fw-bold text-white fs-7">Режим работы телеметрии и Flush в БД</h6>
</div>
<span class="badge bg-primary-subtle text-primary border border-primary-subtle" id="tc-current-mode-status">TELEMETRY</span>
</div>
<div class="card-body p-3">
<div class="row g-3">
<div class="col-12">
<label class="form-label text-muted fs-8 mb-1">Режим телеметрии (telemetry_mode):</label>
<select class="form-select form-select-sm bg-dark text-white border-primary" id="tc-field-telemetry-mode">
<option value="telemetry">telemetry — Стандартный режим (Flush каждые 30с или при 100МБ)</option>
<option value="tc">tc — Режим реального времени ТС (Flush каждые 5с, автовозврат через 5 мин)</option>
</select>
</div>
<div class="col-sm-6">
<label class="form-label text-muted fs-8 mb-1">Период Flush в стандартном режиме:</label>
<div class="input-group input-group-sm">
<input class="form-control bg-dark text-white border-secondary" id="tc-field-standard-flush-interval" min="1" step="1" type="number" value="30.0"/>
<span class="input-group-text bg-dark text-muted border-secondary">с</span>
</div>
</div>
<div class="col-sm-6">
<label class="form-label text-muted fs-8 mb-1">Период Flush в режиме ТС (реальное время):</label>
<div class="input-group input-group-sm">
<input class="form-control bg-dark text-white border-secondary" id="tc-field-tc-flush-interval" min="0.5" step="0.5" type="number" value="5.0"/>
```

```
<span class="input-group-text bg-dark text-muted border-second
ary">c</span>
</div>
</div>
<div class="col-sm-6">
<label class="form-label text-muted fs-8 mb-1">Таймаут возврат
а из режима ТС в стандартный:</label>
<div class="input-group input-group-sm">
<input class="form-control bg-dark text-white border-secondary
" id="tc-field-tc-mode-timeout" min="10" step="10" type="numbe
r" value="300.0"/>
<span class="input-group-text bg-dark text-muted border-second
ary">c</span>
</div>
</div>
<div class="col-sm-6">
<label class="form-label text-muted fs-8 mb-1">Макс. размер ба
зы данных SQLite:</label>
<div class="input-group input-group-sm">
<input class="form-control bg-dark text-white border-secondary
" id="tc-field-max-db-size" min="1" step="5" type="number" val
ue="100.0"/>
<span class="input-group-text bg-dark text-muted border-second
ary">МБ</span>
</div>
</div>
</div>
</div>
</div>
<!-- Section 1: Main Engine Parameters -->
<div class="card bg-dark bg-opacity-50 border-secondary border
-opacity-25 shadow-sm mb-3" data-panel-id="panel-telemetry-con
fig-parametry-sbora-sensorov-i-nag" id="panel-telemetry-config
-parametry-sbora-sensorov-i-nag">
<div class="card-header bg-dark bg-opacity-75 py-2 px-3 border
-secondary border-opacity-25 d-flex align-items-center gap-2">
<i class="bi bi-speedometer2 text-info"></i>
<h6 class="mb-0 fw-bold text-white fs-7">Параметры сбора сенсо
```

```
ров и нагрузки</h6>
</div>
<div class="card-body p-3">
<div class="row g-3">
<div class="col-sm-6">
<label class="form-label text-muted fs-8 mb-1">Профиль сбора (
mode):</label>
<select class="form-select form-select-sm bg-dark text-white b
order-secondary" id="tc-field-mode">
<option value="minimal">Minimal (Легковесный CPU/RAM/Диск)</op
tion>
<option selected="" value="hybrid">Hybrid (Базовый + тяжелые п
ериодически)</option>
<option value="full">Full (Полный непрерывный стек)</option>
</select>
</div>
<div class="col-sm-6">
<label class="form-label text-muted fs-8 mb-1">Режим фильтраци
и процессов:</label>
<select class="form-select form-select-sm bg-dark text-white b
order-secondary" id="tc-field-process-mode">
<option selected="" value="top_n">Top-N процессов (по нагрузке
)</option>
<option value="all">Все запущенные процессы</option>
</select>
</div>
<div class="col-sm-6">
<label class="form-label text-muted fs-8 mb-1">Быстрый базовый
 интервал (сек):</label>
<div class="input-group input-group-sm">
<input class="form-control bg-dark text-white border-secondary
" id="tc-field-interval" min="0.1" step="0.1" type="number" va
lue="5.0"/>
<span class="input-group-text bg-dark text-muted border-second
ary">c</span>
</div>
</div>
<div class="col-sm-6">
```

```
<label class="form-label text-muted fs-8 mb-1">Интервал тяжелых сенсоров (сек):</label>
<div class="input-group input-group-sm">
<input class="form-control bg-dark text-white border-secondary" id="tc-field-heavy-interval" min="1" step="1" type="number" value="60.0"/>
<span class="input-group-text bg-dark text-muted border-secondary">c</span>
</div>
</div>
<div class="col-sm-6">
<label class="form-label text-muted fs-8 mb-1">Количество Top-процессов:</label>
<input class="form-control form-control-sm bg-dark text-white border-secondary" id="tc-field-top-processes" max="500" min="1" step="1" type="number" value="10"/>
</div>
<div class="col-sm-6">
<label class="form-label text-muted fs-8 mb-1">Интервал агрегации (Rollup, сек):</label>
<div class="input-group input-group-sm">
<input class="form-control bg-dark text-white border-secondary" id="tc-field-agg-interval" min="10" step="1" type="number" value="3600"/>
<span class="input-group-text bg-dark text-muted border-secondary">c</span>
</div>
</div>
<div class="col-12 pt-2 border-top border-secondary border-opacity-25">
<div class="form-check form-switch mb-1">
<input checked="" class="form-check-input" id="tc-field-low-priority" role="switch" type="checkbox"/>
<label class="form-check-label text-white fs-8" for="tc-field-low-priority">Пониженный приоритет процесса телеметрии (BELOW_NORMAL_PRIORITY_CLASS)</label>
</div>
<div class="form-check form-switch">
```

```
<input checked="" class="form-check-input" id="tc-field-heavy-auto-switch" role="switch" type="checkbox"/>
<label class="form-check-label text-white fs-8" for="tc-field-heavy-auto-switch">Авто-переключение тяжелого режима в гибридный по истечении лимита</label>
</div>
</div>
</div>
</div>
</div>
<!-- Section 2: SQLite DB & Buffer Management -->
<div class="card bg-dark bg-opacity-50 border-secondary border-opacity-25 shadow-sm mb-3" data-panel-id="panel-telemetry-config-baza-dannykh-sqlite-i-buferizats" id="panel-telemetry-config-baza-dannykh-sqlite-i-buferizats">
<div class="card-header bg-dark bg-opacity-75 py-2 px-3 border-secondary border-opacity-25 d-flex align-items-center gap-2">
<i class="bi bi-database text-info"></i>
<h6 class="mb-0 fw-bold text-white fs-7">База данных SQLite и буферизация</h6>
</div>
<div class="card-body p-3">
<div class="row g-3">
<div class="col-sm-6">
<label class="form-label text-muted fs-8 mb-1">Срок хранения записей (дней):</label>
<div class="input-group input-group-sm">
<input class="form-control bg-dark text-white border-secondary" id="tc-field-retention-days" min="1" step="1" type="number" value="7"/>
<span class="input-group-text bg-dark text-muted border-secondary">дн</span>
</div>
</div>
<div class="col-sm-6">
<label class="form-label text-muted fs-8 mb-1">Интервал автоочистки БД (сек):</label>
<div class="input-group input-group-sm">
```

```
<input class="form-control bg-dark text-white border-secondary" id="tc-field-db-cleanup-interval" min="5" step="1" type="number" value="300.0"/>
<span class="input-group-text bg-dark text-muted border-secondary">c</span>
</div>
</div>
<div class="col-sm-6">
<label class="form-label text-muted fs-8 mb-1">Период сброса буфера (сек):</label>
<div class="input-group input-group-sm">
<input class="form-control bg-dark text-white border-secondary" id="tc-field-flush-interval" min="1" step="1" type="number" value="5.0"/>
<span class="input-group-text bg-dark text-muted border-secondary">c</span>
</div>
</div>
<div class="col-sm-6">
<label class="form-label text-muted fs-8 mb-1">Режим буферизации:</label>
<select class="form-select form-select-sm bg-dark text-white border-secondary" id="tc-field-buffer-mode">
<option selected="" value="memory">Memory (Память с периодическим сбросом)</option>
<option value="direct">Direct (Прямая запись без буфера)</option>
<option value="file">File (JSONL аварийный буфер)</option>
</select>
</div>
<div class="col-sm-6">
<label class="form-label text-muted fs-8 mb-1">Размер буфера (записей):</label>
<input class="form-control form-control-sm bg-dark text-white border-secondary" id="tc-field-buffer-size" min="1" step="1" type="number" value="50"/>
</div>
<div class="col-12 pt-2 border-top border-secondary border-opa
```

```
city-25">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="tc-field-auto-vacuum" role="switch" type="checkbox"/>
<label class="form-check-label text-white fs-8" for="tc-field-auto-vacuum">Автоматический VACUUM SQLite базы данных при очистке</label>
</div>
</div>
</div>
</div>
</div>
<!-- Section 3: Heavy Collectors -->
<div class="card bg-dark bg-opacity-50 border-secondary border-opacity-25 shadow-sm mb-3" data-panel-id="panel-telemetry-config-tyazhelye-periodicheskie-kollekto" id="panel-telemetry-config-tyazhelye-periodicheskie-kollekto">
<div class="card-header bg-dark bg-opacity-75 py-2 px-3 border-secondary border-opacity-25 d-flex align-items-center gap-2">
<i class="bi bi-cpu text-warning"></i>
<h6 class="mb-0 fw-bold text-white fs-7">Тяжелые периодические коллекторы (Heavy Collectors)</h6>
</div>
<div class="card-body p-3">
<div class="row g-2">
<div class="col-sm-6">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="tc-hc-hw-sensors" role="switch" type="checkbox"/>
<label class="form-check-label text-white fs-8" for="tc-hc-hw-sensors">Аппаратные датчики (LHM)</label>
</div>
</div>
<div class="col-sm-6">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="tc-hc-smart" role="switch" type="checkbox"/>
<label class="form-check-label text-white fs-8" for="tc-hc-sma
```

```
rt">SMART накопителей</label>
</div>
</div>
<div class="col-sm-6">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="tc-hc-ping" rol
e="switch" type="checkbox"/>
<label class="form-check-label text-white fs-8" for="tc-hc-pin
g">Сетевой пинг и Fast.com</label>
</div>
</div>
<div class="col-sm-6">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="tc-hc-wmi" role
="switch" type="checkbox"/>
<label class="form-check-label text-white fs-8" for="tc-hc-wmi
">WMI инвентарь оборудования</label>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- Column 2: Sensors Matrix & W64/ETW Monitoring -->
<div class="col-lg-6">
<!-- Section 4: Individual Sensors Grid -->
<div class="card bg-dark bg-opacity-50 border-secondary border
-opacity-25 shadow-sm mb-3" data-panel-id="panel-telemetry-con
fig-individualnye-sensory-i-inter" id="panel-telemetry-config-
individualnye-sensory-i-inter">
<div class="card-header bg-dark bg-opacity-75 py-2 px-3 border
-secondary border-opacity-25 d-flex align-items-center justify
-content-between">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-diagram-3 text-success"></i>
<h6 class="mb-0 fw-bold text-white fs-7">Индивидуальные сенсор
ы и интервалы</h6>
</div>
```

```
<span class="badge bg-secondary-subtle text-secondary fs-8" id
="tc-sensors-count-badge">9 сенсоров</span>
</div>
<div class="card-body p-3">
<div class="table-responsive">
<table class="table table-dark table-sm table-hover align-midd
le mb-0 fs-8 border-secondary border-opacity-25" id="tc-sensor
s-table">
<thead>
<tr class="text-secondary border-bottom border-secondary borde
r-opacity-25">
<th style="width: 40px;">Вкл</th>
<th>Сенсор</th>
<th style="width: 130px;">Интервал (сек)</th>
<th>Метрики</th>
</tr>
</thead>
<tbody id="tc-sensors-tbody">
<!-- Rendered dynamically by main.js -->
</tbody>
</table>
</div>
</div>
</div>
<!-- Section 5: W64 & ETW Events -->
<div class="card bg-dark bg-opacity-50 border-secondary border
-opacity-25 shadow-sm mb-3" data-panel-id="panel-telemetry-con
fig-w64-i-etw-monitoring-sistemnykh" id="panel-telemetry-confi
g-w64-i-etw-monitoring-sistemnykh">
<div class="card-header bg-dark bg-opacity-75 py-2 px-3 border
-secondary border-opacity-25 d-flex align-items-center gap-2">
<i class="bi bi-shield-check text-danger"></i>
<h6 class="mb-0 fw-bold text-white fs-7">W64 и ETW мониторинг
системных событий</h6>
</div>
<div class="card-body p-3">
<div class="row g-2">
<div class="col-sm-6">
```

```
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="tc-w64-enabled"
 role="switch" type="checkbox"/>
<label class="form-check-label text-white fs-8" for="tc-w64-en
abled">W64 сборщик активен</label>
</div>
</div>
<div class="col-sm-6">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="tc-w64-file" ro
le="switch" type="checkbox"/>
<label class="form-check-label text-white fs-8" for="tc-w64-fi
le">Мониторинг файлов</label>
</div>
</div>
<div class="col-sm-6">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="tc-w64-proc" ro
le="switch" type="checkbox"/>
<label class="form-check-label text-white fs-8" for="tc-w64-pr
oc">Мониторинг процессов</label>
</div>
</div>
<div class="col-sm-6">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="tc-w64-reg" rol
e="switch" type="checkbox"/>
<label class="form-check-label text-white fs-8" for="tc-w64-re
g">Мониторинг реестра</label>
</div>
</div>
<div class="col-sm-6">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="tc-w64-net" rol
e="switch" type="checkbox"/>
<label class="form-check-label text-white fs-8" for="tc-w64-ne
t">Сетевые события</label>
</div>
```

```
</div>
<div class="col-sm-6">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="tc-w64-evlog" r
ole="switch" type="checkbox"/>
<label class="form-check-label text-white fs-8" for="tc-w64-ev
log">Журнал событий Windows</label>
</div>
</div>
<div class="col-sm-6">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="tc-w64-etw-proc
" role="switch" type="checkbox"/>
<label class="form-check-label text-white fs-8" for="tc-w64-et
w-proc">ETW трассировка процессов</label>
</div>
</div>
<div class="col-sm-6">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="tc-w64-etw-disk
" role="switch" type="checkbox"/>
<label class="form-check-label text-white fs-8" for="tc-w64-et
w-disk">ETW трассировка диска</label>
</div>
</div>
</div>
</div>
</div>
</div>
</div>
</form>
</div>
<!-- 2. RAW JSON EDITOR TAB -->
<div class="tab-pane fade" id="tc-tab-json" role="tabpanel">
<div class="card bg-dark bg-opacity-50 border-secondary border
-opacity-25 shadow-sm" data-panel-id="panel-telemetry-config-p
ryamoe-redaktirovanie-json" id="panel-telemetry-config-pryamoe
-redaktirovanie-json">
```

```
<div class="card-header bg-dark bg-opacity-75 py-2 px-3 border-secondary border-opacity-25 d-flex align-items-center justify-content-between flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-filetype-json text-warning"></i>
<h6 class="mb-0 fw-bold text-white fs-7">Прямое редактирование JSON</h6>
<span class="badge bg-secondary-subtle text-secondary fs-8" id="tc-json-status">Готов</span>
</div>
<div class="d-flex align-items-center gap-2">
<button class="btn btn-sm btn-outline-secondary d-flex align-items-center gap-1" id="tc-btn-format-json" title="Авто-форматирование с отступами">
<i class="bi bi-text-indent-left"></i>
<span>Форматировать</span>
</button>
<button class="btn btn-sm btn-outline-secondary d-flex align-items-center gap-1" id="tc-btn-copy-json" title="Скопировать весь JSON в буфер">
<i class="bi bi-clipboard"></i>
<span>Копировать</span>
</button>
</div>
</div>
<div class="card-body p-2">
<textarea class="form-control font-monospace bg-dark text-info border-secondary border-opacity-25 fs-8" id="tc-raw-json-editor" rows="22" spellcheck="false" style="tab-size: 2; line-height: 1.45;"></textarea>
<div class="text-danger fs-8 mt-2 d-none d-flex align-items-center gap-1" id="tc-json-error">
<i class="bi bi-exclamation-triangle-fill"></i>
<span id="tc-json-error-text">Ошибка парсинга</span>
</div>
</div>
</div>
</div>
```

```
</div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Telemetry Config Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт редактора файла конфигурации телеметрии
 Windows.
 *   Обеспечивает интерактивное визуальное и raw-JSON редактир
ование параметров
 *   с мгновенной синхронизацией и применением на лету.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/telemetry_config_tab/m
ain.js?v=20261006_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/telemetry_config_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

(function () {
  'use strict';

  const API_BASE = '/api/windows/telemetry/config';
  let currentConfig = {};
  let currentRawJson = '';
  let activeTab = 'gui'; // 'gui' | 'json'

```

```
  const SENSOR_META = {
    cpu: { name: 'Процессор (CPU)', icon: 'bi-cpu', defaultInterval: 5.0, defaultMetrics: ['temperature', 'load', 'clocks'] },
    gpu: { name: 'Видеокарта (GPU)', icon: 'bi-gpu-card', defaultInterval: 10.0, defaultMetrics: ['temperature', 'load', 'memory', 'power'] },
    ram: { name: 'Оперативная память (RAM)', icon: 'bi-memory', defaultInterval: 10.0, defaultMetrics: ['usage', 'swap'] },
    disk: { name: 'Диски и разделы (Disk)', icon: 'bi-hdd', defaultInterval: 30.0, defaultMetrics: ['usage', 'io'] },
    network: { name: 'Сетевые адаптеры (Network)', icon: 'bi-ethernet', defaultInterval: 10.0, defaultMetrics: ['throughput', 'connections'] },
    sensors: { name: 'LHM Сенсоры и вентиляторы', icon: 'bi-fan', defaultInterval: 10.0, defaultMetrics: ['temperature', 'fan', 'voltage'] },
    internet: { name: 'Скорость Fast.com / Latency', icon: 'bi-globe2', defaultInterval: 3600.0, defaultMetrics: ['ping', 'download', 'upload', 'bufferbloat', 'latency_loaded'] },
    storage: { name: 'SMART здоровье накопителей', icon: 'bi-hdd-network', defaultInterval: 32000.0, defaultMetrics: ['smart_attributes', 'temperature', 'wear_level'] },
    device_flapping: { name: 'Детектор PnP Flapping', icon: 'bi-usb-symbol', defaultInterval: 30.0, defaultMetrics: ['connect_events', 'disconnect_events', 'flapping_count'] },
  };

  /**
   * Инициализация при монтировании вкладки
   */
  async function init() {
    bindEvents();
    await loadConfig();
  }

  /**
   * Привязка обработчиков событий
```

```
   */
  function bindEvents() {
    // Кнопка сохранения
    const btnSave = document.getElementById('tc-cfg-btn-save')
;
    if (btnSave) {
      btnSave.addEventListener('click', saveActiveConfig);
    }

    // Кнопка перечитывания
    const btnReload = document.getElementById('tc-cfg-btn-relo
ad');
    if (btnReload) {
      btnReload.addEventListener('click', async () => {
        showAlert('Перечитывание конфигурации с диска...', 'in
fo');
        await loadConfig();
      });
    }

    // Кнопка сброса к значениям по умолчанию
    const btnReset = document.getElementById('tc-cfg-btn-reset
');
    if (btnReset) {
      btnReset.addEventListener('click', async () => {
        if (confirm('Сбросить конфигурацию телеметрии к исходн
ым настройкам по умолчанию?')) {
          await resetConfigToDefault();
        }
      });
    }

    // Кнопка копирования пути к файлу
    const btnCopyPath = document.getElementById('tc-cfg-copy-p
ath-btn');
    if (btnCopyPath) {
      btnCopyPath.addEventListener('click', () => {
        const pathEl = document.getElementById('tc-cfg-file-pa
```

```
th');
        if (pathEl) {
          navigator.clipboard.writeText(pathEl.innerText.trim(
)).then(() => {
            showAlert('Путь к файлу скопирован в буфер обмена'
, 'success', 2500);
          });
        }
      });
    }

    // Переключение вкладок GUI / JSON
    const tabGuiBtn = document.getElementById('tc-tab-gui-btn'
);
    const tabJsonBtn = document.getElementById('tc-tab-json-bt
n');
    if (tabGuiBtn && tabJsonBtn) {
      tabGuiBtn.addEventListener('shown.bs.tab', () => {
        activeTab = 'gui';
        syncJsonToGui();
      });
      tabJsonBtn.addEventListener('shown.bs.tab', () => {
        activeTab = 'json';
        syncGuiToJson();
      });
    }

    // Форматирование JSON
    const btnFormatJson = document.getElementById('tc-btn-form
at-json');
    if (btnFormatJson) {
      btnFormatJson.addEventListener('click', () => {
        const editor = document.getElementById('tc-raw-json-ed
itor');
        if (!editor) return;
        try {
          const parsed = JSON.parse(editor.value);
          editor.value = JSON.stringify(parsed, null, 2);
```

```
          hideJsonError();
          showAlert('JSON отформатирован', 'info', 2000);
        } catch (e) {
          showJsonError(e.message);
        }
      });
    }

    // Копирование JSON
    const btnCopyJson = document.getElementById('tc-btn-copy-j
son');
    if (btnCopyJson) {
      btnCopyJson.addEventListener('click', () => {
        const editor = document.getElementById('tc-raw-json-ed
itor');
        if (editor) {
          navigator.clipboard.writeText(editor.value).then(()
=> {
            showAlert('JSON скопирован в буфер обмена', 'succe
ss', 2500);
          });
        }
      });
    }

    // Live-валидация JSON при вводе
    const jsonEditor = document.getElementById('tc-raw-json-ed
itor');
    if (jsonEditor) {
      jsonEditor.addEventListener('input', () => {
        try {
          JSON.parse(jsonEditor.value);
          hideJsonError();
        } catch (e) {
          showJsonError(e.message);
        }
      });
    }
```

```
149
150     // Закрытие алерта
151     const alertCloseBtn = document.getElementById('tc-cfg-aler
t-close');
152     if (alertCloseBtn) {
153       alertCloseBtn.addEventListener('click', () => {
154         const alertBox = document.getElementById('tc-cfg-alert
');
155         if (alertBox) alertBox.classList.add('d-none');
156       });
157     }
158   }
159
160   /**
161    * Загрузка конфигурации с бэкенда
162    */
163   async function loadConfig() {
164     try {
165       const response = await fetch(API_BASE, { cache: 'no-cach
e' });
166       if (!response.ok) {
167         throw new Error(`HTTP error ${response.status}`);
168       }
169       const data = await response.json();
170       if (data.success) {
171         currentConfig = data.config || {};
172         updateMeta(data.config_path, data.last_modified, curre
ntConfig.mode, currentConfig.telemetry_mode);
173         renderGuiForm(currentConfig);
174         renderRawJson(currentConfig);
175         showAlert('Конфигурация успешно загружена', 'success',
 2500);
176       } else {
177         showAlert(data.message || 'Ошибка загрузки конфигураци
и', 'danger');
178       }
179     } catch (err) {
180       console.error('Ошибка загрузки конфигурации телеметрии:'
```

```
, err);
181       showAlert(`Не удалось загрузить конфигурацию: ${err.mess
age}`, 'danger');
182     }
183   }
184
185   /**
186    * Обновление метаинформации о файле
187    */
188   function updateMeta(path, mtime, mode, telemetryMode) {
189     const pathEl = document.getElementById('tc-cfg-file-path')
;
190     if (pathEl && path) pathEl.innerText = path;
191
192     const mtimeEl = document.getElementById('tc-cfg-mtime');
193     if (mtimeEl && mtime) {
194       try {
195         const d = new Date(mtime);
196         mtimeEl.innerText = d.toLocaleString('ru-RU');
197       } catch (_) {
198         mtimeEl.innerText = mtime;
199       }
200     }
201
202     const tMode = (telemetryMode || 'telemetry').toUpperCase()
;
203     const modeBadge = document.getElementById('tc-cfg-active-m
ode-badge');
204     if (modeBadge) {
205       modeBadge.innerText = `Режим: ${tMode}`;
206       modeBadge.className = `badge border fs-8 ${
207         tMode === 'TC' ? 'bg-warning-subtle text-warning borde
r-warning-subtle' :
208         'bg-primary-subtle text-primary border-primary-subtle'
209       }`;
210     }
211
212     const currentModeStatus = document.getElementById('tc-curr
```

```
ent-mode-status');
213     if (currentModeStatus) {
214       currentModeStatus.innerText = tMode === 'TC' ? 'TC (РЕАЛ
ЬНОЕ ВРЕМЯ)' : 'СТАНДАРТНЫЙ (TELEMETRY)';
215       currentModeStatus.className = `badge border ${tMode ===
'TC' ? 'bg-warning-subtle text-warning border-warning-subtle'
: 'bg-primary-subtle text-primary border-primary-subtle'}`;
216     }
217   }
218
219   /**
220    * Отрисовка значений в элементах GUI формы
221    */
222   function renderGuiForm(cfg) {
223     setVal('tc-field-telemetry-mode', cfg.telemetry_mode || 't
elemetry');
224     setVal('tc-field-standard-flush-interval', cfg.standard_fl
ush_interval_seconds ?? 30.0);
225     setVal('tc-field-tc-flush-interval', cfg.tc_flush_interval
_seconds ?? 5.0);
226     setVal('tc-field-tc-mode-timeout', cfg.tc_mode_timeout_sec
onds ?? 300.0);
227     setVal('tc-field-max-db-size', cfg.max_db_size_mb ?? 100.0
);
228
229     setVal('tc-field-mode', cfg.mode || 'hybrid');
230     setVal('tc-field-process-mode', cfg.process_mode || 'top_n
');
231     setVal('tc-field-interval', cfg.interval_seconds ?? 5.0);
232     setVal('tc-field-heavy-interval', cfg.heavy_interval_secon
ds ?? 60.0);
233     setVal('tc-field-top-processes', cfg.top_processes ?? 10);
234     setVal('tc-field-agg-interval', cfg.aggregation_interval_s
econds ?? 3600);
235     setChecked('tc-field-low-priority', cfg.low_priority !== f
alse);
236     setChecked('tc-field-heavy-auto-switch', cfg.heavy_mode_au
to_switch_enabled !== false);
```

```
237
238     setVal('tc-field-retention-days', cfg.retention_days ?? 7)
    ;
239     setVal('tc-field-db-cleanup-interval', cfg.db_cleanup_inte
    rval_seconds ?? 300.0);
240     setVal('tc-field-flush-interval', cfg.flush_interval_secon
    ds ?? 5.0);
241     setVal('tc-field-buffer-mode', cfg.buffer_mode || 'memory'
    );
242     setVal('tc-field-buffer-size', cfg.buffer_size ?? 50);
243     setChecked('tc-field-auto-vacuum', cfg.auto_vacuum_enabled
     !== false);
244
245     const hc = cfg.heavy_collectors || {};
246     setChecked('tc-hc-hw-sensors', hc.hardware_sensors !== fal
    se);
247     setChecked('tc-hc-smart', hc.storage_smart !== false);
248     setChecked('tc-hc-ping', hc.network_ping !== false);
249     setChecked('tc-hc-wmi', hc.inventory_wmi === true);
250
251     const w64 = cfg.w64_collector || {};
252     setChecked('tc-w64-enabled', w64.enabled !== false);
253     setChecked('tc-w64-file', w64.enable_file_monitoring !== f
    alse);
254     setChecked('tc-w64-proc', w64.enable_process_monitoring !
    = false);
255     setChecked('tc-w64-reg', w64.enable_registry_monitoring !
    = false);
256     setChecked('tc-w64-net', w64.enable_network_monitoring !==
     false);
257     setChecked('tc-w64-evlog', w64.enable_event_log_monitoring
     !== false);
258     setChecked('tc-w64-etw-proc', w64.enable_process_trace !==
     false);
259     setChecked('tc-w64-etw-disk', w64.enable_disk_trace !== fa
    lse);
260
261     renderSensorsTable(cfg.sensors || {});
```

```
  }

  /**
   * Отрисовка таблицы сенсоров
   */
  function renderSensorsTable(sensorsConfig) {
    const tbody = document.getElementById('tc-sensors-tbody');
    if (!tbody) return;

    tbody.innerHTML = '';
    const sensorKeys = Object.keys(SENSOR_META);

    // Добавляем также сенсоры, которые есть в конфиге, но не в SENSOR_META
    Object.keys(sensorsConfig).forEach((k) => {
      if (!sensorKeys.includes(k)) sensorKeys.push(k);
    });

    const countBadge = document.getElementById('tc-sensors-count-badge');
    if (countBadge) countBadge.innerText = `${sensorKeys.length} сенсоров`;

    sensorKeys.forEach((key) => {
      const sMeta = SENSOR_META[key] || {
        name: key,
        icon: 'bi-activity',
        defaultInterval: 10.0,
        defaultMetrics: [],
      };
      const sConf = sensorsConfig[key] || {
        enabled: true,
        interval_seconds: sMeta.defaultInterval,
        metrics: sMeta.defaultMetrics,
      };

      const tr = document.createElement('tr');
      tr.className = 'border-bottom border-secondary border-op
```

```
acity-10';

      const isEnabled = sConf.enabled !== false;
      const intervalSec = sConf.interval_seconds ?? sMeta.defa
ultInterval;
      const metricsList = Array.isArray(sConf.metrics) ? sConf
.metrics.join(', ') : '';

      tr.innerHTML = `
        <td class="text-center">
          <div class="form-check form-switch mb-0">
            <input class="form-check-input tc-sensor-enabled"
type="checkbox" role="switch" data-sensor="${key}" ${isEnabled
 ? 'checked' : ''}>
          </div>
        </td>
        <td>
          <div class="d-flex align-items-center gap-2">
            <i class="bi ${sMeta.icon} text-info"></i>
            <div>
              <span class="fw-semibold text-white">${sMeta.nam
e}</span>
              <span class="text-secondary d-block fs-9 font-mo
nospace">${key}</span>
            </div>
          </div>
        </td>
        <td>
          <div class="input-group input-group-sm" style="max-w
idth: 120px;">
            <input type="number" step="0.5" min="0.1" class="f
orm-control form-control-sm bg-dark text-white border-secondar
y tc-sensor-interval" data-sensor="${key}" value="${intervalSe
c}">
            <span class="input-group-text bg-dark text-muted b
order-secondary fs-9">c</span>
          </div>
        </td>
```

```
        <td>
          <span class="text-secondary fs-9 font-monospace" tit
le="${metricsList}">${metricsList || '—'}</span>
        </td>
      `;

      tbody.appendChild(tr);
    });
  }

  /**
   * Сбор параметров из элементов GUI формы в объект
   */
  function gatherFormConfig() {
    const cfg = JSON.parse(JSON.stringify(currentConfig));

    cfg.telemetry_mode = getVal('tc-field-telemetry-mode') ||
'telemetry';
    cfg.standard_flush_interval_seconds = parseFloat(getVal('t
c-field-standard-flush-interval')) || 30.0;
    cfg.tc_flush_interval_seconds = parseFloat(getVal('tc-fiel
d-tc-flush-interval')) || 5.0;
    cfg.tc_mode_timeout_seconds = parseFloat(getVal('tc-field-
tc-mode-timeout')) || 300.0;
    cfg.max_db_size_mb = parseFloat(getVal('tc-field-max-db-si
ze')) || 100.0;

    cfg.mode = getVal('tc-field-mode') || 'hybrid';
    cfg.process_mode = getVal('tc-field-process-mode') || 'top
_n';
    cfg.interval_seconds = parseFloat(getVal('tc-field-interva
l')) || 5.0;
    cfg.heavy_interval_seconds = parseFloat(getVal('tc-field-h
eavy-interval')) || 60.0;
    cfg.top_processes = parseInt(getVal('tc-field-top-processe
s'), 10) || 10;
    cfg.aggregation_interval_seconds = parseInt(getVal('tc-fie
ld-agg-interval'), 10) || 3600;
```

```
    cfg.low_priority = getChecked('tc-field-low-priority');
    cfg.heavy_mode_auto_switch_enabled = getChecked('tc-field-heavy-auto-switch');

    cfg.retention_days = parseInt(getVal('tc-field-retention-days'), 10) || 7;
    cfg.db_cleanup_interval_seconds = parseFloat(getVal('tc-field-db-cleanup-interval')) || 300.0;
    cfg.flush_interval_seconds = parseFloat(getVal('tc-field-flush-interval')) || 5.0;
    cfg.buffer_mode = getVal('tc-field-buffer-mode') || 'memory';
    cfg.buffer_size = parseInt(getVal('tc-field-buffer-size'), 10) || 50;
    cfg.auto_vacuum_enabled = getChecked('tc-field-auto-vacuum');

    cfg.heavy_collectors = cfg.heavy_collectors || {};
    cfg.heavy_collectors.hardware_sensors = getChecked('tc-hc-hw-sensors');
    cfg.heavy_collectors.storage_smart = getChecked('tc-hc-smart');
    cfg.heavy_collectors.network_ping = getChecked('tc-hc-ping');
    cfg.heavy_collectors.inventory_wmi = getChecked('tc-hc-wmi');

    cfg.w64_collector = cfg.w64_collector || {};
    cfg.w64_collector.enabled = getChecked('tc-w64-enabled');
    cfg.w64_collector.enable_file_monitoring = getChecked('tc-w64-file');
    cfg.w64_collector.enable_process_monitoring = getChecked('tc-w64-proc');
    cfg.w64_collector.enable_registry_monitoring = getChecked('tc-w64-reg');
    cfg.w64_collector.enable_network_monitoring = getChecked('tc-w64-net');
    cfg.w64_collector.enable_event_log_monitoring = getChecked
```

```
('tc-w64-evlog');
    cfg.w64_collector.enable_process_trace = getChecked('tc-w64-etw-proc');
    cfg.w64_collector.enable_disk_trace = getChecked('tc-w64-etw-disk');

    cfg.sensors = cfg.sensors || {};
    document.querySelectorAll('.tc-sensor-enabled').forEach((cb) => {
      const sKey = cb.getAttribute('data-sensor');
      if (!sKey) return;
      cfg.sensors[sKey] = cfg.sensors[sKey] || {};
      cfg.sensors[sKey].enabled = cb.checked;
    });

    document.querySelectorAll('.tc-sensor-interval').forEach((inp) => {
      const sKey = inp.getAttribute('data-sensor');
      if (!sKey) return;
      cfg.sensors[sKey] = cfg.sensors[sKey] || {};
      cfg.sensors[sKey].interval_seconds = parseFloat(inp.value) || 5.0;
    });

    return cfg;
  }

  /**
   * Отрисовка текста в редакторе сырого JSON
   */
  function renderRawJson(cfg) {
    const editor = document.getElementById('tc-raw-json-editor');
    if (editor) {
      editor.value = JSON.stringify(cfg, null, 2);
      hideJsonError();
    }
  }
```

```
404
405   /**
406    * Синхронизация формы в JSON при переходе на вкладку JSON
407    */
408   function syncGuiToJson() {
409     try {
410       const updated = gatherFormConfig();
411       currentConfig = updated;
412       renderRawJson(updated);
413     } catch (e) {
414       console.warn('Ошибка сбора формы:', e);
415     }
416   }
417
418   /**
419    * Синхронизация JSON в форму при переходе на вкладку GUI
420    */
421   function syncJsonToGui() {
422     const editor = document.getElementById('tc-raw-json-editor
');
423     if (!editor) return;
424     try {
425       const parsed = JSON.parse(editor.value);
426       currentConfig = parsed;
427       renderGuiForm(parsed);
428       hideJsonError();
429     } catch (e) {
430       showJsonError(`Невозможно отобразить визуальную форму: $
{e.message}`);
431     }
432   }
433
434   /**
435    * Сохранение активной конфигурации (из GUI или JSON)
436    */
437   async function saveActiveConfig() {
438     const btnSave = document.getElementById('tc-cfg-btn-save')
;
```

```
    const originalText = btnSave ? btnSave.innerHTML : '';
    if (btnSave) {
      btnSave.disabled = true;
      btnSave.innerHTML = '<span class="spinner-border spinner-border-sm me-1" role="status"></span><span>Сохранение...</span>';
    }

    try {
      if (activeTab === 'json') {
        const editor = document.getElementById('tc-raw-json-editor');
        if (!editor) throw new Error('Редактор JSON не найден');
        const rawJson = editor.value;

        // Предварительная валидация
        JSON.parse(rawJson);

        const response = await fetch(`${API_BASE}/raw`, {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ raw_json: rawJson }),
        });

        const resData = await response.json();
        if (!response.ok || !resData.success) {
          throw new Error(resData.detail || resData.message || `Ошибка сервера ${response.status}`);
        }

        currentConfig = resData.config;
        renderGuiForm(currentConfig);
        updateMeta(resData.config_path, new Date().toISOString(), currentConfig.mode, currentConfig.telemetry_mode);
        showAlert('✅ Конфигурация телеметрии успешно сохранена и применена на лету!', 'success');
      } else {
```

```
470         const updatedConfig = gatherFormConfig();
471         const response = await fetch(API_BASE, {
472           method: 'POST',
473           headers: { 'Content-Type': 'application/json' },
474           body: JSON.stringify({ config: updatedConfig }),
475         });
476 
477         const resData = await response.json();
478         if (!response.ok || !resData.success) {
479           throw new Error(resData.detail || resData.message ||
 `Ошибка сервера ${response.status}`);
480         }
481 
482         currentConfig = resData.config;
483         renderRawJson(currentConfig);
484         updateMeta(resData.config_path, new Date().toISOString
(), currentConfig.mode, currentConfig.telemetry_mode);
485         showAlert('✅ Параметры телеметрии успешно сохранены и
применены на лету!', 'success');
486       }
487     } catch (err) {
488       console.error('Ошибка сохранения конфигурации:', err);
489       showAlert(`❌ Ошибка сохранения: ${err.message}`, 'danger
');
490       if (activeTab === 'json') {
491         showJsonError(err.message);
492       }
493     } finally {
494       if (btnSave) {
495         btnSave.disabled = false;
496         btnSave.innerHTML = originalText;
497       }
498     }
499   }
500 
501   /**
502    * Сброс конфигурации к значениям по умолчанию
503    */
```

```
  async function resetConfigToDefault() {
    try {
      const response = await fetch(`${API_BASE}/reset`, {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
      });
      const data = await response.json();
      if (data.success) {
        currentConfig = data.config || {};
        updateMeta(data.config_path, new Date().toISOString(),
 currentConfig.mode, currentConfig.telemetry_mode);
        renderGuiForm(currentConfig);
        renderRawJson(currentConfig);
        showAlert('  Конфигурация телеметрии сброшена к значен
иям по умолчанию!', 'warning');
      } else {
        showAlert(data.message || 'Ошибка сброса конфигурации'
, 'danger');
      }
    } catch (err) {
      showAlert(`Не удалось сбросить конфигурацию: ${err.messa
ge}`, 'danger');
    }
  }

  /**
   * Показать плашку ошибки валидации JSON
   */
  function showJsonError(msg) {
    const errBox = document.getElementById('tc-json-error');
    const errText = document.getElementById('tc-json-error-tex
t');
    const statusBadge = document.getElementById('tc-json-statu
s');
    if (errBox && errText) {
      errText.innerText = msg;
      errBox.classList.remove('d-none');
    }
```

```
    if (statusBadge) {
      statusBadge.innerText = 'Ошибка синтаксиса';
      statusBadge.className = 'badge bg-danger-subtle text-dan
ger border border-danger-subtle fs-8';
    }
  }

  /**
   * Скрыть плашку ошибки валидации JSON
   */
  function hideJsonError() {
    const errBox = document.getElementById('tc-json-error');
    const statusBadge = document.getElementById('tc-json-statu
s');
    if (errBox) errBox.classList.add('d-none');
    if (statusBadge) {
      statusBadge.innerText = 'Синтаксис корректен';
      statusBadge.className = 'badge bg-success-subtle text-su
ccess border border-success-subtle fs-8';
    }
  }

  /**
   * Отображение Toast / Alert сообщения
   */
  function showAlert(msg, type = 'info', autoHideMs = 4000) {
    const alertBox = document.getElementById('tc-cfg-alert');
    const alertMsg = document.getElementById('tc-cfg-alert-msg
');
    const alertIcon = document.getElementById('tc-cfg-alert-ic
on');
    if (!alertBox || !alertMsg) return;

    alertMsg.innerText = msg;
    alertBox.className = `alert alert-${type} alert-dismissibl
e fade show mb-3 py-2 px-3 shadow-sm border`;

    if (alertIcon) {
```

```
      alertIcon.className =
        type === 'success' ? 'bi bi-check-circle-fill text-success fs-5' :
        type === 'danger' ? 'bi bi-exclamation-triangle-fill text-danger fs-5' :
        type === 'warning' ? 'bi bi-exclamation-circle-fill text-warning fs-5' :
        'bi bi-info-circle-fill text-info fs-5';
    }

    alertBox.classList.remove('d-none');

    if (autoHideMs > 0) {
      setTimeout(() => {
        if (!alertBox.classList.contains('d-none')) {
          alertBox.classList.add('d-none');
        }
      }, autoHideMs);
    }
  }

  // Вспомогательные утилиты для работы с элементами DOM
  function setVal(id, val) {
    const el = document.getElementById(id);
    if (el) el.value = val;
  }

  function getVal(id) {
    const el = document.getElementById(id);
    return el ? el.value : '';
  }

  function setChecked(id, val) {
    const el = document.getElementById(id);
    if (el) el.checked = Boolean(val);
  }

  function getChecked(id) {
```

```
    const el = document.getElementById(id);
    return el ? Boolean(el.checked) : false;
  }

  // Запуск при загрузке DOM
  if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', init);
  } else {
    init();
  }
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/telemetry_research_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/telemetry_research_tab/
index.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/telemetry_research_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<div class="container-fluid p-2" id="telemetry-research-tab-co
ntainer">
<!-- Header & Control Toolbar -->
<div class="card shadow-sm border-secondary-subtle mb-3" data-
panel-id="panel-telemetry-research-issledovaniya-telemetrii-i-
obor" id="panel-telemetry-research-issledovaniya-telemetrii-i-
obor">
<div class="card-body p-3">
<div class="d-flex flex-wrap align-items-center justify-conten
t-between gap-2 mb-3">
```

```
<div class="d-flex align-items-center gap-2">
<div class="fs-2">□ </div>
<div>
<h5 class="mb-0 fw-bold">Исследования телеметрии и оборудовани
я</h5>
<small class="text-light text-opacity-75">Глубокий анализ корр
еляций, проверка гипотез производительности, аномалии и Health
 Score</small>
</div>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<div class="input-group input-group-sm" style="width: auto;">
<span class="input-group-text bg-dark text-muted border-second
ary"><i class="bi bi-folder2-open"></i></span>
<select class="form-select form-select-sm bg-dark text-white b
order-secondary" id="tr-select-source" style="max-width: 260px
;">
<option value="">Все доступные логи телеметрии</option>
</select>
</div>
<button class="btn btn-sm btn-primary d-flex align-items-cente
r gap-1 shadow-sm" id="tr-btn-run" title="Запустить полное исс
ледование">
<i class="bi bi-play-circle-fill"></i>
<span>Исследовать</span>
</button>
<button class="btn btn-sm btn-outline-success d-flex align-ite
ms-center gap-1 shadow-sm" id="tr-btn-dashboard" title="Открыт
ь автономный HTML-дашборд в новой вкладке">
<i class="bi bi-file-earmark-bar-graph"></i>
<span>HTML-дашборд</span>
</button>
<button class="btn btn-sm btn-outline-secondary d-flex align-i
tems-center gap-1 shadow-sm" id="tr-btn-json" title="Скачать J
SON отчет">
<i class="bi bi-download"></i>
<span>JSON</span>
</button>
```

```
</div>
</div>
<!-- KPI Metrics Cards -->
<div class="row g-2 text-white">
<!-- Health Score -->
<div class="col-6 col-md-3">
<div class="p-2 rounded bg-dark bg-opacity-50 border border-secondary d-flex align-items-center gap-3">
<div class="fs-3 text-info"><i class="bi bi-heart-pulse-fill"></i></div>
<div class="flex-grow-1">
<div class="text-muted small">Индекс здоровья</div>
<div class="d-flex align-items-baseline gap-2">
<span class="h5 mb-0 fw-bold" id="tr-val-health">--</span>
<span class="small text-muted">/100</span>
</div>
<div class="progress mt-1" style="height: 4px;">
<div class="progress-bar bg-success" id="tr-progress-health" style="width: 0%;"></div>
</div>
</div>
</div>
</div>
<!-- Records Analyzed -->
<div class="col-6 col-md-3">
<div class="p-2 rounded bg-dark bg-opacity-50 border border-secondary d-flex align-items-center gap-3">
<div class="fs-3 text-primary"><i class="bi bi-database-check"></i></div>
<div>
<div class="text-muted small">Записей исследовано</div>
<div class="h5 mb-0 fw-bold" id="tr-val-records">0</div>
<div class="small text-muted" id="tr-val-timespan">Временной срез</div>
</div>
</div>
</div>
<!-- Hypotheses Confirmed -->
```

```
<div class="col-6 col-md-3">
<div class="p-2 rounded bg-dark bg-opacity-50 border border-secondary d-flex align-items-center gap-3">
<div class="fs-3 text-warning"><i class="bi bi-lightbulb-fill"></i></div>
<div>
<div class="text-muted small">Проблемных гипотез</div>
<div class="h5 mb-0 fw-bold" id="tr-val-hypotheses">0</div>
<div class="small text-muted">Из проверенных</div>
</div>
</div>
</div>
<!-- Anomalies & Incidents -->
<div class="col-6 col-md-3">
<div class="p-2 rounded bg-dark bg-opacity-50 border border-secondary d-flex align-items-center gap-3">
<div class="fs-3 text-danger"><i class="bi bi-exclamation-triangle-fill"></i></div>
<div>
<div class="text-muted small">Аномалии и сбои</div>
<div class="h5 mb-0 fw-bold" id="tr-val-anomalies">0</div>
<div class="small text-muted" id="tr-val-devices">Устройства в норме</div>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- Summary Alert Banner -->
<div class="alert alert-info border-info d-flex align-items-center gap-2 mb-3 py-2 px-3 small d-none" id="tr-summary-banner">
<i class="bi bi-info-circle-fill fs-5"></i>
<div class="flex-grow-1" id="tr-summary-text"></div>
</div>
<!-- Tabs Navigation -->
<ul class="nav nav-tabs border-secondary mb-3" id="trResearchT
```

```
abs" role="tablist">
<li class="nav-item" role="presentation">
<button class="nav-link active fw-medium py-2 px-3" data-bs-target="#tr-tab-hypotheses" data-bs-toggle="tab" id="tr-tab-hypotheses-btn" role="tab" type="button">
        Проверка гипотез
      </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link fw-medium py-2 px-3" data-bs-target="#tr-tab-correlations" data-bs-toggle="tab" id="tr-tab-correlations-btn" role="tab" type="button">
        Матрица корреляций
      </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link fw-medium py-2 px-3" data-bs-target="#tr-tab-charts" data-bs-toggle="tab" id="tr-tab-charts-btn" role="tab" type="button">
        Графики метрик
      </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link fw-medium py-2 px-3" data-bs-target="#tr-tab-anomalies" data-bs-toggle="tab" id="tr-tab-anomalies-btn" role="tab" type="button">
        Аномалии и устройства
      </button>
</li>
<li class="nav-item" role="presentation">
<button class="nav-link fw-medium py-2 px-3" data-bs-target="#tr-tab-records" data-bs-toggle="tab" id="tr-tab-records-btn" role="tab" type="button">
        Записи телеметрии
      </button>
</li>
</ul>
<!-- Tab Contents -->
```

```
<div class="tab-content" id="trResearchTabContent">
<!-- 1. Hypotheses Tab -->
<div class="tab-pane fade show active" id="tr-tab-hypotheses"
role="tabpanel">
<div class="row g-3" id="tr-hypotheses-list">
<div class="col-12 text-center text-muted py-4">Нажмите «Иссле
довать» для проверки системных гипотез...</div>
</div>
</div>
<!-- 2. Correlations Tab -->
<div class="tab-pane fade" id="tr-tab-correlations" role="tabp
anel">
<div class="card bg-dark border-secondary" data-panel-id="pane
l-telemetry-research-vzaimosvyazi-mezhdu-sistemnymi-m" id="pan
el-telemetry-research-vzaimosvyazi-mezhdu-sistemnymi-m">
<div class="card-header border-secondary d-flex align-items-ce
nter justify-content-between">
<span class="fw-bold small text-light">Взаимосвязи между систе
мными метриками (Пирсон)</span>
<span class="badge bg-secondary" id="tr-correlations-count">0
пар</span>
</div>
<div class="table-responsive">
<table class="table table-dark table-hover table-striped mb-0
align-middle small">
<thead>
<tr class="text-muted border-secondary">
<th>Метрика А</th>
<th>Метрика Б</th>
<th>Коэффициент (r)</th>
<th>Выборка</th>
<th>Интерпретация</th>
</tr>
</thead>
<tbody id="tr-correlations-tbody">
<tr><td class="text-center text-muted py-3" colspan="5">Нет да
нных корреляций</td></tr>
</tbody>
```

```
</table>
</div>
</div>
</div>
<!-- 3. Charts Tab -->
<div class="tab-pane fade" id="tr-tab-charts" role="tabpanel">
<div class="row g-3" id="tr-charts-container">
<div class="col-12 text-center text-muted py-4">Графики появятся после запуска исследования...</div>
</div>
</div>
<!-- 4. Anomalies & Devices Tab -->
<div class="tab-pane fade" id="tr-tab-anomalies" role="tabpanel">
<div class="row g-3">
<!-- Anomalies Table -->
<div class="col-12 col-lg-7">
<div class="card bg-dark border-secondary h-100" data-panel-id="panel-telemetry-research-1" id="panel-telemetry-research-1">
<div class="card-header border-secondary fw-bold small text-light d-flex justify-content-between">
<span>Зафиксированные аномалии</span>
<span class="badge bg-danger" id="tr-anomalies-badge">0</span>
</div>
<div class="table-responsive">
<table class="table table-dark table-hover table-striped mb-0 align-middle small">
<thead>
<tr class="text-muted border-secondary">
<th>Время</th>
<th>Метрика</th>
<th>Уровень</th>
<th>Значение / Порог</th>
<th>Описание</th>
</tr>
</thead>
<tbody id="tr-anomalies-tbody">
<tr><td class="text-center text-muted py-3" colspan="5">Аномал
```

```
ий не обнаружено</td></tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- Devices Summary -->
<div class="col-12 col-lg-5">
<div class="card bg-dark border-secondary h-100" data-panel-id="panel-telemetry-research-2" id="panel-telemetry-research-2">
<div class="card-header border-secondary fw-bold small text-light d-flex justify-content-between">
<span>Состояние оборудования</span>
<span class="badge bg-info" id="tr-devices-badge">OK</span>
</div>
<div class="card-body p-3 small" id="tr-devices-content">
<p class="text-muted">Данные об оборудовании загружаются...</p>
</div>
</div>
</div>
</div>
</div>
<!-- 5. Records Tab -->
<div class="tab-pane fade" id="tr-tab-records" role="tabpanel">
<div class="card bg-dark border-secondary" data-panel-id="panel-telemetry-research-3" id="panel-telemetry-research-3">
<div class="card-header border-secondary d-flex align-items-center justify-content-between gap-2 flex-wrap">
<div class="d-flex align-items-center gap-2">
<input class="form-control form-control-sm bg-dark text-white border-secondary" id="tr-records-search" placeholder="Поиск по записям..." style="width: 250px;" type="text"/>
<button class="btn btn-sm btn-outline-secondary" id="tr-records-search-btn"><i class="bi bi-search"></i></button>
</div>
<div class="d-flex align-items-center gap-2">
```

```
<span class="small text-muted" id="tr-records-pagination-info">Стр. 1 из 1</span>
<div class="btn-group btn-group-sm">
<button class="btn btn-outline-secondary" id="tr-records-prev-btn"><i class="bi bi-chevron-left"></i></button>
<button class="btn btn-outline-secondary" id="tr-records-next-btn"><i class="bi bi-chevron-right"></i></button>
</div>
</div>
</div>
<div class="table-responsive" style="max-height: 500px;">
<table class="table table-dark table-hover table-striped mb-0 align-middle small font-monospace">
<thead>
<tr class="text-muted border-secondary">
<th>#</th>
<th>Timestamp</th>
<th>CPU %</th>
<th>RAM %</th>
<th>GPU %</th>
<th>Диск (R/W)</th>
<th>Событие / Детали</th>
</tr>
</thead>
<tbody id="tr-records-tbody">
<tr><td class="text-center text-muted py-3" colspan="7">Записи не загружены</td></tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Telemetry Research Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/telemetry_research_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/telemetry_research_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

let trCurrentReport = null;
let trChartsInstances = [];
let trCurrentPage = 1;
const trPageSize = 25;
let trTotalPages = 1;
let trSearchQuery = '';

/**
 * Инициализация вкладки исследования телеметрии.
 */
export async function initTelemetryResearchTab() {
  bindEvents();
```

```
  await loadSources();
  await runResearch();
}

/**
 * Привязка событий элементов управления.
 */
function bindEvents() {
  const btnRun = document.getElementById('tr-btn-run');
  if (btnRun) {
    btnRun.onclick = () => runResearch();
  }

  const btnDashboard = document.getElementById('tr-btn-dashboa
rd');
  if (btnDashboard) {
    btnDashboard.onclick = () => openHtmlDashboard();
  }

  const btnJson = document.getElementById('tr-btn-json');
  if (btnJson) {
    btnJson.onclick = () => downloadJsonReport();
  }

  const selectSource = document.getElementById('tr-select-sour
ce');
  if (selectSource) {
    selectSource.onchange = () => runResearch();
  }

  const searchBtn = document.getElementById('tr-records-search
-btn');
  const searchInput = document.getElementById('tr-records-sear
ch');
  if (searchBtn && searchInput) {
    searchBtn.onclick = () => {
      trSearchQuery = searchInput.value.trim();
      trCurrentPage = 1;
```

```
      loadRecords();
    };
    searchInput.onkeydown = (e) => {
      if (e.key === 'Enter') {
        trSearchQuery = searchInput.value.trim();
        trCurrentPage = 1;
        loadRecords();
      }
    };
  }

  const prevBtn = document.getElementById('tr-records-prev-btn
');
  const nextBtn = document.getElementById('tr-records-next-btn
');
  if (prevBtn) {
    prevBtn.onclick = () => {
      if (trCurrentPage > 1) {
        trCurrentPage--;
        loadRecords();
      }
    };
  }
  if (nextBtn) {
    nextBtn.onclick = () => {
      if (trCurrentPage < trTotalPages) {
        trCurrentPage++;
        loadRecords();
      }
    };
  }

  const recordsTabBtn = document.getElementById('tr-tab-record
s-btn');
  if (recordsTabBtn) {
    recordsTabBtn.addEventListener('shown.bs.tab', () => {
      loadRecords();
    });
```

```
  }
}

/**
 * Загрузка списка доступных источников телеметрии.
 */
async function loadSources() {
  const select = document.getElementById('tr-select-source');
  if (!select) return;

  try {
    const res = await fetch('/apps/telemetry_research/sources'
);
    if (!res.ok) return;
    const files = await res.json();
    select.innerHTML = '<option value="">Все доступные логи те
леметрии</option>';
    files.forEach((f) => {
      const opt = document.createElement('option');
      opt.value = f.path;
      const sizeKb = (f.size_bytes / 1024).toFixed(1);
      opt.textContent = `${f.name} (${sizeKb} KB)`;
      select.appendChild(opt);
    });
  } catch (err) {
    console.warn('[TelemetryResearch] Ошибка загрузки источник
ов:', err);
  }
}

/**
 * Запуск глубокого исследования телеметрии.
 */
async function runResearch() {
  const btnRun = document.getElementById('tr-btn-run');
  const selectSource = document.getElementById('tr-select-sour
ce');
  const sourcePath = selectSource ? selectSource.value : null;
```

```
136
137   if (btnRun) {
138     btnRun.disabled = true;
139     btnRun.innerHTML = '<span class="spinner-border spinner-bo
rder-sm" role="status"></span> Анализ...';
140   }
141
142   try {
143     const res = await fetch('/apps/telemetry_research/run-rese
arch', {
144       method: 'POST',
145       headers: { 'Content-Type': 'application/json' },
146       body: JSON.stringify({ source_path: sourcePath || null }
),
147     });
148
149     if (!res.ok) {
150       throw new Error(`Ошибка сервера: ${res.status}`);
151     }
152
153     const report = await res.json();
154     trCurrentReport = report;
155
156     renderKPI(report);
157     renderHypotheses(report.hypotheses);
158     renderCorrelations(report.correlations);
159     renderCharts(report.base_report.charts);
160     renderAnomalies(report.base_report.anomalies, report.base_
report.device_summary);
161
162     if (window.toast) {
163       window.toast.success('Исследование завершено', `Проанали
зировано ${report.base_report.records_analyzed} записей.`);
164     }
165   } catch (err) {
166     console.error('[TelemetryResearch] Ошибка анализа:', err);
167     if (window.toast) {
168       window.toast.error('Сбой исследования', err.message);
```

```
    }
  } finally {
    if (btnRun) {
      btnRun.disabled = false;
      btnRun.innerHTML = '<i class="bi bi-play-circle-fill"></i> <span>Исследовать</span>';
    }
  }
}

/**
 * Отрисовка сводных KPI карточек.
 */
function renderKPI(report) {
  const base = report.base_report;
  const valHealth = document.getElementById('tr-val-health');
  const progHealth = document.getElementById('tr-progress-health');
  const valRecords = document.getElementById('tr-val-records');
  const valTimespan = document.getElementById('tr-val-timespan');
  const valHypotheses = document.getElementById('tr-val-hypotheses');
  const valAnomalies = document.getElementById('tr-val-anomalies');
  const valDevices = document.getElementById('tr-val-devices');

  if (valHealth) valHealth.textContent = `${base.health_score}`;
  if (progHealth) {
    progHealth.style.width = `${base.health_score}%`;
    progHealth.className = `progress-bar ${
      base.health_score >= 80 ? 'bg-success' : base.health_score >= 60 ? 'bg-warning' : 'bg-danger'
    }`;
  }
```

```
198
199   if (valRecords) valRecords.textContent = `${base.records_ana
lyzed}`;
200   if (valTimespan && base.time_range) {
201     valTimespan.textContent = `${base.time_range.start ? base.
time_range.start.slice(11, 19) : ''} - ${
202       base.time_range.end ? base.time_range.end.slice(11, 19)
: ''
203     }`;
204   }
205
206   const confirmedCount = (report.hypotheses || []).filter((h)
=> h.confirmed).length;
207   if (valHypotheses) valHypotheses.textContent = `${confirmedC
ount} / ${report.hypotheses ? report.hypotheses.length : 0}`;
208
209   if (valAnomalies) valAnomalies.textContent = `${(base.anomal
ies || []).length}`;
210   if (valDevices) {
211     valDevices.textContent = base.device_summary.error_count >
 0
212       ? `Ошибок устройств: ${base.device_summary.error_count}`
213       : 'Оборудование в норме';
214   }
215
216   // Summary alert banner
217   const banner = document.getElementById('tr-summary-banner');
218   const bannerText = document.getElementById('tr-summary-text'
);
219   if (banner && bannerText && report.investigation_summary) {
220     banner.classList.remove('d-none');
221     bannerText.textContent = report.investigation_summary;
222   }
223 }
224
225 /**
226  * Отрисовка списка гипотез.
227  */
```

```
function renderHypotheses(hypotheses) {
  const container = document.getElementById('tr-hypotheses-list');
  if (!container) return;

  if (!hypotheses || hypotheses.length === 0) {
    container.innerHTML = '<div class="col-12 text-center text-muted py-4">Гипотезы не проверены</div>';
    return;
  }

  container.innerHTML = '';
  hypotheses.forEach((h) => {
    const col = document.createElement('div');
    col.className = 'col-12 col-md-6';

    const cardBorder = h.confirmed ? 'border-warning' : 'border-success-subtle';
    const statusBadge = h.confirmed
      ? '<span class="badge bg-warning text-dark"><i class="bi bi-exclamation-triangle"></i> Подтверждена</span>'
      : '<span class="badge bg-success"><i class="bi bi-check-circle"></i> В норме</span>';

    const evidenceItems = h.evidence.map((ev) => `<li>${ev}</li>`).join('');

    col.innerHTML = `
      <div class="card bg-dark ${cardBorder} h-100 shadow-sm">
        <div class="card-header bg-transparent border-secondary d-flex justify-content-between align-items-center">
          <span class="fw-bold text-light">${h.title}</span>
          ${statusBadge}
        </div>
        <div class="card-body p-3 small">
          <p class="text-muted mb-2">${h.description}</p>
          <div class="mb-2">
            <strong class="text-secondary">Факты телеметрии:</
```

```
strong>
            <ul class="text-light text-opacity-75 ps-3 mb-1">${evidenceItems}</ul>
          </div>
          ${
            h.recommendation
              ? `<div class="p-2 rounded bg-body-tertiary bg-opacity-10 text-info mt-2">
                   <strong>  Рекомендация:</strong> ${h.recommendation}
                 </div>`
              : ''
          }
        </div>
        <div class="card-footer bg-transparent border-secondary text-muted small d-flex justify-content-between">
          <span>ID: <code>${h.hypothesis_id}</code></span>
          <span>Уверенность: <strong>${Math.round(h.confidence * 100)}%</strong></span>
        </div>
      </div>
    `;
    container.appendChild(col);
  });
}

/**
 * Отрисовка матрицы корреляций.
 */
function renderCorrelations(correlations) {
  const tbody = document.getElementById('tr-correlations-tbody');
  const countBadge = document.getElementById('tr-correlations-count');
  if (!tbody) return;

  if (!correlations || correlations.length === 0) {
    tbody.innerHTML = '<tr><td colspan="5" class="text-center
```

```
text-muted py-3">Корреляции не рассчитаны</td></tr>';
    if (countBadge) countBadge.textContent = '0 пар';
    return;
  }

  if (countBadge) countBadge.textContent = `${correlations.length} пар`;

  tbody.innerHTML = correlations
    .map((c) => {
      const absVal = Math.abs(c.coefficient);
      let coeffBadgeClass = 'bg-secondary';
      if (absVal >= 0.7) coeffBadgeClass = 'bg-danger';
      else if (absVal >= 0.5) coeffBadgeClass = 'bg-warning text-dark';
      else if (absVal >= 0.3) coeffBadgeClass = 'bg-info text-dark';

      return `
        <tr>
          <td><code class="text-info">${c.metric_a}</code></td>
          <td><code class="text-info">${c.metric_b}</code></td>
          <td><span class="badge ${coeffBadgeClass} font-monospace">${c.coefficient > 0 ? '+' : ''}${c.coefficient.toFixed(3)}</span></td>
          <td>${c.sample_size} точек</td>
          <td class="text-light text-opacity-75">${c.interpretation}</td>
        </tr>
      `;
    })
    .join('');
}

/**
 * Отрисовка интерактивных графиков Chart.js.
```

```
 */
function renderCharts(charts) {
  const container = document.getElementById('tr-charts-container');
  if (!container) return;

  // Очищаем старые инстансы графиков
  trChartsInstances.forEach((inst) => {
    try {
      inst.destroy();
    } catch {}
  });
  trChartsInstances = [];

  if (!charts || charts.length === 0) {
    container.innerHTML = '<div class="col-12 text-center text-muted py-4">Нет доступных графиков</div>';
    return;
  }

  container.innerHTML = '';

  charts.forEach((ch, idx) => {
    const col = document.createElement('div');
    col.className = 'col-12 col-xl-6';

    const canvasId = `tr-chart-canvas-${idx}`;
    col.innerHTML = `
      <div class="card bg-dark border-secondary h-100 shadow-sm">
        <div class="card-header bg-transparent border-secondary d-flex justify-content-between align-items-center">
          <span class="fw-bold text-light">${ch.title}</span>
          <span class="badge bg-secondary small">${ch.chart_type}</span>
        </div>
        <div class="card-body p-2" style="position: relative; height: 280px;">
```

```
          <canvas id="${canvasId}"></canvas>
        </div>
      </div>
    `;
    container.appendChild(col);

    const ctx = document.getElementById(canvasId);
    if (ctx && typeof Chart !== 'undefined') {
      const labels = ch.labels.map((l) => (l.length > 19 ? l.s
lice(11, 19) : l));
      const datasets = ch.datasets.map((ds) => ({
        label: ds.label,
        data: ds.data,
        borderColor: ds.color || '#3b82f6',
        backgroundColor: (ds.color || '#3b82f6') + '22',
        borderWidth: 2,
        fill: ch.chart_type === 'area',
        tension: 0.25,
        pointRadius: ds.data.length > 50 ? 0 : 3,
      }));

      const newInst = new Chart(ctx, {
        type: ch.chart_type === 'bar' ? 'bar' : 'line',
        data: { labels, datasets },
        options: {
          responsive: true,
          maintainAspectRatio: false,
          animation: { duration: 400 },
          scales: {
            x: {
              grid: { color: 'rgba(255,255,255,0.06)' },
              ticks: { color: '#9ca3af', font: { size: 10 } },
            },
            y: {
              grid: { color: 'rgba(255,255,255,0.06)' },
              ticks: { color: '#9ca3af', font: { size: 10 } },
            },
          },
```

```
          plugins: {
            legend: {
              labels: { color: '#e5e7eb', boxWidth: 12, font: { size: 11 } },
            },
          },
        },
      });
      trChartsInstances.push(newInst);
    }
  });
}

/**
 * Отрисовка аномалий и состояния устройств.
 */
function renderAnomalies(anomalies, deviceSummary) {
  const tbody = document.getElementById('tr-anomalies-tbody');
  const anomBadge = document.getElementById('tr-anomalies-badge');
  const devBadge = document.getElementById('tr-devices-badge');
  const devContent = document.getElementById('tr-devices-content');

  if (anomBadge) anomBadge.textContent = anomalies ? `${anomalies.length}` : '0';

  if (tbody) {
    if (!anomalies || anomalies.length === 0) {
      tbody.innerHTML = '<tr><td colspan="5" class="text-center text-success py-3"><i class="bi bi-check-circle"></i> Аномалий в метриках не зафиксировано</td></tr>';
    } else {
      tbody.innerHTML = anomalies
        .map((a) => {
          const sevClass = a.severity === 'critical' ? 'bg-danger' : a.severity === 'high' ? 'bg-warning text-dark' : 'bg-in
```

```
fo text-dark';
          return `
            <tr>
              <td class="text-muted">${a.timestamp ? a.timesta
mp.slice(11, 19) : '--'}</td>
              <td><strong>${a.metric}</strong></td>
              <td><span class="badge ${sevClass}">${a.severity
}</span></td>
              <td><code class="text-warning">${a.value}</code>
 / ${a.threshold}</td>
              <td class="text-light text-opacity-75">${a.messa
ge}</td>
            </tr>
          `;
        })
        .join('');
    }
  }

  if (deviceSummary && devContent) {
    const hasIssues = deviceSummary.error_count > 0 || deviceS
ummary.flapping_devices.length > 0;
    if (devBadge) {
      devBadge.className = `badge ${hasIssues ? 'bg-danger' :
'bg-success'}`;
      devBadge.textContent = hasIssues ? 'Внимание' : 'Норма';
    }

    let flapHtml = '<span class="text-success">Нет</span>';
    if (deviceSummary.flapping_devices && deviceSummary.flappi
ng_devices.length > 0) {
      flapHtml = deviceSummary.flapping_devices.map((d) => `<s
pan class="badge bg-warning text-dark me-1">${d}</span>`).join
('');
    }

    devContent.innerHTML = `
      <div class="mb-2"><strong>Всего инцидентов:</strong> ${d
```

```
eviceSummary.total_events}</div>
      <div class="mb-2"><strong>Ошибок и сбоев:</strong> <span
 class="${deviceSummary.error_count > 0 ? 'text-danger fw-bold
' : 'text-success'}">${deviceSummary.error_count}</span></div>
      <div class="mb-2"><strong>Флаппирующие устройства:</stro
ng> ${flapHtml}</div>
    `;
  }
}

/**
 * Загрузка записей телеметрии с пагинацией.
 */
async function loadRecords() {
  const tbody = document.getElementById('tr-records-tbody');
  const infoEl = document.getElementById('tr-records-paginatio
n-info');
  const selectSource = document.getElementById('tr-select-sour
ce');
  const sourcePath = selectSource ? selectSource.value : null;

  if (tbody) {
    tbody.innerHTML = '<tr><td colspan="7" class="text-center
text-muted py-3"><span class="spinner-border spinner-border-sm
"></span> Загрузка...</td></tr>';
  }

  try {
    const url = new URL('/apps/telemetry_research/records', wi
ndow.location.origin);
    if (sourcePath) url.searchParams.set('source_path', source
Path);
    if (trSearchQuery) url.searchParams.set('query', trSearchQ
uery);
    url.searchParams.set('page', trCurrentPage);
    url.searchParams.set('page_size', trPageSize);

    const res = await fetch(url.toString());
```

```
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();

    trTotalPages = data.total_pages || 1;
    if (infoEl) {
      infoEl.textContent = `Стр. ${data.page} из ${trTotalPages} (всего ${data.total})`;
    }

    if (!data.items || data.items.length === 0) {
      tbody.innerHTML = '<tr><td colspan="7" class="text-center text-muted py-3">Записей не найдено</td></tr>';
      return;
    }

    tbody.innerHTML = data.items
      .map((r, i) => {
        const num = (trCurrentPage - 1) * trPageSize + i + 1;
        const ts = r.timestamp || r.time || '--';
        const cpu = r.cpu ? `${r.cpu.total_percent || r.cpu.load_percent || '--'}%` : '--';
        const ram = r.memory ? `${r.memory.percent || '--'}%` : '--';
        const gpu = r.gpu ? `${r.gpu.load_percent || '--'}%` : '--';
        let disk = '--';
        if (r.disk_io) {
          const rMb = ((r.disk_io.read_bytes_per_sec || 0) / 1048576).toFixed(1);
          const wMb = ((r.disk_io.write_bytes_per_sec || 0) / 1048576).toFixed(1);
          disk = `${rMb} / ${wMb} MB/s`;
        }
        const eventInfo = r.event_type || r.friendly_name || JSON.stringify(r).slice(0, 50);

        return `
          <tr>
```

```
            <td>${num}</td>
            <td class="text-muted">${ts}</td>
            <td>${cpu}</td>
            <td>${ram}</td>
            <td>${gpu}</td>
            <td>${disk}</td>
            <td class="text-truncate" style="max-width: 250px;
" title="${eventInfo}">${eventInfo}</td>
          </tr>
        `;
      })
      .join('');
  } catch (err) {
    if (tbody) {
      tbody.innerHTML = `<tr><td colspan="7" class="text-cente
r text-danger py-3">Ошибка: ${err.message}</td></tr>`;
    }
  }
}

/**
 * Открытие автономного HTML дашборда.
 */
function openHtmlDashboard() {
  const selectSource = document.getElementById('tr-select-sour
ce');
  const sourcePath = selectSource ? selectSource.value : '';
  const url = sourcePath
    ? `/apps/telemetry_research/dashboard?source_path=${encode
URIComponent(sourcePath)}`
    : '/apps/telemetry_research/dashboard';
  window.open(url, '_blank');
}

/**
 * Экспорт JSON отчета.
 */
function downloadJsonReport() {
```

```
  if (!trCurrentReport) {
    if (window.toast) window.toast.warning('Нет данных', 'Снач
ала выполните исследование.');
    return;
  }
  const blob = new Blob([JSON.stringify(trCurrentReport, null,
 2)], { type: 'application/json' });
  const url = URL.createObjectURL(blob);
  const a = document.createElement('a');
  a.href = url;
  a.download = `telemetry_research_${Date.now()}.json`;
  a.click();
  URL.revokeObjectURL(url);
}

// Автозапуск при подключении модуля
if (document.readyState === 'loading') {
  document.addEventListener('DOMContentLoaded', initTelemetryR
esearchTab);
} else {
  initTelemetryResearchTab();
}

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/throttling_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/throttling_tab/index.ht
ml" class="w-100 h-100 border-0"></iframe>

 * File: index.html
 * Project: ai-windows-admin
 * Package: windows/api/webgui/throttling_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ================================================================
==================
-->
<style>
  .throttling-tab-wrapper {
    padding: 0.75rem;
    color: var(--text-color);
  }
  .diag-meter-card {
    background: var(--surface-1);
    border: 1px solid var(--border-color);
    border-radius: var(--radius-md, 8px);
    padding: 0.75rem 1rem;
```

```
  }
  .diag-panel {
    background: var(--surface-1);
    border: 1px solid var(--border-color);
    border-radius: 8px;
    overflow: hidden;
  }
  .diag-panel-header {
    background: var(--surface-2);
    padding: 8px 14px;
    border-bottom: 1px solid var(--border-color);
    font-weight: 600;
    font-size: 0.85rem;
    color: var(--nav-active);
  }
  .about-sys-spec-table {
    width: 100%;
    border-collapse: separate;
    border-spacing: 0;
  }
  .about-sys-spec-table td {
    padding: 8px 12px;
    border-bottom: 1px solid var(--border-color);
    vertical-align: middle;
  }
  .about-sys-spec-key {
    color: var(--text-muted);
    width: 220px;
    font-size: 0.82rem;
    font-weight: 500;
  }
  .about-sys-spec-val {
    color: var(--text-color);
    font-weight: 500;
    font-size: 0.82rem;
  }
</style>
<div class="throttling-tab-wrapper">
```

```
<!-- Top Header & Status -->
<div class="card shadow-sm border-secondary-subtle mb-3" data-panel-id="panel-throttling-3-kachestvo-yadra-i-trottling-ke" id="panel-throttling-3-kachestvo-yadra-i-trottling-ke">
<div class="card-body p-3">
<div class="d-flex flex-wrap align-items-center justify-content-between gap-2">
<div class="d-flex align-items-center gap-2">
<div class="fs-4">□ </div>
<div>
<h5 class="mb-0 fw-bold">3. Качество ядра и Троттлинг (Kernel Quality &amp;amp; Hardware Health)</h5>
<small class="text-muted">DPC/ISR задержки драйверов, аппаратный сброс частот PROCHOT/Power Limit, PCIe линии и аварии BSOD</small>
</div>
</div>
<div class="d-flex align-items-center gap-2 flex-wrap">
<select class="form-select form-select-sm poll-freq-select" data-poll-id="throttling" id="diag-throttling-poll-freq" title="Частота опроса: Троттлинг и ядро">
<option value="start">Только при старте</option>
<option selected="" value="manual">По запросу</option>
<option value="3">Каждые 3 сек</option>
<option value="5">Каждые 5 сек</option>
<option value="10">Каждые 10 сек</option>
<option value="30">Каждые 30 сек</option>
<option value="60">Каждые 60 сек</option>
<option value="120">Каждые 120 сек</option>
</select>
<button class="btn btn-sm btn-outline-info rounded-pill px-3 py-1 fw-semibold d-flex align-items-center gap-1" id="btn-diag-throttling-refresh">
<i class="bi bi-arrow-clockwise"></i> Обновить
          </button>
</div>
</div>
</div>
```

```
</div>
<!-- KPI Latency and Status Meters -->
<div class="row g-2 mb-3">
<div class="col-12 col-sm-6 col-lg-3">
<div class="diag-meter-card">
<div class="d-flex justify-content-between align-items-center small text-muted">
<span>DPC Latency (Отложенные вызовы)</span>
<i class="bi bi-activity text-info"></i>
</div>
<div class="fs-4 fw-bold mt-1 text-info" id="diag-dpc-latency-val">0.0%</div>
<div class="small" id="diag-dpc-latency-sub"><span class="badge bg-success-subtle text-success border border-success">Optimal</span></div>
</div>
</div>
<div class="col-12 col-sm-6 col-lg-3">
<div class="diag-meter-card">
<div class="d-flex justify-content-between align-items-center small text-muted">
<span>ISR Latency (Прерывания драйверов)</span>
<i class="bi bi-cpu text-warning"></i>
</div>
<div class="fs-4 fw-bold text-warning mt-1" id="diag-isr-latency-val">0.0%</div>
<div class="small text-muted">Нагрузка драйверов на ЦП</div>
</div>
</div>
<div class="col-12 col-sm-6 col-lg-3">
<div class="diag-meter-card">
<div class="d-flex justify-content-between align-items-center small text-muted">
<span>Аппаратный троттлинг (PROCHOT)</span>
<i class="bi bi-thermometer-sun text-danger"></i>
</div>
<div class="fs-4 fw-bold mt-1 text-danger" id="diag-throttling-status-val">Inactive</div>
```

```
<div class="small text-muted" id="diag-throttling-status-sub">Терморежим в норме</div>
</div>
</div>
<div class="col-12 col-sm-6 col-lg-3">
<div class="diag-meter-card">
<div class="d-flex justify-content-between align-items-center small text-muted">
<span>Время непрерывной работы</span>
<i class="bi bi-clock text-primary"></i>
</div>
<div class="fs-4 fw-bold text-body mt-1" id="diag-uptime-val">--</div>
<div class="small text-muted">System Uptime</div>
</div>
</div>
</div>
<!-- GPU PCIe Bus & Kernel BSOD Crash Dumps -->
<div class="row g-2">
<div class="col-12 col-lg-6">
<div class="diag-panel h-100" data-panel-id="panel-throttling-1" id="panel-throttling-1">
<div class="diag-panel-header d-flex justify-content-between align-items-center">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-gpu-card text-success"></i>
<span class="text-body">Режим шины PCIe графического ускорителя (GPU Link)</span>
</div>
<span class="badge bg-success-subtle text-success border border-success">PCIe Prober</span>
</div>
<div class="p-3">
<table class="about-sys-spec-table">
<tbody>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-gpu-card me-1.5 text-success"></i>Видеокарта</td>
```

```
<td class="about-sys-spec-val text-info" id="diag-gpu-pcie-nam
e">GeForce GPU</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-speedometer2 me
-1.5 text-warning"></i>Скорость шины (Link Speed)</td>
<td class="about-sys-spec-val" id="diag-gpu-pcie-speed">PCIe 3
.0 / 4.0 (8.0 GT/s)</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-arrows-expand m
e-1.5 text-primary"></i>Ширина линий (Link Width)</td>
<td class="about-sys-spec-val" id="diag-gpu-pcie-width">x16 La
nes (Full Width)</td>
</tr>
<tr>
<td class="about-sys-spec-key"><i class="bi bi-shield-check me
-1.5 text-success"></i>Статус пропускной способности</td>
<td class="about-sys-spec-val text-success" id="diag-gpu-pcie-
status">Штатный режим без деградации</td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
<div class="col-12 col-lg-6">
<div class="diag-panel h-100" data-panel-id="panel-throttling-
2" id="panel-throttling-2">
<div class="diag-panel-header d-flex justify-content-between a
lign-items-center">
<div class="d-flex align-items-center gap-2">
<i class="bi bi-bug-fill text-danger"></i>
<span class="text-body">Журнал аварийных дампов ядра (Minidump
 / BSOD)</span>
</div>
<span class="badge bg-secondary" id="diag-bsod-count">0 сбоев<
/span>
```

```
</div>
<div class="p-2" id="diag-bsod-container" style="max-height: 260px; overflow-y: auto;">
<div class="text-center py-4 text-success small">
<i class="bi bi-check-circle-fill fs-5 d-block mb-1 text-success"></i>
            Аварийных дампов падения ядра (BSOD) в Minidump не обнаружено
          </div>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Throttling Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/throttling_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/throttling_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

(function () {
  'use strict';

  let autoRefreshTimer = null;

  function escapeHtml(str) {
    if (str === null || str === undefined) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
```

```
      .replace(/'/g, '&#039;');
  }

  function setText(id, text) {
    const el = document.getElementById(id);
    if (el) el.textContent = text !== null && text !== undefin
ed ? String(text) : '--';
  }

  async function apiFetch(url) {
    if (window.api && typeof window.api.fetch === 'function')
{
      return await window.api.fetch(url);
    }
    const res = await fetch(url);
    if (!res.ok) throw new Error(`HTTP ${res.status}: ${res.st
atusText}`);
    return await res.json();
  }

  async function fetchKernelThrottling() {
    try {
      const data = await apiFetch('/api/v1/system/diagnostics/
throttling');
      if (!data) return;

      setText('diag-dpc-latency-val', `${data.dpc_latency_pct.
toFixed(2)}%`);
      const dpcSub = document.getElementById('diag-dpc-latency
-sub');
      if (dpcSub) {
        if (data.dpc_status === 'severe') {
          dpcSub.innerHTML = '<span class="badge bg-danger-sub
tle text-danger border border-danger">Критическая латентность
DPC</span>';
        } else if (data.dpc_status === 'elevated') {
          dpcSub.innerHTML = '<span class="badge bg-warning-su
btle text-warning border border-warning">Повышенная задержка D
```

```
PC</span>';
        } else {
          dpcSub.innerHTML = '<span class="badge bg-success-subtle text-success border border-success">Оптимально (< 1.5%)</span>';
        }
      }

      setText('diag-isr-latency-val', `${data.interrupt_latency_pct.toFixed(2)}%`);
      setText('diag-uptime-val', data.uptime_formatted || '--');

      const thVal = document.getElementById('diag-throttling-status-val');
      const thSub = document.getElementById('diag-throttling-status-sub');
      if (thVal && thSub) {
        if (data.thermal_throttling_detected) {
          thVal.textContent = 'PROCHOT Active';
          thVal.className = 'fs-4 fw-bold text-danger mt-1';
          thSub.textContent = 'Обнаружен термический сброс частот CPU';
        } else if (data.power_limit_throttling_detected) {
          thVal.textContent = 'Power Limit PL1/PL2';
          thVal.className = 'fs-4 fw-bold text-warning mt-1';
          thSub.textContent = 'Ограничение по энергопотреблению';
        } else {
          thVal.textContent = 'Inactive (Норма)';
          thVal.className = 'fs-4 fw-bold text-success mt-1';
          thSub.textContent = 'Терморежим и питание в норме';
        }
      }

      // GPU PCIe Link
      if (data.gpu_pcie_link) {
        setText('diag-gpu-pcie-name', data.gpu_pcie_link.gpu_n
```

```
ame || 'GPU Accelerator');
91        setText('diag-gpu-pcie-speed', data.gpu_pcie_link.curr
ent_link_speed || 'PCIe 3.0 / 4.0');
92        setText('diag-gpu-pcie-width', data.gpu_pcie_link.curr
ent_link_width || 'x16 Lanes');
93        setText('diag-gpu-pcie-status', data.gpu_pcie_link.sta
tus || 'Штатный режим без деградации');
94      }
95
96      // BSOD Minidump
97      const bsodContainer = document.getElementById('diag-bsod
-container');
98      if (bsodContainer && data.last_bsod_crashes) {
99        setText('diag-bsod-count', `${data.last_bsod_crashes.l
ength} сбоев`);
100       if (data.last_bsod_crashes.length === 0) {
101         bsodContainer.innerHTML = `
102           <div class="text-center py-4 text-success small">
103             <i class="bi bi-check-circle-fill fs-5 d-block m
b-1 text-success"></i>
104             Аварийных дампов падения ядра (BSOD) в Minidump
не обнаружено
105           </div>
106         `;
107       } else {
108         bsodContainer.innerHTML = data.last_bsod_crashes.map
(c => `
109           <div class="p-2 mb-1.5 rounded bg-black bg-opacity
-40 border border-danger-subtle d-flex align-items-center just
ify-content-between">
110             <div>
111               <strong class="text-danger"><i class="bi bi-fi
le-earmark-binary me-1"></i>${escapeHtml(c.file_name)}</strong
>
112               <div class="small text-muted font-monospace">$
{escapeHtml(c.timestamp)} (${c.size_kb} KB)</div>
113             </div>
114             <span class="badge bg-danger">Crash Dump</span>
```

```
            </div>
          `).join('');
        }
      }
    } catch (err) {
      console.warn('[ThrottlingTab] fetchKernelThrottling erro
r:', err);
    }
  }

  const POLL_ID = 'throttling';

  function getFrequency() {
    try {
      const saved = localStorage.getItem(`poll_freq_${POLL_ID}
`);
      if (saved) return saved;
    } catch (_) {}
    return 'manual';
  }

  function setFrequency(freq) {
    try {
      localStorage.setItem(`poll_freq_${POLL_ID}`, freq);
    } catch (_) {}
    applyPoller(freq, false);
  }

  function stopPolling() {
    if (window.unregisterTabPoller) {
      window.unregisterTabPoller(`tab-throttling_${POLL_ID}`);
    }
    if (autoRefreshTimer) {
      clearInterval(autoRefreshTimer);
      autoRefreshTimer = null;
    }
  }

```

```
  function applyPoller(freq, runInitial = false) {
    stopPolling();
    if (freq === 'start' || freq === 'manual') {
      if (runInitial && (window.isTabActive ? window.isTabActi
ve('tab-throttling') : true)) {
        fetchKernelThrottling();
      }
      return;
    }

    const intervalSec = parseInt(freq, 10);
    if (isNaN(intervalSec) || intervalSec <= 0) return;

    const intervalMs = intervalSec * 1000;
    const pollerId = `tab-throttling_${POLL_ID}`;

    if (window.registerTabPoller) {
      window.registerTabPoller('tab-throttling', fetchKernelTh
rottling, intervalMs, { pollerId, immediate: runInitial });
    } else {
      if (runInitial && (window.isTabActive ? window.isTabActi
ve('tab-throttling') : true)) {
        fetchKernelThrottling();
      }
      autoRefreshTimer = setInterval(() => {
        if (window.isTabActive ? window.isTabActive('tab-throt
tling') : true) {
          fetchKernelThrottling();
        }
      }, intervalMs);
    }
  }

  function bindEvents() {
    const btnRefresh = document.getElementById('btn-diag-throt
tling-refresh');
    if (btnRefresh) {
      btnRefresh.onclick = () => fetchKernelThrottling();
```

```
184     }
185
186     const select = document.getElementById('diag-throttling-po
ll-freq');
187     if (select) {
188       select.onchange = (e) => {
189         const newFreq = e.target.value;
190         setFrequency(newFreq);
191         if (newFreq !== 'manual' && newFreq !== 'start') {
192           fetchKernelThrottling();
193         }
194         if (window.showToast) {
195           const label = select.options[select.selectedIndex]?.
text || newFreq;
196           window.showToast(`Частота опроса троттлинга: ${label
}`, 'info');
197         }
198       };
199     }
200   }
201
202   async function initThrottlingTab() {
203     bindEvents();
204     const currentFreq = getFrequency();
205     const select = document.getElementById('diag-throttling-po
ll-freq');
206     if (select) select.value = currentFreq;
207     applyPoller(currentFreq, true);
208   }
209
210   window.initThrottlingTab = initThrottlingTab;
211   window.activateThrottlingTab = () => {
212     applyPoller(getFrequency(), true);
213   };
214
215   if (document.readyState === 'loading') {
216     document.addEventListener('DOMContentLoaded', () => {
217       if (window.isTabActive ? window.isTabActive('tab-throttl
```

```
ing') : false) {
        initThrottlingTab();
      } else {
        bindEvents();
      }
    });
  } else {
    if (window.isTabActive ? window.isTabActive('tab-throttlin
g') : false) {
      initThrottlingTab();
    } else {
      bindEvents();
    }
  }
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/tts_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/tts_tab/index.html" cla
ss="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/tts_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- TTS Settings Tab -->
<div class="container-fluid">
<div class="alert alert-info">
<h5 data-i18n="tts.title">  Настройка TTS (Синтез речи)</h5>
<p data-i18n="tts.description">Установите голос и движок по ум
олчанию для озвучки описания медиафайлов, а также протестируйт
е их звучание.</p>
</div>
<div class="row g-3">
<!-- LEFT: Settings Configuration -->
```

```
<div class="col-5">
<div class="card mb-3" data-panel-id="panel-tts-1" id="panel-tts-1">
<div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
<span>  Параметры озвучки</span>
</div>
<div class="card-body">
<form id="admin-tts-form" onsubmit="event.preventDefault();">
<!-- Active Voice Badge -->
<div class="mb-3" id="admin-active-voice-container" style="display: none;">
<span class="badge bg-success d-inline-flex align-items-center gap-1 p-2">
<i class="bi bi-check-circle-fill"></i>
<span>Активный голос: <strong id="admin-active-voice-name">...</strong></span>
</span>
</div>
<!-- Engine Selection -->
<div class="mb-3">
<label class="form-label small fw-bold" for="admin-tts-engine">Система синтеза:</label>
<select class="form-select form-select-sm" id="admin-tts-engine">
<option value="browser">Web Speech API (Браузерная)</option>
<option value="gtts">Google TTS (Базовая)</option>
<option selected="" value="edge-tts">Microsoft Edge TTS (Нейросетевая)</option>
<option value="silero">Silero TTS (Локальная оффлайн)</option>
</select>
</div>
<!-- Voice Selection -->
<div class="mb-3" id="admin-voice-select-container">
<label class="form-label small fw-bold" for="admin-tts-voice">Голос / Спикер:</label>
<select class="form-select form-select-sm" id="admin-tts-voice">
```

```
<!-- Dynamically populated -->
</select>
</div>
<!-- Save Default Button -->
<div class="d-grid">
<button class="btn btn-success btn-sm" id="btn-admin-tts-save"
 type="button">
<i class="bi bi-floppy"></i> Применить по умолчанию
              </button>
</div>
</form>
</div>
</div>
</div>
<!-- RIGHT: Text to Speech Playground -->
<div class="col-7">
<div class="card" data-panel-id="panel-tts-2" id="panel-tts-2"
>
<div class="card-header bg-dark text-white">
<span>  Тестирование озвучки</span>
</div>
<div class="card-body">
<div class="mb-3">
<label class="form-label small fw-bold" for="admin-tts-text">Т
екст для проверки:</label>
<textarea class="form-control form-control-sm" id="admin-tts-t
ext" rows="5">Привет! Это проверка работы модульной системы си
нтеза речи. Выберите систему и голос слева, затем нажмите кноп
ку «Озвучить» для воспроизведения.</textarea>
</div>
<div class="d-flex align-items-center gap-3">
<button class="btn btn-primary btn-sm px-4" id="btn-admin-tts-
speak" type="button">
<i class="bi bi-play-fill"></i> Озвучить текст
            </button>
<!-- Loading Spinner -->
<div class="status-badge small text-muted d-none" id="admin-tt
s-status">
```

```
<div class="spinner-border spinner-border-sm text-primary" role="status"></div>
<span class="ms-1" id="admin-status-text">Генерация аудио...</span>
</div>
</div>
<!-- Audio Player Container -->
<div class="mt-4 border-top pt-3 d-none" id="admin-tts-player-container">
<label class="form-label small fw-bold">Результат генерации:</label>
<audio class="w-100" controls="" id="admin-tts-audio-player"></audio>
</div>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ================================================================
 * Process Name: Windows Tts Tab - Main Script
 * ================================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/tts_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/tts_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ================================================================
 */

// TTS Tab Configuration and Testing Logic
// Called via window.initTtsTab() when tab is loaded

'use strict';

function ttsFetch(url, options = {}) {
  return window.api
    ? window.api.fetch(url, options)
    : fetch(url, options).then(r => {
        if (!r.ok) throw new Error(`HTTP ${r.status}`);
        return r.json();
      });
```

```
}

function ttsNotify(msg, type = 'info') {
  if (typeof showNotification === 'function') {
    showNotification(msg, type);
  } else {
    console.log(`[${type}] ${msg}`);
  }
}

// Available voices dictionary
const ADMIN_VOICE_MAP = {
  'browser': [], // Populated dynamically from window.speechSynthesis
  'gtts': [
    { value: 'ru', label: 'Русский (Google TTS)' },
    { value: 'en', label: 'English (Google TTS)' },
    { value: 'he', label: '□ □ □ □ □ (Google TTS)' },
    { value: 'de', label: 'Deutsch (Google TTS)' },
    { value: 'fr', label: 'Français (Google TTS)' },
    { value: 'es', label: 'Español (Google TTS)' }
  ],
  'edge-tts': [
    { value: 'ru-RU-DmitryNeural', label: 'Дмитрий (RU - Мужской)' },
    { value: 'ru-RU-SvetlanaNeural', label: 'Светлана (RU - Женский)' },
    { value: 'en-US-JennyNeural', label: 'Jenny (US - Female)' },
    { value: 'en-US-GuyNeural', label: 'Guy (US - Male)' },
    { value: 'en-US-AriaNeural', label: 'Aria (US - Female)' },
    { value: 'en-GB-SoniaNeural', label: 'Sonia (UK - Female)' },
    { value: 'en-GB-RyanNeural', label: 'Ryan (UK - Male)' },
    { value: 'he-IL-AvriNeural', label: 'Avri (HE - □ □ □ □)' },
    { value: 'he-IL-HilaNeural', label: 'Hila (HE - □ □ □ □)' },
    { value: 'de-DE-KatjaNeural', label: 'Katja (DE - Female)'
```

```
 },
    { value: 'de-DE-ConradNeural', label: 'Conrad (DE - Male)'
 },
    { value: 'fr-FR-DeniseNeural', label: 'Denise (FR - Female
)' },
    { value: 'fr-FR-HenriNeural', label: 'Henri (FR - Male)' }
,
    { value: 'es-ES-ElviraNeural', label: 'Elvira (ES - Female
)' },
    { value: 'es-ES-AlvaroNeural', label: 'Alvaro (ES - Male)'
 }
  ],
  'silero': [
    { value: 'eugene', label: 'Евгений (Silero RU)' },
    { value: 'aidar', label: 'Айдар (Silero RU)' },
    { value: 'baya', label: 'Бая (Silero RU)' },
    { value: 'kseniya', label: 'Ксения (Silero RU)' },
    { value: 'xenia', label: 'Ксения v2 (Silero RU)' },
    { value: 'random', label: 'Рандомный (Silero RU)' }
  ]
};

function getFriendlyVoiceName(system, voice) {
  if (system === 'edge-tts') {
    const found = ADMIN_VOICE_MAP['edge-tts'].find(v => v.valu
e === voice);
    return found ? `Edge: ${found.label}` : `Edge (${voice})`;
  }
  if (system === 'silero') {
    const found = ADMIN_VOICE_MAP['silero'].find(v => v.value
=== voice);
    return found ? found.label : `Silero (${voice})`;
  }
  if (system === 'gtts') {
    const found = ADMIN_VOICE_MAP['gtts'].find(v => v.value ==
= voice);
    return found ? found.label : `Google TTS (${voice})`;
  }
```

```
  if (system === 'browser') {
    return `Браузерный (${voice || 'По умолчанию'})`;
  }
  return `${system} — ${voice}`;
}

async function initTtsTab() {
  const engineSelect = document.getElementById('admin-tts-engine');
  const voiceSelect = document.getElementById('admin-tts-voice');
  const voiceContainer = document.getElementById('admin-voice-select-container');
  const btnSpeak = document.getElementById('btn-admin-tts-speak');
  const btnSave = document.getElementById('btn-admin-tts-save');
  const ttsStatus = document.getElementById('admin-tts-status');
  const statusText = document.getElementById('admin-status-text');
  const playerContainer = document.getElementById('admin-tts-player-container');
  const audioPlayer = document.getElementById('admin-tts-audio-player');
  const activeVoiceContainer = document.getElementById('admin-active-voice-container');
  const activeVoiceName = document.getElementById('admin-active-voice-name');

  if (!engineSelect || !voiceSelect) {
    console.error('TTS Settings Tab DOM elements not found.');
    return;
  }

  function updateActiveBadge(system, voice) {
    if (system && voice) {
      activeVoiceName.textContent = getFriendlyVoiceName(syste
```

```
m, voice);
      activeVoiceContainer.style.display = 'block';
    } else {
      activeVoiceContainer.style.display = 'none';
    }
  }

  // Populate voices based on selected engine
  function populateVoices() {
    const engine = engineSelect.value;
    const voices = ADMIN_VOICE_MAP[engine] || [];

    voiceSelect.innerHTML = '';

    if (voices.length === 0) {
      voiceContainer.style.display = 'none';
    } else {
      voiceContainer.style.display = 'block';
      voices.forEach(voice => {
        const opt = document.createElement('option');
        opt.value = voice.value;
        opt.textContent = voice.label;
        voiceSelect.appendChild(opt);
      });
    }
  }

  // Load browser speech synthesis voices if available
  function loadBrowserVoices() {
    if ('speechSynthesis' in window) {
      const voices = window.speechSynthesis.getVoices();
      ADMIN_VOICE_MAP['browser'] = voices
        .map(v => ({ value: v.name, label: `[${v.lang}] ${v.na
me}` }));

      if (ADMIN_VOICE_MAP['browser'].length === 0) {
        ADMIN_VOICE_MAP['browser'].push({ value: 'default', la
bel: 'Браузер по умолчанию' });
```

```
      }

      if (engineSelect.value === 'browser') {
        populateVoices();
      }
    }
  }

  if ('speechSynthesis' in window) {
    window.speechSynthesis.onvoiceschanged = loadBrowserVoices
;
    loadBrowserVoices();
  }

  engineSelect.addEventListener('change', populateVoices);
  populateVoices(); // Initial run

  // Fetch current user settings
  try {
    const settings = await ttsFetch('/auth/settings');
    if (settings) {
      if (settings.tts_system) {
        engineSelect.value = settings.tts_system;
        populateVoices();
      }
      if (settings.tts_voice) {
        voiceSelect.value = settings.tts_voice;
      }
      updateActiveBadge(settings.tts_system, settings.tts_voic
e);
    }
  } catch (e) {
    console.error('Failed to load active TTS settings:', e);
  }

  // Speak button handler
  btnSpeak.addEventListener('click', async () => {
    const text = document.getElementById('admin-tts-text').val
```

```
ue.trim();
    if (!text) {
      ttsNotify('Пожалуйста, введите текст для озвучки', 'warn
ing');
      return;
    }

    const engine = engineSelect.value;
    const voice = voiceSelect.value;

    if (engine === 'browser') {
      if ('speechSynthesis' in window) {
        window.speechSynthesis.cancel();
        const cleanText = text.replace(/<[^>]*>/g, '').replace
(/[*_`#]/g, '');
        const utterance = new SpeechSynthesisUtterance(cleanTe
xt);
        const voices = window.speechSynthesis.getVoices();
        const selectedVoice = voices.find(v => v.name === voic
e);
        if (selectedVoice) {
          utterance.voice = selectedVoice;
          if (selectedVoice.lang) {
            utterance.lang = selectedVoice.lang;
          }
        }

        ttsStatus.classList.remove('d-none');
        statusText.textContent = 'Воспроизведение браузером...
';

        utterance.onend = () => ttsStatus.classList.add('d-non
e');
        utterance.onerror = () => ttsStatus.classList.add('d-n
one');

        window.speechSynthesis.speak(utterance);
        playerContainer.classList.add('d-none');
```

```
      } else {
        ttsNotify('Ваш браузер не поддерживает Web Speech API', 'danger');
      }
      return;
    }

    // Server-side synthesis
    ttsStatus.classList.remove('d-none');
    statusText.textContent = 'Синтез аудиофайла на сервере...';
    btnSpeak.disabled = true;
    playerContainer.classList.add('d-none');

    try {
      const queryParams = new URLSearchParams({
        text: text,
        system: engine,
        voice: voice
      });

      const audioUrl = `/api/tts/synthesize?${queryParams.toString()}`;

      audioPlayer.src = audioUrl;
      audioPlayer.load();

      audioPlayer.oncanplaythrough = () => {
        ttsStatus.classList.add('d-none');
        btnSpeak.disabled = false;
        playerContainer.classList.remove('d-none');
        audioPlayer.play().catch(err => console.log('Audio autoplay blocked or failed:', err));
      };

      audioPlayer.onerror = (err) => {
        console.error('Audio element error:', err);
        statusText.textContent = 'Ошибка загрузки аудиофайла';
```

```
        btnSpeak.disabled = false;
        setTimeout(() => ttsStatus.classList.add('d-none'), 3000);
      };

    } catch (err) {
      console.error(err);
      statusText.textContent = 'Ошибка сети';
      btnSpeak.disabled = false;
      setTimeout(() => ttsStatus.classList.add('d-none'), 3000);
    }
  });

  // Save Settings handler
  btnSave.addEventListener('click', async () => {
    const engine = engineSelect.value;
    const voice = voiceSelect.value;

    btnSave.disabled = true;
    const originalText = btnSave.innerHTML;
    btnSave.innerHTML = '<span class="spinner-border spinner-border-sm" role="status" aria-hidden="true"></span> Сохранение...';

    try {
      const payload = {
        tts_enabled: 1,
        tts_system: engine,
        tts_voice: voice
      };

      const response = await fetch('/auth/settings', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify(payload)
      });

```

```
      if (response.ok) {
        updateActiveBadge(engine, voice);
        ttsNotify('Настройки TTS успешно применены по умолчани
ю!', 'success');
      } else {
        const errData = await response.json().catch(() => ({})
);
        throw new Error(errData.detail || 'Ошибка сохранения н
астроек');
      }
    } catch (err) {
      console.error(err);
      ttsNotify('Не удалось сохранить настройки: ' + err.messa
ge, 'danger');
    } finally {
      btnSave.disabled = false;
      btnSave.innerHTML = originalText;
    }
  });
}

window.initTtsTab = initTtsTab;

```

google_signin.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Google Signin View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса google_signin.

Usage Examples:
    Web Route:
      /windows/api/webgui/user/components/google_signin.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/user/components/google_
signin.html" class="w-100 h-100 border-0"></iframe>

File: google_signin.html
Project: ai-windows-admin
Package: windows/api/webgui/user/components
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!-- Google Sign-In Component -->
<div id="google-signin-container" class="d-flex align-items-ce
nter gap-2">
  <button id="google-signin-btn" class="btn btn-outline-light
btn-sm">
    <i class="bi bi-google"></i>
    <span data-i18n="auth.googleSignIn">Войти через Google</sp
an>
  </button>
```

```
  <div id="user-profile" class="d-none flex-grow-1">
    <div class="d-flex align-items-center gap-2">
      <img id="user-avatar" src="" class="rounded-circle" widt
h="32" height="32">
      <span id="user-name" class="text-muted small"></span>
      <button id="logout-btn" class="btn btn-sm btn-danger">
        <i class="bi bi-box-arrow-right"></i> Выйти
      </button>
    </div>
  </div>
</div>

<script>
// Google Sign-In Component Logic
(async () => {
  const googleSignInBtn = document.getElementById('google-sign
in-btn');
  const logoutBtn = document.getElementById('logout-btn');
  const userProfile = document.getElementById('user-profile');
  const userAvatar = document.getElementById('user-avatar');
  const userName = document.getElementById('user-name');

  // Check authentication status
  async function checkAuthStatus() {
    try {
      const response = await fetch('/auth/check');
      const data = await response.json();

      if (data.authenticated) {
        // User is authenticated
        userProfile.classList.remove('d-none');
        googleSignInBtn.classList.add('d-none');

        if (data.name) {
          userName.textContent = data.name;
        }
        if (data.email) {
          userName.textContent = data.email.split('@')[0];
```

```
        }
        if (data.picture) {
          userAvatar.src = data.picture;
        }
      } else {
        // User is not authenticated
        userProfile.classList.add('d-none');
        googleSignInBtn.classList.remove('d-none');
      }
    } catch (error) {
      console.error('Auth check failed:', error);
      userProfile.classList.add('d-none');
      googleSignInBtn.classList.remove('d-none');
    }
  }

  // Handle Google Sign-In click
  googleSignInBtn?.addEventListener('click', () => {
    window.location.href = '/auth/google?next=' + encodeURICom
ponent(window.location.pathname + window.location.search);
  });

  // Handle logout click
  logoutBtn?.addEventListener('click', async () => {
    try {
      await fetch('/auth/logout', { method: 'POST' });
      // Reload page to update UI
      window.location.reload();
    } catch (error) {
      console.error('Logout failed:', error);
    }
  });

  // Initial auth check
  checkAuthStatus();
})();
</script>
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/user/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/user/index.html" class=
"w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/user
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!DOCTYPE html>

<html lang="ru">
<head>
<meta charset="utf-8"/>
<meta content="width=device-width, initial-scale=1.0" name="vi
ewport"/>
<title>AI Assistant</title>
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/
css/bootstrap.min.css" rel="stylesheet"/>
```

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.10.0/font/bootstrap-icons.css" rel="stylesheet"/>
<link href="/html/css/main.css?v=20260924_v1" rel="stylesheet" />
<script src="/html/js/toast.js?v=20260924_v1"></script>
<script src="https://cdn.jsdelivr.net/npm/i18next@23.11.5/i18next.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/marked/marked.min.js"></script>
</head>
<body>
<div class="container-fluid mt-2" id="user-interface">
<div class="d-flex align-items-center justify-content-between mb-3 main-nav-container rounded px-2 py-1">
<ul class="nav nav-tabs flex-nowrap flex-grow-1 border-0 me-2 align-items-center" id="mainTabs">
<!-- Группа: Стандартные -->
<li class="nav-item dropdown">
<button class="nav-link dropdown-toggle active d-flex align-items-center gap-1" id="standardTabsDropdown" type="button">
<i class="bi bi-grid-fill text-primary me-1"></i>
<span data-i18n="tabs.groupStandard">Стандартные</span>
</button>
<div class="dropdown-menu shadow border-secondary-subtle">
<button class="dropdown-item active d-flex align-items-center gap-2" data-bs-target="#tab-chat" data-i18n="tabs.chat" data-tab="tab-chat" type="button">
            Чат
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-voice" data-i18n="tabs.voice" data-tab="tab-voice" type="button">
            Голос и аудио
          </button>
</div>
</li>
<!-- Разделитель -->
<li class="nav-item d-flex align-items-center">
```

```
<div class="nav-group-divider"></div>
</li>
<!-- Группа: Расширенные -->
<li class="nav-item dropdown">
<button class="nav-link dropdown-toggle d-flex align-items-center gap-1" id="extendedTabsDropdown" type="button">
<i class="bi bi-sliders2 text-info me-1"></i>
<span data-i18n="tabs.groupExtended">Расширенные</span>
</button>
<div class="dropdown-menu shadow border-secondary-subtle">
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-rag" data-i18n="tabs.rag" data-tab="tab-rag" type="button">
            RAG
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-tts" data-i18n="tabs.tts" data-tab="tab-tts" type="button">
            Настройка TTS
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-skills" data-i18n="tabs.skills" data-tab="tab-skills" type="button">
            Навыки
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-news" data-i18n="tabs.news" data-tab="tab-news" type="button">
            Новости
          </button>
<button class="dropdown-item d-flex align-items-center gap-2" data-bs-target="#tab-user-directories" data-i18n="tabs.userDirectories" data-tab="tab-user-directories" type="button">
            Директории и файлы
          </button>
</div>
</li>
<!-- Разделитель -->
```

```
<li class="nav-item d-flex align-items-center">
<div class="nav-group-divider"></div>
</li>
<!-- Группа: Плагины -->
<li class="nav-item dropdown" id="pluginsNavDropdownWrapper">
<button class="nav-link dropdown-toggle d-flex align-items-cen
ter gap-1" id="pluginsTabsDropdown" type="button">
<i class="bi bi-puzzle-fill text-warning me-1"></i>
<span data-i18n="tabs.plugins">Плагины</span>
</button>
<div class="dropdown-menu shadow border-secondary-subtle" id="
nav-plugins-dropdown-menu">
<div class="nav-enabled-plugins-list" id="nav-enabled-plugins-
list">
<!-- Dynamic enabled plugins will be injected here -->
</div>
<hr class="dropdown-divider my-1"/>
<button class="dropdown-item d-flex align-items-center gap-2"
data-bs-target="#tab-plugins" data-i18n="tabs.managePlugins" d
ata-tab="tab-plugins" type="button">
            Управление
          </button>
</div>
</li>
<!-- Разделитель -->
<li class="nav-item d-flex align-items-center">
<div class="nav-group-divider"></div>
</li>
<!-- Группа: Сервисы -->
<li class="nav-item dropdown">
<button class="nav-link dropdown-toggle d-flex align-items-cen
ter gap-1" id="userServicesDropdown" type="button">
<i class="bi bi-grid-3x3-gap-fill text-success me-1"></i>
<span data-i18n="tabs.groupUserServices">Сервисы</span>
</button>
<div class="dropdown-menu shadow border-secondary-subtle">
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1" id="nav-btn-check-mail" type="button">
```

```
<i class="bi bi-envelope-check text-primary"></i>
<span>Почта</span>
</button>
<button class="dropdown-item d-flex align-items-center gap-2 py-1" id="nav-btn-search-gdocs" type="button">
<i class="bi bi-file-earmark-text text-info"></i>
<span>Google Docs</span>
</button>
<button class="dropdown-item d-flex align-items-center gap-2 py-1" id="nav-btn-gsheets" type="button">
<i class="bi bi-file-earmark-spreadsheet text-success"></i>
<span>Google Таблицы</span>
</button>
<button class="dropdown-item d-flex align-items-center gap-2 py-1" id="nav-btn-news-digest" type="button">
<i class="bi bi-newspaper text-warning"></i>
<span>Новости</span>
</button>
<button class="dropdown-item d-flex align-items-center gap-2 py-1" id="nav-btn-facebook-post" type="button">
<i class="bi bi-facebook text-primary"></i>
<span>Facebook</span>
</button>
</div>
</li>
</ul>
<div class="d-flex align-items-center gap-2 flex-shrink-0 ms-auto pb-1" id="nav-auth-container">
<!-- Google OAuth Register / Sign-in Button (Shown when not logged in) -->
<button class="btn btn-sm btn-light text-dark fw-semibold d-flex align-items-center gap-2 px-3 py-1 shadow-sm rounded-pill btn-google-login" data-i18n-title="auth.googleSignInTitle" id="nav-google-btn" style="border: 1px solid rgba(0,0,0,0.15);" title="Войти или зарегистрироваться через Google">
<svg height="18" viewbox="0 0 24 24" width="18"><path d="M22.56 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h5.92c-.26 1.37-1.04 2.53-2.21 3.31v2.77h3.57c2.08-1.92 3.28-4.74 3.28-8.09z" fill="#4
```

```
285F4"></path><path d="M12 23c2.97 0 5.46-.98 7.28-2.66l-3.57-2.77c-.98.66-2.23 1.06-3.71 1.06-2.86 0-5.29-1.93-6.16-4.53H2.18v2.84C3.99 20.53 7.7 23 12 23z" fill="#34A853"></path><path d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s.13-1.43.35-2.09V7.06H2.18C1.43 8.55 1 10.22 1 12s.43 3.45 1.18 4.94l2.85-2.22.81-.63z" fill="#FBBC05"></path><path d="M12 5.38c1.62 0 3.06.56 4.21 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 1 3.99 3.47 2.18 7.06l3.66 2.84c.87-2.6 3.3-4.52 6.16-4.52z" fill="#EA4335"></path></svg>
<span data-i18n="auth.googleSignUp">Регистрация через Google</span>
</button>
<!-- Authenticated User Menu Dropdown (Shown when logged in) -->
<div class="dropdown d-none" id="user-menu-dropdown-wrapper">
<button aria-expanded="false" class="btn btn-sm btn-outline-secondary d-flex align-items-center gap-2 px-2 py-1 rounded-pill dropdown-toggle" data-bs-toggle="dropdown" id="user-settings-btn" title="Пользовательское меню" type="button">
<img class="rounded-circle d-none" height="22" id="nav-user-avatar" src="" style="object-fit: cover;" width="22"/>
<i class="bi bi-person-circle" id="nav-user-icon"></i>
<span class="small fw-semibold" data-i18n="userSettings.button" id="nav-user-label">Пользователь</span>
</button>
<ul aria-labelledby="user-settings-btn" class="dropdown-menu dropdown-menu-end shadow border-secondary-subtle py-1" style="min-width: 220px;">
<li class="px-3 py-2 border-bottom border-secondary-subtle">
<div class="fw-bold text-truncate" id="user-menu-display-name">Пользователь</div>
<div class="small text-muted text-truncate" id="user-menu-display-email" style="font-size: 0.75rem;">user@gmail.com</div>
</li>
<li><h6 class="dropdown-header small text-uppercase text-muted fw-semibold mt-1">Сервисы</h6></li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
```

```
y-1" id="user-menu-btn-mail" type="button">
<i class="bi bi-envelope-check text-primary"></i>
<span>Почта</span>
</button>
</li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1" id="user-menu-btn-docs" type="button">
<i class="bi bi-file-earmark-text text-info"></i>
<span>Google Docs</span>
</button>
</li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1" id="user-menu-btn-sheets" type="button">
<i class="bi bi-file-earmark-spreadsheet text-success"></i>
<span>Google Таблицы</span>
</button>
</li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1" id="user-menu-btn-news" type="button">
<i class="bi bi-newspaper text-warning"></i>
<span>Новости</span>
</button>
</li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1" id="user-menu-btn-facebook" type="button">
<i class="bi bi-facebook text-primary"></i>
<span>Facebook</span>
</button>
</li>
<li><hr class="dropdown-divider my-1"/></li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1" data-bs-target="#userSettingsModal" data-bs-toggle="modal
" type="button">
```

```
<i class="bi bi-gear-fill text-secondary"></i>
<span data-i18n="userSettings.title">Настройки профиля</span>
</button>
</li>
<li>
<button class="dropdown-item d-flex align-items-center gap-2 p
y-1 text-danger btn-user-logout" type="button">
<i class="bi bi-box-arrow-right"></i>
<span>Выйти</span>
</button>
</li>
</ul>
</div>
<!-- Quick Settings Gear Button -->
<button class="btn btn-sm btn-outline-secondary d-flex align-i
tems-center justify-content-center p-0 rounded-circle" data-bs
-target="#userSettingsModal" data-bs-toggle="modal" data-i18n-
title="userSettings.title" id="user-settings-gear-btn" style="
width: 32px; height: 32px;" title="Настройки">
<i class="bi bi-gear-fill"></i>
</button>
</div>
</div>
<div class="tab-content" id="mainTabContent">
<!-- Chat Tab -->
<div class="tab-pane fade show active" id="tab-chat"></div>
<!-- RAG Tab -->
<div class="tab-pane fade" id="tab-rag"></div>
<!-- Telegram RAG Tab -->
<div class="tab-pane fade" id="tab-telegram-rag"></div>
<!-- News Tab -->
<div class="tab-pane fade" id="tab-news"></div>
<!-- TTS Tab -->
<div class="tab-pane fade" id="tab-tts"></div>
<!-- Voice Tab -->
<div class="tab-pane fade" id="tab-voice"></div>
<!-- Skills Tab -->
<div class="tab-pane fade" id="tab-skills"></div>
```

```
<!-- Plugins Tab -->
<div class="tab-pane fade" id="tab-plugins"></div>
<!-- User Directories Tab -->
<div class="tab-pane fade" id="tab-user-directories"></div>
</div>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
<script src="/html/js/chatService.js?v=20260924_v1"></script>
<script src="/html/user/main.js?v=20260924_v1" type="module"></script>
<!-- Chat Logic & RAG Architecture Modal -->
<div aria-hidden="true" aria-labelledby="chat-logic-modal-label" class="modal fade" id="chat-logic-modal" tabindex="-1">
<div class="modal-dialog modal-xl modal-dialog-scrollable">
<div class="modal-content bg-dark text-white border-secondary">
<div class="modal-header border-secondary d-flex justify-content-between align-items-center">
<h5 class="modal-title d-flex align-items-center gap-2" id="chat-logic-modal-label">
<i class="bi bi-diagram-3-fill text-info"></i>  Архитектура и алгоритм диспетчеризации ответов чата
        </h5>
<button aria-label="Close" class="btn-close btn-close-white" data-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body p-3">
<div class="p-2 border border-secondary rounded bg-black d-flex justify-content-center align-items-center" style="overflow-x: auto; min-height: 520px;">
<img alt="Схема логики чата" src="/html/assets/chat_and_rag_logic.svg?v=20260817_1" style="max-width: 100%; height: auto; border-radius: 6px;"/>
</div>
</div>
<div class="modal-footer border-secondary">
<button class="btn btn-sm btn-secondary" data-bs-dismiss="moda
```

```
l" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
<!-- User Settings Modal -->
<div aria-hidden="true" aria-labelledby="userSettingsModalLabe
l" class="modal fade" id="userSettingsModal" tabindex="-1">
<div class="modal-dialog modal-dialog-centered">
<div class="modal-content shadow border-secondary-subtle">
<div class="modal-header border-secondary-subtle">
<h5 class="modal-title d-flex align-items-center gap-2" id="us
erSettingsModalLabel">
<i class="bi bi-person-gear text-primary"></i>
<span data-i18n="userSettings.title">Пользовательские настройк
и</span>
</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-3">
<!-- User Account & Google OAuth Card -->
<div class="card bg-body-tertiary border-secondary-subtle mb-3
 p-3" data-panel-id="panel-user-1" id="user-settings-auth-card
">
<div class="d-flex align-items-center gap-3">
<img class="rounded-circle border border-2 border-primary" hei
ght="48" id="user-settings-avatar" src="" style="object-fit: c
over;" width="48"/>
<div class="flex-grow-1 min-w-0">
<div class="d-flex align-items-center gap-2">
<strong class="text-truncate" id="user-settings-name">Пользова
тель</strong>
<span class="badge bg-primary" id="user-settings-role-badge">U
ser</span>
</div>
<div class="small text-muted text-truncate" id="user-settings-
email">email@gmail.com</div>
```

```
<div class="small mt-1 text-muted" id="user-settings-google-badge">
<i class="bi bi-google text-danger me-1"></i>Google OAuth
              </div>
</div>
<button class="btn btn-outline-danger btn-sm btn-user-logout" title="Выйти" type="button">
<i class="bi bi-box-arrow-right"></i>
</button>
</div>
</div>
<!-- Guest / Connect Google OAuth Card -->
<div class="card bg-body-tertiary border-secondary-subtle mb-3 p-3 text-center d-none" data-panel-id="panel-user-2" id="user-settings-guest-card">
<div class="text-muted small mb-2">
<i class="bi bi-shield-lock me-1"></i> Авторизация и синхронизация аккаунта
          </div>
<button class="btn btn-outline-danger btn-sm w-100 d-flex align-items-center justify-content-center gap-2 btn-google-login" type="button">
<i class="bi bi-google"></i>
<span>Войти через Google (OAuth)</span>
</button>
<div class="small text-muted mt-2" style="font-size: 0.75rem;">
            Синхронизация с Google Drive, Docs, Calendar и Contacts
          </div>
</div>
<!-- Language Selection -->
<div class="mb-3">
<label class="form-label fw-semibold small d-flex align-items-center gap-2" for="modal-lang-selector">
<i class="bi bi-translate text-info"></i>
<span data-i18n="userSettings.language">Язык интерфейса</span>
</label>
```

```
<select class="form-select lang-selector" id="modal-lang-selector">
<option value="ru">Русский</option>
<option value="en">English</option>
<option value="he">□ □ □ □ □ </option>
</select>
</div>
<!-- Theme Selection -->
<div class="mb-2">
<label class="form-label fw-semibold small d-flex align-items-center gap-2" for="modal-theme-selector">
<i class="bi bi-palette text-warning"></i>
<span data-i18n="userSettings.theme">Тема оформления</span>
</label>
<select class="form-select theme-selector" id="modal-theme-selector">
<option data-i18n="userSettings.themeLight" value="light">□ Светлая</option>
<option data-i18n="userSettings.themeDark" value="dark">□ Тёмная</option>
<option data-i18n="userSettings.themeSystem" value="system">□ Системная</option>
</select>
</div>
</div>
<div class="modal-footer border-secondary-subtle py-2">
<button class="btn btn-secondary btn-sm" data-bs-dismiss="modal" data-i18n="userSettings.close" type="button">Закрыть</button>
</div>
</div>
</div>
</div>
</body>
</html>
```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows User - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/user/main.js?v=2026100
1_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/user
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

// User Interface Main JS
import { initI18n, switchLang, applyTranslations } from '../js
/i18n.js';
import { initTheme, setTheme, getThemeMode, getResolvedTheme }
 from '../js/theme.js';
import { initUserSettings, refreshUserProfile } from '../js/us
erSettings.js';

// Make switchLang and theme functions available globally
window.switchLang = switchLang;
window.applyTranslations = applyTranslations;
window.setTheme = setTheme;
```

```
window.getThemeMode = getThemeMode;
window.getResolvedTheme = getResolvedTheme;
window.initUserSettings = initUserSettings;
window.refreshUserProfile = refreshUserProfile;

// Helper to open plugin from top navbar dropdown
window.openPluginFromDropdown = function(pluginName) {
  if (!pluginName) return;
  document.querySelectorAll('#mainTabs .dropdown-menu.show').forEach((m) => {
    m.classList.remove('show');
    m.closest('.dropdown')?.querySelector('.dropdown-toggle')?.classList.remove('show');
  });

  const normalized = pluginName.toLowerCase().replace(/-/g, '_');

  if (normalized === 'telegram_channel_rag' || normalized === 'telegram_rag') {
    const pane = document.getElementById('tab-telegram-rag') || document.getElementById('tab-telegram_rag');
    if (pane && typeof window.switchTab === 'function') {
      window.switchTab('tab-telegram-rag');
      return;
    }
  }

  if (normalized === 'news_feed' || normalized === 'news' || normalized === 'smart_news' || normalized === 'smart_feed_news') {
    const pane = document.getElementById('tab-news');
    if (pane && typeof window.switchTab === 'function') {
      window.switchTab('tab-news');
      return;
    }
  }

```

```
  if (typeof window.switchTab === 'function') {
    window.switchTab('tab-plugins');
  }

  const select = () => {
    if (typeof window.selectPlugin === 'function') {
      window.selectPlugin(pluginName);
    }
  };

  select();
  setTimeout(select, 150);
  setTimeout(select, 400);
};

// Global API module
window.api = {
  async fetch(url, options = {}) {
    const response = await fetch(url, options);
    if (!response.ok) {
      let msg = response.statusText;
      try {
        const data = await response.json();
        if (data && data.detail) {
          if (typeof data.detail === 'string') {
            msg = data.detail;
          } else if (Array.isArray(data.detail)) {
            msg = data.detail.map(d => d.msg || JSON.stringify
(d)).join(', ');
          } else {
            msg = JSON.stringify(data.detail);
          }
        }
      } catch {}
      throw new Error(`${response.status} ${msg}`);
    }
    return response.json();
  }
```

```
};

// UI Notifications
function showNotification(message, type = 'info') {
  const notification = document.createElement('div');
  notification.className = `alert alert-${type} position-fixed top-0 end-0 m-3 shadow`;
  notification.style.zIndex = '9999';
  notification.style.maxWidth = '400px';
  notification.textContent = message;
  document.body.appendChild(notification);

  setTimeout(() => {
    notification.remove();
  }, 5000);
}
window.showNotification = showNotification;

// Tab Switch Handler
function onTabSwitched(targetId) {
  const cleanId = targetId.startsWith('#') ? targetId.slice(1) : targetId;
  if (cleanId === 'tab-chat') {
    const msgInput = document.getElementById('message-input');
    if (msgInput) msgInput.focus();
  } else if (cleanId === 'tab-rag' && typeof window.initRagTab === 'function') {
    console.log('[UserInterface] Switching to RAG tab...');
    window.initRagTab();
  } else if ((cleanId === 'tab-telegram-rag' || cleanId === 'tab-telegram_rag') && typeof window.initTelegramRagTab === 'function') {
    console.log('[UserInterface] Switching to Telegram RAG tab...');
    window.initTelegramRagTab();
  } else if (cleanId === 'tab-tts' && typeof window.initTtsTab === 'function') {
    console.log('[UserInterface] Switching to TTS tab...');
```

```
    window.initTtsTab();
  } else if (cleanId === 'tab-voice' && typeof window.initVoic
eTab === 'function') {
    console.log('[UserInterface] Switching to Voice tab...');
    window.initVoiceTab();
  } else if (cleanId === 'tab-skills' && typeof window.initSki
llsTab === 'function') {
    console.log('[UserInterface] Switching to Skills tab...');
    window.initSkillsTab();
  } else if (cleanId === 'tab-news' && typeof window.initNewsT
ab === 'function') {
    console.log('[UserInterface] Switching to News tab...');
    window.initNewsTab();
  } else if (cleanId === 'tab-plugins' && typeof window.initPl
uginsTab === 'function') {
    console.log('[UserInterface] Switching to Plugins tab...')
;
    window.initPluginsTab();
  } else if (cleanId === 'tab-user-directories' && typeof wind
ow.initUserDirectoriesTab === 'function') {
    console.log('[UserInterface] Switching to User Directories
 tab...');
    window.initUserDirectoriesTab();
  }
}

// Карта конфигураций вкладок
const USER_TAB_DEFS = {
  'chat': { html: '/html/chat/index.html', js: '/html/chat/mai
n.js' },
  'rag': { html: '/html/rag_tab/index.html', js: '/html/rag_ta
b/main.js' },
  'telegram-rag': { html: '/html/telegram_rag_tab/index.html',
 js: '/html/telegram_rag_tab/main.js' },
  'news': { html: '/html/news_tab/index.html', js: '/html/news
_tab/main.js' },
  'tts': { html: '/html/tts_tab/index.html', js: '/html/tts_ta
b/main.js' },
```

```
  'voice': { html: '/html/voice_tab/index.html', js: '/html/voice_tab/main.js' },
  'skills': { html: '/html/skills_tab/index.html', js: '/html/skills_tab/main.js' },
  'plugins': { html: '/html/plugins_tab/index.html', js: '/html/plugins_tab/main.js' },
  'user-directories': { html: '/html/user_directories_tab/index.html', js: '/html/user_directories_tab/main.js' },
};

const loadedUserTabs = new Set();
const loadingUserTabs = new Map();

async function ensureTabLoaded(tabName) {
  if (loadedUserTabs.has(tabName)) return true;
  if (loadingUserTabs.has(tabName)) return loadingUserTabs.get(tabName);

  const def = USER_TAB_DEFS[tabName];
  if (!def) return false;

  const cb = Date.now();
  const appendCb = (u, cb) => u ? (u.includes('?') ? `${u}&_t=${cb}` : `${u}?v=${cb}`) : '';
  const promise = (async () => {
    try {
      await loadTabContent(tabName, appendCb(def.html, cb), appendCb(def.js, cb));
      loadedUserTabs.add(tabName);
      return true;
    } finally {
      loadingUserTabs.delete(tabName);
    }
  })();

  loadingUserTabs.set(tabName, promise);
  return promise;
}
```

```
186
187 // Programmatic tab switcher
188 async function switchTab(targetId) {
189   if (!targetId) return;
190   const cleanId = targetId.startsWith('#') ? targetId.slice(1)
  : targetId;
191   const tabId = cleanId.startsWith('tab-') ? cleanId : `tab-${
cleanId}`;
192   const tabName = tabId.replace(/^tab-/, '');
193
194   // 1. Update active state on dropdown items & toggles
195   document.querySelectorAll('#mainTabs .dropdown-item').forEac
h((item) => {
196     const itemTarget = item.getAttribute('data-tab') || item.g
etAttribute('data-bs-target')?.replace('#', '');
197     if (itemTarget === tabId || itemTarget === cleanId) {
198       item.classList.add('active');
199     } else {
200       item.classList.remove('active');
201     }
202   });
203
204   document.querySelectorAll('#mainTabs .dropdown').forEach((dr
opdown) => {
205     const toggle = dropdown.querySelector('.dropdown-toggle');
206     const hasActiveChild = dropdown.querySelector('.dropdown-i
tem.active');
207     if (toggle) {
208       if (hasActiveChild) {
209         toggle.classList.add('active');
210       } else {
211         toggle.classList.remove('active');
212       }
213     }
214   });
215
216   // 2. Switch tab-pane (scoped to top-level container to pres
erve nested subtabs)
```

```
  const mainTabContent = document.getElementById('mainTabContent');
  if (mainTabContent) {
    Array.from(mainTabContent.children).forEach((pane) => {
      if (pane.classList.contains('tab-pane')) {
        pane.classList.remove('show', 'active');
      }
    });
  } else {
    document.querySelectorAll('#mainTabContent > .tab-pane, body > .container-fluid > .tab-content > .tab-pane').forEach((pane) => {
      pane.classList.remove('show', 'active');
    });
  }
  const targetPane = document.getElementById(tabId) || document.getElementById(cleanId);
  if (targetPane) {
    targetPane.classList.add('show', 'active');
  }

  // 3. Загружаем содержимое вкладки по требованию, если ещё не загружено
  await ensureTabLoaded(tabName);

  // 4. Notify lifecycle callback
  onTabSwitched(tabId);
}
window.switchTab = switchTab;
window.switchToTab = switchTab;

// Setup dropdowns navigation
function setupDropdownTabs() {
  const mainTabs = document.getElementById('mainTabs');
  if (!mainTabs) return;

  // 1. Dropdown Toggle Buttons
  mainTabs.querySelectorAll('.dropdown-toggle').forEach((btn)
```

```
=> {
    btn.onclick = (e) => {
      e.preventDefault();
      e.stopPropagation();

      const dropdown = btn.closest('.dropdown');
      const menu = dropdown?.querySelector('.dropdown-menu');
      const isAlreadyOpen = menu?.classList.contains('show');

      // Close all dropdowns
      document.querySelectorAll('#mainTabs .dropdown-menu.show
').forEach((m) => {
        m.classList.remove('show');
        m.closest('.dropdown')?.querySelector('.dropdown-toggl
e')?.classList.remove('show');
      });

      // Toggle clicked dropdown
      if (!isAlreadyOpen && menu) {
        menu.classList.add('show');
        btn.classList.add('show');
      }
    };
  });

  // 2. Dropdown Item Buttons (Tab Switchers & Action Buttons)
  mainTabs.querySelectorAll('.dropdown-item').forEach((item) =
> {
    item.onclick = async (e) => {
      const targetId = item.getAttribute('data-tab') || item.g
etAttribute('data-bs-target')?.replace('#', '');
      const pluginName = item.getAttribute('data-plugin');

      // Close dropdown menu
      item.closest('.dropdown-menu')?.classList.remove('show')
;
      item.closest('.dropdown')?.querySelector('.dropdown-togg
le')?.classList.remove('show');
```

```
281
282       if (pluginName && typeof window.openPluginFromDropdown =
    == 'function') {
283         e.preventDefault();
284         e.stopPropagation();
285         window.openPluginFromDropdown(pluginName);
286       } else if (targetId) {
287         e.preventDefault();
288         e.stopPropagation();
289         await switchTab(targetId);
290       }
291     };
292   });
293
294   // 3. Document Click to Close Dropdowns
295   if (!document.body.dataset.dropdownOutsideBound) {
296     document.body.dataset.dropdownOutsideBound = 'true';
297     document.addEventListener('click', (e) => {
298       if (!e.target.closest('#mainTabs .dropdown')) {
299         document.querySelectorAll('#mainTabs .dropdown-menu.sh
    ow').forEach((menu) => {
300           menu.classList.remove('show');
301           menu.closest('.dropdown')?.querySelector('.dropdown-
    toggle')?.classList.remove('show');
302         });
303       }
304     });
305   }
306 }
307
308 async function startUserInterface() {
309   console.log('User interface initializing...');
310   try {
311     initTheme();
312     setupDropdownTabs();
313     const el = document.getElementById('user-interface');
314     if (el) el.style.display = 'block';
315     await initInterface();
```

```
    console.log('User interface ready');
  } catch (err) {
    console.error('Error during startUserInterface initializat
ion:', err);
    const el = document.getElementById('user-interface');
    if (el) el.style.display = 'block';
  }
}

if (document.readyState === 'loading') {
  document.addEventListener('DOMContentLoaded', () => {
    setupDropdownTabs();
    startUserInterface();
  });
} else {
  setupDropdownTabs();
  startUserInterface();
}

async function initInterface() {
  // Initialize theme
  initTheme();

  // Initialize i18n
  await initI18n();

  // Initialize User Settings & Google OAuth
  await initUserSettings();

  // Определение и загрузка ТОЛЬКО активной вкладки
  const hash = location.hash.replace('#', '');
  const initialTabId = hash || 'tab-chat';
  const initialTabName = initialTabId.replace(/^tab-/, '');

  console.log(`[UserInterface] Loading initial active tab: ${i
nitialTabName}`);
  await ensureTabLoaded(initialTabName);
  await switchTab(initialTabId);
```

```
352
353   // Apply translations
354   applyTranslations();
355
356   // Synchronize plugin tabs visibility
357   try {
358     let pluginsResp = await fetch('/api/plugins?scope=user');
359     if (!pluginsResp.ok) {
360       pluginsResp = await fetch('/api/admin/plugins?scope=user
');
361     }
362     if (pluginsResp.ok) {
363       const pluginsData = await pluginsResp.json();
364       if (window.syncPluginTabsVisibility && pluginsData && pl
uginsData.plugins) {
365         window.syncPluginTabsVisibility(pluginsData.plugins);
366       }
367     }
368   } catch (err) {
369     console.error('Ошибка синхронизации видимости плагинов:',
err);
370   }
371
372   setupDropdownTabs();
373 }
374
375 async function loadTabContent(tabName, url, jsOverrideSrc) {
376   try {
377     const container = document.getElementById(`tab-${tabName}`
);
378     if (container && !container.innerHTML.trim()) {
379       container.innerHTML = `
380         <div class="d-flex justify-content-center align-items-
center" style="height: 200px;">
381           <div class="spinner-border text-primary" role="statu
s">
382             <span class="visually-hidden">Загрузка...</span>
383           </div>
```

```
        </div>
      `;
    }

    const response = await fetch(url);
    if (!response.ok) throw new Error(`HTTP ${response.status}
`);
    const html = await response.text();
    if (!container) return;
    container.innerHTML = html;

    // Load JS for the tab if provided or present
    const cacheBuster = Date.now();
    const scriptSrc = jsOverrideSrc || `/html/${tabName}/main.
js?v=${cacheBuster}`;

    await new Promise((resolve) => {
      const script = document.createElement('script');
      if (tabName === 'instructions' || tabName === 'rag') {
        script.type = 'module';
      }
      script.src = scriptSrc;
      script.onload = () => {
        console.log(`[User] Loaded JS for ${tabName}`);
        resolve();
      };
      script.onerror = (err) => {
        console.debug(`[User] Note: No JS loaded for ${tabName
} from ${scriptSrc}`);
        resolve();
      };
      document.body.appendChild(script);
    });

    const camelName = tabName.replace(/-([a-z])/g, (_, c) => c
.toUpperCase());
    const initFuncName = 'init' + camelName.charAt(0).toUpperC
ase() + camelName.slice(1) + 'Tab';
```

```
    if (typeof window[initFuncName] === 'function') {
      console.log(`[User] Calling ${initFuncName} for ${tabNam
e}`);
      try {
        await window[initFuncName]();
      } catch (err) {
        console.error(`[User] Error executing ${initFuncName}:
`, err);
      }
    }
    applyTranslations();
  } catch (e) {
    console.error(`[User] Error loading tab ${tabName}:`, e);
    const container = document.getElementById(`tab-${tabName}`
);
    if (container) {
      container.innerHTML = `<div class="alert alert-danger">O
шибка загрузки: ${e.message}</div>`;
    }
  }
}
```

debounce-patch.js

```
/**
 * ============================================================
 * Process Name: Windows Users Tab - Debounce-Patch Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля debounce-patch.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/users_tab/debounce-patch.js?v=20261001_v1" type="module"></script>
 *
 * File: debounce-patch.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/users_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

/**
 * Users Tab Debounce Patch
 *
 * Применяет улучшенный дебаунс к поиску пользователей,
 * используя RequestManager для снижения нагрузки на сервер
 */

import { createUserSearchDebounce } from '../js/debounce-integration.js';

// Ждем загрузки основного скрипта
```

```
if (window.initUsersTab) {
  const originalInitUsersTab = window.initUsersTab;

  window.initUsersTab = async function() {
    // Вызываем оригинальную функцию
    await originalInitUsersTab.apply(this, arguments);

    console.log('[UsersDebouncePatch] Patching users tab search...');

    // После инициализации, патчим поиск
    setTimeout(() => {
      const searchInput = document.getElementById('users-search-input');

      if (searchInput && window.globalRequestManager) {
        // Сохраняем оригинальный обработчик
        const listeners = searchInput._eventListeners?.input || [];

        // Удаляем старые обработчики
        searchInput.replaceWith(searchInput.cloneNode(true));
        const newSearchInput = document.getElementById('users-search-input');

        // Добавляем новый дебаунсированный обработчик
        newSearchInput.addEventListener('input', (e) => {
          const val = e.target.value;
          const searchClear = document.getElementById('users-search-clear');

          if (searchClear) {
            searchClear.classList.toggle('d-none', !val);
          }

          // Используем улучшенный дебаунс
          window.globalRequestManager.debounce(
            'user-search-optimized',
```

```
            async () => {
              // Обновляем state поиска (симуляция)
              if (window.updateUserSearch) {
                window.updateUserSearch(val.trim());
              }
            },
            500
          ).catch(e => console.error('Search debounce error:', e));
        });

        // Обновляем кнопку очистки
        const searchClear = document.getElementById('users-search-clear');
        if (searchClear) {
          searchClear.onclick = () => {
            newSearchInput.value = '';
            searchClear.classList.add('d-none');
            if (window.updateUserSearch) {
              window.updateUserSearch('');
            }
          };
        }

        console.log('[UsersDebouncePatch] Users tab search patched with RequestManager debounce');
      }
    }, 500);
  };
}

// Альтернативный подход: Если users tab уже загружена, патчим сразу
if (document.getElementById('users-search-input')) {
  console.log('[UsersDebouncePatch] Users tab already loaded, applying patch immediately...');

  const searchInput = document.getElementById('users-search-in
```

```
put');
  const searchClear = document.getElementById('users-search-clear');

  // Создаем новый дебаунсированный обработчик
  const debouncedSearch = createUserSearchDebounce(async (query) => {
    console.log(`[UsersDebouncePatch] Executing debounced search for: ${query}`);
    // Обновляем UI через существующую функцию если она есть
    if (window.updateUserSearch) {
      await window.updateUserSearch(query);
    }
  });

  // Заменяем обработчик
  searchInput.replaceWith(searchInput.cloneNode(true));
  const newSearchInput = document.getElementById('users-search-input');

  newSearchInput.addEventListener('input', (e) => {
    const val = e.target.value;

    if (searchClear) {
      searchClear.classList.toggle('d-none', !val);
    }

    debouncedSearch(val.trim());
  });

  if (searchClear) {
    searchClear.addEventListener('click', () => {
      newSearchInput.value = '';
      searchClear.classList.add('d-none');
      debouncedSearch('');
    });
  }

```

```
  console.log('[UsersDebouncePatch] Search debouncing initialized');
}

export {};

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/users_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/users_tab/index.html" c
lass="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/users_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Users Management Tab -->
<div class="container-fluid py-2" id="users-tab-root">
<!-- Top Banner & Stats -->
<div class="d-flex flex-wrap align-items-center justify-conten
t-between gap-3 mb-3 p-3 bg-body-tertiary border rounded shado
w-sm">
<div>
<h5 class="mb-1 text-primary d-flex align-items-center gap-2">
<i class="bi bi-people-fill text-primary"></i>
<span>Управление пользователями</span>
```

```
</h5>
<p class="mb-0 text-muted small">Управление учетными записями, ролями, доступом к системе и сбросом паролей.</p>
</div>
<div class="d-flex flex-wrap gap-2">
<button class="btn btn-outline-warning btn-sm d-flex align-items-center gap-1" id="btn-check-orphaned-dirs" title="Проверить мертвые директории пользователей, которые уже удалены из системы">
<i class="bi bi-folder-x"></i> Проверить мертвые директории
      </button>
<button class="btn btn-primary btn-sm d-flex align-items-center gap-1" id="btn-open-create-user">
<i class="bi bi-person-plus-fill"></i> Добавить пользователя
      </button>
<button class="btn btn-outline-secondary btn-sm d-flex align-items-center gap-1" id="btn-refresh-users" title="Обновить список">
<i class="bi bi-arrow-clockwise"></i> Обновить
      </button>
</div>
</div>
<!-- Summary Metric Cards -->
<div class="row g-3 mb-3">
<div class="col-6 col-md-3">
<div class="card bg-body-tertiary border h-100 shadow-sm" data-panel-id="panel-users-0" id="panel-users-0">
<div class="card-body p-3 d-flex align-items-center justify-content-between">
<div>
<div class="text-muted small">Всего пользователей</div>
<h4 class="mb-0 fw-bold" id="stat-total-users">0</h4>
</div>
<div class="bg-secondary bg-opacity-25 rounded p-2 text-primary">
<i class="bi bi-people fs-4"></i>
</div>
</div>
```

```
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-body-tertiary border h-100 shadow-sm" data-panel-id="panel-users-0" id="panel-users-0">
<div class="card-body p-3 d-flex align-items-center justify-content-between">
<div>
<div class="text-muted small">Активные аккаунты</div>
<h4 class="mb-0 text-success fw-bold" id="stat-active-users">0</h4>
</div>
<div class="bg-success bg-opacity-25 rounded p-2 text-success">
<i class="bi bi-check-circle fs-4"></i>
</div>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-body-tertiary border h-100 shadow-sm" data-panel-id="panel-users-0" id="panel-users-0">
<div class="card-body p-3 d-flex align-items-center justify-content-between">
<div>
<div class="text-muted small">Администраторы</div>
<h4 class="mb-0 text-warning fw-bold" id="stat-admin-users">0</h4>
</div>
<div class="bg-warning bg-opacity-25 rounded p-2 text-warning">
<i class="bi bi-shield-lock fs-4"></i>
</div>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-body-tertiary border h-100 shadow-sm" data
```

```
-panel-id="panel-users-0" id="panel-users-0">
<div class="card-body p-3 d-flex align-items-center justify-co
ntent-between">
<div>
<div class="text-muted small">Telegram привязано</div>
<h4 class="mb-0 text-info fw-bold" id="stat-tg-users">0</h4>
</div>
<div class="bg-info bg-opacity-25 rounded p-2 text-info">
<i class="bi bi-telegram fs-4"></i>
</div>
</div>
</div>
</div>
</div>
<!-- Search & Filter Controls -->
<div class="card bg-body-tertiary border mb-3 shadow-sm" data-
panel-id="panel-users-1" id="panel-users-1">
<div class="card-body p-3">
<div class="row g-2 align-items-center">
<div class="col-12 col-md-6 col-lg-5">
<div class="input-group input-group-sm">
<span class="input-group-text"><i class="bi bi-search"></i></s
pan>
<input class="form-control" id="users-search-input" placeholde
r="Поиск по имени, email, Telegram..." type="text"/>
<button class="btn btn-outline-secondary d-none" id="users-sea
rch-clear" type="button">□ </button>
</div>
</div>
<div class="col-6 col-md-3 col-lg-3">
<select class="form-select form-select-sm" id="users-filter-ro
le">
<option value="">Все роли</option>
<option value="admin">Администраторы (admin)</option>
<option value="user">Пользователи (user)</option>
<option value="guest">Гости (guest)</option>
</select>
</div>
```

```
<div class="col-6 col-md-3 col-lg-2">
<select class="form-select form-select-sm" id="users-filter-st
atus">
<option value="">Все статусы</option>
<option value="active">Только активные</option>
<option value="inactive">Заблокированные</option>
</select>
</div>
<div class="col-12 col-lg-2 text-end text-muted small" id="use
rs-count-badge">
          Загрузка...
        </div>
</div>
</div>
</div>
<!-- User Alert/Status message -->
<div class="d-none alert alert-dismissible fade show" id="user
s-alert-box" role="alert">
<span id="users-alert-message"></span>
<button aria-label="Close" class="btn-close" onclick="document
.getElementById('users-alert-box').classList.add('d-none')" ty
pe="button"></button>
</div>
<!-- Users Table Card -->
<div class="card shadow-sm border" data-panel-id="panel-users-
2" id="panel-users-2">
<div class="card-body p-0">
<div class="table-responsive">
<table class="table table-hover mb-0 align-middle">
<thead class="table-light small text-uppercase">
<tr>
<th class="text-center" style="width: 50px;">ID</th>
<th>Пользователь</th>
<th class="text-center" style="width: 140px;">Роль</th>
<th class="text-center" style="width: 130px;">Статус</th>
<th style="width: 150px;">Telegram</th>
<th class="text-center" style="width: 120px;">Пароль</th>
<th style="width: 160px;">Регистрация / Вход</th>
```

```
<th class="text-center" style="width: 170px;">Действия</th>
</tr>
</thead>
<tbody id="users-table-body">
<tr>
<td class="text-center py-4 text-muted" colspan="8">
<div class="spinner-border spinner-border-sm text-primary me-2
" role="status"></div>
                Загрузка списка пользователей...
             </td>
</tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
<!-- ========================================================
================ -->
<!-- MODAL: Create User -->
<!-- ========================================================
================ -->
<div aria-hidden="true" aria-labelledby="modal-create-user-tit
le" class="modal fade" id="modal-create-user" tabindex="-1">
<div class="modal-dialog modal-dialog-centered">
<div class="modal-content shadow-lg">
<div class="modal-header">
<h5 class="modal-title d-flex align-items-center gap-2" id="mo
dal-create-user-title">
<i class="bi bi-person-plus text-primary"></i> Добавить нового
 пользователя
        </h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<form id="form-create-user">
<div class="modal-body p-3">
<div class="mb-3">
```

```
<label class="form-label small fw-bold">Email <span class="tex
t-danger">*</span></label>
<input class="form-control form-control-sm" id="create-user-em
ail" placeholder="user@example.com" required="" type="email"/>
</div>
<div class="mb-3">
<label class="form-label small fw-bold">Отображаемое имя <span
 class="text-danger">*</span></label>
<input class="form-control form-control-sm" id="create-user-na
me" placeholder="Иван Иванов" required="" type="text"/>
</div>
<div class="mb-3">
<label class="form-label small fw-bold">Пароль (опционально)</
label>
<div class="input-group input-group-sm">
<input class="form-control font-monospace" id="create-user-pas
sword" placeholder="Оставьте пустым или задайте пароль" type="
text"/>
<button class="btn btn-outline-secondary" id="btn-gen-create-p
assword" title="Сгенерировать безопасный пароль" type="button"
>
<i class="bi bi-dice-5"></i>
</button>
</div>
<div class="form-text text-muted small">Если пароль не задан,
пользователь сможет авторизоваться через Google или Telegram.<
/div>
</div>
<div class="mb-3 p-3 bg-body-tertiary border rounded">
<div class="form-check form-switch d-flex align-items-center j
ustify-content-between p-0 m-0">
<label class="form-check-label fw-bold" for="create-user-is-ad
min">
<i class="bi bi-shield-lock-fill text-warning me-1"></i> Права
 Администратора
              </label>
<input class="form-check-input ms-0 fs-5" id="create-user-is-a
dmin" role="switch" type="checkbox"/>
```

```
</div>
<div class="small text-muted mt-1" id="create-role-hint">Роль:
 <strong class="text-info" id="create-role-label">Пользователь
 (user)</strong></div>
</div>
<div class="row g-2 mb-2">
<div class="col-6">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="create-user-is-
active" role="switch" type="checkbox"/>
<label class="form-check-label small" for="create-user-is-acti
ve">Активен</label>
</div>
</div>
<div class="col-6">
<div class="form-check form-switch">
<input checked="" class="form-check-input" id="create-user-ver
ified" role="switch" type="checkbox"/>
<label class="form-check-label small" for="create-user-verifie
d">Email подтвержден</label>
</div>
</div>
</div>
</div>
<div class="modal-footer">
<button class="btn btn-secondary btn-sm rounded-pill px-3" dat
a-bs-dismiss="modal" type="button">Отмена</button>
<button class="btn btn-primary btn-sm rounded-pill px-3" id="b
tn-submit-create-user" type="submit">Создать пользователя</but
ton>
</div>
</form>
</div>
</div>
</div>
<!-- ========================================================
================ -->
<!-- MODAL: Edit User -->
```

```
<!-- ============================================================
================ -->
<div aria-hidden="true" aria-labelledby="modal-edit-user-title
" class="modal fade" id="modal-edit-user" tabindex="-1">
<div class="modal-dialog modal-dialog-centered">
<div class="modal-content shadow-lg">
<div class="modal-header">
<h5 class="modal-title d-flex align-items-center gap-2" id="mo
dal-edit-user-title">
<i class="bi bi-pencil-square text-warning"></i> Редактировани
е пользователя #<span id="edit-user-id-badge"></span>
</h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<form id="form-edit-user">
<input id="edit-user-id" type="hidden"/>
<div class="modal-body p-3">
<div class="mb-3">
<label class="form-label small fw-bold">Email <span class="tex
t-danger">*</span></label>
<input class="form-control form-control-sm" id="edit-user-emai
l" required="" type="email"/>
</div>
<div class="mb-3">
<label class="form-label small fw-bold">Имя <span class="text-
danger">*</span></label>
<input class="form-control form-control-sm" id="edit-user-name
" required="" type="text"/>
</div>
<div class="mb-3 p-3 bg-body-tertiary border rounded">
<div class="form-check form-switch d-flex align-items-center j
ustify-content-between p-0 m-0">
<label class="form-check-label fw-bold" for="edit-user-is-admi
n">
<i class="bi bi-shield-lock-fill text-warning me-1"></i> Права
 Администратора
                </label>
```

```
<input class="form-check-input ms-0 fs-5" id="edit-user-is-admin" role="switch" type="checkbox"/>
</div>
<div class="small text-muted mt-1" id="edit-role-hint">Роль: <strong class="text-info" id="edit-role-label">Пользователь (user)</strong></div>
</div>
<div class="row g-2 mb-2">
<div class="col-6">
<div class="form-check form-switch">
<input class="form-check-input" id="edit-user-is-active" role="switch" type="checkbox"/>
<label class="form-check-label small" for="edit-user-is-active">Активен</label>
</div>
</div>
<div class="col-6">
<div class="form-check form-switch">
<input class="form-check-input" id="edit-user-verified" role="switch" type="checkbox"/>
<label class="form-check-label small" for="edit-user-verified">Email подтвержден</label>
</div>
</div>
</div>
</div>
<div class="modal-footer">
<button class="btn btn-secondary btn-sm rounded-pill px-3" data-bs-dismiss="modal" type="button">Отмена</button>
<button class="btn btn-warning btn-sm rounded-pill px-3" id="btn-submit-edit-user" type="submit">Сохранить изменения</button>
</div>
</form>
</div>
</div>
</div>
<!-- ============================================================
```

```
================ -->
<!-- MODAL: Set / Reset Password -->
<!-- ============================================================
================ -->
<div aria-hidden="true" aria-labelledby="modal-password-user-t
itle" class="modal fade" id="modal-password-user" tabindex="-1
">
<div class="modal-dialog modal-dialog-centered">
<div class="modal-content shadow-lg">
<div class="modal-header">
<h5 class="modal-title d-flex align-items-center gap-2" id="mo
dal-password-user-title">
<i class="bi bi-key-fill text-info"></i> Смена пароля пользова
теля
        </h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<form id="form-password-user">
<input id="pwd-user-id" type="hidden"/>
<div class="modal-body p-3">
<p class="small text-muted mb-3">
            Установка нового пароля для <strong id="pwd-user-t
arget"></strong>
</p>
<div class="mb-3">
<label class="form-label small fw-bold">Новый пароль <span cla
ss="text-danger">*</span></label>
<div class="input-group">
<input class="form-control font-monospace" id="pwd-input-val"
minlength="4" placeholder="Введите пароль" required="" type="t
ext"/>
<button class="btn btn-outline-info" id="btn-gen-pwd-val" titl
e="Сгенерировать безопасный пароль" type="button">
<i class="bi bi-dice-5"></i> Сгенерировать
              </button>
</div>
</div>
```

```
</div>
<div class="modal-footer">
<button class="btn btn-secondary btn-sm rounded-pill px-3" dat
a-bs-dismiss="modal" type="button">Отмена</button>
<button class="btn btn-info btn-sm rounded-pill px-3" id="btn-
submit-pwd-user" type="submit">Установить пароль</button>
</div>
</form>
</div>
</div>
</div>
<!-- ========================================================
================ -->
<!-- MODAL: User Details & Profile Settings -->
<!-- ========================================================
================ -->
<div aria-hidden="true" aria-labelledby="modal-details-user-ti
tle" class="modal fade" id="modal-details-user" tabindex="-1">
<div class="modal-dialog modal-dialog-centered modal-lg">
<div class="modal-content shadow-lg">
<div class="modal-header">
<h5 class="modal-title d-flex align-items-center gap-2" id="mo
dal-details-user-title">
<i class="bi bi-person-badge text-primary"></i> Профиль и наст
ройки пользователя
        </h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-3" id="details-user-content">
<div class="text-center py-3 text-muted">Загрузка данных...</d
iv>
</div>
<div class="modal-footer">
<button class="btn btn-secondary btn-sm rounded-pill px-3" dat
a-bs-dismiss="modal" type="button">Закрыть</button>
</div>
</div>
```

```
</div>
</div>
<!-- ============================================================
================ -->
<!-- MODAL: Delete Confirmation -->
<!-- ============================================================
================ -->
<div aria-hidden="true" aria-labelledby="modal-delete-user-tit
le" class="modal fade" id="modal-delete-user" tabindex="-1">
<div class="modal-dialog modal-dialog-centered">
<div class="modal-content border-danger shadow-lg">
<div class="modal-header border-danger">
<h5 class="modal-title text-danger d-flex align-items-center g
ap-2" id="modal-delete-user-title">
<i class="bi bi-exclamation-triangle-fill"></i> Подтверждение
удаления
        </h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-3">
<p class="mb-2">Вы действительно хотите удалить пользователя <
strong class="text-danger" id="delete-user-name-target"></stro
ng>?</p>
<p class="small text-muted mb-0">Это действие удалит учетную з
апись, настройки и сессии пользователя. Операция необратима.</
p>
<input id="delete-user-id" type="hidden"/>
</div>
<div class="modal-footer">
<button class="btn btn-secondary btn-sm rounded-pill px-3" dat
a-bs-dismiss="modal" type="button">Отмена</button>
<button class="btn btn-danger btn-sm rounded-pill px-3" id="bt
n-confirm-delete-user" type="button">Удалить пользователя</but
ton>
</div>
</div>
</div>
```

```
</div>
<!-- ============================================================
================ -->
<!-- MODAL: Orphaned User Directories Management -->
<!-- ============================================================
================ -->
<div aria-hidden="true" aria-labelledby="modal-orphaned-dirs-t
itle" class="modal fade" id="modal-orphaned-dirs" tabindex="-1
">
<div class="modal-dialog modal-dialog-centered modal-lg">
<div class="modal-content shadow-lg border-warning">
<div class="modal-header border-warning">
<h5 class="modal-title text-warning d-flex align-items-center
gap-2" id="modal-orphaned-dirs-title">
<i class="bi bi-folder-x"></i> Мертвые директории пользователе
й
        </h5>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-3">
<div class="alert alert-info border-0 small mb-3">
<i class="bi bi-info-circle text-info me-1"></i>
          Обнаружены директории в хранилище <code>data/users/<
/code>, которые не привязаны ни к одному активному пользовател
ю в базе данных или остались после тестов.
        </div>
<!-- Summary Stats Banner -->
<div class="d-flex align-items-center justify-content-between
p-2 mb-3 bg-body-tertiary border rounded">
<div class="small">
<span class="text-muted">Найдено директорий:</span> <strong cl
ass="text-warning" id="orphaned-dirs-count">0</strong>
<span class="mx-2 text-secondary">|</span>
<span class="text-muted">Занято на диске:</span> <strong class
="text-info" id="orphaned-dirs-size">0 B</strong>
<span class="mx-2 text-secondary">|</span>
<span class="text-muted">Файлов:</span> <strong id="orphaned-d
```

```
irs-files">0</strong>
</div>
<button class="btn btn-outline-secondary btn-sm py-0 px-2 roun
ded-pill" id="btn-refresh-orphaned-dirs" title="Пересканироват
ь">
<i class="bi bi-arrow-clockwise"></i> Обновить
          </button>
</div>
<div class="table-responsive" id="orphaned-dirs-table-wrapper"
 style="max-height: 380px; overflow-y: auto;">
<table class="table table-hover table-sm align-middle mb-0">
<thead class="table-light small text-uppercase sticky-top">
<tr>
<th>Папка / ID</th>
<th>Причина</th>
<th class="text-center" style="width: 80px;">Файлов</th>
<th class="text-center" style="width: 100px;">Размер</th>
<th style="width: 140px;">Изменено</th>
<th class="text-center" style="width: 90px;">Действие</th>
</tr>
</thead>
<tbody id="orphaned-dirs-tbody">
<tr>
<td class="text-center py-4 text-muted" colspan="6">
<div class="spinner-border spinner-border-sm text-warning me-2
" role="status"></div>
                  Сканирование хранилища...
                </td>
</tr>
</tbody>
</table>
</div>
</div>
<div class="modal-footer justify-content-between">
<button class="btn btn-secondary btn-sm rounded-pill px-3" dat
a-bs-dismiss="modal" type="button">Закрыть</button>
<button class="btn btn-danger btn-sm rounded-pill px-3 d-flex
align-items-center gap-1" id="btn-cleanup-all-orphaned" type="
```

```
button">
<i class="bi bi-trash3-fill"></i> Удалить все мертвые директор
ии
        </button>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Users Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/users_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/users_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// ============================================================
// Webinterface: Users Tab Logic
// Module: webinterface/users_tab/main.js
// Author: hypo69
// Copyright: © 2026 hypo69
// ============================================================

'use strict';

(function () {
```

```
  const state = {
    users: [],
    stats: {
      total: 0,
      active: 0,
      admins: 0,
      telegram: 0
    },
    filterRole: '',
    filterStatus: '',
    searchQuery: '',
    editingUserId: null,
    savingUserId: null,
    isLoading: false,
    initialized: false
  };

  // Helper API fetch
  async function apiFetch(url, options = {}) {
    if (window.api && typeof window.api.fetch === 'function')
{
      return window.api.fetch(url, options);
    }
    const res = await fetch(url, options);
    if (!res.ok) {
      let errMsg = `HTTP ${res.status}`;
      try {
        const errData = await res.json();
        if (errData && errData.detail) {
          errMsg = typeof errData.detail === 'string' ? errDat
a.detail : JSON.stringify(errData.detail);
        }
      } catch (_) {}
      throw new Error(errMsg);
    }
    return res.json();
  }

```

```
  // Show notification
  function showStatusAlert(msg, type = 'info') {
    const alertBox = document.getElementById('users-alert-box'
);
    const alertMsg = document.getElementById('users-alert-mess
age');
    if (!alertBox || !alertMsg) return;

    alertBox.className = `alert alert-${type} alert-dismissibl
e fade show mb-3`;
    alertMsg.innerHTML = msg;
    alertBox.classList.remove('d-none');

    setTimeout(() => {
      alertBox.classList.add('d-none');
    }, 5000);
  }

  // Generate secure password
  function generatePassword(length = 12) {
    const charset = 'abcdefghijkmnopqrstuvwxyzABCDEFGHJKLMNPQR
STUVWXYZ23456789!@#$%&*';
    let pwd = '';
    const randomValues = new Uint32Array(length);
    window.crypto.getRandomValues(randomValues);
    for (let i = 0; i < length; i++) {
      pwd += charset[randomValues[i] % charset.length];
    }
    return pwd;
  }

  // Escape HTML
  function escapeHtml(str) {
    if (!str) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
```

```
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
  }

  // Format datetime
  function formatDate(isoStr) {
    if (!isoStr) return '<span class="text-muted">—</span>';
    try {
      const d = new Date(isoStr);
      if (isNaN(d.getTime())) return escapeHtml(isoStr);
      return `<span title="${escapeHtml(isoStr)}">${d.toLocale
DateString('ru-RU')} <small class="text-muted">${d.toLocaleTim
eString('ru-RU', { hour: '2-digit', minute: '2-digit' })}</sma
ll></span>`;
    } catch (_) {
      return escapeHtml(isoStr);
    }
  }

  // Load Users List from Backend
  async function loadUsers() {
    state.isLoading = true;
    const tableBody = document.getElementById('users-table-bod
y');
    const countBadge = document.getElementById('users-count-ba
dge');

    if (tableBody && (!state.users || state.users.length === 0
)) {
      tableBody.innerHTML = `
        <tr>
          <td colspan="8" class="text-center py-4 text-muted">
            <div class="spinner-border spinner-border-sm text-
primary me-2" role="status"></div>
            Загрузка списка пользователей...
          </td>
        </tr>`;
    }
```

```
132
133     try {
134       const params = new URLSearchParams();
135       if (state.searchQuery) params.append('q', state.searchQu
ery);
136       if (state.filterRole) params.append('role', state.filter
Role);
137       if (state.filterStatus) params.append('status', state.fi
lterStatus);
138
139       const url = `/api/admin/users?${params.toString()}`;
140       const data = await apiFetch(url);
141
142       state.users = data.users || [];
143       if (data.stats) {
144         state.stats = data.stats;
145       }
146
147       renderStats();
148       renderTable();
149     } catch (err) {
150       console.error('[UsersTab] Error loading users:', err);
151       showStatusAlert(`Ошибка загрузки пользователей: ${err.me
ssage}`, 'danger');
152       if (tableBody) {
153         tableBody.innerHTML = `
154           <tr>
155             <td colspan="8" class="text-center py-4 text-dange
r">
156               <i class="bi bi-exclamation-octagon fs-4 d-block
 mb-1"></i>
157               Не удалось загрузить пользователей: ${escapeHtml
(err.message)}
158             </td>
159           </tr>`;
160       }
161     } finally {
162       state.isLoading = false;
```

```
    }
  }

  // Render Metric Cards
  function renderStats() {
    const totalEl = document.getElementById('stat-total-users'
);
    const activeEl = document.getElementById('stat-active-user
s');
    const adminEl = document.getElementById('stat-admin-users'
);
    const tgEl = document.getElementById('stat-tg-users');

    if (totalEl) totalEl.textContent = state.stats.total || 0;
    if (activeEl) activeEl.textContent = state.stats.active ||
 0;
    if (adminEl) adminEl.textContent = state.stats.admins || 0
;
    if (tgEl) tgEl.textContent = state.stats.telegram || 0;
  }

  // Render Users Table
  function renderTable() {
    const tableBody = document.getElementById('users-table-bod
y');
    const countBadge = document.getElementById('users-count-ba
dge');
    if (!tableBody) return;

    if (countBadge) {
      countBadge.textContent = `Показано: ${state.users.length
} из ${state.stats.total || state.users.length}`;
    }

    if (state.users.length === 0) {
      tableBody.innerHTML = `
        <tr>
          <td colspan="8" class="text-center py-5 text-muted">
```

```
            <i class="bi bi-person-x fs-2 d-block mb-2 text-secondary"></i>
            Пользователи не найдены.
          </td>
        </tr>`;
      return;
    }

    tableBody.innerHTML = state.users.map((user) => {
      const isRoot = user.id === 1;
      const isAdmin = Boolean(user.is_admin || user.role === 'admin');
      const isActive = Boolean(user.is_active);
      const isEmailVerified = Boolean(user.is_email_verified);
      const hasPassword = Boolean(user.has_password);
      const isRowEditing = (state.editingUserId === user.id);
      const isSaving = (state.savingUserId === user.id);

      // User Initials or Avatar
      let avatarHtml = '';
      if (user.picture) {
        avatarHtml = `<img src="${escapeHtml(user.picture)}" class="rounded-circle me-2 flex-shrink-0" style="width:36px;height:36px;object-fit:cover;" alt="avatar">`;
      } else {
        const initial = (user.name || user.email || 'U').charAt(0).toUpperCase();
        const bgClass = isAdmin ? 'bg-warning text-dark' : 'bg-primary text-white';
        avatarHtml = `<div class="rounded-circle ${bgClass} d-flex align-items-center justify-content-center me-2 flex-shrink-0 fw-bold" style="width:36px;height:36px;font-size:14px;">${escapeHtml(initial)}</div>`;
      }

      // Password column
      const pwdHtml = hasPassword
        ? `<span class="badge bg-success-subtle text-success b
```

```
order border-success-subtle" title="Пароль установлен"><i clas
s="bi bi-key-fill"></i> Задан</span>`
222         : `<span class="badge bg-warning-subtle text-warning b
order border-warning-subtle" title="Пароль не установлен (OAut
h/TG)"><i class="bi bi-dash-circle"></i> Нет</span>`;
223
224       // Email verified badge
225       const emailVerifiedBadge = isEmailVerified
226         ? `<i class="bi bi-patch-check-fill text-success ms-1"
 title="Email подтвержден"></i>`
227         : `<i class="bi bi-question-circle text-muted ms-1" ti
tle="Email не подтвержден"></i>`;
228
229       // If this row is in inline-edit mode:
230       if (isRowEditing) {
231         return `
232           <tr data-user-id="${user.id}" class="table-active bo
rder-primary">
233             <td class="text-center text-muted fw-bold align-mi
ddle">${user.id}</td>
234             <td class="align-middle">
235               <div class="d-flex align-items-center">
236                 ${avatarHtml}
237                 <div class="w-100">
238                   <div class="input-group input-group-sm mb-1"
>
239                     <span class="input-group-text text-muted">
<i class="bi bi-person"></i></span>
240                     <input type="text" class="form-control inl
ine-edit-name" value="${escapeHtml(user.name || '')}" placehol
der="Имя пользователя">
241                   </div>
242                   <div class="input-group input-group-sm">
243                     <span class="input-group-text text-muted">
<i class="bi bi-envelope"></i></span>
244                     <input type="email" class="form-control in
line-edit-email" value="${escapeHtml(user.email || '')}" place
holder="Email">
```

```
                  </div>
                </div>
              </div>
            </td>
            <td class="text-center align-middle">
              <select class="form-select form-select-sm inline
-edit-role" ${isRoot ? 'disabled' : ''}>
                <option value="user" ${user.role === 'user' ?
'selected' : ''}>  User</option>
                <option value="admin" ${user.role === 'admin'
? 'selected' : ''}>  Admin</option>
                <option value="guest" ${user.role === 'guest'
? 'selected' : ''}>  Guest</option>
              </select>
            </td>
            <td class="text-center align-middle">
              <select class="form-select form-select-sm inline
-edit-status" ${isRoot ? 'disabled' : ''}>
                <option value="1" ${isActive ? 'selected' : ''
}>  Активен</option>
                <option value="0" ${!isActive ? 'selected' : '
'}>  Блок</option>
              </select>
            </td>
            <td class="align-middle">
              <div class="input-group input-group-sm">
                <span class="input-group-text text-info"><i cl
ass="bi bi-telegram"></i></span>
                <input type="text" class="form-control inline-
edit-tg" value="${escapeHtml(user.telegram_username || '')}" p
laceholder="Username">
              </div>
            </td>
            <td class="text-center align-middle">${pwdHtml}</t
d>
            <td class="small align-middle">
              <div><span class="text-muted">Создан:</span> ${f
ormatDate(user.created_at)}</div>
```

```
            </td>
            <td class="text-center align-middle">
              <div class="d-flex justify-content-center gap-1"
>
                <button class="btn btn-success btn-sm btn-save
-inline" data-id="${user.id}" title="Сохранить изменения (Ente
r)" ${isSaving ? 'disabled' : ''}>
                  ${isSaving ? '<span class="spinner-border sp
inner-border-sm"></span>' : '<i class="bi bi-check-lg"></i>'}
                </button>
                <button class="btn btn-outline-secondary btn-s
m btn-cancel-inline" data-id="${user.id}" title="Отмена (Esc)"
 ${isSaving ? 'disabled' : ''}>
                  <i class="bi bi-x-lg"></i>
                </button>
              </div>
            </td>
          </tr>
        `;
      }

      // Normal view mode with DIRECT editable table fields:
      const roleSelectHtml = `
        <select class="form-select form-select-sm bg-dark bord
er-secondary user-table-role-select text-center ${isAdmin ? 't
ext-warning fw-bold' : 'text-info'}"
          data-user-id="${user.id}"
          ${isRoot ? 'disabled title="Нельзя изменить роль Roo
t"' : 'title="Изменить роль прямо в таблице"'}>
          <option value="user" ${user.role === 'user' ? 'selec
ted' : ''} class="text-info">  User</option>
          <option value="admin" ${isAdmin ? 'selected' : ''} c
lass="text-warning">  Admin</option>
          <option value="guest" ${user.role === 'guest' ? 'sel
ected' : ''} class="text-secondary">  Guest</option>
        </select>
      `;

```

```
      const statusSelectHtml = `
        <select class="form-select form-select-sm bg-dark border-secondary user-table-status-select text-center ${isActive ? 'text-success' : 'text-danger'}"
          data-user-id="${user.id}"
          ${isRoot ? 'disabled title="Нельзя деактивировать Root"' : 'title="Изменить статус прямо в таблице"'}>
          <option value="1" ${isActive ? 'selected' : ''} class="text-success">  Активен</option>
          <option value="0" ${!isActive ? 'selected' : ''} class="text-danger">  Блок</option>
        </select>
      `;

      // Telegram column
      let tgHtml = '<span class="text-muted small">—</span>';
      if (user.telegram_username) {
        tgHtml = `<a href="https://t.me/${escapeHtml(user.telegram_username)}" target="_blank" class="text-info text-decoration-none small d-flex align-items-center gap-1" title="Открыть профиль Telegram">
          <i class="bi bi-telegram"></i> @${escapeHtml(user.telegram_username)}
        </a>`;
      } else if (user.telegram_id) {
        tgHtml = `<span class="text-muted small" title="Telegram ID"><i class="bi bi-telegram text-info"></i> ID: ${escapeHtml(user.telegram_id)}</span>`;
      }

      return `
        <tr data-user-id="${user.id}">
          <td class="text-center text-muted fw-bold align-middle">${user.id}</td>
          <td class="align-middle">
            <div class="d-flex align-items-center justify-content-between">
              <div class="d-flex align-items-center">
```

```
                ${avatarHtml}
                <div>
                  <div class="fw-bold text-white user-cell-nam
e" title="Дважды кликните для быстрого редактирования" style="
cursor: pointer;">
                    ${escapeHtml(user.name || 'Без имени')}
                    <i class="bi bi-pencil-fill text-muted ms-
1 opacity-25 hover-opacity-100" style="font-size:10px;"></i>
                  </div>
                  <div class="small text-muted d-flex align-it
ems-center user-cell-email" title="Дважды кликните для быстрог
о редактирования" style="cursor: pointer;">
                    ${escapeHtml(user.email)} ${emailVerifiedB
adge}
                  </div>
                </div>
              </div>
            </div>
          </td>
          <td class="text-center align-middle" style="min-widt
h: 130px;">${roleSelectHtml}</td>
          <td class="text-center align-middle" style="min-widt
h: 125px;">${statusSelectHtml}</td>
          <td class="align-middle" style="min-width: 140px;">
            <div class="d-flex align-items-center justify-cont
ent-between">
              <div class="user-cell-tg" title="Дважды кликните
 для быстрого редактирования" style="cursor: pointer;">${tgHtm
l}</div>
            </div>
          </td>
          <td class="text-center align-middle">${pwdHtml}</td>
          <td class="small align-middle">
            <div><span class="text-muted">Создан:</span> ${for
matDate(user.created_at)}</div>
            ${user.last_login ? `<div><span class="text-muted"
>Вход:</span> ${formatDate(user.last_login)}</div>` : ''}
          </td>
```

```
          <td class="text-center align-middle">
            <div class="btn-group btn-group-sm" role="group">
              <button class="btn btn-outline-warning btn-inlin
e-edit" data-id="${user.id}" title="Редактировать поля строки
прямо в таблице">
                <i class="bi bi-pencil-square"></i>
              </button>
              <button class="btn btn-outline-info btn-pwd-user
" data-id="${user.id}" data-name="${escapeHtml(user.name || us
er.email)}" title="Сменить пароль">
                <i class="bi bi-key-fill"></i>
              </button>
              <button class="btn btn-outline-light btn-details
-user" data-id="${user.id}" title="Профиль и настройки">
                <i class="bi bi-info-circle-fill"></i>
              </button>
              <button class="btn btn-outline-secondary btn-edi
t-user-modal" data-id="${user.id}" title="Редактировать в мода
льном окне">
                <i class="bi bi-sliders"></i>
              </button>
              <button class="btn btn-outline-danger btn-delete
-user" data-id="${user.id}" data-name="${escapeHtml(user.name
|| user.email)}" ${isRoot ? 'disabled title="Нельзя удалить Ro
ot"' : 'title="Удалить"'}>
                <i class="bi bi-trash-fill"></i>
              </button>
            </div>
          </td>
        </tr>`;
    }).join('');

    attachTableEvents();
  }

  // Attach Table Action Button Events
  function attachTableEvents() {
    // Direct Table Role Selector Change
```

```
    document.querySelectorAll('.user-table-role-select').forEach(sel => {
      sel.addEventListener('change', async (e) => {
        const userId = parseInt(e.target.getAttribute('data-user-id'), 10);
        const newRole = e.target.value;
        const isAdmin = (newRole === 'admin') ? 1 : 0;
        await handleUpdateField(userId, { role: newRole, is_admin: isAdmin }, `Роль изменена на "${newRole}"`);
      });
    });

    // Direct Table Status Selector Change
    document.querySelectorAll('.user-table-status-select').forEach(sel => {
      sel.addEventListener('change', async (e) => {
        const userId = parseInt(e.target.getAttribute('data-user-id'), 10);
        const newStatus = parseInt(e.target.value, 10);
        await handleUpdateField(userId, { is_active: newStatus }, `Статус изменен на "${newStatus ? 'Активен' : 'Заблокирован'}"`);
      });
    });

    // Double-click row or name/email/tg to enter inline-edit mode
    document.querySelectorAll('#users-table-body tr').forEach(row => {
      const userId = parseInt(row.getAttribute('data-user-id'), 10);
      if (!userId) return;

      const nameEl = row.querySelector('.user-cell-name');
      const emailEl = row.querySelector('.user-cell-email');
      const tgEl = row.querySelector('.user-cell-tg');

      [nameEl, emailEl, tgEl].forEach(el => {
```

```
        if (el) {
          el.addEventListener('dblclick', () => {
            state.editingUserId = userId;
            renderTable();
          });
        }
      });
    });

    // Inline Edit Button
    document.querySelectorAll('.btn-inline-edit').forEach(btn
=> {
      btn.addEventListener('click', (e) => {
        const userId = parseInt(e.currentTarget.getAttribute('
data-id'), 10);
        state.editingUserId = userId;
        renderTable();
      });
    });

    // Inline Cancel Button
    document.querySelectorAll('.btn-cancel-inline').forEach(bt
n => {
      btn.addEventListener('click', () => {
        state.editingUserId = null;
        renderTable();
      });
    });

    // Inline Save Button
    document.querySelectorAll('.btn-save-inline').forEach(btn
=> {
      btn.addEventListener('click', async (e) => {
        const userId = parseInt(e.currentTarget.getAttribute('
data-id'), 10);
        await saveInlineRow(userId);
      });
    });
```

```
436
437     // Enter & Escape key handling in inline inputs
438     document.querySelectorAll('.inline-edit-name, .inline-edit
-email, .inline-edit-tg').forEach(inp => {
439       inp.addEventListener('keydown', async (e) => {
440         const tr = inp.closest('tr');
441         const userId = tr ? parseInt(tr.getAttribute('data-use
r-id'), 10) : null;
442         if (!userId) return;
443 
444         if (e.key === 'Enter') {
445           e.preventDefault();
446           await saveInlineRow(userId);
447         } else if (e.key === 'Escape') {
448           e.preventDefault();
449           state.editingUserId = null;
450           renderTable();
451         }
452       });
453     });
454 
455     // Modal Edit User Button
456     document.querySelectorAll('.btn-edit-user-modal').forEach(
btn => {
457       btn.addEventListener('click', (e) => {
458         const userId = parseInt(e.currentTarget.getAttribute('
data-id'), 10);
459         openEditModal(userId);
460       });
461     });
462 
463     // Change Password Button
464     document.querySelectorAll('.btn-pwd-user').forEach(btn =>
{
465       btn.addEventListener('click', (e) => {
466         const userId = parseInt(e.currentTarget.getAttribute('
data-id'), 10);
467         const userName = e.currentTarget.getAttribute('data-na
```

```
me');
        openPasswordModal(userId, userName);
      });
    });

    // View Details Button
    document.querySelectorAll('.btn-details-user').forEach(btn
 => {
      btn.addEventListener('click', (e) => {
        const userId = parseInt(e.currentTarget.getAttribute('
data-id'), 10);
        openDetailsModal(userId);
      });
    });

    // Delete User Button
    document.querySelectorAll('.btn-delete-user').forEach(btn
=> {
      btn.addEventListener('click', (e) => {
        const userId = parseInt(e.currentTarget.getAttribute('
data-id'), 10);
        const userName = e.currentTarget.getAttribute('data-na
me');
        openDeleteModal(userId, userName);
      });
    });
  }

  // Save Inline Row Changes
  async function saveInlineRow(userId) {
    const row = document.querySelector(`tr[data-user-id="${use
rId}"]`);
    if (!row) return;

    const nameInput = row.querySelector('.inline-edit-name');
    const emailInput = row.querySelector('.inline-edit-email')
;
    const roleSelect = row.querySelector('.inline-edit-role');
```

```
    const statusSelect = row.querySelector('.inline-edit-statu
s');
    const tgInput = row.querySelector('.inline-edit-tg');

    const name = nameInput ? nameInput.value.trim() : '';
    const email = emailInput ? emailInput.value.trim() : '';
    const role = roleSelect ? roleSelect.value : 'user';
    const isActive = statusSelect ? parseInt(statusSelect.valu
e, 10) : 1;
    const tg = tgInput ? tgInput.value.trim().replace(/^@/, ''
) : '';

    if (!email) {
      showStatusAlert('Email не может быть пустым', 'warning')
;
      return;
    }
    if (!name) {
      showStatusAlert('Имя пользователя не может быть пустым',
 'warning');
      return;
    }

    const payload = {
      name: name,
      email: email,
      role: role,
      is_admin: (role === 'admin') ? 1 : 0,
      is_active: isActive,
      telegram_username: tg
    };

    state.savingUserId = userId;
    const saveBtn = row.querySelector('.btn-save-inline');
    if (saveBtn) saveBtn.disabled = true;

    try {
      const res = await apiFetch(`/api/admin/users/${userId}`,
```

```
 {
531         method: 'PATCH',
532         headers: { 'Content-Type': 'application/json' },
533         body: JSON.stringify(payload)
534       });
535 
536       state.editingUserId = null;
537       showStatusAlert(`Пользователь #${userId} (<strong>${esca
peHtml(name)}</strong>) успешно сохранён!`, 'success');
538       loadUsers();
539     } catch (err) {
540       showStatusAlert(`Ошибка сохранения: ${err.message}`, 'da
nger');
541     } finally {
542       state.savingUserId = null;
543       if (saveBtn) saveBtn.disabled = false;
544     }
545   }
546 
547   // Update a Single Field or Subset of Fields
548   async function handleUpdateField(userId, payload, successMsg
 = 'Данные сохранены') {
549     try {
550       const res = await apiFetch(`/api/admin/users/${userId}`,
 {
551         method: 'PATCH',
552         headers: { 'Content-Type': 'application/json' },
553         body: JSON.stringify(payload)
554       });
555 
556       // Update local state user
557       const user = state.users.find(u => u.id === userId);
558       if (user && res.user) {
559         Object.assign(user, res.user);
560       }
561       showStatusAlert(`Пользователь #${userId}: ${successMsg}`
, 'success');
562       loadUsers();
```

```
    } catch (err) {
      showStatusAlert(`Ошибка обновления: ${err.message}`, 'da
nger');
      loadUsers();
    }
  }

  // Open Edit Modal
  function openEditModal(userId) {
    const user = state.users.find(u => u.id === userId);
    if (!user) return;

    const isEditAdmin = Boolean(user.is_admin || user.role ===
 'admin');
    document.getElementById('edit-user-id').value = user.id;
    document.getElementById('edit-user-id-badge').textContent
= user.id;
    document.getElementById('edit-user-email').value = user.em
ail || '';
    document.getElementById('edit-user-name').value = user.nam
e || '';

    const editAdminSw = document.getElementById('edit-user-is-
admin');
    const editRoleLabel = document.getElementById('edit-role-l
abel');
    if (editAdminSw) {
      editAdminSw.checked = isEditAdmin;
      editAdminSw.disabled = (userId === 1);
      if (editRoleLabel) {
        editRoleLabel.textContent = isEditAdmin ? 'Администрат
op (admin)' : 'Пользователь (user)';
        editRoleLabel.className = isEditAdmin ? 'text-warning'
 : 'text-info';
      }
    }

    document.getElementById('edit-user-is-active').checked = B
```

```
oolean(user.is_active);
592     document.getElementById('edit-user-verified').checked = Bo
olean(user.is_email_verified);
593
594     const modalEl = document.getElementById('modal-edit-user')
;
595     const modal = bootstrap.Modal.getOrCreateInstance(modalEl)
;
596     modal.show();
597   }
598
599   // Open Password Modal
600   function openPasswordModal(userId, userName) {
601     document.getElementById('pwd-user-id').value = userId;
602     document.getElementById('pwd-user-target').textContent = `
${userName} (ID #${userId})`;
603     document.getElementById('pwd-input-val').value = '';
604
605     const modalEl = document.getElementById('modal-password-us
er');
606     const modal = bootstrap.Modal.getOrCreateInstance(modalEl)
;
607     modal.show();
608   }
609
610   // Open Details Modal
611   async function openDetailsModal(userId) {
612     const modalEl = document.getElementById('modal-details-use
r');
613     const contentEl = document.getElementById('details-user-co
ntent');
614     const modal = bootstrap.Modal.getOrCreateInstance(modalEl)
;
615     modal.show();
616
617     contentEl.innerHTML = `
618       <div class="text-center py-4 text-muted">
619         <div class="spinner-border spinner-border-sm text-prim
```

```
ary me-2" role="status"></div>
        Загрузка профиля пользователя...
      </div>`;

    try {
      const data = await apiFetch(`/api/admin/users/${userId}`);
      const u = data.user || {};
      const s = data.settings || {};
      const perms = data.permissions || [];

      contentEl.innerHTML = `
        <div class="row g-3">
          <div class="col-md-6">
            <h6 class="text-primary border-bottom pb-1 mb-2">👤 Основные данные</h6>
            <table class="table table-sm table-borderless small mb-0">
              <tr><td class="text-muted" style="width:120px;">ID:</td><td class="fw-bold">${u.id}</td></tr>
              <tr><td class="text-muted">Email:</td><td>${escapeHtml(u.email)} ${u.is_email_verified ? '<span class="badge bg-success">Verified</span>' : '<span class="badge bg-secondary">Unverified</span>'}</td></tr>
              <tr><td class="text-muted">Имя:</td><td>${escapeHtml(u.name || '—')}</td></tr>
              <tr><td class="text-muted">Роль:</td><td><span class="badge bg-info text-dark">${escapeHtml(u.role || 'user')}</span> ${u.is_admin ? '<span class="badge bg-warning text-dark">Admin</span>' : ''}</td></tr>
              <tr><td class="text-muted">Статус:</td><td>${u.is_active ? '<span class="text-success">Активен</span>' : '<span class="text-danger">Заблокирован</span>'}</td></tr>
              <tr><td class="text-muted">Пароль:</td><td>${u.has_password ? '<span class="text-success">Установлен</span>' : '<span class="text-muted">Не задан</span>'}</td></tr>
              <tr><td class="text-muted">Создан:</td><td>${escapeHtml(u.created_at || '—')}</td></tr>
```

```
641               <tr><td class="text-muted">Последний вход:</td><
td>${escapeHtml(u.last_login || '—')}</td></tr>
642             </table>
643           </div>
644           <div class="col-md-6">
645             <h6 class="text-info border-bottom pb-1 mb-2">  Te
legram & Настройки</h6>
646             <table class="table table-sm table-borderless smal
l mb-0">
647               <tr><td class="text-muted" style="width:130px;">
Telegram ID:</td><td>${u.telegram_id ? escapeHtml(u.telegram_i
d) : '<span class="text-muted">Не привязан</span>'}</td></tr>
648               <tr><td class="text-muted">TG Username:</td><td>
${u.telegram_username ? `@${escapeHtml(u.telegram_username)}`
: '<span class="text-muted">—</span>'}</td></tr>
649               <tr><td class="text-muted">Тема UI:</td><td>${es
capeHtml(s.theme || 'dark')}</td></tr>
650               <tr><td class="text-muted">Язык:</td><td>${escap
eHtml(s.language || 'ru')}</td></tr>
651               <tr><td class="text-muted">TTS голос:</td><td>${
escapeHtml(s.tts_voice || 'ru-RU-DmitryNeural')} (${escapeHtml
(s.tts_system || 'edge-tts')})</td></tr>
652               <tr><td class="text-muted">Модель чата:</td><td>
${escapeHtml(s.model || 'По умолчанию')}</td></tr>
653             </table>
654           </div>
655           <div class="col-12 mt-3">
656             <h6 class="text-warning border-bottom pb-1 mb-2">
      Разрешения системы</h6>
657             <div class="d-flex flex-wrap gap-1">
658               ${perms.length > 0 ? perms.map(p => `<span class
="badge bg-secondary">${escapeHtml(p)}</span>`).join('') : '<s
pan class="text-muted small">Нет явных разрешений</span>'}
659             </div>
660           </div>
661           ${s.system_instruction ? `
662           <div class="col-12 mt-2">
663             <h6 class="border-bottom pb-1 mb-1">  Пользовател
```

```
ьская системная инструкция</h6>
            <pre class="bg-body-tertiary p-2 rounded border sm
all font-monospace" style="max-height:120px;overflow-y:auto;">
${escapeHtml(s.system_instruction)}</pre>
          </div>` : ''}
        </div>`;
    } catch (err) {
      contentEl.innerHTML = `<div class="text-danger py-3">Оши
бка загрузки данных: ${escapeHtml(err.message)}</div>`;
    }
  }

  // Open Delete Modal
  function openDeleteModal(userId, userName) {
    document.getElementById('delete-user-id').value = userId;
    document.getElementById('delete-user-name-target').textCon
tent = `${userName} (ID #${userId})`;

    const modalEl = document.getElementById('modal-delete-user
');
    const modal = bootstrap.Modal.getOrCreateInstance(modalEl)
;
    modal.show();
  }

  // Initialize Event Listeners
  function initListeners() {
    // Open Create User Modal
    const btnOpenCreate = document.getElementById('btn-open-cr
eate-user');
    if (btnOpenCreate) {
      btnOpenCreate.addEventListener('click', () => {
        const form = document.getElementById('form-create-user
');
        if (form) form.reset();
        document.getElementById('create-user-is-admin').checke
d = false;
        const createRoleLabel = document.getElementById('creat
```

```
e-role-label');
692         if (createRoleLabel) {
693           createRoleLabel.textContent = 'Пользователь (user)';
694           createRoleLabel.className = 'text-info';
695         }
696         document.getElementById('create-user-is-active').check
ed = true;
697         document.getElementById('create-user-verified').checke
d = true;
698         const modalEl = document.getElementById('modal-create-
user');
699         const modal = bootstrap.Modal.getOrCreateInstance(moda
lEl);
700         modal.show();
701       });
702     }
703
704     // Dynamic Switch Hints
705     const createAdminSw = document.getElementById('create-user
-is-admin');
706     const createRoleLabel = document.getElementById('create-ro
le-label');
707     if (createAdminSw && createRoleLabel) {
708       createAdminSw.addEventListener('change', (e) => {
709         createRoleLabel.textContent = e.target.checked ? 'Адми
нистратор (admin)' : 'Пользователь (user)';
710         createRoleLabel.className = e.target.checked ? 'text-w
arning' : 'text-info';
711       });
712     }
713
714     const editAdminSw = document.getElementById('edit-user-is-
admin');
715     const editRoleLabel = document.getElementById('edit-role-l
abel');
716     if (editAdminSw && editRoleLabel) {
717       editAdminSw.addEventListener('change', (e) => {
718         editRoleLabel.textContent = e.target.checked ? 'Админи
```

```
стратор (admin)' : 'Пользователь (user)';
        editRoleLabel.className = e.target.checked ? 'text-warning' : 'text-info';
      });
    }

    // Refresh Users Button
    const btnRefresh = document.getElementById('btn-refresh-users');
    if (btnRefresh) {
      btnRefresh.addEventListener('click', () => {
        loadUsers();
      });
    }

    // Search Input with Debounce
    const searchInput = document.getElementById('users-search-input');
    const searchClear = document.getElementById('users-search-clear');
    let debounceTimeout = null;

    if (searchInput) {
      searchInput.addEventListener('input', (e) => {
        const val = e.target.value;
        if (searchClear) {
          searchClear.classList.toggle('d-none', !val);
        }
        clearTimeout(debounceTimeout);
        debounceTimeout = setTimeout(() => {
          state.searchQuery = val.trim();
          loadUsers();
        }, 300);
      });
    }

    if (searchClear) {
      searchClear.addEventListener('click', () => {
```

```
        if (searchInput) searchInput.value = '';
        searchClear.classList.add('d-none');
        state.searchQuery = '';
        loadUsers();
      });
    }

    // Role Filter
    const filterRole = document.getElementById('users-filter-role');
    if (filterRole) {
      filterRole.addEventListener('change', (e) => {
        state.filterRole = e.target.value;
        loadUsers();
      });
    }

    // Status Filter
    const filterStatus = document.getElementById('users-filter-status');
    if (filterStatus) {
      filterStatus.addEventListener('change', (e) => {
        state.filterStatus = e.target.value;
        loadUsers();
      });
    }

    // Generator button for create modal
    const btnGenCreatePwd = document.getElementById('btn-gen-create-password');
    if (btnGenCreatePwd) {
      btnGenCreatePwd.addEventListener('click', () => {
        const pwdInput = document.getElementById('create-user-password');
        if (pwdInput) pwdInput.value = generatePassword(12);
      });
    }

```

```
    // Generator button for password change modal
    const btnGenPwdVal = document.getElementById('btn-gen-pwd-val');
    if (btnGenPwdVal) {
      btnGenPwdVal.addEventListener('click', () => {
        const pwdInput = document.getElementById('pwd-input-val');
        if (pwdInput) pwdInput.value = generatePassword(12);
      });
    }

    // Create User Form Submit
    const formCreate = document.getElementById('form-create-user');
    if (formCreate) {
      formCreate.addEventListener('submit', async (e) => {
        e.preventDefault();
        const submitBtn = document.getElementById('btn-submit-create-user');
        if (submitBtn) submitBtn.disabled = true;

        const isAdmin = document.getElementById('create-user-is-admin').checked ? 1 : 0;
        const payload = {
          email: document.getElementById('create-user-email').value.trim(),
          name: document.getElementById('create-user-name').value.trim(),
          password: document.getElementById('create-user-password').value.trim(),
          role: isAdmin ? 'admin' : 'user',
          is_admin: isAdmin,
          is_active: document.getElementById('create-user-is-active').checked ? 1 : 0,
          is_email_verified: document.getElementById('create-user-verified').checked ? 1 : 0
        };

```

```
        try {
          await apiFetch('/api/admin/users', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify(payload)
          });
          const modalEl = document.getElementById('modal-creat
e-user');
          const modal = bootstrap.Modal.getInstance(modalEl);
          if (modal) modal.hide();

          showStatusAlert(`Пользователь ${escapeHtml(payload.n
ame)} успешно создан!`, 'success');
          loadUsers();
        } catch (err) {
          showStatusAlert(`Ошибка создания пользователя: ${err
.message}`, 'danger');
        } finally {
          if (submitBtn) submitBtn.disabled = false;
        }
      });
    }

    // Edit User Form Submit (Modal)
    const formEdit = document.getElementById('form-edit-user')
;
    if (formEdit) {
      formEdit.addEventListener('submit', async (e) => {
        e.preventDefault();
        const submitBtn = document.getElementById('btn-submit-
edit-user');
        if (submitBtn) submitBtn.disabled = true;

        const userId = parseInt(document.getElementById('edit-
user-id').value, 10);
        const isAdmin = document.getElementById('edit-user-is-
admin').checked ? 1 : 0;
        const payload = {
```

```
          email: document.getElementById('edit-user-email').value.trim(),
          name: document.getElementById('edit-user-name').value.trim(),
          role: isAdmin ? 'admin' : 'user',
          is_admin: isAdmin,
          is_active: document.getElementById('edit-user-is-active').checked ? 1 : 0,
          is_email_verified: document.getElementById('edit-user-verified').checked ? 1 : 0
        };

        try {
          await apiFetch(`/api/admin/users/${userId}`, {
            method: 'PUT',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify(payload)
          });
          const modalEl = document.getElementById('modal-edit-user');
          const modal = bootstrap.Modal.getInstance(modalEl);
          if (modal) modal.hide();

          showStatusAlert(`Данные пользователя #${userId} успешно обновлены!`, 'success');
          loadUsers();
        } catch (err) {
          showStatusAlert(`Ошибка обновления: ${err.message}`, 'danger');
        } finally {
          if (submitBtn) submitBtn.disabled = false;
        }
      });
    }

    // Change Password Form Submit
    const formPwd = document.getElementById('form-password-user');
```

```
    if (formPwd) {
      formPwd.addEventListener('submit', async (e) => {
        e.preventDefault();
        const submitBtn = document.getElementById('btn-submit-pwd-user');
        if (submitBtn) submitBtn.disabled = true;

        const userId = parseInt(document.getElementById('pwd-user-id').value, 10);
        const newPassword = document.getElementById('pwd-input-val').value.trim();

        try {
          await apiFetch(`/api/admin/users/${userId}/password`, {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ password: newPassword })
          });
          const modalEl = document.getElementById('modal-password-user');
          const modal = bootstrap.Modal.getInstance(modalEl);
          if (modal) modal.hide();

          showStatusAlert(`Пароль для пользователя #${userId} успешно установлен!`, 'success');
          loadUsers();
        } catch (err) {
          showStatusAlert(`Ошибка установки пароля: ${err.message}`, 'danger');
        } finally {
          if (submitBtn) submitBtn.disabled = false;
        }
      });
    }

    // Delete User Confirm Button
    const btnConfirmDelete = document.getElementById('btn-conf
```

```
irm-delete-user');
    if (btnConfirmDelete) {
      btnConfirmDelete.addEventListener('click', async () => {
        btnConfirmDelete.disabled = true;
        const userId = parseInt(document.getElementById('delet
e-user-id').value, 10);

        try {
          await apiFetch(`/api/admin/users/${userId}`, { metho
d: 'DELETE' });
          const modalEl = document.getElementById('modal-delet
e-user');
          const modal = bootstrap.Modal.getInstance(modalEl);
          if (modal) modal.hide();

          showStatusAlert(`Пользователь #${userId} успешно уда
лён.`, 'success');
          loadUsers();
        } catch (err) {
          showStatusAlert(`Ошибка удаления пользователя: ${err
.message}`, 'danger');
        } finally {
          btnConfirmDelete.disabled = false;
        }
      });
    }

    // Orphaned Directories Check Button
    const btnCheckOrphaned = document.getElementById('btn-chec
k-orphaned-dirs');
    if (btnCheckOrphaned) {
      btnCheckOrphaned.addEventListener('click', () => {
        const modalEl = document.getElementById('modal-orphane
d-dirs');
        if (modalEl) {
          const modal = new bootstrap.Modal(modalEl);
          modal.show();
          loadOrphanedDirs();
```

```
        }
      });
    }

    // Orphaned Directories Refresh Button
    const btnRefreshOrphaned = document.getElementById('btn-refresh-orphaned-dirs');
    if (btnRefreshOrphaned) {
      btnRefreshOrphaned.addEventListener('click', () => {
        loadOrphanedDirs();
      });
    }

    // Cleanup All Orphaned Directories Button
    const btnCleanupAll = document.getElementById('btn-cleanup-all-orphaned');
    if (btnCleanupAll) {
      btnCleanupAll.addEventListener('click', async () => {
        if (!confirm('Вы уверены, что хотите удалить ВСЕ обнаруженные мертвые директории пользователей? Это действие необратимо.')) {
          return;
        }
        btnCleanupAll.disabled = true;
        try {
          const res = await apiFetch('/api/admin/users/orphaned-dirs/clean', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({})
          });
          showStatusAlert(`Успешно удалено мертвых директорий: ${res.total_deleted} (освобождено: ${formatBytes(res.freed_bytes)})`, 'success');
          loadOrphanedDirs();
        } catch (err) {
          showStatusAlert(`Ошибка очистки мертвых директорий: ${err.message}`, 'danger');
```

```
        } finally {
          btnCleanupAll.disabled = false;
        }
      });
    }
  }

  // Format bytes helper
  function formatBytes(bytes, decimals = 1) {
    if (!bytes || bytes <= 0) return '0 B';
    const k = 1024;
    const dm = decimals < 0 ? 0 : decimals;
    const sizes = ['B', 'KB', 'MB', 'GB', 'TB'];
    const i = Math.floor(Math.log(bytes) / Math.log(k));
    return parseFloat((bytes / Math.pow(k, i)).toFixed(dm)) +
' ' + sizes[i];
  }

  // Load Orphaned Directories from Backend
  async function loadOrphanedDirs() {
    const tbody = document.getElementById('orphaned-dirs-tbody
');
    const countEl = document.getElementById('orphaned-dirs-cou
nt');
    const sizeEl = document.getElementById('orphaned-dirs-size
');
    const filesEl = document.getElementById('orphaned-dirs-fil
es');
    const btnCleanupAll = document.getElementById('btn-cleanup
-all-orphaned');

    if (tbody) {
      tbody.innerHTML = `
        <tr>
          <td colspan="6" class="text-center py-4 text-muted">
            <div class="spinner-border spinner-border-sm text-
warning me-2" role="status"></div>
            Сканирование хранилища data/users/...
```

```
          </td>
        </tr>`;
    }

    try {
      const data = await apiFetch('/api/admin/users/orphaned-dirs');
      const dirs = data.orphaned_dirs || [];

      if (countEl) countEl.textContent = data.total || 0;
      if (sizeEl) sizeEl.textContent = formatBytes(data.total_size_bytes || 0);
      if (filesEl) filesEl.textContent = data.total_files || 0;

      if (btnCleanupAll) {
        btnCleanupAll.disabled = (dirs.length === 0);
      }

      if (!tbody) return;

      if (dirs.length === 0) {
        tbody.innerHTML = `
          <tr>
            <td colspan="6" class="text-center py-4 text-success">
              <i class="bi bi-check-circle-fill fs-4 d-block mb-1"></i>
              Мертвых директорий не обнаружено. Все папки соответствуют активным пользователям.
            </td>
          </tr>`;
        return;
      }

      tbody.innerHTML = dirs.map(d => {
        const badgeClass = d.is_test_dir ? 'bg-secondary' : 'bg-warning text-dark';
```

```
        return `
          <tr data-dir-name="${escapeHtml(d.name)}">
            <td class="font-monospace fw-bold text-light">
              <i class="bi bi-folder text-warning me-1"></i>${escapeHtml(d.name)}
            </td>
            <td>
              <span class="badge ${badgeClass} small">${escapeHtml(d.reason)}</span>
            </td>
            <td class="text-center">${d.files_count}</td>
            <td class="text-center">${formatBytes(d.size_bytes)}</td>
            <td class="small">${formatDate(d.modified_at)}</td>
            <td class="text-center">
              <button class="btn btn-outline-danger btn-sm py-0 px-2 btn-delete-single-orphaned" data-dir="${escapeHtml(d.name)}" title="Удалить директорию">
                <i class="bi bi-trash"></i>
              </button>
            </td>
          </tr>`;
      }).join('');

      // Add click listeners to single delete buttons
      tbody.querySelectorAll('.btn-delete-single-orphaned').forEach(btn => {
        btn.addEventListener('click', async () => {
          const dirName = btn.getAttribute('data-dir');
          if (!dirName) return;
          if (!confirm(`Удалить мертвую директорию "${dirName}"?`)) return;

          btn.disabled = true;
          try {
            const res = await apiFetch('/api/admin/users/orphaned-dirs/clean', {
```

```
              method: 'POST',
              headers: { 'Content-Type': 'application/json' },
              body: JSON.stringify({ dirs: [dirName] })
            });
            showStatusAlert(`Директория "${dirName}" успешно у
далена.`, 'success');
            loadOrphanedDirs();
          } catch (err) {
            showStatusAlert(`Ошибка удаления директории ${dirN
ame}: ${err.message}`, 'danger');
            btn.disabled = false;
          }
        });
      });
    } catch (err) {
      console.error('[UsersTab] Error loading orphaned dirs:',
 err);
      if (tbody) {
        tbody.innerHTML = `
          <tr>
            <td colspan="6" class="text-center py-4 text-dange
r">
              <i class="bi bi-exclamation-octagon fs-4 d-block
 mb-1"></i>
              Ошибка сканирования: ${escapeHtml(err.message)}
            </td>
          </tr>`;
      }
    }
  }

  // Global Init Function
  function initUsersTab() {
    console.log('[UsersTab] Initializing user management tab w
ith inline editing...');
    if (!state.initialized) {
      initListeners();
      state.initialized = true;
```

```
    }
    loadUsers();
  }

  // Register globally
  window.initUsersTab = initUsersTab;
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/user_assistant_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/user_assistant_tab/inde
x.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/user_assistant_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!-- User Assistant Application Tab -->
<style>
  .assistant-container {
    padding: 1rem;
  }
  .assistant-card {
    background: var(--surface-1) !important;
    border: 1.5px solid var(--border-color) !important;
    border-radius: var(--radius-md, 8px);
```

```
    padding: var(--kpi-padding, 0.75rem 1rem);
    display: flex;
    flex-direction: column;
    gap: 0.25rem;
    box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.08)
);
  }
  .assistant-label {
    font-size: var(--kpi-label-size, 0.72rem);
    text-transform: uppercase;
    letter-spacing: 0.05em;
    font-weight: 600;
    color: var(--text-muted);
  }
  .assistant-value {
    font-size: var(--kpi-val-size, 1.35rem);
    font-weight: 700;
    color: var(--text-color);
    font-family: var(--font-mono, monospace);
  }
  .assistant-panel {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: var(--radius-md, 8px);
    overflow: hidden;
    box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.08)
);
  }
  .assistant-panel-header {
    background: var(--surface-2) !important;
    border-bottom: 1px solid var(--border-color) !important;
    padding: 0.6rem 0.9rem;
    font-weight: 600;
    font-size: 0.85rem;
    color: var(--text-color);
    display: flex;
    align-items: center;
    justify-content: space-between;
```

```
  }
  .assistant-list-item {
    padding: 0.75rem 1rem;
    border-bottom: 1px solid var(--border-subtle, rgba(128, 128, 128, 0.2));
    transition: background 0.15s ease;
    color: var(--text-color);
  }
  .assistant-list-item:hover {
    background: rgba(56, 189, 248, 0.06);
  }
</style>
<div class="assistant-container">
<!-- Header Alert & Status -->
<div class="card shadow-sm border-secondary-subtle p-3 mb-3 d-flex justify-content-between align-items-center flex-wrap gap-3" data-panel-id="panel-user-assistant-user-assistant-agenda-mail-cal" id="panel-user-assistant-user-assistant-agenda-mail-cal">
<div>
<h5 class="mb-1 fw-bold d-flex align-items-center gap-2">
<span class="fs-4">□ </span> User Assistant (Agenda, Mail &amp; Calendar)
      </h5>
<p class="mb-0 small text-muted">
        Персональный ассистент продуктивности: расписание дня, сортировка входящих писем, события календаря и поиск документов.
      </p>
</div>
<div class="d-flex align-items-center gap-2">
<span class="badge rounded-pill bg-success-subtle text-success border border-success px-3 py-2" id="assistant-status-badge">
        ● Готов к работе
      </span>
<button class="btn btn-sm btn-outline-info rounded-pill px-3" id="btn-assistant-refresh">
<i class="bi bi-arrow-clockwise me-1"></i> Обновить
```

```
      </button>
<button class="btn btn-sm btn-outline-warning rounded-pill px-3" id="btn-assistant-config">
<i class="bi bi-gear-fill me-1"></i> Настройки (config.json)
      </button>
</div>
</div>
<!-- Top Metrics Bar -->
<div class="row g-3 mb-3">
<div class="col-md-3 col-sm-6">
<div class="assistant-card">
<span class="assistant-label">Событий на сегодня</span>
<span class="assistant-value text-info" id="assistant-events-count">0</span>
</div>
</div>
<div class="col-md-3 col-sm-6">
<div class="assistant-card">
<span class="assistant-label">Непрочитанных писем</span>
<span class="assistant-value text-warning" id="assistant-unread-count">0</span>
</div>
</div>
<div class="col-md-3 col-sm-6">
<div class="assistant-card">
<span class="assistant-label">Недавних файлов</span>
<span class="assistant-value text-success" id="assistant-files-count">0</span>
</div>
</div>
<div class="col-md-3 col-sm-6">
<div class="assistant-card">
<span class="assistant-label">Рабочее пространство</span>
<span class="assistant-value text-body" id="assistant-workspace-status" style="font-size: 1.05rem;">Активно</span>
</div>
</div>
</div>
```

```
<!-- Main Content Panels Grid -->
<div class="row g-3">
<!-- Calendar & Agenda Panel -->
<div class="col-lg-6">
<div class="assistant-panel h-100">
<div class="assistant-panel-header">
<span>  Повестка дня и ближайшие события</span>
<span class="badge bg-secondary" id="assistant-events-badge">0
 событий</span>
</div>
<div class="p-0" id="assistant-events-list" style="max-height:
 450px; overflow-y: auto;">
<div class="p-3 text-muted text-center small">Загрузка повестк
и дня...</div>
</div>
</div>
</div>
<!-- Email Triage Panel -->
<div class="col-lg-6">
<div class="assistant-panel h-100">
<div class="assistant-panel-header">
<span>  Непрочитанные сообщения</span>
<span class="badge bg-secondary" id="assistant-mail-badge">0 п
исем</span>
</div>
<div class="p-0" id="assistant-mail-list" style="max-height: 4
50px; overflow-y: auto;">
<div class="p-3 text-muted text-center small">Загрузка почты..
.</div>
</div>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows User Assistant Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/user_assistant_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/user_assistant_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// ============================================================
// Process Name: User Assistant Web Tab Module
// ============================================================
// Description:
//   Client-side JavaScript controller for the User Assistant
//   administrative web interface tab.
//
// File: main.js
// Project: ai-windows-admin
// Package: src.api.webinterface.user_assistant_tab
```

```
// Author: hypo69
// Copyright: © 2026 hypo69
// ============================================================
==================

(function() {
  let isAssistantInitialized = false;

  async function fetchAgenda() {
    try {
      const res = await fetch('/api/v1/assistant/agenda');
      if (!res.ok) return;
      const data = await res.json();

      const eventsCountEl = document.getElementById('assistant
-events-count');
      const unreadCountEl = document.getElementById('assistant
-unread-count');
      const filesCountEl = document.getElementById('assistant-
files-count');
      const eventsBadge = document.getElementById('assistant-e
vents-badge');
      const mailBadge = document.getElementById('assistant-mai
l-badge');
      const eventsList = document.getElementById('assistant-ev
ents-list');
      const mailList = document.getElementById('assistant-mail
-list');

      const events = data.events || [];
      const emails = data.unread_emails || [];
      const files = data.recent_files || [];

      if (eventsCountEl) eventsCountEl.innerText = events.leng
th;
      if (unreadCountEl) unreadCountEl.innerText = emails.leng
th;
      if (filesCountEl) filesCountEl.innerText = files.length;
```

```
      if (eventsBadge) eventsBadge.innerText = `${events.length} событий`;
      if (mailBadge) mailBadge.innerText = `${emails.length} писем`;

      if (eventsList) {
        if (events.length === 0) {
          eventsList.innerHTML = '<div class="p-3 text-muted text-center small">Нет запланированных событий на сегодня.</div>';
        } else {
          eventsList.innerHTML = events.map(e => `
            <div class="assistant-list-item d-flex justify-content-between align-items-center">
              <div>
                <div class="fw-semibold text-light">${e.summary || 'Без названия'}</div>
                <div class="text-secondary small">  ${e.start || 'Весь день'} ${e.location ? '  ' + e.location : ''}</div>
              </div>
            </div>
          `).join('');
        }
      }

      if (mailList) {
        if (emails.length === 0) {
          mailList.innerHTML = '<div class="p-3 text-muted text-center small">Нет новых непрочитанных сообщений.</div>';
        } else {
          mailList.innerHTML = emails.map(m => `
            <div class="assistant-list-item d-flex justify-content-between align-items-center">
              <div>
                <div class="fw-semibold text-light">${m.subject || 'Без темы'}</div>
                <div class="text-secondary small">  ${m.from || 'Неизвестный'} &bull; ${m.snippet || ''}</div>
```

```
              </div>
            </div>
          `).join('');
        }
      }
    } catch (e) {
      console.error('[UserAssistantTab] Failed to fetch agenda:', e);
    }
  }

  function initUserAssistantTab() {
    fetchAgenda();
    if (!isAssistantInitialized) {
      const refreshBtn = document.getElementById('btn-assistant-refresh');
      if (refreshBtn) refreshBtn.addEventListener('click', fetchAgenda);

      const configBtn = document.getElementById('btn-assistant-config');
      if (configBtn) {
        configBtn.addEventListener('click', () => {
          if (typeof window.openAppConfigModal === 'function') {
            window.openAppConfigModal('user_assistant');
          }
        });
      }
      isAssistantInitialized = true;
    }
  }

  window.initUserAssistantTab = initUserAssistantTab;
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/user_directories_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/user_directories_tab/in
dex.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/user_directories_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- User Directories and Workspace Storage Tab -->
<div class="container-fluid py-2" id="user-directories-tab-roo
t">
<!-- Top Banner & Quick Controls -->
<div class="d-flex flex-wrap align-items-center justify-conten
t-between gap-2 mb-3 p-3 bg-body-tertiary border rounded shado
w-sm">
<div>
<h5 class="mb-1 text-primary d-flex align-items-center gap-2">
<i class="bi bi-folder2-open text-primary"></i>
```

```
<span>Пользовательские директории и хранилище</span>
</h5>
<p class="mb-0 text-muted small">Мониторинг файловых хранилищ пользователей (<code>data/users/&lt;user_id&gt;/</code>), анал из дисковых квот, аудит деревьев файлов и очистка осиротевших каталогов.</p>
</div>
<div class="d-flex flex-wrap gap-2 align-items-center">
<button class="btn btn-outline-warning btn-sm d-flex align-ite ms-center gap-1.5 rounded-pill px-3" id="btn-open-orphaned-dir s-modal" title="Проверить и удалить неиспользуемые папки">
<i class="bi bi-folder-x"></i> Мертвые директории (<span id="b adge-orphaned-count">0</span>)
      </button>
<button class="btn btn-primary btn-sm d-flex align-items-cente r gap-1.5 rounded-pill px-3" id="btn-refresh-user-dirs" title= "Обновить данные хранилища">
<i class="bi bi-arrow-clockwise"></i> Обновить
      </button>
</div>
</div>
<!-- Metric Cards -->
<div class="row g-3 mb-3">
<div class="col-6 col-md-3">
<div class="card bg-body-tertiary border h-100 shadow-sm" data -panel-id="panel-user-directories-0" id="panel-user-directorie s-0">
<div class="card-body p-3 d-flex align-items-center justify-co ntent-between">
<div>
<div class="text-muted small">Пользователей с хранилищем</div>
<h4 class="mb-0 fw-bold text-primary" id="metric-total-users"> 0</h4>
</div>
<div class="bg-primary bg-opacity-25 rounded p-2 text-primary" >
<i class="bi bi-people-fill fs-4"></i>
</div>
```

```
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-body-tertiary border h-100 shadow-sm" data-panel-id="panel-user-directories-0-b" id="panel-user-directories-0-b">
<div class="card-body p-3 d-flex align-items-center justify-content-between">
<div>
<div class="text-muted small">Занято на диске</div>
<h4 class="mb-0 fw-bold text-info" id="metric-total-storage">0 B</h4>
</div>
<div class="bg-info bg-opacity-25 rounded p-2 text-info">
<i class="bi bi-hdd-fill fs-4"></i>
</div>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="card bg-body-tertiary border h-100 shadow-sm" data-panel-id="panel-user-directories-0" id="panel-user-directories-0">
<div class="card-body p-3 d-flex align-items-center justify-content-between">
<div>
<div class="text-muted small">Всего файлов</div>
<h4 class="mb-0 fw-bold text-success" id="metric-total-files">0</h4>
</div>
<div class="bg-success bg-opacity-25 rounded p-2 text-success">
<i class="bi bi-file-earmark-text-fill fs-4"></i>
</div>
</div>
</div>
</div>
```

```
<div class="col-6 col-md-3">
<div class="card bg-body-tertiary border h-100 shadow-sm" data-panel-id="panel-user-directories-0-b" id="panel-user-directories-0-b">
<div class="card-body p-3 d-flex align-items-center justify-content-between">
<div>
<div class="text-muted small">Осиротевшие каталоги</div>
<h4 class="mb-0 fw-bold text-warning" id="metric-orphaned-size">0 B</h4>
</div>
<div class="bg-warning bg-opacity-25 rounded p-2 text-warning">
<i class="bi bi-trash3-fill fs-4"></i>
</div>
</div>
</div>
</div>
</div>
<!-- Search & Filter Controls -->
<div class="card bg-body-tertiary border mb-3 shadow-sm" data-panel-id="panel-user-directories-1" id="panel-user-directories-1">
<div class="card-body p-2.5">
<div class="row g-2 align-items-center">
<div class="col-12 col-md-5">
<div class="input-group input-group-sm">
<span class="input-group-text"><i class="bi bi-search"></i></span>
<input class="form-control" id="user-dirs-search-input" placeholder="Поиск по имени пользователя, email или имени файла..." type="text"/>
<button class="btn btn-outline-secondary d-none" id="user-dirs-search-clear" type="button">□ </button>
</div>
</div>
<div class="col-6 col-md-3">
<select class="form-select form-select-sm" id="user-dirs-ext-f
```

```
ilter">
<option value="">Все типы файлов</option>
<option value="pdf">Документы PDF (.pdf)</option>
<option value="txt">Текстовые (.txt)</option>
<option value="md">Markdown (.md)</option>
<option value="json">Данные JSON (.json)</option>
<option value="csv">Таблицы CSV (.csv)</option>
<option value="png">Изображения (.png, .jpg)</option>
<option value="mp3">Аудио (.mp3, .wav)</option>
</select>
</div>
<div class="col-6 col-md-4 text-end text-muted small" id="user
-dirs-status-label">
          Загрузка...
        </div>
</div>
</div>
</div>
<!-- Alert Message Box -->
<div class="d-none alert alert-dismissible fade show" id="user
-dirs-alert-box" role="alert">
<span id="user-dirs-alert-message"></span>
<button aria-label="Close" class="btn-close" onclick="document
.getElementById('user-dirs-alert-box').classList.add('d-none')
" type="button"></button>
</div>
<!-- Two-Panel Split Layout (Users List & Files Explorer) -->
<div class="row g-3">
<!-- Left Column: Users List & Storage Quotas -->
<div class="col-12 col-lg-4">
<div class="card border shadow-sm h-100" data-panel-id="panel-
user-directories-polzovateli" id="panel-user-directories-polzo
vateli">
<div class="card-header bg-body-tertiary d-flex align-items-ce
nter justify-content-between py-2 px-3">
<span class="fw-semibold small"><i class="bi bi-people me-1 te
xt-primary"></i> Пользователи</span>
<span class="badge bg-secondary rounded-pill" id="users-badge-
```

```
count">0</span>
</div>
<div class="card-body p-0" style="max-height: 540px; overflow-y: auto;">
<div class="list-group list-group-flush" id="user-dirs-users-list">
<div class="text-center py-4 text-muted small">
<div class="spinner-border spinner-border-sm text-primary me-2" role="status"></div>
              Загрузка каталогов пользователей...
            </div>
</div>
</div>
</div>
</div>
<!-- Right Column: Selected User File Tree & Inspector -->
<div class="col-12 col-lg-8">
<div class="card border shadow-sm h-100" data-panel-id="panel-user-directories-vyberite-polzovatelya-sleva" id="panel-user-directories-vyberite-polzovatelya-sleva">
<div class="card-header bg-body-tertiary d-flex align-items-center justify-content-between py-2 px-3 flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<span class="fw-semibold text-truncate small" id="explorer-user-title">
<i class="bi bi-folder-fill text-warning me-1"></i> Выберите пользователя слева
            </span>
<span class="badge bg-info-subtle text-info border border-info-subtle d-none" id="explorer-user-path-badge"></span>
</div>
<!-- Subfolder Filters Buttons -->
<div class="btn-group btn-group-sm" id="subfolder-filter-group" role="group">
<button class="btn btn-outline-primary active" data-subfolder="" type="button">Все папки</button>
<button class="btn btn-outline-primary" data-subfolder="files" type="button">  files</button>
```

```
<button class="btn btn-outline-primary" data-subfolder="rag" type="button">  rag</button>
<button class="btn btn-outline-primary" data-subfolder="profile" type="button">  profile</button>
<button class="btn btn-outline-primary" data-subfolder="temp" type="button">  temp</button>
</div>
</div>
<!-- Files Table -->
<div class="card-body p-0" style="max-height: 480px; overflow-y: auto;">
<div class="table-responsive">
<table class="table table-hover table-sm align-middle mb-0" style="font-size: 0.84rem;">
<thead class="table-light small text-uppercase sticky-top">
<tr>
<th style="width: 35px;"></th>
<th>Имя файла / Путь</th>
<th class="text-center" style="width: 90px;">Раздел</th>
<th class="text-center" style="width: 95px;">Размер</th>
<th style="width: 140px;">Изменен</th>
<th class="text-center" style="width: 110px;">Действия</th>
</tr>
</thead>
<tbody id="user-files-table-body">
<tr>
<td class="text-center py-5 text-muted" colspan="6">
<i class="bi bi-arrow-left-circle fs-3 d-block mb-2 text-secondary"></i>
                    Выберите пользователя из списка слева для просмотра файлов и директорий
                  </td>
</tr>
</tbody>
</table>
</div>
</div>
<!-- Explorer Footer Stats -->
```

```
<div class="card-footer bg-body-tertiary d-flex align-items-center justify-content-between py-2 px-3 small text-muted">
<div id="explorer-footer-stats">
            Файлов: <strong>0</strong> | Общий размер: <strong>0 B</strong>
</div>
<div id="explorer-footer-quota">
            Квота: <strong>-</strong>
</div>
</div>
</div>
</div>
</div>
</div>
<!-- ========================================================== -->
<!-- MODAL: File Preview -->
<!-- ========================================================== -->
<div aria-hidden="true" aria-labelledby="modal-file-preview-title" class="modal fade" id="modal-file-preview" tabindex="-1">
<div class="modal-dialog modal-dialog-centered modal-lg">
<div class="modal-content shadow-lg border-primary">
<div class="modal-header border-primary py-2.5">
<h6 class="modal-title d-flex align-items-center gap-2 text-truncate" id="modal-file-preview-title">
<i class="bi bi-file-earmark-text text-primary"></i> Предпросмотр файла
        </h6>
<button aria-label="Close" class="btn-close" data-bs-dismiss="modal" type="button"></button>
</div>
<div class="modal-body p-3">
<!-- File Info Bar -->
<div class="d-flex align-items-center justify-content-between p-2 mb-2 bg-body-tertiary border rounded small">
<div>
<span class="text-muted">Путь:</span> <code id="preview-file-p
```

```
ath"></code>
</div>
<div>
<span class="badge bg-secondary me-1" id="preview-file-size"><
/span>
<span class="badge bg-dark border" id="preview-file-type"></sp
an>
</div>
</div>
<!-- File Content Container -->
<div class="bg-dark text-white rounded p-3 font-monospace" id=
"preview-content-wrapper" style="max-height: 440px; overflow-y
: auto; font-size: 0.82rem; white-space: pre-wrap; word-break:
 break-all;">
<div class="text-center py-4 text-muted">
<div class="spinner-border spinner-border-sm text-primary me-2
"></div> Загрузка содержимого...
          </div>
</div>
</div>
<div class="modal-footer justify-content-between py-2">
<button class="btn btn-outline-danger btn-sm rounded-pill px-3
" id="btn-preview-delete-file" type="button">
<i class="bi bi-trash3 me-1"></i> Удалить файл
        </button>
<div class="d-flex gap-2">
<button class="btn btn-outline-secondary btn-sm rounded-pill p
x-3" data-bs-dismiss="modal" type="button">Закрыть</button>
<a class="btn btn-primary btn-sm rounded-pill px-3" download="
" href="#" id="btn-preview-download-file" target="_blank">
<i class="bi bi-download me-1"></i> Скачать
          </a>
</div>
</div>
</div>
</div>
</div>
<!-- ============================================================
```

```
================ -->
<!-- MODAL: Orphaned User Directories Management -->
<!-- ============================================================
================ -->
<div aria-hidden="true" aria-labelledby="modal-orphaned-user-d
irs-title" class="modal fade" id="modal-orphaned-user-dirs" ta
bindex="-1">
<div class="modal-dialog modal-dialog-centered modal-lg">
<div class="modal-content shadow-lg border-warning">
<div class="modal-header border-warning py-2.5">
<h6 class="modal-title text-warning d-flex align-items-center
gap-2" id="modal-orphaned-user-dirs-title">
<i class="bi bi-folder-x"></i> Осиротевшие (мертвые) директори
и пользователей
        </h6>
<button aria-label="Close" class="btn-close" data-bs-dismiss="
modal" type="button"></button>
</div>
<div class="modal-body p-3">
<div class="alert alert-info border-0 small mb-3">
<i class="bi bi-info-circle text-info me-1"></i>
          Директории в папке <code>data/users/</code>, которые
 не принадлежат ни одной активной учетной записи в базе данных
.
        </div>
<div class="d-flex align-items-center justify-content-between
p-2 mb-3 bg-body-tertiary border rounded small">
<div>
            Найдено папок: <strong class="text-warning" id="mo
dal-orphaned-count">0</strong> |
            Занято места: <strong class="text-info" id="modal-
orphaned-size">0 B</strong> |
            Файлов: <strong id="modal-orphaned-files">0</stron
g>
</div>
<button class="btn btn-outline-secondary btn-xs rounded-pill p
x-2 py-0.5" id="btn-modal-refresh-orphaned">
<i class="bi bi-arrow-clockwise"></i> Обновить
```

```
          </button>
</div>
<div class="table-responsive" style="max-height: 360px; overfl
ow-y: auto;">
<table class="table table-hover table-sm align-middle mb-0" st
yle="font-size: 0.84rem;">
<thead class="table-light small text-uppercase sticky-top">
<tr>
<th>Каталог / ID</th>
<th>Причина</th>
<th class="text-center" style="width: 80px;">Файлов</th>
<th class="text-center" style="width: 95px;">Размер</th>
<th style="width: 140px;">Изменен</th>
<th class="text-center" style="width: 85px;">Действие</th>
</tr>
</thead>
<tbody id="modal-orphaned-dirs-tbody">
<tr>
<td class="text-center py-4 text-muted" colspan="6">
<div class="spinner-border spinner-border-sm text-warning me-2
"></div> Сканирование...
                </td>
</tr>
</tbody>
</table>
</div>
</div>
<div class="modal-footer justify-content-between py-2">
<button class="btn btn-secondary btn-sm rounded-pill px-3" dat
a-bs-dismiss="modal" type="button">Закрыть</button>
<button class="btn btn-danger btn-sm rounded-pill px-3 d-flex
align-items-center gap-1.5" id="btn-modal-clean-all-orphaned"
type="button">
<i class="bi bi-trash3-fill"></i> Удалить все мертвые директор
ии
        </button>
</div>
</div>
```

```
</div>
</div>

```

main.js

```
/**
 * ================================================================
 * Process Name: Windows User Directories Tab - Main Script
 * ================================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/user_directories_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/user_directories_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ================================================================
 */

/**
 * user_directories_tab/main.js — Модуль управления пользовательскими директориями и хранилищем
 * AI Windows Admin
 */

let state = {
  users: [],
  selectedUserId: null,
  selectedUserName: '',
  currentSubfolder: '',
  searchQuery: '',
```

```
  extFilter: '',
  files: [],
  previewFile: null,
  orphanedDirs: [],
};

/**
 * Инициализация вкладки пользовательских директорий
 */
export async function initUserDirectoriesTab() {
  const root = document.getElementById('user-directories-tab-root');
  if (!root) return;

  setupEventListeners();
  await refreshData();
}

/**
 * Настройка обработчиков событий элементов управления
 */
function setupEventListeners() {
  // Кнопка обновления
  const refreshBtn = document.getElementById('btn-refresh-user-dirs');
  if (refreshBtn) {
    refreshBtn.onclick = () => refreshData();
  }

  // Поиск с дебаунсом
  const searchInput = document.getElementById('user-dirs-search-input');
  const searchClear = document.getElementById('user-dirs-search-clear');
  if (searchInput) {
    let timeout = null;
    searchInput.oninput = () => {
      clearTimeout(timeout);
```

```
      state.searchQuery = searchInput.value.trim();
      if (searchClear) {
        searchClear.classList.toggle('d-none', !state.searchQu
ery);
      }
      timeout = setTimeout(() => {
        applyFilters();
      }, 300);
    };
  }

  if (searchClear && searchInput) {
    searchClear.onclick = () => {
      searchInput.value = '';
      state.searchQuery = '';
      searchClear.classList.add('d-none');
      applyFilters();
    };
  }

  // Фильтр по расширениям
  const extSelect = document.getElementById('user-dirs-ext-fil
ter');
  if (extSelect) {
    extSelect.onchange = () => {
      state.extFilter = extSelect.value;
      if (state.selectedUserId) {
        loadUserFiles(state.selectedUserId);
      }
    };
  }

  // Фильтры подпапок
  const subfolderGroup = document.getElementById('subfolder-fi
lter-group');
  if (subfolderGroup) {
    subfolderGroup.addEventListener('click', (e) => {
      const btn = e.target.closest('button[data-subfolder]');
```

```
      if (!btn) return;
      subfolderGroup.querySelectorAll('button').forEach(b => b.classList.remove('active'));
      btn.classList.add('active');
      state.currentSubfolder = btn.dataset.subfolder || '';
      if (state.selectedUserId) {
        loadUserFiles(state.selectedUserId);
      }
    });
  }

  // Кнопки модального окна осиротевших каталогов
  const openOrphanedBtn = document.getElementById('btn-open-orphaned-dirs-modal');
  if (openOrphanedBtn) {
    openOrphanedBtn.onclick = () => openOrphanedModal();
  }

  const refreshOrphanedBtn = document.getElementById('btn-modal-refresh-orphaned');
  if (refreshOrphanedBtn) {
    refreshOrphanedBtn.onclick = () => loadOrphanedDirs();
  }

  const cleanAllOrphanedBtn = document.getElementById('btn-modal-clean-all-orphaned');
  if (cleanAllOrphanedBtn) {
    cleanAllOrphanedBtn.onclick = () => cleanOrphanedDirs();
  }

  // Кнопка удаления файла из предпросмотра
  const previewDeleteBtn = document.getElementById('btn-preview-delete-file');
  if (previewDeleteBtn) {
    previewDeleteBtn.onclick = () => {
      if (state.previewFile && state.selectedUserId) {
        deleteFile(state.selectedUserId, state.previewFile.relative_path);
```

```
      }
    };
  }
}

/**
 * Полное обновление данных: сводка и список пользователей
 */
async function refreshData() {
  await Promise.all([
    loadSummary(),
    loadUsersList(),
    loadOrphanedBadge(),
  ]);
}

/**
 * Загрузка сводной статистики хранилища
 */
async function loadSummary() {
  try {
    const res = await fetch('/api/admin/user-directories/summa
ry');
    if (!res.ok) return;
    const data = await res.json();
    if (data.status === 'ok' && data.summary) {
      const s = data.summary;
      const elUsers = document.getElementById('metric-total-us
ers');
      const elStorage = document.getElementById('metric-total-
storage');
      const elFiles = document.getElementById('metric-total-fi
les');
      const elOrphaned = document.getElementById('metric-orpha
ned-size');

      if (elUsers) elUsers.textContent = s.total_users || 0;
      if (elStorage) elStorage.textContent = s.total_storage_f
```

```
ormatted || '0 B';
166       if (elFiles) elFiles.textContent = s.total_files_count |
| 0;
167       if (elOrphaned) elOrphaned.textContent = s.orphaned_form
atted || '0 B';
168     }
169   } catch (err) {
170     console.warn('Ошибка загрузки сводки хранилища:', err);
171   }
172 }
173 
174 /**
175  * Загрузка бейджа осиротевших каталогов
176  */
177 async function loadOrphanedBadge() {
178   try {
179     const res = await fetch('/api/admin/user-directories/orpha
ned');
180     if (!res.ok) return;
181     const data = await res.json();
182     if (data.status === 'ok') {
183       const badge = document.getElementById('badge-orphaned-co
unt');
184       if (badge) badge.textContent = data.total || 0;
185     }
186   } catch (err) {
187     console.warn('Ошибка проверки осиротевших каталогов:', err
);
188   }
189 }
190 
191 /**
192  * Загрузка списка пользователей с метриками хранилища
193  */
194 async function loadUsersList() {
195   const usersListEl = document.getElementById('user-dirs-users
-list');
196   const countBadge = document.getElementById('users-badge-coun
```

```
t');
197   const statusLabel = document.getElementById('user-dirs-statu
s-label');
198
199   try {
200     if (statusLabel) statusLabel.textContent = 'Обновление спи
ска пользователей...';
201     const res = await fetch('/api/admin/user-directories/users
');
202     if (!res.ok) throw new Error(`HTTP ${res.status}`);
203     const data = await res.json();
204
205     state.users = data.users || [];
206     if (countBadge) countBadge.textContent = state.users.lengt
h;
207     if (statusLabel) statusLabel.textContent = `Загружено поль
зователей: ${state.users.length}`;
208
209     renderUsersList();
210
211     // Если уже был выбран пользователь — перезагрузим его фай
лы, иначе выберем первого
212     if (state.selectedUserId && state.users.some(u => u.id ===
 state.selectedUserId)) {
213       selectUser(state.selectedUserId);
214     } else if (state.users.length > 0) {
215       selectUser(state.users[0].id);
216     }
217   } catch (err) {
218     console.error('Ошибка загрузки пользователей:', err);
219     if (usersListEl) {
220       usersListEl.innerHTML = `
221         <div class="text-center py-4 text-danger small">
222           <i class="bi bi-exclamation-triangle fs-4 d-block mb
-1"></i>
223           Не удалось загрузить список пользователей: ${err.mes
sage}
224         </div>
```

```
      `;
    }
  }
}

/**
 * Рендеринг списка пользователей в левой колонке
 */
function renderUsersList() {
  const usersListEl = document.getElementById('user-dirs-users-list');
  if (!usersListEl) return;

  const query = state.searchQuery.toLowerCase();
  const filteredUsers = state.users.filter(u => {
    if (!query) return true;
    return (
      (u.name && u.name.toLowerCase().includes(query)) ||
      (u.email && u.email.toLowerCase().includes(query)) ||
      String(u.id).includes(query)
    );
  });

  if (filteredUsers.length === 0) {
    usersListEl.innerHTML = `
      <div class="text-center py-4 text-muted small">
        Пользователи не найдены
      </div>
    `;
    return;
  }

  usersListEl.innerHTML = filteredUsers.map(u => {
    const isSelected = u.id === state.selectedUserId;
    const activeClass = isSelected ? 'active border-primary' : '';
    const roleBadge = u.is_admin
      ? '<span class="badge bg-warning text-dark me-1">admin</
```

```
span>'
      : '<span class="badge bg-secondary me-1">user</span>';

    const diskWarning = u.quota_percent > 85 ? 'bg-danger' : u.quota_percent > 60 ? 'bg-warning' : 'bg-primary';

    return `
      <a href="javascript:void(0)"
         class="list-group-item list-group-item-action p-2.5 user-select-item ${activeClass}"
         data-user-id="${u.id}">
        <div class="d-flex w-100 justify-content-between align-items-center mb-1">
          <div class="d-flex align-items-center gap-1.5 text-truncate">
            <span class="badge bg-dark border text-info" style="font-size: 0.72rem;">#${u.id}</span>
            <strong class="text-truncate small ${isSelected ? 'text-white' : ''}">${escapeHtml(u.name)}</strong>
          </div>
          <div>${roleBadge}</div>
        </div>
        <div class="text-truncate small text-muted mb-1.5" style="font-size: 0.76rem;">
          ${escapeHtml(u.email || '-')}
        </div>
        <div class="d-flex justify-content-between align-items-center small mb-1" style="font-size: 0.74rem;">
          <span class="text-muted"><i class="bi bi-files me-1"></i>${u.files_count} файлов</span>
          <span class="fw-semibold text-info">${u.size_formatted}</span>
        </div>
        <div class="progress" style="height: 4px;" title="Использовано квоты: ${u.quota_percent}%">
          <div class="progress-bar ${diskWarning}" role="progressbar" style="width: ${u.quota_percent}%;"></div>
        </div>
```

```
      </a>
    `;
  }).join('');

  // Обработчики клика по элементам пользователей
  usersListEl.querySelectorAll('.user-select-item').forEach(item => {
    item.onclick = () => {
      const uid = parseInt(item.dataset.userId, 10);
      selectUser(uid);
    };
  });
}

/**
 * Выбор пользователя и загрузка его файлового проводника
 */
function selectUser(userId) {
  state.selectedUserId = userId;
  const user = state.users.find(u => u.id === userId);
  state.selectedUserName = user ? user.name : `Пользователь #${userId}`;

  // Обновление подсветки в списке слева
  const usersListEl = document.getElementById('user-dirs-users-list');
  if (usersListEl) {
    usersListEl.querySelectorAll('.user-select-item').forEach(item => {
      const uid = parseInt(item.dataset.userId, 10);
      item.classList.toggle('active', uid === userId);
      item.classList.toggle('border-primary', uid === userId);
    });
  }

  // Обновление заголовка проводника
  const titleEl = document.getElementById('explorer-user-title');
```

```
  const pathBadge = document.getElementById('explorer-user-path-badge');
  if (titleEl) {
    titleEl.innerHTML = `<i class="bi bi-folder-fill text-warning me-1"></i> Хранилище: <strong>${escapeHtml(state.selectedUserName)}</strong> (#${userId})`;
  }
  if (pathBadge && user) {
    pathBadge.classList.remove('d-none');
    pathBadge.textContent = `data/users/${userId}/`;
  }

  loadUserFiles(userId);
}

/**
 * Загрузка файлов выбранного пользователя
 */
async function loadUserFiles(userId) {
  const tbody = document.getElementById('user-files-table-body');
  const footerStats = document.getElementById('explorer-footer-stats');
  const footerQuota = document.getElementById('explorer-footer-quota');

  if (tbody) {
    tbody.innerHTML = `
      <tr>
        <td colspan="6" class="text-center py-4 text-muted">
          <div class="spinner-border spinner-border-sm text-primary me-2"></div>
          Загрузка файлов пользователя...
        </td>
      </tr>
    `;
  }

```

```
  try {
    let url = `/api/admin/user-directories/users/${userId}/tree?`;
    const params = new URLSearchParams();
    if (state.currentSubfolder) params.append('subfolder', state.currentSubfolder);
    if (state.searchQuery) params.append('q', state.searchQuery);
    if (state.extFilter) params.append('extension', state.extFilter);

    const res = await fetch(url + params.toString());
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();

    state.files = data.items || [];

    if (footerStats) {
      footerStats.innerHTML = `Файлов: <strong>${data.total_items || 0}</strong> | Общий размер: <strong>${data.total_size_formatted || '0 B'}</strong>`;
    }

    const user = state.users.find(u => u.id === userId);
    if (footerQuota && user) {
      footerQuota.innerHTML = `Квота: <strong>${user.size_formatted} / ${user.quota_formatted}</strong> (${user.quota_percent}%)`;
    }

    renderUserFilesTable();
  } catch (err) {
    console.error('Ошибка загрузки файлов:', err);
    if (tbody) {
      tbody.innerHTML = `
        <tr>
          <td colspan="6" class="text-center py-4 text-danger small">
```

```
            Ошибка загрузки файлов: ${err.message}
          </td>
        </tr>
      `;
    }
  }
}

/**
 * Рендеринг таблицы файлов пользователя
 */
function renderUserFilesTable() {
  const tbody = document.getElementById('user-files-table-body
');
  if (!tbody) return;

  if (state.files.length === 0) {
    tbody.innerHTML = `
      <tr>
        <td colspan="6" class="text-center py-5 text-muted sma
ll">
          <i class="bi bi-inbox fs-3 d-block mb-1 text-seconda
ry"></i>
          В выбранном разделе нет файлов
        </td>
      </tr>
    `;
    return;
  }

  tbody.innerHTML = state.files.map(f => {
    const isDir = f.is_dir;
    const icon = isDir
      ? '<i class="bi bi-folder-fill text-warning fs-6"></i>'
      : getFileIcon(f.extension);

    const subfolderBadge = f.subfolder === 'root'
      ? '<span class="badge bg-secondary-subtle text-secondary
```

```
 border">root</span>'
      : `<span class="badge bg-info-subtle text-info border bo
rder-info-subtle">${escapeHtml(f.subfolder)}</span>`;

    return `
      <tr>
        <td class="text-center">${icon}</td>
        <td>
          <div class="fw-semibold text-truncate text-white" st
yle="max-width: 260px;" title="${escapeHtml(f.relative_path)}"
>
            ${escapeHtml(f.name)}
          </div>
          <div class="text-muted small font-monospace" style="
font-size: 0.72rem;">
            ${escapeHtml(f.relative_path)}
          </div>
        </td>
        <td class="text-center">${subfolderBadge}</td>
        <td class="text-center font-monospace small">${f.size_
formatted}</td>
        <td class="small text-muted font-monospace" style="fon
t-size: 0.76rem;">${f.modified_formatted}</td>
        <td class="text-center">
          <div class="btn-group btn-group-sm" role="group">
            ${!isDir ? `
              <button class="btn btn-xs btn-outline-info btn-p
review-file"
                      data-path="${escapeHtml(f.relative_path)
}"
                      title="Предпросмотр файла">
                <i class="bi bi-eye"></i>
              </button>
              <a href="/api/admin/user-directories/users/${sta
te.selectedUserId}/file/download?path=${encodeURIComponent(f.r
elative_path)}"
                 class="btn btn-xs btn-outline-success"
                 title="Скачать файл"
```

```
                download>
              <i class="bi bi-download"></i>
            </a>
          ` : ''}
          <button class="btn btn-xs btn-outline-danger btn-delete-file"
                  data-path="${escapeHtml(f.relative_path)}"
                  data-is-dir="${isDir}"
                  title="Удалить">
            <i class="bi bi-trash3"></i>
          </button>
        </div>
      </td>
    </tr>
  `;
}).join('');

// Навешивание событий предпросмотра
tbody.querySelectorAll('.btn-preview-file').forEach(btn => {
  btn.onclick = () => {
    const path = btn.dataset.path;
    openFilePreview(state.selectedUserId, path);
  };
});

// Навешивание событий удаления
tbody.querySelectorAll('.btn-delete-file').forEach(btn => {
  btn.onclick = () => {
    const path = btn.dataset.path;
    const isDir = btn.dataset.isDir === 'true';
    const promptText = isDir
      ? `Вы уверены, что хотите удалить директорию "${path}" со всем содержимым?`
      : `Вы уверены, что хотите удалить файл "${path}"?`;
    if (confirm(promptText)) {
      deleteFile(state.selectedUserId, path);
    }
```

```
    };
  });
}

/**
 * Иконка по расширению файла
 */
function getFileIcon(ext) {
  const e = (ext || '').toLowerCase();
  switch (e) {
    case 'pdf':
      return '<i class="bi bi-file-earmark-pdf-fill text-danger fs-6"></i>';
    case 'txt':
    case 'md':
    case 'log':
      return '<i class="bi bi-file-earmark-text-fill text-info fs-6"></i>';
    case 'json':
    case 'xml':
    case 'yaml':
    case 'yml':
      return '<i class="bi bi-file-earmark-code-fill text-warning fs-6"></i>';
    case 'csv':
    case 'tsv':
    case 'xlsx':
      return '<i class="bi bi-file-earmark-spreadsheet-fill text-success fs-6"></i>';
    case 'png':
    case 'jpg':
    case 'jpeg':
    case 'webp':
    case 'gif':
    case 'svg':
      return '<i class="bi bi-file-earmark-image-fill text-primary fs-6"></i>';
    case 'mp3':
```

```
    case 'wav':
    case 'ogg':
      return '<i class="bi bi-file-earmark-music-fill text-info fs-6"></i>';
    case 'py':
    case 'js':
    case 'sh':
    case 'ps1':
      return '<i class="bi bi-filetype-py text-warning fs-6"></i>';
    default:
      return '<i class="bi bi-file-earmark-fill text-secondary fs-6"></i>';
  }
}

/**
 * Открытие модального окна предпросмотра файла
 */
async function openFilePreview(userId, path) {
  const modalEl = document.getElementById('modal-file-preview');
  if (!modalEl) return;

  const titleEl = document.getElementById('modal-file-preview-title');
  const pathEl = document.getElementById('preview-file-path');
  const sizeEl = document.getElementById('preview-file-size');
  const typeEl = document.getElementById('preview-file-type');
  const contentWrapper = document.getElementById('preview-content-wrapper');
  const downloadLink = document.getElementById('btn-preview-download-file');

  if (titleEl) titleEl.textContent = `Предпросмотр: ${path.split('/').pop()}`;
  if (pathEl) pathEl.textContent = `data/users/${userId}/${path}`;
```

```
  if (sizeEl) sizeEl.textContent = '...';
  if (typeEl) typeEl.textContent = '...';
  if (contentWrapper) {
    contentWrapper.innerHTML = `
      <div class="text-center py-4 text-muted">
        <div class="spinner-border spinner-border-sm text-prim
ary me-2"></div> Загрузка содержимого...
      </div>
    `;
  }
  if (downloadLink) {
    downloadLink.href = `/api/admin/user-directories/users/${u
serId}/file/download?path=${encodeURIComponent(path)}`;
  }

  const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
  modal.show();

  try {
    const res = await fetch(`/api/admin/user-directories/users
/${userId}/file/preview?path=${encodeURIComponent(path)}`);
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    state.previewFile = data;

    if (sizeEl) sizeEl.textContent = data.size_formatted;
    if (typeEl) typeEl.textContent = data.mime_type || data.ex
tension;

    if (contentWrapper) {
      if (data.is_image) {
        contentWrapper.innerHTML = `
          <div class="text-center p-2">
            <img src="${data.content}" class="img-fluid rounde
d border border-secondary shadow" style="max-height: 400px; ob
ject-fit: contain;">
          </div>
        `;
```

```
      } else {
        contentWrapper.textContent = data.content || '(Файл пу
ст)';
      }
    }
  } catch (err) {
    console.error('Ошибка предпросмотра файла:', err);
    if (contentWrapper) {
      contentWrapper.innerHTML = `<div class="text-danger p-3"
>Ошибка загрузки содержимого: ${err.message}</div>`;
    }
  }
}

/**
 * Удаление файла или папки
 */
async function deleteFile(userId, path) {
  try {
    const res = await fetch(`/api/admin/user-directories/users
/${userId}/file?path=${encodeURIComponent(path)}`, {
      method: 'DELETE',
    });
    if (!res.ok) {
      const errData = await res.json().catch(() => ({}));
      throw new Error(errData.detail || `HTTP ${res.status}`);
    }

    showAlert(`Файл "${path}" успешно удален`, 'success');

    // Закрываем модальное окно предпросмотра, если оно открыт
о
    const previewModalEl = document.getElementById('modal-file
-preview');
    if (previewModalEl) {
      const modal = bootstrap.Modal.getInstance(previewModalEl
);
      if (modal) modal.hide();
```

```
602     }
603
604     // Перезагружаем файлы и сводку
605     await loadUserFiles(userId);
606     await loadSummary();
607     await loadUsersList();
608   } catch (err) {
609     console.error('Ошибка удаления файла:', err);
610     showAlert(`Ошибка удаления файла: ${err.message}`, 'danger
');
611   }
612 }
613
614 /**
615  * Открытие модального окна управления осиротевшими директория
ми
616  */
617 async function openOrphanedModal() {
618   const modalEl = document.getElementById('modal-orphaned-user
-dirs');
619   if (!modalEl) return;
620   const modal = bootstrap.Modal.getOrCreateInstance(modalEl);
621   modal.show();
622   await loadOrphanedDirs();
623 }
624
625 /**
626  * Загрузка списка осиротевших директорий
627  */
628 async function loadOrphanedDirs() {
629   const tbody = document.getElementById('modal-orphaned-dirs-t
body');
630   const countEl = document.getElementById('modal-orphaned-coun
t');
631   const sizeEl = document.getElementById('modal-orphaned-size'
);
632   const filesEl = document.getElementById('modal-orphaned-file
s');
```

```
633
634   if (tbody) {
635     tbody.innerHTML = `
636       <tr>
637         <td colspan="6" class="text-center py-4 text-muted">
638           <div class="spinner-border spinner-border-sm text-wa
rning me-2"></div> Сканирование...
639         </td>
640       </tr>
641     `;
642   }
643
644   try {
645     const res = await fetch('/api/admin/user-directories/orpha
ned');
646     if (!res.ok) throw new Error(`HTTP ${res.status}`);
647     const data = await res.json();
648
649     state.orphanedDirs = data.orphaned_dirs || [];
650
651     if (countEl) countEl.textContent = data.total || 0;
652     if (sizeEl) sizeEl.textContent = data.total_size_formatted
 || '0 B';
653     if (filesEl) filesEl.textContent = data.total_files || 0;
654
655     const badge = document.getElementById('badge-orphaned-coun
t');
656     if (badge) badge.textContent = data.total || 0;
657
658     renderOrphanedDirsTable();
659   } catch (err) {
660     console.error('Ошибка загрузки осиротевших директорий:', e
rr);
661     if (tbody) {
662       tbody.innerHTML = `
663         <tr>
664           <td colspan="6" class="text-center py-3 text-danger
small">
```

```
            Ошибка сканирования: ${err.message}
          </td>
        </tr>
      `;
    }
  }
}

/**
 * Рендеринг таблицы осиротевших каталогов
 */
function renderOrphanedDirsTable() {
  const tbody = document.getElementById('modal-orphaned-dirs-tbody');
  if (!tbody) return;

  if (state.orphanedDirs.length === 0) {
    tbody.innerHTML = `
      <tr>
        <td colspan="6" class="text-center py-4 text-success small">
          <i class="bi bi-check-circle fs-4 d-block mb-1"></i>
          Осиротевших каталогов не обнаружено. Все директории привязаны к активным пользователям!
        </td>
      </tr>
    `;
    return;
  }

  tbody.innerHTML = state.orphanedDirs.map(d => `
    <tr>
      <td>
        <div class="fw-bold font-monospace text-warning small">${escapeHtml(d.dir_name)}</div>
        <div class="text-muted small" style="font-size: 0.72rem;">${escapeHtml(d.full_path)}</div>
      </td>
```

```
      <td><span class="badge bg-secondary-subtle text-secondar
y border">${escapeHtml(d.reason)}</span></td>
      <td class="text-center font-monospace">${d.files_count}<
/td>
      <td class="text-center font-monospace small">${d.size_fo
rmatted}</td>
      <td class="small text-muted font-monospace" style="font-
size: 0.76rem;">${d.modified_formatted}</td>
      <td class="text-center">
        <button class="btn btn-xs btn-outline-danger btn-clean
-single-orphaned"
                data-dir="${escapeHtml(d.dir_name)}"
                title="Удалить этот каталог">
          <i class="bi bi-trash3"></i>
        </button>
      </td>
    </tr>
  `).join('');

  tbody.querySelectorAll('.btn-clean-single-orphaned').forEach
(btn => {
    btn.onclick = () => {
      const dirName = btn.dataset.dir;
      if (confirm(`Удалить осиротевший каталог "${dirName}"?`)
) {
        cleanOrphanedDirs([dirName]);
      }
    };
  });
}

/**
 * Очистка осиротевших каталогов (конкретных или всех)
 */
async function cleanOrphanedDirs(dirsList = null) {
  if (!dirsList && !confirm('Вы уверены, что хотите удалить ВС
Е осиротевшие каталоги пользователей? Это действие необратимо.
')) {
```

```
    return;
  }

  try {
    const payload = dirsList ? { dirs: dirsList } : {};
    const res = await fetch('/api/admin/user-directories/orpha
ned/clean', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(payload),
    });

    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();

    showAlert(`Удалено каталогов: ${data.deleted_count}, освоб
ождено места: ${data.freed_formatted}`, 'success');

    await loadOrphanedDirs();
    await loadSummary();
  } catch (err) {
    console.error('Ошибка очистки осиротевших каталогов:', err
);
    showAlert(`Ошибка очистки каталогов: ${err.message}`, 'dan
ger');
  }
}

/**
 * Применение клиентских фильтров поиска
 */
function applyFilters() {
  renderUsersList();
  if (state.selectedUserId) {
    loadUserFiles(state.selectedUserId);
  }
}

```

```
/**
 * Вывод сообщения alert
 */
function showAlert(message, type = 'info') {
  const alertBox = document.getElementById('user-dirs-alert-box');
  const alertMsg = document.getElementById('user-dirs-alert-message');
  if (!alertBox || !alertMsg) return;

  alertBox.className = `alert alert-${type} alert-dismissible fade show mb-3 shadow-sm`;
  alertMsg.textContent = message;
  alertBox.classList.remove('d-none');

  setTimeout(() => {
    alertBox.classList.add('d-none');
  }, 6000);
}

/**
 * Экранирование HTML
 */
function escapeHtml(text) {
  if (!text) return '';
  const map = {
    '&': '&amp;',
    '<': '&lt;',
    '>': '&gt;',
    '"': '&quot;',
    "'": '&#039;',
  };
  return String(text).replace(/[&<>"']/g, m => map[m]);
}

// Экспорт по умолчанию
export default initUserDirectoriesTab;

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/user_tts/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/user_tts/index.html" cl
ass="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/user_tts
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!DOCTYPE html>
<html lang="ru">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-s
cale=1.0">
  <title data-i18n="title">Эксперименты с TTS (Синтез речи)</t
itle>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dis
```

```
t/css/bootstrap.min.css" rel="stylesheet">
  <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.10.0/font/bootstrap-icons.css" rel="stylesheet">
  <link href="/html/css/main.css?v=20260915_light_v2" rel="stylesheet">
  <script src="https://cdn.jsdelivr.net/npm/i18next@23.11.5/i18next.min.js"></script>
  <style>
    /* Custom styles for TTS page */
    :root {
      --tts-nav-bg: var(--surface-2);
    }

    body {
      background: var(--bg-color, #fbfbfb);
      color: var(--text-color, #18181b);
      min-height: 100vh;
    }

    .tts-container {
      max-width: 800px;
      margin: 2rem auto;
      padding: 0 1rem;
    }

    .tts-card {
      background: var(--surface-1);
      border: 1px solid var(--border-color);
      border-radius: 12px;
      padding: 2rem;
      box-shadow: 0 4px 12px rgba(0, 0, 0, 0.08);
    }

    h1 {
      color: var(--text-color);
      font-weight: 600;
      margin-bottom: 0.5rem;
    }
```

```
65
66      .subtitle {
67        color: var(--text-muted);
68        font-size: 0.95rem;
69        margin-bottom: 2rem;
70      }
71
72      .form-label {
73        color: var(--text-muted);
74        font-weight: 500;
75        margin-bottom: 0.5rem;
76      }
77
78      .form-control, .form-select {
79        background: var(--surface-2);
80        border: 1px solid var(--border-color);
81        color: var(--text-color);
82        border-radius: 8px;
83        padding: 0.75rem 1rem;
84      }
85
86      .form-control:focus, .form-select:focus {
87        background: var(--surface-2);
88        border-color: var(--nav-active);
89        color: var(--text-color);
90        box-shadow: 0 0 0 2px rgba(88, 166, 255, 0.25);
91      }
92
93      textarea.form-control {
94        resize: vertical;
95        min-height: 120px;
96        font-family: system-ui, -apple-system, sans-serif;
97      }
98
99      .btn-tts {
100       background: var(--nav-active, #1f6feb);
101       color: #fff;
102       border: none;
```

```
      border-radius: 8px;
      padding: 0.85rem 1.5rem;
      font-weight: 500;
      transition: background-color 0.2s;
    }

    .btn-tts:hover {
      background: #388bfd;
      color: #fff;
    }

    .btn-tts:disabled {
      opacity: 0.6;
      cursor: not-allowed;
    }

    .btn-apply {
      background: #238636;
      color: #fff;
      border: none;
      border-radius: 8px;
      padding: 0.85rem 1.5rem;
      font-weight: 500;
      transition: background-color 0.2s;
    }

    .btn-apply:hover {
      background: #2ea043;
      color: #fff;
    }

    .btn-apply:disabled {
      opacity: 0.6;
      cursor: not-allowed;
    }

    .active-badge {
      background: rgba(35, 134, 54, 0.15);
```

```
      border: 1px solid rgba(35, 134, 54, 0.3);
      color: #56d364;
      padding: 0.5rem 1rem;
      border-radius: 20px;
      font-size: 0.85rem;
      display: inline-flex;
      align-items: center;
      gap: 0.5rem;
      margin-bottom: 1.5rem;
    }

    .audio-player-container {
      margin-top: 2rem;
      padding-top: 2rem;
      border-top: 1px solid var(--border-color);
      display: none;
    }

    audio {
      width: 100%;
      border-radius: 8px;
    }

    .status-badge {
      display: inline-flex;
      align-items: center;
      gap: 0.5rem;
      font-size: 0.9rem;
      color: var(--text-muted);
      margin-top: 1rem;
    }

    .spinner-border {
      width: 1rem;
      height: 1rem;
      border-width: 0.15em;
    }

```

```
    .back-link {
      display: inline-flex;
      align-items: center;
      gap: 0.5rem;
      color: var(--text-muted);
      text-decoration: none;
      font-size: 0.95rem;
      margin-top: 2rem;
      transition: color 0.2s;
    }

    .back-link:hover {
      color: var(--nav-active);
    }
  </style>
</head>
<body>

<div class="tts-container">
  <div class="tts-card">
    <h1><i class="bi bi-mic"></i> <span data-i18n="title">Эксп
ерименты с TTS (Синтез речи)</span></h1>
    <p class="subtitle" data-i18n="subtitle">Сравните качество
 синтеза речи различных моделей: Web Speech, Google, Edge и Si
lero.</p>

    <div id="active-voice-container" class="mb-3" style="displ
ay: none;">
      <span class="active-badge">
        <i class="bi bi-check-circle-fill"></i>
        <span><span data-i18n="label_active_voice">Активный го
лос:</span> <strong id="active-voice-name">...</strong></span>
      </span>
    </div>

    <form id="tts-form">
      <div class="mb-4">
        <label for="tts-text" class="form-label" data-i18n="fo
```

```
rm.label_text">Текст для озвучки</label>
212         <textarea id="tts-text" class="form-control" placehold
er data-i18n-placeholder="form.placeholder">Привет! Это демонс
трация работы различных систем синтеза речи. Какая из них звуч
ит лучше всего?</textarea>
213       </div>
214
215       <div class="row">
216         <div class="col-md-6 mb-4">
217           <label for="tts-engine" class="form-label" data-i18n
="form.label_engine">Система синтеза</label>
218           <select id="tts-engine" class="form-select">
219             <option value="browser" data-i18n="options.browser
">Web Speech API (Браузерная)</option>
220             <option value="gtts" data-i18n="options.gtts">Goog
le TTS (Базовая)</option>
221             <option value="edge-tts" selected data-i18n="optio
ns.edge-tts">Microsoft Edge TTS (Нейросетевая)</option>
222             <option value="silero" data-i18n="options.silero">
Silero TTS (Локальная оффлайн)</option>
223           </select>
224         </div>
225
226         <div class="col-md-6 mb-4" id="voice-select-container"
>
227           <label for="tts-voice" class="form-label" data-i18n=
"form.label_voice">Голос / Спикер</label>
228           <select id="tts-voice" class="form-select">
229             <!-- Populated dynamically via JS -->
230           </select>
231         </div>
232       </div>
233
234       <div class="d-flex gap-2">
235         <button type="submit" class="btn btn-tts flex-grow-1"
id="btn-speak" data-i18n="button_speak">Озвучить текст</button
>
236         <button type="button" class="btn btn-apply" id="btn-sa
```

```
ve-default" data-i18n="button_apply">Применить изменения</butt
on>
      </div>

      <div class="status-badge" id="tts-status" style="display
: none;">
        <div class="spinner-border text-success" role="status"
></div>
        <span id="status-text" data-i18n="status_generating">Г
енерация аудиофайла на сервере...</span>
      </div>
    </form>

    <div class="audio-player-container" id="player-container">
      <label class="form-label" data-i18n="label_result">Резул
ьтат генерации</label>
      <audio id="audio-player" controls></audio>
    </div>
  </div>

  <a href="/" class="back-link">
    <i class="bi bi-arrow-left"></i>
    <span data-i18n="back_link">Вернуться в медиатеку</span>
  </a>
</div>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist
/js/bootstrap.bundle.min.js"></script>
<script type="module">
  import { initI18n, switchLang, applyTranslations } from '/ht
ml/js/i18n.js';

  window.switchLang = switchLang;
  window.applyTranslations = applyTranslations;

  // Initialize i18n
  const savedLang = localStorage.getItem('app_language') || 'r
u';
```

```
  await initI18n(savedLang);

  const engineSelect = document.getElementById('tts-engine');
  const voiceSelect = document.getElementById('tts-voice');
  const voiceContainer = document.getElementById('voice-select
-container');
  const ttsForm = document.getElementById('tts-form');
  const btnSpeak = document.getElementById('btn-speak');
  const ttsStatus = document.getElementById('tts-status');
  const statusText = document.getElementById('status-text');
  const playerContainer = document.getElementById('player-cont
ainer');
  const audioPlayer = document.getElementById('audio-player');
  const activeVoiceContainer = document.getElementById('active
-voice-container');
  const activeVoiceName = document.getElementById('active-voic
e-name');

  function getFriendlyVoiceName(system, voice) {
    if (system === 'edge-tts') {
      const found = voiceMap['edge-tts'].find(v => v.value ===
 voice);
      return found ? translateVoiceLabel(found.labelKey) : `Ed
ge (${voice})`;
    }
    if (system === 'silero') {
      const found = voiceMap['silero'].find(v => v.value === v
oice);
      return found ? translateVoiceLabel(found.labelKey) : `Si
lero (${voice})`;
    }
    if (system === 'gtts') {
      const found = voiceMap['gtts'].find(v => v.value === voi
ce);
      return found ? translateVoiceLabel(found.labelKey) : `Go
ogle TTS (${voice})`;
    }
    if (system === 'browser') {
```

```
      return `Браузерный (${voice || 'По умолчанию'})`;
    }
    return `${system} — ${voice}`;
  }

  function updateActiveBadge(system, voice) {
    if (system && voice) {
      activeVoiceName.textContent = getFriendlyVoiceName(syste
m, voice);
      activeVoiceContainer.style.display = 'block';
    } else {
      activeVoiceContainer.style.display = 'none';
    }
  }

  // Available voices dictionary with translated labels
  const voiceMap = {
    'browser': [], // Populated dynamically from window.speech
Synthesis
    'gtts': [
      { value: 'ru', labelKey: 'voices.ru_standard' },
      { value: 'en', labelKey: 'voices.en_standard' },
      { value: 'he', labelKey: 'voices.he_standard' },
      { value: 'de', labelKey: 'voices.de_standard' },
      { value: 'fr', labelKey: 'voices.fr_standard' },
      { value: 'es', labelKey: 'voices.es_standard' }
    ],
    'edge-tts': [
      { value: 'ru-RU-DmitryNeural', labelKey: 'voices.dmitry'
 },
      { value: 'ru-RU-SvetlanaNeural', labelKey: 'voices.svetl
ana' },
      { value: 'en-US-JennyNeural', labelKey: 'voices.jenny' }
,
      { value: 'en-US-GuyNeural', labelKey: 'voices.guy' },
      { value: 'en-US-AriaNeural', labelKey: 'voices.aria' },
      { value: 'en-GB-SoniaNeural', labelKey: 'voices.sonia' }
,
```

```
      { value: 'en-GB-RyanNeural', labelKey: 'voices.ryan' },
      { value: 'he-IL-AvriNeural', labelKey: 'voices.avri' },
      { value: 'he-IL-HilaNeural', labelKey: 'voices.hila' },
      { value: 'de-DE-KatjaNeural', labelKey: 'voices.katja' }
,
      { value: 'de-DE-ConradNeural', labelKey: 'voices.conrad'
 },
      { value: 'fr-FR-DeniseNeural', labelKey: 'voices.denise'
 },
      { value: 'fr-FR-HenriNeural', labelKey: 'voices.henri' }
,
      { value: 'es-ES-ElviraNeural', labelKey: 'voices.elvira'
 },
      { value: 'es-ES-AlvaroNeural', labelKey: 'voices.alvaro'
 }
    ],
    'silero': [
      { value: 'eugene', labelKey: 'voices.eugene' },
      { value: 'aidar', labelKey: 'voices.aidar' },
      { value: 'baya', labelKey: 'voices.baya' },
      { value: 'kseniya', labelKey: 'voices.kseniya' },
      { value: 'xenia', labelKey: 'voices.xenia' },
      { value: 'random', labelKey: 'voices.random' }
    ]
  };

  // Translate voice labels
  function translateVoiceLabel(labelKey) {
    return i18next?.t(labelKey) || labelKey;
  }

  // Load browser native voices
  function loadBrowserVoices() {
    if ('speechSynthesis' in window) {
      const voices = window.speechSynthesis.getVoices();
      voiceMap['browser'] = voices
        .map(v => ({ value: v.name, labelKey: `[${v.lang}] ${v
.name}` }));
```

```
357
358       // Fallback if no voices found in browser
359       if (voiceMap['browser'].length === 0) {
360         voiceMap['browser'].push({ value: 'default', labelKey:
 'voices.default' });
361       }
362
363       if (engineSelect.value === 'browser') {
364         updateVoiceSelect();
365       }
366     }
367   }
368
369   if ('speechSynthesis' in window) {
370     window.speechSynthesis.onvoiceschanged = loadBrowserVoices
;
371     loadBrowserVoices();
372   }
373
374   function updateVoiceSelect() {
375     const engine = engineSelect.value;
376     const voices = voiceMap[engine] || [];
377
378     voiceSelect.innerHTML = '';
379
380     if (voices.length === 0) {
381       voiceContainer.style.display = 'none';
382     } else {
383       voiceContainer.style.display = 'block';
384       voices.forEach(voice => {
385         const opt = document.createElement('option');
386         opt.value = voice.value;
387         opt.textContent = translateVoiceLabel(voice.labelKey);
388         voiceSelect.appendChild(opt);
389       });
390     }
391   }
392
```

```
  engineSelect.addEventListener('change', updateVoiceSelect);
  updateVoiceSelect(); // Initial populate

  // On page load, fetch current settings to pre-select saved voice/engine
  try {
    const settingsResponse = await fetch('/auth/settings');
    if (settingsResponse.ok) {
      const settings = await settingsResponse.json();
      if (settings.tts_system) {
        engineSelect.value = settings.tts_system;
        updateVoiceSelect();
      }
      if (settings.tts_voice) {
        voiceSelect.value = settings.tts_voice;
      }
      updateActiveBadge(settings.tts_system, settings.tts_voice);
    }
  } catch (e) {
    console.error('Failed to load user settings:', e);
  }

  // Update translations when language changes
  const originalSwitchLang = window.switchLang;
  window.switchLang = async function(lang) {
    await originalSwitchLang(lang);
    updateVoiceSelect();
  };

  ttsForm.addEventListener('submit', async (e) => {
    e.preventDefault();
    const text = document.getElementById('tts-text').value.trim();
    if (!text) return;

    const engine = engineSelect.value;
    const voice = voiceSelect.value;
```

```
428
429     // 1. Browser Web Speech synthesis
430     if (engine === 'browser') {
431       if ('speechSynthesis' in window) {
432         window.speechSynthesis.cancel();
433         const cleanText = text.replace(/<[^>]*>/g, '').replace
    (/[*_`#]/g, '');
434         const utterance = new SpeechSynthesisUtterance(cleanTe
    xt);
435         const voices = window.speechSynthesis.getVoices();
436         const selectedVoice = voices.find(v => v.name === voic
    e);
437         if (selectedVoice) {
438           utterance.voice = selectedVoice;
439           if (selectedVoice.lang) {
440             utterance.lang = selectedVoice.lang;
441           }
442         }
443
444         ttsStatus.style.display = 'inline-flex';
445         statusText.textContent = i18next?.t('status_generating
    ') || 'Воспроизведение речи браузером...';
446
447         utterance.onend = () => {
448           ttsStatus.style.display = 'none';
449         };
450         utterance.onerror = () => {
451           ttsStatus.style.display = 'none';
452         };
453
454         window.speechSynthesis.speak(utterance);
455         playerContainer.style.display = 'none';
456       } else {
457         const msg = i18next?.t('errors.no_api') || 'Ваш браузе
    р не поддерживает Web Speech API.';
458         window.showToast?.(msg, 'warning') || alert(msg);
459       }
460       return;
```

```
461     }
462 
463     // 2. Server-side TTS synthesis (gtts, edge-tts, silero)
464     ttsStatus.style.display = 'inline-flex';
465     statusText.textContent = i18next?.t('status_generating') |
| 'Генерация аудиофайла на сервере...';
466     btnSpeak.disabled = true;
467     playerContainer.style.display = 'none';
468 
469     try {
470       const queryParams = new URLSearchParams({
471         text: text,
472         system: engine,
473         voice: voice
474       });
475 
476       // Load audio stream from server
477       const audioUrl = `/api/tts/synthesize?${queryParams.toSt
ring()}`;
478 
479       // Force preload of the audio
480       audioPlayer.src = audioUrl;
481       audioPlayer.load();
482 
483       audioPlayer.oncanplaythrough = () => {
484         ttsStatus.style.display = 'none';
485         btnSpeak.disabled = false;
486         playerContainer.style.display = 'block';
487         audioPlayer.play();
488       };
489 
490       audioPlayer.onerror = (err) => {
491         console.error(err);
492         statusText.textContent = i18next?.t('errors.server_err
or') || 'Ошибка синтеза речи на сервере';
493         btnSpeak.disabled = false;
494         setTimeout(() => { ttsStatus.style.display = 'none'; }
, 3000);
```

```
      };

    } catch (err) {
      console.error(err);
      statusText.textContent = i18next?.t('errors.network_erro
r') || 'Сбой сети при запросе к серверу';
      btnSpeak.disabled = false;
      setTimeout(() => { ttsStatus.style.display = 'none'; },
3000);
    }
  });

  const btnSaveDefault = document.getElementById('btn-save-def
ault');
  btnSaveDefault.addEventListener('click', async () => {
    const engine = engineSelect.value;
    const voice = voiceSelect.value;

    btnSaveDefault.disabled = true;
    try {
      const payload = {
        tts_enabled: 1,
        tts_system: engine,
        tts_voice: voice
      };

      const response = await fetch('/auth/settings', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify(payload)
      });

      if (response.ok) {
        updateActiveBadge(engine, voice);
        alert('Изменения успешно применены! Этот голос теперь
установлен по умолчанию.');
      } else {
        alert('Не удалось сохранить настройки. Убедитесь, что
```

```
вы вошли в систему.');
      }
    } catch (err) {
      console.error(err);
      alert('Ошибка подключения к серверу при сохранении настроек.');
    } finally {
      btnSaveDefault.disabled = false;
    }
  });
</script>
</body>
</html>
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/voice_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/voice_tab/index.html" c
lass="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/voice_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Voice Diarization & Audio Analyzer Tab -->
<div class="container-fluid py-2">
<div class="alert alert-danger d-flex justify-content-between
align-items-center flex-wrap gap-2 mb-3 shadow-sm">
<div>
<h5 class="mb-1"><i class="bi bi-mic-fill me-2"></i><span data
-i18n="voice.title">  Разбор голосовых сообщений и диаризация
</span></h5>
<p class="mb-0 small text-secondary" data-i18n="voice.subtitle
">Запись с микрофона или загрузка аудиофайла: разделение по со
```

```
беседникам, стенограмма и сводка встреч.</p>
</div>
<div class="d-flex align-items-center gap-2">
<span class="badge bg-dark font-monospace p-2">Gemini Multimodal Audio</span>
</div>
</div>
<div class="row g-3">
<!-- LEFT COLUMN: Recording & Audio Upload Controls -->
<div class="col-lg-5 col-md-12">
<!-- Microphone Live Recording Card -->
<div class="card shadow-sm border-0 mb-3" data-panel-id="panel-voice-zapis-s-mikrofona" id="panel-voice-zapis-s-mikrofona">
<div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
<span class="fw-semibold"><i class="bi bi-broadcast me-2 text-danger"></i><span data-i18n="voice.recordTitle">Запись с микрофона</span></span>
<span class="badge bg-danger font-monospace d-none" id="vtab-record-timer">00:00</span>
</div>
<div class="card-body text-center p-4">
<div class="my-3" id="vtab-mic-visual">
<div class="d-inline-flex align-items-center justify-content-center rounded-circle p-4 border border-3 border-danger shadow" id="vtab-mic-circle" style="width: 100px; height: 100px; transition: all 0.3s;">
<i class="bi bi-mic-fill fs-1 text-danger" id="vtab-mic-icon"></i>
</div>
</div>
<h6 class="fw-bold mb-1" data-i18n="voice.clickToRecord" id="vtab-record-status-label">Нажмите для начала записи</h6>
<p class="text-muted small mb-3">Говорите в микрофон — система автоматически распознает разных спикеров.</p>
<div class="d-flex justify-content-center gap-2">
<button class="btn btn-danger px-4 py-2 rounded-pill fw-bold shadow-sm" id="btn-vtab-record-toggle">
```

```
<i class="bi bi-record-fill me-1"></i> <span data-i18n="voice.startRecord" id="vtab-record-btn-text">Начать запись</span>
</button>
<button class="btn btn-outline-secondary px-3 py-2 rounded-pill d-none" id="btn-vtab-record-cancel">
              Отмена
            </button>
</div>
</div>
</div>
<!-- File Upload Card -->
<div class="card shadow-sm border-0 mb-3" data-panel-id="panel-voice-zagruzka-audiofayla" id="panel-voice-zagruzka-audiofayla">
<div class="card-header bg-dark text-white d-flex justify-content-between align-items-center">
<span class="fw-semibold"><i class="bi bi-cloud-arrow-up me-2 text-info"></i><span data-i18n="voice.uploadTitle">Загрузка аудиофайла</span></span>
</div>
<div class="card-body">
<div class="border border-2 border-dashed rounded p-4 text-center cursor-pointer mb-2 bg-body-secondary bg-opacity-25" id="vtab-dropzone" style="border-style: dashed !important;">
<i class="bi bi-file-earmark-music fs-1 text-info mb-2 d-block"></i>
<h6 class="fw-bold mb-1" data-i18n="voice.dropzoneText">Перетащите аудиофайл сюда или нажмите для выбора</h6>
<p class="text-muted small mb-2">Форматы: .mp3, .wav, .ogg, .m4a, .webm, .aac</p>
<input accept="audio/*,.mp3,.wav,.ogg,.m4a,.webm,.aac" class="d-none" id="vtab-file-input" type="file"/>
<button class="btn btn-outline-info btn-sm px-3 rounded-pill mt-1" onclick="document.getElementById('vtab-file-input').click();" type="button">
<i class="bi bi-folder2-open me-1"></i> <span data-i18n="voice.browseAudio">Выбрать аудиофайл</span>
</button>
```

```
</div>
</div>
</div>
<!-- Audio Analysis Settings -->
<div class="card shadow-sm border-0" data-panel-id="panel-voice-parametry-raspoznavaniya" id="panel-voice-parametry-raspoznavaniya">
<div class="card-header bg-dark text-white">
<span class="fw-semibold"><i class="bi bi-sliders me-2"></i>Параметры распознавания</span>
</div>
<div class="card-body">
<div class="mb-2">
<label class="form-label small fw-bold" for="vtab-model-select">Модель:</label>
<select class="form-select form-select-sm" id="vtab-model-select">
<option selected="" value="gemini-3.1-flash">gemini-3.1-flash (Рекомендуется для аудио)</option>
<option value="gemini-2.0-flash">gemini-2.0-flash</option>
<option value="gemini-1.5-flash">gemini-1.5-flash</option>
<option value="gemini-1.5-pro">gemini-1.5-pro</option>
</select>
</div>
<div class="mb-2">
<label class="form-label small fw-bold" for="vtab-lang-select">Язык отчёта и сводки:</label>
<select class="form-select form-select-sm" id="vtab-lang-select">
<option selected="" value="ru">Русский (Russian)</option>
<option value="en">English</option>
<option value="he">□ □ □ □ □ (Hebrew)</option>
</select>
</div>
<div>
<label class="form-label small fw-bold" for="vtab-api-key">Gemini API Key (опционально):</label>
<input autocomplete="new-password" class="form-control form-co
```

```
ntrol-sm" data-1p-ignore="true" data-lpignore="true" id="vtab-
api-key" placeholder="Из .env по умолчанию" type="password"/>
</div>
</div>
</div>
</div>
<!-- RIGHT COLUMN: Audio Playback & Diarization Result -->
<div class="col-lg-7 col-md-12">
<div class="card shadow-sm border-0 mb-3" data-panel-id="panel
-voice-rezultat-analiza-i-stenogramm" id="vtab-result-card">
<div class="card-header bg-dark text-white d-flex justify-cont
ent-between align-items-center">
<span class="fw-semibold"><i class="bi bi-file-text me-2 text-
warning"></i><span data-i18n="voice.resultTitle">Результат ана
лиза и стенограмма</span></span>
<div class="d-flex align-items-center gap-2">
<button class="btn btn-success btn-sm d-none" id="vtab-btn-sav
e-rag">
<i class="bi bi-floppy me-1"></i> <span data-i18n="voice.saveT
oRag">Сохранить в RAG</span>
</button>
</div>
</div>
<div class="card-body p-3" id="vtab-result-body">
<!-- Audio Player Preview -->
<div class="mb-3 d-none p-2 bg-body-tertiary border rounded" i
d="vtab-player-container">
<div class="small fw-semibold mb-1 text-muted"><i class="bi bi
-play-circle me-1 text-danger"></i>Исходная аудиозапись: <span
 class="text-body font-monospace" id="vtab-current-filename"><
/span></div>
<audio class="w-100" controls="" id="vtab-audio-element" style
="max-height: 38px;"></audio>
</div>
<!-- Loading State -->
<div class="d-none text-center py-5" id="vtab-loading">
<div class="spinner-border text-danger mb-3" role="status" sty
le="width: 3rem; height: 3rem;"></div>
```

```
<h6 class="fw-bold text-primary">Идет распознавание и диаризация...</h6>
<p class="text-muted small">Gemini слушает аудио, разделяет собеседников и готовит сводку.</p>
</div>
<!-- Empty Initial State -->
<div class="text-center py-5 text-muted" id="vtab-empty-state">
<i class="bi bi-soundwave fs-1 d-block mb-2 opacity-50"></i>
<h6 class="fw-bold" data-i18n="voice.emptyStateTitle">Запишите голос или загрузите аудиофайл</h6>
<p class="small mb-0">Здесь появится краткая сводка, список ключевых решений и реплики каждого собеседника.</p>
</div>
<!-- Analysis Content Container -->
<div class="d-none" id="vtab-analysis-content"></div>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Voice Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/voice_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/voice_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// ── VOICE_TAB MAIN.JS ──────────────────────────────────────

(function () {
  let isInitialized = false;
  let tabMediaRecorder = null;
  let tabAudioChunks = [];
  let tabRecordingStartTime = 0;
  let tabTimerInterval = null;
  let lastAnalysisData = null;
  let lastAudioFilename = 'voice_recording';

```

```
  function initVoiceTab() {
    const recordBtn = document.getElementById('btn-vtab-record-toggle');
    if (!recordBtn) return;
    if (isInitialized) return;
    isInitialized = true;
    setupEventListeners();
  }

  window.initVoiceTab = initVoiceTab;

  function setupEventListeners() {
    const recordBtn = document.getElementById('btn-vtab-record-toggle');
    const cancelBtn = document.getElementById('btn-vtab-record-cancel');
    const fileInput = document.getElementById('vtab-file-input');
    const dropzone = document.getElementById('vtab-dropzone');
    const saveRagBtn = document.getElementById('vtab-btn-save-rag');

    if (recordBtn) {
      recordBtn.addEventListener('click', toggleRecording);
    }
    if (cancelBtn) {
      cancelBtn.addEventListener('click', cancelRecording);
    }
    if (fileInput) {
      fileInput.addEventListener('change', (e) => {
        if (e.target.files && e.target.files.length > 0) {
          const file = e.target.files[0];
          analyzeAudioBlob(file, file.name);
          fileInput.value = '';
        }
      });
    }
    if (dropzone) {
```

```
      ['dragenter', 'dragover'].forEach(eventName => {
        dropzone.addEventListener(eventName, (e) => {
          e.preventDefault();
          e.stopPropagation();
          dropzone.classList.add('border-info', 'bg-info', 'bg
-opacity-10');
        }, false);
      });

      ['dragleave', 'drop'].forEach(eventName => {
        dropzone.addEventListener(eventName, (e) => {
          e.preventDefault();
          e.stopPropagation();
          dropzone.classList.remove('border-info', 'bg-info',
'bg-opacity-10');
        }, false);
      });

      dropzone.addEventListener('drop', (e) => {
        const dt = e.dataTransfer;
        if (dt && dt.files && dt.files.length > 0) {
          const file = dt.files[0];
          analyzeAudioBlob(file, file.name);
        }
      }, false);
    }
    if (saveRagBtn) {
      saveRagBtn.addEventListener('click', saveResultToRag);
    }
  }

  async function toggleRecording() {
    if (tabMediaRecorder && tabMediaRecorder.state === 'record
ing') {
      stopRecording();
    } else {
      await startRecording();
    }
```

```
  }

  async function startRecording() {
    if (!navigator.mediaDevices || !navigator.mediaDevices.getUserMedia) {
      alert('Доступ к микрофону не поддерживается в вашем браузере.');
      return;
    }

    try {
      const stream = await navigator.mediaDevices.getUserMedia({ audio: true });
      tabAudioChunks = [];

      let options = {};
      if (MediaRecorder.isTypeSupported('audio/webm;codecs=opus')) {
        options = { mimeType: 'audio/webm;codecs=opus' };
      } else if (MediaRecorder.isTypeSupported('audio/ogg;codecs=opus')) {
        options = { mimeType: 'audio/ogg;codecs=opus' };
      }

      tabMediaRecorder = new MediaRecorder(stream, options);

      tabMediaRecorder.ondataavailable = (e) => {
        if (e.data && e.data.size > 0) {
          tabAudioChunks.push(e.data);
        }
      };

      tabMediaRecorder.onstop = () => {
        stream.getTracks().forEach(t => t.stop());
        if (tabAudioChunks.length > 0) {
          const mime = tabMediaRecorder.mimeType || 'audio/webm';
          const blob = new Blob(tabAudioChunks, { type: mime }
```

```
);
          const fname = `voice_${new Date().toISOString().slice(0, 19).replace(/[:T]/g, '_')}.webm`;
          analyzeAudioBlob(blob, fname);
        }
      };

      tabMediaRecorder.start(250);
      tabRecordingStartTime = Date.now();

      // UI state update
      const recordBtn = document.getElementById('btn-vtab-record-toggle');
      const cancelBtn = document.getElementById('btn-vtab-record-cancel');
      const btnText = document.getElementById('vtab-record-btn-text');
      const statusLabel = document.getElementById('vtab-record-status-label');
      const timerBadge = document.getElementById('vtab-record-timer');
      const micCircle = document.getElementById('vtab-mic-circle');

      if (recordBtn) {
        recordBtn.className = 'btn btn-danger px-4 py-2 rounded-pill fw-bold shadow-sm';
      }
      if (btnText) btnText.textContent = 'Остановить и разобрать';
      if (cancelBtn) cancelBtn.classList.remove('d-none');
      if (statusLabel) statusLabel.textContent = 'Идет запись речи...';
      if (timerBadge) {
        timerBadge.classList.remove('d-none');
        timerBadge.textContent = '00:00';
      }
      if (micCircle) {
```

```
        micCircle.classList.add('bg-danger', 'bg-opacity-25');
        micCircle.style.transform = 'scale(1.1)';
      }

      clearInterval(tabTimerInterval);
      tabTimerInterval = setInterval(() => {
        const elapsedSec = Math.floor((Date.now() - tabRecordi
ngStartTime) / 1000);
        const m = String(Math.floor(elapsedSec / 60)).padStart
(2, '0');
        const s = String(elapsedSec % 60).padStart(2, '0');
        if (timerBadge) timerBadge.textContent = `${m}:${s}`;
      }, 500);

    } catch (err) {
      console.error('Microphone error:', err);
      alert(`Не удалось включить микрофон: ${err.message}`);
    }
  }

  function stopRecording() {
    resetRecordingUi();
    if (tabMediaRecorder && tabMediaRecorder.state === 'record
ing') {
      tabMediaRecorder.stop();
    }
  }

  function cancelRecording() {
    resetRecordingUi();
    if (tabMediaRecorder) {
      if (tabMediaRecorder.state === 'recording') {
        tabMediaRecorder.onstop = null;
        tabMediaRecorder.stop();
      }
      tabAudioChunks = [];
    }
  }
```

```
194
195   function resetRecordingUi() {
196     clearInterval(tabTimerInterval);
197     const cancelBtn = document.getElementById('btn-vtab-record
-cancel');
198     const btnText = document.getElementById('vtab-record-btn-t
ext');
199     const statusLabel = document.getElementById('vtab-record-s
tatus-label');
200     const timerBadge = document.getElementById('vtab-record-ti
mer');
201     const micCircle = document.getElementById('vtab-mic-circle
');
202
203     if (cancelBtn) cancelBtn.classList.add('d-none');
204     if (btnText) btnText.textContent = 'Начать запись';
205     if (statusLabel) statusLabel.textContent = 'Нажмите для на
чала записи';
206     if (timerBadge) timerBadge.classList.add('d-none');
207     if (micCircle) {
208       micCircle.classList.remove('bg-danger', 'bg-opacity-25')
;
209       micCircle.style.transform = 'scale(1)';
210     }
211   }
212
213   async function analyzeAudioBlob(audioBlob, filename) {
214     lastAudioFilename = filename;
215     const playerContainer = document.getElementById('vtab-play
er-container');
216     const audioElement = document.getElementById('vtab-audio-e
lement');
217     const filenameLabel = document.getElementById('vtab-curren
t-filename');
218     const loadingState = document.getElementById('vtab-loading
');
219     const emptyState = document.getElementById('vtab-empty-sta
te');
```

```
    const analysisContent = document.getElementById('vtab-analysis-content');
    const saveRagBtn = document.getElementById('vtab-btn-save-rag');

    // Show audio player
    if (playerContainer && audioElement) {
      const audioUrl = URL.createObjectURL(audioBlob);
      audioElement.src = audioUrl;
      if (filenameLabel) filenameLabel.textContent = filename;
      playerContainer.classList.remove('d-none');
    }

    if (emptyState) emptyState.classList.add('d-none');
    if (analysisContent) analysisContent.classList.add('d-none');
    if (loadingState) loadingState.classList.remove('d-none');
    if (saveRagBtn) saveRagBtn.classList.add('d-none');

    const model = document.getElementById('vtab-model-select')?.value || 'gemini-3.1-flash';
    const lang = document.getElementById('vtab-lang-select')?.value || 'ru';
    const apiKey = document.getElementById('vtab-api-key')?.value || '';

    const formData = new FormData();
    formData.append('file', audioBlob, filename);
    formData.append('model', model);
    formData.append('language', lang);
    if (apiKey) formData.append('api_key', apiKey);

    try {
      const res = await fetch('/api/audio/diarize', {
        method: 'POST',
        body: formData,
      });

```

```
      const json = await res.json();
      if (!res.ok) throw new Error(json.detail || 'Analysis error');

      lastAnalysisData = json.data || {};
      renderResults(lastAnalysisData);
      if (saveRagBtn) saveRagBtn.classList.remove('d-none');
    } catch (err) {
      console.error('[VoiceTab] Diarization failed:', err);
      if (analysisContent) {
        analysisContent.innerHTML = `
          <div class="alert alert-danger p-3 mb-0">
            <h6 class="fw-bold mb-1"><i class="bi bi-exclamation-triangle-fill me-2"></i>Ошибка анализа аудио</h6>
            <p class="mb-0 small">${escapeHtml(err.message)}</p>
          </div>
        `;
        analysisContent.classList.remove('d-none');
      }
    } finally {
      if (loadingState) loadingState.classList.add('d-none');
    }
  }

  function renderResults(data) {
    const analysisContent = document.getElementById('vtab-analysis-content');
    if (!analysisContent) return;

    const summary = data.summary || '';
    const keyPoints = data.key_points || [];
    const actionItems = data.action_items || [];
    const transcript = data.transcript || [];
    const speakers = data.speakers || [];

    const speakerColors = [
      '#3b82f6', '#10b981', '#f59e0b', '#ec4899', '#8b5cf6', '
```

```
#06b6d4'
    ];

    let keyPointsHtml = '';
    if (keyPoints.length > 0) {
      keyPointsHtml = `
        <div class="card bg-body-tertiary border-0 mb-3 shadow-sm">
          <div class="card-body p-3">
            <h6 class="fw-bold text-info mb-2"><i class="bi bi-key-fill me-2"></i>Ключевые темы и решения</h6>
            <ul class="mb-0 ps-3 small text-body-secondary">
              ${keyPoints.map(kp => `<li class="mb-1">${escapeHtml(kp)}</li>`).join('')}
            </ul>
          </div>
        </div>
      `;
    }

    let actionItemsHtml = '';
    if (actionItems.length > 0) {
      actionItemsHtml = `
        <div class="card bg-body-tertiary border-0 mb-3 shadow-sm">
          <div class="card-body p-3">
            <h6 class="fw-bold text-success mb-2"><i class="bi bi-check2-square me-2"></i>Задачи и договоренности (Action Items)</h6>
            <ul class="list-unstyled mb-0 small">
              ${actionItems.map(ai => `
                <li class="d-flex align-items-start gap-2 mb-1">
                  <span class="badge bg-success bg-opacity-25 text-success font-monospace">TODO</span>
                  <span>${escapeHtml(ai)}</span>
                </li>
              `).join('')}
```

```
            </ul>
          </div>
        </div>
      `;
    }

    let transcriptHtml = '';
    if (transcript.length > 0) {
      transcriptHtml = `
        <div class="card bg-body-tertiary border-0 shadow-sm">
          <div class="card-header bg-dark text-white d-flex ju
stify-content-between align-items-center py-2">
            <span class="fw-semibold small"><i class="bi bi-ch
at-quote-fill me-2 text-warning"></i>Стенограмма по собеседник
ам (${speakers.length || 2} участников)</span>
          </div>
          <div class="card-body p-3 d-flex flex-column gap-2"
style="max-height: 400px; overflow-y: auto;">
            ${transcript.map((t, idx) => {
              const sp = t.speaker || `Собеседник ${idx % 2 +
1}`;
              const col = speakerColors[idx % speakerColors.le
ngth];
              const ts = t.timestamp ? `<span class="badge bg-
dark text-muted font-monospace">${escapeHtml(t.timestamp)}</sp
an>` : '';
              return `
                <div class="p-2 rounded bg-body border-start b
order-3 shadow-sm" style="border-left-color: ${col} !important
;">
                  <div class="d-flex align-items-center justif
y-content-between mb-1">
                    <span class="fw-bold small" style="color:
${col};"><i class="bi bi-person-fill me-1"></i>${escapeHtml(sp
)}</span>
                    ${ts}
                  </div>
                  <div class="small text-body">${escapeHtml(t.
```

```
text)}</div>
                </div>
              `;
            }).join('')}
          </div>
        </div>
      `;
    }

    analysisContent.innerHTML = `
      ${summary ? `
        <div class="alert alert-primary mb-3 shadow-sm">
          <h6 class="fw-bold mb-1"><i class="bi bi-card-text m
e-2"></i>Краткая сводка (Executive Summary)</h6>
          <p class="mb-0 small">${escapeHtml(summary)}</p>
        </div>
      ` : ''}
      ${keyPointsHtml}
      ${actionItemsHtml}
      ${transcriptHtml}
    `;

    analysisContent.classList.remove('d-none');
  }

  async function saveResultToRag() {
    if (!lastAnalysisData) return;

    try {
      const res = await fetch('/api/audio/save-to-rag', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({
          title: `Голосовая запись: ${lastAudioFilename}`,
          summary: lastAnalysisData.summary || '',
          transcript: lastAnalysisData.transcript || [],
          key_points: lastAnalysisData.key_points || [],
          action_items: lastAnalysisData.action_items || [],
```

```
          markdown_report: lastAnalysisData.markdown_report || '',
        }),
      });

      const json = await res.json();
      if (!res.ok) throw new Error(json.detail || 'Save error');

      alert(`✅ ${json.message || 'Разговор успешно сохранен в базу знаний RAG!'}`);
    } catch (err) {
      console.error('[VoiceTab] Failed to save in RAG:', err);
      alert(`Ошибка сохранения в RAG: ${err.message}`);
    }
  }

  function escapeHtml(text) {
    if (!text) return '';
    return String(text)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
  }

  if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', initVoiceTab);
  } else {
    initVoiceTab();
  }
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/website_monitor_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/website_monitor_tab/ind
ex.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/website_monitor_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Website Intelligence Monitor Application Tab -->
<style>
  .webmon-container {
    padding: 0.5rem 0.75rem;
  }
  .webmon-card {
    background: var(--surface-1) !important;
    border: 1.5px solid var(--border-color) !important;
    border-radius: var(--radius-md, 8px);
    padding: var(--kpi-padding, 0.45rem 0.75rem);
```

```
    display: flex;
    flex-direction: column;
    justify-content: space-between;
    min-height: var(--kpi-min-height, 58px);
    gap: 0.2rem;
    box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.08)
);
  }
  .webmon-label {
    font-size: var(--kpi-label-size, 0.68rem);
    text-transform: uppercase;
    letter-spacing: 0.05em;
    font-weight: 600;
    color: var(--text-muted);
    line-height: 1.1;
  }
  .webmon-value {
    font-size: var(--kpi-val-size, 1.25rem);
    font-weight: 700;
    color: var(--text-color);
    font-family: var(--font-mono, monospace);
    line-height: 1.15;
  }
  .webmon-panel {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: var(--radius-md, 8px);
    overflow: hidden;
    box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.08)
);
  }
  .webmon-panel-header {
    background: var(--surface-2) !important;
    border-bottom: 1px solid var(--border-color) !important;
    padding: 0.45rem 0.85rem;
    font-weight: 600;
    font-size: 0.82rem;
    color: var(--text-color);
```

```
    display: flex;
    align-items: center;
    justify-content: space-between;
  }
  .webmon-table {
    width: 100%;
    font-size: 0.78rem;
    font-family: var(--font-mono, monospace);
    border-collapse: collapse;
    background: transparent;
  }
  .webmon-table th {
    background: var(--surface-2);
    padding: 5px 8px;
    color: var(--text-muted);
    font-weight: 600;
    border-bottom: 1px solid var(--border-color);
  }
  .webmon-table td {
    padding: 4px 8px;
    border-bottom: 1px solid var(--border-subtle, rgba(128, 128, 128, 0.2));
    color: var(--text-color);
  }
</style>
<div class="webmon-container">
<!-- Header Alert & Status (Compact Strip) -->
<div class="card shadow-sm border-secondary-subtle py-1.5 px-3 mb-2 d-flex justify-content-between align-items-center flex-wrap gap-2" data-panel-id="panel-website-monitor-website-intelligence" id="panel-website-monitor-website-intelligence">
<div class="d-flex align-items-center gap-2 flex-wrap">
<span class="fs-5">□ </span>
<span class="fw-bold text-body small" style="font-size: 0.92rem;">Website Intelligence</span>
<span class="badge bg-body-secondary border border-secondary text-info fw-normal" style="font-size: 0.7rem;">Traffic Monitor</span>
```

```
<span class="text-secondary small d-none d-lg-inline" style="font-size: 0.76rem;">| GA4 Realtime, Google Search Console CTR &amp;amp; аномалии</span>
</div>
<div class="d-flex align-items-center gap-1.5 flex-wrap">
<span class="badge rounded-pill bg-success-subtle text-success border border-success px-2 py-1" id="webmon-status-badge" style="font-size: 0.72rem;">
        • Активен
      </span>
<button class="btn btn-xs btn-outline-info rounded-pill px-2.5 py-1" id="btn-webmon-refresh" style="font-size: 0.75rem;">
<i class="bi bi-arrow-clockwise me-1"></i> Обновить
      </button>
<button class="btn btn-xs btn-outline-warning rounded-pill px-2 py-1" id="btn-webmon-config" style="font-size: 0.75rem;">
<i class="bi bi-gear-fill me-1"></i> Настройки
      </button>
</div>
</div>
<!-- Top Metrics Bar (Compact Density Grid) -->
<div class="row g-2 mb-2">
<div class="col-6 col-md-3">
<div class="webmon-card">
<span class="webmon-label">Онлайн (Realtime)</span>
<div class="d-flex align-items-baseline justify-content-between mt-1">
<span class="webmon-value text-success" id="webmon-realtime-users">0</span>
<small class="text-muted" style="font-size: 0.68rem;">Пользователей</small>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="webmon-card">
<span class="webmon-label">Доступность / Ping</span>
<div class="d-flex align-items-baseline justify-content-betwee
```

```
n mt-1">
<span class="webmon-value text-info" id="webmon-availability" style="font-size: 0.95rem;">100% (24ms)</span>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="webmon-card">
<span class="webmon-label">GSC CTR</span>
<div class="d-flex align-items-baseline justify-content-between mt-1">
<span class="webmon-value text-warning" id="webmon-ctr-score">4.2%</span>
<small class="text-muted" style="font-size: 0.68rem;">Search Console</small>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="webmon-card">
<span class="webmon-label">Индекс здоровья</span>
<div class="d-flex align-items-baseline justify-content-between mt-1">
<span class="webmon-value text-body" id="webmon-health-score">98/100</span>
</div>
</div>
</div>
</div>
<!-- Main Content Panels Grid -->
<div class="row g-3">
<!-- Top Pages Panel -->
<div class="col-lg-6">
<div class="webmon-panel h-100">
<div class="webmon-panel-header">
<span>  Популярные страницы (GA4)</span>
<span class="badge bg-secondary" id="webmon-pages-badge">Топ-10</span>
```

```
</div>
<div class="table-responsive">
<table class="webmon-table">
<thead>
<tr>
<th>Путь страницы</th>
<th class="text-end">Просмотры</th>
<th class="text-end">Пользователи</th>
</tr>
</thead>
<tbody id="webmon-pages-table-body">
<tr><td class="text-center text-muted p-3" colspan="3">Загрузка популярных страниц...</td></tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- Traffic Sources & Diagnostics Panel -->
<div class="col-lg-6">
<div class="webmon-panel h-100">
<div class="webmon-panel-header">
<span>  Каналы трафика и AI Диагностика</span>
<span class="badge bg-info text-dark">GA4 &amp;amp; Anomalies</span>
</div>
<div class="p-3">
<div class="mb-3" id="webmon-traffic-channels">
<div class="text-muted small text-center">Загрузка источников...</div>
</div>
<hr class="border-secondary my-2"/>
<h6 class="fw-bold text-info mb-1" id="webmon-diag-title">Анализ трендов трафика...</h6>
<p class="small text-muted mb-0" id="webmon-diag-summary">Ожидание нормализованного отчёта...</p>
</div>
</div>
```

```
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Website Monitor Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/website_monitor_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/website_monitor_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// ============================================================
// Process Name: Website Intelligence Monitor Web Tab Module
// ============================================================
// Description:
//   Client-side JavaScript controller for the Website Intelligence Monitor
//   administrative web interface tab.
//
// File: main.js
// Project: ai-windows-admin
```

```
// Package: src.api.webinterface.website_monitor_tab
// Author: hypo69
// Copyright: © 2026 hypo69
// ============================================================
==================

(function() {
  let isWebmonInitialized = false;

  async function fetchRealtime() {
    try {
      const res = await fetch('/api/v1/website-monitor/realtim
e');
      if (!res.ok) return;
      const data = await res.json();

      const usersEl = document.getElementById('webmon-realtime
-users');
      if (usersEl) usersEl.innerText = data.active_users || 0;
    } catch (e) {
      console.error('[WebsiteMonitorTab] Failed to fetch realt
ime:', e);
    }
  }

  async function fetchSummary() {
    try {
      const res = await fetch('/api/v1/website-monitor/summary
');
      if (!res.ok) return;
      const data = await res.json();

      const availEl = document.getElementById('webmon-availabi
lity');
      const ctrEl = document.getElementById('webmon-ctr-score'
);
      const healthEl = document.getElementById('webmon-health-
score');
```

```
      const diagTitle = document.getElementById('webmon-diag-title');
      const diagSummary = document.getElementById('webmon-diag-summary');

      if (data.technical) {
        if (availEl) availEl.innerText = `${data.technical.availability_pct || 100}% (${data.technical.avg_latency_ms || 0} ms)`;
      }
      if (data.search_console) {
        if (ctrEl) ctrEl.innerText = `${data.search_console.avg_ctr || 0}%`;
      }
      if (healthEl) healthEl.innerText = `${data.health_score || 100}/100`;

      if (diagTitle) diagTitle.innerText = `Статус: ${data.status || 'OK'}`;
      if (diagSummary) diagSummary.innerText = data.summary || 'Все веб-сервисы и каналы трафика функционируют стабильно.';
    } catch (e) {
      console.error('[WebsiteMonitorTab] Failed to fetch summary:', e);
    }
  }

  async function fetchPages() {
    try {
      const res = await fetch('/api/v1/website-monitor/pages?limit=8');
      if (!res.ok) return;
      const pages = await res.json();

      const tbody = document.getElementById('webmon-pages-table-body');
      if (tbody) {
        if (!Array.isArray(pages) || pages.length === 0) {
```

```
          tbody.innerHTML = '<tr><td colspan="3" class="text-center text-muted p-3">Нет данных о просмотрах.</td></tr>';
          return;
        }
        tbody.innerHTML = pages.map((p, idx) => `
          <tr class="webmon-page-row" data-idx="${idx}" style="cursor: pointer;" title="Нажмите для AI-анализа страницы">
            <td class="text-info text-truncate" style="max-width: 240px;" title="${p.page_path || '/'}">${p.page_path || '/'}</td>
            <td class="text-end text-light">${p.screen_page_views || p.views || 0}</td>
            <td class="text-end text-secondary">${p.active_users || p.users || 0}</td>
          </tr>
        `).join('');

        tbody.querySelectorAll('.webmon-page-row').forEach(row => {
          row.onclick = () => {
            const idx = parseInt(row.getAttribute('data-idx'), 10);
            const p = pages[idx];
            if (!p) return;

            if (window.AITableModal) {
              window.AITableModal.show({
                icon: '□ ',
                title: `Веб-страница ${p.page_path || '/'}`,
                subtitle: `Просмотров: ${p.screen_page_views || p.views || 0} | Пользователей: ${p.active_users || p.users || 0}`,
                tableType: 'website',
                badges: [
                  { text: `${p.screen_page_views || 0} views`, class: 'badge bg-info text-dark' },
                  { text: `${p.active_users || 0} users`, class: 'badge bg-success' }
```

```
                ],
                metadata: [
                  { label: 'URL / Путь страницы', value: p.pag
e_path || '/' },
                  { label: 'Просмотры страниц', value: String(
p.screen_page_views || p.views || 0) },
                  { label: 'Активные пользователи', value: Str
ing(p.active_users || p.users || 0) },
                  { label: 'Показатель отказов', value: p.boun
ce_rate ? `${p.bounce_rate}%` : 'N/A' },
                  { label: 'Средняя длительность', value: p.av
g_session_duration ? `${p.avg_session_duration} сек` : 'N/A' }
                ],
                rawTitle: 'Аналитика веб-страницы',
                rawContent: JSON.stringify(p, null, 2),
                requestData: {
                  url: p.page_path,
                  views: p.screen_page_views || p.views,
                  users: p.active_users || p.users
                }
              });
            }
          };
        });
      }
    } catch (e) {
      console.error('[WebsiteMonitorTab] Failed to fetch pages
:', e);
    }
  }

  async function fetchTrafficSources() {
    try {
      const res = await fetch('/api/v1/website-monitor/traffic
-sources');
      if (!res.ok) return;
      const channels = await res.json();

```

```
      const container = document.getElementById('webmon-traffi
c-channels');
      if (container && Array.isArray(channels)) {
        container.innerHTML = channels.slice(0, 5).map(c => `
          <div class="d-flex justify-content-between align-ite
ms-center py-1 border-bottom border-secondary-subtle small">
            <span class="text-light">${c.channel_group || c.na
me || 'Organic'}</span>
            <span class="badge bg-secondary">${c.sessions || c
.users || 0} сессий</span>
          </div>
        `).join('');
      }
    } catch (e) {
      console.error('[WebsiteMonitorTab] Failed to fetch traff
ic sources:', e);
    }
  }

  function initWebsiteMonitorTab() {
    fetchRealtime();
    fetchSummary();
    fetchPages();
    fetchTrafficSources();

    if (!isWebmonInitialized) {
      const refreshBtn = document.getElementById('btn-webmon-r
efresh');
      if (refreshBtn) {
        refreshBtn.addEventListener('click', () => {
          fetchRealtime();
          fetchSummary();
          fetchPages();
          fetchTrafficSources();
        });
      }

      const configBtn = document.getElementById('btn-webmon-co
```

```
nfig');
      if (configBtn) {
        configBtn.addEventListener('click', () => {
          if (typeof window.openAppConfigModal === 'function')
 {
            window.openAppConfigModal('website_monitor');
          }
        });
      }
      isWebmonInitialized = true;
    }
  }

  window.initWebsiteMonitorTab = initWebsiteMonitorTab;
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/wikipedia_research_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/wikipedia_research_tab/
index.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/wikipedia_research_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!-- Wikipedia Research & Multi-Model Benchmark Laboratory Tab
 -->
<style>
  .wiki-lab-container {
    padding: 1rem;
  }
  .wiki-card {
    background: var(--surface-1) !important;
    border: 1.5px solid var(--border-color) !important;
```

```
    border-radius: var(--radius-md, 8px);
    padding: 1.25rem;
    margin-bottom: 1rem;
    box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.08)
);
  }
  .wiki-card-title {
    font-size: 1rem;
    font-weight: 700;
    color: var(--text-color);
    margin-bottom: 0.75rem;
    display: flex;
    align-items: center;
    gap: 0.5rem;
  }
  .wiki-badge-pos {
    background: var(--status-success-bg, rgba(34, 197, 94, 0.1
5));
    color: var(--status-success-text, #166534);
    border: 1px solid var(--status-success-border, #22c55e);
    padding: 2px 8px;
    border-radius: var(--radius-xs, 4px);
    font-weight: 600;
  }
  .wiki-badge-neg {
    background: var(--status-danger-bg, rgba(239, 68, 68, 0.15
));
    color: var(--status-danger-text, #991b1b);
    border: 1px solid var(--status-danger-border, #ef4444);
    padding: 2px 8px;
    border-radius: var(--radius-xs, 4px);
    font-weight: 600;
  }
  .wiki-badge-neu {
    background: var(--status-warning-bg, rgba(234, 179, 8, 0.1
5));
    color: var(--status-warning-text, #92400e);
    border: 1px solid var(--status-warning-border, #eab308);
```

```
    padding: 2px 8px;
    border-radius: var(--radius-xs, 4px);
    font-weight: 600;
  }
  .wiki-table {
    width: 100%;
    font-size: 0.85rem;
    border-collapse: collapse;
    color: var(--text-color);
  }
  .wiki-table th {
    background: var(--surface-2);
    padding: 8px 12px;
    color: var(--text-muted);
    font-weight: 600;
    border-bottom: 1px solid var(--border-color);
    text-align: left;
  }
  .wiki-table td {
    padding: 8px 12px;
    border-bottom: 1px solid var(--border-subtle, rgba(128, 12
8, 128, 0.2));
  }
  .wiki-table tr:hover {
    background: rgba(56, 189, 248, 0.06);
  }
  .wiki-checkbox-grid {
    display: grid;
    grid-template-columns: repeat(auto-fill, minmax(130px, 1fr
));
    gap: 0.5rem;
  }
</style>

<div class="wiki-lab-container">
  <!-- Header -->
  <div class="d-flex justify-content-between align-items-cente
r mb-3">
```

```
    <div>
      <h4 class="fw-bold text-body mb-0">🌐 Wikipedia Research & Model Laboratory</h4>
      <small class="text-muted">Кросс-языковой сравнительный анализ контента Википедии и бенчмарк интерпретации моделей ИИ</small>
    </div>
    <div class="d-flex gap-2">
      <button class="btn btn-outline-info btn-sm" id="btn-wiki-health" title="Проверить статус API">
        <i class="bi bi-heart-pulse"></i> Статус
      </button>
    </div>
  </div>

  <!-- Experiment Configuration Panel -->
  <div class="row g-3 mb-3">
    <div class="col-lg-8">
      <div class="wiki-card">
        <div class="wiki-card-title">
          <i class="bi bi-sliders text-info"></i> Параметры исследования
        </div>

        <!-- Topic Search -->
        <div class="mb-3">
          <label class="form-label text-body small fw-bold">Тема для исследования (Wikipedia Topic):</label>
          <div class="input-group">
            <span class="input-group-text bg-body-secondary border-secondary text-info"><i class="bi bi-search"></i></span>
            <input type="text" class="form-control" id="wiki-topic-input" placeholder="Например: Israel–Gaza war, Quantum computing, Artificial intelligence" value="Israel–Gaza war">
            <button class="btn btn-primary px-3" id="btn-run-exp-a" title="Запустить кросс-языковое сравнение одной моделью">
              <i class="bi bi-translate"></i> Exp A: Языки
```

```
            </button>
            <button class="btn btn-success px-3" id="btn-run-exp-b" title="Запустить кросс-модельный бенчмарк">
              <i class="bi bi-cpu"></i> Exp B: Модели
            </button>
          </div>
        </div>

        <!-- Languages Selector -->
        <div class="mb-3">
          <label class="form-label text-body small fw-bold">Языковые разделы Википедии:</label>
          <div class="wiki-checkbox-grid" id="wiki-languages-list">
            <div class="form-check">
              <input class="form-check-input wiki-lang-check" type="checkbox" value="en" id="lang-en" checked>
              <label class="form-check-label text-body small" for="lang-en">English (en)</label>
            </div>
            <div class="form-check">
              <input class="form-check-input wiki-lang-check" type="checkbox" value="ru" id="lang-ru" checked>
              <label class="form-check-label text-body small" for="lang-ru">Русский (ru)</label>
            </div>
            <div class="form-check">
              <input class="form-check-input wiki-lang-check" type="checkbox" value="he" id="lang-he" checked>
              <label class="form-check-label text-body small" for="lang-he">□ □ □ □ □ (he)</label>
            </div>
            <div class="form-check">
              <input class="form-check-input wiki-lang-check" type="checkbox" value="de" id="lang-de" checked>
              <label class="form-check-label text-body small" for="lang-de">Deutsch (de)</label>
            </div>
```

```
155             <div class="form-check">
156               <input class="form-check-input wiki-lang-check"
type="checkbox" value="fr" id="lang-fr" checked>
157               <label class="form-check-label text-body small"
for="lang-fr">Français (fr)</label>
158             </div>
159             <div class="form-check">
160               <input class="form-check-input wiki-lang-check"
type="checkbox" value="uk" id="lang-uk">
161               <label class="form-check-label text-body small"
for="lang-uk">Українська (uk)</label>
162             </div>
163           </div>
164         </div>
165
166         <!-- Models & Analysis Dimensions -->
167         <div class="row">
168           <div class="col-md-6">
169             <label class="form-label text-body small fw-bold">
AI Модели (Experiment B):</label>
170             <div class="d-flex flex-wrap gap-2" id="wiki-model
s-list">
171               <div class="form-check form-check-inline">
172                 <input class="form-check-input wiki-model-chec
k" type="checkbox" value="gemini" id="mod-gemini" checked>
173                 <label class="form-check-label text-body small
" for="mod-gemini">Gemini</label>
174               </div>
175               <div class="form-check form-check-inline">
176                 <input class="form-check-input wiki-model-chec
k" type="checkbox" value="foundry" id="mod-foundry" checked>
177                 <label class="form-check-label text-body small
" for="mod-foundry">Foundry Local</label>
178               </div>
179               <div class="form-check form-check-inline">
180                 <input class="form-check-input wiki-model-chec
k" type="checkbox" value="ollama" id="mod-ollama" checked>
181                 <label class="form-check-label text-body small
```

```
" for="mod-ollama">Ollama</label>
              </div>
              <div class="form-check form-check-inline">
                <input class="form-check-input wiki-model-chec
k" type="checkbox" value="onnx" id="mod-onnx">
                <label class="form-check-label text-body small
" for="mod-onnx">ONNX DirectML</label>
              </div>
            </div>
          </div>
          <div class="col-md-6">
            <label class="form-label text-body small fw-bold">
Базовая модель для Exp A:</label>
            <select class="form-select form-select-sm" id="wik
i-primary-model">
              <option value="gemini" selected>Google Gemini</o
ption>
              <option value="foundry">Microsoft Foundry Local<
/option>
              <option value="ollama">Ollama Local LLM</option>
              <option value="onnx">ONNX DirectML</option>
            </select>
          </div>
        </div>
      </div>
    </div>

    <!-- Quick Insights & Stats -->
    <div class="col-lg-4">
      <div class="wiki-card h-100">
        <div class="wiki-card-title">
          <i class="bi bi-lightbulb text-warning"></i> Аналити
ческие инсайты
        </div>
        <div id="wiki-insights-container" class="text-secondar
y small">
          <em>Запустите Experiment A или Experiment B для отоб
ражения выводов и кросс-сравнения...</em>
```

```
        </div>
      </div>
    </div>
  </div>

  <!-- Loading State -->
  <div id="wiki-loading-spinner" class="text-center my-4 d-none">
    <div class="spinner-border text-primary" role="status"></div>
    <div class="text-muted mt-2 small" id="wiki-loading-text">Выполняется сбор статей и AI-анализ...</div>
  </div>

  <!-- Results Section -->
  <div id="wiki-results-section" class="d-none">
    <div class="wiki-card">
      <div class="wiki-card-title" id="wiki-results-title">
        <i class="bi bi-bar-chart-fill text-info"></i> Результаты эксперимента
      </div>

      <div class="table-responsive">
        <div id="wiki-results-table-container"></div>
      </div>
    </div>
  </div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Wikipedia Research Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/wikipedia_research_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/wikipedia_research_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// Wikipedia Research & Multi-Model Benchmark Laboratory WebGUI JS
(function () {
  'use strict';

  console.log('[WikiLab] Initializing Wikipedia Research Tab JS...');

  async function checkHealth() {
    try {
      const res = await fetch('/api/v1/wikipedia-research/health');
```

```
      if (res.ok) {
        const data = await res.json();
        alert(`Wikipedia Research API статус: ${data.status.to
UpperCase()} (приложение: ${data.app})`);
      } else {
        alert(`Ошибка API: HTTP ${res.status}`);
      }
    } catch (e) {
      alert(`Не удалось связаться с API: ${e.message}`);
    }
  }

  function getSelectedLanguages() {
    const checkboxes = document.querySelectorAll('.wiki-lang-c
heck:checked');
    const langs = [];
    checkboxes.forEach((cb) => langs.push(cb.value));
    return langs.length > 0 ? langs : ['en', 'ru'];
  }

  function getSelectedModels() {
    const checkboxes = document.querySelectorAll('.wiki-model-
check:checked');
    const models = [];
    checkboxes.forEach((cb) => models.push(cb.value));
    return models.length > 0 ? models : ['gemini', 'foundry'];
  }

  function formatSentimentBadge(val) {
    if (val === undefined || val === null) return '<span class
="text-muted">n/a</span>';
    const num = parseFloat(val);
    if (num > 0.05) {
      return `<span class="wiki-badge-pos">+${num.toFixed(2)}<
/span>`;
    } else if (num < -0.05) {
      return `<span class="wiki-badge-neg">${num.toFixed(2)}</
span>`;
```

```
    }
    return `<span class="wiki-badge-neu">${num.toFixed(2)}</span>`;
  }

  function setLoading(loading, text = 'Выполняется сбор и анализ...') {
    const spinner = document.getElementById('wiki-loading-spinner');
    const loadingText = document.getElementById('wiki-loading-text');
    const resultsSec = document.getElementById('wiki-results-section');
    if (spinner) {
      if (loading) {
        spinner.classList.remove('d-none');
        if (loadingText) loadingText.textContent = text;
        if (resultsSec) resultsSec.classList.add('d-none');
      } else {
        spinner.classList.add('d-none');
        if (resultsSec) resultsSec.classList.remove('d-none');
      }
    }
  }

  function renderInsights(insights) {
    const container = document.getElementById('wiki-insights-container');
    if (!container) return;
    if (!insights || insights.length === 0) {
      container.innerHTML = '<div class="text-muted small">Нет выявленных значимых расхождений.</div>';
      return;
    }
    let html = '<ul class="list-unstyled mb-0 d-flex flex-column gap-2">';
    insights.forEach((ins) => {
      html += `<li class="d-flex align-items-start gap-2 text-
```

```
light"><i class="bi bi-check-circle-fill text-info mt-1"></i>
<span>${ins}</span></li>`;
    });
    html += '</ul>';
    container.innerHTML = html;
  }

  async function runExperimentA() {
    const topicInput = document.getElementById('wiki-topic-inp
ut');
    const topic = topicInput ? topicInput.value.trim() : 'Isra
el–Gaza war';
    const languages = getSelectedLanguages();
    const model = document.getElementById('wiki-primary-model'
)?.value || 'gemini';

    setLoading(true, `Эксперимент A: сбор статей по теме "${to
pic}" и анализ моделью ${model}...`);

    try {
      const response = await fetch('/api/v1/wikipedia-research
/experiment/languages', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ topic, languages, model }),
      });

      if (!response.ok) {
        throw new Error(`HTTP ${response.status}: ${await resp
onse.text()}`);
      }

      const report = await response.json();
      renderExperimentAResults(report);
    } catch (err) {
      alert(`Ошибка выполнения Эксперимента A: ${err.message}`
);
    } finally {
```

```
      setLoading(false);
    }
  }

  function renderExperimentAResults(report) {
    const titleEl = document.getElementById('wiki-results-titl
e');
    if (titleEl) {
      titleEl.innerHTML = `<i class="bi bi-translate"></i> Exp
eriment A: Кросс-языковое сравнение темы «${report.topic}» (Мо
дель: ${report.model_used})`;
    }

    renderInsights(report.cross_language_insights);

    const tableContainer = document.getElementById('wiki-resul
ts-table-container');
    if (!tableContainer) return;

    let html = `
      <table class="wiki-table">
        <thead>
          <tr>
            <th>Язык</th>
            <th>Статья</th>
            <th>Тональность</th>
            <th>Субъективность</th>
            <th>Критика</th>
            <th>Похвала</th>
            <th>Неопределенность</th>
            <th>Спорные тезисы</th>
            <th>Фрейминг</th>
            <th>Слов</th>
          </tr>
        </thead>
        <tbody>
    `;

```

```
    (report.metrics_summary_table || []).forEach((row) => {
      html += `
        <tr>
          <td><strong class="text-info">${row.lang.toUpperCase()}</strong></td>
          <td><a href="${row.url}" target="_blank" class="text-light text-decoration-underline">${row.title}</a></td>
          <td>${formatSentimentBadge(row.sentiment)}</td>
          <td><span class="text-warning">${row.subjectivity.toFixed(2)}</span></td>
          <td><span class="text-danger">${row.criticism.toFixed(2)}</span></td>
          <td><span class="text-success">${row.praise.toFixed(2)}</span></td>
          <td><span class="text-primary">${row.uncertainty.toFixed(2)}</span></td>
          <td class="text-center"><span class="badge bg-secondary">${row.claims_count}</span></td>
          <td><span class="badge bg-dark border border-secondary text-info">${row.framing}</span></td>
          <td class="text-muted small">${row.word_count.toLocaleString()}</td>
        </tr>
      `;
    });

    html += `</tbody></table>`;
    tableContainer.innerHTML = html;
  }

  async function runExperimentB() {
    const topicInput = document.getElementById('wiki-topic-input');
    const topic = topicInput ? topicInput.value.trim() : 'Israel-Gaza war';
    const languages = getSelectedLanguages();
    const models = getSelectedModels();

```

```
    setLoading(true, `Эксперимент B: сопоставление интерпретаций моделями [${models.join(', ')}]...`);

    try {
      const response = await fetch('/api/v1/wikipedia-research/experiment/models', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ topic, languages, models }),
      });

      if (!response.ok) {
        throw new Error(`HTTP ${response.status}: ${await response.text()}`);
      }

      const report = await response.json();
      renderExperimentBResults(report);
    } catch (err) {
      alert(`Ошибка выполнения Эксперимента B: ${err.message}`);
    } finally {
      setLoading(false);
    }
  }

  function renderExperimentBResults(report) {
    const titleEl = document.getElementById('wiki-results-title');
    if (titleEl) {
      titleEl.innerHTML = `<i class="bi bi-cpu"></i> Experiment B: Мульти-модельный бенчмарк интерпретации «${report.topic}»`;
    }

    renderInsights(report.model_alignment_insights);

    const tableContainer = document.getElementById('wiki-resul
```

```
ts-table-container');
    if (!tableContainer) return;

    let html = `
      <table class="wiki-table">
        <thead>
          <tr>
            <th>Языковой раздел</th>
    `;

    report.models.forEach((mod) => {
      html += `<th>Модель: ${mod}</th>`;
    });

    html += `
          </tr>
        </thead>
        <tbody>
    `;

    report.languages.forEach((lang) => {
      html += `<tr><td><strong class="text-warning">${lang.toU
pperCase()}</strong></td>`;
      report.models.forEach((mod) => {
        const sentiment = report.sentiment_grid?.[lang]?.[mod]
;
        const res = report.matrix?.[lang]?.[mod];
        const framing = res?.metrics?.framing_tone || 'n/a';
        html += `
          <td>
            <div class="d-flex flex-column gap-1">
              <div>${formatSentimentBadge(sentiment)}</div>
              <small class="text-muted" style="font-size: 0.75
rem;">(${framing})</small>
            </div>
          </td>
        `;
      });
```

```
      html += `</tr>`;
    });

    html += `</tbody></table>`;
    tableContainer.innerHTML = html;
  }

  function bindEvents() {
    const btnHealth = document.getElementById('btn-wiki-health
');
    if (btnHealth) btnHealth.onclick = checkHealth;

    const btnExpA = document.getElementById('btn-run-exp-a');
    if (btnExpA) btnExpA.onclick = runExperimentA;

    const btnExpB = document.getElementById('btn-run-exp-b');
    if (btnExpB) btnExpB.onclick = runExperimentB;
  }

  window.initWikipediaResearchTab = function () {
    console.log('[WikiLab] initWikipediaResearchTab triggered'
);
    bindEvents();
  };

  if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', bindEvents);
  } else {
    bindEvents();
  }
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/windows_admin_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/windows_admin_tab/index
.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/windows_admin_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!-- Windows System Administrator Application Tab -->
<style>
  .winadmin-container {
    padding: 0.5rem 0.75rem;
  }
  .winadmin-table {
    width: 100%;
    font-size: 0.78rem;
    font-family: var(--font-mono, monospace);
    border-collapse: collapse;
```

```
  }
  .winadmin-table th {
    background: var(--surface-2);
    padding: 5px 8px;
    color: var(--text-muted);
    font-weight: 600;
    border-bottom: 1px solid var(--border-color);
  }
  .winadmin-table td {
    padding: 4px 8px;
    border-bottom: 1px solid var(--border-subtle, rgba(51, 65, 85, 0.4));
  }
  .winadmin-filter-btn.active {
    background-color: #3b82f6 !important;
    color: #fff !important;
    border-color: #3b82f6 !important;
  }
  .badge-hidden {
    background-color: var(--bs-secondary-bg-subtle, rgba(100, 116, 139, 0.2));
    color: var(--bs-secondary-text-emphasis, #94a3b8);
    border: 1px solid var(--bs-border-color, rgba(148, 163, 184, 0.3));
  }
  .badge-admin {
    background-color: var(--bs-warning-bg-subtle, rgba(234, 88, 12, 0.2));
    color: var(--bs-warning-text-emphasis, #f59e0b);
    border: 1px solid var(--bs-warning-border-subtle, rgba(245, 158, 11, 0.4));
  }
  .badge-disabled {
    background-color: var(--bs-danger-bg-subtle, rgba(239, 68, 68, 0.2));
    color: var(--bs-danger-text-emphasis, #f87171);
    border: 1px solid var(--bs-danger-border-subtle, rgba(239, 68, 68, 0.4));
```

```
    font-weight: bold;
  }
</style>
<div class="winadmin-container theme-sysadmin">
<!-- Top Metrics Bar (Compact Density Grid) -->
<div class="row g-2 mb-2">
<div class="col-6 col-md-3">
<div class="winadmin-card">
<span class="winadmin-label">Хост / Домен</span>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<span class="winadmin-value text-truncate" id="winadmin-hostna
me" style="font-size: 0.95rem;">-</span>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="winadmin-card">
<span class="winadmin-label">Active Directory</span>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<span class="winadmin-value text-info text-truncate" id="winad
min-ad-status" style="font-size: 0.9rem;">Локальный хост</span
>
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="winadmin-card">
<span class="winadmin-label">Пользователи (Все / Онлайн)</span
>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<span class="winadmin-value text-success" id="winadmin-user-co
unt">0</span>
<span class="badge bg-dark border border-secondary text-muted
small" id="winadmin-user-breakdown" style="font-size: 0.7rem;"
>0 скрытых</span>
```

```
</div>
</div>
</div>
<div class="col-6 col-md-3">
<div class="winadmin-card">
<span class="winadmin-label">События безопасности</span>
<div class="d-flex align-items-baseline justify-content-betwee
n mt-1">
<span class="winadmin-value text-warning" id="winadmin-events-
count">0</span>
</div>
</div>
</div>
</div>
<!-- Main Panels Row 1: Users & Security Log -->
<div class="row g-3 mb-3">
<!-- User Accounts & Sessions Panel -->
<div class="col-lg-7">
<div class="winadmin-panel h-100" data-panel-id="panel-winadmi
n-users" id="panel-winadmin-users">
<div class="winadmin-panel-header d-flex justify-content-betwe
en align-items-center flex-wrap gap-2">
<div class="d-flex align-items-center gap-2">
<span><i class="bi bi-people-fill me-1"></i> Учетные записи и
сессии Windows</span>
<span class="badge bg-secondary" id="winadmin-users-badge">0 а
ккаунтов</span>
</div>
<!-- Filter Buttons Toolbar -->
<div class="btn-group btn-group-sm" id="winadmin-user-filters"
 role="group">
<button class="btn btn-outline-secondary py-0 px-2 winadmin-fi
lter-btn active" data-filter="all" type="button">Все</button>
<button class="btn btn-outline-secondary py-0 px-2 winadmin-fi
lter-btn" data-filter="active" type="button">В сети</button>
<button class="btn btn-outline-secondary py-0 px-2 winadmin-fi
lter-btn" data-filter="hidden" type="button">Скрытые</button>
<button class="btn btn-outline-secondary py-0 px-2 winadmin-fi
```

```
lter-btn" data-filter="admins" type="button">Админы</button>
</div>
</div>
<div class="p-0 table-responsive" style="max-height: 380px; ov
erflow-y: auto;">
<table class="winadmin-table">
<thead>
<tr>
<th>Учетная запись</th>
<th>Группы / SID</th>
<th>Профиль / Диск</th>
<th>Ресурсы</th>
<th>Статус</th>
<th style="text-align: right;">Действие</th>
</tr>
</thead>
<tbody id="winadmin-users-tbody">
<tr><td class="text-center text-muted p-3" colspan="6">Загрузк
а пользователей и метрик...</td></tr>
</tbody>
</table>
</div>
</div>
</div>
<!-- Security Events Audit Log Panel -->
<div class="col-lg-5">
<div class="winadmin-panel h-100" data-panel-id="panel-winadmi
n-security-events" id="panel-winadmin-security-events">
<div class="winadmin-panel-header">
<span><i class="bi bi-shield-exclamation me-1"></i> Общий журн
ал безопасности</span>
<span class="badge bg-secondary" id="winadmin-events-badge">0
записей</span>
</div>
<div class="p-0 table-responsive" style="max-height: 380px; ov
erflow-y: auto;">
<table class="winadmin-table">
<thead>
```

```
<tr>
<th>Время</th>
<th>Event ID</th>
<th>Уровень</th>
<th>Источник / Описание</th>
</tr>
</thead>
<tbody id="winadmin-events-tbody">
<tr><td class="text-center text-muted p-3" colspan="4">Загрузка журнала событий...</td></tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
</div>

```

main.js

```
/**
 * ============================================================
 * Process Name: Windows Windows Admin Tab - Main Script
 * ============================================================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/windows_admin_tab/main.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/windows_admin_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
 */

// ============================================================
// Process Name: Windows System Administrator Web Tab Module
// ============================================================
// Description:
//   Клиентский контроллер вкладки Windows System Administrator:
//   - Отображение всех локальных и доменных учетных записей Windows
//   - Обнаружение скрытых пользователей (SpecialAccounts UserList)
```

```
//   - Подробная телеметрия ресурсов (память, CPU, процессы, профили на диске)
//   - Фильтрация (Все, В сети, Скрытые, Администраторы)
//   - Интерактивное досье с поддержкой AI-аудита прав и безопасности
//
// File: main.js
// Project: ai-windows-admin
// Package: src.api.webgui.windows_admin_tab
// Author: hypo69
// Copyright: © 2026 hypo69
// ============================================================

(function() {
  let isWinAdminInitialized = false;
  let currentUserFilter = 'all';

  async function fetchStatus() {
    try {
      const res = await fetch('/api/sysadmin/status');
      if (!res.ok) return;
      const data = await res.json();

      const hostEl = document.getElementById('winadmin-hostname');
      const userCountEl = document.getElementById('winadmin-user-count');
      const userBreakdownEl = document.getElementById('winadmin-user-breakdown');
      const adStatusEl = document.getElementById('winadmin-ad-status');
      const eventsCountEl = document.getElementById('winadmin-events-count');

      if (hostEl) hostEl.innerText = `${data.hostname || 'LOCAL'} / ${data.domain || 'WORKGROUP'}`;
      if (userCountEl) {
```

```
        userCountEl.innerText = `${data.total_accounts || data.user_count || 0} (${data.active_users || 0} онлайн)`;
      }
      if (userBreakdownEl) {
        userBreakdownEl.innerText = `${data.hidden_users || 0} скрытых`;
      }
      if (adStatusEl) adStatusEl.innerText = data.ad_connected ? 'Подключен' : 'Локальный хост';
      if (eventsCountEl) eventsCountEl.innerText = data.event_count || 0;
    } catch (e) {
      console.error('[WinAdminTab] Failed to fetch status:', e);
    }
  }

  async function fetchAccounts(filterType = currentUserFilter) {
    try {
      const res = await fetch(`/api/sysadmin/accounts?filter_type=${encodeURIComponent(filterType)}`);
      if (!res.ok) return;
      const data = await res.json();
      const accounts = data.accounts || [];

      const tbody = document.getElementById('winadmin-users-tbody');
      const badge = document.getElementById('winadmin-users-badge');
      if (badge) badge.innerText = `${accounts.length} аккаунтов (${data.total || accounts.length} всего)`;

      if (tbody) {
        if (accounts.length === 0) {
          tbody.innerHTML = '<tr><td colspan="6" class="text-center text-muted p-3">Учетные записи по выбранному фильтру не найдены</td></tr>';
```

```
          return;
        }

        tbody.innerHTML = accounts.map((u, idx) => {
          // Вычисление бейджей
          const badges = [];
          if (u.is_admin) badges.push('<span class="badge bg-warning-subtle text-warning border border-warning-subtle me-1">Admin</span>');
          if (u.is_hidden) badges.push('<span class="badge bg-secondary-subtle text-secondary border border-secondary-subtle me-1">Скрытый</span>');
          if (!u.enabled) badges.push('<span class="badge bg-danger-subtle text-danger border border-danger-subtle me-1">Отключен</span>');

          // Статус
          let statusBadge = '<span class="badge bg-secondary-subtle text-secondary border border-secondary-subtle">Офлайн</span>';
          if (u.is_logged_in) {
            statusBadge = '<span class="badge bg-success-subtle text-success border border-success-subtle">● В сети</span>';
          } else if (!u.enabled) {
            statusBadge = '<span class="badge bg-danger-subtle text-danger border border-danger-subtle me-1">Отключен</span>';
          }

          // Ресурсы
          const resInfo = u.process_count > 0
            ? `<span class="text-info">${u.process_count} проц.</span> <span class="text-muted">(${u.memory_rss_mb} MB)</span>`
            : '<span class="text-muted">-</span>';

          // Профиль
          const profInfo = u.profile_size_mb > 0
```

```
            ? `<span title="${u.profile_path}">${u.profile_size_mb} MB</span>`
            : (u.profile_path ? `<span class="text-muted" title="${u.profile_path}">Локальный</span>` : '<span class="text-muted">-</span>');

          // Группы
          const groupsStr = (u.groups && u.groups.length > 0) ? u.groups.join(', ') : 'Users';

          return `
            <tr class="winadmin-user-row" data-idx="${idx}" style="cursor: pointer;" title="Нажмите для просмотра полного досье пользователя и AI-аудита">
              <td>
                <div class="d-flex align-items-center gap-1.5">
                  <i class="bi bi-person-badge text-info"></i>
                  <div>
                    <span class="fw-semibold text-white">${u.name}</span>
                    <div class="mt-0.5">${badges.join('')}</div>
                  </div>
                </div>
              </td>
              <td>
                <div class="text-truncate text-secondary" style="max-width: 140px;" title="${groupsStr}">${groupsStr}</div>
                <div class="font-monospace text-muted small text-truncate" style="max-width: 130px;" title="${u.sid}">${u.sid.slice(-10) || '-'}</div>
              </td>
              <td class="small">
                ${profInfo}
              </td>
              <td class="small font-monospace">
                ${resInfo}
```

```
              </td>
              <td>${statusBadge}</td>
              <td style="text-align: right;">
                <button class="btn btn-xs btn-outline-info py-0 px-2 btn-inspect-user me-1" data-user="${u.name}" title="Досье и метрики">
                  Досье
                </button>
                ${u.is_logged_in ? `
                  <button class="btn btn-xs btn-outline-danger py-0 px-1.5 btn-disconnect-user" data-user="${u.name}" title="Отключить сессию">
                    Выход
                  </button>
                ` : ''}
              </td>
            </tr>
          `;
        }).join('');

        // Клик по строке для открытия досье
        tbody.querySelectorAll('.winadmin-user-row').forEach(row => {
          row.onclick = (evt) => {
            if (evt.target.closest('.btn-disconnect-user')) return;
            const idx = parseInt(row.getAttribute('data-idx'), 10);
            const u = accounts[idx];
            if (!u) return;

            openUserDossier(u);
          };
        });

        // Кнопки "Досье"
        tbody.querySelectorAll('.btn-inspect-user').forEach(btn => {
```

```
          btn.onclick = (evt) => {
            evt.stopPropagation();
            const username = btn.getAttribute('data-user');
            const u = accounts.find(a => a.name === username);
            if (u) openUserDossier(u);
          };
        });

        // Кнопки "Выход"
        tbody.querySelectorAll('.btn-disconnect-user').forEach
(btn => {
          btn.onclick = async (evt) => {
            evt.stopPropagation();
            const user = btn.getAttribute('data-user');
            if (confirm(`Отключить сессию пользователя ${user}
?`)) {
              await fetch(`/api/sysadmin/users/${user}/disconn
ect`, { method: 'POST' });
              fetchAccounts();
              fetchStatus();
            }
          };
        });
      }
    } catch (e) {
      console.error('[WinAdminTab] Failed to fetch accounts:',
 e);
    }
  }

  function openUserDossier(u) {
    const modal = window.AIModalDialog || window.AITableModal;
    if (!modal) return;

    const metadata = [
      { label: 'Имя пользователя', value: u.name },
      { label: 'Полное имя', value: u.full_name || 'Не указано
' },
```

```
197       { label: 'Описание', value: u.description || 'Отсутствуе
т' },
198       { label: 'SID пользователя', value: u.sid || 'N/A' },
199       { label: 'Тип аккаунта', value: `${u.account_type} (${u.
is_admin ? 'Администратор' : 'Обычный'})` },
200       { label: 'Скрытая учетная запись', value: u.is_hidden ?
'Да (SpecialAccounts / Built-in)' : 'Нет' },
201       { label: 'Учетная запись включена', value: u.enabled ? '
Да' : 'Отключена' },
202       { label: 'Членство в группах', value: (u.groups && u.gro
ups.length > 0) ? u.groups.join(', ') : 'Users' },
203       { label: 'Путь к профилю', value: u.profile_path || 'Не
создан' },
204       { label: 'Размер профиля на диске', value: `${u.profile_
size_mb} MB` },
205       { label: 'Последняя смена пароля', value: u.password_las
t_set || 'N/A' },
206       { label: 'Неверных попыток пароля', value: String(u.bad_
password_count || 0) },
207       { label: 'Последний вход в систему', value: u.last_logon
 || 'Не зафиксирован' },
208       { label: 'Активных процессов', value: `${u.process_count
} (RAM: ${u.memory_rss_mb} MB, CPU: ${u.cpu_percent}%)` }
209     ];
210
211     const badges = [
212       { text: u.is_logged_in ? 'Online' : 'Offline', class: u.
is_logged_in ? 'badge bg-success-subtle text-success border bo
rder-success-subtle' : 'badge bg-secondary-subtle text-seconda
ry border border-secondary-subtle' }
213     ];
214     if (u.is_admin) badges.push({ text: 'Администратор', class
: 'badge bg-warning-subtle text-warning border border-warning-
subtle' });
215     if (u.is_hidden) badges.push({ text: 'Скрытый', class: 'ba
dge bg-secondary-subtle text-secondary border border-secondary
-subtle' });
216     if (!u.enabled) badges.push({ text: 'Отключен', class: 'ba
```

```
dge bg-danger-subtle text-danger border border-danger-subtle'
});
217
218     modal.show({
219       icon: '□ ',
220       title: `Учетная запись: ${u.name}`,
221       subtitle: `${u.full_name ? u.full_name + ' | ' : ''}SID:
 ${u.sid}`,
222       panelId: 'panel-winadmin-users',
223       tableType: 'user_account',
224       badges: badges,
225       metadata: metadata,
226       rawTitle: 'Полная конфигурация и метрики пользователя',
227       rawContent: JSON.stringify(u, null, 2),
228       requestData: {
229         username: u.name,
230         is_admin: u.is_admin,
231         is_hidden: u.is_hidden,
232         groups: u.groups,
233         process_count: u.process_count,
234         memory_rss_mb: u.memory_rss_mb,
235         profile_path: u.profile_path
236       }
237     });
238   }
239
240   async function fetchEvents() {
241     try {
242       const res = await fetch('/api/sysadmin/events?hours=24')
;
243       if (!res.ok) return;
244       const data = await res.json();
245       const events = data.events || [];
246
247       const tbody = document.getElementById('winadmin-events-t
body');
248       const badge = document.getElementById('winadmin-events-b
adge');
```

```
249       if (badge) badge.innerText = `${events.length} записей`;
250
251       if (tbody) {
252         if (events.length === 0) {
253           tbody.innerHTML = '<tr><td colspan="4" class="text-c
enter text-muted p-3">События не зафиксированы</td></tr>';
254           return;
255         }
256         const displayEvents = events.slice(0, 30);
257         tbody.innerHTML = displayEvents.map((e, idx) => `
258           <tr class="winadmin-event-row" data-idx="${idx}" sty
le="cursor: pointer;" title="Нажмите для AI-диагностики событи
я">
259             <td class="font-monospace text-muted small">${e.ti
mestamp?.slice(11, 19) || ''}</td>
260             <td class="font-monospace fw-semibold text-info">$
{e.event_id}</td>
261             <td><span class="badge ${e.level === 'Warning' ? '
bg-warning-subtle text-warning' : (e.level === 'Error' ? 'bg-d
anger-subtle text-danger' : 'bg-info-subtle text-info')}">${e.
level}</span></td>
262             <td class="small text-light">${e.source}: ${e.desc
ription}</td>
263           </tr>
264         `).join('');
265
266         tbody.querySelectorAll('.winadmin-event-row').forEach(
row => {
267           row.onclick = () => {
268             const idx = parseInt(row.getAttribute('data-idx'),
 10);
269             const e = displayEvents[idx];
270             if (!e) return;
271
272             const modal = window.AIModalDialog || window.AITab
leModal;
273             if (modal) {
274               modal.show({
```

```
                icon: '□ ',
                title: `Событие #${e.event_id} (${e.source})`,
                subtitle: `Уровень: ${e.level} | ${e.timestamp || ''}`,
                panelId: 'panel-winadmin-security-events',
                tableType: 'security_event',
                badges: [
                  { text: e.level || 'Info', class: e.level === 'Warning' ? 'badge bg-warning-subtle text-warning border border-warning-subtle' : (e.level === 'Error' ? 'badge bg-danger-subtle text-danger border border-danger-subtle' : 'badge bg-info-subtle text-info border border-info-subtle') }
                ],
                metadata: [
                  { label: 'Event ID', value: String(e.event_id) },
                  { label: 'Источник / Provider', value: e.source },
                  { label: 'Уровень важности', value: e.level },
                  { label: 'Время регистрации', value: e.timestamp }
                ],
                rawTitle: 'Текст сообщения журнала',
                rawContent: e.description || '',
                requestData: {
                  event_id: e.event_id,
                  source: e.source,
                  level: e.level,
                  description: e.description
                }
              });
            }
          };
        });
      }
    } catch (e) {
      console.error('[WinAdminTab] Failed to fetch events:', e
```

```
);
304     }
305   }
306
307   function initWindowsAdminTab() {
308     console.log('[WinAdminTab] Initializing Windows Sysadmin U
ser Metrics & Security tab...');
309     fetchStatus();
310     fetchAccounts('all');
311     fetchEvents();
312
313     if (!isWinAdminInitialized) {
314       const refreshBtn = document.getElementById('btn-winadmin
-refresh');
315       const configBtn = document.getElementById('btn-winadmin-
config');
316
317       if (refreshBtn) {
318         refreshBtn.onclick = () => {
319           fetchStatus();
320           fetchAccounts(currentUserFilter);
321           fetchEvents();
322         };
323       }
324
325       // Фильтры пользователей
326       const filterGroup = document.getElementById('winadmin-us
er-filters');
327       if (filterGroup) {
328         filterGroup.querySelectorAll('.winadmin-filter-btn').f
orEach(btn => {
329           btn.onclick = () => {
330             filterGroup.querySelectorAll('.winadmin-filter-btn
').forEach(b => b.classList.remove('active'));
331             btn.classList.add('active');
332             currentUserFilter = btn.getAttribute('data-filter'
) || 'all';
333             fetchAccounts(currentUserFilter);
```

```
          };
        });
      }

      if (configBtn) {
        configBtn.onclick = () => {
          if (typeof window.openAppConfigModal === 'function') {
            window.openAppConfigModal('windows_sysadmin', 'Windows System Administrator');
          }
        };
      }

      isWinAdminInitialized = true;
    }
  }

  window.initWindowsAdminTab = initWindowsAdminTab;
})();

```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон веб-интерфейса модуля резервного копирования и управл
ения данными Windows.

Usage Examples:
    Web Route:
      /windows/api/webgui/windows_backup_tab/index.html

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/windows_backup_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->

<!-- Windows Backup & Storage Management Tab -->
<style>
  .wb-container {
    padding: 0.65rem 0.85rem;
  }
  .wb-card {
    background: var(--surface-1) !important;
    border: 1.5px solid var(--border-color) !important;
    border-radius: var(--radius-md, 8px);
    padding: var(--kpi-padding, 0.5rem 0.8rem);
    display: flex;
    flex-direction: column;
```

```
    justify-content: space-between;
    height: 100%;
    min-height: var(--kpi-min-height, 64px);
    transition: border-color 0.15s ease, box-shadow 0.15s ease
;
    box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.08)
);
  }
  .wb-card:hover {
    border-color: var(--kpi-accent, var(--nav-active, #0284c7)
) !important;
    box-shadow: var(--shadow-sm, 0 2px 8px rgba(0, 0, 0, 0.15)
);
  }
  .wb-card-label {
    font-size: var(--kpi-label-size, 0.7rem);
    text-transform: uppercase;
    letter-spacing: 0.05em;
    color: var(--text-muted);
    font-weight: 600;
    display: flex;
    justify-content: space-between;
    align-items: center;
    line-height: 1.1;
  }
  .wb-card-value {
    font-size: var(--kpi-val-size, 1.3rem);
    font-weight: 700;
    color: var(--text-color);
    font-family: var(--font-mono, monospace);
    line-height: 1.2;
  }
  .wb-card-sub {
    font-size: var(--kpi-sub-size, 0.72rem);
    color: var(--text-muted);
    line-height: 1.1;
  }

```

```
  .wb-panel {
    background: var(--surface-1) !important;
    border: 1px solid var(--border-color) !important;
    border-radius: var(--radius-md, 8px);
    overflow: hidden;
    box-shadow: var(--shadow-xs, 0 1px 3px rgba(0, 0, 0, 0.08)
);
    margin-bottom: 0.75rem;
  }
  .wb-panel-header {
    background: var(--surface-2) !important;
    border-bottom: 1px solid var(--border-color) !important;
    padding: 0.45rem 0.85rem;
    font-weight: 600;
    font-size: 0.82rem;
    color: var(--text-color);
    display: flex;
    align-items: center;
    justify-content: space-between;
  }
  .wb-table {
    margin-bottom: 0;
    font-size: 0.8rem;
    background: transparent;
  }
  .wb-table th {
    background: var(--surface-2);
    color: var(--text-muted);
    font-weight: 600;
    font-size: 0.72rem;
    text-transform: uppercase;
    border-bottom: 1px solid var(--border-color);
    padding: 0.45rem 0.65rem;
  }
  .wb-table td {
    border-bottom: 1px solid var(--border-subtle, rgba(128, 12
8, 128, 0.2));
    padding: 0.45rem 0.65rem;
```

```
    vertical-align: middle;
    color: var(--text-color);
  }
  .wb-nav-tabs {
    border-bottom: 1px solid var(--border-color);
  }
  .wb-nav-tabs .nav-link {
    color: var(--text-muted);
    font-size: 0.82rem;
    font-weight: 500;
    border: 1px solid transparent;
    border-top-left-radius: 6px;
    border-top-right-radius: 6px;
    padding: 0.45rem 1rem;
    transition: all 0.15s ease;
  }
  .wb-nav-tabs .nav-link:hover {
    color: var(--text-color);
    border-color: var(--border-color) var(--border-color) tran
sparent;
  }
  .wb-nav-tabs .nav-link.active {
    color: var(--nav-active, #0284c7);
    background-color: var(--surface-2);
    border-color: var(--border-color) var(--border-color) tran
sparent;
    font-weight: 600;
  }
</style>

<div class="wb-container">
  <!-- Top Header Bar -->
  <div class="d-flex align-items-center justify-content-betwee
n mb-2.5 flex-wrap gap-2">
    <div>
      <h6 class="mb-0 fw-bold text-body d-flex align-items-cen
ter gap-2">
        <span> </span> Резервное копирование и управление дан
```

```
ными Windows
      </h6>
      <div class="small text-muted" style="font-size: 0.72rem;
">
        Пользовательские папки, системные библиотеки (.library
-ms), служба File History, теневые копии томов (VSS), аудит уд
аления файлов и RAG-поиск по архивам
      </div>
    </div>
    <div class="d-flex align-items-center gap-2">
      <button class="btn btn-sm btn-outline-info d-flex align-
items-center gap-1" id="wb-btn-open-create-lib" title="Создать
 новую системную библиотеку Windows">
        <i class="bi bi-folder-plus"></i>
        <span>Создать библиотеку</span>
      </button>
      <button class="btn btn-sm btn-primary d-flex align-items
-center gap-1" id="wb-btn-trigger-backup" title="Принудительны
й запуск архивации через fhexec">
        <i class="bi bi-play-fill"></i>
        <span>Запустить бэкап</span>
      </button>
      <button class="btn btn-sm btn-outline-secondary d-flex a
lign-items-center gap-1" id="wb-btn-refresh" title="Обновить в
се данные вкладки">
        <i class="bi bi-arrow-clockwise"></i>
        <span>Обновить</span>
      </button>
    </div>
  </div>

  <!-- KPI Stats Row -->
  <div class="row g-2 mb-2.5">
    <div class="col-6 col-md-3">
      <div class="wb-card">
        <div class="wb-card-label">
          <span>Готовность защиты</span>
          <span class="badge bg-secondary" id="wb-badge-status
```

```
">-</span>
        </div>
        <div class="wb-card-value text-info" id="wb-val-health
">- / 100</div>
        <div class="wb-card-sub" id="wb-sub-health">Готовность
 защиты данных</div>
      </div>
    </div>
    <div class="col-6 col-md-3">
      <div class="wb-card">
        <div class="wb-card-label">
          <span>Служба File History</span>
          <span> </span>
        </div>
        <div class="wb-card-value text-white" id="wb-val-servi
ce">-</div>
        <div class="wb-card-sub text-truncate" id="wb-sub-serv
ice">fhsvc статус</div>
      </div>
    </div>
    <div class="col-6 col-md-3">
      <div class="wb-card">
        <div class="wb-card-label">
          <span>Библиотеки / VSS</span>
          <span> </span>
        </div>
        <div class="wb-card-value text-success" id="wb-val-lib
raries">-</div>
        <div class="wb-card-sub" id="wb-sub-libraries">Библиот
ек / VSS снимков</div>
      </div>
    </div>
    <div class="col-6 col-md-3">
      <div class="wb-card">
        <div class="wb-card-label">
          <span>Аудит удаления (24ч)</span>
          <button class="btn btn-xs btn-outline-primary py-0 p
x-1" id="wb-btn-enable-auditpol" title="Включить auditpol /sub
```

```
category:File System" style="font-size: 0.65rem;">Вкл</button>
        </div>
        <div class="wb-card-value text-danger" id="wb-val-dele
tions">0</div>
        <div class="wb-card-sub" id="wb-sub-deletions">Event 4
663 / 4660</div>
      </div>
    </div>
  </div>

  <!-- Sub Navigation Tabs -->
  <ul class="nav wb-nav-tabs mb-2" id="wbSubTabs" role="tablis
t">
    <li class="nav-item" role="presentation">
      <button class="nav-link active" id="wb-tab-storage-btn"
data-bs-toggle="tab" data-bs-target="#wb-pane-storage" type="b
utton" role="tab">
        <i class="bi bi-folder2-open me-1 text-info"></i>Папки
 и библиотеки
      </button>
    </li>
    <li class="nav-item" role="presentation">
      <button class="nav-link" id="wb-tab-backup-btn" data-bs-
toggle="tab" data-bs-target="#wb-pane-backup" type="button" ro
le="tab">
        <i class="bi bi-shield-lock me-1 text-warning"></i>Рез
ервное копирование и VSS
      </button>
    </li>
    <li class="nav-item" role="presentation">
      <button class="nav-link" id="wb-tab-audit-btn" data-bs-t
oggle="tab" data-bs-target="#wb-pane-audit" type="button" role
="tab">
        <i class="bi bi-trash3 me-1 text-danger"></i>Аудит уда
ления файлов
      </button>
    </li>
  </ul>
```

```
220
221   <!-- Sub Panes Content -->
222   <div class="tab-content" id="wbTabContent">
223
224     <!-- ========================================== -->
225     <!-- Tab 1: Пользовательские папки и библиотеки -->
226     <!-- ========================================== -->
227     <div class="tab-pane fade show active" id="wb-pane-storage
" role="tabpanel">
228       <!-- User Shell Folders Storage & Relocation Management
-->
229       <div class="wb-panel">
230         <div class="wb-panel-header">
231           <div class="d-flex align-items-center gap-2">
232             <i class="bi bi-hdd-network-fill text-info"></i>
233             <span>Пользовательские директории и перенос на вто
ричные диски</span>
234           </div>
235           <div class="d-flex align-items-center gap-2">
236             <span class="badge bg-primary-subtle text-primary
border border-primary-subtle" id="wb-total-user-size">Объем: 0
 ГБ</span>
237             <button class="btn btn-xs btn-outline-info py-0 px
-2" id="wb-btn-refresh-user-folders" title="Пересчитать размер
 пользовательских папок" style="font-size: 0.72rem;">
238               <i class="bi bi-arrow-repeat me-1"></i>Пересчита
ть
239             </button>
240           </div>
241         </div>
242
243         <div class="p-2">
244           <!-- Disks summary row -->
245           <div class="mb-2 p-2 rounded bg-body-secondary borde
r border-secondary" style="font-size: 0.76rem;" id="wb-drives-
summary">
246             <div class="text-muted small mb-1 fw-bold">  Обна
руженные физические накопители:</div>
```

```
            <div class="d-flex flex-wrap gap-2" id="wb-drives-list">
              <span class="text-muted">Загрузка информации о дисках...</span>
            </div>
          </div>

          <div class="table-responsive" style="max-height: 280px; overflow-y: auto;">
            <table class="table table-hover wb-table">
              <thead>
                <tr>
                  <th>Директория</th>
                  <th>Текущий путь</th>
                  <th>Диск</th>
                  <th>Размер</th>
                  <th>Файлов</th>
                  <th class="text-end">Действие</th>
                </tr>
              </thead>
              <tbody id="wb-user-folders-tbody">
                <tr>
                  <td colspan="6" class="text-center text-muted p-3">Подсчет размера пользовательских папок...</td>
                </tr>
              </tbody>
            </table>
          </div>
        </div>
      </div>

      <!-- System Libraries (.library-ms) -->
      <div class="wb-panel mb-0">
        <div class="wb-panel-header">
          <div class="d-flex align-items-center gap-2">
            <i class="bi bi-folder-symlink text-info"></i>
            <span>Системные библиотеки Windows (.library-ms)</span>
```

```
          </div>
          <div class="d-flex align-items-center gap-2">
            <button class="btn btn-xs btn-outline-info py-0 px-2" id="wb-btn-quick-add-folder" style="font-size: 0.72rem;" title="Добавить папку в существующую библиотеку">
              <i class="bi bi-plus-circle me-1"></i>Добавить папку
            </button>
            <span class="badge bg-body-secondary border text-muted" id="wb-lib-count">0 шт.</span>
          </div>
        </div>
        <div class="table-responsive" style="max-height: 260px; overflow-y: auto;">
          <table class="table table-hover wb-table">
            <thead>
              <tr>
                <th>Библиотека</th>
                <th>Папки / Источники</th>
                <th>Папка сохранения</th>
                <th>Закреплена</th>
              </tr>
            </thead>
            <tbody id="wb-libraries-body">
              <tr>
                <td colspan="4" class="text-center py-3 text-muted">Загрузка библиотек...</td>
              </tr>
            </tbody>
          </table>
        </div>
      </div>
    </div>

    <!-- ========================================== -->
    <!-- Tab 2: Резервное копирование и VSS         -->
    <!-- ========================================== -->
    <div class="tab-pane fade" id="wb-pane-backup" role="tabpa
```

```
nel">
      <div class="row g-2">
        <!-- Left Column: File History Configuration & Recommendations -->
        <div class="col-12 col-lg-6">
          <!-- File History Config -->
          <div class="wb-panel">
            <div class="wb-panel-header">
              <div class="d-flex align-items-center gap-2">
                <i class="bi bi-clock-history text-primary"></i>
                <span>Настройка File History (История файлов)</span>
              </div>
            </div>
            <div class="p-2.5" style="font-size: 0.8rem;" id="wb-fh-details">
              <div class="mb-2">
                <div class="text-muted small">Целевой каталог бэкапа:</div>
                <div class="fw-semibold text-break font-monospace small text-info" id="wb-fh-target">Не настроено</div>
              </div>
              <div class="mb-2">
                <div class="text-muted small">Файл конфигурации:</div>
                <div class="text-break font-monospace small text-muted" id="wb-fh-config">-</div>
              </div>
              <div class="row g-2 mb-1">
                <div class="col-6">
                  <div class="text-muted small">Интервал архивации:</div>
                  <div class="fw-bold text-body" id="wb-fh-interval">-</div>
                </div>
                <div class="col-6">
                  <div class="text-muted small">Срок хранения
```

```
версий:</div>
                  <div class="fw-bold text-body" id="wb-fh-retention">-</div>
                </div>
              </div>
              <div class="mt-2 pt-2 border-top border-secondary">
                <div class="text-muted small">Последняя успешная архивация:</div>
                <div class="fw-semibold text-warning" id="wb-fh-last-backup">-</div>
              </div>
            </div>
          </div>

          <!-- Recommendations and Health Checklist -->
          <div class="wb-panel mb-0">
            <div class="wb-panel-header">
              <div class="d-flex align-items-center gap-2">
                <i class="bi bi-shield-check text-success"></i>
                <span>Рекомендации безопасности данных</span>
              </div>
            </div>
            <div class="p-2.5" style="max-height: 200px; overflow-y: auto;">
              <ul class="list-unstyled mb-0" id="wb-recommendations-list" style="font-size: 0.78rem;">
                <li class="text-muted text-center py-2">Загрузка рекомендаций...</li>
              </ul>
            </div>
          </div>
        </div>

        <!-- Right Column: VSS Snapshots & RAG Search -->
        <div class="col-12 col-lg-6">
          <!-- VSS Snapshots -->
```

```
          <div class="wb-panel">
            <div class="wb-panel-header">
              <div class="d-flex align-items-center gap-2">
                <i class="bi bi-camera text-warning"></i>
                <span>Теневые копии томов VSS (Volume Shadow C
opies)</span>
              </div>
              <span class="badge bg-body-secondary border text
-muted" id="wb-vss-count">0 шт.</span>
            </div>
            <div class="table-responsive" style="max-height: 1
80px; overflow-y: auto;">
              <table class="table table-hover wb-table">
                <thead>
                  <tr>
                    <th>ID снимка</th>
                    <th>Исходный том</th>
                    <th>Время создания</th>
                    <th>Состояние</th>
                  </tr>
                </thead>
                <tbody id="wb-vss-body">
                  <tr>
                    <td colspan="4" class="text-center py-3 te
xt-muted">Загрузка VSS снимков...</td>
                  </tr>
                </tbody>
              </table>
            </div>
          </div>

          <!-- RAG Search in File History Archive -->
          <div class="wb-panel mb-0">
            <div class="wb-panel-header">
              <div class="d-flex align-items-center gap-2">
                <i class="bi bi-search-heart text-info"></i>
                <span>RAG поиск по архивам бэкапа</span>
              </div>
```

```
              <button class="btn btn-xs btn-outline-primary py-0 px-2" id="wb-btn-rag-sync" title="Индексировать новые версии файлов" style="font-size: 0.72rem;">
                <i class="bi bi-arrow-repeat me-1"></i>Синхронизация
              </button>
            </div>
            <div class="p-2" style="font-size: 0.8rem;">
              <div class="input-group input-group-sm mb-2">
                <input type="text" class="form-control" id="wb-rag-query" placeholder="Поиск документа или фрагмента текста..." />
                <button class="btn btn-primary" id="wb-btn-rag-search"><i class="bi bi-search"></i></button>
              </div>
              <div id="wb-rag-results" style="max-height: 160px; overflow-y: auto;">
                <div class="text-muted text-center py-2 small">Введите запрос для поиска по архивным копиям</div>
              </div>
            </div>
          </div>
        </div>
      </div>
    </div>

    <!-- ========================================== -->
    <!-- Tab 3: Аудит удаления файлов               -->
    <!-- ========================================== -->
    <div class="tab-pane fade" id="wb-pane-audit" role="tabpanel">
      <div class="wb-panel mb-0">
        <div class="wb-panel-header">
          <div class="d-flex align-items-center gap-2">
            <i class="bi bi-trash3-fill text-danger"></i>
            <span>Журнал аудита изменения и удаления файлов (Event 4663 / 4660)</span>
          </div>
```

```
          <div class="d-flex align-items-center gap-2">
            <button class="btn btn-xs btn-outline-info py-0 px-2" id="wb-btn-configure-sacl" title="Настроить SACL Delete для папки" style="font-size: 0.72rem;">
              <i class="bi bi-folder-check me-1"></i>Настроить SACL папки
            </button>
            <span class="badge bg-danger-subtle text-danger border border-danger-subtle" id="wb-deletions-badge">0 удалений</span>
          </div>
        </div>
        <div class="p-0 table-responsive" style="max-height: 380px; overflow-y: auto;">
          <table class="table table-hover wb-table">
            <thead>
              <tr>
                <th>Время</th>
                <th>Event ID</th>
                <th>Файл / Папка</th>
                <th>Процесс-инициатор</th>
                <th>Пользователь</th>
              </tr>
            </thead>
            <tbody id="wb-deletions-tbody">
              <tr><td colspan="5" class="text-center text-muted p-3">Поиск событий удаления файлов...</td></tr>
            </tbody>
          </table>
        </div>
      </div>
    </div>

  </div>

  <!-- Modal for Relocating Folder -->
  <div class="modal fade" id="wb-relocate-modal" tabindex="-1" aria-hidden="true">
```

```
460     <div class="modal-dialog modal-dialog-centered">
461       <div class="modal-content">
462         <div class="modal-header py-2">
463           <h6 class="modal-title fw-bold text-info d-flex alig
    n-items-center gap-2">
464             <i class="bi bi-box-arrow-right"></i>
465             <span>Перенос пользовательской директории</span>
466           </h6>
467           <button type="button" class="btn-close" data-bs-dism
    iss="modal" aria-label="Close"></button>
468         </div>
469         <div class="modal-body" style="font-size: 0.85rem;">
470           <input type="hidden" id="wb-relocate-folder-id" />
471           <div class="mb-2">
472             <label class="text-muted small">Переносимая папка:
    </label>
473             <div class="fw-bold text-body fs-6" id="wb-relocat
    e-folder-name">-</div>
474             <div class="font-monospace small text-muted text-b
    reak" id="wb-relocate-folder-path">-</div>
475           </div>
476           <div class="mb-2">
477             <label class="text-muted small">Текущий размер:</l
    abel>
478             <div class="fw-semibold text-warning" id="wb-reloc
    ate-folder-size">-</div>
479           </div>
480           <div class="mb-2">
481             <label class="text-muted small mb-1" for="wb-reloc
    ate-target-drive">Целевой диск для переноса:</label>
482             <select class="form-select form-select-sm" id="wb-
    relocate-target-drive">
483               <option value="">Выберите диск с достаточным мес
    том...</option>
484             </select>
485             <div class="form-text text-muted small" id="wb-rel
    ocate-drive-hint"></div>
486           </div>
```

```
          <div class="mb-3">
            <label class="text-muted small mb-1" for="wb-relocate-target-path">Целевая директория (папка):</label>
            <div class="input-group input-group-sm">
              <input type="text" class="form-control form-control-sm font-monospace" id="wb-relocate-target-path" placeholder="D:\Users\Username\Pictures" />
              <button class="btn btn-outline-info d-flex align-items-center gap-1" type="button" id="wb-btn-browse-folder" title="Выбрать целевую папку через системный проводник Windows">
                <i class="bi bi-folder2-open"></i>
                <span>Обзор...</span>
              </button>
            </div>
            <div class="form-text text-muted small">Можно выбрать готовую папку через проводник или отредактировать путь вручную.</div>
          </div>
          <div class="form-check mb-2">
            <input class="form-check-input" type="checkbox" id="wb-relocate-delete-source" />
            <label class="form-check-label text-body small" for="wb-relocate-delete-source">
              Очистить исходную папку после успешного копирования и перенастройки
            </label>
          </div>
          <div class="alert alert-info py-2 px-3 small mb-0">
            <i class="bi bi-info-circle me-1"></i>
            Перенос скопирует файлы, обновит реестр Windows (User Shell Folders) и системную библиотеку Windows, благодаря чему служба File History и Проводник продолжат работать без сбоев.
          </div>
        </div>
        <div class="modal-footer py-1.5">
          <button type="button" class="btn btn-sm btn-outline-
```

```
secondary" data-bs-dismiss="modal">Отмена</button>
          <button type="button" class="btn btn-sm btn-primary d-flex align-items-center gap-1" id="wb-btn-execute-relocate">
            <i class="bi bi-check-circle"></i>
            <span>Выполнить перенос</span>
          </button>
        </div>
      </div>
    </div>
  </div>

  <!-- Modal for Creating Library -->
  <div class="modal fade" id="wb-create-lib-modal" tabindex="-1" aria-hidden="true">
    <div class="modal-dialog modal-dialog-centered">
      <div class="modal-content">
        <div class="modal-header py-2">
          <h6 class="modal-title fw-bold text-info d-flex align-items-center gap-2">
            <i class="bi bi-folder-plus"></i>
            <span>Создать библиотеку Windows (.library-ms)</span>
          </h6>
          <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
        </div>
        <div class="modal-body" style="font-size: 0.85rem;">
          <div class="mb-2">
            <label class="text-muted small" for="wb-create-lib-name">Название библиотеки:</label>
            <input type="text" class="form-control form-control-sm" id="wb-create-lib-name" placeholder="Например: Архивы Проектов" />
          </div>
          <div class="mb-2">
            <label class="text-muted small" for="wb-create-lib-folders">Пути к папкам (по одной строке на папку):</label>
            <textarea class="form-control form-control-sm font
```

```
-monospace" id="wb-create-lib-folders" rows="3" placeholder="C
:\Data\Backups&#10;D:\Projects"></textarea>
          </div>
          <div class="form-check mb-2">
            <input class="form-check-input" type="checkbox" id
="wb-create-lib-pinned" checked />
            <label class="form-check-label text-body small" fo
r="wb-create-lib-pinned">
              Закрепить в панели навигации Проводника
            </label>
          </div>
        </div>
        <div class="modal-footer py-1.5">
          <button type="button" class="btn btn-sm btn-outline-
secondary" data-bs-dismiss="modal">Отмена</button>
          <button type="button" class="btn btn-sm btn-info d-f
lex align-items-center gap-1" id="wb-btn-execute-create-lib">
            <i class="bi bi-check-lg"></i>
            <span>Создать</span>
          </button>
        </div>
      </div>
    </div>
  </div>

  <!-- Modal for Adding Folder to Library -->
  <div class="modal fade" id="wb-add-folder-modal" tabindex="-
1" aria-hidden="true">
    <div class="modal-dialog modal-dialog-centered">
      <div class="modal-content">
        <div class="modal-header py-2">
          <h6 class="modal-title fw-bold text-info d-flex alig
n-items-center gap-2">
            <i class="bi bi-folder-plus"></i>
            <span>Добавить папку в библиотеку</span>
          </h6>
          <button type="button" class="btn-close" data-bs-dism
iss="modal" aria-label="Close"></button>
```

```
        </div>
        <div class="modal-body" style="font-size: 0.85rem;">
          <div class="mb-2">
            <label class="text-muted small" for="wb-add-folder
-lib-select">Выберите целевую библиотеку:</label>
            <select class="form-select form-select-sm" id="wb-
add-folder-lib-select">
              <option value="">Загрузка библиотек...</option>
            </select>
          </div>
          <div class="mb-2">
            <label class="text-muted small" for="wb-add-folder
-path">Полный путь к физической папке:</label>
            <input type="text" class="form-control form-contro
l-sm font-monospace" id="wb-add-folder-path" placeholder="D:\B
ackupFolder" />
          </div>
          <div class="form-check mb-2">
            <input class="form-check-input" type="checkbox" id
="wb-add-folder-default-save" />
            <label class="form-check-label text-body small" fo
r="wb-add-folder-default-save">
              Сделать основной папкой сохранения по умолчанию
            </label>
          </div>
        </div>
        <div class="modal-footer py-1.5">
          <button type="button" class="btn btn-sm btn-outline-
secondary" data-bs-dismiss="modal">Отмена</button>
          <button type="button" class="btn btn-sm btn-primary
d-flex align-items-center gap-1" id="wb-btn-execute-add-folder
">
            <i class="bi bi-check-lg"></i>
            <span>Добавить</span>
          </button>
        </div>
      </div>
    </div>
```

```
  </div>
</div>

```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Windows Backup Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля резервног
о копирования и данных Windows.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/windows_backup_tab/mai
n.js?v=20261004_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/windows_backup_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

// Windows Backup & Storage Manager Frontend Logic
(function () {
  let _userFoldersData = null;
  let _librariesData = [];

  async function initWindowsBackupTab() {
    console.log('[WindowsBackup] Initializing tab...');
    bindEvents();
    await loadBackupData();
  }
  window.initWindowsBackupTab = initWindowsBackupTab;
```

```
32
33  function bindEvents() {
34    // Поддержка переключения подвкладок
35    const subTabButtons = document.querySelectorAll('#wbSubTab
s button[data-bs-toggle="tab"]');
36    subTabButtons.forEach(btn => {
37      btn.onclick = (e) => {
38        e.preventDefault();
39        const targetSelector = btn.getAttribute('data-bs-targe
t');
40        if (!targetSelector) return;
41
42        subTabButtons.forEach(b => b.classList.remove('active'
));
43        btn.classList.add('active');
44
45        const panes = document.querySelectorAll('#wbTabContent
 .tab-pane');
46        panes.forEach(p => p.classList.remove('show', 'active'
));
47
48        const targetPane = document.querySelector(targetSelect
or);
49        if (targetPane) {
50          targetPane.classList.add('show', 'active');
51        }
52      };
53    });
54
55    const btnRefresh = document.getElementById('wb-btn-refresh
');
56    if (btnRefresh) {
57      btnRefresh.onclick = () => loadBackupData();
58    }
59
60    const btnTrigger = document.getElementById('wb-btn-trigger
-backup');
61    if (btnTrigger) {
```

```
      btnTrigger.onclick = async () => {
        btnTrigger.disabled = true;
        btnTrigger.innerHTML = '<span class="spinner-border sp
inner-border-sm me-1"></span> Запуск...';
        try {
          const res = await fetch('/api/v1/windows-backup/file
-history/trigger', { method: 'POST' });
          const data = await res.json();
          if (res.ok) {
            alert('Архивация File History успешно запущена: '
+ (data.message || 'OK'));
          } else {
            alert('Ошибка запуска: ' + (data.detail || JSON.st
ringify(data)));
          }
        } catch (e) {
          alert('Сетевая ошибка при запуске архивации: ' + e.m
essage);
        } finally {
          btnTrigger.disabled = false;
          btnTrigger.innerHTML = '<i class="bi bi-play-fill"><
/i> <span>Запустить бэкап</span>';
          await loadBackupData();
        }
      };
    }

    // Modal Create Library
    const btnOpenCreateLib = document.getElementById('wb-btn-o
pen-create-lib');
    if (btnOpenCreateLib) {
      btnOpenCreateLib.onclick = () => {
        const modalEl = document.getElementById('wb-create-lib
-modal');
        if (modalEl) {
          if (window.bootstrap?.Modal) {
            new window.bootstrap.Modal(modalEl).show();
          } else {
```

```
            modalEl.classList.add('show');
            modalEl.style.display = 'block';
          }
        }
      };
    }

    const btnExecCreateLib = document.getElementById('wb-btn-execute-create-lib');
    if (btnExecCreateLib) {
      btnExecCreateLib.onclick = async () => {
        const name = document.getElementById('wb-create-lib-name')?.value?.trim();
        const foldersRaw = document.getElementById('wb-create-lib-folders')?.value || '';
        const isPinned = document.getElementById('wb-create-lib-pinned')?.checked ?? true;

        if (!name) {
          alert('Укажите название библиотеки');
          return;
        }

        const folders = foldersRaw
          .split('\n')
          .map(f => f.trim())
          .filter(f => f.length > 0);

        btnExecCreateLib.disabled = true;
        btnExecCreateLib.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span> Создание...';

        try {
          const res = await fetch('/api/v1/windows-backup/libraries', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ name, folders, is_pinned: i
```

```
sPinned })
          });
          const data = await res.json();
          if (res.ok) {
            alert(`Библиотека "${name}" успешно создана!`);
            const modalEl = document.getElementById('wb-create
-lib-modal');
            if (modalEl && window.bootstrap?.Modal) {
              const modalInst = window.bootstrap.Modal.getInst
ance(modalEl);
              if (modalInst) modalInst.hide();
            }
            await loadLibraries();
          } else {
            alert('Ошибка: ' + (data.detail || JSON.stringify(
data)));
          }
        } catch (e) {
          alert('Ошибка сети: ' + e.message);
        } finally {
          btnExecCreateLib.disabled = false;
          btnExecCreateLib.innerHTML = '<i class="bi bi-check-
lg"></i> <span>Создать</span>';
        }
      };
    }

    // Modal Add Folder to Library
    const btnQuickAddFolder = document.getElementById('wb-btn-
quick-add-folder');
    if (btnQuickAddFolder) {
      btnQuickAddFolder.onclick = () => {
        const select = document.getElementById('wb-add-folder-
lib-select');
        if (select) {
          select.innerHTML = _librariesData.length > 0
            ? _librariesData.map(l => `<option value="${escape
Html(l.name)}">${escapeHtml(l.name)} (${l.folder_count || 0} п
```

```
апок)</option>`).join('')
            : '<option value="">Нет доступных библиотек</option>';
        }
        const modalEl = document.getElementById('wb-add-folder-modal');
        if (modalEl) {
          if (window.bootstrap?.Modal) {
            new window.bootstrap.Modal(modalEl).show();
          } else {
            modalEl.classList.add('show');
            modalEl.style.display = 'block';
          }
        }
      };
    }

    const btnExecAddFolder = document.getElementById('wb-btn-execute-add-folder');
    if (btnExecAddFolder) {
      btnExecAddFolder.onclick = async () => {
        const libName = document.getElementById('wb-add-folder-lib-select')?.value;
        const folderPath = document.getElementById('wb-add-folder-path')?.value?.trim();
        const isDefault = document.getElementById('wb-add-folder-default-save')?.checked ?? false;

        if (!libName || !folderPath) {
          alert('Выберите библиотеку и укажите путь к папке');
          return;
        }

        btnExecAddFolder.disabled = true;
        btnExecAddFolder.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span> Добавление...';

        try {
```

```
          const res = await fetch(`/api/v1/windows-backup/libr
aries/${encodeURIComponent(libName)}/folders`, {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ folder_path: folderPath, is
_default_save: isDefault })
          });
          const data = await res.json();
          if (res.ok) {
            alert(`Папка успешно добавлена в библиотеку "${lib
Name}"!`);
            const modalEl = document.getElementById('wb-add-fo
lder-modal');
            if (modalEl && window.bootstrap?.Modal) {
              const modalInst = window.bootstrap.Modal.getInst
ance(modalEl);
              if (modalInst) modalInst.hide();
            }
            await loadLibraries();
          } else {
            alert('Ошибка: ' + (data.detail || JSON.stringify(
data)));
          }
        } catch (e) {
          alert('Ошибка сети: ' + e.message);
        } finally {
          btnExecAddFolder.disabled = false;
          btnExecAddFolder.innerHTML = '<i class="bi bi-check-
lg"></i> <span>Добавить</span>';
        }
      };
    }

    // RAG Search & Sync
    const btnRagSync = document.getElementById('wb-btn-rag-syn
c');
    if (btnRagSync) {
      btnRagSync.onclick = async () => {
```

```
        btnRagSync.disabled = true;
        btnRagSync.innerHTML = '<span class="spinner-border sp
inner-border-sm me-1"></span> Синхр...';
        try {
          const res = await fetch('/api/v1/windows-backup/file
-history/rag/sync', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ force_rebuild: false })
          });
          const data = await res.json();
          if (res.ok) {
            alert(`Синхронизация RAG завершена!\nПроиндексиров
ано: ${data.total_indexed || 0} версий файлов.`);
          } else {
            alert('Ошибка синхронизации: ' + (data.detail || J
SON.stringify(data)));
          }
        } catch (e) {
          alert('Ошибка сети: ' + e.message);
        } finally {
          btnRagSync.disabled = false;
          btnRagSync.innerHTML = '<i class="bi bi-arrow-repeat
 me-1"></i>Синхронизация';
        }
      };
    }

    const btnRagSearch = document.getElementById('wb-btn-rag-s
earch');
    const inputRagQuery = document.getElementById('wb-rag-quer
y');
    const performRagSearch = async () => {
      const query = inputRagQuery?.value?.trim();
      if (!query) return;

      const resultsContainer = document.getElementById('wb-rag
-results');
```

```
      if (resultsContainer) {
        resultsContainer.innerHTML = '<div class="text-center
text-muted py-2"><span class="spinner-border spinner-border-sm
 me-1"></span>Поиск в архиве...</div>';
      }

      try {
        const res = await fetch('/api/v1/windows-backup/file-h
istory/rag/search', {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify({ query, top_k: 5 })
        });
        const items = await res.json();
        if (resultsContainer) {
          if (!items || items.length === 0) {
            resultsContainer.innerHTML = '<div class="text-mut
ed text-center py-2 small">Ничего не найдено по запросу</div>'
;
            return;
          }
          resultsContainer.innerHTML = items.map(item => `
            <div class="p-1.5 mb-1 rounded bg-dark border bord
er-secondary" style="font-size: 0.75rem;">
              <div class="d-flex justify-content-between align
-items-center mb-0.5">
                <span class="fw-bold text-info text-truncate"
title="${escapeHtml(item.file_name)}">${escapeHtml(item.file_n
ame)}</span>
                <span class="badge bg-primary-subtle text-prim
ary">${(item.score * 100).toFixed(0)}%</span>
              </div>
              <div class="text-muted font-monospace small text
-truncate" title="${escapeHtml(item.original_path)}">${escapeH
tml(item.original_path)}</div>
              ${item.snippet ? `<div class="text-white-50 mt-1
 small text-truncate">${escapeHtml(item.snippet)}</div>` : ''}
            </div>
```

```
          `).join('');
        }
      } catch (e) {
        if (resultsContainer) {
          resultsContainer.innerHTML = `<div class="text-dange
r small py-1">Ошибка поиска: ${escapeHtml(e.message)}</div>`;
        }
      }
    };

    if (btnRagSearch) btnRagSearch.onclick = performRagSearch;
    if (inputRagQuery) {
      inputRagQuery.onkeydown = (e) => {
        if (e.key === 'Enter') {
          e.preventDefault();
          performRagSearch();
        }
      };
    }

    // Audit buttons
    const btnEnableAudit = document.getElementById('wb-btn-ena
ble-auditpol');
    if (btnEnableAudit) {
      btnEnableAudit.onclick = async () => {
        if (confirm('Включить системный аудит файловой системы
 (auditpol /set /subcategory:File System)?')) {
          try {
            const res = await fetch('/api/v1/system/file-audit
/policy', {
              method: 'POST',
              headers: { 'Content-Type': 'application/json' },
              body: JSON.stringify({ enable_success: true, ena
ble_failure: true })
            });
            const data = await res.json();
            alert(data.success ? 'Аудит файловой системы успеш
но активирован!' : `Ошибка: ${data.error || 'Сбой'}`);
```

```
            await loadBackupData();
          } catch (err) {
            alert(`Ошибка выполнения: ${err.message}`);
          }
        }
      };
    }

    const btnConfigureSacl = document.getElementById('wb-btn-configure-sacl');
    if (btnConfigureSacl) {
      btnConfigureSacl.onclick = async () => {
        const targetPath = prompt('Укажите путь к папке для настройки SACL аудита удаления:', 'c:\\Users\\onela\\AppData\\Local\\AI-Windows-Admin');
        if (targetPath) {
          try {
            const res = await fetch('/api/v1/system/file-audit/folder-sacl', {
              method: 'POST',
              headers: { 'Content-Type': 'application/json' },
              body: JSON.stringify({ path: targetPath, principal: 'Everyone', enable: true })
            });
            const data = await res.json();
            alert(data.Success ? `SACL успешно настроен для ${targetPath}!` : `Ошибка: ${data.Error || 'Сбой'}`);
            await loadFileDeletions();
          } catch (err) {
            alert(`Ошибка: ${err.message}`);
          }
        }
      };
    }

    const btnRefreshFolders = document.getElementById('wb-btn-refresh-user-folders');
    if (btnRefreshFolders) {
```

```
      btnRefreshFolders.onclick = () => loadUserFoldersOverview();
    }

    // Modal Relocate Target Drive Change
    const selectRelocateDrive = document.getElementById('wb-relocate-target-drive');
    if (selectRelocateDrive) {
      selectRelocateDrive.onchange = () => {
        const driveVal = selectRelocateDrive.value;
        const folderId = document.getElementById('wb-relocate-folder-id')?.value;
        const folder = (_userFoldersData?.folders || []).find(f => f.folder_id === folderId);
        const inputPath = document.getElementById('wb-relocate-target-path');
        if (driveVal && folder && inputPath) {
          const d = driveVal.replace('/', '\\').replace(/\\+$/, '');
          const parts = folder.current_path.split('\\');
          const folderSubname = parts[parts.length - 1] || folder.name;
          const userMatch = folder.current_path.match(/Users\\([^\\]+)/i);
          const username = userMatch ? userMatch[1] : 'User';
          inputPath.value = `${d}\\Users\\${username}\\${folderSubname}`;
        }
      };
    }

    // Modal Relocate Browse Folder Button
    const btnBrowseFolder = document.getElementById('wb-btn-browse-folder');
    if (btnBrowseFolder) {
      btnBrowseFolder.onclick = async () => {
        const currentPathVal = document.getElementById('wb-relocate-target-path')?.value || '';
```

```
        const currentDriveVal = document.getElementById('wb-relocate-target-drive')?.value || '';
        const initialPath = currentPathVal || (currentDriveVal ? currentDriveVal + '\\' : '');

        btnBrowseFolder.disabled = true;
        btnBrowseFolder.innerHTML = '<span class="spinner-border spinner-border-sm me-1"></span> Выбор...';

        try {
          const res = await fetch(`/api/v1/windows-backup/browse-folder?initial_path=${encodeURIComponent(initialPath)}`, {
            method: 'POST'
          });
          const data = await res.json();
          if (res.ok && data.selected_path) {
            const chosenPath = data.selected_path;
            const inputPath = document.getElementById('wb-relocate-target-path');
            if (inputPath) inputPath.value = chosenPath;

            // Синхронизируем выпадающий список дисков
            const driveLetter = chosenPath.match(/^[a-zA-Z]:/)?.[0]?.toUpperCase();
            if (driveLetter && selectRelocateDrive) {
              for (let opt of selectRelocateDrive.options) {
                if (opt.value && opt.value.toUpperCase().startsWith(driveLetter)) {
                  selectRelocateDrive.value = opt.value;
                  break;
                }
              }
            }
          }
        } catch (err) {
          console.error('[WindowsBackup] Browse folder error:', err);
        } finally {
```

```
          btnBrowseFolder.disabled = false;
          btnBrowseFolder.innerHTML = '<i class="bi bi-folder2
-open"></i> <span>Обзор...</span>';
        }
      };
    }

    // Modal Relocate Execution
    const btnExecRelocate = document.getElementById('wb-btn-ex
ecute-relocate');
    if (btnExecRelocate) {
      btnExecRelocate.onclick = async () => {
        const folderId = document.getElementById('wb-relocate-
folder-id')?.value;
        const targetDrive = document.getElementById('wb-reloca
te-target-drive')?.value;
        const targetPath = document.getElementById('wb-relocat
e-target-path')?.value?.trim();
        const deleteSource = document.getElementById('wb-reloc
ate-delete-source')?.checked || false;

        if (!folderId || (!targetDrive && !targetPath)) {
          alert('Пожалуйста, выберите целевой диск или укажите
 целевую папку для переноса.');
          return;
        }

        const destDisplay = targetPath || targetDrive;
        if (!confirm(`Подтвердите перенос папки в "${destDispl
ay}".\nВсе файлы будут скопированы, а системные пути и библиот
еки перенастроены.`)) {
          return;
        }

        btnExecRelocate.disabled = true;
        btnExecRelocate.innerHTML = '<span class="spinner-bord
er spinner-border-sm me-1"></span> Перенос...';

```

```
        try {
          const res = await fetch('/api/v1/windows-backup/user-folders/relocate', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({
              folder_id: folderId,
              target_drive_letter: targetDrive || null,
              target_path: targetPath || null,
              delete_source_after: deleteSource
            })
          });

          const data = await res.json();
          if (res.ok && data.success) {
            alert(data.message || 'Перенос успешно завершен!');
            const modalEl = document.getElementById('wb-relocate-modal');
            if (modalEl && window.bootstrap?.Modal) {
              const modalInstance = window.bootstrap.Modal.getInstance(modalEl);
              if (modalInstance) modalInstance.hide();
            }
            await loadBackupData();
          } else {
            alert('Ошибка переноса: ' + (data.detail || data.message || JSON.stringify(data)));
          }
        } catch (e) {
          alert('Сетевая ошибка при переносе: ' + e.message);
        } finally {
          btnExecRelocate.disabled = false;
          btnExecRelocate.innerHTML = '<i class="bi bi-check-circle"></i> <span>Выполнить перенос</span>';
        }
      };
    }
```

```
  }

  async function loadBackupData() {
    await Promise.allSettled([
      loadHealthReport(),
      loadLibraries(),
      loadFileHistoryStatus(),
      loadUserFoldersOverview(),
      loadVssSnapshots(),
      loadFileDeletions()
    ]);
  }

  async function loadHealthReport() {
    try {
      const res = await fetch('/api/v1/windows-backup/health')
;
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();

      const elHealth = document.getElementById('wb-val-health'
);
      const elBadge = document.getElementById('wb-badge-status
');
      const elSub = document.getElementById('wb-sub-health');

      const score = data.health_score ?? 0;
      if (elHealth) elHealth.textContent = `${score} / 100`;
      if (elBadge) {
        elBadge.textContent = score >= 80 ? 'Готов' : score >=
 50 ? 'Внимание' : 'Критично';
        elBadge.className = `badge bg-${score >= 80 ? 'success
' : score >= 50 ? 'warning' : 'danger'}`;
      }
      if (elSub) {
        elSub.textContent = data.service_running ? 'Служба акт
ивна' : 'Служба остановлена';
      }
```

```
481
482       const recList = document.getElementById('wb-recommendati
    ons-list');
483       if (recList && Array.isArray(data.recommendations)) {
484         if (data.recommendations.length === 0) {
485           recList.innerHTML = '<li class="text-success small">
    <i class="bi bi-check-circle me-1"></i>Все проверки безопаснос
    ти данных пройдены успешно.</li>';
486         } else {
487           recList.innerHTML = data.recommendations.map(r => `
488             <li class="mb-1.5 text-warning d-flex align-items-
    start gap-1">
489               <i class="bi bi-exclamation-triangle-fill text-w
    arning flex-shrink-0 mt-0.5"></i>
490               <span>${escapeHtml(r)}</span>
491             </li>
492           `).join('');
493         }
494       }
495     } catch (e) {
496       console.warn('[WindowsBackup] Error loading health:', e)
    ;
497     }
498   }
499
500   async function loadLibraries() {
501     const tbody = document.getElementById('wb-libraries-body')
    ;
502     const countBadge = document.getElementById('wb-lib-count')
    ;
503     const valLibs = document.getElementById('wb-val-libraries'
    );
504
505     try {
506       const res = await fetch('/api/v1/windows-backup/librarie
    s');
507       if (!res.ok) throw new Error(`HTTP ${res.status}`);
508       const libs = await res.json();
```

```
509       _librariesData = libs || [];
510
511       if (countBadge) countBadge.textContent = `${libs.length} шт.`;
512       if (valLibs) valLibs.textContent = `${libs.length}`;
513
514       if (!tbody) return;
515       if (!libs || libs.length === 0) {
516         tbody.innerHTML = '<tr><td colspan="4" class="text-center py-3 text-muted">Библиотеки не найдены</td></tr>';
517         return;
518       }
519
520       tbody.innerHTML = libs.map(lib => {
521         const foldersHtml = (lib.folders || []).map(f => {
522           const path = typeof f === 'string' ? f : f.folder_path;
523           const exists = typeof f === 'object' ? f.exists : true;
524           return `
525             <div class="text-break font-monospace small d-flex align-items-center gap-1 ${exists ? 'text-white' : 'text-danger'}">
526               <i class="bi ${exists ? 'bi-folder-check text-info' : 'bi-folder-x text-danger'}"></i>
527               <span>${escapeHtml(path)}</span>
528               ${!exists ? '<span class="badge bg-danger-subtle text-danger py-0 px-1" style="font-size:0.6rem;">Недоступна</span>' : ''}
529             </div>
530           `;
531         }).join('') || '<span class="text-muted small">Нет папок</span>';
532
533         return `
534           <tr>
535             <td class="fw-bold text-white">
536               <i class="bi bi-collection me-1 text-primary"></
```

```
i>${escapeHtml(lib.name)}
            </td>
            <td>${foldersHtml}</td>
            <td class="text-break font-monospace small text-in
fo">
              ${lib.default_save_folder ? escapeHtml(lib.defau
lt_save_folder) : '<span class="text-muted">—</span>'}
            </td>
            <td>
              <span class="badge ${lib.is_pinned ? 'bg-primary
' : 'bg-secondary'}">${lib.is_pinned ? 'Да' : 'Нет'}</span>
            </td>
          </tr>
        `;
      }).join('');
    } catch (e) {
      if (tbody) tbody.innerHTML = `<tr><td colspan="4" class=
"text-center py-3 text-danger">Ошибка: ${escapeHtml(e.message)
}</td></tr>`;
    }
  }

  async function loadFileHistoryStatus() {
    try {
      const res = await fetch('/api/v1/windows-backup/file-his
tory/status');
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const data = await res.json();

      const elService = document.getElementById('wb-val-servic
e');
      const elSubService = document.getElementById('wb-sub-ser
vice');
      const elTarget = document.getElementById('wb-fh-target')
;
      const elConfig = document.getElementById('wb-fh-config')
;
      const elInterval = document.getElementById('wb-fh-interv
```

```
al');
564       const elRetention = document.getElementById('wb-fh-reten
tion');
565       const elLast = document.getElementById('wb-fh-last-backu
p');
566
567       const isRunning = data.is_service_running;
568       if (elService) {
569         elService.textContent = isRunning ? 'Работает' : 'Оста
новлена';
570         elService.className = `wb-card-value ${isRunning ? 'te
xt-success' : 'text-danger'}`;
571       }
572       if (elSubService) {
573         elSubService.textContent = `Запуск: ${data.service_sta
rt_type || 'Manual'}`;
574       }
575
576       const cfg = data.config;
577       if (cfg) {
578         if (elTarget) elTarget.textContent = cfg.target_path |
| cfg.target_url || 'Не настроено';
579         if (elConfig) elConfig.textContent = cfg.config_file_p
ath || 'Стандартный профиль';
580         if (elInterval) elInterval.textContent = cfg.backup_in
terval_minutes ? `${cfg.backup_interval_minutes} мин.` : 'По у
молчанию (1 ч)';
581         if (elRetention) elRetention.textContent = cfg.retenti
on_policy || 'Всегда';
582         if (elLast) elLast.textContent = cfg.last_backup_time
|| 'Нет записей';
583       } else {
584         if (elTarget) elTarget.textContent = 'Конфигурация отс
утствует';
585         if (elConfig) elConfig.textContent = '—';
586         if (elInterval) elInterval.textContent = '—';
587         if (elRetention) elRetention.textContent = '—';
588         if (elLast) elLast.textContent = '—';
```

```
      }
    } catch (e) {
      console.warn('[WindowsBackup] Error loading File History
 status:', e);
    }
  }

  async function loadVssSnapshots() {
    const tbody = document.getElementById('wb-vss-body');
    const countBadge = document.getElementById('wb-vss-count')
;

    try {
      const res = await fetch('/api/v1/windows-backup/vss/snap
shots');
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      const snapshots = await res.json();

      if (countBadge) countBadge.textContent = `${snapshots.le
ngth} шт.`;

      if (!tbody) return;
      if (!snapshots || snapshots.length === 0) {
        tbody.innerHTML = '<tr><td colspan="4" class="text-cen
ter py-3 text-muted">Теневые копии (VSS) не найдены</td></tr>'
;
        return;
      }

      tbody.innerHTML = snapshots.map(s => `
        <tr>
          <td class="font-monospace small text-truncate" style
="max-width: 150px;" title="${escapeHtml(s.snapshot_id || '')}
">
            ${escapeHtml(s.snapshot_id || '—')}
          </td>
          <td class="fw-semibold text-info">${escapeHtml(s.ori
ginal_volume || '—')}</td>
```

```
          <td class="small text-muted">${escapeHtml(s.creation_time || '—')}</td>
          <td><span class="badge bg-success">${escapeHtml(s.status || 'OK')}</span></td>
        </tr>
      `).join('');
    } catch (e) {
      if (tbody) tbody.innerHTML = `<tr><td colspan="4" class="text-center py-3 text-danger">Ошибка: ${escapeHtml(e.message)}</td></tr>`;
    }
  }

  async function loadFileDeletions() {
    try {
      const res = await fetch('/api/v1/system/file-audit/deletions?hours=24&limit=100&deletions_only=true');
      if (!res.ok) return;
      const data = await res.json();
      const events = data.events || [];

      const tbody = document.getElementById('wb-deletions-tbody');
      const badge = document.getElementById('wb-deletions-badge');
      const valDeletions = document.getElementById('wb-val-deletions');

      if (badge) badge.innerText = `${events.length} удалений`;
      if (valDeletions) valDeletions.innerText = events.length;

      if (tbody) {
        if (events.length === 0) {
          tbody.innerHTML = '<tr><td colspan="5" class="text-center text-muted p-3">События удаления (4663/4660) не зафиксированы или аудит не включен</td></tr>';
```

```
          return;
        }
        tbody.innerHTML = events.slice(0, 30).map((e, idx) =>
`
          <tr class="wb-deletion-row" data-idx="${idx}" style=
"cursor: pointer;" title="Нажмите для детального анализа удале
ния">
            <td class="font-monospace text-muted small">${e.ti
mestamp?.slice(11, 19) || ''}</td>
            <td class="font-monospace fw-semibold ${e.event_id
 === 4660 ? 'text-danger' : 'text-warning'}">${e.event_id}</td
>
            <td class="font-monospace text-truncate text-white
" style="max-width: 260px;" title="${escapeHtml(e.object_name)
}">${escapeHtml(e.object_name.split('\\').pop() || e.object_na
me)}</td>
            <td class="small text-info text-truncate" style="m
ax-width: 140px;" title="${escapeHtml(e.process_name)}">${esca
peHtml(e.process_name ? e.process_name.split('\\').pop() : 'N/
A')}</td>
            <td class="small text-muted">${escapeHtml(e.subjec
t_user_name || 'SYSTEM')}</td>
          </tr>
        `).join('');

        tbody.querySelectorAll('.wb-deletion-row').forEach(row
 => {
          row.onclick = () => {
            const idx = parseInt(row.getAttribute('data-idx'),
 10);
            const e = events[idx];
            if (!e) return;

            if (window.AITableModal) {
              window.AITableModal.show({
                icon: '',
                title: `Удаление: ${e.object_name.split('\\').
pop() || e.object_name}`,
```

```
                subtitle: `Event ID: ${e.event_id} | ${e.timestamp}`,
                tableType: 'security_event',
                badges: [
                  { text: `Event ${e.event_id}`, class: e.event_id === 4660 ? 'badge bg-danger' : 'badge bg-warning text-dark' },
                  { text: e.status || 'Success', class: 'badge bg-success' }
                ],
                metadata: [
                  { label: 'Целевой объект', value: e.object_name },
                  { label: 'Тип объекта', value: e.object_type || 'File' },
                  { label: 'Процесс-инициатор', value: e.process_name || 'Не указан' },
                  { label: 'PID процесса', value: e.process_id ? String(e.process_id) : 'N/A' },
                  { label: 'Пользователь', value: `${e.subject_domain_name}\\${e.subject_user_name}` },
                  { label: 'Дескриптор HandleId', value: e.handle_id || 'N/A' },
                  { label: 'Маска доступа', value: e.access_mask || 'DELETE (0x10000)' }
                ],
                rawTitle: 'Полное сообщение Security Log',
                rawContent: e.raw_message || JSON.stringify(e, null, 2),
                requestData: e
              });
            }
          };
        });
      }
    } catch (e) {
      console.error('[WindowsBackup] Failed to fetch deletion events:', e);
```

```
691     }
692   }
693
694   async function loadUserFoldersOverview() {
695     const tbody = document.getElementById('wb-user-folders-tbo
dy');
696     const totalSizeBadge = document.getElementById('wb-total-u
ser-size');
697     const drivesList = document.getElementById('wb-drives-list
');
698
699     try {
700       const res = await fetch('/api/v1/windows-backup/user-fol
ders/overview');
701       if (!res.ok) throw new Error(`HTTP ${res.status}`);
702       const data = await res.json();
703       _userFoldersData = data;
704
705       if (totalSizeBadge) {
706         totalSizeBadge.textContent = `Общий объем: ${data.tota
l_user_size_gb} ГБ (${data.total_user_size_mb} МБ)`;
707       }
708
709       if (drivesList) {
710         if (!data.drives || data.drives.length === 0) {
711           drivesList.innerHTML = '<span class="text-muted">Дис
ки не обнаружены</span>';
712         } else {
713           drivesList.innerHTML = data.drives.map(d => `
714             <div class="px-2 py-1 rounded border ${d.is_system
_drive ? 'border-info bg-dark' : 'border-secondary bg-dark'} d
-flex align-items-center gap-1.5">
715               <i class="bi ${d.is_system_drive ? 'bi-hdd-fill
text-info' : 'bi-hdd text-success'}"></i>
716               <span class="fw-bold">${escapeHtml(d.drive_lette
r)}</span>
717               <span class="text-muted">(${escapeHtml(d.fstype)
})</span>:
```

```
              <span class="text-success fw-semibold">${d.free_space_gb} ГБ своб.</span>
              <span class="text-muted small">из ${d.total_space_gb} ГБ</span>
              ${d.is_system_drive ? '<span class="badge bg-info-subtle text-info py-0 px-1" style="font-size: 0.65rem;">System</span>' : ''}
            </div>
          `).join('');
        }
      }

      if (!tbody) return;
      if (!data.folders || data.folders.length === 0) {
        tbody.innerHTML = '<tr><td colspan="6" class="text-center py-3 text-muted">Пользовательские папки не найдены</td></tr>';
        return;
      }

      tbody.innerHTML = data.folders.map(f => {
        const suitableDrives = (data.available_target_drives || []).filter(d => {
          const folderDrive = f.drive_letter.replace('\\', '');
          const targetDrive = d.drive_letter.replace('\\', '');
          return folderDrive !== targetDrive && d.free_space_gb >= (f.size_gb + 1.0);
        });

        const canRelocate = suitableDrives.length > 0;

        return `
          <tr>
            <td class="fw-bold text-white">
              <i class="bi bi-folder2-open me-1 text-primary"></i>${escapeHtml(f.name)}
```

```
            </td>
            <td class="font-monospace text-muted small text-truncate" style="max-width: 250px;" title="${escapeHtml(f.current_path)}">
              ${escapeHtml(f.current_path)}
            </td>
            <td>
              <span class="badge ${f.drive_letter.startsWith('C') ? 'bg-secondary' : 'bg-primary'}">${escapeHtml(f.drive_letter)}</span>
            </td>
            <td class="fw-semibold font-monospace ${f.size_gb > 1.0 ? 'text-warning' : 'text-info'}">
              ${f.size_gb >= 0.01 ? `${f.size_gb} ГБ` : `${f.size_mb} МБ`}
            </td>
            <td class="small text-muted font-monospace">${f.file_count}</td>
            <td class="text-end">
              <button class="btn btn-xs ${canRelocate ? 'btn-outline-warning' : 'btn-outline-secondary'} py-0 px-2 wb-btn-relocate-modal"
                      data-folder-id="${escapeHtml(f.folder_id)}"
                      ${!canRelocate ? 'title="Нет подходящих дисков с достаточным местом"' : 'title="Перенести на другой диск"'}
                      style="font-size: 0.72rem;">
                <i class="bi bi-box-arrow-right me-1"></i>Перенести
              </button>
            </td>
          </tr>
        `;
      }).join('');

      tbody.querySelectorAll('.wb-btn-relocate-modal').forEach(btn => {
```

```
        btn.onclick = () => {
          const fid = btn.getAttribute('data-folder-id');
          openRelocateModal(fid);
        };
      });

    } catch (e) {
      if (tbody) tbody.innerHTML = `<tr><td colspan="6" class="text-center py-3 text-danger">Ошибка загрузки: ${escapeHtml(e.message)}</td></tr>`;
    }
  }

  function openRelocateModal(folderId) {
    if (!_userFoldersData) return;
    const folder = (_userFoldersData.folders || []).find(f => f.folder_id === folderId);
    if (!folder) return;

    const elId = document.getElementById('wb-relocate-folder-id');
    const elName = document.getElementById('wb-relocate-folder-name');
    const elPath = document.getElementById('wb-relocate-folder-path');
    const elSize = document.getElementById('wb-relocate-folder-size');
    const selectDrive = document.getElementById('wb-relocate-target-drive');
    const elHint = document.getElementById('wb-relocate-drive-hint');

    if (elId) elId.value = folder.folder_id;
    if (elName) elName.textContent = folder.name;
    if (elPath) elPath.textContent = folder.current_path;
    if (elSize) elSize.textContent = `${folder.size_gb >= 0.01 ? `${folder.size_gb} ГБ` : `${folder.size_mb} МБ`} (${folder.file_count} файлов)`;
```

```
796
797     const inputPath = document.getElementById('wb-relocate-tar
    get-path');
798     if (inputPath) inputPath.value = '';
799
800     if (selectDrive) {
801       selectDrive.innerHTML = '<option value="">-- Выберите ди
    ск --</option>';
802       const folderDrive = folder.drive_letter.replace('\\', ''
    ).toUpperCase();
803       let firstSuitableDrive = null;
804
805       (_userFoldersData.drives || []).forEach(d => {
806         const dLetter = d.drive_letter.replace('\\', '').toUpp
    erCase();
807         if (dLetter === folderDrive) return;
808
809         const isEnough = d.free_space_gb >= (folder.size_gb +
    1.0);
810         const opt = document.createElement('option');
811         opt.value = d.drive_letter;
812         opt.textContent = `${d.drive_letter} (${d.fstype}) — С
    вободно: ${d.free_space_gb} ГБ из ${d.total_space_gb} ГБ ${isE
    nough ? '  Достаточно места' : '  Мало места'}`;
813         if (!isEnough) {
814           opt.disabled = true;
815         } else if (!firstSuitableDrive) {
816           firstSuitableDrive = d.drive_letter;
817         }
818         selectDrive.appendChild(opt);
819       });
820
821       if (firstSuitableDrive) {
822         selectDrive.value = firstSuitableDrive;
823         if (inputPath) {
824           const d = firstSuitableDrive.replace('/', '\\').repl
    ace(/\\+$/, '');
825           const parts = folder.current_path.split('\\');
```

```
          const folderSubname = parts[parts.length - 1] || fol
der.name;
          const userMatch = folder.current_path.match(/Users\\
([^\\]+)/i);
          const username = userMatch ? userMatch[1] : 'User';
          inputPath.value = `${d}\\Users\\${username}\\${folde
rSubname}`;
        }
      }
    }

    if (elHint) {
      elHint.textContent = 'Требуется свободного места: миниму
м ' + (folder.size_gb + 1.0).toFixed(2) + ' ГБ (размер + 1 ГБ
резерва).';
    }

    const modalEl = document.getElementById('wb-relocate-modal
');
    if (modalEl) {
      if (window.bootstrap?.Modal) {
        new window.bootstrap.Modal(modalEl).show();
      } else {
        modalEl.classList.add('show');
        modalEl.style.display = 'block';
      }
    }
  }

  function escapeHtml(str) {
    if (str === null || str === undefined) return '';
    return String(str)
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#039;');
  }
```

```
858 })();
859
```

index.html

```
<!--
================================================================
===============
Process Name: Windows Web Interface - Index View
================================================================
===============
Description:
  Шаблон клиентского веб-интерфейса index.

Usage Examples:
    Web Route:
      /windows/api/webgui/windows_users_tab/index.html

    Tab Frame Integration:
      <iframe src="/windows/api/webgui/windows_users_tab/index
.html" class="w-100 h-100 border-0"></iframe>

File: index.html
Project: ai-windows-admin
Package: windows/api/webgui/windows_users_tab
Author: hypo69
Copyright: © 2026 hypo69
Updated: 2026-10-10 17:35:00
================================================================
===============
-->
<!DOCTYPE html>

<html lang="ru">
<head>
<meta charset="utf-8"/>
<meta content="width=device-width, initial-scale=1.0" name="vi
ewport"/>
<title>Пользователи Windows</title>
</head>
<body>
```

```
<div class="container-fluid p-3">
<h4><i class="bi bi-people-fill text-primary"></i> Пользователи Windows</h4>
<p class="text-muted">Управление локальными учетными записями операционной системы Windows.</p>
<div class="card border-secondary mb-3" data-panel-id="panel-windows-users-1" id="panel-windows-users-1">
<div class="card-header bg-dark text-white">
<i class="bi bi-list-ul"></i> Список пользователей
      </div>
<div class="card-body">
<div id="windows-users-list">
<p class="text-muted">Загрузка...</p>
</div>
</div>
</div>
<div class="card border-secondary" data-panel-id="panel-windows-users-2" id="panel-windows-users-2">
<div class="card-header bg-dark text-white">
<i class="bi bi-people-fill"></i> Управление группами
      </div>
<div class="card-body">
<div id="windows-groups-list">
<p class="text-muted">Загрузка...</p>
</div>
</div>
</div>
</div>
<script type="module">
    import { initWindowsUsersTab } from '/html/windows_users_tab/main.js?v=20260920_v1';
    window.initWindowsUsersTab = initWindowsUsersTab;
  </script>
</body>
</html>
```

main.js

```
/**
 * ============================================================
==================
 * Process Name: Windows Windows Users Tab - Main Script
 * ============================================================
==================
 * Description:
 *   Клиентский скрипт управления интерфейсом модуля main.
 *
 * Usage Examples:
 *   HTML Integration:
 *     <script src="/windows/api/webgui/windows_users_tab/main
.js?v=20261001_v1" type="module"></script>
 *
 * File: main.js
 * Project: ai-windows-admin
 * Package: windows/api/webgui/windows_users_tab
 * Author: hypo69
 * Copyright: © 2026 hypo69
 * Updated: 2026-10-10 17:35:00
 * ============================================================
==================
 */

// Windows Users Tab Module
// ============================================================
==================

export async function initWindowsUsersTab() {
  console.log('[WindowsUsersTab] Initializing...');

  try {
    // Load Windows users
    const usersContainer = document.getElementById('windows-us
ers-list');
    const groupsContainer = document.getElementById('windows-g
```

```
roups-list');

    if (usersContainer) {
      usersContainer.innerHTML = '<div class="spinner-border t
ext-primary" role="status"><span class="visually-hidden">Loadi
ng...</span></div>';
    }

    if (groupsContainer) {
      groupsContainer.innerHTML = '<div class="spinner-border
text-primary" role="status"><span class="visually-hidden">Load
ing...</span></div>';
    }

    // Fetch Windows users
    try {
      const response = await fetch('/api/windows-users');
      if (response.ok) {
        const data = await response.json();
        if (data.status === 'ok' && data.users) {
          if (usersContainer) {
            usersContainer.innerHTML = `
              <div class="table-responsive">
                <table class="table table-dark table-sm table-
bordered">
                  <thead>
                    <tr>
                      <th>Имя</th>
                      <th>Enabled</th>
                      <th>Описание</th>
                      <th>SID</th>
                    </tr>
                  </thead>
                  <tbody>
                    ${data.users.map(u => `
                      <tr>
                        <td><strong>${escapeHtml(u.Name || '')
}</strong></td>
```

```
                        <td>${u.Enabled ? '<span class="badge bg-success">Да</span>' : '<span class="badge bg-danger">Нет</span>'}</td>
                        <td>${escapeHtml(u.Description || '')}</td>
                        <td class="text-muted small">${escapeHtml(u.SID || '')}</td>
                      </tr>
                    `).join('')}
                  </tbody>
                </table>
              </div>
            `;
          }
        }
      } catch (err) {
        console.error('[WindowsUsersTab] Error loading users:', err);
        if (usersContainer) {
          usersContainer.innerHTML = `<div class="alert alert-danger">Ошибка загрузки: ${escapeHtml(err.message)}</div>`;
        }
      }

      // Fetch Windows groups
      try {
        const response = await fetch('/api/windows-users/groups');
        if (response.ok) {
          const data = await response.json();
          if (data.status === 'ok' && data.groups) {
            if (groupsContainer) {
              groupsContainer.innerHTML = `
                <div class="table-responsive">
                  <table class="table table-dark table-sm table-bordered">
                    <thead>
```

```
                    <tr>
                      <th>Группа</th>
                      <th>SID</th>
                      <th>Описание</th>
                    </tr>
                  </thead>
                  <tbody>
                    ${data.groups.map(g => `
                      <tr>
                        <td><strong>${escapeHtml(g.Name || '')}</strong></td>
                        <td class="text-muted small">${escapeHtml(g.SID || '')}</td>
                        <td>${escapeHtml(g.Description || '')}</td>
                      </tr>
                    `).join('')}
                  </tbody>
                </table>
              </div>
            `;
          }
        }
      }
    } catch (err) {
      console.error('[WindowsUsersTab] Error loading groups:', err);
      if (groupsContainer) {
        groupsContainer.innerHTML = `<div class="alert alert-danger">Ошибка загрузки групп: ${escapeHtml(err.message)}</div>`;
      }
    }

  } catch (err) {
    console.error('[WindowsUsersTab] Initialization error:', err);
  }
```

```
}

function escapeHtml(text) {
  if (!text) return '';
  return text
    .replace(/&/g, '&amp;')
    .replace(/</g, '&lt;')
    .replace(/>/g, '&gt;')
    .replace(/"/g, '&quot;')
    .replace(/'/g, '&#039;');
}
```